Polyfold Theory, proof by proof

The formal proof of Hofer–Wysocki–Zehnder's Polyfold and Fredholm Theory, Parts I–III, Chapters 1–16, in words

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Part I. The sc-calculus (HWZ Chapter 1, §§1.1–1.4)

Chapter 1 of Part I fixes the vocabulary of the whole theory: sc-Banach spaces, sc-operators, sc-subspaces and quotients, sc-Fredholm operators and their stability under sc^+-perturbations, partial quadrants, sc^0 and sc^1 maps, the tangent construction and the inductive definitions of sc^k and sc-smooth, the relation between sc-differentiability and classical differentiability on the levels (Propositions 1.2.1–1.2.4 (preprint 1.16–1.19) and Corollary 1.2.5 (preprint 1.20), quoted from the 2010 paper), the chain rule, the degeneracy index of a point of a partial quadrant and boundary recognition. The formal proof establishes all of it, Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) under the text's own hypothesis (C^{l+1} from U_{m+l} to F_m for $l \leq k$), by the 2010 paper's induction (S_k) on elementary maps (§I.17). So are the three items the text states without proof or example: Example 1.1.3 (preprint 1.2), the weighted Sobolev spaces (§I.22), the remark after Definition 1.1.15 (preprint 1.14) that Df need not be norm continuous (§I.24), and Remark 1.1.17 (preprint 1.15) on partial cones (§I.25).

What is proved, and what is not. Every numbered statement of Chapter 1 is proved, and so are the three items the first revision of this part left open. The three gaps of the chapter are all closed. G.1, Example 1.1.3 (preprint 1.2), is proved with exactly the printed hypotheses, $\delta_0 = 0$ and (δ_m) strictly increasing, and level 0 is $L^2(\mathbb{R} \times S^1)$ (§I.22). G.2, Remark 1.1.17 (preprint 1.15), holds for closed convex partial cones with nonempty interior: relatively open subsets of such a cone are sc-domains, so §§1.2–1.3 apply to them, and the tangent cone is again a partial cone (§I.25). G.3, the remark after Definition 1.1.15 (preprint 1.14) that $x \mapsto Df(x)$ need not be continuous for the operator norm, is proved by a sequence-space example, since the text gives none (§I.24). No row of the chapter is narrower than the text, and no hypothesis is added to a numbered statement. The formalization goes beyond the text in two places: the index on every level of an sc-Fredholm operator (§§I.6, I.9), and the converse of Proposition 1.1.10 (preprint 1.9), which makes "Fredholm on level 0 and regularizing" a criterion for sc-Fredholm operators (§I.23). What is not done lies outside the statements of the text. The shift map of the 2010 paper, which the exposition's §6 uses on the weighted Sobolev spaces, is not built. And the sequence-space example of §I.24 is proved sc^1, which is all the remark needs, not sc-smooth.

Two decisions shape everything. First, an sc-Banach space is an *indexed family* of Banach spaces E_m with injective compact inclusions $E_{m+1} \to E_m$, not one space with a chain of subspaces (§I.1). "The point x of E_0 lies in E_m" then means that x is in the image of E_m, and every map has level maps $f_m : E_m \to F_m$ determined by injectivity. Second, the text's standing hypothesis "U relatively open in a partial quadrant" enters the proofs of §§1.2–1.3 only through two facts about each level set U_m: it is the intersection of an open and a convex set, and derivatives within it are unique. These two facts make up the notion of an *sc-domain* (§I.10), which the tangent construction and the level shift preserve by a two-line argument, so

the inductions of §1.2 run on sc-domains, and the text's tuples (U, C, E) are a special case.

Four proofs take a different road from the text's. The chain rule replaces the two integrals of §1.4.2 by the one-dimensional mean value inequality along a segment, and does not need the first integral at all (§I.12). The compactness argument behind Propositions 1.2.1–1.2.3 (preprint 1.16–1.18), the subsequence $h_n \to h$ in E_m for unit vectors $h_n \in E_{m+1}$, is isolated once as the *compact-open lemma*, and its C^k version replaces the multilinear maps $\Gamma(x)$ and the differentiation under the integral sign of the 2010 paper (§I.13). Theorem 1.3.5 (preprint 1.25), which the text quotes from another paper, is proved by the argument of §2.4 (preprint §2.3), for sc^1 germs (§I.19). And the induction (S_k) behind Proposition 1.2.4 (preprint 1.19) runs on the order s over all elementary maps $d^j f(\pi_0 z)(\pi_1 z, \dots, \pi_j z)$ at once, rather than on a description of the 2^{k+1} components of $T^{k+1} f$ (§I.17). Smaller departures are flagged where they occur: the complement in Proposition 1.1.11 (preprint 1.10) is built from a right inverse rather than a lifted basis; the regularizing step of §1.4.1 is a bootstrap; $k = k'$ in Lemma 1.3.4 (preprint 1.24) is derived from the lemma, not proved first. What Mathlib lacked, the Riesz–Schauder theorem, the stability of Fredholm operators under compact perturbations and the invariance of the Fredholm index, is proved in §I.7; the index statements of the exposition's Lemma 2.D and Proposition 1.1.11 (preprint 1.10) follow in §I.6 and §I.9. The lemmas that the exposition adds to make the text's remarks precise are proved where they belong: uniqueness of the sc-structure in finite dimensions (§I.1), sc-splittings as sc-projections (§I.3), the smooth points of a quotient (§I.5), the shift and tangent of a tuple (§I.10), compact-open continuity and $\mathrm{sc}^1 = C^1$ on quadrants in finite dimensions (§I.11).

Map.

- `Chapter1/ScBanach`: scales and sc-Banach spaces, the constant structure, sums and shifts; sc-operators, sc-isomorphisms, sc^+-operators; sc-subspaces, sc-splittings, Proposition 1.1.7 (preprint 1.6); quotients, Proposition 1.1.8 (preprint 1.7).
- `Chapter1/FredholmPerturbation`: Fredholm operators between Banach spaces; Riesz–Schauder; compact perturbations.
- `Chapter1/FredholmIndex`: the index of a linear map, its additivity, its invariance under small and compact perturbations; the index on the levels of an sc-Fredholm operator; the index in Proposition 1.1.11 (preprint 1.10).
- `Chapter1/ExpositionLemmas`: the exposition's Lemmas 1.A(2)–(3), 2.B, 2.C, 3.F, and $\mathrm{sc}^1 = C^1$ in finite dimensions on quadrants.
- `Chapter1/ScFredholm`: Definition 1.1.9 (preprint 1.8), the index, the level-wise Fredholm property; Propositions 1.1.10 and 1.1.11 (preprint 1.9 and 1.10).
- `Chapter1/ScSmooth`: level sets, the tangent TE and TU, level maps, sc^0; sc^1 with its derivative family, uniqueness of Df, sc^k and sc-smooth; partial quadrants, tuples, sc-domains.
- `Chapter1/ChainRule`: Theorem 1.3.1 (preprint 1.21) and its iterates.
- `Chapter1/Properties`: the compact-open continuity of Df; the base cases of Propositions 1.2.1, 1.2.4 (preprint 1.16, 1.19) and Corollary 1.2.5 (preprint 1.20).
- `Chapter1/Levels`: the compact-open lemma; the level operators $Df(x) : E_m \to F_m$; Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) on all levels; Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) under the stronger hypothesis "C^{k+1} on every level".
- `Chapter1/LevelsWeak`: Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) as stated: the derivative tower, elementary maps, their sc^1 step and tangent maps, the sum rule for sc^k maps, the induction (S_k).
- `Chapter1/Degeneracy`: Definition 1.3.3 (preprint 1.23) and Lemma 1.3.4 (preprint 1.24).
- `Chapter2/DegeneracyIndex`: Theorem 1.3.5 (preprint 1.25), by the argument of §2.4 (preprint §2.3).

- `Chapter1/ScFredholmCriterion`: regularizing operators; the converse of Proposition 1.1.10 (preprint 1.9); sc-operators from level-preserving bounded maps; restriction to sc-subspaces.
- `Chapter1/ScOneNotNormContinuous`: the sequence sc-Banach space X and an sc^1 map whose derivative is not norm continuous.
- `Chapter1/CylinderWeakDerivatives–CylinderScBanach`: Example 1.1.3 (preprint 1.2): weak derivatives on the cylinder, the spaces $H^{m,\delta}$, Fourier series on a box, Rellich's theorem, the compact inclusion, density of test functions.
- `Chapter1/PartialCones`: Remark 1.1.17 (preprint 1.15).

I.1 Scales and sc-Banach spaces (Definition 1.1.1 (preprint 1.1))

The scale. A **scale of Banach spaces** consists of real Banach spaces E_m, one for each $m \in \mathbb{N}$, together with continuous linear maps $\mathrm{incl}_m : E_{m+1} \to E_m$ that are injective and compact operators `Chapter1/ScBanach · ScScale`. Compactness is Mathlib's: the preimage of some neighbourhood of 0 is mapped into a compact set `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator`. The composite inclusion $\iota_m : E_m \to E_0$ is defined by recursion, $\iota_0 = \mathrm{id}$ and $\iota_{m+1} = \iota_m \circ \mathrm{incl}_m$ `Chapter1/ScBanach · ScScale.toBase`; it is injective `Chapter1/ScBanach · ScScale.toBase_injective`, and the composite $E_{m+k} \to E_m$ is defined likewise `Chapter1/ScBanach · ScScale.inclK` and is compatible with ι `Chapter1/ScBanach · ScScale.toBase_inclK`. The text's E_m is here the image $\iota_m(E_m) \subseteq E_0$; the text's $E_m \supset E_{m+1}$ is the inclusion of images, and "$x \in E_m$" for $x \in E_0$ is "$x \in \mathrm{range}\,\iota_m$". The **smooth points** E_∞ form the subspace of E_0 of points that lift to every level, $E_\infty = \bigcap_m \mathrm{range}\,\iota_m$ `Chapter1/ScBanach · ScScale.smooth` `Chapter1/ScBanach · ScScale.mem_smooth_iff`, and $E_\infty \cap E_m$ is its preimage in E_m `Chapter1/ScBanach · ScScale.smoothLvl`; a smooth point of E_m lifts to every E_{m+k} `Chapter1/ScBanach · ScScale.exists_inclK_of_mem_smoothLvl`.

The density axiom. An **sc-Banach space** is a scale in which $E_\infty \cap E_m$ is dense in E_m for every m `Chapter1/ScBanach · ScBanach`. This is condition (2) of Definition 1.1.1 (preprint 1.1) exactly as the text states it. The weaker statement that E_{m+k} has dense image in E_m follows `Chapter1/ScBanach · ScBanach.denseRange_inclK` `Chapter1/ScBanach · ScBanach.denseRange_incl`; it does not obviously imply the axiom, which is why the strong form is the definition.

Why a family. Nothing in Chapter 1 uses more than the family: the text's identification of E_{m+1} with a subset of E_m is the injectivity of incl_m, a field of the structure, and its "the inclusion is continuous" is part of compactness. The gain is that each level is a Banach space in its own right, with its own norm, and Mathlib's analysis applies on each level without carrying a filtration along.

The constant structure. For a finite-dimensional normed space V the family $E_m = V$ with identity inclusions is a scale: the identity of a finite-dimensional space is a compact operator `Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · isCompactOperator_id_iff_finiteDimensional` `Chapter1/ScBanach · ScBanach.constScale`. Here ι_m is the identity `Chapter1/ScBanach · ScBanach.constScale_toBase`, every point is smooth `Chapter1/ScBanach · ScBanach.constScale_smoothLvl`, and density is trivial, so V is an sc-Banach space `Chapter1/ScBanach · ScBanach.const` `Chapter1/ScBanach · ScBanach.const_toBase`.

Only the constant structure (Lemma 1.A(2)–(3) of the exposition). The text remarks that a finite-dimensional space has precisely one sc-structure. For families this reads: if E is an sc-Banach space with E_0 finite-dimensional, the inclusions identify E with the constant structure on E_0. First, every point is smooth: E_∞ is a subspace of the finite-dimensional E_0, hence closed `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.closed_of_finiteDimensional`, and dense by the axiom at level 0, hence everything `Chapter1/ExpositionLemmas · ScBanach.smooth_eq_top_of_finiteDimensional`. So every $x \in E_0$ lifts to ev-

ery level, and $\iota_m : E_m \to E_0$, injective by construction, is bijective `Chapter1/ExpositionLemmas · ScBanach.toBase_bijective_of_finiteDimensional`. A continuous linear bijection between Banach spaces is an isomorphism (open mapping theorem) `Mathlib/Analysis/Normed/Operator/Banach · ContinuousLinearEquiv.ofBijective`; the isomorphisms ι_m commute with the inclusions by the very definition $\iota_{m+1} = \iota_m \circ \mathrm{incl}_m$, so they form an sc-isomorphism of E onto the constant structure on E_0 `Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional_apply`. In particular all the norms $|\cdot|_m$ are equivalent to $|\cdot|_0$. The second remark, that on an infinite-dimensional space the constant structure fails the compactness axiom, becomes: if an inclusion $\mathrm{incl}_m : E_{m+1} \to E_m$ of a scale is onto, then E_m is finite-dimensional. Indeed incl_m is then a continuous linear bijection, hence an isomorphism e, and $\mathrm{id}_{E_m} = \mathrm{incl}_m \circ e^{-1}$ is compact as a compact operator composed with a bounded one `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.comp_clm`; a Banach space with compact identity is finite-dimensional `Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · FiniteDimensional.of_isCompactOperator_id Chapter1/ExpositionLemmas · ScScale.finiteDimensional_of_surjective_incl`.

Direct sums. For scales E, F, the family $E_m \times F_m$ with the product inclusions is a scale: the product of two compact operators is compact, because the preimage of a product of neighbourhoods is mapped into the product of two compact sets `Chapter1/ScBanach · ScBanach.prodScale`. Its ι_m is the pair of the two `Chapter1/ScBanach · ScBanach.prodScale_toBase`, and its smooth points are the pairs of smooth points `Chapter1/ScBanach · ScBanach.prodScale_smoothLvl`; since a product of dense sets is dense, $E \oplus F$ is an sc-Banach space `Chapter1/ScBanach · ScBanach.prod Chapter1/ScBanach · ScBanach.prod_smoothLvl`.

The level shift. The family E_{m+1} with the inclusions incl_{m+1} is a scale E^1 `Chapter1/ScBanach · ScBanach.shiftScale`. Its composite inclusion to its base E_1, followed by ι_1, is ι_{m+1} `Chapter1/ScBanach · ScBanach.shiftScale_toBase`, and its smooth points at level m are the smooth points of E at level $m+1$ `Chapter1/ScBanach · ScBanach.shiftScale_smoothLvl`, so E^1 is an sc-Banach space `Chapter1/ScBanach · ScBanach.shift Chapter1/ScBanach · ScBanach.shift_toBase Chapter1/ScBanach · ScBanach.shift_smoothLvl`; E^k is the k-fold iterate `Chapter1/ScBanach · ScBanach.shiftK`.

Level sets. For $U \subseteq E_0$ the level set $U_m \subseteq E_m$ is the preimage $\iota_m^{-1}(U)$ `Chapter1/ScSmooth · ScBanach.lvlSet`; $U_0 = U$ `Chapter1/ScSmooth · ScBanach.lvlSet_zero`, $U_{m+1} = \mathrm{incl}_m^{-1}(U_m)$ `Chapter1/ScSmooth · ScBanach.lvlSet_succ`, and level sets preserve inclusions, intersections, openness, closedness and convexity `Chapter1/ScSmooth · ScBanach.lvlSet_mono Chapter1/ScSmooth · ScBanach.lvlSet_inter Chapter1/ScSmooth · ScBanach.isOpen_lvlSet Chapter1/ScSmooth · ScBanach.isClosed_lvlSet Chapter1/ScSmooth · ScBanach.convex_lvlSet`. This is the text's filtration $A_m = A \cap E_m$ of a subset. The level sets of $U_1 \subseteq E_1$, taken in the shifted space E^1, are the U_{m+1} `Chapter1/Levels · ScBanach.shift_lvlSet`, which is the text's $(A^1)_m = A_{1+m}$.

I.2 Sc-operators, sc-isomorphisms, sc⁺-operators (Definitions 1.1.4–1.1.5 (preprint 1.3–1.4))

An **sc-operator** $T : E \to F$ is a family of continuous linear maps $T_m : E_m \to F_m$ commuting with the inclusions, $\mathrm{incl}_m^F \circ T_{m+1} = T_m \circ \mathrm{incl}_m^E$ `Chapter1/ScBanach · ScOperator`. Then $\iota_m T_m = T_0 \iota_m$ by induction `Chapter1/ScBanach · ScOperator.toBase_comm`, so the whole family is determined by the base map T_0 `Chapter1/ScBanach · ScOperator.ext_base`: this is the sense in which the text's single linear map T "with $T(E_m) \subset F_m$ and $T : E_m \to F_m$ continuous" is the same thing. The text's formulation is also available as a predicate on a continuous linear $T_0 : E_0 \to F_0$, "for every m there is a continuous linear T_m with $\iota_m T_m = T_0 \iota_m$" `Chapter1/ScBanach · IsScOperator`, and the two notions pass into each other `Chapter1/ScBanach · ScOperator.isScOperator Chapter1/ScBanach · IsScOperator.toScOperator Chap-`

ter1/ScBanach · IsScOperator.toScOperator_zero. The identity is an sc-operator, sc-operators compose and add Chapter1/ScBanach · ScOperator.id Chapter1/ScBanach · ScOperator.comp Chapter1/ScBanach · ScOperator.add, and they map smooth points to smooth points, since a lift of x to level j is carried by T_j to a lift of $T_0 x$ Chapter1/ScBanach · ScOperator.mapsTo_smoothLvl.

An **sc-isomorphism** is a family of continuous linear equivalences $E_m \simeq F_m$ commuting with the inclusions Chapter1/ScBanach · ScIso; it has an underlying sc-operator Chapter1/ScBanach · ScIso.toScOperator, an inverse Chapter1/ScBanach · ScIso.symm, composites Chapter1/ScBanach · ScIso.trans and the identity Chapter1/ScBanach · ScIso.refl, and ι_m commutes with it and with its inverse Chapter1/ScBanach · ScIso.toBase_comm Chapter1/ScBanach · ScIso.symm_toBase_comm. The text defines an sc-isomorphism as a bijective sc-operator whose inverse is an sc-operator; the continuity of the inverse on each level, which there follows from the open mapping theorem, is here part of the data. The two are equivalent: a bijective sc-operator is an sc-isomorphism by the open mapping theorem on each level Chapter3/LinearizationChartsSc · ScIso.ofBijective. Likewise the continuity in "maps E_m into F_m continuously" is automatic for a bounded $T : E_0 \to F_0$, by the closed graph theorem Chapter1/ScFredholmCriterion · isScOperator_of_mapsTo (§I.23).

An **sc$^+$-operator** $E \to F$ is an sc-operator $E \to F^1$ into the shifted space, i.e. continuous linear maps $S_m : E_m \to F_{m+1}$ commuting with the inclusions Chapter1/ScBanach · ScPlus. The text's first condition, $S(E_m) \subset F_{m+1}$, is contained in the second and is dropped. The inclusion $F^1 \to F$, level by level $\mathrm{incl}_m : F_{m+1} \to F_m$, is itself an sc-operator Chapter1/ScBanach · ScBanach.shiftIncl, and composing gives the sc-operator $E \to F$ underlying S Chapter1/ScBanach · ScPlus.toScOperator Chapter1/ScBanach · ScPlus.toScOperator_apply. Its level-m map is the compact incl_m after the bounded S_m, hence compact Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.comp_clm Chapter1/ScBanach · ScPlus.isCompactOperator: the text's remark that sc$^+$-operators are sc-compact. In terms of images, S raises the level: if $x \in \operatorname{range} \iota_m$ then $S_0 x \in \operatorname{range} \iota_{m+1}$ Chapter1/ScFredholm · ScPlus.toScOperator_mem_range_toBase_succ.

I.3 Sc-subspaces and sc-splittings (Definition 1.1.6 (preprint 1.5))

For a subspace $A \subseteq E_0$ its **level-m part** $A_m \subseteq E_m$ is the preimage of A under ι_m Chapter1/ScBanach · ScBanach.levelSub Chapter1/ScBanach · ScBanach.mem_levelSub; $A_0 = A$ Chapter1/ScBanach · ScBanach.levelSub_zero, the inclusions map A_{m+1} into A_m Chapter1/ScBanach · ScBanach.incl_mem_levelSub, and A_m is closed when A is, as the preimage of a closed set under a continuous map Chapter1/ScBanach · ScBanach.isClosed_levelSub.

A is an **sc-subspace** if A is closed and, for every m, A_m lies in the closure of $A_m \cap E_\infty$ Chapter1/ScBanach · ScBanach.IsScSubspace Chapter1/ScBanach · ScBanach.IsScSubspace.isClosed. The text asks that (A_m) be an sc-structure on A; of the conditions in that phrase, the closedness of A_m in E_m was just proved, and the compactness of the restricted inclusions $A_{m+1} \to A_m$ is automatic, since the restriction of a compact operator to a subspace, with values in a closed subspace, is compact. This is how the scale structure on a closed A is built: levels A_m, inclusions the restrictions of incl_m, injective because incl_m is, compact by the remark Chapter1/ScBanach · ScBanach.ofSubspaceScale. Its composite inclusion is that of E on the underlying points Chapter1/ScBanach · ScBanach.ofSubspaceScale_coe_toBase, and its smooth points are the points of A_m that are smooth in E Chapter1/ScBanach · ScBanach.mem_ofSubspaceScale_smoothLvl Chapter1/ScBanach · ScBanach.ofSubspaceScale_smoothLvl. The density condition of `IsScSubspace` is then exactly the density axiom for this scale, so an sc-subspace is an sc-Banach space with the induced structure Chapter1/ScBanach · ScBanach.ofSubspace.

$E = F \oplus G$ is an **sc-splitting** if F and G are sc-subspaces and, for every m, F_m and G_m are complementary subspaces of E_m `Chapter1/ScBanach · ScBanach.IsScSplitting`. The text asks for a *topological* direct sum $E_m = F_m \oplus G_m$. Since F_m and G_m are closed and complementary, the projection onto F_m along G_m is continuous by the closed graph theorem, which is Mathlib's "closed complemented from complementary closed subspaces" `Chapter1/ScBanach · ScBanach.IsScSplitting.closedComplemented`; so nothing is lost by asking only for complementarity. The splitting is symmetric in F and G `Chapter1/ScBanach · ScBanach.IsScSplitting.symm`.

Sc-splittings are sc-projections (Lemma 2.B of the exposition). The proof of Proposition 1.1.7 (preprint 1.6) in the text passes from "$P \circ P = P$, continuous on every level" to "a sc-projection" and an sc-complement $(1 - P)(E)$ without comment. Both directions are proved. (i) Given an sc-splitting $E = A \oplus B$, on each level A_m and B_m are complementary and closed, hence topological complements `Mathlib/Analysis/Normed/Module/Complemented · Submodule.IsCompl.isTopCompl_of_isClosed Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.isTopCompl`, so the projection P_m of E_m onto A_m along B_m is continuous `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL`. These projections commute with the inclusions: write $x = P_{m+1}x + (x - P_{m+1}x)$ with the first term in A_{m+1} and the second in B_{m+1} `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_eq_self_sub_projectionL`; incl_m carries them into A_m and B_m (§I.3), where P_m is the identity, resp. zero `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_eq_self_iff Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_apply_eq_zero_iff`, so $P_m(\mathrm{incl}_m x) = \mathrm{incl}_m(P_{m+1}x)$. Hence the P_m form an sc-operator P `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj_apply` with $P_m^2 = P_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj_idem`, $P_m(E_m) = A_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.range_scProj` and $\ker P_m = B_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.ker_scProj`. (ii) Conversely let P be an sc-operator with $P_0^2 = P_0$. Idempotence passes to every level by injectivity of ι_m, since $\iota_m P_m P_m x = P_0 P_0 \iota_m x = P_0 \iota_m x = \iota_m P_m x$ `Chapter1/ExpositionLemmas · ScOperator.idem_lvl`. The level-m part of $P_0(E_0)$ is $P_m(E_m)$: if $\iota_m x = P_0 y$ then $P_0 \iota_m x = \iota_m x$, i.e. $\iota_m P_m x = \iota_m x$, so $x = P_m x$ `Chapter1/ExpositionLemmas · ScOperator.levelSub_range_eq`; the level-m part of $\ker P_0$ is $\ker P_m$ (§I.6, `Chapter1/ScFredholm · ScOperator.ker_lvl`). $P_0(E_0) = \{x : P_0 x = x\}$ is closed, and $\ker P_0$ is closed. Density of smooth points: $x \in P_m(E_m)$ satisfies $x = P_m x$ and is the limit of $P_m z_n$ for smooth $z_n \to x$, and the $P_m z_n$ are smooth points of $P_m(E_m)$ (§I.2); $x \in \ker P_m$ satisfies $x = x - P_m x$ and is the limit of $z_n - P_m z_n$, smooth points of $\ker P_m$. Finally $P_m(E_m)$ and $\ker P_m$ are complementary because P_m is idempotent `Mathlib/LinearAlgebra/Projection · LinearMap.IsIdempotentElem.isCompl`. So $E = P_0(E_0) \oplus \ker P_0$ is an sc-splitting `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_of_idem`, and together: $E = A \oplus B$ is an sc-splitting iff an idempotent sc-operator has range A and kernel B `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_iff_exists_scProj`. The text's $(1-P)(E)$ is $\ker P$. The formal Proposition 1.1.7 (preprint 1.6) (§I.4) builds its complement directly and does not use this.

I.4 Proposition 1.1.7 (preprint 1.6): finite-dimensional sc-subspaces

Statement. Let $F \subseteq E_0$ be finite-dimensional. Then F is an sc-subspace if and only if $F \le E_\infty$ `Chapter1/ScBanach · ScBanach.proposition_1_1_7_iff`; and if $F \le E_\infty$ there is a G with $E = F \oplus G$ an sc-splitting `Chapter1/ScBanach · ScBanach.proposition_1_1_7`.

The equivalence. If F is an sc-subspace, the density condition at level 0 says $F \subseteq \overline{F \cap E_\infty}$. But $F \cap E_\infty$ is a subspace of the finite-dimensional F, hence finite-dimensional, hence closed `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.closed_of_finiteDimensional`; so $F \subseteq F \cap E_\infty$, i.e. $F \le E_\infty$. Conversely, if $F \le E_\infty$ then F is closed (finite-dimensional) and every point of F_m is already a smooth point of F_m,

so the density condition holds trivially.

The complement. Mathlib provides a continuous linear projection $p : E_0 \to F$ with $p|_F = \mathrm{id}$, from the closed complementation of finite-dimensional subspaces (Hahn–Banach) `Mathlib/Analysis/LocallyConvex/HahnBanach · Submodule.ClosedComplemented.of_finiteDimensional`. This is the text's $P(x) = \sum \lambda_i(x) e_i$. Put $G := \ker p$. Since $F \le E_\infty$, the restriction of ι_m to F_m is a bijection onto F (injective by injectivity of ι_m, surjective because every point of F lifts to level m) `Chapter1/ScBanach · ScBanach.levelRestrict Chapter1/ScBanach · ScBanach.levelRestrict_bijective Chapter1/ScBanach · ScBanach.levelEquiv`; through this linear isomorphism $F_m \cong F$ the projection lifts to a continuous linear $P_m : E_m \to F_m$, $P_m = (\iota_m|_{F_m})^{-1} \circ p \circ \iota_m$ (continuous because F_m is finite-dimensional) `Chapter1/ScBanach · ScBanach.finProj`. It fixes F_m `Chapter1/ScBanach · ScBanach.finProj_apply_of_mem`, and $P_m x = 0$ iff $p(\iota_m x) = 0$ `Chapter1/ScBanach · ScBanach.finProj_eq_zero_iff`, so $G_m = \ker P_m$. Now F is an sc-subspace by the first part, and G is closed as a kernel. For the density of $G_m \cap E_\infty$ in G_m: given $g \in G_m$, let $Q := 1 - P_m$ on E_m; then $Qg = g$, Q maps E_m into G_m, and Q maps smooth points to smooth points, since $P_m x$ is smooth ($F \le E_\infty$) and smooth points form a subspace `Chapter1/ScBanach · ScScale.smoothLvl_submodule_closed_sub`. Hence $g = Qg$ lies in the closure of $Q(E_\infty \cap E_m) \subseteq G_m \cap E_\infty$, because $E_\infty \cap E_m$ is dense and Q is continuous. This is the text's $g_n = (1 - P) f_n \to g$. Finally F_m and $G_m = \ker P_m$ are complementary because P_m is a projection onto F_m `Mathlib/LinearAlgebra/Projection · isCompl_of_proj`.

I.5 Proposition 1.1.8 (preprint 1.7): quotients

Statement. Let $A \subseteq E_0$ be closed. The quotients E_m/A_m form an sc-Banach space E/A `Chapter1/ScBanach · ScBanach.quotient`, the quotient maps form an sc-operator $E \to E/A$ `Chapter1/ScBanach · ScBanach.quotientMap`, and, as a statement, there are an sc-Banach space Q and an sc-operator $q : E \to Q$ with every q_m surjective and $\ker q_m = A_m$ `Chapter1/ScBanach · ScBanach.proposition_1_1_8`. The text assumes A is an sc-subspace; only closedness is used.

The inclusions. A_m is closed in E_m (§I.3), so E_m/A_m is a Banach space. The map $E_{m+1}/A_{m+1} \to E_m/A_m$ is induced by incl_m: it is well defined because incl_m maps A_{m+1} into A_m `Chapter1/ScBanach · ScBanach.quotIncl Chapter1/ScBanach · ScBanach.quotIncl_mk`. It is injective: if $\mathrm{incl}_m x \in A_m$ then $\iota_{m+1} x = \iota_m(\mathrm{incl}_m x) \in A$, so $x \in A_{m+1}$ `Chapter1/ScBanach · ScBanach.quotIncl_injective`. This injectivity is the text's identification of E_m/A_m with the subspace $\{x + A : x \in E_m\}$ of E/A, and it holds precisely because A_m is the *full* preimage of A. It is compact: the quotient map $E_m \to E_m/A_m$ after the compact incl_m is compact, and the induced map on E_{m+1}/A_{m+1} sends the image of a neighbourhood of 0 (which is a neighbourhood, since quotient maps are open) into the same compact set `Chapter1/ScBanach · ScBanach.quotIncl_isCompactOperator`. So the quotients form a scale `Chapter1/ScBanach · ScBanach.quotientScale`, whose composite inclusion sends the class of x to the class of $\iota_m x$ `Chapter1/ScBanach · ScBanach.quotientScale_toBase_mk`. The text's argument with a sequence $x_k + A_{m+1}$ of norm ≤ 1 and representatives of norm ≤ 2 is the sequential form of the same fact.

Density. The quotient map $E_m \to E_m/A_m$ is continuous and surjective, so the image of the dense set $E_\infty \cap E_m$ is dense. A class $[x]$ with x smooth is a smooth point of the quotient scale: a lift y of x to level j gives the lift $[y]$ of $[x]$. So the smooth points of E_m/A_m are dense `Chapter1/ScBanach · ScBanach.quotient`. This is the inclusion "$\{x + A : x \in E_\infty\} \subseteq (E/A)_\infty$" of the text's proof, which is all its density claim needs.

The smooth points of the quotient (Lemma 2.C of the exposition). The text also asserts that the two sets are equal. Let A be an sc-subspace and $x \in E_0$ with, for every j, a point $x_j \in E_j$ such that $\iota_j x_j - x \in A$.

We find a smooth y with $y - x \in A$ Chapter1/ExpositionLemmas · ScBanach.exists_smooth_mem_coset. *The differences.* $d_j := \mathrm{incl}_j x_{j+1} - x_j \in E_j$ has $\iota_j d_j = \iota_{j+1}x_{j+1} - \iota_j x_j = (\iota_{j+1}x_{j+1} - x) - (\iota_j x_j - x) \in A$, so $d_j \in A_j$. *A uniform bound for the inclusions.* Put $P_j := \prod_{i<j} \max(1, \|\mathrm{incl}_i\|) \geq 1$ Chapter1/ExpositionLemmas · ScBanach.inclBound Chapter1/ExpositionLemmas · ScBanach.one_le_inclBound. By induction, $\|\iota_j v\| \leq P_j \|v\|$ for $v \in E_j$ Chapter1/ExpositionLemmas · ScBanach.norm_toBase_le and $\|\mathrm{incl}_J^{J+k} v\| \leq P_{J+k}\|v\|$ for the composite inclusion $E_{J+k} \to E_J$ Chapter1/ExpositionLemmas · ScBanach.norm_inclK_le, each step multiplying by $\|\mathrm{incl}_i\| \leq \max(1, \|\mathrm{incl}_i\|)$ Chapter1/ExpositionLemmas · ScBanach.norm_incl_le. This replaces the exposition's $M_j = \max_{i \leq j} \|E_j \to E_i\|$, and avoids comparing levels $J + k$ with j. *Smooth corrections.* Since A is an sc-subspace, $A_j \cap E_\infty$ is dense in A_j, so there are $c_j \in A_j \cap E_\infty$ with $\|d_j - c_j\| \leq 2^{-j}/P_j$ Mathlib/Topology/MetricSpace/Pseudo/Defs · Metric.mem_closure_iff. Then $b_j := d_j - c_j \in A_j$ has $P_j\|b_j\| \leq 2^{-j}$. *The series.* $\sum_j \iota_j b_j$ converges absolutely in E_0, its terms being bounded by 2^{-j} Mathlib/Analysis/Normed/Group/InfiniteSum · Summable.of_norm_bounded; let S be its sum and $y := x_0 + S$. Its partial sums lie in A, which is closed, so $S \in A$ (the additive form of Mathlib/Topology/Algebra/InfiniteSum/NatInt · HasProd.tendsto_prod_nat), and $y - x = (x_0 - x) + S \in A$. *$y$ is smooth.* Fix J. The tail $t := \sum_k \mathrm{incl}_J^{J+k} b_{J+k}$ converges in E_J (terms bounded by $2^{-J}2^{-k}$), and $\iota_J t = \sum_k \iota_{J+k} b_{J+k}$ because ι_J is continuous and linear Mathlib/Topology/Algebra/InfiniteSum/Module · ContinuousLinearMap.map_tsum and $\iota_J \circ \mathrm{incl}_J^{J+k} = \iota_{J+k}$ (§I.1). The head telescopes: $\sum_{j<J} \iota_j d_j = \iota_J x_J - x_0$ (the additive form of Mathlib/Algebra/BigOperators/Group/Finset/Basic · Finset.prod_range_div), so $\sum_{j<J} \iota_j b_j = (\iota_J x_J - x_0) - \sum_{j<J} \iota_j c_j$, and the last sum is smooth, hence $\iota_J w$ for some $w \in E_J$. Splitting S into head and tail (the additive form of Mathlib/Topology/Algebra/InfiniteSum/NatInt · Multiplicable.prod_mul_tprod_nat_add) gives $y = \iota_J(x_J - w + t)$. As J was arbitrary, $y \in E_\infty$. *As a statement about E/A.* The class of x is a smooth point of E/A iff for every j it is the image of a class $[z]$, $z \in E_j$, i.e. iff $\iota_j z - x \in A$ for some z (§I.5 and Mathlib/LinearAlgebra/Quotient/Defs · Submodule.Quotient.eq); by the above this happens iff $x + A$ contains a smooth point Chapter1/ExpositionLemmas · ScBanach.mk_mem_quotient_smooth_iff. This is the text's equality.

I.6 Definition 1.1.9 (preprint 1.8): sc-Fredholm operators

Level kernels and ranges. For an sc-operator T, the kernel of T_m is the level-m part of the kernel of T_0 Chapter1/ScFredholm · ScOperator.ker_lvl, and the range of T_m lies in the level-m part of the range of T_0 Chapter1/ScFredholm · ScOperator.range_lvl_le; both from $\iota_m T_m = T_0 \iota_m$ and injectivity.

The definition. An **sc-Fredholm splitting** of $T : E \to F$ consists of subspaces $K, X \subseteq E_0$ and $C, Y \subseteq F_0$ such that $E = K \oplus X$ and $F = C \oplus Y$ are sc-splittings, $K = \ker T_0$, $Y = \mathrm{range}\, T_0$, K and C are finite-dimensional, and T_m restricts to a bijection $X_m \to Y_m$ for every m Chapter1/ScFredholm · ScFredholmSplitting; T is **sc-Fredholm** if such a splitting exists Chapter1/ScFredholm · IsScFredholm. The text's condition (3), "$T : X \to Y$ is an sc-isomorphism", is here the bijectivity on each level; the continuity of the inverses $Y_m \to X_m$ is automatic by the open mapping theorem between the closed subspaces X_m and Y_m, so nothing is lost.

The index. The index of a splitting is $\dim K - \dim C$ Chapter1/ScFredholm · ScFredholmSplitting.index, and the index of T is $\dim \ker T_0 - \dim(F_0/\,\mathrm{range}\, T_0)$ Chapter1/ScFredholm · scIndex. They agree: $K = \ker T_0$, and C is a complement of $Y = \mathrm{range}\, T_0$ in F_0, so $C \cong F_0/Y$ Mathlib/LinearAlgebra/Projection · quotientEquivOfIsCompl Chapter1/ScFredholm · ScFredholmSplitting.index_eq. So the index does not depend on the splitting, which the text does not state.

The remarks after the definition. K is a finite-dimensional sc-subspace, hence $K \leq E_\infty$ by Proposition 1.1.7 (preprint 1.6) Chapter1/ScFredholm · ScFredholmSplitting.K_le_smooth. $\ker T_m = K_m$ Chapter1/

ScFredholm · ScFredholmSplitting.ker_lvl and range $T_m = Y_m$: one inclusion from above, the other because T_m maps X_m onto Y_m Chapter1/ScFredholm · ScFredholmSplitting.range_lvl. The text's "$F = C \oplus T(E_m)$" (read F_m) is the level-m part of the splitting $F = C \oplus Y$ together with range $T_m = Y_m$. Finally, every T_m is a Fredholm operator in Mathlib's sense: its kernel K_m injects into the finite-dimensional K, its range Y_m is closed, and Y_m has the finite-dimensional complement C_m (which injects into C), so its range has finite codimension Mathlib/RingTheory/Finiteness/Cofinite · FG.cofg_of_isCompl Chapter1/ScFredholm · ScFredholmSplitting.isFredholm, using the criterion of §I.7. This is the fact the text uses in the first line of §1.4.1.

The index on every level (Lemma 2.D of the exposition). C is a finite-dimensional sc-subspace, so $C \leq F_\infty$ by Proposition 1.1.7 (preprint 1.6) Chapter1/FredholmIndex · ScFredholmSplitting.C_le_smooth. For any subspace $A \leq E_\infty$, ι_m restricts to a bijection $A_m \to A$ (injective as ι_m is, surjective because every point of A lifts), so $\dim A_m = \dim A$ Chapter1/FredholmIndex · ScBanach.finrank_levelSub_of_le_smooth. Hence $\dim \ker T_m = \dim K_m = \dim K$, and $F_m / \operatorname{range} T_m = F_m / Y_m \cong C_m$ (complements, Mathlib/LinearAlgebra/Projection · quotientEquivOfIsCompl) has dimension $\dim C$. So T_m is Fredholm with index $\dim K - \dim C$ in the sense of §I.7 Chapter1/FredholmIndex · ScFredholmSplitting.fredholmIndex_lvl, which by the previous paragraph is the index of T for every splitting Chapter1/FredholmIndex · IsScFredholm.fredholmIndex_lvl; on level 0 this is the definition, $\operatorname{ind}(T)$ being the Fredholm index of T_0 Chapter1/FredholmIndex · scIndex_eq_fredholmIndex. The text states the splittings on each level but not the index.

I.7 Fredholm operators between Banach spaces: Riesz–Schauder and compact perturbations

Mathlib has a notion of Fredholm operator Mathlib/Analysis/Normed/Operator/Fredholm/Basic · IsFredholm: a continuous linear $u : E \to F$ that is a strict map, with closed range, finite-dimensional kernel, range of finite codimension Mathlib/RingTheory/Finiteness/Cofinite · CoFG and closed complemented kernel. It does not have the two theorems that §1.4.1 quotes as classical, that $1 +$ compact is Fredholm and that compact perturbations preserve the property. Both are proved here, for real Banach spaces.

The criterion. A continuous linear map with closed range between Banach spaces is a strict map Mathlib/Topology/Maps/Strict/Basic · IsStrictMap: by the open mapping theorem its corestriction to the range is an open quotient map Mathlib/Topology/Maps/Strict/Module · isStrictMap_iff_isOpenQuotientMap_rangeRestrict Chapter1/FredholmPerturbation · isStrictMap_of_isClosed_range. Hence a continuous linear map with finite-dimensional kernel, closed range and range of finite codimension is Fredholm, the kernel being closed complemented because it is finite-dimensional Chapter1/FredholmPerturbation · isFredholm_of_banach. A finite-rank operator is compact: the image of the unit ball is bounded in the finite-dimensional range, so its closure is compact Chapter1/FredholmPerturbation · isCompactOperator_of_finiteDimensional_range.

Riesz–Schauder: the kernel. Let $c : E \to E$ be compact and $N := \ker(1 + c)$. On N the operator c is $-\mathrm{id}$, and c restricts to a compact operator of the closed subspace N Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.restrict; so id_N is compact, and N is finite-dimensional Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · FiniteDimensional.of_isCompactOperator_id Chapter1/FredholmPerturbation · finiteDimensional_ker_one_add.

The range is closed. Take a closed complement G of N (which exists since N is finite-dimensional). $1 + c$ is bounded below on G: otherwise there are unit vectors $g_n \in G$ with $(1 + c)g_n \to 0$; the cg_n lie in a compact set Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.image_closedBall_subset_

compact, so along a subsequence $cg_n \to y$ Mathlib/Topology/Sequences · IsCompact.tendsto_subseq, hence $g_n = (1+c)g_n - cg_n \to -y$; then $-y \in G$ (closed), $\|-y\| = 1$, and $(1+c)(-y) = 0$, so $-y \in G \cap N = 0$, a contradiction Chapter1/FredholmPerturbation · exists_bound_below_on_compl. A bounded-below operator from a Banach space has closed range (it is antilipschitz and uniformly continuous), and $(1+c)(G) = (1+c)(E)$ since $1+c$ kills N Chapter1/FredholmPerturbation · isClosed_range_one_add.

The range has finite codimension. Suppose $R := \operatorname{range}(1+c)$ has infinite codimension. Pick, for every proper subspace, a vector outside it Chapter1/FredholmPerturbation · pick Chapter1/FredholmPerturbation · pick_notMem, and build the chain $M_0 = R$, $M_{j+1} = M_j + \mathbb{R}\operatorname{pick}(M_j)$ Chapter1/FredholmPerturbation · chain. Each M_j is R plus a finite-dimensional subspace Chapter1/FredholmPerturbation · chain_eq_sup, hence closed Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.isClosed_sup_finiteDimensional Chapter1/FredholmPerturbation · chain_isClosed, and proper, since R has infinite codimension Mathlib/RingTheory/Finiteness/Cofinite · FG.cofg_of_codisjoint Chapter1/FredholmPerturbation · chain_ne_top; the chain is increasing Chapter1/FredholmPerturbation · chain_mono. Riesz's lemma inside a subspace: for closed $M \subsetneq M'$ there is a unit vector of M' at distance $\geq 1/2$ from M (take $x \in M' \setminus M$, a point $y \in M$ with $\|x-y\| < 2\operatorname{dist}(x, M)$ Mathlib/Topology/MetricSpace/HausdorffDistance · infDist_lt_iff, and normalize $x-y$) Chapter1/FredholmPerturbation · exists_unit_far. This gives unit vectors $u_j \in M_{j+1}$ at distance $\geq 1/2$ from M_j. For $j < l$, $u_j + (1+c)u_l - (1+c)u_j \in M_l$, and $u_l - [u_j + (1+c)u_l - (1+c)u_j] = -(cu_l - cu_j)$, so $\|cu_l - cu_j\| \geq 1/2$. But the cu_j lie in a compact set and so have a convergent subsequence, two of whose terms are within $1/2$ of each other: a contradiction Chapter1/FredholmPerturbation · coFG_range_one_add. Hence $1+c$ is Fredholm Chapter1/FredholmPerturbation · isFredholm_one_add.

Compact perturbations. Let $u : E \to F$ be Fredholm and k compact. Mathlib gives a quasi-inverse $v : F \to E$ of u, with $vu - 1$ and $uv - 1$ of finite rank Mathlib/Analysis/Normed/Operator/Fredholm/Basic · isFredholm_iff_exists_isQuasiInverse; a finite-rank difference from the identity is compact Chapter1/FredholmPerturbation · isCompactOperator_sub_one_of_equiv. Then $v(u+k) = 1 + c_1$ and $(u+k)v = 1 + c_2$ with $c_1 = (vu - 1) + vk$ and $c_2 = (uv - 1) + kv$ compact. The kernel of $u+k$ lies in the finite-dimensional kernel of $1 + c_1$; the range of $u+k$ contains the range of $1 + c_2$, which is closed and of finite codimension, so $\operatorname{range}(u+k)$ has finite codimension Mathlib/RingTheory/Finiteness/Cofinite · CoFG.of_le and is closed, as a subspace containing a closed subspace of finite codimension Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.isClosed_mono_of_finiteDimensional_quotient. By the criterion, $u+k$ is Fredholm Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator.

The index. Mathlib has no index; it is built here, first for linear maps over a field. For $u : E \to F$ put $\operatorname{ind}(u) := \dim\ker u - \dim(F/\operatorname{range} u) \in \mathbb{Z}$ Chapter1/FredholmIndex · linIndex, meaningful when u has *finite index*, i.e. finite-dimensional kernel and cokernel Chapter1/FredholmIndex · HasFiniteIndex. *Additivity.* If $u : E \to F$ and $v : F \to G$ have finite index, so has vu, and $\operatorname{ind}(vu) = \operatorname{ind}(u) + \operatorname{ind}(v)$ Chapter1/FredholmIndex · HasFiniteIndex.comp_and_linIndex Chapter1/FredholmIndex · HasFiniteIndex.comp Chapter1/FredholmIndex · linIndex_comp. The proof is the dimension count along the exact sequence $0 \to \ker u \to \ker vu \to \operatorname{range} u \cap \ker v \to F/\operatorname{range} u \to G/\operatorname{range} vu \to G/\operatorname{range} v \to 0$, done as four applications of rank–nullity Mathlib/LinearAlgebra/FiniteDimensional/Lemmas · LinearMap.finrank_range_add_finrank_ker and two of the third isomorphism theorem Mathlib/LinearAlgebra/Isomorphisms · Submodule.quotientQuotientEquivQuotient: to u restricted to $\ker vu$ (kernel $\ker u$, range $\operatorname{range} u \cap \ker v$), to the quotient map $F \to F/\operatorname{range} u$ restricted to $\ker v$, and to v followed by $G \to G/\operatorname{range} vu$ (kernel $\operatorname{range} u + \ker v$); finiteness of the new kernel and cokernel comes out of the same maps Chapter1/FredholmIndex · finite_of_finite_ker_range Chapter1/FredholmIndex · finite_of_finite_quotient. Conversely, if vu and uv have finite index then so has u, since $\ker u \subseteq \ker vu$ and $\operatorname{range} uv \subseteq \operatorname{range} u$ Chapter1/FredholmIndex · finiteDimensional_ker_of_comp Chapter1/FredholmIndex · finiteDimensional_coker_of_comp Chap-

ter1/FredholmIndex · HasFiniteIndex.of_comp_of_comp. *Index zero.* A linear equivalence has index 0 Chapter1/FredholmIndex · HasFiniteIndex.equiv Chapter1/FredholmIndex · linIndex_equiv. If $w - 1$ has finite-dimensional range R, then $\ker w \subseteq R$, $\operatorname{range} w + R = E$ and $R \cap \operatorname{range} w = w(R)$, so $\dim \ker w$ and $\dim(E/\operatorname{range} w)$ both equal $\dim R - \dim w(R)$ and $\operatorname{ind}(w) = 0$ Chapter1/FredholmIndex · HasFiniteIndex.of_finiteRange_sub_id_and_linIndex Chapter1/FredholmIndex · linIndex_eq_zero_of_finiteRange_sub_id. If a is bijective and r has finite rank, $a + r = a(1 + a^{-1}r)$ has index $0 + 0$ by additivity Chapter1/FredholmIndex · HasFiniteIndex.of_bijective_add_and_linIndex.

Banach spaces. For $u : E \to F$ continuous linear, the **Fredholm index** is $\operatorname{ind}(u)$ Chapter1/FredholmIndex · fredholmIndex; a Fredholm operator has finite index Chapter1/FredholmIndex · IsFredholm.hasFiniteIndex. *Small perturbations.* Let v be a quasi-inverse of u and $\|v\|\,\|s\| < 1$. Then $\|vs\|, \|sv\| < 1$ Mathlib/Analysis/Normed/Operator/Basic · ContinuousLinearMap.opNorm_comp_le, so $1 + vs$ and $1 + sv$ are invertible by the Neumann series Mathlib/Analysis/SpecificLimits/Normed · isUnit_one_sub_of_norm_lt_one Chapter1/FredholmIndex · bijective_one_add_of_norm_lt. Since $vu - 1$ and $uv - 1$ have finite rank Chapter1/FredholmIndex · finiteDimensional_range_sub_id_of_equiv, $v(u + s) = (1 + vs) + (vu - 1)$ and $(u + s)v = (1 + sv) + (uv - 1)$ are bijections plus finite-rank maps, hence of index 0 Chapter1/FredholmIndex · hasFiniteIndex_comp_add_of_norm_lt; the same with $s = 0$ for vu and uv. By the converse above, $u + s$, u and v have finite index, and additivity gives $\operatorname{ind}(u+s) + \operatorname{ind}(v) = 0 = \operatorname{ind}(u) + \operatorname{ind}(v)$, so $\operatorname{ind}(u+s) = \operatorname{ind}(u)$ Chapter1/FredholmIndex · fredholmIndex_add_of_norm_lt. *Compact perturbations.* Let u be Fredholm and k compact, and $\varphi(t) := \operatorname{ind}(u + tk)$ for $t \in \mathbb{R}$. At each t_0, $u + t_0 k$ is Fredholm (the previous paragraph, $t_0 k$ being compact Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.smul), with a quasi-inverse v_0; for t so close to t_0 that $\|v_0\|\,\|(t - t_0)k\| < 1$, $\varphi(t) = \operatorname{ind}((u + t_0 k) + (t - t_0)k) = \varphi(t_0)$. So φ is locally constant Mathlib/Topology/LocallyConstant/Basic · IsLocallyConstant.iff_eventually_eq on the connected $\mathbb{R}$, hence constant Mathlib/Topology/LocallyConstant/Basic · IsLocallyConstant.apply_eq_of_preconnectedSpace, and $\operatorname{ind}(u + k) = \varphi(1) = \varphi(0) = \operatorname{ind}(u)$ Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator. This replaces the citation of Kato IV.5.26 in the exposition's (F2).

I.8 Proposition 1.1.10 (preprint 1.9): sc-Fredholm operators are regularizing

Statement. Let T have an sc-Fredholm splitting, $e \in E_0$, and $T_0 e \in \operatorname{range} \iota_m$. Then $e \in \operatorname{range} \iota_m$ Chapter1/ScFredholm · proposition_1_1_10.

Proof (the text's). Write $T_0 e = \iota_m y$. Since $F_m = C_m \oplus Y_m$, $y = c + y'$ with $c \in C_m$ and $y' \in Y_m$, and $y' = T_m x'$ for some $x' \in X_m$ by the bijectivity of $T_m : X_m \to Y_m$. Then $T_0(e - \iota_m x') = \iota_m c$, so $\iota_m c$ lies in C and in $Y = \operatorname{range} T_0$; these are complementary on level 0, so $\iota_m c = 0$. Hence $e - \iota_m x' \in \ker T_0 = K \leq E_\infty$ (§I.6), so it lifts to some $z \in E_m$, and $e = \iota_m(z + x')$. The step "$c = 0$" is where $C \cap Y = 0$ enters, as the exposition notes. The converse, Fredholm on level 0 and regularizing implies sc-Fredholm, is §I.23.

I.9 Proposition 1.1.11 (preprint 1.10): compact perturbations of sc-Fredholm operators

Statement. If $T : E \to F$ is sc-Fredholm and $S : E \to F$ is an sc$^+$-operator, then $T + S$ is sc-Fredholm Chapter1/ScFredholm · proposition_1_1_11_preprint. Here S is read as the sc-operator $E \to F$ underlying it

(§I.2), and $T + S$ is the sum of sc-operators.

The bootstrap. The text says: if $y = (T + S)e \in Y_m$ then, since T is regularizing, $e \in E_m$. But $Te = y - Se$ with Se only one level up. The formal proof iterates: if $(T + S)_0 e \in \operatorname{range} \iota_{m+1}$, then a fortiori $(T + S)_0 e \in \operatorname{range} \iota_m$ `Chapter1/ScFredholm · ScBanach.range_toBase_succ_le`, so $e \in \operatorname{range} \iota_m$ by induction, so $S_0 e \in \operatorname{range} \iota_{m+1}$ (§I.2), so $T_0 e = (T + S)_0 e - S_0 e \in \operatorname{range} \iota_{m+1}$, and Proposition 1.1.10 (preprint 1.9) gives $e \in \operatorname{range} \iota_{m+1}$ `Chapter1/ScFredholm · ScPlus.regularizing_add`. So $T + S$ is regularizing.

The kernel. Each $(T + S)_m = T_m + S_m$ is Fredholm, as a compact perturbation (§I.7) of the Fredholm operator T_m (§I.6) `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`. Let $K' := \ker(T + S)_0$, finite-dimensional. Every $x \in K'$ is smooth: $(T + S)_0 x = 0 \in \operatorname{range} \iota_m$ for every m, so $x \in \operatorname{range} \iota_m$ by the bootstrap. By Proposition 1.1.7 (preprint 1.6) there is an sc-splitting $E = K' \oplus X'$ `Chapter1/ScBanach · ScBanach.proposition_1_1_7`. (The text's $K_m = K_{m+1}$ is the same bootstrap.)

The range. Let $Y' := \operatorname{range}(T + S)_0$, closed since $(T + S)_0$ is Fredholm. The range of $(T + S)_m$ is exactly Y'_m: one inclusion from §I.6; conversely if $\iota_m y = (T + S)_0 e$ then e lifts to level m by the bootstrap, and $y = (T + S)_m e'$ by injectivity of ι_m. Y' is an sc-subspace: for $y = (T + S)_m e \in Y'_m$, approximate e by smooth points $e_n \in E_m$; then $(T+S)_m e_n \to y$ and the $(T+S)_m e_n$ are smooth points of Y'_m (sc-operators preserve smooth points, §I.2). This is the text's argument "$y_n = (T + S)e_n \in Y_\infty$, $y_n \to y$".

The complement. F_0/Y' is finite-dimensional. Let $p : F_0 \to F_0/Y'$ be the quotient map. The image $p(F_\infty)$ is a subspace of the finite-dimensional F_0/Y', dense because F_∞ is dense in F_0 and p is a continuous surjection, hence closed, hence everything. Instead of lifting a basis, take a linear right inverse g of $p|_{F_\infty}$ and let $\sigma := F_\infty \hookrightarrow F_0$ composed with g, so $p \circ \sigma = \mathrm{id}$; put $C' := \operatorname{range} \sigma$, a finite-dimensional subspace of F_∞. On every level Y'_m and C'_m are complementary: disjoint because an element of $C'_m \cap Y'_m$ is $\sigma(q)$ with $p\sigma(q) = q = 0$; spanning because for $z \in F_m$ the point $c := \sigma(p(\iota_m z))$ is smooth, so it lifts to level m, and $z - c \in Y'_m$. So $F = C' \oplus Y'$ is an sc-splitting (C' is an sc-subspace by Proposition 1.1.7 (preprint 1.6) `Chapter1/ScBanach · ScBanach.proposition_1_1_7_iff`).

The bijections. $(T + S)_m$ maps X'_m into Y'_m; it is injective on X'_m because its kernel on level m is K'_m, which meets X'_m in 0; and it is onto Y'_m because every $e \in E_m$ splits as $k + x$ with $k \in K'_m$, $x \in X'_m$ and $(T + S)_m e = (T + S)_m x$. This is condition (3) of Definition 1.1.9 (preprint 1.8), which the text does not verify; the exposition supplies it with the open mapping theorem, the formalization needs only the bijections (§I.6). Hence $T + S$ is sc-Fredholm `Chapter1/ScFredholm · proposition_1_1_11_preprint`.

The index. The text does not say that the index is preserved; the exposition does. The index of $T+S$ is the Fredholm index of $(T+S)_0 = T_0+S_0$ (§I.6); T_0 is Fredholm and S_0 compact, so by the invariance under compact perturbations (§I.7) $\operatorname{ind}(T + S) = \operatorname{ind}(T)$ `Chapter1/FredholmIndex · proposition_1_1_11_index`. On every level, $T+S$ being sc-Fredholm, its level map $(T+S)_m$ is Fredholm of index $\operatorname{ind}(T+S) = \operatorname{ind}(T)$ by the index on every level (§I.6) `Chapter1/FredholmIndex · proposition_1_1_11_index_lvl`.

I.10 Partial quadrants, tuples and sc-domains (Definition 1.1.12 (preprint 1.11))

The quadrant. In $\mathbb{R}^n \oplus W$, with $\mathbb{R}^n$ carrying its constant structure, the quadrant on level m is the set of (a, w) with $a_i \geq 0$ for all i `Chapter1/ScSmooth · quadrant`; it is convex `Chapter1/ScSmooth · quadrant_convex`, closed `Chapter1/ScSmooth · quadrant_isClosed`, has non-empty interior (the point $(1, \dots, 1, 0)$) `Chapter1/`

ScSmooth · quadrant_interior_nonempty, and the level sets of the level-0 quadrant are the level quadrants Chapter1/ScSmooth · lvlSet_quadrant.

Partial quadrants. $C \subseteq E_0$ is a **partial quadrant** if $C = T_0^{-1}(\text{quadrant})$ for some n, some sc-Banach space W and some sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ Chapter1/ScSmooth · IsPartialQuadrant. The text asks in addition that C be closed and convex; both follow. Indeed the level set C_m is T_m^{-1} of the level-m quadrant Chapter1/ScSmooth · IsPartialQuadrant.lvlSet_eq, hence convex Chapter1/ScSmooth · IsPartialQuadrant.convex_lvlSet, closed Chapter1/ScSmooth · IsPartialQuadrant.isClosed_lvlSet and with non-empty interior Chapter1/ScSmooth · IsPartialQuadrant.interior_lvlSet_nonempty; for $m = 0$ this is the text's "closed convex" Chapter1/ScSmooth · IsPartialQuadrant.convex Chapter1/ScSmooth · IsPartialQuadrant.isClosed. A convex set with non-empty interior has unique differentiability at every point of its closure Mathlib/Analysis/Calculus/TangentCone/Real · uniqueDiffWithinAt_convex, so C_m has unique differentiability at each of its points Chapter1/ScSmooth · IsPartialQuadrant.uniqueDiffWithinAt_lvlSet; this is the exposition's Lemma 3.D, that derivatives on a quadrant are unique, in Mathlib's language. The whole space is a partial quadrant with $n = 0$ Chapter1/ScSmooth · isPartialQuadrant_univ.

Tuples. (U, C, E) is a tuple, U a **relatively open subset of the partial quadrant** C, if C is a partial quadrant and $U = O \cap C$ for an open $O \subseteq E_0$ Chapter1/ScSmooth · IsRelOpenInQuadrant.

Sc-domains. $U \subseteq E_0$ is an **sc-domain** if for every m the level set U_m is the intersection of an open and a convex subset of E_m, and U_m has unique differentiability at each of its points Chapter1/ScSmooth · IsScDomain. A relatively open subset of a partial quadrant is an sc-domain: $U_m = O_m \cap C_m$ with O_m open and C_m convex, and unique differentiability of C_m at a point survives intersection with a neighbourhood Chapter1/ScSmooth · IsRelOpenInQuadrant.isScDomain. The property passes to the tangent: $(TU)_m = U_{m+1} \times E_m$ (§I.11) is $(O \times E_m) \cap (C \times E_m)$, and unique differentiability of a product is the product of the two Chapter1/ScSmooth · IsScDomain.tangentSet; and to the level shift, since the level sets of U_1 in E^1 are the U_{m+1} Chapter1/Levels · IsScDomain.shift. This is what the proofs of §§1.2–1.3 use in place of the text's claim that $T(U, C, E)$ is again a tuple (Definition 1.1.14 (preprint 1.13)). Every theorem of §§1.2–1.3 below is stated for sc-domains; the tuples are a special case.

The shift and the tangent of a tuple (Lemmas 3.A(2) and 3.B of the exposition). The text's claim is nevertheless true, and is proved with Chapter 2, where retracts need it. If $T : E \to \mathbb{R}^n \oplus W$ presents C, its levels T_{m+1} form an sc-isomorphism $T^1 : E^1 \to \mathbb{R}^n \oplus W^1$ Chapter2/LevelPolyfold · ScIso.shift, and $C_1 = T_1^{-1}(\text{quadrant})$ by the level description of a partial quadrant above, so C_1 is a partial quadrant in E^1 Chapter2/LevelPolyfold · IsPartialQuadrant.shift; with $U = O \cap C$, $U_1 = O_1 \cap C_1$ is relatively open in it Chapter2/LevelPolyfold · IsRelOpenInQuadrant.shift. For the tangent, $T_{m+1} \times T_m$ maps $(TE)_m = E_{m+1} \times E_m$ onto $(\mathbb{R}^n \times W_{m+1}) \times (\mathbb{R}^n \times W_m)$, and the reassociation $((a, w), q) \mapsto (a, (w, q))$ is an sc-isomorphism onto $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$ Chapter2/TangentBundle · tangentQuadrantIso; the composite presents $TC = \{(x, h) : x \in C_1\}$, since the quadrant condition only looks at the $\mathbb{R}^n$-coordinates of $T_1 x$ Chapter2/TangentBundle · IsPartialQuadrant.tangentSet. And $TU = (O_1 \times E_0) \cap TC$ is relatively open in TC Chapter2/TangentBundle · IsRelOpenInQuadrant.tangentSet. The identity $T(U^1) = (TU)^1$ is §I.11's Chapter1/Levels · ScBanach.shift_tangentSet.

I.11 Sc⁰ maps, the tangent, sc¹ maps, sc^k (Definitions 1.1.13–1.1.15 and the unnumbered definition after them (preprint Definitions 1.12–1.15))

Level maps. For $f : E_0 \to F_0$ and $x \in E_m$, the level map $f_m(x)$ is the inverse image under ι_m of $f(\iota_m x)$, when $f(\iota_m x) \in$ range ι_m, and a junk value otherwise `Chapter1/ScSmooth · levelMap`; so $\iota_m f_m(x) = f(\iota_m x)$ whenever the right side lifts `Chapter1/ScSmooth · toBase_levelMap`, $f_0 = f$ `Chapter1/ScSmooth · levelMap_zero`, and $f_m(x)$ is characterized by $\iota_m y = f(\iota_m x)$ `Chapter1/ScSmooth · levelMap_eq_of_toBase_eq`. Maps are total functions on E_0; only the values on the domain U will matter.

Sc⁰. f is sc^0 **on** U if for every m and $x \in U_m$ the point $f(\iota_m x)$ lifts to F_m, and f_m is continuous on U_m `Chapter1/ScSmooth · IsSc0` `Chapter1/ScSmooth · IsSc0.toBase_levelMap`. Then f is continuous on U `Chapter1/ScSmooth · IsSc0.continuousOn_base`, f_m maps U_m into V_m when f maps U into V `Chapter1/ScSmooth · IsSc0.levelMap_mapsTo`, the property passes to subsets `Chapter1/ScSmooth · IsSc0.mono` and depends only on the values on U `Chapter1/ScSmooth · IsSc0.congr`. Level maps compose: $(g \circ f)_m = g_m \circ f_m$ on U_m `Chapter1/ScSmooth · IsSc0.levelMap_comp`, so compositions of sc^0 maps are sc^0 `Chapter1/ScSmooth · IsSc0.comp`. The identity is sc^0 `Chapter1/ScSmooth · isSc0_id`, and an sc-operator is sc^0 on all of E_0 with level maps T_m `Chapter1/ScSmooth · ScOperator.isSc0` `Chapter1/ScSmooth · ScOperator.levelMap_eq`. The text's definition says "$f(U_m) \subset V_m$" with V undefined; V is U', and the target tuple enters only as $f(U) \subseteq U'$.

The tangent. $TE := E^1 \oplus E$, with levels $E_{m+1} \times E_m$ `Chapter1/ScSmooth · ScBanach.tangent` `Chapter1/ScSmooth · ScBanach.tangent_toBase`; $TU \subseteq (TE)_0 = E_1 \times E_0$ is the set of (x, h) with $x \in U_1$ `Chapter1/ScSmooth · ScBanach.tangentSet`, with level-m part $U_{m+1} \times E_m$ `Chapter1/ScSmooth · ScBanach.mem_lvlSet_tangentSet` `Chapter1/ScSmooth · ScBanach.lvlSet_tangentSet`, monotone in U `Chapter1/ScSmooth · ScBanach.tangentSet_mono`. The level-1 part of TU is $T(U_1)$ in $T(E^1) = (TE)^1$, the identity $T(U^1) = (TU)^1$ that the inductions of §1.2 use `Chapter1/Levels · ScBanach.shift_tangentSet`.

Sc¹. Given a family $D : E_1 \to \mathcal{L}(E_0, F_0)$, the tangent map is $T_D f(x, h) = (f_1(x), D(x)h)$ `Chapter1/ScSmooth · tangentMapWith`. f is sc^1 **on** U **with derivatives** D if (1) f is sc^0 on U; (2) for every $x \in U_1$ the map $f \circ \iota_1 : E_1 \to F_0$ has derivative $D(x) \circ \iota_1$ at x within U_1 `Mathlib/Analysis/Calculus/FDeriv/Defs · HasFDerivWithinAt`; (3) $T_D f$ is sc^0 on TU `Chapter1/ScSmooth · IsSc1With`. Condition (2) is the text's limit $|f(x+h) - f(x) - Df(x)h|_0 / |h|_1 \to 0$ over $h \in E_1$ with $x + h \in U_1$, in Mathlib's words. f is sc^1 **on** U if some D works `Chapter1/ScSmooth · IsSc1`; `scDeriv` is a chosen such D `Chapter1/ScSmooth · scDeriv` `Chapter1/ScSmooth · IsSc1.isSc1With_scDeriv` and `tangentMap` the tangent map built from it `Chapter1/ScSmooth · tangentMap` `Chapter1/ScSmooth · IsSc1.tangentMap_isSc0`. If f maps U into V then $T_D f$ maps TU into TV `Chapter1/ScSmooth · IsSc1With.tangentMapWith_mapsTo` `Chapter1/ScSmooth · IsSc1.tangentMap_mapsTo`; level 0 of (3) says $(x, h) \mapsto D(x)h$ is continuous on $U_1 \times E_0$ `Chapter1/ScSmooth · IsSc1With.continuousOn_deriv`; and both notions depend only on the values of f on U `Chapter1/ScSmooth · IsSc1With.congr` `Chapter1/ScSmooth · IsSc1.congr`.

Uniqueness of $Df(x)$. The text writes "the" operator $Df(x)$ and "the" tangent map. If D and D' both satisfy (2) at $x \in U_1$ and U_1 has unique differentiability at x, then $D(x) \circ \iota_1 = D'(x) \circ \iota_1$ `Mathlib/Analysis/Calculus/FDeriv/Basic · UniqueDiffWithinAt.eq`; since ι_1 has dense range (§I.1) and both operators are continuous, $D(x) = D'(x)$ `Mathlib/Topology/Algebra/Module/ContinuousLinearMap/Basic · ext_on` `Chapter1/ScSmooth · IsSc1With.eq_of_uniqueDiffWithinAt`. Hence on a set U whose U_1 has unique differentiability everywhere (an sc-domain), the chosen derivative agrees on U_1 with any admissible family `Chapter1/ScSmooth · IsSc1.scDeriv_eqOn` and the chosen tangent map agrees on TU with any $T_D f$ `Chapter1/ScSmooth · IsSc1.tangentMap_eqOn`. This is the exposition's Lemma 3.E, and the formal proofs invoke it wherever the text says "the tangent map". The uniform version of (3) at level 0: for $x \in U_1$, a compact

$K \subseteq E_0$ and $c > 0$ there is $\delta > 0$ such that $\|D(x')w - D(x)w\| \le c$ for all $x' \in U_1$ within δ of x and all $w \in K$; this is the continuity of $(x', w) \mapsto D(x')w - D(x)w$ on $U_1 \times E_0$ turned uniform over the compact K `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually` `Chapter1/Properties · IsSc1With.exists_delta_deriv`, and it is the text's remark that $x \mapsto Df(x)$ is continuous for the compact-open topology.

Compact-open continuity as an equivalence (Lemma 3.F of the exposition). For a metric space X, $s \subseteq X$ and $A : X \to \mathcal{L}(Y, Z)$: $(x, h) \mapsto A(x)h$ is continuous on $s \times Y$ if and only if for every $x \in s$, compact $K \subseteq Y$ and $\varepsilon > 0$, eventually for x' near x in s, $\|A(x')w - A(x)w\| < \varepsilon$ for all $w \in K$ `Chapter1/ExpositionLemmas · continuousOn_apply_iff_uniformOnCompact`. (□) As above, on the subtype s: for each $w \in K$, $(x', w') \mapsto A(x')w' - A(x)w'$ is continuous with value 0 at (x, w), so it is small near (x, w), and compactness of K makes the neighbourhood of x uniform in w `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually`; neighbourhoods in the subtype are neighbourhoods within s `Mathlib/Topology/NhdsWithin · eventually_nhds_subtype_iff`. (□) Continuity within $s \times Y$ at (x, w) is tested on sequences `Mathlib/Order/Filter/AtTopBot/CountablyGenerated · tendsto_iff_seq_tendsto`: let $(x_n, w_n) \to (x, w)$ with $x_n \in s$. The set $K = \{w\} \cup \{w_n\}$ is compact `Mathlib/Topology/Compactness/Compact · Filter.Tendsto.isCompact_insert_range`, so $\|A(x_n)w_n - A(x)w_n\| \to 0$ by the hypothesis, and $A(x)w_n \to A(x)w$ by continuity of $A(x)$; adding, $A(x_n)w_n \to A(x)w$. For an sc^1 map, level 0 of condition (3) is the left side for Df on U_1, so $x \mapsto Df(x)$ is compact-open continuous on U_1 `Chapter1/ExpositionLemmas · IsSc1With.deriv_uniformOnCompact`. The text's other half, that norm continuity can fail, is proved by a sequence-space example in §I.24 `Chapter1/ScOneNotNormContinuous · exists_isSc1_not_continuousOn_deriv`, not by the shift map of the exposition's §6.

Sc^k and sc-smooth. By recursion on k: sc^0 is sc^0, and f is sc^{k+1} on U if it is sc^1 on U and the chosen tangent map is sc^k on TU `Chapter1/ScSmooth · IsSck`; f is **sc-smooth** on U if it is sc^k for every k `Chapter1/ScSmooth · IsScSmooth`. An sc^k map is sc^0 `Chapter1/ScSmooth · IsSck.isSc0`, sc^{k+1} implies sc^k `Chapter1/ScSmooth · IsSck.of_succ` `Chapter1/ScSmooth · IsSck.of_le`, and f is sc-smooth iff it is sc^1 with sc-smooth tangent map `Chapter1/ScSmooth · isScSmooth_iff`. On an sc-domain the notion depends only on the values on U, by induction with the uniqueness of the tangent map `Chapter1/ChainRule · IsSc1.tangentMap_congr` `Chapter1/ChainRule · IsSck.congr`. **Finite dimensions.** The text's remark that sc^1 maps between finite-dimensional spaces are the C^1 maps holds on every sc-domain, in particular on relatively open subsets of quadrants, with boundary. Let V, W carry their constant structures and $U \subseteq V$ be an sc-domain. All level sets of U are U itself `Chapter1/ExpositionLemmas · ScBanach.lvlSet_const` and all $f \circ \iota_m$ are f. If f is sc^1, then $f \circ \iota_1 = f$ is C^1 within $U_1 = U$ by Proposition 1.2.1 (preprint 1.16) (§I.14, `Chapter1/Properties · IsSc1With.contDiffOn_base`); if f is C^1 within U, it is C^1 on every level, so sc^1 by Corollary 1.2.5 (preprint 1.20) at $k = 0$ (§I.17, `Chapter1/Properties · corollary_1_2_5_zero`) `Chapter1/ExpositionLemmas · isSc1_iff_contDiffOn_const`. Mathlib's "C^1 within U" means, on a set with unique derivatives, differentiable within U at every point with $x \mapsto$ fderivWithin $f\ U\ x$ continuous on U `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_fderivWithin` `Chapter1/ExpositionLemmas · isSc1_iff_fderivWithin_const`: at a boundary point of a quadrant this is the one-sided derivative along U, unique because the quadrant has interior (§I.10). Likewise sc-smooth is C^∞ within U, by Proposition 1.2.3 (preprint 1.18) at $m = 0$ (§I.16) and the sc-smooth case of Corollary 1.2.5 (preprint 1.20) `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`. For the text's tuples: `Chapter1/ExpositionLemmas · isSc1_iff_contDiffOn_of_isRelOpenInQuadrant` `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant`. The open case is also checked in the test file `Test · isSc1_iff_contDiffOn_of_finiteDimensional` `Test · isScSmooth_iff_contDiffOn_of_finiteDimensional`. By §I.1 every sc-structure on a finite-dimensional space is the constant one up to sc-isomorphism.

I.12 Theorem 1.3.1 (preprint 1.21): the chain rule

Statement. Let $U \subseteq C \subseteq E$ and $V \subseteq D \subseteq F$ be relatively open subsets of partial quadrants, f sc^1 on U with $f(U) \subseteq V$, and g sc^1 on V. Then $g \circ f$ is sc^1 on U and $T(g \circ f) = Tg \circ Tf$ on TU `Chapter1/ChainRule · theorem_1_3_1`. The same holds for sc-domains `Chapter1/ChainRule · theorem_1_3_1'`, and the existence part needs still less: if g is sc^1 on V with derivatives Dg, f is sc^1 on U with derivatives Df, $f(U) \subseteq V$, and V_1 is open-in-convex, then $g \circ f$ is sc^1 on U with derivatives $x \mapsto Dg(f_1(x)) \circ Df(x)$, with no hypothesis on U `Chapter1/ChainRule · IsSc1With.comp`. The text's third tuple $W \subset Q \subset G$ plays no role.

The derivative. Fix $x \in U_1$, let $y_0 := f_1(x) \in V_1$ and K a compact set containing $Df(x)(\iota_1 \bar{B}_1)$, which exists because ι_1 is compact `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.image_closedBall_subset_compact`. Write $V_1 = O \cap C$ with O open and C convex, and choose $\rho > 0$ with the ρ-ball around y_0 inside O. Given $c > 0$, two tolerances are fixed. First, by the uniform smallness of $[Dg(y) - Dg(y_0)](w + b)$ for $y \in V_1$ near y_0, $w \in K$ and b small, which is the continuity of $(y, b, w) \mapsto [Dg(y) - Dg(y_0)](w + b)$ on $V_1 \times F_0 \times K$ made uniform over K `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually Chapter1/ChainRule · IsSc1With.exists_delta`, there is δ with this expression of norm $\leq c/2$. Second, $\varepsilon := \min(\delta/2, (c/2)/(\|Dg(y_0)\| + 1))$. For $z \in U_1$ close to x three things hold eventually: $\|f_1(z) - y_0\|_1 < \min(\rho, \delta)$ (continuity of f_1 on U_1), and $r := f(\iota z) - f(\iota x) - Df(x)\iota(z - x)$ has $\|r\| \leq \varepsilon \|z - x\|_1$ (condition (2) for f `Mathlib/Analysis/Calculus/FDeriv/Defs · hasFDerivWithinAt_iff_isLittleO`). Let $k := f_1(z) - y_0 \in F_1$. The segment $y_0 + tk$, $t \in [0, 1]$, lies in V_1: inside the ρ-ball, hence in O, and in C by convexity, since both endpoints are in C. The text states this inclusion without comment; it is the one place where the structure of V enters.

The mean value inequality. The text's second integral is replaced by: for $\varphi(t) := g(\iota(y_0 + tk)) - t\, Dg(y_0)(\iota k)$ on $[0, 1]$, the derivative within $[0, 1]$ is $[Dg(y_0 + tk) - Dg(y_0)](\iota k)$ (condition (2) for g along the segment, composed with the affine curve), and if its norm is $\leq C$ on $[0, 1]$ then $\|\varphi(1) - \varphi(0)\| \leq C$ `Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasDerivWithin_le Chapter1/ChainRule · IsSc1With.norm_sub_le_of_segment`. Now write $\iota k = f(\iota z) - f(\iota x) = s(w_1 + b)$ with $s := \|z - x\|_1$, $w_1 := Df(x)\iota(u)$ for the unit vector $u := s^{-1}(z - x)$, so $w_1 \in K$, and $b := s^{-1} r$, so $\|b\| \leq \varepsilon \leq \delta/2 < \delta$. For every t, $y_0 + tk$ is within δ of y_0, so $\|[Dg(y_0 + tk) - Dg(y_0)](\iota k)\| \leq (c/2)\, s$, and the mean value inequality gives

$$\|g(f(\iota z)) - g(f(\iota x)) - Dg(y_0)(\iota k)\| \leq \tfrac{c}{2} \|z - x\|_1.$$

(When $z = x$ both sides vanish.) The text's first integral is not needed: $\|Dg(y_0) r\| \leq \|Dg(y_0)\|\, \varepsilon \|z - x\|_1 \leq \tfrac{c}{2} \|z - x\|_1$ directly, because $Dg(y_0)$ is bounded on F_0. Adding the two estimates, since $\iota k = Df(x)\iota(z - x) + r$,

$$\|g(f(\iota z)) - g(f(\iota x)) - Dg(y_0) Df(x) \iota(z - x)\| \leq c \|z - x\|_1$$

eventually, which is condition (2) for $g \circ f$ at x with the operator $Dg(y_0) \circ Df(x)$ `Chapter1/ChainRule · IsSc1With.hasFDerivWithinAt_comp`. The text's subsequence argument for the uniform convergence of the second integrand is the compactness step above; the text's appeal to Proposition 1.2.3 (preprint 1.18) to make $g : V_1 \to G_0$ of class C^1 is not used.

The tangent map. $g \circ f$ is sc^0 as a composite. The tangent map of $g \circ f$ with the composed derivatives equals $T_{Dg} g \circ T_D f$ on TU, since $(g \circ f)_1 = g_1 \circ f_1$ on U_1 `Chapter1/ChainRule · IsSc1With.tangentMapWith_comp`; the right side is a composite of sc^0 maps ($T_D f$ maps TU into TV), hence sc^0: condition (3) `Chapter1/ChainRule · IsSc1With.comp`. For the chosen derivatives, $T(g \circ f) = Tg \circ Tf$ on TU follows from the

uniqueness of the derivative on U_1 (§I.11), which is where the quadrant structure of U enters `Chapter1/ChainRule · theorem_1_3_1' Chapter1/ChainRule · theorem_1_3_1`.

Iterating. If U, V are sc-domains, f is sc^k on U with $f(U) \subseteq V$ and g is sc^k on V, then $g \circ f$ is sc^k on U: induction on k, the step being the theorem for the sc^1 parts, the induction hypothesis for $Tg \circ Tf$ on the sc-domains TU, TV, and the identity $T(g \circ f) = Tg \circ Tf$ with the independence of sc^k from the values off TU `Chapter1/ChainRule · IsSck.comp Chapter1/ChainRule · IsSck.comp_of_isRelOpenInQuadrant`. Sc-smooth maps compose to sc-smooth maps `Chapter1/ChainRule · IsScSmooth.comp`. This is the exposition's Corollary 4.A.

I.13 The compact-open lemma and the level operators $Df(x) : E_m \to F_m$

The proofs of Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) in the 2010 paper all rest on one mechanism, which the formal proof isolates.

The lemma, $k = 0$. Let $A : X \to \mathcal{L}(Y, Z)$ with $(x, y) \mapsto A(x)y$ continuous on $s \times Y$, and $\iota : Y' \to Y$ a compact operator. Then $x \mapsto A(x) \circ \iota$ is continuous on s into $\mathcal{L}(Y', Z)$ with the operator norm `Chapter1/Levels · continuousOn_clm_comp_of_isCompactOperator`. Proof: the image of the unit ball of Y' under ι lies in a compact K; by uniform continuity on compact sets `Chapter1/Levels · exists_delta_of_continuousOn`, for $x' \in s$ near x and all $w \in K$, $\|A(x')w - A(x)w\| \le \varepsilon/2$, so $\|(A(x') - A(x)) \circ \iota\| \le \varepsilon/2$ on unit vectors. This is the 2010 paper's subsequence argument (unit vectors $h_n \in E_{m+1}$ have a subsequence converging in E_m) in uniform form.

The lemma, C^k. Let s be open-in-convex with unique differentiability at every point, and $(x, y) \mapsto A(x)y$ of class C^k on $s \times Y$. Then $x \mapsto A(x) \circ \iota$ is C^k on s `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator`. Induction on k over all Z and A. For $k + 1$: the partial derivative in x, $B(x)y := D_x(A(x)y)$, is C^k in (x, y) `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn`, linear and continuous in y for $x \in s$ (differentiate the linearity identities and use uniqueness of derivatives), so $B : s \to \mathcal{L}(Y, \mathcal{L}(X, Z))$; by the induction hypothesis $x \mapsto B(x) \circ \iota$ is C^k, and so is its flip $G'(x) \in \mathcal{L}(X, \mathcal{L}(Y', Z))$. G' is the derivative of $x \mapsto A(x) \circ \iota$: for $z \in s$ near x the segment from x to z lies in s (open-in-convex), and for a unit vector $v \in Y'$ the mean value inequality applied to $\tau \mapsto A(x + \tau(z - x))(\iota v) - \tau B(x)(\iota v)(z - x)$ `Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasDerivWithin_le`, with the uniform continuity of B on $\{x\} \times \iota(\bar{B}_1)$, bounds $\|A(z)\iota v - A(x)\iota v - B(x)(\iota v)(z - x)\| \le c\|z - x\|$ uniformly in v. So $x \mapsto A(x) \circ \iota$ is differentiable within s with a C^k derivative, hence C^{k+1} `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`. The 2010 paper's Proposition 2.3 carries the same information in the multilinear maps $\Gamma(x)(h_1, \dots, h_k, h)$ and a differentiation under the integral sign; the lemma replaces both.

The level operators. Let f be sc^1 on U with derivatives D, and $x \in E_{m+1}$. The **level-m operator** $Df(x) : E_m \to F_m$ is the continuous linear T with $\iota_m \circ T = D(\iota x) \circ \iota_m$, when such T exists (it is unique by injectivity), and 0 otherwise `Chapter1/Levels · levelDeriv Chapter1/Levels · toBase_levelDeriv`; on level 0 it is $D(x)$ itself `Chapter1/Levels · levelDeriv_zero`. For $x \in U_{m+1}$ it exists: the level-m tangent map sends $(x, v) \in U_{m+1} \times E_m$ to a pair whose second component $T_m(v)$ satisfies $\iota_m T_m(v) = D(\iota x)\iota_m v$; T_m is linear (by injectivity of ι_m) and continuous (condition (3) on level m) `Chapter1/Levels · IsSc1With.exists_levelDeriv Chapter1/Levels · IsSc1With.toBase_levelDeriv`. So the level-m tangent map is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1With.levelMap_tangentMapWith`, and condition (3)

says exactly that $(x,h) \mapsto Df(x)h$ is continuous on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1With.continuousOn_levelDeriv`: this is condition (2) of Proposition 1.2.1 (preprint 1.16). By the compact-open lemma with $\iota = \mathrm{incl}_m : E_{m+1} \to E_m$, the map $x \mapsto Df(x) \circ \mathrm{incl}_m$ is continuous $U_{m+1} \to \mathcal{L}(E_{m+1}, F_m)$ in the operator norm `Chapter1/Levels · IsSc1With.continuousOn_levelDeriv_clm`: the continuity of df in condition (1) of Proposition 1.2.1 (preprint 1.16). The base case, $x \mapsto D(x) \circ \iota_1$ continuous $U_1 \to \mathcal{L}(E_1, F_0)$, is also proved directly `Chapter1/Properties · IsSc1With.continuousOn_deriv_clm`, and with it $f \circ \iota_1 : U_1 \to F_0$ is C^1 `Chapter1/Properties · IsSc1With.contDiffOn_base`.

Level-raising: differentiability of $f : U_{m+1} \to F_m$. Let U be an sc-domain. For $x \in U_{m+1}$, the map $f_m \circ \mathrm{incl}_m : U_{m+1} \to F_m$ has derivative $Df(x) \circ \mathrm{incl}_m$ at x within U_{m+1} `Chapter1/Levels · IsSc1With.hasFDerivWithinAt_level`. This is the 2010 paper's integral representation $f(x+k) - f(x) = \int_0^1 Df(x+\tau k)k\,d\tau$. For $z \in U_{m+1}$ near x and $k := z - x$, the segment $x + \tau k$ lies in U_{m+1} (open-in-convex). The integrand $g(\tau) := Df(x+\tau k)(\mathrm{incl}_m k)$ is continuous into F_m by condition (3) on level m, so its Bochner integral over $[0,1]$ exists in F_m. Its image under ι_m is the integral of $D(\iota(x+\tau k))\iota_{m+1}k$ in F_0, which by the fundamental theorem of calculus for the C^1 curve $\tau \mapsto f(\iota(x+\tau k))$ (condition (2) for f, composed with the affine curve) `Mathlib/MeasureTheory/Integral/IntervalIntegral/FundThmCalculus · integral_eq_sub_of_hasDerivAt_of_le` equals $f(\iota z) - f(\iota x) = \iota_m(f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x))$. By injectivity, $f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x) = \int_0^1 g$ in F_m. Then

$$\left\| f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x) - Df(x)\mathrm{incl}_m k \right\|_m \le \sup_\tau \|(Df(x+\tau k) - Df(x)) \circ \mathrm{incl}_m\|\, \|k\|_{m+1},$$

and the supremum tends to 0 by the norm continuity just proved. The 2010 paper forms the integral in F_0 and remarks that the integrand is continuous into F_m; here it is formed in F_m and pushed down. Consequently $f_m \circ \mathrm{incl}_m : U_{m+1} \to F_m$ is C^1 within U_{m+1} `Chapter1/Levels · IsSc1With.contDiffOn_level`, which is condition (1) of Proposition 1.2.1 (preprint 1.16) in full.

The level operator is the derivative of f_m. If $f_m : U_m \to F_m$ happens to be differentiable at $\mathrm{incl}_m x$ within U_m with derivative L, then $L = Df(x)$: both $\iota_m \circ L \circ \mathrm{incl}_m$ and $D(\iota x) \circ \iota_{m+1}$ are derivatives of $f \circ \iota_{m+1}$ at x within U_{m+1}, hence equal by uniqueness; and since incl_m has dense range, $\iota_m \circ L = D(\iota x) \circ \iota_m$ on all of E_m `Mathlib/Topology/Algebra/Module/ContinuousLinearMap/Basic · ext_on` `Chapter1/Levels · IsSc1With.levelDeriv_eq`. This is the exposition's Lemma 3.H and the key to Proposition 1.2.4 (preprint 1.19).

I.14 Proposition 1.2.1 (preprint 1.16): sc¹ in terms of the levels

The level conditions. For a family $D_m(x) \in \mathcal{L}(E_m, F_m)$, $x \in E_{m+1}$, the **level conditions** are: (a) for every m and $x \in U_{m+1}$, $f_m \circ \mathrm{incl}_m$ has derivative $D_m(x) \circ \mathrm{incl}_m$ at x within U_{m+1}; (b) for every m, $(x,h) \mapsto D_m(x)h$ is continuous on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1Levels`. These are the text's (1) and (2) with $m-1$ renamed m and the extensions $Df(x)$ made an explicit family. From them, $x \mapsto D_m(x) \circ \mathrm{incl}_m$ is norm continuous (compact-open lemma) `Chapter1/Levels · IsSc1Levels.continuousOn_clm`, so $f : U_{m+1} \to F_m$ is C^1 on an sc-domain `Chapter1/Levels · IsSc1Levels.contDiffOn`: the rest of the text's (1).

Statement. For an sc-domain U and f sc^0 on U: f is sc^1 on U iff some family D_m satisfies the level conditions `Chapter1/Levels · proposition_1_2_1`.

(□). The level operators of §I.13 satisfy (a) (level-raising) and (b) (condition (3)) `Chapter1/Levels · IsSc1With.isSc1Levels`.

(□), and what the text hides. The 2010 paper calls this direction clear. It is, once one knows that the operators $D_m(x)$ for different m are restrictions of one operator, which the text's notation $Df(x)$ assumes. Proof: for $x \in U_{m+1}$, both $\iota_m \circ D_m(x) \circ \mathrm{incl}_m$ and $D_0(\iota x) \circ \iota_{m+1}$ are derivatives of $f \circ \iota_{m+1}$ at x within U_{m+1} (compose (a) at level m with ι_m, and (a) at level 0 with the inclusion $E_{m+1} \to E_1$), so they agree by uniqueness, and since incl_m has dense range, $\iota_m \circ D_m(x) = D_0(\iota x) \circ \iota_m$ `Chapter1/Levels · IsSc1Levels.toBase_apply`. Then f is sc^1 with derivatives D_0: condition (2) is (a) at level 0; condition (3) at level m reads, after the compatibility, $(x, h) \mapsto (f_{m+1}(x), D_m(x)h)$ on $U_{m+1} \times E_m$, which is continuous by (b) and the continuity of f_{m+1} `Chapter1/Levels · IsSc1Levels.isSc1With`.

The "in particular". If f is sc^1 with derivatives D and $x \in U_1$ is a smooth point, then $D(x) : E_0 \to F_0$ is an sc-operator: lift x to $y \in U_{m+1}$; the level-m tangent map at (y, v) has second component $T_m(v)$ with $\iota_m T_m(v) = D(x)\iota_m v$, linear and continuous in v, which is the level-m map required by `IsScOperator` `Chapter1/Properties · IsSc1With.scOperator_deriv`.

I.15 Proposition 1.2.2 (preprint 1.17): lifting the index

Statement. If U is an sc-domain in E and f is sc^k on U, then $f_1 : E_1 \to F_1$ is sc^k on U_1, as a map between the shifted spaces E^1 and F^1 `Chapter1/Levels · IsSck.shift`; sc-smooth maps lift to sc-smooth maps `Chapter1/Levels · IsScSmooth.shift`. The target V of the text plays no role.

Proof. The level maps of f_1, computed in the shifted spaces, are the level maps of f shifted by one `Chapter1/Levels · IsSc0.levelMap_shift`, so an sc^0 map lifts to an sc^0 map, with no hypothesis on U `Chapter1/Levels · IsSc0.shift`. For sc^1 with derivatives D: the derivatives of f_1 are the level-1 operators $Df(x) : E_1 \to F_1$, $x \in U_2$; condition (2) for f_1 is the level-raising statement of §I.13 at $m = 1$, and condition (3) is the shift of condition (3) for f, since $T(U_1) = (TU)^1$ (§I.11) and the level tangent maps are $(x, h) \mapsto (f_{m+2}(x), Df(x)h)$ `Chapter1/Levels · IsSc1With.shift`. For sc^{k+1}: the tangent map of f_1 agrees on $T(U_1)$ with the level-1 map of Tf (uniqueness of derivatives on the sc-domain U_1), which is sc^k by the induction hypothesis applied to Tf on TU `Chapter1/Levels · IsSck.shift`. This is the 2010 paper's argument, with the compatibility $T(U^1) = (TU)^1$ and the restriction of Df proved rather than assumed.

I.16 Proposition 1.2.3 (preprint 1.18): sc^k maps are C^k from U_{m+k} to F_m

Statement. Let U be an sc-domain and f sc^k on U. Then $f_m \circ (E_{m+k} \to E_m) : U_{m+k} \to F_m$ is C^k within U_{m+k} for every m `Chapter1/Levels · IsSck.contDiffOn_levels`; $f : U_{m+l} \to F_m$ is C^l for $l \le k$ `Chapter1/Levels · IsSck.contDiffOn_levels_of_le`; in particular $f \circ \iota_k : U_k \to F_0$ is C^k `Chapter1/Levels · IsSck.contDiffOn_base`; and an sc-smooth map is C^k from U_{m+k} to F_m for all m, k `Chapter1/Levels · IsScSmooth.contDiffOn_levels`.

Proof. The 2010 paper proves the case $m = 0$ and reduces to it with Proposition 1.2.2 (preprint 1.17) applied m times. The formal proof handles all levels at once, by induction on k over all sc-domains, with a compatible level-lowering operator $\rho : E_n \to E_m$, $n = m + k$, $\iota_m \rho = \iota_n$, as a parameter `Chapter1/Levels · IsSck.contDiffOn_levels'`; this avoids identifying $(E^m)_k$ with E_{m+k}. For $k = 0$, ρ is the identity and the claim is the continuity of f_m. For $k + 1$: write $\rho = \mathrm{incl}_m \circ \rho'$ with $\rho' : E_{m+k+1} \to E_{m+1}$ `Chapter1/Levels · ScBanach.incl_shift_inclK Chapter1/Levels · ScBanach.shift_inclK_mem_lvlSet`. By the level-raising statement of §I.13, $f_m \circ \rho$ has derivative $x \mapsto Df(\rho' x) \circ \rho$ on U_{m+k+1}, where $Df(\cdot)$ is the level-m oper-

ator. It remains to see that this derivative is C^k into $\mathcal{L}(E_{m+k+1}, F_m)$. The induction hypothesis for the sc^k map Tf on the sc-domain TU, with the level-lowering operator of TE (which is the pair of those of E^1 and E `Chapter1/Levels · ScBanach.tangent_inclK`), says that the level-m tangent map composed with it, $(x,h) \mapsto (f_{m+1}(\rho' x), Df(\rho' x)(E_{m+k} \to E_m)h)$, is C^k on $U_{m+k+1} \times E_{m+k}$; its second component is $(x,h) \mapsto A(x)h$ with $A(x) := Df(\rho' x) \circ (E_{m+k} \to E_m)$. The compact-open lemma with $\iota = \mathrm{incl}_{m+k} : E_{m+k+1} \to E_{m+k}$ makes $x \mapsto A(x) \circ \mathrm{incl}_{m+k} = Df(\rho' x) \circ \rho$ of class C^k `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator`. A map that is differentiable within an sc-domain with a C^k derivative is C^{k+1} `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`.

I.17 Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20): sc^{k+1} from classical smoothness on the levels

Statement. Let U be an sc-domain and f sc^0 on U. If for every $m \geq 0$ and $0 \leq l \leq k$ the map $f_m \circ (E_{m+l} \to E_m) : U_{m+l} \to F_m$ is C^{l+1} within U_{m+l}, then f is sc^{k+1} on U `Chapter1/LevelsWeak · proposition_1_2_4_text`. This is the text's statement; its "f induces a map $f : U_{m+l} \to F_m$" is the sc^0 hypothesis (values of level m on U_m; the continuity is then the case $l = 0$). The proof is the 2010 paper's (its Proposition 2.4) and lives in `Chapter1/LevelsWeak`; an earlier, shorter proof under a stronger hypothesis is kept and described at the end of this section.

Why the hypothesis does not induct. The obvious induction would show that f is sc^1 and that Tf satisfies the hypothesis with $k-1$. It does not. The map $Tf : (TU)_{m+l} \to (TF)_m$ has the second component $(x,h) \mapsto Df(x)h$ on $U_{m+l+1} \times E_{m+l} \to F_m$, and $Df(x)$ on E_{m+l} is the derivative of $f : U_{m+l} \to F_m$ (§I.13), a C^{l+1} map; so this component is only C^l, one order short of the C^{l+1} that the hypothesis for Tf would ask. What survives the induction is a description of the components of the iterated tangent maps: they are built from the derivatives $d^j f(x)$ of f itself, evaluated at points and vectors of various levels, and the regularity of f on the coarser levels supplies the missing order. The 2010 paper's inductive statement (S_k) carries exactly this description. The formal proof organizes it in six steps: (1) the derivatives $d^j f(x)$ on all level pairs, with their compatibility, form a *derivative tower*; (2) an *elementary map* is $d^j f$ evaluated at sc-operators; (3) it is sc^0, and sc^1 with an explicit derivative, by the level criterion of Proposition 1.2.1 (preprint 1.16); (4) its tangent map is the pair of an elementary map and a sum of elementary maps; (5) by induction on s, every elementary map with $j + s \leq k+1$ is sc^s; (6) f is the elementary map with $j = 0$.

The derivative tower. A derivative tower of order $k+1$ for f on U `Chapter1/LevelsWeak · IsDerivTower` consists of base-level maps $\Phi_i(x)(h_1, \dots, h_i) \in F_0$ $(x, h_a \in E_0)$ and, for $x \in U_{m+l}$, continuous i-linear maps $\Psi_{i,m,l}(x)$ on E_{m+l} with values in F_m, given for $i \leq l+1$ and $i \leq k+1$, such that: $\Phi_0 = f$; $\iota_m \Psi_{i,m,l}(x)(h_1, \dots, h_i) = \Phi_i(\iota_{m+l} x)(\iota_{m+l} h_1, \dots, \iota_{m+l} h_i)$, i.e. all the Ψ are one map seen on different levels; $\Psi_{i,m,l}$ is continuous on U_{m+l}; and for $i \leq l$, $i \leq k$ it is differentiable within U_{m+l} at every point, with derivative $\xi \mapsto \Psi_{i+1,m,l}(x)(\xi, \cdot)$. The ranges of i are exactly what "$f : U_{m+l} \to F_m$ is C^{l+1}" provides; the compatibility is the text's use of one symbol $d^j f(x)$ on all levels. Two values with the same images in E_0 agree, by injectivity of ι_m `Chapter1/LevelsWeak · IsDerivTower.eq_of_toBase`.

The tower of a map satisfying the hypothesis. $\Psi_{i,m,l}$ is Mathlib's i-th derivative within U_{m+l} of $f_m \circ (E_{m+l} \to E_m)$ `Chapter1/LevelsWeak · levelDerivTower` `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin`, and Φ_i is the i-th derivative of $f : U_{i-1} \to F_0$ at the lifts of its arguments to E_{i-1} (arbitrary off the image of E_{i-1}) `Chapter1/LevelsWeak · baseDerivTower`. For $l > k$ the

hypothesis is used with order $k+1$: $f : U_{m+l} \to F_m$ is the C^{k+1} map $f : U_{m+k} \to F_m$ restricted along the inclusion $E_{m+l} \to E_{m+k}$, and U_{m+l} is the preimage of U_{m+k} `Chapter1/LevelsWeak · ScBanach.inclLE` `Chapter1/LevelsWeak · ScBanach.preimage_inclLE_lvlSet`; so $f : U_{m+l} \to F_m$ is $C^{\min(l,k)+1}$ for all m, l `Chapter1/LevelsWeak · contDiffOn_levels_min`. Continuity of $\Psi_{i,m,l}$ for $i \le \min(l,k)+1$ is the continuity of the iterated derivatives of a $C^{\min(l,k)+1}$ map `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn.continuousOn_iteratedFDerivWithin`; for $i \le \min(l,k)$ the i-th derivative is C^1 `Mathlib/Analysis/Calculus/ContDiff/Comp · ContDiffWithinAt.iteratedFDerivWithin_right`, and its derivative is the $(i+1)$-st derivative with the new vector in front `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin_succ_apply_left`. Both use the uniqueness of derivatives on the level sets of the sc-domain. The compatibility `Chapter1/LevelsWeak · levelDerivTower_toBase` is proved in three steps, for $i \le l+1$ (so that $i-1 \le m+l$): (i) composing on the left with the continuous linear ι_m commutes with iterated derivatives within a set `Mathlib/Analysis/Calculus/ContDiff/Basic · ContinuousLinearMap.iteratedFDerivWithin_comp_left`; (ii) on U_{m+l}, $\iota_m \circ f_m \circ (E_{m+l} \to E_m)$ equals $(f : U_{i-1} \to F_0) \circ (E_{m+l} \to E_{i-1})$, since f is sc^0 `Chapter1/ScSmooth · IsSc0.toBase_levelMap`, and a derivative within a set depends only on the values on the set `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin_congr`; (iii) composing on the right with the continuous linear inclusion $E_{m+l} \to E_{i-1}$ commutes as well, the preimage of U_{i-1} being U_{m+l} `Mathlib/Analysis/Calculus/ContDiff/Basic · ContinuousLinearMap.iteratedFDerivWithin_comp_right`. With the lifts in the definition of Φ_i `Chapter1/LevelsWeak · baseDerivTower_toBase` this gives the tower `Chapter1/LevelsWeak · isDerivTower_of_contDiffOn_levels`.

Shifted spaces. E^p is the sc-Banach space with $(E^p)_l = E_{p+l}$ by definition `Chapter1/LevelsWeak · ScBanach.shiftAdd` (the p-fold iterate of the shift of §I.1 has these levels only up to rewriting $p+l$). The identifications $E \cong E^0$ and $(E^p)^1 \cong E^{p+1}$ are sc-isomorphisms whose level maps are identities transported along $l = 0+l$, resp. $p+(l+1) = p+1+l$ `Chapter1/LevelsWeak · ScBanach.castCLE` `Chapter1/LevelsWeak · ScBanach.shiftAddZeroIso` `Chapter1/LevelsWeak · ScBanach.shiftAddShiftIso`; they are used only through their compatibility with the inclusions into the base level `Chapter1/LevelsWeak · ScBanach.toBase_shiftAddShiftIso_symm`. For $b \le a$ the inclusion $E_a \to E_b$ is compatible with the ι's `Chapter1/LevelsWeak · ScBanach.toBase_inclLE`.

Elementary maps. Elementary data on an sc-Banach space X `Chapter1/LevelsWeak · ElemData` are an sc-operator $\pi_0 : X \to E^p$ (the base point), i sc-operators $\pi_a : X \to E^{n_a}$ (the vectors) and a target shift m. The elementary map is

$$g(z) = d^i f(\pi_0 z)(\pi_1 z, \dots, \pi_i z) := \Phi_i(\pi_0 z)(\pi_1 z, \dots, \pi_i z),$$

lifted to $F_m = (F^m)_0$ `Chapter1/LevelsWeak · ElemData.map`, on the domain $W = \pi_0^{-1}(U_p)$ `Chapter1/LevelsWeak · ElemData.dom`, whose level-l set is $\pi_0^{-1}(U_{p+l})$ `Chapter1/LevelsWeak · ElemData.mem_lvlSet_dom`. The exposition's map of type $(p, l; n_1, \dots, n_j)$ on $U^p \oplus E^{n_1} \oplus \cdots \oplus E^{n_j}$ is the case $X = E^p \oplus E^{n_1} \oplus \cdots$ with the projections as slots. The conditions are `Chapter1/LevelsWeak · ElemData.Good`

$$m+i \le p, \qquad m+i \le n_a+1 \ (1 \le a \le i), \qquad i \le k.$$

The second is the text's $j+l-1 \le n_i$. The first is $j+l \le k+1$ for the base point $x \in U_{k+1}$, which the text derives from $j+l-1 \le n_i \le k$ (and states separately, as $l \le k+1$, only for $j=0$ in the exposition). The third bounds the order of derivative that will be used. The text's upper bound $n_i \le k$ is not needed.

On level l, the base point and the vectors pushed down to a common level q ($q \le p+l$, $q \le n_a+l$) are $\rho_0 = (E_{p+l} \to E_q) \circ \pi_0$ and $\rho_a = (E_{n_a+l} \to E_q) \circ \pi_a$ `Chapter1/LevelsWeak · ElemData.ρ₀` `Chapter1/LevelsWeak · ElemData.ρ`. Seen in F_0, g at a point of level l is Φ_i at the base-level images of $\pi_0 z$ and $\pi_a z$

Chapter1/LevelsWeak · ElemData.map_toBase, and its level-l map is $z \mapsto \Psi_{i,m+l,\lambda}(\rho_0 z)(\rho_1 z, \dots, \rho_i z)$ for any λ with $i \le \lambda+1$, $m+l+\lambda \le p+l$ and $m+l+\lambda \le n_a+l$ Chapter1/LevelsWeak · ElemData.levelMap_eq. The first two conditions make $\lambda = i-1$ admissible, and $i \le k$ gives $i \le k+1$, so the tower value exists. Hence g is sc^0: its values at level-l points lie in F_{m+l}, and its level-l map is continuous on W_l as the evaluation of the continuous $\Psi_{i,m+l,i-1} \circ \rho_0$ at the continuous $(\rho_1, \dots, \rho_i)$ Chapter1/LevelsWeak · ElemData.isSc0.

The sc^1 step. For $z \in W_{l+1}$ the candidate derivative $Dg(z) : X_l \to F_{m+l}$ is the derivative (3.5) of the exposition Chapter1/LevelsWeak · ElemData.deriv Chapter1/LevelsWeak · ElemData.deriv_apply:

$$Dg(z)\zeta = \Psi_{i+1,m+l,i}(\rho_0 z)(\rho_0\zeta, \rho_1 z, \dots, \rho_i z) + \sum_{a=1}^{i} \Psi_{i,m+l,i-1}(\rho_0 z)(\rho_1 z, \dots, \rho_a\zeta, \dots, \rho_i z),$$

the first term on the uniform level $m+l+i$, the sum on $m+l+i-1$. The conditions put every argument where it belongs: $\rho_0 z$ and $\rho_a z$ (with z of level $l+1$) in E_{m+l+i}, because $m+i \le p+1$ and $m+i \le n_a+1$; $\rho_0\zeta$ (with ζ of level l) in E_{m+l+i}, because $m+i \le p$; $\rho_a\zeta$ in $E_{m+l+i-1}$, because $m+i-1 \le n_a$; and $\Psi_{i+1,m+l,i}$ exists because $i+1 \le k+1$, which is where $i \le k$ enters. On the base level, $Dg(z)\zeta$ is $\Phi_{i+1}(\pi_0 z)(\pi_0\zeta, \pi z) + \sum_a \Phi_i(\pi_0 z)(\pi z$ with $\pi_a z := \pi_a\zeta)$ Chapter1/LevelsWeak · ElemData.deriv_toBase. The level criterion of Proposition 1.2.1 (preprint 1.16) (§I.14) asks two things.

Differentiability of $g : W_{l+1} \to F_{m+l}$ with derivative $Dg(z)$ restricted to X_{l+1}. On W_{l+1}, the level-l map of g is $z' \mapsto \Psi_{i,m+l,i}(\rho_0 z')(\rho_1 z', \dots, \rho_i z')$ with the slots taken at level $l+1$ (the formula above with $\lambda = i$) Chapter1/LevelsWeak · ElemData.levelMap_incl_eq. In the tower, $\Psi_{i,m+l,i}$ is differentiable with derivative $\Psi_{i+1,m+l,i}$ ($i \le i$ and $i \le k$); the slots are continuous linear; and the evaluation $(A, h) \mapsto A(h_1, \dots, h_i)$ of continuous multilinear maps is differentiable with derivative $(B, \eta) \mapsto B(h) + \sum_a A(h$ with $h_a := \eta_a)$ Chapter1/LevelsWeak · hasFDerivAt_cmm_eval, proved from Mathlib's derivative of a multilinear map in its arguments Mathlib/Analysis/Calculus/FDeriv/Analytic · ContinuousMultilinearMap.hasFDerivAt and the linearity in A (the evaluation written as a flipped multilinear map Chapter1/LevelsWeak · cmm_eval_eq). The chain rule gives a derivative whose first term is that of Dg (the base point restricted from level $l+1$ to level l is the same vector Chapter1/LevelsWeak · ElemData.ρ₀_incl) and whose sum terms are values of $\Psi_{i,m+l,i}$; these equal the $\Psi_{i,m+l,i-1}$ terms of Dg because they have the same images in E_0 (the tower's compatibility) Chapter1/LevelsWeak · ElemData.hasFDerivWithinAt_level.

Continuity of $(z, \zeta) \mapsto Dg(z)\zeta$ on $W_{l+1} \times X_l$: $\Psi_{i+1,m+l,i}$ and $\Psi_{i,m+l,i-1}$ are continuous on their level sets, and the slots, the insertion of $\rho_a\zeta$ and the evaluation are continuous Chapter1/LevelsWeak · ElemData.continuousOn_deriv.

So g satisfies the level conditions Chapter1/LevelsWeak · ElemData.isSc1Levels Chapter1/Levels · IsSc1Levels, and when W is an sc-domain it is sc^1 with derivative Dg at level 0 Chapter1/Levels · IsSc1Levels.isSc1With Chapter1/LevelsWeak · ElemData.isSc1With. This is the text's "Γ is sc^1", with the level criterion in place of the direct check of Definition 1.1.15 (preprint 1.14) that the exposition carries out.

The tangent map of an elementary map. On $TX = X^1 \oplus X$ a slot $\pi : X \to E^n$ gives two slots: π on the first factor, $TX \to E^{n+1}$ through $(E^n)^1 \cong E^{n+1}$ Chapter1/LevelsWeak · ScOperator.tangentX, and π on the second factor, $TX \to E^n$ Chapter1/LevelsWeak · ScOperator.tangentH. Three kinds of elementary data on TX arise from g. The first, g', has all slots on the first factor, with levels $p+1$, n_a+1 and target shift $m+1$ Chapter1/LevelsWeak · ElemData.shiftData: it is g one level up. The second, g_0, has the base point π_0 on the first factor (level $p+1$), $i+1$ vectors, namely π_0 on the second factor (level p) followed by the π_a on the first (levels n_a+1), and target shift m Chapter1/LevelsWeak · ElemData.derivData: it is the term $d^{i+1}f(x)(\delta x, h)$. The third, g_a for $1 \le a \le i$, is like g' but with π_a moved to the second factor (level n_a) and target

shift m Chapter1/LevelsWeak · ElemData.slotData: it is the term $d^i f(x)(h \text{ with } h_a := \delta h_a)$. All satisfy the conditions, which is the text's "one checks that the new indices satisfy the required inequalities": for g', $m+1+i \le p+1$ and $m+1+i \le n_a+2$ Chapter1/LevelsWeak · ElemData.Good.shiftData; for g_0, $m+i+1 \le p+1$ (for the base point and for the new first vector, of level p) and $m+i+1 \le n_a+2$, and $i+1 \le k$, which has to be assumed Chapter1/LevelsWeak · ElemData.Good.derivData; for g_a, $m+i \le p+1$, $m+i \le n_a+1$ for the moved slot and $m+i \le n_b+2$ for the others Chapter1/LevelsWeak · ElemData.Good.slotData. The domain of all three (they share the base point π_0 on the first factor) is TW Chapter1/LevelsWeak · ElemData.tangentSet_dom.

If W is an sc-domain and $i+1 \le k$, the tangent map of g agrees on TW with the pair (g' carried along $(F^m)^1 \cong F^{m+1}$, $g_0 + \sum_a g_a$) Chapter1/LevelsWeak · ElemData.tangentMap_eq0n. Proof: by the uniqueness of the derivative on the sc-domain W, the tangent map is (g on level 1, Dg) Chapter1/ScSmooth · IsSc1.tangentMap_eq0n; the first components agree in F_0, both being Φ_i at the base-level images of the level-1 slots Chapter1/LevelsWeak · ElemData.map_toBase Chapter1/LevelsWeak · ScBanach.toBase_shiftAddShiftIso_symm; the second components agree in F_0 term by term, by the base-level formula for Dg Chapter1/LevelsWeak · ElemData.deriv_toBase.

Sums. The constant map 0 is sc^k, being the zero sc-operator Chapter1/LevelsWeak · ScOperator.zeroOp Chapter1/LevelsWeak · IsSck.zero. If g_1, g_2 are sc^k on an sc-domain, g_1+g_2 is the addition $Y \oplus Y \to Y$ Chapter1/LevelsWeak · ScOperator.addOp, an sc-operator and hence sc-smooth Chapter2/TangentBundle · ScOperator.isScSmooth, composed with the pairing (g_1, g_2), which is sc^k Chapter2/StrongBundle · IsSck.scPair; so it is sc^k by the chain rule (§I.12, Chapter1/ChainRule · IsSck.comp) Chapter1/LevelsWeak · IsSck.add. Finite sums follow by induction over the index set Chapter1/LevelsWeak · IsSck.sum. Scalar multiples are not needed: the text's "linear combinations" are sums.

The induction (S_k)**.** *Claim:* for every s, every sc-Banach space X and all elementary data on X satisfying the conditions, with sc-domain W and $i+s \le k+1$, the elementary map is sc^s on W Chapter1/LevelsWeak · ElemData.isSck. The induction is on s, for all X, i and data at once. For $s=0$ this is the sc^0 statement. For $s+1$: g is sc^1 by the sc^1 step, and it remains to show that its tangent map is sc^s. If $s=0$, the tangent map of an sc^1 map is sc^0 Chapter1/ScSmooth · IsSc1.tangentMap_isSc0. If $s \ge 1$, then $i+1 \le k$ (from $i+s+1 \le k+1$), so the tangent map agrees on TW with (g' carried along $(F^m)^1 \cong F^{m+1}$, $g_0 + \sum_a g_a$); TW is an sc-domain Chapter1/ScSmooth · IsScDomain.tangentSet, equal to the domain of the three new data Chapter1/LevelsWeak · IsScDomain.of_eq Chapter1/LevelsWeak · IsSck.of_eq_set, and sc^s passes to maps that agree on the domain Chapter1/ChainRule · IsSck.congr. The pair is sc^s if its two components are Chapter2/StrongBundle · IsSck.scPair. Now g' and the g_a have i vectors and $i+s \le k+1$, and g_0 has $i+1$ vectors and $(i+1)+s \le k+1$, so all are sc^s by the induction hypothesis (applied on $X' = TX$); g' composed with the sc-isomorphism $(F^m)^1 \cong F^{m+1}$, an sc-operator Chapter1/ScBanach · ScIso.toScOperator, is sc^s by the chain rule; and $g_0 + \sum_a g_a$ by the sum rule.

This is the text's induction, run on s over all elementary maps rather than on the description of $T^{k+1}f$. The 2010 paper proves that every term of $\pi \circ T^{k+1}f$ is sc^1, hence $T^{k+1}f$ is, and that the terms of $T^{k+2}f$ are again elementary with admissible indices; the formal proof shows that an elementary map is sc^{s+1} because its tangent map is made of elementary maps that are sc^s. The computations are the same — the derivative (3.5), the three kinds of new terms and their indices — and the formal version never names $T^{k+1}f$ or its 2^{k+1} components.

Proposition 1.2.4 (preprint 1.19). f is the elementary map on $X = E$ with no vectors, $p = m = 0$ and $\pi_0 : E \cong E^0$ Chapter1/LevelsWeak · ElemData.base; its domain is U Chapter1/LevelsWeak · ElemData.base_dom and the conditions hold trivially Chapter1/LevelsWeak · ElemData.base_good. With the tower of f (step 1) and $s = k+1$, $0+(k+1) \le k+1$, this elementary map is sc^{k+1}; composed with the sc-isomorphism

$F^0 \cong F$ it agrees with f on U, since $\Phi_0 = f$; so f is sc^{k+1} Chapter1/LevelsWeak · proposition_1_2_4_text. If the hypothesis holds for every l, then f is sc^{k+1} for every k, i.e. sc-smooth Chapter1/LevelsWeak · IsScSmooth.of_contDiffOn_levels_text.

Corollary 1.2.5 (preprint 1.20). For a finite-dimensional target V with its constant structure, the level maps of $f : E_0 \to V$ are $f \circ \iota_m$ Chapter1/ScSmooth · levelMap_eq_of_toBase_eq Chapter1/ScBanach · ScBanach.const_toBase. Suppose $f \circ \iota_l$ is C^{l+1} within U_l for $0 \le l \le k$. Then f is sc^0 with no further hypothesis: f is continuous on $U = U_0$ (the case $l = 0$), so each $f \circ \iota_m$ is continuous on U_m. And the hypothesis of Proposition 1.2.4 (preprint 1.19) holds: $f_m \circ (E_{m+l} \to E_m) = (f \circ \iota_l) \circ (E_{m+l} \to E_l)$, a C^{l+1} map composed with a continuous linear one, on U_{m+l}, the preimage of U_l Chapter1/LevelsWeak · ScBanach.preimage_inclLE_lvlSet. So f is sc^{k+1} Chapter1/LevelsWeak · corollary_1_2_5_text. If $f \circ \iota_l$ is C^{l+1} within U_l for every l, then f is sc-smooth (for $k = 0$ the corollary gives sc^1, hence sc^0) Chapter1/LevelsWeak · IsScSmooth.of_contDiffOn_levels_finiteDimensional_text; and if f is C^∞ on U, it is sc-smooth Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional (proved with the stronger version below, since C^∞ on U restricts to C^{k+1} on every U_m). The two sc-smooth statements are the exposition's; the second is what the theory uses (bump functions, Chapter 5).

The earlier version, under a stronger hypothesis. `Chapter1/Levels` proves Proposition 1.2.4 (preprint 1.19) under the hypothesis that every level map $f_m : U_m \to F_m$ is C^{k+1} within U_m Chapter1/Levels · proposition_1_2_4. For $k \ge 1$ this asks more than the text (C^{k+1} instead of C^1 on $U_m \to F_m$ itself), and in exchange the induction needs no (S_k); for $k = 0$ the hypotheses coincide.

$k = 0$. Suppose every f_m is C^1 within U_m. Put $Df(x) :=$ the derivative of $f = f_0$ within U at $\iota_1 x$, for $x \in E_1$ Mathlib/Analysis/Calculus/FDeriv/Defs · fderivWithin. Condition (2) holds because $f \circ \iota_1$ is the restriction of the differentiable f along the continuous linear ι_1. For condition (3) at level m: the derivative $df_m(x)$ of f_m within U_m at $x \in U_m$ is compatible with Df, $\iota_m \circ df_m(x) = Df(\iota_m x) \circ \iota_m$, because $\iota_m \circ f_m = f \circ \iota_m$ on U_m and derivatives within U_m are unique Mathlib/Analysis/Calculus/FDeriv/Basic · UniqueDiffWithinAt.eq; so the level-m tangent map is $(x, h) \mapsto (f_{m+1}(x), df_m(\mathrm{incl}_m x)h)$, continuous on $U_{m+1} \times E_m$ because f_m is C^1 (its derivative is continuous into $\mathcal{L}(E_m, F_m)$, and evaluation is continuous) Chapter1/Properties · proposition_1_2_4_zero. This is the text's proof of (S_0).

$k + 1$ *from* k. Suppose every f_m is C^{k+2}. Then f is sc^1 by the case $k = 0$, with the derivative Df above, and we must show Tf is sc^{k+1} on TU. By the induction hypothesis (for the sc-domain TU and the sc^0 map Tf) it suffices that the level maps of Tf are C^{k+1}. The level-m map of Tf is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ with the level operator (§I.13), and the level operator at $x \in U_{m+1}$ is the derivative of f_m at $\mathrm{incl}_m x$ (Lemma 3.H, Chapter1/Levels · IsSc1With.levelDeriv_eq). So the level-m map of Tf is $(x, h) \mapsto (f_{m+1}(x), df_m(\mathrm{incl}_m x)h)$ on $U_{m+1} \times E_m$, whose components are C^{k+1}: the first as f_{m+1} is, the second as the derivative of a C^{k+2} map is C^{k+1} and evaluation is bilinear Chapter1/Levels · proposition_1_2_4. This is where the stronger hypothesis is used, and where the text's hypothesis would fail (see "Why the hypothesis does not induct").

Sc-smooth, and the corollary. If f maps each U_m into F_m and every f_m is C^∞, then f is sc-smooth Chapter1/Levels · IsScSmooth.of_contDiffOn_levels. For a finite-dimensional target: C^1 on every U_m gives sc^1 Chapter1/Properties · corollary_1_2_5_zero, C^{k+1} on every U_m gives sc^{k+1} Chapter1/Levels · corollary_1_2_5, in particular C^{k+1} on $U = U_0$ does, since restricting along ι_m preserves C^{k+1} Chapter1/Levels · corollary_1_2_5_base.

I.18 Definition 1.3.3 (preprint 1.23) and Lemma 1.3.4 (preprint 1.24): the degeneracy index

Definitions. For $p = (a, w) \in \mathbb{R}^n \oplus W$ let $\#(p)$ be the number of indices i with $a_i = 0$ `Chapter1/Degeneracy · quadrantDeg`. For an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ presenting a partial quadrant C and $x \in E_0$, let $d_T(x) := \#(T_0 x)$ `Chapter1/Degeneracy · degIndexWith`. The **degeneracy index** $d_C(x)$ is $d_T(x)$ for a presentation T of C chosen once and for all, and 0 if C is not a partial quadrant `Chapter1/Degeneracy · degIndex`; by Lemma 1.3.4 (preprint 1.24) below it equals $d_T(x)$ for every presentation and every $x \in C$ `Chapter1/Degeneracy · degIndex_eq`, and on the quadrant itself it is $\#$ `Chapter1/Degeneracy · degIndex_quadrant`. The text's interior points, boundary points and corners are the points with $d_C = 0$, 1, ≥ 2; Chapter 2 makes the boundary a definition.

Lemma 1.3.4 (preprint 1.24), core form. Let $S : \mathbb{R}^n \times W_0 \to \mathbb{R}^{n'} \times W_0'$ be a continuous linear equivalence with $q \in Q \iff Sq \in Q'$ for the two quadrants. Then $\#(p) = \#(Sp)$ for every $p \in Q$ `Chapter1/Degeneracy · quadrantDeg_eq_of_equiv`. Only level 0 and linear algebra are used; the sc-structure plays no role. Proof. For $p \in Q$ let $E_p := \{(b, v) : b_i = 0 \text{ whenever } p_i = 0\}$, the kernel of the surjection $(b, v) \mapsto (b_i)_{i \in I(p)}$ `Chapter1/Degeneracy · coordMap` `Chapter1/Degeneracy · coordMap_surjective` `Chapter1/Degeneracy · mem_ker_coordMap`, so that $\dim(V/E_p) = \#(p)$ `Chapter1/Degeneracy · finrank_quotient_ker_coordMap`: this makes the text's "codimension of E_x" precise. For $u \in E_p$, $p + tu \in Q$ for all t near 0: coordinate by coordinate, if $p_i = 0$ then $u_i = 0$ and the coordinate stays 0; if $p_i > 0$ it stays positive for small t by continuity `Chapter1/Degeneracy · eventually_add_smul_mem_quadrant`. Hence $Sp + tSu \in Q'$ for t near 0, so for each i with $(Sp)_i = 0$ we have $0 \leq t(Su)_i$ for all t near 0, of both signs, which forces $(Su)_i = 0$ `Chapter1/Degeneracy · eq_zero_of_eventually_nonneg_mul`; so $S(E_p) \subseteq E'_{Sp}$ `Chapter1/Degeneracy · map_ker_coordMap_le`. The same argument for S^{-1} gives the reverse inclusion (the text concludes equality from one inclusion "since S is an isomorphism", which for infinite-dimensional W needs this second step), so $S(E_p) = E'_{Sp}$ and S induces a linear isomorphism $V/E_p \cong V'/E'_{Sp}$; comparing dimensions, $\#(p) = \#(Sp)$.

Lemma 1.3.4 (preprint 1.24). If T and T' both present C, then $S := T_0' \circ T_0^{-1}$ carries the quadrant onto the quadrant (both are the image of C), and the core form at $p = T_0 x$ gives $d_T(x) = d_{T'}(x)$ for $x \in C$ `Chapter1/Degeneracy · lemma_1_3_4`. The text proves first that $k = k'$; here this is a consequence: $T_0^{-1}(0) \in C$ and $\#(0) = n$ in $\mathbb{R}^n \oplus W$ `Chapter1/Degeneracy · quadrantDeg_zero`, so the lemma at this point gives $n = n'$ `Chapter1/Degeneracy · lemma_1_3_4_dim`.

I.19 Theorem 1.3.5 (preprint 1.25): boundary recognition

Statement. Let $U \subseteq C \subseteq E$ and $U' \subseteq C' \subseteq E'$ be relatively open subsets of partial quadrants, f sc^1 on U with $f(U) \subseteq U'$, g sc^1 on U' with $g(U') \subseteq U$, $g \circ f = \mathrm{id}$ on U and $f \circ g = \mathrm{id}$ on U'. Then $d_C(x) = d_{C'}(f(x))$ for every $x \in U$ `Chapter2/DegeneracyIndex · theorem_1_3_5`. The text's "germ of a local sc-diffeomorphism" is undefined in Chapter 1 and is read this way, with U, U' as small as one likes; the text has sc-smooth where sc^1 suffices, so the formal statement is stronger. The text gives no proof, referring to "[20], Theorem 1.19"; the formal proof is the argument of §2.4 (preprint §2.3) (Lemma 2.4.9 (preprint 2.29), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)) specialized to the identity retraction, and lives in the Chapter 2 file because it uses the retraction vocabulary. Smooth points are not needed; level-1 points suffice.

The subspaces E_x. For a presentation T of C and $x \in E_0$, let $E_x := \{v \in E_0 : (T_0 v)_i = 0 \text{ whenever } (T_0 x)_i = 0\}$, the kernel of a continuous linear map to a finite-dimensional space `Chapter2/`

DegeneracyIndex · minimalSubspaceWith; its codimension is $d_T(x)$ Chapter2/DegeneracyIndex · finrank_quotient_minimalSubspaceWith. Two facts about the index: it is upper semicontinuous, $d_T(y) \le d_T(x)$ for y near x (a nonzero coordinate stays nonzero) Chapter2/DegeneracyIndex · eventually_degIndexWith_le; and every $x \in C$ is the limit of points of C on level 1 with the same index, namely $T_0^{-1}(a, w_n)$ with $w_n \in W_1$ converging to w in W_0, which exist by density Chapter2/DegeneracyIndex · exists_tendsto_lvl1_degIndexWith. These are the exposition's Lemma 5.A, with level 1 in place of smooth.

The sign argument. Let f be sc^1 on $U \subseteq C$ with values in C', $x \in U_1$, and $v \in E_1$ with $\iota_1 v \in E_x$. Then $x + tv \in U_1$ for t near 0 (the quadrant coordinates as in §I.18, and the openness of U_1) Chapter2/DegeneracyIndex · eventually_add_smul_mem_lvlSet, so $f(\iota(x + tv)) \in C'$ and, for each j with $(T_0' f(x))_j = 0$, the function $t \mapsto (T_0' f(\iota(x + tv)))_j$ is ≥ 0 with a zero at $t = 0$: a local minimum, so its derivative at 0, which is $(T_0' Df(x)\iota_1 v)_j$ by condition (2), vanishes Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.hasDerivAt_eq_zero Chapter2/DegeneracyIndex · IsSc1With.deriv_mem_minimalSubspace_of_lvl1. So $Df(x)$ maps $E_x \cap \iota_1(E_1)$ into $E'_{f(x)}$. Since f is differentiable only along E_1, the density of $E_x \cap \iota_1(E_1)$ in E_x is needed to conclude $Df(x)(E_x) \subseteq E'_{f(x)}$; it holds because E_x is the kernel of a continuous linear map to a finite-dimensional space and $\iota_1(E_1)$ is dense Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace. The text asserts the corresponding density ("the closure of $(T_x^R O)_\infty$ is $T_x^R O$") without proof.

One inequality at level-1 points. With g the inverse, the chain rule gives $Dg(f_1(x)) \circ Df(x) = \mathrm{id}$ on E_0 at $x \in U_1$ Chapter2/DegeneracyIndex · scDeriv_comp_eq_self, so $Df(x)$ is injective and $E_x \cong Df(x)(E_x) \le E'_{f(x)}$ sits inside a subspace of codimension $d_{C'}(f(x))$, while $Dg(f_1(x))$ maps $E'_{f(x)}$ into E_x by the sign argument for g; comparing codimensions, $d_C(x) \le d_{C'}(f(x))$ Chapter2/DegeneracyIndex · degIndexWith_le_of_germ. Applying this to g at $f_1(x)$ gives the reverse inequality, so $d_C(x) = d_{C'}(f(x))$ for $x \in U_1$ Chapter2/DegeneracyIndex · theorem_1_3_5_lvl1.

General points. For $x \in U$, approximate x by level-1 points $x_n \in U$ of C with $d_C(x_n) = d_C(x)$; then $f(x_n) \to f(x)$ (continuity of f on U), so $d_{C'}(f(x_n)) \le d_{C'}(f(x))$ eventually by semicontinuity, and $d_{C'}(f(x_n)) = d_C(x_n) = d_C(x)$ by the level-1 case; hence $d_C(x) \le d_{C'}(f(x))$. The symmetric argument for g at $f(x)$ gives equality Chapter2/DegeneracyIndex · theorem_1_3_5.

I.20 What the formalization taught us

- **The chain rule needs one integral less than the text writes, and one hypothesis more than it states.** The term $Dg(f(x))[f(x+h) - f(x) - Df(x)h]$ is $o(|h|_1)$ because $Dg(f(x))$ is bounded, with no integral Chapter1/ChainRule · IsSc1With.hasFDerivWithinAt_comp; the segment from $f(x)$ to $f(x+h)$ lies in V_1 only because V_1 is open-in-convex and $f : U_1 \to F_1$ is continuous, which the text does not say. The whole proof uses, of the quadrant structure of U, only the uniqueness of derivatives, and that only for $T(g \circ f) = Tg \circ Tf$ Chapter1/ChainRule · theorem_1_3_1'.
- **"The" derivative $Df(x)$ is a theorem, not a notation.** It is unique because E_1 is dense in E_0 and U_1 has unique differentiability Chapter1/ScSmooth · IsSc1With.eq_of_uniqueDiffWithinAt; without it the tangent map, hence sc^2, is undefined. The same density argument shows that the extensions $Df(x) : E_{m-1} \to F_{m-1}$ of Proposition 1.2.1 (preprint 1.16), written with one symbol in the text, are restrictions of one operator Chapter1/Levels · IsSc1Levels.toBase_apply, which is what makes the "clear" direction of that proposition work.
- **One compactness argument carries §1.2.** The compact-open lemma Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator turns joint continuity (or C^k-ness) of $(x, h) \mapsto Df(x)h$ into norm conti-

nuity (or C^k-ness) of $x \mapsto Df(x)|_{E_{m+1}}$; Propositions 1.2.1, 1.2.2 and 1.2.3 (preprint 1.16, 1.17 and 1.18) are this lemma plus bookkeeping of levels.

- **Proposition 1.2.4 (preprint 1.19) does not induct on its own hypothesis.** If f satisfies it with k, then Tf in general does not satisfy it with $k-1$: the second component $(x,h) \mapsto Df(x)h : U_{m+l+1} \times E_{m+l} \to F_m$ is one order short. The stronger hypothesis "C^{k+1} on every level" makes the induction two lines `Chapter1/Levels · proposition_1_2_4`; the text's needs the 2010 paper's (S_k), which remembers f through the elementary maps $d^j f$ on mixed levels `Chapter1/LevelsWeak · ElemData.isSck` `Chapter1/LevelsWeak · proposition_1_2_4_text`. Written out, (S_k) needs three things the text leaves implicit: one $d^j f(x)$ on all levels `Chapter1/LevelsWeak · isDerivTower_of_contDiffOn_levels`, the base-point condition $j + l \le k+1$ in place of the bound $n_i \le k$ (which is not needed) `Chapter1/LevelsWeak · ElemData.Good`, and the derivative of the multilinear evaluation `Chapter1/LevelsWeak · hasFDerivAt_cmm_eval`.
- **Regularizing is a bootstrap.** In §1.4.1 the step "$e \in E_m$ by Proposition 1.9" needs m applications, since Se is only one level above e `Chapter1/ScFredholm · ScPlus.regularizing_add`.
- **The index is an addition, and an easy one once additivity is there.** Neither Proposition 1.1.11 (preprint 1.10) nor the remark after Definition 1.1.9 (preprint 1.8) mentions the index of $T + S$ or of the level operators; with the additivity of the index `Chapter1/FredholmIndex · linIndex_comp` and a quasi-inverse, the index is locally constant along $u + tk$, hence invariant `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`, and the sc-statements follow on every level `Chapter1/FredholmIndex · proposition_1_1_11_index_lvl`.
- **The text's quick remarks are all true, and some need real arguments.** "Precisely one sc-structure" in finite dimensions `Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional`; "$(E/A)_\infty$ consists of the classes of E_∞", a Mittag-Leffler series `Chapter1/ExpositionLemmas · ScBanach.exists_smooth_mem_coset`; "it is a sc-projection" `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_iff_exists_scProj`; "sc^1 is C^1 in finite dimensions", which at the boundary of a quadrant means derivatives within U `Chapter1/ExpositionLemmas · isSc1_iff_fderivWithin_const`.
- **Mathlib had Fredholm operators but not Riesz–Schauder.** The theorem that $1 +$ compact is Fredholm, and the stability under compact perturbations, had to be proved `Chapter1/FredholmPerturbation · isFredholm_one_add` `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`; the finite codimension of the range comes from a chain of closed subspaces and Riesz's lemma `Chapter1/FredholmPerturbation · coFG_range_one_add`.
- **The degeneracy index knows nothing of the sc-structure.** Lemma 1.3.4 (preprint 1.24) holds for any linear equivalence of the base levels carrying one quadrant onto the other `Chapter1/Degeneracy · quadrantDeg_eq_of_equiv`, and $k = k'$ is its value at the origin `Chapter1/Degeneracy · lemma_1_3_4_dim`. Boundary recognition needs only sc^1 and only level-1 points, not smooth ones `Chapter2/DegeneracyIndex · theorem_1_3_5`.
- **Example 1.1.3 (preprint 1.2) holds with exactly the printed hypotheses, and costs a small Sobolev theory.** $\delta_0 = 0$ and strict monotonicity are all that is used; $\delta_m \ge 0$ follows from them `Chapter1/CylinderScBanach · GapP1Sobolev.nonneg_of_strictMono`. Strict monotonicity is used once, for the tail estimate behind compactness `Chapter1/CylinderCompactInclusions · GapP1Sobolev.tail_le`. Mathlib has no Sobolev spaces, so Sobolev's compact embedding is proved in the one case needed, on a bounded cylinder, by Fourier series on a box `Chapter1/CylinderRellich · GapP1Sobolev.rellich`.
- **The text's remark that Df need not be norm continuous needs an example, and a sequence space suffices.** The factor k in the derivative of $\cos(kt)x_k$ is absorbed by one level of the weighted c_0 scale `Chapter1/ScOneNotNormContinuous · ScOneExample.isSc1With`, while unit vectors at high frequency keep Df from being norm continuous `Chapter1/ScOneNotNormContinuous · ScOneExample.two_le_norm_D_sub`. The shift map of the 2010 paper is not needed.

- **"Regularizing" characterizes sc-Fredholm operators among those Fredholm on level 0.** The text does not state the converse of Proposition 1.1.10 (preprint 1.9); its §1.4.1 proves it in one case. As a criterion it is what the later chapters need for their unargued "is sc-Fredholm" `Chapter1/ScFredholmCriterion · isScFredholm_iff` `Chapter1/ScFredholmCriterion · IsScFredholm.restrict`. The continuity in Definition 1.1.4 (preprint 1.3) is automatic, by the closed graph theorem `Chapter1/ScFredholmCriterion · isScOperator_of_mapsTo`.
- **Remark 1.1.17 (preprint 1.15) needs less than it assumes.** For the calculus of §§1.2–1.3, a partial cone needs only convexity and nonempty interior; closedness and the cone property play no role `Chapter1/PartialCones · IsPartialCone.isScDomain_of_convex`. What does not carry over is the degeneracy index and everything built on it, which use quadrant coordinates.

I.21 The 2021 edition: Remark 1.1.2 and Proposition 1.1.11 with the index

The Springer edition (2021) changes Chapter 1 in two places. Remark 1.1.2 (p. 5) is new, and the statement of Proposition 1.1.11 (p. 9; preprint 1.10) now includes the index.

Remark 1.1.2. The remark makes "Banach space" mean a *Banachable* space, a vector space whose topology is induced by some complete norm, and says that all norms inducing the topology are equivalent "in view of the open mapping theorem". The formalization models one vector space with two norms as two normed spaces E_1, E_2 and a linear equivalence $e : E_1 \to E_2$, the identity of the underlying space. It records the remark's content in two readings. *(1) Same topology, equivalent norms* `Chapter1/Book2021 · remark_1_1_2`. If e and e^{-1} are continuous, they are bounded, $\|ex\| \le \|f\|\,\|x\|$ and $\|x\| = \|g(ex)\| \le \|g\|\,\|ex\|$ for the continuous linear maps f, g they define, and $c = \max(\|f\|, \|g\|) + 1$ serves for both inequalities. No completeness enters. *(2) The open mapping theorem* `Chapter1/Book2021 · remark_1_1_2_openMapping`. If both norms are complete and e is continuous, the bijective continuous linear map e is a linear homeomorphism by the open mapping theorem `Mathlib/Analysis/Normed/Operator/Banach · ContinuousLinearEquiv.ofBijective`. So e^{-1} is continuous, and (1) applies. As we read it, the sentence of the remark is true without the open mapping theorem. The theorem is what makes two complete norms equivalent when one of them dominates the other.

Proposition 1.1.11 (preprint 1.10), 2021 statement. Let $T : E \to F$ be sc-Fredholm and $S : E \to F$ an sc$^+$-operator. Then $T + S$ is sc-Fredholm, $\mathrm{ind}(T + S) = \mathrm{ind}(T)$, and every level map $(T + S)_m : E_m \to F_m$ is Fredholm of index $\mathrm{ind}(T)$ `Chapter1/Book2021 · proposition_1_1_11`. Nothing new has to be proved, because the formal statement is the conjunction of three earlier theorems: - the sc-Fredholm property of §I.9 `Chapter1/ScFredholm · proposition_1_1_11_preprint`; - the index on level 0, from the invariance of the Fredholm index under compact perturbations `Chapter1/FredholmIndex · proposition_1_1_11_index`; - the index on every level `Chapter1/FredholmIndex · proposition_1_1_11_index_lvl`.

The 2021 proof in Appendix 1.4.1 is the earlier one and does not argue the index. The index statement, the exposition's addition in the earlier editions, is thus now the book's own. The level statement remains the formalization's.

I.22 Example 1.1.3 (preprint 1.2): the weighted Sobolev spaces $H^{m,\delta_m}(\mathbb{R}\times S^1)$

Statement. Let $\delta_0 = 0 < \delta_1 < \delta_2 < \cdots$. The spaces $E_m = H^{m,\delta_m}(\mathbb{R}\times S^1)$ form an sc-Banach space whose level m is H^{m,δ_m}, whose inclusions are the identity on the underlying functions, and whose level 0 is $L^2(\mathbb{R}\times S^1)$ with its own norm `Chapter1/CylinderScBanach · GapP1Sobolev.example_1_1_3` `Chapter1/CylinderScBanach · GapP1Sobolev.example_1_1_3_lvl` `Chapter1/CylinderScBanach · GapP1Sobolev.level0_isometry` `Chapter1/CylinderScBanach · GapP1Sobolev.level0_surjective`. These are the text's hypotheses as printed. Nonnegativity of the weights, which every estimate below uses, follows from $\delta_0 = 0$ and monotonicity `Chapter1/CylinderScBanach · GapP1Sobolev.nonneg_of_strictMono`. The text names the two ingredients, Sobolev's compact embedding on bounded domains and the strict monotonicity of (δ_m), and leaves the rest. Mathlib has neither Sobolev spaces nor the compact embedding, so the formal proof builds both; it takes thirteen files.

The cylinder and weak derivatives. $\mathbb{R}\times S^1$ is the strip $\mathbb{R}\times(0,1]$ with Lebesgue measure `Chapter1/CylinderWeakDerivatives · GapP1Sobolev.cyl`. The circle enters through the test functions: a test function is smooth, 1-periodic in t, and zero for $|s|$ large `Chapter1/CylinderWeakDerivatives · GapP1Sobolev.IsTest`. A function g is the weak derivative $\partial_s^a\partial_t^b u$ if $\int u\,\partial_s^a\partial_t^b\varphi = (-1)^{a+b}\int g\,\varphi$ for every test function φ `Chapter1/CylinderWeakDerivatives · GapP1Sobolev.HasWeakDeriv`. This is the distributional derivative on the cylinder, and it is unique almost everywhere `Chapter1/CylinderFourier · GapP1Sobolev.HasWeakDeriv.ae_eq`, because an L^2 function orthogonal to all test functions vanishes `Chapter1/CylinderFourier · GapP1Sobolev.ae_eq_zero_of_forall_test`; the proof of that uses the Fourier coefficients below.

The spaces. With the weight $e^{\delta|s|}$ `Chapter1/CylinderSobolevSpaces · GapP1Sobolev.wt`, $u \in H^{m,\delta}$ means that for all $a+b \le m$ the weak derivative $\partial_s^a\partial_t^b u$ exists and $e^{\delta|s|}\partial_s^a\partial_t^b u \in L^2$ `Chapter1/CylinderSobolevSpaces · GapP1Sobolev.MemH`. These u form a subspace of L^2 `Chapter1/CylinderSobolevSpaces · GapP1Sobolev.Hsub`, and $H^{m,\delta}$ carries the norm

$$\|u\|_{m,\delta}^2 = \sum_{a+b\le m} \|e^{\delta|s|}\partial_s^a\partial_t^b u\|_{L^2}^2$$

`Chapter1/CylinderSobolevSpaces · GapP1Sobolev.H` `Chapter1/CylinderSobolevSpaces · GapP1Sobolev.norm_sq_H`. The text fixes no norm; this one is among the usual equivalent ones. By definition $u \mapsto (e^{\delta|s|}\partial^p u)_{|p|\le m}$ is an isometry into a finite product of copies of L^2 `Chapter1/CylinderSobolevSpaces · GapP1Sobolev.isometry_ΦH`. Its range is closed, because weak derivatives pass to L^2 limits: if $e^{\delta|s|}\partial^p u_n \to w_p$, then $e^{-\delta|s|}w_p$ is the weak derivative ∂^p of $e^{-\delta|s|}w_0$, by continuity of the pairing with a test function `Chapter1/CylinderSobolevComplete · GapP1Sobolev.isClosed_range_ΦH`. So $H^{m,\delta}$ is complete `Chapter1/CylinderSobolevComplete · GapP1Sobolev.completeSpace_H`. For $m \le m'$ and $\delta \le \delta'$ the identity of L^2 maps $H^{m',\delta'}$ into $H^{m,\delta}$ with norm at most 1, since $e^{\delta|s|} \le e^{\delta'|s|}$ `Chapter1/CylinderSobolevComplete · GapP1Sobolev.inclH` `Chapter1/CylinderSobolevComplete · GapP1Sobolev.valCLM_inclH`; it is injective, because both sides are subspaces of L^2 `Chapter1/CylinderSobolevComplete · GapP1Sobolev.inclH_injective`.

Fourier series on a box. On a box $(a, a+T]\times(0,1]$ the coefficients $\hat F_{jk} = T^{-1}\int e^{-2\pi i(js/T+kt)}F$ `Chapter1/CylinderFourier · GapP1Sobolev.bc` satisfy Parseval's identity, $\sum_{j,k}|\hat F_{jk}|^2 = T^{-1}\int_{\text{box}}|F|^2$ `Chapter1/CylinderFourier · GapP1Sobolev.hasSum_bc`. Mathlib has Parseval on one circle; the box is a product, and the identity follows by summing the t-Parseval identity of each slice $F(s,\cdot)$ and applying the s-Parseval identity to each t-coefficient. For a function vanishing near the ends of the box, the coefficients of a weak derivative are those of the function times $2\pi ij/T$, respectively $2\pi ik$ `Chapter1/CylinderFourier · GapP1Sobolev.`

bc_ds Chapter1/CylinderFourier · GapP1Sobolev.bc_dt: test the weak derivative against a cut-off times the exponential.

Rellich on a bounded cylinder. For $R > 0$ and C, the set of truncations $1_{|s|\le R}v$ of the $v \in L^2$ with $\|v\|, \|\partial_s v\|, \|\partial_t v\| \le C$ is totally bounded in L^2 Chapter1/CylinderRellich · GapP1Sobolev.rellich Chapter1/CylinderRellich · GapP1Sobolev.H1ball Chapter1/CylinderRellich · GapP1Sobolev.cutR. This is the case of Sobolev's compact embedding that the example needs. *Proof.* Take a smooth cut-off χ equal to 1 on $[-R, R]$ and supported in a box of length $T = 2R + 4$ Chapter1/CylinderRellich · GapP1Sobolev.exists_cutoff'. Then $w = \chi v$ has weak derivatives bounded by a constant times C (Leibniz), and by Parseval and the coefficient formulas, $\sum_{j,k}(j^2 + k^2)|\hat{w}_{jk}|^2 \le M$ with M depending only on C and R Chapter1/CylinderRellich · GapP1Sobolev.summable_freq_bc. The coefficients outside the square $|j|, |k| \le N$ therefore carry L^2 mass at most M/N^2 Chapter1/CylinderRellich · GapP1Sobolev.tsum_sub_le Chapter1/CylinderRellich · GapP1Sobolev.exists_N_eta. And $\|1_{|s|\le R}v\| \le \|w\|_{\text{box}}$ Chapter1/CylinderRellich · GapP1Sobolev.norm_cutR_sq_le. So, up to ε, $1_{|s|\le R}v$ is determined by finitely many bounded complex coordinates, and a set approximated in this way is totally bounded Chapter1/CylinderRellich · GapP1Sobolev.totallyBounded_image_of_coords Chapter1/CylinderRellich · GapP1Sobolev.totallyBounded_of_approx.

The inclusion is compact. For $\delta < \delta'$ the inclusion $H^{m+1,\delta'} \to H^{m,\delta}$ is a compact operator Chapter1/CylinderCompactInclusions · GapP1Sobolev.isCompactOperator_inclH. *Proof.* Let u lie in the unit ball of $H^{m+1,\delta'}$ and $|p| \le m$. The derivative $v = \partial^p u$ has weak derivatives $\partial_s v$, $\partial_t v$ of order at most $m+1$, and all three have L^2 norm at most $\|u\| < 1$, since $e^{\delta'|s|} \ge 1$ Chapter1/CylinderCompactInclusions · GapP1Sobolev.dvCLM_mem_H1ball. Split the component $e^{\delta|s|}v$ of u in $H^{m,\delta}$:

$$e^{\delta|s|}v = 1_{|s|\le R}\, e^{\delta|s|} \cdot 1_{|s|\le R}v \;+\; (1 - 1_{|s|\le R})\, e^{\delta|s|}e^{-\delta'|s|} \cdot e^{\delta'|s|}v.$$

The second term has norm at most $e^{-(\delta'-\delta)R}$, since the multiplier is bounded by that Chapter1/CylinderCompactInclusions · GapP1Sobolev.tail_le and $\|e^{\delta'|s|}v\| < 1$. Here the strict inequality $\delta < \delta'$ enters, as the text says. Given ε, choose R with $e^{-(\delta'-\delta)R} < \varepsilon$. The first term lies in the image of the totally bounded set of Rellich's theorem under multiplication by the bounded function $1_{|s|\le R}e^{\delta|s|}$ Chapter1/CylinderCompactInclusions · GapP1Sobolev.cutWt Chapter1/CylinderCompactInclusions · GapP1Sobolev.abs_cutWt_le. So the set of the p-th components is within ε of a totally bounded set for every ε, hence totally bounded. The product of the closures over the finitely many p is compact, and the isometry $u \mapsto (e^{\delta|s|}\partial^p u)_p$ pulls total boundedness back Chapter1/CylinderCompactInclusions · GapP1Sobolev.totallyBounded_of_isometry.

Test functions lie in every level. A smooth function periodic in t has its classical derivatives as weak derivatives, by integration by parts in s (a test function has bounded support) and in t (periodicity) Chapter1/CylinderIntegrationByParts · GapP1Sobolev.SmPer.hasWeakDeriv. A test function and all its derivatives are bounded and supported in a bounded strip, so they are in L^2 with every weight: $\varphi \in H^{m,\delta}$ for all m and δ Chapter1/CylinderIntegrationByParts · GapP1Sobolev.IsTest.mem_Hsub.

Test functions are dense in every level Chapter1/CylinderTestDensity · GapP1Sobolev.dense_test. This is the density of smooth points that Definition 1.1.1 (preprint 1.1) asks for. *Proof.* Two steps. - *Cut off in s.* For a smooth $\chi(s)$ whose derivatives up to order m are bounded, $u \mapsto \chi u$ maps $H^{m,\delta}$ to itself Chapter1/CylinderMultipliers · GapP1Sobolev.mulH, and its weak derivatives are given by the Leibniz rule in s Chapter1/CylinderLeibniz · GapP1Sobolev.leibniz Chapter1/CylinderMultipliers · GapP1Sobolev.dv_mulH. With $\chi_n(s) = \rho(s/(n+1))$, ρ a smooth bump equal to 1 on $[-1, 1]$ Chapter1/CylinderCutoff · GapP1Sobolev.chi, $\chi_n u \to u$ in $H^{m,\delta}$ Chapter1/CylinderCutoff · GapP1Sobolev.tendsto_cut. The term $(\chi_n - 1)e^{\delta|s|}\partial^p u$ tends to 0 by dominated convergence, and the terms with derivatives of χ_n carry a factor $(n+1)^{-j}$. - *Approx-*

imate a function of bounded support. If the weak derivatives of $v \in H^{m,\delta}$ vanish for $|s| > \rho$, take a box containing $[-\rho,\rho] \times (0,1]$, the partial Fourier sums $S_N v$ over $|j|,|k| \le N$ `Chapter1/CylinderTrigPolynomials · GapP1Sobolev.trig`, and a smooth window χ equal to 1 on $[-\rho,\rho]$ and zero near the ends of the box. Then $\chi \cdot \operatorname{Re} S_N v$ is a test function. Its derivatives are χ times the partial Fourier sums of the derivatives of v, plus terms with derivatives of χ `Chapter1/CylinderTrigPolynomials · GapP1Sobolev.pD_trig` `Chapter1/CylinderTrigPolynomials · GapP1Sobolev.bc_eexp`. Each partial sum converges in L^2 of the box by Parseval `Chapter1/CylinderTrigPolynomials · GapP1Sobolev.tendsto_box_sq`, and the weight is bounded on the box. So $\chi \cdot \operatorname{Re} S_N v \to \chi v = v$ in $H^{m,\delta}$ `Chapter1/CylinderTestDensity · GapP1Sobolev.approx_supp`.

Given u and ε, the first step gives $\chi_n u$ within $\varepsilon/2$, of bounded support, and the second a test function within $\varepsilon/2$ of it.

Assembly. The levels H^{m,δ_m} with these inclusions form a scale `Chapter1/CylinderScBanach · GapP1Sobolev.scaleH`. A test function φ lies in every level, and its images under the inclusions are the same function `Chapter1/CylinderScBanach · GapP1Sobolev.toBase_toH`, so it is a smooth point. As test functions are dense in each level, the scale is an sc-Banach space. For $\delta = 0$ the norm of $H^{0,0}$ is the L^2 norm, and every L^2 function lies in $H^{0,0}$, so level 0 is $L^2(\mathbb{R} \times S^1)$ itself `Chapter1/CylinderScBanach · GapP1Sobolev.level0_isometry` `Chapter1/CylinderScBanach · GapP1Sobolev.level0_surjective`.

I.23 Definition 1.1.4 and Proposition 1.1.10 (preprint 1.3 and 1.9): the sc-Fredholm criterion

The text uses two facts about sc-operators without proving them. An operator that maps every level into the corresponding level is continuous on the levels. And an operator that is Fredholm on level 0 and regularizing is sc-Fredholm: this is what the text's "is sc-Fredholm" amounts to in several later places, among them §1.4.1 and Propositions 3.1.5 and 3.6.5 (preprint 3.5 and 3.52). Both are proved here.

Sc-operators from level-preserving maps. Let $T_0 : E_0 \to F_0$ be bounded and map the image of every E_m into the image of F_m. Then T_0 is an sc-operator `Chapter1/ScFredholmCriterion · isScOperator_of_mapsTo`. *Proof.* By injectivity of ι_m, the level map $g : E_m \to F_m$ is well defined and linear. Its graph is closed: if $x_n \to x$ in E_m and $g x_n \to y$ in F_m, then applying the continuous maps ι_m and T_0 gives $\iota_m y = T_0 \iota_m x = \iota_m g x$, so $y = gx$. By the closed graph theorem g is continuous. So "T maps E_m into F_m continuously" in Definition 1.1.4 (preprint 1.3) is "T maps E_m into F_m". In the same way a bijective sc-operator is an sc-isomorphism: the open mapping theorem on each level makes the inverse continuous `Chapter3/LinearizationChartsSc · ScIso.ofBijective`.

Regularizing operators. An sc-operator T is *regularizing* if $T_0 e \in F_m$ implies $e \in E_m$, for all m `Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing`. Proposition 1.1.10 (preprint 1.9) says that sc-Fredholm operators are regularizing `Chapter1/ScFredholmCriterion · IsScFredholm.isRegularizing`. The kernel of a regularizing operator consists of smooth points, since $T_0 e = 0$ lies in every F_m `Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing.ker_le_smooth`. And if T_0 is onto, so is every T_m: given $y \in F_m$, choose $T_0 x = \iota_m y$, lift x to E_m, and conclude by injectivity of ι_m `Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing.surjective`.

The criterion. If T_0 is Fredholm and T is regularizing, then T is sc-Fredholm `Chapter1/ScFredholmCriterion · isScFredholm_of_isRegularizing`. *Proof.* This is the construction of §1.4.1 with the two properties used there as hypotheses. One builds the splittings of Definition 1.1.9 (preprint 1.8).
- *The kernel and its complement.* $K = \ker T_0$ is finite-dimensional (T_0 is Fredholm) and consists of smooth points (regularizing). By Proposition 1.1.7 (preprint 1.6) (§I.4) it is an sc-subspace with an

sc-complement X, $E = K \oplus X$. - *The range.* $Y = \operatorname{range} T_0$ is closed, since T_0 is Fredholm. On level m it is $T(E_m)$: if $\iota_m y = T_0 e$, then $e \in E_m$ by regularity. Its smooth points are the images of smooth points, which are dense in E_m, so they are dense in Y_m, and Y is an sc-subspace. - *The cokernel.* The quotient F_0/Y is finite-dimensional. The smooth points of F have dense image in it, so, since a finite-dimensional subspace is closed, they map onto it. Choose a linear right inverse of $F_0 \to F_0/Y$ with values in the smooth points. Its range C is a finite-dimensional space of smooth points, and $F_m = Y_m \oplus C_m$ on every level. - *Bijectivity.* T_m maps X_m onto Y_m, because every $y \in Y_m$ is $T_0 e$ with $e \in E_m$ and the K-component of e is smooth. It is injective on X_m, because $X \cap K = 0$.

With the index of §I.6 on each level, T_m is Fredholm of the index of T_0 for every m `Chapter1/ScFredholmCriterion · isScFredholm_of_isRegularizing_index`. Together with Proposition 1.1.10 (preprint 1.9):

$$T \text{ sc-Fredholm} \iff T_0 \text{ Fredholm and } T \text{ regularizing}$$

`Chapter1/ScFredholmCriterion · isScFredholm_iff`. The text does not state the converse.

Restriction to sc-subspaces. If T maps an sc-subspace A into an sc-subspace B, it restricts to an sc-operator $A \to B$ `Chapter1/ScFredholmCriterion · ScOperator.restrict`. A point of A lies on level m of A exactly when it lies on level m of E `Chapter1/ScFredholmCriterion · ScBanach.ofSubspace_mem_range_toBase_iff`, so the restriction of a regularizing operator is regularizing `Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing.restrict`. Hence, by the criterion, the restriction of an sc-Fredholm operator is sc-Fredholm as soon as it is Fredholm on level 0, with the same index on every level `Chapter1/ScFredholmCriterion · ScOperator.IsRegularizing.isScFredholm_restrict` `Chapter1/ScFredholmCriterion · IsScFredholm.restrict`. Part III uses this for Propositions 3.1.5 and 3.6.5 (preprint 3.5 and 3.52).

I.24 The remark after Definition 1.1.15 (preprint 1.14): an sc¹ map whose derivative is not norm continuous

Statement. There are sc-Banach spaces E, F, a relatively open U in a partial quadrant of E, and an sc^1 map $f : U \to F$ such that, for *every* derivative family D of f, $x \mapsto D(x)$ is not continuous from U_1 into $\mathcal{L}(E_0, F_0)$ with the operator norm `Chapter1/ScOneNotNormContinuous · exists_isSc1_not_continuousOn_deriv`. With the compact-open continuity of §I.11 this is the whole remark. The text gives no example, and the exposition's §6 would use the shift map on the weighted Sobolev spaces; the example here uses sequences.

The sequence space. c_0 is the space of real sequences tending to 0 with the sup norm `Chapter1/ScOneNotNormContinuous · ScOneExample.c0`. A bounded multiplier c acts on it as a diagonal operator `Chapter1/ScOneNotNormContinuous · ScOneExample.diag`. The weight $w_k = 1/(k+1)$ `Chapter1/ScOneNotNormContinuous · ScOneExample.w` gives the operator $\operatorname{diag}(w)$ `Chapter1/ScOneNotNormContinuous · ScOneExample.inclOp`, which is injective and compact `Chapter1/ScOneNotNormContinuous · ScOneExample.inclOp_injective` `Chapter1/ScOneNotNormContinuous · ScOneExample.inclOp_isCompactOperator`: its truncations to the first n coordinates have finite rank, hence are compact `Chapter1/ScOneNotNormContinuous · ScOneExample.isCompactOperator_diag_wTrunc`, and they converge to it in norm, at distance $\sup_{k \ge n} w_k = 1/(n+1)$ `Chapter1/ScOneNotNormContinuous · ScOneExample.tendsto_diag_wTrunc`. The scale X has every level equal to c_0, read as the coordinates $y_k = (k+1)^m x_k$ of a sequence x, so that $|x|_m = \sup_k (k+1)^m |x_k|$ and the inclusion $X_{m+1} \to X_m$ is $\operatorname{diag}(w)$ `Chapter1/ScOneNotNormContinuous · ScOneExample.XScale`. Finitely supported sequences lift to every level `Chapter1/`

ScOneNotNormContinuous · ScOneExample.mem_smoothLvl_of_finSupp and are dense in c_0, so X is an sc-Banach space Chapter1/ScOneNotNormContinuous · ScOneExample.X. Then $E = \mathbb{R} \oplus X$, with the constant structure on $\mathbb{R}$ Chapter1/ScOneNotNormContinuous · ScOneExample.E, and $U = E_0$, the whole space, a partial quadrant.

The map. $f(t,x)_k = \cos(kt)\, x_k$ Chapter1/ScOneNotNormContinuous · ScOneExample.f Chapter1/ScOneNotNormContinuous · ScOneExample.cosOp. It is the same formula on every level, since diagonal operators commute with the weights, and it is continuous there, so f is sc^0 Chapter1/ScOneNotNormContinuous · ScOneExample.isSc0_f. The candidate derivative at $(t,x) \in E_1$ is

$$D(t,x)(\tau,\xi)_k = -k \sin(kt)\, x_k\, \tau + \cos(kt)\, \xi_k$$

Chapter1/ScOneNotNormContinuous · ScOneExample.D. In level-1 coordinates $y_k = (k+1)x_k$ the first term is $-k\, w_k \sin(kt)\, y_k\, \tau$, a bounded multiplier ($k w_k \le 1$) applied to $y \in c_0$ Chapter1/ScOneNotNormContinuous · ScOneExample.sinMul Chapter1/ScOneNotNormContinuous · ScOneExample.natCast_mul_w_le. One level absorbs the factor k. $f : E_1 \to F_0$ has derivative $D(t,x)$ at every point Chapter1/ScOneNotNormContinuous · ScOneExample.hasFDerivAt_f, and the tangent map $(t,x,\tau,\xi) \mapsto (f(t,x), D(t,x)(\tau,\xi))$ is again given by the same formulas on every level of TE, hence sc^0 Chapter1/ScOneNotNormContinuous · ScOneExample.continuous_tangentLevel. So f is sc^1 with derivative D Chapter1/ScOneNotNormContinuous · ScOneExample.isSc1With Chapter1/ScOneNotNormContinuous · ScOneExample.isSc1.

Discontinuity. The points $(\pi/(n+1), 0)$ tend to 0 in E_1. Test $D(\pi/(n+1),0) - D(0,0)$ on the unit vector $(0, e_{n+1})$ of E_0: the $(n+1)$-st coordinate is $\cos\pi - \cos 0 = -2$. So $\|D(\pi/(n+1),0) - D(0,0)\| \ge 2$ for all n Chapter1/ScOneNotNormContinuous · ScOneExample.two_le_norm_D_sub, and D is not norm continuous at 0 Chapter1/ScOneNotNormContinuous · ScOneExample.not_continuousAt_D. Every derivative family of f equals D on E_1, because derivatives are unique there (§I.11), so no derivative family is norm continuous Chapter1/ScOneNotNormContinuous · ScOneExample.not_continuousOn_of_isSc1With. The family is compact-open continuous, as every sc^1 derivative is (§I.11): on a compact set of ξ's the high coordinates are uniformly small. The example is proved sc^1 only; it is presumably sc-smooth, which the remark does not need.

I.25 Remark 1.1.17 (preprint 1.15): partial cones

Statement. The remark proposes closed convex partial cones P with nonempty interior in place of partial quadrants, and asks which arguments carry over. A *partial cone* is a closed convex $P \subseteq E_0$ with $\lambda P \subseteq P$ for $\lambda \ge 0$ and nonempty interior Chapter1/PartialCones · IsPartialCone. For O open, $O \cap P$ is an sc-domain Chapter1/PartialCones · IsPartialCone.isScDomain, so every result of §§1.2–1.3, which are stated for sc-domains (§I.10), holds on relatively open subsets of partial cones. The tangent set $TP = P_1 \times E_0$ is again a partial cone in TE Chapter1/PartialCones · IsPartialCone.tangentSet, as the remark says.

Proof. Only convexity and nonempty interior are needed for the sc-domain property Chapter1/PartialCones · IsPartialCone.isScDomain_of_convex. - *Open in convex.* On level m, $(O \cap P)_m = O_m \cap P_m$, with O_m open and $P_m = \iota_m^{-1}(P)$ convex, as the preimage of a convex set under a linear map. - *Unique derivatives.* A convex set with nonempty interior has unique derivatives at each of its points (Mathlib's `uniqueDiffOn_convex`). The interior of P_m is nonempty: the interior of P is a nonempty open set of E_0, so it contains a smooth point, which lifts to E_m, and the preimage of the interior of P is an open subset of P_m Chapter1/PartialCones · exists_toBase_mem_of_isOpen Chapter1/PartialCones · interior_lvlSet_nonempty_of_interior. Uniqueness on the open-in-convex set $O_m \cap P_m$ follows.

Neither closedness nor the cone property enters. For the tangent cone: $P_1 \times E_0$ is closed and convex,

invariant under nonnegative scaling, and its interior contains (interior of P_1) $\times E_0$.

A partial quadrant is a partial cone `Chapter1/PartialCones · IsPartialQuadrant.isPartialCone`: the preimage under a linear isomorphism of $[0,\infty)^n \times W$ is an intersection of closed half-spaces, and its interior contains the preimage of $(0,\infty)^n \times W$. The chain rule on partial cones is an instance of Theorem 1.3.1 (preprint 1.21) on sc-domains `Chapter1/PartialCones · remark_1_1_17_chainRule`. The degeneracy index, tameness and faces, from §1.3 on and in the later chapters, use coordinates of a quadrant, and stay with quadrants, as in the text.

Part II. Differential geometry based on retracts (HWZ Chapter 2, §§2.1, 2.3–2.7 (preprint §§2.1–2.6))

Chapter 2 builds a differential geometry whose local models are not open sets of Banach spaces but images of sc-smooth retractions, sets whose "dimension" may jump. The formal proof establishes the chapter in its order: retractions and retracts and the independence of their tangent from the retraction (§§II.2–II.4), M-polyfolds with their filtration, metrizability and tangent bundle (§§II.5–II.9), sub-M-polyfolds (§II.10), the degeneracy index with the boundary recognition argument (§§II.11–II.13), tame retracts and the faces of tame M-polyfolds (§§II.14–II.16), products and fibred products (§II.17), and strong bundles with their pull-backs (§§II.18–II.20). The 2021 additions follow in §§II.22–II.37, and the items the first revision left open in §§II.38–II.48.

What is proved, and what is not. All fifteen gaps of the chapter are closed: G.4–G.15 from the 2014 text and G.76–G.78 from the 2021 additions. The items the first revision of this part left open are proved, in §§II.38–II.46: - the classical Proposition 2.1.2 (preprint 2.2), by conjugating r to its derivative (G.4, §II.38); - the linear, Banach and sc-structure of T_xX with Proposition 2.3.19 (3)–(4) (preprint 2.21) in T_aX and the fibrewise linearity of Tf (G.7, §II.39); - the openness of $p : TX \to X_1$ (G.8, §II.40); - Remark 2.4.8 (preprint 2.28) and the second statement of Theorem 2.4.12 (preprint 2.31) (G.9, G.11, §II.41); - Lemma 2.4.9 (preprint 2.29) in its formulation by sc-smooth paths (G.10, §II.42); - the extended faces of a partial quadrant from C alone (G.12, §II.43); - Definition 2.5.8 (preprint 2.38) inside T_xX (G.13, §II.44); - strong bundle isomorphisms and the category $\mathcal{SBR}$ (G.14, §II.45); - Michael's Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57) (G.15, §II.46).

G.5, G.6 and G.76–G.78 were closed in §§II.31–II.37. One thing remains unformalized, and it is not a statement: the categories $\mathcal{R}$ and $\mathcal{BR}$ of §2.1 are only named. Bundle retractions and bundle maps themselves are formalized (§II.32), and $\mathcal{SBR}$ is formalized, with forgetful functors that land in the category of sc-smooth retracts (§II.45).

The narrower rows. Four rows are narrower than the text, all in the 2021 appendix §2.7.4, and all for one reason. - (2.44) of Definition 2.7.11 does not hold as we read it for every neighbourhood V; it holds for V centred at x, and the centred neighbourhoods are cofinal, which is what "taking V sufficiently small" provides (V2.84; §II.28). - In Theorem 2.7.13 with Definition 2.7.14 (V2.89), and in Proposition 2.7.12 with the characterization after Theorem 2.7.13 (V2.103), "onto θ" does not hold as we read it for every representative. In general the image of $\Phi_{(x,V,\theta)}$ is the open subset $\theta \cap V^x$ of θ, which is θ for centred V. The Lean counterexample is the slab triple in

$[0,\infty)^3$: Φ is not onto θ there `Chapter2/TameBoundaryExample · TameBoundaryExample.phi_not_surjective Chapter2/TameBoundaryNonCentred · MPolyfold.not_surjOn_phiTot`. - Proposition 2.7.12 itself holds as printed for every pair of representatives. Row V2.104 is narrower only in the characterization, where the image is $\theta \cap V^x$ (§II.37).

Three 2021 rows are now in book form, stated as one theorem each (§II.47): Remark 2.1.6 (V2.68), Definitions 2.5.18–2.5.19 with condition (3) in T_yX (V2.78), and Proposition 2.7.12 (V2.87). Two statements are read rather than taken literally. In Proposition 2.1.2 (preprint 2.2) the printed normalization $\psi(0) = x$ is read as $\psi(x) = 0$ (§II.38). And the second sentence of Remark 2.4.8 (preprint 2.28), that C_xO is in general not a partial quadrant, does not hold as we read it: at every smooth point of every sc-smooth retract, tame or not, C_xO is a partial quadrant in T_xO `Chapter2/PartialConeQuadrant · remark_2_4_8_isPartialQuadrant`. What can fail without tameness is only that its dimension equals $d_O(x)$ (§II.41).

Three choices shape the formal route. First, a retraction is a total map $r : E_0 \to E_0$ with an explicit domain U, and the tangent TO of a retract is described as the *fixed-point set* of Tr in TU (§II.2); this description, which the text's proofs use without stating it, makes Propositions 2.1.11 and 2.1.13 (preprint 2.4 and 2.6) one-line consequences of the chain rule. Second, an M-polyfold is a space with *one* atlas, and every notion is defined through the maximal atlas of all charts compatible with it (§II.5); this needs the fact, not in the text, that compatibility is a local property, proved from the chain rule and the locality of sc-smoothness (§II.1). Third, where the text computes in the model $C = [0,\infty)^k \oplus W$, the formal statements carry a presentation $T : E \cong \mathbb{R}^n \oplus W$ of the partial quadrant and its coordinate functionals $\lambda_i = (T\,\cdot)_i$; the subspaces E_x and cones C_x are defined through T, and are shown not to depend on it (§II.14).

Six proofs take a different road from the text's (or the exposition's). Theorem 2.3.10 (preprint 2.16) (metrizability of the levels) is proved without Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57) (both proved since, §II.46), which need a regularity of X_m the text does not check: a partition of unity of X pulled back to X_m gives an explicit metric, and the same argument proves Smirnov's theorem, which Mathlib lacks (§II.7). Proposition 2.4.10 (preprint 2.30) is proved from the sign arguments at points on level 1, not from Lemma 2.4.9 (preprint 2.29) at smooth points (§II.13); the lemma itself is §II.42. Theorem 2.4.12 (preprint 2.31) is proved for every chart of every M-polyfold by a direct isomorphism of quotients, without reducing to a chart that realizes the index (§II.13). Proposition 2.5.3 (preprint 2.35) uses only the second condition of tameness (§II.14). Proposition 2.5.6's (preprint 2.36) partial quadrant is proved for tame retracts without the equality case of Theorem 2.4.12 (preprint 2.31), the smooth vectors of its proof being written down explicitly (§II.14); the equality case itself, without its hypothesis, is §II.41. The second sentence of Proposition 2.5.16 (preprint 2.44) is proved by removing all coordinates of σ at once rather than by the exposition's induction on $\#\sigma$ (§II.16).

Map.

- `Chapter2/Retracts`: restriction and locality of sc^k maps; retractions, retracts, Proposition 2.1.11 (preprint 2.4), Definitions 2.1.12, 2.1.14 (preprint 2.5, 2.7), Propositions 2.1.13, 2.3.1 (preprint 2.6, 2.10), Theorem 2.1.15 (preprint 2.8); local models.
- `Chapter2/TangentBundle`: the tangent of a retract as a retract, the tangent functor, locality of Tf; the tangent bundle TX, Proposition 2.3.15 (1)–(2) (preprint 2.19), TX as an M-polyfold.
- `Chapter2/MPolyfold`: charts, atlases, M-polyfolds, Definition 2.3.8 (preprint 2.14), the levels X_m, Lemma 2.3.9 (preprint 2.15), sub-M-polyfolds (Definition 2.3.18 (preprint 2.20)), a retract as an M-polyfold.
- `Chapter2/LevelPolyfold`: X^1 as an M-polyfold; the locality of Definition 2.3.8 (preprint 2.14).
- `Chapter2/Metrizable`: Smirnov's theorem, Theorem 2.3.10 (preprint 2.16), Proposition 2.3.15 (3)

(preprint 2.19), Proposition 2.3.19 (3)–(4) (preprint 2.21).

- `Chapter2/TangentMap`: the tangent map $Tf : TX \to TY$ and the chain rule.
- `Chapter2/SubMPolyfold`: Proposition 2.3.19 (1)–(2) (preprint 2.21), Lemma 2.4.6 (preprint 2.26).
- `Chapter2/DegeneracyIndex`: E_x, the two-sided sign argument, the core of Theorem 2.4.12 (preprint 2.31), Theorem 1.3.5 (preprint 1.25), Definition 2.4.1 (preprint 2.22), Lemma 2.4.3 (preprint 2.23), Proposition 2.4.4 (preprint 2.24), Corollary 2.4.13 (preprint 2.32).
- `Chapter2/Boundary`: the boundary, the one-sided sign argument, Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31).
- `Chapter2/Tame`: the definitions of §2.5 (preprint §2.4) and of §2.6's (preprint §2.5) local objects.
- `Chapter2/TameRetracts`: Propositions 2.5.3, 2.5.6 (preprint 2.35, 2.36), the local geometry of faces, Lemma 2.5.12 (preprint 2.40), Corollary 2.5.13 (preprint 2.41), Propositions 2.5.14, 2.5.15 (preprint 2.42, 2.43) (the retraction onto a face).
- `Chapter2/TameFaces`: independence of the presentation; Proposition 2.5.15 (preprint 2.43) (faces are tame), Proposition 2.5.16 (preprint 2.44) (first sentence), Theorem 2.5.17 (preprint 2.45).
- `Chapter2/PartialCone`: Proposition 2.5.6 (preprint 2.36), $T^R_x O$ an sc-subspace and $C_x O$ a partial quadrant.
- `Chapter2/CornerGerms`: Proposition 2.5.16 (preprint 2.44) (second sentence), the corners f_σ.
- `Chapter2/FibredProduct`: products of M-polyfolds (the exposition's Lemma 4.D), Proposition 2.5.20 (preprint 2.46).
- `Chapter2/StrongBundle`, `Chapter2/StrongBundlePullback`: §2.6 (preprint §2.5).
- `Chapter2/SmoothRetractImage`: Proposition 2.1.2 (preprint 2.2).
- `Chapter2/TangentSpaceLinear`, `Chapter2/TangentMapLinear`, `Chapter2/SubMPolyfoldTangent`, `Chapter2/SubMPolyfoldTangentSc`: the linear, Banach and sc-structure of $T_x X$ (Definition 2.3.14 (preprint 2.18)), fibrewise linearity of Tf, Proposition 2.3.19 (3)–(4) (preprint 2.21) in $T_a X$.
- `Chapter2/TangentProjOpen`: Proposition 2.3.15 (2) (preprint 2.19), the openness of p.
- `Chapter2/PartialConeQuadrant`: Flor's theorem in the case needed, Remark 2.4.8 (preprint 2.28), the second statement of Theorem 2.4.12 (preprint 2.31).
- `Chapter2/TangentConePaths`: Lemma 2.4.9 (preprint 2.29).
- `Chapter2/ExtendedFaces`: the extended faces of a partial quadrant from C alone.
- `Chapter2/ReducedTangentIntrinsic`, `Chapter2/TangentConeQuadrant`, `Chapter2/TangentSpaceChartIndependent`: Definition 2.5.8 (preprint 2.38) in $T_x X$.
- `Chapter2/StrongBundleRestrict`, `Chapter3/StrongBundleMaximalAtlas`, `Chapter3/StrongBundleIsoSBR`: transitivity of atlas equivalence, restriction of strong bundle charts, the maximal strong bundle atlas, strong bundle isomorphisms and $\mathcal{SBR}$.
- `Chapter2/MichaelRefinement`: Lemma 2.7.3 and Proposition 2.7.4 (preprint 2.56 and 2.57).
- `Chapter2/BookFormLocalCompleteness2021`, `Chapter2/BookFormTameSub2021`, `Chapter2/BookFormTameBoundary2021`: the 2021 statements in book form.
- `Chapter2/SmoothPaths`: smooth points of connected open sets are path-connected.
- `Chapter2/StrongBundleOne`: the M-polyfold $Y[1]$, $P[1]$, the chart independence of $Y_{m,k}$, sc$^+$-sections into $Y[1]$.

II.1 Restriction and locality of sc^k maps

The text restricts sc-smooth maps to smaller sets ("$r : U \to U \cap U'$ is sc-smooth") and glues local sc-smoothness without comment. Both need an argument, because the tangent map of an sc^1 map is built

from a *chosen* derivative family (Part I, §I.11), and a chosen derivative on U and one on $U' \subseteq U$ need not be the same objects.

Restriction. If f is sc^1 on U with derivative family D and $U' \subseteq U$, the same D works on U' `Chapter2/Retracts · IsSc1With.of_subset` `Chapter2/Retracts · IsSc1.of_subset`. When U' is an sc-domain (unique differentiability on every level, Part I, §I.10), derivatives at points of U'_1 are unique, so the tangent maps built on U' and on U agree on TU' `Chapter2/Retracts · IsSc1.tangentMap_eqOn_of_subset`. By induction on k, using that TU' is again an sc-domain and that the tangent maps agree there, an sc^k map restricts to sc^k on every smaller sc-domain `Chapter2/Retracts · IsSck.of_subset`, and so does an sc-smooth map `Chapter2/Retracts · IsScSmooth.of_subset`; sc-smoothness depends only on the values on U `Chapter2/Retracts · IsScSmooth.congr`. Relatively open subsets of quadrants, and their intersections with open sets, are sc-domains `Chapter2/Retracts · IsRelOpenInQuadrant.inter_isOpen` `Chapter2/Retracts · IsScDomain.inter_isOpen`, and $T(U \cap W) = TU \cap (W_1 \times E_0)$ `Chapter2/Retracts · ScBanach.tangentSet_inter_isOpen`.

Locality. sc^0 is local: continuity on each level is a local property `Chapter2/Retracts · IsSc0.of_locally`. For sc^1, suppose every $x \in U$ has an open $W_x \ni x$ with f sc^1 on $U \cap W_x$. Two such local structures give the same derivative at a common point, by uniqueness on the sc-domain `Chapter2/Retracts · IsSc1With.eq_of_eq_near`. Glue: at a point $x \in U_1$ take the derivative of the structure on $U \cap W_{\iota_1 x}$. This family D agrees near every point with a local derivative family; the three conditions of Definition 1.1.15 (preprint 1.14) (the sc^0 property, differentiability on U_1, and the sc^0 property of the tangent map on TU) are local and hold `Chapter2/Retracts · IsSc1.of_locally`. The induction on k applies the same argument to the tangent maps on TU `Chapter2/Retracts · IsSck.of_locally`, and gives the sc-smooth case `Chapter2/Retracts · IsScSmooth.of_locally`. The identity is sc^1 with derivative the identity `Chapter2/Retracts · isSc1With_id`, its tangent map is the identity on TU `Chapter2/Retracts · tangentMap_id_eqOn`, and it is sc-smooth on every sc-domain `Chapter2/Retracts · isSck_id` `Chapter2/Retracts · isScSmooth_id`.

II.2 Retractions, retracts and their tangent (Definitions 2.1.1, 2.1.3, 2.1.12 (preprint 2.1, 2.3, 2.5); Proposition 2.1.11 (preprint 2.4))

Definition 2.1.1 (preprint 2.1). An *sc-smooth retraction* on a partial quadrant $C \subseteq E_0$ is a set U relatively open in C and a map $r : E_0 \to E_0$ with $r(U) \subseteq U$, r sc-smooth on U and $r(r(x)) = r(x)$ for $x \in U$ `Chapter2/Retracts · ScRetraction`. Since U is relatively open, it is an sc-domain and r is sc^1 `Chapter2/Retracts · ScRetraction.isScDomain` `Chapter2/Retracts · ScRetraction.isSc1`. The *retract* is $O = r(U)$ `Chapter2/Retracts · ScRetraction.retract`. It is contained in U and is exactly the fixed-point set of r in U: if $x = r(y)$ then $r(x) = r(r(y)) = x$ `Chapter2/Retracts · ScRetraction.retract_subset` `Chapter2/Retracts · ScRetraction.apply_eq_self_of_mem_retract` `Chapter2/Retracts · ScRetraction.mem_retract_iff`. On level 1, r_1 fixes the points of O_1 `Chapter2/Retracts · ScRetraction.levelMap_one_eq_self`, and $O_m \subseteq U_m$ `Chapter2/Retracts · ScRetraction.lvlSet_retract_subset`.

Definition 2.1.3 (preprint 2.3). (O, C, E) is an *sc-smooth retract* if O is the retract of some sc-smooth retraction on C `Chapter2/Retracts · IsScRetract` `Chapter2/Retracts · ScRetraction.isScRetract`. Every relatively open $U \subseteq C$ is one, with $r = \mathrm{id}$ `Chapter2/Retracts · ScRetraction.id` `Chapter2/Retracts · IsRelOpenInQuadrant.isScRetract`. A *local model* is a triple (E, C, O) with C a partial quadrant and O a retract in C `Chapter2/Retracts · LocalModel`; these are the models of M-polyfold charts.

The tangent of a retraction. Put $Tr(TU) := \{Tr(p) : p \in TU\} \subseteq E_1 \times E_0$ `Chapter2/Retracts · ScRetraction.tangent`. The chain rule applied to $r \circ r = r$ (Part I, Theorem 1.3.1 (preprint 1.21)) gives $Tr \circ Tr =$

Tr on TU Chapter2/Retracts · ScRetraction.tangentMap_tangentMap Chapter1/ChainRule · theorem_1_3_1, since Tr maps TU into TU Chapter2/Retracts · ScRetraction.tangentMap_mapsTo. Hence the **fixed-point description**: $p \in Tr(TU)$ iff $p \in TU$ and $Tr(p) = p$ Chapter2/Retracts · ScRetraction.mem_tangent_iff. Base points of $Tr(TU)$ lie in O_1 Chapter2/Retracts · ScRetraction.fst_mem_of_mem_tangent. For $x \in O_1$ the fixed-point condition reads $Dr(x)h = h$ Chapter2/Retracts · ScRetraction.mem_tangent_iff_of_mem, and the second component of $Tr \circ Tr = Tr$ at (x, h) gives $Dr(x) \circ Dr(x) = Dr(x)$ Chapter2/Retracts · ScRetraction.scDeriv_comp_self — the remark after Definition 2.1.3 (preprint 2.3) that $Dr(x)$ is a projection. So for $x \in O_1$, $(x, h) \in Tr(TU)$ iff h is in the range of $Dr(x)$ Chapter2/Retracts · ScRetraction.mem_tangent_iff_mem_range, and $Tr(TU) = \{(r(x), Dr(x)h) : x \in U_1, h \in E_0\}$ Chapter2/Retracts · ScRetraction.mem_tangent_iff_exists.

Proposition 2.1.11 (preprint 2.4). *If $r(U) \subseteq r'(U')$ then $r' \circ r = r$ on U*: for $y \in U$, $r(y) \in r'(U')$ is a fixed point of r' Chapter2/Retracts · ScRetraction.comp_eq_self_of_retract_subset. *And then $Tr(TU) \subseteq Tr'(TU')$*: let $p \in Tr(TU)$, so $p \in TU$ and $Tr(p) = p$. Its base point lies in $O_1 \subseteq U'_1$, so $p \in TU'$. The map r sends U into U' (into $O \subseteq U'$), so the chain rule applies to $r' \circ r$ on U: $T(r' \circ r)(p) = Tr'(Tr(p)) = Tr'(p)$; and $T(r' \circ r) = Tr$ on TU because $r' \circ r = r$ on U (tangent maps depend only on the values on U). So $Tr'(p) = Tr(p) = p$, which is the fixed-point condition for r' Chapter2/Retracts · ScRetraction.tangent_subset_of_retract_subset. Applied in both directions, two retractions with the same retract have the same tangent image Chapter2/Retracts · proposition_2_1_11. This is the text's proof, with its step "$Tr'(x, h) = (x, h)$" made the definition of membership.

Definition 2.1.12 (preprint 2.5). TO is $Tr(TU)$ for a retraction chosen (by choice) among those onto O, and the empty set if O is not a retract Chapter2/Retracts · retractTangent; by Proposition 2.1.11 (preprint 2.4) it equals $Tr(TU)$ for *every* retraction onto O Chapter2/Retracts · retractTangent_eq. The tangent space at $x \in O_1$ is the fibre $T_xO = \{h : (x, h) \in TO\}$ Chapter2/Retracts · tangentSpace, the range of $Dr(x)$ Chapter2/Retracts · tangentSpace_eq_range, a linear subspace Chapter2/Retracts · tangentSpace_eq_coe_submodule (closed, as the range of a continuous projection).

II.3 Sc-smooth maps between retracts (Proposition 2.1.13 (preprint 2.6), Definition 2.1.14 (preprint 2.7), Theorem 2.1.15 (preprint 2.8), Proposition 2.3.1 (preprint 2.10))

Proposition 2.1.13 (preprint 2.6). Let r on U and s on V be retractions onto the same O. Since s takes values in O, which r fixes, $(f \circ r) \circ s = f \circ s$ on V Chapter2/Retracts · ScRetraction.comp_comp_eqOn. (1) If $f \circ r$ is sc-smooth on U, then $f \circ r \circ s$ is sc-smooth on V by the chain rule (s maps V into $O \subseteq U$), hence so is $f \circ s$ Chapter2/Retracts · proposition_2_1_13_one Chapter1/ChainRule · IsScSmooth.comp. (2) For $p \in TO$: $p \in TV$ and $Ts(p) = p$ (fixed-point description for s, after Proposition 2.1.11 (preprint 2.4)); so $T(f \circ r)(p) = T(f \circ r)(Ts(p)) = T((f \circ r) \circ s)(p) = T(f \circ s)(p)$ Chapter2/Retracts · proposition_2_1_13_two. (3) If f maps O into $O' = r'(U')$, then $r' \circ f \circ r = f \circ r$ on U; the chain rule gives $Tr'(T(f \circ r)(p)) = T(f \circ r)(p)$ for $p \in TU$, and $T(f \circ r)(p) \in TU'$, so $T(f \circ r)(p) \in TO'$ by the fixed-point description Chapter2/Retracts · proposition_2_1_13_three.

Definition 2.1.14 (preprint 2.7). f is *sc-smooth from (O, C, E) to (O', C', E')* if $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth on U for every retraction r onto O Chapter2/Retracts · IsScSmoothRetractMap. For a retract O this is equivalent to "$f(O) \subseteq O'$ and $f \circ r$ sc-smooth for one r", by Proposition 2.1.13 (1) (preprint 2.6) Chapter2/Retracts · isScSmoothRetractMap_iff_exists. Quantifying over all retractions makes the definition in-

dependent of choices without further argument; the target triple enters only through $f(O) \subseteq O'$. The *tangent map* Tf is $T(f \circ r)$ for the chosen retraction `Chapter2/Retracts · retractTangentMap`; on TO it equals $T(f \circ r)$ for every r onto O, by Proposition 2.1.13 (2) (preprint 2.6) `Chapter2/Retracts · retractTangentMap_eqOn`, it maps TO into TO' by (3) `Chapter2/Retracts · IsScSmoothRetractMap.tangentMap_mapsTo`, and $Tf(x,h) = (f(x), D(f\circ r)(x)h)$ `Chapter2/Retracts · retractTangentMap_apply`. An sc-smooth map between retracts is continuous on O (level 0 of the sc^0 property of $f\circ r$, and $f = f\circ r$ on O) `Chapter2/Retracts · IsScSmoothRetractMap.continuousOn` and maps points of O on level m to points on level m `Chapter2/Retracts · IsScSmoothRetractMap.mem_range_toBase`.

Theorem 2.1.15 (preprint 2.8). Let $O = r(U)$, $O' = r'(U')$ be retracts, f sc-smooth from O to O' and g from O' to O''. Since $f \circ r$ takes values in O', which r' fixes, $(g \circ r') \circ (f \circ r) = (g \circ f) \circ r$ on U `Chapter2/Retracts · ScRetraction.comp_comp_comp_eqOn`. The left side is sc-smooth by the chain rule ($f \circ r$ maps U into $O' \subseteq U'$, and $g \circ r'$ is sc-smooth on U'), so $g \circ f$ is sc-smooth from O to O'' `Chapter2/Retracts · IsScSmoothRetractMap.comp`. For the tangent maps, at $p \in TO$: $T(g \circ f)(p) = T((g \circ f) \circ r)(p) = T(g \circ r')(T(f \circ r)(p)) = Tg(Tf(p))$, the last step because $Tf(p) \in TO'$ `Chapter2/Retracts · retractTangentMap_comp`. Together `Chapter2/Retracts · theorem_2_1_15`. The text gives no proof ("follows from Theorem 1.21"); this is the one it has in mind, with $r' \circ f \circ r = f \circ r$ made explicit. Further: the identity is sc-smooth `Chapter2/Retracts · isScSmoothRetractMap_id`, a retraction is sc-smooth as a map $(U,C,E) \to (O,C,E)$ and the inclusion $O \to U$ as a map $(O,C,E) \to (U,C,E)$ `Chapter2/Retracts · ScRetraction.isScSmoothRetractMap` `Chapter2/Retracts · ScRetraction.isScSmoothRetractMap_inclusion`, and sc-smoothness depends only on $f|_O$ `Chapter2/Retracts · IsScSmoothRetractMap.congr`.

Proposition 2.3.1 (preprint 2.10). (1) For $W \subseteq E_0$ open, $U \cap r^{-1}(W)$ is relatively open in C because r is continuous on U `Chapter2/Retracts · ScRetraction.relOpen_inter_preimage`; the restriction of r to it is a retraction `Chapter2/Retracts · ScRetraction.restrict` (sc-smooth by §II.1) with retract $O \cap W$ `Chapter2/Retracts · ScRetraction.restrict_retract`. So $(O \cap W, C, E)$ is a retract `Chapter2/Retracts · proposition_2_3_1_one` `Chapter2/Retracts · LocalModel.restrict`; "an open subset O' of O'' is $O \cap W$ in the subspace topology. (2) If $V = r(U)$ and s is sc-smooth from (V,C,E) to itself with $s \circ s = s$ on V, then $\rho = s \circ r$ is sc-smooth on U (that is what sc-smoothness of s on the retract V means), maps U into $s(V) \subseteq V \subseteq U$, and is idempotent: $\rho(\rho(x)) = s(r(s(r(x)))) = s(s(r(x))) = s(r(x))$ because r fixes $s(r(x)) \in V$. Its retract is $s(V)$ `Chapter2/Retracts · proposition_2_3_1_two`. As the text's proof.

Restriction and locality between retracts. An sc-smooth map between retracts restricts to $O \cap W$: a retraction onto $O \cap W$ composed with r is handled by the chain rule `Chapter2/Retracts · IsScSmoothRetractMap.restrict`; the target may be shrunk to any set containing the image `Chapter2/Retracts · IsScSmoothRetractMap.mono_target`. Conversely, if every point of O has an open W with f sc-smooth on $(O \cap W, C, E)$, then f is sc-smooth on (O,C,E): for a retraction r onto O and $y \in U$, the set W around $r(y)$ pulls back under the continuous r to an open $u \ni y$; on $U \cap u$ the map $f \circ r$ equals $(f \circ r') \circ r$ with r' the restriction of r onto $O \cap W$, which is sc-smooth, and the locality of §II.1 concludes `Chapter2/Retracts · IsScSmoothRetractMap.of_locally`. The text uses this silently every time it says "the definition does not depend on the choice of charts".

II.4 The tangent functor on retracts

The tangent of a partial quadrant. If $C = T_0^{-1}(\text{quadrant})$ for $T : E \cong \mathbb{R}^n \oplus W$, then $TC = C_1 \times E_0$ is a partial quadrant in $TE = E^1 \oplus E$: compose the tangent of T, an sc-isomorphism $TE \cong T(\mathbb{R}^n \oplus W)$, with the re-association $((a,w),(b,v)) \mapsto (a,(w,(b,v)))$ onto $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$ `Chapter2/TangentBundle · tangentQuadrantIso` `Chapter2/TangentBundle · IsPartialQuadrant.tangentSet`. TU is relatively open

in TC Chapter2/TangentBundle · IsRelOpenInQuadrant.tangentSet (this is Lemma 3.B of the exposition of Chapter 1; Part I, V1.36). Sc-operators are sc-smooth on every sc-domain Chapter2/TangentBundle · ScOperator.isScSmooth.

Tr is a retraction. Tr is sc-smooth on TU (Part I: the tangent map of an sc-smooth map is sc-smooth) Chapter2/Retracts · LocalModel.tangent_isScSmooth, maps TU into itself and is idempotent; so it is an sc-smooth retraction on TC Chapter2/TangentBundle · ScRetraction.tangentRetraction with retract TO Chapter2/TangentBundle · ScRetraction.tangentRetraction_retract. Hence (TO, TC, TE) is a retract Chapter2/TangentBundle · IsScRetract.tangentSet Chapter2/TangentBundle · LocalModel.tangent, base points of TO lie in O_1 and $(x, 0) \in TO$ for $x \in O_1$ Chapter2/TangentBundle · retractTangent_fst_mem Chapter2/TangentBundle · mk_zero_mem_retractTangent.

Tf is sc-smooth. If f is sc-smooth from O to O' then Tf is sc-smooth from (TO, TC, TE) to (TO', TC', TE'): with the retraction Tr onto TO, $Tf \circ Tr = T(f \circ r) \circ Tr = T(f \circ r \circ r) = T(f \circ r)$ on TU, which is sc-smooth Chapter2/TangentBundle · IsScSmoothRetractMap.tangent. Tf depends only on $f|_O$ Chapter2/TangentBundle · retractTangentMap_congr, and a map equal to the identity on O has tangent map the identity on TO Chapter2/TangentBundle · retractTangentMap_id. With Theorem 2.1.15 (preprint 2.8) this is the tangent functor $T : \mathcal{R} \to \mathcal{R}$ of the text; the bundle structure of Definition 2.1.16 (preprint 2.9) is §II.32 (Tr a bundle retraction, Tf a bundle map), the category $\mathcal{BR}$ is not formalized.

Locality of the tangent. For W open, $T(O \cap W) = TO \cap (W_1 \times E_0)$: with the restricted retraction, a fixed point of Tr over a base point in W_1 is a fixed point of the restricted Tr, the two tangent maps agreeing on the smaller domain (§II.1) Chapter2/TangentBundle · retractTangent_inter_isOpen Chapter2/TangentBundle · retractTangent_mono_inter. And Tf computed on $(O \cap W, C, E)$ agrees with Tf computed on (O, C, E) on $T(O \cap W)$ Chapter2/TangentBundle · retractTangentMap_inter_isOpen. These two facts are what Definitions 2.3.12–2.3.14 (preprint 2.17–2.18) use silently when the chain rule is applied on a triple overlap.

II.5 Charts, atlases and M-polyfolds (Definitions 2.3.2–2.3.8 (preprint 2.11–2.14))

Definition 2.3.2 (preprint 2.11). A *chart* on X is a local model (O, C, E) and a partial equivalence φ between X and E_0 with open source V and target O, continuous on V, with inverse continuous on O Chapter2/MPolyfold · MChart. Images of open subsets of V are open subsets $O \cap W$ of O Chapter2/MPolyfold · MChart.exists_isOpen_image_eq, hence retracts (Proposition 2.3.1 (1) (preprint 2.10)) Chapter2/MPolyfold · MChart.isScRetract_image, and charts restrict to open subsets of their sources Chapter2/MPolyfold · MChart.restrict. The overlap $\varphi(V \cap V')$ Chapter2/MPolyfold · MChart.overlap is a retract Chapter2/MPolyfold · MChart.isScRetract_overlap, the transition map is $\psi \circ \varphi^{-1}$ Chapter2/MPolyfold · MChart.transition, mapping $\varphi(V \cap V')$ into $\psi(V \cap V')$ Chapter2/MPolyfold · MChart.transition_mapsTo, and two charts are *compatible* if both transition maps are sc-smooth between the overlap retracts Chapter2/MPolyfold · MChart.Compatible. Compatibility is symmetric, reflexive, and passes to restrictions Chapter2/MPolyfold · MChart.Compatible.symm Chapter2/MPolyfold · MChart.compatible_self Chapter2/MPolyfold · MChart.Compatible.restrict.

Compatibility is local. Let φ, ψ be charts and S a family of charts each compatible with φ and with ψ, whose sources cover $V \cap V'$. Then φ and ψ are compatible Chapter2/MPolyfold · MChart.compatible_of_locally. By the locality between retracts (§II.3) it suffices to show that $\psi \circ \varphi^{-1}$ is sc-smooth near every point $\varphi(x)$ of the overlap. Take $\chi \in S$ around x and the triple overlap $s = V \cap V' \cap V''$, with $\varphi(s) = \varphi(V \cap V') \cap W$

for an open W. On $\varphi(s)$, $\psi \circ \varphi^{-1} = (\psi \circ \chi^{-1}) \circ (\chi \circ \varphi^{-1})$ Chapter2/MPolyfold · MChart.transition_eq_comp_on; the two factors are sc-smooth on the restricted retracts $\varphi(s)$ and $\chi(s)$, and Theorem 2.1.15 (preprint 2.8) composes them. The reverse transition is the same argument with the roles exchanged. This is the step that makes "the equivalence class of atlases" a usable notion; the text does not discuss it.

Definition 2.3.3 (preprint 2.12). An *atlas* is a set of pairwise compatible charts covering X Chapter2/MPolyfold · MAtlas; two atlases are *equivalent* if their union is an atlas Chapter2/MPolyfold · MAtlas.Equiv, i.e. if every chart of one is compatible with every chart of the other Chapter2/MPolyfold · MAtlas.equiv_iff. The relation is reflexive and symmetric Chapter2/MPolyfold · MAtlas.Equiv.refl Chapter2/MPolyfold · MAtlas.Equiv.symm, and transitive by the locality of compatibility (§II.45) Chapter2/StrongBundleRestrict · MAtlas.Equiv.trans.

Definition 2.3.4 (preprint 2.13). An *M-polyfold* structure on a Hausdorff paracompact space is an atlas Chapter2/MPolyfold · MPolyfold. Its *charts* are all charts compatible with every chart of the atlas Chapter2/MPolyfold · MPolyfold.IsChart Chapter2/MPolyfold · MPolyfold.isChart_of_mem. By locality of compatibility (with S the atlas), any two of them are compatible Chapter2/MPolyfold · MPolyfold.IsChart.compatible; restrictions of charts are charts Chapter2/MPolyfold · MPolyfold.IsChart.restrict; every point lies in one Chapter2/MPolyfold · MPolyfold.exists_isChart_mem_source; and equivalent atlases have the same charts, again by locality Chapter2/MPolyfold · MPolyfold.isChart_congr. So the maximal atlas represents the equivalence class, and every later definition (levels, index, tangent bundle) quantifies over it.

Definition 2.3.8 (preprint 2.14). $f : X \to Y$ is *sc-smooth on the open set* U if it is continuous on U and for all charts φ of X, ψ of Y and every open $V \subseteq U \cap \operatorname{dom}\varphi$ with $f(V) \subseteq \operatorname{dom}\psi$, the representation $\psi \circ f \circ \varphi^{-1}$ is sc-smooth from $(\varphi(V), C, E)$ into (O', C', E') Chapter2/MPolyfold · MPolyfold.IsScSmoothOn; *sc-smooth* means sc-smooth on X Chapter2/MPolyfold · MPolyfold.IsScSmoothMap. Continuity is a field: the text's pairs of charts with $f(V) \subseteq V'$ exist around every point only for continuous f. The notion passes to open subsets and to maps agreeing on U Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.mono Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.congr. *One pair of charts suffices*: if every $x \in U$ has charts φ, ψ and an open $V \ni x$ with $\psi \circ f \circ \varphi^{-1}$ sc-smooth on $\varphi(V)$, then f is sc-smooth on U Chapter2/LevelPolyfold · MPolyfold.IsScSmoothOn.of_exists_charts. For arbitrary charts φ', ψ' and V', locality reduces to a neighbourhood $\varphi'(V \cap V')$ of each point, where $\psi' \circ f \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1})$, three sc-smooth maps between retracts composed by Theorem 2.1.15 (preprint 2.8). This is the text's "the definition does not depend on the choice of charts". The identity is sc-smooth Chapter2/MPolyfold · MPolyfold.isScSmoothMap_id, and compositions are sc-smooth by the same three-factor argument with a chart of the middle space Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.comp Chapter2/MPolyfold · MPolyfold.IsScSmoothMap.comp.

II.6 The filtration X_m and the M-polyfold X^1 (Lemma 2.3.9 (preprint 2.15))

Levels. x is on level m if $\varphi(x) \in \iota_m(E_m)$ for some chart φ around x Chapter2/MPolyfold · MPolyfold.levelSet. *Independence of the chart*: if ψ is another chart around x, the transition map $\varphi \circ \psi^{-1}$ is sc-smooth between the overlap retracts, hence maps points on level m to points on level m (§II.3), and $\varphi(x) = (\varphi \circ \psi^{-1})(\psi(x))$ Chapter2/MPolyfold · MPolyfold.mem_levelSet_iff. $X_0 = X$, $X_{m+1} \subseteq X_m$, and $X_\infty = \bigcap_m X_m$ Chapter2/MPolyfold · MPolyfold.levelSet_zero Chapter2/MPolyfold · MPolyfold.levelSet_succ_subset Chapter2/MPolyfold · MPolyfold.smoothSet. For $x \in V_m$ the lift of $\varphi(x)$ to E_m Chapter2/MPolyfold · MPolyfold.levelLift satisfies $\iota_m(\text{lift}) = \varphi(x)$ and is compatible with the inclusions Chapter2/MPolyfold · MPolyfold.toBase_levelLift Chapter2/MPolyfold · MPolyfold.levelLift_eq_incl. X_m carries the topology generated by the sets $\{x \in V : \text{lift}(x) \in W\}$, W open in E_m Chapter2/MPolyfold · MPolyfold.levelBasis Chapter2/

MPolyfold · MPolyfold.levelTopology; that this family is a basis is not needed.

Lemma 2.3.9 (preprint 2.15). The preimage under $X_{m+1} \to X_m$ Chapter2/MPolyfold · MPolyfold.levelIncl of a generating set $\{\text{lift}_m \in W\}$ is the generating set $\{\text{lift}_{m+1} \in \text{incl}_m^{-1}(W)\}$, by the compatibility of lifts with inclusions; so the inclusion is continuous Chapter2/MPolyfold · lemma_2_3_9. This is the text's proof.

The space X_m. As a type, X_m is the subtype of points on level m with this topology Chapter2/LevelPolyfold · MPolyfold.Level. The inclusion $X_m \to X$ is continuous: for V open in X and $x \in X_m$, a chart φ around x gives $\varphi(V \cap \operatorname{dom}\varphi) = O \cap W$, and the generating set $\{\text{lift}_m \in W_m\}$ is a neighbourhood of x inside the preimage of V Chapter2/LevelPolyfold · MPolyfold.continuous_levelVal Chapter2/LevelPolyfold · MPolyfold.isOpen_level_basic. Since this inclusion is injective and continuous into the Hausdorff X, X_m is Hausdorff (an instance in `LevelPolyfold`, the text's argument with disjoint neighbourhoods).

The level shift of the local data. An sc-isomorphism shifts to $E^1 \to F^1$ Chapter2/LevelPolyfold · ScIso.shift, so C_1 is a partial quadrant in E^1 and U_1 relatively open in it Chapter2/LevelPolyfold · IsPartialQuadrant.shift Chapter2/LevelPolyfold · IsRelOpenInQuadrant.shift. A retraction r induces the retraction r_1 on U_1 — sc-smooth by Proposition 1.2.2 (preprint 1.17) (Part I, §I.15), idempotent by the injectivity of ι_1 Chapter2/LevelPolyfold · ScRetraction.shift — with retract O_1 Chapter2/LevelPolyfold · ScRetraction.shift_retract Chapter2/LevelPolyfold · IsScRetract.shift Chapter2/LevelPolyfold · LocalModel.shift. An sc-smooth map between retracts is sc^0 on O, maps O_m into O'_m and is continuous there Chapter2/LevelPolyfold · IsScSmoothRetractMap.isSc0 Chapter2/LevelPolyfold · IsScSmoothRetractMap.mapsTo_levelMap Chapter2/LevelPolyfold · IsScSmoothRetractMap.continuousOn_levelMap, and its level-1 map is sc-smooth from O_1 to O'_1: with the retraction r_1, $f_1 \circ r_1 = (f \circ r)_1$, sc-smooth by Proposition 1.2.2 (preprint 1.17) Chapter2/LevelPolyfold · IsScSmoothRetractMap.shift.

The M-polyfold X^1. For a chart φ of X, $x \mapsto \text{lift}_1(x)$ on V_1 with model (O_1, C_1, E^1) is a chart of X_1 Chapter2/LevelPolyfold · MPolyfold.levelChart (with a junk value for the inverse off O_1, since X_1 might be empty Chapter2/LevelPolyfold · MPolyfold.levelChartInv); its overlap with another level chart is the level-1 part of the overlap Chapter2/LevelPolyfold · MPolyfold.levelChart_overlap, its transition map the level-1 map of the transition map Chapter2/LevelPolyfold · MPolyfold.levelChart_transition, and so any two level charts are compatible by the level shift of sc-smooth maps Chapter2/LevelPolyfold · MPolyfold.levelChart_compatible. They form an atlas Chapter2/LevelPolyfold · MPolyfold.levelAtlas, and X_1 with it is the M-polyfold X^1 Chapter2/LevelPolyfold · MPolyfold.shift Chapter2/LevelPolyfold · MPolyfold.shift_isChart. Its paracompactness, an instance argument, is supplied by Theorem 2.3.10 (preprint 2.16) (§II.7). Only $m = 1$ is constructed; the levels of X^1 are those of X with their topology and charts, so the iterates are the X_m (§II.32; gap G.6, closed).

II.7 Metrizability: Smirnov's theorem and Theorem 2.3.10 (preprint 2.16) (Theorem 2.7.1 (preprint 2.55))

The text proves Theorem 2.3.10 (preprint 2.16) with Smirnov's theorem (2.55), Michael's characterization of paracompactness (Lemma 2.7.3 (preprint 2.56)) and Proposition 2.7.4 (preprint 2.57) (a regular space with a locally finite closed cover by paracompact subspaces is paracompact). Proposition 2.7.4 (preprint 2.57) needs X_m to be regular, which the text does not check. The formal proof needs neither 2.56 nor 2.57.

The metrization lemma. Let Y have open subspaces V_i, each metrizable with a metric d_i, and contin-

uous $\rho_i : Y \to [0,\infty)$ with $\operatorname{supp}\rho_i \subseteq V_i$, the supports locally finite and $\rho_i(y) > 0$ for some i at every y Chapter2/Metrizable · metrizableSpace_of_partition. Put $\delta_i(x,y) := \min(d_i(x,y),1)$ for $x,y \in V_i$ and 1 otherwise Chapter2/Metrizable · truncDist — nonnegative, at most 1, symmetric, zero only on the diagonal, and satisfying the triangle inequality inside V_i Chapter2/Metrizable · truncDist_nonneg Chapter2/Metrizable · truncDist_le_one Chapter2/Metrizable · truncDist_triangle Chapter2/Metrizable · truncDist_eq_zero — and the *cone metric*

$$e_i(x,y) := |\rho_i(x) - \rho_i(y)| + \min(\rho_i(x),\rho_i(y))\,\delta_i(x,y), \qquad D(x,y) := \sum_i e_i(x,y).$$

The sum is finite (only the i with x or y in $\operatorname{supp}\rho_i$ contribute). $D(x,x) = 0$ and D is symmetric. *Triangle inequality:* termwise, by an elementary inequality for $a,b,c \geq 0$ and δ's bounded by 1, which needs the triangle inequality for δ_i only when $\min(\rho_i(x),\rho_i(z)) > 0$ and $\rho_i(y) > 0$, i.e. when all three points lie in V_i Chapter2/Metrizable · cone_triangle. *Separation:* if $D(x,y) = 0$, take i with $\rho_i(x) > 0$; then $\rho_i(y) = \rho_i(x)$ and $\delta_i(x,y) = 0$, so $x = y$. *Topology:* $y \mapsto D(x,y)$ is continuous (locally a finite sum of continuous functions; the term $\min(\rho_i(x),\rho_i(y))\delta_i(x,y)$ is continuous where $y \in V_i$ and vanishes near points outside $\operatorname{supp}\rho_i$), so D-balls are open; conversely, if s is open and $x \in s$, take i with $\rho_i(x) > 0$ and a d_i-ball of radius $r \leq 1$ around x inside s; then $D(x,y) < \frac{1}{2}\rho_i(x)r$ forces $\rho_i(y) > \frac{1}{2}\rho_i(x)$ (so $y \in V_i$) and $\frac{1}{2}\rho_i(x)\delta_i(x,y) < \frac{1}{2}\rho_i(x)r$, so $d_i(x,y) < r$ and $y \in s$. Mathlib then builds a metric space structure with this distance and the given topology Mathlib/Topology/MetricSpace/Defs · MetricSpace.ofDistTopology.

Pulled back, and Smirnov. If $\pi : Y \to X$ is continuous into a paracompact Hausdorff X and every point of X has an open U with $\pi^{-1}(U)$ metrizable, then Y is metrizable: a partition of unity of X subordinate to these U Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate, composed with π, satisfies the hypotheses of the lemma with $V_i = \pi^{-1}(U_i)$ Chapter2/Metrizable · metrizableSpace_of_pullback. With $\pi = \mathrm{id}$ this is **Smirnov's theorem** (Theorem 2.7.1 (preprint 2.55)) Chapter2/Metrizable · metrizableSpace_of_locally. Metrizable spaces are paracompact (Stone; in Mathlib through the instance for pseudo-emetric spaces) Chapter2/Metrizable · paracompactSpace_of_metrizableSpace.

Theorem 2.3.10 (preprint 2.16). *X is metrizable*: chart domains are homeomorphic to subsets O of the Banach space E_0 Chapter2/Metrizable · MChart.sourceHomeomorph Chapter2/Metrizable · MChart.metrizableSpace_source, and Smirnov applies Chapter2/Metrizable · MPolyfold.metrizableSpace. $V_m \cong O_m$: the lift $x \mapsto \mathrm{lift}_m(x)$ is a topological embedding of $V \cap X_m$ into E_m Chapter2/Metrizable · MPolyfold.isEmbedding_levelLift. Its injectivity is that of φ; the induced topology is finer than that of X_m because generating sets of X_m inside V are preimages of open sets; it is coarser because a generating set $\{\mathrm{lift}_m^{\psi} \in W\}$ of another chart, intersected with V, is the preimage under $\mathrm{lift}_m^{\varphi}$ of the open set $\{z \in \varphi(V \cap V')_m : (\psi\circ\varphi^{-1})_m(z) \in W\}$, open since the level-$m$ map of the transition is continuous on the level-m part of the overlap retract. *X_m is metrizable*: apply the pulled-back Smirnov to the continuous inclusion $X_m \to X$ and the chart domains V, whose preimages $V \cap X_m$ embed in E_m Chapter2/Metrizable · MPolyfold.metrizableSpace_level; hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_level. X_∞: the text does not define its topology; it is taken to be the initial topology of the inclusions $X_\infty \to X_m$ Chapter2/Metrizable · MPolyfold.SmoothLevel Chapter2/Metrizable · MPolyfold.smoothIncl Chapter2/Metrizable · MPolyfold.smoothLevel_topology_eq. It embeds in the countable product $\prod_m X_m$, which is metrizable Chapter2/Metrizable · metrizableSpace_pi_nat, so X_∞ is metrizable Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel. Together Chapter2/Metrizable · theorem_2_3_10. *No hypothesis is added.* As a by-product every subset of an M-polyfold is metrizable, hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype, which discharges the paracompactness of sub-M-polyfolds (§II.10).

II.8 The tangent bundle TX (Definitions 2.3.12–2.3.14 (preprint 2.17–2.18), Proposition 2.3.15 (preprint 2.19))

Tuples and transport. A tangent tuple is (x, φ, h) with $x \in X_1$, φ a chart of the structure around x and $(\mathrm{lift}_1 x, h) \in TO$ `Chapter2/TangentBundle · MPolyfold.TangentPre`; its vector is $(\mathrm{lift}_1 x, h)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.vec Chapter2/TangentBundle · MPolyfold.TangentPre.vec_mem`. For a vector of TO over a point of an open $\varphi(s)$, membership in TO and in $T(\varphi(s))$ agree (locality of the tangent, §II.4) `Chapter2/TangentBundle · MPolyfold.TangentPre.mem_retractTangent_image_iff`. The transport of the tuple to a chart ψ around x is $T(\psi \circ \varphi^{-1})(\mathrm{vec})$, the tangent map computed on the overlap retract `Chapter2/TangentBundle · MPolyfold.TangentPre.transport`. Transport to φ itself does nothing, by $T(\mathrm{id}) = \mathrm{id}$ on the overlap $\varphi(V)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_self Chapter2/TangentBundle · MChart.transition_self_eqOn`; the transported vector lies in TO' `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_mem` and has base point $\mathrm{lift}_1^{\psi}(x)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_fst`.

The chain rule for transports. Transporting to ψ and then by $T(\chi \circ \psi^{-1})$ is transporting to χ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_transport`. On the triple overlap $s = V \cap V' \cap V''$ the transition maps are sc-smooth from $\varphi(s)$ to $\psi(s)$ and from $\psi(s)$ to $\chi(s)$ `Chapter2/TangentBundle · MChart.Compatible.transition_image Chapter2/TangentBundle · MChart.image_eq_overlap_inter`, $\chi \circ \varphi^{-1} = (\chi \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1})$ there, and Theorem 2.1.15 (preprint 2.8) gives $T(\chi \circ \varphi^{-1}) = T(\chi \circ \psi^{-1}) \circ T(\psi \circ \varphi^{-1})$ on $T(\varphi(s))$. The three tangent maps here are computed on $\varphi(s)$ and $\psi(s)$, the transports on the full overlaps; the locality of Tf (§II.4) identifies them. The text never checks this.

Definition 2.3.12 (preprint 2.17). Two tuples are *equivalent* if their points agree and the transport of the first to the chart of the second is the second `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel`. Equivalent tuples transport to the same vector in every chart (chain rule for transports) `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.transport`; hence the relation is reflexive, symmetric and transitive `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.refl Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.symm Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.trans`, an equivalence relation `Chapter2/TangentBundle · MPolyfold.TangentPre.setoid`.

Definition 2.3.14 (preprint 2.18). TX is the quotient `Chapter2/TangentBundle · MPolyfold.TangentSpace Chapter2/TangentBundle · MPolyfold.TangentSpace.mk`, with $p : TX \to X^1$, $[(x, \varphi, h)] \mapsto x$ `Chapter2/TangentBundle · MPolyfold.tangentProj`, and $T_x X = p^{-1}(x)$ `Chapter2/TangentBundle · MPolyfold.tangentFiber`. For a chart φ: $TV = p^{-1}(V_1)$ `Chapter2/TangentBundle · MPolyfold.tangentChartSet` and $T\varphi : TV \to TO$ sends a class to the transport of a representative to φ, well defined by the previous paragraph `Chapter2/TangentBundle · MPolyfold.tangentChartMap Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk`; $T\varphi[(x, \varphi, h)] = (\varphi(x), h)$ `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_self`, $T\varphi$ lands in TO over the lift of the base point `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mem Chapter2/TangentBundle · MPolyfold.tangentChartMap_fst`, and it is injective on TV: equal images have equal base points (injectivity of φ and ι_1) and then the representatives are equivalent `Chapter2/TangentBundle · MPolyfold.tangentChartMap_injOn`. Its inverse on TO is $a \mapsto [(\varphi^{-1}(\iota_1 a_1), \varphi, a_2)]$ `Chapter2/TangentBundle · MPolyfold.TangentPre.ofVec Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_ofVec Chapter2/TangentBundle · MPolyfold.mk_ofVec_tangentChartMap`, and in another chart $T\psi \circ (T\varphi)^{-1} = T(\psi \circ \varphi^{-1})$ on $T(\varphi(V \cap V'))$ `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_ofVec_of_mem Chapter2/TangentBundle · MPolyfold.mk_ofVec_mem_tangentChartSet_iff`. The topology is generated by the sets $\{\alpha \in TV : T\varphi(\alpha) \in W\}$, W open `Chapter2/TangentBundle · MPolyfold.tangentBasis Chapter2/TangentBundle · MPolyfold.isOpen_tangent_basic`; TV itself is open `Chapter2/TangentBundle · MPolyfold.isOpen_tangentChartSet`. Here $T_x X$ is identified with $T_{\varphi(x)} O$ by $T\varphi$; its

own vector space structure, for which $T\varphi$ is a linear isomorphism in every chart, is §II.39 Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberLinearEquiv.

Proposition 2.3.15 (preprint 2.19). (2) *p is continuous*: the preimage of a generating set $\{\mathrm{lift}_1^{\psi} \in W\}$ of X_1 is the generating set of TX given by ψ and $W \times E_0$ Chapter2/TangentBundle · MPolyfold.continuous_tangentProj; so is $TX \to X$ Chapter2/TangentBundle · MPolyfold.continuous_levelVal_tangentProj. Openness of p is §II.40 Chapter2/TangentProjOpen · MPolyfold.isOpenMap_tangentProj. (1) *TX is Hausdorff* (the instance `T2Space` in `TangentBundle`): two classes over the same point lie in one TV and have distinct images under the injective $T\varphi$, separated by open sets of $E_1 \times E_0$ whose preimages are basic; classes over distinct points are separated by preimages of disjoint open sets of X under the continuous $TX \to X$. (3) *TX is metrizable*: apply the pulled-back Smirnov theorem to $TX \to X$ and the chart domains V, whose preimages TV are homeomorphic to subsets of $E_1 \times E_0$ through the charts of §II.8's next paragraph Chapter2/Metrizable · MPolyfold.metrizableSpace_tangentSpace Chapter2/Metrizable · MChart.metrizableSpace_source; hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_tangentSpace Chapter2/Metrizable · proposition_2_3_15_three.

TX as an M-polyfold. For a chart φ of X, $(TV, T\varphi, (TO, TC, TE))$ is a chart of TX Chapter2/TangentBundle · MPolyfold.tangentChart Chapter2/TangentBundle · MPolyfold.tangentChartInv; its overlap with another is $T(\varphi(V \cap V'))$ Chapter2/TangentBundle · MPolyfold.tangentChart_overlap and its transition map is $T(\psi \circ \varphi^{-1})$ — formula (5) Chapter2/TangentBundle · MPolyfold.tangentChart_transition — which is sc-smooth between the tangent retracts by the tangent functor (§II.4); so the charts are compatible Chapter2/TangentBundle · MPolyfold.tangentChart_compatible, form an atlas Chapter2/TangentBundle · MPolyfold.tangentAtlas and make TX an M-polyfold Chapter2/TangentBundle · MPolyfold.tangentPolyfold Chapter2/TangentBundle · MPolyfold.tangentPolyfold_isChart. *$p : TX \to X^1$ is sc-smooth*: in the charts $T\varphi$ and the level chart of φ it is the projection $TO \to O_1$, $(a, h) \mapsto a$, which is sc-smooth between the retracts (TO, TC, TE) and (O_1, C_1, E^1) because, with the retraction Tr onto TO, $\mathrm{pr}_1 \circ Tr = r_1 \circ \mathrm{pr}_1$ on TU, a composite of the sc-operator pr_1 Chapter2/TangentBundle · ScBanach.tangentFst and the sc-smooth r_1 Chapter2/TangentBundle · MPolyfold.isScSmoothRetractMap_tangent_fst; one pair of charts per point suffices (§II.5) Chapter2/TangentBundle · MPolyfold.tangentProj_isScSmooth.

II.9 The tangent map of an sc-smooth map between M-polyfolds

The text uses Tf for maps between M-polyfolds from §2.4 (preprint §2.3) on without defining it. *The chain rule on chart images, with locality*: for F sc-smooth on $\varphi(s_1)$ into a retract O_1 mapping $\varphi(t)$ into O_1', G sc-smooth on O_1' and H sc-smooth on $\varphi(s_2)$ with $H = G \circ F$ on $\varphi(t)$ ($t \subseteq s_1 \cap s_2$ open), $TH = TG \circ TF$ on $T(\varphi(t))$, the three tangent maps computed on their own domains Chapter2/TangentMap · MChart.retractTangentMap_comp_image — Theorem 2.1.15 (preprint 2.8) on $\varphi(t)$ plus the locality of Tf. *Levels*: an sc-smooth map maps X_m into Y_m, because its coordinate representations do Chapter2/TangentMap · MPolyfold.IsScSmoothOn.mapsTo_levelSet Chapter2/TangentMap · MPolyfold.IsScSmoothMap.mapsTo_levelSet. *The image vector*: for a tuple (x, φ, h) and a chart ψ around $f(x)$, $T(\psi \circ f \circ \varphi^{-1})(\mathrm{vec})$, computed on $\varphi(V \cap f^{-1}(V'))$ Chapter2/TangentMap · MPolyfold.TangentPre.fdom Chapter2/TangentMap · MPolyfold.TangentPre.ftransport, lies in TO' over $\psi(f(x))$ Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_mem Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_fst; changing ψ changes it by T of the transition map Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_transport, changing φ by transport Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_eq_of_transport, and equivalent tuples give the same image Chapter2/TangentMap · MPolyfold.TangentPre.Rel.ftransport — all by the chain rule with locality. The image tuple $(f(x), \psi, \cdot)$ Chapter2/TangentMap · MPolyfold.TangentPre.fimageWith therefore has a class independent of ψ and of the representative Chapter2/TangentMap · MPolyfold.TangentPre.mk_fimageWith_eq Chapter2/

TangentMap · MPolyfold.TangentPre.mk_fimageWith_eq_of_rel.

Tf. With a chosen ψ, $Tf[(x,\varphi,h)]$ is the class of the image tuple Chapter2/TangentMap · MPolyfold.tangentMap, computable in any chart around $f(x)$ Chapter2/TangentMap · MPolyfold.tangentMap_mk. It covers f Chapter2/TangentMap · MPolyfold.levelVal_tangentProj_tangentMap, maps T_xX into $T_{f(x)}Y$ Chapter2/TangentMap · MPolyfold.tangentMap_mapsTo_fiber, and in charts $T\psi \circ Tf = T(\psi \circ f \circ \varphi^{-1}) \circ T\varphi$ Chapter2/TangentMap · MPolyfold.tangentChartMap_tangentMap. $T(\mathrm{id}) = \mathrm{id}$ Chapter2/TangentMap · MPolyfold.tangentMap_id, and **the chain rule** $T(g \circ f) = Tg \circ Tf$ holds: with charts ψ around $f(x)$ and χ around $g(f(x))$ and the open set $t = \mathrm{fdom}(\varphi, f, \psi) \cap f^{-1}(\mathrm{fdom}(\psi, g, \chi))$, the chain rule with locality gives $T(\chi \circ g \circ f \circ \varphi^{-1}) = T(\chi \circ g \circ \psi^{-1}) \circ T(\psi \circ f \circ \varphi^{-1})$ on $T(\varphi(t))$ Chapter2/TangentMap · MPolyfold.tangentMap_comp. Tf is continuous: near a class in TV, the preimage of a generating set of TY contains the generating set of TX given by φ and the preimage under the continuous $T(\psi \circ f \circ \varphi^{-1})$ Chapter2/TangentMap · MPolyfold.continuous_tangentMap; and Tf is sc-smooth $TX \to TY$, its representation in the charts $T\varphi$, $T\psi$ being $T(\psi \circ f \circ \varphi^{-1})$, sc-smooth by the tangent functor Chapter2/TangentMap · MPolyfold.tangentMap_isScSmooth.

II.10 Sub-M-polyfolds (Definition 2.3.18 (preprint 2.20), Proposition 2.3.19 (preprint 2.21))

Definition 2.3.18 (preprint 2.20). $A \subseteq X$ is a *sub-M-polyfold* if every $a \in A$ has an open $V \ni a$ and a map r with $r(V) \subseteq V$, sc-smooth on V, idempotent on V, with $r(V) = A \cap V$ Chapter2/MPolyfold · MPolyfold.IsSubMPolyfold; X is one Chapter2/MPolyfold · MPolyfold.isSubMPolyfold_univ. The data at a point are packaged Chapter2/SubMPolyfold · MPolyfold.SubData Chapter2/SubMPolyfold · MPolyfold.IsSubMPolyfold.data: r maps V into A, fixes $A \cap V$, and is continuous on V Chapter2/SubMPolyfold · MPolyfold.SubData.apply_mem Chapter2/SubMPolyfold · MPolyfold.SubData.apply_eq_self Chapter2/SubMPolyfold · MPolyfold.SubData.continuousOn.

Proposition 2.3.19 (preprint 2.21), the structure. *Shrinking.* For a chart φ, put $W := (V \cap \mathrm{dom}\,\varphi) \cap r^{-1}(V \cap \mathrm{dom}\,\varphi)$ Chapter2/SubMPolyfold · MPolyfold.SubData.shrink: open (continuity of r) Chapter2/SubMPolyfold · MPolyfold.SubData.isOpen_shrink, containing every $a \in A \cap V \cap \mathrm{dom}\,\varphi$ Chapter2/SubMPolyfold · MPolyfold.SubData.mem_shrink, with $r(W) \subseteq W$ and $r(W) = W \cap A$ Chapter2/SubMPolyfold · MPolyfold.SubData.mapsTo_shrink Chapter2/SubMPolyfold · MPolyfold.SubData.image_shrink. *The retract.* $\rho := \varphi \circ r \circ \varphi^{-1}$ is sc-smooth from $\varphi(W)$ to itself — Definition 2.3.8 (preprint 2.14) for r in the charts φ, φ — idempotent, and $\rho(\varphi(W)) = \varphi(W \cap A)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_isScSmooth Chapter2/SubMPolyfold · MPolyfold.SubData.rho_idem Chapter2/SubMPolyfold · MPolyfold.SubData.rho_image; so $\varphi(W \cap A)$ is a retract by Proposition 2.3.1 (2) (preprint 2.10) Chapter2/SubMPolyfold · MPolyfold.SubData.isScRetract_image. *The chart.* With the subspace topology on A, φ restricted to $W \cap A$ is a chart of A with model $(\varphi(W \cap A), C, E)$ Chapter2/SubMPolyfold · MPolyfold.subChart Chapter2/SubMPolyfold · MPolyfold.subChartInv. *Compatibility.* The overlap of two such charts is $\varphi((W \cap W') \cap A)$, a subset of $\varphi(V \cap V')$ Chapter2/SubMPolyfold · MPolyfold.subChart_overlap. An sc-smooth map between retracts is sc-smooth on every sub-retract $O_1 \subseteq O$: for a retraction s onto O_1, $f \circ s = (f \circ r) \circ s$, sc-smooth by the chain rule Chapter2/SubMPolyfold · IsScSmoothRetractMap.mono_source — this is the text's "$(\psi' \circ \psi^{-1}) \circ s'' = (\varphi' \circ \varphi^{-1}) \circ s''$". The transition maps of the sub-charts are restrictions of those of X, hence sc-smooth Chapter2/SubMPolyfold · MPolyfold.subChart_compatible. They form an atlas Chapter2/SubMPolyfold · MPolyfold.subAtlas, and A is an M-polyfold Chapter2/SubMPolyfold · MPolyfold.subPolyfold Chapter2/SubMPolyfold · MPolyfold.subPolyfold_isChart. The structure requires A to be paracompact (an instance argument); it is, as a subset of the metrizable X (§II.7), and for closed A also elementarily Chapter2/SubMPolyfold · MPolyfold.paracompactSpace_of_isClosed.

(1) and (2). The inclusion $A \to X$ is sc-smooth: in the sub-chart and φ it is the identity of $\varphi(W \cap A)$ into O; one pair of charts per point suffices `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_inclusion_isScSmooth`; it is a homeomorphism onto its image by the choice of topology. For any data (V, r), the map $i^{-1} \circ r$ (a total map with a junk value off V) `Chapter2/SubMPolyfold · MPolyfold.SubData.retractInto` `Chapter2/SubMPolyfold · MPolyfold.SubData.retractInto_val` is sc-smooth on V: near $x \in V$ take a chart φ' around x, a chart φ around $r(x)$ and $V_0 = V \cap \operatorname{dom} \varphi' \cap r^{-1}(\operatorname{dom} \varphi)$; on $\varphi'(V_0)$ the representation in φ' and the sub-chart on φ is $\varphi \circ r \circ \varphi'^{-1}$, sc-smooth by Definition 2.3.8 (preprint 2.14) for r `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_retraction_isScSmooth`.

Proposition 2.3.19 (3), (4) (preprint 2.21), in a chart. Let r be a retraction onto O, s sc-smooth from (O, C, E) to itself with $s \circ s = s$, and $t := s \circ r$, a retraction onto $s(O)$ (Proposition 2.3.1 (2) (preprint 2.10)) `Chapter2/Metrizable · ScRetraction.compRetraction` `Chapter2/Metrizable · ScRetraction.compRetraction_retract`. For $a \in s(O)_1$ put $P := Dr(a)$, $Q := Dt(a) = D(s \circ r)(a)$ — the text's "$Ds(a)$". Both are idempotent (§II.2). From $r \circ t = t$ and $t \circ r = t$ on U (s takes values in O, which r fixes; and $r \circ r = r$), the chain rule at the point a `Chapter2/DegeneracyIndex · scDeriv_comp_apply` gives $PQ = Q$ and $QP = Q$. If a is smooth, P and Q are sc-operators (Part I, Proposition 1.2.1 (preprint 1.16)), and so is $P - Q$; identities between base maps of sc-operators hold on every level `Chapter2/Metrizable · ScOperator.lvl_comp_of_base` `Chapter2/Metrizable · ScOperator.lvl_comp_eq_zero_of_base` `Chapter2/Metrizable · ScOperator.lvl_eq_add_of_base`. Then Q and $P - Q$ are commuting idempotent sc-operators with $Q(P - Q) = 0$, their ranges are sc-subspaces (Part I, §I.3: sc-splittings and sc-projections `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_of_idem`), and on every level the level parts of QE and $(P - Q)E$ are disjoint with sum the level part of PE. Since $T_a O = PE$ and $T_a s(O) = QE$ (§II.2), this is the sc-splitting $T_a O = T_a A \oplus (Dr(a) - Dt(a))E$ `Chapter2/Metrizable · proposition_2_3_19_three`. For (4), s is sc-smooth from O onto $s(O)$ (shrinking the target), and $Ts(a, h) = (a, Qh)$ for $h \in T_a O$ because $Qh = QPh$; as Q maps PE onto $QE = Q(PE)$, $Ts(a)$ maps $T_a O$ onto $T_a s(O)$ `Chapter2/Metrizable · proposition_2_3_19_four`. The transport to $T_a X$ is §II.39 `Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_3_linear` `Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_4`.

II.11 The degeneracy index of an M-polyfold (Definitions 2.4.1, 2.4.5 (preprint 2.22, 2.25); Lemmas 2.4.3, 2.4.6 (preprint 2.23, 2.26); Proposition 2.4.4 (preprint 2.24))

A retract is contained in its quadrant, and chart values lie in C `Chapter2/DegeneracyIndex · LocalModel.O_subset_C` `Chapter2/DegeneracyIndex · MChart.apply_mem_C` `Chapter2/DegeneracyIndex · LocalModel.isPartialQuadrant`.
Definition 2.4.1 (preprint 2.22). $d(x, \varphi) := d_C(\varphi(x))$ `Chapter2/DegeneracyIndex · MPolyfold.chartDegIndex`, and $d_X(x)$ is the least $d(x, \varphi)$ over the charts of the structure around x `Chapter2/DegeneracyIndex · MPolyfold.degIndex`; $d_X(x) \le d(x, \varphi)$ for every such chart and the minimum is attained `Chapter2/DegeneracyIndex · MPolyfold.degIndex_le` `Chapter2/DegeneracyIndex · MPolyfold.exists_isChart_degIndex_eq`.
Lemma 2.4.3 (preprint 2.23). d_C is upper semicontinuous: coordinates positive at x stay positive nearby, so the vanishing set can only shrink `Chapter2/DegeneracyIndex · eventually_degIndexWith_le`; composed with the continuous chart, $d(\cdot, \varphi)$ is upper semicontinuous on V `Chapter2/DegeneracyIndex · MPolyfold.eventually_chartDegIndex_le`, and with a chart realizing $d_X(x)$, $d_X(y) \le d(y, \varphi) \le d(x, \varphi) = d_X(x)$ near x `Chapter2/DegeneracyIndex · lemma_2_4_3`. Hence $\{d_X < k\}$ is open and $\{d_X \ge k\}$ closed `Chapter2/Boundary · MPolyfold.isOpen_setOf_degIndex_lt` `Chapter2/Boundary · MPolyfold.isClosed_setOf_le_degIndex`.

Proposition 2.4.4 (preprint 2.24). An sc-diffeomorphism of open sets is f sc-smooth on U, g sc-smooth on

V, mapping into each other and mutually inverse Chapter2/DegeneracyIndex · MPolyfold.IsScDiffeoOn Chapter2/DegeneracyIndex · MPolyfold.IsScDiffeoOn.symm. For a chart ψ of Y, $\psi \circ f$ on $U \cap f^{-1}(\mathrm{dom}\,\psi)$, with inverse $g \circ \psi^{-1}$, is a chart of X Chapter2/DegeneracyIndex · MPolyfold.transportChart; it is compatible with every chart φ of the atlas of X, because its transition maps with φ are $\psi \circ f \circ \varphi^{-1}$ and $\varphi \circ g \circ \psi^{-1}$, sc-smooth by Definition 2.3.8 (preprint 2.14) for f and for g Chapter2/DegeneracyIndex · MPolyfold.transportChart_isChart. Its index at x is $d(f(x), \psi)$ Chapter2/DegeneracyIndex · MPolyfold.chartDegIndex_transportChart. With ψ realizing $d_Y(f(x))$: $d_X(x) \le d_Y(f(x))$ Chapter2/DegeneracyIndex · MPolyfold.degIndex_le_of_isScDiffeoOn, and with g the reverse Chapter2/DegeneracyIndex · proposition_2_4_4. The text's "from the definitions one deduces immediately" is this construction.

Definition 2.4.5 (preprint 2.25). $\partial X = \{d_X \ge 1\}$ Chapter2/DegeneracyIndex · MPolyfold.boundary; without boundary means $d_X \equiv 0$, equivalently $\partial X = \emptyset$ Chapter2/DegeneracyIndex · MPolyfold.IsWithoutBoundary Chapter2/DegeneracyIndex · MPolyfold.isWithoutBoundary_iff. ∂X and the corners $\{d_X \ge 2\}$ are closed, the interior $\{d_X = 0\}$ is open Chapter2/Boundary · MPolyfold.isClosed_boundary Chapter2/Boundary · MPolyfold.corners Chapter2/Boundary · MPolyfold.isClosed_corners Chapter2/Boundary · MPolyfold.isOpen_interior, and sc-diffeomorphisms preserve ∂X by Proposition 2.4.4 (preprint 2.24) Chapter2/Boundary · MPolyfold.IsScDiffeoOn.mem_boundary_iff.

Lemma 2.4.6 (preprint 2.26). For a sub-M-polyfold A and $a \in A$: take a chart φ of X realizing $d_X(a)$ and the data at a; the sub-chart built on φ has the same quadrant C and the same value $\varphi(a)$, so $d_A(a) \le d_C(\varphi(a)) = d_X(a)$ Chapter2/SubMPolyfold · lemma_2_4_6.

A retract as an M-polyfold. A nonempty retract O, as a subtype, has the single chart id Chapter2/MPolyfold · LocalModel.selfChart, compatible with itself, and is an M-polyfold Chapter2/MPolyfold · LocalModel.toMPolyfold (Hausdorff and paracompact as a subspace of E_0); so is a nonempty relatively open $U \subseteq C$ Chapter2/MPolyfold · IsRelOpenInQuadrant.toMPolyfold Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.selfChart_isChart — the text's X_C.

II.12 The subspaces E_x, the cones C_x and the sign arguments

Coordinates. For a presentation T of C and $x \in E_0$, the coordinates vanishing at x form a continuous linear surjection $E_0 \to \mathbb{R}^{I(x)}$, $v \mapsto ((Tv)_i)_{i \in I(x)}$ Chapter2/DegeneracyIndex · coordProj Chapter2/DegeneracyIndex · coordMapL Chapter2/DegeneracyIndex · coordMapL_surjective. Its kernel is E_x Chapter2/DegeneracyIndex · minimalSubspaceWith Chapter2/DegeneracyIndex · mem_minimalSubspaceWith, closed Chapter2/DegeneracyIndex · isClosed_minimalSubspaceWith, and by the first isomorphism theorem $E_0/E_x \cong \mathbb{R}^{I(x)}$, so $\dim(E_0/E_x) = d_C(x)$ Chapter2/DegeneracyIndex · finrank_quotient_minimalSubspaceWith. The cone $C_x = \{v : (Tv)_i \ge 0 \text{ for } i \in I(x)\}$ Chapter2/Tame · partialQuadrantAtWith is convex and closed and contains E_x Chapter2/Tame · convex_partialQuadrantAtWith Chapter2/Boundary · isClosed_partialQuadrantAtWith Chapter2/Tame · minimalSubspaceWith_subset_partialQuadrantAtWith. *Approximation*: if $Tx = (a, w)$, the points $T^{-1}(a, w')$ with $w' \in W_1$ (resp. W_∞) converge to x as $w' \to w$ and have the same vanishing coordinates, so x is a limit of points on level 1 (resp. smooth points) of the same index, in C iff x is Chapter2/DegeneracyIndex · exists_tendsto_lvl1_degIndexWith Chapter2/TameRetracts · exists_tendsto_smooth_degIndexWith.

Density on level 1. For a continuous linear $L : E_0 \to V$, V finite-dimensional, the level-1 points of $\ker L$ are dense in $\ker L$ Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase. The image S of E_1 is dense in E_0; L maps S onto the finite-dimensional $L(S)$, which has a linear, hence continuous, right inverse g with values in S. For $v \in \ker L$ take $u_k \in S$ with $u_k \to v$; then $L(u_k) \to 0$, $g(L(u_k)) \to 0$, and $u_k - g(L(u_k)) \in S \cap \ker L$ converges to v. This is the text's "the closure of $(T^R_x O)_\infty$ is $T^R_x O$" in the form

the proofs need; the level-1 points of C_x are dense in C_x likewise Chapter2/Boundary · partialQuadrantAtWith_subset_closure.

The two-sided sign argument. Let f be sc¹ on U relatively open in C, with values in C', $x \in U_1$ and $v \in E_1$ with $\iota_1 v \in E_x$. Then $x+tv \in U_1$ for $|t|$ small: the coordinates in $I(x)$ stay 0, the others stay positive, and U_1 is relatively open Chapter2/DegeneracyIndex · eventually_add_smul_mem_lvlSet. For j with $(T'f(x))_j = 0$ the real function $g(t) = (T'f(\iota_1(x+tv)))_j$ is ≥ 0 near 0 (values in C'), vanishes at 0, and has derivative $(T'Df(x)v)_j$ at 0 (Definition 1.1.15's (preprint 1.14) derivative along E_1, composed with the path); a local minimum has zero derivative Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.hasDerivAt_eq_zero. So $Df(x)v \in E'_{f(x)}$ Chapter2/DegeneracyIndex · IsSc1With.deriv_mem_minimalSubspace_of_lvl1, and by density and continuity of $Df(x)$ on E_0, $Df(x)$ maps E_x into $E'_{f(x)}$ Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace.

The one-sided sign argument. For $v \in E_1$ with $\iota_1 v \in C_x$, $x + tv \in U_1$ for small $t \geq 0$ Chapter2/Boundary · eventually_add_smul_mem_lvlSet_of_cone, the same g is ≥ 0 on $[0, \varepsilon)$ with $g(0) = 0$, so its one-sided derivative is ≥ 0 Chapter2/Boundary · nonneg_of_hasDerivWithinAt_Ici; hence $Df(x)v \in C'_{f(x)}$ Chapter2/Boundary · IsSc1With.deriv_mem_partialQuadrant_of_lvl1, and $Df(x)$ maps C_x into $C'_{f(x)}$ Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant. These are the derivative-level content of Lemma 2.4.9 (1) and (2) (preprint 2.29); the lemma itself, about sc-smooth paths, is proved from them in §II.42 Chapter2/TangentConePaths · GapP1Lemma249.lemma_2_4_9_cone.

The chain rule at a point. For sc¹ maps f, g on relatively open sets with $f(U) \subseteq U'$, $Dg(f(x))Df(x) = D(g \circ f)(x)$ at $x \in U_1$ Chapter2/DegeneracyIndex · scDeriv_comp_apply; if $g \circ f = \mathrm{id}$ on U then $g_1(f_1(x)) = x$ and $Dg(f(x))Df(x) = \mathrm{id}$ Chapter2/DegeneracyIndex · levelMap_comp_eq_self Chapter2/DegeneracyIndex · scDeriv_comp_eq_self.

II.13 Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)

Definition 2.4.7 (preprint 2.27), Remark 2.4.8 (preprint 2.28). For a retraction r, a presentation T and $x \in O_1$: $T^R_x O := Dr(x)E_0 \cap E_x$ and $C_x O := Dr(x)E_0 \cap C_x$ Chapter2/Tame · reducedTangentSpaceWith Chapter2/Tame · partialConeWith, defined at every point on level 1; $T^R_x O \subseteq C_x O$ Chapter2/Tame · reducedTangentSpaceWith_subset_partialConeWith. The remark's further claim, that $C_x O$ is in general not a partial quadrant in $T_x O$, does not hold as we read it: at a smooth point it always is one (§II.41). The number $\dim(T_x O / T^R_x O)$ is $\dim(A/(A \cap B))$ for $A = T_x O$, $B = E_x$ Chapter2/Boundary · quotDim; since $A \to E_0/E_x$ has kernel $A \cap E_x$, it is at most $d_C(x)$ and finite Chapter2/Boundary · quotDim_inf_minimalSubspace_le Chapter2/Boundary · quotDim_reducedTangentSpaceWith_le.

Germs between retracts. A germ of sc-diffeomorphisms between $O = r(U)$ and $O' = r'(U')$ is a pair F, G with $F \circ r$, $G \circ r'$ sc¹ (sc¹ suffices here; the text has sc-smooth), $F(O) \subseteq O'$, $G(O') \subseteq O$, inverse to each other on O, O' Chapter2/Boundary · ScRetraction.IsDiffeoGerm; an sc-smooth map between retracts with an sc-smooth inverse is one Chapter2/Boundary · IsScSmoothRetractMap.isDiffeoGerm, and two compatible charts give one between the overlap retracts, with the restricted retractions Chapter2/Boundary · MChart.Compatible.isDiffeoGerm. Since $(G \circ r') \circ (F \circ r) = r$ on U Chapter2/Boundary · ScRetraction.IsDiffeoGerm.comp_eq, the chain rule at $x \in O_1$, $y = (F \circ r)_1(x) \in O'_1$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.image_mem Chapter2/

Boundary · ScRetraction.IsDiffeoGerm.image_image, gives

$$D(G\circ r')(y)\,D(F\circ r)(x)=Dr(x),\qquad D(F\circ r)(x)\,D(G\circ r')(y)=Dr'(y)$$

Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_inv_deriv Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_deriv_inv. So $D:=D(F\circ r)(x)$ maps T_xO into T_yO' Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mem_range, and by the sign arguments for $F\circ r$ (with values in $O'\subseteq C'$) it maps E_x into E'_y and C_x into C'_y Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mapsTo_minimalSubspace Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mapsTo_partialQuadrant.

Proposition 2.4.10 (preprint 2.30). $D(T^R_xO)=T^R_yO'$ and $D(C_xO)=C_yO'$, with $Tf(x)=D$ Chapter2/Boundary · proposition_2_4_10_reduced Chapter2/Boundary · proposition_2_4_10_cone. "$\subseteq$": D maps T_xO into T_yO' and E_x (resp. C_x) into E'_y (resp. C'_y). "$\supseteq$": for $w\in T^R_yO'$, the vector $D'w$, $D'=D(G\circ r')(y)$, lies in T_xO and in E_x by the same facts for the inverse germ, and $D(D'w)=Dr'(y)w=w$ because w is in the range of the projection $Dr'(y)$. This proof does not pass through Lemma 2.4.9 (preprint 2.29).

The core of Theorem 2.4.12 (preprint 2.31). Along a germ F, $\dim(T_xO/T^R_xO)\le d_{C'}(F(x))$ for $x\in O_1$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.quotDim_le. With $P=Dr(x)$, $P'=Dr'(y)$: D maps $T_x=PE$ into $T_y=P'E'$ and D' maps back, $D'D=P$ is the identity on T_x and $DD'=P'$ the identity on T_y. The map $\Phi:T_x\to T_y/(T_y\cap E'_y)$, $v\mapsto[Dv]$, is onto ($[w]=\Phi(D'w)$) with kernel $T_x\cap E_x$ ($Dv\in E'_y$ implies $v=D'Dv\in E_x$). Hence $T_x/T^R_x\cong T_y/T^R_y$, which embeds in E'_0/E'_y, of dimension $d_{C'}(y)$. The same argument with U_0 relatively open in C in place of the retract O is the form used for Corollary 2.4.13 (preprint 2.32) and Theorem 1.3.5 (preprint 1.25) Chapter2/DegeneracyIndex · degIndexWith_le_of_germ: there $D'D=\mathrm{id}$, so $E_0/E_x\cong T_y/T^R_y\hookrightarrow E'_0/E'_y$ and $d_C(x)\le d_{C'}(F(x))$.

Theorem 2.4.12 (preprint 2.31). Let φ be a chart of an M-polyfold around x with $\varphi(x)=\iota_1 z$, r a retraction onto its model and T a presentation. For every other chart ψ around x, the overlaps are $O\cap W$ and $O'\cap W'$, the transition maps form a germ between the restricted retractions, and the restricted retraction has the same derivative at z Chapter2/Boundary · ScRetraction.scDeriv_restrict Chapter2/Boundary · reducedTangentSpaceWith_restrict; so the core gives $\dim(T_zO/T^R_zO)\le d(x,\psi)$ Chapter2/Boundary · MPolyfold.quotDim_le_chartDegIndex. Taking ψ realizing $d_X(x)$: $\dim(T_zO/T^R_zO)\le d_X(x)$ Chapter2/Boundary · theorem_2_4_12. For a retract O as an M-polyfold with the chart id this is the text's statement $\dim(T_xO/T^R_xO)\le d_O(x)$, at points on level 1 Chapter2/Boundary · theorem_2_4_12_retract. The text instead reduces to a chart realizing $d_O(x)$ and counts linearly independent $\mathbb{R}^d$-components; the quotient isomorphism makes the reduction unnecessary. The second statement (equality makes C_xO a partial quadrant) is §II.41, proved without the hypothesis: at every smooth point of every retract, C_xO is a partial quadrant of dimension $\dim(T_xO/T^R_xO)$ Chapter2/PartialConeQuadrant · remark_2_4_8_isPartialQuadrant.

Corollary 2.4.13 (preprint 2.32). Let U be nonempty and relatively open in C, and X_U its M-polyfold Chapter2/DegeneracyIndex · corollary_2_4_13. The chart id gives $d_{X_U}\le d_C$ Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.degIndex_le. At a point x on level 1, a chart ψ realizing $d_{X_U}(x)$ is compatible with id, so $F=\psi$ (on $U_0=\mathrm{dom}\,\psi$, relatively open in C) and $G=\psi^{-1}$ form a germ from U_0 to the retract $\psi(V)$, and the core gives $d_C(x)\le d_{C'}(\psi(x))=d_{X_U}(x)$ Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.degIndex_le_degIndex_of_lvl1. For a general x, approximate by level-1 points $\sigma_i\in U$ with $d_C(\sigma_i)=d_C(x)$ (§II.12); by Lemma 2.4.3 (preprint 2.23), $d_{X_U}(\sigma_i)\le d_{X_U}(x)$ eventually, so $d_C(x)=d_C(\sigma_i)\le d_{X_U}(\sigma_i)\le d_{X_U}(x)$. The text approximates by smooth points; level 1 suffices. **Theorem 1.3.5 (preprint 1.25)** (Part I, §I.19) is proved in the same file by the same argument for a germ between relatively open subsets of two quadrants Chapter2/DegeneracyIndex · theorem_1_3_5_lvl1

Chapter2/DegeneracyIndex · theorem_1_3_5.

II.14 Tame retractions; Propositions 2.5.3 and 2.5.6 (preprint 2.35 and 2.36)

Definition 2.5.1 (preprint 2.33). C_e and E_e are the cone and subspace of §II.12 for the presentation T. They do not depend on it: for two presentations T, T' of C and $e \in C$, $v \in E_e$ (for T) gives $e+tv \in C$ for $|t|$ small Chapter2/TameFaces · eventually_add_smul_mem_of_minimalSubspace; for j with $(T'e)_j = 0$, $(T'(e+tv))_j = t(T'v)_j \geq 0$ for $t = \pm\delta/2$, so $(T'v)_j = 0$ Chapter2/TameFaces · minimalSubspaceWith_le; by symmetry the two subspaces agree Chapter2/TameFaces · minimalSubspaceWith_eq. This is the argument of Lemma 1.3.4 (preprint 1.24).

Definition 2.5.2 (preprint 2.34). r is *tame for* T if (1) $d_C(r(x)) = d_C(x)$ on U and (2) at every smooth $x \in O_1$ some $A \leq E_x$ makes $E = Dr(x)E_0 \oplus A$ an sc-splitting Chapter2/Tame · ScRetraction.IsTameWith; *tame* if tame for every presentation Chapter2/Tame · ScRetraction.IsTame, and by the independence of E_x tameness for one presentation suffices Chapter2/TameFaces · ScRetraction.IsTameWith.isTame. A *tame retract* is the retract of a tame retraction Chapter2/Tame · IsTameRetract Chapter2/Tame · IsTameRetract.isScRetract. Identities of relatively open sets are tame ($T_xU = E$, $A = 0$) Chapter2/Tame · ScRetraction.id_isTame, and restrictions of tame retractions to $U \cap r^{-1}(V)$ are tame (same derivatives) Chapter2/TameFaces · ScRetraction.IsTameWith.restrict.

Proposition 2.5.3 (preprint 2.35). *$Dr(x)E_x \subseteq E_x$ for every retraction at every $x \in O_1$*: the two-sided sign argument for r, with $r(x) = x$ Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace. At a smooth x, $Dr(x)$ is an sc-operator Chapter2/TameRetracts · ScRetraction.isScOperator_scDeriv, so $E = Dr(x)E \oplus (1 - Dr(x))E$ is an sc-splitting Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv. *The proposition* Chapter2/TameRetracts · proposition_2_5_3: with $A \leq E_x$ from tameness (2), $E_0 = Dr(x)E_0 \oplus A$ algebraically (the level-0 part of the sc-splitting); write $v = t+a$; then $(1-Dr(x))v = (1-Dr(x))a = a - Dr(x)a \in E_x$, because $a \in E_x$ and $Dr(x)a \in E_x$. Only condition (2) is used; the text uses condition (1) for $Dr(x)E_x \subseteq E_x$. In coordinates: $\lambda_j(Dr(x)v) = \lambda_j(v)$ for $j \in I(x)$ Chapter2/TameRetracts · coord_scDeriv_eq.

Proposition 2.5.6 (preprint 2.36). *(13) at smooth points.* If $V = A' \oplus B$ and $B \leq K$, then $A'/(A' \cap K) \cong V/K$ Chapter2/TameRetracts · quotDim_eq_finrank_quotient_of_isCompl; with $A' = T_xO$, $B = A$ and $K = E_x$ this gives $\dim(T_xO/T_x^RO) = \dim(E_0/E_x) = d_C(x)$ Chapter2/TameRetracts · quotDim_reducedTangentSpaceWith_eq — the text's equations (15)–(16). *A tame chart realizes the index* Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith: let φ be a chart of any M-polyfold whose model has a tame retraction r. At y with $\varphi(y)$ smooth, $d(y,\varphi) = \dim(T/T^R) \leq d_X(y)$ by Theorem 2.4.12 (preprint 2.31), and $d_X(y) \leq d(y,\varphi)$ by definition. At a general y, with $\varphi(y) = o$, approximate o by smooth points σ_i of C with the same vanishing coordinates; then $r(\sigma_i)$ are smooth points of O (an sc^0 map preserves smoothness Chapter2/TameRetracts · IsSc0.mem_smooth Chapter2/TameRetracts · exists_smoothLvl_one_of_mem_smooth) converging to $r(o) = o$, with $d_C(r(\sigma_i)) = d_C(\sigma_i) = d_C(o)$ by tameness (1); so the points $\varphi^{-1}(r(\sigma_i))$ converge to y and have $d_X = d(\cdot,\varphi) = d_C(o)$, and Lemma 2.4.3 (preprint 2.23) gives $d_C(o) \leq d_X(y)$. For charts of the structure modelled on a tame retract Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract. Applied to the retract itself as an M-polyfold Chapter2/TameRetracts · ScRetraction.toLocalModel Chapter2/TameRetracts · LocalModel.degIndex_toMPolyfold_of_isTameWith: (14) $d_O = d_C$ on O Chapter2/TameRetracts · proposition_2_5_6_degIndex and (13) $\dim(T_xO/T_x^RO) = d_O(x) = d_C(x)$ at smooth points Chapter2/TameRetracts · proposition_2_5_6. This is the text's proof, generalized from the retract to any chart; Proposition 2.5.14 (preprint 2.42) needs the

general form.

Splitting off finitely many coordinates. Let $B \le A$ be sc-subspaces of E, $\ell_1, \dots, \ell_d$ continuous functionals on E_0 with $B = A \cap \bigcap_j \ker \ell_j$, and $e_1, \dots, e_d \in A$ smooth with $\ell_i(e_j) = \delta_{ij}$. Each e_j has a lift to every level m, compatible with the inclusions `Chapter2/PartialCone · smoothLift` `Chapter2/PartialCone · incl_smoothLift`. On level m the maps $v \mapsto ((\ell_j v)_j,\ v - \sum_j \ell_j(v) e_j)$ and $(a,u) \mapsto u + \sum_j a_j e_j$ are continuous linear $A_m \to \mathbb{R}^d \times B_m$ and back `Chapter2/PartialCone · splitFwd` `Chapter2/PartialCone · splitBwd` — the bracket lies in B because every ℓ_i kills it `Chapter2/PartialCone · sub_sum_mem` — and mutually inverse `Chapter2/PartialCone · splitBwd_splitFwd` `Chapter2/PartialCone · splitFwd_splitBwd`; they commute with the inclusions, so they form an sc-isomorphism $A \cong \mathbb{R}^d \oplus B$ `Chapter2/PartialCone · splitIso`, A and B with the sc-structures induced from E `Chapter1/ScBanach · ScBanach.ofSubspace`. It carries $\{v \in A : \ell_j(v) \ge 0\ \forall j\}$ onto $[0,\infty)^d \oplus B$, so that set is a partial quadrant in A `Chapter2/PartialCone · isPartialQuadrant_ofSubspace`. This is the map S of the proof of Theorem 2.4.12 (preprint 2.31), with the e_j smooth as the exposition requires.

Proposition 2.5.6 (preprint 2.36), the remaining statements `Chapter2/PartialCone · proposition_2_5_6_partialCone`. Let r be tame for T, $x \in O_1$ smooth, $P = Dr(x)$ and $I(x) = \{i_1, \dots, i_d\}$. $T_xO = PE$ is an sc-subspace, the range of the sc-projection P (Proposition 2.5.3 (preprint 2.35) above) `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`. The vectors $\hat{e}_j := P(T^{-1}(e_{i_j}, 0))$ `Chapter2/TameRetracts · coordVec` lie in T_xO; they are smooth, because $T^{-1}(e_{i_j}, 0)$ is smooth `Chapter2/PartialCone · coordVec_mem_smooth` and P is an sc-operator at the smooth point x `Chapter2/TameRetracts · ScRetraction.isScOperator_scDeriv` `Chapter1/ScBanach · ScOperator.mapsTo_smoothLvl`; and $\lambda_{i_k}(\hat{e}_j) = \lambda_{i_k}(T^{-1}(e_{i_j}, 0)) = \delta_{jk}$ by Proposition 2.5.3 (preprint 2.35) in coordinates `Chapter2/TameRetracts · coord_scDeriv_eq`. By definition $T_x^R O = T_xO \cap \bigcap_k \ker \lambda_{i_k}$ `Chapter2/TameRetracts · coordL`. $T_x^R O$ *is an sc-subspace.* It is closed, as an intersection of closed subspaces `Chapter2/DegeneracyIndex · isClosed_minimalSubspaceWith`. For the density of smooth points, let $u \in T_x^R O \cap E_m$ with $Tu = (a,w)$ on level m, and put $\phi(w') := P(T^{-1}(a,w'))$, a continuous map $W_m \to E_m$. For smooth w' the point $\phi(w')$ is smooth, lies in T_xO, and lies in E_x, because $T^{-1}(a,w')$ has the same vanishing coordinates in $I(x)$ as u and $PE_x \subseteq E_x$ `Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace`. And $\phi(w) = Pu = u$, since $u \in T_xO$ and P is a projection `Chapter2/Retracts · ScRetraction.scDeriv_comp_self`. As W_∞ is dense in W_m, u lies in the closure of $\phi(W_\infty) \subseteq T_x^R O \cap E_\infty$ (this is the exposition's Lemma 3.A (1) in the case at hand). *The quadrant.* The splitting with $A = T_xO$, $B = T_x^R O$, $\ell_j = \lambda_{i_j}$ and $e_j = \hat{e}_j$ makes $\{v \in T_xO : \lambda_{i_j}(v) \ge 0\ \forall j\}$ a partial quadrant in T_xO, and this set is $C_xO = T_xO \cap C_x$. The text deduces the claim from the equality case of Theorem 2.4.12 (preprint 2.31), which needs smooth lifts $\hat{v}^j$ of a basis of $T_xO / T_x^R O$ (the exposition chooses them by a density argument); for a tame retraction they are the explicit $\hat{e}_j$, and the general equality case, proved in §II.41, is not needed here.

II.15 Faces: local geometry, Lemma 2.5.12 (preprint 2.40), Corollary 2.5.13 (preprint 2.41), Proposition 2.5.14 (preprint 2.42)

Definitions 2.5.7, 2.5.10 (preprint 2.37, 2.39). X is *tame* if an atlas equivalent to its atlas has only tame models `Chapter2/Tame · MPolyfold.IsTame`; charts of an equivalent atlas are charts of the structure `Chapter2/TameRetracts · MPolyfold.isChart_of_mem_equiv`, so every point lies in a chart of the structure with a tame model `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`. A *face* is the closure of the connected component in $\{d_X = 1\}$ of a point with $d_X = 1$ `Chapter2/Tame · MPolyfold.IsFace`, closed `Chapter2/Tame · MPolyfold.IsFace.isClosed` and inside ∂X `Chapter2/Boundary · MPolyfold.IsFace.subset_boundary`; X is *face-structured* if every x

lies in exactly $d_X(x)$ faces Chapter2/Tame · MPolyfold.IsFaceStructured.

Coordinates of faces. $I(z) = \{i : \lambda_i(z) = 0\}$ Chapter2/TameRetracts · zeroSet with $d_C = \#I$ Chapter2/TameRetracts · degIndexWith_eq_card_zeroSet; the extended face f_i is the hyperplane $\{\lambda_i = 0\}$, closed Chapter2/TameRetracts · coordHyper Chapter2/TameRetracts · isClosed_coordHyper; $I(z) \subseteq I(e)$ for z near e Chapter2/TameRetracts · eventually_zeroSet_subset. *A connected piece of $\{d_C = 1\}$ lies in one face* (exposition Lemma 4.A): if $S \subseteq C$ is preconnected with $d_C = 1$ on S and $\lambda_i(p) = 0$ at some $p \in S$, then $S \subseteq f_i$ Chapter2/TameRetracts · subset_coordHyper_of_isPreconnected. Points of S have $I = \{i\}$ or $\lambda_i > 0$ Chapter2/TameRetracts · zeroSet_eq_singleton, so S is covered by the open sets $u = \{\lambda_l > 0 \text{ for } l \neq i\}$ and $v = \{\lambda_i > 0\}$, which are disjoint on S (a point in both would have $d_C = 0$); preconnectedness puts S in one of them Mathlib/Topology/Connected/Clopen · isPreconnected_iff_subset_of_disjoint, and $p \in u$.

Rays stay in their face (exposition Lemma 4.B). If f is sc¹, $x + tv \in U_1$ for small $t \geq 0$ and a functional L vanishes on $f(x + tv)$ for $t \in [0, \varepsilon)$, then $L(Df(x)v) = 0$ Chapter2/TameRetracts · IsSc1With.apply_deriv_eq_zero_of_ray. Now let r be tame, $x \in O_1$ smooth, $i \in I(x)$, $\lambda_i(y) = 0$ and $\lambda_l(y) > 0$ for the other $l \in I(x)$ Chapter2/TameRetracts · ray_zeroSet. For small $\tau > 0$, $x + \tau y \in U$ and $I(x + \tau y) = \{i\}$, so $d_C(r(x + \tau y)) = 1$ by tameness (1), and $I(r(x + \tau y)) \subseteq I(x)$ by continuity of r at $x = r(x)$. The image of $(0, \tau_0]$ under $\tau \mapsto r(x + \tau y)$ is preconnected with $d_C = 1$, so it lies in one f_j with $j \in I(x)$. Differentiating $\lambda_j(r(x + \tau y)) = 0$ at $\tau = 0$ gives $\lambda_j(Dr(x)y) = 0$, and $\lambda_j(Dr(x)y) = \lambda_j(y)$ by Proposition 2.5.3 (preprint 2.35) Chapter2/TameRetracts · coordL; as $\lambda_l(y) > 0$ for $l \in I(x) \setminus \{i\}$, $j = i$. This is the step of Lemma 2.5.12 (preprint 2.40) that the text omits, and it needs tameness (2).

The local structure of a face (exposition Lemma 4.C) Chapter2/TameRetracts · local_face_structure. Let $S = \{z \in O : d_C(z) = 1\}$ Chapter2/TameRetracts · ScRetraction.faceStratum and $e \in O$ with $\lambda_i(e) = 0$. (a) Choose ε so that $B = B_\varepsilon(e) \cap C \subseteq U$ and $I(z), I(r(z)) \subseteq I(e)$ on B. (b) $A_i \cap B = \{z \in B : I(z) = \{i\}\}$ is convex Chapter2/TameRetracts · convex_ball_inter_zeroSet_eq, hence connected, and r maps it into S; a smooth $u = r(T^{-1}(a, w')) \in O \cap B$ with $I(u) = I(e)$ and the ray of Lemma 4.B from u in the direction $T^{-1}(b, 0)$, $b = 1$ on $I(e) \setminus \{i\}$ and 0 elsewhere Chapter2/TameRetracts · coordVec Chapter2/TameRetracts · rayDir Chapter2/TameRetracts · exists_lvl1_coordVec, produces a point $p \in S$ with $\lambda_i(p) = 0$ in $r(A_i \cap B)$; so $r(A_i \cap B)$ lies in the component of S through p. (c) A point $e' \in f_i \cap B$ is the limit of points of $A_i \cap B$ (perturb the coordinates of $I(e') \setminus \{i\}$ upward), so $r(e')$ lies in the closure of that component.

Lemma 2.5.12 (preprint 2.40). A *face of the retract* is $\overline{\hat{F}} \cap O$ for a component $\hat{F}$ of S Chapter2/TameRetracts · ScRetraction.IsRetractFace. A component of S through a point with $\lambda_i = 0$ lies in f_i Chapter2/TameRetracts · faceStratum_degIndexWith Chapter2/TameRetracts · connectedComponentIn_faceStratum_subset, and a component of S in f_i whose closure contains e is the component of Lemma 4.C at e Chapter2/TameRetracts · connectedComponentIn_eq_of_mem_closure. *The face through $x \in S$, $\lambda_i(x) = 0$, is the component of $f_i \cap O$ through x* Chapter2/TameRetracts · closure_connectedComponentIn_faceStratum: it is connected, lies in $f_i \cap O$, is closed in it, and is relatively open in it by Lemma 4.C(c); a preconnected subset meeting a relatively clopen set lies in it Chapter2/TameRetracts · subset_of_relClopen_of_isPreconnected. Conversely, the component of $f_i \cap O$ through any $e \in O$ with $\lambda_i(e) = 0$ is the face through the point p of Lemma 4.C at e Chapter2/TameRetracts · connectedComponentIn_coordHyper_eq. Together: the faces of the retract are the components of the sets $f_i \cap O$ Chapter2/TameRetracts · lemma_2_5_12. If the components of $f_i \cap O$ and $f_j \cap O$ through x coincide, a point z of the face in S has $I(z) = \{i\}$ and is a limit of points with $I = \{j\}$, which near z forces $j = i$ Chapter2/TameRetracts · lemma_2_5_12_injective. The faces of the M-polyfold O (Definition 2.5.10 (preprint 2.39), with $d_O = d_C$ by Proposition 2.5.6 (preprint 2.36)) are exactly the faces of the retract, transferred along the subtype inclusion Chapter2/TameRetracts · image_val_connectedComponentIn Chapter2/TameRetracts · image_val_closure Chapter2/TameRetracts · LocalModel.isFace_toMPolyfold_iff.

Corollary 2.5.13 (preprint 2.41). The map $i \mapsto$ (component of $f_i \cap O$ through x) sends $I(x)$ onto the faces through x (Lemma 2.5.12 (preprint 2.40)) and is injective (its second statement), so x lies in exactly $\#I(x) = d_C(x)$ faces of the retract `Chapter2/TameRetracts · corollary_2_5_13_retract`, and the M-polyfold O is face-structured `Chapter2/TameRetracts · LocalModel.isFaceStructured_toMPolyfold` `Chapter2/TameRetracts · corollary_2_5_13`.

Proposition 2.5.14 (preprint 2.42). Let X be tame. *The global bound* `Chapter2/TameRetracts · MPolyfold.faces_encard_le`: in a tame chart φ around x, $o = \varphi(x)$, Lemma 4.C gives for each $i \in I(o)$ a component $\hat{F}_i$ of S containing all points of O near o with $I = \{i\}$. A face $F = \overline{G}$ through x has points $y \in G$ near x; $d_C(\varphi(y)) = d_X(y) = 1$ (the tame chart realizes d_X) and $I(\varphi y) \subseteq I(o)$, so $I(\varphi y) = \{i\}$ for some $i \in I(o)$, and the preconnected set $\varphi^{-1}(\hat{F}_i) \subseteq \{d_X = 1\}$ meets G, hence lies in G. Distinct faces give distinct i; so at most $d_X(x)$ faces pass through x. *The local statement*: the tame chart realizes d_X on its domain, the M-polyfold O has $d_O = d_C$ (Proposition 2.5.6 (preprint 2.36)), and O is face-structured (Corollary 2.5.13 (preprint 2.41)) `Chapter2/TameRetracts · proposition_2_5_14`. The neighbourhood V of the text is represented by O.

II.16 Faces are tame sub-M-polyfolds: Propositions 2.5.15, 2.5.16 (preprint 2.43, 2.44) and Theorem 2.5.17 (preprint 2.45)

The projection onto a face. $P_i := T^{-1} \circ (a \mapsto a \text{ with } a_i = 0) \circ T$ is an sc-operator `Chapter2/TameRetracts · killCoordL` `Chapter2/TameRetracts · killCoordModel` `Chapter2/TameRetracts · killCoord` which kills the i-th coordinate and keeps the others `Chapter2/TameRetracts · killCoord_coord` `Chapter2/TameRetracts · killCoord_coord_self`, fixes f_i `Chapter2/TameRetracts · killCoord_eq_self` and maps C into C `Chapter2/TameRetracts · killCoord_mem`.

Proposition 2.5.15 (preprint 2.43), the retraction. Let F be a face of the retract, through $x \in S$ with $I(x) = \{i\}$, so $F \subseteq f_i$ `Chapter2/TameRetracts · proposition_2_5_15`. For each $e \in F$, Lemma 4.C gives ε_e with $B_e = B_{\varepsilon_e}(e) \cap C \subseteq U$ and $r(z) \in F$ for $z \in B_e \cap f_i$ (the component of Lemma 4.C at e is the one of F). Put $U^* := \bigcup_{e \in F}(B_{\varepsilon_e}(e) \cap P_i^{-1}(B_{\varepsilon_e}(e))) \cap C$: relatively open in C, contains F (as $P_i e = e$), and for $z \in U^*$ the point $P_i z$ lies in some $B_e \cap f_i$, so $r(P_i z) \in F$. Then $s := r \circ P_i$ is sc-smooth on U^* (chain rule), maps U^* into $F \subseteq U^*$, and fixes F; so it is a retraction onto F, and $U^* \subseteq U$. The preimages under P_i are needed because P_i need not be norm-decreasing for the norm of E_0; the exposition's "$|P_1 z - e|_0 \leq |z - e|_0$" holds only for special norms. *F is a sub-M-polyfold of O*: in the chart id of O, s restricted to $O \cap U^*$ is an sc-smooth retraction onto F `Chapter2/TameRetracts · LocalModel.selfChart_symm_apply` `Chapter2/TameRetracts · LocalModel.isSubMPolyfold_of_isFace`.

The face as a model space. Insert and delete a coordinate: $\mathbb{R}^k \to \mathbb{R}^{k+1}$, $b \mapsto (b_1, \dots, 0_i, \dots, b_k)$, and back `Chapter2/TameFaces · insertCoordL` `Chapter2/TameFaces · removeCoordL`, as sc-operators on $\mathbb{R}^k \oplus W$ `Chapter2/TameFaces · insertCoordModel` `Chapter2/TameFaces · removeCoordModel`; composed with T they give $\iota : \mathbb{R}^k \oplus W \to E$ onto f_i and $\pi : E \to \mathbb{R}^k \oplus W$ `Chapter2/TameFaces · faceIncl` `Chapter2/TameFaces · faceProj`, with $\pi\iota = 1$, $\iota\pi = 1$ on f_i `Chapter2/TameFaces · faceProj_faceIncl` `Chapter2/TameFaces · faceIncl_faceProj`, $\iota(C') \subseteq C$ and $\pi(C) \subseteq C'$ for $C' = [0, \infty)^k \oplus W$ `Chapter2/TameFaces · faceIncl_mem` `Chapter2/TameFaces · faceProj_mem`, and $d_C(y) = d_{C'}(\pi y) + 1$ on f_i `Chapter2/TameFaces · degIndexWith_eq_quadrantDeg_faceProj_add_one`. The derivative of an sc-operator is itself `Chapter2/TameFaces · ScOperator.scDeriv_eq`, and the chain rule at a point holds between arbitrary sc-domains `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain`.

Proposition 2.5.15 (preprint 2.43), the tame model `Chapter2/TameFaces · proposition_2_5_15_tameModel`. With

U^* and s as above, put $U' := \iota^{-1}(U^*)$ (relatively open in C') and $r' := \pi \circ r \circ \iota$ on U'. Since $r \circ \iota$ maps U' into $F \subseteq f_i$, where $\iota\pi = 1$, r' is an sc-smooth retraction with retract $\pi(F)$, and ι maps it back into F. *Tameness (1)*: for $z \in U'$, $d_{C'}(r'z) + 1 = d_C(r(\iota z)) = d_C(\iota z) = d_{C'}(z) + 1$. *Tameness (2)*: at a smooth $x \in \pi(F)$, ιx is a smooth point of O and $Dr'(x)v = \pi(Dr(\iota x)(\iota v))$ by the chain rule; for j with $x_j = 0$ the coordinate $j' = i.\mathrm{succAbove}(j)$ vanishes at ιx, and Proposition 2.5.3 (preprint 2.35) for r gives $\lambda_{j'}(Dr(\iota x)\iota v) = \lambda_{j'}(\iota v) = v_j$; so $\lambda'_j \circ Dr'(x) = \lambda'_j$, and $(1 - Dr'(x))E'$ is an sc-complement of $T_x\pi(F)$ inside E'_x. The exposition instead intersects the complement A of T_xO with f_1; that is not needed. And $d_{C'}(\pi y) + 1 = d_C(y)$ on F.

Proposition 2.5.15 (preprint 2.43), faces are tame `Chapter2/TameFaces · LocalModel.proposition_2_5_15_isTame`. For a face F of a tame retract O (paracompact, being closed), the chart $\pi : F \to \pi(F)$ `Chapter2/TameFaces · MPolyfold.faceChart` `Chapter2/TameFaces · MPolyfold.faceChartInv` is compatible with the sub-charts of F built on the chart id of O, its transition maps being the sc-operators ι and π `Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart`; a chart compatible with all sub-charts built on one chart of X is a chart of the induced structure, by locality of compatibility `Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible`. Its model is the tame retract $(\pi(F), C', E')$, so F is tame (a space all of whose points lie in tame charts is tame `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart`), and the tame chart realizes the index: $d_F(x) = d_{C'}(\pi x) = d_C(x) - 1 = d_O(x) - 1$ (equation (14) twice). The text calls this last statement obvious.

Proposition 2.5.16 (preprint 2.44), first sentence `Chapter2/TameFaces · MPolyfold.proposition_2_5_16`. In a tame chart φ around x: $\varphi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O (Proposition 2.5.14 (preprint 2.42) with Corollary 2.5.13 (preprint 2.41)), and each face of O is a tame sub-M-polyfold with $d_F + 1 = d_O$, where $d_O(\varphi x) = d_X(x)$.

The corner as a model space. For $\sigma \subseteq \{1, \dots, n\}$ list the coordinates outside σ as $j \mapsto l_j$ `Chapter2/CornerGerms · cornerIdx` `Chapter2/CornerGerms · cornerIdx_injective` `Chapter2/CornerGerms · exists_cornerIdx`. Deleting the coordinates in σ and inserting zeros there `Chapter2/CornerGerms · cornerRemoveL` `Chapter2/CornerGerms · cornerInsertL` are mutually inverse on $\{a : a_l = 0,\ l \in \sigma\}$ `Chapter2/CornerGerms · cornerRemoveL_cornerInsertL` `Chapter2/CornerGerms · cornerInsertL_cornerRemoveL`; as sc-operators on $\mathbb{R}^n \oplus W$ and $\mathbb{R}^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerInsertModel` `Chapter2/CornerGerms · cornerRemoveModel`, composed with T, they give $\iota_\sigma : \mathbb{R}^{n-\#\sigma} \oplus W \to E$ onto the corner $f_\sigma = \bigcap_{i\in\sigma} f_i$ and $\pi_\sigma : E \to \mathbb{R}^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerIncl` `Chapter2/CornerGerms · cornerProj`, with $\pi_\sigma\iota_\sigma = 1$ and $\iota_\sigma\pi_\sigma = 1$ on f_σ `Chapter2/CornerGerms · cornerProj_cornerIncl` `Chapter2/CornerGerms · cornerIncl_cornerProj`, $\iota_\sigma(C') \subseteq C$ and $\pi_\sigma(C) \subseteq C'$ for $C' = [0,\infty)^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerIncl_mem` `Chapter2/CornerGerms · cornerProj_mem`, and $d_C(y) = d_{C'}(\pi_\sigma y) + \#\sigma$ on f_σ: the vanishing coordinates of y are those of σ together with the images l_j of the vanishing coordinates of $\pi_\sigma y$, disjointly `Chapter2/CornerGerms · degIndexWith_eq_quadrantDeg_cornerProj_add_card`. The sc-projection $P_\sigma = \iota_\sigma \circ \pi_\sigma$ `Chapter2/CornerGerms · cornerKill` sets the coordinates in σ to zero `Chapter2/CornerGerms · cornerKill_coord_mem`, fixes f_σ `Chapter2/CornerGerms · cornerKill_eq_self` and maps C into C `Chapter2/CornerGerms · cornerKill_mem`. For $\sigma = \{i\}$ these are the ι, π and P_i of Proposition 2.5.15 (preprint 2.43).

The retraction onto a corner `Chapter2/CornerGerms · corner_retraction`. Let r be tame for T, $o \in O$ and $\sigma \subseteq I(o)$. For each $i \in \sigma$, Lemma 4.C at o `Chapter2/TameRetracts · local_face_structure` gives a face F_i of the retract with $o \in F_i \subseteq f_i$, containing a point with $d_C = 1$, and a ball around o in which r maps the points of $C \cap f_i$ into F_i. Distinct i give distinct faces: a point $p \in F_i$ with $d_C(p) = 1$ has $I(p) = \{i\}$, so $F_i = F_j$ forces $i = j$ `Chapter2/TameRetracts · zeroSet_eq_singleton`. Take a ball B around o inside all these balls with $B \cap C \subseteq U$, an open W_s with $W_s \cap U = r^{-1}(B) \cap U$ (continuity of r), and put $V := B \cap P_\sigma^{-1}(B \cap W_s)$,

an open set containing o (since $P_\sigma o = o = r(o)$). On $V \cap C$ the map $s := r \circ P_\sigma$ is sc-smooth by the chain rule. For $z \in V \cap C$, $P_\sigma z \in C \cap f_\sigma \cap B$, so $r(P_\sigma z) \in F_i \subseteq f_i$ for every $i \in \sigma$: $s(z) \in O \cap f_\sigma$. Moreover $s(z) \in B$ (as $P_\sigma z \in W_s$), $P_\sigma s(z) = s(z)$, and $s(z) \in W_s$ (as $r(s(z)) = s(z) \in B$); so s maps $V \cap C$ into itself. On $f_\sigma \cap O$, P_σ and r are the identity, so $s \circ s = s$: s is an sc-smooth retraction with retract $O \cap V \cap f_\sigma$, and this set equals $O \cap V \cap \bigcap_{i\in\sigma} F_i$ (a point of $O \cap V \cap f_\sigma$ is fixed by r, which maps it into each F_i; conversely $F_i \subseteq f_i$). The exposition proceeds by induction on $\#\sigma$, identifying the germ of $F_1 \cap F_2$ with a face of F_1; the formal proof removes all coordinates of σ at once.

The tame model of a corner `Chapter2/CornerGerms · corner_tameModel`. Let s be an sc-smooth retraction on $U_s \subseteq U$, equal to $r \circ P_\sigma$ there, with retract $F \subseteq O \cap f_\sigma$; write $U_s = W_0 \cap C$ with W_0 open, and put $U' := \iota_\sigma^{-1}(W_0) \cap C'$ and $r' := \pi_\sigma \circ r \circ \iota_\sigma$ on U'. For $z \in U'$, $r(\iota_\sigma z) = s(\iota_\sigma z) \in F$ (as $P_\sigma \iota_\sigma = \iota_\sigma$), so r' maps U' into $\pi_\sigma(F) \subseteq U'$; it is sc-smooth by the chain rule and idempotent because $\iota_\sigma \pi_\sigma = 1$ on F and $r = 1$ on $F \subseteq O$; its retract is $\pi_\sigma(F)$. *Tameness (1):* $d_{C'}(r'z) + \#\sigma = d_C(r(\iota_\sigma z)) = d_C(\iota_\sigma z) = d_{C'}(z) + \#\sigma$ (the index formula on f_σ twice, and tameness (1) of r) `Chapter1/Degeneracy · degIndex_eq`. *Tameness (2):* at a smooth x of the retract, $\iota_\sigma x$ is a smooth point of O, and $Dr'(x)v = \pi_\sigma(Dr(\iota_\sigma x)\, \iota_\sigma v)$ by the chain rule between sc-domains `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain` `Chapter2/TameFaces · ScOperator.scDeriv_eq`; for j with $x_j = 0$ the coordinate l_j vanishes at $\iota_\sigma x$ `Chapter2/CornerGerms · coord_cornerIncl_idx`, and Proposition 2.5.3 (preprint 2.35) gives $\lambda_{l_j}(Dr(\iota_\sigma x)\, \iota_\sigma v) = \lambda_{l_j}(\iota_\sigma v) = v_j$ `Chapter2/TameRetracts · coord_scDeriv_eq`; so $(1 - Dr'(x))E'$ lies in E'_x and is an sc-complement of $T_x \pi_\sigma(F)$ `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`. And $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$ for $y \in F$. This is the proof of the tame model of Proposition 2.5.15 (preprint 2.43) with σ in place of $\{i\}$.

Proposition 2.5.16 (preprint 2.44), second sentence `Chapter2/CornerGerms · MPolyfold.proposition_2_5_16_corner`. Let X be tame and $x \in X$, and take a chart $\varphi : V \to O$ of the structure around x with a tame retraction r onto O, tame for a presentation T `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`. The face germs at x are the faces F_i, $i \in I(\varphi x)$, of the retract (Corollary 2.5.13 (preprint 2.41): there are $d_X(x)$ of them), and σ is a set of such indices. The retraction onto the corner gives the pairwise distinct faces $F_i \ni \varphi(x)$, $i \in \sigma$, an open $B \ni \varphi(x)$ and s with retract $O \cap B \cap \bigcap_{i\in\sigma} F_i$. Put $A := \{y \in V : \varphi(y) \in B,\ \varphi(y) \in F_i\ \forall i \in \sigma\}$; its germ at x is the germ F^σ_x. Since $\varphi(A)$ is the retract of s, A is a sub-M-polyfold by the local criterion of Theorem 2.5.17 (preprint 2.45) `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local` (paracompact as every subset `Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype`). The chart $\pi_\sigma \circ \varphi$ of A onto the retract $\pi_\sigma(\varphi(A))$ of r' `Chapter2/TameFaces · MPolyfold.faceChart` is compatible with the sub-charts of A built on φ, its transitions being ι_σ and π_σ `Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart`, hence a chart of the induced structure `Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible`; its model is tame `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame`, so A is tame `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart`. Both tame charts realize the index `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`, so for $a \in A$: $d_A(a) + \#\sigma = d_{C'}(\pi_\sigma \varphi(a)) + \#\sigma = d_C(\varphi(a)) = d_X(a)$; in particular $d_A(x) = d_X(x) - \#\sigma$, and the codimension, the number of quadrant coordinates removed in the tame model, is $\#\sigma$. The text states the proposition without proof.

Theorem 2.5.17 (preprint 2.45). *The interior of a face.* $F \cap \{d_X = 1\} = G$ for $F = \overline{G}$ `Chapter2/TameRetracts · MPolyfold.theorem_2_5_17_interior`: at $y \in F$ with $d_X(y) = 1$, in a tame chart with $I(\varphi y) = \{i\}$, all points of $\{d_X = 1\}$ near y lie in $\varphi^{-1}(\hat{F})$ for the single component $\hat{F}$ of Lemma 4.C at $\varphi(y)$, a connected subset of $\{d_X = 1\}$ containing y; it meets G (as $y \in \overline{G}$), hence lies in G. *A local criterion for sub-M-polyfolds* `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local`: if near each $a \in A$, in a chart φ, $\varphi(A \cap V)$ agrees on an open $B \ni \varphi(a)$ with the retract of an sc-smooth retraction s contained in O, then A is a sub-M-polyfold, with the retraction $\varphi^{-1} \circ s \circ \varphi$ on $V' = \varphi^{-1}(B \cap W_0 \cap s^{-1}(B))$ — shrunk

so that s maps it into itself, a step the exposition's "transport by φ^{-1}" omits. *G is a sub-M-polyfold* Chapter2/TameFaces · MPolyfold.theorem_2_5_17_isSubMPolyfold_interior: at $y \in G$, the local face $\overline{\widehat{F}} \cap O$ agrees with $\varphi(G)$ on the ball of Lemma 4.C (both consist of the points of S there), and Proposition 2.5.15 (preprint 2.43) gives its retraction. *The face-structured case.* Near $x \in F$, in a tame chart, $\varphi(F)$ agrees on a ball with a face of the retract Chapter2/TameFaces · MPolyfold.exists_localFace_of_isFaceStructured: the map $i \mapsto F_i$ (the closure of the component of $\varphi^{-1}(\widehat{F}_i)$) from $I(\varphi x)$ to the faces through x is onto (as in Proposition 2.5.14 (preprint 2.42)), and since there are exactly $d_X(x) = \#I(\varphi x)$ faces through x, it is injective; so the points of G near x lie in $\varphi^{-1}(\widehat{F}_j)$ for the one j with $F = F_j$. Hence F is a sub-M-polyfold Chapter2/TameFaces · MPolyfold.theorem_2_5_17_isSubMPolyfold_face, and with the charts $\pi \circ \varphi$ onto the restrictions of the tame retractions r' it is tame with $d_X(x) = d_F(x) + 1$ Chapter2/TameFaces · MPolyfold.theorem_2_5_17_isTame_face. The local and global face counts of the theorem are Proposition 2.5.14 (preprint 2.42) and the definition of face-structured.

II.17 Products and fibred products: Lemma 4.D and Proposition 2.5.20 (preprint 2.46)

Products of partial quadrants. For presentations $T : E \cong \mathbb{R}^n \oplus W$ and $T' : E' \cong \mathbb{R}^{n'} \oplus W'$ of C and C', concatenating the coordinates ($\mathbb{R}^n \times \mathbb{R}^{n'} \cong \mathbb{R}^{n+n'}$ Chapter2/FibredProduct · finAppendL) gives a presentation $E \oplus E' \cong \mathbb{R}^{n+n'} \oplus (W \oplus W')$ Chapter2/FibredProduct · prodPresentation of $C \times C'$ Chapter2/FibredProduct · prod_eq_prodPresentation, so $C \times C'$ is a partial quadrant Chapter2/FibredProduct · IsPartialQuadrant.prod and $U \times U'$ is relatively open in it Chapter2/FibredProduct · IsRelOpenInQuadrant.prod. The vanishing coordinates of (x, x') are those of x followed by those of x', so $d_{C\times C'}(x, x') = d_C(x) + d_{C'}(x')$ Chapter2/FibredProduct · degIndexWith_prodPresentation Chapter2/FibredProduct · degIndex_prod, and the minimal subspace is $E_x \oplus E'_{x'}$ Chapter2/FibredProduct · minimalSubspaceWith_prodPresentation.

Products of retractions. $f \times f'$ is sc-smooth when f and f' are Chapter2/FibredProduct · IsScSmooth.prodMap, with derivative $Df(x) \times Df'(x')$ Chapter2/FibredProduct · scDeriv_prodMap; so $r \times r'$ is an sc-smooth retraction Chapter2/FibredProduct · ScRetraction.prod with retract $O \times O'$ Chapter2/FibredProduct · ScRetraction.prod_retract Chapter2/FibredProduct · IsScRetract.prod, and products of sc-smooth maps between retracts are sc-smooth Chapter2/FibredProduct · IsScSmoothRetractMap.prodMap. Products of sc-subspaces and of sc-splittings are again such Chapter2/FibredProduct · IsScSubspace.prod Chapter2/FibredProduct · IsScSplitting.prod; hence if r, r' are tame for T, T', then $r \times r'$ is tame for the product presentation Chapter2/FibredProduct · ScRetraction.IsTameWith.prod: (1) is the additivity of the index, and for (2) the complement of $T_{(x,x')}(O \times O') = T_xO \oplus T_{x'}O'$ is $A \oplus A'$ inside $E_x \oplus E'_{x'}$.

Lemma 4.D (the product M-polyfold). $X \times Y$ is paracompact, being metrizable by Theorem 2.3.10 (preprint 2.16) Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod. The product charts Chapter2/FibredProduct · MChart.prod have the product retracts as models Chapter2/FibredProduct · LocalModel.prod, their overlaps are products Chapter2/FibredProduct · MChart.overlap_prod, and products of compatible charts are compatible Chapter2/FibredProduct · MChart.Compatible.prod; so they form an atlas of an M-polyfold structure on $X \times Y$ Chapter2/FibredProduct · MPolyfold.prod, in which products of charts of the factors are charts Chapter2/FibredProduct · MPolyfold.IsChart.prod. In a product chart the index is additive Chapter2/FibredProduct · MPolyfold.chartDegIndex_prod; choosing charts that realize $d_X(x)$ and $d_Y(y)$ gives $d_{X\times Y}(x, y) \le d_X(x) + d_Y(y)$ Chapter2/FibredProduct · MPolyfold.degIndex_prod_le. If X and Y are tame, products of tame charts are tame charts, so $X \times Y$ is tame, and since a tame chart realizes the index (§II.14), $d_{X\times Y}(x, y) = d_X(x) + d_Y(y)$ Chapter2/FibredProduct · MPolyfold.isTame_prod. The text's "$d_{X\times Y} = d_X + d_Y$"

is thus a statement about tame spaces. The projections are sc-smooth `Chapter2/FibredProduct · MPolyfold.isScSmoothMap_fst` `Chapter2/FibredProduct · MPolyfold.isScSmoothMap_snd` and so is the pairing $z \mapsto (f(z), g(z))$ of sc-smooth maps `Chapter2/FibredProduct · MPolyfold.IsScSmoothOn.prodMk`.

Proposition 2.5.20 (preprint 2.46). A *local sc-diffeomorphism* $f : X \to Z$ is read as: every point has an open neighbourhood U that f maps sc-diffeomorphically onto an open subset `Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo`; such an f is continuous and sc-smooth `Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo.continuous` `Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo.isScSmoothMap`. The fibred product $X_f \times_g Y = \{(x, y) : f(x) = g(y)\}$ `Chapter2/FibredProduct · MPolyfold.fibredProduct` is closed in $X \times Y$ `Chapter2/FibredProduct · MPolyfold.isClosed_fibredProduct`, hence paracompact, which the text does not mention. *Sub-M-polyfold.* Near (x, y), with $f|_U : U \to W$ an sc-diffeomorphism and $V = g^{-1}(W)$, the map $R(x', y') = ((f|_U)^{-1}(g(y')), y')$ on $U \times V$ is sc-smooth, idempotent, with image the fibred product near (x, y); in a product chart it is an sc-smooth retraction `Chapter2/FibredProduct · MPolyfold.fibredSubDataL`, and so the fibred product is a sub-M-polyfold `Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct` (symmetrically when g is the local sc-diffeomorphism `Chapter2/FibredProduct · MPolyfold.fibredSubDataR` `Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct'`). *π_2 is a local sc-diffeomorphism.* On the fibred product near (x, y), π_2 is sc-smooth with inverse $y' \mapsto ((f|_U)^{-1}(g(y')), y')$, a pairing of sc-smooth maps with values in the fibred product, hence sc-smooth into the sub-M-polyfold by Proposition 2.3.19 (2) (preprint 2.21) `Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_snd_fibredProduct`; symmetrically for π_1 when g is a local sc-diffeomorphism `Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_fst_fibredProduct`. The exposition factors this inverse through $y' \mapsto (x, y')$, which is not sc-smooth unless x is a smooth point (a constant map is sc^0 only when its value lies in X_∞); the formula above has the same values and avoids it. *The index.* Proposition 2.4.4 (preprint 2.24) along π_2 gives $d_{X_f \times_g Y}(x, y) = d_Y(y)$ for arbitrary Y `Chapter2/FibredProduct · MPolyfold.degIndex_fibredProduct_eq_snd`, and along π_1, when g is a local sc-diffeomorphism, $= d_X(x)$ `Chapter2/FibredProduct · MPolyfold.degIndex_fibredProduct_eq_fst`. *Tameness.* If Y is tame, the charts $\psi \circ \pi_2$, ψ a tame chart of Y restricted to $g^{-1}(W)$, are charts of the fibred product with tame models `Chapter2/FibredProduct · MPolyfold.isTame_fibredProduct`. Together: Proposition 2.5.20 (preprint 2.46) `Chapter2/FibredProduct · MPolyfold.proposition_2_5_20`, and in its last statement $d = d_X = d_Y$ for arbitrary X, Y and, for tame X, Y, $2d = d_{X \times Y}$ by Lemma 4.D `Chapter2/FibredProduct · MPolyfold.proposition_2_5_20_of_isLocalScDiffeo`.

II.18 Strong bundle retractions (Definitions 2.6.1–2.6.3 (preprint 2.47–2.49))

$U \lhd F$. The double filtration is $U_m \times F_k$ `Chapter2/Tame · doubleLevel`. $C \times F_0$ is a partial quadrant in $E \oplus F$, through $(\mathbb{R}^n \oplus W) \oplus F \cong \mathbb{R}^n \oplus (W \oplus F)$ `Chapter2/StrongBundle · prodQuadrantIso` `Chapter2/StrongBundle · IsPartialQuadrant.prod_univ`, and $U \times F_0$ is relatively open in it `Chapter2/StrongBundle · IsRelOpenInQuadrant.prod_univ` — so "sc-smooth on $(U \lhd F)[i] = U \oplus F^{i}$" makes sense (exposition Lemma 1.B of Chapter 1). Pairings of sc^k maps are sc^k, with the tangent map rearranged by an sc-operator `Chapter2/StrongBundle · scPair` `Chapter2/StrongBundle · tangentProdSwap` `Chapter2/StrongBundle · IsSck.scPair` `Chapter2/StrongBundle · IsScSmooth.scPair`, and the projections and inclusions of products are sc-operators `Chapter2/StrongBundle · ScBanach.prodFstOp` `Chapter2/StrongBundle · ScBanach.prodSndOp` `Chapter2/StrongBundle · ScBanach.prodInlOp` `Chapter2/StrongBundle · ScBanach.prodShiftInclOp`.

Definition 2.6.1 (preprint 2.47). A strong bundle map maps $U \times F_0$ into $U' \times F_0'$, has first component independent of h and second linear in h, maps $U_m \times F_k$ into $U'_m \times F'_k$ for $k \leq m + 1$,

is sc-smooth on $U \times F_0$, and its fibre-lifted map on $U \times F_1$ Chapter2/Tame · shiftFiber is sc-smooth for $E \oplus F^1$ Chapter2/Tame · IsStrongBundleMap; a strong bundle retraction is an idempotent one Chapter2/Tame · IsStrongBundleRetraction. *Lemma 5.A.* The base $r(u) = R(u,0)_1$ Chapter2/Tame · IsStrongBundleMap.base Chapter2/StrongBundle · IsStrongBundleRetraction.apply_fst is an sc-smooth retraction on U: sc-smooth as R composed with $u \mapsto (u,0)$ and the first projection, idempotent by the first component of $R \circ R = R$ Chapter2/StrongBundle · IsStrongBundleRetraction.baseRetraction. For $u \in O$, $R(u,h) = (u, \Gamma(u,h))$, so the second component of $R \circ R = R$ reads $\Gamma(u,\Gamma(u,h)) = \Gamma(u,h)$: $\Gamma(u,\cdot)$ is a projection Chapter2/StrongBundle · IsStrongBundleRetraction.fiber_idem. R is an sc-smooth retraction on $U \times F_0$ in $C \times F_0$ Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetraction and, through the lift of the fibre component to F_1 (possible since R preserves the double filtration at $(0,1)$ Chapter2/StrongBundle · IsStrongBundleRetraction.snd_mem_range Chapter2/StrongBundle · IsStrongBundleRetraction.toBase_shiftFiber_snd), on $U \times F_1$ for $E \oplus F^1$ Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetractionOne.

Definition 2.6.2 (preprint 2.48). K is a local strong bundle retract if $K = R(U \times F_0)$ Chapter2/Tame · IsStrongBundleRetract, with $K_{m,k}$, $K[i]_m$ and $K[1] \subseteq E_0 \times F_1$ Chapter2/Tame · strongLevel Chapter2/Tame · strongLevelShift Chapter2/Tame · strongShiftOne and the projection p Chapter2/Tame · strongProj. $K = \{(u,h) : u \in O, \Gamma(u,h) = h\}$ Chapter2/StrongBundle · IsStrongBundleRetraction.mem_image_iff. *Lemma 5.B.* $K[1]$ is the image of R on $U \times F_1$ Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetractionOne_retract, so K and $K[1]$ are retracts Chapter2/StrongBundle · IsStrongBundleRetract.isScRetract Chapter2/StrongBundle · IsStrongBundleRetract.isScRetract_one; the base $p(K)$ is O Chapter2/StrongBundle · IsStrongBundleRetract.base Chapter2/StrongBundle · IsStrongBundleRetract.base_eq; and $p : K[i] \to O$ is sc-smooth, because $p \circ R = r \circ \mathrm{pr}_1$ is Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst_one, as is the inclusion $K[1] \to K[0]$ Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl.

Definition 2.6.3 (preprint 2.49). An sc-smooth section is sc-smooth from (O,C,E) into $(K, C \times F_0, E \oplus F)$ with $f(x)_1 = x$ Chapter2/Tame · IsScSmoothSection; an sc⁺-section is sc-smooth into $K[1]$ Chapter2/Tame · IsScPlusSection Chapter2/Tame · IsScPlusSection.mem. Composed with the inclusion $K[1] \to K[0]$ (Theorem 2.1.15 (preprint 2.8)), an sc⁺-section is an sc-smooth section Chapter2/StrongBundle · IsScPlusSection.isScSmoothSection.

II.19 Strong bundles over M-polyfolds (Definitions 2.6.4–2.6.7 (preprint 2.50–2.53))

Definition 2.6.4 (preprint 2.50). For $P : Y \to X$ with linear structures on the fibres, a strong bundle chart is a chart φ of X (of the structure — the text says only "a homeomorphism onto O"), a strong bundle retraction R on $U \subseteq C$ with base retract O, and a homeomorphism $\Phi : P^{-1}(V) \to K$ with $\Phi(y)_1 = \varphi(P(y))$, linear on fibres Chapter2/StrongBundle · StrongBundleChart Chapter2/StrongBundle · StrongBundleChart.K. K is a local strong bundle retract and a retract on $[0]$ and $[1]$ Chapter2/StrongBundle · StrongBundleChart.isStrongBundleRetract Chapter2/StrongBundle · StrongBundleChart.isScRetract Chapter2/StrongBundle · StrongBundleChart.isScRetract_one Chapter2/StrongBundle · StrongBundleChart.model, the source $P^{-1}(V)$ is open Chapter2/StrongBundle · StrongBundleChart.isOpen_source, $\Phi(P^{-1}(V)) = K$ and $p(K) = O$ Chapter2/StrongBundle · StrongBundleChart.image_source Chapter2/StrongBundle · StrongBundleChart.base_K. "Bounded on the fibres" needs no hypothesis: the restriction of the continuous Φ to a fibre is continuous Chapter2/StrongBundle · StrongBundleChart.continuous_fiber. Two charts are compatible if the transitions $\Phi' \circ \Phi^{-1}$ Chapter2/StrongBundle · StrongBundleChart.overlap Chapter2/

StrongBundle · StrongBundleChart.transition and back are sc-smooth between the overlaps on $[0]$ and, fibre-lifted, on $[1]$, and carry fibre components in F_1 to F_1' Chapter2/StrongBundle · StrongBundleChart.Compatible Chapter2/StrongBundle · StrongBundleChart.Compatible.symm; the last condition, implicit in the text, is what makes $(\Phi' \circ \Phi^{-1})[1]$ the restriction of $\Phi' \circ \Phi^{-1}$. Atlases and their equivalence Chapter2/StrongBundle · StrongBundleAtlas Chapter2/StrongBundle · StrongBundleAtlas.Equiv, an equivalence relation (§II.45) Chapter2/StrongBundleRestrict · StrongBundleAtlas.equivalence; Banachable fibres Chapter2/StrongBundle · IsBanachable.

Definition 2.6.5 (preprint 2.51). A strong bundle is a continuous surjection with Banachable fibres and a strong bundle atlas, Y Hausdorff paracompact Chapter2/StrongBundle · StrongBundle. Each Φ is a chart of Y with model $(K, C \times F_0, E \oplus F)$ Chapter2/StrongBundle · StrongBundleChart.toMChart, compatibility of strong bundle charts on $[0]$ is compatibility of these charts Chapter2/StrongBundle · StrongBundleChart.Compatible.toMChart, and they form the atlas of the M-polyfold $Y[0]$ Chapter2/StrongBundle · StrongBundle.atlas0 Chapter2/StrongBundle · StrongBundle.toMPolyfold0 Chapter2/StrongBundle · StrongBundle.isChart_toMChart. $P : Y[0] \to X$ is sc-smooth: in the charts Φ and φ it is $p : K \to O$ (Lemma 5.B), and one pair of charts per point suffices Chapter2/StrongBundle · StrongBundle.isScSmoothMap_proj. The double filtration $Y_{m,k}$ is defined through all the charts of the atlas around a point Chapter2/StrongBundle · StrongBundle.doubleLevel.

$Y_{m,k}$ does not depend on the chart Chapter2/StrongBundleOne · StrongBundle.mem_doubleLevel_iff. Let y lie in the domain of a chart Φ of the atlas with $\Phi(y) \in E_m \times F_k$, $k \le m+1$, and let Φ' be another chart around y; then $\Phi'(y) = (\Phi' \circ \Phi^{-1})(\Phi(y))$ Chapter2/StrongBundlePullback · StrongBundleChart.transition_apply. *The base component:* $\Phi'(y)_1 = \varphi'(\varphi^{-1}(\Phi(y)_1))$, and the base transition, sc-smooth and so sc^0, maps points of level m to points of level m Chapter2/MPolyfold · MChart.transition_apply. *The fibre component, $k \le m$:* $\Phi(y)$ lies on level k of $E \oplus F$ (levels decrease Chapter2/StrongBundleOne · StrongBundle.range_toBase_mono), and the $[0]$-transition, sc^0 on the overlap retract Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap, keeps it on level k. *The case $k = m+1$:* the fibre component of $\Phi(y)$ lies in F_1, so its lift $\Phi(y)[1]$ to $E_0 \times F_1$ Chapter2/StrongBundleOne · StrongBundleChart.liftOne Chapter2/StrongBundleOne · StrongBundleChart.toBase_liftOne is a point of level m of $E \oplus F^1$ (level m of F^1 is F_{m+1} Chapter1/ScBanach · ScBanach.shiftScale_toBase); the $[1]$-transition, sc^0 for $E \oplus F^1$ on the overlap Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap_one, keeps it on level m, and its value is the lift $\Phi'(y)[1]$, because the F_1-condition of compatibility puts the fibre component of $\Phi'(y)$ in F_1'. So $\Phi'(y) \in E_m' \times F_{m+1}'$. The case $k \le m$ uses the $[0]$-transition only; the case $k = m+1$ uses exactly the F_1-condition made explicit in Definition 2.6.4 (preprint 2.50).

The set $Y[1]$ and its topology. $Y_{0,1}$ is the set of points whose fibre component lies in F_1 in every chart of the atlas around them Chapter2/StrongBundleOne · StrongBundle.oneSet; by the F_1-condition one chart suffices Chapter2/StrongBundleOne · StrongBundle.mem_oneSet_iff. $Y[1]$ is $Y_{0,1}$ as a type Chapter2/StrongBundleOne · StrongBundle.One with the topology generated by the sets $\{y \in \mathrm{dom}\,\Phi : \Phi(y)[1] \in W\}$, W open in $E_0 \times F_1$, as for X_m in §2.3 (preprint §2.2) Chapter2/StrongBundleOne · StrongBundle.isOpen_one_basic. The inclusion $Y[1] \to Y$ is continuous Chapter2/StrongBundleOne · StrongBundle.continuous_oneVal: inside the domain of a chart Φ, an open subset of Y is the preimage under Φ of an open $W \subseteq E_0 \times F_0$ (Φ is a homeomorphism onto K Chapter2/MPolyfold · MChart.exists_isOpen_image_eq), and its points in $Y[1]$ form the basic set of $(1 \times \iota)^{-1}(W)$. A continuous injection into the Hausdorff space Y makes $Y[1]$ Hausdorff Mathlib/Topology/Separation/Hausdorff · T2Space.of_injective_continuous.

The charts $\Phi[1]$ Chapter2/StrongBundleOne · StrongBundle.oneChart. $y \mapsto \Phi(y)[1]$ maps $P^{-1}(V) \cap Y[1]$ into $K[1]$ Chapter2/StrongBundleOne · StrongBundleChart.liftOne_mem, with inverse $z \mapsto \Phi^{-1}(z_1, \iota z_2)$, a point of $Y_{0,1}$ Chapter2/StrongBundleOne · StrongBundle.symm_mem_oneSet Chapter2/StrongBundleOne · StrongBundle.oneChartInv Chapter2/StrongBundleOne · StrongBundle.liftOne_symm. It is continuous by the definition of the topology. Its

inverse is continuous: a basic open set $\{\Phi'(y)[1] \in W\}$ pulls back to the set of $z \in K[1]$ in the $[1]$-part of the overlap with $(\Phi' \circ \Phi^{-1})[1](z) \in W$; the overlap is $K \cap W_0$ for an open W_0 `Chapter2/StrongBundlePullback · StrongBundleChart.overlap_eq_inter` and the $[1]$-transition is continuous on it, so this set is open in $K[1]$. The overlap of $\Phi[1]$ and $\Phi'[1]$ is the $[1]$-part of the overlap of Φ and Φ' `Chapter2/StrongBundleOne · StrongBundle.oneChart_overlap`, and their transition map is the $[1]$-transition $(\Phi' \circ \Phi^{-1})[1]$ `Chapter2/StrongBundleOne · StrongBundle.oneChart_transition`, sc-smooth in both directions by Definition 2.6.4 (preprint 2.50); so the charts $\Phi[1]$ are compatible `Chapter2/StrongBundleOne · StrongBundle.oneChart_compatible` and form an atlas of $Y[1]$ `Chapter2/StrongBundleOne · StrongBundle.atlas1`. $Y[1]$ is metrizable: P restricted to $Y[1]$ is continuous into the paracompact Hausdorff X, and over the domain V of a chart of X its preimage is the source of a chart $\Phi[1]$, homeomorphic to a retract and so metrizable `Chapter2/Metrizable · MChart.metrizableSpace_source`; the pull-back form of Smirnov's theorem applies `Chapter2/Metrizable · metrizableSpace_of_pullback` `Chapter2/StrongBundleOne · StrongBundle.metrizableSpace_one`. Hence $Y[1]$ is paracompact `Chapter2/StrongBundleOne · StrongBundle.paracompactSpace_one`, which an M-polyfold must be and the text does not mention, and $Y[1]$ **is an M-polyfold** `Chapter2/StrongBundleOne · StrongBundle.toMPolyfold1` whose charts include the $\Phi[1]$ `Chapter2/StrongBundleOne · StrongBundle.isChart_oneChart`.

$P[1]$ and the inclusion $Y[1] \to Y[0]$. In the charts $\Phi[1]$ and φ, $P[1]$ is the projection $K[1] \to O$ of Lemma 5.B `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst_one`, and one pair of charts per point suffices `Chapter2/LevelPolyfold · MPolyfold.IsScSmoothOn.of_exists_charts`; so $P[1] : Y[1] \to X$ **is sc-smooth** `Chapter2/StrongBundleOne · StrongBundle.isScSmoothMap_proj1`. In the charts $\Phi[1]$ and Φ the inclusion $Y[1] \to Y[0]$ is $(u, h) \mapsto (u, \iota h)$ `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl`, so it is sc-smooth `Chapter2/StrongBundleOne · StrongBundle.isScSmoothMap_oneVal` (not stated in the text; it gives that an sc⁺-section is an sc-smooth section).

Definition 2.6.6 (preprint 2.52). A section has $P \circ f = \mathrm{id}$ `Chapter2/StrongBundle · IsSection`; it is sc-smooth if sc-smooth into $Y[0]$ `Chapter2/StrongBundle · StrongBundle.IsScSmoothSection`, and an sc⁺-section if in every chart its fibre values lie in F_1 and its fibre-lifted representative `Chapter2/StrongBundle · StrongBundle.localRepOne` is sc-smooth into $K[1]$ `Chapter2/StrongBundle · StrongBundle.IsScPlusSection`. *This is the text's definition* `Chapter2/StrongBundleOne · StrongBundle.isScPlusSection_iff`: f is an sc⁺-section iff it is a section and $f = \iota \circ g$ for an sc-smooth $g : X \to Y[1]$, ι the inclusion. Forward: the values of f lie in $Y_{0,1}$ (one chart suffices), which defines g; on the domain of a chart φ of X under a strong bundle chart Φ, $g = \Phi[1]^{-1} \circ (\text{the local representative}) \circ \varphi$, which gives the continuity of g, and the representative of g in the charts φ, $\Phi[1]$ is the chart-wise representative, sc-smooth by hypothesis. Backward: the fibre values lie in F_1 because g takes values in $Y_{0,1}$, and the chart-wise representative is the representative of g in φ, $\Phi[1]$, sc-smooth by Definition 2.3.8 (preprint 2.14) (a chart with empty domain is handled separately).

Definition 2.6.7 (preprint 2.53). $f^*Y = \{(z, y) : P(y) = f(z)\}$ `Chapter2/StrongBundle · pullbackSpace` with the projections `Chapter2/StrongBundle · pullbackProj` `Chapter2/StrongBundle · pullbackProj'`, $P \circ P' = f \circ P_f$ `Chapter2/StrongBundle · pullback_comm`; P_f is continuous and f^*Y closed in $Z \times Y$ `Chapter2/StrongBundle · continuous_pullbackProj` `Chapter2/StrongBundle · isClosed_pullbackSpace`.

II.20 Proposition 2.6.8 (preprint 2.54): the pull-back strong bundle

The pulled-back retraction. For a retraction r' on U' and $g : O' \to O$ sc-smooth between retracts, $R_g(w, h) := (r'(w), \Gamma(g(r'(w)), h))$ `Chapter2/StrongBundle · pullbackRetraction`, where $g(r'(w)) \in O \subseteq U$ `Chapter2/StrongBundle · pullback_mem_O` `Chapter2/StrongBundle · pullback_mem_U`. It is a strong bundle retraction `Chapter2/StrongBundle · isStrongBundleRetraction_pullback`: it maps $U' \times F_0$ into itself; its first component does

not depend on h and its second is linear in h; it preserves the double filtration (r' and $g \circ r'$ preserve levels, and R preserves the double filtration); on $[0]$ it is the pairing of $r' \circ \mathrm{pr}_1$ with the second component of $R \circ ((g \circ r') \circ \mathrm{pr}_1, \mathrm{pr}_2)$, sc-smooth by the chain rule and the pairing lemma, and likewise on $[1]$; and $R_g \circ R_g = R_g$ because $r' \circ r' = r'$ and $\Gamma(o, \cdot)$ is a projection for $o = g(r'(w)) \in O$ (Lemma 5.A). Its image is $K_g = \{(w,h) : w \in O', (g(w), h) \in K\}$ `Chapter2/StrongBundle · pullback_image` `Chapter2/StrongBundle · isStrongBundleRetract_pullback`.

The fibres. $P_f^{-1}(z) \to P^{-1}(f(z))$, $(z,y) \mapsto y$, is a bijection `Chapter2/StrongBundlePullback · pullbackFiberEquiv`; transporting the linear structure along it `Chapter2/StrongBundlePullback · pullbackFiber_addCommGroup` `Chapter2/StrongBundlePullback · pullbackFiber_module` makes it a linear homeomorphism `Chapter2/StrongBundlePullback · pullbackFiberCLE`, so the fibres of f^*Y are Banachable.

The charts. The data: a strong bundle chart c of Y, a chart ψ of Z with $f(\mathrm{dom}\,\psi) \subseteq \mathrm{dom}\,\varphi$, a retraction r_Z onto the model of ψ, and the sc-smoothness of $g = \varphi \circ f \circ \psi^{-1}$ `Chapter2/StrongBundlePullback · PullbackChartData` `Chapter2/StrongBundlePullback · PullbackChartData.g` `Chapter2/StrongBundlePullback · PullbackChartData.g_smooth`. The chart $\Phi_f(z,y) = (\psi(z), \mathrm{pr}_F \Phi(y))$ maps $P_f^{-1}(\mathrm{dom}\,\psi)$ into K_g — because $\Phi(y) = (g(\psi(z)), \cdot) \in K$ `Chapter2/StrongBundlePullback · PullbackChartData.g_apply` `Chapter2/StrongBundlePullback · PullbackChartData.mem_Kg` — with inverse $(w,h) \mapsto (\psi^{-1}(w), \Phi^{-1}(g(w), h))$ `Chapter2/StrongBundlePullback · PullbackChartData.symm_mem_pullback` `Chapter2/StrongBundlePullback · PullbackChartData.invFun`; both are continuous, it covers ψ and is linear on fibres, so it is a strong bundle chart of f^*Y `Chapter2/StrongBundlePullback · PullbackChartData.toChart` `Chapter2/StrongBundlePullback · PullbackChartData.image_eq_Kg`.

Compatibility. Sc-operators and pairings are sc-smooth between retracts `Chapter2/StrongBundlePullback · ScOperator.isScSmoothRetractMap` `Chapter2/StrongBundlePullback · IsScSmoothRetractMap.scPair` `Chapter2/StrongBundlePullback · isScRetract_univ`, and overlaps of strong bundle charts are retracts on $[0]$ and $[1]$ `Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap` `Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap_one`. On the overlap of two pulled-back charts the transition map is

$$(w,h) \mapsto (\psi_2 \psi_1^{-1}(w),\ \mathrm{pr}_F(\Phi_2 \Phi_1^{-1})(g_1(w), h))$$

`Chapter2/StrongBundlePullback · PullbackChartData.transitionFormula` `Chapter2/StrongBundlePullback · PullbackChartData.transition_eqOn`: its first component is the transition of Z composed with the projection, its second the fibre component of the transition of Y composed with $(w,h) \mapsto (g_1(w), h)$ `Chapter2/StrongBundlePullback · PullbackChartData.gPair`, which maps the overlap into the overlap of Y's charts `Chapter2/StrongBundlePullback · PullbackChartData.gPair_mem`. Each factor is sc-smooth between retracts, and Theorem 2.1.15 (preprint 2.8) and the pairing lemma give the sc-smoothness on $[0]$ `Chapter2/StrongBundlePullback · PullbackChartData.transition_isScSmooth`; the same with F^1 on $[1]$ `Chapter2/StrongBundlePullback · PullbackChartData.transitionOne_eqOn` `Chapter2/StrongBundlePullback · PullbackChartData.transitionOne_isScSmooth`; fibre components in F_1 go to F_1 because the transitions of Y have this property `Chapter2/StrongBundlePullback · PullbackChartData.transition_snd_mem_range`. So any two pulled-back charts are compatible `Chapter2/StrongBundlePullback · PullbackChartData.compatible_toChart`.

Proposition 2.6.8 (preprint 2.54). f^*Y is paracompact: Z and $Y[0]$ are metrizable (Theorem 2.3.10 (preprint 2.16)), so $Z \times Y$ and its subspace f^*Y are metrizable, hence paracompact `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` — a requirement of Definition 2.6.5 (preprint 2.51) that the text does not mention. The pulled-back charts, built around every point from a chart of Y's atlas and a chart of Z restricted to $f^{-1}(\mathrm{dom}\,\varphi)$, cover f^*Y and form an atlas `Chapter2/StrongBundlePullback · pullbackAtlas`. With the continuous surjection P_f and its Banachable fibres this is a

strong bundle structure on $f^*Y \to Z$ Chapter2/StrongBundlePullback · proposition_2_6_8 ("the easy proof is left to the reader").

II.21 What the formalization taught us

- **Theorem 2.3.10 (preprint 2.16) needs no added hypothesis, and not the text's route.** The text's proof applies Proposition 2.7.4 (preprint 2.57) to X_m without checking that X_m is regular. A partition of unity of X, pulled back along the continuous inclusion, metrizes X_m directly Chapter2/Metrizable · metrizableSpace_of_pullback, and the same lemma proves Smirnov's theorem Chapter2/Metrizable · metrizableSpace_of_locally, Proposition 2.3.15 (3) (preprint 2.19) Chapter2/Metrizable · proposition_2_3_15_three and the paracompactness of f^*Y Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace. With it every paracompactness hypothesis carried by the constructions is discharged Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype.
- **Compatibility is local, and the text needs it.** That the charts compatible with an atlas are compatible with each other is not in the text; it is the chain rule plus the locality of sc-smoothness Chapter2/MPolyfold · MChart.compatible_of_locally, and every chart-independence claim of §2.3 (preprint §2.2) rests on it.
- TO **is a fixed-point set.** Proposition 2.1.11 (preprint 2.4) and Proposition 2.1.13 (preprint 2.6) are one-line consequences of $p \in Tr(TU) \iff Tr(p) = p$ Chapter2/Retracts · ScRetraction.mem_tangent_iff, which the text's proofs use without stating.
- **The tangent bundle needs the locality of** Tf**.** That Definition 2.3.12's (preprint 2.17) relation is an equivalence relation needs the chain rule on the triple overlap and the agreement of Tf computed on an open subset with Tf computed on the whole retract Chapter2/TangentBundle · retractTangentMap_inter_isOpen Chapter2/TangentBundle · MPolyfold.TangentPre.transport_transport.
- **Level** 1 **suffices for §2.4 (preprint §2.3).** Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31) and Corollary 2.4.13 (preprint 2.32) hold at points on level 1 and for sc[1] germs Chapter2/Boundary · proposition_2_4_10_reduced Chapter2/Boundary · theorem_2_4_12 Chapter2/DegeneracyIndex · corollary_2_4_13; the text's "closure of $(T^R_x O)_\infty$" is the density of level-1 points in the kernel of a finite-rank map Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase.
- **Proposition 2.5.3 (preprint 2.35) needs only tameness (2), and Lemma 2.5.12 (preprint 2.40) needs (2) through it.** $Dr(x)E_x \subseteq E_x$ holds for every retraction Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace; that the retraction keeps points of a face in that face is the ray lemma Chapter2/TameRetracts · ray_zeroSet, which uses Proposition 2.5.3 (preprint 2.35).
- **Three slips in the exposition.** "$|P_1 z - e|_0 \le |z - e|_0$" in the exposition's proof of Proposition 2.5.15 (preprint 2.43) is false for a general norm on E_0; the formal neighbourhood intersects balls with their P_i-preimages Chapter2/TameRetracts · proposition_2_5_15. The "transport by φ^{-1}" in Theorem 2.5.17 (preprint 2.45) needs a shrinking so that the retraction maps its domain into itself Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local. In Proposition 2.5.20 (preprint 2.46) the inverse of π_2 cannot be factored through the constant-in-x map $y' \mapsto (x, y')$, which is not sc-smooth at a non-smooth x Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_snd_fibredProduct.
- **Tameness is needed only for the additivity of the index under products.** $d_{X\times Y} \le d_X + d_Y$ in general Chapter2/FibredProduct · MPolyfold.degIndex_prod_le, with equality for tame factors Chapter2/FibredProduct · MPolyfold.isTame_prod; the identities $d = d_X = d_Y$ of Proposition 2.5.20 (preprint 2.46) hold without tameness Chapter2/FibredProduct · MPolyfold.proposition_2_5_20_of_isLocalScDiffeo.

- **Definition 2.6.4 (preprint 2.50) needs two explicit conditions.** The base homeomorphism of a strong bundle chart must be a chart of X, and the $[1]$-transition must preserve fibre components in F_1 `Chapter2/StrongBundle · StrongBundleChart.Compatible`. The second condition is exactly what makes $Y_{0,1}$, and so the M-polyfold $Y[1]$, independent of the chart `Chapter2/StrongBundleOne · StrongBundle.mem_oneSet_iff` `Chapter2/StrongBundleOne · StrongBundle.mem_doubleLevel_iff`.
- **Proposition 2.5.6 (preprint 2.36) does not need the equality case of Theorem 2.4.12 (preprint 2.31).** For a tame retraction the smooth vectors of the splitting are $Dr(x)T^{-1}(e_{i_j}, 0)$, dual to the coordinates by Proposition 2.5.3 (preprint 2.35), and $T^R_x O$ is an sc-subspace because $Dr(x)$ preserves E_x `Chapter2/PartialCone · proposition_2_5_6_partialCone`.
- **Corners at once, not by induction.** The intersections F^σ_x of Proposition 2.5.16 (preprint 2.44) are retracts of $r \circ P_\sigma$, by Lemma 4.C for each $i \in \sigma$; no identification of the germ of $F_1 \cap F_2$ with a face of F_1 is needed `Chapter2/CornerGerms · corner_retraction`.
- **$C_x O$ is always a partial quadrant.** The second sentence of Remark 2.4.8 (preprint 2.28) does not hold as we read it. At every smooth point of every sc-smooth retract, $C_x O$ is a partial quadrant in $T_x O$ `Chapter2/PartialConeQuadrant · remark_2_4_8_isPartialQuadrant`. The reason is finite-dimensional: $Dr(x)$ induces a nonnegative idempotent matrix on E/E_x, and the cone of such a matrix is simplicial (Flor) `Chapter2/PartialConeQuadrant · flor_nonnegRange_simplicial`. What tameness adds is only that the quadrant has the full dimension $d_C(x)$. The second statement of Theorem 2.4.12 (preprint 2.31) needs no hypothesis.
- **The linear structure of $T_x X$ is chart independent because the fibre transitions are restrictions of derivatives.** Definition 2.3.14 (preprint 2.18) adds in one chart and does not check independence. It holds because $T\psi \circ (T\varphi)^{-1}$ is the restriction of $D(\psi \circ \varphi^{-1} \circ r)(\varphi(x))$ `Chapter2/TangentSpaceLinear · MPolyfold.exists_transition_clm`. At a smooth point this restriction is an sc-isomorphism of the models, which makes the sc-structure of $T_x X$ chart independent too `Chapter2/TangentSpaceChartIndependent · MPolyfold.tangentFiberScIso`.
- **The openness of $\mathrm{pr}_1 : TO \to O_1$ has a one-line proof.** Appendix 2.7.3 states it without proof. The section $o' \mapsto Tr(o', h)$ through a given tangent vector is continuous because Tr is sc^0 `Chapter2/TangentProjOpen · ScRetraction.exists_isOpen_tangent_section`.
- **Michael's lemma needs a shrinking.** The closures of a locally finite open refinement lie in members of the cover only after the cover is shrunk by regularity `Chapter2/MichaelRefinement · GapP1Michael.exists_closed_refinement`. The converse direction needs no separation at all `Chapter2/MichaelRefinement · GapP1Michael.paracompactSpace_of_closed_refinement`.
- **Proposition 2.1.2 (preprint 2.2) is a conjugation.** $\varphi(y) = P(r(y) - x) + (1 - P)(y - r(y))$ has derivative 1 at x and satisfies $\varphi \circ r = P \circ \varphi$, so the inverse function theorem replaces the remainder estimates of Appendix 2.7.1 `Chapter2/SmoothRetractImage · GapP1Prop212.retractChart_comp`.
- **"An equivalence class of atlases" needs transitivity, which rests on the locality of compatibility.** The text does not check this, for M-polyfolds or for strong bundles `Chapter2/StrongBundleRestrict · MAtlas.Equiv.trans` `Chapter2/StrongBundleRestrict · StrongBundleAtlas.Equiv.trans`.

II.22 The 2021 edition: Proposition 2.1.13 and local completeness (§2.1)

The Springer edition (2021) adds to §2.1 a theory of *local completeness*: Proposition 2.1.4, Definition 2.1.5, Remark 2.1.6, Corollary 2.1.7, Proposition 2.1.8, Lemma 2.1.9 and Proposition 2.1.10, pp. 22–25. It also changes one statement, Proposition 2.1.13.

Proposition 2.1.13 (preprint 2.6), the 2021 statement. For sc-smooth retractions r, s onto the same retract

O, a retraction ρ onto O', and f with $f(O) \subseteq O'$, three things hold Chapter2/Book2021 · proposition_2_1_13: - (1) $f \circ r$ is sc-smooth iff $f \circ s$ is; - (2) if $f \circ r$ is sc-smooth, $T(f \circ r) = T(f \circ s)$ on TO; - (3) if $f \circ r$ is sc-smooth, $T(f \circ r)$ maps TO into TO'.

The 2021 statement of (3) carries the hypothesis "$f \circ r$ sc-smooth", without which $T(f \circ r)$ is not defined. The formal statement of §II.3 had read (3) this way from the start Chapter2/Retracts · proposition_2_1_13_three, so the 2021 theorem is the conjunction of the three earlier ones. The first is used in both directions Chapter2/Retracts · proposition_2_1_13_one Chapter2/Retracts · proposition_2_1_13_two.

Complete metrics on a subspace. "W admits a metric d defining its topology for which K is complete" is a metric structure on W whose topology is the given one, with K complete for it Chapter2/LocalTopology · AdmitsCompleteMetricOn. A metric space admits its own metric Chapter2/LocalTopology · admitsCompleteMetricOn_of_isComplete, and such a W is metrizable Chapter2/LocalTopology · AdmitsCompleteMetricOn.metrizableSpace. The property passes to smaller sets Chapter2/LocalTopology · AdmitsCompleteMetricOn.mono and along embeddings: an embedding $j : T' \to T$ pulls the metric back, and a closed subset of a complete set is complete Chapter2/LocalTopology · isComplete_inter_of_isClosed Chapter2/LocalTopology · AdmitsCompleteMetricOn.of_isEmbedding.

Definition 2.1.5. X is *locally complete* if every x has open $V \ni x$ and W with $\mathrm{cl}_X(V) \subseteq W$ such that W admits a metric defining its topology for which $\mathrm{cl}_X(V)$ is complete Chapter2/LocalTopology · IsLocallyComplete. The text's $V(x) \subseteq W(x)$ follows from $\mathrm{cl}_X(V) \subseteq W$. The text also assumes X Hausdorff; the formal definition does not, because it follows. - *W arbitrarily small* Chapter2/LocalTopology · IsLocallyComplete.exists_subset. Given a neighbourhood N of x, take a metric ball V' around x in W whose closure in W lies in $W \cap N$. Replace W by the open $W \cap \operatorname{int} N$ and V by $V \cap V'$; the metric restricts. - *Regular* Chapter2/LocalTopology · IsLocallyComplete.regularSpace. A neighbourhood N of x contains the closed neighbourhood $\mathrm{cl}_X(V)$ for a $W \subseteq N$; this is where the global condition $\mathrm{cl}_X(V) \subseteq W$ enters. - T_1 Chapter2/LocalTopology · IsLocallyComplete.t1Space. Points are closed in the metrizable W. For $y \neq x$, a neighbourhood of x missing y comes from W when $y \in W$; otherwise W itself serves. - *Hausdorff* Chapter2/LocalTopology · IsLocallyComplete.t2Space: regular and T_1.

Witnesses: a complete metric space ($W = V = X$) Chapter2/LocalTopology · isLocallyComplete_of_completeSpace, $\mathbb{R}$ Chapter2/LocalTopology · isLocallyComplete_real, and a metric space whose points have open $V \subseteq W$ with $\mathrm{cl}(V) \subseteq W$ and $\mathrm{cl}(V)$ complete for the given metric Chapter2/LocalTopology · isLocallyComplete_of_metric.

Remark 2.1.6. If $\mathrm{cl}_X(V) \subseteq W$, the closure of V in the subspace W is $\mathrm{cl}_X(V)$ Chapter2/LocalTopology · remark_2_1_6. The modified notion asks only $V \subseteq W$, with completeness of $\mathrm{cl}_W(V)$ Chapter2/LocalTopology · IsLocallyCompleteRel. Local completeness implies it, by the identity just proved Chapter2/LocalTopology · IsLocallyComplete.isLocallyCompleteRel. The remark's "weaker" is formalized as this implication; that the modified notion is *strictly* weaker is §II.31 (gap G.76, closed).

Proposition 2.1.4. Let $O = r(U)$ be a retract with the metric of E_0. *Closed pieces of O:* for ε small, $O \cap \bar{B}_\varepsilon(x)$ is closed in E_0 Chapter2/LocalTopology · ScRetraction.exists_isClosed_retract_inter_closedBall. Choose δ with $\bar{B}_\delta(x) \cap C \subseteq U$ (U is relatively open in the closed C). A limit v of points $v_k \in O \cap \bar{B}_\varepsilon(x)$ lies in $C \cap \bar{B}_\varepsilon(x) \subseteq U$, and $v_k = r(v_k) \to r(v)$ by continuity of r on U, so $v = r(v) \in O$. *The neighbourhoods* Chapter2/LocalTopology · IsScRetract.exists_isComplete_closure: V the open $\varepsilon/2$-ball and W the open ε-ball in O, inside any given neighbourhood. The closure of V in O lies in $O \cap \bar{B}_{\varepsilon/2}(x)$, closed in the Banach space E_0 and therefore complete. This proves the proposition Chapter2/LocalTopology · proposition_2_1_4. The text's set V' with $r(V') \subseteq V'$ is not needed, because the closed balls already give closed pieces of

O.

Corollary 2.1.7. By Proposition 2.1.4, with the metric d_O restricted to W `Chapter2/LocalTopology · isLocallyComplete_of_metric`, a retract is locally complete `Chapter2/LocalTopology · corollary_2_1_7`. So is E_0, the retract of the identity `Chapter2/LocalTopology · isLocallyComplete_univ_retract`.

Proposition 2.1.8. *Embeddings with locally closed image* `Chapter2/LocalTopology · IsLocallyComplete.of_isEmbedding`. Let $j : Y \to X$ be an embedding with $j(Y) = U \cap Z$, U open and Z closed. For $y \in Y$, take W, V around $j(y)$ with $W \subseteq U$. The preimages $j^{-1}(W)$ and $j^{-1}(V)$ serve for y, for two reasons. First, the closure of $j^{-1}(V)$ corresponds to $\mathrm{cl}_X(V) \cap j(Y)$, and $j(Y) \cap W = Z \cap W$ is closed in W. Second, the embedding $j^{-1}(W) \to W$ has closed image, so the metric of W pulls back with the completeness of the closure. Parts (1)–(3) follow: - (1) a locally closed subspace `Chapter2/LocalTopology · proposition_2_1_8_1`; - (2) a homeomorphic space, the image being everything `Chapter2/LocalTopology · proposition_2_1_8_2`; - (3) an open subspace `Chapter2/LocalTopology · proposition_2_1_8_3`.

The open-cover statement `Chapter2/LocalTopology · proposition_2_1_8_4`. Let $y \in Y_i$, with $U \ni y$ open, $U \subseteq Y_i$ and $\mathrm{cl}_Y(U) \subseteq Y_i$ (condition (b)), and W', V' for y in the locally complete Y_i. Put $V = V' \cap U$ and $W = W'$, both open in Y because Y_i is. Then $\mathrm{cl}_Y(V) \subseteq \mathrm{cl}_Y(U) \subseteq Y_i$ by (b), so $\mathrm{cl}_Y(V)$ is the closure of V in Y_i and lies in W. The metric of $W' \subseteq Y_i$ is transported to $W \subseteq Y$, the identity being a homeomorphism. No Hausdorff hypothesis is used anywhere.

Lemma 2.1.9 and Proposition 2.1.10. *First countable* `Chapter2/LocalTopology · lemma_2_1_9`: the neighbourhood filter of x in X is the image of that of x in the open, metrizable W, and the latter is countably generated. *Completely regular* `Chapter2/LocalTopology · IsLocallyComplete.completelyRegularSpace`. For x outside a closed A, take $W \subseteq X \setminus A$ (arbitrarily small W). In the metrizable W, a continuous $g : W \to [0,1]$ has $g(x) = 0$ and $g = 1$ off V. Extended by 1 on $X \setminus \mathrm{cl}_X(V)$, it is continuous on the two open sets W and $X \setminus \mathrm{cl}_X(V)$, which cover X because $\mathrm{cl}_X(V) \subseteq W$. Mathlib's complete regularity uses this normalization ($f(x) = 0$, $f = 1$ on A); the text's β is $1 - f$. With Hausdorff, the space is $T_{3\frac{1}{2}}$ `Chapter2/LocalTopology · proposition_2_1_10`. The text argues continuity with sequences; the formal proof works with filters and open covers, so first countability is not needed for it.

II.23 The 2021 edition: local path-connectedness and local contractibility (§2.2)

§2.2 of the 2021 edition, pp. 30–33, is new.

Lemma 2.2.1. Let U be relatively open in a partial quadrant C and connected. For $k \in \mathbb{N}$, $U_k = U \cap E_k$ carries the topology of E_k `Chapter1/ScSmooth · ScBanach.lvlSet`. For $k = \infty$ the text does not say which topology U_∞ carries. The formalization takes the topology of all the norms: U_∞ is the set of compatible sequences $(v_m) \in \prod_m E_m$ with $v_0 \in U$ `Chapter2/LocalTopology · ScBanach.lvlSetInfty`, with the product topology. A smooth point x gives its sequence of lifts `Chapter2/LocalTopology · ScBanach.smoothSeq` `Chapter2/LocalTopology · ScBanach.toBase_smoothSeq` `Chapter2/LocalTopology · ScBanach.smoothSeq_mem` `Chapter2/LocalTopology · ScBanach.smoothSeq_zero`. The image of U_∞ in E_0 is $U \cap E_\infty$ `Chapter2/LocalTopology · ScBanach.image_lvlSetInfty`.

The proof uses three facts: - two points of U_∞ whose base points span a segment inside U are joined in U_∞ by the segment, which is continuous into every level `Chapter2/LocalTopology · ScBanach.joinedIn_lvlSetInfty_of_segment`; - smooth points are dense in every level C_m

Chapter2/LocalTopology · IsPartialQuadrant.exists_smooth_near, the text's "U_∞ is dense in every U_k"; - every $z \in U$ has a ball with $B_\varepsilon(z) \cap C \subseteq U$, a convex set Chapter2/LocalTopology · IsRelOpenInQuadrant.exists_ball_inter_subset.

The case $k = \infty$ Chapter2/LocalTopology · lemma_2_2_1_infty. Fix a point a of U_∞, and say that $z \in U$ is good if near z every point of U_∞ is joined to a. Joining propagates inside a convex $B_\varepsilon(z) \cap C \subseteq U$ by segments, and smooth points are dense, so the set of good points is open and closed in U. It is nonempty, so it is all of U because U is connected. *The case $k \in \mathbb{N}$* Chapter2/LocalTopology · lemma_2_2_1: project the path-connected U_∞ to U_k and join points of U_k to nearby smooth points by segments. The text's proof replaces a path in U_0 by a polygon through smooth points, using compactness of the path; the clopen argument avoids both the path and the polygon.

Lemma 2.2.2. For $x \in O$ and $V = V' \cap U$ with $V' \ni x$ open in E_0, the set $W = V \cap r^{-1}(V)$ has these properties Chapter2/LocalTopology · lemma_2_2_2: - it is relatively open in C, by continuity of r on U; - it contains x, since $r(x) = x$; - $W \cap O = V \cap O$, because points of O are fixed; - $r(W) \subseteq W$, since $r(r(y)) = r(y) \in V$; - $r(W) = O \cap V$.

Definition 2.2.3 and Theorem 2.2.4. Mathlib's local path-connectedness is the text's: every open neighbourhood contains a path-connected open neighbourhood Chapter2/LocalTopology · definition_2_2_3. A retract is locally path-connected Chapter2/LocalTopology · theorem_2_2_4. This was proved earlier for Chapter 7 (2017, p. 183, recalls it without proof) Chapter7/LocConnected · IsScRetract.locallyPathConnectedSpace: points o, p of O near each other are joined in O by $\tau \mapsto r(o + \tau(p - o))$, the segment lying in a convex piece of U. The text goes through Lemma 2.2.2 and the component of $W(x)$ containing x.

Definition 2.2.5 and Proposition 2.2.6. *Almost locally contractible* Chapter2/LocalTopology · IsAlmostLocallyContractible: for z and open $V \ni z$, an open W with $z \in W \subseteq V$ and a continuous $\Phi : [0,1] \times W \to Z$ with values in V, $\Phi(0, \cdot) = \mathrm{id}$, $\Phi(t, z) = z$ and $\Phi(1, \cdot) \equiv z$. For a retract Chapter2/LocalTopology · proposition_2_2_6, choose ε with $B_\varepsilon(z) \cap C \subseteq U$ and $r(B_\varepsilon(z) \cap C) \subseteq V$, by continuity of r at $z = r(z)$. Then $\Phi(t, y) = r((1-t)y + tz)$ on $W = B_\varepsilon(z) \cap O$ is the text's homotopy: the segment stays in the convex $B_\varepsilon(z) \cap C$, and r fixes y and z. The model E_0 itself is a witness Chapter2/LocalTopology · isAlmostLocallyContractible_univ.

II.24 The 2021 edition: paracompactness and M-polyfolds (§2.3, Appendix 2.7.2)

The footnote to Definition 2.3.4 (preprint 2.13). The 2021 footnote says that paracompactness does not include the Hausdorff property; it asks only that every open cover have a locally finite open refinement. In the formalization, this paracompactness* is defined with covers as sets of open sets Chapter2/LocalTopology · IsLocallyFiniteSetFamily Chapter2/LocalTopology · IsParacompactStar and coincides with Mathlib's paracompactness, which does not include the Hausdorff property either Chapter2/LocalTopology · isParacompactStar_iff. So "Hausdorff paracompact" *is exactly the hypothesis of the M-polyfold structure (§II.5)* Chapter2/LocalTopology · definition_2_3_4, *and $\mathbb{R}$ is paracompact* Chapter2/LocalTopology · isParacompactStar_real. The footnote changes nothing for M-polyfolds, whose Hausdorff property is required separately.

The convention of Appendix 2.7.2. The appendix (p. 72; 2017 Appendix 2.6.2, p. 51) states the convention with a locally finite open *subcovering* Chapter2/LocalTopology · IsParacompactBySubcover. Read literally, it excludes $\mathbb{R}$ Chapter2/LocalTopology · not_isParacompactBySubcover_real. The cover by the intervals $(-n-1, n+1)$

has no locally finite subcover: a subcover is infinite, since $\mathbb{R}$ is unbounded, and the infinitely many intervals of it all contain 0. So the wording does not hold as we read it, and we read it as "refinement", as in the footnote (possible misprint).

Lemmas 2.7.2 and 2.3.5. In a regular space every open $U \ni x$ contains an open $V \ni x$ with $\mathrm{cl}(V) \subseteq U$, because closed neighbourhoods form a basis `Chapter2/LocalTopology · lemma_2_7_2`. The text states this for regular Hausdorff spaces. A Hausdorff paracompact space is normal, hence regular, and the shrinking can be done inside any given neighbourhood `Chapter2/LocalTopology · exists_isOpen_closure_subset_of_paracompact`. This is Lemma 2.3.5 for M-polyfolds, $V(x)$ "arbitrarily small" `Chapter2/LocalTopology · lemma_2_3_5`.

Proposition 2.3.6. A chart domain is homeomorphic to a retract, hence locally complete by Corollary 2.1.7 and Proposition 2.1.8 (2) `Chapter2/LocalTopology · MChart.isLocallyComplete_source`. The chart domains form an open cover of the M-polyfold X, and condition (b) of the open-cover part of Proposition 2.1.8 holds by Lemma 2.3.5. So X is locally complete `Chapter2/LocalTopology · MPolyfold.isLocallyComplete`, with $W(x)$ arbitrarily small `Chapter2/LocalTopology · proposition_2_3_6`. The text argues directly in a chart, with Proposition 2.1.4 and Lemma 2.3.5. "The topology on level 0" in the text is the topology of X.

Proposition 2.3.7. Every M-polyfold is locally path-connected `Chapter2/LocalTopology · proposition_2_3_7`. As for Theorem 2.2.4, this was already proved for Chapter 7 `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace`, by transporting the local path-connectedness of retracts through charts.

II.25 The 2021 edition: tame sub-M-polyfolds (Definitions 2.5.18, 2.5.19)

Definition 2.5.18. The structure induced on a sub-M-polyfold Y (§II.10) is *tame* if the M-polyfold Y, paracompact as a subspace, is tame `Chapter2/TameBoundary · MPolyfold.HasTameInducedStructure`.

Definition 2.5.19. Condition (3) asks for an sc-complement of T_yY in T_yX inside the reduced tangent space T^R_yX. Here (3) is stated in a chart; with the linear structure of T_yX (gap G.7, closed, §II.39) it is restated in T_yX in §II.47. For retracts $Q \subseteq O$ in C, a presentation T of C, and a level-1 point z of Q, there must be an sc-subspace A of E with $A \subseteq T_zO \cap E_z$ and $(T_zO)_m = (T_zQ)_m \oplus A_m$ on every level `Chapter2/TameBoundary · MPolyfold.HasReducedComplementAt`. (E_z is the subspace of §II.12 and $T_zO \cap E_z$ the reduced tangent space of Definition 2.4.7 (preprint 2.27).) This is literally Chapter 4's general-position condition, restated so that Chapter 2 does not import Chapter 4. $A = 0$ serves for $Q = O$ `Chapter2/TameBoundary · MPolyfold.hasReducedComplementAt_self`.

The data at a point consist of sub-M-polyfold data (V, r), so that $r(V) = Y \cap V$, together with two conditions `Chapter2/TameBoundary · MPolyfold.TameSubData`: - $d_X(r(x)) = d_X(x)$ on V; - at every smooth $y \in Y \cap V$, the chart form of (3) for some chart around y, with Q the chart image of Y near y.

Y is a tame sub-M-polyfold of the tame X if every point of Y has such data `Chapter2/TameBoundary · MPolyfold.IsTameSubMPolyfold`; it is then a sub-M-polyfold `Chapter2/TameBoundary · MPolyfold.IsTameSubMPolyfold.isSubMPolyfold`. Satisfiability: in a tame X, $Y = X$ with $r = \mathrm{id}$ has tame induced structure, the sub-charts built on tame charts having the same tame models `Chapter2/TameBoundary · MPolyfold.hasTameInducedStructure_univ`. It is a tame sub-M-polyfold, with $Q = O$ and $A = 0$ in every chart `Chapter2/TameBoundary · MPolyfold.isTameSubMPolyfold_univ`. The independence of (3) of the chart, the remark that a tame sub-M-polyfold is tame, and the two examples in $[0, \infty)^2 \times \mathbb{R}$ are §II.35 (gap G.77, closed).

II.26 The 2021 edition: faces of open sets and the corner axioms (§2.7.4, Propositions 2.7.5, 2.7.7)

Appendix §2.7.4 is new in the 2021 edition. It builds from a tame M-polyfold X a space bX of "points with a chosen face" and a map $B : bX \to X$. The formalization proves all of it for an abstract $d : X \to \mathbb{N}$ satisfying *corner axioms*, then specializes to the degeneracy index of a tame M-polyfold. The same proofs serve for the object spaces of tame ep-groupoids in 2021 §7.6.5.

Faces of open sets. For an open V and $p \in V$ with $d(p) = 1$, the face of V through p is the closure in V of the connected component of $V \cap \{d = 1\}$ through p `Chapter2/TameBoundary · TameBoundary.faceAt` `Chapter2/TameBoundary · TameBoundary.IsFaceIn`. $F_X(y, V)$ is the set of faces of V through y `Chapter2/TameBoundary · TameBoundary.facesThrough`. For an M-polyfold the faces of X itself are those of Definition 2.5.10 (preprint 2.39) `Chapter2/TameBoundary · MPolyfold.isFaceIn_univ_iff`. The text's faces of V are those of V with its induced structure, whose degeneracy index is $d_X|V$ (2021 Proposition 2.4.4), so defining them from d_X is the same thing.

The corner axioms `Chapter2/TameBoundary · TameBoundary.IsCornerIndex`. First, d is upper semicontinuous (Lemma 2.4.3 (preprint 2.23)). Second, every x has an open D such that every $y \in D$ has *local pieces* in arbitrarily small neighbourhoods N: an open $B \ni y$ in N and $d(y)$ preconnected sets $\ell_i \subseteq N \cap \{d = 1\}$ accumulating at y and covering $B \cap \{d = 1\}$ `Chapter2/TameBoundary · TameBoundary.LocalPieces`. Third, the pieces are separated in D: no preconnected subset of $D \cap \{d = 1\}$ meets two of them. In a tame chart the pieces are the images under r of the sets $\{z \in C : |z - \varphi(y)| < \varepsilon,\ I(z) = \{i\}\}$, $i \in I(\varphi(y))$, from the local face structure of §II.15 `Chapter2/TameBoundary · MPolyfold.exists_localPieces_chart`. So the degeneracy index of a tame M-polyfold satisfies the axioms `Chapter2/TameBoundary · MPolyfold.isCornerIndex_degIndex`, and $d \equiv 0$ does trivially `Chapter2/TameBoundary · TameBoundary.isCornerIndex_zero`. Faces of nested opens are compatible: a face of V meets an open $U \subseteq V$ in a union of faces of U `Chapter2/TameBoundary · TameBoundary.faceAt_inter_eq`. A face θ of U lies in a unique face of V, the union θ' of the faces of V containing it `Chapter2/TameBoundary · TameBoundary.faceExt` `Chapter2/TameBoundary · TameBoundary.faceExt_unique`. Every face of V through a point of U comes from a face of U `Chapter2/TameBoundary · TameBoundary.exists_faceExt_eq`.

Proposition 2.7.5. *Face-structured neighbourhoods exist* `Chapter2/TameBoundary · TameBoundary.exists_nbhd_forall_isFaceStructuredOpen`. Take D from the axioms. For an open $V \subseteq D$ and $y \in V$, the local pieces of y inside V lie on faces of V, and every face through y contains a piece. Distinct pieces lie on distinct faces: a face through two pieces would give a preconnected subset of $D \cap \{d = 1\}$ meeting both, which separation forbids. So y lies on exactly $d(y)$ faces of V. Then $V = D \cap \{d \le d(x)\}$, open by semicontinuity, gives (1) and (2). For (3), on any open $V \ni x$, take the set B of the local pieces of x inside V: a face of V through a point of B contains one of the pieces of x, which accumulate at x, and faces are closed in V `Chapter2/TameBoundary · TameBoundary.exists_nbhd_faces_mem`. Together these prove the proposition `Chapter2/TameBoundary · TameBoundary.proposition_2_7_5` `Chapter2/TameBoundary · MPolyfold.proposition_2_7_5`.

Definition 2.7.6 and Proposition 2.7.7. The definitions: - an open V is *face-structured* if every $y \in V$ lies on exactly $d(y)$ faces of V `Chapter2/TameBoundary · TameBoundary.IsFaceStructuredOpen` `Chapter2/TameBoundary · TameBoundary.faceStructuredOpens`; - N_X is the set of the nonempty face-structured V with $d|V$ bounded `Chapter2/TameBoundary · TameBoundary.faceStructuredNbhds`; - $N_{X,x}$ is the set of those $V \in N_X$ with $x \in V$ and $d \le d(x)$ on V `Chapter2/TameBoundary · TameBoundary.faceStructuredNbhdsAt`.

$N_X = \bigcup_x N_{X,x}$, because a $V \in N_X$ contains a point where the bounded $d|V$ is maximal `Chapter2/TameBoundary · TameBoundary.faceStructuredNbhds_eq_iUnion`. An open subset of a face-structured set is face-

structured: its faces through a point are the traces of those of the larger set, by the local pieces Chapter2/TameBoundary · TameBoundary.IsFaceStructuredOpen.mono. Hence Proposition 2.7.7: - $N_{X,x}$ is nonempty (Proposition 2.7.5) Chapter2/TameBoundary · TameBoundary.faceStructuredNbhdsAt_nonempty, so N_X covers X Chapter2/TameBoundary · TameBoundary.sUnion_faceStructuredNbhds; - nonempty open subsets of members of N_X are in N_X Chapter2/TameBoundary · TameBoundary.mem_faceStructuredNbhds_of_subset; - nonempty intersections of members of N_X are in N_X Chapter2/TameBoundary · TameBoundary.inter_mem_faceStructuredNbhds; - $N_{X,x}$ is closed under intersections Chapter2/TameBoundary · TameBoundary.inter_mem_faceStructuredNbhdsAt.

For tame M-polyfolds this is Chapter2/TameBoundary · MPolyfold.proposition_2_7_7.

Quadrants. In $[0,\infty)^n \oplus W$, d counts the vanishing half-line coordinates Chapter2/TameBoundaryExample · TameBoundaryExample.dq_eq. $\{d = 1\}$ is the union of the strata $\{p_k = 0,\ p_l > 0\ (l \neq k)\}$ Chapter2/TameBoundaryExample · TameBoundaryExample.part Chapter2/TameBoundaryExample · TameBoundaryExample.dq_eq_one_iff. A preconnected subset of $\{d = 1\}$ lies in one stratum Chapter2/TameBoundaryExample · TameBoundaryExample.subset_part, so a face through a point of the k-th stratum lies in $\{p_k = 0\}$ Chapter2/TameBoundaryExample · TameBoundaryExample.faceAt_subset_coord_eq_zero. A point of an open V with $y_k = 0$ lies on the face of V through a nearby point of the k-th stratum Chapter2/TameBoundaryExample · TameBoundaryExample.exists_part_mem_faceAt. Hence every open subset of a quadrant is face-structured Chapter2/TameBoundaryExample · TameBoundaryExample.isFaceStructuredOpen_quadrant.

II.27 The 2021 edition: L_X, F_X, bX and $U(x,V,\theta)$ (Definitions 2.7.8–2.7.11)

Definition 2.7.8. There is a morphism $(x,V) \to (y,W)$ iff $V \subseteq W$, x lies on $d(x)$ faces of W, and each of them contains y Chapter2/TameBoundary · TameBoundary.IsMor. The objects are the pairs (x,V) with $d(x) \geq 1$ and $V \in N_{X,x}$ Chapter2/TameBoundary · TameBoundary.LObj. There is at most one morphism between two objects, so L_X is a preorder, a thin category. *Composition* Chapter2/TameBoundary · TameBoundary.IsMor.trans follows the text's argument. Given $(x,V) \to (y,W) \to (z,Q)$, every face γ of Q through x extends a face θ of W through x Chapter2/TameBoundary · TameBoundary.exists_faceExt_eq. That face contains y, so $y \in \gamma$, and then $z \in \gamma$.

Definition 2.7.9. The functor sends (x,V) to $F_X(x,V)$ and a morphism to $\theta \mapsto \theta'$, the face of W containing θ Chapter2/TameBoundary · TameBoundary.boundaryFunctor Chapter2/TameBoundary · MPolyfold.LocalBoundaryCategory Chapter2/TameBoundary · MPolyfold.boundaryStructureFunctor. It is the text's "face θ' of W through x and y with $\theta \cap \theta'$ open in θ" Chapter2/TameBoundary · TameBoundary.faceExt_unique. Identities and composition hold because extending twice is extending once. $F_X(\varphi)$ is injective, the text's injection (2.43) Chapter2/TameBoundary · TameBoundary.boundaryFunctor_map_injective. It is bijective when the point does not change, since both sets have $d(x)$ elements Chapter2/TameBoundary · TameBoundary.boundaryFunctor_map_bijective.

Definition 2.7.10. An admissible triple is an object (x,V) with a face $\theta \in F_X(x,V)$ Chapter2/TameBoundary · TameBoundary.BTriple. $(x,V,\theta) \sim (x',V',\theta')$ iff $x = x'$ and some $V'' \in N_{X,x}$, $V'' \subseteq V \cap V'$, has a face through x sent to θ and to θ' Chapter2/TameBoundary · TameBoundary.BRel. Reflexivity, symmetry and transitivity hold Chapter2/TameBoundary · TameBoundary.BRel.refl Chapter2/TameBoundary · TameBoundary.BRel.symm Chapter2/TameBoundary · TameBoundary.BRel.trans; for transitivity, intersect the two V'' and use the bijectivity at a fixed point. bX is the quotient Chapter2/TameBoundary · TameBoundary.bSetoid Chapter2/TameBoundary · TameBoundary.BSpace Chapter2/TameBoundary · MPolyfold.TameBoundarySpace, and $B[x,V,\theta] = x$

Chapter2/TameBoundary · TameBoundary.bMap Chapter2/TameBoundary · MPolyfold.tameBoundaryMap. The image of B is $\{d \geq 1\} = \partial X$ Chapter2/TameBoundary · TameBoundary.range_bMap Chapter2/TameBoundary · MPolyfold.range_tameBoundaryMap. *Fibres.* $B^{-1}(x)$ has exactly $d(x)$ points Chapter2/TameBoundary · TameBoundary.encard_bMap_preimage Chapter2/TameBoundary · MPolyfold.encard_tameBoundaryMap_preimage: for a fixed $V \in N_{X,x}$, the classes over x correspond to the $d(x)$ faces of V through x, because every triple over x is equivalent to one with this V and distinct faces give distinct classes. This is the set-level content of Theorem 2.7.13 (4).

Definition 2.7.11. $U(x, V, \theta)$ is the set of classes with a representative (y, V', θ') such that $y \in \theta$, there is a morphism $(y, V') \to (x, V)$, and it sends θ' to θ Chapter2/TameBoundary · TameBoundary.uSet. The formalization reads membership as "for some representative". $\Phi_{(x,V,\theta)}$ is B restricted to it, with values in θ Chapter2/TameBoundary · TameBoundary.phi. It is injective: two classes over the same y in $U(x, V, \theta)$ are sent to the same face θ of V, and F_X of the morphism to (x, V) is injective Chapter2/TameBoundary · TameBoundary.phi_injective.

The image. Put $V^x = \{y \in V : \text{every face of } V \text{ through } y \text{ contains } x\}$ Chapter2/TameBoundary · TameBoundary.reach, an open set Chapter2/TameBoundary · TameBoundary.isOpen_reach. For $y \in V$ there is a morphism $(y, V') \to (x, V)$ exactly when $y \in V^x$. So the image of $\Phi_{(x,V,\theta)}$ is $\theta \cap V^x$ Chapter2/TameBoundary · TameBoundary.bMap_image_uSet. If V is centred at x, i.e. every face of V contains x Chapter2/TameBoundary · TameBoundary.IsCenteredAt, then $V^x = V$ and Φ is a bijection onto θ Chapter2/TameBoundary · TameBoundary.phi_bijective_of_isCenteredAt. V^x is itself centred at x Chapter2/TameBoundary · TameBoundary.isCenteredAt_reach and lies in $N_{X,x}$ Chapter2/TameBoundary · TameBoundary.reach_mem_faceStructuredNbhdsAt. So the centred neighbourhoods are cofinal, which is what the text's "take V sufficiently small" (p. 85) provides. The text calls $\Phi_{(x,V,\theta)}$ a canonical bijection for every triple (2.44); §II.28 shows that some such restriction is needed.

II.28 The 2021 edition: (2.44) and an example in $[0, \infty)^3$

Let $X = [0, \infty)^3 \oplus W$, W any sc-Banach space; the trivial W gives the setting $[0, \infty)^3$. Its degeneracy index satisfies the corner axioms Chapter2/TameBoundaryExample · TameBoundaryExample.dq Chapter2/TameBoundaryExample · TameBoundaryExample.isCornerIndex_dq. Take: - $x = (0, 0, 1)$, a corner point with $d(x) = 2$; - $V = \{p_0 < 1/2,\ 1/2 < p_1 + p_2 < 3/2\}$, a convex open slab around the segment from x to $(0, 1, 0)$ Chapter2/TameBoundaryExample · TameBoundaryExample.slab.

On V, $d \leq 2$. V is face-structured, like every open subset of a quadrant (§II.26), so $V \in N_{X,x}$ Chapter2/TameBoundaryExample · TameBoundaryExample.slab_mem_faceStructuredNbhdsAt. The open segment $\{(0, t, 1 - t)\}$ lies in $\{d = 1\}$, and the face θ of V through it contains its endpoints x and $(0, 1, 0)$ Chapter2/TameBoundaryExample · TameBoundaryExample.slabFace. (x, V, θ) is an admissible triple Chapter2/TameBoundaryExample · TameBoundaryExample.counterTriple. The point $(0, 1, 0)$ also lies on the face of V through the nearby points of the stratum $\{p_2 = 0\}$. That face lies in $\{p_2 = 0\}$, so it misses x Chapter2/TameBoundaryExample · TameBoundaryExample.exists_face_not_mem. Hence $(0, 1, 0) \notin V^x$, and $(0, 1, 0) \in \theta$ is not in the image $\theta \cap V^x$ of $\Phi_{(x,V,\theta)}$ Chapter2/TameBoundaryExample · TameBoundaryExample.pt_010_not_mem_image. So $\Phi_{(x,V,\theta)}$ is not onto θ Chapter2/TameBoundaryExample · TameBoundaryExample.phi_not_surjective, and not every Φ is a bijection Chapter2/TameBoundaryExample · TameBoundaryExample.not_forall_bijective_phi. As we read it, (2.44) does not hold for every $V \in N_{X,x}$. It holds for centred V (§II.27). The example does not depend on reading membership in $U(x, V, \theta)$ as "for some representative", because Φ misses $(0, 1, 0)$ already on the level of points.

II.29 The 2021 edition: the topology of bX (Lemma 2.7.15, Propositions 2.7.12, 2.7.16, Theorem 2.7.13 (1))

The topology. $\mathcal{U} \subseteq bX$ is in T_{bX} if every class $p \in \mathcal{U}$ has a representative (x, V, θ) and a set $O \ni p$, $O \subseteq \mathcal{U} \cap U(x, V, \theta)$, with $B(O)$ open in θ `Chapter2/TameBoundaryTopology · TameBoundary.IsBOpen`. We read O as containing p: otherwise $O = \emptyset$ would make every set open. Two tools: - a triple can be shrunk into any open $N \ni x$, keeping its class and shrinking its U-set `Chapter2/TameBoundaryTopology · TameBoundary.BTriple.exists_shrink`; - equivalent triples have a common refinement whose U-set lies in both `Chapter2/TameBoundaryTopology · TameBoundary.BRel.exists_common`.

If two triples with the same V have meeting U-sets, their faces agree `Chapter2/TameBoundaryTopology · TameBoundary.eq_of_mem_uSet`. These give the alternative characterization: $\mathcal{U} \in T_{bX}$ iff every $p \in \mathcal{U}$ has a representative with $U(x, V, \theta) \subseteq \mathcal{U}$ `Chapter2/TameBoundaryTopology · TameBoundary.isBOpen_iff`. In that form the axioms of a topology are immediate. For intersections, two representatives of p have a common refinement with U-set inside both `Chapter2/TameBoundaryTopology · TameBoundary.bTopology` `Chapter2/TameBoundaryTopology · TameBoundary.isOpen_iff_isBOpen` `Chapter2/TameBoundaryTopology · TameBoundary.isOpen_iff_forall_uSet`. This proves Lemma 2.7.15 `Chapter2/TameBoundaryTopology · TameBoundary.lemma_2_7_15` `Chapter2/TameBoundaryTopology · MPolyfold.lemma_2_7_15`. The text's proof of the intersection axiom uses the sc-smoothness of transition maps; this one uses only faces.

Proposition 2.7.12, topological part. The sets $U(x, V, \theta)$ are open `Chapter2/TameBoundaryTopology · TameBoundary.isOpen_uSet` and form a basis `Chapter2/TameBoundaryTopology · TameBoundary.isTopologicalBasis_uSet`, and B is continuous `Chapter2/TameBoundaryTopology · TameBoundary.continuous_bMap`. *The germ of a face:* if θ is a face of $U \subseteq V$ through y, then near y the extended face θ' of V coincides with θ `Chapter2/TameBoundaryTopology · TameBoundary.exists_nbhd_faceExt_inter_subset`. Hence Φ is open: $B(\mathcal{U} \cap U(x, V, \theta))$ is open in θ for open $\mathcal{U}$ `Chapter2/TameBoundaryTopology · TameBoundary.exists_isOpen_bMap_image`. With injectivity and continuity, every $\Phi_{(x,V,\theta)}$ is an open embedding into θ `Chapter2/TameBoundaryTopology · TameBoundary.isOpenEmbedding_phi`, and a homeomorphism onto θ for V centred at x `Chapter2/TameBoundaryTopology · TameBoundary.phiHomeomorph`. For two triples, the image of $U(x, V, \theta) \cap U(x', V', \theta')$ is open in θ and in θ', and the transition map is the identity on points of X, both maps being restrictions of B `Chapter2/TameBoundaryTopology · TameBoundary.proposition_2_7_12` `Chapter2/TameBoundaryTopology · MPolyfold.proposition_2_7_12`. Not part of this section: the sc-diffeomorphism half of Proposition 2.7.12. For centred representatives it is proved with the M-polyfold structure of bX (§II.36), for every representative in §II.37 (gap G.78, closed).

Proposition 2.7.16. *Hausdorff.* Two classes over different points of the Hausdorff X are separated by preimages under the continuous B. Two classes over the same x have representatives with a common centred V and different faces, and their U-sets are disjoint (Hausdorff instance in `Chapter2/TameBoundaryTopology.lean`). *Metrizable* `Chapter2/TameBoundaryTopology · TameBoundary.proposition_2_7_16`. For $x \in \partial X$ and a centred $W \in N_{X,x}$, write (x, W, θ) for the triple of a face θ of W through x `Chapter2/TameBoundaryTopology · TameBoundary.tripleOf`. The sets $U(x, W, \theta)$ cover $B^{-1}(W)$ `Chapter2/TameBoundaryTopology · TameBoundary.exists_mem_uSet_tripleOf`; they are open, pairwise disjoint and metrizable through Φ. So $B^{-1}(W)$ embeds into $X \times \mathbb{R}^{F_X(x,W)}$ by B and the indicators of these clopen pieces, and is metrizable `Chapter2/TameBoundaryTopology · TameBoundary.metrizableSpace_preimage`. The pull-back form of Smirnov's theorem along B (§II.7) `Chapter2/Metrizable · metrizableSpace_of_pullback` then metrizes bX. For tame M-polyfolds, bX is Hausdorff and metrizable `Chapter2/TameBoundaryTopology · MPolyfold.proposition_2_7_16`. The text's route instead builds a locally finite refinement from one of the cover (V_x) of ∂X, proves

paracompactness, and applies Nagata–Smirnov.

Theorem 2.7.13 (1) and the topological characterization. A topology in which every $U(x, V, \theta)$ is open, and every Φ maps open sets to relatively open subsets of θ, is T_{bX} `Chapter2/TameBoundaryTopology · TameBoundary.eq_of_isOpen_uSet`. With the results above this gives the topological part of the theorem `Chapter2/TameBoundaryTopology · TameBoundary.theorem_2_7_13_topology` `Chapter2/TameBoundaryTopology · MPolyfold.theorem_2_7_13`: - bX is metrizable; - the U-sets are open and the Φ are open embeddings; - T_{bX} is unique with these properties; - B is continuous, with image ∂X and fibres of $d_X(x)$ points.

The tame boundary of Definition 2.7.14, as a continuous map, is B `Chapter2/TameBoundaryTopology · MPolyfold.tameBoundary`. Parts (2)–(4) (the tame M-polyfold structure, B sc-smooth, d_{bX}) and the iterated boundaries are §II.36. *Witness.* The quadrant $[0, \infty)^n \oplus W$ is a tame M-polyfold `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold` `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_isTame`. Its corner has exactly n preimages in bX `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_encard_preimage`, so $bX \neq \emptyset$ for $n \geq 1$ `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_tameBoundarySpace_nonempty`. It is a tame sub-M-polyfold of itself `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_univ`.

II.30 What the 2021 additions taught us

- **Local completeness needs no Hausdorff hypothesis.** The global condition $\mathrm{cl}_X(V) \subseteq W$ gives regularity, and the metrizable W gives T_1 `Chapter2/LocalTopology · IsLocallyComplete.t2Space`. Proposition 2.1.8, Lemma 2.1.9 and complete regularity hold without it.
- **One permanence lemma.** Parts (1)–(3) of Proposition 2.1.8 are one statement about embeddings with locally closed image `Chapter2/LocalTopology · IsLocallyComplete.of_isEmbedding`. Proposition 2.3.6 is then the open-cover part applied to chart domains.
- **"Subcovering" in Appendix 2.7.2 does not hold as we read it.** With it, $\mathbb{R}$ would not be paracompact `Chapter2/LocalTopology · not_isParacompactBySubcover_real`. The footnote to Definition 2.3.4 has "refinement", which is Mathlib's notion `Chapter2/LocalTopology · isParacompactStar_iff` (possible misprint).
- **U_∞ needs a topology.** Lemma 2.2.1 for $k = \infty$ is proved for the topology of all the norms `Chapter2/LocalTopology · lemma_2_2_1_infty`, by a clopen argument rather than polygons.
- **§2.7.4 is combinatorics of d.** Every statement of the appendix about faces, L_X, bX and its topology follows from the upper semicontinuity of d and separated local pieces `Chapter2/TameBoundary · TameBoundary.IsCornerIndex`, which is why it applies to ep-groupoids as well.
- **(2.44) needs V centred.** $\Phi_{(x,V,\theta)}$ has image $\theta \cap V^x$ `Chapter2/TameBoundary · TameBoundary.bMap_image_uSet`, and in $[0, \infty)^3$ it is not onto for a thin slab `Chapter2/TameBoundaryExample · TameBoundaryExample.phi_not_surjective`. Centred neighbourhoods are cofinal `Chapter2/TameBoundary · TameBoundary.reach_mem_faceStructuredNbhdsAt`.
- **The topology of bX needs no sc-calculus.** Lemma 2.7.15 and Proposition 2.7.16 are proved from faces and the pull-back form of Smirnov's theorem `Chapter2/TameBoundaryTopology · TameBoundary.metrizableSpace_preimage`. The text uses the sc-smoothness of transition maps and a paracompactness argument.

II.31 The 2021 edition, continued: Remark 2.1.6 is strict (the K-topology)

§II.22 proved the implication of Remark 2.1.6. That the modified notion is *strictly* weaker needs an example, which the text does not give.

A criterion. If every point has an open neighbourhood W that is completely metrizable, the modified definition holds with $V = W$, the closure of W in itself being W `Chapter2/LocalTopologyExample · isLocallyCompleteRel_of_forall_isCompletelyMetrizable`. A space embedded in $\mathbb{R}$ onto the complement of a sequence (a_n) is completely metrizable: $y \mapsto (y, ((y - a_n)^{-1})_n)$ is a closed embedding into $\mathbb{R} \times \mathbb{R}^{\mathbb{N}}$, with image $\{(u, z) : z_n(u - a_n) = 1 \text{ for all } n\}$ `Chapter2/LocalTopologyExample · isCompletelyMetrizableSpace_of_isEmbedding_compl_range`. This replaces the theorem that G_δ subsets of complete metric spaces are completely metrizable, which Mathlib lacks.

The K-topology. Let $K = \{1/(n + 1)\}$ `Chapter2/LocalTopologyExample · kSeq` and give $\mathbb{R}$ the topology generated by the usual open sets and $\mathbb{R} \setminus K$ `Chapter2/LocalTopologyExample · kTop` `Chapter2/LocalTopologyExample · RealKTop`. It is finer than the usual topology, so Hausdorff. The open sets $\mathbb{R} \setminus K$ and $\mathbb{R} \setminus \{0\}$ cover it, carry the usual subspace topology, and are complements of sequences; by the criterion the modified definition holds `Chapter2/LocalTopologyExample · RealKTop.isLocallyCompleteRel`. But 0 and the closed set K cannot be separated: every closed neighbourhood of 0 contains irrational points arbitrarily close to some $1/(n + 1)$, hence meets K `Chapter2/LocalTopologyExample · RealKTop.not_regularSpace`. A locally complete space is regular (§II.22), so this one is not locally complete `Chapter2/LocalTopologyExample · RealKTop.not_isLocallyComplete`.

Remark 2.1.6, strictness. Some Hausdorff space satisfies the modified definition and not Definition 2.1.5 `Chapter2/LocalTopologyExample · remark_2_1_6_strict`. So the remark holds as we read it, and the example shows that the global condition $\mathrm{cl}_X(V) \subseteq W$ is what makes a locally complete space regular.

II.32 The 2021 edition: bundle retractions and the levels of X^1 (Definition 2.1.16, after Definition 2.3.8)

Definition 2.1.16 (preprint 2.9). A retraction R of $E \oplus F$ is a bundle retraction covering the retraction r of E if its domain is $U \oplus F$, U the domain of r, and $R(u, v) = (r(u), \rho(u)v)$ there, with $\rho(u)$ a continuous linear map `Chapter2/TangentLift · ScRetraction.IsBundleRetractionOver`. A map Φ is a bundle map on B covering a if $\Phi(u, v) = (a(u), \phi(u)v)$ on B `Chapter2/TangentLift · IsBundleMapOn`; then the first component of Φ is a of the first component `Chapter2/TangentLift · IsBundleMapOn.fst_eq`. The text writes "$p' \circ \Phi = p$", which we read as $p' \circ \Phi = a \circ p$ (possible misprint). Sc-smoothness and $R \circ R = R$ are part of the data, as for every retraction (§II.2). The tangent retraction is a bundle retraction covering the level-1 retraction r^1: $Tr(u, h) = (r(u), Dr(u)h)$ `Chapter2/TangentLift · ScRetraction.tangentRetraction_isBundleRetractionOver`. The tangent map of an sc-smooth map between retracts is a bundle map covering f^1 `Chapter2/TangentLift · IsScSmoothRetractMap.retractTangentMap_isBundleMapOn`. The categories $\mathcal{R}$ and $\mathcal{BR}$ and their forgetful functors only name these objects and maps; they are not formalized. The category $\mathcal{SBR}$ of §2.6 is (§II.45).

The levels of X^1 (gap G.6). §II.6 built the M-polyfold X^1 on X_1; its levels are those of X shifted by one as sets. Their topologies agree too. In a chart, the level-m lift of a point is moved to another chart by the level-m map of the transition map `Chapter2/TangentLift · MPolyfold.levelLift_transition`, and the

level-m lift of the level chart φ^1 of X^1 is the level-$(m+1)$ lift of φ, since $(E^1)_m = E_{m+1}$ Chapter2/TangentLift · MPolyfold.levelLift_levelChart. The identity of points is therefore continuous from $(X^1)_m$ to X_{m+1} and back, chart by chart, for an arbitrary chart of X^1 through level maps of transition maps Chapter2/TangentLift · MPolyfold.shiftLevelMap Chapter2/TangentLift · MPolyfold.continuous_shiftLevelMap Chapter2/TangentLift · MPolyfold.shiftLevelInv Chapter2/TangentLift · MPolyfold.continuous_shiftLevelInv. So $(X^1)_m \cong X_{m+1}$ Chapter2/TangentLift · MPolyfold.shiftLevelHomeomorph, with the charts of X_{m+1} Chapter2/TangentLift · MPolyfold.shift_level_eq. Composing these homeomorphisms, each iterate $(X^1)^1 \cdots$ is X_m with its topology and charts, which is the text's statement after Definition 2.3.8.

II.33 The 2021 edition: Theorem 2.3.16 and Proposition 2.3.17, $T(X^i) \equiv (TX)^i$

In a local model. For an sc¹ map f, the tangent map of the level-1 map f^1 is the level-1 map of Tf on $T(U^1) = (TU)^1$ Chapter2/TangentLift · tangentMap_levelMap_one Chapter2/TangentLift · toBase_tangentMap_levelMap_one. For a retract, $T(O^1) = (TO)^1$ Chapter2/TangentLift · retractTangent_shift_of_isScRetract, and for sc-smooth maps between retracts $T(f^1)$ is again the level-1 map of Tf Chapter2/TangentLift · toBase_retractTangentMap_shift.

Theorem 2.3.16, bundle form. For an M-polyfold X, the tangent charts belong to the structure of TX (§II.8), $p : TX \to X^1$ is sc-smooth, and in a tangent chart p is the first projection onto the level chart φ^1. Every model (TO, TC, TE) is a bundle retract, Tr covering r^1, and the transition map of two tangent charts is a bundle map covering the level-1 map of the transition map of the charts Chapter2/TangentLift · theorem_2_3_16. "M-polyfold bundle" is read through the models and transition maps; the linear structure of the fibres T_xX and the linearity of Tf on them are §II.39 (gap G.7, closed).

Proposition 2.3.17 for $i = 1$. A tangent vector q of X^1 at $x \in (X^1)_1 = X_2$, read in the level chart φ^1 of a chosen chart φ of X around x Chapter2/TangentLift · MPolyfold.someChart, is $(\varphi^1(x), h')$ with $h' \in E_1$; it gives the tuple (x, φ, h') of TX Chapter2/TangentLift · MPolyfold.TangentPre.lower. Read in any chart ψ of X, this tuple is the inclusion of q read in ψ^1, because $T(\psi \circ \varphi^{-1})$ on $(TO)^1$ is the tangent map of $(\psi \circ \varphi^{-1})^1$ Chapter2/TangentLift · MPolyfold.TangentPre.transport_lower. So it defines an inclusion $\iota : T(X^1) \to TX$ Chapter2/TangentLift · MPolyfold.tangentLower. Its values lie on level 1 of TX, giving $\Phi : T(X^1) \to (TX)^1$ Chapter2/TangentLift · MPolyfold.tangentShiftMap, a bijection Chapter2/TangentLift · MPolyfold.tangentShiftEquiv and a homeomorphism Chapter2/TangentLift · MPolyfold.tangentShiftHomeomorph. In every chart ψ around the base point, $T\psi(\Phi(\alpha))$ is the inclusion of $T(\psi^1)(\alpha)$ Chapter2/TangentLift · MPolyfold.tangentChartMap_tangentLower, so Φ does not depend on the chosen chart; in the charts $T(\varphi^1)$ and $(T\varphi)^1$ it is the identity of $T(O^1) = (TO)^1$ Chapter2/TangentLift · MPolyfold.levelChart_tangentShiftMap. Hence Φ is an sc-diffeomorphism Chapter2/TangentLift · MPolyfold.tangentShiftEquiv_isScDiffeoOn, it covers the identity of $(X^1)^1$, and the level-1 map $p^1 : (TX)^1 \to (X^1)^1$ is sc-smooth Chapter2/TangentLift · MPolyfold.tangentProjShift Chapter2/TangentLift · proposition_2_3_17.

$i = 2$. A bijection of M-polyfolds sc-smooth in both directions raises to an sc-diffeomorphism of the lifts Chapter2/TangentLift · MPolyfold.levelOneEquiv Chapter2/TangentLift · MPolyfold.levelOneEquiv_isScDiffeoOn. Composing Φ for X^1 with the raise of Φ^{-1} gives $T(X^2) \cong (TX)^2$ Chapter2/TangentLift · MPolyfold.tangentShiftEquivTwo, covering the identity of $(X^1)^2 = (X^2)^1$ Chapter2/TangentLift · MPolyfold.tangentProjShiftTwo Chapter2/TangentLift · proposition_2_3_17_two.

Every i. To speak of X^i as a function of i, an M-polyfold is bundled with its space Chapter2/

TangentLiftAll · BundledMPolyfold; then $X^0 = X$, $X^{i+1} = (X^i)^1$ Chapter2/TangentLiftAll · BundledMPolyfold.lift, and TX is again bundled Chapter2/TangentLiftAll · BundledMPolyfold.tangent. Sc-diffeomorphisms and sc-smooth maps are packaged with their index raises Chapter2/TangentLiftAll · MPolyfold.ScEquiv Chapter2/TangentLiftAll · MPolyfold.ScEquiv.shift Chapter2/TangentLiftAll · MPolyfold.ScMap, with Φ and p as instances Chapter2/TangentLiftAll · MPolyfold.tangentShiftScEquiv Chapter2/TangentLiftAll · MPolyfold.tangentProjScMap. By recursion, $\Phi_0 = \mathrm{id}$ and $\Phi_{i+1} = (\Phi_i)^1 \circ \Phi$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv; likewise $J_i : (X^1)^i \to (X^i)^1$ Chapter2/TangentLiftAll · BundledMPolyfold.shiftLiftEquiv and p^i Chapter2/TangentLiftAll · BundledMPolyfold.tangentProjLift. The inclusions $X^i \to X$ Chapter2/TangentLiftAll · BundledMPolyfold.incl are injective, and $T(X^i) \to TX$ is the composite of the inclusions ι Chapter2/TangentLiftAll · BundledMPolyfold.tangentIncl. With these, Φ_i and J_i are sc-diffeomorphisms, the projections are sc-smooth, $J_i \circ p^i \circ \Phi_i = p$, Φ_i commutes with the inclusions into TX and with those of the two towers $T(X^{i+1}) \to T(X^i)$ and $(TX)^{i+1} \to (TX)^i$, J_i is the identity on points, and these properties determine Φ_i Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_unique Chapter2/TangentLiftAll · proposition_2_3_17_all. Φ_1 is the Φ above Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_one. Because $\iota \circ T(f^1) = Tf \circ \iota$ for every sc-smooth f Chapter2/TangentLiftAll · MPolyfold.tangentLower_tangentMap, Φ_i is natural in X Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_natural. A bundled M-polyfold exists, the half-line Chapter2/TangentLiftAll · BundledMPolyfold.nonempty. Linearity of Φ_i on the fibres is not stated; in charts Φ_i is a composite of index raises of identities. The linear structure of T_xX in which it could be stated is §II.39. Lemma 4.4.4 for every i uses this identification (Chapter 4).

II.34 The 2021 edition: Remark 2.4.2, the diagonal half-line

The text's example is $\Delta = \{(s,s,s) : s \geq 0\}$ in $C = [0,\infty)^3$; here $C = [0,\infty)^3 \oplus W$ for any sc-Banach space W, with $W = 0$ the text's case. The constant linear maps between $\mathbb{R}$ and $\mathbb{R}^3$ are sc-operators Chapter2/TangentLift · ScOperator.ofConst, and replacing the three coordinates by their average Chapter2/TangentLift · diagonalAverage is an sc-smooth retraction of C onto $\Delta \oplus W$ Chapter2/TangentLift · diagonalRetraction Chapter2/TangentLift · mem_diagonalRetraction_retract_iff, the orthogonal projection onto the diagonal; the text gives no retraction. Its retract is a local model Chapter2/TangentLift · diagonalModel and an M-polyfold Chapter2/TangentLift · diagonalPolyfold.

$(s,w) \mapsto ((s,s,s),w)$ Chapter2/TangentLift · diagonalEmbed, with inverse the first coordinate Chapter2/TangentLift · diagonalProject, is an sc-diffeomorphism from the half-line M-polyfold $[0,\infty) \oplus W$ onto $\Delta \oplus W$ Chapter2/TangentLift · diagonalEmbed_isScDiffeoOn. In C, $d_C((s,s,s),w)$ is 3 at $s = 0$ and 0 otherwise Chapter2/TangentLift · degIndex_diagonal_quadrant; on the M-polyfold, d_Δ is 1 at $s = 0$ and 0 otherwise, by the invariance of the index under sc-diffeomorphisms (Proposition 2.4.4 (preprint 2.24), §II.11) applied to the half-line Chapter2/TangentLift · diagonalPolyfold_degIndex. One chart, defined on all of $\Delta \oplus W$, has the model $[0,\infty) \oplus W$, and in it the chart index is d_Δ Chapter2/TangentLift · diagonalHalfLineChart Chapter2/TangentLift · diagonalHalfLineChart_spec Chapter2/TangentLift · remark_2_4_2. So the defining chart realizes d_C, the more economical one d_Δ, as the remark says.

II.35 The 2021 edition: tame sub-M-polyfolds, continued (Definition 2.5.19 and the text after it)

§II.25 stated condition (3) of Definition 2.5.19 in one chart around each smooth point, with one presentation of its partial quadrant.

Chart independence. A point is smooth iff its image in one chart is a smooth point of the model Chapter2/TameSub · MPolyfold.mem_smoothSet_iff_of_isChart. Condition (3) in a chart is the general-position condition of Chapter 4 for the retraction of the model and the chart image of Y, word for word. Chapter 4's chart change for that condition (MPolyfold.isGeneralPositionRetractAt_chart_change, Chapter 4) moves it to every chart $\varphi \ni y$, every presentation of the partial quadrant of φ, every level-1 representative, and the chart image of any other sub-M-polyfold retraction onto Y near y Chapter2/TameSub · MPolyfold.TameSubData.hasReducedComplementAt_of_isChart, in particular of the same one Chapter2/TameSub · MPolyfold.TameSubData.complement_forall_charts. (3) is a statement in charts here; its form in T_yX, equivalent to it, is §II.47.

A tame sub-M-polyfold is tame. Sc-splittings compose: if $E = M \oplus B$ is an sc-splitting and $M_m = N_m \oplus A_m$ on every level for sc-subspaces $N, A \subseteq M$, then $E = N \oplus (A \oplus B)$ is one Chapter2/TameSub · ScBanach.levelSub_sup_of_le_of_isScSplitting Chapter2/TameSub · ScBanach.isScSplitting_sup_of_complementIn. In a chart φ with a tame model, whose tame retraction is t, the chart image $\varphi(W \cap Y)$ is the retract of $(\varphi r \varphi^{-1}) \circ t$. That retraction is tame: the index condition comes from (2) of Definition 2.5.19 and $d_X = d_C \circ \varphi$ in a tame chart; the sc-complement of the tangent space of $\varphi(W \cap Y)$ inside E_x is $A \oplus B$, with A from (3) in the chart φ (chart independence) and B the complement of T_xO from the tameness of t Chapter2/TameSub · MPolyfold.TameSubData.isTameRetract_image. So the sub-charts of Y built on tame charts of X have tame models, and Y has a tame induced structure (Definition 2.5.18) Chapter2/TameSub · MPolyfold.IsTameSubMPolyfold.hasTameInducedStructure. The text calls this not difficult to see and gives no proof.

The examples. In $X = [0,\infty)^n \oplus W$, the fixed set of an idempotent sc-operator L mapping the quadrant into itself Chapter2/TameSub · TameSubExample.fixSet is a sub-M-polyfold, L restricted to the quadrant being a retraction Chapter2/TameSub · TameSubExample.isSubMPolyfold_fixSet; if L keeps the $\mathbb{R}^n$-component, L preserves d_X (condition (2)) and (3) holds in the identity chart, so the fixed set is a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.isTameSubMPolyfold_fixSet. For $n = 2$, $A = \{(x, y, 0)\}$ Chapter2/TameSub · TameSubExample.setA is the case $L(a, w) = (a, 0)$: a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.isTameSubMPolyfold_setA with tame induced structure Chapter2/TameSub · TameSubExample.hasTameInducedStructure_setA. $B = \{(x, x, 0)\}$ Chapter2/TameSub · TameSubExample.setB is a sub-M-polyfold Chapter2/TameSub · TameSubExample.isSubMPolyfold_setB. No retraction onto B with (1) and (2) is defined near the corner: for small $t > 0$ the point $(t, 0, 0)$ has $d_X = 1$, while d_X takes only the values 0 and 2 on B Chapter2/TameSub · TameSubExample.not_mem_tameSubData_setB. So B is not a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.not_isTameSubMPolyfold_setB, and (3) plays no role. Yet $t \mapsto (t, t, 0)$ is an sc-diffeomorphism from the half-line onto B with its induced structure Chapter2/TameSub · TameSubExample.halfLineToB_isScDiffeoOn, which is therefore tame Chapter2/TameSub · TameSubExample.hasTameInducedStructure_setB. For $W = \mathbb{R}$ these are the text's statements Chapter2/TameSub · TameSubExample.examples_A_B.

II.36 The 2021 edition: the tame M-polyfold bX (Theorem 2.7.13 (2)–(4), Definition 2.7.14, (2.45)–(2.47))

Local faces in a tame chart. Every point of a tame M-polyfold has tame charts with domain inside a given neighbourhood Chapter2/TameBoundaryStructure · MPolyfold.exists_tame_chart_subset, and near a point of a face θ of a face-structured open V the face is, in such a chart, the zero set of one coordinate, as in Theorem 2.5.17 (preprint 2.45) (§II.16) Chapter2/TameBoundaryStructure · MPolyfold.exists_localFace_of_faceIn.

Chart data. A chart datum consists of a triple (x, V, θ) with V centred at x (§II.27), a chart φ of X, an open V_1 inside its domain, an sc-isomorphism T of its model space with $\mathbb{R}^{k+1} \oplus W$ and a coor-

dinate i vanishing on $T\varphi(\theta \cap V_1)$, such that the face projection π of Proposition 2.5.15 (preprint 2.43) maps $\varphi(\theta \cap V_1)$ onto a tame retract of $[0,\infty)^k \oplus W$, and $d_X = d \circ \pi \circ \varphi + 1$ on $\theta \cap V_1$ Chapter2/TameBoundaryStructure · MPolyfold.BChartData. Its chart is $p \mapsto \pi(\varphi(B(p)))$ on $U(x,V,\theta) \cap B^{-1}(V_1)$ Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart. Two such charts have the transition map $\pi' \circ (\varphi' \circ \varphi^{-1}) \circ \iota$, ι the face inclusion, which is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart_transition_isScSmooth, so they are compatible Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart_compatible. By the local faces, every point of θ lies in the V_1 of a datum with the triple (x,V,θ), and every class has a centred representative Chapter2/TameBoundaryStructure · TameBoundary.BSpace.exists_centred_rep; so the charts cover bX Chapter2/TameBoundaryStructure · MPolyfold.exists_bChartData and form an atlas Chapter2/TameBoundaryStructure · MPolyfold.bAtlas.

Theorem 2.7.13 (2)–(4). T_{bX} is metrizable (§II.29), hence paracompact Chapter2/TameBoundaryStructure · MPolyfold.paracompactSpace_tameBoundarySpace, and with the atlas bX is an M-polyfold Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold, tame because the models are tame retracts Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_isTame (2). In these charts B is the face inclusion ι followed by φ^{-1}, so B is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryMap_isScSmooth (3). The index of the face model is one less than that of the quadrant: $d_X(B(p)) = d_{bX}(p) + 1$ Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap, which is (4) as printed Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap_sub_one.

The characterization, centred representatives. A face θ of a face-structured open V is a sub-M-polyfold of X Chapter2/TameBoundaryStructure · MPolyfold.isSubMPolyfold_of_isFaceIn, which we take as "the structure induced from V" Chapter2/TameBoundaryStructure · MPolyfold.facePolyfold. For V centred at x, $\Phi_{(x,V,\theta)}$ Chapter2/TameBoundaryStructure · MPolyfold.phiTot is an sc-diffeomorphism of $U(x,V,\theta)$ onto θ, because in the charts above it is the face inclusion Chapter2/TameBoundaryStructure · MPolyfold.phi_isScDiffeoOn. Every M-polyfold structure on bX with the topology T_{bX} for which all these Φ are sc-diffeomorphisms is equivalent to this one Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_unique, which has the property Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_satisfies. The sc-half of Proposition 2.7.12 for centred representatives follows Chapter2/TameBoundaryStructure · MPolyfold.proposition_2_7_12_scDiffeo, and one statement collects (2)–(4), the characterization and its uniqueness Chapter2/TameBoundaryStructure · MPolyfold.theorem_2_7_13_structure.

Definition 2.7.14 and the iterated boundaries. The tame boundary is bX with this structure and the sc-smooth B Chapter2/TameBoundaryStructure · MPolyfold.definition_2_7_14. To iterate b, a tame M-polyfold is bundled with its space Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle, so that bX is again such a bundle Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.b. Then $b^0X = X$, $b^{k+1}X = b^k(bX)$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iter, and $B^0 = \mathrm{id}$, $B^{k+1} = B_X \circ B^k_{bX}$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iterMap, which unwinds to the text's $B^k = B_X \circ B_{bX} \circ \cdots \circ B_{b^{k-1}X}$. By (3) and (4) and induction, B^k is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iterMap_isScSmooth, $d_X(B^k(p)) = d_{b^kX}(p) + k$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.degIndex_iterMap, and the image of B^k is $\{d_X \geq k\}$, by induction from the image ∂X of B (§II.29) and the index formula Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.range_iterMap.

Witness. For the tame quadrant $[0,\infty)^n \oplus W$, $n \geq 1$, chart data exist Chapter2/TameBoundaryStructure · MPolyfold.quadrantPolyfold_bChartData_nonempty, and the n points of bX over the corner have $d_{bX} = n - 1$ Chapter2/TameBoundaryStructure · MPolyfold.quadrantPolyfold_degIndex_tameBoundary.

II.37 The 2021 edition: Proposition 2.7.12 and the characterization for every representative

Centred refinements. Every triple s has an equivalent triple s' centred at its point with $U(s') \subseteq U(s)$ `Chapter2/TameBoundaryNonCentred · TameBoundary.BTriple.exists_centred_subset`: replace V by V^x (§II.27). On the image $\theta \cap V^x$ of $\Phi_{(x,V,\theta)}$ `Chapter2/TameBoundaryNonCentred · MPolyfold.reachSet`, an open subset of θ `Chapter2/TameBoundaryNonCentred · MPolyfold.isOpen_reachSet` equal to θ for centred V `Chapter2/TameBoundaryNonCentred · MPolyfold.reachSet_eq_univ_of_isCenteredAt`, the inverse is a total map `Chapter2/TameBoundaryNonCentred · TameBoundary.phiInv` `Chapter2/TameBoundaryNonCentred · TameBoundary.phiInv_spec`.

Sc-smoothness. Φ is sc-smooth on $U(x, V, \theta)$, as B followed by the inclusion into the sub-M-polyfold θ `Chapter2/TameBoundaryNonCentred · MPolyfold.phiTot_isScSmoothOn`. Near each point y of $\theta \cap V^x$, Φ^{-1} is $\Phi_{s'}^{-1}$ for a centred refinement s' at y, composed with the inclusion of an open piece of θ into the face of s'; that inclusion is sc-smooth because a map into a sub-M-polyfold is sc-smooth as soon as it is sc-smooth into X. This replaces the open sc-embeddings of faces in the text's argument `Chapter2/TameBoundaryNonCentred · MPolyfold.phiInv_isScSmoothOn`. So Φ is an sc-diffeomorphism of $U(x, V, \theta)$ onto the open set $\theta \cap V^x$ `Chapter2/TameBoundaryNonCentred · MPolyfold.phi_isScDiffeoOn_nonCentred`.

Proposition 2.7.12 as printed. For every pair of representatives, the images of $U \cap U'$ lie in $\theta \cap V^x$ and $\theta' \cap V'^{x'}$ and are open there, and the transition map is the composite of an inverse and a forward map, hence an sc-diffeomorphism, with inverse the reverse transition map and the identity on points of X `Chapter2/TameBoundaryNonCentred · MPolyfold.proposition_2_7_12_scDiffeo_nonCentred`.

The characterization. An M-polyfold structure on bX in which every $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism onto $\theta \cap V^x$ is equivalent to that of §II.36 `Chapter2/TameBoundaryNonCentred · MPolyfold.tameBoundaryPolyfold_unique_allReps`, which has the property `Chapter2/TameBoundaryNonCentred · MPolyfold.tameBoundaryPolyfold_satisfies_allReps`. With "onto θ" it does not hold as we read it for every representative: for the slab triple of §II.28, Φ does not map $U(x, V, \theta)$ onto θ `Chapter2/TameBoundaryNonCentred · MPolyfold.not_surjOn_phiTot`. The strongest version that holds is the one onto $\theta \cap V^x$, which for centred V is the printed statement (§II.36).

What these additions taught us. - **"Weaker" is strict**, through a classical non-regular space `Chapter2/LocalTopologyExample · remark_2_1_6_strict`; the global condition of Definition 2.1.5 is what gives regularity. - **The levels of X^1 are those of X**, chart by chart `Chapter2/TangentLift · MPolyfold.shiftLevelHomeomorph`, so the iterates realize the X_m of the text. - **$T(X^i) \equiv (TX)^i$ is one statement for all i** once M-polyfolds are bundled with their spaces `Chapter2/TangentLiftAll · proposition_2_3_17_all`; fibre linearity of Tf is §II.39. - **Condition (3) of Definition 2.5.19 is Chapter 4's general-position condition**, and its chart change is Chapter 4's `Chapter2/TameSub · MPolyfold.TameSubData.hasReducedComplementAt_of_isChart`. - **bX is built from face charts of centred triples** `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold`; non-centred representatives need the image $\theta \cap V^x$ `Chapter2/TameBoundaryNonCentred · MPolyfold.phi_isScDiffeoOn_nonCentred`.

II.38 Proposition 2.1.2 (preprint 2.2): classical smooth retractions have submanifold images

Statement. Let $r : U \to U$ be a C^∞ retraction of an open subset U of a Banach space E, $O = r(U)$ and $x \in O$. With $P = Dr(x)$, $R = \operatorname{range} P$ and $N = \ker P$, the space splits as $E = R \oplus N$ into closed subspaces. There are open $V \ni x$, $W \ni 0$ and a C^∞ diffeomorphism $\psi : V \to W$ with $\psi(x) = 0$

and $\psi(O \cap V) = R \cap W$ `Chapter2/SmoothRetractImage · GapP1Prop212.proposition_2_1_2`. The text prints the normalization as $\psi(0) = x$ for a map $\psi : V \to W$; we read it as $\psi(x) = 0$.

Proof. The text estimates remainders in Appendix 2.7.1. The formal proof conjugates r to the linear projection P instead. - *P is a projection.* From $r \circ r = r$ near x and $r(x) = x$, the chain rule gives $P^2 = P$ `Chapter2/SmoothRetractImage · GapP1Prop212.isIdempotentElem_fderiv`. So R and N are closed and complementary. - *The conjugating map.* Put

$$\varphi(y) = P(r(y) - x) + (1 - P)(y - r(y))$$

`Chapter2/SmoothRetractImage · GapP1Prop212.retractChart`. It is C^∞ on U `Chapter2/SmoothRetractImage · GapP1Prop212.contDiffOn_retractChart`, with $\varphi(x) = 0$. Its derivative at x is $P \cdot P + (1 - P)(1 - P) = P + (1 - P) = 1$ `Chapter2/SmoothRetractImage · GapP1Prop212.hasFDerivAt_retractChart`. - *Conjugation.* $\varphi(r(y)) = P(r(y) - x)$, because $r(r(y)) = r(y)$, and $P\varphi(y) = P(r(y) - x)$, because $P(1 - P) = 0$. So $\varphi \circ r = P \circ \varphi$ on U `Chapter2/SmoothRetractImage · GapP1Prop212.retractChart_comp`. - *Inverse function theorem.* Mathlib's inverse function theorem makes φ a C^∞ diffeomorphism from an open $V_0 \ni x$ onto an open set. Shrink V_0 to the open set V of points $y \in V_0$ with $r(y) \in V_0$ at which $D\varphi(y)$ is invertible, and let $\psi = \varphi|_V$ and $W = \psi(V)$. - *The image of O.* If $y \in O \cap V$, then $\psi(y) = \varphi(r(y)) = P\varphi(y) \in R$. Conversely, if $\psi(y) \in R$, then $\varphi(r(y)) = P\varphi(y) = \varphi(y)$; both y and $r(y)$ lie in V_0, where φ is injective, so $r(y) = y$ and $y \in O$.

II.39 Definition 2.3.14 (preprint 2.18), Proposition 2.3.19 (3)–(4) (preprint 2.21) and Theorem 2.3.16: the linear structure of T_xX

Definition 2.3.14 (preprint 2.18) makes T_xX a vector space by adding representatives in one chart, and the text says after it that T_xX inherits the Banach structure of $T_{\varphi(x)}O$, and the sc-structure at a smooth x. That the result does not depend on the chart is not argued. It holds because the fibre transitions are linear.

The models. For a retract O and $a \in O_1$, the tangent space $T_aO = Dr(a)E_0$ is a closed subspace, the range of a continuous projection `Chapter2/TangentSpaceLinear · tangentSubmodule Chapter2/TangentSpaceLinear · isClosed_tangentSubmodule`. For a chart φ around $x \in X_1$, the model of the fibre is $T_{\varphi(x)}O$ `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberModel`, and $T\varphi$ restricts to a bijection $T_xX \to T_{\varphi(x)}O$ with inverse $h \mapsto [(x, \varphi, h)]$ `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberEquiv`.

The transitions are linear. For two charts φ, ψ around x, the fibre transition $T\psi \circ (T\varphi)^{-1}$ is the restriction to $T_{\varphi(x)}O$ of the bounded operator $D(\psi \circ \varphi^{-1} \circ r)(\varphi(x))$ `Chapter2/TangentSpaceLinear · MPolyfold.exists_transition_clm Chapter2/TangentSpaceLinear · MPolyfold.transitionCLM`. This comes from the description of the tangent map of a map between retracts, $(a, h) \mapsto (g(a), Dg(a)h)$ (Theorem 2.1.15 (preprint 2.8), §II.3). The transitions in the two directions are inverse to each other on the models `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberTransition_transition`.

The vector space. Transport the vector space structure of $T_{\varphi_0(x)}O$ along $T\varphi_0$ for one chosen chart φ_0 around x `Chapter2/TangentSpaceLinear · MPolyfold.fiberChart`. For every other chart φ, $T\varphi = (\text{transition}) \circ T\varphi_0$ is then linear, so $T\varphi : T_xX \to T_{\varphi(x)}O$ is a linear isomorphism for every chart `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberLinearEquiv`. This is the chart independence of the definition. The text's formula $\lambda[(x, \varphi, h_1)] + \mu[(x, \varphi, h_2)] = [(x, \varphi, \lambda h_1 + \mu h_2)]$ holds in every chart `Chapter2/TangentSpaceLinear · MPolyfold.definition_2_3_14`.

The Banach space. Give T_xX the subspace topology of TX. Then $T\varphi$ is a homeomorphism onto $T_{\varphi(x)}O$, since the tangent charts are homeomorphisms (§II.8) `Chapter2/TangentSpaceLinear · MPolyfold.continuous_tangentFiberEquiv` `Chapter2/TangentSpaceLinear · MPolyfold.continuous_tangentFiberEquiv_symm`. So it is a topological linear isomorphism onto a Banach space `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberContinuousLinearEquiv`. The norm pulled back through a chart induces the topology and is complete `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberNormedAddCommGroup` `Chapter2/TangentSpaceLinear · MPolyfold.completeSpace_tangentFiber`. Two charts give equivalent norms, because the transition is an isomorphism of Banach spaces `Chapter2/TangentSpaceLinear · MPolyfold.tangentFiberTransitionEquiv`.

The sc-structure at a smooth point. If x is smooth, so is $\varphi(x)$, and $Dr(\varphi(x))$ is an sc-projection. Its range $T_{\varphi(x)}O$ is therefore an sc-subspace of E `Chapter2/TangentSpaceLinear · MPolyfold.isScSubspace_tangentFiberModel`. The transition operator is an sc-operator at a smooth point `Chapter2/TangentSpaceLinear · MPolyfold.isScOperator_transitionCLM` and maps level m of one model into level m of the other `Chapter2/TangentSpaceLinear · MPolyfold.transitionCLM_mem_range_toBase`. With its inverse, it restricts to an sc-isomorphism of the models `Chapter2/TangentSpaceChartIndependent · MPolyfold.tangentFiberScIso` (§II.44). So the sc-structures through two charts agree.

Tangent maps are linear on the fibres (Theorem 2.3.16). For f sc-smooth near x, Tf restricted to $T_xX \to T_{f(x)}Y$ is, in charts φ and ψ, the bounded operator $D(\psi \circ f \circ \varphi^{-1} \circ r)(\varphi(x))$, and this operator is an sc-operator if x is smooth `Chapter2/TangentMapLinear · MPolyfold.exists_clm_tangentMapOnFiber`. Hence Tf is continuous and linear on the fibres `Chapter2/TangentMapLinear · MPolyfold.tangentMapOnCLM` `Chapter2/TangentMapLinear · MPolyfold.tangentMapOnCLM_apply` `Chapter2/TangentMapLinear · MPolyfold.tangentMapCLM`. Together with §II.33 this is Theorem 2.3.16 with fibre linearity `Chapter2/TangentMapLinear · MPolyfold.theorem_2_3_16_linear`. The text leaves this implicit in "Tf is a bundle map".

Proposition 2.3.19 (3)–(4) (preprint 2.21) in T_aX. Let $A \subseteq X$ be a sub-M-polyfold, $a \in A_1$, and $r : V \to V$ an sc-smooth retraction with $r(V) = A \cap V$. The tangent of the inclusion, $Ti : T_aA \to T_aX$ `Chapter2/SubMPolyfoldTangent · MPolyfold.subInclusionCLM`, and the tangent of $i^{-1} \circ r$, $T_aX \to T_aA$ `Chapter2/SubMPolyfoldTangent · MPolyfold.subRetractionCLM`, are continuous linear maps. Since $i^{-1} \circ r \circ i = \mathrm{id}$ near a, the chain rule gives $T(i^{-1} \circ r) \circ Ti = \mathrm{id}$ `Chapter2/SubMPolyfoldTangent · MPolyfold.subRetractionCLM_subInclusionCLM`. So Ti is injective, $Ti \circ T(i^{-1} \circ r)$ is a continuous projection of T_aX, and

$$T_aX = Ti(T_aA) \oplus \ker T(i^{-1} \circ r)$$

is a splitting into closed subspaces `Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_3_linear`. The projection is $Tr(a)$, so its range is $Tr(a)T_aX = Ti(T_aA)$, and $T(i^{-1} \circ r)$ is onto: this is (4) `Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_4`. At a smooth a the splitting is an sc-splitting. In a chart, the images of the two summands are the ranges of the idempotent sc-operators $L \circ Dr(a)$ and $Dr(a) - L \circ Dr(a)$, where L is the chart representative of $Tr(a)$. Both are sc-subspaces, complementary on every level of $T_{\varphi(a)}O$ `Chapter2/SubMPolyfoldTangentSc · MPolyfold.proposition_2_3_19_3_sc`. This is the text's proof, which writes $Dt(a)$ for $L \circ Dr(a)$, carried to T_aX. The chart statements of §II.10 remain.

II.40 Proposition 2.3.15 (2) (preprint 2.19): $p : TX \to X_1$ is open

Statement. For every M-polyfold, the projection $p : TX \to X^1$ is open `Chapter2/TangentProjOpen · MPolyfold.isOpenMap_tangentProj`. With the continuity of §II.8 this is (2) in full `Chap-`

ter2/TangentProjOpen · proposition_2_3_15_two. Appendix 2.7.3 reduces it to the openness of $\mathrm{pr}_1 : TO \to O_1$ and states that without proof.

The local step Chapter2/TangentProjOpen · ScRetraction.exists_isOpen_tangent_section. Let $O = r(U)$, $(o,h) \in TO$, and W open in $E_1 \times E_0$ with $(o,h) \in W$. The map $o' \mapsto Tr(o',h) = (r(o'), Dr(o')h)$ is continuous on U_1, because Tr is sc^0. At a point $o' \in O_1$ it equals $(o', Dr(o')h)$, a point of TO, because $r(o') = o'$ and $Dr(o')h \in T_{o'}O$. At o it equals (o,h), because $Dr(o)$ is the identity on T_oO. By continuity there is an open $N \ni o$ in E_1 such that every $o' \in N \cap O_1$ carries the vector $Dr(o')h$ with $(o', Dr(o')h) \in TO \cap W$. So $\mathrm{pr}_1(TO \cap W) \supseteq N \cap O_1$.

The global step. An open $S \subseteq TX$ around a class α is, in the tangent chart $T\varphi$ of a representative of α, a set $TO \cap W$. The open set N of the local step gives the basic open set $\{y \in V : \varphi(y) \in N\}$ of X^1 (§II.6), which contains $p(\alpha)$ and lies in $p(S)$. This makes the hypothesis "p open" of Chapter 7's smooth regularity, and of the results of §§7.4 and 8.1 that assume it, a theorem.

II.41 Remark 2.4.8 (preprint 2.28) and Theorem 2.4.12 (preprint 2.31), second statement: C_xO is a partial quadrant

Statement. Let r be an sc-smooth retraction with retract O in a partial quadrant $C = T^{-1}([0,\infty)^n \oplus W)$, and $x \in O$ smooth. Then T_xO and T_x^RO are sc-subspaces of E, and C_xO is a partial quadrant in the sc-Banach space T_xO Chapter2/PartialConeQuadrant · remark_2_4_8_isPartialQuadrant Chapter2/PartialConeQuadrant · remark_2_4_8_partialConeWith. Its dimension is $\dim(T_xO/T_x^RO) \le d_C(x)$. No tameness is assumed. The remark's second sentence, that in general C_xO is not a partial quadrant, therefore does not hold as we read it; the text gives no example, and by this theorem none exists. Under the hypothesis of Theorem 2.4.12 (preprint 2.31), $\dim(T_xO/T_x^RO) = d_O(x)$, and the theorem's second statement follows, without the hypothesis being needed for the quadrant. What tameness adds is only the equality of the dimension with $d_C(x)$ (§II.14).

Notation. Let $I(x) = \{i : \lambda_i(x) = 0\}$, E_x and C_x as in §II.12, and $P = Dr(x)$, so that $T_xO = PE$. Let $\Lambda : E \to \mathbb{R}^{I(x)}$, $\Lambda v = (\lambda_i v)_{i \in I(x)}$, with kernel E_x, and $\sigma : \mathbb{R}^{I(x)} \to E$, $\sigma a = T^{-1}(\tilde{a}, 0)$ with $\tilde{a}$ the extension by zero, a right inverse of Λ.

Step 1: $C_xO = P(C_x)$. The two sign arguments of §II.12, applied to r itself at $x = r(x)$, give $PE_x \subseteq E_x$ and $PC_x \subseteq C_x$ Chapter2/PartialConeQuadrant · ScRetraction.scDeriv_mapsTo_partialQuadrant. As P is the identity on T_xO, $C_xO = T_xO \cap C_x = P(C_x)$ Chapter2/PartialConeQuadrant · partialConeWith_eq_image_scDeriv.

Step 2: a nonnegative idempotent matrix. Let $M = \Lambda P \sigma$ on $\mathbb{R}^{I(x)}$ Chapter2/PartialConeQuadrant · coneMatrix. Since $v - \sigma\Lambda v \in E_x$ and P maps E_x into $\ker\Lambda$, $\Lambda P = M\Lambda$ Chapter2/PartialConeQuadrant · coordMapL_scDeriv. Hence $M^2 = \Lambda P \sigma \Lambda P \sigma = \Lambda P P \sigma = M$ Chapter2/PartialConeQuadrant · coneMatrix_idem. And M maps the orthant into itself, since σ maps it into C_x and $PC_x \subseteq C_x$ Chapter2/PartialConeQuadrant · coneMatrix_mapsTo_orthant. Moreover $\Lambda(T_xO) = \mathrm{range}\, M$, and a $v \in T_xO$ lies in C_x exactly when $\Lambda v \ge 0$.

Step 3: Flor's theorem, in the case needed. For a nonnegative idempotent matrix M, the cone $K = \mathrm{range}\, M \cap [0,\infty)^\iota$, which equals $M([0,\infty)^\iota)$ Chapter2/PartialConeQuadrant · image_orthant_eq_nonnegRange, is simplicial: it is the set of nonnegative combinations of linearly independent columns $c_1, \dots, c_s$ of M Chapter2/PartialConeQuadrant · flor_nonnegRange_simplicial. Write $B(j)$ for the support of column j. - From $M^2 = M \ge 0$, $M_{im} \ge M_{ij}M_{jm}$ Chapter2/PartialConeQuadrant · matEntry_mul_le. So $m \in B(j)$ implies $B(m) \subseteq B(j)$ Chapter2/PartialConeQuadrant · colSupp_subset. - Call j *minimal* if $B(j)$ is nonempty and min-

imal among the nonempty column supports. A minimal column has a private index, one that lies in no other minimal support except an equal one Chapter2/PartialConeQuadrant · IsMinCol.exists_private. Pick $i \in B(j)$. Then $M_{ij} = \sum_m M_{im} M_{mj} > 0$ gives $m \in B(j)$ with $i \in B(m) \subseteq B(j)$, so $B(m) = B(j)$ by minimality, and m is private. - *Generation.* Every $y \in K$ is a nonnegative combination of minimal columns, one per minimal support. The proof is by induction on the size of the support of y. From $y = My$, a coordinate $y_i > 0$ yields m with $y_m > 0$, $i \in B(m)$ and $B(m) \subseteq \operatorname{supp} y$, and then a minimal j with $B(j) \subseteq B(m)$. With $t = \min_{l \in B(j)} y_l / M_{lj}$, the vector $y - t\, c_j$ is nonnegative, lies in range M, and has smaller support Chapter2/PartialConeQuadrant · exists_nonneg_comb_coneGen. - *Independence.* Evaluating a vanishing combination at the private indices shows that the chosen columns are linearly independent Chapter2/PartialConeQuadrant · linearIndependent_coneGen.

So $\Lambda(C_x O) \cong C_x O / T_x^R O$ is a simplicial cone, at every $x \in O_1$ Chapter2/PartialConeQuadrant · remark_2_4_8_quotient_simplicial.

Step 4: the quadrant. At a smooth x, $T_x^R O$ is an sc-subspace, because the sc-projection P preserves E_x Chapter2/PartialConeQuadrant · ScRetraction.isScSubspace_reducedTangentSpaceWith. This is the argument of Proposition 2.5.6 (preprint 2.36) in §II.14, which did not use tameness. The vectors $\hat{e}_s = P\sigma(c_s) \in T_x O$ are smooth, with $\Lambda \hat{e}_s = c_s$. With a linear left inverse ρ of $t \mapsto \sum_s t_s c_s$, the functionals $\ell_s = (\rho \circ \Lambda)_s$ satisfy $\ell_s(\hat{e}_t) = \delta_{st}$, and $T_x^R O = T_x O \cap \bigcap_s \ker \ell_s$. For $v \in T_x O$: $v \in C_x \iff \Lambda v \in K \iff \ell_s(v) \geq 0$ for all s. The sc-isomorphism $T_x O \cong \mathbb{R}^s \oplus T_x^R O$ of §II.14 Chapter2/PartialCone · splitIso therefore carries $C_x O$ onto $[0,\infty)^s \oplus T_x^R O$ Chapter2/PartialCone · isPartialQuadrant_ofSubspace.

The quadrant has dimension $s = \operatorname{rank} M = \dim(T_x O / T_x^R O)$. For a tame retraction M is the identity (Proposition 2.5.3 (preprint 2.35)), so $s = d_C(x)$.

II.42 Lemma 2.4.9 (preprint 2.29): $T_x^R O$ and $C_x O$ through sc-smooth paths

Statement. Let $x \in O$ be smooth. (1) $T_x^R O$ is the closure of the set of derivatives $\dot{\alpha}(0)$ of sc-smooth paths $\alpha : (-\varepsilon, \varepsilon) \to O$ with $\alpha(0) = x$ Chapter2/TangentConePaths · GapP1Lemma249.lemma_2_4_9_reduced Chapter2/TangentConePaths · GapP1Lemma249.twoSidedPathDerivs. (2) $C_x O$ is the closure of the set of derivatives of sc-smooth paths $\alpha : [0, \varepsilon) \to O$ with $\alpha(0) = x$ Chapter2/TangentConePaths · GapP1Lemma249.lemma_2_4_9_cone Chapter2/TangentConePaths · GapP1Lemma249.oneSidedPathDerivs. (3) $T_x O = C_x O - C_x O$, at every $x \in O_1$ Chapter2/TangentConePaths · GapP1Lemma249.lemma_2_4_9_tangent. The proof is the text's.

Paths. The intervals $(-\varepsilon, \varepsilon)$ and $[0, \varepsilon)$ are sc-domains in $\mathbb{R}$ with the constant structure Chapter2/TangentConePaths · GapP1Lemma249.isScDomain_Ioo Chapter2/TangentConePaths · GapP1Lemma249.isScDomain_Ico. The derivative $\dot{\alpha}(0)$ is $D\alpha(0)1$ Chapter2/TangentConePaths · GapP1Lemma249.pathDeriv, the classical two-sided or one-sided derivative Chapter2/TangentConePaths · GapP1Lemma249.hasDerivWithinAt_pathDeriv.

Derivatives of paths lie in the cones. From $r \circ \alpha = \alpha$ the chain rule gives $Dr(x)\dot{\alpha}(0) = \dot{\alpha}(0)$, so $\dot{\alpha}(0) \in T_x O$. For $i \in I(x)$ the coordinate $\lambda_i \alpha(t) \geq 0$ vanishes at $t = 0$. For a two-sided path it has a local minimum there, so its derivative is 0, and $\dot{\alpha}(0) \in E_x$ Chapter2/TangentConePaths · GapP1Lemma249.pathDeriv_mem_minimalSubspaceWith. For a one-sided path the derivative from the right is ≥ 0, and $\dot{\alpha}(0) \in C_x$ Chapter2/TangentConePaths · GapP1Lemma249.pathDeriv_mem_partialQuadrantAtWith. Both cones are closed, so the closures lie in them.

Paths through every smooth vector. For a smooth v, the path $\alpha(t) = r(x + tv)$ is sc-smooth for t small, lies

in O, starts at x, and has $\dot\alpha(0) = Dr(x)v$ Chapter2/TangentConePaths · GapP1Lemma249.path_retract_segment. If $v \in T^R_x O$ is smooth, $x+tv$ stays in C for small $|t|$, since $v \in E_x$, and $Dr(x)v = v$. If $v \in C_x O$ is smooth, $x + tv \in C$ for small $t \geq 0$. So every smooth vector of $T^R_x O$, respectively $C_x O$, is a path derivative.

Density. The smooth vectors of $C_x O$ are dense in $C_x O$ Chapter2/TangentConePaths · GapP1Lemma249.partialConeWith_subset_closure. Write $v = Pu \in C_x O$ and let a be the $\mathbb{R}^n$-component of $T(Pu)$. The map $w' \mapsto P\,T^{-1}(a, w')$ on W_0 is continuous. It sends smooth points of W to smooth points of $C_x O$, because P is an sc-operator at the smooth point x and $T^{-1}(a, w')$ lies in C_x. It sends the W-component of $T(Pu)$ to $P^2 u = v$. Smooth points of W are dense, so v is a limit of smooth points of $C_x O$. For $T^R_x O$ the density is that of the smooth points of an sc-subspace, since $T^R_x O$ is one (§II.41) Chapter2/PartialConeQuadrant · ScRetraction.isScSubspace_reducedTangentSpaceWith. With the two previous steps this proves (1) and (2).

(3). Let $Pk \in T_x O$, and let a be the $\mathbb{R}^n$-component of Tk. Put $k^- = T^{-1}(\max(-a, 0), 0)$, coordinatewise. Then $k + k^-$ and k^- have all coordinates $\lambda_i \geq 0$, so $P(k + k^-)$ and Pk^- lie in $C_x O$ by the one-sided sign argument, and $Pk = P(k + k^-) - Pk^-$. The converse inclusion holds because $C_x O \subseteq T_x O$. Smoothness of x is not used.

II.43 The extended faces of a partial quadrant, intrinsically (§2.5 before Definition 2.5.1 (preprint §2.4 before Definition 2.33))

The text defines the extended faces of a partial quadrant $C \cong [0, \infty)^k \oplus W$ from C alone, through the components of $\{d_C = 1\}$. The library works with a presentation T, $C = T^{-1}([0, \infty)^n \oplus W)$, and the coordinate hyperplanes $f_i = \{\lambda_i = 0\}$. This subsection proves that the two agree, so the presentation enters the proofs only.

The components of $\{d_C = 1\}$. The set $\{e \in C : d_C(e) = 1\}$ Chapter2/ExtendedFaces · GapP1ExtendedFaces.stratumOne is the disjoint union of the sets $A_i = \{\lambda_i = 0,\ \lambda_j > 0 \text{ for } j \neq i\}$ Chapter2/ExtendedFaces · GapP1ExtendedFaces.faceOpen. Each A_i is convex, hence connected Chapter2/ExtendedFaces · GapP1ExtendedFaces.convex_faceOpen, and relatively open in $\{d_C = 1\}$, since the conditions $\lambda_j > 0$ are open. So the connected component of a point of A_i is A_i Chapter2/ExtendedFaces · GapP1ExtendedFaces.connectedComponentIn_eq, and the components are exactly the A_i Chapter2/ExtendedFaces · GapP1ExtendedFaces.faceComponents_eq. The A_i are distinct and nonempty: the point with $\lambda_i = 0$, the other coordinates equal to 1 and W-component 0 lies in A_i only Chapter2/ExtendedFaces · GapP1ExtendedFaces.facePoint_mem. So there are exactly n components Chapter2/ExtendedFaces · GapP1ExtendedFaces.ncard_faceComponents.

Extended faces and half-spaces. The span of A_i is f_i, since A_i is a nonempty relatively open subset of f_i Chapter2/ExtendedFaces · GapP1ExtendedFaces.span_faceOpen. So the extended faces are the f_i Chapter2/ExtendedFaces · GapP1ExtendedFaces.extendedFaces_eq, and there are exactly n of them Chapter2/ExtendedFaces · GapP1ExtendedFaces.ncard_extendedFaces. The closed half-space containing C with boundary f_i is $C + f_i = \{\lambda_i \geq 0\}$ Chapter2/ExtendedFaces · GapP1ExtendedFaces.add_coordFace Chapter2/ExtendedFaces · GapP1ExtendedFaces.isClosed_add_coordFace. For $e \in C$, the faces containing e are the f_i with $\lambda_i(e) = 0$, so $d_C(e) = \#\mathcal{F}(e)$ Chapter2/ExtendedFaces · GapP1ExtendedFaces.ncard_facesAt. And $\bigcap_{f \in \mathcal{F}(e)} (C + f) = C_e$, the partial quadrant of Definition 2.5.1 (preprint 2.33) as the library defines it Chapter2/ExtendedFaces · GapP1ExtendedFaces.iInter_facesAt.

II.44 Definition 2.5.8 (preprint 2.38): $T^R_x X$ and $C_x X$ inside $T_x X$

Statement. For a chart φ around $x \in X_1$ and a presentation of its quadrant, $T^R_x X$ and $C_x X$ are the preimages of $T^R_{\varphi(x)} O$ and $C_{\varphi(x)} O$ under $T\varphi : T_x X \to T_{\varphi(x)} O$ `Chapter2/ReducedTangentIntrinsic · MPolyfold.fiberCoord` `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber` `Chapter2/ReducedTangentIntrinsic · MPolyfold.partialConeFiber`. They do not depend on the chart or the presentation `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber_eq` `Chapter2/ReducedTangentIntrinsic · MPolyfold.partialConeFiber_eq`. At a smooth x, $C_x X$ is a partial quadrant in $T_x X$, and $d_{C_x X}(0_x) = \dim(T_x X / T^R_x X)$. In a tame chart both equal $d_X(x)$ `Chapter2/TangentConeQuadrant · MPolyfold.definition_2_5_8_degIndex`. The sc-structure of $T_x X$ and the quadrant do not depend on the chart `Chapter2/TangentSpaceChartIndependent · MPolyfold.definition_2_5_8_scStructure`.

Chart independence. For two charts, the transition maps form a germ of sc-diffeomorphisms between the overlap retracts `Chapter2/ReducedTangentIntrinsic · MPolyfold.exists_isDiffeoGerm_transition`. Its derivative at $\varphi(x)$ is the fibre transition of §II.39. Proposition 2.4.10 (preprint 2.30) (§II.13) says that this derivative maps T^R onto T^R and C onto C. So the preimages agree. This holds at every point on level 1, not only at smooth points. The quotient dimension in a tame chart at a smooth point is $d_X(x)$ by Proposition 2.5.6 (preprint 2.36) `Chapter2/ReducedTangentIntrinsic · MPolyfold.finrank_quotient_reducedTangentFiber_eq_degIndex`.

The quadrant and its degree. For a partial quadrant C in an sc-Banach space F, $d_C(0) = \dim(F_0/(C \cap -C))$ `Chapter2/TangentConeQuadrant · degIndex_zero_eq_finrank_quotient`. The degree at the apex therefore depends only on the set. Inside $T_x X$, $T^R_x X = C_x X \cap -C_x X$ `Chapter2/TangentConeQuadrant · MPolyfold.mem_reducedTangentFiber_iff_lineality`, because $E_x = C_x \cap -C_x$ `Chapter2/TangentConeQuadrant · mem_minimalSubspaceWith_iff_lineality`. At a smooth x, in any chart, tame or not, the image of $C_x X$ in $T_{\varphi(x)} O$ `Chapter2/TangentConeQuadrant · MPolyfold.partialConeFiberModel` is a partial quadrant, by §II.41 `Chapter2/TangentConeQuadrant · MPolyfold.isPartialQuadrant_partialConeFiberModel`. Its degree at 0 is $\dim(T_x X / T^R_x X)$ `Chapter2/TangentConeQuadrant · MPolyfold.degIndex_partialConeFiberModel_eq_finrank`.

The sc-structure through two charts. An sc-operator with an sc-operator inverse between sc-subspaces restricts to an sc-isomorphism of them `Chapter2/TangentSpaceChartIndependent · ScBanach.subspaceIso`. Applied to the fibre transition and its inverse, which preserve levels at a smooth point (§II.39), this gives an sc-isomorphism $T_{\varphi(x)} O \cong T_{\psi(x)} O'$ `Chapter2/TangentSpaceChartIndependent · MPolyfold.tangentFiberScIso`. It carries $T\varphi(\alpha)$ to $T\psi(\alpha)$ `Chapter2/TangentSpaceChartIndependent · MPolyfold.tangentFiberScIso_comp` and the model of $C_x X$ in φ onto the model in ψ `Chapter2/TangentSpaceChartIndependent · MPolyfold.tangentFiberScIso_image_partialConeFiberModel`. So the identity of $T_x X$ is an sc-isomorphism between the two transported structures, and it preserves $C_x X$.

II.45 Strong bundles: atlas equivalence, restriction, the maximal atlas, isomorphisms and $\mathcal{SBR}$ (Definitions 2.3.3, 2.6.1, 2.6.3–2.6.5 (preprint 2.12, 2.47, 2.49–2.51))

Atlas equivalence is an equivalence relation. The text speaks of "an equivalence class" of atlases, for M-polyfolds (Definition 2.3.3 (preprint 2.12)) and for strong bundles (Definition 2.6.5 (preprint 2.51)), without checking transitivity. It holds by the locality of compatibility (§II.1). If $\mathcal{A} \sim \mathcal{A}'$ and $\mathcal{A}' \sim \mathcal{A}''$, a chart of $\mathcal{A}$ and a chart of $\mathcal{A}''$ are compatible near each point of their overlap, through a chart of $\mathcal{A}'$ there, and so they are compatible `Chapter2/StrongBundleRestrict · MAtlas.Equiv.`

trans Chapter2/StrongBundleRestrict · MAtlas.equivalence. For strong bundles the same argument runs on $[0]$ and on $[1]$. The condition that the $[1]$-transition preserves fibre components in F_1 (§II.21) passes through the factorization Chapter2/StrongBundleRestrict · StrongBundleChart.compatible_of_locally Chapter2/StrongBundleRestrict · StrongBundleAtlas.equiv_iff Chapter2/StrongBundleRestrict · StrongBundleAtlas.Equiv.trans Chapter2/StrongBundleRestrict · StrongBundleAtlas.equivalence.

Restriction of a strong bundle chart. Let s be open in the domain of the base chart φ, with $\varphi(s) = O \cap W$ for an open W. The chart restricts over $\varphi|_s$, with the strong bundle retraction restricted to $(U \cap r^{-1}(W)) \lhd F$ Chapter2/StrongBundleRestrict · IsStrongBundleRetraction.restrict Chapter2/StrongBundleRestrict · StrongBundleChart.restrict. Its retract is $\{(u,h) \in K : u \in \varphi(s)\}$ Chapter2/StrongBundleRestrict · StrongBundleChart.restrict_K, and its domain $\{u \in U : r(u) \in \varphi(s)\}$ does not depend on W Chapter2/StrongBundleRestrict · StrongBundleChart.restrict_U. Compatibility passes to restrictions Chapter2/StrongBundleRestrict · StrongBundleChart.Compatible.restrict. The overlaps are $\Phi(P^{-1}(t)) = K \cap (W \times F)$ Chapter2/StrongBundleRestrict · StrongBundleChart.image_preimage_eq, open in K through the base chart, so continuity of P is never used.

The maximal atlas. The later chapters speak of "a strong bundle chart of P". Two such charts are compatible Chapter3/StrongBundleMaximalAtlas · StrongBundle.IsChart.compatible. All of them form an atlas equivalent to the given one Chapter3/StrongBundleMaximalAtlas · StrongBundle.maximalAtlas Chapter3/StrongBundleMaximalAtlas · StrongBundle.maximalAtlas_equiv. A chart of P is a chart of some atlas of the class Chapter3/StrongBundleMaximalAtlas · StrongBundle.isChart_iff_exists_equiv, and the charts depend only on the class Chapter3/StrongBundleMaximalAtlas · StrongBundle.isChart_congr. Restrictions of charts of P are charts of P Chapter3/StrongBundleMaximalAtlas · StrongBundle.IsChart.restrict.

Strong bundle isomorphisms and $\mathcal{SBR}$ (Definition 2.6.1 (preprint 2.47), pp. 66–67). Strong bundle isomorphisms between the $U \lhd F$ contain the identity and are closed under inverse and composition Chapter3/StrongBundleIsoSBR · IsStrongBundleIso.refl Chapter3/StrongBundleIsoSBR · IsStrongBundleIso.symm Chapter3/StrongBundleIsoSBR · IsStrongBundleIso.trans. In the text's wording, they are the invertible strong bundle maps with a strong bundle inverse Chapter3/StrongBundleIsoSBR · exists_isStrongBundleIso_iff. The category $\mathcal{SBR}$ Chapter3/StrongBundleIsoSBR · SBR has as objects the local strong bundle retracts $(K, C \lhd F, E \lhd F)$. Its morphisms are the maps $K \to K'$ that extend to a map $\Phi : E \lhd F \to E' \lhd F'$ which is linear on the fibres, preserves the double filtration, and has $\Phi[0]$ and $\Phi[1]$ sc-smooth between the retracts Chapter3/StrongBundleIsoSBR · SBR.IsHom Chapter3/StrongBundleIsoSBR · SBR.Hom. These are closed under identity and composition Chapter3/StrongBundleIsoSBR · SBR.IsHom.id Chapter3/StrongBundleIsoSBR · SBR.IsHom.comp. Equality of morphisms is equality on K, so the category laws are those of maps. A strong bundle map of Definition 2.6.1 is a morphism Chapter3/StrongBundleIsoSBR · IsStrongBundleMap.isHom, and a strong bundle isomorphism is an isomorphism Chapter3/StrongBundleIsoSBR · IsStrongBundleIso.toIso Chapter3/StrongBundleIsoSBR · SBR.isoOfInverse. The forgetful functors forget$[i] : K \mapsto K[i]$, $\Phi \mapsto \Phi[i]$ Chapter3/StrongBundleIsoSBR · SBR.forget0 Chapter3/StrongBundleIsoSBR · SBR.forget1 land in the category of sc-smooth retracts Chapter3/StrongBundleIsoSBR · ScRetr. The text has them land in $\mathcal{BR}$, which is only named (§II.32), so $K[i]$ is recorded without its bundle structure.

II.46 Lemma 2.7.3 and Proposition 2.7.4 (preprint 2.56 and 2.57): closed locally finite refinements

Statement. A regular Hausdorff space is paracompact if and only if every open cover has a locally finite refinement by closed sets Chapter2/MichaelRefinement · GapP1Michael.lemma_2_7_3. A regular Hausdorff

space covered by a locally finite family of closed paracompact subspaces is paracompact Chapter2/MichaelRefinement · GapP1Michael.proposition_2_7_4. Both hold as printed, with Mathlib's paracompactness (locally finite open refinements).

Paracompact implies closed refinements Chapter2/MichaelRefinement · GapP1Michael.exists_closed_refinement. The text takes the closures of a locally finite open refinement. Such a closure need not lie in a member of the cover, so first shrink: by regularity each x has an open W_x with $\overline{W_x}$ inside a member $U_{i(x)}$. Take a locally finite open refinement (V_j) of (W_x). Its closures are locally finite and each lies in some $\overline{W_x} \subseteq U_{i(x)}$.

Closed refinements imply paracompact Chapter2/MichaelRefinement · GapP1Michael.paracompactSpace_of_closed_refinement. This direction holds in every topological space, without separation. Let (U_i) be an open cover. Take a locally finite closed refinement (A_b), with $A_b \subseteq U_{i(b)}$. For each x let $B(x)$ be the complement of the union of the A_b not containing x (open, by local finiteness), intersected with the $U_{i(b)}$ for the finitely many $A_b \ni x$. Take a locally finite closed refinement (C_k) of $(B(x))_x$, and put $A'_b = X \setminus \bigcup\{C_k : C_k \cap A_b = \emptyset\}$. Then A'_b is open, contains A_b, and $A'_b \cap U_{i(b)}$ is an open refinement. It is locally finite: a neighbourhood of x meets only finitely many C_k, and each $C_k \subseteq B(x_k)$ meets only the A_b with $x_k \in A_b$, finitely many.

Proposition 2.7.4. Given the closed paracompact Y_i and an open cover, refine the cover on each Y_i by a locally finite closed family (C^i_b), using the lemma in the subspace Y_i. The family of all C^i_b, indexed by pairs (i, b) instead of the text's unions $\bigcup_i C^i_b$, is a locally finite closed refinement of the cover, and the lemma applies.

The library's proof of Theorem 2.3.10 (preprint 2.16) still goes around these results (§II.7), since the text applies Proposition 2.7.4 to X_m without checking that X_m is regular.

II.47 The 2021 statements in book form: Remark 2.1.6, Definitions 2.5.18–2.5.19, Proposition 2.7.12, Theorem 2.7.13

For four 2021 statements, the earlier subsections proved the parts separately. Each is now one theorem in the book's shape.

Remark 2.1.6. The closure identity, the implication from local completeness to the modified notion, and its strictness (the K-topology, §II.31) together Chapter2/BookFormLocalCompleteness2021 · remark_2_1_6_book. The text does not say whether "weaker" means strictly weaker, so both readings are included.

Definitions 2.5.18–2.5.19 with (3) in T_yX. With the linear structure of §II.39 and T^R_yX of §II.44, condition (3) can be stated as the text states it. $T_yY \subseteq T_yX$ is read as the image of Ti Chapter2/BookFormTameSub2021 · MPolyfold.subTangentRange. In a chart φ of X and the sub-chart of Y built on it, $T\varphi \circ Ti$ is the chart map of the sub-chart Chapter2/BookFormTameSub2021 · MPolyfold.fiberCoord_subInclusionCLM. So $T\varphi$ maps T_yY onto the tangent space of the chart image of Y Chapter2/BookFormTameSub2021 · MPolyfold.coe_map_range_subInclusionCLM. Condition (3) in T_yX asks for a subspace $A \subseteq T^R_yX$ with $T_yX = T_yY \oplus A$ whose chart image is an sc-subspace complementary on every level Chapter2/BookFormTameSub2021 · MPolyfold.HasReducedScComplement. It is equivalent to the chart form of §II.25 Chapter2/BookFormTameSub2021 · MPolyfold.hasReducedScComplement_iff. Definition 2.5.19 with this (3) Chapter2/BookFormTameSub2021 · MPolyfold.TameSubData2021 Chapter2/BookFormTameSub2021 · MPolyfold.IsTameSubMPolyfold2021 is therefore the chart definition Chapter2/BookFormTameSub2021 · MPolyfold.isTameSubMPolyfold2021_iff. It implies the tame induced structure of Definition 2.5.18, and $Y = X$ satisfies it Chapter2/BookFormTameSub2021 · MPolyfold.definition_2_

`5_19_book`. Through the equivalence, the two examples of §II.35 apply to it.

Proposition 2.7.12 and Theorem 2.7.13. Proposition 2.7.12 as printed, for every pair of representatives `Chapter2/BookFormTameBoundary2021 · MPolyfold.proposition_2_7_12_book`. Theorem 2.7.13 (1)–(4) with the characterization, the uniqueness of topology and structure, and range $B = \partial X$ of Definition 2.7.14, in one statement `Chapter2/BookFormTameBoundary2021 · MPolyfold.theorem_2_7_13_book`. In the characterization the image of $\Phi_{(x,V,\theta)}$ is $\theta \cap V^x$, which is θ for centred V, as described in the opening of this part.

II.48 §2.2 in M-polyfolds: smooth points of connected open sets are path-connected

Lemma 2.2.1 proves that the levels U_k, $k \leq \infty$, of a connected relatively open $U \subseteq C$ are path-connected. Chapters 6 and 7 use the M-polyfold version without proof: for $U \subseteq X$ open and connected, $U \cap X_\infty$ is connected. It is proved here in two steps. - *Local.* Every $u \in U$ has an open neighbourhood $N \subseteq U$ in which any two smooth points are joined in $U \cap X_\infty$ by a path continuous for the topology of X_∞ `Chapter2/SmoothPaths · MPolyfold.exists_nhds_joinedIn_smooth`. The path is the retracted segment $t \mapsto \varphi^{-1}(r(\varphi(x) + \rho(t)(\varphi(y) - \varphi(x))))$ in a small ball, which is an sc-smooth path. - *Global.* For U open and connected, $U \cap X_\infty$ is path-connected in X_∞ `Chapter2/SmoothPaths · MPolyfold.isPathConnected_smoothVal_preimage`. Fix a smooth $x_0 \in U$. The set of points of U all of whose nearby smooth points are joined to x_0 is open. By the local step and the density of X_∞, it is also closed in U, so it is U.

Part III. Sc-Fredholm theory and the implicit function theorem (HWZ Chapter 3, §§3.1–3.8)

Chapter 3 is the reason for the first two chapters: it defines the sections of strong bundles to which an implicit function theorem applies, the *sc-Fredholm sections*, and proves that theorem. The engine is a fixed point problem with finitely many parameters, $w = B(a, w)$, $a \in \widetilde{C} \subseteq \mathbb{R}^n$, $w \in W$, where $B(a, \cdot)$ contracts with an arbitrarily small constant near 0 *on every level* W_m of the scale. Its solution $w = \delta(a)$ is the *solution germ*; the germ implicit function theorem (Theorem 3.3.6 (preprint 3.33)) says that δ is as regular as B. Around it the chapter builds: germs and their calculus (§III.1), sections of strong bundles in charts (§III.2), fillers (§§III.3–III.4), basic germs and their linearization (§III.5), sc-Fredholm germs and sections with the stability theorem 3.15 (§III.6); the germ implicit function theorem itself (§§III.7–III.11); local regularity and compactness, Theorems 3.5.1 and 3.1.13 (preprint 3.40 and 3.12) (§§III.12–III.13); the finite-dimensional geometry of good position and the inverse and implicit function theorems in partial quadrants, which the formalization proved first and the rest uses (§§III.14–III.17); the basic class and local solvability (§§III.18–III.20); the linearization of an sc-Fredholm section at a zero, good position, and the graph description of the solution set of Theorem 3.6.8 (preprint 3.54) (§§III.21–III.23); the tangent germ property, its invariance, and the manifold structure of solution sets (§III.24); and Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61) of the basic germ criterion (§III.25). Sections III.27–III.39 were added after the part was first assembled, for the Lean written since: tameness under sc-diffeomorphisms (§III.27), Corollary 3.4.3 (preprint 3.39) (§III.28), good complements whose projection maps the quadrant into itself and local solvability with such a complement (§III.29), the index in Proposition 3.6.5 (preprint 3.52) (§III.30), the linearization at a zero in charts and its chart independence (§III.31), Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros (§§III.32–III.33), Theorem 3.6.8 (2) (preprint 3.54) (§§III.34–III.36), the implicit function theorems 3.54, 3.23, 3.19, 3.24, 3.56, 3.57 assembled (§III.37), the basic germ criterion, Theorem 3.7.1 (preprint 3.58) (§III.38), and what this addendum taught us (§III.39). Sections III.40–III.46 were added in a second addendum: Theorem 3.1.21 (preprint 3.18) (§III.40); Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses, by an implicit function theorem at the nearby zero (§§III.41–III.43); the dimension of the solution set as the Fredholm index and its local constancy (§III.44); the kernel as tangent space, good position at interior zeros, and the filler condition of §III.33 as a theorem (§III.45); and what the second addendum taught us (§III.46). Section III.47 repairs a tameness hypothesis, and §III.48 records the switch of Definition 3.1.11 (preprint 3.11) to the text's germ reading; the passages above that use the filling or the sc$^+$-section have been rewritten for it. Section III.49 proves Definition 6.2.1 and Proposition 6.2.2 (preprint 6.1 and 6.2), and §§III.50–III.55 compare the part with the 2021 edition. Sections III.56–III.66

were added with the waves of November 2026, which closed the remaining gaps of the chapter: Theorem 3.2.3 (preprint 3.27) under the tangent germ property alone (§III.56); the implicit function theorems with the 2021 definitions (§III.57); Remark 3.3.8 (preprint 3.36) (§III.58); the extreme-ray lemmas of the appendix (§III.59); Definition 3.8.14, Lemmas 3.8.20 and 3.8.21 (preprint 3.71, 3.76, 3.77) (§III.60); Lemma 3.8.13 (preprint 3.70) in a general presentation and $d_M = d_X$ (§III.61); the intrinsic linearization and $\ker f'(y) = T_yS$ (§III.62); Theorem 3.7.1 (preprint 3.58) for a general sc-splitting (§III.63); an sc-smooth family of projections whose rank jumps (§III.64); Theorem 3.6.8 (preprint 3.54) with untamed 2021 germ data, a counterexample (§III.65); and what is left and what was learnt (§III.66).

What is proved, and what is not. All of Chapter 3 is now formalized. The sixteen gaps the comparison listed for the chapter (G.16–G.30 and G.79) are closed, G.79 by a counterexample, and no hypothesis remains that the text neither states nor takes in its proof.

The germ implicit function theorem. It is proved with the text's hypotheses (Theorems 3.3.2–3.3.7 (preprint 3.29–3.34) and Theorem 3.35 (preprint; not in the 2021 edition)), and in the quantitative form of Remark 3.3.8 (preprint 3.36) (§III.58, G.19). What the formalization adds is precision about *where* the solution germ is regular. A germ is formalized as a fixed sequence of sets, and δ is sc^k on the germ shifted by $s_k = 2^k - 1$ levels. So "if f is sc-smooth so is its solution germ" holds for every k on a germ depending on k (§III.11).

Germs, sections and their linearization. - Definition 3.1.11 (preprint 3.11) is formalized as the text writes it, with the filling and the sc^+-section sc-smooth *germs* (§§III.6, III.48). - Proposition 3.1.10 (preprint 3.10) holds on every level (§III.5, G.21). - Proposition 3.6.5 (preprint 3.52) gives an sc-Fredholm operator in every chart (§§III.30, III.57, G.22). - Corollary 3.4.3 (preprint 3.39) is §III.28 (G.20), and Theorem 3.1.21 (preprint 3.18) is §III.40 (G.16).

Theorem 3.2.3 (preprint 3.27) is proved under the tangent germ property alone (§III.56). The two regularity conditions with which it was first proved (§III.24) are theorems: re-graphing over a projecting good complement gives the one, re-centring the other. The closedness of the good complement in Definition 3.2.1 (preprint 3.25) and Lemma 3.6.9 (preprint 3.55), once listed as an added hypothesis, follows from the text's sc-complement. Two steps of the text's proof do not hold as we read it, each with a Lean counterexample (§III.56).

The implicit function theorems. Theorem 3.6.8 (preprint 3.54) is proved with the text's hypotheses (§§III.41–III.43, G.23). With it come Theorems 3.1.22 and 3.1.26 (preprint 3.19 and 3.23), Corollary 3.1.27 (preprint 3.24) and Theorems 3.6.10 and 3.6.11 (preprint 3.56 and 3.57) (§§III.37, III.43, G.17, G.18). The dimension of the solution set is the Fredholm index (§III.44), and $\ker f'(y) = T_yS$ holds in T_yX for the intrinsic linearization (§III.62, G.30). All of these are also stated in the book's form with the 2021 definitions (§III.57).

"*Tame strong bundle*". The phrase is read as germ data and good position in a tame chart (§III.47). This is the wording of the 2017 text and the preprint, and the chart the 2021 proof takes on p. 137; it is not derived from the 2021 text as printed. The reading is needed: with the 2021 germ data as printed, in any compatible chart, Theorem 3.6.8 does not hold as we read it. The witness is a tame strong bundle over $\mathbb{R}$ whose solution set near a zero in good position is a closed half-line (§III.65, G.79). The reading is satisfiable: the graph section of a smooth map of a trivial bundle has tame germ data (§III.47).

The basic germ criterion, Theorem 3.7.1 (preprint 3.58), is proved as printed, for a general sc-splitting (§III.63, G.24). The proof reduces it to the coordinates of the text's proof (§III.38).

The appendix. - Lemmas 3.8.1–3.8.3 and 3.8.9–3.8.11 (preprint 3.62–3.64 and 3.66–3.68) are §III.59 (G.25, G.26). - Definition 3.8.14 (preprint 3.71) agrees with the C^k the formalization uses (§III.60, G.27). - Lemma 3.8.20 (preprint 3.76) is proved with Lemma 3.8.21 (preprint 3.77) in the general form its use needs (§III.60, G.28). - Lemma 3.8.13 (preprint 3.70) and Proposition 3.8.12 (preprint 3.69) hold for a general presentation of the partial quadrant, which gives $d_M = d_X$ in Theorem 3.2.3 (§III.61, G.29).

From Chapter 6. Definition 6.2.1 and Proposition 6.2.2 (preprint 6.1 and 6.2) are proved with the text's hypotheses for paths on $\mathbb{R}$ (§III.49).

The four narrower rows. Four comparison rows are narrower than the text. In each, the printed statement does not hold as we read it, and a Lean counterexample shows it: - V3.73: the re-centring step in the proof of Theorem 3.2.3, for every pair of data. The lemma is proved with the first graph map C^1 on its whole domain (§§III.35, III.56). - V3.120: Lemma 3.8.6 (2) at $N = 0$ (§III.54). - V3.123: Lemma 3.8.1 at $P = \{0\}$ (§III.59). - V3.130: Theorem 3.6.8 with the 2021 germ data as printed (§III.65).

Seven structural choices. (1) Germs are fixed sequences of sets, and the calculus the text uses on them silently is proved (§III.1). (2) Contraction germs carry explicit contraction regions instead of the text's balls, because the regions of the iterated germ $f^{(1)}$ must be unbounded in the tangent directions (§III.7). (3) "A strong bundle chart" in the definitions of §3.1 is a chart *compatible* with the atlas (§III.2). (4) Tameness, which the text assumes throughout, is used only in Theorem 3.6.8 (preprint 3.54), where it cannot be dropped (§III.65), and only for the chart of the germ data at the zero: through Proposition 2.5.3 (preprint 2.35), $(1 - Dr(0))E \subseteq E_0$, and through the tameness of the model of the chart $\psi \circ \varphi$ (§III.27). The theorems assume germ data and good position in tame charts, as Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) do; the first formalization assumed every compatible chart tame, which cannot be met (§III.47). (5) The coordinate change of Theorem 3.4.2 (preprint 3.38) is linear, which makes the transport of solution sets in §3.5 linear algebra. (6) Classical regularity in the appendix is Mathlib's C^k within a set, and the iterations of the text's Lemmas 3.8.20 and 3.8.26 (preprint 3.76 and 3.82) are replaced by bootstrapping a derivative formula; the text's Definition 3.8.14 (preprint 3.71) is shown to be this notion, and Lemma 3.8.20 and Lemma 3.8.21 in the general form its use needs are proved as well (§III.60). (7) In Definition 3.1.11 (preprint 3.11) the filling g and the sc$^+$-section s are germs on germs of neighbourhoods of their own, and the basic germ is shrunk to the germ G'' of the points that ψ^{-1} sends into both, so that the nearby zeros produced by the implicit function theorem are points where g and s are sc^1 (§§III.6, III.48).

Map.

- `Chapter3/Germ`, `Chapter3/GermLevels`: Definitions 3.1.1, 3.1.2 (preprint 3.1, 3.2); the calculus of germs; Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs.
- `Chapter3/StrongBundleSections`: the section algebra, compatible charts, Definitions 3.1.3, 3.1.6, 3.1.11, 3.1.16 (preprint 3.3, 3.6, 3.11, 3.14), Theorem 3.1.18 (preprint 3.15).
- `Chapter3/Filler`, `Chapter3/FillerLinearAlgebra`, `Chapter3/FillerLinearization`: Definition 3.1.4 (preprint 3.4), Lemmas 3.1.19, 3.1.20 (preprint 3.16, 3.17), Proposition 3.1.5 (preprint 3.5), Remark 3.6.2 (preprint 3.49).
- `Chapter3/BasicGermLinearization`: Lemma 3.1.9 (preprint 3.9) at 0, Proposition 3.1.10 (preprint 3.10).
- `Chapter3/ContractionGerm`, `Chapter3/GermIFT`: Definition 3.3.1 (preprint 3.28), Theorems 3.3.2, 3.3.3 (preprint 3.29, 3.30), Lemma 3.3.4 (preprint 3.31), the induction (Lemma 3.3.5 (preprint 3.32), Theorem 3.3.6 (preprint 3.33)), Corollary 3.3.7 (preprint 3.34), Theorem 3.35 (preprint; not in the 2021 edition).

- `Chapter3/LocalCompactness`, `Chapter3/SectionCompactness`: the parametrized fixed point theorem, Definition 3.1.7 (preprint 3.7), Theorem 3.5.1 (preprint 3.40), Theorem 3.1.13 (preprint 3.12), Corollary 3.1.15 (preprint 3.13).
- `Chapter3/GoodPosition`, `Chapter3/GoodPositionGraph`: Definition 3.1.24 (preprint 3.21), Proposition 3.1.25 (preprint 3.22), Lemmas 3.5.9, 3.8.8 (preprint 3.45, 3.65), Proposition 3.8.12 (preprint 3.69), Lemma 3.8.13 (preprint 3.70).
- `Chapter3/QuadrantInverse`, `Chapter3/QuadrantIFT`: Theorems 3.8.15, 3.8.22 (preprint 3.72, 3.78) with their lemmas.
- `Chapter3/BasicClass`, `Chapter3/SolutionLinearization`, `Chapter3/LocalSolvability`: Theorem 3.4.2 (preprint 3.38), Corollary 3.5.3 (preprint 3.41), Lemmas 3.5.5, 3.5.8, 3.5.10 (preprint 3.42, 3.44, 3.46), Theorem 3.5.6 (preprint 3.43), Theorem 3.5.11 (preprint Corollary 3.47).
- `Chapter3/FredholmGermLinearization`, `Chapter3/GoodPositionGerm`, `Chapter3/GoodPositionOpen`, `Chapter3/GlobalIFT`: Proposition 3.6.5 (preprint 3.52), Definition 3.6.6 (preprint 3.53), Lemma 3.6.9 (preprint 3.55), Theorem 3.6.8 (preprint 3.54) (the linear part and the graph description), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57) from the tangent germ property.
- `Chapter3/TangentGerm`: Definition 3.2.1 (preprint 3.25), Proposition 3.2.2 (preprint 3.26), Theorem 3.2.3 (preprint 3.27).
- `Chapter3/BasicGermCriterion`, `Chapter3/InverseFamily`: Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) (abstract form), Lemma 3.7.4 (preprint 3.61).
- Added later (§§III.27–III.39): `Chapter2/TameInvariance` (tameness under sc-diffeomorphisms); `Chapter3/BasicClassComposition` (Corollary 3.4.3 (preprint 3.39), the calculus of strong bundle isomorphisms); `Chapter3/GoodComplementProjection`, `Chapter3/LocalSolvabilityClosed` (good complements with a projection, Theorems 3.5.6, 3.5.11 (preprint 3.43, 3.47) with a closed projecting complement); `Chapter3/FredholmGermIndex` (Proposition 3.6.5 (preprint 3.52) with its index); `Chapter3/LinearizationCharts` (the linearization at a zero in charts, chart independence, Definition 3.6.6 (preprint 3.53) from a non-centred chart); `Chapter3/GlobalIFTFull`, `Chapter3/GlobalIFTNearbyGraph`, `Chapter3/GlobalIFTNearbyKernel`, `Chapter3/GlobalIFTNearbySurj`, `Chapter3/GlobalIFTNearbyPerturb` (Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros); `Chapter3/GlobalIFTCenter`, `Chapter3/SolutionGermRegularity` (Theorem 3.6.8 (2) (preprint 3.54) at the centre); `Chapter3/SolutionGermRecentre` (the re-centring, regular data, Proposition 3.8.17 (preprint 3.73) / Lemma 3.8.21 (preprint 3.77) with weak hypotheses); `Chapter3/GlobalIFTTheorems` (Theorem 3.2.3 (preprint 3.27) with locally constant dimension, Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19), Corollary 3.1.27 (preprint 3.24), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57)); `Chapter3/BasicGermCriterionProof`, `Chapter3/BasicGermCriterionIndex` (Theorem 3.7.1 (preprint 3.58)).
- Added in the second addendum (§§III.40–III.46): `Chapter3/ParamPerturbation` (Theorem 3.1.21 (preprint 3.18), the pull-back bundle, inert parameters, the linearization of the perturbed section); `Chapter3/GlobalIFTNearbyOntoLemmas`, `Chapter3/GlobalIFTNearbyOntoGerm`, `Chapter3/GlobalIFTNearbyOntoXi`, `Chapter3/GlobalIFTNearbyOntoCurve`, `Chapter3/GlobalIFTNearbyOnto` (Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses, Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19) with them); `Chapter3/GlobalIFTIndex` (the index of the linearization in every chart, dimension = index, the index locally constant, the kernel as tangent space, good position at interior zeros, the filler condition as a theorem).
- Added with the repair of the tameness hypothesis (§III.47): `Chapter3/TameGerms` (tame charts, Definitions 3.6.3, 3.6.4, 3.6.6 (preprint 3.50, 3.51, 3.53) in a tame chart, good position read in a tame chart, Theorem 3.1.18 (preprint 3.15) for tame germ data); `Chapter3/TameVacuity` (a non-tame retraction

onto a product with a point).

- Added with the germ reading of Definition 3.1.11 (preprint 3.11) (§III.48): `Chapter3/GermVsChart` (a germ sc-smooth on no neighbourhood); `Chapter3/GermFilling`, `Chapter3/GermFillingChart` (Definitions 3.1.6 and 3.1.11 (preprint 3.6 and 3.11) on germs, the bridges to the whole-chart notions); `Chapter3/GermCalculus`, `Chapter3/GermCalculusParam`, `Chapter3/GermPushforward`, `Chapter3/GermFillerLinearization` (sums, chain rules with whole-chart maps, the fill defect, push-forwards and products of germs, Proposition 3.1.5 (preprint 3.5) for filling germs); `Chapter3/BasicGermShrink`, `Chapter3/BasicGermShrinkIFT` (the shrunk basic germ G'' and Corollary 3.5.3 (preprint 3.41) on it); and the files `Chapter3/…G` (`FredholmSectionsG`, `TameGermsG`, `GoodPositionGermG`, `FredholmGermLinearizationG`, `GlobalIFTNearbyGraphG`, `GlobalIFTTheoremsG`, …), where the canonical names of Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4, 3.6.6 (preprint 3.11, 3.14, 3.50, 3.51, 3.53) and of the theorems built on them now live. Their original files (`StrongBundleSections`, `GlobalIFT`, …) keep the whole-chart versions, named `…WholeChart` / `…_wholeChart`.
- Added in the fourth addendum (§§III.47, III.49): `Chapter3/TameGermData` (condition (3) of Definition 3.1.4 (preprint 3.4) as a condition on the chart; a tame atlas whose chart carries no germ data); `Chapter3/IndexAlongPaths`, `Chapter3/IndexAlongPathsDevice` (Definition 6.2.1 (preprint 6.1), bump functions on $\mathbb{R}^n \times X$, the section along a path, the slice of a section of the pull-back, Proposition 6.2.2 (preprint 6.2)).
- Added in the fifth addendum (§III.49): `Chapter3/RestrictOpen` (the restriction of M-polyfolds and strong bundles to open sets); `Chapter3/RestrictOpenGerms` (germ data and the index on the restriction); `Chapter3/IndexLocallyConstantLocal` (sc$^+$-sections through the fibre points from a chart over the whole base, the index along the zero set without bump functions); `Chapter3/IndexAlongPathsLocal` (Definition 6.2.1 (preprint 6.1) with local sections, Proposition 6.2.2 (preprint 6.2) with the text's hypotheses); `Chapter3/IndexAlongPathsLocalBump` (the two readings of Definition 6.2.1 (preprint 6.1) agree under bump functions).
- Added with the 2021 edition (§§III.50–III.55): `Chapter3/Book2021Germs` (the 2021 filling and germs, the corner filling, the step on p. 138, Theorem 3.5.1 (preprint 3.40) with the sequential conclusion); `Chapter3/Book2021Items` (cones, Lemmas 3.8.4–3.8.6, Corollary 3.8.7); `Chapter3/Counterexamples2021` (the set Σ of Theorem 3.2.3 (preprint 3.27), Lemma 3.8.6 (2) at $N = 0$).
- For §§III.56–III.65: `Chapter3/TangentGermProjecting`, `Chapter3/GoodComplementClosed`, `Chapter3/RecentreCounterexample2021` (Theorem 3.2.3 (preprint 3.27) under the tangent germ property alone, the closed good complement, the re-centring step); `Chapter3/BasicGermLinearizationLevel` (Proposition 3.1.10 (preprint 3.10) on every level); `Chapter3/MainTheorems2021`, `Chapter3/BookFormIFT2021` (the implicit function theorems with the 2021 definitions); `Chapter3/SolutionGermQuantitative`; `Chapter3/ExtremeRays`, `Chapter3/GoodComplementIntersection`, `Chapter3/GoodComplementIntersectionSc` (Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11); `Chapter3/QuadrantClassCk`, `Chapter3/QuadrantInverseSmooth`, `Chapter3/IteratedTangentMaps`; `Chapter3/GraphDegeneracyIndex`, `Chapter3/TangentGermDegeneracy` (general presentations, $d_M = d_X$); `Chapter3/LinearizationChartChange`, `Chapter3/LinearizationIntrinsic`, `Chapter3/SolutionTangentKernel` (the intrinsic linearization, $\ker f'(y) = T_y S$); `Chapter3/ScIsoConjugation`, `Chapter3/BasicGermCriterionSplit`, `Chapter3/BasicGermCriterionTransport`, `Chapter3/BasicGermCriterionGeneral` (Theorem 3.7.1 (preprint 3.58) as printed); `Chapter3/TameGermTransfer` (tame germ data at interior points); `Chapter3/UntamedJump`, `Chapter3/UntamedCounterexample`, `Chapter3/UntamedBundle`, `Chapter3/UntamedBundleFrame`, `Chapter3/UntamedBundleTame`, `Chapter3/UntamedBundleTheorem` (the jumping projections and the counterexample for untamed germ data).

Throughout, P and Q are finite-dimensional normed spaces with the constant sc-structure (the text's $\mathbb{R}^n$ and $\mathbb{R}^N$), $\widetilde{C} \subseteq P$ a set containing 0, convex, with nonempty interior (the text's $[0,\infty)^k \oplus \mathbb{R}^{n-k}$), W an sc-Banach space, $E = P \oplus W$ with levels $P \times W_m$ `Chapter3/ContractionGerm · ScBanach.paramProd`. Because all levels of P coincide, the condition "$\widetilde{C} \times W$ is quadrant-like" is just "$0 \in \widetilde{C}$, convex, nonempty interior" `Chapter3/ContractionGerm · isQuadrantLike_const_iff` `Chapter3/ContractionGerm · IsQuadrantLike.paramProd`.

III.1 Germs and their calculus (Definitions 3.1.1, 3.1.2 (preprint 3.1, 3.2))

Definition 3.1.1 (preprint 3.1). An *sc-germ of neighbourhoods* around $0 \in C$ is a family of sets $U_m \subseteq E_m$, each of the form $O \cap C_m$ with O open in E_m, containing 0, and nested along the inclusions: $\iota U_{m+1} \subseteq U_m$ `Chapter3/Germ · ScNbhdGerm`. Its *tangent germ* has levels $U_{m+1} \times E_m$ `Chapter3/Germ · ScNbhdGerm.tangent`. The text's example, a relatively open $U \ni 0$ with $U_m = U \cap E_m$, is `ofSet` `Chapter3/Germ · ScNbhdGerm.ofSet`. The germ calculus needs one property of C: every level C_m is convex with nonempty interior and $0 \in C$ ("quadrant-like") `Chapter3/Germ · IsQuadrantLike`. Partial quadrants have it `Chapter3/Germ · IsPartialQuadrant.isQuadrantLike`, and it passes to TC `Chapter3/Germ · IsQuadrantLike.tangentSet`. It makes every U_m a set of unique differentiability `Chapter3/Germ · ScNbhdGerm.uniqueDiffWithinAt`, the intersection of an open and a convex set `Chapter3/Germ · ScNbhdGerm.relOpenConvex`.

A germ is a *fixed* sequence of sets. The text says "the size of the sets U_m does not matter", i.e. it reads germs up to shrinking; the formalization cannot, and must say on which sequence each statement holds. One trivial remark becomes important in §III.9: the re-indexed sequence $(U_{m+1})_m$ is in general not a germ, since U_{m+1} is relatively open in C_{m+1} and not in C_m; it is one when all levels coincide, i.e. for germs in the finite-dimensional parameter space P.

Definition 3.1.2 (preprint 3.2). f is an *sc⁰-germ* on $G = (U_m)$ if $f(U_m) \subseteq F_m$ and the level map $f_m : U_m \to F_m$ is continuous for every m `Chapter3/Germ · IsSc0Germ`. It is an *sc¹-germ with derivative family* D if it is sc⁰, differentiable at every $x \in U_1$ within U_1 in the sense of Definition 1.1.15 (preprint 1.14) (increments in E_1, remainder measured in F_0) with derivative $D(x)$, and its tangent map $(x,h) \mapsto (f(x), D(x)h)$ is an sc⁰-germ on the tangent germ `Chapter3/Germ · IsSc1GermWith`; an sc¹-germ if some such D exists `Chapter3/Germ · IsSc1Germ`, in which case a derivative family is chosen `Chapter3/Germ · scDerivGerm` and gives the tangent germ `Chapter3/Germ · tangentMapGerm`. On quadrant-like C the derivative is unique on U_1 `Chapter3/Germ · IsSc1GermWith.eq_of_uniqueDiffWithinAt`, so the choice does not matter there `Chapter3/Germ · IsSc1Germ.tangentMapGerm_eqOn`. Then sc^k is defined by recursion — sc⁰ for $k = 0$, and sc¹ with sc^k tangent germ for $k+1$ `Chapter3/Germ · IsSckGerm` — and sc-smooth means sc^k for all k `Chapter3/Germ · IsScSmoothGerm`. These are the notions of Part I, §I.11, with $U \cap E_m$ replaced by U_m, and the bridge is exact: an sc^k map on a relatively open $U \ni 0$ is an sc^k-germ on `ofSet U` `Chapter3/Germ · IsSck.isSckGerm` `Chapter3/Germ · IsScSmooth.isScSmoothGerm`.

The calculus of germs. The text restricts, composes and pairs germs without comment. All of it is proved, with the proofs of Part I, §§I.11–I.12. (1) sc^k-germs depend only on the values on U_0 `Chapter3/Germ · IsSckGerm.congr` and restrict to smaller germs `Chapter3/Germ · IsSckGerm.mono` — by induction on k, using that the tangent germs of the smaller germ and of the larger one agree on the smaller tangent germ `Chapter3/Germ · IsSc1Germ.tangentMapGerm_mono`. (2) *The chain rule.* A composition $g \circ f$ makes sense level by level only if f maps U_m into the domain V_m of g for every m `Chapter3/Germ · ScNbhdGerm.MapsToGerm`; then sc⁰-germs compose `Chapter3/Germ · IsSc0Germ.comp`, the derivative of $g \circ f$ at $x \in U_1$ is $Dg(f(x)) \circ Df(x)$ — proved as in §I.12 with the mean value inequality along segments in V_1 and the uniform smallness of $Dg(y) - Dg(y_0)$ on compact sets `Chapter3/Germ · IsSc1GermWith.hasFDerivWithinAt_comp` — so $g \circ f$ is sc¹ with

$T(g\circ f)=Tg\circ Tf$ on the tangent germ `Chapter3/Germ · IsSc1GermWith.comp` `Chapter3/Germ · IsSc1Germ.comp`, and by induction on k it is sc^k `Chapter3/Germ · IsSckGerm.comp` `Chapter3/Germ · IsScSmoothGerm.comp`. (3) An sc-operator T is an sc¹-germ with derivative T itself `Chapter3/Germ · ScOperator.isSc1GermWith`, its tangent map is the sc-operator (T,T) `Chapter3/Germ · ScOperator.tangentOp`, and so it is an sc-smooth germ on every germ `Chapter3/Germ · ScOperator.isSckGerm` `Chapter3/Germ · ScOperator.isScSmoothGerm`; composing with one on the outside preserves sc^k `Chapter3/Germ · IsSckGerm.operator_comp`. (4) The pairing (f,g) of two sc^k-germs is an sc^k-germ `Chapter3/Germ · pairMap` `Chapter3/Germ · IsSckGerm.pair`, through the reshuffle $TF\oplus TG\cong T(F\oplus G)$ `Chapter3/Germ · tangentProdComm`. (5) Where the text says "near 0" it shrinks germs: for an sc⁰-germ f with $f(0)=0$ and a germ G' the sets $\{x\in U_m : f_m(x)\in U'_m\}$ form the largest sub-germ on which f maps into G' `Chapter3/GermFilling · ScNbhdGerm.preimage` `Chapter3/GermFilling · ScNbhdGerm.preimage_mapsToGerm`.

Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs. The text applies these to germs ("recalling Proposition 1.17" in Lemma 3.1.9 (preprint 3.9); Lemma 3.3.4 (preprint 3.31); Corollary 3.3.7 (preprint 3.34)) but states them for relatively open sets. For an sc¹-germ with derivative family D and $x\in U_{m+1}$, $D(x)$ restricts to a bounded level operator $E_m\to F_m$ `Chapter3/GermLevels · IsSc1GermWith.exists_levelDeriv` `Chapter3/GermLevels · IsSc1GermWith.toBase_levelDeriv`, the level-m tangent map is $(x,h)\mapsto(f_{m+1}(x),D_m(x)h)$ `Chapter3/GermLevels · IsSc1GermWith.levelMap_tangentMapWith`, jointly continuous `Chapter3/GermLevels · IsSc1GermWith.continuousOn_levelDeriv` and continuous in x in the operator norm (the compact-open lemma of §I.13) `Chapter3/GermLevels · IsSc1GermWith.continuousOn_levelDeriv_clm`; and $f : U_{m+1}\to F_m$ is differentiable within U_{m+1} with derivative $D_m(x)$, by the integral representation $f(x+k)-f(x)=\int_0^1 D(x+\tau k)k\,d\tau$ in F_m (Proposition 1.2.1 (preprint 1.16)) `Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level`. The shifted sequence $(U_{m+1})_m$ is a germ in E^1 around $0\in C^1$ `Chapter3/GermLevels · ScNbhdGerm.shift`, and an sc^k-germ on G is an sc^k-germ $G^1\to F^1$ (Proposition 1.2.2 (preprint 1.17)) `Chapter3/GermLevels · IsSc0Germ.shift` `Chapter3/GermLevels · IsSc1GermWith.shift` `Chapter3/GermLevels · IsSckGerm.shift`. An sc^k-germ is C^k from U_{m+k} to F_m (Proposition 1.2.3 (preprint 1.18)), by the proof of §I.16 with U_m for $U\cap E_m$ `Chapter3/GermLevels · IsSckGerm.contDiffOn_levels'` `Chapter3/GermLevels · IsSckGerm.contDiffOn_levels`. sc^k depends on a germ only through its sets `Chapter3/GermLevels · IsSckGerm.of_U_eq`.

III.2 Sections of strong bundles in charts (Definition 3.1.3 (preprint 3.3))

Let $P : Y\to X$ be a strong bundle over an M-polyfold (Part II, §II.19). A strong bundle chart $(\Phi,P^{-1}(V),K,U\lhd F)$ has a model $K=R(U\lhd F)$ for a strong bundle retraction R. Its base and fibre parts are read off: $r(u)=(R(u,0))_1$ `Chapter3/Filler · StrongRetr.r`, $\rho(u)h=(R(u,h))_2$ `Chapter3/Filler · StrongRetr.ρ`, and $R(u,h)=(r(u),\rho(u)h)$ because the base component of a strong bundle map does not depend on the fibre variable `Chapter3/Filler · StrongRetr.apply_eq`. Comparing components of $R\circ R=R$ gives $r\circ r=r$ `Chapter3/Filler · StrongRetr.r_r` and $\rho(r(u))\rho(u)=\rho(u)$ `Chapter3/Filler · StrongRetr.ρ_r_ρ`; for $x\in O=r(U)$, $r(x)=x$ `Chapter3/Filler · StrongRetr.r_eq_self` and $\rho(x)$ is a projection `Chapter3/Filler · StrongRetr.ρ_ρ`; R fixes the points of K `Chapter3/Filler · StrongRetr.apply_eq_self_of_mem_K`, so $\rho(x)h=h$ for $(x,h)\in K$ `Chapter3/Filler · StrongRetr.ρ_eq_self_of_mem_K`. $\rho(u)$ is linear by Definition 2.6.1 (preprint 2.47) and continuous because R is sc⁰ on level 0 `Chapter3/Filler · StrongRetr.ρ_continuous`; it is a bounded operator `Chapter3/Filler · StrongRetr.ρCLM` — the text's "$\rho(u):F\to F$ is a bounded linear operator", without comment.

Principal parts and the section algebra. For a section f (a map with $P\circ f=\mathrm{id}$), $\Phi\circ f\circ\varphi^{-1}(z)=(z,\mathbf{f}(z))$; $\mathbf{f}$ is its *principal part* `Chapter3/StrongBundleSections · StrongBundleChart.principal` `Chapter3/`

StrongBundleSections · StrongBundleChart.apply_section. The fibrewise sum $f + s$ is defined with the vector space structure of the fibres $P^{-1}(x)$ Chapter3/StrongBundleSections · sectionAdd and is a section Chapter3/StrongBundleSections · IsSection.sectionAdd. Since Φ is linear on the fibres Chapter3/StrongBundleSections · StrongBundleChart.snd_add and maps the zero of $P^{-1}(x)$ to $(\varphi(x), 0)$ Chapter3/StrongBundleSections · StrongBundleChart.snd_fiberZero, the principal part of $f + s$ is $\mathbf{f} + \mathbf{s}$ Chapter3/StrongBundleSections · StrongBundleChart.principal_sectionAdd and $f(x) = 0$ iff $\mathbf{f}(\varphi(x)) = 0$ Chapter3/StrongBundleSections · StrongBundleChart.isZeroAt_iff Chapter3/StrongBundleSections · IsZeroAt. The fibres of K over O are linear: $(u, a), (u, b) \in K$ give $(u, a + b) \in K$, because membership is $\rho(u)a = a$ Chapter3/StrongBundleSections · StrongBundleChart.add_mem_K. *The sum rule*, used silently in "$f + s$ is an sc-smooth section", is pairing (Part I) followed by the addition sc-operator $F \oplus F \to F$ Chapter3/StrongBundleSections · ScBanach.addOp Chapter3/StrongBundleSections · IsScSmooth.add Chapter3/StrongBundleSections · IsScSmooth.sub Chapter3/StrongBundleSections · IsScSmooth.neg; a map sc-smooth into F^1 is sc-smooth into F (compose with the inclusion) Chapter3/StrongBundleSections · IsScSmooth.toBase_shift.

Which charts. "A strong bundle chart" in Definitions 3.1.3, 3.1.6 and 3.1.11 (preprint 3.3, 3.6 and 3.11) must mean a chart *compatible* with the atlas of P Chapter3/StrongBundleSections · StrongBundle.IsChart Chapter3/StrongBundleSections · StrongBundle.isChart_of_mem: a chart of the atlas need not be centred at the point in question, and the notions must not depend on the atlas within its class. A compatible chart is a chart of the M-polyfold $Y[0]$ Chapter3/StrongBundleSections · StrongBundle.IsChart.toMChart. The price is that representatives of sections must be sc-smooth in every compatible chart. For an sc-smooth section, $z \mapsto (z, \mathbf{f}(z))$ is sc-smooth $O \to K$ by Definition 2.3.8 (preprint 2.14) as it stands Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.isScSmoothRetractMap. For an sc⁺-section, sc-smooth into $Y[1]$, the principal part takes values in F_1 in every compatible chart Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.principal_mem_range, and $z \mapsto (z, \mathbf{s}_1(z))$ is sc-smooth $O \to K[1]$: in a chart Φ' of the atlas this is the definition, and in the given chart it is the composition with the change of chart $(\Phi \circ \Phi'^{-1})[1]$ on $K[1]$, glued over the atlas by locality Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.isScSmoothRetractMap_one. This is precisely the compatibility on $[1]$ of Definition 2.6.4 (preprint 2.50), used here for the first time.

Definition 3.1.3 (preprint 3.3). (f, x_0) is an *sc-smooth section germ* if x_0 is smooth and in some compatible strong bundle chart with $\varphi(x_0) = 0$ the map $\mathbf{f} \circ r$ is an sc-smooth germ on some sc-germ of neighbourhoods G of 0 with $G_0 \subseteq U$, whose levels may shrink with m Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSectionGerm — the text's germ reading, as for Definition 3.1.11 (preprint 3.11) (§III.48); until the fourth addendum the germ was the chart's $U \cap E_m$, which is stricter (§III.48). The text's "continuous section on a neighbourhood V" is the section on the chart domain; near x_0 continuity is a consequence. An sc-smooth section is an sc-smooth section germ at every smooth point with a centred chart: its representative composed with r is sc-smooth on the relatively open U, and §III.1's bridge applies on the germ $U \cap E_m$ Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.isScSmoothSectionGerm. No theorem assumes the definition.

III.3 Fillers (Definitions 3.1.4, 3.1.6 (preprint 3.4, 3.6); Lemmas 3.1.19, 3.1.20 (preprint 3.16, 3.17); Remark 3.6.2 (preprint 3.49))

Definition 3.1.4 (preprint 3.4). For the principal parts $f, g : E_0 \to F_0$ (with $O = r(U)$, $K = R(U \lhd F)$, $0 \in U$), *g fills f* if (1) $f = g$ on O near 0; (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in O; (3) the map $y \mapsto [1 - \rho(r(y))]g(y)$ is sc¹ on U, and its derivative at 0 restricts to a topological linear

isomorphism $\ker Dr(0) \to \ker \rho(0)$ `Chapter3/Filler · IsFillerOf`. The text's condition (3) presupposes the linearization; requiring the map to be sc¹ loses nothing, since it is sc-smooth when g and R are. The sc-smoothness of g itself is added in Definition 3.1.6 (preprint 3.6). **Definition 3.1.6 (preprint 3.6)**: a *filled version* of **f** is a germ $(g, 0)$, and it is formalized as one, a *filling germ* `Chapter3/GermFilling · IsFillingGerm`: g is an sc-smooth germ (§III.1) on a germ G with $G_0 \subseteq U$, whose levels may shrink with m; (1) and (2) hold near 0; and in (3) the fill defect $k(y) = [1 - \rho(r(y))]g(y)$ `Chapter3/GermFilling · fillDefect` is an sc¹-germ on G with a derivative family Dk whose value $Dk(0)$ — unique (§III.1) — restricts to a topological isomorphism $\ker Dr(0) \to \ker \rho(0)$. The first formalization took a filler sc-smooth on the whole U of the chart `Chapter3/StrongBundleSections · IsFilledVersion`, on the ground that a chart can always be restricted to $U \cap W$. That is the special case \$G = \$ `ofSet U` `Chapter3/GermFillingChart · IsFilledVersion.isFillingGerm` `Chapter3/GermFillingChart · IsFillingGerm.isFilledVersion`, and it is narrower: restricting a chart on level 0 never makes the levels shrink, and there are sc-smooth germs with shrinking levels that are sc-smooth on no neighbourhood of 0 `Chapter3/GermVsChart · exists_isScSmoothGerm_not_isScSmooth` (§III.48). The lemmas below about fillers on U are the whole-chart ones; their germ forms, which the canonical notions use, are cited next to them.

Lemma 3.1.19 (preprint 3.16). For a section s of $K \to O$ (that is, $(x, s(x)) \in K$ for $x \in O$), the extension $\tilde{s} = s \circ r$ satisfies $\tilde{s} = s$ on O (as $r(x) = x$ there), and $R(r(u), s(r(u))) = (r(u), s(r(u)))$, i.e. $\rho(r(u))\tilde{s}(u) = \tilde{s}(u)$, because $s(r(u))$ is a point of K and R fixes K `Chapter3/Filler · lemma_3_1_19`. The text's computation uses $s(x) \in K$ without saying so; it is the hypothesis that s is a section. **Lemma 3.1.20 (preprint 3.17).** If g fills f and s is a section of $K \to O$, then $g + s \circ r$ fills $f + s$ `Chapter3/Filler · lemma_3_1_20`: (1) on O, $g + s \circ r = f + s$; (2) by Lemma 3.1.19 (preprint 3.16), $(g + \tilde{s})(y) = \rho(r(y))(g + \tilde{s})(y)$ is equivalent to $g(y) = \rho(r(y))g(y)$; (3) by the same identity $[1 - \rho(r(y))](g + \tilde{s})(y) = [1 - \rho(r(y))]g(y)$ on U, so the two maps of (3) coincide on U and have the same derivative at 0 (derivatives on U_1 are unique). *For a filling germ* g on G and s sc-smooth on U the proof is the same: the identity of (3) holds on U, hence on G_0, so the derivative family Dk of the fill defect of g is one of $g + s \circ r$, a filling germ on the same G `Chapter3/GermCalculus · IsFillingGerm.add_section`; for s itself a germ on G_s the filling lives on $G \cap r^{-1}(G_s)$ `Chapter3/GermCalculus · IsFillingGerm.add_section_germ`. The regularity statements of both lemmas — $\tilde{s}$ sc⁺, $g + \tilde{s}$ sc-smooth — are proved where they are used, in Theorem 3.1.18 (preprint 3.15) (§III.6). Condition (1) only involves f on O `Chapter3/StrongBundleSections · IsFillerOf.congr_left`.

Remark 3.6.2 (preprint 3.49). On a smaller neighbourhood W of 0 condition (1) holds on all of $O \cap W$ and (2) for all $y \in U \cap W$: intersect the two neighbourhoods given by (1) and (2) `Chapter3/FillerLinearization · IsFillerOf.remark_3_6_2`.

III.4 The linearization of a filler (Proposition 3.1.5 (preprint 3.5))

The text derives, "by linearization at $y = 0$", the matrix of $Dg(0)$ in the splittings $E = Dr(0)E \oplus \ker Dr(0)$, $F = \rho(0)F \oplus \ker \rho(0)$, and from it Proposition 3.1.5 (preprint 3.5). The formal proof has two halves: the derivative identities, and the linear algebra.

The derivative identities. r is sc-smooth on U `Chapter3/FillerLinearization · StrongRetr.r_isScSmooth` with $r(0) = 0$ since $0 \in O$ `Chapter3/FillerLinearization · StrongRetr.r_zero`, and the chain rule applied to $r \circ r = r$ makes $Dr(0)$ idempotent `Chapter3/FillerLinearization · StrongRetr.scDeriv_r_idem`; $\rho(0)$ is idempotent `Chapter3/FillerLinearization · StrongRetr.ρCLM_idem`. The fibre part $(u, k) \mapsto \rho(u)k$ is sc¹ on $U \oplus F$ `Chapter3/FillerLinearization · StrongRetr.fiber_isSc1`, and its derivative at $(0, 0)$ is $(h, k) \mapsto \rho(0)k$: in the fibre direction because $k \mapsto \rho(0)k$ is linear, in the base direction because $\rho(u)0 = 0$;

both directional derivatives are computed on the dense level 1 and extended by continuity Chapter3/FillerLinearization · StrongRetr.scDeriv_fiber_zero. By the chain rule for $(u,k) \mapsto \rho(u)k$ composed with $y \mapsto (r(y), g(y))$ at the point $(r(0), g(0)) = (0,0)$ — this is where $g(0) = 0$ enters — the derivative of $y \mapsto [1-\rho(r(y))]g(y)$ at 0 is $(1-\rho(0))Dg(0)$ Chapter3/FillerLinearization · IsFillerOf.scDeriv_sub_eq Chapter3/FillerLinearization · scDeriv_comp_apply'. Next, $[1-\rho(r(y))]g(r(y)) = 0$ near 0, from $g = f$ on O and $\rho(x)f(x) = f(x)$ on O (f a section of K) and the continuity of r; linearizing gives $(1-\rho(0))Dg(0)Dr(0) = 0$ Chapter3/FillerLinearization · IsFillerOf.deriv_comp_r_eq_zero, i.e. $\rho(0)Dg(0)Dr(0) = Dg(0)Dr(0)$ Chapter3/FillerLinearization · IsFillerOf.ρ_deriv_comp_r. Condition (3) becomes: $(1-\rho(0))Dg(0)$ restricts to a topological isomorphism $\ker Dr(0) \to \ker \rho(0)$ Chapter3/FillerLinearization · IsFillerOf.exists_kerEquiv. Finally $f \circ r = g \circ r$ near 0, so $D(f\circ r)(0) = Dg(0)Dr(0)$ — the text's "$Tg(0) = Tf(0)$ on T_0O" Chapter3/FillerLinearization · IsFillerOf.scDeriv_section_comp_r.

The linear algebra. Let p, q be bounded idempotents on Banach spaces E, F and $L : E \to F$ bounded with $(1-q)Lp = 0$ and $(1-q)L : \ker p \to \ker q$ a topological isomorphism C. Then $A := L|_{\operatorname{range} p} : \operatorname{range} p \to \operatorname{range} q$ is well defined Chapter3/FillerLinearAlgebra · fillerRestrict. Two facts carry everything: if $(1-q)Lw = 0$ then $w \in \operatorname{range} p$ (injectivity of C, applied to the $\ker p$-component of w) Chapter3/FillerLinearAlgebra · fillerRestrict_mem_range_of; and $(1-q)Lz = y$ is solvable in $\ker p$ for $y \in \ker q$ (surjectivity of C) Chapter3/FillerLinearAlgebra · fillerRestrict_exists_of. From them: A is onto iff L is Chapter3/FillerLinearAlgebra · fillerRestrict_surjective_iff; the inclusion $\ker A \to \ker L$ is a bijection — the text's $\ker Dg(0) = \ker Tf(0) \oplus \{0\}$ Chapter3/FillerLinearAlgebra · fillerRestrict_kerEquiv; $\operatorname{range} A = \operatorname{range} L \cap \operatorname{range} q$ Chapter3/FillerLinearAlgebra · fillerRestrict_mem_range_iff and $\operatorname{range} q/\operatorname{range} A \cong F/\operatorname{range} L$ Chapter3/FillerLinearAlgebra · fillerRestrict_cokerEquiv. So the indices are equal as $\dim\ker - \dim\operatorname{coker}$ with no finiteness assumption Chapter3/FillerLinearAlgebra · fredholmIndex_fillerRestrict. Closedness of the ranges transfers both ways: $\operatorname{range} A$ is closed in $\operatorname{range} q$ when $\operatorname{range} L$ is Chapter3/FillerLinearAlgebra · fillerRestrict_isClosed_range_of, and conversely $\operatorname{range} L$ is the preimage of $\operatorname{range} A$ under the continuous $M(y) = y - LC^{-1}(1-q)y$ Chapter3/FillerLinearAlgebra · isClosed_range_of_fillerRestrict — this is where "topological" in (3) is used. Hence A is Fredholm iff L is Chapter3/FillerLinearAlgebra · fillerRestrict_isFredholm_iff, and the three statements together Chapter3/FillerLinearAlgebra · proposition_3_1_5_preprint. The text's proof factorizes $\begin{pmatrix} A & B \\ 0 & C \end{pmatrix} = \begin{pmatrix} 1 & BC^{-1} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$ and uses the additivity of the index; the isomorphisms of kernels and cokernels avoid both.

Proposition 3.1.5 (preprint 3.5). For a filler g of the section f of $K \to O$, g sc-smooth with $g(0) = 0$: with $p = Dr(0)$, $q = \rho(0)$, $L = Dg(0)$ the hypotheses of the linear algebra hold by the identities above, and $Tf(0) := Dg(0)|_{Dr(0)E} : Dr(0)E \to \rho(0)F$ is onto iff $Dg(0)$ is, Fredholm iff $Dg(0)$ is, with equal index Chapter3/FillerLinearization · IsFillerOf.proposition_3_1_5. (The text writes the target of the lower right entry as "$(1-D\rho(0))F$"; it is $(1-\rho(0))F$.) *For a filling germ* (§III.3) the proof is the same with any derivative family Dg of g on any germ inside U: the fill defect is an sc-smooth germ on the germ of g, by the chain rule with the whole-chart outer map R (§III.48) Chapter3/GermCalculus · fillDefect_isScSmoothGerm, and the value at 0 of its derivative family is $(1-\rho(0))Dg(0)$ Chapter3/GermFillerLinearization · fillDefect_deriv_zero_eq_rho, which replaces `scDeriv_sub_eq`; so $Tf(0) = Dg(0)|_{Dr(0)E}$ is onto, Fredholm, of the index of $Dg(0)$ iff $Dg(0)$ is Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_1_5. $Dg(0)$ does not depend on the family or the germ Chapter3/GoodPositionGermG · IsSc1GermWith.deriv_zero_eq.

III.5 Basic germs, Lemma 3.1.9 (preprint 3.9) and Proposition 3.1.10 (preprint 3.10)

Definition 3.1.7 (preprint 3.7) (and 3.37). $f : P \times W_0 \to Q \times W_0$ is a *basic germ* on a germ G around 0 in $\widetilde{C} \oplus W$, with contraction map B, if f is an sc-smooth germ with $f(0) = 0$ whose W-component is $w - B(a, w)$ on U_0, and B is an sc^0-contraction germ in the sense of Definition 3.3.1 (preprint 3.28) (§III.7) `Chapter3/LocalCompactness · IsBasicGerm`. ($B = w - f_W$ is then itself an sc-smooth germ `Chapter3/LocalSolvability · IsBasicGerm.B_isScSmoothGerm`.)

Lemma 3.1.9 (preprint 3.9). The text: for every $\varepsilon > 0$ and m, $|D_2B(a,w)\zeta|_m \le \varepsilon|\zeta|_m$ when $(a, w) \in E_{m+1}$ is close to 0 in E_m; in particular $D_2B(0,0) = 0$. Its proof: for $\zeta \in W_{m+1}$ the difference quotient of $t \mapsto B(a, w + t\zeta)$ is bounded by the contraction estimate, and the derivative is the limit (this needs $B : U_{m+1} \to W_m$ differentiable, Proposition 1.2.1 (preprint 1.16) for germs — "recalling Proposition 1.17" in the text); then density of W_{m+1} in W_m. Only sc^1 is used. The formalization proves the three forms the chapter needs. (a) On level m in the working form of §III.7, with the constant $\frac{1}{2}$: at $(a, w) \in U_{m+1}$ with $a \in N(m+1) \cap \widetilde{C}$, $w \in O(m+1)$ and $\iota w \in O(m)$, $\|D_2B(a,w)\|_{L(W_m)} \le \frac{1}{2}$ `Chapter3/ContractionGerm · ContractionGerm.norm_levelDeriv_snd_le`; a linear map of norm $\le \frac{1}{2}$ then has no nonzero fixed point `Chapter3/ContractionGerm · ContractionGerm.eq_zero_of_norm_le_half`. (b) $D_2B(0,0) = 0$, from the difference quotients along $t \mapsto (0, t\zeta)$, bounded by $\varepsilon|\zeta|$ for every ε, then density `Chapter3/BasicGermLinearization · IsSc0ContractionGerm.deriv_zero_snd`. (c) On level 0 with any ε: $|D_2B(x)\zeta|_0 \le \varepsilon|\zeta|_0$ for $x \in U_1$ small *on level* 1 `Chapter3/SolutionLinearization · IsSc0ContractionGerm.exists_norm_deriv_snd_le` — the curve $t \mapsto x + t(0, \zeta)$, $\zeta \in W_1$, must stay in U_1, where B is differentiable.

Proposition 3.1.10 (preprint 3.10). Let $\widetilde{C}$ be a partial quadrant, h a basic germ, s an sc^1-germ into $(Q \oplus W)^1$ (an sc$^+$-germ) with derivative family Ds, and $f = h + \iota s$ sc^1 with derivative family Df. Then $Df(0)$ is an sc-operator, sc-Fredholm of index $\dim P - \dim Q$ `Chapter3/BasicGermLinearization · proposition_3_1_10`. *Proof.* The derivative of an sc^1-germ at the smooth point 0 is an sc-operator, its level-m map being the fibre component of the level-m tangent map at $(0, \cdot)$ (the remark after Proposition 1.2.1 (preprint 1.16)) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`. The operator (32) of the text, $T_0(\delta a, \delta w) = (0, \delta w)$ `Chapter3/BasicGermLinearization · ScBanach.projW0p`, is sc-Fredholm with kernel $P \oplus 0$ and cokernel $Q \oplus 0$ `Chapter3/BasicGermLinearization · ScBanach.projWSplitting` `Chapter3/BasicGermLinearization · ScBanach.isScFredholm_projW0p`, of index $\dim P - \dim Q$ `Chapter3/BasicGermLinearization · ScBanach.scIndex_projW0p`. The text says "$Df(0)$ is an sc$^+$-perturbation of (32)"; the formal proof builds the sc$^+$-operator $S = Df(0) - T_0$ level by level `Chapter3/BasicGermLinearization · proposition_3_1_10_scPlus`: its Q-component is sc$^+$ because Q has all levels equal; $\mathrm{pr}_W Dh(0)(\delta a, 0) = -D_1B(0)\delta a = -DB(0)(\delta a, 0)$ has values in W_∞ ($(\delta a, 0)$ is smooth and $DB(0)$ an sc-operator) and its level maps $E_m \to W_{m+1}$ factor through $DB(0)_{m+1}$; $\iota Ds(0)$ is sc$^+$ because $Ds(0)$ is an sc-operator into $(Q \oplus W)^1$; and $\mathrm{pr}_W Dh(0)(0, \delta w) = \delta w$ because $D_2B(0) = 0$ (Lemma 3.1.9 (preprint 3.9) (b)). The identities $Df(0) = Dh(0) + \iota Ds(0)$ and $\mathrm{pr}_W Dh(0) = \mathrm{pr}_W - DB(0)$ are derivative identities within U_1, extended to E_0 because two bounded operators agreeing on the dense E_1 are equal `Chapter3/BasicGermLinearization · ContinuousLinearMap.ext_of_comp_toBase_one`. Proposition 1.1.11 (preprint 1.10) with its index statement concludes `Chapter1/ScFredholm · proposition_1_1_11_preprint` `Chapter1/FredholmIndex · proposition_1_1_11_index`. For a basic germ alone ($s = 0$) `Chapter3/FredholmGermLinearization · ScOperator.zero` `Chapter3/FredholmGermLinearization · proposition_3_1_10_basic`, with its level-0 consequence: $Dh(0)$ is Fredholm of index $\dim P - \dim Q$ `Chapter3/FredholmGermLinearization · proposition_3_1_10_basic_level_zero`.

The second statement of Proposition 3.1.10 (preprint 3.10) ($Df(a,w)$ Fredholm on every level m for (a,w) small) is proved on level 0 for the germs of Theorem 3.4.2 (preprint 3.38), at points of U_1 small on level 1 (§III.19). On every level m it is proved by the text's argument run on level m (gap G.21) `Chapter3/BasicGermLinearizationLevel · proposition_3_1_10_level`. At $x \in U_{m+1}$ small on level m the level-m operator splits as $Df(x)_m = T_0 + S(x) + K(x)$ `Chapter3/BasicGermLinearizationLevel · IsBasicGerm.levelDeriv_add_eq`. Here $T_0(\delta a, \delta w) = (0, \delta w)$ is Fredholm. $S(x) = (0, -D_2B(x)_m)$ is small by Lemma 3.1.9 (preprint 3.9) on level m. $K(x)$, the finite-rank part plus the sc$^+$-part through $F_{m+1} \to F_m$, is compact `Chapter3/BasicGermLinearizationLevel · IsBasicGerm.isCompactOperator_levelK`. The index is stable under the small perturbation $S(x)$. Both statements of the proposition together are `Chapter3/BasicGermLinearizationLevel · proposition_3_1_10_full`.

III.6 Sc-Fredholm germs and sections; Theorem 3.1.18 (preprint 3.15) (Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4 (preprint 3.11, 3.14, 3.50, 3.51))

Definition 3.1.11 (preprint 3.11). A *strong bundle isomorphism* $\Psi : U \lhd F \to U' \lhd F'$ is a pair of strong bundle maps (Definition 2.6.1 (preprint 2.47)) inverse to each other on $U \oplus F$ and $U' \oplus F'$ `Chapter3/StrongBundleSections · IsStrongBundleIso`; its base map is ψ. A principal part $k : U \to F$ is *conjugated to a basic germ* if there are a strong bundle isomorphism with $\psi(0) = 0$ onto U' relatively open in $\widetilde{C} \oplus W$ ($\widetilde{C}$ a partial quadrant of a finite-dimensional P), a basic germ h on a germ G with $G_0 \subseteq U'$, and $(\Psi(u, k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0 `Chapter3/StrongBundleSections · IsConjugateToBasicGerm`; this depends only on k on U `Chapter3/StrongBundleSections · IsConjugateToBasicGerm.congr`. (f, x_0) is an *sc-Fredholm germ* if in a compatible chart centred at x_0 the principal part has a filling germ g (§III.3), and there is a local sc$^+$-section s — an sc-smooth germ into F^1 on a germ G_s with $(G_s)_0 \subseteq U$ `Chapter3/GermFilling · IsScPlusGerm` — with $\iota s(0) = g(0)$ such that $g - \iota s$ is conjugated to a basic germ `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm`. This is the text's reading: g and s are germs, on germs of neighbourhoods of their own whose levels may shrink with the level, while Ψ is an isomorphism of the whole local bundles and the conjugation identity is asked near 0 only, so it depends only on the germ of $g - \iota s$ `Chapter3/GermFillingChart · IsConjugateToBasicGerm.congr_nhds`. The first formalization asked g and s to be sc-smooth on $U \cap E_m$ for every m `Chapter3/StrongBundleSections · StrongBundle.IsScFredholmGermWholeChart`. That notion implies the germ reading `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGermWholeChart.isScFredholmGerm` and is exactly the germ reading with g and s sc-smooth on U `Chapter3/GermFillingChart · StrongBundle.isScFredholmGerm_iff_germG_wholeChart`; whether every germ datum can be traded for a whole-chart one is open (§III.48). Definitions 3.1.16, 3.6.3, 3.6.4, 3.6.6 (preprint 3.14, 3.50, 3.51, 3.53) and every theorem of this part built on them are stated for the germ reading; the whole-chart notions and theorems are kept under the names `…WholeChart`, `…_wholeChart`. The text's $C' = T\psi(0)C$, used in the proof of Theorem 3.6.8 (preprint 3.54), is not part of the data; it is a theorem (§III.23).

Definition 3.1.16 (preprint 3.14). f is an *sc-Fredholm section* if it is sc-smooth, regularizing ($x \in X_m$ and $f(x) \in Y_{m,m+1}$ imply $x \in X_{m+1}$) and an sc-Fredholm germ at every smooth point `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection`. The text assumes a tame bundle; nothing in Definitions 3.1.3–3.1.16 (preprint 3.3–3.14) or in Theorem 3.1.18 (preprint 3.15) uses tameness, so it is not assumed. (Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51) restate 3.11, 3.14.) A zero of a regularizing section is a smooth point: the zero section lies in $Y_{m,m+1}$ over X_m `Chapter3/SectionCompactness · StrongBundle.fiberZero_mem_doubleLevel`, so a zero on level m is on level $m+1$, and by induction on every level `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`.

Theorem 3.1.18 (preprint 3.15). If f is an sc-Fredholm section and s an sc$^+$-section, then $f+s$ is an sc-Fredholm section `Chapter3/FredholmSectionsG · theorem_3_1_18_preprint`. *Proof.* The text says "$f+s$ is an sc-smooth section which is also regularizing" and checks the germ property; each clause is a proof.

Sc-smooth. In a chart the representative of $f+s$ is $z \mapsto (z, \mathbf{f}(z)+\mathbf{s}(z))$. The map $z \mapsto (z, \mathbf{f}(z))$ is sc-smooth $O \to K$, $z \mapsto (z, \mathbf{s}_1(z))$ is sc-smooth $O \to K[1]$ (§III.2), and composing the latter with $K[1] \subseteq K$ and adding fibre components (sum rule) gives an sc-smooth map with values in K, since its fibres are linear `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.add_isScSmoothRetractMap`; continuity of $f+s$ on each chart domain is read off the same representative `Chapter3/StrongBundleSections · StrongBundle.continuousOn_sectionAdd` `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.continuousOn_source`. So $f+s$ is sc-smooth `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.sectionAdd`.

Regularizing. For $x \in X_m$, the representative of s maps O_m into $K[1]_m$, so $\mathbf{s}(\varphi(x)) \in F_{m+1}$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.snd_mem_range`. If $(f+s)(x) \in Y_{m,m+1}$ then $\mathbf{f}(\varphi(x)) = (\mathbf{f}+\mathbf{s})(\varphi(x)) - \mathbf{s}(\varphi(x)) \in F_{m+1}$, i.e. $f(x) \in Y_{m,m+1}$ (Φ is linear on fibres `Chapter3/StrongBundleSections · StrongBundleChart.snd_sub`), and $x \in X_{m+1}$ `Chapter3/FredholmSectionsG · StrongBundle.regularizing_sectionAdd`.

The germ property at a smooth point x `Chapter3/FredholmSectionsG · StrongBundle.IsScFredholmGerm.sectionAdd`. Take the chart, filling germ g on G_g, sc$^+$-germ t_0 on G_{t_0} with $\iota t_0(0) = g(0)$ and conjugation Ψ for (f,x). Let $t = \mathbf{s}$ be the principal part of s in this chart; it is a section of $K \to O$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.mem_K` with lift t_1 to F_1, $\iota t_1 = t$ on O `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.toBase_principalLift`. Then: $\tilde{t} = t \circ r$ is sc-smooth on the whole U, and $t_1 \circ r$ is sc-smooth into F^1 on U (sc-smoothness on the retract O, Definition 2.1.14 (preprint 2.7)) — s is a section of the bundle, not a germ; $g + \tilde{t}$ is a filling germ of $\mathbf{f}+\mathbf{s}$ on the same G_g (Lemma 3.1.20 (preprint 3.17) for germs, §III.3 `Chapter3/GermCalculus · IsFillingGerm.add_section`, with the filler condition (1) transported along $\mathbf{f}+\mathbf{s}$ on O `Chapter3/FredholmSectionsG · IsFillingGerm.congr`); $t_1 \circ r + t_0$ is an sc$^+$-germ on the same G_{t_0}, a germ plus a whole-chart map (§III.48), with value $(g+\tilde{t})(0)$ at 0 `Chapter3/FredholmSectionsG · StrongBundle.sectionAdd_chartData_germ`; and $(g+\tilde{t}) - \iota(t_1 \circ r + t_0) = g - \iota t_0$ *on* U (where $\iota t_1(r(u)) = t(r(u))$ because $r(u) \in O$), which is conjugated to a basic germ by the same Ψ. This last identity holds on U only, which is why the conjugation condition is stated for $u \in U$. The germs of the data do not change, so nothing of g or t_0 away from G_g, G_{t_0} is used; the whole-chart proof `Chapter3/StrongBundleSections · StrongBundle.IsScFredholmGermWholeChart.sectionAdd` is the same with the sum rule on U in place of the sum of a germ and a whole-chart map. □

III.7 Contraction germs: Definition 3.3.1 (preprint 3.28) and its working form

Definition 3.3.1 (preprint 3.28). For a germ G around 0 in $\widetilde{C} \oplus W$, $B : P \times W_0 \to W_0$ is an *sc^0-contraction germ* (i.e. $f(a,w) = w - B(a,w)$ is one `Chapter3/ContractionGerm · contractionMap`) if B is an sc^0-germ on G, $B(0,0) = 0$, and for every $\varepsilon > 0$ and m there is $r > 0$ such that every (a,w) with $a \in \widetilde{C}$, $|a| < r$, $|w|_m < r$ lies in U_m and $|B_m(a,w) - B_m(a,w')|_m \le \varepsilon |w - w'|_m$ for such points `Chapter3/ContractionGerm · IsSc0ContractionGerm`. (The text: "for all (a,w), (a,w') on level m sufficiently close to $(0,0)$"; that these points are in the domain U_m is presupposed.) B_m is the level-m map `Chapter3/ContractionGerm · levelB`, with $B_m(0) = 0$ `Chapter3/ContractionGerm · levelB_zero`.

The working form. The induction of §III.10 builds from f a contraction germ $f^{(1)}$ whose fibre is $TW =$

$W^1 \oplus W$ and whose map is affine in the second factor; its natural contraction region is unbounded there, which balls cannot express. So the proofs work with explicit regions: a *contraction germ in working form* consists of B and G as above with open neighbourhoods $N(0) \supseteq N(1) \supseteq \cdots$ of 0 in P, open $O(m) \ni 0$ and closed $K(m) \subseteq O(m)$ in W_m with $\iota K(m+1) \subseteq K(m)$, such that for $a \in N(m) \cap \widetilde{C}$: $\{a\} \times O(m) \subseteq U_m$, $B_m(a,\cdot)$ maps $K(m)$ into $K(m)$ and is a $\frac{1}{2}$-contraction on $O(m)$ `Chapter3/ContractionGerm · ContractionGerm`.

Lemma: every sc⁰-contraction germ has a working form `Chapter3/GermIFT · IsSc0ContractionGerm.toContractionGerm`. Apply Definition 3.3.1 (preprint 3.28) with $\varepsilon = \frac{1}{2}$ to get radii r_m `Chapter3/GermIFT · IsSc0ContractionGerm.rad` `Chapter3/GermIFT · IsSc0ContractionGerm.rad_spec`. Make them decrease fast enough for the inclusions: $r'_0 = r_0$, $r'_{m+1} = \min(r_{m+1}, r'_m/(\|\iota_m\| + 1))$ `Chapter3/GermIFT · IsSc0ContractionGerm.rad'` `Chapter3/GermIFT · IsSc0ContractionGerm.norm_incl_le_of_le`. Put $O(m) = \{|w|_m < r'_m\}$, $K(m) = \{|w|_m \le r'_m/2\}$; then $|\iota w|_m \le \|\iota_m\| r'_{m+1}/2 \le r'_m/2$ for $w \in K(m+1)$. Since B_m is continuous at 0 with $B_m(0) = 0$ `Chapter3/GermIFT · IsSc0ContractionGerm.levelB_zero_eq`, there is $\rho^0_m > 0$ with $|B_m(x)|_m < r'_m/4$ for $x \in U_m$, $|x|_m < \rho^0_m$ `Chapter3/GermIFT · IsSc0ContractionGerm.exists_cont_rad` `Chapter3/GermIFT · IsSc0ContractionGerm.contRad`. Let ρ_m decrease with $\rho_m \le \min(\rho^0_m, r'_m)$ `Chapter3/GermIFT · IsSc0ContractionGerm.prad` `Chapter3/GermIFT · IsSc0ContractionGerm.prad_anti` and $N(m) = \{|a| < \rho_m\}$. For $a \in N(m) \cap \widetilde{C}$ and $w \in K(m)$, $|B_m(a,w)|_m \le |B_m(a,w) - B_m(a,0)|_m + |B_m(a,0)|_m \le \frac{1}{2} \cdot \frac{r'_m}{2} + \frac{r'_m}{4} = \frac{r'_m}{2}$.

III.8 Existence of the solution germ: the parametrized fixed point theorem and Theorem 3.3.2 (preprint 3.29)

The parametrized Banach fixed point theorem. The text invokes "the parametrized version of Banach's fixed point theorem" in Theorems 3.3.2 and 3.5.1 (preprint 3.29 and 3.40) without stating it; Mathlib has the theorem without the parameter. Let X be a complete metric space, $K \subseteq X$ closed and nonempty, and $g_l : K \to K$ ($l \in \Lambda'$, a subset of a topological space) $\frac{1}{2}$-contractions such that $l \mapsto g_l(x)$ is continuous for every $x \in K$. Then the fixed point $\phi(l) \in K$ is unique in K and continuous in l: from $d(\phi(l), \phi(l_0)) \le \frac{1}{2} d(\phi(l), \phi(l_0)) + d(g_l\phi(l_0), g_{l_0}\phi(l_0))$ one gets $d(\phi(l), \phi(l_0)) \le 2d(g_l(\phi(l_0)), g_{l_0}(\phi(l_0)))$, which tends to 0 by continuity at the single point $\phi(l_0)$ `Chapter3/LocalCompactness · exists_fixedPoint_continuousOn`.

Theorem 3.3.2 (preprint 3.29). In the working form, for $a \in N(m) \cap \widetilde{C}$ the map $B_m(a,\cdot) : K(m) \to K(m)$ is a $\frac{1}{2}$-contraction of a closed subset of W_m, with a fixed point $\delta_m(a) \in K(m)$ `Chapter3/ContractionGerm · ContractionGerm.exists_fixed` `Chapter3/ContractionGerm · ContractionGerm.fixLvl`, which is the only fixed point in the larger set $O(m)$ (two fixed points in $O(m)$ satisfy $|w - w'|_m \le \frac{1}{2}|w - w'|_m$) `Chapter3/ContractionGerm · ContractionGerm.fixed_unique` `Chapter3/ContractionGerm · ContractionGerm.eq_fixLvl`. For $a \in N(m+1) \cap \widetilde{C}$, $\iota\delta_{m+1}(a) \in K(m) \subseteq O(m)$ is a fixed point of $B_m(a,\cdot)$ because B commutes with the inclusions; hence $\iota\delta_{m+1}(a) = \delta_m(a)$ `Chapter3/ContractionGerm · ContractionGerm.incl_fixLvl`. So $\delta := \delta_0$ `Chapter3/ContractionGerm · ContractionGerm.sol` maps $V_m := N(m) \cap \widetilde{C}$ into W_m with level-m map δ_m `Chapter3/ContractionGerm · ContractionGerm.levelMap_sol`. Each δ_m is continuous on V_m by the parametrized fixed point theorem ($g_a = B_m(a,\cdot)$ on $K(m)$, continuous in a because B_m is) `Chapter3/ContractionGerm · ContractionGerm.continuousOn_fixLvl`. Write $\mathrm{germ}(s) = (N(m+s) \cap \widetilde{C})_m$ `Chapter3/ContractionGerm · ContractionGerm.germ` — legitimately a germ because P is finite-dimensional — and call δ a *solution germ* on a germ V if it is an sc⁰-germ there with $\delta(0) = 0$, $(a, \delta_m(a)) \in U_m$ on V_m and $B(a, \delta(a)) = \delta(a)$ on V_0 `Chapter3/ContractionGerm · ContractionGerm.IsSolutionGerm`. Then: δ is a solution germ on $\mathrm{germ}(0)$, and every $w \in O(0)$ with $B(a,w) = w$, $a \in N(0) \cap \widetilde{C}$, equals $\delta(a)$ `Chapter3/ContractionGerm · theorem_3_3_2` `Chapter3/ContractionGerm · ContractionGerm.eq_sol`. The property passes to every

germ(s) Chapter3/ContractionGerm · ContractionGerm.isSolutionGerm_germ. In the text's form, from Definition 3.3.1 (preprint 3.28) with balls: the working form of §III.7 gives the solution germ, unique among solutions w with $|w|_0 < r_0'$ Chapter3/GermIFT · theorem_3_3_2_of_isSc0ContractionGerm. The text's "uniquely determined" leaves this region implicit (Remark 3.3.8 (preprint 3.36) names it, $|w|_0 \le \tau_0$).

III.9 The solution germ is sc¹: Theorem 3.3.3 (preprint 3.30)

The text cites [21, Theorem 2.3], "the hard part of the regularity theory", and gives no proof. The formal proof shows that δ is sc¹ on the germ shifted by one level, $V_m = N(m+1) \cap \widetilde{C}$, and the shift is forced by the argument.

The operators. Let D be the derivative family of B. For $a \in N(m+1) \cap \widetilde{C}$ the point $(a, \delta_{m+1}(a))$ lies in U_{m+1}, and the level-m operator of §III.1 there is $L_m(a) := DB(a, \delta_{m+1}(a)) : P \times W_m \to W_m$ Chapter3/ContractionGerm · ContractionGerm.Lop, with fibre part $T_m(a) = D_2B(a, \delta_{m+1}(a))$ Chapter3/ContractionGerm · ContractionGerm.Top and $L_m(a)(h,\zeta) = L_m(a)(h,0) + T_m(a)\zeta$ Chapter3/ContractionGerm · ContractionGerm.Lop_apply. By Lemma 3.1.9 (preprint 3.9) (a), $\|T_m(a)\| \le \frac{1}{2}$ Chapter3/ContractionGerm · ContractionGerm.norm_Top_le, so $1 - T_m(a)$ is invertible (Neumann series) Chapter3/ContractionGerm · ContractionGerm.isUnit_one_sub_Top, and the candidate derivative is $A_m(a) := (1 - T_m(a))^{-1} L_m(a)(\cdot, 0) \in L(P, W_m)$ Chapter3/ContractionGerm · ContractionGerm.Aop. It satisfies the *key identity* $L_m(a)(h, A_m(a)h) = A_m(a)h$ Chapter3/ContractionGerm · ContractionGerm.Lop_Aop, and is the unique solution ζ of $L_m(a)(h,\zeta) = \zeta$ Chapter3/ContractionGerm · ContractionGerm.eq_Aop. Because the level operators commute with the inclusions Chapter3/ContractionGerm · ContractionGerm.toBase_Lop, so do the A_m: $\iota A_m(a)h = A_0(a)h$ Chapter3/ContractionGerm · ContractionGerm.toBase_Aop.

Differentiability at $a_0 \in N(2) \cap \widetilde{C}$ Chapter3/ContractionGerm · ContractionGerm.hasFDerivWithinAt_sol. Let $x_1 = (a_0, \delta_1(a_0)) \in U_1$, $A_1 = A_1(a_0)$, $A_0 = A_0(a_0)$. For a near a_0 put $h = a - a_0$ and $k = (h, A_1 h) \in E_1$, so $|k|_1 \le (1 + \|A_1\|)|h|$. *This is where $a_0 \in N(2)$ is needed*: A_1 exists only because $(a_0, \delta_2(a_0)) \in U_2$, and without it the increment $A_0 h$ need not lie in W_1, so the sc¹-differentiability of B at x_1, which controls only increments in E_1, could not be used. For h small, $x_1 + k \in U_1$ and $y := \delta(a_0) + A_0 h \in O(0)$. By the sc¹ property, $B(a,y) = B(a_0, \delta(a_0)) + D(x_1)\iota k + \text{rem} = \delta(a_0) + L_0(a_0)(h, A_0 h) + \text{rem} = y + \text{rem}$ with $|\text{rem}|_0 = o(|h|)$, using the key identity on level 0. So y is an approximate fixed point of $B(a,\cdot)$, and the $\frac{1}{2}$-contraction on $O(0)$ gives $|\delta(a) - y|_0 \le \frac{1}{2}|\delta(a) - y|_0 + |\text{rem}|_0$, i.e. $|\delta(a) - \delta(a_0) - A_0 h|_0 \le 2|\text{rem}|_0 = o(|h|)$.

The tangent map is sc⁰. On level m of the tangent of germ(1), i.e. on $(N(m+2) \cap \widetilde{C}) \times P$, the tangent map is $(a,h) \mapsto (\delta_{m+1}(a), A_m(a)h)$ Chapter3/ContractionGerm · ContractionGerm.tangent_toBase_sol. Its first component is continuous (§III.8); the second because $(x,v) \mapsto D_m(x)v$ is jointly continuous and $\zeta(a) = A_m(a)h$ satisfies $\zeta(a) = L_m(a)(h,0) + T_m(a)\zeta(a)$, whence $|\zeta(a) - \zeta(a')|_m \le 2|L_m(a)(h,0) - L_m(a')(h,0)|_m + 2|(T_m(a) - T_m(a'))\zeta(a')|_m$ Chapter3/ContractionGerm · ContractionGerm.continuousOn_Aop. With the derivative family $a \mapsto A_0(a)$ on germ(1) Chapter3/ContractionGerm · ContractionGerm.solDeriv, δ is an sc¹-germ there Chapter3/ContractionGerm · ContractionGerm.isSc1GermWith_sol **— Theorem 3.3.3 (preprint 3.30)**, on the shifted germ Chapter3/ContractionGerm · theorem_3_3_3. At parameters $a_0 \in N(1) \setminus N(2)$ the argument does not apply, and whether δ is differentiable there is neither needed nor addressed.

III.10 The contraction germ $f^{(1)}$: Lemma 3.3.4 (preprint 3.31)

Lemma 3.3.4 (preprint 3.31). The text: if f is of class sc^k, $k \geq 1$, and δ of class sc^j, then $f^{(1)}(v,b,u,w) = (u - B(v,u), w - DB(v,\delta(v))(b,w))$ is an sc⁰-contraction germ of class sc^{min(k−1, j)} whose solution germ is $T\delta$ — "as one verifies by comparing Tf with $f^{(1)}$", and "from the uniqueness". The formal version: if B is sc^{k+1} with derivative family D and δ is sc^j on germ(s), then $f^{(1)}$ is a contraction germ in working form of class sc^{min(k, j)} with solution germ $T\delta$ on $(N(2+s) \cap \widetilde{C}) \times P$ `Chapter3/GermIFT · ContractionGerm.lemma_3_3_4`. The parameters are $(v,b) \in \widetilde{C} \times P \subseteq P \times P$ (quadrant-like `Chapter3/GermIFT · ContractionGerm.isQuadrantLike_next`), the unknowns $(u,w) \in TW$, with levels $W_{m+1} \times W_m$.

Step 1: the germ and the regions. The germ of $f^{(1)}$ is $U_m^{(1)} = \{((v,b),(u,w)) : (v,u) \in U_{m+1},\ v \in N(m+1+s)\}$ `Chapter3/GermIFT · ContractionGerm.nextG`, and the regions are $N^{(1)}(m) = N(m+2+s) \times P$, $O^{(1)}(m) = O(m+1) \times W_m$, $K^{(1)}(m) = K(m+1) \times W_m$ `Chapter3/GermIFT · ContractionGerm.next`. They are unbounded in b and w, necessarily: the solution must be $(\delta(v), D\delta(v)b)$ for all $b \in P$.

Step 2: the map. With the reshuffling sc-operator $\sigma((v,b),(u,w)) = ((v,u),(b,w))$ `Chapter3/GermIFT · tangentShuffle` and $\Gamma((v,b),(u,w)) = ((v,\delta(v)),(b,w))$, where $v \mapsto \delta(v)$ is regarded as the *shifted* germ into W^1 `Chapter3/GermIFT · ContractionGerm.nextΓ`, formula (38) is $B^{(1)} = (\mathrm{pr}_1 \circ TB \circ \sigma, \mathrm{pr}_2 \circ TB \circ \Gamma)$ `Chapter3/GermIFT · ContractionGerm.nextB` `Chapter3/GermIFT · ContractionGerm.nextB_apply`. On level m, for $v \in N(m+1) \cap \widetilde{C}$, $B_m^{(1)}((v,b),(u,w)) = (B_{m+1}(v,u), L_m(v)(b,w))$ `Chapter3/GermIFT · ContractionGerm.levelB_next`: the point $(v, \delta_{m+1}(v))$ lies in U_{m+1}, which is where the parameter region $N(m+1+\cdot)$ comes from.

Step 3: contraction and invariance. For $(v,b) \in N^{(1)}(m)$ and $(u,w),(u',w') \in O^{(1)}(m)$, the two components contract by $\frac{1}{2}$: $B_{m+1}(v,\cdot)$ on $O(m+1)$, and $T_m(v)$ with $\|T_m(v)\| \leq \frac{1}{2}$ (Lemma 3.1.9 (preprint 3.9) (a)); $K(m+1) \times W_m$ is mapped into itself; $B^{(1)}(0) = 0$ `Chapter3/GermIFT · ContractionGerm.nextB_zero`.

Step 4: the class of $B^{(1)}$. TB is an sc^k-germ on the tangent germ; σ, the projections $(v,b,u,w) \mapsto v$ into P^1 and $\mapsto (b,w)$ `Chapter3/GermIFT · projV` `Chapter3/GermIFT · projBW` and the identification $P^1 \oplus W^1 = (P \oplus W)^1$ `Chapter3/GermIFT · shiftParamOp` are sc-operators; Γ is the pairing of (b,w) with $(v,\delta(v))$, which is sc^j because δ is and so is its shift into W^1 (Proposition 1.2.2 (preprint 1.17) for germs) `Chapter3/GermIFT · ContractionGerm.nextΓ_isSckGerm`; and Γ maps $U_m^{(1)}$ into $U_{m+1} \times E_m$ level by level `Chapter3/GermIFT · ContractionGerm.levelMap_nextΓ` `Chapter3/GermIFT · ContractionGerm.nextΓ_mapsToGerm`, as does σ `Chapter3/GermIFT · ContractionGerm.tangentShuffle_mapsToGerm`. The chain rule and the pairing rule of §III.1 give $B^{(1)}$ sc^j `Chapter3/GermIFT · ContractionGerm.nextB_isSckGerm`.

Step 5: the solution germ is $T\delta$. For $v \in N(2+s) \cap \widetilde{C}$ and $b \in P$, the point $(\delta_1(v), A_0(v)b) \in O(1) \times W_0$ is a fixed point of $B^{(1)}(v,b,\cdot)$ — the first component by $B_1(v,\delta_1(v)) = \delta_1(v)$, the second by the key identity of §III.9 — so by the uniqueness of Theorem 3.3.2 (preprint 3.29) *in* $O^{(1)}(0)$ it is the solution germ of $f^{(1)}$ at (v,b) `Chapter3/GermIFT · ContractionGerm.next_sol`, and since $A_0(v) = D\delta(v)$ (Theorem 3.3.3 (preprint 3.30)) this is $T\delta(v,b)$ `Chapter3/GermIFT · ContractionGerm.next_sol_eq_tangentMapWith`. The text's "from the uniqueness" is true because the regions of Step 1 contain $T\delta(v,b)$ for every b.

III.11 The induction: Lemma 3.3.5 (preprint 3.32), Theorem 3.3.6 (preprint 3.33), Corollary 3.3.7 (preprint 3.34), Theorem 3.35 (preprint)

The text proves Lemma 3.3.5 (preprint 3.32) by iterating $f \mapsto f^{(1)}$ and Theorem 3.3.6 (preprint 3.33) by contradiction (if δ were sc^j but not sc^{j+1}, $f^{(j)}$ would be sc¹, so $T^j\delta$ would be sc¹ by Theorem 3.3.3 (preprint 3.30)). The formal proof is the same argument as one induction on j, with the statement quantified over *all* contraction germs, since $f^{(1)}$ has parameter space $P \times P$ and fibre TW and the induction hypothesis must apply to it. It also keeps track of the germ.

Claim (S_j): for every contraction germ in working form, over any finite-dimensional parameter space, of class sc^k with $k \geq j$, the solution germ is an sc^j-germ on $\mathrm{germ}(s)$ *for some* s `Chapter3/GermIFT · ContractionGerm.sol_isSckGerm`. For $j = 0$ take $s = 0$ (Theorem 3.3.2 (preprint 3.29)). *Step* $j \to j+1$ `Chapter3/GermIFT · ContractionGerm.isSckGerm_succ_of_next`: let B be sc^k, $k = k' + 1 \geq j + 1$. (1) By (S_j) for f, δ is sc^j on $\mathrm{germ}(s)$. (2) By Lemma 3.3.4 (preprint 3.31) with this s, $f^{(1)}$ is a contraction germ of class sc^{min(k', j)} = sc^j. (3) By (S_j) for $f^{(1)}$, its solution germ is sc^j on its $\mathrm{germ}(s')$, whose level m is $(N(m + s' + 2 + s) \cap \widetilde{C}) \times P$. (4) By Theorem 3.3.3 (preprint 3.30) δ is sc¹ on $\mathrm{germ}(1)$, hence on $\mathrm{germ}(s + s' + 1)$ (restriction), whose tangent germ has level m equal to the set in (3). (5) On that set the solution germ of $f^{(1)}$ is $T\delta$ (Lemma 3.3.4 (preprint 3.31), Step 5); the identification of $T\widetilde{C} = \widetilde{C} \times P$ inside TP with $\widetilde{C} \times P$ inside $P \times P$ is the identity on every level, an sc-isomorphism `Chapter3/GermIFT · constTangentOp'` through which the chain rule transports the germ. (6) So δ is sc¹ with sc^j tangent germ: sc^{j+1} on $\mathrm{germ}(s + s' + 1)$. Since the proof does the same for every contraction germ, s and s' are the same function of j, so $s_0 = 0$, $s_{j+1} = 2s_j + 1$: $s_j = 2^j - 1$.

Theorem 3.3.6 (preprint 3.33) follows from (S_k): there is s such that δ is a solution germ and an sc^k-germ on $\mathrm{germ}(s)$ `Chapter3/GermIFT · theorem_3_3_6`; if B is sc-smooth this holds for every k `Chapter3/GermIFT · theorem_3_3_6_smooth`; and from Definition 3.3.1 (preprint 3.28) with balls through the working form `Chapter3/GermIFT · theorem_3_3_6_of_isSc0ContractionGerm`. Lemma 3.3.5 (preprint 3.32) is contained in the induction: its step from j to $j+1$ is items (2)–(3), without the iterated germs $f^{(j)}$. **Corollary 3.3.7 (preprint 3.34)**: if B is sc-smooth, for m and k take s for k; by Proposition 1.2.3 (preprint 1.18) for germs, δ maps $V_{m,k} := N(m + k + s) \cap \widetilde{C}$ into W_m by a C^k map `Chapter3/GermIFT · corollary_3_3_7` `Chapter3/GermLevels · IsSckGerm.contDiffOn_levels`, the inclusions of the constant structure being the identity `Chapter3/GermIFT · ScBanach.const_inclK`. **Theorem 3.35 (preprint)**: for $\widetilde{C}$ a partial quadrant (which is quadrant-like), the same `Chapter3/GermIFT · theorem_3_35_preprint` `Chapter3/GermIFT · theorem_3_35_smooth_preprint`.

What the shrinking means. Definition 3.1.2 (preprint 3.2) calls δ sc-smooth on a germ if it is sc^k *on that germ* for every k. The proof gives, for each k, the germ $\mathrm{germ}(2^k - 1)$; these shrink with k, their intersection is in general not a neighbourhood of 0, and no diagonal choice gives one germ, because being sc^k on a germ involves every level of it. So Theorem 3.3.6 (preprint 3.33) and the first sentence of Theorem 3.35 (preprint) hold for germs read up to shrinking, and the last sentences of Corollary 3.3.7 (preprint 3.34) and Theorem 3.35 (preprint) hold in the sense of Corollary 3.3.7 (preprint 3.34) — for every k on a germ depending on k — and are proved neither by the text nor here as `IsScSmoothGerm` on one germ. In the text's reading of germs the two senses coincide. Whether the shrinking is a property of contraction germs or only of the method is not decided: in the toy model of the exposition (Chapter3.md §3) the solution germ is sc-smooth on the original germ. Remark 3.3.8 (preprint 3.36), the quantitative form with radii ε_i shrinking with the level, is §III.58.

III.12 Local regularity and compactness: Theorem 3.5.1 (preprint 3.40)

Theorem 3.5.1 (preprint 3.40) (text): for $f = h + s$ on a relatively open $U \ni 0$ in C, h a basic germ, s an sc$^+$-germ with $s(0) = 0$, and $S = \{f = 0\}$, there are relatively open neighbourhoods $U \supseteq O(0) \supseteq O(1) \supseteq \cdots$ of 0 on level 0 with $\mathrm{cl}(S \cap O(m)) \subseteq C_m$. Of f only the W-component matters: S lies in the set of solutions of $w = B(a, w) - \mathbf{s}(a, w)$, $\mathbf{s} = P_W \circ s$. The formal statement uses only that B is an sc^0-contraction germ (not sc-smooth), that $\mathbf{s}$ is an sc^0-germ into W^1 with $\mathbf{s}(0) = 0$ (the sc$^+$ property level by level), and that $\widetilde{C}$ is closed (true for partial quadrants; the theorem is proved for convex $\widetilde{C}$, and closedness makes the solution set closed in E_0, where the text takes closures in C_0). It is *stronger* than the text's in the way Theorem 3.1.13 (preprint 3.12) needs: for any set S of solutions in U_0 there are nested open $O(i) \ni 0$ in E_0 and compact $K_i \subseteq E_i$ with $\mathrm{cl}_{E_0}(S \cap O(i)) = \iota_i(K_i)$ `Chapter3/LocalCompactness · theorem_3_5_1_compactness`; in particular $\mathrm{cl}(S \cap O(i)) \subseteq E_i$ `Chapter3/LocalCompactness · theorem_3_5_1_regularity`.

Step 0: the level-m contractions with a parameter. Let r_m be the radii of Definition 3.3.1 (preprint 3.28) for $\varepsilon = \frac{1}{2}$ and $c_m = \min(r_m, r_0/(\|\iota_m\| + 1))$, $\iota_m : W_m \to W_0$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.crad`, so that $|\iota_m y|_0 < r_0$ whenever $|y|_m \le c_m/2$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.norm_toBase_lt_rad` (the text's $\tau_1' < \tau_0/c_1$). Points with $a \in \widetilde{C}$, $|a|, |w|_m < r_m$ lie in U_m and satisfy the contraction estimate `Chapter3/LocalCompactness · IsSc0ContractionGerm.mem_G_of_lt` `Chapter3/LocalCompactness · IsSc0ContractionGerm.contr_of_lt`. Choose $\rho_m \in (0, c_m]$ with $|B_m(a, 0)|_m \le c_m/8$ for $|a| < \rho_m$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_cprad` `Chapter3/LocalCompactness · IsSc0ContractionGerm.cprad`. On $\Sigma_m = \{(a, z) : a \in \widetilde{C}, |a| < \rho_m, |z|_m \le c_m/8\}$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.Sigma` the maps $w \mapsto B_m(a, w) - z$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap` send the closed ball $\bar{B}_m(c_m/2)$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.cBall` into itself ($\frac{1}{2} \cdot \frac{c_m}{2} + \frac{c_m}{8} + \frac{c_m}{8} = \frac{c_m}{2}$) `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_mapsTo`, are $\frac{1}{2}$-contractions `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_contr` and depend continuously on (a, z) `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_continuousOn`. The parametrized fixed point theorem gives a continuous $\delta_m : \Sigma_m \to \bar{B}_m(c_m/2)$ with $\delta_m(a, z) = B_m(a, \delta_m(a, z)) - z$, unique in the ball `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_delta` `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta` `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta_fixed`.

Step 1: the lifts — the induction the text leaves to the reader. Let $\mathrm{Sol}(\tau)$ be the solutions $x = (a, w)$ with $a \in \widetilde{C}$, $|a| \le \tau$, $|w|_0 \le \tau$ `Chapter3/LocalCompactness · solSet`. *Claim:* for every m there are $\tau_m > 0$ and a map $L_m : E_0 \to E_m$ such that for every $x \in \mathrm{Sol}(\tau_m)$: $\iota_m L_m x = x$, $L_m x \in U_m$, $|(L_m x)_W|_m \le c_m/2$, and L_m is continuous on $\mathrm{Sol}(\tau_m)$ *for the level-*0 *topology of the source* `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_lift`. For $m = 0$, $L_0 = \mathrm{id}$. Given L_m: by continuity of $\mathbf{s}_m : U_m \to W_{m+1}$ at 0 (where $\mathbf{s}_m(0) = 0$ `Chapter3/LocalCompactness · levelMap_s_zero`) there is η with $|\mathbf{s}_m(y)|_{m+1} < c_{m+1}/8$ for $|y|_m < \eta$; by continuity of L_m at 0 there is θ with $|L_m x|_m < \eta$ for $x \in \mathrm{Sol}(\tau_m)$, $|x|_0 < \theta$. Put $\tau_{m+1} = \min(\tau_m, \rho_{m+1}/2, \theta/2)$ and, for $x = (a, w) \in \mathrm{Sol}(\tau_{m+1})$, $z = \mathbf{s}_m(L_m x)$, so $(a, z) \in \Sigma_{m+1}$, and $L_{m+1} x := (a, \delta_{m+1}(a, z))$. *It lifts x:* $w' = \iota_{m+1}\delta_{m+1}(a, z)$ solves $w' = B(a, w') - \mathbf{s}(x)$ (apply ι_{m+1}, which commutes with B, and use $\iota_m L_m x = x$); so does w; both are smaller than r_0 on level 0 by the choice of c_{m+1}, so the level-0 contraction gives $w' = w$ — the text's (43), on every level. *Continuity:* $x \mapsto (a, \mathbf{s}_m(L_m x))$ is continuous from the level-0 topology into Σ_{m+1}, and δ_{m+1} is continuous. The induction step needs the continuity of L_m: the bound $|\mathbf{s}_m(L_m x)|_{m+1} \le c_{m+1}/8$ requires $L_m x$ small on level m, whereas x is small only on level 0. In the text's level-1 step $L_0 = \mathrm{id}$ and this does not show; existence, bound and continuity of the lifts have to be proved together.

Step 2: compactness. $\mathrm{Sol}(\tau)$ is closed in E_0 ($\widetilde{C}$ closed, B and $\mathbf{s}$ continuous on level 0) `Chapter3/`

`LocalCompactness · IsSc0ContractionGerm.isClosed_solSet`. For $x \in \mathrm{Sol}(\tau_{m+1})$ the W-component is ι_{m+1} of a vector of norm $\le c_{m+1}/2$, and $\iota_{m+1} : W_{m+1} \to W_0$ is compact `Chapter3/LocalCompactness · ScBanach.toBase_succ_isCompactOperator`; with $|a| \le \tau_{m+1}$ in the finite-dimensional P, a sequence of lifts has a subsequence converging in E_{m+1} (parameters in a compact ball, W_0-parts in a compact image, and the continuity of the lift for the level-0 topology) `Chapter3/LocalCompactness · IsSc0ContractionGerm.isCompact_lift_image`.

Step 3: the neighbourhoods. $O(i)$ are the balls of radius ϱ_i, $\varrho_0 = \tau_1$, $\varrho_{i+1} = \min(\varrho_i, \tau_{i+2})$ — one level ahead of the text's $O(m)$, because compactness in E_i comes from boundedness in E_{i+1}. Then $\mathrm{cl}(S \cap O(i)) \subseteq \mathrm{Sol}(\tau_{i+1})$, every point of it is ι_i of the image under $E_{i+1} \to E_i$ of its lift $L_{i+1}x$, and K_i is the compact set of these images intersected with the closed preimage of $\mathrm{cl}(S \cap O(i))$.

For a basic germ plus an sc$^+$-germ (S the zero set of $h + \iota s$; its Q-component only shrinks S) this is the **local Theorem 3.1.13 (preprint 3.12)** `Chapter3/LocalCompactness · theorem_3_1_13_local`, and **Corollary 3.1.15 (preprint 3.13) in local form**: solutions $x_k \to 0$ in E_0 lie in $\mathrm{Sol}(\tau_m)$ for k large, and $L_m x_k \to L_m 0 = 0$ in E_m `Chapter3/LocalCompactness · corollary_3_1_15_local`.

III.13 Theorem 3.1.13 (preprint 3.12) and Corollary 3.1.15 (preprint 3.13) for sections

Theorem 3.1.13 (preprint 3.12). Let f be a section of a strong bundle with (f, x_0) an sc-Fredholm germ and $f(x_0) = 0$. Then there are nested open $O(i) \ni x_0$ in X such that $\mathrm{cl}_X\{x \in O(i) \mid f(x) = 0\}$ is the image of a compact subset of X_i (with the level topology of Part II, §II.6) under the inclusion $X_i \to X$ `Chapter3/SectionCompactnessG · theorem_3_1_13_preprint`. Of "f an sc-smooth section of a tame strong bundle" only $P \circ f = \mathrm{id}$ is used; the regularity enters through the germ data, and of the filling germ g only its continuity on level 0 near 0. The text calls the theorem "an immediate consequence of Theorem 3.40"; the proof makes explicit six steps.

(i) *Normalizations.* $f(x_0) = 0$ gives $\mathbf{f}(0) = 0$, so $g(0) = 0$ (filler (1)) and $s(0) = 0$ ($\iota s(0) = g(0)$). (ii) *The zeros correspond.* $f(x) = 0 \iff \mathbf{f}(\varphi(x)) = 0$ (§III.2) $\iff g(\varphi(x)) = 0$ (filler (1)) near x_0; zeros of g in U near 0 lie in O (filler (2)); and $g(u) = 0 \iff (h + t)(\psi(u)) = 0$, $t(\psi(u)) := (\Psi(u, s(u)))_2$, because $\Psi(u, k) = (\psi(u), \Gamma(u, k))$ `Chapter3/SectionCompactness · IsStrongBundleMap.apply_eq` with $\Gamma(u, \cdot)$ linear `Chapter3/SectionCompactness · IsStrongBundleMap.snd_sub` and injective `Chapter3/SectionCompactness · IsStrongBundleIso.snd_eq_zero_iff`. (iii) *t is an sc$^+$-germ with $t(0) = 0$:* the base map of a strong bundle map is sc-smooth `Chapter3/SectionCompactness · IsStrongBundleMap.base_isScSmooth`, $\psi'(0) = 0$ `Chapter3/SectionCompactness · IsStrongBundleIso.base_zero'`, and the fibre component of $\Psi[1]$ lifts that of Ψ to F_1' `Chapter3/SectionCompactness · IsStrongBundleMap.toBase_shiftFiber_snd`. Theorem 3.5.1 (preprint 3.40) needs t only as an sc^0-germ into F'^1 on the germ of the basic germ. With s a germ on G_s, t is one on the germ of the points of the germ of h that ψ^{-1} maps into G_s level by level `Chapter3/GermFilling · ScNbhdGerm.preimage` (germ composition with the whole-chart ψ^{-1} and $\Psi[1]$, §III.48), and the basic germ restricts to that germ `Chapter3/BasicGermShrink · IsBasicGerm.mono` (a form of the shrinking of §III.48). (iv) *Local compactness for a subset.* The local theorem for an arbitrary subset of the solution set of $h + \iota t$ `Chapter3/SectionCompactness · theorem_3_1_13_local_subset` is applied on the restricted basic germ to $\psi\varphi$ of the zeros near x_0; its $O'(i)$ are intersected with a small ball, so that E_0-closures stay inside $\psi(U)$ and the region of (ii) (the closure of a subset of the solution set is still compact). (v) *Back to X.* The E_0-closure consists of zeros of $h + \iota t$, a limit of zeros in the domain of continuity being a zero `Chapter3/SectionCompactness · eq_of_mem_closure_of_continuousOn`, hence of zeros of g in U, which lie in O. Here g must be continuous at the limits $\psi^{-1}(u')$: the

level-0 neighbourhood of the filler and conjugation identities is chosen inside the open set O_g with $(G_g)_0 = O_g \cap C$, so the limits lie in $(G_g)_0$, where the sc⁰-germ g is continuous. The whole-chart theorem `Chapter3/SectionCompactness · theorem_3_1_13_wholeChart` took t sc-smooth on all of U' and g continuous on U; nothing of either away from a neighbourhood of 0 was used. An X-closure of a subset of the chart domain could a priori leave it; compactness in E_0 shows it does not. (vi) *Levels.* A compact subset of E_i is carried to a compact subset of X_i because ψ^{-1} is sc⁰ and the inverse chart $O_i \to X_i$ `Chapter3/SectionCompactness · MPolyfold.levelChartSymm` is continuous for the level topology of X_i, the transition maps being sc⁰ `Chapter3/SectionCompactness · MPolyfold.continuousOn_levelChartSymm` — the paragraph after Definition 2.3.8 (preprint 2.14), for every level.

Corollary 3.1.15 (preprint 3.13). If $f(x_k) = 0$ and $x_k \to x_0$, fix m and take $O(m)$ and the compact $K_m \subseteq X_m$; for k large $x_k \in O(m)$, so $x_k = \iota y_k$ with $y_k \in K_m$, and $x_0 = \iota y_0$. X_m is not known to be first countable here (its metrizability, Theorem 2.3.10 (preprint 2.16), is not used), so subsequences are avoided: a sequence in a compact set whose images under a continuous injection converge, converges `Chapter3/SectionCompactness · tendsto_of_isCompact_of_tendsto_map`. So $y_k \to y_0$ in X_m `Chapter3/SectionCompactnessG · corollary_3_1_15_preprint`. The text states the corollary for an arbitrary limit x while the hypothesis concerns the germ at x_0; for an sc-Fredholm section and any limit x of zeros, x is a zero (the principal part is continuous in a chart `Chapter3/SectionCompactness · StrongBundle.IsScSmoothSection.isZeroAt_of_tendsto`), hence a smooth point (regularizing), hence the germ property holds at x and the first part applies `Chapter3/Chapter5BridgeG · corollary_3_1_15_section`.

III.14 Good position (Definition 3.1.24 (preprint 3.21), Lemma 3.8.8 (preprint 3.65), Proposition 3.1.25 (preprint 3.22), Lemma 3.5.9 (preprint 3.45), Proposition 3.8.12 (preprint 3.69))

Definition 3.1.24 (preprint 3.21). P is a *good complement* of the finite-dimensional N with respect to C if $E_0 = N \oplus P$ algebraically and for some $\varepsilon > 0$, $|p|_0 \le \varepsilon |n|_0$ implies $(n \in C \iff n + p \in C)$ `Chapter3/GoodPosition · IsGoodComplement`; N is *in good position* if $N \cap C$ has nonempty interior in N and N has a good complement `Chapter3/GoodPosition · IsInGoodPosition`. The text asks for an sc-complement (or, for non-smooth N, a topological one); none of Proposition 3.1.25 (preprint 3.22), Lemmas 3.5.9, 3.8.8 (preprint 3.45, 3.65) or Proposition 3.8.12 (preprint 3.69) uses more than the algebraic splitting, so only that is required. Where closedness is needed (Lemma 3.6.9 (preprint 3.55), Definition 3.2.1 (preprint 3.25)) it is asked there; it is not an added hypothesis, since the text's sc-complement is closed (§III.56).

Set-up in coordinates. After a presentation $E \cong \mathbb{R}^n \oplus W$, $C = \{\ell_i \ge 0\}$ for continuous functionals $\ell_1, \dots, \ell_n$ jointly onto $\mathbb{R}^n$ `Chapter3/GoodPosition · coordFun` `Chapter3/GoodPosition · coordFun_surj`. Inside N, $Q := N \cap C$ is the cone $\{\ell_i \ge 0\}$ `Chapter3/GoodPosition · quadCone`, closed `Chapter3/GoodPosition · quadCone_isClosed`. Let Σ be the indices i with $\ell_i(a) = 0$ for some nonzero $a \in Q$ `Chapter3/GoodPosition · vanIdx`; by definition, for $i \notin \Sigma$ and $x \in Q \setminus 0$, $\ell_i(x) > 0$ `Chapter3/GoodPosition · pos_of_not_mem_vanIdx`. Let $N \cap W = \{x \in N : \ell_i(x) = 0 \ \forall i\}$ `Chapter3/GoodPosition · nullSpace` and $Q_\Sigma = (\ell_i)_{i \in \Sigma} : N \to \mathbb{R}^\Sigma$ `Chapter3/GoodPosition · sigmaCoord`.

Lemma 3.8.8 (preprint 3.65). A good complement satisfies $\ell_i(p) = 0$ for $p \in P$, $i \in \Sigma$: take $a \in Q \setminus 0$ with $\ell_i(a) = 0$, scale p so that $|p| \le \varepsilon |a|$; $a \in C$ gives $a + p \in C$, so $\ell_i(p) = \ell_i(a + p) \ge 0$; the same for $-p$ `Chapter3/GoodPosition · lemma_3_8_8`. *Consequences.* Q_Σ is onto: for $k \in \mathbb{R}^\Sigma$ pick v with $\ell_i(v) = k_i$ and split $v = n + p$; then $Q_\Sigma(n) = k$ (the text's (55), preprint p. 125) `Chapter3/GoodPosition · sigmaCoord_surjective`.

If an interior point y of Q has $\ell_i(y) = 0$, then $\ell_i = 0$ on N ($y \pm tu \in Q$) `Chapter3/GoodPosition · restrFun_eq_zero_of_interior`.

Proposition 3.1.25 (preprint 3.22), the normal form. There are k, a subspace N_0 and a linear isomorphism $e : N \to \mathbb{R}^k \times N_0$ with $Q = e^{-1}([0,\infty)^k \times N_0)$, $k = \dim N - \dim(N \cap W)$, the $\mathbb{R}^k$-component of e vanishing exactly on $N \cap W$, and the vanishing coordinates of $e(x)$ as many as the vanishing $\ell_i(x)$ for $x \in Q \setminus N \cap W$ `Chapter3/GoodPosition · QuadrantNormalForm` `Chapter3/GoodPosition · exists_quadrantNormalForm`. Three cases. *Case A*, $Q = N$: every ℓ_i is ≥ 0 on a vector space, hence 0; $k = 0$ `Chapter3/GoodPosition · quadrantNormalForm_of_eq_univ` (a case the text's case distinction does not single out). *Case B*, $Q \neq N$, $\Sigma = \emptyset$: an interior point y is nonzero with all $\ell_i(y) > 0$; for $u \in N$ the set $\{t : u + ty \in Q\}$ is closed, contains all large t and is bounded below; at its minimum t^* some $\ell_i(u + t^*y) = 0$, so $u + t^*y = 0$: N is the line $\mathbb{R}y$ and Q a half-line `Chapter3/GoodPosition · quadrantNormalForm_of_vanIdx_empty` (the text's Lemma 3.8.11 (preprint 3.68), second case). *Case C*, $\Sigma \neq \emptyset$: first $Q = \{x \in N : \ell_i(x) \geq 0 \text{ for } i \in \Sigma\}$ — move from such an x towards an interior point y along $x + sy$; at the first parameter s^* with $x + s^*y \in Q$, if $s^* > 0$ the point $z = x + s^*y$ is nonzero (else $\ell_i(x) < 0$ for $i \in \Sigma$), so it has $\ell_i(z) > 0$ for all i ($\ell_i(z) \geq s^*\ell_i(y) > 0$ in Σ, by definition of Σ outside), and finitely many strict inequalities survive a small step back, contradicting minimality `Chapter3/GoodPosition · mem_quadCone_of_vanIdx`. This replaces the text's extreme rays (Lemmas 3.8.1–3.8.3 (preprint 3.62–3.64), cited from [21], and 3.66–3.68); those lemmas are now proved as well, and are still not used (§III.59). Then a linear right inverse g of Q_Σ gives $e(x) = (Q_\Sigma x, x - gQ_\Sigma x) \in \mathbb{R}^\Sigma \times \ker Q_\Sigma$ `Chapter3/GoodPosition · splitEquiv`, and $\ker Q_\Sigma = N \cap W$ `Chapter3/GoodPosition · quadrantNormalForm_of_vanIdx`. For a partial quadrant in good position this applies in a presentation `Chapter3/GoodPosition · IsInGoodPosition.quadrantNormalForm`, and choosing a basis of N_0 makes $(N, N \cap C)$ isomorphic to $(\mathbb{R}^{\dim N}, [0,\infty)^k \oplus \mathbb{R}^{\dim N - k})$ through an sc-isomorphism of constant structures `Chapter3/GoodPosition · scIsoOfLinearEquiv` `Chapter3/GoodPosition · preimage_eq_of_normalForm`: $N \cap C$ **is a partial quadrant in** N `Chapter3/GoodPosition · proposition_3_1_25`.

Lemma 3.5.9 (preprint 3.45) (linear form). A continuous linear isomorphism $S : E_0 \to E_0'$ with $x \in C \iff Sx \in C'$ carries a good complement P of N to one of $S(N)$, with $\varepsilon' = \varepsilon/(\|S\|\|S^{-1}\| + 1)$: for $|Sp| \leq \varepsilon'|Sn|$, $|p| \leq \|S^{-1}\|\varepsilon'\|S\||n| \leq \varepsilon|n|$ `Chapter3/GoodPosition · IsGoodComplement.map`; and the interior condition is transported by the homeomorphism $S|_N$ `Chapter3/GoodPosition · IsInGoodPosition.map`. The text states it for $T\varphi(0)$; since the coordinate change of Theorem 3.4.2 (preprint 3.38) is linear, this is the form §§III.18–III.23 need.

Proposition 3.8.12 (preprint 3.69). In the model $\mathbb{R}^n \oplus W$, for $x \in N \cap C$: $d_{N\cap C}(x) = d_C(x)$ if $x \notin N \cap W$, and $d_{N\cap C}(x) + \dim(N \cap W) = \dim N$ if $x \in N \cap W$ `Chapter3/GoodPosition · proposition_3_8_12` — read off the normal form ($d_{N\cap C}$ does not depend on the presentation, Lemma 1.3.4 (preprint 1.24)).

III.15 The degeneracy index along a graph: Lemma 3.8.13 (preprint 3.70)

Lemma 3.8.13 (preprint 3.70). In the model, for N in good position with good complement Y and $\tau : N \to E$ with values in Y, C^1 on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$, and $d_{N\cap C}(v) = d_C(v + \tau(v))$ if $v \notin N \cap W$ `Chapter3/GoodPositionGraph · lemma_3_8_13`. The text distinguishes $\dim N = 1$ (a contradiction argument with $\tau(v) = o(|v|)$) and $\dim N \geq 2$ (the sign of a directional derivative). The formal proof unifies them.

Ray decomposition. From the normal form, $u_i = e^{-1}(\mathbf{e}_i, 0)$ are finitely many nonzero points of Q,

and every $v \in Q \setminus N \cap W$ has some i and $s > 0$ with $v - su_i \in Q$ and $s|u_i| \le c|v|$ (take $s = \lambda_i > 0$ for $e(v) = (\lambda, n_0)$) `Chapter3/GoodPositionGraph · exists_rayDecomp` — the text's basis e^j of $\widetilde{N}$. *Smallness.* $D\tau(0) = 0$ and continuity of $D\tau$ on the convex Q make τ η-Lipschitz on $Q \cap B(\rho)$ for every η `Chapter3/GoodPositionGraph · exists_lipschitz_small` `Chapter3/GoodPositionGraph · quadCone_convex`. *The margin.* Choose $\eta \le \varepsilon$ with $\eta\|\ell_j\||u_i| < \ell_j(u_i)$ for all i and $j \notin \Sigma$ — a positive lower bound of finitely many positive numbers `Chapter3/GoodPositionGraph · exists_pos_le_of_forall_pos`. *Conclusion* `Chapter3/GoodPositionGraph · lemma_3_8_13_core`. For $|v| < \rho/(1+c)$: $|\tau(v)| \le \varepsilon|v|$, so $v + \tau(v) \in C$ (good complement). For $j \in \Sigma$, $\ell_j(\tau(v)) = 0$ (Lemma 3.8.8 (preprint 3.65)). For $v \notin N \cap W$, $w = v - su_i \in Q$, the segment $[w, v]$ lies in $Q \cap B(\rho)$ — which is why the radius is $\rho/(1+c)$, the text's "close to 0" — and for $j \notin \Sigma$,

$$\ell_j(v + \tau(v)) = \ell_j(w + \tau(w)) + s\ell_j(u_i) + \ell_j(\tau(v) - \tau(w)) \ge 0 + s\ell_j(u_i) - \|\ell_j\|\eta s|u_i| > 0.$$

So v and $v + \tau(v)$ have the same vanishing coordinates, and Proposition 3.8.12 (1) (preprint 3.69) gives the lemma in the model, with the coordinate functionals `Chapter3/GoodPositionGraph · coordCLM`. (The level Y_1 of the text's τ plays no role.) For a partial quadrant with an arbitrary presentation the same proof runs with the coordinate functionals of the presentation, with "$v \notin N \cap W$" read as $-v \notin C$ (§III.61).

III.16 The quadrant inverse function theorem (Theorem 3.8.15 (preprint 3.72), Proposition 3.8.17 (preprint 3.73), Lemmas 3.8.18, 3.8.21 (preprint 3.74, 3.77), Corollary 3.8.19 (preprint 3.75))

Interior points. For the model quadrant, x is interior iff all its half-line coordinates are positive `Chapter3/QuadrantInverse · interior_quadrant`, so $d_C(x) = 0$ exactly at interior points `Chapter3/QuadrantInverse · degIndex_eq_zero_iff`. A linear isomorphism S with $S(C) = C$ preserves d_C (Lemma 1.3.4 (preprint 1.24) for the presentation $T \circ S$) `Chapter3/QuadrantInverse · ScIso.ofCLE` `Chapter3/QuadrantInverse · degIndex_map_eq`. A linear map carrying a set with nonempty interior onto itself is onto `Chapter3/QuadrantInverse · surjective_of_image_eq` — the text's "$\mathbb{R}^n = C - C$".

Proposition 3.8.17 (preprint 3.73). The hypotheses (`InvData`): U relatively open and convex in the partial quadrant C of a finite-dimensional V; $f : U \to C$ of class C^j, $j \ge 1$; $\|Df(x) - 1\| \le \frac{1}{2}$; and x interior iff $f(x)$ interior — the only consequence of $d_C(x) = d_C(f(x))$ the proof uses `Chapter3/QuadrantInverse · InvData`. The text's $0 \in U$, $f(0) = 0$ are not needed. Conclusion: $f(U)$ is relatively open in C, f is injective on U, with an inverse of class C^j on $f(U)$ `Chapter3/QuadrantInverse · proposition_3_8_17`.

Lemma 3.8.18 (preprint 3.74). $x \mapsto x - f(x)$ is a $\frac{1}{2}$-contraction on the convex U (mean value inequality) `Chapter3/QuadrantInverse · InvData.norm_sub_sub_le`, which gives (70): $\frac{1}{2}|x - x'| \le |f(x) - f(x')| \le \frac{3}{2}|x - x'|$ `Chapter3/QuadrantInverse · InvData.half_norm_le` `Chapter3/QuadrantInverse · InvData.norm_sub_le`, and injectivity `Chapter3/QuadrantInverse · InvData.injOn`. *Openness*: if $B(x', 3r) \cap C \subseteq U$ then $B(f(x'), r) \cap C \subseteq f(U)$ `Chapter3/QuadrantInverse · InvData.ball_inter_subset_image`. The set $A = f(\bar{B}(x', 2r) \cap C)$ is compact ($\bar{B}(x', 2r) \cap C \subseteq U$ is compact in finite dimensions). Interior points first: for an interior $y_1 \in B(y', r)$ choose an interior $x_0 \in U$ with $f(x_0) \in B(y', r)$ — the text takes any interior $x_0 \in B(x', 3r) \cap C$, which does not suffice — and follow the segment y_t from $f(x_0)$ to y_1, which stays interior. $\{t : y_t \in A\}$ is closed and contains 0; at its maximum $t^* < 1$, $y_{t^*} = f(x^*)$ with x^* interior and $|x^* - x'| < 2r$, and at the interior point x^*, $Df(x^*)$ is a unit `Chapter3/QuadrantInverse · InvData.isUnit_deriv` and U a neighbourhood `Chapter3/QuadrantInverse · InvData.U_mem_nhds`, so the classical inverse function theorem maps neighbourhoods onto

neighbourhoods `Chapter3/QuadrantInverse · InvData.map_nhds_eq`, pushing t^* further — a contradiction. The compactness of A replaces the text's Cauchy sequence and also gives the remaining points by density. So $f(U)$ is relatively open `Chapter3/QuadrantInverse · InvData.image_relOpen`. **Corollary 3.8.19 (preprint 3.75)**: f and its inverse `Chapter3/QuadrantInverse · InvData.inv` (2-Lipschitz by (70) `Chapter3/QuadrantInverse · InvData.norm_inv_sub_le`) are continuous and inverse to each other `Chapter3/QuadrantInverse · corollary_3_8_19`.

Lemma 3.8.21 (preprint 3.77). The inverse g is differentiable within $f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$ `Chapter3/QuadrantInverse · InvData.hasFDerivWithinAt_inv`: with $x_0 = g(y_0)$, $x_0 + \delta = g(y_0 + h)$, the quotient $|g(y_0 + h) - g(y_0) - Df(x_0)^{-1}h|/|h|$ is $|\delta|/|h|$ (bounded by (70)) times the remainder quotient of f at x_0. The lemma's extra hypotheses (relatively open image, homeomorphism, invertible derivative) are consequences of those of Proposition 3.8.17 (preprint 3.73). *Regularity (the text: Lemma 3.8.20 (preprint 3.76) on the iterated tangent maps $T^k f$).* $x \mapsto [Df(x)]^{-1}$ is C^{j-1} (inversion is smooth on units) `Chapter3/QuadrantInverse · InvData.contDiffOn_inverse_deriv`, so $Dg = [Df]^{-1} \circ g$ and $g \in C^i$, $i < j$, give $g \in C^{i+1}$: g is C^j `Chapter3/QuadrantInverse · InvData.contDiffOn_inv`. Lemma 3.8.20 (preprint 3.76) is not needed here; it is proved for the comparison, with Lemma 3.8.21 (preprint 3.77) in the general form the text's iteration needs (§III.60).

Theorem 3.8.15 (preprint 3.72). U a relatively open neighbourhood of 0 in C, f of class C^j with $f(U) \subseteq C$, $f(0) = 0$, $Df(0)C = C$, $d_C(x) = d_C(f(x))$ on U: then f is a C^j-diffeomorphism between relatively open neighbourhoods U', V' of 0 `Chapter3/QuadrantInverse · theorem_3_8_15`. *Proof.* $Df(0)$ is onto, hence invertible, with $Df(0)^{-1}C = C$; $\tilde{f} = Df(0)^{-1}f$ maps into C, $D\tilde{f}(0) = 1$, and preserves d_C ($d_C \circ Df(0)^{-1} = d_C$), hence interior points. On $U' = B(0,\rho) \cap C$ with $\|D\tilde{f} - 1\| \le \frac{1}{2}$, Proposition 3.8.17 (preprint 3.73) applies, and $V' = Df(0)\tilde{f}(U')$ is relatively open since $Df(0)$ is a linear homeomorphism preserving C. The text's hypothesis (2) is used only through interior points; some such hypothesis is needed ($f(x,y) = (x, y + x^2)$ on $[0,\infty)^2$ satisfies (1) but $f(C)$ is not a neighbourhood of 0 in C).

III.17 The implicit function theorem in partial quadrants (Theorem 3.8.22 (preprint 3.78), Lemmas 3.8.23–3.8.26 (preprint 3.79–3.82))

Theorem 3.8.22 (preprint 3.78). U a relatively open neighbourhood of 0 in the partial quadrant C of a finite-dimensional V, $f : U \to F$ of class C^j, $j \ge 1$, with $f(0) = 0$, $Df(0)$ onto, $K = \ker Df(0)$ in good position with good complement Y. Then there are a relatively open $V' \ni 0$ in $K \cap C$, $\tau : K \to Y$ of class C^j on V' and $\varepsilon, \sigma > 0$ with (1) $\tau(0) = 0$, $D\tau(0) = 0$, and $k + \tau(k) \in U$, $f(k + \tau(k)) = 0$ for $k \in V'$; (2) $|\tau(k)| \le \sigma$ for $|k| \le \varepsilon$; (3) if $k \in V'$, $|k| \le \varepsilon$, $y \in Y$, $|y| \le \sigma$, $k + y \in U$ and $f(k + y) = 0$, then $y = \tau(k)$ `Chapter3/QuadrantIFT · theorem_3_8_22`. The two bold clauses are not in the text; the first is the content of the theorem (its proof has $\tau(k) = B(k, \tau(k))$), the second is needed for $f(k + y)$ to make sense.

Normalization (71)–(72). With $L = (Df(0)|_Y)^{-1}$, in coordinates $x = k + y \in K \oplus Y$, $Lf(k + y) = y - B(k, y)$ with $B(k, y) := y - Lf(k + y)$, $B(0,0) = 0$, $DB(0,0) = 0$. The data after normalization: on $K \times Y$ a convex set C_p with nonempty interior, a relatively open $U_p \ni 0$, B of class C^j on U_p with $B(0) = 0$, $DB(0) = 0$, the good-position estimate and the interior condition `Chapter3/QuadrantIFT · ContractionData`. Only convexity and interior are used; the quadrant enters through good position.

Lemmas 3.8.23, 3.8.24 (preprint 3.79, 3.80): the constants `Chapter3/QuadrantIFT · Radii` `Chapter3/QuadrantIFT · ContractionData.exists_radii`. With ε the good-position constant and $c_0 = \varepsilon/(2(1 + \varepsilon)) \le \frac{1}{2}$ `Chapter3/QuadrantIFT · Radii.c₀`, continuity of DB at 0 gives a, b with $\varepsilon a \le b$, the region

$R = \{|k| \le a, |y| \le b\} \cap C_p \subseteq U_p$ Chapter3/QuadrantIFT · Radii.R Chapter3/QuadrantIFT · Radii.R_sub_U and $\|DB\| \le c_0$ on R. R is convex Chapter3/QuadrantIFT · Radii.R_convex, so the mean value inequality gives $|B(q) - B(p)| \le c_0|q - p|$ on R Chapter3/QuadrantIFT · Radii.norm_sub_le Chapter3/QuadrantIFT · Radii.norm_B_sub_le_snd and $|B(p)| \le c_0|p|$ Chapter3/QuadrantIFT · Radii.norm_B_le. For $k \in V_0 = \{k \in K \cap C : |k| < a\}$ Chapter3/QuadrantIFT · Radii.V₀ and $|y| \le \varepsilon|k|$: good position puts $(k, y) \in C_p$, $|y| \le \varepsilon a \le b$, so $(k, y) \in R$ and $|B(k, y)| \le c_0(1 + \varepsilon)|k| = \frac{\varepsilon}{2}|k|$. So $B(k, \cdot)$ maps $X_k = \bar{B}_Y(\varepsilon|k|)$ into itself Chapter3/QuadrantIFT · Radii.mapsTo_ball and has a fixed point $\tau(k)$ there Chapter3/QuadrantIFT · Radii.exists_fixed Chapter3/QuadrantIFT · Radii.τ; any fixed point y with $(k, y) \in R$ equals $\tau(k)$ Chapter3/QuadrantIFT · Radii.eq_τ. (In Lemma 3.8.24 (preprint 3.80) the maximum is over $B((1 + \varepsilon)a) \cap C$, where DB is defined.)

Lemma 3.8.25 (preprint 3.81): continuity. $|\tau(k)| \le \frac{\varepsilon}{2}|k|$ Chapter3/QuadrantIFT · Radii.norm_τ_le. If $|k_0| \le 2|k|$, then $(k, \tau(k_0)) \in R$ and $|\tau(k) - \tau(k_0)| \le \frac{1}{2}|\tau(k) - \tau(k_0)| + c_0|k - k_0|$, so $|\tau(k) - \tau(k_0)| \le \varepsilon|k - k_0|$ — the text's estimate, whose argument needs $(k, \tau(k_0))$ in the domain. If $|k_0| > 2|k|$, then $|k - k_0| \ge \frac{1}{2}|k_0|$ and $|\tau(k) - \tau(k_0)| \le \frac{\varepsilon}{2}(|k| + |k_0|) \le \frac{3\varepsilon}{2}|k - k_0|$. In all cases the constant is $1 + 2\varepsilon$ Chapter3/QuadrantIFT · Radii.norm_τ_sub_le, and τ is continuous Chapter3/QuadrantIFT · Radii.continuousOn_τ. Since X_k depends on k, the parametrized fixed point theorem of §III.8 does not apply, and continuity is proved by hand, as in the text.

Lemma 3.8.26 (preprint 3.82): regularity. With $D_1B = DB \circ \mathrm{inl}$, $D_2B = DB \circ \mathrm{inr}$ Chapter3/QuadrantIFT · Radii.D₁ Chapter3/QuadrantIFT · Radii.D₂, $\|D_2B\| \le \frac{1}{2}$ on R Chapter3/QuadrantIFT · Radii.norm_D₂_le, $M = (1 - D_2B)^{-1}$ (Neumann) Chapter3/QuadrantIFT · Radii.M and $\Phi = MD_1B$ Chapter3/QuadrantIFT · Radii.Φ: from the fixed-point identities and the little-o composition of B with the Lipschitz $k' \mapsto (k', \tau(k'))$, τ is differentiable within V_0 with $D\tau(k) = \Phi(k, \tau(k))$ Chapter3/QuadrantIFT · Radii.hasFDerivWithinAt_τ Chapter3/QuadrantIFT · Radii.fderivWithin_τ. Φ is C^{j-1} on R Chapter3/QuadrantIFT · Radii.contDiffOn_Φ, so $\tau \in C^i$, $i < j$, gives $D\tau = \Phi \circ (\mathrm{id}, \tau) \in C^i$ and $\tau \in C^{i+1}$: τ is C^j on all of V_0 Chapter3/QuadrantIFT · Radii.contDiffOn_τ. The text iterates $f \mapsto f^{(1)}$ (the finite-dimensional model of Lemma 3.3.4 (preprint 3.31)) and concludes "on a possibly smaller neighbourhood"; in finite dimensions there are no levels to shift and no shrinking is needed. *Assembly* Chapter3/QuadrantIFT · ContractionData.exists_solution: $V' = V_0$, $\sigma = b$, ε below a; (1) $Lf(k + \tau(k)) = \tau(k) - B(k, \tau(k)) = 0$ and L is injective; (3) $f(k + y) = 0$ means $B(k, y) = y$ with $(k, y) \in R$.

III.18 Weak stability of basic germs: Theorem 3.4.2 (preprint 3.38)

Theorem 3.4.2 (preprint 3.38) (text): for a basic germ f and the principal part s of an sc⁺-section with $s(0) = 0$ there is a strong bundle isomorphism Φ covering an sc-diffeomorphism φ such that $(\Phi_*(f + s))^1$, with the levels raised by one, is a basic germ, and $n - N = n' - N'$. The text's proof: with $A = P_W \circ D_2 s(0)$, $1 + A$ is sc-Fredholm of index 0, with sc-splittings $W = C \oplus X = R \oplus Z$, $C = \ker(1 + A)$, $R = \mathrm{range}(1 + A)$; $P_W \circ s(a, w) = Aw + S(a, w)$ with S a small contraction in w on every level $m \ge 1$; and with $L = (1 + A)|_X : X \to R$ the equation becomes $x - L^{-1}P_2\bar{B}(a, c + x) = 0$ in the new parameters (a, c) and the new fibre variable x. The formal proof follows it with two observations: the coordinate change is *linear*, and the index raise is a *change of sc-structure*.

The linear part of s. For an sc¹-germ s into W^1, $A = D_2 s(0)$ is an sc⁺-operator whose level-m map is the level operator of Proposition 1.2.1 (preprint 1.16) at $0 \in U_{m+1}$ Chapter3/BasicClass · scPlusOfDeriv. The remainder $S(a, w) = s(a, w) - Aw$ on level $m + 1$ Chapter3/BasicClass · scPlusRemainder is, for a and w small depending on m and ε, an ε-contraction in w: $D_2S(0) = 0$ on level $m + 1$, the derivative of $s : U_{m+1} \to W_{m+1}$ is continuous in the operator norm (§III.1), and the mean value inequality on a ball

for fixed a concludes Chapter3/BasicClass · scPlusRemainder_contraction. The partial derivative of S in a does not vanish and plays no role. The text gives this one sentence ("S is … an arbitrary small contraction on every level $m \geq 1$").

The data and the splitting. The data of the theorem are the germ G, the W-component s of the sc$^+$-germ with $s(0) = 0$, and an sc-Fredholm splitting $W = C \oplus X = R \oplus Z$ of $T = 1 + A$ Chapter3/BasicClass · StabilityData. Such data exist: the identity is sc-Fredholm of index 0 (the trivial splitting) Chapter3/BasicClass · isScFredholm_id Chapter3/BasicClass · scIndex_id, so $1 + A$ is sc-Fredholm by Proposition 1.1.11 (preprint 1.10) Chapter3/BasicClass · StabilityData.exists_of_scPlus. With the projections $P_1 : W_m \to X_m$ along C_m, $P_2 : W_m \to R_m$ along Z_m Chapter3/BasicClass · StabilityData.projX Chapter3/BasicClass · StabilityData.projR, $L = T|_X : X_m \to R_m$ is bijective Chapter3/BasicClass · StabilityData.Lm_bijective, a topological isomorphism by the open mapping theorem Chapter3/BasicClass · StabilityData.Lequiv, and $L^{-1}P_2$ is an sc-operator $W \to X$ Chapter3/BasicClass · StabilityData.LinvPOp. The finite-dimensional C, $Z \subseteq W_\infty$ lift to every level Chapter3/BasicClass · ScBanach.finLift.

The coordinate changes. On the index-raised spaces, the new source is $E' = (P \oplus C) \oplus X^1$ Chapter3/BasicClass · StabilityData.E' and the new target $F' = (Q \oplus Z) \oplus X^1$ Chapter3/BasicClass · StabilityData.F' — the text's $\tau : Z \to C$, making the new finite-dimensional target $\mathbb{R}^N \oplus C$, is not applied. The base change $\psi(a, w) = ((a, (1 - P_1)w), P_1 w)$ Chapter3/BasicClass · StabilityData.split and its inverse $\psi^{-1}((a, c), x) = (a, c + x)$ Chapter3/BasicClass · StabilityData.glue are sc-operators inverse to each other Chapter3/BasicClass · StabilityData.glue_split Chapter3/BasicClass · StabilityData.split_glue; so are the fibre change $\Psi(q, w) = ((q, (1 - P_2)w), L^{-1}P_2 w)$ Chapter3/BasicClass · StabilityData.fiber and $\Psi^{-1}((q, z), x) = (q, z + (1 + A)x)$ Chapter3/BasicClass · StabilityData.fiberInv Chapter3/BasicClass · StabilityData.fiber_fiberInv Chapter3/BasicClass · StabilityData.fiberInv_fiber, and Ψ vanishes only at 0 Chapter3/BasicClass · StabilityData.fiber_eq_zero_iff. No general strong bundle isomorphism is needed.

The push-forward. The perturbed germ is $f + \iota s$ with the inclusion $W^1 \to Q \oplus W$ Chapter3/BasicClass · ScBanach.inrOp Chapter3/BasicClass · StabilityData.fsum, sc-smooth by the pairing and operator rules Chapter3/BasicClass · StabilityData.fsum_isScSmoothGerm. Its germ of neighbourhoods on E' is $\psi(U_{m+1})$, the preimage of U_{m+1} under ψ^{-1} Chapter3/BasicClass · StabilityData.pushGerm Chapter3/BasicClass · StabilityData.glue_mapsToGerm, and the push-forward is $(\Phi_*(f + s))^1 = \Psi \circ (f + \iota s)^1 \circ \psi^{-1}$ with the level-shifted operators Chapter3/BasicClass · ScOperator.shiftOp Chapter3/BasicClass · StabilityData.pushMap. It is sc-smooth: $(f + \iota s)^1$ is sc-smooth on the shifted germ (Proposition 1.2.2 (preprint 1.17) for germs) and Ψ, ψ^{-1} are sc-operators Chapter3/BasicClass · StabilityData.pushMap_isScSmoothGerm. With $\bar{B} = B - S$ Chapter3/BasicClass · StabilityData.Bbar, the W-component of $(f + \iota s)^1$ at a level-1 point is $(1 + A)w - \bar{B}(a, w)$ Chapter3/BasicClass · StabilityData.snd_levelMap_fsum, so the X-component of the push-forward is $x - \hat{B}((a, c), x)$ with $\hat{B}((a, c), x) = L^{-1}P_2\bar{B}(a, c + x)$ Chapter3/BasicClass · StabilityData.pushB Chapter3/BasicClass · StabilityData.snd_pushMap — the text's "desired formula". $\hat{B}$ is an sc^0-contraction germ on the index-raised structure: on level m of E', i.e. level $m + 1$ of W, B contracts by Definition 3.1.7 (preprint 3.7) and S by the remainder estimate, and $L^{-1}P_2$ is bounded Chapter3/BasicClass · StabilityData.levelMap_pushB Chapter3/BasicClass · StabilityData.pushB_isSc0ContractionGerm. So **the push-forward is a basic germ with contraction map** $\hat{B}$ Chapter3/BasicClass · theorem_3_4_2. Its zeros correspond to those of $f + \iota s$ under ψ, since Ψ is a linear isomorphism Chapter3/BasicClass · StabilityData.pushMap_eq_zero_iff. *The index*: $\mathrm{ind}(1 + A) = \mathrm{ind}(1) = 0$ (Proposition 1.1.11 (preprint 1.10) with its index), i.e. $\dim C = \dim Z$, so $n' - N' = (n + \dim C) - (N + \dim Z) = n - N$ Chapter3/BasicClass · StabilityData.finrank_Cs_eq_finrank_Zs.

The same on level 0. For §§III.19–III.20 the coordinate changes are also needed before the index raise: ψ, ψ^{-1}, Ψ, Ψ^{-1} on every level `Chapter3/BasicClass · StabilityData.split₀` `Chapter3/BasicClass · StabilityData.glue₀` `Chapter3/BasicClass · StabilityData.fiber₀`, the germ $\psi(U_m)$ `Chapter3/BasicClass · StabilityData.pushGerm₀`, and $g = \Phi_*(f+s) = \Psi \circ (f + \iota s) \circ \psi^{-1}$ on level 0 `Chapter3/BasicClass · StabilityData.pushMap₀`, sc-smooth `Chapter3/BasicClass · StabilityData.pushMap₀_isScSmoothGerm`, whose level-1 map is the index-raised push-forward `Chapter3/BasicClass · StabilityData.pushMap_eq_levelMap_pushMap₀`.

III.19 The linearization at nearby solutions: Corollary 3.5.3 (preprint 3.41)

Corollary 3.5.3 (preprint 3.41). For the data of Theorem 3.4.2 (preprint 3.38) with $f := h$ basic and $D(h+\iota s)(0)$ surjective, there is $r > 0$ such that every zero x of $h + \iota s$ in U_0 with $|x|_0 < r$ is $\iota(x_1)$ with $x_1 \in U_1$, and $D(h+\iota s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$ `Chapter3/SolutionLinearization · StabilityData.corollary_3_5_3`. The text's sc⁺-map s has an $\mathbb{R}^N$-component; it can be absorbed into the basic germ (as in §III.23), so the data carry only the W-component.

The derivative on U_1. For $x \in U_1$, $D(h+\iota s)(x) = Dh(x) + \iota Ds(x)$ — uniqueness of the derivative within U_1, then density of E_1 `Chapter3/SolutionLinearization · StabilityData.Dfs'_eq` — and $\mathrm{pr}_W Dh(x) = \mathrm{pr}_W - DB(x)$, from $\mathrm{pr}_W h = \mathrm{pr}_W - B$ on U_0 `Chapter3/SolutionLinearization · StabilityData.snd_Df_eq`; $D_2B(0) = 0$ `Chapter3/SolutionLinearization · StabilityData.DB_zero_snd`. *The decomposition.* $D(h+\iota s)(x) = D(h+\iota s)(0) + S(x) + K(x)$ with $S(x)(\alpha,\zeta) = (0, -D_2B(x)\zeta)$ `Chapter3/SolutionLinearization · StabilityData.Sop` `Chapter3/SolutionLinearization · StabilityData.Dfs'_eq_add` of norm $\le \varepsilon$ for $x \in U_1$ small on level 1, by Lemma 3.1.9 (preprint 3.9) (c) `Chapter3/SolutionLinearization · StabilityData.exists_norm_Sop_le`, and $K(x)$ `Chapter3/SolutionLinearization · StabilityData.Kop` compact: the sum of finite-rank operators through Q and through P and of $\iota(Ds(x) - Ds(0))$ through the compact inclusion $W_1 \to W_0$ `Chapter3/SolutionLinearization · StabilityData.Kop_isCompactOperator`.

Fredholm with index $n - N$ *at every small* $x \in U_1$. $D(h+\iota s)(0)$ is Fredholm of index $n - N$: $Dh(0)$ is (Proposition 3.1.10 (preprint 3.10) for the basic germ, §III.5) and $\iota Ds(0)$ is compact `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`. A small perturbation of a Fredholm operator is Fredholm: if v is a quasi-inverse of u and $\|v\|\|s\| < 1$, then $(1+vs)^{-1}v$ is a quasi-inverse of $u+s$ (Neumann series) `Chapter3/SolutionLinearization · isFredholm_add_of_norm_lt`, with the same index `Chapter3/SolutionLinearization · isFredholm_fredholmIndex_add_of_norm_lt`; a compact perturbation does not change the index `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`. So $D(h+\iota s)(x)$ is Fredholm of index $n-N$ for *every* $x \in U_1$ small on level 1, not only at solutions `Chapter3/SolutionLinearization · StabilityData.exists_isFredholm_Dfs'`.

Surjectivity: the compactness argument. Let N be a closed complement of $\ker D(h+\iota s)(0)$. Then $\ker D(h+\iota s)(x) \cap N = 0$ for $x \in U_1$ small on level 1 `Chapter3/SolutionLinearization · StabilityData.exists_ker_Dfs'_inf_eq_bot`. Otherwise there are $x_k \to 0$ in E_1 and unit vectors $(\alpha_k, \zeta_k) \in N$ in the kernels; the W-component of the equation reads $\zeta_k = D_1B(x_k)\alpha_k + D_2B(x_k)\zeta_k - \iota Ds(x_k)(\alpha_k,\zeta_k)$. α_k has a convergent subsequence (P finite-dimensional); $Ds(x_k)(\alpha_k,\zeta_k)$ is bounded in W_1 (the derivatives near 0 are uniformly bounded `Chapter3/SolutionLinearization · IsSc1GermWith.exists_norm_deriv_le`), so its image in W_0 has a convergent subsequence; $D_1B(x_k)\alpha_k \to D_1B(0)\alpha$ by the joint continuity of the tangent map `Chapter3/SolutionLinearization · IsSc1GermWith.tendsto_deriv`; and $\|D_2B(x_k)\| \to 0$. So ζ_k converges, $(\alpha_k,\zeta_k) \to (\alpha, z_0) \in N$ of norm 1 in $\ker D(h+\iota s)(0)$ — a contradiction. (The text uses the Neumann series $\zeta_k = \sum (D_2B(x_k))^l z_k$; here $|\zeta_k - z_k| \le \varepsilon_k \to 0$ suffices.) Then

$\dim\ker D(h+\iota s)(x) \le \dim\ker D(h+\iota s)(0) = n - N = \mathrm{ind}$, and an operator of finite index with $\dim\ker \le \mathrm{ind}$ is onto `Chapter3/SolutionLinearization · HasFiniteIndex.surjective_of_finrank_ker_le` — instead of the text's "$Df(x)|_N$ has index 0 and is injective". *Zeros are on level* 1 *and small there*: by the lifts of Theorem 3.5.1 (preprint 3.40), continuous at 0 (§III.12); the text's detour through "bounded on level 2, hence convergent on level 1" is not needed.

III.20 Local solvability: Lemma 3.5.10 (preprint 3.46), Theorem 3.5.6 (preprint 3.43), Lemmas 3.5.5, 3.5.8 (preprint 3.42, 3.44), Theorem 3.5.11 (preprint Corollary 3.47)

Lemma 3.5.10 (preprint 3.46) (Appendix 3.8.3 (preprint 3.8.2)). If $K' = \{(\alpha, L\alpha) : \alpha \in \widetilde{K}\}$, the image of the graph map `Chapter3/LocalSolvability · graphMap` (with $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ iff $\alpha \in \widetilde{C}$ `Chapter3/LocalSolvability · graphMap_mem_prod_iff`), has a good complement Y' with respect to $\widetilde{C} \oplus W$, then $\widetilde{K}$ has a good complement Z with respect to $\widetilde{C}$ such that $Z \times W$ is a good complement of K' `Chapter3/LocalSolvability · lemma_3_5_10`. *Proof.* $Z' := \mathrm{pr}_P(Y')$ satisfies $P = \widetilde{K} + Z'$; choose $Z \subseteq Z'$ with $P = \widetilde{K} \oplus Z$ and a linear right inverse $\rho : Z' \to Y'$ of the projection (bounded, finite dimensions); for $|z| \le \varepsilon'|\alpha|$, $\varepsilon' = \varepsilon/(\|\rho\| + 1)$, $y = \rho(z) = (z, w) \in Y'$ has $|y| \le \varepsilon|(\alpha, L\alpha)|$, so $(\alpha, L\alpha) \in C'$ iff $(\alpha, L\alpha) + y \in C'$, i.e. $\alpha \in \widetilde{C}$ iff $\alpha + z \in \widetilde{C}$. The text's intermediate complement Y_0 is not needed. Good position of K' gives good position of $\widetilde{K}$ `Chapter3/LocalSolvability · lemma_3_5_10_isInGoodPosition`.

Projecting solutions into the quadrant. A gap in the text's proof of Theorem 3.5.6 (3) (preprint 3.43): a solution $a = k + z$ ($k \in \widetilde{K}$, $z \in Z$) of $H(a) = 0$ is matched with the solution of Theorem 3.8.22 (preprint 3.78) by its uniqueness clause, which needs $k \in \widetilde{C}$ — not verified in the text. It holds because z is small compared with k: from $H(a) = 0 = H(0)$ and $DH(0)k = 0$, $|DH(0)z| = o(|a|)$; $DH(0)$ is injective on Z, hence bounded below there `Chapter3/LocalSolvability · exists_norm_le_of_injOn`; so $|z| \le \varepsilon|k|$ near 0, and the good-complement property gives $k \in \widetilde{C}$ from $k + z = a \in \widetilde{C}$ `Chapter3/LocalSolvability · exists_proj_mem_of_eq_zero`.

The data of Theorem 3.5.6 (preprint 3.43). The text states it for $g = \Phi_*(f + s)$; the formal statement is for any sc-smooth germ g on G with $g(0) = 0$ whose index-raised germ g^1 on the shifted germ is a basic germ with contraction map B `Chapter3/LocalSolvability · ShiftedBasicData`. Its solution germ δ (of the working form of B) takes values in W_1 `Chapter3/LocalSolvability · ShiftedBasicData.δ`, with $\mathrm{pr}_W g(a, \delta(a)) = 0$ on level 0 `Chapter3/LocalSolvability · ShiftedBasicData.snd_g_sol` and uniqueness in the contraction region `Chapter3/LocalSolvability · ShiftedBasicData.eq_δ`. Put $H(a) = \mathrm{pr}_Q\, g(a, \delta(a))$ `Chapter3/LocalSolvability · ShiftedBasicData.H`. *Regularity of* H (the text: "H restricted to U_i' is of class C^{i-1}"): for every $j \ge 1$ there is a relatively open $V \ni 0$ in $\widetilde{C}$ on which H is C^j — by Corollary 3.3.7 (preprint 3.34) for g^1, δ lifts to a C^j map into W_{2j}, and $g : E_{2j} \to F_j$ is C^j by Proposition 1.2.3 (preprint 1.18) for germs `Chapter3/LocalSolvability · ShiftedBasicData.H_contDiffOn`. A single C^1 neighbourhood V_1 carries the derivative data `Chapter3/LocalSolvability · ShiftedBasicData.Deriv` `Chapter3/LocalSolvability · ShiftedBasicData.exists_deriv`: $D\delta(0) : P \to W_1$, $L = \iota D\delta(0)$, and $DH(0) = \mathrm{pr}_Q Dg(0)(\alpha, L\alpha)$ `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.DH₀` `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.hasFDerivWithinAt_H`, by the chain rule for $a \mapsto g(a, \delta(a))$ `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.hasFDerivWithinAt_comp`.

Lemma 3.5.8 (preprint 3.44). "Recalling from Lemma 3.9 that $D_2B(0) = 0$": Lemma 3.1.9 (preprint 3.9) gives this for g^1, on level 1; the kernel of $Dg(0)$ is computed on level 0. So first the W-component

of $Dg(0)(0,\zeta)$ is ζ for $\zeta \in W_1$, by Lemma 3.1.9 (preprint 3.9) (b) for g^1, and then for all $\zeta \in W_0$ by density Chapter3/LocalSolvability · ShiftedBasicData.Deriv.snd_Dg_inr. Differentiating $\mathrm{pr}_W g(a,\delta(a)) = 0$ at 0 gives $\mathrm{pr}_W Dg(0)(\alpha, L\alpha) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.snd_Dg_graph. Hence $Dg(0)(\alpha,\zeta) = 0$ iff $\zeta = L\alpha$ and $DH(0)\alpha = 0$: $\ker Dg(0) = \{(\alpha, D\delta(0)\alpha) : \alpha \in \ker DH(0)\}$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.ker_Dg_eq, and $DH(0)$ is onto when $Dg(0)$ is Chapter3/LocalSolvability · ShiftedBasicData.Deriv.DH₀_surjective.

Theorem 3.5.6 (preprint 3.43) Chapter3/LocalSolvability · theorem_3_5_6. Hypotheses: $\widetilde{C}$ a partial quadrant, $Dg(0)$ onto, $K' = \ker Dg(0)$ in good position to $\widetilde{C} \oplus W$, and *the regularity of Lemma 3.5.5 (preprint 3.42) as an explicit hypothesis*: for every $\varepsilon > 0$ there is $r > 0$ such that every zero x of g in U_0 with $|x|_0 < r$ has W-component $\iota(w_1)$, $|w_1|_1 < \varepsilon$. *The choices* Chapter3/LocalSolvability · ShiftedBasicData.Sol Chapter3/LocalSolvability · ShiftedBasicData.exists_sol: $\widetilde{K} = \ker DH(0)$ with K' its graph (Lemma 3.5.8 (preprint 3.44)); the good complement Z of $\widetilde{K}$ with $Z \oplus W$ good for K' (Lemma 3.5.10 (preprint 3.46)); and the solution $\sigma : V' \to Z$ of $H(k+\sigma(k)) = 0$ with its uniqueness constants, from Theorem 3.8.22 (preprint 3.78) applied to H (§III.17). Then $\tau(y) = (\sigma(v), \delta(v+\sigma(v)) - Lv)$ for $y = (v, Lv)$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ on $V = \pi^{-1}(V')$, relatively open in $K' \cap C'$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.V Chapter3/LocalSolvability · ShiftedBasicData.Sol.V_relOpen, with values in $Z \oplus W$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ_mem_Y, so that $y + \tau(y) = (a, \delta(a))$ with $a = v + \sigma(v)$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.coe_add_τ. (3), first half: $y + \tau(y)$ lies in U_0 and is a zero, since $\mathrm{pr}_W g(a,\delta(a)) = 0$ and $\mathrm{pr}_Q g(a,\delta(a)) = H(a) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.sol_mem_and_eq_zero. Regularity: τ is C^1 on V Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ_contDiffOn with W-component on level 1 Chapter3/LocalSolvability · ShiftedBasicData.Sol.snd_τ_mem_range, and $D\tau(0)\eta = -D\delta(0)\pi(\eta) + D\delta(0)\pi(\eta) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.hasFDerivWithinAt_τ Chapter3/LocalSolvability · ShiftedBasicData.Sol.fderivWithin_τ — (2). (3), second half Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_eq_add_τ: a zero $x = (a, w)$ near 0 has, by the regularity hypothesis, w on level 1 and small there, so $w = \delta(a)$ by the uniqueness of the solution germ on level 1, so $H(a) = 0$; write $a = k + z$ with the projection along Z Chapter3/LocalSolvability · ShiftedBasicData.Sol.projK; then $k \in \widetilde{K} \cap \widetilde{C}$ (the gap above) and $z = \sigma(k)$ by the uniqueness clause of Theorem 3.8.22 (preprint 3.78); so $x = y + \tau(y)$ for $y = (k, Lk)$. *Clause (1), "τ is an sc-smooth germ"*, is proved level-wise: for every $k \geq 1$ there are a relatively open $V_k \subseteq V$ containing 0 and a C^k map $\tau_k : V_k \to E_k = P \oplus W_k$ with $\iota_k \tau_k = \tau$ on V_k Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_contDiffOn_level — the reading of Corollary 3.3.7 (preprint 3.34) and Definition 3.2.1 (ii) (preprint 3.25). The proof applies Theorem 3.8.22 (preprint 3.78) to H at regularity k on the neighbourhood of its regularity lemma, and identifies the C^k solution σ_k with σ near 0 through the uniqueness clause (3) of Theorem 3.8.22 (preprint 3.78) — which needs σ continuous at 0 with $\sigma(0) = 0$, so that $|\sigma(y)|$ is below the uniqueness constant for y small; the text says nothing about why the solutions produced for the different k coincide. The W-component $\iota\delta(a) - L\pi(y)$, $a = \pi y + \sigma(\pi y)$, lifts to W_k through the C^k lift of δ into W_{2k}, with L the derivative of that lift at 0 composed with the inclusion (uniqueness of the derivative of $\iota\delta$ at 0). The full statement is Chapter3/LocalSolvability · theorem_3_5_6_sc_smooth; the statement without (1) is kept for Theorem 3.6.8 (preprint 3.54).

Lemma 3.5.5 (preprint 3.42). For the push-forward $g = \Phi_*(f+s)$ of Theorem 3.4.2 (preprint 3.38) — packaged as the data of Theorem 3.5.6 (preprint 3.43), g on level 0 with index-raised basic germ $(\Phi_*(f+s))^1$ Chapter3/LocalSolvability · StabilityData.toShiftedBasicData — the regularity hypothesis holds: a zero x' of g in $\psi(U_0)$ with $|x'|_0$ small has $\psi^{-1}(x')$ a zero of $f + \iota s$, which by the lifts of Theorem 3.5.1 (preprint 3.40), continuous at 0, lies on level 1 and is small there; ψ is linear on level 1 Chapter3/

LocalSolvability · StabilityData.lemma_3_5_5.

Theorem 3.5.11 (preprint Corollary 3.47) Chapter3/LocalSolvability · theorem_3_5_11_sc_smooth Chapter3/LocalSolvability · theorem_3_5_11. For a basic germ f and the data of Theorem 3.4.2 (preprint 3.38) with $D(f+\iota s)(0)$ surjective and kernel K in good position to $C = \widetilde{C} \oplus W$: a good complement Y, V relatively open in $K \cap C$, $\sigma : V \to Y$ of class C^1 with values on level 1, $\sigma(0) = 0$, $D\sigma(0) = 0$, the graph consisting of zeros of $f + \iota s$, an open $U'' \ni 0$ in which every zero lies on the graph, and for every k a C^k lift of σ into E_k near 0. *Proof.* Theorem 3.5.6 (preprint 3.43) for $g = \Phi_*(f+s)$, with Lemma 3.5.5 (preprint 3.42) for its regularity hypothesis. Since the coordinate change is linear, $Dg(0) = \Psi \circ D(f+\iota s)(0) \circ \psi^{-1}$ (chain rule for sc-operators) Chapter3/LocalSolvability · StabilityData.scDerivGerm_pushMap₀, so surjectivity and the kernels correspond, $\ker Dg(0) = \psi(K)$; good position of $\psi(K)$ follows by Lemma 3.5.9 (preprint 3.45) (finite dimensionality of K from Lemma 3.5.8 (preprint 3.44)); and the conclusion is transported back by ψ^{-1}: $Y = \psi^{-1}(Y')$, $V = \psi^{-1}(V')$, $\sigma = \psi^{-1}\tau\psi$. The text transports the solution set back with the nonlinear φ and justifies the good position with the degeneracy index (Lemma 3.8.13 (preprint 3.70), Theorem 3.8.15 (preprint 3.72)); with ψ linear neither is needed. The transport uses that $\widetilde{C} \oplus C$ is a partial quadrant for finite-dimensional C Chapter3/LocalSolvability · IsPartialQuadrant.const_prod_univ.

III.21 The linearization at a zero (Proposition 3.6.5 (preprint 3.52)) and good position of a germ (Definition 3.6.6 (preprint 3.53))

Proposition 3.6.5 (preprint 3.52) (text): if f is an sc-Fredholm section of a tame strong bundle and $f(x) = 0$, then $Tf(x) : T_xX \to Y_x$ is an sc-Fredholm operator. Formalized in the chart of Definition 3.1.11 (preprint 3.11) centred at a zero x_0, with the level-0 Fredholm property Chapter3/FredholmGermLinearizationG · proposition_3_6_5_preprint Chapter3/FredholmGermLinearizationG · proposition_3_6_5_local: $Dg(0) : E_0 \to F_0$ and $Tf(x_0) = Dg(0)|_{T_0O} : T_0O \to K_0$ are Fredholm with $\operatorname{ind} Tf(x_0) = \operatorname{ind} Dg(0)$. Here $Dg(0)$ is the value at 0 of a derivative family Dg of the filling germ on a germ G with $G_0 \subseteq U$, the same for every family and every germ (§III.4); the statement exposes G and Dg, where the whole-chart statement Chapter3/FredholmGermLinearization · proposition_3_6_5_wholeChart has the derivative on U, a junk value for a germ. (The sc-Fredholm property of $Tf(x)$ would need $T_0O = Dr(0)E$ and $K_0 = \rho(0)F$ as sc-Banach spaces, not set up. The statements quantify the index as "$n - N$ for some natural numbers n, N", which every integer satisfies; the identification with $\dim P - \dim Q$ of the conjugating basic germ is made inside the proof, by `proposition_3_1_10_basic_level_zero`, and not exported.)

Step (a): the derivative of the conjugation Chapter3/FredholmGermLinearizationG · IsConjugateToBasicGerm.scDeriv_isFredholm. Let $k = g - \iota s$, an sc¹-germ on $G_g \cap G_s$ (§III.48) with $k(0) = 0$ ($s(0) = g(0)$ in Definition 3.1.11 (preprint 3.11), $g(0) = \mathbf{f}(0) = 0$ by the filler) and $(\Psi(u, k(u)))_2 = h(\psi(u))$ near 0. Linearizing needs: the chain rule for the germ $h \circ \psi$ on a germ small enough that ψ maps it into the germ of h (§III.1, `ScNbhdGerm.preimage`); the derivative of $u \mapsto \Gamma(u)k(u)$ at 0, where $\Gamma(u) = (\Psi(u,\cdot))_2$ is a bounded operator Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM and $D(\mathrm{pr}_{F'} \circ \Psi)(0,0)(h,k) = \Gamma(0)k$ (as in §III.4) Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_zero, so the derivative is $\Gamma(0)Dk(0)$ because $k(0) = 0$ Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_comp — for the germ k by the chain rule with the whole-chart outer map $\mathrm{pr}_{F'} \circ \Psi$, the two derivative families agreeing at 0 on $G_k \cap \psi^{-1}(G)$ Chapter3/FredholmGermLinearizationG · IsStrongBundleIso.fiberCLM_comp_deriv_zero_eq_germ, so no shrinking of the basic germ is needed at 0; and the invertibility of

$D\psi(0)$ and $\Gamma(0)$ (inverses $D\psi'(0)$, $\Gamma'(0)$, chain rule for $\psi' \circ \psi = \mathrm{id}$). Then $\Gamma(0)Dk(0) = Dh(0)D\psi(0)$, so $Dk(0) = \Gamma(0)^{-1}Dh(0)D\psi(0)$ is Fredholm with the index of $Dh(0)$: Fredholm operators and their index are stable under composition with isomorphisms on both sides `Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv`, and $Dh(0)$ is Fredholm by Proposition 3.1.10 (preprint 3.10) for the basic germ `Chapter3/FredholmGermLinearization · proposition_3_1_10_basic_level_zero`. *Step (b)*: $Dg(0) = Dk(0) + \iota Ds(0)$ — for germs this is algebra at 0: $g - \iota s$ has the derivative family $Dg - \iota Ds$ on $G_g \cap G_s$ `Chapter3/FredholmGermLinearizationG · IsSc1GermWith.sub_toBase_shift Chapter3/FredholmGermLinearizationG · IsSc1GermWith.deriv_zero_eq_add_scPlus` (on U it was the uniqueness of the derivative within U_1 and density) — a compact perturbation since $\iota Ds(0)$ is the level-0 map of an sc$^+$-operator (s is an sc^1-germ into F^1 and 0 is smooth); the index is unchanged (Part I, §I.9). *Step (c)*: Proposition 3.1.5 (preprint 3.5) (§III.4) transfers both to $Tf(x_0)$; it needs $g(0) = 0$.

Definition 3.6.6 (preprint 3.53) (text): (f, x) with $f(x) = 0$ is in good position if (1) $f'(x) : T_xX \to Y_x$ is onto and (2) if $d_X(x) \geq 1$, $\ker f'(x)$ is in good position to the partial cone C_xX in T_xX. In the chart: Definition 3.1.24 (preprint 3.21) *inside a subspace* $T \subseteq E_0$ — $N \subseteq T$, interior of $N \cap C$ in N nonempty, and a good complement $P \subseteq T$ of N in T `Chapter3/GoodPositionGerm · IsInGoodPositionIn` (for $T = E_0$ it is Definition 3.1.24 (preprint 3.21) `Chapter3/GoodPositionGerm · isInGoodPositionIn_top_iff`); and the germ is in good position if $Tf(0) = Dg(0)|_{T_0O}$ is onto and $\ker Dg(0)$ is in good position to $T_0O \cap C$ inside $T_0O = Dr(0)E$ `Chapter3/GoodPositionGerm · IsGoodPositionAt Chapter3/GoodPositionGerm · IsGoodPositionAt.surjective_fillerRestrict`. Here $\ker Dg(0) \subseteq T_0O$, because $(1 - \rho(0))Dg(0)$ is injective on $\ker Dr(0)$ (filler (3)) and $(1-\rho(0))Dg(0)Dr(0) = 0$ `Chapter3/GoodPositionGerm · IsFillerOf.ker_deriv_le_range`, and it is the kernel of $Tf(0)$ `Chapter3/GoodPositionGerm · IsFillerOf.map_ker_fillerRestrict` — the text's "$\ker Tf(0) \oplus \{0\} = \ker Dg(0)$ by the properties of the filler". The case distinction $d_X(x) \geq 1$ is dropped: at an interior point the partial cone is the whole tangent space and (2) holds for every subspace. For a filling germ the condition is one on the operator $Dg(0)$ (the text's $f'(x)$ is defined through the filler at the centre only), which the chart notion takes as a parameter `Chapter3/GoodPositionGermG · IsGoodPositionAtGerm` — the notion above is the case of the derivative on U `Chapter3/GoodPositionGermG · isGoodPositionAt_iff_germ` — and the two kernel facts hold with the same proofs `Chapter3/GoodPositionGermG · IsFillingGerm.ker_deriv_le_range Chapter3/GoodPositionGermG · IsFillingGerm.map_ker_fillerRestrict`. For a section, (f, x_0) is in good position if it has the data of Definition 3.1.11 (preprint 3.11) at x_0 and a derivative family Dg of the filling germ on a germ inside U, with the germ in good position at $Dg(0)$ in that chart `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm`; as $Dg(0)$ does not depend on the family, "for some family" and "for every family" agree `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm.forall_isGoodPositionAtGerm`; it is then an sc-Fredholm germ `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm.isScFredholmGerm`. The definition is in a chart, not in T_xX; the transport of C_xX to T_xX (Definition 2.5.8 (preprint 2.38)) is not formalized.

III.22 Good position along a graph: Lemma 3.6.9 (preprint 3.55)

Lemma 3.6.9 (preprint 3.55) (text, Appendix 3.8.4 (preprint 3.8.3)): for N in good position to C with good complement Y, $V \subseteq N \cap C$ relatively open and $\tau : V \to Y_1$ of class C^1 with $\tau(0) = 0$, $D\tau(0) = 0$, for v near 0: (1) $v + \tau(v) \in C_1$; (2) Y is a topological complement of $N_v = \{n + D\tau(v)n\}$; (3) for some $\gamma_v > 0$, $n \in C_x \iff n + y \in C_x$ for $n \in N_v$, $y \in Y$, $|y|_0 \leq \gamma_v |n|_0$, $x = v + \tau(v)$. *Formalized* `Chapter3/GoodPositionOpen · lemma_3_6_9_preprint` in the model $\mathbb{R}^n \oplus W$, with the core in coordinates for any C cut out by functionals `Chapter3/GoodPositionOpen · lemma_3_6_9_core`: for N in good position with good complement Y, Y **closed**, and $\tau : N \to E$ with values in Y, C^1 on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$ and Y is a good complement of N_v (the subspace $\{n + Ln\}$

for $L = D\tau(v)$ `Chapter3/GoodPositionOpen · graphSpace`) with respect to the partial quadrant $C_{v+\tau(v)}$.

The added hypothesis. The step "$D\tau(v)n \in Y$ for all $n \in N$" needs Y closed: the derivative within $N \cap C$ at a point of unique differentiability of a map with values in a closed subspace takes values there, because the tangent cone of a subset of a closed subspace lies in it `Chapter3/GoodPositionOpen · tangentConeAt_subset_of_subset_closed Chapter3/GoodPositionOpen · fderivWithin_mem_of_mapsTo`; for a non-closed algebraic complement it fails. The text's $Y = Y_0$ is a topological complement, hence closed. With Y closed, (2) is linear algebra: $E_0 = N_v \oplus Y$ whenever $E_0 = N \oplus Y$ and L takes values in Y `Chapter3/GoodPositionOpen · graphSpace_isCompl`, and N_v is closed (finite-dimensional) `Chapter3/GoodPositionOpen · graphSpace_isClosed`.

The estimate. (1) is Lemma 3.8.13's (preprint 3.70) smallness: $|\tau(v)| \le \varepsilon|v|$ near 0 and good position. For (3), take $\varepsilon_0 = \min(\varepsilon, 1)$ and the neighbourhood where $\|D\tau(v)\| \le \varepsilon_0/4$; then $\gamma = \varepsilon_0/4$ works *for every* v, since $\gamma(1+\varepsilon_0/4) + \varepsilon_0/4 \le \varepsilon_0$ (the text: $\gamma_v(1+\varepsilon) + \varepsilon < \gamma$, with γ_v chosen after v). If $v \in N \cap W$, $C_x = C$ and the estimate transfers the good-complement property from N to N_v. If $v \notin N \cap W$, no normal form is needed: if $\ell_i(v) = 0$ then $i \in \Sigma$, so ℓ_i vanishes on Y (Lemma 3.8.8 (preprint 3.65)), and $\ell_i(v + \tau(v)) = 0$ iff $\ell_i(v) = 0$ (Lemma 3.8.13 (preprint 3.70)); so $\ell_i(h + y) = \ell_i(h)$ for every coordinate i that defines C_x, and (3) holds for every γ. The text's cases $\dim N = 1$, $\dim N > 1$ are not needed, and the sc-level of the values of τ plays no role. The formal τ is given on all of $N \cap C$ rather than on a neighbourhood V; the conclusion is local.

III.23 Theorem 3.6.8 (preprint 3.54): the linear part and the graph description of the solution set

Theorem 3.6.8 (preprint 3.54) (text): for an sc-Fredholm section f of a tame strong bundle with $f(x) = 0$ and (f, x) in good position there is an open $V \ni x$ such that $S = \{y \in V : f(y) = 0\}$ satisfies (1) (f, y) is in good position at every $y \in S$, and (2) S is a sub-M-polyfold with the structure of a manifold with boundary with corners. The proof: the linear part (surjectivity and good position transported from T_xX to the chart and the push-forward); (3), the graph description of S from Theorem 3.5.11 (preprint Corollary 3.47) for $\Psi_*(g) = h + t$; then Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) for (1), and Theorem 3.2.3 (preprint 3.27) for (2). Formalized: the linear part and (3).

The linear part `Chapter3/GoodPositionGermG · theorem_3_6_8_linear`. In the chart, with the germ in good position and $(1 - Dr(0))E \subseteq E_0$ (the minimal subspace at the corner 0; Proposition 2.5.3 (preprint 2.35) for the tame retraction r `Chapter3/GoodPositionGermG · theorem_3_6_8_linear_of_isTameWith Chapter2/TameRetracts · proposition_2_5_3`): $Dg(0)$ is onto (Proposition 3.1.5 (preprint 3.5) for the filling germ, §III.4, from the surjectivity of $Tf(0)$), and $\ker Dg(0)$ is in good position to C in E; only the filler at 0 enters, and r is whole-chart, so $Dr(0)$ is a genuine derivative. The latter is the *transport lemma*: for a bounded idempotent π with $(1-\pi)E_0$ inside the minimal subspace at 0, good position inside πE_0 gives good position in E_0, with the good complement $P \oplus (1-\pi)E_0$ and the constant $\varepsilon/(\|\pi\| + 1)$ `Chapter3/GoodPositionGerm · IsInGoodPositionIn.isInGoodPosition`. Two points the text passes over: for $p = z + k$ with $|p| \le \varepsilon'|n|$ the T_0O-component is $z = Dr(0)p$, so $|z| \le \|Dr(0)\|\varepsilon'|n|$ — the splitting must be topological; and $\ker Dr(0) \subseteq E_0$ is used only as "the $\mathbb{R}^n$-coordinates of k vanish, so $n+z+k \in C \iff n+z \in C$". *This inclusion is the only place where tameness enters.*

$C' = T\psi(0)C$. Not part of the data of a conjugation (§III.6); it is a theorem: for an sc-diffeomorphism ψ between relatively open subsets of partial quadrants with $\psi(0) = 0$, $x \in C \iff D\psi(0)x \in C'$, by the one-sided sign argument for ψ and ψ^{-1} (Part II, §II.12) `Chapter2/Boundary · IsSc1With.mapsTo_`

partialQuadrant, since $C_0 = C$ at the corner Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero Chapter3/GoodPositionGerm · scDeriv_mem_iff_of_inverse.

The push-forward. $\Psi_*(g)(u') = (\Psi(\psi^{-1}(u'), g(\psi^{-1}(u'))))_2$ Chapter3/GermPushforward · pushforward is an sc-smooth germ on the germ of the points that ψ^{-1} maps into the germ of g Chapter3/GermPushforward · IsStrongBundleIso.pullGerm Chapter3/GermPushforward · IsStrongBundleIso.pushforward_isScSmoothGerm (chain rules with the whole-chart Ψ and ψ^{-1}, §III.48), with $\Psi_*(g)(0) = 0$ and $D(\Psi_*g)(0) = \Gamma(0)Dg(0)D\psi(0)^{-1}$ for its explicit derivative family Chapter3/GermPushforward · IsStrongBundleIso.pushforwardDeriv_zero ($Dk(0) = \Gamma(0)Dg(0)$ for $k(u) = (\Psi(u, g(u)))_2$) Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_linearization; so it is onto Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_surjective, its kernel $D\psi(0)(\ker Dg(0))$ is in good position to C' (Lemma 3.5.9 (preprint 3.45)) Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_ker_isInGoodPosition, and it is Fredholm (transported from $Dg(0)$, Proposition 3.6.5 (preprint 3.52)) — the hypotheses of Theorem 3.5.11 (preprint Corollary 3.47), for *any* conjugating isomorphism Chapter3/GoodPositionGermG · theorem_3_6_8_pushforward, and in bundle form Chapter3/GoodPositionGermG · theorem_3_6_8_pushforward_section. The text instead obtains the Fredholm property of $D(h+t)(0)$ from Propositions 3.1.10 and 1.1.11 (preprint 3.10 and 1.10).

The graph description Chapter3/GlobalIFTG · theorem_3_6_8_graph. Let the charts of the bundle have tame base retractions (`htame`, the hypothesis of the first formalization; it cannot be met, and it is replaced by good position in a tame chart, §III.47), $f(x_0) = 0$, (f, x_0) in good position. The proof, in the order of the Lean.

1. *The chart and the identities.* Take the chart of Definition 3.6.6 (preprint 3.53) centred at x_0, filling germ g on G_g, sc⁺-germ t_0 on G_s and conjugation Ψ covering ψ, with inverse Ψ' covering ψ'. *Shrink the basic germ* to the germ G'' of the points of G that ψ' sends into $G_g \cap G_s$ level by level: same Ψ, h, B and conjugation identity Chapter3/BasicGermShrink · ConjugationData.shrink (§III.48); below, G denotes G''. There is an open $N_1 \ni 0$ on which $\mathbf{f} = g$ on O, zeros of g in U lie in O, and $\Psi(u, (g - \iota t_0)(u))_2 = h(\psi(u))$ (filler (1), (2) and the conjugation).
2. *The sc⁺-germ t.* $t(u') = \Psi[1](\psi'(u'), t_0(\psi'(u')))_2$ is an sc-smooth germ into $(Q \oplus W)^1$ on G: $t_0 \circ \psi'$ is a germ on the preimage of G_s, which contains $G = G''$, and the whole-chart $\Psi[1]$ is composed on the outside (§III.48). (With t_0 sc-smooth on U, t was sc-smooth on all of U'.) On $U \cap N_1$, $\Psi_*(g)(\psi(u)) = h(\psi(u)) + \iota t(\psi(u))$, by the linearity of the fibre maps.
3. *Absorbing the $\mathbb{R}^N$-component.* The data of Theorem 3.4.2 (preprint 3.38) carry only the W-component of an sc⁺-germ. Let t_Q, t_W be the two components of t, through the sc-operators $(q, w) \mapsto (q, 0)$ and $(q, w) \mapsto w$ Chapter3/GlobalIFT · ScBanach.qPartOp Chapter3/GlobalIFT · ScBanach.wPartOp. Then $h' = h + (t_Q, 0)$ is again a basic germ with the same contraction B (Definition 3.1.7 (preprint 3.7) constrains only the W-component), and with the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) for $s = t_W$ (they exist, §III.18) one has $\mathcal{S}.\mathrm{fsum}\, h' = h' + \iota t_W = h + \iota t$.
4. *The hypotheses of Theorem 3.5.11 (preprint Corollary 3.47).* Tameness gives $(1 - Dr(0))E \subseteq E_0$ (Proposition 2.5.3 (preprint 2.35)), so the push-forward step applies: $D(\Psi_*g)(0)$ is surjective with kernel in good position. Since $\mathcal{S}.\mathrm{fsum}\, h' = \Psi_*(g)$ only on a neighbourhood W' of 0, the two derivatives at 0 are identified by the uniqueness of the derivative within G_1'' and density of E_1 Chapter3/GlobalIFT · scDeriv_eq_of_eqOn_nhds; $\Psi_*(g)$ is a germ on the preimage of G_g, which contains G''.
5. *Theorem 3.5.11 (preprint Corollary 3.47) for $h + t$* gives Y, V_0, σ (good complement, C^1, level 1, $\sigma(0) = 0$, $D\sigma(0) = 0$), the graph consisting of zeros, and U'' in which every zero lies on the graph.
6. *The neighbourhoods* (the text does not say which V). Let $\Omega = U'' \cap W'$ and $V' = V_0 \cap (1 + \sigma)^{-1}(\Omega)$,

relatively open in $K \cap C'$ because $v \mapsto v + \sigma(v)$ is continuous. Let $V_x = \{y \in \operatorname{dom}\varphi : \varphi(y) \in N_1,\ \psi\varphi(y) \in \Omega \cap G_0\}$, open and containing x_0 by continuity of $\psi \circ \varphi$. $D\sigma(0) = 0$ persists on the smaller V' (unique differentiability at 0 of V').

7. *The conclusions.* $\psi \circ \varphi$ is injective on V_x (φ is, and $\psi' \circ \psi = \mathrm{id}$ on U). For $y \in V_x$: $f(y) = 0 \iff \mathbf{f}(\varphi y) = 0 \iff g(\varphi y) = 0 \iff (h + \iota t)(\psi\varphi y) = 0$ (items 1–2). If $f(y) = 0$, then $\psi\varphi(y)$ is a zero in $U'' \cap G_0$, hence $v + \sigma(v)$ with $v \in V_0$, and $v \in V'$ since $v + \sigma(v) \in \Omega$. Conversely, for $v \in V'$, $u = \psi'(v + \sigma(v))$ lies in $U \cap N_1$, $g(u) = 0$, so $u \in O$ and $u = \varphi(y)$ for some y, which lies in V_x and is a zero.

So $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$. When this section was written, (1) and (2) were not formalized; they are now: the kernel and good position at the nearby zeros in §§III.32–III.33 (the surjectivity of $f'(y)$ under an added hypothesis), the data of Definition 3.2.1 (preprint 3.25) at every zero in §§III.34–III.36, and the theorem in §III.37. The graph description also records, since then, that Y is closed, that the kernel is a smooth subspace and the level-wise lifts of σ (Theorem 3.5.11 (1) (preprint Corollary 3.47)), and it is a corollary of the version with a good complement whose projection maps C' into itself (§III.29).

Theorems 3.6.10 and 3.6.11 (preprint 3.56 and 3.57), the step from the tangent germ property. With $S = \{f = 0\}$ `Chapter3/GlobalIFT · solutionSet`: if $S \neq \emptyset$ has at every point regular data of Definition 3.2.1 (preprint 3.25) (§III.24), then S is a sub-M-polyfold whose induced structure is tame and an n-dimensional manifold with boundary with corners (Theorem 3.2.3 (preprint 3.27)) `Chapter3/GlobalIFT · theorem_3_6_11_of_tangentGerm`; if X has no boundary, the induced structure is without boundary, since $d_S \leq d_X = 0$ (Lemma 2.4.6 (preprint 2.26)) `Chapter2/SubMPolyfold · lemma_2_4_6` `Chapter3/GlobalIFT · theorem_3_6_10_of_tangentGerm`. The first step of the text's proof — good position at every zero gives the tangent germ property — was not formalized when this section was written; the theorems with the text's hypotheses are in §III.37.

III.24 The tangent germ property: Definition 3.2.1 (preprint 3.25) and Theorem 3.2.3 (preprint 3.27)

Definition 3.2.1 (preprint 3.25). The data at $x \in M$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart`: a chart $\varphi : (U, x) \to (O, 0)$ onto a tame retract (O, C, E); an n-dimensional smooth $N \subseteq E$ in good position to C with a good complement Y, Y **closed**; a relatively open $V \ni 0$ in $N \cap C$; $\delta : V \to E$ continuous with values in Y; (i) $\varphi(M \cap U) = \{v + \delta(v) : v \in V\}$; (ii) $\delta(0) = 0$, $D\delta(0) = 0$ and δ an sc^∞-germ, read as: for every m, k a relatively open $V_{m,k} \ni 0$ in $N \cap C$ with $\delta(V_{m,k}) \subseteq E_m$ and $\delta : V_{m,k} \to E_m$ of class C^k `Chapter3/TangentGerm · IsScInftyGermAt`. This k-dependent reading is what the germ implicit function theorem delivers (§III.11); the reading "sc-smooth on one fixed germ" implies it by Proposition 1.2.3 (preprint 1.18) for germs `Chapter3/TangentGerm · IsScSmoothGerm.isScInftyGermAt`, as does sc-smoothness on V as a set `Chapter3/TangentGerm · IsScSmooth.isScInftyGermAt`, so the theorem covers all three. M has the property if $M \subseteq X_\infty$ and every point has such data `Chapter3/TangentGerm · MPolyfold.HasTangentGermProperty`. With these data: $N \cap C$ is a partial quadrant in N (Proposition 3.1.25 (preprint 3.22)) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isPartialQuadrant`, and N, Y are topological complements — this is where Y closed is used — so the projection $\pi : E \to N$ along Y is continuous `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isTopCompl` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π`; with $\Gamma(v) = v + \delta(v)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ`, $\pi\Gamma = \mathrm{id}$ and $\varphi(y) = \Gamma(\pi\varphi(y))$ for $y \in M \cap U$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.exists_eq_Γ`. "Y closed"

is not an added hypothesis: the text's good complement is an sc-complement, hence closed, and the two notions of datum agree (§III.56).

Proposition 3.2.2 (preprint 3.26) Chapter3/TangentGerm · proposition_3_2_2. For an sc-diffeomorphism $\psi : X \to X'$ (sc-smooth on all of X with sc-smooth inverse ψ' on all of X') and $\psi(M) = M'$, M' has the n-dimensional tangent germ property iff M has it. One direction suffices, applied to ψ and ψ' Chapter3/TangentGerm · proposition_3_2_2_image: $\psi(M) \subseteq X'_\infty$ because sc-smooth maps preserve the levels, and the data at $\psi(x)$ are the transported chart $\varphi \circ \psi^{-1}$ Chapter2/DegeneracyIndex · transportChart, whose model is that of φ Chapter3/TangentGerm · MPolyfold.transportChart_model_0, with the same N, Y, V, δ Chapter3/TangentGerm · MPolyfold.TangentGermChart.transport: condition (i) is transported because $\varphi\psi^{-1}(\psi(M) \cap \psi(U)) = \varphi(M \cap U)$. As in the text; tameness of X plays no role.

Two gaps in the text's proof of Theorem 3.2.3 (preprint 3.27), and the conditions that close them Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular. (a) The text asserts that δ is an sc^∞-germ near every $v_0 \in V$, by linearizing at v_0 and claiming that a certain map $N' \to N$ is an isomorphism. That needs δ differentiable at v_0, known only near 0, and fails: in $X = \mathbb{R}^2$ take M the graph of g with $g(s) = s^2$ near 0 and $g(s) = 4 + (s-2)^{1/3}$ near 2, a smooth embedded curve with a vertical tangent at $(2, 4)$; with $\varphi = \mathrm{id}$, N the horizontal axis, Y the vertical one and $\delta = g$, the data at 0 satisfy Definition 3.2.1 (preprint 3.25), M has the tangent germ property (at $(2, 4)$ use a rotated chart), but δ is not differentiable at 2 and the chart $\pi\varphi$ on all of $M \cap U$ is not smoothly compatible with the chart at $(2, 4)$. So condition (a): δ is sc-smooth on V as a map on a set. (b) The text's $\Sigma = \{v + y \in C : v \in V, y \in Y\}$ is claimed relatively open in C, so that $t(v + w) = v + \delta(v)$ is defined near 0 in O. This does not hold as we read it: $E = \mathbb{R}^3$, $C = [0, \infty)^3$, $N = \mathbb{R}(1, 1, 1)$, $Y = \{x + y - z = 0\}$ is a good complement, and $\pi(x, y, z) = (x + y - z)(1, 1, 1)$, so $\pi z \in C$ iff $x + y \geq z$; Σ is not open at $(1, 0, 1)$ and points $(\varepsilon, 0, \varepsilon + \varepsilon^2) \notin \Sigma$ approach 0. So condition (b): $\pi(O) \subseteq C$; then $\Sigma \cap O = \{z \in O : \pi z \in V\}$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.Sig is relatively open in O Chapter3/TangentGerm · MPolyfold.TangentGermChart.Sig_eq. (A good complement with $\pi'(C) \subseteq N \cap C$ always exists, via the normal form, and $M \cap U$ can be rewritten as a graph along it with the quadrant inverse function theorem; with the re-centring of §III.36 this makes (a) and (b) theorems rather than hypotheses, §III.56, where both examples are also Lean counterexamples.)

Theorem 3.2.3 (preprint 3.27) Chapter3/TangentGerm · theorem_3_2_3_of_isRegular. If $M \neq \emptyset$ has at every point data satisfying (a), (b), then (since §III.56 the conditions are not needed: such data exist at every point):

M is a sub-M-polyfold Chapter3/TangentGerm · theorem_3_2_3_isSubMPolyfold, with the text's retraction. $t(v + w) = v + \delta(v)$, i.e. $t = \Gamma \circ \pi$ on $\Sigma \cap O$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.t, is idempotent with image $\Gamma(V)$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_t Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_eq_self_of_mem. It is sc-smooth: π is a continuous linear map into a finite-dimensional space, hence sc-smooth (Corollary 1.2.5 (preprint 1.20)) Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_isScSmooth; $\Gamma = \iota + \delta$ is sc-smooth on V by (a) and because the inclusion of a smooth finite-dimensional subspace is sc-smooth, its level maps being the linear lifts $N \to E_m$ Chapter3/TangentGerm · finSmoothLift Chapter3/TangentGerm · isScSmooth_subtype_val Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_isScSmooth; and for a retraction R onto $\Sigma \cap O$, $t \circ R = \Gamma \circ \pi \circ R$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_isScSmoothRetractMap. The text says nothing about either. On $U_0 = \varphi^{-1}(\Sigma \cap O)$, open Chapter3/TangentGerm · MPolyfold.TangentGermChart.U₀ Chapter3/TangentGerm · MPolyfold.TangentGermChart.isOpen_U₀, the map $s = \varphi^{-1} \circ t \circ \varphi$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.s is an sc-smooth retraction Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_s Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_isScSmoothOn onto $M \cap U_0$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.image_s — Definition 2.3.18 (preprint 2.20) at x.

The charts $\pi \circ \varphi$. On $M \cap U_0$, $y \mapsto \pi\varphi(y)$ is a chart with model $(V, N \cap C, N)$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.model, a tame retract (the identity retraction) Chapter3/TangentGerm · MPolyfold.TangentGermChart.model_isTameRetract, with inverse $v \mapsto \varphi^{-1}(v + \delta(v))$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.chartInv Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart, a finite-dimensional corner chart of dimension n Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_isCorner containing x Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_chart_source. It is compatible with every chart $\varphi'|_{W\cap M}$ of the induced structure of Proposition 2.3.19 (preprint 2.21): the transition maps are $\varphi' \circ \varphi^{-1} \circ \Gamma$ and $\pi \circ \varphi \circ \varphi'^{-1}$, compositions of sc-smooth maps between retracts Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_isScSmoothRetractMap Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_isScSmoothRetractMap Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_compatible_subChart. So it is a chart of the induced structure — **Theorem 3.2.3 (2) (preprint 3.27)** Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_isChart Chapter2/SubMPolyfold · subPolyfold; these charts are pairwise compatible by locality Chapter3/TangentGerm · MPolyfold.tangentCharts_compatible and form an atlas Chapter3/TangentGerm · MPolyfold.germAtlas equivalent to the induced one Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_atlas_equiv, with tame models: **the induced structure is tame** Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_isTame, and **sc-smoothly equivalent to an n-dimensional manifold with boundary with corners** Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_isManifoldWithCorners. "Manifold with boundary with corners" belongs to Chapter 4; here it is an equivalent atlas of charts on relatively open subsets of n-dimensional partial quadrants Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners, whose transition maps are classically smooth because sc-smooth maps between finite-dimensional constant structures are C^∞ Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const. Tameness of X is not used: the text's "since O is a tame retract, there exists … a tame sc-smooth retraction r" is Proposition 2.3.1 (1) (preprint 2.10) for any retract. *The degeneracy index*: since $\pi\varphi$ has a tame model, $d_M(y) = d_{N\cap C}(\pi\varphi(y))$ (Proposition 2.5.6 (preprint 2.36)) Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_degIndex; Lemma 3.8.13 (preprint 3.70) in a general presentation turns this into $d_M(y) = d_X(y)$ near the centre, off the lineality space (§III.61).

III.25 Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61): towards the basic germ criterion

Lemmas 3.7.2 and 3.7.3 (preprint 3.59 and 3.60), abstract form. In the proof of Theorem 3.7.1 (preprint 3.58) the family $L(b) = P_Y \circ Df(b)|_X : X \to Y$ is shown to be injective for b near 0 (Lemma 3.7.2 (preprint 3.59)) and to satisfy uniform lower bounds $|L(b)h|_m \geq c_m|h|_m$ (Lemma 3.7.3 (preprint 3.60)) — the hypotheses of Lemma 3.7.4 (preprint 3.61). Both are sequence arguments, and all they use of the compactness condition (2) of Theorem 3.7.1 (preprint 3.58) is a property of the family itself: if $b_j \to b$, (h_j) is bounded in X_m and $L(b_j)h_j \to 0$ in Y_m, then (h_j) has a convergent subsequence in X_m Chapter3/BasicGermCriterion · FamilyCompactness. (For $L = P_Y \circ Df|_X$ it follows from (2), since $Df(b_j)h_j - L(b_j)h_j = (1 - P_Y)Df(b_j)h_j$ lies in the finite-dimensional smooth complement A and is bounded on every level.) The two lemmas are formalized for an abstract family of sc-operators with jointly continuous level maps. *Lemma 3.7.2 (preprint 3.59)*: if $L(0)$ is injective, so is $L(b)$ for b near 0 — otherwise $b_j \to 0$ and unit vectors h_j with $L(b_j)h_j = 0$ give, by the compactness property, a limit h with $|h| = 1$ and $L(0)h = 0$ Chapter3/BasicGermCriterion · lemma_3_7_2. *Lemma 3.7.3 (preprint 3.60)*: on a compact parameter set K on which every $L(b)_m$ is injective there is $c_m > 0$ with $c_m|h|_m \leq |L(b)h|_m$ — otherwise $b_j \in K$ and unit h_j with $|L(b_j)h_j|_m \to 0$ give, after subsequences, $b_j \to b \in K$, $h_j \to h$, $|h|_m = 1$, $L(b)h = 0$ Chapter3/BasicGermCriterion · lemma_3_7_3. The text's step "$P_Y \circ Df(b)|_X$ is an injective sc-Fredholm operator of index 0, hence an sc-isomorphism" is needed only to feed Lemma 3.7.4 (preprint 3.61). The setting of Theorem

3.7.1 (preprint 3.58) (the splitting as coordinates, the complement A, the index computation), the strong bundle map $\Phi((b,x),(c,y)) = ((b,x),(c,L(b)^{-1}y))$ and the contraction estimate from condition (3) were not formalized when this section was written; they are now, in the coordinates of the text's proof, and there the hypotheses of the two lemmas are derived from those of the theorem (§III.38).

Lemma 3.7.4 (preprint 3.61) (quoted by the text from [25, Proposition 4.8] without proof). Let V be a germ of relatively open neighbourhoods of 0 in a partial quadrant $\widetilde{C}$ of a finite-dimensional G, $V \oplus E$ the product germ with levels $V_m \times E_m$ `Chapter3/InverseFamily · ScNbhdGerm.prodFull`, and $L(v) : E \to F$ sc-isomorphisms such that $\Phi(v,h) = L(v)h$ `Chapter3/InverseFamily · famMap` is an sc-smooth germ on $V \oplus E$ and $c_m|h|_m \le |L(v)h|_m$ for $v \in V_0$. Then $\Lambda(v,k) = L(v)^{-1}k$ `Chapter3/InverseFamily · famInv` is an sc-smooth germ on $V \oplus F$ `Chapter3/InverseFamily · lemma_3_7_4`. Operator-norm continuity of $v \mapsto L(v)$ is not assumed.

sc⁰. On level m, with $h_0 = L(v_0)^{-1}k_0$, $|L(v)^{-1}k - L(v_0)^{-1}k_0|_m \le c_m^{-1}(|k-k_0|_m + |(L(v_0)-L(v))h_0|_m)$, and $v \mapsto L(v)h_0$ is continuous `Chapter3/InverseFamily · famInv_isSc0Germ`.

Differentiability `Chapter3/InverseFamily · famInv_hasFDerivWithinAt`. The candidate is $D\Lambda(v,k)(\delta v,\delta k) = L(v)^{-1}(\delta k - D_1\Phi(v,L(v)^{-1}k)\delta v)$ `Chapter3/InverseFamily · famInvDeriv`. With $h = \Lambda(v,k)$ and $w = D\Lambda(v,k)(\delta v,\delta k)$, $\Lambda(v',k') - \Lambda(v,k) - w = L(v')^{-1}[(L(v)-L(v'))w] - L(v')^{-1}r$, where $r = \Phi(v',h) - \Phi(v,h) - D_1\Phi(v,h)\delta v = o(|\delta v|)$ is the remainder of Φ along $(\delta v, 0)$ with h fixed, so that only the finite-dimensional variable moves. The term $(L(v)-L(v'))w$ is $o(|(\delta v,\delta k)|_1)$ *because the increment is measured on level* 1: $w/|(\delta v,\delta k)|_1$ lies in the compact set $D\Lambda(v,k)(\iota(B_1))$ ($\iota : F_1 \to F_0$ compact), and $(L(v)-L(v'))u \to 0$ uniformly for u in a compact set — a tube lemma for jointly continuous maps vanishing on $\{x_0\} \times K$ `Chapter3/InverseFamily · exists_forall_norm_le_of_isCompact`. Strong continuity plus the compactness of $F_1 \to F_0$ replace the operator-norm continuity; the differentiability of Λ as a map $(V \oplus F)_0 \to E_0$ would need the latter and is not claimed.

The tangent map and the induction. $T\Lambda((v,k),(\delta v,\delta k)) = (\Lambda(v,k), \Lambda(v,\delta k - (T\Phi((v,\Lambda(v,k)),(\delta v,0)))_2))$, a composition of Λ, $T\Phi$ and sc-operators (the projections `Chapter3/InverseFamily · tangentBaseOp` `Chapter3/InverseFamily · tangentInlOp` `Chapter3/InverseFamily · tangentFiberOp` and subtraction `Chapter3/InverseFamily · ScBanach.subOp`); it is sc⁰ `Chapter3/InverseFamily · famInv_tangent_isSc0`, so Λ is sc¹ `Chapter3/InverseFamily · famInv_isSc1GermWith`; and if Φ is sc^{k+1}, then Λ (sc^k by induction) and $T\Phi$ (sc^k) make $T\Lambda$ sc^k, so Λ is sc^{k+1} `Chapter3/InverseFamily · famInv_isSckGerm`. The sc¹ step uses no quadrant hypothesis; the induction uses the composition rules for germs.

III.26 What the formalization taught us

1. **The germ implicit function theorem holds with the text's hypotheses**, but on shrinking germs: the solution germ is sc^k on the germ shifted by $2^k - 1$ levels. In the text's reading of germs (up to shrinking) this is the theorem; read with fixed germs, "if f is sc-smooth so is its solution germ" holds for every k on a germ depending on k. Every later "sc-smooth germ" (Definition 3.2.1 (ii) (preprint 3.25), Theorem 3.5.6 (1) (preprint 3.43), Theorem 3.5.11 (1) (preprint Corollary 3.47)) must be read the same way. Whether the shrinking is intrinsic is open.
2. **Theorem 3.3.3 (preprint 3.30) has a short proof**, which the text cites from [21]: an approximate fixed point and the $\frac{1}{2}$-contraction on level 0, at parameters where the level-1 operators exist.
3. **The contraction regions must be explicit** for "from the uniqueness we conclude $\delta^{(1)} = T\delta$" (Lemma 3.3.4 (preprint 3.31)) to be true.

4. **Several "immediate consequences" are proofs of their own**: the compactness of Theorem 3.1.13 (preprint 3.12) (not in Theorem 3.5.1 (preprint 3.40)), the induction of Theorem 3.5.1 (preprint 3.40) (lifts, bounds and continuity together), the sc-smoothness and regularizing property in Theorem 3.1.18 (preprint 3.15), the sc^+ property in Proposition 3.1.10 (preprint 3.10), the projection of solutions into the quadrant in Theorem 3.5.6 (preprint 3.43).
5. **Three added hypotheses**: Y closed in Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25); and the two regularity conditions of Theorem 3.2.3 (preprint 3.27), each with a counterexample to the step of the text's proof it replaces. (None of the three remains: §III.56.)
6. **Tameness is nearly unused**: not in §3.1 at all, and in §3.6 only through $(1-Dr(0))E \subseteq E_0$ and, for the tangent germ property of the solution set, the tameness of the model of the chart $\psi \circ \varphi$ (§III.27).
7. **Linear coordinate changes** in Theorem 3.4.2 (preprint 3.38) make the text's degeneracy-index argument for Theorem 3.5.11 (preprint Corollary 3.47) unnecessary.
8. **The extreme-ray lemmas and the iterated tangent maps are not needed**: Proposition 3.1.25 (preprint 3.22) by a segment argument, the regularity in Theorems 3.8.15 and 3.8.22 (preprint 3.72 and 3.78) by bootstrapping, without shrinking. (They are proved as well since, §§III.59–III.60.)
9. **What remained of Theorem 3.6.8 (preprint 3.54)** when this list was written — the re-centring of good position at the nearby zeros, and the level-wise regularity of the solution map in the data of Definition 3.2.1 (preprint 3.25) at every zero — is done in §§III.31–III.37, with one added hypothesis for Theorem 3.6.8 (1) (preprint 3.54); see §III.39.

III.27 Tameness is carried by sc-diffeomorphisms, and by open subsets

The text never states that tameness is invariant under sc-diffeomorphisms, or that an open piece of a tame retract is a tame retract, but it uses both. In the proof of Theorem 3.6.8 (preprint 3.54) the chart $\psi \circ \varphi$ onto $\psi(O)$ must be a chart of the tame M-polyfold X for Definition 3.2.1 (preprint 3.25) to apply; in the proof of Proposition 3.2.2 (preprint 3.26) it writes "$X' = \psi(X)$ is tame"; and wherever a chart is restricted to a smaller domain (the shrinking of the data of Definition 3.2.1 (preprint 3.25) in §§III.36–III.37) the model of the restricted chart must again be a tame retract. Both facts are proved here, with Definition 2.5.2 (preprint 2.34) as the formalization states it (Part II).

The objects. An sc-diffeomorphism between relatively open subsets $U \subseteq C$, $U' \subseteq C'$ of partial quadrants is a pair $\psi : U \to U'$, $\psi' : U' \to U$ of sc-smooth maps, each mapping its domain into the other's, inverse to each other on U and on U' `Chapter2/TameInvariance · IsScDiffeoQuadrant` (its inverse is again one `Chapter2/TameInvariance · IsScDiffeoQuadrant.symm`). A retraction $r : U \to U$ is *tame* for a presentation T of C if (1) $d_C(r(x)) = d_C(x)$ for $x \in U$ and (2) at every smooth point x of $O = r(U)$ on level 1 the tangent space $T_xO = Dr(x)E$ has an sc-complement A contained in the minimal subspace E_x `Chapter2/Tame · ScRetraction.IsTameWith`; it is tame if it is tame for every presentation `Chapter2/Tame · ScRetraction.IsTame`, and (O, C, E) is a tame retract if O is the image of a tame retraction `Chapter2/Tame · IsTameRetract`.

The transported retraction. For an sc-smooth retraction r with domain U, put $r' = \psi \circ r \circ \psi' : U' \to U'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport`. It maps U' into U' because $\psi'(U') \subseteq U$, $r(U) \subseteq U$ and $\psi(U) \subseteq U'$; it is sc-smooth by the chain rule applied twice; and it is idempotent, $r'(r'(u')) = \psi(r(\psi'\psi(r(\psi'u')))) = \psi(r(r(\psi'u'))) = r'(u')$, since $\psi'\psi = \mathrm{id}$ on U and r is idempotent. Its image is $\psi(O)$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport_retract`: $r'(u') = \psi(r(\psi'u'))$ with $r(\psi'u') \in O$; conversely, for $z \in O$, $r'(\psi z) = \psi(r(z)) = \psi(z)$.

Condition (1). The degeneracy index is preserved by ψ, $d_{C'}(\psi(u)) = d_C(u)$ for $u \in U$: this is Theorem

1.3.5 (preprint 1.25) `Chapter2/TameInvariance · IsScDiffeoQuadrant.degIndex_eq` `Chapter2/DegeneracyIndex · theorem_1_3_5`. Then for $u' \in U'$: $d_{C'}(r'(u')) = d_C(r(\psi' u')) = d_C(\psi' u') = d_{C'}(\psi\psi' u') = d_{C'}(u')$, the middle step being condition (1) for r.

The derivative of r'. On level 0, $D\psi'(\psi(x)) \circ D\psi(x) = 1$ for $x \in U_1$ and $D\psi(\psi'(y)) \circ D\psi'(y) = 1$ for $y \in U_1'$, by the chain rule for $\psi' \circ \psi = \mathrm{id}$ and $\psi \circ \psi' = \mathrm{id}$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.deriv_inv_deriv` `Chapter2/TameInvariance · IsScDiffeoQuadrant.deriv_deriv_inv` `Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`. If $y \in U_1'$ and $\psi'(y) \in O$, then with $x = \psi_1'(y)$,

$$Dr'(y) = D\psi(x) \circ Dr(x) \circ D\psi'(y)$$

`Chapter2/TameInvariance · IsScDiffeoQuadrant.scDeriv_transport`. The chain rule is applied to $\psi \circ (r \circ \psi')$ and then to $r \circ \psi'$ `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain`; the outer derivative of ψ is taken at the level-1 point of $r \circ \psi'$ over y, and this point is x because r fixes $\psi'(y) \in O$. Without $\psi'(y) \in O$ the formula would involve Dr at $\psi'(y)$ and $D\psi$ at the different point $r(\psi'(y))$.

Sc-isomorphisms carry sc-splittings. Let $D : E \to E'$ be an sc-operator with an sc-operator inverse D'. Its level maps are topological linear isomorphisms $E_m \cong E'_m$ `Chapter2/TameInvariance · scIsoLevel` (inverse to each other by the injectivity of the inclusions into level 0); they carry $A_m = A \cap E_m$ onto $(DA)_m$ `Chapter2/TameInvariance · levelSub_map_eq` and smooth points to smooth points `Chapter2/TameInvariance · mem_smoothLvl_of_isScOperator`. Hence an sc-subspace A (closed, with the smooth points of A_m dense in A_m for every m) is carried to the sc-subspace DA: $DA = (D')^{-1}(A)$ is closed, and the image of the closure of the smooth points of A_m lies in the closure of the image `Chapter2/TameInvariance · IsScSubspace.map`. An sc-splitting $E = F \oplus A$ (sc-subspaces, complementary on every level) is carried to the sc-splitting $E' = DF \oplus DA$, since an isomorphism of each level carries complements to complements `Chapter2/TameInvariance · IsScSplitting.map`.

Condition (2) `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameWith_transport`. Let T, T' be presentations of C, C', r tame for T, and y a smooth point of $\psi(O)$ on level 1. Then $x = \psi_1'(y)$ lies on level 1 of O (if $y = \psi(z)$ with $z \in O$, then $\psi'(y) = z$) and is smooth, because sc-smooth maps carry smooth points to smooth points. Tameness of r at x gives an sc-splitting $E = T_xO \oplus A$ with $A \subseteq E_x$. Because x and y are smooth points, $D\psi(x)$ and $D\psi'(y)$ are sc-operators (the remark after Proposition 1.2.1 (preprint 1.16)) `Chapter1/Properties · IsSc1With.scOperator_deriv` — at a point that is only on level 1 they need not be — and they are inverse to each other. From the formula for $Dr'(y)$ and $D\psi'(y)D\psi(x) = 1$: $T_y\psi(O) = \operatorname{range} Dr'(y) = D\psi(x)(\operatorname{range} Dr(x)) = D\psi(x)(T_xO)$. So $E' = T_y\psi(O) \oplus D\psi(x)A$ is an sc-splitting. Finally $D\psi(x)A \subseteq E'_y$: the derivative of an sc¹-map between relatively open subsets of partial quadrants maps the minimal subspace E_x into $E'_{\psi(x)}$ (the one-sided sign argument, and density) `Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace`, which needs $\psi(U) \subseteq C'$, and $\psi(x) = y$. Since T' was arbitrary and the presentation T of C is available, the transported retraction of a tame retraction is tame `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTame_transport`, and **the image $(\psi(O), C', E')$ of a tame retract under an sc-diffeomorphism is a tame retract**, with retraction $\psi \circ r \circ \psi^{-1}$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image`.

Open subsets. For an open $W \subseteq E_0$, the restriction of r to $U \cap r^{-1}(W)$ is an sc-smooth retraction onto $O \cap W$ (Proposition 2.3.1 (1) (preprint 2.10)) `Chapter2/Retracts · ScRetraction.restrict`. It is tame if r is `Chapter3/SolutionGermRecentre · ScRetraction.restrict_isTame`: condition (1) is inherited pointwise; for (2), at a smooth point x of $O \cap W$ on level 1 the derivative of the restricted retraction is $Dr(x)$, because the two maps agree on the smaller relatively open domain, which contains x since $r(x) = x \in W$, and the

derivative at a level-1 point depends only on the map near the point `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen`; so the complement A of r at x serves. Hence $O \cap W$ is a tame retract whenever O is `Chapter3/SolutionGermRecentre · IsTameRetract.inter_isOpen`. For a chart φ and an open s inside its source, $\varphi(s) = O \cap W$ for an open W `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq`, so the model of the restricted chart is a tame retract when that of φ is `Chapter3/SolutionGermRecentre · MChart.isTameRetract_image`.

These are used in §III.34 (the chart $\psi \circ \varphi$ of Theorem 3.6.8 (preprint 3.54) has the tame model $\psi(O)$) and in §§III.36–III.37 (the shrunk data of Definition 3.2.1 (preprint 3.25) keep tame models). The text's sentence "$X' = \psi(X)$ is tame" in Proposition 3.2.2 (preprint 3.26) is not needed there (§III.24): the tameness Definition 3.2.1 (preprint 3.25) asks for is that of the chart models, which the transport of charts leaves unchanged.

III.28 Corollary 3.4.3 (preprint 3.39): composition with a given conjugation

Corollary 3.4.3 (preprint 3.39) (text): if the sc-germ g is equivalent, by a strong bundle isomorphism Φ, to a basic germ and s is an sc$^+$-germ, there is a strong bundle map Ψ such that $((\Psi \circ \Phi)_*(g+s))^1$ is a basic germ. The text's proof: $\Phi_*(g) = h$ is basic, $t = \Phi_*(s)$ is an sc$^+$-germ with $t(0) = 0$, $\Phi_*(g+s) = h + t$, and Theorem 3.4.2 (preprint 3.38) gives Ψ with $(\Psi_*(h+t))^1$ basic; take $\Psi \circ \Phi$. Four things are silent: that Theorem 3.4.2's (preprint 3.38) coordinate change is a strong bundle *isomorphism*; that the index raise of a strong bundle isomorphism is one between the shifted structures; that strong bundle isomorphisms and push-forwards compose; and that $\Phi_*(g+s) = h + t$ holds only where the conjugation identity holds. The formalization proves all four and states the corollary with the composite explicit.

Linear strong bundle isomorphisms. Two sc-operators $S : E \to E'$, $T : F \to F'$ give the sc-operator $S \oplus T$ `Chapter3/BasicClassComposition · ScOperator.prodMap`, and if $S(U) \subseteq U'$ the map $(u,k) \mapsto (Su, Tk)$ is a strong bundle map $U \lhd F \to U' \lhd F'$ (Definition 2.6.1 (preprint 2.47) as formalized `Chapter2/Tame · IsStrongBundleMap`): its base component does not depend on k, it is linear in k, it preserves the double filtration because sc-operators preserve the levels, it is the sc-operator $S \oplus T$ on $(U \lhd F)[0]$, and on $(U \lhd F)[1]$ it is $S \oplus T^1$ with T^1 the shifted operator `Chapter3/BasicClassComposition · ScOperator.isStrongBundleMap_prodMap`. With inverse sc-operators S' (inverse to S on U, U') and T' it is a strong bundle isomorphism `Chapter3/BasicClassComposition · ScOperator.isStrongBundleIso_prodMap`.

Theorem 3.4.2's (preprint 3.38) coordinate change as an isomorphism. In §III.18 the change of coordinates of Theorem 3.4.2 (preprint 3.38) is the pair of linear sc-isomorphisms `$□ = $ split`, `$□ = $ fiber` (inverses `glue`, `fiberInv`) on the index-raised spaces `Chapter3/BasicClass · StabilityData.split` `Chapter3/BasicClass · StabilityData.fiber`. Put $\Lambda((a,w),(q,w')) = (\psi(a,w), \Psi(q,w'))$ `Chapter3/BasicClassComposition · StabilityData.linIso`. For every U relatively open in the shifted quadrant $(\widetilde{C} \oplus W)^1$, $\psi(U) = \mathrm{glue}^{-1}(U)$ is relatively open in $(\widetilde{C} \oplus C) \oplus X^1$ (`glue` is continuous and preserves the quadrant coordinates) `Chapter3/BasicClassComposition · StabilityData.isRelOpen_glue_preimage`, and $\Lambda : U \lhd (\mathbb{R}^N \oplus W)^1 \to \psi(U) \lhd ((\mathbb{R}^N \oplus Z) \oplus X^1)$ is a strong bundle isomorphism by the preceding paragraph `Chapter3/BasicClassComposition · StabilityData.linIso_isStrongBundleIso`. Before the index raise, the same construction with the level-0 operators `split₀`, `fiber₀` gives Λ_0 `Chapter3/BasicClassComposition · StabilityData.linIso₀`, a strong bundle isomorphism $U \lhd (\mathbb{R}^N \oplus W) \to \psi(U) \lhd ((\mathbb{R}^N \oplus Z) \oplus X)$ — the text's "strong bundle map covering ψ" `Chapter3/BasicClassComposition · StabilityData.linIso₀_isStrongBundleIso` — and its index raise is Λ, because `split` and `fiber` are `split₀` and `fiber₀` on the levels $m+1$ `Chapter3/BasicClassComposition · StabilityData.strongShift_`

linIso$_0$.

The index raise. For $\Psi : E_0 \times F_0 \to E_0' \times F_0'$ let Ψ^1 be its level-1 map on $E_1 \times F_1$ Chapter3/BasicClassComposition · strongShift. For a strong bundle map it exists on $U_1 \oplus F_1$ and satisfies $\iota\Psi^1(p) = \Psi(\iota p)$, because Ψ maps $U_1 \oplus F_1$ into $U_1' \oplus F_1'$ (the filtration with $k = 1 \le m + 1$) Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_strongShift; its base map is the level-1 map of the base map of Ψ Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_base_strongShift. **Ψ^1 is a strong bundle map** $U^1 \lhd F^1 \to U'^1 \lhd F'^1$ of the shifted structures Chapter3/BasicClassComposition · IsStrongBundleMap.shift: independence of the base component and fibre linearity are read through the injective inclusions; the double filtration $(U^1 \lhd F^1)_{m,k} = U_{m+1} \oplus F_{k+1}$ is preserved because $k + 1 \le (m + 1) + 1$ whenever $k \le m + 1$, so the filtration property of Ψ applies; $(U^1 \lhd F^1)[0]$ is the shift of $(U \lhd F)[0]$, so Ψ^1 is sc-smooth there by Proposition 1.2.2 (preprint 1.17) Chapter1/Levels · IsScSmooth.shift; and $(U^1 \lhd F^1)[1]$, with levels $E_{m+1} \oplus F_{m+2}$, is the shift of $(U \lhd F)[1]$, on which $(\Psi^1)[1]$ agrees with $(\Psi[1])^1$ (the lift of the fibre component to F_2' is the same element, by the injectivity of $F_2' \to F_1' \to F_0'$), so Proposition 1.2.2 (preprint 1.17) applies again. For a strong bundle isomorphism with inverse Ψ', Ψ'^1 is the inverse of Ψ^1, again checked after the inclusion into level 0 Chapter3/BasicClassComposition · IsStrongBundleIso.shift. The text uses this whenever it writes $((\Psi \circ \Phi)_*(\cdot))^1$.

Composition. The base map of $\Lambda \circ \Phi$ is the composite of the base maps on U Chapter3/BasicClassComposition · IsStrongBundleMap.base_comp, and $(\Lambda \circ \Phi)[1] = \Lambda[1] \circ \Phi[1]$ on $U \oplus F_1$ Chapter3/BasicClassComposition · IsStrongBundleMap.shiftFiber_comp; so the composite is a strong bundle map — fibre linearity because over u it is the fibre map of Λ over $\varphi(u)$ after that of Φ over u, sc-smoothness on $[0]$ and $[1]$ by the chain rule Chapter3/BasicClassComposition · IsStrongBundleMap.comp — and the composite of strong bundle isomorphisms is one, with the inverses composed in the opposite order Chapter3/BasicClassComposition · IsStrongBundleIso.comp. Push-forwards compose: $(\Lambda \circ \Phi)_* g = \Lambda_*(\Phi_* g)$ on the target of Λ, since the base of $(\Lambda \circ \Phi)^{-1}$ is $\varphi^{-1} \circ \lambda^{-1}$ and $\Phi(\varphi^{-1}(w), \cdot)$ lies over w Chapter3/BasicClassComposition · IsStrongBundleIso.pushforward_comp Chapter3/GermPushforward · pushforward. The index raise commutes with composition, $(\Lambda \circ \Phi)^1 = \Lambda^1 \circ \Phi^1$ on $U_1 \oplus F_1$ Chapter3/BasicClassComposition · IsStrongBundleMap.strongShift_comp.

Corollary 3.4.3 (preprint 3.39) as formalized Chapter3/BasicClassComposition · corollary_3_4_3. Let U be relatively open in a partial quadrant C of E with $0 \in U$, $g : U \to F$ sc-smooth (the text's "sc-germ"; sc-smoothness is what makes $(g + s)^1$ take values on level 1), d conjugation data of g (§III.30: a strong bundle isomorphism $\Phi : U \lhd F \to U' \lhd (\mathbb{R}^N \oplus W)$ covering φ with $\varphi(0) = 0$ and a basic germ h on a germ G inside U', with $(\Phi(u, g(u)))_2 = h(\varphi(u))$ for $u \in U$ near 0; the text's "equivalent by a strong bundle isomorphism to the basic germ $\Phi_* g$"), and $s : U \to F^1$ sc-smooth into F^1 with $s(0) = 0$ (the text's sc$^+$-germ $s : O(C, 0) \to (F, 0)$). The proof, in the order of the Lean:

1. *The sc$^+$-germ $t = \Phi[1]_* s$.* Put $t(u') = (\Phi[1](\varphi^{-1}u', s(\varphi^{-1}u')))_2$ Chapter3/BasicClassComposition · ConjugationData.pushPlus. It is sc-smooth from U' into $(\mathbb{R}^N \oplus W)^1$: the pairing of φ^{-1} with $s \circ \varphi^{-1}$ maps U' into $U \oplus F^1$, and $\Phi[1]$ is sc-smooth there by Definition 2.6.1 (preprint 2.47); restricted to the germ $G \subseteq U'$ it is an sc-smooth germ. And $t(0) = 0$, from $\varphi^{-1}(0) = 0$, $s(0) = 0$ and $\Phi(0, 0)_2 = 0$ (fibre linearity).
2. *Absorbing the $\mathbb{R}^N$-component.* As in Theorem 3.6.8 (preprint 3.54) (§III.23, step 3), $h' = h + (t_{\mathbb{R}^N}, 0)$ Chapter3/BasicClassComposition · ConjugationData.absorb Chapter3/GlobalIFT · ScBanach.qPartOp is a basic germ with the same contraction map: it is an sc-smooth germ (pairing and the sum operator), $h'(0) = h(0) + 0 = 0$, and its W-component is that of h. The W-component t_W is an sc-smooth germ into W^1 vanishing at 0, so the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) exist for $s = t_W$ on

the germ G of h Chapter3/BasicClass · StabilityData.exists_of_scPlus, and $\mathcal{S}.\mathrm{fsum}\, h' = h + \iota t$ Chapter3/BasicClass · StabilityData.fsum.

3. $\Phi_*(g+s) = h + t$ *near* 0. For $u \in U$ at which the conjugation identity holds, $\Phi_*(g+\iota s)(\varphi(u)) = (\Phi(u, g(u)+\iota s(u)))_2$ because $\varphi^{-1}\varphi(u) = u$; by fibre linearity this is $(\Phi(u, g(u)))_2 + (\Phi(u, \iota s(u)))_2 = h(\varphi(u)) + \iota t(\varphi(u))$, where $\iota(\Phi[1](u, s(u)))_2 = (\Phi(u, \iota s(u)))_2$ Chapter3/SectionCompactness · IsStrongBundleMap.toBase_shiftFiber_snd. So $\Phi_*(g+s) = \mathcal{S}.\mathrm{fsum}\, h'$ at the points $\varphi(u)$, $u \in U$ near 0 — not everywhere, as the text's "clearly" suggests.
4. *The composite.* $\Gamma = \Lambda \circ \Phi^1$ Chapter3/BasicClassComposition · ConjugationData.compIso, with inverse $\Phi'^1 \circ \Lambda^{-1}$ Chapter3/BasicClassComposition · ConjugationData.compIsoInv, is a strong bundle isomorphism $U^1 \lhd F^1 \to \psi(U'^1) \lhd ((\mathbb{R}^N \oplus Z) \oplus X^1)$ Chapter3/BasicClassComposition · ConjugationData.compDomain: the index raise of Φ (relatively open U_1' in the shifted quadrant) followed by Λ, composed by the composition lemma. Its base map is $u \mapsto \psi(\varphi_1(u))$, which vanishes at 0; $\Gamma_* = \Lambda_* \circ \Phi^1_*$ (composition of push-forwards); and the germ of $\mathcal{S}.\mathrm{pushMap}\, h'$ lies in the target, because `glue` maps the level-1 part of G into U_1'.
5. *The basic germ.* $\mathcal{S}.\mathrm{pushMap}\, h' = \mathrm{fiber} \circ (\mathcal{S}.\mathrm{fsum}\, h')^1 \circ \mathrm{glue}$ Chapter3/BasicClass · StabilityData.pushMap is a basic germ by Theorem 3.4.2 (preprint 3.38) (§III.18) Chapter3/BasicClass · theorem_3_4_2.
6. *The conjugation identity on level* 1. For $u \in U^1$ near 0: $(\Gamma(u, (g+s)^1(u)))_2 = \mathcal{S}.\mathrm{pushMap}\, h'(\gamma(u))$. Since $\gamma(u) = \mathrm{split}(\varphi_1(u))$ and $\mathrm{glue} \circ \mathrm{split} = \mathrm{id}$, both sides are `fiber` applied to elements of $(\mathbb{R}^N \oplus W)_1$, which are compared after the injective inclusion into level 0: $\iota(\Phi^1(u, (g+s)^1(u)))_2 = (\Phi(\iota u, (g+s)(\iota u)))_2$ Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_strongShift_snd, which is $\mathcal{S}.\mathrm{fsum}\, h'(\varphi(\iota u))$ by step 3, and this is ι of $(\mathcal{S}.\mathrm{fsum}\, h')^1(\varphi_1(u))$. "Near 0" means: ιu lies in the neighbourhood where the conjugation identity holds (continuity of $\iota : E_1 \to E_0$), and $\varphi_1(u)$ lies in the level-1 part of the germ G, a relatively open condition at 0 by the continuity of φ on level 1.

The conclusion of `corollary_3_4_3` collects these: $\mathcal{S}$ with $\mathcal{S}.G = G$ and $\mathcal{S}.s = t_W$, $t(0) = 0$, h' basic, $\Phi_*(g+s) = \mathcal{S}.\mathrm{fsum}\, h'$ near 0, Γ a strong bundle isomorphism onto a relatively open subset of $(\widetilde{C} \oplus C) \oplus X^1$ with $\gamma(0) = 0$ and $\Gamma_* = \Lambda_* \circ \Phi^1_*$, the basic germ $\mathcal{S}.\mathrm{pushMap}\, h'$ inside the target, and the identity of step 6. In the form of Definition 3.1.11 (preprint 3.11): $(g+s)^1 : U^1 \to F^1$ **is conjugated to a basic germ** on the shifted structures E^1, F^1 Chapter3/BasicClassComposition · corollary_3_4_3_isConjugate, with parameter space $\mathbb{R}^n \oplus C$, target $\mathbb{R}^N \oplus Z$ and fibre X^1 (the text's $\tau : Z \to C$ is not applied, as in §III.18). The text's form "$\Gamma = \Psi \circ \Phi$, then raise the index" is the same map: $\Lambda_0 \circ \Phi$ is a strong bundle isomorphism on level 0 and $(\Lambda_0 \circ \Phi)^1 = \Lambda_0^1 \circ \Phi^1 = \Lambda \circ \Phi^1$ on $U_1 \oplus F_1$ Chapter3/BasicClassComposition · ConjugationData.compIso_eq_strongShift. No hypothesis is added beyond the two readings ("sc-germ" as sc-smooth on U; "equivalent by a strong bundle isomorphism" as conjugation data). The first reading is the whole-chart one of §III.6, narrower than the text's germs; no theorem of the germ reading uses Corollary 3.4.3 (preprint 3.39) (the facts at the nearby zeros come from the chart at x_0 through the linear coordinate change of Theorem 3.4.2 (preprint 3.38), §III.43).

III.29 Good complements with a projection into the quadrant; Theorems 3.5.6, 3.5.11 and 3.6.8 (3) (preprint 3.43, 3.47 and 3.54) with a closed projecting complement

Two steps of the text depend on the choice of a good complement without saying so. In the proof of Theorem 3.2.3 (preprint 3.27) the retraction $t(v+w) = v + \delta(v)$ is defined on $\Sigma = \{v + y \in C : v \in V,\ y \in Y\}$, which the text calls relatively open in C; §III.24 shows that this fails for the good

complement $Y = \{x + y - z = 0\}$ of $N = \mathbb{R}(1,1,1)$ in $C = [0,\infty)^3$, and makes the remedy a hypothesis: condition (b) of the regularity conditions, $\pi(O) \subseteq C$ for the projection π onto N along Y `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`. In the proof of Theorem 3.5.6 (preprint 3.43) (Lemma 3.5.10 (preprint 3.46)) a good complement Z of $\widetilde{K} = \ker DH(0)$ is produced by projecting a good complement of K'. This section shows that for solution sets the complement can be chosen once and for all, at the bottom of the chain Theorem 3.8.22 (preprint 3.78) □ Theorem 3.5.6 (preprint 3.43) □ Theorem 3.5.11 (preprint Corollary 3.47) □ Theorem 3.6.8 (3) (preprint 3.54), so that its projection maps the partial quadrant into itself; condition (b) then holds for the chart of §III.34 without any hypothesis (Findings, Chapter 3, items 130–133). Along the way the closedness of the complement, which Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25) need (items 49, 81), is recorded in the statements (item 93).

The projection property. For subspaces N, Y of a vector space and a set C, *N projects C into itself along Y* if $z \in C$, $k \in N$ and $z - k \in Y$ imply $k \in C$ `Chapter3/GoodComplementProjection · ProjectsInto`. The definition avoids naming the projection; when $E = N \oplus Y$ it says exactly that the projection π onto N along Y maps C into C `Chapter3/GoodComplementProjection · ProjectsInto.projectionOnto_mem` `Chapter3/GoodComplementProjection · projectsInto_of_projectionOnto_mem` ($z - \pi z \in Y$, and conversely $z - k \in Y$, $k \in N$ force $\pi z = k$). It is carried by a linear isomorphism S with $x \in C \iff Sx \in C'$: given $z' \in C'$, $k \in N$ and $z' - Sk = Sy$ with $y \in Y$, the point $S^{-1}z'$ lies in C and $S^{-1}z' - k = y$, so $k \in C$ and $Sk \in C'$ `Chapter3/GoodComplementProjection · ProjectsInto.map`, in particular for continuous linear isomorphisms `Chapter3/GoodComplementProjection · ProjectsInto.mapL` — the analogue of Lemma 3.5.9 (preprint 3.45) (§III.14).

The finite-dimensional lemma. *If the finite-dimensional subspace N is in good position to the partial quadrant C of the sc-Banach space E, then N has a closed good complement Y such that the projection onto N along Y maps C into $N \cap C$* `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto`. The text has no such statement; Definition 3.1.24 (preprint 3.21) and Proposition 3.1.25 (preprint 3.22) do not need it. The proof works in the coordinates of §III.14: after a presentation, $C = \{\ell_i \geq 0\}$ for continuous functionals $\ell_1, \dots, \ell_n$ jointly onto $\mathbb{R}^n$, the cone $Q = N \cap C$ is $\{x \in N : \ell_i(x) \geq 0\}$, Σ is the set of indices vanishing at some nonzero point of Q, and for a set S of indices $Q_S = (\ell_i)_{i \in S} : N \to \mathbb{R}^S$ `Chapter3/GoodComplementProjection · coordOn` (for $S = \Sigma$ it is the map Q_Σ of Proposition 3.1.25 (preprint 3.22) `Chapter3/GoodComplementProjection · sigmaCoord_eq_coordOn`).

Step 1: the choice of coordinates `Chapter3/GoodComplementProjection · exists_coordSet`. Wanted is a set S of indices with (1) Q_S onto, (2) $Q = \{x \in N : \ell_i(x) \geq 0 \text{ for } i \in S\}$, (3) every ℓ_i, $i \notin S$, positive on $Q \setminus 0$. Property (3) holds for $S = \Sigma$ by the definition of Σ `Chapter3/GoodPosition · pos_of_not_mem_vanIdx`, and (1) for $S = \Sigma$ is the consequence of Lemma 3.8.8 (preprint 3.65) proved in §III.14 `Chapter3/GoodPosition · sigmaCoord_surjective`. Three cases, those of the normal form. If $Q = N$, take $S = \Sigma$; (2) is trivial. If $\Sigma \neq \emptyset$, take $S = \Sigma$; (2) is the segment argument of Proposition 3.1.25 (preprint 3.22) `Chapter3/GoodPosition · mem_quadCone_of_vanIdx`. In the remaining case, $\Sigma = \emptyset$ and $Q \neq N$, the choice $S = \Sigma$ fails, and here the sketch of Finding 83, which takes $S = \Sigma$ throughout, misses a case (item 131). With $S = \emptyset$, (2) would say $Q = N$; and the formula of Step 4 below would reduce to an arbitrary projection onto N, which need not map C into C: for $C = [0,\infty)^2$ and $N = \mathbb{R}(1,1)$ every nonzero point of Q has both coordinates positive, so $\Sigma = \emptyset$; the line $\mathbb{R}(1,2)$ is a good complement of N (for $|s| \leq \varepsilon|t|$ the coordinates of $t(1,1) + s(1,2)$ have the sign of t), and the projection along it maps $(0,1) = -(1,1) + (1,2)$ to $-(1,1) \notin C$. The formal proof keeps one coordinate. Some $x_1 \in N$ lies outside C, so there is at least one coordinate, j_0 say, and $S = \{j_0\}$. An interior point y of Q is nonzero — if 0 were interior, Q would contain a ball of N around 0 and, being closed under positive scaling, all of N `Chapter3/GoodComplementProjection · mem_quadCone_of_zero_mem_interior` `Chapter3/GoodPosition · smul_mem_quadCone` — so $\ell_{j_0}(y) > 0$ by (3), and $Q_{\{j_0\}}$ is

onto: $k \mapsto (k/\ell_{j_0}(y))\, y$ is a preimage. Property (2) for $S = \{j_0\}$ is the next step.

Step 2: the segment argument for any S `Chapter3/GoodComplementProjection · mem_quadCone_of_coordSet`. Suppose (3) holds for S, some $i_0 \in S$ is positive at a point $y \in Q$, and $x \in N$ has $\ell_i(x) \geq 0$ for $i \in S$. The set $T = \{s \geq 0 : x + sy \in Q\}$ is nonempty (for $i \in S$, $\ell_i(x) + s\ell_i(y) \geq 0$ for all $s \geq 0$; for $i \notin S$, $\ell_i(y) > 0$, so large s work), closed and bounded below; let $s^* = \min T$. If $s^* > 0$, the point $z = x + s^* y$ is nonzero, since $\ell_{i_0}(z) \geq s^* \ell_{i_0}(y) > 0$; so by (3) all ℓ_i, $i \notin S$, are positive at z and, by continuity, at $x + sy$ for s near s^*; the coordinates in S stay ≥ 0 for all $s \geq 0$; so a point $s_1 = \max(0, s^* - \delta/2) < s^*$ lies in T, contradicting minimality. Hence $s^* = 0$ and $x \in Q$. This is the argument of Proposition 3.1.25 (preprint 3.22), Case C (§III.14), with the positivity outside S as the only input; with $S = \{j_0\}$ and y interior it gives (2) in the third case of Step 1.

Step 3: a lower bound `Chapter3/GoodComplementProjection · exists_coord_ge`. A coordinate positive on $Q \setminus 0$ is bounded below by $c|x|$ on Q, $c > 0$: its minimum on the compact set $Q \cap \{|x| = 1\}$ is positive, and Q is a cone.

Step 4: the projection and the estimate `Chapter3/GoodComplementProjection · exists_proj_quadCone`. Take S from Step 1, a continuous projection $p_N : E_0 \to N$ (Hahn–Banach, N finite-dimensional), a linear right inverse g of Q_S, and

$$\pi(z) = p_N(z) + g\big((\ell_i(z - p_N z))_{i \in S}\big).$$

Then π is continuous, $\pi = \mathrm{id}$ on N (the bracket vanishes), and $\ell_i(\pi z) = \ell_i(p_N z) + \ell_i(z - p_N z) = \ell_i(z)$ for $i \in S$, so for $z \in C$ the coordinates of πz in S are ≥ 0 and $\pi z \in Q$ by (2): $\pi(C) \subseteq N \cap C$. The kernel of π is a complement of N, and it is a good complement. For $p \in \ker \pi$ the coordinates in S vanish ($\ell_i(p) = \ell_i(\pi p) = 0$). Let c_i be the constants of Step 3 for $i \notin S$ and $\varepsilon' = \min_i c_i/(\|\ell_i\| + 1)$ `Chapter3/GoodComplementProjection · exists_pos_le_of_finite`. Let $x \in N$, $p \in \ker \pi$, $|p| \leq \varepsilon'|x|$. If $x \in C$: for $i \in S$, $\ell_i(x + p) = \ell_i(x) \geq 0$; for $i \notin S$, $\ell_i(x + p) \geq c_i|x| - \|\ell_i\|\,|p| \geq c_i|x| - \|\ell_i\|\varepsilon'|x| \geq 0$. If $x + p \in C$: the coordinates of x in S are those of $x + p$, hence ≥ 0, and $x \in Q$ by (2). The reverse implication needs no estimate.

Step 5: partial quadrants. For a partial quadrant presented by an sc-isomorphism T the functionals are the coordinate functions of T, continuous `Chapter3/GoodComplementProjection · continuous_coordFun` and jointly onto `Chapter3/GoodPosition · coordFun_surj`; good position supplies a good complement P with its constant and the interior point. Step 4 then gives a continuous projection $\pi : E_0 \to N$ with $\pi(C) \subseteq C$ whose kernel is a good complement `Chapter3/GoodComplementProjection · exists_proj_goodComplement` ($\ker \pi$ is a topological complement of N, as the kernel of a continuous projection). With $Y = \ker \pi$, closed as the kernel of a continuous map, the projection property is immediate: $z \in C$, $k \in N$, $z - k \in Y$ give $k = \pi k = \pi z \in C$ `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto`.

Theorem 3.5.6 (preprint 3.43) with a prescribed good complement. In the text's proof the complement Z of $\widetilde{K} = \ker DH(0)$ in $\mathbb{R}^{n'}$ comes from a good complement of $K' = \{(\alpha, L\alpha) : \alpha \in \widetilde{K}\}$ through Lemma 3.5.10 (preprint 3.46) (§III.20). Nothing later uses that origin. *For every good complement Z of $\widetilde{K}$ (constant ε), $Z \oplus W$ is a good complement of K' with respect to $\widetilde{C} \oplus W$, with constant $\varepsilon/(\|L\| + 1)$* `Chapter3/GoodComplementProjection · IsGoodComplement.prod_top_graph`. The splitting: $K' \cap (Z \oplus W) = 0$ because $\widetilde{K} \cap Z = 0$, and $(a, w) = (\alpha, L\alpha) + (z, w - L\alpha)$ for $a = \alpha + z$. The estimate: with the maximum norm, $|(\alpha, L\alpha)| \leq (\|L\| + 1)|\alpha|$, so $|(z, w)| \leq \varepsilon/(\|L\| + 1) \cdot |(\alpha, L\alpha)|$ gives $|z| \leq \varepsilon|\alpha|$, and $(\alpha, L\alpha) + (z, w) \in \widetilde{C} \oplus W$ iff $\alpha + z \in \widetilde{C}$ iff $\alpha \in \widetilde{C}$ iff $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ `Chapter3/LocalSolvability · graphMap_mem_prod_iff`. This is the second half of the proof of Lemma 3.5.10 (preprint 3.46) for an arbitrary Z; Lemma 3.5.10 (preprint 3.46) itself is needed only for the good position of $\widetilde{K}$. The projection property passes the same way: if

$z \in \widetilde{C} \oplus W$ and $z - (\alpha, L\alpha) \in Z \oplus W$, then $z_1 - \alpha \in Z$, so $\alpha \in \widetilde{C}$ when $(\widetilde{K}, Z)$ has the property, and $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ `Chapter3/GoodComplementProjection · ProjectsInto.prod_top_graph`; the projection along $Z \oplus W$ is $(a, w) \mapsto (\pi_Z a, L\pi_Z a)$, and $\widetilde{C} \oplus W$ constrains only the first component.

So the data of the proof of Theorem 3.5.6 (preprint 3.43) (§III.20: the derivative data, the graph description $K' = \operatorname{graph}(L|_{\widetilde{K}})$ of Lemma 3.5.8 (preprint 3.44), a complement Z, and the solution σ of $H(k + \sigma(k)) = 0$ from Theorem 3.8.22 (preprint 3.78) with its uniqueness constants `Chapter3/LocalSolvability · ShiftedBasicData.Sol`) exist with any prescribed good complement Z of $\widetilde{K}$ `Chapter3/GoodComplementProjection · ShiftedBasicData.exists_sol_of_goodComplement`: K' is the graph `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.ker_Dg_eq`, so $\widetilde{K}$ is in good position (Lemma 3.5.10 (preprint 3.46)) `Chapter3/LocalSolvability · lemma_3_5_10_isInGoodPosition`, $Z \oplus W$ is a good complement of K' by the lemma above, and Theorem 3.8.22 (preprint 3.78) is applied to H with Z `Chapter3/QuadrantIFT · theorem_3_8_22`. Choosing Z by the finite-dimensional lemma in $\mathbb{R}^{n'}$ (constant sc-structure, partial quadrant $\widetilde{C}$) gives data whose Z projects $\widetilde{C}$ into $\widetilde{K} \cap \widetilde{C}$ `Chapter3/GoodComplementProjection · ShiftedBasicData.exists_sol_proj`.

Theorem 3.5.6 (preprint 3.43), closed and projecting. With the hypotheses of Theorem 3.5.6 (preprint 3.43) (§III.20: $\widetilde{C}$ a partial quadrant, $Dg(0)$ onto, $K' = \ker Dg(0)$ in good position, the regularity of Lemma 3.5.5 (preprint 3.42) as a hypothesis), the conclusions of the level-wise form `Chapter3/LocalSolvability · theorem_3_5_6_sc_smooth` hold with a good complement $Y = Z \oplus W$ that is **closed** `Chapter3/LocalSolvabilityClosed · theorem_3_5_6_closed` — Z is finite-dimensional, hence closed, and $Z \times W_0$ is closed — and, with the data of the last paragraph, that has the **projection property** for $\widetilde{C} \oplus W$ `Chapter3/GoodComplementProjection · theorem_3_5_6_proj` (rewrite K' as the graph and apply the passage of the property to $Z \oplus W$). The other conclusions (V, τ with values in Y and W-components on level 1, C^1, $\tau(0) = 0$, $D\tau(0) = 0$, the graph of zeros, the C^k lifts τ_k, the neighbourhood U'' in which every zero lies on the graph) are those of §III.20, read off the same data `Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_contDiffOn_level` `Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_eq_add_τ`.

Theorem 3.5.11 (preprint Corollary 3.47), closed and projecting `Chapter3/LocalSolvabilityClosed · theorem_3_5_11_proj`. For a basic germ f, the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) with $D(f + \iota s)(0)$ onto and kernel K in good position to $\widetilde{C} \oplus W$: the conclusions of the level-wise Theorem 3.5.11 (preprint Corollary 3.47) `Chapter3/LocalSolvability · theorem_3_5_11_sc_smooth` hold with Y closed and K projecting $\widetilde{C} \oplus W$ into itself along Y. The proof is that of §III.20 with the two new properties carried along. Let S be the linear coordinate change $\psi(a, w) = ((a, (1 - P_1)w), P_1 w)$ of Theorem 3.4.2 (preprint 3.38) on level 0, a continuous linear isomorphism with inverse the gluing map. The push-forward $g = \Phi_*(f + s)$, packaged as data of Theorem 3.5.6 (preprint 3.43) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`, has $Dg(0) = \Psi \circ D(f + \iota s)(0) \circ S^{-1}$ `Chapter3/LocalSolvability · StabilityData.scDerivGerm_pushMap₀`, so $\ker Dg(0) = S(K)$, $Dg(0)$ is onto, and K is finite-dimensional (Lemma 3.5.8 (preprint 3.44) for g and the injectivity of S^{-1}). S carries $\widetilde{C} \oplus W$ onto $(\widetilde{C} \oplus C_s) \oplus X$, where C_s is the finite-dimensional space of the splitting of Theorem 3.4.2 (preprint 3.38) (§III.18) and $\widetilde{C} \oplus C_s$ is a partial quadrant of $\mathbb{R}^n \oplus C_s$ `Chapter3/LocalSolvability · IsPartialQuadrant.const_prod_univ`; so $S(K)$ is in good position (Lemma 3.5.9 (preprint 3.45), §III.14). Theorem 3.5.6 (preprint 3.43), closed and projecting, for g, with Lemma 3.5.5 (preprint 3.42) for its regularity hypothesis `Chapter3/LocalSolvability · StabilityData.lemma_3_5_5`, gives Y', V', τ. Transported back: $Y = S^{-1}(Y')$ is a good complement of $K = S^{-1}(S(K))$ (Lemma 3.5.9 (preprint 3.45)) `Chapter3/GoodPosition · IsGoodComplement.map`, closed as the image of a closed set under a linear homeomorphism, and projecting, by the transport of the projection property `Chapter3/GoodComplementProjection · ProjectsInto.mapL`; $V = (S|_K)^{-1}(V')$ and $\sigma = S^{-1} \circ \tau \circ S|_K$. The W-component of $\sigma(y)$ is $c + w$ with $c \in C_s \subseteq W_\infty$ `Chapter3/BasicClass · StabilityData.Cs_le_smooth` and w the level-1 X-component of τ, so it lies on level 1;

$D\sigma(0) = S^{-1} \circ D\tau(0) \circ S|_K = 0$, by the chain rule within V, which has unique differentiability at 0 as a relatively open subset of the convex cone $K \cap C$ with interior; the graph points are zeros because the linear fibre map of Theorem 3.4.2 (preprint 3.38) is injective; the lifts are ψ^{-1} on level k applied to τ_k; and $U'' := S^{-1}(U'')$. The form with only the closedness is a corollary `Chapter3/LocalSolvabilityClosed · theorem_3_5_11_closed`.

Theorem 3.6.8 (3) (preprint 3.54), closed and projecting `Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj`. The graph description of §III.23 — for a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the chart φ centred at x_0, the conjugating sc-diffeomorphism ψ, the germ $h + t$ with surjective linearization and kernel K in good position, Y, V', σ, and the open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$ — is proved with Theorem 3.5.11 (preprint Corollary 3.47) closed and projecting in place of the plain Theorem 3.5.11 (preprint Corollary 3.47), and the statement now records five more facts. (i) The inverse ψ' of ψ, sc-smooth on U', with $\psi'\psi = \mathrm{id}$ on U and $\psi\psi' = \mathrm{id}$ on U' (the base maps of the conjugating strong bundle isomorphism and its inverse); §III.34 needs it to make $\psi \circ \varphi$ a chart. (ii) Y is closed, and (iii) K projects $C' = \widetilde{C} \oplus W$ into itself along Y, both from Theorem 3.5.11 (preprint Corollary 3.47) — the shrinking of V' in step 6 of §III.23 ($V' = V_0 \cap O_1$ with O_1 open, chosen so that $v + \sigma(v)$ stays in $\Omega = U'' \cap W'$) does not touch Y. (iv) K is a smooth subspace `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth` (proved in §III.34). (v) Every $v + \sigma(v)$, $v \in V'$, is a zero of $h + t$ in the germ, and for every $k \geq 1$ the C^k lift σ_k of Theorem 3.5.11 (1) (preprint Corollary 3.47) survives the shrinking: on $V_k \cap O_1$, relatively open in $K \cap C'$, containing 0 and contained in V', it is still C^k and still lifts σ. The theorem of §III.23, with the same statement except the projection property, is a corollary `Chapter3/GlobalIFTG · theorem_3_6_8_graph`.

Where the formal route departs from the text. The text never chooses the good complement; Definition 3.2.1 (preprint 3.25) lets it be any good complement, and for an arbitrary subset with the tangent germ property the repair of the claim "Σ is relatively open" would need a re-graphing over a better complement, which is not done ($\pi(O) \subseteq C$ remains a hypothesis of Theorem 3.2.3 (preprint 3.27), §III.24). For solution sets the complement is not given but produced, at Theorem 3.8.22 (preprint 3.78), and every link of the chain accepts a prescribed complement; so the repair is a choice made at the bottom of the chain, not a change in the proof of Theorem 3.2.3 (preprint 3.27) (item 130). The closedness of Y is part of the text's sc-complement and is recorded because Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25) use it at the nearby zeros (items 49, 81, 93).

III.30 Proposition 3.6.5 (preprint 3.52) with its index

Proposition 3.6.5 (preprint 3.52) (text): the linearization $Tf(x) : T_xX \to Y_x$ of an sc-Fredholm section at a zero is an sc-Fredholm operator; in the proof (and in Theorem 3.6.8 (preprint 3.54)) its index is the index $n - N$ of the basic germ $h : O(([0,\infty)^k \oplus \mathbb{R}^{n-k}) \oplus W, 0) \to (\mathbb{R}^N \oplus W, 0)$ to which $g - s$ is conjugated. The statements of §III.21 state the index as "$n - N$ for some natural numbers n, N", which every integer is. The reason is that the conjugation of Definition 3.1.11 (preprint 3.11) is an existential proposition `Chapter3/StrongBundleSections · IsConjugateToBasicGerm`, so the dimensions of the basic germ cannot be named outside it. Here the witnesses become data.

Conjugation data `Chapter3/FredholmGermIndex · ConjugationData`: finite-dimensional normed spaces P_n (the text's $\mathbb{R}^n$) and Q (the text's $\mathbb{R}^N$), an sc-Banach space W, a partial quadrant $\widetilde{C} \subseteq P_n$, a relatively open $U' \subseteq \widetilde{C} \oplus W$, a strong bundle isomorphism $\Psi : U \lhd F \to U' \lhd (Q \oplus W)$ with inverse Ψ' and $\psi(0) = 0$ for its base map, a germ G with $G_0 \subseteq U'$, a basic germ h on G with contraction map B, and

$(\Psi(u,k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0. A map is conjugated to a basic germ iff such data exist Chapter3/FredholmGermIndex · isConjugateToBasicGerm_iff_nonempty.

The derivative of a conjugated map Chapter3/FredholmGermIndexG · ConjugationData.scDeriv_isFredholm_index. Let U be relatively open in the partial quadrant C, $0 \in U$, k an sc¹-germ with derivative family Dk on a germ G_k with $(G_k)_0 \subseteq U$, $k(0) = 0$, and d conjugation data of k. Then $Dk(0) : E_0 \to F_0$ is Fredholm of index $\dim P_n - \dim Q$; for k sc-smooth on U this is the whole-chart form Chapter3/FredholmGermIndex · ConjugationData.scDeriv_isFredholm_index_wholeChart. The proof is that of §III.21, step (a), with the dimensions kept:

(a) $Dh(0)$ is Fredholm of index $\dim P_n - \dim Q$ (Proposition 3.1.10 (preprint 3.10) for the basic germ on level 0) Chapter3/FredholmGermLinearization · proposition_3_1_10_basic_level_zero.

(b) $\Gamma(0)Dk(0) = Dh(0)D\psi(0)$, with $\Gamma(0) = (\Psi(0,\cdot))_2$ the fibre operator of Ψ at 0 Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM. The left side is the derivative at 0 of $u \mapsto (\Psi(u,k(u)))_2$, because $k(0) = 0$ Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_comp. For the right side, let N_1 be an open neighbourhood of 0 on which the conjugation identity holds; the germ $\{U \cap N_1\}$ has a subgerm G_1 mapped by ψ into G level by level Chapter3/GermFilling · ScNbhdGerm.preimage; on G_1 the chain rule for germs gives the derivative family $Dh(\psi_1(x))D\psi(x)$ of $h \circ \psi$ Chapter3/Germ · IsSc1GermWith.comp; and $u \mapsto (\Psi(u,k(u)))_2$, an sc¹-germ on G_k by the chain rule with the whole-chart outer map (§III.48), agrees with $h \circ \psi$ on $G_1 \cap G_k$, so the two derivative families agree at 0 (uniqueness of the derivative of a germ at points of U_1) Chapter3/Germ · IsSc1Germ.scDerivGerm_eqOn, with $\psi_1(0) = 0$.

(c) $D\psi(0)$ and $D\psi'(0)$ are inverse to each other (chain rule for $\psi' \circ \psi = \mathrm{id}$ on U and $\psi \circ \psi' = \mathrm{id}$ on U') Chapter2/DegeneracyIndex · scDeriv_comp_eq_self, and so are $\Gamma(0)$ and the fibre operator $\Gamma'(0)$ of Ψ' at 0: $\Psi'(\Psi(0,y)) = (0,y)$ with $\Psi(0,y) = (0,\Gamma(0)y)$ because $\psi(0) = 0$, and symmetrically.

(d) Hence $Dk(0) = \Gamma(0)^{-1} \circ Dh(0) \circ D\psi(0)$, a Fredholm operator composed with isomorphisms on both sides, of the same index Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv.

Consequently **the index $n - N$ does not depend on the conjugation**: two conjugation data d, d' of the same k give $\dim d.P_n - \dim d.Q = \operatorname{ind} Dk(0) = \dim d'.P_n - \dim d'.Q$ Chapter3/FredholmGermIndexG · ConjugationData.index_eq. This is the text's remark after Theorem 3.4.2 (preprint 3.38) that "the Fredholm index is invariant under strong bundle isomorphisms", for the conjugations of one germ.

Proposition 3.6.5 (preprint 3.52) in the chart, with the index Chapter3/FredholmGermIndexG · proposition_3_6_5_local_index. Hypotheses: U relatively open in C, R a strong bundle retraction on $U \lhd F$ with base retraction r, $0 \in U$ and $0 \in O = r(U)$; g a filling germ of the principal part $\hat{f}$ (§III.3) and Dg a derivative family of g on a germ G with $G_0 \subseteq U$; $\hat{f}(0) = 0$ and $(x, \hat{f}(x)) \in K$ for $x \in O$; s an sc⁺-germ with derivative family Ds and $\iota s(0) = g(0)$; and conjugation data d of $k = g - \iota s$. Conclusion: $Dg(0) : E_0 \to F_0$ and $Tf(0) = Dg(0)|_{T_0O} : T_0O \to K_0$ Chapter3/FillerLinearAlgebra · fillerRestrict are Fredholm with index $\dim d.P_n - \dim d.Q$. Proof: $g(0) = \hat{f}(0) = 0$ (filler condition (1) at 0), so k is an sc¹-germ on $G \cap G_s$ with the family $Dg - \iota Ds$ and $k(0) = 0$ Chapter3/FredholmGermLinearizationG · IsSc1GermWith.sub_toBase_shift, and $Dk(0)$ is Fredholm of index $\dim d.P_n - \dim d.Q$ by the preceding theorem. Since 0 is a smooth point, $Ds(0) : E \to F^1$ is an sc-operator (the remark after Proposition 1.2.1 (preprint 1.16), for germs) Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero, so $\iota Ds(0)$ is the level-0 map of an sc⁺-operator and compact. $Dg(0) = Dk(0) + \iota Ds(0)$ is then the value at 0 of the identity $Dg = (Dg - \iota Ds) + \iota Ds$ Chapter3/FredholmGermLinearizationG · IsSc1GermWith.deriv_zero_eq_add_scPlus; on U it was the uniqueness of derivatives within U_1 and the density of E_1 Chapter3/

BasicGermLinearization · ContinuousLinearMap.ext_of_comp_toBase_one. A compact perturbation of a Fredholm operator is Fredholm of the same index Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator. Proposition 3.1.5 (preprint 3.5) (§III.4) transfers the Fredholm property and the index to $Tf(0)$ Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_1_5.

Proposition 3.6.5 (preprint 3.52), with the index Chapter3/FredholmGermIndexG · proposition_3_6_5_index. Let f be a section of the strong bundle $P : Y \to X$ with (f, x_0) an sc-Fredholm germ and $f(x_0) = 0$. In the strong bundle chart of Definition 3.1.11 (preprint 3.11) centred at x_0, with the filling germ g of $\widehat{f}$, its germ G and a derivative family Dg, and the local sc$^+$-germ s, conjugation data of $g - s$ exist, $Dg(0)$ and $Tf(x_0)$ are Fredholm, and **for every** conjugation data d of $g - s$ their index is $\dim d.P_n - \dim d.Q$. Proof: unpack Definition 3.1.11 (preprint 3.11); $0 \in O$ because the chart is centred at x_0; $\widehat{f}(0) = 0$ from $f(x_0) = 0$; $(x, \widehat{f}(x)) \in K$ because the principal part of a section lies in the retract K of the chart; data d_0 exist by the equivalence above and give the Fredholm property; and the local theorem applied to an arbitrary d gives the index. No hypothesis is added. As in §III.21, only the level-0 Fredholm property is proved (T_0O and K_0 are not set up as sc-Banach spaces). The statements of §III.21 stand next to these, with the same proofs and without the dimensions; the whole-chart forms are kept as well Chapter3/FredholmGermIndex · proposition_3_6_5_index_wholeChart.

III.31 The linearization at a zero in charts, its chart independence, and Definition 3.6.6 (preprint 3.53) from a non-centred chart

Definition 3.6.6 (preprint 3.53) is stated in T_xX, with the linearization $f'(x) : T_xX \to Y_x$ and the partial cone C_xX of Definition 2.5.8 (preprint 2.38). The text defines $f'(x)$ only at the *centre* of a chart, before Definition 3.1.4 (preprint 3.4), as the restriction of $D(\mathbf{f} \circ r)(0)$ to T_0O, identifies it with $Dg(0)|_{T_0O}$ for a filler g, and appeals to Proposition 2.4.10 (preprint 2.30) for the independence of T^R_xX and C_xX of the chart. That $f'(x)$ itself, and the two conditions of Definition 3.6.6 (preprint 3.53), do not depend on the chart is used — the proof of Theorem 3.6.8 (preprint 3.54) repeats "as above" the argument at x_0 in a chart centred at a nearby zero y — but not proved (Finding 110). The formalization (§III.21) states Definition 3.6.6 (preprint 3.53) in the chart of Definition 3.1.11 (preprint 3.11) centred at the zero; this section proves the local form of the missing independence at a zero, in charts, and the assembly step that Theorem 3.6.8 (1) (preprint 3.54) needs: Definition 3.6.6 (preprint 3.53) at a zero y from data in a chart which is *not* centred at y (the chart at x_0). Throughout, z is a point on level 1 (the text has smooth points), and tameness is not used.

Linear algebra of conjugate operators. Let $L : T \to K$ and $L' : T' \to K'$ be bounded operators between normed spaces and $A : T \cong T'$, $\Gamma : K \cong K'$ topological linear isomorphisms with $L'A = \Gamma L$. Then $L' = \Gamma L A^{-1}$ Chapter3/LinearizationCharts · eq_comp_of_conj, so L is onto iff L' is Chapter3/LinearizationCharts · surjective_iff_of_conj; $\ker L' = A(\ker L)$ Chapter3/LinearizationCharts · ker_eq_map_of_conj; for complete spaces L is Fredholm iff L' is, with the same index, because Fredholm operators and their index are stable under composition with isomorphisms on both sides Chapter3/LinearizationCharts · isFredholm_iff_of_conj Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv; and if $D \subseteq K$ is dense and contained in the range of L, then $\Gamma(D)$ is dense in K' and contained in the range of L' Chapter3/LinearizationCharts · dense_image_of_conj. The isomorphisms that occur are restrictions of bounded operators $a : E_0 \to E_0'$, $a' : E_0' \to E_0$ which map a subspace T into T' and T' into T and are inverse to each other there Chapter3/LinearizationCharts · restrictEquiv. A subspace $N \subseteq E_0$ is trans-

ported by $A : T \cong T'$ to $A(N) = \{Ax : x \in T \cap N\} \subseteq T'$ `Chapter3/LinearizationCharts · transportSub`, with $y \in A(N) \iff A^{-1}y \in N$ for $y \in T'$ `Chapter3/LinearizationCharts · mem_transportSub_iff` and $A^{-1}(A(N)) = N$ for $N \subseteq T$ `Chapter3/LinearizationCharts · transportSub_symm`.

Good position inside a subspace is transported `Chapter3/LinearizationCharts · IsInGoodPositionIn.transport`. Good position inside T (§III.21, `Chapter3/GoodPositionGerm · IsInGoodPositionIn`) asks: $N \subseteq T$, the interior of $N \cap C$ in N nonempty, and a complement $P \subseteq T$ of N *in* T with the ε-condition of Definition 3.1.24 (preprint 3.21). If $A : T \cong T'$ respects the cones ($x \in C \iff Ax \in C'$ for $x \in T$) and N is in good position to C inside T, then $A(N)$ is in good position to C' inside T'. *The interior*: A restricts to a homeomorphism $N \cong A(N)$ which carries $N \cap C$ onto $A(N) \cap C'$, so it carries interior points to interior points. *The complement*: $A(P)$; $A(N) \cap A(P) = 0$ and $A(N) + A(P) = T'$ by pulling back with A^{-1}. *The constant*: with $a = \|A\|$, $b = \|A^{-1}\|$, take $\varepsilon' = \varepsilon/(ab+1)$; for $y = Ax$, $q = Ap$ with $|q| \le \varepsilon'|y|$, $|p| \le b|q| \le b\varepsilon' a|x| \le \varepsilon|x|$, so $x \in C \iff x + p \in C$, i.e. $y \in C' \iff y + q \in C'$ — the constant of Lemma 3.5.9 (preprint 3.45), no finite dimensionality needed (Finding 115). Applied to A and A^{-1} this gives an equivalence `Chapter3/LinearizationCharts · isInGoodPositionIn_transport_iff`.

Two derivative tools at a level-1 point. (i) *Uniqueness*: if $f_1 = f_2$ on a relatively open $V \subseteq U_1 \cap U_2$ containing z, then $Df_1(z) = Df_2(z)$: both are derivatives within V_1 (level 1), V is an sc-domain with unique differentiability, and two operators $E_0 \to F_0$ that agree after composition with the dense inclusion $E_1 \to E_0$ are equal `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen`. (ii) *The fibre derivative at a point of the zero section* `Chapter3/LinearizationCharts · scDeriv_fiber_of_vanishing`: let Φ be sc¹ on a set $D \subseteq E \oplus F$ containing $\{(u,0) : u \in S\}$ (S relatively open, $z \in S_1$) and the whole fibre $\{z\} \times F$, with $\Phi(u,0) = 0$ on S and $\Phi(z,k) = \Gamma k$ for a bounded Γ. Then $D\Phi(z,0)(h,k) = \Gamma k$. Proof: along $k_1 \mapsto (z,k_1)$, $k_1 \in F_1$, the composite is $\Gamma \circ \iota$, so $D\Phi(z,0)$ restricted to the fibre direction is Γ after composition with the inclusion, hence equal to Γ by density; along $u_1 \mapsto (u_1,0)$, $u_1 \in S_1$, the composite vanishes, and unique differentiability of S_1 at z gives zero in the base direction. This is the generalization from 0 to z of the identities of §III.4 and §III.21; it is the reason why only the fibre part survives at a zero (Finding 112).

The linearization at a point of the retract, without a filler. For the principal part $\mathbf{f}$ of a section in a strong bundle chart with retraction $R(u,h) = (r(u), \rho(u)h)$ and a level-1 point z of $O = r(U)$, put

$$\mathrm{Lin}(z) = D(\mathbf{f} \circ r)(z)|_{T_zO}, \qquad T_zO = Dr(z)E_0$$

`Chapter3/LinearizationCharts · linAt`; $Dr(z)$ is idempotent `Chapter3/LinearizationCharts · scDeriv_r_idem_at`. Three facts make it the right object.

Its values lie in $K_z = \rho(z)F$ at a zero `Chapter3/LinearizationCharts · rho_scDeriv_comp_r` (the text's "$\rho(0)Tf(0) = Tf(0)$", now at any zero). Since $(r(u), \mathbf{f}(r(u))) \in K$, $\rho(r(u))\mathbf{f}(r(u)) = \mathbf{f}(r(u))$ on U `Chapter3/Filler · ρ_eq_self_of_mem_K`; so $\mathbf{f} \circ r$ equals $\Psi_2(u) = (R(r(u), \mathbf{f}(r(u))))_2$, and their derivatives at z agree by (i). The chain rule for $\Psi_2 = (\text{fibre part of } R) \circ (r, \mathbf{f} \circ r)$ at z, where $(r, \mathbf{f} \circ r)$ passes through $(z,0)$ (because $r(z) = z$ and $\mathbf{f}(z) = 0$), and (ii) for the fibre map $(u,k) \mapsto \rho(u)k$, which vanishes on the zero section and is $\rho(z)$ on the fibre over z, give $D(\mathbf{f} \circ r)(z)v = \rho(z)D(\mathbf{f} \circ r)(z)v$. The fibre K_z is the fibre of K over z `Chapter3/LinearizationCharts · mem_rangeS_ρCLM_iff`.

The bridge to fillers `Chapter3/LinearizationCharts · scDeriv_comp_r_eq_of_eventuallyEq` (the text's "$Tg(0) = Tf(0)$ on T_0O", now at any point): if g is sc¹ and agrees with $\mathbf{f}$ on O near z, say on $O \cap W$ with W open, then $\mathbf{f} \circ r = g \circ r$ on the relatively open $U \cap r^{-1}(W)$, so by (i) and the chain rule (r maps z to itself on level 1) $D(\mathbf{f} \circ r)(z) = Dg(z)Dr(z)$. Hence $\mathrm{Lin}(z) = Dg(z)$ on T_zO `Chapter3/LinearizationCharts · linAt_`

eq_scDeriv, by idempotence of $Dr(z)$. At a centre, for a filled version, $\mathrm{Lin}(0)$ is the $Tf(0) = Dg(0)|_{T_0O}$ of Proposition 3.1.5 (preprint 3.5) Chapter3/LinearizationCharts · IsFilledVersion.fillerRestrict_eq_linAt Chapter3/FillerLinearAlgebra · fillerRestrict. At $z \neq 0$ the full $Dg(z) : E \to F$ is not related to $\mathrm{Lin}(z)$ (condition (3) of the filler holds at 0 only), so the relevant kernel is $\ker Dg(z) \cap T_zO = \ker \mathrm{Lin}(z)$ (Finding 111). *For a filling germ g* on G the bridge needs two things the whole-chart one did not: $Dg(z)$ exists only for z on level 1 of G, and the chain rule for $g \circ r$ is the inner one on the germ $r^{-1}(G)$, which is defined when $r(0) = 0$, i.e. $0 \in O$ Chapter3/LinearizationChartsG · scDeriv_comp_r_eq_germ. Under these two conditions $\mathrm{Lin}(z) = Dg(z)$ on T_zO Chapter3/LinearizationChartsG · linAt_eq_germDeriv Chapter3/LinearizationChartsG · StrongBundle.linAt_eq_germDeriv_chart, and at the centre, which lies in every germ, $\mathrm{Lin}(0) = Tf(0)$ Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt. Both hold wherever the bridge is used: the chart is centred at a zero, and the points $z \neq 0$ are the level-1 lifts of nearby zeros, which lie in the germ of g by the shrinking of the basic germ (§§III.32, III.48).

Fibres of strong bundle charts and of transitions. For a strong bundle chart $\Phi : P^{-1}(V) \to K$ covering φ: points of K lie over O Chapter3/LinearizationCharts · StrongBundleChart.fst_mem_0; $\Phi^{-1}(x, \cdot)$ is additive and homogeneous on the fibre of K over x, since Φ is a fibrewise linear bijection of $P^{-1}(\varphi^{-1}(x))$ onto that fibre Chapter3/LinearizationCharts · StrongBundleChart.symm_add Chapter3/LinearizationCharts · StrongBundleChart.symm_smul; the fibre of K is closed under scalars and contains $(u, \rho(u)k)$ Chapter3/LinearizationCharts · StrongBundleChart.mem_K_ρ. For a second chart Φ' the transition $\Theta = \Phi' \circ \Phi^{-1}$ covers the base transition $\tau = \varphi' \circ \varphi^{-1}$ Chapter3/LinearizationCharts · StrongBundleChart.transition_fst, is linear on the fibres Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_add Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_smul, maps the zero section to the zero section Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_zero and K into K' Chapter3/LinearizationCharts · StrongBundleChart.transition_mem_K, and is inverted by the reverse transition Chapter3/LinearizationCharts · StrongBundleChart.transition_transition. Over a point x of the base overlap, $\Gamma(x)k = (\Theta(x, \rho(x)k))_2$ is a bounded operator $F \to F'$ — linear by the fibre linearity, continuous as a composition of $\rho(x)$, Φ^{-1} and Φ' Chapter3/LinearizationCharts · StrongBundleChart.transitionFiberCLM — which on K_x is the transition itself Chapter3/LinearizationCharts · StrongBundleChart.transitionFiberCLM_apply_of_mem. The text's "bounded linear on the fibres" (Definition 2.6.4 (preprint 2.50)) is about Φ only; the linearity of Φ^{-1} on the fibres of K is this lemma (Finding 113). Finally, if $\varphi(V \cap V') = O \cap W$ with W open, the overlap of K is $K \cap (W \times F)$ Chapter3/LinearizationCharts · StrongBundleChart.overlap_eq_inter_prod.

The chart change at a zero Chapter3/LinearizationCharts · linAt_chart_change. *Statement.* Let c, c' be strong bundle charts whose transition Θ is sc-smooth on $[0]$ (as a map between the overlap retracts of K and K'), f a section whose representatives $z \mapsto (z, \mathbf{f}(z))$ are sc-smooth $O \to K$ and $O' \to K'$, y a zero of f in both chart domains, and z, z' level-1 points over $\varphi(y)$, $\varphi'(y)$. Then there are topological isomorphisms $A : T_zO \cong T_{z'}O'$ and $\Gamma : K_z \cong K'_{z'}$, with $\Gamma \circ \mathrm{Lin}(z) = \mathrm{Lin}'(z') \circ A$, and $x \in C_z \iff Ax \in C'_{z'}$, for $x \in T_zO$, where C_z is the partial quadrant at $\varphi(y)$ of a presentation of C Chapter2/Tame · partialQuadrantAtWith. The proof, in the order of the Lean:

1. *The base transition as a germ of sc-diffeomorphisms.* The base charts are compatible (both are charts of X, Chapter2/MPolyfold · IsChart.compatible). Choose open W, W' with $\varphi(V \cap V') = O \cap W$ and $\varphi'(V \cap V') = O' \cap W'$ Chapter2/MPolyfold · exists_isOpen_image_eq and restrict the base retractions to $U \cap r^{-1}(W)$, $U' \cap r'^{-1}(W')$ (Proposition 2.3.1 (1) (preprint 2.10)); their retracts are the overlaps. Then τ, $\tau' = \varphi \circ \varphi'^{-1}$ form a germ of sc-diffeomorphisms between them Chapter2/Boundary · ScRetraction.IsDiffeoGerm, and $\tau \circ r_W$ maps z to z' on level 1. The derivatives of the restricted retraction and of $\mathbf{f} \circ r$ at z are those on the full domains, by (i) (Finding 114).

2. *The transition near the zero section.* Restrict the strong bundle retraction to $W \times F$; its retract is the overlap of K, so $\Theta \circ R_W$ is sc-smooth and its fibre part Φ is sc¹. $\Phi(z,k) = \Gamma(z)k$ because $R(z,k) = (z, \rho(z)k)$; $\Phi(u,0) = 0$ because $R(u,0) = (r(u),0)$ and Θ preserves the zero section.
3. *The section in the two charts.* For u in the restricted domain, $r(u) = \varphi(y_1)$ for some $y_1 \in V \cap V'$, and both $\mathbf{f}' \circ r' \circ \tau \circ r_W$ and $\Phi \circ (r, \mathbf{f} \circ r)$ take at u the value $(\Phi'(f(y_1)))_2$ (the principal parts are the fibre parts of $\Phi \circ f \circ \varphi^{-1}$ and $\Phi' \circ f \circ \varphi'^{-1}$, `Chapter3/StrongBundleSections · StrongBundleChart.apply_section'`). By (i) their derivatives at z agree.
4. *The key identity.* The chain rule on both sides, with $(r, \mathbf{f} \circ r)$ passing through $(z,0)$ since $\mathbf{f}(z) = 0$, and (ii) for Φ give
$$D(\mathbf{f}' \circ r')(z')\, D(\tau \circ r_W)(z)v = \Gamma(z)\, D(\mathbf{f} \circ r)(z)v.$$
The derivative of $x \mapsto \Gamma(x)\mathbf{f}(z)$ in the base direction has dropped out *because* $\mathbf{f}(z) = 0$; away from zeros the linearization is not chart independent (Finding 112).
5. *The isomorphism A.* $A_0 = D(\tau \circ r_W)(z)$ maps T_zO into $T_{z'}O'$ and $A_0' = D(\tau' \circ r'_{W'})(z')$ maps back `Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mem_range`; on these subspaces they are inverse `Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_inv_deriv` (with the idempotence of $Dr(z)$); A is the induced isomorphism (`restrictEquiv`).
6. *The isomorphism Γ.* $\Gamma(z)$ maps K_z into $K'_{z'}$ (the transition maps K into K' and covers τ), the fibre operator $\Gamma'(z')$ of the reverse transition maps back, and they are inverse on these fibres (the reverse transition inverts the transition).
7. *Conclusions.* For $x \in T_zO$, $\mathrm{Lin}(z)x = D(\mathbf{f} \circ r)(z)x$ and $\mathrm{Lin}'(z')(Ax) = D(\mathbf{f}' \circ r')(z')A_0x$, so step 4 is $\Gamma\,\mathrm{Lin}(z) = \mathrm{Lin}'(z')A$. The cones: Proposition 2.4.10 (preprint 2.30) for the partial cone, applied to the germ of step 1 `Chapter2/Boundary · proposition_2_4_10_cone`, gives $A_0(T_zO \cap C_z) = T_{z'}O' \cap C'_{z'}$ for the formalized partial cone `Chapter2/Tame · partialConeWith`, which is the cone condition on T_zO.

Hypotheses. Of the compatibility of Definition 2.6.4 (preprint 2.50) only the sc-smoothness of Θ on $[0]$ is used — not on $[1]$, not the reverse transition (Finding 116). For two strong bundle charts of P it holds because both are charts of the M-polyfold $Y[0]$, hence compatible `Chapter3/LinearizationCharts · StrongBundle.IsChart.transition_isScSmoothRetractMap`; and for an sc-smooth section the representatives are sc-smooth in every strong bundle chart of P, so no hypothesis is added.

Chart independence `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`. For two strong bundle charts of P around a zero y of an sc-smooth section, with level-1 points z, z' over $\varphi(y)$, $\varphi'(y)$: the linearizations $L = \mathrm{Lin}(z) : T_zO \to K_z$ and $L' = \mathrm{Lin}'(z') : T_{z'}O' \to K'_{z'}$ (corestricted to the fibres by the first fact above) are simultaneously surjective, simultaneously Fredholm with the same index, and $\ker L$ is in good position to C_z inside T_zO iff $\ker L'$ is in good position to $C'_{z'}$ inside $T_{z'}O'$. The four spaces are complete, being ranges of the idempotents $Dr(z)$, $\rho(z)$ and their primed versions `Chapter3/FillerLinearAlgebra · isClosed_range_of_idem`; the rest is the conjugation algebra, with $\ker L' = A(\ker L)$ and the transport of good position.

Definition 3.6.6 (preprint 3.53) at a zero, from a non-centred chart `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`. *Statement.* Let f be an sc-smooth section, y a zero with (f,y) an sc-Fredholm germ (Definition 3.1.11 (preprint 3.11)), c a strong bundle chart of P around y (not necessarily centred at y), z on level 1 over $\varphi(y)$, and g sc¹ on U agreeing with $\mathbf{f}$ on O near $\varphi(y)$ — the case in view was the chart at x_0 with a whole-chart filled version. Assume (a) the range of $Dg(z)|_{T_zO}$ contains $\rho(z)\hat{k}$ for every smooth $\hat{k} \in F_\infty$, and (b) $\ker Dg(z) \cap T_zO$ is in good position to $C_{\varphi(y)}$ inside T_zO. Then (f,y) is in good position in the whole-chart sense `Chapter3/GoodPositionGerm · StrongBundle.IsGoodPositionGermWholeChart`, in the chart c' centred at y given by Definition

3.1.11 (preprint 3.11), with its filled version g'. *Proof.* The chart change between c at z and c' at 0 gives A, Γ with $\Gamma\,\mathrm{Lin}(z) = \mathrm{Lin}'(0)A$; the bridges give $\mathrm{Lin}(z) = Dg(z)$ on T_zO and $\mathrm{Lin}'(0) = Dg'(0)$ on T_0O' (filler condition (1)). *Surjectivity of $Tf(y) = Dg'(0)|_{T_0O'}$ onto K_0'.* By Proposition 3.6.5 (preprint 3.52) in the centred chart c' `Chapter3/FredholmGermLinearizationG · proposition_3_6_5_local` it is Fredholm, so its range S is closed in the closed subspace K_0', hence closed in F'. The map $k \mapsto \Gamma(\rho(z)k)$ is continuous $F \to F'$; by (a) and the conjugation it maps F_∞ into S; it maps F onto K_0' (preimages by Γ^{-1}, which lands in K_z where $\rho(z)$ is the identity); since F_∞ is dense in F, every point of K_0' is a limit of points of S, hence in S. So only a dense subset of K_z in the range is needed (Finding 117 (b)). *The kernel.* (b) is transported by A to good position inside T_0O' with respect to $C_0' = C'$ (the partial quadrant at the corner is C' itself `Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero`); and the transported subspace is $\ker Dg'(0)$: $\ker Dg'(0) \subseteq T_0O'$ by condition (3) of the filler g' *at its centre* `Chapter3/GoodPositionGerm · IsFillerOf.ker_deriv_le_range`, and for $w \in T_0O'$, $Dg'(0)w = 0 \iff \Gamma(Dg(z)A^{-1}w) = 0 \iff A^{-1}w \in \ker Dg(z)$. This is where the hypothesis is on $\ker Dg(z) \cap T_zO$ and not on $\ker Dg(z)$: at $z \neq 0$ condition (3) is not available and $\ker Dg(z)$ need not lie in T_zO (Finding 117 (a)). Variants: with all of K_z in the range `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_chart`, and with the hypotheses on $\mathrm{Lin}(z)$ itself, taking $g = \mathbf{f} \circ r$, which agrees with $\mathbf{f}$ on O because r is the identity there `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_linAt` `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj`.

In the germ reading the assembly is the same, with g a germ with derivative family Dg, z on level 1 of its germ, and the bridge for germs at z and at 0; the conclusion is good position at $Dg'(0)$ for the filling germ g' at y. The dense form needs the closed range of $Dg'(0)|_{T_0O'}$, which is now a hypothesis at chart level `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart_dense`, supplied at section level by Proposition 3.6.5 (preprint 3.52) for germs in a chart centred at y `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense`. The forms with all of K_z in the range need no Proposition 3.6.5 (preprint 3.52) at all: $\mathrm{Lin}'(0) = \Gamma\,\mathrm{Lin}(z)A^{-1}$ is then onto K_0' directly `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart` (the whole-chart variant had gone through the dense one).

What remains is to supply (a) and (b) in the chart at x_0, at the zeros $z_0 = \varphi(y)$ near 0: (b) with the text's hypotheses (§III.32), (a) first under an added hypothesis (§III.33) and then with the text's hypotheses (§III.43).

III.32 Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros: Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) on the graph, and the kernel

The text proves property (4) of Theorem 3.6.8 (preprint 3.54) — at every zero $y \in V$, with $\psi\varphi(y) = v + \sigma(v)$, "ker $f'(y)$ is in good position to C_yX and $f'(y)$ is surjective" — in one sentence: it follows from Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55). Unpacked, three steps are needed beyond the graph description of §III.23. First, Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) have to be applied at the graph points, Lemma 3.6.9 (preprint 3.55) on the relatively open V' of Theorem 3.5.11 (preprint Corollary 3.47) and for a general presentation of C' (Finding 92). Second, Lemma 3.6.9 (preprint 3.55) is about the tangent space $N_v = \{n + D\sigma(v)n\}$ of the graph, not about a kernel; the identification $N_v = \ker D(h+t)(v+\sigma(v))$ is not commented on (Finding 91 (a), 101). Third, the statements live in the push-forward chart $\psi \circ \varphi$, about $D(h+t)$ on E', while Definition 3.6.6 (preprint 3.53) at y (via §III.31)

asks for statements about $Dg(z_0)$ on $T_{z_0}O$ in the chart at x_0 (Findings 102–104). This section proves all of it for the kernel; the surjectivity of $f'(y)$ is §III.33. No hypothesis beyond those of the graph description is used.

Lemma 3.6.9 (preprint 3.55) for a relatively open domain and a general presentation `Chapter3/GlobalIFTFull · lemma_3_6_9_relOpen`. *Statement.* Let $C = \{x : \ell_i(x) \geq 0\}$ be a partial quadrant presented by an sc-isomorphism T, with the coordinate functionals $\ell_i(x) = (Tx)_i$ `Chapter3/GlobalIFTFull · coordCLMWith`, N finite-dimensional in good position to C with a good complement Y, Y **closed**, V relatively open in $N \cap C$ with $0 \in V$, and $\sigma : N \to E_0$ with values in Y, C^1 on V, $\sigma(0) = 0$, $D\sigma(0) = 0$ (within V). Then there is $\rho > 0$ such that for $v \in V$ with $|v| < \rho$ and $z = v + \sigma(v)$: $z \in C$; Y is a good complement of $N_v = \{n + D\sigma(v)n\}$ `Chapter3/GoodPositionOpen · graphSpace` with respect to the partial quadrant C_z; and N_v is in good position to C_z. *Proof.* $D = N \cap C$ is convex with nonempty interior in N (good position), so D and $V = D \cap O$ have unique differentiability. *The extension.* The text's Lemma 3.6.9 (preprint 3.55) (§III.22, `Chapter3/GoodPositionOpen · lemma_3_6_9_core`) wants a map on the whole cone. Take a ball $B(0,r) \subseteq O$ and a smooth bump b equal to 1 on $\bar{B}(0,r/2)$ and 0 outside $B(0,r)$, and $\Phi(v) = b(v)v$. Φ maps D into V: D is closed under nonnegative scaling, and $|b(v)v| < r$. So $\tau = \sigma \circ \Phi$ is C^1 on the cone, agrees with σ on $B(0,r/2)$, and there its derivatives within the cone are those of σ within V (a neighbourhood of the point inside O does not change derivatives within a set); in particular $\tau(0) = 0$, $D\tau(0) = 0$. *Lemma 3.6.9 (preprint 3.55) in coordinates* for τ, with the good-complement estimate of (N, Y) rewritten in the ℓ_i and the joint surjectivity of the coordinates ($T^{-1}(k,0)$ has coordinates k), gives ρ_1, the membership $z \in C$, and the good complement Y of N_v with respect to C_z. *The interior*, which Lemma 3.6.9 (preprint 3.55) does not state but good position requires: by the continuity of $D\sigma$ at 0 (a C^1 map on a set of unique differentiability) there is ρ_2 with $\|D\sigma(v)\| < \varepsilon$, the good-complement constant of (N, Y). $L = D\sigma(v)$ takes values in Y — this is where Y closed is used `Chapter3/GoodPositionOpen · fderivWithin_mem_of_mapsTo` — so $n \mapsto n + Ln$ is injective ($N \cap Y = 0$), a linear homeomorphism $N \cong N_v$ in finite dimensions. It carries the interior of D to an open subset of N_v contained in C_z: for w in the interior of D, $w \in C$ and $|Lw| \leq \varepsilon|w|$, so $w + Lw \in C$ by the good-complement property, and $C \subseteq C_z$. Take $\rho = \min(\rho_1, r/2, \rho_2)$.

Corollary 3.5.3 (preprint 3.41) at the graph points `Chapter3/GlobalIFTFull · StabilityData.exists_surjective_Dfs'_graph`. For the data of Theorem 3.4.2 (preprint 3.38) with $D(h+s)(0)$ surjective and a continuous σ on $V \ni 0$, $\sigma(0) = 0$, whose graph points $v + \sigma(v)$ are zeros in the germ: there is $\rho > 0$ such that for $v \in V$, $|v| < \rho$, the zero $v + \sigma(v)$ lies on level 1, at a point x_1 of the level-1 germ, and $D(h+s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$. Corollary 3.5.3 (preprint 3.41) `Chapter3/SolutionLinearization · StabilityData.corollary_3_5_3` gives a radius r for the zeros, and continuity of $v \mapsto v + \sigma(v)$ at 0 gives ρ.

Property (4) in the push-forward chart `Chapter3/GlobalIFTFullG · theorem_3_6_8_nearby`. *Statement.* For a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the data of the graph description (§III.23, `Chapter3/GlobalIFTG · theorem_3_6_8_graph`: chart, ψ, $h+t$, K, closed Y, V', σ, V_x), a presentation T of $C' = \widetilde{C} \oplus W$, and an open $V_{x,1} \subseteq V_x$ containing x_0 such that for every zero $y \in V_{x,1}$, with $\psi\varphi(y) = v + \sigma(v)$, $v \in V'$: the point lies on level 1 and $D(h+t)$ there is surjective Fredholm of index $n - N$; $v + \sigma(v) \in C'$; Y is a good complement of N_v with respect to $C'_{v+\sigma(v)}$, and N_v is in good position to it. Only f a section is assumed, not an sc-Fredholm section. *Proof.* Take the radii ρ_1 (Corollary 3.5.3 (preprint 3.41) at the graph points) and ρ_2 (Lemma 3.6.9 (preprint 3.55), relatively open form), $\rho = \min(\rho_1, \rho_2)$, and the projection π onto K along Y, continuous because Y is closed and K finite-dimensional. Put $V_{x,1} = V_x \cap (\psi\varphi)^{-1}B(0, \rho/(\|\pi\| + 1))$, open since $\psi \circ \varphi$ is continuous on V_x. For a zero $y \in V_{x,1}$, $\psi\varphi(y) = v + \sigma(v)$ by the graph description, $v = \pi(\psi\varphi(y))$ because $\sigma(v) \in Y$, so

$|v| \le \|\pi\|\,|\psi\varphi(y)| < \rho$, and both lemmas apply at v.

Derivative identities at a nearby zero. Four lemmas move these statements from E' and N_v to E and $\ker Dg(z_0)$.

Uniqueness at a point of C_1 `Chapter3/GlobalIFTNearbyGraph · scDeriv_eq_of_eqOn_nhds_at`: two maps with sc¹-derivatives D_1, D_2 at $x \in C_1$ that agree on $W_1 \cap C_1$ for an open $W_1 \ni x$ of E_1 have $D_1 = D_2$ — uniqueness of the derivative within the quadrant on level 1, then density of E_1. The text writes $\Psi_*(g) = h + t$ as an identity; formally they agree only on $U' \cap W'$, and this lemma identifies their derivatives at every nearby point (Finding 104).

The fibre derivative of a strong bundle map at $(u_0, 0)$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleMap.scDeriv_fiber_at`: for a strong bundle map Ψ with fibre operators $\Gamma(u) = (\Psi(u,\cdot))_2$ `Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM` and $u_0 \in U_1$, $D(\mathrm{pr}_{F'} \circ \Psi)(u_0,0)(h,k) = \Gamma(u_0)k$; hence the derivative of $u \mapsto \Gamma(u)k(u)$ at a zero u_0 of k is $\Gamma(u_0)Dk(u_0)$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleMap.scDeriv_fiber_comp_at` — §III.21's identity at 0, moved to u_0.

The chain rule for the push-forward at a zero `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.scDeriv_pushforward_at`: for a strong bundle isomorphism Ψ covering ψ with inverse Ψ' covering ψ', and a zero $u_0 \in U_1$ of the sc-smooth g, $\Psi_*(g) = \kappa \circ \psi'$ with $\kappa(u) = (\Psi(u, g(u)))_2$, so

$$D(\Psi_* g)(\psi(u_0)) = \Gamma(u_0) \circ Dg(u_0) \circ D\psi'(\psi(u_0)).$$

$D\psi'(\psi(u_0))$ and $D\psi(u_0)$ are inverse on level 0 `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.scDeriv_base_inv_base`, and $\Gamma(u_0)$ is injective because Ψ' inverts Ψ on the fibres `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.fiberCLM_injective`; hence $\ker Dg(u_0) = D\psi'(\psi(u_0))(\ker D(\Psi_* g)(\psi(u_0)))$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.ker_scDeriv_eq_map`. *For a germ g* with derivative family Dg the formula is an identity of the explicit derivative family of $\Psi_*(g)$ (§III.23), valid at every level-1 zero u_0 and for every family `Chapter3/GlobalIFTNearbyGraphG · IsStrongBundleIso.pushforwardDeriv_at`, and the kernel formula follows as before `Chapter3/GlobalIFTNearbyGraphG · IsStrongBundleIso.ker_deriv_eq_map_germ`; that g is sc¹ at u_0 then means that u_0 lies on level 1 of the germ of g.

Corresponding partial quadrants `Chapter3/GlobalIFTNearbyGraph · scDeriv_mem_partialQuadrantAtWith_iff`: for an sc-diffeomorphism $\psi : U \to U'$ between relatively open subsets of partial quadrants with inverse ψ' and $z \in U_1'$, $x \in C_z' \iff D\psi'(z)x \in C_{\psi'(z)}$: the one-sided sign argument `Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant` for ψ' at z gives one direction, for ψ at $\psi'(z)$ the other, and $D\psi(\psi'(z))D\psi'(z) = 1$. At the centre this was "$C' = T\psi(0)C$" (§III.23).

Tangent vectors of a family of points of a retract `Chapter3/GlobalIFTNearbyGraph · scDeriv_apply_toBase_deriv_eq_of_fixed`: let ρ be sc¹ on U and $\Theta_1 : N \to E_1$ differentiable at v within V (unique differentiability at v), with values in U_1 and $\rho(\iota\Theta_1(w)) = \iota\Theta_1(w)$ for $w \in V$. Then $D\rho(\Theta_1(v))$ fixes $\iota(D\Theta_1(v)n)$ for every n: differentiate $\rho \circ \iota \circ \Theta_1 = \iota \circ \Theta_1$ at v. For ρ a retraction onto O: the tangent vectors of a C^1 family of points of O lie in $T_{z_0}O$.

$N_v = \ker D(h+s)(v+\sigma(v))$ (Finding 101). *The inclusion* `Chapter3/GlobalIFTNearbyGraph · StabilityData.graphSpace_le_ker_Dfs'`. For the solution graph σ of Theorem 3.5.11 (preprint Corollary 3.47) on V, a C^1-lift $\sigma_1 : V_1 \to E_1$ of σ on a relatively open $V_1 \subseteq V$ (Theorem 3.5.11 (1) (preprint Corollary 3.47), §III.20), $v \in V_1$ and the level-1 point x_1 over $v + \sigma(v)$ in the germ: $N_v \subseteq \ker D(h+s)(x_1)$. *Proof.* $h + s$ is differentiable on level 1 only, so the graph is lifted: the kernel K is a smooth sub-

space Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth, hence lifts linearly to E_1 Chapter3/BasicClass · finLift, and $\Gamma_1 = \text{lift} + \sigma_1$ is a C^1 map $V_1 \to E_1$ with $\iota\Gamma_1(w) = w + \sigma(w)$ and $\Gamma_1(v) = x_1$. $V_1 = V \cap O_1$ Chapter3/GlobalIFTNearbyGraph · relOpen_cone_inter has unique differentiability at its points Chapter3/GlobalIFTNearbyGraph · uniqueDiffWithinAt_relOpen_cone, and $D\sigma(v) = \iota \circ D\sigma_1(v)$ Chapter3/GlobalIFTNearbyGraph · fderivWithin_eq_toBase_comp. The composite $(h+s)\circ\iota\circ\Gamma_1$ vanishes on V_1, and near v (on $V_1 \cap \Gamma_1^{-1}$ of the open set defining the level-1 germ) Γ_1 maps into the level-1 germ, where $h+s$ has the derivative $D(h+s)(x_1)$ composed with ι Chapter3/SolutionLinearization · StabilityData.isSc1GermWith_Dfs'. The chain rule and uniqueness give $D(h+s)(x_1)(n + D\sigma(v)n) = 0$ for $n \in K$. *Equality* Chapter3/GlobalIFTNearbyGraph · StabilityData.graphSpace_eq_ker_Dfs' under the conclusions of Corollary 3.5.3 (preprint 3.41) at x_1 (Fredholm, index $\dim P - \dim Q$, surjective) and with the closed good complement Y: $D\sigma(v)$ takes values in Y, so $\dim N_v = \dim K$ Chapter3/GlobalIFTNearbyGraph · finrank_graphSpace_of_compl; for a surjective Fredholm operator $\dim\ker = \text{ind}$ Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective, so $\dim K = \text{ind}\, D(h+s)(0) = \dim P - \dim Q = \text{ind}\, D(h+s)(x_1) = \dim\ker D(h+s)(x_1)$, and an inclusion of finite-dimensional spaces of the same dimension is an equality.

Good position restricts to a subspace containing the kernel Chapter3/GlobalIFTNearbyGraph · IsInGoodPosition.isInGoodPositionIn_of_le: if N is in good position to C in E_0 and $N \subseteq T$, a good complement P restricts to the good complement $P \cap T$ of N in T, with the same constant ($n + p \in T$ and $n \in T$ give $p \in T$).

The graph description with the filler data exposed Chapter3/GlobalIFTNearbyGraphG · theorem_3_6_8_chart_graph. This is the body of the graph description (§III.23) stated in the chart, with the filling germ g of $\mathbf{f}$ on G_g and a derivative family Dg, the sc⁺-germ t_0 on G_s, the conjugation Ψ (base ψ, inverse ψ'), the basic germ on a germ G that ψ' maps into $G_g \cap G_s$ level by level ($G \subseteq G''$, §III.48), and the tameness consequence $(1 - Dr(0))E \subseteq E_0$ as hypotheses, so that the later steps can use g at the points z_0. Its conclusions: the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) on G itself and the basic germ $h' = h + (t_{\mathbb{R}^N}, 0)$ with $\mathcal{S}.\text{fsum}\, h' = \Psi_*(g)$ on $U' \cap W'$; $D(h+t)(0)$ surjective with kernel K in good position; Y (closed), V', σ with its C^k-lifts; an open $N_1 \ni 0$ on which the filler identities (1), (2) and the conjugation identity hold, with $\psi'(U' \cap W') \subseteq N_1$; and an open $N_2 \subseteq N_1$ with $\psi(U \cap N_2)$ inside the germ and $U' \cap W'$, every zero of g in $U \cap N_2$ in O, $g(u) = 0 \iff (h+t)(\psi(u)) = 0$ on $O \cap N_2$, and $\psi(\{u \in O \cap N_2 : g(u) = 0\}) = \{v + \sigma(v) : v \in V'\}$. The proof is that of §III.23 (items 1–7), in the chart; $D(h+t)(0) = D(\Psi_* g)(0)$ for the explicit family is part of the conclusion.

The kernel at the nearby zeros, in the chart at x_0 Chapter3/GlobalIFTNearbyKernelG · theorem_3_6_8_chart_kernel. *Statement.* In the data of the preceding theorem there is an open $N_3 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_3$ of g there are $v \in V'$ with $\psi(z_0) = v + \sigma(v)$, the level-1 point $z_{0,1}$ over z_0, **on level 1 of the germ G_g of the filling**, and $x_1 = \psi_1(z_{0,1})$ in the level-1 germ over $v + \sigma(v)$, with ($Dg(z_{0,1})$ the value of the derivative family): (i) $D(h+t)(x_1)$ surjective Fredholm of index $n - N$; (a) $N_v = \ker D(h+t)(x_1)$; (b) $D(\Psi_* g)(x_1) = D(h+t)(x_1)$ and $\ker Dg(z_{0,1}) = D\psi'(x_1)(\ker D(h+t)(x_1))$, and $Dg(z_{0,1})$ is surjective; (c) $\ker Dg(z_{0,1}) \subseteq T_{z_0}O = Dr(z_{0,1})E$; (d) $\ker Dg(z_{0,1})$ is in good position to C_{z_0} inside $T_{z_0}O$. *Proof.*

1. *The neighbourhood.* Take the C^1-lift σ_1 on $V_1 = V' \cap O_1$ (Theorem 3.5.11 (1) (preprint Corollary 3.47) with $k = 1$), the radii ρ_1 (Corollary 3.5.3 (preprint 3.41) at the graph points), ρ_2 (Lemma 3.6.9 (preprint 3.55) relatively open, for the presentation of $C' = \widetilde{C} \oplus W$), ρ_3 with $B(0, \rho_3) \subseteq O_1$, and $\rho = \min(\rho_1, \rho_2, \rho_3)$. With the projection π onto K along Y, let $N_3 = N_2 \cap O_3$ where $O_3 \cap U = \psi^{-1}(B(0, \rho/(\|\pi\| + 1))) \cap U$ (continuity of ψ on U). For a zero $z_0 \in O \cap N_3$ of g, $\psi(z_0) = v + \sigma(v)$ with $v \in V'$, and as in `theorem_3_6_8_nearby` $|v| < \rho$, so $v \in V_1$ and both lemmas apply.
2. *(i), and the level-1 points.* Corollary 3.5.3 (preprint 3.41) at v gives x_1 on level 1 of $\mathcal{S}.G =$

$G \subseteq G''$ (§III.48, `Chapter3/BasicGermShrinkIFT · ConjugationData.corollary_3_5_3_shrink`); x_1 lies in U_1', so $z_{0,1} = \psi_1'(x_1)$ is a level-1 point over $\psi'(\psi(z_0)) = z_0$, and $\psi_1(z_{0,1}) = x_1$ `Chapter2/DegeneracyIndex · levelMap_comp_eq_self`. Since ψ_1' maps G_1'' into $(G_g)_1$, $z_{0,1}$ lies in $(G_g)_1$ `Chapter3/BasicGermShrink · ConjugationData.mem_of_levelMap_base_mem`: this is the step that needs the shrinking of the basic germ, since g is differentiable only there. (With g sc-smooth on U, any level-1 point would do.)

3. *(a)* is $N_v = \ker D(h+t)(x_1)$ above, fed with the lift and (i).
4. $D(\Psi_* g)(x_1) = D(h+t)(x_1)$: the two maps agree on $U' \cap W'$, so the uniqueness lemma at x_1 applies on the open set of E_1' cut out by $W' \cap U_o$ (with $U' = U_o \cap C'$) and the open set defining the level-1 germ G_1; $\Psi_*(g)$ is an sc¹-germ on the preimage of G_g, which contains G `Chapter3/GermPushforward · IsStrongBundleIso.pushforward_isSc1GermWith` (whole-chart: sc-smooth on U' `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_linearization`).
5. *(b)*: the kernel formula is the chain rule for the push-forward at the zero $z_{0,1}$ of g, in the germ form above. *Surjectivity of $Dg(z_{0,1})$*: given k, solve $D(h+t)(x_1)w = \Gamma(z_0)k$ by (i); then $\Gamma(z_0)Dg(z_{0,1})D\psi'(x_1)w = \Gamma(z_0)k$, and $\Gamma(z_0)$ is injective. This is the surjectivity of the *filled* $Dg(z_0) : E \to F$, not of $Tf(z_0) = Dg(z_0)|_{T_{z_0}O} : T_{z_0}O \to K_{z_0}$ (§III.33).
6. *(c), the tangency to O* (Finding 102). Condition (3) of Definition 3.1.4 (preprint 3.4), which gave $\ker Dg(0) \subseteq T_0 O$ at the centre, is not available at z_0. Instead: the transported retraction $r' = \psi \circ r \circ \psi'$ on U' `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport` fixes every graph point, because each $w + \sigma(w)$, $w \in V'$, is $\psi(u)$ for a zero $u \in O$ of g (the graph description), and $r(u) = u$. The lifted graph map $\Gamma_1 = \text{lift} + \sigma_1$ maps V_1 into U_1', so the tangent-vector lemma for r' gives $Dr'(x_1)$ fixing $n + D\sigma(v)n$ for every $n \in K$, i.e. $Dr'(x_1)$ is the identity on N_v. The derivative of the transported retraction is $Dr'(x_1) = D\psi(z_{0,1})Dr(z_{0,1})D\psi'(x_1)$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.scDeriv_transport` (§III.27); applying $D\psi'(x_1)$ and using that it inverts $D\psi(z_{0,1})$, $Dr(z_{0,1})$ fixes $D\psi'(x_1)m$ for every $m \in N_v$. By (a) and (b), these are exactly the elements of $\ker Dg(z_{0,1})$, so $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$.
7. *(d)* (Finding 103). Lemma 3.6.9 (preprint 3.55) gives N_v in good position to $C'_{v+\sigma(v)}$ in E'. The isomorphism $S = D\psi'(x_1) : E_0' \cong E_0$ (inverse $D\psi(z_{0,1})$, `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.derivInvEquiv`) carries $C'_{v+\sigma(v)}$ onto C_{z_0} by the corresponding-quadrants lemma, so Lemma 3.5.9 (preprint 3.45) `Chapter3/GoodPosition · IsInGoodPosition.map` gives $S(N_v) = \ker Dg(z_{0,1})$ (by (a), (b)) in good position to C_{z_0} in E; by (c) and the restriction lemma it is in good position inside $T_{z_0}O$.

Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros, the kernel, for the section `Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_nearby_kernel`. *Statement.* For a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the data of the graph description, with the chart c centred at x_0, its filling germ g with a derivative family and the sc⁺-germ t_0 exposed (and K smooth, Y closed, the lifts), and an open $V_{x,2} \subseteq V_x$ containing x_0, such that for every zero $y \in V_{x,2}$: $\mathbf{f} = g$ on O near $\varphi(y)$, $g(\varphi(y)) = 0$, and with $z_0 = \varphi(y)$ the conclusions (i), (a)–(d) of the chart statement hold, with $z_{0,1}$ on level 1 of the germ of g. *Proof.* Tameness gives $(1 - Dr(0))E \subseteq E_0$ (Proposition 2.5.3 (preprint 2.35), `Chapter2/TameRetracts · proposition_2_5_3`, at the smooth corner point 0 of the tame base retraction; §III.23). The conjugation data are first shrunk to G'' (§III.48), so that the chart statements apply; they give N_1, N_2, N_3, with $\psi(U \cap N_2) \subseteq G_0''$; put $V_x = \varphi^{-1}(N_2)$ and $V_{x,2} = \varphi^{-1}(N_3)$ within the chart domain, both open by continuity of φ. On V_x, $\mathbf{f}(\varphi(y)) = g(\varphi(y))$ (filler (1) on N_1), so $f(y) = 0 \iff g(\varphi(y)) = 0 \iff (h+t)(\psi\varphi(y)) = 0$ `Chapter3/StrongBundleSections · StrongBundleChart.isZeroAt_iff`; $\psi \circ \varphi$ is injective on V_x; and the image of the zero

set is the graph. For a zero $y \in V_{x,2}$, $\varphi(y) \in O \cap N_3$ is a zero of g, and the kernel statement applies. The hypotheses are those of the graph description: **the kernel part of Theorem 3.6.8 (1) (preprint 3.54) needs no added hypothesis**, and of f only that it is a section (the regularity enters through the filling germ g, at 0 and at the lifts $z_{0,1}$ of the nearby zeros, which lie in its germ).

III.33 Theorem 3.6.8 (1) (preprint 3.54) under the filler condition at the nearby zeros, and the λ-extension

At a zero y near x_0, with $z_0 = \varphi(y)$ in the chart at x_0, §III.32 gives the kernel condition (b) of the assembly step of §III.31 and the surjectivity of the *filled* $Dg(z_0) : E \to F$. Condition (a) asks for more: that $Tf(z_0) = Dg(z_0)|_{T_{z_0}O}$ maps onto $K_{z_0} = \rho(z_0)F$ (or onto a dense part of it). The text obtains it from Corollary 3.5.3 (preprint 3.41). This section shows exactly what is missing, proves Theorem 3.6.8 (1) (preprint 3.54) under the **added hypothesis** that closes the gap, and records the formalized part of an attempt to close it without that hypothesis (Findings 100, 105–107).

The block form at a zero. With respect to $E = T_{z_0}O \oplus \ker Dr(z_0)$ and $F = K_{z_0} \oplus \ker \rho(z_0)$ ($Dr(z_0)$ and $\rho(z_0)$ are idempotent),

$$Dg(z_0) = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}, \qquad A = Tf(z_0), \quad C = (1-\rho(z_0))Dg(z_0)|_{\ker Dr(z_0)} : \ker Dr(z_0) \to \ker \rho(z_0).$$

The lower left block vanishes at a zero: $Dg(z_0)Dr(z_0) = D(\mathbf{f} \circ r)(z_0)$ (the bridge, §III.31) takes values in K_{z_0} (the values of the linearization at a zero, §III.31). If $Dg(z_0)$ is onto, C is onto, and (Finding 100) $b \mapsto [Bb]$ induces an isomorphism $\ker C \cong \operatorname{coker} A$: so $Tf(z_0)$ **is onto if and only if** C **is injective** — condition (3) of Definition 3.1.4 (preprint 3.4) at z_0. Definition 3.1.4 (preprint 3.4) imposes it at 0 only, and it does not propagate: $u \mapsto Dg(u)$ is continuous for the strong operator topology, not in norm, and a strongly continuous family of surjections may pass from an isomorphism to a non-injective operator. In index terms: Corollary 3.5.3 (preprint 3.41) gives $\operatorname{ind} Dg(z_0) = \dim K$ at all nearby zeros, $\operatorname{ind} Dg(z_0) = \operatorname{ind} A + \operatorname{ind} C = \dim\ker A$ when C is onto, and $\operatorname{ind}(f,y) = \operatorname{ind} A = \dim K - \dim\operatorname{coker} A$; so $\operatorname{coker} Tf(z_0) = 0$ is the local constancy $\operatorname{ind}(f,y) = \operatorname{ind}(f,x_0)$ along the solution set — the index stability of Proposition 6.2.2 (preprint 6.2) of the text, whose proof uses the same step ("F' surjective" for the intrinsic linearization, from the implicit function theorem for the filled one). The text's "(4) follows from Corollary 3.41 and Lemma 3.55" therefore has a gap for the surjectivity of $f'(y)$; the same sentence stands in the book's arXiv version (Theorem 3.13 there, with Definition 3.13, the filling, still imposing (3) at 0 only; Finding 157). Formalized of this paragraph is the direction used:

Proposition 3.1.5 (1) (preprint 3.5) at a zero of a filler `Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective`. *Statement.* Let U be relatively open in a partial quadrant, R a strong bundle retraction on $U \lhd F$, $\mathbf{f}$ with $\mathbf{f} \circ r$ sc¹ and $(x, \mathbf{f}(x)) \in K$ on O, g sc¹, z on level 1 with $\iota z \in O$, $\mathbf{f}(\iota z) = 0$, $\mathbf{f} = g$ on O near ιz. If $Dg(z)$ is surjective and $(1-\rho(z))Dg(z)$ is injective on $\ker Dr(z)$ (in the form: $Dr(z)b = 0$ and $\rho(z)Dg(z)b = Dg(z)b$ imply $b = 0$), then for every $k \in F$, $\rho(z)k = Dg(z)x$ for some $x \in T_zO$. For a germ g, with $Dg(z)$ the value of a derivative family at a level-1 point of its germ, the proof is the same with the bridge for germs (§III.31) `Chapter3/GlobalIFTNearbySurjG · exists_range_deriv_eq_ρ_of_normal_injective_germ`. *Proof.* Write $Dg(z)e = \rho(z)k$, $a = Dr(z)e$, $b = e - a \in \ker Dr(z)$ (idempotence). $Dg(z)a = D(\mathbf{f}\circ r)(z)e$ lies in K_z (bridge and values at a zero), so $\rho(z)Dg(z)a = Dg(z)a$; then $\rho(z)Dg(z)b = \rho(z)(\rho(z)k - Dg(z)a) = \rho(z)k - Dg(z)a = Dg(z)b$ ($\rho(z)$ is linear and idempotent

on the fibre). So $b = 0$, $e = a \in T_z O$. The converse direction and the index identity above are not formal statements; they explain why the hypothesis is the right one.

The added hypothesis `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled`. (f, x_0) is in good position *with a filler satisfying condition (3) at the nearby zeros* if x_0 is smooth and there are the data of Definition 3.6.6 (preprint 3.53) in the chart form (a strong bundle chart centred at x_0, tame since §III.47, a filling germ g of $\mathbf{f}$ with a derivative family Dg on a germ G inside U, a local sc$^+$-germ s with $s(0) = g(0)$ and $g - s$ conjugated to a basic germ, the germ in good position at $Dg(0)$) together with an open $N \ni 0$ such that at every level-1 point z_1 *of* G with $\iota z_1 \in O \cap N$ and $g(\iota z_1) = 0$, $(1 - \rho(\iota z_1))Dg(z_1)$ is injective on $\ker Dr(z_1)$. The condition is imposed only where Dg is defined, which is where the lifts of the nearby zeros lie (§III.32); the whole-chart form imposed it at every level-1 point over $O \cap N$ and implies it `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilledWholeChart.isGoodPositionGermFilled`. It implies Definition 3.6.6 (preprint 3.53) `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled.isGoodPositionGerm` (forget N). According to the docstring, the fillers of the applications satisfy (3) at every point; Definition 3.1.4 (preprint 3.4) as stated does not ask for it. It is recorded as an added hypothesis in the docstrings and in the Findings (items 100, 153).

Theorem 3.6.8 (1) (preprint 3.54) `Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_one`. *Statement.* Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$, and (f, x_0) in good position with a filler satisfying condition (3) at the nearby zeros. Then there is an open $V_x \ni x_0$ such that (f, y) is in good position (Definition 3.6.6 (preprint 3.53), `IsGoodPositionGerm`) at every zero $y \in V_x$. *Proof.*

1. In the chart c at x_0 with the filling germ g: $(1 - Dr(0))E \subseteq E_0$ from tameness (Proposition 2.5.3 (preprint 2.35)); the germ G of the condition is intersected with the germ of g, and the basic germ is shrunk to the points that ψ' maps into this intersection and into the germ of s (§III.48); then the graph description in the chart and the kernel statement of §III.32 give N_1, N_2, N_3. Put $V_x = \varphi^{-1}(N_3 \cap N)$ in the chart domain, with N the neighbourhood of the added hypothesis.
2. For a zero $y \in V_x$: $z_0 = \varphi(y) \in O \cap N_3$, $\mathbf{f}(z_0) = 0$, and $g(z_0) = \mathbf{f}(z_0) = 0$ by filler (1) on N_1; $\mathbf{f} = g$ on O near z_0. The kernel statement gives the level-1 point $z_{0,1}$, on level 1 of G, $Dg(z_{0,1})$ surjective, $\ker Dg(z_{0,1}) \subseteq T_{z_0} O$, and $\ker Dg(z_{0,1})$ in good position to C_{z_0} inside $T_{z_0} O$.
3. y is a smooth point, because f is regularizing `Chapter3/TameGermsG · IsScFredholmSection.mem_smoothSet_of_isZeroAt`, so (f, y) is an sc-Fredholm germ (Definition 3.1.16 (preprint 3.14)); and $\mathbf{f} \circ r$ is sc^1 for an sc-smooth section `Chapter3/GlobalIFTNearbySurj · StrongBundle.IsScSmoothSection.isSc1_principal_comp_r`.
4. Apply the assembly step of §III.31 in the chart c at $z_{0,1}$ with g, in its germ form, the closed range in the tame chart at y coming from Proposition 3.6.5 (preprint 3.52) for germs `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense`. *Condition (a)*: the added hypothesis at $z_{0,1}$ (with $\iota z_{0,1} = z_0 \in O \cap N$, $g(z_0) = 0$) is the injectivity of C, and Proposition 3.1.5 (1) (preprint 3.5) at z_0 turns the surjectivity of $Dg(z_{0,1})$ into $\rho(z_0)k \in Dg(z_{0,1})(T_{z_0} O)$ for every k, smooth or not. *Condition (b)*: $\ker Dg(z_{0,1}) \cap T_{z_0} O = \ker Dg(z_{0,1})$ by the tangency (c), and the good position is (d).

So Theorem 3.6.8 (1) (preprint 3.54) holds under condition (3) of the filler at the nearby zeros; the kernel half holds with the text's hypotheses (§III.32). When this section was written, **the surjectivity of $f'(y)$ at the nearby zeros with Definition 3.1.4 (preprint 3.4) as stated was open in the formalization**; it is proved in §§III.41–III.43 by another argument, and the condition (3) at the nearby zeros turns out to be a consequence of good position (§III.45). The global theorems are not affected: they assume good position at every zero, which supplies at each x_0 the nearby good position that the regularity argument needs

(§III.36–§III.37, Finding 156).

The λ-extension: what the perturbation argument of Proposition 6.2.2 (preprint 6.2) gives. The text's device for index stability (Proposition 6.2.2 (preprint 6.2)) is to add finitely many parameters λ multiplying sc⁺-sections, so that the extended problem is surjective, and to apply the implicit function theorem to it. In the chart at x_0 it can be set up without new bundle isomorphisms: for a finite-dimensional $\Lambda_0 \subseteq F_\infty$, the sc⁺-germ $\tilde{e}(\lambda, u) = \rho(r(u))(\lambda_1(\pi_K \psi(u)) + \hat{\jmath}(\lambda_2))$ (Finding 105). Two points are not addressed by the text, and the first is formalized.

The obstruction at the corner `Chapter3/GlobalIFTNearbyPerturb · mem_lineality_of_goodComplement_prod`. Theorem 3.5.11 (preprint Corollary 3.47) for the extended problem needs its kernel in good position to the extended partial quadrant at the corner. Let $C \subseteq V$ contain 0 and be closed under nonnegative scaling, $N \subseteq V$ a subspace, and $P \subseteq \mathbb{R} \times V$ a subspace with the ε-condition of Definition 3.1.24 (preprint 3.21) for $\mathbb{R} \times N$ `Chapter3/GlobalIFTNearbyPerturb · prodSub` with respect to $\mathbb{R} \times C$ `Chapter3/GlobalIFTNearbyPerturb · prodCone`. Then every $p = (\mu, y) \in P$ has $y \in C$ and $-y \in C$. *Proof.* $n = (1, 0)$ lies in $\mathbb{R} \times N$ and in $\mathbb{R} \times C$, $|n| = 1$; for $t = \pm\varepsilon/|p|$, $|tp| \le \varepsilon|n|$, so $n + tp \in \mathbb{R} \times C$, i.e. $ty \in C$, and scaling by $1/|t|$ gives $\pm y \in C$. Hence a good complement projects into the lineality space $C \cap (-C)$, and $\mathbb{R} \times N$ is in good position only if every $v \in V$ is $n + y$ with $n \in N$, $y \in C \cap (-C)$ `Chapter3/GlobalIFTNearbyPerturb · sup_lineality_eq_top_of_goodComplement_prod`. For the naive extension (parameters multiplying smooth vectors with a factor vanishing at 0) the kernel is $\mathbb{R} \times K$, so good position fails unless K maps onto all boundary coordinates — the text's "has a complement contained in T^R" is exactly the statement that the complement lies in the lineality space.

The fix: a tilted kernel `Chapter3/GlobalIFTNearbyPerturb · isInGoodPosition_extendedKernel`. In $(\Lambda \times P_n) \oplus W$ with the partial quadrant $(\Lambda \times \widetilde{C}) \oplus W$ (the parameters λ unconstrained), let K, Y' be subspaces of $P_n \oplus W$ and $j : \Lambda \to P_n \oplus W$ linear with $K \cap j(\Lambda) = 0$, $(K + j(\Lambda)) \oplus Y' = P_n \oplus W$, Y' in the lineality space of $\widetilde{C} \oplus W$ ($v + y \in \widetilde{C} \oplus W \iff v \in \widetilde{C} \oplus W$ for $y \in Y'$), and $K + j(\Lambda)$ containing an interior point of $\widetilde{C} \oplus W$. Then the kernel $\widetilde{N} = \{((\lambda, a), w) : (a, w) + j(\lambda) \in K\}$ of $((\lambda, a), w) \mapsto D(h + t)(0)((a, w) + j(\lambda))$ `Chapter3/GlobalIFTNearbyPerturb · extendedKernel Chapter3/GlobalIFTNearbyPerturb · extProj Chapter3/GlobalIFTNearbyPerturb · extParam` is in good position. *Proof.* Choose a complement Λ_c of $\ker j$; every element of $j(\Lambda)$ is $j(\mu)$ with $\mu \in \Lambda_c$. *Interior*: if $k_0 + j(\mu_0)$ is interior, $n_0 = ((-\mu_0, k_0 + j(\mu_0)))$ lies in $\widetilde{N}$ and its $(P_n \oplus W)$-part is interior, an open condition. *The complement* $\Lambda_c \times Y'$ `Chapter3/GlobalIFTNearbyPerturb · extendedCompl`: an element of both has $(a, w) \in (K + j(\Lambda)) \cap Y' = 0$, then $j(\lambda) \in K \cap j(\Lambda) = 0$, $\lambda \in \ker j \cap \Lambda_c = 0$; and every $x = ((\lambda, a), w)$ splits by decomposing $(a, w) + j(\lambda) = k + j(\mu) + y$. *The constant*: since Y' lies in the lineality space, adding an element of $\Lambda_c \times Y'$ never changes membership in the quadrant, so the ε-condition holds for every ε (the proof takes $\varepsilon = 1$). The directions $j(\Lambda) \subseteq T_0(\psi(O)) \cap E'_\infty$ with $K + j(\Lambda) + E'_0 = E'$ exist by tameness ($T_0 O + E_0 = E$) and would be realized by the sc⁺-perturbations $\rho(r(u)) Dg(0) D\psi^{-1}(0) j(\lambda)$.

Why the argument does not close the gap (Findings 105–106). Carried through (not formalized: the extended basic germ and its sc⁺-perturbation, its derivative at the zeros, Theorem 3.5.11 (preprint Corollary 3.47) for it, and the tangency argument), the argument gives for each finite-dimensional $\Lambda_0 \subseteq F_\infty$ a neighbourhood $V(\Lambda_0)$ of x_0 with $\rho(z_0)\Lambda_0 \subseteq \operatorname{range} Tf(z_0)$ at every zero in $V(\Lambda_0)$. The assembly step at a fixed zero y needs $\rho(z_0)\hat{k}$ in the range for *all* smooth $\hat{k}$, and $V(\Lambda_0)$ comes from Theorem 3.5.11 (preprint Corollary 3.47) for the Λ_0-extended problem with no control in Λ_0: the radii of Theorems 3.5.1, 3.5.6, 3.5.11 (preprint 3.40, 3.43, 3.47) would have to be uniform in the finite-dimensional parameter space, which the statements do not provide; $\operatorname{coker} Tf(z_0)$ is finite-dimensional and spanned by smooth vectors, but the spanning set depends on y. The formalization of this route was stopped there; §§III.41–III.43 take another one.

III.34 Theorem 3.6.8 (2) (preprint 3.54) at the centre: the data of Definition 3.2.1 (preprint 3.25) for the solution set at x_0

The text ends the proof of Theorem 3.6.8 (preprint 3.54) with $\psi\circ\varphi(\{y \in V : f(y) = 0\}) = \{v+\sigma(v) : v \in V'\}$ and concludes that the solution set has the d-dimensional tangent germ property, $d = \dim\ker f'(x)$, so that Theorem 3.2.3 (preprint 3.27) applies. Definition 3.2.1 (preprint 3.25) asks for much more than the graph identity (§III.24): a chart of X centred at the point onto a *tame* retract, a *smooth* finite-dimensional subspace N in good position with a *closed* good complement Y, a relatively open $V \ni 0$ in $N \cap C$, and $\delta : V \to Y$ continuous with the graph identity, $\delta(0) = 0$, $D\delta(0) = 0$ within $N \cap C$, and the sc$^\infty$-germ property in the k-dependent reading `Chapter3/TangentGerm · MPolyfold.TangentGermChart Chapter3/TangentGerm · IsScInftyGermAt`. This section assembles all of it at the centre x_0 from the graph description of §III.29, adding the three facts the text leaves silent: the chart is $\psi\circ\varphi$ and it is a chart of X; its model $\psi(O)$ is a tame retract; and $K = \ker D(h+t)(0)$ consists of smooth points. With the projecting complement of §III.29 the data also satisfy condition (b) of the regularity conditions. The data at the other zeros and condition (a) come in §§III.36–III.37.

The sc$^\infty$-germ property from the level-wise lifts. Theorem 3.5.11 (1) (preprint Corollary 3.47) gives, for every $k \geq 1$, a relatively open $V_k \ni 0$ in the cone and a C^k map $\sigma_k : V_k \to E_k$ with $\iota_k\sigma_k = \sigma$; Definition 3.2.1 (ii) (preprint 3.25) asks, for every level m and every k, for a relatively open $V_{m,k} \ni 0$ on which σ takes values in E_m and is C^k into E_m. Given m and k, take $j = \max(m,k)+1$, $V_{m,k} = V_j$, and compose σ_j with the inclusion $E_j \to E_m$, a continuous linear map compatible with the inclusions into E_0 `Chapter3/SolutionGermRegularity · exists_clm_toBase_eq` (an iterated composition of the level inclusions): it is a lift of σ into E_m, of class C^j, hence C^k, and it coincides with the level map of σ on V_j by the injectivity of the inclusions `Chapter3/SolutionGermRegularity · isScInftyGermAt_of_lifts`. So the level-wise reading of Theorem 3.5.11 (1) (preprint Corollary 3.47) (§III.20) is exactly what Definition 3.2.1 (ii) (preprint 3.25) needs (item 94 (a)).

The kernel is smooth `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth`. For a basic germ f with contraction map B and the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) (sc$^+$-germ s into W^1), every $x = (a,w) \in \ker D(f+\iota s)(0)$ is a smooth point of $\mathbb{R}^n \oplus W$. The text takes the kernel of an sc-Fredholm operator to be smooth without comment. *The kernel equation.* $D(f+\iota s)(0) = Df(0) + \iota Ds(0)$ `Chapter3/SolutionLinearization · StabilityData.Dfs'_eq`, and the W-component of $Df(0)x$ is $w - DB(0)x$ `Chapter3/SolutionLinearization · StabilityData.snd_Df_eq`. Lemma 3.1.9 (preprint 3.9) at $(0,0)$ gives $DB(0)(0,w) = 0$ `Chapter3/BasicGermLinearization · IsSc0ContractionGerm.deriv_zero_snd`, so $DB(0)x = DB(0)(a,0)$, and the W-component of the kernel equation reads

$$w = DB(0)(a,0) - \iota Ds(0)x.$$

The bootstrap. $DB(0)$ is an sc-operator (the derivative of an sc^1-germ at the smooth point 0) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`, and so is $Ds(0) : E \to W^1$, which raises the level by one. By induction on m, $w \in W_m$: if $w = \iota y$ with $y \in W_m$, then $(a,0)$ lies on level $m+1$ (the parameter space has all levels equal), so $DB(0)(a,0) \in W_{m+1}$, and $(a,y) \in E_m$ gives $Ds(0)x \in W_{m+1}$; hence $w \in W_{m+1}$. So $x \in E_\infty$.

The chart $\psi\circ\varphi$. An sc-smooth map g on a relatively open U is an sc-smooth map between retracts $(O,C,E) \to (O',C',E')$ for every retract $O \subseteq U$ mapped into O': for every sc-smooth retraction R onto O, $g \circ R$ is sc-smooth by the chain rule `Chapter3/GlobalIFTCenter · IsScSmooth.isScSmoothRetractMap_of_subset`. For a chart $\varphi : V \to O$ of X and an sc-diffeomorphism $\psi : U \to U'$ between relatively open

subsets of partial quadrants with inverse ψ' (§III.27) `Chapter2/TameInvariance · IsScDiffeoQuadrant` and $O \subseteq U$, given that $\psi(O)$ is a retract of C', the map $\psi \circ \varphi$ is a chart onto the model $(\psi(O), C', E')$, with inverse $\varphi^{-1} \circ \psi'$ and the source of φ `Chapter3/GlobalIFTCenter · MChart.compDiffeo` (continuity of both maps: ψ, ψ' are sc^0, hence continuous on level 0). *It is a chart of the M-polyfold structure* `Chapter3/GlobalIFTCenter · MChart.compDiffeo_isChart`: for a chart φ' of the atlas, the overlap of $\psi\varphi$ with φ' is ψ of the overlap of φ with φ', and the two transition maps are $(\varphi' \circ \varphi^{-1}) \circ \psi'$ and $\psi \circ (\varphi \circ \varphi'^{-1})$; both are compositions of sc-smooth maps between retracts — the transitions of φ and φ', compatible because both are charts of X, and ψ' resp. ψ restricted to the overlap retracts by the lemma above.

The tame model. For the base retraction r of the centred strong bundle chart, tame by the hypothesis on the charts, and the sc-diffeomorphism ψ of the conjugation, $(\psi(O), C', E')$ is a tame retract with the retraction $\psi \circ r \circ \psi'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image` (§III.27); the text takes this for granted when it calls $\psi \circ \varphi$ a chart of the tame M-polyfold (item 95).

Condition (b) from the choice of complement. For data of Definition 3.2.1 (preprint 3.25) whose complement Y has the projection property for C, $\pi(O) \subseteq C$: the retract lies in C, and π maps C into C `Chapter3/GlobalIFTCenter · MPolyfold.TangentGermChart.π_mem_of_projectsInto`. So the regularity conditions reduce to condition (a), the sc-smoothness of δ on V in the set sense `Chapter3/GlobalIFTCenter · MPolyfold.TangentGermChart.isRegular_of_projectsInto`.

Theorem 3.6.8 (2) (preprint 3.54) at the centre `Chapter3/GlobalIFTCenterG · theorem_3_6_8_tangentGermChart_proj`. *Let f be a section of a strong bundle whose charts compatible with the atlas have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position. Then there are an open $V_x \ni x_0$ and a dimension d such that $S_{V_x} = \{y \in V_x : f(y) = 0\}$ has at x_0 the data of Definition 3.2.1 (preprint 3.25) of dimension d, with $\pi(O) \subseteq C$ for their projection and retract.* The proof fills the fields of the data one by one from the graph description, closed and projecting (§III.29) `Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj`:

1. *The chart.* ψ and ψ' form an sc-diffeomorphism $U \to U'$ (item (i) of §III.29); $O = r(U) \subseteq U$ since the retraction maps U into U `Chapter3/Filler · StrongRetr.r_mem`; $\psi(O)$ is a tame retract (above, with the retract of the base retraction identified with $O = r(U)$ `Chapter3/StrongBundleSections · StrongBundleChart.r_image`); the chart is $\chi = \psi \circ \varphi$, a chart of X, containing x_0, with $\chi(x_0) = \psi(0) = 0$.
2. *The subspace.* $N = K = \ker D(h + t)(0)$, finite-dimensional because $D(h + t)(0)$ is Fredholm `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`, smooth (above), in good position to C' (the hypothesis of Theorem 3.5.11 (preprint Corollary 3.47) established in §III.23); $d := \dim K$.
3. *The complement and the domain.* Y closed, a good complement; $V = V'$, relatively open in $K \cap C'$, containing 0; $\delta = \sigma$, continuous (it is C^1), with values in Y.
4. *The graph identity.* $\chi(S_{V_x} \cap \text{source}) = \psi\varphi(S_{V_x})$, since V_x lies in the source of φ; this is the graph identity of Theorem 3.6.8 (3) (preprint 3.54).
5. *Regularity at* 0. The sc^∞-germ property from the lifts (above); $\sigma(0) = 0$; and $D\sigma(0) = 0$ within the whole cone $K \cap C'$, from $D\sigma(0) = 0$ within $V' = O \cap (K \cap C')$ with O open, since derivatives within a set do not change when the set is intersected with a neighbourhood of the point.
6. *Condition (b).* From the projection property of Y, by the lemma above.

Without the last item this is the plain form `Chapter3/GlobalIFTCenterG · theorem_3_6_8_tangentGermChart`. *What the statement gives about d.* The dimension is existentially quantified in the statement; the proof takes $d = \dim\ker D(h + t)(0)$, the dimension of the kernel of the linearization of the push-forward, and the statement does not identify d with $\dim\ker f'(x_0)$ (the docstring's "$n = \dim\ker f'(x_0)$" is not part of the statement). The local constancy of §III.37 makes d intrinsic in another way: it is the dimension of

every datum of Definition 3.2.1 (preprint 3.25) for the solution set at points of the chart domain.

What is missing at this stage, and where it is supplied. Condition (a), the sc-smoothness of σ on V' in the set sense, is not provided by Theorem 3.5.11 (1) (preprint Corollary 3.47), which gives the sc$^\infty$-germ property at 0 only (item 82); it is obtained in §III.36 by re-centring the data at the nearby zeros, which needs the data of this section at every zero near x_0, hence good position there (Theorem 3.6.8 (1) (preprint 3.54), §§III.32–III.33). The solution set here is the one in V_x; §III.37 restricts the chart so that it describes the global solution set.

III.35 The re-centring lemma

The text's proof of Theorem 3.2.3 (preprint 3.27) begins (preprint p. 84) by showing that the map δ of Definition 3.2.1 (preprint 3.25) is an sc$^\infty$-germ near *every* point $v_0 \in V$, not only near 0: take the point $x_0 = \varphi^{-1}(v_0 + \delta(v_0))$ of M, the data (ψ, N', Y', V', τ) of Definition 3.2.1 (preprint 3.25) at x_0, and consider $w \mapsto \pi \circ \varphi \circ \psi^{-1}(w + \tau(w))$; its linearization at 0 is claimed to be an isomorphism $N' \to N$, so it is a local diffeomorphism near 0, and $\delta \circ (\pi\varphi\psi^{-1}(1+\tau)) = (1-\pi)\varphi\psi^{-1}(1+\tau)$ shows that δ is regular near v_0. For an arbitrary set with the tangent germ property the claim does not hold as we read it (§III.56 has the Lean counterexample): δ need not even be differentiable at v_0 (the curve with a vertical tangent of §III.24), and that is why condition (a) of `IsRegular` had to be added to Theorem 3.2.3 (preprint 3.27). What the argument really needs is that δ be C^1 at v_0. The formal *re-centring lemma* is the text's argument under that hypothesis, for two graph descriptions of the same set: one in a chart χ_1 by a map σ_1 of class C^1 on all of its domain, one in a chart χ_2 centred at the point over v_0 by a map σ_2 with the level-wise C^k-lifts of Theorem 3.5.11 (1) (preprint Corollary 3.47) at 0. Its conclusion is that σ_1 is C^k into every level $E_{1,m}$ near v_0, and that the two subspaces have the same dimension. Section III.36 applies it to the shrunk data of Definition 3.2.1 (preprint 3.25); for the solution set of an sc-Fredholm section, §III.37 supplies the second description at every nearby zero. Four ingredients are proved first: the inverse function machinery of Proposition 3.8.17 (preprint 3.73) with the hypotheses its proofs use, a local inverse theorem for C^k homeomorphisms between relatively open subsets of convex sets, Sard's lemma in fixed dimension, and two facts about relatively open subsets of convex sets.

The inverse function machinery with the hypotheses its proofs use `Chapter3/SolutionGermRecentre · InvDataW`. On a finite-dimensional space V, let $U \subseteq V$ be convex with unique differentiability at each of its points, $f : V \to V$ of class C^j on U with $j \geq 1$, $\|Df(x) - 1\| \leq \frac{1}{2}$ for $x \in U$ (derivatives within U), and let the image $f(U)$ have unique differentiability at each of its points. These are the data of Proposition 3.8.17 (preprint 3.73) as formalized in §III.16 (`InvData`, `Chapter3/QuadrantInverse · InvData`) with the partial quadrant removed: `InvData` asks for a partial quadrant C, U relatively open in C and convex, $f(U) \subseteq C$, and that x be an interior point of C iff $f(x)$ is; from the last hypothesis it derives that $f(U)$ is relatively open in C `Chapter3/QuadrantInverse · InvData.image_relOpen`, and the only use of that openness in the regularity proof is the unique differentiability of $f(U)$. The weak data keep convexity and the derivative bound and replace the quadrant hypotheses by unique differentiability of U and of $f(U)$. The proofs are those of §III.16, retold here because they now stand on their own.

The bi-Lipschitz estimate (Lemma 3.8.18 (preprint 3.74), (70)). $f - \mathrm{id}$ has derivative $Df - 1$ of norm $\leq \frac{1}{2}$ on the convex U, so by the mean value inequality it is a $\frac{1}{2}$-contraction there: $|(f(x') - x') - (f(x) - x)| \leq \frac{1}{2}|x' - x|$ `Chapter3/SolutionGermRecentre · InvDataW.norm_sub_sub_le`. Writing $x' - x = (f(x') - f(x)) - ((f(x') - x') - (f(x) - x))$ gives $\frac{1}{2}|x' - x| \leq |f(x') - f(x)|$ `Chapter3/SolutionGermRecentre · InvDataW.`

half_norm_le, hence f is injective on U Chapter3/SolutionGermRecentre · InvDataW.injOn. *The derivative is invertible*: $Df(x) = 1 - (1 - Df(x))$ with $\|1 - Df(x)\| \le \frac{1}{2} < 1$ is a unit by the Neumann series Chapter3/SolutionGermRecentre · InvDataW.derivUnit Chapter3/SolutionGermRecentre · InvDataW.isUnit_deriv. *The inverse* g on $f(U)$ is the inverse of f on U (a choice of preimage, unique by injectivity) Chapter3/SolutionGermRecentre · InvDataW.inv Chapter3/SolutionGermRecentre · InvDataW.inv_apply; the estimate gives $|g(y') - g(y)| \le 2|y' - y|$ Chapter3/SolutionGermRecentre · InvDataW.norm_inv_sub_le, so g is Lipschitz, hence continuous on $f(U)$ Chapter3/SolutionGermRecentre · InvDataW.continuousOn_inv.

Lemma 3.8.21 (preprint 3.77) Chapter3/SolutionGermRecentre · InvDataW.hasFDerivWithinAt_inv: g is differentiable within $f(U)$ at every $y \in f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$. With $x = g(y)$: as $y' \to y$ in $f(U)$, $g(y') \to x$ in U (continuity), so $f(g(y')) - f(x) - Df(x)(g(y') - x) = o(|g(y') - x|)$, and $|g(y') - x| \le 2|y' - y|$ turns this into $o(|y' - y|)$; applying the bounded operator $Df(x)^{-1}$ and using $f(g(y')) = y'$ gives $g(y') - g(y) - Df(x)^{-1}(y' - y) = o(|y' - y|)$. No hypothesis on $f(U)$ enters here.

Proposition 3.8.17 (preprint 3.73), the regularity of the inverse Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inv: g is C^j on $f(U)$. First, $x \mapsto [Df(x)]^{-1}$ is C^{j-1} on U: the derivative within U of a C^j map is C^{j-1} there because U has unique differentiability, and inversion is smooth near the units Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inverse_deriv. Then by induction on $i \le j$: g is C^0 (continuous); if g is C^i with $i < j$, then g has the derivative $y \mapsto [Df(g(y))]^{-1}$ at every point of $f(U)$, which is C^i as the composite of the C^{j-1} map $x \mapsto [Df(x)]^{-1}$ with g, and a map with a C^i derivative on a set of unique differentiability is C^{i+1} there Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn. This last step is the one place where the unique differentiability of $f(U)$ is used. It is the bootstrapping of §III.16 in place of the text's iterated tangent maps (Lemma 3.8.20 (preprint 3.76)).

Relatively open subsets of convex sets. Let $D \subseteq V_1$ be convex with nonempty interior and $A = O \cap D$ with O open. Then A has unique differentiability at each of its points (a convex set with interior has it, and intersecting with an open set keeps it) Chapter3/SolutionGermRecentre · uniqueDiffOn_relOpen_convex; and if A is nonempty, it has nonempty interior: for $a \in A$ and an interior point w of D, the points $a + t(w - a)$ with $0 < t$ small lie in O by continuity and in the interior of D, because the open segment from a point of the closure of a convex set to an interior point lies in the interior Chapter3/SolutionGermRecentre · interior_nonempty_of_relOpen_convex.

Sard's lemma in fixed dimension Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty. If $f : V_1 \to V_2$ between finite-dimensional spaces is differentiable on s and $f(s)$ has nonempty interior, then $\dim V_2 \le \dim V_1$. Suppose $\dim V_1 < \dim V_2$. There is a linear surjection $p : V_2 \to V_1$ (send the first $\dim V_1$ vectors of a basis of V_2 to a basis of V_1 and the rest to 0) Chapter3/SolutionGermRecentre · exists_linearMap_surjective. The map $f \circ p$ on $p^{-1}(s) \subseteq V_2$ has the derivative $Df(p(x)) \circ p$, an endomorphism of V_2 whose range lies in the range of $Df(p(x))$, of dimension $\le \dim V_1 < \dim V_2$; so it is not injective and its determinant vanishes. By Mathlib's change-of-variables estimate, a set on which a map is differentiable with vanishing Jacobian determinant has an image of Haar measure zero Mathlib/MeasureTheory/Function/Jacobian · MeasureTheory.addHaar_image_eq_zero_of_det_fderivWithin_eq_zero. But $f(s) \subseteq (f \circ p)(p^{-1}(s))$ since p is onto, and the nonempty open interior of $f(s)$ has positive Haar measure: a contradiction. The lemma uses only differentiability (not continuity of the derivative) and only the finite dimension of both spaces.

A C^k homeomorphism with invertible derivative has a C^k inverse Chapter3/SolutionGermRecentre · exists_contDiffOn_inverse. Let $D_1 \subseteq V_1$, $D_2 \subseteq V_2$ be convex with nonempty interior, A relatively open in D_1, β of class C^k on A with $k \ge 1$, $\beta(A)$ relatively open in D_2, α continuous on $\beta(A)$ with $\alpha(\beta(w)) = w$

for $w \in A$, and $D\beta(w_0) = L$ (derivative within A) a linear isomorphism $V_1 \to V_2$ at some $w_0 \in A$. Then there is a relatively open $B' \subseteq \beta(A)$ in D_2 containing $\beta(w_0)$ on which α is C^k. *Proof.* Normalize: $\hat\beta = L^{-1} \circ \beta$ is C^k on A with $D\hat\beta(w_0) = 1$. Its derivative is continuous within A (C^k, $k \geq 1$, on a set of unique differentiability), so $\|D\hat\beta(w) - 1\| \leq \frac{1}{2}$ for $w \in A$ in a ball $B(w_0, r_1)$; take $r \leq r_1$ so small that $B(w_0, r) \subseteq O$, where $A = O \cap D_1$. The set $U = D_1 \cap B(w_0, r)$ is convex, lies in A, is relatively open in D_1 (hence has unique differentiability), and the derivatives of $\hat\beta$ within U and within A agree there (locality of the derivative within a set), so $\|D\hat\beta - 1\| \leq \frac{1}{2}$ on U. Its image $\beta(U) = \beta(A) \cap \alpha^{-1}(B(w_0, r))$ — because α inverts β on A — is relatively open in D_2 by the continuity of α, so $\beta(U)$, and with it $\hat\beta(U) = L^{-1}(\beta(U))$, has unique differentiability. So $(U, \hat\beta, k)$ are weak inverse data, and $\hat\beta^{-1}$ is C^k on $\hat\beta(U)$ (Proposition 3.8.17 (preprint 3.73) above). On $B' = \beta(U)$, $\alpha = \hat\beta^{-1} \circ L^{-1}$, since $\alpha(\beta(w)) = w = \hat\beta^{-1}(\hat\beta(w))$; hence α is C^k on B'. The relative openness of $\beta(U)$ comes from the continuity of the given inverse α, not from a degeneracy index; that is the point of the weak data.

The re-centring lemma `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`. Let χ_1, χ_2 be charts of the M-polyfold X `Chapter2/MPolyfold · MPolyfold.IsChart.compatible`, with models (O_i, C_i, E_i), and $S \subseteq X$. Suppose:

- (in χ_1) $K_1 \subseteq E_1$ is a finite-dimensional smooth subspace in good position to C_1, Y_1 a closed algebraic complement of K_1, V_1 relatively open in $K_1 \cap C_1$, $\sigma_1 : K_1 \to E_{1,0}$ with values in Y_1 and of class C^1 on V_1; $V_{x,1}$ is open in X inside the domain of χ_1 and $\chi_1(S \cap V_{x,1}) = \{v + \sigma_1(v) \mid v \in V_1\}$;
- (in χ_2) the same kind of data K_2, Y_2, $V_2 \ni 0$, σ_2 with values in Y_2, continuous on V_2, $\sigma_2(0) = 0$, $D\sigma_2(0) = 0$ within V_2, and for every $k \geq 1$ a relatively open $V_{2,k} \subseteq V_2$ containing 0 and a map $\sigma_{2,k} : K_2 \to E_{2,k}$ of class C^k on $V_{2,k}$ with $\iota_k \sigma_{2,k} = \sigma_2$ there (the lifts of Theorem 3.5.11 (1) (preprint Corollary 3.47)); $V_{x,2}$ open inside the domain of χ_2 with $\chi_2(S \cap V_{x,2}) = \{w + \sigma_2(w) \mid w \in V_2\}$;
- $y \in V_{x,1} \cap V_{x,2}$ with $\chi_2(y) = 0$, and $v_0 \in V_1$ with $\chi_1(y) = v_0 + \sigma_1(v_0)$.

Then $\dim K_1 = \dim K_2$, and for every level m and every k there is a relatively open $B' \subseteq V_1$ in $K_1 \cap C_1$ containing v_0 on which σ_1 takes values in $E_{1,m}$ and its level map $\sigma_1 : B' \to E_{1,m}$ is of class C^k.

The hypotheses are exactly what each step uses: the closedness of Y_i makes the projection π_i onto K_i along Y_i continuous; good position gives $K_i \cap C_i$ a nonempty interior in K_i (for unique differentiability and for Sard); smoothness of K_i gives the linear lifts of K_i to every level `Chapter3/BasicClass · ScBanach.finLift`. Note that V_1 need not contain 0 and σ_1 need not vanish anywhere: χ_1 is an arbitrary chart in which S is a C^1 graph.

Step 1: notation and the points of S. Write $\Gamma_1(v) = v + \sigma_1(v)$, $\Gamma_2(w) = w + \sigma_2(w)$, and π_1, π_2 for the continuous projections along Y_1, Y_2, so that $\pi_i \Gamma_i = \mathrm{id}$. `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels` For $v \in V_1$ the point $\chi_1^{-1}(\Gamma_1(v))$ lies in $S \cap V_{x,1}$ and χ_1 maps it back to $\Gamma_1(v)$, which therefore lies in O_1; likewise for $w \in V_2$. Since $\sigma_2(0) = 0$, $\Gamma_2(0) = 0$, and both $\chi_1^{-1}(\Gamma_1(v_0))$ and $\chi_2^{-1}(\Gamma_2(0))$ are y.

Step 2: the transition and the retractions onto the overlaps. The charts are compatible, so the transition $T = \chi_1 \circ \chi_2^{-1}$ is an sc-smooth map from the overlap retract $\chi_2(V_{\chi_2} \cap V_{\chi_1})$ to $\chi_1(V_{\chi_1} \cap V_{\chi_2})$, and $T' = \chi_2 \circ \chi_1^{-1}$ in the other direction `Chapter2/MPolyfold · MChart.transition` `Chapter2/MPolyfold · MChart.overlap`. The overlaps are sc-retracts `Chapter2/MPolyfold · MChart.isScRetract_overlap`: choose sc-smooth retractions r_2, r_1 onto them, with domains U_{r_2}, U_{r_1}. Then $T \circ r_2$ is sc-smooth on U_{r_2}, $T' \circ r_1$ on U_{r_1}, $T(\chi_2(q)) = \chi_1(q)$ on the domain of χ_2, and $T' \circ T = \mathrm{id}$ on the overlap. Here the formal proof departs from the text, which treats $\varphi \circ \psi^{-1}$ as a map one can differentiate along the graph; a transition is only an sc-smooth map between retracts,

and every derivative below is taken of $T \circ r_2$ or $T' \circ r_1$, on relatively open subsets of partial quadrants.

Step 3: the parameter sets and the maps β, α. Let $A = \{w \in V_2 \mid \chi_2^{-1}(\Gamma_2(w)) \in V_{x,1}\}$ and $B_s = \{v \in V_1 \mid \chi_1^{-1}(\Gamma_1(v)) \in V_{x,2}\}$: the parameters whose points of S lie in both chart neighbourhoods. By continuity of $w \mapsto \chi_2^{-1}(\Gamma_2(w))$ on V_2 (and of its counterpart), A is relatively open in $K_2 \cap C_2$, B_s in $K_1 \cap C_1$; $0 \in A$ and $v_0 \in B_s$. Define

$$\beta(w) = \pi_1(T(\Gamma_2(w))), \qquad \alpha(v) = \pi_2(T'(\Gamma_1(v))).$$

For $w \in A$, with $q = \chi_2^{-1}(\Gamma_2(w)) \in S \cap V_{x,1} \cap V_{x,2}$: $T(\Gamma_2(w)) = \chi_1(q)$ lies on the graph of σ_1, say $= \Gamma_1(v)$ with $v \in V_1$; applying π_1 gives $\beta(w) = v$, so $T(\Gamma_2(w)) = \Gamma_1(\beta(w))$, $\beta(w) \in B_s$, $\alpha(\beta(w)) = w$, and $\Gamma_2(w)$ lies in the overlap. Symmetrically, for $v \in B_s$: $T'(\Gamma_1(v)) = \Gamma_2(\alpha(v))$, $\alpha(v) \in A$, $\beta(\alpha(v)) = v$. Moreover $\beta(0) = v_0$, and α is continuous on B_s ($\pi_2 \circ \chi_2 \circ \chi_1^{-1} \circ \Gamma_1$). Consequently, for every $A' \subseteq A$, $\beta(A') = \{v \in B_s \mid \alpha(v) \in A'\}$, which is relatively open in $K_1 \cap C_1$ whenever A' is relatively open in $K_2 \cap C_2$. This is the text's "local diffeomorphism" in its topological half, obtained without any derivative. `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`

Step 4: the lifted graph maps. For $j \geq 1$ let $A_j = A \cap V_{2,j}$ and $\Gamma_{2,j}(w) = \lambda_j(w) + \sigma_{2,j}(w) \in E_{2,j}$, where $\lambda_j : K_2 \to E_{2,j}$ is the linear lift `Chapter3/BasicClass · ScBanach.finLift`. Then A_j is relatively open and contains 0, $\Gamma_{2,j}$ is C^j on A_j with $\iota_j \Gamma_{2,j} = \Gamma_2$ there, $\Gamma_{2,j}$ maps A_j into the level-j part of the domain of r_2, and $r_2(\Gamma_2(w)) = \Gamma_2(w)$ for $w \in A_j$ (the graph points lie in the overlap, where r_2 is the identity). Its derivative within A_j at 0 is λ_j: indeed $\iota_j \circ D\sigma_{2,j}(0) = D\sigma_2(0)$ within A_j, which equals the derivative within V_2 (the two sets agree near 0) and so vanishes; ι_j is injective, so $D\sigma_{2,j}(0) = 0$. Here the unique differentiability of A_j at 0 is that of a relatively open subset of the convex cone $K_2 \cap C_2$ with interior `Chapter3/SolutionGermRecentre · uniqueDiffOn_relOpen_convex`.

Step 5: β is C^j on A_j. On A_j, $\beta = \pi_1 \circ ((T \circ r_2) \circ \iota_j) \circ \Gamma_{2,j}$, because r_2 fixes $\Gamma_2(w) = \iota_j \Gamma_{2,j}(w)$. The map $(T \circ r_2) \circ \iota_j$ is C^j on the level-j part of U_{r_2} — Proposition 1.2.3 (preprint 1.18): an sc^j map is C^j from level j to level 0 `Chapter1/Levels · IsSck.contDiffOn_base` — and π_1 is continuous linear; so β is C^j on A_j. This is where the lifts are indispensable: σ_2 itself is only continuous into level 0, and $T \circ r_2$ is not differentiable on level 0.

Step 6: the chain rule at 0. Take $j = 2$; then β is C^2, in particular C^1, on A_2, with derivative $D\beta = D\beta(0)$ within A_2 at 0. Read the lift on level 1: $\Gamma_{2,1} = \iota_{1,2} \circ \Gamma_{2,2}$, with $\iota_1 \Gamma_{2,1} = \Gamma_2$ on A_2, values in the level-1 part of U_{r_2}, derivative $\iota_{1,2}\lambda_2$ at 0, and $\iota_1 \iota_{1,2} \lambda_2(n) = n$ for $n \in K_2$. Let $\mathcal{D} = D(T \circ r_2)(\Gamma_{2,1}(0))$, the sc-derivative at the level-1 point over 0. The map $w \mapsto T(r_2(\iota_1 \Gamma_{2,1}(w)))$ has, by the chain rule for the sc^1 map $T \circ r_2$ (differentiable from level 1 to level 0) and the differentiable $\Gamma_{2,1}$, the derivative $n \mapsto \mathcal{D}n$ at 0 within A_2. On A_2 this map equals $\Gamma_1 \circ \beta$ (Step 3 and $r_2 \circ \Gamma_2 = \Gamma_2$), and $\Gamma_1 \circ \beta$ has, at 0, the derivative $n \mapsto D\beta\, n + D\sigma_1(v_0) D\beta\, n$ by the chain rule — here the C^1-property of σ_1 *at* v_0, within V_1, is used, together with $\beta(A_2) \subseteq V_1$ and $\beta(0) = v_0$. By uniqueness of the derivative within A_2 at 0,

$$\mathcal{D}n = D\beta\, n + D\sigma_1(v_0)\, D\beta\, n \qquad (n \in K_2).$$

Step 7: $D\beta(0)$ is injective. Let $D\beta\, n = 0$; then $\mathcal{D}n = 0$. The composite $(T' \circ r_1) \circ (T \circ r_2)$ is sc-smooth on U_{r_2} ($T \circ r_2$ maps U_{r_2} into the overlap retract of χ_1, hence into U_{r_1}), and it equals r_2 there: $r_2(u)$ lies in the overlap, $T(r_2(u))$ in the other overlap where r_1 is the identity, and $T' \circ T = \mathrm{id}$ on the overlap. By the chain rule for sc^1 maps between relatively open subsets of partial quadrants `Chapter2/DegeneracyIndex · scDeriv_comp_apply`, $D((T' \circ r_1) \circ (T \circ r_2))(\Gamma_{2,1}(0))n = D(T' \circ r_1)(\cdot)\, \mathcal{D}n = 0$.

Since the composite agrees with r_2 on the relatively open U_{r_2}, its derivative at the level-1 point is $Dr_2(\Gamma_{2,1}(0))$ Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen. And $Dr_2(\Gamma_{2,1}(0))n = n$: the family $w \mapsto \iota_1\Gamma_{2,1}(w)$ consists of fixed points of r_2, so the derivative of r_2 fixes its tangent vectors at 0, which are the vectors $\iota_1\iota_{1,2}\lambda_2(n) = n$ Chapter3/GlobalIFTNearbyGraph · scDeriv_apply_toBase_deriv_eq_of_fixed — the tangent space K_2 of the graph lies in the tangent space of the retract. So $n = 0$. Hence $D\beta(0)$ is injective and $\dim K_2 \le \dim K_1$.

Step 8: surjectivity by counting dimensions. β is differentiable on A_2, and $\beta(A_2)$ is a nonempty relatively open subset of the convex cone $K_1 \cap C_1$ (Step 3; it contains $\beta(0) = v_0$), which has nonempty interior in K_1 by good position; so $\beta(A_2)$ has nonempty interior Chapter3/SolutionGermRecentre · interior_nonempty_of_relOpen_convex, and Sard's lemma in fixed dimension gives $\dim K_1 \le \dim K_2$ Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty. So $\dim K_1 = \dim K_2$, the injective $D\beta(0)$ is bijective, and $L = D\beta(0) : K_2 \to K_1$ is a linear isomorphism. This is the first conclusion of the lemma, and also the reason why the dimension of a set with two such descriptions cannot jump.

Step 9: the inverse, and the regularity of σ_1. Fix m and k and take $j = m + k + 2$. By Step 5, β is C^j, hence $C^{\max(k,1)}$, on A_j, $\beta(A_j) \subseteq B_s$, and its derivative within A_j at 0 is again L (the derivatives within A_j and within A_2 agree at 0, both sets being relatively open near 0 in the same cone). Now apply the local inverse theorem above with $D_1 = K_2 \cap C_2$, $D_2 = K_1 \cap C_1$ (convex with interior), $A = A_j$, the continuous inverse α on $\beta(A_j)$ and $w_0 = 0$ Chapter3/SolutionGermRecentre · exists_contDiffOn_inverse: there is a relatively open $B' \subseteq \beta(A_j)$ in $K_1 \cap C_1$ containing v_0 on which α is $C^{\max(k,1)}$, with $\alpha(B') \subseteq A_j$. Next the level-m map

$$\Psi(w) = (T \circ r_2)_m(\iota_{m,k+2}\,\Gamma_{2,j}(w)) - \lambda_m(\beta(w)) \in E_{1,m}, \qquad w \in A_j,$$

where $\iota_{m,k+2} : E_{2,m+k+2} \to E_{2,m}$ and λ_m is the lift of K_1 to level m. It is C^k on A_j: the sc^{k+2} map $T \circ r_2$ is C^{k+2} from level $m+k+2$ to level m (Proposition 1.2.3 (preprint 1.18) Chapter1/Levels · IsSck.contDiffOn_levels), $\Gamma_{2,j}$ is C^j into level $j = m+k+2$, and $\lambda_m \circ \beta$ is C^j. Including into level 0, $\iota_m\Psi(w) = T(r_2(\Gamma_2(w))) - \beta(w) = \Gamma_1(\beta(w)) - \beta(w) = \sigma_1(\beta(w))$. Hence for $v \in B'$, $v = \beta(\alpha(v))$ and $\sigma_1(v) = \iota_m\Psi(\alpha(v))$: σ_1 takes values in $E_{1,m}$ on B', its level-m map is $\Psi \circ \alpha$ Chapter1/ScSmooth · levelMap_eq_of_toBase_eq, which is C^k on B' as a composite of C^k maps, and $B' \subseteq B_s \subseteq V_1$. This is the text's identity $\delta \circ (\pi\varphi\psi^{-1}(1+\tau)) = (1-\pi)\varphi\psi^{-1}(1+\tau)$, written on level m through the lifts.

Comparison with the text. Four things in the formal argument are not in the sketch of preprint p. 84. (a) The isomorphism claim needs σ_1 differentiable at v_0, which for an arbitrary set with the tangent germ property fails; here σ_1 is assumed C^1 on all of V_1, and §III.36 arranges this by shrinking V_1 to the neighbourhood of 0 on which the germ property makes δ of class C^1. (b) Only the *injectivity* of the linearization comes from the chain rule, and it uses that the tangent space K_2 of the second graph lies in the tangent space of the overlap retract (Step 7). The symmetric argument for surjectivity would differentiate $T' \circ \Gamma_1 = \Gamma_2 \circ \alpha$ at v_0; but $T' \circ r_1$ is differentiable only from level 1 to level 0, so this would need σ_1 differentiable *into level* 1 at v_0, which is known near the centre of χ_1 (Theorem 3.5.11 (1) (preprint Corollary 3.47)) and not at v_0. Surjectivity therefore comes from the dimension count by Sard's lemma (Step 8), which at the same time shows that the dimension is locally constant. (c) The regularity of $\beta^{-1} = \alpha$ is not the quadrant inverse function theorem 3.72 (§III.16): that theorem asks for $d_C(x) = d_C(\beta(x))$ (formally: interior points to interior points), which here would be a statement about the intrinsic degeneracy index of S — not available. It is Lemma 3.8.21 (preprint 3.77) and the bootstrapping of Proposition 3.8.17 (preprint 3.73) under the weak data, with the relative openness of $\beta(U)$ supplied by the continuity of the inverse α. (d) The neighbourhood B' depends on (m,k) through the lift $\Gamma_{2,m+k+2}$; this is harmless,

because sc-smoothness in the set sense is a local property of the level maps (§III.36).

III.36 Regular data by re-centring, and Theorem 3.2.3 (preprint 3.27) with locally constant dimension

This section turns the re-centring lemma of §III.35 into condition (a) of `IsRegular` (§III.24) and adapts Theorem 3.2.3 (preprint 3.27) to data whose dimension varies from point to point. The tool is the shrinking of the data of Definition 3.2.1 (2) (preprint 3.25): the chart is restricted to an open subset of its domain and the parameter set V to a smaller relatively open set, with the same N, Y and δ. The data at a point x are shrunk into the neighbourhood of 0 where δ is C^1; then, at every point of the shrunk graph, the data at that point (which a set with the tangent germ property has everywhere) provide the second graph description that the re-centring lemma needs.

Open subsets of tame retracts (told in §III.27). The restriction of a tame retraction $r : U \to U$ to $U \cap r^{-1}(W)$, W open (Proposition 2.3.1 (1) (preprint 2.10)) `Chapter2/Retracts · ScRetraction.restrict`, is tame: the degeneracy-index condition of Definition 2.5.2 (preprint 2.34) is inherited pointwise, and at a smooth point x of $O \cap W$ the sc-complement of T_xO of the original retraction serves, because the derivative of the restricted retraction at x is $Dr(x)$ — the two maps agree on a relatively open neighbourhood `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen` `Chapter3/SolutionGermRecentre · ScRetraction.restrict_isTame`. So $O \cap W$ is a tame retract `Chapter3/SolutionGermRecentre · IsTameRetract.inter_isOpen`, and the model $\varphi(s)$ of a chart restricted to an open s of its domain is tame when the model of φ is, since $\varphi(s) = O \cap W$ for some open W `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq` `Chapter3/SolutionGermRecentre · MChart.isTameRetract_image`. The text uses this silently whenever it shrinks a chart of a tame M-polyfold.

Shrinking the data `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink`. Let $D = (\varphi, N, Y, V, \delta)$ be data of Definition 3.2.1 (2) (preprint 3.25) for M at x of dimension n `Chapter3/TangentGerm · MPolyfold.TangentGermChart`, s an open subset of the domain of φ containing x, $V' \subseteq V$ relatively open in $N \cap C$ containing 0, and M' a set with $\varphi(M' \cap s) = \{v + \delta(v) \mid v \in V'\}$. Then the chart φ restricted to s `Chapter2/MPolyfold · MChart.restrict` — a chart of the structure `Chapter2/MPolyfold · MPolyfold.IsChart.restrict` with a tame model by the paragraph above — together with the same N, Y, δ and the parameter set V', are data of Definition 3.2.1 (2) (preprint 3.25) for M' at x of dimension n. Every field is inherited: good position, the closed good complement, $\delta(0) = 0$, $D\delta(0) = 0$ and the germ property of δ concern N, Y, δ and the cone $N \cap C$ only, which are unchanged, and continuity on V' follows from continuity on V. The set M' may differ from M; this is how §III.37 passes between the solution set in a neighbourhood and the global solution set. The shrunk data have the source s, the model $\varphi(s)$, the same chart values, the same N, δ and projection π, and the parameter set V' `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_source` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_model_0` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_apply` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_N` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_V` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_δ` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_π`. They are regular `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular` as soon as δ is sc-smooth on V' in the set sense and the original data satisfy condition (b), $\pi(O) \subseteq C$: condition (b) passes to the smaller retract $\varphi(s) \subseteq O$ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`.

The shrinking data `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_shrinkData`. For an open

$W \ni x$ in X and an open $O'' \ni 0$ in N, let

$$s = \{y \in \operatorname{dom}\varphi \cap W \mid \pi\varphi(y) \in O''\}, \qquad V' = \{v \in V \cap O'' \mid \varphi^{-1}(v + \delta(v)) \in W\}.$$

Then s is open (the preimage of O'' under the continuous $\pi \circ \varphi$ on the open $\operatorname{dom}\varphi \cap W$) and contains x ($\pi\varphi(x) = \pi(0) = 0$); V' is relatively open in $N \cap C$ because $v \mapsto \varphi^{-1}(v + \delta(v))$ is continuous on V `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_continuousOn`, and contains 0 ($\varphi^{-1}(0) = x \in W$); $s \subseteq W$, $V' \subseteq V \cap O''$; and $\varphi(M \cap s) = \{v + \delta(v) \mid v \in V'\}$. For the last identity: a point $y \in M \cap s$ has $\varphi(y) = \Gamma(v)$ with $v = \pi\varphi(y) \in V$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.exists_eq_Γ`, $v \in O''$ because $y \in s$, and $\varphi^{-1}(\Gamma(v)) = y \in W$; conversely for $v \in V'$ the point $\varphi^{-1}(\Gamma(v))$ lies in M and in the domain `Chapter3/TangentGerm · MPolyfold.TangentGermChart.symm_Γ_mem`, in W by definition of V', and $\pi\varphi$ of it is $v \in O''$ since $\pi\Gamma = \mathrm{id}$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_Γ`.

The lifts from the germ property. Definition 3.2.1 (ii) (preprint 3.25), in the reading of §III.24 `Chapter3/TangentGerm · IsScInftyGermAt`, gives for every m, k a relatively open $V_{m,k} \ni 0$ on which δ takes values in E_m and is C^k into E_m. With $m = k$: for every $k \geq 1$ the level map of δ on $V_{k,k} \cap V$ is a C^k-lift δ_k of δ into E_k — the hypothesis of the re-centring lemma on the centred side `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_lift`. With $(m, k) = (0, 1)$: δ is C^1 on a relatively open neighbourhood of 0 in $N \cap C$ `Chapter3/TangentGerm · IsScInftyGermAt.contDiffOn_one`. And $D\delta(0) = 0$ within V, since the data give it within $N \cap C$ and V agrees with $N \cap C$ near 0 `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.fderivWithin_V_zero`.

Condition (a) of `IsRegular` by re-centring `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`. Let X be an M-polyfold (Hausdorff, paracompact), $M \subseteq X$, D data of Definition 3.2.1 (2) (preprint 3.25) for M at x of dimension n satisfying condition (b), $\pi(O) \subseteq C$, and $U \ni x$ open such that every point of $M \cap U$ has data of Definition 3.2.1 (2) (preprint 3.25) for M of *some* dimension. Then there are data D' for M at x of dimension n which are regular (`IsRegular`: (a) and (b)), whose chart domain lies in U, and such that every datum of Definition 3.2.1 (2) (preprint 3.25) for M at a point of M in the chart domain of D' has dimension n. (The proof constructs D' as the shrinking of D, with the same N, Y, δ; the statement records only the regularity, the domain and the dimension.)

Step 1: shrinking into the C^1 neighbourhood. By the germ property, δ is C^1 on $V_1 = O_1 \cap (N \cap C)$ with O_1 open and $0 \in O_1$. The shrinking data for $W = U$ and $O'' = O_1$ give $s \subseteq U$ and $V' \subseteq V \cap O_1$, so $V' \subseteq V_1$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.V_subset`, with $\varphi(M \cap s) = \Gamma(V')$; let D_1 be the shrunk data for M itself. Then δ is C^1 on all of D_1's parameter set V': this is hypothesis (a) of the re-centring lemma on the side of D_1, and the reason for the shrinking.

Step 2: the re-centring at every point. Let $y \in M \cap s$ and let D'' be any data of Definition 3.2.1 (2) (preprint 3.25) for M at y, of dimension n'. Put $v_0 = \pi\varphi(y)$; then $v_0 \in V'$ and $\varphi(y) = \Gamma(v_0)$. The re-centring lemma applies with $\$\square_1 = \$$ the chart of D_1 (graph of δ over V' on its whole domain s, δ of class C^1 on V') and $\$\square_2 = \$$ the chart of D'', centred at y (graph of δ'' over V'' on its whole domain, $\delta''(0) = 0$, $D\delta''(0) = 0$ within V'', the lifts from the germ property of D''); all the remaining hypotheses — smooth finite-dimensional N, closed good complement, good position, relatively open parameter sets — are fields of the two data `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`. It gives $n = \dim N = \dim N'' = n'$, and for every m, k a relatively open $B' \subseteq V'$ in $N \cap C$ containing v_0 on which δ takes values in E_m and is C^k into E_m.

Step 3: δ is sc-smooth on V' in the set sense. V' is a relatively open subset of the partial quadrant $N \cap C$ of N with the constant sc-structure `Chapter3/TangentGerm · MPolyfold.TangentGermChart.V_isRelOpenInQuadrant`, all

of whose levels are V'. By Proposition 1.2.4 (preprint 1.19) in the form "maps every level into the corresponding level, and the level maps are C^∞" Chapter1/Levels · IsScSmooth.of_contDiffOn_levels, it suffices that for every m: (i) $\delta(V') \subseteq E_m$, and (ii) the level map $\delta : V' \to E_m$ is C^k for every k. For $v \in V'$ the point $y = \varphi^{-1}(\Gamma(v))$ lies in $M \cap s \subseteq M \cap U$, so by hypothesis there are data at y of some dimension, and Step 2 applies with $v_0 = \pi\varphi(y) = v$. For (i), the set B' for $(m, 0)$ contains v, and δ maps it into E_m. For (ii), fix k; around every $v \in V'$ the set $B' = O_B \cap (N \cap C)$ for (m, k) contains v, and the level map is C^k on $O_B \cap V' \subseteq B'$; a map that is C^k near every point of a set is C^k on the set Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_of_locally_contDiffOn. Here the dependence of B' on (m, k) (§III.35, Step 9) does no harm: the property is checked one (m, k) at a time, and locally.

Step 4: conclusion. With (a) from Step 3 and (b) inherited from D, the shrunk data D_1 are regular Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular; their chart domain is $s \subseteq U$; and by Step 2 every datum at a point of $M \cap s$ has dimension n.

What the statement gives for an arbitrary subset. The hypotheses of the theorem are about M and D only; no solution set, section or Fredholm property enters. If M has the n-dimensional tangent germ property Chapter3/TangentGerm · MPolyfold.HasTangentGermProperty, every point of M has data of dimension n, so the hypothesis "every point of $M \cap U$ has data of some dimension" holds with $U = X$. Hence, read off the statement (this is not a separate Lean theorem): *for any M with the tangent germ property, any data at x that satisfy condition (b) can be shrunk to regular data at x, of the same dimension.* Condition (a) of `IsRegular` is therefore not an independent hypothesis of Theorem 3.2.3 (preprint 3.27): if at every point of M some n-dimensional data satisfy (b), the regular data that Theorem 3.2.3 (preprint 3.27) asks for exist at every point Chapter3/TangentGerm · theorem_3_2_3_of_isRegular. (The inclusion $M \subseteq X_\infty$ of Definition 3.2.1 (1) (preprint 3.25) is not used.) The counterexample to the text's step "δ is an sc$^\infty$-germ near every v_0" (§III.24) is not contradicted: there the data are regular only after shrinking V away from the point with a vertical tangent, which is what Step 1 does. What remains a genuine restriction is condition (b), which fails for some good complements (§III.24) and is obtained for solution sets by the choice of the complement (§III.29, §III.34).

The dimension is locally constant. The last clause of the theorem says more than the text: around every point where regular data exist, all data of Definition 3.2.1 (2) (preprint 3.25) at nearby points of M have the same dimension. The dimension of a set with the tangent germ property at varying dimensions is therefore locally constant along M — but not necessarily globally constant, and Theorem 3.2.3 (preprint 3.27) as formalized fixes n (§III.24). For the solution set of an sc-Fredholm section, whose dimension $\dim \ker f'(x)$ can change between components, the fixed-dimension theorem would need S split first; instead Theorem 3.2.3 (preprint 3.27) is restated for locally constant dimension.

Manifolds with corners of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCornersLoc. An M-polyfold structure is that of a manifold with boundary with corners of locally constant dimension if it has an equivalent atlas all of whose charts have models (O, C, E) with O relatively open in the partial quadrant C and E_0 finite-dimensional — the definition of §III.24 Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners without the requirement that every chart have dimension n. A manifold with corners of dimension n is one of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCorners.isManifoldWithCornersLoc.

Theorem 3.2.3 (preprint 3.27) with locally constant dimension Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular. Let X be an M-polyfold, $M \subseteq X$ nonempty, and suppose every $x \in M$ has regular data of Definition 3.2.1 (2) (preprint 3.25) for M of some dimension $n(x)$. Then M is a sub-M-polyfold of X; its induced structure (Proposition 2.3.19 (preprint 2.21)) is tame and that of a manifold with boundary

with corners of locally constant dimension; and for every $x \in M$, all regular data d at x of any dimension and every base point $a_0 \in M$ (the value of the chart inverse off its domain), the chart $y \mapsto \pi\varphi(y)$ of d is a chart of the induced structure whose domain contains x. The proof is that of Theorem 3.2.3 (preprint 3.27) (§III.24) with the dimension quantified pointwise; every step there is either pointwise or does not use n.

Sub-M-polyfold Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_isSubMPolyfold. Definition 2.3.18 (preprint 2.20) is pointwise: at $a \in M$ take regular data d (of whatever dimension); the open set $U_0 = \varphi^{-1}(\Sigma \cap O)$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.U₀ Chapter3/TangentGerm · MPolyfold.TangentGermChart.isOpen_U₀ Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_U₀ and the sc-smooth retraction $s = \varphi^{-1} \circ t \circ \varphi$ onto $M \cap U_0$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.s Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_isScSmoothOn Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_s Chapter3/TangentGerm · MPolyfold.TangentGermChart.image_s are those of §III.24, exactly as in the fixed-dimension proof Chapter3/TangentGerm · theorem_3_2_3_isSubMPolyfold.

The charts and the atlas. Let the charts be all maps $\pi\varphi$ of all regular data at all points of M, of all dimensions, with all base points Chapter3/GlobalIFTTheorems · MPolyfold.tangentChartsLoc Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart. Each is a chart of the induced structure Chapter2/SubMPolyfold · subPolyfold: it is compatible with every chart $\varphi'|_{W\cap M}$ of the induced atlas, by the lemma of §III.24, whose proof never compares dimensions Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_compatible_subChart Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_isChart. Any two of them are compatible, by the locality of compatibility: both are compatible with every chart of the induced atlas, whose domains cover M Chapter2/SubMPolyfold · MPolyfold.subAtlas Chapter2/MPolyfold · MChart.compatible_of_locally Chapter3/GlobalIFTTheorems · MPolyfold.tangentChartsLoc_compatible. The argument never mentions the dimension of either chart. They form an atlas: every $y \in M$ has regular data, and the chart of these data, with the base point a fixed element of M (here $M \neq \emptyset$ is used), contains y Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_chart_source Chapter3/GlobalIFTTheorems · MPolyfold.germAtlasLoc. This atlas is equivalent to the induced atlas of Proposition 2.3.19 (preprint 2.21), since each of its charts is a chart of the induced structure Chapter2/MPolyfold · MAtlas.equiv_iff Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_atlas_equiv.

Tame, and a manifold with corners. The model of each chart $\pi\varphi$ is $(V, N\cap C, N)$ with the identity retraction, a tame retract Chapter3/TangentGerm · MPolyfold.TangentGermChart.model_isTameRetract; so the induced structure is tame, witnessed by the equivalent atlas just built Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_isTame. Each model is relatively open in the partial quadrant $N \cap C$ of the finite-dimensional N Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_isCorner; so the induced structure is that of a manifold with boundary with corners of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_2_3_loc_isManifoldWithCornersLoc. The assembled statement collects these, with the chart property of every $\pi\varphi$ Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular.

As for Theorem 3.2.3 (preprint 3.27), the hypotheses are the regularity conditions at every point; by the observation above, condition (a) there can be replaced by condition (b) plus shrinking. Theorem 3.6.11 (preprint 3.57) (§III.37) applies this version to the solution set, where regular data exist at every zero by §§III.34–III.36 and the dimension $\dim\ker f'(x)$ is locally constant by Step 2 of the theorem above; the fixed-dimension statement for each part S_n of the solution set is obtained from Theorem 3.2.3 (preprint 3.27) itself.

III.37 The implicit function theorems assembled: Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19), Corollary 3.1.27 (preprint 3.24), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57)

The text derives all its implicit function theorems from Theorem 3.6.8 (preprint 3.54): (1) good position at every zero near x, (2) the sub-M-polyfold structure of the solution set near x, through the tangent germ property and Theorem 3.2.3 (preprint 3.27). In the formalization the pieces are: the data of Definition 3.2.1 (preprint 3.25) at the centre, with condition (b) (§III.34); good position at the nearby zeros, under an added hypothesis (§III.33); the re-centring, which turns data at all nearby points into regular data and shows that the dimension is locally constant (§§III.35–III.36); and Theorem 3.2.3 (preprint 3.27), with fixed dimension (§III.24) or with locally constant dimension `Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular` (§III.36). This section assembles them. Throughout, f is a section of a strong bundle $P : Y \to X$ over an M-polyfold, "the charts have tame base retractions" means that the base retraction of every strong bundle chart compatible with the atlas is tame (the formal reading of "tame strong bundle", §III.23), $S = \{x \in X : f(x) = 0\}$ is the global solution set `Chapter3/GlobalIFT · solutionSet`, and good position is Definition 3.6.6 (preprint 3.53) in the chart form `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm` (§III.21).

The data for the global solution set `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8_tangentGermChart_solutionSet`. *If the charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) is in good position, then S has at x_0 data of Definition 3.2.1 (preprint 3.25) of some dimension d with $\pi(O) \subseteq C$.* §III.34 gives data D for $S_{V_x} = \{y \in V_x : f(y) = 0\}$. Restrict its chart to the open set $s = \text{source} \cap V_x \ni x_0$, keeping N, Y, V, δ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink` `Chapter2/MPolyfold · MChart.restrict`: the graph identity holds for the global S because $S \cap s = S_{V_x} \cap \text{source}$. The restricted model $\varphi(s)$ is a tame retract (an open subset of a tame retract, §III.36), and condition (b) passes to it, since $\varphi(s) \subseteq O$ `Chapter2/MPolyfold · MChart.image_subset_model` and the projection is unchanged `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_π`.

Theorem 3.6.8 (2) (preprint 3.54), from the nearby good position `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8_two`. *Let the charts have tame base retractions, $f(x_0) = 0$, (f, x_0) in good position, and let every zero of f in an open $V_{x_0} \ni x_0$ be in good position. Then there is a dimension d and regular data of Definition 3.2.1 (preprint 3.25) of dimension d for S at x_0 (condition (a): δ sc-smooth on V in the set sense; condition (b): $\pi(O) \subseteq C$) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`, on a chart whose source lies in V_{x_0}; and every datum of Definition 3.2.1 (preprint 3.25) for S at a point of S in that source has dimension d.* Only that f is a section is assumed; the regularity of f enters through the good position at x_0 and at the nearby zeros, which carry the sc-Fredholm germ data. *Proof.* Take the data D at x_0 above. Every $y \in S \cap V_{x_0}$ is a zero in good position, so the same theorem at y gives data of Definition 3.2.1 (preprint 3.25) for S at y, of some dimension. These are the hypotheses of the re-centring theorem of §III.36 `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally` for D (which satisfies (b)) and the open set V_{x_0}: it shrinks D to regular data D' with source in V_{x_0}, of the same dimension d, along which every datum has dimension d. This is the text's "proceeding as above we can construct a map σ" at every zero, made into a proof: the regularity condition (a), an added hypothesis of Theorem 3.2.3 (preprint 3.27) for general sets (§III.24), is a theorem for solution sets (items 150–151).

Theorem 3.6.8 (1) and (2) (preprint 3.54) `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8`. *Let f be an sc-Fredholm section, the charts with tame base retractions, $f(x_0) = 0$, and let (f, x_0) be in good position* ***with a filler that satisfies condition (3) of Definition 3.1.4 (preprint 3.4) at the zeros of the filler near*** 0 `Chapter3/`

GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled. *Then there is an open $V_x \ni x_0$ such that (1) (f, y) is in good position at every zero $y \in V_x$, and (2) the conclusions of Theorem 3.6.8 (2) (preprint 3.54) above hold with $V_{x_0} = V_x$.* (1) is §III.33 Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_one; (2) is the previous theorem, fed with (1), with the strengthened good position weakened to good position Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled.isGoodPositionGerm and the section property of the sc-Fredholm section. The filler condition was an **added hypothesis** when this form was proved: Definition 3.1.4 (preprint 3.4) imposes condition (3) at 0 only, and the surjectivity of $f'(y)$ at the nearby zeros is equivalent to it there (§III.33; items 100, 105–106); the kernel half of (1) needs no added hypothesis (§III.32). Since the second addendum the condition is a consequence of good position (§III.45), and Theorem 3.6.8 (preprint 3.54) is proved with the text's hypotheses (§III.43). Part (3) of the text's proof, the graph description, is §III.29 Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj, and part (4), the kernel at the nearby zeros in the chart at x_0, is §III.32 Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_nearby_kernel. The text's neighbourhood V is the V_x of (3), shrunk in (2) to the chart source of the re-centred data.

Theorem 3.1.26 (preprint 3.23), boundary case, from the nearby good position Chapter3/GlobalIFTTheoremsG · theorem_3_1_26_of_nearby. *Let f be an sc-Fredholm section, the charts with tame base retractions, $f(x) = 0$, (f, x) in good position ($f'(x)$ onto with kernel in good position, Definition 3.6.6 (preprint 3.53)), and every zero in an open $V_{x_0} \ni x$ in good position. Then there are an open U with $x \in U \subseteq V_{x_0}$ and a dimension d such that every zero in U is a smooth point, $S(f, U) = \{y \in U : f(y) = 0\}$ is a sub-M-polyfold of X, and its induced M-polyfold structure (Proposition 2.3.19 (preprint 2.21)) Chapter2/SubMPolyfold · subPolyfold is tame and equivalent to that of a d-dimensional manifold with boundary with corners* Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners. *Proof.* Theorem 3.6.8 (2) (preprint 3.54) at x gives regular data D of dimension d with source $U \subseteq V_{x_0}$ and the local constancy of the dimension; let $S' = S(f, U)$. At every $y \in S'$, Theorem 3.6.8 (2) (preprint 3.54) at y, with the open neighbourhood $U \cap V_{x_0}$ in place of V_{x_0}, gives regular data D_y for S with source in $U \cap V_{x_0}$, of a dimension d_y; since $y \in S \cap U$, the local constancy along D gives $d_y = d$. Because the source of D_y lies in U, $S' \cap \text{source} = S \cap \text{source}$, so D_y restricted to its own source is a datum for S' Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink, still regular Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular. So every point of S' has regular data of dimension d, $S' \ni x$ is nonempty, and Theorem 3.2.3 (preprint 3.27) with fixed dimension (§III.24) Chapter3/TangentGerm · theorem_3_2_3_of_isRegular gives the sub-M-polyfold with its tame induced structure equivalent to a d-dimensional manifold with boundary with corners. The zeros are smooth points because an sc-Fredholm section is regularizing Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt. The text's "a uniquely determined structure … the same as the one defined in Proposition 2.21" is the induced structure itself, with the equivalent atlas of the charts $\pi \circ \varphi$; uniqueness is not stated separately.

Theorem 3.1.26 (preprint 3.23) Chapter3/GlobalIFTTheoremsG · theorem_3_1_26. *The same under the added filler hypothesis at x instead of the nearby good position, with in addition: every zero in U is in good position.* Theorem 3.6.8 (1) (preprint 3.54) (§III.33) supplies the open V_x of nearby good position, and the previous theorem applies with $V_{x_0} = V_x$.

Theorem 3.1.22 (preprint 3.19), interior case. The first clause, "$f'(x_0)$ is sc-Fredholm", is Proposition 3.6.5 (preprint 3.52) in the chart at x_0 Chapter3/GlobalIFTTheoremsG · theorem_3_1_22_linearization Chapter3/FredholmGermLinearizationG · proposition_3_6_5_preprint, applied to the sc-Fredholm germ contained in the good position: $Dg(0)$, for a derivative family Dg of the filling germ (§III.21), and $Tf(x_0) = Dg(0)|_{T_0O}$ are Fredholm on level 0 with the same index. The statement inherits the index clause of that proposition, "$= n - N$ for some natural numbers n, N", which carries no information; the index as the $n - N$ of the conjugating basic germ is §III.30 Chapter3/FredholmGermIndexG · proposition_3_6_5_index. *The theorem* Chap–

ter3/GlobalIFTTheoremsG · theorem_3_1_22: *let $d_X \equiv 0$* Chapter2/DegeneracyIndex · MPolyfold.IsWithoutBoundary*, f an sc-Fredholm section, the charts with tame base retractions, $f(x_0) = 0$, and (f, x_0) in good position with the filler condition at the nearby zeros. Then there are an open $U \ni x_0$ and d such that every zero in U is in good position (in particular $f'(y)$ is onto) and a smooth point, and $S(f, U)$ is a sub-M-polyfold whose induced structure is equivalent to a d-dimensional manifold with boundary with corners and has degeneracy index 0 everywhere. Proof.* Theorem 3.1.26 (preprint 3.23) gives everything but the last clause; for that, Lemma 2.4.6 (preprint 2.26), $d_S(y) \le d_X(y)$ Chapter2/SubMPolyfold · lemma_2_4_6, and $d_X = 0$. So the interior case is the boundary case plus Lemma 2.4.6 (preprint 2.26); $d_X \equiv 0$ is used for nothing else. *Two readings and two omissions* (item 154), all resolved in the second addendum. (a) The text's hypothesis is "$f'(x_0)$ surjective"; good position adds condition (2) of Definition 3.6.6 (preprint 3.53), which the text calls an empty condition when $d_X(x) = 0$. Formally condition (2) is stated in the chart for the chart's partial quadrant, so this form takes good position as in the boundary case; that the condition is void at $d_X = 0$ in a tame chart, and Theorem 3.1.22 (preprint 3.19) with "$f'(x_0)$ surjective", are §III.45. The filler condition at the nearby zeros was the added hypothesis of Theorem 3.6.8 (1) (preprint 3.54); the form with the text's hypotheses is §III.43. (b) "The dimension agrees with the Fredholm index": this statement quantifies d existentially; the proof takes $d = \dim\ker D(h+t)(0)$ (§III.34), and $\operatorname{ind} D(h+t)(0) = \dim P - \dim Q = \operatorname{ind} f'(x_0)$ holds for the conjugation data of the proof (§§III.18, III.30); the graph description with the conjugation exposed, and $\dim\ker = \operatorname{ind}$ for the surjective $D(h+t)(0)$, give the dimension as the index in §III.44. (c) "$\ker f'(y) = T_y S(f, U)$": in the chart at x_0, $\ker D(h+t)(z) = N_v$, the tangent space of the graph (§III.32); the chart form of the statement is §III.45, and the identification with $T_y S$ as a subspace of $T_y X$, with the intrinsic linearization, is §III.62.

Theorem 3.6.11 (preprint 3.57), with the text's hypotheses Chapter3/GlobalIFTTheoremsG · theorem_3_6_11. *Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, with (f, x) in good position at every zero x, and let $S \neq \emptyset$. Then S is a sub-M-polyfold of X, its induced structure is tame and equivalent to that of a manifold with boundary with corners of locally constant dimension* Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCornersLoc *(an equivalent atlas of charts on relatively open subsets of partial quadrants of finite-dimensional spaces), and at every $x \in S$ there are regular data of Definition 3.2.1 (preprint 3.25) of a dimension $d(x)$ such that every datum at a point of S in their chart source has dimension $d(x)$. Proof.* At every $x \in S$, Theorem 3.6.8 (2) (preprint 3.54) with $V_{x_0} = X$ — the hypothesis "every zero near x in good position" is part of the global hypothesis — gives the regular data and the local constancy. Theorem 3.2.3 (preprint 3.27) with locally constant dimension (§III.36) Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular then gives the sub-M-polyfold, the tameness and the corner atlas. **No hypothesis is added**: good position at every zero is the text's hypothesis, and it supplies at every point the nearby good position that the re-centring needs, so the gap of Theorem 3.6.8 (1) (preprint 3.54) does not enter (item 156). $S \neq \emptyset$ is needed for the formal charts, whose inverses need a point of S as a value off the chart (for $S = \emptyset$ the statement says nothing). "Tame strong bundle" is read as above. The text states no dimension; the theorem says that the dimension is locally constant, which the text's tangent germ property of a fixed dimension d near each point implies but does not state (item 152).

Corollary 3.1.27 (preprint 3.24) Chapter3/GlobalIFTTheoremsG · corollary_3_1_27 is Theorem 3.6.11 (preprint 3.57) without the last clause: if at every zero $f'(x)$ is onto with kernel in good position, the solution set is a sub-M-polyfold with tame induced structure equivalent to a manifold with boundary with corners of locally constant dimension.

Theorem 3.6.10 (preprint 3.56), $\partial X = \emptyset$, with the text's hypotheses Chapter3/GlobalIFTTheoremsG · theorem_3_6_10. *Under the hypotheses of Theorem 3.6.11 (preprint 3.57) and $d_X \equiv 0$, S is a sub-M-polyfold whose induced structure is equivalent to a manifold with corners of locally constant dimension and has degeneracy index 0 every-*

where: Theorem 3.6.11 (preprint 3.57) and Lemma 2.4.6 (preprint 2.26), as for Theorem 3.1.22 (preprint 3.19). (The statement keeps only these two conclusions; the tameness of the induced structure and the regular data are those of Theorem 3.6.11 (preprint 3.57).) The text's hypothesis "$f'(x)$ surjective at every zero" is read as good position, which is the text's Definition 3.6.6 (preprint 3.53) for $d_X(x) = 0$; that condition (2) is void in a tame chart at $d_X = 0$, and Theorem 3.6.10 (preprint 3.56) with "$f'(x)$ surjective" and the tameness of the induced structure, are §III.45.

The parts of fixed dimension `Chapter3/GlobalIFTTheoremsG · theorem_3_6_11_dim`. Under the hypotheses of Theorem 3.6.11 (preprint 3.57) (without $S \neq \emptyset$), let S_n be the set of points of S at which S has data of Definition 3.2.1 (preprint 3.25) of dimension n `Chapter3/GlobalIFTTheorems · solutionSetDim`, and U_n the union of the sources of the regular data of dimension n along which the dimension is constantly n `Chapter3/GlobalIFTTheorems · solutionSetDimOpen`, an open set `Chapter3/GlobalIFTTheorems · isOpen_solutionSetDimOpen`. *Then S is the union of the S_n, the S_n are pairwise disjoint, $S_n = S \cap U_n$ and $S_n = S \cap F_n$ for a closed F_n (so S_n is open and closed in S), and each nonempty S_n is a sub-M-polyfold whose induced structure is tame and equivalent to an n-dimensional manifold with boundary with corners. Proof.* At every $x \in S$, Theorem 3.6.8 (2) (preprint 3.54) with $V_{x_0} = X$ gives regular data D_x of a dimension $d(x)$ with the local constancy. *The dimension at a point is well defined*: two data at x of dimensions n, m both have the dimension $d(x)$ by the local constancy along D_x, since x lies in its source; so the S_n cover S (by D_x) and are pairwise disjoint. $S_n = S \cap U_n$: for $x \in S_n$, $d(x) = n$ and D_x is one of the data whose sources make up U_n; conversely, if $x \in S$ lies in the source of regular data of dimension n along which the dimension is n, then $d(x) = n$ and D_x witnesses $x \in S_n$. *S_n is closed in S*: $S_n = S \setminus \bigcup_{m \neq n} U_m$, since a point of S lies in exactly one S_m; take F_n the complement of that open union. *Each nonempty S_n*: at $x \in S_n$, restrict D_x (of dimension n) to a datum for S_n — $S_n \cap$ source $= S \cap$ source because every point of S in the source has dimension n — still regular `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`, and apply Theorem 3.2.3 (preprint 3.27) with fixed dimension n `Chapter3/TangentGerm · theorem_3_2_3_of_isRegular`.

Where the formal route departs from the text. Two steps of the text's proof are replaced (items 150, 157): the regularity of the solution germ away from the centre, which the proof of Theorem 3.2.3 (preprint 3.27) asserts for any set and which fails for general sets (§III.24), is proved for solution sets by re-centring with the C^1-property of σ on V' and the level-wise lifts at the nearby zero (§§III.35–III.36); and the relative openness of Σ, which does not hold as we read it for an arbitrary good complement, is obtained by choosing the complement (§III.29). With these, the global theorems 3.56, 3.57 and Corollary 3.1.27 (preprint 3.24) are proved with the text's hypotheses, and the local Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) under the hypothesis that the zeros near the point are in good position, which is Theorem 3.6.8 (1) (preprint 3.54); this section proves that one under the filler condition at the nearby zeros, which Definition 3.1.4 (preprint 3.4) does not impose, and §III.43 with the text's hypotheses. Not stated here: the uniqueness of the manifold structure separately from the equivalence of atlases; the dimension d of these statements is existentially quantified (only as the common dimension of all data of Definition 3.2.1 (preprint 3.25) for S near the point) — its identification with $\dim \ker f'(x) = \operatorname{ind} f'(x)$ is §III.44, and $\ker f'(y) = T_y S$ is §III.45 in the chart and §III.62 in $T_y X$.

III.38 The basic germ criterion: Theorem 3.7.1 (preprint 3.58)

Theorem 3.7.1 (preprint 3.58) (text): let U be a relatively open neighbourhood of 0 in a partial quadrant $C \subseteq E$ and $f : U \to F$ sc-smooth, with (1) $Df(x)$ sc-Fredholm at the smooth points, of constant index; (2) an sc-splitting $E = B \oplus X$, B finite-dimensional containing $\ker Df(0)$, with a compactness property: for $b_j \to b$ in $B \cap U$ small and (η_j) bounded in X_m with $Df(b_j)\eta_j = y_j + z_j$, $y_j \to 0$ in F_m, (z_j)

bounded in F_{m+1}, (η_j) has a convergent subsequence in X_m; (3) for every m and $\varepsilon > 0$, $|[D_2 f(b,0) - D_2 f(b,x)]h|_m \le \varepsilon |h|_m$ for b small and $x \in X_{m+1}$ small on level m. Then $f - f(0)$ is conjugated near 0 to a basic germ. The proof: $Y = Df(0)X$, $F = A \oplus Y$, P the projection onto Y; Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) and an index argument make $L(b) = P \circ Df(b)|_X$ sc-isomorphisms with uniform lower bounds; Lemma 3.7.4 (preprint 3.61) makes $\Phi((b,x),(c,y)) = ((b,x),(c,L(b)^{-1}y))$ a strong bundle isomorphism; and $H(b,x) = x - L(b)^{-1}Ph(b,x)$, $h = f - f(0)$, is a contraction by (3).

The coordinates. The formalization works in the coordinates of the text's proof: $E = \mathbb{R}^n \oplus X$ with $\mathbb{R}^n$ a finite-dimensional normed space P (constant sc-structure), $C = \widetilde{C} \oplus X$ with $\widetilde{C}$ a partial quadrant of P, $F = \mathbb{R}^N \oplus Y$ with $\mathbb{R}^N = Q$ finite-dimensional, and "$P\circ$" the Y-component. Condition (3) becomes `Chapter3/BasicGermCriterionProof · DerivCondition`: for every m and $\varepsilon > 0$ there is $r > 0$ with $|Df(b,0)(0,h) - Df(b,x)(0,h)|_m \le \varepsilon |h|_m$ for $h \in X_m$, $b \in \widetilde{C}$, $|b| < r$, $x \in X_{m+1}$, $|x|_m < r$, $(b,0)$ and (b,x) in U_{m+1} (the level operators of Proposition 1.2.1 (preprint 1.16)). Condition (2) becomes `Chapter3/BasicGermCriterionIndex · CompactnessCondition`: there is $r > 0$ such that for every m, every $b_j \to b_0$ in $\widetilde{C}$ with $(b_j,0) \in U$ and $|b_0| < r$, every bounded $(\eta_j) \subseteq X_m$ with $Df(b_j,0)(0,\eta_j) = y_j + \iota z_j$, $y_j \to 0$ in F_m and (z_j) bounded in F_{m+1}, (η_j) has a subsequence converging in X_m. (The smallness of b is read as uniform in m, as the order of the text's quantifiers suggests; Lemma 3.7.3 (preprint 3.60) needs the property on every level on one neighbourhood.)

Theorem 3.7.1 (preprint 3.58) as formalized `Chapter3/BasicGermCriterionIndex · theorem_3_7_1`. Let U be relatively open in $\widetilde{C} \oplus X$ with $0 \in U$ and $f : U \to Q \oplus Y$ sc-smooth such that (1) for every $b \in \widetilde{C}$ with $(b,0) \in U$ the linearization $Df(b,0)$, an sc-operator `Chapter3/BasicGermCriterionIndex · derivParamOp`, is sc-Fredholm with the sc-index of $Df(0)$; (2) $h \mapsto (Df(0)(0,h))_2$ is bijective $X_0 \to Y_0$, and the compactness condition holds; (3) the derivative condition holds. Then there are an open $V_o \ni 0$ in P and sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_2$ for $b \in V_o \cap \widetilde{C}$, such that the push-forward $k(b,x) = ((f - f(0))_1(b,x), L(b)^{-1}(f - f(0))_2(b,x))$ is a basic germ on $U \cap (V_o \oplus X)$ with contraction map $H(b,x) = x - L(b)^{-1}(f(b,x) - f(0))_2$, Φ is a strong bundle isomorphism covering the identity, and $f - f(0)$ is conjugated to a basic germ (Definition 3.1.11 (preprint 3.11)). The hypotheses differ from the text's in three places: (1) is asked only at the points $(b,0)$, $b \in \widetilde{C}$, which is all the proof uses; "constant index" is equality with the index at 0; and the text's "$B \supseteq \ker Df(0)$, $Y = Df(0)X$" is replaced by its consequence, the bijectivity of $P \circ Df(0)|_X$ on level 0. The reduction of a general sc-splitting $E = B \oplus X$, $F = A \oplus Y$ to these coordinates is §III.63.

The proof has two halves, as in the text.

First half: Theorem 3.7.1 (preprint 3.58) from the family of sc-isomorphisms.

Theorem 3.7.1 (preprint 3.58), given the family `Chapter3/BasicGermCriterionProof · theorem_3_7_1_of_family`. In addition to the coordinates, f sc-smooth on U and (3), suppose: V_o open with $0 \in V_o$ and $(b,0) \in U$ for $b \in V_o \cap \widetilde{C}$; sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_2$ on level 0 for $b \in V_o \cap \widetilde{C}$; and for every m a constant $c_m > 0$ with $c_m |h|_m \le |L(b)h|_m$ for $b \in V_o \cap \widetilde{C}$, $h \in X_m$. Then, with $U' = U \cap (V_o \oplus X)$, the conclusions above hold.

Step 1: $(b,h) \mapsto L(b)h$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus X$ `Chapter3/BasicGermCriterionProof · famMap_isScSmooth_of_deriv` (the text: "since B is a finite-dimensional sc-smooth space, the map $(b,x) \mapsto P \circ Df(b)x$ is sc-smooth"). It is $\mathrm{pr}_Y \circ Tf \circ J$ with the sc-operator $J(b,h) = ((b,0),(0,h))$ into the tangent space `Chapter3/BasicGermCriterionProof · tangentInjOp`; J maps into the tangent set because $(b,0) \in U$ lies on every level; and the tangent map of the sc-smooth f is sc-smooth.

Step 2: $(b,y) \mapsto L(b)^{-1}y$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus Y$ `Chapter3/BasicGermCriterionProof · famInv_isScSmooth_`

of. This is Lemma 3.7.4 (preprint 3.61) (§III.25) Chapter3/InverseFamily · lemma_3_7_4 for the germ $\{V_o \cap \widetilde{C}\}$ with the uniform lower bounds, translated between maps on sets and germs on the special germ of a set: an sc^k-germ on $\{U\}$ is an sc^k-map on U Chapter3/BasicGermCriterionProof · IsSckGerm.isSck_ofSet Chapter3/BasicGermCriterionProof · IsScSmoothGerm.isScSmooth_ofSet, by induction on k: the sc^0 case is a matter of definitions Chapter3/BasicGermCriterionProof · IsSc0Germ.isSc0_ofSet; for the step, the tangent germ of $\{U\}$ is the germ of the tangent set, and the derivative family of the germ agrees with the derivative of the map on the tangent set; the converse is `IsSck.isSckGerm` (§III.1).

Step 3: the strong bundle isomorphism Φ. For a family $M(b) : Y \to X$ of sc-isomorphisms put $\Phi_M((b,z),(c,y)) = ((b,z),(c,M(b)y))$ Chapter3/BasicGermCriterionProof · fiberIsoMap. If $(b,y) \mapsto M(b)y$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus Y$ and the parameters of U' lie in $V_o \cap \widetilde{C}$, Φ_M is a strong bundle map $U' \lhd (Q \oplus Y) \to U' \lhd (Q \oplus X)$ covering the identity Chapter3/BasicGermCriterionProof · fiberIsoMap_isStrongBundleMap: it is linear on the fibres; it preserves the filtration because $M(b)$ maps Y_k to X_k; on $[0]$ it is sc-smooth as the pairing of the projection with $(c, M(b)y)$, the latter the composite of the sc-operator $((b,z),(c,y)) \mapsto (b,y)$ Chapter3/BasicGermCriterionProof · paramFiberOp with $(b,y) \mapsto M(b)y$ Chapter3/BasicGermCriterionProof · fiberIsoMap_isScSmooth; on $[1]$, $\Phi_M[1]$ is Φ_{M^1} for the shifted family $M^1(b) = M(b) : Y^1 \to X^1$ Chapter3/BasicGermCriterionProof · shiftFam Chapter3/BasicGermCriterionProof · shiftFiber_fiberIsoMap, and $(b,y) \mapsto M^1(b)y$ is sc-smooth by Proposition 1.2.2 (preprint 1.17) Chapter3/BasicGermCriterionProof · famMap_shiftFam_isScSmooth — the text's "if we raise the index by one, the map is also sc-smooth by Proposition 1.17". With $M = L^{-1}$ (Step 2) and $M = L$ (Step 1) the two maps are inverse to each other, so $\Phi = \Phi_{L^{-1}}$ is a strong bundle isomorphism Chapter3/BasicGermCriterionProof · fiberIsoMap_isStrongBundleIso.

Step 4: the push-forward and the contraction map. With $h = f - f(0)$ Chapter3/BasicGermCriterionProof · shiftedGerm, $\Phi_* h = k$, $k(b,x) = (h_1(b,x), L(b)^{-1}h_2(b,x))$ Chapter3/BasicGermCriterionProof · pushGerm358, and the X-component of k is $x - H(b,x)$ with $H(b,x) = x - L(b)^{-1}h_2(b,x)$ Chapter3/BasicGermCriterionProof · contrMap358 Chapter3/BasicGermCriterionProof · pushGerm358_snd. Both are sc-smooth on U': sums, pairings and the composite of $(b,y) \mapsto L(b)^{-1}y$ with $(b,x) \mapsto (b, h_2(b,x))$. On level m, $H_m(b,w) = w - L(b)^{-1}_m((f_m(b,w))_2 - (f_m(0))_2)$ Chapter3/BasicGermCriterionProof · levelMap_contrMap358, and $|L(b)^{-1}_m| \le c_m^{-1}$ Chapter3/BasicGermCriterionProof · norm_symm_le.

Step 5: the contraction estimate Chapter3/BasicGermCriterionProof · contrMap358_contraction. For every m and $\varepsilon > 0$ there is $r > 0$ such that $(a,w) \in U'_m$ and $|H(a,w) - H(a,w')|_m \le \varepsilon |w - w'|_m$ for $a \in \widetilde{C}$, $|a| < r$, $w, w' \in X_m$ small on level m. Choose r below the radius r_3 of (3) for εc_m, below the radius of a ball around 0 in V_o, and so small that $(a,z) \in U_m$ whenever $|a| < r$ and $|z|_m < r$ ($U = V \cap C$ with V open; the level-m norm of z controls its level-0 norm through the norm of the inclusion) Chapter3/BasicGermCriterionProof · mem_lvlSet_of_small. First let $w_1, w_1' \in X_{m+1}$ with $|w_1|_m, |w_1'|_m < r$. On the convex set $S = \{z \in X_{m+1} : |z|_m < r\}$ the map $g(z) = f_m(a, \iota z)$ has the derivative $Df(a,z)(0,\iota\,\cdot)$ (Proposition 1.2.1 (preprint 1.16) on level m) Chapter1/Levels · IsSc1With.hasFDerivWithinAt_level, and by (3) $|Df(a,z)(0,v) - Df(a,0)(0,v)|_m \le \varepsilon c_m |v|_m$. **The mean value inequality with the derivative bound in the weaker norm** Chapter3/BasicGermCriterionProof · norm_sub_sub_le_of_hasFDerivWithinAt_segment — the one-dimensional mean value inequality for $t \mapsto g(w_1' + t(w_1 - w_1')) - T\iota(w_1' + t(w_1 - w_1'))$ on $[0,1]$, which measures the increment in the norm of X_m rather than of X_{m+1}, as condition (3) requires — gives

$$|g(w_1) - g(w_1') - T\iota(w_1 - w_1')|_m \le \varepsilon c_m |\iota(w_1 - w_1')|_m, \qquad T = Df(a,0)(0,\cdot).$$

The Y-component of T is $L(a)_m$ Chapter3/BasicGermCriterionProof · snd_levelDeriv_eq_L (the level-m operator of Proposition 1.2.1 (preprint 1.16) restricts the level-0 one). Then $H(a,w_1) - H(a,w_1') = L(a)^{-1}[L(a)(w_1 - w_1') - (f(a,w_1) - f(a,w_1'))_2]$, whose norm is at most $c_m^{-1}\varepsilon c_m|w_1 - w_1'|_m$. This is the text's integral $\int_0^1 D_2 f(b, tx + (1-t)x')(x - x')\,dt$, made precise. For general $w, w' \in X_m$: approximate them by sequences from X_{m+1} (dense in X_m), small on level m for large indices, and pass to the limit with the continuity of H_m on U_m' — the text's "by the density of X_{m+1} in X_m".

Step 6: the basic germ and the conjugation. k is a basic germ on the germ $\{U'\}$ with contraction map H (Definition 3.1.7 (preprint 3.7) as formalized Chapter3/LocalCompactness · IsBasicGerm): it is an sc-smooth germ (Step 4), $k(0) = 0$, its X-component is $x - H(b,x)$, and H is an sc^0-contraction germ — sc^0, $H(0) = 0$ since $L(0)^{-1}0 = 0$, and the contraction estimate on every level with every ε (Step 5). The conjugation identity $(\Phi(u, h(u)))_2 = k(u)$ holds on all of U', because Φ covers the identity; with Φ from Step 3 these are the data of Definition 3.1.11 (preprint 3.11) for $h = f - f(0)$.

Second half: the family from conditions (1) and (2).

The family. At the smooth point $(b,0)$ the linearization $Df(b,0)$ is an sc-operator Chapter3/BasicGermCriterionIndex · derivParamOp Chapter1/Properties · IsSc1With.scOperator_deriv whose level maps are the level operators of Proposition 1.2.1 (preprint 1.16) Chapter3/BasicGermCriterionIndex · derivParamOp_lvl; $L(b) = \mathrm{pr}_Y \circ Df(b,0) \circ \mathrm{inr}_X$ is an sc-operator $X \to Y$ (set to 0 where $(b,0) \notin U$) Chapter3/BasicGermCriterionIndex · derivFamOp; and $(b,h) \mapsto L(b)h$ is jointly continuous on $O \times X_m$ for parameters with $(b,0) \in U$, by the continuity of the level operators (Proposition 1.2.1 (2) (preprint 1.16)) Chapter3/BasicGermCriterionIndex · continuousOn_levelDeriv_param Chapter3/BasicGermCriterionIndex · continuousOn_derivFamOp.

The compactness property of the family Chapter3/BasicGermCriterionIndex · familyCompactness_derivFamOp. Let O be a set of parameters in $\widetilde{C}$ with $(b,0) \in U$ and $|b| < r$ (r from (2)), $b_j \to b_0$ in O, (h_j) bounded in X_m, $L(b_j)h_j \to 0$. Write $Df(b_j,0)(0,h_j) = (0, L(b_j)h_j) + ((Df(b_j,0)(0,h_j))_Q, 0)$. The first summand tends to 0 in F_m. The second lies in $Q \oplus 0$, where all levels coincide, and is bounded: a jointly continuous family of operators is uniformly bounded near a parameter Chapter3/BasicGermCriterionIndex · exists_bound_of_continuousOn_clm (continuity at $(b_0,0)$, then scaling), so $|Df(b_j,0)(0,\cdot)|$ is bounded for j large. Condition (2) applied to the tails of the sequences gives a convergent subsequence: this is the compactness property of the abstract Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) (§III.25) Chapter3/BasicGermCriterion · FamilyCompactness. The text's "$z_j = Df(b_j)h_j \in A$ is bounded on every level since A is a smooth subspace" omits the uniform operator bound.

Lemma 3.7.2 (preprint 3.59). Let $V' = \{b : (b,0) \in V\}$ ($U = V \cap C$) and $O = V' \cap B(0,r) \cap \widetilde{C}$, which contains 0. $L(0)$ is injective on X_0 by (2), so Lemma 3.7.2 (preprint 3.59) Chapter3/BasicGermCriterion · lemma_3_7_2 gives $r_1 > 0$ with $L(b)$ injective on X_0 for $b \in O$, $|b| < r_1$, hence on every level (the inclusions are injective and commute with $L(b)$) Chapter3/BasicGermCriterionIndex · ScOperator.injective_lvl.

Lemma 3.7.3 (preprint 3.60). Take a closed ball $\bar{B}(0, r_1')$ inside $V' \cap B(0,r) \cap B(0,r_1)$ and $K = \bar{B}(0,r_1') \cap \widetilde{C}$, compact (since $\widetilde{C}$ is closed) and inside O. On K every $L(b)_m$ is injective, the family is jointly continuous and has the compactness property (restricted from O Chapter3/BasicGermCriterionIndex · FamilyCompactness.mono), so Lemma 3.7.3 (preprint 3.60) Chapter3/BasicGermCriterion · lemma_3_7_3 gives $c_m > 0$ with $c_m|h|_m \le |L(b)h|_m$ for $b \in K$.

The index argument. The text: "$P \circ Df(b)|_X$ is an injective sc-Fredholm operator of index 0, hence an sc-isomorphism". The formal proof uses a block computation instead. **The index of a block operator** Chapter3/BasicGermCriterionIndex · fredholmIndex_block: for a Fredholm $T : P \oplus X \to Q \oplus Y$ with P, Q

finite-dimensional and the block $L = \mathrm{pr}_Y \circ T \circ \mathrm{inr}_X$ of finite index, $\operatorname{ind} T = \dim P + \operatorname{ind} L - \dim Q$. Indeed $T - \mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ is $(p,x) \mapsto T(p,0) + ((T(0,x))_Q, 0)$, of finite rank, hence compact, so $\mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ is Fredholm with the index of T `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator` `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`; and the indices of pr_X and inr_Y are $\dim P$ and $-\dim Q$ `Chapter3/BasicGermCriterionIndex · hasFiniteIndex_snd_and_linIndex` `Chapter3/BasicGermCriterionIndex · hasFiniteIndex_inr_and_linIndex`, which add under composition `Chapter1/FredholmIndex · linIndex_comp`. At $b = 0$, $L(0)$ is bijective on level 0, of index 0, so $\operatorname{ind} Df(0) = \dim P - \dim Q$ on level 0, which is the sc-index (an sc-Fredholm operator has the same index on every level) `Chapter1/FredholmIndex · IsScFredholm.fredholmIndex_lvl`. **Surjectivity from the index** `Chapter3/BasicGermCriterionIndex · surjective_of_fredholmIndex_block`: if $\operatorname{ind} T = \dim P - \dim Q$ and the block L is injective, then L has finite index — its cokernel is a quotient of the cokernel of the Fredholm operator $\mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ — and index 0 by the block formula, so it is surjective. For $b \in K$ and every m, $Df(b,0)_m$ is Fredholm of index $\operatorname{ind} Df(0) = \dim P - \dim Q$ by (1) and the level-wise index, and $L(b)_m$ is injective; so $L(b)_m$ is bijective, a topological isomorphism by the open mapping theorem, and the $L(b)$, $b \in K$, are sc-isomorphisms. The sc-Fredholm property of the projection P and the composition rule for sc-Fredholm operators are not needed.

Conclusion. Extend $b \mapsto L(b)$ outside K by $L(0)$, put $V_o = B(0, r_1')$ (so $V_o \cap \widetilde{C} \subseteq K$), and apply the first half with the lower bounds of Lemma 3.7.3 (preprint 3.60). The neighbourhood replacement $U' = U \cap (V_o \oplus X)$ is the text's "replacing, if necessary, U by a smaller neighbourhood".

III.39 What the addendum taught us

1. **The global implicit function theorems hold with the text's hypotheses.** Theorem 3.6.11 (preprint 3.57), Corollary 3.1.27 (preprint 3.24) and Theorem 3.6.10 (preprint 3.56) are proved for an sc-Fredholm section of a strong bundle whose charts have tame base retractions, with good position at every zero `Chapter3/GlobalIFTTheoremsG · theorem_3_6_11` `Chapter3/GlobalIFTTheoremsG · corollary_3_1_27` `Chapter3/GlobalIFTTheoremsG · theorem_3_6_10` (§III.37). Theorem 3.6.10 (preprint 3.56) reads "$f'(x)$ surjective" as good position, which the text's Definition 3.6.6 (preprint 3.53) makes the same thing at points with $d_X(x) = 0$; that the kernel condition is void in the chart form was not formalized when this was written; it is now (§III.45).
2. **Two steps of the text's proof had to be replaced, and both repairs are specific to solution sets.** The regularity of the solution germ away from the centre, which the proof of Theorem 3.2.3 (preprint 3.27) asserts for any set and which fails in general (§III.24), follows for two graph descriptions of one set from the C^1-property of the first and the level-wise lifts of the second, by the re-centring lemma `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels` (§III.35). The relative openness of Σ, which does not hold as we read it for an arbitrary good complement, holds for a complement whose projection maps the partial quadrant into itself, and the proof of Theorem 3.6.8 (preprint 3.54) can produce such a complement from the start `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto` (§III.29). Read off the statement of the regularity theorem `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`, the first repair works for every set with the tangent germ property whose data satisfy the second regularity condition (§III.36).
3. **The proof of the local Theorem 3.6.8 (1) (preprint 3.54) has a gap, and it is the same gap as in the book's arXiv version.** At a nearby zero, Corollary 3.5.3 (preprint 3.41) gives the surjectivity of the filled linearization; the surjectivity of $f'(y)$ needs condition (3) of Definition 3.1.4 (preprint 3.4) at that zero, which the definition imposes at the centre only `Chapter3/GlobalIFTNearbySurj · exists_range_`

scDeriv_eq_ρ_of_normal_injective. The local theorems were therefore first proved under that condition at the nearby zeros Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_one (§III.33); the kernel half needs nothing (§III.32). The global theorems do not need it, because their hypothesis already gives good position at every zero. The second addendum proves the surjectivity with Definition 3.1.4 (preprint 3.4) as stated (§§III.41–III.43).

4. **Definition 3.6.6 (preprint 3.53) presupposes a chart independence the text does not prove.** It holds at zeros, because only the fibre part of a chart transition survives there Chapter3/LinearizationCharts · linAt_chart_change (§III.31), and it is what lets the good position found in the chart at x_0 be transported to the chart centred at a nearby zero.
5. **Three silent steps are theorems of their own**: tameness is carried by sc-diffeomorphisms and by open subsets Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image (§III.27); the kernel of the linearization of a basic germ plus an sc^+-germ consists of smooth points Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth (§III.34); and the index raise of a strong bundle isomorphism is one Chapter3/BasicClassComposition · IsStrongBundleIso.shift (§III.28).
6. **The dimension of the solution set is locally constant, not constant**, and the dimension count that proves the re-centring is also what shows it (Sard's lemma in fixed dimension Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty, §III.35). Theorem 3.2.3 (preprint 3.27) is therefore also proved with a dimension that may vary Chapter3/GlobalIFTTheorems · theorem_3_2_3_loc_of_isRegular (§III.36), and the solution set of Theorem 3.6.11 (preprint 3.57) splits into open and closed parts of fixed dimension Chapter3/GlobalIFTTheoremsG · theorem_3_6_11_dim.
7. **Proposition 3.6.5's (preprint 3.52) index needs the conjugation as data.** With the witnesses of the conjugation named, the index of $Tf(x_0)$ is the $n - N$ of every conjugating basic germ, and that number does not depend on the conjugation Chapter3/FredholmGermIndexG · proposition_3_6_5_index Chapter3/FredholmGermIndexG · ConjugationData.index_eq (§III.30).
8. **The quadrant inverse function theorem is not the right tool away from the centre**: it needs the degeneracy index to be preserved, which for the solution set is unknown; Lemma 3.8.21's (preprint 3.77) bootstrapping with the hypotheses its proof uses suffices Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inv (§III.35).
9. **The basic germ criterion is proved in the coordinates of its proof**, with condition (1) needed only along the finite-dimensional directions and the text's index argument replaced by a block index formula Chapter3/BasicGermCriterionIndex · fredholmIndex_block Chapter3/BasicGermCriterionIndex · theorem_3_7_1 (§III.38); the form as printed reduces to it (§III.63).
10. **What was left of Chapter 3** after this addendum: the surjectivity in Theorem 3.6.8 (1) (preprint 3.54) with Definition 3.1.4 (preprint 3.4) as stated; in Theorem 3.1.22 (preprint 3.19), the dimension as the Fredholm index and $\ker f'(y) = T_y S$; Proposition 3.6.5 (preprint 3.52) as an sc-Fredholm operator on $T_x X$; Theorem 3.7.1 (preprint 3.58) for a general sc-splitting; Theorem 3.1.21 (preprint 3.18); Lemma 3.8.13 (preprint 3.70) for a general presentation of the partial quadrant; and Theorem 3.2.3 (preprint 3.27) for sets whose data violate the second regularity condition. The second addendum (§§III.40–III.46) settles the first three items, the third in charts, and Theorem 3.1.21 (preprint 3.18). The rest is closed since: §§III.56, III.57, III.61, III.62, III.63.

III.40 Theorem 3.1.21 (preprint 3.18): parametrized perturbations

The text states Theorem 3.1.21 (preprint 3.18) after a five-line argument: the section $\tilde{f}(r,x) = ((r,x), f(x))$ of the pull-back bundle $\pi^*P \to \mathbb{R}^n \times X$, $\pi(r,x) = x$, "is a sc-Fredholm section as is readily verified", $\tilde{s}(r,x) = ((r,x), \sum_i r_i s_i(x))$ is an sc$^+$-section, and Theorem 3.1.18 (preprint 3.15) adds them. The formalization follows this route. What it has to build first is everything the five lines take for granted: $\mathbb{R}^n$ as an M-polyfold with enough charts, the pull-back bundle as a strong bundle, the sc-calculus for a scalar factor, the product chart with its filler, and the extension of a basic germ by n inert parameters (Findings 160–167). Tameness is not used anywhere, as for Theorem 3.1.18 (preprint 3.15).

$\mathbb{R}^n$ and $\mathbb{R}^n \times X$. $\mathbb{R}^n$ carries the constant sc-structure `Chapter3/ParamPerturbation · rnSc` and the local model $(\mathbb{R}^n, \mathbb{R}^n, \mathbb{R}^n)$ `Chapter3/ParamPerturbation · rnModel`. For every a the translation $r \mapsto r - a$ is a chart `Chapter3/ParamPerturbation · rnChart`; two translation charts are compatible, because their transition maps are translations and a C^∞ map between finite-dimensional spaces is sc-smooth for the constant structures `Chapter3/ParamPerturbation · isScSmoothRetractMap_const_of_contDiff Chapter3/ParamPerturbation · rnChart_compatible`. The M-polyfold $\mathbb{R}^n$ has the identity chart as its atlas `Chapter3/ParamPerturbation · rnPolyfold`, and every translation chart is a chart of it `Chapter3/ParamPerturbation · isChart_rnChart`. These charts are needed: Definition 3.1.11 (preprint 3.11) asks for a chart *centred* at the smooth point (r_0, x), and the product of the identity of $\mathbb{R}^n$ with a chart of X is centred only at $r_0 = 0$. $\mathbb{R}^n$ is tame `Chapter3/ParamPerturbation · rnPolyfold_isTame` and every point lies on every level `Chapter3/ParamPerturbation · rnPolyfold_levelSet`; so (r,x) is on level m of the product iff x is `Chapter3/ParamPerturbation · prod_rn_mem_levelSet_iff`, smooth iff x is `Chapter3/ParamPerturbation · prod_rn_mem_smoothSet_iff`, and $\mathbb{R}^n \times X$ is tame when X is `Chapter3/ParamPerturbation · prod_rn_isTame` — which the text needs, since Definition 3.1.16 (preprint 3.14) is stated for tame bundles.

Two rules of the sc-calculus. Scalar multiplication $\mathbb{R} \oplus F \to F$ is continuous and bilinear on every level, hence sc-smooth by Proposition 1.2.4 (preprint 1.19) `Chapter3/ParamPerturbation · isScSmooth_smul`; composed with a pairing, $x \mapsto a(x)v(x)$ is sc-smooth for sc-smooth $a : U \to \mathbb{R}$ and $v : U \to F$ `Chapter3/ParamPerturbation · IsScSmooth.smul`; finite sums of sc-smooth maps are sc-smooth `Chapter3/ParamPerturbation · IsScSmooth.finsetSum`. The coordinate $r \mapsto r_i$ is an sc-operator `Chapter3/ParamPerturbation · rnCoordOp`. Chapter 1 states neither rule.

The pull-back bundle. Proposition 2.6.8 (preprint 2.54) is formalized as an existence statement `Chapter2/StrongBundlePullback · proposition_2_6_8`; here it is made a structure, the strong bundle π^*P with the atlas of all pulled-back charts `Chapter3/ParamPerturbation · StrongBundle.pullback`, whose total space is paracompact by Theorem 2.3.10 (preprint 2.16) `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace`. A pulled-back chart is made from a chart ψ of the base Z around z and a strong bundle chart c of P around $\pi(z)$, with π mapping the domain of ψ into that of c `Chapter3/ParamPerturbation · exists_pullbackChartData`; such a chart is a chart of π^*P as soon as c is merely compatible with the atlas of P `Chapter3/ParamPerturbation · PullbackChartData.isChart_toChart`. A map $\tau : Z \to Y$ over π is a section $z \mapsto (z, \tau(z))$ of π^*Y `Chapter3/ParamPerturbation · pullbackLift Chapter3/ParamPerturbation · isSection_pullbackLift`, whose value in a pulled-back chart is $(\psi(z), \mathrm{pr}_F \Phi(\tau(z)))$ `Chapter3/ParamPerturbation · PullbackChartData.toChart_pullbackLift`. For $\tau = \sigma \circ \pi$ the representative is $w \mapsto (w, \hat{\sigma}(g(w)))$ with $g = \varphi \circ \pi \circ \psi^{-1}$, which maps the retract of ψ into that of φ `Chapter3/ParamPerturbation · PullbackChartData.g_mem_0 Chapter3/ParamPerturbation · PullbackChartData.toChart_pullbackLift_comp`; so the pull-back of an sc-smooth section along an sc-smooth map is sc-smooth `Chapter3/ParamPerturbation · StrongBundle.IsScSmoothSection.pullback` (Finding 167).

The three sections. $F(r,x) = f(x) + \sum_i r_i s_i(x)$ is computed in the fibre $P^{-1}(x)$, from the

fibre combination $\sum_i r_i s_i(x)$ Chapter3/ParamPerturbation · fiberComb, and read in the fibre of π^*Y over (r,x) Chapter3/ParamPerturbation · paramSection; $\tilde{f}$ Chapter3/ParamPerturbation · paramLift and $\tilde{s}$ Chapter3/ParamPerturbation · paramComb are the text's sections, and $F = \tilde{f} + \tilde{s}$ in the fibres of π^*Y Chapter3/ParamPerturbation · paramSection_eq_sectionAdd.

$\tilde{s}$ is an sc⁺-section Chapter3/ParamPerturbation · StrongBundle.isScPlusSection_paramComb. The atlas of π^*P contains the charts pulled back through *every* chart ψ of $\mathbb{R}^n \times X$, not only product charts, so the check must work for an arbitrary ψ. In a pulled-back chart, by the linearity of Φ on the fibre over $x = \pi(z)$, $\tilde{s}(z)$ has the value $(\psi(z), \sum_i r_i \hat{s}_i(x))$ Chapter3/ParamPerturbation · PullbackChartData.toChart_paramComb, so the principal part is $w \mapsto \sum_i \lambda_i(w)\hat{s}_i(g(w))$ with $\lambda(w) = (\psi^{-1}(w))_1$. The map λ is sc-smooth on the retract of ψ: it is the projection $\mathbb{R}^n \times X \to \mathbb{R}^n$ (an sc-smooth map of M-polyfolds) read in ψ and in the identity chart of $\mathbb{R}^n$ Chapter3/ParamPerturbation · isScSmooth_chart_coord. Each $\hat{s}_i \circ g$ is sc-smooth into F^1, s_i being sc⁺. The product rule and the sum rule conclude.

The product chart and its filler. For a strong bundle chart c of P and $r_0 \in \mathbb{R}^n$, the product chart Chapter3/ParamPerturbation · PullbackChartData.ofProd is $\psi = (r \mapsto r - r_0) \times \varphi$ on $\mathbb{R}^n \times V$, with the retraction $1 \times r$ of $\mathbb{R}^n \times U$ onto $\mathbb{R}^n \times O$; there $g = \varphi \circ \pi \circ \psi^{-1}$ is the projection $(a,u) \mapsto u$. *Filler conditions.* If g fills $\hat{f}$ in c, then $(a,u) \mapsto g(u)$ fills $(a,u) \mapsto \hat{f}(u)$ in the product chart Chapter3/ParamPerturbation · PullbackChartData.ofProd_isFillerOf. Conditions (1) and (2) are those of g, because the fibre projection of the product retraction at (a,u) is $\rho(r(u))$ — which uses $\varphi(\varphi^{-1}(r(u))) = r(u)$ on O. Condition (3): $D(1 \times r)(0) = 1 \times Dr(0)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_scDeriv_r, so its kernel is $0 \times \ker Dr(0)$; the fibre projection at 0 is $\rho(0)$ (from $r(0) = 0$); and $[1 - \rho(r(u))]g(u)$ does not depend on a, so its derivative at 0 on $0 \times \ker Dr(0)$ is that of g. *For a filling germ g* on G the same holds on the product germ $\mathbb{R}^n \times G$, with levels $\mathbb{R}^n \times G_m$, unbounded in the parameter direction, which Definition 3.1.1 (preprint 3.1) allows Chapter3/GermCalculusParam · ScNbhdGerm.univProd: $(a,u) \mapsto g(u)$ is a germ on it with the derivative family $Dg(u) \circ \mathrm{pr}_2$ Chapter3/GermCalculusParam · IsSc1GermWith.univProd, and the fill defect in the product chart is the pull-back along pr_2 of that of g, so condition (3) is the kernel identification just made, without a uniqueness argument Chapter3/GermCalculusParam · PullbackChartData.ofProd_isFillingGerm. *Principal parts* ($a = r - r_0$ the chart coordinate): $\tilde{s}$ has $(a,u) \mapsto \sum_i (a + r_0)_i \hat{s}_i(u)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_principal_paramComb and F has $(a,u) \mapsto \hat{f}(u) + \sum_i (a + r_0)_i \hat{s}_i(u)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_principal_paramSection, the text's $(r,u) \mapsto \hat{f}(u) + \sum_i r_i \hat{s}_i(u)$.

Inert parameters. Let Λ be finite-dimensional. The product $1_\Lambda \times \Phi$, $((a,u),k) \mapsto ((a,\varphi(u)),\Gamma(u,k))$, of a map Φ of $U \lhd F$ with the identity of Λ Chapter3/ParamPerturbation · paramMap is a strong bundle map when Φ is: on $[0]$ and $[1]$ it is $1 \times \Phi$ up to the reassociations of the factors, which are sc-operators Chapter3/ParamPerturbation · ScBanach.assocOp Chapter3/ParamPerturbation · ScBanach.assocInvOp Chapter3/ParamPerturbation · IsStrongBundleMap.paramMap; and a strong bundle isomorphism, with inverse $1_\Lambda \times \Phi'$, when Φ is Chapter3/ParamPerturbation · IsStrongBundleIso.paramMap. The Λ-extension of a germ of neighbourhoods is $U''_m = \{((a,\lambda),w) \mid (a,w) \in U_m\}$ Chapter3/ParamPerturbation · ScNbhdGerm.paramExtend, unbounded in λ, which Definition 3.1.1 (preprint 3.1) allows. *The Λ-extension of a basic germ is basic* Chapter3/ParamPerturbation · IsBasicGerm.paramExtend: if $h(a,w) = (h_1(a,w), w - B(a,w))$ is basic on $\widetilde{C} \oplus W$, then $h'((a,\lambda),w) = h(a,w)$ is basic on $(\widetilde{C} \times \Lambda) \oplus W$, with $B'((a,\lambda),w) = B(a,w)$: it is the composite of h with the projection forgetting λ Chapter3/ParamPerturbation · paramForgetOp, and the contraction estimate of Definition 3.3.1 (preprint 3.28) holds with the same radius since $|a| \le |(a,\lambda)|$. *The Λ-extension of a conjugation* Chapter3/ParamPerturbation · IsConjugateToBasicGerm.paramExtend: if k is conjugated to h by Ψ, then $(\lambda,u) \mapsto k(u)$ is conjugated to h' by $\Psi''((\lambda,u),k) = ((((\psi(u))_1,\lambda),(\psi(u))_2),\Gamma(u,k))$,

the composite Chapter3/BasicClassComposition · IsStrongBundleIso.comp of $1_\Lambda \times \Psi$ with the linear reassociation $(\lambda,(a,w)) \mapsto ((a,\lambda),w)$ Chapter3/ParamPerturbation · paramSwapOp Chapter3/ParamPerturbation · paramSwapInvOp, a strong bundle isomorphism as a pair of sc-isomorphisms Chapter3/BasicClassComposition · ScOperator.isStrongBundleIso_prodMap. The parameter quadrant is $\widetilde{C} \times \Lambda$ in $P \times \Lambda$, put second so as to reuse the product of a partial quadrant with a whole space; the order is immaterial (Finding 163).

$\tilde{f}$ is an sc-Fredholm section Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmSection.param. *Sc-smooth*: it is the pull-back of f along the projection. *Regularizing* Chapter3/ParamPerturbationG · StrongBundle.regularizing_paramLift: if (r,x) is on level m and $\tilde{f}(r,x)$ on level $(m,m+1)$ of π^*Y, then $x \in X_m$ and $f(x) \in Y_{m,m+1}$ — tested in the pulled-back charts, whose fibre parts are those of P — so $x \in X_{m+1}$ because f is regularizing, and (r,x) is on level $m+1$. *The germ at a smooth point* (r_0,x) Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmGerm.param: x is smooth, so Definition 3.1.11 (preprint 3.11) at x gives a centred chart c, a filling germ g on G_g, an sc$^+$-germ t_0 on G_{t_0} with $t_0(0) = g(0)$ and a conjugation of $g - t_0$ to a basic germ. In the product chart at (r_0,x), which is centred there, the principal part of $\tilde{f}$ is $(a,u) \mapsto \hat{f}(u)$, its filling germ $(a,u) \mapsto g(u)$ on $\mathbb{R}^n \times G_g$ (the filler conditions above), the sc$^+$-germ to subtract $(a,u) \mapsto t_0(u)$ on $\mathbb{R}^n \times G_{t_0}$ Chapter3/GermCalculusParam · IsScSmoothGerm.univProd, and $(a,u) \mapsto (g-t_0)(u)$ is conjugated to the $\mathbb{R}^n$-extension of the basic germ; the conjugation concerns $g-t_0$ near 0 in U, and $1 \times \Psi$ and the extended basic germ are whole-chart objects, so no germ calculus enters it. This is the "readily verified".

Theorem 3.1.21 (preprint 3.18) Chapter3/ParamPerturbationG · theorem_3_1_21_preprint. *Statement.* For a strong bundle $P : Y \to X$, an sc-Fredholm section f and sc$^+$-sections $s_1, \dots, s_n$, the section F is an sc-Fredholm section of π^*P. *Proof.* $F = \tilde{f} + \tilde{s}$; $\tilde{f}$ is sc-Fredholm and $\tilde{s}$ is sc$^+$; Theorem 3.1.18 (preprint 3.15) Chapter3/FredholmSectionsG · theorem_3_1_18_preprint. The paracompactness of $\mathbb{R}^n \times X$ and of π^*Y (Definitions 2.3.4, 2.6.5 (preprint 2.13, 2.51)) enters as instance arguments; the version without them supplies both by Theorem 2.3.10 (preprint 2.16) Chapter3/ParamPerturbationG · theorem_3_1_21' Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod. The text's sketch of the germ of F — filled version $(r,u) \mapsto g(u) + \sum_i r_i \hat{s}_i(r(u))$, sc$^+$-part $t_0(u) + \sum_i r_i \hat{s}_i(r(u))$, conjugation $\Psi'(r,u,k) = (r - r_0, \Psi(u,k))$ — is exactly what Theorem 3.1.18 (preprint 3.15) produces from the germ data of $\tilde{f}$ and the principal part of $\tilde{s}$ Chapter3/FredholmSectionsG · StrongBundle.IsScFredholmGerm.sectionAdd; the translation $r - r_0$ sits in the chart rather than in the conjugation (Finding 164).

The linearization (after Theorem 3.1.22 (preprint 3.19) in the text, used for the finite-dimensional reduction and for transversality). In the product chart at (r_0,x), for a chart c with $0 \in O$, the derivative of $\hat{F} \circ (1 \times r)$ at 0 is

$$(\delta a, \delta u) \mapsto D(\hat{G} \circ r)(0)\delta u + \sum_i \delta a_i \hat{s}_i(0), \qquad \hat{G} = \hat{f} + \sum_i r_{0,i} \hat{s}_i$$

Chapter3/ParamPerturbationG · scDeriv_paramSection: the product rule at $a = 0$, where the terms $\sum_i a_i D(\hat{s}_i \circ r)$ vanish. The tangent space of the product retract at 0 lies in $\mathbb{R}^n \times T_0O$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_mem_tangent, and in the chart form of the linearization (§III.31) $\mathrm{Lin}_F(0)(\delta a, \delta u) = \mathrm{Lin}_{f+\sum r_{0,i}s_i}(0)\delta u + \sum_i \delta a_i \hat{s}_i(0)$ Chapter3/ParamPerturbationG · linAt_paramSection. At $r_0 = 0$ this is the text's formula $\tilde{f}'(0,x)(h,u) = f'(x)u + \sum_i h_i e_i$ Chapter3/ParamPerturbationG · linAt_paramSection_zero. The formula is the derivative at the centre of a chart and does not use $F(r_0,x) = 0$; at a general (r_0,x) its first term is the linearization of $f + \sum r_{0,i}s_i$, not $f'(x)$ (Finding 165). The text's "sc^1-sections s_i" (p. 101 (preprint p. 75)) must be sc$^+$ for Theorem 3.1.21 (preprint 3.18) to apply (Finding 166).

III.41 Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses: the plan and the elementary lemmas

§III.33 showed what the text's argument for property (4) of Theorem 3.6.8 (preprint 3.54) leaves open: at a zero $z_0 = \varphi(y)$ near 0, in the chart at x_0, the filled operator $Dg(z_0)$ is onto (Corollary 3.5.3 (preprint 3.41)), but $f'(y) = Dg(z_0)|_{T_{z_0}O}$ is onto only if condition (3) of the filler holds at z_0, which Definition 3.1.4 (preprint 3.4) does not ask; and the perturbation argument of the text's Proposition 6.2.2 (preprint 6.2) gives $\rho(z_0)\Lambda_0 \subseteq$ range $f'(y)$ only on a neighbourhood depending on the finite-dimensional Λ_0. This section and the next two prove the surjectivity anyway, with the text's hypotheses, by a new argument found while formalizing (Findings 180–189). The text's statement is true; its proof, and that of the book's arXiv version, is incomplete.

The plan. By the assembly step of §III.31 it suffices to show, in the chart at x_0 and for every zero z_0 of g in O near 0, that $\rho(z_0)k \in Dg(z_0)(T_{z_0}O)$ for every smooth $k \in F_\infty$; the closed range of $f'(y)$ (Proposition 3.6.5 (preprint 3.52) at y) does the rest. Fix z_0 and k. Suppose $u(\mu)$, $0 \le \mu < \mu_0$, is a curve through $u(0) = z_0$ of zeros of the *perturbed filler* $g + \mu\rho(r(\cdot))k$, right differentiable at 0 into E_1 with velocity ξ. The curve lies in O (the first lemma below), so $r(u(\mu)) = u(\mu)$ and $Dr(z_0)\xi = \xi$, i.e. $\xi \in T_{z_0}O$; differentiating $g(u(\mu)) = -\mu\rho(r(u(\mu)))k$ at $\mu = 0$ gives $Dg(z_0)\xi = -\rho(z_0)k$. So one curve for each pair (z_0, k) suffices. It is produced by an implicit function theorem *at the nearby zero*, in the normal form of Theorem 3.4.2 (preprint 3.38): its infinite-dimensional part is Theorem 3.3.3 (preprint 3.30) for a contraction germ extended by the parameter μ, applied at the centre, whose regions in the parameter a do not depend on k (§III.42); its finite-dimensional part is a one-sided implicit function theorem at the image of z_0, in the direction of an interior vector of the partial quadrant that good position provides (this section and §III.43). The set of zeros to which the argument applies is cut out by conditions on the data at x_0 alone. This section proves the elementary lemmas, with no sc-structure involved.

Zeros of the perturbed filler lie in O `Chapter3/GlobalIFTNearbyOntoLemmas · StrongRetr.mem_O_of_perturbed_zero`. *Statement.* Let R be a strong bundle retraction on $U \lhd F$, $u \in U$, $\mu \in \mathbb{R}$, $k \in F$, and let condition (2) of Definition 3.1.4 (preprint 3.4) hold at u ($g(u) = \rho(r(u))g(u)$ implies $u \in O$). If $g(u) + \mu\rho(r(u))k = 0$, then $u \in O$. *Proof.* $r(u) \in U$ `Chapter3/Filler · StrongRetr.r_mem` lies in O, $\rho(r(u))$ is linear `Chapter3/Filler · StrongRetr.ρ_isLinear` and idempotent on the fibre over a point of O `Chapter3/Filler · StrongRetr.ρ_ρ`; so $\rho(r(u))g(u) = -\mu\rho(r(u))\rho(r(u))k = -\mu\rho(r(u))k = g(u)$, and condition (2) applies. □

An inward direction from good position `Chapter3/GlobalIFTNearbyOntoLemmas · exists_inward_of_isInGoodPosition`. *Statement.* Let $C = \{x : \ell_i(x) \ge 0\}$ be a partial quadrant presented by an sc-isomorphism T, with coordinate functionals $\ell_i = (T\cdot)_i$, $z \in C$, and $N \ne 0$ a subspace in good position to the partial quadrant $C_z = \{v : \ell_i(v) \ge 0$ for the i with $\ell_i(z) = 0\}$. Then there are $v \in N$ and $\eta, t_0 > 0$ with $z + t(v + \zeta) \in C$ for all $t \in [0, t_0]$ and $|\zeta| < \eta$. *Proof.* Let v be an interior point of $N \cap C_z$ in N (good position). *For an active index i,* $\ell_i(v) > 0$. Otherwise $\ell_i(v) = 0$; for $u \in N$ and small t, $v \pm tu \in C_z$, so $\pm t\ell_i(u) \ge 0$ and ℓ_i vanishes on N. Let P be a good complement with constant ε and $p \in P$; choose $n_0 \ne 0$ in $N \cap C_z$ (it has interior and $N \ne 0$) and scale it so that $|p| \le \varepsilon|n_0|$. Good position gives $n_0 \pm p \in C_z$, so $\ell_i(\pm p) = \ell_i(n_0 \pm p) \ge 0$ and $\ell_i(p) = 0$. Then ℓ_i vanishes on $N \oplus P = E$, which contradicts the surjectivity of the coordinates of a presentation. *The estimates.* A finite family of positive numbers has a positive lower bound `Chapter3/GlobalIFTNearbyOntoLemmas · exists_pos_le_of_forall_pos_fin`: $\ell_i(v) \ge m$ for the active i and $\ell_i(z) \ge m$ for the others. For the active i, $\ell_i(z + t(v + \zeta)) = t(\ell_i(v) + \ell_i(\zeta)) \ge t(m - \|\ell_i\|\,|\zeta|) \ge 0$ when $|\zeta|$ is small; for the others, $\ell_i(z + t(v + \zeta)) \ge m - t\|\ell_i\|(|v| + |\zeta|) \ge 0$ when t is small. □ The lemma needs $N \ne 0$: for the zero

subspace good position is an empty condition, and the case $\dim K = 0$ of Theorem 3.6.8 (1) (preprint 3.54) is treated separately (§III.43). The note behind the formalization phrased it as "N meets the interior of C_z"; the cone condition is what transports along the linear coordinate change (Finding 185).

Inward directions absorb any vector after rescaling `Chapter3/GlobalIFTNearbyOntoLemmas · inward_add_smul`. If $a_0 + t(\alpha+\beta) \in \widetilde{C}$ for $t \in [0,t_0]$ and $|\beta| < \eta$, then for every a_1 and $s \geq (|a_1|+1)/\eta$ the vector $\alpha_0 = a_1 + s\alpha$ satisfies $a_0 + t(\alpha_0+\beta) \in \widetilde{C}$ for $t \in [0, t_0/s]$ and $|\beta| < 1$: indeed $a_0 + t(\alpha_0+\beta) = a_0 + ts(\alpha + (a_1+\beta)/s)$ with $ts \leq t_0$ and $|(a_1+\beta)/s| < (|a_1|+1)/s \leq \eta$.

A one-sided curve of zeros `Chapter3/GlobalIFTNearbyOntoLemmas · exists_curve_of_inward` (the finite-dimensional implicit function theorem at a boundary point, in an inward direction). *Statement.* Let $\widetilde{C} \subseteq P$ be convex, $\Omega \subseteq \mathbb{R}\times P$ open, $D = \Omega \cap (\mathbb{R}\times\widetilde{C})$, $a_0 \in \widetilde{C}$ with $(0,a_0) \in \Omega$, $H : \mathbb{R}\times P \to Q$ of class C^1 on D, $H(0,a_0) = 0$, $D_aH(0,a_0)$ onto (the derivative within D, restricted to $0\times P$), and α_0 with $DH(0,a_0)(1,\alpha_0) = 0$ and $a_0 + t(\alpha_0+\beta) \in \widetilde{C}$ for $t \in [0,t_0]$, $|\beta| < \eta$. Then there are $\mu_0 > 0$ and $a : \mathbb{R} \to P$ with $a(0) = a_0$, right derivative α_0 at 0, and $(\mu, a(\mu)) \in D$, $H(\mu,a(\mu)) = 0$ for $0 \leq \mu < \mu_0$. *Proof. Unique derivatives.* The inward property puts a ball around $a_0 + t_0\alpha_0$ into $\widetilde{C}$, so $\mathbb{R}\times\widetilde{C}$ is convex with nonempty interior and D has unique derivatives; derivatives within D are well defined. *The contraction.* Let S be a right inverse of $M_0 = D_aH(0,a_0)$ and Z its range; SM_0 is the identity on Z. For $\mu \geq 0$ and $\zeta \in Z$ with $|\zeta| \leq c\mu$, $c = \eta/2$, the point $(\mu, a_0 + \mu\alpha_0 + \zeta)$ lies in D and close to $(0,a_0)$ `Chapter3/GlobalIFTNearbyOntoLemmas · dist_curve_le`, and $a_0 + \mu\alpha_0 + \zeta = a_0 + \mu(\alpha_0 + \zeta/\mu)$ is in $\widetilde{C}$ by the inward property ($|\zeta/\mu| \leq c < \eta$). Put $T_\mu(\zeta) = \zeta - SH(\mu, a_0+\mu\alpha_0+\zeta)$. Its derivative in ζ is $1 - SD_aH(\mu, a_0+\mu\alpha_0+\zeta)|_Z$, which tends to $1 - SM_0|_Z = 0$ as $(\mu,\zeta) \to 0$ by the continuity of DH at $(0,a_0)$; the mean value inequality on the convex ball makes T_μ a $\frac{1}{2}$-contraction for small μ. The differentiability of H at $(0,a_0)$ in the direction $(1,\alpha_0)$, with $DH(0,a_0)(1,\alpha_0) = 0$, gives $|T_\mu(0)| = |SH(\mu, a_0+\mu\alpha_0)| = o(\mu)$; so $|T_\mu(0)| \leq c\mu/2$ for small μ and T_μ maps the ball of radius $c\mu$ into itself. *The curve.* The fixed point $\zeta(\mu)$ has $SH(\mu, a_0+\mu\alpha_0+\zeta(\mu)) = 0$, hence $H = 0$ there (S is injective), and $|\zeta(\mu)| \leq 2|T_\mu(0)| = o(\mu)$; so $a(\mu) = a_0 + \mu\alpha_0 + \zeta(\mu)$ has right derivative α_0 at 0. □

Two facts on right derivatives. If k is continuous at 0 from the right, $\mu \mapsto \mu k(\mu)$ has the right derivative $k(0)$ at 0 `Chapter3/GlobalIFTNearbyOntoLemmas · hasDerivWithinAt_smul_right_of_continuousWithinAt`; and right derivatives at 0 on $[0,\infty)$ are unique `Chapter3/GlobalIFTNearbyOntoLemmas · eq_of_hasDerivWithinAt_Ici`.

III.42 The λ-extended contraction germ and the perturbation in the normal form

Throughout, the normal form is the data of Theorem 3.4.2 (preprint 3.38) from §III.20 `Chapter3/LocalSolvability · ShiftedBasicData`: a germ $g : O(\widetilde{C}\oplus W, 0) \to Q\oplus W$, sc-smooth, whose index raise $g^1 = (G_1, w - B(a,w))$ is a basic germ on the shifted structures; at the chart at x_0 it is obtained from the stability data of the graph description by Theorem 3.4.2 (preprint 3.38) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`. (In this section g and e denote the normal form and its perturbation; the filler of the chart at x_0 returns in §III.43.) The norms of w below are those of W_{m+1}, the shifted level m.

The perturbation on the shifted structure. For an sc$^+$-perturbation $e : O(\widetilde{C}\oplus W, 0) \to (Q\oplus W)^1$, its W-component on level 1 `Chapter3/GlobalIFTNearbyOntoGerm · perturbEW` is sc-smooth on the index-raised germ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.eW_isScSmoothGerm`, by Proposition 1.2.2 (preprint 1.17) for germs applied to the level-raised map `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelMap_e_isScSmoothGerm` `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.eW_eq_on`. Its level maps are those of e read

on the shifted levels Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_eW.

Contraction regions that do not depend on the perturbation Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.Radii. A system of regions consists of radii ν_m in P and ρ_m in W_{m+1} such that for $a \in \widetilde{C}$, $|a| < \nu_m$ and $|w|, |w'| < \rho_m$: (a, w) lies in the domain of g^1 on level m, $|B(a,w) - B(a,w')| \le \frac{1}{4}|w - w'|$, and $|B(a,0)| \le \rho_m/8$; ν decreasing Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.Radii.ν_anti_le and the ρ_m compatible with the inclusions. *They exist below any prescribed radii* δ_m Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.exists_radii: on each level, Definition 3.3.1 (preprint 3.28) with $\varepsilon = \frac{1}{4}$ gives a radius on which $B(a,\cdot)$ is a $\frac{1}{4}$-contraction, and since $B(0,0) = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_B_zero and B is continuous at 0 on each level, $|B(a,0)| \le \rho_m/8$ for $|a|$ small; the radii are then shrunk level by level to be monotone and below δ_m. Nothing here involves a perturbation.

The λ-extended germ. On $(\mathbb{R} \times \widetilde{C}) \oplus W^1$ take the germ of neighbourhoods $\{((\mu, a), w) \mid (a, w) \in U^1_m\}$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbGerm, with the projections forgetting μ and keeping μ as sc-operators Chapter3/GlobalIFTNearbyOntoGerm · perturbProj Chapter3/GlobalIFTNearbyOntoGerm · perturbMuProj and $\mathbb{R} \times \widetilde{C}$ quadrant-like Chapter3/GlobalIFTNearbyOntoGerm · isQuadrantLike_perturb, and put

$$B_\mu(a, w) = B(a, w) - \mu\, Pe(a, w)$$

Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB, P the projection onto W, with $B_\mu(0,0) = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB_zero. It is sc-smooth on the germ: B composed with the projection, minus the scalar product of the μ-coordinate with Pe composed with the projection, by the product rule with a scalar factor of §III.40 Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB_isScSmoothGerm Chapter3/ParamPerturbation · IsScSmooth.smul; its level maps are $B_m - \mu(Pe)_m$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_perturbB. Given constants $c_m \ge 0$ with $|Pe(a,w)| \le c_m$ and $|Pe(a,w) - Pe(a,w')| \le c_m|w - w'|$ on the regions, define the μ-radii $\mu_m = \min_{j \le m} \min(\frac{1}{4}, \rho_j/8)/(c_j + 1)$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu, positive and decreasing Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_pos Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_anti, with $\mu_m c_m \le \frac{1}{4}$ and $\mu_m c_m \le \rho_m/8$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_mul_le.

Lemma: B_μ is a contraction germ in the working form Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction, with the parameter regions $(-\mu_m, \mu_m) \times \{|a| < \nu_m\}$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.mem_perturbContraction_N_iff, the open balls $\{|w| < \rho_m\}$ and the closed balls $\{|w| \le \rho_m/2\}$ (Definition 3.A, §III.7). *The contraction*: $|B_\mu(a,w) - B_\mu(a,w')| \le (\frac{1}{4} + |\mu|c_m)|w - w'| \le \frac{1}{2}|w - w'|$. *The closed balls are invariant*: for $|w| \le \rho_m/2$, $|B_\mu(a,w)| \le |B(a,w) - B(a,0)| + |B(a,0)| + |\mu||Pe(a,w)| \le \frac{1}{4} \cdot \frac{\rho_m}{2} + \frac{\rho_m}{8} + |\mu|c_m \le \frac{\rho_m}{2}$. The regions in a and w are those of B and do not depend on e; only the μ-intervals do. The general construction of a working form from Definition 3.3.1 (preprint 3.28) Chapter3/GermIFT · toContractionGerm is *not* used: its radii in a would depend on e, and the finite-dimensional step needs the nearby zero inside the level-3 region for every perturbation (Finding 182).

Theorem 3.3.3 (preprint 3.30) for B_μ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction_isSc1GermWith. The solution germ $\delta_\mu(\mu, a)$ is sc¹ on the germ re-indexed by one level, $((-\mu_{m+1}, \mu_{m+1}) \times \{|a| < \nu_{m+1}\}) \cap (\mathbb{R} \times \widetilde{C})$, with the derivative family of the proof of Theorem 3.3.3 (preprint 3.30) Chapter3/ContractionGerm · theorem_3_3_3. *At* $\mu = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction_sol_zero: a fixed point $w \in W_1$ of $B(a,\cdot)$ with $|w| < \rho_0$, $|a| < \nu_0$, $a \in \widetilde{C}$, is $\delta_\mu(0, a)$,

by the uniqueness of the fixed point in the open ball of level 0.

The perturbation in the normal form. Now the chart at x_0 returns: the strong bundle retraction R on $U \lhd F$, the conjugation Ψ covering ψ with inverse Ψ', and the linear maps glue, split, fiber of the coordinate change of Theorem 3.4.2 (preprint 3.38) (§III.18). The perturbation $u \mapsto \rho(r(u))k$ of the filler, pushed to the normal form, is $\hat{e}_k(y) = \Xi(y,k)$ with the *joint* map

$$\Xi(y,\kappa) = \operatorname{fiber}\Big(\Psi[1](\psi^{-1}(\operatorname{glue} y),\ (R[1](r(\psi^{-1}(\operatorname{glue} y)),\kappa))_F)\Big)_{F'}$$

`Chapter3/GlobalIFTNearbyOntoXi · xiMap` on $\psi(U) \oplus F^1$ `Chapter3/GlobalIFTNearbyOntoXi · xiDom` `Chapter3/GlobalIFTNearbyOntoXi · mem_xiDom`, a relatively open set (glue$^{-1}(U')$ is relatively open in the quadrant of the normal form) `Chapter3/GlobalIFTNearbyOntoXi · glue₀_preimage_relOpen` `Chapter3/GlobalIFTNearbyOntoXi · xiDom_relOpen`. Ξ *is sc-smooth into the index-raised target* `Chapter3/GlobalIFTNearbyOntoXi · xiMap_isScSmooth`: it is the composite of the linear glue, $\psi^{-1} \times 1$, the map $(u,\kappa) \mapsto (u,(R[1](r(u),\kappa))_F)$, the strong bundle map $\Psi[1]$ on the index-raised bundle (Definition 2.6.1 (preprint 2.47)), a projection and the linear fiber. Ξ *is linear in* κ `Chapter3/GlobalIFTNearbyOntoXi · xiMap_smul` `Chapter3/GlobalIFTNearbyOntoXi · xiMap_zero`, also on every level `Chapter3/GlobalIFTNearbyOntoXi · levelMap_xiMap_smul`, since R and Ψ are linear on the fibres. For a smooth k with level-1 representative k_1, $\hat{e}_k = \Xi(\cdot,k_1)$ `Chapter3/GlobalIFTNearbyOntoXi · ehat` is sc-smooth into the index-raised target `Chapter3/GlobalIFTNearbyOntoXi · ehat_isScSmooth` (a constant map with smooth value is sc-smooth `Chapter3/GlobalIFTNearbyOntoXi · isScSmooth_const_of_smooth`), an sc$^+$-perturbation on the germ of Theorem 3.4.2 (preprint 3.38) `Chapter3/GlobalIFTNearbyOntoXi · pushGerm₀_U_subset` `Chapter3/GlobalIFTNearbyOntoXi · ehat_isScSmoothGerm`, with level maps $(\hat{e}_k)_m(y) = \Xi_m(y,k_{m+1})$ `Chapter3/GlobalIFTNearbyOntoXi · levelMap_ehat`.

Lemma: bounds uniform in k `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_bounds`. *For every level m there are $\delta > 0$ and $M \geq 0$ such that for $|a| < \delta$, $|\kappa|_{m+2} < \delta$ and a convex set S of $w \in W_{m+1}$ with $|w| < \delta$ on S (inside the domain of Ξ on level $m+1$): $|\Xi_m((a,w),\kappa)| \leq 1$ and $|\Xi_m((a,w),\kappa) - \Xi_m((a,w'),\kappa)| \leq M|w-w'|$ on S. Proof.* By Proposition 1.2.3 (preprint 1.18) `Chapter1/Levels · IsScSmooth.contDiffOn_levels`, Ξ is C^1 from level $m+1$ to level m of its index-raised target, which is the shifted level m of the normal form: on the shifted structure the perturbation is C^1 on every level, which is what contraction estimates on the shifted levels need. *This is where the index raise of Theorem 3.4.2 (preprint 3.38) is used* (Finding 183). Ξ_m vanishes at $\kappa = 0$, so by continuity of Ξ_m and of its derivative at 0 both are small on a small ball, and the mean value inequality on the convex S gives the Lipschitz bound. □ *The constants for* $\hat{e}_k$ `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds`: there are radii $\delta_m > 0$, depending on Ξ only, such that for every system of regions with $\nu_m, \rho_m \leq \delta_m$ and every smooth k, the constants $c_m = (2|k|_{m+2}/\delta_m)(M_m+1)$ bound $(P\hat{e}_k)_m$ and its Lipschitz constant in w on the regions. The proof is the *scaling* $\Xi_m(y,\kappa) = (2|\kappa|/\delta_m)\,\Xi_m(y,(\delta_m/2|\kappa|)\kappa)$, which needs only the linearity of Ξ in κ; the note behind the formalization used the linearity of $D_y\Xi$ in κ instead.

III.43 The curve of perturbed zeros, the surjectivity in the chart at x_0, and Theorem 3.6.8 (1) (preprint 3.54)

The reduction at a nearby zero (in the normal form, with a perturbation e, regions R and constants c as in §III.42). Let $z = (a_0,w_0)$ be a level-1 zero of g in the domain of the germ with $a_0 \in \widetilde{C}$, $|a_0| < \nu_3$ and $|w_0|_1 < \rho_0$. Then w_0 is a fixed point of $B(a_0,\cdot)$ `Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.B_eq_of_zero` (because $g^1 = (G_1, w - B)$), so

$w_0 = \delta_\mu(0,a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.sol_zero_eq. On the domain $D_3 = (-\mu_3,\mu_3)\times(\{|a| < \nu_3\}\cap\widetilde{C})$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDom, which has unique derivatives Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.uniqueDiffOn_curveDom, put $(\mu,a)\mapsto(a,\delta_\mu(\mu,a))\in E_1$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveLift Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveLift_mem and

$$\widetilde{G}(\mu,a) = (g+\mu\,\iota e)(a,\delta_\mu(\mu,a))$$

Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveMap. *Its W-component vanishes* Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveMap_snd: $\delta_\mu(\mu,a)$ is the fixed point of $B_\mu(a,\cdot)$, and the W-component of $g+\mu\iota e$ is $w-B_\mu(a,w)$ (read on level 1, Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.snd_g_toBase). So the whole equation is carried by $H = (\widetilde{G})_Q$. *$\widetilde{G}$ is C^1* on D_3 Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.contDiffOn_curveMap: δ_μ is C^1 into W_2 by Theorem 3.3.3 (preprint 3.30) on the re-indexed germ (Proposition 1.2.1 (preprint 1.16) on level 1), and g, e are C^1 from level 1 to level 0. *Its derivative at $(0,a_0)$* Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.hasFDerivWithinAt_curveMap is

$$D\widetilde{G}(0,a_0)(\dot\mu,\alpha) = Dg(z)(\alpha,\ \iota D\delta_\mu(0,a_0)(\dot\mu,\alpha)) + \dot\mu\,\iota e(z)$$

Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.solD Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv: the chain rule, with the derivative of the solution germ on level 1 Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level, the derivative of g at z Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.hasFDerivWithinAt_Dg', and the product rule for $\mu\,\iota e$, whose derivative at $\mu = 0$ in the base directions vanishes. Its W-component vanishes too Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv_snd.

Lemma: $D_aH(0,a_0)$ is onto, by a dimension count Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.surjective_DH_and_ker_eq. *Statement.* Let moreover $\ker Dg(z)$ be finite-dimensional of dimension $\dim P - \dim Q$. Then $M_0 = D_aH(0,a_0) : P\to Q$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.DH is onto, and $\ker Dg(z)$ is the graph of $L = \iota D_a\delta_\mu(0,a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.solL over $\ker M_0$. *Proof.* At $\dot\mu = 0$ the derivative formula reads $Dg(z)(\alpha,L\alpha) = (M_0\alpha,0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv_inr Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.Dg'_graph, so the graph of L over $\ker M_0$ lies in $\ker Dg(z)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.map_graph_le_ker. The graph map is injective, so $\dim\ker M_0 \le \dim\ker Dg(z) = \dim P - \dim Q$; rank–nullity gives $\operatorname{rank} M_0 \ge \dim Q$, so M_0 is onto, and then $\dim\ker M_0 = \dim\ker Dg(z)$ and the inclusion is an equality. □ The obvious route, eliminating the W-block with $(1-D_2B(z))^{-1}$, needs $\|D_2B(z)\|\le\frac12$ on the level where $Dg(z)$ is known to be onto, i.e. the surjectivity on the shifted structure; the dimension count uses only the kernel dimension on level 0 (Finding 184).

Proposition: the curve of perturbed zeros Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.exists_perturbed_curve. *Statement.* Under the hypotheses of the lemma, let $v\in\ker Dg(z)$ with $a_0+t(v_1+\beta)\in\widetilde{C}$ for $t\in[0,t_0]$, $|\beta|<\eta$ (v_1 the P-component). Then there are $\mu_0>0$ and a curve $u'(\mu) = (a(\mu),\delta_\mu(\mu,a(\mu)))$, $0\le\mu<\mu_0$, on level 2, in the domain of the germ with $a(\mu)\in\widetilde{C}$, with $u'(0)=z$ (on level 1), right differentiable at 0 as a map into E_2, and $(g+\mu\iota e)(u'(\mu)) = 0$. *Proof.* (i) By the lemma, M_0 is onto and $v_1\in\ker M_0$ (as $v=(v_1,Lv_1)$). (ii) Choose a_1 with $M_0a_1 = -D_\mu H(0,a_0)$, and $\alpha_0 = a_1 + sv_1$ with s from the rescaling lemma of §III.41: α_0 is inward at a_0 (with $|\beta|<1$) and

$DH(0,a_0)(1,\alpha_0) = D_\mu H(0,a_0) + M_0 a_1 + sM_0 v_1 = 0$. (iii) H is C^1 on D_3 with $H(0,a_0) = 0$, and $\widetilde{C}$ is convex; the one-sided curve lemma (§III.41) on $\$\square = \$$ the open region of level 3 gives $a(\mu)$ with right derivative α_0 and $H(\mu, a(\mu)) = 0$. (iv) The level-1 map $(\mu, a) \mapsto \delta_\mu(\mu, a) \in W_2$ is differentiable at $(0, a_0)$ within D_3 (Theorem 3.3.3 (preprint 3.30), `Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level`), so by the chain rule for $\mu \mapsto (\mu, a(\mu))$ the curve u' is right differentiable at 0 into E_2. (v) At $\mu = 0$, $\delta_\mu(0, a_0) = w_0$; and $(g + \mu\iota e)(u'(\mu)) = \widetilde{G}(\mu, a(\mu)) = (H(\mu, a(\mu)), 0) = 0$. □

The normal form at a nearby point. For the chart at x_0: at every level-1 point y of $\psi(U_1)$ in the germ, $Dg(y) = \text{fiber} \circ D(h + t)(\text{glue}\, y) \circ \text{glue}$ for the normal form g of $h + t$ `Chapter3/GlobalIFTNearbyOnto · StabilityData.scDerivGerm_pushMap₀_at` (the coordinate change is linear), and the kernel of fiber $\circ\, T \circ$ glue is $\text{split}(\ker T)$ `Chapter3/GlobalIFTNearbyOnto · StabilityData.ker_fiber₀_comp_glue₀`, split being injective `Chapter3/GlobalIFTNearbyOnto · StabilityData.split₀_injective`. Corollary 3.4.3 (preprint 3.39) is not used: the facts at the nearby zero come from the chart at x_0 through these linear maps (Finding 187).

Theorem A: the surjectivity in the chart at x_0 `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_chart_onto`. *Statement.* Take the data of the graph description in the chart at x_0 (§III.29, `Chapter3/GlobalIFTNearbyGraphG · theorem_3_6_8_chart_graph`): the strong bundle retraction $R = (r, \rho)$ on U, a filling germ g of $\mathbf{f}$ on G_g with a derivative family Dg, $\mathbf{f}(0) = 0$, the conjugation Ψ covering ψ, $\psi(0) = 0$, the stability data on a germ that ψ' maps into G_g level by level (inside G'', §III.48) with the basic germ h', $D(h' + \iota t)(0)$ onto with kernel K in good position, a closed good complement Y of K, the C^1 graph $\sigma : V' \to Y$ with $\sigma(0) = 0$, $D\sigma(0) = 0$ and its C^k-lifts, an open W_o on which $h' + \iota t$ is the push-forward of g, an open $N_2 \ni 0$ on which ψ maps into the germ and into $U' \cap W_o$, and on which ψ maps the zeros of g in O onto the graph. Assume condition (2) of Definition 3.1.4 (preprint 3.4) on $U \cap N_2$ and $K \neq 0$. Then there is an open $N_4 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_4$ of g, every level-1 point $z_{0,1}$ over z_0 and every smooth k, there is $x \in T_{z_0}O = Dr(z_{0,1})E_0$ with $Dg(z_{0,1})x = \rho(z_0)k$. *Proof*, in the order of the Lean:

1. *Data at x_0, independent of k.* Lemma 3.6.9 (preprint 3.55) on the graph with its radius ρ_2 `Chapter3/GlobalIFTFull · lemma_3_6_9_relOpen` (§III.32); the kernel at the nearby zeros on an open $N_3 \subseteq N_2$ `Chapter3/GlobalIFTNearbyKernelG · theorem_3_6_8_chart_kernel` (§III.32); the normal form $\mathcal{D}$ of Theorem 3.4.2 (preprint 3.38) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`; the radii δ_m of the bounds of Ξ `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds` and contraction regions below them `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.exists_radii`; and the radius r_{42} of Lemma 3.5.5 (preprint 3.42) for the bound ρ_0 `Chapter3/LocalSolvability · StabilityData.lemma_3_5_5`: zeros of the normal form on level 0 smaller than r_{42} have their W-component on level 1, smaller than ρ_0 there.
2. *The neighbourhood.* Let π be the projection onto K along Y (a topological complement: Y closed, K finite-dimensional), $c_s = \|\text{split}\| + 1$, and $r' = \min(\rho_2/(\|\pi\| + 1), r_{42}/c_s, \nu_3/c_s)$. By the continuity of ψ on U, $\psi^{-1}(B(0, r')) \cap U = O_4 \cap U$ for an open $O_4 \ni 0$; put $N_4 = N_3 \cap O_4$. Nothing here depends on k.
3. *At a zero $z_0 \in O \cap N_4$.* The kernel theorem at z_0 gives $v \in V'$ with $\psi(z_0) = v + \sigma(v)$, the level-1 point x_1 over it, $D(h' + \iota t)(x_1)$ onto and Fredholm of index $\dim P - \dim Q$, its kernel N_v, and $\ker Dg(z_{0,1}) = D\psi^{-1}(x_1)N_v \subseteq T_{z_0}O$. Since $|v| = |\pi(\psi(z_0))| \leq \|\pi\| r' < \rho_2$, Lemma 3.6.9 (preprint 3.55) at v gives $x_1 \in C'$ and N_v in good position to C'_{x_1}.
4. *The zero of the normal form.* $z = \text{split}(x_1)$ lies in the germ on level 1, with $a_0 = (x_1)_P \in \widetilde{C}$; $|\text{split}\, \psi(z_0)| < c_s r'$, so $|a_0| < \nu_3$, and Lemma 3.5.5 (preprint 3.42) gives $|w_0|_1 < \rho_0$; g vanishes at z because $h' + \iota t$ vanishes at the graph point.

5. *Its kernel.* $Dg(z) = \text{fiber} \circ D(h' + \iota t)(x_1) \circ \text{glue}$, so $\ker Dg(z) = \text{split}(N_v)$, finite-dimensional of dimension $\dim\ker D(h' + \iota t)(x_1) = \dim P - \dim Q$ (for a surjective Fredholm operator the kernel dimension is the index, `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`).
6. *The inward direction.* $N_v \neq 0$: its dimension is $\dim P - \dim Q$, which is also $\dim K$ ($D(h' + \iota t)(0)$ is onto and Fredholm of that index, `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`), and $K \neq 0$. The inward lemma at x_1 (§III.41) gives $v_v \in N_v$ with $x_1 + t(v_v + \zeta) \in C'$; then split $v_v \in \ker Dg(z)$, and its P-component is inward at a_0 in the quadrant of the normal form for $|\beta| < \eta/(\|\text{glue}\| + 1)$, since $\text{glue}(\beta, 0)$ has norm at most $\|\text{glue}\|\,|\beta|$.
7. *The perturbation and the curve.* Take the level-1 representative k_1 of k; $\hat{e}_k$ is an sc$^+$-perturbation (§III.42) with constants c_m on the regions; the proposition above gives μ_0 and the curve u' of zeros of $g + \mu\iota\hat{e}_k$ through z.
8. *Back in the chart.* $u(\mu) = \psi^{-1}(\text{glue}\, u'(\mu))$ on level 1: ψ^{-1} is C^1 from level 2 to level 1 (Proposition 1.2.3 (preprint 1.18), `Chapter1/Levels · IsScSmooth.contDiffOn_levels`), so u is right differentiable at 0 with some velocity ξ_1, $u(0) = z_{0,1}$, and its base points $u_\mu = \psi^{-1}(\text{glue}\, \iota u'(\mu))$ lie in U.
9. *The perturbed equation.* By continuity at $\mu = 0$, $\text{glue}\, \iota u'(\mu) \in W_o$ and $u_\mu \in N_2$ for $0 \leq \mu < \mu_2$ `Chapter3/GlobalIFTNearbyOnto · exists_Ico_of_eventually_nhdsWithin_Ici`. There the push-forward identity of g, the description of $\hat{e}_k$ as the Ψ-fibre image of $\rho(r(u_\mu))k$ transported by the linear fiber, and the injectivity of the fibre maps of Ψ turn $(g + \mu\iota\hat{e}_k)(u'(\mu)) = 0$ into $g(u_\mu) + \mu\rho(r(u_\mu))k = 0$.
10. *Tangency.* By the first lemma of §III.41 and condition (2) on N_2, $u_\mu \in O$.
11. *Differentiation.* r is sc^1 on U, and g is an sc^1-germ on G_g; the curve $u(\mu)$ stays on level 1 of G_g, because $\text{glue}\, \iota u'(\mu)$ lies on level 1 of the germ of the stability data and ψ_1' maps that into $(G_g)_1$ `Chapter3/BasicGermShrink · ConjugationData.levelMap_inclK_mem_of_subset_shrinkGerm`. So both are differentiable from level 1 to level 0 at $z_{0,1}$ along the curve, g by the chain rule within $(G_g)_1$ with derivative $Dg(z_{0,1})$ (with g sc-smooth on U no such condition was needed). Along the curve, $r(u_\mu) = u_\mu$, so the right derivatives at 0 agree, $Dr(z_{0,1})\xi_0 = \xi_0$ ($\xi_0 = \iota\xi_1$; uniqueness of right derivatives, §III.41): $\xi_0 \in T_{z_0}O$. The map $\mu \mapsto \rho(r(u_\mu))k$ is continuous at 0 (R is continuous on level 0), so $\mu \mapsto \mu\rho(r(u_\mu))k$ has the right derivative $\rho(r(z_0))k = \rho(z_0)k$; differentiating $g(u_\mu) = -\mu\rho(r(u_\mu))k$ gives $Dg(z_{0,1})\xi_0 = -\rho(z_0)k$. Take $x = -\xi_0$. □

Where the hypotheses enter. Tameness and good position at x_0 through the graph description, Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) (steps 1, 3) and the inward direction at boundary zeros (step 6); condition (2) of the filler through the tangency (step 10); $K \neq 0$ for the inward direction. The sc-Fredholm property at y is not used here; it enters only through the closed range in the assembly step. Condition (3) of the filler at the nearby zeros is not used. *Why there is no uniformity problem.* The only k-dependent quantities are the constants c_m and the μ-intervals; N_4 is fixed in step 2 before k is chosen. The perturbation argument of §III.33 applied the implicit function theorem at x_0 to the Λ_0-extended problem, whose neighbourhood depends on Λ_0; here it is applied at the nearby zero (Finding 181).

Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'`. *Statement.* Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$, and (f, x_0) in good position. Then there is an open $V_x \ni x_0$ such that (f, y) is in good position at every zero $y \in V_x$. *Proof.* Destructure the good position at x_0: the centred chart c, the filling germ g with its derivative family, the sc$^+$-germ, the conjugation, shrunk to G'' (§III.48), and the good position in the chart. Tameness gives $(1 - Dr(0))E \subseteq E_0$ by Proposition 2.5.3 (preprint 2.35) `Chapter2/TameRetracts · proposition_2_5_3`; the graph description in the chart `Chapter3/GlobalIFTNearbyGraphG · theorem_3_6_8_chart_graph` and the kernel at the nearby zeros `Chapter3/GlobalIFTNearbyKernelG · theorem_3_6_8_chart_kernel` give $N_1 \supseteq N_2 \supseteq N_3$. A zero y of f in the chart domain with $\varphi(y) \in N_2$ is a zero of g ($\mathbf{f} = g$ on $O \cap N_1$).

Case $K = 0$. Put $V_x = \varphi^{-1}(N_2)$ in the chart domain. A zero $y \in V_x$ maps by $\psi\varphi$ onto a graph point $v + \sigma(v)$ with $v \in K = 0$, so $\psi(\varphi(y)) = \sigma(0) = 0 = \psi(0)$, $\varphi(y) = 0$ (as $\psi^{-1}\psi = 1$ on U), and $y = x_0$ by the injectivity of the chart: the only zero is x_0, in good position by hypothesis. *Case* $K \neq 0$. Theorem A gives N_4 (condition (2) on $U \cap N_2$ is the filler's, §III.29); put $V_x = \varphi^{-1}(N_3 \cap N_4)$ in the chart domain. For a zero $y \in V_x$: $z_0 = \varphi(y) \in O$ is a zero of g; the kernel theorem gives the level-1 point $z_{0,1}$, $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$ and its good position inside $T_{z_0}O$; y is smooth `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`, so (f, y) is an sc-Fredholm germ; $\mathbf{f} = g$ on O near z_0. The assembly step in its germ form `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense` (§III.31; whole-chart `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`) applies at $z_{0,1} \in (G_g)_1$ with (a) from Theorem A — $\rho(z_0)\hat{k}$ in the range of $Dg(z_{0,1})|_{T_{z_0}O}$ for every smooth $\hat{k}$ — and (b) from the kernel theorem, $\ker Dg(z_{0,1}) \cap T_{z_0}O = \ker Dg(z_{0,1})$. □ **No hypothesis is added**: the statement has exactly the hypotheses of the kernel theorem of §III.32 plus the sc-Fredholm property (Finding 189).

Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) with the text's hypotheses. Theorem 3.6.8 (1) and (2) (preprint 3.54) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8'`: (1) as above on V_x, and (2), the regular data of Definition 3.2.1 (preprint 3.25) at x_0 of a dimension d that every nearby datum shares, by §III.37 `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8_two` fed with (1). Theorem 3.1.26 (preprint 3.23) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_26'`: (1) supplies the open set of nearby good position that the boundary case from the nearby good position needs `Chapter3/GlobalIFTTheoremsG · theorem_3_1_26_of_nearby`, and every zero of the resulting U is in good position. Theorem 3.1.22 (preprint 3.19) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_22'`: Theorem 3.1.26 (preprint 3.23) and Lemma 2.4.6 (preprint 2.26), $d_S \leq d_X = 0$ `Chapter2/SubMPolyfold · lemma_2_4_6`. The first forms under the filler condition at the nearby zeros (§§III.33, III.37) are kept; that condition is now a theorem (§III.45).

III.44 The dimension of the solution set is the Fredholm index, and the index is locally constant

Theorem 3.1.22 (preprint 3.19) says that the solution set is a manifold "whose dimension agrees with the Fredholm index", and the proof of Theorem 3.6.8 (preprint 3.54) takes $d = \dim\ker f'(x)$. In the statements of §§III.34–III.37 the dimension was existential. This section identifies it, at every zero and in every chart, and deduces the local constancy of the index along the solution set (Findings 170–171).

"$f'(y)$ is onto, Fredholm, of index d", chart-intrinsically `Chapter3/GlobalIFTIndex · StrongBundle.IsSurjLinOfIndex`: in every strong bundle chart around y compatible with the atlas, at the level-1 point z over $\varphi(y)$, every corestriction $L : T_zO \to K_z$ of the linearization $\mathrm{Lin}(z)$ (§III.31) is onto and Fredholm with $\mathrm{ind}\, L = \dim\ker L = d$. "$f'(y)$ is onto" alone `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt`: in *some* chart, $\mathrm{Lin}(z)$ maps onto the fibre $\{k \mid (\varphi(y), k) \in K\}$; by the chart independence it is then onto in every chart.

Proposition 3.6.5 (preprint 3.52) in every chart `Chapter3/GlobalIFTIndexG · StrongBundle.linAt_index_of_conjugationData`. *Statement.* Let f be an sc-smooth section, x_0 a zero, c a strong bundle chart centred at x_0 with a filling germ g of $\mathbf{f}$ and a derivative family Dg on a germ inside U, an sc⁺-germ s with $s(0) = g(0)$, and conjugation data d_c of $g - s$ to a basic germ `Chapter3/FredholmGermIndex · ConjugationData` with finite-dimensional spaces P_n, Q. Then in every strong bundle chart c' around x_0, at the level-1 point z' over $\varphi'(x_0)$, every corestriction L' of $\mathrm{Lin}'(z')$ is Fredholm of index $\dim P_n - \dim Q$,

and onto if the germ is in good position at $Dg(0)$ in c. *Proof.* Proposition 3.6.5 (preprint 3.52) in c (§III.30, `Chapter3/FredholmGermIndexG · proposition_3_6_5_local_index`): $Tf(0)$ is Fredholm of index $\dim P_n - \dim Q$. $Tf(0) = Dg(0)|_{T_0O}$ is $\mathrm{Lin}(0)$ by the bridge for germs at the centre, which lies in every germ `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt` (whole-chart: `Chapter3/LinearizationCharts · IsFilledVersion.fillerRestrict_eq_linAt`), up to the identification of the fibres $\rho(0)F$ and $\rho(\iota 0)F$, which are equal subspaces but different types in Lean `Chapter3/GlobalIFTIndex · exists_rangeS_pCLM_equiv`. The chart independence (§III.31, `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`) carries the Fredholm property, the index and the surjectivity from c at 0 to c' at z'; the four spaces are complete as ranges of idempotents `Chapter3/FillerLinearAlgebra · isClosed_range_of_idem`. □ At a zero in good position this gives $\mathrm{IsSurjLinOfIndex}(f, x_0, \dim P_n - \dim Q)$ `Chapter3/GlobalIFTIndexG · StrongBundle.isSurjLinOfIndex_of_conjugationData`, because for a surjective Fredholm operator the kernel dimension is the index `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`; and the d with $\mathrm{IsSurjLinOfIndex}(f, x_0, d)$ equals $\dim P_n - \dim Q$ for every conjugation in the chart of Definition 3.6.6 (preprint 3.53) `Chapter3/GlobalIFTIndexG · StrongBundle.IsGoodPositionGerm.exists_isSurjLinOfIndex` (a conjugation exists, `Chapter3/FredholmGermIndex · isConjugateToBasicGerm_iff_nonempty`). The first clause of Theorem 3.1.22 (preprint 3.19) with the index `Chapter3/GlobalIFTIndexG · theorem_3_1_22_linearization_index`: for an sc-Fredholm section and a zero in good position, in the chart of Definition 3.6.6 (preprint 3.53), $Dg(0)$ and $Tf(x_0)$ are Fredholm, $Tf(x_0)$ is onto, and $\mathrm{ind}\, Dg(0) = \mathrm{ind}\, Tf(x_0) = \dim\ker Tf(x_0) = \dim d.P_n - \dim d.Q$ for every conjugation d — the old formulation's "$n - N$ for some natural numbers" made precise.

The conjugation exposed. The graph description and the tangent germ data at the centre are built from the chart of Definition 3.6.6 (preprint 3.53) by destructuring the existential good position, so their statements did not record which conjugation they used. Their proofs now prove versions that expose it `Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj_conj` `Chapter3/GlobalIFTCenterG · theorem_3_6_8_tangentGermChart_proj_conj`: the filling germ with its germ and derivative family Dg, the sc⁺-germ, the good position at $Dg(0)$ in the chart and conjugation data d_c, with $\dim K = \dim d_c.P_n - \dim d_c.Q$ for $K = \ker D(h+t)(0)$, because $D(h+t)(0)$ is onto and Fredholm of index $n - N$ (§III.18, `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`). The earlier statements are corollaries.

Dimension = index. *At the centre* `Chapter3/GlobalIFTIndexG · theorem_3_6_8_tangentGermChart_index`: for an sc-smooth section of a strong bundle whose charts have tame base retractions and a zero x_0 in good position, the data of Definition 3.2.1 (preprint 3.25) at x_0 for $\{y \in V_x : f(y) = 0\}$ (with $\pi(O) \subseteq C$) have a dimension d with $\mathrm{IsSurjLinOfIndex}(f, x_0, d)$: the exposed $d = \dim K = \dim d_c.P_n - \dim d_c.Q$ and the previous paragraph. The same for the global solution set `Chapter3/GlobalIFTIndexG · theorem_3_6_8_tangentGermChart_solutionSet_index` (restriction of the chart, §III.37) and for the regular data of Theorem 3.6.8 (2) (preprint 3.54) `Chapter3/GlobalIFTIndexG · theorem_3_6_8_two_index` (the re-centring keeps the dimension, `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`). *At a nearby zero* `Chapter3/GlobalIFTIndexG · isSurjLinOfIndex_of_forall_dim`: if y is a zero in good position at which every datum of Definition 3.2.1 (preprint 3.25) for the solution set has dimension d, then $\mathrm{IsSurjLinOfIndex}(f, y, d)$ — the data at y from the centre theorem at y have dimension $\mathrm{ind}\, f'(y)$, and that dimension is d.

The index is locally constant along the solution set `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index_locally_constant`. *Statement.* For an sc-Fredholm section of a strong bundle whose charts have tame base retractions, a zero x_0 in good position: there are an open $V_x \ni x_0$ and d with $\mathrm{IsSurjLinOfIndex}(f, y, d)$ at every zero $y \in V_x$; in particular $\mathrm{ind}\, f'(y) = \mathrm{ind}\, f'(x_0)$. *Proof.* Theorem 3.6.8 (1) (preprint 3.54) (§III.43) gives an open $V_{x,1}$ of nearby good position; Theorem 3.6.8 (2) (preprint 3.54) with the index on it gives regular data D of dimension d with source in $V_{x,1}$ along which every datum has dimension d; on the

source of D, every zero is in good position with all its data of dimension d, and the previous paragraph applies. □ The proof does not repeat the kernel computation of the matrix argument (Finding 188): "the dimension is locally constant" (from the re-centring and Sard's lemma in fixed dimension, §III.35) and "the dimension is the index" give "the index is locally constant". Along the solution set this is the index stability of the text's Proposition 6.2.2 (preprint 6.2), whose proof needs exactly the surjectivity of §III.43. *Theorem 3.6.8 (preprint 3.54) with the index* `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index`: on an open $V_x \ni x_0$ every zero is in good position with IsSurjLinOfIndex(f, y, d), and the solution set has at x_0 regular data of dimension d whose chart domain *is* V_x, every datum at a point of the solution set in V_x having dimension d.

Theorems 3.1.26, 3.1.22 and 3.6.11 (preprint 3.23, 3.19 and 3.57) with the index. Theorem 3.1.26 (preprint 3.23) `Chapter3/GlobalIFTIndexG · theorem_3_1_26_index`: for an sc-Fredholm section with tame base retractions and a zero x in good position, an open $U \ni x$ and d such that every zero $y \in U$ is in good position with IsSurjLinOfIndex(f, y, d), the zeros in U are smooth, and $S(f, U)$ is a sub-M-polyfold with tame induced structure equivalent to a d-dimensional manifold with boundary with corners — the proof of the boundary case from the nearby good position (§III.37) with Theorem 3.6.8 (2) (preprint 3.54) with the index. Theorem 3.1.22 (preprint 3.19) `Chapter3/GlobalIFTIndexG · theorem_3_1_22_index`: for $d_X \equiv 0$ the same, with degeneracy index 0 everywhere (Lemma 2.4.6 (preprint 2.26)): "a smooth manifold whose dimension agrees with the Fredholm index", now with the tameness of the induced structure. Theorem 3.6.11 (preprint 3.57) `Chapter3/GlobalIFTIndexG · theorem_3_6_11_index`: under the hypotheses of §III.37 (good position at every zero, $S \neq \emptyset$), at every $x \in S$ there are regular data of a dimension d, shared by every datum at the points of S in their chart domain, with IsSurjLinOfIndex(f, y, d) at every zero y of that domain.

III.45 The kernel as tangent space, good position at interior zeros, and the filler condition as a theorem

"$\ker f'(y) = T_y S(f, U)$", in a chart `Chapter3/GlobalIFTIndexG · theorem_3_6_8_ker_eq_tangent`. *Statement.* For an sc-smooth section of a strong bundle whose charts have tame base retractions and a zero x_0 in good position, there are the strong bundle chart c centred at x_0 of Definition 3.6.6 (preprint 3.53), the conjugating sc-diffeomorphism $\psi : U \to U' \subseteq C'$ with inverse ψ' and $\psi(0) = 0$, the finite-dimensional $K = \ker D(h + t)(0)$, the solution graph $\sigma : V' \to E'$ (C^1, $\sigma(0) = 0$, $D\sigma(0) = 0$, V' relatively open in $K \cap C'$), an open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$, and an open $V_{x,2} \subseteq V_x$ containing x_0, such that at every zero $y \in V_{x,2}$, with $\psi(\varphi(y)) = v + \sigma(v)$, and every level-1 point z over $\varphi(y)$:

$$D\psi(z)(\ker \mathrm{Lin}(z)) = N_v = \{n + D\sigma(v)n \mid n \in K\},$$

the tangent space at v of the graph, which is the solution set in the chart $\psi \circ \varphi$ of its tangent germ data. *Proof.* From the kernel statement at the nearby zeros (§III.32, `Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_nearby_kernel`): z lies on level 1 of the germ of g, $\ker Dg(z) = D\psi'(\psi z)N_v$, $\ker Dg(z) \subseteq T_z O$, and $\mathrm{Lin}(z) = Dg(z)$ on $T_z O$ (the bridge for germs, which needs exactly z in the germ and $0 \in O$, `Chapter3/LinearizationChartsG · linAt_eq_germDeriv`; whole-chart `Chapter3/LinearizationCharts · linAt_eq_scDeriv`), so $\ker \mathrm{Lin}(z) = \ker Dg(z)$; and $D\psi(z)D\psi'(\psi z) = 1$ on level 0 by the chain rule for $\psi \circ \psi' = 1$ on U'. □ The intrinsic statement, with $T_y S$ as a subspace of $T_y X$ (Definition 2.5.8 (preprint 2.38)), is not formalized: $T_y X$ has no linear structure in the formalization (Chapter 2 gap G.13); the chart form is the statement (Finding 172).

Good position at a zero with $d_X = 0$ is surjectivity. *Linear algebra* Chapter3/GlobalIFTIndex · isInGoodPositionIn_of_subset: a subspace N of $T \subseteq E_0$ is in good position inside T to every cone C containing T — $N \cap C = N$ has interior in N, and any algebraic complement of N in T is a good complement with any constant, both sides of the condition being true. *The statement* Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_linAt_surj_of_degIndex_eq_zero: let f be an sc-smooth section, y a zero with (f, y) an sc-Fredholm germ and $d_X(y) = 0$, c a strong bundle chart around y with tame base retraction, and $\mathrm{Lin}(z)$ onto K_z at the level-1 point z over $\varphi(y)$; then (f, y) is in good position. *Proof.* The model of c is a tame retract, so $d_C(\varphi(y)) = d_X(y) = 0$ by the strong form of Proposition 2.5.6 (preprint 2.36) Chapter2/TameRetracts · degIndex_eq_chartDegIndex_of_isTameRetract; no coordinate of the presentation vanishes at $\varphi(y)$, so the partial quadrant at $\varphi(y)$ is the whole space, and the kernel condition of the assembly step holds by the linear algebra. The assembly step with all of K_z in the range Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj concludes. □ For an sc-Fredholm section with tame base retractions, "$f'(y)$ onto in some chart" therefore gives good position at a zero with $d_X(y) = 0$ Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt (y is smooth, so the germ is sc-Fredholm); conversely good position gives "$f'(y)$ onto" in the chart of Definition 3.6.6 (preprint 3.53) Chapter3/GlobalIFTIndexG · StrongBundle.IsGoodPositionGerm.isLinSurjAt; so at such a zero the two are equivalent Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_iff_isLinSurjAt — the text's remark after Theorem 3.6.8 (preprint 3.54). In a non-tame chart $d_C(\varphi(y))$ can exceed $d_X(y)$, and the cone would have to be transported from a chart realizing $d_X(y)$; the standing tameness hypothesis makes that unnecessary (Finding 173).

Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) with "$f'(x)$ surjective". Theorem 3.1.22 (preprint 3.19) Chapter3/GlobalIFTIndexG · theorem_3_1_22_of_surjective: for $d_X \equiv 0$, an sc-Fredholm section with tame base retractions, $f(x_0) = 0$ and $f'(x_0)$ onto in some chart: the equivalence gives good position at x_0, and Theorem 3.1.22 (preprint 3.19) with the index (§III.44) applies. Theorem 3.6.10 (preprint 3.56) with the tameness and the regular data Chapter3/GlobalIFTIndexG · theorem_3_6_10_tame: for $d_X \equiv 0$ and good position at every zero, $S \neq \emptyset$: Theorem 3.6.11 (preprint 3.57) with the index (§III.44) and Lemma 2.4.6 (preprint 2.26) — the solution set is a sub-M-polyfold with tame induced structure, a manifold of locally constant dimension without boundary, with regular data at every zero of a dimension equal to the index at the zeros of their chart domain. With "$f'(x)$ surjective at every zero" Chapter3/GlobalIFTIndexG · theorem_3_6_10_of_surjective: the equivalence at every zero, then the previous statement.

The converse of the matrix argument, and the filler condition as a theorem. *The converse* Chapter3/GlobalIFTIndex · normal_injective_of_exists_range_scDeriv_eq_ρ. Let z be a level-1 point over a point z_0 of the retract with $\ker Dg(z) \subseteq T_{z_0}O$, and suppose $\rho(z_0)k \in Dg(z)(T_{z_0}O)$ for every $k \in F$. Then $Dr(z)b = 0$ and $\rho(z_0)Dg(z)b = Dg(z)b$ imply $b = 0$: choose $a \in T_{z_0}O$ with $Dg(z)a = \rho(z_0)Dg(z)b = Dg(z)b$; then $a - b \in \ker Dg(z) \subseteq T_{z_0}O$, so $b \in T_{z_0}O$; writing $b = Dr(z)w$, idempotence gives $b = Dr(z)b = 0$ Chapter3/LinearizationCharts · scDeriv_r_idem_at. The surjectivity of $Dg(z)$ is not needed for this direction. With the direction of §III.33 Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective, under $Dg(z)$ onto and $\ker Dg(z) \subseteq T_{z_0}O$ (and g agreeing with **f** on O near a zero z_0 of **f**): $\rho(z_0)F \subseteq Dg(z)(T_{z_0}O)$ iff condition (3) of Definition 3.1.4 (preprint 3.4) holds at z_0 Chapter3/GlobalIFTIndex · exists_range_scDeriv_eq_ρ_iff_normal_injective — the "iff" of Finding 100, both directions formalized. Both are linear algebra in the operator $Dg(z)$ and hold for a germ at the points of its germ Chapter3/GlobalIFTIndexG · normal_injective_of_exists_range_deriv_eq_ρ_germ Chapter3/GlobalIFTIndexG · exists_range_deriv_eq_ρ_iff_normal_injective_germ. *The filler condition* Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm: for an sc-Fredholm section with tame base retractions and a zero x_0 in good position, (f, x_0) is in good position with the filler condition at

the nearby zeros `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled`. *Proof.* Take the chart and filler of the kernel statement (§III.32) and the open $V_{x,1}$ of nearby good position (§III.43); let N be an open neighbourhood of 0 on which $\mathbf{f} = g$ on O and $O \cap N = \varphi(V_{x,2} \cap V_{x,1})$ (`Chapter2/MPolyfold · exists_isOpen_image_eq`). At a level-1 zero z_1 of g over a point of $O \cap N$, on level 1 of the germ of g (the condition asks no more, §III.33): $z_0 = \varphi(y)$ for a zero y of f in $V_{x,2} \cap V_{x,1}$; y is in good position, so $\mathrm{Lin}(z_1)$ is onto K_{z_0} in the chart c (§III.44); the kernel statement gives $\ker Dg(z_1) \subseteq T_{z_0}O$; and the converse gives condition (3) at z_0. □ So the added hypothesis of the first forms of Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19) (§§III.33, III.37) holds whenever their other hypotheses do: it was never a restriction, but proving that needed the surjectivity of §III.43 (Finding 174).

III.46 What the second addendum taught us

1. **The text's Theorem 3.6.8 (1) (preprint 3.54) is true, and the gap of its proof can be closed.** The surjectivity of $f'(y)$ at the zeros near a zero in good position does not follow from Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55), as the text (and the book's arXiv version) says; it follows from an implicit function theorem *at the nearby zero* for the perturbed filler $g + \mu\rho(r(\cdot))k$, one smooth direction k at a time, whose curve of zeros is tangent to the retract by condition (2) of the filler `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_chart_onto` `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'` (§§III.41–III.43). With it, Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) hold with the text's hypotheses.
2. **The uniformity problem disappears when the contraction regions are written down.** Theorem 3.3.3 (preprint 3.30) is applied at the centre to the k-extended contraction germ, whose regions in a are those of the unperturbed germ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction`; only the μ-interval depends on k, and the bounds of the perturbation scale linearly in k `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds`. The black-box use of Theorem 3.5.11 (preprint Corollary 3.47) for an extended problem gives neighbourhoods depending on the extension.
3. **The index raise of Theorem 3.4.2 (preprint 3.38) is what makes the perturbation C^1 on the shifted levels** (§III.42), and the surjectivity of the reduced finite-dimensional problem comes from a dimension count on level 0 `Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.surjective_DH_and_ker_eq`, not from inverting $1 - D_2B$.
4. **The index is locally constant along the solution set** `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index_locally_constant`, the special case of the text's Proposition 6.2.2 (preprint 6.2), and the dimension of the solution set is the Fredholm index at every zero, in every chart `Chapter3/GlobalIFTIndexG · theorem_3_1_22_index` (§III.44); $\ker f'(y)$ is the tangent space of the solution set in the chart of the proof `Chapter3/GlobalIFTIndexG · theorem_3_6_8_ker_eq_tangent` (§III.45).
5. **The filler condition at the nearby zeros was never a restriction** `Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm`, and good position at a zero with $d_X = 0$ is surjectivity for a tame strong bundle `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_iff_isLinSurjAt` (§III.45), so Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) hold with "$f'(x)$ surjective".
6. **Theorem 3.1.21's (preprint 3.18) "readily verified" is a construction**: translation charts of $\mathbb{R}^n$, the pull-back bundle as a structure, the product rule for a scalar factor, and the extension of a basic germ by inert parameters `Chapter3/ParamPerturbation · IsBasicGerm.paramExtend` `Chapter3/ParamPerturbationG · theorem_3_1_21_preprint` (§III.40).

7. **What is left of Chapter 3**: Proposition 3.6.5 (preprint 3.52) as an sc-Fredholm operator on T_xX; $\ker f'(y) = T_yS$ with $T_yS \subseteq T_yX$; Theorem 3.7.1 (preprint 3.58) for a general sc-splitting; Lemma 3.8.13 (preprint 3.70) for a general presentation of the partial quadrant; Theorem 3.2.3 (preprint 3.27) for sets whose data violate the second regularity condition; and Proposition 6.2.2 (preprint 6.2) beyond solution sets (since proved, with the text's hypotheses for paths on $\mathbb{R}$, §III.49). The only added hypotheses are the closedness of good complements and the regularity conditions of Theorem 3.2.3 (preprint 3.27) for general subsets. All of this is closed since, and neither hypothesis remains (§§III.56–III.57, III.61–III.63).

III.47 The tameness hypothesis repaired

The implicit function theorems of §§III.23 and III.32–III.45 were first formalized with the hypothesis, `htame` in the Lean, that the base retraction of **every** strong bundle chart compatible with the atlas is tame; the phrase "a strong bundle whose charts have tame base retractions" in those sections describes it. No strong bundle over a nonempty base satisfies it, so the statements that assumed it were vacuously true (Finding 190). This section records why, and what replaced it. The proofs told above are unchanged; the hypotheses and the charts of the conclusions changed, and the names of the theorems did not.

Why the hypothesis fails. Compatibility of strong bundle charts (Definition 2.6.4 (preprint 2.50)) constrains the transition maps between the retracts K, not the retraction whose image K is. From a compatible chart with base retraction r on $U \subseteq C$, the chart $(1, \varphi)$ onto $\{1\} \times O$, with model $\mathbb{R} \oplus E$, quadrant $[0,\infty) \times C$ and base retraction $r'(a,u) = (1, r(u))$ on $[0,\infty) \times U$, has the transition maps of the first chart composed with affine maps, so it is compatible with every chart the first one is; but r' lowers the degeneracy index at $(0,u)$. The last point is machine-checked. The constant retraction $(a,w) \mapsto (1,0)$ of $[0,\infty) \oplus W$ `Chapter3/TameVacuity · ScRetraction.halfLineConst` has retract $\{(1,0)\}$ `Chapter3/TameVacuity · ScRetraction.halfLineConst_retract`, and its product with any sc-smooth retraction r of a nonempty U is not tame `Chapter3/TameVacuity · ScRetraction.halfLineConst_prod_not_isTame`: the degeneracy index of a product is the sum of the indices `Chapter2/FibredProduct · degIndex_prod`, so it is $1 + d_C(u)$ at $((0,0),u)$, $u \in r(U)$, and $d_C(u)$ at the image $((1,0),u)$, against condition (1) of Definition 2.5.2 (preprint 2.34). That the chart $(1,\varphi)$ is compatible is not formalized.

The new hypotheses. The text asks for the tameness of particular charts: the chart of an sc-Fredholm germ has "a tame strong local bundle" $K \to O$ (Definitions 3.1.6 and 3.6.3 (preprint 3.6 and 3.50)), and the proof of Proposition 3.6.5 (preprint 3.52) starts from "a tame strong bundle chart". A *tame chart* is a compatible strong bundle chart whose base retraction is tame `Chapter3/TameGerms · StrongBundle.IsTameChart`. *Tame germ data* at x are the data of Definition 3.1.11 (preprint 3.11) (§III.6) in a tame chart `Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame`; *good position in a tame chart* adds Definition 3.6.6 (preprint 3.53) in that chart (§III.21) `Chapter3/TameGermsG · StrongBundle.IsGoodPositionGermTame`; and an sc-Fredholm section *with tame germ data* has them at every smooth point `Chapter3/TameGermsG · StrongBundle.IsScFredholmSectionTame`. Each implies the notion without tameness `Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame.isScFredholmGerm` `Chapter3/TameGermsG · StrongBundle.IsGoodPositionGermTame.isGoodPositionGerm`. They can be met: tameness depends only on the domain of a retraction and its values there `Chapter3/TameGerms · ScRetraction.IsTame.congr`, and the identity retraction is tame `Chapter2/Tame · ScRetraction.id_isTame`, so a compatible chart whose base retraction is the identity of U is tame `Chapter3/TameGerms · StrongBundle.isTameChart_of_eqOn_id`.

The restated theorems. The proofs used the tameness for three charts only (Finding 191): the chart

of the germ data at x_0, for $(1 - Dr(0))E \subseteq E_0$ in the linear part (§III.23) `Chapter3/GlobalIFTG · theorem_3_6_8_graph_proj_conj`; the chart $\psi \circ \varphi$ built from it, whose model is tame because r is (§§III.27, III.34) `Chapter3/GlobalIFTCenterG · theorem_3_6_8_tangentGermChart_proj_conj`; and the chart in which $f'(y)$ is onto at a zero with $d_X(y) = 0$ (§III.45). So each theorem now takes (f, x_0) in good position in a tame chart in place of good position and `htame`; where it concludes something at the other zeros it takes an sc-Fredholm section with tame germ data; its conclusions are good position in tame charts, and the charts it produces are tame. This applies to Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'`, to its first form, whose filler condition is now stated in a tame chart `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled Chapter3/GlobalIFTNearbySurjG · theorem_3_6_8_one`, to Theorems 3.1.26 and 3.1.22 (preprint 3.23 and 3.19) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_26' Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_22'`, to the global theorems, with good position in a tame chart at every zero `Chapter3/GlobalIFTTheoremsG · theorem_3_6_11 Chapter3/GlobalIFTTheoremsG · theorem_3_6_10 Chapter3/GlobalIFTTheoremsG · corollary_3_1_27`, to the forms with the index `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index`, and to the filler condition as a theorem `Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm`.

Reading the conclusion in the tame chart. At a nearby zero y the good position is proved in the chart at x_0 and carried to a chart centred at y (§III.31). That chart was the one Definition 3.1.11 (preprint 3.11) supplies at y, of which nothing says that it is tame. The chart change now gives the good position in **every** chart carrying germ data at y `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionAt_of_chart_dense`, the statement of §III.31 being its corollary `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`. A good position given in any chart therefore holds in the tame chart of the germ data `Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame`: in the chart of the given good position $Tf(0)$ maps onto $K_0 \supseteq \rho(0)F_\infty$, $\ker Dg(0) \subseteq T_0O$ and $C_0 = C$, which are the hypotheses of the chart change at $z = 0$; in the germ reading the point $z = 0$ lies in every germ, $Tf(0)$ is onto all of K_0, and the form of the chart change with all of K_0 in the range needs no Proposition 3.6.5 (preprint 3.52) (§III.31) `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart`. The same step gives good position from "$f'(y)$ surjective" at a zero with $d_X(y) = 0$: the surjectivity, given in one chart, holds in every chart `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt.linAt_surjective`, in particular in the tame chart of the germ data at y, where $d_C = d_X = 0$ `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`.

Theorems 3.1.18 and 3.1.21 (preprint 3.15 and 3.18) keep tame germ data. The proof of Theorem 3.1.18 (preprint 3.15) (§III.6) does not change the chart, so $f + s$ has germ data in the tame charts of f `Chapter3/TameGermsAddG · StrongBundle.IsScFredholmGermTame.sectionAdd Chapter3/TameGermsAddG · theorem_3_1_18_tame`. For Theorem 3.1.21 (preprint 3.18) (§III.40) the germ of $\tilde{f}$ at (r_0, x) is taken in the product chart $(r \mapsto r - r_0) \times \varphi$, whose base retraction $1 \times r$ on $\mathbb{R}^n \times U$ is tame when r is: the identity retraction of $\mathbb{R}^n$ is tame, so $d(a, u) = d_C(u)$, and the minimal subspaces and tangent spaces of the product are products `Chapter2/FibredProduct · ScRetraction.IsTameWith.prod Chapter3/ParamPerturbation · PullbackChartData.ofProd_isTame`. Hence $\tilde{f}$ has tame germ data `Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmGermTame.param Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmSectionTame.param`, and so has $F(r, x) = f(x) + \sum_i r_i s_i(x)$ `Chapter3/ParamPerturbationG · theorem_3_1_21_tame`, which is what the implicit function theorems need when applied to F.

Not done (Finding 199): the compatibility of the chart $(1, \varphi)$, that is, the vacuity of `htame` at the level of charts; and the passage from an atlas of tame charts to tame germ data (Definition 3.1.11 (preprint 3.11) □ Definition 3.6.3 (preprint 3.50)). The text builds the tame chart into Definition 3.6.3 (preprint 3.50), and so do the theorems. The passage does not hold in general (§III.65).

Germ data and the charts of a tame atlas (Findings 350–353, fourth addendum). The passage was to be made by re-centring a tame chart of the atlas at the point and transporting the filler and the conjugation to it. That strategy fails, because a filling is bound to its chart. *Condition (3) of Definition 3.1.4 (preprint 3.4) is a condition on the chart*: a filling germ in a chart with strong bundle retraction $R = (r, \rho)$ gives a topological isomorphism $\ker Dr(0) \cong \ker \rho(0)$, the linearization of the fill defect $y \mapsto [1 - \rho(r(y))]g(y)$ on $\ker Dr(0)$ `Chapter3/TameGermData · IsFillingGerm.nonempty_kerEquiv`; so in a chart with $\ker Dr(0) \neq 0$ and $\rho(0)$ injective no map fills anything `Chapter3/TameGermData · not_isFillingGerm_of_ker`. *A tame chart with no filling.* For the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ take the model $\mathbb{R} \oplus \mathbb{R}^n$ without boundary, the base retraction $r(a, u) = (0, u)$, the product of the constant retraction of $\mathbb{R}$ onto 0 with the identity `Chapter3/TameGermData · TrivialBundle.Reven`, the chart $x \mapsto (0, x)$, and $R((a, u), h) = ((0, u), h)$ `Chapter3/TameGermData · TrivialBundle.evenBundleChart`. It is compatible with the translation charts `Chapter3/TameGermData · TrivialBundle.isChart_evenBundleChart` and tame — the degeneracy index of $\mathbb{R} \oplus \mathbb{R}^n$ is 0 everywhere and $\mathbb{R} \oplus 0$ is an sc-complement of $T_x O = 0 \oplus \mathbb{R}^n$ `Chapter3/TameGermData · TrivialBundle.Reven_isTame` `Chapter3/TameGermData · TrivialBundle.isTameChart_evenBundleChart` — but $\ker Dr(0) = \mathbb{R} \oplus 0$ while $\rho(0) = 1$, so no map fills the principal part of any section in it `Chapter3/TameGermData · TrivialBundle.not_isFillingGerm_evenBundleChart`. *A tame atlas without germ data in its chart.* With the one-chart atlas $\{\text{this chart}\}$ `Chapter3/TameGermData · TrivialBundle.bundleEven` every chart of the atlas is tame `Chapter3/TameGermData · TrivialBundle.bundleEven_atlas_isTameChart`, the graph section of a C^∞ map has tame germ data — in the compatible translation charts `Chapter3/TameGermData · TrivialBundle.isScFredholmSectionTame_graphSection_even` — and in the chart of the atlas no section has a filling `Chapter3/TameGermData · TrivialBundle.tame_atlas_no_germData_in_atlas_chart`. So Definition 3.6.3 (preprint 3.50) must be the existential over all compatible tame charts, as formalized; read as "in a chart of the tame atlas" it would make every section of this bundle non-Fredholm. Whether germ data in some compatible chart give germ data in some compatible tame chart, for a tame atlas, was left open here (Finding 353, gap G.79): the transition between two charts is defined on the retracts only, while the filling lives off them, so it cannot simply be pushed forward. The answer is no, even for a tame strong bundle over a tame M-polyfold without boundary (§III.65); it is yes when the chart of the data is centred at an interior point of its partial quadrant (§III.65).

III.48 Definition 3.1.11 (preprint 3.11) on germs

The text states Definitions 3.1.4, 3.1.6 and 3.1.11 (preprint 3.4, 3.6 and 3.11) for germs: the filled version is "a germ $(g, 0)$", the sc$^+$-section a local one, and a germ of neighbourhoods (Definition 3.1.1 (preprint 3.1)) is a sequence U_m that may shrink with m. The first formalization asked g and s to be sc-smooth on the whole chart domain U, on the ground that a chart can be restricted. This section records why that was narrower, and the two devices on which the proofs in the text's germ reading rest: the calculus of germs with whole-chart maps, and the shrinking of the basic germ (Findings 260–334).

Germs are not restrictions. On $\mathbb{R}$ with the constant structure, $f(x) = \sum_n c_n (x - a_n)^{n+1} |x - a_n|$, $a_n = 1/(n+2)$, $c_n = (4(n+2))^{-(n+2)}$, is C^k on $U_k = (-1/(k+2), 1/(k+2))$ — the terms $n \leq k$ have their singular points outside, the tail is C^{k+2} — and C^{n+2} on no open set containing a_n. So f is an sc-smooth germ on (U_m) and sc-smooth on no open $U \ni 0$ `Chapter3/GermVsChart · exists_isScSmoothGerm_not_isScSmooth`. Restricting a chart on level 0 never makes the levels shrink, so the whole-chart reading is a restriction of the class of sections; for a section sc-smooth on X the two readings of Definition 3.1.11 (preprint 3.11) differ exactly in the regularity of g and s away from their germs (§III.6). Whether every germ datum can be traded for a whole-chart one is open: replacing the sc$^+$-germ by a whole-chart sc$^+$-section costs one level (the index raise of Theorem 3.4.2 (preprint 3.38)), and on level 0 no argument

was found in either direction (Finding 264). No theorem depends on the answer, since the theorems are proved in the germ reading.

The calculus with whole-chart maps. The text forms $g + s$, $g - s$, $[1 - \rho(r(y))]g(y)$ and $\Psi_*(g)$ for germs without comment. Sums, differences, scalar multiples and compositions with sc-operators of sc¹-germs are sc¹-germs with the explicit derivative families, on $G_1 \cap G_2$ for germs on two germs `Chapter3/GermCalculus · IsSc1GermWith.add` `Chapter3/GermCalculus · IsScSmoothGerm.add_inter`. The chain rule of §III.1 composes two germs *at* 0, but R and Ψ are applied at $(r(y), g(y))$, which is $(0, 0)$ only when $g(0) = 0$ and $0 \in O$; so the chain rule is proved again with the outer map sc¹ on a set V and the inner one a germ mapping G_0 into V, with no condition at a base point `Chapter3/GermCalculus · IsSc1With.comp_germ` `Chapter3/GermCalculus · IsScSmooth.comp_germ`, and with a whole-chart inner map ψ, $\psi(0) = 0$, on the preimage germ `Chapter3/GermCalculus · IsScSmoothGerm.comp_preimage`. Hence the fill defect of a germ is a germ on the same germ (§III.4); the push-forward $\Psi_*(g)$ is a germ on the germ of the points that ψ^{-1} maps into the germ of g, with $D(\Psi_* g)(0) = \Gamma(0) Dg(0) D\psi(0)^{-1}$ (§III.23); a germ plus a whole-chart map is a germ on the same germ (Theorem 3.1.18 (preprint 3.15), §III.6); and $(a, u) \mapsto g(u)$ is a germ on $\mathbb{R}^n \times G$ (Theorem 3.1.21 (preprint 3.18), §III.40). The strong bundle isomorphism Ψ and the basic germ h remain what they were.

A basic germ restricts. Definitions 3.1.7 and 3.3.1 (preprint 3.7 and 3.28) pass to every smaller germ with the same h and B: the contraction field after shrinking its radius, since every level of a germ is a relatively open neighbourhood of 0, and the sc-smoothness by the uniqueness of derivatives in the quadrant `Chapter3/BasicGermShrink · IsBasicGerm.mono`.

The shrunk basic germ G''. The text does not relate the germ G of the basic germ to the germs G_g, G_s of g and s, and nothing forces $\psi^{-1}(G_m) \subseteq (G_g)_m$. For conjugation data with base map ψ, inverse ψ' and basic germ on G, let $G''_m = \{x \in G_m \mid \psi'_m(x) \in (G_g)_m \cap (G_s)_m\}$, the preimage germ along the sc⁰-germ $\psi'|_G$ (ψ' is sc-smooth on all of U' and $\psi'(0) = 0$) `Chapter3/BasicGermShrink · ConjugationData.shrinkGerm`. The conjugation data shrunk to G'' have the same Ψ, h, B and conjugation identity `Chapter3/BasicGermShrink · ConjugationData.shrink` `Chapter3/BasicGermShrink · ConjugationData.basic_shrinkGerm`; a point of G''_m is sent by ψ'_m into $(G_g)_m \cap (G_s)_m$ `Chapter3/BasicGermShrink · ConjugationData.levelMap_mem_inter_of_mem_shrinkGerm`, and read on the E-side, $u \in U_m$ with $\psi_m(u) \in G''_m$ lies in both germs `Chapter3/BasicGermShrink · ConjugationData.mem_of_levelMap_base_mem`. *Corollary 3.5.3 (preprint 3.41) on G''*: every zero x of the normal form with $|x|_0$ small is ιx_1 with x_1 on level 1 of G'' — it is small on level 1 (Theorem 3.5.1 (preprint 3.40)), and the contraction field puts a level-1 ball around 0 inside G''_1 — so $\psi'_1(x_1)$ lies on level 1 of G_g and G_s `Chapter3/BasicGermShrinkIFT · ConjugationData.corollary_3_5_3_shrink`. This is the device of every argument at the nearby zeros: the level-1 lift $z_{0,1}$ of a zero near x_0 lies in the germ of g, where $Dg(z_{0,1})$ is defined (§§III.32, III.33, III.43, III.45). It relies on the lifts being small on level 1; a point small on level 0 only need not lie in G''_1. At the centre no shrinking is needed: 0 lies in every germ, and $Dg(0)$ is the same for all germs and families (§III.4).

What the proofs use of g and s. At 0: $Dg(0)$ and $Ds(0)$ (Propositions 3.1.5 and 3.6.5 (preprint 3.5 and 3.52), Definition 3.6.6 (preprint 3.53), the linear part of Theorem 3.6.8 (preprint 3.54); §§III.4, III.21, III.23, III.30). On level 0 near 0: continuity of g and of the sc⁺-part (Theorem 3.1.13 (preprint 3.12), §III.13; the local step of Theorem 5.2.2 (preprint 5.5), §V.6). At the level-1 lifts of the nearby zeros: Dg, through G'' (Theorem 3.6.8 (1) (preprint 3.54), §§III.32–III.33, III.43, III.45). Nowhere g or s on $U \cap E_m$ away from their germs. So every theorem of Chapter 3 holds in the germ reading with the text's hypotheses and its former conclusions, and no hypothesis is added; the statements that named the derivative of g on U at 0 (Proposition 3.6.5 (preprint 3.52), the linearization statements of Theorems 3.1.22 and 3.6.8 (preprint 3.19

and 3.54)) now name the germ, a derivative family Dg and $Dg(0)$, and Corollary 5.2.5 (preprint 5.8) in a chart has an sc⁺-germ on the shifted germ for its sc⁺-part (§V.7).

The switch. The canonical names now denote the germ reading: Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4 and 3.6.6 (preprint 3.11, 3.14, 3.50, 3.51 and 3.53) with the tame, filled and sc variants of good position `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm` `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection` `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm`, and the theorems built on them. The whole-chart notions and theorems keep their statements under the names `…WholeChart` and `…_wholeChart`, and each whole-chart notion implies its germ twin `Chapter3/TameGermsG · StrongBundle.IsScFredholmSectionWholeChart.isScFredholmSection` `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGermWholeChart.isGoodPositionGerm`. The non-vacuity witnesses are whole-chart data, hence germ data. Chapter 5 is stated with the germ notions: the local step of Theorem 5.2.2 (preprint 5.5) and the spanning sections changed locally, and Corollary 5.2.5 (preprint 5.8) in a chart changed its statement (§§V.6, V.12, V.7).

III.49 Definition 6.2.1 (preprint 6.1) and Proposition 6.2.2 (preprint 6.2): the index along paths

The text's Chapter 6 opens with the index of an sc-Fredholm germ at a smooth point which need not be a zero (Definition 6.2.1 (preprint 6.1), pp. 237–240 (preprint pp. 200–201)) and proves that it is constant along sc-smooth paths of smooth points (Proposition 6.2.2 (preprint 6.2), p. 240 (preprint p. 201)). Its proof is the device that §III.44 used along the zero set: stabilize by finitely many sc⁺-sections and apply the implicit function theorem. This section retells both: first with sc⁺-sections on all of X, which needs bump functions (Findings 340–348), then as the text does it, with locally defined sections on the restriction of the bundle to a chart domain and without bump functions (Findings 349, 360–367).

Definition 6.2.1 (preprint 6.1) `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt`. *f has index i at x* if there is an sc⁺-section s of P with $(f+s)(x)=0$ and $(f+s)'(x)$ Fredholm of index i in every strong bundle chart around x (the chart-intrinsic statement of §III.44). The text's $f-s$ with $s(x)=f(x)$ is $f+(-s)$; the text's section is local, this one global; the reading with local sections, and the agreement of the two readings under bump functions, close this section. *Well defined.* At a zero of an sc-Fredholm section the index in `IsLinOfIndex` is unique, because in a chart $\mathrm{Lin}(z)$ is a bounded operator into K_z (§III.44), so the quantification over corestrictions is not vacuous `Chapter3/IndexAlongPaths · StrongBundle.IsLinOfIndex.unique`. Adding an sc⁺-section s with $s(x)=0$ at a zero x of f does not change the index: the germ datum of f at x, with conjugation data dc, gives the index $\dim P_n - \dim Q$ of $f'(x)$ (Proposition 3.6.5 (preprint 3.52)), and Theorem 3.1.18 (preprint 3.15) gives germ data of $f+s$ in the same chart whose conjugation data are dc up to their values on U, since $(g+\hat{s}\circ r)-(\hat{s}_1\circ r+t_0)=g-t_0$ `Chapter3/IndexAlongPaths · StrongBundle.isLinOfIndex_sectionAdd_of_isZeroAt`. For two sections s, s' of the definition, $s'-s$ is an sc⁺-section vanishing at x `Chapter3/IndexAlongPaths · isZeroAt_sectionDiff_of_isZeroAt_sectionAdd` and $(f+s)+(s'-s)=f+s'$ `Chapter3/IndexAlongPaths · sectionAdd_sectionAdd_sectionDiff`, so the indices agree `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt.unique`. The text argues instead that $(t-s)'(x)$ is an sc⁺-operator, hence compact on every level, and invokes Proposition 1.1.11 (preprint 1.10); the project has no lemma that the linearization of an sc⁺-section is compact, and the germ data make it unnecessary. At a zero the index of the germ is that of $f'(x)$ `Chapter3/IndexAlongPaths · StrongBundle.isIndexAt_of_isZeroAt` `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt.isLinOfIndex_of_isZeroAt`. *Existence.* If X admits bump functions, every smooth point has an index: $f(x)$ is a smooth point of Y, Lemma 5.3.3 (preprint 5.11) gives an sc⁺-section s_0 with $s_0(x)=f(x)$, and for $s=-s_0$ the germ data of the sc-Fredholm section $f+s$

at its zero x (Theorem 3.1.18 (preprint 3.15)) give the index Chapter3/IndexAlongPaths · StrongBundle.exists_isIndexAt.

Along the zero set, without good position (Findings 240–242). Let X admit bump functions, f be an sc-Fredholm section with tame germ data and $f(x_0) = 0$; neither good position nor surjectivity at x_0 is assumed. In a chart centred at x_0, finitely many smooth vectors span a complement of the range of $\mathrm{Lin}(0)$, closed of finite codimension by Proposition 3.6.5 (preprint 3.52), and Lemma 5.3.3 (preprint 5.11) realizes them as allowable sc⁺-sections $s_1, \dots, s_k$ Chapter3/IndexLocallyConstantG · exists_spanning_sections_germ; with further sections that tilt the kernel into good position at a corner (the conditions (i)–(ii) of §V.19), the parametrized section $F(\lambda, x) = f(x) + \sum_i \lambda_i s_i(x)$ of Theorem 3.1.21 (preprint 3.18) is in good position at $(0, x_0)$ in a tame chart Chapter3/IndexLocallyConstantG · exists_sections_isParamGP_germ Chapter3/IndexLocallyConstantG · isGoodPositionGermTame_paramSection_of_isParamGP. So the index of F' is one integer d at the zeros of F near $(0, x_0)$ (§III.44), among them $(0, y)$ for every zero y of f near x_0. At $(0, y)$, in the product chart, $\mathrm{Lin}_F(0)(\delta\lambda, \delta u) = \mathrm{Lin}_f(0)\delta u + \sum_i \delta\lambda_i \widehat{s}_i(0)$, a finite-rank extension, so $d = k + \mathrm{ind}\, f'(y)$ Chapter3/IndexLocallyConstant · fredholmIndex_eq_of_param Chapter3/IndexLocallyConstantG · StrongBundle.isLinOfIndex_of_paramSection_isLinOfIndex. Hence $f'(y)$ has index $d - k$ in every chart at every zero y near x_0 Chapter3/IndexLocallyConstantG · index_locally_constant_of_admitsBumpFunctions, also as an sc-Fredholm operator on every level Chapter3/IndexLocallyConstantG · theorem_3_6_8_index_locally_constant_general. This is the text's device for Proposition 6.2.2 (preprint 6.2) in the case of a zero.

The local form. For X with bump functions, f with tame germ data and an sc⁺-section s with $(f + s)(x_0) = 0$, the index of the germ is one integer at the zeros of $f + s$ near x_0: $f + s$ has tame germ data (§III.47) and the previous paragraph applies to it Chapter3/IndexAlongPaths · isIndexAt_locally_constant_of_isZeroAt_sectionAdd. Hence along a path continuous at t_0 on which $f + s$ vanishes near t_0 the index is constant near t_0 Chapter3/IndexAlongPaths · isIndexAt_eventually_const_of_path, and on a preconnected parameter space a locally constant index is constant: by uniqueness the index is a locally constant function of t Chapter3/IndexAlongPaths · isIndexAt_const_of_forall_eventually. What remains is a section s, depending on t_0, that vanishes along the path — the text's $s(t, x)$ on $[0, 1] \times X$ — and the passage from the product back to X.

Bump functions on $\mathbb{R}^n \times X$. The device works on the product of Theorem 3.1.21 (preprint 3.18) (§III.40), which must admit bump functions when X does. A C^∞ function on $\mathbb{R}^n$ is sc-smooth on the M-polyfold $\mathbb{R}^n$ Chapter3/IndexAlongPaths · isScSmoothMap_rn_real_of_contDiff, products of sc-smooth functions are sc-smooth by the product rule in every chart Chapter3/IndexAlongPaths · MPolyfold.IsScSmoothMap.mul, and $(r, x) \mapsto \beta_1(r)\beta_2(x)$ has support in $\mathrm{supp}\,\beta_1 \times \mathrm{supp}\,\beta_2$ Chapter3/IndexAlongPaths · tsupport_prod_mul_subset; so $\mathbb{R}^n \times X$ admits bump functions Chapter3/IndexAlongPaths · MPolyfold.admitsBumpFunctions_prod_rn. Then for the pull-back $\tilde{f} = f \circ \pi$ and an sc⁺-section S of $\pi^* P$ with $\tilde{f} + S$ vanishing along the graph $t \mapsto (t, \varphi(t))$ near t_0, the index of $(\tilde{f} + S)'(t, \varphi(t))$ in every chart is one integer for t near t_0: $\tilde{f} + S$ has tame germ data (§§III.40, III.47), and the local constancy along its zero set applies Chapter3/IndexAlongPaths · isLinOfIndex_paramLift_sectionAdd_eventually_const. The proposition thereby reduces to two steps, the section along the path (E) and the index count on a slice (S) Chapter3/IndexAlongPaths · proposition_6_2_2_of_sections_of_slice.

(E): the text's section along the path Chapter3/IndexAlongPathsDevice · exists_isScPlusSection_pullback_of_path. For an sc-smooth section f (sc-Fredholm is not needed), an sc-smooth path $\varphi : \mathbb{R}^n \to X$ and t_0, there is an sc⁺-section S of $\pi^* P$ with $\tilde{f} + S = 0$ along the graph near t_0. Take a strong bundle chart c around $\varphi(t_0)$. First the path is reparametrized by a C^∞ map σ of $\mathbb{R}^n$ equal to the identity near t_0 with values in $\varphi^{-1}(\mathrm{dom}\,\varphi_c)$, $\sigma(t) = t_0 + b(t - t_0)(t - t_0)$ for a smooth bump b Chapter3/IndexAlongPathsDevice · exists_contDiff_eq_id_near_mapsTo; C^∞ self-maps of $\mathbb{R}^n$ are sc-smooth Chapter3/IndexAlongPathsDevice · isScSmoothMap_

rn_of_contDiff, so $\psi = \varphi \circ \sigma$ is an sc-smooth path that stays in the chart. The representative $t \mapsto \Phi(f(\psi(t)))$ is sc-smooth from the constant structure $\mathbb{R}^n$ into $E \oplus F$; Proposition 1.2.2 (preprint 1.17) makes it sc-smooth from $(\mathbb{R}^n)^1 = \mathbb{R}^n$ into $(E \oplus F)^1 = E^1 \oplus F^1$, and its F^1-component $\hat{e}_1$ is the curve of smooth points that the text's $s(t,x) = R(x, f(\varphi(t)))$ needs in F^1 — no new analysis is required. In the product chart $(r \mapsto r - t_0) \times \varphi_c$ of π^*P the local section is $\tau(a,u) = -R[1](u, \hat{e}_1(a + t_0))_2$, sc-smooth into $K[1]$ because the bundle retraction is sc-smooth on $[1]$; it is cut off by a bump of X equal to 1 near $\varphi(t_0)$ with support in the chart domain (Proposition 5.5.3 (preprint 5.30)) and extended by 0 (§V.9). Along the graph $\Phi(S(t,\varphi(t)))_2 = -R(\varphi_c(\varphi(t)), \Phi(f(\varphi(t)))_2)_2 = -\Phi(f(\varphi(t)))_2$, since $\Phi(f(\varphi(t))) \in K$ is fixed by R, so $\tilde{f} + S$ vanishes there for t near t_0, where $\sigma(t) = t$ and the bump is 1. The bump function and the reparametrization make the text's local section global; the text needs neither.

(S): the slice. The slice $S(t,\cdot)$ of a section of π^*P, read in Y through the fibres $(\pi^*Y)_{(t,x)} = P^{-1}(x)$, is a section of P Chapter3/IndexAlongPathsDevice · pullbackSlice Chapter3/IndexAlongPathsDevice · IsSection.pullbackSlice; $G(t,x) = 0$ iff the slice vanishes at x Chapter3/IndexAlongPathsDevice · isZeroAt_pullback_iff_pullbackSlice, and the slice of $\tilde{f} + S$ is $f + S(t,\cdot)$ Chapter3/IndexAlongPathsDevice · pullbackSlice_sectionAdd_paramLift. *The slice of an sc⁺-section is sc⁺* Chapter3/IndexAlongPathsDevice · StrongBundle.IsScPlusSection.pullbackSlice: for a chart c of the atlas of P, the product chart $(r \mapsto r - t) \times \varphi_c$ is a chart of the atlas of π^*P, and the local representative of $S(t,\cdot)$ is that of S composed with the sc-operator $u \mapsto (0,u)$ Chapter3/IndexAlongPathsDevice · ScBanach.prodSndLeftOp and projected. *The linearization on the slice direction*: in the product chart at (t,x), whose base retraction is $(a,u) \mapsto (a, r(u))$ Chapter3/IndexAlongPathsDevice · PullbackChartData.r_ofProd, $\mathrm{Lin}_G(0)(0,\delta u) = \mathrm{Lin}_{G(t,\cdot)}(0)\delta u$, by the chain rule for $u \mapsto \widehat{G}(0, r(u))$ Chapter3/IndexAlongPathsDevice · StrongBundle.linAt_ofProd_slice. *The index count*: if G is an sc-Fredholm section of π^*P whose slice $g = G(t,\cdot)$ is an sc-Fredholm section with $g(x) = 0$, and $G'(t,x)$ is Fredholm of index i in every chart, then $g'(x)$ is Fredholm of index $i - n$ in every chart: $\mathrm{Lin}_G(0)(\delta a, \delta u) = \mathrm{Lin}_g(0)\delta u + \mathrm{Lin}_G(0)(\delta a, 0)$ by linearity and the slice identity, the last term has rank at most n, and the parameter count of the zero-set case applies, the chart independence coming from the germ datum of g at x Chapter3/IndexAlongPathsDevice · StrongBundle.isLinOfIndex_pullbackSlice_of_isLinOfIndex. For $G = \tilde{f} + S$ ($n = 1$) this is $\mathrm{ind}(f,x) = i - 1$, with $S(t,\cdot)$ as the section of Definition 6.2.1 (preprint 6.1) Chapter3/IndexAlongPathsDevice · StrongBundle.isIndexAt_of_pullback_isLinOfIndex.

Proposition 6.2.2 (preprint 6.2) Chapter3/IndexAlongPathsDevice · proposition_6_2_2. *Let X admit sc-smooth bump functions, f be an sc-Fredholm section with tame germ data and $\varphi : \mathbb{R} \to X$ an sc-smooth path. Then* $\mathrm{ind}(f, \varphi(t))$ *is one integer for all t*; in particular $\mathrm{ind}(f,\varphi(0)) = \mathrm{ind}(f,\varphi(1))$. For each t_0, (E) gives S; the index of $\tilde{f} + S$ is constant near t_0 along the graph; (S) turns it into $\mathrm{ind}(f,\varphi(t)) + 1$; and $\mathbb{R}$ is connected. The paracompactness of $\mathbb{R} \times X$ and of π^*Y is supplied by Theorem 2.3.10 (preprint 2.16) Chapter3/IndexAlongPathsDevice · proposition_6_2_2'. *Against the text.* (i) The path is defined on $\mathbb{R}$, not on $[0,1]$: $\mathbb{R} \times X$ is the product of Theorem 3.1.21 (preprint 3.18) and has no boundary, and the text's "tame M-polyfold $[0,1] \times X$" is not used; a path on $[0,1]$ that extends sc-smoothly to $\mathbb{R}$ is covered, and by the reparametrization in (E) only a neighbourhood of each t_0 matters. (ii) The hypotheses are tame germ data, the reading of "tame strong bundle" of §III.47, and bump functions, which the text does not assume ("we do not need a sc-smooth partition of unity"): they are used for the stabilizing sections (Lemma 5.3.3 (preprint 5.11)) and for the cut-off of the section along the path, both globalizations of the text's locally defined sections. Without them the device runs on the restriction of P to a chart domain, as the text intends; that is the rest of this section. (iii) Definition 6.2.1 (preprint 6.1) is proved well defined without Proposition 1.1.11 (preprint 1.10). (iv) The two points the text's proof does not address — the tilting sections for the kernel condition at a corner and the uniformity in the zero — are those of the zero-set case above.

Without bump functions: the restriction to an open set (Findings 349, 360). The text restricts the bundle to a chart domain without comment ("in a chart, $s(y) = R(y, f(x))$"; "we do not need a sc-smooth partition of unity", preprint p. 202). For an open $V \subseteq X$, a chart of X whose domain lies in V is transported to a chart of the subtype V with the same model `Chapter3/RestrictOpen · MChart.restrictTo`, compatibility is transported `Chapter3/RestrictOpen · MChart.Compatible.restrictTo`, and the charts of X inside V form the atlas of the M-polyfold V `Chapter3/RestrictOpen · MPolyfold.restrictOpen`, whose levels are the traces of those of X `Chapter3/RestrictOpen · MPolyfold.mem_levelSet_restrictOpen_iff`; an sc-smooth map with values in V is sc-smooth into V `Chapter3/RestrictOpen · MPolyfold.IsScSmoothMap.codRestrict_restrictOpen`. The restriction $P|V$ lives on the pull-back $\iota^*Y = \{(v, y) \mid P(y) = v\}$ of Definition 2.6.7 (preprint 2.53), as the restricted sections of Definition 5.3.13 (preprint 5.20) do (§V.23), and its fibres are those of P. A strong bundle chart of P with base domain inside V is transported with the same F, U, R, K `Chapter3/RestrictOpen · StrongBundleChart.restrictTo` `Chapter3/RestrictOpen · StrongBundleChart.Compatible.restrictTo`, and these charts form the atlas of $P|V$ `Chapter3/RestrictOpen · StrongBundle.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.restrictTo_mem_restrictOpen_atlas`; every chart of P restricts to $\operatorname{dom}\varphi \cap V$ `Chapter3/RestrictOpen · StrongBundleChart.restrictInter`. The restriction $f|_V$ of an sc-smooth, sc$^+$ or regularizing section is again one `Chapter3/RestrictOpen · StrongBundle.IsScSmoothSection.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.IsScPlusSection.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.regularizing_restrictOpen`. Two choices of the formalization: the paracompactness of V and of ι^*Y is supplied by Theorem 2.3.10 (preprint 2.16), and V is nonempty, for the junk values of the transported inverse charts.

Germ data and the index on $P|V$ (Findings 361, 362). The data of Definition 3.1.11 (preprint 3.11) on a relatively open U restrict to every relatively open $U_0 \subseteq U$ containing 0: the germs of the filling and of the sc$^+$-part are intersected with U_0, and $Dr(0)$ and $\rho(0)$ do not change, so the kernel isomorphism of condition (3) is the same `Chapter3/RestrictOpenGerms · IsFillingGerm.of_subset` `Chapter3/RestrictOpenGerms · IsScPlusGerm.of_subset`; the strong bundle isomorphism Ψ of the conjugation is restricted to $U_0 \lhd F \to \psi(U_0) \lhd F'$, where $\psi(U_0) = U' \cap \psi'^{-1}(U_0)$ is relatively open by the continuity of ψ', and the basic germ is shrunk to $G \cap \psi'^{-1}(U_0)$ as for G'' (§III.48), with the same P_n, Q, h, B `Chapter3/RestrictOpenGerms · ConjugationData.ofSubset` `Chapter3/RestrictOpenGerms · IsConjugateToBasicGerm.of_subset`. So a germ datum at $x \in V$ survives in its chart restricted to $\operatorname{dom}\varphi \cap V$ `Chapter3/RestrictOpenGerms · StrongBundleChart.germData_restrictInter`, which is tame when the chart is `Chapter3/RestrictOpenGerms · StrongBundle.IsTameChart.restrictInter`, and an sc-Fredholm section with tame germ data restricts to one on $P|V$ `Chapter3/RestrictOpenGerms · StrongBundle.IsScFredholmSectionTame.restrictOpen`. *The index.* At a zero $x \in V$, $f'(x)$ has index i in every chart of P iff $(f|_V)'(x)$ has index i in every chart of $P|V$ `Chapter3/RestrictOpenGerms · StrongBundle.IsLinOfIndex.restrictOpen` `Chapter3/RestrictOpenGerms · StrongBundle.IsLinOfIndex.of_restrictOpen`. The charts are not compared — a chart of $P|V$ need not come from one of P —: in either bundle the index is $\dim P_n - \dim Q$ for the germ datum (§III.44), whose restriction keeps P_n and Q `Chapter3/RestrictOpenGerms · StrongBundle.isLinOfIndex_restrictOpen_of_conjugationData`, and uniqueness identifies it with i. The same argument passes from $P|V$ to $P|W$ for an open $W \subseteq V$, where the chart restricted to $\operatorname{dom}\varphi \cap W$ is common to both and the two linearizations agree in it `Chapter3/IndexAlongPathsLocal · sectionRestrict` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsScPlusSection.restrictOpen_of_subset` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsLinOfIndex.restrictOpen_of_subset`; Definition 6.2.1 (preprint 6.1) restricts accordingly `Chapter3/RestrictOpenGerms · StrongBundle.IsIndexAt.restrictOpen` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAt.restrictOpen_of_subset`.

The stabilizing sections of a chart, without cut-off (Finding 363). In the zero-set argument above, bump functions enter only through Lemma 5.3.3 (preprint 5.11), as sc$^+$-sections through given smooth points of the fibres. That property — through every smooth point e of the fibre over a smooth point x there

is an sc⁺-section s with $s(x) = e$ — is isolated `Chapter3/IndexLocallyConstantLocal · StrongBundle.HasScPlusSectionsThrough`. Bump functions give it by Lemma 5.3.3 (preprint 5.11) `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`, and so does a chart of the atlas over the whole base, with no cut-off `Chapter3/IndexLocallyConstantLocal · StrongBundle.hasScPlusSectionsThrough_of_chart`: $\hat{e} = \Phi(e)_2$ is a smooth point of F, with a lift $\hat{e}_1 \in F^1$, and the section $y \mapsto \Phi^{-1}(\varphi(y), R(\varphi(y), \hat{e}))$ — the text's $R(\cdot, e)$ — has the local representative $z \mapsto R[1](z, \hat{e}_1)_2$, sc-smooth into $K[1]$ because the bundle retraction is sc-smooth on $[1]$; it is the cut-off section of Lemma 5.3.3 (preprint 5.11) with $\beta = 1$ (§V.9), and its value at x is e since $\Phi(e) \in K$ is fixed by R. The restriction $P|V$ to a chart domain V has such a chart: the chart itself, transported to V. The three steps of the zero-set argument — the spanning sections, the tilting sections that put the parametrized section in good position, the local constancy — are repeated with this property in place of bump functions `Chapter3/IndexLocallyConstantLocal · exists_spanning_sections_of_sections` `Chapter3/IndexLocallyConstantLocal · exists_sections_isParamGP_of_sections` `Chapter3/IndexLocallyConstantLocal · index_locally_constant_of_sections`; the support and norm bounds of Lemma 5.3.3 (preprint 5.11), which the index argument does not use, are dropped. *The index along the zero set with the text's hypotheses* `Chapter3/IndexLocallyConstantLocal · index_locally_constant_tame`: *let f be an sc-Fredholm section with tame germ data of any strong bundle and x_0 a zero; then there are an open $V \ni x_0$ and $i \in \mathbb{Z}$ with $f'(y)$ Fredholm of index i in every chart at every zero $y \in V$.* Take a chart around x_0 with domain V: $f|_V$ has tame germ data on $P|V$, which has sc⁺-sections through every smooth fibre point, so the index of $f|_V$ is one integer at its zeros near x_0, and at each of them it is the index on P. In sc form, with the sc-Fredholm property of $\mathrm{Lin}(z)$ at every zero in every chart `Chapter3/LinearizationChartsScG · StrongBundle.linAt_isScFredholm`, $\mathrm{Lin}(z)$ is sc-Fredholm of index i on every level `Chapter3/IndexLocallyConstantLocal · theorem_3_6_8_index_locally_constant_general_tame`. This is the localization that Finding 343 described; the gap ind $C(z) = 0$ of Finding 243 remains only without tame germ data.

Definition 6.2.1 (preprint 6.1) with local sections (Finding 364) `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAtLocal`. *f has index i at x* if there are an open $V \ni x$ and an sc⁺-section s of $P|V$ with $(f|_V + s)(x) = 0$ and $(f|_V + s)'(x)$ Fredholm of index i in every chart of $P|V$: the global definition at the beginning of this section, on $P|V$. *Well defined.* Two local sections, on V and on V', both restrict to $V \cap V'$ (the restriction of Definition 6.2.1 (preprint 6.1) to a smaller open set, above), where the uniqueness on $P|V \cap V'$ applies `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAtLocal.unique`. A global section gives a local one, with $V = X$ `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAt.isIndexAtLocal`, and at a zero the index is that of $f'(x)$ `Chapter3/IndexAlongPathsLocal · StrongBundle.isIndexAtLocal_of_isZeroAt`. *Existence, without bump functions.* On the domain V of a chart around x, the section through $f(x)$ of the previous paragraph, negated, is an sc⁺-section s of $P|V$ with $(f|_V + s)(x) = 0$, and Theorem 3.1.18 (preprint 3.15) gives the germ datum of $f|_V + s$ at x, hence an index `Chapter3/IndexAlongPathsLocal · StrongBundle.exists_isIndexAtLocal`. As for the global reading, a locally constant local index along a map from a preconnected space is constant `Chapter3/IndexAlongPathsLocal · isIndexAtLocal_const_of_forall_eventually`.

Proposition 6.2.2 (preprint 6.2) with the text's hypotheses (Finding 365) `Chapter3/IndexAlongPathsLocal · proposition_6_2_2_local`. *Let f be an sc-Fredholm section with tame germ data of a strong bundle $P : Y \to X$ — no bump functions — and $\varphi : \mathbb{R} \to X$ an sc-smooth path. Then* $\mathrm{ind}(f, \varphi(t))$*, read with local sections, is one integer for all t;* any two values of the index at $\varphi(0)$ and at $\varphi(1)$ agree `Chapter3/IndexAlongPathsLocal · proposition_6_2_2_local'`. The proof is that of (E), (S) and Proposition 6.2.2 (preprint 6.2) above, run near each t_0 on $P|V$, V the domain of a chart c around $\varphi(t_0)$ `Chapter3/IndexAlongPathsLocal · isIndexAtLocal_eventually_const_of_path`. The path is reparametrized into V as in

(E), and is then an sc-smooth path of the M-polyfold V. The section along the path is the one of (E) with the cut-off as a datum Chapter3/IndexAlongPathsLocal · exists_isScPlusSection_pullback_of_path_of_bump, taken equal to 1, which is possible because the chart $c|_V$ of $P|V$ covers its base Chapter3/IndexAlongPathsLocal · exists_isScPlusSection_pullback_of_path_of_chart. The product chart of $c|_V$ covers $\mathbb{R} \times V$, so the pull-back has sc$^+$-sections through every smooth fibre point without cut-off Chapter3/IndexAlongPathsLocal · hasScPlusSectionsThrough_pullback_rn_of_chart, and the index of $\tilde{f} + S$ is constant along the graph near t_0 Chapter3/IndexAlongPathsLocal · isLinOfIndex_paramLift_sectionAdd_eventually_const_of_sections. The slice relation (S), applied to $P|V$ unchanged, turns it into $\mathrm{ind}(f|_V, \varphi(t)) = i - 1$ near t_0, a local index of f, and the connectedness of $\mathbb{R}$ ends the proof. *Against the text*, what remains is the domain of the path: it is $\mathbb{R}$, and a path sc-smooth on $[0,1]$ only is not covered, the project having no M-polyfold $[0,1]$. The hypotheses on f are the text's, tame germ data being the reading of "tame strong bundle" of §III.47.

The two readings agree under bump functions (Finding 367) Chapter3/IndexAlongPathsLocalBump · StrongBundle.isIndexAtLocal_iff_isIndexAt. If X admits bump functions, a local section of Definition 6.2.1 (preprint 6.1) gives a global one with the same index Chapter3/IndexAlongPathsLocalBump · StrongBundle.IsIndexAtLocal.isIndexAt: the local section s on V is restricted to $W = \mathrm{dom}\,\varphi \cap V$ for a chart φ around x, read in that chart as $\tau : O \to F^1$, and cut off by a bump β supported in W and equal to 1 near x; the cut-off of Lemma 5.3.3 (preprint 5.11) is made here in a chart of P that need not belong to the atlas Chapter3/IndexAlongPathsLocalBump · StrongBundle.cutoffSection_mem_oneSet_of_isChart Chapter3/IndexAlongPathsLocalBump · StrongBundle.isScPlusSection_cutoffSection_of_isChart. Then $f + \beta\tau$ vanishes at x, and its linearization there has the index of $(f|_W + s)'(x)$: the two sections agree where $\beta = 1$, so the two linearizations coincide in the chart restricted to that set, and the germ datum of $f + \beta\tau$ at x carries the index to every chart. So the global definition of the beginning of this section and the text's local one are the same notion whenever the hypotheses of Chapter 5 hold, and only the text's is available without them.

III.50 The 2021 edition: the filling without tameness (Definitions 3.1.4, 3.6.1 (preprint 3.4, 3.48)) and a corner example

The 2021 filling. Definition 3.1.4 (p. 91) and its repetition Definition 3.6.1 (p. 134) of the Springer edition ask the strong bundle retraction $R(u,h) = (r(u), \rho(u)h)$ to cover an *sc-smooth* retraction r with $r(0) = 0$. The 2017 text and the preprint ask for a *tame* one. Otherwise the conditions are those of §III.3: - (1) $\hat{f} = g$ on O near 0; - (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in O; - (3) the linearization at 0 of $k(y) = [1 - \rho(r(y))]g(y)$, restricted to $\ker Dr(0)$, is a topological linear isomorphism onto $\ker \rho(0)$.

The formal definition is a structure with $r(0) = 0$ as a field and (1)–(3) in the germ reading of §III.48 Chapter3/Book2021Germs · IsFilling2021 Chapter3/Book2021Germs · HasFilling2021. The 2017 filling adds the tameness of r Chapter3/Book2021Germs · IsFilling2017 Chapter3/Book2021Germs · isFilling2017_iff.

The project's filling is the 2021 one. The germ filling of §III.3 never asked r to be tame Chapter3/GermFilling · IsFillingGerm. A 2021 filling is exactly a germ filling with $r(0) = 0$ Chapter3/Book2021Germs · isFilling2021_iff Chapter3/Book2021Germs · IsFilling2021.isFillingGerm. Conversely, a germ filling is a 2021 filling as soon as $0 \in O$, because points of O are fixed by r Chapter3/Book2021Germs · IsFillingGerm.isFilling2021; so is a whole-chart filled version Chapter3/Book2021Germs · IsFilledVersion.isFilling2021. In a chart centred at a point of X, $0 = \varphi(x_0)$ lies in O, so the text's standing assumption $r(0) = 0$ costs nothing there. A 2017 filling is a 2021 filling Chapter3/

Book2021Germs · IsFilling2017.isFilling2021. Both notions are satisfiable: over the identity strong bundle retraction, which is tame, every sc-smooth g fills itself Chapter3/Book2021Germs · isFilling2017_id, for instance $g = 0$ on $\mathbb{R}$ Chapter3/Book2021Germs · isFilling2017_rn.

A 2021 filling that is not a 2017 filling. Take the following data: - $E = \mathbb{R} \oplus \mathbb{R}$, $C = U = [0, \infty) \times \mathbb{R}$, $F = \mathbb{R}$; - $R((a, b), h) = ((0, b), 0)$ Chapter3/Book2021Germs · CornerFilling.Rc, a strong bundle retraction Chapter3/Book2021Germs · CornerFilling.isStrongBundleRetraction_Rc, so $r(a, b) = (0, b)$, $O = \{0\} \times \mathbb{R}$ and $\rho \equiv 0$ Chapter3/Book2021Germs · CornerFilling.ρCLM_Rc; - the zero section $\hat{f} = 0$ of $K \to O$, and $g(a, b) = a$ Chapter3/Book2021Germs · CornerFilling.gc Chapter3/Book2021Germs · CornerFilling.isScSmooth_gc.

The derivatives at 0 are $Dr(0)(a, b) = (0, b)$ and $Dg(0)(a, b) = a$ Chapter3/Book2021Germs · CornerFilling.scDeriv_r Chapter3/Book2021Germs · CornerFilling.scDeriv_g. Since $\rho \equiv 0$, the fill defect k is g itself Chapter3/Book2021Germs · CornerFilling.fillDefect_gc. Now check the three conditions: - (1) $g = 0$ on O; - (2) $g(y) = \rho(r(y))g(y) = 0$ forces $y \in O$; - (3) $\ker Dr(0) = \mathbb{R} \oplus 0$ Chapter3/Book2021Germs · CornerFilling.mem_ker_scDeriv_r, and $Dk(0)$ maps it by $(a, 0) \mapsto a$ onto $\ker \rho(0) = \mathbb{R}$ Chapter3/Book2021Germs · CornerFilling.kerMap Chapter3/Book2021Germs · CornerFilling.kerMap_bijective.

So g is a filled version Chapter3/Book2021Germs · CornerFilling.isFilledVersion_gc. With $0 \in O$ Chapter3/Book2021Germs · CornerFilling.zero_mem_O, it is a 2021 filling Chapter3/Book2021Germs · CornerFilling.isFilling2021_gc. But r is not tame: $d_C(1, 0) = 0$, while $d_C(r(1, 0)) = d_C(0, 0) = 1$, against Definition 2.5.2 (1) (preprint 2.33 (1)) Chapter3/Book2021Germs · CornerFilling.not_isTame_r. So g is not a 2017 filling Chapter3/Book2021Germs · CornerFilling.not_isFilling2017_gc, and the 2021 notion is strictly more general Chapter3/Book2021Germs · exists_isFilling2021_not_isFilling2017.

Sc-Fredholm germs and sections. With the 2021 filling in place of the filling of §III.6: - the 2021 sc-Fredholm germ (Definition 3.1.11 = 3.6.3) Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2021; - the 2021 sc-Fredholm section (Definition 3.1.16 = 3.6.4) Chapter3/Book2021Germs · StrongBundle.IsScFredholmSection2021; - the 2017 germ, with the 2017 filling Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2017.

In a chart centred at x_0, $0 \in O$ Chapter3/Book2021Germs · StrongBundle.zero_mem_O_of_chart. So the 2021 germ is the project's germ Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2021_iff Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm, and the 2021 section the project's section Chapter3/Book2021Germs · StrongBundle.isScFredholmSection2021_iff Chapter3/TameGermsG · StrongBundle.IsScFredholmSection. The 2017 germ is the germ with tame germ data of §III.47 Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2017_iff Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame, and 2017 implies 2021 Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2017.isScFredholmGerm2021. The graph section of a C^∞ map $\mathbb{R}^n \to \mathbb{R}^k$ satisfies both Chapter3/Book2021Germs · isScFredholmSection2021_graphSection Chapter3/Book2021Germs · isScFredholmGerm2017_graphSection. Whether every 2021 germ is a 2017 germ is the question of Finding 353, the passage from germ data in some compatible chart to germ data in a tame one (gap G.79). It is not: §III.65 gives a 2021 germ of a section of a tame strong bundle that is not a 2017 germ Chapter3/UntamedBundleTheorem · UntamedCorner.isScFredholmGerm2021_not_2017.

III.51 The 2021 edition: the theorems whose proofs do not use tameness

The project's proofs of the following results never used the tameness of r; each 2021 statement is the earlier theorem transported through the equivalences of §III.50: - **Proposition 3.1.5 (preprint 3.5)** (§III.4).

If g is a 2021 filling of $\hat{f}$ with $\hat{f}(0) = 0$, then $0 \in O$ by $r(0) = 0$, and $g(0) = \hat{f}(0) = 0$ by (1). So the germ form of the proposition applies Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_1_5. $Dg(0)$ maps $T_0O = \operatorname{range} Dr(0)$ into $K_0 = \operatorname{range} \rho(0)$. The restriction $T\hat{f}(0) : T_0O \to K_0$ is onto iff $Dg(0)$ is, and is Fredholm iff $Dg(0)$ is, with the same index Chapter3/Book2021Germs · proposition_3_1_5. - **Theorem 3.1.13 and Corollary 3.1.15 (preprint 3.12 and 3.13)** (§III.13) Chapter3/SectionCompactnessG · theorem_3_1_13_preprint Chapter3/SectionCompactnessG · corollary_3_1_15_preprint Chapter3/Book2021Germs · theorem_3_1_13 Chapter3/Book2021Germs · corollary_3_1_15. - **Theorem 3.1.18 (preprint 3.15)** (§III.6) Chapter3/FredholmSectionsG · theorem_3_1_18_preprint Chapter3/Book2021Germs · theorem_3_1_18. - **Theorem 3.1.21 (preprint 3.18)** (§III.40) Chapter3/ParamPerturbationG · theorem_3_1_21_preprint Chapter3/Book2021Germs · theorem_3_1_21. - **Proposition 3.6.5 (preprint 3.52)** (§§III.21, III.30) Chapter3/FredholmGermLinearizationG · proposition_3_6_5_preprint Chapter3/Book2021Germs · proposition_3_6_5.

So the 2021 change of the filling does not affect these statements, and the 2021 text's version of them is verified as printed.

III.52 The 2021 edition: Theorem 3.6.8 (preprint 3.54), the step on p. 138

The proof of Theorem 3.6.8 (pp. 137–138) starts from a germ in good position (Definition 3.6.6, §III.21). There $\ker T\tilde{f}(0)$ is in good position to the partial cone $C_0O = T_0O \cap C$ in T_0O. The proof passes to the good position of $\ker Dg(0)$ to C in E, the hypothesis of the implicit function theorem in the quadrant (§III.17). The passage uses an sc-complement of T_0O inside the subspace E_0 of C at 0, namely $\ker Dr(0)$. Proposition 2.5.3 (preprint 2.35) supplies it for a tame retract. In the linear part of §III.23 this is the hypothesis $(1 - Dr(0))E \subseteq E_0$ Chapter3/GoodPositionGermG · theorem_3_6_8_linear. The 2021 text keeps the sentence that the retract is tame by assumption, while its Definition 3.6.1 no longer assumes it.

The step does not go through without tameness. In the example of §III.50, every other hypothesis of the linear part holds: - g is a 2021 filling of $\hat{f} = 0$, $0 \in O$, $\hat{f}(0) = 0$, and $\hat{f}$ is a section of K; - g is sc^1 with derivative $Dg(0)$, and $Dg(0)$ is onto; - the germ is in good position in the chart Chapter3/GoodPositionGermG · IsGoodPositionAtGerm Chapter3/Book2021Germs · CornerFilling.isGoodPositionAtGerm_gc: $T\tilde{f}(0) : T_0O \to K_0 = 0$ is onto, and its kernel $T_0O = 0 \oplus \mathbb{R}$ is all of $T_0O \cap C$. Here $\ker Dg(0) = \operatorname{range} Dr(0)$ Chapter3/Book2021Germs · CornerFilling.ker_Lg_eq_range_Lr.

But $(1 - Dr(0))(1, 0) = (1, 0)$ is not in $E_0 = 0 \oplus \mathbb{R}$ Chapter3/Book2021Germs · CornerFilling.not_range_le_minimalSubspace. And $\ker Dg(0) = 0 \oplus \mathbb{R}$ is not in good position to C in E Chapter3/Book2021Germs · CornerFilling.not_isInGoodPosition_gc, for two reasons: - a complement P contains a point p with first coordinate 1, from writing $(1, 0) = n + p$; - the first coordinate vanishes at $(0, 1) \in \ker Dg(0) \cap C$, so by Lemma 3.8.8 (preprint 3.65) every good complement lies in $\{a = 0\}$ Chapter3/GoodPosition · lemma_3_8_8.

The whole is one statement Chapter3/Book2021Germs · theorem_3_6_8_tame_step. So this step of the proof does not hold as we read it for the 2021 definitions. The conclusion of the theorem still holds in this example: its solution set $O \cong \mathbb{R}$ is a manifold. Since $X = \mathbb{R}$ there has no boundary, even the proof of Theorem 3.6.10 (preprint 3.56), where $\partial X = \emptyset$, passes through this step.

The project states Theorem 3.6.8 and what rests on it (Theorems 3.1.22, 3.1.26, Corollary 3.1.27, Theorems 3.6.10 and 3.6.11 (preprint 3.19, 3.23, 3.24, 3.56 and 3.57)) with tame germ data, the 2017 notion (§III.47). For the untamed 2021 data, read with the definitions as printed, Theorem 3.6.8 does not hold as we read it (gap G.79, settled): §III.65 builds on the corner of §III.50 a tame strong bundle over $\mathbb{R}$ with a

2021 sc-Fredholm section whose solution set near a zero in good position is a closed half-line `Chapter3/UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not`. In that example the step on p. 138 is exactly where the proof needs the tame chart.

III.53 The 2021 edition: Theorem 3.5.1 (preprint 3.40) with the sequential conclusion

The 2021 statement of Theorem 3.5.1 (p. 118) adds a second conclusion. If points (a_k, w_k) of the solution set S converge to 0 in E_0, then for every m they eventually lie in C_m and converge to 0 there. The formal statement has both conclusions `Chapter3/Book2021Germs · theorem_3_5_1`, in the following setting: - h a basic germ on an sc-germ G of neighbourhoods of 0 in $C = \widetilde{C} \times W$, with $\widetilde{C}$ closed; - $U = G_0$, s an sc$^+$-germ on G with $s(0) = 0$, and $S = \{x \in U : h(x) + s(x) = 0\}$.

The neighbourhoods. Take the open sets $O'(m)$ of the local Theorem 3.1.13 of §III.12 `Chapter3/LocalCompactness · theorem_3_1_13_local` and put $O(m) = O'(m) \cap G_0$. These are nested, relatively open in C, contain 0 and lie in U. The closure of $S \cap O(m)$ is taken in E_0. It lies in C, which is closed and contains G_0, and it lies in the image of E_m, because it is the image of the compact set K_m of §III.12.

The sequences. A zero of $h + s$ solves $w = B(a, w) - \mathbf{s}(a, w)$ with $\mathbf{s}$ the W-component of s `Chapter3/Book2021Germs · IsBasicGerm.solves`. $\mathbf{s}$ is an sc^0-germ into W^1 with $\mathbf{s}(0) = 0$. So the local Corollary 3.1.15 of §III.12 applies `Chapter3/LocalCompactness · corollary_3_1_15_local`. For every m there are $y_k \in E_m$ with $\iota_m(y_k) = x_k$ for k large and $y_k \to 0$ in E_m; these are the lifts $L_m x_k$. "Converges on level m" is thus stated with level-m lifts, as in Corollary 3.1.15 (preprint 3.13).

III.54 The 2021 edition: cones and Lemmas 3.8.4–3.8.6, Corollary 3.8.7 (Appendix 3.8.1.3)

The appendix on "cones and partial quadrants in infinite dimensions" (pp. 145–149) is new in the 2021 edition. Its notions: - a *closed convex cone* K in a finite-dimensional space is a closed convex set with $K \cap (-K) = \{0\}$ and $tK \subseteq K$ for $t \geq 0$ `Chapter3/Book2021Items · IsClosedConvexCone`; - *a generates an extreme ray* if $a \in K \setminus \{0\}$, and $y \in K$, $a - y \in K$ imply $y \in \mathbb{R}_+ a$ `Chapter3/Book2021Items · IsExtremeRayGen`.

A closed convex cone is pointed `Chapter3/Book2021Items · IsClosedConvexCone.eq_zero`. $[0, \infty)$ is one, generated by the extreme ray of 1 `Chapter3/Book2021Items · isClosedConvexCone_Ici` `Chapter3/Book2021Items · isExtremeRayGen_one`.

The setting is the good position of §III.14: $C = T^{-1}([0, \infty)^n \oplus W)$ for a presentation T (the text's case is $T = \mathrm{id}$), and N a finite-dimensional subspace in good position. Within it: - $N \cap C$ is the cone of N `Chapter3/GoodPosition · quadCone`; - Γ, the union of the sets $\sigma_a = \{i : a_i = 0\}$ over $0 \neq a \in N \cap C$, is the vanishing index set `Chapter3/GoodPosition · vanIdx`; - $P : N \to \mathbb{R}^\Gamma$ is the coordinate projection `Chapter3/GoodPosition · sigmaCoord`, whose kernel is $N \cap (\mathbb{R}^{\Gamma^c} \oplus W)$.

The quadrant form of a pointed cone. If $N \cap C$ is pointed, a linear isomorphism $e : N \cong \mathbb{R}^{\dim N}$ carries $N \cap C$ onto $[0, \infty)^{\dim N}$ `Chapter3/Book2021Items · exists_quadrantEquiv_of_pointed`. It comes from the normal form of §III.14, $N \cap C \cong [0, \infty)^k \times N_0$ `Chapter3/GoodPosition · IsInGoodPosition.quadrantNormalForm`, in which pointedness forces $N_0 = 0$. The vectors $e^{-1}(e_j)$ generate extreme rays `Chapter3/Book2021Items · isExtremeRayGen_symm_single`. A nonzero $a \in N \cap C$ has a coordinate $e(a)_j > 0$

with $a - e(a)_j e^{-1}(e_j) \in N \cap C$ Chapter3/Book2021Items · exists_sub_single_mem. As in §III.14, this replaces the Krein–Milman-type Lemmas 3.8.1–3.8.3 (preprint 3.62–3.64): the cone is a quadrant at once. Those lemmas are now proved too, by Krein–Milman (§III.59).

Lemma 3.8.4 Chapter3/Book2021Items · lemma_3_8_4. Let $N^\perp$ be a good complement and $N \cap C$ a closed convex cone. Parts (1) and (3) say that $y_i = 0$ for $y \in N^\perp$ and $i \in \sigma_a$, a a generator, respectively $i \in \Gamma$. They hold without the cone hypothesis, because they are Lemma 3.8.8 (preprint 3.65) (§III.14) Chapter3/GoodPosition · lemma_3_8_8. For (2), $\Gamma = \bigcup_{a \text{ generator}} \sigma_a$, the inclusion $\supseteq$ is the definition. For $\subseteq$, let $0 \neq a \in N \cap C$ with $a_i = 0$, and take j and $b = e^{-1}(e_j)$ as above. Then $0 \leq (a - e(a)_j b)_i = -e(a)_j b_i \leq 0$ forces $b_i = 0$, so $i \in \sigma_b$. The text writes a as a sum of extreme-ray generators (Lemma 3.8.1); here only pointedness is used.

Lemma 3.8.5 Chapter3/Book2021Items · lemma_3_8_5. The text's statement is: either $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, in which case $\dim N = 1$ and $\Gamma = \emptyset$, or $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$. The proof assumes $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) \neq \{0\}$, and the formal statement is the dichotomy it shows. P is onto Chapter3/GoodPosition · sigmaCoord_surjective. - $\Gamma \neq \emptyset$. A point x of N with $x_i = 0$ for $i \in \Gamma$ lies in $N \cap C$, and so does $-x$ Chapter3/GoodPosition · mem_quadCone_of_vanIdx, so $x = 0$. - $\Gamma = \emptyset$ *and* $N \neq 0$. The kernel is all of N. If $\dim N \geq 2$, the generators b_0, b_1 have all coordinates positive (no index vanishes on a nonzero point of the cone). Then $b_0 - t b_1 \in N \cap C$ for small $t > 0$, but its e-coordinate j_1 is $-t < 0$. - $N = 0$. The kernel is $\{0\}$, and the first alternative holds.

Read literally, the first alternative does not hold as we read it for $N = 0$ Chapter3/Book2021Items · lemma_3_8_5_not_literal. In $E = \mathbb{R}$, $C = [0, \infty)$, the subspace $N = 0$ is in good position, $N \cap C = \{0\}$ is a closed convex cone, and $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$; but $\dim N = 0$ and $N \ncong \mathbb{R}$. The dichotomy holds there through its other alternative.

Lemma 3.8.6. (1) If $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$, then P is a linear isomorphism carrying $N \cap C$ onto $[0, \infty)^\Gamma$ Chapter3/Book2021Items · lemma_3_8_6. This needs no cone hypothesis: P is injective by hypothesis and onto. A point whose Γ-coordinates are ≥ 0 is in $N \cap C$ by the segment argument of §III.14 when $\Gamma \neq \emptyset$; when $\Gamma = \emptyset$, $N = 0$. (2) If $N \cap C$ is a closed convex cone, $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$ and $N \neq 0$, then $\dim N = 1$ by Lemma 3.8.5. A linear isomorphism $N \cong \mathbb{R}$ then carries $N \cap C$ onto $[0, \infty)$ Chapter3/Book2021Items · lemma_3_8_6_two. As printed, (2) does not hold as we read it for $N = 0$, by the same example Chapter3/Counterexamples2021 · Counterexamples2021.not_lemma_3_8_6_two (the narrower row V3.120). With $N \neq 0$ its hypotheses are met by $N = E = \mathbb{R}$, $C = [0, \infty)$ Chapter3/Book2021Items · halfLine_witness.

Corollary 3.8.7. If $N \cap C$ is a closed convex cone, it is a partial quadrant in N, and $(N, N \cap C) \cong (\mathbb{R}^{\dim N}, [0, \infty)^{\dim N})$ Chapter3/Book2021Items · corollary_3_8_7. The first part is Proposition 3.1.25 (preprint 3.22) (§III.14), which needs no cone hypothesis Chapter3/GoodPosition · proposition_3_1_25. The second is the quadrant form above, which the text also derives on p. 149.

III.55 What the 2021 additions taught us

- **The 2021 filling is the formalization's.** The germ filling never asked for tameness. The 2021 text drops it, so the formal definitions and the printed ones now agree Chapter3/Book2021Germs · isFilling2021_iff Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2021_iff. The 2017 notion is strictly narrower Chapter3/Book2021Germs · exists_isFilling2021_not_isFilling2017.
- **One proof step needs the dropped hypothesis.** All the theorems of Chapter 3 that the project proved without tameness hold for the 2021 definitions as printed. The passage of Theorem 3.6.8 from good position in $T_0 O$ to good position in E (p. 138) does not hold as we read it without a tame

chart `Chapter3/Book2021Germs · theorem_3_6_8_tame_step`. The implicit function theorems remain verified with tame germ data; for untamed data Theorem 3.6.8 does not hold as we read it (§III.65; G.79 settled).

- **Krein–Milman is not needed.** A pointed cone in good position is a quadrant `Chapter3/Book2021Items · exists_quadrantEquiv_of_pointed`. Lemma 3.8.4 (2) follows by one subtraction, and (1) and (3) need no cone hypothesis. (Lemmas 3.8.1–3.8.3, which use it, are proved in §III.59 for the comparison.)
- $N = 0$. Lemma 3.8.5, read as an implication, and Lemma 3.8.6 (2) need $N \neq 0$ `Chapter3/Book2021Items · lemma_3_8_5_not_literal`. The dichotomy the proof shows holds for all N.

III.56 Theorem 3.2.3 (preprint 3.27) under the tangent germ property alone; the good complement is closed

§III.24 proved Theorem 3.2.3 (preprint 3.27) for data with two regularity conditions, (a) δ sc-smooth on V as a map on a set and (b) $\pi(O) \subseteq C$, and with the good complement Y closed. None of the three is a hypothesis any more.

The good complement of Definition 3.2.1 (preprint 3.25) is closed. Definition 3.2.1 as written `Chapter3/GoodComplementClosed · MPolyfold.TangentGermChart2021` asks for N in good position in the sense of Definition 3.1.24 (preprint 3.21), whose good complement is an *sc-complement*; it defines δ on V only, and reads "sc$^\infty$-germ" with k-dependent neighbourhoods, the weakest reading (§III.24). An sc-complement is closed `Chapter3/GoodPositionSc · IsScGoodComplement.isClosed` and makes N smooth by Proposition 1.1.7 (preprint 1.6) `Chapter3/GoodPositionSc · IsScGoodComplement.le_smooth`. So the book's datum is the project's datum, with δ extended by 0 off V `Chapter3/GoodComplementClosed · MPolyfold.TangentGermChart2021.toTangentGermChart`; conversely a closed good complement of a smooth N is an sc-complement `Chapter3/GoodComplementClosed · MPolyfold.TangentGermChart.toTangentGermChart2021`. The two tangent germ properties coincide `Chapter3/GoodComplementClosed · MPolyfold.hasTangentGermProperty2021_iff`, and Lemma 3.6.9 (preprint 3.55) with the book's good complement is the lemma of §III.22, whose closedness hypothesis the sc-complement supplies `Chapter3/GoodComplementClosed · lemma_3_6_9`. The "Y closed" of §§III.22, III.24 and III.32 is therefore the text's own hypothesis.

Condition (b) is a choice of the complement. Let $(\varphi, N, Y, V, \delta)$ be a datum at x. In a presentation of C by functionals ℓ_i, let Σ be the indices that vanish on some nonzero point of $N \cap C$ (§III.14). The interior of $N \cap C$ in N is $\{\ell_i > 0,\ i \in \Sigma\}$ `Chapter3/TangentGermProjecting · interior_quadCone`, and the ℓ_i, $i \in \Sigma$, vanish on every good complement (Lemma 3.8.8 (preprint 3.65)) `Chapter3/TangentGermProjecting · IsGoodComplement.coordFun_eq_zero`.

1. *A projecting complement.* There is a continuous projection $\pi' : E \to N$ with $\pi'(C) \subseteq N \cap C$ whose kernel Y' is a closed good complement `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.exists_projData` `Chapter3/GoodComplementProjection · exists_proj_goodComplement` (§III.29).
2. *The re-graphing map* $g(v) = \pi'(v + \delta(v))$, $V \to N \cap C$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.regraphMap`. It has $g(0) = 0$ and $Dg(0) = 1$, because $D\delta(0) = 0$ and $\pi' = 1$ on N `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.hasFDerivWithinAt_regraphMap_zero`. It preserves the coordinates ℓ_i, $i \in \Sigma$, since these vanish on Y and on Y'; hence $g(v)$ is interior iff v is `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.regraphMap_interior_iff`.
3. *The quadrant inverse function theorem applies.* Its hypothesis "$d_C(x) = d_C(g(x))$" is used only as "interior iff interior" (§III.16), which step 2 gives. On a small convex ball $B(0,r) \cap N \cap C$, $\|Dg - 1\| \leq \frac{1}{2}$,

so g satisfies the hypotheses of Proposition 3.8.17 (preprint 3.73), of class C^k on a ball depending on k, since δ is C^k into E_m only near 0 on neighbourhoods depending on (m,k) `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.exists_invData_ball`. The inverse is the same map on all these balls.

4. *The re-graphed datum.* Shrink the chart domain to $\{y : \pi\varphi(y) \in B(0,r)\}$. Keep the chart and N, take the complement $Y' = \ker\pi'$, $V' = g(V)$ and $\delta' = (1-\pi')\circ(1+\delta)\circ g^{-1}$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.regraph`. It describes the same set, $\varphi(M\cap U) = \{w+\delta'(w) : w \in V'\}$, because $g(u)+\delta'(g(u)) = u+\delta(u)$ `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.regraphδ_regraphMap`, and its projection is π' `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.π_eq_of_ker`.

So every datum at x can be replaced, on a smaller chart domain, by one whose projection maps O into C `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.exists_projecting` — condition (b).

Condition (a) follows. Given (b), the re-centring theorem of §III.36 shrinks a datum to one where δ is sc-smooth on V, provided every nearby point of M has a datum `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally` — which the tangent germ property gives. Hence every datum can be shrunk to regular data `Chapter3/TangentGermProjecting · MPolyfold.TangentGermChart.exists_isRegular`, and under the tangent germ property every point has regular data `Chapter3/TangentGermProjecting · MPolyfold.HasTangentGermProperty.exists_isRegular`.

Theorem 3.2.3 (preprint 3.27) as stated `Chapter3/TangentGermProjecting · theorem_3_2_3`. *If $M \neq \emptyset$ has the n-dimensional tangent germ property, then M is a sub-M-polyfold, its induced structure is tame and sc-smoothly equivalent to an n-dimensional manifold with boundary with corners, and the charts $\pi\circ\varphi$ of regular data are charts of it.* This is §III.24 fed with the regular data. The same holds with the dimension depending on the point `Chapter3/TangentGermProjecting · theorem_3_2_3_loc`, and with the data of Definition 3.2.1 as written `Chapter3/TangentGermProjecting · theorem_3_2_3_2021def`. Statement (2), "$\pi\circ\varphi$ is a chart", is stated for regular data, which exist at every point; for an arbitrary datum it would need the step (a′) below.

The two steps of the text's proof, as Lean counterexamples. Each of them does not hold as we read it; the examples of §III.24 are now machine-checked.

- (a′) *"δ is an sc$^\infty$-germ near every $v_0 \in V$."* The text re-centres at $x_0 = \varphi^{-1}(v_0+\delta(v_0))$ and claims that $k(w) = \pi\varphi\psi^{-1}(w+\tau(w))$ has an invertible linearization at 0. In $X = \mathbb{R}^2$ let $M = \{(t^3,t)\}$, a smooth embedded curve with a vertical tangent at the origin. It has the 1-dimensional tangent germ property `Chapter3/RecentreCounterexample2021 · Recentre2021.CE.hasTangentGermPropertyLit_M`. At $(1,1)$ take $N = \mathbb{R}(3,1)$, $Y = \mathbb{R}(0,1)$, $V = N$ and $\delta(v) = (\sqrt[3]{1+v_1} - 1 - v_1/3)(0,1)$ `Chapter3/RecentreCounterexample2021 · Recentre2021.CE.germData_bad`. This δ is smooth near 0 with $D\delta(0) = 0$, but not differentiable at $v_0 = -\frac{1}{3}(3,1)$, the point over the origin `Chapter3/RecentreCounterexample2021 · Recentre2021.CE.not_germNear`. So the claim does not hold as we read it `Chapter3/RecentreCounterexample2021 · Recentre2021.not_theorem_3_2_3_germNearEveryPoint`, and neither does the re-centring step for every pair of data `Chapter3/RecentreCounterexample2021 · Recentre2021.not_theorem_3_2_3_recentreStep`. This is why row V3.73 is narrower: the re-centring lemma is proved with the first graph map C^1 on its whole domain (§III.35).
- (b′) *"$\Sigma = \{v+y \in C : v \in V, y \in Y\}$ is relatively open in C."* In $\mathbb{R}^3$ with $C = [0,\infty)^3$, $N = \mathbb{R}(1,1,1)$ has the good sc-complement $Y = \{x+y-z = 0\}$ `Chapter3/Counterexamples2021 · Counterexamples2021.isScGoodComplement_cube`. On Σ, $x+y-z \geq 0$, so the points $(a,0,a+a^2) \in C$, $a>0$, lie outside Σ and tend to 0: for no $V \subseteq N\cap C$ is Σ a relatively open neighbourhood of 0 `Chapter3/Counterexamples2021 · Counterexamples2021.theorem_3_2_3_sigma_not_relOpen`.

The theorem itself holds; only the route changes. Step (b′) is replaced by the choice of the projecting complement and the re-graphing, and step (a′) by the re-centring along solution data, which needs the nearby data that the tangent germ property supplies.

III.57 Theorems 3.6.8 and 3.1.22 (preprint 3.54 and 3.19) in the book's form, with the 2021 definitions

§§III.37 and III.43–III.45 prove the implicit function theorems with the project's notions. Two files restate them with the 2021 definitions literally, so that a reader checks only those.

The dictionary `Chapter3/MainTheorems2021 · StrongBundle.IsScFredholmGermTame2021`. The 2021 germ data (Definition 3.6.3 = 3.1.11, with the 2021 filling of §III.50) in a *tame* strong bundle chart are the project's tame germ data. In a tame chart the 2021 and the 2017 fillings agree, so these are also the 2017 germ data `Chapter3/MainTheorems2021 · StrongBundle.isScFredholmGermTame2021_iff_2017`. The section with such data at every smooth point `Chapter3/MainTheorems2021 · StrongBundle.IsScFredholmSectionTame2021` and good position in a tame chart `Chapter3/MainTheorems2021 · StrongBundle.IsGoodPositionGermTame2021` complete the dictionary. Reading "tame strong bundle" as germ data in a tame chart is the reading of §III.47 (row V3.103); §III.65 shows that it cannot be dropped.

The theorems. Each is a corollary of the project's form by the dictionary, using the primed theorems of §III.43, which carry no added filler hypothesis: Theorem 3.6.8 (1), (2) `Chapter3/MainTheorems2021 · theorem_3_6_8_2021def`, Theorem 3.1.26 `Chapter3/MainTheorems2021 · theorem_3_1_26_2021def`, Theorem 3.1.22 `Chapter3/MainTheorems2021 · theorem_3_1_22_2021def`, Corollary 3.1.27 `Chapter3/MainTheorems2021 · corollary_3_1_27_2021def`, Theorems 3.6.10 and 3.6.11 `Chapter3/MainTheorems2021 · theorem_3_6_10_2021def` `Chapter3/MainTheorems2021 · theorem_3_6_11_2021def`.

Theorem 3.6.8 (preprint 3.54) with property (4) of its proof `Chapter3/BookFormIFT2021 · theorem_3_6_8_book`. *Let f be an sc-Fredholm section with tame germ data, $f(x) = 0$, and (f, x) in good position in a tame chart. There are an open $V \ni x$ and d such that (1) every zero in V is in good position; (4) at every zero $y \in V$, $f'(y)$ is onto in every strong bundle chart, with index d; (2) $S = \{y \in V : f(y) = 0\}$ is a sub-M-polyfold whose induced structure is that of a d-dimensional manifold with boundary with corners.* The proof is the index form of Theorem 3.1.26 (§III.44) `Chapter3/GlobalIFTIndexG · theorem_3_1_26_index`. **Its last step** `Chapter3/BookFormIFT2021 · theorem_3_6_8_tangentGermProperty`: there are an open $U \ni x$ and $d = \operatorname{ind} f'(x) = \dim \ker f'(x)$ such that $S(f, U)$ has the d-dimensional tangent germ property *as written in Definition 3.2.1*, and Theorem 3.2.3 applies to it. *Proof.* Theorem 3.6.8 (1) gives an open V_x of nearby good position `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'`. Theorem 3.6.8 (2) with the index gives data of Definition 3.2.1 of dimension d at every zero in a smaller open set `Chapter3/GlobalIFTIndexG · theorem_3_6_8_two_index`. These are the project's data, which are the book's by §III.56 `Chapter3/GoodComplementClosed · MPolyfold.hasTangentGermProperty2021_iff`. Theorem 3.2.3 for regular data (§III.24) `Chapter3/TangentGerm · theorem_3_2_3_of_isRegular` closes the argument.

Theorem 3.1.22 (preprint 3.19) `Chapter3/BookFormIFT2021 · theorem_3_1_22_book`. *Let $d_X \equiv 0$, f an sc-Fredholm section with tame germ data, $f(x_0) = 0$. Then $f'(x_0)$ is sc-Fredholm in every strong bundle chart around x_0* `Chapter3/LinearizationChartsScG · StrongBundle.linAt_isScFredholm`. *If it is onto, there are $U \ni x_0$ and d with $f'(y)$ onto of index $d = \dim \ker f'(y)$ at every zero $y \in U$, and $S(f, U)$ a sub-M-polyfold with the structure of a d-dimensional manifold without boundary* `Chapter3/GlobalIFTIndexG · theorem_3_1_22_of_surjective`. The sc-Fredholm property in every chart is Proposition 3.6.5 (preprint 3.52) in sc form, carried between charts by the chart change at a zero (§§III.30–III.31, gap G.22) `Chapter3/FredholmGermLinearizationScG · proposition_`

3_6_5_sc. The last clause, $\ker f'(y) = T_y S(f, U)$, is §III.62.

III.58 Remark 3.3.8 (preprint 3.36): the quantitative solution germ

Remark 3.3.8 (preprint 3.36) Chapter3/SolutionGermQuantitative · remark_3_3_8. *Let $f(a,w) = w - B(a,w)$ be an sc^0-contraction germ which is an sc-smooth germ with $f(0) = 0$, on a quadrant-like $\widetilde{C}$. There are non-increasing sequences $\varepsilon_i, \tau_i > 0$ such that (1) the solution germ δ is continuous on $\{a \in \widetilde{C} : |a| \le \varepsilon_0\}$ with values in $\{|w|_0 \le \tau_0\}$, $\delta(0) = 0$; (2) every solution of $f(a,w) = 0$ with $|a| \le \varepsilon_0$, $|w|_0 \le \tau_0$ is $w = \delta(a)$; (3) $|a| \le \varepsilon_i$ implies $\delta(a) \in W_i$ and $|\delta(a)|_i \le \tau_i$; (4) $\delta : \{|a| \le \varepsilon_i\} \to W_i$ is of class C^i.* "Decreasing" is read as non-increasing.

Proof. 1. *Level* 0. The contraction region of Theorem 3.3.2 (preprint 3.29) at 0 gives τ_0' and the solution germ δ, continuous, with uniqueness in the region Chapter3/SolutionGermQuantitative · ContractionGerm.exists_radius. Put $\tau_i = \tau_0'/2^{i+1}$; uniqueness gives (2) Chapter3/ContractionGerm · ContractionGerm.eq_sol. 2. *Level i.* Corollary 3.3.7 (preprint 3.34) with $m = k = i$ (§III.11) gives a relatively open $V_i \ni 0$ on which δ maps into W_i by a C^i map Chapter3/GermIFT · corollary_3_3_7. The level-i fixed points are continuous, so a radius e_i with $\bar{B}(0,e_i) \cap \widetilde{C} \subseteq V_i$ and $|\delta(a)|_i \le \tau_i$ for $|a| \le e_i$ exists. At $i = 0$ the level-0 map is δ itself Chapter3/SolutionGermQuantitative · ContractionGerm.fixLvl_zero_eq. 3. *Monotone radii.* $\varepsilon_i = \min(e_0, \dots, e_i)$, the running minima Chapter3/SolutionGermQuantitative · exists_antitone_le.

Like Corollary 3.3.7, the remark lives on balls that shrink with i, the shrinking of §III.11 made explicit.

III.59 Extreme rays: Lemmas 3.8.1–3.8.3 and 3.8.9–3.8.11 (preprint 3.62–3.64 and 3.66–3.68)

The formalization proved Proposition 3.1.25 (preprint 3.22) and Lemmas 3.8.4–3.8.6 without extreme rays (§§III.14, III.54). The text's lemmas on extreme rays are now proved as well, for the comparison; nothing else uses them.

Definitions. A closed convex cone P in a finite-dimensional N is closed, convex, with $P \cap (-P) = \{0\}$ and $\mathbb{R}_+ P = P$ Chapter3/Book2021Items · IsClosedConvexCone. The ray of x is $\mathbb{R}_+ x$ Chapter3/ExtremeRays · rayOf; it is an extreme ray if $x \in P \setminus 0$ and $y, x - y \in P$ imply $y \in \mathbb{R}_+ x$ Chapter3/ExtremeRays · IsExtremeRay Chapter3/ExtremeRays · extremeRays. This is the generator notion of §III.54 Chapter3/ExtremeRays · IsClosedConvexCone.isExtremeRay_rayOf_iff. A quadrant is a cone carried onto $[0,\infty)^n$ by a linear isomorphism $N \cong \mathbb{R}^n$ Chapter3/ExtremeRays · IsQuadrantIn.

A strictly positive functional Chapter3/ExtremeRays · IsClosedConvexCone.exists_pos_functional. *There are $\varphi \in N^*$ and $c > 0$ with $\varphi(x) \ge c|x|$ on P.* 1. For a unit vector $z \in P$, $-z \notin P$, and Hahn–Banach separation of $-z$ from the closed convex P gives a functional positive at z and ≥ 0 on P Chapter3/ExtremeRays · IsClosedConvexCone.exists_functional_pos. 2. Each is positive on an open neighbourhood of z. Finitely many neighbourhoods cover the compact unit sphere of P. The sum φ of their functionals is positive on that sphere, and its minimum there is the constant c.

So the base $P \cap \{\varphi = 1\}$ is compact and convex.

Lemma 3.8.1 (preprint 3.62) Chapter3/ExtremeRays · lemma_3_8_1. *If $P \neq \{0\}$, P is the closed convex hull of its extreme rays.* 1. Mathlib's Krein–Milman theorem makes the compact convex base the closed convex hull of its extreme points. 2. An extreme point of the base spans an extreme ray of P Chapter3/

ExtremeRays · IsClosedConvexCone.rayOf_mem_extremeRays. 3. A nonzero $x \in P$ is $\varphi(x)$ times the point $x/\varphi(x)$ of the base, so it lies in the closed convex hull of the rays.

For $P = \{0\}$ the statement does not hold as we read it Chapter3/ExtremeRays · not_lemma_3_8_1_zero: $\{0\}$ has no extreme ray, and the closed convex hull of the empty family is empty. With 0 adjoined the lemma holds for every closed convex cone Chapter3/ExtremeRays · lemma_3_8_1_zero_union. This is the narrower row V3.123; Lemma 3.8.11 below uses the form with 0 adjoined.

Lemma 3.8.2 (preprint 3.63) Chapter3/ExtremeRays · lemma_3_8_2. *A closed convex cone with nonempty interior is a quadrant iff it has exactly* $\dim N$ *extreme rays.* - (□) The extreme rays of $[0,\infty)^n$ are the $\mathbb{R}_+ e_i$ Chapter3/ExtremeRays · extremeRays_nonnegOrthant, and the count is transported by the isomorphism. - (□) Pick generators $a_1, \dots, a_n$. By Lemma 3.8.1 and the nonempty interior they span N; by the count they are a basis. The cone is the preimage of the orthant under the dual coordinates.

Lemma 3.8.3 (preprint 3.64), in the model $C = [0,\infty)^n \oplus W$. *If $N \cap C$ is a closed convex cone and a generates an extreme ray, then* $\dim N - 1 \le \#\sigma_a$, $\sigma_a = \{i : a_i = 0\}$, *with equality if N is in good position.* 1. Let $\Pi : N \to \mathbb{R}^{\sigma_a}$ be the coordinate projection, the text's map (3.34) Chapter3/ExtremeRays · zeroIdxProj. 2. Its kernel lies in $\mathbb{R}a$ (the text's (3.33)) Chapter3/ExtremeRays · ker_zeroIdxProj_le. For z in the kernel, $la + z$ and $la - z$ lie in $N \cap C$ when $l = \sum_{i \notin \sigma_a} |z_i|/a_i + 1$. They add up to $2la$, so extremality puts z on the line $\mathbb{R}a$. 3. Rank–nullity gives the inequality Chapter3/ExtremeRays · lemma_3_8_3. 4. Under good position, the coordinates in σ_a vanish on the good complement (Lemma 3.8.8 (preprint 3.65)), so Π is onto and equality holds Chapter3/ExtremeRays · lemma_3_8_3_goodPosition. Only the algebraic splitting is used.

Lemmas 3.8.9–3.8.11 (preprint 3.66–3.68). Let W be $\{0\} \oplus W$ Chapter3/GoodComplementIntersection · wPart, $\widetilde{N}$ an algebraic complement of $N \cap W$ in N, Σ the union of the σ_a over nonzero $a \in N \cap C$ Chapter3/GoodComplementIntersection · bigSigma, and $\mathbb{R}^{\Sigma^c} \oplus W$ the subspace where the coordinates in Σ vanish Chapter3/GoodComplementIntersection · sigmaCompl. - *Lemma 3.8.9: $\widetilde{N}$ is in good position, with good complement $(N \cap W) \oplus N^\perp$.* The proof uses only that $N^\perp$ is closed Chapter3/GoodComplementIntersection · lemma_3_8_9_of_isClosed: 1. $E_0 = \widetilde{N} \oplus ((N \cap W) \oplus N^\perp)$ by linear algebra. 2. The good-complement constant is $\varepsilon/(1 + \|\mathrm{pr}\|)$, with pr the continuous projection onto N along $N^\perp$ — the text's c_1. 3. The interior of $\widetilde{N} \cap C$ comes from that of $N \cap C$ by translating along $N \cap W$, which does not change membership in C.

So the lemma holds for the text's sc-complement Chapter3/GoodComplementIntersection · lemma_3_8_9. With the text's notions on both sides, $(N \cap W) \oplus N^\perp$ is again an sc-complement Chapter3/GoodComplementIntersectionSc · lemma_3_8_9_sc: a finite-dimensional smooth A plus an sc-subspace is an sc-subspace Chapter3/GoodComplementIntersectionSc · ScBanach.isScSubspace_sup_of_le_smooth, and the splitting holds on every level Chapter3/GoodComplementIntersectionSc · ScBanach.isCompl_levelSub_of_isScSplitting. - *Lemma 3.8.10: $C \cap \widetilde{N}$ is a closed convex cone, and a generator a of an extreme ray has* $\dim \widetilde{N} - 1 = \#\sigma_a = d_C(a)$ Chapter3/GoodComplementIntersection · lemma_3_8_10. The cone property needs only $\widetilde{N} \cap W = 0$ Chapter3/GoodComplementIntersection · isClosedConvexCone_of_disjoint. The formula is Lemma 3.8.3 for $\widetilde{N}$. - *Lemma 3.8.11: either $\widetilde{N} \cap (\mathbb{R}^{\Sigma^c} \oplus W) = 0$, or $\widetilde{N} \subseteq \mathbb{R}^{\Sigma^c} \oplus W$,* $\dim \widetilde{N} = 1$ *and $\Sigma = \emptyset$* Chapter3/GoodComplementIntersection · lemma_3_8_11. By the text's argument: 1. Let $0 \neq x$ lie in the intersection, and let b generate an extreme ray. The coordinates of b outside Σ are positive and those of x in Σ vanish, so with $\lambda = \sum_{i \notin \Sigma} |x_i|/b_i + 1$ both $\lambda b + x$ and $\lambda b - x$ lie in the cone Chapter3/GoodComplementIntersection · large_smul_add_mem. They add up to $2\lambda b$, so extremality puts x on $\mathbb{R}b$ Chapter3/GoodComplementIntersection · exists_eq_smul_of_isExtremeRayGen. 2. So every extreme ray lies on $\mathbb{R}x$. By Lemma 3.8.1 with 0 adjoined, $C \cap \widetilde{N} \subseteq \mathbb{R}x$. 3. The nonempty interior gives $\widetilde{N} = \mathbb{R}x$, of dimension 1. 4. An extreme ray exists, and Lemma 3.8.10 gives $\#\sigma = 0$, so $\Sigma = \emptyset$.

III.60 Definition 3.8.14 (preprint 3.71), Lemma 3.8.21 (preprint 3.77) in general form, and Lemma 3.8.20 (preprint 3.76)

Definition 3.8.14 (preprint 3.71): class C^k on partial quadrants. - $f : U \to \mathbb{R}^m$ on a relatively open $U \subseteq C$ is C^1 if it has a continuous family of one-sided derivatives `Chapter3/QuadrantClassCk · IsClassC1On`. The derivative condition is the text's limit over $x+h \in U$ `Chapter3/QuadrantClassCk · isClassC1On_iff_tendsto`. - The tangent map is $Tf(x,h) = (f(x), Df(x)h)$ on $U \oplus \mathbb{R}^n$ `Chapter3/QuadrantClassCk · bookTangentMap`. f is C^k if f and Tf are C^{k-1} `Chapter3/QuadrantClassCk · IsClassCOn`, and smooth if it is C^k for all k `Chapter3/QuadrantClassCk · IsClassSmoothOn`.

Agreement. On a set of unique differentiability every derivative family agrees with the derivative within the set `Chapter3/QuadrantClassCk · IsClassC1On.eqOn_fderivWithin`, so Tf is well defined. 1. C^1 is Mathlib's C^1 within U `Chapter3/QuadrantClassCk · isClassC1On_iff_contDiffOn`. 2. C^{k+2} is C^{k+1} of Tf `Chapter3/QuadrantClassCk · contDiffOn_add_two_iff_tangentMap`, so by induction C^k is Mathlib's C^k within U `Chapter3/QuadrantClassCk · isClassCOn_iff_contDiffOn Chapter3/QuadrantClassCk · isClassSmoothOn_iff_contDiffOn`. This holds in particular on relatively open subsets of partial quadrants `Chapter3/QuadrantClassCk · isClassCOn_iff_contDiffOn_of_isRelOpenInQuadrant`. 3. The text's C^1 and smooth are sc¹ and sc-smooth for the constant structure `Chapter3/QuadrantClassCk · isSc1_iff_isClassC1On Chapter3/QuadrantClassCk · isScSmooth_iff_isClassSmoothOn`.

So the C^k used in §§III.16–III.17 is the text's notion (row V3.126).

Lemma 3.8.21 (preprint 3.77), general form `Chapter3/QuadrantInverseSmooth · lemma_3_8_21`. *Let U be relatively open in a partial quadrant $C \subseteq \mathbb{R}^n$, $f : U \to C$ of class C^1, $V = f(U)$ relatively open, $f : U \to V$ a homeomorphism and $Df(x)$ invertible on U. Then $g = f^{-1}$ is differentiable within V with $Dg(y) = [Df(g(y))]^{-1}$, and is C^1 on V.* The text adds "satisfying the assumptions of Proposition 3.8.17"; its proof of that proposition applies the lemma to the iterated tangent maps $T^k f$, which do not satisfy $\|D(T^k f) - 1\| \le \frac{1}{2}$, so the form used is this general one. 1. *Derivative.* Mathlib's derivative of a local left inverse needs only the invertibility of $Df(g(y))$ and the continuity of g; the text's bi-Lipschitz estimate is not needed. Relatively open subsets of a quadrant have unique derivatives `Chapter3/QuadrantInverseSmooth · IsRelOpenInQuadrant.uniqueDiffOn_const Chapter3/QuadrantInverseSmooth · hasFDerivWithinAt_inverse_of_isUnit`. 2. *Bootstrap.* $x \mapsto [Df(x)]^{-1}$ is C^{j-1} `Chapter3/QuadrantInverseSmooth · contDiffOn_inverse_deriv_comp`, and $Dg = [Df]^{-1} \circ g$, so $f \in C^j$ gives $g \in C^j$ `Chapter3/QuadrantInverseSmooth · lemma_3_8_21_contDiffOn` and f smooth gives g smooth `Chapter3/QuadrantInverseSmooth · lemma_3_8_21_smooth`.

The printed form is a special case `Chapter3/QuadrantInverseSmooth · lemma_3_8_21_printed` (row V3.62, now stronger).

Lemma 3.8.20 (preprint 3.76) `Chapter3/IteratedTangentMaps · lemma_3_8_20`. *Under the hypotheses of Proposition 3.8.17, every $T^k f : T^k U \to T^k V$ is a smooth homeomorphism of relatively open subsets of the partial quadrant $T^k C$, with smooth inverse.* 1. *The data.* The induction runs over bundled data, a space, a quadrant, a set and a map `Chapter3/IteratedTangentMaps · QuadMapData`. Their tangent is $(V \times V, C \times V, U \times V, Tf)$ `Chapter3/IteratedTangentMaps · QuadMapData.tangent`; the product of a partial quadrant with a whole space is one `Chapter3/IteratedTangentMaps · constProdIso`. 2. *The property inherited.* The hypotheses of the general Lemma 3.8.21 — partial quadrant, relatively open domain and image, smooth, continuous inverse, invertible derivative `Chapter3/IteratedTangentMaps · QuadMapData.IsSmoothHomeo` — pass from f to Tf `Chapter3/IteratedTangentMaps · QuadMapData.IsSmoothHomeo.tangent`. - The derivative of Tf at (x,h) is $(a,b) \mapsto (Df(x)a, D^2 f(x)(a,h) + Df(x)b)$ `Chapter3/IteratedTangentMaps · hasFDerivWithinAt_tangentMap`. It is injective because $Df(x)$ is, hence invertible in finite dimensions `Chapter3/IteratedTangentMaps · isUnit_fderivWithin_`

`tangentMap`. - The inverse is the text's $\Phi(y,l) = (g(y), [Df(g(y))]^{-1}l)$. 3. *Iteration.* By induction on k, $T^k f$ is $(\text{tangent}^{[k]} d).f$ `Chapter3/IteratedTangentMaps · QuadMapData.iterTangent`, and Lemma 3.8.21 applies at each stage.

The lemma as printed, with the hypotheses of Proposition 3.8.17 for every order, is a special case `Chapter3/IteratedTangentMaps · lemma_3_8_20_printed`. §III.16 does not need the lemma: there the bootstrap of step 2 above replaces the iteration.

III.61 Lemma 3.8.13 and Proposition 3.8.12 (preprint 3.70 and 3.69) in a general presentation; $d_M = d_X$ in Theorem 3.2.3 (preprint 3.27)

§§III.14–III.15 proved Proposition 3.8.12 and Lemma 3.8.13 in the model $C = [0,\infty)^n \oplus W$. Their uses are in charts, whose partial quadrant C has an arbitrary presentation $T : E \cong \mathbb{R}^n \oplus W$.

The reading of "$v \notin N \cap W$". In the model, $\{0\} \oplus W$ is the lineality space $C \cap (-C)$, so for $v \in C$ the condition reads $-v \notin C$, which makes sense for every partial quadrant. In a presentation, $v \in C \cap (-C)$ iff all coordinates of Tv vanish `Chapter3/GraphDegeneracyIndex · mem_lineality_iff` `Chapter3/GraphDegeneracyIndex · mem_nullSpace_coordFun_iff`.

Proposition 3.8.12 (preprint 3.69), general `Chapter3/GraphDegeneracyIndex · proposition_3_8_12_general`. For N in good position to any partial quadrant C: 1. $d_{N\cap C}(x) = d_C(x)$ if $-x \notin C$; 2. $d_{N\cap C}(x) + \dim(N \cap C \cap (-C)) = \dim N$ otherwise.

Lemma 3.8.13 (preprint 3.70), general `Chapter3/GraphDegeneracyIndex · lemma_3_8_13_general`. Under the hypotheses of §III.15 for any partial quadrant: 1. near 0, $v + \tau(v) \in C$; 2. $d_{N\cap C}(v) = d_C(v + \tau(v))$ if $-v \notin C$.

Both proofs are the model proofs, run with the coordinate functionals $x \mapsto (Tx)_i$ `Chapter3/GraphDegeneracyIndex · coordCLMOf` in place of the model coordinates. The cores were already stated for jointly onto functionals (§III.15). A local form, for τ given only on a relatively open $V \ni 0$, extends τ by a radial cut-off `Chapter3/GraphDegeneracyIndex · lemma_3_8_13_local`.

$d_M = d_X$ `Chapter3/TangentGermDegeneracy · MPolyfold.theorem_3_2_3_degIndex_eq`. *Let M have regular data at every point and $(\varphi, N, Y, V, \delta)$ regular data at x. There is $\rho > 0$ such that $d_M(y) = d_X(y)$ for every $y \in M$ in the chart domain with $v = \pi\varphi(y)$ of norm $< \rho$ and $-v \notin C$.* 1. The chart $\pi \circ \varphi$ gives $d_M(y) = d_{N\cap C}(v)$ (§III.24) `Chapter3/TangentGerm · MPolyfold.theorem_3_2_3_degIndex`. 2. The tame chart φ gives $d_X(y) = d_C(\varphi(y))$ (Proposition 2.5.6 (preprint 2.36)), and $\varphi(y) = v + \delta(v)$. 3. The local Lemma 3.8.13 equates the two.

Some restriction is needed. For $X = [0,\infty)^2$ and M the diagonal, $d_M(0) = 1$ and $d_X(0) = 2$; there $v = 0$ lies in the lineality space $\{0\}$.

With Lemma 3.8.13 in general form, Lemma 3.6.9 (preprint 3.55) holds in every presentation as well (row V3.80); its proof in coordinates (§III.22, §III.32) was already written for functionals cut out by a presentation.

III.62 The intrinsic linearization, and $\ker f'(y) = T_yS$ (Theorem 3.1.22 (preprint 3.19), last clause)

§§III.31 and III.45 stated the linearization at a zero and "ker $f'(y) = T_yS$" in a chart. With the linear structure of T_yX from Chapter 2 (Definition 2.5.19 in book form), both are now intrinsic.

The fibre map of a chart. For a strong bundle chart c and y in its domain, the fibre Y_y maps linearly into F by the second component of c `Chapter3/LinearizationIntrinsic · StrongBundleChart.fiberLin`. This map is injective `Chapter3/LinearizationIntrinsic · StrongBundleChart.fiberLin_injective`, its range is the fibre of K over $\varphi(y)$ `Chapter3/LinearizationIntrinsic · StrongBundleChart.mem_range_fiberLin`, and between two charts it changes by the fibre part of the transition `Chapter3/LinearizationIntrinsic · StrongBundleChart.transitionFiberCLM_fiberLin`.

The intrinsic linearization. In the chart, the linearization at a zero on level 1 is $Tf(z)$ composed with the tangent of the chart `Chapter3/LinearizationIntrinsic · StrongBundleChart.linChart`. At a zero its values lie in the fibre of K, the range of the fibre map `Chapter3/LinearizationIntrinsic · StrongBundleChart.linChart_mem`. So it pulls back to $f'(y) : T_yX \to Y_y$ `Chapter3/LinearizationIntrinsic · StrongBundle.linIntr`, with $fibre map □ f'(y) = $ the chart linearization `Chapter3/LinearizationIntrinsic · StrongBundle.fiberLin_linIntr`.

Chart independence `Chapter3/LinearizationIntrinsic · StrongBundle.linIntr_chart_independent`. At a zero only the fibre part of a chart transition survives in the linearization `Chapter3/LinearizationChartChange · linAt_chart_change_key` (§III.31), and it matches the change of the fibre maps.

ker $f'(y)$ **in the chart** `Chapter3/SolutionTangentKernel · StrongBundle.coe_map_ker_linIntr`. The tangent of the chart carries ker $f'(y)$ onto the kernel of the chart linearization $T_zO \to F$, because the fibre map is injective.

ker $f'(y) = T_yS$ **in** T_yX `Chapter3/SolutionTangentKernel · StrongBundle.ker_linIntr_eq_subTangentRange`. *Let f have tame germ data, let every zero in an open U be in good position in a tame chart, and let $S = S(f,U)$ be a sub-M-polyfold. Then at every $y \in S$ on level* 1, ker $f'(y)$ *is the image of T_yS in T_yX* `Chapter2/BookFormTameSub2021 · MPolyfold.subTangentRange`. *Proof.* 1. In the chart of the tangent germ data at y, T_yS is the kernel of the chart linearization (§III.45) `Chapter5/GeneralPosition · exists_tangentSpace_eq_linKer`. 2. The inclusion $S \subseteq X$ has tangent map with that image `Chapter2/BookFormTameSub2021 · MPolyfold.coe_map_range_subInclusionCLM`. 3. The previous paragraph identifies the kernels.

Theorem 3.1.22 (preprint 3.19), with its last clause `Chapter3/SolutionTangentKernel · theorem_3_1_22_book_ker`. *Theorem 3.1.22 as in §III.57, and in addition* ker $f'(y) = T_yS(f,U)$ *in T_yX at every $y \in S(f,U)$ on level* 1, *for the intrinsic $f'(y)$ through any strong bundle chart.* Gap G.30 is closed with it.

III.63 Theorem 3.7.1 (preprint 3.58) for a general sc-splitting

§III.38 proved the basic germ criterion in the coordinates of the text's proof, $E = \mathbb{R}^n \oplus X$, $F = \mathbb{R}^N \oplus Y$. The theorem as printed has a partial quadrant $C \subseteq E$ and an sc-splitting $E = B \oplus X$ with $X \subseteq C$.

The statement `Chapter3/BasicGermCriterionGeneral · theorem_3_7_1_general`. *Let U be a relatively open neighbourhood of* 0 *in a partial quadrant $C \subseteq E$ and $f : U \to F$ sc-smooth, with* 1. *at every smooth point $x \in U$, $Df(x)$ sc-Fredholm* `Chapter3/BasicGermCriterionGeneral · derivOpAt`, *with index independent of x;* 2. *an sc-splitting $E = B \oplus X$, B finite-dimensional containing* ker $Df(0)$, *$X \subseteq C$, with the compactness property*

Chapter3/BasicGermCriterionGeneral · CompactnessCondition371; 3. *the estimate for the derivative along* X Chapter3/BasicGermCriterionGeneral · DerivCondition371.

Then $f - f(0)$ *is conjugated near* 0 *to a basic germ.*

Three readings, none of them a restriction (row V3.89, restated): - (3) is printed for $h \in E_m$; it is read for $h \in X_m$, since $D_2 f$ is the derivative along X. - In (2) one radius serves all levels, the text's order of quantifiers; $B \subseteq E_\infty$ is finite-dimensional. - A smooth point is a point of $U \cap E_\infty$.

The proof is a reduction to §III.38. 1. *The target splitting* Chapter3/BasicGermCriterionGeneral · exists_targetSplitting371. As in the text, $Y = Df(0)X$ has the finite-dimensional complement $A = C_0 \oplus Df(0)B \subseteq F_\infty$, with C_0 a smooth complement of the range of the sc-Fredholm $Df(0)$. A has an sc-complement Y' by Proposition 1.1.7 (preprint 1.6). 2. *Coordinates.* The sc-splittings give sc-isomorphisms $S : E \cong B \oplus X$ and $T : F \cong A \oplus Y'$ Chapter3/BasicGermCriterionSplit · ScBanach.splitScIso. Since $X \subseteq C$, S carries C to $(B \cap C) \oplus X$ Chapter3/BasicGermCriterionSplit · splitScIso_preimage, and $B \cap C$ is a partial quadrant of B. Indeed X, a subspace inside C, lies in the lineality space $C \cap (-C)$, and adding such a point does not change membership in C Chapter3/BasicGermCriterionSplit · IsPartialQuadrant.add_mem_iff_of_mem; so X is a good complement of B with any ε, and B is in good position Chapter3/BasicGermCriterionSplit · isInGoodPosition_of_isScSplitting Chapter3/BasicGermCriterionSplit · isPartialQuadrant_of_isScSplitting Chapter3/BasicGermCriterionSplit · isPartialQuadrant_split. The preimage of U is relatively open Chapter3/BasicGermCriterionGeneral · isRelOpenInQuadrant_symm_preimage. 3. *The conjugate* $g = T \circ f \circ S^{-1}$ Chapter3/ScIsoConjugation · scIsoConjMap is sc-smooth Chapter3/ScIsoConjugation · isScSmooth_conjMap, with $Dg = T\,Df\,S^{-1}$ on level 0 and on every level Chapter3/ScIsoConjugation · scDeriv_conjMap Chapter3/ScIsoConjugation · levelDeriv_conjMap. So condition (1) passes to g Chapter3/ScIsoConjugation · isScFredholm_of_conj. - Condition (2): the compactness property passes with the radius $\min(r, \delta)$ Chapter3/BasicGermCriterionGeneral · compactnessCondition_conj. - Condition (3): the estimate passes with ε rescaled by $\|T_m\| + 1$ Chapter3/BasicGermCriterionGeneral · derivCondition_conj. - The block $h \mapsto (Dg(0)(0, h))_{Y'}$ is bijective on level 0 Chapter3/BasicGermCriterionGeneral · bijective_splitScIso_snd. 4. *Conclusion.* The coordinate theorem of §III.38 conjugates $g - g(0)$ to a basic germ Chapter3/BasicGermCriterionIndex · theorem_3_7_1. The conjugation moves back through S and T Chapter3/BasicGermCriterionTransport · IsConjugateToBasicGerm.of_scIso.

Gap G.24 is closed with it.

III.64 An sc-smooth family of projections whose rank jumps

The counterexample of §III.65 needs a strong bundle whose fibre dimension jumps: an sc-smooth family of projections π_t of an sc-Banach space W, with $\pi_t = 0$ for $t \le 0$ and π_t of rank one for $0 < t < 1$. Such a family is sc-smooth but not norm-continuous; the sc-smooth maps of the earlier sections all came from Proposition 1.2.4 (preprint 1.19), which needs C^1 on equal levels and cannot produce one.

A criterion Chapter3/UntamedJump · Jump.Family. A *jump family* on V consists of: - operators $A_m(t) : V_m \to V_m$ and $A'_m(t) : V_{m+1} \to V_m$, compatible with the inclusions; - $(t, h) \mapsto A_m(t)h$ continuous on each level; - $t \mapsto A_m(t)|_{V_{m+1}}$ norm-differentiable with derivative $A'_m(t)$; - $(s, t, h) \mapsto sA'(t)h$ sc-smooth from $\mathbb{R}^2 \oplus V^1$ to V.

Then $(t, h) \mapsto A(t)h$ *is sc-smooth* Chapter3/UntamedJump · Jump.Family.isScSmooth_map. *Proof.* By induction on k, over all jump families at once Chapter3/UntamedJump · Jump.Family.isSck_map. The tangent map is

$T\pi(t,h;\delta t,\delta h) = (A(t)h, \delta t\, A'(t)h + A(t)\delta h)$. Its second summand is the given sc-smooth map; its third is again a jump family, on the shifted space `Chapter3/UntamedJump · Jump.Family.shift`.

The moving rank-one projection `Chapter3/UntamedJump · Jump.family`. 1. *The scale.* Weighted c_0, with $|x|_m = \sup_k (k+1)^m |x_k|$, and positions $K_i = 16^i - 1$ `Chapter3/UntamedJump · Jump.pos`, where the weights are powers of 16. 2. *The profile.* Λ, built from Mathlib's smooth transition and supported in $[\frac{1}{2}, 2]$ `Chapter3/UntamedJump · Jump.prof`. Put $e(t) = \sum_i \Lambda(2^i t) e_{K_i}$, $\ell_t = \sum_i \Lambda(2^i t)\delta_{K_i}$ and $\pi_t = \ell_t \otimes e(t)$. For $0 < t < 1$ exactly two consecutive coefficients are nonzero, equal to $(\cos\theta, \sin\theta)$. 3. *The derivatives on levels.* From source level $b+g$ to target level b, the j-th t-derivative is a tridiagonal operator `Chapter3/UntamedJump · Jump.T`, with coefficients $2^{ij}16^{-ig}$ times derivatives of products of Λ. It is norm-differentiable in t when $j+2 \le 4g$ `Chapter3/UntamedJump · Jump.hasDerivAt_T` and continuous at $g = 0$ `Chapter3/UntamedJump · Jump.continuous_T_zero`: one level pays for one derivative. 4. *The criterion's last hypothesis* holds by Proposition 1.2.4 `Chapter3/UntamedJump · Jump.isScSmooth_deriv`.

The projections `Chapter3/UntamedJump · Jump.proj` have the following properties: - $\pi_t = 0$ for $t \le 0$ `Chapter3/UntamedJump · Jump.proj_of_nonpos`; - π_t is of rank one for $0 < t < 1$ `Chapter3/UntamedJump · Jump.exists_proj_eq` `Chapter3/UntamedJump · Jump.proj_ne_zero`; - $\pi_t^2 = \pi_t$ for $t < 1$ `Chapter3/UntamedJump · Jump.proj_proj`; - a vector fixed by π_t for $t \le 0$ is 0 `Chapter3/UntamedJump · Jump.eq_zero_of_proj_eq`.

III.65 Theorem 3.6.8 (preprint 3.54) with untamed 2021 germ data: the counterexample

The 2021 filling (§III.50) drops the tameness of the retraction, and Definition 3.6.3 asks for germ data in *some* strong bundle chart. §III.52 showed that the step on p. 138 needs a tame chart. The question left there (Finding 353, gap G.79) was whether germ data in an arbitrary compatible chart can be moved to a tame chart, or else whether the theorem fails. It fails. *Read with the 2021 definitions as printed, Theorem 3.6.8 does not hold as we read it* `Chapter3/UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not`.

The local model (namespace `UntamedCorner`). Let π_t be the projections of §III.64, v a smooth vector with $\pi_t v \neq 0$ for $0 < t < 1$, and ν a functional with $\nu \neq 0$ on $\operatorname{ran}\pi_t \setminus 0$. - $E = \mathbb{R}^2 \oplus W \ni ((a,t),w)$, $C = [0,\infty)\times\mathbb{R}\times W$, $U = C \cap \{t < 1\}$ and $F = \mathbb{R}\oplus W$. - $R(((a,t),w),(x,h)) = (((0,t),0),(0,\pi_t h))$ is a strong bundle retraction `Chapter3/UntamedCounterexample · UntamedCorner.isStrongBundleRetraction_R`. It covers $r(a,t,w) = (0,t,0)$, so $O \cong \mathbb{R}$, and r is not tame `Chapter3/UntamedCounterexample · UntamedCorner.not_isTame_baseRetr`: $d_C = 1$ on O, but 0 at $(1,t,0)$. The fibre of $K \to O$ is 0 for $t \le 0$ and a line for $0 < t < 1$. - The filling is $g((a,t),w) = (a + \nu(w)^2, w + \pi_t v)$.

It is a 2021 filling `Chapter3/UntamedCounterexample · UntamedCorner.isFilling2021_g`: 1. g is the section on O. 2. If $g(u) = \rho(r(u))g(u)$, then $a + \nu(w)^2 = 0$ and $\pi_t w = w$. Since $a \ge 0$, this forces $a = 0$, $\nu(w) = 0$, $w \in \operatorname{ran}\pi_t$, so $w = 0$ and $u \in O$. **Here the boundary $a \ge 0$ of the untamed chart is used.** 3. The fill defect has $Dk(0)((a,t),w) = (a,w)$, an isomorphism $\ker Dr(0) = \{t = 0\} \to \ker\rho(0) = F$ `Chapter3/UntamedCounterexample · UntamedCorner.kerEquiv`.

The rest of the germ data. - g is a basic germ, with $B = -\pi_t v$ constant in w, and $s = 0$ `Chapter3/UntamedCounterexample · UntamedCorner.isBasicGerm_g` `Chapter3/UntamedCounterexample · UntamedCorner.isConjugateToBasicGerm_g`. - $Dg(0)((a,t),w) = (a,w)$ is onto `Chapter3/UntamedCounterexample · UntamedCorner.surjective_scDeriv_g`, with kernel the t-axis, $T_0 O$. So the germ is in good position in its chart `Chapter3/UntamedCounterexample · UntamedCorner.isGoodPositionAtGerm_g`.

The solution set. $g(u) = 0$ iff $w = -\pi_t v$ and $a = -\nu(\pi_t v)^2$. In C this forces $a = 0$ and $\pi_t v = 0$, so $S = \{((0,t),0) : t \le 0\}$ Chapter3/UntamedCounterexample · UntamedCorner.solutionSet_eq, a closed half-line in $O \cong \mathbb{R}$ ending at the zero.

Why S is not a sub-M-polyfold. A smooth retraction ϱ of $(-\varepsilon, \varepsilon)$ onto $(-\varepsilon, 0]$ would have $\varrho'(0) = 1$, so $\varrho(t) > 0$ for small $t > 0$ Chapter3/UntamedCounterexample · UntamedCorner.not_exists_retraction. The retract O with its one chart is a tame M-polyfold Chapter3/UntamedCounterexample · UntamedCorner.polyO Chapter3/UntamedCounterexample · UntamedCorner.isTame_polyO, with $d_X \equiv 0$ Chapter3/UntamedCounterexample · UntamedCorner.degIndex_polyO, and S is not a sub-M-polyfold of it Chapter3/UntamedCounterexample · UntamedCorner.not_isSubMPolyfold_solutionSet. The local statements are collected in Chapter3/UntamedCounterexample · UntamedCorner.theorem_3_6_8_untamed_local.

The strong bundle. $K \to O$ is made a strong bundle over O Chapter3/UntamedBundle · UntamedCorner.bundle, with the chart R above Chapter3/UntamedBundle · UntamedCorner.chart0 and the section $x \mapsto (x, g(x))$ Chapter3/UntamedBundle · UntamedCorner.sec, sc-smooth Chapter3/UntamedBundle · UntamedCorner.isScSmoothSection_sec, whose zero set is S Chapter3/UntamedBundle · UntamedCorner.solutionSet_sec. Each printed hypothesis of Theorem 3.6.8 holds: - *The bundle is tame* in both readings of the phrase "tame strong bundle", which the text does not define separately: - the base is tame; - a tame chart covers all of O, namely $R'(t,(x,h)) = (t,(0,\pi_t h))$, which covers the identity Chapter3/UntamedBundleTame · UntamedCorner.isStrongBundleRetraction_RT Chapter3/UntamedBundleTame · UntamedCorner.chartTame, is compatible with the chart of R Chapter3/UntamedBundleTame · UntamedCorner.compatible_chart0_chartTame, and is tame Chapter3/UntamedBundleTame · UntamedCorner.isTameChart_chartTame; - so the bundle is tame Chapter3/UntamedBundleTheorem · StrongBundle.IsTameBundle Chapter3/UntamedBundleTheorem · UntamedCorner.isTameBundle_bundle. - *At the zero x_0 ($t = 0$),* the section is a 2021 sc-Fredholm germ in the chart of R Chapter3/UntamedBundle · UntamedCorner.isScFredholmGerm2021_sec_zero and is in good position Chapter3/UntamedBundle · UntamedCorner.isGoodPositionGerm_sec_zero. Here $d_X(x_0) = 0$ Chapter3/UntamedBundleTheorem · UntamedCorner.isWithoutBoundary_polyO and the fibre over x_0 is 0 Chapter3/UntamedBundle · UntamedCorner.subsingleton_fiber_zero, so every reading of Definition 3.6.6 holds. - *At the other points* there are germ data in tame charts, from frames in which the section is constant Chapter3/UntamedBundleFrame · UntamedCorner.Frame Chapter3/UntamedBundleFrame · UntamedCorner.Frame.chart Chapter3/UntamedBundleFrame · UntamedCorner.Frame.compatible_chart0 Chapter3/UntamedBundleFrame · UntamedCorner.Frame.isScFredholmGermInterior_sec: - the zero frame over $t < 0$ Chapter3/UntamedBundleFrame · UntamedCorner.frame0; - over $0 < t < 1$ the frame $s \mapsto s\pi_t v$ with coframe $h \mapsto \nu(h)/\nu(\pi_t v)$ Chapter3/UntamedBundleFrame · UntamedCorner.frame1.

The re-centred charts are centred at interior points of their quadrants, hence tame Chapter3/UntamedBundleFrame · UntamedCorner.isScFredholmGermTame_sec_of_ne. So the section is sc-Fredholm in the sense of 2021 Definitions 3.1.16 = 3.6.4 Chapter3/UntamedBundleFrame · UntamedCorner.isScFredholmSection2021_sec.

The conclusion fails. No neighbourhood of x_0 meets the zero set in a sub-M-polyfold Chapter3/UntamedBundle · UntamedCorner.not_isSubMPolyfold_zeroSet. There are no tangent germ data at x_0 either Chapter3/UntamedBundle · UntamedCorner.isEmpty_tangentGermChart: in a tame chart $d_C(0) = d_X(x_0) = 0$, so the data would be a graph over a neighbourhood of 0 in N, and a continuous injective function on an interval has no interior maximum. - The facts above are collected in one statement Chapter3/UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_example. - The conclusion of Theorem 3.6.8 fails for this section, even with (1) weakened to good position in any chart Chapter3/UntamedBundleTheorem · UntamedCorner.not_theorem_3_6_8_conclusion. - As a universal statement over tame

strong bundles with 2021 sc-Fredholm sections, the theorem does not hold as we read it `Chapter3/UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not`, and neither do Theorems 3.1.22 and 3.1.26, whose hypothesis $d_X \equiv 0$ the example meets `Chapter3/UntamedBundleTheorem · UntamedCorner.theorem_3_1_22_untamed_not`.

This is the narrower row V3.130.

Untamed germ data cannot be moved to a tame chart. - The germ at x_0 has no tame germ data `Chapter3/UntamedBundleTheorem · UntamedCorner.not_isScFredholmGermTame_sec_zero`. Otherwise the section would have tame germ data everywhere, and the good position would hold in the tame chart (§III.47) `Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame`. Theorem 3.6.8 with tame data (§III.43) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8'` would then give tangent germ data at x_0, which do not exist. - So "every 2021 sc-Fredholm germ of a tame strong bundle has tame germ data" does not hold `Chapter3/UntamedBundleTheorem · UntamedCorner.isScFredholmGerm2021_not_tame`, and a 2021 germ need not be a 2017 germ `Chapter3/UntamedBundleTheorem · UntamedCorner.isScFredholmGerm2021_not_2017` (row V3.131).

This answers the question of §§III.47 and III.50 in the negative.

The interior case, and the reading that is needed. Germ data in a compatible chart centred at a point *interior* to its partial quadrant, $d_C(\varphi(x_0)) = 0$, do give tame germ data: the chart shrunk into the interior is tame and carries the restricted data `Chapter3/TameGermTransfer · StrongBundle.isScFredholmGermTame_of_chart_interior`. A section with such data at every point is a section with tame germ data `Chapter3/TameGermTransfer · StrongBundle.isScFredholmSectionTame_of_interior`. The corner filling of §III.50 is centred on the boundary `Chapter3/TameGermTransfer · CornerFilling.zero_not_mem_interior`, as is the chart of R here, where $d_X(x_0) = 0 < 1 = d_C(\varphi(x_0))$. So the interior result is sharp.

Remark 3.6.7's "if $d_X(x) = 0$ we do not need additional assumptions" presumably means $d_C(\varphi(x)) = 0$ in the chart of the germ data. Theorem 3.6.8 holds in the reading of §III.47 — germ data and good position in a tame chart, the wording of the 2017 text and the preprint, and the chart the 2021 proof itself takes on p. 137 (§§III.43, III.57; rows V3.85, V3.94, V3.103). This reading is not derived from the 2021 text as printed; it is satisfiable, since the graph section of a smooth map has tame germ data (§III.47).

III.66 What is not formalized, and what the formalization taught us

What is not formalized. Nothing of Chapter 3. The sixteen gaps of the chapter are closed: - G.16 (Theorem 3.1.21, §III.40); - G.17, G.18 (Theorems 3.1.22, 3.1.26, Corollary 3.1.27, §§III.37, III.43, III.57); - G.19 (Remark 3.3.8, §III.58); - G.20 (Corollary 3.4.3, §III.28); - G.21 (Proposition 3.1.10 on every level, §III.5); - G.22 (Proposition 3.6.5 as an sc-Fredholm operator, §§III.30, III.57); - G.23 (Theorem 3.6.8 (1) with the text's hypotheses, §III.43); - G.24 (Theorem 3.7.1, §III.63); - G.25, G.26 (Lemmas 3.8.1–3.8.3, 3.8.9–3.8.11, §III.59); - G.27, G.28 (Definition 3.8.14, Lemma 3.8.20, §III.60); - G.29 (Lemma 3.8.13 in general presentations, $d_M = d_X$, §III.61); - G.30 (dimension = index, $\ker f'(y) = T_yS$, §§III.44, III.62); - G.79 (untamed germ data, settled by the counterexample, §III.65).

Four comparison rows remain narrower than the text, each because the printed statement does not hold as we read it, with a Lean counterexample: - V3.73, the re-centring step for every pair of data (§III.56, `Recentre2021.not_theorem_3_2_3_recentreStep`); - V3.120, Lemma 3.8.6 (2) at $N = 0$ (§III.54, `Counterexamples2021.not_lemma_3_8_6_two`); - V3.123, Lemma 3.8.1 at $P = \{0\}$ (§III.59, `not_lemma_3_8_1_zero`); - V3.130, Theorem 3.6.8 with the 2021 germ data as printed (§III.65, `UntamedCorner.theorem_3_6_8_untamed_not`).

Side questions that are not items of the comparison stay as they were: whether the shrinking of solution germs is intrinsic (§III.11), whether every germ datum can be traded for a whole-chart one (§III.48), and the compatibility of the chart $(1, \varphi)$ of §III.47.

What the formalization taught us. 1. **No hypothesis of Chapter 3 had to be added after all.** The closedness of good complements is the text's sc-complement (§III.56). The two regularity conditions of Theorem 3.2.3 are theorems: condition (b) is a choice of complement, reached by re-graphing with the quadrant inverse function theorem, whose degeneracy hypothesis is needed only at interior points and holds there because the good complements share the vanishing coordinates; condition (a) follows by re-centring along the data at nearby points (§III.56). The two proof steps they replace do not hold as we read it, and both are now Lean counterexamples. 2. **"Tame strong bundle" must mean germ data in a tame chart.** The 2021 edition moved tameness from the filling to the phrase "tame strong bundle". In the weaker reading — a tame base and a tame atlas, with germ data anywhere — Theorem 3.6.8 does not hold as we read it (§III.65). The mechanism is a fibre that jumps: an sc-smooth but not norm-continuous family of projections (§III.64), which no finite-dimensional or norm-continuous example can show. The boundary of the untamed chart is used in condition (2) of the filling; at points interior to the chart's quadrant the untamed data are tame (§III.65). 3. **Proposition 3.8.12 and Lemma 3.8.13 are intrinsic** once "$v \notin N \cap W$" is read as "$-v \notin C$" (§III.61). Then $d_M = d_X$ near the centre of a datum, off the lineality space, and not at it (the diagonal of the quarter plane). 4. **The extreme-ray lemmas hold, with one degenerate case.** Lemma 3.8.1 needs $P \neq \{0\}$ or 0 adjoined (§III.59); Lemma 3.8.6 (2) needs $N \neq 0$ (§III.54). The formal route through good position (§III.14) remains shorter. 5. **Lemma 3.8.21 must be used in a form more general than its statement**, because the iteration of Lemma 3.8.20 leaves the hypotheses of Proposition 3.8.17 (§III.60). The text's Definition 3.8.14 is Mathlib's C^k within a set. 6. **The basic germ criterion reduces to its coordinates** by an sc-isomorphic change of both spaces; the text's hypothesis $X \subseteq C$ puts X in the lineality space of C, which makes $B \cap C$ a partial quadrant of B (§III.63). 7. $f'(y)$ **is intrinsic.** With the fibre maps of the charts, the linearization at a zero is a map $T_y X \to Y_y$ independent of the chart, and its kernel is $T_y S$ (§III.62).

Part IV. Manifolds with boundary with corners, and sc-differential forms (HWZ Chapter 4, §§4.1–4.5)

Chapter 4 asks when an M-polyfold — a space glued from images of sc-smooth retractions — is secretly an ordinary manifold with boundary with corners, and why the fibres of a family of such manifolds are good for almost every parameter. The pivot is the *sc⁺ condition*: a retraction that gains a level. At a smooth point its derivative is compact, so its retract has finite-dimensional tangent spaces (Lemma 4.1.7 (preprint 4.7)), and near a point where the tangent space is in good position to the partial quadrant the retract is the graph of a map over that tangent space (Proposition 4.1.9 (preprint 4.9)). Theorem 4.1.2 (preprint 4.2) characterizes the manifolds with corners among M-polyfolds by these two properties; Theorem 4.2.7 (preprint 4.18) applies the characterization to submanifolds in good position; Theorem 4.2.11 (preprint 4.22) describes a submanifold near a point of general position; Theorem 4.3.1 (preprint 4.23) combines that description with Sard's theorem on the strata of the corners. The last section, rewritten in the 2021 edition, builds sc-differential forms, their exterior derivative, Stokes' theorem and a Poincaré lemma, with vector fields and their bracket in Appendix 4.5. This Part retells: finite-dimensional sc-spaces and corner charts (§IV.1); sc⁺-maps (§IV.2); sc⁺-retractions, Lemmas 4.1.5 and 4.1.7 (preprint 4.5 and 4.7) (§IV.3); Proposition 4.1.8 (preprint 4.8) with its chart form and Remark 4.1.3 (preprint 4.3) (§IV.4); charts centred at a smooth point, which the text uses and never constructs (§IV.5); Proposition 4.1.9 (preprint 4.9) with Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12), by a direct proof (§§IV.6–IV.9); Theorem 4.1.2 (preprint 4.2) (§IV.10); smooth finite-dimensional submanifolds and Remark 4.2.2 (preprint 4.14) (§IV.11); Proposition 4.2.3 (preprint 4.15) (§IV.12); good and general position in charts and their chart independence (§IV.13); Lemma 4.2.8 (preprint 4.19) and Theorem 4.2.7 (preprint 4.18) (§IV.14); Remark 4.2.9 (preprint 4.20) (§IV.15); the linear algebra at the corner of a centred chart (§IV.16); Theorem 4.2.11 (preprint 4.22) (§§IV.17–IV.18) and Remark 4.3.2 (preprint 4.24) (§IV.19); Sard's theorem (§IV.20); the finite-dimensional core of Theorem 4.3.1 (preprint 4.23) (§§IV.21–IV.22); the fibre over a regular value (§IV.23); Theorem 4.3.1 (preprint 4.23) (§IV.24); what is not formalized (§IV.25); what the formalization taught us (§IV.26); Remark 4.1.3's (preprint 4.3) example (§IV.27); and the 2021 edition's Theorems 4.2.11 and 4.3.1, vector fields and the Lie bracket (§§IV.28–IV.29). The later subsections close the remaining gaps of §§4.1–4.3: Lemma 4.1.11 (preprint 4.10) (§IV.30); Remark 4.2.2 (preprint 4.14) in both wordings and Question 4.2.4 (§IV.31); the local faces of Theorem 4.2.11 (1) (preprint 4.22) (§IV.32); the chart independence of stratified regularity (§IV.33); and Lemma 4.5.1 between open subsets with the pull-back bundle $TX|X^2$ (§IV.34). They then retell §4.4 and Appendix 4.5 of the 2021 edition: the Whitney sum and Definition 4.4.1 (preprint 4.25) (§IV.35); the exterior derivative, $d^2 = 0$ and Definition 4.4.6 (preprint 4.27) (§§IV.36–IV.37); the for-

mula of p. 197 (§IV.38); pull-back, wedge product and the graded differential algebra (§IV.39); forms on ∂X and the relative complex (§IV.40); integration and Stokes' theorem on quadrants and on Mathlib manifolds with corners (§§IV.41–IV.42); Theorem 4.4.7 (preprint 4.28) for sc-forms (§IV.43); and the Poincaré Lemmas 4.5.2 and 4.4.8 (§§IV.44–IV.45). Last, the wedge product on the forms on ∂X and on the relative complex (§IV.46).

What is proved, and what is not. Theorems 4.1.2, 4.2.7, 4.2.11 and 4.3.1 (preprint 4.2, 4.18, 4.22 and 4.23) are proved with the text's hypotheses, and Theorem 4.1.2 (preprint 4.2) needs no tameness at all.

§§4.1–4.3. - Proposition 4.1.9 (preprint 4.9) is proved with two hypotheses the text leaves implicit: the good complement Y is closed, and p is the projection onto T_0O *along* Y rather than $Dt(0)$. The proof is a direct argument (§§IV.6–IV.9). Where the proposition is applied (Theorems 4.1.2, 4.2.7, 4.2.11 (preprint 4.2, 4.18, 4.22), Remark 4.2.9 (preprint 4.20)), the closed complement is chosen, so nothing is added there. - The text's own route goes through Lemma 4.1.11 (preprint 4.10). That lemma is now proved as well, as an sc-Fredholm section in the sense of 2021 Definition 3.1.16 on all of U (gap G.32 closed, §IV.30). Proposition 4.1.9 does not use it. - One step in the proof of Lemma 4.2.8 (preprint 4.19) does not hold as we read it. It is repaired, and the lemma holds (§IV.14). - Theorem 4.2.11 (preprint 4.22): part (4), which the text leaves to "the previous arguments", is proved (§IV.18). Part (1) is proved in the text's sense, with the local faces of the neighbourhood U rather than the faces of the chart model (gap G.35 closed, §IV.32). - Theorem 4.3.1 (preprint 4.23): "regular value" has to be read as the stratified notion of its proof. This notion is now shown not to depend on the product chart (gap G.36 closed, §IV.33). The finite cover of the proof is replaced by a countable one (§IV.24). - Remark 4.2.2 (preprint 4.14) is proved in the preprint wording ($d_X = 0$ gives a manifold) and in the 2021 wording ($d_A = 0$ gives a manifold, $d_A \leq 1$ a manifold with boundary), without tameness (gap G.33 closed, §IV.31). - Question 4.2.4 (preprint: the open question after Proposition 4.15) is a question, so nothing in it is to be formalized (G.34). Its settled last sentence is proved: the "T" is not homeomorphic to an sc$^+$-retract (§IV.31). - Remark 4.1.3's (preprint 4.3) "$X^1 \neq X$ is possible", for which the preprint and 2017 texts give no example, is proved by one: an M-polyfold whose retraction loses a level in its first derivative, with a new criterion for sc-smoothness (gap G.31 closed, §IV.27). The example the 2021 edition adds holds as we read it and is not formalized. - Proved although the text does not construct them: charts and strong bundle charts centred at a smooth point (§IV.5), an sc$^+$-retraction onto the fibre A_λ (§IV.23), Theorem 3.8.22 (preprint 3.78) in C^∞ form (§IV.21), and Sard's theorem in all dimensions (§IV.20).

§4.4 and Appendix 4.5 (2021 edition). - Vector fields, Lemma 4.4.4, Proposition 4.4.5 and Lemma 4.5.1 are proved (§§IV.28–IV.29). They now also cover local sc-diffeomorphisms between open subsets, and the bracket is a section of the pull-back bundle $TX|X^2 \to X^2$ (gap G.92 closed, §IV.34). - Gaps G.37–G.40 and G.80 are closed. The formalized parts are: - sc-differential forms on the Whitney sum (Definition 4.4.1 (preprint 4.25)), the system (4.4) and $\Omega^*_\infty(X)$ (§IV.35); - the exterior derivative, chart independent, with $d^2 = 0$, commuting with (4.4), and H^*_{sc} (Definition 4.4.6 (preprint 4.27)) (§§IV.36–IV.37); - pull-back, the wedge product (classical normalization, since the text does not define it) and the graded differential algebra (§IV.39); - forms on ∂X, read through the tame boundary, with j^* and the relative complex (§IV.40), and the wedge product on both, on the relative complex the product of the mapping cone of j^* (§IV.46); - Theorem 4.4.7 (preprint 4.28), Stokes' theorem for sc-forms with the sum over the faces (§§IV.41–IV.43). It has one added hypothesis, not derived from the text: the boundary being "a union of smooth faces" is read as face-structured. It is satisfied, for instance, without boundary and by the quadrants. - the Poincaré Lemmas 4.5.2 and 4.4.8 (§§IV.44–IV.45). - One row is narrower. The formula for $d\omega$ on p. 197, with the printed sign $(-1)^{\ell+k}$, does not hold as we read it (the counterexample `Chapter4/ScExteriorDerivative2021 · invariantFormula_printedSign_counterexample`). It holds with $(-1)^{\ell+j+1}$ for the text's

bracket (V4.83, §IV.38).

Not formalized (§IV.25): - the pull-back of forms on ∂X along a general sc-smooth map beyond the faces of finite-dimensional sub-M-polyfolds, which no statement of Parts I–II uses (gap G.80 is closed); - the argument that an example for Remark 4.1.3 needs a retraction that loses a level, and the 2021 example; - the classical manifold structure behind Definition 4.1.1 (preprint 4.1) beyond the C^∞ transitions.

Six structural choices. (1) *sc⁺ is a property of maps.* The text defines it for retractions and operators and uses it for maps; here a map is sc⁺ when its lift into F^1 is sc-smooth, and the calculus of such maps is proved first (§IV.2). (2) *Tangent spaces of X are read in charts.* T_xX carries no linear structure in the formalization (gap G.7 of Chapter 2), so T_aA, good position, general position and stratified regularity are stated in a chart, and proved independent of the chart (§IV.13; for stratified regularity §IV.33). (3) *Centred charts are re-centred charts.* A chart with $\psi(x) = 0$ is the chart $\varphi - \varphi(x)$ on the set where the partial quadrant C agrees with $C_{\varphi(x)}$, which exists only at smooth points (§IV.5). (4) *Proposition 4.1.9 (preprint 4.9) on fixed sets.* The sc-diffeomorphism $p : O \cap W \to V$ is proved with $\Gamma = p^{-1}$ sc-smooth on a *fixed* relatively open V, by contraction arguments on level 1, a frame formula for $D\Gamma$ and a C^k-bootstrap along the finite-dimensional tangent space. (5) *Stratified Sard.* Sard's theorem is applied stratum by stratum in the chart of Theorem 4.2.11 (preprint 4.22), and the null sets of critical values of countably many charts are collected. (6) *Forms are read in charts, and integrals on Mathlib manifolds.* The Whitney sum $\oplus_k TX$ is never built as an M-polyfold: a form is a family of alternating forms on the tangent spaces whose chart representatives on $\oplus_k TO$ are sc-smooth (§IV.35). Forms on ∂X are forms on the tame boundary bX that agree over $B : bX \to X$ (§IV.40). Integration happens on a finite-dimensional sub-M-polyfold made a Mathlib manifold with corners by its corner structure, with signed charts (§§IV.41–IV.43).

Map.

- `Chapter4/ScPlus`: sc⁺-maps, Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6), Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7), Proposition 4.1.8 (preprint 4.8) and its chart form, Remark 4.1.3 (preprint 4.3), finite-dimensional sc-spaces.
- `Chapter4/Recentre`: the partial quadrant C_e and its neighbourhood $G(e)$, translations, re-centred retracts, charts and strong bundle charts.
- `Chapter4/ScPlusGraph`: the mean value estimate for sc⁺-retractions, Proposition 4.1.9 (preprint 4.9) with Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12), the frame formula and the bootstrap.
- `Chapter4/Characterization`: Definition 4.1.1 (preprint 4.1), corner charts, Theorem 4.1.2 (preprint 4.2).
- `Chapter4/Submanifolds`: Definitions 4.2.1, 4.2.5, 4.2.6, 4.2.10 (preprint 4.13, 4.16, 4.17, 4.21), Remark 4.2.2 (preprint 4.14), Proposition 4.2.3 (preprint 4.15), Theorem 4.2.7 (preprint 4.18), Lemma 4.2.8 (preprint 4.19), Remark 4.2.9 (preprint 4.20), the chart independence of good and general position.
- `Chapter4/LocalStructure`: Theorem 4.2.11 (1)–(4) (preprint 4.22), Remark 4.3.2 (preprint 4.24), the linear algebra of a centred chart, Lemmas 7.A and 7.B of the exposition.
- `Vendor/MilnorSard/…`: Sard's theorem, copied from our Milnor project (`math_faithful`) with the namespace renamed.
- `Chapter4/FamiliesLocal`, `Chapter4/Families`: the strata and Sard on them, Theorem 3.8.22 (preprint 3.78) in C^∞ form, the fibre retraction, the sc⁺-retraction onto A_λ and its general position, Theorem 4.3.1 (preprint 4.23).
- Added in the fourth addendum (§IV.27): `Chapter4/RemarkFourThreeExample` (parameter families of level operators with one level of loss, their sc-smoothness, the reduction of Remark 4.1.3's

(preprint 4.3) example), `Chapter4/RemarkFourThreeFamily` (the family of moving projections, the M-polyfold of a retract, the example).

- The 2021 edition (§§IV.28–IV.29, IV.34): `Chapter4/Book2021`, `Chapter4/VectorFields`, `Chapter4/BracketLevel`, `Chapter4/VectorFieldsGlobal`, `Chapter4/BookFormLieBracket2021`, `Chapter4/LieBracketOpenSubsets` (open subsets), `Chapter4/LieBracketBundle` (the pull-back bundle $TX|X^2$).
- Gaps of §§4.1–4.3: `Chapter4/RetractionSectionFredholm`, `Chapter4/RetractionSectionSplitting`, `Chapter4/LocalStrongBundle` (Lemma 4.1.11, §IV.30); `Chapter4/RemarkLocalForm`, `Chapter4/SubmanifoldNearInterior`, `Chapter4/SubmanifoldNearBoundary`, `Chapter4/ScPlusRetractQuestion` (Remark 4.2.2, Question 4.2.4, §IV.31); `Chapter4/LocalFaces`, `Chapter4/GeneralPositionLocalFaces` (§IV.32); `Chapter4/StratifiedRegularValues` (§IV.33).
- sc-differential forms (§§IV.35–IV.40): `Chapter4/ScForms2021` (Whitney sum, Definition 4.4.1, the system (4.4)), `Chapter4/ScExteriorDerivative2021` (the local d, the formula of p. 197), `Chapter4/ScFormsNatural2021`, `Chapter4/ScFormsGlobal2021`, `Chapter4/ScFormsSquare2021`, `Chapter4/ScFormsLower2021` (d on an M-polyfold, $d^2 = 0$, H_{sc}), `Chapter4/ScFormsPullback2021`, `Chapter4/ScFormsWedge2021`, `Chapter4/ScFormsWedge2021b` (pull-back, wedge, graded algebra), `Chapter4/BoundaryForms2021`, `Chapter4/A3BoundaryForms2021`, `Chapter4/A3BoundaryForms2021b` (forms on ∂X, the relative complex), `Chapter4/BoundaryFormsPullback2021` (forms on ∂X pulled back to faces of a sub-M-polyfold).
- Integration and Stokes (§§IV.41–IV.43): `Chapter4/EuclideanForms2021`, `Chapter4/EuclideanStokes2021`, `Chapter4/ManifoldForms2021`, `Chapter4/ManifoldIntegral2021`, `Chapter4/ManifoldBoundary2021`, `Chapter4/ManifoldStokes2021`, `Chapter4/ManifoldFaceForms2021`, `Chapter4/StokesFaces`, `Chapter4/StokesFaceCharts`, `Chapter4/StokesFaceOrientation`, `Chapter4/StokesFaceIntegral`, and in Chapter 9 `Chapter9/MathlibFormsBridge2021`, `Chapter9/StokesFaceSum`.
- The Poincaré lemma (§§IV.44–IV.45): `Chapter4/Poincare2021` (parametric integrals), `Chapter4/Poincare2021b` (the homotopy operator, Lemmas 4.5.2 and 4.4.8), `Chapter4/OpenForms`, `Chapter4/Poincare448Shift`, `Chapter4/Poincare448`.
- The wedge product on ∂X and on the relative complex (§IV.46): `Chapter4/BoundaryWedge`, `Chapter4/RelativeWedge`.

Throughout, $C = T^{-1}([0,\infty)^n \oplus W)$ is a partial quadrant presented by an sc-isomorphism $T : E \cong \mathbb{R}^n \oplus W$, with coordinates $\lambda_1, \dots, \lambda_n$; $I(e) = \{i : \lambda_i(e) = 0\}$, $d_C(e) = \#I(e)$, $C_e = \{\lambda_i \geq 0, i \in I(e)\}$ and $E_e = \{\lambda_i = 0, i \in I(e)\}$ (Definition 2.5.1 (preprint 2.33), §II.12). For a retract $O = r(U)$ and a level-1 point z, $T_zO = Dr(z)E_0$ is the tangent space of Definition 2.3.12 (preprint 2.17) and $T^R_zO = T_zO \cap E_z$ the reduced tangent space.

IV.1 Finite-dimensional sc-spaces and corner charts (Definition 4.1.1 (preprint 4.1))

Finite-dimensional sc-spaces. Definition 4.1.1 (preprint 4.1) models its charts on finite-dimensional spaces "with the constant sc-structure"; the text remarks after Definition 1.1.1 (preprint 1.1) that a finite-dimensional space has only that one. This is proved. If E_0 is finite-dimensional, every level E_m is, since it injects into E_0 `Chapter4/ScPlus · ScBanach.finiteDimensional_lvl`. The range of $E_m \to E_0$ is then a finite-dimensional, hence closed, subspace; it contains the smooth points, which are dense (Definition 1.1.1 (2) (preprint 1.1)); so it is all of E_0 `Chapter4/ScPlus · ScBanach.toBase_surjective`. The inclusions are

therefore bijective, and they form an sc-isomorphism of E onto E_0 with the constant structure Chapter4/ScPlus · ScBanach.scIsoConst; in particular every point of E_0 is smooth Chapter4/Characterization · ScBanach.mem_smooth_of_finiteDimensional. The density axiom is what makes this work: $E_0 = \mathbb{R}$, $E_m = 0$ for $m \geq 1$ has compact injective inclusions and levels that differ (Findings, Chapter 4, item 6).

Definition 4.1.1 (preprint 4.1). A chart of X is a *corner chart* if its local model is (U, C, V) with V a finite-dimensional real normed space carrying the constant sc-structure, C a partial quadrant in V, U relatively open in C, and the identity retraction Chapter4/Characterization · MChart.IsCornerChart. Its model is then relatively open in a partial quadrant Chapter4/Characterization · MChart.IsCornerChart.isRelOpenInQuadrant, finite-dimensional Chapter4/Characterization · MChart.IsCornerChart.finiteDimensional, a tame retract — the identity retraction is tame Chapter2/Tame · ScRetraction.id_isTame Chapter4/Characterization · MChart.IsCornerChart.isTameRetract — and an sc$^+$-retract by Remark 4.1.3 (preprint 4.3) (§IV.4) Chapter4/Characterization · MChart.IsCornerChart.isScPlusRetract; every point of its model space is smooth Chapter4/Characterization · MChart.IsCornerChart.mem_smooth. *X has a corner structure* if it has an atlas of corner charts equivalent to its own atlas Chapter4/Characterization · MPolyfold.HasCornerStructure. The dimension of V may vary from chart to chart, as in the text. The text's "sc-smoothly compatible charts", "φ an sc-diffeomorphism", is read as compatibility with the atlas of X, as for tameness (Definition 2.5.7 (preprint 2.37)).

The definition is local Chapter4/Characterization · MPolyfold.hasCornerStructure_iff: X has a corner structure iff every point lies in a corner chart that is a chart of the M-polyfold structure. One direction takes a chart of the equivalent atlas through the point. For the other, the corner charts that are charts of X form an atlas: they cover X by hypothesis, any two charts of X are compatible Chapter2/MPolyfold · MPolyfold.IsChart.compatible, and the atlas is equivalent to that of X because each of its members is a chart of X. *A corner structure makes X tame* Chapter4/Characterization · MPolyfold.HasCornerStructure.isTame: the same atlas is an atlas of tame charts.

The transition maps are C^∞. For a chart whose model is relatively open in a partial quadrant, every overlap $\chi(V \cap V')$ is relatively open Chapter4/Characterization · MChart.isRelOpenInQuadrant_overlap. For compatible corner charts, the transition map is sc-smooth between the overlap retracts; composed with the identity retraction of the overlap it is sc-smooth on a relatively open subset of a partial quadrant in a finite-dimensional space with the constant structure, and such maps are exactly the C^∞ maps with derivatives within the set Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const Chapter4/Characterization · MChart.IsCornerChart.contDiffOn_transition. That is the text's "classically smooth"; "the atlas defines the structure of a smooth manifold with boundary with corners" is not formalized beyond it, because Mathlib's manifolds with corners have a single model space and here the partial quadrants vary from chart to chart.

Arbitrary finite-dimensional models. The constant structure is no restriction. A partial quadrant of a finite-dimensional E is one of E_0 with the constant structure Chapter4/Characterization · IsPartialQuadrant.toConst Chapter4/Characterization · IsRelOpenInQuadrant.toConst, and the identity of E_0 is an sc-diffeomorphism between the two models Chapter4/Characterization · IsRelOpenInQuadrant.isScDiffeoQuadrant_toConst. Composing a chart with it (the construction `compDiffeo` of Chapter 3) Chapter3/GlobalIFTCenter · MChart.compDiffeo gives, with the same map and domain, a corner chart of the same dimension Chapter4/Characterization · MChart.exists_isCornerChart_of_finiteDimensional. Hence X has a corner structure iff it has an equivalent atlas of charts onto relatively open subsets of partial quadrants in sc-Banach spaces with finite-dimensional E_0 Chapter4/Characterization · MPolyfold.hasCornerStructure_iff_finiteDimensional — verbatim the definition of the manifolds with corners of locally constant dimension of Theorem 3.2.3 (preprint 3.27) (§III.36) — and the manifolds with corners of

dimension n of Theorem 3.2.3 (preprint 3.27) are the corner structures all of whose charts have dimension n `Chapter4/Characterization · MPolyfold.isManifoldWithCorners_iff` `Chapter4/Characterization · MPolyfold.IsManifoldWithCorners.hasCornerStructure`.

IV.2 Sc⁺-maps

The text uses "sc⁺-map" in the proof of Lemma 4.1.7 (preprint 4.7) and composes sc⁺-retractions with sc-smooth maps in Lemma 4.1.5 (preprint 4.5), but defines sc⁺ only for retractions (Definition 4.1.4 (preprint 4.4)) and operators (Definition 1.1.5 (preprint 1.4)). The formalization defines it for maps on any set, as the neighbouring notions of the project are (sc⁺-operators are sc-operators into F^1, sc⁺-sections sc-smooth maps into $K[1]$).

The lift. For $x \in E_1 \subseteq E_0$, $1_0^1 x$ is the point of E_1 over x, with a junk value off E_1 `Chapter4/ScPlus · ScBanach.liftOne` `Chapter4/ScPlus · ScBanach.toBase_liftOne` `Chapter4/ScPlus · ScBanach.liftOne_toBase`.

Definition. f is *sc⁺ on* U if $f(U) \subseteq F_1$ and the lift $x \mapsto 1_0^1 f(x)$, a map $U \to F^1$, is sc-smooth `Chapter4/ScPlus · IsScPlus`. Then $f(U_m) \subseteq F_{m+1}$: the lift is sc⁰, so it maps U_m into $(F^1)_m = F_{m+1}$ `Chapter4/ScPlus · IsScPlus.mem_range_succ`. On an sc-domain the canonical lift can be replaced by any sc-smooth $g : U \to F^1$ with $\iota \circ g = f$, which is how the text phrases it `Chapter4/ScPlus · IsScPlus.iff_exists`. The calculus: an sc⁺-map is sc-smooth, being $\iota\circ$ (lift) with the sc-operator $\iota : F^1 \to F$ `Chapter4/ScPlus · IsScPlus.isScSmooth`; the property depends only on the values on U `Chapter4/ScPlus · IsScPlus.congr` and passes to smaller sc-domains `Chapter4/ScPlus · IsScPlus.mono`; *sc⁺ □ sc-smooth is sc⁺*, the lift of $g \circ f$ being (lift of g) $\circ f$, sc-smooth by the chain rule `Chapter1/ChainRule · IsScSmooth.comp` `Chapter4/ScPlus · IsScPlus.comp_isScSmooth`; *sc-smooth □ sc⁺ is sc⁺*, the lift of $g \circ f$ being $g^1\circ$ (lift of f) with $g^1 : V^1 \to G^1$ the map g induces on level 1, sc-smooth by Proposition 1.2.2 (preprint 1.17) `Chapter1/Levels · IsScSmooth.shift` `Chapter4/ScPlus · IsScSmooth.comp_isScPlus`; and an sc⁺-operator is an sc⁺-map `Chapter4/ScPlus · ScPlus.isScPlus`.

The derivative at a smooth point `Chapter4/ScPlus · IsScPlus.exists_scPlus_deriv`. Let f be sc⁺ on an sc-domain U and $x \in U_1$ smooth. The lift g is sc-smooth into F^1, so its derivative at the smooth point x is an sc-operator $E \to F^1$ (the remark after Proposition 1.2.1 (preprint 1.16)), that is an sc⁺-operator $S : E \to F$. Since $f = \iota \circ g$ on U, the chain rule and the uniqueness of the derivative on an sc-domain give $Df(x) = \iota\circ Dg(x)$, the level-0 operator of S. This is the text's step "$Dr(o)$ is a sc⁺-operator, since r is a sc⁺-map". Consequently $Df(x)$ is compact, and so is every level operator of S `Chapter4/ScPlus · IsScPlus.isCompactOperator_deriv`.

A compact projection has finite-dimensional range `Chapter4/ScPlus · finiteDimensional_range_of_isCompactOperator_of_idem`: if $P = P \circ P$ is compact, its range is the closed set $\{v : Pv = v\}$, P restricts to a compact operator of the range which is the identity, and a normed space with compact identity is finite-dimensional (Riesz). The text takes this for granted.

IV.3 Sc⁺-retractions and sc⁺-retracts (Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6), Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7))

Definition 4.1.4 (preprint 4.4). An sc-smooth retraction $r : U \to U$ is an *sc⁺-retraction* if it is an sc⁺-map on U `Chapter4/ScPlus · ScRetraction.IsScPlus`. The text's condition $r(U_m) \subseteq U_{m+1}$ is then a consequence, not a hypothesis `Chapter4/ScPlus · ScRetraction.IsScPlus.mem_lvlSet_succ`. On an open subset V of an M-polyfold, an sc⁺-retraction is a map s, sc-smooth on V, with $s(V) \subseteq V$, $s\circ s = s$ on V, $s(V) \subseteq X_1$, and $1_0^1 \circ s : V \to$

X^1 sc-smooth Chapter4/ScPlus · MPolyfold.IsScPlusRetractionOn; the junk value of 1_0^1 off X_1 is quantified over and plays no role.

Lemma 4.1.5 (preprint 4.5) Chapter4/ScPlus · lemma_4_1_5. *If t is an sc⁺-retraction onto O and s an sc-smooth retraction (of a possibly different relatively open set) onto O, then s is sc⁺.* The text's proof: s maps its domain into O, which lies in the domain of t, and t fixes O, so $s = t \circ s$, which is sc⁺ □ sc-smooth. (The text writes "$s(V) = 0$" for $s(V) = O$.)

Definition 4.1.6 (preprint 4.6). (O, C, E) is an *sc⁺-retract* if it is the retract of an sc⁺-retraction Chapter4/ScPlus · IsScPlusRetract; it is then a retract Chapter4/ScPlus · IsScPlusRetract.isScRetract, every sc-smooth retraction onto it is sc⁺ by Lemma 4.1.5 (preprint 4.5) Chapter4/ScPlus · IsScPlusRetract.isScPlus, and so the property is intrinsic Chapter4/ScPlus · isScPlusRetract_iff_forall.

Lemma 4.1.7 (preprint 4.7) (i) Chapter4/ScPlus · lemma_4_1_7_i. *The retract of an sc⁺-retraction consists of smooth points.* By induction on m: a point $y \in O$ lies in E_m; then $y \in U_m$, and $y = r(y) \in E_{m+1}$ by the level gain. (ii) Chapter4/ScPlus · lemma_4_1_7_ii. *At a level-1 point o over a point of O, $Dr(o)E_0$ is finite-dimensional.* o is smooth by (i), so $Dr(o)$ is compact (§IV.2); it is a projection because $r \circ r = r$ gives $Dr(o) \circ Dr(o) = Dr(o)$ at points of O Chapter2/Retracts · ScRetraction.scDeriv_comp_self; and a compact projection has finite-dimensional range. In the form with the tangent space of Definition 2.3.12 (preprint 2.17) Chapter4/ScPlus · lemma_4_1_7_ii_tangentSpace: every $o \in O$ has a smooth lift $o_1 \in E_1$ Chapter2/TameRetracts · exists_smoothLvl_one_of_mem_smooth, and $T_{o_1}O$ is the range of $Dr(o_1)$ Chapter2/Retracts · tangentSpace_eq_coe_submodule, finite-dimensional. Tangent spaces are indexed by level-1 points, which is why the statement mentions the lift.

IV.4 Proposition 4.1.8 (preprint 4.8), its chart form, and Remark 4.1.3 (preprint 4.3)

Proposition 4.1.8 (preprint 4.8) Chapter4/ScPlus · proposition_4_1_8. *A retract (O, C, E) is an sc⁺-retract iff $O \subseteq E_\infty$ and the lift 1_0^1 is sc-smooth from (O, C, E) to (O_1, C_1, E^1)* in the sense of Definition 2.1.14 (preprint 2.7). (□) $O \subseteq E_\infty$ is Lemma 4.1.7 (i) (preprint 4.7); and by the criterion of Definition 2.1.14 (preprint 2.7) through one retraction Chapter2/Retracts · isScSmoothRetractMap_iff_exists, the lift is sc-smooth between the retracts because $1_0^1 \circ r$ is the lift of the sc⁺-map r. (□) For a retraction r onto O, the composite $U \to O \to O_1 \to E^1$ is sc-smooth (Definition 2.1.14 (preprint 2.7), the field `comp_isScSmooth` of Chapter2/Retracts · IsScSmoothRetractMap) and is the lift of r, whose values lie in E_1 because $O \subseteq E_\infty$. The text's proof is complete; it ends "the lemma is proved".

The chart form. Theorem 4.1.2 (preprint 4.2) needs the statement for the charts of an M-polyfold, which the text does not state. The map $1_0^1 : X \to X^1$ is defined on X_1, with a junk value x_0 off it Chapter4/ScPlus · MPolyfold.toLevelOne Chapter4/ScPlus · MPolyfold.levelVal_toLevelOne Chapter4/ScPlus · MPolyfold.toLevelOne_levelVal. In the chart φ of X and the level chart $(V_1, \varphi, (O_1, C_1, E^1))$ of X^1 Chapter2/LevelPolyfold · MPolyfold.levelChart it is the lift $O \to O_1$ Chapter4/ScPlus · MPolyfold.levelChart_toLevelOne. A point of a chart domain is smooth iff its image is Chapter4/ScPlus · MPolyfold.mem_smoothSet_iff. So *the model of a chart $\varphi : V \to O$ is an sc⁺-retract iff $V \subseteq X_\infty$ and 1_0^1 is sc-smooth on V* Chapter4/ScPlus · MPolyfold.proposition_4_1_8_chart: (□) continuity on V comes from the lift read in the charts, and sc-smoothness is checked in the pair of charts φ, `levelChart` φ Chapter2/LevelPolyfold · IsScSmoothOn.of_exists_charts, where it is Proposition 4.1.8 (preprint 4.8); (□) the same pair of charts turns the sc-smoothness of 1_0^1 on V into that of the lift on O. Hence "$X_0 = X_\infty$ and 1_0^1 sc-smooth" holds iff every chart of X is modelled on an

sc⁺-retract Chapter4/ScPlus · MPolyfold.isScSmoothMap_toLevelOne_iff, and already if some cover of X by such charts exists Chapter4/Characterization · MPolyfold.isScSmoothMap_toLevelOne_of_cover — sc-smoothness being local.

Remark 4.1.3 (preprint 4.3). The inclusion $\iota : (O_1, C_1, E^1) \to (O, C, E)$ is sc-smooth between retracts Chapter4/ScPlus · isScSmoothRetractMap_toBase_one, so $1^0_1 : X^1 \to X$ is always sc-smooth, being ι in the charts `levelChart` φ and φ Chapter4/ScPlus · MPolyfold.remark_4_1_3_levelVal. If every point lies in X_1 and 1^1_0 is sc-smooth, it is an sc-diffeomorphism with inverse 1^0_1 Chapter4/ScPlus · MPolyfold.remark_4_1_3_isScDiffeo. For a finite-dimensional V with the constant structure, $V^1 = V$ definitionally, the lift is the identity, and every retract in V is an sc⁺-retract, its retraction being its own lift Chapter4/ScPlus · remark_4_1_3_const. The remark's "it is possible that $X^1 \neq X$ as M-polyfolds although the sets agree" is given without an example; one is constructed in §IV.27.

IV.5 Charts centred at a smooth point

Theorem 4.1.2 (ii) (preprint 4.2) asks for charts with $\psi(x) = 0$, the proofs of Theorems 4.2.7 and 4.2.11 (preprint 4.18 and 4.22) and of Lemma 5.3.3 (preprint 5.11) use them, and the text never says where they come from. Translating a chart $\varphi : V \to O \subseteq C$ by $e = \varphi(x)$ fails twice: translation by e preserves every level only if e is smooth, and $C - e$ is not a partial quadrant unless $e \in C \cap (-C)$. The construction (Findings, items 1, 2) uses the partial quadrant C_e and the neighbourhood on which it agrees with C.

The partial quadrant C_e and the neighbourhood $G(e)$. Let $G(e) = \{v : \lambda_i(v) > 0$ whenever $\lambda_i(e) \neq 0\}$ Chapter4/Recentre · recentreNbhd, open Chapter4/Recentre · isOpen_recentreNbhd and containing e when $e \in C$ Chapter4/Recentre · mem_recentreNbhd. On it C and C_e agree: $G(e) \cap C = G(e) \cap C_e$, because on $G(e)$ the coordinates not vanishing at e are positive anyway Chapter4/Recentre · recentreNbhd_inter_eq. C_e is invariant under translation by e Chapter4/Recentre · add_mem_partialQuadrantAtWith_iff, so $C_e - e = C_e$ Chapter4/Recentre · image_sub_partialQuadrantAtWith. *C_e is a partial quadrant* — Definition 2.5.1 (preprint 2.33) asserts it without proof: the vanishing indices $I(e)$ Chapter4/Recentre · zeroIdx, $d = \#I(e)$ of them Chapter4/Recentre · recentreDim, are put first, $\mathbb{R}^n \cong \mathbb{R}^{I(e)} \times \mathbb{R}^{\text{rest}}$ Chapter4/Recentre · splitCoordsL, which gives an sc-isomorphism $\mathbb{R}^n \oplus W \cong \mathbb{R}^{I(e)} \oplus (\mathbb{R}^{\text{rest}} \oplus W)$ Chapter4/Recentre · splitCoordsIso and the presentation T_e of C_e Chapter4/Recentre · recentreIso Chapter4/Recentre · partialQuadrantAtWith_eq_preimage Chapter4/Recentre · isPartialQuadrant_partialQuadrantAtWith: the coordinates not vanishing at e join the linear factor. *Re-centring keeps the tame data near e.* For $v \in G(e)$ the vanishing coordinates of $v - e$ for T_e are those of v for T Chapter4/Recentre · recentreIso_fst_sub Chapter4/Recentre · zero_of_mem_recentreNbhd, so $d_{C_e}(v-e) = d_C(v)$ Chapter4/Recentre · degIndexWith_recentreIso Chapter4/Recentre · degIndex_sub_recentre, and the minimal subspaces and partial quadrants agree: $E^{T_e}_{v-e} = E^T_v$ Chapter4/Recentre · minimalSubspaceWith_recentreIso, $(C_e)_{v-e} = C_v$ Chapter4/Recentre · partialQuadrantAtWith_recentreIso.

Translations. Translation by a smooth vector a is sc-smooth on every sc-domain: its level maps are $x \mapsto x + a_m$, a_m the lift of a to E_m, and the criterion of Proposition 1.2.4 (preprint 1.19) applies Chapter1/Levels · IsScSmooth.of_contDiffOn_levels Chapter4/Recentre · isScSmooth_add_const Chapter4/Recentre · isScSmooth_sub_const. For U relatively open in C with $U \subseteq G(e)$, the translate $U - e$ is relatively open in C_e Chapter4/Recentre · relOpen_translate and, for smooth e, $v \mapsto v - e$ is an sc-diffeomorphism $U \to U - e$ Chapter4/Recentre · isScDiffeoQuadrant_translate.

Re-centred retracts. A retraction r restricted to $U \cap r^{-1}(W) \cap W$, W open, is a retraction whose domain lies in W Chapter4/Recentre · ScRetraction.restrictIn, onto $r(U) \cap W$ Chapter4/Recentre · ScRetraction.restrictIn_retract, with the same derivative Chapter4/Recentre · ScRetraction.scDeriv_restrictIn, and tame

if r is Chapter4/Recentre · ScRetraction.IsTameWith.restrictIn Chapter4/Recentre · ScRetraction.restrictIn_isTame. Transporting the restriction to $G(e)$ along the translation Chapter2/TameInvariance · IsScDiffeoQuadrant.transport shows that $((O \cap G(e)) - e, C_e, E)$ is a retract, with retraction $u \mapsto r(u+e) - e$ Chapter4/Recentre · isScRetract_recentre, and tame if (O, C, E) is Chapter2/TameInvariance · IsScDiffeoQuadrant.isTame_transport Chapter4/Recentre · isTameRetract_recentre — the transport of tameness uses exactly the invariance of d and of the minimal subspaces above.

Re-centred charts. For a chart φ, a presentation of its partial quadrant and a smooth e, the domain $V \cap \varphi^{-1}(G(e))$ Chapter4/Recentre · MChart.recentreSource is open, and the chart $\varphi_e = \varphi - e$ on it is φ restricted Chapter2/MPolyfold · MChart.restrict and composed with the translation Chapter4/Recentre · MChart.recentre, modelled on $((O \cap G(e)) - e, C_e, E)$. It is a chart of the M-polyfold, its transition maps being those of φ composed with translations Chapter3/GlobalIFTCenter · MChart.compDiffeo_isChart Chapter4/Recentre · MChart.recentre_isChart; $\varphi_e(x) = 0$ when $\varphi(x) = e$ Chapter4/Recentre · MChart.recentre_apply_self; its model is tame when that of φ is Chapter4/Recentre · MChart.isTameRetract_recentre; and it computes the same degeneracy index Chapter4/Recentre · MChart.chartDegIndex_recentre. So *every smooth point x of a chart domain is the centre of a chart of X* inside that domain, tame if the original is Chapter4/Recentre · MPolyfold.exists_centredChart, and *in a tame M-polyfold every smooth point is the centre of a tame chart* Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart Chapter4/Recentre · MPolyfold.IsTame.exists_centredChart. At the centre every coordinate of C_e vanishes, so the minimal subspace there is $\{\lambda = 0\}$, the text's "$\{0\} \oplus W$", which is what §§IV.14, IV.16 and IV.17 use.

Re-centred strong bundle charts (for Lemma 5.3.3 (preprint 5.11)). Only the base is translated: $(u, h) \mapsto (u \mp e, h)$ Chapter4/Recentre · subFst Chapter4/Recentre · addFst is sc-smooth on $E \oplus F$ and on $E \oplus F^1$ because $(e, 0)$ is smooth in both Chapter4/Recentre · mem_smooth_prod_zero Chapter4/Recentre · isScSmooth_addFst Chapter4/Recentre · isScSmooth_subFst. A strong bundle retraction restricts to $(U \cap r^{-1}(G) \cap G) \lhd F$ Chapter4/Recentre · IsStrongBundleRetraction.restrictIn and translates to $R_e(u, h) = (r(u+e) - e, \Gamma(u+e, h))$ Chapter4/Recentre · IsStrongBundleRetraction.translate; with $\Phi_e(y) = (\Phi(y)_1 - e, \Phi(y)_2)$ this is the re-centred chart Chapter4/Recentre · StrongBundleChart.recentre. Its overlaps are translates of open parts of the old ones Chapter4/Recentre · StrongBundleChart.recentre_overlap Chapter4/Recentre · StrongBundleChart.overlap_recentre_eq, its transition maps on $[0]$ and $[1]$ are the old ones composed with translations by $(e, 0)$, so compatibility passes to it Chapter4/Recentre · StrongBundleChart.Compatible.recentre, and it is a strong bundle chart Chapter4/Recentre · StrongBundle.IsChart.recentre. Every smooth point of the base is the centre of one Chapter4/Recentre · StrongBundle.IsChart.exists_centred. No hypothesis beyond the smoothness of the point is needed.

IV.6 Proposition 4.1.9 (preprint 4.9): the data, and the mean value estimate

The text. Let $t : U \to U$ be an sc$^+$-retraction onto $O \ni 0$ with T_0O in good position to C, Y a good complement and $p = Dt(0)$ "the sc-projection". Then p maps a neighbourhood of 0 in O sc-diffeomorphically onto one in $T_0O \cap C$. The proof: the section $\mathbf{f}(u) = (1-p)(u - t(u))$ of $U \lhd Y$ is sc-Fredholm (Lemma 4.1.11 (preprint 4.10)), Theorem 3.5.11 (preprint Corollary 3.47) describes its zeros near 0 as a graph $\{a + \delta(a)\}$, Lemma 4.1.12 (preprint 4.11) says that every small zero is on the graph, Lemma 4.1.13 (preprint 4.12) that the graph lies in O.

What changes, and why. (a) $Dt(0)$ is a projection onto T_0O along $\ker Dt(0)$, which nothing relates to Y; Lemma 4.1.13 (preprint 4.12) writes $t(a + \delta(a)) = p\,t(a + \delta(a)) + \delta(a)$ with $\delta(a) \in Y$, which needs the

projection *along* Y. (b) Y must be closed, or that projection is not continuous. (c) Theorem 3.5.11 (preprint Corollary 3.47) gives δ over a complement of its own choosing and sc-smooth only as a germ — C^k into E_m on neighbourhoods shrinking with m and k (§III.20) — while the claim is an sc-diffeomorphism between fixed open sets. The formalization therefore takes p to be the projection along a closed Y and proves the proposition directly, with the sc$^+$ property doing the work of the Fredholm theory (Findings, items 20–28).

The data `Chapter4/ScPlusGraph · GraphData`: the sc$^+$-retraction t with $0 \in O$; N with $N = T_0O$, which equals $Dt(0)E_0$ `Chapter4/ScPlusGraph · GraphData.N_eq_range` and is finite-dimensional by Lemma 4.1.7 (preprint 4.7) `Chapter4/ScPlusGraph · GraphData.finiteDimensional`; N in good position; a good complement Y, closed; and $p : E_0 \to N$ continuous linear with $p = 1$ on N, $p = 0$ on Y. The tangent space consists of smooth points: $Dt(0)$ is an sc$^+$-operator and N its fixed vectors `Chapter4/ScPlusGraph · GraphData.smooth`. $Y = \ker p$ `Chapter4/ScPlusGraph · GraphData.mem_Y_iff`. Because $N \subseteq E_\infty$ is finite-dimensional, it lifts continuously to every level `Chapter3/TangentGerm · finSmoothLift` `Chapter4/ScPlusGraph · GraphData.liftN`, and p, $q = 1 - p$ act on every level E_m `Chapter4/ScPlusGraph · GraphData.pL` `Chapter4/ScPlusGraph · GraphData.qL`. The identities the text uses hold for this p on every level: $(1 - p)Dt(0) = 0$, because the range of $Dt(0)$ is N `Chapter4/ScPlusGraph · GraphData.qL_lvlDeriv_zero`, and $Dt(0) = 1$ on N `Chapter4/ScPlusGraph · GraphData.lvlDeriv_zero_of_mem_N`. The good complement has a constant ε: for $n \in N$, $y \in Y$, $|y|_0 \le \varepsilon|n|_0$, $n \in C \iff n + y \in C$ `Chapter4/ScPlusGraph · GraphData.eps` `Chapter4/ScPlusGraph · GraphData.eps_spec`. Its level-1 version is $\varepsilon' = \varepsilon/(\|\iota_1\|\|\mathrm{lift}_N\| + 1)$ `Chapter4/ScPlusGraph · GraphData.eps'`: $|y|_1 \le \varepsilon'|a|_1$ implies $|y|_0 \le \varepsilon|a|_0$ for $a \in N$ `Chapter4/ScPlusGraph · GraphData.norm_toBase_le_eps_mul`.

The level data of an sc$^+$-retraction. The lift $t^1 : U \to E^1$ `Chapter4/ScPlusGraph · ScRetraction.liftOne` is sc-smooth `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.isScSmooth_liftOne`, sc^1 with derivative family Dt^1 `Chapter4/ScPlusGraph · ScRetraction.liftDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.isSc1With_liftOne`. By Proposition 1.2.1 (preprint 1.16) for t^1, the map $t : U_{k+1} \to E_{k+1}$ `Chapter4/ScPlusGraph · ScRetraction.lvlMap` is differentiable within U_{k+1} with a derivative $Dt(x) \in \mathcal{L}(E_{k+1})$ `Chapter4/ScPlusGraph · ScRetraction.lvlDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.hasFDerivWithinAt_lvlMap` that depends continuously on x in the operator norm `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.continuousOn_lvlDeriv`; t is continuous on U_{k+1} `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.continuousOn_lvlMap`, fixes O_{k+1} `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.lvlMap_eq_self`, is t itself through the inclusions `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_lvlMap`, and its level derivatives are restrictions of Dt `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_liftDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_lvlDeriv`, projections at points of O `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.lvlDeriv_lvlDeriv`.

The mean value estimate (the text's (76)–(77)). The abstract form `Chapter4/ScPlusGraph · exists_norm_sub_sub_le_of_hasFDerivWithinAt`: if s is differentiable within a convex set S with derivative D continuous at $u_0 \in S$, then for every $\theta > 0$ there is $r > 0$ with $|s(z') - s(z) - D(u_0)(z' - z)| \le \theta|z' - z|$ for $z, z' \in S \cap B(u_0, r)$ — the mean value inequality for $s - D(u_0)$. For an sc$^+$-retraction `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.exists_mvt`: at $u_0 \in U_{k+1}$ a ball meets C_{k+1} inside U_{k+1} `Chapter4/ScPlusGraph · ScRetraction.exists_ball_inter_subset` `Chapter4/ScPlusGraph · ScRetraction.lvlSet_U_subset_lvlSet_C`, the set $C_{k+1} \cap B(u_0, r_0)$ is convex, and the abstract form and the continuity of Dt give r with $C_{k+1} \cap B(u_0, r) \subseteq U_{k+1}$, $\|Dt(z) - Dt(u_0)\| \le \theta$ and the estimate with constant θ there. The text integrates along segments in U_{k+1}, which need not be convex; the convex set $C_{k+1} \cap B$ is the repair.

A shortcut does not work: inverting $p \circ t$ on the finite-dimensional quadrant $N \cap C$ by Proposition 3.8.17 (preprint 3.73). In $\mathbb{R}^3$ with $C = [0, \infty)^2 \times \mathbb{R}$, $\Gamma(a) = (a_1, a_2, a_2^2)$ and $t(x) = \Gamma(x_1 + (x_3 - x_2^2)^2, x_2)$, t is a retraction with $N = \mathbb{R}^2 \times 0$, but $p\,t(x_1, x_2, 0) = (x_1 + x_2^4, x_2)$ misses $(0, y_2)$ for $y_2 > 0$.

IV.7 Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12): uniqueness and existence on level 1

Lemma 4.1.12 (preprint 4.11), the contraction estimate `Chapter4/ScPlusGraph · GraphData.exists_contraction_radius`. For $\theta > 0$ the mean value estimate at $u_0 = 0$ gives $r > 0$ with the estimates of §IV.6 on $C_{k+1} \cap B(0,r)$ and, for $u, u' \in O_{k+1}$ with norms below r, $|(1-p)(u'-u)|_{k+1} \le \|1-p\|\theta|u'-u|_{k+1}$: since $u = t(u)$, $u' = t(u')$ and $(1-p)Dt(0) = 0$, $(1-p)(u'-u) = (1-p)[t(u') - t(u) - Dt(0)(u'-u)]$.
Uniqueness `Chapter4/ScPlusGraph · GraphData.eq_of_p_eq`: if $p(u) = p(u')$ and $|(1-p)(u'-u)| \le c|u'-u|$ with $c < 1$, then $u' - u \in Y$, so $(1-p)(u'-u) = u'-u$ and $u' = u$. This is what the text's Lemma 4.1.12 (preprint 4.11) is used for; its statement, "$y = \delta(a)$", presupposes the δ of Theorem 3.5.11 (preprint Corollary 3.47).

Lemma 4.1.13 (preprint 4.12), existence by a contraction `Chapter4/ScPlusGraph · GraphData.exists_graph_point`. Suppose the estimates hold on $C_1 \cap B(0,r)$ with $\theta < 1$, $\|1-p\|\theta < 1$ and $\|1-p\|\theta(1+\varepsilon') \le \varepsilon'$. As $t(0) = 0$ and $t : U_1 \to E_1$ is continuous at 0, there is δ with $|t(u)|_1 < r$ for $|u|_1 < \delta$; put $r' = \min(r, \delta)$ and $\rho = r'/(1+\varepsilon')$. For $a \in N \cap C$ with $|a|_1 < \rho$:

1. The set $B_a = \{y \in Y_1 : |y|_1 \le \varepsilon'|a|_1\}$ is closed, hence complete, because Y is closed — the first use of that hypothesis.
2. For $y \in B_a$, $a + y \in C_1$ by the good complement property, $|y|_0 \le \varepsilon|a|_0$ coming from $|y|_1 \le \varepsilon'|a|_1$; and $|a+y|_1 \le (1+\varepsilon')|a|_1 < r'$. So $a + y$ lies in U_1, where t is defined — a point the text does not check.
3. $\Phi_a(y) = (1-p)t(a+y)$ maps B_a into itself: $\Phi_a(y) = (1-p)[t(a+y) - t(0) - Dt(0)(a+y)]$, of norm at most $\|1-p\|\theta(1+\varepsilon')|a|_1 \le \varepsilon'|a|_1$, and it lies in Y.
4. Φ_a is a contraction with constant $\|1-p\|\theta < 1$, by the estimate between $a+y$ and $a+y'$.
5. Its fixed point y (Banach's theorem on the complete B_a) gives $u = a + y$ with $(1-p)(t(u) - u) = 0$, i.e. $w - u \in N$ for $w = t(u)$. Now $w \in O$, $|w|_1 < r$ by the choice of δ, $t(w) = w$, and $Dt(0)(w-u) = w - u$. The estimate between u and w reads $|t(w) - t(u) - Dt(0)(w-u)|_1 = |w-u|_1 \le \theta|w-u|_1$, so $w = u$: $u = t(u) \in O$ — the text's argument "$a_1 = a$".

So $u \in O_1$, $p(u) = a$, $u - a \in Y$, $|u-a|_1 \le \varepsilon'|a|_1$ and $|u|_1 < r$. The text's "$|a(\tau)|_1 \le \varepsilon_1$ if $|a|_0 < \varepsilon$" is the level-1 continuity of t at 0 used in step 5.

IV.8 The step $p(O \cap W) \subseteq C$, and the neighbourhoods

p maps points of O near 0 into C `Chapter4/ScPlusGraph · GraphData.p_mem_C`, which the conclusion "$p : U_0 \to V \subseteq T_0O \cap C$" needs and the text never checks. If $u \in O_1$ has $|(1-p)u|_1 \le \kappa|u|_1$ with $\kappa \le \frac{1}{2}$, then $|u|_1 \le |pu|_1 + \frac{1}{2}|u|_1$, so $|u|_1 \le 2|pu|_1 \le 2\|\mathrm{lift}_N\||pu|_0$, and $|(1-p)u|_0 \le \|\iota_1\|\kappa|u|_1 \le 2\|\iota_1\|\kappa\|\mathrm{lift}_N\||pu|_0$; if that constant is at most ε, the good complement property gives $p(u) \in C \iff p(u) + (1-p)u = u \in C$, and $u \in O \subseteq C$. The hypothesis on u is the contraction estimate between 0 and u.

The neighbourhoods `Chapter4/ScPlusGraph · GraphData.Nbhd` `Chapter4/ScPlusGraph · GraphData.exists_nbhd`. For every $\theta_0 > 0$ the proof chooses, in this order:

1. $\theta = \min(\theta_0, \theta_1, \theta_2)$ with $\theta_1 = \varepsilon'/(2(\|1-p\|+1)(1+\varepsilon'))$ (so that $\|1-p\|\theta \le \frac{1}{2}$ and $\|1-p\|\theta(1+\varepsilon') \le \varepsilon'$, for Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12)) and $\theta_2 = \varepsilon/(2\|\iota_1\|\|\mathrm{lift}_N\|(\|1-p\|+1)+1)$ (for $p(O \cap W) \subseteq C$);

2. r from the contraction estimate on level 1, and ρ and the graph points $u(a)$ from Lemma 4.1.13 (preprint 4.12);
3. δ_0 from the continuity of the lift $t^1 : U \to E_1$ at 0 on level 0: $|t(u)|_1 < r$ for $u \in U$, $|u|_0 < \delta_0$;
4. $\rho' = \min(\rho, r/(1+\varepsilon'), \delta_0/(\|\iota_1\|(1+\varepsilon')+1))$, the open $V' = \{a \in N : |a|_1 < \rho'\}$ of N, $V = V' \cap C$, and $W = B_0(0,\delta_0) \cap p^{-1}(V')$, open in E_0;
5. $\Gamma(a) = u(a)$ for $a \in V$.

Then $\Gamma(a) \in O \cap W$ (on level 0, $|\Gamma(a)|_0 \le \|\iota_1\|(1+\varepsilon')|a|_1 < \delta_0$), $p\Gamma(a) = a$, $\Gamma(a) - a \in Y$, $|\Gamma(a)|_1 < r$. A point $u_0 \in O \cap W$ satisfies $u_0 = t(u_0)$, hence $|u_0|_1 < r$ by step 3 — *this is how a neighbourhood open on level* 0 *is controlled on level* 1, the sc⁺ property a second time; the contraction estimate between 0 and u_0 and the first constraint on θ then give $p(u_0) \in C$, so $p(u_0) \in V$, and uniqueness gives $\Gamma(p(u_0)) = u_0$. Also $\|Dt(z) - Dt(0)\|_{\mathcal{L}(E_1)} \le \theta_0$ on $C_1 \cap B(0,r)$, and $|(1-p)(u'-u)|_1 \le \frac{1}{2}|u'-u|_1$ on $O_1 \cap B_1(0,r)$. Consequently $\Gamma(V) = O \cap W$ Chapter4/ScPlusGraph · GraphData.Nbhd.image_V, $\Gamma(0) = 0$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_zero, and p is injective on $O \cap W$ Chapter4/ScPlusGraph · GraphData.Nbhd.injOn_p.

IV.9 The regularity of Γ on the fixed V: the frame formula and the bootstrap

Continuity into every level. $\Gamma(V) \subseteq O \subseteq E_\infty$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_mem_smooth, so Γ has lifts $\Gamma_m : V \to E_m$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl with values in O_m Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl_mem_retract. Γ_1 is Lipschitz: with $u = \Gamma(a)$, $u' = \Gamma(a')$, $|u'-u|_1 \le |p(u'-u)|_1 + |(1-p)(u'-u)|_1 \le \|\mathrm{lift}_N\||a'-a| + \frac{1}{2}|u'-u|_1$, so $|u'-u|_1 \le 2\|\mathrm{lift}_N\||a'-a|$ Chapter4/ScPlusGraph · GraphData.Nbhd.norm_Γ_lvl_one_sub_le Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ_lvl_one. And $\Gamma_{k+1} = t^1 \circ \Gamma_k$ on level k, because $\Gamma(a) = t(\Gamma(a))$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl_succ; the lift t^1 is continuous from U_k into E_{k+1}, so by induction Γ is continuous into every level Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ_lvl Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ.

The frame at a point $u \in O$ near 0. Let $A(u) = p \circ Dt(u)|_N : N \to N$ Chapter4/ScPlusGraph · GraphData.A, $M(u) = A(u)^{-1}$ Chapter4/ScPlusGraph · GraphData.M and $B_u = Dt(u) \circ M(u)$, the *frame* Chapter4/ScPlusGraph · GraphData.frame, with its level-$(k+1)$ version Chapter4/ScPlusGraph · GraphData.frameL. For $n \in N$, $Dt(u)n - n = \iota_1(Dt_1(u) - Dt_1(0))\mathrm{lift}(n)$ Chapter4/ScPlusGraph · GraphData.scDeriv_sub_eq. Choose θ_0 with $\|p\|\|\iota_1\|\|\mathrm{lift}_N\|\theta_0 \le \frac{1}{2}$ and $\|1-p\|\theta_0 < 1$ Chapter4/ScPlusGraph · GraphData.FrameSmall Chapter4/ScPlusGraph · GraphData.exists_frameSmall, and let $\|Dt_1(u) - Dt_1(0)\| \le \theta_0$. Then (L1) $|A(u)n - n| \le \frac{1}{2}|n|$, so $A(u)$ is injective Chapter4/ScPlusGraph · GraphData.A_injective, hence an isomorphism of the finite-dimensional N Chapter4/ScPlusGraph · GraphData.Aeq, with $A(u)M(u) = 1$ Chapter4/ScPlusGraph · GraphData.A_M and $p \circ B_u = 1$ Chapter4/ScPlusGraph · GraphData.p_frame. (L2) $T_uO \cap Y = 0$: a vector $v = Dt(u)v$ in Y satisfies $v = (1-p)v = (1-p)(Dt(u) - Dt(0))v$ on level 1, so $|v|_1 \le \|1-p\|\theta_0|v|_1$ Chapter4/ScPlusGraph · GraphData.eq_zero_of_mem_Y_of_scDeriv_eq. Hence for $u \in O_1$: $Dt(u)x - B_u(p\,Dt(u)x)$ lies in $T_uO \cap Y$ and vanishes, $(1 - B_up)Dt(u) = 0$ Chapter4/ScPlusGraph · GraphData.scDeriv_sub_frame; $Dt(u)B_u = B_u$ Chapter4/ScPlusGraph · GraphData.scDeriv_frame; and both hold on every level Chapter4/ScPlusGraph · GraphData.p_toBase_frameL Chapter4/ScPlusGraph · GraphData.lvlDeriv_sub_frameL. On the points of $\Gamma(V)$ the derivative bound holds by the choice of r Chapter4/ScPlusGraph · GraphData.Nbhd.deriv_bound_Γ.

The abstract step. On a set S of a normed space, let s satisfy the mean value estimate with derivative D_0 at u_0 and constant θ, let q, B be continuous linear with $q \circ B = 1$ and $(1-Bq)D_0 = 0$, and $\theta\|1-Bq\| \le \frac{1}{2}$. For fixed points u, u' of s in S, $(1-Bq)(u'-u) = (1-Bq)[s(u') - s(u) - D_0(u'-u)]$, of norm at

most $\frac{1}{2}|u'-u|$, so $|u'-u| \le 2\|B\||q(u'-u)|$ Chapter4/ScPlusGraph · norm_sub_le_of_projection. And *every continuous right inverse Γ of q with values in the fixed points of s in S, through u_0, is differentiable at $z_0 = q(u_0)$ with derivative B* Chapter4/ScPlusGraph · hasFDerivWithinAt_of_projection: $\Gamma(z) - \Gamma(z_0) - B(z-z_0) = (1 - Bq)(\Gamma(z) - u_0) = (1 - Bq)[(\Gamma(z) - u_0) - D_0(\Gamma(z) - u_0)]$, which is $o(|\Gamma(z) - u_0|)$ by the differentiability of s at u_0, and $|\Gamma(z) - u_0| \le 2\|B\||z - z_0|$ by the Lipschitz estimate.

The frame formula Chapter4/ScPlusGraph · GraphData.Nbhd.hasFDerivWithinAt_Γ_lvl. At $a_0 \in V$ and on level $k+1$, apply the abstract step with $s = t$, $u_0 = \Gamma_{k+1}(a_0)$, $q = p \circ \iota_{k+1}$, B the level-$(k+1)$ frame at u_0, $\theta = 1/(2(\|1 - Bq\| + 1))$, and $S = C_{k+1} \cap B(u_0, r)$ from the mean value estimate *at* u_0; Γ_{k+1} stays in $B(u_0, r)$ near a_0 by continuity. So

$$D\Gamma(a) = Dt(\Gamma(a))|_N \circ \left(p \circ Dt(\Gamma(a))|_N\right)^{-1} \quad \text{into every level } E_{k+1}.$$

The estimate is used at $\Gamma(a)$ with the projection $1 - B_{\Gamma(a)}p$, not at 0 with $1 - p$: this is why the regularity holds at every point of the fixed V.

The C^k-bootstrap. By Proposition 1.2.3 (preprint 1.18) for the tangent map of the lift t^1, $(x, w) \mapsto Dt_{j+1}(x)w$ is C^k on $U_{j+k+1} \times E_{j+k+1}$, read through the inclusions into level $j+1$ Chapter4/ScPlusGraph · GraphData.contDiffOn_lvlDeriv_apply. If Γ_{j+k+1} and $w : V \to N$ are C^k, then so is $a \mapsto Dt_j(\Gamma_{j+1}(a))\,\mathrm{lift}(w(a))$ Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_lvlDeriv_Γ; in particular $a \mapsto A(\Gamma(a))$ is C^k when Γ_{k+1} is Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_A_Γ, and so is its inverse, inversion being smooth on the invertible operators Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_M_Γ. Now induct on k, for all levels at once Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl: Γ_{j+1} is C^0 (continuity); if every $\Gamma_{j'+1}$ is C^k, the derivative $a \mapsto \mathrm{frame}_j(\Gamma_{j+1}(a))$ applied to n is $Dt_j(\Gamma_{j+1}(a))\mathrm{lift}(M(\Gamma_1(a))n)$, which is C^k because Γ_{j+k+1} is (the hypothesis at $j' = j+k$) and $M(\Gamma_1(\cdot))n$ is (the hypothesis at $j' = k$, through A); so Γ_{j+1} is C^{k+1}. Derivatives of t are only ever applied to the smooth vectors of the finite-dimensional N, so the loss of levels in Tt is absorbed by Γ being C^k into *every* level and does not accumulate. Hence Γ is C^∞ into every level Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl_succ_infty Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl_infty and sc-smooth on V by the criterion of Proposition 1.2.4 (preprint 1.19) for a finite-dimensional source Chapter4/ScPlusGraph · GraphData.Nbhd.isScSmooth_Γ.

The sc-diffeomorphism. $(O \cap W, C, E)$ is a retract (Proposition 2.3.1 (1) (preprint 2.10)) Chapter2/Retracts · proposition_2_3_1_one Chapter4/ScPlusGraph · GraphData.Nbhd.isScRetract_inter_W; $(V, N \cap C, N)$ is a local model, $N \cap C$ being a partial quadrant of N (Proposition 3.1.25 (preprint 3.22)) and V relatively open in it Chapter4/ScPlusGraph · GraphData.Nbhd.V_isRelOpenInQuadrant Chapter4/ScPlusGraph · GraphData.Nbhd.isScRetract_V. p is sc-smooth $E \to N$ as a continuous linear map into a finite-dimensional space (Corollary 1.2.5 (preprint 1.20)) Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional Chapter4/ScPlusGraph · GraphData.p_isScSmooth, so p and Γ are sc-smooth between the retracts Chapter4/ScPlusGraph · GraphData.Nbhd.p_isScSmoothRetractMap Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_isScSmoothRetractMap.

Proposition 4.1.9 (preprint 4.9) Chapter4/ScPlusGraph · proposition_4_1_9. *Statement.* Let t be an sc$^+$-retraction onto $O \ni 0$, $N = T_0O$ in good position to C, Y a closed good complement of N and $p : E_0 \to N$ the projection along Y. There are an open $W \ni 0$ in E_0, a relatively open $V \ni 0$ of $N \cap C$ and $\Gamma : N \to E_0$, sc-smooth on V, with $p \circ \Gamma = 1$ on V, $\Gamma(a) - a \in Y$, $\Gamma(V) = O \cap W$, $\Gamma \circ p = 1$ on $O \cap W$, $\Gamma(0) = 0$, and p, Γ sc-smooth between $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *Proof.* θ_0 from the frame condition, the neighbourhoods of §IV.8 for it, and the regularity above. The finite dimension of N, an instance argument needed to form N as an sc-Banach space, is Lemma 4.1.7 (preprint 4.7) Chapter4/

ScPlusGraph · finiteDimensional_tangentSpace_zero.

IV.10 Theorem 4.1.2 (preprint 4.2): the characterization

Statement Chapter4/Characterization · MPolyfold.theorem_4_1_2_general. For every M-polyfold X and every junk value $x_0 \in X_1$ of 1^1_0, the following are equivalent. (i) X has a corner structure. (ii) Every point of X is smooth, $1^1_0 : X \to X^1$ is sc-smooth, and every x lies in a chart ψ of X with $\psi(x) = 0$ whose model is a tame retract and whose tangent space $N = T_0O$ is finite-dimensional and in good position to C. The text's form, for tame X, is the same statement with an unused hypothesis Chapter4/Characterization · MPolyfold.theorem_4_1_2. Tameness is used neither in the text's proof nor here, and (i) implies it (§IV.1).

Charts composed with an sc-diffeomorphism between retracts. The proof of (ii) □ (i) composes a chart with p, which is an sc-diffeomorphism between retracts, not between relatively open sets, so the construction of Chapter 3 does not apply. For a chart φ onto (O, C, E) and mutually inverse f, g sc-smooth between (O, C, E) and (O', C', E'), $f \circ \varphi$ is a chart onto (O', C', E') Chapter4/Characterization · MChart.compRetract, and a chart of the M-polyfold when φ is Chapter4/Characterization · MChart.compRetract_isChart: for a chart φ' of X the transition maps are $(\varphi' \circ \varphi^{-1}) \circ g$ and $f \circ (\varphi \circ \varphi'^{-1})$, compositions of maps sc-smooth between retracts (Theorem 2.1.15 (preprint 2.8)) Chapter2/Retracts · IsScSmoothRetractMap.comp. This is the compatibility the text's "we obtain a smoothly compatible atlas" skips (Findings, item 55).

(i) □ (ii). Corner charts are modelled on sc⁺-retracts, so every point is smooth and 1^1_0 is sc-smooth by Proposition 4.1.8 (preprint 4.8) on a cover (§IV.4). The text stops here; (ii) also asks for a centred tame chart with T_0O finite-dimensional and in good position (Findings, item 53). Let x lie in a corner chart χ. Its image is smooth (§IV.1), so the re-centred chart $\psi = \chi - \chi(x)$ exists; its model is a relatively open subset of $C_{\chi(x)}$ in the same finite-dimensional space Chapter4/Characterization · image_sub_const_eq_preimage_add' Chapter4/Characterization · MChart.IsCornerChart.exists_centred. With the identity retraction it is a tame retract; the tangent space of a relatively open subset of a partial quadrant is the whole space, because the derivative of the identity retraction is the identity Chapter4/Characterization · tangentSpace_relOpen; and the whole space is in good position to every partial quadrant, with the complement 0, the interior of a partial quadrant being nonempty Chapter4/Characterization · isInGoodPosition_top Chapter4/Characterization · MChart.IsCornerChart.exists_centred_tame.

(ii) □ (i). *The data of the corner chart* Chapter4/Characterization · CornerChartData for a chart ψ and a finite-dimensional N: a closed good complement Y, the projection p along it, W, V, Γ with p, Γ mutually inverse and sc-smooth between $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *They exist* when the model of ψ is an sc⁺-retract containing 0 and $N = T_0O$ is in good position Chapter4/Characterization · exists_cornerChartData: here no complement is prescribed, so one can be chosen — the normal form of Proposition 3.1.25 (preprint 3.22) gives a continuous projection onto N whose kernel is a good complement, closed, even mapping C into C Chapter3/GoodComplementProjection · exists_proj_goodComplement — and Proposition 4.1.9 (preprint 4.9) is applied to an sc⁺-retraction onto O (Definition 4.1.6 (preprint 4.6)). The issue "$p = Dt(0)$" of §IV.6 does not arise, and nothing is added to Theorem 4.1.2 (preprint 4.2) (Findings, item 54). *The chart* $p \circ \psi$ on $\psi^{-1}(W)$ Chapter4/Characterization · CornerChartData.chart is ψ restricted to $\psi^{-1}(W)$ composed with p by `compRetract`, modelled on $(V, N \cap C, N)$ Chapter4/Characterization · CornerChartData.model; it is a corner chart Chapter4/Characterization · CornerChartData.chart_isCornerChart, a chart of X Chapter4/Characterization · CornerChartData.chart_isChart, contains x when $\psi(x) = 0$ Chapter4/Characterization · CornerChartData.mem_chart_source and sends it to 0 Chapter4/

Characterization · CornerChartData.chart_apply_eq_zero. *At every point* Chapter4/Characterization · MPolyfold.theorem_4_1_2_chart: if all points are smooth and 1^1_0 is sc-smooth, the model of every chart is an sc⁺-retract (§IV.4), so for a chart ψ with $\psi(x) = 0$ and $N = T_0O$ finite-dimensional and in good position the corner chart $p \circ \psi$ exists. With such a chart at every point, X has a corner structure by the local form of Definition 4.1.1 (preprint 4.1) Chapter4/Characterization · MPolyfold.hasCornerStructure_of_centred — whether or not the charts ψ are tame.

IV.11 Smooth finite-dimensional submanifolds; the chart retraction; Remark 4.2.2 (preprint 4.14)

Definition 4.2.1 (preprint 4.13). $A \subseteq X$ is a *smooth finite-dimensional submanifold* if every $a \in A$ has an open $V \ni a$ and an sc⁺-retraction s on V with $s(V) = A \cap V$ Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold; the data at a point are a structure Chapter4/Submanifolds · MPolyfold.SubmanifoldData Chapter4/Submanifolds · MPolyfold.isSmoothSubmanifold_iff. An sc⁺-retraction is an sc-smooth retraction, so the data are those of Definition 2.3.18 (preprint 2.20) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.toSubData, A is a sub-M-polyfold Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.isSubMPolyfold and carries the induced structure of Proposition 2.3.19 (preprint 2.21) Chapter2/SubMPolyfold · MPolyfold.subPolyfold. Its points lie on level 1, as $a = s(a) \in X_1$ Chapter4/Submanifolds · MPolyfold.SubmanifoldData.mem_levelSet_one.

The sc⁺ condition read in a chart (Lemma 6.A of the exposition). The text never reads the sc⁺ condition in charts, and all of §§4.2–4.3 rests on it (Findings, item 60). Let φ be a chart of X and $W = V \cap \operatorname{dom}\varphi \cap s^{-1}(V \cap \operatorname{dom}\varphi)$, the open set of the proof of Proposition 2.3.19 (preprint 2.21) Chapter2/SubMPolyfold · MPolyfold.SubData.shrink. (a) The lift of $\varphi \circ s \circ \varphi^{-1}$ into E^1 is sc-smooth from $(\varphi(W), C, E)$ to (O_1, C_1, E^1) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isScSmoothRetractMap_lift: it is the sc-smoothness of $1^1_0 \circ s$ read in the charts φ of X and `levelChart` φ of X^1, where 1^1_0 is the lift. (b) For every retraction r onto O: $\varphi(W) = O \cap G$ for an open G; $\rho = \varphi \circ s \circ \varphi^{-1}$ is sc-smooth on the retract $\varphi(W)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_isScSmooth and idempotent there Chapter2/SubMPolyfold · MPolyfold.SubData.rho_idem, with image $\varphi(W \cap A)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_image; so $t = \rho \circ r$ on $U \cap r^{-1}(G)$ — the restriction of r Chapter2/Retracts · ScRetraction.restrict composed with ρ Chapter2/Metrizable · ScRetraction.compRetraction — is a retraction onto $\varphi(W \cap A)$, and it is sc⁺: its values $\varphi(s(x))$ lie in E_1 because $s(x) \in X_1$, and its lift is the map of (a) composed with r Chapter4/Submanifolds · MPolyfold.SubmanifoldData.exists_chartRetraction. This is the text's $t = \varphi \circ s \circ \varphi^{-1} \circ r$ of the proof of Theorem 4.2.7 (preprint 4.18), with W in place of "$V = \varphi^{-1}(O)$". Hence the model $\varphi(W \cap A)$ of the chart of A induced by φ is an sc⁺-retract Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isScPlusRetract_image, and $\varphi(a)$ is smooth for $a \in A \cap V \cap \operatorname{dom}\varphi$ (Lemma 4.1.7 (i) (preprint 4.7)) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.apply_mem_smooth.

Tangent spaces of sub-retracts. The tangent space of a retract is a germ: for an open K, $T_z(Q \cap K) = T_zQ$ Chapter4/Submanifolds · tangentSpace_inter_isOpen. For retracts $Q \subseteq O = r(U)$, $T_zQ \subseteq T_zO$: a retraction s onto Q satisfies $s = r \circ s$, so $Ds(z) = Dr(z)Ds(z)$ Chapter4/Submanifolds · tangentSpace_subset_of_subset. And if $F \circ r$ is sc-smooth and F maps the sub-retract Q into a retract Q', then $D(F \circ r)(z)$ maps T_zQ into $T_{F(z)}Q'$ (Proposition 2.1.13 (3) (preprint 2.6) for $F|_Q$) Chapter4/Submanifolds · scDeriv_mem_tangentSpace_of_subset.

Remark 4.2.2 (preprint 4.14) in a chart Chapter4/Submanifolds · MPolyfold.SubmanifoldData.remark_4_2_2. The text: $T_aA = Ts(a)(T_aX)$ is the image of the sc⁺-operator $Ts(a)$, hence finite-dimensional. In a chart, at $a \in A \cap V \cap \operatorname{dom}\varphi$ and the level-1 point z over $\varphi(a)$: z is smooth (Lemma 4.1.7 (i) (preprint 4.7) for the

chart retraction t); the tangent map of ρ at z maps $T_zO = T_z(\varphi(W))$ onto $T_z(\varphi(W \cap A))$ (Proposition 2.3.19 (4) (preprint 2.21) for ρ) `Chapter2/Metrizable · proposition_2_3_19_four`; $T_z(\varphi(W \cap A)) \subseteq T_zO$; and it is the range of $Dt(z)$, finite-dimensional by Lemma 4.1.7 (ii) (preprint 4.7). The sc⁺-operator is $Dt(z)$, not $Ts(a)$.

Points with $d_X = 0$. The remark adds that near points with $d_X(a) = 0$, A is a classical manifold. Every subspace is in good position to the whole space, any algebraic complement being good `Chapter4/Submanifolds · isInGoodPosition_univ`, and at a point with $d_C(e) = 0$ the partial quadrant C_e is the whole space `Chapter4/Submanifolds · partialQuadrantAtWith_eq_univ_of_degIndex_eq_zero`. So at $a \in A$ with $d_X(a) = 0$, in a chart realizing $d_X(a) = 0$ `Chapter2/DegeneracyIndex · MPolyfold.exists_isChart_degIndex_eq`, A is in good position (Definition 4.2.5 (preprint 4.16), §IV.13) `Chapter4/Submanifolds · MPolyfold.isGoodPositionAt_of_degIndex_eq_zero`; and if $d_X = 0$ on all of A, re-centring at each point gives charts modelled in $C_{\varphi(a)} = E$, in which $T_0(\psi(W \cap A))$ is in good position, so the induced structure has a corner structure by the core of Theorem 4.2.7 (preprint 4.18) (§IV.14) — without tameness of X `Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_degIndex_eq_zero`. The local statement for the open piece $A \cap \{d_X = 0\}$ of an arbitrary A, and the 2021 wording with d_A in place of d_X, are §IV.31.

The open question after Proposition 4.2.3 (preprint 4.15), Question 4.2.4 in the 2021 edition — is a sub-M-polyfold with $A = A_\infty$ and finite, locally constant $\dim T_aA$ a smooth finite-dimensional submanifold; is the "T" $\{(x,0) : |x| < 1\} \cup \{(0,y) : 0 \le y < 1\}$ homeomorphic to a retract $O = O_\infty$ — is a question, and nothing in it is to be formalized (gap G.34). Its last sentence, that this is impossible for sc⁺-retracts, is proved in §IV.31: an sc⁺-retract without boundary is locally a graph over its tangent space, and the "T" is not a topological manifold at its branch point.

IV.12 Proposition 4.2.3 (preprint 4.15)

(1) $A \subseteq X_\infty$ `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.subset_smoothSet`: in a chart at a, $\varphi(a)$ lies in the retract of the sc⁺ chart retraction (Lemma 4.1.7 (i) (preprint 4.7)). The text does not distinguish this from "$A = A_\infty$ for the induced structure", which holds because the levels of the induced structure are the traces of those of X, computed in the induced charts `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_isChart` `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.mem_levelSet_subPolyfold_iff` `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_one`.

(2) *A is locally closed* `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_two`. The text's definition is garbled — "a point $b \in V(a)$ belongs to A if $U \cap A \neq \emptyset$ for all open neighbourhoods U of b", with U also the letter of the domain — and is read as $V(a) \cap \overline{A} \subseteq A$, one of the equivalent forms of Mathlib's `IsLocallyClosed` (the intersection of an open and a closed set). Proof: for $b \in V \cap \overline{A}$, b is in the closure of $A \cap V$ (V open), so the filter of neighbourhoods of b within $A \cap V$ is nontrivial; s is continuous at b and equal to the identity on $A \cap V$, so $s(b)$ and b are both limits of the identity along that filter, and $s(b) = b$ (X Hausdorff); hence $b \in s(V) \subseteq A$. A filter replaces the text's sequence $a_k \to b$, so no metrizability is needed.

(3) $\partial A = \{d_A \ge 1\} \subseteq \partial X$ `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_three`, by $d_A \le d_X$ (Lemma 2.4.6 (preprint 2.26)) `Chapter2/SubMPolyfold · lemma_2_4_6`. The text's argument for $d_X(a) = 0$ is the proof of that lemma; it needs the chart re-centred at a, since only $C_{\varphi(a)}$, not C, is the whole space.

(4) $1^1_0 : A \to A^1$ *is sc-smooth* `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_2_3_four`. The

text says only that "the postulated sc⁺-retraction is the identity on the retract A". The proof: every point of A lies in an induced chart `Chapter2/SubMPolyfold · MPolyfold.subChart` whose model $\varphi(W \cap A)$ is an sc⁺-retract (§IV.11), and Proposition 4.1.8 (preprint 4.8) on a cover applies (§IV.4).

IV.13 Good and general position in charts, and their chart independence (Definitions 4.2.5, 4.2.6, 4.2.10 (preprint 4.16, 4.17, 4.21))

Definition 4.2.5 (preprint 4.16) in a chart. For retracts $Q \subseteq O$, a presentation T and a level-1 point z: T_zQ is *in good position to C_z inside T_zO* `Chapter4/Submanifolds · IsGoodPositionRetractAt` — in the sense of Definition 3.1.24 (preprint 3.21) inside a subspace `Chapter3/GoodPositionGerm · IsInGoodPositionIn`: $T_zQ \subseteq T_zO$, $T_zQ \cap C_z$ has interior points in T_zQ, and T_zQ has a complement inside T_zO with a constant ε. The condition depends only on the two tangent spaces `Chapter4/Submanifolds · IsGoodPositionRetractAt.congr`. For the data d of Definition 4.2.1 (preprint 4.13), a chart φ and $Q = \varphi(W \cap A)$ it is the chart condition `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.IsGoodPositionChart`, and A *is in good position at a* if it holds for some data at a, some chart at a, some presentation and the level-1 point over $\varphi(a)$ `Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt`. Explicitly, for retractions r onto O and t onto $\varphi(W \cap A)$, the condition says that $Dt(z)E$ is in good position to C_z inside $Dr(z)E$ `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isGoodPositionChart_iff`. The text states Definition 4.2.5 (preprint 4.16) for tame X so that C_aX is a partial quadrant (Proposition 2.5.6 (preprint 2.36)); the chart form does not need it.

Definition 4.2.10 (preprint 4.21) in a chart. *T_zQ has in T_zO an sc-complement contained in T^R_zO* `Chapter4/Submanifolds · IsGeneralPositionRetractAt`: an sc-subspace $B \subseteq T_zO \cap E_z$ of E with $(T_zO)_m = (T_zQ)_m \oplus B_m$ on every level, the sc-structure of T_zO being the one induced from E; again it depends only on the tangent spaces `Chapter4/Submanifolds · IsGeneralPositionRetractAt.congr`. The chart condition `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.IsGeneralPositionChart` and *general position at a* `Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt` are defined as for good position. **Definition 4.2.6 (preprint 4.17)**: A is *tame* if its induced structure is `Chapter4/Submanifolds · MPolyfold.IsTameSubmanifold`.

Transport by a germ of sc-diffeomorphisms. Let F be a germ of sc-diffeomorphisms between retracts $O = r(U)$ and $O' = r'(U')$ with inverse G `Chapter2/Boundary · IsDiffeoGerm`, mapping sub-retracts Q, Q' into each other. At $z \in Q_1$, $\mathcal{A} = D(F \circ r)(z)$ restricts to an isomorphism $T_zO \cong T_{F(z)}O'$ with inverse $D(G \circ r')(F(z))$, carrying the partial cones onto each other (Proposition 2.4.10 (preprint 2.30)) `Chapter2/Boundary · proposition_2_4_10_cone` and T_zQ onto $T_{F(z)}Q'$ (§IV.11 in both directions). Good position inside a subspace is transported by such an isomorphism (Lemma 3.5.9 (preprint 3.45)) `Chapter3/LinearizationCharts · IsInGoodPositionIn.transport`, so *good position of Q at z gives good position of Q' at $F(z)$* `Chapter4/Submanifolds · IsGoodPositionRetractAt.transport`. For general position z must be smooth: then $\mathcal{A}$ is an sc-operator, it carries T^R_zO onto $T^R_{F(z)}O'$ (Proposition 2.4.10 (preprint 2.30)) `Chapter2/Boundary · proposition_2_4_10_reduced`, and the image of the sc-complement B is an sc-complement inside $T^R_{F(z)}O'$, level by level `Chapter4/Submanifolds · IsGeneralPositionRetractAt.transport`.

The chart change `Chapter4/Submanifolds · MPolyfold.chart_change_of_transport`. Let Φ be a property of $(Q \subseteq O, T, z)$ that depends only on the tangent spaces and is carried by germs of sc-diffeomorphisms at smooth points. If Φ holds for $\varphi(G \cap A) \subseteq O$ at z (smooth), it holds for $\varphi'(G' \cap A) \subseteq O'$ at z', for any presentations: on $H = G \cap G'$ the transition $\varphi' \circ \varphi^{-1}$ is a germ of sc-diffeomorphisms between the overlap retracts `Chapter2/Boundary · IsScSmoothRetractMap.isDiffeoGerm` carrying $\varphi(H \cap A)$ onto $\varphi'(H \cap A)$ `Chapter4/Submanifolds · MChart.image_inter_eq_inter`, and the tangent spaces at z are those of the open pieces (the germ property). Applied to the two properties `Chapter4/Submanifolds · MPolyfold.isGoodPositionRetractAt_chart_`

change Chapter4/Submanifolds · MPolyfold.isGeneralPositionRetractAt_chart_change, with G, G' the open sets of two submanifold charts: *if A is in good position at a, the chart condition holds for every data, chart, presentation and lift* Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt.isGoodPositionChart, and good position is equivalent to the condition for all of them Chapter4/Submanifolds · MPolyfold.isGoodPositionAt_iff_forall; likewise for general position Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt.isGeneralPositionChart Chapter4/Submanifolds · MPolyfold.isGeneralPositionAt_iff_forall, the smoothness of the point holding on A. The text takes this independence for granted (Findings, item 66).

IV.14 Lemma 4.2.8 (preprint 4.19) and Theorem 4.2.7 (preprint 4.18)

Lemma 4.2.8 (preprint 4.19), the complement kept explicit Chapter4/Submanifolds · isGoodComplement_sup_range. Let π be a bounded idempotent with $(1-\pi)E_0$ inside the minimal subspace at 0, $E_0^{\min} = \{\lambda = 0\}$, and $Y \subseteq \pi E_0$ a complement of N inside πE_0 with constant ε. Then $Y \oplus (1-\pi)E_0$ is a good complement of N in E_0 with constant $\varepsilon/(\|\pi\|+1)$: for $y \in Y$, $k \in (1-\pi)E_0$ with $|y+k| \le \frac{\varepsilon}{\|\pi\|+1}|n|$, $y = \pi(y+k)$, so $|y| \le \varepsilon|n|$ and $n \in C \iff n+y \in C$; and adding $k \in E_0^{\min}$ changes no coordinate, so $n+y \in C \iff n+y+k \in C$. The splitting: $N \oplus Y = \pi E_0$ and $\pi E_0 \oplus (1-\pi)E_0 = E_0$.
Lemma 4.2.8 (preprint 4.19) Chapter4/Submanifolds · lemma_4_2_8: *in the situation of the lemma, if N is in good position to C inside πE_0, then N is in good position to C in E_0, with good complements $Y \oplus (1-\pi)E_0$.* The good-position half is the transport lemma of the proof of Theorem 3.6.8 (preprint 3.54) Chapter3/GoodPositionGerm · IsInGoodPositionIn.isInGoodPosition (§III.23).

One step of the text's proof does not hold as we read it. It concludes $n+y \in C$ from $n+y+z \in C$ "since $z \in Z \subset C$". This needs $-z \in C$: in $C = [0,\infty)$, $z = 1$ and $n+y = -\frac{1}{2}$ give $n+y+z \in C$, $n+y \notin C$. What holds, and what is used, is $Z \subseteq E_0^{\min} \subseteq C \cap (-C)$ — Proposition 2.5.3 (preprint 2.35) at the corner of the centred chart. The text's norm $|y|_0 + |z|_0$ on $Y \oplus Z$ is not the norm of E; the constant $\varepsilon/(\|\pi\|+1)$ compares them (Findings, item 69).

Good position read in a tame centred chart Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt.isInGoodPosition_of_tame. If A is in good position at a and ψ is a chart of X with $\psi(a) = 0$ modelled on a tame retract $O = r(U)$, then $N = T_0(\psi(W \cap A))$ is in good position to C in E: by chart independence the chart condition holds in ψ, i.e. N is in good position to C_0 inside $T_0O = \operatorname{range} Dr(0)$; at the corner of the centred chart $C_0 = C$ Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero; Proposition 2.5.3 (preprint 2.35) for the tame r at the smooth point 0 gives $(1 - Dr(0))E_0 \subseteq E_0^{\min}$ Chapter2/TameRetracts · proposition_2_5_3; and $Dr(0)$ is idempotent. Lemma 4.2.8 (preprint 4.19) with $\pi = Dr(0)$ concludes. Tameness of X enters Theorem 4.2.7 (preprint 4.18) here and nowhere else.

The corner charts of A Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_centred_isInGoodPosition. If every $a \in A$ lies in a chart ψ of X with $\psi(a) = 0$ in which $T_0(\psi(W \cap A))$ is in good position to C in E, the induced structure has a corner structure. Proof (for nonempty A): every point of A is smooth and $1_0^1 : A \to A^1$ is sc-smooth (Proposition 4.2.3 (4) (preprint 4.15)); the chart of A induced by ψ is a chart of the induced structure, centred at a, modelled on $\psi(W \cap A)$; its tangent space at 0 is the range of $Dt(0)$ for the sc⁺ chart retraction, finite-dimensional (Lemma 4.1.7 (preprint 4.7)), and in good position by hypothesis; Theorem 4.1.2 (ii) (preprint 4.2) □ (i) for the M-polyfold A (§IV.10) concludes.

Theorem 4.2.7 (preprint 4.18) Chapter4/Submanifolds · MPolyfold.theorem_4_2_7. *Statement.* Let X be tame and A a smooth finite-dimensional submanifold in good position at every point. Then the induced structure on A has a corner structure, and A is tame. *Proof* (the text's route through Theorem 4.1.2 (preprint 4.2)): at $a \in A$ take a tame chart centred at a (§IV.5) — the "sc-smooth tame chart $\varphi : (V, a) \to (O, 0)$" the text

takes for granted; good position read in it (above) gives the hypothesis of the previous paragraph; and a corner structure makes A tame (§IV.1). What the text skips, besides the step of Lemma 4.2.8 (preprint 4.19) repaired above: the centred chart; the set W in place of "$V = \varphi^{-1}(O)$" (§IV.11); the sc$^+$ condition in charts; the closed complement and the projection along it in place of "$p = Dt(0)$" (§IV.10); and the compatibility of the charts $p \circ \varphi$ (§IV.10). The text's "$V(a) = \varphi^{-1}(V_0) \cap A$" should be $(p \circ \varphi)^{-1}(V_0) \cap A$; its "Proposition 4.2" is Theorem 4.1.2 (preprint 4.2) (Findings, item 68). No hypothesis is added.

IV.15 Remark 4.2.9 (preprint 4.20): good position is an open condition

Statement `Chapter4/Submanifolds · MPolyfold.remark_4_2_9`. For tame X, if A is in good position at a, there is an open $U \ni a$ with A in good position at every point of $U \cap A$. The text omits "tame" (Definition 4.2.5 (preprint 4.16) is for tame X, and Lemma 4.2.8 (preprint 4.19) is used) and argues "since the tangent spaces of A move only slowly, Lemma 3.55 implies…". The formal proof makes this precise.

The tangent spaces along a graph `Chapter4/Submanifolds · tangentSpace_eq_range_graph`. If the retract Q is the image of $\Gamma : (V, N \cap C, N) \to (Q, C, E)$, sc-smooth between retracts with inverse p, then at $v \in V$ and the level-1 point y over $\Gamma(v)$: Γ is differentiable within V with derivative its sc-derivative `Chapter4/Submanifolds · hasFDerivWithinAt_scDeriv_const`, $T_yQ = D\Gamma(v)N$, and $p \circ D\Gamma(v) = 1$ — the tangent maps of the germ of sc-diffeomorphisms Γ, p.

Proof of Remark 4.2.9 (preprint 4.20). Take a tame chart ψ centred at a with tame retraction r, the sc$^+$ chart retraction t onto $\psi(W \cap A)$, and $N = T_0$, in good position to C in E (§IV.14). The normal form of Proposition 3.1.25 (preprint 3.22) gives a continuous projection π onto N with closed kernel a good complement `Chapter3/GoodComplementProjection · exists_proj_goodComplement`, and Proposition 4.1.9 (preprint 4.9) gives W_1, V, Γ over $N \cap C$. Write $\Gamma(v) = v + \sigma(v)$ with $\sigma(v) \in \ker \pi$. Then σ is C^1 on V, $\sigma(0) = 0$, and $D\sigma(0) = 0$: $D\Gamma(0)m$ lies in $T_0 = N$, so $D\Gamma(0)m = \pi D\Gamma(0)m = m$. The tangent space at $\Gamma(v)$ is the graph $\{n + D\sigma(v)n\}$ `Chapter3/GoodPositionOpen · graphSpace`. Lemma 3.6.9 (preprint 3.55), with the closed complement $\ker \pi$ `Chapter3/GlobalIFTFull · lemma_3_6_9_relOpen`, gives $\rho > 0$ such that for $|v| < \rho$ this graph is in good position to $C_{\Gamma(v)}$ in E. It lies in $T_{\Gamma(v)}O$ (Remark 4.2.2 (preprint 4.14) in the chart), and good position in E of a subspace of $T_{\Gamma(v)}O$ gives good position inside $T_{\Gamma(v)}O$, with the complement cut down by the modular law `Chapter3/GlobalIFTNearbyGraph · IsInGoodPosition.isInGoodPositionIn_of_le`. That is Definition 4.2.5 (preprint 4.16) at $b = \psi^{-1}(\Gamma(v))$, read in ψ. The neighbourhood is $U = W \cap \psi^{-1}(W_1 \cap \{|\pi u| < \rho\})$.

IV.16 Linear algebra at the corner of a centred chart; Lemmas 7.A and 7.B of the exposition

In a chart centred at 0, every coordinate vanishes at 0, so the minimal subspace there is $E_0^{\min} = \{\lambda = 0\}$, the text's "$\{0\} \oplus W$" `Chapter4/LocalStructure · mem_minimalSubspaceWith_zero`, and it lies in every E_x `Chapter4/LocalStructure · minimalSubspaceWith_zero_le`. Adding a vector of $E_0^{\min}$ changes no coordinate `Chapter4/LocalStructure · coord_add_of_mem_zero`, hence neither the degeneracy index `Chapter4/LocalStructure · degIndexWith_add_of_mem_zero` nor membership in C `Chapter4/LocalStructure · add_mem_iff_of_mem_zero`.

Closed complements. If N is finite-dimensional, K closed and $N + K$ the whole space, N has a closed complement $Y \subseteq K$: take a continuous projection q onto the finite-dimensional $N \cap K$ and $Y = K \cap \ker q$

Chapter4/LocalStructure · exists_isClosed_isCompl_le. The projection along a closed complement of a finite-dimensional subspace is continuous, with kernel exactly the complement Chapter4/LocalStructure · exists_proj_of_isCompl.

Good position from $N+E_0^{\min}=E_0$. Let N be finite-dimensional with $N+E_0^{\min}=E_0$. The coordinates restricted to N Chapter4/LocalStructure · coordsOn map N onto $\mathbb{R}^n$ Chapter4/LocalStructure · coordsOn_surjective, so $N\cap C$ has interior points, those with all coordinates positive Chapter4/LocalStructure · interior_nonempty_of_sup_eq_top. A complement inside $E_0^{\min}$ is good with every ε Chapter4/LocalStructure · isGoodComplement_of_le_zero. Hence N is in good position, with a closed good complement $Y\subseteq E_0^{\min}$ Chapter4/LocalStructure · exists_goodComplement_le_zero. **The degeneracy index on the slice** Chapter4/LocalStructure · degIndex_slice: $N\cap C$ is the partial quadrant of N presented by $v\mapsto(\lambda(v),v-s(\lambda(v)))$, s a linear section of $\lambda|_N$, and $d_{N\cap C}(v)=d_C(v)$ — both count the vanishing coordinates. This replaces the text's splitting $T_0A=N\oplus(T_0A\cap W)$, $E=N\oplus W$ (Findings, item 83).

Lemma 7.A: an sc-complement inside an sc-subspace Chapter4/LocalStructure · exists_scComplement_le. Let $N\subseteq E_\infty$ be finite-dimensional, R an sc-subspace and $M=N+R$. Then N has in M an sc-complement $B\subseteq R$: $K=N\cap R$ is finite-dimensional and smooth, so it has an sc-complement G in E (Proposition 1.1.7 (preprint 1.6)) Chapter1/ScBanach · ScBanach.proposition_1_1_7; put $B=R\cap G$. Then $R=K\oplus B$, $N+R=N+B$, $N\cap B\subseteq K\cap G=0$, and $M_m=N\oplus B_m$ on every level because $N\subseteq E_\infty$. B is closed with $B_m=(1-P)R_m$, P the sc-projection onto K, so the smooth points of B are dense in every level: for $x\in B_m$ and smooth $u\in R_m$ near x, $u-Pu\in B$ is smooth and near $x-Px=x$.

General position from $T_yQ+E_0^{\min}=E_0$ Chapter4/LocalStructure · isGeneralPositionRetractAt_of_sup_eq_top. Let r be tame, $Q\subseteq O$ a retract, $y\in Q_1$ smooth, and T_yQ finite-dimensional, smooth, with $T_yQ+E_0^{\min}=E_0$. As $E_0^{\min}\subseteq E_y$, intersecting with $T_yO\supseteq T_yQ$ gives $T_yO=T_yQ+(T_yO\cap E_0^{\min})\subseteq T_yQ+T_y^RO$ (modular law). T_y^RO is an sc-subspace by Proposition 2.5.6 (preprint 2.36) for the tame r Chapter2/PartialCone · proposition_2_5_6_partialCone, and Lemma 7.A gives the sc-complement inside it. This is the step of Theorem 4.2.11 (3) (preprint 4.22) "implying that T_aA has in T_aO a sc-complement contained in T_a^RO" (Findings, item 85).

Lemma 7.B: general position implies good position Chapter4/LocalStructure · IsGeneralPositionRetractAt.isGoodPositionRetractAt. For r tame and a smooth $z\in O_1$: if T_zQ has in T_zO an sc-complement $B\subseteq T_z^RO$, then T_zQ is in good position to C_z inside T_zO. $B\subseteq E_z$ changes none of the coordinates λ_i, $i\in I(z)$, so it is a good complement with every ε. The coordinates $\lambda_{I(z)}$ restricted to T_zO have kernel T_z^RO, whose codimension is $d_C(z)=\#I(z)$ (Proposition 2.5.6 (preprint 2.36)) Chapter2/TameRetracts · quotDim_reducedTangentSpaceWith_eq, so they map T_zO onto $\mathbb{R}^{I(z)}$; they kill B, so they map T_zQ onto $\mathbb{R}^{I(z)}$, and the preimage of the open positive orthant gives interior points of $T_zQ\cap C_z$. The text never states this step, which Theorem 4.2.11 (4) (preprint 4.22) and Theorem 4.3.1 (preprint 4.23) need for tameness and corner structures.

IV.17 Theorem 4.2.11 (1)–(3) (preprint 4.22): local structure in general position

$T_0Q+E_0^{\min}=E_0$ from general position at the centre Chapter4/LocalStructure · sup_minimalSubspaceWith_eq_top. If T_0Q has in T_0O an sc-complement $B\subseteq T_0^RO\subseteq E_0^{\min}$ and r is tame, then $E_0=T_0O+(1-Dr(0))E_0$ with $(1-Dr(0))E_0\subseteq E_0^{\min}$ (Proposition 2.5.3 (preprint 2.35)), so $E_0=T_0Q+B+(1-Dr(0))E_0\subseteq T_0Q+E_0^{\min}$. (The text: "$E=T_0O\oplus Z=T_0A\oplus\widetilde{A}\oplus Z$, so T_0A has in E an sc-complement

Y contained in W"; only $T_0A + W = E$ is used afterwards.)

$T_yQ + E_0^{\min} = E_0$ **along the graph** `Chapter4/LocalStructure · sup_minimalSubspaceWith_eq_top_graph`. If $\Gamma : V \to Q \cap W$ is the graph map of Proposition 4.1.9 (preprint 4.9) for a projection p onto N whose kernel lies in $Y \subseteq E_0^{\min}$, and $N + E_0^{\min} = E_0$, then at y over $\Gamma(v)$, $T_yQ = D\Gamma(v)N$ (§IV.15) and $D\Gamma(v)n - n \in \ker p \subseteq E_0^{\min}$ because $p \circ D\Gamma(v) = 1$; so $T_yQ + E_0^{\min} = N + E_0^{\min} = E_0$. This is the text's "by means of the particular form of the map $\alpha : h \mapsto h + D\delta(v)h$, also $T_aA + W = E$", which needs $D\delta(v)h \in W$.

Theorem 4.2.11 (preprint 4.22) in a tame centred chart `Chapter4/LocalStructure · theorem_4_2_11_chart`. Let r be tame onto $O \ni 0$, t an sc⁺-retraction onto $Q \subseteq O$ with $0 \in Q$, $N = T_0Q$, and suppose Q is in general position at 0. Then: $N + E_0^{\min} = E_0$ (above); N is in good position with a closed good complement $Y \subseteq E_0^{\min}$ (§IV.16) and the projection p along it, with kernel exactly Y; Proposition 4.1.9 (preprint 4.9) for this Y gives W, V, Γ; (2) $d_{N\cap C}(v) = d_C(\Gamma(v))$ for $v \in V$, because $d_{N\cap C}(v) = d_C(v)$ on the slice and $\Gamma(v) - v \in Y \subseteq E_0^{\min}$ changes no coordinate; (3) at every level-1 point y over $Q \cap W$, Q is in general position: y is smooth (Lemma 4.1.7 (i) (preprint 4.7)), T_yQ is the range of $Dt(y)$, finite-dimensional (Lemma 4.1.7 (ii) (preprint 4.7)) and smooth — the range of $Dt(y)$ is an sc-subspace at a smooth point `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`, finite-dimensional sc-subspaces are smooth `Chapter1/ScBanach · ScBanach.proposition_1_1_7_iff` — and $T_yQ + E_0^{\min} = E_0$ along the graph gives general position (§IV.16).

Theorem 4.2.11 (preprint 4.22) `Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_preprint`. *Statement.* Let X be tame, A a smooth finite-dimensional submanifold, in general position at $x \in A$. There are a chart ψ of X with $\psi(x) = 0$, a tame retraction r onto its model for a presentation with n half-line coordinates, and an open $U \ni x$ inside the domain of ψ, such that: (1) $d_X(x) = n$, and $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O; (2) $d_X(a) = d_A(a)$ for $a \in A \cap U$; (3) A is in general position at every point of $A \cap U$; (4) for every $\sigma \subseteq \{1, \dots, n\}$ the intersection of local faces F_σ is a tame sub-M-polyfold inside U with $d_{F_\sigma} + \#\sigma = d_X$, and $A \cap F_\sigma$ is a tame smooth finite-dimensional submanifold of F_σ, in general position in F_σ at each of its points. *Proof.* x is smooth (Proposition 4.2.3 (1) (preprint 4.15)), so there is a tame chart ψ centred at x (§IV.5), with a tame retraction r and a presentation T; the chart retraction t of A in ψ is sc⁺ (§IV.11), and general position at x read in ψ (chart independence, §IV.13) is general position of $t(U)$ at 0. Theorem 4.2.11 (preprint 4.22) in the chart gives Y, p, W_1, V, Γ; put $U = W \cap \psi^{-1}(W_1)$, W the open set of the submanifold chart.

1. *(1) is a statement about the chart.* At the centre all n coordinates vanish, so $d_C(0) = n$; the tame chart realizes d_X `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`, so $d_X(x) = n$. The face count is Corollary 2.5.13 (preprint 2.41) for the M-polyfold O `Chapter2/TameRetracts · isFaceStructured_toMPolyfold` `Chapter2/TameRetracts · LocalModel.degIndex_toMPolyfold_of_isTameWith`. "Precisely d local faces contained in U" is read here for the faces $O \cap \{\lambda_i = 0\}$ of the chart model; their identification with the local faces of the neighbourhood U itself (Definition 2.5.10 for U), and (1) in that sense, are §IV.32 (Findings, item 86).
2. *(2).* The text: "$p : A \to (T_0A \cap C) \cap V$ is a sc-diffeomorphism … by Proposition 2.24, $d_A(a) = d_{T_0A\cap C}(p(a))$". The sc-diffeomorphism is the corner chart $p \circ \psi$ of A: the data of §IV.10 for the chart of A induced by ψ, with the complement $Y \subseteq E_0^{\min}$, give a chart of the induced structure modelled on $(V, N \cap C, N)$ with the identity retraction. That model is tame, so the chart realizes d_A `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract`; ψ realizes d_X; and $d_{N\cap C}(p\psi(a)) = d_C(\Gamma(p\psi(a))) = d_C(\psi(a))$ by (2) of the chart form (Findings, item 84).
3. *(3)* is (3) of the chart form, read back through the chart: at $b \in U \cap A$ the level-1 point over $\psi(b)$ is

smooth, and the chart condition in ψ is general position at b.

4. *(4)* is §IV.18.

The text's further claims "$\delta(0) = 0$ and $D\delta(0) = 0$" hold (§IV.15) but are not used.

IV.18 Theorem 4.2.11 (4) (preprint 4.22): the faces, and $A \cap F_\sigma$

The text: "in view of the arguments in the proof of Proposition 2.43, the sets F_σ are tame M-polyfolds. Moreover, $A \cap F_\sigma$ are smooth submanifolds of F_σ. The previous arguments, but now applied to $A \cap F_\sigma$ conclude the proof." Four things are missing (Findings, item 89): Proposition 2.5.15 (preprint 2.43) is about one face, and intersections need Proposition 2.5.16 (preprint 2.44) (second sentence); "smooth submanifold of F_σ" needs an sc⁺-retraction *in the M-polyfold* F_σ; general position in F_σ needs an argument; and tameness of $A \cap F_\sigma$ needs Theorem 4.2.7 (preprint 4.18) in F_σ and good position.

Killing coordinates. For $\sigma \subseteq \{1, \dots, n\}$, $a \mapsto a \cdot 1_\sigma$ keeps the coordinates in σ `Chapter4/LocalStructure · maskL`, and for a linear section s of $\lambda|_N$, $\kappa_\sigma(v) = v - s(\lambda(v)1_\sigma)$ `Chapter4/LocalStructure · killOn` has $\lambda_i(\kappa_\sigma v) = 0$ for $i \in \sigma$ and $\lambda_j(\kappa_\sigma v) = \lambda_j(v)$ otherwise `Chapter4/LocalStructure · coordsOn_killOn`, fixes the vectors of the stratum $f_\sigma = \{\lambda_i = 0, i \in \sigma\}$ `Chapter4/LocalStructure · killOn_eq_self`, is idempotent `Chapter4/LocalStructure · killOn_idem` and maps $N \cap C$ into itself `Chapter4/LocalStructure · killOn_mem`. A continuous linear map into a finite-dimensional space is an sc-operator for the constant structure `Chapter4/LocalStructure · opToConst`.

The sc⁺-retraction onto $A \cap F_\sigma$ in the chart `Chapter4/LocalStructure · exists_scPlus_retraction_corner`. In the situation of the chart form of Theorem 4.2.11 (preprint 4.22), for every open $B \ni 0$ there are an open $G \ni 0$, $G \subseteq W \cap B$, and an sc⁺-retraction $t_\sigma = \Gamma \circ \kappa_\sigma \circ p \circ t$ on $U \cap t^{-1}(G)$ onto $Q_\sigma = Q \cap G \cap f_\sigma$: (a) it is sc⁺, as sc-smooth □ sc⁺ (§IV.2), p, κ_σ and Γ being sc-smooth; (b) it is a retraction, because $t\Gamma = \Gamma$, $p\Gamma = 1$ and $\kappa_\sigma^2 = \kappa_\sigma$; (c) its image is $Q \cap G \cap f_\sigma$, because $\Gamma(v) - v \in Y \subseteq E_0^{\min}$ changes no coordinate, so $\lambda(\Gamma(v)) = \lambda(v)$; (d) at a point y of Q_σ the tangent vectors of Q lying in f_σ are tangent to Q_σ: for such h, $k = ph$ has $\lambda(k) = \lambda(h)$, so $\kappa_\sigma k = k$ and $Dt_\sigma(y)h = D\Gamma(w)\kappa_\sigma pDt(y)h = D\Gamma(w)k = h$, $w = p(y)$, by the chain rule and $T_yQ = D\Gamma(w)N$.

The corner model `Chapter4/LocalStructure · exists_cornerModel_retraction`. For a retraction S with retract in f_σ, $\pi_\sigma \circ S \circ \iota_\sigma$ on $\iota_\sigma^{-1}(S.U)$, with the corner inclusion and projection of Chapter 2 `Chapter2/CornerGerms · cornerIncl` `Chapter2/CornerGerms · cornerProj`, is a retraction of $[0,\infty)^{n-\#\sigma} \oplus W$ onto $\pi_\sigma(S(U))$, sc⁺ if S is.

Theorem 4.2.11 (4) (preprint 4.22) in the chart `Chapter4/LocalStructure · theorem_4_2_11_four_chart`. For every σ and every open $B_0 \ni 0$ there are faces $F_i \ni 0$ of O ($i \in \sigma$) and an open $G \ni 0$, $G \subseteq W \cap B_0$, such that: *F_σ near 0 is a tame retract* — $O \cap G \cap f_\sigma$ is the retract of the face retraction of Proposition 2.5.16 (preprint 2.44) `Chapter2/CornerGerms · corner_retraction` restricted to G, whose corner model is the retract of a tame retraction r' with $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$ `Chapter2/CornerGerms · corner_tameModel`; *$A \cap F_\sigma$ near 0 is an sc⁺-retract* — $Q \cap G \cap f_\sigma$ is the retract of t_σ, and its corner model that of the sc⁺-retraction t', inside the retract of r'; *general position in F_σ* — at a level-1 point y' of $\pi_\sigma(Q \cap G \cap f_\sigma)$, with $y = \iota_\sigma(y')$: $T_yQ + E_0^{\min} = E_0$ (§IV.17) gives for every $b \in \mathbb{R}^{n-\#\sigma}$ a $v \in T_yQ$ with $\lambda(v) = \iota(b)$, zero on σ; $v \in f_\sigma$, so v is tangent to Q_σ, and $\pi_\sigma v$ is tangent to $\pi_\sigma(Q_\sigma)$ at y' (§IV.11), with coordinates b; so the tangent space of the corner model plus its minimal subspace at 0 is everything, and general position follows (§IV.16), in the corner model with the identity presentation.

sc⁺-retractions in a chart are sc⁺-retractions on open subsets `Chapter4/LocalStructure · MPolyfold.`

`isScPlusRetractionOn_of_chart` — the converse of §IV.11, which the text does not state. For a chart χ and an sc⁺-retraction τ of its model with $\tau(U) \subseteq O$, $s = \chi^{-1} \circ \tau \circ \chi$ is an sc⁺-retraction on the open set $\chi^{-1}(\tau.U)$, with image $\chi^{-1}(\tau(U))$: sc-smoothness of s and of $1^1_0 \circ s$ are read in the charts χ of X and `levelChart` χ of X^1, where they are τ and its lift; the values lie in X_1 since $\tau(U) \subseteq E_\infty$ (Lemma 4.1.7 (i) (preprint 4.7)).

(4) on the M-polyfold, for one σ `Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_four_of_chart`. In the tame chart ψ centred at x with the chart data of §IV.17: the faces F_i and G of the chart form, with G inside the domain of the submanifold chart, give $F_\sigma = \psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ `Chapter4/LocalStructure · MChart.cornerPreimage`, which contains x and lies in W. It is a sub-M-polyfold, a local property `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local`; its face chart $\pi_\sigma \circ \psi$, compatible with the induced charts `Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart`, is a chart of the induced structure `Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible` modelled on the tame corner model, so F_σ is tame `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart` with $d_{F_\sigma} + \#\sigma = d_X$. $A \cap F_\sigma$ is, in the face chart, the retract of the sc⁺-retraction t', which the previous paragraph turns into an sc⁺-retraction on an open subset of F_σ: it is a smooth finite-dimensional submanifold of F_σ. It is in general position at each of its points by the chart form read in the face chart. Its tameness is Theorem 4.2.7 (preprint 4.18) in the tame M-polyfold F_σ, good position following from general position (Lemma 7.B). F_σ is a germ, chosen inside U — the text's "contained in U" — and G depends on σ. Paracompactness of F_σ and $A \cap F_\sigma$, instance arguments of the induced structures, holds for every subset `Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype`.

The proof of Theorem 4.3.1 (preprint 4.23) uses $\dim(A \cap F_\sigma) = \dim A - \#\sigma$, which Theorem 4.2.11 (preprint 4.22) does not state; it is proved in the model in §IV.21.

IV.19 Remark 4.3.2 (preprint 4.24)

If $T_zO = T_zQ + B$ with $B \subseteq K$, the image of T_zQ spans T_zO/K, so $\dim(T_zO/K) \leq \dim T_zQ$ `Chapter4/LocalStructure · quotDim_le_finrank_of_eq_sup`. **In a chart** `Chapter4/LocalStructure · remark_4_3_2_chart`: at a smooth point z of a tame retract, $\dim(T_zO/T^R_zO) = d_C(z)$ (Proposition 2.5.6 (preprint 2.36)), and if T_zQ has in T_zO an sc-complement inside T^R_zO, then $d_C(z) \leq \dim T_zQ$. **On the M-polyfold**: for tame X and smooth x, in a tame chart at x the reduced tangent space has codimension $d_X(x)$ `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart` `Chapter4/LocalStructure · MPolyfold.remark_4_3_2_codim` — the remark's first sentence, which needs X tame and x smooth; if A is in general position at a, then $d_X(a) \leq \dim T_aA$ in any tame chart at a `Chapter4/LocalStructure · MPolyfold.remark_4_3_2`; and a submanifold in general position with $T_aA = 0$ does not meet ∂X at a `Chapter4/LocalStructure · MPolyfold.remark_4_3_2_not_mem_boundary`. The remark's "$x \in A_\lambda$ can only belong to a corner if $d_X(x) \leq \dim A_\lambda$" holds for every submanifold in general position, so it applies to the A_λ of Theorem 4.3.1 (preprint 4.23) once their general position is proved (§IV.24).

IV.20 Sard's theorem, vendored

The proof of Theorem 4.3.1 (preprint 4.23) applies "Sard's theorem" to maps $A \cap F^\circ_\sigma \to \mathbb{R}^m$ between spaces of arbitrary dimensions. Mathlib has the cases $\dim E \leq \dim F$ only. The Milnor project of this repository proves the general theorem by Milnor's induction; its import closure is copied into `Polyfold/Vendor/MilnorSard/` with the namespace renamed and no statement or proof changed (Findings, item

100). The proof, in Milnor's steps:

- *The objects.* Crit $= \{x \in U : Df(x) \text{ not onto}\}$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.critSet`, and C_k, the points where all derivatives of orders $1, \dots, k$ vanish `Vendor/MilnorSard/Basic · MilnorSard.Sard.vanishSet`, with Crit $\subseteq$ (Crit $\setminus C_1$) $\cup \bigcup_{1 \le i < k} (C_i \setminus C_{i+1}) \cup C_k$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.critSet_subset_union`. The statement in dimension n `Vendor/MilnorSard/Basic · MilnorSard.Sard.SardStatement`.
- *Straightening.* A smooth w with $Dw(\bar{x}) \neq 0$ is straightened by $H(x) = (w(x), Px)$, a local diffeomorphism by the inverse function theorem `Vendor/MilnorSard/InverseFunction · MilnorSard.isLocalDiffeomorphAt_of_isInvertible_mfderiv` `Vendor/MilnorSard/Straighten · MilnorSard.Sard.Straightening` `Vendor/MilnorSard/Straighten · MilnorSard.Sard.exists_straightening`.
- *Step 1,* $f(\text{Crit} \setminus C_1)$ null, locally `Vendor/MilnorSard/Step1 · MilnorSard.Sard.step1_local`: in straightened coordinates $g = f \circ H^{-1}$ maps hyperplanes to hyperplanes, the critical points of g on a hyperplane are those of its restriction, the induction hypothesis in dimension $n-1$ applies to the restrictions, and Fubini concludes.
- *Step 2,* $f(C_k \setminus C_{k+1})$ null, locally `Vendor/MilnorSard/Step2 · MilnorSard.Sard.step2_local`: a k-th derivative w vanishes on C_k with $Dw \neq 0$; after straightening, C_k lies in a hyperplane, on which every point of C_k is critical, and the induction hypothesis applies. Local null sets glue along a countable cover `Vendor/MilnorSard/Step2 · MilnorSard.Sard.measure_image_eq_zero_of_locally`.
- *Step 3,* $f(C_k \cap K)$ null for $(k+1)\dim F > \dim E$ and K compact convex `Vendor/MilnorSard/Step3 · MilnorSard.Sard.measure_image_vanishSet_inter_eq_zero`: Taylor's bound $|f(y) - f(x)| \le c|y - x|^{k+1}$ for $x \in C_k$ `Vendor/MilnorSard/Step3 · MilnorSard.Sard.norm_sub_le_of_iteratedFDeriv_eq_zero` makes f Hölder of exponent $k+1$ there, so $f(C_k \cap K)$ has Hausdorff dimension at most $\dim E/(k+1) <$ $\dim F$, hence measure zero `Vendor/MilnorSard/Basic · MilnorSard.Sard.addHaar_eq_zero_of_dimH_lt_finrank` — Milnor's cube count is the standard proof of that estimate.
- *The induction* `Vendor/MilnorSard/Sard · MilnorSard.Sard.sardStatement`, with the zero-dimensional target treated separately (no critical points) `Vendor/MilnorSard/Sard · MilnorSard.Sard.critSet_eq_empty_of_subsingleton`.

Sard's theorem `Vendor/MilnorSard/Sard · MilnorSard.sard`. *For E, F finite-dimensional real normed spaces in the same universe, a Haar measure μ on F, $U \subseteq E$ open and f of class C^∞ on U, $\mu(f(\text{Crit})) = 0$;* and the complement of the critical values is dense `Vendor/MilnorSard/Sard · MilnorSard.dense_compl_image_critSet`.

IV.21 The strata, Sard on the strata, and Theorem 3.8.22 (preprint 3.78) in C^∞ form

The finite-dimensional model of the proof of Theorem 4.3.1 (preprint 4.23): a finite-dimensional V (in the application T_0A of the chart of Theorem 4.2.11 (preprint 4.22)) with linear coordinates $\ell : V \to \mathbb{R}^n$ (those of the tame chart, onto), the partial quadrant $C = \{\ell \ge 0\}$ `Chapter4/FamiliesLocal · quadC`, a relatively open $U = O \cap C$ and a map f of class C^∞ on U (in the application the projection to $\mathbb{R}^m$ composed with the graph map Γ).

The strata. The stratum subspace of σ is $\{\ell_i = 0, i \in \sigma\}$ `Chapter4/FamiliesLocal · stratSub` `Chapter4/FamiliesLocal · mem_stratSub`, the tangent space of $A \cap F_\sigma$; $\sigma(v) = \{i : \ell_i(v) = 0\}$ is the set of faces through v `Chapter4/FamiliesLocal · zeroSetOf`, and v lies in the stratum subspace of $\sigma(v)$ `Chapter4/FamiliesLocal · mem_stratSub_zeroSetOf`. **The dimension count** `Chapter4/FamiliesLocal · finrank_stratSub`: if ℓ is onto, the stratum subspace of σ has dimension $\dim V - \#\sigma$ — it is the kernel of $v \mapsto (\ell_i(v))_{i \in \sigma}$, which is onto $\mathbb{R}^\sigma$. This is

the count "$A \cap F_\sigma^\circ$ is a smooth manifold … having dimension $\dim(A) - \#\sigma$", which the text attributes to "Theorem 4.23" (it means Theorem 4.2.11 (preprint 4.22), which does not state it).

Stratified critical points `Chapter4/FamiliesLocal · stratCritSet`: the $v \in U$ at which $Df(v)$ restricted to the stratum subspace of $\sigma(v)$ is not onto — the critical points of the text's $p_\sigma : A \cap F_\sigma^\circ \to \mathbb{R}^m$ for $\sigma = \sigma(v)$. **Sard on the strata** `Chapter4/FamiliesLocal · measure_image_stratCritSet`: *the stratified critical values form a null set for every Haar measure.* The open stratum of σ, $\{k \in O : \ell_i(k) = 0\ (i \in \sigma),\ \ell_j(k) > 0\ (j \notin \sigma)\}$, is an open subset of the stratum subspace and lies in U; f restricted to it is C^∞, with derivative $Df(v)$ restricted to the stratum subspace; so a stratified critical point v with $\sigma = \sigma(v)$ is a critical point of f on the open stratum of σ, and Sard's theorem applies to each of the finitely many strata. The union of the open strata is U, and a point is stratified-regular iff it is a regular point of the p_σ for its own σ; so the stratified critical values are the union of the critical values of all the p_σ.

Theorem 3.8.22 (preprint 3.78) in C^∞ form. Theorem 3.8.22 (preprint 3.78) (§III.17) is stated for one j, with j-dependent constants in its proof; the fibre retraction below needs one solution that is C^∞. In the normalized form `Chapter4/FamiliesLocal · ContractionData.exists_solution_smooth`: if the contraction data of the proof hold for every $j \geq 1$, the radii of Lemmas 3.8.23–3.8.24 (preprint 3.79–3.80) `Chapter3/QuadrantIFT · ContractionData.exists_radii` and the fixed point $\tau(k)$ are those for $j = 1$, independent of j, and only the regularity statement depends on j `Chapter3/QuadrantIFT · Radii.contDiffOn_τ`; so one τ is C^j for every j. **Theorem 3.8.22 (preprint 3.78), C^∞** `Chapter4/FamiliesLocal · theorem_3_8_22_smooth`: for U relatively open in a partial quadrant C of a finite-dimensional V, $0 \in U$, f of class C^∞ on U with $f(0) = 0$ and $Df(0)$ onto, $K = \ker Df(0)$ in good position to C with good complement Y: there are a relatively open $V' \ni 0$ of $K \cap C$ and $\tau : K \to Y$, C^∞ on V', with $\tau(0) = 0$, $D\tau(0) = 0$, $k + \tau(k) \in U$ and $f(k + \tau(k)) = 0$, and the uniqueness of $\tau(k)$ among small y — the proof of Theorem 3.8.22 (preprint 3.78) `Chapter3/QuadrantIFT · theorem_3_8_22` (the splitting $V \cong K \times Y$, $L = Df(0)$ an isomorphism $Y \to F$, $B(k, y) = y - L^{-1} f(k + y)$) with the C^∞ normalized form in place of the C^j one.

IV.22 The fibre over a stratified regular point

Partial quadrants of the model. $\{\ell_i \geq 0, i \in \sigma\}$ `Chapter4/FamiliesLocal · quadAt` is, for $\sigma = \sigma(v_0)$, the partial quadrant C_{v_0} of Definition 2.5.1 (preprint 2.33). With a linear section s of ℓ, $v \mapsto (\ell(v), v - s(\ell(v)))$ is a linear isomorphism $V \cong \mathbb{R}^n \times \ker \ell$ `Chapter4/FamiliesLocal · sliceEquiv`, which presents both as partial quadrants `Chapter4/FamiliesLocal · isPartialQuadrant_quadC Chapter4/FamiliesLocal · isPartialQuadrant_quadAt`; C is convex `Chapter4/FamiliesLocal · convex_quadC` with nonempty interior `Chapter4/FamiliesLocal · interior_quadC_nonempty`, so derivatives within $O \cap C$ are unique `Chapter4/FamiliesLocal · uniqueDiffWithinAt_quadC`.

The kernel at a stratified regular point is in good position `Chapter4/FamiliesLocal · exists_goodComplement_stratSub` — the argument of Lemma 5.26 (preprint; not in the 2021 edition), which the text needs here and does not give. Let $K + \{\ell_i = 0, i \in \sigma\} = V$ (for $K = \ker Df(v_0)$, $\sigma = \sigma(v_0)$, this is stratified regularity). The stratum subspace is closed, so K has a closed complement Y inside it (§IV.16). Adding a vector of Y changes none of the coordinates ℓ_i, $i \in \sigma$, so Y is a good complement with $\varepsilon = 1$. For the interior points: choose x with $\ell(x) = (1, \dots, 1)$ and write $x = k + y$; then $\ell_i(k) = 1$ for $i \in \sigma$, and the open set $\{k' \in K : \ell_i(k') > 0, i \in \sigma\}$ contains k and lies in $K \cap \{\ell_i \geq 0, i \in \sigma\}$.

The fibre retraction `Chapter4/FamiliesLocal · exists_fibre_retraction`. *Let ℓ be onto, f C^∞ on $O \cap C$ and $v_0 \in O \cap C$ not stratified-critical. There are an open $O' \ni v_0$, $O' \subseteq O$, and ρ, C^∞ on $O' \cap C$, mapping $O' \cap C$ into itself and into $f^{-1}(f(v_0))$, equal to the identity on the fibre there, with $D\rho(v_0)$ of range* $\ker Df(v_0)$ *and equal to the identity on it.* Proof:

1. Stratified regularity gives $Df(v_0)$ onto (its restriction to the stratum already is) and $K+\text{stratum} = V$ for $K = \ker Df(v_0)$; so K is in good position to C_{v_0} with a closed good complement Y in the stratum subspace.
2. Near 0, $v_0 + C_{v_0}$ and C agree: on $G_0 = \{w : \ell_j(v_0+w) > 0, j \notin \sigma\}$, a point v_0+w is in C iff $w \in C_{v_0}$, since $\ell_i(v_0) = 0$ for $i \in \sigma$ — the re-centring of §IV.5 in the model. So $U_1 = (O - v_0) \cap G_0 \cap C_{v_0}$ is relatively open in C_{v_0}, contains 0, and $w \mapsto v_0 + w$ maps it into U.
3. $g(w) = f(v_0+w) - f(v_0)$ is C^∞ on U_1, $g(0) = 0$, $Dg(0) = Df(v_0)$ onto; Theorem 3.8.22 (preprint 3.78) in C^∞ form gives the fibre near v_0 as the graph $\{v_0 + k + \tau(k)\}$ of a C^∞ map $\tau : V' \to Y$, $V' \subseteq K \cap C_{v_0}$, unique for small k and y.
4. With $q(v)$ the projection of $v - v_0$ onto K along Y `Chapter4/LocalStructure · exists_proj_of_isCompl`, put $\rho(v) = v_0 + q(v) + \tau(q(v))$ on $O' \cap C$, O' a small ball around v_0 chosen so that q maps it into the uniqueness region and $\rho(O' \cap C) \subseteq O'$. The projection along Y keeps the coordinates in σ, so q maps C near v_0 into C_{v_0}; ρ lands on the fibre; and on the fibre $\rho(v) = v$ by uniqueness.
5. $D\tau(0) = 0$ gives $D\rho(v_0) = q$, whose range is K and which is the identity on K.

IV.23 Stratified regularity, the sc⁺-retraction onto A_λ, and its general position

The family. For tame X, $\mathbb{R}^m \times X$ with the product charts is tame `Chapter3/ParamPerturbation · prod_rn_isTame` (§III.40 for $\mathbb{R}^m$ as an M-polyfold `Chapter3/ParamPerturbation · rnPolyfold` with its translation charts `Chapter3/ParamPerturbation · rnChart`). The fibre of $A \subseteq \mathbb{R}^m \times X$ over λ is $A_\lambda = \{x : (\lambda, x) \in A\}$ `Chapter4/Families · fibreSet`. $\mathbb{R}^m$ is presented as a partial quadrant without half-line coordinates `Chapter4/Families · rnPres` `Chapter4/Families · rnPres_eq`, and the product chart $\chi = (r \mapsto r - a) \times \varphi$ has the product presentation `Chapter2/FibredProduct · prodPresentation`, whose minimal subspace at z is $E_z = \mathbb{R}^m \times (E_X)_{\varphi(z_2)}$.

Stratified regular points `Chapter4/Families · IsStratRegularPoint`. $z \in A$ is a stratified regular point if, for some data of Definition 4.2.1 (preprint 4.13) at z, some $a \in \mathbb{R}^m$, some chart φ of X at z_2, some presentation of its partial quadrant and the level-1 point y over $\chi(z)$, every $\mu \in \mathbb{R}^m$ is the first component of a tangent vector v of the chart model of A at y with $v_2 \in E_{\varphi(z_2)}$: the projection maps $T_zA \cap E_z$ — the tangent space of the stratum through z — onto $\mathbb{R}^m$. λ is a *stratified regular value* if every point of A over λ is a stratified regular point `Chapter4/Families · IsStratRegularValue`. This is the text's "regular value of every p_σ". The condition does not depend on these choices (§IV.33). *The naive notion does not suffice*: in $\mathbb{R} \times [0,\infty)$, $A = \{(x,x) : 0 \le x < 1\} \cup (-1,1) \times \{5\}$ and $\varepsilon = \frac{1}{2}$ satisfy the hypotheses of Theorem 4.3.1 (preprint 4.23) (at $(0,0)$, $T_zA = \mathbb{R}(1,1)$ has the complement $\mathbb{R} \times 0 = T_z^R Z$), Dp is onto everywhere, but $A_0 = \{0, 5\}$ is zero-dimensional and meets ∂X, so it is not in general position (Remark 4.3.2 (preprint 4.24)); $(0,0)$ is stratified-critical, $T_zA \cap E_z = 0$.

The slice $e \mapsto (c, e)$: the inclusion $h \mapsto (0,h)$ `Chapter4/Families · ScBanach.prodInrOp` followed by the translation by the smooth vector $(c,0)$ `Chapter4/Families · mem_smooth_rn_zero` is sc-smooth `Chapter4/Families · isScSmooth_slice` with derivative $h \mapsto (0,h)$ `Chapter4/Families · scDeriv_slice`. On a finite-dimensional source the sc-derivative is the classical one `Chapter4/Families · hasFDerivWithinAt_scDeriv_const'`.

The sc⁺-retraction onto A_λ `Chapter4/Families · exists_slice_retraction` — Definition 4.2.1 (preprint 4.13) requires it and the text does not construct it. In the model $\mathbb{R}^m \oplus E$ of a product chart, let t be an sc⁺-retraction (the chart model of A), described near a point as the graph $\Gamma : V \to t(U) \cap W$ with inverse p (Proposition 4.1.9 (preprint 4.9)), and ρ a C^∞ retraction of $O' \cap (\mathbb{R}^m \times C) \subseteq V$ onto the points v with $\Gamma(v)_1 = c$

(the fibre retraction of §IV.22 for $f = (\Gamma)_1$). Then

$$\tau(e) = \big(\Gamma(\rho(p(t(c,e))))\big)_2$$

is an sc⁺-retraction of E (partial quadrant C) onto $\{e : (c,e) \in t(U) \cap W,\ p(c,e) \in O'\}$: it is sc⁺ because t is sc⁺ and the slice, p, ρ (sc-smooth as a C^∞ map on a finite-dimensional set, Corollary 1.2.5 (preprint 1.20)), Γ and the second projection `Chapter2/StrongBundle · ScBanach.prodSndOp` are sc-smooth (§IV.2); and it is a retraction, because for $e' = \tau(e)$, $\Gamma(w) = (c, e')$ with $w = \rho(\cdot)$ on the fibre, so $t(c,e') = \Gamma(w)$, $p\Gamma(w) = w = \rho(w)$ and $\tau(e') = e'$. *Its tangent space*: if $D\rho(v_0) = 1$ on the h with $(D\Gamma(v_0)h)_1 = 0$, then at the level-1 point over $\Gamma(v_0)_2$ the tangent space of the retract of τ contains $(D\Gamma(v_0)h)_2$ for all such h, by the chain rule $D\tau(y)w = (D\Gamma(v_0)D\rho(v_0)p\,Dt(c,y)(0,w))_2$, Dt fixing $T(t(U)) = D\Gamma(v_0)N$ and $p \circ D\Gamma(v_0) = 1$.

General position of the fibre `Chapter4/Families · isGeneralPositionRetractAt_slice`. Let r_X be tame, the chart model of A in general position at $z = (c,y)$ with respect to id $\times\, r_X$ and the product presentation, z regular on its stratum (every μ is the first component of some $a \in T_zA \cap E_z$), and τ an sc⁺-retraction onto $Q \subseteq O$ whose tangent space at the smooth y contains the second components of the tangent vectors of A at z with first component 0. Then Q is in general position at y. For $w \in T_yO$ write $(0,w) = a + b$ with $a \in T_zA$ and $b \in T^R_zZ = \mathbb{R}^m \times T^R_yO$ (general position of A at z); correct a by $a_2 \in T_zA \cap E_z$ with the same first component as a (regularity on the stratum); then $(a - a_2)_2 \in T_yQ$ and $(b + a_2)_2 \in T^R_yO$, a_2 lying in E_z, so $w \in T_yQ + T^R_yO$. Proposition 2.5.6 (preprint 2.36) makes T^R_yO an sc-subspace and Lemma 7.A gives the sc-complement (§IV.16). This is the text's "$T_{z'}(A \cap F^\circ_\sigma) = T_{z'}((\{\lambda\} \times A_\lambda) \cap F^\circ_\sigma) \oplus \xi_{z'}$ … in view of the assumption (2)", made precise: regularity on the stratum is what puts a_2 in E_z.

IV.24 Theorem 4.3.1 (preprint 4.23): families and Sard's theorem

The conclusions at a point `Chapter4/Families · FibreLocal`: $z = (\lambda, x)$ is a stratified regular point, A_λ is near x the image of an sc⁺-retraction on an open subset of X, and A_λ is in general and in good position at x.

The local step `Chapter4/Families · exists_local_family`. *For X tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold in general position at every point, every $z_0 \in A$ has an open neighbourhood U and a Lebesgue-null* CV $\subseteq \mathbb{R}^m$ *such that the conclusions hold at every $z \in A \cap U$ with $z_1 \notin$ CV.* Proof:

1. *The chart.* $z_{0,2}$ is smooth `Chapter3/ParamPerturbation · prod_rn_mem_smoothSet_iff`, so there is a tame chart φ of X centred at it (§IV.5); the product chart $\chi = (r \mapsto r - \lambda_0) \times \varphi$ is a chart of $\mathbb{R}^m \times X$ `Chapter3/ParamPerturbation · isChart_rnChart` `Chapter2/FibredProduct · MPolyfold.IsChart.prod` centred at z_0, with the tame retraction id $\times\, r_X$ `Chapter2/FibredProduct · ScRetraction.prod` `Chapter2/FibredProduct · ScRetraction.IsTameWith.prod` and coordinates those of φ ($\mathbb{R}^m$ has none).
2. *Theorem 4.2.11 (preprint 4.22) in the chart.* The chart retraction t of A in χ is sc⁺ (§IV.11), and general position at z_0 read in χ is general position at 0; Theorem 4.2.11 (preprint 4.22) in the chart (§IV.17) gives $N = T_0(\chi A)$ with $N + E^{\min}_0 = E_0$, so the coordinates ℓ are onto on N, a closed good complement $Y \subseteq E^{\min}_0$, the projection p, and $\Gamma : V \to \chi(A)$ over $V = O_V \cap \{\ell \geq 0\} \subseteq N$, with $\ell \circ \Gamma = \ell$.
3. *Sard.* $f(v) = \Gamma(v)_1 + \lambda_0$, the projection P read in the chart, is sc-smooth on V as the first projection `Chapter2/StrongBundle · ScBanach.prodFstOp` composed with Γ, hence C^∞ (§IV.1); CV $= f($stratified critical points of $f)$ is null (§IV.21). $U = W_A \cap \chi^{-1}(W)$, with W_A the open set of the submanifold chart (§IV.11) and W that of Proposition 4.1.9 (preprint 4.9).

4. *A point $z \in A \cap U$ with $z_1 \notin$ CV.* $\chi(z) \in t(U) \cap W$, so $v_0 = p(\chi z) \in V$ with $\Gamma(v_0) = \chi(z)$ and $f(v_0) = z_1$; hence v_0 is not stratified-critical. Since $\ell(D\Gamma(v)h) = \ell(h)$, $D\Gamma(v_0)$ maps the stratum subspace of $\sigma(v_0)$ onto $T_yA \cap E_y$, $y = \Gamma(v_0)$; so z is a stratified regular point, in the chart χ.
5. *The fibre.* The fibre retraction ρ of f at v_0 (§IV.22), the sc$^+$-retraction τ onto the chart image of A_{z_1} (§IV.23), the sc$^+$-retraction $\varphi^{-1} \circ \tau \circ \varphi$ on an open subset of X (§IV.18), whose image is A_{z_1} near z_2; the tangent space of τ's retract contains the vertical tangent vectors of A, since $D\rho(v_0) = 1$ on $\ker Df(v_0)$; general position of A_{z_1} at z_2 (§IV.23); good position from general position (Lemma 7.B).

The cover `Chapter4/Families · theorem_4_3_1_of_subset`. *For X tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with compact closure, in general position at every point, and any set $S \subseteq \mathbb{R}^m$ of parameters, there is $\Sigma \subseteq S$ with $S \setminus \Sigma$ null, consisting of stratified regular values, with the conclusions at every point of every A_λ, $\lambda \in \Sigma$.* The text covers $\{z \in A : |p(z)| \le \varepsilon\}$ "with the finitely many such neighborhoods $U(z_1), \dots, U(z_k)$". That set need not be compact: in $\mathbb{R} \times \mathbb{R}$, $A = (-2\varepsilon, 2\varepsilon) \times \{2\} \cup \{(\varepsilon(1-x), x) : 0 < x < 1\}$ has compact closure and compact non-empty fibres over $|\lambda| \le \varepsilon$, but $\{|p| \le \varepsilon\}$ contains the open segment, which accumulates at $(\varepsilon, 0) \notin A$, and neighbourhoods of radius less than half the distance to $(\varepsilon, 0)$ have no finite subfamily covering it. A countable subcover suffices: $\mathbb{R}^m \times X$ is metrizable (Theorem 2.3.10 (preprint 2.16)) `Chapter2/Metrizable · MPolyfold.metrizableSpace`, so the compact $\overline{A}$, and with it A, is separable, hence second countable; countably many of the neighbourhoods of the local step cover A, the countable union of their null sets is null, and $\Sigma = S \setminus \bigcup_i \mathrm{CV}(z_i)$ (Findings, item 103). Some countability is necessary: an uncountable discrete A can have critical values of positive measure; here it comes from the compactness of $\overline{A}$, which the text assumes.

Theorem 4.3.1 (preprint 4.23) `Chapter4/Families · theorem_4_3_1_preprint`. *Statement.* Let X be tame, $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon > 0$, and (1) for $|\lambda| \le \varepsilon$ ($|\lambda|^2 = \sum \lambda_i^2$) the set $A \cap \{z_1 = \lambda\}$ is compact and non-empty, (2) A is in general position at every point. Then there is $\Sigma \subseteq \{|\lambda| < \varepsilon\}$ with null complement in $\{|\lambda| < \varepsilon\}$, consisting of stratified regular values, such that for $\lambda \in \Sigma$ the fibre A_λ is compact, a smooth finite-dimensional submanifold of X, in general position at each of its points, with a corner structure (Definition 4.1.1 (preprint 4.1)), and tame. *Proof.* The cover for $S = \{|\lambda| < \varepsilon\}$. A_λ is the image of the compact $A \cap \{z_1 = \lambda\}$ under the projection to X, hence compact. It is a smooth finite-dimensional submanifold and in general position by the conclusions at its points; it is in good position at every point (Lemma 7.B), so Theorem 4.2.7 (preprint 4.18) gives the corner structure and tameness. Hypothesis (2), as printed ("there exists a sc-complement T_zA in T_zZ which is contained in T_z^RZ"), is read as Definition 4.2.10 (preprint 4.21), as the proof uses it. Not used: the tameness of the induced structure of A, the non-emptiness in (1), $\varepsilon > 0$ (Findings, item 101). Every stratified regular value is a regular value — $T_zA \cap E_z \subseteq T_zA$ — so Σ consists of regular values in the text's sense; but the conclusion is not true at every ordinary regular value (§IV.23).

IV.25 What is not formalized

- **The pull-back of forms on ∂X along a general sc-smooth map** of tame M-polyfolds. The text pulls back forms on ∂X along a map only in the proof of Theorem 6.7.1, for p_1 on the faces of the solution set, and that case is formalized: the pull-back to the faces of a finite-dimensional sub-M-polyfold S of a second tame M-polyfold, along an sc-smooth $G : S \to X$ with $d_S = d_X \circ G$ `Chapter4/BoundaryFormsPullback2021 · MPolyfold.isChartFaceForm_faceAlongData` `Chapter4/BoundaryFormsPullback2021 · MPolyfold.faceAlongData_formsD` (Part VI). No statement of Parts I–II uses the

general case, which is estimated at 600–1,500 lines. Gap G.80 is closed: the wedge product on the relative complex is §IV.46 (Findings, Chapter 4, items 158, 167–173, 177, 178).

- **Remark 4.1.3 (preprint 4.3).** The example of §IV.27 closes gap G.31. Not formalized: the argument that an example cannot come from a retraction whose level maps are C^1, and the example the 2021 edition adds (the space of its Subsection 1.4.4), which holds as we read it (Findings, Chapter 4, item 159).
- **Question 4.2.4 (preprint: the open question after Proposition 4.15)** is a question, so nothing is to be formalized (G.34). Its settled last sentence is §IV.31.
- The classical manifold structure behind Definition 4.1.1 (preprint 4.1) beyond the C^∞ transitions (§IV.1).

Closed since the first version of this Part: - Lemma 4.1.11 (preprint 4.10) (G.32, §IV.30); - the local form of Remark 4.2.2 (preprint 4.14) (G.33, §IV.31); - the local faces in Theorem 4.2.11 (1) (preprint 4.22) (G.35, §IV.32); - the chart independence of stratified regularity (G.36, §IV.33); - §4.4 apart from the item above (G.37–G.40, §§IV.35–IV.45); - Lemma 4.5.1 between open subsets and the pull-back bundle $TX|X^2$ (G.92, §IV.34); - the wedge product on the forms on ∂X and on the relative complex (G.80, §IV.46).

IV.26 What the formalization taught us

1. *Centred charts are a construction.* They exist only at smooth points and live in the partial quadrant $C_{\varphi(x)}$, on the set where it agrees with C; four theorems of the chapter and Lemma 5.3.3 (preprint 5.11) rely on them, and the text never builds them (§IV.5).
2. *Proposition 4.1.9 (preprint 4.9) is a statement about fixed sets, and its proof is not the implicit function theorem.* The text's route, as we read it, gives a germ over a different complement from the one needed. The direct proof uses only the sc$^+$ property — a mean value estimate on every level and the level-1 continuity of t — and its regularity argument rests on two observations: the estimate is applied at $\Gamma(a)$ with the frame $B_{\Gamma(a)}$, not at 0, and derivatives of t only meet the finite-dimensional smooth tangent space, so the loss of levels does not accumulate (§§IV.6–IV.9). The projection must be the one along the closed good complement, not $Dt(0)$.
3. *Theorem 4.1.2 (preprint 4.2) does not need tameness*, and its compatibility claim needs charts composed with sc-diffeomorphisms between retracts (§IV.10).
4. *The sc$^+$ condition has to be read in charts*, in both directions: a submanifold's chart retraction is sc$^+$ (§IV.11), and an sc$^+$-retraction in a chart is one on an open set (§IV.18). Every statement of §§4.2–4.3 passes through these two facts.
5. *One step in the proof of Lemma 4.2.8 (preprint 4.19) does not hold as we read it* ($C = [0,\infty)$, $z = 1$, $n + y = -\frac{1}{2}$), and the repair is the observation that the complement of T_0O lies in $C \cap (-C)$ at the corner of a centred chart (§IV.14).
6. *General position implies good position* (§IV.16), a step the text uses in Theorem 4.2.11 (4) (preprint 4.22) and Theorem 4.3.1 (preprint 4.23) without stating it.
7. *Theorem 4.3.1 (preprint 4.23) needs stratified regular values and a countable cover*, and the fibre's manifold structure needs the implicit function theorem at corner points with the kernel in good position — the argument of Lemma 5.26 (preprint) — and Theorem 3.8.22 (preprint 3.78) in C^∞ form (§§IV.21–IV.24). Sard's theorem had to be supplied in all dimensions (§IV.20).
8. *In a tame M-polyfold the good position of Theorem 4.1.2 (2) (preprint 4.2) is automatic, and for $d \leq 1$ tameness is not needed.* The case that would block good position, a boundary coordinate vanishing on the tangent space, is excluded by the graph argument of 2021 Lemma 9.2.13. This makes the

sentence of Remark 4.2.2 (preprint 4.14) on $d_A = 1$ a theorem without hypotheses (§IV.31).

9. *The formula for $d\omega$ on p. 197 does not hold as we read it with the printed sign.* The text's bracket is the negative of the classical one, so the classical sign fails too; $(-1)^{\ell+j+1}$ is the sign that works (§IV.38). The formal d is defined by the local alternating sum of base derivatives, and it is chart independent by the symmetry of second derivatives (§§IV.36–IV.37).
10. *Forms on ∂X need a reading, and the reading closes.* ∂X is not an M-polyfold; forms on it are forms on bX that agree over B. That d preserves agreement needs no corner strata as M-polyfolds, only the map $r \circ P_i \circ P_j$ into two faces at once (§IV.40).
11. *Stokes' theorem needs the faces as manifolds, signed charts and an added reading.* "A union of smooth faces" is read as face-structured. $[0,1]$ with the quadrant model has no oriented atlas, so integrals use signed charts. The face orientation is the outward vector first (§§IV.41–IV.43).
12. *The Poincaré lemma needs a smooth centre other than 0.* In the proof of Lemma 4.4.8 the convex set $B_\delta(\varphi x) \cap C$ need not contain 0. The homotopy about a smooth centre works with the same proof, and its sc-smoothness is a theorem on parametric integrals of its own (§§IV.44–IV.45).
13. *The text uses $\wedge$ without defining it.* With the classical normalization the Leibniz sign $(-1)^k$ holds as we read it, and associativity and graded commutativity are identities between permutation sums that need no alternation (§IV.39).

IV.27 Remark 4.1.3's (preprint 4.3) example: a retraction that loses a level

Remark 4.1.3 (preprint 4.3) says that "$X^1 \neq X$" is possible as M-polyfolds although the underlying sets agree, and gives no example. By Proposition 4.1.8 (preprint 4.8) in charts (§IV.4) an example is an sc-retract $O = r(U)$ with $O = O_\infty$ that is not an sc$^+$-retract. It cannot come from a retraction whose level maps $U_m \to E_m$ are C^1 — the maps that Proposition 1.2.4 (preprint 1.19), Corollary 1.2.5 (preprint 1.20) and their composites produce: then $T_oO = Dr(o)E_0$ is finite-dimensional by a Baire category argument (O is a countable union of the compact sets $\mathrm{cl}_0\{x \in O : |x|_1 \leq n\}$), and the topologies of O in E_0 and E_1 agree near o (Findings 141; an argument, not formalized). So the retraction must lose a level in its first derivative, as the splicings of HWZ do, and its sc-smoothness needs a criterion the project did not have (Findings 140–146).

A criterion with loss. Let V be finite-dimensional and $L(p)_m : Z_m \to W_m$, $p \in V$, be bounded operators compatible with the inclusions, such that $(p,z) \mapsto L(p)_m z$ is continuous on $V \times Z_m$ — strong continuity, not continuity of $p \mapsto L(p)_m$ in the operator norm — and $p \mapsto A_m(p) = L(p)_m \circ \iota$ is C^∞ as a map into the operators $Z_{m+1} \to W_m$: one level of loss buys all the p-derivatives `Chapter4/RemarkFourThreeExample · IsLossFamily` `Chapter4/RemarkFourThreeExample · famLoss` `Chapter4/RemarkFourThreeExample · famDeriv`. *Then $(p,z) \mapsto L(p)z$ is sc-smooth on $V \oplus Z$* `Chapter4/RemarkFourThreeExample · lossFamMap` `Chapter4/RemarkFourThreeExample · IsLossFamily.isScSmooth`. The map is sc^0 by the strong continuity `Chapter4/RemarkFourThreeExample · IsLossFamily.isSc0`, and sc^1 with derivative $(\pi,\zeta) \mapsto \partial_p A_0(p)[\pi]z + L(p)\zeta$ at a level-1 point (p,z) `Chapter4/RemarkFourThreeExample · IsLossFamily.differentiable_famLoss` `Chapter4/RemarkFourThreeExample · IsLossFamily.isSc1With`: the increment in p is taken in the operator norm of A_0, which needs $z \in Z_1$, and the derivative family is continuous on the tangent levels by the C^∞ dependence of $\partial_p A_m$ `Chapter4/RemarkFourThreeExample · IsLossFamily.continuous_famDeriv`. The tangent map is again a map of the same kind: after the sc-isomorphism that reorders (p,z,π,ζ) as $((p,\pi),(z,\zeta))$ `Chapter4/RemarkFourThreeExample · famSwap` it is the map of the *tangent family* $\tilde{L}(p,\pi)(z,\zeta) = (L(p)z, \partial_p A(p)[\pi]z + L(p)\zeta)$ on $V \times V$ `Chapter4/`

RemarkFourThreeExample · famTangent Chapter4/RemarkFourThreeExample · tangentMapWith_lossFamMap_eq, and $\tilde{L}$ is a loss family Chapter4/RemarkFourThreeExample · IsLossFamily.tangent. Induction on k gives sc^k for every k Chapter4/RemarkFourThreeExample · isSck_lossFamMap. Only one level of loss is allowed: maps whose j-th derivative costs j levels, such as the map $(t,x) \mapsto (\cos(kt)x_k)_k$ of the example after Definition 1.1.15 (preprint 1.14), are not covered.

The reduction. For a loss family of projections ($L(p)_0^2 = L(p)_0$), $r(p,z) = (p, L(p)z)$ is an sc-smooth retraction of $V \oplus Z$ Chapter4/RemarkFourThreeExample · IsLossFamily.isScSmooth_famRetr Chapter4/RemarkFourThreeExample · IsLossFamily.retraction; if the ranges of the $L(p)_0$ are smooth points, $O = O_\infty$ Chapter4/RemarkFourThreeExample · IsLossFamily.retract_subset_smooth; and if the lift $O \to E_1$ is discontinuous, Proposition 4.1.8 (preprint 4.8) says that O is not an sc^+-retract Chapter4/RemarkFourThreeExample · IsLossFamily.not_isScPlusRetract. These are the data of the example Chapter4/RemarkFourThreeExample · RemarkFourThreeData Chapter4/RemarkFourThreeExample · exists_retract_smooth_not_scPlus_of_data.

The family. On the weighted sequence space X of the example after Definition 1.1.15 (preprint 1.14) ($|x|_m = \sup_k (k+1)^m |x_k|$; Chapter1/ScOneNotNormContinuous · ScOneExample.X) the moving vector must pass between coordinates of comparable weight, or the level-m operators are unbounded, so the transitions must be fast and the gain per step geometric: the indices are $n_k = 2^k$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.nk. With nodes $1/(k+1)$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.node and angles $\theta_k(t)$ turning from 0 (for $t \geq 1/(k+1)$) to $\pi/2$ (for $t \leq 1/(k+2)$) through Mathlib's smooth transition, flat at both ends Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition, the coefficients $b_k = \sin\theta_{k-1}\cos\theta_k$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb vanish except $b_k = \cos\theta_k$ and $b_{k+1} = \sin\theta_k$ at the active block $k = \lfloor 1/t \rfloor - 1$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_eq_zero_of_pos Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_kt Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_kt_succ, and all vanish for $t \leq 0$. $L(t) = \sum_k$ (block k) Chapter4/RemarkFourThreeFamily · RemarkFourThree.blk Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam is a sum with two non-zero terms, which on level 0 is the projection onto $b_k e_{n_k} + b_{k+1} e_{n_{k+1}}$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_of_pos; on level m its entries carry the weight ratios $((a+1)/(b+1))^m \leq 2^m$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.elemL. It is a loss family Chapter4/RemarkFourThreeFamily · RemarkFourThree.isLossFamily_fam: compatible with the inclusions Chapter4/RemarkFourThreeFamily · RemarkFourThree.toBase_fam; uniformly bounded, $\|L(t)_m\| \leq 3 + 2^m$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.norm_fam_le; strongly continuous, from the bound and the norm continuity after the dense inclusion by an $\varepsilon/3$ argument Chapter4/RemarkFourThreeFamily · RemarkFourThree.continuous_apply_of_bound; and $t \mapsto L(t)_m \circ \iota$ is C^∞ in the operator norm, by Mathlib's differentiation of series Mathlib/Analysis/Calculus/SmoothSeries · contDiff_tsum: j derivatives of block k are bounded by a polynomial in k of degree $2j$ (Leibniz's rule and the derivatives of the flat transition), while one level of loss gains a factor $(2 + 2^m)2^{-k}$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.contDiff_fam_comp. Every $L(t)_0$ is a projection Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_idem with finitely supported, hence smooth, range Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_mem_smooth. The lift is discontinuous: the points $o_k = (1/(k+1), (1/(k+1))e_{n_k})$ lie in O Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_node_smul_6v and tend to 0 in E_0, while their level-1 coordinates $(2^k+1)/(k+1)\, e_{n_k}$ have norm at least 1 Chapter4/RemarkFourThreeFamily · RemarkFourThree.not_continuousOn_liftOne. So the data exist Chapter4/RemarkFourThreeFamily · RemarkFourThree.remarkFourThreeData, and there is an sc-retract $O = O_\infty$ that is not an sc^+-retract Chapter4/RemarkFourThreeFamily · remark_4_1_3_retract_example.

The M-polyfold. A retract with its single chart $O \to O$ is an M-polyfold Chapter4/

RemarkFourThreeFamily · retractChart Chapter4/RemarkFourThreeFamily · retractPolyfold; if $O \subseteq E_\infty$ and O is not an sc⁺-retract, every point of it is smooth and $1^1_0 : X \to X^1$ is not sc-smooth, for every junk value (Proposition 4.1.8 (preprint 4.8) in charts, §IV.4) Chapter4/RemarkFourThreeFamily · retractPolyfold_smooth_not_toLevelOne. *Remark 4.1.3's (preprint 4.3) example* Chapter4/RemarkFourThreeFamily · remark_4_1_3_example: there is an M-polyfold with $X_0 = X_\infty$ for which 1^1_0 is not sc-smooth, hence (by the remark's first part, §IV.4) not an sc-diffeomorphism: $X^1 \neq X$ as M-polyfolds. In this example X and X^1 even carry different topologies on the same set, since $o_k \to 0$ in X and not in X^1. "$X^1 \neq X$" is read as "1^1_0 is not sc-smooth"; the M-polyfold is a single retract, which is all the remark needs; its parameter space is $\mathbb{R}$ and its partial quadrant all of $\mathbb{R} \oplus X$, so there is no boundary.

IV.28 The Springer 2021 edition: Theorems 4.2.11 and 4.3.1, vector fields and the Lie bracket

The 2021 edition changes two statements of §§4.2–4.3 and rewrites §4.4, adding sc-smooth vector fields (Definition 4.4.3, Lemma 4.4.4), a proof of Proposition 4.4.5 (preprint 4.26) in a new Appendix 4.5, and a Poincaré lemma. The files are `Chapter4/Book2021` (the two theorems), `Chapter4/VectorFields` (vector fields and the bracket) and `Chapter4/BracketLevel` (an example on the level of the bracket). The forms of §4.4 and the Poincaré lemma are §§IV.35–IV.45 (Findings, Chapter 4, items 150–159).

Theorem 4.2.11 (preprint 4.22), 2021 form Chapter4/Book2021 · MPolyfold.theorem_4_2_11. *Statement.* As in §IV.17–§IV.18, with (1) strengthened: the $d = d_X(x)$ local faces lie in U *and contain* x. Formally: a tame chart ψ centred at x with tame retraction R and presentation $T : E \cong \mathbb{R}^n \oplus W$, and an open $U \ni x$ in its domain, such that $d_X(x) = n$, $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O, and for every $i < n$ there are a face $F_i \ni 0$ of the retract in the i-th coordinate hyperplane and an open $G \ni 0$ with $x \in \psi^{-1}(G \cap F_i) \subseteq U$; (2)–(4) unchanged. *Proof.* Everything except the new clause is Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_preprint, which already counts the faces *through* $\psi(x)$ — the 2021 reading of (1). For the new clause apply (4) to $\sigma = \{i\}$: it gives a family $(F_j)_{j\in\{i\}}$ of faces through 0 in the coordinate hyperplanes and an open $G \ni 0$ with $x \in \psi^{-1}(G \cap \bigcap_j F_j) \subseteq U$; since the index set has one element, the intersection over it is F_i itself (an extensionality step on the preimage set). The local faces here are those of the chart model; the faces of the neighbourhood U itself are §IV.32.

Theorem 4.3.1 (preprint 4.23), 2021 form Chapter4/Book2021 · theorem_4_3_1. *Statement.* X tame, $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon > 0$, (1) $A \cap \{z_1 = \lambda\}$ compact for $|\lambda| \leq \varepsilon$ — no longer "non-empty" — and (2) general position at every point. Then there is a *measurable* $\Sigma \subseteq B = \{|\lambda| < \varepsilon\}$ with $B \setminus \Sigma$ null, of stratified regular values Chapter4/Families · IsStratRegularValue, such that each A_λ, $\lambda \in \Sigma$, is compact, a smooth finite-dimensional submanifold in general position at its points, with a corner structure, and tame. *Proof.* The cover of §IV.24 for $S = B$ Chapter4/Families · theorem_4_3_1_of_subset gives $\Sigma_0 \subseteq B$ with $B \setminus \Sigma_0$ null and the local conclusions at every point of every fibre. B is measurable (a strict sublevel set of a continuous function). Let N be a measurable null set containing $B \setminus \Sigma_0$ (Mathlib's measurable hull Mathlib/MeasureTheory/Measure/MeasureSpaceDef · MeasureTheory.toMeasurable) and $\Sigma = B \setminus N$: measurable, inside Σ_0, with $B \setminus \Sigma \subseteq N$ null. For $\lambda \in \Sigma$, the fibre A_λ is the image of the compact set $A \cap \{z_1 = \lambda\}$ (hypothesis (1), $|\lambda| < \varepsilon$) under the second projection, hence compact; it is a submanifold and in general and good position by the local conclusions; Theorem 4.2.7 (preprint 4.18) Chapter4/Submanifolds · MPolyfold.theorem_4_2_7 gives the corner structure and tameness. Neither non-emptiness, nor the tameness of A, nor $\varepsilon > 0$ is used.

Vector fields in a local model (Definition 4.4.3). For a local model (O, C, E) a map $A : E_1 \to E_0$ is

a *vector field* if $x \mapsto (x, A(x))$ is an sc-smooth map between retracts from (O^1, C^1, E^1) to the tangent (TO, TC, TE) `Chapter4/VectorFields · IsRetractVectorField` — the text's sc-smooth section of $TX \to X^1$, read in a chart: its domain is X^1. The zero map is one on every sc-smooth retract: $(x, 0)$ lies in TO `Chapter2/TangentBundle · mk_zero_mem_retractTangent` and $x \mapsto (x, 0)$ is the pair of the retraction and the zero operator `Chapter4/VectorFields · isRetractVectorField_zero`. Consequences: $(x, A(x)) \in TO$ for $x \in O_1$ `Chapter4/VectorFields · IsRetractVectorField.mem_retractTangent`; for a retraction r onto O, $\tilde{A} = A \circ r^1$ is sc-smooth on U^1, as the second component of the section composed with the shifted retraction `Chapter4/VectorFields · IsRetractVectorField.isScSmooth_comp`; and $Dr(x)A(x) = A(x)$ for $x \in O_1$, formula (4.8), because T_xO is the fixed space of $Dr(x)$ `Chapter4/VectorFields · IsRetractVectorField.scDeriv_apply`.

Vector fields on an M-polyfold (Definition 4.4.3). $A : X^1 \to TX$ is an sc-smooth vector field if it is sc-smooth for the M-polyfold structures of X^1 and of TX and $p(A(x)) = x$ `Chapter4/VectorFields · MPolyfold.IsScVectorField`. The zero section takes at x the zero tangent vector in a chart chosen around x `Chapter4/VectorFields · MPolyfold.zeroTangentPre` `Chapter4/VectorFields · MPolyfold.zeroVectorField`; the tangent transition maps are linear in the fibre, so in every chart ψ around x it reads $(\psi(x), 0)$ `Chapter4/VectorFields · MPolyfold.zeroTangentPre_transport` `Chapter4/VectorFields · MPolyfold.tangentChartMap_zeroVectorField`. Hence it is continuous (the preimage of a basic open set of TX is a basic open set of X^1) `Chapter4/VectorFields · MPolyfold.continuous_zeroVectorField` and, in each chart, the sc-smooth map $z \mapsto (z, 0)$: the zero section is an sc-smooth vector field `Chapter4/VectorFields · MPolyfold.isScVectorField_zero`.

Raising the index (Lemma 4.4.4, local form, $i = 1$). *For a retraction r onto O, $T(O^1) = (TO)^1$* `Chapter4/VectorFields · retractTangent_shift`: a point (x, h) of $E_2 \oplus E_1$ lies in the tangent of the shifted retract iff $Tr^1(x, h) = (x, h)$, and the base point of $Tr^1(x, h)$ in $E_1 \oplus E_0$ is $Tr(x, h)$, because the derivative of the shifted retraction is the level-1 restriction of Dr `Chapter1/ScSmooth · IsSc1.scDeriv_eqOn`; the inclusion of levels being injective, the two fixed-point conditions agree. *If A is a vector field on (O, C, E), its level map $A_1 : E_2 \to E_1$ is a vector field on (O^1, C^1, E^1)* `Chapter4/VectorFields · IsRetractVectorField.shift`: shifting the sc-smooth map $x \mapsto (x, A(x))$ gives an sc-smooth map from O^2 into $(TO)^1$ `Chapter2/LevelPolyfold · IsScSmoothRetractMap.shift`, which by the identity just proved is $T(O^1)$, and its level-1 values are $(x, A_1(x))$ `Chapter1/ScSmooth · toBase_levelMap`. Larger i by iteration; the M-polyfold form, through the identification $T(X^1) \cong (TX)^1$ of Proposition 2.3.17 (gap G.6, closed), is in §IV.29, for $i = 1$ and for every i.

The second derivative and its symmetry. For f sc² on U, $z \in U_2$ and $a, b \in E_1$, $D^2f(z)(a, b)$ is defined as the second component of $D(Tf)(z, a)(b, 0)$ — the derivative of $z \mapsto Df(z)a$ in the direction b `Chapter4/VectorFields · scSecondDeriv`. (The text writes $D^2r(x)(A(x), B(x))$ with $A(x), B(x) \in E_1$, where $r : U_2 \to E_0$ has a second derivative only in directions of E_2.) The vertical part of $D(Tf)$ is $D(Tf)(z, a)(0, l) = (0, Df(z)l)$ `Chapter4/VectorFields · scDeriv_tangentMap_inr`. *Symmetry* `Chapter4/VectorFields · scSecondDeriv_symm`: for U relatively open in a partial quadrant, $D^2f(z)(a, b) = D^2f(z)(b, a)$. (i) On E_2: $f \circ \iota : U_2 \to F_0$ is C^2 within U_2 (sc² gives C^2 on level 2), every point of U_2 lies in the closure of the interior of U_2 `Chapter4/VectorFields · IsRelOpenInQuadrant.mem_closure_interior_lvlSet`, and Mathlib's symmetry of second derivatives within a set applies `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffWithinAt.isSymmSndFDerivWithinAt`; for $v, w \in E_2$ the iterated derivative within U_2 is $D^2f(z)(\iota v, \iota w)$ `Chapter4/VectorFields · fderivWithin_fderivWithin_eq`, by the derivative of Df along level 2 `Chapter4/VectorFields · hasFDerivWithinAt_two` `Chapter4/VectorFields · hasFDerivWithinAt_scDeriv_apply`. (ii) On E_1: $(a, b) \mapsto D^2f(z)(a, b)$ is continuous ($D(Tf)$ is continuous on level 0), $E_2 \times E_2$ is dense in $E_1 \times E_1$, and two continuous functions equal on a dense set are equal.

The bracket in a local model (§4.5.1). For a retraction r onto O and $x \in E_2$, $[A, B](x) = D\tilde{A}(x)B_1(x) -$

$D\tilde{B}(x)A_1(x)$ with $\tilde{A} = A \circ r^1$, $\tilde{B} = B \circ r^1$ Chapter4/VectorFields · retractLieBracket; $[A,B] = -[B,A]$ Chapter4/VectorFields · retractLieBracket_antisymm. On O^2, $(A \circ r^1)_1 = A_1$, since r^1 fixes O^1 Chapter4/VectorFields · levelMap_comp_shift_eq. *The differentiated identity* Chapter4/VectorFields · IsRetractVectorField.diff_identity: for $x \in O_2$ and b with $Dr^1(x)b = b$, $D^2r(x)(A(x),b) + Dr(x)D\tilde{A}(x)b = D\tilde{A}(x)b$. The map $\alpha : y \mapsto (r^1(y), \tilde{A}(y))$ is sc-smooth on U^1 with values in TO (as $r^1(y) \in O^1$), so $Tr \circ \alpha = \alpha$ on all of U^1, TO being the fixed set of Tr. Differentiate at x in the direction b by the chain rule Chapter1/ChainRule · theorem_1_3_1'; the derivative of α is $(Dr^1(x)b, D\tilde{A}(x)b) = (b, D\tilde{A}(x)b)$, and $\alpha_1(x) = (x, A_1(x))$ since r^1 fixes x. Split $(b, D\tilde{A}(x)b) = (b,0) + (0, D\tilde{A}(x)b)$: the first part gives $D^2r(x)(A(x),b)$ by definition, the second the vertical part $Dr(x)D\tilde{A}(x)b$; compare second components. *Tangency* Chapter4/VectorFields · retractLieBracket_mem_tangent: by Lemma 4.4.4, $B_1(x)$ and $A_1(x)$ are fixed by $Dr^1(x)$; apply the identity to $\tilde{A}$ in the direction $B_1(x)$ and to $\tilde{B}$ in the direction $A_1(x)$, subtract, and cancel the second derivatives by symmetry: $Dr(x)[A,B](x) = [A,B](x)$.

Proposition 4.4.5 (preprint 4.26), local form Chapter4/VectorFields · proposition_4_4_5_local. *Statement.* For vector fields A, B on (O,C,E) with retraction r: $(x,[A,B](x)) \in TO$ for $x \in O_2$, and $x \mapsto (x,[A,B](x))$ is an sc-smooth map from (O^2,C^2,E^2) to (TO,TC,TE). *Proof.* The first clause is tangency, T_xO being the fixed space of $Dr(x)$ at $x \in O_1$. For sc-smoothness between retracts it suffices to give an sc-smooth extension to U^2 that agrees on O^2 Chapter2/Retracts · isScSmoothRetractMap_iff_exists. First component: the inclusion $E^2 \to E^1$ composed with r^2. Second component: $y \mapsto D\tilde{A}(r^2y)(\tilde{B})_1(y) - D\tilde{B}(r^2y)(\tilde{A})_1(y)$. Each term is the second component of the tangent map $T\tilde{A}$, which is sc-smooth because $\tilde{A}$ is, composed with the sc-smooth pair $(r^2,(\tilde{B})_1)$ Chapter4/VectorFields · isScSmooth_scDeriv_apply; and $B_1 \circ r^2 = (\tilde{B})_1$ on U^2 Chapter4/VectorFields · levelMap_shift_shift_eq, so the extension is $[A,B] \circ r^2$. The statement on an M-polyfold, with Lemma 4.5.1, whose proof the text leaves to the reader, is in §IV.29 (gap G.38, closed).

The level of the bracket: an example Chapter4/BracketLevel · lieBracket_level_counterexample. The text concludes from tangency that the bracket lies in "the fiber $(T_xO)_1$ whenever $x \in O_2$" (p. 200) and that for $x \in O_2$ it "belongs to the level 1" (p. 201). Read as $[A,B](x) \in E_1$, this does not hold as we read it. *Generalities.* For $O = E$ (no boundary, identity retraction Chapter4/BracketLevel · idRetraction), $TO = TE$ Chapter4/BracketLevel · retractTangent_univ; every sc-operator $S : E^1 \to E$ is a vector field Chapter4/BracketLevel · isRetractVectorField_scOperator, and the bracket of two is the commutator $S_0T_1 - T_0S_1$, the derivative of a linear map being itself Chapter4/BracketLevel · retractLieBracket_scOperator. *The space.* E is the weighted sequence space with $|x|_m = \sup_k (k+1)^m |x_k|$ Chapter1/ScOneNotNormContinuous · ScOneExample.X, level m being c_0 in the coordinates $(k+1)^m x_k$. *The operators.* $(Sx)_k = (-1)^k(k+1)x_k$ Chapter4/BracketLevel · BracketLevel.opS, on every level the diagonal $\mathrm{diag}((-1)^k)$ in the weighted coordinates; $(Tx)_k = (k+1)x_{k+1}$ Chapter4/BracketLevel · BracketLevel.opT, on level m the weighted shift with weights $((k+1)/(k+2))^{m+1} \le 1$ Chapter4/BracketLevel · BracketLevel.wshift Chapter4/BracketLevel · BracketLevel.tw; both are vector fields Chapter4/BracketLevel · BracketLevel.isRetractVectorField_opS Chapter4/BracketLevel · BracketLevel.isRetractVectorField_opT. *The computation.* $[S,T]x = ((-1)^k(k+1)(2k+3)x_{k+1})_k$ — the sign $(-1)^k$ makes the two terms add instead of cancelling; with polynomial weights the commutator of a non-oscillating diagonal operator and a shift would lose only one level. For $x_k = (k+1)^{-3}$, a point of E_2 Chapter4/BracketLevel · BracketLevel.xPt, the level-1 coordinates $(k+1)[S,T](x)_k$ have absolute value $P^3 + P^2 \ge 1/4$, $P = (k+1)/(k+2) \ge 1/2$, so they do not tend to 0 and $[S,T]x \notin E_1$ Chapter4/BracketLevel · BracketLevel.bracket_not_mem_level_one. The universal statement therefore fails Chapter4/BracketLevel · not_forall_lieBracket_mem_level_one. Here $T_xO = E_0$, so $(T_xO)_1 = E_1$ and the sentence of p. 200 is covered too. What holds, and what Proposition 4.4.5 states, is the shifted statement $x \in O_{m+2} \Rightarrow [A,B](x) \in E_m$: $[A,B]$ is a tangent vector at the point x of level 1 of X^1 — the

local form above. A first example on a weighted ℓ^2 space ($E_m = \ell^2$ with weights 4^{mk}, $S = \mathrm{diag}(2^k)$, $(Tx)_k = 2^k x_{k+1}$, $x_j = 4^{-j}/j$) is sketched in the header of `Chapter4/VectorFields` and not formalized, the project having no weighted ℓ^2 sc-structure.

What remains. Nothing of this subsection: Lemma 4.4.4 and Proposition 4.4.5 on an M-polyfold and Lemma 4.5.1 are §IV.29 and §IV.34; Definitions 4.4.1–4.4.2, d and $d \circ d = 0$, Definition 4.4.6, Theorem 4.4.7 and Lemmas 4.4.8 and 4.5.2 are §§IV.35–IV.45. The example the 2021 edition adds to Remark 4.1.3 (preprint 4.3), the space of its Subsection 1.4.4, for which 1^1_0 is not sc-smooth because Theorem 4.1.2 (preprint 4.2) would make it a manifold of constant local dimension, holds as we read it and is not formalized (Findings, Chapter 4, item 159); gap G.31 is closed by the example of §IV.27.

IV.29 The Springer 2021 edition: vector fields and the bracket on an M-polyfold (Lemmas 4.4.4, 4.5.1, Proposition 4.4.5)

The local forms of §IV.28 are carried to M-polyfolds in `Chapter4/VectorFieldsGlobal`, on top of the identification $\Phi : T(X^1) \cong (TX)^1$ of Part I's Proposition 2.3.17 `Chapter2/TangentLift · MPolyfold.tangentShiftEquiv Chapter2/TangentLift · proposition_2_3_17`. The plan: Lemma 4.5.1 in a local model, by the computation of §4.5.2; principal parts in charts and the independence of the chart; the bracket $X^2 \to TX$; Lemma 4.5.1 on M-polyfolds; Lemma 4.4.4 for $i = 1$ and for every i (gap G.38 closed; Findings, Chapter 4, items 160–164).

Lemma 4.5.1, local form (the computation of §4.5.2) `Chapter4/VectorFieldsGlobal · retractLieBracket_natural`. *Statement.* Let r, r' be retractions onto O, O', f an sc-smooth map from the retract O to the retract O', $g = f \circ r$, and A, B vector fields on O, A', B' on O' with $Dg(x)A(x) = A'(f(x))$ and $Dg(x)B(x) = B'(f(x))$ for $x \in O_1$. Then $Dg(x)[A,B](x) = [A',B'](f(x))$ for $x \in O_2$. The text states it for a local sc-diffeomorphism $\varphi : X \to X$ with the relations at points of X_2, and leaves the proof to the reader (p. 201); §4.5.2 is the computation, and it differentiates the relations, so they are assumed on the open set O_1 of level 1. *Proof.* (i) *The relation differentiated* `Chapter4/VectorFieldsGlobal · IsRetractVectorField.diff_related`: the map $y \mapsto Dg(r^1 y)\tilde{A}(y) - \tilde{A}'(g_1(y))$, $\tilde{A} = A \circ r^1$, $\tilde{A}' = A' \circ r'^1$, vanishes on U^1 (since $r^1 y \in O_1$ and $g \circ r = g$); differentiating at $x \in O_2$ in a direction b with $Dr^1(x)b = b$, as in §IV.28's differentiated identity, gives $D^2 g(x)(A(x),b) + Dg(x)D\tilde{A}(x)b = D\tilde{A}'(g_2(x))Dg^1(x)b$. (ii) *The relation on level* 1 `Chapter4/VectorFieldsGlobal · IsRetractVectorField.related_shift`: the derivative of the shifted map g^1 at $x \in O_2$ is the level-1 restriction of $Dg(x)$ `Chapter1/ScSmooth · IsSc1.scDeriv_eqOn`, so $Dg^1(x)B_1(x) = B'_1(g_2(x))$; and $g_2 = f_2$ on O_2 `Chapter4/VectorFieldsGlobal · levelMap_levelMap_comp_eq`. (iii) Apply (i) to A with $b = B_1(x)$ and to B with $b = A_1(x)$ (both fixed by $Dr^1(x)$, §IV.28), rewrite the right sides by (ii), subtract, and cancel $D^2 g(x)(A(x),B(x)) - D^2 g(x)(B(x),A(x))$ by symmetry `Chapter4/VectorFields · scSecondDeriv_symm`: $Dg(x)[A,B](x) = D\tilde{A}'(f_2 x)B'_1(f_2 x) - D\tilde{B}'(f_2 x)A'_1(f_2 x) = [A',B'](f(x))$. §4.5.2 ends with "The proof of Proposition 4.5.1 is complete"; there is no Proposition 4.5.1, presumably Lemma 4.5.1 is meant.

On an open part. A transition map of charts is defined only on an open part $O \cap W$ of a model. A vector field on O is one on $O \cap W = r'(U')$ for the restricted retraction r' `Chapter4/VectorFieldsGlobal · IsRetractVectorField.restrict`, and at points of $(O \cap W)_2$ the two brackets agree, since r' is r on the smaller open domain U' and a derivative on an open subdomain is the restriction of the derivative `Chapter4/VectorFieldsGlobal · retractLieBracket_restrict`. So for f sc-smooth on $O \cap W$ with $Tf(z,A(z)) = (f(z),A'(f(z)))$ and $Tf(z,B(z)) = (f(z),B'(f(z)))$ on $(O \cap W)_1$, also $Tf(x,[A,B](x)) = (f(x),[A',B'](f(x)))$ on $(O \cap W)_2$ `Chapter4/VectorFieldsGlobal · retractLieBracket_natural_inter`.

Principal parts and the independence of the chart. Tangent charts change by $T\psi = T(\psi \circ \varphi^{-1}) \circ T\varphi$ over the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_change` (formula (2.9)). Choose a retraction onto the model of each chart `Chapter4/VectorFieldsGlobal · MPolyfold.chartRetraction Chapter4/VectorFieldsGlobal · MPolyfold.chartRetraction_retract`. The *principal part* of a vector field A in φ is the second component of $T\varphi(A(\varphi^{-1}z))$ `Chapter4/VectorFieldsGlobal · MPolyfold.principalPart`, so $T\varphi(A(y)) = (\varphi(y), A_\varphi(\varphi(y)))$ for y of level 1 in the domain of φ `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_eq_principalPart`; read through the charts φ^1 of X^1 and $T\varphi$ of TX, the sc-smoothness of A says that A_φ is a vector field on the model `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.isRetractVectorField_principalPart`. The principal parts in two charts are related by the transition map on level 1 of the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.principalPart_related`. The bracket of the principal parts `Chapter4/VectorFieldsGlobal · MPolyfold.chartLieBracket` satisfies the local Proposition 4.4.5 `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.chartLieBracket_spec`, and — the use of Lemma 4.5.1 that p. 201 names — the brackets in φ and ψ are related by $T(\psi\varphi^{-1})$ on level 2 of the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.chartLieBracket_change`: the overlap is $O \cap W$ for an open W, and the local form on an open part applies.

Proposition 4.4.5 (preprint 4.26) on an M-polyfold `Chapter4/VectorFieldsGlobal · proposition_4_4_5`. *Statement.* For sc-smooth vector fields A, B on X, the bracket $[A,B] : X^2 \to TX$ is sc-smooth and $p([A,B](x)) = x$. The text's "sc-smooth section of $TX|X^2 \to X^{2}$" is read as an sc-smooth map $X^2 \to TX$ covering the inclusion $X^2 \to X^1$; the pull-back bundle itself, and the bracket as its section in the text's form, are §IV.34. *Construction.* At $x \in X^2$ take the chart φ chosen around x; the level lift of x in the chart $(\varphi^1)^1$ of X^2 lies in O_2 and is $\varphi(x)$ on level 1 `Chapter4/VectorFieldsGlobal · MPolyfold.levelLift_levelChart_spec`, so $(\varphi(x), [A_\varphi, B_\varphi](\varphi(x)))$ lies in TO and defines a tangent vector at x `Chapter4/VectorFieldsGlobal · MPolyfold.lieBracketPre Chapter4/VectorFieldsGlobal · MPolyfold.lieBracket` over x `Chapter4/VectorFieldsGlobal · MPolyfold.tangentProj_lieBracket`. *Independence of the chart* `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_lieBracket`: in every chart ψ around x, $T\psi([A,B](x)) = (\psi(x), [A_\psi, B_\psi](\psi(x)))$, by the change of tangent charts and the relation of the brackets in φ and ψ. *Proof of the statement.* On the domain of $(\psi^1)^1$, $T\psi \circ [A,B]$ is the section $z \mapsto (z, [A_\psi, B_\psi](z))$ composed with the chart `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_lieBracket_eq`. That section is continuous and sc-smooth by the local proposition, so $[A,B]$ is continuous (the preimage of a basic open set of TX is open in each chart domain of X^2) `Chapter4/VectorFieldsGlobal · MPolyfold.continuous_lieBracket`, and sc-smooth in the charts $(\psi^1)^1$ of X^2 and $T\psi$ of TX.

Lemma 4.5.1 on M-polyfolds `Chapter4/VectorFieldsGlobal · lemma_4_5_1`. *Statement.* Let $f : X \to Y$ be sc-smooth, with level maps $f_1 : X^1 \to Y^1$ `Chapter4/VectorFieldsGlobal · MPolyfold.liftLevelOneMap` and $f_2 : X^2 \to Y^2$ `Chapter4/VectorFieldsGlobal · MPolyfold.levelTwoMap`, and A, B on X, A', B' on Y sc-smooth vector fields with $Tf \circ A = A' \circ f_1$ and $Tf \circ B = B' \circ f_1$ on X^1. Then $Tf \circ [A,B] = [A',B'] \circ f_2$ on X^2. This is more general in the map than the text's local sc-diffeomorphism $X \to X$; one between open subsets of X is §IV.34. *Proof.* Fix $x \in X^2$, a chart φ around x and ψ around $f(x)$. Both sides lie over $f(x)$; in $T\psi$ the right side is $(\psi f(x), [A'_\psi, B'_\psi](\psi f(x)))$ by the independence of the chart, and on level 1, $(\psi f \varphi^{-1})_1(\varphi(x)) = \psi(f(x))$ `Chapter4/VectorFieldsGlobal · MPolyfold.levelMap_coord_levelLift`. The left side, through φ, is $T(\psi f \varphi^{-1})(\varphi(x), [A_\varphi, B_\varphi](\varphi(x)))$. The map $\psi f \varphi^{-1}$ is sc-smooth on the open part of the model of φ that $\varphi(f^{-1}(\text{domain of } \psi))$ cuts out, and the hypotheses read in the charts say that it relates A_φ, B_φ to A'_ψ, B'_ψ on level 1 there; the local form on an open part gives the equality.

Lemma 4.4.4 on an M-polyfold, $i = 1$ `Chapter4/VectorFieldsGlobal · lemma_4_4_4`. *Statement.* For an sc-smooth vector field A, $A^{(1)} = \Phi^{-1} \circ A_1 : X^2 \to T(X^1)$ `Chapter4/VectorFieldsGlobal · MPolyfold.raiseIndex`, $A_1 : X^2 \to (TX)^1$ the level-1 map of A `Chapter4/VectorFieldsGlobal · MPolyfold.vectorFieldLevelOne`, is an

sc-smooth vector field on the M-polyfold X^1, and $\Phi(A^{(1)}(x)) = A(x)$ in $(TX)^1$. *Proof.* A_1 is sc-smooth, as the level-1 map of the sc-smooth A; Φ^{-1} is sc-smooth (Proposition 2.3.17), so $A^{(1)}$ is; Φ commutes with the projections to X^2, so $p(A^{(1)}(x)) = x$ follows from $p(A(x)) = x$; and $\Phi \circ \Phi^{-1}$ is the identity. Every i follows by applying the lemma to X^1, X^2, …, as the next paragraph does. A witness for the hypotheses of the section: the zero section `Chapter4/VectorFields · MPolyfold.isScVectorField_zero`.

Lemma 4.4.4 on an M-polyfold, every i `Chapter2/TangentLiftAll · lemma_4_4_4_all`. To state the lemma once for all i, the M-polyfolds are bundled with their spaces `Chapter2/TangentLiftAll · BundledMPolyfold`, so that the lifts $X^0 = X$, $X^{i+1} = (X^i)^1$ form a family `Chapter2/TangentLiftAll · BundledMPolyfold.lift`. Part I's Proposition 2.3.17 for every i `Chapter2/TangentLiftAll · proposition_2_3_17_all` gives the sc-diffeomorphisms $\Phi_i : T(X^i) \to (TX)^i$ `Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv` and $J_i : (X^1)^i \to (X^i)^1$ `Chapter2/TangentLiftAll · BundledMPolyfold.shiftLiftEquiv`, and Φ_i commutes with the inclusions into TX. *Statement.* With $A^{(0)} = A$ and $A^{(i+1)} = (A^{(i)})^{(1)}$, the case $i = 1$ on X^i `Chapter2/TangentLiftAll · BundledMPolyfold.raiseIndexAll`: for every i, $A^{(i)}$ is an sc-smooth vector field on X^i; its image in TX at $x \in X^{i+1}$ is $A(x)$ `Chapter2/TangentLiftAll · BundledMPolyfold.tangentIncl_raiseIndexAll`; and $\Phi_i \circ A^{(i)} \circ J_i$ is the i-th index raise of A, viewed as an sc-smooth map $X^1 \to TX$ `Chapter2/TangentLiftAll · BundledMPolyfold.vectorFieldScMap` `Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_raiseIndexAll` — the diagram printed before the lemma. *Proof.* The first claim is the case $i = 1$, applied i times. The second by induction: the case $i = 1$ for X^i says that $A^{(i+1)}(x)$ and $A^{(i)}(x)$ agree after $T(X^{i+1}) \to T(X^i)$. For the third, both sides are compared after the inclusion $(TX)^i \to TX$, which is injective: on the left it is the inclusion $T(X^i) \to TX$ after Φ_i, so the second claim applies; on the right it is A on the underlying point. *Witness:* the zero vector field, for every i `Chapter2/TangentLiftAll · BundledMPolyfold.lemma_4_4_4_all_zero` (Findings, Chapter 4, item 164).

What remains. Nothing: Lemma 4.5.1 between open subsets and the pull-back bundle $TX|X^2 \to X^2$ are §IV.34 (gap G.92 closed), the forms of §4.4 §§IV.35–IV.45.

IV.30 Lemma 4.1.11 (preprint 4.10): the section $u \mapsto (1-p)(u-t(u))$ is sc-Fredholm

The text proves Proposition 4.1.9 (preprint 4.9) through this lemma and the implicit function theorem. The formalization proves the proposition directly (§§IV.6–IV.9), so the lemma is not used there. It is proved here on its own, in the 2021 notion of an sc-Fredholm section.

Statement `Chapter4/RetractionSectionFredholm · lemma_4_1_11`. Take the data of Proposition 4.1.9 (preprint 4.9): an sc$^+$-retraction $t : U \to U$ of a relatively open $U \subseteq C$ onto $O = t(U) \ni 0$, $N = T_0O$ in good position, a good complement Y with $E = N \oplus Y$ an sc-splitting, and p the projection onto N along Y. Then the section $u \mapsto (u, (1-p)(u-t(u)))$ of the local strong bundle $U \lhd Y \to U$ is an sc-Fredholm section in the sense of 2021 Definition 3.1.16: it is sc-smooth, it is regularizing, and it is an sc-Fredholm germ at every smooth point. It holds on all of U, not only near 0. That $E = N \oplus Y$ is an sc-splitting is the text's "good sc-complement" (Definition 3.1.24 asks for an sc-complement), not an added hypothesis. N is smooth and finite-dimensional by Lemma 4.1.7 (preprint 4.7) (§IV.3).

The bundle and the section. The local strong bundle $U \lhd Y$ lives over the M-polyfold U, which has one chart and the identity retraction. It is a strong bundle with one strong bundle chart `Chapter4/LocalStrongBundle · LocalBundle.bundle`. The section `Chapter4/LocalStrongBundle · LocalBundle.sec` is sc-smooth `Chapter4/LocalStrongBundle · LocalBundle.isScSmoothSection_sec`: t is sc-smooth, and $1 - p$ is an sc-operator

$E \to Y$ Chapter4/RetractionSectionSplitting · qOp.

Regularizing. Let $u \in U_m$ with $(1-p)(u - t(u)) \in Y_{m+1}$. Write

$$u = (1-p)(u - t(u)) + t(u) + p(u - t(u)).$$

Here $t(u) \in E_{m+1}$ because t is sc$^+$, and $p(u - t(u)) \in N \subseteq E_\infty$. So $u \in E_{m+1}$.

The germ at a smooth point e. Go into the strong bundle chart centred at e Chapter4/LocalStrongBundle · LocalBundle.centred. Its retraction is pointwise the identity, so the principal part fills itself and no filler has to be built Chapter4/LocalStrongBundle · isFilling2021_of_R_eq_self. In this chart the principal part is $g(v) = (1-p)(v + e - t(v+e))$ Chapter4/LocalStrongBundle · LocalBundle.principal_centred. The text's sc$^+$-section is $s(v) = (1-p)(e - t(v+e))$. It is sc$^+$ because t is, read as a map into Y^1. Then $g - s = (1-p)v$, and it remains to conjugate this germ to a basic germ.

The splitting $E = (N \oplus B) \oplus W$. Let λ be the coordinates of the partial quadrant. Their values on the smooth points of Y already fill $\lambda(Y)$: the smooth points are dense and $\lambda(Y)$ is finite-dimensional Chapter4/RetractionSectionSplitting · map_le_map_smooth. So there are smooth $e_1, \dots, e_d \in Y$ and functionals ℓ_j with $\ell_i(e_j) = \delta_{ij}$ such that, on Y, the ℓ_j vanish exactly where λ does Chapter4/RetractionSectionSplitting · exists_splitData. Put $B = \operatorname{span}(e_j)$, a smooth subspace inside Y_∞, and $W = Y \cap \bigcap_j \ker \ell_j$ Chapter4/RetractionSectionSplitting · splitW. W is an sc-subspace Chapter4/RetractionSectionSplitting · isScSubspace_splitW, and $Y = B \oplus W$. Adding a vector of W changes no coordinate. So near 0 the partial quadrant is $((N \oplus B) \cap C) \oplus W$, and the quadrant on $N \oplus B$ comes from the corner coordinates, not from good position Chapter4/RetractionSectionSplitting · isPartialQuadrant_preimage_quadrant. In the coordinates $(a+b, w)$, $a \in N$, $b \in B$, $w \in W$, the germ $(1-p)v$ becomes $(a+b, w) \mapsto (b, w)$. This is a basic germ with contraction part 0 Chapter4/RetractionSectionSplitting · isConjugateToBasicGerm_qOp.

The proof follows the text. We read it differently from the printed text in a few places, all recorded in Findings, Chapter 4, item 57: the sign of $D\mathbf{f}(u)$; $Dt(u)$, not $D\mathbf{f}(u)$, is the sc$^+$-operator; "a perturbation of the identity" is said of an operator into Y; $L(Y)$ is meant where $L(T_0O)$ is printed; and the text omits the interior points of $(T_0O \oplus B) \cap C$.

IV.31 Remark 4.2.2 (preprint 4.14) in both wordings, and Question 4.2.4 (preprint: the question after Proposition 4.15)

The preprint wording, $d_X(a) = 0$. Let A be a smooth finite-dimensional submanifold. The open piece $A \cap \{d_X = 0\}$ is again a smooth finite-dimensional submanifold. Its induced structure has a corner structure with degeneracy index 0 everywhere: a manifold without boundary Chapter4/RemarkLocalForm · MPolyfold.remark_4_2_2_interior. The pointwise form is Chapter4/RemarkLocalForm · MPolyfold.remark_4_2_2_near. The piece is open because d_X is upper semicontinuous. The corner structure is the theorem of §IV.11 for $d_X = 0$ Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_degIndex_eq_zero, applied to this piece.

The 2021 wording, $d_A(a) = 0$ Chapter4/SubmanifoldNearInterior · MPolyfold.remark_4_2_2_dA_zero Chapter4/SubmanifoldNearInterior · MPolyfold.remark_4_2_2_dA_zero_near. The 2021 text speaks of d_A, not d_X. Since $d_A \le d_X$ (Lemma 2.4.6 (preprint 2.26)), this covers more points. For example, the parabola $\{(y^2, y)\}$ in $[0,\infty) \times \mathbb{R}$ has $d_A \equiv 0$, while $d_X(0,0) = 1$ (Findings, Chapter 4, items 165, 176). The proof is

one step. A with its induced structure is a smooth finite-dimensional submanifold of itself Chapter4/SubmanifoldNearInterior · MPolyfold.IsSmoothSubmanifold.isSmoothSubmanifold_univ_subPolyfold: the retraction is the identity, and it is sc$^+$ because $A = A_\infty$ and $1^1_0 : A \to A^1$ is sc-smooth (Proposition 4.2.3 (4) (preprint 4.15), §IV.12). The preprint form, applied to A inside A, has d_A in place of d_X. No tameness of X or A is used.

The sentence on $d_A(a) = 1$. The text says, without proof, that near such points A is a manifold with boundary. The key step is a general statement Chapter4/SubmanifoldNearBoundary · MPolyfold.hasCornerStructure_of_degIndex_le_one: *an M-polyfold with $X_0 = X_\infty$, 1^1_0 sc-smooth and $d_X \le 1$ everywhere has a corner structure*, with no tameness. By Theorem 4.1.2 (preprint 4.2) in its centred form Chapter4/Characterization · MPolyfold.hasCornerStructure_of_centred, it suffices to find at every x a chart centred at x with $N = T_0O$ in good position. Take a chart realizing $d_X(x)$ and re-centre it at $e = \varphi(x)$ (§IV.5). The new partial quadrant C_e is cut out by the at most one coordinate that vanishes at e.

- If $d_X(x) = 0$, then $C_e = E$, and every subspace is in good position Chapter4/Submanifolds · isInGoodPosition_univ.
- If $d_X(x) = 1$, then $C_e = \{\lambda \ge 0\}$. If λ does not vanish on N, then N is in good position with a complement inside $\ker \lambda$ Chapter4/SubmanifoldNearBoundary · isInGoodPosition_halfSpace.
- If λ vanishes on N, then T_0O lies in the reduced tangent space. The model is then near 0 a graph over an open subset of N, a chart without boundary, so $d_X(x) = 0$ Chapter9/BoundaryPositionConverse · MPolyfold.degIndex_eq_zero_of_isReducedTangentAt. This is the graph argument of 2021 Lemma 9.2.13, and it contradicts $d_X(x) = 1$.

The same holds for the induced structure of a submanifold with $d_A \le 1$ Chapter4/SubmanifoldNearBoundary · MPolyfold.IsSmoothSubmanifold.hasCornerStructure_of_degIndex_le_one. The set $\{d_A \le 1\}$ is open Chapter4/SubmanifoldNearBoundary · MPolyfold.isOpen_degIndex_le. So it is a smooth submanifold of A whose induced structure is a manifold with boundary Chapter4/SubmanifoldNearBoundary · MPolyfold.remark_4_2_2_dA_one, and the same holds near one point with $d_A(a) = 1$ Chapter4/SubmanifoldNearBoundary · MPolyfold.remark_4_2_2_dA_one_near.

In a tame M-polyfold, good position is automatic. At the centre of a tame chart, $T_0O \cap C$ has interior points in T_0O Chapter4/SubmanifoldNearInterior · ScRetraction.interior_rangeS_scDeriv_zero_nonempty. With Lemma 4.2.8 (preprint 4.19), T_0O is then in good position Chapter4/SubmanifoldNearInterior · ScRetraction.isInGoodPosition_rangeS_of_isTameWith. So *a tame M-polyfold with $X_0 = X_\infty$ and 1^1_0 sc-smooth has a corner structure* Chapter4/SubmanifoldNearInterior · MPolyfold.hasCornerStructure_of_isTame: the good-position hypothesis of Theorem 4.1.2 (2) (preprint 4.2) holds by itself. This gives a second route to the $d_A = 1$ sentence when A is tame Chapter4/SubmanifoldNearInterior · MPolyfold.remark_4_2_2_dA_one_of_isTame.

Question 4.2.4 (preprint: the open question after Proposition 4.15). It asks two things. Is a sub-M-polyfold with $A = A_\infty$ and finite, locally constant $\dim T_aA$ a smooth finite-dimensional submanifold? Is the "T" $T = \{(x,0) : |x| < 1\} \cup \{(0,y) : 0 \le y < 1\}$ Chapter4/ScPlusRetractQuestion · Question424.tSet homeomorphic to an sc-retract $O = O_\infty$ without boundary? These are questions, and nothing in them is to be formalized (gap G.34). The text adds that this is impossible for sc$^+$-retracts. That sentence is proved Chapter4/ScPlusRetractQuestion · Question424.question_4_2_4_scPlus in two steps.

1. *An sc$^+$-retract (O, E, E) is locally a graph over its finite-dimensional tangent space* Chapter9/BoundaryPositionConverse · ScRetraction.IsScPlus.exists_flat. This is the argument of 2021 Lemma 9.2.13 with $C = E$, so that $T_zO \subseteq E_z = E$.
2. *T is not a topological manifold at the branch point* Chapter4/ScPlusRetractQuestion · Question424.not_

`homeomorph_open`. No relatively open neighbourhood of $(0,0)$ in T is homeomorphic to an open subset of a finite-dimensional space $\mathbb{R}^k$. The proof splits by k:

- For $k = 0$: a point is not a neighbourhood in T.
- For $k = 1$: an injective arc leaves its starting point to one side `Chapter4/ScPlusRetractQuestion · Question424.arc_side`, so two of the three arms go to the same side of the image of $(0,0)$, and their images overlap `Chapter4/ScPlusRetractQuestion · Question424.arc_overlap`.
- For $k \geq 2$: a punctured ball is connected, but a punctured neighbourhood of $(0,0)$ in T is not.

IV.32 Theorem 4.2.11 (1) (preprint 4.22): the local faces of the neighbourhood

In the statement of Theorem 4.2.11, the local faces at x are the faces of a small open $U \ni x$: the closures in U of the components of $U \cap \{d_X = 1\}$, as recalled before Proposition 2.5.16 (preprint 2.44) (Definition 2.5.10 for U). In the proof, the text uses the chart faces $O \cap \{\lambda_i = 0\}$ instead. §IV.17 proved (1) for the chart faces. This subsection identifies the two.

The faces of a neighbourhood in a tame centred chart `Chapter4/LocalFaces · MPolyfold.exists_faces_chart`. Let ψ be a chart centred at x, modelled on a tame retract $O = r(U_r)$ with n half-line coordinates, and let $U \ni x$ be open inside its domain. Then there are an open $U' \ni x$ inside U and pairwise distinct faces $\theta_0, \dots, \theta_{n-1}$ of U such that $\theta_i \cap U' = U' \cap \psi^{-1}\{\lambda_i = 0\}$. Moreover, every face $F \ni 0$ of the retract with $F \subseteq \{\lambda_i = 0\}$ satisfies $U' \cap \psi^{-1}(F) = U' \cap \psi^{-1}\{\lambda_i = 0\}$. If $d_X(x) = n$, the faces of U through x are exactly the θ_i.

Proof. Fix i. 1. For small δ, the set $\{z \in C : |z| < \delta,\ I(z) = \{i\}\}$ is convex, so it is connected. 2. Its image under r lies in $\{d = 1\}$, so it lies in one component of $U \cap \{d_X = 1\}$. Let θ_i be the closure of that component, a face of U. 3. Near 0, every point of $O \cap \{\lambda_i = 0\}$ is a limit of such images. This is Lemma 4.C of the exposition `Chapter2/TameRetracts · local_face_structure`. So θ_i contains the chart face near x. 4. Conversely, a connected subset of $\{d = 1\}$ near 0 stays inside one coordinate hyperplane. This is Lemma 4.A `Chapter2/TameRetracts · subset_coordHyper_of_isPreconnected`, and it gives the reverse inclusion. 5. The θ_i are distinct because they lie in distinct hyperplanes.

(1) and (4) with the faces of U `Chapter4/LocalFaces · MPolyfold.theorem_4_2_11_faces`. This is Theorem 4.2.11 (preprint 4.22) with the local faces θ_i of U. Near x, the text's $F_\sigma = \bigcap_{i\in\sigma} \theta_i$ agrees with the set $\psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ of the formal (4) (§IV.18), for every σ. So (4) holds for the germ of F_σ at x.

(1) as printed `Chapter4/GeneralPositionLocalFaces · MPolyfold.theorem_4_2_11_one`. There is an open $U \ni x$ such that: - the faces of U through x number exactly $d_X(x)$, and each is contained in U; - (2) and (3) hold on U; - U is face-structured: every $y \in U$ lies in exactly $d_X(y)$ faces of U.

Proof. Take the neighbourhood of §IV.17 and shrink it. A tame X has at x an open $D \ni x$ all of whose small open subsets are face-structured `Chapter2/TameBoundary · TameBoundary.exists_nbhd_forall_isFaceStructuredOpen`. Intersect with D and apply the identification above.

IV.33 Theorem 4.3.1 (preprint 4.23): stratified regularity does not depend on the chart

The definition of a stratified regular point (§IV.23) asks for the condition in *some* product chart $\chi = (r \mapsto r - a) \times \varphi$ of $\mathbb{R}^m \times X$, with some data of Definition 4.2.1 (preprint 4.13), some presentation and some lift. In that chart the condition is that the projection maps $T_z A \cap (\mathbb{R}^m \times E_{z_2})$ onto $\mathbb{R}^m$ `Chapter4/StratifiedRegularValues · StratRegCond`. Like the conditions of §IV.13, it depends only on the tangent spaces `Chapter4/StratifiedRegularValues · StratRegCond.congr`. The text treats the notion as intrinsic to A and its strata (Findings, Chapter 4, items 102, 108).

Transport. Let F be a germ of sc-diffeomorphisms whose $\mathbb{R}^m$-component is a translation. Then F carries the condition `Chapter4/StratifiedRegularValues · StratRegCond.transport`, for three reasons: - its derivative maps $T_z Q$ onto $T_{F(z)} Q'$ (§IV.11); - by the sign argument of Proposition 2.4.10 (preprint 2.30), it maps E_z into $E'_{F(z)}$; - it keeps the $\mathbb{R}^m$-component of every vector. If the first components of two sc¹ maps differ by a constant, so do the first components of their derivatives `Chapter4/StratifiedRegularValues · scDeriv_fst_eq_of_fst_eq`.

The chart change `Chapter4/StratifiedRegularValues · stratRegCond_chart_change`. The transition of two product charts is $w \mapsto (w_1 + a - a', \varphi' \circ \varphi^{-1}(w_2))$ `Chapter4/StratifiedRegularValues · transition_rnChart_prod_fst`. This is a germ of the kind just treated, so the chart-change argument of §IV.13 applies.

So the condition holds in every product chart once it holds in one `Chapter4/StratifiedRegularValues · IsStratRegularPoint.stratRegCond`. The chart-free forms are `Chapter4/StratifiedRegularValues · isStratRegularPoint_iff_forall` `Chapter4/StratifiedRegularValues · isStratRegularValue_iff_forall`. Since $T_z A$ has no chart-free linear structure here (gap G.7), "intrinsic" is read as "independent of every choice in the product charts".

IV.34 Lemma 4.5.1 between open subsets; the pull-back bundle $TX|X^2 \to X^2$ and Proposition 4.4.5 as printed

Open subsets. An open $U \subseteq X$ is an M-polyfold by Part I's restriction of the atlas `Chapter3/RestrictOpen · MPolyfold.restrictOpen`. Its inclusion is sc-smooth, and TU is the part of TX over U. A vector field A on X restricts to U as $A|_U = (T\iota)^{-1} \circ A \circ \iota_1$ `Chapter4/LieBracketOpenSubsets · MPolyfold.restrictVectorField`. This restriction is an sc-smooth vector field of U `Chapter4/LieBracketOpenSubsets · MPolyfold.IsScVectorField.restrictOpen` with $T\iota(A|_U(x)) = A(x)$ `Chapter4/LieBracketOpenSubsets · MPolyfold.tangentMap_restrictVectorField`.

The bracket is local: $T\iota[A|_U, B|_U](x) = [A, B](x)$ for $x \in U^2$ `Chapter4/LieBracketOpenSubsets · MPolyfold.tangentMap_lieBracket_restrictOpen`. In other words, the bracket of the restrictions is the restriction of the bracket `Chapter4/LieBracketOpenSubsets · MPolyfold.lieBracket_restrictOpen`. The reason is that both are computed in the same charts by the local formula, and Lemma 4.5.1 (§IV.29) applies to the inclusion.

Lemma 4.5.1 for maps between open subsets `Chapter4/LieBracketOpenSubsets · lemma_4_5_1_open`. Let $U, U' \subseteq X$ be open and $\varphi : U \to U'$ sc-smooth; a local sc-diffeomorphism between open subsets is one such map. Suppose $T\varphi \circ A|_U = A|_{U'} \circ \varphi_1$ and $T\varphi \circ B|_U = B|_{U'} \circ \varphi_1$ on U^1. Then $T\varphi \circ [A|_U, B|_U] = [A|_{U'}, B|_{U'}] \circ \varphi_2$ on U^2. This is §IV.29 for the M-polyfolds U and U'. Read in TX, for $g = \iota' \circ \varphi$, it says $Tg((T\iota)^{-1}[A, B](x)) = [A, B](g(x))$ `Chapter4/LieBracketOpenSubsets · lemma_4_5_1_open_ambient`. No

hypothesis is added. The relations are asked on U^1, as in §IV.29.

The pull-back bundle. The text says that $[A, B]$ is an sc-smooth section of $TX|X^2 \to X^2$ and gives this bundle no structure. We use the sub-M-polyfold

$$TX|X^2 = \{(x, \alpha) \in X^2 \times TX \mid p(\alpha) = x\}$$

`Chapter4/LieBracketBundle · MPolyfold.tangentPullback`. Take a chart φ of X. The product chart of $X^2 \times TX$ is the lifted chart times $T\varphi$ `Chapter4/LieBracketBundle · MPolyfold.tangentPullbackChart`. In it, $p(\alpha) = x$ says $y = \iota z$ for $(z, (y, v))$ `Chapter4/LieBracketBundle · MPolyfold.mem_tangentPullback_iff`, where $\iota : E_2 \to E_1$ is the inclusion. The local retraction is

$$(z, (y, v)) \mapsto (z, Tr(\iota z, v))$$

`Chapter4/LieBracketBundle · tangentPullbackLocal`, for r a retraction onto the model of φ. It is the composite of the linear sc-operator $(z, (y, v)) \mapsto (z, (\iota z, v))$ with id $\times\, Tr$, hence sc-smooth `Chapter4/LieBracketBundle · isScSmoothRetractMap_tangentPullbackLocal`.

So $TX|X^2$ is a sub-M-polyfold in the sense of Definition 2.3.18 (preprint 2.20) `Chapter4/LieBracketBundle · MPolyfold.isSubMPolyfold_tangentPullback`, and an M-polyfold `Chapter4/LieBracketBundle · MPolyfold.tangentPullbackPolyfold`. Its projection to X^2 is sc-smooth `Chapter4/LieBracketBundle · MPolyfold.isScSmoothMap_tangentPullbackProj`, and its fibre over x is T_xX `Chapter4/LieBracketBundle · MPolyfold.tangentPullbackFiberEquiv`.

Proposition 4.4.5 (preprint 4.26) as printed `Chapter4/LieBracketBundle · proposition_4_4_5_section`. The map $x \mapsto (x, [A, B](x))$ `Chapter4/LieBracketBundle · MPolyfold.lieBracketSection` is an sc-smooth section of $TX|X^2 \to X^2$. A map into the sub-M-polyfold is sc-smooth when its two components are `Chapter4/LieBracketBundle · MPolyfold.isScSmoothMap_tangentPullback_mk`, and both components are sc-smooth by §IV.29. The form "in every chart, the usual formula" is collected in one statement `Chapter4/BookFormLieBracket2021 · MPolyfold.proposition_4_4_5_book`.

Lemma 4.5.1 on sections `Chapter4/LieBracketBundle · lemma_4_5_1_section`. For $f : X \to Y$ sc-smooth, the bundle map $(f_2, Tf) : TX|X^2 \to TY|Y^2$ `Chapter4/LieBracketBundle · MPolyfold.tangentPullbackMap` is sc-smooth `Chapter4/LieBracketBundle · MPolyfold.isScSmoothMap_tangentPullbackMap`. Under the hypotheses of §IV.29, $(f_2, Tf) \circ [A, B] = [A', B'] \circ f_2$ (Findings, Chapter 4, item 163).

IV.35 The Whitney sum $\oplus_k TX$, Definition 4.4.1 (preprint 4.25), and the system (4.4)

The 2021 text defines a k-form on X as an sc-smooth map $\oplus_k TX \to \mathbb{R}$ on the Whitney sum, linear and alternating in the fibres. The Whitney sum is not constructed in the text, and T_xX has no linear structure in the formalization (gap G.7). So $\oplus_k TX$ is read in its charts $\oplus_k T\varphi$.

The Whitney sum of a local model `Chapter4/ScForms2021 · ScBanach.whitney`. For an sc-Banach space E, the space $E^1 \oplus E^{\oplus k}$ has the fibre power $E^{\oplus k}$ `Chapter4/ScForms2021 · ScBanach.fiberPow`. For a retraction r onto O, the map

$$r_k(x, h_1, \dots, h_k) = (r(x), Dr(x)h_1, \dots, Dr(x)h_k)$$

is a retraction `Chapter4/ScForms2021 · ScRetraction.whitney` `Chapter4/ScForms2021 · ScRetraction.whitney_retract`. Its image $\oplus_k TO$ is a retract `Chapter4/ScForms2021 · IsScRetract.whitney` `Chapter4/ScForms2021 · retractWhitney`. An

sc-smooth map f between retracts induces the sc-smooth $\oplus_k Tf$ Chapter4/ScForms2021 · retractWhitneyMap Chapter4/ScForms2021 · IsScSmoothRetractMap.whitney. This is the chain rule of Proposition 2.1.13 (preprint 2.6), applied in each fibre.

Definition 4.4.1 (preprint 4.25) in charts. A family ω of alternating k-forms $\omega(x)$ on the tangent spaces T_xX, $x \in X^1$, has in a chart φ the representative

$$\omega_\varphi(y)(h_1, \dots, h_k) = \omega(x)(T\varphi^{-1}h_1, \dots)$$

Chapter4/ScForms2021 · MPolyfold.chartForm Chapter4/ScForms2021 · MPolyfold.chartForm_eq. ω is an *sc-form* if every representative is sc-smooth on $\oplus_k TO$ Chapter4/ScForms2021 · MPolyfold.IsScForm Chapter4/ScForms2021 · IsRetractScForm. Two representatives differ by $\oplus_k T$ of the transition map Chapter4/ScForms2021 · MPolyfold.chartForm_transition. So it suffices to check the condition on a cover by charts Chapter4/ScForms2021 · MPolyfold.isScForm_of_cover, and the condition does not quantify over all charts in a way that could be vacuous. The sc-forms of degree k form a subspace $\Omega^k(X)$ Chapter4/ScForms2021 · MPolyfold.scForms. It contains the constant functions in degree 0 Chapter4/ScForms2021 · MPolyfold.constForm_mem_scForms.

The system (4.4) and Definition 4.4.2. A form on X^i restricts to a form on X^{i+1} along the inclusion. On tangent fibres the inclusion is a continuous linear map Chapter4/ScForms2021 · MPolyfold.lowerFiberCLM, and the restriction is Chapter4/ScForms2021 · MPolyfold.formLower. It maps sc-forms to sc-forms Chapter4/ScForms2021 · MPolyfold.formLower_mem_scForms. The M-polyfolds X^i are the lifts of the bundled M-polyfold (§IV.29). They give the spaces $\Omega^k(X^i)$ Chapter4/ScForms2021 · BundledMPolyfold.forms and the maps $\Omega^k(X^i) \to \Omega^k(X^j)$, $i \le j$ Chapter4/ScForms2021 · BundledMPolyfold.formsMap. These compose Chapter4/ScForms2021 · BundledMPolyfold.formsMap_comp, so they form a directed system Chapter4/ScForms2021 · BundledMPolyfold.directedSystem_forms. $\Omega^k_\infty(X)$ is its direct limit Chapter4/ScForms2021 · BundledMPolyfold.formsInfty, with the canonical maps Chapter4/ScForms2021 · BundledMPolyfold.formsInftyOf Chapter4/ScForms2021 · BundledMPolyfold.formsInftyOf_formsMap. $\Omega^*_\infty(X) = \bigoplus_k \Omega^k_\infty(X)$ is the graded space Chapter4/ScForms2021 · BundledMPolyfold.formsInftyGraded.

IV.36 The exterior derivative in a local model, and its naturality

The local d. Let Ω be an sc-smooth map on $\oplus_k TU$, alternating in the fibres. At a point $(y, h_0, \dots, h_k)$ on level 1 define

$$d\Omega(y)(h_0, \dots, h_k) = \sum_{j=0}^{k} (-1)^j \, D\Omega(y, h_0, \dots, \widehat{h_j}, \dots, h_k)(h_j, 0)$$

Chapter4/ScExteriorDerivative2021 · ambientD Chapter4/ScExteriorDerivative2021 · ambientD_apply. Each term is the derivative of Ω in the base direction h_j with the fibre fixed. The sc-derivative of Ω in the direction $(h, 0)$ is determined by the level-1 derivative and the density of E_2 in E_1 Chapter4/ScExteriorDerivative2021 · scDeriv_relation. So $d\Omega(y)$ is the alternation of the base derivative Chapter4/ScExteriorDerivative2021 · ambientD_whitneyMk, an alternating $(k+1)$-form. $d\Omega$ is again sc-smooth Chapter4/ScExteriorDerivative2021 · IsScSmooth.ambientD Chapter4/ScExteriorDerivative2021 · IsAmbientScForm.ambientD. On a retract, d of Ω is d of $\Omega \circ r_k$, restricted to the retract Chapter4/ScExteriorDerivative2021 · retractD Chapter4/ScExteriorDerivative2021 · IsRetractScForm.retractD.

Naturality Chapter4/ScFormsNatural2021 · ambientD_comp_whitneyMap. For f sc-smooth, $d(\Omega \circ \oplus_k Tf) = d\Omega \circ \oplus_{k+1} Tf^1$. The derivative of $\Omega(f(y), Df(y)h_\bullet)$ in the direction h_j has two parts: - a term $D\Omega(\dots)(Df(y)h_j, 0)$; - terms with the second derivative $D^2f(y)(h_j, h_i)$.

The second-derivative terms are symmetric in (i, j) (§IV.28) Chapter4/VectorFields · scSecondDeriv_symm. They cancel in the alternating sum Chapter4/ScFormsNatural2021 · sum_update_symm_eq_zero. The derivatives involved are computed by the chain rule for sc-derivatives Chapter4/ScFormsNatural2021 · scDeriv_comp and in the fibres of the Whitney sum Chapter4/ScFormsNatural2021 · scDeriv_whitney_fiber Chapter4/ScFormsNatural2021 · scDeriv_whitneyMap_whitneyMk. On retracts, d commutes with $\oplus_k Tf$ Chapter4/ScFormsNatural2021 · retractD_comp_retractWhitneyMap, and it is local: it does not change when the retract is shrunk to an open part Chapter4/ScFormsNatural2021 · retractD_eq_of_subset.

IV.37 d on an M-polyfold, $d \circ d = 0$, and Definition 4.4.6 (preprint 4.27)

The global d. In a chart φ, d of the representative is Chapter4/ScFormsGlobal2021 · MPolyfold.chartD. By naturality, the representatives in two charts are related by $\oplus_{k+1} T$ of the transition map Chapter4/ScFormsGlobal2021 · MPolyfold.chartD_transition. So

$$d\omega(x)(\alpha_0, \dots, \alpha_k) = d(\omega_\varphi)(\varphi(x))(T\varphi^1 \alpha_0, \dots)$$

for one chosen chart at x defines a form on X^1 Chapter4/ScFormsGlobal2021 · MPolyfold.extD Chapter4/ScFormsGlobal2021 · MPolyfold.extD_apply. Its representative in every chart is d of the representative Chapter4/ScFormsGlobal2021 · MPolyfold.chartForm_extD. It maps $\Omega^k(X)$ to $\Omega^{k+1}(X^1)$ Chapter4/ScFormsGlobal2021 · MPolyfold.extD_mem_scForms, and it is linear Chapter4/ScFormsGlobal2021 · MPolyfold.extD_add Chapter4/ScFormsGlobal2021 · MPolyfold.extD_smul Chapter4/ScFormsGlobal2021 · MPolyfold.extD□ . The 2021 text writes $d : \Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$. The preprint's $\Omega^k(X^{i+1}) \to \Omega^{k+1}(X^i)$ presumably means the same map, with the indices exchanged (Findings, Chapter 4, item 120).

$d \circ d = 0$. In a local model, $dd\Omega(y)(h_0, \dots, h_{k+1})$ is a double alternating sum of second derivatives of Ω in base directions Chapter4/ScFormsSquare2021 · scDeriv_ambientD_whitneyMk. These are symmetric (§IV.28), so the sum cancels in pairs Chapter4/ScFormsSquare2021 · sum_removeNth_symm_eq_zero Chapter4/ScFormsSquare2021 · ambientD_ambientD. The same holds on retracts Chapter4/ScFormsSquare2021 · retractD_shift_retractD and on an M-polyfold Chapter4/ScFormsSquare2021 · MPolyfold.extD_extD. The text asserts $d^2 = 0$ (Findings, Chapter 4, item 121).

Passing to Ω^*_∞. d commutes with the restriction to X^{i+1} Chapter4/ScFormsLower2021 · MPolyfold.extD_formLower. It gives maps $\Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$ Chapter4/ScFormsLower2021 · BundledMPolyfold.formsD that commute with the system (4.4) Chapter4/ScFormsLower2021 · BundledMPolyfold.formsD_formsStep Chapter4/ScFormsLower2021 · BundledMPolyfold.formsD_formsMap. Hence d is defined on the limit Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyD Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyD_of and on the graded space Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyGradedD, with $d \circ d = 0$ there as well Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyD_comp_formsInftyD Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyGradedD_comp_self.

Definition 4.4.6 (preprint 4.27). The image of d lies in its kernel Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyDRange Chapter4/ScFormsLower2021 · BundledMPolyfold.formsInftyDRange_le_ker. $H^k_{sc}(X)$ is the quotient Chapter4/ScFormsLower2021 · BundledMPolyfold.scDeRham, a real vector space, with classes Chapter4/ScFormsLower2021 · BundledMPolyfold.scDeRhamMk. Two closed forms have the same class iff they differ by an exact form Chapter4/ScFormsLower2021 · BundledMPolyfold.scDeRhamMk_eq_iff. The relative version is §IV.40.

IV.38 The formula of p. 197 for $d\omega$, and its sign

The text gives $d\omega(A_0^{(1)}, \dots, A_k^{(1)})$ by "the familiar formula". It is the sum over j of $(-1)^j D(\omega(A_0, \dots, \hat{A}_j, \dots, A_k)) \cdot A_j^{(1)}$, with the summation index printed as i, plus the sum over $\ell < j$ of $(-1)^{\ell+k}\omega([A_\ell, A_j], \dots)$. Here $[A, B] = DA \cdot B - DB \cdot A$ (p. 200). The formula is checked in the classical setting, a real normed space, to which it reduces in a chart.

The text's bracket `Chapter4/ScExteriorDerivative2021 · bookBracket` is the negative of Mathlib's Lie bracket `Chapter4/ScExteriorDerivative2021 · bookBracket_eq_neg_lieBracket`. Write the formula with a sign function on the pairs $\ell < j$ `Chapter4/ScExteriorDerivative2021 · invariantFormula`. With the sign $(-1)^{\ell+j+1}$ `Chapter4/ScExteriorDerivative2021 · correctSign`, it is Mathlib's exterior derivative for vector fields differentiable at the point `Chapter4/ScExteriorDerivative2021 · extDeriv_eq_invariantFormula`.

With the printed sign the formula does not hold as we read it `Chapter4/ScExteriorDerivative2021 · invariantFormula_printedSign_counterexample`. Take $k = 1$, $\omega = dy$ on $\mathbb{R}^2$, $A_0 = (1, 0)$ and $A_1(p) = (0, p_1)$: - with the printed sign $(-1)^{\ell+k}$ `Chapter4/ScExteriorDerivative2021 · printedSign`, the right side at 0 is 2, while $d\omega = 0$; - the field $A_1' = 0$ agrees with A_1 at 0, and with it the right side is 0, so the printed formula is not tensorial; - the classical sign $(-1)^{\ell+j}$ `Chapter4/ScExteriorDerivative2021 · item158Sign`, proposed earlier in Findings, gives 2 on the same example, because the text's bracket is minus the classical one.

The printed sign may go with another bracket convention. The index i of the first sum is read as j (Findings, Chapter 4, items 158, 167). This is the one narrower row of the chapter (V4.83). The formal d of §§IV.36–IV.37 is defined by the local formula, so the sign does not enter it.

IV.39 The pull-back f^*, the wedge product, and the graded differential algebra

The pull-back (p. 198; the 2017 text and the preprint state only that f induces f^*). Let $f : X \to Y$ be sc-smooth. On tangent fibres, Tf is a continuous linear map `Chapter4/ScFormsPullback2021 · MPolyfold.tangentFiberMap`. Define $f^*\omega(x)(\alpha_\bullet) = \omega(f(x))(Tf\alpha_\bullet)$ `Chapter4/ScFormsPullback2021 · MPolyfold.formPullback`. In charts it is $\omega_\psi \circ \oplus_k T(\psi f \varphi^{-1})$ `Chapter4/ScFormsPullback2021 · MPolyfold.chartForm_formPullback`, so it maps $\Omega^k(Y)$ to $\Omega^k(X)$ `Chapter4/ScFormsPullback2021 · MPolyfold.formPullback_mem_scForms`. It satisfies $df^*\omega = (f^1)^* d\omega$ `Chapter4/ScFormsPullback2021 · MPolyfold.extD_formPullback`, by naturality of the local d (§IV.36). The text leaves this identity to the reader; it holds as we read it. Pull-back is functorial `Chapter4/ScFormsPullback2021 · MPolyfold.formPullback_comp` `Chapter4/ScFormsPullback2021 · MPolyfold.formPullback_id` and commutes with (4.4) `Chapter4/ScFormsPullback2021 · MPolyfold.formLower_formPullback`. On the lifts, the maps $(f^i)^*$ `Chapter4/ScFormsPullback2021 · BundledMPolyfold.formsPullback` pass to the limit `Chapter4/ScFormsPullback2021 · BundledMPolyfold.formsInftyPullback`. There $df^* = f^* d$ `Chapter4/ScFormsPullback2021 · BundledMPolyfold.formsInftyD_comp_formsInftyPullback`, pull-back is functorial `Chapter4/ScFormsPullback2021 · BundledMPolyfold.formsInftyPullback_comp`, and it induces a map on H^k_{sc} `Chapter4/ScFormsPullback2021 · BundledMPolyfold.scDeRhamPullback`.

The wedge product. The text uses $\wedge$ through "$(\Omega^*_\infty(X), d)$ is a graded differential algebra" but never

defines it. We take the classical normalization:

$$(a \wedge b)(v) = (k!\,l!)^{-1} \sum_{\sigma \in S_{k+l}} \operatorname{sign}\sigma \; a(v_{\sigma(1)}, \dots, v_{\sigma(k)})\, b(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)})$$

`Chapter4/ScFormsWedge2021 · wedgeFun` `Chapter4/ScFormsWedge2021 · altWedge` `Chapter4/ScFormsWedge2021 · coe_altWedge`. Write $\delta A(v) = \sum_j (-1)^j A(v_j)(v_0, \dots, \widehat{v_j}, \dots)$ for the alternating sum that defines d. Two identities hold:

$$\delta(A \wedge b) = \delta A \wedge b, \qquad \delta(a \wedge B) = (-1)^k a \wedge \delta B$$

`Chapter4/ScFormsWedge2021 · deltaFun_wedgeFun_left` `Chapter4/ScFormsWedge2021 · deltaFun_wedgeFun_right`. They are identities between sums over permutations. They use the decomposition $S_{n+1} \cong \{0, \dots, n\} \times S_n$ by insertion, and they need no alternation. With the product rule, they give the local Leibniz rule `Chapter4/ScFormsWedge2021 · ambientD_lwedge`.

On an M-polyfold, the pointwise wedge `Chapter4/ScFormsWedge2021 · MPolyfold.formWedge` maps $\Omega^k(X) \times \Omega^l(X)$ into $\Omega^{k+l}(X)$ `Chapter4/ScFormsWedge2021 · MPolyfold.formWedge_mem_scForms`. It satisfies

$$d(\omega \wedge \tau) = d\omega \wedge \iota^*\tau + (-1)^k \iota^*\omega \wedge d\tau$$

`Chapter4/ScFormsWedge2021 · MPolyfold.extD_formWedge` and $f^*(\omega \wedge \tau) = f^*\omega \wedge f^*\tau$ `Chapter4/ScFormsWedge2021 · MPolyfold.formPullback_formWedge`. It passes to $\Omega^*_\infty(X)$ `Chapter4/ScFormsWedge2021 · BundledMPolyfold.formsInftyWedge` `Chapter4/ScFormsWedge2021 · BundledMPolyfold.formsInftyD_formsInftyWedge`.

Associativity and graded commutativity are again identities between permutation sums `Chapter4/ScFormsWedge2021b · wedgeFun_assoc` `Chapter4/ScFormsWedge2021b · wedgeFun_comm`. The sign $(-1)^{kl}$ is that of a rotation of $k + l$ places by k. The unit is the constant 1 `Chapter4/ScFormsWedge2021b · MPolyfold.formOne_mem_scForms`. On $\Omega^*_\infty(X)$ `Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyWedge_assoc` `Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyWedge_comm`, they make $\bigoplus_k \Omega^k_\infty(X)$ a graded $\mathbb{R}$-algebra `Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyGAlgebra` that is graded commutative `Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyGraded_of_mul_of_comm`. d is a derivation of degree $+1$ `Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyGradedD_of_mul_of` with $d^2 = 0$ (§IV.37). This is the graded differential algebra of p. 198. f^* is a morphism of graded algebras `Chapter4/ScFormsWedge2021 · BundledMPolyfold.formsInftyPullback_formsInftyWedge` `Chapter4/ScFormsWedge2021b · BundledMPolyfold.formsInftyPullback_formsInftyOne`. The Leibniz sign $(-1)^k$ holds as we read it.

IV.40 Forms on ∂X, j^*, and the relative complex (pp. 198–199)

For a tame X the text says that the inclusion $\partial X \to X$ is sc-smooth on local faces, so differential forms on ∂X "make sense". It gives no definition. ∂X is not an M-polyfold (Findings, Chapter 4, item 122).

The reading. Forms on ∂X are read through the tame boundary $B : bX \to X$ of Theorem 2.7.13, an sc-smooth map from an M-polyfold that is locally the inclusion of a local face. Let $f : X \to Y$ be sc-smooth. A family τ on X *agrees over* f `Chapter4/BoundaryForms2021 · MPolyfold.IsAgreeingOver` if $\tau_p(v) = \tau_{p'}(v')$ whenever $f(p) = f(p')$ and $Tf\, v_j = Tf\, v'_j$ for all j. The base-point condition matters in degree 0. A form on ∂X^i is an sc-form on bX^i that agrees over B^i `Chapter4/BoundaryForms2021 · MPolyfold.IsFaceCompatible`. So on two local faces through a point, the form coincides on vectors with the same image in TX (reading R-a; Findings, Chapter 4, item 173).

The definition Chapter4/BoundaryForms2021 · MPolyfold.bForms also lists the agreement of $d\tau$, which makes $\Omega^*(\partial X)$ the largest d-stable family of agreeing forms Chapter4/BoundaryForms2021 · MPolyfold.le_bForms. With it, d maps $\Omega^k(\partial X^i)$ to $\Omega^{k+1}(\partial X^{i+1})$ Chapter4/BoundaryForms2021 · MPolyfold.extD_mem_bForms.

The agreement of $d\tau$ follows from that of τ Chapter4/A3BoundaryForms2021b · MPolyfold.isFaceCompatible_extD. So the extra condition is automatic, and $\Omega^k(\partial X^i)$ is the set of agreeing forms Chapter4/A3BoundaryForms2021b · MPolyfold.bForms_eq_agreeForms Chapter4/A3BoundaryForms2021b · MPolyfold.mem_bForms_iff. The general statement is for any sc-smooth f with two properties Chapter4/A3BoundaryForms2021 · BundledMPolyfold.isAgreeingOver_extD: - Tf is injective on each tangent space Chapter4/A3BoundaryForms2021 · BundledMPolyfold.IsTangentInjective; - f has a *corner cover* Chapter4/A3BoundaryForms2021 · BundledMPolyfold.HasCornerCover. For $p \neq p'$ with $f(p) = f(p')$, there are sc-smooth $g, g' : Z \to X$ with $f \circ g = f \circ g'$ and $g(z) = p$, $g'(z) = p'$ such that every pair of tangent vectors at p, p' with the same image is $(Tg\, v, Tg'\, v)$.

Given these, $g^*\tau = g'^*\tau$, hence $g^*d\tau = g'^*d\tau$ by $df^* = f^*d$ Chapter4/A3BoundaryForms2021 · MPolyfold.formPullback_extD_eq_of_isAgreeingOver. The two properties are stated on level 0 and transported to every X^i.

For B, TB is the face inclusion in tangent charts Chapter4/A3BoundaryForms2021b · MPolyfold.tangentChartMap_bdryMap, so it is injective Chapter4/A3BoundaryForms2021b · MPolyfold.isTangentInjective_bdryMap. The corner cover is built as follows Chapter4/A3BoundaryForms2021b · MPolyfold.hasCornerCover_bdryMap: 1. At two faces through x, take a tame chart with retraction r and coordinates λ_i, λ_j vanishing on them Chapter4/A3BoundaryForms2021b · MPolyfold.CornerChart. 2. The map $r \circ P_i \circ P_j$, where P_i kills λ_i, sends a relatively open piece of the partial quadrant into both faces. 3. Its two lifts to bX have the same composite with B. 4. Every vector of $T_xF \cap T_xF'$ is in the image of its tangent map Chapter4/A3BoundaryForms2021b · MPolyfold.CornerChart.exists_tangent, since $D(rP_iP_j)(\varphi x)h = Dr(\varphi x)h = h$ for h with $\lambda_i h = \lambda_j h = 0$.

No corner strata are built as M-polyfolds, and the point x need not be smooth. The text writes $d\tau$ without comment; this holds as we read it.

j^*. The pull-back along B, by §IV.39, maps $\Omega^k(X)$ into $\Omega^k(\partial X)$ Chapter4/BoundaryForms2021 · MPolyfold.formsPullback_mem_bForms. It passes to the limits Chapter4/BoundaryForms2021 · MPolyfold.bFormsInfty Chapter4/BoundaryForms2021 · MPolyfold.jStar, with $dj^* = j^*d$ Chapter4/BoundaryForms2021 · MPolyfold.jStar_comp_D. The text states this identity without proof (Findings, Chapter 4, item 121).

The relative complex. In degree 0 it is $\Omega^0_\infty(X)$, and in degree $k+1$ it is $\Omega^{k+1}_\infty(X) \times \Omega^k_\infty(\partial X)$ Chapter4/BoundaryForms2021 · MPolyfold.RelForms. The differential is $d(\omega, \tau) = (d\omega, j^*\omega - d\tau)$ Chapter4/BoundaryForms2021 · MPolyfold.relD. It squares to 0 Chapter4/BoundaryForms2021 · MPolyfold.relD_comp_relD, because $d^2 = 0$ and $dj^* = j^*d$. $H^k_{dR}(X, \partial X)$ is its cohomology Chapter4/BoundaryForms2021 · MPolyfold.relDeRham Chapter4/BoundaryForms2021 · MPolyfold.relDeRhamMk_eq_iff. If $\partial X = \emptyset$, then bX is empty and $H^k_{dR}(X, \partial X) \cong H^k_{sc}(X)$ Chapter4/BoundaryForms2021 · MPolyfold.relDeRhamEquivOfNoBoundary. The text calls the relative complex a differential algebra. The wedge product on it is §IV.46.

IV.41 Integration and Stokes' theorem on quadrants

Theorem 4.4.7 (preprint 4.28) integrates over finite-dimensional sub-M-polyfolds with corners. The integration is built from the bottom: first in $\mathbb{R}^{n+1}$, then on Mathlib manifolds with corners (§IV.42), then for sc-forms (§IV.43).

Top forms in $\mathbb{R}^{n+1}$. For a top-degree form ω, define $\int_s \omega = \int_s \omega(x)(e_0, \dots, e_n)\, dx$ on the standard frame `Chapter4/EuclideanForms2021 · stdFrame` `Chapter4/EuclideanForms2021 · topFormIntegral`. Pull-back `Chapter4/EuclideanForms2021 · formPullback` multiplies the frame value by the determinant `Chapter4/EuclideanForms2021 · alternatingMap_comp_eq_det_mul`. So the change of variables formula holds with the sign of the determinant `Chapter4/EuclideanForms2021 · topFormIntegral_image`.

Faces and their orientation. The quadrant is $Q_S = \{x_i \geq 0, i \in S\}$. Its boundary differs from that of the open quadrant by a null set `Chapter4/EuclideanForms2021 · openQuadrant_ae_eq_quadrant`, and the corners are null `Chapter4/EuclideanForms2021 · image_faceEmbedding_openQuadrant`. The face $\{x_i = 0\}$ is parametrized by $\iota_i : \mathbb{R}^n \to \mathbb{R}^{n+1}$, inserting a zero at place i `Chapter4/EuclideanForms2021 · faceEmbedding`. It carries the pulled-back form `Chapter4/EuclideanForms2021 · facePullback` and the sign $(-1)^{i+1}$ `Chapter4/EuclideanForms2021 · faceSign`. This is the sign of the frame $(-e_i, \text{face frame})$ `Chapter4/EuclideanForms2021 · det_cons_neg_stdFrame_faceFrame`, the induced orientation with the outward vector $-e_i$ first (2021 Definition 9.3.8).

Stokes on a quadrant `Chapter4/EuclideanStokes2021 · stokes_quadrant`. For a C^1 form ω of degree n vanishing off a compact set,

$$\int_{Q_S} d\omega = \sum_{i \in S} (-1)^{i+1} \int_{\mathbb{R}^n_{\geq 0}} \iota_i^* \omega.$$

The engine is Mathlib's divergence theorem on a box `Chapter4/EuclideanStokes2021 · integral_divergence_piQuadrant`. Take the box $[a, b]$ with $a_i = 0$ for $i \in S$, $a_i = -R$ otherwise, and $b_i = R$. The faces $x_i = -R$ and $x_i = R$ contribute nothing, because ω vanishes there. A form defined only on an open U is extended by 0 `Chapter4/EuclideanStokes2021 · stokes_quadrant_inter_open`.

IV.42 Forms, integrals and Stokes' theorem on Mathlib manifolds with corners

Forms and d. A k-form on a manifold M modelled on $[0, \infty)^{n+1}$ is a family of alternating forms on the tangent spaces `Chapter4/ManifoldForms2021 · ManifoldForm`. It has chart representatives `Chapter4/ManifoldForms2021 · chartForm`, which are pull-backs of one another `Chapter4/ManifoldForms2021 · chartForm_eq_formPullback`, and a smoothness class `Chapter4/ManifoldForms2021 · ContMDiffForm`. d is the derivative of the chart representative within the range of the model `Chapter4/ManifoldForms2021 · ManifoldForm.extDeriv`. It commutes with pull-back by chart changes `Chapter4/ManifoldForms2021 · extDerivWithin_formPullback`, so it has the right representative in every chart `Chapter4/ManifoldForms2021 · chartForm_extDeriv`.

Integration. The interval $[0, 1]$, with the quadrant model, has no oriented atlas: a chart at 1 reverses the orientation. So the charts are signed `Chapter4/ManifoldIntegral2021 · signedOr` `Chapter4/ManifoldIntegral2021 · orSign`. The integral in one chart carries the sign of the orientation there `Chapter4/ManifoldIntegral2021 · chartIntegral` `Chapter4/ManifoldIntegral2021 · chartIntegral_eq_integral_sign`. *Integration data* `Chapter4/ManifoldIntegral2021 · IntegrationData` consist of finitely many charts, regions on which the orientation has a constant sign, and C^∞ functions ρ_j summing to 1 on the compact support. They exist `Chapter4/ManifoldIntegral2021 · exists_integrationData`, by a subordinate smooth partition of unity. Any two give the same sum `Chapter4/ManifoldIntegral2021 · IntegrationData.integral_eq`, by the change of variables in the overlaps. This defines $\int_M \omega$ `Chapter4/ManifoldIntegral2021 · ManifoldForm.integral` `Chapter4/ManifoldIntegral2021 · ManifoldForm.integral_eq_integrationData`. The integral is additive, and it changes sign with the orientation `Chapter4/ManifoldIntegral2021 · ManifoldForm.integral_neg_orientation`.

The boundary integral. In one chart, the boundary integral is $\sum_i (-1)^{i+1} \int \iota_i^* \omega_e$ Chapter4/ManifoldStokes2021 · chartBoundaryIntegral. Through integration data this defines $\int_{\partial M} \omega$ Chapter4/ManifoldStokes2021 · ManifoldForm.boundaryIntegral. It does not depend on the data Chapter4/ManifoldStokes2021 · IntegrationData.boundaryIntegral_eq. The proof of independence goes through Stokes: both sums equal $\int_M d\omega$ for an extension. The orientation of the faces has to be compatible across charts. The chart change maps a face to a face Chapter4/ManifoldBoundary2021 · exists_face_chartChange, inward vectors to inward vectors, with $(De_i)_j > 0$. The sign identity faceSign(i) det $A = (Ae_i)_j$ faceSign(j) det $\bar{A}$, where $\bar{A}$ is the face restriction Chapter4/ManifoldStokes2021 · faceSign_mul_det_eq, gives the compatibility Chapter4/ManifoldBoundary2021 · orientation_face_chartChange.

Stokes' theorem Chapter4/ManifoldStokes2021 · stokes_of_compactSpace. For M compact with a continuous orientation and ω of class C^1, $\int_M d\omega = \int_{\partial M} \omega$. The proof has two steps: 1. Stokes in one chart is §IV.41 Chapter4/ManifoldStokes2021 · chartIntegral_extDeriv_eq. 2. With integration data, $d\omega = \sum_j d(\rho_j \omega)$, so the general case is the sum of the chart cases Chapter4/ManifoldStokes2021 · IntegrationData.stokes Chapter4/ManifoldStokes2021 · integral_extDeriv_eq_boundaryIntegral.

Forms on the faces of M. These are face data in the charts Chapter4/ManifoldFaceForms2021 · IsChartFaceForm Chapter4/ManifoldFaceForms2021 · ChartFaceForm. On each chart face there is a smooth form, compatible under chart changes at points of degeneracy index 1, and agreeing at corners on vectors tangent to both faces. Every smooth form restricts to such data Chapter4/ManifoldFaceForms2021 · ChartFaceForm.ofForm, so they are satisfiable. Compactly supported face data extend to a form on M Chapter4/ManifoldFaceForms2021 · exists_extension, by an inclusion–exclusion sum over the faces of box charts. This defines the integral of face data as the boundary integral of an extension Chapter4/ManifoldFaceForms2021 · ChartFaceForm.integral. It also gives $\int_{\partial M} d\tau = 0$ Chapter4/ManifoldFaceForms2021 · ChartFaceForm.integral_d_eq_zero, by Stokes and $d^2 = 0$ Chapter4/ManifoldFaceForms2021 · ManifoldForm.extDeriv_extDeriv.

IV.43 Theorem 4.4.7 (preprint 4.28): Stokes' theorem for sc-forms, with the sum over the faces

Statement Chapter9/StokesFaceSum · MPolyfold.theorem_4_4_7_hyp. Let X be an M-polyfold and $S \subseteq X$ a compact sub-M-polyfold. Assume: - S has a corner structure of dimension $n+1$ (Definition 4.1.1 (preprint 4.1)); - S is face-structured; - S has a continuous tangent orientation o.

Then for $\omega \in \Omega^n_\infty(X)$,

$$\int_S (d\omega)|_S = \sum_F \int_F (\omega|_S)|_F,$$

the sum over the finitely many faces F of S. Each face is a manifold modelled on $[0,\infty)^n$ with the induced orientation, outward vector first.

The text's "boundary ∂N, a union of smooth faces F" is read as: S is face-structured (Definition 2.5.10), so every point lies on exactly $d_S(x)$ faces. This hypothesis is added; it is not derived from the text. It is satisfiable. It holds when S has no boundary Chapter9/StokesFaceSum · MPolyfold.isFaceStructured_of_forall_degIndex_eq_zero, and then the theorem says $\int_S d\omega = 0$. It also holds for the quadrants Chapter11/EpEquivalenceGeometryFaces · isFaceStructured_quadrant. The corner structure and the continuity of o are the meaning of "a finite-dimensional oriented manifold with boundary with corners" in the text. The 2021 text writes $\int_{N'}$, the preprint and 2017 texts $\int_M$ (Findings, Chapter 4, items 123, 171, 172).

Step 1: the restriction of sc-forms to S. The corner structure makes S a Mathlib manifold, and $S \subseteq X_\infty$ Chapter9/MathlibFormsBridge2021 · MPolyfold.subset_smoothSet_of_hasCornerStructureOfDim. An sc-form ω on X^i restricts to a classical form on S Chapter9/MathlibFormsBridge2021 · MPolyfold.scFormRestrict, through the pull-back of §IV.39 along the inclusion lifted to $S \to X^i$. This works because $S = S^1 = S^i$ as sets. In a chart of X the restricted form is ω_φ evaluated on the model derivatives of the corner chart Chapter9/MathlibFormsBridge2021 · MPolyfold.scFormRestrict_apply_modelDeriv. It commutes with (4.4) Chapter9/MathlibFormsBridge2021 · MPolyfold.scFormRestrict_formLower and passes to Ω_∞ Chapter9/MathlibFormsBridge2021 · MPolyfold.formsInftyRestrict. It is C^∞ Chapter9/MathlibFormsBridge2021 · MPolyfold.contMDiffForm_formsInftyRestrict, and $d(\omega|_S) = (d\omega)|_S$ Chapter9/MathlibFormsBridge2021 · MPolyfold.extDeriv_formsInftyRestrict. With §IV.42 this gives $\int_S (d\omega)|_S = \int_{\partial S} \omega|_S$ Chapter9/MathlibFormsBridge2021 · MPolyfold.stokes_formsInftyRestrict. Such S exist for every tame X with a submanifold in good position Chapter9/MathlibFormsBridge2021 · exists_hasCornerStructureOfDim_of_tame.

Step 2: the faces of S. A compact tame M-polyfold has finitely many faces Chapter4/StokesFaces · MPolyfold.finite_setOf_isFace. In a face-structured one, near each of its points and in any chart modelled on a tame retract, a face is exactly a coordinate face Chapter4/StokesFaces · MPolyfold.exists_face_eq_coordHyper. In corner coordinates of S this gives, at $x \in F$, a coordinate i and an open set on which $F = \{y_i = 0\}$ Chapter4/StokesFaceCharts · MPolyfold.CornerCoords.exists_faceData. Removing the coordinate i gives a corner chart of F. So *every face has a corner structure of dimension n* Chapter4/StokesFaceCharts · MPolyfold.hasCornerStructureOfDim_face, as the sub-M-polyfold of Theorem 2.5.17.

Step 3: the induced orientation. Take a chart of a face and the chart of S it comes from. The chart change of two such pairs has derivative the face restriction of the chart change of S Chapter4/StokesFaceOrientation · FaceChartPair.chartChangeDeriv_face. This derivative maps the face to the face and inward vectors to inward vectors, at every point, corners included Chapter4/StokesFaceOrientation · FaceChartPair.inward. The orientation induced on the face reads, in every chart pair, as the sign of o times $(-1)^{i+1}$ Chapter4/StokesFaceOrientation · IsInducedOrientation. It exists, is continuous and is unique Chapter4/StokesFaceOrientation · inducedOrientation Chapter4/StokesFaceOrientation · isContinuous_inducedOrientation Chapter4/StokesFaceOrientation · IsInducedOrientation.unique. On S it is Chapter9/StokesFaceSum · MPolyfold.faceOrientation.

Step 4: the face sum is the boundary integral Chapter4/StokesFaceIntegral · sum_integral_restrictForm_eq_boundaryIntegral. The form $\omega|_F$ is the pull-back along $F \hookrightarrow S$ Chapter4/StokesFaceIntegral · restrictForm. Its representative in a face chart is ι_i^* of that of ω Chapter4/StokesFaceIntegral · FaceChartPair.chartForm_restrictForm. Locally, on a *good region*, the face sum equals the chart boundary integral Chapter4/StokesFaceIntegral · GoodRegion.sum_integral_eq. A good region is a signed region inside the face chart domains and the neighbourhood of Proposition 2.7.5 (3) Chapter9/StokesFaceSum · MPolyfold.goodRegion. Inside it, distinct faces through a point get distinct coordinates, because a point of degeneracy index 1 lies on exactly one face Chapter9/StokesFaceSum · MPolyfold.face_eq_of_degIndex_eq_one. Integration data subordinate to good regions Chapter4/StokesFaceIntegral · ManifoldForm.exists_integrationData_of_regions sum the local identities. With Step 1 the theorem follows.

IV.44 Parametric integrals, and Lemma 4.5.2: the Poincaré lemma on a convex piece of a partial quadrant

Parametric integrals Chapter4/Poincare2021 · isSck_intervalIntegral Chapter4/Poincare2021 · isScSmooth_intervalIntegral. If g is sc^k (sc-smooth) on $[0,1] \times U$, with $[0,1]$ carrying the constant sc-structure,

then $x \mapsto \int_0^1 g(t,x)\,dt$ is sc^k (sc-smooth) on U. Its derivative is the integral of the partial derivative `Chapter4/Poincare2021 · scDeriv_intervalIntegral`. The proof is an induction on k through the tangent map, $T\int g = \int Tg$. The derivative is differentiation under the integral sign, uniform by the compactness of $[0,1]$, within the relatively open U. The level-wise C^{l+1} criteria lose one level too many and are not used.

The homotopy operator (2021 p. 203) `Chapter4/Poincare2021b · homotopyForm`. For a closed $(k+1)$-form ω on a convex D and a centre c, put

$$\tau(y)(v_1,\dots,v_k) = \int_0^1 t^k\,\omega(c+t(y-c))(y-c,v_1,\dots,v_k)\,dt.$$

τ is sc-smooth if c is smooth `Chapter4/Poincare2021b · isScSmooth_homotopyForm`: the integrand is sc-smooth on $[0,1]\times D$ because $y \mapsto c+t(y-c)$ is, on every level. Then $d\tau = \omega$ on D^1 `Chapter4/Poincare2021b · ambientD_homotopyForm`. This is the classical computation: 1. Write $d\tau(y)(v_0,\dots,v_k)$ with the alternating sum δ of §IV.39. 2. Use $d\omega = 0$ to trade the alternating sum of base derivatives for the radial derivative of ω. 3. The integrand becomes $\frac{d}{dt}(t^{k+1}\omega(c+t(y-c))(v_0,\dots,v_k))$, whose integral is $\omega(y)(v_\bullet)$.

The vectors are taken on level 1 at points of D^1, so no density argument is needed.

Lemma 4.5.2 (2021, p. 202). In local form `Chapter4/Poincare2021b · lemma_4_5_2`: let D be relatively open in a partial quadrant and convex, and $c \in D$ smooth. Every closed sc-form of degree $k+1$ on D has a primitive of degree k on D^1. On the M-polyfold D, with forms as in §IV.35 `Chapter4/Poincare2021b · hasPoincare_relOpen`, the bridge is that on a relatively open D the local d of §IV.36 does not depend on the retraction `Chapter4/Poincare2021b · retractD_eq_ambientD`. The printed statement, with $0 \in U$ and centre 0, is the case $c = 0$ `Chapter4/Poincare2021b · hasPoincare_relOpen_of_zero_mem`. The hypotheses are satisfiable: take $D = C = E$ and $c = 0$.

Two readings, neither an added hypothesis. The text's degrees $k \geq 1$, $k-1$ are $k+1$, k here. "$d\tau = \omega$ on U^1" is $d\tau = \iota^*\omega$ with $\iota : U^1 \to U$. Any smooth centre is allowed. Lemma 4.4.8 needs this generality (§IV.45); the text's remark after Lemma 4.5.2, with centre $(0,w_0)$, does not cover the case needed (Findings, Chapter 4, item 177).

IV.45 Lemma 4.4.8 (2021, p. 199): the Poincaré lemma on an M-polyfold

Forms on open subsets. An open $U \subseteq X$ is an M-polyfold (§IV.34). Restriction maps $\Omega^k(X)$ to $\Omega^k(U)$ `Chapter4/OpenForms · MPolyfold.formRestrict` and commutes with d `Chapter4/OpenForms · MPolyfold.extD_formRestrict`. On the limits, for nonempty U as a bundled M-polyfold `Chapter4/OpenForms · BundledMPolyfold.openSub`, there are restrictions $\Omega^k_\infty(X) \to \Omega^k_\infty(U)$ and $\Omega^k_\infty(U) \to \Omega^k_\infty(V)$ for $V \subseteq U$ `Chapter4/OpenForms · BundledMPolyfold.formsInftyResOpen` `Chapter4/OpenForms · BundledMPolyfold.formsInftyResOpenOpen`. The text's "$[\omega] = [\tau]$ on V" of (4.7) `Chapter4/OpenForms · BundledMPolyfold.FormsInftyEqOn` means: for representatives on one level i, the restrictions agree on V^ℓ for some $\ell \geq i$ `Chapter4/OpenForms · BundledMPolyfold.formsInftyEqOn_of_iff`.

Statement `Chapter4/Poincare2021b · lemma_4_4_8`. Every $x \in X$ has open sets $V(x) \subseteq U(x)$ with the following property. For every k, every closed $a \in \Omega^{k+1}_\infty(U(x))$ has $b \in \Omega^k_\infty(V(x))$ with $db = a|_{V(x)}$. The text's $k \geq 1$ is $k+1$ here.

Proof `Chapter4/Poincare448 · lemma_4_4_8_of_local`. Lemma 4.5.2 with a smooth centre `Chapter4/`

Poincare448 · LocalPoincare is §IV.44 Chapter4/Poincare2021b · localPoincare. 1. **The chart and the radii** Chapter4/Poincare448 · exists_poincareChart. At x, take a chart φ with model retraction r, and $\varepsilon, \delta > 0$ such that $B_\varepsilon(\varphi x) \cap C$ lies in the domain of r and $r(B_\delta(\varphi x) \cap C) \subseteq B_\varepsilon(\varphi x)$. Put $U(x) = \varphi^{-1}(B_\varepsilon(\varphi x))$ and $V(x) = \varphi^{-1}(B_\delta(\varphi x))$. Since r is the identity on the model, this is the text's $r(B_\delta \cap r^{-1}(B_\delta))$. 2. **The two maps.** $D = B_\delta(\varphi x) \cap C$ is convex and relatively open, with a smooth point Chapter4/Poincare448 · exists_smooth_mem_relOpen. The map $\rho = \varphi^{-1} \circ r : D \to U(x)$ is sc-smooth Chapter4/Poincare448 · isScSmoothMap_retractChart, and $j = \varphi : V(x) \to D$ is sc-smooth. Their composite $\rho \circ j$ is the inclusion $V(x) \to U(x)$. 3. **Every level.** A representative of a lives on some $U(x)^m$, so the lemma is needed on the m-th lift of D. That lift is the M-polyfold of $D^m \subseteq E^m$ Chapter4/Poincare448Shift · LocalModel.liftEquiv, the text's "working with $C^i \subset E^i$". The Poincaré property is invariant under sc-diffeomorphisms Chapter4/Poincare448Shift · MPolyfold.HasPoincare, so it holds on every lift Chapter4/Poincare448 · hasPoincare_lift. 4. **The primitive** Chapter4/Poincare448 · exists_primitive_of_factor. $\rho^* a$ is closed on D, so it has a primitive τ. Then $b = j^*\tau$ satisfies $db = j^*\rho^* a = (\rho \circ j)^* a = a|_{V(x)}$, by §IV.39 on the direct limits.

Here φx is only a level-0 point of a general partial quadrant, so $0 \notin D$ in general. This is why Lemma 4.5.2 is needed with a smooth centre other than 0 (§IV.44).

IV.46 The wedge product on the forms on ∂X and on the relative complex (gap G.80)

The text (p. 199) calls $\Omega^*_\infty(X, \partial X)$ a differential algebra and gives no product. Two products are formalized.

On the forms on ∂X. A form on ∂X is an sc-form on bX that agrees over B (§IV.40). The wedge product of two forms that agree over an sc-smooth f agrees over f, because at a point it is computed from the values of the two forms on the same tangent vectors Chapter4/BoundaryWedge · MPolyfold.IsAgreeingOver.formWedge Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsWedge. It passes to the limits Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyWedge Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyWedge_of_of and, up to the casts of degrees Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyCast, it satisfies the Leibniz rule Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyD_agreeDFormsInftyWedge, is associative Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyWedge_assoc and has the constant 1 as a unit on both sides Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyOne Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyWedge_agreeDFormsInftyOne Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyCast_agreeDFormsInftyOne_wedge. The inclusion into $\Omega^*_\infty(bX)$ is multiplicative Chapter4/BoundaryWedge · BundledMPolyfold.agreeDFormsInftyIncl_agreeDFormsInftyWedge, and so is the pull-back Chapter4/BoundaryWedge · BundledMPolyfold.agreeInftyPullback_formsInftyWedge. For $f = B$ this is the wedge product on $\Omega^*_\infty(\partial X)$, and $j^*(\omega \wedge \omega') = j^*\omega \wedge j^*\omega'$ Chapter4/BoundaryWedge · MPolyfold.jStar_formsInftyWedge.

On the relative complex: the product of the mapping cone of j^*. For (ω, τ) of degree k and (ω', τ') of degree l,

$$(\omega, \tau)(\omega', \tau') = (\omega \wedge \omega', (-1)^k\, j^*\omega \wedge \tau')$$

Chapter4/RelativeWedge · MPolyfold.relWedge Chapter4/RelativeWedge · MPolyfold.relFst_relWedge Chapter4/RelativeWedge · MPolyfold.relSnd_relWedge_succ Chapter4/RelativeWedge · MPolyfold.relSnd_relWedge_zero. This choice is ours: the text gives no product, and this is the usual product on the mapping cone of a multiplicative map of complexes. - *d is a derivation of degree $+1$:* $d(xy) = (dx)y + (-1)^k x\, dy$ Chapter4/

RelativeWedge · MPolyfold.relD_relWedge. With $d(\omega,\tau) = (d\omega, j^*\omega - d\tau)$ (§IV.40), both sides have first component $d\omega\wedge\omega' + (-1)^k\omega\wedge d\omega'$ and second component $j^*\omega\wedge j^*\omega' - (-1)^k j^*d\omega\wedge\tau' - j^*\omega\wedge d\tau'$, by the Leibniz rules on X and on ∂X and the multiplicativity of j^*. - *Associativity* Chapter4/RelativeWedge · MPolyfold.relWedge_assoc, from the associativity on X and on ∂X and $j^*(\omega\wedge\omega') = j^*\omega\wedge j^*\omega'$. - *The unit.* $(1,0)$ is a unit on the left Chapter4/RelativeWedge · MPolyfold.relOne Chapter4/RelativeWedge · MPolyfold.relCast_relWedge_relOne. On the right, $(\omega,\tau)(1,0) = (\omega,0)$ Chapter4/RelativeWedge · MPolyfold.relWedge_relOne, so it is not a unit there; relative cohomology has no unit in general either. The product is not graded commutative on forms. - *The graded ring.* With Mathlib's graded multiplication Chapter4/RelativeWedge · MPolyfold.relGMul Chapter4/RelativeWedge · MPolyfold.relGNonUnitalNonAssocSemiring Chapter4/RelativeWedge · MPolyfold.relGraded_of_mul_of and associativity Chapter4/RelativeWedge · MPolyfold.relGraded_mul_assoc, $\bigoplus_k \Omega^k_\infty(X,\partial X)$ is an associative ring without unit Chapter4/RelativeWedge · MPolyfold.relGradedNonUnitalRing. The differential on it Chapter4/RelativeWedge · MPolyfold.relGradedD squares to 0 Chapter4/RelativeWedge · MPolyfold.relGradedD_comp_self and is a derivation Chapter4/RelativeWedge · MPolyfold.relGradedD_of_mul_of. The projection to $\Omega^*_\infty(X)$ is multiplicative.

So the relative complex is read as a differential graded algebra without a right unit (V4.91). The pullback of forms on ∂X along a general sc-smooth map is not formalized; the one use the text makes of it is (§IV.25). Gap G.80 is closed (Findings, Chapter 4, item 178).

Part V. The Fredholm package for M-polyfolds (HWZ Chapter 5, §§5.1–5.5 (preprint §§5.1–5.4))

Chapter 5 turns the implicit function theorems of Chapter 3 into the working tools of a Fredholm theory: a way to measure perturbations (auxiliary norms, §5.1), the compactness that keeps the solution sets of small perturbations under control (§5.2), a supply of perturbations built from sc-smooth bump functions (the start of §5.3), and the perturbation theorems themselves — transversality when there is no boundary (Theorem 5.3.5 (preprint 5.13)), general position (Theorem 5.3.10 (preprint 5.18)), good position relative to the boundary (Theorem 5.3.12 (preprint 5.19) and Theorem 5.25 (preprint; not in the 2021 edition)), and the Morse-type Theorem 5.3.15 (preprint 5.22). §5.2 (preprint Appendix 5.4.1) proves Theorem 5.2.2 (preprint 5.5); §5.5 (preprint Appendix 5.4.2) relates sc-smooth bump functions and partitions of unity to the geometry of Banach spaces. This Part retells §5.1 (§§V.1–V.3); §5.2, with the proof that the preprint gives in Appendix 5.4.1 (§§V.4–V.7), the longest retelling being Theorem 5.2.2 (preprint 5.5) (§§V.5–V.6); the start of §5.3 — bump functions, the calculus of sections, Lemma 5.3.3 (preprint 5.11), allowable sections (§§V.8–V.10); Theorem 5.3.5 (preprint 5.13) (§§V.11–V.15); general position, Lemma 5.26 (preprint; not in the 2021 edition) and Theorem 5.3.10 (preprint 5.18) (§§V.16–V.20); Theorem 5.3.12 (preprint 5.19) and Theorem 5.25 (preprint) (§§V.21–V.22); the homotopy lemmas (§V.23); §5.5 (preprint Appendix 5.4.2) (§§V.24–V.26); what is not formalized (§V.27); and what the formalization taught us (§V.28). A fourth addendum adds Corollary 5.2.5 (preprint 5.8) on the shifted strong bundle (§V.29), the metric space of the allowable sections (§V.30) and weak against pointwise bump functions (§V.31); §V.32 compares with the Springer 2021 edition. A fifth addendum adds Remark 5.3.8 (preprint 5.16) with Theorem 5.3.10 (preprint 5.18) relative to a compact part (§V.33), the dense form of Theorem 5.3.12 (preprint 5.19) (§V.34), Remark 5.3.14 (preprint 5.21) (§V.35), Theorem 5.3.15 (preprint 5.22) (§§V.36–V.41), Theorem 5.5.10 (preprint 5.37) for every chart (§V.42), and the theorems that §5.5 cites: Theorem 5.5.5 (preprint 5.32) with the Sobolev remark (§§V.43–V.44), Theorem 5.5.13 (preprint 5.40) (§V.45) and Theorem 5.5.16 (preprint 5.43) (§§V.46–V.47). A sixth addendum (Part III, wave 1) adds the items of the 2021 edition that the preprint does not have, numbered by the 2021 edition with the 2017 numbers in parentheses: weak sequential compactness in reflexive spaces (§V.48); reflexive $(0,1)$-fibres, Lemma 5.1.5, mixed convergence, reflexive auxiliary norms and Proposition 5.1.8 (§§V.49–V.51); the reflexive local compactness property with Proposition 5.2.7, Theorem 5.2.8 and Remarks 5.2.9, 5.2.10 (§§V.52–V.53); and §5.4, Definition 5.4.1 and the ∂-extension Theorem 5.4.2 (§§V.54–V.55).

What is proved, and what is not. Theorems 5.2.2, 5.2.3, 5.2.4, 5.3.5, 5.3.10, 5.3.12, 5.3.15, 5.5.5, 5.5.7,

5.5.10, 5.5.13, 5.5.16 (preprint 5.5, 5.6, 5.7, 5.13, 5.18, 5.19, 5.22, 5.32, 5.34, 5.37, 5.40, 5.43) and Theorem 5.25 (preprint) are proved: Theorems 5.2.2–5.2.4 (preprint 5.5–5.7) without tameness, Theorem 5.3.12 (preprint 5.19) also in its dense form (§V.34; gap G.45), Theorem 5.5.10 (preprint 5.37) for every chart (§V.42). So are Lemmas 5.1.2, 5.3.3, 5.3.16, 5.3.17 (preprint 5.2, 5.11, 5.23, 5.24) and Lemma 5.26 (preprint), Propositions 5.1.3, 5.3.2, 5.5.3, 5.5.8, 5.5.11, 5.5.15 (preprint 5.3, 5.10, 5.30, 5.35, 5.38, 5.42), Remarks 5.3.4, 5.3.8, 5.3.14 (preprint 5.12, 5.16, 5.21) (§§V.10, V.33, V.35; gaps G.44, G.47), and Corollary 5.5.17 (preprint 5.44), now without hypothesis (§V.46). The items of the 2021 edition that the preprint does not have are proved as well (§§V.48–V.55): Definitions 5.1.4, 5.1.6, 5.1.7, 5.2.6 and 5.4.1, Lemma 5.1.5, Proposition 5.1.8, whose proof the text leaves to the reader, Proposition 5.2.7 and the remark after it, in a stronger form that does not use the reflexive local compactness, Theorem 5.2.8, in a stronger form for every continuous section (tameness and the sc-Fredholm property are not used), and the ∂-extension Theorem 5.4.2 of §5.4, which the text only sketches (2017: Definitions 5.2, 5.3, 5.4, 5.6, 5.11, Lemma 5.2, Propositions 5.2, 5.3, Theorems 5.4, 5.9). Remark 5.2.9 restates Theorem 5.2.8. The weak sequential compactness of bounded sets in reflexive spaces, which the text uses without comment, is proved (§V.48). Every definition of the chapter is formalized. Theorem 5.3.15 (preprint 5.22) (gap G.48) is proved with no added hypothesis, Definition 5.3.13 (preprint 5.20) being read in its 2021 form, whose condition (3) supplies the hypothesis $dt(x) \neq 0$ that the preprint's proof uses (§§V.23, V.32); "Morse-type" is read through smooth parametrizations of the solution set and shown independent of the parametrization (§§V.36–V.41). The cited Theorems 5.5.5, 5.5.13, 5.5.16 (preprint 5.32, 5.40, 5.43) (gaps G.50, G.52, G.53) are proved, the last for complete spaces, the text's "Banach" (§§V.43, V.45–V.47). The Sobolev remark (gap G.51) is proved for every domain on level 0; the sc-structure $E_m = W^{1+m,p}(\Omega)$ is built for $\Omega = (0,1)$ and every $p > 1$, and on other domains its compactness and density are hypotheses — the one narrower row of the chapter, V5.76 (§V.44). The step of Theorem 5.3.12 (preprint 5.19) at the boundary zeros that the formal proof avoids (Finding 88, gap G.46) is shown not to hold as we read it by a Lean counterexample (§V.22). Corollary 5.2.5 (preprint 5.8) is proved on the shifted strong bundle $P^1 : Y^1 \to X^1$, which the text uses without defining it, and by iteration for every fixed i, with the filler condition on level 1 that its "it is a priori clear" hides (§V.29; gap G.41). The perturbation theorems take f with *tame germ data* (Definition 3.6.4 (preprint 3.51)), the reading of the tame strong bundle with which §5.3 opens and the hypothesis under which the §3.6 implicit function theorems are proved since their repair (Part III, Findings of Chapter 3, items 190–199); Theorem 5.3.10 (preprint 5.18) also assumes X tame, which its proof uses through Theorems 4.2.11 and 4.3.1 (preprint 4.22 and 4.23). The metric space $(\Gamma^{+,1}_U(P), \rho)$ is built, and the text's "not complete" holds for the trivial bundle but does not hold as we read it without a hypothesis (§V.30; gap G.42). Nothing to formalize: Remark 5.3.6 (preprint 5.14), a comment on earlier work (gap G.43), Remark 5.2.10 (2017: 5.2), a comment on inductive perturbation arguments, and Theorem 5.27 (preprint), which is not in the 2021 edition (gap G.49; its general-position analogue is in §V.33). Not formalized (§V.27): Corollary 5.2.5 (preprint 5.8) as one statement for a family indexed by i, and its tame form; non-completeness of $(\Gamma^{+,1}_U(P), \rho)$ in the text's setting; weak against pointwise bump functions at a non-smooth point; the comparison of the parametrizations of §V.36 with the Mathlib charts of the solution set; the Sobolev sc-structure on domains other than $(0,1)$.

Four structural choices. (1) *The linearization is read in charts.* T_xX carries no linear structure in the formalization (gap G.7 of Chapter 2), so $f'(x)$ is the operator $\mathrm{Lin}(z) : T_zO \to K_z$ of a strong bundle chart (Part III), transversality is surjectivity of Lin in a chart, and general position is stated in a tame chart; its chart independence is proved where it matters (§V.18). (2) *"Generic" is "off a null set".* Every perturbation theorem is proved first in a generic form — for λ outside a Lebesgue null set of a small ball, $s_0 + \sum \lambda_i t_i$ does the job — and the text's dense-set form is derived from it. (3) *The parametrized solution*

set is not built as a manifold. The Sard step is local: at each zero of $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ in good position, Theorem 3.6.8 (2) (preprint 3.54) gives a regular tangent germ chart, Sard's theorem is applied on its finite-dimensional model, and finitely many null sets are collected (§V.14). Where Theorem 4.3.1 (preprint 4.23) needs a submanifold, the *bridge lemma* supplies it (§V.17). (4) *Density is pointwise.* "Within ε" is $N(s(x) - s'(x)) \le \varepsilon$ for all x; this is $\rho(s, s') \le \varepsilon$ for the metric $\rho(s, s') = \sup_x N(s(x) - s'(x))$, which the perturbation theorems never needed and which was built afterwards (§V.30).

Map.

- `Chapter5/AuxNorm`: the charts $\Phi[1]$, the fibre $(Y_{0,1})_x$ and its chart norms, Definition 5.1.1 (preprint 5.1), Lemma 5.1.2 (preprint 5.2), Proposition 5.1.3 (preprint 5.3).
- `Chapter5/Compactness`: Definition 5.2.1 (preprint 5.4), Theorems 5.2.2–5.2.4 (preprint 5.5–5.7), Corollary 5.2.5 (preprint 5.8) in a chart, Remark 5.3.4 (preprint 5.12).
- `Chapter5/BumpFunctions`: $\mathbb{R}$ as an M-polyfold, Definitions 5.3.1, 5.5.2, 5.5.6 (preprint 5.9, 5.29, 5.33), Propositions 5.3.2, 5.5.3, 5.5.8 (preprint 5.10, 5.30, 5.35), Theorem 5.5.7 (preprint 5.34), the calculus of sections, Lemma 5.3.3 (preprint 5.11), $\Gamma^{+,1}_U(P)$.
- `Chapter5/Transversality`: Definitions 5.3.7, 5.3.9 (preprint 5.15, 5.17), Lemma 5.26 (preprint), the parametrized section, the Sard step, Theorem 5.3.5 (preprint 5.13).
- `Chapter5/GeneralPosition`: the bridge lemma, $T_y S = \ker f'(y)$, chart independence and openness of general position, properties (i)–(ii), Theorems 5.3.10, 5.3.12 (preprint 5.18, 5.19) and Theorem 5.25 (preprint).
- `Chapter5/Homotopy`: Definition 5.3.13 (preprint 5.20), Lemmas 5.3.16, 5.3.17 (preprint 5.23, 5.24).
- `Chapter5/PartitionsOfUnity`: the algebra of sc-smooth functions, Definitions 5.5.1, 5.5.4, 5.5.9, 5.5.12, 5.5.14 (preprint 5.28, 5.31, 5.36, 5.39, 5.41), Theorem 5.5.10 (preprint 5.37), Propositions 5.5.11, 5.5.15 (preprint 5.38, 5.42), Corollary 5.5.17 (preprint 5.44), the statements of Theorems 5.5.13, 5.5.16 (preprint 5.40, 5.43).
- `Chapter5/Book2021`: the 2021 forms of Definitions 5.2.1, 5.3.13 and Theorems 5.2.2, 5.3.5, the witness of Definition 5.3.13 (§V.32).
- Added in the fourth addendum (§§V.29–V.31): `Chapter5/StrongBundleShift` (the shifted strong bundle P^1, raising the index of sc-smooth maps and sections), `Chapter5/Corollary58` (the filler condition on every level, Corollary 5.2.5 (preprint 5.8)), `Chapter5/AllowableSectionsMetric` (the metric ρ, completeness and its failure), `Chapter5/BumpWeakPointwise` (bump points, an sc-Banach space as an M-polyfold).
- Added in the fifth addendum (§§V.33–V.47): `Chapter5/CylinderPerturbation2021` (the chart criterion of general position, Theorem 5.3.10 relative to a compact part, the cobordism between admissible sections), `Chapter5/FredholmArcPerturbation` (Remark 5.3.8), `Chapter5/DenseGoodPosition` (the dense form of Theorem 5.3.12), `Chapter5/CompactHomotopyIso` (Remark 5.3.14); for Theorem 5.3.15 `Chapter5/Theorem5315_Ends`, `_GeneralPosition` (the ends and conclusion (1)), `_ParametricMorse`, `_MorseLagrange`, `_MorseFamily` (the parametrized Sard step, Lagrange multipliers), `_MorseParam`, `_MorseTransfer`, `_Morse` (smooth parametrizations, the families $a_i\sigma$, the covering argument), `_LocalParam`, `_LocalTangent`, `_LocalData` (the local data), `_LocalInvariance`, `_LocalInvarianceClause` (independence of the parametrization), `_Final`; `Chapter5/ApproxPropertyEveryChart` (Theorem 5.5.10 for every chart); `Chapter5/LpWeightedProduct`, `Chapter5/BonicFrampton` (Theorem 5.5.5); `Chapter5/SobolevRemarkAbstract–3`, `Chapter5/SobolevCubeVolterra–5` (the Sobolev remark); `Chapter5/Torunczyk` (Theorem 5.5.13); `Chapter5/SeparablePartitionsOfUnity`, `Chapter5/HilbertPartitionsOfUnity` (separable and Hilbert spaces, Corollary 5.5.17 with-

out hypothesis); `Chapter5/LocallyFiniteBases`, `Chapter5/AmirLindenstrauss`, `Chapter5/ProjectionalResolution`, `Chapter5/WCGPartitionsOfUnity` (Theorem 5.5.16).
- Added in the sixth addendum (§§V.48–V.55): `Chapter5/WeakSequentialCompactness` (weak convergence, weak sequential compactness in reflexive spaces), `Chapter5/ReflexiveFibres` (Definitions 5.1.4, 5.1.6, 5.1.7, Lemma 5.1.5, Proposition 5.1.8), `Chapter5/ReflexiveCompactness` (Definition 5.2.6, Proposition 5.2.7, Theorem 5.2.8, the witnesses on the trivial bundle), `Chapter5/BoundaryExtension` (Definition 5.4.1, Theorem 5.4.2); the local extension it uses is `Chapter12/LocalBoundaryExtension` (Part XII, §XII.12).

Throughout, $P : Y \to X$ is a strong bundle over an M-polyfold (both paracompact and Hausdorff), a section is a map $s : X \to Y$ with $P \circ s = \mathrm{id}$, 0_x is the zero of the fibre over x, and $\operatorname{supp} s$ is the closure of $\{s \neq 0\}$ `Chapter5/BumpFunctions · sectionSupport`. A strong bundle chart Φ has base chart $\varphi : V \to O$, $O = r(U)$ a retract, and local bundle $K \to O$; the *principal part* of f is $\widehat{f}(u) = \Phi(f(\varphi^{-1}(u)))_2$, and $\mathrm{Lin}(z)$ is its linearization at a level-1 point z (Part III). $|\lambda|$ is the maximum norm on $\mathbb{R}^m$.

V.1 The fibre $Y_{0,1}$ and the charts $\Phi[1]$

§5.1 works on the part $Y_{0,1}$ of biregularity $(0,1)$, the M-polyfold $Y[1]$ of §2.6 (preprint §2.5) (Part II), and assumes two things the project did not have: charts $\Phi[1]$ for *every* strong bundle chart, and a Banach structure on the fibres of $Y_{0,1}$.

$Y_{0,1}$ in any chart. For a strong bundle chart Φ compatible with the atlas — the text's "strong bundle isomorphism $\Phi : Y|V \to K$" — a point y over $\operatorname{dom}\varphi$ lies in $Y_{0,1}$ iff $\Phi(y)_2 \in F_1$: compare with a chart of the atlas through the transition map, which is a strong bundle map and so preserves biregularity `Chapter5/AuxNorm · StrongBundle.mem_oneSet_iff_of_isChart`. On $P^{-1}(V) \cap Y_{0,1}$ the map $w \mapsto \Phi(w)$ lifted to $K[1]$ is then a partial homeomorphism `Chapter5/AuxNorm · StrongBundle.oneChartOf`, compatible with every chart $\Phi'[1]$ of the atlas because the transition is the $[1]$-transition of Definition 2.6.4 (preprint 2.50) `Chapter5/AuxNorm · StrongBundle.oneChartOf_transition` `Chapter5/AuxNorm · StrongBundle.oneChartOf_compatible`, hence a chart of the M-polyfold $Y[1]$ `Chapter5/AuxNorm · StrongBundle.isChart_oneChartOf`. For a chart of the atlas it is the chart of Part II `Chapter5/AuxNorm · StrongBundle.oneChart_eq_oneChartOf`.

The fibre. The fibres $P^{-1}(x)$ carry linear structures; the points of $P^{-1}(x)$ in $Y_{0,1}$ form a linear subspace $(Y_{0,1})_x$, because the charts are linear on the fibres and $F_1 \subseteq F_0$ is a subspace `Chapter5/AuxNorm · StrongBundle.oneFiber`. Its points are points of $Y[1]$ and conversely `Chapter5/AuxNorm · StrongBundle.toOne` `Chapter5/AuxNorm · StrongBundle.ofOne`; its zero is the point 0_x of $Y[1]$ `Chapter5/AuxNorm · StrongBundle.zeroOne` `Chapter5/AuxNorm · StrongBundle.fiberZero_mem_oneSet`.

The chart norm $|h|_1$. In a chart around x, the fibre component of $\Phi[1]$ is a linear map $(Y_{0,1})_x \to F_1$ `Chapter5/AuxNorm · StrongBundle.liftOneFiber`, injective because Φ is injective on the fibre `Chapter5/AuxNorm · StrongBundle.liftOneFiber_injective`. Its image is the slice $\{h \in F_1 \mid (\varphi(x), h) \in K[1]\}$ `Chapter5/AuxNorm · StrongBundle.mem_range_liftOneFiber`, the fixed point set of the continuous map $h \mapsto R[1](\varphi(x), h)$, hence closed in F_1 `Chapter5/AuxNorm · StrongBundle.isClosed_slice`. So a sequence of the fibre that is Cauchy for $|\Phi[1](\cdot)_2|_1$ converges: its images converge in F_1, the limit stays in the slice, and it is the image of a point of the fibre `Chapter5/AuxNorm · StrongBundle.complete_liftOneFiber`. This is the text's "the fibers Y_x have the structure of Banach spaces". The norm depends on the chart, but only up to equivalence, and locally uniformly: by Lemma 5.1.2's (preprint 5.2) upper bound (§V.2) applied to $g = |\Phi_1[1](\cdot)_2|_1$, the norms of two charts are comparable with one constant on the fibres over a neighbourhood of x `Chapter5/AuxNorm · StrongBundle.exists_norm_liftOne_le_mul`, and so are finitely many charts and one more `Chapter5/`

AuxNorm · StrongBundle.exists_uniform_comparison.

V.2 Auxiliary norms and Lemma 5.1.2 (preprint 5.2)

Definition 5.1.1 (preprint 5.1). A function $N : Y[1] \to \mathbb{R}$ is a *complete norm on the fibre over* x if $N(v) = 0$ only for $v = 0$, $N(tv) = |t|N(v)$, $N(v + w) \le N(v) + N(w)$, and every N-Cauchy sequence of $(Y_{0,1})_x$ converges Chapter5/AuxNorm · StrongBundle.IsCompleteFiberNorm. An *auxiliary norm* is a continuous such N, complete on every fibre, with condition (2): if $P(w_k) \to x$ and $N(w_k) \to 0$, then $w_k \to 0_x$ in $Y[1]$ Chapter5/AuxNorm · StrongBundle.IsAuxNorm. It is non-negative Chapter5/AuxNorm · StrongBundle.IsAuxNorm.nonneg and homogeneous on the fibres Chapter5/AuxNorm · StrongBundle.IsAuxNorm.smul.

Lemma 5.1.2 (preprint 5.2), the upper bound. Let $g : Y[1] \to \mathbb{R}$ be continuous at 0_x and positively homogeneous on the fibres near x, and Φ a chart around x. By continuity at 0_x — read through the chart $\Phi[1]$, whose neighbourhoods of $(\varphi(x), 0)$ are the sets $|p - \varphi(x)|_0 + |h|_1 < \varepsilon$ — $g < 1$ where $|p - \varphi(x)|_0 < \varepsilon/2$ and $|h|_1 \le \varepsilon/2$; homogeneity scales every vector of a nearby fibre to $|h|_1 = \varepsilon/2$, which gives $g(w) \le (2/\varepsilon)|h|_1$ on the fibres over a neighbourhood V of x Chapter5/AuxNorm · StrongBundle.exists_le_mul_norm_liftOne. This is the text's first half of "(i) □ (ii)".

Lemma 5.1.2 (preprint 5.2), the lower bound. For an auxiliary norm, $|h|_1 \le A\,N(w)$ near x Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le. Suppose not: then for every k there is w_k over $\varphi^{-1}(B(\varphi(x), 1/(k+1)))$ with $|h_k|_1 > (k+1)N(w_k)$; rescale to $|h_k|_1 = 1$. Then $N(w_k) \to 0$ and $P(w_k) \to x$ — the points of the shrinking balls converge to x because φ^{-1} is continuous on O, so no metrizability of X is needed — hence $w_k \to 0_x$ by condition (2), and $\Phi[1](w_k) \to (\varphi(x), 0)$ by the continuity of $\Phi[1]$, contradicting $|h_k|_1 = 1$. Together: **in every chart around** x there are a neighbourhood V and $0 < a < A$ with $aN(w) \le |h|_1 \le AN(w)$ over V Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_bounds.

Lemma 5.1.2 (preprint 5.2). For N continuous and a complete norm on every fibre, N is an auxiliary norm iff every x has a chart and such two-sided bounds near x Chapter5/AuxNorm · lemma_5_1_2. "(i) □ (ii)" is the previous paragraph. "(ii) □ (i)", "obvious" in the text: if $P(w_k) \to x$ and $N(w_k) \to 0$, the bound gives $|h_k|_1 \to 0$, so $\Phi[1](w_k) \to (\varphi(x), 0)$, and $w_k \to 0_x$ because $\Phi[1]^{-1}$ is continuous on $K[1]$ (§V.1). In the statement V is a neighbourhood of x on which the bounds hold; the text's V is the domain of Φ, but its own proof gives them "for $P(w)$ close to x".

V.3 Proposition 5.1.3 (preprint 5.3)

Existence. For the charts Φ_i of the atlas and a partition of unity (ρ_i) subordinate to their domains — it exists since X is paracompact and Hausdorff, hence normal Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate — the *glued norm* is $N(w) = \sum_i \rho_i(P(w))\,|\Phi_i[1](w)_2|_1$ Chapter5/AuxNorm · StrongBundle.gluedNorm Chapter5/AuxNorm · StrongBundle.gluedNorm_eq_sum. It is continuous on $Y[1]$: each term vanishes off the closed set $P^{-1}(\operatorname{supp}\rho_i)$, which lies in the open domain of $\Phi_i[1]$, where the term is continuous, and the sum is locally finite Chapter5/AuxNorm · StrongBundle.continuous_gluedNorm. Near a point x only finitely many ρ_i are non-zero, all with x in the domain of Φ_i and summing to 1 Chapter5/AuxNorm · StrongBundle.sum_partition_eq_one; their chart norms are uniformly comparable with the norm of any one chart Φ' around x (§V.1), so N is two-sided comparable with $|\Phi'[1](\cdot)_2|_1$ near x Chapter5/AuxNorm · StrongBundle.gluedNorm_bounds. On each fibre N is a convex combination of norms, hence a norm, and comparable with a complete norm, hence complete

Chapter5/AuxNorm · StrongBundle.complete_of_bounds Chapter5/AuxNorm · StrongBundle.isCompleteFiberNorm_gluedNorm. Lemma 5.1.2 (preprint 5.2) "(ii) □ (i)" concludes Chapter5/AuxNorm · proposition_5_1_3_exists. The text gives this as one sentence.

Comparison. For auxiliary norms N_0, N_1, Lemma 5.1.2 (preprint 5.2) "(i) □ (ii)" for both in the *same* chart around y — which is why "(i) □ (ii)" was proved in every chart — gives $r_y > 0$ with $r_y N_0 \le N_1$ and $r_y N_1 \le N_0$ over a neighbourhood of y. At each point the admissible constants r form a convex set (r admissible and $0 < r' \le r$ give r' admissible), so a partition of unity glues the local constants into a continuous $f > 0$ with values in these sets Mathlib/Analysis/Convex/PartitionOfUnity · exists_continuous_forall_mem_convex_of_local_const, and $f(x)N_0 \le N_1 \le f(x)^{-1}N_0$ on the fibre over x Chapter5/AuxNorm · proposition_5_1_3_compare. Tameness plays no role in §5.1.

V.4 Definition 5.2.1 (preprint 5.4) and proper sections

"$N(y) \le r$" for a point y of Y is read with the text's convention $N(y) = \infty$ off $Y_{0,1}$: it means $y \in Y_{0,1}$ and $N(y) \le r$ Chapter5/Compactness · StrongBundle.AuxLe, a property of the point only Chapter5/Compactness · StrongBundle.AuxLe.congr. A section f is *weakly proper* if $f^{-1}(0)$ is compact and for every auxiliary norm N there is an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc$^+$-section s with support in U and $N(s) \le 1$ Chapter5/Compactness · StrongBundle.IsWeaklyProper; *proper* if instead $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le 1\}$ is compact Chapter5/Compactness · StrongBundle.IsProper.

Proper implies weakly proper for a continuous f Chapter5/Compactness · StrongBundle.IsProper.isWeaklyProper. Take the U of properness. A solution of $f(x) = s(x)$ outside U has $s(x) = 0$, so $f(x) = 0$ and $x \in f^{-1}(0) \subseteq U$ after all; inside U, $f(x) = s(x)$ gives $N(f(x)) = N(s(x)) \le 1$. So $\{f = s\}$ lies in the compact closure, and it is closed because f and s are continuous and Y is Hausdorff. Weakly proper sections have compact solution sets by definition Chapter5/Compactness · StrongBundle.IsWeaklyProper.isCompact_solutionSet. The text's "obviously" hides the continuity of f.

V.5 Theorem 5.2.2 (preprint 5.5): the local form for basic germs

Theorem 5.2.2 (preprint 5.5) says that for sc-Fredholm sections a compact solution set already forces properness. The appendix proves it in the coordinates of a basic germ, and that local statement is the heart of the matter.

The setting. A basic germ (Definition 3.1.7 (preprint 3.7), Part III) is $h : \tilde{C} \oplus W \supseteq U_0 \to \mathbb{R}^N \oplus W$ whose W-component is $w - B(a, w)$, where B is an sc^0-contraction germ: on small balls, $w \mapsto B(a, w)$ contracts with constant $\le 1/2$ on level 0, uniformly in a, and $B(0, 0) = 0$. Part III built the *parametrized fixed point* $\delta(a, z)$, the unique solution w in a closed ball $\bar{B}(r)$ of $w = B(a, w) - z$ for (a, z) in a set Σ of small parameters, continuous in (a, z) on level 0 Chapter3/LocalCompactness · IsSc0ContractionGerm.delta_continuousOn Chapter5/Compactness · IsSc0ContractionGerm.eq_delta_zero.

The local form Chapter5/Compactness · IsSc0ContractionGerm.exists_isCompact_bdd. Assume $\tilde{C}$ closed. There is an open $O \ni 0$, *independent of c*, such that for every c the points $x = (a, w) \in U_0 \cap O$ with $w - B(a, w) \in W_1$ and $|w - B(a, w)|_1 \le c$ lie in a compact set. The proof:

1. *A small ball.* $B(a, w) - w$ is continuous at 0 within U_0 and vanishes there, so for $|(a, w)|_0 < \rho_1$ it has norm $< r/8$. Let $\rho = \min(\rho_1, \rho'/2, r/2)$, with ρ' the radius of the parameter set, and $O = B(0, \rho)$.

2. *The compact set of parameters.* The inclusion $W_1 \to W_0$ is compact, so the image of the closed level-1 ball of radius c lies in a compact $K_c \subseteq W_0$. The set $L = (\tilde{C} \cap \bar{B}(\rho)) \times (K_c \cap \bar{B}(r/8))$ is compact (here $\tilde{C}$ closed is used) and lies in Σ.
3. *The candidate.* The compact set is the image of L under the continuous map $(a, z) \mapsto (a, \delta(a, z))$.
4. *Every bounded point is hit.* Let $x = (a, w) \in U_0 \cap O$ with $w - B(a, w) = \iota(z_1)$, $|z_1|_1 \le c$. Put $p = (a, B(a, w) - w)$. Then $a \in \tilde{C}$ with $|a| < \rho$; $B(a, w) - w = \iota(-z_1)$ lies in K_c and has norm $< r/8$ by step 1; so $p \in L$. And w lies in the ball $\bar{B}(r)$ and solves $w = B(a, w) - (B(a, w) - w)$, so $w = \delta(p)$ by uniqueness `Chapter5/Compactness · IsSc0ContractionGerm.eq_delta_zero`, i.e. x is the image of p.

Only the level-0 contraction is used. For a basic germ f itself the W-component of $f(x)$ is $w - B(a, w)$, so the same O works with the level-1 bound $|f(x)|_1 \le c$ `Chapter5/Compactness · IsBasicGerm.exists_isCompact_bdd`. This is the appendix's (82)–(90), where the text writes "$a \in \mathbb{R}^{n}$" for $a \in \tilde{C}$ (B is defined on $\tilde{C} \oplus W$ only), "$\mathbb{R}^n \oplus Y_1$" for $\mathbb{R}^N \oplus W_1$, and in (89) mixes the $\mathbb{R}^N$- and W-components; the t of its "$h + t$" and its estimates (83), (85) disappear in the next step.

V.6 Theorem 5.2.2 (preprint 5.5): the local step and the covering

Uniform fibre bounds. A strong bundle map Ψ (Definition 2.6.1 (preprint 2.47)) is linear on the fibres and $\Psi[1]$ is continuous; at $(u_0, 0)$, where $\Psi[1](u_0, 0)_2 = 0$ `Chapter5/Compactness · IsStrongBundleMap.shiftFiber_snd_zero`, continuity gives $|\Psi[1](u, k)_2|_1 \le 1$ for $|u - u_0|_0 < \delta$, $|k|_1 \le \eta$, and fibre linearity `Chapter5/Compactness · IsStrongBundleMap.shiftFiber_snd_smul` turns this into $|\Psi[1](u, k)_2|_1 \le \eta^{-1}|k|_1$ for all $k \in F_1$ `Chapter5/Compactness · IsStrongBundleMap.exists_fiber_bound_one`. The text uses this bound silently when it replaces $N(f(x)) \le 1$ by a bound in basic germ coordinates.

The local step `Chapter5/Compactness · exists_nbhd_isCompact_closure_bdd`. Let (f, x_0) be an sc-Fredholm germ (Definition 3.1.11 (preprint 3.11)) — x_0 need not be a zero. Its data, in the text's germ reading (§III.6): a strong bundle chart Φ centred at x_0, a filling germ g of the principal part $\hat{f}$ on a germ inside the domain U of the retraction, an sc$^+$-germ t_0 on a germ G_s, and a strong bundle isomorphism Ψ with base map ψ conjugating $g - t_0$ to a basic germ h: $h(\psi(u)) = \Psi(u, g(u) - t_0(u))_2$ for $u \in U$ near 0. The claim is an open $V \ni x_0$, independent of C, such that for every C the closure in X of $Z_C = \{x \in V \mid |\Phi(f(x))_2|_1 \le C\}$ is compact.

1. *The neighbourhood.* Near 0 the filler agrees with the principal part on O and the conjugation identity holds (a set N_1); t_0 is continuous into F_1 on level 0 of its germ, $(G_s)_0 = O_s \cap C$ with O_s open — not on U — so the neighbourhood is also cut with O_s, and $|t_0(u)|_1 \le |t_0(0)|_1 + 1$ for $u \in U \cap O_s$, $|u|_0 < \delta_2$ (with t_0 sc-smooth on U the cut was not needed); the fibre bound of Ψ holds for $|u|_0 < \delta_3$ with constant M. On the side of the basic germ, the local form (§V.5) gives an open O'; ψ maps a neighbourhood of 0 into a closed ball $\bar{B}(r_0) \cap C'$ inside the domain U' of ψ^{-1} (relative openness of U'). V is the set of points of the chart domain whose image lies in all of these.
2. *The bound transported.* For $x \in Z_C$, $u = \varphi(x) \in O$, $g(u) = \hat{f}(u)$ lies in F_1 with $|g(u)|_1 \le C$, so $|g(u) - t_0(u)|_1 \le C + |t_0(0)|_1 + 1$, and $|h(\psi(u))|_1 = |\Psi[1](u, g(u) - t_0(u))_2|_1 \le M(C + |t_0(0)|_1 + 1) =: c'$. *Subtracting the sc$^+$-part before the conjugation* is what makes the text's t and its estimates unnecessary, and the neighbourhood independent of C.
3. *Compactness on the side of the basic germ.* So $\psi(\varphi(Z_C))$ lies in the compact set $K_{c'}$ of the local form, intersected with the closed ball $\bar{B}(r_0) \cap C'$; its closure A is compact and inside U'.
4. *Back to U.* ψ^{-1} is continuous on U', so $T = \psi^{-1}(A)$ is compact, contains $\varphi(Z_C)$ and lies in U.

5. *Back to O.* The closure of $\varphi(Z_C)$ lies in $T \subseteq U$ and in the closure of O; since O is the fixed point set of the continuous retraction r on U, it is closed in U, so the closure lies in O.
6. *Back to X.* φ^{-1} is continuous on O, so φ^{-1} of that closure is compact and contains Z_C, hence its closure.

The text's "it suffices to find in local coordinates an open neighborhood" leaves steps 4–6 implicit: its set $\{(a,w) \in U \mid |(h+t)(a,w)|_1 \le c\}$ lives in the chart domain U of the filling, which contains points not in the retract.

Theorem 5.2.2 (preprint 5.5) `Chapter5/Compactness · theorem_5_2_2_preprint`. Let f be sc-Fredholm with compact solution set, and N an auxiliary norm. At each zero x the germ (f,x) is sc-Fredholm, so the local step gives a chart Φ and an open $V(x)$; Lemma 5.1.2's (preprint 5.2) lower bound in that chart `Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le` gives $V_1(x)$ and A with $|\Phi(w)_2|_1 \le A\,N(w)$ over $V_1(x)$, so on $V(x) \cap V_1(x)$ the condition $N(f(y)) \le 1$ implies $|\Phi(f(y))_2|_1 \le A$, and $\mathrm{cl}\{y \in V(x) \cap V_1(x) \mid N(f(y)) \le 1\}$ is a closed subset of a compact set. Finitely many of these neighbourhoods cover the compact $f^{-1}(0)$; their union is U, and the closure of $\{y \in U \mid N(f(y)) \le 1\}$ lies in the finite union of the compact closures. That is the text's last sentence, "since the solution set of f is compact, the proof is complete". Tameness is not used, as in the appendix's statement of the theorem.

The equivalence. For an sc-Fredholm section, which is continuous, proper □ weakly proper (§V.4) □ compact solution set □ proper (Theorem 5.2.2 (preprint 5.5)) `Chapter5/Compactness · theorem_5_2_2_iff`.

V.7 Theorems 5.2.3, 5.2.4 (preprint 5.6, 5.7) and Corollary 5.2.5 (preprint 5.8)

Theorem 5.2.3 (preprint 5.6) is Theorem 3.1.13 (preprint 3.12) at a zero: a zero of an sc-Fredholm section is a smooth point (regularizing), where (f,x) is an sc-Fredholm germ, and Theorem 3.1.13 (preprint 3.12) gives nested open $U(i) \ni x$ with $\mathrm{cl}\{y \in U(i) \mid f(y) = 0\}$ the image of a compact subset of X_i `Chapter3/SectionCompactnessG · theorem_3_1_13_preprint` `Chapter5/Compactness · theorem_5_2_3`.

The space X_∞. The text says "X_∞ is a metric space" but never gives it a topology. It is the set of smooth points with the initial topology of the inclusions $X_\infty \to X_m$ `Chapter2/Metrizable · MPolyfold.smoothLevel_topology_eq`, metrizable by Theorem 2.3.10 (preprint 2.16) `Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel`, and a sequence converges in it iff it converges on every level `Chapter5/Compactness · MPolyfold.tendsto_smoothLevel_iff`; the inclusions into X and into each X_i are continuous `Chapter5/Compactness · MPolyfold.continuous_smoothVal` `Chapter5/Compactness · MPolyfold.continuous_smoothIncl`.

Theorem 5.2.4 (preprint 5.7) `Chapter5/Compactness · theorem_5_2_4`. $S = f^{-1}(0)$ lies in X_∞ because f is regularizing. Take a sequence (x_k) in S; a subsequence converges in X to some x. The limit is a zero, since a limit of zeros of a continuous section is a zero `Chapter3/SectionCompactness · StrongBundle.IsScSmoothSection.isZeroAt_of_tendsto`, so it is smooth and (f,x) is an sc-Fredholm germ. Now *Corollary 3.1.15 (preprint 3.13) at the point x* gives the convergence $x_k \to x$ on every level `Chapter3/Chapter5BridgeG · corollary_3_1_15_section`, hence in X_∞. The text cites Theorem 5.2.3 (preprint 5.6), which gives compactness in X_i, not the convergence of a given sequence. Tameness is not used.

Corollary 5.2.5 (preprint 5.8), the compact solution set. The zeros in X_i are the image of the compact subset of X_∞ of Theorem 5.2.4 (preprint 5.7) under the continuous inclusion $X_\infty \to X_i$ `Chapter5/Compactness · isCompact_solutionSet_level`.

Corollary 5.2.5 (preprint 5.8) in a chart. "It is a priori clear that f^i is sc-Fredholm" is not clear, and $Y^i \to X^i$ is not defined in the text. What is formalized is the shift of the germ data. A basic germ f on $\tilde{C} \oplus W$ shifts to a basic germ f^1 on $\tilde{C} \oplus W^1$ with contraction germ B^1: the shifted neighbourhood germ `Chapter5/Compactness · ScNbhdGerm.shiftParam`, the contraction estimates on the shifted levels `Chapter5/Compactness · IsSc0ContractionGerm.shift`, and sc-smoothness of f^1 by Proposition 1.2.2 (preprint 1.17) for germs, which needs $\tilde{C}$ a partial quadrant `Chapter5/Compactness · IsBasicGerm.shift`. If k is conjugated to h by Ψ, then k^1 is conjugated to h^1 by the index raise Ψ^1 `Chapter3/BasicClassComposition · IsStrongBundleIso.shift Chapter5/Compactness · IsConjugateToBasicGerm.shift`. Hence, for an sc-Fredholm section with compact solution set, the zeros in X_1 are compact, and at every smooth point, in the chart of the germ, $(g - s)^1$ is conjugated to a basic germ and s^1 is an sc$^+$-germ on the shifted germ G^1_s (Proposition 1.2.2 (preprint 1.17) for germs, `Chapter3/GermLevels · IsSckGerm.shift`), the text's setting `Chapter5/Compactness · corollary_5_2_5_chart`. In the germ reading of Definition 3.1.11 (preprint 3.11) (§III.48) g is a filling germ and $g - s$ only an sc^0-germ on the intersection of the two germs; the level shift of the conjugation needs no more, since the conjugation identity is used near 0 only `Chapter5/Compactness · IsConjugateToBasicGerm.shift_germ`. The statement changed with the switch: in the whole-chart reading s^1 was an sc$^+$-section of all of $U^1 \lhd F^1$. The filler of g^1 was the gap: its condition (3) on level 1 does not follow from level 0. It is closed, together with the shifted strong bundle and the whole corollary, in §V.29.

V.8 $\mathbb{R}$ as an M-polyfold, and bump functions

$\mathbb{R}$. "An sc-smooth map $f : X \to \mathbb{R}$" needs $\mathbb{R}$ as an M-polyfold: the constant sc-structure `Chapter5/BumpFunctions · realSc` and the single chart id `Chapter5/BumpFunctions · realChart Chapter5/BumpFunctions · realPolyfold Chapter5/BumpFunctions · isChart_realChart`. Sc-smoothness of a map between M-polyfolds can be checked in one pair of charts around each point, continuity included, since the local representatives determine the map near the point `Chapter5/BumpFunctions · MPolyfold.isScSmoothMap_of_local Chapter5/BumpFunctions · isScSmoothMap_real_of_local`; an sc-smooth function read in any chart is sc-smooth on the retract `Chapter5/BumpFunctions · MPolyfold.IsScSmoothMap.real_chart`; constants are sc-smooth `Chapter5/BumpFunctions · isScSmoothMap_real_const`; a C^∞ function $\sigma : \mathbb{R} \to \mathbb{R}$ is an sc-smooth self-map of $\mathbb{R}$ (Corollary 1.2.5 (preprint 1.20) in the constant structure) `Chapter5/BumpFunctions · isScSmoothMap_real_of_contDiff`, so $\sigma \circ f$ is sc-smooth `Chapter5/BumpFunctions · MPolyfold.IsScSmoothMap.comp_contDiff`.

Definitions 5.3.1 and 5.5.2 (preprint 5.9 and 5.29). *X admits sc-smooth bump functions* if every x and open $U \ni x$ have an sc-smooth f with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$ `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctions`. The text has two definitions: Definition 5.3.1 (preprint 5.9) asks "$f \neq 0$", Definition 5.5.2 (preprint 5.29) "not identically zero". The proof of Proposition 5.5.3 (preprint 5.30) starts from "a bump function g with $g(0) = 1$", i.e. $g(x) \neq 0$ at the given point; on an M-polyfold there are no translations to move a bump to x (even on an sc-Banach space, translation by a non-smooth vector does not preserve the levels). So Definition 5.3.1 (preprint 5.9) is read with $f(x) \neq 0$; Definition 5.5.2 (preprint 5.29) as written `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctionsWeak` is implied `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctions.weak`. How far apart the two readings are is §V.31.

Propositions 5.3.2 and 5.5.3 (preprint 5.10 and 5.30). Given x and U, take a bump g with $g(x) \neq 0$ and let $f = \tau(3g/g(x) - 1)$, τ the smooth transition from 0 on $(-\infty, 0]$ to 1 on $[1, \infty)$ `Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition`. Then f is sc-smooth, $0 \leq f \leq 1$, $f(x) = \tau(2) = 1$, $f \equiv 1$ on the open set $\{g/g(x) > 2/3\} \ni x$, and $f \neq 0$ only where $g/g(x) > 1/3$, so $\operatorname{supp} f \subseteq \operatorname{supp} g \subseteq U$ `Chapter5/BumpFunctions · proposition_5_5_3 Chapter5/BumpFunctions · proposition_5_3_2`. The text's σ ("0 for $s \leq 1$ and 1 for $s \geq 1$") is not a function.

V.9 The calculus of sections, and Lemma 5.3.3 (preprint 5.11)

$\Gamma^+(P)$ and its algebra. $\Gamma^+(P)$ is the set of sc⁺-sections (Definition 2.6.6 (preprint 2.52)) `Chapter5/BumpFunctions · StrongBundle.scPlusSections`; their values lie in $Y_{0,1}$, because in a chart their principal parts take values in F_1 `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.mem_oneSet`. The text uses, and does not state, the rules that make $\Gamma^+(P)$ a module over the sc-smooth functions. The zero section is sc⁺ `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_fiberZero`. *The product rule:* for f sc-smooth and s sc⁺, the fibrewise product $f \cdot s$ `Chapter5/BumpFunctions · sectionSmul` is sc⁺: in a chart its principal part on level 1 is $(f \circ \varphi^{-1}) \cdot \hat{s}$, the product of an sc-smooth function and an sc-smooth map into F^1, sc-smooth by the product rule of the sc-calculus `Chapter3/ParamPerturbation · IsScSmooth.smul` `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.smul_left`. Sums are sc⁺, since principal parts add in a chart `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.add`, and by induction finite sums `Chapter5/BumpFunctions · sectionSum` `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_sectionSum`.

Cut-offs. Lemma 5.3.3 (preprint 5.11) builds a section in one chart and "extends it by 0". For a chart Φ of the atlas, $\tau : O \to F_1$ with $z \mapsto (z, \tau(z))$ sc-smooth into $K[1]$, and β sc-smooth with $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$, the section $y \mapsto \Phi^{-1}(\varphi(y), \beta(y)\tau(\varphi(y)))$ on $\operatorname{dom}\varphi$, 0_y outside `Chapter5/BumpFunctions · StrongBundle.cutoffSection`, is sc⁺ `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_cutoffSection`: on $\operatorname{dom}\varphi$ its representative in $\Phi[1]$ is $z \mapsto (z, \beta(\varphi^{-1}(z))\tau(z))$, sc-smooth by the product rule; on the open set $X \setminus \operatorname{supp}\beta$ it is the zero section; the two open sets cover X.

Lemma 5.3.3 (preprint 5.11) `Chapter5/BumpFunctions · lemma_5_3_3`. Let X admit bump functions, N be an auxiliary norm, x smooth, e a smooth point of Y over x, $\varepsilon > 0$ and $U \ni x$ open. Take any chart Φ of the atlas around x — neither a centred chart nor tameness is needed — and $\hat{e} = \Phi(e)_2$, a smooth point of F because e is smooth. The constant $\hat{e}$ is sc-smooth into F^1 `Chapter5/BumpFunctions · isScSmooth_const_of_mem_smooth`, and $R[1]$ is sc-smooth, so $t(z) = R(z, \hat{e})$ gives an sc⁺-section near x with $t(x) = e$. Since N is continuous on $Y[1]$ and t is continuous into $Y[1]$, the set $\{y \mid N(t(y)) < N(e) + \varepsilon\}$ is an open neighbourhood of x. Proposition 5.5.3 (preprint 5.30) gives $\beta : X \to [0,1]$ with $\beta(x) = 1$ and support in that set, in U and in the chart domain, and the cut-off $s = \beta \cdot t$ is sc⁺ with $s(x) = e$ and $\operatorname{supp} s \subseteq U$. Finally, at every y, $N(s(y)) = \beta(y)N(t(y)) \le N(t(y)) < N(e) + \varepsilon$, because N is a norm on the fibre and $0 \le \beta \le 1$; off the support, $N(s(y)) = N(0_y) = 0$. The text states the bound only at x, where it is trivial; its proof gives, and §5.3 needs, the bound everywhere.

V.10 Allowable sections and Remark 5.3.4 (preprint 5.12)

$\Gamma^{+,1}_U(P)$. For an auxiliary norm N and an open U, the allowable sections are the sc⁺-sections with support in U and $N(s(x)) \le 1$ at every point `Chapter5/BumpFunctions · StrongBundle.allowableSections` — a condition at every point because sc⁺-sections take values in $Y_{0,1}$. The text's metric $\rho(s, s') = \sup_x N(s(x) - s'(x))$ is used through its pointwise form: $N(s(x) - s'(x))$, the difference taken in the fibre $(Y_{0,1})_x$ `Chapter5/Transversality · StrongBundle.sectionDistAt`. The metric space itself is §V.30.

Remark 5.3.4 (preprint 5.12) `Chapter5/Compactness · remark_5_3_4`. Let $S_f \subseteq U$ with $\operatorname{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact (the U of properness), and let s vanish outside U with $N(s(x)) \le 1$ everywhere — properties that pass to uniform limits of allowable sections, which is all the remark uses of the completion. At a zero x of $f + s$: if $x \notin U$, then $s(x) = 0$ and $f(x) = 0$, so $x \in S_f \subseteq U$, a contradiction; so $x \in U$, and $f(x) = -s(x)$ gives $N(f(x)) = N(s(x)) \le 1$ `Chapter5/Compactness · StrongBundle.IsAuxNorm.auxLe_of_sectionAdd`. So $S_{f+s} \subseteq \{x \in U \mid N(f(x)) \le 1\}$, whose closure is compact. The remark

claims the solution set compact, which needs it closed; for an allowable s this holds because $f+s$ is sc-Fredholm (Theorem 3.1.18 (preprint 3.15)) `Chapter3/FredholmSectionsG · theorem_3_1_18_preprint` and solution sets of continuous sections are closed `Chapter5/Transversality · isClosed_solutionSet`. This is how part (1) of Theorems 5.3.5, 5.3.10, 5.3.12 (preprint 5.13, 5.18, 5.19) and Theorem 5.25 (preprint) is proved: U from Theorem 5.2.2 (preprint 5.5), then Remark 5.3.4 (preprint 5.12) and closedness.

V.11 Transversality and the parametrized section

Definition 5.3.7 (preprint 5.15). g is transversal if at every zero its linearization is onto in a chart `Chapter5/Transversality · StrongBundle.IsTransversal`, hence in every chart `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt.linAt_surjective Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`. The kernel of $\mathrm{Lin}(z)$ in a chart is a subspace of E_0 inside T_zO `Chapter5/Transversality · StrongBundleChart.linKer`.

The parametrized section. For a section g and sections $t_1,\dots,t_m$, Theorem 3.1.21 (preprint 3.18) (Part III) makes $F(\lambda,y)=g(y)+\sum\lambda_i t_i(y)$ a section of the pull-back of P to $\mathbb{R}^m\times X$, sc-Fredholm when g is and the t_i are sc$^+$, and with tame germ data when g has them `Chapter3/ParamPerturbationG · theorem_3_1_21_tame`. Its *slice* at λ is the section $g+\sum\lambda_i t_i$ of P `Chapter5/Transversality · paramSlice Chapter5/Transversality · paramSlice_eq_sectionAdd`, and (λ,y) is a zero of F iff y is a zero of the slice `Chapter5/Transversality · isZeroAt_paramSection_iff`. The slices are sc-Fredholm, since $\sum\lambda_i t_i$ is sc$^+$ (§V.9, Theorem 3.1.18 (preprint 3.15)) `Chapter5/Transversality · isScFredholmSection_paramSlice`; slices of sc$^+$-sections are sc$^+$ `Chapter5/Transversality · isScPlusSection_paramSlice` with support in the union of the supports `Chapter5/Transversality · sectionSupport_paramSlice_subset` and vanish off it `Chapter5/Transversality · isZeroAt_paramSlice_of_not_mem_support`. *Allowability:* if $N(s_0)\le 1-\varepsilon$ and $N(t_i)\le 1$, the triangle inequality on the fibre gives $N(s_0+\sum\lambda_i t_i)\le 1-\varepsilon+m|\lambda|$ `Chapter5/Transversality · StrongBundle.IsAuxNorm.toOne_sum_le`, which is ≤ 1 when $m|\lambda|\le\varepsilon$ `Chapter5/Transversality · auxLe_paramSlice` — the text's "$N(\sum\lambda_j t_j(y))<\varepsilon/2$", with $|\lambda|$ the maximum norm.

The linear algebra of "regular value ⇒ transversal". Let $M(a,u)=Lu+\sum a_i e_i:\mathbb{R}^m\oplus T\to K$ be onto. If the projection $\ker M\to\mathbb{R}^m$ is onto, L is onto: given $k=Lu+\sum a_i e_i$, choose u' with $(a,u')\in\ker M$, so $k=L(u-u')$ `Chapter5/Transversality · surjective_of_forall_exists_ker_fst`. Conversely, if L is onto, every a lifts to $\ker M$ `Chapter5/Transversality · forall_exists_ker_fst_of_surjective`. In the product chart $(r\mapsto r-\lambda)\times\Phi$ of a chart Φ centred at y, the linearization of F at (λ,y) is $\mathrm{Lin}_F(0)(a,u)=\mathrm{Lin}_{g_\lambda}(0)u+\sum a_i\hat{t}_i(0)$ `Chapter3/ParamPerturbationG · linAt_paramSection`, the fibre of the product chart over its centre being the fibre K_0 of Φ `Chapter5/Transversality · mem_K_ofProd_zero_iff`. So if Lin_F is onto at (λ,y) and its kernel projects onto $\mathbb{R}^m$, the slice g_λ is transversal at y `Chapter5/Transversality · isLinSurjAt_paramSlice_of_ker_fst`. In the other direction, if in a chart centred at a zero y of g the range of $\mathrm{Lin}_g(0)$ and the $\hat{t}_i(0)$ span K_0, then F is transversal at $(0,y)$ `Chapter3/ParamPerturbationG · linAt_paramSection_zero Chapter5/Transversality · isLinSurjAt_paramSection_zero_of_spanning`; and adding sections keeps this, because a combination over a subfamily is one over the family `Chapter5/Transversality · exists_sum_comp_eq Chapter5/Transversality · isLinSurjAt_paramSection_zero_of_comp`. Only "regular value ⇒ transversal" is needed in Theorem 5.3.5 (preprint 5.13).

V.12 Spanning sections

"For every $x\in S_{f+s_0}$ we find finitely many allowable sc$^+$-sections such that the range of $(f+s_0)'(x)$ together with the sections span Y_x": the text does not say how. Two ingredients.

Linear algebra `Chapter5/Transversality · exists_fin_mem_of_dense_of_closedComplemented`. Let $R \subseteq K$ be closed, complemented and of finite codimension, and $D \subseteq K$ a dense subspace. With π a continuous projection onto R, $Q = \ker \pi$ is finite-dimensional, and $(1-\pi)(D)$ is a dense subspace of Q, hence closed, hence all of Q; a basis of Q lifts to finitely many $d_i \in D$, and $K = R + \operatorname{span}\{d_i\}$.

The sections `Chapter5/Transversality · exists_spanning_sections`. Let g be sc-Fredholm, $g(y) = 0, U \ni y$ open. In the chart Φ of the germ, centred at y, $\mathrm{Lin}(0) : T_0O \to K_0$ is $Dg(0)|_{T_0O}$ for a derivative family Dg of the filling germ on its germ (the bridge for germs at the centre, `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt`), hence Fredholm (Proposition 3.6.5 (preprint 3.52) for germs, which exposes the germ and Dg) `Chapter3/FredholmGermLinearizationG · proposition_3_6_5_preprint`, so its range R is closed, complemented and of finite codimension in $K_0 = \rho(0)F$. The smooth vectors $\rho(0)F_\infty$ are dense in K_0 — $\rho(0)$ is continuous and preserves the levels at the smooth point 0, and F_∞ is dense — so finitely many smooth k_i span a complement. Each k_i is $\Phi(e_i)_2$ for a smooth point e_i of the fibre over y `Chapter5/Transversality · mem_smoothSet_toMPolyfold0_of_chart`; Lemma 5.3.3 (preprint 5.11) gives s_i with $s_i(y) = e_i$, support in U and $N(s_i) < N(e_i) + 1$, and $s_i/(N(e_i)+1)$ is allowable, with support unchanged `Chapter5/Transversality · sectionSupport_sectionSmul_const_subset`, N scaled `Chapter5/Transversality · StrongBundle.IsAuxNorm.oneMk_sectionSmul_const` and principal part scaled `Chapter5/Transversality · principal_sectionSmul_const`. The density of the smooth vectors is the point the text omits: Lemma 5.3.3 (preprint 5.11) only reaches smooth points of the fibre.

V.13 Theorem 5.3.5 (preprint 5.13): openness, and the omitted compactness step

No boundary on $\mathbb{R}^m \times X$. "The interior case, Theorem 3.19" is applied to F on $\mathbb{R}^m \times X$, so $\partial(\mathbb{R}^m \times X) = \emptyset$ is needed: $d_{\mathbb{R}^m} \equiv 0$ `Chapter5/Transversality · rnPolyfold_degIndex` and $d_{\mathbb{R}^m \times X} \le d_{\mathbb{R}^m} + d_X$ `Chapter2/FibredProduct · MPolyfold.degIndex_prod_le Chapter5/Transversality · isWithoutBoundary_prod_rn`. At a point with $d = 0$ good position is surjectivity `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`.

Transversality spreads. If g has tame germ data and F is transversal at $(0,x)$, x a zero of g, then $(F,(0,x))$ is in good position, and Theorem 3.6.8 (1) (preprint 3.54) gives an open $V \ni (0,x)$ all of whose zeros are in good position, hence transversal; $U(x) = \{y \mid (0,y) \in V\}$ is the text's neighbourhood `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one' Chapter5/Transversality · exists_isOpen_isLinSurjAt_zero_near`. Likewise, if F is transversal at every $(0,y)$, $y \in S_g$, the union of these sets is an open $V \supseteq \{0\} \times S_g$ all of whose zeros of F are in good position in a tame chart `Chapter5/Transversality · exists_isOpen_goodPosition_of_isLinSurjAt`.

The omitted step. The text passes from the neighbourhood U of $\{0\} \times S_{f+s_0}$ to "$U \subset \{|\lambda| < \delta_0\} \oplus X$" without argument; what is needed is that *every* zero of F with $|\lambda|$ small lies in V. The topological core: if $Z \subseteq \mathbb{R}^m \times X$ is compact and the open V contains $Z \cap \{\lambda = 0\}$, then $Z \setminus V$ is compact, its projection to $\mathbb{R}^m$ is compact and misses 0, and any δ below its distance to 0 works `Chapter5/Transversality · exists_pos_forall_mem_of_isCompact`. The compact Z is the set of zeros of F with $|\lambda| \le \delta_1$: for such λ the perturbation $s_0 + \sum \lambda_i t_i$ is allowable (§V.11), so by Remark 5.3.4 (preprint 5.12) its zeros lie in the compact $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ of Theorem 5.2.2 (preprint 5.5) — Z is a closed subset of $\bar{B}(\delta_1)\times$ (that compact set), closed because F is continuous.

V.14 Theorem 5.3.5 (preprint 5.13): the Sard step

"$\tilde{S} := F^{-1}(0)$ is a smooth manifold"; "taking a regular value λ_0 for the projection $\tilde{S} \to \mathbb{R}^m$, it is easily verified that $f + s_0 + s(\lambda_0, \cdot)$ is transversal". The formalization does not build $\tilde{S}$; it proves the needed statement near each zero `Chapter5/Transversality · exists_null_isLinSurjAt_paramSlice_near`.

Claim. Let g have tame germ data, t_i be sc$^+$, and z_0 a zero of F in good position in a tame chart. There are an open $W \ni z_0$ and a null set $CV \subseteq \mathbb{R}^m$ such that for every zero $(\lambda, y) \in W$ with $\lambda \notin CV$ the slice g_λ is transversal at y.

1. *The chart of $\tilde{S}$.* Theorem 3.6.8 (2) (preprint 3.54) gives at z_0 a regular tangent germ chart D of the solution set of F (Definition 3.2.1 (2) (preprint 3.25)): a chart $D.\varphi$ of $\mathbb{R}^m \times X$ with domain W, a finite-dimensional N, a relatively open $V \subseteq N \cap C$ and an sc-smooth graph map Γ with $D.\varphi^{-1} \circ \Gamma$ parametrizing $\tilde{S} \cap W$ `Chapter3/GlobalIFTTheoremsG · theorem_3_6_8_two`.
2. *The projection.* $q = p \circ D.\varphi^{-1} \circ \Gamma : V \to \mathbb{R}^m$, p the projection to the parameter, is sc-smooth on a finite-dimensional space with the constant structure, hence C^∞ `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`.
3. *Sard on the strata.* V is relatively open in a partial quadrant of N; Sard's theorem on each stratum of the quadrant (Part IV) makes the set CV of stratified critical values of q null `Chapter4/FamiliesLocal · measure_image_stratCritSet`. Since $\partial X = \emptyset$ the points over the zeros are interior and classical Sard would do, but proving that needs the degeneracy index of the induced structure; the stratified form makes the question unnecessary.
4. *The point.* A zero $(\lambda, y) \in W$ is $D.\varphi^{-1}(\Gamma(v))$ for some $v \in V$ with $q(v) = \lambda$. Let Φ be a chart centred at y and $\psi = (r \mapsto r - \lambda) \times \varphi$ the product chart. On a relatively open $V' \subseteq V$ around v, gmap $= \psi \circ D.\varphi^{-1} \circ \Gamma$ is sc-smooth (the transition between the two charts of $\mathbb{R}^m \times X$ restricted to the overlap), and $q = \mathrm{pr}_1 \circ \mathrm{gmap} + \lambda$ there.
5. *Kernel vectors.* If $\lambda \notin CV$, $Dq(v)$ maps the stratum through v onto $\mathbb{R}^m$. The values of $D\,\mathrm{gmap}(v)$ lie in the tangent space $T_0 O$ of the product chart, and in $\ker \mathrm{Lin}_F(0)$: the principal part of F vanishes along $r \circ \mathrm{gmap}$, and the chain rule gives $\mathrm{Lin}_F(0) \circ D\,\mathrm{gmap}(v) = 0$. So for every $a \in \mathbb{R}^m$ there is a kernel vector with first component a.
6. *Conclusion.* F is transversal at (λ, y) (good position near z_0), and §V.11 gives the transversality of the slice at y.

Only range $D\,\mathrm{gmap}(v) \subseteq \ker \mathrm{Lin}_F$ is used, not "$T\tilde{S} = \ker F'$".

V.15 Theorem 5.3.5 (preprint 5.13): the assembly

The generic form `Chapter5/Transversality · theorem_5_3_5_generic`. Let $\partial X = \emptyset$, X admit bump functions, f have tame germ data and compact solution set, N be an auxiliary norm.

1. *U and part (1).* Theorem 5.2.2 (preprint 5.5) gives $U \supseteq S_f$ with $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact `Chapter5/Compactness · theorem_5_2_2_preprint`; for allowable s, Remark 5.3.4 (preprint 5.12) and closedness give S_{f+s} compact (§V.10).
2. *The section $g = f + s_0$.* For s_0 sc$^+$ with $\mathrm{supp}\, s_0 \subseteq U$ and $N(s_0) \le 1 - \varepsilon$, g has tame germ data `Chapter3/TameGermsAddG · theorem_3_1_18_tame`, a compact solution set (part (1)), and $S_g \subseteq U$ (Remark 5.3.4 (preprint 5.12)).
3. *Local families.* At every zero x of g, spanning sections $s_1^x, \dots, s_{k_x}^x$ supported in U (§V.12) make F_x transversal at $(0, x)$, and at $(0, y)$ for the zeros y in an open $U(x)$ (§V.13).

4. *One family.* Finitely many $U(x_j)$ cover the compact S_g; the families are united through an equivalence $\Sigma_j \mathrm{Fin}(k_{x_j}) \simeq \mathrm{Fin}(m)$ into $t_1, \dots, t_m$, allowable and supported in U; adding sections keeps transversality (§V.11), so F, whose slice at $\lambda = 0$ is g `Chapter5/Transversality · paramSlice_zero`, is transversal at every $(0, y)$, $y \in S_g$.
5. *Good position near the zero slice.* An open $V \supseteq \{0\} \times S_g$ of zeros of F in good position (§V.13).
6. *Allowability and compactness.* For $|\lambda| \le \delta_1 = \varepsilon/(m+1)$ the slices $s_0 + \sum \lambda_i t_i$ are allowable, and the zeros of F over $|\lambda| \le \delta_1$ form a compact Z (§V.13).
7. *The omitted step.* $Z \cap \{|\lambda| < \delta_0\} \subseteq V$ for some $\delta_0 > 0$.
8. *Sard.* With $\delta = \min(\delta_0, \delta_1)$, the zeros Z_0 of F over $|\lambda| \le \delta/2$ are compact and lie in V; the Sard step (§V.14) at each point of Z_0 gives an open W_z and a null CV_z; finitely many W_z cover Z_0, and $CV = \bigcup CV_z$ is null.
9. *The conclusions.* For $|\lambda| < \delta/2$, $\lambda \notin CV$: $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable; every zero y of $f + s_\lambda$ gives a zero $(\lambda, y) \in Z_0$, inside some W_z, so $f + s_\lambda$ is transversal at y; S_{f+s_λ} is compact by part (1); and if it is non-empty, Theorem 3.6.10 (preprint 3.56) for a transversal section over X without boundary, applied to $f + s_\lambda$ with tame germ data, makes it a sub-M-polyfold whose induced structure is tame, a manifold without boundary of locally constant dimension, with regular tangent germ data of dimension the Fredholm index at every zero `Chapter3/GlobalIFTIndexG · theorem_3_6_10_of_surjective` `Chapter5/Transversality · StrongBundle.SolutionSetIsManifold`.

The dense form `Chapter5/Transversality · theorem_5_3_5_preprint`. Given an allowable s' and $\varepsilon > 0$, let $\varepsilon' = \min(\varepsilon, 1)$ and $s_0 = (1 - \varepsilon'/2)s'$, so $N(s_0) \le 1 - \varepsilon'/2$. The generic form gives t, δ, CV; since CV is null and the ball $\{|\lambda| < \min(\delta, \varepsilon'/(2m+2))\}$ has positive measure, some λ in it avoids CV. Then $s = s_0 + \sum \lambda_i t_i$ has $N(s(x) - s'(x)) \le (\varepsilon'/2) N(s'(x)) + \sum |\lambda_i| N(t_i(x)) \le \varepsilon'$. The text's "$N(s_0(y) + s(\lambda_0, y)) \le 1 - \varepsilon/2$" is presumably meant as ≤ 1, its "$N(f(x)) \le 1$" in the statement is $N(s(x)) \le 1$, and its "the germ (f, x) is in good position" is $(f + s, x)$.

V.16 Definition 5.3.9 (preprint 5.17) and Lemma 5.26 (preprint)

Definition 5.3.9 (preprint 5.17), in a tame chart `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm`. (f, y), $f(y) = 0$, is in general position if in some tame strong bundle chart around y (Definition 3.6.3 (preprint 3.50)) `Chapter3/TameGerms · StrongBundle.IsTameChart`, at the level-1 point z over $\varphi(y)$, $\mathrm{Lin}(z)$ is onto K_z, and for every presentation T of the partial quadrant the kernel has an sc-complement B in $T_z O$ inside the reduced tangent space $T_z^R O = T_z O \cap E_z$ `Chapter2/Tame · reducedTangentSpaceWith`, on every level: $(T_z O)_m = (\ker \mathrm{Lin}(z))_m \oplus B_m$. The chart is tame because $T_x^R X$ is the reduced tangent space of a tame M-polyfold and the dimension count below needs it. General position includes transversality `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isLinSurjAt`.

The linear algebra of Lemma 5.26 (preprint) `Chapter5/Transversality · isInGoodPositionIn_of_scComplement_le_reduced`. Let r be a tame retraction, z a smooth point of the retract, and $N, B \subseteq T_z O$ with $T_z O = N \oplus B$ and $B \subseteq T_z^R O$. Then N is in good position to the cone C_z inside $T_z O$ (Definition 3.1.24 (preprint 3.21)): B is a good complement, because adding a vector of E_z does not change membership in C_z (the coordinates λ_i, $i \in I(z)$, vanish on E_z); and $N \cap C_z$ has non-empty interior in N, because the coordinates λ_i, $i \in I(z)$, map $T_z O$ onto $\mathbb{R}^{I(z)}$ — $\dim T_z O / T_z^R O = d_C(z)$ in a tame retract (Proposition 2.5.6 (preprint 2.36)) — and vanish on B, so they map N onto $\mathbb{R}^{I(z)}$, and a vector of N with all these coordinates equal to 1 is interior. The text omits the interior condition; this is the argument of Part IV for tangent spaces `Chapter4/LocalStructure · IsGeneralPositionRetractAt.isGoodPositionRetractAt`, for a subspace not given as one.

Lemma 5.26 (preprint) `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGerm`. For f sc-Fredholm and (f, y) in general position: in the chart of Definition 5.3.9 (preprint 5.17) $\mathrm{Lin}(z)$ is onto and its kernel is in good position by the linear algebra; Definition 3.6.6 (preprint 3.53) then follows `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj`. For f with tame germ data, good position holds in the tame chart of the germ data `Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGermTame` — the form the §3.6 theorems take. X itself need not be tame. (The text places the lemma after Theorem 5.25 (preprint) and calls it "clear" after Theorem 5.3.10 (preprint 5.18); it is used in Theorem 5.3.10 (preprint 5.18) and Theorem 5.25 (preprint) here.)

V.17 The bridge lemma, and $T_yS = \ker f'(y)$

Theorem 5.3.10 (preprint 5.18) applies Theorem 4.3.1 (preprint 4.23) to the parametrized solution set $\tilde{S}$, which needs $\tilde{S}$ to be a smooth submanifold in the sense of Definition 4.2.1 (preprint 4.13) — locally the image of an sc⁺-retraction. The implicit function theorem gives tangent germ data (Definition 3.2.1 (preprint 3.25)), and the text never connects the two.

sc⁺ on a finite-dimensional domain `Chapter5/GeneralPosition · isScPlus_of_isScSmooth_const`. On a finite-dimensional space with the constant structure all levels of the domain coincide, so an sc-smooth f is C^∞ into every F_m, and its lift into F^1 is sc-smooth: f is sc⁺.

The bridge lemma. Let D be regular tangent germ data for M at x (Definition 3.2.1 (2) (preprint 3.25)), with graph map $\Gamma : V \to E$ over the finite-dimensional N, projection π onto N, and r a retraction onto the chart model. Then Γ is sc⁺ on V `Chapter5/GeneralPosition · MPolyfold.TangentGermChart.Γ_isScPlus`, so $\tau = \Gamma \circ \pi \circ r$, on the set of points whose image under $\pi \circ r$ lies in V, is an sc-smooth retraction (as in the proof of Theorem 3.2.3 (preprint 3.27)) `Chapter3/TangentGerm · theorem_3_2_3_of_isRegular` which is sc⁺ (sc⁺ ⊡ sc-smooth), with $\tau \circ \Gamma = \Gamma$ on V and retract the chart image of M `Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_scPlusRetraction`. Transported to X it is an sc⁺-retraction on an open set with image $M \cap V$ `Chapter4/LocalStructure · MPolyfold.isScPlusRetractionOn_of_chart Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_isScPlusRetractionOn`. Hence a set with regular tangent germ data at each point is a smooth finite-dimensional submanifold `Chapter5/GeneralPosition · MPolyfold.isSmoothSubmanifold_of_isRegular`. For solution sets: if all zeros of f in an open W are in good position in a tame chart, Theorem 3.6.8 (2) (preprint 3.54) gives regular data at each, on charts inside W, and $S_f \cap W$ is a smooth submanifold `Chapter5/GeneralPosition · StrongBundle.isSmoothSubmanifold_solutionSet_inter`.

$T_yS \subseteq \ker f'(y)$ `Chapter5/GeneralPosition · tangentSpace_subset_linKer`. In a chart, let $Q \subseteq O$ be a retract on which the principal part $\hat{f}$ vanishes, S a retraction onto Q, z a level-1 point of Q. Then $T_zQ = \mathrm{range}\, DS(z) \subseteq T_zO$, and $\hat{f} \circ r \circ S = 0$ gives $D(\hat{f} \circ r)(z) \circ DS(z) = 0$ by the chain rule, i.e. $T_zQ \subseteq \ker \mathrm{Lin}(z)$ `Chapter5/GeneralPosition · StrongBundleChart.mem_linKer_iff`.

Equality, by counting `Chapter5/GeneralPosition · exists_tangentSpace_eq_linKer`. Let f have tame germ data and (f, x_0) be a zero in good position in a tame chart. The index form of Theorem 3.6.8 (preprint 3.54) gives an open $V \ni x_0$ whose zeros are in good position, and regular data D of a fixed dimension d at each zero, with $\dim\ker \mathrm{Lin}(z) = d$ in *every* chart `Chapter3/GlobalIFTIndexG · theorem_3_6_8_index Chapter3/GlobalIFTIndex · StrongBundle.IsSurjLinOfIndex`. On the other side, the tangent space of the bridge retraction τ at the point over $\Gamma(v)$ contains the d vectors $D\Gamma(v)b_i$, b_i a basis of N: $D\tau \circ D\Gamma(v) = D(\tau \circ \Gamma)(v) = D\Gamma(v)$, and $\pi \circ D\Gamma(v) = 1$ makes them independent `Chapter5/GeneralPosition · MPolyfold.`

TangentGermChart.exists_linearIndependent_tangentSpace. "Having d independent tangent vectors" Chapter5/GeneralPosition · HasIndepTangent is carried by germs of sc-diffeomorphisms, since $D(F \circ r)(z)$ maps T_zQ into $T_{F(z)}Q'$ and is injective on T_zO Chapter5/GeneralPosition · HasIndepTangent.transport, so it passes from the chart of D to any chart Φ through the transition maps Chapter4/Submanifolds · MPolyfold.chart_change_of_transport. A d-dimensional subspace of the d-dimensional $\ker \mathrm{Lin}(z)$ — tangent spaces of retracts are germs, so the retract $\varphi(G \cap S_f \cap V)$ of any small open G has the same tangent space — is all of it.

V.18 General position: chart independence and openness

Definition 5.3.9 (preprint 5.17) is Definition 4.2.10 (preprint 4.21) Chapter5/GeneralPosition · exists_isGeneralPositionGerm_iff. Near a zero x_0 in good position (with f of tame germ data) there is an open V such that every zero in V is in good position, $S = S_f \cap V$ is a smooth submanifold (§V.17), and at every zero $y \in V$: (f, y) is in general position iff S is in general position at y (Definition 4.2.10 (preprint 4.21), Part IV). Since $T_zS = \ker \mathrm{Lin}(z)$ in every chart, an sc-complement of $\ker \mathrm{Lin}(z)$ inside T^R_zO *is* an sc-complement of T_zS inside T^R_zO. Definition 4.2.10 (preprint 4.21) is chart independent (Part IV) Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt.isGeneralPositionChart, so Definition 5.3.9 (preprint 5.17) is too, at these zeros.

From submanifold general position to good position Chapter5/GeneralPosition · isGoodPositionGermTame_of_isGeneralPositionAt. If $f'(y)$ is onto and a smooth submanifold $A \subseteq S_f$ is in general position at y, then in the tame chart of the germ data $T_zA \subseteq \ker \mathrm{Lin}(z)$ (§V.17) Chapter5/GeneralPosition · StrongBundleChart.linKer_le and $T_zA + T^R_zO = T_zO$; so $\ker \mathrm{Lin}(z) + T^R_zO = T_zO$, the kernel has an algebraic complement inside T^R_zO Chapter5/GeneralPosition · exists_compl_le_of_sup_eq, and §V.16 gives good position.

Openness Chapter5/GeneralPosition · exists_isOpen_isGeneralPositionGerm. Let X be tame, f with tame germ data, (f, x_0) in general position. By Lemma 5.26 (preprint) it is in good position; the previous paragraph turns its general position into general position of the submanifold $S_f \cap V$ at x_0; Theorem 4.2.11 (3) (preprint 4.22) for tame X (Part IV) makes that open along the submanifold Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_preprint; Definition 4.2.10 (preprint 4.21) depends only on the germ of the submanifold, since tangent spaces of chart models agree on open sets Chapter5/GeneralPosition · MPolyfold.IsGeneralPositionAt.congr_nhds; and the equivalence turns it back into general position of (f, y) at the nearby zeros. The text uses this without proof ("they will also hold at $(0, y)$ for $y \in S_f$ close to x"; "the properties (i)–(ii) hold for all $(\lambda, y) \in \widetilde{S}$").

V.19 Properties (i)–(ii)

In a chart centred at y, with a presentation T, the properties of the proof of Theorem 5.3.10 (preprint 5.18) are Chapter5/GeneralPosition · StrongBundleChart.IsParamGP: (i) $\mathrm{Lin}_g(0)(T_0O) + \mathrm{span}\{\hat{t}_i(0)\} = K_0$ — $F'(0, y)$ onto; (ii) every $w \in T_0O$ is $u + e$ with $e \in E_0$ and $\mathrm{Lin}_g(0)u + \sum a_i\hat{t}_i(0) = 0$ for some a — $\ker F'(0, y)$ transversal to $\mathbb{R}^m \oplus T^R$. Adding sections keeps both, by linear algebra Chapter5/GeneralPosition · StrongBundleChart.IsParamGP.comp.

(i)–(ii) give general position Chapter5/GeneralPosition · isGeneralPositionGerm_paramSection_of_isParamGP. In the tame product chart $(r \mapsto r) \times \Phi$ of a tame chart centred at y Chapter5/GeneralPosition · pc0, whose tangent space at the centre is $\mathbb{R}^m \times T_0O$ Chapter5/GeneralPosition · ofProd_mem_rangeS and whose reduced tangent space is $\mathbb{R}^m \times T^R_0O$, the minimal subspace of $\mathbb{R}^m$ being everything Chapter5/GeneralPosition · minimalSubspaceWith_rnPres: (i) is the surjectivity of $F'(0, y)$; (ii) says $\ker F'(0, y) + T^R =$

$T(\mathbb{R}^m \times O)$ on level 0, so the kernel has an algebraic complement inside T^R, and §V.16 gives good position of $(F,(0,y))$. *Then* $\ker F'(0,y) = T_{(0,y)}\tilde{S}$ (§V.17), which is finite-dimensional and smooth; a finite-dimensional smooth subspace has an sc-complement inside any sc-subspace it complements algebraically `Chapter4/LocalStructure · exists_scComplement_le`, which gives Definition 5.3.9 (preprint 5.17) for F. The text's "at this point the combined system of sections already satisfies (i) and (ii)" compresses these two steps.

General position gives (i)–(ii) in every tame chart centred at y `Chapter5/GeneralPosition · isParamGP_of_isGeneralPositionGerm`: (i) is surjectivity read in the product chart; for (ii), $\tilde{S}$ is in general position at $(0,y)$ (§V.18), and in the product chart its tangent space is $\ker F'(0,y)$, which therefore has a complement inside $\mathbb{R}^m \times T^R$.

Achieving (i)–(ii) `Chapter5/GeneralPosition · exists_sections_isParamGP`. Spanning sections give (i) (§V.12) `Chapter5/GeneralPosition · spanning_of_isLinSurjAt`. For (ii), "take a linear subspace L complementing $T^R_x X$ in $T_x X$ and add sections which at x span $f'(x)L$": the sections must come from Lemma 5.3.3 (preprint 5.11), which needs *smooth* points of the fibre, so L must consist of smooth vectors. The coordinates $\lambda_{I(0)}$ map T_0O onto $\mathbb{R}^{I(0)}$ with kernel T^R_0O (Proposition 2.5.6 (preprint 2.36)), and already map the dense smooth vectors of T_0O onto it (a dense subspace of a finite-dimensional image is all of it); so there are smooth $l_i \in T_0O$ with $\lambda(l_i) = e_i$. $\mathrm{Lin}_g(0)l_i$ is smooth (Lin is an sc-operator at the smooth point), Lemma 5.3.3 (preprint 5.11) gives allowable sections through $\Phi^{-1}(0, \mathrm{Lin}_g(0)l_i)$ supported in U, and with these added every $w \in T_0O$ is $u+e$: subtract $\sum \lambda_i(w) l_i$, whose image under $\mathrm{Lin}_g(0)$ is cancelled by the new sections.

V.20 Theorem 5.3.10 (preprint 5.18)

From a family `Chapter5/GeneralPosition · theorem_5_3_10_of_family`. Let X be tame, f with tame germ data and compact solution set, N auxiliary, U from Theorem 5.2.2 (preprint 5.5). Fix s_0 with $\operatorname{supp} s_0 \subseteq U$, $N(s_0) \le 1-\varepsilon$, and allowable $t_1, \dots, t_m$ with F in general position at every $(0,x)$, $x \in S_g$, $g = f + s_0$.

1. *General position near the zero slice.* $\mathbb{R}^m \times X$ is tame `Chapter3/ParamPerturbation · prod_rn_isTame` and F has tame germ data `Chapter3/ParamPerturbationG · theorem_3_1_21_tame`, so openness (§V.18) gives, around each $(0,x)$, an open set on which the zeros of F are in good and general position; their union $W \supseteq \{0\} \times S_g$.
2. *Compactness and the omitted step* as in §V.13: the zeros Z of F over $|\lambda| \le \delta_1$ are compact, and $Z \cap \{|\lambda| < \delta_0\} \subseteq W$.
3. *The submanifold.* $A = \tilde{S} \cap W \cap \{|\lambda| < \delta\}$ is a smooth finite-dimensional submanifold of $\mathbb{R}^m \times X$ (the bridge lemma, §V.17), in general position at each of its points (Definition 4.2.10 (preprint 4.21), by §V.18 and locality), with compact closure (inside Z).
4. *Theorem 4.3.1 (preprint 4.23).* In its countable-cover form (Part IV) `Chapter4/Families · theorem_4_3_1_of_subset`, it gives a set Σ of parameters with null complement over which the fibre A_λ is a smooth submanifold of X in general position.
5. *Transversality.* The Sard step of §V.14 on the compact zeros over $|\lambda| \le \delta/2$ gives a further null set off which the slices are transversal at their zeros.
6. *The conclusions for λ.* For $|\lambda| < \delta/2$ off both null sets, with $h = f + s_\lambda$: the fibre of A over λ is S_h; h is transversal; S_h is a submanifold in general position, so (h,y) is in good position (§V.18), and then in general position by the equivalence of §V.18 applied at y together with the locality of Definition 4.2.10 (preprint 4.21); Theorem 4.2.7 (preprint 4.18) gives the corner structure and tameness of S_h

Chapter4/Submanifolds · MPolyfold.theorem_4_2_7, and Theorem 4.2.11 (2) (preprint 4.22) gives $d_{S_h} = d_X$ on S_h Chapter4/LocalStructure · MPolyfold.theorem_4_2_11_preprint. These are the conclusions collected in Chapter5/GeneralPosition · StrongBundle.SolutionSetInGeneralPosition.

The generic form Chapter5/GeneralPosition · theorem_5_3_10_param. At every zero x of g choose a tame chart centred at x with a presentation; sections with (i)–(ii) there (§V.19) make F_x general at $(0, x)$, hence (openness, §V.18) at $(0, y)$ for the zeros y in an open $U(x)$. Finitely many $U(x_j)$ cover S_g; the united family t keeps (i)–(ii) at every $(0, y)$ — read (i)–(ii) off general position in the chart of x_j, add sections, go back — so F is in general position at every $(0, y)$, and the previous theorem applies. "Replacing $f + s_0$ by f and N by cN" is not needed; the proof keeps s_0.

The dense form Chapter5/GeneralPosition · theorem_5_3_10_dense is derived exactly as for Theorem 5.3.5 (preprint 5.13) (§V.15).

The text's $\tilde{S}$ omits the equation $F = 0$, and its "$p^{-1}(\{|\lambda| \le \varepsilon\})$ is compact" is the omitted step of §V.13.

V.21 Theorem 5.25 (preprint)

Chapter5/GeneralPosition · theorem_5_25_preprint. Let X be tame with bump functions, f proper with tame germ data. Properness gives a compact solution set; Theorem 5.3.10 (preprint 5.18) with $s_0 = 0$ gives U, t, δ and a null CV Chapter5/GeneralPosition · StrongBundle.IsAuxNorm.fiberZero_eq Chapter5/GeneralPosition · sectionAdd_fiberZero; the ball $\{|\lambda| < \delta\}$ has positive measure, so some λ avoids CV, and $s = \sum \lambda_i t_i$ is allowable with $f + s$ in general position at every zero and all the conclusions of Theorem 5.3.10 (preprint 5.18); Lemma 5.26 (preprint) turns general into good position in the tame chart of the germ data (§V.16). The text's proof says "in view of Theorem 3.54 … in general position", which is the openness of §V.18 (Theorem 3.6.8 (preprint 3.54) is about good position), and writes $x \in S_f$ for $x \in S_{f+s_0}$. Here U is the neighbourhood Theorem 5.3.10 (preprint 5.18) produces; the text lets the reader choose "an associated" one.

V.22 Theorem 5.3.12 (preprint 5.19)

Chapter5/GeneralPosition · theorem_5_3_12_param. Let X admit bump functions, f have tame germ data and compact solution set, and (f, x) be in good position at the zeros $x \in \partial X$. The text's "as before" applies the implicit function theorem to F at the boundary zeros $(0, x)$, where all t_i vanish and $\ker F'(0, x) = \mathbb{R}^m \times \ker f'(x)$; this is in good position to $\mathbb{R}^m \times C_x$ only if $\ker f'(x) + E_x \supseteq T_x X$, a general position condition that good position does not imply (Finding 88, below). The formal proof avoids those points:

1. *Near the boundary zeros.* Theorem 3.6.8 (1) (preprint 3.54) at each zero of f on ∂X gives an open set of zeros in good position Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8_one'; their union V_{bd}, and an open V_1 with $S_f \cap \partial X \subseteq V_1$ and $\operatorname{cl} V_1 \subseteq V_{bd}$.
2. *Away from them.* The zeros of f outside V_1 are not on ∂X, so interior ($d = 0$). Spanning sections at these zeros are taken with supports in $U \setminus (\partial X \cup \operatorname{cl} V_1)$, an open set containing them; finitely many families cover the compact $S_f \setminus V_1$, and F is onto at every $(0, y)$, $y \in S_f$ outside V_1.
3. *Good position of F off the boundary.* At the interior points good position is surjectivity Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt; Theorem 3.6.8 (1) (preprint 3.54) for F gives an open $W \supseteq \{0\} \times (S_f \setminus V_1)$ of zeros in good position, off $\mathbb{R}^m \times \partial X$.
4. *Compactness and Sard* as in §V.15 for the zeros of F over small λ outside $\mathbb{R}^m \times V_1$.

5. *The conclusions.* For small λ off the null set, $h = f + s_\lambda$. On V_1 all t_i vanish, so $h = f$ there, and the good position of (f, y) at the zeros in V_1 passes to (h, y) because good position depends only on the germ of the section `Chapter3/Chapter5BridgeG · StrongBundle.IsGoodPositionGermTame.congr_nhds`. Off V_1 the zeros of h are interior, h is transversal there by the Sard step, hence in good position. Theorem 3.6.11 (preprint 3.57) makes S_h, if non-empty, a sub-M-polyfold whose induced structure is tame and a manifold with boundary with corners `Chapter3/GlobalIFTTheoremsG · theorem_3_6_11`.

$s_\lambda = \sum \lambda_i t_i$ has support disjoint from ∂X, so it vanishes on the open neighbourhood $X \setminus \operatorname{supp} s_\lambda$ of ∂X, and it is "arbitrarily small" as $\lambda \to 0$. The text's tameness of X is not used, and its "(f, x) in good position for every $x \in S_{f+s}$" is $(f + s, x)$. The dense form is §V.34.

Finding 88 in Lean (gap G.46). The smallest instance `Chapter3/Counterexamples2021 · Counterexamples2021.theorem_5_3_12_goodPosition_not_stable`: the line $\mathbb{R}(1, 1)$ is in good position to the corner $[0, \infty)^2$, with good complement $\mathbb{R}(1, -1)$ `Chapter3/Counterexamples2021 · Counterexamples2021.isGoodComplement_diag_antidiag` `Chapter3/Counterexamples2021 · Counterexamples2021.isInGoodPosition_diag`, also with the sc-complement of the definition as written `Chapter3/Counterexamples2021 · Counterexamples2021.isInGoodPositionSc_diag`; but $\mathbb{R} \times \mathbb{R}(1, 1)$ is not in good position to $\mathbb{R} \times [0, \infty)^2$, not even with an algebraic complement `Chapter3/Counterexamples2021 · Counterexamples2021.not_isInGoodPosition_prod_diag`. A good complement Y would have $(a, 0) + y \in \mathbb{R} \times [0, \infty)^2$ for $y \in Y$ and $|a|$ large, so the second components of Y lie in $[0, \infty)^2 \cap -[0, \infty)^2 = 0$, $Y \subseteq \mathbb{R} \times 0$, and $(0, (1, 0))$ is not in $\mathbb{R} \times \mathbb{R}(1, 1) + Y$. So the step "as before" — the implicit function theorem for good position applied to F at the boundary zeros — does not hold as we read it; the theorem does, by the route above.

V.23 Definition 5.3.13 (preprint 5.20), and Lemmas 5.3.16 and 5.3.17 (preprint 5.23 and 5.24)

Definition 5.3.13 (preprint 5.20) `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy`. X tame, f with tame germ data and compact solution set, $t : X \to \mathbb{R}$ sc-smooth with $t(X) = [0, 1]$, and for every $\tau \in [0, 1]$: $X_\tau = t^{-1}(\tau)$ a sub-M-polyfold (Definition 2.5.7 (2017: 2.19)) with tame induced structure, on which the restriction $f|X_\tau$ — the section $a \mapsto (a, f(a))$ of the pull-back of P along the inclusion `Chapter5/Homotopy · restrictSection` `Chapter3/ParamPerturbation · StrongBundle.pullback`, the inclusion being sc-smooth (Proposition 2.3.19 (preprint 2.21)) — has tame germ data. The text never says what $Y|X_t$ is; the pull-back of Proposition 2.6.8 (preprint 2.54) is the strong bundle structure available. The end problems are the restrictions to $t^{-1}(0)$ and $t^{-1}(1)$ (or problems isomorphic to them, Remark 5.3.14 (preprint 5.21), §V.35); their solution sets are $S_f \cap X_\tau$ `Chapter5/Homotopy · isZeroAt_restrictSection_iff`, closed in the compact S_f, hence compact `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy.isCompact_solutionSet_restrict`.

Lemma 5.3.16 (preprint 5.23) and the hidden hypothesis. The proof of Theorem 5.3.15 (preprint 5.22) fixes a critical point x of $t|S$ and says "since x is a smooth point, we find a one-dimensional smooth linear sc-subspace $Z \subset T_x X$ such that $T_x X = Z \oplus \ker(dt(x))$". For $dt(x) = 0$, $\ker dt(x) = T_x X$ and the only complement is 0: a non-zero complement exists only if $\ell \neq 0$ `Chapter5/Homotopy · ne_zero_of_isCompl_ker`. Theorem 5.3.15 (preprint 5.22) does not assume that t is a submersion at the critical points of $t|S$, so this is a hidden hypothesis of the preprint. The 2021 condition (3) supplies it (§V.32), and Theorem 5.3.15 is proved in §§V.36–V.41. When $\ell \neq 0$, some vector e of any spanning set (e.g. the dense smooth vectors) has $\ell(e) \neq 0$, and $\mathbb{R}e$ complements $\ker \ell$ `Chapter5/Homotopy · exists_isCompl_ker_of_ne_zero`. Lemma 5.3.16 (preprint 5.23) is then linear algebra: $\ell|_Z$ is an isomorphism onto $\mathbb{R}$, so $b = (\ell|_Z)^{-1} \circ \tau$ is the unique

$b : S \to Z$ with $\ell \circ b = \tau$, and adding a with values in $\ker \ell$ does not change $\ell \circ b$ `Chapter5/Homotopy · lemma_5_3_16_preprint`.

The calculus. At a level-1 point z, $D(a \cdot w)(z)v = Da(z)v \cdot w(z) + a(z)Dw(z)v$ for sc-smooth $a : U \to \mathbb{R}$, $w : U \to F$ `Chapter5/Homotopy · scDeriv_smul_apply`, and $D(\ell(\rho(\cdot) - p))(z) = \ell \circ D\rho(z)$ `Chapter5/Homotopy · scDeriv_clm_sub_apply`. The function $g = \beta \cdot \ell(\varphi(\cdot) - \varphi(x))$ on $\operatorname{dom}\varphi$, 0 outside `Chapter5/Homotopy · chartLinCutoff`, is sc-smooth when β is with $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$ `Chapter5/Homotopy · isScSmoothMap_chartLinCutoff`.

Lemma 5.3.17 (preprint 5.24). *The construction* `Chapter5/Homotopy · exists_scPlusSection_linAt`: for a chart Φ around the smooth point x, a smooth point e over x, a continuous linear ℓ on the model space and $U \ni x$, let s_e be the sc⁺-section through e of Lemma 5.3.3 (preprint 5.11), supported in $U \cap \operatorname{dom}\varphi$, β a bump with $\beta \equiv 1$ near x and support in the chart domain, and $s = g \cdot s_e$ with g as above. Then s is sc⁺ (§V.9), $s(x) = g(x)e = 0$ `Chapter5/Homotopy · principal_sectionSmul`, $\operatorname{supp} s \subseteq U$ `Chapter5/Homotopy · sectionSupport_sectionSmul_subset`, and in the chart, by the product rule, $\mathrm{Lin}_s(z)v = D(g \circ \varphi^{-1} \circ r)(z)v \cdot \hat{s}_e(\varphi(x)) + g(x)\mathrm{Lin}_{s_e}(z)v = \ell(Dr(z)v)\Phi(e)_2 = \ell(v)\Phi(e)_2$ for $v \in T_zO$, since $g \circ \varphi^{-1} = \ell(\cdot - \varphi(x))$ near $\varphi(x)$ `Chapter5/Homotopy · chartLinCutoff_of_mem` and $Dr(z) = 1$ on T_zO. *The lemma* `Chapter5/Homotopy · lemma_5_3_17`: for a finite-dimensional $K \subseteq T_zO$ (the text's $T_xS = \ker f'(x)$) and a linear $b : K \to T_zO$ with $\mathrm{Lin}_f(z)(bv) = \lambda(v)\Phi(e)_2$, extend $-\lambda$ (linear on the finite-dimensional K) to a continuous linear functional ℓ on the model space (Hahn–Banach) and apply the construction: $\mathrm{Lin}_f(z)(bv) + \mathrm{Lin}_s(z)v = \lambda(v)\Phi(e)_2 - \lambda(v)\Phi(e)_2 = 0$. The text's graph of S over T_xS, its complement L with $j : L \to \mathbb{R}$, and its case "if this operator is the zero operator we can take $s = 0$" are not needed. That $f'(x) \circ b_\tau$ has its image in the line of a *smooth* point is a hypothesis; the text's justification ("the image of any vector in T_xS belongs to Y_∞") is not formalized; the proof of Theorem 5.3.15 in §V.41 does not need it, because it takes the vector of the section's value at a smooth point directly.

V.24 Bump functions in local models: Theorem 5.5.7 (preprint 5.34) and Proposition 5.5.8 (preprint 5.35)

Definition 5.5.6 (preprint 5.33) `Chapter5/BumpFunctions · LocalModel.HasBumpProperty`. For $x \in O$ and W open in E_0 with $\mathrm{cl}(O \cap W) \subseteq O$, an sc-smooth f on O with $f(x) \neq 0$ (as in Definition 5.3.1 (preprint 5.9)) and support in O inside W.

A retract is closed near its points `Chapter5/BumpFunctions · closure_inter_ball_subset`. If $B(z,r) \subseteq W'$ and $U = C \cap W'$ is the domain of a retraction ρ onto O, then $O \cap \mathrm{cl}\,B(z, r/2) \subseteq O$: a limit of points of O in the closed half ball lies in C (C closed) and in $B(z,r)$, hence in U, where ρ is continuous, and it is a fixed point of ρ.

Theorem 5.5.7 (preprint 5.34) `Chapter5/BumpFunctions · theorem_5_5_7`. "□ ": every chart's model has the property — given x, W, a bump of X at $\varphi^{-1}(x)$ supported in $\varphi^{-1}(O \cap W)$, read in the chart. "□ ", "clearly" in the text, hides a support condition: a local bump f_0 with support in $U(x)$, $\mathrm{cl}_E(U(x)) \subseteq O$, gives $f = f_0 \circ \varphi$ on the chart domain, but $\mathrm{cl}_X\{f \neq 0\}$ may leave the domain. For $X = E$ a Hilbert space, the chart $\varphi(u) = u/(1 - |u|_0^2)$ from the open unit ball onto $O = E$, $U(x) = O$ and $f_0 \equiv 1$, the extension by 0 is the indicator of the ball. The proof chooses $U_1 \ni x$ with $\mathrm{cl}_X U_1 \subseteq U \cap \operatorname{dom}\varphi$ (regularity of X) and r with $B(\varphi(x), r) \cap O \subseteq \varphi(U_1)$, takes the local bump in $W = B(\varphi(x), r/2)$ — admissible by the previous paragraph — and extends $f_0 \circ \varphi$ by 0: its support lies in $\mathrm{cl}_X U_1 \subseteq U$, and it is sc-smooth because it vanishes on the open complement of its support and is sc-smooth in the chart (§V.8).

Proposition 5.5.8 (preprint 5.35). If $\kappa\|v\|^2 \le b(v,v)$ for a continuous bilinear b on E_0, then $f(v) = \tau(1 - b(v-x, v-x)/\rho)$ is C^∞ on E_0, equal to 1 at x and supported in a small ball; it is sc-smooth on E by Corollary 1.2.5 (preprint 1.20) from the base level `Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional`, and so is its composition with any retraction onto O `Chapter5/BumpFunctions · LocalModel.hasBumpProperty_of_quadratic`. "E_0 is a Hilbert space" is stated as $\|v\|^2 = b(v,v)$ `Chapter5/BumpFunctions · proposition_5_5_8`: an inner product structure on E_0 would be a second normed structure next to that of the sc-scale. Translating by a non-smooth x is legitimate here because f is C^∞ on E_0 itself; for a merely sc-smooth function it would not be. With Theorem 5.5.7 (preprint 5.34): an M-polyfold covered by charts on sc-Hilbert spaces admits sc-smooth bump functions, the remark before Definition 5.3.1 (preprint 5.9) `Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert`.

V.25 Partitions of unity: Theorem 5.5.10 (preprint 5.37) and Proposition 5.5.11 (preprint 5.38)

The algebra of sc-smooth functions. Sums, differences and products of sc-smooth functions are sc-smooth, on retracts `Chapter5/PartitionsOfUnity · IsScSmoothRetractMap.real_add` `Chapter5/PartitionsOfUnity · IsScSmoothRetractMap.real_mul` and on X `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_add` `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_sub` `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_mul`, and so are finite sums and products `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsetSum` `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsetProd`. Sc-smoothness is local `Chapter5/PartitionsOfUnity · isScSmoothMap_real_of_eventuallyEq`, so a locally finite sum, which near each point is a finite sum, is sc-smooth `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsum`; and a function supported in a chart domain with sc-smooth representative is sc-smooth on X `Chapter5/PartitionsOfUnity · isScSmoothMap_real_of_chart`.

Definitions 5.5.1 and 5.5.9 (preprint 5.28 and 5.36). A partition of unity is Mathlib's (continuous $\rho_i \ge 0$, locally finite supports, $\sum \rho_i = 1$), sc-smooth if every ρ_i is `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothPartitionOfUnity`; X admits sc-smooth partitions of unity if every open cover has a subordinate one `Chapter5/PartitionsOfUnity · MPolyfold.AdmitsScPartitionsOfUnity`. The approximation property: for $V = O \cap W$ with $\mathrm{cl}\, V \subseteq O$, continuous $f : O \to [0,1]$ supported in V and $\varepsilon > 0$, an sc-smooth $g : O \to [0,1]$ supported in V with $|f - g| \le \varepsilon$ `Chapter5/PartitionsOfUnity · LocalModel.HasApproxProperty`. Partitions of unity give bump functions: the function attached to U of a partition subordinate to $\{U, X \setminus \{x\}\}$ is 1 at x `Chapter5/PartitionsOfUnity · MPolyfold.AdmitsScPartitionsOfUnity.admitsBumpFunctions` — the easy half of the open problem the text states after Definition 5.5.2 (preprint 5.29).

"(i) ⇒ (ii)" `Chapter5/PartitionsOfUnity · hasApproxProperty_restrict`. The text calls $f \circ \varphi$ "sc-smooth" (f is only continuous) and asserts $\mathrm{cl}_X(\varphi^{-1}(V)) \subseteq U$; with the chart of the unit ball above, $V = O$ and $f \equiv 1$, the extension by 0 is not continuous. Since (ii) asks only for *some* atlas, the proof restricts each chart to an open s with $\mathrm{cl}_X(s) \subseteq \mathrm{dom}\, \varphi$. Then for closed $A \subseteq \varphi(s)$ the set $\varphi^{-1}(A)$ is closed in X `Chapter5/PartitionsOfUnity · MChart.isClosed_inter_preimage`, so $F = f \circ \varphi$ extended by 0 is continuous. Cover X by $\{F < \varepsilon/2\}$ and the sets $U_x = \{|F - F(x)| < \varepsilon/2\} \cap \varphi^{-1}(V)$, take a subordinate sc-smooth partition of unity (β_0, β_x), and let $\hat{g} = \sum_x \beta_x F(x)$: a locally finite sum of sc-smooth functions with values in $[0,1]$ `Chapter5/PartitionsOfUnity · PartitionOfUnity.finsum_mul_mem_Icc`, supported in $\varphi^{-1}(V)$ `Chapter5/PartitionsOfUnity · PartitionOfUnity.tsupport_finsum_mul_subset`, and within $\varepsilon/2$ of F, since every $F(x)$ with $\beta_x(y) \ne 0$ is `Chapter5/PartitionsOfUnity · PartitionOfUnity.abs_sub_finsum_mul_le`. Every chart has the property, by a different route (§V.42).

"(ii) □ (i)". Every point of a chart domain has small open neighbourhoods N with $\mathrm{cl}_X N \subseteq \mathrm{dom}\,\varphi$ and $\mathrm{cl}\,\varphi(N) \subseteq O$ — half a ball inside the domain of a retraction, closed by §V.24 `Chapter5/PartitionsOfUnity · MChart.exists_small_nhds` — and the approximation property transports to continuous functions on X supported in such N `Chapter5/PartitionsOfUnity · LocalModel.HasApproxProperty.exists_approx`. *Sc-smooth Urysohn functions* `Chapter5/PartitionsOfUnity · exists_isScSmoothMap_urysohn`: for disjoint closed s, t, cover t by small neighbourhoods W_x inside $X \setminus s$, take continuous $f_x = 1$ near x supported in W_x, a locally finite refinement, approximations $\hat{g}_x$ within $1/2$ — so $\hat{g}_x \geq 1/2$ where $f_x = 1$ — and $F = \tau(4\sum \hat{g}_x - 1)$, sc-smooth, 0 on s and 1 on t. Then Mathlib's bump coverings produce a covering indexed by the given cover with sc-smooth functions `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate_of_prop`, and $\rho_i = f_i \prod_{j<i}(1 - f_j)$, locally a finite product, is the partition of unity `Mathlib/Topology/PartitionOfUnity · BumpCovering.toPartitionOfUnity` `Chapter5/PartitionsOfUnity · admitsScPartitionsOfUnity_of_approx`. This replaces the text's refinement "with the same index set", which does not exist in general (the cover $\{X\}$); metrizability is not needed.

Theorem 5.5.10 (preprint 5.37) `Chapter5/PartitionsOfUnity · theorem_5_5_10` combines the two directions, "atlas" read as a family of charts covering X.

Proposition 5.5.11 (preprint 5.38) `Chapter5/PartitionsOfUnity · proposition_5_5_11` is not "an immediate consequence of Theorem 5.37": that theorem produces some atlas, not the given model, and applied to O would need partitions of unity on an open subset of C. The proof repeats "(i) □ (ii)" in E_0: $A = \{x \in O \mid f(x) \geq \varepsilon/2\}$ is closed in E_0 because $\mathrm{cl}\,V \subseteq O$; $E_0 \setminus A$ and balls $B(x, r_x) \subseteq W$, $x \in V$, on which f varies by less than $\varepsilon/2$, cover E_0; a C^∞ partition of unity of E_0 gives $g = \sum \beta_x f(x)$, C^∞ on E_0, hence sc-smooth (Corollary 1.2.5 (preprint 1.20)), composed with a retraction onto O.

V.26 Banach space geometry: Definitions 5.5.4, 5.5.12, 5.5.14 (preprint 5.31, 5.39, 5.41), Proposition 5.5.15 (preprint 5.42), Corollary 5.5.17 (preprint 5.44)

A Banach space admits a C^k-bump function if some C^k function, not identically zero, has bounded support `Chapter5/PartitionsOfUnity · HasContDiffBump`, and C^k partitions of unity if every open cover has a subordinate C^k one `Chapter5/PartitionsOfUnity · AdmitsContDiffPartitionsOfUnity`. $c_0(\Gamma)$ is the subspace of $\ell^\infty(\Gamma)$ of functions tending to 0 along the cofinite filter `Chapter5/PartitionsOfUnity · cZero`, equivalently with $\{|f(\gamma)| > \varepsilon\}$ finite for every ε `Chapter5/PartitionsOfUnity · mem_cZero_iff`, normed by the supremum `Chapter5/PartitionsOfUnity · norm_cZero`; coordinate-wise C^k embeddings are as in the text `Chapter5/PartitionsOfUnity · IsCoordinatewiseContDiffEmbedding`, and Toruńczyk's Theorem 5.5.13 (preprint 5.40) is recorded as a statement `Chapter5/PartitionsOfUnity · TorunczykStatement`, proved in §V.45. WCG means a weakly compact set with dense span `Chapter5/PartitionsOfUnity · IsWCG`; reflexive, used in Proposition 5.5.15 (preprint 5.42) and not defined in the text, means $V \to V''$ onto `Chapter5/PartitionsOfUnity · IsReflexive`.

Proposition 5.5.15 (preprint 5.42). (i) For reflexive V the weak closure of the closed unit ball is weakly compact — Kakutani's theorem, in Mathlib as Banach–Alaoglu in V'' through the embedding of (V, weak) into $(V'', \text{weak-*})$ `Mathlib/Analysis/Normed/Module/DoubleDual · NormedSpace.isCompact_closure_of_isBounded` — and its span contains the ball, hence is V `Chapter5/PartitionsOfUnity · span_closedBall_eq_top` `Chapter5/PartitionsOfUnity · proposition_5_5_15_reflexive`. (ii) For separable V, a dense sequence (y_n) scaled by $1/((n+1)(|y_n|+1))$ is a null sequence; with 0 it is norm compact, hence weakly compact, and its span contains every y_n `Chapter5/PartitionsOfUnity · proposition_5_5_15_separable`. The text's "$\{0\} \cup \{x_n/n\}$" has

its index range missing.

Corollary 5.5.17 (preprint 5.44) `Chapter5/PartitionsOfUnity · corollary_5_5_17`. Let $\|v\|^2 = b(v,v)$ on E_0, and assume Theorem 5.5.16 (preprint 5.43) for E_0 and $k = \infty$, recorded as a statement `Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement`. A Hilbert space is reflexive: the parallelogram law holds, so the norm comes from an inner product `Mathlib/Analysis/InnerProductSpace/OfNorm · InnerProductSpace.ofNorm`, and every $\phi \in V''$ is evaluation at the Riesz representative of $x \mapsto \phi(\langle x, \cdot\rangle)$ `Chapter5/PartitionsOfUnity · isReflexive_of_hilbert`; hence WCG by Proposition 5.5.15 (i) (preprint 5.42). It has the C^∞-bump $\tau(1 - b(v,v))$ `Chapter5/PartitionsOfUnity · hasContDiffBump_of_hilbert`. Theorem 5.5.16 (preprint 5.43) gives C^∞ partitions of unity of E_0, and Proposition 5.5.11 (preprint 5.38) — not Theorem 5.5.10 (preprint 5.37), as the text says — the approximation property. With Theorem 5.5.10 (preprint 5.37), an M-polyfold on such charts admits sc-smooth partitions of unity `Chapter5/PartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert`. The hypothesis is removed in §V.46: E_0 is separable, and Theorem 5.5.16 (preprint 5.43) is proved for separable spaces there and for every WCG space in §V.47.

V.27 What is not formalized

Of the chapter's gaps, G.41, G.42, G.44–G.48 and G.50–G.53 are closed, G.41 for each fixed i and G.42 with a correction; G.43 and G.49 have nothing to formalize. What remains:

- **Corollary 5.2.5 (preprint 5.8) as one statement for all i.** It is proved for each fixed i (§V.29; gap G.41 closed in the fourth addendum). Not formalized: a family P^i indexed by i, and the tame form of the conclusion, which needs the tameness of the index raises of tame charts. The identification of the iterated base $(X^1)^1$ with X_2 as a topological space is proved since (§II.32; gap G.6, closed). Condition (3) of a filling on level 1 does not follow from level 0 alone — on $E_m = \{x \mid \sum e^{2mk} x_k^2 < \infty\}$ the right shift T makes $1 - T/2$ invertible on E_0 while $(1-T/2)^{-1} e_0 = \sum 2^{-n} e_n \notin E_1$ — so the corollary's proof needs the regularity argument of §V.29, through Lemma 3.1.20 (preprint 3.17) `Chapter3/Filler · lemma_3_1_20`.
- **The non-completeness of $(\Gamma^{+,1}_U(P), \rho)$ in the text's setting**, beyond the trivial bundle, and the identification of the completion with the sections of Remark 5.3.4 (preprint 5.12) (§V.30; gap G.42).
- **Weak against pointwise bump functions at a non-smooth point** (§V.31).
- **Theorem 5.3.15 (preprint 5.22)**, proved (§§V.36–V.41; gap G.48), leaves the comparison of the smooth parametrizations through which "Morse-type" is read with the charts of the Mathlib manifold structure on the solution set. That the complement Z of Lemma 5.3.16 (preprint 5.23) can be chosen smooth is not formalized; the formal proof does not use Lemmas 5.3.16 and 5.3.17 (preprint 5.23 and 5.24), proved in §§V.23 and V.32.
- **Remark 5.3.8 (preprint 5.16)** is formalized on $\mathbb{R}^1 \times X$ with the time window $[0,1]$, not on $[0,1] \times X$, which would need a tame structure on the product with an interval (§V.33).
- **Theorem 5.27 (preprint; not in the 2021 edition)** (gap G.49, nothing to formalize in the 2021 edition; the preprint leaves the proof to the reader). Its general-position analogue is §V.33. A good-position form at the corner points $\{i\} \times \partial X$ would need the step that Finding 88 shows not to hold as we read it `Chapter3/Counterexamples2021 · Counterexamples2021.not_isInGoodPosition_prod_diag`; no counterexample to the theorem itself is built (Findings, Chapter 5, items 104, 186).
- **The Sobolev sc-structure on other domains** (row V5.76, narrower): compactness of the inclusions $W^{k+1,p}(\Omega) \to W^{k,p}(\Omega)$ and density of E_∞ are proved for $\Omega = (0,1)$ and every $p > 1$; for cubes in $\mathbb{R}^n$, $n \geq 2$, and other domains they are hypotheses (§V.44). The bump claims hold for every Ω.

- **The 2021 remark that the notions of Definition 5.2.1 differ for general sc-smooth sections** (§V.32).
- **Remark 5.3.6 (preprint 5.14)** (gap G.43): nothing to formalize; the formalization assumes neither separable sc-Hilbert fibres nor further properties of auxiliary norms.

V.28 What the formalization taught us

The tameness hypothesis. The first formal versions of Theorem 5.3.5 (preprint 5.13) and Lemma 5.26 (preprint) assumed, as the §3.6 theorems then did, that the base retraction of *every* compatible strong bundle chart — and of every chart of the pull-backs to $\mathbb{R}^k \times X$ — is tame. No strong bundle over a nonempty base satisfies this (a compatible chart $(1, \varphi)$ with base retraction $(a, u) \mapsto (1, r(u))$ on $[0, \infty) \times U$ is never tame), so those statements were vacuous. The repair is the text's own Definitions 3.6.3, 3.6.4, 3.6.6 (preprint 3.50, 3.51, 3.53) in a tame chart: the perturbation theorems take f with tame germ data, the parametrized sections inherit them through the tame product chart, and Definition 5.3.9 (preprint 5.17) lives in a tame chart, where Lemma 5.26's (preprint) count $\dim T_zO/T_z^RO = d_C(z)$ holds.

What "it is easily verified" costs. The perturbation theorems of §5.3 are sketches. Their formal proofs needed the density of smooth vectors (§V.12), the compactness of the zeros over a small parameter ball (§V.13), a local, stratified Sard step (§V.14), the bridge from tangent germ data to Definition 4.2.1 (preprint 4.13) (§V.17), $T_yS = \ker f'(y)$ by a dimension count, which gives the chart independence of Definition 5.3.9 (preprint 5.17) and the openness of general position (§§V.17–V.18), and, in Theorem 5.3.12 (preprint 5.19), a route around the implicit function theorem at boundary points where its hypothesis fails (§V.22).

Theorem 5.3.15 without a trivialization. The text's proof moves the solution set S along a trivialization $\Phi_\lambda : S \to S_\lambda$ and differentiates the equation twice. The formal proof takes the perturbations of the form $a_i\sigma_{g(i)}$, so that every S_λ is a level set inside one fixed solution manifold of $f + \sum \mu_j\sigma_j$, and replaces the trivialization by Lagrange multipliers and the bordered Hessian (§§V.38–V.39). The "parameterized version of Sard's theorem" is the inverse function theorem plus Sard's lemma in equal dimensions (§V.38). When the families of several critical points are united, the transversality obtained at one of them must survive the others; the bounded coordinates $\arctan(q_i)/2$ and the constant 1 make it do so, because a multiplier of the larger family restricts to a non-zero multiplier of the smaller one (§V.39; Findings, Chapter 5, item 183).

The corner case is no case. The boundary of the solution set might have corners where $t|S$ is not regular. It cannot: at a zero on ∂X, t takes the value 0 or 1 and is an extremum, so $dt(x)$ vanishes on the two-sided directions $E_x \supseteq T_x^RX$; general position and $dt(x) \neq 0$ then make $dt(x)$ non-zero on $\ker f'(x)$, at every degeneracy index (§V.41).

The hidden hypothesis is the 2021 condition. The hypothesis $dt(x) \neq 0$ that §V.23 found missing from the preprint is Definition 5.3.13 (3) of the 2021 edition, and with it Theorem 5.3.15 needs nothing beyond the text (§§V.32, V.36).

Finding 88 is machine-checked. Good position is not stable under $\mathbb{R} \times \cdot$ (§V.22), so the implicit function theorem cannot be applied "as before" at the boundary zeros of the parametrized section; Theorem 5.3.12, its dense form and the cobordisms of §V.33 are proved on routes that avoid those points.

E_0 is always separable. The inclusion $E_1 \to E_0$ of an sc-Banach space is compact with dense range, so E_0 is separable (§V.46). In the sc-calculus, Theorem 5.5.16 (preprint 5.43) is needed only for separable spaces, where it is elementary, and Corollary 5.5.17 (preprint 5.44) holds without it. The general

WCG case was proved anyway, without Krein's theorem, the Eberlein–Šmulian theorem or a transfinite recursion: Mazur's theorem and finite character of a closure operator suffice (§V.47).

Every chart, by shells. The text's derivation of the approximation property in a chart uses $\mathrm{cl}_X(W) \subseteq U$, which does not follow from $\mathrm{cl}_C(V) \subseteq O$ (§V.25; Findings, Chapter 5, item 111). Exhausting the chart domain by shells $\{1/(n+2) \le d \le 1/(n+1)\}$ and summing local approximations gives the property for every chart (§V.42).

The Sobolev remark lives on level 0. Its bump claims use only $E_0 = W^{1,4}(\Omega)$ or $W^{1,3/2}(\Omega)$: a C^k bump on E_0 is sc^k on any sc-structure. The sc-structure $E_m = W^{1+m,p}(\Omega)$ itself — Rellich compactness and density of smooth functions — is a separate matter, proved here for $(0,1)$ (§V.44).

V.29 Corollary 5.2.5 (preprint 5.8) on the shifted strong bundle

Corollary 5.2.5 (preprint 5.8) says that, for an sc-Fredholm section f with compact solution set, f^i is an sc-Fredholm section of $P^i : Y^i \to X^i$ with compact solution set, and that "it is a priori clear that f^i is sc-Fredholm". The text defines $Y[i]$ for $i = 0, 1$ only, never Y^i, and "a priori clear" hides the filler condition on level 1 (§V.7). Both are supplied here (Findings, Chapter 5, items 140–146).

The reading of Y^1. $Y^1 = Y_{1,1}$ with $(Y^1)_{m,k} = Y_{m+1,k+1}$ — the convention of E^1 and of $(U \lhd F)^1 = U^1 \lhd F^1$ — not $Y[1] = Y_{0,1}$ and not $Y_{1,2}$. It is forced by "f^i is obtained by raising the index": an sc-smooth section maps X_1 into $Y_{1,1}$, and in general not into $Y_{1,2}$.

Fibres of a strong bundle. Through a chart, $P^{-1}(x)$ is linearly homeomorphic to $K_{\varphi(x)} = \{h \mid \Gamma(\varphi(x), h) = h\}$ `Chapter5/StrongBundleShift · StrongBundleChart.fiberK` `Chapter5/StrongBundleShift · StrongBundleChart.fiberCLE`, the fixed points of a continuous projection, a closed subspace of the fibre model `Chapter5/StrongBundleShift · StrongBundleChart.isClosed_fiberK`; so every fibre is a Banach space `Chapter5/StrongBundleShift · StrongBundleChart.isBanachable_fiber`, which Definition 2.6.5 (preprint 2.51) asks of a strong bundle.

The levels of X^1, and raising the index of maps. $(X^1)_m = X_{m+1}$ and $(X^1)_\infty = X_\infty$ as sets `Chapter5/StrongBundleShift · MPolyfold.mem_shift_levelSet_iff` `Chapter5/StrongBundleShift · MPolyfold.mem_shift_smoothSet_iff` (their topologies agree too, §II.32; gap G.6, closed). An sc-smooth map $f : X \to Z$ between M-polyfolds induces an sc-smooth $f_1 : X^1 \to Z^1$: continuity, and in the level charts the local representatives are the level-1 maps of those of f, sc-smooth by Proposition 1.2.2 (preprint 1.17) for retracts `Chapter5/StrongBundleShift · MPolyfold.levelChart_apply_levelOne` `Chapter5/StrongBundleShift · MPolyfold.IsScSmoothMap.levelOne`.

The shifted strong bundle `Chapter5/StrongBundleShift · StrongBundle.shift`. Its total space is $Y_{1,1}$, the level 1 of the M-polyfold $Y[0]$ with its level topology, Hausdorff and paracompact by Theorem 2.3.10 (preprint 2.16) `Chapter5/StrongBundleShift · StrongBundle.ShiftSpace`; the projection is $P^1 = P|_{Y_{1,1}}$ `Chapter5/StrongBundleShift · StrongBundle.shiftProj`, continuous and onto `Chapter5/StrongBundleShift · StrongBundle.continuous_shiftProj` `Chapter5/StrongBundleShift · StrongBundle.surjective_shiftProj`; the fibre $P^{-1}(x) \cap Y_{1,1}$ is a linear subspace — in a chart it is the preimage of F_1 under the linear fibre map `Chapter5/StrongBundleShift · StrongBundle.shiftFiberSubmodule` — carries the topology of F_1, not that of $P^{-1}(x)$, and is a Banach space, a closed subspace of F_1 through a chart `Chapter5/StrongBundleShift · StrongBundle.banachable_shift`. The atlas consists of the *index raises* of the strong bundle charts `Chapter5/StrongBundleShift · StrongBundleChart.shift` `Chapter5/StrongBundleShift · StrongBundle.shiftAtlas`: base chart the level chart $\varphi^1 : V_1 \to O_1$, fibre model F^1, retraction R on $U^1 \lhd F^1$

Chapter5/StrongBundleShift · IsStrongBundleRetraction.shift, bundle chart the level chart of Φ in $Y[0]$. Compatible charts have compatible index raises Chapter5/StrongBundleShift · StrongBundleChart.Compatible.shift: on $[0]$ the transitions are level charts of compatible charts, on $[1]$ the transition $(\Phi'^1 \circ (\Phi^1)^{-1})[1]$ is the level-1 map of $(\Phi' \circ \Phi^{-1})[1]$ (Proposition 1.2.2 (preprint 1.17) for retracts) Chapter5/StrongBundleShift · StrongBundleChart.shift_transition_one, and the F_1-condition of Definition 2.6.4 (preprint 2.50) holds because the $[1]$-transition is sc^0 Chapter5/StrongBundleShift · StrongBundleChart.shift_compatible_range. Checks: $(Y^1)_{m,k} = Y_{m+1,k+1}$ Chapter5/StrongBundleShift · StrongBundle.mem_shift_doubleLevel_iff, the M-polyfold $Y^1[0]$ is $Y[0]^1$ Chapter5/StrongBundleShift · StrongBundle.isChart_shift_toMPolyfold0_iff, and the index raise of every chart of P is a chart of P^1 Chapter5/StrongBundleShift · StrongBundle.isChart_shift.

The section f^1. $f^1 = f|_{X_1}$ Chapter5/StrongBundleShift · StrongBundle.sectionShift is a section of P^1 Chapter5/StrongBundleShift · StrongBundle.isSection_sectionShift, sc-smooth by the raising of the index of maps Chapter5/StrongBundleShift · StrongBundle.IsScSmoothSection.shift, with the zeros of f in X_1 as its zeros Chapter5/StrongBundleShift · StrongBundle.solutionSet_sectionShift, and regularizing, from $(X^1)_m = X_{m+1}$ and $(Y^1)_{m,m+1} = Y_{m+1,m+2}$ Chapter5/StrongBundleShift · StrongBundle.regularizing_sectionShift. In the index raise of a chart the principal part of f^1 is the level-1 map of that of f Chapter5/Corollary58 · StrongBundleChart.principal_sectionShift, and the raised chart is centred where the chart is Chapter5/Corollary58 · StrongBundleChart.shift_φ_eq_zero.

Condition (3) on every level, at every centre Chapter5/Corollary58 · IsFillingGerm.mem_range_toBase_of_fillDefect. *Let (g, s) be a datum of Definition 3.1.11 (preprint 3.11) in a chart — g a filling germ of $\hat{f}$ with fill defect $k = [1 - \rho(r)]g$ and derivative family Dk, s an sc$^+$-germ with $s(0) = g(0)$, $g - s$ conjugated to a basic germ. If $v \in \ker Dr(0)$ and $Dk(0)v \in F_m$, then $v \in E_m$* — at the centre of the datum, which need not be a zero. At a zero this is the regularity of §III.45 Chapter3/LinearizationChartsScG · IsFillingGerm.mem_range_toBase_of_scDeriv_r_eq_zero. The reduction to a zero needs neither a global sc$^+$-section nor a bump function: $g(0) = \iota s(0)$ is a smooth point of F Chapter5/Corollary58 · IsScSmoothGerm.toBase_zero_mem_smooth; $\hat{s}(z) = \Gamma(z, -g(0))$ is sc-smooth on U Chapter3/GlobalIFTNearbyOntoXi · isScSmooth_const_of_smooth and a section of $K \to O$, so by Lemma 3.1.20 (preprint 3.17) $g' = g + \hat{s} \circ r$ fills $\hat{f} + \hat{s}$ with the *same* fill defect Chapter3/GermCalculus · IsFillingGerm.add_section, and $g'(0) = g(0) - \Gamma(0, g(0)) = 0$. With the sc$^+$-germ $s' = s + \Gamma^1(r(\cdot), -s(0))$, $g' - s' = g - s$ on U, so Proposition 3.6.5 (preprint 3.52) in sc form makes $Dg'(0)$ sc-Fredholm Chapter3/FredholmGermLinearizationScG · proposition_3_6_5_local_sc; and $Dk(0) = (1 - \rho(0))Dg'(0)$ Chapter3/GermFillerLinearization · fillDefect_deriv_zero_eq_rho, so the case of a zero applies to g'.

The filling on level 1 Chapter5/Corollary58 · IsFillingGerm.germ_shift. The fill defect of g^1 is the level-1 map of k, an sc^1-germ with derivative family $(Dk)_1$ by Proposition 1.2.2 (preprint 1.17) for germs Chapter3/GermLevels · IsSc1GermWith.shift; $\iota\, Dr^1(0) = Dr(0)\, \iota$, by the uniqueness of the derivative of $r^1 = r_1$ on the shifted germ Chapter5/Corollary58 · toBase_scDeriv_r_shift, and $\iota\, \rho^1(0) = \rho(0)\, \iota$ Chapter5/Corollary58 · toBase_ρCLM_shift. So $Dk^1(0) = Dk(0)|_{E_1}$ maps $\ker Dr^1(0)$ to $\ker \rho^1(0)$; it is injective by condition (3) on level 0, onto by the regularity just proved with $m = 1$, and a topological isomorphism by the open mapping theorem. Conditions (1) and (2) of the filling shift directly Chapter5/Corollary58 · StrongBundleChart.shift_filling_eq_on Chapter5/Corollary58 · StrongBundleChart.shift_filling_mem_of_fixed, and g^1 is a filling germ of the principal part of f^1 in Φ^1 Chapter5/Corollary58 · StrongBundleChart.shift_isFillingGerm.

Corollary 5.2.5 (preprint 5.8). At every smooth point of X^1, the datum (g, s) of f in a chart Φ gives the datum (g^1, s^1) of f^1 in Φ^1: g^1 fills, s^1 is an sc$^+$-germ on the shifted germ, $s^1(0) = g^1(0)$, and $g^1 - s^1 = (g - s)_1$ near 0 is conjugated to a basic germ by the index raise of the conjugation (§V.7) Chapter5/Corollary58 · isScFredholmGerm_sectionShift. *For an sc-Fredholm section f, f^1 is an sc-Fredholm section*

of P^1 Chapter5/Corollary58 · corollary_5_2_5_isScFredholmSection; *if* f *has compact solution set, so has* f^1, by the compactness on level 1 (§V.7) Chapter5/Corollary58 · corollary_5_2_5. Since this holds for every strong bundle, it applies to P^1 and f^1: $f^2 = (f^1)^1$ is an sc-Fredholm section of $P^2 = (P^1)^1$ with compact solution set Chapter5/Corollary58 · corollary_5_2_5_two, and so on for every fixed i. The unpacked data in the shifted chart are kept for reference Chapter5/Corollary58 · corollary_5_2_5_data. No hypothesis beyond the text's is added, and tameness is not used. *Not done*: a family P^i indexed by i (a dependent type of strong bundles) and the tame form of the conclusion — germ data of f^1 in tame charts of P^1 — which needs the presentation independence of the degeneracy index for C^1 and that $A \cap E_1$ is an sc-complement of T_xO^1 when A is one of T_xO; there is no counterexample either.

V.30 The metric ρ on $\Gamma_U^{+,1}(P)$

The text: $\Gamma_U^{+,1}(P)$ "becomes a metric space" for $\rho(s,s') = \sup_x N(s(x) - s'(x))$, and "is not complete" (p. 217 (preprint p. 174)). The first is proved here; the second turns out to need a hypothesis (Findings, Chapter 5, items 150–153).

The metric. $\rho(s,s')$ is the supremum of the pointwise distances of §V.10 Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist. For an auxiliary norm the fibre norm is symmetric and subadditive (Definition 5.1.1 (1) (preprint 5.1)) Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.neg Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.sub_le Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.sub_triangle, so for allowable sections $N(s(x) - s'(x)) \le N(s(x)) + N(s'(x)) \le 2$ Chapter5/AllowableSectionsMetric · StrongBundle.sectionDistAt_le_two: the supremum is finite and $\rho \le 2$ Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_le_two. $\rho(s,s) = 0$, symmetry and the triangle inequality come from the fibre norms Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_self Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_comm Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_triangle, and $\rho(s,s') = 0$ forces $s = s'$ because N vanishes only at 0_x Chapter5/AllowableSectionsMetric · StrongBundle.eq_of_allowableDist_eq_zero. So ρ is a metric Chapter5/AllowableSectionsMetric · StrongBundle.allowableMetric, carried by a type synonym of $\Gamma_U^{+,1}(P)$, since the sections $X \to Y$ carry the product topology Chapter5/AllowableSectionsMetric · StrongBundle.AllowableSpace. Only Definition 5.1.1 (1) (preprint 5.1) is used. The density statements of Theorems 5.3.5 and 5.3.10 (preprint 5.13 and 5.18) (§§V.15, V.20) are $\rho(s,s') \le \varepsilon$ Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_le_iff.

Not complete, by an example Chapter5/AllowableSectionsMetric · TrivialBundle.not_completeSpace_allowableSections. Take the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ of the non-vacuity witnesses of Chapter 3, the fibre norm $N(x,v) = |v|$ — an auxiliary norm, including the condition on the topology of $Y[1]$ Chapter5/AllowableSectionsMetric · TrivialBundle.fibNorm Chapter5/AllowableSectionsMetric · TrivialBundle.isAuxNorm_fibNorm — and $U = \mathbb{R}$. Graphs of C^∞ maps are sc⁺-sections Chapter5/AllowableSectionsMetric · TrivialBundle.isScPlusSection_graphSection, so $s_j(x) = (x, \sqrt{x^2 + 1/(j+1)^2}/(2\sqrt{1+x^2}))$ is allowable, with $N \le 1/2$ Chapter5/AllowableSectionsMetric · TrivialBundle.approx Chapter5/AllowableSectionsMetric · TrivialBundle.graphSection_approx_mem, and $\rho(s_i,s_j) \le (\varepsilon_i + \varepsilon_j)/2$ with $\varepsilon_j = 1/(j+1)$, by $|\sqrt{x^2+a^2} - |x|| \le a$ Chapter5/AllowableSectionsMetric · TrivialBundle.abs_sqrt_sub_abs_le Chapter5/AllowableSectionsMetric · TrivialBundle.abs_approx_sub_limitFn. A ρ-limit s would have, by pointwise convergence, the fibre part $|x|/(2\sqrt{1+x^2})$ Chapter5/AllowableSectionsMetric · TrivialBundle.limitFn, which is not differentiable at 0 Chapter5/AllowableSectionsMetric · TrivialBundle.not_differentiableAt_limit; but the fibre part of an sc⁺-section of the trivial bundle is C^1, by Proposition 1.2.3 (preprint 1.18) in the chart $\Phi = \mathrm{id}$ Chapter5/

AllowableSectionsMetric · TrivialBundle.differentiable_snd_of_isScPlusSection.

Complete, in two cases. For $U = \emptyset$ every allowable section is the zero section, so the space is a point, for every bundle and auxiliary norm Chapter5/AllowableSectionsMetric · StrongBundle.subsingleton_allowableSections_empty Chapter5/AllowableSectionsMetric · StrongBundle.completeSpace_allowableSections_empty. Over the point $\mathbb{R}^0$ with fibre $\mathbb{R}^k$ every section is the graph of a constant, hence sc$^+$ Chapter5/AllowableSectionsMetric · TrivialBundle.isScPlusSection_of_isSection_point, and the space is the closed unit ball of $\mathbb{R}^k$ (or $\{0\}$), complete for every U Chapter5/AllowableSectionsMetric · TrivialBundle.completeSpace_allowableSections_point. So the text's "not complete" does not hold as we read it without a hypothesis and is read as "not complete in general". In the text's setting — U a neighbourhood of the solution set of an sc-Fredholm section on an M-polyfold with bump functions — one expects non-completeness as soon as U contains a non-isolated point over which $Y_{0,1}$ is nonzero, by the same mechanism (bump functions times a non-smooth uniform limit); this is not formalized, and neither is the identification of the metric completion with the sections of Remark 5.3.4 (preprint 5.12) (§V.10).

V.31 Weak and pointwise bump functions

§V.8 reads Definition 5.3.1 (preprint 5.9) pointwise, $f(x) \neq 0$, because Proposition 5.5.3 (preprint 5.30) needs it, and keeps Definition 5.5.2's (preprint 5.29) "not identically zero" as the weak reading. How far apart are they? (Findings, Chapter 5, items 160–163.)

Bump points. x is a *bump point* if every open $U \ni x$ carries an sc-smooth f with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$ Chapter5/BumpWeakPointwise · MPolyfold.HasBumpAt. The pointwise reading says that every point is a bump point Chapter5/BumpWeakPointwise · MPolyfold.admitsBumpFunctions_iff_forall_hasBumpAt, and the weak reading holds as soon as the bump points are dense: a bump at a bump point inside U is not identically zero and is supported in U Chapter5/BumpWeakPointwise · MPolyfold.admitsBumpFunctionsWeak_of_dense. (The converse, weak $\Box$ dense bump points, is a nested-neighbourhoods argument with Cantor's theorem in a chart; not formalized.)

On an sc-Banach space. An sc-Banach space E is an M-polyfold with the identity chart and model (E, E, E) Chapter5/BumpWeakPointwise · scChart Chapter5/BumpWeakPointwise · scPolyfold, whose sc-smooth functions are the sc-smooth maps $E \to \mathbb{R}$ Chapter5/BumpWeakPointwise · isScSmoothMap_scPolyfold_iff. *There the weak reading gives a bump point at every smooth point* Chapter5/BumpWeakPointwise · MPolyfold.AdmitsBumpFunctionsWeak.hasBumpAt_of_mem_smooth: given $x \in E_\infty$ and $B(x, \varepsilon) \subseteq U$, a weak bump supported in $B(x, \varepsilon/2)$ is non-zero on a non-empty open set, which contains a smooth point y because E_∞ is dense; $a = y - x$ is smooth, so $v \mapsto f(v + a)$ is sc-smooth Chapter4/Recentre · isScSmooth_add_const, non-zero at x, with support in $B(x, \varepsilon) \subseteq U$. The argument stops at E_∞: translation by a vector $a \notin E_m$ is not even sc^0, since it sends $0 \in E_m$ out of E_m Chapter5/BumpWeakPointwise · not_isSc0_add_const. On M-polyfolds built on sc-Hilbert spaces both readings hold at every point Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert. *Open*: whether the weak reading gives a bump function non-zero at a non-smooth point, on an sc-Banach space or an M-polyfold; a counterexample would need an E_0 without C^1 bump functions.

V.32 The Springer 2021 edition: Definition 5.2.1, Theorems 5.2.2 and 5.3.5, Definition 5.3.13

The 2021 edition adds two compactness properties to Definition 5.2.1 (preprint 5.4) and extends Theorem 5.2.2 (preprint 5.5) to them, concludes in Theorem 5.3.5 (preprint 5.13) that the perturbed solution sets are compact sub-M-polyfolds, and changes Definition 5.3.13 (preprint 5.20): t need not be onto, the fibres X_τ are no longer asked to be Fredholm problems, and $dt(x)$ is asked onto at every smooth point. Chapter 5 has no item that is new in the 2021 edition. The file is `Chapter5/Book2021.lean` (Findings, Chapter 5, items 170–179).

Definition 5.2.1, 2021 form `Chapter5/Book2021 · StrongBundle.IsWeaklyProperOnU` `Chapter5/Book2021 · StrongBundle.IsWeaklyProperLtInfty` `Chapter5/Book2021 · StrongBundle.IsProperLtInfty`. *Statement.* (2) weakly proper: compact solution set, and for every auxiliary norm N an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc$^+$-section s supported in U with $N(s(y)) \le 1$ for $y \in U$; (4) weakly proper$_{<\infty}$: the same for every such s with $N(s) \le C$ on U for some constant C; (5) proper$_{<\infty}$: compact solution set, and for every N an open $U \supseteq f^{-1}(0)$ with $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le C\}$ compact for every $C > 0$, where $N(f(x)) \le C$ includes $f(x) \in Y_{0,1}$ `Chapter5/Compactness · StrongBundle.AuxLe`. *The bound on U is the old bound everywhere* `Chapter5/Book2021 · StrongBundle.isWeaklyProperOnU_iff`: a section supported in U vanishes off U `Chapter5/Book2021 · StrongBundle.isZeroAt_of_not_mem_of_support_subset`, and the zero of a fibre has $N \le r$ for every $r \ge 0$ `Chapter5/Book2021 · StrongBundle.IsAuxNorm.auxLe_fiberZero`; so (2) is the `IsWeaklyProper` of §V.4 `Chapter5/Compactness · StrongBundle.IsWeaklyProper`, and (1), (3) are unchanged `Chapter5/Compactness · StrongBundle.IsProper`.

The a priori dependencies (p. 209). Proper$_{<\infty}$ gives proper with $C = 1$ `Chapter5/Book2021 · StrongBundle.IsProperLtInfty.isProper`; weakly proper$_{<\infty}$ gives weakly proper, since the bound 1 is a constant `Chapter5/Book2021 · StrongBundle.IsWeaklyProperLtInfty.isWeaklyProperOnU`; weakly proper includes a compact solution set `Chapter5/Book2021 · StrongBundle.IsWeaklyProperOnU.isCompact_solutionSet`. For a continuous f, proper gives weakly proper by §V.4 `Chapter5/Book2021 · StrongBundle.IsProper.isWeaklyProperOnU`, and proper$_{<\infty}$ gives weakly proper$_{<\infty}$ `Chapter5/Book2021 · StrongBundle.IsProperLtInfty.isWeaklyProperLtInfty`: let $N(s) \le C$ on U. A solution x of $f = s$ lies in U — otherwise $s(x) = 0$, so $x \in f^{-1}(0) \subseteq U$ — and there $N(f(x)) = N(s(x)) \le \max(C, 1)$; so $\{f = s\}$, closed because f and s are continuous, lies in the compact $\mathrm{cl}\{x \in U \mid N(f(x)) \le \max(C, 1)\}$ `Chapter5/Book2021 · StrongBundle.AuxLe.mono`. The text's remark that the notions differ for general sc-smooth sections is not formalized.

Theorem 5.2.2 (preprint 5.5), 2021 form `Chapter5/Book2021 · theorem_5_2_2`. *Statement.* An sc-Fredholm section with compact solution set is proper$_{<\infty}$. *Proof.* As for §V.6 `Chapter5/Compactness · theorem_5_2_2_preprint`. At a zero x, the local step gives a chart c, an open $V \ni x$ and for every bound b a compact set containing the closure of the points of V whose fibre part has level-1 norm $\le b$ `Chapter5/Compactness · exists_nbhd_isCompact_closure_bdd`; near x the chart norm is at most $A \cdot N$ `Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le`. So on a smaller V the closure of $\{N(f) \le C\}$ lies in the compact set for $b = A \cdot C$, and V does not depend on C. Finitely many such V cover $f^{-1}(0)$, and the closure of the finite union lies in the finite union of the compact closures. □ The text's first display writes $N(f(x)) \le d$ for the points z of the set, presumably meant as $N(f(z)) \le d$; the later display has y throughout. *The five properties are equivalent* for sc-Fredholm sections `Chapter5/Book2021 · theorem_5_2_2_tfae`: the cycle (5) □ (3) □ (2) □ (1) □ (5) and (5) □ (4) □ (2), with f continuous because it is sc-smooth.

Theorem 5.3.5 (preprint 5.13), 2021 form `Chapter5/Book2021 · theorem_5_3_5`. *Statement.* The conclusions of §V.15 `Chapter5/Transversality · theorem_5_3_5_preprint`, and moreover S_{f+s} is a compact sub-M-polyfold

for the transversal perturbations s. *Proof.* `theorem_5_3_5_preprint` gives S_{f+s} compact and, if non-empty, a manifold, a sub-M-polyfold in particular; the empty set is a sub-M-polyfold `Chapter5/Book2021 · isSubMPolyfold_solutionSet_of_manifold`. □ The perturbations are the allowable sections, bounded by $N(s) \leq 1$ `Chapter5/BumpFunctions · StrongBundle.allowableSections`; the printed "$N(f(x)) \leq 1$ for all x" is read as $N(s(x)) \leq 1$, presumably what is meant, as in §V.15.

Definition 5.3.13 (preprint 5.20), 2021 form `Chapter5/Book2021 · StrongBundle.IsSubmersiveGenCompactHomotopy`. *Statement.* X tame; f an sc-Fredholm section with tame germ data and compact solution set; $t : X \to \mathbb{R}$ sc-smooth with values in $[0, 1]$; for $i = 0, 1$ the end $t^{-1}(i)$ a sub-M-polyfold with tame induced structure over which the restriction of f to the pull-back bundle has tame germ data `Chapter5/Book2021 · StrongBundle.IsCompactFredholmRestriction` `Chapter5/Homotopy · restrictSection`; and (3) $dt(x)$ onto at every smooth x. Since T_xX has no linear structure in the project, (3) is read in a chart φ around x with a retraction R onto its model and the level-1 point z over $\varphi(x)$: every $a \in \mathbb{R}$ is the derivative of $t \circ \varphi^{-1} \circ R$ at z on some $DR(z)v \in T_zO$ `Chapter5/Book2021 · MPolyfold.HasSurjDerivAt`. *Relation to §V.23.* The data of the earlier reading `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy` together with (3) give the new data `Chapter5/Book2021 · StrongBundle.IsGenCompactHomotopy.isSubmersiveGenCompactHomotopy` (t onto $[0, 1]$ gives values in $[0, 1]$, and the fibre conditions at $\tau = 0, 1$ are the end conditions). Every fibre $t^{-1}(\tau)$ still has a compact solution set: it is closed, and the solution set of the restriction is the trace of $f^{-1}(0)$ `Chapter5/Book2021 · StrongBundle.IsSubmersiveGenCompactHomotopy.isCompact_solutionSet_restrict`.

A witness, and the converse. Over an empty end the end conditions hold: $\emptyset$ is a tame sub-M-polyfold whose atlas has no charts, and every section over it is tame sc-Fredholm `Chapter5/Book2021 · StrongBundle.isCompactFredholmRestriction_of_eq_empty` `Chapter5/Book2021 · isScSmoothRetractMap_of_domain_empty`. On the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ take the graph section of the identity, with solution set $\{0\}$ `Chapter5/Book2021 · GenCompactHomotopyWitness.solutionSet_graphSection_id` and tame germ data `Chapter3/NonVacuityG · TrivialBundle.isScFredholmSectionTame_graphSection`, and $t(x) = 1/2 + \arctan(x)/\pi$ `Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime`, a C^∞ function with values in $(0, 1)$ `Chapter5/Book2021 · GenCompactHomotopyWitness.contDiff_arctanTime` `Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime_mem_Ioo`. Both ends are empty `Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime_preimage_eq_empty`, and $t'(x) = 1/(\pi(1+x^2)) \neq 0$ `Chapter5/Book2021 · GenCompactHomotopyWitness.hasFDerivAt_arctanTime`, so $dt(x)$ is onto in the identity chart `Chapter5/Book2021 · GenCompactHomotopyWitness.hasSurjDerivAt_arctanTime`. So the 2021 definition is satisfiable `Chapter5/Book2021 · GenCompactHomotopyWitness.isSubmersiveGenCompactHomotopy_witness`; the same data are not a generalized compact homotopy in the earlier sense, as 0 is not a value of t `Chapter5/Book2021 · GenCompactHomotopyWitness.not_isGenCompactHomotopy_witness`.

Lemma 5.3.16 (preprint 5.23) under (3) `Chapter5/Book2021 · lemma_5_3_16`. *Statement.* If $dt(x)$ is onto, then in a chart the derivative ℓ on T_zO is non-zero, $\ker \ell$ has a one-dimensional complement Z, and the conclusions of Lemma 5.3.16 hold. *Proof.* Take v with $\ell(DR(z)v) = 1$; so $\ell \neq 0$. A vector on which ℓ is non-zero spans a complement of $\ker \ell$ `Chapter5/Homotopy · exists_isCompl_ker_of_ne_zero`, and §V.23 applies `Chapter5/Homotopy · lemma_5_3_16_preprint`. □ This is the hypothesis $dt(x) \neq 0$ that §V.23 found missing from Theorem 5.3.15 (preprint 5.22): the 2021 condition (3) supplies it. That Z can be chosen smooth is not formalized; Theorem 5.3.15 itself is proved without it (§§V.36–V.41; gap G.48).

Theorem 5.5.5 (preprint 5.32). The 2017 and 2021 statements print the exponent p that the preprint omits in (1) and (2); the theorem is proved as printed there, with the exponent (§V.43; gap G.50).

V.33 Remark 5.3.8 (preprint 5.16): Theorem 5.3.10 relative to a compact part, and the cobordism

Remark 5.3.8 joins two transversal sections f_0, f_1 by an sc-Fredholm arc $(t,x)\mapsto f_t(x)$ with compact solution set and says that a small perturbation vanishing near $t=0,1$ makes the arc transversal, its solution set a compact cobordism; "the reader will be able to carry out this construction". Two things are needed: Theorem 5.3.10 (preprint 5.18) for a section that is already in general position off a compact part of its zeros, and the arc itself. The arc lives on $[0,1]\times X$ in the text; it is read on $\mathbb{R}^1\times X$ with the time window $[0,1]$ (reading D2 of Part VI), so that no tame structure on $[0,1]\times X$ is needed (Findings, Chapter 5, item 186; gap G.44).

A chart criterion `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_of_sup_eq`. If in a tame chart centred at a zero y the linearization $\mathrm{Lin}_f(0)$ is onto and $\ker\mathrm{Lin}_f(0)+T_0^RO=T_0O$ on level 0, then (f,y) is in general position. This is the second half of the argument of §V.19, now for any section: the kernel has an algebraic complement inside T_0^RO `Chapter5/GeneralPosition · exists_compl_le_of_sup_eq`, §V.16 gives good position, the kernel is then the tangent space of the solution set, finite-dimensional and smooth (§V.17), and it has an sc-complement inside T_0^RO `Chapter4/LocalStructure · exists_scComplement_le`. Conversely general position gives $\ker\mathrm{Lin}_f(0)+T_0^RO=T_0O$ in every tame chart centred at y `Chapter5/CylinderPerturbation2021 · isParamGP_elim0_of_isGeneralPositionGerm`. Two consequences: if (g,y) is in general position, the family $F(\lambda,x)=g(x)+\sum\lambda_it_i(x)$ is in general position at $(0,y)$ whatever sections are added `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramSection_of_isGeneralPositionGerm`, and the lift $\tilde g(r,x)=g(x)$ to $\mathbb{R}^m\times X$ is in general position at every (r_0,y), since in the product chart $\mathrm{Lin}_{\tilde g}(0)(a,u)=\mathrm{Lin}_g(0)u$ `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramLift`. General position, like good position, depends only on the germ of the section `Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_congr_nhds`.

The perturbation step relative to an open set `Chapter5/CylinderPerturbation2021 · exists_null_isGeneralPositionGerm_paramSlice`. Let G have tame germ data, N be an auxiliary norm, W open with $\mathrm{cl}\{x\in W\mid N(G(x))\le 1\}$ compact, and $t_1,\dots,t_m$ sc$^+$-sections with $N(t_i)\le 1$ such that F is in general position at $(0,y)$ for every zero y of G. Then there are $\delta>0$ and a null set CV such that for $|\lambda|<\delta$, $\lambda\notin CV$, $N(\sum\lambda_it_i)\le 1$ and $G+\sum\lambda_it_i$ is in general position at every zero in W. This is the mechanism of §V.20 with $s_0=0$ and only the zeros in W controlled: openness of general position near the zero slice, compactness of the zeros over a small parameter ball inside the margin, Theorem 4.3.1 (preprint 4.23) and the Sard step.

Theorem 5.3.10 relative to a compact part `Chapter5/CylinderPerturbation2021 · theorem_5_3_10_relative_hyp`. Let X be tame with sc-smooth bump functions, G with tame germ data, N auxiliary, $K\subseteq W$ with K compact and W open, such that (G,x) is in general position at every zero $x\notin K$ and $\mathrm{cl}\{x\in W\mid N(G(x))\le 1\}$ is compact. Then for every $\varepsilon>0$ there is an sc$^+$-section p supported in W with $N(p)\le\varepsilon$ and $(G+p,x)$ in general position at every zero. *Proof.* Sections supported in W make F general at $(0,y)$ for the zeros y near the compact $K\cap S_G$, by (i)–(ii) of §V.19 at finitely many centres `Chapter5/CylinderPerturbation2021 · exists_sections_isGeneralPositionGerm_near`; at the other zeros F is general because G is (the chart criterion). The relative step, for the norm N/η with $\eta=\min(\varepsilon,1)$, gives $p=\sum\lambda_it_i$ with the zeros in W in general position. A zero outside W has a neighbourhood on which p vanishes, since $\operatorname{supp}p\subseteq W$ is closed, so $G+p=G$ there; such a zero is a zero of G not in K, in general position, and germ locality concludes. □ The text's theorem is the case $K=S_G$, $W=U$; here S_G need not be compact.

The margin Chapter5/FredholmArcPerturbation · exists_margin_of_isCompact. For a compact set K of zeros of an sc-Fredholm section f and an open $V \supseteq K$, some open W with $K \subseteq W \subseteq V$ has $\mathrm{cl}\{x \in W \mid N(f(x)) \leq 1\}$ compact: the local step of Theorem 5.2.2 (§V.6) at the points of K, finitely many of them.

Remark 5.3.8 relative to a time function Chapter5/FredholmArcPerturbation · remark_5_3_8_relative. Let X be tame with sc-smooth bump functions, F with tame germ data, N auxiliary, $\tau : X \to \mathbb{R}$ continuous, $c < a$, $b < d$. Suppose (F, x) is in general position at every zero with $\tau(x) \notin (a, b)$, and the zeros with $\tau \in [c, d]$ form a compact set. Then for every $\varepsilon > 0$ there are $c < c' < a$, $b < d' < d$ and an sc$^+$-section p with $N(p) \leq \varepsilon$, vanishing where $\tau \notin (c', d')$, such that $F + p$ is in general position at every zero and its zeros with $\tau \in [c, d]$ are compact. *Proof.* The zeros over $[a, b]$ are compact; the margin gives W around them inside $\tau^{-1}(c', d')$, and Theorem 5.3.10 relative to that compact set gives p supported in W. □

The arc Chapter5/FredholmArcPerturbation · remark_5_3_8_arc. For $F(t, x) = f_t(x)$ a section of the pull-back of P to $\mathbb{R}^1 \times X$ with tame germ data, in general position at its zeros with $t \notin (a, b)$, $0 < a, b < 1$, and with compact zeros over $[0, 1]$, the instance $\tau = t$, $[c, d] = [0, 1]$ gives an sc$^+$-section s with $N(s) \leq \varepsilon$, $s(t, \cdot) = 0$ for $t \notin (c', d')$, and $F + s$ in general position at every zero with compact zeros over $[0, 1]$. "f_0, f_1 transversal" is read as general position of F at the zeros near $t = 0, 1$; for X without boundary and F stationary near the ends it is the transversality of f_0, f_1. Its orientation, Remark 6.6.19 (preprint 6.42), is formalized in Part VI (gap G.66).

The cobordism between admissible sections Chapter5/CylinderPerturbation2021 · exists_cobordism, the form Theorem 6.7.1 uses (Part VI). Let $U \supseteq S_f$ be open with $\mathrm{cl}\{x \in U \mid N(f(x)) \leq 2\}$ compact Chapter5/CylinderPerturbation2021 · exists_adapted_margin, and s_0, s_1 allowable with $f + s_i$ in general position at every zero. There is an sc$^+$-section S of the pull-back to $\mathbb{R}^1 \times X$ with $S(t, \cdot) = s_0$ for $t \leq 1/8$, $S(t, \cdot) = s_1$ for $t \geq 7/8$, $S = 0$ over $X \setminus U$, $N(S) \leq 2$, $\tilde{f} + S$ in general position at every zero, and compact zeros over every $[a, b]$. *Proof.* $S = \hat{s} + p$, where $\hat{s} = (1 - \chi)\tilde{s}_0 + \chi\tilde{s}_1$ with the C^∞ cut-off χ, 0 for $t \leq 3/16$ and 1 for $t \geq 13/16$ Chapter5/CylinderPerturbation2021 · cylCut, an sc$^+$-section Chapter5/CylinderPerturbation2021 · isScPlusSection_cylInterp; near the ends $\tilde{f} + \hat{s}$ is the lift of $f + s_i$, general by the lift lemma; p comes from the relative theorem for $G = \tilde{f} + \hat{s}$, the pulled-back auxiliary norm Chapter5/CylinderPerturbation2021 · StrongBundle.IsAuxNorm.pullback, $W = (1/8, 7/8) \times U$ and K the zeros over $[3/16, 13/16]$. □ The homotopy version is Chapter6/ProperHomotopy2021 · exists_homotopy_cobordism.

V.34 Theorem 5.3.12 (preprint 5.19): the dense form

Theorems 5.3.5 and 5.3.10 have dense forms (§§V.15, V.20); Theorem 5.3.12 is printed with "arbitrarily small", and §V.22 proves that form. The dense form is proved too (gap G.45).

The generic form around a given section Chapter5/DenseGoodPosition · theorem_5_3_12_param_rel. Let f have tame germ data, N be auxiliary, $U \supseteq S_f$ open with $\mathrm{cl}\{x \in U \mid N(f(x)) \leq 1\}$ compact, (f, x) in good position at the zeros $x \in \partial X$, and s_0 an sc$^+$-section supported in U, with $N(s_0) \leq 1 - \varepsilon$ and support disjoint from ∂X. Then there are allowable $t_1, \ldots, t_m$ supported in $U \setminus \partial X$, $\delta > 0$ and a null CV such that for $|\lambda| < \delta$, $\lambda \notin CV$, the section $s_0 + \sum \lambda_i t_i$ is allowable, its support misses ∂X, and $f + s_0 + \sum \lambda_i t_i$ is transversal and in good position at every zero. *Proof.* The proof of §V.22 with $g = f + s_0$ in place of f. Off the support of s_0, $g = f$ Chapter5/DenseGoodPosition · sectionAdd_eq_of_not_mem_support, so near ∂X the zeros of g are those of f, in good position, and germ locality Chapter3/Chapter5BridgeG · StrongBundle.IsGoodPositionGermTame.congr_nhds carries this to g; g has tame germ data (§V.15, step 2), and its zeros lie in the compact margin (Remark 5.3.4 (preprint 5.12), §V.10). Steps 1–5 of §V.22 then apply to g; on the slices the value of a sum at a zero is read off Chapter5/DenseGoodPosition · paramSlice_

apply_of_isZeroAt. □ For a section, "vanishing near ∂X" and "support disjoint from ∂X" are the same Chapter5/DenseGoodPosition · eventually_isZeroAt_nhdsSet_iff.

The dense form Chapter5/DenseGoodPosition · theorem_5_3_12_dense. Let X admit bump functions, f have tame germ data and compact solution set, N be auxiliary, and (f, x) be in good position at the zeros on ∂X. There is an open $U \supseteq S_f$ with S_{f+s} compact for every $s \in \Gamma_U^{+,1}(P)$, and for every $s' \in \Gamma_U^{+,1}(P)$ vanishing near ∂X and every $\varepsilon > 0$ an $s \in \Gamma_U^{+,1}(P)$ vanishing near ∂X with $N(s(x) - s'(x)) \le \varepsilon$ for all x, $f + s$ transversal and in good position at every zero, and S_{f+s} compact and, if non-empty, a tame sub-M-polyfold with the structure of a manifold with boundary with corners. *Proof.* U from Theorem 5.2.2 (preprint 5.5), compactness by Remark 5.3.4 (§V.10). With $\varepsilon' = \min(\varepsilon, 1)$ and $s_0 = (1 - \varepsilon'/2)s'$, $N(s_0) \le 1 - \varepsilon'/2$ and $\operatorname{supp} s_0 \subseteq \operatorname{supp} s'$ misses ∂X; the generic form gives t, δ, CV; the ball of radius $\min(\delta, \varepsilon'/(2m+2))$ is not null, so it contains a $\lambda \notin CV$, and $s = s_0 + \sum \lambda_i t_i$ has $N(s - s') \le \varepsilon'/2 + m|\lambda| \le \varepsilon'$, exactly as in §V.15. □

V.35 Remark 5.3.14 (preprint 5.21): isomorphic end problems

Definition 5.3.13 asks $(f|X_i, P|Y_i) = (f_i, P_i)$; Remark 5.3.14 proposes to read the equalities up to isomorphism and to make the isomorphisms part of the data. Formalized as follows (gap G.47; Findings, Chapter 5, item 100).

Isomorphic problems Chapter5/CompactHomotopyIso · StrongBundle.IsProblemIso. (P, f) and (P', f') are isomorphic if there is a strong bundle isomorphism (h, Φ), with inverse (g, Ψ), carrying f to f': $\Phi \circ f = f' \circ h$. Strong bundle isomorphisms are those of the formalization of Chapter 8 Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleIso (the file imports them, a forward import that builds). The relation is reflexive, symmetric and transitive Chapter5/CompactHomotopyIso · StrongBundle.IsProblemIso.refl Chapter5/CompactHomotopyIso · StrongBundle.IsProblemIso.symm Chapter5/CompactHomotopyIso · StrongBundle.IsProblemIso.trans. A strong bundle map sends the zero of the fibre over x to the zero over $h(x)$ Chapter5/CompactHomotopyIso · StrongBundle.IsStrongBundleMap.map_fiberZero, so the base map of the inverse carries $S_{f'}$ onto S_f Chapter5/CompactHomotopyIso · StrongBundle.IsProblemIso.image_solutionSet, and compactness of the solution sets is invariant Chapter5/CompactHomotopyIso · StrongBundle.IsProblemIso.isCompact_solutionSet_iff.

Ends up to isomorphism. For a sub-M-polyfold $A \subseteq X$, the restricted problem — $f|A$ in the pull-back bundle, as in §V.23 — is isomorphic to a given problem Chapter5/CompactHomotopyIso · StrongBundle.EndIso; end isomorphisms compose with isomorphisms of the given problem Chapter5/CompactHomotopyIso · StrongBundle.EndIso.trans and give compact end solution sets Chapter5/CompactHomotopyIso · StrongBundle.EndIso.isCompact_solutionSet. A *generalized compact homotopy between* (P_0, f_0) *and* (P_1, f_1) is a generalized compact homotopy in the 2021 sense (§V.32) with end isomorphisms at $t^{-1}(0)$ to (P_0, f_0) and at $t^{-1}(1)$ to (P_1, f_1) Chapter5/CompactHomotopyIso · StrongBundle.IsGenCompactHomotopyBetween. It is invariant under isomorphism of the end problems Chapter5/CompactHomotopyIso · StrongBundle.IsGenCompactHomotopyBetween.congr_left Chapter5/CompactHomotopyIso · StrongBundle.IsGenCompactHomotopyBetween.congr_right, its end solution sets are compact Chapter5/CompactHomotopyIso · StrongBundle.IsGenCompactHomotopyBetween.isCompact_solutionSet_left Chapter5/CompactHomotopyIso · StrongBundle.IsGenCompactHomotopyBetween.isCompact_solutionSet_right, and every generalized compact homotopy is one between its own restrictions, with identity isomorphisms Chapter5/CompactHomotopyIso · StrongBundle.IsSubmersiveGenCompactHomotopy.isGenCompactHomotopyBetween — so the notion is satisfiable whenever Definition 5.3.13 is (§V.32).

V.36 Theorem 5.3.15 (preprint 5.22): the statement, and "Morse-type"

Theorem 5.3.15 says: let P be a strong bundle over a tame M-polyfold with sc-smooth bump functions, (f,t) a generalized compact homotopy (Definition 5.3.13) whose ends $X_i = t^{-1}(i)$ have no boundary and carry problems in general position, with $\partial X = X_0 \sqcup X_1$, and N an auxiliary norm. Then there is an open $U \supseteq S_f$ with S_{f+s} compact for $s \in \Gamma_U^{+,1}(P)$, and for every $\varepsilon > 0$ an $s_0 \in \Gamma_U^{+,1}(P)$ with $N(s_0) \le \varepsilon$, vanishing near ∂X, such that (1) $(f + s_0, x)$ is in general position at every zero and (2) $t : S_{f+s_0} \to [0,1]$ has only Morse-type critical points. The proof (pp. 223–228) finds that $d(t|S)$ has no zeros near ∂S, builds at each critical point x of $t|S$ sections with Lemmas 5.3.16 and 5.3.17 that make $(\lambda, y) \mapsto d(t\circ\Phi_\lambda)(y)$ transversal to the zero section near $(0,x)$, through a trivialization $\Phi_\lambda : S \to S_\lambda$ and the second derivative of the equation, unites the sections of finitely many critical points, and concludes with "a parameterized version of Sard's theorem". The formal proof keeps this outline and changes one device: Lagrange multipliers on level sets of the solution manifold of $f + \sum \mu_j \sigma_j$ replace Φ_λ (§§V.38–V.40). §§V.36–V.41 retell it (gap G.48; Findings, Chapter 5, items 101, 178, 183).

The ends `Chapter5/Theorem5315_Ends · StrongBundle.IsGeneralPositionAtEnds`. For $i = 0, 1$ and every proof that $t^{-1}(i)$ is a sub-M-polyfold: the induced M-polyfold has no boundary, and the restriction of f to the pull-back bundle (§V.23) is in general position at each of its zeros. The text's (f_i, P_i) are these restrictions (Definition 5.3.13 (2) as read in §V.32), or problems isomorphic to them (§V.35).

Smooth parametrizations `Chapter5/Theorem5315_MorseParam · MPolyfold.IsSmoothParam`. "Morse-type critical points" is not defined in the text, and S_{f+s_0} is a manifold with boundary with corners. It is read through parametrizations. A *smooth parametrization* of $A \cap G$, $G \subseteq X$ open, by a set Ω of a normed space E consists of p, continuous on Ω with $p(\Omega) = A \cap G$, and $q : X \to E$ with sc-smooth components and $q \circ p = \mathrm{id}$ on Ω, such that $h \circ p$ is C^∞ on Ω for every sc-smooth $h : X \to \mathbb{R}$. A point x of A is a *regular point or a Morse-type critical point of* $t|A$ `Chapter5/Theorem5315_MorseParam · MPolyfold.IsMorseOrRegularPoint` if some parametrization by a set $\Omega \subseteq \mathbb{R}^n$ with unique derivatives, $x = p(u_0)$, has the property: if $d(t\circ p)(u_0) = 0$, then Ω is a neighbourhood of u_0 and the Hessian of $t\circ p$ at u_0 is injective. So critical points are interior and non-degenerate, and boundary points must be regular. *t has only Morse-type critical points on A* if every point of A is such a point `Chapter5/Theorem5315_MorseParam · MPolyfold.HasOnlyMorseCriticalPoints`. Any finite-dimensional parameter space and open Ω will do `Chapter5/Theorem5315_MorseParam · MPolyfold.isMorseOrRegularPoint_of_param Chapter5/Theorem5315_MorseParam · MPolyfold.IsSmoothParam.comp_equiv`. In a chart the notion is the classical one `Chapter5/Theorem5315_ParametricMorse · HasOnlyMorseCriticalPointsOn`.

Independence of the parametrization. For two parametrizations p, p' through one point, the transition $\theta = q' \circ p$ is C^∞ on Ω, $\theta(u) = u'$ and $p' \circ \theta = p$ near u `Chapter5/Theorem5315_LocalInvariance · MPolyfold.IsSmoothParam.transition`; the chain rule within the parameter sets `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.hasFDerivWithinAt_transition` gives $D(q \circ p')(u') \circ D(q' \circ p)(u) = 1$ `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.fderivWithin_transition_comp`. Hence $d(t \circ p')(u') \neq 0$ implies $d(t \circ p)(u) \neq 0$, at boundary points too `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.fderivWithin_ne_zero_congr`; Ω' a neighbourhood of u' makes Ω one of u, by the inverse function theorem for the transition `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.mem_nhds_congr`; and at an interior critical point the second-order chain rule `Chapter5/Theorem5315_MorseLagrange · fderiv_fderiv_comp_apply`, whose first-order term vanishes there, conjugates the two Hessians by the invertible $D\theta(u)$ `Chapter5/Theorem5315_LocalInvariance · MPolyfold.IsSmoothParam.hessian_injective_congr`. So the whole clause transfers `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsSmoothParam.morseClause_congr`: if x is a regular or Morse-type critical point of $t|A$, then in *every* smooth parametrization of A around x by a set with unique derivatives

a critical point is interior with injective Hessian `Chapter5/Theorem5315_LocalInvarianceClause · MPolyfold.IsMorseOrRegularPoint.morseClause`. The comparison with the charts of the Mathlib manifold structure that the project builds on solution sets is not formalized.

Theorem 5.3.15 `Chapter5/Theorem5315_Final · theorem_5_3_15`. *Statement.* X with sc-smooth bump functions, (f,t) a generalized compact homotopy in the 2021 sense `Chapter5/Book2021 · StrongBundle.IsSubmersiveGenCompactHomotopy` — so X tame, f with tame germ data and compact solution set, t sc-smooth with values in $[0,1]$ and $dt(x)$ onto at every smooth x — with the end hypotheses above, $\partial X = t^{-1}(0) \cup t^{-1}(1)$, and N auxiliary. Then: an open $U \supseteq S_f$ with S_{f+s} compact for $s \in \Gamma^{+,1}_U(P)$, and for every $\varepsilon > 0$ an $s_0 \in \Gamma^{+,1}_U(P)$ with $N(s_0(x)) \le \varepsilon$ for all x, $s_0 = 0$ near ∂X, (1) $(f + s_0, x)$ in general position at every $x \in S_{f+s_0}$ and (2) t with only Morse-type critical points on S_{f+s_0}. No hypothesis is added; the bump functions on the X_i are not assumed. The hypotheses are satisfiable: the witness of §V.32 has empty ends and $\partial\mathbb{R} = \emptyset = X_0 \cup X_1$ `Chapter5/Book2021 · GenCompactHomotopyWitness.isSubmersiveGenCompactHomotopy_witness` `Chapter5/Theorem5315_Ends · theorem_5_3_15_generalPosition_witness`. *Proof.* $s_0 = s_1 + s_2$: s_1 gives (1) (§V.37); s_2 comes from the covering argument for $f + s_1$ (§§V.38–V.40) `Chapter5/Theorem5315_Morse · theorem_5_3_15_hyp`, whose local data are derived from the implicit function theorem and Definition 5.3.13 (3) (§V.41) `Chapter5/Theorem5315_LocalData · StrongBundle.hasMorseLocalData`. □ The hidden hypothesis $dt(x) \neq 0$ of the preprint (§V.23) is the 2021 condition (3) (§V.32); of $\partial X = X_0 \sqcup X_1$ the proof uses $\partial X \subseteq X_0 \cup X_1$ for (1) and $t(\partial X) \subseteq \{0,1\}$ for (2).

V.37 Theorem 5.3.15 (preprint 5.22): compactness and general position, from the ends

"In view of the assumption that (f_i, X_i) are already in general position" the text perturbs away from ∂X. What this needs is general position of (f,x) at the zeros $x \in \partial X$.

From the restriction to X `Chapter5/Theorem5315_Ends · isGeneralPositionGerm_of_restrict`. Let f have tame germ data, $A \subseteq X$ be a sub-M-polyfold without boundary with $f|A$ sc-smooth in the pull-back bundle, and $a \in A$ a zero with $(f|A)'(a)$ onto. Then (f,a) is in general position in X. *Proof.* Take the tame chart c of the germ data, centred at a. The sub-chart of A built from c `Chapter5/Theorem5315_Ends · subChartOf` and the chart of the pull-back bundle over it `Chapter5/Theorem5315_Ends · subPullbackData` make $\varphi\circ\iota\circ\psi^{-1}$ the inclusion $\varphi(W \cap A) \subseteq O$, and the principal part of $f|A$ is $\widehat{f}$ on $\varphi(W \cap A)$ `Chapter5/Theorem5315_Ends · principal_restrictSection_subPullback`. Surjectivity of $(f|A)'(a)$ is chart independent, and the derivative of a retraction is the identity on the tangent space of a smaller retract `Chapter5/Theorem5315_Ends · scDeriv_apply_eq_self_of_retract_subset`, so every k in the fibre is $D(\widehat{f}\circ r)(z)v$ for some v tangent to $\varphi(W \cap A)$ `Chapter5/Theorem5315_Ends · exists_scDeriv_eq_of_isLinSurjAt_restrict`. Since A has no boundary, these tangent vectors lie in E_z: the reduced tangent space is chart independent `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber_eq`, and in a chart of A at a point of index 0 the point is interior, so $T^R_a A = T_a A$ `Chapter5/Theorem5315_Ends · mem_minimalSubspaceWith_of_isWithoutBoundary`. So $\mathrm{Lin}_f(0)$ is onto already on a subspace of $T^R_0 O$, $\ker \mathrm{Lin}_f(0) + T^R_0 O = T_0 O$, and the chart criterion (§V.33) concludes. □ Of the end hypotheses only the surjectivity of $f_i'(x)$ and $d_{X_i} \equiv 0$ are used; the kernel complement in X_i and the degeneracy index of x in X are not.

At the boundary `Chapter5/Theorem5315_Ends · StrongBundle.IsSubmersiveGenCompactHomotopy.isGeneralPositionGerm_of_mem_boundary`. If $\partial X \subseteq t^{-1}(0) \cup t^{-1}(1)$, a zero $x \in \partial X$ lies in an end, a sub-M-polyfold without boundary on which the restriction is in general position, hence onto; the previous lemma gives (f,x) in general position.

Compactness and conclusion (1) `Chapter5/Theorem5315_Ends · theorem_5_3_15_generalPosition`, the hypotheses of §V.36. *Proof* `Chapter5/Theorem5315_GeneralPosition · theorem_5_3_15_generalPosition_hyp`. U from Theorem 5.2.2 (preprint 5.5); compactness of S_{f+s} for $s \in \Gamma^{+,1}_U(P)$ by Remark 5.3.4 (§V.10). General position at the zeros on ∂X holds at the zeros of an open $W_b \supseteq S_f \cap \partial X$ (openness, §V.18) `Chapter5/Theorem5315_GeneralPosition · exists_isOpen_isGeneralPositionGerm_of_forall`; Theorem 5.3.10 relative to the compact part $K = S_f \setminus W_b$ (§V.33), with the perturbation supported in $U \setminus \partial X$, gives s_0 small, vanishing near ∂X, with $(f + s_0, x)$ in general position at every zero. □

V.38 Theorem 5.3.15 (preprint 5.22): the parametrized Sard step, and Lagrange multipliers

The text's last step is "a parameterized version of Sard's theorem" (p. 228). Here it is proved in a chart from the inverse function theorem and Sard's lemma in equal dimensions, which Mathlib has.

Parametric transversality in equal dimensions. A surjective $L : \Lambda \times E \to F$ with $\dim E = \dim F$ is completed by a linear $T : \Lambda \times E \to \Lambda$, a projection onto $\ker L \cong \Lambda$, to an isomorphism (L, T) `Chapter5/Theorem5315_ParametricMorse · exists_clm_prod_equiv`. *The local step* `Chapter5/Theorem5315_ParametricMorse · exists_isOpen_measure_image_fst_eq_zero`: near a point where $G'(p_0)$ is onto, (G, T) is a local diffeomorphism by the inverse function theorem; in its coordinates the zeros of G form a graph over Λ, a parameter λ of a zero at which $\partial_z G$ is not injective is a critical value of a map between spaces of the same dimension $\dim \Lambda$, and Sard's lemma in equal dimensions makes these parameters null. *The theorem* `Chapter5/Theorem5315_ParametricMorse · ae_forall_injective_of_surjective`: if $G : \Lambda \times E \to F$ is C^1 on an open W, $\dim E = \dim F$, and $G'(p)$ is onto at every zero $p \in W$, then for almost every λ, at every zero $(\lambda, z) \in W$, $\partial_z G(\lambda, z)$ is injective. Countably many local steps cover W.

Morse functions for almost every parameter `Chapter5/Theorem5315_ParametricMorse · ae_hasOnlyMorseCriticalPointsOn`. If $g : \Lambda \times E \to \mathbb{R}$ is C^2 on the open W and $(\lambda, z) \mapsto \partial_z g(\lambda, z) \in E^*$ — the text's $(\lambda, y) \mapsto d(t \circ \Phi_\lambda)(y)$ in a chart — has a surjective derivative at its zeros, then for almost every λ, $g(\lambda, \cdot)$ has only Morse-type critical points on its slice: apply the previous theorem to $G = \partial_z g$, with $\dim E^* = \dim E$ `Chapter5/Theorem5315_ParametricMorse · finrank_clm_real` and $\partial_z G$ the Hessian `Chapter5/Theorem5315_ParametricMorse · fderiv_slice_eq`. The hypothesis is satisfiable: for a C^2 function h and almost every linear form ℓ, $h + \ell$ has only Morse-type critical points, since $D(\partial_z g)(\ell, z)(\delta\ell, 0) = \delta\ell$ `Chapter5/Theorem5315_ParametricMorse · ae_hasOnlyMorseCriticalPointsOn_add_linear`.

Lagrange multipliers. For a constraint $h : V \to F_c$, F_c finite-dimensional, and $T : V \to \mathbb{R}$, the Lagrangian is $L(v, c) = T(v) - c(h(v))$, $c \in F_c^*$ `Chapter5/Theorem5315_MorseLagrange · lagrangian`. (v, c) is critical iff $h(v) = 0$ and $dT(v) = c \circ dh(v)$ `Chapter5/Theorem5315_MorseLagrange · fderiv_lagrangian_eq_zero_iff`; when $dh(v)$ is onto, a functional vanishing on $\ker dh(v)$ factors through $dh(v)$ `Chapter5/Theorem5315_MorseLagrange · exists_clm_comp_eq_of_ker_le`, so a critical point of T on $\{h = 0\}$ has a multiplier. Near a regular point the level set is the image of a C^∞ map γ on an open set of $\ker dh(v)$, with an affine left inverse and $\operatorname{im} d\gamma(u) = \ker dh(\gamma(u))$ at every point `Chapter5/Theorem5315_MorseLagrange · exists_levelSet_param` (the classical implicit function theorem). The multiplier is *non-degenerate* if the Hessian of L at (v, c) is injective `Chapter5/Theorem5315_MorseLagrange · IsLagrangeNondegenerate`. *The bordered Hessian* `Chapter5/Theorem5315_MorseLagrange · injective_hessian_comp_of_lagrange`: if γ parametrizes $\{h = 0\}$ near $\gamma(u)$, c is a multiplier there and the Lagrangian is non-degenerate, the Hessian of $T \circ \gamma$ at u is injective. By the second-order chain rule and $dh \circ D^2\gamma = -D^2h(d\gamma, d\gamma)$ (from $h \circ \gamma = 0$), the Hessian of $T \circ \gamma$ is the v-Hessian of L on $\ker dh$; a vector a in its kernel gives the kernel vector $(d\gamma(u)a, \delta c)$ of the Hessian of

L, with δc from the factorization through dh, so $d\gamma(u)a = 0$ and $a = 0$.

The family of level sets. For $\mu,\ B_1, \dots, B_d : V \to F_c$, the constraints $h_\lambda = \mu - \sum \lambda_i B_i$ `Chapter5/Theorem5315_MorseFamily · famConstraint` and the gradient of the Lagrangian of the family, $(\lambda, v, c) \mapsto D_{(v,c)}(T - c(h_\lambda))$ `Chapter5/Theorem5315_MorseFamily · famGradient`. *The transversality (5.13)–(5.14) in Lagrange form* `Chapter5/Theorem5315_MorseFamily · surjective_fderiv_famGradient`: if $d\mu(v_0)$ is onto and the forms $\sum \delta_i\, c(dB_i(v_0)\,\cdot)$, over the δ with $\sum \delta_i B_i(v_0) = 0$, restrict onto the dual of $\ker d\mu(v_0)$, then the derivative of the gradient at $(0, v_0, c)$ is onto. The gradient has a component in V^* and one in F_c. The direction δv reaches every value of the F_c-component, $d\mu(v_0)$ being onto `Chapter5/Theorem5315_MorseFamily · fderiv_famGradient_point`; the direction δc gives $-\delta c \circ d\mu(v_0)$ in V^*, all functionals vanishing on $\ker d\mu(v_0)$ `Chapter5/Theorem5315_MorseFamily · fderiv_famGradient_mult`; and the direction $\delta\lambda$ with $\sum \delta_i B_i(v_0) = 0$ gives the forms of the hypothesis, onto the dual of $\ker d\mu(v_0)$ `Chapter5/Theorem5315_MorseFamily · fderiv_famGradient_param`. Only first derivatives of μ and B_i at v_0 enter; the text differentiates the equation $f(\Phi_\lambda(z)) + \sum \lambda_i s_i(\Phi_\lambda(z)) = 0$ twice.

The Sard step for the family `Chapter5/Theorem5315_MorseFamily · ae_isLagrangeNondegenerate_family`. Let T, μ, B_i be C^∞ on the open Ω, $d\mu(v_0)$ onto, and the hypothesis above hold for every multiplier c at v_0 for $\lambda = 0$. Then there are $\varepsilon > 0$ and an open $\Omega' \ni v_0$ such that for almost every λ with $|\lambda| < \varepsilon$, at every $v \in \Omega'$ with $h_\lambda(v) = 0$, $dh_\lambda(v)$ is onto and every multiplier at v is non-degenerate. Surjective operators into a finite-dimensional space form an open set `Chapter5/Theorem5315_MorseFamily · isOpen_setOf_surjective_clm` and the multiplier of a regular level set depends continuously on the point, so the transversality holds near every $(0, v_0, c)$; the parametric Sard theorem above, applied to the Lagrangian of the family on $\Lambda \times (V \times F_c^*)$, gives the rest. This is the text's open neighbourhood $U(x)$ of p. 227.

V.39 Theorem 5.3.15 (preprint 5.22): the families $a_i\sigma$, and transversality at one centre

The solution manifold of a family. For sc⁺-sections $\sigma_1, \dots, \sigma_k$ let $M = \{(\mu, y) \mid f(y) + \sum \mu_j \sigma_j(y) = 0\} \subseteq \mathbb{R}^k \times X$ `Chapter5/Theorem5315_Morse · famZero`. The sections are taken of the form $s_i = a_i \cdot \sigma_{g(i)}$, a_i sc-smooth functions and $g : \{1, \dots, d\} \to \{1, \dots, k\}$ — the form $g \cdot s_e$ of the section in the proof of Lemma 5.3.17 (§V.23). Then y is a zero of $f + \sum \lambda_i a_i \sigma_{g(i)}$ iff $(\Lambda(\lambda, y), y) \in M$, $\Lambda(\lambda, y)_j = \sum_{g(i)=j} \lambda_i a_i(y)$ `Chapter5/Theorem5315_MorseParam · famShift` `Chapter5/Theorem5315_Morse · fiberComb_smul_eq` `Chapter5/Theorem5315_Morse · isZeroAt_paramSlice_smul_iff`. So S_λ lies in the fixed M, and λ enters through the sc-smooth map Λ `Chapter5/Theorem5315_MorseParam · isScSmoothMap_famShift`, sc-smooth into $\mathbb{R}^k$ because its components are `Chapter5/Theorem5315_MorseParam · isScSmoothMap_rn_of_components`. No trivialization Φ_λ and no second derivative of f are needed.

The local statement `Chapter5/Theorem5315_MorseParam · ae_isMorseOrRegularPoint_famSlice`. Let M be smoothly parametrized by $p : \Omega \to M \cap G$, Ω open in a finite-dimensional space, $p(v_0) = (0, y_0)$; let t and $a_1, \dots, a_d$ be sc-smooth and $S_\lambda = \{y \mid (\Lambda(\lambda, y), y) \in M\}$. Suppose $\mu \circ p$ has a surjective derivative at v_0 (μ the projection to $\mathbb{R}^k$), and for every c with $d(t\circ p)(v_0) = c \circ d(\mu\circ p)(v_0)$ the forms $\sum \delta_i c_{g(i)}\, d(a_i \circ p)(v_0)$, over the δ with $\Lambda(\delta, y_0) = 0$, restrict onto the dual of $\ker d(\mu \circ p)(v_0)$. Then there are $\varepsilon > 0$ and an open $O \ni y_0$ such that for almost every λ with $|\lambda| < \varepsilon$ every point of $S_\lambda \cap O$ is a regular point or a Morse-type critical point of $t|S_\lambda$. *Proof.* In the parameter space of p, with $T = t \circ \mathrm{pr}_2 \circ p$ and $B_i = (a_i \circ \mathrm{pr}_2 \circ p) e_{g(i)}$, S_λ near y_0 is the level set $\{h_\lambda = 0\}$; the Sard step for the family (§V.38) applies. At a point of the level set, the level set parametrization composed with p is a smooth parametrization of S_λ `Chapter5/Theorem5315_MorseParam · MPolyfold.IsSmoothParam.of_levelSet`; a critical point of t on it has a multiplier, non-degenerate,

so the bordered Hessian is injective, and the point is interior because Ω' is open. □

Multipliers and critical points. A Lagrange multiplier of t on M at $(0, y)$ makes y a critical point of $t|S_f$ `Chapter5/Theorem5315_MorseTransfer · MPolyfold.IsSmoothParam.fderivWithin_eq_zero_of_multiplier Chapter5/Theorem5315_Morse · not_isRegularPointOn_of_multiplier`. So at a regular point y of $t|S_f$ there is no multiplier, and the hypothesis of the local statement holds vacuously. Regular points form an open subset `Chapter5/Theorem5315_MorseTransfer · MPolyfold.IsSmoothParam.exists_isOpen_regular` and satisfy the clause of §V.36, also for a set that coincides with A near the point `Chapter5/Theorem5315_MorseTransfer · MPolyfold.IsSmoothParam.isMorseOrRegularPoint_of_regular_of_eq Chapter5/Theorem5315_Morse · MPolyfold.IsRegularPointOn`.

The functions. At a critical point x, with one section σ_x and the parametrization (p_x, q_x) of $M_x = \{f+\mu\sigma_x = 0\} \subseteq \mathbb{R}\times X$, the functions are $a_0 = 1$ and $a_i = \arctan(q_{x,i}(0,\cdot))/2$, the bounded coordinates of the parametrization: $|a_i| \le 1$ `Chapter5/Theorem5315_Morse · abs_arctan_div_two_le_one`, and $\arctan(\cdot)/2$ is C^∞ `Chapter5/Theorem5315_Morse · contDiff_arctan_div_two` with non-vanishing derivative.

The transversality survives in the larger family `Chapter5/Theorem5315_MorseTransfer · exists_spanning_of_group`. Let $M_1 \subseteq \mathbb{R} \times X$ be the j-th one-parameter slice of $M \subseteq \mathbb{R}^k \times X$ `Chapter5/Theorem5315_MorseTransfer · inclJ`, both smoothly parametrized near $(0, y)$, the functions $a_0 = 1$, $a_i = \arctan(q_{1,i}(0,\cdot))/2$ built from the left inverse q_1 of the parametrization of M_1, and every multiplier of t on M_1 at $(0, y)$ non-zero. Then for every multiplier c of t on M at $(0, y)$, $c_j \neq 0$, and the forms $\sum \delta_i c_j\, d(a_i \circ p)$, $\sum \delta_i a_i(y) = 0$, restrict onto the dual of $\ker d(\mu \circ p)$: the hypothesis of the local statement. Restricted to the slice, c gives the multiplier c_j on M_1, non-zero; the differentials of the coordinates $q_{1,i}$, and so of the a_i, restrict onto the dual of the tangent space of the solution set at y; and the constant $a_0 = 1$, whose differential vanishes, adjusts any δ to meet $\sum \delta_i a_i(y) = 0$. The multipliers on M_x at $(0, y)$ are non-zero exactly when $(0, y)$ is not critical for t on M_x, an open condition (above) that defines $U(x)$. This is the text's "by construction, the smooth map has, at every point $(0, y)$ satisfying $d(t|S)(y) = 0$, a linearization which is surjective" (p. 227); when the families of several centres are united it needs this argument (Findings, Chapter 5, item 183).

V.40 Theorem 5.3.15 (preprint 5.22): the covering argument

The local data. For a section g `Chapter5/Theorem5315_Morse · StrongBundle.HasMorseLocalData`: (A1) for all sc⁺-sections $\sigma_1, \dots, \sigma_k$ and every zero $y \notin \partial X$, the solution set of $g + \sum \mu_j\sigma_j$ has a smooth parametrization around $(0, y)$ by an open set on which μ is a submersion `Chapter5/Theorem5315_Morse · HasFamilyParam`; (A2) every zero $x \notin \partial X$ has, inside every open $W \ni x$ and with arbitrarily small N, an sc⁺-section σ for which $(0, x)$ is a regular point of t on the solution set of $g + \mu\sigma$ `Chapter5/Theorem5315_Morse · HasRegularFamilyParam`; (A3) the zeros on ∂X are regular points of $t|S_g$.

The covering argument `Chapter5/Theorem5315_Morse · exists_morse_perturbation_hyp`. Let X be tame, f with tame germ data and compact S_f, in general position at every zero, N auxiliary, t sc-smooth, $V \supseteq S_f$ open, and (A1)–(A3) hold for f. Then there is an open U, $S_f \subseteq U \subseteq V$, with S_{f+s} compact for $s \in \Gamma^{+,1}_U(P)$, and for every $\varepsilon > 0$ an $s_0 \in \Gamma^{+,1}_U(P)$ with $N(s_0) \le \varepsilon$, vanishing near ∂X, $(f + s_0, x)$ in general position at every zero, and only Morse-type critical points of t on S_{f+s_0}. *Proof.*

1. *The critical set.* The critical points C of $t|S_f$ form a closed, hence compact, subset of S_f (regular points form an open set, §V.39), and $C \cap \partial X = \emptyset$ by (A3).
2. *The centres.* At $x \in C$, (A2) gives a small σ_x supported in $V \setminus \partial X$ near x and a parametrization of M_x with $(0, x)$ regular for t; the regular points form an open $U(x)$. Finitely many $U(x_1), \dots, U(x_k)$

cover C.

3. *The family.* $\sigma_j = \sigma_{x_j}$, the functions 1 and $\arctan(q_{x_j,i}(0,\cdot))/2$ of §V.39, and the sections $s_i = a_i\sigma_{g(i)}$, allowable and supported in $V \setminus \partial X$.
4. *General position.* f is in general position at every zero, so the family is in general position at every $(0, y)$ (§V.33), and the relative step gives a null set off which the small slices are in general position at every zero `Chapter5/CylinderPerturbation2021 · exists_null_isGeneralPositionGerm_paramSlice` — Theorem 5.3.10's mechanism.
5. *The local statements.* Every zero y of f has an open O_y and a null set. For $y \in \partial X$ the perturbations vanish near y, so $S_\lambda = S_f$ there, and the regularity of (A3) persists on an open set. For $y \notin \partial X$, (A1) parametrizes M at $(0, y)$ with μ a submersion; if y lies in some $U(x_j)$, the hypothesis of §V.39's local statement comes from the survival of the transversality, and otherwise y is regular and the hypothesis is vacuous. The local statement gives O_y and the null set.
6. *Size.* For small $|\lambda|$, $N(\sum \lambda_i s_i) \le \varepsilon$ and the sum is allowable.
7. *Compactness.* Finitely many O_y cover S_f, and the zeros of the slices for small $|\lambda|$ lie in their union: the compactness argument of §V.13 in the margin of Theorem 5.2.2.
8. *The parameter.* Off the finitely many null sets a small λ exists; $s_0 = \sum \lambda_i s_i$. Every zero of $f + s_0$ lies in some O_y, where it is regular or a Morse-type critical point, and $(f + s_0, \cdot)$ is in general position at every zero. □

With the ends `Chapter5/Theorem5315_Morse · theorem_5_3_15_hyp`: the hypotheses of §V.36 and (A1)–(A3) for every perturbed section $f+s$ (s sc$^+$, vanishing near ∂X, with compact solution set and general position at every zero). Let $\eta = \min(\varepsilon, 1)/2$. §V.37 gives U and s_1 with $N(s_1) \le \eta$, vanishing near ∂X, $f_1 = f + s_1$ in general position at every zero; S_{f_1} is compact and inside U. The covering argument for f_1 inside $V = U$ gives s_2 with $N(s_2) \le \eta$. Then $f + (s_1 + s_2) = f_1 + s_2$ `Chapter5/Theorem5315_Morse · sectionAdd_sectionAdd_eq`, $N(s_1 + s_2) \le \min(\varepsilon, 1)$ `Chapter5/CylinderPerturbation2021 · StrongBundle.IsAuxNorm.auxLe_sectionAdd`, its support lies in U, and it vanishes near ∂X `Chapter5/Theorem5315_Morse · isZeroAt_sectionAdd_of_isZeroAt`. □

V.41 Theorem 5.3.15 (preprint 5.22): the local data from the implicit function theorem

(A1)–(A3) hold for every section g with tame germ data in general position at every zero, when X has bump functions and t is as in Definition 5.3.13 `Chapter5/Theorem5315_LocalData · StrongBundle.hasMorseLocalData`: t sc-smooth with values in $[0, 1]$, $dt(x)$ onto at every smooth x (the 2021 condition (3), read in a chart `Chapter5/Book2021 · MPolyfold.HasSurjDerivAt`), and $\partial X \subseteq t^{-1}(0) \cup t^{-1}(1)$.

Parametrizations from tangent germ data `Chapter5/Theorem5315_LocalParam · MPolyfold.TangentGermChart.exists_isSmoothParam`. For regular tangent germ data D of M at x (Definition 3.2.1 (2) (preprint 3.25)), $p = D.\varphi^{-1} \circ \Gamma$ on $\Omega = V \cap O_E$, O_E open in the finite-dimensional N, is a smooth parametrization of M with $p(0) = x$ and $q = \beta \cdot (\pi \circ D.\varphi)$, β a bump function equal to 1 near x and supported in the chart domain, the cut-off of §V.23 `Chapter5/Homotopy · chartLinCutoff`; $h \circ p$ is C^∞ because Γ is sc-smooth on a finite-dimensional domain (§V.14, step 2). *At a zero in general position* `Chapter5/Theorem5315_LocalData · StrongBundle.exists_isSmoothParam_solutionSet`, the index form of Theorem 3.6.8 (§V.17) gives D; the parametrization is by a set of $\mathbb{R}^n$ with unique derivatives (V is relatively open in a partial quadrant), and by an open set when $d_X(y) = 0$, since the degeneracy index of the tame chart is $d_X(y)$ and at index 0 the partial quadrant is the whole space `Chapter5/Theorem5315_LocalData · MPolyfold.IsSmoothParam.exists_isOpen_of_mem_nhds`.

Kernel vectors are tangent vectors. For a smooth parametrization p of A and a chart φ with retraction R around $p(u_0)$, let S be a retraction whose retract lies in $\varphi(A)$ and passes through $z = \varphi(p(u_0))$. Every $w \in T_z$ of that retract has a v, namely $D(q \circ \varphi^{-1} \circ S)(z)w$, with $d(g \circ p)(u_0)v = D(g \circ \varphi^{-1} \circ R)(z)w$ for all sc-smooth g, at interior and boundary points alike `Chapter5/Theorem5315_LocalTangent · MPolyfold.IsSmoothParam.exists_fderivWithin_eq_scDeriv`; the chosen sc-derivative is any bounded operator that restricts to the level-1 derivative `Chapter5/Theorem5315_LocalTangent · scDeriv_eq_of_hasFDerivWithinAt`. At a zero in general position, every strong bundle chart carries a retraction S with retract in $\varphi(S_f)$ through $\varphi(y)$ and range $DS(z) = \ker \mathrm{Lin}(z)$ `Chapter5/Theorem5315_LocalTangent · StrongBundle.exists_retract_range_eq_linKer`, by the equivalence of §V.18 `Chapter5/GeneralPosition · exists_isGeneralPositionGerm_iff`. Together: every $w \in \ker \mathrm{Lin}(z)$ is tangent to every smooth parametrization of S_f through y `Chapter5/Theorem5315_LocalTangent · StrongBundle.exists_fderivWithin_eq_scDeriv_of_mem_linKer`. Since condition (3) is stated in an existential chart, one change of chart for the derivative of a function is needed `Chapter5/Theorem5315_LocalTangent · MPolyfold.exists_scDeriv_eq_chart_change`.

(A1) `Chapter5/Theorem5315_LocalData · StrongBundle.hasFamilyParam`. The family $F(\mu, y) = g(y) + \sum \mu_j \sigma_j(y)$ is in general position at $(0, y)$ (§V.33), so its solution set has a parametrization around $(0, y)$ — open for $y \notin \partial X$ — whose tangent vectors include $\ker F'(0, y)$ in the product chart `Chapter5/Theorem5315_LocalData · StrongBundle.exists_isSmoothParam_famZero`. In the product chart the derivative of μ_j is $(\delta a, \delta u) \mapsto \delta a_j$ `Chapter5/Theorem5315_LocalData · scDeriv_pc0_fst_apply`. Since $g'(y)$ is onto, every $\delta\mu$ has a w with $g'(y)w = -\sum \delta\mu_j \hat{\sigma}_j(y)$, i.e. $(\delta\mu, w) \in \ker F'(0, y)$; so $d(\mu \circ p)(v_0)$ is onto.

(A2) `Chapter5/Theorem5315_LocalData · StrongBundle.exists_hasRegularFamilyParam`. Let $x \notin \partial X$ be a zero, $W \ni x$ open, $\eta > 0$. In a tame chart centred at x, $dt(x) \neq 0$ and E_∞ is dense in E_0, so there is a smooth v_1 with $dt(x)(Dr(0)v_1) \neq 0$. $w_1 = Dr(0)v_1$ and $k = \mathrm{Lin}(0)w_1$ are smooth points, because $Dr(0)$ and $D(\hat{g} \circ r)(0)$ are sc-operators at the smooth point 0 and sc-operators map smooth points to smooth points `Chapter5/Theorem5315_LocalData · IsScOperator.apply_mem_smooth`. Lemma 5.3.3 (preprint 5.11) gives an sc⁺-section through $\Phi^{-1}(0, k)$ supported in W, rescaled by $\theta = \eta/(N(e) + 1)$ to σ; then $(-1/\theta, w_1) \in \ker F'(0, x)$, and the derivative of $t \circ \mathrm{pr}_2$ in the product chart `Chapter5/Theorem5315_LocalData · scDeriv_pc0_snd_apply` is $dt(x)w_1 \neq 0$ on it: $(0, x)$ is a regular point of t on the solution set of $g + \mu\sigma$. Neither the open mapping theorem nor density in the fibre is needed. This is where the text's $b(e_i)$, in an algebraic complement Z (Lemma 5.3.16), is replaced: Lemma 5.3.3 reaches only smooth points of the fibre, and $-c\,g'(x)w_1$ is one because w_1 is smooth.

(A3) at every corner `Chapter5/Theorem5315_LocalData · StrongBundle.isRegularPointOn_of_extremum`. *The extremum lemma* `Chapter5/Theorem5315_LocalTangent · scDeriv_eq_zero_of_extremum`: if U is relatively open in a partial quadrant $C = T^{-1}([0, \infty)^n \oplus W)$ and an sc¹ function τ on U has a minimum or a maximum at a level-1 point z, then $D\tau(z)e = 0$ for every $e \in E_z$. A direction $e \in E_z$ keeps at 0 the coordinates that vanish at z, so $z + se \in C$ for small $|s|$ of both signs; for e of level 1, $s \mapsto \tau(z + se)$ is differentiable with an extremum at 0; the level-1 vectors are dense in E_z `Chapter5/Theorem5315_LocalTangent · mem_closure_levelOne_minimalSubspaceWith` and $D\tau(z)$ is continuous on E_0. At a zero $x \in \partial X$, $t(x) \in \{0, 1\}$ and $0 \leq t \leq 1$, so t has an extremum, and $dt(x)$ vanishes on $E_x \supseteq T^R_x X$, whatever the codimension of $T^R_x X$. General position gives $T_x X = \ker g'(x) + T^R_x X$ `Chapter5/CylinderPerturbation2021 · isParamGP_elim0_of_isGeneralPositionGerm` and $dt(x) \neq 0$, so $dt(x)$ does not vanish on $\ker g'(x)$, the tangent space of the parametrization: x is a regular point of $t|S_g$. No hypothesis on the degeneracy index $d_X(x)$ is needed; of $\partial X = X_0 \sqcup X_1$ only $t(\partial X) \subseteq \{0, 1\}$ enters here, and "the X_i have no boundary" only in §V.37.

The theorem. With (A1)–(A3) for every perturbed section, `theorem_5_3_15_hyp` (§V.40) is the theorem without hypothesis (§V.36). Theorem 5.3.15 holds as printed, with Definition 5.3.13 read in its 2021 form.

V.42 Theorem 5.5.10 (preprint 5.37): every chart has the approximation property

§V.25 proved "(i) □ (ii)" for restricted charts, because the text's $\mathrm{cl}_X(W) \subseteq U$ does not follow from $\mathrm{cl}_C(V) \subseteq O$ (Findings, Chapter 5, item 111). Every chart has the property, by a different route `Chapter5/ApproxPropertyEveryChart · hasApproxProperty_of_isChart`, and Theorem 5.5.10 holds with "every chart" `Chapter5/ApproxPropertyEveryChart · theorem_5_5_10_every_chart`.

Tools on X. With sc-smooth partitions of unity, disjoint closed sets have sc-smooth Urysohn functions `Chapter5/ApproxPropertyEveryChart · MPolyfold.AdmitsScPartitionsOfUnity.exists_urysohn`, and a continuous function with values in $[0,1]$ and support closed in an open set is approximated within ε by sc-smooth functions with closed support in that set `Chapter5/ApproxPropertyEveryChart · MPolyfold.AdmitsScPartitionsOfUnity.exists_approx` — the argument of §V.25 on X itself.

The shells. Let $f : O \to [0,1]$ be continuous with support in $V = O \cap W$, $\mathrm{cl}\, V \subseteq O$, and $F = f \circ \varphi$ on the chart domain; F extended by 0 need not be continuous (§V.25). Write the chart domain as $\{d > 0\}$ with d continuous `Chapter5/ApproxPropertyEveryChart · exists_continuous_pos_iff`. Take sc-smooth κ_n, 1 on $\{d \geq 1/(n+1)\}$ and 0 on $\{d \leq 1/(n+2)\}$, and the weights $\lambda_n = \kappa_n \prod_{j<n}(1-\kappa_j)$, which telescope to 1 on the chart domain and are locally finite. Where $\lambda_n \neq 0$, $\kappa_{n+1} = 1$, so $F = F\kappa_{n+1}$ there; $F\kappa_{n+1}$ has support closed in X, and its approximation $\hat{g}_n$ has closed support in the chart domain. $g = \sum_n \lambda_n \hat{g}_n$ is finite near each point, hence sc-smooth, with values in $[0,1]$ `Chapter5/ApproxPropertyEveryChart · sum_mul_mem_Icc`, and within ε of F as a convex combination of ε-approximations `Chapter5/ApproxPropertyEveryChart · abs_sub_sum_mul_le`. Read in the chart, g gives the approximation on O. □ The restricted charts of §V.25 are kept.

V.43 Theorem 5.5.5 (preprint 5.32): Bonic–Frampton

The statement `Chapter5/BonicFrampton · BonicFrampton.theorem_5_5_5`. On the real $L^p(\mu)$ of any measure space, $p \geq 1$, $F(f) = \|f\|^p$ is (1) C^∞ for p an even integer, (2) C^{p-1} for p an odd integer, (3) $C^{[p]}$ for p not an integer — as printed (gap G.50; Findings, Chapter 5, items 117, 120, 179).

Weighted products `Chapter5/LpWeightedProduct · BonicFrampton.wMul`. For $w \in L^q$ with $q^{-1} + m/p = 1$, $(h_1, \dots, h_m) \mapsto \int w \prod h_i$ is a continuous m-linear form of norm $\leq \|w\|_q$, by the generalized Hölder inequality `Chapter5/LpWeightedProduct · BonicFrampton.eLpNorm_prod_le` `Chapter5/LpWeightedProduct · BonicFrampton.norm_wMul_le`. *The even case* `Chapter5/LpWeightedProduct · BonicFrampton.theorem_5_5_5_even`: $\|f\|^{2k} = \int f^{2k}$ `Chapter5/LpWeightedProduct · BonicFrampton.norm_rpow_eq_integral` is the diagonal of the $2k$-linear form with weight $1 \in L^\infty$, hence C^∞.

The general case. Let g_j be the j-th derivative of $|t|^p$, of the form $a_j \max(t,0)^{p-j} + b_j \max(-t,0)^{p-j}$ `Chapter5/BonicFrampton · BonicFrampton.gK` `Chapter5/BonicFrampton · BonicFrampton.hasDerivAt_gK`. Two one-variable estimates for such homogeneous kernels carry the analysis: $|g(s+t) - g(s)| \leq \varepsilon|s|^r + C_\varepsilon|t|^r$ for $r > 0$, from the continuity of z^r at 1 and homogeneity `Chapter5/BonicFrampton · BonicFrampton.kern_sub_le`, and the first-order remainder bound $|t|(\varepsilon|s|^{r-1} + C_\varepsilon|t|^{r-1})$ by the mean value theorem `Chapter5/BonicFrampton · BonicFrampton.taylor1_le` `Chapter5/BonicFrampton · BonicFrampton.gK_taylor_le`. For $j < p$, $g_j(f) \in L^{p/(p-j)}$ `Chapter5/BonicFrampton · BonicFrampton.memLp_gK`, and the exponents form Hölder triples `Chapter5/BonicFrampton · BonicFrampton.holderTriple_qE`. The forms $T_j(f) = \int g_j(f) \prod h_i$ `Chapter5/BonicFrampton · BonicFrampton.Tj` are continuous in f (the first estimate, integrated) `Chapter5/`

BonicFrampton · BonicFrampton.continuous_Tj, and the derivative of T_j is T_{j+1} curried (the second estimate with Hölder $L^p \times L^{p/(p-j-1)} \to L^{p/(p-j)}$) Chapter5/BonicFrampton · BonicFrampton.hasFDerivAt_Tj. So (T_j) is a Taylor series of $\|\cdot\|^p$ up to every integer $n < p$ Chapter5/BonicFrampton · BonicFrampton.hasFTaylorSeriesUpTo_normRpow, and $\|\cdot\|^p$ is C^n Chapter5/BonicFrampton · BonicFrampton.contDiff_normRpow_of_lt. Case (2) is $n = 2k < 2k+1$, case (3) is $n = [p] < p$. No calculus of Nemytskii operators is needed beyond these two estimates.

V.44 The Sobolev remark after Theorem 5.5.5

After Theorem 5.5.5 the text says that the usual norms of $W^{k,p}(\Omega)$ have the differentiability of $\|\cdot\|_{L^p}$, that $\beta(\|x\|^p)$ is a bump function of L^p, that $W^{1,4}(\Omega)$ with $E_m = W^{1+m,4}(\Omega)$ admits sc-smooth bump functions, while on $W^{1,3/2}(\Omega)$ the straightforward bump is only sc¹, and it asks whether an sc-smooth one exists (pp. 230–231). Formalized, with the sc-structure built on one domain (row V5.76, narrower; gap G.51; Findings, Chapter 5, items 118, 185).

$W^{k,p}(\Omega)$ Chapter5/SobolevRemarkSpaces · SobolevRemark.Wsub. For $\Omega \subseteq \mathbb{R}^n$ and $p \geq 1$: the tuples $(v_\alpha)_{|\alpha|\leq k}$ in the ℓ^p-product of copies of $L^p(\Omega)$ with $\int_\Omega v_0\, \partial^\alpha \varphi = (-1)^{|\alpha|} \int_\Omega v_\alpha \varphi$ for all test functions φ on Ω. Each identity is a continuous functional Chapter5/SobolevRemarkSpaces · SobolevRemark.wk, so the subspace is closed Chapter5/SobolevRemarkSpaces · SobolevRemark.isClosed_Wsub, a Banach space with $\|u\|^p = \sum_\alpha \|D^\alpha u\|_{L^p}^p$ Chapter5/SobolevRemarkSpaces · SobolevRemark.norm_rpow_W; weak derivatives are unique on open Ω Chapter5/SobolevRemarkSpaces · SobolevRemark.Dop_zero_injective.

The norms and the bumps. Any norm with $\|u\|^p = \sum_i \|T_i u\|_{L^p}^p$, T_i continuous linear, has the differentiability of Theorem 5.5.5 Chapter5/SobolevRemarkAbstract · SobolevRemark.theorem_5_5_5_sum, which is the first claim Chapter5/SobolevRemarkSpaces · SobolevRemark.remark_5_5_5_sobolev. For β smooth with compact support, $\beta(\|x\|^p)$ has bounded support Chapter5/SobolevRemarkAbstract · SobolevRemark.bump_comp_rpow Chapter5/SobolevRemarkAbstract · SobolevRemark.hasContDiffBump_of_rpow; "non-vanishing β" is read as $\beta(0) \neq 0$, as in the Hilbert example of p. 229. This gives bumps of L^p Chapter5/SobolevRemarkAbstract · SobolevRemark.hasContDiffBump_Lp and of $W^{k,p}(\Omega)$ Chapter5/SobolevRemarkSpaces · SobolevRemark.remark_5_5_5_sobolev_bump, C^∞, C^{p-1} or $C^{[p]}$ in the three cases.

From level 0 to the sc-structure. A C^∞ bump on E_0 gives the sc-smooth bump property of every local model on E: rescale and translate it to x as in Proposition 5.5.8 (§V.24); it is C^∞ on E_0, hence sc-smooth (Corollary 1.2.5 (preprint 1.20)), and composed with the retraction Chapter5/SobolevRemarkAbstract · LocalModel.hasBumpProperty_of_hasContDiffBump Chapter5/SobolevRemarkAbstract · MPolyfold.admitsBumpFunctions_of_hasContDiffBump. Hence every sc-Banach space with $E_0 \cong W^{1,4}(\Omega)$ admits sc-smooth bump functions, as do its local models and the M-polyfolds modelled on it Chapter5/SobolevRemarkScStructures · SobolevRemark.remark_5_5_5_W14; on $W^{1,3/2}(\Omega)$ the bump $\beta(\|x\|^{3/2})$ is $C^1 = C^{[3/2]}$ on E_0, hence sc¹ Chapter5/SobolevRemarkAbstract · exists_isSck_bump Chapter5/SobolevRemarkScStructures · SobolevRemark.remark_5_5_5_W1_3_2. These use only the level 0: they hold for every sc-structure on these spaces and every Ω. The closing question is a question, not a claim.

The sc-structure $E_m = W^{1+m,p}(\Omega)$ Chapter5/SobolevRemarkScStructures · SobolevRemark.sobolevSc, with the inclusions $W^{k+1,p} \to W^{k,p}$ Chapter5/SobolevRemarkSpaces · SobolevRemark.inclW, injective on open Ω Chapter5/SobolevRemarkSpaces · SobolevRemark.inclW_injective, takes the compactness of the inclusions and the density of E_∞ as hypotheses; $\Omega = \emptyset$ is the trivial instance Chapter5/SobolevRemarkScStructures · SobolevRemark.sobolevSc_hyps_empty. Both are proved for $\Omega = (0,1) \subset \mathbb{R}^1$ and every $p > 1$ Chapter5/SobolevCubeDensity · SobolevCube.cubeSc:

1. *The Volterra operator* $Vg(x) = \int_0^x g$ is compact on $L^p(0,1)$ Chapter5/SobolevCubeVolterra · SobolevCube.isCompactOperator_V: by Hölder $|Vg(x) - Vg(y)| \le \|g\|\,|x-y|^{1-1/p}$ Chapter5/SobolevCubeVolterra · SobolevCube.abs_Vf_sub_le, so the image of the unit ball is uniformly close to step functions on a grid of N intervals Chapter5/SobolevCubeVolterra · SobolevCube.dist_V_le, hence totally bounded.
2. *Rellich.* A function with weak derivative 0 is constant Chapter5/SobolevCubeWeakDerivatives · SobolevCube.eq_cst_of_weakDeriv_zero, so every component of $u \in W^{k,p}$ is a constant plus V of the next one Chapter5/SobolevCubeWeakDerivatives · SobolevCube.eq_cst_add_V Chapter5/SobolevCubeSpaces · SobolevCube.comp_eq_cst_add_V, with the constant bounded by the norm Chapter5/SobolevCubeRellich · SobolevCube.abs_cst_le; the inclusion $W^{k+1,p} \to W^{k,p}$ is compact Chapter5/SobolevCubeRellich · SobolevCube.isCompactOperator_inclW_cube.
3. *Density.* Polynomials are dense Chapter5/SobolevCubeWeakDerivatives · SobolevCube.denseRange_poly; approximating the top derivative and integrating gives smooth approximations on every level Chapter5/SobolevCubeDensity · SobolevCube.exists_smooth_approx, so E_∞ is dense Chapter5/SobolevCubeDensity · SobolevCube.dense_smoothLvl_cube.

On this structure the claims hold for the text's sc-structure Chapter5/SobolevCubeDensity · SobolevCube.remark_5_5_5_W14_cube Chapter5/SobolevCubeDensity · SobolevCube.remark_5_5_5_W14_instance Chapter5/SobolevCubeDensity · SobolevCube.remark_5_5_5_W1_3_2_cube. Cubes in $\mathbb{R}^n$, $n \ge 2$, and other bounded domains — where, as we read it, a regular boundary is presupposed — are not done; hence the row is narrower.

V.45 Theorem 5.5.13 (preprint 5.40): Toruńczyk

The statement recorded in §V.26 Chapter5/PartitionsOfUnity · TorunczykStatement — C^k partitions of unity iff a coordinate-wise C^k homeomorphic embedding into some $c_0(\Gamma)$ — holds for every real Banach space and every $k \in \mathbb{N} \cup \{\infty\}$ Chapter5/Torunczyk · theorem_5_5_13 (gap G.52). The coordinates of $c_0(\Gamma)$ are 1-Lipschitz Chapter5/Torunczyk · abs_cz_sub_le and only finitely many exceed a given ε Chapter5/Torunczyk · finite_cz_gt.

(□) Chapter5/Torunczyk · torunczyk_forward. For each n take a C^k partition of unity $(\rho_{n,c})_{c\in V}$ subordinate to the balls $B(c, 1/(n+1))$, and $h(x) = (\rho_{n,c}(x)/(n+1))_{(n,c)}$. It lies in $c_0(\mathbb{N}\times V)$, since the partitions are locally finite and the weights tend to 0; it is continuous, because up to ε only finitely many n matter; its coordinates are C^k; and it is an embedding: if $h(y)$ is close to $h(x)$, some coordinate (n,c) large at x is still non-zero at y, so y and x share a ball of radius $1/(n+1)$.

(□) Chapter5/Torunczyk · torunczyk_backward, through C^k Urysohn functions Chapter5/Torunczyk · exists_contDiff_urysohn_of_embedding and Mathlib's bump coverings Chapter5/Torunczyk · admitsContDiffPartitionsOfUnity_of_urysohn. With a C^∞ bump θ, 1 on $[-1/2, 1/2]$ and 0 outside $(-3/5, 3/5)$ Chapter5/Torunczyk · thetaB, the *lattice bumps* are $b_{n,\sigma}(x) = \prod_\gamma \theta((n+1)h(x)(\gamma) - \sigma(\gamma))$ for finitely supported $\sigma : \Gamma \to \mathbb{Z}$ Chapter5/Torunczyk · latticeBump. Near a point all but finitely many factors are 1, so they are C^k Chapter5/Torunczyk · contDiff_latticeBump (locally finite sums and products of C^k functions are C^k Chapter5/Torunczyk · contDiff_finsum_of_locallyFinite); for fixed n they are locally finite in σ Chapter5/Torunczyk · locallyFinite_latticeBump; at x, the bump of the rounding σ of $(n+1)h(x)$ equals 1 Chapter5/Torunczyk · exists_latticeBump_eq_one, and its support lies in $\{y \mid \|h(y) - h(x)\| \le 2/(n+1)\}$ Chapter5/Torunczyk · tsupport_latticeBump_subset_ball. For disjoint closed s, t, the bumps whose support misses s or misses t cover V, h being an embedding; Michael's products, which multiply the bumps of level n by η Chapter5/Torunczyk · etaB of the sums of the levels $m < n$, make them locally finite across n,

and the quotient of the sum over the bumps on the side of t by the sum of all is the Urysohn function.

V.46 Theorem 5.5.16 (preprint 5.43) for separable spaces, and Corollary 5.5.17 (preprint 5.44) without hypothesis

E_0 is separable `Chapter5/SeparablePartitionsOfUnity · ScBanach.separableSpace_zero`, for every sc-Banach space: the inclusion $E_1 \to E_0$ is a compact operator with dense range; the range of a compact operator lies in $\bigcup_n nK$ with K compact, so it is separable `Chapter5/SeparablePartitionsOfUnity · IsCompactOperator.isSeparable_range`, and a space with a dense separable subset is separable. So in the sc-calculus Theorem 5.5.16 is needed only for separable spaces (Findings, Chapter 5, item 184).

The separable case. From a C^k bump, a C^k function $\psi : V \to [0,1]$, 1 on $B(0,1)$, supported in $B(0,R)$ `Chapter5/SeparablePartitionsOfUnity · HasContDiffBump.exists_one_on_ball`. For disjoint closed s, t: a countable cover of t by balls $B(x_n, r_n)$ whose R-fold enlargements miss s (Lindelöf), $\psi_n = \psi((\cdot - x_n)/r_n)$, and the telescoping products $f_n = \psi_n \prod_{j<n}(1-\psi_j)$ with $\sum_{n<N} f_n = 1 - \prod_{n<N}(1-\psi_n)$ `Chapter5/SeparablePartitionsOfUnity · sum_range_mul_prod_one_sub` give a C^k Urysohn function `Chapter5/SeparablePartitionsOfUnity · exists_contDiff_urysohn_of_separable`, and bump coverings give C^k partitions of unity `Chapter5/SeparablePartitionsOfUnity · admitsContDiffPartitionsOfUnity_of_separable`: Theorem 5.5.16 for separable spaces, the Bonic–Frampton construction `Chapter5/SeparablePartitionsOfUnity · godefroyTroyanskiWhitfieldZizlerStatement_of_separable`.

Corollary 5.5.17 (preprint 5.44) `Chapter5/SeparablePartitionsOfUnity · corollary_5_5_17_unconditional`. If $\|v\|^2 = b(v,v)$ on E_0, the local model has the sc-smooth approximation property, with no further hypothesis: E_0 is separable, has the C^∞ bump $\tau(1 - b(v,v))$ (§V.26), hence C^∞ partitions of unity, and Proposition 5.5.11 (preprint 5.38) applies. The M-polyfold form `Chapter5/SeparablePartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional` drops the hypothesis of §V.26.

Hilbert spaces of any density `Chapter5/HilbertPartitionsOfUnity · admitsContDiffPartitionsOfUnity_of_hilbertSpace`. For a Hilbert basis (b_γ), $x \mapsto (\|x\|^2, (\langle b_\gamma, x\rangle)_\gamma)$ maps into c_0 with C^∞ coordinates, and it is a homeomorphic embedding by a Kadec–Klee estimate in ℓ^2 `Chapter5/HilbertPartitionsOfUnity · l2_inverse_estimate`: if $|v_\gamma - u_\gamma| \le \delta$ for all γ and $|\|v\|^2 - \|u\|^2| \le \delta$, then $\|v-u\|^2 \le n\delta^2 + 2\delta + 2n\delta(2\|u\| + \delta) + 4\,\mathrm{tail}_F(u)$, with F a set of n coordinates carrying most of u. Toruńczyk (□) concludes `Chapter5/HilbertPartitionsOfUnity · godefroyTroyanskiWhitfieldZizlerStatement_of_hilbertSpace`.

V.47 Theorem 5.5.16 (preprint 5.43): every WCG Banach space

`Chapter5/WCGPartitionsOfUnity · theorem_5_5_16`. A weakly compactly generated Banach space with a C^k bump admits C^k partitions of unity, for every k (gap G.53). The recorded statement `Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement` leaves completeness out; the theorem assumes it, the text's "Banach". The route follows Deville–Godefroy–Zizler, reorganized so that neither Krein's theorem nor the Eberlein–Šmulian theorem nor a transfinite recursion is needed.

Bases `Chapter5/LocallyFiniteBases · IsRelC0Base`. A *base relative to Q* is a locally c_0-finite family of C^k functions with values in $[-1,1]$ whose cozero sets are Q-small. A base gives partitions of unity: $x \mapsto (f_i(x))_i$ is a coordinate-wise C^k embedding into $c_0(I)$, and Toruńczyk applies `Chapter5/`

LocallyFiniteBases · admitsContDiffPartitionsOfUnity_of_isRelC0Base; conversely $f_{(c,n)} = \rho_{n,c}/(n+1)$ Chapter5/LocallyFiniteBases · exists_isRelC0Base_of_admits. Bases transfer along operators Chapter5/LocallyFiniteBases · IsRelC0Base.comp, and a bump gives a cut-off Chapter5/LocallyFiniteBases · HasContDiffBump.exists_cutoff. *The gluing lemma* Chapter5/LocallyFiniteBases · isRelC0Base_glue: if X has a C^k bump, Δ_i, Q_i are operators with $\|\Delta_i\| \le C$, $(\|\Delta_i x\|)_i \in c_0$ for every x, every $x \neq 0$ is close to some $Q_i x$ with $\Delta_i x \neq 0$, and each Q_i carries a base relative to Q_i, then X has a base. The new functions are $(m+1)^{-1}(N+1)^{-1}\psi((m+1)\Delta_i y)\,\chi((N+1)(y-Q_i y))\,g_{ij}(y)$ and $(N+1)^{-1}\chi((N+1)y)$, with $\psi = 1-\chi$.

Projections from Lindenstrauss pairs Chapter5/AmirLindenstrauss · IsLPair. For K weakly compact with dense span, a *Lindenstrauss pair* (A, D): A a $\mathbb{Q}$-subspace, D a set of functionals norming A, and $A \cap K$ dense in K for the topology of D. It gives a norm-one projection onto $\operatorname{cl} A$ along $D_\perp$ Chapter5/AmirLindenstrauss · IsLPair.exists_proj, by weak compactness (the finite intersection property) and Mazur's theorem; the convexity of K (Krein) is not used. A monotone closure operator $S \mapsto (A(S), D(S))$, an ω-iteration of finitary operations with global choices of norming functionals and witnesses Chapter5/AmirLindenstrauss · lclA Chapter5/AmirLindenstrauss · lclD, yields a pair Chapter5/AmirLindenstrauss · isLPair_lcl of finite character Chapter5/AmirLindenstrauss · lclA_finite and of cardinality at most $\max(\#S, \aleph_0)$ Chapter5/AmirLindenstrauss · card_lcl.

The resolution of the identity Chapter5/ProjectionalResolution · priP. With e enumerating a dense set by the ordinals below μ, P_α is the projection of the closure of $\{e_\beta \mid \beta < \alpha\}$. The projections commute, $P_\mu = 1$ Chapter5/ProjectionalResolution · priP_top, and the family is continuous at limit ordinals Chapter5/ProjectionalResolution · PRICont Chapter5/ProjectionalResolution · priP_cont because the closure has finite character — so no recursion over the ordinals is needed. For such a family the increments of x are in c_0 Chapter5/ProjectionalResolution · pri_finite, and x is approximated by some $P_{\alpha+1}x$ with a non-zero increment, through the least β with $P_\beta x = x$ Chapter5/ProjectionalResolution · pri_approx.

The induction Chapter5/WCGPartitionsOfUnity · admitsContDiffPartitionsOfUnity_of_wcg_card, strong induction on a cardinal κ bounding a dense set. For $\kappa \le \aleph_0$ the space is separable (§V.46). Otherwise the pieces $P_{\alpha+1}X$ are WCG Chapter5/WCGPartitionsOfUnity · IsWCG.of_proj, carry a C^k bump Chapter5/WCGPartitionsOfUnity · HasContDiffBump.submodule, and have dense sets of smaller cardinality Chapter5/WCGPartitionsOfUnity · exists_dense_card_le, so by induction they admit partitions of unity, hence bases; with $\Delta_\alpha = P_{\alpha+1} - P_\alpha$ and $Q_\alpha = P_{\alpha+1}$ the gluing lemma gives a base of X, and partitions of unity.

V.48 Weak sequential compactness in reflexive spaces

The text uses without comment that a bounded sequence in a reflexive Banach space has a weakly convergent subsequence (proof of Proposition 5.2.7, "after perhaps taking a subsequence"), and that the compact inclusion $F_1 \to F_0$ maps weakly convergent sequences to norm convergent ones (proof of Lemma 5.1.5). Mathlib has the sequential Banach–Alaoglu theorem only for the dual of a separable space, so both are proved here. A space is *reflexive* if the canonical map $F \to F^{**}$ is onto Chapter5/PartitionsOfUnity · IsReflexive (the definition of §V.26); finite-dimensional spaces are reflexive Chapter5/WeakSequentialCompactness · isReflexive_of_finiteDimensional. *Weak convergence* $u_k \rightharpoonup h$ is $g(u_k) \to g(h)$ for every $g \in F^*$ Chapter5/WeakSequentialCompactness · WeakTendsto. Weakly convergent sequences are bounded, by uniform boundedness Chapter5/WeakSequentialCompactness · WeakTendsto.exists_bound, and the norm is weakly lower semicontinuous, $\|h\| - \varepsilon < \|u_k\|$ eventually Chapter5/WeakSequentialCompactness · WeakTendsto.eventually_norm_gt.

Weak sequential compactness `Chapter5/WeakSequentialCompactness · exists_subseq_weakTendsto`: in a reflexive F every bounded sequence has a weakly convergent subsequence; also with a bound that holds only eventually `Chapter5/WeakSequentialCompactness · exists_subseq_weakTendsto_of_eventually`. *Step 1.* The closed span M of the sequence is separable, so countably many norm-one functionals g_d norm the points of M. *Step 2.* The values $(g_d(u_k))_d$ lie in a cube $\prod_d[-C, C]$, compact and metrizable; a subsequence makes every $g_d(u_k)$ converge. *Step 3.* The images $J(u_k)$ in the bidual lie in a weak-*compact ball (Banach–Alaoglu). Every weak* cluster point is $J(p)$ for some $p \in M$, by reflexivity and Hahn–Banach, and the g_d determine p; so the subsequence has one cluster point, and a sequence in a compact set with one cluster point converges. **Two consequences.** A compact operator maps weakly convergent sequences to norm convergent ones `Chapter5/WeakSequentialCompactness · WeakTendsto.tendsto_of_isCompactOperator`. In a reflexive space, a bounded sequence whose image under an injective bounded operator converges to the image of h converges weakly to h `Chapter5/WeakSequentialCompactness · weakTendsto_of_bounded_of_tendsto`: every subsequence has a weakly convergent subsequence, whose limit is h by injectivity.

V.49 Definition 5.1.4 (2017: 5.2; not in the preprint) and Lemma 5.1.5 (2017: 5.2; not in the preprint)

Definition 5.1.4 (restated) `Chapter5/ReflexiveFibres · StrongBundle.HasReflexiveFibers`: $P : Y \to X$ has reflexive $(0, 1)$-fibres if every $x \in X$ lies in the domain of a strong bundle chart compatible with the atlas whose model has F_1 reflexive. The text asks for an equivalent atlas of such charts; this is read as "the reflexive compatible charts cover X", which is the same, since chart domains are of the form $P^{-1}(V)$. It holds if every chart of the given atlas is reflexive `Chapter5/ReflexiveFibres · StrongBundle.hasReflexiveFibers_of_atlas`. In a chart c the fibre part of a point $y \in Y_{0,1}$ is $h \in F_1$ `Chapter5/ReflexiveFibres · StrongBundleChart.liftOneSnd`.

Lemma 5.1.5 `Chapter5/ReflexiveFibres · lemma_5_1_5`: let $y_k \in Y_{0,1}$ with $P(y_k) \to x$ on level 0, and Ψ, Ψ' charts around x with F_1, F_1' reflexive; then the fibre parts h_k converge weakly in F_1 iff the h_k' converge weakly in F_1'. The strong form `Chapter5/ReflexiveFibres · lemma_5_1_5_limit` identifies the limits, which the text's proof uses implicitly: if $h_k \rightharpoonup h$ (and only F_1' needs to be reflexive), there is one $y \in Y_{0,1}$ over x with Ψ-part h, $y_k \to y$ in Y, and h_k' converges weakly to the Ψ'-part of y `Chapter5/ReflexiveFibres · StrongBundle.weakTendsto_transfer`. *Step 1, the limit point.* $h_k \rightharpoonup h$ in F_1 gives $h_k \to h$ in F_0, through the compact inclusion (§V.48). The limit (a, h) lies in the local model K, because the retraction is continuous on $U \lhd F$ and fixes K `Chapter5/ReflexiveFibres · StrongBundleChart.mem_K_of_tendsto`; it is a point y of $Y_{0,1}$ with $y_k \to y$ `Chapter5/ReflexiveFibres · StrongBundle.exists_limit_of_weakTendsto`. *Step 2, the other chart.* The h_k' are bounded in F_1', by the local comparison of the norms $|\cdot|_1$ of two charts (Lemma 5.1.2), and converge in F_0' to the fibre part of y, because $y_k \to y$ in Y and Ψ' is continuous; so they converge weakly in the reflexive F_1' (§V.48) `Chapter5/ReflexiveFibres · StrongBundle.weakTendsto_liftOneSnd_of_tendsto`. Fibre parts are read in a chart only for large k.

V.50 Definitions 5.1.6 and 5.1.7 (2017: 5.3 and 5.4; not in the preprint): mixed convergence and reflexive auxiliary norms

Definition 5.1.6 `Chapter5/ReflexiveFibres · StrongBundle.MixedTendsto`: $y_k \in Y_{0,1}$ is mixed convergent to $y \in Y_{0,1}$ if $P(y_k) \to P(y)$ in X_0 and, in some chart around $P(y)$ with F_1 reflexive, the fibre parts converge weakly. The text's "in view of this lemma we can define" is proved: the fibre parts then

converge weakly in every compatible reflexive chart around $P(y)$ Chapter5/ReflexiveFibres · StrongBundle.MixedTendsto.weakTendsto_of_isChart, and $y_k \to y$ in Y Chapter5/ReflexiveFibres · StrongBundle.MixedTendsto.tendsto_oneVal, both by Lemma 5.1.5. *Mixed convergent subsequences* Chapter5/ReflexiveFibres · StrongBundle.exists_mixedTendsto_subseq: if $P(y_k) \to x$ and $N(y_k) \le r$ for an auxiliary norm N, a subsequence is mixed convergent; the fibre parts in a reflexive chart are bounded by Lemma 5.1.2, and §V.48 applies.

Definition 5.1.7 Chapter5/ReflexiveFibres · StrongBundle.IsReflexiveAuxNorm: a reflexive auxiliary norm is an auxiliary norm with $N(y) \le \liminf_k N(y_k)$ whenever y_k is mixed convergent to y. The text prints $\liminf_{k\to 0}$; presumably $k \to \infty$. The lim inf is taken in the extended reals, so that an unbounded sequence has $\liminf = \infty$. Equivalent forms: if $N(y_k) \le r$ infinitely often then $N(y) \le r$ Chapter5/ReflexiveFibres · StrongBundle.IsReflexiveAuxNorm.le_of_frequently_le, and a criterion through eventual bounds Chapter5/ReflexiveFibres · StrongBundle.le_liminf_coe_of_eventually.

V.51 Proposition 5.1.8 (2017: 5.2; not in the preprint): reflexive auxiliary norms exist

Proposition 5.1.8 Chapter5/ReflexiveFibres · proposition_5_1_8: with reflexive $(0,1)$-fibres a reflexive auxiliary norm exists Chapter5/ReflexiveFibres · proposition_5_1_8_exists, and for an auxiliary norm N_0 and a reflexive one N_1 there is a continuous $f : X \to (0,\infty)$ with $f(x)N_0 \le N_1 \le f(x)^{-1}N_0$ on the fibre over x Chapter5/ReflexiveFibres · proposition_5_1_8_compare. The text leaves the proof to the reader. *Existence.* The norm of Proposition 5.1.3 (preprint 5.3) built over the reflexive compatible charts, $N(w) = \sum_i \rho_i(P(w))\,|\Phi_i[1](w)|_1$ for a partition of unity (ρ_i) subordinate to their domains, is an auxiliary norm by Lemma 5.1.2 (preprint 5.2). *Lower semicontinuity* Chapter5/ReflexiveFibres · StrongBundle.le_liminf_gluedNorm: near $P(y)$ only finitely many terms are nonzero; $\rho_i(P(y_k)) \to \rho_i(P(y))$; in each chart the fibre parts converge weakly, by the chart independence of §V.50, and the norm is weakly lower semicontinuous (§V.48). So N is a reflexive auxiliary norm Chapter5/ReflexiveFibres · StrongBundle.isReflexiveAuxNorm_gluedNorm. *Comparison.* A reflexive auxiliary norm is an auxiliary norm, and Proposition 5.1.3 (preprint 5.3) compares any two. Two tools for later chapters: $(g \circ P)\cdot N$ is again a (reflexive) auxiliary norm for continuous $g > 0$ Chapter5/ReflexiveFibres · StrongBundle.IsReflexiveAuxNorm.comp_mul, and with reflexive fibres every auxiliary norm lies between two reflexive ones Chapter5/ReflexiveFibres · StrongBundle.exists_reflexiveAuxNorm_sandwich.

V.52 Definition 5.2.6 and Proposition 5.2.7 (2017: 5.6 and 5.3; not in the preprint)

Definition 5.2.6 Chapter5/ReflexiveCompactness · StrongBundle.HasReflexiveLocalCompactness: f has the reflexive local compactness property if for every reflexive auxiliary norm N and every x there is an open $U(x) \ni x$ with $\mathrm{cl}_X\{q \in U(x) \mid N(f(q)) \le 1\}$ compact. The text prints $N(f(x)) \le 1$ inside the set; presumably $N(f(q))$. $N(f(q)) \le 1$ includes $f(q) \in Y_{0,1}$, as in §V.32. The standing assumptions of the text (reflexive fibres, f sc-smooth and regularizing) are hypotheses of the statements that use it.

Proposition 5.2.7 (stronger) Chapter5/ReflexiveCompactness · proposition_5_2_7. With the text's hypotheses: (1) every $x \in X_0 \setminus X_1$ has a neighbourhood on which $N(f(q)) > 1$; (2) so has every x with $N(f(x)) > 1$. *(2) for every continuous section* Chapter5/ReflexiveCompactness · StrongBundle.exists_nhds_not_auxLe. If not, x is a limit of points x_k with $N(f(x_k)) \le 1$; X is metrizable (Theorem 2.3.10), so a sequence suffices. A

subsequence of $f(x_k)$ is mixed convergent to some y (§V.50), and $f(x_k) \to y$ in Y, so $y = f(x)$ by continuity of f; lower semicontinuity gives $N(f(x)) \le 1$. *(1)*: for a regularizing f, $f(x) \notin Y_{0,1}$ at $x \notin X_1$, so $N(f(x)) = \infty$ `Chapter5/ReflexiveCompactness · StrongBundle.not_auxLe_of_not_mem_levelSet`, and (2) applies. The reflexive local compactness property is not used, and (2) needs only continuity of f. In other words, $\{x \mid N(f(x)) \le 1\}$ is closed `Chapter5/ReflexiveCompactness · StrongBundle.isClosed_setOf_auxLe`.

The remark after Proposition 5.2.7 (stronger) `Chapter5/ReflexiveCompactness · StrongBundle.closure_auxLe_subset`: the closure of $\{x \in U \mid N(f(x)) \le 1\}$ lies in $\{x \in \mathrm{cl}_X U \mid N(f(x)) \le 1\}$, for every U and no compactness assumption, since the second set is closed. The text prints $\{x \in X \mid \dots\}$ in the closure; presumably $x \in U$, which is what the argument above it uses.

V.53 Theorem 5.2.8 (2017: 5.4; not in the preprint) and Remarks 5.2.9, 5.2.10 (2017: 5.1, 5.2)

Theorem 5.2.8 (stronger) `Chapter5/ReflexiveCompactness · theorem_5_2_8`. Hypotheses as in the text: X tame, reflexive $(0,1)$-fibres, f sc-Fredholm with compact solution set S and the reflexive local compactness property, N a reflexive auxiliary norm, and U_∂ an open neighbourhood of $S_\partial = S \cap \partial X$ in ∂X with $\mathrm{cl}\{x \in U_\partial \mid N(f(x)) \le 1\}$ compact; "open in ∂X" is read as $U_\partial = W \cap \partial X$ with W open. Conclusion: an open $U \supseteq S$ with $U \cap \partial X = U_\partial$ and $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact. Tameness and the sc-Fredholm property are not used; the theorem holds for every continuous section of a strong bundle with reflexive fibres over any M-polyfold `Chapter5/ReflexiveCompactness · theorem_5_2_8_of_continuous`. *Step 1, the boundary part.* Finitely many $U(\bar{x}_1), \dots, U(\bar{x}_\ell)$ of Definition 5.2.6 cover the compact $\mathrm{cl}\{x \in U_\partial \mid N(f(x)) \le 1\}$; with the open set $\{N \circ f > 1\}$ (§V.52) they form $\tilde{U}$, which contains U_∂. *Step 2, the interior part.* $S \setminus \tilde{U}$ is compact and misses ∂X; finitely many $U(x) \setminus \partial X$ cover it, and their union with $\tilde{U}$ is an open U'. *Step 3, compactness.* $\{x \in U' \mid N(f(x)) \le 1\}$ lies in the finite union of the compact sets $\mathrm{cl}\{q \in U(x_i) \mid N(f(q)) \le 1\}$, because on $\{N \circ f > 1\}$ there is nothing to add. *Step 4, the boundary.* $U = U' \cap (W \cup (X \setminus \partial X))$ contains S, has $U \cap \partial X = U_\partial$, and its controlled set is a closed subset of the compact union. Witness for all hypotheses at once: the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ with the section id (solution set $\{0\}$, $U_\partial = \emptyset$) `Chapter5/ReflexiveCompactness · TrivialBundle.theorem_5_2_8_witness`.

Remark 5.2.9 restates Theorem 5.2.8 in words: if (U_∂, N) controls compactness on the boundary, some (U, N) with $U \cap \partial X = U_\partial$ controls it on X. **Remark 5.2.10**, that the theorem is crucial for inductive perturbation arguments, has no mathematical content to formalize.

V.54 Definition 5.4.1 (2017: 5.11; not in the preprint): sc⁺-sections over the boundary

Definition 5.4.1 (restated) `Chapter5/BoundaryExtension · StrongBundle.IsBoundaryScPlusSection`: a section s of $W|\partial X \to \partial X$ is an sc⁺-section over ∂X if its restrictions to the local faces at every $x \in \partial X$ are sc⁺-sections. Read chart-wise: on ∂X, s is a section with values in $W_{0,1}$, and for every tame strong bundle chart Φ around x, every tame presentation T of the corner of its model and every $i \in I(\varphi(x))$, the local representative $\Phi[1] \circ s \circ \varphi^{-1}$ restricted to the face germ $O \cap \{t_i = 0\}$ at $\varphi(x)$ is an sc⁺-section of $K \to O$ near $\varphi(x)$ (Part XII, §XII.12). That $s(x) \in W_{0,1}$ is stated outright, so that $N(s(x))$ in Theorem 5.4.2 is defined at every boundary point. The definition quantifies over all tame charts and presentations; its witness is the zero section, for every strong bundle `Chapter5/BoundaryExtension · StrongBundle.`

`isBoundaryScPlusSection_fiberZero`. The text's $\mathrm{supp}(s)$ is the closure of $\{x \in \partial X \mid s(x) \neq 0_x\}$ `Chapter5/BoundaryExtension · StrongBundle.boundarySupport`.

V.55 Theorem 5.4.2 (2017: 5.9; not in the preprint): the ∂-extension theorem

Theorem 5.4.2 `Chapter5/BoundaryExtension · theorem_5_4_2`. Let $P : W \to X$ be a strong bundle over a tame M-polyfold admitting sc-smooth partitions of unity, s an sc⁺-section over ∂X, N an auxiliary norm, $\tilde{U}$ an open neighbourhood of $\mathrm{supp}(s)$, and $f : X \to [0, \infty)$ continuous, supported in $\tilde{U}$, with $N(s(x)) < f(x)$ on $\mathrm{supp}(s)$. Then there is an sc⁺-section $\tilde{s}$ with (1) $N(\tilde{s}) \leq f$, (2) $\mathrm{supp}(\tilde{s}) \subset \tilde{U}$, (3) $\tilde{s}|\partial X = s$. The text gives a sketch: Proposition 12.3.7 locally, glued by a partition of unity. "Strong bundle over a tame M-polyfold" is the project's tame strong bundle, whose base is tame and every point of which lies in a tame strong bundle chart; this fixes the spaces in the lowest universe, and the core below takes only the chart hypothesis. *Step 1, the local step* `Chapter5/BoundaryExtension · StrongBundle.exists_local_extension`. At $x \in \partial X$, in a tame chart, Proposition 12.3.7 (Part XII, §XII.12) extends the local representative of s near $\varphi(x)$; a boundary point of X is a boundary point of the model `Chapter5/BoundaryExtension · StrongBundle.degIndex_ne_zero_of_mem_boundary`. Cut off by a bump function equal to 1 near x (Proposition 5.5.3 (preprint 5.30)), the extension becomes a global sc⁺-section agreeing with s on ∂X near x, by the locality of sc⁺ `Chapter5/BoundaryExtension · StrongBundle.isScPlusSection_of_locally_p3b`. *Step 2, local bounds* `Chapter5/BoundaryExtension · StrongBundle.exists_local_bounded`. Every x has an open $U \ni x$ and an sc⁺-section σ with $N(\sigma) \leq f$ on U and $\sigma = s$ on $\partial X \cap U$. At $x \in \mathrm{supp}(s)$, σ is the local extension, and $N(\sigma) < f$ near x by continuity of $N \circ \sigma$ and of f. Elsewhere $\sigma = 0$ `Chapter5/BoundaryExtension · StrongBundle.aux_fiberZero`: off ∂X, and on $\partial X \setminus \mathrm{supp}(s)$, where s vanishes on a neighbourhood. *Step 3, gluing* `Chapter5/BoundaryExtension · StrongBundle.exists_boundary_extension`. With an sc-smooth partition of unity (ρ_i) subordinate to these U_i, $\tilde{s} = \sum_i \rho_i \sigma_i$ is sc⁺ by locality; $N(\tilde{s}) \leq \sum_i \rho_i N(\sigma_i) \leq f$ by the triangle inequality and homogeneity of N; on ∂X, $\tilde{s} = \sum_i \rho_i s = s$ since $\sum_i \rho_i = 1$. (2) follows from (1): where $f = 0$, $N(\tilde{s}) = 0$, so $\mathrm{supp}(\tilde{s}) \subset \mathrm{supp}(f) \subset \tilde{U}$. That $\tilde{U}$ is open and contains $\mathrm{supp}(s)$ is kept from the text and not used.

Part VI. Linearizations, orientations and invariants (HWZ Chapter 6, §§6.2–6.8 (preprint §§6.1–6.7))

Chapter 6 builds the orientation theory of sc-Fredholm sections and, at its end, the invariant of oriented proper sections. At a smooth point x it collects the linearizations of f into a convex set $\mathrm{Lin}(f,x)$ of Fredholm operators (§6.2 (preprint §6.1)); it fixes the conventions for determinant lines and for the isomorphism attached to a four-term exact sequence (§6.3 (preprint §6.2)); it defines the determinant $\det(T)=\lambda(\ker T)\otimes\lambda^*(\mathrm{coker}\,T)$ of a Fredholm operator, compares it with the determinants of PT for good left-projections P and of the stabilizations T_φ, and builds the determinant line bundle $\mathrm{DET}(E,F)$ over the Fredholm operators (§§6.4–6.5 (preprint §§6.3–6.4)); it then restricts that bundle to $\mathrm{Lin}(f,x)$, propagates its orientations along sc-smooth paths of X, first in small neighbourhoods (Theorem 6.6.11 (preprint 6.35)) and then globally, as a functor on the path category (Theorem 6.6.13 (preprint 6.37)), puts a topology on the orientation space (Theorem 6.6.17, new in 2021), and defines orientability and orientations of f, orients the solution set (Theorem 6.6.18 (preprint 6.41)) and the sections $f+s$ over $X\times[0,1]$ (Remark 6.6.19 (preprint 6.42)) (§6.6 (preprint §6.5)); §6.7 (preprint §6.6) defines the admissible perturbations and the invariant $\Psi_{(f,o)}:H^*_{dR}(X,\partial X)\to\mathbb{R}$ and proves that it is well defined and invariant under proper oriented homotopies (Theorem 6.7.1 (preprint 6.43)); and the appendix §6.8 (preprint §6.7) proves Lemma 6.3.3 (preprint 6.5) and Proposition 6.4.11 (preprint 6.12). This Part retells §6.2 (preprint §6.1) (§VI.1); §6.3 (preprint §6.2) — determinant lines, Definition 6.3.1 (preprint 6.3) and Proposition 6.3.2 (preprint 6.4), the isomorphism Φ_E, Lemma 6.3.3 (preprint 6.5) and Proposition 6.3.5 (preprint 6.6) (§§VI.2–VI.3); §6.4 (preprint §6.3) — Definitions 6.4.1–6.4.3 (preprint 6.7–6.9), Lemma 6.10 (preprint; not in the 2021 edition), the isomorphisms γ_T^P, the stabilization and the direct sums (§§VI.4–VI.6); §6.5 (preprint §6.4) — the local determinant bundles and $\mathrm{DET}(E,F)$ (§§VI.7–VI.8); §6.6 (preprint §6.5) — orientations over $\mathrm{Lin}(f,x)$ (§VI.9), Proposition 6.6.3 (preprint 6.28) (§VI.10), Lemmas 6.6.4–6.6.6 (preprint 6.29–6.31) (§§VI.11–VI.13), the push-forward (§§VI.14–VI.15), Lemma 6.6.8 (preprint 6.33), Lemma 6.6.10 (preprint 6.34) and Proposition 6.6.7 (preprint 6.32) (§§VI.16–VI.17), Theorem 6.6.11 (preprint 6.35) and Definition 6.6.12 (preprint 6.36) (§§VI.18–VI.19), the path category (§VI.20), the comparison of charts (§VI.21), the transport and Theorem 6.6.13 (preprint 6.37) (§§VI.22–VI.23), Definitions 6.6.14–6.6.16 (preprint 6.38–6.40) (§VI.24), Theorem 6.6.18 (preprint 6.41) (§VI.25) and Remark 6.6.19 (preprint 6.42) (§VI.26); §6.7 (preprint §6.6), the admissible sections and Theorem 6.7.1 (preprint 6.43) with its full proof (§VI.27); the appendix §6.8 (preprint §6.7) (§§VI.28–VI.29); what is not formalized (§VI.30); what the formalization taught us (§VI.31); the 2021 additions — Lemma 6.4.4, Proposition 6.4.13, Lemmas 6.6.8 and 6.6.9, Theorem 6.6.17 (§§VI.32–VI.33); and the material formalized since the

first edition of this Part: the two conventions of §§6.3–6.4 and the remark that $\mathrm{DET}(E,F)$ is not orientable (§VI.34), Lemma 6.5.3 (preprint 6.19) for general vector bundles (§VI.35), Lemmas 6.6.4 and 6.6.5 (preprint 6.29 and 6.30) as statements about manifolds (§VI.36), paths and families on $[0,1]$ and the facts on paths before Proposition 6.2.2 (preprint 6.2) (§VI.37), and the orientation bundle after Theorem 6.6.17 (§VI.38). Proposition 6.2.2 (preprint 6.2) was formalized with Chapter 3 and is retold in §III.49. Each item carries the number of its comparison item (V6.1–V6.82) in `Exposition/verification/ch06.md`.

What is proved, and what is not. *Theorem 6.7.1 (preprint 6.43) is proved, with the hypotheses of the text only* — X tame with sc-smooth bump functions — and so is the case $\partial X = \emptyset$ after it (§VI.27; rows V6.76, V6.77 restated; gaps G.54, G.55 closed). The proof follows the text's: one admissible perturbation defines Ψ, Stokes' theorem makes it a map on cohomology, and a cobordism S_F over $\mathbb{R}^1 \times X$ compares two perturbations and two ends of a homotopy; the decomposition of ∂S_F is replaced by a cut-off in time. Theorems 6.6.11, 6.6.13, 6.6.17 and 6.6.18 (preprint 6.35, 6.37, —, 6.41) are proved, the last three with the text's hypotheses (§§VI.18, VI.23, VI.33, VI.25); Theorem 6.6.18 (preprint 6.41) in the book's shape, the solution set a compact Mathlib manifold with boundary and corners carrying a continuous tangent orientation which is the Fredholm one, read compatibly in any two charts, and the paragraph after it on perturbed solution sets (gaps G.67–G.69 closed). Remark 6.6.19 (preprint 6.42) gives an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40), the 2021 convention e first, with no sign (§VI.26; gap G.66 closed). Proved besides: Propositions 6.3.2, 6.3.5, 6.4.10, 6.4.11, 6.4.13, 6.4.15, 6.5.4, 6.5.7, 6.5.8, 6.6.1, 6.6.7 (preprint 6.4, 6.6, Lemma 6.11, 6.12, —, 6.15, 6.20, 6.23, 6.24, 6.26, 6.32), Lemmas 6.3.3, 6.4.4, 6.4.14, 6.5.1, 6.5.2, 6.5.5, 6.5.9, 6.6.6, 6.6.8–6.6.9, 6.6.10 (preprint 6.5, —, 6.14, 6.17, 6.18, 6.21, 6.25, 6.31, 6.33, 6.34) and Lemma 6.10 (preprint), Corollary 6.4.12 (preprint 6.13), Lemma 6.5.3 (preprint 6.19) for general topological vector bundles (§VI.35; G.59 closed), Lemma 6.6.4 (preprint 6.29) with property (P) for the slices and Lemma 6.6.5 (preprint 6.30) as a statement about manifolds, with π a submersion and L a smooth vector bundle (§VI.36; G.62, G.63 closed), Proposition 6.6.3 (preprint 6.28) with (1) as printed in 2021 and with (1′), on $\mathbb{R}$ and on $[0,1]$ (§VI.10; G.61 closed), the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), the sc-Fredholm property of $\mathrm{Lin}(f,x)$ on every level (§VI.1; G.64 closed), the facts on paths recalled before Proposition 6.2.2 (preprint 6.2) (§VI.37; G.56 closed), the alternative determinant and the identification (6.25) (preprint (95)) (§VI.34; G.57, G.58 closed), Proposition 6.4.13, last sentence (§VI.33; G.81 closed), property (2) of Theorem 6.6.17 without further hypotheses (§VI.33; G.90 closed), the orientation bundle, the $\mathbb{Z}_2$-action and $\det_f$ after it (§VI.38; G.91 closed), and the classical determinant bundle at the opening of §6.6 (§VI.34). Every definition of the chapter is formalized. **Gaps:** G.54–G.64, G.66–G.69, G.81, G.90, G.91 closed (G.61 with Proposition 6.6.3 (1) in its 2021 wording); G.65 (paths and families on $[0,1]$) in part; no gap is open. **Narrower rows**, with their reasons: V6.48, Theorem 6.6.11 (preprint 6.35) — the statement does not introduce the section f (presumably an omission; it is read for an sc-Fredholm section with tame germ data), and U is not shown contractible in itself, the preprint's (1); the 2021 form of (1), a smaller U' contractible in U, is proved, and (1′) is what the proof uses; V6.80, the remark that $\mathrm{DET}(E,F)$ is not orientable — it does not hold as we read it for finite-dimensional E, F (Lean counterexample `exists_two_lineOrientation_det_of_finiteDimensional`) and is proved for infinite-dimensional Hilbert spaces and under complex structures, not for all Banach spaces; V6.81, the classical case — C^1 maps in place of smooth ones, and a topological, not smooth, line bundle. Several places needed more than a reading. The projection of the proof of Lemma 6.10 (preprint) does not lie below the two given projections as we read it, and the lemma is proved by another construction (§VI.4). The isomorphism Φ_E of formula (94) does not vary continuously in the topology of DET where $\dim\operatorname{coker} T$ changes parity, as we read it, so the propagation uses $(-1)^{\dim\operatorname{coker} T}\Phi_E$ (§VI.14). Proposition 6.6.3 (1) is proved in its 2021 wording, a smaller U' contractible in U; the preprint's "U is

sc-smoothly contractible" is not established by its proof as we read it, and the proofs use (1′) (§§VI.10, VI.18). The form $\bar{\omega} = p_1^*\omega \wedge p_2^* dt$ of the proof of Theorem 6.7.1 (preprint 6.43) has one degree too many for the comparison, as we read it, and $p_1^*\omega$ is used (§VI.27). Seven proofs the text omits are supplied: the orientability of a line bundle over a convex set (§VI.9), the general position of Lemma 6.6.4 (preprint 6.29) (§VI.11), the "simple computation" of Lemma 6.6.8 (preprint 6.33) (§VI.16), the covering argument of Theorem 6.6.13 (preprint 6.37) (§VI.22), the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) (§VI.24), the construction of the topology of Theorem 6.6.17 (§VI.33), and the orientation of the cobordism on its ends in Theorem 6.7.1 (preprint 6.43) (§VI.27).

Six structural choices. (1) *Chart points.* T_xX carries no linear structure in the formalization (gaps G.6, G.7, G.13 of Chapter 2), so $\mathrm{Lin}(f,x)$, $\mathrm{DET}(f,x)$ and Or_x^f are made at a *chart point* (c,z) of x, with operators $T_zO = Dr(z)E \to K_z = \rho(z)F$ on level 0, and they are identified between charts by explicit transition maps (§VI.21); the text's $f-s$ with $s(x) = f(x)$ is $f+s$ with $(f+s)(x) = 0$. (2) *Global sections, from bump functions.* The text assumes sc-smooth partitions of unity from §6.3 (preprint §6.2) on, "to simplify the presentation" (p. 241 (preprint p. 203)). The formalization assumes sc-smooth bump functions, which partitions of unity give (§V.25), and uses them only to make global sc$^+$-sections (Findings, Chapter 6, item 43); Theorem 6.7.1 (preprint 6.43) assumes them in its statement. (3) *Families on $\mathbb{R}^n$, and on $[0,1]^n$.* Paths, homotopies and families are first sc-smooth maps on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ whose conditions are imposed on $[0,1]$, $[0,1]^2$, $[0,1]^n$, and $X \times [0,1]$ is read as $\mathbb{R}^n \times X$ with the pull-back π^*P of the bundle along the projection, the parameter first (decision D2, as for Proposition 6.2.2 (preprint 6.2) in §III.49; Findings, Chapter 6, item 102). Most of these statements now also hold as printed, on the M-polyfolds $[0,1]$, $[0,1]^2$, $[0,1]^n \times X$ (§VI.37); what is still read on $\mathbb{R}$ is listed in gap G.65 (§VI.30). The cobordism of Theorem 6.7.1 (preprint 6.43) lives over $\mathbb{R}^1 \times X$. (4) *The sign $\alpha(T) = (-1)^{\dim \operatorname{coker} T}$* at both end points of every propagation (§VI.14). (5) *(1′), and orientable neighbourhoods as pairs $V \subseteq U$*, since what the proofs use of Proposition 6.6.3 (preprint 6.28) is (1′) rather than contractibility (§§VI.10, VI.19). (6) *Dyadic subdivisions* in place of the finite coverings of the proof of Theorem 6.6.13 (preprint 6.37) (§VI.22). An orientation of a line is a ray (Mathlib's `Module.Ray`); an orientation of a topological line bundle is a choice of rays given near every point by a continuous nowhere-zero section; an orientation of a Mathlib manifold is a continuous tangent orientation (§VI.25).

Map.

- `Chapter6/LinSet`: chart points, T_zO, K_z, sc$^+$-operators, Definition 6.2.1 (preprint 6.1), the index and convexity of $\mathrm{Lin}(f,x)$.
- `Chapter6/DetLine`: $\lambda(E)$, $\lambda^*(E)$, wedges, v^*, $\lambda(\Phi)$, Definition 6.3.1 (preprint 6.3), Proposition 6.3.2 (preprint 6.4).
- `Chapter6/ExactSequence`: four-term exact sequences, the choices of (91), Φ_E, Lemma 6.3.3 (preprint 6.5) with the proof of §6.8.1 (preprint §6.7.1), (92), Proposition 6.3.5 (preprint 6.6).
- `Chapter6/FredholmDet`: Definitions 6.4.1, 6.4.2 (preprint 6.7, 6.8).
- `Chapter6/GoodProjections`: Definition 6.4.3 (preprint 6.9), the order on Π_T, the facts after it, Lemma 6.10 (preprint) and the example on $\mathbb{R}^2$.
- `Chapter6/ProjectionSequence`: Proposition 6.4.10 (preprint Lemma 6.11), γ_T^P, Proposition 6.4.11 (preprint 6.12) with the proof of §6.8.2 (preprint §6.7.2), Corollary 6.4.12 (preprint 6.13).
- `Chapter6/Stabilization`: Lemma 6.4.14 (preprint 6.14), formula (94), Proposition 6.4.15 (preprint 6.15).
- `Chapter6/DirectSumDet`: Definition 6.4.16 (preprint 6.16).
- `Chapter6/LocalDetBundle`: Lemmas 6.5.1, 6.5.2 (preprint 6.17, 6.18), the frames of Proposition 6.5.4 (preprint 6.20), Lemma 6.5.5 (preprint 6.21).

- `Chapter6/DetBundle`: $\mathrm{DET}(E,F)$ as a vector prebundle, Definition 6.5.6 (preprint 6.22), Propositions 6.5.7, 6.5.8 (preprint 6.23, 6.24), Lemma 6.5.9 (preprint 6.25), Proposition 6.6.1 (preprint 6.26).
- `Chapter6/LineBundleOrientation`: orientations of line bundles, the orientation double cover.
- `Chapter6/LinOrientation`: Definition 6.6.2 (preprint 6.27).
- `Chapter6/OrientableNbhd`: sc-smooth homotopies, Proposition 6.6.3 (preprint 6.28).
- `Chapter6/Lemma629`: Lemma 6.6.4 (preprint 6.29).
- `Chapter6/KernelFamily`: the kernels of Lemma 6.6.5 (preprint 6.30).
- `Chapter6/SubspaceBundle`: continuous families of subspaces, their determinant lines, continuation over $[0,1]$.
- `Chapter6/StabilizationContinuity`: families of kernels and their images, the stabilization sign, the example without the sign.
- `Chapter6/ParamKernel`: the stabilized section in a product chart, Lemma 6.6.6 (preprint 6.31).
- `Chapter6/Continuation`: continuation over a simply connected base, the propagation $\Phi_{E_1}^{-1} \circ$ continuation $\circ\, \Phi_{E_0}$.
- `Chapter6/Propagation`: stabilized kernels and the push-forward φ_*; Lemmas 6.6.8 and 6.6.10 (preprint 6.33 and 6.34) for it.
- `Chapter6/StabKernelsExist`: the stabilized kernels exist.
- `Chapter6/PushData`: push-forwards along paths, their existence, halves and reparametrizations.
- `Chapter6/AddSections`: the linear algebra of Lemma 6.6.8 (preprint 6.33).
- `Chapter6/FamilySections`: sections over parameter families.
- `Chapter6/PropagationWellDefined`: Proposition 6.6.7 (preprint 6.32), the homotopy argument, Theorem 6.6.11 (preprint 6.35).
- `Chapter6/ScPaths`: sc-smooth paths in a chart, concatenation.
- `Chapter6/ContinuousOrientation`: Definition 6.6.12 (preprint 6.36), continuous orientations.
- `Chapter6/ScHomotopy`: reparametrization homotopies.
- `Chapter6/PathCategory`: homotopy classes and the path category PX.
- `Chapter6/ChartTransfer`: conjugation of determinants, the transfer between chart points.
- `Chapter6/PushTransfer`: the push-forward does not depend on the chart.
- `Chapter6/SectionOrientation`: orientable neighbourhoods as structures, Or_x^f in a chosen chart, Definitions 6.6.15, 6.6.16 (preprint 6.39, 6.40).
- `Chapter6/PathTransport`: transport along paths from local push-forwards.
- `Chapter6/OrientationFunctor`: Theorem 6.6.13 (preprint 6.37), Definitions 6.6.14–6.6.16 (preprint 6.38–6.40) and their equivalences.
- `Chapter6/SolutionOrientation`: Theorem 6.6.18 (preprint 6.41).
- `Chapter6/ProductOrientation`: Remark 6.6.19 (preprint 6.42).
- `Chapter6/Book2021`, `Chapter6/Book2021Continuity`: Lemma 6.4.4, Proposition 6.4.13, Lemmas 6.6.8 and 6.6.9 of the 2021 edition (§§VI.32–VI.33).
- `Chapter6/OrientationSpace`: the orientation space O_f and Theorem 6.6.17 (§VI.33).
- `Chapter6/LinSetFredholm`: the members of $\mathrm{Lin}(f,x)$ are sc-Fredholm on every level (§VI.1).
- `Chapter6/DetConventions`, `Chapter6/DetNonOrientable`, `…Ell2`, `…Transport`, `Chapter6/ClassicalDetBundle`: the alternative determinant, (6.25), the remark that $\mathrm{DET}(E,F)$ is not orientable, the classical case (§VI.34).
- `Chapter6/DetVectorBundles2021`: Lemma 6.5.3 (preprint 6.19) (§VI.35).
- `Chapter6/SolutionManifoldFamily2021`, `Chapter6/SubmersionNearFamily`, `Chapter6/SlicePropertyP`, `Chapter6/SmoothFiniteDimRanges–3`, `Chapter6/CubePropertyP`, `CubeBoundarySlice`, `CubeBoundarySubmersion`, `CubeCharts`, `CubeFredholm`, `CubeGerms`, `CubeFamilyLemmas`,

`CubeSections`: Lemmas 6.6.4 and 6.6.5 (preprint 6.29 and 6.30) as statements about manifolds, on $\mathbb{R}^n$ and on $[0,1]^n$ (§VI.36).

- `Chapter6/SmoothPathComponents2021`, `Chapter6/PathsInterval`, `Chapter6/PathCategoryInterval`, `Chapter6/UnitIntervalSections`: paths on $[0,1]$, the facts before Proposition 6.2.2 (preprint 6.2), Proposition 6.6.3 (3), (4) on $[0,1]\times X$ (§VI.37).
- `Chapter6/OrientationMainClean`, `Chapter6/OrientationConvention2021`: Theorems 6.6.13, 6.6.17, 6.6.18 with the text's hypotheses; Remark 6.6.19 in the 2021 convention.
- `Chapter6/ManifoldOrientation`, `Chapter6/SolutionSetOrientation`, `Chapter6/SolutionOrientationCharts`, `Chapter6/PerturbedOrientation`, `Chapter6/SolutionSetOpen2021`: orientations of Mathlib manifolds; the solution set as an oriented manifold; two charts; perturbed solution sets (§VI.25).
- `Chapter6/ProductSectionOrientation2021`, `…2021b–f`: the orientation $e\wedge o$ of $\hat{f}$ over $\mathbb{R}^1\times X$ (§VI.26).
- `Chapter6/OrientationBundle`: the orientation bundle, the $\mathbb{Z}_2$-action, $\det_f$ (§VI.38).
- `Chapter5/CylinderPerturbation2021`: margins, admissible sections, the relative form of Theorem 5.3.10 (preprint 5.18), the cobordism data of part (1) (§VI.27).
- `Chapter6/ProperHomotopy2021`: proper oriented homotopies and the cobordism data of part (2) (§VI.27).
- `Chapter6/InvariantSingle2021`: the index parts S_k, the integrals, Ψ_s for one admissible s (§VI.27).
- `Chapter6/InvariantProductEnds2021`, `…2021b`: product ends of the cobordism, Fubini on them (§VI.27).
- `Chapter6/InvariantCobordism2021`: the cut-off identity and S_F (§VI.27).
- `Chapter6/CobordismRelativeForms2021`, `…2021b`, `Chapter4/BoundaryFormsPullback2021`: the relative form on the cobordism (§VI.27).
- `Chapter6/CobordismOrientation2021`, `Chapter6/CobordismOrientationEnds2021`, `…2021b–d`: the orientation of the cobordism and its value on the ends (§VI.27).
- `Chapter6/Theorem671_2021`, `Chapter6/Theorem671Assembly2021`, `Chapter6/Theorem671Final2021`: the assembly of Theorem 6.7.1 (preprint 6.43) and the case $\partial X=\emptyset$ (§VI.27).
- `Chapter6/ScContractible`, `Chapter6/IntervalHomotopy`: Proposition 6.6.3 (1) as printed in 2021, and (1′) on $[0,1]$ (§VI.10); `Chapter6/KernelBundleInterval`: L of Lemma 6.6.5 over $[0,1]$ (§VI.30).

Throughout, E and F are real Banach spaces, $T:E\to F$ is bounded, $R(T)$ is its range and $\operatorname{coker}T=F/R(T)$; T is Fredholm if $\ker T$ and $\operatorname{coker}T$ are finite-dimensional, with index $\dim\ker T-\dim\operatorname{coker}T$. A *line* is a one-dimensional real space. $P:Y\to X$ is a strong bundle over an M-polyfold, both paracompact and Hausdorff, and from §VI.10 on X is tame with sc-smooth bump functions and f is an sc-Fredholm section with tame germ data (Definition 3.6.4 (preprint 3.51)), the reading of Chapter 5 (Part V). A strong bundle chart c has base chart φ, retraction r on an open set U of a partial quadrant C, retract $O=r(U)$, and fibre projections $\rho(u)$; the *principal part* of a section is its fibre coordinate in the chart. $\mathbb{R}^1$ is the M-polyfold of the parameter t, with t the point of $\mathbb{R}^1$ `Chapter6/OrientableNbhd·pt1`, and (τ,t) the point of $\mathbb{R}^2$ `Chapter6/OrientableNbhd·pt2`; κ is Mathlib's smooth transition function, 0 on $(-\infty,0]$, 1 on $[1,\infty)$.

VI.1 Definition 6.2.1 (preprint 6.1): the space $\mathrm{Lin}(f,x)$

§6.2 (preprint §6.1) (pp. 237–240 (preprint pp. 200–201)) works at a smooth point x. Since $f(x)$ is a smooth point of the fibre, there is an sc⁺-section s with $s(x) = f(x)$ — in a chart, $s(y) = R(y, f(x))$ — and $(f-s)'(x) : T_xX \to Y_x$ is sc-Fredholm. Definition 6.2.1 (preprint 6.1) collects these linearizations, perturbed by sc⁺-operators, into $\mathrm{Lin}(f,x)$; the index is the index of any member, and Proposition 6.2.2 (preprint 6.2) says that it is constant along sc-smooth paths.

The chart point (V6.1). A chart point of x is a strong bundle chart c of the atlas whose base chart contains x, with the level-1 point z of the model over $\varphi(x)$ `Chapter6/LinSet · ChartPoint`. In it T_xX is $T_zO = Dr(z)E$, the range of the derivative of the retraction `Chapter6/LinSet · ChartPoint.T`, and Y_x is $K_z = \rho(z)F$, the range of the fibre projection over z `Chapter6/LinSet · ChartPoint.K`; both are closed, as ranges of continuous projections, hence Banach spaces. An sc⁺-operator $T_xX \to Y_x$ is level 0 of an sc⁺-operator between these sc-subspaces `Chapter6/LinSet · ChartPoint.IsScPlusOp`.

Definition 6.2.1 (preprint 6.1) (V6.1). $\mathrm{Lin}(f,x)$, at the chart point p, is the set of the operators $(f+s)'(x) + a : T_zO \to K_z$ for an sc⁺-section s with $(f+s)(x) = 0$ and an sc⁺-operator a `Chapter6/LinSet · ChartPoint.linSet` `Chapter6/LinSet · ChartPoint.mem_linSet_iff`. The linearization of $f+s$ at its zero x is the derivative at z of the principal part composed with r, restricted to T_zO — the $\mathrm{Lin}(z)$ of Part III. Replacing the text's s by $-s$ turns $f-s$ with $s(x) = f(x)$ into $f+s$ with $(f+s)(x) = 0$.

Two sections differ by an sc⁺-operator. For two sections s, t with $(f+s)(x) = (f+t)(x) = 0$, the principal parts of $f+s$ and $f+t$ differ by that of the sc⁺-section $s-t$, whose derivative at z, followed by $\rho(z)$, is level 0 of an sc⁺-operator: the derivative of an sc⁺-map is sc⁺ on the shifted levels. So $\mathrm{Lin}(f,x)$ is $(f+s)'(x)$ plus all the sc⁺-operators, for any one s `Chapter6/LinSet · ChartPoint.exists_isScPlusOp_linAt_eq`, and two members differ by an sc⁺-operator `Chapter6/LinSet · ChartPoint.isScPlusOp_sub_of_mem_linSet`. The paragraph before the definition puts this as "$(f-s)'(x)$ is an sc-Fredholm operator if and only if $(f-s)'(x)$ is"; it is read with $(f-t)'(x)$ in the second place (possible misprint; Findings, Chapter 6, item 41).

Not empty. The text makes its section locally, in a chart. The formalization uses global sc⁺-sections: $\mathrm{Lin}(f,x)$ is not empty when P has sc⁺-sections through the smooth points of its fibres, one through $-f(x)$ giving a member `Chapter6/LinSet · ChartPoint.linSet_nonempty`, and bump functions give such sections `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`. This is the hidden hypothesis of the definition: an sc⁺-section through $f(x)$.

Fredholm, of one index. Every member is Fredholm on level 0, with the index $\mathrm{ind}(f,x)$ of the germ at x (§III.49) `Chapter6/LinSet · ChartPoint.isFredholm_isIndexAt_of_mem_linSet` `Chapter6/LinSet · ChartPoint.exists_index_linSet` `Chapter6/LinSet · ChartPoint.fredholmIndex_of_mem_linSet`. The proof: $f+s$ has germ data at x (an sc-Fredholm germ plus an sc⁺-germ is one), so its linearization at the zero x is Fredholm of the index of the germ, through the conjugation to a basic germ (Proposition 3.6.5 (preprint 3.52), §III.30); adding an sc⁺-operator, compact on level 0, keeps the Fredholm property and the index; and the index of the germ does not depend on the choices. *The members are sc-Fredholm on every level*, as the text says, all of one index `Chapter6/LinSetFredholm · ChartPoint.exists_isScFredholm_of_mem_linSet` `Chapter6/LinSetFredholm · ChartPoint.definition_6_2_1_linSet_isScFredholm`: for $L = (f+s)'(x) + a$, the section $f+s$ is sc-Fredholm with a zero at x, so its linearization T_zOoK_z is sc-Fredholm, between these sc-subspaces with their induced sc-structures (row V3.96); a is level 0 of an sc⁺-operator; and an sc-Fredholm operator plus an sc⁺-operator is sc-Fredholm of the same index on every level (Proposition 1.1.11). The level-0 operator is L, so the indices agree. The determinant lines use level 0 only (gap G.64 closed; Findings, Chapter 6, items 40,

122).

Convex `Chapter6/LinSet · ChartPoint.convex_linSet`. $(1-\tau)(L+a)+\tau(L+b) = L+((1-\tau)a+\tau b)$, and the sc⁺-operators form a vector space. The text notes this on preprint p. 219, where Proposition 6.6.1 (preprint 6.26) needs it.

Proposition 6.2.2 (preprint 6.2) (V3.98). The constancy of the index along sc-smooth paths was formalized with Chapter 3 and is retold in §III.49: for paths on $\mathbb{R}$, with the text's hypotheses — tame germ data, no bump functions, the locally defined sc⁺-sections of Definition 6.2.1 (preprint 6.1). It is not repeated here. The facts on paths recalled before it on preprint p. 201 — an M-polyfold is locally path connected, and two smooth points in one path component are joined by an sc⁺-smooth path — are formalized, with paths on $[0,1]$ (§VI.37; gap G.56 closed; Findings, Chapter 6, items 42, 124).

VI.2 Determinant lines, Definition 6.3.1 (preprint 6.3), Proposition 6.3.2 (preprint 6.4)

$\lambda(E)$ and $\lambda^*(E)$ (V6.2). For a finite-dimensional real space E, $\lambda(E) = \Lambda^{\max}E$ is the top exterior power `Chapter6/DetLine · maxPow` and $\lambda^*(E)$ its dual `Chapter6/DetLine · maxPowDual`; both are lines `Chapter6/DetLine · finrank_maxPow`, $\lambda(\{0\}) = \mathbb{R}$ being the zeroth exterior power. A family $a_1, \dots, a_n$ with $n = \dim E$ has its wedge $a_1 \wedge \cdots \wedge a_n$ `Chapter6/DetLine · wedge`. Two rules carry every computation of the chapter. A change of basis multiplies the wedge by its determinant `Chapter6/DetLine · wedge_eq_det_smul`, also when only one block of a two-block wedge $a \wedge b$ is changed `Chapter6/DetLine · wedge_append_basis_left Chapter6/DetLine · wedge_append_basis_right`; and adding to the vectors of one block vectors of the span of the other does not change the wedge `Chapter6/DetLine · wedge_append_add_left Chapter6/DetLine · wedge_append_add_right`. For $v \neq 0$ in a line, v^* is the functional with $v^*(v) = 1$ `Chapter6/DetLine · dualVec Chapter6/DetLine · dualVec_self`, and that value determines it `Chapter6/DetLine · eq_dualVec_of_apply_eq_one`. For $\dim E = \dim F$ and $\Phi : E \to F$, $\lambda(\Phi)(a_1 \wedge \cdots \wedge a_n) = \Phi a_1 \wedge \cdots \wedge \Phi a_n$ `Chapter6/DetLine · maxPowMap Chapter6/DetLine · maxPowMap_wedge`, with $\lambda(\Psi\Phi) = \lambda(\Psi)\lambda(\Phi)$ `Chapter6/DetLine · maxPowMap_comp`, and $\lambda(\Phi) \neq 0$ iff Φ is bijective `Chapter6/DetLine · maxPowMap_ne_zero_iff`; $\lambda^*(\Phi)$ is its dual `Chapter6/DetLine · maxPowDualMap` (pp. 241–243 (preprint pp. 203–204)).

Definition 6.3.1 (preprint 6.3) (V6.3). $\iota : \lambda(E^*) \to \lambda^*(E)$ is $\iota(e_1^* \wedge \cdots \wedge e_n^*)(a_1 \wedge \cdots \wedge a_n) = \det(e_i^*(a_j))$ `Chapter6/DetLine · iotaDual Chapter6/DetLine · iotaDual_wedge_wedge`. The formula is alternating and multilinear in both families, so it descends to the exterior powers; that is the text's well-definedness. ι is a nonzero map of lines `Chapter6/DetLine · iotaDual_ne_zero`, and for a basis with its dual basis $\iota(e_1^* \wedge \cdots \wedge e_n^*) = (e_1 \wedge \cdots \wedge e_n)^*$ `Chapter6/DetLine · iotaDual_wedge_dualBasis`, since the matrix is the identity. The text's matrix $(e_i^*(a_j))$ is the transpose of the one of Mathlib's pairing, with the same determinant (preprint p. 204).

Proposition 6.3.2 (preprint 6.4) (V6.4) `Chapter6/DetLine · proposition_6_3_2`. For an isomorphism $\Phi : E \to F$, $\lambda^*(\Phi) \circ \iota = \iota \circ \lambda(\Phi^*)$. Both sides, evaluated at $f_1^* \wedge \cdots \wedge f_n^*$ and $a_1 \wedge \cdots \wedge a_n$, give $\det(f_i^*(\Phi a_j))$: the left side by the definition of $\lambda^*(\Phi)$, the right side because $(\Phi^* f_i^*)(a_j) = f_i^*(\Phi a_j)$ (p. 243 (preprint p. 205)).

VI.3 Four-term exact sequences: Φ_E, Lemma 6.3.3 (preprint 6.5), Proposition 6.3.5 (preprint 6.6)

The data of (91) (V6.5). Let $\mathbf{E} : 0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \xrightarrow{\gamma} D \to 0$ be an exact sequence of finite-dimensional spaces `Chapter6/ExactSequence · FourTermExact`. A *choice* `Chapter6/ExactSequence · FourTermExact.Choice` consists of a complement Z of $\alpha(A)$ in B, a complement V of $\beta(B)$ in C, bases a of A, d of D, b' of Z, and the γ-preimages c in V of the basis d. Then $\alpha(a), b'$ is a basis of B, and $c, \beta(b')$ is one of C, because β maps Z isomorphically onto $\beta(B)$ and γ maps V isomorphically onto D. Formula (91) assigns to the choice the value $(\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b'_1 \wedge \cdots \wedge b'_{m-n}) \otimes (c_1 \wedge \cdots \wedge c_l \wedge \beta(b'_1) \wedge \cdots \wedge \beta(b'_{m-n}))^*$ as the image of $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, with the text's two ordering conventions `Chapter6/ExactSequence · FourTermExact.Choice.value`. $\Phi_E : \lambda(A) \otimes \lambda^*(D) \to \lambda(B) \otimes \lambda^*(C)$ is the linear map of lines defined by one fixed choice `Chapter6/ExactSequence · FourTermExact.canonicalChoice` `Chapter6/ExactSequence · FourTermExact.phi`; it is nonzero `Chapter6/ExactSequence · FourTermExact.phi_ne_zero`, hence an isomorphism `Chapter6/ExactSequence · FourTermExact.phiEquiv` (preprint p. 206).

Lemma 6.3.3 (preprint 6.5) (V6.6) `Chapter6/ExactSequence · FourTermExact.lemma_6_3_3`. The text: Φ_E does not depend on the choices. The formalization proves the strongest form: Φ_E, defined from one choice, maps the source element $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$ of *any other* choice to the right-hand side of (91) computed with that choice. This is the form every later computation uses — γ_T^P is always evaluated in bases adapted to the problem at hand (Findings, Chapter 6, item 1). The proof is §6.8.1 (preprint §6.7.1) (§VI.28).

Proposition 6.3.5 (preprint 6.6) (V6.7). With the contraction $\bar{\iota}(v \otimes b \otimes \varphi) = \varphi(v)\, b$ `Chapter6/ExactSequence · contractDual`, (92) defines $\Psi_E(c \otimes h) = \bar{\iota}(c \otimes \Phi_E(h))$ `Chapter6/ExactSequence · FourTermExact.psi`. The proposition `Chapter6/ExactSequence · FourTermExact.proposition_6_3_5`: Ψ_E maps the wedge of the basis $c, \beta(b')$ of C, tensored with $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, to $\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b'_1 \wedge \cdots \wedge b'_{m-n}$. By Lemma 6.3.3 (preprint 6.5) in its strong form, Φ_E of the source is the value of (91) for this choice, and the dual wedge evaluated on the wedge is 1. The wedge in C is taken in the order of (91); the statement exchanges the names of its two blocks (possible misprint; Findings, Chapter 6, item 2) (preprint p. 207).

VI.4 Determinants of Fredholm operators, good left-projections, and Lemma 6.10 (preprint)

Definition 6.4.1 (preprint 6.7) (V6.8). $\det(T) = \lambda(\ker T) \otimes \lambda^*(F/R(T))$ `Chapter6/FredholmDet · linDet` `Chapter6/FredholmDet · fredholmDet`, a line for T Fredholm `Chapter6/FredholmDet · finrank_fredholmDet`. Equal operators have identified determinants `Chapter6/FredholmDet · linDetEquivOfEq`: PQT and PT are the same operator when $P \le Q$, but not the same expression, and Proposition 6.4.11 (preprint 6.12) compares their determinants. The alternative definition $\lambda(\ker T) \otimes \lambda(\mathrm{coker}(T)^*)$ mentioned after the definition is naturally isomorphic to $\det(T)$ through ι (§VI.34; gap G.57 closed) (preprint p. 208).

Definition 6.4.2 (preprint 6.8) (V6.9). An orientation of T is a ray of $\det(T)$ `Chapter6/FredholmDet · FredholmOrientation`; there is one `Chapter6/FredholmDet · nonempty_fredholmOrientation`, and exactly two `Chapter6/FredholmDet · exists_two_fredholmOrientation` (preprint p. 208).

Definition 6.4.3 (preprint 6.9) and the facts after it (V6.10). A *good left-projection* for T is a bounded projection $P : F \to F$ with $\dim F/R(P) < \infty$ and $R(PT) = R(P)$ `Chapter6/GoodProjections · IsGoodLeftProj`;

Π_T is their set Chapter6/GoodProjections · goodLeftProjs. $P \le Q$ iff $P = PQ = QP$ Chapter6/GoodProjections · ProjLE, a partial order Chapter6/GoodProjections · ProjLE.refl Chapter6/GoodProjections · ProjLE.antisymm Chapter6/GoodProjections · ProjLE.trans. A good left-projection is Fredholm of index 0 — $\ker P$ is a complement of $R(P)$, of dimension $\dim F/R(P)$ Chapter6/GoodProjections · IsGoodLeftProj.isFredholm Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_eq_zero — so PT is Fredholm with $\operatorname{ind}(PT) = \operatorname{ind}(T)$ when T is, by the index of a composite Chapter6/GoodProjections · IsGoodLeftProj.isFredholm_comp Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_comp. For $P, P' \in \Pi_T$, $R(P) \cap R(P')$ has finite codimension Chapter6/GoodProjections · IsGoodLeftProj.finiteDimensional_quotient_inf. And $\Pi_T \neq \emptyset$ for T Fredholm Chapter6/GoodProjections · exists_isGoodLeftProj — the projection onto $R(T)$ along a finite-dimensional complement is one — which the text uses from Proposition 6.4.10 (preprint Lemma 6.11) on without stating it (Findings, Chapter 6, item 3) (preprint p. 208).

Lemma 6.10 (preprint) (V6.11) Chapter6/GoodProjections · lemma_6_10_preprint. *For $P, P' \in \Pi_T$ there is $P'' \in \Pi_T$ with $P'' \le P$ and $P'' \le P'$*, for any bounded T. The text's P'' projects onto $R(P) \cap R(P')$ along a complement assembled from $\ker P$, $\ker P'$ and a further piece, and $P'' = P''P$ needs $\ker P \subseteq \ker P''$. As we read it, this does not hold in general: on $\mathbb{R}^2$ with $T = 1$, let P project onto the x-axis along the y-axis and P' along the diagonal; $R(P) \cap R(P')$ is the x-axis, and a projection onto it below both would equal both Chapter6/GoodProjections · lemma_6_10_text_reading_preprint. The lemma holds with a smaller range. *Step 1.* $K = \ker P + \ker P'$ is finite-dimensional, each kernel having the dimension of a finite-dimensional quotient. *Step 2.* K has a closed complement N (finite-dimensional subspaces are complemented, by Hahn–Banach), and $R'' = R(P) \cap R(P') \cap N$ is closed, of finite codimension, and meets K only in 0. *Step 3.* R'' has a complement $C \supseteq K$, topological because the quotient by R'' is finite-dimensional; P'' is the projection onto R'' along C. *Step 4.* $\ker P \subseteq C = \ker P''$ gives $P''P = P''$, and $R'' \subseteq R(P)$ gives $PP'' = P''$; the same for P'. *Step 5.* $R(P''T) = P''(R(PT)) = P''(R(P)) = R''$, since P'' is the identity on $R'' \subseteq R(P)$. Nothing uses that T is Fredholm. The 2017 version drops the lemma and remarks that a common lower bound need not exist; with the order of Definition 6.4.3 (preprint 6.9), used by both versions, it always exists, as we read it, so the preprint's route to Lemma 6.5.9 (preprint 6.25) is sound and is the one formalized (Findings, Chapter 6, item 4) (preprint pp. 208–209).

VI.5 Proposition 6.4.10 (preprint Lemma 6.11), γ_T^P, Proposition 6.4.11 (preprint 6.12) and Corollary 6.4.12 (preprint 6.13)

Proposition 6.4.10 (preprint Lemma 6.11) (V6.12). For $P \in \Pi_T$, the sequence $\mathbf{E}(T, P) : 0 \to \ker T \to \ker(PT) \to F/R(P) \to \operatorname{coker} T \to 0$, with the inclusion Chapter6/ProjectionSequence · projSeqJ, $x \mapsto Tx + R(P)$ Chapter6/ProjectionSequence · projSeqPhi and $y + R(P) \mapsto (I - P)y + R(T)$ Chapter6/ProjectionSequence · projSeqPi, is exact Chapter6/ProjectionSequence · proposition_6_4_10, a four-term exact sequence of §VI.3 Chapter6/ProjectionSequence · projSeq. The four checks use only $P^2 = P$ and $R(PT) = R(P)$. At $\ker(PT)$: if $PTx = 0$ and $Tx \in R(P)$, then $Tx \in R(P) \cap \ker P = 0$. At $F/R(P)$: the last map is well defined because $(I - P)R(P) = 0$; it kills $[Tx]$ for $x \in \ker(PT)$, since $(I - P)Tx = Tx$; and if $(I - P)y = Tx$, then $PTx = 0$ and $y - Tx = Py \in R(P)$. Onto: $Py = PTe$ for some e, and then $(I - P)(y - Te) = y - Te$, which has the class of y in $\operatorname{coker} T$ (preprint p. 209).

The isomorphism γ_T^P (V6.13). $\gamma_T^P : \det(T) \to \det(PT)$ is $\Phi_{\mathbf{E}(T,P)}$ followed by the identification of $\lambda^*(F/R(P))$ with $\lambda^*(F/R(PT))$ Chapter6/ProjectionSequence · gammaProj Chapter6/ProjectionSequence · gammaProj_tmul — the text's "recalling $R(P) = R(PT)$" (Findings, Chapter 6, item 5). By Lemma 6.3.3 (preprint 6.5) it is given on wedges by (91) for any choice Chapter6/

ProjectionSequence · gammaProj_wedge_tmul (preprint p. 210).

Proposition 6.4.11 (preprint 6.12) (V6.14). *For $P \le Q$ in Π_T, QT is Fredholm, $P \in \Pi_{QT}$, and $\gamma^P_{QT} \circ \gamma^Q_T = \gamma^P_T$,* up to the identification of $\det(P(QT))$ with $\det(PT)$ Chapter6/ProjectionSequence · proposition_6_4_11. $P \in \Pi_{QT}$ because $R(PQT) = R(PT) = R(P)$ Chapter6/ProjectionSequence · IsGoodLeftProj.of_projLE. The identity is a computation in adapted bases Chapter6/ProjectionSequence · comp_comp_eq_of_projLE, the proof of §6.8.2 (preprint §6.7.2), retold in §VI.29 (preprint p. 211).

Corollary 6.4.12 (preprint 6.13) (V6.15) Chapter6/ProjectionSequence · corollary_6_4_12. For $P \le S$ and $P \le R$ in Π_T, $(\gamma^P_{RT})^{-1} \circ \gamma^P_{ST} = \gamma^R_T \circ (\gamma^S_T)^{-1}$. Proposition 6.4.11 (preprint 6.12) twice gives $\gamma^P_{ST}\gamma^S_T = \gamma^P_T = \gamma^P_{RT}\gamma^R_T$; rearranging gives the claim. So the left side, the comparison of two local frames through a common lower bound, does not depend on the lower bound (preprint p. 211).

VI.6 The stabilization: Lemma 6.4.14 (preprint 6.14), formula (94), Proposition 6.4.15 (preprint 6.15), Definition 6.4.16 (preprint 6.16)

Lemma 6.4.14 (preprint 6.14) (V6.16). For T Fredholm and $\varphi : \mathbb{R}^n \to F$ with $T_\varphi(e,r) = Te + \varphi r$ onto, the sequence $0 \to \ker T \xrightarrow{j} \ker T_\varphi \xrightarrow{\pi} \mathbb{R}^n \xrightarrow{[\varphi]} F/R(T) \to 0$, with $j(e) = (e,0)$ Chapter6/Stabilization · stabJ, $\pi(e,r) = r$ Chapter6/Stabilization · stabPi and $[\varphi](r) = \varphi(r) + R(T)$ Chapter6/Stabilization · stabClass, is exact Chapter6/Stabilization · lemma_6_4_14 Chapter6/Stabilization · stabSeq. j is injective; $\ker\pi \cap \ker T_\varphi$ is $\{(e,0) \mid Te = 0\} = j(\ker T)$; for $(e,r) \in \ker T_\varphi$, $\varphi r = -Te \in R(T)$, and conversely $\varphi r = -Te$ puts (e,r) in $\ker T_\varphi$; $[\varphi]$ is onto because T_φ is. A stabilizing φ exists for every Fredholm T Chapter6/Stabilization · exists_surjective_coprod: send a basis of $\mathbb{R}^n$ to representatives of a basis of the cokernel. The proof writes $j(e) = (0,e)$, read as $(e,0)$ (possible misprint; Findings, Chapter 6, item 8) (pp. 251–256 (preprint pp. 211–212)).

Formula (94) (V6.17). $\psi : \lambda^*(\mathbb{R}^n) \to \lambda^*(\{0\})$ maps $(e_1 \wedge \cdots \wedge e_n)^*$ to 1^* for the standard basis Chapter6/Stabilization · stdWedge Chapter6/Stabilization · stabOne Chapter6/Stabilization · stabPsi. Then $\psi((c_1 \wedge \cdots \wedge c_n)^*) = \det[c_1,\dots,c_n]^{-1}\, 1^*$ for every basis Chapter6/Stabilization · stabPsi_dualVec: $c_1 \wedge \cdots \wedge c_n = \det[c]\, e_1 \wedge \cdots \wedge e_n$, and the dual vectors scale inversely (p. 256 (preprint p. 212)).

Proposition 6.4.15 (preprint 6.15) (V6.18). Since T_φ is onto, $\det(T_\varphi) = \lambda(\ker T_\varphi) \otimes \lambda^*(\{0\})$, and $\iota_\varphi = (1 \otimes \psi) \circ \Phi_{\mathbf{E}}$, for the sequence $\mathbf{E}$ of Lemma 6.4.14 (preprint 6.14), is an isomorphism $\det(T) \to \det(T_\varphi)$ Chapter6/Stabilization · stabIota Chapter6/Stabilization · stabIota_bijective. The formula Chapter6/Stabilization · proposition_6_4_15: for a basis a of $\ker T$, vectors $\bar c_1, \dots, \bar c_m$ with $[\varphi](\bar c_i)$ a basis of $F/R(T)$, and $\bar b_i = (b_i, r_i)$ completing $(a,0)$ to a basis of $\ker T_\varphi$, $\iota_\varphi(h) = \det(\bar c, r)^{-1}\, ((a_1,0) \wedge \cdots \wedge (a_k,0) \wedge \bar b_1 \wedge \cdots \wedge \bar b_l) \otimes 1^*$. These data are a choice of (91) for $\mathbf{E}$ — Z the span of the $\bar b_i$, V the span of the $\bar c_j$ — so Lemma 6.3.3 (preprint 6.5) gives $\Phi_{\mathbf{E}}(h) = ((a,0) \wedge \bar b) \otimes (\bar c \wedge r)^*$, and (94) turns the second factor into $\det(\bar c, r)^{-1} 1^*$. To form $\det(\bar c, r)$ one needs $m + l = n$: the statement takes it as a hypothesis, and it is proved to hold always Chapter6/Stabilization · stab_card_eq; that the r_i and $\bar c_j$ form a basis of $\mathbb{R}^n$, which the text notices, is proved — the r_i span $\pi(\ker T_\varphi) = \ker[\varphi]$, and the $\bar c_j$ a complement of it Chapter6/Stabilization · isCompl_range_stabPi_span. The statement calls ψ "φ" (possible misprint). The identification (95) and the maps $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ after the proposition are formalized separately (§VI.34; gap G.58 closed); the statements here use only (94) (Findings, Chapter 6, items 9, 10) (preprint p. 213).

Definition 6.4.16 (preprint 6.16) (V6.19). For Fredholm T, S, the operator $T \oplus S$ has

$\ker(T \oplus S) \cong \ker T \times \ker S$ and $\operatorname{coker}(T \oplus S) \cong \operatorname{coker} T \times \operatorname{coker} S$, through the inclusions of the factors `Chapter6/DirectSumDet · kerInl` `Chapter6/DirectSumDet · kerInr` `Chapter6/DirectSumDet · cokerInl` `Chapter6/DirectSumDet · cokerInr` `Chapter6/DirectSumDet · kerProdEquiv` `Chapter6/DirectSumDet · cokerProdEquiv`; the dimensions add `Chapter6/DirectSumDet · finrank_ker_prodMap` `Chapter6/DirectSumDet · finrank_coker_prodMap`, so $T \oplus S$ is Fredholm `Chapter6/DirectSumDet · isFredholm_prodMap`. The map $\det(T) \otimes \det(S) \to \det(T \oplus S)$, $h \otimes h' \mapsto (a \wedge a') \otimes (d \wedge d')^*$, is defined by the formula in one choice of bases `Chapter6/DirectSumDet · directSumVec` `Chapter6/DirectSumDet · directSumDet`, satisfies it for every choice `Chapter6/DirectSumDet · directSumVec_eq_smul` `Chapter6/DirectSumDet · definition_6_4_16` — a change of bases multiplies source and target by the same product of determinants, the block wedges by the block determinants — and is an isomorphism `Chapter6/DirectSumDet · directSumDet_bijective`. "Natural" is read as the formula for every choice, as in Lemma 6.3.3 (preprint 6.5) (Findings, Chapter 6, item 11) (p. 258 (preprint p. 214)).

VI.7 Local determinant bundles: Lemmas 6.5.1–6.5.3 (preprint 6.17–6.19), Proposition 6.5.4 (preprint 6.20), Lemma 6.5.5 (preprint 6.21)

Lemma 6.5.1 (preprint 6.17) (V6.20). The Fredholm operators form a subspace of $L(E,F)$ `Chapter6/LocalDetBundle · FredholmOp`. *For a bounded T, $P \in \Pi_T$ and a topological complement Y of* $\ker(PT)$*, there is $\varepsilon > 0$ such that for $\|S - T\| < \varepsilon$, $P \in \Pi_S$ and $PS|_Y : Y \to R(P)$ is a topological isomorphism* `Chapter6/LocalDetBundle · lemma_6_5_1`. *Step 1.* $PT|_Y$ is injective, since $Y \cap \ker(PT) = 0$, and onto $R(P)$, since $PT(E) = PT(Y)$ and $R(PT) = R(P)$; as a bijection between the Banach spaces Y and $R(P)$ (both closed) it is an isomorphism by the open mapping theorem. *Step 2.* $\|PS|_Y - PT|_Y\| \le \|P\|\,\|S - T\|$ `Chapter6/LocalDetBundle · projRestrict`, and the invertible operators form an open set. *Step 3.* $R(PS) \supseteq PS(Y) = R(P)$, so $P \in \Pi_S$. That T is Fredholm is not used (Findings, Chapter 6, item 20); a topological complement is an algebraic complement with a continuous projection (p. 258 (preprint p. 214)).

Lemma 6.5.2 (preprint 6.18) (V6.21). A *local determinant chart* is the data (T, P, Y, ε) of Lemma 6.5.1 (preprint 6.17) with its conclusions `Chapter6/LocalDetBundle · LocalDetChart`. For $k \in \ker(PT)$, $y(k,S) = -(PS|_Y)^{-1}PSk$ `Chapter6/LocalDetBundle · LocalDetChart.ySol` is continuous in (k,S), inversion being continuous, and $k + y(k,S) \in \ker(PS)$. The map $(S,k) \mapsto (S, k + y(k,S))$ `Chapter6/LocalDetBundle · LocalDetChart.kerMap` is onto $\bigcup_{\|S-T\|<\varepsilon} \{S\} \times \ker(PS)$, inverted by the continuous projection onto $\ker(PT)$ along Y `Chapter6/LocalDetBundle · LocalDetChart.kerProj`: for $w \in \ker(PS)$ write $w = k + y$ with $y \in Y$; then $PSy = -PSk$, so $y = y(k,S)$ `Chapter6/LocalDetBundle · LocalDetChart.lemma_6_5_2`. "Is a trivial vector bundle" is stated as this trivialization and its inverse (Findings, Chapter 6, item 21) (preprint p. 215).

Lemma 6.5.3 (preprint 6.19) and Proposition 6.5.4 (preprint 6.20) (V6.22). For a basis k of $\ker(PT)$, $k_i(S) = k_i + y(k_i, S)$ is a continuous frame of the kernels `Chapter6/LocalDetBundle · LocalDetChart.kerFrame`, and with a constant basis e of $F/R(P) = F/R(PS)$, $\omega(S) = (k_1(S) \wedge \cdots \wedge k_n(S)) \otimes (e_1 \wedge \cdots \wedge e_l)^*$ `Chapter6/LocalDetBundle · LocalDetChart.frame` is a nowhere-zero frame `Chapter6/LocalDetBundle · LocalDetChart.frame_ne_zero` of $\mathrm{DET}(T,P,\varepsilon) = \bigcup_S \det(PS)$. The frames of two charts with the same P differ by a continuous nowhere-zero factor `Chapter6/LocalDetBundle · proposition_6_5_4`: the determinant of the matrix expressing one kernel frame in the other, continuous because both frames are continuous in S, times the constant change of the bases of $F/R(P)$. Lemma 6.5.3 (preprint 6.19) — exterior powers, duals, sums and tensor products of topological vector bundles — is applied here only to trivial bundles, and Proposition 6.5.4 (preprint 6.20) is proved with the explicit frames; the lemma in general is §VI.35 (gap G.59 closed); "in a natural way" is read as this relation between frames (Findings, Chapter 6, item 22)

(preprint p. 216).

Lemma 6.5.5 (preprint 6.21) (V6.23). *For $Q \leq P$, γ^Q_{PS} maps the P-frame at S to a continuous nowhere-zero multiple of the Q-frame, on the common ball of any two local determinant charts* `Chapter6/LocalDetBundle · LocalDetChart.gammaProj_frame_eq` `Chapter6/LocalDetBundle · lemma_6_5_5`. The text's sketch — extend the kernel sections of PS to sections spanning $\ker(QS)$ — is the proof. For a basis $f_1, \dots$ of $R(P) \cap \ker Q$, $z_m(S) = (PS|_Y)^{-1} f_m$ is continuous and $PS z_m(S) = f_m$; the $k_i(S)$ and $z_m(S)$ span $\ker(QPS) = \ker(QS)$; and in the choice of (91) for $\mathbf{E}(PS, Q)$ these vectors go with the *constant* classes $[f_m]$. So $\gamma^Q_{PS}(\omega_P(S))$ is $(k(S) \wedge z(S)) \otimes (e \wedge [f])^*$, continuous in S, and its ratio to the Q-frame of the other chart is the determinant of a continuous change of basis. The lemma is stated for two charts with different centres and radii, the form Lemma 6.5.9 (preprint 6.25) uses (Findings, Chapter 6, item 23) (pp. 259–261 (preprint pp. 216–217)).

VI.8 $\mathrm{DET}(E, F)$: Definition 6.5.6 (preprint 6.22), Propositions 6.5.7 and 6.5.8 (preprint 6.23 and 6.24), Lemma 6.5.9 (preprint 6.25)

The prebundle (V6.24). The fibre over a Fredholm S is $\det(S)$ `Chapter6/DetBundle · DetFiber`. A local determinant chart gives the pretrivialization $(S, h) \mapsto (S, \omega(S)^*(\gamma^P_S h))$ over its ball, with model fibre $\mathbb{R}$ `Chapter6/DetBundle · LocalDetChart.pretriv` `Chapter6/DetBundle · LocalDetChart.coord` — the text's $\gamma_{(T,P,\varepsilon)}$ read in the frame of Proposition 6.5.4 (preprint 6.20) (Findings, Chapter 6, item 24). Every Fredholm operator is the centre of one `Chapter6/DetBundle · exists_localDetChart`: $\Pi_T \neq \emptyset$ (§VI.4) and the finite-dimensional $\ker(PT)$ has a topological complement. The charts form a Mathlib vector prebundle `Chapter6/DetBundle · detPrebundle`, which needs exactly that the coordinate changes be continuous, Lemma 6.5.9 (preprint 6.25) below; its topology and bundle structure are those of $\mathrm{DET}(E, F)$ (preprint p. 217).

Lemma 6.5.9 (preprint 6.25) (V6.28), proved first. *The transition between two charts is $(S, t) \mapsto (S, g(S)t)$ with g continuous and nowhere zero on the common ball* `Chapter6/DetBundle · lemma_6_5_9_local` `Chapter6/DetBundle · lemma_6_5_9`. The proof is the text's: a common lower bound $Q \leq P, P'$ (Lemma 6.10 (preprint), §VI.4); Corollary 6.4.12 (preprint 6.13) to pass from γ^P to $\gamma^{P'}$ through Q; and Lemma 6.5.5 (preprint 6.21) for each chart, which turns both frames into multiples of Q-frames (preprint p. 218).

Definition 6.5.6 (preprint 6.22), Propositions 6.5.7 and 6.5.8 (preprint 6.23 and 6.24). $\mathcal{B}$ is the collection of sets lying over the ball of some chart whose image in ball $\times \mathbb{R}$ is open (V6.25) `Chapter6/DetBundle · detBasis`. It is a basis of the topology of the prebundle (V6.26) `Chapter6/DetBundle · proposition_6_5_7`: its members are open, the charts being homeomorphisms onto ball $\times \mathbb{R}$, and every open set is the union of its intersections with chart domains, which are members; so the topology $\mathcal{B}$ generates is that one. $\mathrm{DET}(E, F)$ is a topological vector bundle with model fibre $\mathbb{R}$, and every $\gamma_{(T,P,\varepsilon)}$, read in its frame, is a trivialization (V6.27) `Chapter6/DetBundle · proposition_6_5_8` (preprint p. 218). The remark that $\mathrm{DET}(E, F)$ is not orientable, which does not hold as we read it for finite-dimensional E, F, and the classical case at the opening of §6.6 (preprint §6.5) are §VI.34 (gap G.60 closed; Findings, Chapter 6, items 26, 135).

VI.9 Orientations of line bundles, Proposition 6.6.1 (preprint 6.26), Definition 6.6.2 (preprint 6.27)

Orientations of line bundles (V6.29). An orientation of a topological line bundle is a choice of rays of the fibres given near every point by a continuous nowhere-zero section `Chapter6/`

LineBundleOrientation · IsLineOrientation Chapter6/LineBundleOrientation · LineOrientation. Two orientations that agree on one fibre agree over a preconnected base Chapter6/LineBundleOrientation · LineOrientation. eq_of_apply_eq: the set where they agree is open and closed. The rays of the fibres form a two-sheeted covering Chapter6/LineBundleOrientation · OrCover Chapter6/LineBundleOrientation · isCoveringMap_orCover, and an orientation is a continuous section of it; over a simply connected, locally path-connected base the lifting criterion extends every orientation of one fibre Chapter6/LineBundleOrientation · exists_ lineOrientation_apply_eq, and over a nonempty such base there are exactly two orientations Chapter6/ LineBundleOrientation · exists_two_lineOrientation. The text gives "since the base space is contractible" as the reason, without the theorem it needs (Findings, Chapter 6, item 25) (p. 263 (preprint p. 219)).

Proposition 6.6.1 (preprint 6.26) (V6.30). Over a nonempty convex set C of Fredholm operators Chapter6/ DetBundle · fredholmIncl, the restriction $\mathrm{DET}(E,F)|_C$ Chapter6/DetBundle · DetOn is a topological line bundle with exactly two orientations Chapter6/DetBundle · proposition_6_6_1: a nonempty convex set of a normed space is contractible, hence simply connected, and locally path-connected. The proposition is stated for every nonempty convex C; $\mathrm{Lin}(f,x)$ is one (§VI.1) (preprint p. 219).

Definition 6.6.2 (preprint 6.27) (V6.31). At a chart point the members of $\mathrm{Lin}(f,x)$ are Fredholm Chapter6/LinOrientation · ChartPoint.isFredholm_of_mem_linSet; $\mathrm{DET}(f,x)$ is the pull-back of $\mathrm{DET}(T_zO,K_z)$ to $\mathrm{Lin}(f,x)$ Chapter6/LinOrientation · ChartPoint.detLin, and Or_x^f is the set of its orientations Chapter6/ LinOrientation · ChartPoint.Or, with exactly two elements Chapter6/LinOrientation · ChartPoint.exists_two_or. For each $L \in \mathrm{Lin}(f,x)$, $o \mapsto o(L)$ is a bijection onto the two orientations of $\det(L)$ Chapter6/ LinOrientation · ChartPoint.orEquivRay: injective because an orientation is determined by one value over the connected base, onto because each ray extends. The category $\mathcal{O}_f$ is read through its objects, the two-element sets, with bijections as morphisms; between two two-element sets there are bijections Chapter6/ OrientationFunctor · nonempty_equiv_of_two, each determined by one value Chapter6/OrientationFunctor · equiv_ eq_of_apply_eq_of_two. No negation is set up on the tensor products, so $\{o_x,-o_x\}$ is stated as "exactly two elements" (Findings, Chapter 6, item 44) (preprint pp. 219–220). The comparison of Or_x^f in two charts is §VI.21.

VI.10 Proposition 6.6.3 (preprint 6.28): the neighbourhoods of the propagation

The text: every smooth point x has an open neighbourhood U which (1) is sc-smoothly contractible, (2) has $\{y \in \mathrm{cl}(U) \mid f(y)=0\}$ compact, (3) carries, for every sc-smooth path $\varphi : [0,1] \to U$, an sc$^+$-section $s : U \times [0,1] \to Y$ with $s(\varphi(t),t) = f(\varphi(t))$, and (4) carries sc$^+$-sections $q : U \times [0,1] \to Y$ through every smooth point of every fibre over U_∞, at every t_0. The proof takes local strong bundle coordinates, $P : K \to O$ with $x = 0$. This is the preprint's wording of (1). The 2021 text (p. 263) prints (1) with a second neighbourhood: $U' \subset U$ is sc-smoothly contractible in U to the point x; that form is proved at the end of this section.

The statement formalized (V6.32) Chapter6/OrientableNbhd · proposition_6_6_3. *With sc-smooth bump functions, every smooth x has open $U \subseteq W$ in the domain of one chart of the atlas with* (1′) *any two sc-smooth paths in U with the same end points are sc-smoothly homotopic in W with the end points fixed* Chapter6/ OrientableNbhd · IsScHomotopyRel; (2) *$\{y \in \mathrm{cl}(U) \mid f(y) = 0\}$ is compact*; (3) *for every sc-smooth family $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n) \subseteq W$ there is an sc$^+$-section S of $\pi^*P \to \mathbb{R}^n \times X$ with $(f+S)(q,\Phi(q)) = 0$ for q near $[0,1]^n$, depending on $q \in [0,1]^n$ only through $\Phi(q)$*; (4) *π^*P has sc$^+$-sections through all smooth fibre points*. A path "in U" is one with $\gamma([0,1]) \subseteq U$.

The neighbourhoods. Take a chart c of the atlas at x, $u_0 = \varphi(x)$, the retraction r on $U_c \subseteq C$ with $O = r(U_c)$. Choose δ with $B_\delta \cap C \subseteq U_c$ (the domain of the retraction is relatively open); δ_2 with $r(B_{\delta_2} \cap U_c) \subseteq B_\delta$ (continuity of r at u_0 on level 0, $r(u_0) = u_0$); a neighbourhood N of x whose closure meets the zeros of f in a compact set `Chapter6/OrientableNbhd · exists_nhds_isCompact_zeros` — at a zero by the local compactness of Theorem 5.2.3 (preprint 5.6) (§V.7), elsewhere because the zero set is closed and misses a neighbourhood of x; a closed neighbourhood $T \subseteq N$ of x and δ_3 with $\varphi^{-1}(B_{\delta_3} \cap O) \subseteq T$. With $\delta' = \min(\delta, \delta_2, \delta_3)$, $W = \varphi^{-1}(B(u_0, \delta) \cap O)$ and $U = \varphi^{-1}(B(u_0, \delta') \cap O)$.

(1′). Let γ_0, γ_1 be sc-smooth paths in U with the same end points. Since paths are maps on $\mathbb{R}$ that lie in U only for $t \in [0,1]$, first take a C^∞ map $\sigma : \mathbb{R} \to \mathbb{R}$ equal to the identity near $[0,1]$ with values in the open set $\gamma_0^{-1}(U) \cap \gamma_1^{-1}(U)$ `Chapter6/OrientableNbhd · exists_contDiff_eq_id_near_Icc_mapsTo`. Then $\Phi(\tau, t) = \varphi^{-1}(r((1-\kappa(\tau))\,\varphi(\gamma_0(\sigma(t))) + \kappa(\tau)\,\varphi(\gamma_1(\sigma(t)))))$. The convex combination lies in the convex set $B_{\delta'} \cap C$, hence in U_c; r maps it into $B_\delta \cap O$, so Φ takes values in W; it is sc-smooth as a composite of sc-smooth maps in the chart; at $\tau = 0, 1$ it is γ_0, γ_1 on $[0,1]$, because σ is the identity there and r fixes O; and the end points stay fixed, because γ_0 and γ_1 agree there. The text's contraction $\hat{r}(t, y) = r(ty)$ contracts $r(V)$ for a convex V; for $U = r(V')$ it needs $r(V') \subseteq V'$, which is not stated, and a convex domain of the retraction is not available in general. As we read it, the preprint's (1) is not established for U (Findings, Chapter 6, item 45); the 2021 wording of (1) is proved below. What Theorem 6.6.11 (preprint 6.35) uses is (1′).

(2). $\mathrm{cl}(U) \subseteq T \subseteq \mathrm{cl}(N)$, since $U \subseteq \varphi^{-1}(B_{\delta_3} \cap O) \subseteq T$ and T is closed; so the zeros in $\mathrm{cl}(U)$ are a closed subset of a compact set.

(3), sections along families `Chapter6/OrientableNbhd · exists_isScPlusSection_pullback_of_family`. This is the text's $s(y) = R(y, f(x))$ made to depend on the parameter. An sc-smooth map on the finite-dimensional $\mathbb{R}^n$ takes values in the smooth points, and the principal part $q \mapsto \hat{f}(\Phi(q))$ lifts to a C^∞ map into the level-1 fibre model. In the chart, the section over (q, y) whose fibre coordinate is the projection onto $K_{\varphi(y)}$ of that lift of $-\hat{f}(\Phi(q))$ — after a reparametrization of the parameter, equal to the identity near $[0,1]^n$, that keeps the family in the chart domain — is an sc$^+$-section of the pull-back over the chart domain, and at $y = \Phi(q)$ it is $-f(\Phi(q))$. Cut off by an sc-smooth function on X supported in the chart domain and equal to 1 near a compact set containing $\Phi([0,1]^n)$ `Chapter6/OrientableNbhd · exists_isScSmoothMap_eq_one_nhdsSet`, it becomes a global sc$^+$-section, with $f + S = 0$ at the points $(q, \Phi(q))$ where the cut-off is 1 `Chapter6/OrientableNbhd · exists_isScPlusSection_pullback_of_mapsTo_of_bump`. For $q \in [0,1]^n$ it is the projection of $-\hat{f}(\Phi(q))$, so it depends on q only through $\Phi(q)$. The text's (3) is for paths; it is stated for families because the homotopies of Theorem 6.6.11 (preprint 6.35) need it (Findings, Chapter 6, item 46).

(4) `Chapter6/OrientableNbhd · hasScPlusSectionsThrough_pullback_rn_of_bump`. Products of bump functions on $\mathbb{R}^n$ and on X are bump functions on $\mathbb{R}^n \times X$, and they give the sections. $\mathbb{R}^n \times X$ `Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod` and the total space of π^*P `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` are paracompact, as the project's M-polyfolds and strong bundles must be. *Hidden hypotheses:* bump functions, for the global sections of (3) and (4), where the text assumes partitions of unity from §6.3 (preprint §6.2) on. The construction uses neither that x is smooth nor that X is tame (pp. 263–264 (preprint pp. 220–221)). (3) and (4) also hold as printed, with sections over the M-polyfolds $[0,1] \times X$ and $[0,1)^n \times X$ (§VI.37).

(1), as printed in 2021 `Chapter6/ScContractible · proposition_6_6_3_one`. *For every smooth x and every open $U \ni x$ there are an open $U' \ni x$ in U and an sc-smooth H on the product M-polyfold $[0,1] \times X$ with values in U on $[0,1] \times U'$, $H(0,\cdot) = x$, $H(1,\cdot) = <built - infunctionid>$ on U', and $H(t,x) = x$* `Chapter6/ScContractible · IsScContraction` `Chapter6/ScContractible · exists_isScContraction`. In a chart φ with retraction r

and $u_0 = \varphi(x)$, take $H(t,y) = \varphi^{-}1r(u_0 + t(\varphi(y) - u_0))$ and $U' = \varphi^{-}1(B_\varepsilon(u_0))$, the ball chosen so that $B_\varepsilon \cap C$ lies in the domain of r and $\varphi^{-}1r(B_\varepsilon \cap C) \subseteq U$. The segment stays in the convex set $B_\varepsilon \cap C$; at $t = 1$ it is $\varphi(y)$, which r fixes; at $t = 0$ it is u_0. The smoothness of x is used: the constant u_0 must be a smooth point of E for H to be sc-smooth. The text's $\hat{r}(t,y) = r(ty)$ is this map for $x = 0$. The form with parameter $\mathbb{R}$ is `Chapter6/ScContractible · exists_isScContractionRn`. `proposition_6_6_3_one` is the proposition with (1) in this form and with (1′), (2), (3), (4), for $U' \subseteq U \subseteq W$ in one chart (V6.32, restated; gap G.61, closed; Findings, Chapter 6, item 137).

(1′) on $[0,1]$ `Chapter6/IntervalHomotopy · exists_isScHomotopyI`. For paths on the M-polyfold $[0,1]$ no reparametrization σ is needed. For sc-smooth $\gamma_0, \gamma_1 : [0,1] \to U$ with the same end points, the segment $H(\tau,t) = \varphi^{-}1r(\varphi\gamma_0(t) + \tau(\varphi\gamma_1(t) - \varphi\gamma_0(t)))$ is sc-smooth on the M-polyfold $[0,1]^2$: in the charts t and $1-t$ of $[0,1]$ its representative is $(v,a,b) \mapsto r(a + s(v)(b-a))$ with $s(v) = v$ or $1 - v$ `Chapter6/IntervalHomotopy · chartSegment` `Chapter6/IntervalHomotopy · isScSmoothOn_chartSegment`. The parameter has to stay in $[0,1]$, since outside it the segment leaves the partial quadrant on which r is defined. The edge conditions hold exactly, and they are the hypotheses of the invariance of the transport along $[0,1]$-paths `Chapter6/PathsInterval · orTransportI_eq_of_homotopy` (§VI.37). Together, (1) with parameter $[0,1]$, (1′) for $[0,1]$-paths, (2) and (3), for one U: `Chapter6/IntervalHomotopy · proposition_6_6_3_interval` (Findings, Chapter 6, item 138).

VI.11 Lemma 6.6.4 (preprint 6.29): general position along a family

The text (preprint pp. 221–222): along a path φ with a section $s(\cdot,t)$, the section $F(t,y,\lambda) = f(y) - s(y,t) + \sum \lambda_i s_i(y,t)$ of the bundle over $[0,1] \times \mathbb{R}^k \times X$; Lemma 6.6.4 (preprint 6.29) asserts that there are $s_1, \dots, s_k$ with property (P): for every t, the slice F_t is in general position to the boundary of $X \times \mathbb{R}^k$ at $(\varphi(t), 0)$. The proof is the sentence "using the results from the transversality theory".

The statement formalized (V6.33) `Chapter6/Lemma629 · lemma_6_6_4`. *Over a tame X with bump functions, for f with tame germ data, an sc-smooth $\Phi : \mathbb{R}^n \to X$ and an sc⁺-section S of π^*P with $(f+S)(q,\Phi(q)) = 0$ near $[0,1]^n$, there are finitely many sc⁺-sections $s_1, \dots, s_k$ of π^*P such that $F(\lambda,(q,y)) = f(y) + S(q,y) + \sum \lambda_i s_i(q,y)$ is in general position at $(0,(q,\Phi(q)))$ for all q near $[0,1]^n$.*

The proof is that of Theorem 5.3.10 (preprint 5.18) (§V.20). The points $(0,(q,\Phi(q)))$ are zeros of F `Chapter6/Lemma629 · isZeroAt_paramSection_zero`, and for q in a compact neighbourhood of $[0,1]^n$ they form a compact set. At each of them finitely many sections make the parametrized section satisfy properties (i)–(ii) of §V.19; general position is open (§V.18); finitely many neighbourhoods cover the compact set, and the sections of all of them together do the job `Chapter6/Lemma629 · exists_sections_isGeneralPositionGerm_of_isCompact`. *Hidden hypotheses:* X tame, bump functions, tame germ data, the standing hypotheses of §6.6 (preprint §6.5) and of Theorem 5.3.10 (preprint 5.18). The conclusion is the general position of F at the points $(t,\varphi(t),0)$, which the text derives from (P) and which Lemmas 6.6.5 and 6.6.6 (preprint 6.30 and 6.31) use. (P) itself, the general position of each slice F_t, follows from it by a chart computation — the tangent vectors of the family of zeros lie in $\ker F'$ and $\mathbb{R}^n$ has no boundary — and is §VI.36 (gap G.63 closed; Findings, Chapter 6, item 47).

VI.12 Lemma 6.6.5 (preprint 6.30): kernels as continuous families of subspaces

The text (preprint p. 222): near $(t, \varphi(t), 0)$ the solution set of F is a manifold with boundary with corners, $\pi(t, y, \lambda) = t$ restricted to it is a submersion, and the kernels L_t form a smooth vector bundle L over $[0, 1]$. What the propagation needs is less: the kernels form a continuous family, so that an orientation of one of them can be continued to the others.

The kernels (V6.34). At a zero with surjective linearization and germ data of index d, $\dim\ker = d$ `Chapter6/KernelFamily · StrongBundle.IsSurjLinOfIndex.finrank_linKer`. Near a zero in good position the kernel is the tangent space of the solution set (Theorem 3.6.8 (preprint 3.54), §III.45), the range of the derivative of an sc-retraction onto the solution set read in the chart. So along a continuous family y_q of zeros in good position, read in one chart at continuously varying level-1 points z_q, near every q_0 the vectors $DS(z_q)b_i$, for a basis b of the kernel at q_0 and that retraction S, are a continuous frame of the kernels `Chapter6/KernelFamily · exists_local_frame_linKer` — continuous because $(z, v) \mapsto DS(z)v$ is, independent near q_0 because independence is open, and spanning because the dimension is constant. The kernels form a continuous family of subspaces of the model space, of dimension the index `Chapter6/KernelFamily · exists_subspaceFamily_linKer`. For an sc-smooth family of zeros $\mathbb{R}^n \to X$ read in a chart `Chapter6/KernelFamily · hasFDerivAt_scDeriv_rn`, differentiating $\widehat{f}(\Phi(q)) = 0$ shows that the derivative lies in the kernel of the linearization `Chapter6/KernelFamily · fderiv_mem_linKer`. For the propagation the bundle L is used as a continuous family of subspaces; the solution set as a manifold with boundary with corners near $(t, \varphi(t), 0)$, the submersion π and the smooth structure of L are §VI.36 (gap G.62 closed; Findings, Chapter 6, items 48, 126).

Continuous families of subspaces (V6.35). A family of n-dimensional subspaces L_b of a normed space, spanned near every point by a continuous frame `Chapter6/SubspaceBundle · SubspaceFamily`, has the line bundle $b \mapsto \lambda(L_b)$ `Chapter6/SubspaceBundle · SubspaceFamily.det`, with the wedges of the frames as trivializations `Chapter6/SubspaceBundle · SubspaceFamily.Frame.pretriv`; the coordinate change between two frames is the determinant of the matrix expressing one in the other, continuous because that matrix solves a linear system with continuous coefficients `Chapter6/SubspaceBundle · SubspaceFamily.Frame.coordChange_continuousOn`. Families and orientations restrict along continuous maps `Chapter6/SubspaceBundle · SubspaceFamily.comp` `Chapter6/SubspaceBundle · LineOrientation.comp`. Over $[0, 1]$ an orientation of $\lambda(L_0)$ extends uniquely to an orientation of $\lambda(L)$ (§VI.9), and its value at 1 is the continuation `Chapter6/SubspaceBundle · SubspaceFamily.transport` `Chapter6/SubspaceBundle · SubspaceFamily.transport_eq`, injective `Chapter6/SubspaceBundle · SubspaceFamily.transport_injective` and compatible with restriction `Chapter6/SubspaceBundle · SubspaceFamily.transport_comp`; over a simply connected, locally path-connected base the continuation along a path depends only on its end points. This is the text's "an orientation of the lines … determines by continuation an orientation of all the other lines" (pp. 264–266 (preprint pp. 222–223)), and the bundles $\widetilde{L}$ of pp. 266–273 (preprint pp. 224–227). "Vector bundle" is read as a family of subspaces of one model space with continuous local frames.

Families of kernels and their images (V6.36). The kernels of a norm-continuous family of surjective Fredholm operators over a connected base form a continuous family `Chapter6/StabilizationContinuity · exists_subspaceFamily_ker`: their dimension is the index, and near q_0 the kernel of A_q is the graph over $\ker A_{q_0}$ given by Lemma 6.5.2 (preprint 6.18) with $P = 1$. The images of a continuous family under a jointly continuous family of linear maps injective on its members form one `Chapter6/StabilizationContinuity · SubspaceFamily.exists_map`, and continuous orientations are carried along `Chapter6/`

StabilizationContinuity · SubspaceFamily.isLineOrientation_map. Cutting a continuous family by the kernel of a fixed ℓ that is onto on every member gives one Chapter6/StabilizationContinuity · SubspaceFamily.exists_inf_ker. "As τ varies, the bundle varies continuously" (pp. 265–272 (preprint pp. 223–226)) is read in the operator norm, and for maps into a model space as joint continuity.

VI.13 The stabilized section in a product chart, and Lemma 6.6.6 (preprint 6.31)

The text (pp. 264–266 (preprint pp. 222–223)): with $T = (f - s(\cdot,t))'(\varphi(t))$, the kernel L_t of $F_t'(\varphi(t),0)$ fits into the sequence (96) $0 \to \ker T \to L_t \to \mathbb{R}^k \to \operatorname{coker} T \to 0$, and Lemma 6.6.6 (preprint 6.31) says that it is exact.

The product chart (V6.37). The parametrized section F lives on $\mathbb{R}^k \times (\mathbb{R}^n \times X)$; it is read in the product chart $(\lambda \mapsto \lambda) \times ((q \mapsto q) \times \varphi)$ at $(0,(q,z))$, whose tangent space and fibre are products Chapter6/ParamKernel · PullbackChartData.ofProd_scDeriv_r_apply Chapter6/ParamKernel · PullbackChartData.ofProd_mem_tangent_iff Chapter6/ParamKernel · PullbackChartData.mem_K_ofProd_iff. There the linearization of F is $(\lambda,(\sigma,v)) \mapsto D(\hat{g}_q \circ r)(z)v + \sum \lambda_i \hat{s}_i(q,\cdot)(z) + D_q\sigma$, with $g_q = f + S(q,\cdot)$ and D_q the derivative in the parameter Chapter6/ParamKernel · StrongBundle.scDeriv_ofProd_slice_apply Chapter6/ParamKernel · StrongBundle.scDeriv_paramSection_apply Chapter6/ParamKernel · StrongBundleChart.paramDerivAt Chapter6/ParamKernel · StrongBundle.scDeriv_paramSection_ofProd_ofProd. With the projection to σ and the inclusion of the slice $\sigma = 0$ Chapter6/ParamKernel · sliceProj Chapter6/ParamKernel · sliceIncl, the kernel cut by $\sigma = 0$ is the image of $\ker(T \oplus \varphi)$ Chapter6/ParamKernel · linKer_inf_ker_sliceProj_eq_map Chapter6/ParamKernel · exists_sliceOps, where $T = g_q'(x)$ lies in $\mathrm{Lin}(f,x)$ Chapter6/ParamKernel · mem_linSet_sectionAdd_pullbackSlice and $\varphi(\lambda) = \sum \lambda_i s_i(q,x)$. *$T \oplus \varphi$ is onto when F is in general position at $(0,(q,\Phi(q)))$* Chapter6/ParamKernel · surjective_coprod_of_isLinSurjAt: given w, the surjectivity of F' gives (λ,σ,v) with image w; the derivative of the curve of zeros, $(0,\sigma,D\Phi\,\sigma)$, lies in $\ker F'$ (§VI.12); subtracting it gives $(\lambda,0,v - D\Phi\,\sigma)$ with the same image, a preimage under $T \oplus \varphi$. The kernel is then a closed subspace of the product with the right inclusion Chapter6/ParamKernel · exists_sliceKernel. $\ker F_t'(\varphi(t),0)$ is read as the kernel of the linearization of the whole F cut by $\sigma = 0$ (Findings, Chapter 6, item 49) (pp. 263–266 (preprint pp. 221–223)).

Lemma 6.6.6 (preprint 6.31) (V6.38) Chapter6/ParamKernel · lemma_6_6_6. With these operators, (96) $0 \to \ker T \to \ker(T \oplus \varphi) \to \mathbb{R}^k \to K_z/R(T) \to 0$ is the stabilization sequence of T and φ (Lemma 6.4.14 (preprint 6.14), §VI.6), hence exact. The text has c onto by (P); here it is onto by the general position of F (§VI.11) and the derivative of the family of zeros, which absorbs the $\mathbb{R}^n$-directions (preprint p. 223).

VI.14 Continuation, the sign, and the push-forward

The text (pp. 265–269 (preprint pp. 223–224)): Φ_{E_t} of (94) identifies $\det(T_t)$ with $\lambda(L_t)$; an orientation o_0 of $\det(T_0)$ gives one of $\lambda(L_0)$, continuation carries it to $\lambda(L_1)$, and $\Phi_{E_1}^{-1}$ carries it to $\det(T_1)$: this defines $o_1 = \varphi_* o_0$.

Continuation (V6.39). Over a simply connected, locally path-connected base the continuation of a ray of $\lambda(L_{q_0})$ to $\lambda(L_{q_1})$ is the value at q_1 of the unique orientation extending it Chapter6/Continuation · SubspaceFamily.cont Chapter6/Continuation · SubspaceFamily.cont_eq. It is the identity from a point to itself Chapter6/Continuation · SubspaceFamily.cont_self, composes Chapter6/Continuation · SubspaceFamily.

cont_trans, is a bijection of rays Chapter6/Continuation · SubspaceFamily.contEquiv, is compatible with pull-back Chapter6/Continuation · SubspaceFamily.cont_comp, and along $[0,1]$ it is the continuation of §VI.12 Chapter6/Continuation · SubspaceFamily.cont_eq_transport. For a stabilization $T \oplus \varphi$ that is onto, $\det(T \oplus \varphi) \cong \lambda(\ker(T \oplus \varphi)) \otimes \lambda^*(\{0\}) \cong \lambda(\ker(T \oplus \varphi))$ Chapter6/Continuation · detEquivOfSurjective, and through ι_φ (§VI.6) and an identification of the kernel with a member of L the rays of $\det(T)$ and of $\lambda(L)$ correspond Chapter6/Continuation · stabRayEquiv. The propagation is Φ_{E_0}, then continuation, then $\Phi_{E_1}^{-1}$ on rays Chapter6/Continuation · propagate, with the identity on a constant family Chapter6/Continuation · propagate_self, composition Chapter6/Continuation · propagate_trans and pull-back Chapter6/Continuation · propagate_comp. It is stated over any simply connected, locally path-connected base, so that one statement serves the paths and the squares; the fixed identification $\lambda^*(\mathbb{R}^k) \cong \lambda^*(\{0\})$ of (94) occurs at both ends.

The sign (V6.40). In a local determinant chart with projection P, the chart coordinate of $\iota_\psi(x) \in \det(S \oplus \psi)$ is $(-1)^{k \dim F/R(P) + \dim \operatorname{coker} S}\, D(S,\psi)$ times that of x Chapter6/StabilizationContinuity · coord_stabIota, with D Chapter6/StabilizationContinuity · stabD continuous Chapter6/StabilizationContinuity · continuousOn_stabD and nowhere zero Chapter6/StabilizationContinuity · stabD_ne_zero near the centre. The first summand of the exponent is constant on a chart, the second is not. So where $\dim \operatorname{coker} T$ changes parity along a family, the rays carried by Φ_E jump. With $\alpha(T) = (-1)^{\dim \operatorname{coker} T}$ Chapter6/StabilizationContinuity · stabSign: *for a continuous orientation o of* DET*, a norm-continuous family T_q and a continuous stabilization φ_q with $T_q \oplus \varphi_q$ onto, the rays $\alpha(T_q)\,\Phi(o(T_q))$ of $\lambda(\ker(T_q \oplus \varphi_q))$ form an orientation* Chapter6/StabilizationContinuity · isLineOrientation_stabRay. Without the sign they need not Chapter6/StabilizationContinuity · not_isLineOrientation_stabRay_uncorrected: $E = F = \mathbb{R}$, $k = 1$, $T_q = q/(1+|q|)$ and $\varphi(r) = r_0$, where $\dim \operatorname{coker} T_q$ is 1 at $q = 0$ and 0 elsewhere. As we read it, the orientations carried by Φ_E of (94) along a continuous family are not continuous where $\dim \operatorname{coker} T$ changes parity, and Lemmas 6.6.8, 6.6.10 (preprint 6.33, 6.34) and the homotopy argument before Theorem 6.6.11 (preprint 6.35) use that they are. The propagation uses $\alpha(T)\Phi_E$, which on each operator carries the text's orientation or its opposite (Findings, Chapter 6, item 50) (p. 256 (preprint p. 212), as used on pp. 265–272 (preprint pp. 223–226)).

The push-forward (V6.41). At a chart point p of a point y Chapter6/Propagation · chartPt, a section S with $(f+S)(y) = 0$ and sections $s_1, \dots, s_k$ give $T = (f+S)'(y) \in \operatorname{Lin}(f,y)$ Chapter6/Propagation · ChartPoint.stabT Chapter6/Propagation · ChartPoint.stabT_mem_linSet and $\varphi(\lambda) = \sum \lambda_i \hat{s}_i(z)$ Chapter6/Propagation · ChartPoint.stabPhi, with appended sections giving the appended stabilization Chapter6/Propagation · ChartPoint.stabPhi_finAppend; $T \oplus \varphi$ is Fredholm Chapter6/Propagation · isFredholm_stabCoprod, and the sign is an involution Chapter6/Propagation · stabSign_inv Chapter6/Propagation · toPerm_stabSign_symm. A family of *stabilized kernels* Chapter6/Propagation · IsStabKernels over a parameter space Q consists of a global sc$^+$-section S of the pull-back with $f + S = 0$ at the points $(q, y(q))$, sections $s_1, \dots, s_k$, a continuous lift of $q \mapsto \varphi(y(q))$ to level 1 of the chart, and a continuous family Λ of subspaces of $E_0 \times \mathbb{R}^k$ such that at every chart point of $y(q)$ in the chart $T_q \oplus \varphi_q$ is onto and Λ_q is the image of its kernel. It restricts along continuous maps Chapter6/Propagation · IsStabKernels.comp. Its *push-forward* between two parameters a, b Chapter6/Propagation · IsStabKernels.push: $o \in \operatorname{Or}^f_{y(a)}$ gives the ray $o(T_a)$, then $\alpha(T_a)\Phi_{E_a}(o(T_a))$, a ray of $\lambda(\Lambda_a)$; continuation carries it to $\lambda(\Lambda_b)$; $\alpha(T_b)\Phi_{E_b}^{-1}$ carries it to $\det(T_b)$; and $\operatorname{Or}^f_{y(b)} \cong \{\text{rays of } \det(T_b)\}$ (§VI.9) gives the image. It depends only on the data Chapter6/Propagation · IsStabKernels.push_congr. With $\operatorname{orRay}_q(o)$, the ray of $\lambda(\Lambda_q)$ read from o at q Chapter6/Propagation · IsStabKernels.orRay, $\varphi_* = \operatorname{orRay}_b^{-1} \circ \operatorname{cont}_{ab} \circ \operatorname{orRay}_a$ Chapter6/Propagation · IsStabKernels.push_eq_orRay. *Along a path with continuously varying data and a fixed point, continuation carries the rays read at one parameter to those read at the next* Chapter6/Propagation · IsStabKernels.cont_orRay_path: the operators T_q, φ_q vary continuously; their kernels form a continuous family (§VI.12);

the sign makes the rays $\alpha\Phi(o(T_q))$ an orientation; and the inclusion into $E_0 \times \mathbb{R}^k$ carries that orientation to $\lambda(\Lambda)$. The push-forward is compatible with pull-back `Chapter6/Propagation · IsStabKernels.push_comp` and with concatenation `Chapter6/Propagation · IsStabKernels.push_trans`. For given end-point operators the result differs from the text's by $\alpha(T_a)\alpha(T_b)$; the sections enter pointwise, and the continuity in the parameter is that of Λ (Findings, Chapter 6, item 50) (preprint p. 224).

VI.15 The stabilized kernels exist; push-forwards along paths

Lemma 6.6.5 (preprint 6.30) for the push-forward (V6.42) `Chapter6/StabKernelsExist · exists_isStabKernels`. *For an sc-smooth $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n)$ in the domain of a chart, an sc⁺-section S with $f + S = 0$ along Φ near $[0,1]^n$, and sections $s_1, \dots, s_k$ as in Lemma 6.6.4 (preprint 6.29), the stabilized kernels of the slices form a continuous family over $[0,1]^n$.* The family $q \mapsto \varphi(\Phi(q))$ has a continuous lift to level 1 `Chapter6/StabKernelsExist · exists_continuousOn_lift_chart`, the values of an sc-smooth map on $\mathbb{R}^n$ being smooth points. The zeros $(0,(q,\Phi(q)))$ of F are in general position, hence in good position; their kernels form a continuous family (§VI.12); cutting it by $\{d\sigma = 0\}$, a fixed functional onto on every member because the derivative of the family of zeros lies in the kernel, keeps it continuous (§VI.12); and §VI.13 identifies the cut with $\ker(T_q \oplus \varphi_q)$. The kernels also form a smooth vector bundle (§VI.36); the continuation uses only their continuity.

Push-forwards along paths (V6.43). The parameter points of $[0,1]$ `Chapter6/PushData · pt1_mem_Icc`, level-1 lifts at smooth points `Chapter6/PushData · exists_lift_of_smooth` and the chart points they give `Chapter6/PushData · chartPtOf` are the bookkeeping. *$e : \mathrm{Or}^f_{\gamma(0)} \to \mathrm{Or}^f_{\gamma(1)}$ is a push-forward along an sc-smooth path γ in a chart* `Chapter6/PushData · IsPush` if there are global sc⁺-sections $S, s_1, \dots, s_k$ of $\pi^*P \to \mathbb{R} \times X$ with $(f+S)(t,\gamma(t)) = 0$ for t near $[0,1]$, and a family of stabilized kernels of their slices along γ whose push-forward is e. Equal data give equal push-forwards `Chapter6/PushData · IsStabKernels.push_eq_of_eq`. Pulled back along an affine reparametrization the sections still vanish along the reparametrized path `Chapter6/PushData · isZeroAt_sectionAdd_reparamSection` and give a push-forward along it `Chapter6/PushData · isPush_reparam`; so a push-forward along γ is the composite of push-forwards along its two halves, each reparametrized to $[0,1]$ `Chapter6/PushData · IsPush.split`. *Along every sc-smooth path in a set with property (3) of Proposition 6.6.3 (preprint 6.28) there is a push-forward* `Chapter6/PushData · exists_isPush`: (3) gives S, Lemma 6.6.4 (preprint 6.29) gives $s_1, \dots, s_k$, and the stabilized kernels exist. Global sections of the pull-back stand in place of the text's sections over $U \times [0,1]$, and (P) is used only to construct a push-forward (preprint p. 224).

VI.16 Lemma 6.6.8 (preprint 6.33) and sections over families

Lemma 6.6.8 (preprint 6.33) (V6.45) `Chapter6/Propagation · IsStabKernels.push_append`. *Appending sc⁺-sections $\bar{s}_1, \dots, \bar{s}_d$ after $s_1, \dots, s_k$ does not change the push-forward.* The text proves it with a homotopy $\tilde{F}(\tau,\cdot)$ and "one verifies by a simple computation". The computation is this. For $\ell_1, \dots, \ell_d \in \ker(T \oplus (\varphi,\psi))$ whose ψ-coordinates are the standard basis of $\mathbb{R}^d$, the map $w \mapsto (w,0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$ `Chapter6/AddSections · wedgeExt` `Chapter6/AddSections · wedgeExt_wedge` does not depend on the ℓ_j — two choices differ by vectors of the image of $\ker(T \oplus \varphi)$, the span of the first block `Chapter6/AddSections · wedgeExt_congr` — and is bijective `Chapter6/AddSections · wedgeExt_bijective`. With the appended stabilization `Chapter6/AddSections · appendStab` and the kernel inclusion $(v,\lambda) \mapsto (v,\lambda,0)$ `Chapter6/AddSections · kerAppend`, *the stabilization isomorphism of (φ,ψ) is that of φ followed by $w \mapsto (w,0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$, exactly, with no sign* `Chapter6/AddSections · detEquiv_stabIota_append`: a choice of Proposition 6.4.15 (preprint 6.15) for (φ,ψ) is one for φ with the ℓ_j appended to the

kernel basis and the standard basis vectors to the r_i, and with the appended sections last the matrix $(\bar{c}, r)$ becomes block triangular with an identity block, so the determinant in (94) is unchanged. Along a family the ℓ_j are chosen continuously `Chapter6/AddSections · SubspaceFamily.exists_continuous_lifts` — the larger kernels project onto $\mathbb{R}^d$ because $T \oplus \varphi$ is already onto — and the wedge extension carries continuous orientations along `Chapter6/AddSections · isLineOrientation_wedgeExt`, through the kernel inclusion `Chapter6/AddSections · kerEquivMap`. So the stabilized rays `Chapter6/AddSections · stabRayEquiv_append` and the propagations `Chapter6/AddSections · propagate_append` agree, and with them the push-forwards. The homotopy of the text's proof is not needed (Findings, Chapter 6, item 51) (pp. 266–271 (preprint pp. 224–225)).

Sections over families (V6.46). For g affine, $(q, y) \mapsto S(g(q), y)$ `Chapter6/FamilySections · reparamSection` is again an sc⁺-section of the pull-back `Chapter6/FamilySections · StrongBundle.IsScPlusSection.reparam`, through the sc-smooth product map $g \times 1$ `Chapter6/FamilySections · rnProdMapOp`, with the expected slices `Chapter6/FamilySections · pullbackSlice_reparamSection`. Sc⁺-sections may be checked in one chart around each point `Chapter6/FamilySections · StrongBundle.isScPlusSection_of_local`. Adding sections keeps the general position of F at $(0, y)$ `Chapter6/FamilySections · StrongBundle.isGeneralPositionGerm_paramSection_of_comp`: the linearization only gains columns, and the properties of Definition 5.3.9 (preprint 5.17) survive. These are the sections $s(y, \tau, t)$ over a square and the text's "adding more sections, the properties will still hold" (pp. 266–272 (preprint pp. 224–226)).

VI.17 Lemma 6.6.10 (preprint 6.34) and Proposition 6.6.7 (preprint 6.32)

Lemma 6.6.10 (preprint 6.34) (V6.47) `Chapter6/Propagation · IsStabKernels.push_square`. *For a family of stabilized kernels over the square $[0,1]^2$ with the end points x_0, x_1 fixed and the end-point operators varying continuously, the push-forwards along the edges $\tau = 0$ and $\tau = 1$ agree.* The edges are the restrictions to affine segments of the square `Chapter6/PropagationWellDefined · sqEdge` `Chapter6/PropagationWellDefined · sqPath`. On the edge $t = 0$ the point is x_0 and the operators vary continuously, so by §VI.14 continuation along it carries $\mathrm{orRay}_{(0,0)}(o)$ to $\mathrm{orRay}_{(1,0)}(o)$; the same on $t = 1$. The square is simply connected, so continuation from $(0,0)$ to $(1,1)$ is the same through $(0,1)$ and through $(1,0)$. Writing both push-forwards as $\mathrm{orRay}^{-1} \circ \mathrm{cont} \circ \mathrm{orRay}$ gives the claim. The text's convex homotopy s^τ is part of the square of Proposition 6.6.7 (preprint 6.32) below, and the continuity of the end-point orientations along τ is that of §VI.14, with the sign (preprint p. 226).

Proposition 6.6.7 (preprint 6.32) (V6.44) `Chapter6/PropagationWellDefined · proposition_6_6_7`. *Two push-forwards along the same sc-smooth path in the domain of a chart agree.* Let the data be $(S^0; s^0_1, \dots, s^0_{k^0})$ and $(S^1; s^1_1, \dots, s^1_{k^1})$. The text combines Lemma 6.6.8 (preprint 6.33) (more sections) and Lemma 6.6.10 (preprint 6.34) (a homotopy of the data); the formalization does both over one square (Findings, Chapter 6, item 52). *Step 1, the square.* $\Phi(\tau, t) = \gamma(t)$, constant in τ, and $S^\tau = (1-\tau)S^0 + \tau S^1$, a section of the pull-back over $\mathbb{R}^2 \times X$ by §VI.16; $f + S^\tau$ vanishes along Φ since both $f + S^0$ and $f + S^1$ do and the fibres are linear `Chapter6/PropagationWellDefined · isZeroAt_sectionAdd_combo`. *Step 2, the sections.* With $\widehat{s}^i_j = s^i_j$ for $j < k^i$ and 0 otherwise, the sections $(1-\tau)\widehat{s}^0_j + \tau \widehat{s}^1_j$, followed by the sections that Lemma 6.6.4 (preprint 6.29) gives on the square. The parametrized section is in general position along the square — adding sections keeps it (§VI.16) — so the stabilized kernels exist over $[0,1]^2$ (§VI.15). *Step 3, the edges* $\tau = 0, 1$. At $\tau = 0$ the sections s^0 are the first k^0 of the family and at $\tau = 1$ the sections s^1 are the first k^1, so Lemma 6.6.8 (preprint 6.33) compares each datum with the edge of the square without reordering `Chapter6/PropagationWellDefined · IsStabKernels.push_eq_of_prefix`, the edges being data of the same kind `Chapter6/PropagationWellDefined · IsStabKernels.congr_data` `Chapter6/PropagationWellDefined · IsStabKernels.`

push_congr_data. *Step 4, the edges $t = 0, 1$.* There the point is fixed and the operators are affine in τ Chapter6/PropagationWellDefined · linAt_principal_combo, hence continuous Chapter6/PropagationWellDefined · continuous_principal_pullbackSlice. *Step 5.* Lemma 6.6.10 (preprint 6.34) equates the push-forwards along the edges $\tau = 0$ and $\tau = 1$, and Step 3 identifies them with the two given ones. *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5) (X tame, bump functions, tame germ data). "Does not depend on the choices" is read as the agreement of any two data (pp. 266–272 (preprint pp. 224–226)).

VI.18 Theorem 6.6.11 (preprint 6.35), and paths in a chart

Theorem 6.6.11 (preprint 6.35) (V6.48). U has the *local propagation property* in a chart c Chapter6/PropagationWellDefined · HasLocalPropagation if U lies in the domain of c, along every sc-smooth path in U there is a push-forward, and any two push-forwards along paths in U with the same end points agree. Chapter6/PropagationWellDefined · theorem_6_6_11: *over a tame X with bump functions, for f with tame germ data, every smooth point x has an open $U \ni x$ and a chart c with the local propagation property.* The statement names the bundle and the point but not the section (possible omission); it is read for an sc-Fredholm section with tame germ data (Findings, Chapter 6, item 53). *Step 1.* $U \subseteq W$ and c from Proposition 6.6.3 (preprint 6.28); push-forwards exist along paths in W, a fortiori in U, by (3) (§VI.15). *Step 2, homotopy invariance* Chapter6/PropagationWellDefined · push_eq_of_homotopy. For an sc-smooth $\Phi : \mathbb{R}^2 \to X$ with $\Phi([0,1]^2) \subseteq W$ and the end points fixed, (3) for $n = 2$ gives S, Lemma 6.6.4 (preprint 6.29) gives sections, and the stabilized kernels exist over the square. The edge $t \mapsto (\tau, t)$ is an affine map, so the restriction to it is a push-forward datum along $\Phi(\tau, \cdot)$ (§§VI.14–VI.15). Along $t = 0, 1$ the point is constant and so is the section $S(q, \cdot)$, because it depends on q only through $\Phi(q)$; the end-point operators are then constant, and Lemma 6.6.10 (preprint 6.34) applies. With Proposition 6.6.7 (preprint 6.32), any push-forward along $\Phi(0, \cdot)$ equals any along $\Phi(1, \cdot)$. *Step 3, the reparametrization.* The homotopy of (1′) is γ_0 and γ_1 at its edges only for $t \in [0, 1]$, while a push-forward datum asks for zeros of $f + S$ for t near $[0, 1]$. The proof therefore passes through $\gamma_i \circ \kappa$, which maps all of $\mathbb{R}$ into $\gamma_i([0, 1])$, with three homotopies whose edges are exact for all t: $H(\sigma, t) = \gamma_0((1 - \sigma)t + \sigma\kappa(t))$ from γ_0 to $\gamma_0 \circ \kappa$; the homotopy $(\tau, t) \mapsto \Phi(\tau, \kappa(t))$ from $\gamma_0 \circ \kappa$ to $\gamma_1 \circ \kappa$; and the analogue of H for γ_1. Step 2 applied three times gives the agreement. U is not shown to be sc-smoothly contractible (gap G.61); what the proof uses is (1′) (pp. 271–273 (preprint pp. 226–227)). For paths on the M-polyfold $[0, 1]$ the theorem is §VI.37.

Paths in a chart (V6.49). Sc-smooth maps that agree near every point of a set are sc-smooth together Chapter6/ScPaths · MPolyfold.isScSmoothMap_of_eventuallyEq; C^∞ maps of the parameter compose Chapter6/ScPaths · pt1_apply Chapter6/ScPaths · contDiff_pt1_comp. *For an open U in the domain of a chart and $x \in U$ there is an open $V \ni x$ in U whose smooth points are pairwise joined by sc-smooth paths in U* Chapter6/ScPaths · exists_nhds_scPath: in the chart, $t \mapsto \varphi^{-1}(r(u_1 + \kappa(t)(u_2 - u_1)))$, for u_1, u_2 in a small ball around $\varphi(x)$, as in (1′). The concatenation Chapter6/ScPaths · concatPath is $\gamma_1(\kappa(3s))$ for $s \le 1/2$ and $\gamma_2(\kappa(3s - 2))$ for $s > 1/2$ Chapter6/ScPaths · concatPath_of_le Chapter6/ScPaths · concatPath_of_gt. It is constant on $[1/3, 2/3]$, with value $\gamma_1(1) = \gamma_2(0)$, so the two pieces glue sc-smoothly Chapter6/ScPaths · isScSmoothMap_concatPath; it has the right end points Chapter6/ScPaths · concatPath_zero Chapter6/ScPaths · concatPath_one and halves Chapter6/ScPaths · concatPath_half₁ Chapter6/ScPaths · concatPath_half₂. Reparametrizing by a C^∞ map keeps sc-smoothness Chapter6/ScPaths · isScSmoothMap_reparam, and push-forwards compose along concatenations Chapter6/ContinuousOrientation · HasLocalPropagation.eq_trans. Paths are joined here locally, inside a chart; the global facts recalled on preprint p. 201 are §VI.37 (gap G.56 closed) (pp. 272–274 (preprint pp. 227–228)).

VI.19 Definition 6.6.12 (preprint 6.36): orientable neighbourhoods and continuous orientations

Definition 6.6.12 (preprint 6.36) (V6.50). The text: an orientable neighbourhood of (f, x) is an open, sc-smoothly contractible U with the local propagation property, and a continuous orientation over it is a choice $y \mapsto o_y$ carried into itself by the propagation. Since Proposition 6.6.3 (preprint 6.28) gives (1′), an orientable neighbourhood is a pair $V \subseteq U$ `Chapter6/SectionOrientation · IsOrientableNbhd`: U open with the local propagation property in a chart, $V \ni x$ open with any two smooth points joined by an sc-smooth path in U. A *continuous orientation* is a choice $y \mapsto o_y$ on V_∞, read in the chart, carried into itself by the push-forwards along the paths in U `Chapter6/ContinuousOrientation · IsContinuousOrientation`. *There are exactly two* `Chapter6/ContinuousOrientation · exists_two_isContinuousOrientation`: given o_x, set o_y to be the push-forward of o_x along any path in U from x to y, well defined by the local propagation property; push-forwards compose along concatenations, so this is a continuous orientation; the two values of o_x give two different ones; and a continuous orientation is determined by its value at x. *Every smooth point has an orientable neighbourhood* `Chapter6/ContinuousOrientation · exists_orientableNbhd` `Chapter6/SectionOrientation · exists_isOrientableNbhd`, by Theorem 6.6.11 (preprint 6.35) and §VI.18. *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5) (Findings, Chapter 6, item 54) (preprint p. 227).

VI.20 The path category PX

Reparametrization homotopies (V6.51). The values of an sc-smooth path at parameters are smooth points `Chapter6/ScHomotopy · mem_smoothSet_of_isScSmoothMap`, and C^∞ maps of $\mathbb{R}^2$ compose `Chapter6/ScHomotopy · contDiff_pt2_comp`. *For a C^∞ θ with $\theta(0) = 0$, $\theta(1) = 1$, $\gamma \circ \theta$ is sc-smoothly homotopic to γ with the end points fixed* `Chapter6/ScHomotopy · isScHomotopyRel_reparam`, by $H(\tau, t) = \gamma((1-\tau)t + \tau\theta(t))$, and inside every set containing $\gamma([0,1])$ when $\theta([0,1]) \subseteq [0,1]$ `Chapter6/ScHomotopy · reparam_mem`. With the halves of a path `Chapter6/ScHomotopy · leftHalf` `Chapter6/ScHomotopy · rightHalf`, the concatenation of the two halves of γ is such a reparametrization `Chapter6/ScHomotopy · halvesReparam` `Chapter6/ScHomotopy · concatPath_halves` (preprint p. 228).

The category (V6.52). The tools are the value $\kappa(1/2)$ `Chapter6/PathCategory · smoothTransition_half`, constant maps at smooth points `Chapter6/PathCategory · isScSmoothMap_const_of_mem_smoothSet`, maps defined by cases on closed pieces that agree near the seams `Chapter6/PathCategory · isScSmoothMap_ite` `Chapter6/PathCategory · isScSmoothMap_ite₃` and the middle third of a concatenation `Chapter6/PathCategory · concatPath_of_le_two_thirds` `Chapter6/PathCategory · concatPath_of_one_third_le`. Sc-smooth homotopy with the end points fixed `Chapter6/PathCategory · ScHomotopic` is an equivalence relation on the sc-smooth paths from x to y `Chapter6/PathCategory · ScHomotopic.refl` `Chapter6/PathCategory · ScHomotopic.symm` `Chapter6/PathCategory · ScHomotopic.trans`. Transitivity needs care: a homotopy meets its edge conditions only on $[0,1]$ (decision D2), and two homotopies glued in τ need not agree off $[0,1]$, so all paths and homotopies are first reparametrized by κ in t (Findings, Chapter 6, item 71). The paths with smooth end points `Chapter6/PathCategory · ScPath` `Chapter6/PathCategory · ScPath.source_mem_smoothSet` `Chapter6/PathCategory · ScPath.target_mem_smoothSet` have the constant paths `Chapter6/PathCategory · ScPath.const`, concatenation `Chapter6/PathCategory · ScPath.concat` and reversal `Chapter6/PathCategory · ScPath.reverse`. Concatenation is well defined on classes `Chapter6/PathCategory · ScPath.concat_congr`, has the constant paths as units `Chapter6/PathCategory · ScPath.const_concat` `Chapter6/PathCategory · ScPath.concat_const`, is associative `Chapter6/PathCategory · ScPath.concat_assoc` — the two bracketings differ by an explicit reparametrization `Chapter6/PathCategory · assocReparam`, joined by a three-piece homotopy `Chapter6/PathCategory · isScHomotopyRel_triHomotopy` — and the reversed paths

are inverses Chapter6/PathCategory · ScPath.concat_reverse Chapter6/PathCategory · ScPath.reverse_concat. The classes Chapter6/PathCategory · ScPathClass with their composition Chapter6/PathCategory · ScPathClass.comp make the smooth points a category PX Chapter6/PathCategory · PathCat Chapter6/PathCategory · PathCat.category, indeed a groupoid Chapter6/PathCategory · PathCat.groupoid, with the morphism of a path Chapter6/PathCategory · PathCat.homMk, composition and identities on representatives Chapter6/PathCategory · PathCat.comp_mk Chapter6/PathCategory · PathCat.id_eq_mk, equality of morphisms as homotopy Chapter6/PathCategory · PathCat.mk_eq_mk_iff, every morphism represented Chapter6/PathCategory · PathCat.exists_eq_mk and inverses by reversal Chapter6/PathCategory · PathCat.inv_mk. The text does not name the identities; they are the classes of the constant paths. The second piece $\psi(\beta(s))$ of the composition on preprint p. 228 leaves $[0,1]$, since $\beta([1/2,1]) = [1,2]$; it is read as $\psi(\beta(s) - 1)$ (possible misprint; Findings, Chapter 6, item 55). Mathlib's $[\varphi] \gg [\psi]$ is the text's $[\psi] * [\varphi]$ (pp. 272–274 (preprint pp. 227–228)). The category with paths on $[0,1]$, as printed, is equivalent to this one (§VI.37).

VI.21 Comparing charts: the transfer, and Or_x^f in a chosen chart

The proof of Theorem 6.6.13 (preprint 6.37) carries an orientation along orientable neighbourhoods of different points of a path, which may lie in different charts. The text takes T_xX, Y_x, $\mathrm{Lin}(f,x)$ and Or_x^f to be the same whichever chart is used (preprint p. 219); here they are made at chart points (§VI.1), so the charts have to be compared (Findings, Chapter 6, item 70).

Conjugation of determinants (V6.53). For isomorphisms A, Γ with $S'A = \Gamma S$, A maps $\ker S$ onto $\ker S'$ Chapter6/ChartTransfer · conjKerEquiv and Γ induces $\operatorname{coker} S \cong \operatorname{coker} S'$ Chapter6/ChartTransfer · conjCokerEquiv; together they give $\det(S) \cong \det(S')$ Chapter6/ChartTransfer · conjDet, on wedges $(a \wedge \dots) \otimes (d \wedge \dots)^* \mapsto (Aa \wedge \dots) \otimes ([\Gamma d] \wedge \dots)^*$ Chapter6/ChartTransfer · conjDet_wedge_tmul, compatible with composition, equal data and the identity Chapter6/ChartTransfer · conjDet_trans Chapter6/ChartTransfer · conjDet_congr Chapter6/ChartTransfer · conjDet_refl. Good left-projections are conjugated, $\Gamma P \Gamma^{-1} \in \Pi_{S'}$ Chapter6/ChartTransfer · IsGoodLeftProj.conj, and the γ^P commute with the conjugation Chapter6/ChartTransfer · gammaProj_conjDet; read in local determinant charts the conjugation has a continuous nowhere-zero coordinate Chapter6/ChartTransfer · coord_conjDet_coordInv, so continuous orientations of DET are carried along Chapter6/ChartTransfer · isLineOrientation_conj Chapter6/ChartTransfer · LineOrientation.conj Chapter6/ChartTransfer · LineOrientation.conj_apply, with the same rules Chapter6/ChartTransfer · LineOrientation.conj_refl Chapter6/ChartTransfer · LineOrientation.conj_trans Chapter6/ChartTransfer · LineOrientation.conj_congr.

The transfer between two chart points (V6.53). For chart points p, p' of x, A is the derivative at z of the transition of the base charts Chapter6/ChartTransfer · ChartPoint.baseDeriv, with the cocycle rules Chapter6/ChartTransfer · ChartPoint.baseDeriv_self Chapter6/ChartTransfer · ChartPoint.baseDeriv_trans, an sc-operator Chapter6/ChartTransfer · ChartPoint.isScOperator_baseDeriv; Γ is the fibre part of the transition of the strong bundle charts Chapter6/ChartTransfer · ChartPoint.fiberMap Chapter6/ChartTransfer · ChartPoint.fiberMap_self Chapter6/ChartTransfer · ChartPoint.fiberMap_trans Chapter6/ChartTransfer · ChartPoint.isScOperator_fiberMap. They restrict to isomorphisms $T_zO \cong T_{z'}O'$ and $K_z \cong K_{z'}$ Chapter6/ChartTransfer · ChartPoint.transferT Chapter6/ChartTransfer · ChartPoint.transferK, again with the cocycle rules Chapter6/ChartTransfer · ChartPoint.transferT_self_apply Chapter6/ChartTransfer · ChartPoint.transferT_trans_apply. At a zero the linearizations in the two charts are related by them, the chain rule applied to the principal parts Chapter6/ChartTransfer · ChartPoint.scDeriv_principal_baseDeriv, and sc⁺-operators conjugate to sc⁺-operators Chapter6/ChartTransfer · ChartPoint.IsScPlusOp.conj; so $L \in \mathrm{Lin}_p(f,x)$ iff $\Gamma L A^{-1} \in \mathrm{Lin}_{p'}(f,x)$ Chapter6/ChartTransfer · ChartPoint.mem_linSet_transfer_iff. Conjugation of orientations over this bijection of

the bases is the transfer $\mathrm{Or}_x^f(p) \cong \mathrm{Or}_x^f(p')$ Chapter6/ChartTransfer · ChartPoint.orTransfer Chapter6/ChartTransfer · ChartPoint.orTransfer_apply, with the cocycle rules Chapter6/ChartTransfer · ChartPoint.orTransfer_self Chapter6/ChartTransfer · ChartPoint.orTransfer_trans.

The push-forward does not depend on the chart (V6.54). The sign is invariant, conjugation preserving dim coker Chapter6/PushTransfer · stabSign_conj; the stabilization is natural — $(\Gamma S A^{-1}) \oplus \Gamma\varphi$ is the conjugate of $S \oplus \varphi$, with the conjugated kernel, onto iff $S \oplus \varphi$ is Chapter6/PushTransfer · coprod_conj Chapter6/PushTransfer · ker_coprod_conj Chapter6/PushTransfer · surjective_coprod_conj — and so is its isomorphism on rays Chapter6/PushTransfer · conj_stab_key Chapter6/PushTransfer · stabRayEquiv_conjDet; continuation is natural under jointly continuous families of injective linear maps Chapter6/PushTransfer · SubspaceFamily.cont_map. The principal parts of the sections transform by Γ Chapter6/PushTransfer · ChartPoint.fiberMap_principal, so the operators T and φ of a datum in one chart are the conjugates of those in the other Chapter6/PushTransfer · ChartPoint.conj_of_linAt Chapter6/PushTransfer · ChartPoint.conj_of_principal, and A varies continuously along a path Chapter6/PushTransfer · continuous_baseDeriv. A family of stabilized kernels in one chart is therefore carried by $A \times 1$ to one in the second chart, with the same sections Chapter6/PushTransfer · IsStabKernels.exists_transfer, the rays read from an orientation corresponding under the transfer Chapter6/PushTransfer · IsStabKernels.orRay_transfer; a push-forward in one chart, conjugated by the transfers at the end points, is a push-forward in the other Chapter6/PushTransfer · IsPush.transfer, and the local propagations of two charts agree Chapter6/PushTransfer · HasLocalPropagation.push_compat. The text does not compare charts (pp. 264–274 (preprint pp. 222–228)).

Or_x^f **in a chosen chart** (V6.55). A chart of the atlas at x is chosen for every x Chapter6/SectionOrientation · orChart Chapter6/SectionOrientation · isChart_orChart Chapter6/SectionOrientation · mem_orChart, and Or_x^f is the set of the orientations read there Chapter6/SectionOrientation · Orx, with two elements Chapter6/OrientationFunctor · exists_two_orx. It is identified with the orientations read in any other chart at x by the transfer Chapter6/SectionOrientation · toChartOr Chapter6/SectionOrientation · toChartOr_orChart, compatibly with composition Chapter6/OrientationFunctor · toChartOr_trans, and a map $y \mapsto o_y$ is read in the chart of an orientable neighbourhood through it Chapter6/SectionOrientation · restrictOr. A chosen chart stands for the text's chart-free Or_x^f (preprint pp. 220, 228).

VI.22 Transport along paths from local push-forwards

The proof of Theorem 6.6.13 (preprint 6.37) (p. 273 (preprint p. 228)): for a path φ, "using the compactness of $[0,1]$ and a finite covering" by orientable neighbourhoods, φ_* is the composite of the local propagations; "by a covering argument" it depends only on $[\varphi]$; and "it is obvious that Γ_f is unique". The formalization separates this argument from orientations: it is proved for any family of types over X_∞ (Findings, Chapter 6, item 72).

Local transport data (V6.56) Chapter6/PathTransport · LocalTransport: a family of *good* open sets covering X_∞, all fibres in bijection, and for every good set U a push-forward along every sc-smooth path in U, such that push-forwards in one good set compose and do not depend on the good set containing the path. Then the push-forward along a loop in a good set is the identity Chapter6/PathTransport · LocalTransport.push_eq_refl, in a good set it depends only on the end points Chapter6/PathTransport · LocalTransport.push_eq_push, and only on the path on $[0,1]$ Chapter6/PathTransport · LocalTransport.push_congr.

The dyadic transport. At level 0 it is the push-forward in a chosen good set containing $\gamma([0,1])$ (an arbitrary bijection if there is none); at level $n+1$ it is the composite of the transports at level n along the two halves Chapter6/PathTransport · LocalTransport.transportN. On a path in a good set every level gives the push-

forward Chapter6/PathTransport · LocalTransport.transportN_eq_push, by the composition rule. The pieces of level N lie in good sets Chapter6/PathTransport · LocalTransport.IsFine for some N: the preimages of the good sets cover the compact $[0,1]$, so there is a Lebesgue number δ Chapter6/PathTransport · LocalTransport.IsLebesgueNumber Chapter6/PathTransport · LocalTransport.exists_isLebesgueNumber, and $2^N\delta > 1$ suffices Chapter6/PathTransport · LocalTransport.isFine_of_isLebesgueNumber. From such an N on the levels agree Chapter6/PathTransport · LocalTransport.transportN_eq_of_isFine, since at the next level each piece is split inside its good set. The *transport* is the value at such a level Chapter6/PathTransport · LocalTransport.transport.

Its properties. It is the push-forward on a path in a good set Chapter6/PathTransport · LocalTransport.transport_eq_push — property (□) — depends only on the path on $[0,1]$ Chapter6/PathTransport · LocalTransport.transport_congr, is the identity on paths constant on $[0,1]$ Chapter6/PathTransport · LocalTransport.transport_const, and is the composite along the two halves Chapter6/PathTransport · LocalTransport.transport_halves. *Homotopy invariance* Chapter6/PathTransport · LocalTransport.transport_eq_of_isScHomotopyRel is the covering argument. For a homotopy $\Phi : \mathbb{R}^2 \to X$ the preimages of the good sets cover $[0,1]^2$, with a Lebesgue number δ Chapter6/PathTransport · LocalTransport.exists_isSqLebesgueNumber. *The square lemma* Chapter6/PathTransport · LocalTransport.transportN_square: for a rectangle of $[0,1]^2$ with sides $< 2^n\delta$, the transports at level n along its vertical, horizontal and diagonal segments Chapter6/PathTransport · vSeg Chapter6/PathTransport · hSeg Chapter6/PathTransport · dSeg commute around its boundary — left side then top equals bottom then right side. By induction on n: for $n = 0$ the rectangle lies in one good set, and both composites are the push-forward along the diagonal; for $n + 1$ the rectangle is cut into four quadrants and the four commutative squares are pasted, the interior edges cancelling Chapter6/PathTransport · equiv_paste_squares. Applied to $[0,1]^2$ at a level with $2^n\delta > 1$, with the sides $\tau \mapsto \Phi(\tau, 0)$, $\tau \mapsto \Phi(\tau, 1)$ constant on $[0,1]$ and of trivial transport, it gives the invariance. The transport is invariant under reparametrization Chapter6/PathTransport · LocalTransport.transport_reparam (§VI.20) and takes concatenations to composites Chapter6/PathTransport · LocalTransport.transport_concatPath, the halves of a concatenation being reparametrizations of the two paths. *Uniqueness* Chapter6/PathTransport · LocalTransport.transport_unique: a family of bijections along the sc-smooth paths that is homotopy invariant, takes concatenations to composites and is the push-forward on paths in good sets, is the transport — $G(\gamma) = G(\text{left half}) \circ G(\text{right half})$ through the reparametrization, and induction on the dyadic level. Dyadic subdivisions of $[0,1]$ and of $[0,1]^2$ stand in place of arbitrary finite coverings (preprint p. 228).

VI.23 Theorem 6.6.13 (preprint 6.37): the functor Γ_f

The text: *for a strong bundle over a tame M-polyfold and an sc-Fredholm section f there is a uniquely determined functor $\Gamma_f : PX \to \mathcal{O}_f$, $x \mapsto \mathrm{Or}^f_x$, with (□): for every smooth x, every sc-smoothly contractible open $U \ni x$ on which f has the local propagation property, and every sc-smooth path $\varphi : [0,1] \to U$, the propagation φ_* and $\Gamma_f([\varphi])$ coincide.*

The local data of f (V6.57). U is a propagation neighbourhood if it is open with the local propagation property in some chart Chapter6/OrientationFunctor · IsPropagationNbhd. Along a path in such a U the push-forward, read in the chosen charts of the end points through the transfer (§VI.21), is orPush Chapter6/OrientationFunctor · orPush; it equals every push-forward along the path in every chart in which the path lies in a set with the local propagation property Chapter6/OrientationFunctor · orPush_eq — by the chart independence of §VI.21 and the cocycle rule of the transfer — it does not depend on the neighbourhood Chapter6/OrientationFunctor · orPush_compat, and it composes Chapter6/OrientationFunctor · orPush_trans. With Theorem 6.6.11 (preprint 6.35), which makes the propagation

neighbourhoods cover X_∞, and §VI.9, which puts every Or_x^f in bijection with every other, these are local transport data `Chapter6/OrientationFunctor · orLocalTransport`, and their transport satisfies (□) `Chapter6/OrientationFunctor · orLocalTransport_transport_eq`.

The functor. $\Gamma_f([\varphi])$ is the transport along φ, well defined on classes by homotopy invariance `Chapter6/OrientationFunctor · orFunctorMap` `Chapter6/OrientationFunctor · orFunctorMap_mk`; it takes identities to identities `Chapter6/OrientationFunctor · orFunctorMap_id` and compositions to composites `Chapter6/OrientationFunctor · orFunctorMap_comp`, by the constancy and concatenation rules of §VI.22; and it is a functor $PX \to \mathrm{Type}$ `Chapter6/OrientationFunctor · orFunctor`. Property (□) for a family G on the morphisms of PX, on the members of a family of open sets `Chapter6/OrientationFunctor · IsStarPropertyOn` and on all of them `Chapter6/OrientationFunctor · IsStarProperty`, reads $G([\varphi]) = \mathrm{toChartOr}_{\varphi(1)}^{-1} \circ e \circ \mathrm{toChartOr}_{\varphi(0)}$ for every chart c, every open U with the local propagation property in c, every sc-smooth path φ in U with smooth end points and every push-forward e along φ in c. Γ_f has it `Chapter6/OrientationFunctor · orFunctorMap_isStarProperty`, so the property is satisfiable. *Uniqueness*: a family with the rules for identities and composites and with (□) on the members of a family of such open sets covering X_∞ is Γ_f `Chapter6/OrientationFunctor · orFunctorMap_unique_of_cover` — restrict the local data to the cover `Chapter6/OrientationFunctor · LocalTransport.restrict` and apply the uniqueness of the transport — in particular with (□) everywhere `Chapter6/OrientationFunctor · orFunctorMap_unique`.

Theorem 6.6.13 (preprint 6.37) `Chapter6/OrientationFunctor · theorem_6_6_13_hyp`. *Over a tame X with bump functions, for f with tame germ data and P with sc⁺-sections through the smooth fibre points, there is exactly one family of bijections $\Gamma_f([\varphi]) : \mathrm{Or}_x^f \to \mathrm{Or}_y^f$ on the morphisms of PX taking identities to identities and compositions to composites, with property (□). Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5), and sc⁺-sections through every smooth fibre point, so that every Or_x^f has two elements (bump functions give them). (□) is asked for every open set with the local propagation property, not only for sc-smoothly contractible ones (the preprint's (1), §VI.10), and the uniqueness needs it only on a covering family — for instance on the sc-smoothly contractible ones, wherever they cover X_∞. $\mathcal{O}_f$ is read as types and bijections, and "uniquely determined" for the maps on morphisms: on objects there is no choice (preprint p. 228). With the text's hypotheses only — X tame with bump functions, f with tame germ data — the sc⁺-sections through the smooth fibre points come from the bump functions (§VI.1) and the paracompactness of $\mathbb{R}^n \times X$ and of the pull-backs is proved, so the theorem holds without the further hypothesis `Chapter6/OrientationMainClean · theorem_6_6_13` (Findings, Chapter 6, item 122).

VI.24 Definitions 6.6.14–6.6.16 (preprint 6.38–6.40): orientability and orientations

Definition 6.6.14 (preprint 6.38) (V6.58). *f is orientable if for every pair of smooth points x, y in the same path component, $\Gamma_f([\varphi])$ does not depend on the class $[\varphi] : x \to y$* `Chapter6/OrientationFunctor · IsOrientableSection`. "In the same path component" is read as "joined by an sc-smooth path" `Chapter6/OrientationFunctor · scPathSetoid`; for points not so joined there is nothing to check (Findings, Chapter 6, item 73). The reading is justified in §VI.37: for smooth points, one connected component, one path component and one sc-smooth path component are the same. *Equivalently there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for every $[\varphi] : x \to y$* `Chapter6/OrientationFunctor · isOrientableSection_iff_exists`: from left to right, choose a base point in each class of points joined by sc-smooth paths and set $o_y = \Gamma_f([\varphi])o_{\mathrm{base}}$ for any $[\varphi]$, which does not depend on $[\varphi]$; from right to left, $\Gamma_f([\varphi])$ and $\Gamma_f([\psi])$ are bijections between two-element sets with the same value at o_x, hence equal (§VI.9) (preprint p. 229).

Definition 6.6.15 (preprint 6.39) (V6.59). *f is orientable if there is $y \mapsto o_y$ on X_∞ such that every smooth point has an orientable neighbourhood over which it is one of the two continuous orientations* `Chapter6/SectionOrientation · IsOrientableSection'`. The definition writes "$y \in U_\infty$" before any U is introduced; it is read as $y \in X_\infty$ (possible misprint; Findings, Chapter 6, item 74). *This holds iff there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi]$* `Chapter6/OrientationFunctor · isOrientableSection'_iff_exists`. From right to left: on an orientable neighbourhood $V \subseteq U$ the push-forwards along paths in U are values of Γ_f, by (□), so the restriction is a continuous orientation. From left to right: elements carried into each other by the transport along the paths in the members of an open covering are carried into each other along every path `Chapter6/OrientationFunctor · LocalTransport.transport_apply_eq_of_local` — the transport is computed on dyadic pieces inside orientable neighbourhoods. *So Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) are equivalent* `Chapter6/OrientationFunctor · isOrientableSection_iff`. The text calls them equivalent without proof (Findings, Chapter 6, item 73) (preprint p. 229).

Definition 6.6.16 (preprint 6.40) (V6.60). *An orientation of f is a map $y \mapsto o_y$ on X_∞ whose restriction to every orientable neighbourhood of every smooth point is a continuous orientation* `Chapter6/SectionOrientation · IsSectionOrientation`. Equivalently $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi] : x \to y$ `Chapter6/OrientationFunctor · isSectionOrientation_iff`, by the argument of Definition 6.6.15 (preprint 6.39); so f has an orientation iff it is orientable `Chapter6/OrientationFunctor · exists_isSectionOrientation_iff`, two orientations that agree at a point agree on the smooth paths joined to it by sc-smooth paths `Chapter6/OrientationFunctor · IsSectionOrientation.apply_eq_iff`, and there are exactly two on each such class `Chapter6/OrientationFunctor · IsSectionOrientation.exists_two`. An orientable neighbourhood is the pair of §VI.19 (Findings, Chapter 6, item 75) (preprint p. 229).

VI.25 Theorem 6.6.18 (preprint 6.41): the natural orientation of the solution set

The text: *over a tame X, let f be a proper sc-Fredholm section in general position to the boundary at every zero, and o^f an orientation of f; then $S = f^{-1}(0)$ is a smooth compact manifold with boundary with corners with a natural orientation.* The proof: S is such a manifold "from the transversality discussion"; $\ker f'(x) = T_xS$ and, $f'(x)$ being onto, T_xS is oriented by o^f_x; and "one verifies that the differential geometric local prolongation of the orientation on S is the same as the Fredholm one".

The first assertion (V6.61) `Chapter6/SolutionOrientation · theorem_6_6_18_solutionSet`. *For X tame and a proper f with tame germ data in general position (Definition 5.3.9 (preprint 5.17)) at every zero, S is compact, a smooth submanifold in general position with the induced structure of a tame manifold with boundary with corners* — the conclusions of Theorem 5.3.10 (preprint 5.18) (§V.20), with good position from general position by Lemma 4.2.8 (preprint 4.19) (§IV.14) `Chapter6/SolutionOrientation · MPolyfold.IsGeneralPositionAt.isGoodPositionAt`. In a chart, $T_xS = \ker f'(x)$ `Chapter6/SolutionOrientation · tangentSpace_solutionSet_eq_linKer` (§V.17). "Proper" is the text's hypothesis; the proof uses only the compactness of S, which for sc-Fredholm sections is equivalent to it (Theorem 5.2.2 (preprint 5.5), §V.6), so nothing of the hypothesis is lost (Findings, Chapter 6, item 90).

The tangent orientation. At a zero, $f'(x)$ — the linearization of $f + s$ with $s = 0$ `Chapter6/SolutionOrientation · ChartPoint.linZero` — lies in $\mathrm{Lin}(f, x)$ `Chapter6/SolutionOrientation · ChartPoint.linZero_mem_linSet` and is onto `Chapter6/SolutionOrientation · ChartPoint.linZero_surjective`, with kernel T_xS in the chart `Chapter6/SolutionOrientation · ChartPoint.map_ker_linZero`. So an element o_x of Or^f_x gives a ray of $\det f'(x) \cong \lambda(\ker f'(x)) = \lambda(T_xS)$, the *tangent orientation* `Chapter6/SolutionOrientation · ChartPoint.`

tangentOr, read in a chart at a zero Chapter6/SolutionOrientation · tangentOrAt. The sign of a surjective operator is 1 Chapter6/SolutionOrientation · stabSign_of_surjective, and the stabilization by $\mathbb{R}^0$ of a surjective operator is $\det(T) \cong \lambda(\ker T)$ on rays Chapter6/SolutionOrientation · stabRayEquiv_zero, so the tangent orientation is the stabilized ray of §VI.14 for the empty family of sections Chapter6/SolutionOrientation · ChartPoint.stabRayEquiv_eq_tangentOr.

"The local prolongation is the Fredholm one." Along an sc-smooth path γ of zeros in the domain of a chart, the datum $S = 0$, $k = 0$ is a family of stabilized kernels Chapter6/SolutionOrientation · isStabKernels_zero, its kernels being those of $f'(\gamma_q)$ Chapter6/SolutionOrientation · SubspaceFamily.congrL; they form a continuous family of subspaces, by Lemma 6.6.5 (preprint 6.30) (§VI.12) Chapter6/SolutionOrientation · exists_tangentFamily Chapter6/SolutionOrientation · tangentFamily Chapter6/SolutionOrientation · tangentFamily_L, and the rays read from an orientation are the tangent orientations Chapter6/SolutionOrientation · IsStabKernels.orRay_eq_tangentOr. So along a path of zeros the push-forward is continuation between tangent orientations Chapter6/SolutionOrientation · cont_tangentOr, and *the tangent orientations given by an orientation of* DET *along the path form an orientation of the line bundle* $\lambda(\Lambda)$ Chapter6/SolutionOrientation · isLineOrientation_tangentOr. Intermediate parameters are reached by the paths $s \mapsto \gamma(t\kappa(s))$, which stay in $\gamma([0,t])$ for all $s \in \mathbb{R}$. For an orientation o of f and an sc-smooth path of zeros in an orientable neighbourhood $V \subseteq U$, the tangent orientations of o along it form an orientation of $\lambda(\Lambda)$ Chapter6/SolutionOrientation · isLineOrientation_tangentOr_of_isSectionOrientation, and together this is the theorem Chapter6/SolutionOrientation · theorem_6_6_18_hyp (Findings, Chapter 6, item 91).

Two charts (V6.61). The tangent orientations are read at chart points, so the text's "natural" needs that they do not depend on the chart. *For $o \in \mathrm{Or}_x^f$ at a zero with $f'(x)$ onto and two charts c, c' at x, the orientation of* $\ker f'(x)$ *read in c' is the image of the one read in c under the tangent map $D(\tau \circ r)(z)$ of the transition* Chapter6/SolutionOrientationCharts · tangentOrAt_toChartOr. The tangent map restricts to an isomorphism of the two kernels Chapter6/SolutionOrientationCharts · ChartPoint.tangentKerEquiv, it conjugates the two linearizations, and the transfer of §VI.21, which is the conjugation of determinants, maps the stabilized ray for the empty family to the stabilized ray; on a surjective operator the determinant is $\lambda(\ker)$ and the conjugation is λ of the tangent map. (Gap G.68 closed.)

Theorem 6.6.18 (preprint 6.41) in the book's shape Chapter6/OrientationMainClean · theorem_6_6_18. *Over a tame X, for a proper f with tame germ data in general position at every zero and an orientation o of f (Definition 6.6.16 (preprint 6.40)), S is a compact manifold with boundary with corners in general position with the natural orientation given by o* Chapter6/OrientationMainClean · IsNaturalSolutionOrientation: at every zero o_x orients $T_xS = \ker f'(x)$; the orientations read in two charts correspond under the tangent map of the transition; and along every sc-smooth path of zeros in an orientable neighbourhood they form an orientation of the kernel line bundle. The three clauses are the three paragraphs above.

S as an oriented manifold (V6.61). Mathlib has manifolds with corners but no orientations of them. A *tangent orientation* of a Mathlib manifold M is an orientation of every tangent space, and it is *continuous* if, read in the charts of the maximal atlas, it is locally constant Chapter6/ManifoldOrientation · TangentOrientation.IsContinuous; the tangent maps of chart changes Chapter6/ManifoldOrientation · chartChangeDeriv have continuous, nowhere-zero Jacobian determinant, so this is well posed. An *oriented atlas* is a covering family of charts whose changes have positive Jacobian determinant Chapter6/ManifoldOrientation · IsOrientedAtlas; a continuous tangent orientation gives one whenever the positive charts cover M Chapter6/ManifoldOrientation · isOrientedAtlas_positiveCharts. The solution set is a compact Mathlib manifold with corners, modelled on the quadrant of $\mathbb{R}^d$, d the index Chapter9/MathlibSolutionSets · theorem_6_6_18_isManifold, from the corner structure of Theorem 3.1.26 (Part III).

Theorem `Chapter6/SolutionSetOrientation · theorem_6_6_18_oriented`: *with the bump functions of §6.6 (preprint §6.5), S carries a continuous tangent orientation o_S which at every zero x, in every chart c of X at x, is o_x: the tangent map ι of $S \hookrightarrow X$ at x is an isomorphism $T_xS \cong \ker f'(x)$, and a basis b of T_xS is positive for o_S iff $\iota b_1 \wedge \cdots \wedge \iota b_d$ is positive for the tangent orientation of o_x. Step 1, the isomorphism.* The tangent map read in the corner coordinates κ of S and the chart c is injective with values in $\ker f'(x)$, and both have dimension d `Chapter6/SolutionSetOrientation · solKerEquiv`. *Step 2, the orientation at a point.* $o_S(x)$ is the pull-back of the tangent orientation along it, read in the corner chart of S at x and the chosen chart of X at x `Chapter6/SolutionSetOrientation · solOr` `Chapter6/SolutionSetOrientation · solOr_spec`. *Step 3, the chart of X plays no role*: two charts of X change the isomorphism by the tangent map of the transition (chain rule), which also relates the two tangent orientations (two charts, above) `Chapter6/SolutionSetOrientation · solReadOr_chartChange`. *Step 4, continuity.* Along a path of S in corner coordinates the wedges $\iota_q b_1 \wedge \cdots \wedge \iota_q b_d$ of a fixed basis are a continuous section of the kernel line bundle — the derivative of the inverse corner chart is continuous `Chapter6/SolutionSetOrientation · isLineOrientation_solKer` — and the tangent orientations of o along the path are an orientation of the same bundle (the prolongation paragraph); two orientations of a line bundle over an interval that agree at one point agree everywhere, so o_S is locally constant in the corner charts `Chapter6/SolutionSetOrientation · isContinuous_solOr`. (Gap G.67 closed.) *Hidden hypothesis:* the bump functions, which give the orientable neighbourhoods on which o is continuous; the text assumes partitions of unity from §6.3 (preprint §6.2) on, which give them, and they hold on Hilbert models `Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert`. "A natural orientation" of the manifold is read as a continuous tangent orientation, not as an oriented atlas: for the quadrant model the two differ — $[0,1]$ has no atlas of quadrant charts with positive changes, since a chart at 0 increases and one at 1 decreases, and in dimension 0 every atlas is oriented while a tangent orientation is a sign at every point (Findings, Chapter 6, item 127). The same holds for an open part of the solution set, without properness: if every zero in an open V is in general position with index d, $S \cap V$ is a Hausdorff manifold with corners with the continuous tangent orientation of o `Chapter6/SolutionSetOpen2021 · theorem_6_6_18_oriented_open`; Theorem 6.7.1 (preprint 6.43) uses it for the index parts of S_{f+s} and for the cobordism S_F (§VI.27).

The paragraph after the theorem (V6.61). For f not in general position the text perturbs by an sc$^+$-section s supported near S and orients S_{f+s} from o^f, using $\mathrm{Lin}(f+s,x) = \mathrm{Lin}(f,x)$. This identity holds at every chart point `Chapter6/PerturbedOrientation · ChartPoint.linSet_eq_of_sectionAdd`: $(f+s)+t$ ranges over the same sections as $f+(s+t)$, and the sc$^+$-sections form a group. So $\mathrm{DET}(f+s,x) = \mathrm{DET}(f,x)$, the push-forwards of $f+s$ are those of f (the sections S of the push-forward data absorb s), and an orientation of f is one of $f+s$ `Chapter6/PerturbedOrientation · isSectionOrientation_sectionAdd` `Chapter6/PerturbedOrientation · SectionOrientation.sectionAdd`; for $f+s$ proper and in general position, S_{f+s} has the natural orientation coming from o^f `Chapter6/PerturbedOrientation · theorem_6_6_18_sectionAdd`, and this applies to the perturbations of Theorem 5.3.10 (preprint 5.18) in its pointwise reading `Chapter6/PerturbedOrientation · theorem_6_6_18_perturbation`. (Gap G.69 closed.) *Readings:* tame germ data, the standing hypothesis of Chapter 5; the orientation o exists iff f is orientable (§VI.24) (Findings, Chapter 6, items 92, 93, 123, 127) (pp. 274–276 (preprint pp. 229–230)).

VI.26 Remark 6.6.19 (preprint 6.42): the orientation of $\hat{f}$

The text: for an oriented (f,o) and an sc$^+$-section s of $Y \to X \times [0,1]$, $\hat{f}(x,t) = f(x) + s(x,t)$ has a natural orientation from o and the standard orientation of $[0,1]$: for $L = (f-t)'(x)$ oriented by $h = (a_1 \wedge \cdots \wedge a_n) \otimes (b_1 \wedge \cdots \wedge b_l)^*$, orient $\tilde{L}(a,b) = La$ by $\tilde{h} = (a_1 \wedge \cdots \wedge a_n \wedge e) \otimes (b_1 \wedge \cdots \wedge b_l)^*$, e the standard orientation of $[0,1]$ (the preprint's order; the 2021 edition puts e first); $\tilde{L}$ lies in the linearization

space of $\hat{f}$ at (x,t) (p. 276 (preprint p. 230)).

The linear algebra (V6.62). For Fredholm L, $\tilde{L} = L \oplus 0 : E \times \mathbb{R} \to F$ `Chapter6/ProductOrientation · extOp` is Fredholm `Chapter6/ProductOrientation · isFredholm_extOp` with $\ker \tilde{L} = \ker L \times \mathbb{R}$ — one dimension more `Chapter6/ProductOrientation · finrank_ker_extOp`, the new direction $e = (0,1)$ `Chapter6/ProductOrientation · extE` — and the same range `Chapter6/ProductOrientation · range_extOp` and cokernel `Chapter6/ProductOrientation · extCoker`. The map $h \mapsto \tilde{h}$, $(a_1 \wedge \cdots \wedge a_n) \otimes d^* \mapsto ((a_1,0) \wedge \cdots \wedge (a_n,0) \wedge e) \otimes d^*$, is an isomorphism $\det(L) \cong \det(\tilde{L})$ `Chapter6/ProductOrientation · extDet` `Chapter6/ProductOrientation · extDet_wedge_tmul`, the wedge extension of §VI.16 with one vector. In local determinant charts its coordinate is $(-1)^{\dim F/R(P) + \dim \operatorname{coker} L}$ times a continuous nowhere-zero function `Chapter6/ProductOrientation · coord_extDet`. Over the convex $\mathrm{Lin}(f,x)$, where $\dim \operatorname{coker} L$ can change parity, the orientation defined by $\tilde{h}$ at one L could therefore differ from the one defined at another, as we read it. With the sign of §VI.14, $(-1)^{\dim \operatorname{coker} L}\, h \mapsto \tilde{h}$ `Chapter6/ProductOrientation · extDetN` has a coordinate continuous up to a sign constant on a chart `Chapter6/ProductOrientation · coord_extDetN`, agrees with the text's map when L is onto `Chapter6/ProductOrientation · extDetN_eq_of_surjective`, and pulls continuous orientations back along $L \mapsto \tilde{L}$ `Chapter6/ProductOrientation · isLineOrientation_extPull` `Chapter6/ProductOrientation · LineOrientation.extPull` (Findings, Chapter 6, item 95). Transported along isomorphisms A, Γ, $\hat{L} = \Gamma^{-1}(L \oplus 0)A$ `Chapter6/ProductOrientation · extConjOp` gets the isomorphism $h \mapsto \tilde{h}$ followed by the conjugation of §VI.21 `Chapter6/ProductOrientation · extConjDet`; and for a preconnected C and a simply connected, locally path-connected $\hat{C}$ containing the $\hat{L}$, an orientation o of DET over a nonempty C gives exactly one orientation $\hat{o}$ over $\hat{C}$ with $\hat{o}(\hat{L})$ the image of $o(L)$ for every $L \in C$ `Chapter6/ProductOrientation · existsUnique_extConjOr`.

In charts. $X \times [0,1]$ is read as $\mathbb{R} \times X$ (decision D2), with the parameter first; s is an sc⁺-section S of π^*P, and $\hat{f} = \tilde{f} + S$ with $\tilde{f}(t,y) = f(y)$. At the product chart point (t,z) `Chapter6/ProductOrientation · ChartPoint.prodPt` the tangent space is $\mathbb{R} \times T_z O$ `Chapter6/ProductOrientation · ChartPoint.mem_prodPt_T_iff` and the fibre K_z `Chapter6/ProductOrientation · ChartPoint.mem_prodPt_K_iff`. The text's $\tilde{L}(a,b) = La$ puts the $\mathbb{R}$-factor second, so $\hat{L} = \Gamma^{-1}(L \oplus 0)A$ with A the swap of the factors and $\Gamma = 1$ `Chapter6/ProductOrientation · ChartPoint.prodT` `Chapter6/ProductOrientation · ChartPoint.prodK` `Chapter6/ProductOrientation · ChartPoint.prodLin`. Sc⁺-operators lift to the product `Chapter6/ProductOrientation · ChartPoint.IsScPlusOp.prodPt`, sections of P lift to t-constant sections of π^*P `Chapter6/ProductOrientation · constLift`, and so $\hat{L} \in \mathrm{Lin}(\hat{f},(t,x))$ for every $L \in \mathrm{Lin}(f,x)$ `Chapter6/ProductOrientation · ChartPoint.prodLin_mem_linSet` — the derivative in t of a t-constant lift vanishes, and S only contributes an sc⁺-operator. *An orientation o of* $\mathrm{DET}(f,x)$ *gives exactly one orientation* $\hat{o}$ *of* $\mathrm{DET}(\hat{f},(t,x))$ *with* $\hat{o}(\hat{L})$ *the image of* $o(L)$ *for every* L `Chapter6/ProductOrientation · ChartPoint.existsUnique_prodOr` `Chapter6/ProductOrientation · ChartPoint.prodOr` `Chapter6/ProductOrientation · ChartPoint.prodOr_apply`, and $o \mapsto \hat{o}$ is injective `Chapter6/ProductOrientation · ChartPoint.prodOr_injective` (Findings, Chapter 6, item 96). *Hidden hypotheses:* x smooth and $\mathrm{Lin}(f,x)$ nonempty (bump functions give it, §VI.1); only the orientation of $\mathrm{DET}(f,x)$ at the one point is used, not an orientation of f. That these pointwise orientations, in the 2021 convention, form an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40) is the last paragraph below.

The 2021 convention, e **first** (V6.62). The 2021 edition prints $\tilde{h} = (e \wedge a_1 \wedge \cdots \wedge a_n) \otimes (b_1 \wedge \cdots \wedge b_l)^*$, with e first; the preprint puts it last. Moving e across n vectors costs $(-1)^n$ `Chapter6/OrientationConvention2021 · wedge_snoc_eq_smul_wedge_cons`. The 2021 map $h \mapsto \tilde{h}$ `Chapter6/OrientationConvention2021 · extDetFirst` `Chapter6/OrientationConvention2021 · extDetFirst_wedge_tmul` is $(-1)^{\mathrm{ind}\, L}$ times the sign-normalized map of the previous paragraph `Chapter6/OrientationConvention2021 · extDetFirst_eq_indSign_smul_extDetN`, because $(-1)^{\dim \ker L} = (-1)^{\mathrm{ind}\, L}(-1)^{\dim \operatorname{coker} L}$ `Chapter6/OrientationConvention2021 · kerSign_eq_indSign_mul_stabSign`. The index

is locally constant on the Fredholm operators Chapter6/OrientationConvention2021 · eventually_fredholmIndex_eq Chapter6/OrientationConvention2021 · eventually_indSign_eq, and a line orientation times a locally constant sign is a line orientation, so *the text's own map pulls continuous orientations back, over any set of Fredholm operators* Chapter6/OrientationConvention2021 · isLineOrientation_extPullFirst. With the 2021 convention no sign has to be changed: the factor $(-1)^{\dim \operatorname{coker} L}$ of the preprint's map and the $(-1)^{\dim \ker L}$ of moving e to the front combine to a locally constant sign. In charts, *an orientation o of* $\mathrm{DET}(f,x)$ *gives exactly one orientation $\hat{o}$ of* $\mathrm{DET}(\hat{f},(t,x))$ *with $\hat{o}(\hat{L}) = e \wedge o(L)$ for every $L \in \mathrm{Lin}(f,x)$* Chapter6/OrientationConvention2021 · ChartPoint.existsUnique_prodOrFirst Chapter6/OrientationConvention2021 · ChartPoint.prodOrFirst, and on $\mathrm{Lin}(f,x)$ the 2021 map is the preprint's times the one constant sign $(-1)^{\mathrm{ind}(f,x)}$ Chapter6/OrientationConvention2021 · ChartPoint.exists_extConjDetFirst_eq_sign_smul (Findings, Chapter 6, item 122).

The orientation of $\hat{f}$ (V6.62) Chapter6/ProductSectionOrientation2021f · SectionOrientation.prodFirst. *Over a tame X with bump functions, for f with tame germ data, an orientation o of f (Definition 6.6.16 (preprint 6.40)) and an sc⁺-section S of $\pi^*P \to \mathbb{R}^1 \times X$, the map $(t,x) \mapsto e \wedge o_x$ is an orientation of $\hat{f} = \tilde{f} + S$ in the sense of Definition 6.6.16 (preprint 6.40).* The text's "natural orientation", with its remark that the choices made in the construction are compatible, is this statement. *Step 1, the map.* $(t,x) \mapsto e \wedge o_x$ is defined in the product chart point over any chart point of x Chapter6/ProductSectionOrientation2021f · prodFirstFun Chapter6/ProductSectionOrientation2021e · prodOrOf, and its reading in any chart of $\mathbb{R}^1 \times X$ is computed through the product chart Chapter6/ProductSectionOrientation2021f · toChartOr_prodFirstFun; the chart of X plays no role, since the transfer of §VI.21 between two product chart points is the transfer between the chart points of X with e put in front Chapter6/ProductSectionOrientation2021b · ChartPoint.orTransfer_prodPt_prodOrFirst Chapter6/ProductSectionOrientation2021f · prodOrOf_orTransfer. *Step 2, the lifted data.* A push-forward datum of f along a path γ — sections S, $s_1,\dots,s_k$ and stabilized kernels Λ (§§VI.14–VI.15) — lifts to one of $\tilde{f}$ along $q \mapsto (\tau(q),\gamma(q))$: the sections are lifted constant in t Chapter6/ProductSectionOrientation2021c · liftProd Chapter6/ProductSectionOrientation2021c · isScPlusSection_liftProd, the linearization of a t-constant lift is $(\sigma,v) \mapsto g'(x)v$, and the stabilized kernels become $\mathbb{R} \times \Lambda$ Chapter6/ProductSectionOrientation2021c · SubspaceFamily.prodLine Chapter6/ProductSectionOrientation2021c · IsStabKernels.prodLift. *Step 3, no sign.* The stabilization isomorphism of $\hat{L}$ maps $e \wedge h$ to $\hat{e} \wedge J(\cdot)$ of the stabilization of h, J the inclusion $v \mapsto (0,v)$ of the old kernel Chapter6/ProductSectionOrientation2021c · stabRayEquiv_extConjDetFirst Chapter6/ProductSectionOrientation2021 · detEquiv_stabIota_extDetFirst: in adapted bases, the appended vector e is first on both sides of formula (91), so the computation is that of Proposition 6.4.15 (preprint 6.15) with one more kernel vector. Continuation commutes with $\hat{e} \wedge J(\cdot)$, proved directly in frames Chapter6/ProductSectionOrientation2021c · SubspaceFamily.cont_prodLine, and the rays read from the product orientation in $\lambda((\mathbb{R} \times \Lambda)_q)$ are $\hat{e} \wedge J$ of those read from o Chapter6/ProductSectionOrientation2021d · ChartPoint.prodOrFirst₀ Chapter6/ProductSectionOrientation2021e · IsStabKernels.orRay_prodLift. Hence *the lifted push-forward maps $e \wedge o$ to $e\wedge$ (the push-forward of o)* Chapter6/ProductSectionOrientation2021e · IsPush.prodLift. *Step 4.* An orientation of f is carried by every push-forward along paths in orientable neighbourhoods (§VI.24); by Step 3 so is $e \wedge o$ along the lifted paths, and every path of $\mathbb{R}^1 \times X$ in a product chart is such a lift, so $e \wedge o$ is an orientation of $\tilde{f}$ Chapter6/ProductSectionOrientation2021f · isSectionOrientation_prodFirstFun Chapter6/ProductSectionOrientation2021f · SectionOrientation.prodFirst₀, and of $\tilde{f}+S$ by the paragraph after Theorem 6.6.18 (preprint 6.41) (§VI.25). (Gap G.66 closed; Findings, Chapter 6, items 96, 97, 122.) *Readings:* $X \times [0,1]$ is read as $\mathbb{R}^1 \times X$ with the parameter first (decision D2; gap G.65 in part), and the hypotheses — X tame with sc-smooth bump functions, tame germ data — are those of Theorem 6.6.18 (preprint 6.41) in §VI.25. On a product end of the solution set of $\hat{f}$ this orientation is the product orientation, the

time vector first (§VI.27).

VI.27 §6.7 (preprint §6.6) and Theorem 6.7.1 (preprint 6.43): the invariant $\Psi_{(f,o)}$

§6.7 (preprint §6.6) (p. 276 (preprint p. 230)) recalls, for a proper oriented (f,o) over a tame X with sc-smooth bump functions and an auxiliary norm N, the neighbourhood U of S_f of Theorem 5.3.10 (preprint 5.18) and the sc$^+$-perturbations s supported in U with $N(s) \le 1$ for which $f+s$ is in general position at its zeros; it calls them *admissible for the pair* (N,U). Theorem 6.7.1 (preprint 6.43) (p. 277 (preprint p. 231)): *there is a well-defined map assigning to a proper oriented (f,o) a linear map $\Psi_{(f,o)} : H^*_{dR}(X,\partial X) \to \mathbb{R}$ such that* (1) *for every auxiliary norm N and every s admissible for (N,U), S_{f+s} is an oriented compact manifold with boundary with corners and $\Psi_{(f,o)}([\omega,\tau]) = \int_{S_{f+s}} \omega - \int_{\partial S_{f+s}} \tau$, the integrals being zero when the dimensions do not agree;* (2) *$\Psi_{(f_0,o_0)} = \Psi_{(f_1,o_1)}$ for a proper oriented homotopy between (f_0,o_0) and (f_1,o_1).* The proof (pp. 277–279 (preprint pp. 231–233)) defines Ψ from one admissible s_0, joins two admissible sections by a homotopy s_t whose solution set S_F over $X \times [0,1]$ is an oriented compact manifold with boundary with corners, decomposes $\partial S_F = S_0 \cup S_1 \cup \tilde{S}$, and compares the two sides by Stokes' theorem applied to an extension $\bar{\omega}$ of ω and to τ; part (2) is "the same homotopy argument". For $\partial X = \emptyset$ (p. 279 (preprint p. 233)), $\Psi([\omega]) = \int_{S_{f+s}} \omega$ for $[\omega] \in H^n_{dR}(X)$, n the index, and zero in the other degrees.

The theorem formalized (V6.76) `Chapter6/Theorem671Final2021 · theorem_6_7_1`. *Let $P : Y \to X$ be a strong bundle over a tame M-polyfold with sc-smooth bump functions. There is Ψ assigning to every proper sc-Fredholm section f with tame germ data and orientation o (Definition 6.6.16 (preprint 6.40)) linear maps $\Psi_{(f,o)} : H^k_{dR}(X,\partial X) \to \mathbb{R}$, $k \ge 0$, such that* (1) *for every auxiliary norm N there are open $U \supseteq S_f$ with $\mathrm{cl}\{x \in U \mid N(f(x)) \le C\}$ compact for every $C > 0$; for every such U, S_{f+s} is compact for every allowable s, admissible sections exist, and for every admissible s, S_{f+s} is compact and the union of its index parts S_k, each S_k a compact manifold with boundary and corners of dimension k oriented by o, and $\Psi_{(f,o)}([\omega,\tau]) = \int_{S_k} \omega - \int_{\partial S_k} \tau$ in degree k;* (2) *$\Psi_{(f_0,o_0)} = \Psi_{(f_1,o_1)}$ in every degree for a proper oriented homotopy.* No hypothesis beyond the text's is used: the proof below assumes nothing of its own steps. The readings (Findings, Chapter 6, items 128–133, 136):

- *R-a, the cohomology.* $H^k_{dR}(X,\partial X)$ is the relative cohomology of Part IV: pairs (ω,τ) of an sc-form on X and a form on the faces of X, with $d(\alpha,\beta) = (d\alpha, j^*\alpha - d\beta)$ `Chapter4/BoundaryForms2021 · MPolyfold.RelForms` `Chapter4/BoundaryForms2021 · MPolyfold.relD`. The integrals are integrals of Mathlib forms over the Mathlib manifolds S_k and their faces (Part IX).
- *R-b, the neighbourhood U with a margin.* The text takes U from Theorem 5.3.10 (preprint 5.18), for one N. It is read existentially and for every N, with the margin "$\mathrm{cl}\{x \in U \mid N(f(x)) \le C\}$ compact for every $C > 0$", which the cobordism below needs for $C = 2$ (item 128).
- *R-c, "zero if the dimensions do not agree".* The index of f need not be constant. Degree k is paired with the part S_k of S_{f+s} where the index is k (item 129).
- *R-d, the proper oriented homotopy*, which the text does not define. It is read through Remark 6.6.19 (preprint 6.42): an sc$^+$-section S of $\pi^*P \to \mathbb{R}^1 \times X$ with $S(0,\cdot) = 0$, $f_1 = f_0 + S(1,\cdot)$ and o_1 the orientation o_0 carried to f_1, compact zeros of $\tilde{f}_0 + S$ over $t \in [0,1]$, and the properness clause of Definition 5.2.1 (3) on that slab `Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopy` `Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopyBetween`; then f_0, f_1 are

proper Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopy.isProper_zero Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopy.isProper_one, and $S = 0$ is one Chapter6/ProperHomotopy2021 · witness_isProperOrientedHomotopyBetween (item 130).

- *R-e, the extension of ω.* The proof's $\bar{\omega} = p_1^*\omega \wedge p_2^*dt$ has degree $k+1$, one more than the dimension k of ∂S_F, while the comparison needs $\int_{S_0} \omega$ and $\int_{S_1} \omega$; as we read it, the argument goes through with $p_1^*\omega$ and $p_1^*\tau$ in their place, and that is what is formalized (item 131).
- *R-f, signs.* The boundary orientation is outward vector first, which gives Stokes' theorem with the sign $+$, as the text uses it; S_F is oriented by $e \wedge o$, time first (item 132). The cut-off route below does not need the signs of the pieces of ∂S_F.

Step 1: the neighbourhood and the admissible sections (V6.76). A proper f has compact S_f (Theorem 5.2.2 (preprint 5.5), §V.6). *For an auxiliary norm N and every open $V \supseteq S_f$ there is an open $S_f \subseteq U \subseteq V$ with* $\mathrm{cl}\{x \in U \mid N(f(x)) \le C\}$ *compact for every $C > 0$* Chapter5/CylinderPerturbation2021 · exists_adapted_margin: for compact S_f, Theorem 5.2.2 (preprint 5.5) gives an open $U_0 \supseteq S_f$ with this property for every $C > 0$ (§V.6), and $U = U_0 \cap V$ keeps it. With such a U, *S_{f+s} is compact for every s allowable for (N, U)* Chapter6/Theorem671_2021 · isCompact_solutionSet_sectionAdd_of_margin: off U the zeros of $f+s$ are zeros of f, which lie in U; in U, $f(x) = -s(x)$ gives $N(f(x)) \le 1$; so S_{f+s} is a closed part of a compact set (Remark 5.3.4 (preprint 5.12)). *Admissible* is the text's notion: s is allowable for (N, U) — supported in U, $N(s) \le 1$ — and $f+s$ is in general position (Definition 5.3.9 (preprint 5.17)) at every zero Chapter6/InvariantSingle2021 · StrongBundle.IsAdmissible (gap G.55 closed). They exist by a relative form of Theorem 5.3.10 (preprint 5.18) Chapter5/CylinderPerturbation2021 · theorem_5_3_10_relative_hyp: *for G with tame germ data, N an auxiliary norm, $K \subseteq W$ with K compact and W open, G in general position at every zero outside K and* $\mathrm{cl}\{x \in W \mid N(G(x)) \le 1\}$ *compact, there is for every $\varepsilon > 0$ an sc$^+$-section p supported in W with $N(p) \le \varepsilon$ and $G + p$ in general position at every zero.* Its proof is that of §V.20: finitely many sections make $G(x) + \sum \lambda_i s_i(x)$ general at the points $(0, y)$ near the compact $K \cap S_G$ Chapter5/CylinderPerturbation2021 · exists_sections_isGeneralPositionGerm_near, and at the other zeros, where G is already general Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramSection_of_isGeneralPositionGerm; Sard's theorem and Theorem 4.3.1 give a regular value λ of norm as small as needed Chapter5/CylinderPerturbation2021 · exists_null_isGeneralPositionGerm_paramSlice; off W the perturbation vanishes near the zero, and general position there is that of G Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_congr_nhds. The text's theorem is the case $K = S_G$, $W = U$; here S_G need not be compact, which the cobordisms need. With $K = S_f$ it gives an admissible section for (N, V) inside any V Chapter5/CylinderPerturbation2021 · exists_admissible Chapter6/InvariantSingle2021 · StrongBundle.exists_isAdmissible_nbhd, and, applied to the maximum of finitely many norms Chapter5/CylinderPerturbation2021 · IsAuxNorm.max Chapter5/CylinderPerturbation2021 · allowableSections_max_subset, one section admissible for two pairs (N_0, U_0), (N_1, U_1) at once Chapter5/CylinderPerturbation2021 · exists_admissible_subset Chapter5/CylinderPerturbation2021 · exists_admissible_two — the step that compares two norms.

Step 2: the index parts as oriented manifolds (V6.76). For admissible s, S_k is the set of zeros of $f+s$ at which f has index k Chapter6/InvariantSingle2021 · StrongBundle.solPart. The index is locally constant (Proposition 6.2.2 (preprint 6.2), §III.49), so S_k is open and closed in S_{f+s} Chapter6/InvariantSingle2021 · StrongBundle.isClosed_solPart, compact Chapter6/InvariantSingle2021 · StrongBundle.isCompact_solPart, and $S_{f+s} = \bigcup_k S_k$ Chapter6/InvariantSingle2021 · StrongBundle.solutionSet_eq_iUnion_solPart. Theorem 6.6.18 (preprint 6.41) for an open part (§VI.25), applied to $f+s$ with the orientation o carried to it (§VI.25, the paragraph after the theorem), makes S_k a compact Mathlib manifold of dimension k Chapter6/InvariantSingle2021 · isManifold_solPart with a continuous tangent orientation which is the Fredholm one at every point Chapter6/InvariantSingle2021 · solPartOr Chapter6/

Chapter6/InvariantSingle2021 · solPartIsOrientedManifold. For a constant index n, $S_n = S_{f+s}$ and $S_k = \emptyset$ for $k \neq n$ Chapter6/InvariantSingle2021 · StrongBundle.solPart_eq_solutionSet Chapter6/InvariantSingle2021 · StrongBundle.solPart_eq_empty_of_ne: the text's clause "zero if the dimensions do not agree".

Step 3: Ψ_s for one admissible section (V6.76). On relative forms of degree k, $\Psi_s(\omega,\tau) = \int_{S_k}\omega - \int_{\partial S_k}\tau$ Chapter6/InvariantSingle2021 · solIntegral Chapter6/InvariantSingle2021 · solBoundaryIntegral Chapter6/InvariantSingle2021 · psiForms: ω restricted to S_k is a Mathlib form Chapter9/MathlibFormsBridge2021 · MPolyfold.formsInftyRestrict, and τ is read on the faces of S_k, which lie in faces of X (Theorem 4.2.11, §IV) Chapter9/SolutionFaceForms2021 · MPolyfold.solFaceForm. *Ψ_s vanishes on exact relative forms* Chapter6/InvariantSingle2021 · psiForms_relD: for $d(\alpha,\beta) = (d\alpha, j^*\alpha - d\beta)$,

$$\int_S d\alpha - \int_{\partial S} j^*\alpha + \int_{\partial S} d\beta = 0,$$

by Stokes' theorem on the compact manifold with corners S_k Chapter9/MathlibFormsBridge2021 · MPolyfold.stokes_formsInftyRestrict, the face form of $j^*\alpha$ being α restricted to the faces Chapter9/SolutionFaceForms2021 · MPolyfold.solFaceForm_jStar_integral, and $\int_{\partial S} d\beta = 0$ since the face form of $d\beta$ is d of the face form Chapter9/SolutionFaceForms2021 · MPolyfold.solFaceForm_d and exact forms on the compact faces integrate to zero Chapter4/ManifoldFaceForms2021 · ChartFaceForm.integral_d_eq_zero_of_compactSpace. So Ψ_s descends to $H^k_{dR}(X,\partial X)$ Chapter6/InvariantSingle2021 · psiAt Chapter6/InvariantSingle2021 · psiAt_mk. It depends on s only, not on (N,U) Chapter6/InvariantSingle2021 · psiAt_congr_admissible. In degree 0 it is a signed count, $\Psi_s([\omega]) = \sum_{x\in S_0}\varepsilon(x)\,\omega(x)$ Chapter6/InvariantSingle2021 · psiAt_mk_zero_eq_sum Chapter6/InvariantSingle2021 · ManifoldForm.integral_eq_sum_of_dim_zero, and it is 0 when $S_{f+s} = \emptyset$ Chapter6/InvariantSingle2021 · psiAt_of_eq_empty. *An integrand.* A form a on S_k with $a = \omega|_{S_k} + d\gamma$, γ a compactly supported C^∞ form extending $-\tau|_{\partial S}$ ($a = \omega|_{S_0}$ in degree 0), is an *integrand* of (ω,τ) Chapter6/InvariantCobordism2021 · IsPsiIntegrand; integrands exist, by extension of face data on the compact S_k Chapter6/InvariantCobordism2021 · exists_isPsiIntegrand, and $\Psi_s([\omega,\tau]) = \int_{S_k} a$, since $\int_S d\gamma = \int_{\partial S} j^*\gamma = -\int_{\partial S}\tau$ Chapter6/InvariantCobordism2021 · IsPsiIntegrand.psiAt_eq. This is the form $\omega - d\beta$ with $j^*\beta = \tau$ of the text's computation, written on S_k.

Step 4: the cobordism data of part (1) (V6.76). The text joins s_0 and s_1 by an sc⁺-homotopy s_t (Remark 5.3.8 (preprint 5.16)) with $F(x,t) = f(x) + s_t(x)$ in general position. Here $X\times[0,1]$ is $\mathbb{R}^1\times X$ with the parameter first (decision D2), and the homotopy is stationary near the ends. *Cobordism data* for f, σ_0, σ_1 are an sc⁺-section $\widehat{S}$ of π^*P with $\widehat{S}(t,\cdot) = \sigma_0$ for $t \le 1/8$ and σ_1 for $t \ge 7/8$, $f+\sigma_i$ in general position at every zero, $F = \tilde{f} + \widehat{S}$ in general position at every zero, and the zeros of F over every $[a,b]$ compact Chapter6/ProperHomotopy2021 · IsCobordismData. *For U with the margin and s_0, s_1 admissible for (N,U), cobordism data for f, s_0, s_1 exist, with $\widehat{S} = 0$ off U and $N(\widehat{S}) \le 2$* Chapter5/CylinderPerturbation2021 · exists_cobordism Chapter6/ProperHomotopy2021 · exists_isCobordismData. *Proof.* Let χ be a smooth cut-off, 0 for $t \le 3/16$ and 1 for $t \ge 13/16$ Chapter5/CylinderPerturbation2021 · cylCut, and $\hat{s} = (1-\chi)\tilde{s}_0 + \chi\tilde{s}_1$ Chapter5/CylinderPerturbation2021 · cylInterp Chapter5/CylinderPerturbation2021 · isScPlusSection_cylInterp, with $N(\hat{s}) \le 1$. $G = \tilde{f} + \hat{s}$ is $(f+s_0)\tilde{}$ for $t < 3/16$ and $(f+s_1)\tilde{}$ for $t > 13/16$, where it is in general position at its zeros, since the lift of a section in general position is Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramLift Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_paramLift_add_of_eq. Let K be the zeros of G over $[3/16,13/16]$, compact because they lie in $[3/16,13/16]\times \mathrm{cl}\{x\in U \mid N(f(x)) \le 1\}$, and $W = (1/8,7/8)\times U$. Where $N(G)\le 1$ on W, $N(f)\le N(G)+N(\hat{s})\le 2$; this is where the margin $C=2$ is used. The relative Theorem 5.3.10 (preprint 5.18) (Step 1) for G, the pulled-back norm, K, W and $\varepsilon = 1$ gives p, and $\widehat{S} = \hat{s} + p$: it equals s_0 for $t \le 1/8$ and s_1 for $t \ge 7/8$, vanishes off U, has $N(\widehat{S}) \le 2$,

and $\tilde{f}+\hat{S}$ is in general position at every zero; its zeros over $[a,b]$ lie in $[a,b]\times\mathrm{cl}\{x\in U \mid N(f(x))\le 2\}$, hence are compact Chapter5/CylinderPerturbation2021 · isCompact_slab_paramLift_add. □

Step 5: the cobordism data of part (2) (V6.76) Chapter6/ProperHomotopy2021 · exists_homotopy_cobordism. *For a proper homotopy S from f_0 to $f_1 = f_0 + S(1,\cdot)$, an auxiliary norm N, a neighbourhood W of the zeros over the slab with $\mathrm{cl}\{w \in W \mid t \in [0,1],\ N(F(w)) \le 2\}$ compact* Chapter6/ProperHomotopy2021 · IsProperOrientedHomotopy.exists_slab_nbhd*, s_0 admissible for f_0 and (N,U_0) and s_1 for f_1 and (N,U_1) with $\{0\}\times U_0$, $\{1\}\times U_1 \subseteq W$, there are cobordism data for f_0, s_0 and $S(1,\cdot)+s_1$.* So $\tilde{f}_0+\hat{S}$ is f_0+s_0 for $t\le 1/8$ and f_1+s_1 for $t\ge 7/8$. *Proof.* $\hat{S} = S\circ(\chi\times 1)+w_0\tilde{s}_0+w_1\tilde{s}_1+p$, with a time change χ, 0 for $t\le 3/8$ and 1 for $t\ge 5/8$ Chapter6/ProperHomotopy2021 · homTime, a weight w_0 that switches s_0 off over $[3/16,5/16]$, before the homotopy moves, and a weight w_1 that switches s_1 on over $[11/16,13/16]$, after it has stopped Chapter6/ProperHomotopy2021 · homWeightZero Chapter6/ProperHomotopy2021 · homWeightOne Chapter6/ProperHomotopy2021 · homInterp, and p from the relative Theorem 5.3.10 (preprint 5.18) on $(1/8,7/8)\times\chi^{-1}(W)$, as in Step 4. The zeros lie over the projection of $\mathrm{cl}\{w\in W \mid t\in[0,1],\ N(F(w))\le 2\}$, which is compact. □ The slab neighbourhood and the supports of s_0, s_1 in it are hypotheses of this step only; they can be met inside any given neighbourhoods of S_{f_0} and S_{f_1} Chapter6/ProperHomotopy2021 · exists_homotopy_cobordism_data, so nothing is added to the theorem.

Step 6: the cobordism and its ends (V6.76). Let $F=\tilde{f}_0+\hat{S}$ for cobordism data. The index-$(k+1)$ part S_F^{k+1} of its solution set Chapter6/InvariantCobordism2021 · cobPart is, by Theorem 6.6.18 (preprint 6.41) for an open part over the tame $\mathbb{R}^1\times X$, a $(k+1)$-dimensional submanifold with a corner structure Chapter6/InvariantCobordism2021 · isSubmanifoldOfDim_cobPart Chapter6/InvariantCobordism2021 · hasCornerStructureOfDim_cobPart, σ-compact Chapter6/InvariantCobordism2021 · sigmaCompactSpace_cobPart, with the smooth time function t Chapter6/InvariantCobordism2021 · cobTime whose slabs $\{t\in[c,d]\}$ are compact Chapter6/InvariantCobordism2021 · isCompact_cobSlab. *The ends.* Over $t\le 1/8$ the solution set of F is $\{t\le 1/8\}\times S_{f+\sigma_0}$, and over $t\ge 7/8$ it is $\{t\ge 7/8\}\times S_{f+\sigma_1}$ Chapter6/InvariantProductEnds2021 · solutionSet_left_eq_prod Chapter6/InvariantProductEnds2021 · solutionSet_right_eq_prod, since F is the t-constant lift there. *The index on the ends*: at a zero (t,x) where F is the lift of $g=f+\sigma_i$ and in general position, $F'(t,x)$ has index $k+1$ exactly when $g'(x)$ has index k Chapter6/InvariantProductEnds2021b · isLinOfIndex_end_iff: the lifted linearization is $(\sigma,v)\mapsto g'(x)v$, with one more kernel dimension, and the index of F' is unique. So S_F^{k+1} over $(0,1/8)$ is $(0,1/8)\times S_k(f+\sigma_0)$, and over $(7/8,1)$ it is $(7/8,1)\times S_k(f+\sigma_1)$ Chapter6/InvariantCobordism2021 · cobPart_end_left Chapter6/InvariantCobordism2021 · cobPart_end_right Chapter6/InvariantCobordism2021 · cobEndLeft Chapter6/InvariantCobordism2021 · cobEndRight. *As manifolds*: a *product end* of a $(k+1)$-manifold M over a k-manifold N is an open part identified with $(a,b)\times N$ such that the product charts $(t,y)\mapsto(t-a,e(y))$ belong to the maximal atlas of M Chapter6/InvariantProductEnds2021 · ProductEnd. The corner coordinates of $(a,b)\times A$ are the products of those of A with the time coordinate Chapter6/InvariantProductEnds2021b · MPolyfold.CornerCoords.prodRn Chapter6/InvariantProductEnds2021b · isChart_prodEndChart, and they agree with the charts of the identification Chapter6/InvariantProductEnds2021b · endChart_eqOnSource, so the two ends are product ends Chapter6/InvariantProductEnds2021b · productEndOfSub Chapter6/InvariantProductEnds2021b · solProductEnd. *Fubini on an end*: if M carries the product orientation $e\wedge o_N$ on the end, then for h supported in (a,b), $\int_M h(t)\,dt\wedge p_1^*\alpha = (\int_a^b h)\int_N\alpha$ Chapter6/InvariantProductEnds2021 · ProductEnd.IsEndProductOrientation Chapter6/InvariantProductEnds2021 · ProductEnd.integral_end_fubini.

Step 7: the orientation of S_F (V6.76). The text orients S_F "from the orientation o of f and the standard orientation of $[0,1]$". This is Remark 6.6.19 (preprint 6.42) as an orientation of F (§VI.26): $e\wedge o_0$ is an orientation of $F=\tilde{f}_0+\hat{S}$, and Theorem 6.6.18 (preprint 6.41) for an open part turns

it into a continuous tangent orientation of S_F^{k+1} Chapter6/CobordismOrientation2021 · cobOr Chapter6/CobordismOrientation2021 · isContinuous_cobOr. What the comparison needs is its value on the ends: *on* $(0,1/8)\times S_k(f+\sigma_0)$ *and* $(7/8,1)\times S_k(f+\sigma_1)$ *it is the product orientation* $e\wedge o_S$*, time first, with no sign* Chapter6/CobordismOrientationEnds2021d · isEndProductOrientation_cobOr — the step the text leaves to "taking the orientations into account". *Proof.* Let (t,x) be a point of an end, $g=f+\sigma_i$. (a) *The linearization.* F and the t-constant lift $\tilde g$ agree near (t,x), and two sc-Fredholm sections that agree near a common zero have the same linearization there Chapter6/CobordismOrientationEnds2021c · ChartPoint.linZero_congr_of_eventuallyEq; that of $\tilde g$ in the product chart point is $\hat L(\sigma,v)=g'(x)v$ Chapter6/CobordismOrientationEnds2021c · ChartPoint.linZero_prodPt_paramLift, so $\ker F'(t,x)=\mathbb{R}\times\ker g'(x)$. (b) *The orientation of the kernel.* For a surjective T and $\hat T=\Gamma^{-1}(T\oplus 0)A$, the kernel ray of the 2021 image $e\wedge r$ of a ray r of $\det T$ is $\hat e\wedge J(\cdot)$ of the kernel ray of r Chapter6/CobordismOrientationEnds2021b · surjRay_extConjDetFirst (the stabilization by $\mathbb{R}^0$ of §VI.25 with one more vector in front); in coordinates $\hat\iota(s,w)=(s,\iota w)$ of the kernels, the pulled-back orientation is the standard one of $\mathbb{R}^{k+1}$ iff the one of $\ker T$ is the standard one of $\mathbb{R}^k$ Chapter6/CobordismOrientationEnds2021b · pullOr_surjRay_extConjDetFirst_iff Chapter6/CobordismOrientationEnds2021 · pullOr_wedgeExtLeft_eq_std_iff, the time coordinate being first on both sides. In chart points this is Chapter6/CobordismOrientationEnds2021c · ChartPoint.pullOr_tangentOrOf_prodOrFirst₀_iff, and the tangent orientation of $e\wedge o$ in a product chart is that of the product orientation of o read in the chart of X Chapter6/CobordismOrientationEnds2021c · tangentOr_toChartOr_prodFirst. (c) *The tangent map.* In the corner coordinates of the product end, the tangent map of $S_F\hookrightarrow\mathbb{R}^1\times X$ is $(s,w)\mapsto(s,T\iota(w))$, $T\iota$ that of $S\hookrightarrow X$ Chapter6/CobordismOrientationEnds2021 · MPolyfold.derivIn_prodRn Chapter6/CobordismOrientationEnds2021c · modelDeriv_prodRn. (d) By (a)–(c) the readings of the orientation of S_F in the product coordinates and of o_S in the coordinates of S correspond Chapter6/CobordismOrientationEnds2021d · solReadOrOpen_prodFirst_iff, which is the product form on the end. □ The orientation on the end is that of $S_k(g)$ for the orientation of $g=f_0+\sigma_i$ carried from o_0. *Matching the ends.* In part (1), $o_1=o_0$ and both ends carry o_0 carried to $f+s_i$. In part (2) the right end carries o_0 carried to $f_0+S(1,\cdot)+s_1=f_1+s_1$, while $\Psi_{s_1}(f_1,o_1)$ uses $o_1=o_0$ carried to f_1, then carried to f_1+s_1; the two have the same values at every point of every chart Chapter6/CobordismOrientationEnds2021d · IsInducedSectionOrientation Chapter6/CobordismOrientationEnds2021d · IsInducedSectionOrientation.sectionAdd, because $\mathrm{Lin}(g+s,x)=\mathrm{Lin}(g,x)$ (§VI.25). Hence the orientation of the cobordism is of product form on both ends, for both parts Chapter6/CobordismOrientationEnds2021d · cobordismOrientation_of_induced Chapter6/CobordismOrientation2021 · cobordismOrientation_of_ends Chapter6/CobordismOrientationEnds2021d · cobordismOrientationHyp (Findings, Chapter 6, item 134).

Step 8: the relative form on the cobordism (V6.76). Fix a closed relative form (ω,τ) of degree k and integrands $a_i=\omega|_{S_i}+d\gamma_i$ on the ends $S_i=S_k(f+\sigma_i)$ (Step 3). The text's form on S_F is $p_1^*\omega-d\beta$ with $j^*\beta=p_1^*\tau$ on ∂S_F. *Why* $p_1^*\tau$ *lives on the faces of* S_F, which the text does not say: the degeneracy index of S_F is that of $\mathbb{R}^1\times X$ (Theorem 4.2.11 (2)) Chapter6/CobordismRelativeForms2021b · degIndex_cobPart, and that of $\mathbb{R}^1\times X$ at (t,x) is $d_X(x)$ Chapter4/BoundaryFormsPullback2021 · MPolyfold.CornerCoords.zero_coord_face_along, so p_1 maps the faces of S_F into faces of X and τ pulls back along it Chapter4/BoundaryFormsPullback2021 · MPolyfold.faceAlongData Chapter6/CobordismRelativeForms2021b · MPolyfold.pullOver Chapter6/CobordismRelativeForms2021b · MPolyfold.faceRestriction_pullOver (item 136 (1)). *The form* Chapter6/CobordismRelativeForms2021b · exists_closed_form_of_ends: *on a part* A' *of* $\mathbb{R}^1\times X$ *which is* $T_0\times A_0$ *over the open time set* T_0 *and* $T_1\times A_1$ *over* T_1*, with compact slabs, and cut-offs* ρ_0*,* ρ_1 *in time supported over* T_0*,* T_1 *with* $\rho_0+\rho_1=1$ *outside a compact interval, there is a closed* C^∞ *form* $A=p_1^*\omega|_{A'}+d\gamma$ *with vanishing face restriction which reads as the pull-back of* $\omega|_{A_i}+d\gamma_i$ *where* $\rho_i=1$. Here γ extends $-p_1^*\tau$ on the

faces and is $p_1^*\gamma_i$ where $\rho_i = 1$: $\gamma = \rho_0\, p_1^*\gamma_0 + \rho_1\, p_1^*\gamma_1 + (1-\rho_0-\rho_1)\gamma_{\text{mid}}$, γ_{mid} an extension of the face data cut off in time Chapter6/CobordismRelativeForms2021 · exists_glued_extension. A is closed since $dp_1^*\omega = p_1^*d\omega = 0$, and its face restriction is $p_1^*j^*\omega + d(-p_1^*\tau) = p_1^*(j^*\omega - d\tau) = 0$. $p_1^*\omega$ and $p_1^*\gamma_i$ read as pull-backs on the product ends Chapter6/InvariantProductEnds2021b · MPolyfold.isEndPullback_formsInftyRestrict_snd Chapter6/CobordismRelativeForms2021 · ProductEnd.isEndPullback_of_pullAt, and so does $dp_1^*\gamma_i$ Chapter6/CobordismRelativeForms2021 · ProductEnd.IsEndPullback.extDeriv. *The inner ends* (item 136 (2)). The gluing needs room in time on both sides of the region where ρ_i falls from 1 to 0, inside the product end. The cobordism data fix the slices only for $t \le 1/8$ and $t \ge 7/8$, so $\rho_0 = 1 - \text{cutRise}(1/16, 1/8)$ and $\rho_1 = \text{cutRise}(7/8, 15/16)$ are used, and A reads as $p_1^*a_i$ on the *inner ends* $(0, 1/16) \times S_0$ and $(15/16, 1) \times S_1$ Chapter6/CobordismRelativeForms2021b · cobordismRelativeFormsInner_succ Chapter6/CobordismRelativeForms2021b · cobEndLeftIn Chapter6/CobordismRelativeForms2021b · cobEndRightIn; the orientation is still of product form there Chapter6/CobordismRelativeForms2021 · ProductEnd.IsEndProductOrientation.of_subEnd. In degree 0, $A = p_1^*\omega|_{S_F}$, a closed function on the 1-dimensional S_F^1 vanishing at its boundary points Chapter6/CobordismRelativeForms2021b · cobordismRelativeFormsInner_zero. So the relative form exists for all cobordism data Chapter6/CobordismRelativeForms2021b · cobordismRelativeFormsInner. The statement on the full ends Chapter6/InvariantCobordism2021 · CobordismRelativeForms would need a smooth extension across $t = 1/8$, and is not used.

Step 9: the cut-off identity (V6.76) Chapter6/InvariantCobordism2021 · integral_eq_of_cutCobordism. In place of the decomposition $\partial S_F = S_0 \cup S_1 \cup \tilde{S}$, which would need face integrals over $\tilde{S} \subseteq \partial X \times (0,1)$ and a Fubini theorem for face forms, the proof cuts off in time (item 133). *Let the $(k+1)$-manifold M, with a continuous orientation and a smooth time function t with compact slabs, have cut ends over N_0 on $t^{-1}(a_0, b_0)$ and over N_1 on $t^{-1}(a_1, b_1)$ — product ends with the product orientation, on which the closed form A with vanishing face restriction reads as $p_1^*\alpha_i$* Chapter6/InvariantCobordism2021 · ProductEnd.IsCutEnd Chapter6/InvariantCobordism2021 · HasCutEnd *— or an empty N_i and no points over (a_i, b_i). Then $\int_{N_0}\alpha_0 = \int_{N_1}\alpha_1$. Proof.* Let $g_i(t) = \text{cutRise}(a_i, b_i)$, 0 before the first third of (a_i, b_i) and 1 after the second, so that g_i' is supported in (a_i, b_i) with $\int g_i' = 1$ Chapter6/InvariantCobordism2021 · cutRise Chapter6/InvariantCobordism2021 · integral_deriv_cutRise, and $g = g_0 - g_1$. gA is supported over the compact slab $[a_0, b_1]$, so Stokes' theorem applies: $\int_M d(gA) = \int_{\partial M} gA = 0$, the face restriction of A being 0 Chapter6/InvariantCobordism2021 · integral_extDeriv_smulFun_eq_zero_of_faces. Since $dA = 0$, $d(g_iA) = g_i'(t)\,dt \wedge A$ is supported in the i-th end Chapter6/InvariantCobordism2021 · tsupport_extDeriv_smulFun_subset, where it is $g_i'(t)\,dt \wedge p_1^*\alpha_i$, and by Fubini on the end $\int_M d(g_iA) = \int_{N_i}\alpha_i$ Chapter6/InvariantCobordism2021 · integral_extDeriv_smulFun_cutRise. Subtracting, $\int_{N_0}\alpha_0 - \int_{N_1}\alpha_1 = \int_M d(gA) = 0$. □ *Applied to S_F^{k+1}* with the inner ends Chapter6/CobordismRelativeForms2021b · hasCutEnd_leftIn, the form A of Step 8 and $\alpha_i = a_i$: by Step 3,

$$\Psi_{s_0}(f_0, o_0)([\omega, \tau]) = \int_{S_0} a_0 = \int_{S_1} a_1 = \Psi_{s_1}(f_1, o_1)([\omega, \tau])$$

Chapter6/CobordismRelativeForms2021b · psiAt_eq_of_cobordism_of_orientation Chapter6/InvariantCobordism2021 · psiAt_eq_of_cobordism. This is the text's identity $\int_{S_0}\omega - \int_{\partial S_0}\tau = \int_{S_1}\omega - \int_{\partial S_1}\tau$.

Step 10: the assembly (V6.76). (A) *For U with the margin and s_0, s_1 admissible for (N, U), $\Psi_{s_0} = \Psi_{s_1}$* Chapter6/Theorem671_2021 · PsiIndepAdmissible Chapter6/CobordismRelativeForms2021b · psiAt_eq_of_admissible_of_orientation: Steps 4 and 9 with $o_1 = o_0$. (B) *For a proper oriented homotopy and s_i admissible for f_i, $\Psi_{s_0}(f_0, o_0) = \Psi_{s_1}(f_1, o_1)$* Chapter6/Theorem671_2021 · PsiIndepHomotopy Chapter6/CobordismRelativeForms2021b · psiAt_eq_of_homotopy_of_orientation: Steps 5 and 9. *The map.* For each (f, o) choose a norm, a neighbourhood with the margin and an admissible section Chapter6/Theorem671_

2021 · AdmissibleData Chapter6/Theorem671_2021 · nonempty_admissibleData, and let $\Psi_{(f,o)}$ be Ψ_s for the chosen s Chapter6/Theorem671_2021 · psiInvariant. *(1)*: for any N, U with the margin and s admissible for (N,U), take s_2 admissible for both (N,U) and the chosen pair (Step 1); (A) for the chosen pair, Ψ_{s_2} not depending on the pair, and (A) for (N,U) give $\Psi_{(f,o)} = \Psi_s$ Chapter6/Theorem671_2021 · psiInvariant_eq_psiAt Chapter6/Theorem671_2021 · psiInvariant_spec. *(2)*: by (1) at both ends and (B) for sections admissible inside the neighbourhoods of Step 5 Chapter6/Theorem671_2021 · psiInvariant_homotopy. So the theorem follows from (A) and (B) alone Chapter6/Theorem671_2021 · theorem_6_7_1_of_indep, (A) and (B) follow from the orientation of the cobordism Chapter6/CobordismRelativeForms2021b · theorem_6_7_1_of_orientation, and that is Step 7 Chapter6/Theorem671Final2021 · theorem_6_7_1. The earlier form of the theorem with the two cobordism statements as hypotheses Chapter6/Theorem671Assembly2021 · theorem_6_7_1_hyp Chapter6/Theorem671Assembly2021 · CobordismOrientationHyp is kept; both hypotheses are now discharged, the second by its inner-end form. (Gap G.54 closed.)

The case $\partial X = \emptyset$ (V6.77) Chapter6/Theorem671Final2021 · theorem_6_7_1_noBoundary. *For X without boundary there is Φ assigning to every proper oriented (f,o) linear maps $\Phi_{(f,o)} : H^k_{dR}(X) \to \mathbb{R}$ such that, for every admissible s and n the index if it is constant, $S_{f+s} = S_n$, $\Phi([\omega]) = \int_{S_{f+s}} \omega$ in degree n and $\Phi = 0$ in the other degrees, invariant along proper oriented homotopies.* Without boundary the face forms vanish and $H^k_{dR}(X,\partial X) \cong H^k_{dR}(X)$ Chapter4/BoundaryForms2021 · MPolyfold.relDeRhamEquivOfNoBoundary, the boundary integral is 0 Chapter6/InvariantSingle2021 · psiAt_mk_of_isWithoutBoundary, and Φ is Ψ composed with this isomorphism Chapter6/Theorem671_2021 · psiInvariantNoBoundary Chapter6/Theorem671_2021 · psiInvariantNoBoundary_mk; off degree n, $S_k = \emptyset$ and $\Phi = 0$ Chapter6/Theorem671_2021 · psiInvariantNoBoundary_of_ne Chapter6/Theorem671_2021 · theorem_6_7_1_noBoundary_of_indep. $H^n_{dR}(X)$ is read as the cohomology of sc-forms of Part IV (Chapter 4, item 173).

Not vacuous Chapter6/Theorem671_2021 · theorem_6_7_1_witness. On the point $X = \mathbb{R}^0$ with the zero bundle and $f = 0$ the hypotheses hold, f is a proper tame sc-Fredholm section with an orientation, and on the class of the constant 0-form 1, $\Psi_s([1]) = \pm 1$ for every admissible s: S_{f+s} is the point, of index 0, counted with its sign. Every Ψ with property (1) agrees with Ψ_s, so it is not 0 either.

Readings and what is left. The hypotheses are the text's: X tame with sc-smooth bump functions (the first sentence of the theorem); "sc-Fredholm" is the tame form of Definition 3.6.4 (preprint 3.51); "proper" is Definition 5.2.1 (3); "oriented" is Definition 6.6.16 (preprint 6.40). The readings are R-a to R-f above. Nothing of the proof is assumed. What the formalization adds to the text: the margin of U and its use for $C = 2$; the relative Theorem 5.3.10 (preprint 5.18), for sets of zeros that are not compact; the index parts; the cut-off in place of the decomposition of ∂S_F; the inner ends; the reason why p_1 maps faces to faces; and the computation that $e \wedge o$ is the product orientation on the ends, with no sign (Findings, Chapter 6, items 100, 101, 103, 128–134, 136) (pp. 276–279 (preprint pp. 230–233)).

VI.28 Appendix §6.8.1 (preprint §6.7.1): the proof of Lemma 6.3.3 (preprint 6.5)

The text's proof (preprint pp. 234–235) has three steps: a change of the basis of the complement of $\alpha(A)$ multiplies both wedges of (91) by its determinant; a change of the bases of A and D multiplies the source by $\det(\tau)/\det(\varepsilon)$; and a change of complements adds to the basis of the complement of $\alpha(A)$ vectors of $\alpha(A)$, and to the γ-preimages vectors of $\beta(B)$, which leaves the wedges unchanged.

The formalized proof (V6.63) does the three steps at once. For two choices χ, χ' there is a scalar μ with $\text{source}(\chi') = \mu\,\text{source}(\chi)$ and $\text{value}(\chi') = \mu\,\text{value}(\chi)$ Chapter6/ExactSequence · FourTermExact.Choice.

`exists_smul_source_value`. *Step 1.* Project the basis b'' of the new complement Z' onto Z along $\alpha(A)$; the projections form a basis of Z, related to b' by some σ, and since $\beta \circ \alpha = 0$ they have the same β-images as b''. *Step 2.* In B, $\alpha(a') \wedge b'' = \alpha(a') \wedge p(b'')$, adding vectors of the span of the first block; with $a' = \tau a$ this is $\det\tau \, \det\sigma \; \alpha(a) \wedge b'$. *Step 3.* In C, the γ-preimages c' of $d' = \varepsilon d$ in V' differ from the γ-preimages in V by vectors of $\ker\gamma = \beta(B)$, the span of the second block, so the wedge of the basis of C `Chapter6/ExactSequence · FourTermExact.Choice.wedgeC` is $\det\varepsilon \, \det\sigma$ times that of χ. *Step 4.* So the value scales by $\det\tau \, \det\sigma/(\det\varepsilon \, \det\sigma) = \det\tau/\det\varepsilon$, as the source does, and Lemma 6.3.3 (preprint 6.5) follows from the fixed choice that defines Φ_E. The step "independent of the choices of the basis if and only if $\det(\tau) = \det(\varepsilon)$" is read as this homogeneity of (91) in the bases of A and D, τ being the σ just introduced; the corrections p_i lie in $\beta(B)$, written $\beta(C)$, and are combinations of the β-images, whose names are exchanged with those of the γ-preimages in the last step (possible misprints; Findings, Chapter 6, item 1).

VI.29 Appendix §6.8.2 (preprint §6.7.2): the proof of Proposition 6.4.11 (preprint 6.12)

(V6.64.) Let $P \le Q$ in Π_T. *Step 1, adapted bases.* Take a basis a of $\ker T$ and classes $[d_i]$ forming a basis of $\operatorname{coker} T$ with $d_i \in \ker Q$, possible because $R(QT) = R(Q)$; complete a by b to a basis of $\ker(QT)$, and (a,b) by $\tilde{b}$ to a basis of $\ker(PQT) = \ker(PT)$. *Step 2, the three values.* By (91) with these choices (Lemma 6.3.3 (preprint 6.5), §VI.3), $\gamma_T^Q(h_0) = (a \wedge b) \otimes ([d]_Q \wedge [Tb]_Q)^*$; $\gamma_{QT}^P \gamma_T^Q(h_0) = (a \wedge b \wedge \tilde{b}) \otimes ([d]_P \wedge [Tb]_P \wedge [QT\tilde{b}]_P)^*$, the classes $[d]_P$, $[Tb]_P$ being preimages of $[d]_Q$, $[Tb]_Q$ because $Pd_i = PQd_i = 0$ and $PTb \in R(P) \subseteq R(Q)$; and $\gamma_T^P(h_0) = (a \wedge b \wedge \tilde{b}) \otimes ([d]_P \wedge [Tb]_P \wedge [T\tilde{b}]_P)^*$ `Chapter6/ProjectionSequence · gammaProj_wedge_tmul`. *Step 3, the spanning.* $[T\tilde{b}]_P - [QT\tilde{b}]_P = [(I-Q)T\tilde{b}]_P$, and $(I-Q)T\tilde{b} \in \ker Q$. If $Qg = 0$, then $g = \sum c_i d_i + Tx$ with $x \in \ker(QT)$ — write $[g] = \sum c_i [d_i]$ in $\operatorname{coker} T$ and use $Qd_i = 0$ — so the classes of the d_i and of the Tb_j span what the text calls $\ker \tilde{Q}$ `Chapter6/ProjectionSequence · mem_span_append_of_ker`. *Step 4.* The last block of the third value differs from that of the second by vectors of the span of the first two blocks, which does not change the wedge (§VI.2), and with the identification of $\det(P(QT))$ with $\det(PT)$ the two values agree `Chapter6/ProjectionSequence · comp_comp_eq_of_projLE`. The spanning is proved directly in F; the name α_q in the last part is read as α^P (possible misprint; Findings, Chapter 6, items 6, 7) (pp. 280–286 (preprint pp. 235–237)).

VI.30 What is not formalized

Theorem 6.7.1 (preprint 6.43), the admissible sections, the orientation of the solution set and of $\hat{f}$, Lemma 6.5.3 (preprint 6.19), Lemmas 6.6.4 and 6.6.5 (preprint 6.29 and 6.30) as statements about manifolds, the remarks of §§6.3–6.6 and after Theorem 6.6.17, and the facts on paths are now formalized, and so is Proposition 6.6.3 (1) in its 2021 wording (gaps G.54–G.64, G.66–G.69, G.81, G.90, G.91 closed). What remains:

- **Proposition 6.6.3 (1) in the preprint's wording** (gap G.61 is closed): "U is sc-smoothly contractible". As we read it, the proof does not establish it for its U: it needs $r(V') \subseteq V'$, which is not stated. No counterexample is known to us. The 2021 wording, a smaller U' contractible in U, is proved, and the proofs use (1′) (§§VI.10, VI.18, VI.19, VI.23).
- **Paths and families on** $[0,1]$ (gap G.65, in part): most statements hold on $[0,1]$, $[0,1]^2$, $[0,1]^n \times X$ as printed (§§VI.36, VI.37), and so does Proposition 6.6.3 (1′) for paths on $[0,1]$ (§VI.10). For $n = 1$

the bundle L of Lemma 6.6.5 is a smooth vector bundle over the manifold with boundary $[0,1]$ itself, the pull-back along the inclusion `Chapter6/KernelBundleInterval · lemma_6_6_5_smoothBundle_interval` `Chapter6/KernelBundleInterval · SmoothSubspaceFamily.comp` `Chapter6/KernelBundleInterval · contMDiff_unitIntervalToOpens`. Still read on $\mathbb{R}$ (decision D2): L over a neighbourhood of $[0,1]^n$ for $n \geq 2$ (V6.34; Mathlib has no manifold structure on the cube, estimated at 300–600 lines), the stabilized kernels and their continuation for families with sections over $[0,1]^n \times X$ (V6.42, 400–700 lines), and Remark 6.6.19 (preprint 6.42) over $\mathbb{R}^1 \times X$ (V6.62, 1,500–2,500 lines), with the cobordisms of Theorem 6.7.1 (preprint 6.43).

- **The narrower rows.** V6.48, Theorem 6.6.11 (preprint 6.35): the section f, which the statement does not introduce, is supplied (presumably an omission), and the neighbourhood is not shown contractible in itself (the preprint's (1)). V6.80, the remark that $\mathrm{DET}(E,F)$ is not orientable: it does not hold as we read it for finite-dimensional E, F (`exists_two_lineOrientation_det_of_finiteDimensional`), and it is proved only for spaces isomorphic to an infinite-dimensional Hilbert space or to $H \times H$ with $H \cong \mathbb{R} \times H$, and under complex structures with an odd-dimensional kernel; infinite-dimensional Banach spaces without such structures are not covered (§VI.34). V6.81, the classical case: C^1 in place of smooth, and the smoothness of the line bundle is not formalized (§VI.34). V6.32, Proposition 6.6.3, is restated.
- **The chart-free objects** T_xX, Y_x, $\mathrm{Lin}(f,x)$, Or_x^f (gaps G.6, G.7, G.13 of Chapter 2): they are made in chart points and identified between charts (§VI.21).

VI.31 What the formalization taught us

The sign of (94). The main change of substance is a sign. The coordinate of Φ_E in a local determinant chart carries the factor $(-1)^{\dim \operatorname{coker} S}$, which jumps where the parity of the cokernel dimension changes; the orientations it carries along a continuous family are then not continuous, as we read it, and an example on $\mathbb{R}$ shows it (§VI.14). The continuity statements of pp. 265–272 (preprint pp. 223–226) are not written in coordinates in the text, and it is in coordinates that the sign appears. The normalized $\alpha(T)\Phi_E$ at both ends of every propagation changes nothing on surjective operators — the tangent orientations of Theorem 6.6.18 (preprint 6.41) are the text's — and the same sign is used in Remark 6.6.19 (preprint 6.42) (§VI.26).

Lemma 6.10 (preprint) and the 2017 remark. The common lower bound of two good left-projections exists, but not with the range $R(P) \cap R(P')$: a projection onto a smaller, finite-codimensional piece of it, along a complement containing both kernels, is below both (§VI.4). This keeps the preprint's construction of $\mathrm{DET}(E,F)$ — one lower bound, Corollary 6.4.12 (preprint 6.13), Lemma 6.5.5 (preprint 6.21) — intact.

Contractibility was never needed. Proposition 6.6.3 (1) (preprint 6.28) in the preprint's wording is not established for the neighbourhood its proof builds, as we read it; the 2021 wording, with a smaller U', holds (§VI.10). Every later use of contractibility is a use of (1′): two paths in U with the same end points are homotopic in a larger region where sections along families exist. Orientable neighbourhoods become pairs $V \subseteq U$, and property ($\square$) of Theorem 6.6.13 (preprint 6.37) holds on every set with the local propagation property (§§VI.10, VI.19, VI.23).

Paths on $\mathbb{R}$ cost reparametrizations. With paths and homotopies on $\mathbb{R}$ whose conditions hold on $[0,1]$ (decision D2), the homotopy of (1′) has the right edges only on $[0,1]$ while push-forward data need zeros near $[0,1]$; transitivity of homotopy, associativity of concatenation and Theorem 6.6.11 (preprint 6.35) all pass through the smooth transition function κ (§§VI.18, VI.20).

Charts must be compared. The text's chart-free Or_x^f becomes a set read in a chosen chart, and the invariance of the local propagation under a change of chart — the naturality of the stabilization isomorphism, of the sign and of continuation — is a file of its own (§VI.21).

"By a covering argument" and "one verifies". The covering argument of Theorem 6.6.13 (preprint 6.37) is a dyadic transport with a square lemma, proved for any family of types (§VI.22); the "simple computation" of Lemma 6.6.8 (preprint 6.33) is an exact identity of stabilization isomorphisms, with no sign, once the appended sections come last (§VI.16); and the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), stated without proof, needs the transport computed on dyadic pieces inside orientable neighbourhoods (§VI.24).

Theorem 6.7.1 is proved as the text proves it, with two changes of route. The invariant is defined from one admissible perturbation, made a map on cohomology by Stokes' theorem, and compared along a cobordism, as in the text. The decomposition $\partial S_F = S_0 \cup S_1 \cup \tilde{S}$ is replaced by a cut-off in time: $\int d(gA) = 0$ for a closed form A with vanishing face restriction, while $d(g_i A)$ lives on the product ends, where Fubini gives the integrals over S_0 and S_1. This avoids face integrals over $\tilde{S} \subseteq \partial X \times (0,1)$ and the signs of the pieces of ∂S_F. And the relative form is built on inner ends, because gluing $p_1^*\gamma_i$ to an extension needs room in time inside the product end (§VI.27).

The extension $\bar{\omega}$ has one degree too many, as we read it. The proof extends ω to $X \times [0,1]$ by $p_1^*\omega \wedge p_2^* dt$, a form of degree $k+1$ on the $(k+1)$-dimensional S_F, whose integrals over the k-dimensional pieces of ∂S_F vanish. The comparison goes through with $p_1^*\omega$ and $p_1^*\tau$ (§VI.27, reading R-e).

The orientation of the cobordism is a computation, and has no sign. "Taking the orientations into account" is: the linearization of F on an end is that of a t-constant lift, its kernel is $\mathbb{R} \times \ker$, the 2021 map puts e in front of the kernel orientation, and the tangent map of the end is $(s,w) \mapsto (s, T\iota(w))$; with the time vector first on both sides, $e \wedge o$ is the product orientation $e \wedge o_{S_i}$ (§VI.27). The 2021 convention $\tilde{h} = e \wedge h$ also removes the sign that the preprint's $h \mapsto \tilde{h}$ needed: it differs from the continuous map by $(-1)^{\mathrm{ind}\, L}$, which is locally constant (§VI.26).

Hidden data of the theorem. The neighbourhood U needs a margin, $\mathrm{cl}\{N(f) \leq 2\}$ compact, for the cobordism between two admissible sections; Theorem 5.3.10 (preprint 5.18) is needed relative to a compact part of a zero set that need not be compact; the degree of a form is paired with the part of S_{f+s} of that index, since the index need not be constant; the proper oriented homotopy, which the text does not define, is read through Remark 6.6.19 (preprint 6.42); and p_1 maps the faces of S_F to faces of X because the degeneracy index of $\mathbb{R}^1 \times X$ is that of X (§VI.27).

$\mathrm{DET}(E,F)$ **and dimension.** The remark that $\mathrm{DET}(E,F)$ is not orientable does not hold as we read it for finite-dimensional E, F: there $L(E,F)$ is convex and Proposition 6.6.1 (preprint 6.26) gives two orientations (`exists_two_lineOrientation_det_of_finiteDimensional`). It holds for infinite-dimensional Hilbert spaces, through a loop of surjective operators whose kernels form a Möbius band; in finite dimension the cokernels form a second Möbius band and the two cancel in det (§VI.34).

Orientations of manifolds with corners. Mathlib has none. An oriented atlas, the usual definition, does not fit the quadrant model: $[0,1]$ has no atlas of quadrant charts with positive changes, and in dimension 0 every atlas is oriented. A continuous tangent orientation does, and it is the form in which Theorem 6.6.18 (preprint 6.41), Stokes' theorem and the cobordism argument use orientations (§§VI.25, VI.27).

(P) follows from the general position of F. The text derives the general position of F along the family from property (P) of the slices. The formalization goes the other way: general position of F is proved by

the method of Theorem 5.3.10 (preprint 5.18), and (P) follows from it by a chart computation, since the tangent vectors of the family of zeros lie in $\ker F'$ and have reduced X-components; on the cube $[0,1]^n$ the two are equivalent (§VI.36).

VI.32 The Springer 2021 edition: Lemma 6.4.4, Proposition 6.4.13, Lemmas 6.6.8 and 6.6.9

The 2021 edition adds index equalities to Lemma 6.4.4 (2017: 6.3), makes the hypothesis of Proposition 6.4.13 (2017: 6.6) explicit, and splits preprint Lemma 6.33 (2017: 6.15) into Lemmas 6.6.8 and 6.6.9. Neither Lemma 6.4.4 nor Proposition 6.4.13 has a preprint counterpart. The file is `Chapter6/Book2021.lean` (Findings, Chapter 6, items 110–115).

Lemma 6.4.4 (2017: 6.3) `Chapter6/Book2021 · lemma_6_4_4`. *Statement.* Let E, F be Banach spaces, $T_0 : E \to F$ Fredholm and $P_0 \in \Pi_{T_0}$ `Chapter6/GoodProjections · IsGoodLeftProj`. There is $\varepsilon > 0$ such that for every T and every projection P with $\|T - T_0\| < \varepsilon$, $\|P - P_0\| < \varepsilon$: T is Fredholm, $P \in \Pi_T$, and $\operatorname{ind}(P \circ T) = \operatorname{ind}(T) = \operatorname{ind}(T_0) = \operatorname{ind}(P_0 \circ T_0)$, all indices of maps $E \to F$ `Chapter1/FredholmIndex · fredholmIndex`. *Proof.* (a) With a quasi-inverse v of T_0, every T with $\|T - T_0\| < 1/(\|v\| + 1)$ is Fredholm of the index of T_0 `Chapter3/SolutionLinearization · isFredholm_add_of_norm_lt` `Chapter1/FredholmIndex · fredholmIndex_add_of_norm_lt`. (b) Instead of the text's map $\Phi_{(P,T)}(x,h) = PTx + (I-P)h$, take the intertwiner $J(P) = PP_0 + (1-P)(1-P_0)$ `Chapter6/Book2021 · projIntertwiner`. It is continuous in P `Chapter6/Book2021 · continuous_projIntertwiner`, $J(P_0) = 1$ `Chapter6/Book2021 · projIntertwiner_self`, so $J(P)$ is a unit for P near P_0; and $J(P)P_0 = PJ(P)$ for every projection P `Chapter6/Book2021 · projIntertwiner_mul`, so $P = JP_0J^{-1}$. (c) $J^{-1}T$ is near T_0, so by Lemma 6.5.1 (preprint 6.17) $P_0 \in \Pi_{J^{-1}T}$ `Chapter6/LocalDetBundle · lemma_6_5_1`. Then $R(P) = JR(P_0)$ has finite codimension, $(1-P)F$ complements it, and $R(PT) = JR(P_0J^{-1}T) = JR(P_0) = R(P)$: $P \in \Pi_T$. (d) The indices of $P \circ T$ and $P_0 \circ T_0$ are those of T and T_0 `Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_comp`. □ As a map onto $R(P)$ the operator $P \circ T$ would have index $\dim\ker(PT)$; the text's equalities hold for the map $E \to F$. E and F are taken in the universe `Type`, that of (a).

A common complement `Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq`. *Statement.* Two finite-dimensional subspaces K_1, K_2 of the same dimension of a real vector space have a common algebraic complement. *Proof.* Induction on $n = \dim(K_1 + K_2) - \dim K_1$ `Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq_aux`. For $n = 0$, $K_1 = K_2$. Otherwise neither contains the other; take $a \in K_2 \setminus K_1$, $b \in K_1 \setminus K_2$. Then $a+b$ lies in neither, $K_i + \mathbb{R}(a+b)$ has dimension $\dim K_i + 1$ `Chapter6/Book2021 · finrank_sup_span_singleton_of_notMem` and their sum is still $K_1 + K_2$, so the induction gives a common complement X' of $K_1 + \mathbb{R}(a+b)$ and $K_2 + \mathbb{R}(a+b)$, and $X' + \mathbb{R}(a+b)$ complements K_1 and K_2 `Chapter6/Book2021 · isCompl_sup_span_singleton`. □ The text takes X closed; that is not needed for the algebraic part below, and a closed one is §VI.33.

Proposition 6.4.13 (2017: 6.6), the algebraic part `Chapter6/Book2021 · proposition_6_4_13`. *Statement.* Let T be Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$ (the 2021 hypothesis $\Pi_T \cap \mathcal{P}_H \neq \emptyset$ is the existence of such projections). There are linear isomorphisms $A_T : \ker(PT) \to \ker(QT)$, $B_T : F/R(P) \to F/R(Q)$ with $A_T \circ j_P = j_Q$, $\Phi_Q \circ A_T = B_T \circ \Phi_P$ and $\pi_Q \circ B_T = \pi_P$, for the maps of $E(T,P)$, $E(T,Q)$ (§VI.5) `Chapter6/ProjectionSequence · projSeqJ` `Chapter6/ProjectionSequence · projSeqPhi` `Chapter6/ProjectionSequence · projSeqPi`. *Proof (the text's).* PT and QT are Fredholm `Chapter6/GoodProjections · IsGoodLeftProj.isFredholm_comp` of index $\operatorname{ind}(T)$, and $F/R(P) = F/R(Q)$, so $\dim\ker(PT) = \dim\ker(QT)$; let X be a common complement. QT maps X isomorphically onto

$R(Q)$; for $e \in \ker(PT)$ let $x(e) \in X$ solve $QTx = QTe$ and put $A_T(e) = e - x(e) \in \ker(QT)$. It is injective ($A_T(e) = 0$ puts e in $X \cap \ker(PT) = 0$), the identity on $\ker T$, and onto by the dimensions. For $f \in F$ write $(1-P)f = Tx + \tilde{f}$ with $x \in X$, $\tilde{f} \in (1-Q)F$ (solve $QTx = Q(1-P)f$), and put $B_T(f+H) = \tilde{f} + H$. The three identities are direct computations; B_T is injective by a direct computation, hence an isomorphism, the two quotients having the same finite dimension (the text uses the five lemma). □ The last sentence, that A_T and B_T depend continuously on T, is §VI.33, with a closed X (gap G.81, closed).

Lemmas 6.6.8 and 6.6.9 `Chapter6/Book2021 · lemma_6_6_8` `Chapter6/Book2021 · lemma_6_6_9`. *Statement.* For stabilized kernels of the data $(f, s_1, \dots, s_k)$ along a family `Chapter6/Propagation · IsStabKernels` and of the data with d sections appended, the push-forwards of orientations agree: for zero sections (Lemma 6.6.8, the parameters μ as dummy variables) and for arbitrary sections $\bar{s}_1, \dots, \bar{s}_d$ (Lemma 6.6.9). *Proof.* Both are §VI.16 `Chapter6/Propagation · IsStabKernels.push_append`, Lemma 6.6.8 with $\bar{s}_j = 0$. □ The statement of preprint Lemma 6.33 (2017: 6.15) is Lemma 6.6.9, and the first step of its proof is Lemma 6.6.8; the formalization of §VI.16 never needed that step.

Already covered. The other 2021 changes of the chapter concern statements verified before: Proposition 6.4.15 with its hypotheses made explicit (§VI.6) `Chapter6/Stabilization · proposition_6_4_15`, Proposition 6.6.3 for a tame X and sections valued in $Y \times [0,1]$ (§VI.10, without tameness) `Chapter6/OrientableNbhd · proposition_6_6_3`, and Lemma 6.6.5 with the solution set near $(t, \varphi(t), 0)$ in the points (t, y, λ), whose kernel family (§VI.12) is unaffected `Chapter6/KernelFamily · exists_subspaceFamily_linKer`.

VI.33 The Springer 2021 edition, continued: Proposition 6.4.13 (last sentence) and Theorem 6.6.17

The last sentence of Proposition 6.4.13 (2017: 6.6), the continuity of A_T and B_T in T, is proved with a closed common complement. Theorem 6.6.17 is new in the 2021 edition: a topology on the orientation space O_f of pp. 274–275, whose proof the text leaves to the reader. It is proved with properties (1)–(3); property (2) uses a hypothesis on X, connected smooth neighbourhoods, which every M-polyfold satisfies (gap G.90, closed). The files are `Chapter6/Book2021Continuity.lean` and `Chapter6/OrientationSpace.lean` (Findings, Chapter 6, items 116–121).

A common closed complement `Chapter6/Book2021Continuity · exists_isClosed_isTopCompl_isTopCompl_of_finrank_eq`. *Statement.* Two finite-dimensional subspaces K_1, K_2 of the same dimension of a real normed space E have a common closed complement X: $E = K_1 \oplus X = K_2 \oplus X$ topologically. *Proof.* Let X_0 be a common algebraic complement (§VI.32) `Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq` and Z a closed complement of the finite-dimensional $K_1 + K_2$. Then $X = Z + X_0 \cap (K_1 + K_2)$ complements K_i: inside $K_1 + K_2$, $X_0 \cap (K_1 + K_2)$ complements K_i, and Z complements $K_1 + K_2$ `Chapter6/Book2021Continuity · isCompl_sup_inf_of_isCompl`. X is closed, a closed subspace plus a finite-dimensional one, and an algebraic complement of a finite-dimensional subspace that is closed is a topological one. □

Proposition 6.4.13 (2017: 6.6), the last sentence `Chapter6/Book2021Continuity · proposition_6_4_13_continuity`. *Statement.* Let T be Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$. There are a closed $X \subseteq E$ and $\varepsilon > 0$ such that on $\|S - T\| < \varepsilon$ the maps $S \mapsto A_S = 1 - \xi_S \circ S$ `Chapter6/Book2021Continuity · kerTransition` and $S \mapsto B_S = (1-P) - S\xi_S(1-P)$ `Chapter6/Book2021Continuity · cokerTransition`, where $\xi_S = (QS|_X)^{-1} \circ Q$ `Chapter6/Book2021Continuity · solveOp` `Chapter6/Book2021Continuity · projCod`, are continuous into $L(E)$, $L(F)$ in the operator norm, and jointly with the vector; and for every such S: $P, Q \in \Pi_S$,

$E = \ker(PS) \oplus X = \ker(QS) \oplus X$ topologically, $e - A_S e \in X$ and $QSA_S e = 0$ (6.15)–(6.16) `Chapter6/Book2021Continuity · kerTransition_apply`, $QB_S f = 0$ and $Sx + B_S f = (1-P)f$ for some $x \in X$ (6.18)–(6.19) `Chapter6/Book2021Continuity · cokerTransition_apply Chapter6/Book2021Continuity · solveOp_mem`, and A_S, B_S induce isomorphisms $\ker(PS) \to \ker(QS)$, $F/R(P) \to F/R(Q)$ filling the diagram of $E(S,P)$, $E(S,Q)$ `Chapter6/Book2021Continuity · diagram_of_solveOp`. *Proof.* Take X from the previous statement for $\ker(PT)$, $\ker(QT)$, which have the same dimension (§VI.32). (a) By Lemma 6.5.1 (preprint 6.17) `Chapter6/LocalDetBundle · lemma_6_5_1` there is ε with $P, Q \in \Pi_S$ and $QS|_X : X \to R(Q)$ an isomorphism for $\|S - T\| < \varepsilon$ `Chapter6/LocalDetBundle · projRestrict`. (b) $S \mapsto QS|_X$ is continuous `Chapter6/LocalDetBundle · continuous_projRestrict` and inversion is continuous at isomorphisms, X being complete; so ξ_S is continuous in S `Chapter6/Book2021Continuity · continuousAt_solveOp`, and so are A_S, B_S, which are built from ξ_S, S and P by composition. (c) $QS\xi_S = Q$ `Chapter6/Book2021Continuity · apply_solveOp_of_projRestrict_eq` gives (6.15)–(6.19). (d) $\ker(QS) \cap X = 0$ because $QS|_X$ is injective, and $\ker(QS) + X = E$ because $e - \xi_S S e \in \ker(QS)$ `Chapter6/Book2021Continuity · disjoint_ker_comp_of_projRestrict_eq`; the projection $1 - \xi_S S$ onto $\ker(QS)$ along X is continuous `Chapter6/Book2021Continuity · isTopCompl_ker_comp_of_projRestrict_eq`; the same for P, $R(P) = R(Q)$. (e) The diagram is §VI.32 for S with this X. □ *Satisfiable:* $T = 1$ on $\mathbb{R}$ and $P = Q$ any element of Π_T `Chapter6/Book2021Continuity · proposition_6_4_13_continuity_hyps`. "Depend continuously" is read as the continuity of A_S on E and B_S on F; these determine the maps between the varying kernels $\ker(PS)$, $\ker(QS)$ and on F/H, and the joint continuity is continuity on the kernel bundle $\bigcup_S \{S\} \times \ker(PS)$ of Remark 6.4.5 (gap G.81, closed).

The orientation space `Chapter6/OrientationSpace · OrSpace Chapter6/OrientationSpace · orProj`. *Definition.* O_f is the set of pairs (x, o) with $x \in X_\infty$ and $o \in \mathrm{Or}^f_x$, read in the chosen chart (§VI.21) `Chapter6/SectionOrientation · Orx`, and $\pi_f(x,o) = x$; X_∞ carries the topology of Theorem 2.3.10 (§II.7) `Chapter2/Metrizable · MPolyfold.SmoothLevel`. For an sc-smooth path φ (a map $\mathbb{R} \to X$, decision D2) the pull-back $\varphi^* O_f$ is the subspace of $[0,1] \times O_f$ of the (t,p) with $\varphi(t) = \pi_f(p)$ `Chapter6/OrientationSpace · OrPullback`, and a continuous section over an open U is a continuous $\sigma : U_\infty \to O_f$ with $\pi_f \circ \sigma = \mathrm{id}$ `Chapter6/OrientationSpace · IsContSectionOver`. The three properties of the theorem: (1) π_f is a local homeomorphism; (2) every $x \in X_\infty$ has an open U in X over whose smooth part there are exactly two continuous sections `Chapter6/OrientationSpace · HasTwoContSections`; (3) for every sc-smooth path φ and continuous section $t \mapsto (\varphi(t), o_t)$ of $\varphi^* O_f$, for every t_0 and t near t_0, o_t is the push-forward of o_{t_0} along the segment $\varphi|[t_0, t]$ `Chapter6/OrientationSpace · segPath Chapter6/OrientationSpace · isScSmoothMap_segPath` in a set with the local propagation property `Chapter6/OrientationFunctor · IsPropagationNbhd Chapter6/OrientationFunctor · orPush`, which does not depend on the set `Chapter6/OrientationFunctor · orPush_compat Chapter6/OrientationSpace · IsPropagationContinuous`. An *orientation topology* is a Hausdorff topology with (1)–(3) `Chapter6/OrientationSpace · IsOrientationTopology`. This is our reading of the text's "local propagation of o_{t_0}".

Sc-smooth paths in X_∞ `Chapter6/OrientationSpace · continuous_smoothMk_comp Chapter6/OrientationSpace · eventually_exists_scPath_smoothLevel`. *Statement.* (a) An sc-smooth $\Gamma : \mathbb{R}^n \to X$ is continuous into X_∞. (b) If N is a neighbourhood of x in X_∞, every $y \in X_\infty$ near x is joined to x by an sc-smooth path with all its points in N. *Proof.* (a) Its values are smooth points `Chapter6/ScHomotopy · mem_smoothSet_of_isScSmoothMap`; in a chart, Γ is continuous into every level m `Chapter6/OrientationSpace · continuousOn_levelLift_comp Chapter6/OrientationSpace · rnChart_toBase`, which is continuity into X_∞. (b) In a chart at x, take the retracted segment $t \mapsto r((1-t)u_0 + tu_1)$ `Chapter6/OrientationSpace · chartSegPath`, sc-smooth `Chapter6/OrientationSpace · isScSmoothMap_chartSegPath`; for y near x in X_∞ its points are near x in every level, hence in N. □ The neighbourhood version for open sets of X is `Chapter6/OrientationSpace · exists_nhds_scPath_of_isOpen`, from §VI.18 `Chapter6/ScPaths · exists_nhds_scPath`.

The topology Chapter6/OrientationSpace · orTopology. *Definition.* Fix local transport data L (§VI.22) Chapter6/PathTransport · LocalTransport. A *continuous orientation* over an open U is a family $y \mapsto o_y$ on U_∞ carried along by the transport of L along every sc-smooth path in U Chapter6/OrientationSpace · IsContOrOn; its sheet is $\{(y, o_y) : y \in U_\infty\}$ Chapter6/OrientationSpace · orSec Chapter6/OrientationSpace · orBasic. A set $W \subseteq O_f$ is open if $\{y \in U_\infty : (y, o_y) \in W\}$ is open in X_∞ for every continuous orientation o over an open U. *Properties, for every L.* (a) Two continuous orientations that agree at a point agree near it Chapter6/OrientationSpace · IsContOrOn.eventually_iff; so the sheets are open Chapter6/OrientationSpace · isOpen_orBasic and π_f is continuous Chapter6/OrientationSpace · continuous_orProj. (b) Every point of O_f lies on a sheet Chapter6/OrientationSpace · exists_isContOrOn: push the orientation from x along the paths of §VI.18 in a good set of L. (c) On a sheet π_f is a homeomorphism onto U_∞, with inverse $y \mapsto (y, o_y)$ Chapter6/OrientationSpace · continuousOn_orSec Chapter6/OrientationSpace · eq_orSec_of_mem_orBasic Chapter6/OrientationSpace · orSec_mem_orBasic Chapter6/OrientationSpace · orChartHomeo; with (b), property (1) Chapter6/OrientationSpace · isLocalHomeomorph_orProj. (d) Hausdorff Chapter6/OrientationSpace · t2Space_orTopology: points over different x are separated by π_f, X_∞ being metrizable Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel; the two points over one x lie on two sheets which, by (a), are disjoint over a neighbourhood of x. □ The text gives no construction ("in view of Proposition 6.6.3 and Theorem 6.6.11").

Property (2), under connected smooth neighbourhoods Chapter6/OrientationSpace · hasTwoContSections_orTopology. *Statement.* Suppose every smooth point has, inside every open $W \ni x$, an open $U \ni x$ with U_∞ preconnected in X_∞ Chapter6/OrientationSpace · MPolyfold.HasConnectedSmoothNbhds, and every Or_y^f has two elements. Then the topology has property (2) for every L, the two sections being $y \mapsto (y, \pm o_y)$. *Proof.* Take a sheet of o over $V \ni x$ (property (b)) and such a $U \subseteq V$. Every point over y is (y, o_y) or $(y, -o_y)$ Chapter6/OrientationSpace · eq_orSec_or_eq_orSec, and both sheets are open; so for a continuous section the sets where it equals (y, o_y) and where it equals $(y, -o_y)$ are open and complementary in U_∞, which is preconnected. □ The hypothesis holds for $\mathbb{R}^n$ (convex balls) Chapter6/OrientationSpace · rnPolyfold_hasConnectedSmoothNbhds and for every M-polyfold on a subsingleton Chapter6/OrientationSpace · MPolyfold.hasConnectedSmoothNbhds_of_subsingleton. It holds on every M-polyfold (below; gap G.90 closed).

Theorem 6.6.17 Chapter6/OrientationSpace · theorem_6_6_17_hyp. *Statement.* Let X be tame with bump functions and f an sc-Fredholm section with tame germ data and sc⁺-sections through the smooth fibre points, the hypotheses under which Γ_f is built (§VI.23) Chapter6/OrientationFunctor · theorem_6_6_13_hyp Chapter6/OrientationFunctor · orLocalTransport. If X has connected smooth neighbourhoods, O_f carries exactly one Hausdorff topology with (1), (2), (3). Without that hypothesis, O_f carries exactly one Hausdorff topology with (1) and (3) Chapter6/OrientationSpace · existsUnique_orientationTopology. *Proof.* Existence: the topology for the transport data of Γ_f. It has (1) and is Hausdorff (above), has (2) by the previous statement, every Or_y^f having two elements Chapter6/OrientationFunctor · exists_two_orx, and has (3) Chapter6/OrientationSpace · isPropagationContinuous_orTopology: near t_0 a continuous section of $\varphi^* O_f$ stays in one sheet Chapter6/OrientationSpace · eq_orSec_iff Chapter6/OrientationSpace · eventually_segPath_mem, and a sheet is carried along the segments of φ by the transport. Uniqueness: let t be any topology with (1) and (3) Chapter6/OrientationSpace · eq_orTopology_of. (i) A t-continuous local section s over an open $N \subseteq X_\infty$ that agrees with a continuous orientation o at y_0 agrees with it near y_0 Chapter6/OrientationSpace · eventually_eq_orSec_of_continuousOn: join y near y_0 to y_0 by a path in N (statement (b) above), which is continuous into X_∞ (statement (a)); then s along the path is a continuous section of the pull-back, and by (3) the set of parameters where it equals o is open and closed in $[0, 1]$. (ii) Through every point a local inverse of π_f for t is such a section. By (i) the t-open sets and the open sets of the topology coincide. □ Neither Hausdorffness nor (2) is used in the uniqueness. The text's "uniquely defined" is read as "exactly one".

Property (2) without the hypothesis (V6.74). *Every M-polyfold has connected smooth neighbourhoods* `Chapter6/OrientationMainClean · MPolyfold.hasConnectedSmoothNbhds`: inside an open $W \ni x$, the connected component U of x in W is open, M-polyfolds being locally connected (Proposition 2.3.7), and $U \cap X_\infty$ is path connected in X_∞ `Chapter2/SmoothPaths · MPolyfold.isPathConnected_smoothVal_preimage`, since nearby smooth points are joined by paths in X_∞ `Chapter2/SmoothPaths · MPolyfold.exists_nhds_joinedIn_smooth` and U is connected. So property (2) holds without the hypothesis, and *Theorem 6.6.17 holds with (1), (2), (3) under the text's hypotheses* — X tame with bump functions, f with tame germ data `Chapter6/OrientationMainClean · theorem_6_6_17` (gap G.90 closed; Findings, Chapter 6, items 120, 122). The remarks after the theorem — the orientation bundle, orientability as a global continuous section, the $\mathbb{Z}_2$-action and $\det_f$ — are §VI.38 (gap G.91 closed).

VI.34 §§6.4–6.6 (preprint §§6.3–6.5), remarks: the alternative determinant, the identification (6.25) (preprint (95)), the remark that $\mathrm{DET}(E,F)$ is not orientable, and the classical case

Four remarks of the text, without proofs, are formalized here; the chapter's statements do not use them (Findings, Chapter 6, items 10, 26, 135). The files are `Chapter6/DetConventions.lean`, `Chapter6/DetNonOrientable.lean`, `…Ell2.lean`, `…Transport.lean` and `Chapter6/ClassicalDetBundle.lean`.

The alternative determinant (V6.78; after Definition 6.4.1 (preprint 6.7), p. 246 (preprint p. 208)). *Some authors use* $\lambda(\ker T)\otimes\lambda(\mathrm{coker}(T)^*)$*; it is naturally isomorphic to* $\det(T)$. The map $\iota:\lambda(A^*)\to\lambda^*(A)$ of Definition 6.3.1 (preprint 6.3) is a nonzero map of lines (§VI.2), hence an isomorphism `Chapter6/DetConventions · iotaDualEquiv`, natural by Proposition 6.3.2 (preprint 6.4). With $A=\mathrm{coker}\,T$, $\mathrm{id}\otimes\iota$ is the isomorphism `Chapter6/DetConventions · altLinDetEquiv` `Chapter6/DetConventions · altFredholmDetEquiv`, and on a cokernel basis d with its dual basis it is $(a_1\wedge\dots)\otimes(d_1^*\wedge\cdots\wedge d_m^*)\mapsto(a_1\wedge\dots)\otimes(d_1\wedge\cdots\wedge d_m)^*$ `Chapter6/DetConventions · altLinDetEquiv_tmul_dualBasis`, since the matrix of the pairing is the identity. (Gap G.57 closed.)

The identification (6.25) (preprint (95)) and the maps after Proposition 6.4.15 (preprint 6.15) (V6.79). $\mathbb{R}\cong\lambda(\mathbb{R}^n)$, $1\mapsto e_1\wedge\cdots\wedge e_n$, has inverse $c_1\wedge\cdots\wedge c_n\mapsto\det[c_1,\dots,c_n]$, the matrix with columns c_j `Chapter6/DetConventions · stdLine` `Chapter6/DetConventions · stdLine_symm_wedge`, by the change-of-basis rule of §VI.2; its transpose $\lambda^*(\mathbb{R}^n)\cong\mathbb{R}^*$ maps $(e_1\wedge\cdots\wedge e_n)^*$ to 1^* `Chapter6/DetConventions · stdLineDual_dualVec`, the isomorphism (94) of Proposition 6.4.15 (preprint 6.15). The maps $V\otimes\lambda(\mathbb{R}^n)\to V$ and $V\otimes\lambda^*(\mathbb{R}^n)\to V$, $v\otimes(e_1\wedge\cdots\wedge e_n)\mapsto v$ and $v\otimes(e_1\wedge\cdots\wedge e_n)^*\mapsto v$, are isomorphisms `Chapter6/DetConventions · tensorStdLine_tmul_stdWedge` `Chapter6/DetConventions · tensorStdLineDual_tmul_dualVec` for every vector space V; the text asks V finite-dimensional, which is not needed (stronger). (Gap G.58 closed.)

The remark that $\mathrm{DET}(E,F)$ **is not orientable** (V6.80; end of §6.5, p. 262 (preprint p. 218)). The text: $\mathrm{DET}(E,F)$ is known to be non-orientable, but orientable over certain subsets (Proposition 6.6.1 (preprint 6.26)); no hypothesis on E, F, and no proof or reference. "Not orientable" is read as: no orientation in the sense of §VI.9. *For finite-dimensional E, F the remark does not hold as we read it*: every operator is Fredholm, the base $L(E,F)$ is convex, and $\mathrm{DET}(E,F)$ has exactly two orientations by Proposition 6.6.1 (preprint 6.26) `Chapter6/DetNonOrientable · exists_two_lineOrientation_det_of_finiteDimensional` — the Lean counterexample. *It holds under hypotheses.* (a) *A loop criterion*: a line bundle with a continuous nowhere-zero section along a map from a preconnected space that takes the values v and $-v$ in one fibre has no orientation `Chapter6/DetNonOrientable · isEmpty_lineOrientation_of_loop` — an orientation would have

to agree with the section everywhere or nowhere. (b) *A section of* DET *from kernel bases*: for surjective Fredholm A_x, norm-continuous in x, and continuous bases v_x of the kernels, $(v_{x,1} \wedge \cdots \wedge v_{x,n}) \otimes 1^*$ is a continuous section of $\mathrm{DET}(E,F)$ along $x \mapsto A_x$ `Chapter6/DetNonOrientable · continuous_kerDetMk`: in a local determinant chart with $P = 1$ its coordinate is the determinant of the projections of the $v_{x,i}$ to the kernel at the centre. (c) Hence, *if such a family has* $A_{x_1} = A_{x_0}$, $v_{x_1} = -v_{x_0}$ *and odd* n, $\mathrm{DET}(E,F)$ *has no orientation* `Chapter6/DetNonOrientable · isEmpty_lineOrientation_det_of_kernel_loop`, since $(-v_1) \wedge \cdots \wedge (-v_n) = -(v_1 \wedge \cdots \wedge v_n)$. (d) *If* E *and* F *carry bounded* J, J' *with* $J^2 = -1$, $J'^2 = -1$ *and there is a surjective* $T : E \to F$ *with kernel of odd finite dimension,* $\mathrm{DET}(E,F)$ *has no orientation* `Chapter6/DetNonOrientable · isEmpty_lineOrientation_det_of_complexStructure`: with $R_\theta = \cos\theta + \sin\theta\, J$ and R'_θ likewise, the loop $A_\theta = R'_\theta T R_{-\theta}$, $\theta \in [0,\pi]$, consists of surjective operators with kernels $R_\theta(\ker T)$ and bases $R_\theta v$; at $\theta = \pi$, $R_\pi = -1$, so $A_\pi = T$ and $R_\pi v = -v$, and (c) applies. The kernels form a Möbius band. (e) *Consequences*: $\mathrm{DET}(H \times H, H \times H)$ has no orientation when H has a surjective operator with kernel of odd finite dimension `Chapter6/DetNonOrientable · isEmpty_lineOrientation_det_prod`, for instance $H = \ell^2$ `Chapter6/DetNonOrientableHilbert · isEmpty_lineOrientation_det_ellTwo`; orientability depends only on the isomorphism classes of E and F `Chapter6/DetNonOrientableTransport · nonempty_lineOrientation_det_iff_of_equiv`; so $\mathrm{DET}(H,H)$ has no orientation for every infinite-dimensional real Hilbert space H `Chapter6/DetNonOrientableTransport · isEmpty_lineOrientation_det_hilbert`, nor $\mathrm{DET}(E,F)$ for E, F isomorphic to such an H or to $H \times H$ with $H \cong \mathbb{R} \times H$ `Chapter6/DetNonOrientableTransport · isEmpty_lineOrientation_det_of_equiv_hilbert` `Chapter6/DetNonOrientableTransport · isEmpty_lineOrientation_det_of_equiv_prod`. Infinite dimension enters through the surjective operator with a kernel; on a finite-dimensional space the same loop also has a Möbius band of cokernels, and the two cancel in det. *Not claimed*: infinite-dimensional Banach spaces without such structures. The remark presumably intends infinite-dimensional E, F of the kind met in the applications (Findings, Chapter 6, item 135). (Gap G.60 closed, with this corrected form; row V6.80 narrower.)

The classical case (V6.81; opening of §6.6, p. 262). The text: for a smooth $f : U \to F$ on an open $U \subseteq E$ with Fredholm derivatives, the determinant lines of the $Df(x)$ form a continuous, even smooth, line bundle, with two orientations when U is contractible. For f of class C^1, $x \mapsto Df(x)$ is a continuous map into the Fredholm operators `Chapter6/ClassicalDetBundle · derivFredholm`, the pull-back of $\mathrm{DET}(E,F)$ along it is a topological line bundle over U `Chapter6/ClassicalDetBundle · ClassicalDet` `Chapter6/ClassicalDetBundle · classicalDet_vectorBundle`, and for contractible U it has exactly two orientations, every orientation of one line extending to exactly one of the bundle `Chapter6/ClassicalDetBundle · classicalDet_two_orientations`: U is then simply connected, and locally path connected as an open subset of a Banach space (§VI.9). *Narrower* (V6.81): C^1 in place of smooth, which is weaker, and the bundle is topological; its smoothness, which would need smooth frames of the local determinant charts, is not formalized.

VI.35 Lemma 6.5.3 (preprint 6.19): determinant bundles of vector bundles

The text (p. 260 (preprint p. 216)): *for finite-dimensional topological vector bundles* E, F *over* X, *the bundles* $\lambda(E)$, $\lambda^*(F)$, $E \oplus F$ *and* $E \otimes F$ *have in a natural way the structure of topological vector bundles*, a standard lemma whose proof is left to the reader. §VI.7 applied it only to trivial bundles (V6.22). The file is `Chapter6/DetVectorBundles2021.lean` (Findings, Chapter 6, item 125).

The statement formalized (V6.22) `Chapter6/DetVectorBundles2021 · DetBundle.lemma_6_5_3`. *For Mathlib topological vector bundles* E, F *with finite-dimensional normed model fibres over a topological space, the fibrewise* $\lambda(E_b)$,

$\lambda^*(E_b)$, $E_b \times F_b$ *and* $E_b \otimes F_b$ *are Mathlib topological vector bundles, with trivializations induced by those of* E *and* F. *Proof,* $\lambda(E)$. A trivialization e of E and a basis $f_1, \dots, f_n$ of the model fibre give the frame $b \mapsto e^{-1}(b)f_1, \dots, e^{-1}(b)f_n$ over the base set of e, whose wedge $w_e(b)$ is a continuous nowhere-zero section of $\lambda(E)$; the trivialization is $x \mapsto w_e(b)^*(x)$ `Chapter6/DetVectorBundles2021 · DetBundle.lamPrebundle`. By the change-of-basis rule of §VI.2 the coordinate changes are the determinants of those of E `Chapter6/DetVectorBundles2021 · DetBundle.coord_wedgeAt_eq_det`, continuous because the coordinate changes of E are; this is what a Mathlib vector prebundle asks, and it gives the topology and the bundle structure. $\lambda^*(E)$: $\xi \mapsto \xi(w_e(b))$, with the inverse determinants `Chapter6/DetVectorBundles2021 · DetBundle.lamDualPrebundle`. $E \oplus F$ is Mathlib's fibrewise product with its vector bundle structure. $E \otimes F$: a pair of trivializations gives $u \otimes w \mapsto (i \mapsto (eu)_i\, e_1 w)$ into F_1^n, $n = \dim F_0$ `Chapter6/DetVectorBundles2021 · DetBundle.tenPrebundle`, with the matrices of $A \otimes C$ as coordinate changes for those A, C of E, F `Chapter6/DetVectorBundles2021 · DetBundle.tenMap_tenInv_eq`. □ *Readings.* "In a natural way" is read as: the trivializations of the new bundles are induced by those of E and F, so that local frames give local frames. The model fibre of $E \otimes F$ is F_1^n, since Mathlib puts no norm on $F_0 \otimes F_1$; this does not change the bundle. Proposition 6.5.4 (preprint 6.20) is still proved with the explicit frames of §VI.7. (Gap G.59 closed; row V6.22 restated.)

VI.36 Lemmas 6.6.4 and 6.6.5 (preprint 6.29 and 6.30) as statements about manifolds: property (P), the solution set near the family, π a submersion, L a smooth bundle

§§VI.11–VI.12 proved what the propagation uses: the general position of F along the family, and the kernels as a continuous family of subspaces. The text says more: property (P) for every slice F_t; the solution set of F near $(t, \varphi(t), 0)$ is a manifold with boundary with corners on which $\pi(t, y, \lambda) = t$ is a submersion; its fibres S_t are manifolds with tangent spaces L_t; and L is a smooth vector bundle (pp. 264–265 (preprint p. 222)). All of it is now formalized, on $\mathbb{R}^n$ (decision D2) and on the cube $[0,1]^n$ itself. The setting is that of §VI.11: $G = \tilde{f} + S$ with $G(q, \Phi(q)) = 0$ near $[0,1]^n$, and $F(\lambda, (q, y)) = G(q, y) + \sum \lambda_i s_i(q, y)$ (Findings, Chapter 6, items 47, 48, 126).

Property (P) for the slices (V6.33) `Chapter6/SlicePropertyP · lemma_6_6_4_propertyP`. *Under the hypotheses of Lemma 6.6.4 (preprint 6.29) there are finitely many sc⁺-sections* $s_1, \dots, s_k$ *such that for every* q *near* $[0,1]^n$ *the slice* $F_q(\lambda, y) = G(q, y) + \sum \lambda_i s_i(q, y)$ *is in general position at* $(0, \Phi(q))$ *— property (P) — and* F *is in general position at* $(0, (q, \Phi(q)))$. The second clause is §VI.11. The first is derived from it, the reverse of the text's order `Chapter6/SlicePropertyP · isGeneralPositionGerm_slice`. *The chart computation* `Chapter6/SlicePropertyP · isParamGP_slice_of_isParamGP`: in a tame chart centred at $\Phi(q)$ and the product chart $(r \mapsto r - q) \times c$, properties (i)–(ii) of §V.19 for G and the s_i at $(q, \Phi(q))$ give (i)–(ii) for the slice at $\Phi(q)$. The $\mathbb{R}^n$-directions are absorbed by the derivative of the curve of zeros $q' \mapsto (q', \Phi(q'))$, which lies in $\ker G'$ (§VI.12); and its X-component lies in the reduced tangent space, because every corner coordinate of $\varphi \circ \Phi$ is ≥ 0 near q and vanishes at q, so its derivative vanishes `Chapter6/SlicePropertyP · fderiv_mem_minimalSubspaceWith_of_centre` — $\mathbb{R}^n$ has no boundary. *The fibres*: the zeros of F_q are the fibre S_q of the solution set of F over q; the slice is tame sc-Fredholm `Chapter6/SlicePropertyP · isScFredholmSectionTame_slice`, and given (P), S_q is near $(0, \Phi(q))$ a Mathlib manifold with boundary and corners `Chapter6/SlicePropertyP · lemma_6_6_5_fibre_isManifold`, by Theorem 3.1.26 (Part III) for F_q. (Gap G.63 closed; row V6.33 restated.)

The solution set near the family, and π (V6.34, V6.42). *For* q *near* $[0,1]^n$, $(0, (q, \Phi(q)))$ *has an open neighbourhood in which the solution set of* F *is a Mathlib manifold with boundary and corners* `Chapter6/SolutionManifoldFamily2021 · lemma_6_6_5_isManifold`, *a smooth submanifold of dimension the index* `Chapter6/`

SolutionManifoldFamily2021 · lemma_6_6_5_isSubmanifoldOfDim (Theorems 3.1.26 and 3.6.8 applied to F). *At the points of the family $d\pi$ maps the tangent space onto $\mathbb{R}^n$* Chapter6/SolutionManifoldFamily2021 · lemma_6_6_5_submersion: in the product chart the tangent space is $\ker F'$, $d\pi$ is the projection to the parameter, and the derivative of the curve of zeros lies in $\ker F'$ with parameter component the identity. *Off the family* Chapter6/SubmersionNearFamily · lemma_6_6_5_submersion_nhds: surjectivity of $d\pi$ on the kernels is open along the zeros Chapter6/SubmersionNearFamily · exists_nhds_surjective_linKer, because the zeros near a point in good position are the image of an sc$^+$-retraction, so the level-1 chart points vary continuously along them, the kernels have a continuous frame (§VI.12), and a determinant that is 1 at one point is nonzero nearby. So π is a submersion on a neighbourhood of every point of the family. (Gap G.62 closed; rows V6.34, V6.42 restated.)

L is a smooth vector bundle (V6.34) Chapter6/SmoothKernelFrames · lemma_6_6_5_smoothBundle. *In the setting of §VI.15 (with (P)), there are an open connected $V \supseteq [0,1]^n$ and a smooth family of subspaces over the manifold V whose fibre at q is $L_q = \ker F'(0,(q,z_q)) \cap \{dq = 0\}$, a Mathlib C^∞ vector bundle. Step 1.* An sc-smooth map on $\mathbb{R}^n$ read in a chart is C^∞ into every level on the open sets it maps into the chart Chapter6/SmoothFiniteDimRanges · IsScSmooth.contDiffOn_levelMap_const. *Step 2.* The kernel vectors of an sc-Fredholm linearization are smooth points Chapter6/SmoothFiniteDimRanges · IsSc1With.apply_mem_smooth_of_finiteDimensional. *Step 3, a smooth frame* Chapter6/SmoothKernelFrames · exists_local_smooth_frame_linKer: with an sc-retraction S onto the solution set (Theorem 3.6.8) and a basis b of one kernel, $e_i(q) = DS(z_q)b_i$ is C^∞ into every level, since $q \mapsto (z_q, b_i)$ is sc-smooth into the tangent bundle and the tangent map of S is sc-smooth; it is a frame near the base point, as in §VI.12. *Step 4.* A family of subspaces of a normed space with smooth local frames is a smooth vector bundle Chapter6/SmoothSubspaceFamilies · SmoothSubspaceFamily.contMDiffVectorBundle, and cutting it by $\{dq = 0\}$, onto which the projection is surjective, keeps it smooth Chapter6/SmoothSubspaceFamilies · SmoothSubspaceFamily.infKer. The continuation of orientations still uses only continuity. (Rows V6.34, V6.42 restated.)

On the cube itself (V6.33, V6.34). The same statements hold for families on $[0,1]^n$, boundary and corners included, with sections over the M-polyfold with corners $[0,1]^n \times X$ Chapter6/CubeSections · cubePolyfold, whose degeneracy index counts the coordinates in $\{0,1\}$ Chapter6/CubeSections · cubePolyfold_degIndex. The pull-back of a tame sc-Fredholm section to $[0,1]^n \times X$ is tame sc-Fredholm Chapter6/CubeFredholm · StrongBundle.IsScFredholmSectionTame.paramCube, with charts of $[0,1]^n$ centred at every point Chapter6/CubeCharts · UnitCube.cChart and basic germs extended by the mixed quadrants Chapter6/CubeGerms · IsConjugateToBasicGerm.quadExtend. Lemma 6.6.4 (preprint 6.29) holds there Chapter6/CubeFamilyLemmas · lemma_6_6_4_cube Chapter6/CubeFamilyLemmas · lemma_6_6_4_cube_family, and near the family the solution set is a manifold with boundary and corners and a submanifold of dimension the index Chapter6/CubeFamilyLemmas · lemma_6_6_5_isManifold_cube Chapter6/CubeFamilyLemmas · lemma_6_6_5_isSubmanifoldOfDim_cube. On the cube, *(P) at q is equivalent to the general position of F at $(0,(q,\Phi(q)))$*, boundary and corners included Chapter6/CubePropertyP · isGeneralPositionGerm_cubeSlice_iff Chapter6/CubePropertyP · lemma_6_6_4_propertyP_cube: the tangent vectors of the family of zeros in reduced directions of $[0,1]^n$ have reduced X-components, by a one-sided minimum argument Chapter6/CubeBoundarySlice · mem_minimalSubspaceWith_of_hasFDerivWithinAt. At these points π is a submersion in the corner sense: $d\pi$ maps $\ker F'$ onto $\mathbb{R}^n$ and its reduced part onto the reduced tangent space of $[0,1]^n$ Chapter6/CubeBoundarySubmersion · lemma_6_6_5_submersion_cube.

VI.37 Paths and families on $[0,1]$; the facts on paths before Proposition 6.2.2 (preprint 6.2)

Decision D2 reads the paths, homotopies and families of the chapter on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$. Most statements now also hold as printed, for maps on the M-polyfolds $[0,1]$, $[0,1]^2$, $[0,1]^n$ (gap G.65, in part; Findings, Chapter 6, items 102, 122, 124).

$[0,1]$, $[0,1]^2$ and $X \times [0,1]$. $[0,1]$ is a tame M-polyfold with boundary `Chapter3/UnitIntervalPolyfold · unitIntervalPolyfold`, and every sc-smooth path on it is the reparametrization by $\sin^2(\pi r/2)$ of an sc-smooth path on $\mathbb{R}$ `Chapter3/UnitIntervalPolyfold · MPolyfold.IsScSmoothMap.exists_rn_reparam`; this transfers the statements on $\mathbb{R}$. $X \times [0,1]$ is tame for tame X, with $d(x,t) = d_X(x) + d_{[0,1]}(t)$ `Chapter6/PathsInterval · isTame_prod_unitInterval`, and $[0,1]^2$ is an M-polyfold with corners `Chapter6/PathsInterval · unitSquarePolyfold`.

Theorems 6.6.11, 6.6.13 and 6.6.17 for paths on $[0,1]$ (V6.48, V6.57, V6.71). The push-forward along $\varphi : [0,1] \to U$ is read along the path $\varphi \circ \sin^2(\pi r/2)$ on $\mathbb{R}$, independently of the reparametrization `Chapter6/PathsInterval · theorem_6_6_11_interval`; the transport of Theorem 6.6.13 (preprint 6.37) is invariant under sc-smooth homotopies $[0,1]^2 \to X$ with fixed end points `Chapter6/PathsInterval · orTransportI_eq_of_homotopy`, and Theorem 6.6.17 holds with property (3) for paths on $[0,1]$ `Chapter6/PathsInterval · theorem_6_6_17_interval`.

The path category on $[0,1]$ (V6.52). Sc-smooth homotopies $[0,1]^2 \to X$ with fixed end points `Chapter6/PathCategoryInterval · ScHomotopicI` are an equivalence relation, and two paths on $[0,1]$ are homotopic in this sense iff their reparametrized paths on $\mathbb{R}$ are homotopic in the sense of §VI.20 `Chapter6/PathCategoryInterval · scHomotopicI_iff`. The composition is the text's $\gamma(s) = \varphi(\beta(s))$ on $[0,1/2]$ and $\psi(\beta(s) - 1)$ on $[1/2,1]$, with $\beta(s) = \kappa(3s) + \kappa(3s-2)$ `Chapter6/PathCategoryInterval · betaConcat` `Chapter6/PathCategoryInterval · ScPathI.concat` `Chapter6/PathCategoryInterval · ScPathI.concat_toFun_of_le` `Chapter6/PathCategoryInterval · ScPathI.concat_toFun_of_ge`. The classes form a category `Chapter6/PathCategoryInterval · PathCatI` `Chapter6/PathCategoryInterval · PathCatI.category`, a groupoid `Chapter6/PathCategoryInterval · PathCatI.groupoid`, and $[\varphi] \mapsto [\varphi \circ \sin^2(\pi r/2)]$ is a full and faithful functor to the category of §VI.20, bijective on objects `Chapter6/PathCategoryInterval · PathCatI.toPathCat_isEquivalence`, so the two readings give isomorphic categories.

Proposition 6.6.3 (3) and (4) on $[0,1] \times X$ and $[0,1]^n \times X$ (V6.32). *With bump functions, for an sc-smooth path $\varphi : [0,1] \to X$ in the domain of a strong bundle chart there is an sc$^+$-section S of the pull-back of P to the M-polyfold $[0,1] \times X$ whose fibre part is the text's $-\beta(y)R(y, f(\varphi(t)))$, β an sc-smooth cut-off equal to 1 near $\varphi([0,1])$, with $(f+S)(t,\varphi(t)) = 0$ for every $t \in [0,1]$* `Chapter6/UnitIntervalSections · proposition_6_6_3_three_interval`. The construction works in any product chart $\chi \times \varphi_c$ of $Q \times X$ `Chapter6/UnitIntervalSections · PullbackChartData.ofProdChart` `Chapter6/UnitIntervalSections · exists_isScPlusSection_prodChart`; on $[0,1]$ the principal part lifts to level 1, the parameter space being finite-dimensional, and the two charts t, $1-t$ of $[0,1]$ are glued with a partition of unity in t. $[0,1] \times X$ admits sc-smooth bump functions when X does, which gives (4) `Chapter6/UnitIntervalSections · proposition_6_6_3_four_interval`. The same holds for families on $[0,1]^n$, with sections over $[0,1]^n \times X$ `Chapter6/CubeSections · proposition_6_6_3_three_cube` `Chapter6/CubeSections · proposition_6_6_3_four_cube` `Chapter6/CubeSections · admitsBumpFunctions_prod_cube`. The sign is that of $f + s$ (structural choice (1)).

The facts on paths before Proposition 6.2.2 (preprint 6.2) (V6.49) `Chapter6/SmoothPathComponents2021 · MPolyfold.paths_before_proposition_6_2_2`. The text recalls (p. 240 (preprint

p. 201)): *an M-polyfold is locally path connected, and two smooth points in one connected component (preprint: path component) are joined by an sc⁺-smooth path $\varphi : [0,1] \to X$. Step 1, the local step* `Chapter6/SmoothPathComponents2021 · MPolyfold.exists_nhds_scJoinedIn`: every u in an open U has an open $N \ni u$ in U whose smooth points are pairwise joined by sc-smooth paths in U — the retracted segment of a chart of X, without a strong bundle chart. Local path connectedness is proved from the charts, which are retracts of open sets of partial quadrants `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace`. *Step 2, the global step* `Chapter6/SmoothPathComponents2021 · MPolyfold.scJoinedIn_of_isConnected`: in a connected open U any two smooth points are joined by an sc-smooth path in U, by a chain argument over the neighbourhoods of Step 1, with the concatenation of paths (§VI.18), using that U is connected. *Step 3, sc⁺* `Chapter6/SmoothPathComponents2021 · MPolyfold.IsScSmoothMap.exists_scPlus_lift`: $[0,1]$ is modelled on sc⁺-retracts, so the inclusion $[0,1] \to [0,1]^1$ is sc-smooth, and an sc-smooth path on $[0,1]$ lifts to an sc-smooth path into X^1. For smooth points, being in one connected component, in one path component and being joined by an sc-smooth path on $[0,1]$ are equivalent `Chapter6/SmoothPathComponents2021 · mem_connectedComponent_iff_exists_scPath Chapter6/SmoothPathComponents2021 · MPolyfold.mem_pathComponent_iff_exists_scPath`, which justifies reading "in the same path component" in Definitions 6.6.2 and 6.6.14 (preprint 6.27 and 6.38) as "joined by an sc-smooth path" (§VI.24); and two smooth points of one connected component have the same index `Chapter6/SmoothPathComponents2021 · proposition_6_2_2_connectedComponent_2021def`. "Sc⁺-smooth path" is read as an sc-smooth path that lifts to X^1; every sc-smooth path on $[0,1]$ does. (Gap G.56 closed.)

What is still read on $\mathbb{R}$ (gap G.65, in part; Proposition 6.6.3 (1′) holds on $[0,1]$, §VI.10): the smooth bundle L of Lemma 6.6.5 (preprint 6.30) over a neighbourhood of $[0,1]^n$ in $\mathbb{R}^n$ (V6.34), the stabilized kernels and their continuation (V6.42), and Remark 6.6.19 (preprint 6.42) over $\mathbb{R}^1 \times X$ (V6.62), with the cobordisms of Theorem 6.7.1 (preprint 6.43) (§VI.27).

VI.38 The remarks after Theorem 6.6.17: the orientation bundle, orientability, $\mathbb{Z}_2$ and $\det_f$

The text (p. 275, new in the 2021 edition): with the topology of Theorem 6.6.17, $\pi_f : O_f \to X_\infty$ is the orientation bundle of f; f is orientable iff π_f has a global continuous section; $\mathbb{Z}_2$ acts continuously on O_f by $(x, o_x) \mapsto (x, -o_x)$; and $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f \to X_\infty$ is the determinant bundle of f. The file is `Chapter6/OrientationBundle.lean` (V6.82; Findings, Chapter 6, item 121).

Sections (V6.82). For local transport data whose fibres have two elements, with the topology of §VI.33: *$y \mapsto (y, o_y)$ is continuous iff the transport along every sc-smooth path carries o along* `Chapter6/OrientationBundle · continuous_orSec_iff` — near each point a continuous section stays in one sheet, and a sheet is carried along by the transport (§VI.33) — and every section of π_f has this form `Chapter6/OrientationBundle · exists_eq_orSec`. For the transport of Γ_f: *$y \mapsto o_y$ is an orientation of f (Definition 6.6.16 (preprint 6.40)) iff $y \mapsto (y, o_y)$ is continuous* `Chapter6/OrientationBundle · isSectionOrientation_iff_continuous`, by §VI.24, and so *f is orientable iff π_f has a global continuous section* `Chapter6/OrientationBundle · isOrientableSection_iff_exists_continuous_section`.

The $\mathbb{Z}_2$-action. $(x, o) \mapsto (x, -o)$, $-o$ the other element of Or_x^f, maps each sheet onto the sheet of the opposite continuous orientation, so it is a homeomorphism `Chapter6/OrientationBundle · orNegHomeomorph`, an involution without fixed points over π_f; it is the action of $\mathbb{Z}^\times \cong \mathbb{Z}_2$ `Chapter6/OrientationBundle · orMulAction`, continuous `Chapter6/OrientationBundle · continuous_orUnitsAct`.

$\det_f$. A Mathlib vector bundle core with fibre $\mathbb{R}$ over X_∞, with a trivialization for every continuous

orientation (U, o) and coordinate changes ± 1 according as two continuous orientations agree — locally constant by §VI.33 (a) `Chapter6/OrientationBundle · detCore` `Chapter6/OrientationBundle · detBundleCore`. Its fibre over x is $\mathbb{R} \otimes_{\mathbb{Z}_2} \mathrm{Or}_x^f$: $r \otimes a$ and $r' \otimes a'$ have the same class iff $(r', a') = (r, a)$ or $(-r, -a)$ `Chapter6/OrientationBundle · detMk_eq_iff`, and $r \otimes a$ has coordinate r in the trivialization of (U, o) if $a = o_x$, and $-r$ if not `Chapter6/OrientationBundle · localTriv_detMk`. (Gap G.91 closed; row V6.82 restated.)

Part VII. Ep-groupoids (HWZ Chapter 7, §§7.1–7.3, 7.5, 7.6 (2017 §§7.1–7.5))

Chapter 7 opens Part II of the book. Part II has no 2014 preprint, so its numbers and pages are those of the book (Springer 2021, pp. 287–356), with the 2017 text's in parentheses (2017 pp. 225–256; see `BookConcordance.md`). The chapter introduces ep-groupoids, the sc-smooth version of étale proper Lie groupoids. §7.1 defines groupoids, isotropy groups and ep-groupoids (Definitions 7.1.1–7.1.3 (2017: 7.1–7.3)), the local sc-diffeomorphism $\widehat{\varphi}$ of a morphism and its tangent (Definition 7.1.4 (2017: 7.4)), the levels and the degeneracy index of the orbit space (Definition 7.1.5 (2017: 7.5)), tame ep-groupoids and the category $\mathcal{EP}$ (Definitions 7.1.6, 7.1.7 (2017: 7.6, 7.7)). It proves that π is open, that $|X|$ has countable bases of neighbourhoods, the local closure relation and the finiteness of the isotropy groups (Propositions 7.1.8–7.1.12 (2017: 7.1–7.4), Lemma 7.1.11 (2017: 7.1)). It states the natural representation of G_x (Theorem 7.1.13 (2017: 7.1)), discusses the translation groupoid $G_x \ltimes U(x)$ and the action $\widehat{\Phi}$ (pp. 293–297 (2017 pp. 231–233)), extends the natural representation to finite sets of isomorphic objects (Proposition 7.1.15 (2017: 7.5), Definition 7.1.18 (2017: 7.8)), identifies $|U(x)|$ with $G_x \backslash U(x)$ (Proposition 7.1.19 (2017: 7.6)) and defines local uniformizers and ep-subgroupoids (Definitions 7.1.20, 7.1.21 (2017: 7.9, 7.10), Proposition 7.1.22 (2017: 7.7)). §7.2 treats effective ep-groupoids, the subgroups T_x, the reduced morphism space $\mathbf{X}^R$ and the reduced ep-groupoid X_R (Definitions 7.2.1–7.2.15 (2017: 7.11–7.13), Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3), Propositions 7.2.4, 7.2.10 (2017: 7.8, 7.9) and Proposition 7.10 (2017; rewritten in 2021 as Theorem 7.2.14)). §7.3 proves that $|X|$ is locally metrizable, regular and Hausdorff (Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2–7.3.5 (2017: 7.4–7.7), Corollaries 7.3.6, 7.3.8 (2017: 7.1, 7.2), Definition 7.3.7 (2017: 7.14)) and lifts ep-groupoids to level 1 (Proposition 7.3.9 (2017: 7.11)). §7.5 (2017 §7.4) defines paracompact orbit spaces and sc-smooth partitions of unity (Definitions 7.5.1, 7.5.5 (2017: 7.15, 7.16), Remarks 7.5.2, 7.5.6 (2017: 7.1, 7.2)), proves Proposition 7.5.3 (2017: 7.12) and Theorems 7.5.4, 7.5.7 (2017: 7.3, 7.4), and characterizes the closed subsets of $|X|$ (Definition 7.5.8 (2017: 7.17), Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14)). The appendix §7.6 (2017 §7.5) proves Theorem 7.1.13 (2017: 7.1) as Theorem 7.6.1 (2017: 7.5) (Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9)) and Theorem 7.5.7 (2017: 7.4) as Theorem 7.6.4 (2017: 7.6) (Lemma 7.6.5 (2017: 7.10)), and ends with Corollary 7.6.6 (2017: 7.3). This Part retells §7.1 in §§VII.1–VII.12 — Definitions 7.1.1, 7.1.2 (2017: 7.1, 7.2) (§VII.1), Definition 7.1.3 (2017: 7.3) and properness (§VII.2), Definitions 7.1.4–7.1.6 (2017: 7.4–7.6) (§VII.3), Definition 7.1.7 (2017: 7.7) and the remark after it (§VII.4), Propositions 7.1.8–7.1.10 (2017: 7.1–7.3) (§VII.5), Lemma 7.1.11 (2017: 7.1) and Proposition 7.1.12 (2017: 7.4) (§VII.6), Theorem 7.1.13 (2017: 7.1) (§VII.7), pp. 293–297 (2017 pp. 231–233) (§VII.8), the uniqueness of natural representations (§VII.9), Proposition 7.1.15 (2017: 7.5) and Definition 7.1.18 (2017: 7.8) (§VII.10), Proposition 7.1.19 (2017: 7.6) and Definition 7.1.20 (2017: 7.9) (§VII.11), Definition 7.1.21 (2017: 7.10) and Proposition 7.1.22 (2017: 7.7) (§VII.12); §7.2 in §§VII.13–VII.17 — Definition 7.2.1 (2017: 7.11) and Lemmas 7.2.2, 7.2.3 (2017:

7.2, 7.3) (§VII.13), Proposition 7.2.4 (2017: 7.8) (§VII.14), T_x and Definition 7.2.7 (2017: 7.12) (§VII.15), Proposition 7.2.10 (2017: 7.9) (§VII.16), X_R, Proposition 7.10 (2017; rewritten in 2021 as Theorem 7.2.14) and Definition 7.2.15 (2017: 7.13) (§VII.17); §7.3 in §§VII.18–VII.22 — Theorem 7.3.1 (2017: 7.2) and Lemmas 7.3.2, 7.3.3 (2017: 7.4, 7.5) (§VII.18), Lemmas 7.3.4, 7.3.5 (2017: 7.6, 7.7) and Corollary 7.3.6 (2017: 7.1) (§VII.19), Michael's theorem of Part I (§VII.20), Definition 7.3.7 (2017: 7.14) and Corollary 7.3.8 (2017: 7.2) (§VII.21), Proposition 7.3.9 (2017: 7.11) (§VII.22); §7.5 (2017 §7.4) in §§VII.23–VII.25 — Definition 7.5.1 (2017: 7.15), Remark 7.5.2 (2017: 7.1) and Proposition 7.5.3 (2017: 7.12) (§VII.23), Theorem 7.5.4 (2017: 7.3), Definition 7.5.5 (2017: 7.16), Remark 7.5.6 (2017: 7.2) and the statement of Theorem 7.5.7 (2017: 7.4) (§VII.24), Definition 7.5.8 (2017: 7.17) and Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14) (§VII.25); the appendix in §§VII.26–VII.29 — the proof of Theorem 7.1.13 (2017: 7.1) (§VII.26), Lemma 7.6.5 (2017: 7.10) (§VII.27), the proof of Theorem 7.5.7 (2017: 7.4) (§VII.28) and Corollary 7.6.6 (2017: 7.3) (§VII.29); the witnesses and counterexamples (§VII.30); what is not formalized (§VII.31); and what the formalization taught us (§VII.32). Each item carries the number of its comparison item (V7.1–V7.68) in `Exposition/verification/ch07.md`. The additions of the 2021 edition (V7.69–V7.104) are retold in §§VII.33–VII.43: §7.1 (§VII.33), §7.2 (§§VII.34–VII.35), Appendices 7.6.3 and 7.6.4 (§VII.36), §7.4, new in the 2021 edition (§§VII.37–VII.39), Appendix 7.6.5 (§VII.40), what is not formalized and what they taught us (§VII.41), Theorem 7.6.9 in full, the boundary ep-groupoid (§VII.42), and §7.4 on every lift X^i (§VII.43). The work that closed the remaining gaps is retold in §§VII.44–VII.50: $T\varphi$ as an sc-isomorphism, the facts (3), (4), $|X_\infty|$ and the density of X_∞ (§VII.44), the Part I facts of §7.4 as theorems (§VII.45), Theorem 7.6.7 and Corollary 7.6.8 for every i (§VII.46), Zhou's retract of Appendix 7.6.3 (§VII.47), the 2021 forms of Propositions 7.2.10, 7.2.16, Theorem 7.2.14 and Corollary 7.6.6 as printed (§VII.48), an ep-groupoid whose orbit space is not paracompact (§VII.49), and Theorem 7.6.9 for every representative and in the book's form (§VII.50).

What is proved, and what is not. Every statement of the chapter is formalized, and every gap of the chapter is closed: G.70 ($T\varphi$, §VII.44), G.71 (the M-polyfold $\mathbf{X}^R$, §§VII.34, VII.48), G.72 ($|X_\infty|$, §VII.44), G.73 (the density of X_∞, §VII.44), G.82 (Theorem 7.6.7 for every i, §VII.46), G.83 (Zhou's retract, §VII.47), G.84 (§7.4 on every lift, §VII.43) and G.85 (the Part I facts of §7.4, §VII.45). Theorems 7.1.13–7.6.4 (2017: 7.1–7.6) are proved, and so are Propositions 7.1.8–7.2.4 and 7.3.9–7.5.10 (2017: 7.1–7.8 and 7.11–7.14), Lemmas 7.1.11–7.6.5 (2017: 7.1–7.10) and Corollaries 7.3.6–7.6.6 (2017: 7.1–7.3); Corollary 7.6.6 (2017: 7.3) as printed, Part I's Theorem 5.5.16 (2017: 5.14) of Godefroy–Troyanski–Whitfield–Zizler being now a theorem (§§VII.29, VII.48). Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) are proved in their 2021 form: the natural charts of $\mathbf{X}^R$ with no hypothesis (Proposition 7.2.10), paracompactness for every ep-groupoid (Proposition 7.2.12), and the reduced ep-groupoid under the 2021 text's hypothesis that T_R is Hausdorff (Theorem 7.2.14, Proposition 7.2.16) (§§VII.34, VII.48); for every ep-groupoid, as the 2017 text states them, they do not hold as we read it, and $\mathbb{Z}/2$ acting on a dimension-jumping retract is the counterexample `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.exists_not_t2Space_reducedMor` (§VII.16). The 2021 additions to §§7.1, 7.2, the new §7.4 and Appendices 7.6.3–7.6.5 are proved (§§VII.33–VII.50). Remark 7.5.6 (2017: 7.2) is an assessment, with nothing to formalize (row V7.57). One statement is formalized in a narrower form: in Theorem 7.6.9 (row V7.98), "Φ is a homeomorphism onto θ" holds as we read it only for V centred at x; for the other representatives Φ is an sc-diffeomorphism onto the open set $\theta \cap V^x$, and both are proved (§§VII.40, VII.50). The open question of p. 340 (2017 p. 249), whether property (5) follows from (1)–(4), is settled negatively: the Čech groupoid of a Prüfer-type surface is an ep-groupoid whose orbit space is not paracompact `Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_epGroupoid_not_hasParacompactOrbitSpace` (§VII.49). Five places of the 2017 text needed more than a reading. The remark after Definition 7.1.7 (2017: 7.7), that a functor sc-smooth on objects is sc-

smooth on morphisms, does not hold as we read it, and a counterexample on the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$ shows it; it holds for functors continuous on morphisms, which is what its argument uses and what the 2021 text asks (§§VII.4, VII.33). The proof of Proposition 7.2.10 (2017: 7.9) gives the charts of $\mathbf{X}^R$ and their sc-smooth transitions but, as we read it, neither Hausdorffness nor paracompactness; Hausdorffness is the hypothesis of the 2021 Theorem 7.2.14, and paracompactness is proved (§§VII.16, VII.34). The natural representation of Theorem 7.1.13 (2017: 7.1) is taken on a connected neighbourhood, which the text chooses on 2017 p. 231 but does not put in the statement; connectedness makes Φ a homomorphism and Γ a functor, and the local path connectedness of M-polyfolds that supplies it is proved (§VII.7). The tameness of X^1 in Proposition 7.3.9 (2017: 7.11) needs that the index lift of a tame retraction is tame, which Part I left open; it is proved (§VII.22). And several omitted steps are supplied: the uniqueness of natural representations behind "adapting restrictions" (§VII.9), the conjugation $\varphi T_x \varphi^{-1} = T_y$ that the composition of X_R needs (§VII.15), the passage from compact to component compact sets in Corollary 7.3.8 (2017: 7.2) (§VII.21), the support bound that makes the sum in the proof of Theorem 7.5.7 (2017: 7.4) locally finite (§§VII.27–VII.28) and Michael's theorem on unions of closed paracompact sets (§VII.20). Proposition 7.1.15 (2017: 7.5) is proved by a construction without induction (§VII.10), Proposition 7.5.3 (2017: 7.12) by Smirnov's theorem pulled back to $|X^1|$ (§VII.23), Proposition 7.5.10 (2) (2017: 7.14) without Lemma 2.7.3 (2017: 2.7) (§VII.25), Lemma 7.3.4 (2017: 7.6) from the closedness of points and regularity (§VII.19), and the normalization of Theorem 7.5.7 (2017: 7.4) by bump coverings (§VII.28). Nothing of the chapter is left unformalized (§VII.31).

Seven structural choices. (1) *A groupoid is a structure of maps.* Objects and morphisms are types; s, t, u, ι are maps and the composition m is a total map whose value on a non-composable pair is a junk value never used; the groupoid laws are required on composable pairs. A bridge to Mathlib's groupoids makes "full" and "faithful" Mathlib's. (2) *Properness is Mathlib's.* Definition 7.1.3 (3) (2017: 7.3) is the properness, in Mathlib's sense, of t restricted to $s^{-1}(\mathrm{cl}_X V)$, for some open $V \ni x$, per point; on metrizable spaces it is equivalent to the sequential form the text uses (§VII.2). (3) *Condition (4) in two forms.* It is stated literally, on the fibred product of Part I's Proposition 2.5.20 (2017: 2.15; preprint 2.46, §II.17), and used in the equivalent *sheet form*: for every local inverse σ of s, the map $\psi \mapsto m(\sigma(t\psi), \psi)$ is sc-smooth. (4) *Natural representations are data.* "An open neighbourhood $U(x)$ equipped with the natural G_x-action" is always a datum (U, Φ, Γ) with U open, connected and containing x, never a hypothesis "for every such $U(x)$". (5) *$G \times U$ is a product with a discrete M-polyfold*, the finite disjoint union of copies of U, on which sc-smoothness is sc-smoothness on each slice (§VII.8). (6) *The reduced morphisms* $\mathbf{X}^R$ are a quotient whose M-polyfold structure is built under the hypothesis that it is Hausdorff, the 2021 text's hypothesis on T_R; its paracompactness is proved, for every ep-groupoid (§§VII.16, VII.34). (7) *Michael's theorem without Hausdorff.* Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17) (gap G.15 there) are proved for regular spaces, and Mathlib's paracompactness, like Definition 7.5.1 (2017: 7.15), does not include Hausdorff (§VII.20).

Map.

- `Chapter7/Groupoid`: Definitions 7.1.1, 7.1.2 (2017: 7.1, 7.2); the orbit space, saturations, the conjugation of isotropy groups, regular objects, full subcategories, translation groupoids, functors and the bridge to Mathlib.
- `Chapter7/EpGroupoid`: Definition 7.1.3 (2017: 7.3) with the fibred-product and sheet forms of (4), Definition 7.1.4 (2017: 7.4), the levels, Definitions 7.1.5–7.5.8 (2017: 7.5–7.17) as structures and predicates, the trivial ep-groupoid and its witnesses.
- `Chapter7/ProperMaps`: proper maps between metrizable spaces, the sequential criterion, proper restrictions $t : s^{-1}(Q) \to X$.

- `Chapter7/LocalDiffeo`: local sc-diffeomorphisms, their levels, degeneracy index and tameness; Part I's partitions of unity on $\mathbb{R}^n$.
- `Chapter7/Basic`: the remark after Definition 7.1.7 (2017: 7.7), Propositions 7.1.8–7.1.12 (2017: 7.1–7.4), Lemma 7.1.11 (2017: 7.1), the conjugation $G_x \cong G_y$.
- `Chapter7/Discrete`: finiteness from proper locally injective maps, discrete M-polyfolds, $G \times X$ and its sheets.
- `Chapter7/LocConnected`: local path connectedness of retracts and M-polyfolds, uniqueness of sections on connected sets.
- `Chapter7/NaturalRep`: Theorem 7.1.13 (2017: 7.1) = Theorem 7.6.1 (2017: 7.5) with Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9).
- `Chapter7/NaturalAction`: the consequences of Theorem 7.1.13 (2017: 7.1) on pp. 293–297 (2017 pp. 231–233) and the uniqueness of natural representations.
- `Chapter7/NaturalRepFinite`: Proposition 7.1.15 (2017: 7.5) by the star construction, Definition 7.1.18 (2017: 7.8).
- `Chapter7/Uniformizer`: Proposition 7.1.19 (2017: 7.6), Definition 7.1.20 (2017: 7.9) and the claims after it.
- `Chapter7/SubGroupoid`: Definition 7.1.21 (2017: 7.10), the M-polyfold $\mathbf{A}$, Proposition 7.1.22 (2017: 7.7).
- `Chapter7/Effective`: Definition 7.2.1's (2017: 7.11) consequences, Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3), Proposition 7.2.4 (2017: 7.8), T_x.
- `Chapter7/Reduced`: Definition 7.2.7 (2017: 7.12) made precise, Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017), Definition 7.2.15 (2017: 7.13).
- `Chapter7/OrbitTopology`: Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2–7.3.5 (2017: 7.4–7.7), Corollaries 7.3.6, 7.3.8 (2017: 7.1, 7.2).
- `Chapter7/Michael`: Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17), unions of closed paracompact sets.
- `Chapter7/Lift`: Proposition 7.3.9 (2017: 7.11), the map $|X^1| \to |X|$, Definition 7.5.1's (2017: 7.15) consequences, Remark 7.5.2 (2017: 7.1), Proposition 7.5.3 (2017: 7.12).
- `Chapter7/ProperSubsets`: Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14).
- `Chapter7/PartitionsOfUnity`: Theorem 7.5.4 (2017: 7.3), Lemma 7.6.5 (2017: 7.10), Theorem 7.5.7 (2017: 7.4) = Theorem 7.6.4 (2017: 7.6), Corollary 7.6.6 (2017: 7.3).
- `Chapter7/NonVacuity`: the translation groupoid of a finite group action, $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection, the trivial $\mathbb{Z}/3$-action, the counterexamples.
- `Chapter7/Book2021Basic`: the 2021 forms of the remark after Definition 7.1.7 and of Proposition 7.1.15, the data (7.3), Proposition 7.1.14, Lemma 7.1.16, Corollary 7.1.17 (§VII.33).
- `Chapter7/ReducedCriterion`: the 2021 §7.2 from Definition 7.2.5 on, LUC, Theorem 7.2.22, Appendices 7.6.3 and 7.6.4 (§§VII.34–VII.36).
- `Chapter7/Zhou`: §7.4, the index lifts, the Part I facts as hypotheses, (ZHOU), (SLUC), (VSLUC), (LR), (SLR) (§§VII.37–VII.39).
- `Chapter7/TameBoundaryEp`: Appendix 7.6.5, the boundary groupoid of a tame ep-groupoid, Theorem 7.6.9 (topological part) (§VII.40).
- `Chapter7/TameBoundaryEpStructure`: Theorem 7.6.9 in full, on Part I's Theorem 2.7.13 (2)–(4): the boundary ep-groupoid and the ep-functor B (§VII.42).
- `Chapter7/ZhouLevels`: §7.4 on every lift X^i: Propositions 7.4.9, 7.4.16–7.4.18, and Corollary 7.4.11, Theorems 7.4.13, 7.4.24 for every X^i (§VII.43).
- `Chapter7/TangentMorphism`, `Chapter7/TangentMorphismLinear`: Definition 7.1.4, $T\varphi$ as a bijec-

tion, a bounded linear isomorphism and an sc-isomorphism (§VII.44).

- `Chapter7/SubGroupoidTangent`, `Chapter7/SubGroupoidTangentSc`: the facts (3), (4) on A (§VII.44).
- `Chapter7/SmoothPointsDense`: the density of X_∞ in every X_m, the filtration of $|X|$ (§VII.44).
- `Chapter7/ZhouFacts`, `Chapter7/SmoothConnectedReps`, `Chapter7/SmoothRegularAssembly`: the Part I facts of §7.4 for every M-polyfold, and §7.4, Appendix 7.6.4 without them (§VII.45).
- `Chapter7/RaisedEpGroupoid`: Theorem 7.6.7 and Corollary 7.6.8 for every i, as isomorphisms of ep-groupoids (§VII.46).
- `Chapter7/LineSobolevScale`, `LineTranslates`, `FlatGluing`, `ZhouRetraction`, `ZhouGroupoid`, `ZhouSigma`: Zhou's retract of Appendix 7.6.3: the scale $H^{m,\delta_m}(\mathbb{R})$, the sc-smooth retraction, the ep-groupoid and its non-effective part (§VII.47); `Chapter7/ZhouRetract`: the same computation for the analogue (§VII.36).
- `Chapter7/ReducedNaturalCharts`, `Chapter7/HilbertPartitionsOfUnityEp`: Proposition 7.2.10, Theorem 7.2.14, Proposition 7.2.16 in the 2021 form; Corollary 7.6.6 as printed (§VII.48).
- `Chapter7/PropertyFiveResearch`: an ep-groupoid with Hausdorff, non-paracompact orbit space (§VII.49).
- `Chapter7/TameBoundaryFull`, `Chapter7/BookFormTameBoundaryEp2021`: Theorem 7.6.9 for every representative, and in the book's form (§VII.50).

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid: X is the object M-polyfold and $\mathbf{X}$ the morphism M-polyfold, both Hausdorff and paracompact, as every M-polyfold of the project (Definition 2.3.4 (2017: 2.8), §II.5); $s, t : \mathbf{X} \to X$ are source and target, $m(\varphi, \psi) = \varphi \circ \psi$ the composition, $u(x) = 1_x$ the unit and $\iota(\varphi) = \varphi^{-1}$ the inverse. $|X|$ is the orbit space with the quotient topology and $\pi : X \to |X|$ the projection; G_x is the isotropy group of x, cl_X the closure in X, and $\mathbf{U}(x)$ the set of morphisms with source and target in $U(x)$. For a natural representation (U, Φ, Γ) at x, $g * y = \Phi(g)(y)$. A *sheet* is an open set $\mathbf{W} \subseteq \mathbf{X}$ on which s is an sc-diffeomorphism onto an open set, its inverse a *local inverse* of s. Part I results are cited in the same way, by their 2021 numbers with the 2017 numbers in parentheses; the Lean names carry the 2021 numbers (Lemma 2.7.3 and Proposition 2.7.4, 2017 Lemma 2.7 and Proposition 2.17, are `lemma_2_7_3` and `proposition_2_7_4`; Theorem 5.5.10, 2017 Theorem 5.12, is `theorem_5_5_10`), and the Lean names of Chapter 7 carry the 2021 numbers too, with `_2017` for the statements that exist only in 2017. Findings are those of `Exposition/Findings.md`, "Chapter 7".

VII.1 Definitions 7.1.1 and 7.1.2 (2017: 7.1 and 7.2): groupoids, the orbit space, isotropy

Definition 7.1.1 (2017: 7.1) (V7.1). A groupoid is a structure of maps s, t, m, u, ι on a type of objects and a type of morphisms `Chapter7/Groupoid · GroupoidStr`, with associativity, units and inverses required on composable pairs; the morphisms $x \to y$ are a set `Chapter7/Groupoid · GroupoidStr.mor`, and so are the morphisms between points of a set A `Chapter7/Groupoid · GroupoidStr.morOn`. The composition of a non-composable pair is a junk value; no statement uses it. The text's list of consequences — $(\varphi^{-1})^{-1} = \varphi$, $1_x^{-1} = 1_x$, $(\varphi \circ \psi)^{-1} = \psi^{-1} \circ \varphi^{-1}$, uniqueness of inverses — is derived from the laws `Chapter7/Groupoid · GroupoidStr.inv_inv` `Chapter7/Groupoid · GroupoidStr.inv_unit` `Chapter7/Groupoid · GroupoidStr.inv_comp_rev` `Chapter7/Groupoid · GroupoidStr.eq_inv_of_comp_eq_unit`. Objects are isomorphic if there is a morphism between them `Chapter7/Groupoid · GroupoidStr.Isomorphic`, an equivalence relation `Chapter7/Groupoid · GroupoidStr.orbitSetoid` whose quotient is $|X|$ `Chapter7/Groupoid · GroupoidStr.OrbitSpace` with projection π `Chapter7/Groupoid · GroupoidStr.π`; $\pi(x) = \pi(y)$ iff $x \cong y$ `Chapter7/Groupoid · GroupoidStr.π_eq_π_iff`,

$\pi(s\varphi) = \pi(t\varphi)$ Chapter7/Groupoid · GroupoidStr.π_src_eq_π_tgt and π is onto Chapter7/Groupoid · GroupoidStr.surjective_π. The saturation $\pi^{-1}(\pi(A))$, used from Definition 7.1.21 (2017: 7.10) on, is defined here Chapter7/Groupoid · GroupoidStr.saturation, with saturated sets Chapter7/Groupoid · GroupoidStr.IsSaturated Chapter7/Groupoid · GroupoidStr.isSaturated_iff. The bridge to Mathlib's groupoids is a type synonym of the objects Chapter7/Groupoid · GroupoidStr.Ob with the groupoid instance Chapter7/Groupoid · GroupoidStr.instGroupoid, whose isomorphism classes are those of the orbit relation Chapter7/Groupoid · GroupoidStr.orbitSetoid_eq (Findings, Chapter 7, item 1) (pp. 287–288 (2017 pp. 225–226)).

Definition 7.1.2 (2017: 7.2) (V7.2). G_x, the morphisms $x \to x$ Chapter7/Groupoid · GroupoidStr.Isotropy, is a group under composition with $gh = g \circ h$ Chapter7/Groupoid · GroupoidStr.Isotropy.val_mul Chapter7/Groupoid · GroupoidStr.Isotropy.val_one Chapter7/Groupoid · GroupoidStr.Isotropy.val_inv, the text's order. Mathlib's vertex group multiplies in the other order; it is identified with the opposite group Chapter7/Groupoid · GroupoidStr.isotropyEquivVertexGroup, and G_x with the automorphisms of x in the bridge Chapter7/Groupoid · GroupoidStr.isotropyEquivAut (Findings, Chapter 7, item 2) (p. 288 (2017 p. 226)).

VII.2 Definition 7.1.3 (2017: 7.3), and properness

Definition 7.1.3 (2017: 7.3) (V7.3) Chapter7/EpGroupoid · EpGroupoid. An ep-groupoid is a groupoid with M-polyfold structures on the objects and on the morphisms such that (1) s and t are local sc-diffeomorphisms, (2) u and ι are sc-smooth, (3) every x has an open neighbourhood $V(x)$ with $t : s^{-1}(\mathrm{cl}_X V(x)) \to X$ proper, and (4) m is sc-smooth on $\mathbf{X}\,{}_s\times_t \mathbf{X}$. "In particular all structure maps are sc-smooth": s and t are Chapter7/EpGroupoid · EpGroupoid.isScSmoothMap_src Chapter7/EpGroupoid · EpGroupoid.isScSmoothMap_tgt, and continuous Chapter7/EpGroupoid · EpGroupoid.continuous_src Chapter7/EpGroupoid · EpGroupoid.continuous_tgt Chapter7/EpGroupoid · EpGroupoid.continuous_unit Chapter7/EpGroupoid · EpGroupoid.continuous_inv (pp. 288–290 (2017 pp. 226–227)).

Condition (3). It is existential, one open $V(x)$ per point, as the text writes it. The universal reading, "for every open $V \ni x$", fails: in the pair groupoid of $\mathbb{Z}$ (objects $\mathbb{Z}$, one morphism between any two objects) t is proper over $s^{-1}(\{x\})$ but not over $s^{-1}(\mathbb{Z})$ Chapter7/NonVacuity · Examples.pair_isProperMap Chapter7/NonVacuity · Examples.pair_not_isProperMap_univ Chapter7/NonVacuity · Examples.pair_not_forall_isProperMap; every smaller open neighbourhood of x works as soon as one does (§VII.25) (Findings, Chapter 7, items 3, 103).

Condition (4). The text cites "Lemma 2.8 in [38]" for the M-polyfold structure of the fibred product; this is an external reference, and the statement used is Part I's Proposition 2.5.20 (2017: 2.15) (§II.17): since s is a local sc-diffeomorphism, $\mathbf{X}\,{}_s\times_t \mathbf{X}$ is a sub-M-polyfold of $\mathbf{X} \times \mathbf{X}$ Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct, and (4) is stated on it Chapter7/EpGroupoid · IsScSmoothComp. The proofs use an equivalent *sheet form* Chapter7/EpGroupoid · IsScSmoothCompSheet: for every sc-diffeomorphism $s : \mathbf{W} \to V$ with inverse σ, the map $\psi \mapsto m(\sigma(t\psi), \psi)$ is sc-smooth on $t^{-1}(V)$. *Step 1.* The graph of $\sigma \circ t$ over $t^{-1}(V)$ is an open piece of the fibred product, and the second projection is an sc-diffeomorphism from it onto $t^{-1}(V)$, with inverse $\psi \mapsto (\sigma(t\psi), \psi)$. *Step 2.* So the literal (4) gives the sheet form by composition, and conversely the fibred product is covered by such graphs Chapter7/EpGroupoid · isScSmoothComp_iff_sheet. Every ep-groupoid satisfies the sheet form Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet. The "for every σ" of the sheet form carries no vacuity risk, being equivalent to the literal condition (Findings, Chapter 7, item 4).

Properness in sequential form (V7.4). The text uses properness through sequences: in Proposition 7.1.10 (2017: 7.3), Lemma 7.1.11 (2017: 7.1), Proposition 7.1.22 (2017: 7.7), Proposition 7.3.9 (2017: 7.11), Proposition 7.5.9 (2017: 7.13) and Lemma 7.6.2 (2017: 7.8). For a continuous map between metrizable spaces,

properness is equivalent to: every sequence whose images converge has a convergent subsequence `Chapter7/ProperMaps · isProperMap_iff_exists_tendsto_subseq`. *Step 1, from properness.* The images and their limit form a compact set, whose preimage is compact, hence sequentially compact `Chapter7/ProperMaps · exists_tendsto_subseq_of_isProperMap`. *Step 2, to properness.* Preimages of compact sets are sequentially compact, hence compact in a metrizable space, and a continuous map with compact fibres of compacta into a compactly generated space is proper (Mathlib's characterization) `Chapter7/ProperMaps · isProperMap_of_exists_tendsto_subseq`. For the restriction of t to $s^{-1}(Q)$ the criterion reads: every sequence with $s(\varphi_k) \in Q$ and $t(\varphi_k)$ convergent has a subsequence converging to a morphism over Q `Chapter7/ProperMaps · isProperMap_domRestrict_iff`, over $\mathrm{cl}\,Q$ when Q is closed `Chapter7/ProperMaps · isProperMap_domRestrict_iff_of_isClosed` `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff`, and it is the compactness of $s^{-1}(Q) \cap t^{-1}(K)$ for compact K `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff_isCompact_preimage`. Three consequences recur. Properness passes to smaller closures, so any open $V' \subseteq V(x)$ around x will do `Chapter7/ProperMaps · isProperMap_domRestrict_closure_mono`; the image $t(s^{-1}(Q))$ is closed for closed Q, a proper map being closed `Chapter7/ProperMaps · isClosed_image_preimage_of_isProperMap_domRestrict`; and the fibres over points are finite when t is locally injective `Chapter7/ProperMaps · finite_setOf_of_isProperMap_domRestrict`. If t is proper over $s^{-1}(Q)$ and the units exist, Q is closed: a limit x of $x_n \in Q$ is the source of a limit of a subsequence of the units 1_{x_n} `Chapter7/ProperMaps · isClosed_of_isProperMap_domRestrict`. The hidden hypothesis is metrizability, which M-polyfolds have (Theorem 2.3.10 (2017: 2.2), §II.7) (Findings, Chapter 7, item 3).

VII.3 Definition 7.1.4 (2017: 7.4), the levels, Definitions 7.1.5 and 7.1.6 (2017: 7.5 and 7.6)

Definition 7.1.4 (2017: 7.4) (V7.5). For a morphism $\varphi : x \to y$ the text takes open $\mathbf{U}(\varphi)$, $U(x)$, $U(y)$ with $s : \mathbf{U}(\varphi) \to U(x)$ and $t : \mathbf{U}(\varphi) \to U(y)$ sc-diffeomorphisms and sets $\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} : U(x) \to U(y)$. Such a sheet exists at every morphism `Chapter7/EpGroupoid · EpGroupoid.Sheet` `Chapter7/EpGroupoid · EpGroupoid.exists_sheet`: (1) gives open neighbourhoods of φ on which s, resp. t, is an sc-diffeomorphism onto an open set, and on their intersection both are, an sc-diffeomorphism restricting to one on every open subset. A sheet is chosen at every morphism `Chapter7/EpGroupoid · EpGroupoid.sheetAt` `Chapter7/EpGroupoid · EpGroupoid.mem_sheetAt`, and $\widehat{\varphi}$ is defined from it `Chapter7/EpGroupoid · EpGroupoid.Sheet.hat` `Chapter7/EpGroupoid · EpGroupoid.hat`: an sc-diffeomorphism `Chapter7/EpGroupoid · EpGroupoid.Sheet.isScDiffeoOn_hat` `Chapter7/EpGroupoid · EpGroupoid.isScDiffeoOn_hat`, a composite of sc-diffeomorphisms, with $\widehat{\varphi}(x) = y$ `Chapter7/EpGroupoid · EpGroupoid.hat_src`. It depends on the sheet only as a germ. $T\varphi = T\widehat{\varphi}(x)$ is read through the chain rule as $Tt \circ (Ts|_{T_\varphi \mathbf{X}})^{-1}$ with Part I's tangent maps `Chapter7/EpGroupoid · EpGroupoid.tangentMor`; that it equals $T\widehat{\varphi}(s(\varphi))$, is a bounded linear isomorphism on level 0 for φ on level 1, and an sc-isomorphism for smooth φ, is proved in §VII.44, with the linear structure of T_xX that Part I now provides (gaps G.7, G.70, closed). The display "$\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} \circ U(x) \to U(y)$" is read with : for the last $\circ$, and "$T\varphi : T_{s(\varphi)}X \to T_{t(\varphi)}$" with the missing X (possible misprints; Findings, Chapter 7, items 5, 110) (2017 p. 227).

The levels and the degeneracy index (V7.6). A local sc-diffeomorphism preserves the levels and the degeneracy index: in a sheet it is an sc-diffeomorphism, and Part I's Proposition 2.4.4 (2017: 2.7) (§II.11) applies `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.mem_levelSet_iff` `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.degIndex_eq`. Hence $\varphi \in \mathbf{X}_m$ iff $s(\varphi) \in X_m$ iff $t(\varphi) \in X_m$ `Chapter7/EpGroupoid · EpGroupoid.mem_levelSet_src_iff` `Chapter7/EpGroupoid · EpGroupoid.mem_levelSet_tgt_iff`, and $d_X(s\varphi) = d_{\mathbf{X}}(\varphi) = d_X(t\varphi)$ `Chapter7/EpGroupoid · EpGroupoid.degIndex_src` `Chapter7/EpGroupoid · EpGroupoid.degIndex_tgt`. The text's $ml(x)$,

the largest m with $x \in X_m$, is the maximal level `Chapter7/LocalDiffeo · MPolyfold.maxLevel`, preserved too `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.maxLevel_eq`. $|X_k| = \pi(X_k)$ is defined `Chapter7/EpGroupoid · EpGroupoid.orbitLevel`; the set $|X_\infty|$, the identity $|X_\infty| = \bigcap_k |X_k|$ and the density of X_∞ in X_k are §VII.44 (gaps G.72, G.73, closed; Findings, Chapter 7, items 6, 112, 116, 153) (pp. 289–291 (2017 pp. 227–228)).

Definition 7.1.5 (2017: 7.5) (V7.7). $d_{|X|}(|x|) = d_X(x)$ is well defined by the previous paragraph `Chapter7/EpGroupoid · EpGroupoid.degIndexOrbit` `Chapter7/EpGroupoid · EpGroupoid.degIndexOrbit_π` (2017 p. 228).

Definition 7.1.6 (2017: 7.6) (V7.8). X is tame if the object M-polyfold is tame `Chapter7/EpGroupoid · EpGroupoid.IsTame`, "tame" being the existence of an equivalent tame atlas (Definition 2.5.7 (2017: 2.19), §II.14) `Chapter2/Tame · MPolyfold.IsTame` — not "every chart is tame", which would be unsatisfiable. *Then* **X** *is tame.* At a morphism, a sheet $s : \mathbf{W} \to V$ composed with a tame chart of X around $s(\varphi)$ is a tame chart of **X**, since tameness is carried by sc-diffeomorphisms (§III.27) `Chapter7/LocalDiffeo · MPolyfold.IsScDiffeoOn.exists_tame_chart`; these charts are compatible with the atlas of **X** and form a tame atlas `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isTame`; conversely a surjective local sc-diffeomorphism transfers tameness down `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isTame_iff_of_surjective`. The witnesses are the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn` and the reflection groupoid of §VII.30 (Findings, Chapter 7, items 7, 106) (2017 p. 228).

VII.4 Definition 7.1.7 (2017: 7.7), and the remark after it

Definition 7.1.7 (2017: 7.7) (V7.9). A functor between groupoids is a pair of maps commuting with s, t, m on composable pairs and u `Chapter7/Groupoid · GroupoidFunctor`; it commutes with ι, by the uniqueness of inverses `Chapter7/Groupoid · GroupoidFunctor.mor_inv`, has identities and composites `Chapter7/Groupoid · GroupoidFunctor.id` `Chapter7/Groupoid · GroupoidFunctor.comp`, a Mathlib functor `Chapter7/Groupoid · GroupoidFunctor.toFunctor` and a map of orbit spaces `Chapter7/Groupoid · GroupoidFunctor.orbitMap`. An ep-functor is sc-smooth on objects and on morphisms `Chapter7/EpGroupoid · EpFunctor` `Chapter7/EpGroupoid · EpFunctor.id` `Chapter7/EpGroupoid · EpFunctor.comp` `Chapter7/EpGroupoid · EpFunctor.ext`, and $\mathcal{EP}$ is the category of bundled ep-groupoids and ep-functors `Chapter7/EpGroupoid · EP` (Findings, Chapter 7, item 8) (2017 p. 228).

The remark after Definition 7.1.7 (2017: 7.7) (V7.10). The text says that it suffices to ask sc-smoothness between the object spaces, because near φ one has $f(\gamma) = (s|\mathbf{U}(\psi))^{-1}(f(s(\gamma)))$ for γ in a neighbourhood $\mathbf{U}(\varphi)$ with $f(\mathbf{U}(\varphi)) \subseteq \mathbf{U}(\psi)$, $\psi = f(\varphi)$. Choosing that neighbourhood uses the continuity of f on morphisms, and without it the claim does not hold as we read it. *The counterexample.* On the trivial action of $\mathbb{Z}/3$ on $\mathbb{R}$ (objects $\mathbb{R}$, morphisms $\mathbb{Z}/3 \times \mathbb{R}$, $s = t$ the projection), let f be the identity on objects and $(g, y) \mapsto (g, y)$ for $y \le 0$, $(g, y) \mapsto (g^2, y)$ for $y > 0$ `Chapter7/NonVacuity · Examples.remarkFunctor`. Since $g \mapsto g^2$ is an automorphism of $\mathbb{Z}/3$, this is a functor; it is sc-smooth on objects `Chapter7/NonVacuity · Examples.isScSmoothMap_remarkFunctor_obj` but not continuous on morphisms, jumping at $(g, 0)$ for $g \neq 1$ `Chapter7/NonVacuity · Examples.not_continuous_remarkFunctor_mor`, hence not sc-smooth there `Chapter7/NonVacuity · Examples.not_isScSmoothMap_remarkFunctor_mor`, and no ep-functor has these components `Chapter7/NonVacuity · Examples.not_exists_epFunctor_remarkFunctor` `Chapter7/NonVacuity · Examples.remark_definition_7_1_7_counterexample`. *The remark for functors continuous on morphisms.* Take a sheet $\mathbf{U}(\psi)$ of s at $\psi = f(\varphi)$, with inverse σ, and $\mathbf{U}(\varphi) = f^{-1}(\mathbf{U}(\psi))$, open by continuity. For $\gamma \in \mathbf{U}(\varphi)$, $f(\gamma)$ lies in $\mathbf{U}(\psi)$ and has source $f(s\gamma)$, so $f(\gamma) = \sigma(f(s\gamma))$ `Chapter7/Basic · EpGroupoid.mor_eq_srcInv_obj_src`, a composite of sc-smooth maps `Chapter7/Basic · EpGroupoid.isScSmoothMap_mor_of_continuous`. The sheet at φ

and the neighbourhoods $U(s\varphi)$, $U(s\psi)$ of the text are not used. Since every ep-functor is continuous on morphisms `Chapter7/Basic · EpFunctor.continuous_mor`, the remark becomes an equivalence for functors sc-smooth on objects `Chapter7/Basic · EpGroupoid.isScSmoothMap_mor_iff_continuous`, and a functor sc-smooth on objects and continuous on morphisms is an ep-functor `Chapter7/Basic · EpFunctor.ofContinuous`. *The 2021 sentence* (p. 291) asks continuity on level 0 between the morphism spaces, which is the hypothesis of the form just proved, so the 2021 remark holds as we read it (§VII.33); the counterexample concerns the 2017 sentence only `Chapter7/NonVacuity · EpGroupoid.Examples.remark_definition_7_1_7_counterexample` (Findings, Chapter 7, items 9, 104) (pp. 290–292 (2017 pp. 228–229)).

VII.5 Propositions 7.1.8–7.1.10 (2017: 7.1–7.3): the topology of $|X|$

Proposition 7.1.8 (2017: 7.1) (V7.11) `Chapter7/Basic · EpGroupoid.proposition_7_1_8`. *π is open.* $|X|$ carries the quotient topology, so π is continuous `Chapter7/EpGroupoid · EpGroupoid.continuous_π`, and $\pi(U)$ is open iff its saturation $\pi^{-1}(\pi(U))$ is. *Step 1.* The saturation of U is $t(s^{-1}(U))$: $y \cong x \in U$ iff some φ has $s\varphi \in U$ and $t\varphi = y$ `Chapter7/Basic · EpGroupoid.preimage_π_image_eq`. *Step 2.* $s^{-1}(U)$ is open since s is continuous, and t is an open map, being a local sc-diffeomorphism `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isOpenMap`. The text argues with the sheets $V(\varphi)$, which is the openness of t unpacked. So π is an open quotient map `Chapter7/Basic · EpGroupoid.proposition_7_1_8_isOpenQuotientMap` and carries neighbourhood filters to neighbourhood filters `Chapter7/Basic · EpGroupoid.proposition_7_1_8_map_nhds` (Findings, Chapter 7, item 10) (2017 p. 229).

Proposition 7.1.9 (2017: 7.2) (V7.12) `Chapter7/Basic · EpGroupoid.proposition_7_1_9`. *Every $z \in |X|$ has a countable basis of open neighbourhoods, chosen monotone.* X is metrizable (Theorem 2.3.10 (2017: 2.2)); for $z = \pi(x)$ the images of the balls $B(x, 1/(k+1))$ are open by Proposition 7.1.8 (2017: 7.1) and decreasing, and they form a basis because π maps the neighbourhood filter of x onto that of z `Chapter7/Basic · EpGroupoid.proposition_7_1_9_of_hasAntitoneBasis`. The statement is Mathlib's antitone basis indexed by $\mathbb{N}$ (radii $1/(k+1)$, the index starting at 0), and $|X|$ is first countable `Chapter7/Basic · EpGroupoid.proposition_7_1_9_firstCountableTopology` (Findings, Chapter 7, item 10) (2017 p. 229).

Proposition 7.1.10 (2017: 7.3) (V7.13) `Chapter7/Basic · EpGroupoid.proposition_7_1_10`. *Every x has an open U with $\pi(\mathrm{cl}_X A) = \mathrm{cl}_{|X|}\pi(A)$ for every $A \subseteq U$.* The text proves "$\supseteq$" with a sequence $\varphi_k : a_k \to b_k$, $b_k \to b$, $a_k \in A$, of which properness extracts a convergent subsequence. The formalization proves more, for every U with t proper over $s^{-1}(\mathrm{cl}_X U)$ and every $A \subseteq \mathrm{cl}_X U$, and without sequences `Chapter7/Basic · EpGroupoid.proposition_7_1_10_of_isProperMap`. *Step 1.* Since π is open and continuous, $\pi^{-1}(\mathrm{cl}\,\pi(A)) = \mathrm{cl}\,\pi^{-1}(\pi(A)) = \mathrm{cl}\,t(s^{-1}(A))$. *Step 2.* $t(s^{-1}(\mathrm{cl}A))$ is closed, as the image of the closed $s^{-1}(\mathrm{cl}A) \subseteq s^{-1}(\mathrm{cl}U)$ under a proper, hence closed, map (§VII.2), and contains $t(s^{-1}(A))$; so $\mathrm{cl}\,\pi(A) \subseteq \pi(t(s^{-1}(\mathrm{cl}A))) = \pi(\mathrm{cl}A)$. *Step 3.* "$\subseteq$" is the continuity of π. "For every z and every representative x" is "for every x" (Findings, Chapter 7, item 11) (pp. 291–293 (2017 pp. 229–230)).

VII.6 Lemma 7.1.11 (2017: 7.1), Proposition 7.1.12 (2017: 7.4), and the conjugation $G_x \cong G_y$

Lemma 7.1.11 (2017: 7.1) (V7.14) `Chapter7/Basic · EpGroupoid.lemma_7_1_11`. *If $t : s^{-1}(\mathrm{cl}_X U(x_0)) \to X$ is proper, only finitely many ψ have $s(\psi) = x$ and $t(\psi) \in U(x_0)$.* **The text's proof** argues by contradiction with distinct $\psi_n : s(\psi_n) \to x$, $s(\psi_n) \in U(x_0)$; properness gives a convergent subsequence, t is a local

sc-diffeomorphism near the limit, and $t(\psi_n) = t(\psi_m) = x$ forces $\psi_n = \psi_m$. The proof thus treats the inverses of the morphisms of the statement, with s and t exchanged, and writes "$t : V(\psi) \to V(y)$" for $V(x)$ and "$s(\psi) \in s^{-1}(\mathrm{cl}_X(U(x_0)))$"; parentheses are unbalanced in the statement and the proof (possible misprints; Findings, Chapter 7, items 12, 96). **The formalized proof** does the proof's form first and has no sequences. *Step 1.* The fibre of the proper restriction of t over x is compact. *Step 2.* It is discrete, t being locally injective as a local sc-diffeomorphism `Chapter7/Discrete · MPolyfold.IsLocalScDiffeo.isLocallyInjective` `Chapter7/Discrete · IsLocallyInjective.isDiscrete_preimage_singleton`. *Step 3.* A compact discrete set is finite `Chapter7/Discrete · IsProperMap.finite_preimage_singleton_of_isLocallyInjective` `Chapter7/Discrete · finite_setOf_mem_of_isProperMap_domRestrict` `Chapter7/Discrete · MPolyfold.finite_setOf_mem_of_isLocalScDiffeo`. This gives the finiteness of $\{\psi \mid s\psi \in \mathrm{cl}_X U(x_0),\ t\psi = x\}$ `Chapter7/Basic · EpGroupoid.lemma_7_1_11_swap_of_isProperMap` `Chapter7/Basic · EpGroupoid.lemma_7_1_11_swap`. *Step 4.* The statement's form follows by ι, which exchanges s and t and is injective, and holds even for targets in $\mathrm{cl}_X U(x_0)$ `Chapter7/Basic · EpGroupoid.lemma_7_1_11_of_isProperMap`. The openness of $U(x_0)$ and $x_0 \in U(x_0)$ are kept as hypotheses but not used (2017 p. 230).

Proposition 7.1.12 (2017: 7.4) (V7.15) `Chapter7/Basic · EpGroupoid.proposition_7_1_12`. *G_x is finite.* Lemma 7.1.11 (2017: 7.1) with $x_0 = x$ and $U(x_0) = V(x)$ of Definition 7.1.3 (3) (2017: 7.3): G_x is contained in the finite set of morphisms with source x and target in $V(x)$ `Chapter7/Discrete · MPolyfold.finite_isotropy_of_isLocalScDiffeo`. The proposition is an instance, so the cardinality of G_x is its order as a finite group (Findings, Chapter 7, item 13) (2017 p. 230).

The conjugation (V7.16). For $\varphi : x \to y$, $g \mapsto \varphi \circ g \circ \varphi^{-1}$ is an isomorphism $G_x \to G_y$ `Chapter7/Groupoid · GroupoidStr.isotropyConj` `Chapter7/Groupoid · GroupoidStr.isotropyConj_val` `Chapter7/Basic · EpGroupoid.conjIsotropy` `Chapter7/Basic · EpGroupoid.conjIsotropy_val`, so isomorphic objects have isomorphic isotropy groups of the same order `Chapter7/Basic · EpGroupoid.nonempty_isotropy_mulEquiv` `Chapter7/Basic · EpGroupoid.nat_card_isotropy_eq`. "There is no canonical identification" is given its content: two choices φ, φ' give isomorphisms that differ by conjugation with $\varphi'\varphi^{-1} \in G_y$, an inner automorphism `Chapter7/Basic · EpGroupoid.isotropyConj_eq_isotropyConj_conj` (Findings, Chapter 7, item 13) (2017 p. 230).

VII.7 Theorem 7.1.13 (2017: 7.1), and local path connectedness

Theorem 7.1.13 (2017: 7.1) (V7.17) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13`. *For every object x and every open $V \ni x$ there are an open neighbourhood $U \subseteq V$ of x, a homomorphism $\Phi : G_x \to \mathrm{Diff}_{sc}(U)$ and an sc-smooth $\Gamma : G_x \times U \to \mathbf{X}$ with (1) $\Gamma(g, x) = g$, (2) $s(\Gamma(g, y)) = y$ and $t(\Gamma(g, y)) = \Phi(g)(y)$, (3) every morphism $h : y \to z$ between points of U is $\Gamma(g, y)$ for a unique g.* The data form a *natural representation* at x `Chapter7/EpGroupoid · EpGroupoid.NaturalRep`: U open, connected and containing x; $\Phi(g)$ an sc-diffeomorphism of U with inverse $\Phi(g^{-1})$ and $\Phi(gh) = \Phi(g) \circ \Phi(h)$ on U; Γ sc-smooth on each slice $\{g\} \times U$, which is sc-smoothness on $G_x \times U$ (§VII.8), with (1)–(3). The first consequences: $\Phi(g)$ maps U into U `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.mapsTo_Φ`, $\Phi(1)$ is the identity on U `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Φ_one`, every $\Phi(g)$ fixes x `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Φ_self`, $\Gamma(g, y)$ is a morphism between points of U `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Γ_mem_morOn`, $\Phi(g^{-1})\Phi(g) = 1$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_inv_Φ`, and (3) is a bijection $G_x \times U \to \mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.bijOn_Γ`. The action $g * y = \Phi(g)(y)$ is a group action on U `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.smul_val`. The proof is the text's Theorem 7.6.1 (2017: 7.5) in the appendix (§VII.26). *Hidden hypothesis: U* connected. The text chooses "the connected open neighborhood $U(x)$" on p. 293 (2017 p. 231) but does not put it into the statement; as we read it, the homomorphism property of Φ, the functoriality of Γ and the uniqueness of (Φ, Γ) all need it (Findings, Chapter 7, item 20) (2017 p. 231).

Local path connectedness (V7.18). Connected neighbourhoods exist because M-polyfolds are locally path connected, which the text uses without statement (2017 p. 231, Lemma 7.6.3 (2017: 7.9), Corollary 7.3.8 (2017: 7.2)). *Step 1, a retract.* For $o \in O = r(U)$, U open in a partial quadrant C, and a neighbourhood W of o, there is $\varepsilon > 0$ such that every $p \in O$ with $\|p - o\| < \varepsilon$ is joined to o inside $O \cap W$ by the path $\tau \mapsto r(o + \tau(p - o))$: the segment lies in the convex set $B_\varepsilon(o) \cap C \subseteq U$, r is continuous on level 0 and fixes o and p `Chapter7/LocConnected · ScRetraction.exists_ball_joinedIn`. So an sc-smooth retract is locally path connected `Chapter7/LocConnected · IsScRetract.locallyPathConnectedSpace`. *Step 2, an M-polyfold.* Through any chart around a point, path components of open sets are neighbourhoods `Chapter7/LocConnected · MChart.pathComponentIn_mem_nhds`, and X is locally path connected `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace`, hence locally connected `Chapter7/LocConnected · MPolyfold.locallyConnectedSpace`. *Step 3, consequences.* Open connected neighbourhoods form a basis `Chapter7/LocConnected · MPolyfold.hasBasis_isOpen_isConnected` `Chapter7/LocConnected · MPolyfold.hasBasis_isOpen_isPathConnected` `Chapter7/LocConnected · MPolyfold.exists_isOpen_isConnected_subset`; components, also of open sets, are open `Chapter7/LocConnected · MPolyfold.isOpen_connectedComponent` `Chapter7/LocConnected · MPolyfold.isOpen_connectedComponentIn`; an open set is connected iff path connected `Chapter7/LocConnected · MPolyfold.isConnected_iff_isPathConnected`; every quotient of X, in particular $|X|$, is locally path connected `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace_quotient`; and finitely many points of a Hausdorff locally connected space have pairwise disjoint connected open neighbourhoods inside a given open set `Chapter7/LocConnected · exists_pairwiseDisjoint_isOpen_isConnected`. *Step 4, the tool for connectedness.* Two continuous sections of a locally injective map into a Hausdorff space that agree at one point of a preconnected set agree on all of it `Chapter7/LocConnected · eqOn_sections_of_isPreconnected`, by Mathlib's statement for separated maps `Mathlib/Topology/SeparatedMap · eqOn_of_comp_eqOn`. For s this says that a local inverse of s on a connected U is determined by one value; it is what makes Φ a homomorphism (§VII.26) and Γ a functor (§VII.8). The paths are continuous on level 0; sc-smooth paths between smooth points are Chapter 6's (§VI.18). This is the first fact of gap G.56 of Part I, now formalized (Findings, Chapter 7, items 21, 114).

VII.8 Pages 231–233: the sheets $\mathbf{U}(g)$, $G_x \ltimes U(x)$, the functor Γ, and $\hat{\Phi}$

Everything on these pages is derived from one natural representation (U, Φ, Γ); the text's "using the proof of Theorem 7.1 one verifies" is not needed. No possible misprint was found on pp. 293–297 (2017 pp. 231–233) (Findings, Chapter 7, item 22).

The splitting $\mathbf{U}(x) = \bigsqcup_g \mathbf{U}(g)$ (V7.19). Put $\mathbf{U}(g) = \Gamma(g, U)$. *Step 1, open.* The image of a continuous section of a local homeomorphism over an open set is open `Chapter7/NaturalAction · isOpen_image_of_section`; $\Gamma(g, \cdot)$ is a section of s by (2) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_image_Γ`, containing $\Gamma(g, y)$ for every $y \in U$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_mem_image`. *Step 2, connected, disjoint, covering.* $\mathbf{U}(g)$ is connected as a continuous image of U `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isConnected_image_Γ`; two sheets are disjoint by the uniqueness in (3) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_inj` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.pairwise_disjoint_image_Γ`; they cover $\mathbf{U}(x)$ by the existence in (3) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.iUnion_image_Γ`, which is therefore open `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_morOn`. *Step 3, components.* Open, connected, disjoint and covering, they are the connected components of $\mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.connectedComponentIn_morOn`, and there are $\#G_x$ of them `Chapter7/NaturalAction · EpGroupoid.NaturalRep.injective_image_Γ`. *Step 4.* $s, t : \mathbf{U}(g) \to U$ are sc-diffeomorphisms, with inverses $\Gamma(g, \cdot)$ and $\Gamma(g, \cdot) \circ \Phi(g)^{-1}$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_src_image` `Chapter7/`

NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_tgt_image. Every $\varphi \in \mathbf{U}(x)$ has a sheet index h Chapter7/NaturalAction · EpGroupoid.NaturalRep.index with $\varphi = \Gamma(h, s\varphi)$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_index and $t\varphi = \Phi(h)(s\varphi)$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.tgt_eq_Φ_index, locally constant on $\mathbf{U}(x)$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.eventually_index_eq. Once U is connected, the text's "choose $U(x)$ so small that $\mathbf{U}(x)$ splits into $\#G_x$-many components" needs no shrinking. "$\mathbf{U}(x)$ fits into an arbitrarily small open neighbourhood of G_x" belongs to the construction: Lemma 7.6.2 (2017: 7.8) holds for arbitrary open $N_g \ni g$ (§VII.26) Chapter7/NaturalRep · EpGroupoid.lemma_7_6_2 (2017 p. 231).

The M-polyfold $G \times U$ (V7.20). "A natural M-polyfold structure as the finite disjoint union of copies of U" is the product with a discrete M-polyfold. *Step 1.* A discrete space is an M-polyfold with the charts $\{g\} \to \mathbb{R}^0$ Chapter7/Discrete · discreteChart Chapter7/Discrete · discretePolyfold Chapter7/Discrete · isChart_discreteChart, pairwise compatible because their transitions are the identity of the one-point space or the empty map Chapter7/Discrete · discreteChart_compatible; every point is smooth Chapter7/Discrete · discretePolyfold_smoothSet, and a map into it is sc-smooth iff continuous Chapter7/Discrete · isScSmoothMap_discrete_iff. *Step 2.* In $G \times X$ the slices $\{g\} \times X$ are open Chapter7/Discrete · isOpen_sheet, $x \mapsto (g, x)$ is sc-smooth Chapter7/Discrete · isScSmoothMap_sheet and the projection of a slice is an sc-diffeomorphism Chapter7/Discrete · isScDiffeoOn_snd_sheet, with $d_{G\times X}(g, x) = d_X(x)$ Chapter7/Discrete · degIndex_prod_discrete. *Step 3.* Hence a map on $G \times U$ is sc-smooth iff it is on each slice Chapter7/Discrete · isScSmoothOn_prod_discrete_iff Chapter7/Discrete · isScSmoothOn_univ_prod_discrete_iff Chapter7/Discrete · isScSmoothMap_prod_discrete_iff, the text's sheet-by-sheet reading of "Γ sc-smooth on $G \times U$". For a natural representation the objects are the open-subset M-polyfold U Chapter7/NaturalAction · EpGroupoid.NaturalRep.objPolyfold, the morphisms $G_x \times U$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.morPolyfold, and $(g, y) \mapsto g*y$ is sc-smooth Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScSmoothMap_smul (Findings, Chapter 7, item 23) (2017 p. 232).

The translation groupoid $G_x \ltimes U(x)$ **is an ep-groupoid** (V7.21). The groupoid has objects U, morphisms $G\times U$, $s(g, y) = y$, $t(g, y) = g*y$, $(g, h*z)\circ(h, z) = (gh, z)$ Chapter7/Groupoid · GroupoidStr.translation Chapter7/EpGroupoid · EpGroupoid.NaturalRep.translationGroupoid. *(1).* s is the projection, a local sc-diffeomorphism by Step 2 above Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isLocalScDiffeo_src; t is the projection after the shear $(g, y) \mapsto (g, g * y)$, an sc-diffeomorphism with inverse $(g, y) \mapsto (g, g^{-1} * y)$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.shearHomeomorph Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isLocalScDiffeo_tgt. *(3), for every* V. t is the shear followed by the projection $G \times U \to U$, proper because G_x is finite Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isProperMap. *(4), sheet form.* For a local inverse σ of s, $\sigma(t\psi) \circ \psi = (k(\psi)\psi_1, \psi_2)$ with k locally constant, sc-smooth on each slice Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isScSmoothCompSheet. So $G_x \ltimes U(x)$ is an ep-groupoid Chapter7/NaturalAction · EpGroupoid.NaturalRep.translationEpGroupoid, built from (Φ, Γ) alone (2017 p. 232).

The functor $\Gamma : G_x \ltimes U(x) \to X$ (V7.22). *Step 1, composition.* $y \mapsto \Gamma(g, h*y)\circ\Gamma(h, y)$ and $y \mapsto \Gamma(gh, y)$ are continuous sections of s over U — the first by (4) in the sheet form, for the sheet $\mathbf{U}(g)$ — with the same value $g \circ h$ at x by (1); on the connected U they agree (§VII.7) Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_mul. *Step 2.* In the same way $\Gamma(1, y) = 1_y$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_one, and $\Gamma(g^{-1}, g*y) = \Gamma(g, y)^{-1}$ by the uniqueness of inverses Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_inv. *Step 3.* So Γ, the inclusion on objects, is a functor Chapter7/NaturalAction · EpGroupoid.NaturalRep.functor, full and faithful by (3) Chapter7/NaturalAction · EpGroupoid.NaturalRep.full Chapter7/NaturalAction · EpGroupoid.NaturalRep.faithful, onto the full subcategory on U Chapter7/Groupoid · GroupoidStr.fullSub, and an ep-functor Chapter7/NaturalAction · EpGroupoid.NaturalRep.epFunctor. *Hidden hypothesis:* U connected, without which the sheet index of $\Gamma(g, h * y) \circ \Gamma(h, y)$ is only locally constant (2017 p. 232).

The action $\hat{\Phi}$ on $\mathbf{U}(x)$ (V7.23). The text defines $\hat{\Phi}(g)(\varphi) = \Gamma(ghg^{-1}, g * s(\varphi))$ for $\varphi \in \mathbf{U}(h)$. The formalization defines the total map $\hat{\Phi}(g)(\varphi) = \Gamma(g, t\varphi) \circ \varphi \circ \Gamma(g, s\varphi)^{-1}$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat` and recovers the text's formula from Γ_{mul} and Γ_{inv} `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_Γ`, together with $\Phi(g)\Phi(h)\Phi(g)^{-1} = \Phi(ghg^{-1})$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_conj`. Then $\hat{\Phi}(g)$ maps $\mathbf{U}(h)$ onto $\mathbf{U}(ghg^{-1})$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_mem_image` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.image_Φhat` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.index_Φhat`; the two diagrams are $s \circ \hat{\Phi}(g) = \Phi(g) \circ s$ and $t \circ \hat{\Phi}(g) = \Phi(g) \circ t$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.src_Φhat` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.tgt_Φhat`; $\hat{\Phi}(1) = 1$ and $\hat{\Phi}(gh) = \hat{\Phi}(g)\hat{\Phi}(h)$ on $\mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_one` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_mul`; and $\hat{\Phi}(g)$ is an sc-diffeomorphism of $\mathbf{U}(x)$ with inverse $\hat{\Phi}(g^{-1})$, sc-smooth by (4) in the sheet form and the sc-smoothness of ι `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_Φhat` (pp. 294–297 (2017 pp. 232–233)).

VII.9 Uniqueness and restriction of natural representations

(V7.24.) The text uses without statement that natural representations can be shrunk and compared: "adapting the corresponding restrictions" (Lemma 7.2.3 (2017: 7.3)), "$U(x)$ can be chosen so small" (T_x, 2017 p. 241). Let $R = (U, \Phi, \Gamma)$ and $R' = (U', \Phi', \Gamma')$ be natural representations at x and C the connected component of x in $U \cap U'$, open and connected (§VII.7) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_connectedComponentIn_inter` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isConnected_connectedComponentIn_inter`. *Step 1.* $\Gamma(g, \cdot)$ and $\Gamma'(g, \cdot)$ are continuous sections of s over C with the value g at x, so they agree on C `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_eqOn`; applying t, $\Phi(g) = \Phi'(g)$ on C `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_eqOn`. *Step 2.* C is invariant: $\Phi(g)(C)$ is connected, contains x and lies in $U \cap U'$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.mapsTo_Φ_connectedComponentIn`. *Step 3.* So a natural representation restricts to every invariant connected open neighbourhood of x inside U `Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrict` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.restrict` `Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_restrict`, in particular to C, where the restrictions of R and R' coincide `Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrictInter` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrictInter_eq`, and natural representations exist inside every open $V \ni x$ `Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_shrink`. *Step 4.* If $\Phi(g) = 1$ for one natural representation, then $\Phi'(g) = 1$ near x for every one `Chapter7/NaturalAction · EpGroupoid.NaturalRep.eventually_Φ_eq_of_actsTrivially`. Uniqueness holds on the component C; $U \cap U'$ itself need not be connected (Findings, Chapter 7, item 24).

VII.10 Proposition 7.1.15 (2017: 7.5) and Definition 7.1.18 (2017: 7.8): the star construction

Proposition 7.1.15 (2017: 7.5) (V7.25) `Chapter7/NaturalRepFinite · EpGroupoid.proposition_7_1_15_2017`. *Let Y be a finite set of mutually isomorphic objects and V a neighbourhood of Y. There is a functor f assigning to every $y \in Y$ a connected open $U(y) \subseteq V$, mutually disjoint, and to every $\psi : y \to y'$ between points of Y an sc-diffeomorphism $f(\psi) : U(y) \to U(y')$, such that the morphisms between points of $\bigcup U(y)$ split into components $\mathbf{U}(\psi)$ with (1) the natural action of G_y on $U(y)$, (2) $s : \mathbf{U}(\psi) \to U(y)$, $t : \mathbf{U}(\psi) \to U(y')$ sc-diffeomorphisms, (3) $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$.* The datum is a structure `Chapter7/EpGroupoid · EpGroupoid.NaturalRepFinite` (Definition 7.1.18 (2017: 7.8), V7.26), "decomposes into the connected components" being

read as: the $\mathbf{U}(\psi)$ are open, connected, pairwise disjoint and cover; they are then the components Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.connectedComponentIn_morOn (pp. 295–305 (2017 pp. 233–236)).

The text's proof (pp. 297–305 (2017 pp. 234–236)) is an induction on $\#Y$. $\#Y = 1$ is Theorem 7.1.13 (2017: 7.1). For $Y = \{y_1, y_2\}$ it fixes $\gamma_1 : y_1 \to y_2$ with a sheet and $\hat{\gamma}_1 = t \circ (s|\cdot)^{-1}$, takes a G_{y_1}-invariant connected $U(y_1)$ with $\hat{\gamma}_1(U(y_1))$ inside a G_{y_2}-invariant $W(y_2)$, puts $U(y_2) = \hat{\gamma}_1(U(y_1))$ and shows it invariant; the other morphisms $y_1 \to y_2$ are $g \circ \gamma_1$, $g \in G_{y_2}$, with components $\Phi_g(\mathbf{U}(\gamma_1))$, $\Phi_g(\varphi) = \Gamma(g, t\varphi) \circ \varphi$, and the morphisms $y_2 \to y_1$ get the inverse components. For $Y = Y' \cup \{y_{k+1}\}$ it fixes $\delta_j : y_{k+1} \to y_j$, shrinks as in the case of two objects, and adjusts the components of the morphisms of Y' to $(s|\mathbf{U}'(\gamma))^{-1}(U(y_i))$, showing $t(\mathbf{U}(\gamma)) = U(y_j)$.

The formalized proof (V7.25) is a *star*: everything is transported from one base object, and nothing is shrunk afterwards. *Step 1, transport along a sheet.* Let $R = (U, \Phi, \Gamma)$ be a natural representation at x and $\mathbf{S}_1, \mathbf{S}_2$ sheets of morphisms out of x with local inverses $a_i = (s|\mathbf{S}_i)^{-1}$ and $U \subseteq s(\mathbf{S}_i)$, $\hat{\gamma}_i = t \circ a_i$. The images $\hat{\gamma}_i(U)$ are open and connected Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportSet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isOpen_transportSet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isConnected_transportSet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_hat_transportSet, $\hat{\gamma}_i^{-1}$ is the inverse there Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.back, and the conjugates $\hat{\gamma}_2 \circ \Phi(g) \circ \hat{\gamma}_1^{-1}$ are sc-diffeomorphisms compatible with composition Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap_conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap_one. The transported morphisms are $\Theta_g(z) = a_2(\Phi(g)u) \circ \Gamma(g, u) \circ a_1(u)^{-1}$, $u = \hat{\gamma}_1^{-1}(z)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport, with source z and target $\hat{\gamma}_2(\Phi(g)u)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.src_transport Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.tgt_transport, equal to $\gamma_2 \circ g \circ \gamma_1^{-1}$, $\gamma_i = a_i(x)$, at the point $\hat{\gamma}_1(x)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport_hat_self, sc-smooth by (4) in the sheet form, used twice, and the sc-smoothness of ι Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScSmoothOn_transport. *Step 2, the bijection.* A morphism k from $\hat{\gamma}_1(U)$ to $\hat{\gamma}_2(U)$ pulls back to the morphism $a_2(\cdot)^{-1} \circ k \circ a_1(\cdot)$ between points of U, which is $\Gamma(g, u)$ for a unique g by Theorem 7.1.13 (3) (2017: 7.1); so $k = \Theta_g(sk)$ for a unique g Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.existsUnique_transport Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport_injective. *Step 3, a natural representation at y.* For $\gamma : x \to y$, $h \mapsto \gamma^{-1} h \gamma$ is an isomorphism $G_y \to G_x$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportConj, and $U(y) = \hat{\gamma}(U)$, $\Phi_y(h) = \hat{\gamma} \circ \Phi(\gamma^{-1} h \gamma) \circ \hat{\gamma}^{-1}$, $\Gamma_y(h, \cdot) = \Theta_{\gamma^{-1} h \gamma}$ is a natural representation at y Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_U Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_Φ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_Γ, after shrinking R (same Φ, Γ) into s of a sheet of γ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.exists_spoke. *Step 4, the star.* Fix one x isomorphic to all of Y, morphisms $\gamma_y : x \to y$ with sheets $\mathbf{K}_y$, and by Hausdorff separation of the finite Y pairwise disjoint open $W_y \ni y$; take $U_0 = \bigcap_y (s(\mathbf{K}_y) \cap \hat{\gamma}_y^{-1}(W_y \cap V))$ and a natural representation inside U_0 (Theorem 7.1.13 (2017: 7.1)) Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.exists_star. Then $U(y) = \hat{\gamma}_y(U(x)) \subseteq W_y \cap V$ are pairwise disjoint. *Step 5, the functor.* $\psi \mapsto g_\psi = \gamma_{y'}^{-1} \circ \psi \circ \gamma_y \in G_x$ is a bijection from the morphisms $y \to y'$ onto G_x compatible with composition, units and inverses Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_comp Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_unit Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_inv Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_isotropy; $f(\psi) = \hat{\gamma}_{y'} \circ \Phi(g_\psi) \circ \hat{\gamma}_y^{-1}$ Chapter7/NaturalRepFinite · EpGroupoid.

NatRepFinite.Star.F is an sc-diffeomorphism Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.isScDiffeoOn_F and a functor, since $\psi \mapsto g_\psi$ is one and Φ is a homomorphism Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.F_comp Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.F_unit. *Step 6, the components.* $\mathbf{U}(\psi) = \Theta_{g_\psi}(U(y))$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.0 Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.sheet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.mem_sheet are pairwise disjoint Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.pairwiseDisjoint_sheet and cover the morphisms between points of $\bigcup U(y)$, by Step 2 Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.morOn_eq; s and t are sc-diffeomorphisms on them, as images of sections Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_src_image_of_section Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_tgt_image_of_section, and $t \circ (s|\mathbf{U}(\psi))^{-1} = f(\psi)$. *Step 7, (1).* The natural representation of G_y on $U(y)$ is the transported one of Step 3, whose Φ_y is f on G_y Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.toNaturalRepFinite; by §VII.9 it agrees near y with every other one. The text instead chooses $U(y)$ invariant under a natural representation of G_y given beforehand. The text's case of two objects already has this form ($U(y_2) = \hat{\gamma}_1(U(y_1))$); the adjustment of the components of Y' in the inductive step has no counterpart. "A neighbourhood V of Y" is a neighbourhood of the set Chapter7/NaturalRepFinite · EpGroupoid.proposition_7_1_15_of_mem_nhdsSet, reduced to an open one; $Y = \emptyset$ is allowed Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.emptyNaturalRepFinite, and enlarging V keeps a datum Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.mono. "Mutually isomorphic" is used only through one x isomorphic to all of Y. Among the possible misprints of the proof (Findings, Chapter 7, item 26): "Φ_g" for $g \in G_{y_1}$ where $t(\varphi) \in U(y_2)$ requires G_{y_2}; "$\mathbf{U}(\gamma_g) := \Psi_g(\mathbf{U}(\gamma_1))$" for Φ_g; the induction step "let $k \geq 3$", which must start at $k \geq 2$ to reach $\#Y = 3$; "$\varphi = \Gamma(g, s(\varphi)) \circ \psi$" for $\Gamma(g, t(\psi)) \circ \psi$; "$y : \mathbf{U}(\delta_j) \to U(y_j)$" for t; and "$g \in G_{y_i}$" for G_{y_j} on 2017 p. 236 (Findings, Chapter 7, items 25, 26).

VII.11 Proposition 7.1.19 (2017: 7.6) and Definition 7.1.20 (2017: 7.9): $|U(x)| = G_x\backslash U(x)$, local uniformizers

Proposition 7.1.19 (2017: 7.6) (V7.27) Chapter7/Uniformizer · EpGroupoid.proposition_7_1_19. *For a natural representation at x, $\tau : G_x\backslash U(x) \to |U(x)| = \pi(U(x))$, $G_x * y \mapsto |y|$, is a homeomorphism.* $G_x\backslash U(x)$ is the orbit space of the action $g * y$ with the quotient topology. **The text's proof**: τ is well defined and onto; it is injective because a morphism $\varphi : y \to y'$ is $\Gamma(g, y)$ by Theorem 7.1.13 (3) (2017: 7.1), so $y' = g * y$; $|U(x)|$ is open by Proposition 7.1.8 (2017: 7.1); and $\tau \circ \sigma = \pi|U(x)$ for the open quotient map σ, so τ is continuous and open. **The formalized proof.** *Step 1.* $|y| = |y'|$ iff $y' = g * y$ for some g Chapter7/Uniformizer · EpGroupoid.NaturalRep.π_eq_π_iff_exists_smul. *Step 2.* τ as a map into $|X|$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.orbitQuotientMap is continuous by the quotient topology and injective by Step 1. *Step 3, open.* The image of an open Q is $\pi(\sigma^{-1}(Q))$, and $\sigma^{-1}(Q)$ is open in the open $U(x)$, so Proposition 7.1.8 (2017: 7.1) applies; the openness of σ is not needed. So τ is an open embedding into $|X|$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_orbitQuotientMap with range $\pi(U(x))$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_orbitQuotientMap Chapter7/Uniformizer · EpGroupoid.proposition_7_1_19_apply_mk. *The second form.* The functor $G_x \ltimes U(x) \to X$ of §VII.8 is fully faithful Chapter7/Uniformizer · EpGroupoid.proposition_7_1_19_fullyFaithful, and the same argument for the orbit space of the translation groupoid, whose classes are the same orbits, gives an open embedding with range $\pi(U(x))$ Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_orbitMap Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_functor_orbitMap Chapter7/Uniformizer · EpGroupoid.proposition_7_1_19_orbitMap, with $|U(x)|_{G_x \ltimes U(x)} \cong G_x\backslash U(x)$ Chapter7/

Uniformizer · EpGroupoid.translationOrbitHomeomorph (Findings, Chapter 7, item 27) (pp. 299–306 (2017 pp. 236–237)).

Definition 7.1.20 (2017: 7.9) (V7.28). A local uniformizer around x is a fully faithful functor $\Psi_x : G_x \ltimes U(x) \to X$, the inclusion on objects, inducing a homeomorphism of $|U(x)|_{G_x \ltimes U(x)}$ onto the open $\pi(U(x))$ Chapter7/EpGroupoid · EpGroupoid.LocalUniformizer. "Embedding" is not a separate field: the injectivity on objects and morphisms, which the text notes after the definition, follows from the inclusion on objects and faithfulness Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.injective_obj Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.injective_mor; for the uniformizers built from natural representations it is also topological, Ψ_x being an open embedding on objects and on morphisms Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_obj Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_mor Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_functor_mor. "There exists a local uniformizer around every object": every natural representation is one Chapter7/Uniformizer · EpGroupoid.NaturalRep.toLocalUniformizer, and one exists inside every open $V \ni x$ Chapter7/Uniformizer · EpGroupoid.exists_localUniformizer; its orbit map is a homeomorphism onto the open $|U(x)|$ Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.isOpen_image Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.orbitHomeomorph. "$\Psi_x \circ G_x \ltimes U(x) \to X$" is read with :, and "a a local uniformizer" has a doubled article (possible misprints; Findings, Chapter 7, item 28) (2017 p. 237).

VII.12 Definition 7.1.21 (2017: 7.10), the M-polyfold A, and Proposition 7.1.22 (2017: 7.7)

Definition 7.1.21 (2017: 7.10) (V7.29). An ep-subgroupoid is the full subcategory on a set A of objects which is a sub-M-polyfold (Definition 2.3.18 (2017: 2.12), §II.10) and saturated Chapter7/EpGroupoid · EpGroupoid.IsEpSubgroupoid. Witnesses: X itself Chapter7/EpGroupoid · EpGroupoid.isEpSubgroupoid_univ, every saturated open set Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_of_isOpen, the saturation of every open set, open by Proposition 7.1.8 (2017: 7.1) Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_saturation, and every sub-M-polyfold of a trivial ep-groupoid Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_trivial (Findings, Chapter 7, items 40, 105) (2017 p. 237).

The facts (1)–(4) on A (V7.30). By Part I's Proposition 2.3.19 (2017: 2.6) (§II.10), A has an M-polyfold structure Chapter2/SubMPolyfold · MPolyfold.subPolyfold Chapter7/SubGroupoid · EpGroupoid.subObjPolyfold with (1) the inclusion sc-smooth and a homeomorphism onto its image Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_val_subObj and (2) $i^{-1} \circ r$ sc-smooth for every sc-smooth retraction r onto $A \cap V$ Chapter7/SubGroupoid · EpGroupoid.isScSmoothOn_retractInto_subObj. The tangent facts (3), (4) — T_aA a closed, sc-complemented subspace of T_aX, and $T_aA = Tr(a)T_aX$ — are proved in §VII.44 with the linear structure of tangent spaces Chapter7/SubGroupoidTangent · EpGroupoid.fact_3_subObj Chapter7/SubGroupoidTangent · EpGroupoid.fact_4_subObj; in a chart they are Part I's Chapter2/Metrizable · proposition_2_3_19_three, and nothing below uses them (gap G.70, closed; Findings, Chapter 7, items 40, 154) (2017 pp. 237–238).

A is a sub-M-polyfold (V7.31). The text takes, for $\varphi \in \mathbf{A}$ with $a = s(\varphi)$, a sheet $s : \mathbf{U}(\varphi) \to U(a)$ and a retraction r of $U(a)$ onto $U(a) \cap A$, and shows that $\mathbf{r} = (s|\mathbf{U}(\varphi))^{-1} \circ r \circ s$ retracts $\mathbf{U}(\varphi)$ onto $\mathbf{U}(\varphi) \cap \mathbf{A}$. *Step 1, saturation.* For saturated A, $t(\varphi) \in A$ iff $s(\varphi) \in A$ Chapter7/SubGroupoid · EpGroupoid.src_mem_of_tgt_mem_of_isSaturated, so $\mathbf{A} = s^{-1}(A) = t^{-1}(A)$ Chapter7/SubGroupoid · EpGroupoid.morOn_eq_preimage_src Chapter7/SubGroupoid · EpGroupoid.morOn_eq_preimage_tgt. *Step 2, a more general statement.* The preimage of a sub-M-polyfold under a local sc-diffeomorphism is one: the retraction is pulled back

along a sheet, after shrinking $U(a)$ to $(V \cap V_0) \cap r^{-1}(V \cap V_0)$ so that r maps it into itself Chapter7/SubGroupoid · isSubMPolyfold_preimage_of_isLocalScDiffeo Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_preimage_src Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_preimage_tgt Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_morOn. So $\mathbf{A}$ is an M-polyfold Chapter7/SubGroupoid · EpGroupoid.subMorPolyfold with facts (1), (2) Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_val_subMor Chapter7/SubGroupoid · EpGroupoid.isScSmoothOn_retractInto_subMor; its paracompactness, which the project's M-polyfolds need, is that of a subspace of a metrizable space. The text's "$s(\mathbf{r}(\psi)) = r(s(\psi)) \in A$ and $t(\mathbf{r}(\psi)) \in U(a)$" is not used: the second claim does not hold in general as we read it, $t(\mathbf{r}(\psi))$ lying near $t(\varphi)$ rather than near a, and $s(\mathbf{r}(\psi)) \in A$ with saturation gives $t(\mathbf{r}(\psi)) \in A$ (possible misprint or extraction artefact for $s(\mathbf{r}(\psi)) \in U(a)$; Findings, Chapter 7, item 41) (2017 p. 238).

Proposition 7.1.22 (2017: 7.7) (V7.32) Chapter7/SubGroupoid · EpGroupoid.proposition_7_1_22. *For a saturated sub-M-polyfold A, the full subcategory on A is an ep-groupoid; A and $\mathbf{A}$ are sub-M-polyfolds, and all structure maps are the restrictions of those of X.* The text checks the conditions of Definition 7.1.3 (2017: 7.3) in a few lines; the formalization supplies the unstated step that *a map into a sub-M-polyfold which is sc-smooth into the ambient M-polyfold is sc-smooth into the sub-M-polyfold*: near a point it is $(i^{-1} \circ r) \circ i \circ f$, by fact (2) Chapter7/SubGroupoid · isScSmoothOn_subPolyfold_of_val. *(1).* A sheet of s maps $\mathbf{U}(\varphi) \cap \mathbf{A}$ onto $U(a) \cap A$; with the unstated step, s and t restrict to local sc-diffeomorphisms Chapter7/SubGroupoid · EpGroupoid.isLocalScDiffeo_fullSub_src Chapter7/SubGroupoid · EpGroupoid.isLocalScDiffeo_fullSub_tgt. *(2).* u and ι are restrictions Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_fullSub_unit Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_fullSub_inv. *(3).* The text uses a $U(a)$ with the natural G_a-action and t proper over $s^{-1}(\mathrm{cl}_X U(a))$ and notes that $U(a) \cap A$ is invariant; the invariance is not used, and any $V(a)$ of Definition 7.1.3 (3) (2017: 7.3) works with $V(a) \cap A$: a sequence with sources in $\mathrm{cl}_A(V(a) \cap A) \subseteq \mathrm{cl}_X V(a)$ and convergent targets in A has a subsequence converging in $\mathbf{X}$ (§VII.2), whose source lies in A by saturation and in $\mathrm{cl}_A(V(a) \cap A)$ since this set is closed in A Chapter7/SubGroupoid · EpGroupoid.exists_isProperMap_fullSub. *(4), literally.* The inclusion $\mathbf{A}\,{}_s\!\times_t \mathbf{A} \to \mathbf{X}\,{}_s\!\times_t \mathbf{X}$ is sc-smooth between the fibred products of Proposition 2.5.20 (2017: 2.15), by the unstated step applied twice, and m of A is m of X composed with it Chapter7/SubGroupoid · EpGroupoid.fullSub_comp_val Chapter7/SubGroupoid · EpGroupoid.isScSmoothComp_fullSub. So the full subcategory is an ep-groupoid Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid whose structure maps are the restrictions Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid_src_val Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid_comp_val, and the inclusion is a full and faithful ep-functor Chapter7/SubGroupoid · EpGroupoid.subEpFunctor Chapter7/SubGroupoid · EpGroupoid.subEpFunctor_full Chapter7/SubGroupoid · EpGroupoid.subEpFunctor_faithful, which the text does not state. *Hidden hypotheses:* none; the paracompactness of A and $\mathbf{A}$ is proved (Findings, Chapter 7, item 42) (pp. 306–308 (2017 pp. 238–239)).

VII.13 Definition 7.2.1 (2017: 7.11), Lemmas 7.2.2 and 7.2.3 (2017: 7.2 and 7.3)

Definition 7.2.1 (2017: 7.11) (V7.33). X is *effective* if for every x, every $g \in G_x \setminus \{1_x\}$ and every open neighbourhood $\mathbf{U}(g)$ there is $\psi \in \mathbf{U}(g)$ with $s(\psi) \neq t(\psi)$ Chapter7/EpGroupoid · EpGroupoid.IsEffective; x is *regular* if $G_x = \{1_x\}$ Chapter7/Groupoid · GroupoidStr.IsRegular, and X_{reg} is the set of regular objects Chapter7/Groupoid · GroupoidStr.regularSet. "If $s(\varphi)$ is regular, so is $t(\varphi)$": conjugation with φ is an isomorphism of isotropy groups (§VII.6) Chapter7/Groupoid · GroupoidStr.isRegular_tgt_iff Chapter7/Effective · EpGroupoid.isRegular_tgt_of_isRegular_src, so X_{reg} is saturated Chapter7/Effective · EpGroupoid.isSaturated_regularSet. The definition is satisfiable non-vacuously, by $\mathbb{Z}/2$ acting on $\mathbb{R}$ by reflection (§VII.30) Chapter7/NonVacuity · Examples.isEffective_reflection_nonvacuous (Findings, Chapter 7, items 50, 101)

(2017 p. 239).

Lemma 7.2.2 (2017: 7.2) (V7.34) `Chapter7/Effective · EpGroupoid.lemma_7_2_2`. X_{reg} *is open.* For regular x and a natural representation at x, every morphism $\varphi : a \to b$ between points of $U(x)$ is $\Gamma(1_x, a)$ by Theorem 7.1.13 (3) (2017: 7.1), and $\Gamma(1_x, a) = 1_a$ (§VII.8); so $b = a$, $\varphi = 1_a$, and G_a is trivial. The formalization proves the stronger statement that $U(x) \subseteq X_{reg}$ for *every* natural representation at a regular x `Chapter7/Effective · EpGroupoid.NaturalRep.subset_regularSet`. *Hidden hypothesis:* $\Gamma(1, a) = 1_a$, which needs $U(x)$ connected. "Examples show that the regular set can be empty" is the trivial $\mathbb{Z}/3$-action (§VII.30) (Findings, Chapter 7, item 51) (p. 308 (2017 p. 239)).

Lemma 7.2.3 (2017: 7.3) (V7.35) `Chapter7/Effective · EpGroupoid.lemma_7_2_3`. *Equivalent: (1) X is effective; (2) every open $V(x)$ contains a natural representation with $\Phi : G_x \to \mathrm{Diff}_{sc}(U(x))$ injective.* Injectivity is read as "$\Phi(g) = \mathrm{id}$ on U only for $g = 1$" `Chapter7/Effective · EpGroupoid.NaturalRep.injective_Φ_iff`, the triviality of the kernel of $G_x \to \mathrm{Perm}(U)$ `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ` `Chapter7/Effective · EpGroupoid.NaturalRep.mem_kerΦ` `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ_eq_bot_iff` `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ_eq_bot_iff_injective`. **The text's proof.** (1) □ (2) by contradiction: if $\Phi(h) = 1$ with $h \neq 1$, effectiveness gives $\varphi_k \to h$ with $s(\varphi_k) \neq t(\varphi_k)$; $\varphi_k = \Gamma(h_k, s\varphi_k)$ with $h_k \neq h$, after a subsequence $h_k = g$, and in the limit $h = \Gamma(g, x) = g$. (2) □ (1): $\Phi(g) \neq 1$ gives a with $\Phi(g)(a) \neq a$, and $\Gamma(g, a)$ has $s \neq t$; "by taking smaller and smaller neighborhoods $U(x)$ and adapting the corresponding restrictions of Γ" one finds such morphisms converging to g. **The formalized proof.** *(1) □ (2), for every natural representation, without sequences* `Chapter7/Effective · EpGroupoid.IsEffective.eq_one`. If $\Phi(h) = 1$ on U, the sheet $\Gamma(h, U)$ is an open neighbourhood of h on which $s = t$, since $t(\Gamma(h, y)) = \Phi(h)(y) = y$; effectiveness forces $h = 1$. So effectiveness is the injectivity of every natural representation `Chapter7/Effective · EpGroupoid.isEffective_iff_forall_naturalRep`, and natural representations exist inside every $V(x)$ (§VII.9). *(2) □ (1).* Fix one natural representation $R_0 = (U_0, \Phi_0, \Gamma_0)$, $g \neq 1$ and an open $\mathbf{U}(g)$. *Step 1.* Apply (2) once, to $V = \{y \in U_0 \mid \Gamma_0(g, y) \in \mathbf{U}(g)\}$, open and containing x. *Step 2.* The natural representation R that (2) gives inside V has $\Phi(g) \neq 1$, so some $a \in U$ has $\Phi(g)(a) \neq a$. *Step 3.* R need not be a restriction of R_0, as we read "adapting the corresponding restrictions"; it agrees with R_0 on the component of x in $U \cap U_0 = U$, i.e. on U (§VII.9) `Chapter7/Effective · EpGroupoid.NaturalRep.Γ_eqOn_of_subset` `Chapter7/Effective · EpGroupoid.NaturalRep.Φ_eqOn_of_subset`. *Step 4.* So $\Gamma_0(g, a) = \Gamma(g, a) \in \mathbf{U}(g)$ has source a and target $\Phi(g)(a) \neq a$. One application of (2) suffices (Findings, Chapter 7, item 52) (pp. 307–309 (2017 pp. 239–240)).

VII.14 Proposition 7.2.4 (2017: 7.8)

Proposition 7.2.4 (2017: 7.8) (V7.36) `Chapter7/Effective · EpGroupoid.proposition_7_2_4`. *If X is effective, X_{reg} is open and dense.* **The text's proof**: in a $U(x_0)$ take x with $\#G_x$ minimal; if $\#G_x > 1$, Lemma 7.2.3 (2017: 7.3) gives $U(x) \subseteq U(x_0)$ with Φ injective and, for $g \neq 1$, a point a with $\Phi(g)(a) \neq a$; G_a is isomorphic to $H = \{h \in G_x \mid \Phi(h)(a) = a\}$, so $\#G_x \leq \#G_a = \#H \leq \#G_x$, $H = G_x$, a contradiction. **The formalized proof.** *Step 1.* For a natural representation at x and $a \in U$, $h \mapsto \Gamma(h, a)$ is an isomorphism from the stabilizer of a onto G_a, by Theorem 7.1.13 (3) (2017: 7.1) and the functoriality of Γ `Chapter7/Effective · EpGroupoid.NaturalRep.stabilizerEquivIsotropy`. *Step 2.* Given a nonempty open O, take $x \in O$ minimizing $\#G_x$ over O itself, so the text's $U(x_0)$ is not needed. *Step 3.* If $G_x \neq 1$, take a natural representation inside O (§VII.9), injective by Lemma 7.2.3 (2017: 7.3), $g \neq 1$ and $a \in U$ with $\Phi(g)(a) \neq a$; then $\#G_a = \#\mathrm{Stab}(a) < \#G_x$, contradicting minimality. So O meets X_{reg}; openness is Lemma 7.2.2 (2017: 7.2). Only $\#G_a < \#G_x$ is used. *The converse* holds too, and is not in the text `Chapter7/Effective · EpGroupoid.isEffective_of_dense_regularSet` `Chapter7/Effective · EpGroupoid.isEffective_`

iff_dense_regularSet: if X_{reg} is dense and $g \neq 1$, a regular a near x with $\Gamma(g,a)$ in a given $\mathbf{U}(g)$ has $\Gamma(g,a) \neq 1_a = \Gamma(1,a)$, so $s \neq t$ there (Findings, Chapter 7, item 53) (pp. 308–311 (2017 pp. 240–241)).

VII.15 The subgroups T_x, and Definition 7.2.7 (2017: 7.12)

T_x (V7.37). $g \in G_x$ *acts trivially* if $\Phi(g) = 1$ for some natural representation Chapter7/EpGroupoid · EpGroupoid.ActsTrivially, and T_x is the set of such g Chapter7/EpGroupoid · EpGroupoid.trivialSet Chapter7/Groupoid · EpGroupoid.one_mem_trivialSet. The text: since G_x is finite, $U(x)$ can be chosen so small that the kernel of Φ is T_x, a normal subgroup as a kernel. Since sc-smooth maps are not analytic, $\Phi(g) = \mathrm{id}$ near x does not give $\Phi(g) = \mathrm{id}$ on a connected $U(x)$, so the kernel can depend on $U(x)$; the text's "small" is made precise in three statements. *Step 1, for every representation.* $g \in T_x$ iff $\Phi(g) = \mathrm{id}$ near x, for any given natural representation (§VII.9, Step 4) Chapter7/Effective · EpGroupoid.NaturalRep.actsTrivially_iff. *Step 2, intrinsically.* $g \in T_x$ iff $s = t$ on a neighbourhood of g in $\mathbf{X}$ Chapter7/Effective · EpGroupoid.mem_trivialSet_iff_eventually: one direction is the sheet $\Gamma(g,U)$; for the other, a continuous section of s through g along which $s = t$ near x gives $\Phi(g) = \mathrm{id}$ near x Chapter7/Effective · EpGroupoid.mem_trivialSet_of_section. *Step 3, small neighbourhoods.* For one R_0, the common fixed set in U_0 of the finitely many elements of T_x is a neighbourhood N of x, and $\ker\Phi = T_x$ for *every* natural representation inside N Chapter7/Effective · EpGroupoid.exists_mem_nhds_trivialSet_iff Chapter7/Effective · EpGroupoid.exists_naturalRep_trivialSet_iff Chapter7/Effective · EpGroupoid.exists_naturalRep_kerΦ_eq. *Step 4.* So T_x is a kernel, a normal subgroup Chapter7/Effective · EpGroupoid.trivialSubgroup Chapter7/Effective · EpGroupoid.mem_trivialSubgroup Chapter7/Effective · EpGroupoid.trivialSubgroup_normal, its own normal closure Chapter7/Effective · EpGroupoid.normalClosure_trivialSet, and by Step 2 Definition 7.2.1 (2017: 7.11) says exactly $T_x = \{1_x\}$ for all x Chapter7/Effective · EpGroupoid.isEffective_iff_trivialSubgroup_eq_bot. *At nearby points.* For $g \in \ker\Phi$, $\Gamma(g,y) \in T_y$ for $y \in U$, which the proof of Proposition 7.2.10 (2017: 7.9) uses Chapter7/Effective · EpGroupoid.NaturalRep.mem_trivialSet_of_val_eq_Γ. *Along a morphism, not stated in the text.* For $\varphi : x \to y$ and $g \in T_x$, $z \mapsto \theta(z) \circ \Gamma(g, s\theta(z)) \circ \theta(z)^{-1}$, $\theta = (t|\mathbf{U}(\varphi))^{-1}$, is a continuous section of s through $\varphi g \varphi^{-1}$ — by (4) in the sheet form, twice — along which $s = t$ near y Chapter7/Effective · EpGroupoid.exists_conj_section; by Step 2, $\varphi g \varphi^{-1} \in T_y$ Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet_iff, and $\varphi T_x \varphi^{-1} = T_y$ Chapter7/Effective · EpGroupoid.map_isotropyConj_trivialSubgroup. This is what the descent of the composition to $\mathbf{X}^R$ needs. "Diff$_{sc}(U(x)$" lacks a parenthesis (possible misprint or extraction artefact). Witnesses: $T_x = 1$ for the trivial ep-groupoid Chapter7/Effective · EpGroupoid.trivialSubgroup_trivial and the reflection, $T_x = G_x \cong \mathbb{Z}/3$ for the trivial $\mathbb{Z}/3$-action (§VII.30) (Findings, Chapter 7, item 54) (2017 p. 241).

Definition 7.2.7 (2017: 7.12) (V7.38). $\mathbf{X}^T = \bigcup_x T_x$ Chapter7/EpGroupoid · EpGroupoid.nonEffectivePart; $\varphi \sim \varphi'$ iff $s(\varphi) = s(\varphi') = x$ and $\varphi = \varphi' \circ g$ with $g \in T_x$ Chapter7/EpGroupoid · EpGroupoid.ReducedRel; $\mathbf{X}^R$ is the set of classes Chapter7/EpGroupoid · EpGroupoid.ReducedMor. *Step 1, $\mathbf{X}^T$ intrinsically.* By §VII.15 Step 2, $\mathbf{X}^T$ is the interior of $\{\varphi \mid s\varphi = t\varphi\}$ Chapter7/Reduced · EpGroupoid.mem_nonEffectivePart_iff Chapter7/Reduced · EpGroupoid.nonEffectivePart_eq_interior Chapter7/Reduced · EpGroupoid.val_mem_nonEffectivePart_iff, open Chapter7/Reduced · EpGroupoid.isOpen_nonEffectivePart and closed under units, inverses, composition and conjugation Chapter7/Reduced · EpGroupoid.unit_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.inv_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.comp_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.conj_mem_nonEffectivePart. So $\varphi \sim \varphi'$ iff $\varphi = \varphi' \circ k$ with $k \in \mathbf{X}^T$ Chapter7/Reduced · EpGroupoid.reducedRel_iff, a form without dependent types. *Step 2, an equivalence relation,* because T_x is a subgroup Chapter7/Reduced · EpGroupoid.reducedRel_refl Chapter7/Reduced · EpGroupoid.ReducedRel.symm Chapter7/Reduced · EpGroupoid.ReducedRel.trans Chapter7/Reduced · EpGroupoid.reducedSetoid Chapter7/Reduced · EpGroupoid.reducedMor_eq_quotient. *Step 3, the left*

form. By the conjugation $\varphi T_x \varphi^{-1} = T_y$, $\varphi \sim \psi$ iff $\varphi = k \circ \psi$ with $k \in T_{t\psi}$ `Chapter7/Reduced · EpGroupoid.reducedRel_comp_left` `Chapter7/Reduced · EpGroupoid.ReducedRel.exists_comp_left`; the relation respects s, t, ι and composition on both sides `Chapter7/Reduced · EpGroupoid.ReducedRel.src_eq` `Chapter7/Reduced · EpGroupoid.ReducedRel.tgt_eq` `Chapter7/Reduced · EpGroupoid.ReducedRel.inv` `Chapter7/Reduced · EpGroupoid.ReducedRel.comp_left` `Chapter7/Reduced · EpGroupoid.ReducedRel.comp_right`. "The subset $\mathbf{X}^R$ of reduced morphisms" is a quotient, as the next sentence of the text says (possible misprint; Findings, Chapter 7, item 55) (2017 p. 241).

VII.16 Proposition 7.2.10 (2017: 7.9): the M-polyfold $\mathbf{X}^R$

Proposition 7.2.10 (2017: 7.9) (V7.39) `Chapter7/Reduced · EpGroupoid.proposition_7_2_10`. $\mathbf{X}^R$ *has a natural M-polyfold structure.* **The text's proof**: $\mathbf{X}^R$ carries the quotient topology; on a sheet $s : U(\varphi_0) \to U(s\varphi_0)$ the projection π' is continuous and injective; its image $U([\varphi_0])$ is open, by the map $\delta(\psi') = \psi' \circ \Gamma(g, s\psi')$ near an equivalent ψ, with $U(s\psi)$ chosen with $\ker \Phi = T_{s\psi}$; the same argument shows $\Psi_{\varphi_0} : U(\varphi_0) \to U([\varphi_0])$ open; and the transitions are, near φ with $\psi = \varphi \circ g$, the sc-smooth δ. **The formalized proof.** *Step 1, the topology.* The projection $\pi^R : \mathbf{X} \to \mathbf{X}^R$ `Chapter7/Reduced · EpGroupoid.reducedπ` `Chapter7/Reduced · EpGroupoid.reducedπ_eq_iff` `Chapter7/Reduced · EpGroupoid.surjective_reducedπ` is a quotient map `Chapter7/Reduced · EpGroupoid.isQuotientMap_reducedπ` `Chapter7/Reduced · EpGroupoid.continuous_reducedπ`. *Step 2, π^R is open* `Chapter7/Reduced · EpGroupoid.isOpenMap_reducedπ`. The text's δ composes on the right with a section of s; its continuity holds by (4) applied to inverses. The left form is used instead: for $\varphi \sim \psi$, $\varphi = \Gamma(g, t\psi) \circ \psi$ near ψ with $g \in T_{t\psi}$ and a natural representation with $\ker \Phi = T_{t\psi}$ (§VII.15); $\delta(\psi') = \Gamma(g, t\psi') \circ \psi'$ is sc-smooth by (4) in the sheet form `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScSmoothOn_comp_Γ_tgt` and $\psi' \sim \delta(\psi')$ since $\Gamma(g, t\psi') \in T_{t\psi'}$. So the saturation of an open set is open. *Step 3, sheets.* On every sheet $s : \mathbf{W} \to V$, π^R is injective — equivalent morphisms have the same source — and hence an open embedding `Chapter7/Reduced · EpGroupoid.injOn_reducedπ` `Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ` `Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ_image_Γ`; "small" sheets are not needed. *Step 4, charts.* For a chart ψ of $\mathbf{X}$ with domain in a sheet $\mathbf{W}$, $\sigma = (s|\mathbf{W})^{-1}$, the chart of $\mathbf{X}^R$ is $[\varphi'] \mapsto \psi(\sigma(s[\varphi']))$ on $\pi^R(\operatorname{dom}\psi)$ `Chapter7/Reduced · EpGroupoid.reducedChart`; every point lies in one `Chapter7/Reduced · EpGroupoid.exists_chart_sheet`. *Step 5, transitions.* On an overlap, an element of the second sheet equivalent to φ' has the source $s(\varphi')$, so the text's δ equals $\sigma_2 \circ s$ and the transition is $\psi_2 \circ \sigma_2 \circ s \circ \psi_1^{-1}$ `Chapter7/Reduced · EpGroupoid.reducedChart_transition`, sc-smooth; what needs the text's argument is that the overlap is open, i.e. Step 2 `Chapter7/Reduced · EpGroupoid.overlap_reducedChart` `Chapter7/Reduced · EpGroupoid.reducedChart_compatible`. The charts form an atlas `Chapter7/Reduced · EpGroupoid.reducedCharts` `Chapter7/Reduced · EpGroupoid.reducedAtlas`, all without hypotheses. *Step 6, Hausdorff and paracompact.* An M-polyfold is a Hausdorff paracompact space (Definition 2.3.4 (2017: 2.8)). The proof gives neither as we read it, and $\mathbf{X}^R$ need not be Hausdorff (the example at the end of this paragraph, formalized); so the M-polyfold $\mathbf{X}^R$ is built under the hypothesis that $\mathbf{X}^R$ is Hausdorff, the 2021 text's hypothesis on T_R (§VII.34); paracompactness, a second hypothesis in the first reading, is now an instance, proved for every ep-groupoid as Proposition 7.2.12 `Chapter7/ReducedCriterion · EpGroupoid.paracompactSpace_reducedMor` `Chapter7/Reduced · EpGroupoid.reducedPolyfold` `Chapter7/Reduced · EpGroupoid.isChart_reducedChart`, and then π^R is a local sc-diffeomorphism, with inverse $\sigma \circ s$ on $\pi^R(\mathbf{W})$ `Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedπ` `Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedπ` `Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedπ` (gap G.71, closed, §§VII.34, VII.48). Steps 1–5 are the 2021 Proposition 7.2.10, with no hypothesis `Chapter7/ReducedNaturalCharts · EpGroupoid.proposition_7_2_10_natural_charts`. *The hypotheses are satisfiable.* For effective X, $\sim$ is equality `Chapter7/Reduced · EpGroupoid.reducedRel_iff_eq_of_isEffective`, π^R a homeomorphism `Chapter7/Reduced · EpGroupoid.reducedHomeomorph`, and $\mathbf{X}^R$ is Hausdorff, paracompact

and regular Chapter7/Reduced · EpGroupoid.t2Space_reducedMor_of_isEffective Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_of_isEffective Chapter7/Reduced · EpGroupoid.t3Space_reducedMor_of_isEffective; this covers the trivial ep-groupoid of every M-polyfold Chapter7/Reduced · EpGroupoid.t2Space_reducedMor_trivial Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_trivial. Paracompactness also follows from regularity of $\mathbf{X}^R$ when $\mathbf{X}$ is second countable Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_of_t3; these partial results of the first reading are superseded by Proposition 7.2.12, which needs no hypothesis. Four possible misprints of the proof: "$\psi' = (s|U(\psi))^{-1}(s(\psi))$" for $s(\psi')$, "$U(\varphi) \subset \Psi_{\varphi_0}(U([\varphi_0]) \cap U([\psi_0]))$" for $\Psi_{\varphi_0}^{-1}$, "T_x agrees with the kernel of the $G_{s(\varphi)}$-action" for $T_{s(\varphi)}$, and a garbled sentence on the smoothness of the source and target maps (Findings, Chapter 7, items 56, 57, 60, 111) (2017 pp. 242–243). *The Hausdorff hypothesis cannot be dropped* Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.exists_not_t2Space_reducedMor. Let $\mathbb{Z}/2$ act by $(t, z) \mapsto (t, \pm z)$ on the retract O of Part I's Remark 4.2.2 (2017: 4.3) example, a point over each $t \leq 0$ and a line over each $t > 0$; the action is the restriction of an sc-operator Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.isScSmoothMap_smul. Over $t < 0$ both elements act trivially, so $(-1, (t, 0)) \in \mathbf{X}^T$ Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.mem_nonEffectivePart; at $(0, 0)$ points of O with $t > 0$ tend to 0 and are moved by -1, so $(-1, (0, 0)) \notin \mathbf{X}^T$ Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_mem_nonEffectivePart. The classes of $(\pm 1, (-1/(n+1), 0))$ coincide and converge to both $[(-1, 0)] \neq [(1, 0)]$ Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_t2Space_reducedMor. So Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017), for every ep-groupoid, do not hold as we read them; the 2021 text takes the Hausdorff hypothesis (Findings, Chapter 7, item 107; found on paper while reading, proved in the round-trip audit).

VII.17 The reduced ep-groupoid X_R, Proposition 7.10 (2017), Definition 7.2.15 (2017: 7.13)

X_R (V7.40). Objects X, morphisms $\mathbf{X}^R$, $s([\varphi]) = s(\varphi), t([\varphi]) = t(\varphi)$, with unit, inverse and composition descended by §VII.15 Chapter7/Reduced · EpGroupoid.reducedSrc Chapter7/Reduced · EpGroupoid.reducedTgt Chapter7/Reduced · EpGroupoid.reducedUnit Chapter7/Reduced · EpGroupoid.reducedInv Chapter7/Reduced · EpGroupoid.reducedComp Chapter7/Reduced · EpGroupoid.reducedComp_π, s and t onto Chapter7/Reduced · EpGroupoid.surjective_reducedSrc Chapter7/Reduced · EpGroupoid.surjective_reducedTgt, a groupoid Chapter7/Reduced · EpGroupoid.reducedGroupoidStr; the composition of a non-composable pair is a junk value, as in §VII.1 (2017 p. 243).

Proposition 7.10 (2017) (V7.40) Chapter7/Reduced · EpGroupoid.proposition_7_10_2017. *Under the hypotheses of §VII.16, X_R is an ep-groupoid, and it is effective. (1).* For a sheet $\mathbf{W}$ of Definition 7.1.4 (2017: 7.4), $s, t : \pi^R(\mathbf{W}) \to X$ are $s \circ (\pi^R|\mathbf{W})^{-1}$ and $t \circ (\pi^R|\mathbf{W})^{-1}$, the text's $s(\Psi_\varphi(\varphi')) = s(\varphi')$ Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedSrc Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedTgt Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedSrc Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedTgt. *(2).* $u = \pi^R \circ u_X$ and $\iota = \pi^R \circ \iota_X \circ (\pi^R|\mathbf{W})^{-1}$ on $\pi^R(\mathbf{W})$ Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedUnit Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedInv. *(3), "proved similarly".* For the same $V(x)$, $t \circ \pi^R = t_X$ on $s^{-1}(\mathrm{cl}V(x))$, with π^R a continuous surjection onto the corresponding set of $\mathbf{X}^R$; a map whose composite with a continuous surjection is proper is proper (Mathlib) Chapter7/Reduced · EpGroupoid.exists_isProperMap_reduced. *(4), "reduced by our local charts",* in the sheet form: for an sc-diffeomorphism $s : \mathbf{W}' \to V'$ of $\mathbf{X}^R$ with inverse σ', $\kappa = \sigma_2 \circ s \circ \sigma'$ is the inverse of s on an open set of $\mathbf{X}$ — a step the text leaves implicit — and $m(\sigma'(t\psi), \psi) = \pi^R(m(\kappa(t\psi), \psi))$ Chapter7/Reduced · EpGroupoid.isScSmoothCompSheet_reduced. So X_R is an ep-groupoid Chapter7/Reduced · EpGroupoid.reducedEpGroupoid and $X \to X_R$ an ep-functor Chapter7/Reduced · EpGroupoid.reductionFunctor. *Effective.* The text's point a with

$\Phi(g)(a) \neq a$ is replaced by the intrinsic form of T_x: if $s = t$ near $[g]$ in $\mathbf{X}^R$, then $s = t$ near g in $\mathbf{X}$, π^R being continuous, so $g \in T_x$ and $[g] = [1_x]$ Chapter7/Reduced · EpGroupoid.isEffective_reducedEpGroupoid. For effective X, $X_R \cong X$ in $\mathcal{EP}$ Chapter7/Reduced · EpGroupoid.reductionInv Chapter7/Reduced · EpGroupoid.reductionIso. In the 2021 form, Theorem 7.2.14 with Proposition 7.2.16, only the Hausdorff hypothesis remains Chapter7/ReducedNaturalCharts · EpGroupoid.theorem_7_2_14_reduced (§VII.48) (Findings, Chapter 7, items 58, 60) (2017 p. 244).

The isotropy of X_R, and Definition 7.2.15 (2017: 7.13) (V7.41). $g \mapsto [g]$ is a surjective homomorphism from G_x onto the isotropy of x in X_R Chapter7/Reduced · EpGroupoid.reducedIsotropyHom Chapter7/Reduced · EpGroupoid.surjective_reducedIsotropyHom with kernel T_x — the text's "$[1_x] = T_x$" Chapter7/Reduced · EpGroupoid.ker_reducedIsotropyHom — so that isotropy is G_x/T_x Chapter7/Reduced · EpGroupoid.quotientTrivialSubgroupEquiv. The effective isotropy group $G_x^{\mathrm{eff}} = G_x/T_x$ Chapter7/EpGroupoid · EpGroupoid.EffIsotropy is defined as the quotient by the normal closure of T_x, which is G_x/T_x by §VII.15 Chapter7/Effective · EpGroupoid.effIsotropyEquiv, and it is the isotropy group of X_R Chapter7/Reduced · EpGroupoid.effIsotropyEquivReduced. "For and object" (possible misprint; Findings, Chapter 7, item 59) (2017 p. 244).

VII.18 Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2 and 7.3.3 (2017: 7.4 and 7.5)

Theorem 7.3.1 (2017: 7.2) (V7.42) Chapter7/OrbitTopology · EpGroupoid.theorem_7_3_1. *$|X|$ is locally metrizable, regular and Hausdorff; if $|X|$ is paracompact, it is metrizable.* "Locally metrizable" is read as: every point has an open neighbourhood which is metrizable in the subspace topology. The three properties are Lemmas 7.3.2, 7.3.5 and 7.3.4 (2017: 7.4, 7.7 and 7.6). The metrizability Chapter7/OrbitTopology · EpGroupoid.theorem_7_3_1_metrizable is the text's "Nagata/Smirnov" in Part I's form (§II.7): a paracompact Hausdorff locally metrizable space is metrizable Chapter2/Metrizable · metrizableSpace_of_locally. The hypothesis "$|X|$ paracompact" is satisfiable, by the trivial ep-groupoid Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial (Findings, Chapter 7, item 70) (2017 p. 244).

Lemma 7.3.2 (2017: 7.4) (V7.43) Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_2. *Every $z \in |X|$ has an open metrizable neighbourhood; one can take $|U(x)|$ for a natural representation at x.* **The text's proof**: U is metrizable with a metric d; $\rho(y, y') = \max_g d(g * y, g * y')$ is an equivalent G_x-invariant metric; $\hat{\rho}([y], [y']) = \min\{\rho(g * y, h * y') \mid g, h \in G_x\}$ defines the quotient topology on $G_x\backslash U$; and Proposition 7.1.19 (2017: 7.6) carries it to $|U|$. **The formalized proof** follows it, for any finite group acting by homeomorphisms on a metrizable space Chapter7/OrbitTopology · metrizableSpace_orbitRel_quotient. *Step 1.* ρ is the maximum over the finite G_x Chapter7/OrbitTopology · maxDist, invariant Chapter7/OrbitTopology · maxDist_smul, and $d \leq \rho$. *Step 2.* $\hat{\rho}$ is computed as $\min_h \rho(y, h * y')$, the same number by invariance Chapter7/OrbitTopology · quotDist Chapter7/OrbitTopology · quotDist_mk; it is symmetric and satisfies the triangle inequality by invariance, and $\hat{\rho}([y], [y']) = 0$ forces $y' \in G_x * y$, orbits of a finite group being closed. *Step 3, the topology.* A set of $G_x\backslash U$ is open in the quotient topology iff it contains a $\hat{\rho}$-ball around each of its points: $\{y' \mid \rho(y, y') < \varepsilon\} = \bigcap_g g^{-1}(B_d(g * y, \varepsilon))$ is open, and $d \leq \rho$ gives the converse. This is Mathlib's construction of a metric space with a prescribed topology; ρ itself is never installed as a metric on U. So $G_x\backslash U(x)$ is metrizable Chapter7/OrbitTopology · EpGroupoid.metrizableSpace_orbitRel_quotient_naturalRep and so is $|U(x)|$, its homeomorphic image (§VII.11) Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_2_naturalRep (Findings, Chapter 7, item 71) (pp. 319–325 (2017 pp. 244–245)).

Lemma 7.3.3 (2017: 7.5) (V7.44) Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_3. *Every point of $|X|$ has a monotone countable basis of open neighbourhoods.* This is Proposition 7.1.9 (2017: 7.2) (§VII.5), proved there from

the metrizability of X and the openness of π; the text derives it from Lemma 7.3.2 (2017: 7.4) (Findings, Chapter 7, item 72) (2017 p. 245).

VII.19 Lemmas 7.3.4 and 7.3.5 (2017: 7.6 and 7.7), Corollary 7.3.6 (2017: 7.1), subsets of $|X|$

The text proves Lemma 7.3.4 (2017: 7.6) before Lemma 7.3.5 (2017: 7.7); the formalization proves Lemma 7.3.5 (2017: 7.7) first and derives Lemma 7.3.4 (2017: 7.6) from it (Findings, Chapter 7, item 72).

Lemma 7.3.5 (2017: 7.7) (V7.46) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_5`. $|X|$ *is regular.* **The text's proof**: for closed $A \not\ni z = \pi(x)$ choose $O \ni x$ with $\pi(O) \cap A = \emptyset$ and t proper over $s^{-1}(\mathrm{cl}O)$, and $x \in O' \subseteq \mathrm{cl}O' \subseteq O$; every $y \in \pi^{-1}(A)$ has a $V(y)$ which no morphism from $\mathrm{cl}O'$ reaches, else properness gives φ with $s\varphi \in \mathrm{cl}O'$ and $t\varphi = y$; then $\pi(\bigcup V(y)) \supseteq A$ and $\pi(O')$ are disjoint. **The formalized proof** uses a closed neighbourhood basis `Mathlib/Topology/Separation/Regular · RegularSpace.of_exists_mem_nhds_isClosed_subset`. *Step 1.* For a neighbourhood N of $\pi(x)$, take a closed neighbourhood O' of x inside $\pi^{-1}(N) \cap V(x)$, X being regular and $V(x)$ from Definition 7.1.3 (3) (2017: 7.3). *Step 2.* $\pi(O')$ is closed by Proposition 7.1.10 (2017: 7.3), $O' \subseteq \mathrm{cl}_X V(x)$ `Chapter7/Basic · EpGroupoid.proposition_7_1_10_of_isProperMap`, a neighbourhood of $\pi(x)$ by Proposition 7.1.8 (2017: 7.1), and inside N. The same properness replaces the text's sets $V(y)$ (2017 p. 246).

Lemma 7.3.4 (2017: 7.6) (V7.45) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_3_4`. $|X|$ *is Hausdorff.* **The text's proof** is by contradiction: $x_k \to x$, $x'_k \to x'$ with morphisms $\varphi_k : x_k \to x'_k$, of which properness extracts a subsequence converging to a morphism $x \to x'$. **The formalized proof** is two lines. *Step 1, points are closed* `Chapter7/OrbitTopology · EpGroupoid.t1Space_orbitSpace`: the point $\{x\}$ is compact, hence a proper subset `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_of_isCompact`, and $\pi(\{x\})$ is closed by Proposition 7.5.10 (1) (2017: 7.14) `Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_10_one` (§VII.25), which uses the same properness as the text's sequence. *Step 2.* A regular T_1 space is Hausdorff, with Lemma 7.3.5 (2017: 7.7) (pp. 324–326 (2017 pp. 245–246)).

Corollary 7.3.6 (2017: 7.1) (V7.47) `Chapter7/OrbitTopology · EpGroupoid.corollary_7_3_6`. *A second countable* $|X|$ *is metrizable*: Urysohn's metrization theorem, an instance of Mathlib, $|X|$ being regular and Hausdorff by Lemmas 7.3.4, 7.3.5 (2017: 7.6, 7.7) (Findings, Chapter 7, item 73) (2017 p. 247).

Subsets of $|X|$ (V7.48). "The induced topology on a subset of $|X|$ is again locally metrizable, regular and Hausdorff": local metrizability passes to subspaces `Chapter7/OrbitTopology · EpGroupoid.exists_metrizable_nhds_subtype`, regularity and Hausdorffness are Mathlib's subspace instances, and so a paracompact subset of $|X|$ is metrizable, by Smirnov's theorem in the subspace `Chapter7/OrbitTopology · EpGroupoid.metrizableSpace_of_paracompactSpace_subtype` (Findings, Chapter 7, item 73) (2017 p. 247).

VII.20 Michael's theorem: Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17)

(V7.50.) Corollary 7.3.8 (2017: 7.2) needs that a finite union of closed paracompact subsets of the regular $|X|$ is paracompact, and the text's route to Proposition 7.5.3 (2017: 7.12) uses Lemma 2.7.3 (2017: 2.7). Part I's Lemma 2.7.3 (2017: 2.7) (Michael: a regular Hausdorff space is paracompact iff every open cover has a locally finite closed refinement) and Proposition 2.7.4 (2017: 2.17) (a regular Haus-

dorff space that is the union of a locally finite family of closed paracompact subspaces is paracompact), 2017 p. 51, were gap G.15 of Part I; they are formalized here, for regular spaces and without Hausdorff. Michael's Lemma 1 has four conditions on a regular space: (a) paracompact; (b) every open cover has a σ-locally finite open refinement; (c) every open cover has a locally finite refinement by arbitrary sets; (d) every open cover has a locally finite closed refinement `Chapter7/Michael · michael_tfae`. *(d) □ (a), in every topological space* `Chapter7/Michael · paracompactSpace_of_closed_refinement`. Let (A_b) be a closed locally finite refinement of the cover (U_i), with $A_b \subseteq U_{f(b)}$; cover the space by open sets each meeting finitely many A_b and take a closed locally finite refinement (B_c) of that cover; then $V_b = (\bigcup\{B_c \mid B_c \cap A_b = \emptyset\})^c \cap U_{f(b)}$ is open (a locally finite union of closed sets is closed), contains A_b, and each B_c meets only the finitely many V_b with $B_c \cap A_b \neq \emptyset$; so (V_b) is a locally finite open refinement. *(c) □ (d) in a regular space* `Chapter7/Michael · closed_refinement_of_refinement`: refine first a cover by open sets whose closures lie in the cover, and take closures. *(b) □ (c), in every space* `Chapter7/Michael · refinement_of_sigma_refinement`: $t_{n,b} \setminus \bigcup_{k<n} \bigcup_{b'} t_{k,b'}$. Hence (c) □ (a) and (b) □ (a) in a regular space `Chapter7/Michael · paracompactSpace_of_refinement` `Chapter7/Michael · paracompactSpace_of_sigma_refinement`, and (a) □ (d) by Mathlib's shrinking lemma, with the same index and $v_i \subseteq u_i$ `Chapter7/Michael · exists_closed_precise_refinement` `Chapter7/Michael · exists_closed_precise_refinement_set`. **Lemma 2.7.3 (2017: 2.7)** is the equivalence of (a) and (d) `Chapter7/Michael · lemma_2_7_3`. **Proposition 2.7.4 (2017: 2.17)** `Chapter7/Michael · proposition_2_7_4` follows the text: closed locally finite refinements $(C^i_j)_j$ of the traces $(Y_i \cap U_j)_j$ by Lemma 2.7.3 (2017: 2.7) in each Y_i, which is regular because Y is; $C_j = \bigcup_i C^i_j$ is locally finite in Y because (Y_i) is and a locally finite family in a closed subspace is locally finite in the space `Chapter7/Michael · locallyFinite_image_val`; and (d) □ (a) in Y. The subspace forms are what Chapter 7 uses: a locally finite family of closed paracompact subsets has a paracompact union `Chapter7/Michael · paracompactSpace_iUnion`, so do finite families and pairs `Chapter7/Michael · paracompactSpace_iUnion_of_finite` `Chapter7/Michael · paracompactSpace_biUnion_of_finite` `Chapter7/Michael · paracompactSpace_union`, and a closed subset of a paracompact subspace is paracompact `Chapter7/Michael · paracompactSpace_of_isClosed_subset`. *The Hausdorff property is never used*; regularity enters through (c) □ (d) and (a) □ (d). Part I's 2017 p. 51 says "subcovering" in the definition of paracompactness, read as "refinement", as on 2017 p. 249 (Findings, Chapter 7, items 75, 114).

VII.21 Definition 7.3.7 (2017: 7.14) and Corollary 7.3.8 (2017: 7.2)

Definition 7.3.7 (2017: 7.14) (V7.49). $K \subseteq |X|$ is component compact if it meets every connected component of $|X|$ in a compact set `Chapter7/EpGroupoid · EpGroupoid.IsComponentCompact` (2017 p. 247).

Corollary 7.3.8 (2017: 7.2) (V7.51) `Chapter7/OrbitTopology · EpGroupoid.corollary_7_3_8`. *A component compact K has an open neighbourhood U with* $\mathrm{cl}_{|X|}U$ *paracompact; then* $\mathrm{cl}U$ *and U are metrizable.* **The text's proof**: work in a connected component of K, so that K may be assumed compact; every $z \in K$ has an open $U(z)$ with metrizable, hence paracompact, closure; finitely many $U(z_i)$ cover K; $U = \bigcup \mathrm{cl}U(z_i)$ is paracompact, Hausdorff, regular and locally metrizable, hence metrizable, and so is $U = \bigcup U(z_i)$. **The formalized proof.** *Step 1, the compact case, inside a given open $W \supseteq K$* `Chapter7/OrbitTopology · EpGroupoid.corollary_7_3_8_of_isCompact`. Every z has an open $U(z)$ with $\mathrm{cl}U(z) \subseteq W$ metrizable, by Lemma 7.3.2 (2017: 7.4) and the regularity of $|X|$ `Chapter7/OrbitTopology · EpGroupoid.exists_isOpen_metrizableSpace_closure`; finitely many cover K; $U = \bigcup U(z_i)$ has $\mathrm{cl}U = \bigcup \mathrm{cl}U(z_i)$, paracompact by Michael (§VII.20), which the text does not cite, and metrizable by §VII.19. *Step 2, the components are open.* "We work in a connected component of K" is read in a component C of $|X|$, where $K \cap C$ is compact. This needs the components of $|X|$ to be open, which the text does not state: $|X|$ is a quotient of the locally path connected X (§VII.7), hence locally connected `Chapter7/OrbitTopology · EpGroupoid.locallyPathConnectedSpace_orbitSpace`. *Step 3, back*

to K, not in the text. Step 1 in each component C, with $W = C$, gives $U_C \supseteq K \cap C$ with $\mathrm{cl}U_C \subseteq C$ paracompact. The components are open and pairwise disjoint, so $(\mathrm{cl}U_C)_C$ is locally finite, $\mathrm{cl}(\bigcup U_C) = \bigcup \mathrm{cl}U_C$, and this locally finite union of closed paracompact sets is paracompact (Proposition 2.7.4 (2017: 2.17), §VII.20) `Chapter7/Michael · paracompactSpace_iUnion`. The text's U names both $\bigcup \mathrm{cl}U(z_i)$ and $\bigcup U(z_i)$ (possible misprint); here U is the open one (Findings, Chapter 7, item 74) (2017 p. 247).

VII.22 Proposition 7.3.9 (2017: 7.11): index lifting

Proposition 7.3.9 (2017: 7.11) (V7.52) `Chapter7/Lift · EpGroupoid.proposition_7_3_9`. *If $X = (X, \mathbf{X})$ is an ep-groupoid, so is $X^1 = (X^1, \mathbf{X}^1)$; if X is tame, so is X^1.* X^1 is the M-polyfold of Part I with the levels shifted by one (§II.6). **The structure maps of X^1.** The text restricts s, t, u, ι, m to level 1 without comment. Sc-smooth maps preserve the levels, so s, t, u, ι restrict `Chapter7/LocalDiffeo · MPolyfold.levelOneMap Chapter7/Lift · EpGroupoid.liftSrc Chapter7/Lift · EpGroupoid.liftTgt Chapter7/Lift · EpGroupoid.liftUnit Chapter7/Lift · EpGroupoid.liftInv`; a composite of composable morphisms on level 1 is on level 1, since $ml(t(\varphi \circ \psi)) = ml(t\varphi) = ml(\varphi)$ (§VII.3) `Chapter7/Lift · EpGroupoid.comp_mem_levelSet Chapter7/Lift · EpGroupoid.liftComp`; the groupoid is `Chapter7/Lift · EpGroupoid.liftGroupoidStr`. **The text's proof** says that (1), (2) and (4) hold trivially, and proves (3) by sequences: for $x \in X_1$ and $V(x)$ with t proper over $s^{-1}(\mathrm{cl}_X V(x))$, $W(x) = V(x) \cap X_1$ is open in X_1 and $\mathrm{cl}_{X_1} W(x) \subseteq \mathrm{cl}_X V(x)$; for (φ_k) with $(t\varphi_k)$ in a compact subset of X_1, after a subsequence (a) $\varphi_k \to \varphi$ in $\mathbf{X}_0$, (b) s, t converge in X_0, (c) $t(\varphi_k) \to t(\varphi)$ in X_1, so (d) $\varphi \in \mathbf{X}_1$, and $\varphi_k = (t|\mathbf{U}(\varphi))^{-1}(t\varphi_k) \to \varphi$ in $\mathbf{X}^1$. **The formalized proof.** *(1).* The level-1 lift of a local sc-diffeomorphism is one: an sc-diffeomorphism on an open set lifts with its inverse `Chapter7/LocalDiffeo · MPolyfold.IsScDiffeoOn.levelOne Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.levelOne Chapter7/Lift · EpGroupoid.isLocalScDiffeo_liftSrc Chapter7/Lift · EpGroupoid.isLocalScDiffeo_liftTgt`. *(2).* Part I's lifting of the index for sc-smooth maps, Proposition 1.2.2 (preprint 1.17) in charts (§I.15) `Chapter5/StrongBundleShift · MPolyfold.IsScSmoothMap.levelOne Chapter7/Lift · EpGroupoid.isScSmoothMap_liftUnit Chapter7/Lift · EpGroupoid.isScSmoothMap_liftInv`. *(4).* The text's "trivially", as we read it, uses $(\mathbf{X}\,{}_s\!\times_t \mathbf{X})^1 = \mathbf{X}^1\,{}_s\!\times_t \mathbf{X}^1$ as M-polyfolds, which is not proved here; it is avoided through the sheet form (§VII.2). A local inverse σ_1 of s on level 1 agrees near each point with the level-1 lift of a local inverse σ of s, both inverting s near that point, and $\psi \mapsto m(\sigma(t\psi), \psi)$ lifts by the lifting of sc-smooth maps on open sets `Chapter7/LocalDiffeo · MPolyfold.IsScSmoothOn.levelOne Chapter7/Lift · EpGroupoid.isScSmoothCompSheet_lift Chapter7/Lift · EpGroupoid.isScSmoothComp_lift`. *(3), the text's argument* `Chapter7/Lift · EpGroupoid.exists_isProperMap_lift`. "$(t\varphi_k)$ in a compact subset of X_1" is replaced by "$t\varphi_k$ converges in X_1", the sequential criterion for a closed set (§VII.2) `Chapter7/ProperMaps · isProperMap_domRestrict_iff_of_isClosed`, X^1 and $\mathbf{X}^1$ being metrizable (Theorem 2.3.10 (2017: 2.2)) `Chapter2/Metrizable · MPolyfold.metrizableSpace_level`; $\mathrm{cl}_{X_1} W(x) \subseteq \mathrm{cl}_X V(x)$ is the continuity of $X_1 \to X$; (a) is properness on level 0, (c) the hypothesis, (d) that t is a local sc-diffeomorphism and preserves levels; step (b) is not needed; and "$(t|\mathbf{U}(\varphi))^{-1}$ is sc^0" is used as the continuity on level 1 of its level-1 lift. So X^1 is an ep-groupoid `Chapter7/Lift · EpGroupoid.liftOne`, paracompact on both levels `Chapter2/Metrizable · MPolyfold.paracompactSpace_level`. The display (7.5), "$s^{-1}(\mathrm{cl}_{X_1}(W(x)) \to X_1$", has an unbalanced parenthesis (possible misprint; Findings, Chapter 7, items 80, 96).

Tameness of X^1 `Chapter7/Lift · EpGroupoid.isTame_liftOne`. The text calls it trivial. As we read it, it needs that the index lift of a tame retraction is tame, which Part I left open (Findings, Chapter 5, item 145); it is proved `Chapter7/Lift · MPolyfold.IsTame.levelOne`. *Step 1, the shifted presentation.* For a tame r with a presentation $T : E \cong \mathbb{R}^n \oplus W$ of the partial quadrant C, the presentation $T^1 : E^1 \cong \mathbb{R}^n \oplus W^1$ of C^1 has the same $\mathbb{R}^n$-coordinates `Chapter7/Lift · shiftPres Chapter7/Lift · shift_presentation`, so

$d_{C^1} = d_C$ on C^1 `Chapter7/Lift · degIndex_shift` and the subspace E_x for T^1 is $E_x \cap E_1$ `Chapter7/Lift · mem_minimalSubspaceWith_shift`. *Step 2, the tangent.* $Dr^1(x) = Dr(x)|_{E_1}$ (Proposition 1.2.2 (preprint 1.17)), whose range is $T_xO \cap E_1$ since $Dr(x)$ is a projection. *Step 3, the complement.* An sc-complement A of T_xO gives the sc-complement $A \cap E_1$ of T_xO^1, an sc-splitting shifting to one `Chapter7/Lift · isScSplitting_levelSub_one` `Chapter7/Lift · isScSubspace_levelSub_one`. *Step 4.* So r^1 is tame for T^1 `Chapter7/Lift · isTameWith_shift` `Chapter7/Lift · isTameRetract_shift`, hence tame by Part I's independence of the presentation `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame`, and the level charts of tame charts form a tame atlas of X^1 (Findings, Chapter 7, item 81) (pp. 326–328 (2017 pp. 247–248)).

VII.23 Definition 7.5.1 (2017: 7.15), Remark 7.5.2 (2017: 7.1), and Proposition 7.5.3 (2017: 7.12)

Definition 7.5.1 (2017: 7.15) (V7.53). An ep-groupoid has paracompact orbit space if in addition (5) $|X|$ is paracompact `Chapter7/EpGroupoid · EpGroupoid.HasParacompactOrbitSpace`; the text stipulates that paracompact does not include Hausdorff, and Mathlib's notion does not. Since $|X|$ is Hausdorff and locally metrizable (Theorem 7.3.1 (2017: 7.2)), (5) is equivalent to the metrizability of $|X|$ `Chapter7/Lift · EpGroupoid.hasParacompactOrbitSpace_iff_metrizableSpace`, and it holds when $|X|$ is second countable (Corollary 7.3.6 (2017: 7.1)) `Chapter7/Lift · EpGroupoid.hasParacompactOrbitSpace_of_secondCountableTopology`. *The sentence before Remark 7.5.2 (2017: 7.1)*, "in case X has a second countable topology, (1)–(4) imply (5)", is read with X the object space `Chapter7/Lift · EpGroupoid.remark_7_5_2`: $|X|$ is then second countable, π being open and onto (Proposition 7.1.8 (2017: 7.1)), hence metrizable by Corollary 7.3.6 (2017: 7.1), hence paracompact. Remark 7.5.2 (2017: 7.1) itself, on the second countability required in [38], has no formal content. The open question the text states here, whether (5) is a consequence of (1)–(4), is settled negatively in §VII.49: the Čech groupoid of a Prüfer-type surface is an ep-groupoid with Hausdorff, non-paracompact orbit space `Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_epGroupoid_not_hasParacompactOrbitSpace` (Findings, Chapter 7, items 82, 113, 152) (p. 340 (2017 p. 249)).

The map $|X^1| \to |X|$. The text writes $|X_1|$ for the orbit space of X^1 and treats it as a subset of $|X|$ ("$A_x \cap |X_1|$", 2017 p. 250). Here $|X^1|$ keeps its own quotient topology: the inclusion $X^1 \to X$ is a functor `Chapter7/Lift · EpGroupoid.liftIncl` and induces $|X^1| \to |X|$ `Chapter7/Lift · EpGroupoid.orbitIncl` `Chapter7/Lift · EpGroupoid.orbitIncl_π`, continuous `Chapter7/Lift · EpGroupoid.continuous_orbitIncl` and injective, because a morphism between points of X_1 lies in $\mathbf{X}_1$ ($ml(s\varphi) = ml(t\varphi)$, §VII.3) `Chapter7/Lift · EpGroupoid.orbitIncl_injective`, with range $|X_1| = \pi(X_1)$ `Chapter7/Lift · EpGroupoid.range_orbitIncl`; the preimage of $\pi(U)$ is $\pi_1(U \cap X_1)$ `Chapter7/Lift · EpGroupoid.preimage_orbitIncl_image`.

Proposition 7.5.3 (2017: 7.12) (V7.54) `Chapter7/Lift · EpGroupoid.proposition_7_5_3`. *If X has paracompact orbit space, so has X^1.* **The text's proof**: by Lemma 2.7.3 (2017: 2.7) the cover $(|U(x)|)$ of $|X|$ has a locally finite closed refinement (A_x), $A_x \subseteq |U(x)|$; $A_{x,1} = A_x \cap |X_1|$ is a locally finite closed cover of $|X_1|$; G_x acts on $U(x)_1$ and, as in Lemma 7.3.2 (2017: 7.4), an invariant metric passes to $|U(x)|_1 = |U(x)_1|$, so $A_{x,1}$ is metrizable, hence paracompact, and $|X_1|$ is a locally finite union of closed paracompact sets (Proposition 2.7.4 (2017: 2.17)). **The formalized proof** proves more, that $|X^1|$ is metrizable `Chapter7/Lift · EpGroupoid.metrizableSpace_liftOne_orbitSpace`, by Smirnov's theorem pulled back along $|X^1| \to |X|$ (Part I): a continuous injection into a paracompact Hausdorff space whose points have open neighbourhoods with metrizable preimages has a metrizable source `Chapter2/Metrizable · metrizableSpace_of_pullback`. *Step 1, the restricted action.* For a natural representation at x, G_x acts on $U(x)_1 = U(x) \cap X_1$ by the level-1 lifts of the $\Phi(g)$, homeomorphisms of $U(x)_1$

Chapter7/Lift · EpGroupoid.NaturalRep.liftU Chapter7/Lift · EpGroupoid.NaturalRep.liftΦ Chapter7/Lift · EpGroupoid.NaturalRep.liftMulAction Chapter7/Lift · EpGroupoid.NaturalRep.continuous_liftSmul. "$U(x)_1$ has the natural G_x-action" is read as this restricted action only: $U(x)_1$ need not be connected in X^1, so it is not a natural representation of X^1. *Step 2, the fibres.* Points of $U(x)_1$ have the same image in $|X^1|$ iff they are in one G_x-orbit, by Theorem 7.1.13 (3) (2017: 7.1) and §VII.3 Chapter7/Lift · EpGroupoid.NaturalRep.liftOne_π_eq_iff. *Step 3, the metric.* $G_x\backslash U(x)_1$ is metrizable by the text's $\hat{\rho}$, the argument of Lemma 7.3.2 (2017: 7.4) repeated for a finite group acting by homeomorphisms on a metrizable space Chapter7/Lift · lvMaxDist Chapter7/Lift · lvQuotDist Chapter7/Lift · metrizableSpace_orbitRel_quotient_of_finite. *Step 4, Proposition 7.1.19 (2017: 7.6) on level* 1. $G_x\backslash U(x)_1 \to |X^1|$ is an open embedding Chapter7/Lift · EpGroupoid.NaturalRep.liftOrbitQuotientMap Chapter7/Lift · EpGroupoid.NaturalRep.isOpenEmbedding_liftOrbitQuotientMap onto the preimage of $|U(x)|$ Chapter7/Lift · EpGroupoid.NaturalRep.range_liftOrbitQuotientMap, so that preimage is metrizable Chapter7/Lift · EpGroupoid.NaturalRep.metrizableSpace_preimage_orbitIncl. *Step 5.* $|X|$ is paracompact by hypothesis and Hausdorff by Lemma 7.3.4 (2017: 7.6), and every point has the open neighbourhood $|U(x)|$; the pulled-back Smirnov theorem gives $|X^1|$ metrizable, hence paracompact (Stone). The regularity of $|X^1|$ and Michael's theorem are not needed on this route, though both are available (§§VII.19, VII.20) (Findings, Chapter 7, item 83) (pp. 340–342 (2017 pp. 249–250)).

VII.24 Theorem 7.5.4 (2017: 7.3), Definition 7.5.5 (2017: 7.16), Remark 7.5.6 (2017: 7.2), and the statement of Theorem 7.5.7 (2017: 7.4)

Theorem 7.5.4 (2017: 7.3) (V7.55) Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_4. *If $|X|$ is paracompact, every open cover (O_λ) of $|X|$ has a subordinate partition of unity (σ_λ), and every cover (U_λ) of X by saturated open sets has a subordinate continuous partition of unity (β_λ) taking equal values on isomorphic objects. Step 1.* The first statement is Mathlib's existence of subordinate partitions of unity on a paracompact normal space Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate. *Hidden hypothesis:* normality, which needs $|X|$ Hausdorff — Lemma 7.3.4 (2017: 7.6) — besides paracompact. *Step 2.* For the second, $O_\lambda = \pi(U_\lambda)$ is open (Proposition 7.1.8 (2017: 7.1)), and $\beta_\lambda = \sigma_\lambda \circ \pi$ Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_apply has $\beta_\lambda(s\varphi) = \beta_\lambda(t\varphi)$ Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_src. *Step 3, supports.* $\operatorname{supp}(\sigma_\lambda \circ \pi) \subseteq \pi^{-1}(\operatorname{supp}\sigma_\lambda) \subseteq \pi^{-1}(\pi(U_\lambda)) = U_\lambda$, since U_λ is saturated Chapter7/PartitionsOfUnity · EpGroupoid.tsupport_orbitPullback_subset Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_isSubordinate. *Step 4, local finiteness.* If an open $V \ni \pi(x)$ meets finitely many supports of the σ_λ, the saturated open $\pi^{-1}(V)$ meets finitely many supports of the β_λ Chapter7/PartitionsOfUnity · EpGroupoid.exists_isSaturated_nhds_finite_orbitPullback. "Subordinate" is Mathlib's, $\operatorname{supp}\sigma_\lambda \subseteq O_\lambda$; "some of the $\sigma_\lambda \equiv 0$" says only that the index set is Λ (Findings, Chapter 7, item 90) (2017 p. 250).

Definition 7.5.5 (2017: 7.16) (V7.56). X has sc-smooth partitions of unity if every cover of X by saturated open sets has a subordinate sc-smooth partition of unity (β_λ) with (1) every point having a saturated open neighbourhood meeting finitely many supports and (2) $\beta_\lambda(s\varphi) = \beta_\lambda(t\varphi)$ Chapter7/EpGroupoid · EpGroupoid.HasScPartitionsOfUnity, the partitions being Part I's Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothPartitionOfUnity with supports the closed supports and index types in the universe of X. The hypothesis is satisfiable: for the trivial ep-groupoid it reduces to Part I's sc-smooth partitions of unity Chapter7/EpGroupoid · EpGroupoid.hasScPartitionsOfUnity_trivial, which $\mathbb{R}^n$ has Chapter7/LocalDiffeo · rnPolyfold_admitsScPartitionsOfUnity (Findings, Chapter 7, item 91) (pp. 341–343 (2017 pp. 250–251)).

Remark 7.5.6 (2017: 7.2) (V7.57). Apart from sc-Hilbert spaces, the existence of sc-smooth partitions of unity is "a zoo"; the remark is an assessment, with nothing to formalize, and the sc-Hilbert case is Corollary 7.6.6, now as printed (§§VII.29, VII.48) (2017 p. 251).

Theorem 7.5.7 (2017: 7.4) (V7.58) `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_7`. *If $|X|$ is paracompact and the object M-polyfold admits sc-smooth partitions of unity, X has sc-smooth partitions of unity.* The proof is the text's Theorem 7.6.4 (2017: 7.6) (§§VII.27–VII.28). The hypotheses are satisfiable, by the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_7_hypotheses_rn` `Chapter7/PartitionsOfUnity · EpGroupoid.hasScPartitionsOfUnity_trivial_rn_of_theorem_7_5_7` (Findings, Chapter 7, items 94, 105) (2017 p. 251).

VII.25 Definition 7.5.8 (2017: 7.17), Propositions 7.5.9 and 7.5.10 (2017: 7.13 and 7.14)

Definition 7.5.8 (2017: 7.17) (V7.59). $Q \subseteq X$ is *proper* if $t : s^{-1}(Q) \to X$ is proper `Chapter7/EpGroupoid · EpGroupoid.IsProperSubset`, and an open U has the *properness property* if $\mathrm{cl}_X U$ is proper `Chapter7/EpGroupoid · EpGroupoid.HasPropernessProperty`; every x has such a U, by Definition 7.1.3 (3) (2017: 7.3) `Chapter7/EpGroupoid · EpGroupoid.exists_hasPropernessProperty`. In metrizable spaces Q is proper iff $s^{-1}(Q) \cap t^{-1}(K)$ is compact for every compact K `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iff_isCompact`, and for closed Q iff the sequential criterion holds `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iff_exists_tendsto_subseq`. Basic facts: a closed subset of a proper set is proper `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.of_subset`; $\emptyset$, finite unions and compact sets are proper `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_empty` `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.union` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_biUnion` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iUnion` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_of_isCompact`; an open set inside one with the properness property has it `Chapter7/ProperSubsets · EpGroupoid.HasPropernessProperty.isProperSubset` `Chapter7/ProperSubsets · EpGroupoid.HasPropernessProperty.mono`, and every neighbourhood of x contains an open, also an open connected, neighbourhood of x with the properness property `Chapter7/ProperSubsets · EpGroupoid.exists_hasPropernessProperty_subset` `Chapter7/ProperSubsets · EpGroupoid.exists_hasPropernessProperty_isConnected_subset` (Findings, Chapter 7, item 92) (2017 p. 251).

Proposition 7.5.9 (2017: 7.13) (V7.60) `Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_9`. *A proper subset is closed.* The text: for $x_n \to x$ in Q, the units 1_{x_n} lie in the compact $(t|s^{-1}(Q))^{-1}(\{x_n\} \cup \{x\})$, a subsequence converges to some φ with $s(\varphi) \in Q$, and $s(\varphi) = x$. The formalization is this argument with the compact preimage (§VII.2) `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.isClosed` (2017 p. 251).

Proposition 7.5.10 (2017: 7.14) (V7.61). *(1) If Q is proper, $\pi(Q)$ is closed. (2) If $|X|$ is paracompact, $A \subseteq |X|$ is closed iff $A = \pi(Q)$ for a proper Q.* **(1)** `Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_10_one`. The text uses sequences $\varphi_k : x_k \to y_k$ with $x_k \in Q$, $y_k \to y$. The formalization uses the closed-map property: a set of $|X|$ is closed iff its preimage is `Chapter7/ProperSubsets · EpGroupoid.isClosed_iff_preimage_π`, and $\pi^{-1}(\pi(Q)) = t(s^{-1}(Q))$ is the image of the closed $s^{-1}(Q)$ (Proposition 7.5.9 (2017: 7.13)) under the proper, hence closed, t `Chapter7/ProperSubsets · EpGroupoid.isClosed_saturation_of_isProperSubset`. The proof's "$x = s(\varphi) \in Q$" reuses the name x (possible misprint).

(2) `Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_10_two`. One direction is (1). **The text's proof** of the other chooses for $z \in A$ a representative x_z and a neighbourhood $U(x_z)$ with the natural action whose closure Q_z is proper; by Lemma 2.7.3 (2017: 2.7) the closed, hence paracompact, A has a locally finite

cover by closed $A_i \subseteq \pi(Q_{z_i})$; $P_i = (\pi|\mathrm{cl}_X U(x_{z_i}))^{-1}(A_i)$ and $Q = \bigcup P_i$ has $\pi(Q) = A$; it is proper because only finitely many A_i meet a neighbourhood of $\pi(y)$, so the sources lie in finitely many P_i. Lemma 2.7.3 (2017: 2.7) assumes a regular Hausdorff space. **The formalized proof** needs only the paracompactness of $|X|$. *Step 1.* Representatives x_z are chosen by the quotient, and $V(x_z)$ is any open neighbourhood with the properness property; the natural action is not used. *Step 2, an open refinement.* The open sets $\pi(V(x_z))$ cover the closed A; Mathlib gives an open locally finite precise refinement (v_z) of this cover of A, from paracompactness alone `Mathlib/Topology/Compactness/Paracompact · precise_refinement_set`. *Step 3, closures.* $P_z = \mathrm{cl}_X V(x_z) \cap \pi^{-1}(\mathrm{cl}(v_z) \cap A)$ is a closed subset of the proper $\mathrm{cl}_X V(x_z)$, hence proper, with $\pi(P_z) \subseteq \mathrm{cl}(v_z) \cap A$; and $\pi(P_z) \supseteq v_z \cap A$, since $v_z \subseteq \pi(V(x_z))$. As the v_z cover A, $\pi(\bigcup P_z) = A$. *Step 4, properness of $Q = \bigcup P_z$.* The closures $\mathrm{cl}(v_z)$ of a locally finite family are locally finite, and a union of proper sets whose images form a locally finite family is proper: a compact K has compact $\pi(K)$, which meets only finitely many $\mathrm{cl}(v_z)$ `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iUnion_of_locallyFinite`. Neither Lemma 2.7.3 (2017: 2.7) nor the regularity of $|X|$ is used, and no version under a regularity hypothesis is needed; the closed refinement of Michael's theorem would give the text's route as well (§VII.20). The text writes $\mathrm{cl}_X(U_{x_{z_i}})$ for $\mathrm{cl}_X(U(x_{z_i}))$ (possible misprint; Findings, Chapter 7, item 92) (pp. 342–344 (2017 pp. 251–252)).

VII.26 Appendix §7.6.1 (2017 §7.5.1): the proof of Theorem 7.1.13 (2017: 7.1) (Theorem 7.6.1 (2017: 7.5), Lemmas 7.6.2 and 7.6.3 (2017: 7.8 and 7.9))

Theorem 7.6.1 (2017: 7.5) (V7.62) `Chapter7/NaturalRep · EpGroupoid.theorem_7_6_1` is Theorem 7.1.13 (2017: 7.1) with $\Phi(g) = t_g \circ s_g^{-1}$; it is stated as the same theorem. **The text's proof** (pp. 344–346 (2017 pp. 253–254)): for every $g \in G_{x_0}$ choose contractible open N_g^t, N_g^s on which t and s are sc-diffeomorphisms onto an open $U_0 \ni x_0$, disjoint for distinct g since G_{x_0} is finite; $N_g = N_g^t \cap N_g^s$, $s_g = s|N_g$, $t_g = t|N_g$; Lemmas 7.6.2 and 7.6.3 (2017: 7.8 and 7.9) give U; $\Phi(g) = t_g \circ s_g^{-1}$, a homomorphism "since the neighborhoods N_g are disjoint and since the structure maps are continuous"; $\Gamma(g, y) = s_g^{-1}(y)$. **The formalized proof.** *Step 1, the sheets* `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.exists_sheets`. G_{x_0} is finite (Proposition 7.1.12 (2017: 7.4)) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.finite_isotropy`; Hausdorff separation of the finite set $G_{x_0} \subseteq \mathbf{X}$ gives pairwise disjoint open $N_g \ni g$, shrunk so that s is an sc-diffeomorphism from N_g onto its open image with inverse $\sigma_g = s_g^{-1}$; t is one too, which is recorded but not used. Only connectedness will be needed, not contractibility. *Step 2, $U \subseteq V$.* The proof does not mention V; it starts at $U_0 = V \cap \bigcap_g s(N_g)$. *Step 3, Lemma 7.6.2 (2017: 7.8), then Lemma 7.6.3 (2017: 7.9)* (below). *Step 4, the homomorphism* `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isScSmoothOn_hat`. On the connected U, $y \mapsto \sigma_{gh}(y)$ and $y \mapsto \sigma_g(\Phi(h)y) \circ \sigma_h(y)$ are continuous sections of s — the second by (4) in the sheet form — agreeing at x_0, so they agree on U (§VII.7) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eqOn_σ_mul` and $\Phi(gh) = \Phi(g)\Phi(h)$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat_mul`. In the same way σ_1 is the unit section u on U `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eqOn_σ_one` and $\Phi(1) = \mathrm{id}$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat_one`, which the text does not mention and which makes $\Phi(g)$ an sc-diffeomorphism with inverse $\Phi(g^{-1})$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isScDiffeoOn_hat`. As we read it, "since the N_g are disjoint and the structure maps are continuous" shows only that the sheet index is locally constant; connectedness of U makes

it constant. *Step 5, (3).* A morphism h between points of U lies in some $N_{g'}$ by Lemma 7.6.2 (2017: 7.8), and s is injective there, so $h = \sigma_{g'}(sh)$; disjointness of the N_g gives uniqueness `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eq_of_mem_N`. *Step 6.* The natural representation is built on every open connected invariant $U \subseteq \bigcap_g s(N_g)$ with the conclusion of Lemma 7.6.2 (2017: 7.8) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.mkRep`, and Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9) provide one `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.exists_naturalRep_sheets`. "$\Phi : G_x \to \mathrm{Diff}_{sc}(U)$" is read with G_{x_0} (possible misprint; Findings, Chapter 7, item 29).

Lemma 7.6.2 (2017: 7.8) (V7.63) `Chapter7/NaturalRep · EpGroupoid.lemma_7_6_2`. *There is an open $U_1 \subseteq U_0$ of x_0 such that every morphism with source and target in U_1 lies in some N_g* — for any open $N_g \ni g$ and any neighbourhood U_0; the sheet property is not used. **The text's proof**: otherwise there are $h_k \notin \bigcup N_g$ with $s(h_k), t(h_k) \to x_0$; by properness $h_{k_l} \to h$, necessarily $h \in G_{x_0}$, so $h \in N_g$ for some g, a contradiction. **The formalized proof** follows it. *Step 1.* The sequence is taken along a countable decreasing basis of x_0, X being metrizable. *Step 2.* "By the properness assumption" is Definition 7.1.3 (3) (2017: 7.3) at x_0, which applies to the h_k with $s(h_k) \in \mathrm{cl}V(x_0)$, i.e. to all large k (§VII.2). *Step 3.* "Necessarily $h \in G_{x_0}$": $t(h) = x_0$ comes with the properness, $s(h) = x_0$ from the continuity of s and the uniqueness of limits. *Step 4.* N_g is an open neighbourhood of the limit, so eventually $h_{k_l} \in N_g$. "The properness assumption of ep-polyfolds" is read as that of ep-groupoids (possible misprint; Findings, Chapter 7, item 30) (pp. 344–346 (2017 pp. 253–254)).

Lemma 7.6.3 (2017: 7.9) (V7.64) `Chapter7/NaturalRep · EpGroupoid.lemma_7_6_3`. *There is an open $U_2 \subseteq U_1$ of x_0 such that $U = \bigcup_g t_g s_g^{-1}(U_2)$ lies in U_1 and is invariant under all $t_g s_g^{-1}$.* **The text's proof**: choose U_2 so small that U and $t_g s_g^{-1}(U)$ lie in U_1; for $x = t_h s_h^{-1}(u)$, the point $v = t_g s_g^{-1} t_h s_h^{-1}(u) \in U_1$ is the target of a morphism $u \to v$, which by Lemma 7.6.2 (2017: 7.8) is $s_{g'}^{-1}(u)$, so $v = t_{g'} s_{g'}^{-1}(u) \in U$. **The formalized proof.** *Step 1.* Each $t_g s_g^{-1}$ is continuous at its fixed point x_0 and G_{x_0} is finite, so a U_2 with $t_h s_h^{-1}(U_2) \subseteq U_1$ and $t_g s_g^{-1} t_h s_h^{-1}(U_2) \subseteq U_1$ for all g, h exists; it is taken *connected* (§VII.7). *Step 2.* U is open, t being open and $s_g^{-1}(U_2) = N_g \cap s^{-1}(U_2)$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.unionHat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isOpen_image_hat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isOpen_unionHat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.mem_unionHat_iff`, and connected, a union of connected sets through $x_0 = t_g s_g^{-1}(x_0)$, so no passage to a component is needed `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isConnected_unionHat`. *Step 3, invariance.* The morphism $u \to v$ is $\sigma_g(x) \circ \sigma_h(u)$, composable because $x \in U_1 \subseteq s(N_g)$; it lies in some $N_{g'}$ by Lemma 7.6.2 (2017: 7.8), and "it has necessarily the form $v = t_{g'} s_{g'}^{-1}(u)$" uses that s is injective on $N_{g'}$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.image_σ` (Findings, Chapter 7, item 31) (2017 p. 254).

VII.27 Appendix §7.6.2 (2017 §7.5.2): Lemma 7.6.5 (2017: 7.10)

Theorem 7.6.4 (2017: 7.6) (V7.65) `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_6_4` is Theorem 7.5.7 (2017: 7.4), stated again (Findings, Chapter 7, item 95) (2017 p. 255).

Lemma 7.6.5 (2017: 7.10) (V7.66) `Chapter7/PartitionsOfUnity · EpGroupoid.lemma_7_6_5`. *Let $|X|$ be paracompact and X admit sc-smooth partitions of unity; let $U(x) \subseteq \mathrm{cl}_X U(x) \subseteq V(x)$ be open, $V(x)$ with the natural G_x-action and $U(x)$ invariant. A continuous $\beta : X \to [0,1]$ supported in $U(x)$ and invariant on $V(x)$ lies below an sc-smooth $f \geq 0$ supported in $V(x)$ and invariant on $V(x)$.* The formalization adds the conclusion $\mathrm{supp} f \subseteq \mathrm{cl}_X U(x)$, which the proof gives and Theorem 7.5.7 (2017: 7.4) needs, and omits the hypothesis $x \in U(x)$, which the proof does not use and Theorem 7.5.7 (2017: 7.4) cannot supply; the paracompactness of $|X|$

and the continuity of β are kept, unused. **The text's proof**: open $W(z)$, $z \in V(x)$, with numbers a_z — $\mathrm{cl}W(z) \subseteq U(x)$ and $\beta < a_z$ on $W(z)$ for $z \in U(x)$, otherwise $W(z) \subseteq V(x)$, $\beta|W(z) = 0$, $a_z = 0$; an sc-smooth partition of unity (σ_z) of $V(x)$ subordinate to them; $\gamma = \sum \sigma_z a_z \geq \beta$, supported in $\mathrm{cl}U(x)$; and $f(y) = |G_x|^{-1} \sum_g \gamma(g * y)$. **The formalized proof.** *Step 1, a cover of X.* The hypothesis is that X, not the open subset $V(x)$, admits sc-smooth partitions of unity, so the cover is completed by $X \setminus \mathrm{cl}U(x)$, where $\beta = 0$ and $a = 0$; and it is coarsened to two sets, $U(x)$ with $a = 1$ (as $\beta \leq 1$) and $X \setminus \mathrm{supp}\,\beta$ with $a = 0$. The text's $\mathrm{cl}W(z) \subseteq U(x)$ and $\beta < a_z$ are not needed. *Step 2, γ.* γ is the function attached to $U(x)$ of a partition of unity subordinate to $\{U(x), X \setminus \mathrm{supp}\,\beta\}$: sc-smooth, ≥ 0, supported in $U(x)$, and $\gamma = 1 \geq \beta$ on $\mathrm{supp}\,\beta$, where the other function vanishes. *Step 3, the average.* $f = |G_x|^{-1} \sum_g \gamma \circ \Phi(g)$ on $V(x)$ and 0 elsewhere. Each $\gamma \circ \Phi(g)$, extended by 0, is supported in $\Phi(g)^{-1}(\mathrm{cl}U(x)) = \mathrm{cl}U(x) \subseteq V(x)$ by the invariance of $U(x)$, hence sc-smooth on X `Chapter7/PartitionsOfUnity · isScSmoothMap_real_of_tsupport_subset` `Chapter7/PartitionsOfUnity · isScSmoothMap_real_of_isScSmoothOn_nhds`. *Step 4.* $\beta \leq f$ by the invariance of β: $\beta(y) = \beta(g*y) \leq \gamma(g*y)$ for every g. f is invariant by the reindexing $g \mapsto gh$, and $\mathrm{supp} f \subseteq \mathrm{cl}U(x)$ (Findings, Chapter 7, item 93) (2017 p. 255).

VII.28 Appendix §7.6.2 (2017 §7.5.2): the proof of Theorem 7.5.7 (2017: 7.4)

The text's proof (pp. 346–347 (2017 pp. 255–256)). For every x choose an open $V(x)$ with (1) an open $W(x)$ with the natural G_x-action, $V(x) \subseteq \mathrm{cl}V(x) \subseteq W(x)$ and $V(x)$ invariant; (2) $t : s^{-1}(\mathrm{cl}V(x)) \to X$ proper; (3) $V(x) \subseteq U_\lambda$ for some λ; (4) $(\pi^{-1}(|V(x)|) \setminus V(x)) \cap W(x) = \emptyset$. Take a locally finite refinement (V_i') of $(|V(x)|)$ with $V_i' \subseteq |V(x_i)|$, open $V_i'' \subseteq \mathrm{cl}V_i'' \subseteq V_i'$ covering $|X|$, $U_i'' = V(x_i) \cap \pi^{-1}(V_i'')$ and a partition of unity (β_i) subordinate to (V_i''). "Approximate $(\beta_i \circ \pi)|V(x_i)$ using Lemma 7.10" gives f_i supported in $V(x_i)$; averaging, extending to the saturation and by 0 gives h_i; $h = \sum h_i > 0$ is locally finite and (h_i/h) is the partition of unity.

The formalized proof (V7.67) builds the same functions h_i, but for a Urysohn lemma, and normalizes with Mathlib's bump coverings, as Part I's Theorem 5.5.10 (2017: 5.12) does (§V.25) `Chapter5/PartitionsOfUnity · admitsScPartitionsOfUnity_of_approx`. The four points where the text needed more than a reading are marked (a)–(d).

Step 1, properties (1)–(3) `Chapter7/PartitionsOfUnity · EpGroupoid.exists_invariant_nhds_of_isOpen`. Given x and an open $O \ni x$, take $V_0 \subseteq O$ with the properness property (§VII.25), $W(x) = U$ of a natural representation inside V_0 (Theorem 7.1.13 (2017: 7.1)), an open $N \ni x$ with $\mathrm{cl}N \subseteq W(x)$, X being regular, and $V(x) = \bigcap_g \Phi(g)^{-1}(N)$: open, invariant, $x \in V(x)$, $\mathrm{cl}V(x) \subseteq \mathrm{cl}N \subseteq W(x)$, and proper over its closure by §VII.25. Elements of one orbit have one image in $|X|$ `Chapter7/PartitionsOfUnity · EpGroupoid.π_Φ_eq`.

Step 2, (4) follows from (1) and is not a further choice. The morphisms between points of $W(x)$ are the $\Gamma(g, \cdot)$ (Theorem 7.1.13 (3) (2017: 7.1)), so a point of $W(x)$ isomorphic to a point of the invariant $V(x)$ is $g * v$ for some $v \in V(x)$, hence lies in $V(x)$ `Chapter7/PartitionsOfUnity · EpGroupoid.mem_of_π_mem_image_of_invariant` `Chapter7/PartitionsOfUnity · EpGroupoid.saturation_diff_inter_eq_empty`.

Step 3, extension to the saturation. A G_x-invariant f on $W(x)$ defines F on $|X|$ by $F(|y|) = f(y)$ for $y \in W(x)$ and 0 elsewhere, well defined because points of $W(x)$ in one orbit differ by the natural action (§VII.11) `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension` `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension_π` `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension_nonneg` `Chapter7/PartitionsOfUnity · EpGroupoid.support_orbitExtension_subset`. (d) *$F \circ \pi$ is sc-smooth* when f is

sc-smooth with $\operatorname{supp} f \subseteq \operatorname{cl} V(x)$ `Chapter7/PartitionsOfUnity · EpGroupoid.isScSmoothMap_orbitExtension`: $F \circ \pi$ vanishes off the saturation of $\operatorname{supp} f$, which is closed because $\operatorname{supp} f \subseteq \operatorname{cl} V(x)$ is proper by (2) `Chapter7/ProperSubsets · EpGroupoid.isClosed_saturation_of_isProperSubset`; near a point $y \cong z$ of that saturation, with $\varphi : z \to y$, $F \circ \pi = f \circ s \circ (t|\mathbf{U}(\varphi))^{-1}$ (Definition 7.1.4 (2017: 7.4)). The text does not mention the closedness.

Step 4, the invariant Urysohn lemma `Chapter7/PartitionsOfUnity · EpGroupoid.exists_isScSmoothMap_orbit_urysohn`: *for disjoint closed $s', t' \subseteq |X|$ there is a continuous $F : |X| \to [0,1]$ with $F \circ \pi$ sc-smooth, $F = 0$ on s' and $F = 1$ on t'.* For every $x \in \pi^{-1}(t')$ Step 1 gives $V(x) \subseteq \operatorname{cl} V(x) \subseteq W(x) \subseteq \pi^{-1}(|X| \setminus s')$. The open sets $|V(x)|$ cover the closed t'; take an open locally finite precise refinement (V'_x) of this cover of t' `Mathlib/Topology/Compactness/Paracompact · precise_refinement_set` and a bump covering (b_x) of t' subordinate to it `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate`, whose $\operatorname{supp} b_x \subseteq V'_x$ takes the place of the text's V''_x; put $U''_x = V(x) \cap \pi^{-1}(V'_x)$. (b) *The cut-off.* $c_x = b_x \circ \pi$ on $V(x)$, 0 elsewhere, is continuous and supported in U''_x: a point of $\operatorname{cl} V(x) \cap \pi^{-1}(\operatorname{supp} b_x)$ lies in $W(x)$ and is isomorphic to a point of $V(x)$, as $\operatorname{supp} b_x \subseteq |V(x)|$, hence lies in $V(x)$ by (4). (a) *Lemma 7.6.5 (2017: 7.10)* is applied with $W(x)$ in the role of $V(x)$ and U''_x in the role of $U(x)$: its hypothesis $\operatorname{cl} U''_x \subseteq W(x)$, with the natural action on $W(x)$, holds by (1), and U''_x is invariant. The text's "approximate $(\beta_i \circ \pi)|V(x_i)$ using Lemma 7.10" reads as $V(x_i)$ in the role of $V(x)$ (possible misprint). It gives an invariant sc-smooth $f_x \geq c_x$ with $\operatorname{supp} f_x \subseteq \operatorname{cl} U''_x$, and F_x its extension (Step 3), using $\operatorname{cl} U''_x \subseteq \operatorname{cl} V(x)$. (c) *Local finiteness.* $\operatorname{supp}(F_x \circ \pi) \subseteq \pi^{-1}(\operatorname{cl} V'_x)$ by the support bound of Lemma 7.6.5 (2017: 7.10), and $(\operatorname{cl} V'_x)$ is locally finite, so $G = \sum_x F_x$ is a locally finite sum. The text does not state this bound: Lemma 7.6.5 (2017: 7.10) as stated gives only $\operatorname{supp} f_x \subseteq W(x)$, and the saturations of the $W(x)$ need not be locally finite. *The function.* $G \circ \pi$ is sc-smooth, $G = 0$ on s' since every $W(x)$ misses $\pi^{-1}(s')$, and $G \geq 1$ on t', since at every point of t' some b_x equals 1, and then $f_x \geq c_x = 1$ over it; $F = \tau \circ G$ with Mathlib's smooth transition function τ `Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition`, 0 on $(-\infty, 0]$ and 1 on $[1, \infty)$.

Step 5, the normalization `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_7`. Given a saturated open cover (U_λ), the $\pi(U_\lambda)$ cover $|X|$ `Chapter7/PartitionsOfUnity · EpGroupoid.iUnion_image_π_eq_univ`. Mathlib's existence of bump coverings subordinate to an open cover with all functions satisfying a predicate for which Urysohn's lemma holds `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate_of_prop`, for the class of the g with $g \circ \pi$ sc-smooth and Step 4 as its input, gives (b_λ) on the paracompact normal $|X|$, subordinate to $(\pi(U_\lambda))$, each $b_\lambda \circ \pi$ sc-smooth. The partition $\rho_\lambda = b_\lambda \prod_{\mu<\lambda}(1 - b_\mu)$ `Mathlib/Topology/PartitionOfUnity · BumpCovering.toPartitionOfUnity` is locally a finite product of such functions, so $\rho_\lambda \circ \pi$ is sc-smooth, and $\beta_\lambda = \rho_\lambda \circ \pi$, Theorem 7.5.4's (2017: 7.3) pull-back (§VII.24), has the properties (1), (2) of Definition 7.5.5 (2017: 7.16). This replaces the text's (h_i/h) and the passage from the index set of the refinement back to Λ, which the text leaves implicit. "We average using the G_{x_i}-action": f_x is already invariant by Lemma 7.6.5 (2017: 7.10), and no second average is taken. "$t : s^{-1}(\mathrm{cl}_X(V(x)) \to X$" has an unbalanced parenthesis (possible misprint; Findings, Chapter 7, items 94, 96) (pp. 346–347 (2017 pp. 255–256)).

VII.29 Corollary 7.6.6 (2017: 7.3)

Corollary 7.6.6 (2017: 7.3) (V7.68) `Chapter7/PartitionsOfUnity · EpGroupoid.corollary_7_6_6`. *If $|X|$ is paracompact and the object M-polyfold has an atlas whose local models have Hilbert spaces E_0, every saturated open cover of X has a subordinate sc-smooth partition of unity of X.* The text cites Part I's Proposition 5.5.8 (2017: 5.6), Theorem 5.5.10 (2017: 5.12) and Corollary 5.5.17 (2017: 5.2), the ingredients of Part

I's theorem that such an M-polyfold admits sc-smooth partitions of unity (§§V.25–V.26) Chapter5/PartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert, and Theorem 7.5.7 (2017: 7.4) applies. "An atlas consisting of charts whose local models have Hilbert spaces" is existential, one chart per point; "every compatible chart has a Hilbert E_0" would be unsatisfiable. In the first reading, as in Part I, the theorem of Godefroy–Troyanski–Whitfield–Zizler (Part I's Theorem 5.5.16 (2017: 5.14)) was an explicit hypothesis Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement (gap G.53), which holds for finite-dimensional spaces Chapter7/LocalDiffeo · godefroyTroyanskiWhitfieldZizlerStatement_of_finiteDimensional Chapter7/LocalDiffeo · admitsContDiffPartitionsOfUnity_of_finiteDimensional. The hypotheses are satisfiable on $\mathbb{R}^1$ Chapter7/LocalDiffeo · rnPolyfold_one_hilbert_hypothesis Chapter7/LocalDiffeo · rnPolyfold_one_admitsScPartitionsOfUnity_of_hilbert Chapter7/PartitionsOfUnity · EpGroupoid.corollary_7_6_6_hypotheses_rn. *The corollary as printed.* Part I now proves Theorem 5.5.16 Chapter5/WCGPartitionsOfUnity · theorem_5_5_16; the case used here is that of a separable space Chapter5/SeparablePartitionsOfUnity · godefroyTroyanskiWhitfieldZizlerStatement_of_separable, E_0 being separable for every sc-Banach space, so an M-polyfold with Hilbert E_0 admits sc-smooth partitions of unity with no hypothesis Chapter5/SeparablePartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional, and Corollary 7.6.6 holds as printed Chapter7/HilbertPartitionsOfUnityEp · EpGroupoid.corollary_7_6_6_unconditional, with the same witness on $\mathbb{R}^1$ Chapter7/HilbertPartitionsOfUnityEp · EpGroupoid.corollary_7_6_6_hypotheses_rn_unconditional (gap G.53, closed; §VII.48) (Findings, Chapter 7, items 95, 106) (p. 347 (2017 p. 256)).

VII.30 Witnesses and counterexamples

Every hypothesis of the chapter that is a predicate on ep-groupoids or on their data is shown satisfiable, and for the readings that do not hold a counterexample is given (Findings, Chapter 7, items 100–107, 152, 156).

The trivial ep-groupoid of an M-polyfold Z, with only units, exists on every M-polyfold Chapter7/EpGroupoid · EpGroupoid.trivial. It is tame for tame Z and for $\mathbb{R}^n$ Chapter7/EpGroupoid · EpGroupoid.isTame_trivial Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn, effective with $X_{reg} = Z$ Chapter7/EpGroupoid · EpGroupoid.isEffective_trivial Chapter7/EpGroupoid · EpGroupoid.regularSet_trivial, every closed set is proper and every open set has the properness property Chapter7/EpGroupoid · EpGroupoid.isProperSubset_trivial Chapter7/EpGroupoid · EpGroupoid.hasPropernessProperty_trivial, $|X| \cong Z$ Chapter7/EpGroupoid · EpGroupoid.trivialOrbitHomeomorph is paracompact Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial, and it has natural representations on every connected open neighbourhood Chapter7/EpGroupoid · EpGroupoid.trivialNaturalRep Chapter7/EpGroupoid · EpGroupoid.trivialNaturalRepFinite. Its witnesses are vacuous for the isotropy, which the next ones are not.

The translation groupoid $K \ltimes U$ of a finite discrete group acting on an M-polyfold by sc-smooth maps, with morphisms the product with the discrete M-polyfold Chapter7/NonVacuity · Examples.actionMorPolyfold, is an ep-groupoid Chapter7/NonVacuity · Examples.actionEpGroupoid: s is a projection, t a projection after the shear Chapter7/NonVacuity · Examples.actionShear Chapter7/NonVacuity · Examples.action_isLocalScDiffeo_src Chapter7/NonVacuity · Examples.action_isLocalScDiffeo_tgt, (4) holds in the sheet form Chapter7/NonVacuity · Examples.action_isScSmoothCompSheet and t is proper on all of $\mathbf{X}$ Chapter7/NonVacuity · Examples.action_isProperMap_tgt. Its isotropy is the stabilizer Chapter7/NonVacuity · Examples.isotropyEquivStabilizer Chapter7/NonVacuity · Examples.isRegular_action_iff Chapter7/NonVacuity · Examples.regularSet_action, its orbit space U/K Chapter7/NonVacuity · Examples.action_π_eq_π_iff Chapter7/NonVacuity · Examples.actionOrbitHomeomorph, its proper sets the closed sets Chapter7/NonVacuity · Examples.isProperSubset_action Chapter7/NonVacuity · Examples.isProperSubset_

action_iff Chapter7/NonVacuity · Examples.hasPropernessProperty_action, and its orbit space is paracompact for second countable U Chapter7/NonVacuity · Examples.hasParacompactOrbitSpace_action.

$\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection Chapter7/NonVacuity · Examples.isScSmoothMap_units_smul Chapter7/NonVacuity · Examples.reflectionEpGroupoid is effective Chapter7/NonVacuity · Examples.isEffective_reflection with a non-trivial isotropy element $(-1, 0)$ Chapter7/NonVacuity · Examples.reflectionIsotropyNegOne, so Definition 7.2.1 (2017: 7.11) is met non-vacuously Chapter7/NonVacuity · Examples.isEffective_reflection_nonvacuous; $X_{reg} = \mathbb{R} \setminus \{0\}$ Chapter7/NonVacuity · Examples.regularSet_reflection, $G_0 \cong \mathbb{Z}/2$ Chapter7/NonVacuity · Examples.reflectionIsotropyZeroEquiv, $T_x = 1$ Chapter7/NonVacuity · Examples.trivialSubgroup_reflection; it is tame and has paracompact orbit space Chapter7/NonVacuity · Examples.isTame_reflection Chapter7/NonVacuity · Examples.hasParacompactOrbitSpace_reflection.

The trivial action of $\mathbb{Z}/3$ on $\mathbb{R}$ Chapter7/NonVacuity · Examples.trivialZ3EpGroupoid has $G_x \cong \mathbb{Z}/3$ Chapter7/NonVacuity · Examples.trivialZ3IsotropyEquiv, $T_x = G_x$ Chapter7/NonVacuity · Examples.trivialSubgroup_trivialZ3 Chapter7/NonVacuity · Examples.trivialZ3TrivialSubgroupEquiv, is not effective Chapter7/NonVacuity · Examples.not_isEffective_trivialZ3, and its regular set is empty Chapter7/NonVacuity · Examples.regularSet_trivialZ3: the text's "examples show that the regular set can be empty" (p. 308 (2017 p. 239)). It carries the counterexample to the remark after Definition 7.1.7 (2017: 7.7) (§VII.4).

The pair groupoid of $\mathbb{Z}$ Chapter7/NonVacuity · Examples.pairGroupoidStr Chapter7/NonVacuity · Examples.pairEpGroupoid is an ep-groupoid in which properness fails over all of $\mathbb{Z}$ Chapter7/NonVacuity · Examples.pair_exists_not_isProperMap, refuting the universal reading of Definition 7.1.3 (3) (2017: 7.3) (§VII.2). The abstract finiteness statements of §VII.6 are met by the trivial action of a finite discrete group, whose isotropy set has $|G|$ elements Chapter7/Discrete · isotropy_snd_witness.

$\mathbb{Z}/2$ acting on a dimension-jumping retract Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.X, by $(t, z) \mapsto (t, \pm z)$ on the retract of Part I's Remark 4.2.2 (2017: 4.3) example, is a non-effective ep-groupoid whose reduced morphism space $\mathbf{X}^R$ is not Hausdorff Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.exists_not_t2Space_reducedMor: the hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) cannot be dropped (§VII.16; Findings, Chapter 7, item 107). Its non-effective part is computed exactly: $\mathbf{X}^T = (\{1\} \times O) \cup (\{-1\} \times \{t < 0\})$, with $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$ one point Chapter7/ZhouRetract · EpGroupoid.Examples.JumpingRetract.appendix_7_6_3 (§VII.36).

Zhou's retract Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.X, $\mathbb{Z}/2$ acting by $(t, f) \mapsto (t, -f)$ on the sc-smooth retract of $\mathbb{R} \oplus L^2(\mathbb{R})$ of Appendix 7.6.3, for every choice of the data Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.Data, of which one exists Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.Data.std: an ep-groupoid in which T_R is not Hausdorff and LUC and (ZHOU) fail (§VII.47; Findings, Chapter 7, item 156).

The Čech groupoid of a Prüfer-type surface Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.epGroupoid: an ep-groupoid whose orbit space is Hausdorff and not paracompact, so property (5) of Definition 7.5.1 (2017: 7.15) is not a consequence of (1)–(4) (§VII.49; Findings, Chapter 7, item 152).

VII.31 What is not formalized

Nothing of the chapter: every gap is closed, and the one narrower row is a reading (§VII.50). In detail:

- **Closed since the first reading.** $T\varphi$ as a bounded linear isomorphism and an sc-isomorphism (gap G.70), with the tangent facts (3), (4) on A (§VII.44); $|X_\infty|$ and $|X_\infty| = \bigcap_k |X_k|$ (gap G.72) and the

density of X_∞ in X_k (gap G.73) (§VII.44); the M-polyfold $\mathbf{X}^R$ (gap G.71): paracompact for every ep-groupoid, Hausdorff exactly under LUC, which is the 2021 text's hypothesis (§§VII.34, VII.35, VII.48); Corollary 7.6.6 (2017: 7.3) as printed, Part I's gap G.53 being closed (§§VII.29, VII.48).

- **The 2017 statements that do not hold as we read them.** The remark after Definition 7.1.7 (2017: 7.7) for functors that are only sc-smooth on objects (§VII.4; the 2021 remark holds), and Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) for every ep-groupoid (§VII.16; the 2021 forms hold). Each has a counterexample in Lean.
- **The open question of p. 340 (2017 p. 249)**, whether (5) follows from (1)–(4): settled negatively (§VII.49). Remark 7.5.2 (2017: 7.1) has no formal content, and Remark 7.5.6 (2017: 7.2) is an assessment, with nothing to formalize (§§VII.23, VII.24; Findings, Chapter 7, items 82, 113, 152).
- **Not needed.** The contractibility of N^t_g, N^s_g in the proof of Theorem 7.6.1 (2017: 7.5) (p. 344 (2017 p. 253)): connectedness suffices (§VII.26; Findings, Chapter 7, item 29). The identification $(\mathbf{X}\,{}_s\!\times_t \mathbf{X})^1 = \mathbf{X}^1\,{}_s\!\times_t \mathbf{X}^1$, which the sheet form avoids (§VII.22). The comparison with the classical properness of (s,t) on 2017 p. 227 has no formal content.
- **Gaps of Part I taken up**: G.15 (Lemma 2.7.3 (2017: 2.7), Proposition 2.7.4 (2017: 2.17)) is formalized (§VII.20); the local path connectedness of M-polyfolds (§VII.7) and, now, the path connectedness of $U \cap X_\infty$ (§VII.45); the tameness of X^1, left open in Part I, is proved (§VII.22) (Findings, Chapter 7, item 114).

VII.32 What the formalization taught us

Continuity is the content of the remark after Definition 7.1.7 (2017: 7.7). The formula $f(\gamma) = (s|\mathbf{U}(\psi))^{-1}(f(s\gamma))$ is right wherever it applies; what it needs is a neighbourhood of φ mapped into the sheet at $f(\varphi)$, and for that the functor must be continuous on morphisms. A functor of the trivial $\mathbb{Z}/3$-action that swaps two isotropy elements on one side of 0 shows that sc-smoothness on objects alone does not give it, as we read the remark (§VII.4).

Connectedness is the hidden hypothesis of the natural representation. Every proof that Φ is a homomorphism, that Γ is a functor, that $\Gamma(1,a) = 1_a$ (Lemma 7.2.2 (2017: 7.2)), and that two natural representations agree compares two continuous sections of s that agree at x; without connectedness they agree only near x. The text chooses a connected $U(x)$ on 2017 p. 231 and never states it again. Making U connected a field of the datum, and proving that M-polyfolds are locally path connected, settles all of these at once (§§VII.7–VII.9, VII.26).

"Small" neighbourhoods and the kernel of Φ. Since sc-smooth maps are not analytic, the kernel of Φ can depend on $U(x)$, and the text's "for $U(x)$ small, $\ker \Phi = T_x$" needs a precise form: a neighbourhood N of x such that every natural representation inside N has kernel T_x, and the intrinsic description $g \in T_x$ iff $s = t$ near g. That description turns Definition 7.2.1 (2017: 7.11) into $T_x = \{1_x\}$, proves the effectiveness of X_R in one line, and gives the conjugation $\varphi T_x \varphi^{-1} = T_y$ that the composition of X_R needs and the text does not state (§§VII.15, VII.17).

The reduced charts do not give Hausdorffness. The proof of Proposition 7.2.10 (2017: 7.9) builds charts and sc-smooth transitions for $\mathbf{X}^R$ without any hypothesis; an M-polyfold also has to be Hausdorff and paracompact, and these do not follow from the charts as we read it. The formalization separates what is proved unconditionally from the two hypotheses, and shows them satisfiable (§VII.16).

A star instead of an induction. Proposition 7.1.15 (2017: 7.5) becomes a construction transported from one object along chosen morphisms, with nothing shrunk afterwards; the text's inductive adjustment of

components, and most of its possible misprints, have no counterpart (§VII.10).

Properness does more than sequences. Where the text extracts convergent subsequences, the formalization mostly uses that a proper map is closed or that preimages of compact sets are compact: Proposition 7.1.10 (2017: 7.3), Proposition 7.5.10 (1) (2017: 7.14), the closedness of points behind Lemma 7.3.4 (2017: 7.6), and the closed saturations behind the sc-smoothness in Theorem 7.5.7 (2017: 7.4). With compact sets in place of sequences, Proposition 7.5.10 (2) (2017: 7.14) needs only paracompactness, no regularity (§§VII.5, VII.19, VII.25, VII.28).

Paracompact without Hausdorff. Definition 7.5.1 (2017: 7.15) stipulates that paracompact does not include Hausdorff, and Michael's theorem, which the text quotes for regular Hausdorff spaces, holds for regular spaces; one direction holds in every space. This closes gap G.15 of Part I and gives Corollary 7.3.8 (2017: 7.2) its finite and locally finite unions (§§VII.20, VII.21). For Proposition 7.5.3 (2017: 7.12) a shorter route proves more: $|X^1|$ is metrizable (§VII.23).

Lemma 7.6.5 (2017: 7.10) needs its support bound. As stated, Lemma 7.6.5 (2017: 7.10) bounds the support of f by $V(x)$; the proof of Theorem 7.5.7 (2017: 7.4) needs the bound $\operatorname{cl} U(x)$, which the lemma's own proof gives, for the sum $\sum h_i$ to be locally finite, and it applies the lemma to sets U_i'' that need not contain x_i. Stating the lemma with the stronger conclusion and the weaker hypothesis makes the proof of Theorem 7.5.7 (2017: 7.4) go through as written, with $W(x_i)$ in the role of $V(x)$ (§§VII.27–VII.28).

Paracompactness of the orbit space is a genuine condition. The text leaves open whether (5) follows from (1)–(4). It does not: the orbit space of an ep-groupoid is locally a finite quotient of an M-polyfold, so every Hausdorff, locally Euclidean space whose atlas has a Čech groupoid with proper target maps is one, and a Prüfer-type surface is such a space without being paracompact. The object space of the example is not second countable, as it must be by Remark 7.5.2 (2017: 7.1) (§VII.49).

What the tangent of a morphism needs is the linear structure of T_xX. Once Part I gives T_xX its linear structure through charts, Definition 7.1.4 (2017: 7.4) is the chain rule and the fact that the derivative at a smooth point is an sc-operator; nothing in Chapter 7 itself was missing (§VII.44).

VII.33 The 2021 additions to §7.1: the remark after Definition 7.1.7, Propositions 7.1.14, 7.1.15, Lemma 7.1.16, Corollary 7.1.17

The Springer edition (2021) changes the remark after Definition 7.1.7 and Proposition 7.1.15 and adds Proposition 7.1.14, Lemma 7.1.16 and Corollary 7.1.17; Sections VII.33–VII.41 retell the additions of the 2021 edition, with its numbers and pages (V7.69–V7.98).

The remark after Definition 7.1.7, 2021 form (V7.69) `Chapter7/Book2021Basic · EpGroupoid.remark_definition_7_1_7`. *A functor sc-smooth on objects and continuous on morphisms is sc-smooth on morphisms, through $f(\gamma) = (s|U(\psi))^{-1}(f(s(\gamma)))$ on a neighbourhood $U(\varphi)$ mapped into a sheet $U(\psi)$ at $\psi = f(\varphi)$.* The 2021 text adds the continuity on morphisms, the hypothesis under which §VII.4 proved the 2017 remark; the continuity gives the neighbourhood $U(\varphi)$, the sheets give the formula, and sc-smoothness follows as in §VII.4. The functor of the counterexample of §VII.4 is sc-smooth on objects and not continuous on morphisms, so the 2021 hypothesis excludes it `Chapter7/Book2021Basic · EpGroupoid.remark_definition_7_1_7_remarkFunctor` (Findings, Chapter 7, item 117) (p. 291).

The data (7.3) and Proposition 7.1.14 (V7.70) `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_14`. *For data (7.3) — an open $W(x)$ and sheets $U^s(g)$, $U^t(g)$ of s and t at each $g \in G_x$ onto $W(x)$ — and a*

representation (Φ,Γ) *with the properties of Theorem 7.1.13, there is a* Φ*-invariant open* $V(x) \subseteq W(x) \cap U(x)$ *on which* $\Gamma(g,v) = s_g^{-1}(v)$ *and* $\Phi(g)(v) = t(s_g^{-1}(v))$. The data are a structure `Chapter7/Book2021Basic · EpGroupoid.IsotropySheetData` and exist at every object `Chapter7/Book2021Basic · EpGroupoid.exists_isotropySheetData`; "neighbourhoods $U^s(g)$, $U^t(g)$ for g" is read as neighbourhoods of g, without which $\Gamma(g,x) = g = s_g^{-1}(x)$ fails. *The proof.* On the open set N of the $v \in W(x) \cap U(x)$ with $\Gamma(g,v) \in U^s(g) \cap U^t(g)$ for all g, $\Gamma(g,v) = s_g^{-1}(s(\Gamma(g,v))) = s_g^{-1}(v)$; applying t gives the formula for $\Phi(g)$, and $\Gamma(g,v) = t_g^{-1}(\Phi(g)(v))$, the identity (7.4) of the proof, is part of the conclusion. $V(x) = \bigcap_g \Phi(g)^{-1}(N)$, a finite intersection, is open and invariant. Property (3) of Theorem 7.1.13 is a hypothesis as printed and is not used. The same holds for every natural representation `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_14_of_naturalRep` (Findings, Chapter 7, item 118) (p. 296).

Proposition 7.1.15, 2021 form (V7.71) `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_15`. *For a finite set* Y *of mutually isomorphic objects and* $V \in \mathcal{N}(Y)$ *there is a natural representation of* Y *in* V *whose sheets* $\mathbf{U}(\psi) \ni \psi$ *are the components of the morphisms between points of* $\bigcup_y U(y)$*, with* $s : \mathbf{U}(\psi) \to U(s(\psi))$*,* $t : \mathbf{U}(\psi) \to U(t(\psi))$ *sc-diffeomorphisms,* $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$ *and* $t(\psi') = f(\psi)(s(\psi'))$ *on* $\mathbf{U}(\psi)$. This is the star construction of §VII.10, whose structure already had the 2021 form of (2). *The 2017 form of (2).* As printed in 2017, $s,t : \mathbf{U}(\psi) \to U(y)$ with one y for both maps; with $y \in U(y)$, disjoint $U(y)$ and $\psi \in \mathbf{U}(\psi)$ this forces $s(\psi) = t(\psi)$ `Chapter7/Book2021Basic · EpGroupoid.src_eq_tgt_of_printed_2017`, and for $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection with $Y = \{1,-1\}$ the hypotheses hold, a natural representation of the 2021 form exists, and no data of the printed 2017 form exist `Chapter7/Book2021Basic · EpGroupoid.not_printed_2017_reflection`. So the 2017 form does not hold as we read it, and the 2021 form is the one proved (Findings, Chapter 7, items 25, 119) (p. 297).

Lemma 7.1.16 (V7.72) `Chapter7/Book2021Basic · EpGroupoid.lemma_7_1_16`. *If* $\psi'_k \in \mathbf{U}(\psi)$ *converge to* ψ' *and* $s(\psi'_k)$ *or* $t(\psi'_k)$ *converges to a point of* $U(s(\psi)) \cup U(t(\psi))$*, then* $\psi' \in \mathbf{U}(\psi)$ *and the sources and targets converge inside* $U(s(\psi))$, $U(t(\psi))$. The four cases reduce to two, since a limit lying in $U(b)$ of points of $U(a)$ lies in $U(a)$, the $U(y)$ being open and disjoint `Chapter7/Book2021Basic · EpGroupoid.NaturalRepFinite.mem_U_of_tendsto`; then s on $\mathbf{U}(\psi)$ is a homeomorphism onto $U(s(\psi))$ and the limits agree (Findings, Chapter 7, item 120) (p. 300).

Corollary 7.1.17 (4)–(7) (V7.73). *(4) For* $x \in Y$*, the natural representation of* G_x *on* $U(x)$ *has* $\Gamma(g,z) = (s|\mathbf{U}(g))^{-1}(z)$ *and* $f(g) = \Phi(g)$ `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_4` — by a topological argument, the image of the connected $U(x)$ under $\Gamma(g,\cdot)$ lying in one component, since $f(\psi) = f(\psi')$ does not force $\psi = \psi'$ in a non-effective ep-groupoid. *(5)* $f(\varphi)$ is equivariant along the conjugation $g \mapsto \varphi \circ g \circ \varphi^{-1}$ `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_5`. *(6)* The multiplication is an sc-diffeomorphism from $\mathbf{U}(\psi)\,{}_s\!\times_t \mathbf{U}(\varphi)$, an open subset of the fibred-product sub-M-polyfold of Part I's Proposition 2.5.20 (2017: 2.15), onto $\mathbf{U}(\psi \circ \varphi)$ `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_6`: it is $(s|\mathbf{U}(\psi\circ\varphi))^{-1} \circ s \circ \mathrm{pr}_2$ once its image is known to lie in $\mathbf{U}(\psi\circ\varphi)$, which follows, as in the text, from the connectedness of the domain and the continuity of the multiplication, without sequences. *(7)* The $U(y)$ can be taken with the properness property `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_7` (Findings, Chapter 7, item 120) (p. 301).

VII.34 The 2021 §7.2: T_x, the topology T_R, Lemma 7.2.11, Propositions 7.2.12, 7.2.13 and Theorem 7.2.14

Definition 7.2.5 and Lemma 7.2.6 (V7.74) `Chapter7/ReducedCriterion · EpGroupoid.definition_7_2_5` `Chapter7/ReducedCriterion · EpGroupoid.lemma_7_2_6`. T_x *is the maximal subgroup of* G_x *acting trivially on some small*

$W(x)$*; conjugation by* $\varphi : x \to y$ *maps* T_x *onto* T_y. For a natural representation, T_x^V is the group of the g with $\Phi(g) = \mathrm{id}$ on $V \cap U(x)$ `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn` `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.mem_kerOn`; it grows as V shrinks `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_anti`, is normal for invariant V `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_normal` and lies in the intrinsic T_x of §VII.15 `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_le_trivialSubgroup`. "Maximal" is read as two statements: $T_x \supseteq T_x^W$ for every open $W \ni x$, with equality for every small W, by §VII.15 Step 3. Lemma 7.2.6 is the conjugation $\varphi T_x \varphi^{-1} = T_y$ that §VII.15 had proved for the composition of X_R, now an isomorphism `Chapter7/ReducedCriterion · EpGroupoid.trivialSubgroupEquiv` `Chapter7/ReducedCriterion · EpGroupoid.trivialSubgroupEquiv_apply` (Findings, Chapter 7, item 121) (p. 310).

Remark 7.2.8 and Lemma 7.2.9 (V7.75) `Chapter7/ReducedCriterion · EpGroupoid.remark_7_2_8`. T_R *is the finest topology on* $\mathbf{X}^R$ *for which* π *is continuous; in general* $(X, \mathbf{X}^R)$ *cannot be made an ep-groupoid, but it can when* T_R *is Hausdorff.* The finest such topology is the quotient topology of §VII.16, and π is an open quotient map `Chapter7/ReducedCriterion · EpGroupoid.isOpenQuotientMap_reducedπ`; Lemma 7.2.9 is §VII.16 Steps 2 and 3 `Chapter7/Reduced · EpGroupoid.isOpenMap_reducedπ` `Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ`. In Definition 7.2.7 the target map is printed "$s([\varphi]) = t_X(\varphi)$", a possible misprint for $t([\varphi])$ `Chapter7/Reduced · EpGroupoid.reducedTgt`. The remark's "in general not" is what the counterexample of §VII.16 shows (Findings, Chapter 7, items 122, 123) (p. 311).

Lemma 7.2.11 (V7.76). *If* $\varphi_1, \dots, \varphi_k$ *are the other members of the class of* φ_0*, there are neighbourhoods* $U(\varphi_i)$ *with disjoint closures, each mapped by* π *homeomorphically onto one* $U([\varphi_0])$*, which contain every* ψ *with* $[\psi] \in U([\varphi_0])$. The class is finite `Chapter7/ReducedCriterion · EpGroupoid.finite_reducedπ_preimage`, by properness. The conclusion is a predicate `Chapter7/ReducedCriterion · EpGroupoid.IsReducedEvenlyCovered`, with "homeomorphism onto $U([\varphi_0])$" read as: $\pi(U(\varphi_i)) = U([\varphi_0])$ and π an open embedding on $U(\varphi_i)$. *As we read it, the lemma needs* T_R *Hausdorff.* The text's proof of (3) takes a sequence ψ_k with $[\psi_k] \to [\varphi_0]$ and uses that the limit of $[\psi_k]$ is unique. Without that, (1)–(3) at every unit already make $\mathbf{X}^T$ closed `Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_isReducedEvenlyCovered`, and (1)–(3) at every morphism are equivalent to T_R Hausdorff `Chapter7/ReducedCriterion · EpGroupoid.isReducedEvenlyCovered_iff_t2Space`; in the ep-groupoid of §VII.16 they fail at $(1, o_0)$, whose class contains only $(1, o_0)$ while the morphisms $(-1, (-1/(n+1), 0))$ have classes converging to it `Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.reducedπ_preimage_one` `Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isReducedEvenlyCovered`. *Under* T_R *Hausdorff,* the setting of Theorem 7.2.14 where the lemma is used, every open $N \supseteq \pi^{-1}([\varphi_0])$ contains $\pi^{-1}(O)$ for an open $O \ni [\varphi_0]$, which gives (3) for sheets at the finitely many members of the class, chosen with disjoint closures `Chapter7/ReducedCriterion · EpGroupoid.exists_preimage_reducedπ_subset_of_t2Space` `Chapter7/ReducedCriterion · EpGroupoid.isReducedEvenlyCovered_of_t2Space`, and the lemma holds `Chapter7/ReducedCriterion · EpGroupoid.lemma_7_2_11` (Findings, Chapter 7, item 124) (p. 315).

Proposition 7.2.12 (V7.77) `Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_12`. $(\mathbf{X}^R, T_R)$ *is paracompact, for every ep-groupoid.* **The text's proof** refines an open cover through the neighbourhoods of Lemma 7.2.11, which as we read it need T_R Hausdorff. **The formalized proof** needs no hypothesis. *Step 1, descent.* Paracompactness descends along an open continuous surjection p if every point q has a compact K such that every open $N \supseteq K$ contains $p^{-1}(O)$ for some open $O \ni q$ `Chapter7/ReducedCriterion · EpGroupoid.paracompactSpace_of_isOpenMap`: the images of a locally finite precise refinement upstairs form one downstairs, with no separation axiom. *Step 2, the compact set.* For $q = [\varphi]$ and V the open set of Definition 7.1.3 (3) at $s(\varphi)$, $K = \{\psi \in s^{-1}(\mathrm{cl}\, V) \mid (s,t)(\psi) = (s,t)(\varphi)\}$ is compact by properness and contains the class of φ; (s,t) is proper, hence closed, on $s^{-1}(\mathrm{cl}\, V)$, so an open $N \supseteq K$ contains $\pi^{-1}(O)$ for an open $O \ni q$ `Chapter7/ReducedCriterion · EpGroupoid.exists_isCompact_preimage_reducedπ_subset`. So `Paracom–`

pactSpace X.ReducedMor is an instance Chapter7/ReducedCriterion · EpGroupoid.paracompactSpace_reducedMor, and of the two hypotheses of §VII.16 only the Hausdorff one remains, the 2021 text's own (gap G.71, closed) (Findings, Chapter 7, items 111, 125) (p. 317).

Proposition 7.2.13 and Theorem 7.2.14 (V7.78) Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_13 Chapter7/ReducedCriterion · EpGroupoid.theorem_7_2_14. *The natural charts make $(\mathbf{X}^R, T_R)$ an M-polyfold iff T_R is Hausdorff; if T_R is Hausdorff, $(X, \mathbf{X}^R)$ is naturally an ep-groupoid.* An M-polyfold is Hausdorff and paracompact by definition, so Proposition 7.2.13 is read as the equivalence of "an M-polyfold structure with the atlas of §VII.16 exists" and T_R Hausdorff; one direction is the definition, the other is §VII.16 with Proposition 7.2.12. Theorem 7.2.14 is Proposition 7.10 (2017) of §VII.17 under exactly the hypothesis the formalization had taken: X_R Chapter7/Reduced · EpGroupoid.reducedEpGroupoid, with the M-polyfold $\mathbf{X}^R$ Chapter7/Reduced · EpGroupoid.reducedPolyfold, the groupoid structure of §VII.17 and the ep-functor $X \to X_R$; it is effective (Proposition 7.2.16) Chapter7/Reduced · EpGroupoid.isEffective_reducedEpGroupoid. The 2021 text thus agrees with the reading of §§VII.16–VII.17 (gap G.71, closed in the 2021 form); the 2021 statements, with the Hausdorff hypothesis alone, are §VII.48 Chapter7/ReducedNaturalCharts · EpGroupoid.proposition_7_2_16 (Findings, Chapter 7, items 56, 107, 111, 126, 143) (p. 318).

VII.35 Proposition 7.2.17, the local unique continuation property, and Theorem 7.2.22

Proposition 7.2.17 (V7.79) Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_17. *T_R is Hausdorff iff $\mathbf{X}^T$ is closed in $\mathbf{X}$. Step 1, the relation.* $[\varphi] = [\psi]$ iff $t\varphi = t\psi$ and $\psi^{-1} \circ \varphi \in \mathbf{X}^T$ Chapter7/ReducedCriterion · EpGroupoid.reducedπ_eq_iff_comp_inv; in particular $\varphi \in \mathbf{X}^T$ iff $[\varphi]$ is the class of a unit Chapter7/ReducedCriterion · EpGroupoid.mem_nonEffectivePart_iff_reducedπ. *Step 2, closed relation.* $(\varphi, \psi) \mapsto \psi^{-1} \circ \varphi$ is continuous on $\{s\varphi = s\psi\}$ Chapter7/ReducedCriterion · EpGroupoid.continuous_comp_fibredProduct Chapter7/ReducedCriterion · EpGroupoid.continuousOn_comp_inv, so the relation is closed when $\mathbf{X}^T$ is Chapter7/ReducedCriterion · EpGroupoid.isClosed_reducedπ_eq_of_isClosed, and a quotient by an open quotient map is Hausdorff iff its relation is closed (Mathlib). *Step 3, conversely,* if T_R is Hausdorff, $\mathbf{X}^T = \{\varphi \mid [\varphi] = [1_{s(\varphi)}]\}$ is closed, the set where two continuous maps into a Hausdorff space agree Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_t2Space. Properness is not needed. Always, $\mathrm{cl}\,\mathbf{X}^T \subseteq \{s = t\}$ Chapter7/ReducedCriterion · EpGroupoid.isClosed_src_eq_tgt Chapter7/ReducedCriterion · EpGroupoid.closure_nonEffectivePart_subset (Findings, Chapter 7, item 127) (p. 320).

Definition 7.2.18 (= 7.4.1), Remarks 7.2.19 and 7.2.20 (V7.80). *(LUC): every object has a natural representation on which $\hat{g} = \mathrm{id}$ on a nonempty open subset forces $\hat{g} = \mathrm{id}$.* Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.HasLUC Chapter7/ReducedCriterion · EpGroupoid.IsLUC. The choice of $U(x)$ does not matter: if $\mathbf{X}^T$ is closed, every natural representation has the property Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.hasLUC_of_isClosed, so LUC holds iff it holds for all of them Chapter7/ReducedCriterion · EpGroupoid.isLUC_iff_forall_naturalRep, and Remark 7.2.20 (inside every $O(x)$) follows Chapter7/ReducedCriterion · EpGroupoid.remark_7_2_20. Remark 7.2.19, some ep-groupoid is not LUC, is §VII.36 Chapter7/ReducedCriterion · EpGroupoid.Examples.remark_7_2_19. Effective ep-groupoids are LUC, $\mathbf{X}^T$ being the closed set of units Chapter7/ReducedCriterion · EpGroupoid.isClosed_range_unit Chapter7/ReducedCriterion · EpGroupoid.nonEffectivePart_eq_range_unit_of_isEffective Chapter7/ReducedCriterion · EpGroupoid.isLUC_of_isEffective, and so are the trivial ep-groupoid, the reflection and the non-effective trivial $\mathbb{Z}/3$-action Chapter7/ReducedCriterion · EpGroupoid.isLUC_trivial Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_reflection Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_trivialZ3 (Findings, Chapter 7, items 128, 144) (p. 321).

Proposition 7.2.21, Theorem 7.2.22 and Proposition 7.2.23 (V7.81) Chapter7/ReducedCriterion · EpGroupoid.theorem_7_2_22. *T_R Hausdorff, LUC and $\mathbf{X}^T$ closed are equivalent; under LUC, X_R is an ep-groupoid. Proposition 7.2.21, by connectedness.* In a natural representation, $\Gamma(g,y) \in \mathbf{X}^T$ iff $\Phi(g) = \mathrm{id}$ near y Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.eventually_Φ_eq_of_Γ_mem Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.Γ_mem_of_eventually. So the set of $y \in U(x)$ near which $\Phi(g) = \mathrm{id}$ is open, and closed in $U(x)$ when $\mathbf{X}^T$ is closed, hence empty or all of the connected $U(x)$: LUC. Conversely, under LUC, a morphism $\varphi \in \mathrm{cl}\,\mathbf{X}^T$ has $s\varphi = t\varphi = x$, so $g = \varphi \in G_x$; the open sheet $\Gamma(g, U(x)) \ni \varphi$ meets $\mathbf{X}^T$ in some $\Gamma(g,y)$, so $\Phi(g) = \mathrm{id}$ near y, LUC gives $\Phi(g) = \mathrm{id}$ on $U(x)$, and $\varphi \in T_x$ Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_isLUC Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_21. With Proposition 7.2.17 this is Theorem 7.2.22, and Proposition 7.2.23 is Theorem 7.2.14 under LUC Chapter7/ReducedCriterion · EpGroupoid.t2Space_reducedMor_of_isLUC Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_23. For the trivial $\mathbb{Z}/3$-action, $\mathbf{X}^T = \mathbf{X}$ is closed and T_R is Hausdorff Chapter7/ReducedCriterion · EpGroupoid.Examples.nonEffectivePart_trivialZ3 Chapter7/ReducedCriterion · EpGroupoid.Examples.t2Space_reducedMor_trivialZ3, the case the first reading had left open (Findings, Chapter 7, items 60, 128, 129) (pp. 321–323).

VII.36 Appendices 7.6.3 and 7.6.4: Zhou's example, and the reduced groupoid of X^1

Appendix 7.6.3 (V7.82) Chapter7/ReducedCriterion · EpGroupoid.Examples.remark_7_2_19. *Zhou's example: $\mathbb{Z}/2$ acting on the retract of Part I's Subsection 1.4.4 by $(s,f) \mapsto (s,-f)$; $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$ is one point, so T_R is not Hausdorff and LUC fails.* The formalization uses its own example of §VII.16, $\mathbb{Z}/2$ acting by $(t,z) \mapsto (t, \pm z)$ on the retract of Part I's Remark 4.2.2 (2017: 4.3), which has the same features: the action is trivial over $t < 0$ and moves points arbitrarily close to the origin over $t > 0$. The point $(-1, o_0)$ is a limit of $(-1,(t,0))$, $t < 0$, which lie in $\mathbf{X}^T$, and is not in $\mathbf{X}^T$ Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.mem_closure_nonEffectivePart Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isClosed_nonEffectivePart, so by Theorem 7.2.22 LUC fails Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isLUC and T_R is not Hausdorff, as §VII.16 had shown directly Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_t2Space_reducedMor. $\mathbf{X}^T$ *exactly.* Since then the non-effective part of the analogue is computed: $\mathbf{X}^T = (\{1\} \times O) \cup (\{-1\} \times \{(t,z) \in O \mid t < 0\})$ — the element 1 acts trivially everywhere, -1 near exactly the points over $t < 0$, since near a point over $t \geq 0$ there are points of O with $z \neq 0$ — and $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T = \{(-1, o_0)\}$, one point, as the text says of its example Chapter7/ZhouRetract · EpGroupoid.Examples.JumpingRetract.nonEffectivePart_eq Chapter7/ZhouRetract · EpGroupoid.Examples.JumpingRetract.closure_nonEffectivePart_diff Chapter7/ZhouRetract · EpGroupoid.Examples.JumpingRetract.appendix_7_6_3. Zhou's example itself is §VII.47 Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.appendix_7_6_3 (gap G.83, closed). The counterexample found in the first reading is thus consistent with the 2021 text: it is of the kind Remark 7.2.8 announces and Appendix 7.6.3 gives, and it fails LUC as Theorem 7.2.22 requires (Findings, Chapter 7, items 107, 129, 130, 156) (p. 348).

Appendix 7.6.4, Theorem 7.6.7 and Corollary 7.6.8 (V7.83) Chapter7/ReducedCriterion · EpGroupoid.theorem_7_6_7. *Under (SLUC), $(\mathbf{X}^T)^1 = (\mathbf{X}^1)^T$, and the identification of $(X^1)_R$ with $(X_R)^1$ is an sc-diffeomorphism; likewise for every i. (7.24).* A morphism of level 1 in $\mathbf{X}^T$ lies in $(\mathbf{X}^1)^T$, since $s = t$ near it in $\mathbf{X}$ gives $s = t$ near it in $\mathbf{X}^1$ Chapter7/ReducedCriterion · EpGroupoid.mem_nonEffectivePart_liftOne_of_levelVal. *The converse* follows the text: for g acting trivially near x in X^1, $\hat{g} = \mathrm{id}$ on $U(x) \cap X_1$ by LUC for X^1, through the level-one form of Γ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.levelOneΓ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.continuousOn_levelOneΓ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.eventually_of_levelOneΓ_mem Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.levelOneΓ_mem_of_eventually, and on $U(x)$ by density

Chapter7/ReducedCriterion · EpGroupoid.levelVal_mem_nonEffectivePart_of_liftOne. The text uses that $U(x) \cap X_1$ is connected in X^1 and dense in $U(x)$; both were a hypothesis Chapter7/ReducedCriterion · EpGroupoid.HasConnectedDenseLevelOneReps, the density being that of X_1 in X (gap G.73), and both now hold for every ep-groupoid Chapter7/SmoothConnectedReps · EpGroupoid.hasConnectedDenseLevelOneReps (§VII.45), so that $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ needs only LUC for X^1 Chapter7/SmoothConnectedReps · EpGroupoid.nonEffectivePart_liftOne_eq_of_isClosed. Then $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ Chapter7/ReducedCriterion · EpGroupoid.nonEffectivePart_liftOne_eq, the two relations agree on level 1 Chapter7/ReducedCriterion · EpGroupoid.liftOne_reducedπ_eq_iff, and $[\varphi]_1 \mapsto [\varphi]$ Chapter7/ReducedCriterion · EpGroupoid.reducedLevelOneMap is well defined, injective and continuous with range $\pi(\mathbf{X}_1)$. Of (SLUC) only LUC for X^1 is used. Corollary 7.6.8 takes "(ZHOU) implies LUC for X^1" as a hypothesis Chapter7/ReducedCriterion · EpGroupoid.corollary_7_6_8, which §VII.38 discharges Chapter7/Zhou · EpGroupoid.corollary_7_6_8_of_isZhou. All hypotheses hold for the trivial ep-groupoid of a point Chapter7/ReducedCriterion · EpGroupoid.Examples.pointEpGroupoid Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_liftOne_point Chapter7/ReducedCriterion · EpGroupoid.Examples.hasConnectedDenseLevelOneReps_point Chapter7/ReducedCriterion · EpGroupoid.Examples.theorem_7_6_7_point Chapter7/Zhou · EpGroupoid.Examples.corollary_7_6_8_of_isZhou_point. The sc-diffeomorphism, which needs the M-polyfold $(\mathbf{X}^R)^1$, and $i \geq 2$ are §VII.46 (gap G.82, closed) (Findings, Chapter 7, item 131) (pp. 348–351).

VII.37 §7.4 (new in the 2021 edition): index lifts, Lemma 7.4.4, the Part I facts, and Tg

§7.4 has no 2017 counterpart. It introduces the Zhou condition and the unique continuation and regularity properties that the reduced groupoid and the index lifts need.

Index lifts, Proposition 7.4.2 and Remark 7.4.3 (V7.84). The lifts X^i are bundled ep-groupoids Chapter7/Zhou · EP.liftOne Chapter7/Zhou · EP.lift, with $X^0 = X$, $X^{i+1} = (X^i)^1 = (X^1)^i$ Chapter7/Zhou · EP.lift_zero Chapter7/Zhou · EP.lift_succ Chapter7/Zhou · EP.lift_succ'. *Under LUC, the property passes from a connected invariant $U(x)$ to a larger $W(x)$, so $U(x)$ may be taken small or large.* By §VII.35 every natural representation has the property under LUC, so the hypotheses on $U(x)$ are not used Chapter7/Zhou · EpGroupoid.proposition_7_4_2 Chapter7/Zhou · EpGroupoid.remark_7_4_3 (Findings, Chapter 7, item 132) (pp. 328–329).

Lemma 7.4.4 (V7.85) Chapter7/Zhou · EpGroupoid.lemma_7_4_4. *If X^i has LUC, so has every X^k, $k \leq i$.* For $i = 1$ the text shows that the points of $U(x) \cap X_1$ near which $\hat{g} = \mathrm{id}$ in X^1 form an open, closed and nonempty subset of the connected $U(x) \cap X_1$, and concludes by density Chapter7/Zhou · EpGroupoid.NaturalRep.eqOn_of_isClosed_liftOne Chapter7/Zhou · EpGroupoid.isLUC_of_isLUC_liftOne. The connectedness of $U(x) \cap X_1$ in X^1 is used without proof, and the density is that of X_1 in X (gap G.73); both are a hypothesis on each X^k, $k < i$ Chapter7/Zhou · EpGroupoid.HasLevelOneConnectedDenseReps (gap G.85, closed: the hypothesis holds for every ep-groupoid, §VII.45; the lemma is not restated without it), satisfied on subsingletons and by every lift of the point Chapter7/Zhou · EpGroupoid.hasLevelOneConnectedDenseReps_of_subsingleton Chapter7/Zhou · EpGroupoid.Examples.hasLevelOneConnectedDenseReps_lift_pointEP Chapter7/Zhou · EpGroupoid.Examples.lemma_7_4_4_pointEP. In the proof, "$s : W(g) \to W(y)$" for $W(h)$ (possible misprint) (Findings, Chapter 7, item 133) (p. 329).

The Part I facts (V7.86). The proofs of §7.4 use, without stating them, facts about an M-polyfold that Part I did not state; in the first reading they were hypotheses, and §VII.45 proves them for every M-polyfold (gap G.85, closed): X_∞ dense in every X_m (gap G.73) Chapter7/Zhou · MPolyfold.DenseSmooth, $U \cap X_\infty$ connected for connected open U Chapter7/Zhou · MPolyfold.SmoothConnected, maps agreeing near a smooth point in X_∞ have the same tangent there Chapter7/Zhou · MPolyfold.TangentDeterminedSmooth Chapter7/Zhou · MPolyfold.TangentDeterminedSmooth.eq_self, and tangent vectors are approximated by tangent

vectors at nearby smooth points, which follows from the openness of $p : TX \to X^1$ (gap G.8) Chapter7/Zhou · MPolyfold.eqOn_tangentFiber_of_mem_closure. They are bundled as one hypothesis Chapter7/Zhou · MPolyfold.SmoothRegular, with the density of $(T_xX)_\infty$ in T_xX separate Chapter7/Zhou · MPolyfold.DenseSmoothTangent, and consequences Chapter7/Zhou · MPolyfold.DenseSmooth.dense Chapter7/Zhou · MPolyfold.DenseSmooth.subset_closure_inter Chapter7/Zhou · MPolyfold.DenseSmooth.exists_smooth. They hold on subsingletons with all points smooth and on the discrete M-polyfold of a point Chapter7/Zhou · MPolyfold.smoothRegular_of_subsingleton Chapter7/Zhou · MPolyfold.smoothRegular_discrete_unit Chapter7/Zhou · MPolyfold.denseSmoothTangent_discrete_unit (gap G.85) (Findings, Chapter 7, item 134) (pp. 331–340).

The tangent Tg (V7.87). For a smooth x and $g \in G_x$, Chapter 7's tangent of a morphism (Definition 7.1.4, $Tt \circ (Ts|T_g\mathbf{X})^{-1}$, without charts, §VII.3) is, on T_xX, the tangent $T\Phi(g)$ of $\hat{g}$, for every natural representation Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_Γ Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_eq_tΦ Chapter7/Zhou · EpGroupoid.NaturalRep.tΦ_eq_tΦ: $t = \Phi(g) \circ s$ on the sheet $\Gamma(g, \cdot)$. At a fixed point z of $\Phi(g)$, $\Gamma(g, z) \in G_z$ acts near z as g Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ_val Chapter7/Zhou · EpGroupoid.NaturalRep.eventually_Φ_eq_Γ Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_isoΓ, and lies in T_z iff $\Phi(g) = \mathrm{id}$ near z Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ_mem_trivialSubgroup_iff. Tg is a set map on the fibre; (ZHOU) asks only that it be the identity, so the linear structure of T_xX (gap G.7) is not needed (Findings, Chapter 7, item 135) (p. 331).

VII.38 §7.4: (ZHOU), (ZHOU'), unique continuation, (SLUC) and (VSLUC)

The forms below take the Part I facts of §VII.37 as hypotheses; since these hold for every M-polyfold, every result also holds without them, and the restatements are listed in §VII.45.

Definitions 7.4.5 and 7.4.7 (V7.88). *(ZHOU): for every smooth x and $g \in G_x$, $Tg = \mathrm{id}$ on T_xX implies $g \in T_x$* Chapter7/Zhou · EpGroupoid.IsZhou. T_x^∞, the g acting trivially near x in X_∞ Chapter7/Zhou · EpGroupoid.trivialSetInf Chapter7/Zhou · EpGroupoid.trivialSubgroupInf, testable in any natural representation Chapter7/Zhou · EpGroupoid.mem_trivialSetInf_iff Chapter7/Zhou · EpGroupoid.mem_trivialSubgroupInf_iff, contains T_x Chapter7/Zhou · EpGroupoid.trivialSubgroup_le_trivialSubgroupInf. *(ZHOU'): $Tg = \mathrm{id}$ on $(T_xX)_\infty$ implies $g \in T_x^\infty$* Chapter7/Zhou · EpGroupoid.IsZhou'. Both hold when every $T_x = G_x$ Chapter7/Zhou · EpGroupoid.isZhou_of_trivialSubgroup_eq_top Chapter7/Zhou · EpGroupoid.isZhou'_of_trivialSubgroup_eq_top, on subsingletons Chapter7/Zhou · EpGroupoid.isZhou_of_subsingleton Chapter7/Zhou · EpGroupoid.isZhou'_of_subsingleton, for the trivial epgroupoid Chapter7/Zhou · EpGroupoid.isZhou_trivial Chapter7/Zhou · EpGroupoid.isZhou'_trivial and for the trivial $\mathbb{Z}/3$-action, where $T_x \neq 1$ Chapter7/Zhou · EpGroupoid.Examples.isZhou_trivialZ3 Chapter7/Zhou · EpGroupoid.Examples.isZhou'_trivialZ3 (Findings, Chapter 7, items 135, 144) (p. 331).

Lemma 7.4.6 and Proposition 7.4.8 (V7.89) Chapter7/Zhou · EpGroupoid.proposition_7_4_8. *(ZHOU) and (ZHOU') are equivalent. (ZHOU) □ (ZHOU')*: if $Tg = \mathrm{id}$ on the dense $(T_xX)_\infty$, then on T_xX, and (ZHOU) gives $g \in T_x \subseteq T_x^\infty$ Chapter7/Zhou · EpGroupoid.lemma_7_4_6. *(ZHOU') □ (ZHOU)*: the text uses sequences; the formalization shows that the smooth points y of $U(x)$ that $\Phi(g)$ fixes with tangent the identity form an open and closed subset of the connected $U(x)_\infty$ Chapter7/Zhou · EpGroupoid.NaturalRep.mem_tangentU_of_fiber Chapter7/Zhou · EpGroupoid.NaturalRep.tΦ_eq_self_of_eqOn Chapter7/Zhou · EpGroupoid.NaturalRep.fix_and_tangent_of_mem_closure Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_smooth_clopen Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_smooth_tangent; continuity and density give $\Phi(g) = \mathrm{id}$ near x Chapter7/Zhou · eq_of_mem_closure_of_continuousAt Chapter7/Zhou · EpGroupoid.isZhou_of_isZhou'. The Part I facts of §VII.37 are hypotheses, satisfied by the point Chapter7/Zhou · EpGroupoid.Examples.proposition_7_4_8_point.

"$z_k \to Y$" for $z_k \to y$ in the proof (possible misprint) (Findings, Chapter 7, item 136) (pp. 331–332).

Proposition 7.4.10 and Corollary 7.4.11 (V7.90) `Chapter7/Zhou · EpGroupoid.proposition_7_4_10`. *(ZHOU) implies LUC, and then every X^i has LUC.* The text shows that the smooth points near which $\hat{g} = \mathrm{id}$ form an open and closed subset of $U(x)_\infty$, closedness by approximating tangent vectors at a limit point by tangent vectors at nearby smooth points. The formalization isolates the conclusion as unique continuation through X_∞ `Chapter7/Zhou · EpGroupoid.IsSmoothUC` `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_isZhou`, which gives LUC for X, by the density of X_∞, and for X^1, through Proposition 7.2.21, $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ being closed in $\mathbf{X}^1$ `Chapter7/Zhou · EpGroupoid.isLUC_of_isSmoothUC` `Chapter7/Zhou · EpGroupoid.isLUC_liftOne_of_isSmoothUC` `Chapter7/Zhou · EpGroupoid.corollary_7_4_11`. This form for every X^i takes (ZHOU) for every X^i, the conclusion of Proposition 7.4.9, as a hypothesis `Chapter7/Zhou · EpGroupoid.corollary_7_4_11_lift`; with (ZHOU) for X only it is §VII.43 (gap G.84, closed) (Findings, Chapter 7, items 136, 137) (pp. 333–334).

Definition 7.4.12 and Theorem 7.4.13 (V7.91) `Chapter7/Zhou · EpGroupoid.theorem_7_4_13`. *(SLUC), every X^i has LUC* `Chapter7/Zhou · EP.IsSLUC`, *follows from (ZHOU).* Proposition 7.4.9, (ZHOU) for X gives (ZHOU) for every X^i, needs the identification of $(X^i)_\infty$ with X_∞ and of the tangent spaces; this form of the theorem takes (ZHOU) and the Part I facts for every X^i, and Proposition 7.4.9 itself, with the theorem for (ZHOU) on X only, is §VII.43 (gap G.84, closed). They hold for every lift of the point `Chapter7/Zhou · EpGroupoid.Examples.pointEP` `Chapter7/Zhou · EpGroupoid.Examples.lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.smoothRegular_lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.isZhou_lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.isSLUC_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.theorem_7_4_13_pointEP` (Findings, Chapter 7, item 137) (p. 334).

Definition 7.4.14, Lemma 7.4.15, Propositions 7.4.16–7.4.18 (V7.92). *(VSLUC): LUC for X_∞, on G_x-invariant open $V(x) \subseteq X_\infty$* `Chapter7/Zhou · EpGroupoid.IsVSLUC`, read with $V(x)$ open in X_∞, inside a natural representation and invariant. It is equivalent to unique continuation through X_∞, given the Part I facts `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_isVSLUC` `Chapter7/Zhou · EpGroupoid.isVSLUC_of_isSmoothUC`, so it gives LUC (Lemma 7.4.15) `Chapter7/Zhou · EpGroupoid.lemma_7_4_15` for X and X^1 `Chapter7/Zhou · EpGroupoid.proposition_7_4_16` by the argument of Proposition 7.4.10, with connectedness of $U(x)_\infty$ in place of the text's maximality argument; (ZHOU) implies it `Chapter7/Zhou · EpGroupoid.isVSLUC_of_isZhou`. This form of Proposition 7.4.17 takes (VSLUC) for every X^i `Chapter7/Zhou · EpGroupoid.proposition_7_4_17`; Proposition 7.4.16 for every X^i, Proposition 7.4.17 for (VSLUC) on X only and Proposition 7.4.18, which needs natural representations of X^N in terms of those of X, are §VII.43 (gap G.84, closed). Witnesses: `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_subsingleton` `Chapter7/Zhou · EpGroupoid.isVSLUC_of_subsingleton` `Chapter7/Zhou · EpGroupoid.isLUC_of_subsingleton_mor` `Chapter7/Zhou · EpGroupoid.isVSLUC_trivial` `Chapter7/Zhou · EpGroupoid.isSmoothUC_trivial` (Findings, Chapter 7, items 136, 137) (pp. 335–336).

VII.39 §7.4: (LR), (SLR), and the example of Appendix 7.6.3

As in §VII.38, the Part I facts taken as hypotheses below are theorems (§VII.45).

Definitions 7.4.19, 7.4.21 and Lemma 7.4.20 (V7.93). *(LR): inside every neighbourhood of x a natural representation such that every sc-diffeomorphism β of $U(x)$ with $|\beta(y)| = |y|$ equals some $\hat{g}$* `Chapter7/Zhou · EpGroupoid.IsLR`; *(LR'): the same for β onto an open image with $\beta(x) = x$* `Chapter7/Zhou · EpGroupoid.IsLR'`. "Arbitrarily small" is "inside every open $O \ni x$", and $|\beta(y)| = |y|$ is $\pi(\beta(y)) = \pi(y)$. *Lemma 7.4.20: such a β maps $U(x)$ onto $U(x)$* `Chapter7/Zhou · EpGroupoid.lemma_7_4_20`. Two points of $U(x)$ in one orbit are related by the natural action, so where $\beta(y) \in U(x)$, $\beta(y) = \hat{g}(y)$ for some g `Chapter7/Zhou · EpGroupoid.NaturalRep.exists_Φ_eq_of_π_eq`; so $U(x) \cap \beta^{-1}(U(x))$ is covered by the finitely many closed sets $\{\beta = \hat{g}\}$, and connected-

ness of $U(x)$, instead of the text's paths, gives $\beta(U(x)) \subseteq U(x)$ and then equality; the only point of $U(x)$ in the orbit of x is x Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_π_eq. Hence (LR) and (LR') are equivalent Chapter7/Zhou · EpGroupoid.isLR_iff_isLR' (Findings, Chapter 7, item 138) (pp. 337–338).

Proposition 7.4.22, Definition 7.4.23 and Theorem 7.4.24 (V7.94) Chapter7/Zhou · EpGroupoid.proposition_7_4_22. *(ZHOU) implies (LR), and (SLR), every X^i has (LR)* Chapter7/Zhou · EP.IsSLR. The text passes to the reduced ep-groupoid and uses the density of the regular points. The formalization does neither: $U(x)_\infty$ is covered by the closed sets $\{\beta = \hat{g}\}$; the union of their X_∞-interiors is open and dense, and two pieces whose closures meet have $T\hat{g} = T\hat{g}'$ at a common smooth point, so $\hat{g} = \hat{g}'$ by (ZHOU) and LUC Chapter7/Zhou · EpGroupoid.NaturalRep.eqOn_Φ_of_tangent; connectedness of $U(x)_\infty$ and density finish Chapter7/Zhou · EpGroupoid.NaturalRep.exists_eq_Φ_of_isZhou. This form of Theorem 7.4.24 takes (ZHOU) for every X^i, and the form for (ZHOU) on X only is §VII.43 (gap G.84, closed) Chapter7/Zhou · EpGroupoid.theorem_7_4_24 Chapter7/Zhou · EpGroupoid.Examples.isSLR_pointEP (Findings, Chapter 7, item 138) (pp. 338–340).

The example of Appendix 7.6.3 and (ZHOU) (V7.95) Chapter7/Zhou · EpGroupoid.Examples.JumpingRetract.not_isZhou. *The example fails the Zhou condition (p. 331).* For the analogue of §VII.36: it is not LUC, and (ZHOU) implies LUC by Proposition 7.4.10, so it fails (ZHOU). In the first reading this was conditional on the Part I facts for its object M-polyfold; they hold for every M-polyfold (§VII.45), so the analogue fails (ZHOU) with no hypothesis Chapter7/SmoothRegularAssembly · EpGroupoid.Examples.JumpingRetract.not_isZhou_unconditional, and Zhou's example itself fails it by the same two steps Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.not_isZhou (§VII.47; gap G.83, closed) (Findings, Chapter 7, items 139, 156) (p. 331).

VII.40 Appendix 7.6.5: the boundary of a tame ep-groupoid, Theorem 7.6.9 (topological part)

Transport along index-preserving local homeomorphisms (V7.96). The maps (7.25)–(7.28) are defined through sc-diffeomorphisms of sheets; what they need of them is topological. A map that is a homeomorphism of an open W onto an open set and preserves the corner index Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn Chapter7/TameBoundaryEp · TameBoundary.IsDLocalHomeo Chapter7/TameBoundaryEp · TameBoundary.isDHomeoOn_id carries faces, face-structured neighbourhoods, the morphisms of Chapter 2's L_X and F_X and their extensions to the corresponding objects Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.isFaceIn_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.mem_faceStructuredNbhdsAt_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.isMor_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.image_faceExt, so it maps triples (y, V, θ) to triples Chapter7/TameBoundaryEp · TameBoundary.BTriple.map with a class independent of W Chapter7/TameBoundaryEp · TameBoundary.map_cls_eq. For an index-preserving local homeomorphism f, $bf[y, V, \theta] = [f(y), f(V), f(\theta)]$ Chapter7/TameBoundaryEp · TameBoundary.bLift Chapter7/TameBoundaryEp · TameBoundary.bLift_cls is the unique lift through B of this kind Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf.unique Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf.comp Chapter7/TameBoundaryEp · TameBoundary.isLiftOf_id, continuous, open and a local homeomorphism Chapter7/TameBoundaryEp · TameBoundary.continuous_bLift Chapter7/TameBoundaryEp · TameBoundary.isOpenMap_bLift Chapter7/TameBoundaryEp · TameBoundary.isLocalHomeomorph_bLift; B has compact preimages of compact sets Chapter7/TameBoundaryEp · TameBoundary.isCompact_preimage_bMap, and representatives can be centred Chapter7/TameBoundaryEp · TameBoundary.exists_rep_centered. Sc-diffeomorphisms preserve the degeneracy index Chapter7/TameBoundaryEp · MPolyfold.IsScDiffeoOn.isDHomeoOn Chapter7/TameBoundaryEp · MPolyfold.IsLocalScDiffeo.isDLocalHomeo (Findings, Chapter 7, item 140) (pp. 353–355).

(7.25)–(7.29), the boundary groupoid (V7.97) Chapter7/TameBoundaryEp · EpGroupoid.boundaryGroupoid. For a tame ep-groupoid, $\mathbf{X}$ is tame Chapter7/TameBoundaryEp · EpGroupoid.isTame_PMor, and bX, $b\mathbf{X}$ are its tame boundaries Chapter7/TameBoundaryEp · EpGroupoid.cornerObj Chapter7/TameBoundaryEp · EpGroupoid.cornerMor Chapter7/TameBoundaryEp · EpGroupoid.bObj Chapter7/TameBoundaryEp · EpGroupoid.bMor. s, t, ι, u are index-preserving local homeomorphisms Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_src Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_tgt Chapter7/TameBoundaryEp · EpGroupoid.isScDiffeoOn_inv Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_inv Chapter7/TameBoundaryEp · EpGroupoid.isDHomeoOn_unit Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_unit — u as a continuous section of the local homeomorphism s, onto the open set $u(X)$ — so (7.25), (7.27), (7.28) are their lifts Chapter7/TameBoundaryEp · EpGroupoid.bSrc Chapter7/TameBoundaryEp · EpGroupoid.bTgt Chapter7/TameBoundaryEp · EpGroupoid.bInv Chapter7/TameBoundaryEp · EpGroupoid.bUnit, with the printed formulas Chapter7/TameBoundaryEp · EpGroupoid.bSrc_cls Chapter7/TameBoundaryEp · EpGroupoid.bTgt_cls Chapter7/TameBoundaryEp · EpGroupoid.bInv_cls Chapter7/TameBoundaryEp · EpGroupoid.bUnit_cls. *(7.29).* A class of $b\mathbf{X}$ is determined by its image under B and its source Chapter7/TameBoundaryEp · EpGroupoid.eq_of_bMap_eq_of_bSrc_eq, so $m(a,b)$ is the unique class over $\psi\circ\varphi$ with source $s(b)$ Chapter7/TameBoundaryEp · EpGroupoid.exists_bComp Chapter7/TameBoundaryEp · EpGroupoid.bComp Chapter7/TameBoundaryEp · EpGroupoid.bComp_spec; near φ, $\gamma \mapsto (s|\mathbf{W})^{-1}(t\gamma)\circ\gamma$ agrees with a sheet inverse of s composed with s Chapter7/TameBoundaryEp · EpGroupoid.eventuallyEq_comp_sheet, so the printed class $[\psi\circ\varphi, \mathbf{W}\,{}_s\!\circ_t\mathbf{V}, \sigma\,{}_s\!\circ_t\theta]$ is this one Chapter7/TameBoundaryEp · EpGroupoid.bComp_eq. Off the composable pairs m takes a junk value, as in §VII.1. The groupoid laws hold and B is a functor Chapter7/TameBoundaryEp · EpGroupoid.boundaryFunctor (Findings, Chapter 7, item 140) (pp. 354–355).

Theorem 7.6.9, topological part (V7.98) Chapter7/TameBoundaryEp · EpGroupoid.theorem_7_6_9. *For a tame ep-groupoid, (1) bX and $b\mathbf{X}$ are metrizable, (3) both maps B are continuous (and sc-smooth), (4) a point of index d has d preimages, (5) $(bX, b\mathbf{X})$ is a tame ep-groupoid; the topology is characterized by the openness of the sets $U(x,V,\theta)$ and the maps Φ.* Formalized: metrizability Chapter7/TameBoundaryEp · EpGroupoid.metrizableSpace_bObj Chapter7/TameBoundaryEp · EpGroupoid.metrizableSpace_bMor from Chapter 2's tame boundary; the characterization; continuity of B and the fibre count; s, t local homeomorphisms Chapter7/TameBoundaryEp · EpGroupoid.isLocalHomeomorph_bSrc Chapter7/TameBoundaryEp · EpGroupoid.isLocalHomeomorph_bTgt, u, ι continuous Chapter7/TameBoundaryEp · EpGroupoid.continuous_bSrc Chapter7/TameBoundaryEp · EpGroupoid.continuous_bTgt Chapter7/TameBoundaryEp · EpGroupoid.continuous_bInv Chapter7/TameBoundaryEp · EpGroupoid.continuous_bUnit, m continuous on composable pairs Chapter7/TameBoundaryEp · EpGroupoid.continuousOn_bComp, and the properness property of Definition 7.1.3 (3), by the compact preimages under B Chapter7/TameBoundaryEp · EpGroupoid.exists_isProperMap_bTgt. "Φ is a homeomorphism onto θ" holds, as we read it, only for V centred at x, as for (2.44) in Chapter 2; in general Φ is an open embedding, the form used. Witness: the trivial ep-groupoid of $[0,\infty)^n \oplus W$, $n \geq 1$, with n classes over the corner Chapter7/TameBoundaryEp · EpGroupoid.bSrc_trivial Chapter7/TameBoundaryEp · EpGroupoid.trivial_quadrant_witness. The rest — (2), the sc-smoothness in (3) and (5), d_{bX} in (4) — depends on 2021 Theorem 2.7.13 (2) and is §VII.42; the characterization for every representative and the book's form of the theorem are §VII.50 (Findings, Chapter 7, items 140, 141, 145–147, 151, 155) (p. 355).

VII.41 The 2021 additions: what is not formalized, and what they taught us

Every gap of the 2021 additions is closed:

- **Theorem 7.6.7 for every** i (gap G.82, closed): the identification $(X^1)_R \cong (X_R)^1$ is an sc-

diffeomorphism and an isomorphism of ep-groupoids, and $(X^i)_R \cong (X_R)^i$ for every i under (SLUC) (§VII.46; Findings, Chapter 7, item 131).

- **Zhou's retract** (gap G.83, closed): the example itself, with the sc-smoothness the text cites from [60], and its non-effective part computed as printed; the analogue's non-effective part is computed exactly as well (§§VII.36, VII.47; Findings, Chapter 7, items 130, 139, 156).
- X^i**,** $i \geq 2$**, in §7.4** (gap G.84, closed): Propositions 7.4.9 and 7.4.16–7.4.18 and the results for every X^i are §VII.43; of "(ZHOU') is the same for X and X^i" only the direction from X to X^i, the one the proof uses, is formalized (Findings, Chapter 7, items 137, 148–150).
- **The Part I facts** (gap G.85, closed): they hold for every M-polyfold, and every natural representation has $U(x) \cap X_m$ connected and $U(x) \cap X_1$ dense, so §7.4 and Appendix 7.6.4 hold with no added hypothesis (§VII.45; Findings, Chapter 7, item 153). Lemma 7.4.4 is not restated without its hypothesis, which now holds.
- **Theorem 7.6.9** (row V7.98, narrower): "Φ is a homeomorphism onto θ" holds as we read it only for V centred at x; the image is $\theta \cap V^x$ in general, and the theorem is proved for every representative in that form (§VII.50; Findings, Chapter 7, items 141, 151, 155).

The counterexample and the 2021 text agree. The first reading found that $\mathbf{X}^R$ need not be Hausdorff, on a dimension-jumping retract, and took Hausdorffness and paracompactness as hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017). The 2021 text rewrites §7.2 to the same effect: Remark 7.2.8 says the structure fails in general, Theorem 7.2.14 assumes T_R Hausdorff, Theorem 7.2.22 characterizes it by LUC and the closedness of $\mathbf{X}^T$, and Appendix 7.6.3 gives Zhou's example of the same kind. Paracompactness, which the formalization had left open, holds always (§§VII.34–VII.36; gap G.71, closed).

Hausdorffness sits inside Lemma 7.2.11. Its conclusion at every morphism is equivalent to T_R Hausdorff, so it cannot serve, as we read it, as a step towards Hausdorffness; the paracompactness of Proposition 7.2.12 is proved instead from the openness of π and the properness of (s,t) (§VII.34).

Connectedness replaces sequences. Propositions 7.2.21, 7.4.8, 7.4.10, 7.4.22, Lemmas 7.4.4, 7.4.15 and 7.4.20 are proved by showing a set open and closed in a connected $U(x)$, $U(x)_\infty$ or $U(x) \cap X_1$; what this needs of Part I — density and connectedness of the smooth points, locality of tangents, the openness of $p : TX \to X^1$ — becomes visible as explicit hypotheses (§§VII.35–VII.39).

Tg **needs no linear structure.** (ZHOU) asks that a map be the identity; Chapter 7's chart-free tangent of a morphism suffices (§VII.37).

VII.42 Appendix 7.6.5: Theorem 7.6.9 in full (the boundary ep-groupoid)

Theorem 7.6.9 (2)–(5) rest on Part I's Theorem 2.7.13 (2)–(4): for a tame M-polyfold A, the tame boundary bA carries a tame M-polyfold structure `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold` `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_isTame` whose charts are $p \mapsto \pi(\varphi(Bp))$, π a coordinate projection of a tame chart φ and the face read off the triple; $B : bA \to A$ is sc-smooth `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryMap_isScSmooth` and $d_A(Bp) = d_{bA}(p) + 1$ `Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap` `Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap_sub_one`; the structure is characterized by the charts Φ of centred triples, and unique with that property `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_unique` `Chapter2/`

TameBoundaryStructure · MPolyfold.theorem_2_7_13_structure. The text leaves the verification of Theorem 7.6.9 to the reader, with a sketch: for $[\varphi, \mathbf{V}, \theta]$ choose $\mathbf{V}$ with $s : \mathbf{V} \to V$ an sc-diffeomorphism; in the charts $\Phi_{(\varphi,\mathbf{V},\theta)}$ and $\Phi_{(x,V,s(\theta))}$ the map s becomes $s : \theta \to s(\theta)$, an sc-diffeomorphism of faces. The formalization replaces this sc-diffeomorphism by a computation in the charts of bA, which needs only that the map below is sc-smooth (Findings, Chapter 7, items 145–147).

Maps of tame boundaries (V7.99) Chapter7/TameBoundaryEpStructure · MPolyfold.isScSmoothOn_of_bMap. *Let $f : A \to B$ be sc-smooth on an open U and $g : bA \to bB$ continuous on an open $O \subseteq B^{-1}(U)$ with $B \circ g = f \circ B$ on O. Then g is sc-smooth on O. Proof.* Sc-smoothness is local. Near $p \in O$ take the chart $\chi = \pi \circ \varphi \circ B$ of bA at p and $\chi' = \pi' \circ \varphi' \circ B$ of bB at $g(p)$, with φ, φ' tame charts at Bp and $f(Bp)$. On a neighbourhood of $\chi(p)$ in the face, $\chi' \circ g \circ \chi^{-1} = \pi' \circ (\varphi' \circ f \circ \varphi^{-1}) \circ \iota$, with ι the inclusion of the face into the model of φ: indeed $B \circ \chi^{-1} = \varphi^{-1} \circ \iota$ near $\chi(p)$ and $B \circ g = f \circ B$. The right side is a composition of sc-smooth maps, π' and ι being linear. *The lifts.* The lift bf of an sc-smooth index-preserving local homeomorphism (§VII.40) covers f and is continuous, so it is sc-smooth Chapter7/TameBoundaryEpStructure · MPolyfold.isScSmoothMap_bLift. For a local sc-diffeomorphism f and a sheet $f : W \to V$ with inverse σ, a lift of σ takes $bf(c)$ back to c for $Bc \in W$, by the uniqueness of lifts Chapter7/TameBoundaryEpStructure · MPolyfold.isLiftOf_bLift_sheet; so bf is injective on $B^{-1}(W)$, maps it onto $B^{-1}(V)$, and its inverse, covering σ and continuous, is sc-smooth by the same statement: bf is a local sc-diffeomorphism Chapter7/TameBoundaryEpStructure · MPolyfold.isLocalScDiffeo_bLift (Findings, Chapter 7, item 146).

The boundary ep-groupoid (V7.100) Chapter7/TameBoundaryEpStructure · EpGroupoid.boundaryEpGroupoid. For a tame ep-groupoid, bX and $b\mathbf{X}$ are paracompact Chapter7/TameBoundaryEpStructure · EpGroupoid.paracompactSpace_bObj Chapter7/TameBoundaryEpStructure · EpGroupoid.paracompactSpace_bMor and carry the structures of Theorem 2.7.13 (2) for the object and the morphism M-polyfold Chapter7/TameBoundaryEpStructure · EpGroupoid.bPObj Chapter7/TameBoundaryEpStructure · EpGroupoid.bPMor ($\mathbf{X}$ is tame, §VII.40). Since s, t are local sc-diffeomorphisms of $(X, \mathbf{X})$, their lifts (7.25) are local sc-diffeomorphisms Chapter7/TameBoundaryEpStructure · EpGroupoid.isLocalScDiffeo_bSrc Chapter7/TameBoundaryEpStructure · EpGroupoid.isLocalScDiffeo_bTgt; u and ι are sc-smooth index-preserving local homeomorphisms, so their lifts (7.28), (7.27) are sc-smooth Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothMap_bUnit Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothMap_bInv. *The multiplication (7.29).* Condition (4) of Definition 7.1.3 (2017: 7.3) is proved in its sheet form (§VII.1, V7.3): for every sheet σ of $s : b\mathbf{X} \to bX$ over V, $b \mapsto m(\sigma(tb), b)$ is sc-smooth on $t^{-1}(V)$ Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothCompSheet_bComp. Near b take a sheet σ_1 of $s : \mathbf{X} \to X$ through $B(\sigma(tb))$; then $b' \mapsto m(\sigma(tb'), b')$ is continuous and covers $\gamma \mapsto m(\sigma_1(t\gamma), \gamma)$, which is sc-smooth by (4) for $(X, \mathbf{X})$ in sheet form Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet, so the statement on maps of tame boundaries applies; the sheet form gives (4) Chapter7/EpGroupoid · isScSmoothComp_iff_sheet Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothComp_bComp. With the groupoid and the properness of §VII.40 this is an ep-groupoid, tame by Theorem 2.7.13 (2) Chapter7/TameBoundaryEpStructure · EpGroupoid.isTame_boundaryEpGroupoid. *B is an ep-functor* Chapter7/TameBoundaryEpStructure · EpGroupoid.boundaryEpFunctor: a functor (§VII.40), sc-smooth on objects and morphisms, so a morphism of $\mathcal{EP}$ (Definition 7.1.7 (2017: 7.7)) Chapter7/EpGroupoid · EpFunctor. *The diagram of the sketch* Chapter7/TameBoundaryEpStructure · EpGroupoid.phiTot_bSrc: for $(\varphi, \mathbf{V}, \theta)$ with $\mathbf{V}$ in a sheet of s, s maps $U(\varphi, \mathbf{V}, \theta)$ into $U(x, s(\mathbf{V}), s(\theta))$ and $\Phi_{(x,s(\mathbf{V}),s(\theta))} \circ s = s \circ \Phi_{(\varphi,\mathbf{V},\theta)}$ there (Findings, Chapter 7, items 145, 146) (p. 355).

Theorem 7.6.9 (V7.101) Chapter7/TameBoundaryEpStructure · EpGroupoid.theorem_7_6_9_full. *For a tame ep-groupoid: (1) bX, $b\mathbf{X}$ are metrizable; (2) they carry the tame M-polyfold structures above, characterized as in Theorem 2.7.13 — for a triple with V centred at x, $U(x, V, \theta)$ is open and $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism onto the face θ, inverse to the homeomorphism of §VII.40 — and every structure with this property is equivalent to it;*

(3) both maps B are sc-smooth; (4) $d_X(B(p))-1=d_{bX}(p)$ and $d_{\mathbf{X}}(B(a))-1=d_{b\mathbf{X}}(a)$; (5) the boundary ep-groupoid is tame, with the structure maps (7.25)–(7.29), s, t local sc-diffeomorphisms, u, ι sc-smooth, m sc-smooth on the fibred product; and B is an ep-functor. Proof. (1) is §VII.40; (2)–(4) are Theorem 2.7.13 (2)–(4) for the object and the morphism M-polyfold; (5) and the ep-functor are the previous paragraph. *Readings.* "Natural" in (2) is read through the characterization and its uniqueness, for representatives with V centred at x (in Part I every representative is covered since, with image $\theta\cap V^x$, §II.37; gap G.78, closed; for the boundary ep-groupoid, §VII.50); the hypotheses of the two uniqueness statements are satisfied by the structures of (2) themselves. (4) is stated as printed, with the subtraction of natural numbers (B takes values in ∂X, where $d_X\geq 1$); the additive form is Part I's. *Witness* `Chapter7/TameBoundaryEpStructure · EpGroupoid.trivial_quadrant_boundaryEp_witness`: the trivial ep-groupoid of the tame quadrant $[0,\infty)^n\oplus W$, $n\geq 1$, is tame, its boundary ep-groupoid is tame with nonempty bX, $b\mathbf{X}$, and the classes over the corner have $d_{bX}=n-1$ (Findings, Chapter 7, items 145, 147) (p. 355).

VII.43 §7.4 on every lift X^i: Propositions 7.4.9, 7.4.16–7.4.18

The per-level hypotheses below — the Part I facts on every X^i, the density of X_∞ and level-connected natural representations — hold for every ep-groupoid (§VII.45), where the forms without them are listed.

The smooth points of X^1 (V7.102) `Chapter7/ZhouLevels · MPolyfold.continuous_levelDown`. The inclusion $X_{m+1}\to X_m$ is continuous; so the identity $(X^1)_\infty\to X_\infty$ and its inverse are continuous `Chapter7/ZhouLevels · MPolyfold.shiftSmoothMap` `Chapter7/ZhouLevels · MPolyfold.shiftSmoothInv` `Chapter7/ZhouLevels · MPolyfold.continuous_shiftSmoothMap` `Chapter7/ZhouLevels · MPolyfold.continuous_shiftSmoothInv`, a homeomorphism `Chapter7/ZhouLevels · MPolyfold.shiftSmoothHomeomorph`. This is the text's "$X_\infty=X^i_\infty$" in Propositions 7.4.9 and 7.4.16, as topological spaces.

Proposition 7.4.9 (V7.102) `Chapter7/ZhouLevels · EP.proposition_7_4_9`. *If X satisfies (ZHOU), so does every X^i.* The text says (ZHOU') is the same for X and X^i, by $X_\infty=X^i_\infty$ and Proposition 7.4.8. *(ZHOU') passes from X to X^1.* An isotropy element of $x\in X_1$ in X^1 is one in X `Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal`; a natural representation of X^1 at x with $U'(x)\subseteq U(x)\cap X_1$ acts, on $U'(x)$, by the natural action of X `Chapter7/ZhouLevels · EpGroupoid.levelVal_Φ_liftOne`, and the tangent of a smooth morphism of $\mathbf{X}^1$ is, under $T(X^1)\to TX$ (Part I's Proposition 2.3.17 for $i=1$), that of $\mathbf{X}$ `Chapter7/ZhouLevels · EpGroupoid.tangentLower_tangentMor_liftOne`. So $Tg=\mathrm{id}$ on $(T_xX^1)_\infty$ gives $Tg=\mathrm{id}$ on $(T_xX)_\infty$, (ZHOU') for X gives $\hat g=\mathrm{id}$ near x in X_∞, hence near x in $(X^1)_\infty$ `Chapter7/ZhouLevels · EpGroupoid.isZhou'_liftOne`, and by induction on i for every X^i `Chapter7/ZhouLevels · EP.isZhou'_lift`. The converse, from X^i to X, is not used and not formalized. *The proposition.* (ZHOU) for X gives (ZHOU') for X by Lemma 7.4.6, which uses $(T_xX)_\infty$ dense in T_xX; then (ZHOU') for X^i, and (ZHOU) for X^i by Proposition 7.4.8 on X^i, which uses the Part I facts on X^i (§VII.37). Witness `Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_9_pointEP` (Findings, Chapter 7, item 148) (p. 333).

Corollary 7.4.11, Theorems 7.4.13 and 7.4.24 for every X^i (V7.102). With Proposition 7.4.9, the forms of §§VII.38–VII.39 that took (ZHOU) for every X^i hold for (ZHOU) on X only: every X^i has LUC `Chapter7/ZhouLevels · EP.corollary_7_4_11_all`, X has (SLUC) `Chapter7/ZhouLevels · EP.theorem_7_4_13_all` and (SLR) `Chapter7/ZhouLevels · EP.theorem_7_4_24_all`. Witnesses `Chapter7/ZhouLevels · EpGroupoid.Examples.theorem_7_4_13_all_pointEP` `Chapter7/ZhouLevels · EpGroupoid.Examples.theorem_7_4_24_all_pointEP` (pp. 334, 340).

Propositions 7.4.16 and 7.4.17 (V7.103) `Chapter7/ZhouLevels · EP.proposition_7_4_16_all`. *If X has (VSLUC), so does every X^i, and in particular every X^i has LUC; so (VSLUC) implies (SLUC).* For $x\in X_1$ smooth, the text's $V(x)$ is taken inside a natural representation $U'(x)$ of X^1 at x, with $U'(x)\subseteq U(x)$:

$V'(x) = V(x) \cap U'(x)$, an open invariant subset of $(X^1)_\infty = X_\infty$, and $\hat{g}$ is the same map on it Chapter7/ZhouLevels · EpGroupoid.isVSLUC_liftOne Chapter7/ZhouLevels · EP.isVSLUC_lift. The density of X_∞ in every level passes to X^1 Chapter7/ZhouLevels · MPolyfold.DenseSmooth.shift and so to every X^i Chapter7/ZhouLevels · EP.denseSmooth_lift. Lemma 7.4.15 on every X^i (§VII.38), with `SmoothConnected` on every X^i, gives LUC; and (SLUC) follows Chapter7/ZhouLevels · EP.proposition_7_4_17_all. Witnesses Chapter7/ZhouLevels · EpGroupoid.Examples.isVSLUC_pointEP Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_17_all_pointEP (Findings, Chapter 7, item 149) (p. 336).

Proposition 7.4.18 (V7.104) Chapter7/ZhouLevels · EP.proposition_7_4_18. *(VSLUC) and (SLUC) are equivalent.* One direction is Proposition 7.4.17. *(SLUC)* $\Rightarrow$ *(VSLUC)*, the text's proof: take $V(x) = U(x) \cap X_\infty$; an open subset of it on which $\hat{g} = \mathrm{id}$ contains $W' \cap X_\infty$ for a nonempty open W' of some X_N; by density and continuity $\hat{g} = \mathrm{id}$ on W', and LUC for X^N gives $\hat{g} = \mathrm{id}$ on $U(x)$, $U(x) \cap X_N$ being connected "by construction". *The neighbourhood basis.* The inclusion $X_N \to X_0$ through the levels Chapter7/ZhouLevels · MPolyfold.levelDownTo Chapter7/ZhouLevels · MPolyfold.continuous_levelDownTo is continuous, and the sets $O \cap X_\infty$, O open in some X_{N+1}, form a neighbourhood basis of every smooth point Chapter7/ZhouLevels · MPolyfold.exists_level_nhds_smooth. *Natural representations one level up.* An isotropy element of $x \in X_1$ in X is one in X^1 Chapter7/ZhouLevels · EpGroupoid.isotropyLevelMk Chapter7/ZhouLevels · EpGroupoid.isotropyLevelMk_levelVal Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal_mul Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal_inv. If $U(x) \cap X_1$ is connected in X^1, the natural action of X restricts to $U(x) \cap X_1$ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.liftOneΦ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.levelVal_liftOneΦ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.mapsTo_liftOneΦ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.isScDiffeoOn_liftOneΦ and is a natural representation of X^1 at x Chapter7/ZhouLevels · EpGroupoid.NaturalRep.liftOneRep, with the same $\hat{g}$ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.eqOn_of_liftOneRep. The text's "by construction" is a hypothesis: every $U(x) \cap X_m$ connected in X_m Chapter7/ZhouLevels · EpGroupoid.NaturalRep.LevelConnected, which passes to the restriction Chapter7/ZhouLevels · EpGroupoid.NaturalRep.levelConnected_liftOneRep, so LUC for X^N applies to the restriction of $U(x)$ to X^N, built one level at a time Chapter7/ZhouLevels · EP.eqOn_of_level_smooth. Hence, with the density of X_∞ and level-connected natural representations at every smooth point, (SLUC) gives (VSLUC) Chapter7/ZhouLevels · EP.isVSLUC_of_isSLUC. Witnesses Chapter7/ZhouLevels · EpGroupoid.Examples.levelConnected_pointEP Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_18_pointEP (gaps G.73, G.85) (Findings, Chapter 7, item 150) (p. 336).

VII.44 Definition 7.1.4 (2017: 7.4): $T\varphi$ as an sc-isomorphism; the facts (3), (4); $|X_\infty|$ and the density of X_∞

The first reading left these statements of §7.1 as definitions, for want of the linear structure of T_xX and of the density of smooth points in an M-polyfold. Part I now gives T_xX the linear structure of Definition 2.3.14, transported from the tangent space $T_oO = Dr(o)E$ of a retract through any chart (gap G.7, closed), and the remaining steps are the chain rule and a density argument.

$T\varphi$ as a map (V7.5) Chapter7/TangentMorphism · EpGroupoid.definition_7_1_4. *For $\varphi \in \mathbf{X_1}$: $\hat{\varphi}(s(\varphi)) = t(\varphi)$; $T\varphi = T\hat{\varphi}(s(\varphi))$ on $T_{s(\varphi)}X$; $T\varphi$ is a bijection $T_{s(\varphi)}X \to T_{t(\varphi)}X$ with inverse $T(\varphi^{-1})$; $T(\psi\circ\varphi) = T\psi\circ T\varphi$; and in charts it is a bounded linear isomorphism. Step 1, $T\varphi = T\hat{\varphi}$.* On the sheet of Definition 7.1.4, $t = \hat{\varphi}\circ s$, and the chain rule for Part I's tangent maps gives $Tt = T\hat{\varphi}\circ Ts$ on $T_\varphi\mathbf{X}$; so $Tt \circ (Ts|_{T_\varphi\mathbf{X}})^{-1}$, the definition of §VII.3, is $T\hat{\varphi}$ at $s(\varphi)$ Chapter9/Tangent · EpGroupoid.tangentMor_eq_tangentMapOn_hat. *Step 2, the inverse.* The inverse map ι exchanges s and t, so $T(\varphi^{-1})$ inverts $T\varphi$ on the fibres, and $T\varphi$ is a bijection between them Chapter7/TangentMorphism · EpGroupoid.bijOn_tangentMor Chapter7/TangentMorphism · EpGroupoid.invOn_

tangentMor. *Step 3, in charts.* For charts χ around $s(\varphi)$ and ψ around $t(\varphi)$, $T\varphi$ is represented on the fibres by $L = D(\psi \circ \widehat{\varphi} \circ \chi^{-1} \circ r)(\chi(s\varphi))$, a bounded operator Chapter7/TangentMorphism · EpGroupoid.tangentMor_chart, and L and the operator L' of φ^{-1} restrict to mutually inverse maps $T_{\chi(s\varphi)}O \rightleftarrows T_{\psi(t\varphi)}O'$, by Step 2 Chapter7/TangentMorphism · EpGroupoid.tangentMor_chart_iso.

Linearity (V7.5) Chapter7/TangentMorphismLinear · EpGroupoid.definition_7_1_4_linear. *For $\varphi \in \mathbf{X}_1$, $T\varphi : T_{s(\varphi)}X \to T_{t(\varphi)}X$ is additive, homogeneous, continuous and bijective, with inverse $T(\varphi^{-1})$.* The linear structure of T_xX is the one transported by a chart, so in the charts of Step 3 the fibre map $T\varphi$ is the restriction of the bounded linear L to $T_{\chi(s\varphi)}O$; additivity, homogeneity and continuity are those of L, and bijectivity is Step 2. So $T\varphi$ is a continuous linear equivalence of the fibres Chapter7/TangentMorphismLinear · EpGroupoid.tangentMorCLE.

An sc-isomorphism for smooth φ (V7.5) Chapter7/TangentMorphismLinear · EpGroupoid.definition_7_1_4_sc. *If φ is smooth, $T\varphi$ is an sc-isomorphism.* For smooth φ the point $\chi(s\varphi)$ is smooth, and the derivative at a smooth point of an sc-smooth map is an sc-operator (Part I), so L and L' are sc-operators Chapter7/TangentMorphismLinear · EpGroupoid.tangentMor_chart_sc; they restrict to mutually inverse maps of the sc-subspaces $T_{\chi(s\varphi)}O$ and $T_{\psi(t\varphi)}O'$, which are $T_{s(\varphi)}X$ and $T_{t(\varphi)}X$ as sc-Banach spaces. This closes gap G.70; the paragraph's "bounded linear isomorphism" on level 0 for $\varphi \in \mathbf{X}_1$ and "sc-isomorphism" for smooth φ hold as we read them (Findings, Chapter 7, items 5, 110, 154) (p. 289 (2017 p. 227)).

The facts (3), (4) on A (V7.30). For a saturated sub-M-polyfold A of the object M-polyfold, $a \in A_1 \cap V$ and an sc-smooth retraction r of V onto $A \cap V$, the tangent of the inclusion $Ti : T_aA \to T_aX$ Chapter7/SubGroupoidTangent · EpGroupoid.subObjInclusionCLM and that of $i^{-1} \circ r$, $T_aX \to T_aA$ Chapter7/SubGroupoidTangent · EpGroupoid.subObjRetractionCLM, are bounded linear. *(3)* $Ti(T_aA)$ is a closed subspace of T_aX with the closed complement $\ker T(i^{-1} \circ r)$ Chapter7/SubGroupoidTangent · EpGroupoid.fact_3_subObj, and at a smooth a the complement is an sc-complement, read in a chart around a Chapter7/SubGroupoidTangentSc · EpGroupoid.fact_3_subObj_sc. *(4)* $T(i^{-1} \circ r)$ is onto, $Ti \circ T(i^{-1} \circ r) = Tr(a)$, and $Tr(a)T_aX = Ti(T_aA)$, i.e. $T_aA = Tr(a)T_aX$ Chapter7/SubGroupoidTangent · EpGroupoid.fact_4_subObj. *Proof.* Both are Part I's Proposition 2.3.19 (3) in its linear form, applied to the object M-polyfold Chapter2/SubMPolyfoldTangent · MPolyfold.proposition_2_3_19_3_linear: since $r \circ r = r$ near a, $Tr(a)$ is a bounded projection of T_aX with range $Ti(T_aA)$, and $i^{-1} \circ r \circ i =$ id gives the surjectivity (Findings, Chapter 7, items 40, 154) (pp. 305–307 (2017 pp. 237–238)).

The density of X_∞ (V7.6) Chapter7/SmoothPointsDense · MPolyfold.denseSmooth. *For every M-polyfold and every m, X_∞ is dense in X_m, with the topology of X_m.* The text states this on p. 289 without proof; Part I had it for sc-Banach spaces only. *Step 1, a retract.* For an sc-smooth retraction r of an open subset U of a partial quadrant C with retract O, the points of O_∞ are dense in O_m Chapter7/SmoothPointsDense · IsScRetract.lvlSet_subset_closure_smooth: for a partial quadrant C, $C_m \cap E_\infty$ is dense in C_m, since the smooth points of E are dense in E_m and C is defined by finitely many continuous linear inequalities; the same holds for U relatively open in C, the fact Part I uses in the proof of Lemma 2.2.1; and a point $o \in O_m$ is the limit in E_m of smooth points e_n of U, r maps smooth points to smooth points and is continuous on level m, so $r(e_n) \to r(o) = o$ in O_m. *Step 2, the M-polyfold.* The level lift of a chart is an embedding of the chart domain in X_m onto the level m of its retract, so density passes to X_m. For an ep-groupoid, X_∞ is dense in every X_k and $\mathbf{X}_\infty$ in every $\mathbf{X}_k$ Chapter7/SmoothPointsDense · EpGroupoid.p289_dense; with the density of X_∞ in X, Chapter 7's hypothesis of the first reading becomes a theorem (gap G.73, closed; Findings, Chapter 7, items 116, 153).

The filtration of $|X|$ (V7.6) Chapter7/SmoothPointsDense · EpGroupoid.p290_orbit_filtration. $|X| = |X_0| \supset |X_1| \supset \cdots \supset |X_\infty| = \bigcap_k |X_k|$. $|X_\infty| = \pi(X_\infty)$ Chapter7/SmoothPointsDense · EpGroupoid.orbitSmooth.

Since morphisms preserve the levels (§VII.3), X_k is saturated and $\pi^{-1}(|X_k|) = X_k$ Chapter7/SmoothPointsDense · EpGroupoid.preimage_orbitLevel; so the $|X_k|$ decrease Chapter7/SmoothPointsDense · EpGroupoid.orbitLevel_antitone and their intersection is π of $\bigcap_k X_k = X_\infty$ Chapter7/SmoothPointsDense · EpGroupoid.orbitSmooth_eq_iInter. The text does not state that $|X_\infty|$ is dense in $|X|$; it follows from the density of X_∞, π being continuous and onto Chapter7/SmoothPointsDense · EpGroupoid.dense_orbitSmooth. For M-polyfolds the density is also recorded in the form Chapter 9 uses Chapter7/SmoothPointsDense · MPolyfold.denseSmooth (gap G.72, closed; Findings, Chapter 7, items 6, 112) (pp. 289–290 (2017 pp. 227–228)).

VII.45 §7.4: the Part I facts as theorems, and §7.4 without them

§VII.37 listed the facts about M-polyfolds that the proofs of §7.4 use without statement; the first reading took them as one hypothesis, satisfiable on a point. They hold for every M-polyfold Chapter7/SmoothRegularAssembly · MPolyfold.smoothRegular, and every result of §7.4, Appendix 7.6.4 and Proposition 8.1.17 is restated without them (gap G.85, closed; Findings, Chapter 7, item 153).

The four facts (V7.86). *(a) Density.* X_∞ is dense in every X_m (§VII.44). *(b) Connectedness.* For a connected open U, $U \cap X_\infty$ is connected in X_∞ Chapter7/SmoothConnectedReps · MPolyfold.smoothConnected. The local step: every point u of an open U has a neighbourhood $N \subseteq U$ in which any two smooth points are joined inside $U \cap X_\infty$, for the topology of X_∞, by the retracted segment between their images in a chart, Lemma 2.2.1 for one convex piece followed by the retraction. The global step: for a fixed smooth point x_0, the points of U having a neighbourhood whose smooth points are joined to x_0 form an open set, and by the local step so does its complement in U; by density it is nonempty, so it is all of the connected U, and $U \cap X_\infty$ is path connected Chapter2/SmoothPaths · MPolyfold.isPathConnected_smoothVal_preimage. *(c) Openness of* $p : TX \to X^1$ Chapter2/TangentProjOpen · MPolyfold.isOpenMap_tangentProj, Part I's gap G.8, closed. *(d) Locality of tangents.* Two maps sc-smooth on open sets that agree near a smooth point y in X_∞ have the same tangent on T_yX Chapter7/ZhouFacts · MPolyfold.tangentDeterminedSmooth: the sets $O \cap X_\infty$, O open in some X_{N+1}, form a neighbourhood basis of y in X_∞ (§VII.43), so the maps agree on $O \cap X_\infty$, by density and continuity on the open set O of X_{N+1}, and in a chart the derivative at y of an sc^1 map is determined by its values on a level-1 neighbourhood Chapter7/ZhouFacts · IsSc1.scDeriv_eq_of_eqOn_level. The fifth fact, kept apart in §VII.37: *(e)* $(T_xX)_\infty$, read as $T_xX \cap (TX)_\infty$, is dense in T_xX for smooth x Chapter7/ZhouFacts · MPolyfold.denseSmoothTangent: in a chart the tangent vectors at o are $(o, Dr(o)k)$, $Dr(o)$ is an sc-operator at the smooth point o, so $k \in E_\infty$ gives smooth tangent vectors, and they are dense since E_∞ is dense in E and $Dr(o)$ maps E continuously onto T_oO.

Natural representations (V7.83, V7.85, V7.104). *Every natural representation $U(x)$ has $U(x) \cap X_m$ connected in X_m for every m, and $U(x) \cap X_1$ dense in $U(x)$.* The paths of (b) lie in X_∞ with its topology, finer than that of every X_m, so $U(x) \cap X_\infty$ is connected in X_m; it is dense in $U(x) \cap X_m$ by (a), and the closure of a connected set is connected Chapter7/SmoothConnectedReps · EpGroupoid.NaturalRep.levelConnected Chapter7/SmoothConnectedReps · EpGroupoid.NaturalRep.isPreconnected_liftU; the density is that of $X_\infty \subseteq X_1$ in X Chapter7/SmoothConnectedReps · EpGroupoid.NaturalRep.subset_closure_inter_levelSet. So the hypotheses of Appendix 7.6.4 and of Proposition 7.4.18 hold for every ep-groupoid Chapter7/SmoothConnectedReps · EpGroupoid.hasConnectedDenseLevelOneReps Chapter7/SmoothConnectedReps · EpGroupoid.exists_naturalRep_levelConnected; the text's choice of $U(x)$ "by construction" is automatic.

The restatements. With (a)–(e) and the natural representations, the results of §§VII.36–VII.39 and VII.43 hold with only the hypotheses the text states. *(ZHOU).* Lemma 7.4.6 Chapter7/ZhouFacts · EpGroupoid.lemma_7_4_6_of_isZhou and Proposition 7.4.8 Chapter7/SmoothRegularAssembly · EpGroupoid.

proposition_7_4_8_unconditional; Proposition 7.4.10 and the unique continuation through X_∞ Chapter7/SmoothRegularAssembly · EpGroupoid.proposition_7_4_10_of_isZhou Chapter7/SmoothRegularAssembly · EpGroupoid.isSmoothUC_of_isZhou'; Corollary 7.4.11 for X^1 and for every X^i Chapter7/SmoothRegularAssembly · EpGroupoid.corollary_7_4_11_of_isZhou Chapter7/SmoothRegularAssembly · EP.corollary_7_4_11_all_of_isZhou; Proposition 7.4.9 Chapter7/SmoothRegularAssembly · EP.proposition_7_4_9_of_isZhou; Theorem 7.4.13 Chapter7/SmoothRegularAssembly · EP.theorem_7_4_13_all_of_isZhou; Proposition 7.4.22 and Theorem 7.4.24 Chapter7/SmoothRegularAssembly · EpGroupoid.proposition_7_4_22_of_isZhou Chapter7/SmoothRegularAssembly · EP.theorem_7_4_24_all_of_isZhou, all under (ZHOU) for X alone. *(VSLUC).* Lemma 7.4.15 and its unique continuation Chapter7/SmoothConnectedReps · EpGroupoid.isSmoothUC_of_isVSLUC' Chapter7/SmoothConnectedReps · EpGroupoid.lemma_7_4_15_of_isVSLUC; Propositions 7.4.16 and 7.4.17, for X^1 and for every X^i Chapter7/SmoothConnectedReps · EpGroupoid.proposition_7_4_16_of_isVSLUC Chapter7/SmoothConnectedReps · EP.proposition_7_4_16_all_of_isVSLUC Chapter7/SmoothConnectedReps · EP.proposition_7_4_17_of_isVSLUC Chapter7/SmoothConnectedReps · EP.proposition_7_4_17_all_of_isVSLUC, under (VSLUC) for X alone; and Proposition 7.4.18, (VSLUC) $\Leftrightarrow$ (SLUC), with no hypothesis at all Chapter7/SmoothConnectedReps · EP.isVSLUC_of_isSLUC' Chapter7/SmoothConnectedReps · EP.proposition_7_4_18_unconditional. *Appendix 7.6.4.* Theorem 7.6.7 for $i = 1$ as sets under LUC for X^1, and Corollary 7.6.8 under (ZHOU) Chapter7/SmoothConnectedReps · EpGroupoid.theorem_7_6_7_of_isLUC Chapter7/SmoothConnectedReps · EpGroupoid.corollary_7_6_8_of_imp Chapter7/SmoothRegularAssembly · EpGroupoid.corollary_7_6_8_of_isZhou'; the full theorem is §VII.46. *The example.* The analogue of Zhou's example fails (ZHOU) with no hypothesis (§VII.39). Lemma 7.4.4 is not restated; its hypothesis now holds (§VII.37). The forms with the hypotheses are kept, since the hypotheses are defined in files the new proofs import (Findings, Chapter 7, items 133, 134, 136–139, 148–150, 153) (pp. 329–340, 349).

VII.46 Appendix 7.6.4: Theorem 7.6.7 and Corollary 7.6.8 for every i

Theorem 7.6.7 for $i = 1$ (V7.83) Chapter7/RaisedEpGroupoid · EpGroupoid.theorem_7_6_7_one. *If T_R is Hausdorff for X and for X^1 (equivalently, by Theorem 7.2.22, X and X^1 have LUC), the natural identification of $(X^1)_R$ with $(X_R)^1$ — the identity on the common object M-polyfold X^1 and $[\varphi]_1 \mapsto [\varphi]$ on morphisms — is an sc-diffeomorphism on objects and on morphisms, and the two functors are inverse isomorphisms of ep-groupoids.* §VII.36 had the identification as a continuous injection of sets. *Step 1, the objects.* $(X^1)_R$ and $(X_R)^1$ both have the object M-polyfold X^1, by definition. *Step 2, the projection on level* 1. The projection $\pi^R : \mathbf{X} \to \mathbf{X}^R$ is a local sc-diffeomorphism (§VII.16), so its lift $\pi^{R,1} : \mathbf{X}^1 \to (\mathbf{X}^R)^1$ is one Chapter7/RaisedEpGroupoid · EpGroupoid.reducedπOne Chapter7/RaisedEpGroupoid · EpGroupoid.isLocalScDiffeo_reducedπOne. *Step 3, the map.* $[\varphi]_1 \mapsto [\varphi]$ Chapter7/RaisedEpGroupoid · EpGroupoid.reducedRaise is well defined and injective because the two relations agree on level 1: $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ under the Hausdorff hypothesis on X^1 Chapter7/RaisedEpGroupoid · EpGroupoid.nonEffectivePart_liftOne_eq_of_t2 (§VII.36, (7.24) and its converse); it is onto, since the class of a level-1 point of $\mathbf{X}^R$ consists of level-1 morphisms (levels are preserved, §VII.3) Chapter7/RaisedEpGroupoid · EpGroupoid.bijective_reducedRaise. *Step 4, sc-smoothness.* Near $[\varphi]_1$ the map is $\pi^{R,1} \circ (\pi_1^R|U)^{-1}$, with $\pi_1^R : \mathbf{X}^1 \to (\mathbf{X}^1)^R$ the projection of X^1, a local sc-diffeomorphism by §VII.16 for X^1, and U a sheet; its inverse is locally $\pi_1^R \circ (\pi^{R,1}|U)^{-1}$. Both are sc-smooth Chapter7/RaisedEpGroupoid · EpGroupoid.isScSmoothMap_reducedRaise Chapter7/RaisedEpGroupoid · EpGroupoid.isScSmoothMap_reducedRaiseInv, so the map is an sc-diffeomorphism Chapter7/RaisedEpGroupoid · EpGroupoid.isScDiffeoOn_reducedRaise. *Step 5, the functors.* The map commutes with sources and composition Chapter7/RaisedEpGroupoid · EpGroupoid.src_reducedRaise Chapter7/RaisedEpGroupoid · EpGroupoid.reducedRaise_comp, so with the identity on objects it is an ep-functor $(X^1)_R \to (X_R)^1$ Chapter7/RaisedEpGroupoid · EpGroupoid.

raiseReducedFunctor, and its inverse another Chapter7/RaisedEpGroupoid · EpGroupoid.raiseReducedFunctorInv, the two composites being identities. The hypotheses hold on the point Chapter7/RaisedEpGroupoid · EpGroupoid.Examples.theorem_7_6_7_pointEP.

Theorem 7.6.7 for every i, and Corollary 7.6.8 (V7.83) Chapter7/RaisedEpGroupoid · EP.theorem_7_6_7. *Under (SLUC), $(X^i)_R \cong (X_R)^i$ in $\mathcal{EP}$ for every i; under (ZHOU), the same.* Under LUC, T_R is Hausdorff (Theorem 7.2.22), so X_R is an object of $\mathcal{EP}$ Chapter7/RaisedEpGroupoid · EP.reducedOfLUC, and the case $i = 1$ is an isomorphism $(X^1)_R \cong (X_R)^1$ Chapter7/RaisedEpGroupoid · EP.raiseReducedIso. Lifting to level one carries isomorphisms of $\mathcal{EP}$ to isomorphisms Chapter7/RaisedEpGroupoid · EP.liftOneIso. The isomorphism for i is built by recursion Chapter7/RaisedEpGroupoid · EP.raiseReducedIsoLift: the identity for $i = 0$, and for $i + 1$ the composite $(X^{i+1})_R = ((X^i)^1)_R \cong ((X^i)_R)^1 \cong ((X_R)^i)^1 = (X_R)^{i+1}$ of the case $i = 1$ for X^i, which uses LUC for X^i and X^{i+1}, and the lift of the isomorphism for i. This is the composite of identifications of the text's proof. Corollary 7.6.8 follows, since (ZHOU) implies (SLUC) (Theorem 7.4.13, §VII.45) Chapter7/RaisedEpGroupoid · EP.corollary_7_6_8. *Reading.* "The identity on objects" is stated for $i = 1$ only: for variable i the object types of $(X^i)_R$ and $(X_R)^i$ are equal but not by definition, and the statement is an isomorphism in $\mathcal{EP}$ (gap G.82, closed; Findings, Chapter 7, item 131) (pp. 349–351).

VII.47 Appendix 7.6.3: Zhou's retract

Appendix 7.6.3 (p. 348), paraphrased. On $E = L^2(\mathbb{R})$ with the levels $E_m = H^{m,\delta_m}(\mathbb{R})$, $0 = \delta_0 < \delta_1 < \cdots$, and a smooth compactly supported $\beta \geq 0$ with $\int \beta^2 = 1$, the map $r(t, f) = (t, \pi_t f)$ on $\mathbb{R} \oplus E$, with $\pi_t = 0$ for $t \leq 0$ and $\pi_t f = \langle f, \beta_c \rangle \beta_c$, $\beta_c = \beta(\cdot + c)$, $c = e^{1/t}$, for $t > 0$, is an sc-smooth retraction (the text cites [60] and Part I's Subsection 1.4.4). Its retract Y is a point over each $t \leq 0$ and a line over each $t > 0$. $\mathbb{Z}/2$ acts on Y by $(t, f) \mapsto (t, -f)$; the text computes the non-effective part on $\Sigma \cong Y$ and finds one point in $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$. The first reading used an analogue (§VII.36); the example itself is formalized as follows.

The sc-Hilbert space (V7.82) Chapter7/LineSobolevScale · ZhouLine.lineSc. *For $0 = \delta_0 < \delta_1 < \cdots$, $E_m = H^{m,\delta_m}(\mathbb{R})$ is an sc-structure on $L^2(\mathbb{R})$, with level 0 equal to $L^2(\mathbb{R})$* Chapter7/LineSobolevScale · ZhouLine.level0_line. *Step 1, the levels.* $H^{m,\delta}(\mathbb{R})$ is taken inside $L^2(\mathbb{R})$ as the f whose lift $f \circ \mathrm{pr}_1$ to the cylinder $\mathbb{R} \times S^1$ lies in Part I's weighted Sobolev space $H^{m,\delta}(\mathbb{R} \times S^1)$, normed by the lift Chapter7/LineSobolevScale · ZhouLine.H1 Chapter7/LineSobolevScale · ZhouLine.liftH; a function of s alone has no t-derivatives, so the norm is $\sum_{a \leq m} \|e^{\delta|s|} \partial_s^a f\|_{L^2}^2$. Completeness holds because the range of the lift is closed. *Step 2, compactness.* The inclusions $E_{m+1} \to E_m$ are restrictions of Part I's compact inclusions on the cylinder Chapter7/LineSobolevScale · ZhouLine.isCompactOperator_incl1. *Step 3, density.* Averaging over S^1 Chapter7/LineSobolevScale · ZhouLine.avgH, of norm at most 1 on every level and the adjoint of the lift, maps the dense test functions of the cylinder to test functions of $\mathbb{R}$, which are therefore dense in every level Chapter7/LineSobolevScale · ZhouLine.dense_ofTest1. *Reading.* The level is defined through the lift; the equivalence with the one-dimensional definition by weak derivatives is not formalized, and the norms are the same.

Translates (V7.82). The translate $\tau_c \chi = \chi(\cdot + c)$ of a test function lies in every level Chapter7/LineTranslates · ZhouLine.trH, with $\|\tau_c \chi\|_{H^{m,\delta}} \leq C e^{\delta(|c|+R)}$ for χ supported in $[-R, R]$ Chapter7/LineTranslates · ZhouLine.norm_trH_le, while the pairing with a far translate is small, $\|\langle \tau_c \chi, \cdot \rangle\|_{(H^{m,\delta})'} \leq C e^{-\delta(c-R)}$, since $e^{-\delta|s|} \leq e^{-\delta(c-R)}$ on the support of $\tau_c \chi$ Chapter7/LineTranslates · ZhouLine.norm_ellL_tr0_le. Both depend smoothly on c, with derivative given by χ' Chapter7/LineTranslates · ZhouLine.hasDerivAt_trH Chapter7/LineTranslates · ZhouLine.hasDerivAt_ellT.

Flatness at $t = 0$ (V7.82). *If $g = 0$ for $t \leq 0$, g is C^∞ on $t > 0$ and every derivative tends to 0 at 0^+, then*

g is C^∞ Chapter7/FlatGluing · ZhouFlat.contDiff_of_flat. *If Q is C^∞ with every derivative bounded by $Ke^{-\varepsilon c}$, $\varepsilon > 0$, then $t \mapsto Q(e^{1/t})$ for $t > 0$, 0 for $t \le 0$, is C^∞* Chapter7/FlatGluing · ZhouFlat.contDiff_flat_comp: by Faà di Bruno's formula the derivatives of $Q(e^{1/t})$ are sums of derivatives of Q at $e^{1/t}$ times polynomials in $1/t$ and $e^{1/t}$, and the decay $e^{-\varepsilon e^{1/t}}$ beats them as $t \to 0^+$.

The retraction (V7.82) Chapter7/ZhouRetraction · ZhouLine.zhouRetraction. *$r(t, f) = (t, \pi_t f)$ is an sc-smooth retraction of $\mathbb{R} \oplus E$.* The operators π_t on each level Chapter7/ZhouRetraction · ZhouLine.zhouFam form a loss family in the sense of Part I's criterion of Remark 4.1.3 Chapter7/ZhouRetraction · ZhouLine.isLossFamily_zhouFam, which gives the sc-smoothness of r: *compatibility*, π_t commutes with the inclusions of the levels; *continuity*, $(t, f) \mapsto \pi_t f$ is continuous on every level, from a bound on $\|\pi_t\|$ uniform in t and the continuity on the dense test functions; *loss of one level*, $t \mapsto \pi_t \circ \iota$, $\iota : E_{m+1} \to E_m$, is C^∞ into the bounded operators: for $t > 0$ it is $Q(e^{1/t})$ with $Q(c) = \langle \cdot, \beta_c \rangle \beta_c$ as an operator $E_{m+1} \to E_m$, whose derivatives satisfy $\|D^n Q(c)\| \le Ke^{-(\delta_{m+1} - \delta_m)c}$ by the two bounds on translates Chapter7/ZhouRetraction · ZhouLine.norm_iteratedFDeriv_zhouQ_le, and flatness applies with $\varepsilon = \delta_{m+1} - \delta_m > 0$; *idempotence*, $\pi_t \pi_t = \pi_t$ since $\langle \beta_c, \beta_c \rangle = \int \beta^2 = 1$. So the sc-smoothness the text cites from [60] is proved, and only one level of loss is used. *The data* Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.Data are the weights and a test function β with $\int \beta^2 = 1$; the text's $\beta \ge 0$ is not needed. A β as the book requires exists Chapter7/ZhouRetraction · ZhouLine.exists_beta, and $\delta_m = m$ with a normalized bump is a witness of the data Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.Data.std.

The ep-groupoid and its non-effective part (V7.82) Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.appendix_7_6_3. *For $\mathbb{Z}/2$ acting on Y by $(t, f) \mapsto (t, -f)$: $\mathbf{X}^T = (\{1\} \times Y) \cup (\{-1\} \times \{t < 0\})$, $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T = \{(-1, (0, 0))\}$, T_R is not Hausdorff, and LUC and (ZHOU) fail.* The action is the restriction of the sc-operator $(t, f) \mapsto (t, \pm f)$, which commutes with r Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.isScSmoothMap_smul, and the translation groupoid of §VII.30 is an ep-groupoid Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.X. *Step 1, the retract as printed.* Y is $\{(t, 0)\}$ over $t \le 0$ and the line $\mathbb{R}\beta_{e^{1/t}}$ over each $t > 0$ Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.mem_0_iff'. *Step 2, $\mathbf{X}^T$.* 1 acts trivially; -1 acts trivially near a point over $t < 0$, where Y is the half-line of points $(t, 0)$, and near no point over $t \ge 0$, where points with $f \ne 0$ accumulate Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.nonEffectivePart_eq. *Step 3, the closure.* $(-1, (0, 0))$ is the limit of $(-1, (t, 0))$, $t \to 0^-$, and the only point of the closure not in $\mathbf{X}^T$ Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.closure_nonEffectivePart_diff. *Step 4, the consequences.* $\mathbf{X}^T$ is not closed, so T_R is not Hausdorff Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.not_t2Space_reducedMor and LUC fails Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.not_isLUC (Theorem 7.2.22); (ZHOU) implies LUC (Proposition 7.4.10, whose Part I facts hold, §VII.45), so (ZHOU) fails Chapter7/ZhouGroupoid · EpGroupoid.Examples.ZhouRetract.not_isZhou.

Σ as printed (V7.82) Chapter7/ZhouSigma · EpGroupoid.Examples.ZhouRetract.appendix_7_6_3_sigma. The text works on $\Sigma = \{(s, t) \in \mathbb{R}^2 \mid t = 0 \text{ if } s \le 0\}$, with $\mathbb{Z}/2$ written $\{0, 1\}$. $(s, t) \mapsto (s, t\beta(\cdot + e^{1/s}))$ (and $(s, 0)$ for $s \le 0$) is a homeomorphism $\Sigma \cong Y$, with inverse $(s, f) \mapsto (s, \langle f, \beta_{e^{1/s}} \rangle)$, using $\|\beta_c\| = 1$ Chapter7/ZhouSigma · EpGroupoid.Examples.ZhouRetract.sigmaHomeo; pulled back to $\{0, 1\} \times \Sigma$ (with $0 \mapsto 1$, $1 \mapsto -1$), the non-effective part and $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$, the single point $(1, (0, 0))$ in the text's notation, are the printed sets Chapter7/ZhouSigma · EpGroupoid.Examples.ZhouRetract.nonEffectivePart_sigma Chapter7/ZhouSigma · EpGroupoid.Examples.ZhouRetract.closure_nonEffectivePart_sigma. Every claim of the appendix holds as we read it (gap G.83, closed; rows V7.82, V7.95; Findings, Chapter 7, items 130, 139, 156) (pp. 331, 348).

VII.48 Propositions 7.2.10, 7.2.16 and Theorem 7.2.14 in the 2021 form; Corollary 7.6.6 as printed

Proposition 7.2.10, 2021 form (V7.39) `Chapter7/ReducedNaturalCharts · EpGroupoid.proposition_7_2_10_natural_charts`. *$\mathbf{X}^R$ with the quotient topology T_R has the natural charts $\Psi_\varphi = \psi \circ \sigma \circ s$; their domains cover $\mathbf{X}^R$ and any two are sc-smoothly compatible.* This is Steps 1–5 of §VII.16, which need no hypothesis; the 2021 text states the proposition as the charts, and leaves Hausdorffness to Theorem 7.2.14, as the formalization had. The charts form the atlas of §VII.16 `Chapter7/Reduced · EpGroupoid.reducedAtlas` (p. 313).

Theorem 7.2.14 and Proposition 7.2.16 (V7.40) `Chapter7/ReducedNaturalCharts · EpGroupoid.theorem_7_2_14_reduced` `Chapter7/ReducedNaturalCharts · EpGroupoid.proposition_7_2_16`. *If T_R is Hausdorff, $X_R = (X, \mathbf{X}^R)$ is an ep-groupoid with the structure maps of X descended to $\mathbf{X}^R$, the object M-polyfold of X and the natural M-polyfold structure on $\mathbf{X}^R$; it is effective; its isotropy group at x is G_x/T_x; and π is an ep-functor $X \to X_R$.* The proof is §VII.17 — Proposition 7.10 (2017) — with the Hausdorff hypothesis only: paracompactness is Proposition 7.2.12 (§VII.34), an instance, so the M-polyfold $\mathbf{X}^R$ of §VII.16 is built from Hausdorffness alone. Each clause is a component already proved: the groupoid and the structures are those of §§VII.16–VII.17 by definition, effectiveness is §VII.17, the isotropy is §VII.17's G_x/T_x, and the functor is the reduction functor. The hypothesis is the text's own, satisfied by every effective ep-groupoid and by every ep-groupoid with LUC (§VII.35) (rows V7.39, V7.40, now same; Findings, Chapter 7, items 56, 60, 107, 111, 125, 126) (pp. 318–319).

Corollary 7.6.6 as printed (V7.68) `Chapter7/HilbertPartitionsOfUnityEp · EpGroupoid.corollary_7_6_6_unconditional`. *If $|X|$ is paracompact and the object M-polyfold has an atlas whose local models have Hilbert spaces E_0, X has sc-smooth partitions of unity.* The text's proof: Part I's Corollary 5.5.17 and Theorem 5.5.10 give sc-smooth partitions of unity on the object M-polyfold, and Theorem 7.5.7 applies. With Part I's Theorem 5.5.16 now a theorem (§VII.29), the first step needs no hypothesis `Chapter5/SeparablePartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert_unconditional` and the corollary is the text's two lines. The hypotheses are satisfiable on $\mathbb{R}^1$ `Chapter7/HilbertPartitionsOfUnityEp · EpGroupoid.corollary_7_6_6_hypotheses_rn_unconditional` (row V7.68, now same; Findings, Chapter 7, item 95) (p. 347). Remark 7.5.6, which points to this case, is an assessment with nothing to formalize (row V7.57; §VII.24).

VII.49 Property (5) does not follow from (1)–(4): an ep-groupoid with a non-paracompact orbit space

The text (p. 340 (2017 p. 249)). After Definition 7.5.1 the text recalls that $|X|$ is always Hausdorff (Lemma 7.3.4), says that it is presently not known whether (5) is a consequence of (1)–(4), and that it is when X is second countable. **The answer is no** `Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_epGroupoid_not_hasParacompactOrbitSpace`. The orbit space of an ep-groupoid is Hausdorff and locally the quotient of an M-polyfold by a finite group; a non-metrizable surface is such a space, as the orbit space of the Čech groupoid of an atlas. The example is the Čech groupoid of the atlas of a Prüfer-type surface (after Prüfer; Calabi–Rosenlicht 1953).

The surface. One chart $\mathbb{R}^2$ for each a in an uncountable discrete index set ι, injected into $\mathbb{R}$ by c; the chart a is glued to the chart b along the open upper half-planes by $(u, y) \mapsto (u+(a-b)/y, y)$. In the canonical coordinates (X, l), $y = e^l$, $X = u + ae^{-l}$, the gluing is the identity; the coordinate change $\mathrm{chartPt}_a(X, l) = (X - ae^{-l}, e^l)$ `Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.chartPt` is a diffeomorphism of $\mathbb{R}^2$

onto the open upper half-plane, with inverse canon_a Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.canon. So the surface is the open upper half-plane with one closed lower half-plane attached along its own copy of the real axis for each a.

The ep-groupoid Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.epGroupoid. Objects $\iota \times \mathbb{R}^2$ and morphisms $(\iota \sqcup \{(a,b) \mid a \neq b\}) \times \mathbb{R}^2$, disjoint unions of copies of $\mathbb{R}^2$ over discrete sets, hence M-polyfolds, metrizable and paracompact. The sheet of a carries the units of the chart a, and the sheet of (a,b) the morphisms from $(a, \mathrm{chartPt}_{c(a)}(p))$ to $(b, \mathrm{chartPt}_{c(b)}(p))$, p the canonical coordinates. There is at most one morphism between two objects, so composition, units and inverses are determined by sources and targets, and the groupoid laws reduce to that uniqueness. *(1)* On each sheet, s and t are the identity or chartPt, diffeomorphisms onto open sets; a C^∞ map on an open subset of $\mathbb{R}^m$ is sc-smooth there Chapter7/PropertyFiveResearch · isScSmoothOn_rn_rn_of_contDiffOn. *(2), (4)* are checked sheet by sheet, (4) in the sheet form Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.isScSmoothCompSheet. *(3)* With $V(x)$ a unit ball in one chart: a morphism with source over $\mathrm{cl}\, V(x)$ and target in a compact K lies in one of finitely many sheets, and on the sheet (a,b), $a \neq b$, both e^l and $(a-b)e^{-l}$ are bounded, so the canonical coordinates are bounded Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_isProperMap.

Not (5) Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.not_paracompactSpace_orbitSpace. *Step 1.* The point $d_a = |(a,0)|$ is the only point of its orbit in the charts, since the axis $y = 0$ is not glued. The open sets $W_a = |X| \setminus \{d_b \mid b \neq a\}$ cover $|X|$, and $d_a \in W_b$ only for $b = a$; so a locally finite open refinement has, for every a, a member containing d_a, and these members are distinct for distinct a. *Step 2, the density step.* Every open set containing d_a contains $|(a,(0,y))|$ for small $y > 0$, which is a point $|(a_0,q)|$ of a fixed chart a_0, and so a point $|(a_0,r)|$ with $r \in \mathbb{Q}^2$ Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.exists_rat_mem_of_mem. *Step 3.* Each of the countably many points $|(a_0,r)|$ lies in finitely many members of the refinement, so only countably many members contain some d_a, and ι would be countable. For $\iota = \mathbb{R}$ with the discrete topology Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.pruefer this is absurd: the orbit space is not paracompact Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.pruefer_not_hasParacompactOrbitSpace, though Hausdorff, as Lemma 7.3.4 says Chapter7/PropertyFiveResearch · EpGroupoid.Examples.Pruefer.t2Space_orbitSpace_pruefer. *Reading.* "(1)–(4)" are the four conditions of Definition 7.1.3 as formalized, including the existential reading of (3) and the sheet form of (4); no second countability is assumed. The object space has uncountably many components and is not second countable, in line with the text's remark that second countability gives (5) (§VII.23) (Findings, Chapter 7, items 82, 113, 152) (p. 340 (2017 p. 249)).

VII.50 Appendix 7.6.5: Theorem 7.6.9 for every representative, and in the book's form

Theorem 7.6.9 for every representative (V7.101) Chapter7/TameBoundaryFull · EpGroupoid.theorem_7_6_9_allReps. *For a tame ep-groupoid: (1)–(5) and "B is an ep-functor" as in §VII.42, and, for every representative (x,V,θ) of bX and of $b\mathbf{X}$, centred or not: $U(x,V,\theta)$ and $\theta \cap V^x$ are open, $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism of $U(x,V,\theta)$ onto $\theta \cap V^x$, which is all of θ for V centred at x; and every M-polyfold structure with this property for every representative is equivalent to the one of (2).* §VII.42 characterized the structure through centred representatives. The proof applies Part I's Theorem 2.7.13 for every representative — the chart Φ of a non-centred triple is an sc-diffeomorphism onto $\theta \cap V^x$, and the uniqueness holds for that family — to the object and to the morphism M-polyfold (§II.37; gap G.78, closed). No hypothesis is added; tameness is met by the trivial

ep-groupoid of a quadrant (§VII.42) (Findings, Chapter 7, item 151).

Theorem 7.6.9 in the book's form (V7.98) `Chapter7/BookFormTameBoundaryEp2021 · EpGroupoid.theorem_7_6_9_book`. *For a tame ep-groupoid: (1) bX, $b\mathbf{X}$ are metrizable, with the only topologies for which every $U(x, V, \theta)$ is open and every $\Phi_{(x,V,\theta)}$ is open into θ; (2) the structures are tame, every Φ is an open embedding and an sc-diffeomorphism onto the open set $\theta \cap V^x$, onto θ for centred V, and the structures are unique up to equivalence; (3) both maps B are sc-smooth; (4) the index formulas hold; (5) the boundary ep-groupoid is tame with the structure maps (7.25)–(7.29); and B is an ep-functor.* The statement assembles the topological half of §VII.40 and the previous paragraph into the order of the text. *Reading.* After the theorem the text says that Φ is a homeomorphism onto θ. That holds as we read it for V centred at x only; for other representatives the image is $\theta \cap V^x$, a proper subset in general, as in Chapter 2's (2.44); the counterexample is a non-centred triple of the quadrant $[0, \infty)^3 \oplus W$, whose Φ is not onto θ `Chapter2/TameBoundaryNonCentred · MPolyfold.not_surjOn_phiTot`, and every class has a centred representative `Chapter7/TameBoundaryEp · TameBoundary.exists_rep_centered`. This is the one row of the chapter formalized in a narrower form (row V7.98; Findings, Chapter 7, items 141, 155) (p. 355).

Part VIII. Bundles and covering functors (HWZ Chapter 8, §§8.1–8.3, 8.5, 8.6 (2017 §§8.1–8.5))

Chapter 8 continues Part II; as for Chapter 7, its numbers and pages are those of the book (Springer 2021, pp. 357–400), with the 2017 text's in parentheses (2017 pp. 257–289). It treats bundles over ep-groupoids and covering functors. §8.1 equips the tangent $T(X,\mathbf{X}) = (TX, T\mathbf{X})$ of an ep-groupoid with the structure maps (8.1)–(8.9) and proves that it is an ep-groupoid, tame if X is, with an sc-smooth projection functor to X^1 (Theorem 8.1.1 (2017: 8.1), Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8)); it describes the morphisms of $T\mathbf{X}$ by pairs $(T\varphi, a)$ (Remark 8.1.10 (2017: 8.1)), proves that the tangent of an sc-smooth functor is an sc-smooth functor and that $T : \mathcal{EP} \to \mathcal{EP}$ is a functor (Theorems 8.1.11, 8.1.12 (2017: 8.2, 8.3)), defines natural transformations and natural equivalence (Definition 8.1.13 (2017: 8.1)), proves that the tangent of a natural transformation is one (Proposition 8.1.14 (2017: 8.1)), and that the paracompactness of $|X|$ passes to $|TX|$ (Theorem 8.1.15 (2017: 8.4)). §8.2 defines sc-differential forms on ep-groupoids and the sc-de Rham complex (Definitions 8.2.1, 8.2.2 (2017: 8.2, 8.3)). §8.3 defines strong bundles over ep-groupoids (Definition 8.3.1 (2017: 8.4)) with their functorial picture (pp. 375–377 (2017 pp. 271–272)), strong bundle maps and strong bundle functors (Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6)) and section functors (Definition 8.3.4 (2017: 8.7)). §8.5 (2017 §8.4) defines proper covering functors (Definition 8.5.1 (2017: 8.8)), proves that they compose (Proposition 8.5.2 (2017: 8.2)), describes them near a fibre (Theorem 8.5.3 (2017: 8.5), Remark 8.5.4 (2017: 8.2)), defines covering functors of finite groupoids and geometric lifts (Definition 8.5.5 (2017: 8.9)), constructs the proper covering functor of a geometric lift (Proposition 8.5.7 (2017: 8.3)), shows that every proper covering functor is locally of this form (Theorem 8.5.8 (2017: 8.6)), and defines proper strong bundle covering functors (Definition 8.5.9 (2017: 8.10)). The appendix counts the fibres of a proper covering functor (§8.6.1 (2017 §8.5.1): Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11), Proposition 8.6.4 (2017: 8.4)) and describes strong bundle coverings fibrewise (§8.6.2 (2017 §8.5.2)). This Part retells §8.1 in §§VIII.1–VIII.9 — the tangent data (8.1)–(8.9) and the tangents of maps defined on open sets (§VIII.1), Theorem 8.1.1 (2017: 8.1) and Lemma 8.1.2 (2017: 8.1) (§VIII.2), Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5) (§VIII.3), Lemmas 8.1.7 and 8.1.8 (2017: 8.6 and 8.7) (§VIII.4), Lemma 8.1.9 (2017: 8.8) (§VIII.5), Remark 8.1.10 (2017: 8.1) (§VIII.6), Theorems 8.1.11 and 8.1.12 (2017: 8.2 and 8.3) (§VIII.7), Definition 8.1.13 (2017: 8.1), the equivalence relation $\simeq$ and Proposition 8.1.14 (2017: 8.1) (§VIII.8), Theorem 8.1.15 (2017: 8.4) (§VIII.9); §8.3 in §§VIII.10–VIII.14 — strong bundle maps and the pull-back along s (§VIII.10), Definition 8.3.1 (2017: 8.4) and the fibre maps (§VIII.11), the functorial picture (§VIII.12), Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6) and their equivalence (§VIII.13), Definition 8.3.4 (2017: 8.7) (§VIII.14); §8.5 (2017 §8.4) in §§VIII.15–VIII.22 — Definition 8.5.1 (2017: 8.8) and the degeneracy remark (§VIII.15),

Proposition 8.5.2 (2017: 8.2) (§VIII.16), Theorem 8.5.3 (2017: 8.5) and Remark 8.5.4 (2017: 8.2) (§VIII.17), Diff_{sc}, Definition 8.5.5 (2017: 8.9) and the discussion after it (§VIII.18), disjoint unions of M-polyfolds (§VIII.19), Proposition 8.5.7 (2017: 8.3) (§VIII.20), Theorem 8.5.8 (2017: 8.6) (§VIII.21), Definition 8.5.9 (2017: 8.10) (§VIII.22); the appendix in §§VIII.23–VIII.25 — §8.6.1 (2017 §8.5.1) and Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11) (§VIII.23), Proposition 8.6.4 (2017: 8.4) (§VIII.24), §8.6.2 (2017 §8.5.2) (§VIII.25); the witnesses and the counterexample (§VIII.26); what is not formalized, among it §8.2 (§VIII.27); and what the formalization taught us (§VIII.28). Each item carries the number of its comparison item (V8.1–V8.43) in `Exposition/verification/ch08.md`. The additions of the 2021 edition (V8.44–V8.52) are retold in §§VIII.29–VIII.33: §8.1 (§VIII.29), §§8.3–8.4 (§VIII.30), §8.5 (§VIII.31), what is not formalized and what they taught us (§VIII.32), and Proposition 8.1.17 (2) for every $(TX)^i$ (§VIII.33). The work that closed the remaining gaps is retold in §§VIII.34–VIII.40: Lemma 8.1.2 as printed and the M-polyfold of pairs of Remark 8.1.10 (§VIII.34), Definition 8.2.1 (§VIII.35), Definition 8.2.2 and the pull-back of forms (§VIII.36), the Poincaré lemma, Proposition 8.2.3 (§VIII.37), Proposition 8.1.16 without added hypothesis (§VIII.38), Propositions 8.4.1 and 8.4.3 (§VIII.39), and Lemma 8.5.10, Proposition 8.5.7 for $D = \emptyset$, Theorem 8.5.8 in every universe and Proposition 8.1.17 without added hypotheses (§VIII.40).

What is proved, and what is not. Every statement and every definition of the chapter is formalized, and every gap of the chapter is closed: G.74 (Definitions 8.2.1, 8.2.2, §§VIII.35–VIII.36), G.75 (Lemma 8.1.2 as printed, §VIII.34), G.86 (Proposition 8.4.1 (3), §VIII.39), G.87 (the Poincaré lemma, Proposition 8.2.3, §VIII.37), G.88 (Proposition 8.1.16 without added hypothesis, §VIII.38) and G.89 (Lemma 8.5.10, §VIII.40). Theorems 8.1.1–8.5.8 (2017: 8.1–8.6) are proved, Theorem 8.5.8 (2017: 8.6) for ep-groupoids in any one universe, and Theorem 8.5.3 (2017: 8.5) with (5) read with s on both vertical arrows; so are Propositions 8.1.14, 8.5.2 and 8.6.4 (2017: 8.1, 8.2 and 8.4), the 2021 Propositions 8.1.16, 8.1.17, 8.2.3, 8.4.1 and 8.4.3, Proposition 8.5.7 (2017: 8.3) for nonempty D, Lemmas 8.1.2–8.6.3 (2017: 8.1–8.11), Lemma 8.1.2 (2017: 8.1) also as printed, Lemma 8.5.10, Remark 8.1.10 (2017: 8.1) with its M-polyfold of pairs and its triples, and Remark 8.5.4 (2017: 8.2). Three rows are formalized in a narrower form. In Remark 8.1.10 (2017: 8.1) (row V8.12) the map $G(\varphi, h) = (T\varphi, Ts(h))$ is a bijection with $T\varphi$ labelled by φ; read with the bare linear map $T\varphi$, it does not hold as we read it, and the trivial $\mathbb{Z}/3$-action is the counterexample `Chapter8/TangentEpNotInjective · EpGroupoid.Examples.remark_8_1_10_not_injective` (§VIII.6). The degeneracy remark after Definition 8.5.1 (2017: 8.8) (row V8.28) is proved for tame X, Y, a hypothesis the text does not state, and as an inequality in general; equality without tameness is open in the project (§VIII.15). Proposition 8.5.7 (2017: 8.3) (row V8.35) is proved for nonempty D; for $D = \emptyset$ it does not hold as we read it, and a covering functor of the empty groupoid with a geometric lift on $\mathbb{R}$ is the counterexample `Chapter8/GeometricLiftEmpty2021 · CoveringExamples.not_proposition_8_5_7_empty` (§§VIII.20, VIII.40). Three places of the 2017 text needed more than a reading. Theorem 8.5.3 (5) (2017: 8.5), as printed with t on the right vertical arrow, does not hold as we read it: the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$ violates it in every chart at 0, and the square holds with s (§VIII.17). Definition 8.5.5 (2017: 8.9) allows $D = \emptyset$, and then the functor of Proposition 8.5.7 (2017: 8.3) is surjective on objects only if $U(x) = \emptyset$; Proposition 8.5.7 (2017: 8.3) is proved for nonempty D, which holds where Theorem 8.5.8 (2017: 8.6) uses it (§VIII.20). The degeneracy remark after Definition 8.5.1 (2017: 8.8) uses the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, which Part I has for tame M-polyfolds; it is proved for tame X, Y, and as an inequality in general (§VIII.15). Several omitted steps are supplied: the tangents of maps defined on open sets, which the proofs of §8.1 differentiate without a definition (§VIII.1); the tameness of Tr in Lemma 8.1.9 (2017: 8.8), asserted without argument (§VIII.5); that $\simeq$ is an equivalence relation (§VIII.8); the open embedding $G_x \backslash TU(x) \to |TX|$ in the proof of Theorem 8.1.15 (2017: 8.4) (§VIII.9); the ep-groupoid of the functorial picture (§VIII.12); the equivalence of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6) (§VIII.13); and the "one verifies" of Proposition 8.5.7 (2017:

8.3) and of Theorem 8.5.8 (2017: 8.6) (§§VIII.20, VIII.21). Several proofs take another route: Lemmas 8.1.4–8.1.6 (2017: 8.3–8.5), Theorem 8.1.11 (2017: 8.2) and Proposition 8.1.14 (2017: 8.1) by fibre injectivity instead of differentiating identities (§§VIII.3, VIII.7, VIII.8); condition (4) of Definition 7.1.3 (2017: 7.3) for $T\mathbf{X}$ in Chapter 7′s sheet form (§VIII.4); Theorem 8.1.15 (2017: 8.4) by metrizability instead of a closed paracompact cover (§VIII.9); Theorem 8.5.3 (2017: 8.5) through Definition 8.5.1 (3) (2017: 8.8) instead of sheets and natural representations in Y (§VIII.17). Not formalized: a few remarks that nothing uses, and equality in the degeneracy remark without tameness (§VIII.27).

Seven structural choices. (1) *The tangent is Part I's.* TX and $T\mathbf{X}$ are Part I's tangent M-polyfolds and Ts, Tt, Tu, $T\iota$ Part I's tangent maps (§§II.8, II.9); for X in `Type u` they live in `Type (max u 1)`, and so does the tangent functor of Theorem 8.1.12 (2017: 8.3). The fibres T_xX carry Part I's linear structure, transported through charts (Definition 2.3.14; gap G.7, closed since the first reading), which gives $T\varphi$ as a linear isomorphism (§VII.44) and the forms of §8.2 (§VIII.35); the proofs of §8.1 do not use it. (2) *The composition of $T\mathbf{X}$ is defined by choice*: total, with a junk value off composable pairs, as Chapter 7′s composition, and unique on composable pairs. (3) *Fibre injectivity.* s being a local sc-diffeomorphism, two vectors of $T\mathbf{X}$ with the same base point and the same image under Ts are equal; most identities of §8.1 are proved this way instead of by differentiating identities. (4) *Tangents of maps defined on open sets* are built from the charts, as Part I builds Tf, with a junk value off $p^{-1}(U)$. (5) *Condition (4) in sheet form*, as in Chapter 7 (§VII.2): for $T\mathbf{X}$, for the functorial picture of a strong bundle and for the ep-groupoid of a geometric lift; Lemma 8.1.2 (2017: 8.1) is stated in this form. (6) *Strong bundle maps between strong bundles over M-polyfolds are a reading*: Part I defines strong bundle maps only between local models, and the global notion is a map covering a map of the bases, linear on the fibres, sc-smooth on $[0]$ and on $[1]$ (§VIII.10). (7) *Data, not hypotheses.* "A connected and sufficiently small $U(x)$ admitting the natural G_x-action" in Theorems 8.5.3 and 8.5.8 (2017: 8.5 and 8.6) is a datum, a chart of Theorem 8.5.3 (2017: 8.5), as natural representations are in Chapter 7; Definition 8.5.5 (2017: 8.9) is stated with Mathlib's groupoids, C being `SingleObj G`, and its finiteness enters as instance arguments of the results that use it.

Map.

- `Chapter8/Basic`: the tangent data (8.1)–(8.9), Definitions 8.1.13, 8.5.1, 8.5.5 (2017: 8.1, 8.8, 8.9) and the category Diff_{sc}.
- `Chapter8/TangentLocal`: tangents of maps sc-smooth on open sets, of sc-diffeomorphisms between open sets and of local sc-diffeomorphisms.
- `Chapter8/TangentEp`: Theorem 8.1.1 (2017: 8.1), Lemmas 8.1.2–8.1.8 (2017: 8.1–8.7), Remark 8.1.10 (2017: 8.1), Theorems 8.1.11, 8.1.12 (2017: 8.2, 8.3), Proposition 8.1.14 (2017: 8.1).
- `Chapter8/TangentTame`: Lemma 8.1.9 (2017: 8.8), the tangent of a tame retraction, the tameness clauses of Theorems 8.1.1 and 8.1.12 (2017: 8.1 and 8.3).
- `Chapter8/NatTrans`: natural equivalence is an equivalence relation; whiskering.
- `Chapter8/TangentOrbit`: the action of G_x on $TU(x)$, the orbit space of TX, Theorem 8.1.15 (2017: 8.4).
- `Chapter8/StrongBundleMap`: strong bundle maps between strong bundles over M-polyfolds, the pull-back along s.
- `Chapter8/EpStrongBundle`: Definitions 8.3.1–8.3.4 (2017: 8.4–8.7), the functorial picture, Definition 8.5.9 (2017: 8.10), §8.6.2 (2017 §8.5.2).
- `Chapter8/ProperCovering`: the degeneracy remark after Definition 8.5.1 (2017: 8.8), Proposition 8.5.2 (2017: 8.2), §8.6.1 (2017 §8.5.1) with Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11) and Proposition 8.6.4 (2017: 8.4).
- `Chapter8/ProperCoveringLocal`: Theorem 8.5.3 (2017: 8.5) and Remark 8.5.4 (2017: 8.2).

- `Chapter8/SigmaPolyfold`: disjoint unions of M-polyfolds.
- `Chapter8/GeometricLift`: the discussion after Definition 8.5.5 (2017: 8.9), Proposition 8.5.7 (2017: 8.3), Theorem 8.5.8 (2017: 8.6).
- `Chapter8/NonVacuity`: covering functors and a geometric lift; the counterexample to Theorem 8.5.3 (5) (2017: 8.5) as printed.
- `Chapter8/NonVacuityBundle`: witnesses for Definitions 8.3.1–8.5.1 and 8.5.9 (2017: 8.4–8.8 and 8.10).
- `Chapter8/TangentZhou`: Propositions 8.1.16, 8.1.17 and Definition 8.4.2 (§§VIII.29–VIII.30).
- `Chapter8/Book2021`: Definition 8.3.5, Proposition 8.4.1, Example 8.5.6, Definition 8.5.9 in the 2021 form, Lemma 8.5.10 (§§VIII.30–VIII.31).
- `Chapter8/TangentZhouLevels`: Proposition 8.1.17 (2) for every $(TX)^i$, through $T(X^i) \cong (TX)^i$ as ep-groupoids (§VIII.33).
- `Chapter8/TangentFibred`: $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$ and Lemma 8.1.2 as printed (§VIII.34).
- `Chapter8/TangentPairs`, `Chapter8/TangentEpNotInjective`: the M-polyfold of pairs and the triples of Remark 8.1.10, and the bare-map reading (§§VIII.6, VIII.34).
- `Chapter8/EpForms`, `EpFormsObj`, `EpFormsFunctor`: Definitions 8.2.1, 8.2.2, the direct limit, the sc-de Rham complex, pull-backs by functors (§§VIII.35–VIII.36).
- `Chapter8/EpPoincare` and `Chapter11/FormAveraging`, `TranslationAverage`, `SubGroupoidModel`, `EpPoincareLemma`: Proposition 8.2.3 (§VIII.37); the proof uses the push-forward of §11.2, hence the files of Chapter 11.
- `Chapter8/TangentVertical2021`, `Chapter8/TangentZhouReflect2021`: vertical vectors of $T(TX)$ and Proposition 8.1.16 (§VIII.38).
- `Chapter8/BundleOrbitParacompact2021`, `BundleOrbitParacompactOne`, `BundleLocalRegularity2021`: Propositions 8.4.1 and 8.4.3 (§VIII.39).
- `Chapter8/PullbackStrongBundle2021`, `Chapter8/GeometricLiftEmpty2021`: Lemma 8.5.10 and Proposition 8.5.7 for $D = \emptyset$ (§VIII.40); `Chapter7/SmoothRegularAssembly`, `Chapter7/SmoothConnectedReps`: Proposition 8.1.17 without added hypotheses (§VIII.40). `Chapter8/ProperCoveringUniverse` is an earlier copy of Proposition 8.5.7 and Theorem 8.5.8 for one universe, made unnecessary when Definition 8.5.5 itself was stated in any universe.

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid as in Part VII, with source and target s, t, composition $m(\psi, \varphi) = \psi \circ \varphi$, unit $u(x) = 1_x$ and inverse $\iota(\varphi) = \varphi^{-1}$. TX and $T\mathbf{X}$ are Part I's tangent M-polyfolds (§II.8), p the projection of a tangent vector to its base point on level 1, and Tf Part I's tangent map of an sc-smooth f (§II.9). For a vector $v \in T\mathbf{X}$ we write $p(v) \in \mathbf{X}_1$ for its base point, so $p(Ts(v)) = s(p(v))$. A *sheet* of s is an open set $\mathbf{W} \subseteq \mathbf{X}$ on which s is an sc-diffeomorphism onto an open V, its inverse σ a *local inverse* of s; for open $U \subseteq X$, $TU = p^{-1}(U)$. G_x is the isotropy group of x, and a natural representation $(U(x), \Phi, \Gamma)$ is Chapter 7's (§VII.7), with $U(x)$ open and connected. Part I results are cited in the same way, by their 2021 numbers with the 2017 numbers in parentheses (Definition 2.5.2, tame retractions, is 2017 Definition 2.17, §II.14; Proposition 2.5.20, fibred products, is 2017 Proposition 2.15, §II.17; Proposition 2.6.8, the pull-back strong bundle, is 2017 Proposition 2.16, §II.20; Proposition 2.4.4, 2017 Proposition 2.7, is `proposition_2_4_4`, §II.11; Lemma 1.3.4 is 2017 Lemma 1.1, §I.18); the Lean names of Parts I and II carry the 2021 numbers. Findings are those of `Exposition/Findings.md`, "Chapter 8".

VIII.1 The tangent data (8.1)–(8.9), and tangents of maps on open sets

The tangent data (V8.1). The objects of $T(X, \mathbf{X})$ are the points of TX Chapter2/TangentBundle · MPolyfold.TangentSpace Chapter2/TangentBundle · MPolyfold.tangentPolyfold, the morphisms those of $T\mathbf{X}$; a vector $(\varphi, h) \in T_\varphi\mathbf{X}$ is viewed as a morphism $Ts(\varphi)(h) \to Tt(\varphi)(h)$ (8.2), so that $s_T = Ts$ and $t_T = Tt$ (8.3) Chapter8/Basic · EpGroupoid.tSrc Chapter8/Basic · EpGroupoid.tTgt, Part I's tangent maps Chapter2/TangentMap · MPolyfold.tangentMap. The unit is $u_T = Tu$ and the inverse $\iota_T = T\iota$ (8.9) Chapter8/Basic · EpGroupoid.tUnit Chapter8/Basic · EpGroupoid.tInv, and the base point $p : T\mathbf{X} \to \mathbf{X}^1$ is Part I's projection Chapter8/Basic · EpGroupoid.tBase Chapter2/TangentBundle · MPolyfold.tangentProj. *The composition* (8.5), $(\psi, k) \circ (\varphi, h) = (\psi \circ \varphi, Ts(\psi \circ \varphi)^{-1}(Ts(\varphi)(h)))$ for $Ts(\psi)(k) = Tt(\varphi)(h)$ (8.4), needs the inverse of Ts on one fibre. *Step 1, fibre bijectivity.* Since s is a local sc-diffeomorphism, Ts maps every fibre $T_\varphi\mathbf{X}$ bijectively onto $T_{s(\varphi)}X$ (Step 4 of the next paragraph). So two vectors of $T\mathbf{X}$ with the same base point and the same image under Ts are equal Chapter8/TangentEp · EpGroupoid.tangent_ext, and every vector over $s(\varphi)$ is $Ts(v)$ for a v over φ Chapter8/TangentEp · EpGroupoid.exists_tSrc_eq. *Step 2, the choice.* For a composable pair, $s(p\,k) = t(p\,h)$ Chapter8/TangentEp · EpGroupoid.src_tBase_eq, and $k \circ h$ is chosen as a vector over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$ Chapter8/Basic · EpGroupoid.tComp Chapter8/TangentEp · EpGroupoid.tComp_spec Chapter8/TangentEp · EpGroupoid.tBase_tComp Chapter8/TangentEp · EpGroupoid.tSrc_tComp; off composable pairs it is the junk value h, which no statement uses. By Step 1 the choice is the only possible one, which is (8.5) in this form Chapter8/TangentEp · EpGroupoid.tComp_eq. *Step 3, the units.* $u_T = Tu$ is (8.8), $1_a = (1_x, Ts(1_x)^{-1}a)$: $Tu(a)$ lies over 1_x, and $Ts(Tu(a)) = a$ by the chain rule for $s \circ u = \mathrm{id}$, so Step 1 identifies the two Chapter8/TangentEp · EpGroupoid.tUnit_eq. (8.5) writes $T_{\varphi\circ\psi}\mathbf{X}$ for $T_{\psi\circ\varphi}\mathbf{X}$ (possible misprint) (Findings, Chapter 8, items 1, 2) (pp. 357–359 (2017 pp. 257–258)).

Tangents of maps on open sets (V8.2). The proofs of §8.1 differentiate maps that are defined only on open sets, such as a local inverse $(s|\mathbf{U}(\psi))^{-1}$ and the map α of Lemma 8.1.3 (2017: 8.2), and write TU for the part of TX over an open U; Part I defines Tf only for maps sc-smooth on the whole M-polyfold. Here TU is the open set $p^{-1}(U)$ Chapter8/TangentLocal · MPolyfold.tangentPreimage Chapter8/TangentLocal · MPolyfold.isOpen_tangentPreimage, and it is the tangent of the open sub-M-polyfold U when $U \neq \emptyset$ Chapter8/TangentLocal · MPolyfold.isScDiffeoOn_tangentMap_subtypeVal Chapter8/TangentLocal · MPolyfold.isOpenEmbedding_tangentMap_subtypeVal. *Step 1, the construction.* For f sc-smooth on U, Tf on $p^{-1}(U)$ is built from the charts as Part I builds Tf (§II.9), with $U \cap V \cap f^{-1}(V')$ in place of $V \cap f^{-1}(V')$ Chapter8/TangentLocal · MPolyfold.TangentPre.fimageOn Chapter8/TangentLocal · MPolyfold.tangentMapOn; off $p^{-1}(U)$ it takes a junk value, which exists because a nonempty M-polyfold has points on level 1 (E_1 is dense in E_0 and a nonempty retract meets it) Chapter8/TangentLocal · MPolyfold.exists_mem_levelSet_one Chapter8/TangentLocal · MPolyfold.nonempty_tangentSpace. *Step 2, the calculus.* Tf lies over f Chapter8/TangentLocal · MPolyfold.levelVal_tangentProj_tangentMapOn, agrees with Part I's tangent when f is sc-smooth everywhere Chapter8/TangentLocal · MPolyfold.tangentMapOn_eq_tangentMap, satisfies the chain rule Chapter8/TangentLocal · MPolyfold.tangentMapOn_comp and is sc-smooth on $p^{-1}(U)$ Chapter8/TangentLocal · MPolyfold.tangentMapOn_isScSmoothOn. *Step 3, sc-diffeomorphisms between open sets.* The tangent of an sc-diffeomorphism $U \to V$ is an sc-diffeomorphism $p^{-1}(U) \to p^{-1}(V)$ with the tangent of the inverse as its inverse, by the chain rule Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.tangent Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.tangentMap, and it is bijective on every fibre Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.bijOn_tangentFiber; this is the step "$Ts : T\mathbf{U}(\varphi) \to TU(x)$ is an sc-diffeomorphism" of Lemma 8.1.8 (2017: 8.7). *Step 4, local sc-diffeomorphisms.* A sheet of Tf around v is $p^{-1}(U)$ for a sheet U of f around $p(v)$, so Tf is a local sc-diffeomorphism Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.tangentMap, bijective on the fibres Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.bijOn_tangentFiber and surjective when f is Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.surjective_tangentMap. Step 4 is Lemma 8.1.7's (2017: 8.6) "by con-

struction". The chain rule with a globally sc-smooth outer map, $TF \circ Tf = T(F \circ f)$ on $p^{-1}(U)$, is used throughout `Chapter8/TangentEp · MPolyfold.tangentMap_tangentMapOn` (Findings, Chapter 8, item 3) (used on pp. 358–364 (2017 pp. 258–263)).

VIII.2 Theorem 8.1.1 (2017: 8.1), and Lemma 8.1.2 (2017: 8.1)

Theorem 8.1.1 (2017: 8.1) (V8.3) `Chapter8/TangentEp · EpGroupoid.theorem_8_1_1`. $T(X, \mathbf{X}) = (TX, T\mathbf{X})$ *with the structure maps (8.3)–(8.9) is an ep-groupoid; it is tame if X is; and the projections form an sc-smooth functor $(p, \mathbf{p}) : T(X, \mathbf{X}) \to (X, \mathbf{X})^1$.* The text proves it on pp. 363–365 (2017 pp. 263–264) by collecting Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8), and the formalization does the same. The groupoid laws are Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5) (§VIII.3) `Chapter8/TangentEp · EpGroupoid.tangentGroupoidStr`; conditions (1), (2) and (4) of Definition 7.1.3 (2017: 7.3) are Lemma 8.1.7 (2017: 8.6) and condition (3) is Lemma 8.1.8 (2017: 8.7) (§VIII.4); together they make the ep-groupoid `Chapter8/TangentEp · EpGroupoid.tangentEpGroupoid`, and `theorem_8_1_1` records that it has the M-polyfolds TX, $T\mathbf{X}$, the structure maps of the text, and the base point and source of a composite. The tameness sentence is Lemma 8.1.9 (2017: 8.8) (§VIII.5) `Chapter8/TangentTame · EpGroupoid.theorem_8_1_1_tame`. *The projection functor.* $(X, \mathbf{X})^1$ is Chapter 7's lift to level 1 (§VII.22) `Chapter7/Lift · EpGroupoid.liftOne`, and $(p, \mathbf{p})$ is a functor into it `Chapter8/TangentEp · EpGroupoid.tangentProjFunctor`: it commutes with sources and targets because $p \circ Ts = s \circ p$ and $p \circ Tt = t \circ p$, with the composition because $p(k \circ h) = p(k) \circ p(h)$ (Step 2 of §VIII.1, the text's "This is obvious"), and with the units because $Tu(a)$ lies over 1_x; the two projections are sc-smooth by Part I (Findings, Chapter 8, items 2, 9) (2017 pp. 258, 263–264).

Lemma 8.1.2 (2017: 8.1) (V8.4) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_2`. *As printed*: with $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X})$ identified with $T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$, the composition (8.5) is $(\psi, k) \circ (\varphi, h) = Tm(\psi, \varphi)(k, h)$. **The text's proof** varies φ' near φ, writes the partner as $\psi' = F(\varphi') = (s|\mathbf{U}(\psi))^{-1}(t(\varphi'))$, applies s to $m(F(\varphi'), \varphi')$, which gives $s(\varphi')$, and differentiates this identity at φ; since s is a local sc-diffeomorphism, $Tm(k, h) = Ts(\psi \circ \varphi)^{-1} Ts(\varphi)(h)$. **The formalized statement** is the form the proof uses; the identification of the tangent of the fibred product with the fibred product of the tangents, and Tm on it, the lemma as printed, are §VIII.34 (gap G.75, closed). Let $s : \mathbf{W} \to V$ be a sheet with inverse σ (the text's $(s|\mathbf{U}(\psi))^{-1}$), and k, h composable with $p(k) \in \mathbf{W}$. *Step 1.* $g(\varphi') = m(\sigma(t\varphi'), \varphi')$, the text's $m(F(\varphi'), \varphi')$, is sc-smooth on $t^{-1}(V)$: this is Chapter 7's sheet form of condition (4) (§VII.2) `Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet`. *Step 2,* $k = T\sigma(Tt(h))$. $Tt(h) = Ts(k)$, and $T\sigma$ inverts Ts on $p^{-1}(\mathbf{W})$ (Step 3 of §VIII.1) `Chapter8/TangentEp · EpGroupoid.tangentMapOn_srcInv_tSrc`; in particular h lies over $t^{-1}(V)$ `Chapter8/TangentEp · EpGroupoid.mem_tangentPreimage_tgt`. *Step 3,* $k \circ h = Tg(h)$. $Tg(h)$ lies over $g(p\,h) = m(\sigma(s(p\,k)), p\,h) = p(k) \circ p(h)$, and $Ts(Tg(h)) = Ts(h)$ by the chain rule for $s \circ g = s$ on $t^{-1}(V)$, which is the text's identity; by the uniqueness of (8.5) the two vectors agree `Chapter8/TangentEp · EpGroupoid.tComp_eq_tangentMapOn`. The lemma also restates that $k \circ h$ lies over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$. The proof writes $s|\mathbf{U}(\varphi \circ \psi)$ for $s|\mathbf{U}(\psi \circ \varphi)$ (possible misprint) (Findings, Chapter 8, items 5, 111) (2017 p. 259).

VIII.3 Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5): the groupoid laws of $T(X, \mathbf{X})$

Lemma 8.1.3 (2017: 8.2) (V8.5) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_3`. $s_T(\Psi \circ \Phi) = s_T(\Phi)$ *and* $t_T(\Psi \circ \Phi) = t_T(\Psi)$. The first is (8.5) itself (Step 2 of §VIII.1). **The text's proof** of the second takes the map $\alpha = t \circ (s|\mathbf{U}(\psi \circ \varphi))^{-1}$, notes $\alpha \circ s = t \circ (s|\mathbf{U}(\psi))^{-1} \circ t$ near φ, and differentiates both sides at φ in the direction h. **The formalized proof** does the same with the tangents of §VIII.1, on a sheet $s : \mathbf{W} \to V$ with inverse σ around $p(k)$. *Step 1.* By Lemma 8.1.2 (2017: 8.1), $k \circ h = Tg(h)$ with $g(\varphi') = m(\sigma(t\varphi'), \varphi')$. *Step 2.*

On $t^{-1}(V)$, $t\circ g = t\circ\sigma\circ t$ (the text's identity), so $Tt(Tg(h)) = T(t\circ\sigma\circ t)(h) = Tt(T\sigma(Tt(h))) = Tt(k)$, the last step by Step 2 of Lemma 8.1.2 (2017: 8.1) `Chapter8/TangentEp · EpGroupoid.tTgt_tangentMapOn`. This is the only identity of §8.1 that is genuinely differentiated; the others follow from fibre injectivity. The proof has "==", ": $\mathbf{U}(\psi\circ\varphi) \to U(x)$" without the map s, and "$Tt)\varphi)(h)$" (possible misprints) (Findings, Chapter 8, items 4, 6) (pp. 359–361 (2017 pp. 259–260)).

Lemma 8.1.4 (2017: 8.3) (V8.6) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_4`. *The composition of $T\mathbf{X}$ is associative.* **The text's proof** computes both sides of $(\sigma, l)\circ((\psi, k)\circ(\varphi, h))$ with (8.5) and cancels $Ts(\psi\circ\varphi)(Ts(\psi\circ\varphi))^{-1}$. **The formalized proof** compares the two sides by fibre injectivity (§VIII.1): both lie over $p(l)\circ p(k)\circ p(h)$, by the associativity of $\mathbf{X}$, and both have Ts equal to $Ts(h)$, by Lemma 8.1.3 (2017: 8.2) applied twice; composability of the outer pairs is Lemma 8.1.3's (2017: 8.2) target formula (Findings, Chapter 8, item 4) (pp. 360–362 (2017 pp. 260–261)).

Lemma 8.1.5 (2017: 8.4) (V8.7) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_5`. $1_q = (1_x, Ts(1_x)^{-1}q)$ *is a morphism $q \to q$ and a two-sided unit.* $1_q = Tu(q)$ (Step 3 of §VIII.1). *Step 1,* $1_q : q \to q$. $Ts(Tu(q)) = q$ and $Tt(Tu(q)) = q$ are the chain rule for $s\circ u = \mathrm{id}$ and $t\circ u = \mathrm{id}$ `Chapter8/TangentEp · EpGroupoid.tSrc_tUnit` `Chapter8/TangentEp · EpGroupoid.tTgt_tUnit`. The text argues the second through $t\circ(s|\mathbf{U}(1_x))^{-1} = \mathrm{id}$ near x; the identity $t\circ u = \mathrm{id}$ holds everywhere and needs no local inverse. *Step 2, the unit laws.* $k\circ 1_{Ts(k)}$ and k lie over $p(k)\circ 1 = p(k)$ and have the same Ts, and likewise on the left; the uniqueness of (8.5) gives both `Chapter8/TangentEp · EpGroupoid.tComp_tUnit` `Chapter8/TangentEp · EpGroupoid.tUnit_tComp` (Findings, Chapter 8, items 4, 6) (2017 p. 261).

Lemma 8.1.6 (2017: 8.5) (V8.8) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_6`. $\iota_T(\varphi, k) = (\varphi^{-1}, T\iota(\varphi)k)$ *is a two-sided inverse.* As in the text, the chain rule for $s\circ\iota = t$ and $t\circ\iota = s$ makes $\iota_T(v)$ a morphism $t_T(v) \to s_T(v)$ `Chapter8/TangentEp · EpGroupoid.tSrc_tInv` `Chapter8/TangentEp · EpGroupoid.tTgt_tInv`. The text then computes $\iota_T(\varphi, k)\circ(\varphi, k)$ with (8.5); here the uniqueness of (8.5) does it: $\iota_T(v)\circ v$ lies over $\varphi^{-1}\circ\varphi = 1_x$ and has Ts equal to $Ts(v)$, which characterizes $1_{Ts(v)}$, and the other side is the same `Chapter8/TangentEp · EpGroupoid.tInv_tComp` `Chapter8/TangentEp · EpGroupoid.tComp_tInv` (Findings, Chapter 8, item 4) (pp. 361–363 (2017 pp. 261–262)).

VIII.4 Lemmas 8.1.7 and 8.1.8 (2017: 8.6 and 8.7): the differential conditions

Lemma 8.1.7 (2017: 8.6) (V8.9) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_7`. s_T, t_T *are surjective local sc-diffeomorphisms and all structure maps of $T(X, \mathbf{X})$ are sc-smooth. Step 1, s_T and t_T.* "By construction" is Step 4 of the tangents on open sets (§VIII.1): the tangent of a surjective local sc-diffeomorphism is one `Chapter8/TangentEp · EpGroupoid.isLocalScDiffeo_tSrc` `Chapter8/TangentEp · EpGroupoid.isLocalScDiffeo_tTgt` `Chapter8/TangentEp · EpGroupoid.surjective_tSrc` `Chapter8/TangentEp · EpGroupoid.surjective_tTgt`. *Step 2, u_T and ι_T* are Part I's tangents of the sc-smooth u and ι `Chapter8/TangentEp · EpGroupoid.isScSmoothMap_tUnit` `Chapter8/TangentEp · EpGroupoid.isScSmoothMap_tInv`. *Step 3, the multiplication.* **The text** says it is sc-smooth "by inspection", or because $T\mathbf{X}_{s_T}\times_{t_T} T\mathbf{X}$ is the tangent of $\mathbf{X}_s\times_t\mathbf{X}$ and $m_{T\mathbf{X}}$ "is precisely Tm". **The formalized proof** uses Chapter 7's sheet form of condition (4) (§VII.2) `Chapter7/EpGroupoid · IsScSmoothCompSheet`: for a sheet of s_T with inverse σ_T, the map $\Psi \mapsto \sigma_T(t_T\Psi)\circ\Psi$ is sc-smooth on $t_T^{-1}(V_T)$. Near Ψ_0, take a sheet $s : \mathbf{W} \to V$ around the base point of $\sigma_T(t_T\Psi_0)$; on the open set of Ψ with $p(\sigma_T(t_T\Psi)) \in \mathbf{W}$, the map is Tg for the g of Lemma 8.1.2 (2017: 8.1), by Step 3 of Lemma 8.1.2 (2017: 8.1), and Tg is sc-smooth `Chapter8/TangentEp · EpGroupoid.isScSmoothCompSheet_tangent`; sc-smoothness is local `Chapter7/Discrete · MPolyfold.IsScSmoothOn.of_locally`. Chapter 7's equivalence of the two forms gives

(4) on the fibred product itself `Chapter7/EpGroupoid · isScSmoothComp_iff_sheet Chapter8/TangentEp · EpGroupoid.isScSmoothComp_tangent`, without $T(\mathbf{X}_s\times_t\mathbf{X}) = T\mathbf{X}_{Ts}\times_{Tt}T\mathbf{X}$ (gap G.75) (Findings, Chapter 8, item 7) (2017 p. 262).

Lemma 8.1.8 (2017: 8.7) (V8.10) `Chapter8/TangentEp · EpGroupoid.lemma_8_1_8`. *Every $a \in TX$ has an open $V' \ni a$ with $t_T : s_T^{-1}(\operatorname{cl} V') \to TX$ proper*, namely $V' = p^{-1}(V)$ for an open V around $p(a)$ as in Definition 7.1.3 (3) (2017: 7.3) for X. **The text's proof** takes (φ_k, h_k) with sources in $\operatorname{cl} TV$ and convergent targets, gets $\varphi_k \to \varphi$ in $\mathbf{X}$ from the properness of X, then in $\mathbf{X}^1$, and concludes that (φ_k, h_k) converges since $Ts : T\mathbf{U}(\varphi) \to TU(x)$ is an sc-diffeomorphism. **The formalized proof** checks Chapter 7's sequential criterion for properness on metrizable spaces (§VII.2) `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff_of_isClosed`. *Step 1.* Let Φ_k have $s_T(\Phi_k) \in \operatorname{cl} V'$ and $t_T(\Phi_k) \to b$. The base points have $s(p\,\Phi_k) \in \operatorname{cl} V$, p being continuous, and $t(p\,\Phi_k) \to p(b)$; properness of X gives a subsequence with $p(\Phi_{k_j}) \to \varphi_0$ in $\mathbf{X}$ `Chapter7/ProperMaps · MPolyfold.exists_tendsto_subseq_of_isProperMap_domRestrict`. *Step 2.* Take a sheet $t : \mathbf{W} \to V''$ around φ_0. Its tangent is an sc-diffeomorphism $p^{-1}(\mathbf{W}) \to p^{-1}(V'')$ with continuous inverse $(Tt)^{-1}$ (Step 3 of §VIII.1), and $b \in p^{-1}(V'')$; eventually $p(\Phi_{k_j}) \in \mathbf{W}$, so $\Phi_{k_j} = (Tt)^{-1}(t_T\Phi_{k_j}) \to (Tt)^{-1}(b)$. The text's step "$\varphi_k \to \varphi$ in $\mathbf{X}^1$" is not needed: convergence in $\mathbf{X}$ already puts the φ_k in the sheet, and TX, $T\mathbf{X}$ are metrizable, so the sequential criterion applies. The proof writes $t(\varphi_k, h_k) = Ts(\varphi_k)(h_k)$ for $Tt(\varphi_k)(h_k)$ (possible misprint) (Findings, Chapter 8, item 7) (pp. 362–364 (2017 pp. 262–263)).

VIII.5 Lemma 8.1.9 (2017: 8.8): the tangent of a tame ep-groupoid

Lemma 8.1.9 (2017: 8.8) (V8.11) `Chapter8/TangentTame · EpGroupoid.lemma_8_1_9`. *If X is tame, so is TX*; tame means that the object M-polyfold has an equivalent tame atlas (Definition 7.1.6 (2017: 7.6), §VII.3) `Chapter7/EpGroupoid · EpGroupoid.IsTame`. **The text's proof** is one sentence: for a tame chart onto a tame retract $O = r(U)$, "$Tr : TU \to TU$ is also a tame retraction". Part I does not state this, and the text gives no argument; it is proved here from Part I's Definition 2.5.2 (2017: 2.17) (§II.14) `Chapter2/Tame · ScRetraction.IsTame`. The tangent of the retract (O, C, E) is (TO, TC, TE), $TC = C^1 \oplus E$, with $Tr(a,b) = (r(a), Dr(a)b)$ `Chapter2/TangentBundle · ScRetraction.tangentRetraction Chapter2/TangentBundle · LocalModel.tangent`. *Step 1, a presentation of TC.* An sc-isomorphism of TE with $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$, built from a presentation T of C, presents TC as a partial quadrant whose $\mathbb{R}^n$-coordinates at (a,b) are those of a `Chapter8/TangentTame · ScIso.tangentPres Chapter8/TangentTame · tangentSet_eq_tangentPres`; so $d_{TC}(a,b) = d_C(a)$ `Chapter8/TangentTame · degIndex_tangentSet`, and the minimal subspace at (a,b) is $(E_a \cap E_1) \oplus E$ `Chapter8/TangentTame · mem_minimalSubspaceWith_tangentPres`. *Step 2, condition (1)* of Definition 2.5.2 (2017: 2.17) for Tr at (a,b) is condition (1) for r at a, by Step 1 and $Tr(a,b) = (r(a), Dr(a)b)$. *Step 3, condition (2)* at a smooth point $(a,b) \in TO$. With A the sc-complement of $T_aO = Dr(a)E$ in E_a that the tameness of r gives, the complement for Tr is $A' = (A \cap E_1) \oplus A$. With $L = D(Tr)(a,b)$, three facts suffice: L is a projection, since $Tr \circ Tr = Tr$ `Chapter2/Retracts · ScRetraction.scDeriv_comp_self`; its first row is $(Dr(a), 0)$ `Chapter8/TangentTame · ScRetraction.scDeriv_tangentRetraction_fst`; and $L(0,l) = (0, Dr(a)l)$ `Chapter8/TangentTame · ScRetraction.scDeriv_tangentRetraction_inr`. Then $LTE \cap A' = 0$ (if $(k,l) = L(k,l) \in A'$, then $k \in T_aO \cap A = 0$ and $l = Dr(a)l \in T_aO \cap A = 0$) and $LTE + A' = TE$ (split k, then the remainder of l), on every level. The second derivative of r, which enters L, is never needed `Chapter8/TangentTame · ScRetraction.IsTameWith.tangent`. *Step 4, all presentations.* Tameness quantifies over all presentations of TC; Part I's Lemma 1.3.4 (2017: 1.1), the presentation independence of d_C (§I.18), passes from one to all `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame Chapter8/TangentTame · ScRetraction.IsTame.tangentRetraction Chapter8/TangentTame · IsTameRetract.tangent`. *Step 5, from charts to TX.* Around

$\alpha \in TX$ the tangent chart of a chart onto a tame retract around $p(\alpha)$ is modelled on the tame retract $T(O, C, E)$ `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`, so these charts form an equivalent tame atlas `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart Chapter8/TangentTame · MPolyfold.IsTame.tangent`. The tameness clauses of Theorem 8.1.1 (2017: 8.1) and Theorem 8.1.12 (2017: 8.3) follow `Chapter8/TangentTame · EpGroupoid.theorem_8_1_1_tame Chapter8/TangentTame · theorem_8_1_12_tame`, and the hypothesis is satisfiable: the tangent of the trivial ep-groupoid of $\mathbb{R}^n$ is tame `Chapter8/TangentTame · EpGroupoid.isTame_tangentEpGroupoid_trivial_rn` (Findings, Chapter 8, items 8, 104) (2017 p. 263).

VIII.6 Remark 8.1.10 (2017: 8.1): the morphisms as pairs $(T\varphi, a)$

Remark 8.1.10 (2017: 8.1) (V8.12) `Chapter8/TangentEp · EpGroupoid.remark_8_1_10`. The text describes $T\mathbf{X}$ a second way: a morphism $\varphi : x \to y$ on level 1 has a linear isomorphism $T\varphi : T_xX \to T_yX$, the tangent of $t \circ (s|\mathbf{U}(\varphi))^{-1}$ at x; the pairs $(T\varphi, a)$, $a \in T_{s(\varphi)}X$, composed by $(T\psi, h) \circ (T\varphi, l) = (T(\psi \circ \varphi), l)$, form a groupoid, and $G(\varphi, h) = (T\varphi, Ts(h))$ is a bijection compatible with the compositions; moreover $t_T \circ (s_T|T_\varphi\mathbf{X})^{-1} = T\varphi$. $T\varphi$ is Chapter 7's $Tt \circ (Ts|T_\varphi\mathbf{X})^{-1}$ (§VII.3) `Chapter7/EpGroupoid · EpGroupoid.tangentMor`. *Step 1, well defined.* $T\varphi(Ts(h)) = Tt(h)$ for every $h \in T_\varphi\mathbf{X}$, which is the last sentence of the remark `Chapter8/TangentEp · EpGroupoid.tangentMor_tSrc`. *Step 2, functorial.* $T(\psi \circ \varphi) = T\psi \circ T\varphi$ on $T_{s(\varphi)}X$: given a, take h over φ with $Ts(h) = a$ (fibre surjectivity) and k over ψ with $Ts(k) = Tt(h)$; then $k \circ h$ lies over $\psi \circ \varphi$ with $Ts = a$, and $Tt(k \circ h) = Tt(k)$ by Lemma 8.1.3 (2017: 8.2) `Chapter8/TangentEp · EpGroupoid.tangentMor_comp`. *Step 3, the map G.* As a bare map $(\varphi, h) \mapsto (T\varphi, Ts(h))$ need not be injective: distinct morphisms can have the same $T\varphi$, as the three morphisms at a point of the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$ (§VII.30), all with $T\varphi = \mathrm{id}$. So the first entry is read as $T\varphi$ labelled by φ `Chapter8/TangentEp · EpGroupoid.remarkG`; then G is injective by fibre injectivity, its range is the set of pairs (φ, a) with $\varphi \in \mathbf{X}_1$ and a over $s(\varphi)$ by fibre surjectivity, composability of (k, h) is $Ts(k) = T\varphi(Ts(h))$ with $\varphi = p(h)$, and $G(k \circ h) = (p(k) \circ p(h), Ts(h))$, the text's computation. *The bare-map reading.* Read with the bare map $T\varphi$ as first entry, G is not injective, so the remark does not hold as we read it in that reading: on the trivial $\mathbb{Z}/3$-action $T\varphi = \mathrm{id}$ for every morphism `Chapter8/TangentEpNotInjective · EpGroupoid.Examples.tangentMor_trivialZ3`, and lifting one tangent vector along Ts over $(1, x)$ and over (g, x), $g \neq 1$, gives two vectors of $T\mathbf{X}$ with the same image `Chapter8/TangentEpNotInjective · EpGroupoid.Examples.remark_8_1_10_not_injective` (row V8.12, narrower). *Since the first reading.* That $T\varphi$ is a linear isomorphism, and the tangent of $\hat{\varphi}$ at x, is §VII.44 `Chapter7/TangentMorphism · EpGroupoid.definition_7_1_4 Chapter7/TangentMorphismLinear · EpGroupoid.definition_7_1_4_linear Chapter7/TangentMorphismLinear · EpGroupoid.tangentMorCLE` (gap G.70, closed); the M-polyfold structure on the pairs, for which G is an sc-diffeomorphism, and the triples $(a, T\varphi, b)$ are §VIII.34. The remark writes "$T\varphi : T_xX \to T_yY$" for T_yX, and calls $T\varphi$ the linearization "at φ" on 2017 p. 265 where 2017 p. 264 has "at x" (possible misprints) (Findings, Chapter 8, items 10, 112, 113, 131) (pp. 364–366 (2017 pp. 264–265)).

VIII.7 Theorems 8.1.11 and 8.1.12 (2017: 8.2 and 8.3): the tangent functor

Theorem 8.1.11 (2017: 8.2) (V8.13) `Chapter8/TangentEp · EpFunctor.theorem_8_1_11`. *For an sc-smooth functor $(\Phi, \Phi) : X \to Y$, $(T\Phi, T\Phi)$ is an sc-smooth functor $TX \to TY$* `Chapter8/TangentEp · EpFunctor.tangent`. **The text's proof** differentiates $\Phi(m(\psi, \varphi)) = m(\Phi\psi, \Phi\varphi)$ on the fibred product, using Lemma 8.1.2 (2017: 8.1) as printed, and differentiates $s \circ \Phi = \Phi \circ s$ and $t \circ \Phi = \Phi \circ t$; it leaves the units to the reader. **The formalized proof.** *Step 1.* The tangent maps are sc-smooth by Part I, and compatibility

with sources and targets is the chain rule for $s \circ \Phi = \Phi \circ s$, $t \circ \Phi = \Phi \circ t$, as in the text. *Step 2, the composition, by fibre injectivity in TY.* $T\Phi(k \circ h)$ lies over $\Phi(p(k) \circ p(h)) = \Phi(p\,k) \circ \Phi(p\,h)$, and its Ts is $T\Phi(Ts(k \circ h)) = T\Phi(Ts(h)) = Ts(T\Phi(h))$; the composite $T\Phi(k) \circ T\Phi(h)$ has the same base point and the same Ts. No tangent of the fibred product is needed. *Step 3, the units*, not checked in the text, follow from the chain rule for $\Phi \circ u = u \circ \Phi$ (Findings, Chapter 8, items 4, 11) (pp. 365–367 (2017 pp. 265–266)).

Theorem 8.1.12 (2017: 8.3) (V8.14) `Chapter8/TangentEp · theorem_8_1_12`. *T is a functor $\mathcal{EP} \to \mathcal{EP}$, $X \mapsto TX$, $\Phi \mapsto T\Phi$, and TX is tame if X is.* On objects it is Theorem 8.1.1 (2017: 8.1) `Chapter8/TangentEp · EP.tangent`, on morphisms Theorem 8.1.11 (2017: 8.2), and the functor laws are Part I's chain rule and the tangent of the identity `Chapter2/TangentMap · MPolyfold.tangentMap_comp` `Chapter2/TangentMap · MPolyfold.tangentMap_id` `Chapter8/TangentEp · tangentFunctor`; $\mathcal{EP}$ is Chapter 7's category (§VII.4) `Chapter7/EpGroupoid · EP`. Since TX lives in `Type (max u 1)`, the functor goes from $\mathcal{EP}$ in that universe to itself, an artifact of Part I's `TangentSpace`. The tameness sentence is Lemma 8.1.9 (2017: 8.8) (§VIII.5) `Chapter8/TangentTame · theorem_8_1_12_tame` (Findings, Chapter 8, items 2, 11) (2017 p. 266).

VIII.8 Definition 8.1.13 (2017: 8.1), the equivalence relation $\simeq$, and Proposition 8.1.14 (2017: 8.1)

Definition 8.1.13 (2017: 8.1) (V8.15). A natural transformation $\tau : F \to G$ between sc-smooth functors $X \to Y$ is an sc-smooth map $\tau : X \to \mathbf{Y}$ with $\tau(x) : F(x) \to G(x)$ and $\tau(x') \circ F(h) = G(h) \circ \tau(x)$ for every $h : x \to x'$ `Chapter8/Basic · EpNatTrans`; $F \simeq G$ if one exists `Chapter8/Basic · EpFunctor.NaturallyEquivalent` (Findings, Chapter 8, item 12) (2017 p. 266).

The equivalence relation (V8.16) `Chapter8/NatTrans · EpFunctor.naturallyEquivalent_equivalence`. The text states without proof that $\simeq$ is an equivalence relation. Reflexivity is $x \mapsto 1_{F(x)}$ `Chapter8/NatTrans · EpNatTrans.refl`, symmetry $x \mapsto \tau(x)^{-1}$, sc-smooth since ι is, natural by inverting the square of h^{-1} `Chapter8/NatTrans · EpNatTrans.symm`, and transitivity $x \mapsto \tau'(x) \circ \tau(x)$ `Chapter8/NatTrans · EpNatTrans.trans` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.refl` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.symm` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.trans`. *The one analytic point* is the sc-smoothness of $x \mapsto \tau'(x) \circ \tau(x)$ `Chapter8/NatTrans · EpNatTrans.isScSmoothMap_comp_app`. Near x_0, take a sheet $\mathbf{W} \ni \tau'(x_0)$ of s with inverse σ; for $x \in \tau'^{-1}(\mathbf{W})$, $\tau'(x) = \sigma(s\,\tau'(x)) = \sigma(G(x)) = \sigma(t\,\tau(x))$, so $\tau'(x) \circ \tau(x) = m(\sigma(t\psi), \psi)$ at $\psi = \tau(x)$, which is sc-smooth by Chapter 7's sheet form of condition (4), composed with the sc-smooth τ. Whiskering, not in the text, is added `Chapter8/NatTrans · EpNatTrans.whiskerLeft` `Chapter8/NatTrans · EpNatTrans.whiskerRight` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_left` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right` (Findings, Chapter 8, item 12) (2017 p. 267).

Proposition 8.1.14 (2017: 8.1) (V8.17) `Chapter8/TangentEp · proposition_8_1_14`. *If $\tau : \Phi \to \Psi$ is a natural transformation, $T\tau : TX \to T\mathbf{Y}$ is a natural transformation $T\Phi \to T\Psi$; hence $\Phi \simeq \Psi$ implies $T\Phi \simeq T\Psi$* `Chapter8/TangentEp · EpNatTrans.tangent`. *Step 1, as in the text.* $T\tau$ is sc-smooth by Part I, and the chain rule for $s \circ \tau = \Phi$ and $t \circ \tau = \Psi$ makes $T\tau(a)$ a morphism $T\Phi(a) \to T\Psi(a)$ of TY. *Step 2, naturality.* **The text** differentiates the naturality square $\tau(t\alpha) \circ \Phi(\alpha) = \Psi(\alpha) \circ \tau(s\alpha)$ at $\alpha_0 \in \mathbf{X}^1$, again through Lemma 8.1.2 (2017: 8.1) as printed. **The formalized proof** compares $T\tau(Tt(v)) \circ T\Phi(v)$ and $T\Psi(v) \circ T\tau(Ts(v))$ by fibre injectivity in TY: they lie over the two sides of the naturality square at $p(v)$, which are equal, and both have Ts equal to $T\Phi(Ts(v))$, by Lemma 8.1.3 (2017: 8.2) and Step 1. The proof has "$T\tau(x_0(h))$", and labels the lower arrow of the last diagram $T\tau(x_0)h$ for $T\tau(x_0')h'$ (possible misprints, the second perhaps an artefact of the text extraction) (Findings, Chapter 8, items 4, 13) (pp. 367–372 (2017 pp. 267–268)).

VIII.9 Theorem 8.1.15 (2017: 8.4): the orbit space of TX

Theorem 8.1.15 (2017: 8.4) (V8.18) `Chapter8/TangentOrbit · EpGroupoid.theorem_8_1_15`. *If $|X|$ is paracompact, so is $|TX|$*; paracompact orbit space is Chapter 7's Definition 7.5.1 (2017: 7.15) (§VII.23) `Chapter7/EpGroupoid · EpGroupoid.HasParacompactOrbitSpace`. **The text's proof**: $|X^1|$ is paracompact (Proposition 7.5.3 (2017: 7.12)); for $z \in |X^1|$ take $x_z \in X_1$ and $U(x_z)$ with the natural G_{x_z}-action such that $TU(x_z)$ is, as a bundle over $U(x_z)^1$, sc-isomorphic to a local bundle model $TO \to O^1$ with an sc-smooth bundle action, so that $|TU(x_z)|$ and $|U(x_z)^1|$ are metrizable; a locally finite closed refinement (A_z) of the cover $(|U(x_z)^1|)$ gives closed paracompact sets $\tilde{A}_z \subseteq |TX|$ covering it locally finitely, and $|TX|$ is regular Hausdorff, so Part I's Proposition 2.7.4 (2017: 2.17) applies. **The formalized proof** proves more, that $|TX|$ is metrizable `Chapter8/TangentOrbit · EpGroupoid.metrizableSpace_tangent_orbitSpace`, along the route of Proposition 7.5.3 (2017: 7.12) (§VII.23): Smirnov's theorem pulled back along the continuous map $|p| : |TX| \to |X^1|$ (§II.7) `Chapter2/Metrizable · metrizableSpace_of_pullback`. *Step 1, the map $|p|$.* The projection functor of Theorem 8.1.1 (2017: 8.1) induces $|p|$ `Chapter8/TangentOrbit · EpGroupoid.tangentOrbitProj Chapter8/TangentOrbit · EpGroupoid.continuous_tangentOrbitProj`, and $|p|^{-1}(\pi_1(U \cap X_1)) = \pi_T(p^{-1}(U))$ for every $U \subseteq X$, by fibre surjectivity of Ts: a vector over a point isomorphic in X^1 to a point of $U \cap X_1$ is isomorphic in TX to a vector over it `Chapter8/TangentOrbit · EpGroupoid.preimage_tangentOrbitProj_image`. *Step 2, the action of G_x on $TU(x)$.* For a natural representation at $x \in X_1$, $TU(x) = p^{-1}(U(x))$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangentU` carries the action $g \mapsto T\Phi(g)$ by the tangents of the natural action, sc-diffeomorphisms of $TU(x)$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tΦ Chapter8/TangentOrbit · EpGroupoid.NaturalRep.isScDiffeoOn_tΦ Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangentMulAction`, with the tangent $T\Gamma(g)$ of the sheet `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tΓ`. Its linearity on the fibres is not formalized (gap G.7), and the local bundle model $TO \to O^1$ is not used. *Step 3, the open embedding.* Two vectors of $TU(x)$ have the same image in $|TX|$ iff they are in one G_x-orbit `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangent_π_eq_iff`: a morphism $k : v \to w$ of TX lies over a morphism between points of $U(x)$, so $p(k) = \Gamma(g, s(p\,k))$ by Theorem 7.1.13 (3) (2017: 7.1), and $k = T\Gamma(g)(v)$ by fibre injectivity. So $G_x\backslash TU(x) \to |TX|$ is an open embedding onto $|TU(x)|$, the argument of Proposition 7.1.19 (2017: 7.6) for TX, which the text leaves implicit `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.isOpenEmbedding_tangentOrbitQuotientMap Chapter8/TangentOrbit · EpGroupoid.NaturalRep.range_tangentOrbitQuotientMap`. *Step 4, metrizability.* $TU(x)$ is metrizable as a subspace of TX (Part I's Proposition 2.3.15 (3) (2017: 2.5)) `Chapter2/Metrizable · MPolyfold.metrizableSpace_tangentSpace`, and $G_x\backslash TU(x)$ by the metric $\hat{\rho}$ of Lemma 7.3.2 (2017: 7.4), repeated for a finite group acting by homeomorphisms on a metrizable space; hence $|TU(x)|$ is metrizable `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.metrizableSpace_image_tangentU`, and by Step 1 it is $|p|^{-1}(|U(x)_1|)$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.image_tangentU_eq_preimage`, $|U(x)_1|$ being open in $|X^1|$. These are the text's two metrizability claims `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.metrizableSpace_image_liftU`. *Step 5.* $|X^1|$ is metrizable by Proposition 7.5.3 (2017: 7.12) `Chapter7/Lift · EpGroupoid.metrizableSpace_liftOne_orbitSpace`, hence paracompact and Hausdorff; every point of it lies in some $|U(x)_1|$, for a natural representation at a representative (Theorem 7.1.13 (2017: 7.1)) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13`, whose preimage is metrizable by Step 4. The pulled-back Smirnov theorem makes $|TX|$ metrizable, hence paracompact (Stone). The closed sets $\tilde{A}_z$, the regularity of $|TX|$ and Proposition 2.7.4 (2017: 2.17) are not used; the step shared with the text is the metrizability of $|TU(x_z)|$. The theorem is not vacuous `Chapter8/TangentOrbit · EpGroupoid.hasParacompactOrbitSpace_tangent_trivial`. "parcompact" (possible misprint) (Findings, Chapter 8, items 20, 21, 104) (2017 p. 268).

VIII.10 Strong bundle maps, and the pull-back along s

Strong bundle maps and the pull-back (V8.19). §8.3 works with strong bundle maps between strong bundles over M-polyfolds, the structure map $\mu : \mathbf{X}_s\times_P W \to W$ first of all, and with the pull-back of $P : W \to X$ along s; it defines neither notion beyond Part I. *The pull-back* is Part I's Proposition 2.6.8 (2017: 2.16) (§II.20) `Chapter2/StrongBundle · pullbackSpace` `Chapter3/ParamPerturbation · StrongBundle.pullback`, and its paracompactness, which Part I's definition of a strong bundle requires, is proved `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` `Chapter8/StrongBundleMap · StrongBundle.paracompactSpace_pullback`. Added here: its atlas, built from pairs of charts `Chapter8/StrongBundleMap · StrongBundle.pullback_atlas` `Chapter8/StrongBundleMap · PullbackChartData.exists_mem_atlas`; the two projections are sc-smooth `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackProj` `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackProj'`; its universal property on $[0]$, a map into the pull-back being sc-smooth iff its two components are `Chapter8/StrongBundleMap · StrongBundle.isScSmoothOn_to_pullback_iff` `Chapter8/StrongBundleMap · pullbackPair` `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackPair`, sc-smoothness being checked chart by chart `Chapter8/StrongBundleMap · MPolyfold.isScSmoothMap_of_forall_charts`; and over a sheet $s : \mathbf{W} \to V$ with inverse σ the projection $(\varphi, w) \mapsto w$ is an sc-diffeomorphism onto $P^{-1}(V)$, with inverse $w \mapsto (\sigma(Pw), w)$ `Chapter8/StrongBundleMap · pullbackSheetInv` `Chapter8/StrongBundleMap · StrongBundle.isScDiffeoOn_pullbackProj'`, so a local sc-diffeomorphism `Chapter8/StrongBundleMap · StrongBundle.isLocalScDiffeo_pullbackProj'`. *Strong bundle maps.* Part I defines them only between local models (Definition 2.6.1 (2017: 2.22), Definition 2.47 of the preprint, §II.18), and asks the transition maps of a strong bundle atlas to be sc-smooth on $[0]$ and $[1]$. The global notion is a reading `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap`: $\Phi : Y \to Y'$ with $P' \circ \Phi = f \circ P$, linear on the fibres `Chapter8/StrongBundleMap · fiberRestrict`, sc-smooth $Y[0] \to Y'[0]$, mapping $Y_{0,1}$ into $Y'_{0,1}$ and sc-smooth $Y[1] \to Y'[1]$. In the charts the last two conditions say that the local representatives $K[i] \to K'[i]$ are sc-smooth, as Part I asks of transition maps. The preservation of the double filtration, part of Definition 2.6.1 (2017: 2.22), is not assumed but follows `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.mapsTo_doubleLevel`, and so do the continuity of Φ, the boundedness of its fibre maps and the sc-smoothness of the base map $f = P' \circ \Phi \circ 0$ `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.continuous` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.fiberMap` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.isScSmoothMap_base`. Strong bundle maps compose and identities are strong bundle maps `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.comp` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.id`; an isomorphism is a pair of inverse strong bundle maps `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleIso`; and the projection of the pull-back to W is a strong bundle map covering s, the identity on each fibre `Chapter8/StrongBundleMap · StrongBundle.isStrongBundleMap_pullbackProj'` `Chapter8/StrongBundleMap · StrongBundle.fiberMap_pullbackProj'`. Strong bundle maps over open subsets of the base are not formalized (§VIII.27) (Findings, Chapter 8, items 30, 31) (used on pp. 373–396 (2017 pp. 270–285)).

VIII.11 Definition 8.3.1 (2017: 8.4): strong bundles over ep-groupoids, and the fibre maps

Definition 8.3.1 (2017: 8.4) (V8.20) `Chapter8/EpStrongBundle · EpStrongBundle`. *A strong bundle over X is a pair (P, μ): a strong bundle $P : W \to X$ over the object M-polyfold and a strong bundle map $\mu : \mathbf{X}_s\times_P W \to W$ covering t, with (1) μ a surjective local sc-diffeomorphism, linear on the fibres; (2) $\mu(1_x, w) = w$; (3) $\mu(\psi \circ \gamma, w) = \mu(\psi, \mu(\gamma, w))$.* The formalization takes the definition as printed, on Part I's pull-back and with

the strong bundle maps of §VIII.10; the conditions $s(\psi) = P(w)$ are carried by the points of the pull-back, the pairs $(1_{Pw}, w)$ are points of it `Chapter8/EpStrongBundle · EpGroupoid.unitPair`, and the composite $P \circ \mu$ is $t \circ \pi_1$ `Chapter8/EpStrongBundle · EpStrongBundle.P_μ`. The surjectivity in (1) follows from (2) and is kept, as in the text; μ is continuous `Chapter8/EpStrongBundle · EpStrongBundle.continuous_μ`. The definition is met by every strong bundle over an M-polyfold, as a strong bundle over the trivial ep-groupoid with $\mu(z, w) = w$ `Chapter8/NonVacuityBundle · EpStrongBundle.trivial` (Findings, Chapter 8, items 32, 102) (2017 p. 271).

The fibre maps $\mu(\psi, \cdot)$ (V8.21) `Chapter8/EpStrongBundle · EpStrongBundle.fiberMorEquiv`. *For $\psi : x \to y$, $\mu(\psi, \cdot) : W_x \to W_y$ is a linear isomorphism*, stated after the definition without proof. It is proved in the stronger form of a topological linear isomorphism `Chapter8/EpStrongBundle · EpStrongBundle.fiberMor`, whose value at w is $\mu(\psi, w)$ `Chapter8/EpStrongBundle · EpStrongBundle.fiberMor_apply`. *Step 1, linear and bounded.* μ is linear on the fibres of $\mathbf{X}_s\times_P W \to \mathbf{X}$, so $\mu(\psi, \cdot)$ is additive and homogeneous `Chapter8/EpStrongBundle · EpStrongBundle.μ_add` `Chapter8/EpStrongBundle · EpStrongBundle.μ_smul`, and bounded, the fibre maps of a strong bundle map being bounded (§VIII.10). *Step 2, the inverse* is $\mu(\psi^{-1}, \cdot)$: by (3) and (2), $\mu(\psi^{-1}, \mu(\psi, w)) = \mu(1_x, w) = w$, and symmetrically `Chapter8/EpStrongBundle · EpStrongBundle.μ_inv_μ` `Chapter8/EpStrongBundle · EpStrongBundle.μ_μ_inv` (Findings, Chapter 8, item 32) (2017 p. 271).

VIII.12 The functorial picture: the ep-groupoid $(W, \mathbf{W})$

The functorial picture (V8.22) `Chapter8/EpStrongBundle · EpStrongBundle.toEpGroupoid`. *With $\mathbf{W} = \mathbf{X}_s\times_P W$, $s(\psi, w) = w$, $t(\psi, w) = \mu(\psi, w)$, units $(1_x, w)$, inverses $(\psi^{-1}, \mu(\psi, w))$ and composition $(\psi, w') \circ (\gamma, w) = (\psi \circ \gamma, w)$, $(W, \mathbf{W})$ is an ep-groupoid, and $(P, \pi_1) : (W, \mathbf{W}) \to (X, \mathbf{X})$ is an sc-smooth functor.* **The text** says "we can define the ep-groupoid" and gives no proof. **The formalized proof** checks Definition 7.1.3 (2017: 7.3). *Step 1, the groupoid.* The composition is total, with the junk value (γ, w) off composable pairs, and the composite of a composable pair has target $\mu(\psi \circ \gamma, w) = \mu(\psi, w')$ by Definition 8.3.1 (3) (2017: 8.4) `Chapter8/EpStrongBundle · EpStrongBundle.morComp` `Chapter8/EpStrongBundle · EpStrongBundle.morComp_val` `Chapter8/EpStrongBundle · EpStrongBundle.μ_morComp`; the inverse is a morphism $\mu(\psi, w) \to w$ by Definition 8.3.1 (2), (3) (2017: 8.4) `Chapter8/EpStrongBundle · EpStrongBundle.morInv`, and the groupoid laws are those of $\mathbf{X}$ in the first entry `Chapter8/EpStrongBundle · EpStrongBundle.toGroupoidStr`. *Step 2, (1) and (2).* $s_{\mathbf{W}}$ is the projection of the pull-back along the local sc-diffeomorphism s, a surjective local sc-diffeomorphism by §VIII.10, and $t_{\mathbf{W}} = \mu$ is one by Definition 8.3.1 (1) (2017: 8.4); the units and inverses are sc-smooth by the universal property of the pull-back. *Step 3, (3), by sequences.* For $w \in W$ take $V(Pw)$ as in Definition 7.1.3 (3) (2017: 7.3) for X and $V = P^{-1}(V(Pw))$. If (ψ_k, w_k) has $w_k \in \operatorname{cl} V$ and $\mu(\psi_k, w_k) \to w'$, then $s(\psi_k) = P(w_k) \in \operatorname{cl} V(Pw)$ and $t(\psi_k) = P(\mu(\psi_k, w_k)) \to P(w')$, so a subsequence has $\psi_k \to \psi$ by the properness of X, and $w_k = \mu(\psi_k^{-1}, \mu(\psi_k, w_k)) \to \mu(\psi^{-1}, w')$ by the continuity of μ and ι `Chapter8/EpStrongBundle · EpStrongBundle.exists_isProperMap_tgt`. *Step 4, (4)* is proved in Chapter 7's sheet form, through a sheet of s in $\mathbf{X}$ and condition (4) for X `Chapter8/EpStrongBundle · EpStrongBundle.isScSmoothCompSheet_toGroupoidStr`. *Step 5, the functor.* $s_{\mathbf{W}}$, $t_{\mathbf{W}}$ are strong bundle maps covering s, t `Chapter8/EpStrongBundle · EpStrongBundle.isStrongBundleMap_src` `Chapter8/EpStrongBundle · EpStrongBundle.isStrongBundleMap_tgt`, (P, π_1) is an sc-smooth functor `Chapter8/EpStrongBundle · EpStrongBundle.projFunctor`, and $|P| : |W| \to |X|$ is continuous and surjective `Chapter8/EpStrongBundle · EpStrongBundle.continuous_orbitMap_projFunctor` `Chapter8/EpStrongBundle · EpStrongBundle.surjective_orbitMap_projFunctor`, since a point isomorphic to $P(w)$ is the image of a point isomorphic to w, namely $\mu(\psi, w)$ `Chapter8/EpStrongBundle · EpStrongBundle.exists_isomorphic_of_isomorphic`. An `EpGroupoid` has its objects and morphisms in one universe, so from here on W lives in the universe of the objects and morphisms of X (Findings, Chapter 8, items 2, 33) (pp. 375–377 (2017

pp. 271–272)).

VIII.13 Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6), and "these are identical notions"

Definition 8.3.2 (2017: 8.5) (V8.23) `Chapter8/EpStrongBundle · EpStrongBundle.IsMap`. *A strong bundle map $(P,\mu) \to (P',\mu')$ is a strong bundle map $\Phi : W \to W'$ covering the object map of an sc-smooth functor $\phi : X \to X'$, with $\Phi(\mu(\gamma,w)) = \mu'(\phi(\gamma),\Phi(w))$.* As printed, with the strong bundle maps of §VIII.10; the identity is one `Chapter8/NonVacuityBundle · EpStrongBundle.isMap_id` (Findings, Chapter 8, item 34) (2017 p. 273).

Definition 8.3.3 (2017: 8.6) (V8.24) `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor`. *A strong bundle functor is a functor $\Phi : (W,\mathbf{W}) \to (W',\mathbf{W}')$, linear on the fibres, covering an sc-smooth functor ϕ and inducing strong bundle maps $W \to W'$ and $\mathbf{W} \to \mathbf{W}'$.* "Linear on the fibres", "covers ϕ" and "induces strong bundle maps" are read together as two strong bundle maps in the sense of §VIII.10: on objects, for P and P', covering ϕ on objects, which contains $P' \circ \Phi = \phi \circ P$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.projFunctor_comp`; and on morphisms, for the pull-backs $\mathbf{W} \to \mathbf{X}$ and $\mathbf{W}' \to \mathbf{X}'$, covering ϕ on morphisms. The second condition forces the morphism map into the form $(\gamma,w) \mapsto (\phi(\gamma),\Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.mor_val`. The identity is a strong bundle functor `Chapter8/NonVacuityBundle · EpStrongBundle.isStrongBundleFunctor_id` (Findings, Chapter 8, item 34) (2017 p. 274).

The equivalence (V8.25) `Chapter8/EpStrongBundle · EpStrongBundle.isMap_iff_strongBundleFunctor`. *$\Phi : W \to W'$ is a strong bundle map of Definition 8.3.2 (2017: 8.5) iff it is the object map of a strong bundle functor of Definition 8.3.3 (2017: 8.6), and the functor, $\Phi(\gamma,w) = (\phi(\gamma),\Phi(w))$, is unique.* The text states that "these are identical notions" and that the functor is recovered from the map. *Step 1, from the functor to the map.* The object map of a strong bundle functor is a strong bundle map by definition, and its compatibility with μ is the functor's compatibility with targets, read through the form of the morphism map: $\Phi(\mu(\gamma,w)) = \Phi(t(\gamma,w)) = t'(\phi(\gamma),\Phi(w)) = \mu'(\phi(\gamma),\Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.isMap`. *Step 2, from the map to the functor.* $(\gamma,w) \mapsto (\phi(\gamma),\Phi(w))$ lands in the pull-back, since $s(\phi(\gamma)) = \phi(Pw) = P'(\Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.liftMor`; it is a functor by the functor laws of ϕ and the compatibility with μ `Chapter8/EpStrongBundle · EpStrongBundle.IsMap.liftFunctor`. It is a strong bundle map of the pull-backs: linear on the fibres `Chapter8/EpStrongBundle · isLinearMap_fiberRestrict_pullbackMap`, sc-smooth on $[0]$ by the universal property of §VIII.10, and on $[1]$ by the universal property of the pull-back on level $[1]$, which is proved here `Chapter8/EpStrongBundle · StrongBundle.pullbackProj'One` `Chapter8/EpStrongBundle · StrongBundle.isScSmoothMap_to_pullback_one` `Chapter8/EpStrongBundle · EpStrongBundle.IsMap.isStrongBundleFunctor_liftFunctor`. *Step 3, uniqueness.* Every strong bundle functor with object map Φ has this morphism map, by the form of Definition 8.3.3's (2017: 8.6) morphism maps `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.eq_liftFunctor` (Findings, Chapter 8, item 34) (pp. 377–379 (2017 pp. 273–274)).

VIII.14 Definition 8.3.4 (2017: 8.7): section functors

Definition 8.3.4 (2017: 8.7) (V8.26). *(1) An sc-smooth section functor of (P,μ) is an sc-smooth section f of P with $f(y) = \mu(\varphi,f(x))$ for every $\varphi : x \to y$; (2) an sc$^+$-section functor takes values in $W_{0,1}$ and is sc-smooth into $W[1]$; (3) an sc-Fredholm section functor is an sc-smooth section functor that is*

an sc-Fredholm section of P. The sc-smooth section functors form a vector space $\Gamma(P,\mu)$, the sc$^+$-section functors a subspace $\Gamma^+(P,\mu)$. The three notions `Chapter8/EpStrongBundle · EpStrongBundle.IsFunctorialSection` `Chapter8/EpStrongBundle · EpStrongBundle.IsScSmoothSectionFunctor` `Chapter8/EpStrongBundle · EpStrongBundle.IsScPlusSectionFunctor` `Chapter8/EpStrongBundle · EpStrongBundle.IsScFredholmSectionFunctor` and their sets `Chapter8/EpStrongBundle · EpStrongBundle.sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.fredholmSectionFunctors` are as printed; (3) uses Part I's Definition 3.1.16 (2017: 3.8) in its germ reading (§III.48) `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection`, and Part I's sections `Chapter2/StrongBundle · StrongBundle.IsScSmoothSection`. *"Vector space"* is read as closure under Part I's fibrewise zero section, sums and constant multiples `Chapter3/StrongBundleSections · fiberZero` `Chapter3/StrongBundleSections · sectionAdd`; no `Module` structure is built. *Step 1, the analytic point.* Part I does not state that the sum of two sc-smooth sections, or a constant multiple of one, is sc-smooth. It is proved in the charts: a section is sc-smooth as soon as its principal parts are, as maps into the local strong bundle models `Chapter8/EpStrongBundle · StrongBundle.isScSmoothSection_of_principal`, and the principal part of a sum is the sum of the principal parts `Chapter8/EpStrongBundle · StrongBundle.IsScSmoothSection.sectionAdd_of_isScSmoothSection` `Chapter8/EpStrongBundle · StrongBundle.IsScSmoothSection.sectionSmul_const` `Chapter8/EpStrongBundle · StrongBundle.isScSmoothSection_fiberZero`. *Step 2, functoriality* is preserved because each $\mu(\varphi,\cdot)$ is linear (§VIII.11). So Γ contains the zero section and is closed under sums and constant multiples `Chapter8/EpStrongBundle · EpStrongBundle.fiberZero_mem_sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionAdd_mem_sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionSmul_mem_sectionFunctors`, and so is $\Gamma^+ \subseteq \Gamma$, the values in $W_{0,1}$ and the sc-smoothness into $W[1]$ being preserved in the same way `Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors_subset` `Chapter8/EpStrongBundle · EpStrongBundle.fiberZero_mem_plusSectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionAdd_mem_plusSectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionSmul_mem_plusSectionFunctors`; the sc-Fredholm section functors lie in Γ `Chapter8/EpStrongBundle · EpStrongBundle.fredholmSectionFunctors_subset`. All three sets are inhabited by non-zero sections: over a trivial ep-groupoid every section is functorial, and the graph section of a smooth map $\mathbb{R}^n \to \mathbb{R}^k$ lies in Γ and is sc-Fredholm (§VIII.26) (Findings, Chapter 8, items 35, 102, 103) (pp. 378–384 (2017 pp. 274–275)).

VIII.15 Definition 8.5.1 (2017: 8.8), and the degeneracy remark after it

Definition 8.5.1 (2017: 8.8) (V8.27) `Chapter8/Basic · EpFunctor.IsProperCovering`. *An sc-smooth functor $F : Y \to X$ is a proper covering functor if (1) F is a surjective local sc-diffeomorphism on objects; (2) every $F^{-1}(x)$ is finite, and x has a neighbourhood $U(x)$ with $F^{-1}(U(x))$ the disjoint union of neighbourhoods $U(y)$, $y \in F^{-1}(x)$, each mapped sc-diffeomorphically onto $U(x)$; (3) $L_F : \varphi \mapsto (F(\varphi), s(\varphi))$ is an sc-diffeomorphism $\mathbf{Y} \to \mathbf{X}\,{}_s\!\times_F Y$.* In (2) $U(x)$ is open and the local inverses $\sigma(y)$ of F on the $U(y)$ are part of the data; (3) is stated on Part I's fibred-product sub-M-polyfold of $\mathbf{X} \times Y$ (Proposition 2.5.20 (2017: 2.15), Proposition 2.46 of the preprint, §II.17) `Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct` `Chapter8/Basic · EpFunctor.fibredMap`, as Chapter 7 states Definition 7.1.3 (4) (2017: 7.3). The identity functor of every ep-groupoid is a proper covering functor `Chapter8/NonVacuityBundle · EpFunctor.isProperCovering_id` (Findings, Chapter 8, items 50, 102) (2017 p. 275).

The degeneracy remark (V8.28). *For $\varphi \in \mathbf{Y}$, $d_{\mathbf{Y}}(\varphi) = d_{\mathbf{X}\,{}_s\!\times_F Y}(F(\varphi), s(\varphi)) = \frac{1}{2}\, d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi))$.* **The text** argues that L_F is an sc-diffeomorphism and that $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, with $d_{\mathbf{X}}(F(\varphi)) = d_Y(s(\varphi)) = d_{\mathbf{Y}}(\varphi)$, F and s being local sc-diffeomorphisms. **The formalized proof** splits the claims. *Step 1, in general.* $d_{\mathbf{Y}}(\varphi) = d_{\mathbf{X}\,{}_s\!\times_F Y}(L_F(\varphi))$ and $d_{\mathbf{X}}(F(\varphi)) = d_{\mathbf{Y}}(\varphi)$, by Part I's Proposition 2.4.4 (2017: 2.7), the invariance of d under sc-diffeomorphisms (§II.11) `Chapter2/DegeneracyIndex · proposition_2_`

4_4 Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_fibredMap Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_mor. *Step 2, the product formula.* Part I's formalization has $d_{X\times Y} = d_X + d_Y$ for tame factors Chapter2/FibredProduct · MPolyfold.isTame_prod and in general only $d_{X\times Y} \le d_X + d_Y$ Chapter2/FibredProduct · MPolyfold.degIndex_prod_le, and Part I's Proposition 2.5.20 (2017: 2.15) states its formula $d = \frac{1}{2} d_{X\times Y}$ for tame X, Y. So $d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi)) = 2\, d_{\mathbf{Y}}(\varphi)$, and with it the remark's second equality, is proved for tame X and Y ($\mathbf{X}$ is then tame, s being a local sc-diffeomorphism) Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_prod_fibredMap, and in general as the inequality $\le$ Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_prod_fibredMap_le. As we read it, the remark needs X and Y tame, which the text does not state; whether the equality holds without tameness is not settled here (Findings, Chapter 8, item 51) (pp. 379–385 (2017 pp. 275–276)).

VIII.16 Proposition 8.5.2 (2017: 8.2): composites of proper covering functors

Proposition 8.5.2 (2017: 8.2) (V8.29) Chapter8/ProperCovering · EpFunctor.IsProperCovering.proposition_8_5_2. *If $F : Y \to X$ and $H : Z \to Y$ are proper covering functors, so is $F \circ H$.* **The text's proof** says that (1) and (2) "follow easily", and for (3) writes $L_{F\circ H} = \tilde{K} \circ L_H$ with $\tilde{K}(\psi, z) = (F(\psi), z)$, whose inverse is $\tilde{L}(\varphi, z) = (L_F^{-1}(\varphi, H(z)), z)$. **The formalized proof.** *Step 1, (1)* holds since composites of surjective local sc-diffeomorphisms are such, and the fibre of $F \circ H$ over x is the finite union of the fibres of H over the points of $F^{-1}(x)$. *Step 2, (2).* The even covering of F at x has to be shrunk: take $U(x)$ for F and, for each $y \in F^{-1}(x)$, the evenly covered neighbourhood $U'(y)$ of H; the set $U_0 = \bigcap_y F(U(y) \cap U'(y))$, a finite intersection, is an open neighbourhood of x with $(F|U(y))^{-1}(U_0) \subseteq U'(y)$, and over U_0 the sheets of $F \circ H$ are those of H over the $U'(y)$, cut down to the preimages of the sets $(F|U(y))^{-1}(U_0)$ Chapter8/ProperCovering · EpFunctor.IsProperCovering.evenlyCovered_comp. *Step 3, (3)*, through the text's $\tilde{K}$ and $\tilde{L}$, which are inverse to each other Chapter8/ProperCovering · EpFunctor.IsProperCovering.isScDiffeo_fibredMap_comp. Their sc-smoothness, as maps between sub-M-polyfolds of products, uses Part I's Proposition 2.3.19 (2) (2017: 2.6) (a map into a sub-M-polyfold is sc-smooth as soon as it is sc-smooth into the ambient M-polyfold), proved again privately, as in Chapter 7. The proof has "fo" for "for" (possible misprint) (Findings, Chapter 8, item 52) (pp. 384–386 (2017 pp. 276–277)).

VIII.17 Theorem 8.5.3 (2017: 8.5) (Proper Covering I), and Remark 8.5.4 (2017: 8.2)

Theorem 8.5.3 (2017: 8.5) (V8.30) Chapter8/ProperCoveringLocal · EpFunctor.theorem_8_5_3. *Let $F : Y \to X$ be a proper covering functor and $x \in X$. For a connected, sufficiently small open $U(x)$ admitting the natural G_x-action: (0) $F^{-1}(U(x))$ is the disjoint union of connected open $U(y) \ni y$, $y \in F^{-1}(x)$, with $F : U(y) \to U(x)$ sc-diffeomorphisms; (1) $\mathbf{U}(x) = \{\varphi \mid s(\varphi), t(\varphi) \in U(x)\}$ is the disjoint union of the components $\mathbf{U}(g)$, $g \in G_x$, on which s, t are sc-diffeomorphisms onto $U(x)$; (2) the morphisms from $U(y)$ to $U(y')$ form the disjoint union of connected open $\mathbf{U}(\psi_0) \ni \psi_0$, $\psi_0 : y \to y'$, with (3) $s : \mathbf{U}(\psi_0) \to U(y)$ and $t : \mathbf{U}(\psi_0) \to U(y')$ sc-diffeomorphisms, (4) $F : \mathbf{U}(\psi_0) \to \mathbf{U}(F(\psi_0))$ an sc-diffeomorphism, and (5) a commutative square of sc-diffeomorphisms with top arrow F, left arrow s, right arrow t and bottom arrow $F : U(s(\psi_0)) \to U(t(F(\psi_0)))$.* The data form a chart Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart: a natural representation R of G_x on $U(x)$ (Chapter 7's datum, §VII.7, with $U(x)$ open and connected), the sets $U(y)$ and $\mathbf{U}(\psi_0)$, and (0)–(5) as fields; $\mathbf{U}(g)$ is read as the connected component of $\mathbf{U}(x)$ at g, which contains $F(\psi_0)$ by Re-

mark 8.5.4 (2017: 8.2) (a). "Sufficiently small" is proved in two forms: charts with $U(x)$ inside every open $V \ni x$ (`theorem_8_5_3`), and one open $W \ni x$ such that every natural representation on a connected $U(x) \subseteq W$ is the R of a chart `Chapter8/ProperCoveringLocal · EpFunctor.theorem_8_5_3_of_naturalRep`.

The text's proof starts from $U'(x)$ and the $U'(y)$ of Definition 8.5.1 (2) (2017: 8.8), takes connected $V(y) \subseteq U'(y)$ with natural representations of the G_y and, for each $\psi : y \to y'$, sheets $V(\psi)$ with $\widehat{\psi} = t \circ (s|V(\psi))^{-1} : V(y) \to V(y')$ (8.15), so that the morphisms between points of $\bigcup V(y)$ form the disjoint union of the $V(\psi)$. It shrinks to a G_x-invariant connected $U(x)$, puts $U(y) = (F|V(y))^{-1}(U(x))$, shows that $\widehat{\psi}$ maps $U(y)$ onto $U(y')$, and sets $\mathbf{U}(\psi) = (s|V(\psi))^{-1}(U(y))$; (4) comes from $F = (s|\mathbf{U}(g))^{-1} \circ F \circ s$ on $\mathbf{U}(\psi_0)$, and (5) is called obvious. **The formalized proof** uses Definition 8.5.1 (3) (2017: 8.8) instead of natural representations in Y: the sheets $\mathbf{U}(\psi_0)$ are the images of continuous sections of s given by L_F^{-1}. *Step 1, the input* `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data`: $U'(x)$ with the $U'(y)$ and local inverses $\sigma(y)$ of F (Definition 8.5.1 (2) (2017: 8.8)), the continuous inverse L_F^{-1} of the sc-diffeomorphism L_F (Definition 8.5.1 (3) (2017: 8.8)), and a natural representation R with $U(x) \subseteq U'(x)$; one $W = U'(x)$ serves for every such R `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.exists_data`. *Step 2, (0).* $U(y) = U'(y) \cap F^{-1}(U(x)) = \sigma(y)(U(x))$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.U` is open, connected as the image of the connected $U(x)$, and mapped sc-diffeomorphically onto $U(x)$. *Step 3, (1)* involves only R and is Chapter 7's description of $\mathbf{U}(x)$ by the sheets $\Gamma(g, U(x))$ (§VII.8) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.iUnion_image_Γ` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.connectedComponentIn_morOn`. *Step 4, the sheets.* For $\psi_0 : y \to y'$ with $g = F(\psi_0) \in G_x$, put $\tau(z) = L_F^{-1}(\Gamma(g, F(z)), z)$ for $z \in U(y)$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.τ` and $\mathbf{U}(\psi_0) = \tau(U(y))$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.sheet`. τ is a continuous section of s with $\tau(y) = \psi_0$, and a continuous section of a local sc-diffeomorphism over an open set is sc-smooth, being locally a local inverse `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.isScSmoothOn_of_section` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScSmoothOn_τ`; this gives s in (3). *Step 5, t in (3).* $F(t(\tau z)) = t(\Gamma(g, Fz)) = \Phi(g)(Fz) \in U(x)$, so $t \circ \tau$ maps $U(y)$ into $F^{-1}(U(x))$, the disjoint union of the open $U(y'')$; since $U(y)$ is connected and $t(\tau(y)) = y'$, it maps into $U(y')$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.mapsTo_of_isPreconnected` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.tgt_τ_mem`. This is the counterpart of the text's claim that $\widehat{\psi}$ maps $U(y)$ onto $U(y')$; there $t \circ \tau = \sigma(y') \circ \Phi(g) \circ F$, an sc-diffeomorphism with inverse $\sigma(y) \circ \Phi(g)^{-1} \circ F$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.isScDiffeoOn_congr_left` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScDiffeoOn_hat`. *Step 6, (2).* A morphism ψ from $U(y)$ to $U(y')$ has $F(\psi) \in \mathbf{U}(x)$, so $F(\psi) = \Gamma(g, F(s\psi))$ for one $g \in G_x$ by (1), and $\psi = L_F^{-1}(\Gamma(g, F(s\psi)), s\psi)$ lies in the sheet of the morphism ψ_0 out of y with $F(\psi_0) = g$; distinct ψ_0 give disjoint sheets by the injectivity of L_F, and the sheets are the connected components `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.connectedComponentIn_eq_of_iUnion`. *Step 7, (4)* is the text's formula: $F = \Gamma(g, \cdot) \circ F \circ s$ on $\mathbf{U}(\psi_0)$, an sc-diffeomorphism onto $\Gamma(g, U(x)) = \mathbf{U}(g)$ with inverse $\tau \circ \sigma(y) \circ s$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScDiffeoOn_mor_sheet`. *Step 8, (5)* is the functoriality $F(s\psi) = s(F\psi)$ (the field `comm_src`), together with the companion square $F(t\psi) = t(F\psi)$ (the field `comm_tgt`); the arrows are sc-diffeomorphisms by (0), (1), (3), (4). The chart is assembled `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.toChart`, and charts exist inside every V because natural representations do (Theorem 7.1.13 (2017: 7.1)) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13`. The finiteness of $F^{-1}(x)$ and Definition 8.5.1 (1) (2017: 8.8) are not used.

(5) as printed. With t on the right arrow and the corner $U(t(F(\psi_0)))$, the square says $F(s(\psi)) = t(F(\psi))$ on $\mathbf{U}(\psi_0)$. A functor gives $F(s(\psi)) = s(F(\psi))$, and $F(\psi) = \Gamma(g, F(s\psi))$ has $t(F(\psi)) = \Phi(g)(F(s\psi))$,

which differs from $F(s\psi)$ wherever $\Phi(g)$ does not fix $F(s\psi)$. So (5) as printed does not hold as we read it: for the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$, a proper covering functor, every chart at 0 contains, in the sheet of $\psi_0 = (-1, 0)$ `Chapter8/NonVacuity · CoveringExamples.reflectionψ₀`, a $\psi = (-1, z)$ with $F(s\psi) = z \neq -z = t(F\psi)$ `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_3_five_as_printed`, and such charts exist `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_3_five_as_printed_nonvacuous`. The square holds with s on both vertical arrows (the corner $U(s(F(\psi_0))) = U(x)$ is the same set), which suggests t as a possible misprint for s. Possible misprints in the proof: "$g \in G_y$" and "G_y-invariant" for G_x, "$F(z') = s(F(\varphi))$" for $t(F(\varphi))$, "admitting admitting", and "morphismsm" in (2) (Findings, Chapter 8, items 53, 54, 55) (pp. 385–388 (2017 pp. 277–279)).

Remark 8.5.4 (2017: 8.2) (V8.31). *(a) A morphism between points of $F^{-1}(x)$ is mapped into G_x; (b) everything in Theorem 8.5.3 (2017: 8.5) is determined by the choice of $U(x)$.* (a) is functoriality `Chapter8/ProperCoveringLocal · EpFunctor.remark_8_5_4_a`, and defines the element of G_x `Chapter8/ProperCoveringLocal · EpFunctor.isotropyMor`. (b) is read as: two charts with the same $U(x)$ have the same Φ and Γ on $U(x)$, the same $U(y)$ and the same $\mathbf{U}(\psi_0)$ `Chapter8/ProperCoveringLocal · EpFunctor.remark_8_5_4_b`. *Step 1.* On a connected $U(x)$, $\Gamma(g, \cdot)$ is the continuous section of s through g, and two such sections agree (§VII.9) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_eqOn`; $\Phi(g) = t \circ \Gamma(g, \cdot)$ follows. *Step 2.* The $U(y)$ and the $\mathbf{U}(\psi_0)$ are the connected components of $F^{-1}(U(x))$ at y and of the morphisms from $U(y)$ to $U(y')$ at ψ_0, so they are determined by $U(x)$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart.connectedComponentIn_U` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart.connectedComponentIn_sheet`. The remarks on the naturality of Theorem 7.1.13's (2017: 7.1) construction have no further formal content (Findings, Chapter 8, item 56) (pp. 385–387 (2017 pp. 277–278)).

VIII.18 Diff_{sc}, Definition 8.5.5 (2017: 8.9), and the discussion after it

Diff_{sc} and Definition 8.5.5 (2017: 8.9) (V8.32). *Diff_{sc} has the M-polyfolds as objects and the sc-diffeomorphisms as morphisms. For finite groupoids C, with one object, and D, a functor $f : D \to C$ is a covering functor if $\psi \mapsto (f(\psi), s(\psi))$ is a bijection $\mathbf{D} \to C \,{}_s\!\times_f D$; a geometric lift of f is a pair of functors $L_D : D \to \mathrm{Diff}_{sc}$, $L_C : C \to \mathrm{Diff}_{sc}$ with a natural transformation $\tau : L_D \to L_C \circ f$.* Diff_{sc} is the category of bundled M-polyfolds of one universe `Chapter8/Basic · MPolyfoldCat` with Part I's sc-diffeomorphisms, bundled with their inverses `Chapter8/Basic · ScDiffeo` `Chapter8/Basic · ScDiffeo.id` `Chapter8/Basic · ScDiffeo.comp` `Chapter8/Basic · ScDiffeo.ext`. C is Mathlib's `SingleObj G` for a group G, its automorphism group, and D a Mathlib groupoid, both in `Type`; then $C \,{}_s\!\times_f D = G \times D$ `Chapter8/Basic · coveringMap` `Chapter8/Basic · IsCoveringFunctor` `Chapter8/Basic · GeometricLift`. The covering property says that a $g \in G$ lifts uniquely at every y `Chapter8/GeometricLift · IsCoveringFunctor.existsUnique_lift` `Chapter8/GeometricLift · IsCoveringFunctor.lift`. The finiteness ("finitely many objects and morphisms") is not part of the two structures; it is carried as instance arguments by the results that use it, Proposition 8.5.7 (2017: 8.3) and Theorem 8.5.8 (2017: 8.6). $D = \emptyset$ is allowed, as in the text `Chapter8/GeometricLift · isCoveringFunctor_of_isEmpty` (§VIII.20). The definition is not vacuous in either direction: the identity of `SingleObj G` is a covering functor, and for non-trivial G the functor sending everything to 1 is not `Chapter8/NonVacuity · CoveringExamples.isCoveringFunctor_id` `Chapter8/NonVacuity · CoveringExamples.not_isCoveringFunctor_one` (Findings, Chapter 8, items 57, 100) (pp. 387–389 (2017 pp. 279–280)).

The discussion after Definition 8.5.5 (2017: 8.9) (V8.33). *With $U(x) = L_C(\star)$, $U(y) = L_D(y)$, the maps $\tau(y) : U(y) \to U(x)$ and $\varphi_{yy'}(\psi) = L_D(\psi)$ are sc-diffeomorphisms, $\varphi_{yy''}(\psi \circ \gamma) = \varphi_{y'y''}(\psi) \circ \varphi_{yy'}(\gamma)$, G_y acts on $U(y)$ and G_x on $U(x)$, and $\tau(y)(h * z) = f(h) * \tau(y)(z)$.* The text states these as consequences of the definition; they are proved. $U(x)$ `Chapter8/GeometricLift · GeometricLift.Ux` and

$\tau(y)$ Chapter8/GeometricLift · GeometricLift.τy Chapter8/GeometricLift · GeometricLift.isScDiffeoOn_τy; $\varphi_{yy'}$ Chapter8/GeometricLift · GeometricLift.φ Chapter8/GeometricLift · GeometricLift.isScDiffeoOn_φ is functorial because L_D is Chapter8/GeometricLift · GeometricLift.φ_comp Chapter8/GeometricLift · GeometricLift.φ_id Chapter8/GeometricLift · GeometricLift.φ_inv_φ. The actions are $h * z = \varphi_{yy}(h)(z)$ and $g * z = L_C(g)(z)$ Chapter8/GeometricLift · GeometricLift.smul_def Chapter8/GeometricLift · GeometricLift.smul_def', sc-smooth Chapter8/GeometricLift · GeometricLift.isScSmoothMap_smul; the action of G_x is a left action through the composition of `SingleObj G`, in which $g \gg h = hg$. The naturality of τ at a morphism, $\tau(y')(\varphi_{yy'}(\psi)(z)) = f(\psi) * \tau(y)(z)$ Chapter8/GeometricLift · GeometricLift.τy_φ, specializes to the equivariance Chapter8/GeometricLift · GeometricLift.τ_equivariant. Functoriality is stated with Mathlib's $\gamma \gg \psi$ for the text's $\psi \circ \gamma$, and G_y is Mathlib's monoid `End y` (Findings, Chapter 8, item 57) (pp. 388–390 (2017 pp. 280–281)).

VIII.19 Disjoint unions of M-polyfolds

Disjoint unions (V8.34). *In the proof of Proposition 8.5.7 (2017: 8.3), "$Y = \bigsqcup U(y)$ possesses a natural M-polyfold structure".* The text does not describe it. It is read as the structure whose charts are those of the summands, transported along the inclusions Chapter8/SigmaPolyfold · MChart.pushforward Chapter8/SigmaPolyfold · MPolyfold.sigmaChart Chapter8/SigmaPolyfold · MPolyfold.sigma, for any index type. *Step 1, compatibility.* Charts of different summands have disjoint domains, and Part I's compatibility then asks for the sc-smoothness of maps between empty retracts, which holds, so nothing is added Chapter8/SigmaPolyfold · MChart.compatible_of_disjoint; the charts of the union are exactly the transported charts Chapter8/SigmaPolyfold · MPolyfold.isChart_sigmaChart_iff. A transported chart is extended by 0 off its image, so no summand needs to be nonempty. *Step 2, paracompactness* is proved summand by summand, without metrizability Chapter8/SigmaPolyfold · paracompactSpace_sigma. *Step 3, summand-wise properties.* The summands are open and sc-diffeomorphic to the X_i Chapter8/SigmaPolyfold · MPolyfold.isScDiffeoOn_sigmaMk; sc-smoothness, local sc-diffeomorphisms, the degeneracy index and tameness are checked summand by summand Chapter8/SigmaPolyfold · MPolyfold.isScSmoothMap_sigma_iff Chapter8/SigmaPolyfold · MPolyfold.isLocalScDiffeo_sigma_iff Chapter8/SigmaPolyfold · MPolyfold.degIndex_sigma Chapter8/SigmaPolyfold · MPolyfold.isTame_sigma_iff. *Step 4, products.* Products distribute over the union Chapter8/SigmaPolyfold · MPolyfold.isScDiffeoOn_sigmaProdDistrib, and $G \times X \cong \bigsqcup_G X$ for discrete G Chapter8/SigmaPolyfold · isScDiffeoOn_prod_discrete_sigma, which identifies the text's $\mathrm{mor}(y, y') \times U(y)$ with the model of §VIII.20 (Findings, Chapter 8, item 58) (2017 p. 281).

VIII.20 Proposition 8.5.7 (2017: 8.3): the proper covering functor of a geometric lift

Proposition 8.5.7 (2017: 8.3) (V8.35) Chapter8/GeometricLift · proposition_8_5_7. *A covering functor $f : D \to C$ with a geometric lift gives a proper covering functor $F : Y \to X$, with $X = G_x \ltimes U(x)$, $Y = \bigsqcup_y U(y)$, $\mathbf{Y} = \bigsqcup_{y,y'} \mathrm{mor}(y, y') \times U(y)$, $F|U(y) = \tau(y)$ and $F(\psi, z) = (f(\psi), \tau(y)(z))$.* **The text's proof** defines X, Y and F, says that "one verifies" that Y is an ep-groupoid and F a functor, and shows that (8.16), L_F, is injective, surjective and a local sc-diffeomorphism. **The formalized proof** is for nonempty D. *Step 1, X* is Chapter 7's translation groupoid (§VII.30) Chapter7/NonVacuity · EpGroupoid.Examples.actionEpGroupoid Chapter8/GeometricLift · geometricLiftTarget, with G discrete and lifted by `ULift` to the universe of $U(x)$; a finite G makes it an ep-groupoid. *Step 2, Y.* Objects and morphisms are disjoint unions (§VIII.19) Chapter8/GeometricLift · GeometricLift.

YObj Chapter8/GeometricLift · GeometricLift.YMor Chapter8/GeometricLift · GeometricLift.yObjPolyfold Chapter8/GeometricLift · GeometricLift.yMorPolyfold, $\mathrm{mor}(y,y') \times U(y)$ being modelled as $\bigsqcup_{\psi: y \to y'} U(y)$. The composition $(\psi, z') \circ (\gamma, z) = (\psi \circ \gamma, z)$ is total, with a junk value off composable pairs Chapter8/GeometricLift · homComp Chapter8/GeometricLift · GeometricLift.sourceStr. On the summand of ψ, s is the identity of $U(y)$ and t is $\varphi_{yy'}(\psi)$, so both are local sc-diffeomorphisms Chapter8/GeometricLift · GeometricLift.isLocalScDiffeo_ySrc Chapter8/GeometricLift · GeometricLift.isLocalScDiffeo_yTgt. "One verifies" is carried out: t is proper, finitely many summands each being mapped homeomorphically onto a summand, so Definition 7.1.3 (3) (2017: 7.3) holds with $V = Y$ Chapter8/GeometricLift · GeometricLift.isProperMap_yTgt; and (4) holds in the sheet form, since near (ψ_0, z_0) the map $\psi \mapsto \sigma(t\psi) \circ \psi$ is $(\psi_0, z) \mapsto (\gamma_0 \circ \psi_0, z)$, the summand index γ_0 of $\sigma(t\psi)$ being locally constant Chapter8/GeometricLift · GeometricLift.isScSmoothCompSheet_y Chapter8/GeometricLift · geometricLiftSource. *Step 3,* F is a functor by the naturality of τ (§VIII.18) Chapter8/GeometricLift · geometricLiftFunctor. *Step 4, Definition 8.5.1 (1), (2) (2017: 8.8).* F is a local sc-diffeomorphism, each $\tau(y)$ being one on an open summand Chapter8/GeometricLift · geometricLiftFunctor_isLocalScDiffeo, and surjective since $D \neq \emptyset$ Chapter8/GeometricLift · geometricLiftFunctor_surjective; its fibres have one point in each summand and are finite Chapter8/GeometricLift · geometricLiftFunctor_finite_fiber, and (2) holds with $U(x)$ the whole object space of X, the sheets being the summands Chapter8/GeometricLift · geometricLiftFunctor_evenlyCovered. *Step 5, (8.16).* $L_F(\psi, z) = ((f(\psi), \tau(y)z), (y, z))$ is bijective because f is a covering functor. The text deduces the sc-diffeomorphism from "bijection and local sc-diffeomorphism"; here the inverse is shown sc-smooth directly: near $((g_0, q_0), (y_0, z_0))$ it is $((g_0, q), (y_0, z)) \mapsto (\psi_0, z)$, ψ_0 the lift of g_0 at y_0 Chapter8/GeometricLift · GeometricLift.fibredInv Chapter8/GeometricLift · GeometricLift.isScSmoothMap_fibredInv Chapter8/GeometricLift · geometricLiftFunctor_isScDiffeo_fibredMap. *D nonempty.* Definition 8.5.5 (2017: 8.9) allows $D = \emptyset$, and then $Y = \emptyset$ while $U(x)$ is arbitrary: F is surjective on objects iff $D \neq \emptyset$ or $U(x) = \emptyset$ Chapter8/GeometricLift · geometricLiftFunctor_surjective_iff. The text does not say so; the hypothesis is used for the surjectivity only, and holds in Theorem 8.5.8 (2017: 8.6). For $D = \emptyset$ the proposition does not hold as we read it: the empty groupoid has a covering functor to $\mathbb{Z}/2$ with a geometric lift on $\mathbb{R}$, whose functor is not surjective Chapter8/GeometricLiftEmpty2021 · CoveringExamples.not_proposition_8_5_7_empty (§VIII.40). The proposition is not vacuous: $\mathbb{Z}/2$ acting on $\mathbb{R}$ by reflection is a geometric lift of the identity of `SingleObj` $\mathbb{Z}/2$ (§VIII.26). The proof's "$F : X \to Y$" is read as $F : Y \to X$ (possible misprint) (Findings, Chapter 8, items 59, 101, 130) (pp. 389–391 (2017 pp. 281–282)).

VIII.21 Theorem 8.5.8 (2017: 8.6) (Proper Covering II)

Theorem 8.5.8 (2017: 8.6) (V8.36) Chapter8/GeometricLift · EpFunctor.theorem_8_5_8. *Let $F : Y \to X$ be a proper covering functor, $x \in X$, C the full subcategory on x and D the full subcategory on $F^{-1}(x)$. Then $f = F|D : D \to C$ is a covering functor, and for $U(x)$ as in Theorem 8.5.3 (2017: 8.5) there are a geometric lift $A : K \to L$ of f and fully faithful open embeddings $\Phi : L \to X$, $\Psi : K \to Y$ whose images are the full subcategories on $U(x)$ and $F^{-1}(U(x))$, with $F \circ \Psi = \Phi \circ A$.* **The text's proof** puts $L_C(g) = \Phi(g)$, $L_D(y) = U(y)$, $L_D(\psi) = t \circ (s|\mathbf{U}(\psi_0))^{-1}$, $\tau(y) = F|U(y)$, says "one verifies" that this is a geometric lift, takes A from Proposition 8.5.7 (2017: 8.3) and puts $\Phi(g, z) = \Gamma(g, z)$, $\Psi(\psi, z) = (s|\mathbf{U}(\psi_0))^{-1}(z)$. **The formalized proof** follows it, and carries out the "one verifies" through F being injective on each $U(y)$. *Step 1, f.* C is `SingleObj` G_x and D the full subcategory on $F^{-1}(x)$ Chapter8/GeometricLift · GroupoidStr.FullOb Chapter8/GeometricLift · EpFunctor.FiberCat; F sends its morphisms into G_x Chapter8/GeometricLift · EpFunctor.fiberIsotropy Chapter8/GeometricLift · EpFunctor.restrictFiber. f is a covering functor because L_F is a bijection and a morphism ψ with $s(\psi) \in F^{-1}(x)$ and $F(\psi) \in G_x$

has $t(\psi) \in F^{-1}(x)$ `Chapter8/GeometricLift · EpFunctor.covering_restrict`. D is finite by Definition 8.5.1 (2) (2017: 8.8) and Proposition 7.1.12 (2017: 7.4) `Chapter8/GeometricLift · EpFunctor.fintypeFiberCat`, and nonempty by the surjectivity of F `Chapter8/GeometricLift · EpFunctor.nonempty_fiberCat`. *Step 2, the lift.* For a chart of Theorem 8.5.3 (2017: 8.5), let σ_ψ be the inverse of s on $\mathbf{U}(\psi)$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.σSrc`. By (1) and (4) of Theorem 8.5.3 (2017: 8.5), $F(\sigma_\psi(z)) = \Gamma(f(\psi), F(z))$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.mor_σSrc`, so $\tau(y') \circ L_D(\psi) = \Phi(f(\psi)) \circ \tau(y)$, the naturality of τ. The functoriality of L_D follows: $t\sigma_{\gamma \gg \psi}(z)$ and $t\sigma_\psi(t\sigma_\gamma(z))$ lie in the same $U(y'')$ and have the same image $\Phi(f(\psi))\Phi(f(\gamma))F(z)$ under F, which is injective on $U(y'')$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.hat_comp`. The sets $U(x)$, $U(y)$ become objects of Diff_{sc} as open sub-M-polyfolds `Chapter8/GeometricLift · MPolyfoldCat.ofOpen` `Chapter8/GeometricLift · ScDiffeo.ofOpen`, and the geometric lift is assembled `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.geometricLift`; A is its functor of Proposition 8.5.7 (2017: 8.3), a proper covering functor since $D \neq \emptyset$. *Step 3, Φ and Ψ.* Φ is the inclusion on objects and Γ on morphisms `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.Φ`; Ψ is the inclusion on objects and $\Psi(\psi, z) = \sigma_\psi(z)$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.Ψ`, which preserves compositions and units by the unique lifting of Definition 8.5.1 (3) (2017: 8.8): $\Psi(\psi, z') \circ \Psi(\gamma, z)$ and $\Psi(\gamma \gg \psi, z)$ have the same source and the same image under F. "Fully faithful open embedding" is read as full and faithful on the underlying groupoids, and an injective local sc-diffeomorphism on objects and on morphisms `Chapter8/GeometricLift · EpFunctor.IsFullyFaithfulOpenEmbedding` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Φ` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Ψ`; the images are the full subcategories, on objects and on morphisms `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Φ_obj` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Φ_mor` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Ψ_obj` `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Ψ_mor`, the last two by Theorem 8.5.3 (0) and (2) (2017: 8.5). *Step 4, the square.* $F(\Psi(\psi, z)) = F(\sigma_\psi(z)) = \Gamma(f(\psi), F(z)) = \Phi(A(\psi, z))$ `Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.comp_Ψ`. *Readings.* The theorem is stated for ep-groupoids X, Y whose objects and morphisms lie in any one universe, Definition 8.5.5 (2017: 8.9) taking D, G and Diff_{sc} in that universe; the first reading had it in `Type` only, and no proof changed (§VIII.40) `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_8_id`; "a sufficiently small $U(x)$ as in Theorem 8.5" is any chart of Theorem 8.5.3 (2017: 8.5), and such charts exist inside every open $V \ni x$, which `theorem_8_5_8` restates; "a geometric lift $A : K \to L$" is a geometric lift with its functor A of Proposition 8.5.7 (2017: 8.3); and since the morphisms of D are the ψ_0 themselves, "$\psi \in \mathbf{U}(\psi_0)$" in the definition of L_D is read as $\psi = \psi_0$. Possible misprints: "there exists exists", "$\tau(t)$" for $\tau(y)$, and "$A(\psi, z) = (f(z), F(z))$" for $(f(\psi), F(z))$, as in Proposition 8.5.7 (2017: 8.3) (Findings, Chapter 8, items 2, 60, 125) (2017 pp. 283–284).

VIII.22 Definition 8.5.9 (2017: 8.10): proper strong bundle covering functors

Definition 8.5.9 (2017: 8.10) (V8.37) `Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering`. *Let (P, μ) be a strong bundle over Y, (Q, ν) one over X, and $A : W \to V$ a strong bundle map with $Q \circ A = F \circ P$. A is a proper strong bundle covering functor if (1) A is surjective and a local strong bundle isomorphism covering a proper covering functor F; (2) over the even coverings of Definition 8.5.1 (2) (2017: 8.8), $A : W|U(y) \to V|U(x)$ are strong bundle isomorphisms; (3) $(\psi, w) \mapsto ((F(\psi), A(w)), w)$ is a strong bundle isomorphism $\mathbf{W} \to \mathbf{V}\,{}_s\!\times_A W$ covering $L_F : \mathbf{Y} \to \mathbf{X}\,{}_s\!\times_F Y$. Readings.* A is taken in the functorial picture, as a strong bundle functor covering F (Definition 8.3.3 (2017: 8.6); equivalently its object map is a strong bundle map of Definition 8.3.2 (2017: 8.5), §VIII.13). "$A : W|U \to V|U'$ is a strong

bundle isomorphism" means that $F : U \to U'$ is an sc-diffeomorphism and $A : P^{-1}(U) \to Q^{-1}(U')$ an sc-diffeomorphism on $[0]$ and on $[1]$ `Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn`; its inverse is then linear on the fibres, so the fibre maps are topological linear isomorphisms `Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn.fiberEquiv`. The finiteness of $F^{-1}(x)$ in (2) is that of Definition 8.5.1 (2) (2017: 8.8) and is not repeated. *The strong bundle of (3).* The text does not say which strong bundle structure $\mathbf{V}\,{}_s\!\times_A W \to \mathbf{X}\,{}_s\!\times_F Y$ carries. It is read as the pull-back of $P : W \to Y$ along the sc-smooth projection $\mathrm{pr}_2 : \mathbf{X}\,{}_s\!\times_F Y \to Y$ `Chapter8/EpStrongBundle · EpFunctor.fibredSnd` `Chapter8/EpStrongBundle · EpFunctor.isScSmoothMap_fibredSnd`, on Part I's fibred-product sub-M-polyfold. Its points $((\varphi, y), w)$ with $P(w) = y$ correspond to the text's $((\varphi, v), w)$ with $v = A(w)$ `Chapter8/EpStrongBundle · EpStrongBundle.fibredBundleEquiv`; under the correspondence the projection is the text's $((\varphi, v), w) \mapsto (\varphi, P(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.pullbackProj_fibredBundleEquiv`, and the map of (3) is $(\psi, w) \mapsto ((F(\psi), s(\psi)), w)$ `Chapter8/EpStrongBundle · EpStrongBundle.fibredLift`, which the correspondence turns into the text's $((F(\psi), A(w)), w)$ `Chapter8/EpStrongBundle · EpStrongBundle.fibredBundleEquiv_symm_fibredLift` and which covers L_F `Chapter8/EpStrongBundle · EpStrongBundle.pullbackProj_fibredLift`. The definition is satisfiable: the identity of every strong bundle over an ep-groupoid is a proper strong bundle covering functor, with (3) on this reading (§VIII.26). The closing sentence, "all structural statements about proper covering functors lift to the strong bundle situation", is not formalized beyond §8.6.2 (2017 §8.5.2) (Findings, Chapter 8, items 70, 102, 103) (2017 p. 285).

VIII.23 Appendix §8.6.1 (2017 §8.5.1): the morphisms over a fibre, and Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11)

The morphisms over a fibre (V8.38). *For a proper covering functor $F : Y \to X$ and $x \in X$, $\psi \mapsto (F(\psi), s(\psi))$ is a bijection $\mathrm{mor}(F^{-1}(x)) \to G_x \times F^{-1}(x)$, so $\#\mathrm{mor}(F^{-1}(x)) = \#G_x \cdot \#F^{-1}(x)$; $F^{-1}(x)$ splits into isomorphism classes $\Theta_1, \dots, \Theta_{q'}$ and points of one class have isomorphic isotropy groups.* $\mathrm{mor}(F^{-1}(x))$ is the set of morphisms between points of $F^{-1}(x)$. *Step 1, the bijection.* A morphism between points of $F^{-1}(x)$ is mapped into G_x (Remark 8.5.4 (2017: 8.2) (a)) `Chapter8/ProperCovering · EpFunctor.isotropyOfMorOn`, and by Definition 8.5.1 (3) (2017: 8.8) every (g, y) has exactly one lift `Chapter8/ProperCovering · EpFunctor.IsProperCovering.bijective_fibredMap` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.existsUnique_lift`, whose target lies again in $F^{-1}(x)$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.morOnFiberEquiv`. *Step 2, the count.* $\#$ is the cardinality of finite sets: the fibre is finite by Definition 8.5.1 (2) (2017: 8.8), the isotropy groups by Proposition 7.1.12 (2017: 7.4) (§VII.6) `Chapter7/Basic · EpGroupoid.proposition_7_1_12`, hence so is $\mathrm{mor}(F^{-1}(x))$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.finite_morOn_fiber` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_morOn_fiber`. *Step 3, the classes* are indexed by their images in $|Y|$: $\Theta_c = F^{-1}(x) \cap \pi^{-1}(c)$ `Chapter8/ProperCovering · EpFunctor.fiberClass` `Chapter8/ProperCovering · EpFunctor.fiberIndex` `Chapter8/ProperCovering · EpFunctor.fiberClasses` `Chapter8/ProperCovering · EpFunctor.mem_fiberClass_iff`; they cover the fibre and are pairwise disjoint `Chapter8/ProperCovering · EpFunctor.iUnion_fiberClass` `Chapter8/ProperCovering · EpFunctor.pairwiseDisjoint_fiberClass`, with $\Theta_c = \emptyset$ for $c \notin \pi(F^{-1}(x))$, and isomorphic points have conjugate, hence equinumerous, isotropy groups `Chapter8/ProperCovering · EpFunctor.nat_card_isotropy_eq_of_mem_fiberClass`. With this indexing the correspondence $\Theta_j \leftrightarrow \Theta'_j$ of Lemma 8.6.2 (2017: 8.10) is "the same index" (Findings, Chapter 8, item 71) (2017 p. 286).

Lemma 8.6.1 (2017: 8.9) (V8.39) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_6_1`. *$\#G_x \cdot \#\Theta_j = \#\{\psi \mid s(\psi), t(\psi) \in \Theta_j\} = \#G_{y_0} \cdot (\#\Theta_j)^2$, and $\#G_x = \#G_{y_0} \cdot \#\Theta_j$ for $y_0 \in \Theta_j$.* **The text's proof** restricts the bijection to (8.17), $\{\psi \mid s(\psi), t(\psi) \in \Theta_j\} \to G_x \times \Theta_j$, and, choosing

$\psi_y : y_0 \to y$, uses the bijections $\mathrm{mor}(y_0, y_0) \to \mathrm{mor}(y, y')$, $\psi \mapsto \psi_{y'} \circ \psi \circ \psi_y^{-1}$. **The formalized proof** does the same. *Step 1.* The set of the lemma is $\mathrm{mor}(\Theta_j)$ `Chapter8/ProperCovering · EpFunctor.morOn_fiberClass_eq`, and (8.17) is a bijection because $t(\psi) \in \Theta_j$ is automatic for $s(\psi) \in \Theta_j$ and $t(\psi) \in F^{-1}(x)$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.morOnClassEquiv`. *Step 2.* The text's bijections assemble into one, $\mathrm{mor}(\Theta_j) \cong (\Theta_j \times \Theta_j) \times G_{y_0}$ `Chapter8/ProperCovering · EpFunctor.nonempty_morOnClass_equiv_isotropy`. *Step 3.* The third claim divides by $\#\Theta_j \geq 1$. (8.17) writes "$\psi \to (F\psi), s(\psi))$" for $\psi \mapsto (F(\psi), s(\psi))$ (possible misprint) (Findings, Chapter 8, items 71, 72) (pp. 396–398 (2017 pp. 286–287)).

Lemma 8.6.2 (2017: 8.10) (V8.40) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_6_2`. *For isomorphic x, x' the isomorphism classes of $F^{-1}(x)$ and $F^{-1}(x')$ correspond one to one, and isomorphic $y \in F^{-1}(x)$, $y' \in F^{-1}(x')$ lie in corresponding classes Θ_j, Θ'_j for exactly one j.* **The text** argues before the lemma by lifting a morphism $x \to x'$ through Definition 8.5.1 (3) (2017: 8.8). **The formalized proof** does the same: every $y \in F^{-1}(x)$ is the source of a lift of $\varphi : x \to x'$, whose target lies in $F^{-1}(x')$ and is isomorphic to y `Chapter8/ProperCovering · EpFunctor.IsProperCovering.exists_isomorphic_of_isomorphic`. So the index sets in $|Y|$ agree `Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberIndex_eq`, and with the classes indexed by their images in $|Y|$ the unique common index of y and y' is their common image. As a bijection between the sets of classes the correspondence is `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_6_2_equiv` (Findings, Chapter 8, items 71, 73) (2017 p. 287).

Lemma 8.6.3 (2017: 8.11) (V8.41) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_6_3`. *For isomorphic x, x', $\#\Theta_j = \#\Theta'_j$ and $\#F^{-1}(x) = \#F^{-1}(x')$.* As in the text: $\#\Theta_j = \#G_x / \#G_{y_0}$ by Lemma 8.6.1 (2017: 8.9), $\#G_x = \#G_{x'}$, and $\#G_{y_0} = \#G_{y'_0}$ for $y_0 \in \Theta_j$, $y'_0 \in \Theta'_j$, isomorphic points having conjugate isotropy groups; the division is by $\#G_{y_0} \geq 1$. For an index outside both index sets both classes are empty. The fibre is the disjoint union of its classes `Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_fiber_eq_sum` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_fiber_eq_of_isomorphic`. Possible misprints: after Lemma 8.6.2 (2017: 8.10), "the third claim of Lemma 8.10" for that of Lemma 8.6.1 (2017: 8.9), which has the claim $\#G_x = \#G_{y_0} \cdot \#\Theta_j$; and in the lemma, $\sum_{i=1}^{q} \#\Theta_j$ sums over i with summands indexed by j (Findings, Chapter 8, item 73) (2017 p. 287).

VIII.24 Proposition 8.6.4 (2017: 8.4): the number of sheets

Proposition 8.6.4 (2017: 8.4) (V8.42) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.proposition_8_6_4`. *$\#F^{-1}(x)$ is constant on the preimage of every connected component of $|X|$, and $x \mapsto \#F^{-1}(x)$ is an sc-smooth functor $X \to \mathbb{N}$.* **The text** argues that under small perturbations of x "the structure of isomorphism classes stays the same". **The formalized proof** uses only the local constancy of $\#F^{-1}(x)$. *Step 1.* For $x' \in U(x)$, Definition 8.5.1 (2) (2017: 8.8) gives the bijection $y \mapsto (F|U(y))^{-1}(x')$ from $F^{-1}(x)$ onto $F^{-1}(x')$, the $U(y)$ being disjoint and covering $F^{-1}(U(x))$; so $\#F^{-1}$ is locally constant on X `Chapter8/ProperCovering · EpFunctor.IsProperCovering.eventually_ncard_fiber_eq` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.isLocallyConstant_ncard_fiber`. *Step 2.* By Lemma 8.6.3 (2017: 8.11) it descends to $|X|$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberCardOrbit`, where it is locally constant in the quotient topology, a level set upstairs being open `Chapter8/ProperCovering · EpFunctor.IsProperCovering.isLocallyConstant_fiberCardOrbit`, hence constant on connected components. *Step 3, the functor.* $\mathbb{N}$ is Lean's, with 0, as the trivial ep-groupoid of the discrete M-polyfold `Chapter7/EpGroupoid · EpGroupoid.trivial` `Chapter7/Discrete · discretePolyfold`; a morphism φ goes to the unit of $\#F^{-1}(s\varphi) = \#F^{-1}(t\varphi)$, and the object map is sc-smooth because it is locally constant `Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberCardFunctor`. The values are ≥ 1, F being surjective, which matches the text's $\mathbb{N} = \{1, 2, \dots\}$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.one_le_ncard_fiber` (Findings, Chapter 8, item 74) (2017

p. 288).

VIII.25 Appendix §8.6.2 (2017 §8.5.2): the structure of strong bundle coverings

§8.6.2 (2017 §8.5.2) (V8.43) `Chapter8/EpStrongBundle · EpStrongBundle.section_8_5_2`. *For a proper strong bundle covering functor A over F, $A(\varphi, w) = (F(\varphi), A_{s(\varphi)}(w))$ with topological linear isomorphisms $A_y : W_y \to V_{F(y)}$, and $\mu_X(F(\varphi)) \circ A_{s(\varphi)} = A_{t(\varphi)} \circ \mu_Y(\varphi)$: $y \mapsto A_y$ is a natural transformation $\mu_Y \to \mu_X \circ F$ of functors into* Ban. **The text** writes the morphism map in this form and computes $t(A(\varphi, w)) = A(t(\varphi, w))$. **The formalized proof** does the same. *Step 1.* The morphism map has the form $(\varphi, w) \mapsto (F(\varphi), A(w))$, as for every strong bundle functor (§VIII.13) `Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering.mor_val`, and the fibre maps A_y are topological linear isomorphisms by Definition 8.5.9 (1) (2017: 8.10), the local strong bundle isomorphisms having linear inverses `Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering.fiberEquiv`. *Step 2.* A strong bundle functor commutes with the targets, $\nu(F(\varphi), A(w)) = A(\mu(\varphi, w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.μ_comm`, which is the text's computation. The natural transformation is stated as its naturality identity; the fibres $Q^{-1}(F(s\varphi))$ and $Q^{-1}(s(F\varphi))$ are identified through their values in V, and the category Ban is not formalized. Only the strong bundle functor part of Definition 8.5.9 (2017: 8.10) enters the identity (Findings, Chapter 8, item 75) (2017 pp. 288–289).

VIII.26 Witnesses and counterexamples

Every hypothesis of the chapter is shown satisfiable, and each statement that does not hold as we read it has a counterexample (Findings, Chapter 8, items 100–105, 125, 130).

Covering functors (Definition 8.5.5 (2017: 8.9)). The identity of `SingleObj G` is a covering functor for every group G `Chapter8/NonVacuity · CoveringExamples.isCoveringFunctor_id`; for non-trivial G the functor sending every morphism to 1 is not `Chapter8/NonVacuity · CoveringExamples.not_isCoveringFunctor_one`; the empty groupoid gives a covering functor, with a geometric lift for which Proposition 8.5.7 (2017: 8.3) does not hold as we read it `Chapter8/GeometricLiftEmpty2021 · CoveringExamples.not_proposition_8_5_7_empty` (§§VIII.20, VIII.40). Theorem 8.5.8 (2017: 8.6) applies to the identity functor of every ep-groupoid in every universe `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_8_id`.

A geometric lift, and Proposition 8.5.7 (2017: 8.3) applied. $\mathbb{Z}/2 = \mathbb{Z}^\times$ acting on $\mathbb{R}$ `Chapter8/NonVacuity · CoveringExamples.rCat` by $y \mapsto \pm y$ `Chapter8/NonVacuity · CoveringExamples.reflectionDiffeo` `Chapter8/NonVacuity · CoveringExamples.reflectionHom` is a geometric lift of the identity of `SingleObj` $\mathbb{Z}^\times$, with $L_D = L_C$ and $\tau = 1$ `Chapter8/NonVacuity · CoveringExamples.reflectionLift`. Proposition 8.5.7 (2017: 8.3) makes it a proper covering functor onto the translation groupoid $\mathbb{Z}/2 \ltimes \mathbb{R}$, with non-trivial isotropy at 0 `Chapter8/NonVacuity · CoveringExamples.isProperCovering_reflectionLift`.

Identities and trivial strong bundles. The identity functor of every ep-groupoid is a proper covering functor `Chapter8/NonVacuityBundle · EpFunctor.isScDiffeoOn_fibredMap_id` `Chapter8/NonVacuityBundle · EpFunctor.isProperCovering_id`. Every strong bundle over an M-polyfold is a strong bundle over its trivial ep-groupoid with $\mu(z, w) = w$ (Definition 8.3.1 (2017: 8.4)) `Chapter8/NonVacuityBundle · EpStrongBundle.trivial`; the identity is a strong bundle map and a strong bundle functor (Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6)) `Chapter8/NonVacuityBundle · EpStrongBundle.isMap_id` `Chapter8/NonVacuityBundle · EpStrongBundle.`

isStrongBundleFunctor_id, and the identity of every strong bundle over an ep-groupoid is a proper strong bundle covering functor Chapter8/NonVacuityBundle · StrongBundle.isStrongBundleIsoOn_id Chapter8/NonVacuityBundle · EpStrongBundle.isStrongBundleIso_fibredLift_id Chapter8/NonVacuityBundle · EpStrongBundle.isProperStrongBundleCovering_id. Over a trivial ep-groupoid every section is functorial Chapter8/NonVacuityBundle · EpStrongBundle.isFunctorialSection_trivial, so Γ, Γ^+ and the sc-Fredholm section functors are Part I's sc-smooth, sc$^+$- and sc-Fredholm sections Chapter8/NonVacuityBundle · EpStrongBundle.mem_sectionFunctors_trivial Chapter8/NonVacuityBundle · EpStrongBundle.mem_plusSectionFunctors_trivial Chapter8/NonVacuityBundle · EpStrongBundle.mem_fredholmSectionFunctors_trivial. The witnesses use that maps of pull-backs $(z, y) \mapsto (g(z), \Phi(y))$ are strong bundle maps Chapter8/NonVacuityBundle · pullbackMap Chapter8/NonVacuityBundle · StrongBundle.isStrongBundleMap_pullbackMap.

A strong bundle over an ep-groupoid in Type. Part I's trivial strong bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ Chapter3/NonVacuity · TrivialBundle.bundle over the trivial ep-groupoid of $\mathbb{R}^n$ Chapter8/NonVacuityBundle · TrivialBundle.epBundle has an ep-groupoid as functorial picture Chapter8/NonVacuityBundle · TrivialBundle.epBundleEpGroupoid, its identity is a proper strong bundle covering functor Chapter8/NonVacuityBundle · TrivialBundle.epBundle_isMap_id Chapter8/NonVacuityBundle · TrivialBundle.epBundle_isProperStrongBundleCovering, $0 \in \Gamma^+$ Chapter8/NonVacuityBundle · TrivialBundle.fiberZero_mem_plusSectionFunctors_epBundle, and for a smooth $\phi : \mathbb{R}^n \to \mathbb{R}^k$ the graph section lies in Γ and is sc-Fredholm Chapter8/NonVacuityBundle · TrivialBundle.graphSection_mem_sectionFunctors_epBundle Chapter8/NonVacuityBundle · TrivialBundle.graphSection_mem_fredholmSectionFunctors_epBundle, from Part I's results on graph sections Chapter3/NonVacuity · TrivialBundle.isScSmoothSection_graphSection.

Tangent witnesses. The tangent of the trivial ep-groupoid of $\mathbb{R}^n$ is tame Chapter8/TangentTame · EpGroupoid.isTame_tangentEpGroupoid_trivial_rn (Lemma 8.1.9 (2017: 8.8); the trivial ep-groupoid itself is tame by Chapter 7 Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn), and the tangent of the trivial ep-groupoid of any M-polyfold has a paracompact orbit space Chapter8/TangentOrbit · EpGroupoid.hasParacompactOrbitSpace_tangent_trivial (Theorem 8.1.15 (2017: 8.4); its hypothesis is Chapter 7's Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial).

Theorem 8.5.3 (5) (2017: 8.5) as printed. For the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$ (§VII.30) Chapter7/NonVacuity · EpGroupoid.Examples.reflectionEpGroupoid, every chart of Theorem 8.5.3 (2017: 8.5) at 0 has, in the sheet of $\psi_0 = (-1, 0)$, a morphism ψ with $F(s(\psi)) \neq t(F(\psi))$ Chapter8/NonVacuity · CoveringExamples.theorem_8_5_3_five_as_printed, and such charts exist Chapter8/NonVacuity · CoveringExamples.theorem_8_5_3_five_as_printed_nonvacuous (§VIII.17).

Remark 8.1.10 (2017: 8.1), the bare-map reading. On the trivial $\mathbb{Z}/3$-action every $T\varphi$ is the identity, and $(\varphi, h) \mapsto (T\varphi, Ts(h))$ is not injective Chapter8/TangentEpNotInjective · EpGroupoid.Examples.remark_8_1_10_not_injective (§VIII.6).

Forms. The constant 0-forms are forms on every ep-groupoid Chapter8/EpForms · EpGroupoid.constForm_mem_epForms, and on an ep-groupoid with $s = t$ every form of the object M-polyfold is one Chapter11/EpPoincareLemma · EP.epFormsInftyToObj_surjective_of_src_eq_tgt (§§VIII.35, VIII.37).

What the witnesses do not exercise. "A sufficiently small $U(x)$" is a chart datum, and charts exist inside every open $V \ni x$, so part (c) of Theorem 8.5.8 (2017: 8.6), which quantifies over all charts, is not vacuous; no hypothesis of the form "for all compatible charts" is made in the chapter. But every proper covering functor constructed has one-point fibres $F^{-1}(x)$ (identities, and the functor of the reflection lift, whose D has one object), so the even coverings of Definitions 8.5.1 (2) and 8.5.9 (2) (2017: 8.8 and 8.10) are met with one sheet and the counting of §8.6.1 (2017 §8.5.1) only with $q = 1$, $\#\Theta_1 = 1$; no proper strong

bundle covering functor other than identities is constructed, and Γ^+ is witnessed by the zero section only (Findings, Chapter 8, item 105).

VIII.27 What is not formalized

Every gap of the chapter is closed (G.74, G.75, G.86–G.89, §§VIII.34–VIII.40). What remains:

- **The degeneracy remark for non-tame X, Y** (pp. 379–385 (2017 pp. 275–276); row V8.28, narrower): only $d_{\mathbf{X}\times Y} \leq 2\, d_{\mathbf{Y}}$ along the image of L_F; equality without tameness is open in the project, Part I proving the product formula only for tame M-polyfolds (§VIII.15; Findings, Chapter 8, item 51).
- **The statements that do not hold as we read them**, each with a counterexample: Theorem 8.5.3 (5) (2017: 8.5) as printed, where the square with s on both vertical arrows and the companion square with t on both are proved (§VIII.17; Findings, Chapter 8, item 54); Proposition 8.5.7 (2017: 8.3) for $D = \emptyset$ (§§VIII.20, VIII.40; Findings, Chapter 8, items 59, 130); and Remark 8.1.10 (2017: 8.1) in the bare-map reading (§VIII.6; Findings, Chapter 8, item 10).
- **The local bundle model in the proof of Theorem 8.1.15 (2017: 8.4)** (2017 p. 268): the sc-isomorphism of $TU(x_z)$ with $TO \to O^1$ and the linearity of the bundle action of G_{x_z}; not needed by the proof here (§VIII.9; Findings, Chapter 8, items 21, 113).
- $W[1]$ **as an ep-groupoid**: Proposition 8.4.1 (3) needs only its orbit space, which is built directly (§VIII.39; Findings, Chapter 8, item 133).
- **Remarks without formal content**: the naturality remarks of Remark 8.5.4 (2017: 8.2) (b), strong bundle maps over open subsets of the base, the closing sentence of Definition 8.5.9 (2017: 8.10), "the structure of isomorphism classes stays the same" before Proposition 8.6.4 (2017: 8.4), and the category Ban of §8.6.2 (2017 §8.5.2); none is used elsewhere in the chapter, so none is given a gap (Findings, Chapter 8, item 113).

VIII.28 What the formalization taught us

Fibre injectivity replaces calculus on fibred products. The text proves the groupoid laws of $T(X, \mathbf{X})$, Theorem 8.1.11 (2017: 8.2) and Proposition 8.1.14 (2017: 8.1) by differentiating identities on $\mathbf{X}\,{}_s\!\times_t \mathbf{X}$, which needs the tangent of a fibred product. Since s is a local sc-diffeomorphism, a vector of $T\mathbf{X}$ is determined by its base point and its image under Ts; base points are computed in $\mathbf{X}$ and images under Ts by Lemma 8.1.3 (2017: 8.2). Only one identity of §8.1, the target formula of Lemma 8.1.3 (2017: 8.2), is genuinely differentiated, and the identification $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$ is never needed (§§VIII.3, VIII.7, VIII.8).

The tangent calculus of §8.1 lives on open sets. Every proof of §8.1 differentiates a map defined only near a point: a local inverse of s, the map α of Lemma 8.1.3 (2017: 8.2), $\widehat{\psi}$. Part I differentiates only globally sc-smooth maps. Tangents of maps on open sets, with the chain rule and the tangents of local sc-diffeomorphisms, are the infrastructure the chapter rests on, and Lemma 8.1.7's (2017: 8.6) "by construction" is exactly their last property (§VIII.1).

Definition 8.5.1 (3) (2017: 8.8) carries Theorem 8.5.3 (2017: 8.5). The text builds the sheets $\mathbf{U}(\psi_0)$ from natural representations of the groups G_y in Y and never invokes condition (3). With (3), the sheets are images of the continuous sections $z \mapsto L_F^{-1}(\Gamma(g, Fz), z)$ of s, and connectedness of $U(y)$ does the rest; neither the finiteness of the fibre nor condition (1) is used. The same condition counts the morphisms

over a fibre in §8.6.1 (2017 §8.5.1) and gives the unique lifting behind Theorem 8.5.8 (2017: 8.6) (§§VIII.17, VIII.21, VIII.23).

A functor commutes with s, not with "s on one side and t on the other". Theorem 8.5.3 (5) (2017: 8.5) as printed equates $F(s(\psi))$ with $t(F(\psi))$; a non-trivial element of G_x moves points, so the square fails for the reflection of $\mathbb{R}$ in every chart at 0. The two squares that functoriality gives, with s on both sides and with t on both sides, are what the proofs downstream use (§VIII.17).

Empty groupoids and non-tame products. Two statements needed a hypothesis that the text does not state: Proposition 8.5.7 (2017: 8.3) needs $D \neq \emptyset$, because Definition 8.5.5 (2017: 8.9) admits the empty groupoid, over which the functor of Proposition 8.5.7 (2017: 8.3) is not surjective; and the degeneracy remark after Definition 8.5.1 (2017: 8.8) needs tameness, the only form of the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$ that Part I proves. In both cases the hypothesis holds where the text uses the statement (§§VIII.15, VIII.20).

Metrizability is the shorter road to paracompactness, again. As for Proposition 7.5.3 (2017: 7.12), Theorem 8.1.15 (2017: 8.4) follows from Smirnov's theorem pulled back along $|TX| \to |X^1|$: $|TX|$ is metrizable, which is stronger than paracompact, and the closed paracompact sets $\tilde{A}_z$, the regularity of $|TX|$ and Proposition 2.7.4 (2017: 2.17) drop out. What remains of the text's argument is the metrizability of $|TU(x_z)|$, and for it the open embedding $G_x \backslash TU(x) \to |TX|$, which the text leaves implicit (§VIII.9).

Strong bundles over ep-groupoids need more of Part I than Part I states. A global notion of strong bundle map, the atlas and universal property of the pull-back on both levels $[0]$ and $[1]$, the sum of sc-smooth sections, and the strong bundle structure on $\mathbf{V}_{s}\times_{A} W$ are all used in §8.3 and Definition 8.5.9 (2017: 8.10) without being defined or proved in Part I. With them in place, the claims of §8.3 that the text states without proof, the functorial picture and the identity of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6), are routine (§§VIII.10–VIII.14, VIII.22).

Compatibility with morphisms is $s^*\omega = t^*\omega$. Definition 8.2.1 asks $(T\varphi)^*\omega_y = \omega_x$ for every morphism. Since $T\varphi = Tt \circ (Ts|T_{\varphi}\mathbf{X})^{-1}$, this is the equality of the pull-backs of ω by the two global sc-smooth maps $s, t : \mathbf{X} \to X$. In that form every closure property of §8.2 — under d, the wedge product, restriction to X^1, pull-back by functors — is one line, from the naturality of pull-backs in Part I; the text's local maps $t \circ (s|\mathbf{U}(\varphi_0))^{-1}$ are not needed (§VIII.35).

The Poincaré lemma on an ep-groupoid is a statement about equivalences. The text averages a primitive over G_x and extends it to the saturation by morphisms. Both steps are instances of general facts: averaging works for any finite family of self-functors covering the morphisms of a groupoid, and the extension is the push-forward of forms along the equivalence $G_x \ltimes U(x) \to X|\pi^{-1}(\pi(U(x)))$ of §11.2. So the proof of Proposition 8.2.3 rests on Chapter 11 (§VIII.37).

Vertical vectors are the derivative the text takes. Proposition 8.1.16 differentiates (8.16) and uses that $T\tau_X$ is onto. On the vertical vectors of $T(TX)$ the tangent of an isotropy element of TX acts as Tg, and that suffices; no surjectivity is used (§VIII.38).

VIII.29 The 2021 additions to §8.1: Propositions 8.1.16 and 8.1.17, the Zhou condition and LUC for TX

The Springer edition (2021) adds Propositions 8.1.16, 8.1.17, Definition 8.3.5, §8.4, Example 8.5.6, condition (0) of Definition 8.5.9 and Lemma 8.5.10; Sections VIII.29–VIII.33 retell them, with the 2021 numbers

and pages (V8.44–V8.52). The conditions of Chapter 7's §7.4 are those of §§VII.35–VII.39.

Proposition 8.1.16 (V8.44) `Chapter8/TangentZhou · EpGroupoid.proposition_8_1_16`. *If X satisfies (ZHOU), so does TX.* **The text's proof**: an isotropy element of a smooth $b \in TX$ over x is a pair (g,h), $g \in G_x$; differentiating (8.16) and using that $T\tau$ is onto, $T(g,h) = \mathrm{id}$ gives $Tg = \mathrm{id}$, so $g \in T_x$ by (ZHOU), and then $(g,h) \in T_b$. **The formalized proof.** *Step 1.* A smooth point of TX lies over a smooth point of X `Chapter8/TangentZhou · MPolyfold.levelVal_tangentProj_mem_smoothSet`, and the base $p(k)$ of an isotropy element k of b is an isotropy element of $p(b)$. *Step 2.* The step "$T(g,h) = \mathrm{id}$ implies $Tg = \mathrm{id}$" needs the second tangent bundle $T(TX)$ and $T\tau$, which are not formalized; it was the hypothesis `Chapter8/TangentZhou · EpGroupoid.TangentZhouReflect` in the first reading, and holds for every ep-groupoid by §VIII.38 (gap G.88, closed). *Step 3.* (ZHOU) for X puts $p(k)$ in $\mathbf{X}^T$, and if $s = t$ near $p(v)$ then $Ts = Tt$ near v, so $k \in (T\mathbf{X})^T$ `Chapter8/TangentZhou · EpGroupoid.mem_nonEffectivePart_tangent`. As in Chapter 7, Tg is a set map on the fibre, no linear structure is used (§VII.37). For every trivial ep-groupoid the tangent of a morphism is the identity and the hypothesis holds `Chapter8/TangentZhou · EpGroupoid.tangentMor_trivial Chapter8/TangentZhou · EpGroupoid.tangentZhouReflect_trivial Chapter8/TangentZhou · EpGroupoid.Examples.proposition_8_1_16_point` (Findings, Chapter 8, item 115) (p. 369).

Proposition 8.1.17 (V8.45) `Chapter8/TangentZhou · EpGroupoid.proposition_8_1_17_one`. *(1) If X^1 has LUC, so has TX, with $(T\mathbf{X})^T = p^{-1}((\mathbf{X}^1)^T)$. (2) (SLUC) for X gives (SLUC) for TX. (1).* If $v \in (T\mathbf{X})^T$, then $Ts = Tt$ on a neighbourhood O of v, and $s = t$ on the open set $p(O)$, p being open: $p(v) \in (\mathbf{X}^1)^T$. Conversely, $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ by Appendix 7.6.4 (§VII.36), and Step 3 of Proposition 8.1.16 applies `Chapter8/TangentZhou · EpGroupoid.nonEffectivePart_tangent_eq`. So $(T\mathbf{X})^T$ is closed as a preimage of the closed $(\mathbf{X}^1)^T$ (Proposition 7.2.21 for X^1), and TX has LUC by Proposition 7.2.21 for TX. The hypotheses are those of Appendix 7.6.4 — $U(x) \cap X_1$ connected in X^1 and dense in $U(x)$ (the density is gap G.73) — and the openness of $p : T\mathbf{X} \to \mathbf{X}^1$ (gap G.8); both now hold for every ep-groupoid (§VII.45), and the proposition without them is §VIII.40. They are satisfied by the point `Chapter8/TangentZhou · EpGroupoid.isOpenMap_tangentProj_of_subsingleton Chapter8/TangentZhou · EpGroupoid.Examples.proposition_8_1_17_one_point`. *(2)* for $(TX)^0 = TX$ is (1) `Chapter8/TangentZhou · EpGroupoid.proposition_8_1_17_two`; for every $(TX)^i$ it needs $(TX)^i = T(X^i)$, which is §VIII.33 (gap G.88, closed) (Findings, Chapter 8, items 116, 124, 126) (p. 371).

VIII.30 Definition 8.3.5, Proposition 8.4.1 and Definition 8.4.2: auxiliary norms, paracompactness, local regularity

Definition 8.3.5 (V8.46) `Chapter8/Book2021 · EpStrongBundle.IsAuxNorm`. *An auxiliary norm for a strong bundle over an ep-groupoid is a continuous $N : W_{0,1} \to [0,\infty)$, a complete norm on every fibre, with $h_k \to 0_x$ whenever $P(h_k) \to x$ and $N(h_k) \to 0$, and invariant: $N(h) = N(k)$ if $h = \mu(\varphi, k)$.* The first three conditions are Part I's auxiliary norm (Definition 5.1.1) on the underlying strong bundle, so N lives on its level-one space; values in $[0,\infty)$ follow from (1). In (3), $h \in W_{0,1}$ is given and $h = \mu(\varphi,k)$ is an equation in W; it is not vacuous, since $\mu(\varphi,k) \in W_{0,1}$ for $k \in W_{0,1}$ `Chapter8/Book2021 · EpStrongBundle.exists_oneVal_eq_μ`. Over the trivial ep-groupoid every auxiliary norm of Part I is one `Chapter8/Book2021 · EpStrongBundle.exists_isAuxNorm_trivial`, and $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ has one `Chapter8/Book2021 · TrivialBundle.exists_isAuxNorm_epBundle` (Findings, Chapter 8, item 117) (p. 379).

Proposition 8.4.1 (V8.47) `Chapter8/Book2021 · EpStrongBundle.proposition_8_4_1_two_imp_one`. *$|X|$, $|W[0]|$ and $|W[1]|$ are paracompact together. (2) □ (1).* $\mu(\varphi, 0_{s(\varphi)}) = 0_{t(\varphi)}$ `Chapter8/Book2021 · EpStrongBundle.μ_fiberZero`, so the zero section induces $|0| : |X| \to |W[0]|$ `Chapter8/Book2021 · EpStrongBundle.zeroOrbitMap`, continuous `Chapter8/Book2021 · EpStrongBundle.continuous_zeroOrbitMap`, with left inverse $|P|$

Chapter8/Book2021 · EpStrongBundle.leftInverse_zeroOrbitMap, hence a topological embedding Chapter8/Book2021 · EpStrongBundle.isEmbedding_zeroOrbitMap. The text concludes as a closed subset of a paracompact space; the formalized proof goes through metrizability: $|W[0]|$ paracompact is metrizable by Theorem 7.3.1 and Smirnov's theorem (§VII.23), so is its subspace $|X|$, hence paracompact. (1) □ (2) and the statements on $|W[1]|$ are §VIII.39, without bundle charts (gap G.86, closed) (Findings, Chapter 8, items 118, 127, 133) (p. 380).

Definition 8.4.2 (V8.48) Chapter8/TangentZhou · EpStrongBundle.IsLR. *The strong bundle has the local regularity property if every x has arbitrarily small $U(x)$ such that every strong bundle isomorphism B of $W|U(x)$ with $|B(w)| = |w|$ is $w \mapsto g * w$ for some $g \in G_x$,* the natural lift $g * w = \mu(\Gamma(g, P(w)), w)$ Chapter8/TangentZhou · EpStrongBundle.natLift. As for Definition 8.5.9 (§VIII.22), the strong bundle isomorphism is read as a strong bundle map $W \to W$ that is Part I's strong bundle isomorphism on $W|U(x)$, "arbitrarily small" as "inside every open $O \ni x$", and $|B(w)| = |w|$ as equality of orbits in $(W, \mathbf{W})$, as for (LR) (§VII.39). Every strong bundle over a trivial ep-groupoid has it Chapter8/TangentZhou · EpStrongBundle.isLR_trivial. Proposition 8.4.3, effective X with (LR) gives the property, is §VIII.39, with the openness of P from Chapter 10 (Findings, Chapter 8, items 119, 128) (p. 382).

VIII.31 Example 8.5.6, Definition 8.5.9 in the 2021 form, and Lemma 8.5.10

The remark after Definition 8.5.5 and Example 8.5.6 (V8.49) Chapter8/Book2021 · example_8_5_6. *For a covering functor $f : D \to C$, $f : G_d \to G_{f(d)}$ is injective* Chapter8/Book2021 · IsCoveringFunctor.injective_map_end — by the bijectivity required in Definition 8.5.5 — *but need not be surjective*: D with two objects and one morphism between any two Chapter8/Book2021 · Example856D, mapped to $\mathbb{Z}^\times$ as a one-object category with $f(\varphi) = f(\varphi^{-1}) = -1$ Chapter8/Book2021 · example856Functor, is a covering functor whose G_a, G_b are trivial while $G_x \cong \mathbb{Z}/2$ (Findings, Chapter 8, item 120) (p. 388).

Definition 8.5.9, 2021 form (V8.50) Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021. *A strong bundle map $A : W \to V$ over an ep-functor F with $Q \circ A = F \circ P$ is a proper strong bundle covering functor if (0) $\tau(F(\psi), A(w)) = A(\mu(\psi, w))$ and (1)–(3) of the 2017 definition hold.* It is stated for the object map, (0) being the strong bundle map property of Definition 8.3.2 (2017: 8.5); (1)–(3) are those of §VIII.22. The reading of §VIII.22 had taken 2017's "strong bundle map A" as a strong bundle functor, which contains (0); the two are equivalent Chapter8/Book2021 · EpStrongBundle.isProperStrongBundleCovering2021_iff Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering.isProperStrongBundleCovering2021: the functor is determined by its object map, on morphisms $(\psi, w) \mapsto (F(\psi), A(w))$. Identities are witnesses Chapter8/Book2021 · EpStrongBundle.isProperStrongBundleCovering2021_id Chapter8/Book2021 · TrivialBundle.epBundle_isProperStrongBundleCovering2021 (Findings, Chapter 8, items 70, 121) (p. 394).

Lemma 8.5.10 (V8.51) Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.lemma_8_5_10. *$B : W \to F^*V$, $w \mapsto (P(w), A(w))$, is a strong bundle isomorphism over Y for $\mu_{F^*V}(\varphi, (y, v)) = (t(\varphi), \mu_X(F(\varphi), v))$, with (8.23) $B \circ \mu_Y = \mu_{F^*V} \circ B$, and $A = \bar{A} \circ B$.* F^*V is Part I's pull-back, whose paracompactness always holds, and μ_{F^*V} is its formula Chapter8/Book2021 · EpStrongBundle.pullbackμ. *Step 1.* B lands in F^*V Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_val, and (8.23) is (0) Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_μ. *Step 2, bijective.* On the fibre over y, A is the linear isomorphism $A_y : W_y \to V_{F(y)}$ of a local strong bundle isomorphism of Definition 8.5.9 (1) at y, so B is bijective Chapter8/

Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bijective_bundleB. *Step 3, the inverse.* Near every point, $B^{-1}(y,v) = \Psi(v)$ for the inverse Ψ of a local strong bundle isomorphism of Definition 8.5.9 (1), whose base map is injective Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.eq_of_isStrongBundleIsoOn Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv_eq_of, so B and B^{-1} are strong bundle maps Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleB Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.mapsTo_oneSet_bundleBInv Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleBInv. "Strong bundle isomorphism over the ep-groupoid Y" is read as a strong bundle isomorphism $W \to F^*V$ covering the identity, with (8.23); that μ_{F^*V} makes F^*V a strong bundle over the ep-groupoid Y, and B an isomorphism of strong bundles over Y, is §VIII.40 (gap G.89, closed) (Findings, Chapter 8, items 122, 129) (p. 395).

VIII.32 The 2021 additions: what is not formalized, and what they taught us

Every gap of the 2021 additions is closed:

- **Proposition 8.1.16 without `TangentZhouReflect`** (gap G.88, closed): through the vertical vectors of $T(TX)$ (§VIII.38; Findings, Chapter 8, items 115, 126). Proposition 8.1.17 (2) for every $(TX)^i$ is §VIII.33; of $(TX)^i = T(X^i)$ the compatibility with unit, inverse and composition is not formalized, since LUC does not use it (Findings, Chapter 8, item 124).
- **Proposition 8.2.3** (p. 373; gap G.87, closed): the Poincaré lemma, as printed, with the forms of §8.2 (gap G.74, closed) (§§VIII.35–VIII.37; Findings, Chapter 8, items 110, 123, 132).
- **Proposition 8.4.1 and Proposition 8.4.3** (gap G.86, closed): (1) □ (2) □ (3), with $|W[1]|$ built as an orbit space, and the local regularity of strong bundles (§VIII.39; Findings, Chapter 8, items 118, 119, 127, 128, 133).
- **F^*V as a strong bundle over the ep-groupoid Y** in Lemma 8.5.10 (gap G.89, closed) (§VIII.40; Findings, Chapter 8, items 122, 129).

The tangent inherits the Chapter 7 conditions through the base. Both propositions of §8.1 reduce to Chapter 7: a vector of $T\mathbf{X}$ is in $(T\mathbf{X})^T$ as soon as its base is in $(\mathbf{X}^1)^T$, and the openness of p gives the converse. The passage from $T(g,h)$ to Tg needs the second tangent bundle; it goes through the vertical vectors of $T(TX)$ (§VIII.38).

Condition (0) was already in the reading. The 2021 Definition 8.5.9 adds (0); the first reading had read 2017's "strong bundle map" as a strong bundle functor, which contains it, and the two forms are equivalent (§VIII.31).

VIII.33 Proposition 8.1.17 (2) for every $(TX)^i$

Equivalences of ep-groupoids (V8.52) Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv. The text's "naturally $(TX)^i = T(X^i)$" is used only to carry LUC from $T(X^i)$ to $(TX)^i$. For that it suffices to have sc-diffeomorphisms of the object and of the morphism M-polyfolds commuting with source and target; an sc-diffeomorphism is a homeomorphism Chapter8/TangentZhouLevels · MPolyfold.ScEquiv.toHomeomorph. Such equivalences compose and contain the identities Chapter8/TangentZhouLevels · EpGroupoid.

EpEquiv.refl Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.trans, pass to the index raises Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.shift, map $\mathbf{X}^T$ onto $\mathbf{X}'^T$, since $s = t$ near a morphism is preserved Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.nonEffectivePart_eq, and so transport LUC, by the characterization of Proposition 7.2.21 ($\mathbf{X}^T$ closed) Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.isLUC. The compatibility with unit, inverse and composition is not formalized, since LUC does not use it.

$T(X^i) \cong (TX)^i$ (V8.52) Chapter8/TangentZhouLevels · EP.tangentLiftEpEquiv. Part I's Proposition 2.3.17 for $i = 1$, on the object and the morphism M-polyfold, gives an equivalence $T(X^1) \cong (TX)^1$, commuting with s and t by the naturality of the inclusion $T(X^1) \to TX$ Chapter8/TangentZhouLevels · EpGroupoid.tangentShiftEpEquiv. By recursion, composing with the index raise of the previous equivalence, $T(X^i) \cong (TX)^i$ for every i, for the bundled tangent ep-groupoid Chapter8/TangentZhouLevels · EP.tangentEP, as Part I's Φ_i for M-polyfolds Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv Chapter2/TangentLiftAll · proposition_2_3_17_all.

Proposition 8.1.17 (2) (V8.52) Chapter8/TangentZhouLevels · EP.proposition_8_1_17_two_all. *If X has (SLUC), so has TX.* **The text's proof**: (SLUC) gives LUC for every $X^{i+1} = (X^i)^1$; by (1) every $T(X^i)$ has LUC; and $(TX)^i = T(X^i)$. **The formalized proof** is the same: (1) applied to X^i Chapter8/TangentZhouLevels · EP.isLUC_tangent_lift, then the equivalence. The hypotheses of (1) — natural representations with $U(x) \cap (X^i)_1$ connected in $(X^i)^1$ and dense in $U(x)$ (gap G.73), and the openness of $p : T(\mathbf{X}^i) \to (\mathbf{X}^i)^1$ (gap G.8) — are taken on every X^i; both hold for every ep-groupoid (§VII.45), and the proposition without them is §VIII.40 Chapter7/SmoothRegularAssembly · EP.isLUC_tangent_lift_of_isSLUC. They hold for the point ep-groupoid and all its lifts Chapter8/TangentZhouLevels · EpGroupoid.hasConnectedDenseLevelOneReps_of_subsingleton Chapter8/TangentZhouLevels · EpGroupoid.Examples.proposition_8_1_17_two_all_pointEP (gap G.88, closed for (2)) (Findings, Chapter 8, item 124) (p. 371).

VIII.34 Lemma 8.1.2 (2017: 8.1) as printed, and the M-polyfold of pairs of Remark 8.1.10 (2017: 8.1)

§VIII.2 proved Lemma 8.1.2 in the form the construction of $T\mathbf{X}$ uses: the composition (8.5) is defined through the local sc-diffeomorphisms Ts, Tt, and $T\mathbf{X}$ is an ep-groupoid. The text states more: under the identification of $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X})$ with $T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$, the multiplication of $T\mathbf{X}$ is Tm. This is now formalized as printed (V8.4; gap G.75, closed).

The M-polyfold $\mathbf{X}\,{}_s\!\times_t \mathbf{X}$ Chapter8/TangentFibred · EpGroupoid.compPolyfold is the sub-M-polyfold of $\mathbf{X} \times \mathbf{X}$ of Definition 7.1.3 (4); its second projection π_2 is a local sc-diffeomorphism, since s is one Chapter8/TangentFibred · EpGroupoid.isLocalScDiffeo_compSnd, and the multiplication m is sc-smooth by condition (4) Chapter8/TangentFibred · EpGroupoid.compMul.

The identification Chapter8/TangentFibred · EpGroupoid.tangentCompPairs is $v \mapsto (T\pi_1 v, T\pi_2 v)$. *Step 1.* As π_2 is a local sc-diffeomorphism, $T\pi_2$ is bijective on the fibres, and a vector v of $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X})$ is determined by its base point and $T\pi_2 v$. *Step 2.* $T\pi_1(v)$ is the vector over π_1 with $Ts(T\pi_1 v) = Tt(T\pi_2 v)$, by the chain rule for $s \circ \pi_1 = t \circ \pi_2$. So the identification is injective with range the pairs (k, h) with $Ts(k) = Tt(h)$ Chapter8/TangentFibred · EpGroupoid.range_tangentCompPairs. *Step 3, Tm is the composition.* $Tm(v)$ lies over $m(\pi_1 v, \pi_2 v)$ and $Ts(Tm\,v) = Ts(T\pi_2 v)$, by the chain rule for $s \circ m = s \circ \pi_2$; since Ts is injective on the fibres of $\mathbf{X}$, this characterizes the composition (8.5): $Tm(v) = T\pi_1(v) \circ T\pi_2(v)$ Chapter8/TangentFibred · EpGroupoid.lemma_8_1_2_tangentMul. In the text's words: for $Ts(k) = Tt(h)$ there is exactly one v over (k, h), and $k \circ h = Tm(v)$, a vector over $p(k) \circ p(h)$ with $Ts(k \circ h) = Ts(h)$ Chapter8/

TangentFibred · EpGroupoid.lemma_8_1_2_book; the multiplication of the tangent ep-groupoid on its fibred product is Tm transported by the identification Chapter8/TangentFibred · EpGroupoid.lemma_8_1_2_compMul. *Step 4, as M-polyfolds.* The identification is an sc-diffeomorphism onto the sub-M-polyfold $T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$ of $T\mathbf{X}\times T\mathbf{X}$ Chapter8/TangentFibred · EpGroupoid.isScDiffeoOn_tangentCompPairs: it is a bijection which, composed with the local sc-diffeomorphism $T\pi_2$ on the right side, gives the local sc-diffeomorphism $T\pi_2$ on the left, and such a map is a local sc-diffeomorphism Chapter8/TangentFibred · MPolyfold.isLocalScDiffeo_of_comp_eq. **A reading.** The text derives $Ts(k\circ h) = Ts(h)$ by differentiating $s(m(\psi,\varphi)) = s(\varphi)$; the proof above uses the same identity, in the form $s\circ m = s\circ\pi_2$ on the M-polyfold of pairs (p. 359).

Remark 8.1.10, the M-polyfold of pairs (V8.12). §VIII.6 formalized the bijection of the remark between morphisms of $T\mathbf{X}$ and pairs $(T\varphi, a)$ with the label φ in place of the bare map $T\varphi$ (as a bare map the assignment is not injective, §VIII.6). Here $T\varphi$, defined as $Tt\circ(Ts|T_\varphi\mathbf{X})^{-1}$, is the tangent at $s(\varphi)$ of the local sc-diffeomorphism $\hat\varphi = t\circ(s|\mathbf{U}(\varphi))^{-1}$, as the text has it Chapter9/Tangent · EpGroupoid.tangentMor_eq_tangentMapOn_hat, a bijection $T_{s(\varphi)}X\to T_{t(\varphi)}X$ with inverse $T(\varphi^{-1})$ Chapter7/TangentMorphism · EpGroupoid.bijOn_tangentMor, represented in charts by mutually inverse bounded linear maps Chapter7/TangentMorphism · EpGroupoid.tangentMor_chart_iso. The remark also says that the pairs form an M-polyfold for which the bijection is an sc-diffeomorphism. *The pairs* (φ,a), $\varphi\in\mathbf{X}^1$, $a\in T_{s(\varphi)}X$, form the fibred product $\mathbf{X}^1\,{}_s\!\times_p TX$, a sub-M-polyfold of $\mathbf{X}^1\times TX$ by Proposition 2.5.20, $s:\mathbf{X}^1\to X^1$ being a local sc-diffeomorphism Chapter8/TangentPairs · EpGroupoid.tangentPairsPolyfold. *The map* $G(v) = (p(v), Ts(v))$ Chapter8/TangentPairs · EpGroupoid.remarkGInto is an sc-diffeomorphism $T\mathbf{X}\to\mathbf{X}^1\,{}_s\!\times_p TX$ Chapter8/TangentPairs · EpGroupoid.remark_8_1_10_scDiffeo, by the argument of Step 4: it is bijective and commutes with the local sc-diffeomorphisms $Ts : T\mathbf{X}\to TX$ and $\pi_2 : \mathbf{X}^1\,{}_s\!\times_p TX\to TX$. *The triples.* $v\mapsto (Ts\,v, p(v), Tt\,v)$ Chapter8/TangentPairs · EpGroupoid.remarkTriple is injective, with range the triples (a,φ,b) with $a\in T_{s(\varphi)}X$ and $b = T\varphi(a)$ Chapter8/TangentPairs · EpGroupoid.remark_8_1_10_triples, the text's "$(a, T\varphi, b)$" with φ recorded (pp. 364–365). The row stays narrower only for the reading of $T\varphi$ as a label (Findings, Chapter 8, item 131).

VIII.35 Definition 8.2.1 (2017: 8.2): sc-differential forms on ep-groupoids

§VIII.27 of the first reading recorded §8.2 as not formalized, since Part I's forms (§4.4) were not yet available. With them (Part IV), Definition 8.2.1 is formalized as printed (V8.53; gap G.74, closed).

The condition Chapter8/EpForms · EpGroupoid.IsMorCompatible. *An sc-differential k-form on the ep-groupoid* $(X,\mathbf{X})$ *is an sc-differential k-form* ω *on the object M-polyfold, Definition 4.4.1, with* $(T\varphi)^*\omega_y = \omega_x$ *for every morphism* $\varphi : x\to y$, $T\varphi$ being the bounded linear isomorphism $T_xX\to T_yX$ of Definition 7.1.4 (§VIII.6). The forms satisfying it form the vector space $\Omega^k_{ep}(X)$ Chapter8/EpForms · EpGroupoid.epForms. **The key reformulation**: ω is compatible with the morphisms iff $s^*\omega = t^*\omega$ on the morphism M-polyfold $\mathbf{X}$, the pull-backs being Part I's pull-backs by the sc-smooth maps $s,t:\mathbf{X}\to X$ Chapter8/EpForms · EpGroupoid.isMorCompatible_iff_formPullback; for $T\varphi = Tt\circ(Ts)^{-1}$ on the fibre over φ is the definition of §VIII.6. Every closure property follows from this form. *(a)* A form on X is a form on X^1 Chapter8/EpForms · EpGroupoid.formLower_mem_epForms. *(b)* $d\omega$ is compatible with the morphisms: $s^*d\omega = d\,s^*\omega = d\,t^*\omega = t^*d\omega$, by the naturality of Part I's d; so $d:\Omega^k_{ep}(X)\to\Omega^{k+1}_{ep}(X^1)$ Chapter8/EpForms · EpGroupoid.extD_mem_epForms. The text argues with the local sc-diffeomorphisms $f = t\circ(s|\mathbf{U}(\varphi_0))^{-1}$ and $f^*d = df^*$; the global maps s, t give the same in one line. *(c)* Wedge products Chapter8/EpForms · EpGroupoid.formWedge_mem_epForms. *(d)* For an sc-smooth functor $F : X\to Y$, F^* maps $\Omega^k_{ep}(Y)$ to $\Omega^k_{ep}(X)$, as $s_Y\circ F = F\circ s_X$ Chapter8/

EpForms · EpGroupoid.formPullback_mem_epForms, and naturally equivalent functors (Definition 8.1.13) have the same pull-back: for $\theta : F \to F'$, $F = s \circ \theta$ and $F' = t \circ \theta$, so $F^*\omega = \theta^* s^* \omega = \theta^* t^* \omega = F'^* \omega$ Chapter8/EpForms · EpGroupoid.formPullback_eq_of_natTrans. The constant 0-forms are a witness (§VIII.26).

The direct limit. For the index lifts X^i (an ep-groupoid by Proposition 7.3.9), $\Omega^k_{ep}(X^i)$ Chapter8/EpForms · EP.epForms, with the pull-backs by the inclusions as the maps of a directed system Chapter8/EpForms · EP.epFormsMap, and $\Omega^k_{ep,\infty}(X)$ is its direct limit Chapter8/EpForms · EP.epFormsInfty, Mathlib's direct limit of modules. Every class has a representative on some X^i Chapter8/EpForms · EP.exists_epFormsInftyOf_eq, and two forms on X^i, X^j represent the same class iff their pull-backs to some X^l, $l \geq i, j$, agree Chapter8/EpForms · EP.epFormsInftyOf_eq_iff, as the text says. *The exterior derivative* $d[\omega] = [d\omega]$ Chapter8/EpForms · EP.epFormsInftyD Chapter8/EpForms · EP.epFormsInftyD_of is well defined since d commutes with the steps, and $d \circ d = 0$ from Part I Chapter8/EpForms · EP.epFormsInftyD_comp_epFormsInftyD. **Readings.** "An sc-smooth map $\oplus_k TX \to \mathbb{R}$, linear in each argument and skew-symmetric" is Part I's Definition 4.4.1, as the text says ("already introduced in Section 4.4"), so a form is a family of alternating maps on the fibres; compatibility on level i is with the morphisms of X^i on level one, that is of $\mathbf{X}_{i+1}$; and $T_y Y$ in the definition is read as $T_y X$ (Findings, Chapter 8, items 110, 132) (pp. 372–373).

VIII.36 Definition 8.2.2 (2017: 8.3): the sc-de Rham complex, and pull-backs by functors

The complex (V8.54). $\Omega^*_{ep,\infty}(X) = \oplus_k \Omega^k_{ep,\infty}(X)$ Chapter8/EpForms · EP.epFormsInftyGraded with d Chapter8/EpForms · EP.epFormsInftyGradedD and $d \circ d = 0$ Chapter8/EpForms · EP.epFormsInftyGradedD_comp_self is the sc-de Rham complex; its cohomology $\ker d / \operatorname{im} d$ Chapter8/EpForms · EP.epDeRham is that of the same complex as a cochain complex of real vector spaces Chapter8/EpFormsObj · EP.epDeRhamComplex. *It is a subcomplex of Part I's complex of the object M-polyfold.* The text defines d as "the exterior derivative of an sc-differential form on the object M-polyfold". The inclusion $\Omega^k_{ep,\infty}(X) \to \Omega^k_\infty(\operatorname{Ob} X)$, the object M-polyfold of X^i being the lift $(\operatorname{Ob} X)^i$ of Part I, is injective Chapter8/EpFormsObj · EP.epFormsInftyToObj Chapter8/EpFormsObj · EP.epFormsInftyToObj_injective and commutes with d Chapter8/EpFormsObj · EP.formsInftyD_comp_epFormsInftyToObj (p. 373).

Pull-backs by sc-smooth functors. Used by Proposition 8.2.3 and by §11.2 (Part XI), and stated in Theorem 11.2.1 (1), (2): for $F : X \to Y$ with index raises $F^i : X^i \to Y^i$, $F^*[\omega] = [(F^i)^* \omega]$ Chapter8/EpFormsFunctor · EP.epFormsInftyPullback, well defined since the $(F^i)^*$ commute with the steps; it commutes with d Chapter8/EpFormsFunctor · EP.epFormsInftyD_comp_epFormsInftyPullback, and naturally equivalent functors give the same map, by §VIII.35 (d) on every level, applied to the index raises of the natural transformation Chapter8/EpFormsFunctor · EP.epFormsInftyPullback_eq_of_natTrans. The inclusion into the forms of the object M-polyfold commutes with these pull-backs Chapter11/EpPoincareLemma · EP.epFormsInftyToObj_epFormsInftyPullback.

VIII.37 Proposition 8.2.3 (new in the 2021 edition; the 2017 text promised it for later): the Poincaré lemma

Proposition 8.2.3 (V8.55; gap G.87, closed) Chapter11/EpPoincareLemma · EP.proposition_8_2_3. *Every object x of an ep-groupoid X has saturated open neighbourhoods $V \subseteq U$ such that every closed $[\omega] \in \Omega^k_{ep,\infty}(X|U)$, $k \geq 1$, is $d[\sigma]$ on V for some $[\sigma] \in \Omega^{k-1}_{ep,\infty}(X|V)$.* Here $X|U$ is the restriction of Proposition 7.1.22

Chapter11/SubGroupoidModel · EP.sub, $k \geq 1$ is written $k+1$, and "on V" is the pull-back by the inclusion functor $X|V \to X|U$ Chapter11/SubGroupoidModel · EP.subIncl. The proposition holds also inside every saturated open neighbourhood U given in advance Chapter11/EpPoincareLemma · EP.proposition_8_2_3_of_isOpen; the statement above takes $U = X$. **The text's proof**: Lemma 4.4.8 on the object M-polyfold gives a primitive τ on a neighbourhood $V'(x)$ invariant under G_x; the average $\tau' = \frac{1}{\#G_x}\sum_g g^*\tau$ is invariant with $d\tau' = \omega$, and is extended to the saturation by $\sigma(y) = \psi^*\tau'(x')$ for a morphism $\psi : x' \to y$. **The formalized proof** follows these steps; the extension is the push-forward along an equivalence.

Step 1, the object M-polyfold. Lemma 4.4.8 Chapter4/Poincare2021b · lemma_4_4_8, applied to the image of a in the forms of the object M-polyfold of $X|U$, gives an open $V_0 \ni x$ and a primitive among the forms of the object M-polyfold on V_0 Chapter8/EpPoincare · EP.proposition_8_2_3_object.

Step 2, the translation groupoid. Theorem 7.1.13 Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13 gives a natural representation of G_x on an open $U(x) \subseteq V_0$, and with it the translation ep-groupoid $G_x \ltimes U(x)$, the M-polyfold $U(x)$ with only the unit morphisms, the functor $J : U(x) \to G_x \ltimes U(x)$, the identity on objects Chapter11/TranslationAverage · EpGroupoid.NaturalRep.unitsFunctor, the functors $\Phi_g : y \mapsto g * y$ of $U(x)$ Chapter11/TranslationAverage · EpGroupoid.NaturalRep.smulFunctor and the natural transformations $\theta_g : J \to J \circ \Phi_g$, $y \mapsto (g, y)$ Chapter11/TranslationAverage · EpGroupoid.NaturalRep.smulNatTrans. In an ep-groupoid with $s = t$, such as $U(x)$ with its units, every form of the object M-polyfold is a form of the ep-groupoid Chapter11/EpPoincareLemma · EP.epFormsInftyToObj_surjective_of_src_eq_tgt, so the primitive of Step 1, restricted to $U(x)$, is a $\beta \in \Omega^k_{ep,\infty}(U(x))$ with $d\beta = J^*c$, c the pull-back of a to $G_x \ltimes U(x)$; β need not be invariant.

Step 3, averaging. This is done for any $J : W \to T$ which is a bijective local sc-diffeomorphism on objects, with a finite family Φ_g closed under composition up to permutations σ_g of the index set ($\Phi_h \circ \Phi_g = \Phi_{\sigma_g(h)}$), and $\theta_g : J \to J \circ \Phi_g$ whose values exhaust the morphisms of T Chapter11/FormAveraging · EP.exists_primitive_of_average. On every level i the sum $\sum_g (\Phi^i_g)^*\beta$ is invariant under every $(\Phi^i_h)^*$, by the permutations, and is therefore $(J^i)^*\tau$ for a form τ of the ep-groupoid T^i: compatibility with a morphism $\theta_g(y)$ of T is invariance under Φ_g Chapter11/FormAveraging · EP.exists_epForms_sum_pullback. Dividing by $\#G_x$, $J^*(d\tau) = dJ^*\tau = J^*c$, and J^* is injective since J is a surjective local sc-diffeomorphism on objects Chapter11/FormAveraging · EP.injective_epFormsInftyPullback_of_surjective; so $c = d\tau$ with τ the text's τ' Chapter11/TranslationAverage · EpGroupoid.NaturalRep.exists_primitive_translation.

Step 4, extension to the saturation. The functor $G_x \ltimes U(x) \to X|\pi^{-1}\pi(U(x))$ Chapter11/SubGroupoidModel · EP.modelFunctor is an equivalence Chapter11/SubGroupoidModel · EP.isEquivalence_modelFunctor: full and faithful by the uniqueness property of Γ (Theorem 7.1.13), essentially surjective by the definition of the saturation, and a local sc-diffeomorphism on objects. If $E : T \to Z$ is an equivalence and $E^*a = d\tau$, then $a = d(E_*\tau)$, with E_* the push-forward of Definition 11.2.3 Chapter11/EquivalenceFormsInfty · EP.epFormsInftyPush Chapter11/FormAveraging · EP.exists_primitive_of_equivalence; $E_*\tau$ at y is $\psi^*\tau(x')$ for a morphism $\psi : x' \to y$, the text's σ. So $V = \pi^{-1}\pi(U(x))$, saturated, open and contained in U. No hypothesis is added (Findings, Chapter 8, items 123, 132) (pp. 373–374).

VIII.38 Proposition 8.1.16 without added hypothesis: vertical vectors

§VIII.29 proved Proposition 8.1.16 with the step "$T(g,h) = \mathrm{id}$ implies $Tg = \mathrm{id}$" as the hypothesis `TangentZhouReflect`. The step holds for every ep-groupoid Chapter8/TangentZhouReflect2021 · EpGroupoid.tangentZhouReflect, so *(ZHOU) for X gives (ZHOU) for TX* with the text's hypothesis only Chapter8/TangentZhouReflect2021 · EpGroupoid.proposition_8_1_16_2021 (V8.44; gap G.88, closed).

Vertical vectors. For a smooth $\alpha \in T_xX$ (level one of TX) and $w \in T_xX$, the vertical vector $\mathrm{vert}_\alpha w \in T_\alpha(TX)$ `Chapter8/TangentVertical2021 · MPolyfold.vert` is the vector with chart representative $(T\varphi(\alpha), (0,h))$, h the representative of w, in a chart of TX at α `Chapter8/TangentVertical2021 · MPolyfold.vertChart`; it does not depend on the chart, its representative in every chart has the same form `Chapter8/TangentVertical2021 · MPolyfold.tangentChartMap_vert`. $w \mapsto \mathrm{vert}_\alpha w$ is injective `Chapter8/TangentVertical2021 · MPolyfold.vert_injective` and natural: $T(Tf)(\mathrm{vert}_\alpha w) = \mathrm{vert}_{Tf\alpha}(Tf\,w)$ for sc-smooth f `Chapter8/TangentVertical2021 · MPolyfold.tangentMap_tangentMap_vert`. The local form is $D(Tf)(x,h)(0,v) = (0, Df(x)v)$, from the linearity of Tf in the fibre `Chapter8/TangentVertical2021 · IsSc1.scDeriv_tangentMap_vert`.

The proof. Let $k = (g,h)$ be in the isotropy group of a smooth $b \in TX$ over x, with $Tk = \mathrm{id}$, and let $v \in T_xX$. Choose $u \in T_g\mathbf{X}$ with $Ts(u) = v$ (Ts is bijective on the fibres) and put $V = \mathrm{vert}_k u \in T_k(T\mathbf{X})$. By naturality, $T(Ts)V = \mathrm{vert}_b v$ and $T(Tt)V = \mathrm{vert}_b(Tt\,u)$. Tk sends $T(Ts)V$ to $T(Tt)V$, so $Tk = \mathrm{id}$ gives $\mathrm{vert}_b(Tt\,u) = \mathrm{vert}_b v$, hence $Tt(u) = v$ by injectivity, that is $Tg(v) = Tt(u) = v$. This is the text's "differentiate (8.16)" restricted to the vertical directions of $T_b(TX)$, on which Tk acts as Tg; the surjectivity of $T\tau$ is not used. Steps 1 and 3 of §VIII.29 are unchanged (Findings, Chapter 8, items 115, 126) (p. 369).

VIII.39 Propositions 8.4.1 and 8.4.3

Proposition 8.4.1 (V8.47; gap G.86, closed) `Chapter8/BundleOrbitParacompactOne · EpStrongBundle.proposition_8_4_1`. *For a strong bundle $P : W \to X$ over an ep-groupoid, the following are equivalent: (1) $|X|$ is paracompact; (2) $|W[0]|$ is paracompact; (3) $|W[1]|$ is paracompact.* (2) □ (1) is §VIII.30.

(1) □ (2) `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.proposition_8_4_1_one_imp_two`. *Step 1, the local action.* For a natural representation of G_x on $U(x)$, G_x acts on $W|U(x) = P^{-1}(U(x))$ by $g * w = \mu(\Gamma(g, P(w)), w)$, the text's lift on p. 382, by homeomorphisms `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.bundleMulAction`. *Step 2.* Two points of $W|U(x)$ have the same orbit in $|W|$ iff they are G_x-related `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.π_eq_π_iff_exists_smul`, so $G_x\backslash(W|U(x)) \to |W|$ is an open embedding `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.isOpenEmbedding_bundleOrbitQuotientMap` onto $|P|^{-1}(|U(x)|)$ `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.range_bundleOrbitQuotientMap`. *Step 3.* The quotient of a metrizable space by a finite group of homeomorphisms is metrizable, by the metric $\hat\rho$ of Lemma 7.3.2 `Chapter8/BundleOrbitParacompact2021 · metrizableSpace_orbitRel_quotient_finite`; W is metrizable as an M-polyfold, so every $|P|^{-1}(|U(x)|)$ is metrizable. *Step 4.* $|X|$ paracompact is metrizable (Theorem 7.3.1), and Smirnov's theorem, pulled back along $|P|$ as for TX in §VII.23, makes $|W[0]|$ metrizable `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.metrizableSpace_orbitSpace`, hence paracompact `Chapter8/BundleOrbitParacompact2021 · EpStrongBundle.proposition_8_4_1_one_iff_two`. This replaces the text's closed cover and Proposition 2.7.4 by the partition-of-unity form of Smirnov's theorem; no bundle charts are needed, only the continuity of μ.

(1) □ (3). $W[1]$ is not built as an ep-groupoid in the project; $|W[1]|$ is taken as the orbit space of the groupoid $(W[1], \mathbf{X}_s\times_P W[1])$, the quotient of the M-polyfold $W[1]$ by $w \sim \mu(\varphi, w)$ `Chapter8/BundleOrbitParacompactOne · EpStrongBundle.OrbitSpaceOne` `Chapter8/BundleOrbitParacompactOne · EpStrongBundle.πOne_eq_πOne_iff`, its ep-groupoid structure being unused by the statement. The quotient map is open, as in Proposition 7.1.8 `Chapter8/BundleOrbitParacompactOne · EpStrongBundle.isOpenMap_πOne`. The action of G_x on $W[1]|U(x)$ is continuous for the level-one topology, since $\mu : \mathbf{X}_s\times_P W[1] \to W[1]$ is

sc-smooth (part of "strong bundle map") Chapter8/BundleOrbitParacompactOne · EpStrongBundle.continuous_bundleOneSmul, and Steps 2–4 repeat Chapter8/BundleOrbitParacompactOne · EpStrongBundle.isOpenEmbedding_bundleOneOrbitQuotientMap Chapter8/BundleOrbitParacompactOne · EpStrongBundle.metrizableSpace_orbitSpaceOne: (1) □ (3) Chapter8/BundleOrbitParacompactOne · EpStrongBundle.proposition_8_4_1_one_imp_three. Conversely the zero section is an sc$^+$-section, so $|0| : |X| \to |W[1]|$ is continuous with continuous left inverse $|P[1]|$ and closed image Chapter8/BundleOrbitParacompactOne · EpStrongBundle.isClosedEmbedding_zeroOrbitMapOne, and $|X|$ is paracompact as a closed subspace: (3) □ (1) Chapter8/BundleOrbitParacompactOne · EpStrongBundle.proposition_8_4_1_three_imp_one Chapter8/BundleOrbitParacompactOne · EpStrongBundle.proposition_8_4_1_one_iff_three. No hypothesis beyond the text's (Findings, Chapter 8, items 118, 127, 133) (p. 380).

Proposition 8.4.3 (V8.48) Chapter8/BundleLocalRegularity2021 · EpStrongBundle.proposition_8_4_3. *If X is effective and has (LR), the strong bundle has the local regularity property of Definition 8.4.2.* Take $U(x)$ from (LR) for X, and a strong bundle isomorphism Φ of $W|U(x)$ with $|\Phi(w)| = |w|$, over $\beta = g * \cdot$ by (LR). *Step 1.* $|\Phi(w)| = |w|$ gives a morphism $(\psi, w) : w \to \Phi(w)$, $\psi : P(w) \to \beta(P(w))$ Chapter8/BundleLocalRegularity2021 · EpStrongBundle.exists_mor_of_π_eq. *Step 2.* At a regular object two morphisms with the same source and target are equal Chapter8/BundleLocalRegularity2021 · EpGroupoid.eq_of_isRegular_src, so over a regular $P(w)$, $\psi = \Gamma(g, P(w))$ and $\Phi(w) = \mu(\psi, w) = g * w$ directly (the text's $C(w) = w$). *Step 3.* The regular points are open and dense (Proposition 7.2.4), and P is an open map Chapter10/LocalizationLREffectiveBundle · StrongBundle.isOpenMap_proj, so $P^{-1}(X_{reg})$ is dense in $W|U(x)$, and the closed set where $\Phi(w) = g * w$ is all of $W|U(x)$. The uniqueness of g, which the text takes from effectiveness, is not needed (Findings, Chapter 8, items 119, 128) (p. 382).

VIII.40 Lemma 8.5.10, Proposition 8.5.7 for $D = \emptyset$, Theorem 8.5.8 in every universe, Proposition 8.1.17 without added hypotheses

Lemma 8.5.10, F^*V over the ep-groupoid Y (V8.51; gap G.89, closed). §VIII.31 proved the lemma with F^*V as Part I's pull-back and μ_{F^*V} as a formula. *The strong bundle.* For every sc-smooth functor $F : Y \to X$ and strong bundle $Q : V \to X$, (F^*Q, μ_{F^*V}) is a strong bundle over the ep-groupoid Y (Definition 8.3.1: μ_{F^*V} is a surjective strong bundle map covering t, a local sc-diffeomorphism, with the unit and composition laws) Chapter8/PullbackStrongBundle2021 · EpStrongBundle.pullbackBundle2021: it is Chapter 10's pull-back along F (Part X), and its structure map is the text's formula $\mu_{F^*V}(\varphi, (y, v)) = (t(\varphi), \mu_X(F(\varphi), v))$ Chapter8/PullbackStrongBundle2021 · EpStrongBundle.pullbackBundle2021_μ, so the text's sentence "defines the strong bundle" holds Chapter8/PullbackStrongBundle2021 · EpStrongBundle.pullbackBundle2021_isStrongBundle. *The isomorphism.* For a proper strong bundle covering functor, $B(w) = (P(w), A(w))$ and B^{-1} are strong bundle maps of strong bundles over ep-groupoids (Definition 8.3.2) covering the identity functor of Y Chapter8/PullbackStrongBundle2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isMap_bundleB Chapter8/PullbackStrongBundle2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isMap_bundleBInv, the compatibility with μ being (8.23); so B is a strong bundle isomorphism over the ep-groupoid Y Chapter8/PullbackStrongBundle2021 · EpStrongBundle.IsProperStrongBundleCovering2021.lemma_8_5_10_full (Findings, Chapter 8, items 122, 129) (p. 395).

Proposition 8.5.7 for $D = \emptyset$ (V8.35). Definition 8.5.5 allows $D = \emptyset$, and then Proposition 8.5.7 does not hold as we read it Chapter8/GeometricLiftEmpty2021 · CoveringExamples.not_proposition_8_5_7_empty. The functor from the empty category to $\mathbb{Z}^\times$, as a one-object category, is a covering functor, both sides of the bijection of Definition 8.5.5 being empty Chapter8/GeometricLiftEmpty2021 · CoveringExamples.emptyCover; with $L_C(\star) = \mathbb{R}$,

$\mathbb{Z}/2$ acting by reflection, it has a geometric lift `Chapter8/GeometricLiftEmpty2021 · CoveringExamples.emptyLift`. The induced $F : \emptyset \to \mathbb{Z}/2 \ltimes \mathbb{R}$ is not surjective on objects, hence not a proper covering functor. The project's Proposition 8.5.7 carries $D \neq \emptyset$, which the text presumably intends (§VIII.20; Findings, Chapter 8, item 130) (p. 390).

Theorem 8.5.8 in every universe (V8.36). The first reading proved Theorem 8.5.8 for ep-groupoids in `Type` only. Read with Definition 8.5.5 in one universe — D, G and the category Diff_{sc} in which the lift takes values — no proof changes, and X, Y share the universe because Diff_{sc} is one category (§VIII.21). The identity of a one-object category, with its geometric lift, is a witness `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_8_id` (Findings, Chapter 8, item 125) (p. 392).

Proposition 8.1.17 without added hypotheses (V8.45, V8.52). The hypotheses of §VIII.29 and §VIII.33 — natural representations with $U(x) \cap X_1$ connected in X^1 and dense in $U(x)$ (gap G.73), and the openness of $p : T\mathbf{X} \to \mathbf{X}^1$ (gap G.8) — hold for every ep-groupoid (§VII.45): the first `Chapter7/SmoothConnectedReps · EpGroupoid.hasConnectedDenseLevelOneReps`, the second by Part I `Chapter2/TangentProjOpen · MPolyfold.isOpenMap_tangentProj`. With only the second still taken as an assumption the proposition is `Chapter7/SmoothConnectedReps · EpGroupoid.proposition_8_1_17_one_of_isOpenMap`; with none: *(1) if X^1 has LUC, so has TX, and $(T\mathbf{X})^T = p^{-1}((\mathbf{X}^1)^T)$* `Chapter7/SmoothRegularAssembly · EpGroupoid.proposition_8_1_17_one_of_isLUC` `Chapter7/SmoothRegularAssembly · EpGroupoid.nonEffectivePart_tangent_eq_of_isLUC`; *(2) if X has (SLUC), TX has LUC* `Chapter7/SmoothRegularAssembly · EP.proposition_8_1_17_two_of_isSLUC`, *every $T(X^i)$ has LUC* `Chapter7/SmoothRegularAssembly · EP.isLUC_tangent_lift_of_isSLUC`, *and TX has (SLUC)* `Chapter7/SmoothRegularAssembly · EP.proposition_8_1_17_two_all_of_isSLUC`, through $(TX)^i \cong T(X^i)$ of §VIII.33 (Findings, Chapter 8, items 116, 124, 126) (p. 371).

Part IX. Branched ep⁺-subgroupoids (HWZ Chapter 9, §§9.1–9.5 and §9.6.1 (2017 §§9.1–9.5))

Chapter 9 continues Part II; its numbers and pages are those of the book (Springer 2021, pp. 401–458), with the 2017 text's in parentheses (2017 pp. 291–332). It introduces the objects that will carry the solution sets of perturbed sc-Fredholm sections with symmetries: functors $\Theta : X \to \mathbb{Q}^+$ on an ep-groupoid that are, near every point of their support, finite weighted sums of indicator functions of properly embedded smooth finite-dimensional submanifolds. §9.1 gives the definition (Definition 9.1.2 (2017: 9.1)), the local closedness of the orbit support (Lemma 9.1.5 (2017: 9.1)), symmetric local branching structures (Lemma 9.1.7 (2017: 9.2)) and the dimensional decomposition (Propositions 9.1.11 (2017: 9.2) and 9.1.12, the latter new in 2021 with its proof in Appendix §9.6.1). §9.2 attaches to Θ a section T_Θ of formal sums of tangent spaces (Theorem 9.2.1 (2017: 9.1)), proved through good and bad points and a graph representation of the sheets near a bad point (Proposition 9.2.8, Lemmas 9.2.9–9.2.11 (2017: 9.3, 9.6–9.8)); it defines the tangent $T\Theta$ (Definition 9.2.12 (2017: 9.6)) and, through the position of a submanifold relative to ∂X (Lemma 9.2.13 (2017: 9.9)), the boundary $\partial\Theta$ (Proposition 9.2.15, Definition 9.2.16 (2017: 9.4, 9.7)). §9.3 orients Θ (Definition 9.3.2 (2017: 9.8)), compares orientations at good points (Lemma 9.3.4 (2017: 9.10)), gives an example (Example 9.3.5 (2017: 9.1)) and the boundary orientation (Definitions 9.3.6–9.3.10). §9.4 studies essential points and sets of measure zero of the sheets and decomposes an oriented structure (Propositions 9.4.9, 9.4.10 (2017: 9.5, 9.6)). §9.5 integrates sc-differential forms and proves Stokes' theorem: the sets of measure zero in $|\mathrm{supp}(\Theta)|$ and the σ-algebras $L(S)$, $L(\partial S)$ (Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2 (2017: Definitions 9.15–9.17, Lemma 9.16)), the branched integrals (Theorems 9.5.4, 9.5.5 (2017: 9.2, 9.3), Definition 9.5.6, Lemmas 9.5.7, 9.5.9–9.5.11 (2017: 9.17–9.19), Proposition 9.5.13), the boundary measures (Theorems 9.5.15, 9.5.16 (2017: 9.4, 9.5)) and Stokes' theorem (Theorems 9.5.17, 9.5.18 (2017: 9.6, 9.7)). Every gap of §§9.1–9.4 is closed and §9.5 is formalized with the text's hypotheses, except part of Remark 9.5.20; §§IX.26–IX.39 hold the later material.

What is proved, and what is not. All definitions of §§9.1–9.5 and §9.6.1 are formalized. Of the eighteen gaps of the chapter, seventeen are closed (G.93–G.109) and one is formalized in part (G.110, for Remark 9.5.20 only); none is open and none has nothing to formalize. *§§9.1–9.4.* The facts of level-0 sc-calculus that the first version of this part took as hypotheses on a general M-polyfold (the local constancy of the dimension of Remark 9.1.1, the graph representation of Lemmas 9.2.9–9.2.11, invariance of domain, the level-1 continuity of frames) are theorems on every M-polyfold (§IX.27). So Proposition 9.1.12, Theorem 9.2.1 (2017: 9.1), Lemma 9.2.7 (2), Lemma 9.6.4 and Proposition 9.6.5 hold for every branched ep⁺-subgroupoid of every ep-groupoid, Lemma 9.3.4 (2017: 9.10) for every n-dimensional one, and $T\Theta$ is a

branched ep⁺-subgroupoid of TX when $|X|$ is paracompact (§IX.11). Also proved: Remark 9.1.4, Proposition 9.1.9 (2017: 9.1) and Remark 9.1.10 (§IX.26); the global decomposition of Proposition 9.1.11 (§IX.4); the footnote to Definition 9.1.2 (§IX.1); the classification of the orientations of Example 9.3.5 (§IX.29); a tame sheet as a Mathlib manifold with corners, the boundary operation built from the sheet, the continuity of $o_{\partial M}$, the boundary stratum and Figure 9.1 (§IX.30); Lemma 9.4.3 (§IX.18); Lemmas 9.4.4 and 9.4.6 for the sheets of an M-polyfold, with no hypothesis (§IX.28). Lemmas 9.1.5, 9.1.7, 9.2.3, 9.2.5, 9.2.7 (1), 9.3.7, 9.4.2, 9.4.8, 9.6.1, 9.6.2, Propositions 9.2.15, 9.3.9, 9.4.9, 9.4.10 and the remarks with formal content are proved, several in a stronger form. *§9.5.* The Lebesgue null sets of the sheets of an M-polyfold are built (Hausdorff measure in the charts), and Lemma 9.5.2 holds for n-dimensional Θ (§IX.28); Theorems 9.5.4 and 9.5.5 hold as printed for the integral of sc-differential forms, with "oriented, tame" read as: near every point of the support an oriented tame structure describes $\hat{T}_\Theta$ (§§IX.31–IX.32); (9.21), (9.22), Lemmas 9.5.7, 9.5.9–9.5.11 and Proposition 9.5.13 hold as identities of measures (§§IX.31, IX.33); $L(\partial S)$ is built with $(n-1)$-dimensional null sets (§IX.34). Theorems 9.5.15–9.5.18 are proved as printed, with the text's hypotheses only, for every $n \geq 1$ (G.109 closed, V9.67 and V9.68 restated): the class conditions `hcov` and `hcovS` of the first proofs (§§IX.34–IX.35) are derived (§IX.36), the local formula of Theorems 9.5.15 and 9.5.16 holds for every oriented tame structure (§IX.37), and for $n = 1$ the 0-forms are integrated over the finite sets of oriented boundary points (§IX.38). Witnesses with nonempty boundary, with the local unique continuation property, are the closed upper hemisphere ($n = 2$) and the closed half-circle ($n = 1$) (§§IX.35, IX.38). **Formalized in part** (G.110): Remark 9.5.20. Its M-polyfold X, ∂X, $\mathbb{Z}_3 \ltimes X$ and the effective isotropy at ∂X, three read in X and one read in ∂X, are proved; Θ, $\oint_\Theta df$ and the two boundary integrals are not (§§IX.24, IX.39).

Narrower than printed, and why. V9.2: the dimension is measured in a chart. V9.9 (Proposition 9.1.11): the uniqueness among all decompositions does not hold as we read it; it holds among local decompositions, and local finiteness off the support needs a bound on the dimensions (§IX.4). V9.27 (Definition 9.2.16): the uniqueness does not hold as we read it; the characterization is unique with (2) asked on the larger support (§IX.13). V9.32 (Lemma 9.3.4): Θ is n-dimensional; for mixed dimensions the lemma does not hold as we read it (§IX.15). V9.33 (Example 9.3.5): "otherwise different" does not hold as we read it, and "four possibilities" holds for each structure but not for Θ, which has infinitely many orientations, six of them with integer weights (§§IX.15, IX.29). V9.42 (Lemma 9.4.4): the sheets have one dimension (§IX.17). V9.65 (Remark 9.1.10): one dimension n, since a Mathlib manifold has one model space; the composition of the groupoid is sc-smooth only, Mathlib having no orbifolds (§IX.26). V9.67 and V9.68 (Theorems 9.5.15–9.5.18) are restated (§§IX.36–IX.38). **Statements that do not hold as we read them**, each with a Lean counterexample: the uniqueness in Proposition 9.1.11 (§IX.4); Lemma 9.2.13 (2) □ (1) at interior points (§IX.12); the uniqueness in Definition 9.2.16 (§IX.13); Lemma 9.3.4 for mixed dimensions and two claims of Example 9.3.5 (§§IX.15, IX.29); the footnote to Definition 9.1.2 for $\Theta = 0$ (§IX.1); Lemma 9.4.4 and the transfer of null sets in Lemma 9.5.2 across dimensions (§§IX.17, IX.19, IX.28).

Seven structural choices. (1) *Tangent sets, linear algebra in charts.* When this part was first written, Part I's T_xX carried no linear structure (gap G.7, since closed). The tangent space T_yM of a submanifold is the set of tangent vectors fixed by the tangent of the retraction that presents M, and $\mathrm{Gr}(x)$ is generated by such sets; in a chart the image of T_yM is a linear subspace of the right dimension, and every argument that needs linear algebra is carried out there (§IX.6). (2) *Functors as functions.* Θ is a function on objects with $\Theta(s(\varphi)) = \Theta(t(\varphi))$, valued in $\mathbb{Q}$, nonnegativity a field. (3) *Dimension labellings.* The splitting by dimension is done with labellings $\dim : I \to \mathbb{N}$, unique when they exist (§IX.1); since the dimension is locally constant on every M-polyfold, a labelled structure exists at every point of the support (§IX.27). (4) *Slices instead of sequences.* Proposition 9.2.8 is used in the text to count the weights of

a class of tangent sheets by the values of Θ at good points; the formalization counts them on one finite slice transverse to the class (§IX.9). The same device proves Proposition 9.6.5 (§IX.22). (5) *Hypotheses first, then theorems.* The facts of the text's §9.2 that need level-0 sc-calculus inside a partial quadrant were first named hypotheses (`HasGraphChartAt`, `HasLocallyConstantDimAt`, `HasDomainInvarianceAt`, `HasContinuousLevelOneOn`), proved on $\mathbb{R}^m$; they are now theorems on every M-polyfold, with the graph representation in the weaker form of one cone per tangent space, which is all the slice arguments use (§IX.27). The statements under the hypotheses are kept. (6) *Sheet geometry and null sets as structures.* §9.4 is formalized for Mathlib manifolds with corners and, through the structure `SheetGeometry`, for sheets in an M-polyfold (§IX.17); the null sets of §9.5 are a structure `SheetNullSets` (§IX.19). Both are now instantiated on every M-polyfold by the n-dimensional Hausdorff measure of the chart images (§IX.28). (7) *Integration as a structure.* The integral of a form over a measurable part of an oriented sheet enters §9.5 through a structure `SheetIntegration` with the properties the text uses (linearity, countable additivity, vanishing on null sets, the comparison of two sheets, invariance under morphisms); Theorems 9.5.4 and 9.5.5 and the lemmas before them are proved for every such structure (§IX.31), and the structure is then built for sc-differential forms by integrating in graph charts (§IX.32).

Map.

- `Chapter9/Basic`: functors, supports, Definition 9.1.2 (2017: 9.1), Lemma 9.1.5 (2017: 9.1), Definitions 9.1.6, 9.1.8 (2017: 9.2, 9.3), the indicator of a point.
- `Chapter9/Symmetric`: the restriction remark and Lemma 9.1.7 (2017: 9.2).
- `Chapter9/Propositions`: Remark 9.1.1, n-dimensional Θ, Remark 9.1.3 (b), the statements and the local construction of Propositions 9.1.11 and 9.1.12, the counterexample to the uniqueness of 9.1.11, submanifolds of $\mathbb{R}^n$.
- `Chapter9/GoodPoints`: Definitions 9.2.2, 9.2.6, Lemmas 9.2.3, 9.2.5, 9.2.7 (1) (openness), Remark 9.2.4.
- `Chapter9/Tangent`: T_yM, $\mathrm{Gr}(X)$, Theorem 9.2.1 under the independence, Definition 9.2.12.
- `Chapter9/TangentSlicesBasic`, `TangentSlicesRn`, `TangentSlices`: the chart form of T_yM, the graph representation and tangent slices, Lemma 9.2.7 (1) (density) and (2), the case of $\mathbb{R}^m$.
- `Chapter9/AppendixSlices`: one graph chart for all dimensions, Proposition 9.6.5, Proposition 9.1.12 and the independence of Theorem 9.2.1 on $\mathbb{R}^m$ without hypotheses.
- `Chapter9/Appendix`: Lemmas 9.6.1, 9.6.2, 9.6.4, Definition 9.6.3, Proposition 9.1.12 under `TopMultiplicity`.
- `Chapter9/BoundaryPosition`, `BoundaryPositionConverse`: Lemma 9.2.13, Remark 9.2.14, Definition 9.3.6, Lemma 9.3.7.
- `Chapter9/Boundary`: Proposition 9.2.15, Definition 9.2.16, Remark 9.2.17.
- `Chapter9/OrientationTangent`, `OrientationBasic`, `OrientationExample`, `OrientationUniqueness`, `OrientationBoundary`, `OrientationRn` (index `Chapter9/Orientation`): §9.3.
- `Chapter9/BranchingGeometryMeasure`, `BranchingGeometryManifold`, `BranchingGeometry`: §9.4.
- `Chapter9/Measure`: Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2, the remark after Definition 9.5.3, witnesses.
- `Chapter9/MeasureEffective`: the isotropy and set parts of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13.
- `Chapter9/GeneralLevelZero`, `GeneralCorollaries`, `GeneralBoundaryPoints`: the level-0 facts on every M-polyfold and their consequences (§IX.27).
- `Chapter9/DimDecompositionExists`, `DimDecompositionUnique`: the global decomposition of Proposition 9.1.11 (§IX.4).

- `Chapter9/LocallyEquidimensional2021`: the footnote to Definition 9.1.2 and "locally equidimensional" (§IX.1).
- `Chapter9/TangentSubmanifoldSmooth`: TM_i smooth in TX (§IX.11).
- `Chapter9/MathlibManifoldBridge`: tame sheets as Mathlib manifolds with corners (§IX.16).
- `Chapter9/BoundaryOperation`, `BoundaryOrientationContinuity2021`, `BoundaryStratumDense`, `StratumRetraction`, `StratumTangent`, `BoundaryOrientation`, `FaceOrientation`: Definitions 9.3.8, 9.3.10 and Figure 9.1 (§IX.30).
- `Chapter9/PropertyTShrinking2021`: Lemma 9.4.3 (§IX.18).
- `Chapter9/SheetEssentialChart2021`, `SheetChartChange`, `SheetEssential`, `SheetGeometryFree`, `SheetMeasure`, `SheetNull`, `SheetNullBridge`: Lemmas 9.4.4, 9.4.6 and the null sets for M-polyfold sheets, Lemma 9.5.2 (§IX.28).
- `Chapter9/OrbitSubsetMetrizable`, `SupportOrbifold*`, `DescentPolyfold`, `SupportGroupoidNaturality`, `SupportClassicalStructures`, `SupportGroupoidEtale`, `XAxisWitness`: Remark 9.1.4, Proposition 9.1.9, Remark 9.1.10 (§IX.26).
- `Chapter9/OrientationExampleSigns`, `OrientationExampleClasses`, `OrientationExampleSheets`, `OrientationAnyDim`, `BoundaryCharacterization`: Example 9.3.5, Lemma 9.3.4, Definition 9.2.16 (§§IX.13, IX.15, IX.29).
- `Chapter9/BranchedIntegral`, `LocalMeasure`, `PointIntegration`, `SheetIntegrationAbstract`, `MeasureGlue`: branched integration for an abstract sheet integration (§IX.31).
- `Chapter9/ChartIntegral`, `ChartVectors`, `IntegralCovering`, `DensityLocal`, `FormEvalEp`, `FormEvalInfty`, `FrameSign`, `GraphChart`, `GraphChartExists`, `GraphCoord`, `GraphDiffeo`, `ManifoldBridge`, `ManifoldIntegralBridge`, `SheetIntegrationConcrete`, `TameOpen`, `GraphChartTransition`, `ChartAbsIntegral`, `CircleWitness`, `CircleOneSheet`, `DensitySignedMeasure`, `SignedMeasureIntegrals`, `FiniteTheoremText`, `FiniteDominated`, `FiniteTheorem`, `CoveringFromText`, `MultiChart`, `MultiChartCompact`, `MultiChartIntegration`, `MultiChartTheorem`, `ReducedCircle`, `ReducedIntegral`, `ReducedRep`: the integral of sc-forms, Theorems 9.5.4, 9.5.5, (9.22), Proposition 9.5.13 (2), the circle (§§IX.32–IX.33).
- `Chapter9/BoundaryFaceNullSets`, `BoundaryNotNull*`, `BoundarySheets`, `BoundaryFamily`, `BoundaryMeasure`, `FamilyMeasure`, `SheetFamily`: $L(\partial S)$ and Theorems 9.5.15, 9.5.16 (§IX.34).
- `Chapter9/StokesLocalFormulas`, `BoundaryNotNull`, `BoundaryNullSigmaAlgebra`, `FunctionTimesForm`, `LevelRestrict`, `MeasureWholeSheets`, `StokesPartition`, `StokesSheet`, `StokesSheetAllDegrees*`, `BoundaryIntegralStratum`, `FaceChange`, `FaceFrame`, `FaceIntegrand`, `BoundaryIntegralFaces`, `BoundaryLocal`, `HemisphereStokes`, `StokesBoundaryMeasure`, `HemisphereSheet`, `SphereRadialRetraction`, `HemisphereWitness`: Theorems 9.5.17, 9.5.18 and the hemisphere (§IX.35).
- `Chapter9/CornerFinite`, `BoundaryCover`, `StokesTextHypotheses`, `HemisphereBoundaryMeasure`: the class conditions derived, Theorems 9.5.15–9.5.18 with the text's hypotheses for $n \geq 2$ (§IX.36); `AllStructures`: the local formula for every structure (§IX.37).
- `Chapter9/AllStructuresDimOne`, `HalfCircleSheet`, `HalfCircleWitness`, `BoundaryMeasureDimOne`, `FaceIntegralDimOne`, `FinitePointIntegration`, `HemisphereStokesEffective`, `HemisphereStokesLUC`, `EffectiveDimOneLUC`, `StokesDimOne`, `ZeroFormValues`: $n = 1$, 0-forms, finite sets of points, the half-circle, and the local unique continuation property of the witnesses (§IX.38).
- `Chapter9/DimensionJumpRetract`, `DimensionJumpPolyfold`, `CyclicGroupoidIsotropy`: Remark 9.5.20, the M-polyfold, the groupoid and the effective isotropy (§IX.39).

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid as in Part VII, s, t its source and target, $\pi : X \to |X|$ the orbit map, X^∞ the full subcategory of smooth objects, G_x the isotropy group of x with a natural representation (Φ, Γ) on a neighbourhood $U(x)$ (§VII.4). A *submanifold* is a smooth finite-dimensional submanifold in the sense of Part I (§IV.2): the image, near each of its points, of an sc⁺-retraction. For $\Theta : X \to \mathbb{Q}$, $\mathrm{supp}(\Theta) = \{\Theta > 0\}$ and $|\mathrm{supp}(\Theta)| = \pi(\mathrm{supp}(\Theta))$. A local branching structure at x is $(U(x), (M_i)_{i\in I}, (\sigma_i)_{i\in I})$ as in Definition 9.1.2; we write $I_y = \{i \mid y \in M_i\}$ and $M_U = \bigcup_i M_i$.

IX.1 Definition 9.1.2 (2017: 9.1), and the dimension of a submanifold (Remark 9.1.1 (2017: 9.1))

Definition 9.1.2 (2017: 9.1) (V9.1) `Chapter9/Basic · EpGroupoid.IsBranchedEpSubgroupoid` `Chapter9/Basic · EpGroupoid.LocalBranchingStructure`. A branched ep⁺-subgroupoid is a functor $\Theta \geq 0$ `Chapter9/Basic · EpGroupoid.IsFunctorial` with $\mathrm{supp}(\Theta) \subseteq X_\infty$ `Chapter9/Basic · EpGroupoid.supp`, $|\mathrm{supp}(\Theta)|$ paracompact `Chapter9/Basic · EpGroupoid.orbitSupp`, and a local branching structure at every point of the support. The formalization reads the paracompactness of the induced topology as a `ParacompactSpace` instance on the subtype $|\mathrm{supp}(\Theta)|$, and "the inclusion $M_i \to U(x)$ is proper" as `IsProperMap` of the inclusion of subtypes. For a closed M_i in $U(x)$ the latter is automatic `Chapter9/Symmetric · isProperMap_inclusion_iff`. Witnesses: the zero functor `Chapter9/Basic · EpGroupoid.isBranchedEpSubgroupoid_zero` and the indicator of a smooth point of $\mathbb{R}^n$, a 0-dimensional submanifold `Chapter9/Basic · rnPolyfold_isSmoothSubmanifold_singleton` `Chapter9/Basic · EpGroupoid.localBranchingStructure_indicator` `Chapter9/Basic · EpGroupoid.isBranchedEpSubgroupoid_indicator`. The 2017 text refers in (4) to "(ii) and (iii)", the 2021 text to (2) and (3).

The dimension (V9.2). Remark 9.1.1 says that $m \mapsto \dim T_m M$ is locally constant. With $T_m M$ a set of tangent vectors, the dimension is measured in a chart: M has dimension k at a if the tangent space at the chart image of a of the sub-retract presenting M is k-dimensional `Chapter9/Propositions · MPolyfold.HasDimAt`. *Step 1, chart independence.* A germ of sc-diffeomorphisms between retracts restricts to a linear isomorphism of the tangent spaces of sub-retracts `Chapter9/Propositions · finrank_span_tangentSpace_eq`, so the value is the same for every chart and every presentation `Chapter9/Propositions · MPolyfold.finrank_tangentSpace_chart_change` `Chapter9/Propositions · MPolyfold.HasDimAt.finrank_eq`; it is unique `Chapter9/Propositions · MPolyfold.HasDimAt.unique` and exists at every point `Chapter9/Propositions · MPolyfold.IsSmoothSubmanifold.exists_hasDimAt`. *Step 2, invariance.* It is unchanged by intersecting with an open set and by sc-diffeomorphisms `Chapter9/Propositions · MPolyfold.hasDimAt_inter_isOpen_iff` `Chapter9/Propositions · MPolyfold.hasDimAt_image_iff`. *Step 3, local constancy* `Chapter9/AppendixSlices · MPolyfold.HasLocallyConstantDimAt` holds on every M-polyfold `Chapter9/GeneralLevelZero · MPolyfold.hasLocallyConstantDimAt`: in a chart the dimension is the rank of Dt for the chart sc⁺-retraction t presenting the submanifold `Chapter9/GeneralLevelZero · hasDimAt_finrank`, and this rank is locally constant along the retract `Chapter9/GeneralLevelZero · eventually_finrank_eq` (§IX.27). On $\mathbb{R}^m$ it was proved first: $T_y M$ is the range of the projection $Ds(y)$, whose rank is its trace, continuous in y and integer valued `Chapter9/AppendixSlices · MPolyfold.hasLocallyConstantDimAt_rn`. The dimensions of the sheets of one structure enter as labellings; all sheets of one structure pass through x, so a labelling is unique `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dim_unique`, and by local constancy every point of the support has a labelled structure `Chapter9/GeneralCorollaries · hasLabelledBranchingAt_general`.

n-dimensional Θ. The 2021 text says the sheets "can be taken" n-dimensional (2017: are n-dimensional), and this is the formal reading: some local branching structure at each point of the support has only n-dimensional sheets `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.IsOfDim` `Chap-`

ter9/Propositions · EpGroupoid.IsNDimensional. The zero functor is n-dimensional for every n Chapter9/Propositions · EpGroupoid.isNDimensional_zero. **The footnote**, that an n-dimensional Θ is not m-dimensional for $m \neq n$, is proved for Θ with nonempty support Chapter9/LocallyEquidimensional2021 · definition_9_1_2_footnote Chapter9/LocallyEquidimensional2021 · definition_9_1_2_footnote_of_ne_zero. *Step 1.* At a point of the support two structures, one with n-dimensional and one with m-dimensional sheets, give the same tangent sum (Theorem 9.2.1, now unconditional, §IX.10), so some sheet of the first has the tangent space of some sheet of the second Chapter9/LocallyEquidimensional2021 · exists_submanifoldTangent_eq. *Step 2.* Equal tangent spaces have equal dimensions, so $m = n$ Chapter9/LocallyEquidimensional2021 · isOfDim_unique. For $\Theta = 0$ the footnote does not hold as we read it, the zero functor being n-dimensional for every n Chapter9/LocallyEquidimensional2021 · definition_9_1_2_footnote_zero. "Locally equidimensional" is formalized in both readings, which agree: at every point of the support some structure has sheets of one dimension Chapter9/LocallyEquidimensional2021 · IsLocallyEquidimensional, iff at every such point there is an n such that every structure there, restricted to a small enough neighbourhood, has only n-dimensional sheets Chapter9/LocallyEquidimensional2021 · isLocallyEquidimensional_iff, the book's "sufficiently small"; nearby points of the support then have structures of the same dimension Chapter9/LocallyEquidimensional2021 · IsLocallyEquidimensional.exists_isOpen_isNDimensionalAt (gap G.93, closed).

Submanifolds of $\mathbb{R}^n$ (V9.3). The witnesses of the chapter are built from closed submanifolds of $\mathbb{R}^n$. The image of a smooth idempotent s is one, of dimension rank $Ds(a)$ Chapter9/Propositions · rnPolyfold_isSmoothSubmanifold_range Chapter9/Propositions · rnPolyfold_hasDimAt_range. The graph of a smooth $\beta : \mathbb{R} \to \mathbb{R}$ is the image of $(s,t) \mapsto (s, \beta(s))$ Chapter9/Propositions · graphRetraction Chapter9/Propositions · rnPolyfold_isSubmanifoldOfDim_graph2. Finite weighted sums of closed submanifolds, and indicators of locally closed ones, are branched ep⁺-subgroupoids of the trivial ep-groupoid Chapter9/Propositions · EpGroupoid.isBranchedEpSubgroupoid_sum_rn Chapter9/Propositions · EpGroupoid.isNDimensional_sum_rn Chapter9/Propositions · EpGroupoid.isBranchedEpSubgroupoid_indicator_rn, whose orbit space is $\mathbb{R}^n$, so the paracompactness is free Chapter9/Propositions · EpGroupoid.paracompactSpace_orbitSupp_trivial_rn.

IX.2 Remark 9.1.3 (b) (2017: 9.2 (b)), Lemma 9.1.5 (2017: 9.1), Definitions 9.1.6 and 9.1.8 (2017: 9.2 and 9.3)

Remark 9.1.3 (b) (2017: 9.2 (b)) (V9.4) Chapter9/Propositions · EpGroupoid.remark_9_1_3_b. The text takes β_1, β_2 with prescribed signs, the graphs G_1, G_2 with weight 1, and $\Theta(p) = \#\{i \mid p \in G_i\}$ Chapter9/Propositions · EpGroupoid.thetaGraphs Chapter9/Propositions · EpGroupoid.thetaGraphs_eq, and says that the graphs of $\gamma_1 = 0$ and of some γ_2 positive off 0 describe Θ too. *Step 1.* The sign conditions are the structure `Remark913Hyp` Chapter9/Propositions · EpGroupoid.Remark913Hyp, satisfied by $\beta_1(s) = e(s)$, $\beta_2(s) = e(-s)$ with Mathlib's `expNegInvGlue` e Chapter9/Propositions · EpGroupoid.remark913Hyp_expNegInvGlue. *Step 2.* For the second structure to describe the same Θ, γ_2 must be $\beta_1 + \beta_2$ Chapter9/Propositions · EpGroupoid.Remark913Hyp.eq_add_of_thetaGraphs_eq, which is smooth and positive off 0 Chapter9/Propositions · EpGroupoid.Remark913Hyp.γ₂_pos. The two structures are Chapter9/Propositions · EpGroupoid.Remark913Hyp.structureβ and Chapter9/Propositions · EpGroupoid.Remark913Hyp.structureγ. *Step 3.* "Not unique, even close to x" is read as: no sheet of the first structure agrees with a sheet of the second on any neighbourhood of the origin, since each of β_1, β_2 differs from 0 and from $\beta_1 + \beta_2$ at points arbitrarily close to 0.

Lemma 9.1.5 (2017: 9.1) (V9.5) Chapter9/Basic · EpGroupoid.lemma_9_1_5. *$|\mathrm{supp}(\Theta)|$ is locally closed in $|X|$.* **The text's proof** takes a neighbourhood with the natural G_x-action, the properness property and a local branching structure, and shows that the image of $\bigcup_i M_i$ is closed in the image of $U(x)$. **The**

formalized proof. *Step 1.* For a functor the support is saturated, $\pi^{-1}(|\mathrm{supp}(\Theta)|) = \mathrm{supp}(\Theta)$ `Chapter9/Basic · EpGroupoid.preimage_π_orbitSupp`. *Step 2.* On $U(x)$ the support is $\bigcup_i M_i$ `Chapter9/Basic · EpGroupoid.LocalBranchingStructure.mem_supp_iff`, a finite union of sets closed in $U(x)$ `Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.isClosed_M` `Chapter9/Basic · EpGroupoid.LocalBranchingStructure.isClosed_preimage_val_supp`, so $\mathrm{supp}(\Theta)$ is locally closed in X `Chapter9/Basic · EpGroupoid.isLocallyClosed_supp`. *Step 3.* π is open (Chapter 7), and the image of the locally closed saturated set is locally closed. The properness property of the text is not used.

Definitions 9.1.6 and 9.1.8 (2017: 9.2 and 9.3) (V9.6, V9.8) are read literally `Chapter9/Basic · EpGroupoid.IsClosedBranched` `Chapter9/Basic · EpGroupoid.IsCompactBranched` `Chapter9/Basic · EpGroupoid.IsOrbifoldType` `Chapter9/Basic · EpGroupoid.IsManifoldType`; the indicator of a point is compact and of manifold type `Chapter9/Basic · EpGroupoid.isCompactBranched_indicator` `Chapter9/Basic · EpGroupoid.isManifoldType_indicator`. Proposition 9.1.9 and Remark 9.1.10 (natural manifold and orbifold structures on $|\mathrm{supp}(\Theta)|$) and Remark 9.1.4 (metrizability, the compactness criterion) are proved in §IX.26 (gaps G.94, G.95, closed).

IX.3 The restriction remark, and Lemma 9.1.7 (2017: 9.2)

The restriction (V9.7). Before Lemma 9.1.7 the text replaces M_i by $N_i = M_i \cap V(x_0)$ for $V(x_0) \subseteq U(x)$ and notes that some N_i may be empty. Definition 9.1.2 asks $x \in M_i$, which the N_i need not satisfy at x_0; the remark is formalized for an open $V \subseteq U(x)$ containing x itself, which is the case the symmetrization uses `Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.restrict`. The N_i are submanifolds `Chapter9/Symmetric · MPolyfold.IsSmoothSubmanifold.inter_isOpen`, and they are properly embedded in V because they are closed in $U(x)$.

Lemma 9.1.7 (2017: 9.2) (V9.7) `Chapter9/Symmetric · EpGroupoid.lemma_9_1_7` `Chapter9/Symmetric · EpGroupoid.lemma_9_1_7'`. *At $x \in \mathrm{supp}(\Theta)$ there is a local branching structure on the domain of a natural representation of G_x, with a G_x-action on the index set, $g * M_i = M_{g(i)}$ and $\sigma_{g(i)} = \sigma_i$; it can be found inside every open $V \ni x$.* **The text's proof** sets $J = I \times G_x$, $M_{(i,g)} = g * M_i$, $\sigma_{(i,g)} = \sigma_i/|G_x|$ and computes $\sum_{y \in g*M_i} \sigma_{(i,g)} = \frac{1}{|G_x|}\sum_g \Theta(g^{-1} * y) = \Theta(y)$ by functoriality. **The formalized proof** follows it `Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.symmetrize` `Chapter9/Symmetric · EpGroupoid.SymmetricBranchingStructure`. *Step 1.* Restrict the given structure to the domain $U \subseteq V \cap U(x)$ of a natural representation (Theorem 7.1.13, §VII.4). *Step 2.* $\Phi(g)(M_i \cap U)$ is a submanifold properly embedded in U through x, since $\Phi(g)$ is an sc-diffeomorphism of U fixing x `Chapter9/Symmetric · MPolyfold.IsSmoothSubmanifold.image_of_isScDiffeoOn`; the text uses this without saying so. *Step 3.* The weights: $\Theta(\Phi(g)(y)) = \Theta(y)$ `Chapter9/Symmetric · EpGroupoid.IsFunctorial.apply_Φ` and $y \in \Phi(g)(A) \iff \Phi(g)^{-1}(y) \in A$ `Chapter9/Symmetric · EpGroupoid.NaturalRep.mem_image_Φ_iff` turn the text's computation into a sum over G_x with $|G_x| =$ `Nat.card` of the finite group. *Step 4.* G_x acts on J by left multiplication on the second factor `Chapter9/Symmetric · EpGroupoid.mulActionProdLeft`.

IX.4 Proposition 9.1.11 (2017: 9.2): the local construction, and the uniqueness

Proposition 9.1.11 (2017: 9.2) (V9.9) `Chapter9/Propositions · EpGroupoid.Proposition_9_1_11`. *A branched ep⁺-subgroupoid determines a unique sequence $\Theta_0, \Theta_1, \ldots$ of branched ep⁺-subgroupoids with Θ_k k-dimensional and $\Theta = \sum_k \Theta_k$, locally finite.* The formal statement: for every branched Θ there is exactly one family (Θ_k)

with `IsDimDecomposition` Chapter9/Propositions · EpGroupoid.IsDimDecomposition, every Θ_k branched and k-dimensional, at each object finitely many nonzero and Θ their sum, and every such family is locally finite Chapter9/Propositions · EpGroupoid.IsLocallyFiniteFamily.

The local construction. The text splits the indices of a symmetric structure by the dimension of M_i and puts $\Theta_k(y) = \sum_{i\in I_k, y\in M_i} \sigma_i$. Formally, for a labelling dim, Θ_k Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart has the k-dimensional local branching structure $(M_i)_{\dim i=k}$ Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartStructure Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartStructure_isOfDim and $\Theta = \sum_k \Theta_k$ on $U(x)$ Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.eq_sum_dimPart; $\Theta_k = 0$ if no sheet has dimension k Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart_eq_zero, and $\Theta_n = \Theta$ for an n-dimensional structure Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart_eq_of_isOfDim. For a symmetric structure G_x preserves dimension, since $\Phi(g)$ is an sc-diffeomorphism Chapter9/Propositions · EpGroupoid.SymmetricBranchingStructure.dim_smul, so Θ_k is G_x-invariant Chapter9/Propositions · EpGroupoid.SymmetricBranchingStructure.dimPart_Φ. That the local Θ_k does not depend on the structure is Proposition 9.1.12 (§IX.5).

The uniqueness, as we read it, does not hold Chapter9/Propositions · EpGroupoid.not_proposition_9_1_11_rn. On the trivial ep-groupoid of $\mathbb{R}^2$, let L be the x-axis Chapter9/Propositions · EpGroupoid.axis2 and $\Theta = 1_L$. *Family A:* $\Theta_0 = 1_{\{0\}}$, $\Theta_1 = 1_{L\setminus\{0\}}$ Chapter9/Propositions · EpGroupoid.decompA Chapter9/Propositions · EpGroupoid.decompA_isDimDecomposition. *Family B:* $\Theta_0 = 0$, $\Theta_1 = 1_L$ Chapter9/Propositions · EpGroupoid.decompB Chapter9/Propositions · EpGroupoid.decompB_isDimDecomposition. In family A, $L\setminus\{0\}$ is a 1-dimensional submanifold properly embedded in the open set $\mathbb{R}^2\setminus\{0\}$, which is all Definition 9.1.2 asks at its points, and $1_{\{0\}}$ is 0-dimensional; both families satisfy (1) and (2). The proof of the text constructs family B.

The repaired uniqueness. Call a decomposition *local* if near every point of $\mathrm{supp}(\Theta)$ its Θ_k is the k-dimensional part of a local branching structure of Θ, which is what the text's proof produces Chapter9/DimDecompositionUnique · IsLocalDimDecomposition. Family B is local and family A is not Chapter9/DimDecompositionUnique · decompB_isLocalDimDecomposition Chapter9/DimDecompositionUnique · decompA_not_isLocalDimDecomposition. Two local decompositions are equal: near a point of the support both are dimensional parts of structures there, which agree near the point by Proposition 9.1.12, and off the support both vanish Chapter9/DimDecompositionUnique · isDimDecomposition_unique. Proposition 9.1.12 now holds on every ep-groupoid (§IX.5), so the uniqueness among local decompositions has no hypothesis Chapter9/GeneralCorollaries · proposition_9_1_11_unique.

The global decomposition (gap G.96, closed). *Step 1, the definition.* $\Theta_k(y)$ is the sum of the weights of the k-dimensional sheets of a chosen labelled structure at y, and 0 if there is none Chapter9/DimDecompositionExists · HasLabelledBranchingAt Chapter9/DimDecompositionExists · dimComponent. Every sheet passes through its base point, so this is the value at y of the local Θ_k at y. *Step 2, independence and locality.* By Proposition 9.1.12, $\Theta_k(x)$ is the same for every labelled structure at x Chapter9/DimDecompositionExists · dimComponent_eq_self, and on all of $U(x)$ it is the k-dimensional part of any labelled structure at x, restricted to y Chapter9/DimDecompositionExists · dimComponent_eq_of_mem Chapter9/DimDecompositionExists · dimPart_restrictAt. *Step 3, functoriality.* Along a morphism φ, a labelled structure at $s(\varphi)$ is transported to $t(\varphi)$, preserving dimensions Chapter9/DimDecompositionExists · isSubmanifoldOfDim_transport, and Step 2 at the target gives $\Theta_k(s(\varphi)) = \Theta_k(t(\varphi))$ Chapter9/DimDecompositionExists · isFunctorial_dimComponent; the text's symmetric structures are not needed. *Step 4, paracompactness*, which the text does not discuss: $|\mathrm{supp}(\Theta_k)|$ is closed in $|\mathrm{supp}(\Theta)|$ Chapter9/

DimDecompositionExists · isClosed_orbitSupp_dimComponent Chapter9/DimDecompositionExists · paracompactSpace_orbitSupp_dimComponent. *Step 5.* So each Θ_k is a k-dimensional branched ep⁺-subgroupoid Chapter9/DimDecompositionExists · isBranchedEpSubgroupoid_dimComponent Chapter9/DimDecompositionExists · isNDimensional_dimComponent, the family is a local decomposition Chapter9/GeneralCorollaries · isLocalDimDecomposition_dimComponent_general, and for every ep-groupoid there is exactly one local decomposition Chapter9/GeneralCorollaries · existsUnique_localDimDecomposition_general. *Step 6, local finiteness.* At the points of the support only finitely many Θ_k are nonzero nearby, those of the sheets of one structure Chapter9/GeneralCorollaries · dimComponent_locallyFinite_of_mem_general. At points of the closure of the support outside it, the formalization needs a bound on the dimensions of the submanifolds of X Chapter9/GeneralCorollaries · proposition_9_1_11_local_of_le, which holds on $\mathbb{R}^m$ Chapter9/DimDecompositionExists · rn_hasDimAt_le Chapter9/DimDecompositionExists · proposition_9_1_11_local_rn. For the line L the construction gives family B Chapter9/DimDecompositionExists · dimComponent_axis2. Row V9.9 is narrower for these two reasons: "uniquely" is read among local decompositions, and local finiteness off the support uses the bound.

IX.5 Proposition 9.1.12: the statement and its cases

Proposition 9.1.12 (V9.10) Chapter9/Propositions · EpGroupoid.Proposition_9_1_12, new in the 2021 edition. *For two local branching structures of Θ on $U(x)$ there is an open G_x-invariant $V(x) \subseteq U(x)$ on which the dimensional decompositions agree.* The formal conclusion is `DimPartsAgreeNear`: an open $V \ni x$ in both domains on which $\Theta_k = \Theta'_k$ for every k Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear. "G_x-invariant, with the usual properties" is obtained by shrinking V to the domain of a natural representation Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear.exists_naturalRep. *The equidimensional case* is immediate, since then $\Theta_n = \Theta$ for both Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_isOfDim. The general case is proved in Appendix §9.6.1 (§§IX.21–IX.22): under the hypothesis `TopMultiplicity` Chapter9/Appendix · EpGroupoid.proposition_9_1_12_of_topMultiplicity, under the graph representation Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_of_hasGraphChartAt, without hypothesis on every ep-groupoid over $\mathbb{R}^m$ Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_rn, and now without hypothesis on every ep-groupoid Chapter9/GeneralCorollaries · proposition_9_1_12: `TopMultiplicity` holds on every M-polyfold, by the slice argument of §IX.22 run with one cone per tangent space (§IX.27) Chapter9/GeneralCorollaries · topMultiplicity. Mixed structures exist: $\Theta = 1 + \delta_0$ on $\mathbb{R}^1$ has sheets of dimensions 0 and 1 Chapter9/Appendix · exists_mixed_rn_one, and on $\mathbb{R}^2$ two labelled structures of one Θ with sheets of dimensions 1 and 0 have different sheets and agreeing dimensional parts Chapter9/AppendixSlices · EpGroupoid.Remark913Hyp.exists_mixed_rn_two.

IX.6 The category $\mathrm{Gr}(X)$, and the tangent spaces T_yM

T_yM (V9.12). Part I's Remark 4.2.2 (preprint: 4.14) defines $T_yM = Tr(y)(T_yX)$ for an sc⁺-retraction r with image M near y. Here T_yM is the set of tangent vectors at y fixed by the tangent of r Chapter9/Tangent · MPolyfold.submanifoldTangent. *Step 1.* It does not depend on the retraction Chapter9/Tangent · MPolyfold.mem_submanifoldTangent_iff Chapter9/Tangent · MPolyfold.submanifoldTangent_congr, lies in the fibre over y Chapter9/Tangent · MPolyfold.submanifoldTangent_subset_tangentFiber, is local Chapter9/Tangent · MPolyfold.submanifoldTangent_inter_isOpen, and is mapped onto $T_{f(y)}f(M)$ by an sc-diffeomorphism f Chapter9/Tangent · MPolyfold.submanifoldTangent_image. *Step 2, the chart form.* In a chart the second components of the images of T_yM form a set Chapter9/TangentSlicesBasic · MPolyfold.

chartTangent equal to the tangent space of the chart model of M Chapter9/TangentSlicesBasic · MPolyfold.chartTangent_eq, a linear subspace Chapter9/TangentSlicesBasic · MPolyfold.exists_submodule_chartTangent of dimension $\dim_y M$ Chapter9/TangentSlicesBasic · MPolyfold.finrank_chartTangent; a chart change acts on it by a continuous linear map Chapter9/TangentSlicesBasic · MPolyfold.exists_clm_tangentChartMap_snd Chapter9/TangentSlicesBasic · MPolyfold.chartTangent_chart_change, a linear isomorphism Chapter9/TangentSlicesBasic · MPolyfold.exists_linearEquiv_chartTangent. So the generators T_xM_i of $T_\Theta(x)$ are linear subspaces in every chart Chapter9/TangentSlicesBasic · EpGroupoid.LocalBranchingStructure.exists_submodule_of_mem_support, which is what the text means by "finite-dimensional linear subspaces of T_xX".

$\mathrm{Gr}(X)$ (V9.11). $\mathrm{Gr}(x)$ is the additive group of finitely supported $\mathbb{Q}$-valued functions on subsets of T_xX Chapter9/Tangent · EpGroupoid.Gr, a morphism $\varphi : x \to y$ acts by the image under its tangent $T\varphi$ (Chapter 7) Chapter9/Tangent · EpGroupoid.grPush, with $T1_x = \mathrm{id}$ Chapter9/Tangent · EpGroupoid.tangentMor_unit_of_eq and $T\varphi$ injective on the fibre Chapter9/Tangent · EpGroupoid.tangentMor_injOn; the push-forward is functorial Chapter9/Tangent · EpGroupoid.grPush_comp Chapter9/Tangent · EpGroupoid.grPush_unit. The objects (x, a) Chapter9/Tangent · EpGroupoid.GrObj form a category over the full subcategory X^∞ Chapter9/Tangent · EpGroupoid.SmoothCat with the projection π Chapter9/Tangent · EpGroupoid.grProj.

IX.7 Good points: Definitions 9.2.2 and 9.2.6 (2017: 9.4 and 9.5), Lemmas 9.2.3 and 9.2.5 (2017: 9.3 and 9.4), Remark 9.2.4 (2017: 9.4)

Definition 9.2.2 (2017: 9.4) (V9.15) Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.IsGoodPoint. $y \in M_U$ Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.MU is good if there is an open $V(y) \subseteq U(x)$ on which all sheets through y coincide. The text states this for equidimensional Θ; no proof below uses equidimensionality, so the definition and Lemmas 9.2.3 and 9.2.5 are stated for every Θ with a local branching structure.

The intrinsic form Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.isGoodPoint_iff. *For $y \in U(x)$: y is good iff $y \in \mathrm{supp}(\Theta)$ and Θ is constant on $\mathrm{supp}(\Theta)$ near y. Step 1.* Near y take V in the complement of the sheets not through y Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.nbhdAt Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.isOpen_nbhdAt; there $\Theta(z) \le \Theta(y)$ Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.le_of_mem_nbhdAt, with equality iff z lies on all sheets through y Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.eq_iff_of_mem_nbhdAt. *Step 2.* If all sheets through y coincide on V, Θ is $\Theta(y)$ on $\mathrm{supp}(\Theta) \cap V$; conversely, if Θ is constant on the support near y, every point of a sheet through y near y lies on all of them, by Step 1.

Lemma 9.2.3 (2017: 9.3) (V9.16) Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.lemma_9_2_3 follows at once, for any two structures, at possibly different points, whose domains contain y: the intrinsic form does not mention the structure. This is the text's argument, in which $Q(z) = V(y) \cap W(y)$ is presumably $Q(y)$.

Remark 9.2.4 (2017: 9.4) (V9.17). (a) Chapter9/GoodPoints · EpGroupoid.remark_9_2_4_a is Lemma 9.2.3 with the good set of Definition 9.2.6 Chapter9/GoodPoints · EpGroupoid.mem_goodSet_iff. (b) Chapter9/GoodPoints · EpGroupoid.remark_9_2_4_b: near a good y, $\Theta = \Theta(y) \cdot 1_M$ with $M = M_i \cap V(y)$ for any $i \in I_y$, properly embedded in $V(y)$ Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.restrictAt.

Lemma 9.2.5 (2017: 9.4) (V9.18) Chapter9/GoodPoints · EpGroupoid.lemma_9_2_5. *$s(\varphi)$ is good iff $t(\varphi)$ is good.* The text transports local branching structures along morphisms. The formal proof uses the intrinsic form and the sheet of φ: on a sheet $\mathbf{W}$ of s around φ, $\widehat{\varphi} = t \circ (s|\mathbf{W})^{-1}$ is an sc-diffeomorphism between open

neighbourhoods of $s(\varphi)$ and $t(\varphi)$ with $\Theta \circ \hat{\varphi} = \Theta$ `Chapter9/GoodPoints · EpGroupoid.Sheet.apply_hat`, so "Θ constant on the support near the point" passes from one to the other `Chapter9/GoodPoints · EpGroupoid.src_mem_goodSet_of_tgt`.

Definition 9.2.6 (2017: 9.5) (V9.19) `Chapter9/GoodPoints · EpGroupoid.goodSet` `Chapter9/GoodPoints · EpGroupoid.badSet`. Both sets are saturated `Chapter9/GoodPoints · EpGroupoid.preimage_π_image_goodSet` `Chapter9/GoodPoints · EpGroupoid.preimage_π_image_badSet`, by Lemma 9.2.5. The text cites "Remark 9.2.4 (c)" here; the remark has (a) and (b), and Lemma 9.2.5 is what the descent to $|X|$ uses.

IX.8 Lemma 9.2.7 (2017: 9.5): good points are dense and accumulate on every sheet

Lemma 9.2.7 (1) (2017: 9.5 (1)) (V9.20). The openness of the good set in $\mathrm{supp}(\Theta)$ is the definition `Chapter9/GoodPoints · EpGroupoid.lemma_9_2_7_1`. **The text** derives the density from (2). **The formalized proof** needs neither (2) nor equidimensionality `Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_1_dense`. *Step 1.* Given an open $W \subseteq U(x)$ meeting the support, shrink W one sheet at a time so that $M_U \cap W$ either lies in M_j or misses it `Chapter9/TangentSlices · exists_isOpen_subset_inter_eq_empty_or_subset`: if $M_U \cap W \not\subseteq M_j$, a point of $M_U \cap W$ off the closed set M_j has a neighbourhood in W missing M_j, and the sheets already decided stay decided. *Step 2.* On the final W every sheet through a point of $M_U \cap W$ contains $M_U \cap W$, so all sheets through it agree on W and it is good. So the good set is dense `Chapter9/TangentSlices · EpGroupoid.dense_goodSet` and the bad set nowhere dense `Chapter9/TangentSlices · EpGroupoid.isNowhereDense_badSet`.

Lemma 9.2.7 (2) (2017: 9.5 (2)) (V9.21) `Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_2`. *For an n-dimensional structure at x and every i, good points of M_i converge to x.* **The text's proof** fixes i_0, sets $\Sigma_i = M_i \cap M_{i_0} \cap D_k(x)$ and either finds a point of $M_{i_0} \cap D_k(x)$ off all Σ_i or moves a point off the boundaries of those Σ_i with interior; such a point is "clearly" good. **The formalized proof** does this with Step 1 above applied inside M_{i_0}. The point found is good only if $M_{i_0} \subseteq M_i$ near it forces $M_{i_0} = M_i$ near it, which is invariance of domain for submanifolds of equal dimension. It was first the hypothesis `HasDomainInvarianceAt` `Chapter9/TangentSlices · MPolyfold.HasDomainInvarianceAt`, proved on $\mathbb{R}^m$ by the inverse function theorem `Chapter9/TangentSlices · MPolyfold.hasDomainInvarianceAt_rn` `Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_2_rn`. It now holds on every M-polyfold `Chapter9/GeneralLevelZero · hasDomainInvarianceAt`: in a chart both submanifolds are the ranges of chart sc⁺-retractions, and two such retracts with the same tangent space at the point, one inside the other, agree near it `Chapter9/GeneralLevelZero · eq_of_range_eq` (§IX.27). So Lemma 9.2.7 (2) holds for every ep-groupoid `Chapter9/GeneralCorollaries · lemma_9_2_7_2_general` (gap G.98, closed). The statement holds at every x, not only at bad points. In Remark 9.1.3 (b) the origin is bad `Chapter9/TangentSlices · EpGroupoid.Remark913Hyp.zero_mem_badSet` and is approached by good points on each graph `Chapter9/TangentSlices · EpGroupoid.Remark913Hyp.exists_seq_goodSet_structureβ`.

IX.9 Proposition 9.2.8 and Lemmas 9.2.9–9.2.11 (2017: 9.3 and 9.6–9.8): graph charts and tangent slices

The text. At a bad point x the indices with the same $T_x M_i = L$ form a class Ξ. Locally, $x = 0$ in a partial quadrant C; $L \cap C$ has interior in L (Lemma 9.2.9); the M_i, $i \in \Xi$, are graphs $a + B_i(a)$ over a cone around a unit h_0 in that interior, with an sc-complement Z of L (Lemma 9.2.10), through the inversion of a map Φ on the cone (Lemma 9.2.11). Proposition 9.2.8 then splits $\Xi = I^1 \sqcup \cdots \sqcup I^p$ with

sequences of good points $x_k^e \to x$ lying on M_i exactly for $i \in I^e$, and the proof of Theorem 9.2.1 counts $\sum_{i\in\Xi} \sigma_i = \sum_e \Theta(x_k^e)$.

What the count needs. One finite set F near x that meets each M_i, $i \in \Xi$, exactly once and misses the other sheets, so that $\sum_{i\in\Xi} \sigma_i = \sum_{y\in F} \Theta(y)$: a *tangent slice* Chapter9/Tangent · MPolyfold.HasTangentSlicesAt. The sequences of Proposition 9.2.8 are one way to produce such sets; a slice needs only one good direction $h_0 \in L$, off the finitely many other tangent spaces. So Proposition 9.2.8 is formalized in this slice form (V9.22).

The graph representation, first as a hypothesis. Lemmas 9.2.9–9.2.11 rest on sc-calculus at level 0 inside a partial quadrant (sc-complements, the mean value formula for sc⁺-retractions, an inverse function theorem on a cone). In the first version of this part their content in a chart was the hypothesis `IsGraphChartAt`, which is now proved on every M-polyfold (§IX.27; gap G.97, closed): every n-dimensional submanifold through x is, along each admissible ray th of its tangent space, met by the affine fibre through th, with secants uniformly approximated by T_xM Chapter9/TangentSlicesBasic · MPolyfold.IsGraphChartAt Chapter9/TangentSlicesBasic · MPolyfold.HasGraphChartsAt. Rays replace the cones of Lemma 9.2.10, and the admissible directions play the role of C. *From graphs to slices* Chapter9/TangentSlicesBasic · MPolyfold.hasTangentSlicesAt_of_hasGraphChartsAt: with Q a projection onto L, the slice is $\{y \mid Q(\varphi y - \varphi x) = th_0,\ \|\varphi y - \varphi x - th_0\| < \varepsilon t\}$ for small t; it meets each sheet of the class once, by the graph property, and misses the other sheets, since their tangent spaces do not contain h_0 Chapter9/TangentSlicesBasic · exists_mem_ball_forall_not_mem Chapter9/TangentSlicesBasic · eq_zero_of_proj_eq_zero. *On* $\mathbb{R}^m$ the graph representation holds at every point and in every dimension Chapter9/TangentSlicesRn · MPolyfold.isGraphChartAt_rn Chapter9/TangentSlicesRn · MPolyfold.hasGraphChartsAt_rn Chapter9/TangentSlicesRn · MPolyfold.hasTangentSlicesAt_rn: a submanifold is near x the image of a smooth retraction s, and the inverse function theorem for $Q \circ s$ restricted to L gives the graph Chapter9/TangentSlicesRn · exists_mem_graph_of_hasStrictFDerivAt Chapter9/TangentSlicesRn · MPolyfold.rn_exists_retraction.

One chart for all dimensions. For Proposition 9.6.5 and the independence of Theorem 9.2.1 without equidimensionality, the sheets of all dimensions through x must be compared in one chart: `HasGraphChartAt` asks one chart and one set of admissible directions for all of them Chapter9/AppendixSlices · MPolyfold.HasGraphChartAt Chapter9/AppendixSlices · MPolyfold.HasGraphChartAt.hasGraphChartsAt, which holds on $\mathbb{R}^m$ Chapter9/AppendixSlices · MPolyfold.hasGraphChartAt_rn. In it, every class has slices inside every neighbourhood Chapter9/AppendixSlices · MPolyfold.exists_slice_of_isGraphChartAt, which can be chosen to miss finitely many given submanifolds of smaller dimension Chapter9/AppendixSlices · MPolyfold.exists_generic_slice. For Lemma 9.3.4 good points must be found near the slice, which a ray does not provide; the ball form `HasGraphBallAt` asks the graph property over a ball of foot points Chapter9/OrientationRn · MPolyfold.HasGraphBallAt Chapter9/OrientationRn · MPolyfold.HasGraphBallAt.hasGraphChartsAt, again true on $\mathbb{R}^m$ Chapter9/OrientationRn · hasGraphBallAt_rn. On a general M-polyfold, one cone of admissible directions for all submanifolds through a boundary point is more than the slice arguments use: they need a cone for each tangent space separately, `HasClassGraphChartAt` Chapter9/GeneralLevelZero · HasClassGraphChartAt, which holds at every point Chapter9/GeneralLevelZero · hasClassGraphChartAt; the stronger `HasGraphChartAt` and `HasGraphBallAt` hold at interior points Chapter9/GeneralLevelZero · hasGraphChartAt_of_mem_interior Chapter9/GeneralLevelZero · hasGraphBallAt_of_mem_interior (§IX.27).

IX.10 Theorem 9.2.1 (2017: 9.1): the section T_Θ

Theorem 9.2.1 (2017: 9.1) (V9.13) `Chapter9/Tangent · EpGroupoid.theorem_9_2_1`. *There is a section functor* $T_\Theta : X^\infty \to \mathrm{Gr}(X)$, $\pi \circ T_\Theta = \mathrm{Id}$, *with* $T_\Theta(x) = \sum_i \sigma_i T_x M_i$ *for every local branching structure at* $x \in \mathrm{supp}(\Theta)$, *the zero sum off the support, and* $T_\Theta(\varphi) = (T_\Theta(s(\varphi)), \varphi)$. The theorem is proved in two parts. *Part 1, the section, given the independence.* Write $\sum_i \sigma_i T_x M_i$ as an element of $\mathrm{Gr}(x)$ `Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum`. If any two structures at x give the same sum `Chapter9/Tangent · EpGroupoid.HasIndependentTangentSums`, $T_\Theta(x)$ is that sum for a chosen structure, and 0 off the support `Chapter9/Tangent · EpGroupoid.tangentValue` `Chapter9/Tangent · EpGroupoid.tangentValue_eq`. For a morphism φ, a structure at $s(\varphi)$ is transported to $t(\varphi)$ by $\hat{\varphi}$ `Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.transport`, which maps the sheets' tangent sets by $T\varphi$ `Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_transport` `Chapter9/Tangent · EpGroupoid.tangentMor_eq_tangentMapOn_hat`; by the independence $T\varphi_*(T_\Theta(s(\varphi))) = T_\Theta(t(\varphi))$ `Chapter9/Tangent · EpGroupoid.grPush_tangentValue`, which makes T_Θ a functor `Chapter9/Tangent · EpGroupoid.tangentSection`.

Part 2, the independence (V9.14). *At a good point* there is no hypothesis: near it all sheets through it coincide, so the sum is $\Theta(y) \cdot T_y M$ for every structure `Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isGoodPoint` `Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_subset_goodSet`. *Equidimensional structures* `Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_isOfDim`: group the sheets of both structures through x by tangent set; for each class L, a tangent slice F (§IX.9) meeting each sheet of either structure with tangent space L once gives $\sum_{T_x M_i = L} \sigma_i = \sum_{y \in F} \Theta(y) = \sum_{T_x M'_j = L} \sigma'_j$. With sheets of one dimension a slice for one structure is one for the other. *In general* the text reduces to the equidimensional case through the dimensional decomposition, and so does the formalization: if the dimensional parts of two structures agree near x (Proposition 9.1.12), their tangent sums agree dimension by dimension `Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_sum_dimPartStructure` `Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_dimPartsAgreeNear` `Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_dimPartsAgreeNear`. With one graph chart for all dimensions and the local constancy of the dimension, which supplies the labellings, the independence follows `Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_of_hasGraphChartAt_of_hasLocallyConstantDimAt`; restricting a structure to a neighbourhood where the dimensions are constant is `Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.exists_restrict_labelled`. On every ep-groupoid over $\mathbb{R}^m$ both hypotheses hold, so T_Θ exists for every branched Θ `Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_rn`. Now Proposition 9.1.12 holds on every ep-groupoid and labelled structures exist at every point (§IX.27), so the independence, and with it Theorem 9.2.1, holds for every branched ep⁺-subgroupoid of every ep-groupoid `Chapter9/GeneralCorollaries · hasIndependentTangentSums`; the tangent slices of the equidimensional case exist at every point too `Chapter9/GeneralCorollaries · hasTangentSlicesAt`. In Remark 9.1.3 (b) the two structures have the same tangent sum at the bad origin `Chapter9/TangentSlicesRn · EpGroupoid.Remark913Hyp.tangentSum_structureβ_eq_structureγ`. The text writes "it suffices to prove the proposition" (p. 410) for the theorem and closes with "This completes the proof of Theorem 9.2.8" (p. 418), presumably Theorem 9.2.1; in $\sum_{i \in I \mid y \in M_j} \sigma_i T_y M_i$ (p. 418) the index is presumably i.

IX.11 Definition 9.2.12 (2017: 9.6): the tangent $T\Theta$

Definition 9.2.12 (2017: 9.6) (V9.23) `Chapter9/Tangent · EpGroupoid.tangentTheta`. $T\Theta(h) = 0$ over the complement of the support and $T\Theta(h) = \sum_{h \in TM_i} \sigma_i$ over $U(x)$, with $h \in TM_i$ read as $h \in T_{p(h)} M_i$ `Chapter9/Tangent · EpGroupoid.tangentSubmanifold`. The formal definition evaluates $T_\Theta(p(h))$ at the tangent

sets containing h Chapter9/Tangent · EpGroupoid.grEval, so it does not depend on a structure by construction; that it agrees with the text's local formula for every structure Chapter9/Tangent · EpGroupoid.tangentTheta_eq is the text's "the local definitions fit together", and comes from the independence through the identity $\sum_{h\in T_yM_i}\sigma_i=\sum_{h\in T_yM'_j}\sigma'_j$ Chapter9/Tangent · EpGroupoid.sum_tangent_eq_sum_tangent. $T\Theta\geq 0$ Chapter9/Tangent · EpGroupoid.tangentTheta_nonneg and it is a functor on the tangent ep-groupoid of Chapter 8 Chapter9/Tangent · EpGroupoid.isFunctorial_tangentTheta.

Is $T\Theta$ a branched ep⁺-subgroupoid? The text says so without discussion. It needs that the TM_i are smooth submanifolds of TX of smooth points, closed in $TU(x)$, and that $|\mathrm{supp}(T\Theta)|$ is paracompact; these are the fields of `TangentBranchingHyp` Chapter9/Tangent · EpGroupoid.TangentBranchingHyp, under which the theorem holds Chapter9/Tangent · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta Chapter9/Tangent · EpGroupoid.tangentBranchingStructure. *Closedness* is proved Chapter9/TangentSlices · EpGroupoid.isClosed_tangentSubmanifold. *Paracompactness* follows from that of $|X|$ Chapter9/TangentSlices · EpGroupoid.paracompactSpace_orbitSupp_tangentTheta. So only the smoothness of the TM_i remains Chapter9/TangentSlices · EpGroupoid.tangentBranchingHyp_of_isSmoothSubmanifold Chapter9/TangentSlices · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_of_isSmoothSubmanifold. TM_i is a sub-M-polyfold of TX, the image of the retraction Ts Chapter9/AppendixSlices · MPolyfold.IsSmoothSubmanifold.isSubMPolyfold_tangent Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.isSubMPolyfold_tangentSubmanifold. *That Ts is an sc⁺-retraction of TX* (gap G.99, closed): for the sc⁺-retraction s presenting a smooth finite-dimensional submanifold, $g=1^1_0\circ s$ is sc-smooth into X^1 and $Ts=T\iota\circ Tg$ for the inclusion $\iota:X^1\to X$. The tangent of ι is the natural inclusion $T(X^1)\to TX$ of Proposition 2.3.17, in a chart the tangent of the sc-operator $E^1\to E$, which is itself Chapter9/TangentSubmanifoldSmooth · retractTangentMap_shiftIncl Chapter9/TangentSubmanifoldSmooth · tangentLower_tangentMapOn_one. Hence Ts maps into $(TX)_1$ Chapter9/TangentSubmanifoldSmooth · tmap_mem_levelSet, and $1^1_0\circ Ts$ is Tg followed by the sc-diffeomorphism $T(X^1)\cong(TX)^1$, so Ts is an sc⁺-retraction of TX Chapter9/TangentSubmanifoldSmooth · isScPlusRetractionOn_tmap. So TA is a smooth submanifold of TX Chapter9/TangentSubmanifoldSmooth · IsSmoothSubmanifold.isSmoothSubmanifold_tangent, the field holds for every local branching structure Chapter9/TangentSubmanifoldSmooth · LocalBranchingStructure.isSmoothSubmanifold_tangentSubmanifold, and for $|X|$ paracompact $T\Theta$ is a branched ep⁺-subgroupoid of TX on every ep-groupoid Chapter9/TangentSubmanifoldSmooth · tangentBranchingHyp Chapter9/TangentSubmanifoldSmooth · isBranchedEpSubgroupoid_tangentTheta_general. On $\mathbb{R}^m$ this was proved first with the field as the only hypothesis Chapter9/AppendixSlices · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_rn; the tangent of a line in $\mathbb{R}^2$ is a sub-M-polyfold of $T\mathbb{R}^2$ Chapter9/AppendixSlices · rnPolyfold_isSubMPolyfold_tangent_graph2. The full hypothesis has the zero functor as witness Chapter9/Tangent · EpGroupoid.tangentBranchingHyp_zero Chapter9/Tangent · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_zero. The text's introduction (p. 408) calls $T\Theta$ a "branched ep-subgroupoid", Definition 9.2.12 an ep⁺-subgroupoid.

IX.12 Lemma 9.2.13 (2017: 9.9) and Remark 9.2.14 (2017: 9.5): the position of M relative to ∂X

Reduced tangent position in a chart (V9.24). T^R_xX is not a subspace of a vector space in the project; in a chart φ with $\varphi(x)=z$ in a partial quadrant, "$T_xM\subseteq T^R_xX$" is the inclusion of the chart image of T_xM in the reduced tangent space E_z Chapter9/BoundaryPosition · IsReducedTangentAt Chapter9/BoundaryPosition · isReducedTangentAt_iff_subset_reduced, equivalently in the partial quadrant or the partial cone at z Chapter9/BoundaryPosition · isReducedTangentAt_iff_subset_partialQuadrant Chap-

ter9/BoundaryPosition · isReducedTangentAt_iff_subset_partialCone. It does not depend on the chart Chapter9/BoundaryPosition · MPolyfold.isReducedTangentAt_chart_change Chapter9/BoundaryPosition · MPolyfold.isTangentReducedAt_iff_forall, and defines `IsTangentReducedAt` Chapter9/BoundaryPosition · MPolyfold.IsTangentReducedAt. $d_M(x)$ is the degeneracy index of the induced M-polyfold structure Chapter9/BoundaryPosition · MPolyfold.subDegIndex, at most $d_X(x)$ Chapter9/BoundaryPosition · MPolyfold.subDegIndex_le.

Lemma 9.2.13 (2017: 9.9). *For $x \in M$: (1) $x \in \partial X$ and $d_M(x) = 0$ iff (2) $T_xM \subseteq T^R_xX$.* **As printed (2) □ (1) does not hold at interior points**: for a point p of $\mathbb{R}^n$, $\{p\}$ is in reduced tangent position at p and $p \notin \partial\mathbb{R}^n$ Chapter9/BoundaryPosition · MPolyfold.not_lemma_9_2_13_two_one. The lemma is about $x \in M \cap \partial X$, where the text uses it, and is formalized there. *(1)* □ *(2)* Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13 Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13_chart: if $d_M(x) = 0$, Part I's Theorem 2.5.? (preprint: 2.31) gives a chart of M without boundary near x, whose tangent space lies in the reduced tangent space of X; the boundary hypothesis is not used, and the implication holds off ∂X too Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13_of_not_mem_boundary. *(2)* □ *(1)* Chapter9/BoundaryPositionConverse · MPolyfold.lemma_9_2_13_two_one. **The text's proof** builds a complement Y and a map F and asserts that the image is M near 0. **The formalized proof.** *Step 1.* In a chart, M is the image of an sc⁺-retraction s near z, with $T_zM \subseteq E_z$. *Step 2,* "the image is M near 0", which the text does not spell out: the projection p onto T_zM along a complement is injective on M near z, by a level-1 contraction with the mean value inequality Chapter9/BoundaryPositionConverse · ScRetraction.IsScPlus.exists_injOn_proj. *Step 3.* By Mathlib's inverse function theorem for $p \circ s$ on T_zM, $\chi = p(\cdot - z)$ is a chart of $M \cap G$ onto an open subset of $T_zM \subseteq E_z$, with inverse $s \circ (z + \cdot) \circ \Psi^{-1}$, a model without boundary Chapter9/BoundaryPositionConverse · ScRetraction.IsScPlus.exists_flat. *Step 4.* So $d_M(x) = 0$ Chapter9/BoundaryPositionConverse · MPolyfold.subDegIndex_eq_zero_of_isTangentReducedAt Chapter9/BoundaryPositionConverse · MPolyfold.degIndex_eq_zero_of_isReducedTangentAt. The complement Y and the map F are not needed. Together: the equivalence at $x \in \partial X$ Chapter9/BoundaryPositionConverse · MPolyfold.lemma_9_2_13_iff, and at every $x \in M$: $d_M(x) = 0 \iff T_xM \subseteq T^R_xX$ Chapter9/BoundaryPositionConverse · MPolyfold.subDegIndex_eq_zero_iff. The witness is a point of $\mathbb{R}^n$; a witness at a boundary point needs a submanifold API for quadrants, not yet available.

Remark 9.2.14 (2017: 9.5) (V9.25). ∂M is the boundary of the induced M-polyfold Chapter9/BoundaryPosition · MPolyfold.mem_boundary_subPolyfold_iff. *$x \in \partial M$ iff $x \in \partial X$ and $T_xM \not\subseteq T^R_xX$* Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_iff: □ with $d_M(x) \geq 1$ from Lemma 9.2.13 Chapter9/BoundaryPosition · MPolyfold.remark_9_2_14; □ since $\partial M \subseteq \partial X$ Chapter9/BoundaryPosition · MPolyfold.remark_9_2_14_mem_boundary and a point of M in reduced position has $d_M = 0$ Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_not_isTangentReducedAt Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_not_mem_boundary_iff.

IX.13 Proposition 9.2.15 (2017: 9.4), Definition 9.2.16 (2017: 9.7) and Remark 9.2.17 (2017: 9.6): the boundary $\partial\Theta$

The boundary of a sheet. $y \in \partial M$ is characterized through T_yM (Remark 9.2.14), so the set of y with $y \in \partial M_i$ Chapter9/Boundary · MPolyfold.submanifoldBoundary Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_iff is read off the tangent sets: $y \in \partial M$ iff $y \in \partial X$ and T_yM is not in reduced position Chapter9/Boundary · MPolyfold.IsReducedTangentSet Chapter9/Boundary · MPolyfold.isReducedTangentSet_iff. It is local and invariant under sc-diffeomorphisms Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_inter_isOpen_iff Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_image_iff.

Proposition 9.2.15 (2017: 9.4) (V9.26). *(1)* The boundary weight $\sum_{y\in\partial M_i}\sigma_i$ is the part of $T_\Theta(y)$ on tangent sets not in reduced position `Chapter9/Boundary · EpGroupoid.boundaryPart` `Chapter9/Boundary · EpGroupoid.filter_tangentValue_eq`; so, under the independence of Theorem 9.2.1, it is the same for any two structures whose domains contain y `Chapter9/Boundary · EpGroupoid.proposition_9_2_15_1`. *(2)* If $d_X(y)=0$, no tangent set is out of reduced position, so the weight and the formal sum vanish for every structure, with no independence `Chapter9/Boundary · EpGroupoid.proposition_9_2_15_2`. *(3)* $\partial\Theta$, 0 off ∂X and the boundary part of T_Θ on it `Chapter9/Boundary · EpGroupoid.boundaryTheta`, is a functor `Chapter9/Boundary · EpGroupoid.isFunctorial_boundaryTheta`, nonnegative, supported in ∂X and given by the local formula on every structure `Chapter9/Boundary · EpGroupoid.boundaryTheta_eq` `Chapter9/Boundary · EpGroupoid.proposition_9_2_15_3`. (1) and (2) need neither $x\in\operatorname{supp}(\Theta)\cap\partial X$ nor that Θ be branched.

Definition 9.2.16 (2017: 9.7) (V9.27). *$\partial\Theta$ is uniquely characterized by (1) $\operatorname{supp}(\partial\Theta)\subseteq\partial X$ and (2) the local formula at points of $\operatorname{supp}(\partial\Theta)$* `Chapter9/Boundary · EpGroupoid.IsBoundaryCharacterized`. $\partial\Theta$ has (1) and (2) `Chapter9/Boundary · EpGroupoid.isBoundaryCharacterized_boundaryTheta`. **The uniqueness, as we read it, does not hold**: (2) is asked only on $\operatorname{supp}(F)$, so $F=0$ has (1) and (2) for every Θ `Chapter9/Boundary · EpGroupoid.isBoundaryCharacterized_zero`, and for $\Theta=1$ on $[0,\infty)$, where $\partial\Theta(0)=1$ `Chapter9/Boundary · boundaryTheta_halfLine_one`, both 0 and $\partial\Theta$ do `Chapter9/Boundary · not_unique_isBoundaryCharacterized_halfLine`. With (2) asked at the points of $\operatorname{supp}(\Theta)$ and $F=0$ off $\operatorname{supp}(\Theta)$, $F=\partial\Theta$ `Chapter9/Boundary · EpGroupoid.eq_boundaryTheta`. **A reading with the printed (2)** `Chapter9/BoundaryCharacterization · definition_9_2_16_existsUnique`: among the $F\geq 0$ vanishing off $\operatorname{supp}(\Theta)$ with (1) and (2), there is exactly one whose support contains the supports of all the others, and it is $\partial\Theta$. *Step 1.* Such an F agrees with $\partial\Theta$ on its own support, by (2) and the local formula for $\partial\Theta$ `Chapter9/BoundaryCharacterization · IsBoundaryCharacterized.eq_boundaryTheta_on`, so its support lies in $\operatorname{supp}(\partial\Theta)$ `Chapter9/BoundaryCharacterization · IsBoundaryCharacterized.supp_subset`. *Step 2.* Hence $F=\partial\Theta$ iff its support contains $\operatorname{supp}(\partial\Theta)$ `Chapter9/BoundaryCharacterization · IsBoundaryCharacterized.eq_boundaryTheta_iff`, and $\partial\Theta$ itself has (1) and (2). Row V9.27 is narrower for this reason. Before the definition the text writes "In view of Proposition 9.2.13" (2017: "Proposition 9.9", the number of Lemma 9.2.13 there), presumably Proposition 9.2.15, and in it "branched ep⁺-groupoid".

Remark 9.2.17 (2017: 9.6) (V9.28). $\partial\Theta\leq\Theta$ `Chapter9/Boundary · EpGroupoid.boundaryTheta_le`, and $\partial\Theta$ is not Θ restricted to ∂X: for the indicator of the corner 0 of $[0,\infty)$, $\Theta(0)=1$ and $\partial\Theta(0)=0$ `Chapter9/Boundary · boundaryTheta_halfLineCorner`, while for $\Theta=1$ on $[0,\infty)$, whose sheet is all of $[0,\infty)$ `Chapter9/Boundary · MPolyfold.HasCornerStructure.isSmoothSubmanifold_univ` `Chapter9/Boundary · EpGroupoid.isBranchedEpSubgroupoid_one`, $\partial\Theta(0)=1$ `Chapter9/Boundary · EpGroupoid.boundaryTheta_one`. For the indicator of an interior point $\partial\Theta=0$ `Chapter9/Boundary · EpGroupoid.boundaryTheta_indicator_of_mem`. The witnesses satisfy the independence `Chapter9/Boundary · EpGroupoid.hasIndependentTangentSums_indicator_of_mem`. The remark's "In Section 9.5, Definition 9.3.6" refers to §9.3.

IX.14 Orientations: $\widehat{\mathrm{Gr}}(X)$, Definition 9.3.2 (2017: 9.8) and Remark 9.3.3 (2017: 9.8)

Oriented tangent sets (V9.29). An orientation of a finite-dimensional vector space is a class of bases; for a tangent set it is a set of frames, lists of vectors of the set, whose chart images `Chapter9/OrientationTangent · MPolyfold.chartVec` `Chapter9/OrientationTangent · MPolyfold.chartVec_injOn` are the positive bases of an orientation of the chart image `Chapter9/OrientationTangent · MPolyfold.OrTangent` `Chapter9/OrientationTangent · MPolyfold.chartFrames` `Chapter9/OrientationTangent · MPolyfold.IsChartOrientation` `Chapter9/OrientationTangent · MPolyfold.IsOrientedAt`. A chart change acts linearly

Chapter9/OrientationTangent · MPolyfold.exists_clm_chartVec, so this does not depend on the chart Chapter9/OrientationTangent · MPolyfold.IsOrientedAt.isChartOrientation Chapter9/OrientationTangent · MPolyfold.isOrientedAt_iff_forall; every tangent set with linear chart image has exactly two orientations Chapter9/OrientationTangent · MPolyfold.exists_two_orientations, and sc-diffeomorphisms and morphisms push them forward Chapter9/OrientationTangent · MPolyfold.OrTangent.push Chapter9/OrientationTangent · MPolyfold.IsOrientedAt.push_tangentMapOn Chapter9/OrientationTangent · EpGroupoid.isOrientedAt_push_tangentMor. $\widehat{\mathrm{Gr}}(x)$ is generated by oriented tangent sets Chapter9/OrientationTangent · EpGroupoid.GrHat Chapter9/OrientationTangent · EpGroupoid.grHatPush Chapter9/OrientationTangent · EpGroupoid.GrHatObj Chapter9/OrientationTangent · EpGroupoid.grHatProj, and forgetting the orientation is a functor over X^∞ Chapter9/OrientationTangent · EpGroupoid.grForget Chapter9/OrientationTangent · EpGroupoid.grForgetFunctor Chapter9/OrientationTangent · EpGroupoid.grForgetFunctor_comp_grProj, adding the two orientations of a set to the set Chapter9/OrientationTangent · EpGroupoid.grForget_single_add_single. Remark 9.3.1 (2017: 9.7), on an earlier notion of orientation, has no formal content.

Definition 9.3.2 (2017: 9.8) (V9.30) Chapter9/OrientationBasic · EpGroupoid.IsOrientation. An orientation of a submanifold is a continuous choice of orientations of its tangent sets Chapter9/OrientationBasic · MPolyfold.SubmanifoldOrientation, continuity being local constancy of the chart frames; an oriented local branching structure carries one per sheet Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure, with oriented sum $\sum_{y\in M_i}\sigma_i T_y(M_i,o_i)$ Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure.orientedSum forgetting to the tangent sum Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure.grForget_orientedSum. An orientation of Θ assigns to each smooth object an element of $\widehat{\mathrm{Gr}}$, naturally in morphisms, forgetting to T_Θ, and locally an oriented sum; the "lifts T_Θ" clause uses T_Θ, hence the independence of Theorem 9.2.1 Chapter9/OrientationBasic · EpGroupoid.IsOrientation.mk', which on $\mathbb{R}^m$ is free Chapter9/OrientationRn · EpGroupoid.isOrientation_sum_trivial_rn. It defines a section functor Chapter9/OrientationBasic · EpGroupoid.orientedSection Chapter9/OrientationBasic · EpGroupoid.orientedSection_comp_grHatProj Chapter9/OrientationBasic · EpGroupoid.orientedSection_comp_grForgetFunctor_obj. Witnesses: $\Theta = 0$ Chapter9/OrientationBasic · EpGroupoid.isOrientation_zero, and finite weighted sums of oriented closed submanifolds of $\mathbb{R}^n$ Chapter9/OrientationBasic · EpGroupoid.isOrientation_sum_rn.

Remark 9.3.3 (2017: 9.8) (V9.31) Chapter9/OrientationBasic · EpGroupoid.remark_9_3_3. For the indicator of a point p, $T_q = q\cdot(\{p\},+) + (1-q)\cdot(\{p\},-)$, $q\in(0,1)\cap\mathbb{Q}$, are pairwise distinct orientations: a 0-dimensional tangent set has two orientations, and any split of the weight 1 between them is an orientation.

IX.15 Lemma 9.3.4 (2017: 9.10) and Example 9.3.5 (2017: 9.1)

Lemma 9.3.4 (2017: 9.10) (V9.32). *Two orientations of Θ that agree at good points are the same.* **Without equidimensionality, as we read it, the lemma does not hold** Chapter9/OrientationRn · lemma_9_3_4_mixed_dim_rn Chapter9/OrientationUniqueness · lemma_9_3_4_mixed_dim: on $\mathbb{R}^2$ let $\Theta = 1_{\mathbb{R}\times 0} + 1_{(0,0)}$ Chapter9/OrientationUniqueness · mixedTheta Chapter9/OrientationUniqueness · mixedTheta_isBranched. The origin is bad Chapter9/OrientationUniqueness · zero_not_mem_goodSet_mixed, the good points are the other points of the line, and two orientations that orient the line the same way and the point $\{0\}$ oppositely agree at all good points and differ at the origin. The lemma is formalized for n-dimensional Θ Chapter9/OrientationUniqueness · EpGroupoid.IsOrientationOfDim.

The text's proof takes, at a bad x, the sequences of good points of Proposition 9.2.8 for the first structure and compares the orientations of the two structures along them. As we read it, this needs that the same

sequences serve the second structure, which the text does not show, and that frames of different sheets can be compared along a sequence. **The formalized proof** Chapter9/OrientationUniqueness · EpGroupoid.lemma_9_3_4. *Step 1.* Its hypothesis `HasOrientedGoodPointsAt` Chapter9/OrientationUniqueness · MPolyfold.HasOrientedGoodPointsAt asks, for each class of sheets of both structures with one tangent set at x, good points near x on every sheet of the class, on which the orientations of the sheets can be compared; this is Proposition 9.2.8 applied to both families together. Under it, the oriented sums at x agree class by class, which is the lemma. *Step 2,* from the ball graph representation (§IX.9) Chapter9/OrientationRn · MPolyfold.hasOrientedGoodPointsAt_of_hasGraphBallAt: over a ball of foot points in the class tangent space each sheet of the class is a graph, the good points are dense in it (Lemma 9.2.7 (1)), and the relative orientations of the sheets are locally constant. *Step 3,* local constancy: the relative orientation of two frame fields is the sign of a determinant Chapter9/OrientationRn · frameDet Chapter9/OrientationRn · orientation_eq_iff_frameDet Chapter9/OrientationRn · MPolyfold.frames_eq_iff_frameDet, continuous if the chart frames are continuous on level 1 Chapter9/OrientationRn · MPolyfold.continuousOn_chartVec_of_continuousOn Chapter9/OrientationRn · MPolyfold.SubmanifoldOrientation.exists_isOpen_tangent_eq_iff. This continuity was first the hypothesis `HasContinuousLevelOneOn` Chapter9/OrientationRn · MPolyfold.HasContinuousLevelOneOn, true on $\mathbb{R}^m$ Chapter9/OrientationRn · hasContinuousLevelOneOn_rn, and Chapter9/OrientationRn · EpGroupoid.lemma_9_3_4_of_hasGraphBallAt holds under the two hypotheses, on every ep-groupoid over $\mathbb{R}^m$ without them Chapter9/OrientationRn · EpGroupoid.lemma_9_3_4_rn Chapter9/OrientationRn · hasOrientedGoodPointsAt_rn; witness in dimension 0: Chapter9/OrientationUniqueness · hasOrientedGoodPointsAt_rn_zero. *Step 4, no hypothesis* (gap G.100, closed). The level-1 continuity holds along every smooth finite-dimensional submanifold of every M-polyfold Chapter9/GeneralLevelZero · hasContinuousLevelOneOn: on the submanifold the level-1 lift of the chart is $1^1_0 \circ t$ composed with the chart, for the chart sc$^+$-retraction t Chapter9/GeneralLevelZero · levelLift_eq_plusLift, continuous since t is sc$^+$. The ball graph representation holds at interior points, which gave the lemma for Θ supported where $d_X = 0$ Chapter9/GeneralCorollaries · lemma_9_3_4_of_hasGraphBallAt' Chapter9/GeneralCorollaries · lemma_9_3_4_of_degIndex_eq_zero. At boundary points a ball of foot points is not available, but it is not needed: the good direction h_0 off the other tangent spaces is chosen near an admissible direction h_1, so that every h near h_0 is admissible with one radius and one factor Chapter9/GeneralBoundaryPoints · exists_small_cone, and the tangent sheets are graphs over a small ball $\{u \in T \mid \|u - th_0\| < \tau t\}$ for t below a uniform bound Chapter9/GeneralBoundaryPoints · exists_graph_unif_chart. So the good-point input holds at every point Chapter9/GeneralBoundaryPoints · hasOrientedGoodPointsAt, and Lemma 9.3.4 holds for every n-dimensional Θ of every ep-groupoid Chapter9/GeneralBoundaryPoints · lemma_9_3_4_general, every orientation of an n-dimensional Θ being n-dimensional Chapter9/OrientationAnyDim · IsOrientation.isOrientationOfDim Chapter9/OrientationAnyDim · lemma_9_3_4_of_isNDimensional. Row V9.32 stays narrower only because of the mixed-dimensional counterexample above.

Example 9.3.5 (2017: 9.1) (V9.33) Chapter9/OrientationExample · example_9_3_5 Chapter9/OrientationRn · example_9_3_5_rn. Take $\beta^{++}(s) = e(s) + e(-s)$ with $e =$ `expNegInvGlue`, $\beta^{--} = -\beta^{++}$, β^{+-} equal to β^{++} for $s \le 0$ and to β^{--} for $s > 0$, and β^{-+} the other way Chapter9/OrientationExample · betaPP Chapter9/OrientationExample · betaMM Chapter9/OrientationExample · betaPM Chapter9/OrientationExample · betaMP. The two pairs of graphs cover each point equally often Chapter9/OrientationExample · example_9_3_5_count, which defines Θ Chapter9/OrientationExample · example935Theta Chapter9/OrientationExample · example935Theta_isBranched. Orienting the graphs by $\pm(1, \beta')$ Chapter9/OrientationExample · graphOrientation Chapter9/OrientationExample · chartVec_image_graph2 gives $\hat{T}^1$ from G^{++}, G^{--} and $\hat{T}^2$ from G^{+-}, G^{-+} Chapter9/OrientationExample · exampleT1 Chapter9/OrientationExample · exampleT2. Verified as stated: they coincide at the origin Chapter9/OrientationExample · example_9_3_5_eq_origin, differ at $(s, \beta^{++}(s))$ for $s > 0$ Chapter9/

OrientationExample · example_9_3_5_ne_of_pos, and no signs for the second structure reproduce $\hat{T}^1$ near the origin Chapter9/OrientationExample · example_9_3_5_cannot_coincide. **Two claims do not hold as we read them.** "Otherwise different": they also coincide wherever $x < 0$, where $G^{+-} = G^{++}$ and $G^{-+} = G^{--}$ with the same orientations Chapter9/OrientationExample · example_9_3_5_eq_of_neg. "Four possibilities to orient": the four sign choices for the first structure give four different orientations Chapter9/OrientationExample · orientedLift1_injective, and $\hat{T}^2$ is none of them Chapter9/OrientationExample · exampleT2_ne_orientedLift1. Counting both structures, the sign choices give exactly six orientations, the two structures sharing the two constant choices Chapter9/OrientationExampleSigns · orientedLift1_eq_orientedLift2_iff Chapter9/OrientationExampleSigns · orientedLift2_injective Chapter9/OrientationExampleSigns · ncard_signOrientations Chapter9/OrientationExampleSigns · example_9_3_5_six, and with fractional weights there are infinitely many, as in Remark 9.3.3. So "four possibilities" holds for each structure and, for Θ, does not hold as we read it; the full classification is §IX.29 (gap G.101, closed). The good points distinguish $\hat{T}^1$ from $\hat{T}^2$, in line with Lemma 9.3.4 Chapter9/OrientationRn · exists_goodSet_exampleT1_ne_exampleT2.

IX.16 Definition 9.3.6 (2017: 9.10), Lemma 9.3.7, and the boundary orientation ((9.11), Definition 9.3.8, Proposition 9.3.9, Definition 9.3.10)

Definition 9.3.6 (2017: 9.10) (V9.34) Chapter9/BoundaryPosition · EpGroupoid.TameBranchingStructure Chapter9/BoundaryPosition · EpGroupoid.IsTameBranched. A tame structure has a natural representation on its domain, the properness property, and sheets with tame induced structures in the sense of Part I's Definition 2.5.18. Witnesses: $\Theta = 0$ Chapter9/BoundaryPosition · EpGroupoid.isTameBranched_zero and a point of $\mathbb{R}^n$ Chapter9/BoundaryPosition · EpGroupoid.isTameBranched_indicator Chapter9/BoundaryPosition · MPolyfold.hasTameInducedStructure_of_degIndex_eq_zero. "In particular every M_i has an equivalent structure of a manifold with boundary with corners" (gap G.102, closed): for a tame structure with n-dimensional sheets, each M_i with the subspace topology is a Mathlib manifold with corners modelled on $[0,\infty)^n$ Chapter9/MathlibManifoldBridge · TameBranchingStructure.hasCornerStructureOfDim Chapter9/MathlibManifoldBridge · TameBranchingStructure.isManifold. *Step 1.* A tame induced structure gives corner coordinates near each point, charts onto relatively open subsets of partial quadrants of $\mathbb{R}^n$ with smooth transitions Chapter9/MathlibManifoldBridge · CornerCoords Chapter9/MathlibManifoldBridge · IsSubmanifoldOfDim.hasCornerStructureOfDim_of_hasTameInducedStructure. *Step 2.* A Mathlib manifold has one model, so each partial quadrant is moved into the full quadrant by a linear isomorphism Chapter9/MathlibManifoldBridge · exists_cle_partialQuadrant, which gives a charted space Chapter9/MathlibManifoldBridge · cornerChartedSpace with smooth transitions Chapter9/MathlibManifoldBridge · isManifold_cornerChartedSpace. The same bridge serves Part I's solution sets.

Lemma 9.3.7 (V9.35) Chapter9/BoundaryPosition · MPolyfold.lemma_9_3_7, new in the 2021 edition. *For tame Y and submanifolds M_1, M_2 of one dimension with tame induced structures, $T_yM_1 = T_yM_2$ implies $d_{M_1}(y) = d_{M_2}(y)$.* **The text's proof** describes T_yM in a tame chart of M by the cone C_yM and shows that $T_0O \cap C$ is the reduced tangent space. **The formalized proof.** For a submanifold with tame induced structure, $d_M(y)$ is the codimension of the reduced part of the chart image of T_yM, which depends only on that linear space and on the partial quadrant of the chart of Y Chapter9/BoundaryPosition · MPolyfold.subDegIndex_eq_quotDim_of_hasTameInducedStructure (Remark 2.5.9 in every chart). So equal tangent spaces give equal indices Chapter9/BoundaryPosition · MPolyfold.subDegIndex_eq_of_tangentSpace_eq Chapter9/OrientationBoundary · MPolyfold.subDegIndex_eq_of_submanifoldTangent_eq, with neither "Y tame" nor "of the same dimension". The text's "$T_0O \cap C = T_0^R O$" (a cone against a subspace) is read through the linearity of the chart image of T_0M.

Definition 9.3.8 (2017: 9.9) (V9.36) Chapter9/OrientationBoundary · MPolyfold.boundaryFrames. *For an oriented (M, o), $y \in \partial M$ with $d_M(y) = 1$: a basis $(a_1, \dots, a_{n-1})$ of $T_y\partial M$ is positive iff $(e, a_1, \dots, a_{n-1})$ is positive, e outward.* Linear algebra first: the convention orients a hyperplane K Chapter9/OrientationBoundary · exists_boundary_orientation, by the determinant criterion Chapter9/OrientationBoundary · exists_basis_orientation_iff_det_pos Chapter9/OrientationBoundary · det_cons_smul_add. In a tangent set: for $K \subseteq T_yM$ with $(m-1)$-dimensional chart image and $e \notin K$, the frames orient K Chapter9/OrientationBoundary · MPolyfold.isOrientedAt_boundaryFrames, and $e \mapsto ce + k$, $c > 0$, $k \in K$, does not change them Chapter9/OrientationBoundary · MPolyfold.boundaryFrames_eq; both orientations occur, on a graph Chapter9/OrientationBoundary · exists_boundaryFrames_graph. $T_y\partial M$ and e are data in this definition. That $T_y\partial M = T_y^R M$ and the outward half-space are determined by M through the cones C_yM, and that the resulting orientation is continuous along the stratum $\{d_M = 1\}$, is proved in §IX.30 (gap G.103, closed); row V9.36 is now explicit.

(9.11) and Proposition 9.3.9 (V9.37), new in the 2021 edition. Split the oriented sum at y by $\ell = d_{M_i}(y)$ Chapter9/OrientationBoundary · EpGroupoid.degreePart Chapter9/OrientationBoundary · EpGroupoid.degreePart_orientedSum; by Lemma 9.3.7 tangent sets of different degree never coincide Chapter9/OrientationBoundary · MPolyfold.HasSubDegree Chapter9/OrientationBoundary · MPolyfold.SubmanifoldOrientation.hasSubDegree_tangent_iff, so two oriented structures with the same oriented sum at y have the same degree-ℓ parts for every ℓ Chapter9/OrientationBoundary · EpGroupoid.eq_9_11. Applying a boundary operation $T_y(M, o) \mapsto T_y(\partial M, o_{\partial M})$, a function `bd` of the oriented tangent set, to the degree-1 part gives (9.12), which therefore does not depend on the structure Chapter9/OrientationBoundary · EpGroupoid.proposition_9_3_9. **Definition 9.3.10 (2017: 9.11)** (V9.38) is (9.12) with 0 off the support Chapter9/OrientationBoundary · EpGroupoid.boundaryLift; the condition $y \in \partial M_i$ is implied by $d_{M_i}(y) = 1$. The operation `bd` is now built from the sheet, and (9.12) and Proposition 9.3.9 hold with it Chapter9/BoundaryOperation · EpGroupoid.proposition_9_3_9_boundaryOp Chapter9/BoundaryOperation · EpGroupoid.boundaryOrientation; the continuity of $o_{\partial M}$ along ∂M and its extension to the faces (Figure 9.1) are §IX.30. Remark 9.3.11, on the non-tame case, has no formal content.

IX.17 §9.4: essential points and sets of measure zero (Definitions 9.4.1, 9.4.5, 9.4.7, Lemmas 9.4.2–9.4.8 (2017: 9.12–9.14, 9.11–9.15))

For Mathlib manifolds. Definitions 9.4.1 and 9.4.5 are about a smooth finite-dimensional manifold M with boundary and corners, formalized for Mathlib manifolds with corners. *Definition 9.4.1 (1)* (V9.39): $T_z^K M$ is the span of the unit limits $\varphi(z_k)/|\varphi(z_k)|$, $z_k \to z$ in $K \setminus \{z\}$, in a chart φ centred at z. In a chart it is the span of Mathlib's tangent cone of the chart image of K Chapter9/BranchingGeometryManifold · chartImage Chapter9/BranchingGeometryManifold · span_tangentConeAt_chartImage_eq_map Chapter9/BranchingGeometryMeasure · span_unitDirections Chapter9/BranchingGeometryMeasure · unitDirections Chapter9/BranchingGeometryManifold · relTangent, and a chart change maps it by the derivative Chapter9/BranchingGeometryManifold · relTangent_eq_map, which is the text's "independent of the choice of φ". *Definition 9.4.1 (2)*: z is essential if it is a limit of points of K where $T^K M$ is everything Chapter9/BranchingGeometryManifold · relTangent_eq_top_iff Chapter9/BranchingGeometryManifold · fullSet Chapter9/BranchingGeometryManifold · essentialSet Chapter9/BranchingGeometryManifold · essentialSet_eq. For two crossing lines in $\mathbb{R}^2$ the essential set is $\{0\}$ Chapter9/BranchingGeometryManifold · crossLines Chapter9/BranchingGeometryManifold · essentialSet_crossLines Chapter9/BranchingGeometryManifold · nonEssential_crossLines. *Lemma 9.4.2 (2017: 9.11)* (V9.40), "obvious from the definition": K^e is a closure taken in K Chapter9/BranchingGeometryManifold · lemma_9_4_2 Chapter9/BranchingGeometryManifold · lemma_9_

4_2_open. *Definition 9.4.5 (2017: 9.13)* (V9.43) Chapter9/BranchingGeometryManifold · IsMeasureZero: null in every chart; countable unions Chapter9/BranchingGeometryManifold · isMeasureZero_iUnion, a chart cover suffices Chapter9/BranchingGeometryManifold · isMeasureZero_of_forall_mem, and $\mathbb{R}^2$ is not null Chapter9/BranchingGeometryManifold · not_isMeasureZero_univ. *Lemma 9.4.6 (2017: 9.14)* (V9.44) Chapter9/BranchingGeometryManifold · lemma_9_4_6, "proved similarly as Lemma 2.8 in [57]": *Step 1.* At a point where $T_z^K M \neq T_z M$ there is a unit direction v off the span, and points of K near z lie outside a cone around v Chapter9/BranchingGeometryMeasure · eventually_abs_le_of_tangentConeAt. *Step 2.* A cone has positive density at its vertex Chapter9/BranchingGeometryMeasure · exists_pos_measure_cone_lt, so z is not a Lebesgue density point of K; by Lebesgue's density theorem the set of such z is null Chapter9/BranchingGeometryMeasure · addHaar_setOf_span_tangentConeAt_ne_top Chapter9/BranchingGeometryManifold · isMeasureZero_setOf_relTangent_ne_top. *Step 3.* A non-essential point has a neighbourhood in K of such points, by Lemma 9.4.2. The statement holds for every K, not only for $K(i,i')$.

For sheets in an M-polyfold. The sheets M_i are submanifolds of an M-polyfold, which in general carry no Mathlib manifold structure (a tame sheet does, §IX.16). So the essential points and null sets of the sheets are the data of a `SheetGeometry` Chapter9/BranchingGeometry · MPolyfold.SheetGeometry, with Lemma 9.4.4 (`ess_comm`), Lemma 9.4.6 (`isNull_diff_ess`) and the note of Definition 9.4.7 (`submanifoldTangent_eq`) as fields; it was first witnessed by pieces of graphs in $\mathbb{R}^2$, with the essential sets and null sets of the Mathlib part read in the chart $z \mapsto z_0$ Chapter9/BranchingGeometry · graphSheets Chapter9/BranchingGeometry · graphSheetGeometry. It is now built on the n-dimensional submanifolds of every M-polyfold with no hypothesis, Definitions 9.4.1 and 9.4.5 being read in the M-polyfold charts as in the text's proof of Lemma 9.4.4 Chapter9/SheetEssential · sheetGeometry (§IX.28; gap G.105, closed). So Lemma 9.4.8 and Propositions 9.4.9, 9.4.10 hold for every n-dimensional Θ with no sheet-geometry hypothesis Chapter9/SheetGeometryFree · lemma_9_4_8_of_isNDimensional Chapter9/SheetGeometryFree · proposition_9_4_9_of_isNDimensional Chapter9/SheetGeometryFree · proposition_9_4_10_of_isNDimensional.

Lemma 9.4.4 (2017: 9.13) (V9.42). *Essential for $K(i,i') \subseteq M_i$ iff for $K(i,i') \subseteq M_{i'}$*, "left to the reader". **As we read it, it needs sheets of one dimension** Chapter9/BranchingGeometry · lemma_9_4_4_mixed_dim: $\mathbb{R} \times 0$ as a subset of $\mathbb{R}$ is essential at every point, as a subset of $\mathbb{R}^2$ nowhere, since its relative tangent space is a line. For an n-dimensional Θ the text's reduction to $T_0^D N$ in a chart applies, and is now proved for n-dimensional submanifolds of every M-polyfold Chapter9/SheetEssentialChart2021 · lemma_9_4_4_chart and for the sheets of every structure of an n-dimensional Θ Chapter9/SheetMeasure · lemma_9_4_4_of_isNDimensional (§IX.28).

Definition 9.4.7 (2017: 9.14) (V9.45). $i \sim_y i'$ through a chain of links $y \in K(i_{p-1}, i_p)^e$ Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.IsLink Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.YEquiv is an equivalence relation Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.equivalence_yEquiv Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.ySetoid; equivalent sheets have the same tangent set at y Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.submanifoldTangent_eq_of_yEquiv; $\#P_y$ is Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.numClasses.

Lemma 9.4.8 (2017: 9.15) (V9.46) Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.lemma_9_4_8. $\{y \in M_U \mid \#P_y \geq 2\}$ *is null.* "Null in M_U" is null in every sheet through the point Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.IsNullIn. The text decomposes M_U by (I_y, P_y) and concludes that the bad part lies in $\bigcup_{j,j'} K(j,j')^{ne}$. As we read it, $K(j,j')^{ne}$ is null only in M_j and $M_{j'}$; inside a sheet M_i through y the proof uses $K(i,j)^{ne}$ for j not y-equivalent to i: if y had two classes, the class of i omits some $j \in I_y$, so $y \in K(i,j)$ is not essential. Hence $M_U \setminus M_U^*$ is null Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.MStar Chapter9/BranchingGeometry · MPolyfold.

SheetGeometry.isNullIn_diff_MStar.

IX.18 Property (T), Propositions 9.4.9 and 9.4.10 (2017: 9.5 and 9.6)

Property (T) (V9.41) Chapter9/BranchingGeometry · EpGroupoid.OrientedBranchingStructure.PropertyT: the orientation relation of two sheets with the same tangent space is the same at every common point with the same tangent space; the first structure of Example 9.3.5 has it Chapter9/BranchingGeometry · propertyT_example935B1. Under (T) the sign map $K(i,i')^e \to \{\pm 1\}$ is constant Chapter9/BranchingGeometry · EpGroupoid.tangent_eq_iff_of_mem_ess. **Lemma 9.4.3** (V9.41) Chapter9/PropertyTShrinking2021 · lemma_9_4_3 Chapter9/PropertyTShrinking2021 · lemma_9_4_3_propertyT. *Every oriented local branching structure has Property (T) after restriction to a smaller open $U'(x)$* (gap G.104, closed). **The text's proof** argues along sequences of points converging to x. **The formalized proof** finds, for each of the finitely many pairs (i,i'), an open neighbourhood of x on which the pair satisfies (T), and intersects Chapter9/PropertyTShrinking2021 · SubmanifoldOrientation.exists_isOpen_pairT Chapter9/PropertyTShrinking2021 · restrict Chapter9/PropertyTShrinking2021 · propertyT_restrict_iff. *Case 1, different dimensions:* two submanifolds with the same tangent set at a point have the same dimension there Chapter9/PropertyTShrinking2021 · HasDimAt.eq_of_submanifoldTangent_eq, so (T) asks nothing of such a pair. *Case 2, dimension 0:* a point where a submanifold is 0-dimensional is isolated in it, by the chord estimate with vanishing derivative Chapter9/PropertyTShrinking2021 · HasDimAt.exists_isOpen_eq_of_mem. *Case 3, equal tangent spaces at x:* where the tangent spaces agree, the relative orientation is the sign of a frame determinant, locally constant by the level-1 continuity (§IX.15) Chapter9/OrientationRn · MPolyfold.SubmanifoldOrientation.exists_isOpen_tangent_eq_iff. *Case 4, different tangent spaces at x:* they stay different nearby, since a continuous frame field of one, written in the chart, has a nonzero minor against the other Chapter9/PropertyTShrinking2021 · exists_chartTangent_field Chapter9/PropertyTShrinking2021 · exists_frameDet_ne_zero Chapter9/PropertyTShrinking2021 · exists_isOpen_submanifoldTangent_ne, so after shrinking (T) is empty for the pair.

Proposition 9.4.9 (2017: 9.5) (V9.47) Chapter9/BranchingGeometry · EpGroupoid.proposition_9_4_9. *Off a null set $M_U = M_U^* = \bigsqcup_J M_U^J$, and on M_U^J: $\Theta = \sigma_J$, $T_\Theta = \sigma_J T^J$, $\hat{T}_\Theta = a_J(T^J, o_J^a) + b_J(T^J, o_J^b)$, $a_J + b_J = \sigma_J$, $o_J^a = -o_J^b$. Step 1.* The null set is Lemma 9.4.8; the pieces are the level sets of $y \mapsto I_y$ Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.MStarPart. *Step 2.* On M_U^J all sheets of J are y-equivalent, so they have one tangent set T^J and $T_\Theta(y) = \sigma_J T^J$ Chapter9/BranchingGeometry · EpGroupoid.LocalBranchingStructure.tangentValue_eq_sum, which uses the independence of Theorem 9.2.1. *Step 3.* By (T) the orientations of the sheets of J at y fall into two opposite classes; $o_J^a = -o_J^b$ is "$o_J^a \neq o_J^b$ on the same tangent set", since it has exactly two orientations Chapter9/BranchingGeometry · MPolyfold.OrTangent.eq_or_eq_or_eq_of_isOrientedAt Chapter9/BranchingGeometry · MPolyfold.exists_ne_isOrientedAt; if one class is empty its orientation is the other one. The footnote's "measure zero" in M_U refers forward to Definition 9.5.1 and Lemma 9.5.2 (p. 437), consistently as we read it.

Proposition 9.4.10 (2017: 9.6) (V9.48) Chapter9/BranchingGeometry · EpGroupoid.proposition_9_4_10. *Two oriented structures on $U(x)$ with (a) the same weight sums and (b) the same oriented sums agree off a null set, piece by piece, as in (1)–(5).* **The text** takes the union B'' of the two structures, which describes 2Θ, applies Proposition 9.4.9 to B, B', B'' and intersects. **The formalized proof** takes $M_U^\dagger = M_U''^*$ for the union only, not intersected with M_U^* and $M_U'^*$, which (1)–(5) do not need. The pieces A_p are cut out by the finitely many values of $y \mapsto (I_y, I'_y, J_+(y), J'_+(y))$, with $J_\pm(y)$ the two orientation classes at y, so Property (T) is not used. The two structures of Example 9.3.5, all signs $+$, satisfy (a) and (b) Chapter9/BranchingGeometry · example935B1 Chapter9/BranchingGeometry · example935B2 Chapter9/BranchingGeometry · example935_hypotheses_9_4_10 Chapter9/BranchingGeometry · example935_MU_eq; at the origin G^{++} and G^{--} are tangent but

not 0-equivalent, so the set of Lemma 9.4.8 is not empty `Chapter9/BranchingGeometry · example935_two_le_numClasses`.

IX.19 §9.5: sets of measure zero and the σ-algebras $L(S)$, $L(\partial S)$ (Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2 (2017: Definitions 9.15–9.17, Lemma 9.16))

The setting of §9.5. Θ is oriented, tame, compact and n-dimensional, and $S = |\mathrm{supp}(\Theta)|$. Measure zero in a sheet M_i is that of Definition 9.4.5, which the project first had for Mathlib manifolds only (§IX.17). So, as in §9.4, the null sets of the sheets are data, now instantiated on every M-polyfold by the n-dimensional Hausdorff measure of the chart images (§IX.28; gap G.106, closed): a system `SheetNullSets` on a class $\mathcal{S}$ of sheets `Chapter9/Measure · SheetNullSets`, with null sets closed under subsets, countable unions `Chapter9/Measure · SheetNullSets.isNull_iUnion_countable` and finite unions `Chapter9/Measure · SheetNullSets.isNull_union`, local, induced alike by two sheets of $\mathcal{S}$ on their overlap (`isNull_transfer`), and a nonempty sheet not null in itself. A §9.4 sheet geometry with these properties is a `SheetMeasureGeometry` `Chapter9/Measure · MPolyfold.SheetMeasureGeometry` `Chapter9/Measure · MPolyfold.SheetMeasureGeometry.toSheetNullSets`, and then null preimages are §9.4's "null in M_U" `Chapter9/Measure · EpGroupoid.LocalBranchingStructure.hasNullPreimages_iff_isNullIn`. *The covering criterion* `Chapter9/Measure · SheetNullSets.isNull_of_cover`: a subset C of a sheet $N \in \mathcal{S}$ is null in N if near each of its points it is covered by finitely many sheets of $\mathcal{S}$ in which it is null; by locality and the transfer property.

Definition 9.5.1 (2017: 9.15) (V9.49) `Chapter9/Measure · EpGroupoid.IsMeasureZeroSupp`. $A \subseteq |X|$ has measure zero if every $x \in \mathrm{supp}(\Theta)$ has a tame local branching structure (Definition 9.3.6, the text's "suitable $U(x)$") with sheets in $\mathcal{S}$ whose preimages $M_i \cap \pi^{-1}(A)$ are null in M_i `Chapter9/Measure · EpGroupoid.LocalBranchingStructure.HasNullPreimages`. The variants ask this of some local branching structure `Chapter9/Measure · EpGroupoid.IsMeasureZeroSuppAny`, or of every structure with sheets in $\mathcal{S}$ `Chapter9/Measure · EpGroupoid.IsMeasureZeroSuppAll`; the standing assumption is that tame structures with sheets in $\mathcal{S}$ exist `Chapter9/Measure · EpGroupoid.HasTameSheetsIn`. Compactness and orientation are not used.

Lemma 9.5.2 (2017: 9.16) (V9.50) `Chapter9/Measure · EpGroupoid.lemma_9_5_2`. *The notion does not depend on the choice of the local branching structures.* **The text** cites the literature, without proof. **The formalized proof.** *Step 1, the core* `Chapter9/Measure · EpGroupoid.isMeasureZeroSuppAll_of_any`. Suppose some structure at every point has null preimages of A, and let B' be any structure with sheets in $\mathcal{S}$ and M'_j one of its sheets. Each $z \in M'_j \cap \pi^{-1}(A)$ lies in the support, so a structure B at z with null preimages exists. Near z, $M'_j \subseteq \mathrm{supp}(\Theta) \cap U_B = \bigcup_i M_i$, so $M'_j \cap \pi^{-1}(A)$ is covered near z by the finitely many M_i, in which it is null. The covering criterion gives that it is null in M'_j. *Step 2.* A tame structure is a structure `Chapter9/Measure · EpGroupoid.isMeasureZeroSuppAny_of_isMeasureZeroSupp`, and under the standing assumption every point has a tame one with sheets in $\mathcal{S}$; the three notions agree. *Step 3.* The sets of measure zero form a σ-ideal `Chapter9/Measure · EpGroupoid.isSigmaIdeal_isMeasureZeroSuppAll` `Chapter9/Measure · EpGroupoid.isSigmaIdeal_isMeasureZeroSupp`, and $S \neq \emptyset$ is not null, since a sheet through a point of the support is not null in itself `Chapter9/Measure · EpGroupoid.not_isMeasureZeroSuppAny_orbitSupp` `Chapter9/Measure · EpGroupoid.not_isMeasureZeroSupp_orbitSupp`.

The transfer property, as we read it, needs one dimension `Chapter9/Measure · isNull_transfer_mixed_dim`. Step 1 uses that a set null in the sheets M_i is null in M'_j on their overlap. For sheets of different dimensions this does not hold: the axis $\mathbb{R} \times 0$ has Lebesgue measure zero in the sheet $\mathbb{R}^2$ and is not null

in itself, a 1-dimensional sheet; so no single system contains the planar null sets and those of graphs. The lemma is therefore proved for structures whose sheets lie in one class, as do the n-dimensional structures of an n-dimensional Θ. That *every* structure of an n-dimensional Θ has n-dimensional sheets is now proved on every ep-groupoid, from the independence of Theorem 9.2.1 `Chapter9/SheetMeasure · isOfDim_of_isNDimensional`, so for n-dimensional Θ Lemma 9.5.2 holds as printed (§IX.28; row V9.50 explicit). With the Hausdorff null sets the transfer across dimensions does not hold as we read it either: the axis of $\mathbb{R}^2$ has 2-dimensional and not 1-dimensional measure zero `Chapter9/SheetMeasure · isHNull_mixed_dim`.

Definition 9.5.3 (2017: 9.16) and the remark after it (V9.51). *Step 1, a general fact.* For a family $\mathcal{N}$ of subsets, the σ-algebra generated by the Borel sets and $\mathcal{N}$ `Chapter9/Measure · borelNullSigma Chapter9/Measure · borel_le_borelNullSigma Chapter9/Measure · measurableSet_borelNullSigma_of_mem` is, if $\mathcal{N}$ is a σ-ideal `Chapter9/Measure · IsSigmaIdeal`, the family $\{A \mid A \,\Delta\, B \in \mathcal{N} \text{ for a Borel } B\}$ `Chapter9/Measure · borelNullCompletion Chapter9/Measure · measurableSet_borelNullSigma_iff`: that family is a σ-algebra, by $\bigcup_k A_k \,\Delta\, \bigcup_k B_k \subseteq \bigcup_k (A_k \,\Delta\, B_k)$ `Chapter9/Measure · symmDiff_iUnion_subset`, contains the Borel sets and $\mathcal{N}$, and lies in every σ-algebra containing both. *Step 2.* $L(S)$ is this σ-algebra on the subspace S for the sets of measure zero `Chapter9/Measure · EpGroupoid.measureZeroSets Chapter9/Measure · EpGroupoid.lebesgueSigma`, a σ-ideal under the standing assumption `Chapter9/Measure · EpGroupoid.isSigmaIdeal_measureZeroSets`. *Step 3.* The text's remark that $L(S)$ is obtained by adding the sets of measure zero is read as the completion: $K \in L(S)$ iff $K \,\Delta\, B$ has measure zero for a Borel B `Chapter9/Measure · EpGroupoid.remark_9_5_3`; sets of measure zero belong to $L(S)$ `Chapter9/Measure · EpGroupoid.measurableSet_lebesgueSigma_of_isMeasureZeroSupp`. The form $K = B \cup N$ would need Borel null hulls of null sets, which an abstract system does not provide.

Definition 9.5.14 (2017: 9.17) and the paragraph before it (V9.52). $\partial S = |\mathrm{supp}(\partial\Theta)| \subseteq S$ `Chapter9/Measure · EpGroupoid.orbitSupp_boundaryTheta_subset`. $(n-1)$-dimensional measure zero in the boundaries ∂M_i `Chapter9/Boundary · MPolyfold.submanifoldBoundary` is a second system on a class of boundaries, and K has measure zero in ∂S if at every point of $\mathrm{supp}(\partial\Theta)$ a tame structure has boundary preimages null in the ∂M_i `Chapter9/Measure · EpGroupoid.IsMeasureZeroBoundary Chapter9/Measure · EpGroupoid.IsMeasureZeroBoundaryAll`. *"A well-defined notion"* `Chapter9/Measure · EpGroupoid.isMeasureZeroBoundaryAll_of`: the proof of Lemma 9.5.2 applies to the boundaries. By the local formula for $\partial\Theta$ (Proposition 9.2.15), which holds for every structure given the independence of Theorem 9.2.1, a point of $\partial M'_j$ lies in $\mathrm{supp}(\partial\Theta)$, and a point of $\mathrm{supp}(\partial\Theta)$ lies in some ∂M_i of any structure at it, a positive sum having a nonzero term `Chapter9/Measure · EpGroupoid.exists_mem_of_sum_pos`; so the boundary preimage in $\partial M'_j$ is covered near each point by the ∂M_i of a structure with null boundary preimages. $L(\partial S)$ is the σ-algebra generated by the Borel sets of ∂S and these sets `Chapter9/Measure · EpGroupoid.boundaryMeasureZeroSets Chapter9/Measure · EpGroupoid.lebesgueSigmaBoundary`. The $(n-1)$-dimensional null sets themselves, a witness with $\partial S \neq \emptyset$ and the σ-algebra $L(\partial S)$ built with them are §IX.34 (gap G.108, closed).

Witnesses (V9.53). *Tools.* Measure zero of Definition 9.4.5 is local `Chapter9/Measure · isMeasureZero_image_of_local`, holds for subsingletons `Chapter9/Measure · isMeasureZero_of_subsingleton`, and on $\mathbb{R}^2$ is Lebesgue measure zero `Chapter9/Measure · isMeasureZero_rn2_iff`. On the trivial ep-groupoid of $\mathbb{R}^n$ every local branching structure restricts to a tame one, on the domain of a natural representation `Chapter9/Measure · exists_tameBranchingStructure_trivial_rn Chapter9/Measure · hasTameSheetsIn_trivial_rn`, and π is injective `Chapter9/Measure · injective_π_trivial_rn`. *Example 9.3.5.* The graph sheets of §9.4 carry a `SheetMeasureGeometry` `Chapter9/Measure · graphSheetMeasureGeometry Chapter9/Measure · graphNull`; for the Θ of the two graphs G^{++}, G^{--}, which touch at 0 `Chapter9/Measure · theta935 Chapter9/Measure · isBranchedEpSubgroupoid_theta935 Chapter9/Measure · hasTameSheetsIn_theta935 Chapter9/Measure · zero_mem_supp_theta935`, the point $|0|$ has measure zero `Chapter9/Measure · isMeasureZeroSupp_theta935_zero`, S does not `Chapter9/Measure · not_isMeasureZeroSupp_theta935`, and both are in $L(S)$ `Chapter9/Measure · lebesgueSigma_theta935`. *The plane.* For $\Theta = 1$ on $\mathbb{R}^2$,

with the open subsets of $\mathbb{R}^2$ as sheets and Lebesgue null sets Chapter9/Measure · planeSheets Chapter9/Measure · planeNull Chapter9/Measure · planeSheet Chapter9/Measure · isSmoothSubmanifold_univ_rn2 Chapter9/Measure · thetaPlane Chapter9/Measure · isBranchedEpSubgroupoid_thetaPlane Chapter9/Measure · mem_supp_thetaPlane Chapter9/Measure · hasTameSheetsIn_thetaPlane, some member of $L(S)$ is not Borel Chapter9/Measure · exists_mem_lebesgueSigma_not_borel_thetaPlane: every subset of the null axis is in $L(S)$, there are $2^{\mathfrak{c}}$ of them, and only $\mathfrak{c}$ Borel sets.

IX.20 §9.5: the isotropy factors (Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 (2017: Lemmas 9.18, 9.19))

The measures μ_ω of Theorems 9.5.4 and 9.5.5 are local weighted integrals of a form over the sheets, divided by $\#G_x$ or $\#G_x^{\mathrm{eff}}$. This subsection gives what the proofs of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 rest on, the orders of the isotropy groups and the decompositions of the preimages; the measure identities themselves are stated and proved in §IX.31 (Lemmas 9.5.7, 9.5.9–9.5.11, Proposition 9.5.13 (1)) and §IX.33 (Proposition 9.5.13 (2)), and rows V9.54–V9.56 are explicit.

Lemma 9.5.9 (2017: 9.18) (V9.54). **The text's proof** identifies $H = \{g \mid g * y = y\}$ with G_y, writes $G = \bigcup_{r\in R} rH$, chooses $V(y)$ invariant under H and disjoint from its other translates, notes that $\tilde{M}_j = V \cap M_j$ is a branching structure on V, decomposes the preimage of K as $\bigcup_r \varphi_r(\tilde{K})$, and uses $\#R = \#G/\#H$. **The formalized part.** *Step 1,* $\#R \cdot \#G_y = \#G_x$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_isotropy: the stabilizer H of y for the action of G_x on $U(x)$ is isomorphic to G_y by the natural representation (Chapter 7) Chapter9/MeasureEffective · EpGroupoid.NaturalRep.mem_stabilizer_iff_Φ, and R is `Quotient.out` on G_x/H. *Step 2, the slice* Chapter9/MeasureEffective · EpGroupoid.NaturalRep.exists_slice: inside any neighbourhood of y there is an open V with $\varphi_g(V) = V$ for $g \in H$ and $\varphi_g(V) \cap V = \emptyset$ for $g \notin H$: separate y from $\varphi_g(y) \neq y$ for the finitely many $g \notin H$, and make the neighbourhood H-invariant. *Step 3, the preimage.* $|\varphi_g(z)| = |z|$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.π_Φ, so φ_g maps $M_U = \mathrm{supp}(\Theta) \cap U$ to itself, Θ being a functor Chapter9/MeasureEffective · EpGroupoid.NaturalRep.mapsTo_Φ_MU; for $K \subseteq |M_U \cap V|$, $M_U \cap \pi^{-1}(K) = \bigcup_g \varphi_g(M_U \cap V \cap \pi^{-1}(K))$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion, the same with g running over the representatives r Chapter9/MeasureEffective · EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion_quotient, and the union over R is disjoint Chapter9/MeasureEffective · EpGroupoid.NaturalRep.pairwise_disjoint_image_Φ_out. The restriction $\tilde{M}_j$ is §IX.3's `restrictAt`: the project's structures pass through their base point, so at y the sheets through y are kept.

Lemma 9.5.10 (V9.55), new in the 2021 edition. *Step 1.* $\#T_x \cdot \#G_x^{\mathrm{eff}} = \#G_x$ for the subgroup T_x acting trivially near x Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_mul_card_trivialSubgroup. *Step 2.* Under LUC for the given $U(x)$ (Definition 7.2.18, `NaturalRep.HasLUC`), T_x is the kernel of the representation Chapter9/MeasureEffective · EpGroupoid.NaturalRep.kerΦ_eq_trivialSubgroup, hence lies in every stabilizer Chapter9/MeasureEffective · EpGroupoid.NaturalRep.kerΦ_le_stabilizer, and corresponds to T_y under $H \cong G_y$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.stabilizerEquivIsotropy_mem_trivialSubgroup_iff, so $\#T_y = \#T_x$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_trivialSubgroup_eq. *Step 3, (9.23).* Dividing Step 1 of Lemma 9.5.9 by $\#T_x = \#T_y$ gives $\#R \cdot \#G_y^{\mathrm{eff}} = \#G_x^{\mathrm{eff}}$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_effIsotropy.

Lemma 9.5.11 (2017: 9.19) (V9.56). The orders $\#G_x$ and $\#G_x^{\mathrm{eff}}$ are invariant along morphisms, by conjugation with φ (Lemma 7.2.6) Chapter9/MeasureEffective · EpGroupoid.card_isotropy_eq_of_mor Chap-

ter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_of_mor. A functor is constant on orbits Chapter9/MeasureEffective · EpGroupoid.IsFunctorial.eq_of_π_eq, so for structures on U, U' and any map $\hat{\varphi}$ of U onto U' with $|\hat{\varphi}(z)| = |z|$, $|M_U| = |M_{U'}|$ Chapter9/MeasureEffective · EpGroupoid.image_π_MU_eq; the maps φ_g of a natural representation are such maps Chapter9/MeasureEffective · EpGroupoid.NaturalRep.Φ_hyps. The "change of variables" in the integrals is the invariance of the sheet integration under the local sc-diffeomorphism $\hat{\varphi}$, and gives Lemma 9.5.11 as a measure identity in §IX.31.

Proposition 9.5.13 (V9.57), new in the 2021 edition. *(1)* For effective X, $T_x = \{1\}$, so $\#G_x^{\mathrm{eff}} = \#G_x$ Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_of_isEffective and the normalizing factors of the two integrals agree. *(2)* The reduced ep-groupoid X_R (Chapter 7) has the object M-polyfold of X, with Hausdorff and paracompact morphism space as instances (Hausdorff under LUC); functors Chapter9/MeasureEffective · EpGroupoid.isFunctorial_reduced_iff and local branching structures Chapter9/MeasureEffective · EpGroupoid.LocalBranchingStructure.toReduced of X and X_R coincide, and $\#G_x^{\mathrm{eff}} = \#G_x^R = \#(G_x^R)^{\mathrm{eff}}$ Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_reduced. The identity of the integrals, $\oint_{\hat{\Theta}} \omega = \int_{\hat{\Theta}_R} \omega$, is §IX.33 (row V9.57 restated).

IX.21 Appendix §9.6.1: Lemmas 9.6.1 and 9.6.2, Definition 9.6.3 and Lemma 9.6.4

In place of "measure zero." The text's proofs of Lemmas 9.6.1, 9.6.2 and 9.6.4 remove sets of smaller dimension or not containing a sheet near the point, "of measure zero", and move a point slightly. The formalization uses a topological covering lemma instead, with no Lebesgue measure and no Baire category. *Step 1.* If $A \cap W \subseteq N$ for an open $W \ni a$, then $\dim_a A \le \dim_a N$, by the inclusion of the chart tangent spaces Chapter9/Appendix · MPolyfold.HasDimAt.le_of_inter_subset. *Step 2*, the covering lemma Chapter9/Appendix · MPolyfold.exists_mem_forall_notMem: a nonempty open piece of a k-dimensional submanifold is not covered by finitely many submanifolds of smaller dimension closed in an open set, by induction on the family: removing one closed set of smaller dimension leaves a nonempty open piece, by Step 1. *Step 3*, "moving z slightly": a point of the piece off all the given sets Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsGeneric Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.exists_isGeneric.

Lemma 9.6.1 (V9.58) Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_1. *Two structures at x have the same top dimension* Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.topDim. If the first has top dimension k and the second $\ell < k$, a generic point of a k-dimensional sheet of the first has $\Theta > 0$ by the first structure and $\Theta = 0$ by the second. So $\dim_\Theta(x)$ is well defined Chapter9/Appendix · EpGroupoid.localTopDim Chapter9/Appendix · EpGroupoid.localTopDim_eq, $-\infty$ off the support Chapter9/Appendix · EpGroupoid.localTopDim_eq_bot.

Lemma 9.6.2 (V9.59) Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_2. *The unions M_U^k of the top-dimensional sheets agree on $U \cap U'$* Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.topUnion Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.mem_topUnion_iff_of_mem: a generic point of $M_U^k \setminus M_{U'}^k$ would have $\Theta > 0$ by the first structure and, off the lower-dimensional sheets of the second, $\Theta = 0$.

Definition 9.6.3 (V9.60) Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsTopGoodPoint Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsTopBadPoint, as in the text. At points top-good for both structures the top parts agree, with no hypothesis Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_isTopGoodPoint: near such a point the top part is $\sigma \cdot 1_M$ for both, and σ is read off Θ at a generic point of M.

Lemma 9.6.4 (V9.61). *Top-goodness does not depend on the structure.* **The text's proof**: near a top-good z, $\Theta = \sigma \cdot \delta_M + \Theta_0 + \cdots + \Theta_{k-1}$; a point of M near z off the lower-dimensional sets has $\Theta = \sigma$, and "similarly as good points", z is top-good for the second structure. **As we read it, this needs that the top weight of the second structure at y is σ**, an argument the text does not give: the second structure could a priori have, through y, several top-dimensional sheets with tangent spaces that differ. **The formalized proof** makes this step the hypothesis `TopMultiplicity` Chapter9/Appendix · MPolyfold.TopMultiplicity: at a point where Θ equals σ on generic points of a k-dimensional sheet, the top weight of any structure is σ. Under it the lemma holds Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_4 Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.isTopBadPoint_iff, the top parts agree on $U \cap U'$ Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_topMultiplicity, and Proposition 9.1.12 follows by the descending induction of p. 454: subtract the top parts Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.subOn and repeat Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_topMultiplicity Chapter9/Appendix · EpGroupoid.proposition_9_1_12_of_topMultiplicity. Witnesses: $k = 0$ on every M-polyfold Chapter9/Appendix · MPolyfold.topMultiplicity_zero, and $k \geq n$ on $\mathbb{R}^n$, where an n-dimensional submanifold is a neighbourhood of each of its points Chapter9/Appendix · rnPolyfold_mem_nhds_of_hasDimAt Chapter9/Appendix · topMultiplicity_rn_of_le; so Proposition 9.1.12 holds on $\mathbb{R}^0$ and $\mathbb{R}^1$ unconditionally Chapter9/Appendix · proposition_9_1_12_rn_zero Chapter9/Appendix · proposition_9_1_12_rn_one. The step itself is proved from the graph representation in §IX.22 Chapter9/AppendixSlices · MPolyfold.topMultiplicityAt_of_hasGraphChartAt Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.lemma_9_6_4_of_hasGraphChartAt, and now on every M-polyfold Chapter9/GeneralCorollaries · topMultiplicityAt Chapter9/GeneralCorollaries · topMultiplicity, so Lemma 9.6.4 holds for every ep-groupoid Chapter9/GeneralCorollaries · lemma_9_6_4_general Chapter9/GeneralCorollaries · isTopBadPoint_iff_general. In the proof the top-good point is called z and then y.

IX.22 Proposition 9.6.5, the formula on p. 457, and the proof of Proposition 9.1.12

Proposition 9.6.5 (V9.62) Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.proposition_9_6_5. *At a top-bad x, a class Ξ of top-dimensional indices with one tangent space splits as $I^1 \sqcup \cdots \sqcup I^p$ with sequences of top-good points $x_j^e \to x$ lying on M_i exactly for $i \in I^e$.* In (1) the text writes "$I_k = I^1 \sqcup \cdots \sqcup I^p$"; the I^e being subsets of Ξ, it reads as $\Xi = I^1 \sqcup \cdots \sqcup I^p$. **The slice form.** Under `HasGraphChartAt` at x, for every class and every neighbourhood W of x there is a finite top slice in W Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.IsTopSlice: it meets each sheet of the class once, misses the other top-dimensional sheets and all lower-dimensional ones, using a generic slice (§IX.9). Top slices in every W replace the top-good sequences.

The formula on p. 457 Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_sum_slices. $\Theta_k(x) = \sum_\Xi \sum_{y \in \text{slice}} \Theta(y)$; for one class the slice gives $\sum_{i \in \Xi} \sigma_i$ Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.sum_class_eq_sum_of_isTopSlice. The text's $\Theta_k(x_e^k)$ uses points x_e^k not introduced, presumably points x_j^e of the sequences. **The independence of the right-hand side**, which the text asserts while Ξ, I^e and x^e come from the first structure, is an argument the text does not give. It is proved by choosing one slice for the sheets of both structures, off all lower-dimensional sets of both, so that both class weights are the same sum of values of Θ Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.sum_class_eq_of_hasGraphChartAt Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_hasGraphChartAt. The

same grouping proves `TopMultiplicityAt` Chapter9/AppendixSlices · MPolyfold.TopMultiplicityAt Chapter9/AppendixSlices · MPolyfold.topMultiplicity_of_hasGraphChartAt, on $\mathbb{R}^m$ in all dimensions Chapter9/AppendixSlices · MPolyfold.topMultiplicity_rn. A per-dimension graph representation is not enough here: the lower-dimensional sheets through y must be avoided in the chart in which the top sheets are graphs, which is why one chart for all dimensions was asked. One cone per tangent space in that chart suffices, since the slice for a class only uses the cone of that class; with this change the slices exist at every point of every M-polyfold Chapter9/GeneralCorollaries · exists_slice_of_classGraphChart Chapter9/GeneralCorollaries · exists_generic_slice_general, and Proposition 9.6.5 and the formula on p. 457 hold for every ep-groupoid Chapter9/GeneralCorollaries · proposition_9_6_5_general Chapter9/GeneralCorollaries · sum_class_eq Chapter9/GeneralCorollaries · dimPart_topDim_eq.

The proof of Proposition 9.1.12. With the top parts agreeing near x, subtract them from both structures and repeat in lower dimensions Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_hasGraphChartAt Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_of_hasGraphChartAt; on $\mathbb{R}^m$ without hypothesis Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_rn, and on every ep-groupoid Chapter9/GeneralCorollaries · LocalBranchingStructure.dimPartsAgreeNear Chapter9/GeneralCorollaries · proposition_9_1_12. "This completes the proof of Proposition 9.6.5" (p. 457) refers to Proposition 9.1.12, Proposition 9.6.5 being stated without proof.

IX.23 Witnesses, counterexamples, and the hypotheses on $\mathbb{R}^m$

Hypotheses first taken on a general M-polyfold, now theorems. Each was proved on $\mathbb{R}^m$ first, and each now holds on every M-polyfold or every ep-groupoid; the statements under the hypotheses are kept. The independence of Theorem 9.2.1, `HasIndependentTangentSums`: first every Θ over $\mathbb{R}^m$ Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_rn and the indicator of a point Chapter9/Tangent · hasIndependentTangentSums_indicator, now every Θ Chapter9/GeneralCorollaries · hasIndependentTangentSums. Tangent slices `HasTangentSlicesAt`: $\mathbb{R}^m$ Chapter9/TangentSlicesRn · MPolyfold.hasTangentSlicesAt_rn, $\mathbb{R}^0$ Chapter9/Tangent · hasTangentSlicesAt_rn_zero, every M-polyfold Chapter9/GeneralCorollaries · hasTangentSlicesAt. Graph charts: `IsGraphChartAt`, `HasGraphChartsAt`, `HasGraphChartAt`, `HasGraphBallAt` at every point of $\mathbb{R}^m$ (§IX.9); on every M-polyfold the class form at every point Chapter9/GeneralLevelZero · hasClassGraphChartAt and the one-cone and ball forms at interior points Chapter9/GeneralLevelZero · hasGraphChartAt_of_mem_interior. The local constancy of the dimension Chapter9/GeneralLevelZero · MPolyfold.hasLocallyConstantDimAt, invariance of domain Chapter9/GeneralLevelZero · hasDomainInvarianceAt and the level-1 continuity Chapter9/GeneralLevelZero · hasContinuousLevelOneOn: every M-polyfold. `TopMultiplicity`: every M-polyfold Chapter9/GeneralCorollaries · topMultiplicity. `HasOrientedGoodPointsAt`: every point Chapter9/GeneralBoundaryPoints · hasOrientedGoodPointsAt. `TangentBranchingHyp`: every Θ with $|X|$ paracompact Chapter9/TangentSubmanifoldSmooth · tangentBranchingHyp. `SheetGeometry`: the n-dimensional submanifolds of every M-polyfold Chapter9/SheetEssential · sheetGeometry, first graphs in $\mathbb{R}^2$ Chapter9/BranchingGeometry · graphSheetGeometry. Property (T): after shrinking, every oriented structure Chapter9/PropertyTShrinking2021 · lemma_9_4_3, first Example 9.3.5 Chapter9/BranchingGeometry · propertyT_example935B1. `SheetNullSets`: the Hausdorff null sets of every M-polyfold Chapter9/SheetNull · sheetNullSets, first graphs and open subsets of $\mathbb{R}^2$ Chapter9/Measure · graphSheetMeasureGeometry Chapter9/Measure · planeNull. `HasTameSheetsIn`: every structure on $\mathbb{R}^n$ Chapter9/Measure · hasTameSheetsIn_trivial_rn and every tame n-dimensional Θ for the n-dimensional sheets Chapter9/SheetMeasure · hasTameSheetsIn_of_isNDimensional. `Remark913Hyp`: `expNegInvGlue` Chapter9/Propositions · EpGroupoid.remark913Hyp_expNegInvGlue.

Hypotheses and readings that remain, with their witnesses. The class hypotheses (a) and (b) of the first forms of Theorems 9.5.15–9.5.18 are derived from the text's hypotheses (§§IX.36–IX.38); the forms with them are kept. (a) `hcov` (Theorems 9.5.15, 9.5.16): at every point of $\operatorname{supp}(\partial\Theta)$ an oriented tame structure describing $\widehat{T}_\Theta$ whose nonempty boundary sheets lie in the class of the integration; witnesses: $\Theta = 0$ `Chapter9/BoundaryMeasure · hcov_zero`, compact boundary sheets `Chapter9/BoundaryMeasure · hcov_msheets_of_isCompact`, and the closed upper hemisphere `Chapter9/HemisphereWitness · e8w_hcov` (§IX.34). (b) `hcovS` (Theorems 9.5.17, 9.5.18): at every point of $\operatorname{supp}(\Theta)$ such a structure whose sheets lie in one graph chart, for 9.5.18 with the local unique continuation property; witness: the closed upper hemisphere, $m = 1$ `Chapter9/HemisphereWitness · e8w_stokes_hypotheses` `Chapter9/HemisphereStokes · E8W.e8_hemisphere_stokes`, and for the LUC part the same hemisphere (§IX.38). Not derived from the text: (c) the bound on the dimensions of submanifolds for the local finiteness in Proposition 9.1.11 off the support; witness $\mathbb{R}^m$ `Chapter9/DimDecompositionExists · rn_hasDimAt_le` (§IX.4). Readings: (d) "oriented, tame" in §9.5 as `hloc`, near every point of the support an oriented tame structure describing $\widehat{T}_\Theta$; derived when $d_X = 0$ on the support `Chapter9/CoveringFromText · hloc_of_degIndex_eq_zero`, witnessed by the circle `Chapter9/CircleWitness · circle_witness` (§IX.32). (e) (9.22) for $\tau = f \cdot \omega$ read through the evaluation on frames, `IsMul`, witnessed by constants (§IX.33).

Unconditional. On every ep-groupoid: Propositions 9.1.11 (read among local decompositions, local finiteness on the support) and 9.1.12, Theorem 9.2.1, Lemma 9.2.7 (2), Lemma 9.3.4 for n-dimensional Θ, Lemma 9.4.3, Lemmas 9.4.4, 9.4.6 for n-dimensional Θ, Lemma 9.5.2 for n-dimensional Θ, Lemma 9.6.4, Proposition 9.6.5, Remark 9.1.4, Proposition 9.1.9; Definition 9.2.12 with $|X|$ paracompact; Theorems 9.5.4, 9.5.5 and 9.5.15–9.5.18 under the reading (d).

Witnesses for §9.5. The unit circle in $\mathbb{R}^2$, counter-clockwise: Theorems 9.5.4, 9.5.5 with compact nonempty support `Chapter9/FiniteTheoremText · circle_theorem_9_5_4_full`, the branched integral as the integral over S^1 `Chapter9/CircleOneSheet · circle_multiBranchedIntegral`, Proposition 9.5.13 (2) `Chapter9/ReducedCircle · circle_proposition_9_5_13_2`. The half-line $[0,\infty)$ with $\Theta = 1$: $\partial S = \{|0|\} \neq \emptyset$ and $L(\partial S)$ `Chapter9/BoundaryFaceNullSets · halfLine_lebesgueSigmaBoundary`. The closed upper hemisphere: Theorems 9.5.15–9.5.18 with $\partial S \neq \emptyset$ (§§IX.35, IX.36, IX.38). The closed half-circle in $[0,\infty) \times \mathbb{R}$: the same for $n = 1$ (§IX.38). The groupoid $\mathbb{Z}_3 \ltimes X$ of Remark 9.5.20 (§IX.39). The x-axis of $\mathbb{R}^2$: Remark 9.1.10 `Chapter9/XAxisWitness · remark_9_1_10_hypotheses_axis` and the one-chart class `Chapter9/TameOpen · hcov_axis`.

Counterexamples (each to a statement as we read it): the uniqueness in Proposition 9.1.11 `Chapter9/Propositions · EpGroupoid.not_proposition_9_1_11_rn`; the footnote to Definition 9.1.2 for $\Theta = 0$ `Chapter9/LocallyEquidimensional2021 · definition_9_1_2_footnote_zero`; Lemma 9.2.13 (2) ⇏ (1) off ∂X `Chapter9/BoundaryPosition · MPolyfold.not_lemma_9_2_13_two_one`; the uniqueness in Definition 9.2.16 `Chapter9/Boundary · not_unique_isBoundaryCharacterized_halfLine`; Lemma 9.3.4 without equidimensionality `Chapter9/OrientationRn · lemma_9_3_4_mixed_dim_rn`; "otherwise different" and "four possibilities" in Example 9.3.5 `Chapter9/OrientationExample · example_9_3_5_eq_of_neg` `Chapter9/OrientationExampleSigns · example_9_3_5_six` `Chapter9/OrientationExampleClasses · example_9_3_5_infinite`; Lemma 9.4.4 for sheets of different dimensions `Chapter9/BranchingGeometry · lemma_9_4_4_mixed_dim`; the transfer of null sets in Lemma 9.5.2 across dimensions `Chapter9/Measure · isNull_transfer_mixed_dim` `Chapter9/SheetMeasure · isHNull_mixed_dim`.

IX.24 What is not formalized

Gaps G.93–G.109 are closed, G.110 is formalized in part, for Remark 9.5.20 only, and no gap of the chapter is open.

- **G.110 (in part), Remark 9.5.20.** Formalized: the M-polyfold $X \subseteq \mathbb{R} \times \mathbb{C}$, $\partial X = \{(\pm 1, 0)\}$, $\mathbb{Z}_3 \ltimes X$ with the local unique continuation property, ∂X as a sub-ep-groupoid, and the effective isotropy at a point of ∂X, three read in X and one read in ∂X (§IX.39). Not formalized: $\Theta(t, z) = \delta_0(z)$ with its orientation, $\oint_{\hat{\Theta}} df = \frac{1}{3}(\phi(1) - \phi(-1))$ and the two boundary integrals, estimated at 2,340–3,790 lines; the main risks are sc-smooth partitions of unity on the c_0 scale of the example and the tameness of M at the infinite-dimensional ends. Theorems 9.5.17 and 9.5.18 themselves are proved as printed for every $n \geq 1$ (§§IX.36–IX.38).
- **Closed: G.109**, Theorems 9.5.15 and 9.5.16 as printed for every $n \geq 1$ (§§IX.36–IX.38), and with it the Theorems 9.5.17, 9.5.18 part of G.110.
- **Left inside closed gaps.** Local finiteness of the decomposition of Proposition 9.1.11 off the support without a bound on the dimensions (G.96). The form $f \cdot \omega$ for a non-constant sc-smooth f, read through the evaluation (G.107); in dimension 0 the integration is the abstract one `Chapter9/PointIntegration · pointIntegration`. The orientations of the faces of the faces, the last sentence of the caption of Figure 9.1 (G.103). The comparison of the face-by-face null sets with those taken in the ambient charts (G.108), and of the essential points in M-polyfold charts with those in corner charts (G.105), which no statement needs. A witness of Lemma 9.2.13 at a boundary point.
- **Nothing to formalize.** Remarks 9.3.1, 9.3.11 and 9.5.8 (the last has no content beyond Lemma 9.5.11); Remarks 9.5.12 and 9.5.19 and §§9.6.2–9.6.3 discuss open questions.

IX.25 What the formalization taught us

The definition of a branched ep⁺-subgroupoid is local and says nothing about how the local pictures fit together, and three statements that read as uniqueness claims, Proposition 9.1.11, Definition 9.2.16 and Lemma 9.3.4, turned out to depend on exactly that: an indicator can be split into a point and a punctured line, a characterization asked only on its own support is met by zero, and a lower-dimensional sheet through a bad point escapes the good points. In each case the repair is a condition that ties the object to the local branching structures of Θ (local decompositions, the largest support, equidimensionality). The analytic heart of §9.2, the graph representation of tangent sheets near a bad point, is a statement about one chart in which sheets of all dimensions are graphs; separated from the counting it serves, it became a single hypothesis, true on $\mathbb{R}^m$ by the inverse function theorem, from which Proposition 9.1.12, Theorem 9.2.1, Lemma 9.6.4 and Proposition 9.6.5 follow. Two steps the text leaves to "similarly" (Lemma 9.6.4) or asserts (the independence on p. 457) needed that both structures be compared on one common slice. Finally, "measure zero" in §9.6.1 is not needed: the arguments only use that a piece of a k-dimensional submanifold is not covered by finitely many thinner ones, which is topology. In §9.5 the opposite happens: the measure-theoretic part needs only that the null sets of the sheets form a local σ-ideal compatible on overlaps, and that compatibility is exactly where dimensions must agree; and of the isotropy computations that normalize the integrals, everything except the integrals themselves is finite group theory.

The second round of work taught more. *The level-0 hypotheses were one fact.* Every level-0 statement of §9.2 (the cone, the graphs, invariance of domain, the local constancy of rank, the level-1 continuity of frames) follows from one observation: the lift $1_0^1 \circ r$ of an sc⁺-retraction to level 1 is C^1 on the retract, so that Mathlib's inverse function theorem applies on level 1 and the conclusions descend through the inclusion. The project's first hypothesis, one cone for all submanifolds through a boundary point, was more than the text uses; one cone per tangent space is enough, and holds everywhere (§IX.27). *A tacit finiteness in §9.5.* The text integrates ω over measurable subsets of the sheets, which presupposes $\int |\omega| < \infty$.

For the sheets themselves this follows from compactness and the properness of $\operatorname{cl} U(x)$ (§IX.32); for the boundary sheets $\partial_1 M_i$ it was first an added hypothesis of Theorems 9.5.15–9.5.18 (§§IX.34–IX.35), and it follows from the continuity of Part I's chart forms up to the corners and, toward the frontier of $U(x)$, from a comparison of tangent spaces with other structures, in which $\{d = 1\}$ depends on the tangent space alone (§§IX.36–IX.37). The same proof needs facts the text does not state: that the sheets are Lindelöf, so that countably many neighbourhoods suffice in Theorems 9.5.4, 9.5.5; that $M_i \cap \operatorname{supp} \beta_k$ is compact in the proof of Stokes' theorem; that chart changes are differentiable along a finite-dimensional submanifold for the level-0 norms, which makes Definitions 9.4.1 and 9.4.5 chart independent; and, for Proposition 9.5.13 (2), a natural action of G_x^R built from that of G_x under the local unique continuation property. *Some hypotheses are not used.* Proposition 9.1.9 does not use the paracompactness of $|X|$, the continuity of $o_{\partial M}$ needs no tameness, only reduced codimension 1, and the effective clause of Lemma 9.5.11 needs no local unique continuation. *Examples are more rigid than they look.* In Example 9.3.5 every one-dimensional submanifold inside the support through the origin has tangent $\mathbb{R}(1,0)$ and meets every vertical line, and this rigidity determines all orientations: a three-parameter family, of which six have integer weights (§IX.29). *Paracompactness through metrizability.* In Remark 9.1.4 the step from a paracompact neighbourhood to a component compact set goes through metrizability, since a subspace of a paracompact space need not be paracompact (§IX.26).

Remark 9.5.20 needs infinite dimensions. In finite dimensions a finite group that fixes a boundary point and acts trivially on the face acts trivially on the inward normal as well, so the effective isotropy of a boundary point is the same read in X and in ∂X. The remark's example has to be an M-polyfold whose local dimension jumps at its boundary, $0 \to 3 \to 0$ (§IX.39).

IX.26 Remark 9.1.4, Proposition 9.1.9 (2017: 9.1) and Remark 9.1.10 (2017: 9.2)

Remark 9.1.4 (V9.63), new in the 2021 edition `Chapter9/OrbitSubsetMetrizable · EpGroupoid.remark_9_1_4`. *A paracompact $L \subseteq |X|$ is metrizable, and if $|\operatorname{supp}(\Theta)|$ is component compact, the paracompactness in Definition 9.1.2 (1) is automatic.* **The text's argument**, followed exactly. *Step 1.* $|X|$ is locally metrizable, regular and Hausdorff (Theorem 7.3.1), so a paracompact subspace is metrizable by Smirnov's theorem `Chapter9/OrbitSubsetMetrizable · EpGroupoid.remark_9_1_4_metrizable`. *Step 2.* A component compact K has an open paracompact neighbourhood U (Corollary 7.3.8), metrizable by Step 1 `Chapter9/OrbitSubsetMetrizable · EpGroupoid.remark_9_1_4_exists_nhds`. *Step 3,* which the text does not spell out: K is paracompact. A subspace of a paracompact space need not be paracompact, but a subspace of the metrizable U is metrizable, hence paracompact `Chapter9/OrbitSubsetMetrizable · EpGroupoid.metrizableSpace_of_isComponentCompact Chapter9/OrbitSubsetMetrizable · EpGroupoid.remark_9_1_4_paracompactSpace`. *Step 4.* So a functor with the other properties of Definition 9.1.2 and component compact $|\operatorname{supp}(\Theta)|$ is a branched ep⁺-subgroupoid `Chapter9/OrbitSubsetMetrizable · EpGroupoid.isBranchedEpSubgroupoid_of_isComponentCompact`, compact supports in particular `Chapter9/OrbitSubsetMetrizable · EpGroupoid.paracompactSpace_orbitSupp_of_isCompact`.

Proposition 9.1.9 (2) (V9.64), *for Θ of orbifold type, X induces a natural orbifold structure on $Z = |\operatorname{supp}(\Theta)|$.* **The text's proof.** Near $x \in \operatorname{supp}(\Theta)$ all sheets coincide, $\Theta = 1_{M_x}$; the groupoid with objects $\bigsqcup_x M_x$ and morphisms $\bigsqcup M(x,x')$ is étale and proper, the functor $\beta : M \to Z$ induces a homeomorphism $|\beta|$, and other choices give an equivalent structure. **The formalized proof.** *Step 1, one sheet.* Θ takes only the values 0 and 1, so a sheet missing a point of the support near x would make

$0 < \Theta < 1$ there; hence $\mathrm{supp}(\Theta) \cap U(x) = M_i$ for every i Chapter9/SupportOrbifold · EpGroupoid.supp_inter_eq_of_isOrbifoldType. *Step 2.* So $\mathrm{supp}(\Theta)$ is a smooth finite-dimensional submanifold of X Chapter9/SupportOrbifold · EpGroupoid.isSmoothSubmanifold_supp_of_isOrbifoldType, saturated, hence an ep-subgroupoid, whose full subcategory is an ep-groupoid (Proposition 7.1.22) Chapter9/SupportOrbifold · EpGroupoid.isEpSubgroupoid_supp_of_isOrbifoldType. *Step 3.* For a saturated A, the map $|A| \to \pi(A)$ induced by the inclusion is a homeomorphism Chapter9/SupportOrbifold · EpGroupoid.orbitSuppHomeomorph. This gives a choice-free form Chapter9/SupportOrbifold · EpGroupoid.proposition_9_1_9_two. *Step 4, the text's groupoid.* For a choice of local branching structures at all points of the support Chapter9/SupportGroupoidNaturality · SheetChoice, the sets $M_x = \mathrm{supp}(\Theta) \cap U(x)$ form an open cover Chapter9/SupportGroupoidNaturality · sheetCover, and the text's M is the refinement of Proposition 10.1.3 of the full ep-subgroupoid along it Chapter9/SupportGroupoidNaturality · textGroupoid: objects (x, y) with $y \in M_x$, morphisms $((x, x'), \varphi)$ with $s(\varphi) \in M_x$, $t(\varphi) \in M_{x'}$. It is an ep-groupoid, $|\beta|$ is a homeomorphism Chapter9/SupportGroupoidNaturality · textOrbitHomeomorph, and $M \to \mathrm{supp}(\Theta)$ is an equivalence over Z Chapter9/SupportGroupoidNaturality · proposition_9_1_9_two_text. *Step 5, independence.* A refinement of covers gives an equivalence of refinements Chapter9/SupportGroupoidNaturality · OpenCover.isEquivalence_refineFunctor; for two choices c, c' the common refinement by the $M_x \cap M'_{x'}$ gives $M_c \leftarrow M_{c,c'} \to M_{c'}$, a generalized isomorphism of Chapter 10 (Theorem 10.3.10) compatible with the homeomorphisms to Z Chapter9/SupportGroupoidNaturality · proposition_9_1_9_two_independent. "Equivalent orbifold structure" is read as this generalized isomorphism. The construction assumes $\mathrm{supp}(\Theta) \neq \emptyset$; for $\Theta = 0$ there is nothing to construct.

Proposition 9.1.9 (1), *for Θ of manifold type, a natural manifold structure on Z. Step 1.* Between two objects of the support there is at most one morphism, so the isotropy of the ep-subgroupoid is trivial Chapter9/SupportManifold · EpGroupoid.subsingleton_isotropy_of_isManifoldType, and π is injective on the domain of a natural representation Chapter9/SupportManifold · EpGroupoid.injOn_π_of_subsingleton_isotropy, the text's "Ψ_x is injective". *Step 2, descent.* A general lemma: a continuous, open, surjective, locally injective map p from an M-polyfold to a Hausdorff paracompact space, such that $p(a) = p(b)$ is realized near a by an sc-smooth f with $f(a) = b$ and $p \circ f = p$ Chapter9/DescentPolyfold · MPolyfold.IsDescentMap, pushes the charts forward Chapter9/DescentPolyfold · MPolyfold.descChart to compatible charts, an M-polyfold structure on the target Chapter9/DescentPolyfold · MPolyfold.descend for which p is a local sc-diffeomorphism Chapter9/DescentPolyfold · MPolyfold.isLocalScDiffeo_descend. *Step 3.* The orbit map $\mathrm{supp}(\Theta) \to Z$ is such a map, with $f = t \circ (s|U(\varphi))^{-1}$ Chapter9/SupportManifold · EpGroupoid.isDescentMap_orbitSupp, which gives the structure Chapter9/SupportManifold · EpGroupoid.proposition_9_1_9_one. *Step 4, "natural"* is read as uniqueness: any two M-polyfold structures on Z for which the orbit map is a local sc-diffeomorphism agree Chapter9/SupportGroupoidNaturality · EpGroupoid.proposition_9_1_9_one_unique. The paracompactness of $|X|$, a hypothesis of the proposition, is not used.

Remark 9.1.10 (V9.65, narrower). *For Θ n-dimensional with $d_{|X|} = 0$ on Z, the structures are classical. Step 1.* The support is an n-dimensional submanifold of degeneracy index 0 Chapter9/SupportClassicalStructures · EpGroupoid.isSubmanifoldOfDim_supp Chapter9/SupportClassicalStructures · EpGroupoid.degIndex_eq_zero_of_mem_supp, so it has a corner structure of dimension n (§IX.16), which a surjective local sc-diffeomorphism carries to Z Chapter9/SupportClassicalStructures · MPolyfold.HasCornerStructureOfDim.of_isLocalScDiffeo_surjective. *Step 2.* In case (1) Z is a Mathlib manifold, `IsManifold (□ n) ∞` Chapter9/SupportClassicalStructures · EpGroupoid.remark_9_1_10_manifold, and the orbit map is a Mathlib local diffeomorphism Chapter9/SupportGroupoidEtale · EpGroupoid.remark_9_1_10_manifold_proj. *Step 3.* In case (2), for every choice, the object and morphism spaces of M are Mathlib n-manifolds Chapter9/SupportClassicalStructures · EpGroupoid.remark_9_1_10_orbifold, and source, target, unit, inverse and the equiv-

alence to the support are Mathlib local diffeomorphisms Chapter9/SupportGroupoidEtale · EpGroupoid.remark_9_1_10_orbifold_etale, since a local sc-diffeomorphism between such corner structures of degeneracy index 0 is a Mathlib local diffeomorphism Chapter9/SupportGroupoidEtale · MPolyfold.isLocalDiffeomorph_of_isLocalScDiffeo. The row is narrower for two reasons: the dimension n is fixed, a Mathlib manifold having one model space while the text's Z may have components of different dimensions; and the composition of M is sc-smooth, as for every ep-groupoid, but not stated as Mathlib-smooth, Mathlib having no notion of an orbifold. Witness: the x-axis of $\mathbb{R}^2$ Chapter9/XAxisWitness · EpGroupoid.remark_9_1_10_hypotheses_axis.

IX.27 The level-0 facts on every M-polyfold: Remark 9.1.1 (2017: 9.1), Lemmas 9.2.9–9.2.11 (2017: 9.6–9.8), invariance of domain, and the level-1 continuity

These are the facts the first version of this part took as hypotheses (gaps G.93, G.97, G.98, G.100, all closed). **The text** works at level 0 in a chart, with $x = 0$ in a partial quadrant C and M the image of an sc⁺-retraction: $L = T_0M$ meets C in a set with interior in L (Lemma 9.2.9); M is a graph $a + B(a)$ over a cone in $L \cap C$ with $DB(0) = 0$ (Lemma 9.2.10), by inverting $\Phi(a) = a + PB(a)$ (Lemma 9.2.11); $\dim T_mM$ is locally constant (Remark 9.1.1); a point in the interior of the sets it meets is "clearly" good (Lemma 9.2.7 (2)). The text treats B as C^1 at level 0, while the retraction is only sc-smooth.

The level-1 calculus of an sc⁺-retraction. *Step 1.* For an sc⁺-retraction r the lift $1^1_0 \circ r$ into level 1 Chapter9/GeneralLevelZero · plusLift is C^1 on level 1 Chapter9/GeneralLevelZero · hasFDerivWithinAt_plusLift Chapter9/GeneralLevelZero · continuousOn_plusDeriv, strictly differentiable Chapter9/GeneralLevelZero · exists_strict_plusLift, and its derivative brought down by the inclusion ι is Dr Chapter9/GeneralLevelZero · toBase_plusDeriv. All estimates are proved on level 1 and brought down by ι; on the finite-dimensional $T = \operatorname{range} Dr(z)$ the two norms are equivalent. *Step 2, the chord estimate* ($DB(0) = 0$): chords of the retract are close to T Chapter9/GeneralLevelZero · exists_approx Chapter9/GeneralLevelZero · exists_strict_range. *Step 3, Lemma 9.2.9:* Dr maps into the cone of the quadrant Chapter9/GeneralLevelZero · nonneg_scDeriv, so a closed ball of T around an admissible direction h_1, scaled by small s, stays in C Chapter9/GeneralLevelZero · ScRetraction.exists_cone. *Step 4, Lemmas 9.2.10 and 9.2.11:* the projection of the retract onto T approximates the identity of T on small balls, so Mathlib's surjectivity theorem for maps approximating a linear map (`ApproximatesLinearOn`) inverts it over a cone, uniformly in the foot point Chapter9/GeneralLevelZero · exists_graph_unif Chapter9/GeneralLevelZero · exists_graph, and over a ball of T at interior points Chapter9/GeneralLevelZero · exists_graph_ball. *Step 5, invariance of domain:* if two sc⁺-retractions r_1, r_2 have the same tangent space at z, every point b of the first retract near z with $r_2(b)$ in the first retract is fixed by r_2. On level 1, with u the difference of the lifts of $r_2(b)$ and b, the chord estimate for both retractions gives $\|u\|_1 \le c\eta\|u\|_1$ with $c\eta < 1$, so $u = 0$ Chapter9/GeneralLevelZero · eq_of_range_eq. *Step 6, local constancy of the rank:* the idempotents $Dr(z)$, read on level 1, depend continuously on the point in the operator norm, and an idempotent at distance < 1 from another has rank at most that of the other Chapter9/GeneralLevelZero · finrank_range_le_of_norm_sub_lt; applied both ways, the rank is locally constant along the retract Chapter9/GeneralLevelZero · eventually_finrank_eq.

In an M-polyfold. *Step 7.* A submanifold A is, in a chart, the range of a chart sc⁺-retraction t Chapter9/GeneralLevelZero · exists_plusRetraction, whose derivative has range the chart tangent space Chapter9/GeneralLevelZero · chartTangent_eq_range, and on A the level-1 lift of the chart is $1^1_0 \circ t$ Chapter9/GeneralLevelZero · levelLift_eq_plusLift; so the dimension is the rank of Dt Chapter9/GeneralLevelZero · hasDimAt_finrank. *Step 8, consequences.* Remark 9.1.1 Chapter9/GeneralLevelZero · MPolyfold.

hasLocallyConstantDimAt (Step 6), invariance of domain Chapter9/GeneralLevelZero · hasDomainInvarianceAt (Step 5), the level-1 continuity along submanifolds Chapter9/GeneralLevelZero · hasContinuousLevelOneOn (Step 7). *Step 9, the graph representation.* Each tangent space T_xA has its admissible cone of directions Chapter9/GeneralLevelZero · goodDirections, and every submanifold through x is a graph over it Chapter9/GeneralLevelZero · isGraphChartAt_goodDirections. This is the class form `HasClassGraphChartAt`, one cone per tangent space, at every point Chapter9/GeneralLevelZero · hasClassGraphChartAt; at interior points one cone and the ball form serve all submanifolds Chapter9/GeneralLevelZero · exists_chart_mem_interior_of_degIndex_eq_zero Chapter9/GeneralLevelZero · isGraphBallAt_of_mem_interior Chapter9/GeneralLevelZero · hasGraphChartAt_of_mem_interior.

What follows Chapter9/GeneralCorollaries · exists_slice_of_classGraphChart. The slice arguments of §§IX.9–IX.10 and IX.21–IX.22 only use the cone of the class at hand, so they run with the class form: tangent slices at every point Chapter9/GeneralCorollaries · hasTangentSlicesAt, generic slices missing finitely many thinner sets Chapter9/GeneralCorollaries · exists_generic_slice_general, `TopMultiplicity` Chapter9/GeneralCorollaries · topMultiplicity, Proposition 9.1.12 Chapter9/GeneralCorollaries · proposition_9_1_12, Lemma 9.6.4 Chapter9/GeneralCorollaries · lemma_9_6_4_general, Proposition 9.6.5 Chapter9/GeneralCorollaries · proposition_9_6_5_general, the independence of Theorem 9.2.1 Chapter9/GeneralCorollaries · hasIndependentTangentSums, Lemma 9.2.7 (2) Chapter9/GeneralCorollaries · lemma_9_2_7_2_general, Proposition 9.1.11 in the reading of §IX.4 Chapter9/GeneralCorollaries · existsUnique_localDimDecomposition_general, and, with the boundary argument of §IX.15, Lemma 9.3.4 Chapter9/GeneralBoundaryPoints · lemma_9_3_4_general. Proposition 9.2.8 itself stays in slice form (V9.22).

IX.28 Lemmas 9.4.4 and 9.4.6 (2017: 9.13, 9.14) for the sheets of an M-polyfold, and Lemma 9.5.2 (2017: 9.16) with Lebesgue null sets

The sheets of an M-polyfold carry no manifold charts of their own, so Definitions 9.4.1 and 9.4.5 are read in the M-polyfold charts ψ of the ambient space, as the text itself does in the proof of Lemma 9.4.4 (gaps G.105, G.106, closed).

Definition 9.4.1 in M-polyfold charts. T_z^KM is the span of Mathlib's tangent cone of $\psi(K)$ at $\psi(z)$ in E_0 Chapter9/SheetEssentialChart2021 · chartRelTangent, and z is essential if it is a limit of points where $T^KM = TM$ in every chart Chapter9/SheetEssentialChart2021 · IsRelTangentFull Chapter9/SheetEssentialChart2021 · chartEssentialSet. *Step 1.* For $K \subseteq M$, $T_z^KM \subseteq T_zM$, since chords of the chart retract are approximated by its tangent space (§IX.27, Step 2) Chapter9/SheetEssentialChart2021 · chartRelTangent_le. *Step 2.* Conversely $T_zM \subseteq T_z^MM$: for level-1 directions p into the quadrant, $Dr(z)p$ is a limit direction of the retract, and these span the range of $Dr(z)$ since E_1 is dense in E_0 Chapter9/SheetEssentialChart2021 · ScRetraction.range_scDeriv_subset_span_tangentConeAt Chapter9/SheetEssentialChart2021 · chartTangent_subset_chartRelTangent. So every point of M is essential for $M \subseteq M$ and the reading is not degenerate Chapter9/SheetEssentialChart2021 · chartEssentialSet_self. *Step 3, chart independence.* On level 0 a chart change is only continuous, but along a finite-dimensional submanifold it is differentiable for the level-0 norms: the points of the chart image lie on level 1 with level-1 distance controlled by level-0 distance, because the chart retraction is sc^+ Chapter9/SheetChartChange · ScRetraction.exists_lift_lipschitz, and an sc^1 map is differentiable from level 1 to level 0 (Proposition 1.2.1) Chapter9/SheetChartChange · ScRetraction.hasFDerivWithinAt_of_isSc1 Chapter9/SheetChartChange · exists_hasFDerivWithinAt_transition. The derivative maps tangent cones to tangent cones Chapter9/SheetChartChange · exists_clm_tangentCone_chart_change, so "$T_z^KM = T_zM$" in one chart gives it in

all Chapter9/SheetChartChange · isRelTangentFull_of_chart.

Lemma 9.4.4 (V9.42, narrower) Chapter9/SheetEssentialChart2021 · lemma_9_4_4_chart. *For n-dimensional M, M', $K = M \cap M'$ has the same essential points in M and in M'.* The text's reduction to $T_0^D N$ is the pointwise statement, which holds in each chart separately: $T_z^K M = T_z M$ iff $T_z^K M' = T_z M'$, both tangent spaces being n-dimensional and containing T_z^K Chapter9/SheetEssentialChart2021 · isRelTangentFull_iff. At essential points $T_z M = T_z M'$, the note in Definition 9.4.7 Chapter9/SheetEssentialChart2021 · submanifoldTangent_eq_of_mem_chartEssentialSet. For different dimensions the lemma does not hold as we read it (§IX.17).

Definition 9.4.5 in M-polyfold charts. *Step 1.* `HNull d s`: the d-dimensional Hausdorff measure of s vanishes Chapter9/SheetNull · HNull. A subset C of a sheet has n-dimensional measure zero if its separable pieces have n-null chart images in every chart Chapter9/SheetNull · IsHNull; E_0 need not be separable, and a non-separable set has infinite Hausdorff measure, but the sheets are locally separable, so nothing is lost. It is a local σ-ideal Chapter9/SheetNull · isHNull_iUnion Chapter9/SheetNull · isHNull_of_local. *Step 2, why these are Lebesgue null sets.* Near a point of a sheet the tangent projection p onto $T_x M$ satisfies $\|(\psi a - \psi b) - p(\psi a - \psi b)\| \le \frac{1}{2}\|\psi a - \psi b\|$ Chapter9/SheetNull · exists_half_approx, so it is bi-Lipschitz there, and on $\mathbb{R}^n$ the n-dimensional Hausdorff measure is Lebesgue measure Chapter9/SheetNullBridge · hNull_iff_volume_euclidean. On a tame sheet these are exactly the null sets of Definition 9.4.5 in the smooth charts of the manifold with corners of §IX.16 Chapter9/SheetNullBridge · isHNull_iff_isMeasureZero_cornerChartedSpace Chapter9/SheetNullBridge · isHNull_iff_isMeasureZero_sheet. *Step 3, chart independence:* a map Lipschitz at each point sends separable null sets to null sets Chapter9/SheetChartChange · hNull_image_of_pointwise_lipschitz, which with Step 3 above gives Chapter9/SheetChartChange · isHNull_of_forall_exists_chart. *Step 4, positivity:* an n-dimensional sheet is not null in itself, its tangent projection containing a ball of $T_x M$ (§IX.27, the graph over a cone) Chapter9/SheetNull · not_isHNull_of_hasDimAt. *Step 5.* So the null sets form a `SheetNullSets` on the n-dimensional submanifolds with no hypothesis Chapter9/SheetNull · sheetNullSets; the transfer property is immediate, since the definition does not mention the sheet.

Lemma 9.4.6 (V9.44) Chapter9/SheetEssential · lemma_9_4_6_chart. *For $K \subseteq M$, M n-dimensional, $K \setminus K^e$ has measure zero.* The proof of §IX.17 transported through the tangent projection: every limit direction of $p(\psi(A))$ is the projection of a limit direction of $\psi(A)$, chords of M being close to the finite-dimensional tangent space Chapter9/SheetEssential · exists_mem_tangentConeAt_proj; the points of K where $T^K \neq T$ project to points of $p(\psi(K))$ where the tangent cone does not span, a Lebesgue null set by the density theorem Chapter9/SheetEssential · exists_hNull_chart_not_isRelTangentFull Chapter9/SheetEssential · isHNull_setOf_not_isRelTangentFull. The n-dimensional submanifolds carry the `SheetGeometry` and `SheetMeasureGeometry` of §§IX.17, IX.19 with no hypothesis Chapter9/SheetEssential · sheetGeometry Chapter9/SheetEssential · sheetMeasureGeometry.

For n-dimensional Θ, and Lemma 9.5.2 (V9.50, explicit). *Step 1.* Every local branching structure of an n-dimensional Θ has n-dimensional sheets Chapter9/SheetMeasure · isOfDim_of_isNDimensional: at a point of the support, a structure with n-dimensional sheets and any other structure have the same tangent sum (Theorem 9.2.1), so each sheet of the second has the tangent space of a sheet of the first. *Step 2.* So Lemmas 9.4.4 and 9.4.6 hold for the sheets of every structure of Θ Chapter9/SheetMeasure · lemma_9_4_4_of_isNDimensional Chapter9/SheetMeasure · lemma_9_4_6_of_isNDimensional, and Lemma 9.4.8, Propositions 9.4.9, 9.4.10 follow (§IX.17). *Step 3.* Definition 9.5.1 with these null sets Chapter9/SheetMeasure · IsMeasureZeroH, and Lemma 9.5.2 as printed: null preimages for some choice of structures iff for every structure Chapter9/SheetMeasure · lemma_9_5_2_of_isNDimensional, and for tame Θ iff for a choice of tame structures Chapter9/SheetMeasure · lemma_9_5_2_tame; the proof is that of §IX.19 with one class of sheets. *Step 4.* The

sets of measure zero form a σ-ideal Chapter9/SheetMeasure · isSigmaIdeal_isMeasureZeroH, S is not one of them Chapter9/SheetMeasure · not_isMeasureZeroH_orbitSupp, and $L(S)$ is the completion of §IX.19 Chapter9/SheetMeasure · remark_9_5_3_H Chapter9/SheetMeasure · lebesgueSigmaH_eq. Across dimensions the transfer does not hold as we read it Chapter9/SheetMeasure · isHNull_mixed_dim.

IX.29 Example 9.3.5 (2017: 9.1): all orientations of Θ

The text says that Θ has "four different possibilities to orient". With the first claim of §IX.15 settled, the question is what all orientations of Θ are (gap G.101, closed; V9.33, narrower). The support is $G^{++} \cup G^{--}$ Chapter9/OrientationExampleSheets · Ex935.supp_example935Theta, and $\operatorname{supp}(\Theta) \setminus \{0\}$ consists of four open arcs, G^{++} and G^{--} over $s > 0$ and over $s < 0$; $\Theta = 1$ on the arcs and $\Theta(0) = 2$ Chapter9/OrientationExampleSheets · Ex935.example935Theta_eq_one Chapter9/OrientationExampleSheets · Ex935.example935Theta_zero.

The sheets inside the support. Every sheet of a local branching structure of Θ lies in the support and is 1-dimensional (§IX.28). *Step 1, velocities.* A curve in the support through a point z of G_b has velocity in $\mathbb{R}(1, \beta_b'(z_0))$: off the origin it runs in G_b; at the origin $|\gamma_1| = \beta^{++}(\gamma_0)$ is $o(t)$ since β^{++} is flat, so the velocity is horizontal Chapter9/OrientationExampleSheets · Ex935.hasDerivAt_snd_eq_of_mem_Z. Hence every 1-dimensional submanifold inside the support has, at $z \in G_b$, the tangent space of G_b, and at the origin the tangent $\mathbb{R}(1, 0)$ Chapter9/OrientationExampleSheets · Ex935.submanifoldTangent_eq_of_subset_Z. *Step 2, signs.* Its orientation is one of $\pm(1, \beta')$ at every point Chapter9/OrientationExampleSheets · Ex935.tangent_eq_orTan_or, with a locally constant sign Chapter9/OrientationExampleSheets · Ex935.exists_isOpen_tangent_eq_orTan_iff. *Step 3, vertical lines.* A sheet through the origin meets every vertical line $\{s\} \times \mathbb{R}$ for small $|s|$, by the ball graph representation over $T_0M = \mathbb{R} \times 0$ Chapter9/OrientationExampleSheets · Ex935.exists_mem_fst_eq.

The classification Chapter9/OrientationExampleClasses · example_9_3_5_classification. *Step 1, normal form.* At a point y of the support an orientation is $c \cdot (1, \beta') + (\Theta(y) - c) \cdot (-(1, \beta'))$ Chapter9/OrientationExampleClasses · Ex935.eq_single_add_single. *Step 2, constancy along arcs.* Near a point of an arc every sheet of a structure is the arc with a locally constant sign (Step 2 above), so the coefficient a is constant along each arc, $0 \le a \le 1$ Chapter9/OrientationExampleClasses · Ex935.coef_pt_eq_of_pos Chapter9/OrientationExampleClasses · Ex935.coef_pt_eq_of_neg. *Step 3, the origin.* Every sheet of a structure at 0 meets each vertical line near 0 (Step 3 above), and the count $\Theta(s, \beta^{++}(s)) + \Theta(s, \beta^{--}(s)) = 2 = \Theta(0)$ forces it to meet it exactly once. With the constancy of the signs, the weight A of $(1, 0)$ at the origin is $a(+, G^{++}) + a(+, G^{--})$, and also $a(-, G^{++}) + a(-, G^{--})$ Chapter9/OrientationExampleClasses · Ex935.coef_zero_eq. *Step 4, the family.* So every orientation is a member of the family classOrient(a), a indexed by side and branch, with the two sums equal Chapter9/OrientationExampleClasses · Ex935.classOrient Chapter9/OrientationExampleClasses · Ex935.eq_classOrient, and different a give different orientations Chapter9/OrientationExampleClasses · Ex935.classOrient_injective. *Step 5, the converse.* Each member is an orientation: it is the oriented sum of the four graphs $G^{++}, G^{--}, G^{+-}, G^{-+}$, each with both orientations, with weights solving a 2×2 transport problem between the sides $s > 0$ and $s < 0$ Chapter9/OrientationExampleClasses · Ex935.exists_transport Chapter9/OrientationExampleClasses · Ex935.isOrientation_classOrient.

Consequences. Θ has infinitely many orientations, a three-parameter family Chapter9/OrientationExampleClasses · example_9_3_5_infinite; those with integer weights are exactly the six sign choices of §IX.15 Chapter9/OrientationExampleClasses · example_9_3_5_integral Chapter9/OrientationExampleClasses · example_9_3_5_integral_ncard. So "four different possibilities" holds for

each of the two structures, and for Θ it does not hold as we read it.

IX.30 The boundary operation, the continuity of $o_{\partial M}$, the stratum $\{d_M = 1\}$ and Figure 9.1 (Definitions 9.3.8 and 9.3.10 (2017: 9.9 and 9.11))

The text. For a tame M the points with $d_M(y) = 1$ form an open dense part of ∂M with a manifold structure; for (M, o) oriented, a basis $(a_1, \dots, a_{n-1})$ of $T_y\partial M$ is positive iff $(e, a_1, \dots, a_{n-1})$ is positive, e outward; these orientations "fit continuously together", and the orientation of the interior of a face extends to the closed face (Figure 9.1). §IX.16 formalized the linear algebra with $T_y\partial M$ and e as data. Here they are built from M (gap G.103, closed; V9.36–V9.38 explicit).

The boundary operation (`Chapter9/BoundaryOperation`). *Step 1, presented charts.* $T_y^R X$ and $C_y X$ are read in a chart with a presentation of its partial quadrant `Chapter9/BoundaryOperation · MPolyfold.PresentedChart`; a chart change acts on chart vectors over y by a continuous linear map carrying reduced space to reduced space and cone to cone `Chapter9/BoundaryOperation · MPolyfold.PresentedChart.exists_clm`, so "reduced", "inward" and the reduced codimension $\dim(L/L \cap T_y^R X)$ are chart independent `Chapter9/BoundaryOperation · MPolyfold.isReducedVec_iff` `Chapter9/BoundaryOperation · MPolyfold.isInwardVec_iff` `Chapter9/BoundaryOperation · MPolyfold.hasReducedCodim_iff`. *Step 2.* For tame M, $d_M(y)$ is the reduced codimension of T_yM, the content of Lemma 9.3.7's proof `Chapter9/BoundaryOperation · MPolyfold.subDegIndex_eq_quotDim_span`. *Step 3, the cone spans.* Every vector of T_yM is a difference of two vectors of $C_yM = T_yM \cap C_yX$ `Chapter9/BoundaryOperation · MPolyfold.exists_sub_eq_vec_submanifoldTangent`; so if T_yM has reduced codimension 1, C_yM is a half-space of T_yM bounded by $T_y^R M$ `Chapter9/BoundaryOperation · exists_coeff_halfSpace`, outward vectors exist `Chapter9/BoundaryOperation · MPolyfold.IsBoundaryAdmissible.exists_outward`, and two of them satisfy $e' = ce + k$ with $c > 0$, $k \in T_y^R M$ `Chapter9/BoundaryOperation · MPolyfold.IsBoundaryAdmissible.exists_pos`. *Step 4, the operation.* $T_y(M, o) \mapsto (T_y^R M, o_\partial)$, with the frames a such that (e, a) is positive for every outward e `Chapter9/BoundaryOperation · MPolyfold.boundaryOp`, a function of the oriented tangent set with no choice; by Step 3 and §IX.16 it is Definition 9.3.8 for any single outward e `Chapter9/BoundaryOperation · MPolyfold.boundaryOp_eq`. *Step 5, identification.* The chart image of its carrier is the reduced tangent space of the induced chart of M, $T_y\partial M = T_y^R M$ `Chapter9/BoundaryOperation · MPolyfold.chartVec_image_boundaryCarrier`, and "outward" means "not in the partial cone of M" `Chapter9/BoundaryOperation · MPolyfold.isOutward_iff_not_mem_partialCone`. *Step 6.* With it, $\hat{T}_{\partial\Theta}$ is the oriented sum filtered by reduced codimension 1, which needs no quantifier over submanifolds `Chapter9/BoundaryOperation · EpGroupoid.boundaryOrientation`; it agrees with Definition 9.3.10 for tame orientations `Chapter9/BoundaryOperation · EpGroupoid.boundaryLift_boundaryOp_eq_boundaryOrientation`, and Proposition 9.3.9 holds `Chapter9/BoundaryOperation · EpGroupoid.proposition_9_3_9_boundaryOrientation`. The situation occurs at the corner of $[0, \infty)$ `Chapter9/BoundaryOperation · MPolyfold.exists_isBoundaryAdmissible_halfLine`.

Continuity of $o_{\partial M}$ `Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.SubmanifoldOrientation.exists_continuous_boundaryFrames`. The text says the orientations fit together "by definition". *Step 1.* The boundary stratum is the set of points where T_yA has reduced codimension 1 `Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.boundaryStratum`, $\{d_A = 1\}$ for tame A `Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.boundaryStratum_eq_of_tame`. *Step 2.* Let r be the chart sc⁺-retraction at y_0, w a positive frame of $o_{\partial A}$ at y_0, $s \in C_{y_0}A$ off $T_{y_0}\partial A$. For z in the stratum near y_0, $Dr(z)$ maps the partial quadrant and the reduced space at y_0 into those at z `Chapter9/BoundaryOrientationContinuity2021 · ScRetraction.scDeriv_mem_partialQuadrantAtWith` `Chapter9/`

BoundaryOrientationContinuity2021 · ScRetraction.scDeriv_mem_minimalSubspaceWith, so $Dr(z)w$ is a basis of $T_z\partial A$ by a dimension count, and $-Dr(z)s$ is outward. *Step 3.* The sign of $(-Dr(z)s, Dr(z)w)$ against a continuous positive frame of o is a frame determinant, locally constant (§IX.15). So $o_{\partial M}$ is an orientation along the stratum Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.OrientationAlong Chapter9/BoundaryOrientationContinuity2021 · MPolyfold.SubmanifoldOrientation.boundaryOrientation, for every oriented submanifold: tameness is not used, only the reduced codimension 1. Every oriented structure gives one on each sheet Chapter9/BoundaryOrientationContinuity2021 · EpGroupoid.OrientedBranchingStructure.exists_boundaryOrientation.

The stratum. *Step 1, open and dense.* $\{d_Q = 1\} = \partial Q \cap \{d_Q \le 1\}$ is open in ∂Q for every M-polyfold Chapter9/BoundaryStratumDense · MPolyfold.setOf_degIndex_eq_one Chapter9/BoundaryStratumDense · MPolyfold.isOpen_degIndex_le_one, and dense in ∂Q for a corner structure, since in corner coordinates a point with vanishing coordinates is a limit of points with exactly one of them vanishing Chapter9/BoundaryStratumDense · MPolyfold.HasCornerStructureOfDim.boundary_subset_closure_degIndex_eq_one; so for a tame sheet the stratum is open and dense in ∂M Chapter9/BoundaryStratumDense · MPolyfold.exists_isOpen_boundaryStratum_eq Chapter9/BoundaryStratumDense · MPolyfold.submanifoldBoundary_subset_closure_boundaryStratum. *Step 2, a submanifold.* In corner coordinates κ, on the open piece where all coordinates but y_i are positive Chapter9/StratumRetraction · MPolyfold.CornerCoords.facePiece, the map $\kappa^{-1} \circ (y_i \mapsto 0) \circ \kappa$ is an sc-smooth retraction onto the points with $d = 1$ Chapter9/StratumRetraction · MPolyfold.CornerCoords.faceRet. With s the sc⁺-retraction of M, $\iota \circ \kappa^{-1} \circ (y_i \mapsto 0) \circ \kappa \circ s$ is an sc⁺-retraction of an open subset of X onto the stratum, the sc⁺ property coming from $r = s \circ r$ Chapter9/StratumRetraction · MPolyfold.exists_stratum_coords Chapter9/StratumRetraction · MPolyfold.exists_isScPlusRetractionOn_stratumOne Chapter9/StratumRetraction · MPolyfold.isSmoothSubmanifold_stratumOne Chapter9/StratumRetraction · MPolyfold.stratumOne_eq_boundaryStratum. *Step 3, dimension and tangent spaces.* In a chart the image of $T_y\partial_1 M$ is the image of the hyperplane $\{v_i = 0\}$ under the derivative of the inclusion Chapter9/StratumTangent · MPolyfold.chartTangent_stratumOne_eq, so $\partial_1 M$ is $(n-1)$-dimensional Chapter9/StratumTangent · MPolyfold.isSubmanifoldOfDim_stratumOne with $T_y\partial_1 M = T^R_y M$ Chapter9/StratumTangent · MPolyfold.submanifoldTangent_stratumOne. *Step 4, oriented.* $o_{\partial M}$ is an orientation of the submanifold $\partial_1 M$ in the sense of Definition 9.3.2's sheets Chapter9/BoundaryOrientation · MPolyfold.SubmanifoldOrientation.stratumOrientation Chapter9/BoundaryOrientation · MPolyfold.SubmanifoldOrientation.stratumOrientation_tangent. In corner coordinates where only y_i vanishes, $T\iota(v) \in C_y M$ iff $v_i \ge 0$ Chapter9/BoundaryOrientation · MPolyfold.isInwardVec_incFrame_iff, so $T\iota(-e_i)$ is outward, and a frame $T\iota(v_1), \dots, T\iota(v_m)$ of $\{v_i = 0\}$ is positive for $o_{\partial M}$ iff $(T\iota(-e_i), T\iota(v_1), \dots)$ is positive for o Chapter9/BoundaryOrientation · MPolyfold.incFrame_mem_boundaryFramesAt_iff: "outward normal first", Part I's rule for the faces of a manifold with corners.

Figure 9.1 Chapter9/FaceOrientation · MPolyfold.figure_9_1_existsUnique_faceOrientation. For a face-structured tame sheet, Part I reads the closed faces as manifolds with corners Chapter9/StokesFaceSum · MPolyfold.faceChartedSpace with the induced orientation, outward normal first Chapter9/StokesFaceSum · MPolyfold.faceOrientation. *Step 1, linear algebra.* $\det(-e_i, \iota_i v_1, \dots, \iota_i v_m) = (-1)^{i+1} \det v$ for the face embedding ι_i Chapter9/FaceOrientation · det_cons_neg_stdFrame_faceEmbedding, and the basis $(-e_i, \iota_i b)$ Chapter9/FaceOrientation · consFaceBasis has a given orientation iff b has the one Part I's rule assigns Chapter9/FaceOrientation · orientation_consFaceBasis_eq_iff. *Step 2, agreement.* At every point of a face with $d_M = 1$, a basis b of a face chart is positive for Part I's face orientation iff the frame $T\iota(\iota_i b_1), \dots, T\iota(\iota_i b_m)$ is positive for $o_{\partial M}$ Chapter9/FaceOrientation · MPolyfold.faceOrientation_eq_iff_mem_stratumOrientation Chapter9/FaceOrientation · MPolyfold.faceOrientation_eq_iff_mem_stratumOrientation_pair. *Step 3, extension.* The points with $d_M = 1$ are dense in each closed face Chapter9/FaceOrientation · MPolyfold.dense_faceStratum, and two continuous tangent orientations agreeing on a dense set are equal Chapter9/

`FaceOrientation · TangentOrientation.eq_of_eqOn_dense`. So the orientation of the open face extends to exactly one continuous orientation of the closed face, Part I's face orientation. The text's "interiors of the faces" are read as the points with $d_M = 1$, and "extends" as the unique continuous extension. The orientations of the faces of the faces (the caption's last sentence) are not formalized.

IX.31 Branched integration for an abstract sheet integration: (9.20), Lemma 9.5.7 (2017: 9.17), (9.21), Lemmas 9.5.9–9.5.11 (2017: 9.18, 9.19), Theorems 9.5.4, 9.5.5 (2017: 9.2, 9.3), Definition 9.5.6 and Proposition 9.5.13 (1)

The text. For an oriented local branching structure (M_i, o_i), (σ_i) on $U(x)$ with the natural G_x-action and a measurable $K \subseteq |M_U|$ with preimages $K_i \subseteq M_i$, the local measures are $\mu_\omega^U(K) = \frac{1}{\#G_x}\sum_i \sigma_i \int_{K_i} \omega|(M_i, o_i)$ and $\mu_\omega^{\mathrm{eff},U}$ with $\#G_x^{\mathrm{eff}}$ (9.21). They do not depend on the structure (Lemma 9.5.7, via (9.20)), on shrinking U (Lemmas 9.5.9, 9.5.10) or on moving along a morphism (Lemma 9.5.11); covering S by finitely many $|M_{U_p}|$ and summing gives a unique linear $\omega \mapsto \mu_\omega$ into the signed measures on $(S, L(S))$ with the local formula (Theorems 9.5.4, 9.5.5), and $\int_{\hat\Theta} \omega = \mu_\omega(S)$, $\oint_{\hat\Theta} \omega = \mu_\omega^{\mathrm{eff}}(S)$ (Definition 9.5.6). Rows V9.54–V9.57 and V9.66 are explicit or restated (gap G.107, closed).

The integration as a structure. *Step 1.* The measurable subsets of a sheet N are the sets whose trace differs from a Borel trace by a null set of N `Chapter9/SheetIntegrationAbstract · SheetNullSets.IsMeasIn`; two sheets induce the same measurable subsets of their intersection `Chapter9/SheetIntegrationAbstract · SheetNullSets.IsMeasIn.transfer`. *Step 2.* A `SheetIntegration` gives numbers $\int_A \omega|(N, o)$ for the forms ω of a real vector space Ω, the oriented sheets N of a class, and measurable $A \subseteq N$ `Chapter9/SheetIntegrationAbstract · EpGroupoid.SheetIntegration`, with the properties the text uses: linearity in ω, countable additivity, vanishing on null sets, equal (opposite) integrals over a common subset of two sheets where their oriented tangent spaces agree (are opposite), invariance under the sc-diffeomorphisms whose tangent maps are those of morphisms `Chapter9/SheetIntegrationAbstract · EpGroupoid.IsMorTangentOn`, Lindelöf sheets, and Lemma 9.4.6 with the note of Definition 9.4.7. It is built for sc-forms in §IX.32.

(9.20) and Lemma 9.5.7 `Chapter9/SheetIntegrationAbstract · EpGroupoid.SheetIntegration.sum_integ_eq_zero` `Chapter9/SheetIntegrationAbstract · EpGroupoid.SheetIntegration.sum_integ_eq_sum_integ`. *A finite family of oriented sheets with weights whose weighted sum of oriented tangent spaces vanishes at every point of K has $\sum_k w_k \int_{K\cap N_k} \omega|(N_k, o_k) = 0$.* **The text's proof** removes a null set and compares on the pieces of Proposition 9.4.10. **The formalized proof** redoes that decomposition for measurable pieces. *Step 1.* The points where two sheets through a point have different tangent spaces form a null set in every sheet `Chapter9/SheetIntegrationAbstract · EpGroupoid.SheetIntegration.isNull_tangentBad`. *Step 2.* The rest of K is cut into the finitely many measurable pieces on which the set of sheets through y and the pairs with equal oriented tangent space are constant `Chapter9/SheetIntegrationAbstract · EpGroupoid.SheetIntegration.piece` `Chapter9/SheetIntegrationAbstract · EpGroupoid.SheetIntegration.isMeasIn_piece` `Chapter9/SheetIntegrationAbstract · EpGroupoid.SheetIntegration.pairwise_disjoint_piece`. *Step 3.* On a piece all the integrals are $\pm$ the integral over one sheet, and the signed sum of the weights vanishes because a tangent space has two orientations `Chapter9/SheetIntegrationAbstract · EpGroupoid.SheetIntegration.sum_sign_weight_eq_zero`. The families need not pass through one point. For two oriented tame structures describing $\hat T_\Theta$ this gives Lemma 9.5.7 `Chapter9/LocalMeasure · lemma_9_5_7`.

(9.21) and Lemmas 9.5.9–9.5.11. *The local measures.* An oriented tame structure is a tame structure (natural representation, properness, tame sheets) with n-dimensional oriented sheets `Chapter9/LocalMeasure · OrientedTameStructure`; it describes $\hat{T}_\Theta$ if its oriented sum is $\hat{T}_\Theta$ near x `Chapter9/LocalMeasure · Represents`. The weighted sum $\sum_i \sigma_i \int_{A\cap M_i} \omega|(M_i, o_i)$ `Chapter9/LocalMeasure · sheetSum` divided by $\#G_x$ or $\#G_x^{\mathrm{eff}}$ is μ_ω^U, $\mu_\omega^{\mathrm{eff},U}$ `Chapter9/LocalMeasure · localMeasure` `Chapter9/LocalMeasure · localMeasureEff`, linear in ω and additive in the set `Chapter9/LocalMeasure · sheetSum_hasSum`. *Lemma 9.5.9* `Chapter9/LocalMeasure · lemma_9_5_9`: with the slice V of §IX.20 and the decomposition $M_U \cap \pi^{-1}(K) = \bigsqcup_{r\in R} \varphi_r(M_U \cap V \cap \pi^{-1}(K))$, each φ_r preserves the integrals, its tangent maps being those of morphisms `Chapter9/LocalMeasure · NaturalRep.isMorTangentOn_Φ` `Chapter9/LocalMeasure · sheetSum_image`, so the sum over U is $\#R$ times the sum over V `Chapter9/LocalMeasure · sheetSum_preimage_eq_card_mul_slice`; with $\#R \cdot \#G_y = \#G_x$ the normalized measures agree. The text takes the restricted structure on V; by Lemma 9.5.7 any structure at y describing $\hat{T}_\Theta$ serves, and the text's proof uses Lemma 9.5.7 and the naturality of $\hat{T}_\Theta$ here. *Lemma 9.5.10* `Chapter9/LocalMeasure · lemma_9_5_10`: the same with (9.23), $\#R \cdot \#G_y^{\mathrm{eff}} = \#G_x^{\mathrm{eff}}$ under the local unique continuation property. *Lemma 9.5.11* `Chapter9/LocalMeasure · lemma_9_5_11`: the local sc-diffeomorphism $\hat{\varphi} : U \to U'$ of a morphism has, at each point, the tangent map of a morphism, so it carries the structure on U to one on U' with the same sums `Chapter9/LocalMeasure · sheetSum_preimage_eq_of_morDiffeo`, and $\#G_x = \#G_{x'}$ (§IX.20). The effective clause holds without the local unique continuation property, only $\#G_x^{\mathrm{eff}} = \#G_{x'}^{\mathrm{eff}}$ being used. Remark 9.5.8 has no content beyond Lemma 9.5.11.

Theorems 9.5.4 and 9.5.5 `Chapter9/BranchedIntegral · theorem_9_5_4` `Chapter9/BranchedIntegral · theorem_9_5_5`. *Step 1, measurability.* The local sc-diffeomorphism $t \circ (s|U(\varphi))^{-1}$ of a morphism has the tangent maps of the morphisms of its sheet `Chapter9/BranchedIntegral · isMorTangentOn_hat`, and the preimage of $K \in L(S)$ has measurable traces on the sheets, by Lemma 9.5.2 `Chapter9/BranchedIntegral · OrientedTameStructure.isMeas_preimage`. *Step 2, overlaps.* Near two points $z \in U$, $z' \in U'$ over one point of $|X|$ the two local measures agree, combining Lemmas 9.5.9 and 9.5.11 `Chapter9/BranchedIntegral · exists_slice_pair` `Chapter9/BranchedIntegral · exists_local_agreement`; since the sheets are Lindelöf, countably many such neighbourhoods cover a measurable $K \subseteq |M_U| \cap |M_{U'}|$, and the two structures give the same measure of K `Chapter9/BranchedIntegral · overlap`. The text does not mention that countably many pieces suffice. *Step 3, gluing.* Each structure gives a signed measure on $(S, L(S))$, linear in ω `Chapter9/BranchedIntegral · localSignedMeasure`. On a compact space with a σ-algebra containing the Borel sets, linear local expressions on the members of an open cover that agree on overlaps glue to exactly one linear map into signed measures: cut K into the pieces $W_k \setminus \bigcup_{j<k} W_j$ of a finite subcover and add `Chapter9/MeasureGlue · existsUnique_linearMap_signedMeasure_of_local` `Chapter9/MeasureGlue · existsUnique_linearMap_signedMeasure_of_openCover`. This is the text's "defines our desired signed measure". *Step 4.* Hence exactly one linear $\omega \mapsto \mu_\omega$ with the local formula for every oriented tame structure with sheets in the class describing $\hat{T}_\Theta$ and every measurable $K \subseteq |U(x)|$ `Chapter9/BranchedIntegral · existsUnique_branchedMeasure` `Chapter9/BranchedIntegral · branchedMeasure` `Chapter9/BranchedIntegral · branchedMeasureEff`, given such a structure at every point of the support (`hcov` in this abstract form, the class condition of the integration). *Definition 9.5.6* `Chapter9/BranchedIntegral · branchedIntegral` `Chapter9/BranchedIntegral · branchedIntegralEff`, and the identity after Theorem 9.5.5, $\mu^{\mathrm{eff},U} = (\#G_x/\#G_x^{\mathrm{eff}})\,\mu^U$ `Chapter9/BranchedIntegral · OrientedTameStructure.localMeasureEff_eq`. *Proposition 9.5.13 (1)*: for effective X the two integrals agree `Chapter9/BranchedIntegral · proposition_9_5_13_1`. A witness in dimension 0: points as sheets, the integral the signed value of an invariant function `Chapter9/PointIntegration · pointIntegration`.

IX.32 The integral of sc-differential forms over oriented sheets, and Theorems 9.5.4 and 9.5.5 as printed

The text integrates "$\omega|(M_i, o_i)$" over measurable subsets of the sheets without constructing the integral; the sheets are manifolds with boundary and corners and ω an sc-differential form on the ep-groupoid (Definition 8.2.1). Here the integral is built for $n \geq 1$ and shown to be a `SheetIntegration` of §IX.31, first on sheets in one chart, then on sheets covered by countably many charts (gap G.107, closed).

Evaluating forms. *Step 1.* An evaluation of forms is a real vector space Ω with values $\omega(\alpha_1, \dots, \alpha_n)$ on tangent vectors, linear in ω, alternating and multilinear on the tangent space of a submanifold (stated through determinants), invariant under the tangent maps of morphisms, and continuous along continuous frame fields `Chapter9/GraphChart · FormEval`. *Step 2.* The forms $\Omega^n_{ep}(X)$ of Definition 8.2.1 are one: the chart expression of ω is sc-smooth, hence continuous along level-1 lifts, which are continuous along a submanifold since the chart retraction is sc^+ `Chapter9/FormEvalEp · MPolyfold.continuousWithinAt_levelLift` `Chapter9/FormEvalEp · epFormEval`. *Step 3.* The direct limit $\Omega^n_{ep,\infty}(X)$ of Definition 8.2.2 is one too: a submanifold of X in X_∞ is one of every level X^i with the same tangent spaces and continuous frame fields, so the forms of X^i are evaluated on lifted tangent vectors, compatibly with the steps of the limit `Chapter9/FormEvalInfty · IsSmoothSubmanifold.levelOne` `Chapter9/FormEvalInfty · epFormsInftyEval`.

Graph charts. *Step 1.* A graph chart of an n-dimensional submanifold N is a chart φ of the M-polyfold containing N and a continuous linear $p : E_0 \to \mathbb{R}^n$, injective on $\varphi(N)$, whose inverse is differentiable within $p(\varphi(N))$ with derivative a linear isomorphism onto the chart image of T_yN, continuous in y `Chapter9/GraphChart · MPolyfold.GraphChart`. No corner structure of N is used, and derivatives are taken within a set that need not be open. *Step 2.* Near every point of an n-dimensional submanifold there is one: the tangent projection of §IX.28 is bi-Lipschitz on N near the point, and its image is closed `Chapter9/GraphChartExists · ScRetraction.exists_graphData` `Chapter9/GraphChartExists · exists_graphChart`. *Step 3, the integral in a chart.* The density is $\pm\omega(\partial_1, \dots, \partial_n)$ for the coordinate frame, with sign $+$ where the frame is positive for the orientation `Chapter9/ChartIntegral · density`, and $\int_A$ is its Lebesgue integral over the coordinate image of $A \cap N$ `Chapter9/ChartIntegral · chartInteg`; the sign is locally constant on N `Chapter9/FrameSign · GraphChart.exists_isOpen_isPos`. *Step 4, the comparison of two charts.* At a common point of two sheets with $T_yN = T_yN'$ the coordinate change is differentiable within the coordinate image of $N \cap N'$, with derivative expressing one frame in the other `Chapter9/GraphChartTransition · GraphChart.exists_transition`; the change of variables for maps differentiable on a measurable set `Chapter9/GraphChart · integral_image_eq_of_det` `Chapter9/GraphChartTransition · GraphChart.setIntegral_coord_image_eq` gives equal integrals where the oriented tangent spaces agree and opposite ones where they are opposite `Chapter9/ChartIntegral · FormEval.chartInteg_congr` `Chapter9/ChartIntegral · FormEval.chartInteg_neg`. With $N = N'$ this is the independence of the graph chart; no smoothness of $N \cap N'$ is needed. This is how the proof of Lemma 9.5.7 compares sheets. *Step 5.* Invariance under the sc-diffeomorphisms whose tangent maps are those of morphisms `Chapter9/ChartIntegral · FormEval.chartInteg_mapDiffeo`.

The one-chart integration `Chapter9/SheetIntegrationConcrete · FormEval.sheetIntegration`. The class consists of the n-dimensional submanifolds in one graph chart on which every form is absolutely integrable `Chapter9/SheetIntegrationConcrete · FormEval.sheets`, the null sets are the n-dimensional Hausdorff null sets `Chapter9/SheetIntegrationConcrete · FormEval.sheetNull`, and the integral is that of Step 3 `Chapter9/SheetIntegrationConcrete · FormEval.integ`; all fields of a `SheetIntegration` hold, among them the Lindelöf property, through the coordinate homeomorphism `Chapter9/SheetIntegrationConcrete · FormEval.`

hereditarilyLindelofSpace, and the integral vanishes where ω vanishes on the sheet Chapter9/SheetIntegrationConcrete · FormEval.integ_eq_zero_of_ev_eq_zero. With it, §IX.31 gives Theorems 9.5.4, 9.5.5 for structures whose sheets lie in one graph chart Chapter9/TameOpen · theorem_9_5_4_tame Chapter9/TameOpen · theorem_9_5_5_tame Chapter9/TameOpen · tameBranchedIntegral; such structures exist near every point of the support once some oriented tame structure does, by restriction to a smaller domain Chapter9/IntegralCovering · exists_restrict_mem_sheets Chapter9/TameOpen · HasTameInducedStructure.inter_isOpen Chapter9/TameOpen · hcov_of_loc.

Several charts. *Step 1, $\int|\omega|$.* In a graph chart, $\int_A|\omega|$ bounds $|\int_A\omega|$, does not depend on the chart, is invariant under morphisms and countably additive Chapter9/ChartAbsIntegral · abs_chartInteg_le Chapter9/ChartAbsIntegral · chartAbs_congr Chapter9/ChartAbsIntegral · chartAbs_mapDiffeo. *Step 2.* A sheet covered by countably many open V_k with $N\cap V_k$ in the one-chart class has the disjoint Borel pieces $P_k = N\cap V_k\setminus\bigcup_{j<k}V_j$ Chapter9/MultiChart · ChartCover; the cover is finite in the sense $\sum_k\int_{P_k}|\omega| < \infty$ Chapter9/MultiChart · ChartCover.IsFinite, and then $\int_A\omega = \sum_k\int_{A\cap P_k}\omega$ converges absolutely Chapter9/MultiChart · ChartCover.gInteg. *Step 3, independence.* Two covers, or two sheets over a common set where their oriented tangent spaces agree or are opposite, give equal or opposite sums, by a double series over the pieces of the two covers Chapter9/MultiChart · hasSum_of_double Chapter9/MultiChart · ChartCover.gInteg_eq_of. *Step 4.* The class of sheets with a finite cover Chapter9/MultiChart · FormEval.msheets carries a `SheetIntegration` Chapter9/MultiChartIntegration · FormEval.msInteg Chapter9/MultiChartIntegration · FormEval.msheetIntegration that contains the one-chart class and agrees with it there Chapter9/MultiChartIntegration · FormEval.sheets_subset_msheets Chapter9/MultiChartIntegration · FormEval.msInteg_eq_integ; Theorems 9.5.4, 9.5.5 follow for structures with sheets in this class Chapter9/MultiChartTheorem · theorem_9_5_4_multi Chapter9/MultiChartTheorem · theorem_9_5_5_multi Chapter9/MultiChartTheorem · multiBranchedIntegral Chapter9/MultiChartTheorem · multiBranchedIntegralEff. Compact sheets are in it Chapter9/MultiChartCompact · msheets_of_isCompact.

The finiteness the text uses tacitly Chapter9/FiniteTheorem · msheets_of_orientedTameStructure. The text integrates over measurable subsets of the sheets, which presupposes $\int|\omega|<\infty$. For the sheets of an oriented tame structure of a compact n-dimensional Θ it holds. *Step 1.* The support is locally compact, being locally a finite union of finite-dimensional submanifolds; a compact $K'\subseteq\operatorname{supp}(\Theta)$ meets every orbit of the support since $|\operatorname{supp}(\Theta)|$ is compact; and $\operatorname{supp}(\Theta)\cap U\subseteq s(\{\varphi\mid s(\varphi)\in\operatorname{cl}U,\ t(\varphi)\in K'\})$, compact by the properness of $\operatorname{cl}U$ (Definition 7.5.8) Chapter9/FiniteTheorem · exists_isCompact_supp_inter. *Step 2.* An n-dimensional submanifold inside a compact part of the support is covered by finitely many one-chart sheets of structures at points of it Chapter9/FiniteTheorem · msheets_of_subset_isCompact, and $\int|\omega|$ over it is dominated by the finitely many one-chart integrals Chapter9/FiniteDominated · chartAbs_eq_of_borel Chapter9/FiniteDominated · isFinite_of_dominated. *Step 3, Theorems 9.5.4 and 9.5.5 as printed* Chapter9/FiniteTheorem · theorem_9_5_4_full Chapter9/FiniteTheorem · theorem_9_5_5_full: there is exactly one linear $\omega\mapsto\mu_\omega$ into the signed measures on $(S,L(S))$ with $\mu_\omega(K)=\mu_\omega^U(K)$ for every oriented tame structure describing $\hat T_\Theta$ and every measurable $K\subseteq|U(x)|$; the n-dimensionality of Θ follows from the hypothesis Chapter9/FiniteTheorem · isNDimensional_of_loc.

The reading `hloc` of "oriented, tame". The theorems take as hypothesis that every point of the support has an oriented tame structure describing $\hat T_\Theta$. Without boundary it follows from the text's hypotheses Chapter9/CoveringFromText · hloc_of_degIndex_eq_zero: the oriented structure of Definition 9.3.2 has n-dimensional sheets, and restricted to the domain of a natural representation inside an open set with the properness property it is tame, every submanifold having a tame induced structure when $d_X = 0$ Chapter9/CoveringFromText · exists_orientedTameStructure_of_degIndex_eq_zero Chapter9/

FiniteTheoremText · theorem_9_5_4_full_of_degIndex_eq_zero. With boundary, Definition 9.3.6 gives tame sheets of *some* structure and Definition 9.3.2 oriented sheets of *another*; that the oriented structure can be taken tame is the reading of "tame".

Part I's integral, and a witness. Over a tame sheet the integral equals Part I's manifold integral of the restricted form for the corresponding tangent orientation Chapter9/ManifoldBridge · bridgeOrientation Chapter9/ManifoldBridge · isContinuous_bridgeOrientation Chapter9/ManifoldIntegralBridge · FormEval.integ_eq_manifoldIntegral, chart by chart of a partition of unity by the change of variables between graph and corner coordinates Chapter9/ManifoldBridge · FormEval.chartInteg_eq_setIntegral_corner; for classes of $\Omega^n_{ep,\infty}(X)$ represented on any level see Chapter9/LevelRestrict · epFormsInftyEval_integ_eq_integral. The unit circle in $\mathbb{R}^2$, the image of the radial retraction $r \mapsto r/|r|$ of $\mathbb{R}^2 \setminus \{0\}$, oriented counter-clockwise Chapter9/CircleWitness · rnPolyfold_isSubmanifoldOfDim_unitCircle2 Chapter9/CircleWitness · circleOrientation, satisfies all hypotheses with compact nonempty support Chapter9/CircleWitness · circle_witness Chapter9/FiniteTheoremText · circle_theorem_9_5_4_full, and with $U(x) = \mathbb{R}^2$ and the whole circle as one sheet $\int_{\hat\Theta} \omega = \int_{S^1} \omega$ Chapter9/CircleOneSheet · circle_multiBranchedIntegral.

IX.33 (9.22) and Proposition 9.5.13 (2)

(9.22) (V9.69, explicit). *If $\tau = f \cdot \omega$, then $d\mu_\tau = \bar f \, d\mu_\omega$, with $\bar f \circ \pi = f$.* **Reading.** $f \cdot \omega$ is read through the evaluation, $\tau(\alpha) = f(y)\,\omega(\alpha)$ on the frames of the tangent spaces T_yN of the sheets Chapter9/DensityLocal · IsMul, witnessed for constant f Chapter9/DensityLocal · isMul_smul; the form $f \cdot \omega$ for a non-constant sc-smooth f is not built. *Step 1, local, exact.* In a graph chart, $\int_A \tau$ is the integral of f times the density of ω Chapter9/DensityLocal · integ_eq_setIntegral_mul, so $\mu^U_\tau(K)$ is (9.21) for ω with f inserted in the integrands Chapter9/DensityLocal · OrientedTameStructure.localMeasure_eq_sum_setIntegral_mul. *Step 2, the estimate.* If $|\bar f - a| \le \varepsilon$ on K, then $|\mu_\tau(K) - a\,\mu_\omega(K)| \le \varepsilon\, M_\omega(K)$ with $M_\omega(K) = \frac{1}{\#G_x} \sum \sigma_i \int_{K_i} |\omega|$, additive in K Chapter9/DensityLocal · OrientedTameStructure.abs_localMeasure_sub_le Chapter9/DensityLocal · abs_branchedMeasure_sub_le Chapter9/DensitySignedMeasure · OrientedTameStructure.absSheetSum_union. *Step 3, integrals against a signed measure.* Mathlib's integral against a vector measure is, for a signed measure s, $\int_K g\, ds^+ - \int_K g\, ds^-$ for the Jordan decomposition Chapter9/SignedMeasureIntegrals · SignedDensity.setIntegral_eq_jordan. If a second signed measure t satisfies the estimate of Step 2 against s on the measurable subsets of a set, then $t(K) = \int_K g\, ds$, by Riemann–Stieltjes sums over the partition of K by the values of g and $\varepsilon \to 0$ Chapter9/SignedMeasureIntegrals · SignedDensity.eq_setIntegral_of_estimate; on a compact space, estimates near every point suffice Chapter9/SignedMeasureIntegrals · SignedDensity.eq_setIntegral_of_local_estimate. *Step 4.* So for $\bar f$ bounded and $L(S)$-measurable, continuous for instance, $\mu_\tau(K) = \int_K \bar f \, d\mu_\omega$ for every $K \in L(S)$, and the same for the effective measures Chapter9/DensitySignedMeasure · eq_setIntegral_of_sheetSum Chapter9/DensitySignedMeasure · branchedMeasure_eq_setIntegral_mul Chapter9/DensitySignedMeasure · branchedMeasureEff_eq_setIntegral_mul Chapter9/DensitySignedMeasure · branchedMeasure_eq_jordan.

Proposition 9.5.13 (2) (V9.57, restated). *Under the local unique continuation property, $\oint_{\hat\Theta} \omega = \int_{\hat\Theta'} \omega'$ on the reduced ep-groupoid X_R.* **The text's proof**: X and X_R have the same object M-polyfold, $\Theta' = \Theta$, "we can take the same local branching structure", and $\#G^{\mathrm{eff}}_x = \#G'_x$. **The formalized proof.** *Step 1, the natural action of G^R_x*, which "the same local branching structure" needs and the text does not construct. For a natural representation of G_x on $U(x)$ with the property of Definition 7.2.18, $\Phi^R([g]) = \Phi(g)$ and $\Gamma^R([g], y) = [\Gamma(g, y)]$ define one of $G^R_x = G_x/T_x$ on the same $U(x)$ Chapter9/ReducedRep · NaturalRep.toReduced: Φ descends because $T_x = \ker \Phi$ Chapter9/ReducedRep · Φ_eq_of_reducedIsotropyHom_eq, Γ be-

cause $\Gamma(gk,y) = \Gamma(g,y)\circ\Gamma(k,y)$ with $\Gamma(k,y)$ trivial in X_R Chapter9/ReducedRep · reducedπ_Γ_eq_of_reducedIsotropyHom_eq, and the uniqueness in Theorem 7.1.13 (3) holds because $[\Gamma(a,y)] = [\Gamma(b,y)]$ forces $a = bc$ with $\Phi(c) = \mathrm{id}$ near y, and the local unique continuation property gives $c \in T_x$ Chapter9/ReducedRep · reducedIsotropyHom_eq_of_reducedπ_Γ_eq. *Step 2, the same structures.* The properness property, tame and oriented tame structures carry over with the same U, sheets, weights and orientations Chapter9/ReducedRep · HasPropernessProperty.toReduced Chapter9/ReducedRep · OrientedTameStructure.toReduced, and $\#G_x^R = \#G_x^{\mathrm{eff}}$ Chapter9/ReducedRep · card_isotropy_reduced_eq_card_effIsotropy. *Step 3, the same integrals.* The tangent map of $[\varphi]$ is that of φ Chapter9/ReducedRep · tangentMor_reducedπ, so a form of X is one of X_R and a sheet integration on X is one on X_R with the same integrals Chapter9/ReducedRep · SheetIntegration.toReduced; the orientation carries over Chapter9/ReducedRep · IsOrientation.toReduced. *Step 4, the same σ-algebra.* $|X| \cong |X_R|$ and $|\mathrm{supp}(\Theta)|$ in both are identified Chapter9/ReducedIntegral · orbitHomeomorphReduced Chapter9/ReducedIntegral · orbitSuppHomeomorphReduced, carrying $L(S)$ to $L(S)$ Chapter9/ReducedIntegral · measurableSet_image_orbitSuppHomeomorphReduced. *Step 5.* By the uniqueness in Theorem 9.5.4 on X_R, $\mu^{\mathrm{eff}}_{\hat\Theta,\omega}(K) = \mu_{\hat\Theta',\omega'}(K)$ for $K \in L(S)$ Chapter9/ReducedIntegral · proposition_9_5_13_2_measure, and with $K = S$ the proposition Chapter9/ReducedIntegral · proposition_9_5_13_2. The local unique continuation property also makes the morphism space of X_R Hausdorff, so that X_R is an ep-groupoid. No hypothesis is added. Witness: the circle Chapter9/ReducedCircle · circle_proposition_9_5_13_2.

IX.34 Definition 9.5.14 (2017: 9.17) with $(n-1)$-dimensional null sets, and Theorems 9.5.15, 9.5.16 (2017: 9.4, 9.5): the boundary measures

The text. For $K \subseteq |\partial M_U|$ the preimages lie in the ∂M_i, manifolds with boundary and corners, and measure zero is $(n-1)$-dimensional Lebesgue measure zero in every boundary face (p. 448). Then (Theorem 9.5.15) there is a linear map $\omega \mapsto \mu^\partial_\omega$ into the signed measures on $(\partial S, L(\partial S))$, "uniquely characterized" by $\mu^\partial_\omega(K) = \frac{1}{\#G_x}\sum_i \sigma_i \int_{K_i} \omega|(\partial M_i, o_{\partial M_i})$, and Theorem 9.5.16 the same with $\#G_x^{\mathrm{eff}}$ under the local unique continuation property; the proof is "using the same techniques" as Theorem 9.5.4.

The null sets of the boundaries (gap G.108, closed; V9.52, V9.53). *Step 1.* As for the sheets (§IX.28), $(n-1)$-dimensional measure zero is read in the M-polyfold charts with the $(n-1)$-dimensional Hausdorff measure. *Step 2, positivity.* In corner coordinates of dimension $m+1$, near a point whose only vanishing coordinate is y_i, the chart image of $\{y_i = 0\}$ is an open piece of a coordinate hyperplane, not m-null Chapter9/BoundaryNotNull · MPolyfold.CornerCoords.not_hNull_chart_coord_eq_zero; so a nonempty ∂M of a tame n-dimensional sheet is not $(n-1)$-null Chapter9/BoundaryNotNull · MPolyfold.HasCornerStructureOfDim.not_isHNull_boundary Chapter9/BoundaryNotNull · MPolyfold.not_isHNull_submanifoldBoundary. *Step 3.* These null sets form a null-set system on the class of the boundaries of tame sheets with no hypothesis Chapter9/BoundaryNullSigmaAlgebra · MPolyfold.tameBoundaries Chapter9/BoundaryNullSigmaAlgebra · MPolyfold.boundaryNullSets; for $n = 1$ they are the counting null sets Chapter9/BoundaryNullSigmaAlgebra · MPolyfold.boundaryNullSets_one_isNull_iff Chapter9/BoundaryFaceNullSets · SheetNullSets.point. *Step 4, well defined.* For n-dimensional Θ, K has measure zero in ∂S iff near each point of $\mathrm{supp}(\partial\Theta)$ a tame structure has $\partial M_i \cap \pi^{-1}(K)$ null for all i, and then every tame structure has Chapter9/BoundaryNullSigmaAlgebra · EpGroupoid.isMeasureZeroBoundary_boundaryNullSets_iff Chapter9/BoundaryNullSigmaAlgebra · EpGroupoid.isHNull_boundary_of_isMeasureZeroBoundary; Definition 9.5.14 is Chapter9/BoundaryNullSigmaAlgebra · EpGroupoid.lebesgueSigmaBoundaryH. *Step 5, the corners.* A codimension-2 coordinate subspace of $\mathbb{R}^n$ is $(n-1)$-null, so the points with $d_M \geq 2$ are $(n-1)$-null Chapter9/

BoundaryNullSigmaAlgebra · MPolyfold.isHNull_two_le_subDegIndex and ∂M minus the stratum $\{d_M = 1\}$ is null Chapter9/BoundaryNullSigmaAlgebra · MPolyfold.isHNull_submanifoldBoundary_diff_boundaryStratum: the boundary integral lives on the stratum. *Face by face.* The text's notion for a face-structured sheet is formalized as a proper σ-ideal Chapter9/BoundaryFaceNullSets · MPolyfold.IsFaceNull Chapter9/BoundaryFaceNullSets · MPolyfold.not_isFaceNull_face; its comparison with the ambient reading is not formalized, and no statement needs it. *Witness with $\partial S \neq \emptyset$:* $\Theta = 1$ on $[0, \infty)$, tame and 1-dimensional Chapter9/BoundaryFaceNullSets · halfLineTame, with $|0| \in \partial S$ Chapter9/BoundaryFaceNullSets · halfLine_orbitSupp_boundary_nonempty and $L(\partial S)$ the Borel sets Chapter9/BoundaryFaceNullSets · halfLine_lebesgueSigmaBoundary.

The boundary sheets. *Step 1.* For an oriented tame structure of dimension $m + 1$, $\partial_1 M_i = \{d_{M_i} = 1\}$ is an m-dimensional submanifold (§IX.30) with the orientation $o_{\partial M_i}$ Chapter9/BoundarySheets · OrientedTameStructure.boundarySheet Chapter9/BoundarySheets · OrientedTameStructure.boundaryOrient. *Step 2, (9.12) with the stratum.* If the structure describes $\hat{T}_\Theta$, then $\hat{T}_{\partial\Theta}(y) = \sum_{y \in \partial_1 M_i} \sigma_i T_y(\partial_1 M_i, o_{\partial M_i})$ Chapter9/BoundarySheets · OrientedTameStructure.boundaryOrientation_eq_sum; the left side depends on $\hat{T}_\Theta$ only. *Step 3.* Near each of its points a boundary sheet lies in the one-chart class for m-forms Chapter9/BoundarySheets · OrientedTameStructure.exists_boundarySheet_mem_sheets. *Step 4, the traces.* For $K \in L(\partial S)$ the trace of $\pi^{-1}(K)$ on $\partial_1 M_i$ is measurable, by induction over the generators of $L(\partial S)$: Borel generators have Borel traces, null generators null traces by Step 4 above and $\partial_1 M_i \subseteq \partial M_i$ Chapter9/BoundaryMeasure · OrientedTameStructure.isMeasIn_boundarySheet.

The machinery of §IX.31 for sheet families. The boundary sheets are not the sheets of a branching structure, so (9.20), (9.21) and Lemmas 9.5.7–9.5.11 are proved again for an oriented sheet family: a natural representation on $U(x)$ and finitely many m-dimensional oriented submanifolds with rational weights whose union is $D \cap U(x)$ for a fixed set D closed under morphisms Chapter9/SheetFamily · IsMorSaturated Chapter9/SheetFamily · OrientedSheetFamily, described by a field T' carried along by morphisms Chapter9/SheetFamily · IsNaturalOn Chapter9/SheetFamily · OrientedSheetFamily.lemma_9_5_7 Chapter9/SheetFamily · OrientedSheetFamily.lemma_9_5_9 Chapter9/SheetFamily · OrientedSheetFamily.lemma_9_5_11. An oriented tame structure is the instance $D = \operatorname{supp}(\Theta)$ Chapter9/SheetFamily · OrientedTameStructure.toFamily Chapter9/SheetFamily · isNaturalOn_of_isOrientation. The overlaps and the gluing go as in §IX.31 Chapter9/FamilyMeasure · OrientedSheetFamily.overlap Chapter9/FamilyMeasure · existsUnique_familyMeasure. *The boundary family.* D is the set of points of the support where $\hat{T}_\Theta$ has a term of reduced codimension 1 Chapter9/BoundaryFamily · boundaryStratumSet, closed under morphisms Chapter9/BoundaryFamily · isMorSaturated_boundaryStratumSet since tangent maps of morphisms preserve the reduced codimension; for a structure describing $\hat{T}_\Theta$ it is the union of the $\partial_1 M_i$ in $U(x)$, the weights being positive. $\hat{T}_{\partial\Theta}$ is natural Chapter9/BoundaryFamily · isNaturalOn_boundaryOrientation, and the $(\partial_1 M_i, o_{\partial M_i})$, (σ_i) form a family describing it Chapter9/BoundaryFamily · OrientedTameStructure.boundaryFamily Chapter9/BoundaryFamily · OrientedTameStructure.represents_boundaryFamily.

Theorems 9.5.15 and 9.5.16, first form (V9.67; the form with the text's hypotheses is in §§IX.36–IX.38) Chapter9/BoundaryMeasure · theorem_9_5_15_hyp Chapter9/BoundaryMeasure · theorem_9_5_16_hyp. *For every sheet integration whose null sets are the m-dimensional ones, $n = m + 1$, $m \geq 1$, there is exactly one linear $\omega \mapsto \mu^\partial_\omega$ into the signed measures on $(\partial S, L(\partial S))$ with the printed local formula*, and for the integration over countably many graph charts Chapter9/BoundaryMeasure · theorem_9_5_15_msheets Chapter9/BoundaryMeasure · theorem_9_5_16_msheets. ∂S is compact, being closed in S Chapter9/MeasureGlue · EpGroupoid.isCompact_orbitSupp_boundaryTheta, and the gluing is Chapter9/MeasureGlue · EpGroupoid.theorem_9_5_15_of_overlap. The sum runs over the nonempty $\partial_1 M_i$, the empty set need not belong to the class Chapter9/BoundaryMeasure · OrientedSheetFamily.nonemptyPart, and n-dimensionality is stated explicitly. **Added hypothesis `hcov`**, not derived from the

text: at every point of $\operatorname{supp}(\partial\Theta)$ an oriented tame structure describing $\widehat{T}_\Theta$ whose nonempty boundary sheets lie in the class of the integration. For the multi-chart class this is the finiteness of $\int |\omega|$ over $\partial_1 M_i$ near the corners of M_i, which the text uses tacitly; the difficulty lies where $\partial_1 M_i$ approaches the corners, and compact boundary sheets satisfy it `Chapter9/BoundaryMeasure · OrientedTameStructure.boundarySheet_mem_msheets_of_isCompact` `Chapter9/BoundaryMeasure · hcov_msheets_of_isCompact`. Witnesses: $\Theta = 0$ `Chapter9/BoundaryMeasure · hcov_zero`, and with $\operatorname{supp}(\partial\Theta) \neq \emptyset$ the closed upper hemisphere of §IX.35 `Chapter9/HemisphereWitness · e8w_hcov`. `hcov` is derived from the text's hypotheses (§§IX.36–IX.37), and the case $m = 0$ is §IX.38 (gap G.109, closed).

IX.35 Theorems 9.5.17, 9.5.18 (2017: 9.6, 9.7): Stokes' theorem, and the hemisphere

The text, paraphrased (pp. 449–451). For X with sc-smooth partitions of unity, Θ tame, oriented and compact of dimension n, and ω an sc-differential form of degree $n-1$, $\int_{\widehat{\Theta}} d\omega = \mu^{\partial}_{\omega}(\partial S)$; the effective version asks in addition for the local unique continuation property. The proof covers S by finitely many $|U(x_k)|$ carrying oriented branching structures $(M_i, o_i)_{i\in I_k}$, takes an sc-smooth partition of unity $\beta_0, \dots, \beta_\ell$ constant on the orbits and subordinate to the saturations U_k^* and a further saturated U_0^*, puts $\omega_k = \beta_k\omega$, applies the classical Stokes theorem on each sheet, and sums: $\mu_{d\omega}(S) = \sum_k c_k \sum_{i\in I_k} \sigma_i \int_{(M_i,o_i)} d\omega_k = \sum_k c_k \sum_{i\in I_k} \sigma_i \int_{\partial(M_i,o_i)} \omega_k = \sum_k \mu^{\partial}_{\omega_k}(\partial S) = \mu^{\partial}_{\omega}(\partial S)$, with $c_k = 1/\sharp G_{x_k}$ or $1/\sharp G^{\mathrm{eff}}_{x_k}$.

The Lean proof, step by step.

Step 1 (the cover and the partition of unity). From the sc-smooth partitions of unity of Definition 7.5.5 and the compactness of S one gets finitely many structures and $\beta_0, \dots, \beta_\ell$ as in the text, with $U_0^* = \pi^{-1}(|X| \setminus S)$ `Chapter9/StokesPartition · EP.exists_stokes_partition`. The product $\beta\omega$ is a form of $\Omega^m_{ep,\infty}(X)$ `Chapter9/FunctionTimesForm · EP.epFormsInftySmul`, and the restriction to a level commutes with the evaluation used by the integral `Chapter9/LevelRestrict · formsInftyRestrict_epFormsInftyToObj` `Chapter9/LevelRestrict · epFormsInftyEval_integ_eq_integral`.

Step 2 (Stokes on one sheet). The sheets M_i need not be compact. What makes the classical theorem applicable is that $M_i \cap \operatorname{supp}\beta_k$ is compact, which the text does not state; it follows from the properness of $U(x) \to |U(x)|$ and the compactness of Θ `Chapter9/StokesLocalFormulas · EpGroupoid.NaturalRep.exists_ultrafilter_le_nhds` `Chapter9/StokesLocalFormulas · EpGroupoid.TameBranchingStructure.isCompact_inter`. Then $\int_M (d\omega)|_M = \int_{\partial M} \omega|_M$ on a σ-compact oriented sheet with corners when the restriction has compact support `Chapter9/StokesLocalFormulas · MPolyfold.stokes_formsInftyRestrict_of_isCompact` `Chapter9/StokesLocalFormulas · MPolyfold.OrientedSheet.stokes`. A sheet of an oriented tame structure is such a sheet, with the tangent orientation of o_i `Chapter9/StokesSheet · OrientedTameStructure.orientedSheet`, so $\int_{(M_i,o_i)} d(\beta\omega) = \int_{\partial(M_i,o_i)} \beta\omega$ `Chapter9/StokesSheet · EP.integ_d_smul_eq_boundaryIntegral`, and on one structure $\Phi(d(\beta\omega))(S) = c(x)\sum_i \sigma_i \int_{\partial(M_i,o_i)} \beta\omega$ for every Φ with the local formula of Theorem 9.5.4 or 9.5.5 `Chapter9/StokesSheet · EP.stokes_one_structure` (the second and third equalities of the display). The term $k = 0$ vanishes `Chapter9/StokesSheet · EP.stokes_off_support`.

Step 3 (the sum). Summing over k gives the first three equalities of the display `Chapter9/StokesPartition · EP.stokes_of_partition`, hence Theorems 9.5.17 and 9.5.18 with the right-hand side written as the sum of boundary integrals `Chapter9/StokesPartition · EP.theorem_9_5_17_boundary_sum` `Chapter9/StokesPartition · EP.theorem_9_5_18_boundary_sum`. The measure $\mu_{d\omega}$ is local: it vanishes on sets above which the integrals van-

ish Chapter9/MeasureWholeSheets · measure_eq_zero_of_integ_eq_zero Chapter9/MeasureWholeSheets · measure_univ_eq_of_integ_eq_zero. The last two equalities of the display are two properties of $\omega \mapsto \mu^{\partial}_{\omega}(\partial S)$: the local formula for $\beta\omega$ with $\operatorname{supp}\beta \subseteq U(x)^*$, and vanishing when $\operatorname{supp}\beta$ misses $\operatorname{supp}\Theta$. Every linear functional with these two properties satisfies Stokes' theorem Chapter9/StokesPartition · EP.theorem_9_5_17_of_boundary Chapter9/StokesPartition · EP.theorem_9_5_18_of_boundary, and such a functional exists Chapter9/StokesPartition · EP.exists_boundary_functional.

Step 4 (every dimension $n \geq 1$). The statements of Step 3 were first proved for $n \geq 2$, because in degree 0 the evaluation on frames is cut off. The product rule $(\beta\omega)|_N = \beta\,(\omega|_N)$ is proved in every degree Chapter9/StokesSheetAllDegrees · EP.formsInftyRestrict_epFormsInftySmul_all, and the statements hold for $n = 1$ as well Chapter9/StokesSheetAllDegrees · EP.stokes_one_structure_all Chapter9/StokesPartitionAllDegrees · EP.stokes_of_partition_all Chapter9/StokesPartitionAllDegrees · EP.theorem_9_5_17_boundary_sum_all Chapter9/StokesPartitionAllDegrees · EP.theorem_9_5_18_boundary_sum_all Chapter9/StokesPartitionAllDegrees · EP.exists_boundary_functional_all.

Step 5 (the classical boundary integral is the integral over the stratum). The text's $\int_{\partial(M_i,o_i)}$ is Part I's integral over the faces $\{y_l = 0\}$ of corner charts with signs; μ^{∂} of §IX.34 integrates over the stratum $\partial_1 M_i = \{d_{M_i} = 1\}$ with the boundary orientation $o_{\partial M_i}$ of Definition 9.3.8. The text identifies the two without comment. The face frame $T\iota(\iota_l e_1), \dots, T\iota(\iota_l e_m)$ is positive for $o_{\partial M}$ exactly when the sign of the face matches the reading of o in the chart Chapter9/FaceFrame · MPolyfold.faceFrame_mem_stratumOrientation_iff; the change of variables on a face is Chapter9/FaceChange · FormEval.face_change; the open face piece and the face domain differ by a null set, the corners Chapter9/BoundaryIntegralFaces · CornerCoords.faceSet_ae_eq; one term of Part I's integral is an integral over a face piece Chapter9/BoundaryIntegralFaces · FormEval.face_term_eq, and a weighted integral over the stratum splits over the faces Chapter9/BoundaryIntegralFaces · FormEval.integral_density_eq_sum_face. Hence $\int_{\partial(M_i,o_i)} a = \int_{\partial_1 M_i} a|(\partial_1 M_i, o_{\partial M_i})$ when the nonempty boundary sheet lies in the multi-chart class Chapter9/BoundaryIntegralStratum · EP.boundaryIntegral_eq_msInteg.

Step 6 (the boundary measures are local). $\mu^{\partial}_{\omega}(D) = 0$ when the integrals of ω over the boundary sheets above D vanish Chapter9/BoundaryLocal · boundaryMeasure_eq_zero_of_integ_eq_zero, and $\mu^{\partial}_{\omega}(\partial S) = c(x)\sum_i \sigma_i \int_{\partial_1 M_i} \omega$ for one structure when the integrals vanish off a closed saturated subset of $U(x)^*$ Chapter9/BoundaryLocal · boundaryMeasure_univ_eq. For structures at points off $\operatorname{supp}(\partial\Theta)$ the structure is first restricted into a neighbourhood that avoids $\operatorname{supp}(\partial\Theta)$ Chapter9/BoundaryLocal · OrientedTameStructure.exists_restrict_subset, and both sides vanish.

Step 7 (the theorems). Steps 5 and 6 give the two properties of Step 3 for $\omega \mapsto \mu^{\partial}_{\omega}(\partial S)$. The generic statement is Chapter9/StokesBoundaryMeasure · EP.stokes_of_boundaryMeasure; for every functional with the local formula of Theorem 9.5.15 or 9.5.16 it is Chapter9/StokesBoundaryMeasure · EP.theorem_9_5_17_hyp Chapter9/StokesBoundaryMeasure · EP.theorem_9_5_18_hyp; with the boundary measures themselves Chapter9/StokesBoundaryMeasure · EP.boundaryBranchedMeasure Chapter9/StokesBoundaryMeasure · EP.boundaryBranchedMeasureEff it reads $\int_{\hat\Theta} d\omega = \mu^{\partial}_{\omega}(\partial S)$ and $\oint_{\hat\Theta} d\omega = \oint_{\partial S} \omega\, d\mu^{\partial}$ Chapter9/StokesBoundaryMeasure · EP.theorem_9_5_17_msheets Chapter9/StokesBoundaryMeasure · EP.theorem_9_5_18_msheets, for $\omega \in \Omega^m_{ep,\infty}(X)$ with $m \geq 1$. **Added hypothesis `hcovS`**, not derived from the text: at every point of $\operatorname{supp}\Theta$ an oriented tame structure describing $\hat T_\Theta$ whose sheets lie in one graph chart and whose nonempty boundary sheets lie in the multi-chart class; it gives the `hcov` of §IX.34 Chapter9/StokesBoundaryMeasure · EP.hcovB_of_hcovS. It is derived from the text's hypotheses in §IX.36. For Theorem 9.5.18 the hypotheses of Theorem 9.5.5 are kept as in §IX.32.

The witness. X is the trivial ep-groupoid of the half-space $[0,\infty) \times \mathbb{R}^2$, and $\Theta = 1$ on the closed upper hemisphere $\{|p| = 1\}$, a compact 2-dimensional tame sheet whose boundary stratum is

the equator Chapter9/HemisphereSheet · hemi Chapter9/HemisphereSheet · isSubmanifoldOfDim_hemi Chapter9/HemisphereSheet · boundaryStratum_hemi, with its orientation and $\hat{T}_\Theta$ Chapter9/HemisphereWitness · thetaH Chapter9/HemisphereWitness · hemiOrientation Chapter9/HemisphereWitness · TH. X has sc-smooth partitions of unity Chapter9/HemisphereWitness · e8w_hasScPartitionsOfUnity, $\hat{T}_\Theta$ is an orientation Chapter9/HemisphereWitness · e8w_isOrientation, `hcov` and `hcovS` hold Chapter9/HemisphereWitness · e8w_hcov, and $\partial S \neq \emptyset$ Chapter9/HemisphereWitness · e8w_boundary_nonempty; together Chapter9/HemisphereWitness · e8w_stokes_hypotheses. So Theorem 9.5.17 applies with a nonempty boundary Chapter9/HemisphereStokes · E8W.e8_hemisphere_stokes. The witness has $m = 1$ and no corners, branching or isotropy. As a trivial ep-groupoid it has the local unique continuation property, so it is a witness for Theorem 9.5.18 as well (§IX.38).

With the text's hypotheses. `hcovS` is derived (§IX.36), the case $m = 0$ holds with μ^∂ (§IX.38), and the effective version has the witness of §IX.38; row V9.68 is restated. What remains of gap G.110 is Remark 9.5.20 (§IX.39).

IX.36 The class conditions derived: the corner estimate, and Theorems 9.5.15–9.5.18 for $n \geq 2$

The first proofs of §§IX.34–IX.35 assume `hcov` and `hcovS`: the nonempty boundary sheets $\partial_1 M_i$ of the structures lie in the multi-chart class, so that $\int |\omega|$ over them is finite. Both follow from the text's hypotheses. The earlier estimate for this (800–1,200 lines) assumed that the closed faces of a sheet would be built as submanifolds; the route taken needs none.

Step 1, forms are continuous up to the corners. On a tame sheet M, Part I's chart form of $\omega \in \Omega^m_{ep,\infty}(X)$ is a C^∞ form on the Mathlib manifold with corners M, read in its corner charts, which lie in the maximal atlas (§IX.30).

Step 2, the face change of variables with absolute values. The change of variables of §IX.35, Step 5, is redone with absolute values and lower integrals: on a face set of a corner chart κ, $|\omega|$ on the graph-chart frame is $|\det|$ times the absolute value of the face pull-back Chapter9/CornerFinite · EpGroupoid.FormEval.abs_ev_faceFrame. On the preimage of a compact subset of the chart domain the face pull-back is a continuous function on a compact set, so it is integrable Chapter9/CornerFinite · EpGroupoid.integrableOn_facePullback_of_isCompact.

Step 3, the corner estimate Chapter9/CornerFinite · EpGroupoid.FormEval.boundaryStratum_inter_mem_msheets. If $\partial_1 M \cap V \subseteq K \subseteq M$ with K compact, then $\partial_1 M \cap V$ is in the multi-chart class. Finitely many corner charts, each with a compact piece of its domain, cover K. $\partial_1 M \cap V$ is a countable union of face images, so it is Lindelöf and has a cover by countably many graph charts Chapter9/CornerFinite · EpGroupoid.FormEval.exists_chartCover_of_isLindelof. Each piece of that cover is split by corner chart and pulled back by Step 2. For a fixed corner chart the pieces are disjoint and lie in one compact set of face parameters, so the sum of the integrals of $|\omega|$ is at most a finite sum of integrals over compact sets.

Step 4, shrinking a structure. Every oriented tame structure at x restricts to one with the same natural representation, describing the same $\hat{T}_\Theta$, whose $U(x)$ lies in the interior of a closed neighbourhood t Chapter9/BoundaryCover · EpGroupoid.OrientedTameStructure.exists_restrict_eq. Then $\partial_1 M_i \subseteq M_i \cap t \cap K$, where K is the compact set containing $\operatorname{supp}\Theta \cap U(x)$ given by the properness property, and M_i meets a compact subset of $U(x)$ in a compact set because $M_i \to U(x)$ is proper Chapter9/BoundaryCover · EpGroupoid.LocalBranchingStructure.isCompact_M_inter. Step 3 applies Chapter9/BoundaryCover · EP.boundaryStratum_inter_mem_

msheets Chapter9/BoundaryCover · EP.exists_restrict_boundary_mem_msheets. The natural representation is kept, and with it the local unique continuation property.

Step 5, the theorems. So hcov and hcovS follow from hloc, the reading (d) of "oriented, tame" (§IX.23) Chapter9/BoundaryCover · EP.hcov_of_loc Chapter9/BoundaryCover · EP.hcov_of_loc_isLUC Chapter9/BoundaryCover · EP.hcovS_of_loc Chapter9/BoundaryCover · EP.hcovS_of_loc_isLUC. Theorems 9.5.15 and 9.5.16 hold with the local formula for the structures whose nonempty boundary sheets are in the class Chapter9/StokesTextHypotheses · EP.theorem_9_5_15_class Chapter9/StokesTextHypotheses · EP.theorem_9_5_16_class; such structures exist at every point and every structure restricts to one, so the uniqueness is the text's. The canonical measures are Chapter9/StokesTextHypotheses · EP.boundaryMeasure Chapter9/StokesTextHypotheses · EP.boundaryMeasureEff Chapter9/StokesTextHypotheses · EP.boundaryMeasure_spec. **Theorems 9.5.17 and 9.5.18 as printed,** $m \geq 1$ (V9.68) Chapter9/StokesTextHypotheses · EP.theorem_9_5_17 Chapter9/StokesTextHypotheses · EP.theorem_9_5_18. *For X with sc-smooth partitions of unity, Θ compact, branched, with an orientation $\hat{T}_\Theta$ and* hloc*, and $\omega \in \Omega^m_{ep,\infty}(X)$: $\int_{\hat{\Theta}} d\omega = \mu^\partial_\omega(\partial S)$, with the measure of Theorem 9.5.15; with the local unique continuation property of X, the same for $\oint$ and the measure of Theorem 9.5.16* Chapter9/StokesTextHypotheses · EP.hloc_hasLUC. The hemisphere of §IX.35 satisfies these hypotheses with $\partial S \neq \emptyset$ Chapter9/HemisphereBoundaryMeasure · E8W.k1_hemisphere_stokes.

IX.37 The local formula for every oriented tame structure

The text states the formula of Theorem 9.5.15 for every structure. Step 4 of §IX.36 handles a structure after shrinking it; toward the frontier of $U(x)$ the corner charts of M_i give no uniform bound. The formula holds for every structure all the same, by comparison with other structures.

Step 1, a finite cover. $\partial_1 M_i$ lies in a compact subset of $\operatorname{supp}\Theta$, by the properness property. Cover it by finitely many $U(p)'$ of structures as in §IX.36, whose nonempty boundary sheets are in the class.

Step 2, a common tangent space Chapter9/AllStructures · MPolyfold.k1_exists_tangent_eq. At $z \in \partial_1 M_i \cap U(p)'$ some sheet M'_k of the other structure passes through z with $T_z M'_k = T_z M_i$. Otherwise there is an open $O \ni z$ on which, at every common point, the tangent spaces of M_i and of each M'_k differ. Since $M_i \subseteq \operatorname{supp}\Theta$, every point of $M_i \cap O$ lies on some M'_k, and there the relative tangent space of $M_i \cap M'_k$ is not full. Such points form an $(m+1)$-null set, by the argument of Lemma 9.4.6 (§IX.28), while an open piece of an $(m+1)$-dimensional submanifold is not null.

Step 3, the same stratum. $\{d = 1\}$ is defined through the tangent space alone, so $z \in \partial_1 M'_k$ Chapter9/AllStructures · MPolyfold.k1_mem_boundaryStratum_of_tangent_eq. Whether $d_{M_i} = d_X$, which the tame induced structure does not give, does not matter.

Step 4, domination. $\partial_1 M_i$ lies in the union of finitely many sheets of the class, and a submanifold covered by finitely many sheets of the class is in the class Chapter9/AllStructures · EpGroupoid.FormEval.k1_msheets_of_subset_list. So every nonempty boundary sheet of every oriented tame structure is in the class Chapter9/AllStructures · EP.boundarySheet_mem_msheets_of_loc.

Theorems 9.5.15 and 9.5.16 as printed, $m \geq 1$ (V9.67, restated) Chapter9/AllStructures · EP.theorem_9_5_15 Chapter9/AllStructures · EP.theorem_9_5_16. *For $n = m + 1$ with $m \geq 1$, Θ compact, branched, with an orientation $\hat{T}_\Theta$ and* hloc*, there is exactly one linear $\omega \mapsto \mu^\partial_\omega$ from $\Omega^m_{ep,\infty}(X)$ to the signed measures on $(\partial S, L(\partial S))$ with $\mu^\partial_\omega(K) = \frac{1}{\sharp G_x} \sum_i \sigma_i \int_{K_i} \omega|(\partial_1 M_i, o_{\partial M_i})$ for every $x \in \operatorname{supp}(\partial\Theta)$, every oriented tame structure at x describing $\hat{T}_\Theta$ and every measurable K with $|K| \subseteq |U(x)|$;* with $\sharp G_x^{\text{eff}}$ when X has the local unique continuation property. The canonical measure has the formula for every structure Chapter9/AllStructures · EP.

boundaryMeasure_apply_all. The sum runs over the nonempty $\partial_1 M_i$, which contribute nothing in the text when empty, and n-dimensionality follows from hloc.

IX.38 Dimension one: 0-forms, finite sets of boundary points, and the half-circle

For $n = 1$ the boundary sheets are 0-dimensional and $\omega \in \Omega^0_{ep,\infty}(X)$; the multi-chart class of §IX.34 needs $m > 0$.

Step 1, 0-forms as functions. A 0-form is read as a function on objects, its level-0 value at a smooth point, invariant under morphisms Chapter9/ZeroFormValues · EP.zeroFormEval; its restriction to a sheet is this value Chapter9/ZeroFormValues · EP.formsInftyRestrict_zero_eq. The integral over an oriented point p is $\pm\omega(p)$, the sign from the orientation: the point integration of §IX.31 through this evaluation Chapter9/ZeroFormValues · EP.zeroFormIntegration.

Step 2, finite sets of oriented points Chapter9/FinitePointIntegration · EpGroupoid.finPointIntegration. For a finite set N of smooth points, $\int_A \omega = \sum_{p \in N \cap A} \pm\omega(p)$. This is a sheet integration in the sense of §IX.31 for the class of finite sets Chapter9/FinitePointIntegration · EpGroupoid.finPointSheets, with the null sets of dimension 0: the tangent space of a finite set at one of its points is that of the point Chapter9/FinitePointIntegration · EpGroupoid.submanifoldTangent_finite_eq, and every subset of a finite set is measurable. On sets with at most one point it is the point integration Chapter9/FinitePointIntegration · EpGroupoid.finPointInteg_eq_pointInteg. For 0-forms: Chapter9/FinitePointIntegration · EP.finZeroFormIntegration.

Step 3, the boundary sheets are finite. In a corner chart with values in $[0, \infty)$ the face is one point, so near every point a 1-dimensional tame sheet has at most one point of its boundary stratum Chapter9/BoundaryMeasureDimOne · MPolyfold.exists_isOpen_boundaryStratum_subsingleton. Every oriented tame structure restricts to one whose boundary sheets have at most one point Chapter9/BoundaryMeasureDimOne · EpGroupoid.OrientedTameStructure.exists_boundarySheet_subsingleton, and the class conditions follow from hloc Chapter9/BoundaryMeasureDimOne · EpGroupoid.hcov_dim1_of_loc Chapter9/BoundaryMeasureDimOne · EpGroupoid.hcovEff_dim1_of_loc. The covering argument of §IX.37 then shows that every $\partial_1 M_i$ is finite Chapter9/AllStructuresDimOne · EP.boundarySheet_finite_of_loc Chapter9/AllStructuresDimOne · EP.boundarySheet_mem_finPointSheets_of_loc.

Step 4, Part I's boundary integral. For a 1-dimensional sheet with one boundary point p, Part I's $\int_{\partial M} \omega$ is $\pm\omega(p)$: the Lebesgue measure on $\mathbb{R}^0$ is a Dirac measure, and the face sign is that of §IX.35, Step 5, at $m = 0$ Chapter9/FaceIntegralDimOne · EpGroupoid.boundaryIntegral_eq_pointInteg Chapter9/FaceIntegralDimOne · EP.boundaryIntegral_eq_zeroFormIntegration.

Step 5, Stokes. The locality argument of §IX.35, Step 6, is redone for any sheet integration Chapter9/StokesDimOne · EpGroupoid.boundaryMeasure_univ_eqI and combined with the boundary-sum form of §IX.35, Step 4 Chapter9/StokesDimOne · EP.stokes_of_boundaryMeasure_dim1.

The theorems for $n = 1$ (V9.67, V9.68). Theorems 9.5.15 and 9.5.16 with the formula for every oriented tame structure, $\int_{K_i} \omega | (\partial_1 M_i, o) = \sum_{p \in K_i} \pm\omega(p)$ Chapter9/AllStructuresDimOne · EP.theorem_9_5_15_dim1_all Chapter9/AllStructuresDimOne · EP.theorem_9_5_16_dim1_all, first proved for the structures whose boundary sheets are points Chapter9/BoundaryMeasureDimOne · EP.theorem_9_5_15_dim1 Chapter9/BoundaryMeasureDimOne · EP.theorem_9_5_16_dim1 Chapter9/EffectiveDimOneLUC · EP.theorem_9_5_16_dim1_isLUC. Theorems 9.5.17 and 9.5.18: $\int_{\hat{\Theta}} d\omega = \mu^{\partial}_{\omega}(\partial S)$ Chapter9/StokesDimOne · EP.theorem_9_5_17_dim1 Chapter9/StokesDimOne · EP.theorem_9_5_18_dim1 Chapter9/EffectiveDimOneLUC · EP.theorem_9_5_18_dim1_isLUC, with the measures Chapter9/StokesDimOne · EP.

boundaryBranchedMeasureDim1 Chapter9/StokesDimOne · EP.boundaryBranchedMeasureEffDim1, which are those of the $n = 1$ Theorems 9.5.15 and 9.5.16 Chapter9/AllStructuresDimOne · EP.boundaryBranchedMeasureDim1_apply_all Chapter9/AllStructuresDimOne · EP.boundaryBranchedMeasureEffDim1_apply_all. The hypotheses are the text's.

Witnesses with nonempty boundary. $n = 1$. $X = [0,\infty) \times \mathbb{R}$ as a trivial ep-groupoid, and $\Theta = 1$ on the closed half-circle $\{|p| = 1\}$, the image of the sc$^+$-retraction $p \mapsto p/|p|$ of $X \setminus \{0\}$: a compact, tame, 1-dimensional submanifold Chapter9/HalfCircleSheet · K2Arc.isSubmanifoldOfDim_arc Chapter9/HalfCircleSheet · K2Arc.isTameSubMPolyfold_arc whose boundary stratum is its two end points on $\{0\} \times \mathbb{R}$ Chapter9/HalfCircleSheet · K2Arc.boundaryStratum_arc, oriented by a field that is a unit tangent on the circle Chapter9/HalfCircleWitness · K2Arc.arcOrientation. All hypotheses of Theorems 9.5.15–9.5.18 for $n = 1$ hold, with $\operatorname{supp}(\partial\Theta) \neq \emptyset$ Chapter9/HalfCircleWitness · K2Arc.arc_hloc Chapter9/HalfCircleWitness · K2Arc.arc_isLUC Chapter9/HalfCircleWitness · K2Arc.k2_arc_hypotheses, and Stokes' theorem applies Chapter9/HalfCircleWitness · K2Arc.k2_arc_stokes Chapter9/HalfCircleWitness · K2Arc.k2_arc_stokes_eff. $n = 2$, *the effective version.* Every natural representation of a trivial ep-groupoid has the local unique continuation property Chapter9/HemisphereStokesEffective · E8W.hasLUC_trivial, so the hemisphere of §IX.35 has it Chapter9/HemisphereStokesLUC · E8W.e8w_isLUC, and Theorem 9.5.18 applies with $\partial S \neq \emptyset$ Chapter9/HemisphereStokesLUC · E8W.k2_hemisphere_stokes_eff Chapter9/HemisphereStokesEffective · E8W.e8_hemisphere_stokes_eff. Neither witness has corners, branching or nontrivial isotropy; Remark 9.5.20 is about the last (§IX.39).

IX.39 Remark 9.5.20 (2021 pp. 451–452): the example, in part

The text. $X = \{(-1,0)\} \cup ((-1,1) \times B) \cup \{(1,0)\} \subseteq \mathbb{R} \times \mathbb{C}$ with an M-polyfold structure, $\mathbb{Z}_3$ rotating the $\mathbb{C}$-factor, $M = [-1,1] \times \{0\}$, $\Theta(t,z) = \delta_0(z)$ oriented by $[-1,1]$, and $f(t,z) = \phi(t)$. The remark computes $\oint_\Theta df = \frac{1}{3}(\phi(1) - \phi(-1))$ and observes that $\partial X = \{(\pm 1, 0)\}$ is a sub-ep-groupoid whose points have trivial effective isotropy, while in X it is $\mathbb{Z}_3$: the boundary integral of f is $\frac{1}{3}(\phi(1) - \phi(-1))$ read in X, in accordance with Stokes' theorem, and $\phi(1) - \phi(-1)$ read in ∂X.

The M-polyfold Chapter9/DimensionJumpPolyfold · K3.P. The text says only that X can be given an M-polyfold structure. It is given one with one chart, onto an sc-smooth retract of the partial quadrant $[0,\infty)^2 \times (c_0 \oplus c_0)$ in $E = \mathbb{R}^2 \oplus c_0 \oplus c_0$, with the weighted scale of Remark 4.1.3's example (§IV.27). The retraction is $R(p,a,b) = (2p/|p|, \pi_{\tau(p)}a, \pi_{\tau(p)}b)$ Chapter9/DimensionJumpRetract · K3.Rt Chapter9/DimensionJumpRetract · K3.mem_0_iff, with Remark 4.1.3's family π_τ (rank one for $\tau > 0$, zero for $\tau \leq 0$) composed with $\tau(p) = p_0 p_1$ Chapter9/DimensionJumpRetract · IsLossFamily.comp_contDiff Chapter9/DimensionJumpRetract · IsLossFamily.diag. Its retract is the quarter circle of radius 2 carrying a plane over each interior point and a point over each end. The chart is $\psi(t,z) = (2(\cos\theta, \sin\theta), \operatorname{Re} w\,\beta_\tau, \operatorname{Im} w\,\beta_\tau)$ with $\theta = \pi(1+t)/4$ and $w = z/\sqrt{1-|z|^2}$ Chapter9/DimensionJumpPolyfold · K3.chart. The degeneracy index is 1 at $(\pm 1, 0)$ and 0 elsewhere Chapter9/DimensionJumpPolyfold · K3.degIndex_eq; the lower bound at the ends is Theorem 2.4.12, from an inward vector in the tangent space of the retract Chapter9/DimensionJumpPolyfold · K3.scDeriv_end. So $\partial X = \{(-1,0),(1,0)\}$ Chapter9/DimensionJumpPolyfold · K3.boundary_eq.

The groupoid Chapter9/CyclicGroupoidIsotropy · K3.ε. $\mathbb{Z}_3$ acts by $k \cdot (t,z) = (t, \omega^k z)$, sc-smoothly, since in the chart it is the restriction of the rotation of (a,b), an sc-operator commuting with R Chapter9/CyclicGroupoidIsotropy · K3.isScSmoothMap_smul. Its non-effective part is $\{1\} \times X$ Chapter9/CyclicGroupoidIsotropy · K3.nonEffectivePart_ε, which is closed, so $\varepsilon = \mathbb{Z}_3 \ltimes X$ has the local unique continuation property Chapter9/CyclicGroupoidIsotropy · K3.isLUC_ε. The effective isotropy has three elements on M Chapter9/CyclicGroupoidIsotropy · K3.card_effIsotropy_of_snd_eq_zero and one off M Chapter9/

CyclicGroupoidIsotropy · K3.card_effIsotropy_of_snd_ne_zero. ∂X is a saturated sub-M-polyfold Chapter9/CyclicGroupoidIsotropy · K3.isEpSubgroupoid_boundary, and ε_B is the sub-ep-groupoid of Proposition 7.1.22 on it Chapter9/CyclicGroupoidIsotropy · K3.εB. At a point of ∂X the isotropy group has three elements in ε and in ε_B Chapter9/CyclicGroupoidIsotropy · K3.card_isotropy_εB, and the effective isotropy has three in ε and one in ε_B Chapter9/CyclicGroupoidIsotropy · K3.card_effIsotropy_εB: every morphism of ε_B acts trivially near its points. Collected: Chapter9/CyclicGroupoidIsotropy · K3.remark_9_5_20_effIsotropy.

Not formalized (gap G.110, in part). Θ with its orientation and `hloc` at the infinite-dimensional ends; sc-smooth partitions of unity of ε (Corollary 7.6.6 asks for a Hilbert space on level 0, and here E_0 contains c_0); f as a 0-form; $\oint_{\partial\Theta} df = \frac{1}{3}(\phi(1) - \phi(-1))$, by the $n = 1$ theorems of §IX.38 and the local formula at the two ends; the integral of f over $\partial\Theta$ in ε_B; and that ∂X is a smooth submanifold, shown at each of its two points only. Estimated at 2,340–3,790 lines (Findings, Chapter 9, item 120).

Part X. Equivalences and localization (HWZ Chapter 10, §§10.1–10.5 and Appendix §§10.6.1–10.6.3 (2017 §§10.1–10.5))

Chapter 10 continues Part II; its numbers and pages are those of the book (Springer 2021, pp. 461–520), with the 2017 text's in parentheses (2017 pp. 333–374). It studies the sc-smooth functors between ep-groupoids that should become invertible once ep-groupoids are taken as models of polyfolds. §10.1 defines equivalences (Definition 10.1.1 (2017: 10.1)), characterizes them as the equivalences of categories that are local sc-diffeomorphisms on objects (Lemma 10.1.2 (2017: 10.1)), builds the refinement along an open covering (Proposition 10.1.3 (2017: 10.1)), shows that natural equivalence, the two-out-of-three property and the tangent functor respect equivalences (Definition 10.1.4, Propositions 10.1.5, 10.1.6, Theorem 10.1.7 (2017: 10.2, 10.2, 10.3, 10.1)) and, mostly new in 2021, that the properties of Chapter 7 are invariant (Remarks 10.1.8, 10.1.11, 10.1.12, Propositions 10.1.9, 10.1.10, 10.1.13, 10.1.14, 10.1.16–10.1.19, Example 10.1.15). §10.2 constructs the weak fibred product (Definition 10.2.1 (2017: 10.3), Theorem 10.2.2 (2017: 10.2), Remark 10.2.3 (2017: 10.2)). §10.4 carries all this to strong bundles: strong bundle equivalences (Definition 10.4.1 (2017: 10.10)), the pull-back and push-forward of section functors (Propositions 10.4.2–10.4.4, Theorem 10.4.5 (2017: 10.7–10.9, 10.8)), natural transformations of strong bundle functors (Definitions 10.4.8, 10.4.10, Remark 10.4.9 (2017: 10.11, 10.12, 10.4)) and the category $\mathcal{SEP}$ (Definition 10.4.11 (2017: 10.13)). §10.3, treated here after §10.4 in the order of the verified edition, inverts the equivalences: diagrams $X \xleftarrow{F} A \xrightarrow{\Phi} Y$ with F an equivalence, their refinement and equivalence (Definitions 10.3.1, 10.3.3, 10.3.5, Lemmas 10.3.2, 10.3.4, 10.3.6, Definition 10.3.7 (2017: 10.4–10.7, 10.3–10.5)), the composition of classes (Theorem 10.3.8 (2017: 10.3), Appendix §10.6.1), generalized isomorphisms (Definition 10.3.9, Theorem 10.3.10 (2017: 10.8, 10.4), Appendix §10.6.2), the category $\mathcal{EP}(E^{-1})$ with the functors i and $|\cdot|$ (Proposition 10.3.11, Definition 10.3.12, Theorems 10.3.14, 10.3.15 (2017: 10.6, 10.9, 10.5, 10.6)) and the invariance of the properties of Chapter 7 under generalized isomorphisms (Propositions 10.3.16, 10.3.17, Remark 10.3.18, Theorem 10.3.19, Definition 10.3.20, Theorem 10.3.21). §10.5, treated after §10.3, repeats the localization for strong bundles: diagrams $W \xleftarrow{\Phi} W'' \xrightarrow{\Psi} W'$ with Φ a strong bundle equivalence, their classes, the maps $[d]$, $|[D]|$ and the homomorphisms γ (pp. 504–506), the category $\mathcal{SEP}(F^{-1})$ and generalized strong bundle isomorphisms (Definitions 10.5.1, 10.5.2 (2017: 10.14, 10.15), Remark 10.5.3), the push-forward of section functors (Theorem 10.5.4 (2017: 10.9), Remark 10.5.5) and the local regularity property (Proposition 10.5.6, Theorem 10.5.7 (2017: 10.10)). Appendix §10.6.3, new

in 2021, gives the example behind Remark 10.1.8 on a weighted Sobolev scale; §10.1 also introduces the full subcategory $\mathcal{EP}_{lreg}$ (p. 461) and §10.4 the strong bundle versions of Propositions 10.1.9 and 10.1.10 (Propositions 10.4.6, 10.4.7).

What is proved, and what is not. Every gap of Chapter 10 is closed (G.111–G.127): each statement of the chapter is proved, and two, Theorems 10.3.19 and 10.5.7, do not hold as we read them and are proved for an effective target. Nothing in the chapter is open or formalized only in part. *§10.1.* All definitions are formalized. Lemma 10.1.2 (with its converse), Propositions 10.1.3, 10.1.5, 10.1.6, 10.1.13, 10.1.14, 10.1.16–10.1.19, Theorem 10.1.7, Remarks 10.1.11 and 10.1.12 and Example 10.1.15 are proved, several in a stronger form. Proposition 10.1.9 is proved, and Proposition 10.1.10 with (LR) on one side only (§§X.5, X.7; G.113–G.115). The subcategory $\mathcal{EP}_{lreg}$ is §X.27 (G.111). Remark 10.1.8 with the example of Appendix §10.6.3, on the weighted Sobolev scale as printed, is §X.28 (G.112). Two results the text quotes or leaves to the reader are proved: Proposition 10.1.5, and the fullness and faithfulness of TF in Theorem 10.1.7. *§10.2.* All claims of Theorem 10.2.2 are proved, including that the structures are those induced from the triple products (§X.29; G.116), and so is Remark 10.2.3. *§10.4.* All definitions are formalized, and Propositions 10.4.2–10.4.4, Remark 10.4.9 and the equivalence of Definitions 10.4.8 and 10.4.10 are proved. Theorem 10.4.5 is proved with its Fred part (§X.31; G.121); the sc-smoothness of the push-forward, "easily verified" in the text, is proved too. Propositions 10.4.6 and 10.4.7 are §X.32 (G.122), with Definition 8.4.2 read for strong bundle isomorphisms of $W|U(x)$; Proposition 10.4.7 is proved with the property for the target only. *§10.3.* All definitions are formalized, with the proofs of the appendix. Proved: (E1)–(E3), Lemmas 10.3.2, 10.3.4, 10.3.6, Theorems 10.3.8, 10.3.10, 10.3.14, 10.3.15, Propositions 10.3.11, 10.3.16, 10.3.17, Remark 10.3.18 and Theorem 10.3.21 (§§X.16–X.20; G.118 closed by §X.20, G.119 by §X.23). The universal property of the localization (G.117) is §X.30. Theorem 10.3.19 does not hold as we read it, and holds for an effective target (§X.20; G.120, settled). *§10.5.* Formalized: the diagrams, their refinements and classes, $[d]$, $|[D]|$, the square $|\mathcal{D}|$, γ, Definitions 10.5.1 and 10.5.2, and $\mathcal{SEP}(F^{-1})$ as a category through the weak fibred product of strong bundles (§§X.21, X.22). (10.24) with generalized vertical maps and the local expressions q, Q are §X.33 (G.124). Proved: Remark 10.5.3, with both directions of the strong bundle Theorem 10.3.10 (§§X.22, X.34; G.123); Theorem 10.5.4 with its Fred part and as a linear isomorphism (§§X.23, X.31; G.125); Remark 10.5.5, with an example; and Proposition 10.5.6 (§§X.23, X.32; G.126). Theorem 10.5.7 does not hold as we read it, and holds over an effective base of the target (§X.23; G.127, settled). Remark 10.3.13 and the sentences on "smaller models" in Remark 10.2.3 have no formal content. *The narrower row.* There is one: V10.66, Theorem 10.3.19 for an effective target. The effectiveness of Y is an added hypothesis, not derived from the text; without it the theorem does not hold as we read it (`not_theorem_10_3_19`). Its satisfiability witness is the point (§X.20). The strong bundle analogue is part of the restated row V10.67, with the same reason: Theorem 10.5.7 over an effective base of the target, with (LR) of a base or the local regularity property of the target bundle (§X.23).

Six structural choices. (1) *Equivalences as a structure.* `EpFunctor.IsEquivalence` has the four conditions of Definition 10.1.1 as fields, with full and faithful stated on the sets of morphisms; Mathlib's `Functor.IsEquivalence` of the underlying functor is derived (§X.1). (2) *Indexed coverings.* The refinement $X_{\mathcal{U}}$ takes a family $(U_i)_{i\in\iota}$ of open sets; objects and morphisms are open subsets of products with the discrete ι (§X.2). (3) *Iterated fibred products.* The weak fibred product is built from Part I's fibred products, twice, so that its M-polyfold structures come with their universal properties; the admissibility, F or G a local sc-diffeomorphism on objects, is all that the construction uses (§X.9). (4) *Total pull-backs and push-forwards.* Φ^* and Φ_* are defined on all maps, by the zero of the fibre where needed, and the push-forward with one choice of x and σ made for all; all statements are about section functors, where the choice does not matter

(§X.12). (5) *Diagrams one universe up.* A diagram contains its middle ep-groupoid as an object of $\mathcal{EP}$, so the classes of diagrams form a type in the universe of the objects of $\mathcal{EP}$ and $\mathcal{EP}(E^{-1})$ is a large category like $\mathcal{EP}$; compositions are written in Mathlib's order, d.comp e for the text's $[e] \circ [d]$ (§X.16). (6) *The weak fibred product of strong bundles as a pull-back.* The text leaves it to the reader; it is the pull-back of W'' along π_Z, so that one projection is a strong bundle equivalence by construction and the squares in the proofs of the composition commute strictly, by the fibrewise injectivity of strong bundle equivalences; the generalized strong bundle maps, defined as the classes of the equivalence relation generated by having a common refinement, are the text's classes by the transitivity (§§X.21, X.22).

Map.

- `Chapter10/Equivalence`: local sc-diffeomorphisms at a point, Definition 10.1.1, Lemma 10.1.2, Definition 10.1.4, Propositions 10.1.5, 10.1.6, Theorem 10.1.7, the pair groupoid of $\mathbb{Z}$ over the point.
- `Chapter10/EquivalenceRefinement`: open coverings and Proposition 10.1.3, the covering of $\mathbb{Z}/2 \ltimes \mathbb{R}$.
- `Chapter10/EquivalenceReduced`: Remark 10.1.12, Propositions 10.1.13, 10.1.14, 10.1.16, 10.1.18.
- `Chapter10/EquivalenceExamples`: Remark 10.1.11, Example 10.1.15.
- `Chapter10/EquivalenceZhou`: Proposition 10.1.17 and the reduction of Proposition 10.1.19.
- `Chapter10/EquivalenceLR`: transport charts $Q = \widehat{\psi} \circ F$, Propositions 10.1.9, 10.1.10 and 10.1.19.
- `Chapter10/EquivalenceLREffective`: local families over an effective target and the natural equivalence they give.
- `Chapter10/WeakFibredProduct`: Definition 10.2.1 and the first claim of Theorem 10.2.2.
- `Chapter10/WeakFibredProductTheorem`: the projections, Theorem 10.2.2, Remark 10.2.3, witnesses.
- `Chapter10/StrongBundleEquivalence`: Definition 10.4.1, the pull-back and the push-forward, Propositions 10.4.2–10.4.4, Theorem 10.4.5.
- `Chapter10/StrongBundleEquivalenceNat`: Definitions 10.4.8, 10.4.10, 10.4.11, Remark 10.4.9.
- `Chapter10/StrongBundleEquivalenceExamples`: the strong bundle equivalence over $-\mathrm{id}$ and its uses.
- `Chapter10/Localization`: diagrams, Definitions 10.3.1, 10.3.3, 10.3.5, 10.3.7, Lemmas 10.3.2, 10.3.4, 10.3.6, Theorem 10.3.8 with Appendix §10.6.1, the diagonal of Appendix §10.6.2.
- `Chapter10/LocalizationCategory`: (E1), Definitions 10.3.9, 10.3.12, Theorem 10.3.10 with Appendix §10.6.2, Proposition 10.3.11, Theorems 10.3.14, 10.3.15, witnesses.
- `Chapter10/LocalizationProperties`: Proposition 10.3.17, Remark 10.3.18, Definition 10.3.20, Theorem 10.3.21.
- `Chapter10/LocalizationLR`: Proposition 10.3.16, Theorem 10.3.21 for all five properties.
- `Chapter10/LocalizationLRCounterexample`: Theorems 10.3.19 and 10.5.7 as statements, their negations over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$, Theorem 10.3.19 for an effective target.
- `Chapter10/StrongBundleLocalization`: diagrams of strong bundles, their refinements and classes, $[d]$, $|[D]|$, the square $|\mathcal{D}|$, γ, Definitions 10.5.1, 10.5.2, Remark 10.5.3.
- `Chapter10/StrongBundleLocalizationSections`: the push-forward and pull-back, Theorem 10.5.4, the reduction of Proposition 10.5.6.
- `Chapter10/StrongBundleLocalizationExamples`: the relation of the homomorphisms γ, the witnesses over $-\mathrm{id}$, Remark 10.5.5 over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$.
- `Chapter10/StrongBundleWeakFibredProductPullback`: pull-backs of strong bundles along functors, transports along natural transformations, fibrewise inverse lifts.
- `Chapter10/StrongBundleWeakFibredProduct`: the weak fibred product of strong bundles and the

strong bundle Theorem 10.2.2.

- `Chapter10/StrongBundleLocalizationCategory`: the strong bundle Lemma 10.3.4 and Theorem 10.3.8, $\mathcal{SEP}(F^{-1})$ as a category with its localization functor, one direction of the strong bundle Theorem 10.3.10.
- `Chapter10/LocalizationLREffectiveBundle`: Theorem 10.5.7 over an effective base of the target, with (LR) of a base.
- `Chapter10/LocalizationLRBundleBase`: Definition 8.4.2 read for strong bundle isomorphisms of $W|U(x)$, Proposition 10.4.7 on the bases, Theorem 10.5.7 over an effective base from that property.
- `Chapter10/LocalRegularitySubcategory`: the subcategory $\mathcal{EP}_{lreg}$ of $\mathcal{EP}$ and of $\mathcal{EP}(E^{-1})$.
- `Chapter10/ZhouExampleRealScale`: Remark 10.1.8 and Appendix §10.6.3 as printed, on the scale $\mathbb{R} \oplus H^{m,\delta_m}(\mathbb{R})$.
- `Chapter10/ZhouExample`: the same example with the projections of Remark 4.1.3.
- `Chapter10/WeakFibredProductTriple`: the structures of Theorem 10.2.2 induced from the triple products.
- `Chapter10/LocalizationUniversal`: the universal property of $i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$.
- `Chapter10/FredholmEquivalence`: the Fred parts of Theorems 10.4.5 and 10.5.4.
- `Chapter10/SectionFunctorLinear2021`: the push-forwards of Theorems 10.4.5 and 10.5.4 as linear isomorphisms.
- `Chapter10/StrongBundleLocalIso`: local strong bundle isomorphisms between two strong bundles, transports along families of morphisms.
- `Chapter10/StrongBundleLocalRegularity`: Propositions 10.4.6 and 10.5.6.
- `Chapter10/StrongBundleLocalFamilies`: Proposition 10.4.7.
- `Chapter10/StrongBundleDiagram`: the square (10.24) in $\mathcal{EP}(E^{-1})$ and the germ q.
- `Chapter10/StrongBundleDiagramLift`: the lift Q and (10.25).
- `Chapter10/StrongBundleIsoConverse`: the converse half of the strong bundle Theorem 10.3.10.

Throughout, $X = (X, \mathbf{X})$, Y, Z are ep-groupoids as in Part VII, s, t the source and target, $|X|$ the orbit space and $|F| : |X| \to |Y|$ the induced map, G_x the isotropy group, T_x its subgroup acting trivially near x and $\mathbf{X}^T$ the non-effective part (§VII.3). Functors are sc-smooth functors (`EpFunctor`), and natural transformations those of Chapter 8 (`EpNatTrans`).

X.1 Definition 10.1.1 (2017: 10.1) and Lemma 10.1.2 (2017: 10.1)

Definition 10.1.1 (2017: 10.1) (V10.1) `Chapter10/Equivalence · EpFunctor.IsEquivalence` `Chapter10/Equivalence · EP.equivalences`. The four fields are fullness and faithfulness on the sets of morphisms, $|F|$ a homeomorphism, and F a local sc-diffeomorphism on objects. Fullness and faithfulness are those of Mathlib for the underlying functor `Chapter10/Equivalence · GroupoidFunctor.full_iff` `Chapter10/Equivalence · GroupoidFunctor.faithful_iff`, and so is essential surjectivity `Chapter10/Equivalence · GroupoidFunctor.essSurj_iff`. *Step 1, essential surjectivity.* For $y \in Y$, $|F|$ is onto, so $[y] = [F(x)]$ for some x, that is a morphism $F(x) \to y$ exists `Chapter10/Equivalence · EpFunctor.IsEquivalence.essSurj` `Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso`. *Step 2, the morphism sets.* Every $\psi : F(x) \to F(x')$ is $F(\varphi)$ `Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_mor_eq` and F is injective on morphisms with the same ends `Chapter10/Equivalence · EpFunctor.IsEquivalence.mor_injective`. *Step 3, the class E.* The identity is an equivalence `Chapter10/Equivalence · EpFunctor.IsEquivalence.id`, composites are `Chapter10/Equivalence · EpFunctor.IsEquivalence.comp` (the composite of homeomorphisms and of local sc-diffeomorphisms, with $|G \circ F| = |G| \circ |F|$ `Chapter10/Equivalence · GroupoidFunctor.toFunctor_`

comp), so `EP.equivalences` is a multiplicative class. Witness: the functor from the pair groupoid of $\mathbb{Z}$ (one morphism between any two integers) to the point Chapter10/Equivalence · EpGroupoid.Examples.pairToPoint Chapter10/Equivalence · EpGroupoid.Examples.isEquivalence_pairToPoint, a local sc-diffeomorphism because $\mathbb{Z}$ is discrete Chapter10/Equivalence · EpGroupoid.Examples.isLocalScDiffeo_discrete.

Local sc-diffeomorphisms at a point. The local property of Definition 10.1.1 (3) is localized Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt Chapter10/Equivalence · MPolyfold.isLocalScDiffeo_iff_forall Chapter10/Equivalence · MPolyfold.IsLocalScDiffeo.at, with the rules used below: composition Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.comp, agreement near the point Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.congr, and the two cancellation rules: if $g \circ f$ and g are local sc-diffeomorphisms at x and $f(x)$, so is f at x Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.of_comp_left; if $g \circ f$ and f are, so is g at $f(x)$ Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.of_comp_right.

Lemma 10.1.2 (2017: 10.1) (V10.2) Chapter10/Equivalence · EpFunctor.lemma_10_1_2 Chapter10/Equivalence · EpFunctor.isEquivalence_iff. *Step 1, $|F|$ is bijective.* For a functor of groupoids, $|F|$ is injective when F is full ($F(x) \cong F(x')$ lifts to $x \cong x'$) Chapter10/Equivalence · GroupoidFunctor.injective_orbitMap_of_full, surjective exactly when F is essentially surjective Chapter10/Equivalence · GroupoidFunctor.surjective_orbitMap_iff, hence bijective for an equivalence of categories Chapter10/Equivalence · GroupoidFunctor.bijective_orbitMap_of_isEquivalence. *Step 2, $|F|$ is a homeomorphism.* $|F|$ is continuous by the universal property of the quotient Chapter10/Equivalence · EpFunctor.continuous_orbitMap; it is open because F is a local sc-diffeomorphism, hence open, on objects and the orbit maps are open (Proposition 7.1.8 (2017: 7.1)) Chapter7/Basic · EpGroupoid.proposition_7_1_8 Chapter10/Equivalence · EpFunctor.isOpenMap_orbitMap. *Step 3, the converse.* An equivalence is full, faithful and essentially surjective (§X.1), so an equivalence of categories Chapter10/Equivalence · EpFunctor.IsEquivalence.isEquivalence_toFunctor. *Not invertible.* The functor $\mathbb{Z} \to \mathrm{pt}$ above has no left inverse on objects Chapter10/Equivalence · EpGroupoid.Examples.not_exists_leftInverse_pairToPoint.

X.2 Proposition 10.1.3 (2017: 10.1): the refinement $X_{\mathcal{U}}$

The construction Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.objSet Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.morSet Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.groupoidStr. A covering is a family of open sets U_i with every object in some U_i; the trivial covering by X is one Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.univ. The text indexes the covering by itself and writes (U, x); the index type is the formal reading. The objects (i, x), $x \in U_i$, form an open subset of $\iota \times X$ Chapter10/EquivalenceRefinement · isOpen_of_discrete_left, the morphisms (i, φ, j) one of $(\iota \times \iota) \times \mathbf{X}$, and both get the open-subset M-polyfold structures Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.objPolyfold Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.morPolyfold, which need a nonempty set: the object set of X is assumed nonempty (for an empty X there is nothing to prove). Products with a discrete space keep the paracompactness and metrizability Chapter10/EquivalenceRefinement · paracompactSpace_discrete_prod.

$X_{\mathcal{U}}$ is an ep-groupoid Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.epGroupoid. *Step 1, local sc-diffeomorphisms.* On a product with a discrete factor a map that is a local sc-diffeomorphism on each sheet is one Chapter10/EquivalenceRefinement · isLocalScDiffeo_sheet Chapter10/EquivalenceRefinement · isLocalScDiffeo_snd_discrete; $s(i, \varphi, j) = (i, s(\varphi))$ and $t(i, \varphi, j) = (j, t(\varphi))$ are local sc-diffeomorphisms Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isLocalScDiffeo_src Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isLocalScDiffeo_tgt, through the restriction

rules for open subsets Chapter10/EquivalenceRefinement · MPolyfold.isScSmoothOn_restrictOpen_iff Chapter10/EquivalenceRefinement · MPolyfold.isScSmoothMap_restrict Chapter10/EquivalenceRefinement · MPolyfold.isLocalScDiffeoAt_restrict. *Step 2, unit, inverse, multiplication.* They are those of X on each sheet, sc-smooth by the continuity on the discrete factor Chapter10/EquivalenceRefinement · isScSmoothOn_discrete_of_continuousOn Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothMap_unit Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothMap_inv Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothCompSheet. *Step 3, properness.* At (i, x) take a neighbourhood V_0 of x with the properness property of X and an open $V(x)$ with $\mathrm{cl}\, V(x) \subseteq U_i \cap V_0$; then t is proper on $s^{-1}(\mathrm{cl}(\{i\} \times V(x)))$ Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.exists_isProperMap. The proof uses ultrafilters in place of the text's sequences; the index j of a limit is determined by the sheet, which is clopen.

Proposition 10.1.3 (2017: 10.1) (V10.3) Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.proposition_10_1_3. $F(i, x) = x$, $F(i, \varphi, j) = \varphi$ Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.functor is a local sc-diffeomorphism on objects, faithful because a morphism of $X_{\mathcal{U}}$ is determined by its ends and its φ, full because $\psi : x \to x'$ lifts to (i, ψ, j), and essentially surjective because every x lies in some U_i; Lemma 10.1.2 applies. The text writes the morphism as (U, φ, v) in the formula for F; we read (U, φ, V) (possible misprint). *Witness.* The covering $\{x > -1\}$, $\{x < 1\}$ of $\mathbb{Z}/2 \ltimes \mathbb{R}$ Chapter10/EquivalenceRefinement · EpGroupoid.Examples.reflectionCover gives an equivalence Chapter10/EquivalenceRefinement · EpGroupoid.Examples.isEquivalence_reflectionCover that is not injective on objects Chapter10/EquivalenceRefinement · EpGroupoid.Examples.not_injective_reflectionCover_obj.

X.3 Definition 10.1.4 (2017: 10.2), Propositions 10.1.5 and 10.1.6 (2017: 10.2 and 10.3)

Definition 10.1.4 (2017: 10.2) (V10.4) Chapter10/Equivalence · definition_10_1_4. The definition is Chapter 8's Definition 8.1.13 Chapter8/Basic · EpNatTrans Chapter8/Basic · EpFunctor.NaturallyEquivalent; the claims after it, that $\simeq$ is an equivalence relation and compatible with composition, are Chapter 8's Chapter8/NatTrans · EpFunctor.naturallyEquivalent_equivalence Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_left Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right. A natural transformation of ep-groupoids is a natural isomorphism of the underlying functors, every morphism being invertible Chapter10/Equivalence · EpNatTrans.toNatIso.

Proposition 10.1.5 (2017: 10.2) (V10.5) Chapter10/Equivalence · EpFunctor.proposition_10_1_5 Chapter10/Equivalence · EpFunctor.IsEquivalence.of_natTrans. The text quotes it from the literature. *Step 1.* $\tau : X \to \mathbf{X}'$ is continuous Chapter10/Equivalence · EpNatTrans.continuous_app and, since $s \circ \tau = F$ with F and s local sc-diffeomorphisms, a local sc-diffeomorphism Chapter10/Equivalence · EpNatTrans.isLocalScDiffeo_app. *Step 2.* $G = t \circ \tau$ is a composite of local sc-diffeomorphisms Chapter10/Equivalence · EpNatTrans.isLocalScDiffeo_obj. *Step 3.* The underlying functors are naturally isomorphic, so G is an equivalence of categories with F, and Lemma 10.1.2 applies. The symmetric statement follows from the symmetry of $\simeq$.

Proposition 10.1.6 (2017: 10.3) (V10.6) Chapter10/Equivalence · EpFunctor.proposition_10_1_6. *Step 1, categories.* Φ is an equivalence of categories since F and $\Phi \circ F$ are (Mathlib's two-out-of-three). *Step 2, objects in the image.* At $F(x)$, $\Phi \circ F$ and F are local sc-diffeomorphisms at x, so Φ is one at $F(x)$ (the right cancellation rule). *Step 3, transport.* The text's argument stops here; an arbitrary y receives a morphism $\sigma : F(x) \to y$ Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso, and the local sc-diffeomorphism property of a functor passes from $s(\sigma)$ to $t(\sigma)$ Chapter10/Equivalence · EpFunctor.isLocalScDiffeoAt_obj_tgt: near σ, $t \circ \tau = \Phi \circ t \circ \dots$ is written with a local inverse τ of s, and on morphisms Φ is a local sc-

diffeomorphism where it is one on objects `Chapter10/Equivalence · EpFunctor.isLocalScDiffeoAt_mor` `Chapter10/Equivalence · EpFunctor.isLocalScDiffeo_mor`. *Step 4.* Lemma 10.1.2. The text's $|\Phi| : |X| \to |Y|$ in the proof is read $|\Phi| : |Y| \to |Z|$ (possible misprint).

X.4 Theorem 10.1.7 (2017: 10.1): the tangent of an equivalence

Theorem 10.1.7 (2017: 10.1) (V10.7) `Chapter10/Equivalence · EpFunctor.theorem_10_1_7`. TF is Chapter 8's tangent functor `Chapter8/TangentEp · EpFunctor.tangent`; the text proves the homeomorphism and the local sc-diffeomorphism property and leaves full and faithful to the reader. Here all goes through Lemma 10.1.2. *Step 1, local sc-diffeomorphism.* The tangent of a local sc-diffeomorphism is one (Part I). *Step 2, faithful.* A tangent morphism (φ, v) is determined by φ and its source vector `Chapter8/TangentEp · EpGroupoid.tangent_ext`, and F is faithful. *Step 3, full.* For $(\psi, w) : TF(x, a) \to TF(x', a')$, $\psi = F(\varphi)$ by fullness of F; φ lifts to a tangent morphism with source (x, a) `Chapter8/TangentEp · EpGroupoid.exists_tSrc_eq'`, whose image under TF has the same source as (ψ, w), and the targets agree since TF is injective on the tangent fibre at x' `Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.injOn_tangentFiber`. *Step 4, essentially surjective.* For (y, b) take $\psi : F(x) \to y$, lift ψ^{-1} with source (y, b) and use that TF maps T_xX onto $T_{F(x)}Y$ `Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.tangentFiber_subset_image`, then invert `Chapter8/Basic · EpGroupoid.tInv`.

X.5 Remarks 10.1.8 and 10.1.11 (2017: 10.1), Propositions 10.1.9 and 10.1.10 (2017: 10.4)

Also formalized. The full subcategory $\mathcal{EP}_{lreg}$ of p. 461 (G.111) is §X.27. Remark 10.1.8 (new in the 2021 edition), with the example of Appendix §10.6.3 on its weighted Sobolev sc-retract (G.112), is §X.28. Propositions 10.1.9 and 10.1.10 (G.113, G.114) are the next two paragraphs.

Proposition 10.1.9 (V10.62) `Chapter10/EquivalenceLR · EpFunctor.proposition_10_1_9`, for (LR) as in Definition 7.4.19 `Chapter7/Zhou · EpGroupoid.IsLR`. *The transport chart* `Chapter10/EquivalenceLR · EpFunctor.TransportChart` `Chapter10/EquivalenceLR · EpFunctor.exists_transportChart`: for $\psi : F(x) \to y$, $Q = \widehat{\psi} \circ F$ with $\widehat{\psi} = t \circ (s|\mathbf{U}(\psi))^{-1}$ is an sc-diffeomorphism of a neighbourhood of x onto one of y, with morphisms $\psi_u : F(u) \to Q(u)$. *The isomorphism* $g \mapsto \psi \circ F(g) \circ \psi^{-1}$, $G_x \to G_y$ `Chapter10/EquivalenceLR · EpFunctor.TransportChart.conj` `Chapter10/EquivalenceLR · EpFunctor.TransportChart.bijective_conj`; on p. 469 its second factor $\sigma \mapsto \psi \circ \sigma \circ \psi^{-1}$ is printed with target $U(y)$, read G_y (possible misprint). The equivariance (10.2)–(10.3), $Q(g * z) = (\psi \circ F(g) \circ \psi^{-1}) * Q(z)$ `Chapter10/EquivalenceLR · EpFunctor.TransportChart.Q_Φ`, comes from the uniqueness of Γ on connected sets `Chapter10/EquivalenceLR · EpGroupoid.NaturalRep.eqOn_Γ`. *The step the text takes for granted,* $U(y) = Q(U(x))$ "with the natural action": a natural representation at y (Theorem 7.1.13) restricted to $Q(U(x))$ is one, invariant under G_y by the equivariance and the surjectivity of the isomorphism `Chapter10/EquivalenceLR · EpFunctor.TransportChart.exists_naturalRep_push`; the converse direction restricts one at x to $Q^{-1}(U(y))$ `Chapter10/EquivalenceLR · EpFunctor.TransportChart.exists_naturalRep_pull` `Chapter10/EquivalenceLR · EpGroupoid.NaturalRep.eqOn_Φ`. *The conjugation.* The condition of (LR) on one $U(x)$ `Chapter10/EquivalenceLR · EpGroupoid.NaturalRep.HasLR` `Chapter10/EquivalenceLR · EpGroupoid.isLR_iff_hasLR` passes both ways along an equivariant sc-diffeomorphism over a surjection of the groups: $\beta = Q^{-1} \circ \gamma \circ Q = \hat{g}$ gives $\gamma = (\psi \circ F(g) \circ \psi^{-1})^\wedge$ `Chapter10/EquivalenceLR · EpGroupoid.NaturalRep.hasLR_iff_of_equivariant`. *Witness* `Chapter10/EquivalenceLR · EquivalenceLRExamples.nonempty_transportChart_pointEP`.

Proposition 10.1.10 (2017: 10.4) (V10.63) `Chapter10/EquivalenceLR · EpFunctor.proposition_10_1_10_one` `Chapter10/EquivalenceLR · EpFunctor.proposition_10_1_10_one_of_isLR_right` `Chapter10/EquivalenceLR · EpFunctor.`

proposition_10_1_10_two. (1) As in the text: $\beta = G^{-1} \circ \hat{\psi} \circ F$ of (10.4) is an sc-diffeomorphism near x with $|\beta(z)| = |z|$, (LR) for X gives $\beta = \hat{g}$, and $\alpha_z = G(\Gamma(g,z))^{-1} \circ \psi_z$. Only (LR) for X is used; with (LR) for Y, Proposition 10.1.9 supplies it. (2) Differentiating α, $T\alpha(v)$ is a morphism $TF(v) \to TG(v)$, so $|TF| = |TG|$. *Witness* Chapter10/EquivalenceLR · EquivalenceLRExamples.proposition_10_1_10_hypotheses. The families are in general not natural (§X.20).

Remark 10.1.11 (2017: 10.1) (V10.8) Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11. The ep-groupoid is $\mathbb{Z}/2$ acting on $\mathbb{R}^2$ by $z \mapsto -z$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.pointReflectionEpGroupoid, sc-smooth by Chapter10/EquivalenceExamples · EpGroupoid.Examples.isScSmoothMap_units_smul_rn; $G = -\mathrm{id}$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.negFunctor is an equivalence Chapter10/EquivalenceExamples · EpGroupoid.Examples.isEquivalence_negFunctor with $|\mathrm{id}| = |G|$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.orbitMap_id_eq_negFunctor. *Step 1.* $\alpha_z = (-1, z)$ is an sc-smooth family $F(z) \to G(z)$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11_family: the conclusion of Proposition 10.1.10 (1) holds in the example. *Step 2.* A family with $\alpha_0 = 1_0$ and $\alpha_z : z \to -z$ near 0 must take the value $(-1, z)$ at every $z \neq 0$, because the only morphism $z \to -z$ is the group element -1 Chapter10/EquivalenceExamples · EpGroupoid.Examples.rn_eq_zero_of_eq_neg Chapter10/EquivalenceExamples · EpGroupoid.Examples.not_eventually_eq_neg; continuity at 0 fails. Hence no sc-smooth family exists either Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11_scSmooth.

X.6 Remark 10.1.12, Propositions 10.1.13 and 10.1.14 (2017: 10.5), Example 10.1.15 (2017: 10.1)

Remark 10.1.12 (V10.9) Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.remark_10_1_12. *Step 1.* $g \mapsto F(g)$ is a homomorphism $G_x \to G_{F(x)}$ Chapter10/EquivalenceReduced · EpFunctor.isotropyHom Chapter10/EquivalenceReduced · EpFunctor.isotropyHom_val, bijective for an equivalence (fullness and faithfulness at (x,x)) Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.bijective_isotropyHom. *Step 2.* $g \in T_x$ iff $F(g) \in T_{F(x)}$ Chapter10/EquivalenceReduced · EpFunctor.isotropyHom_mem_trivialSet_iff, since F is a local sc-diffeomorphism on objects and on morphisms, and the local actions correspond. *Step 3.* Conjugation by $\psi : F(x) \to y$ Chapter7/Groupoid · GroupoidStr.isotropyConj maps $T_{F(x)}$ onto T_y Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet_iff. The text's $G_x \to Y_y$ is read $G_x \to G_y$ (possible misprint).

Proposition 10.1.13 (V10.10) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_13. *Step 1,* $\mathbf{X}^T$ *along* F. For a local sc-diffeomorphism on objects, $\varphi \in \mathbf{X}^T$ iff $F(\varphi) \in \mathbf{Y}^T$ Chapter10/EquivalenceReduced · EpFunctor.mor_mem_nonEffectivePart Chapter10/EquivalenceReduced · EpFunctor.mem_nonEffectivePart_of_mor, that is $\mathbf{X}^T = F^{-1}(\mathbf{Y}^T) \cap \{s = t\}$ Chapter10/EquivalenceReduced · EpFunctor.nonEffectivePart_eq. *Step 2, (1).* So F_R is well defined Chapter10/EquivalenceReduced · EpFunctor.reducedMap Chapter10/EquivalenceReduced · EpFunctor.reducedMap_π, continuous Chapter10/EquivalenceReduced · EpFunctor.continuous_reducedMap and open, F being open on morphisms and the quotient maps open Chapter10/EquivalenceReduced · EpFunctor.isOpenMap_reducedMap; only the local sc-diffeomorphism property of F is used. *Step 3, closedness.* For an equivalence, $\mathbf{X}^T$ is closed iff $\mathbf{Y}^T$ is Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.isClosed_nonEffectivePart_iff. From Y to X: $\mathbf{X}^T$ is the preimage of a closed set intersected with a closed set. From X to Y: a $\psi \in \mathrm{cl}(\mathbf{Y}^T)$ at y is conjugated, by a local inverse of t at a morphism $\sigma : F(x) \to y$, into a morphism at $F(x)$ in $\mathrm{cl}(F(\mathbf{X}^T))$, with the continuity of the composition at a point Chapter10/EquivalenceReduced · EpGroupoid.continuousAt_comp; closures replace the text's sequences. *Step 4, (2).* Theorem 7.2.22 says $\mathbf{X}^R$ is Hausdorff iff $\mathbf{X}^T$ is closed Chapter7/ReducedCriterion · EpGroupoid.theorem_7_2_22; with Step 3 this gives (2), the route of the text.

Proposition 10.1.14 (2017: 10.5) (V10.11) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_14. With

both reduced morphism spaces Hausdorff the reduced ep-groupoids of Chapter 7 exist Chapter7/Reduced · EpGroupoid.reducedEpGroupoid. F_R is F on objects and F_R of Step 2 on morphisms Chapter10/EquivalenceReduced · EpFunctor.reducedGroupoidFunctor Chapter10/EquivalenceReduced · EpFunctor.reduced; it is a functor since $[\cdot]$ is Chapter10/EquivalenceReduced · EpFunctor.reducedπ_eq_of_mor, sc-smooth since the M-polyfold structure of $\mathbf{X}^R$ is the one for which the quotient map is a local sc-diffeomorphism, full and faithful by Remark 10.1.12, and $|F_R| = |F|$. By Proposition 10.1.13 (2) either Hausdorff hypothesis implies the other; both are kept, as in the text.

Example 10.1.15 (2017: 10.1) (V10.12) Chapter10/EquivalenceExamples · EpGroupoid.Examples.example_10_1_15. $X = \mathbb{Z}/4 \ltimes \mathbb{R}$ with $1, 3$ acting by -1 Chapter10/EquivalenceExamples · EpGroupoid.Examples.z4LineAction Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X, $Y = \mathbb{Z}/4 \ltimes \mathbb{R}^2$ by quarter turns Chapter10/EquivalenceExamples · EpGroupoid.Examples.z4PlaneAction Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y, Φ to the origin Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Functor. *Step 1.* 2 acts trivially on $\mathbb{R}$, not on $\mathbb{R}^2$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X_two_mem_nonEffectivePart Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y_two_not_mem_nonEffectivePart. *Step 2.* Hence $(1, x)$ and $(3, x) = (2, x) \circ (1, x)$ have the same class Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X_reducedπ_one_eq_three, their images not Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y_reducedπ_one_ne_three, so no map of sets f with $f([\varphi]) = [\Phi(\varphi)]$ exists. The text speaks of an induced map of the reduced ep-groupoids; the statement here needs no topology.

X.7 Propositions 10.1.16–10.1.19: (LUC), (ZHOU), (SLUC), (SLR)

Proposition 10.1.16 (V10.13) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_16. (LUC) Chapter7/ReducedCriterion · EpGroupoid.IsLUC is the closedness of $\mathbf{X}^T$ (Proposition 7.2.21, part of Theorem 7.2.22) Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_21, and the closedness passes along equivalences (§X.6, Step 3). The text cites Theorem 7.2.22 here and in Proposition 10.1.13; the citation is the theorem on p. 323.

Proposition 10.1.17 (V10.14) Chapter10/EquivalenceZhou · EpFunctor.proposition_10_1_17. *Step 1, smooth points.* x is smooth iff $F(x)$ is, and iff a $y \cong F(x)$ is Chapter10/EquivalenceZhou · EpGroupoid.mem_smoothSet_src_iff Chapter10/EquivalenceZhou · EpGroupoid.mem_smoothSet_tgt_iff. *Step 2, Y to X.* For $g \in G_x$ with $Tg = \mathrm{id}$ on T_xX: $T(F(g)) \circ TF = TF \circ Tg$ Chapter10/EquivalenceZhou · EpFunctor.tangentMor_mor and TF is onto $T_{F(x)}Y$, so $T(F(g)) = \mathrm{id}$, $F(g) \in T_{F(x)}$ by (ZHOU) for Y, and $g \in T_x$ by Remark 10.1.12 Chapter10/EquivalenceZhou · EpFunctor.IsEquivalence.isZhou_of_isZhou_right. *Step 3, X to Y.* For $k \in G'_y$ with $Tk = \mathrm{id}$ and $\psi : F(x) \to y$, $k' = \psi^{-1}k\psi \in G'_{F(x)}$ has $Tk' = \mathrm{id}$ by the composition rule for T Chapter8/TangentEp · EpGroupoid.tangentMor_comp; $k' = F(g)$ with $g \in G_x$ (Remark 10.1.12), $Tg = \mathrm{id}$ by the injectivity of TF on T_xX, so $g \in T_x$, $k' \in T_{F(x)}$ and $k \in T_y$ Chapter10/EquivalenceZhou · EpFunctor.IsEquivalence.isZhou_of_isZhou. The text argues with natural representations instead; it writes $U(F(y))$ and "$x \in U(x)$" in (b), read $U(F(x))$ and $a \in U(x)$ (possible misprints).

Proposition 10.1.18 (V10.15) Chapter10/EquivalenceReduced · EP.proposition_10_1_18. *Step 1.* The restriction of F to level 1 is an sc-smooth functor $X^1 \to Y^1$ Chapter10/EquivalenceReduced · EpFunctor.liftOne, an equivalence when F is Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.liftOne (orbit spaces of X^1 carry the level-1 topology, and $|F^1|$ is a homeomorphism by Lemma 10.1.2 again). *Step 2.* By induction, every $X^i \to Y^i$ Chapter10/EquivalenceReduced · EP.exists_isEquivalence_lift over Chapter 7's lifts Chapter7/Zhou · EP.lift. *Step 3.* Proposition 10.1.16 at each level. The text cites Proposition 10.1.9 at Step 3; we read Propo-

sition 10.1.16 (possible misprint).

Proposition 10.1.19 (V10.16) `Chapter10/EquivalenceLR · EP.proposition_10_1_19` `Chapter10/EquivalenceZhou · EP.isSLR_iff_of_isLR_iff`. Steps 1–2 as above, then Proposition 10.1.9 at each level (§X.5). The reduction alone holds when the two ep-groupoids agree `Chapter10/EquivalenceZhou · EP.isLR_iff_lift_self`; G.115 is closed.

X.8 Definition 10.2.1 (2017: 10.3): the weak fibred product

Definition 10.2.1 (2017: 10.3) (V10.17) `Chapter10/WeakFibredProduct · WeakFibredProduct.groupoid`. *Objects.* $\{(\phi, z) \mid t(\phi) = G(z)\} \subseteq \mathbf{Y} \times Z$ `Chapter10/WeakFibredProduct · WeakFibredProduct.objRight`, then $\{(x, (\phi, z)) \mid F(x) = s(\phi)\}$ `Chapter10/WeakFibredProduct · WeakFibredProduct.objSet` `Chapter10/WeakFibredProduct · WeakFibredProduct.Obj`, with projections and extensionality `Chapter10/WeakFibredProduct · WeakFibredProduct.Obj.ext`. *Morphisms.* The same with $\mathbf{X}$, $\mathbf{Z}$ and $F \circ s$, $G \circ s$ `Chapter10/WeakFibredProduct · WeakFibredProduct.morRight` `Chapter10/WeakFibredProduct · WeakFibredProduct.morSet` `Chapter10/WeakFibredProduct · WeakFibredProduct.Mor` `Chapter10/WeakFibredProduct · WeakFibredProduct.Mor.ext`. *Structure maps.* The conjugate $G(k) \circ \phi \circ F(h)^{-1}$ `Chapter10/WeakFibredProduct · WeakFibredProduct.conj` has the ends $F(t(h))$, $G(t(k))$ `Chapter10/WeakFibredProduct · WeakFibredProduct.src_conj` `Chapter10/WeakFibredProduct · WeakFibredProduct.tgt_conj`; it is the target's middle entry `Chapter10/WeakFibredProduct · WeakFibredProduct.tgt`, s keeps ϕ `Chapter10/WeakFibredProduct · WeakFibredProduct.src`, the unit is $(1_x, \phi, 1_z)$ `Chapter10/WeakFibredProduct · WeakFibredProduct.unit` and the inverse $(h^{-1}, \mathrm{conj}, k^{-1})$ `Chapter10/WeakFibredProduct · WeakFibredProduct.inv`. The text prints $\iota(h, \psi, k) = (h^{-1}, G(k) \circ \phi \circ F(h)^{-1}, k^{-1})$; we read $\psi = \phi$ (possible misprint). The product of a composable pair is $(h \circ h', \phi', k \circ k')$ `Chapter10/WeakFibredProduct · WeakFibredProduct.compOf`, extended to a total map that returns its first argument off composable pairs `Chapter10/WeakFibredProduct · WeakFibredProduct.comp`. *Groupoid laws.* By the rules $\mathrm{conj}(h \circ h', \phi', k \circ k') = \mathrm{conj}(h, \mathrm{conj}(h', \phi', k'), k)$ `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_comp`, conj of units `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_unit` and of inverses `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_inv`, with ϕ recovered from the target `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_recover` `Chapter10/WeakFibredProduct · WeakFibredProduct.conj_comp_mor`; a morphism is a loop iff its ends agree `Chapter10/WeakFibredProduct · WeakFibredProduct.src_eq_tgt_iff`.

X.9 Theorem 10.2.2 (2017: 10.2): $X \times_Y Z$ is an ep-groupoid

The admissibility `Chapter10/WeakFibredProduct · WeakFibredProduct.Admissible`. The construction uses of F (or of G) only that it is a local sc-diffeomorphism on objects, which every equivalence is `Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Admissible.of_isEquivalence_left` `Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Admissible.of_isEquivalence_right`. *Step 1, sub-M-polyfolds.* Part I's fibred product of an sc-smooth map with a local sc-diffeomorphism is a sub-M-polyfold (Proposition 2.5.20); $t : \mathbf{Y} \to Y$ is one, so $\mathbf{Y}_t \times_G Z$ is a sub-M-polyfold of $\mathbf{Y} \times Z$ `Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_objRight` `Chapter10/WeakFibredProduct · WeakFibredProduct.PA`, and its projection to Z is a local sc-diffeomorphism `Chapter10/WeakFibredProduct · MPolyfold.isLocalScDiffeo_snd_fibredProduct`; then the second fibred product with F or with that projection `Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_objSet` `Chapter10/WeakFibredProduct · WeakFibredProduct.PObj`, and the same for morphisms `Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_morRight` `Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_morSet` `Chapter10/WeakFibredProduct · WeakFibredProduct.PB` `Chapter10/WeakFibredProduct · WeakFibredProduct.PMor`. The text says the sets are sub-M-polyfolds of

the triple products. They are, and the iterated structures are the induced ones (§X.29; G.116); the construction here does not use it. *Step 2, the projections.* x, ϕ, z, h, ϕ, k are sc-smooth Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_x Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_z Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_k, and a map into $X \times_Y Z$ is sc-smooth when its entries are Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothOn_obj_of Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothOn_mor_of. For F admissible, $(x, \phi, z) \mapsto z$ and $(h, \phi, k) \mapsto k$ are local sc-diffeomorphisms ((10.9), (10.10)) Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_z Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_k; for G admissible the projections to X and $\mathbf{X}$ are Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_x Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_h Chapter10/WeakFibredProduct · MPolyfold.isLocalScDiffeo_fst_fibredProduct. *Step 3, s and t.* $\pi_Z \circ s = s_Z \circ \pi_{\mathbf{Z}}$ and $\pi_Z \circ t = t_Z \circ \pi_{\mathbf{Z}}$ with three local sc-diffeomorphisms make s, t ones Chapter10/WeakFibredProduct · MPolyfold.IsLocalScDiffeo.of_comm Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_src Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_tgt; their sc-smoothness uses that of the conjugate Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_conj Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_src Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_tgt. *Step 4, unit, inverse, multiplication* Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_unit Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_inv Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothComp: entries are multiplications of X and Z and projections, sc-smooth on the composable pairs Chapter10/WeakFibredProduct · EpGroupoid.isScSmoothOn_comp_of. *Step 5, properness.* For open $U \ni x$, $V \ni z$ with the properness property in X and Z, $W = \{x' \in U, z' \in V\}$ has t proper on $s^{-1}(\mathrm{cl}\, W)$ Chapter10/WeakFibredProduct · WeakFibredProduct.exists_isProperMap: subsequences of h_n, then of k_n, converge, and $\phi_n = G(k_n)^{-1} \circ (G(k_n) \circ \phi_n \circ F(h_n)^{-1}) \circ F(h_n)$ converges by continuity of the multiplication Chapter10/WeakFibredProduct · EpGroupoid.tendsto_comp Chapter10/WeakFibredProduct · WeakFibredProduct.continuous_conj. The text writes $t : s^{-1}(U) \to X$; Definition 7.1.3 (3) asks for the closure, which is the reading. No admissibility is needed here. *Conclusion* (V10.18) Chapter10/WeakFibredProduct · WeakFibredProduct.epGroupoid.

X.10 Theorem 10.2.2 (2017: 10.2): the projections, and Remark 10.2.3 (2017: 10.2)

The functors Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projLeft Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projRight Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projLeft_obj. $\sigma(x, \phi, z) = \phi$ is a natural transformation $F \circ \pi_X \to G \circ \pi_Z$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.natTrans Chapter10/WeakFibredProductTheorem · WeakFibredProduct.natTrans_app: its naturality is the definition of the target, and it is sc-smooth as a projection. Hence $F \circ \pi_X \simeq G \circ \pi_Z$ whenever $X \times_Y Z$ is defined Chapter10/WeakFibredProductTheorem · WeakFibredProduct.naturallyEquivalent.

F an equivalence (V10.19) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.isEquivalence_projRight. *Step 1, faithful.* A morphism over k is (h, ϕ, k) with $F(h)$ determined by k and the ends, and F is faithful. *Step 2, full.* For $k : z \to z'$ between (x, ϕ, z) and (x', ϕ', z'), $\phi'^{-1} \circ G(k) \circ \phi : F(x) \to F(x')$ is $F(h_0)$ by fullness of F, and (h_0, ϕ, k) lies over k. The text writes the composite as $\phi' \circ G(k) \circ \phi^{-1}$, which does not compose; we read $\phi'^{-1} \circ G(k) \circ \phi$ (possible misprint). *Step 3, essentially surjective.* $G(z)$ receives $\psi : F(x) \to G(z)$ Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso, and (x, ψ, z) lies over z. *Step 4.* Lemma 10.1.2 with Step 2 of §X.9. *The rest of the theorem* Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2: an equivalence preserves the degeneracy

index Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.degIndex_obj, so $d(x,\phi,z) = d_Z(z)$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.degIndex_eq_z, and pulls back tameness Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.isTame_of_isTame and the paracompactness of the orbit space, through $|\pi_Z|$ Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.hasParacompactOrbitSpace. *Witness.* $\mathbb{Z} \to$ pt and $\mathrm{id}_{\mathrm{pt}}$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Examples.theorem_10_2_2_pairToPoint.

G an equivalence, and both (V10.20) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.isEquivalence_projLeft Chapter10/WeakFibredProductTheorem · WeakFibredProduct.degIndex_eq_x Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2_right Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2_both. The symmetric argument with ϕ^{-1}. The admissibility is a proposition, so the ep-groupoids built from F and from G coincide and the two cases combine.

Remark 10.2.3 (2017: 10.2) (V10.21) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.remark_10_2_3. Both projections are equivalences and $\tau = \sigma$ is natural. *Witness.* id and $-\mathrm{id}$ on $\mathbb{Z}/2 \ltimes \mathbb{R}^2$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Examples.remark_10_2_3_negFunctor. The sentences on "smaller models" have no formal content.

X.11 Definition 10.4.1 (2017: 10.10): strong bundle equivalences

The orbit space at level $[1]$ Chapter10/StrongBundleEquivalence · EpStrongBundle.oneOrbitSetoid Chapter10/StrongBundleEquivalence · EpStrongBundle.OneOrbitSpace Chapter10/StrongBundleEquivalence · EpStrongBundle.oneOrbitMap. $W[1]$ is not built as an ep-groupoid; $|W[1]|$ is the quotient of $W_{0,1}$ with its level-1 topology by the orbit relation of W, and a strong bundle functor induces a continuous map of these quotients Chapter10/StrongBundleEquivalence · EpStrongBundle.continuous_oneOrbitMap.

Definition 10.4.1 (2017: 10.10) (V10.22) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence. A strong bundle functor Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor with ϕ an equivalence, Φ an equivalence at level $[0]$, $|\Phi|$ a homeomorphism of $|W[1]|$, and local strong bundle isomorphisms on objects and morphisms Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn. Clause (1) at level $[1]$ is read in this form: fullness and faithfulness at $[1]$ are those at $[0]$ restricted, and the local sc-diffeomorphism property at $[1]$ follows from (2) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_one Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_obj. *Consequences.* $P' \circ \Phi = \phi \circ P$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.covers; $\Phi_x : W_x \to W'_{\phi(x)}$ is a linear isomorphism, the local strong bundle isomorphism at x restricted Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.P_inv Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.fiberEquiv, so Φ is injective on fibres Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.eq_of_obj_eq. *Class.* The identity Chapter10/StrongBundleEquivalence · EpStrongBundle.isStrongBundleEquivalence_id; composites Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.comp, with strong bundle functors Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleFunctor.comp and local strong bundle isomorphisms composing on open sets Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.restrict Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.comp Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.exists_comp; isomorphisms with an inverse strong bundle functor Chapter10/StrongBundleEquivalenceNat · StrongBundle.isStrongBundleIsoOn_

univ_of_inverse Chapter10/StrongBundleEquivalenceNat · EpFunctor.isEquivalence_of_inverse Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.isStrongBundleEquivalence_of_inverse. *Witness.* Over $-\mathrm{id}$ of $\mathbb{R}^n$ Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negMap, the projection of the pull-back of the trivial bundle Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negBundle Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negEpBundle, with inverse Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negInv Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.isStrongBundleMap_negInv, is a strong bundle equivalence Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.isStrongBundleEquivalence_pullbackProj' not over the identity Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExa trivialNeg_obj_ne_id.

X.12 The pull-back and the push-forward, (10.15)–(10.17) and (10.22)

Pull-back (V10.23) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback. $\Phi^* f'(x)$ is the unique $w \in W_x$ with $\Phi(w) = f'(\phi(x))$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.obj_pullback Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_eq, a section Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.P_pullback Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isSection_pullback, functorial when f' is Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pullback ((10.17), by the injectivity of Φ on fibres and of $\mu'(\sigma, \cdot)$ Chapter10/StrongBundleEquivalence · EpStrongBundle.μ_injective_snd).

Push-forward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.preObj Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.preMor Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_def. For x' choose x and $\sigma : \phi(x) \to x'$ (essential surjectivity of ϕ) and set $\Phi_* f(x') = \mu'(\sigma, \Phi(f(x)))$, a section Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.P_pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isSection_pushforward. *(10.22).* For a section functor f and any y, $\sigma' : \phi(y) \to x'$, the value is $\mu'(\sigma', \Phi(f(y)))$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq: $\sigma'^{-1} \circ \sigma : \phi(x) \to \phi(y)$ is $\phi(\tau)$ by fullness, $f(y) = \mu(\tau, f(x))$ by functoriality, and Φ intertwines μ and μ' Chapter10/StrongBundleEquivalence · EpStrongBundle.μ_mk_congr. Hence $\Phi_* f$ is a section functor Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pushforward, and $\Phi^* \Phi_* f = f$, $\Phi_* \Phi^* f' = f'$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_pullback.

(10.15) Chapter10/StrongBundleEquivalence · EpStrongBundle.exists_μ_eq_of_orbitMap_eq Chapter10/StrongBundleEquivalence · EpStrongBundle.orbitMap_eq_of_forall_exists. $|\Phi| = |\Psi|$ iff every $\Psi(w)$ is $\mu'(\sigma, \Phi(w))$ for some σ.

X.13 Propositions 10.4.2–10.4.4 (2017: 10.7–10.9)

Proposition 10.4.2 (2017: 10.7) (V10.24) `Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_2` `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_eq_of_orbitMap_eq`. With $g = \Psi^* f'$, (10.15) gives $\sigma : \phi(x) \to \psi(x)$ with $\mu'(\sigma, \Phi(g(x))) = \Psi(g(x)) = f'(\psi(x))$; functoriality of f' gives $f'(\psi(x)) = \mu'(\sigma, f'(\phi(x))) = \mu'(\sigma, \Phi(\Phi^* f'(x)))$; the injectivity of $\mu'(\sigma, \cdot)$ and of Φ_x gives $g(x) = \Phi^* f'(x)$. sc-smoothness is not used. The text's "$f' : W' \to X'$" is read $f' : X' \to W'$ (possible misprint).

Proposition 10.4.3 (2017: 10.8) (V10.25) `Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_3` `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq_of_orbitMap_eq`. With $\sigma : \phi(x) \to \psi(x)$ from (10.15) at $f(x)$, and (10.22) for Ψ applied to the composite of σ^{-1} with the choice for Φ.

Proposition 10.4.4 (2017: 10.9) (V10.26) `Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_4`. The pull-back identity pointwise from the uniqueness in the definition; the push-forward identity from (10.22) for $\Psi \circ \Phi$.

X.14 Theorem 10.4.5 (2017: 10.8)

Theorem 10.4.5 (2017: 10.8) (V10.27) `Chapter10/StrongBundleEquivalence · EpStrongBundle.theorem_10_4_5`. *Step 1, pull-back.* Near x_0, $\Phi^* f'$ agrees with $\Phi_U^{-1} \circ f' \circ \phi$ for the local strong bundle isomorphism Φ_U `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.eqOn_pullback`, so it is sc-smooth `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pullback` and sc^+ when f' is `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pullback`, smoothness being a local property in charts `Chapter10/StrongBundleEquivalence · MPolyfold.isScSmoothOn_of_forall_charts`. *Step 2, push-forward.* The text says the sc-smoothness is easily verified. Near x_0', with a local inverse τ of t' at the chosen $\sigma_0 : \phi(y_0) \to x_0'$ and a local inverse g_1 of ϕ at y_0, (10.22) gives $\Phi_* f = \mu' \circ (\tau, \Phi \circ f \circ g_1 \circ s' \circ \tau)$ `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.exists_pushforward_local`, sc-smooth `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pushforward`. At level $[1]$ the pair is sc-smooth into the pull-back's level $[1]$ by its universal property on an open set `Chapter10/StrongBundleEquivalence · StrongBundle.isScSmoothOn_to_pullback_one`, so sc^+ is preserved `Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pushforward`. *Step 3.* With §X.12 and §X.13, bijections between Chapter 8's sets of section functors `Chapter8/EpStrongBundle · EpStrongBundle.sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors`, independent of Φ within $|\Phi|$. The text's second display, $\Phi^* : \Gamma(P', \mu') \to \Gamma(P, \mu)$ in the Γ^+ line, is read with Γ^+ (a reading point). *The sc-Fredholm part* (G.121) is §X.31 `Chapter10/FredholmEquivalence · EpStrongBundle.theorem_10_4_5_fred`, and the bijections are linear isomorphisms there too. *Witness* `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.bijOn_pushforward_example`.

X.15 Definitions 10.4.8, 10.4.10 and 10.4.11, Remark 10.4.9 (2017: 10.11–10.13, 10.4)

Definition 10.4.8 (2017: 10.11) (V10.28) `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans` `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatEquiv`. T is data with $T(w) : \Phi(w) \to \Psi(w)$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.mem` and naturality, over an `EpNatTrans` τ. Reflexivity `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.refl`; a witness with $\Phi \neq \Psi$ `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.strongBundleNatEquiv_example`.

Remark 10.4.9 (2017: 10.4) (V10.29). *Step 1.* A morphism of $\mathbf{W}'$ is (σ, w') with $\sigma \in \mathbf{X}'$; $T(w)$ lies over $\tau(P(w))$ and starts at $\Phi(w)$, so $T(w) = (\tau(P(w)), \Phi(w))$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.T_val` `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.T_eq` and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.obj_eq`. The text prints the second line of (10.23) with μ; we read μ'. *Step 2.* T and Ψ on objects are determined by Φ and τ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.unique`, and Ψ on morphisms too, if Ψ is a strong bundle functor `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.unique_functor`. *Step 3.* T is a strong bundle map covering τ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.isStrongBundleMap_T`.

Definition 10.4.10 (2017: 10.12) (V10.30) `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.IsStrongBundleNatTransMap`. Definition 10.4.8 gives it by Remark 10.4.9 `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.isStrongBundleNatTransMap`; conversely the naturality of T follows from (10.23) and the functoriality of Φ, Ψ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.IsStrongBundleNatTransMap.toNatTrans`. $\Phi \simeq \Psi$ implies $|\Phi| = |\Psi|$ `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.orbitMap_eq` `Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatEquiv.orbitMap_eq`.

Definition 10.4.11 (2017: 10.13) (V10.31) `Chapter10/StrongBundleEquivalenceNat · SEP` `Chapter10/StrongBundleEquivalenceNat · SEP.Hom` `Chapter10/StrongBundleEquivalenceNat · SEP.Hom.ext'`. A morphism is a pair (ϕ, Φ) with Φ a strong bundle functor covering ϕ; Φ alone determines ϕ only on the image of P. The strong bundle equivalences form a multiplicative class `Chapter10/StrongBundleEquivalenceNat · SEP.equivalences`, mapped into E by the forgetful functor `Chapter10/StrongBundleEquivalenceNat · SEP.toEP` `Chapter10/StrongBundleEquivalenceNat · SEP.toEP_map_mem_equivalences`. *Witness* `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEP` `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.epSEP` `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEPHom` `Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEPHom_mem_equivalences`.

X.16 §10.3: (E1)–(E3), diagrams and their equivalence (Definitions 10.3.1, 10.3.3, 10.3.5, Lemmas 10.3.2, 10.3.4 (2017: Definitions 10.4–10.6, Lemmas 10.3, 10.4))

(E1)–(E3) (V10.32). (E1): an isomorphism of $\mathcal{EP}$ has an inverse sc-smooth functor, so it is full, faithful, a homeomorphism on orbit spaces and an sc-diffeomorphism on objects `Chapter10/`

LocalizationCategory · EPLocalization.isEquivalence_of_isIso. (E2) and (E3) are §X.1 and Proposition 10.1.5 Chapter10/Equivalence · EpFunctor.IsEquivalence.comp Chapter10/Equivalence · EpFunctor.proposition_10_1_5.

Diagrams (V10.33) Chapter10/Localization · EPLocalization.Diagram. The middle ep-groupoid is an object of $\mathcal{EP}$, so that a diagram, and later its class, lives one universe above the object and morphism types; no class-sized collection occurs. The weak fibred product of §X.9 is made an object of $\mathcal{EP}$ when admissible Chapter10/Localization · EPLocalization.wfp.

Definition 10.3.1 (2017: 10.4) Chapter10/Localization · EPLocalization.Diagram.IsRefinement Chapter10/Localization · EPLocalization.Diagram.IsRefinement.refl. Both examples after the definition: precomposition with an equivalence H Chapter10/Localization · EPLocalization.Diagram.precomp Chapter10/Localization · EPLocalization.Diagram.isRefinement_precomp, with H itself refining and the natural equivalences equalities Chapter10/Localization · EpFunctor.NaturallyEquivalent.of_eq; and naturally equivalent legs, with $H = 1_A$ Chapter10/Localization · EPLocalization.Diagram.isRefinement_of_naturallyEquivalent.

Lemma 10.3.2 (2017: 10.3) (V10.34) Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_2 Chapter10/Localization · EPLocalization.Diagram.IsRefinement.trans. For $H : d' \to d$ and $H' : d'' \to d'$, $H \circ H'$ is an equivalence (§X.1) and $F \circ H \circ H' \simeq F' \circ H' \simeq F''$, by the compatibility of $\simeq$ with composition Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right and its transitivity; the same for Φ.

Definition 10.3.3 (2017: 10.5) and Lemma 10.3.4 (2017: 10.4) (V10.35, V10.36) Chapter10/Localization · EPLocalization.Diagram.IsCommonRefinement Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_4. *Step 1.* For W refining d by H and d' by H', and W' refining d' by K and d'' by K', H' and K are equivalences into A', so the weak fibred product $W \times_{A'} W'$ is defined and its projections are equivalences (Theorem 10.2.2). *Step 2.* The diagram (10.11) over $W \times_{A'} W'$ refines d by $H \circ \pi_W$; it refines d'' by $K' \circ \pi_{W'}$, because $F' \circ H' \circ \pi_W \simeq F' \circ K \circ \pi_{W'}$ by the natural transformation σ of the weak fibred product (§X.10) and Lemma 10.3.2. A refinement is equivalent to the refined diagram Chapter10/Localization · EPLocalization.Diagram.IsRefinement.equiv.

Definition 10.3.5 (2017: 10.6) (V10.37) Chapter10/Localization · EPLocalization.Diagram.Equiv Chapter10/Localization · EPLocalization.Diagram.setoid Chapter10/Localization · EPLocalization.Diagram.equiv_iff. Reflexivity and symmetry are immediate Chapter10/Localization · EPLocalization.Diagram.Equiv.refl Chapter10/Localization · EPLocalization.Diagram.Equiv.symm, transitivity is Lemma 10.3.4 Chapter10/Localization · EPLocalization.Diagram.Equiv.trans. *Witness.* pt $\leftarrow$ $\mathbb{Z}$-pair $\to$ pt, both legs the equivalence of §X.1, is equivalent to 1_{pt} through the refinement by that equivalence, which is not an isomorphism Chapter10/LocalizationCategory · EPLocalization.Examples.equiv_pairToPoint_id.

X.17 Lemma 10.3.6 (2017: 10.5), Definition 10.3.7 (2017: 10.7) and Theorem 10.3.8 (2017: 10.3) with Appendix §10.6.1

Lemma 10.3.6 (2017: 10.5) (V10.38) Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_6. $|d| = |\Phi| \circ |F|^{-1}$, with $|F|^{-1}$ the inverse of the homeomorphism of Definition 10.1.1 (2) Chapter10/Localization · EPLocalization.Diagram.orbitMap Chapter10/Localization · EPLocalization.Diagram.continuous_orbitMap Chapter10/Localization · EPLocalization.Diagram.orbitMap_orbitMap_F Chapter10/Localization · EPLocalization.Diagram.orbitMap_eq_of. For $H : d' \to d$, $|F'| = |F \circ H| = |F| \circ |H|$ Chapter10/Localization · EpFunctor.orbitMap_comp Chapter10/Localization · EpFunctor.orbitMap_id, since naturally equivalent functors have the same orbit map Chapter10/Localization · EpFunctor.NaturallyEquivalent.

`orbitMap_eq`, and likewise $|\Phi'| = |\Phi| \circ |H|$; the factors $|H|$ cancel. Equivalent diagrams have a common refinement, hence the same map `Chapter10/Localization · EPLocalization.Diagram.Equiv.orbitMap_eq`.

Definition 10.3.7 (2017: 10.7) (V10.39) `Chapter10/Localization · EPLocalization.Diagram.orbitMapQ` `Chapter10/Localization · EPLocalization.Diagram.orbitMapQ_mk`. When Φ is an equivalence, $|\Phi| \circ |F|^{-1}$ is a composite of homeomorphisms `Chapter10/Localization · EPLocalization.Diagram.isHomeomorph_orbitMap`.

The composite `Chapter10/Localization · EPLocalization.Diagram.comp` `Chapter10/Localization · EPLocalization.Diagram.id`. For d and $e : Y \xleftarrow{G} B \xrightarrow{\Psi} Z$, the weak fibred product of Φ and G is admissible since G is an equivalence `Chapter10/Localization · EPLocalization.Diagram.admissible`, and π_A is an equivalence (Theorem 10.2.2), so $F \circ \pi_A$ is one. `d.comp e` is the text's $[e] \circ [d]$.

Theorem 10.3.8 (2017: 10.3) (V10.40) `Chapter10/Localization · EPLocalization.Diagram.theorem_10_3_8`, the proof of Appendix §10.6.1. *Step 1, the functor L.* For refinements $H : d' \to d$, $K : e' \to e$ with natural transformations $\tau_{A'} : \Phi' \to \Phi \circ H$ and $\tau_{B'} : G' \to G \circ K$, $L(a', \phi, b') = (H(a'), \tau_{B'}(b') \circ \phi \circ \tau_{A'}(a')^{-1}, K(b'))$ ((10.28), (10.30)) `Chapter10/Localization · EPLocalization.mapMid` `Chapter10/Localization · EPLocalization.mapObj` `Chapter10/Localization · EPLocalization.mapMor` `Chapter10/Localization · EPLocalization.wfpMap` is an sc-smooth functor, entry by entry, with the bookkeeping identity `Chapter10/Localization · GroupoidStr.comp_comp_inv_comp_inv`. In the text's naturality of $\tau_{B'}$ the letter τ_A appears; we read $\tau_{B'}$ (possible misprint). *Step 2, L is an equivalence* `Chapter10/Localization · EPLocalization.isEquivalence_wfpMap`. $\pi_A \circ L = H \circ \pi_{A'}$ holds on the nose `Chapter10/Localization · EPLocalization.projLeft_comp_wfpMap` `Chapter10/Localization · EPLocalization.projRight_comp_wfpMap`; π_A, $\pi_{A'}$ and H are equivalences, and the two-out-of-three rule on the other side from Proposition 10.1.6 `Chapter10/Localization · EpFunctor.IsEquivalence.of_comp_left` applies. The text checks directly that L is full, faithful and a local sc-diffeomorphism. *Step 3, the refinement.* $(F \circ \pi_A) \circ L = F \circ H \circ \pi_{A'} \simeq F' \circ \pi_{A'}$, and $(\Psi \circ \pi_B) \circ L = \Psi \circ K \circ \pi_{B'} \simeq \Psi' \circ \pi_{B'}$ `Chapter10/Localization · EPLocalization.Diagram.IsRefinement.comp`; the text writes the second as "$(\Phi \circ \pi_B) \circ L \simeq \Psi' \circ \pi_{B'}$", read with Ψ (possible misprint). With common refinements on both sides, the class of the composite depends only on the classes `Chapter10/Localization · EPLocalization.Diagram.Equiv.comp`. *Step 4, associativity.* The text identifies $(A \times_Y A') \times_Z A''$ and $A \times_Y (A' \times_Z A'')$; here the reassociation $((a, \phi, a'), \psi, a'') \mapsto (a, \phi, (a', \psi, a''))$ is a functor `Chapter10/Localization · EPLocalization.assocObj` `Chapter10/Localization · EPLocalization.assocMor` `Chapter10/Localization · EPLocalization.assocFunctor`, an equivalence by two out of three `Chapter10/Localization · EPLocalization.projLeft_comp_assocFunctor` `Chapter10/Localization · EPLocalization.isEquivalence_assocFunctor`, and a refinement with identity natural transformations `Chapter10/Localization · EPLocalization.Diagram.isRefinement_assoc` `Chapter10/Localization · EPLocalization.Diagram.equiv_assoc`. *Step 5, units.* $X \xleftarrow{\pi_X} X \times_X A \xrightarrow{\Phi \circ \pi_A} Y$ refines d by π_A, with $F \circ \pi_A \simeq \pi_X$ by $\tau(x, \phi, a) = \phi$ `Chapter10/Localization · EPLocalization.Diagram.isRefinement_id_comp`, and symmetrically on the other side `Chapter10/Localization · EPLocalization.Diagram.isRefinement_comp_id`. The text's proof announces $[1_X] \circ [d] = [d]$ for $d : X \to Y$, which in its order of composition is $[d] \circ [1_X] = [d]$, writes $\tau(a, \phi, a)$ for $\tau(x, \phi, a)$ and "$\psi = \varphi$" for $\psi = \phi$ (possible misprints).

X.18 Definitions 10.3.9 and 10.3.12, Theorem 10.3.10 (2017: Definitions 10.8, 10.9, Theorem 10.4) with Appendix §10.6.2

Definition 10.3.12 (2017: 10.9) (V10.44) `Chapter10/LocalizationCategory · EPLocalization.EPLoc`. The objects are a copy of those of $\mathcal{EP}$ `Chapter10/LocalizationCategory · EPLocalization.EPLoc.toEP_of` `Chapter10/LocalizationCategory · EPLocalization.EPLoc.of_toEP`, the morphisms the quotients of dia-

grams Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk, with $[d] \gg [e] = [d.\mathrm{comp}\, e]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_comp and $\mathrm{id} = [1_X]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.id_eq; the category laws are Theorem 10.3.8. $[d] = [d']$ iff d and d' are equivalent Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_eq_homMk_iff, and every morphism is a $[d]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_surjective.

Definition 10.3.9 (2017: 10.8) (V10.41) Chapter10/LocalizationCategory · EPLocalization.EPLoc.IsGeneralizedIso, the isomorphisms of the category Chapter10/LocalizationCategory · EPLocalization.EPLoc.isGeneralizedIso_iff_isIso.

Theorem 10.3.10 (2017: 10.4) (V10.42) Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_10, the proof of Appendix §10.6.2. *Step 1, Φ an equivalence.* The reversed diagram $Y \xleftarrow{\Phi} A \xrightarrow{F} X$ Chapter10/LocalizationCategory · EPLocalization.Diagram.symm. The composite $X \xleftarrow{F\circ\pi_1} A \times_Y A \xrightarrow{F\circ\pi_2} X$ is refined by $X \xleftarrow{F} A \xrightarrow{F} X$ through the diagonal $a \mapsto (a, 1_{\Phi(a)}, a)$ Chapter10/Localization · EPLocalization.diagObj Chapter10/Localization · EPLocalization.diagMor Chapter10/Localization · EPLocalization.diagFunctor, an equivalence since $\pi_1 \circ \Delta = 1_A$ Chapter10/Localization · EPLocalization.isEquivalence_diagFunctor Chapter10/Localization · EPLocalization.Diagram.isRefinement_comp_symm, and that diagram refines 1_X by F Chapter10/Localization · EPLocalization.Diagram.isRefinement_id; so both composites are units Chapter10/LocalizationCategory · EPLocalization.Diagram.equiv_comp_symm Chapter10/LocalizationCategory · EPLocalization.Diagram.equiv_symm_comp. *Step 2, the converse.* Let $[b] = [Y \xleftarrow{G} B \xrightarrow{\Psi} X]$ be an inverse. A diagram $X \leftarrow C \xrightarrow{\Theta} X$ equivalent to 1_X has Θ an equivalence (Propositions 10.1.5 and 10.1.6 applied to a common refinement) Chapter10/LocalizationCategory · EPLocalization.Diagram.isEquivalence_Φ_of_equiv_id; so $\Psi \circ \pi_B$ and $\Phi \circ \pi_{A'}$ are equivalences. *Step 3, (C1).* Φ is an equivalence of categories Chapter10/LocalizationCategory · EPLocalization.isEquivalence_toFunctor_of_comp. Faithfulness is checked on morphisms (h, ϕ, k) with $k = 1_b$; in the text's injectivity argument $A \times_X B$ is read $A \times_Y B$ and $\mathrm{mor}_Y(\Psi(a), \Psi(a'))$ as $\mathrm{mor}_X(\Psi(b), \Psi(b'))$, and in the essential surjectivity "between $\Phi(a)$ and x" as "and y" (possible misprints). *Step 4, (C2).* The text defines $\Gamma(b') = (b', (s|U)^{-1}(b'), \ldots)$, read with $(s|U)^{-1}(\Psi(b'))$ (possible misprint), and then uses that $\Gamma' : U' \to A$ is a local sc-diffeomorphism; Γ' is a composite with Ψ, so as we read it this needs Ψ to be a local sc-diffeomorphism, the other half of (C2), and the step does not hold as we read it. The formal argument: a local inverse κ of the equivalence $\Phi \circ \pi_{A'}$ at (b, ϕ, a) gives the local sc-smooth right inverse $\pi_{A'} \circ \kappa$ of Φ with value a at $\Phi(a)$; Φ is injective near a by a local section $a' \mapsto (a', \phi(a'), b(a'))$ of the local sc-diffeomorphism $\pi_A : A \times_Y B \to A$ and the local injectivity of $\Psi \circ \pi_B$, s and G Chapter10/LocalizationCategory · MPolyfold.IsLocalScDiffeoAt.exists_injOn; an injective map with such a right inverse is a local sc-diffeomorphism Chapter10/LocalizationCategory · MPolyfold.isLocalScDiffeoAt_of_injOn Chapter10/LocalizationCategory · EPLocalization.isLocalScDiffeo_obj_of_comp. *Step 5.* Lemma 10.1.2 Chapter10/LocalizationCategory · EPLocalization.Diagram.isEquivalence_Φ_of_equiv. *(2).* An inverse is unique, so every inverse is the class of the reversed diagram Chapter10/LocalizationCategory · EPLocalization.EPLoc.isIso_homMk_iff Chapter10/LocalizationCategory · EPLocalization.EPLoc.isIso_homMk Chapter10/LocalizationCategory · EPLocalization.EPLoc.inv_homMk.

X.19 Proposition 10.3.11, Theorems 10.3.14 and 10.3.15 (2017: Proposition 10.6, Theorems 10.5, 10.6)

Proposition 10.3.11 (2017: 10.6) (V10.43) Chapter10/LocalizationCategory · EPLocalization.EPLoc.proposition_10_3_11. $||[d]||$ on the quotient Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitMap Chapter10/

LocalizationCategory · EPLocalization.EPLoc.orbitMap_homMk Chapter10/LocalizationCategory · EPLocalization.EPLoc.continuous_orbitMap Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitMap_id. The computation before the proposition: an object (a, ϕ, b) of $A \times_Y B$ has $\phi : \Phi(a) \to G(b)$, so $|\Phi(a)| = |G(b)|$, and $|d.\mathrm{comp}\, e| = |e| \circ |d|$ Chapter10/Localization · EPLocalization.Diagram.orbitMap_comp.

Theorem 10.3.14 (2017: 10.5) (V10.45) Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_14. *Step 1.* $d_\Phi = X \xleftarrow{1} X \xrightarrow{\Phi} Y$ Chapter10/LocalizationCategory · EPLocalization.Diagram.ofFunctor and i Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor_obj Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor_map. *Step 2, functoriality.* The composite (10.13) $X \xleftarrow{\pi_X} X \times_Y Y \xrightarrow{\Psi \circ \pi_Y} Z$ refines $d_{\Psi\circ\Phi}$ by π_X, since $\Phi \circ \pi_X \simeq \pi_Y$ (§X.10) Chapter10/LocalizationCategory · EPLocalization.Diagram.isRefinement_ofFunctor_comp. The text states the refinement in the other direction; with Definition 10.3.1 (the refining equivalence goes out of the refinement) it is this one, and the equality of classes is the same. *Step 3, inverses.* For $F \in E$, Theorem 10.3.10 with the legs 1 and F Chapter10/LocalizationCategory · EPLocalization.EPLoc.isInvertedBy_localizationFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.inv_localizationFunctor_map, and every generalized map is $i(\Phi) \circ i(F)^{-1}$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_eq. *Witness.* The pair groupoid of $\mathbb{Z}$ and the point as objects of $\mathcal{EP}$ Chapter10/LocalizationCategory · EPLocalization.Examples.pairEP Chapter10/LocalizationCategory · EPLocalization.Examples.pointEP Chapter10/LocalizationCategory · EPLocalization.Examples.pairToPointEP Chapter10/LocalizationCategory · EPLocalization.Examples.isEquivalence_pairToPointEP: the functor is not an isomorphism of $\mathcal{EP}$ Chapter10/LocalizationCategory · EPLocalization.Examples.not_isIso_pairToPointEP, its image is invertible Chapter10/LocalizationCategory · EPLocalization.Examples.isGeneralizedIso_pairToPoint, with the inverse of Theorem 10.3.14 Chapter10/LocalizationCategory · EPLocalization.Examples.inv_localizationFunctor_map_pairToPoint. *The universal property* mentioned on p. 483 (G.117) is §X.30: i is a localization of $\mathcal{EP}$ at E Chapter10/LocalizationUniversal · EPLocalization.EPLoc.isLocalization.

Theorem 10.3.15 (2017: 10.6) (V10.46) Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_15. A functor into Mathlib's `TopCat` by Proposition 10.3.11.

X.20 Propositions 10.3.16, 10.3.17, Remark 10.3.18 (2017: 10.3), Theorem 10.3.19 (2017: 10.7), Definition 10.3.20, Theorem 10.3.21

The reduction Chapter10/LocalizationProperties · EPLocalization.EPLoc.exists_diagram_of_isGeneralizedIso Chapter10/LocalizationProperties · EPLocalization.EPLoc.iff_of_isGeneralizedIso. A generalized isomorphism is some $[d]$, and by Theorem 10.3.10 its Φ is an equivalence; a property invariant under equivalences passes from X to A to Y, as on p. 492.

Proposition 10.3.17 (V10.47) Chapter10/LocalizationProperties · EPLocalization.EPLoc.proposition_10_3_17, by Proposition 10.1.16. **Proposition 10.3.16** (V10.64) Chapter10/LocalizationLR · EPLocalization.EPLoc.proposition_10_3_16, by Proposition 10.1.9. **Theorem 10.3.21** (V10.50) Chapter10/LocalizationProperties · EPLocalization.EPLoc.theorem_10_3_21 Chapter10/LocalizationLR · EPLocalization.EPLoc.theorem_10_3_21_all, by Propositions 10.1.9, 10.1.16, 10.1.19, 10.1.18 and 10.1.17; the text leaves the proof to the reader (G.118, closed).

Theorem 10.3.19 (2017: 10.7) does not hold as we read it (V10.65) Chapter10/LocalizationLRCounterexample · EPLocalization.EPLoc.Theorem_10_3_19 Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexa not_theorem_10_3_19. *The example.* $X = \mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action has (LR): its orbits are points,

so an orbit-preserving β is the identity, the action of 1 Chapter10/LocalizationLRCounterexample · EpGroupoid.Examples.isLR_trivialZ3. $[X \xleftarrow{1} X \xrightarrow{\mathrm{Inv}} X]$ and $[1_X]$ are generalized isomorphisms with the same orbit map that differ Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.theorem_10_3_19_counterexample, the base part of Remark 10.5.5 (§X.23). *The step on p. 494.* The proof arranges $|H| = |H'|$ for equivalences $H, H' : A \to Y$ and uses the families α_a of Proposition 10.1.10 (1) as a natural transformation $H_{\mathcal{U}} \to H'_{\mathcal{U}}$. Here $z \mapsto 1_z : z \to \mathrm{Inv}(z)$ is such a family, but there is no natural transformation $1 \to \mathrm{Inv}$: naturality at the generator g of G_z would give $g = g^{-1}$ Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.not_naturallyEquivalent_id_inv. X is not effective Chapter7/NonVacuity · EpGroupoid.Examples.not_isEffective_trivialZ3. The 2017 Theorem 10.7, without (LR), does not hold as we read it either (G.120, settled).

Theorem 10.3.19 for an effective target (V10.66) Chapter10/LocalizationLRCounterexample · EPLocalization.EPLoc.theorem_10_3_19_of_isEffective, the strongest correct version we prove. *Step 1.* In an effective ep-groupoid a continuous family $v \mapsto k(v) \in G_v$ on an open set is the family of units Chapter10/EquivalenceLREffective · EpGroupoid.eqOn_unit_of_isEffective, so two continuous families $\alpha_z, \alpha'_z : H(z) \to H'(z)$ on an open set agree when H is a local sc-diffeomorphism on objects Chapter10/EquivalenceLREffective · EpFunctor.eqOn_of_isEffective. *Step 2.* The families of Proposition 10.1.10 (1) therefore glue on all of X, and they are natural: at a morphism h, $z \mapsto H'(\sigma(z))^{-1} \circ \alpha(t(\sigma(z))) \circ H(\sigma(z))$, with σ the local inverse of s through h, is another such family Chapter10/EquivalenceLREffective · EpFunctor.naturallyEquivalent_of_isEffective. *Step 3.* For $e = f' \circ f^{-1}$, represented by $Y \xleftarrow{F} A \xrightarrow{\Phi} Y$ with $|\Phi| = |F|$, Step 2 gives $F \simeq \Phi$, so $e = 1_Y$ and $f = f'$, without the weak fibred product and the cover $A_{\mathcal{U}}$ of the text. *Witnesses*, the point Chapter10/LocalizationLRCounterexample · EPLocalization.EPLoc.theorem_10_3_19_of_isEffective_hypotheses Chapter10/EquivalenceLREffective · EquivalenceLREffectiveExamples.naturallyEquivalent_of_isEffective_hypotheses.

Remark 10.3.18 (2017: 10.3) (V10.48) Chapter10/LocalizationProperties · EPLocalization.OrbitChart Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel. Reflexive by 1_X Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.refl, symmetric by the inverse Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.symm and transitive by the composite Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.trans, each with Proposition 10.3.11 Chapter10/LocalizationProperties · EPLocalization.OrbitChart.setoid. *Witness* Chapter10/LocalizationProperties · EPLocalization.OrbitChart.rel_pairEP_pointEP. The metrizability of Z is not used. The non-uniqueness of f is shown by the base classes of the example of Remark 10.5.5 (§X.23; G.119, closed).

Definition 10.3.20 (V10.49) Chapter10/LocalizationProperties · EPLocalization.EPreg Chapter10/LocalizationProperties · EPLocalization.EPLocReg, Mathlib full subcategories, with the point as an object of both Chapter10/LocalizationProperties · EPLocalization.Examples.pointReg Chapter10/LocalizationProperties · EPLocalization.Examples.pointLocReg. Remark 10.3.13 has no formal content.

X.21 §10.5: diagrams of strong bundles, $[d]$, $|[D]|$ and γ

Diagrams and refinements (V10.51) Chapter10/StrongBundleLocalization · SBLocalization.SDiagram Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement. A diagram is a middle object of $\mathcal{SEP}$ with a strong bundle equivalence Φ and a morphism Ψ of $\mathcal{SEP}$. A refinement is a strong bundle equivalence H with $\Phi \circ H \simeq \Phi'$ and $\Psi \circ H \simeq \Psi'$, natural equivalence of morphisms of $\mathcal{SEP}$ being Definition 10.4.8 Chapter10/StrongBundleLocalization · SEP.NatEquiv, an equivalence re-

lation compatible with composition on the left Chapter10/StrongBundleLocalization · SEP.NatEquiv.comp_left. The two examples Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.isRefinement_precomp Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.isRefinement_of_natEquiv and the strong bundle Lemma 10.3.2 Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.trans follow: $H \circ H'$ is a strong bundle equivalence and $\Phi \circ H \circ H' \simeq \Phi' \circ H' \simeq \Phi''$. *The classes.* Having a common refinement Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv is reflexive and symmetric Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.refl Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.symm; its transitivity, the strong bundle Lemma 10.3.4, is proved in §X.22 through the weak fibred product of strong bundles. $[D]$ is the class for the generated equivalence relation Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.cls, which is having a common refinement (§X.22); every invariant below is proved invariant under `Equiv`, hence defined on the classes, and a refinement has the class of the refined diagram Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.equiv Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.cls_eq_cls_of_equiv.

$[d]$ and $|[D]|$ (V10.52) Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.base Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.total Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.base Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.orbitMap. Forgetting the bundles gives the diagram d of the base ep-groupoids, and the total spaces as ep-groupoids $(W, \mathbf{W})$ Chapter10/StrongBundleLocalization · SEP.totalEP give a diagram of total spaces; refinements and equivalences pass to both Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.base Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.base Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.total, since a natural equivalence of strong bundle functors covers one of the base functors and is one of the total-space functors. $|[D]|$ is §10.3's orbit map $|\Psi| \circ |\Phi|^{-1}$ of the diagram of total spaces Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap, continuous Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.continuous_orbitMap and the same for equivalent diagrams by Lemma 10.3.6 Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.orbitMap_eq.

The square $|\mathcal{D}|$ (V10.53) Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap_square Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.orbitMap_square. Every orbit of $|W|$ is $|\Phi(w)|$ for some $w \in W''$, $|\Phi|$ being onto; then $|[D]|(|\Phi(w)|) = |\Psi(w)|$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap_orbitMap_Φ, $|P'(\Psi(w))| = |\psi(P''(w))|$ since Ψ covers ψ, and $|[d]|(|\phi(P''(w))|) = |\psi(P''(w))|$ by Lemma 10.3.6. The printed $|X| \to |X|$ is read with X'.

γ (V10.54) Chapter10/StrongBundleLocalization · EPLocalization.Diagram.exists_isotropyData Chapter10/StrongBundleLocalization · EPLocalization.Diagram.isotropyHom. The data: x'' and a from the essential surjectivity of ϕ, b from $|[d]|(|x|) = |x'|$. γ is the composite of the conjugation by a^{-1}, the inverse of the isomorphism $G_{x''} \to G_{\phi(x'')}$ of the equivalence ϕ (Remark 10.1.12), ψ on isotropy groups and the conjugation by b, with its values Chapter10/StrongBundleLocalization · EPLocalization.Diagram.isotropyHom_spec; for ψ an equivalence each factor is bijective Chapter10/StrongBundleLocalization · EPLocalization.Diagram.bijective_isotropyHom. The germ q and its lift Q of pp. 505–506 are §X.33 (G.124) Chapter10/StrongBundleDiagram · EPLocalization.Diagram.exists_germ_q Chapter10/StrongBundleDiagramLift · SBLocalization.SDiagram.exists_lift_Q.

X.22 Definitions 10.5.1 and 10.5.2 (2017: 10.14 and 10.15), Remark 10.5.3

The weak fibred product of strong bundles: Theorem 10.2.2 for strong bundles (V10.68) `Chapter10/StrongBundleWeakFibredProduct · SBWeakFibredProduct.wfpSEP` `Chapter10/StrongBundleWeakFibredProduct · SBWeakFibredProduct.theorem_10_2_2`. The text introduces refinements and classes "following our earlier construction" (p. 504) and gives no weak fibred product of strong bundles. *The tools.* For a strong bundle W over X and a functor $\pi : A \to X$, the pull-back π^*W over A, with $\mu_\pi(\gamma,(a,w)) = (t(\gamma),\mu(\pi(\gamma),w))$ `Chapter10/StrongBundleWeakFibredProductPullback · EpStrongBundle.pullbackAlong`, whose projection is a strong bundle equivalence when π is an equivalence `Chapter10/StrongBundleWeakFibredProductPullback · EpStrongBundle.isStrongBundleEquivalence_pullbackProjFunctor`; the transport $w \mapsto \mu'(\tau(Pw),\Psi(w))$ of a strong bundle functor along a natural transformation τ `Chapter10/StrongBundleWeakFibredProductPullback · EpStrongBundle.transportFunctor`; and, for a strong bundle equivalence Φ covering ϕ and a strong bundle functor Θ covering $\phi\circ\kappa$, the unique lift Λ covering κ with $\Phi\circ\Lambda = \Theta$, fibrewise $\Lambda(c) = \Phi^{-1}_{\kappa(Pc)}(\Theta(c))$, well defined because Φ is a linear isomorphism on fibres (Proposition 10.4.4) and sc-smooth because it is locally $\Phi^{-1}\circ\Theta$ `Chapter10/StrongBundleWeakFibredProductPullback · EpStrongBundle.invLiftFunctor`. *The construction.* $W\times_{W'}W'' = \pi_Z^*W''$ over $X\times_{X'}X''$; Π_Z is the projection of the pull-back, an equivalence since π_Z is one (Theorem 10.2.2), and Π_X the lift of the transport of $\Psi\circ\Pi_Z$ along the natural transformation $\psi\circ\pi_Z \to \phi\circ\pi_X$ `Chapter10/StrongBundleWeakFibredProduct · SEP.coneHom`, so that $\Phi\circ\Pi_X \simeq \Psi\circ\Pi_Z$ `Chapter10/StrongBundleWeakFibredProduct · SEP.natEquiv_cone`. When Ψ is an equivalence, Π_X is an isomorphism onto π_X^*W followed by its projection, an equivalence `Chapter10/StrongBundleWeakFibredProduct · SBWeakFibredProduct.theorem_10_2_2_right`. *Witness* `Chapter10/StrongBundleWeakFibredProduct · StrongBundleWeakFibredProductExamples.theorem_10_2_2_negSEPHom`: Φ over $-\mathrm{id}$ and $\Psi = 1$, with both projections strong bundle equivalences.

Transitivity: Lemma 10.3.4 for strong bundles (V10.69) `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.lemma_10_3_4` `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.Equiv.trans`. As in §X.16: if w refines D, D' by H, H' and w' refines D', D'' by K, K', the diagram on the weak fibred product of H' and K refines D by $H\circ\Pi_X$ and D'' by $K'\circ\Pi_Z$, both strong bundle equivalences since K is one. Hence $[D] = [D']$ iff D and D' have a common refinement `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.cls_eq_cls_iff`, and the classes of §X.21 are the text's.

The composition: Theorem 10.3.8 for strong bundles (V10.70) `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.comp` `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.Equiv.comp`. For D and $E : W' \xleftarrow{\Phi'} V'' \xrightarrow{\Psi'} W'''$ the middle of the composite is the pull-back of W'' along π_X over the weak fibred product of ψ and ϕ' `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.compMid`, so that its projection $\Pi_{W''}$ is a strong bundle equivalence `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.compLeft`, and $\Pi_{V''}(p,w) = \Phi'^{-1}_z(\mu'(\varphi,\Psi(w)))$ is a lift as above `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.compRight`. If D' refines D by H and E' refines E by K, the lift of $H\circ\Pi_{W''}$ over the functor L of (10.28), (10.30) refines the composite `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.IsRefinement.comp`; associativity is the lift over the reassociation functor of §X.17 `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.isRefinement_assoc` `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_assoc`, and the units come from sections into the middle `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.unitSection` `Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_comp_`

id Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_id_comp. In each proof the square with the projections to the middle of E, or of the last diagram, commutes strictly, because a strong bundle equivalence is injective on fibres. *Witness* Chapter10/StrongBundleLocalizationCategory · StrongBundleLocalizationCategoryExamples.homMk_foldDiagram_comp Chapter10/StrongBundleLocalizationCategory · StrongBundleLocalizationCategoryExamples.nonempty_comp_foldDiagram: for $F = [W' \xleftarrow{P'} (-\mathrm{id})^*W' \xrightarrow{P'} W']$, $F \circ F = 1$, with a nonempty middle.

Definition 10.5.1 (2017: 10.14) (V10.55) Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.idHom Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.localizationPrefunctor. The objects are a one-field structure around $\mathcal{SEP}$, the morphisms the classes, the identities $[1_W]$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.id; with the composition $[E] \circ [D] = [D \circ E]$ Chapter10/StrongBundleLocalizationCategory · SBLocalization.GenSBMap.comp the laws of the previous paragraph make it a category Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.category (G.123, closed). $\Psi \mapsto [1, \Psi]$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.ofHom, first a prefunctor mapping 1_W to $[1_W]$ Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.localizationPrefunctor_map_id, is a functor by $[1, \Psi'] \circ [1, \Psi] = [1, \Psi' \circ \Psi]$ Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_ofHom_comp Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.localizationFunctor, and it inverts the strong bundle equivalences Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.isInvertedBy_localizationFunctor.

Definition 10.5.2 (2017: 10.15) (V10.56) Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.IsGeneralizedIso: some representative has Ψ in F (Φ is in F for every diagram). The reversed diagram Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.symm of such a representative, and $[1, \Psi]$ for Ψ in F, give examples Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.isGeneralizedIso_symm Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.isGeneralizedIso_ofHom. *Witness* Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.negDiagram Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isGeneralizedIso_negDiagram: $[1, P']$ for the projection $P' : (-\mathrm{id})^*W' \to W'$ of §X.11, between two different strong bundles.

Remark 10.5.3 (V10.57) Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.remark_10_5_3. For a representative with Φ, Ψ in F, ϕ and ψ are equivalences and Φ, Ψ are equivalences of the total spaces (Definition 10.4.1); Theorem 10.3.10 makes $[d]$ and the class of the diagram of total spaces generalized isomorphisms, and Lemma 10.3.6 makes $|[D]|$ a homeomorphism. The strong bundle version of Theorem 10.3.10, which the remark builds into Definition 10.5.2, is read as: the generalized strong bundle isomorphisms are exactly the isomorphisms of $\mathcal{SEP}(F^{-1})$ (next paragraph and §X.34).

Theorem 10.3.10 for strong bundles, the first direction (V10.71; the converse is §X.34) Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.theorem_10_3_10 Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.isIso_of_isGeneralizedIso. If Ψ is a strong bundle equivalence, $D \circ D^{-1}$ and $D^{-1} \circ D$, for the reversed diagram D^{-1}, are refined through sections into their middles Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_comp_symm Chapter10/StrongBundleLocalizationCategory · SBLocalization.SDiagram.equiv_symm_comp, so $[D]$ is an isomorphism with inverse $[D^{-1}]$ Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.inv_homMk. *Towards the converse.* $[D] \mapsto [d]$ is a functor to $\mathcal{EP}(E^{-1})$ Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.baseFunctor, so an isomorphism $[D]$ gives an isomorphism $[d]$, and Theorem 10.3.10 for the underlying diagram makes ψ an equivalence Chapter10/StrongBundleLocalizationCategory · SBLocalization.SEPLoc.isEquivalence_base_of_isIso. That Ψ itself is a strong bundle equivalence is §X.34 (G.123,

closed for this half too), by a route that replaces the claims (C1), (C2) of Appendix §10.6.2 Chapter10/StrongBundleIsoConverse · SBLocalization.SEPLoc.isIso_iff_isGeneralizedIso. *Witness* Chapter10/StrongBundleLocalizationCategory · StrongBundleLocalizationCategoryExamples.isIso_homMk_negDiagram: the class over $-\mathrm{id}$ of Definition 10.5.2.

X.23 Theorem 10.5.4 (2017: 10.9), Remark 10.5.5, Proposition 10.5.6

The push-forward along a diagram (V10.58) Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pullback. For Φ, Ψ in F, $D_*f = \Psi_*\Phi^*f$ with §10.4's pull-back and push-forward, and D^*g is the push-forward along the reversed diagram. That they map section functors to section functors, are mutually inverse and do the same on sc^+-section functors Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.mapsTo_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.invOn_pullback_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.bijOn_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.bijOn_pushforward_plus comes from Theorem 10.4.5. *The text's step* Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward_eq_of_isRefinement: for a refinement by H, $\Psi_*\Phi^*f = \Psi_*H_*H^*\Phi^*f = (\Psi\circ H)_*(\Phi\circ H)^*f = \Psi'_*\Phi'^*f$, by Theorem 10.4.5 for H, Proposition 10.4.4, and Propositions 10.4.2, 10.4.3 for the naturally equivalent pairs, which have the same orbit maps. *Through the orbit spaces.* Step 1: $|\mu(\sigma,w)| = |w|$ Chapter10/StrongBundleLocalizationSections · EpStrongBundle.π_μ, so a section functor takes isomorphic objects to one orbit Chapter10/StrongBundleLocalizationSections · EpStrongBundle.π_eq_π_of_isFunctorialSection, and two section functors with $|f(x)| = |g(x)|$ for all x agree: a morphism $(\sigma, f(x)) : f(x) \to g(x)$ has $\sigma \in G_x$, and $f(x) = \mu(\sigma, f(x))$ since f is functorial Chapter10/StrongBundleLocalizationSections · EpStrongBundle.eq_of_forall_π_eq. Step 2: $|(D_*f)(x')| = |[D]|(|f(x)|)$ whenever $|[d]|(|x|) = |x'|$ Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.π_pushforward. Step 3: $|[d]|$ is determined by $|[D]|$ through the square, $|P|$ being onto Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.base_orbitMap_eq_of_orbitMap_eq. Hence diagrams with the same $|D|$ have the same push-forward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward_eq_of_orbitMap_eq and, reversed, the same pull-back Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pullback_eq_of_orbitMap_eq.

Theorem 10.5.4 (2017: 10.9) (V10.59) Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.theorem_10_5_4. $[D]_*$ and $[D]^*$ use a representative with Ψ in F chosen once Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.rep Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pullback; every other such representative has the same $|D|$, an invariant of the class, hence the same push-forward and pull-back Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pushforward_eq Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pullback_eq. The rest is the previous paragraph for the chosen representative, the last assertion with Step 3. The Fred part (G.125) Chapter10/FredholmEquivalence · SBLocalization.GenSBMap.theorem_10_5_4_fred and "isomorphism" as a linear isomorphism Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.pushforwardEquiv are §X.31. *Witnesses* Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.bijOn_pushforward_negDiagram Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.pushforward_foldDiagram: $W' \xleftarrow{P'} (-\mathrm{id})^*W' \xrightarrow{P'} W'$ refines $1_{W'}$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isRefinement_foldDiagram Chapter10/

StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.cls_foldDiagram, so its push-forward is that of $[1_{W'}]$, the identity Chapter10/StrongBundleLocalizationExamples · SBLocalization.SDiagram.pushforward_id.

Remark 10.5.5 (V10.60) Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.remark_10_5_5. The positive half is the last assertion of Theorem 10.5.4 Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.pushforward_inv_eq_pushforward_id; for the negative half the text gives no example. *The invariant* Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.IsotropyRel: $g \in G_x$ and $g' \in G_{x'}$ are related by d if $g = a \circ \phi(k) \circ a^{-1}$ and $g' = b \circ \psi(k) \circ b^{-1}$ for some z, $k \in G_z$, a, b; the relation contains the graph of every γ Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.isotropyRel_isotropyHom, and it does not change under refinement, through the conjugations by the natural transformations Chapter10/StrongBundleLocalizationExamples · EpNatTrans.isotropyHom_eq Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.IsRefinement.isotropyRel_iff, nor under equivalence Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.Equiv.isotropyRel_iff. *The example.* $W = \mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$ over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with $\mu((g,x),w) = w$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3EpBundle, and Inv over $(g,x) \mapsto (g^{-1},x)$, a functor since $\mathbb{Z}/3$ is abelian and a strong bundle equivalence as its own inverse Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3InvHom Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3InvHom_mem_equivalences. Inv is the identity on objects, so $|[\mathrm{Inv}]| = |[1_W]|$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.orbitMap_ofHom_z3InvHom and the base orbit maps agree Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.base_orbitMap_ofHom_z3InvHom. For $[1_W]$ the generator of $G_0 = \mathbb{Z}/3$ is related to itself Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isotropyRel_id; for $[\mathrm{Inv}]$ a related pair is (k,k^{-1}) up to conjugation, which is trivial in $\mathbb{Z}/3$, and the generator is not its own inverse Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.not_isotropyRel_inv. So the base classes differ, and with them the classes. By Remark 10.5.3 the base classes are two distinct generalized isomorphisms with the same orbit map: the non-uniqueness claimed in Remark 10.3.18 (G.119, closed).

Proposition 10.5.6 (V10.61) Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.iff_of_isGeneralizedIso. A representative with Φ, Ψ in F carries a property invariant under strong bundle equivalences from W to W'' to W'. For the local regularity property this is Proposition 10.4.6 Chapter10/StrongBundleLocalRegularity · EpStrongBundle.proposition_10_4_6, with Definition 8.4.2 read as `IsLRLocal`, and Proposition 10.5.6 follows Chapter10/StrongBundleLocalRegularity · SBLocalization.proposition_10_5_6 (G.122, G.126, closed); the text's "see Proposition 10.4.7" is read as Proposition 10.4.6.

Theorem 10.5.7 (2017: 10.10) does not hold as we read it (V10.67) Chapter10/LocalizationLRCounterexample · SBLocalization.Theorem_10_5_7 Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.not_theorem_10_5_7. The strong bundle of Remark 10.5.5 has the local regularity property of Definition 8.4.2, its orbits being points Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.isLR_z3EpBundle, and $[\mathrm{Inv}] \neq [1_W]$ with the same orbit map. The text's proof, "immediately from Proposition 10.4.7", meets the step of Theorem 10.3.19 (§X.20); the 2017 Theorem 10.10, without local regularity, does not hold as we read it either (G.127, settled).

Theorem 10.5.7 over an effective base of the target (V10.67) Chapter10/LocalizationLREffectiveBundle · SBLocalization.theorem_10_5_7_of_isEffective Chapter10/LocalizationLRBundleBase · SBLocalization.theorem_10_5_7_of_isEffective_of_isLRLocal, the strong bundle analogue of the effective Theorem 10.3.19. *Step 1, the bases.* $|\Phi| = |\Psi|$ on total spaces gives $|\phi| = |\psi|$ on the bases Chapter10/LocalizationLREffectiveBundle · SEP.orbitMap_

base_eq, since the projection of a strong bundle is open Chapter10/LocalizationLREffectiveBundle · StrongBundle.isOpenMap_proj. *Step 2, the natural equivalence.* For strong bundle equivalences $F, G : W'' \to W'$ with $|F| = |G|$ and an effective base of W', the base transformation $\tau : \phi_F \to \phi_G$ is that of §X.20, Step 2; then $G(w) = \mu'(\tau(P(w)), F(w))$: both sides are continuous in w and agree over the regular points, where only one morphism $\phi_F(P(w)) \to \phi_G(P(w))$ exists, and these are dense Chapter10/LocalizationLREffectiveBundle · SEP.natEquiv_of_isEffective. The families of Step 2 come from (LR) of a base (Proposition 10.1.10) in the first form, and in the second from Definition 8.4.2 read for strong bundle isomorphisms of $W|U(x)$ Chapter10/LocalizationLRBundleBase · EpStrongBundle.IsLRLocal through Proposition 10.4.7 on the bases Chapter10/LocalizationLRBundleBase · SEP.exists_family_of_isLRLocal (§X.32). *Step 3.* For D, D' with $|[D]| = |[D']|$, the weak fibred product of Φ and Φ' refines both: $\Phi \circ \Pi_X \simeq \Phi' \circ \Pi_Z$ by the strong bundle Theorem 10.2.2 and $\Psi \circ \Pi_X \simeq \Psi' \circ \Pi_Z$ by Step 2. *Hypotheses.* The effectiveness of the base of the target is added in both forms and is not derived from the text; (LR) of a base is added in the first form and is not derived from the text either. The second form needs only the local regularity property of the target bundle, and it implies the formal (LR) of strong bundles Chapter10/LocalizationLRBundleBase · EpStrongBundle.IsLRLocal.isLR. Effectiveness cannot be dropped under this reading: the bundle over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ has the property Chapter10/LocalizationLRBundleBase · LocalizationLRBundleBaseExamples.isLRLocal_z3EpBundle, and the conclusion fails there Chapter10/LocalizationLRBundleBase · LocalizationLRBundleBaseExamples.not_theorem_10_5_7_of_isLRLocal. *Witnesses,* $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ with $[1_W]$ Chapter10/LocalizationLREffectiveBundle · LocalizationLREffectiveBundleExamples.theorem_10_5_7_of_isEffective_hypotheses Chapter10/LocalizationLRBundleBase · LocalizationLRBundleBaseExamples.theorem_10_5_7_of_isEffective_of_isLRLocal_hypotheses, and the local regularity property of a trivial bundle Chapter10/LocalizationLRBundleBase · EpStrongBundle.isLRLocal_trivial.

X.24 Witnesses and the hypotheses

Hypotheses taken, with their witnesses. The nonempty object set in Proposition 10.1.3: the covering of $\mathbb{Z}/2 \ltimes \mathbb{R}$ (§X.2). `Admissible`: every equivalence (§X.9), and the pair groupoid of $\mathbb{Z}$ over the point, and $\pm$id on $\mathbb{Z}/2 \ltimes \mathbb{R}^2$ (§X.10). Proposition 10.1.9 for the lifts, once a hypothesis of Proposition 10.1.19, is proved (§X.5). The two Hausdorff hypotheses of Proposition 10.1.14 are the text's. The effectiveness of the target in Theorem 10.3.19: the point (§X.20). The effectiveness of the base of the target in Theorem 10.5.7, with (LR) of a base or the local regularity property of the target bundle: $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ (§X.23). Definition 8.4.2 read for strong bundle isomorphisms of $W|U(x)$ is a reading, not an added hypothesis; it holds for trivial bundles (§§X.23, X.32). Otherwise §§10.3–10.5 and the new work of §§X.27–X.34 take no hypothesis.

Every definition has a non-identity witness. Equivalences: $\mathbb{Z} \to$ pt, the refinement of $\mathbb{Z}/2 \ltimes \mathbb{R}$, $-$id of $\mathbb{Z}/2 \ltimes \mathbb{R}^2$; the weak fibred product: §X.10; strong bundle equivalences, push-forwards, natural equivalences of strong bundle functors, morphisms of $\mathcal{SEP}$: the bundle over $-$id (§§X.11–X.15); equivalent diagrams, generalized isomorphisms that do not come from isomorphisms, equivalent pairs (X, β) and objects of $\mathcal{EP}_{reg}$: the pair groupoid of $\mathbb{Z}$ and the point (§§X.16–X.20); refinements of diagrams of strong bundles, weak fibred products of strong bundles, composites in $\mathcal{SEP}(F^{-1})$, generalized strong bundle isomorphisms with their inverses and push-forwards: the bundle over $-$id, and distinct classes with the same orbit map: the bundle over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (§§X.22, X.23). The subcategory $\mathcal{EP}_{lreg}$: the point (§X.27); an orbit-preserving sc-diffeomorphism that is no $\hat{g}$, and an ep-groupoid without (LR): η on $\mathbb{Z}/2 \ltimes Y$ of Appendix §10.6.3 (§X.28); the hypotheses of Proposition 10.4.7: $\Phi = \Psi = 1$ on $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ (§X.32). The other results of §§X.29–X.34 add no hypothesis. The hypotheses of Proposition 10.1.10 and of the

effective version of Theorem 10.3.19 are witnessed on the point (§§X.5, X.20).

X.25 What is not formalized

Nothing. Every gap of Chapter 10 is closed: no item is open, none is formalized only in part, and none is set aside as having nothing to formalize. The list below says where each former gap is now proved. The only narrower row is V10.66, Theorem 10.3.19 for an effective target (§X.20).

- §10.1: $\mathcal{EP}_{lreg}$ (G.111, §X.27) `Chapter10/LocalRegularitySubcategory · EP.lreg` `Chapter10/LocalRegularitySubcategory · EP.lreg_iff_of_isEquivalence`; Remark 10.1.8 and Appendix §10.6.3 with the appendix's own projections (G.112, §X.28) `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.remark_10_1_8`. Propositions 10.1.9, 10.1.10 and 10.1.19 (G.113–G.115) are closed by §§X.5, X.7.
- §10.2: the structures induced from the triple products (G.116, §X.29) `Chapter10/WeakFibredProductTriple · WeakFibredProduct.theorem_10_2_2_triple`.
- §10.3: the universal property of the localization (G.117, §X.30) `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.lift` `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.liftMap_comp`. Proposition 10.3.16 and Theorem 10.3.21 (G.118) are closed by §X.20, and the non-uniqueness in Remark 10.3.18 (G.119) by §X.23. Theorem 10.3.19 (G.120) is settled by §X.20: it does not hold as we read it, and holds for an effective target. Remark 10.3.13 has no formal content.
- §10.4: the Fred part of Theorem 10.4.5 (G.121, §X.31) `Chapter10/FredholmEquivalence · EpStrongBundle.theorem_10_4_5_fred`; Propositions 10.4.6 and 10.4.7 (G.122, §X.32), with Definition 8.4.2 read as `IsLRLocal` `Chapter10/StrongBundleLocalRegularity · EpStrongBundle.proposition_10_4_6` `Chapter10/StrongBundleLocalFamilies · EpStrongBundle.proposition_10_4_7`.
- §10.5: the converse half of the strong bundle Theorem 10.3.10 (G.123, §X.34; the other parts are closed by §X.22) `Chapter10/StrongBundleIsoConverse · SBLocalization.SEPLoc.isIso_iff_isGeneralizedIso`; (10.24) with generalized vertical maps, and the local expressions q, Q (G.124, §X.33) `Chapter10/StrongBundleDiagram · SBLocalization.GenSBMap.square_10_24` `Chapter10/StrongBundleDiagram · EPLocalization.Diagram.exists_germ_q` `Chapter10/StrongBundleDiagramLift · SBLocalization.SDiagram.exists_lift_Q`; the Fred part of Theorem 10.5.4 (G.125, §X.31) `Chapter10/FredholmEquivalence · SBLocalization.GenSBMap.theorem_10_5_4_fred`; Proposition 10.5.6 (G.126, §X.32) `Chapter10/StrongBundleLocalRegularity · SBLocalization.proposition_10_5_6`. Theorem 10.5.7 (G.127) is settled by §X.23: it does not hold as we read it, and holds over an effective base of the target.

X.26 What the formalization taught us

The equivalences of §10.1 are the sc-smooth versions of equivalences of categories, and most proofs of the section are categorical arguments plus one local fact: a functor which is a local sc-diffeomorphism on objects at $s(\varphi)$ is one at $t(\varphi)$. The text uses this fact silently in Proposition 10.1.6, where the argument covers only the image of F; with it the proof is two lines. Proposition 10.1.5, quoted without proof, and the fullness and faithfulness of TF, left to the reader, are short once tangent morphisms are known to be determined by their base and source vector. The invariance results new in 2021 split into two kinds: those about $\mathbf{X}^T$ ((LUC), the reduced groupoids, (SLUC)) go through Theorem 7.2.22 and one closure argument; those about natural representations ((LR), (SLR), Proposition 10.1.10) need natural representations carried along $Q = \hat{\psi} \circ F$, which is done by restricting a natural representation at the

target rather than by transporting Γ. The local families of Proposition 10.1.10 are not natural in general, and that is where Theorem 10.3.19 does not hold as we read it: on $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action, which has (LR) because its orbits are points, the identity and the inversion have the same orbit map and are not naturally equivalent; over an effective target the families are unique and glue, and the theorem holds without the weak fibred product of its proof. In §10.2 the weak fibred product needs much less than an equivalence: one local sc-diffeomorphism on objects for the structure, nothing for properness. In §10.3 the two places where the text argues with spaces, "the same space" for the associativity and the direct check that L is an equivalence, become functors and two-out-of-three arguments, and the local half of the hard direction of Theorem 10.3.10 needs a different route, through a local right inverse and local injectivity. In §10.4 the push-forward is well defined on section functors because of fullness (10.22), and once (10.22) is in hand its sc-smoothness is a local formula with two local inverses; Propositions 10.4.2–10.4.4 do not need sc-smoothness at all. In §10.5 the push-forward along a generalized strong bundle isomorphism is best seen through the orbit spaces: a section functor is determined by the orbits of its values, so the push-forward depends only on $|[D]|$, which gives the independence of the representative and the last assertion of Theorem 10.5.4 at once; and the example the text does not give for Remark 10.5.5 needs an invariant finer than the orbit maps, the relation of the isotropy homomorphisms γ, which also settles the non-uniqueness left informal in Remark 10.3.18. The weak fibred product of strong bundles, which the text leaves to the reader, is best taken as a pull-back: one projection is then a strong bundle equivalence by construction, and since strong bundle equivalences are injective on fibres, the squares in the proofs of transitivity, composition and associativity commute strictly, so that the proofs of §10.3 carry over.

What the closing of the gaps shows. First, the example of Appendix §10.6.3 needs no more than the scale and the loss families of Chapter 1: $\pi_{|s|}$ and the map η are differences and sums of loss families, and the failure of (LR) comes down to the observation that in a natural representation of a translation groupoid each $\Phi(g)$ is the action of g itself, by connectedness. Second, the universal property of the localization reduces to one fact: every functor that inverts the equivalences identifies naturally equivalent functors, through the cylinder $X \times_Y Y$, whose projection to X is an equivalence with sections s_τ. Third, Definition 8.4.2 must be read for strong bundle isomorphisms of $W|U(x)$ alone. Asked only of maps defined on all of W, it cannot be applied to the conjugates $H^{-1} \circ B \circ H$ of the proof of Proposition 10.4.6, which live over $U(x)$; read locally, Propositions 10.4.6, 10.4.7 and 10.5.6 follow the proofs of Propositions 10.1.9 and 10.1.10 with transports along families of morphisms in place of $\hat{\psi}$. Fourth, the converse half of the strong bundle Theorem 10.3.10 does not need strong bundle versions of the claims (C1), (C2) of the appendix: the two composites with the inverse class give fibrewise injectivity and local right inverses separately. The Fred parts of Theorems 10.4.5 and 10.5.4, called clear in the text, are the one place where a strong bundle equivalence is not a pull-back along a global isomorphism; the push-forward is handled with the action of the morphisms $\sigma : \phi(y) \to x'$, which is itself a local strong bundle isomorphism.

X.27 The subcategory $\mathcal{EP}_{lreg}$ (2021 p. 461)

The subcategory (V10.74) `Chapter10/LocalRegularitySubcategory · EP.lreg` `Chapter10/LocalRegularitySubcategory · EP.Lreg`. The text introduces, new in 2021, the full subcategory of $\mathcal{EP}$ generated by the ep-groupoids with the local regularity property (Definition 7.4.19). It is an object property of $\mathcal{EP}$ with Mathlib's full subcategory, whose inclusion is fully faithful `Chapter10/LocalRegularitySubcategory · EP.fullyFaithful_ι_lreg`. *Step 1, along equivalences.* For an equivalence $F : X \to Y$, $X \in \mathcal{EP}_{lreg}$ iff $Y \in \mathcal{EP}_{lreg}$, by Proposition 10.1.9 (§X.5) `Chapter10/LocalRegularitySubcategory · EP.lreg_iff_of_isEquivalence`;

this is the closing sentence of the proof of Proposition 10.1.9 on p. 469. *Step 2, isomorphisms of $\mathcal{EP}$.* An isomorphism of $\mathcal{EP}$ is an equivalence by (E1), so the property is closed under isomorphisms `Chapter10/LocalRegularitySubcategory · EP.isClosedUnderIsomorphisms_lreg`. *Step 3, in $\mathcal{EP}(E^{-1})$.* The same object property on the localization `Chapter10/LocalRegularitySubcategory · EPLocalization.EPLoc.lreg` is closed under the isomorphisms there, since an isomorphism is $[X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ with F, Φ equivalences (Theorem 10.3.10) and Step 1 applies twice `Chapter10/LocalRegularitySubcategory · EPLocalization.EPLoc.isClosedUnderIsomorphisms_lreg`; the middle of a diagram out of an object of $\mathcal{EP}_{lreg}$ is in $\mathcal{EP}_{lreg}$ `Chapter10/LocalRegularitySubcategory · EPLocalization.EPLoc.lreg_A_of_lreg`. *Witness*, the point `Chapter10/LocalRegularitySubcategory · EP.lreg_pointEP Chapter10/LocalRegularitySubcategory · EP.Lreg.point`.

X.28 Remark 10.1.8 and Appendix §10.6.3 (2021 pp. 467, 519)

The example as printed (V10.75) `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.remark_10_1_8`. Remark 10.1.8, new in 2021, says that an sc-diffeomorphism β of $U(x)$ with $|\beta(y)| = |y|$ need not be of the form $y \mapsto g * y$, and Appendix §10.6.3 gives the example. *Step 1, the scale and the retraction.* $F = \mathbb{R} \oplus L^2(\mathbb{R})$ with $F_m = \mathbb{R} \oplus H^{m,\delta_m}(\mathbb{R})$ `Chapter7/LineSobolevScale · ZhouLine.lineSc`, and the family π_t of Subsection 1.4.4 `Chapter7/ZhouRetraction · ZhouLine.zhouFam Chapter7/ZhouRetraction · ZhouLine.isLossFamily_zhouFam`. $\pi_{|s|} = \pi_s + \pi_{-s}$ is a loss family `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.isLossFamily_piAbs`, by sums and reflections of loss families `Chapter10/ZhouExample · IsLossFamily.add Chapter10/ZhouExample · IsLossFamily.comp_neg`, so $r(s,f) = (s, \pi_{|s|}f)$ is an sc-smooth retraction `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.Rt₂` with image Y `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.O₂`; $\mathbb{Z}/2$ acts by $(-1) * (s,f) = (s,-f)$, sc-smoothly `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.isScSmoothMap_smul₂`, and $X = \mathbb{Z}/2 \ltimes Y$ `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.X₂`. *Step 2, η.* $\eta(s,f) = (s,f)$ for $s \le 0$ and $(s,-f)$ for $s > 0$ `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.η0_of_nonpos Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.η0_of_pos` is, on Y, the map $(s,f) \mapsto (s, (\pi_{-s} - \pi_s)f)$, with $\pi_{-s} - \pi_s$ a loss family `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.isLossFamily_famη Chapter10/ZhouExample · IsLossFamily.sub`; it is its own inverse `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.η0_η0`, hence an sc-diffeomorphism `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.isScDiffeoOn_η0` fixing 0 `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.η0_o₂`, and $\eta(y)$ is y or $(-1) * y$, so $|\eta(y)| = |y|$ `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.π_η0`. *Step 3, no g.* The points $(\pm\frac{1}{n+1}, \frac{1}{n+1}\beta(\cdot + e^{n+1}))$ lie in Y and tend to 0 `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.ptσ Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.tendsto_ptσ`; on those with $s > 0$, $\eta(y) = -y \ne y$, and on those with $s < 0$, $\eta(y) = y \ne -y$, so on no neighbourhood of 0 is η the action of one g `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.exists_η0_ne_smul`. *Step 4, (LR) fails.* In a natural representation at 0, $\Gamma(g,\cdot)$ has a constant group component on the connected U, so $\Phi(g)$ is $y \mapsto g * y$ there `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.naturalRep_Φ_eq`. η maps U into itself, being the identity or the action of -1 pointwise, so it is an orbit-preserving sc-diffeomorphism of U; (LR) would make it $\Phi(g)$ on U, against Step 3 `Chapter10/ZhouExampleRealScale · EpGroupoid.Examples.ZhouTenSix.not_isLR`. *The variant* with the moving projections of Remark 4.1.3 and η of the opposite sign convention `Chapter10/ZhouExample · ZhouSigma.remark_10_1_8 Chapter10/ZhouExample · ZhouSigma.not_isLR` is kept.

X.29 Theorem 10.2.2 (2017: 10.2): the structures induced from the triple products

The last sentence of the statement `Chapter10/WeakFibredProductTriple · WeakFibredProduct.theorem_10_2_2_triple`. The text says that the object set $L \subseteq X \times \mathbf{Y} \times Z$ and the morphism set $\mathbf{L} \subseteq \mathbf{X} \times \mathbf{Y} \times \mathbf{Z}$ are sub-M-polyfolds and carry the induced structures; §X.9 builds the structures iteratively, $L = X \,{}_F\!\times_s (\mathbf{Y}\,{}_t\!\times_G Z)$ inside $X \times (\mathbf{Y}\,{}_t\!\times_G Z)$. *Step 1, products with a sub-M-polyfold.* For a sub-M-polyfold $A \subseteq N$, $X \times A$ is a sub-M-polyfold of $X \times N$ `Chapter10/WeakFibredProductTriple · MPolyfold.isSubMPolyfold_univ_prod`, and the subtype $X \times A$ with the induced structure is sc-diffeomorphic to $X \times A$ with the product of the structure of A `Chapter10/WeakFibredProductTriple · MPolyfold.prodSubtypeMk` `Chapter10/WeakFibredProductTriple · MPolyfold.isScDiffeoOn_prodSubtype`. *Step 2, transitivity.* A sub-M-polyfold of an M-polyfold sc-diffeomorphic to a sub-M-polyfold $S \subseteq M$ is a sub-M-polyfold of M `Chapter10/WeakFibredProductTriple · MPolyfold.isSubMPolyfold_image_of_isScDiffeoOn`. *Step 3, the triple sets.* The triple-product sets `Chapter10/WeakFibredProductTriple · WeakFibredProduct.objSetTriple` `Chapter10/WeakFibredProductTriple · WeakFibredProduct.morSetTriple` are the images of the iterated ones under the reassociation $(x,(\phi,z)) \mapsto (x,\phi,z)$ `Chapter10/WeakFibredProductTriple · WeakFibredProduct.Obj.toTriple` `Chapter10/WeakFibredProductTriple · WeakFibredProduct.objSetTriple_eq`, hence sub-M-polyfolds by Steps 1, 2 `Chapter10/WeakFibredProductTriple · WeakFibredProduct.isSubMPolyfold_objSetTriple` `Chapter10/WeakFibredProductTriple · WeakFibredProduct.isSubMPolyfold_morSetTriple`. *Step 4.* The reassociations are sc-diffeomorphisms of the iterated structures onto the induced ones, so the two structures agree. Only the admissibility of §X.9 is used (G.116).

X.30 The universal property of the localization (2021 p. 483 and Theorem 10.3.14)

$i : \mathcal{EP} \to \mathcal{EP}(E^{-1})$ is a localization `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.isLocalization` `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.universal_property`. The text introduces $\mathcal{EP}(E^{-1})$ as the category in which the equivalences become invertible universally. *Step 1, the cylinder.* For a natural transformation $\tau : F \to F'$ of functors $X \to Y$, the weak fibred product $X \times_Y Y$ of F and 1_Y receives the sc-smooth section $s_\tau(x) = (x, \tau(x), F'(x))$ `Chapter10/LocalizationUniversal · EPLocalization.secObj` `Chapter10/LocalizationUniversal · EPLocalization.secFunctor`, with $\pi_X \circ s_\tau = 1_X$ and $\pi_Y \circ s_\tau = F'$ `Chapter10/LocalizationUniversal · EPLocalization.secHom_cylLeft` `Chapter10/LocalizationUniversal · EPLocalization.secHom_cylRight`. *Step 2, natural equivalence is inverted.* π_X is an equivalence (Theorem 10.2.2), so for every functor L inverting E, $L(s_\sigma) = L(\pi_X)^{-1}$ for every σ, and $L(F') = L(s_\tau) \circ L(\pi_Y) = L(s_{1_F}) \circ L(\pi_Y) = L(F)$ `Chapter10/LocalizationUniversal · EPLocalization.map_eq_of_naturallyEquivalent`. *Step 3, the lift.* $L'[X \xleftarrow{F} A \xrightarrow{\Phi} Y] = L(F)^{-1} \circ L(\Phi)$ `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.liftMap`. For a refinement by H, $L(F') = L(H) \circ L(F)$ and $L(\Phi') = L(H) \circ L(\Phi)$ by Step 2, and the factors $L(H)$ cancel `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.liftMap_eq_of_isRefinement` `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.liftMap_eq_of_equiv`. *Step 4, functoriality.* The identity `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.liftMap_id`, and composites through the square $L(\pi_A) \circ L(\Phi) = L(\pi_B) \circ L(G)$, which is Step 2 for the natural equivalence of the weak fibred product (§X.10) `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.liftMap_comp` `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.lift`. *Step 5, factorization and uniqueness.* $i \circ L' = L$ `Chapter10/LocalizationUniversal · EPLocalization.EPLoc.fac`, and L' is unique because every generalized map is $i(\Phi) \circ i(F)^{-1}$ (Theorem 10.3.14) `Chapter10/`

LocalizationUniversal · EPLocalization.EPLoc.uniq; this is Mathlib's strict universal property for every target category Chapter10/LocalizationUniversal · EPLocalization.EPLoc.strictUniversalPropertyFixedTarget (G.117).

X.31 Theorems 10.4.5 and 10.5.4 (2017: 10.8 and 10.9): the Fred parts, and linearity

Part I's half. An sc-Fredholm germ stays one along a local strong bundle isomorphism: such an isomorphism Chapter3/FredholmPullbackIso · StrongBundle.LocalIso transports charts Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.transportChart Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.isChart_transportChart and germs Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.isScFredholmGerm, and the sc-Fredholm property of a section is local Chapter3/FredholmPullbackIso · StrongBundle.isScFredholmSection_of_forall.

Theorem 10.4.5, the Fred part (V10.27) Chapter10/FredholmEquivalence · EpStrongBundle.theorem_10_4_5_fred. The text calls it clear. *Step 1.* A local strong bundle isomorphism of Definition 10.4.1 (2) is one of Part I Chapter10/FredholmEquivalence · StrongBundle.IsStrongBundleIsoOn.toLocalIso. *Step 2, Φ^*.* Near x_0, $\Phi^* f'$ is the pull-back of f' along that isomorphism (§X.14, Step 1) Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.exists_localIso_pullback, so it is sc-Fredholm when f' is, in the germ reading of Definition 3.1.18 and with tame germ data Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScFredholmSection_pullback Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScFredholmSectionTame_pullback. *Step 3, the action of a morphism.* For a local section α of s, $w \mapsto \mu(\alpha(P(w)), w)$ Chapter10/FredholmEquivalence · EpStrongBundle.sectionAct is a local strong bundle isomorphism $W|t(T) \to W|s(T)$ along which every section functor is its own pull-back, by functoriality (10.16) Chapter10/FredholmEquivalence · EpStrongBundle.exists_localIso_mor. *Step 4, Φ_*.* The push-forward is not a pull-back along a global isomorphism, ϕ being only essentially surjective. Near x' with $\sigma : \phi(y) \to x'$, $\Phi_* f$ is $\Phi \circ f \circ \phi^{-1}$ near $\phi(y)$, sc-Fredholm by Step 2 for the inverse isomorphism, and near x' it is its own pull-back along the action of σ (Step 3) Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.exists_localIso_pushforward Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScFredholmSection_pushforward Chapter10/FredholmEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScFredholmSectionTame_pushforward. *Step 5.* With §X.12, Φ_* and Φ^* restrict to inverse bijections of the sets of sc-Fredholm section functors (G.121).

Theorem 10.5.4, the Fred part (V10.59) Chapter10/FredholmEquivalence · SBLocalization.GenSBMap.theorem_10_5_4_fred. $D_* = \Psi_* \Phi^*$ preserves the sc-Fredholm property by the above Chapter10/FredholmEquivalence · SBLocalization.SDiagram.mapsTo_pushforward_fred, and so does D^* Chapter10/FredholmEquivalence · SBLocalization.SDiagram.mapsTo_pullback_fred; with §X.23 they are inverse bijections Chapter10/FredholmEquivalence · SBLocalization.SDiagram.bijOn_pushforward_fred, on the chosen representative of the class (G.125).

"Isomorphism" as a linear isomorphism Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.pushforwardEquiv Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.plusPushforwardEquiv. *Step 1.* $\Gamma(P, \mu)$ and $\Gamma^+(P, \mu)$ of Definition 8.3.4 are realized as submodules of $\prod_x W_x$ Chapter10/SectionFunctorLinear2021 · EpStrongBundle.sectionSpace Chapter10/SectionFunctorLinear2021 · EpStrongBundle.plusSectionSpace. *Step 2.* Pull-back and push-forward are additive and homogeneous pointwise, for all sections, without functoriality Chapter10/SectionFunctorLinear2021 · EpStrongBundle.IsStrongBundleEquivalence.pullback_sectionAdd Chapter10/SectionFunctorLinear2021 · EpStrongBundle.IsStrongBundleEquivalence.pushforward_sectionAdd Chapter10/SectionFunctorLinear2021 · EpStrongBundle.IsStrongBundleEquivalence.pushforward_

sectionSmul, hence along diagrams and classes Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.pushforward_sectionAdd. *Step 3.* With the bijections of Theorem 10.5.4, $[D]_*$ is a linear isomorphism with inverse $[D]^*$ Chapter10/SectionFunctorLinear2021 · SBLocalization.GenSBMap.pushforwardEquiv_symm_apply, and the same on Γ^+.

X.32 Propositions 10.4.6, 10.4.7 and 10.5.6 (new in 2021)

The reading of Definition 8.4.2 Chapter10/LocalizationLRBundleBase · EpStrongBundle.IsLRLocal. The text's B is a strong bundle isomorphism of $W|U(x)$ with $|B(w)| = |w|$, and the conclusion is $B(w) = g * w$. We read it for maps defined on $W|U(x)$ only, with no condition off $W|U(x)$ Chapter10/LocalizationLRBundleBase · StrongBundle.IsLocalIsoOn. Read for strong bundle maps defined on all of W, the conjugate $H^{-1} \circ B \circ H$ of the proof of Proposition 10.4.6 would not be in its scope. The local reading implies the global one Chapter10/LocalizationLRBundleBase · EpStrongBundle.IsLRLocal.isLR; we do not know whether the converse holds. The condition on one $U(x)$ is recorded separately Chapter10/StrongBundleLocalRegularity · EpStrongBundle.HasLRLocal Chapter10/StrongBundleLocalRegularity · EpStrongBundle.isLRLocal_iff_hasLRLocal.

Local isomorphisms between two bundles Chapter10/StrongBundleLocalIso · StrongBundle.IsLocalIso₂. The proofs compose local isomorphisms $Y|U \to Y'|U'$ between different bundles and invert them: inverse, composite and restriction Chapter10/StrongBundleLocalIso · StrongBundle.IsLocalIso₂.symm Chapter10/StrongBundleLocalIso · StrongBundle.IsLocalIso₂.comp Chapter10/StrongBundleLocalIso · StrongBundle.IsLocalIso₂.restrict, with the inverse linear on the fibres Chapter10/StrongBundleLocalIso · StrongBundle.IsLocalIsoOn.exists_isLocalIso₂; a strong bundle equivalence restricts to one Chapter10/StrongBundleLocalIso · StrongBundle.IsStrongBundleIsoOn.exists_isLocalIso₂. For an sc-smooth family $m(y) : y \to \tau(y)$ over U with τ an sc-diffeomorphism, $w \mapsto \mu(m(Pw), w)$ is one covering τ, with inverse $w \mapsto \mu(m(\tau^{-1}(Pw))^{-1}, w)$ Chapter10/StrongBundleLocalIso · EpStrongBundle.isLocalIso₂_lrAct Chapter10/StrongBundleLocalIso · EpStrongBundle.lrAct_comp; the natural lift of g is such a map Chapter10/StrongBundleLocalIso · EpStrongBundle.natLift_eq_lrAct.

Proposition 10.4.6 (V10.72) Chapter10/StrongBundleLocalRegularity · EpStrongBundle.proposition_10_4_6. *Step 1, the transport.* For $\psi : F(x) \to y$ and the transport chart $Q = \hat{\psi} \circ F$ of §X.5, $v \mapsto \nu(\psi_{P(v)}, \Phi(v))$, the text's $H_{\hat{\psi}} \circ \Phi$, is a local strong bundle isomorphism $V|S \to W|Q(S)$ covering Q Chapter10/StrongBundleLocalRegularity · EpStrongBundle.IsStrongBundleEquivalence.exists_transport, equivariant along $g \mapsto \psi \circ F(g) \circ \psi^{-1}$ Chapter10/StrongBundleLocalRegularity · EpStrongBundle.IsStrongBundleEquivalence.transport_Γ and preserving and reflecting orbit equality Chapter10/StrongBundleLocalRegularity · EpStrongBundle.IsStrongBundleEquivalence.π_transport_iff. *Step 2, the natural representations.* $U(y) = Q(U(x))$ with the natural action, as in Proposition 10.1.9 Chapter10/EquivalenceLR · EpFunctor.TransportChart.exists_naturalRep_push Chapter10/EquivalenceLR · EpFunctor.TransportChart.exists_naturalRep_pull. *Step 3, the conjugation.* For any such H along a surjection $G_x \to G_y$, $B \mapsto H^{-1} \circ B \circ H$ carries the orbit-preserving strong bundle isomorphisms of $W|U(y)$ to those of $V|U(x)$ and the natural lifts to the natural lifts, both ways Chapter10/StrongBundleLocalRegularity · EpStrongBundle.hasLRLocal_iff_of_equivariant. "$\Phi : W \to W$" in the statement is read $\Phi : V \to W$.

Proposition 10.4.7 (V10.73) Chapter10/StrongBundleLocalFamilies · EpStrongBundle.proposition_10_4_7 Chapter10/StrongBundleLocalFamilies · EpStrongBundle.proposition_10_4_7', with the property for the target only and $U(x)$ inside any given neighbourhood: a stronger form. *Step 1.* $|\Phi| = |\Psi|$ gives $\psi_0 : \psi(x) \to \phi(x)$; $Q = \hat{\psi}_0 \circ \psi$ and the transport $H(v) = \nu(\psi_{P(v)}, \Psi(v))$ of Proposition 10.4.6, Step 1. *Step 2.* $B = H \circ \Phi^{-1}$ is a strong bundle isomorphism of $W|U(\phi(x))$ covering $\gamma = Q \circ \phi^{-1}$, with $\gamma(U) = U$ by Lemma 7.4.20 and

$|B(w)| = |w|$ by $|\Phi| = |\Psi|$. *Step 3.* Definition 8.4.2 gives $B = k*\cdot$, so $\nu(\psi_z, \Psi(v)) = \nu(\Gamma(k, \phi(z)), \Phi(v))$ and $\alpha_z = \psi_z^{-1} \circ \Gamma(k, \phi(z))$. On the bases this is `Chapter10/LocalizationLRBundleBase · SEP.exists_family_of_isLRLocal`; the proof shows that orbit equality on the total spaces determines it on the bases `Chapter10/StrongBundleLocalFamilies · EpStrongBundle.π_base_eq_of_orbitMap_eq`. *Witness*, $\Phi = \Psi = 1$ on $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ `Chapter10/StrongBundleLocalFamilies · GapP2_GBundleFamilyExamples.proposition_10_4_7_hypotheses`.

Proposition 10.5.6 (V10.61) `Chapter10/StrongBundleLocalRegularity · SBLocalization.proposition_10_5_6`. A representative with Φ, Ψ strong bundle equivalences carries the property from W to W'' to W', by Proposition 10.4.6 twice (§X.23) (G.122, G.126).

X.33 (10.24) and the local expressions q, Q (2021 pp. 504–506)

The square (10.24) `Chapter10/StrongBundleDiagram · SBLocalization.SDiagram.square_10_24` `Chapter10/StrongBundleDiagram · SBLocalization.GenSBMap.square_10_24`. The text says the square with vertical arrows P, P' "does not make too much sense", and that with the vertical arrows replaced by "a suitable generalized map version" it commutes, leaving this to the reader. We take $i(P) = [W \xleftarrow{1} W \xrightarrow{P} X]$, with P a morphism of $\mathcal{EP}$ `Chapter10/StrongBundleDiagram · SEP.projEP`. *Step.* $P' \circ \Psi = \psi \circ P''$ on the representative, and i is a functor (Theorem 10.3.14), so $[D]_{\text{total}} \circ i(P') = i(P) \circ [d]$ in $\mathcal{EP}(E^{-1})$.

The germ q `Chapter10/StrongBundleDiagram · EPLocalization.Diagram.exists_germ_q`. For the data (x'', a, b) of γ (§X.21), q is the composite $U(x) \leftarrow U(\phi(x'')) \leftarrow U(x'') \to U(\psi(x'')) \to U(x')$ of $t \circ (s|U(a))^{-1}$, of ϕ^{-1}, of ψ and of $t \circ (s|U(b))^{-1}$. *Step 1.* A functor intertwines natural representations at u and $F(u)$ along $F : G_u \to G_{F(u)}$: both sides are sections of s agreeing at u `Chapter10/StrongBundleDiagram · EpFunctor.mor_Γ` `Chapter10/StrongBundleDiagram · EpFunctor.obj_Φ`. *Step 2.* The conjugations by a and b are equivariant for the conjugation homomorphisms `Chapter10/StrongBundleDiagram · EPLocalization.Diagram.isotropyHom_conj`. *Step 3.* Composing, $q(x) = x'$ and $q(g * z) = \gamma(g) * q(z)$ on natural representations $U(x)$, $U(x')$, the actions in which the equivariance is read.

The lift Q **and (10.25)** `Chapter10/StrongBundleDiagramLift · SBLocalization.SDiagram.exists_lift_Q`. "A strong bundle map $W|O(X, x) \to W'|O(X', x')$" is read as a local strong bundle map on $W|U(x)$, with no condition off it `Chapter10/StrongBundleDiagramLift · StrongBundle.IsLocalMap₂` `Chapter10/StrongBundleDiagramLift · StrongBundle.IsLocalMap₂.comp` `Chapter10/StrongBundleDiagramLift · StrongBundle.IsLocalIso₂.isLocalMap₂`. *Step.* $Q = H_b \circ \Psi \circ H_a^{-1}$ with the transports $H_a(v) = \mu(a_{P(v)}, \Phi(v))$ and $H_b(u) = \mu'(b_{P'(u)}, u)$ of §X.32, Step 1. It covers $q = Q_b \circ \psi \circ Q_a^{-1}$, so $P' \circ Q = q \circ P$, (10.25), and $Q(g * w) = \gamma(g) * Q(w)$ for the natural lifts (G.124). As the text says, q and Q depend on the representative.

X.34 Theorem 10.3.10 (2017: 10.4) for strong bundles: the converse

Isomorphisms of $\mathcal{SEP}(F^{-1})$ **are generalized strong bundle isomorphisms** (V10.71) `Chapter10/StrongBundleIsoConverse · SBLocalization.SEPLoc.equivalences_Ψ_of_isIso` `Chapter10/StrongBundleIsoConverse · SBLocalization.SEPLoc.isIso_homMk_iff` `Chapter10/StrongBundleIsoConverse · SBLocalization.SEPLoc.isIso_iff_isGeneralizedIso`. Remark 10.5.3 says "a similar result can be proved" and builds it into Definition 10.5.2. Let $[E] = [W' \xleftarrow{\Phi'} V'' \xrightarrow{\Psi'} W]$ be the inverse of $[D]$. *Step 1, the base.* ψ is an equivalence (§X.22). *Step 2, fibrewise injectivity from* $[D] \circ [E] = 1$. A strong bundle equivalence H into the middle $W'' \times_{W'} V''$ makes $\Psi' \circ \Pi_{V''} \circ H$ naturally equivalent

to a strong bundle equivalence Chapter10/StrongBundleIsoConverse · SBLocalization.SDiagram.exists_natEquiv_of_equiv. Since $\Pi_{V''}(m)$ is determined by $\Psi(\Pi_{W''}(m))$, Ψ is injective on the fibres over the points $\Pi_{W''}(H(c))$, and on all fibres by transport along morphisms Chapter10/StrongBundleIsoConverse · SEP.FibInj Chapter10/StrongBundleIsoConverse · SBLocalization.SDiagram.fibInj_Ψ_of_equiv. *Step 3, a natural equivalence from* $[E]\circ[D] = 1$. Likewise $\Psi \circ G$ is naturally equivalent to a strong bundle equivalence K for $G = \Pi_{W''} \circ H$. *Step 4, local inverses.* For x choose $\gamma : g(c) \to x$. The transport along γ gives $G' = H_{\hat{\gamma}} \circ G$ (§X.32), and $\Psi \circ G'$ is the transport of K, a local strong bundle isomorphism near c. So $G' \circ (\Psi \circ G')^{-1}$ is a local right inverse of Ψ near $\psi(x)$, at levels $[0]$ and $[1]$. With the fibrewise injectivity and the local injectivity of ψ, Ψ is a strong bundle isomorphism near x Chapter10/StrongBundleIsoConverse · SEP.isStrongBundleIsoOn_of_rightInv Chapter10/StrongBundleIsoConverse · SEP.exists_isStrongBundleIsoOn, and on morphisms through the pull-back projections Chapter10/StrongBundleIsoConverse · SEP.isLocalStrongBundleIso_mor_of_obj. *Step 5, the total space and* $|W[1]|$. The total functor is full Chapter10/StrongBundleIsoConverse · SEP.full_total and an equivalence by Lemma 10.1.2 Chapter10/StrongBundleIsoConverse · SEP.isEquivalence_total. $|\Psi|$ on $|W''[1]|$ is a continuous bijection with the continuous inverse $|G| \circ |K|^{-1}$ Chapter10/StrongBundleIsoConverse · SEP.NatEquiv.oneOrbitMap_eq Chapter10/StrongBundleIsoConverse · SEP.isHomeomorph_oneOrbitMap_of. Together, the saturation step Chapter10/StrongBundleIsoConverse · SEP.equivalences_of_natEquiv_comp: a strong bundle functor covering an equivalence, injective on fibres, with $\Psi \circ G \simeq K$ for a strong bundle equivalence K, is a strong bundle equivalence. This replaces the claims (C1), (C2) of Appendix §10.6.2, which the text does not restate for strong bundles (G.123).

Part XI. Geometry up to equivalences (HWZ Chapter 11, §§11.1–11.6 (2017 §§11.1–11.6))

Chapter 11 closes the ep-groupoid half of Part II; its numbers and pages are those of the book (Springer 2021, pp. 521–566), with the 2017 text's in parentheses (2017 pp. 375–416). It asks which of the notions built on ep-groupoids in Chapters 8 and 9 survive the passage to $\mathcal{EP}(E^{-1})$ of Chapter 10, where generalized isomorphisms are the isomorphisms. §11.1 extends the tangent functor (Theorem 11.1.1 (2017: 11.1)), shows that the degeneracy index is invariant (Proposition 11.1.2 (2017: 11.1)) and treats faces (Definition 11.1.3, Example 11.1.4, Remark 11.1.5, Proposition 11.1.6, Remark 11.1.7, Proposition 11.1.8 (2017: Definition 11.1, Example 11.1, Remark 11.1, Proposition 11.2, Remark 11.2, Proposition 11.3)). §11.2 does the same for sc-differential forms (Theorem 11.2.1 to Proposition 11.2.9 (2017: Theorem 11.2 to Proposition 11.5)). §11.3 pushes forward and pulls back functors $\Theta : X \to \mathbb{Q}^+$ (Definition 11.3.1, Lemma 11.3.2, Theorem 11.3.3 (2017: Definition 11.4, Lemma 11.5, Theorem 11.3)) and continues with the bundles of tangent sets and orientations (Lemma 11.3.4 to Theorem 11.3.9 (2017: Lemma 11.6 to Theorem 11.7)). §11.4 transforms the canonical measures (Theorems 11.4.1, 11.4.2, 11.4.4, 11.4.5, Remark 11.4.3 (2017: Theorems 11.8, 11.9, Remark 11.3)). §11.5 carries auxiliary norms and sc$^+$-multisections along generalized strong bundle isomorphisms (Proposition 11.5.1, Lemma 11.5.2, Proposition 11.5.3 (2017: Proposition 11.6, Lemma 11.7, Proposition 11.7)). §11.6 studies proper covering functors up to equivalence (Lemmas 11.6.1–11.6.3, Definitions 11.6.4, 11.6.5, 11.6.7, Theorem 11.6.6, Proposition 11.6.8, Remark 11.6.9 (2017: Lemmas 11.8, 11.9, Definitions 11.6–11.8, Theorem 11.10, Proposition 11.8, Remark 11.4); Lemma 11.6.1 is new in the 2021 edition).

What is proved, and what is not. All eight gaps of Chapter 11 are closed (G.128–G.135). None is set aside as having nothing to formalize. *§11.1.* Theorem 11.1.1, Proposition 11.1.2, Definition 11.1.3, Example 11.1.4 (for $[0,\infty)^2 \oplus W$ with any W), Remark 11.1.5 (with the result it quotes), Remark 11.1.7 and Proposition 11.1.8 are proved (§§XI.1–XI.3). Proposition 11.1.6 is proved except for its intersection clause, which does not hold as we read it and holds in a union form (§§XI.3, XI.19, XI.20; G.128): the faces of a face structured tame ep-groupoid are tame ep-groupoids, and a generalized isomorphism induces generalized isomorphisms $[d]_F$ of faces. *§11.2.* All of it is proved: Theorem 11.2.1, Lemmas 11.2.2, 11.2.4–11.2.6, Definitions 11.2.3, 11.2.7, (11.5) and Propositions 11.2.8, 11.2.9 (§XI.4; G.129). *§11.3.* Proved: Definition 11.3.1, Lemma 11.3.2 in a stronger form, the push-forward and pull-back along generalized maps, and all six parts of Theorem 11.3.3, (6) for every dimensional decomposition (§§XI.5–XI.7). Also proved: Lemma 11.3.4 with (11.8) and (11.9), Theorem 11.3.5, Definition 11.3.6 with (11.10), and Theorems 11.3.7 and 11.3.8, whose proofs the text leaves to the reader (§XI.8). They were first proved under

Chapter 9's hypothesis that the tangent sums of Theorem 9.2.1 are independent, which Chapter 9 now proves for every Θ. Theorem 11.3.9 is proved for the intrinsic boundary orientation of Definition 9.3.10, independently of the representative and with no hypothesis (§XI.8; G.130). *§11.4.* Formalized (§XI.9; G.131). Theorems 11.4.1 and 11.4.2 and Remark 11.4.3 are proved, with no added hypothesis, for the canonical measures of Chapter 9 built from its concrete sheet integrals. Theorems 11.4.4 and 11.4.5 are proved with the text's hypotheses, for every $n \geq 1$, for the boundary measures of Theorems 9.5.15 and 9.5.16. *§11.5.* Proposition 11.5.1 is proved with conditions (1) and (2) of Definition 8.3.5 (§XI.10; G.132), and Lemma 11.5.2 and Proposition 11.5.3 for sc$^+$-multisections with their local section structures (§XI.11; G.133). *§11.6.* Proved: Lemmas 11.6.1–11.6.3, Definitions 11.6.4, 11.6.5, 11.6.7, all claims of Theorem 11.6.6 and Proposition 11.6.8 (§§XI.12–XI.15). Also proved: the localization $\mathcal{PC}(E^{-1})$ with I (§XI.21; G.134), and the category of proper strong bundle coverings of Remark 11.6.9 with its localization (§XI.22; G.135), both sketched in the text with the details left to the reader. *The narrower row.* V11.6, Proposition 11.1.6: the clause that two faces meet in a face of each does not hold as we read it. The Lean counterexample is `FaceMeetExample.not_facesMeetInFaces`, two faces of $([0,\infty)^3 \oplus W) \setminus K$ meeting in two pieces of an edge. What is proved is that they meet in a union of faces of each (§XI.20). V11.29, Theorems 11.4.4 and 11.4.5, is restated: the class hypothesis `hcov` is derived in Chapter 9 and $n = 1$ is covered (§XI.9).

Six structural choices. (1) *Faces by their object sets.* A face X_θ of X is the saturated set $\pi^{-1}(\theta)$ with the morphisms whose source lies in it; the faces of $|X|$ are Chapter 2's faces for the index $d_{|X|}$ on the open set $|X|$, so Chapter 2's lemmas on faces apply to them (§XI.2). (2) *Total push-forwards.* $F_*\Theta$ is defined for every sc-smooth functor and every map on objects, with one choice of x and of a morphism $F(x) \to y$, and 0 where there is none; the statements say when the choice does not matter (§XI.5). (3) *Classes through representatives.* f_* and f^* on $\mathcal{EP}(E^{-1})$ are computed on a chosen representative with an equivalence as left leg (`rep`), and shown independent of it (§XI.6). (4) *Multisections as functors.* A multisection functor on a strong bundle is a map on the total object space constant along the morphisms of the total ep-groupoid; the local section structures of Definition 13.2.1 are a predicate on top of it (§XI.11). (5) *Functors of proper coverings as pairs.* A functor $F \to F'$ is a structure with the two functors and the two commutation equations, on objects and on morphisms, and $\mathcal{PC}$ is a category with composition associative on the nose (§§XI.13, XI.15). (6) *Sections and boundary operations.* Section functors of $\widehat{Gr}$ are functions on all objects, 0 off X_∞; Chapter 9's boundary lift depends on a boundary operation given as data, so Theorem 11.3.9 is first proved for a representative under a hypothesis relating the boundary operations on the two sides, and then, for the intrinsic boundary orientation of Definition 9.3.10, with no hypothesis (§XI.8).

Map.

- `Chapter11/EpEquivalenceGeometry`: Theorem 11.1.1 (diagrams, refinements, the comparison functor `wfpLift`, the functor T on $\mathcal{EP}(E^{-1})$) and Proposition 11.1.2.
- `Chapter11/EpEquivalenceGeometryFaces`: Definition 11.1.3, Example 11.1.4, Remark 11.1.5, Propositions 11.1.6, 11.1.8, Remark 11.1.7, faces of open sets for Chapter 2's index.
- `Chapter11/BranchedEquivalence`: Definition 11.3.1, Lemma 11.3.2, local branching structures moved by sc-diffeomorphisms, Theorem 11.3.3 for an equivalence.
- `Chapter11/BranchedEquivalenceLocalization`: push-forward and pull-back along diagrams and classes, base change in the weak fibred product, Theorem 11.3.3, the witnesses on $\mathbb{Z}/2 \ltimes \mathbb{R}$.
- `Chapter11/BranchedEquivalenceTangentSection`: section functors of $\widehat{Gr}$, (11.8), (11.9), Lemma 11.3.4, the witness of Example 9.3.5.
- `Chapter11/BranchedEquivalenceTangent`: Chapter 9's tangents, orientations and boundaries under sc-diffeomorphisms between two M-polyfolds, the independence of the tangent sums carried along, Theorems 11.3.7 and 11.3.8.

- `Chapter11/BranchedEquivalenceTangentOrientation`: Theorem 11.3.5, Definition 11.3.6, (11.10), the boundary-transfer hypothesis, Theorem 11.3.9 and its witnesses.
- `Chapter11/StrongBundleUpToEquivalence`: $|D|$ on $|W[1]|$, Proposition 11.5.1, multisection functors, Lemma 11.5.2, Proposition 11.5.3.
- `Chapter11/CoveringEquivalence`: Lemmas 11.6.1–11.6.3, Definitions 11.6.4 and 11.6.5, the identity coverings of the pair groupoid of $\mathbb{Z}$ and the point.
- `Chapter11/CoveringEquivalenceTheorem`: the functor H and Theorem 11.6.6.
- `Chapter11/CoveringEquivalenceDiagram`: the category $\mathcal{PC}$, Definition 11.6.7, Proposition 11.6.8.
- `Chapter11/FaceEpGroupoids`: Proposition 11.1.6, faces as tame ep-groupoids.
- `Chapter11/FaceGeneralizedIso`: the generalized isomorphisms $[d]_F$ of (11.1), Proposition 11.1.8 for them.
- `Chapter11/FaceIntersectionCounterexample`: the intersection clause of Proposition 11.1.6 and its counterexample.
- `Chapter11/FaceIntersectionUnion`: the intersection as a union of faces.
- `Chapter11/FormDescent`: forms along surjective local sc-diffeomorphisms, (11.4).
- `Chapter11/EquivalenceForms`: Lemma 11.2.2, Definition 11.2.3, Lemma 11.2.4 (3) on one level.
- `Chapter11/EquivalenceFormsInfty`: Definition 11.2.3 on $\Omega^*_{ep,\infty}$, (11.5), Lemmas 11.2.4–11.2.6.
- `Chapter11/GeneralizedForms`: Definition 11.2.7, Theorem 11.2.1 (1)–(3).
- `Chapter11/OrbitForms`: Propositions 11.2.8, 11.2.9, Theorem 11.2.1 (4).
- `Chapter11/BoundaryOrientationEquiv`: Theorem 11.3.9 for the intrinsic boundary orientation.
- `Chapter11/FormEvalPullback`, `FormEvalPullbackEquivalence`: evaluations of forms compatible with the pull-back.
- `Chapter11/GraphChartsDiffeo`, `OrientedStructureDiffeo`, `OrientedStructurePairs`, `MultiChartSheetsEquivalence`: structures and sheet integrals moved between two ep-groupoids.
- `Chapter11/MeasureEquivalence`, `MeasureGeneralizedIso`: Theorem 11.4.1, Remark 11.4.3.
- `Chapter11/EffectiveMeasureEquivalence`, `EffectiveMeasureGeneralizedIso`: Theorem 11.4.2.
- `Chapter11/BoundarySheetsDiffeo`, `BoundaryMeasureEquivalence`, `BoundaryMeasureGeneralizedIso`: Theorems 11.4.4 and 11.4.5.
- `Chapter11/AuxiliaryNormTransport2021`: conditions (1) and (2) in Proposition 11.5.1.
- `Chapter11/MultisectionLocalStructure2021`: local section structures, Lemma 11.5.2 and Proposition 11.5.3 for sc$^+$-multisections.
- `Chapter11/ProperCoveringLocalization`: $\mathcal{PC}(E^{-1})$ and I.
- `Chapter11/ProperStrongBundleCoverings`, `StrongBundleCoveringPullback`, `StrongBundleCoveringFibredProduct`, `StrongBundleCoveringLocalization`: the category of Remark 11.6.9, Theorem 11.6.6 for it, its localization.
- `Chapter11/BoundaryMeasureTextHypotheses`: Theorems 11.4.4 and 11.4.5 with the text's hypotheses, $n \geq 2$; `Chapter11/BoundaryMeasureEquivalenceDimOne`, `BoundaryMeasureGeneralizedIsoDimOne`: the same for $n = 1$ (§XI.9).

XI.1 Theorem 11.1.1 (2017: 11.1): the tangent functor on $\mathcal{EP}(E^{-1})$

The tangent of a diagram (V11.1) `Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.tangent`. For $d = [X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ the diagram $Td = [TX \xleftarrow{TF} TA \xrightarrow{T\Phi} TY]$; TF is an equivalence by Theorem 10.1.7. Chapter 8's functoriality of T is used in the form $T(G \circ F) = TG \circ TF$ `Chapter11/EpEquivalenceGeometry · EpFunctor.tangent_comp` and $T1 = 1$ `Chapter11/EpEquivalenceGeometry · EpFunctor.tangent_`

id. *Step 1, refinements.* A refinement $H : d' \to d$ with natural transformations $\sigma : F' \to F \circ H$, $\tau : \Phi' \to \Phi \circ H$ gives TH with $T\sigma$, $T\tau$ (Proposition 8.1.14), so $TH : Td' \to Td$ is a refinement Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.IsRefinement.tangent, and equivalent diagrams have equivalent tangents Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.Equiv.tangent; this is the argument of the text. *Step 2, identities.* T of the identity diagram refines the identity diagram of TX Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.isRefinement_tangent_id. *Step 3, composition, left to the reader.* For functors $P : C \to X$, $Q : C \to Z$ and a natural transformation $\Phi \circ P \to G \circ Q$, $c \mapsto (P(c), \tau(c), Q(c))$ is a functor into the weak fibred product Chapter11/EpEquivalenceGeometry · EPLocalization.wfpLift with the right projections Chapter11/EpEquivalenceGeometry · EPLocalization.projLeft_comp_wfpLift Chapter11/EpEquivalenceGeometry · EPLocalization.projRight_comp_wfpLift. Applied to $T\pi_A$, $T\pi_B$ and T of the transformation of $A \times_Y B$, it gives $T(A \times_Y B) \to TA \times_{TY} TB$; its composite with the projection to TA is the equivalence $T\pi_A$, so it is an equivalence by two out of three, and it refines $Td \circ Te$ by $T(d \circ e)$ Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.isRefinement_tangent_comp. *Step 4, the functor.* `tangentFunctor` on classes Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.tangentFunctor Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.tangentFunctor_obj Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.tangentFunctor_map_homMk, compatible with the localization functor Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.tangentFunctor_map_localizationFunctor; it maps generalized isomorphisms to generalized isomorphisms Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.isGeneralizedIso_tangent. **Theorem 11.1.1** Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.theorem_11_1_1 collects these.

XI.2 Proposition 11.1.2 (2017: 11.1), Definition 11.1.3 (2017: 11.1) and Example 11.1.4 (2017: 11.1)

Proposition 11.1.2 (2017: 11.1) (V11.2) Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.proposition_11_1_2. *Step 1, equivalences.* An equivalence is a local sc-diffeomorphism on objects, and a local sc-diffeomorphism preserves the degeneracy index, so $d_Y(F(x)) = d_X(x)$, and d is constant on orbits Chapter11/EpEquivalenceGeometryFaces · EpFunctor.IsEquivalence.degIndexOrbit_orbitMap Chapter7/EpGroupoid · EpGroupoid.degIndexOrbit. *Step 2, diagrams.* For a diagram with both legs equivalences, $|d| = |\Phi| \circ |F|^{-1}$ preserves $d_{|\cdot|}$ Chapter11/EpEquivalenceGeometry · EPLocalization.Diagram.degIndexOrbit_orbitMap. *Step 3, classes.* A generalized isomorphism has such a representative (Theorem 10.3.10; the proof says "Proposition 10.3.10", a possible misprint) Chapter11/EpEquivalenceGeometry · EPLocalization.EPLoc.degIndexOrbit_orbitMap.

Definition 11.1.3 (2017: 11.1) (V11.3) Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.IsOrbitFace Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.faceObj Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.faceMor Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.IsEpFace Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.IsEpFaceStructured Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.IsWeaklyFaceStructured. The object set of X_θ is saturated Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.isSaturated_faceObj and its morphisms are those between its objects Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.faceMor_eq. A face of $|X|$ is exactly a face, in Chapter 2's sense, of the open set $|X|$ for the index $d_{|X|}$ Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.isOrbitFace_iff_isFaceIn, so the faces through z Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.orbitFacesThrough are Chapter 2's Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.orbitFacesThrough_eq and face-structuredness is Chapter 2's condition for $|X|$ Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.isEpFaceStructured_iff. The number of faces through z is counted in $\mathbb{N} \cup \{\infty\}$, so that infinitely many faces do not count as 0.

Example 11.1.4 (2017: 11.1) (V11.4) Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.example_11_1_4. *The trivial groupoid.* With only identities, π is a homeomorphism Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.isHomeomorph_π_trivial and the face structure of $C = [0,\infty)^2 \oplus W$ is that of the quadrant Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.isFaceStructured_quadrant. *The swap.* $\mathbb{Z}/2 \ltimes C$ with -1 exchanging the two coordinates Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.swapEpGroupoid; its object M-polyfold is C, face structured. In $|X|$ the set $\{d_{|X|} = 1\}$ is the image of one connected stratum, since the swap identifies the two open half-axes Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.setOf_degIndexOrbit_eq_one_swap; $|X|$ has exactly one face Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.isOrbitFace_swap_iff, and $|0|$, of index 2, lies in one face only Chapter11/EpEquivalenceGeometryFaces · EpGroupoid.Examples.not_isEpFaceStructured_swap. The text gives no reason; this is it.

XI.3 Remark 11.1.5 (2017: 11.1), Proposition 11.1.6 (2017: 11.2), Remark 11.1.7 (2017: 11.2), Proposition 11.1.8 (2017: 11.3)

The quoted result (V11.5). The text quotes [56, Proposition 3.15] for the fact that faces correspond under a homeomorphism preserving the index. For the faces of Chapter 2 on open sets: such a homeomorphism maps faces to faces Chapter11/EpEquivalenceGeometryFaces · TameBoundary.isFaceIn_univ_image, every face is the image of one Chapter11/EpEquivalenceGeometryFaces · TameBoundary.exists_isFaceIn_univ_preimage, the orbit map of an equivalence is such a homeomorphism Chapter11/EpEquivalenceGeometryFaces · TameBoundary.isDHomeoOn_univ, and face-structuredness transfers Chapter11/EpEquivalenceGeometryFaces · TameBoundary.isFaceStructuredOpen_univ_iff. *Equivalences.* $|\Phi|$ induces the bijection of faces Chapter11/EpEquivalenceGeometryFaces · EpFunctor.IsEquivalence.bijOn_faces, with $\Phi_*(X_\theta) = \text{saturation}(\Phi(X_\theta)) = Y_{|\Phi|(\theta)}$ Chapter11/EpEquivalenceGeometryFaces · EpFunctor.saturation_image_faceObj and $\Phi^{-1}(Y_{\theta'}) = X_{|\Phi|^{-1}(\theta')}$ Chapter11/EpEquivalenceGeometryFaces · EpFunctor.preimage_faceObj. *Generalized isomorphisms.* $||[d]||$ is a homeomorphism Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.isHomeomorph_orbitMap; `faceMap` and `faceComap` are the image and preimage of faces Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceMap Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceComap, computed on a representative as $\Psi_* \circ \Phi^*$ and $\Phi_* \circ \Psi^*$ Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceMap_homMk Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceComap_homMk, and mutually inverse bijections Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.bijOn_faceMap. **Remark 11.1.5** Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.remark_11_1_5.

Proposition 11.1.6 (2017: 11.2), the part on $[d]_*$ (V11.6) Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.proposition_11_1_6. $[d]_*$ and $[d]^*$ map faces to faces Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.isOrbitFace_faceMap Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.isOrbitFace_faceComap and are inverse Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceComap_faceMap Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.faceMap_faceComap; $||[d]||$ restricts to homeomorphisms $\theta \to [d]_*\theta$. No face structure is used. The faces as tame ep-groupoids and the generalized isomorphisms $[d]_F$ are §XI.19, and the intersection clause §XI.20 (G.128, closed).

Remark 11.1.7 (2017: 11.2) (V11.7) Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.remark_11_1_7. Chapter 10 transfers tameness from Y to X along an equivalence Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.isTame_of_isTame. The converse goes through $X \times_Y Y$ for F and 1_Y: its projection to X is an equivalence, so $X \times_Y Y$ is tame by Chapter 10's direction, and its projection to Y is a surjective local sc-diffeomorphism on objects, by the essential surjectivity

of F, which carries tameness down Chapter11/EpEquivalenceGeometryFaces · EpFunctor.IsEquivalence.isTame_of_isTame_left Chapter11/EpEquivalenceGeometryFaces · EpFunctor.IsEquivalence.isTame_iff. With §XI.2 and the transfer of face-structuredness above, a generalized isomorphism preserves "tame and face structured" both ways.

Proposition 11.1.8 (2017: 11.3) (V11.8) Chapter11/EpEquivalenceGeometryFaces · EPLocalization.EPLoc.proposition_11_1_8. f_* and f^* depend only on $|f|$, by their definition as image and preimage; on a face, $|f|$ and $|g|$ agree. The text's maps $|F| \to |f_*F|$ come from the generalized isomorphisms of (11.1). They are the restrictions of $|f|$, and the second claim, for the generalized isomorphisms $[f]_F$ themselves, is §XI.19 Chapter11/FaceGeneralizedIso · EPLocalization.EPLoc.proposition_11_1_8_faceIso.

XI.4 §11.2: sc-differential forms (Theorem 11.2.1 to Proposition 11.2.9 (2017: Theorem 11.2 to Proposition 11.5))

§11.2 is formalized (G.129), for Chapter 8's forms Ω^k_{ep} of an ep-groupoid and their direct limit $\Omega^*_{ep,\infty}$ over the index lifts X^i. The pattern is that of §11.3: a push-forward along equivalences by a choice of x and $\psi : F(x) \to y$ that does not matter, composition rules, invariance under refinements, and generalized maps through $F_* \circ G^*$. The formal route differs in one point: F_* is defined as the inverse of F^*, and the text's defining formula becomes a theorem.

Descent along a surjective local sc-diffeomorphism. For M-polyfolds and a surjective local sc-diffeomorphism $q : X \to Y$, the tangent spaces of Y are images Chapter11/FormDescent · MPolyfold.IsLocalScDiffeo.exists_tangentFiber_lift, so q^* is injective on families of alternating forms Chapter11/FormDescent · MPolyfold.formPullback_injective_of_isLocalScDiffeo. A family τ on Y with $q^*\tau$ an sc-form is an sc-form: on the image V of a sheet of q, with inverse g, $\tau|V = g^*(q^*\tau)$, the text's (11.4), and sc-smoothness of a form is local Chapter11/FormDescent · MPolyfold.isScForm_of_formPullback. A family η on X taking the same values on vectors with the same image under Tq descends, $\tau_y = \eta_x \circ (Tq(x))^{-1}$, with $q^*\tau = \eta$ Chapter11/FormDescent · MPolyfold.formDescend Chapter11/FormDescent · MPolyfold.formPullback_formDescend.

Lemma 11.2.2 and Definition 11.2.3 (2017: Lemma 11.1, Definition 11.2) (V11.10). *Step 1, the full case.* For F full and a local sc-diffeomorphism on objects, a form of the ep-groupoid X takes the same values on vectors over x, x' with $F(x) = F(x')$ and the same images under TF, since a morphism $x \to x'$ exists and forms of ep-groupoids are invariant under morphisms; if F is also surjective on objects, every form on X descends to a form τ on Y with $F^*\tau$ the given one Chapter11/EquivalenceForms · EpFunctor.exists_epForms_formPullback_eq_of_surjective. *Step 2, equivalences, through the weak fibred product.* For an equivalence F, $A = X \times_Y Y$ of F and 1_Y has the objects (x, ψ, y), $\psi : F(x) \to y$, of the text's definition; $\pi_Y : A \to Y$ is an equivalence surjective on objects and $F \circ \pi_X \simeq \pi_Y$. Pull-backs compose Chapter11/EquivalenceForms · EpFunctor.formPullback_comp_functor, and F^* is injective Chapter11/EquivalenceForms · EpFunctor.IsEquivalence.injOn_formPullback and onto, $F_*\omega$ being the descent of $\pi_X^*\omega$ along π_Y Chapter11/EquivalenceForms · EpFunctor.IsEquivalence.exists_formPullback_eq. So $F^* : \Omega^k_{ep}(Y) \to \Omega^k_{ep}(X)$ is bijective Chapter11/EquivalenceForms · EpFunctor.IsEquivalence.bijective_formPullback, and F_* is its inverse Chapter11/EquivalenceForms · EpFunctor.IsEquivalence.formPush. *Step 3, the text's formula.* $(F_*\omega)_y(T\psi(TF(x)l_1), \dots, T\psi(TF(x)l_k)) = \omega_x(l_1, \dots, l_k)$ for every x, $\psi : F(x) \to y$ and $l_j \in T_xX$ Chapter11/EquivalenceForms · EpFunctor.IsEquivalence.definition_11_2_3, and every $h_1, \dots, h_k \in T_yY$ is of this form Chapter11/EquivalenceForms · EpFunctor.IsEquivalence.exists_lift_tangent, so the formula determines $F_*\omega$. *Step 4, Lemma 11.2.2* Chapter11/EquivalenceForms · EpFunctor.IsEquivalence.lemma_11_2_2: both sides equal $(F_*\omega)_y(h_1, \dots, h_k)$. Before Definition 11.2.3 the text's $\Omega(X)$ for the target is read $\Omega(Y)$.

Lemmas 11.2.4–11.2.6 (2017: 11.2–11.4) and (11.5) (V11.11). *Step 1, levels.* The index raises $F^i : X^i \to Y^i$ of an equivalence are equivalences (Chapter 10) Chapter11/EquivalenceFormsInfty · EP.isEquivalence_liftFunctor, so F_* exists on each level Chapter11/EquivalenceFormsInfty · EP.epFormsPush and commutes with the maps of the directed system; this gives F_* on $\Omega^k_{ep,\infty}$ Chapter11/EquivalenceFormsInfty · EP.epFormsInftyPush, inverse to F^* Chapter11/EquivalenceFormsInfty · EP.epFormsInftyPullback_comp_epFormsInftyPush Chapter11/EquivalenceFormsInfty · EP.epFormsInftyPush_comp_epFormsInftyPullback, Lemma 11.2.4 (3). *Step 2, (11.5).* $d_Y \circ F_* = F_* \circ d_X$, from d commuting with F^* and $F_* = (F^*)^{-1}$ Chapter11/EquivalenceFormsInfty · EP.epFormsInftyD_comp_epFormsInftyPush. *Step 3, Lemma 11.2.4 (1), (2).* $(G \circ F)^* = F^* \circ G^*$ is Chapter 8's Chapter8/EpFormsFunctor · EP.epFormsInftyPullback_comp, and inverting it gives $(G \circ F)_* = G_* \circ F_*$ Chapter11/EquivalenceFormsInfty · EP.lemma_11_2_4_two. *Step 4, Lemma 11.2.5.* Naturally equivalent functors have the same pull-back (Chapter 8) Chapter8/EpFormsFunctor · EP.epFormsInftyPullback_eq_of_natTrans; F_* is the only right inverse of F^* Chapter11/EquivalenceFormsInfty · EP.epFormsInftyPush_eq_of_comp_eq_id, so naturally equivalent equivalences have the same push-forward Chapter11/EquivalenceFormsInfty · EP.lemma_11_2_5_push. *Step 5, Lemma 11.2.6.* For $H : A' \to A$ with $F \circ H \simeq F'$ and $G \circ H \simeq G'$: $F_*G^* = F_*H_*H^*G^* = (F \circ H)_*(G \circ H)^* = F'_*G'^*$ by Steps 3 and 4 Chapter11/EquivalenceFormsInfty · EP.lemma_11_2_6, and the same for G_*F^* Chapter11/EquivalenceFormsInfty · EP.lemma_11_2_6_push. In the proof of Lemma 11.2.6 the text's "Lemma 11.2.2 and 11.2.3" is read as Lemmas 11.2.4 and 11.2.5.

Definition 11.2.7 (2017: 11.3) and Theorem 11.2.1 (1)–(3) (2017: 11.2) (V11.12, V11.9). *Step 1, diagrams.* $d^* = F_* \circ \Phi^*$ and, for Φ an equivalence, $d_* = \Phi_* \circ F^*$ Chapter11/GeneralizedForms · EPLocalization.Diagram.formsPullback Chapter11/GeneralizedForms · EPLocalization.Diagram.formsPush. They are unchanged under refinement by Lemma 11.2.6 Chapter11/GeneralizedForms · EPLocalization.Diagram.formsPullback_eq_of_isRefinement, hence on equivalent diagrams Chapter11/GeneralizedForms · EPLocalization.Diagram.formsPullback_eq_of_equiv Chapter11/GeneralizedForms · EPLocalization.Diagram.formsPush_eq_of_equiv. *Step 2, composites.* A base change square in the weak fibred product Chapter11/GeneralizedForms · EP.epFormsInftyPush_epFormsInftyPullback_of_square gives the pull-back of the composite diagram Chapter11/GeneralizedForms · EPLocalization.Diagram.formsPullback_comp. *Step 3, classes.* f^* for every generalized map, equal to $F_* \circ \Phi^*$ on every representative Chapter11/GeneralizedForms · EPLocalization.EPLoc.formsPullback Chapter11/GeneralizedForms · EPLocalization.EPLoc.formsPullback_homMk, with $(g \circ f)^* = f^* \circ g^*$ Chapter11/GeneralizedForms · EPLocalization.EPLoc.formsPullback_comp, so $\Omega^k_{ep,\infty}$ is a contravariant functor on $\mathcal{EP}(E^{-1})$, beyond the text. For a generalized isomorphism with inverse g, $h_* = g^*$ Chapter11/GeneralizedForms · EPLocalization.EPLoc.formsPush, which by Theorem 10.3.10 is $\Phi_* \circ F^*$ on every representative Chapter11/GeneralizedForms · EPLocalization.EPLoc.formsPush_homMk, the text's definition. *Step 4, Theorem 11.2.1.* (1) $d \circ d = 0$ Chapter11/GeneralizedForms · EPLocalization.EPLoc.theorem_11_2_1_one; (2) $d_X \circ f^* = f^* \circ d_Y$ from (11.5) and Chapter 8 Chapter11/GeneralizedForms · EPLocalization.EPLoc.theorem_11_2_1_two; (3) $h_* \circ d_X = d_Y \circ h_*$, $h_* \circ h^* = 1$, $h^* \circ h_* = 1$ Chapter11/GeneralizedForms · EPLocalization.EPLoc.theorem_11_2_1_three.

Propositions 11.2.8 and 11.2.9 (2017: 11.4 and 11.5), Theorem 11.2.1 (4) (V11.13). *Step 1, local families.* The conclusion of Proposition 10.1.10 (1), an sc-smooth $z \mapsto \alpha_z : F(z) \to G(z)$ near every object Chapter11/OrbitForms · EpFunctor.HasLocalNatIso, holds for equivalences with $|F| = |G|$ when X or Y has the local regularity property Chapter11/OrbitForms · EpFunctor.hasLocalNatIso_of_orbitMap_eq, and passes to level 1 and to the index raises Chapter11/OrbitForms · EpFunctor.HasLocalNatIso.liftOne. *Step 2, the pull-backs agree.* Locally $F = s \circ \alpha$ and $G = t \circ \alpha$, and $s^*\omega = t^*\omega$ for a form of the ep-groupoid Y, so $F^*\omega = G^*\omega$ Chapter11/OrbitForms · EpFunctor.HasLocalNatIso.formPullback_eq Chapter11/OrbitForms · EP.epFormsInftyPullback_eq_of_orbitMap_eq. The text's second computation for $F_* = G_*$ is replaced by inverting. **Proposition 11.2.8** Chapter11/OrbitForms · EP.proposition_11_2_8. *Step 3, generalized isomorphisms.* As in the text, through the weak

fibred product $A \times_X A'$: Proposition 11.2.8 applies to the equivalences $G \circ \pi_A$ and $G' \circ \pi_{A'}$, where the property of Y suffices by Proposition 10.1.9 `Chapter11/OrbitForms · EPLocalization.EPLoc.formsPullback_eq_of_orbitMap_eq`. **Proposition 11.2.9** `Chapter11/OrbitForms · EPLocalization.EPLoc.proposition_11_2_9` and **Theorem 11.2.1 (4)** `Chapter11/OrbitForms · EPLocalization.EPLoc.theorem_11_2_1_four`. The hypotheses are the text's; the local regularity property of the lifts X^i is neither assumed nor needed, the families of Step 1 being restricted to the levels. The property is satisfiable, by the point and by $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (Chapter 10).

XI.5 Definition 11.3.1 (2017: 11.4) and Lemma 11.3.2 (2017: 11.5)

Definition 11.3.1 (2017: 11.4) (V11.14) `Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta` `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta`. $F^*\Theta' = \Theta' \circ F$ `Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta_apply`. $F_*\Theta(y)$ is Θ at a chosen x with $F(x) \cong y$, and 0 if there is none `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_of_not_exists`; a nonzero value implies the existence `Chapter11/BranchedEquivalence · EpFunctor.exists_isomorphic_of_pushforwardTheta_ne_zero`, and nonnegativity is kept `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_nonneg`. The text writes the push-forward as a map $X' \to \mathbb{Q}^+$; we read $Y \to \mathbb{Q}^+$ (possible misprint).

Lemma 11.3.2 (2017: 11.5) (V11.15). *Step 1, the choice does not matter.* Functors are constant on isomorphism classes `Chapter11/BranchedEquivalence · EpGroupoid.IsFunctorial.eq_of_isomorphic`. If F is full and $F(x) \cong y \cong F(x')$, then $F(x) \cong F(x')$ `Chapter11/BranchedEquivalence · EpFunctor.isomorphic_obj` lifts to $x \cong x'$ `Chapter11/BranchedEquivalence · EpFunctor.isomorphic_of_full`, so $F_*\Theta(y) = \Theta(x)$ for every such x `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_eq` `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_obj` `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_eq_of_isomorphic`; this is the text's argument with σ lifting $\psi^{-1} \circ \psi'$. *Step 2, functors.* $F_*\Theta$ is a functor for full F `Chapter11/BranchedEquivalence · EpFunctor.isFunctorial_pushforwardTheta`, $F^*\Theta'$ for every F `Chapter11/BranchedEquivalence · EpFunctor.isFunctorial_pullbackTheta`. *Step 3, (1) and (2).* $F^*F_*\Theta = \Theta$ for full F `Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta_pushforwardTheta`; $F_*F^*\Theta' = \Theta'$ for essentially surjective F `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_pullbackTheta`. *Step 4, (3).* Naturally equivalent full functors have equal push-forwards `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_eq_of_naturallyEquivalent`, naturally equivalent functors equal pull-backs `Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta_eq_of_naturallyEquivalent`. *Step 5, composition.* $(G \circ F)^* = F^* \circ G^*$ `Chapter11/BranchedEquivalence · EpFunctor.pullbackTheta_comp`, $(G \circ F)_* = G_* \circ F_*$ for full functors `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_comp`, $1_* = 1$ `Chapter11/BranchedEquivalence · EpFunctor.pushforwardTheta_id`.

XI.6 The push-forward and pull-back along generalized maps, pp. 536–537

Diagrams (V11.16) `Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pushTheta` `Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pullTheta`. For $d = [X \xleftarrow{F} A \xrightarrow{\Phi} Y]$, $d_* = \Phi_* F^*$ and $d^* = F_*\Phi^*$. *Step 1, refinements.* For a refinement $H : d' \to d$, H is an equivalence and $F' \simeq F \circ H$, $\Phi' \simeq \Phi \circ H$; Lemma 11.3.2 (3) and the composition rules give $d'^* = d^*$, and $d'_* = d_*$ when Φ is an equivalence, which then holds for Φ' as well `Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.IsRefinement.isEquivalence_Φ_iff` `Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pullTheta_eq_of_isRefinement` `Chapter11/`

BranchedEquivalenceLocalization · EPLocalization.Diagram.pushTheta_eq_of_isRefinement. *Step 2, equivalent diagrams* have a common refinement Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pullTheta_eq_of_equiv Chapter11/BranchedEquivalenceLocalization · EPLocalization.Diagram.pushTheta_eq_of_equiv.

Classes. A chosen representative Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.rep Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.homMk_rep Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.equiv_rep_homMk, with an equivalence as right leg for a generalized isomorphism Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isEquivalence_rep_Φ, defines f^* for every generalized map and f_* for generalized isomorphisms Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pullTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta, computed on any representative Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pullTheta_homMk Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_homMk, and mapping functors to functors Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isFunctorial_pullTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isFunctorial_pushTheta. *Composition, "easily verified" in the text.* The composite of $[d]$ and $[e]$ is represented through the weak fibred product $A \times_Y B$ (Theorem 10.3.8). In it, Θ satisfies base change: $(\pi_B)_*(\pi_A)^*\Theta = G^*F_*\Theta$ when F is an equivalence, and symmetrically Chapter11/BranchedEquivalenceLocalization · WeakFibredProduct.pushforwardTheta_projRight_pullbackTheta Chapter11/BranchedEquivalenceLocalization · WeakFibredProduct.pushforwardTheta_projLeft_pullbackTheta; the proof evaluates both sides at (a, ψ, b) and uses that a functor Θ takes equal values on isomorphic objects. Hence $(f \gg g)^* = f^* \circ g^*$ for all generalized maps Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pullTheta_comp and $(f \gg g)_* = g_* \circ f_*$ for generalized isomorphisms Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_comp, with $(f^{-1})_* = f^*$ Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_inv and the inverse laws Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pullTheta_pushTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_pullTheta. Nothing here uses Theorem 10.3.19, which does not hold as we read it (Chapter 10).

XI.7 Theorem 11.3.3 (2017: 11.3)

Moving a local branching structure (V11.17). Given an sc-diffeomorphism $f : W \to W'$ between open subsets of the object spaces of two ep-groupoids, with $\Theta' \circ f = \Theta$ on W, a local branching structure (U, M_i, σ_i) of Θ at x gives one of Θ' at $f(x)$: the open set $f(U \cap W)$ Chapter11/BranchedEquivalence · EpGroupoid.LocalBranchingStructure.mapU Chapter11/BranchedEquivalence · EpGroupoid.LocalBranchingStructure.mem_image_iff_of_mem_mapU, the sheets $f(M_i \cap W)$, first restricted to W, with the same weights Chapter11/BranchedEquivalence · EpGroupoid.LocalBranchingStructure.mapDiffeo. The sheets keep their dimension Chapter11/BranchedEquivalence · EpGroupoid.LocalBranchingStructure.isOfDim_mapDiffeo, because the dimension of a submanifold at a point is invariant under sc-diffeomorphisms Chapter11/BranchedEquivalence · MPolyfold.HasDimAt.of_image Chapter11/BranchedEquivalence · MPolyfold.hasDimAt_image_iff_of_isScDiffeoOn Chapter11/BranchedEquivalence · MPolyfold.IsSubmanifoldOfDim.image_isScDiffeoOn.

The step for an equivalence. *Step 1, supports.* $|\mathrm{supp}(F^*\Theta')| = |F|^{-1}(|\mathrm{supp}(\Theta')|)$ Chapter11/BranchedEquivalence · EpFunctor.orbitSupp_pullbackTheta and, for full F, $|\mathrm{supp}(F_*\Theta)| = |F|(|\mathrm{supp}(\Theta)|)$ Chapter11/BranchedEquivalence · EpFunctor.orbitSupp_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.exists_mem_supp_of_mem_supp_pushforwardTheta; since $|F|$ is a homeomorphism, the paracompactness of 2021 Definition 9.1.2 (1) transfers, a point the proof does not mention. *Step 2, local branching structures.* For F_*: at y with $\psi : F(x) \to y$, Chapter 10's transport chart $\hat{\psi} \circ F$ Chapter10/

EquivalenceLR · EpFunctor.TransportChart is an sc-diffeomorphism near x with $F_*\Theta \circ \hat{\psi} \circ F = \Theta$, and the structure is moved Chapter11/BranchedEquivalence · EpFunctor.exists_localBranchingStructure_pushforwardTheta; for F^*, F itself near x. Hence **branched ep⁺-subgroupoids go to branched ep⁺-subgroupoids** Chapter11/BranchedEquivalence · EpFunctor.isBranchedEpSubgroupoid_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isBranchedEpSubgroupoid_pullbackTheta. *Step 3, properties.* Closedness and compactness from Step 1 Chapter11/BranchedEquivalence · EpFunctor.isClosedBranched_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isClosedBranched_pullbackTheta Chapter11/BranchedEquivalence · EpFunctor.isCompactBranched_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isCompactBranched_pullbackTheta; orbifold-type from Step 2 Chapter11/BranchedEquivalence · EpFunctor.isOrbifoldType_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isOrbifoldType_pullbackTheta; manifold-type by comparing two morphisms $y \to y'$ through the faithfulness of F Chapter11/BranchedEquivalence · EpFunctor.isManifoldType_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isManifoldType_pullbackTheta. In this step the text writes $\psi \circ \varphi' \circ \psi^{-1} : F(x) \to F(x)$ for $\psi : F(x) \to y$ and $\varphi' : y \to y$; we read $\psi^{-1} \circ \varphi' \circ \psi$ (possible misprint). *Step 4, dimension.* n-dimensionality Chapter11/BranchedEquivalence · EpFunctor.isNDimensional_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isNDimensional_pullbackTheta and dimensional decompositions, the push-forward of a sum being the sum of the push-forwards Chapter11/BranchedEquivalence · EpFunctor.exists_pushforwardTheta_eq_apply Chapter11/BranchedEquivalence · EpFunctor.isDimDecomposition_pushforwardTheta Chapter11/BranchedEquivalence · EpFunctor.isDimDecomposition_pullbackTheta.

Theorem 11.3.3 (2017: 11.3) (V11.18). On a representative with both legs equivalences, $f_* = \Phi_* F^*$ and $f^* = F_* \Phi^*$, and each part follows from the step for an equivalence: (1) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_one, (2) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_two, (3) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_three, (4) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_four, (5) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_five, (6) Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.theorem_11_3_3_six, with n-dimensionality Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isNDimensional_pushTheta Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.isNDimensional_pullTheta and the support Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.orbitSupp_pushTheta. (6) speaks of "the" dimensional decomposition of Proposition 9.1.11, whose uniqueness does not hold as we read it Chapter9/Propositions · EpGroupoid.not_proposition_9_1_11_rn; it is proved for every dimensional decomposition.

Witnesses (V11.19). On $\mathbb{Z}/2 \ltimes \mathbb{R}$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.reflectionEP, $\Theta_0 = 1_{\{0\}}$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.reflectionTheta has the local branching structure $U = \mathbb{R}$, $M = \{0\}$ at the point with isotropy $\mathbb{Z}/2$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.localBranchingStructure_reflectionTheta Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.isBranchedEpSubgroupoid_reflectionTheta. Along the refinement of the covering $\{x > -1\}$, $\{x < 1\}$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.reflectionCoverEP, a functor not injective on objects, $F^*\Theta_0$ is a branched ep⁺-subgroupoid Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.isBranchedEpSubgroupoid_pullbackTheta_reflectionCover, equal to 1 at the two objects over 0 Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.pullbackTheta_reflectionCover_two_points, and $F_* F^* \Theta_0 = \Theta_0$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.pushforwardTheta_pullbackTheta_reflectionCover. The generalized isomorphism of the refinement Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.reflectionCoverDiagram Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.isGeneralizedIso_reflectionCoverDiagram pushes Θ_0 to

$F^*\Theta_0$ Chapter11/BranchedEquivalenceLocalization · EpGroupoid.Examples.pushTheta_reflectionCoverDiagram.

XI.8 Lemma 11.3.4 to Theorem 11.3.9 (2017: Lemma 11.6 to Theorem 11.7): sections, orientations, tangents, boundaries

Section functors of $\widehat{Gr}$ (V11.20) Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.IsGrHatSection Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.GrHatSections. $h(x)$ is a finite formal sum of oriented tangent sets at $x \in X_\infty$, natural along morphisms, with no continuity; it is recorded on all objects, 0 off X_∞. Every orientation of Chapter 9 is one Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.IsOrientation.isGrHatSection. *Step 1, (11.8).* $F_*h(y)$ moves $h(x)$ by $T\psi \circ TF(x)$ for a chosen $\psi : F(x) \to y$, the formal sums by the image of carriers and frames Chapter11/BranchedEquivalenceTangent · MPolyfold.OrTangent.map Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection. For an equivalence the value is the same for every ψ Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection_eq: two choices differ by T of a morphism of X, which h absorbs by naturality, as the text says. *Step 2, (11.9).* $F^*h'(x) = TF(x)^{-1}h'(F(x))$, with the inverse of TF on T_xX Chapter11/BranchedEquivalenceTangentSection · MPolyfold.fibreInv Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pullSection. *Step 3, equivalences.* Section functors go to section functors Chapter11/BranchedEquivalenceTangentSection · EpFunctor.isGrHatSection_pushSection Chapter11/BranchedEquivalenceTangentSection · EpFunctor.isGrHatSection_pullSection; $F^*F_* = 1$, $F_*F^* = 1$ Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pullSection_pushSection Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection_pullSection; naturally equivalent equivalences agree Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection_eq_of_natTrans Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pullSection_eq_of_natTrans; composition Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pushSection_comp Chapter11/BranchedEquivalenceTangentSection · EpFunctor.pullSection_comp; base change in the weak fibred product of two equivalences Chapter11/BranchedEquivalenceTangentSection · WeakFibredProduct.pushSection_projRight_pullSection Chapter11/BranchedEquivalenceTangentSection · WeakFibredProduct.pushSection_projLeft_pullSection. *Step 4, Lemma 11.3.4.* On diagrams Chapter11/BranchedEquivalenceTangentSection · EPLocalization.Diagram.pushSection Chapter11/BranchedEquivalenceTangentSection · EPLocalization.Diagram.pullSection, unchanged on equivalent diagrams Chapter11/BranchedEquivalenceTangentSection · EPLocalization.Diagram.pushSection_eq_of_equiv Chapter11/BranchedEquivalenceTangentSection · EPLocalization.Diagram.pullSection_eq_of_equiv; on classes Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pullSection, computed on any representative Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection_homMk Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pullSection_homMk, with the composition laws for generalized isomorphisms through the base change Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection_comp Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pullSection_comp and $(f^{-1})_* = f^*$ Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection_inv. **Lemma 11.3.4** Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.lemma_11_3_4.

Chapter 9 across two M-polyfolds. Chapter 9 moves submanifolds, tangent sets and orientations by sc-diffeomorphisms of one M-polyfold; equivalences need them between open subsets $U \subseteq X$, $U' \subseteq Y$ of two. For such an f: $T_{f(a)}f(A) = Tf(T_aA)$ Chapter11/BranchedEquivalenceTangent · MPolyfold.submanifoldTangent_image_of_isScDiffeoOn, chart vectors in transported charts Chapter11/BranchedEquivalenceTangent · MPolyfold.chartVec_tangentMapOn_of_isScDiffeoOn,

oriented tangent sets Chapter11/BranchedEquivalenceTangent · MPolyfold.IsOrientedAt.map_tangentMapOn, orientations of submanifolds, their frames moved by Tf and their continuity read in the chart $\varphi \circ f^{-1}$ Chapter11/BranchedEquivalenceTangent · MPolyfold.SubmanifoldOrientation.mapDiffeo Chapter11/BranchedEquivalenceTangent · MPolyfold.SubmanifoldOrientation.mapDiffeo_tangent, the reduced tangent condition Chapter11/BranchedEquivalenceTangent · MPolyfold.IsTangentReducedAt.image_of_isScDiffeoOn Chapter11/BranchedEquivalenceTangent · MPolyfold.isTangentReducedAt_image_iff_of_isScDiffeoOn and the boundary of a submanifold Chapter11/BranchedEquivalenceTangent · MPolyfold.mem_submanifoldBoundary_image_iff_of_isScDiffeoOn. For local branching structures moved as in §XI.7: the tangent sums are moved by Tf Chapter11/BranchedEquivalenceTangent · EpGroupoid.LocalBranchingStructure.tangentSum_mapDiffeo; the independence of the tangent sums of Theorem 9.2.1 is local and carried along, by moving structures back by f^{-1}, comparing and moving forward Chapter11/BranchedEquivalenceTangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isScDiffeoOn; $T\Theta'(Tf\,\alpha) = T\Theta(\alpha)$ Chapter11/BranchedEquivalenceTangent · EpGroupoid.tangentTheta_tangentMapOn and $\partial\Theta'(f(x)) = \partial\Theta(x)$ Chapter11/BranchedEquivalenceTangent · EpGroupoid.boundaryTheta_apply_of_isScDiffeoOn; oriented structures Chapter11/BranchedEquivalenceTangent · EpGroupoid.OrientedBranchingStructure.mapDiffeo Chapter11/BranchedEquivalenceTangent · EpGroupoid.OrientedBranchingStructure.orientedSum_mapDiffeo.

Theorem 11.3.5 (2017: 11.4) (V11.21). Along a transport chart $Q = \hat{\psi} \circ F$, $T\psi_u \circ TF = TQ$ on T_uX Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.TransportChart.tangentMor_ψ_tangent_obj, so an oriented branching structure at x representing $\hat{T}_\Theta$, moved by Q, represents $F_*\hat{T}_\Theta$ near y Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.isOrientation_pushSection; for F^*, a structure at $F(x)$ is moved back by a local inverse of F Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.isOrientation_pullSection. This is the text's proof. For generalized isomorphisms Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.EPLoc.theorem_11_3_5_push Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.EPLoc.theorem_11_3_5_pull. The independence of the tangent sums, which Chapter 9 now proves for every Θ Chapter9/GeneralCorollaries · EpGroupoid.hasIndependentTangentSums, was assumed for Θ and is carried to $F_*\Theta$, $F^*\Theta'$ Chapter11/BranchedEquivalenceTangent · EpFunctor.hasIndependentTangentSums_pushforwardTheta Chapter11/BranchedEquivalenceTangent · EpFunctor.hasIndependentTangentSums_pullbackTheta and to $f_*\Theta$, $f^*\Theta'$ Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.hasIndependentTangentSums_pushTheta Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.hasIndependentTangentSums_pullTheta.

Definition 11.3.6 (2017: 11.5) and (11.10) (V11.22) Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.push Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.pull. The structure records Θ, $\hat{T}_\Theta$ and the independence; push-forward and pull-back are oriented branched ep⁺-subgroupoids by Theorems 11.3.3 and 11.3.5; (11.10) Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.push_comp Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.pull_comp from Lemma 11.3.4 and §XI.6. Of the two inverse laws of (11.10), $f^*f_*\hat{\Theta} = \hat{\Theta}$ is proved for $\hat{T}_\Theta$ zero off X_∞ Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.OrientedBranched.pull_push, and $f_*f^*\hat{\Theta}' = \hat{\Theta}'$ for its two components Chapter11/BranchedEquivalenceLocalization · EPLocalization.EPLoc.pushTheta_pullTheta Chapter11/BranchedEquivalenceTangentSection · EPLocalization.EPLoc.pushSection_pullSection.

Theorem 11.3.7 (2017: 11.5) (V11.23). *Step 1, equivalences.* $T(F_*\Theta)(F(x)) = TF(T\Theta(x))$ Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentValue_pushforwardTheta_obj and, for $\psi : F(x) \to y$, $T(F_*\Theta)(y)$ is $T\Theta(x)$ moved by $T\psi \circ TF(x)$ Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentValue_pushforwardTheta; for the pull-back, with a local inverse of F Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentValue_pullbackTheta. Hence $(TF)_*T\Theta = T(F_*\Theta)$, $(TF)^*T\Theta' = T(F^*\Theta')$, where

TF is an equivalence (Theorem 10.1.7) and $(TF)_*$ the push-forward of Definition 11.3.1 on TX Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentTheta_pushforwardTheta Chapter11/BranchedEquivalenceTangent · EpFunctor.tangentTheta_pullbackTheta. *Step 2, generalized isomorphisms.* Tf is $[Td]$ for the chosen representative Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.tangentFunctor_map_eq (Theorem 11.1.1), and Step 1 applies to both legs Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.theorem_11_3_7_push Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.theorem_11_3_7_pull. The text leaves the proof to the reader.

Theorem 11.3.8 (2017: 11.6) (V11.24). By the local step, $\partial\Theta$ of Definition 9.2.16 is moved pointwise by every chart $\hat{\psi} \circ F$: the text's "$d_M(x)$ is preserved" is the invariance of ∂M above. For equivalences Chapter11/BranchedEquivalenceTangent · EpFunctor.boundaryTheta_pushforwardTheta Chapter11/BranchedEquivalenceTangent · EpFunctor.boundaryTheta_pullbackTheta and generalized isomorphisms, (11.11) Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.theorem_11_3_8_push Chapter11/BranchedEquivalenceTangent · EPLocalization.EPLoc.theorem_11_3_8_pull.

Theorem 11.3.9 (2017: 11.7), for a representative (V11.25). Chapter 9's boundary orientation $\hat{T}_{\partial\Theta}$ is the boundary lift of $\hat{T}_\Theta$ by a boundary operation bd on oriented tangent sets, taken as data. A linear map $T_xX \to T_yY$ transfers bd to bd' if it keeps degree 1 and commutes with them there Chapter11/BranchedEquivalenceTangentOrientation · MPolyfold.IsBoundaryTransfer; the identity does Chapter11/BranchedEquivalenceTangentOrientation · MPolyfold.isBoundaryTransfer_id. A fibre map that transfers them commutes with the degree-1 part and the boundary operation Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.mapDomain_boundary_mapDomain_map, so along an equivalence whose maps $T\psi \circ TF(x)$ transfer bd to bd', $F_*\hat{T}_{\partial\Theta} = \hat{T}_{\partial(F_*\Theta)}$ Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.pushSection_boundaryLift, and dually Chapter11/BranchedEquivalenceTangentOrientation · EpFunctor.pullSection_boundaryLift. With Theorem 11.3.8 for $\partial\Theta$, for a representative $X \xleftarrow{F} A \xrightarrow{\Phi} Y$ and boundary operations on X, A, Y corresponding under $TF(a)^{-1}$ and $T\psi \circ T\Phi(a)$ Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.EPLoc.theorem_11_3_9_push Chapter11/BranchedEquivalenceTangentOrientation · EPLocalization.EPLoc.theorem_11_3_9_pull. That Definition 9.3.8's boundary operation satisfies the hypothesis along equivalences is the next paragraph (G.130, closed).

Theorem 11.3.9 for the intrinsic boundary orientation (V11.25) Chapter11/BoundaryOrientationEquiv · EPLocalization.EPLoc.theorem_11_3_9_push_boundaryOrientation Chapter11/BoundaryOrientationEquiv · EPLocalization.EPLoc.theorem_11_3_9_pull_boundaryOrientation, independent of the representative, with no transfer hypothesis (G.130). *Step 1, sc-diffeomorphisms move the boundary operation.* A presented chart ψ at $f(x)$ gives the presented chart $\psi \circ f$ at x with the same model, so Tf preserves T^R_xX, C_xX and the reduced codimension, and with them Definition 9.3.8's boundary operation: $\partial(Tf_*(L,o)) = Tf_*(\partial(L,o))$ Chapter11/BoundaryOrientationEquiv · MPolyfold.boundaryOp_map_tangentMapOn. "$d_M(x)$ is preserved", in the proof of Theorem 11.3.8, is read for the reduced codimension of T_xM, which for tame M is $d_M(x)$ and is preserved by every sc-diffeomorphism, without tameness. *Step 2, equivalences.* Through the transport chart $\hat{\psi} \circ F$ of Chapter 10, the maps $T\psi \circ TF(x)$ move it for every $\psi : F(x) \to y$ Chapter11/BoundaryOrientationEquiv · EpFunctor.isReducedBoundaryTransfer_tangentMor, and so do the fibre inverses $TF(x)^{-1}$ Chapter11/BoundaryOrientationEquiv · EpFunctor.isReducedBoundaryTransfer_fibreInv. *Step 3, the intrinsic orientation.* $\hat{T}_{\partial\Theta}$ of Definition 9.3.10 is a function of $\hat{T}_\Theta$, computed with the boundary operation pointwise. By Steps 1 and 2 the push-forward along a representative commutes with it Chapter11/BoundaryOrientationEquiv · EPLocalization.Diagram.pushSection_boundaryOrientation Chapter11/BoundaryOrientationEquiv · EPLocalization.Diagram.pullSection_boundaryOrientation. With Theorem 11.3.8 for

$\partial\Theta$, $f_*(\partial\hat{\Theta}) = \partial(f_*\hat{\Theta})$ and the same for f^*. The statement holds for every section of $\widehat{Gr}$, oriented or not. *The boundary-lift form.* For Chapter 9's boundary lift with Definition 9.3.8's operation, the boundary-transfer hypothesis above follows from Steps 1 and 2 `Chapter11/BoundaryOrientationEquiv · MPolyfold.IsReducedBoundaryTransfer.isBoundaryTransfer` `Chapter11/BoundaryOrientationEquiv · EpFunctor.isBoundaryTransfer_tangentMor`, given that f carries tangent sets of tame submanifolds to tangent sets of tame submanifolds `Chapter11/BoundaryOrientationEquiv · MPolyfold.CarriesTameTangents`. This form, which also asks that $f_*\hat{\Theta}$ be tame `Chapter11/BoundaryOrientationEquiv · EPLocalization.EPLoc.theorem_11_3_9_push_boundaryOp`, has those two conditions as hypotheses; they hold along the identity `Chapter11/BoundaryOrientationEquiv · MPolyfold.carriesTameTangents_id` and for Example 9.3.5 `Chapter11/BoundaryOrientationEquiv · EpGroupoid.Examples.theorem_11_3_9_push_boundaryOp_example935`. That sc-diffeomorphisms preserve tame induced structures on submanifolds is not formalized, and the intrinsic form does not need it.

Witnesses (V11.26). The orientation $\hat{T}^1$ of Example 9.3.5 is a section functor on $\mathbb{R}^2$ `Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.Examples.exampleT1_mem_grHatSections`, fixed by the identity `Chapter11/BranchedEquivalenceTangentSection · EpGroupoid.Examples.pushSection_id_exampleT1`; $(\Theta, \hat{T}^1)$ on the trivial ep-groupoid of $\mathbb{R}^2$ `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.rn2EP` `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.example935Oriented` is pushed forward along the identity generalized isomorphism `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.isGeneralizedIso_id_rn2EP` `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.example935Oriented_push`, and the boundary-transfer hypothesis holds there `Chapter11/BranchedEquivalenceTangentOrientation · EpGroupoid.Examples.isBoundaryTransfer_id_rn2`.

XI.9 §11.4: integration (Theorems 11.4.1, 11.4.2, 11.4.4, 11.4.5, Remark 11.4.3 (2017: Theorems 11.8, 11.9, Remark 11.3))

§11.4 is formalized (G.131). Theorems 11.4.1, 11.4.2 and Remark 11.4.3 hold with no added hypothesis. Theorems 11.4.4 and 11.4.5 were first proved under the class hypothesis of Chapter 9's boundary measures and for $n \geq 2$; they now hold with the text's hypotheses for every $n \geq 1$. The measures are Chapter 9's: the canonical measures of Theorems 9.5.4 and 9.5.5 built from the concrete sheet integrals `Chapter9/BranchedIntegral · EpGroupoid.theorem_9_5_4` `Chapter9/BranchedIntegral · EpGroupoid.theorem_9_5_5`, and the boundary measures of Theorems 9.5.15 and 9.5.16 on sheets covered by countably many graph charts `Chapter9/BoundaryMeasure · EpGroupoid.theorem_9_5_15_msheets` `Chapter9/BoundaryMeasure · EpGroupoid.theorem_9_5_16_msheets`. "Oriented, tame" is read as in §XI.8, and the independence of the tangent sums holds for every Θ.

What the proof uses of the forms. For an equivalence Φ the identity $\mu^{\Phi^*\hat{\Theta}'}_{\Phi^*\omega'} = |\Phi|^*\mu^{\hat{\Theta}'}_{\omega'}$ uses two facts about the pull-back of forms, recorded as a structure `Chapter11/FormEvalPullback · EpGroupoid.FormEval.IsPullback`: $(\Phi^*\omega')(\alpha_1, \dots, \alpha_n) = \omega'(T\Phi\,\alpha_1, \dots, T\Phi\,\alpha_n)$ on the tangent spaces of finite-dimensional submanifolds, and every form on X is a pull-back (§XI.4). Both hold for Ω^n_{ep} `Chapter11/FormEvalPullbackEquivalence · EpFunctor.IsEquivalence.epFormEval_isPullback` and for the direct limit $\Omega^n_{ep,\infty}$ `Chapter11/FormEvalPullbackEquivalence · EP.epFormsInftyEval_isPullback`.

Theorem 11.4.1 for an equivalence (V11.27) `Chapter11/MeasureEquivalence · EpFunctor.theorem_11_4_1_pull_equiv` `Chapter11/MeasureEquivalence · EpFunctor.theorem_11_4_1_push_equiv`. The text takes $U(x)$, $U(F(x))$ on which F is an sc-diffeomorphism, both with the properness property and the natural actions, and moves the local branching structure by F. *Step 1, between two M-polyfolds.* An sc-diffeomorphism

$f : U \to U'$ between open subsets of two M-polyfolds carries graph charts to graph charts in the transported chart Chapter11/GraphChartsDiffeo · MPolyfold.GraphChart.mapDiffeoCross, null sets to null sets Chapter11/GraphChartsDiffeo · MPolyfold.IsHNull.image_of_isScDiffeoOn_cross, and tame induced structures to tame induced structures Chapter11/GraphChartsDiffeo · MPolyfold.HasTameInducedStructure.image_of_isScDiffeoOn. *Step 2, moving structures.* An oriented tame structure is restricted to a smaller natural representation Chapter11/OrientedStructureDiffeo · EpGroupoid.OrientedTameStructure.restrictRep and moved, $(f(M_i \cap W), f_* o_i)$, (σ_i) Chapter11/OrientedStructureDiffeo · EpGroupoid.OrientedTameStructure.move Chapter11/OrientedStructureDiffeo · EpGroupoid.OrientedTameStructure.represents_move. *Step 3, (11.12) = (11.13).* When $\omega_T(Tf \circ \beta) = \omega_S(\beta)$ on the frames of a sheet Chapter11/OrientedStructureDiffeo · EpGroupoid.FormEval.EvCompatOn, $\int_{f(A)} \omega_T|(f(N), f_* o) = \int_A \omega_S|(N, o)$ Chapter11/OrientedStructureDiffeo · EpGroupoid.FormEval.integ_image_cross, and so the weighted sums of a structure and of its moved structure agree Chapter11/OrientedStructureDiffeo · EpGroupoid.OrientedTameStructure.sheetSum_move; the same holds for sheets covered by several graph charts Chapter11/MultiChartSheetsEquivalence · EpGroupoid.FormEval.image_mem_msheets_cross Chapter11/MultiChartSheetsEquivalence · EpGroupoid.FormEval.msInteg_image_cross. *Step 4, the transport chart.* For $Q = \widehat{\psi} \circ \Phi$ (Chapter 10), Q preserves X_∞ and $\Phi^*\Theta'$ Chapter11/OrientedStructurePairs · EpFunctor.TransportChart.pullbackTheta_Q, moves the orientations Chapter11/OrientedStructurePairs · EpFunctor.TransportChart.pullSection_Q and moves ω' to $\Phi^*\omega'$ Chapter11/OrientedStructurePairs · EpFunctor.TransportChart.evCompatOn_Q. For small $U(x)$ there are natural representations on $U(x)$ and on $Q(U(x))$ Chapter11/OrientedStructurePairs · EpFunctor.TransportChart.exists_rep_push. A structure at y is restricted to $Q(U(x))$ and moved to x by Q^{-1} Chapter11/OrientedStructurePairs · EpGroupoid.OrientedTameStructure.pullStructure Chapter11/OrientedStructurePairs · EpGroupoid.OrientedTameStructure.represents_pullStructure, and one at x is moved to y by Q Chapter11/OrientedStructurePairs · EpGroupoid.OrientedTameStructure.exists_push_structure; this is the text's "$F^*\hat{\Theta}'$, $F_*\hat{\Theta}$ are oriented and tame" Chapter11/MeasureEquivalence · EpFunctor.hcov_pull Chapter11/MeasureEquivalence · EpFunctor.hcov_push. *Step 5, the isotropy factor.* $\sharp G_x = \sharp G_y$ by the isomorphism $g \mapsto \psi \circ \Phi(g) \circ \psi^{-1}$ Chapter10/EquivalenceLR · EpFunctor.TransportChart.bijective_conj. *Step 6, $\mathcal{L}$.* $|\Phi|$ and its inverse map sets of measure zero (Definition 9.5.1) to sets of measure zero Chapter11/MeasureEquivalence · EpFunctor.isMeasureZeroSupp_image Chapter11/MeasureEquivalence · EpFunctor.isMeasureZeroSupp_preimage, so $|\Phi| : |\operatorname{supp} \Phi^*\Theta'| \to |\operatorname{supp} \Theta'|$ is an isomorphism of the σ-algebras $\mathcal{L}$ Chapter11/MeasureEquivalence · EpFunctor.measurable_suppMap Chapter11/MeasureEquivalence · EpFunctor.measurable_suppInv. The text uses this without comment. *Step 7, gluing.* By Steps 3–5 the local formulas of Theorem 9.5.4 for $\mu^{\Phi^*\hat{\Theta}'}_{\Phi^*\omega'}$ and for $\mu^{\hat{\Theta}'}_{\omega'} \circ |\Phi|$ agree on the sets $|U(x)|$, and the measure with these local formulas is unique Chapter9/MeasureGlue · existsUnique_linearMap_signedMeasure_of_openCover Chapter11/MeasureEquivalence · EpFunctor.measure_pull_core. The push-forward identity is the same with $\Phi_*\omega$ any ω' with $\Phi^*\omega' = \omega$. The integrals follow with $K = |\operatorname{supp}|$ Chapter11/MeasureEquivalence · EpFunctor.integral_pull_equiv Chapter11/MeasureEquivalence · EpFunctor.integral_push_equiv.

Theorem 11.4.1 (2017: 11.8) and Remark 11.4.3 (2017: 11.3) (V11.27, V11.28) Chapter11/MeasureGeneralizedIso · EPLocalization.EPLoc.theorem_11_4_1_pull Chapter11/MeasureGeneralizedIso · EPLocalization.EPLoc.theorem_11_4_1_push. As in the text: for equivalences $F : A \to X$, $G : A \to Y$ and $\Theta_A = F^*\Theta_X = G^*\Theta_Y$, the equivalence case along F and along G gives $\mu^{\hat{\Theta}_X}_{\omega_X}(K) = \mu^{\hat{\Theta}_Y}_{\omega_Y}(|G|(|F|^{-1}(K)))$ when $F^*\omega_X = G^*\omega_Y$ Chapter11/MeasureGeneralizedIso · EpFunctor.measure_span_core. With the forms of Definition 11.2.7 on the chosen representative this is the theorem for diagrams Chapter11/MeasureGeneralizedIso · EPLocalization.Diagram.theorem_11_4_1_pull Chapter11/MeasureGeneralizedIso · EPLocalization.Diagram.theorem_11_4_1_push and for generalized isomorphisms. Remark 11.4.3, in the form Chapter 11 uses it: two ep-groupoid models related by a generalized isomorphism give the same branched integrals Chapter11/MeasureGeneralizedIso · EPLocalization.

EPLoc.integral_pull Chapter11/MeasureGeneralizedIso · EPLocalization.EPLoc.integral_push Chapter11/MeasureGeneralizedIso · EPLocalization.EPLoc.remark_11_4_3. The relation on pairs (X, β) in Remark 11.4.3 is Chapter 10's.

Theorem 11.4.2 (new in 2021) (V11.27) Chapter11/EffectiveMeasureGeneralizedIso · EPLocalization.EPLoc.theorem_11_4_2_pull Chapter11/EffectiveMeasureGeneralizedIso · EPLocalization.EPLoc.theorem_11_4_2_push. "The proof is similar to the previous theorem"; the one new step is $\sharp G_x^{eff} = \sharp G_y^{eff}$ along a transport chart Chapter11/EffectiveMeasureEquivalence · EpFunctor.TransportChart.card_effIsotropy_eq. The isomorphism of Step 5 carries $\ker \Phi_R = T_x$ onto $\ker \Phi_{R'} = T_y$ for natural representations on $U(x)$ and on $Q(U(x))$, by the equivariance of Q. Under the local unique continuation property every natural representation has the property used, and the property passes along equivalences (Proposition 10.1.16), so the covering hypotheses of Theorem 9.5.5 pass to $\Phi^*\widehat{\Theta}'$, $\Phi_*\widehat{\Theta}$ Chapter11/EffectiveMeasureEquivalence · EpFunctor.hcovL_pull Chapter11/EffectiveMeasureEquivalence · EpFunctor.hcovL_push. Then the proof of Theorem 11.4.1 Chapter11/EffectiveMeasureEquivalence · EpFunctor.measure_pull_core_eff Chapter11/EffectiveMeasureEquivalence · EpFunctor.theorem_11_4_2_pull_equiv Chapter11/EffectiveMeasureEquivalence · EpFunctor.theorem_11_4_2_push_equiv, and the diagram step, the middle A having the property by Proposition 10.1.16 Chapter11/EffectiveMeasureGeneralizedIso · EpFunctor.measure_span_core_eff. The hypotheses are the text's.

Theorems 11.4.4 and 11.4.5 (2017: 11.9) (V11.29) Chapter11/BoundaryMeasureGeneralizedIso · EPLocalization.EPLoc.theorem_11_4_4_pull Chapter11/BoundaryMeasureGeneralizedIso · EPLocalization.EPLoc.theorem_11_4_4_push Chapter11/BoundaryMeasureGeneralizedIso · EPLocalization.EPLoc.theorem_11_4_5_pull Chapter11/BoundaryMeasureGeneralizedIso · EPLocalization.EPLoc.theorem_11_4_5_push. The text says that there are versions of Theorems 11.4.1 and 11.4.2 for the boundary case, relying on Theorems 9.5.15 and 9.5.16. *Step 1, boundary sheets move.* $\partial_1 f(M) = f(\partial_1 M)$, the reduced codimension being preserved Chapter11/BoundarySheetsDiffeo · EpGroupoid.boundaryStratum_image_cross. The boundary orientation is moved by Tf Chapter11/BoundarySheetsDiffeo · EpGroupoid.OrientedTameStructure.boundarySheet_move Chapter11/BoundarySheetsDiffeo · EpGroupoid.OrientedTameStructure.boundaryOrient_move_tangent, by the boundary operation of §XI.8, Step 1, and the boundary sums of Theorem 9.5.15 agree for a structure and its moved structure Chapter11/BoundarySheetsDiffeo · EpGroupoid.OrientedTameStructure.boundarySheetSum_move, with the change of variables for the multi-chart integrals. *Step 2, the measures.* The boundary measures, the unique maps of Theorems 9.5.15 and 9.5.16 Chapter11/BoundaryMeasureEquivalence · EpGroupoid.boundaryMeasure Chapter11/BoundaryMeasureEquivalence · EpGroupoid.boundaryMeasureEff. $|\Phi|$ is measurable both ways on the boundary supports Chapter11/BoundaryMeasureEquivalence · EpFunctor.measurable_bsuppMap Chapter11/BoundaryMeasureEquivalence · EpFunctor.measurable_bsuppInv, and the gluing argument of Step 7 applies to any pair of measures with the local formulas of Theorem 9.5.15 Chapter11/BoundaryMeasureEquivalence · EpFunctor.measure_pull_core_bd Chapter11/BoundaryMeasureEquivalence · EpFunctor.theorem_11_4_4_pull_equiv Chapter11/BoundaryMeasureEquivalence · EpFunctor.theorem_11_4_5_push_equiv. *Step 3, diagrams* Chapter11/BoundaryMeasureGeneralizedIso · EpFunctor.measure_span_core_bd. *The added hypothesis.* The boundary measures of Chapter 9 carry the class hypothesis `hcov`: the nonempty boundary sheets of the structures lie in the class of integrated sheets, with finite $\int |\omega|$ over $\partial_1 M_i$. It is derived from the text's hypotheses (§IX.36). It passes along equivalences and diagrams Chapter11/BoundaryMeasureEquivalence · EpFunctor.hcovB_pull Chapter11/BoundaryMeasureEquivalence · EpFunctor.hcovB_push Chapter11/BoundaryMeasureGeneralizedIso · EPLocalization.Diagram.hcovB_pullTheta Chapter11/BoundaryMeasureGeneralizedIso · EPLocalization.Diagram.hcovB_pushTheta. Its satisfiability witnesses: compact boundary sheets Chapter9/BoundaryMeasure · EpGroupoid.hcov_msheets_of_isCompact and, with nonempty boundary, the closed upper hemisphere in $[0, \infty) \times \mathbb{R}^2$ Chapter9/HemisphereWitness · E8W.e8w_hcov Chapter9/

StokesBoundaryMeasure · EP.hcovB_of_hcovS. These first forms have $n = m + 1$ with $m > 0$.

Theorems 11.4.4 and 11.4.5 with the text's hypotheses (V11.29, restated; gap G.131, closed). $n \geq 2$ Chapter11/BoundaryMeasureTextHypotheses · EPLocalization.EPLoc.theorem_11_4_4_pull_loc Chapter11/BoundaryMeasureTextHypotheses · EPLocalization.EPLoc.theorem_11_4_4_push_loc Chapter11/BoundaryMeasureTextHypotheses · EPLocalization.EPLoc.theorem_11_4_5_pull_loc Chapter11/BoundaryMeasureTextHypotheses · EPLocalization.EPLoc.theorem_11_4_5_push_loc. For a generalized isomorphism f and $\hat{\Theta}'$ compact, branched, oriented and tame (`hloc`), $|f|(K)$ is measurable and $\mu^{\partial f^*\hat{\Theta}'}_{f^*\omega'}(K) = \mu^{\partial\hat{\Theta}'}_{\omega'}(|f|(K))$; for $\hat{\Theta}$ with the same hypotheses $\mu^{\partial\hat{\Theta}}_{\omega}(K) = \mu^{\partial f_*\hat{\Theta}}_{f_*\omega}(|f|(K))$; the same for the effective measures when both ep-groupoids have the local unique continuation property. They are the versions above with `hloc` in place of `hcov`, the class condition being derived (§IX.36). The measures are those of Theorems 9.5.15 and 9.5.16 with the text's hypotheses, by definition Chapter11/BoundaryMeasureTextHypotheses · EP.boundaryMeasure_eq Chapter11/BoundaryMeasureTextHypotheses · EP.boundaryMeasureEff_eq; on the transported side the class conditions are the terms `hcovB_pull`, `hcovB_push` above. $n = 1$. The value of a 0-form moves with the point: $(F^*\omega)(p) = \omega(F(p))$ Chapter11/BoundaryMeasureEquivalenceDimOne · EP.zeroFormEval_epFormsInftyPullback, and on a transport chart a boundary point moves to its image with the same frame sign Chapter11/BoundaryMeasureEquivalenceDimOne · EpGroupoid.OrientedTameStructure.boundarySheetSum_move_dim1. The gluing of Step 2 is redone for the point sheets of §IX.38 Chapter11/BoundaryMeasureEquivalenceDimOne · EpFunctor.measure_pull_core_bd_dim1 Chapter11/BoundaryMeasureEquivalenceDimOne · EpFunctor.theorem_11_4_4_pull_equiv_dim1 Chapter11/BoundaryMeasureEquivalenceDimOne · EpFunctor.theorem_11_4_4_push_equiv_dim1 Chapter11/BoundaryMeasureEquivalenceDimOne · EpFunctor.theorem_11_4_5_pull_equiv_dim1 Chapter11/BoundaryMeasureEquivalenceDimOne · EpFunctor.theorem_11_4_5_push_equiv_dim1, then for spans Chapter11/BoundaryMeasureGeneralizedIsoDimOne · EpFunctor.measure_span_core_bd_dim1 and for generalized isomorphisms Chapter11/BoundaryMeasureGeneralizedIsoDimOne · EPLocalization.EPLoc.theorem_11_4_4_pull_dim1 Chapter11/BoundaryMeasureGeneralizedIsoDimOne · EPLocalization.EPLoc.theorem_11_4_4_push_dim1 Chapter11/BoundaryMeasureGeneralizedIsoDimOne · EPLocalization.EPLoc.theorem_11_4_5_pull_dim1 Chapter11/BoundaryMeasureGeneralizedIsoDimOne · EPLocalization.EPLoc.theorem_11_4_5_push_dim1, with the measures of the $n = 1$ Theorems 9.5.15 and 9.5.16.

XI.10 Proposition 11.5.1 (2017: 11.6): auxiliary norms

The orbit map on $|W[1]|$. $W_{0,1}$ with the morphisms of W has orbit space $|W[1]|$; the map $|W[1]| \to |W|$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.oneToOrbit is injective Chapter11/StrongBundleUpToEquivalence · SBLocalization.oneToOrbit_injective. A strong bundle functor induces a continuous map on $|W[1]|$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.oneMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.continuous_oneMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.oneToOrbit_oneMap, and a diagram with both legs strong bundle equivalences the map $|D| = |\Psi| \circ |\Phi|^{-1}$ on $|W[1]|$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.oneOrbitMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.continuous_oneOrbitMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.oneOrbitMap_oneMap Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.oneToOrbit_oneOrbitMap, determined by $|D|$ on $|W|$ because $|W[1]| \to |W|$ is injective Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.oneOrbitMap_eq_of_orbitMap_eq.

Proposition 11.5.1 (V11.30) Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.

proposition_11_5_1. A function on $W_{0,1}$ is invariant if it is constant along morphisms Chapter11/StrongBundleUpToEquivalence · SBLocalization.IsOneInvariant Chapter11/StrongBundleUpToEquivalence · SBLocalization.isOneInvariant_iff Chapter11/StrongBundleUpToEquivalence · SBLocalization.isOneInvariant_const; auxiliary norms are, by condition (3) Chapter11/StrongBundleUpToEquivalence · EpStrongBundle.IsAuxNorm.isOneInvariant. *Step 1, the definition.* $[D]^*N'(e) = N'(e')$ for a representative e' of $|D|(|e|)$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushNorm; for invariant N' it does not depend on the representative Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_eq and equals the text's $N'(\Psi(e''))$ for every e'' with $\Phi(e'') \cong e$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_eq_Ψ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_eq_lift. *Step 2, the properties.* It is invariant Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.isOneInvariant_pullNorm and continuous, as the map induced by N' on $|W'[1]|$ composed with $|D|$ and the quotient map Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.continuous_pullNorm; the two are inverse Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushNorm_pullNorm Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_pushNorm and depend only on the orbit map Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullNorm_eq_of_orbitMap_eq Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushNorm_eq_of_orbitMap_eq. *Step 3, classes.* Hence well defined on generalized strong bundle isomorphisms Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.pullNorm Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.pushNorm. *Step 4, conditions (1) and (2) of Definition 8.3.5* (G.132) Chapter11/AuxiliaryNormTransport2021 · SBLocalization.GenSBMap.proposition_11_5_1_book, the statement as printed with no added hypothesis; the text calls it a consequence of the definitions. (1) On a fibre of $W_{0,1}$, $[D]^*N'$ is N' composed with linear isomorphisms: the action $\mu(\sigma,\cdot)$ between fibres at level 1 Chapter11/AuxiliaryNormTransport2021 · SBLocalization.oneFiberMor and a strong bundle equivalence on fibres Chapter11/AuxiliaryNormTransport2021 · SBLocalization.oneFiberObj. A complete norm stays complete along a linear isomorphism Chapter11/AuxiliaryNormTransport2021 · StrongBundle.IsCompleteFiberNorm.of_linearEquiv, so the norm is complete Chapter11/AuxiliaryNormTransport2021 · SBLocalization.SDiagram.isCompleteFiberNorm_pullNorm. (2) For h_k with $P(h_k) \to x$ and $[D]^*N'(h_k) \to 0$: fix $\sigma : \phi(x'') \to x$, a local inverse τ of the target map at σ and a local inverse of ϕ at x'', and lift h_k to e''_k over $g(s(\tau(P(h_k))))$. Condition (2) for N' gives $\Psi(e''_k) \to 0$. A strong bundle equivalence is a local homeomorphism of $W[1]$ near 0 (Definition 10.4.1 (2)) Chapter11/AuxiliaryNormTransport2021 · SBLocalization.tendsto_zeroOne_of_tendsto_obj, so $e''_k \to 0$, and $h_k = \mu(\tau(\cdot), \Phi(e''_k)) \to 0$ Chapter11/AuxiliaryNormTransport2021 · SBLocalization.SDiagram.tendsto_zeroOne_pullNorm. Hence $[D]^*N'$ is an auxiliary norm Chapter11/AuxiliaryNormTransport2021 · SBLocalization.SDiagram.isAuxNorm_pullNorm, and $[D]_*N$ is one as the pull-back along the reversed diagram Chapter11/AuxiliaryNormTransport2021 · SBLocalization.SDiagram.isAuxNorm_pushNorm. The sentence of p. 547 deriving $[D] = [D']$ from the local regularity property is Theorem 10.5.7, which does not hold as we read it (Chapter 10); Step 2 needs only $|[D]| = |[D']|$.

XI.11 Lemma 11.5.2 and Proposition 11.5.3 (2017: 11.7 and 11.7): multisection functors

Functors on W (V11.31) Chapter11/StrongBundleUpToEquivalence · SBLocalization.IsWFunctor Chapter11/StrongBundleUpToEquivalence · SBLocalization.isWFunctor_const. A map Λ on the total space constant along $\mu(\varphi, w)$ takes equal values over isomorphic objects Chapter11/StrongBundleUpToEquivalence · SBLocalization.IsWFunctor.eq_of_π_eq. $\Psi^*\Lambda' = \Lambda' \circ \Psi$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.pullMult

Chapter11/StrongBundleUpToEquivalence · SBLocalization.IsWFunctor.pullMult. **Lemma 11.5.2 (2017: 11.7)** Chapter11/StrongBundleUpToEquivalence · SBLocalization.lemma_11_5_2: for a strong bundle equivalence, $\Phi_*\Lambda(w')$ is Λ at a chosen w with $\Phi(w) \cong w'$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.π_obj_choose, equal for every such w Chapter11/StrongBundleUpToEquivalence · SBLocalization.pushMult_eq, a functor; with the inverse laws Chapter11/StrongBundleUpToEquivalence · SBLocalization.pullMult_pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.pushMult_pullMult.

Proposition 11.5.3 (2017: 11.7) (V11.32) Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.proposition_11_5_3. On diagrams $[D]_* = \Psi_*\Phi^*$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushMult and $[D]^* = \Phi_*\Psi^*$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullMult; $[D]_*\Lambda(w') = \Lambda(w)$ whenever $|D|(|w|) = |w'|$ Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushMult_eq Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullMult_eq, so everything depends on $|D|$ only: functors go to functors Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.isWFunctor_pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.isWFunctor_pullMult, the two are inverse Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullMult_pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushMult_pullMult and agree for equal orbit maps Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pushMult_eq_of_orbitMap_eq Chapter11/StrongBundleUpToEquivalence · SBLocalization.SDiagram.pullMult_eq_of_orbitMap_eq; on classes Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.pushMult Chapter11/StrongBundleUpToEquivalence · SBLocalization.GenSBMap.pullMult. The text's "generalized strong bundle maps" in the last clause is read as generalized strong bundle isomorphisms, for which $[D]_*$ is defined.

sc⁺-multisections: Lemma 11.5.2 and Proposition 11.5.3 as stated (V11.31, V11.32) Chapter11/MultisectionLocalStructure2021 · SBLocalization.lemma_11_5_2_book Chapter11/MultisectionLocalStructure2021 · SBLocalization.GenSBMap.proposition_11_5_3_book, with no added hypothesis (G.133). *Definition 13.2.1.* An sc⁺-multisection is a functor $\Lambda : W \to \mathbb{Q}^+$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsScPlusMultisection with a local section structure around every x_0: on the domain $U(x_0)$ of a natural representation, sc⁺-sections $s_1, \dots, s_k$ with positive rational weights summing to 1 and $\Lambda(w) = \sum_{i:w=s_i(P(w))} \sigma_i$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.HasLocalSectionStructure. Such a representation on any open neighbourhood Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn gives a local section structure, by shrinking to a natural representation (Theorem 7.1.13). The weights and values are in $\mathbb{Q}_{\geq 0}$, which contains the value 0 off the sections. *Step 1, pull-back.* $\Lambda' \circ \Phi$ has the local sections $\Phi_U^{-1} \circ s_i \circ \phi$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.isScPlusMultisection_pullMult. *Step 2, push-forward, the text's construction.* Around x', with $\sigma' : \phi(x) \to x'$, $s_i'(y') = \mu'(\sigma', \Phi(s_i(x)))$ with σ' extended by the local inverse τ of t' at σ' and the local inverse g of ϕ. The sections are defined on $O' = \{y' : s(\tau(y')) \in \phi(U),\ g(s(\tau(y'))) \in O\}$ and take arbitrary values off O' Chapter11/MultisectionLocalStructure2021 · SBLocalization.exists_isLocalRepOn_pushMult. *Step 3, diagrams and classes.* $\Psi_*\Phi^*$ and $\Phi_*\Psi^*$ Chapter11/MultisectionLocalStructure2021 · SBLocalization.SDiagram.isScPlusMultisection_pushMult Chapter11/MultisectionLocalStructure2021 · SBLocalization.SDiagram.isScPlusMultisection_pullMult, with the rest of Proposition 11.5.3 from the previous paragraph. *Witness*, the multisection of the zero section Chapter11/MultisectionLocalStructure2021 · SBLocalization.isScPlusMultisection_zeroMultisection.

XI.12 Lemmas 11.6.1–11.6.3 (2017: Lemmas 11.8, 11.9): the fibres of a proper covering

Lemma 11.6.1 (V11.33) `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.lemma_11_6_1`. For a proper covering functor $F : Y \to X$, $\alpha : \psi \mapsto (F(\psi), s(\psi))$ is an sc-diffeomorphism onto the fibred product (Definition 8.5.1), so every (φ, y) with $F(y) = s(\varphi)$ has a unique lift `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.lift` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.mor_lift` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.src_lift` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.eq_lift` over $t(\varphi)$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.obj_tgt_lift`; lifts with prescribed target are unique as well `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.existsUnique_lift_tgt`. Surjectivity on morphisms comes from surjectivity on objects, and the local sc-diffeomorphism property from α and the source map.

The action and Lemma 11.6.2 (2017: 11.8) (V11.34) `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.lemma_11_6_2`. The fibre Y_x `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.Fiber` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.Fiber.val` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.Fiber.mk` carries $g * y = t(\alpha^{-1}(g, y))$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.fiberSMul` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.fiberSMul_val_of`, an action ((11.15)) `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.instMulActionFiber` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.smul_val_of`: $\alpha^{-1}(g_2 g_1, y)$ and $\alpha^{-1}(g_2, g_1 * y) \circ \alpha^{-1}(g_1, y)$ have the same image under α. In the text's display of $\alpha(\psi_2 \circ \psi_1)$ the second factor is printed $(F(\psi_2), s(\psi_1))$; we read $(F(\psi_1), s(\psi_1))$ (possible misprint). The full subcategory $Y(x)$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.FiberFull` and the translation groupoid `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.Translation` are related by the functor that is the identity on objects and α on morphisms `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.toTranslation`, bijective on objects, full and faithful `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.bijective_toTranslation_obj` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.full_toTranslation` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.faithful_toTranslation`: the "natural identification".

Lemma 11.6.3 (2017: 11.9) (V11.35) `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.lemma_11_6_3`. $B_\varphi(y) = t(\mathrm{lift}(\varphi, y))$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.transportFiber` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.transportFiber_val_of`, with inverse $B_{\varphi^{-1}}$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.transportFiber_inv_transportFiber` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.bijective_transportFiber`, equivariant along conjugation `Chapter7/Basic · EpGroupoid.conjIsotropy` `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.transportFiber_smul`, by uniqueness of lifts. $\tilde{\varphi}$ `Chapter11/CoveringEquivalence · EpFunctor.IsProperCovering.translationMap` is bijective on objects, full and faithful.

XI.13 Definitions 11.6.4 and 11.6.5 (2017: 11.6 and 11.7)

Definition 11.6.4 (2017: 11.6) (V11.36) `Chapter11/CoveringEquivalence · ProperCoveringHom` `Chapter11/CoveringEquivalence · ProperCoveringHom.ext`. Ψ maps fibres into fibres `Chapter11/CoveringEquivalence · ProperCoveringHom.mapsTo_fiber`; identities and composites `Chapter11/CoveringEquivalence · ProperCoveringHom.id` `Chapter11/CoveringEquivalence · ProperCoveringHom.comp`. (2): the fibre condition `Chapter11/CoveringEquivalence · ProperCoveringHom.IsFiberBijective` and equivalences `Chapter11/CoveringEquivalence · ProperCoveringHom.IsEquivalence` `Chapter11/CoveringEquivalence · ProperCoveringHom.`

IsEquivalence.id Chapter11/CoveringEquivalence · ProperCoveringHom.IsEquivalence.comp. *The fibre condition is redundant* Chapter11/CoveringEquivalence · ProperCoveringHom.isFiberBijective_of_isEquivalence Chapter11/CoveringEquivalence · ProperCoveringHom.isEquivalence_iff. *Injectivity.* If $\Psi(y_1) = \Psi(y_2)$ with $F(y_1) = F(y_2) = x$, fullness of Ψ gives $\eta : y_1 \to y_2$ with $\Psi(\eta) = 1$; then $\Phi(F(\eta)) = F'(\Psi(\eta)) = 1 = \Phi(1_x)$ and faithfulness of Φ give $F(\eta) = 1_x$, so η and 1_{y_1} lift the same $(1_x, y_1)$ and $y_2 = y_1$. *Surjectivity.* For $y' \in Y'_{\Phi(x)}$, essential surjectivity gives y and $\eta' : \Psi(y) \to y'$; $F'(\eta') : \Phi(F(y)) \to \Phi(x)$ is $\Phi(\varphi)$ for some $\varphi : F(y) \to x$ by fullness; then $\Psi(\mathrm{lift}(\varphi, y))$ and η' lift the same morphism with the same source, so $\Psi(B_\varphi(y)) = y'$. *Witness.* The functor from the pair groupoid of $\mathbb{Z}$ to the point, between the identity coverings Chapter11/CoveringEquivalence · CoveringEquivalence.Examples.pairToPointCovering, is an equivalence Chapter11/CoveringEquivalence · CoveringEquivalence.Examples.isEquivalence_pairToPointCovering.

Definition 11.6.5 (2017: 11.7) (V11.37) Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans Chapter11/CoveringEquivalence · ProperCoveringHom.NaturallyEquivalent. (11.20) objectwise; the square after the definition Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.naturality_F. Identity, inverse and composite Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.refl Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.symm Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.trans, whiskerings Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.whiskerLeft Chapter11/CoveringEquivalence · ProperCoveringHom.NatTrans.whiskerRight; natural equivalence is an equivalence relation Chapter11/CoveringEquivalence · ProperCoveringHom.naturallyEquivalent_equivalence compatible with composition Chapter11/CoveringEquivalence · ProperCoveringHom.NaturallyEquivalent.comp_left Chapter11/CoveringEquivalence · ProperCoveringHom.NaturallyEquivalent.comp_right. Witness Chapter11/CoveringEquivalence · CoveringEquivalence.Examples.naturallyEquivalent_pairToPointCovering.

XI.14 Theorem 11.6.6 (2017: 11.10)

The functor H (V11.38) Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hObj Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hMor Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hFunctor. On morphisms (11.19), with sources and targets checked through the commutation equations Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.mor_conj Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hMor_src Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.hMor_tgt. Assume Ψ and Φ are local sc-diffeomorphisms on objects and Ψ maps each Y_x onto $Y'_{\Phi(x)}$. *Step (A), objects.* Surjectivity: for (x, φ', x'') lift x'' to y'', then $\varphi' : \Phi(x) \to \xi(x'')$ along F' to ψ' with target $\zeta(y'')$; the source of ψ' lies over $\Phi(x)$, so it is $\Psi(y)$ for some $y \in Y_x$ by the hypothesis on fibres Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.surjective_hObj. The local sc-diffeomorphism property: the projections $P \to Y''$ and $Q \to X''$ are local sc-diffeomorphisms on objects because Ψ and Φ are, F'' is one, and $\pi_{X''} \circ H = F'' \circ \pi_{Y''}$, so H is one by two out of three Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.isLocalScDiffeo_hObj. The text's heading of (A) has P and Q in the reverse order (possible misprint). *Step (B), α.* The map $\psi \mapsto (H(\psi), s(\psi))$ for H is an sc-diffeomorphism onto the fibred product, its inverse built from those of F and F'' Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.isScDiffeo_fibredMap_hFunctor. *Step (C), finite fibres and properness.* A fibre of H sits in the product of fibres of F and F'' Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.finite_fiber_hObj; a proper covering is proper on objects Chapter11/CoveringEquivalenceTheorem · EpFunctor.IsProperCovering.isProperMap, and so is H Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.isProperMap_hObj; a proper local sc-diffeomorphism is evenly covered near a point with finite fibre, by disjoint sheets around the points of the fibre and the closedness

of proper maps Chapter11/CoveringEquivalenceTheorem · MPolyfold.IsLocalScDiffeo.evenlyCovered_of_isProperMap, the sheets cut down to the neighbourhood found Chapter11/CoveringEquivalenceTheorem · MPolyfold.IsScDiffeoOn.restrictTarget. The properness of H is an ultrafilter argument from the properness of F and F'' on objects and the continuity of the inverse of the map α of F'; it replaces the text's argument with sequences. **(1)** Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.isProperCovering_hFunctor_of.

Theorem 11.6.6 (V11.39) Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.theorem_11_6_6. An equivalence (Ψ, Φ) satisfies the hypotheses of (1), the fibre condition by §XI.13. *Step (D), (2).* $\pi_{Y''}$ and $\pi_{X''}$ are equivalences Chapter10/WeakFibredProductTheorem · WeakFibredProduct.isEquivalence_projRight, so the pair Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.projRightHom is an equivalence by §XI.13. *Step (E), (3).* $\tau_P(y, \psi', y'') = \psi'$ and $\tau_Q(x, \varphi', x'') = \varphi'$ satisfy (11.20) Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.natTransProj. *Step (F), tameness.* $P \to Y''$ and $Q \to X''$ are surjective local sc-diffeomorphisms on objects, and $Y'' \to X''$ is one; tameness pulls back along local sc-diffeomorphisms Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isTame. If (ζ, ξ) is an equivalence, the other projection Chapter11/CoveringEquivalenceTheorem · ProperCoveringHom.projLeftHom is one by symmetry. Witness: the equivalence of §XI.13 and the identity of the point Chapter11/CoveringEquivalenceTheorem · CoveringEquivalence.Examples.theorem_11_6_6_pairToPoint, with nonempty fibres Chapter11/CoveringEquivalence · CoveringEquivalence.Examples.nonempty_fiber_id.

XI.15 The category $\mathcal{PC}$, Definition 11.6.7, Proposition 11.6.8 (2017: 11.8, 11.8) and Remark 11.6.9 (2017: 11.4)

$\mathcal{PC}$ (V11.40) Chapter11/CoveringEquivalenceDiagram · PC Chapter11/CoveringEquivalenceDiagram · ProperCoveringHom.comp_assoc Chapter11/CoveringEquivalenceDiagram · PC.equivalences. The weak fibred product as an object, with its projections Chapter11/CoveringEquivalenceDiagram · PC.fibred Chapter11/CoveringEquivalenceDiagram · PC.fibredFst Chapter11/CoveringEquivalenceDiagram · PC.fibredSnd. **Definition 11.6.7 (2017: 11.8)** Chapter11/CoveringEquivalenceDiagram · PC.Diagram Chapter11/CoveringEquivalenceDiagram · PC.Diagram.CommonRefinement Chapter11/CoveringEquivalenceDiagram · PC.Diagram.HasCommonRefinement. Witnesses: the pair and the point Chapter11/CoveringEquivalenceDiagram · PC.Examples.pair Chapter11/CoveringEquivalenceDiagram · PC.Examples.point Chapter11/CoveringEquivalenceDiagram · PC.Examples.equivalences_pairToPoint, the diagrams pair $\leftarrow$ pair $\rightarrow$ point Chapter11/CoveringEquivalenceDiagram · PC.Examples.diagram and pair $\leftarrow$ pair $\times_{\text{point}}$ pair $\rightarrow$ point Chapter11/CoveringEquivalenceDiagram · PC.Examples.fibredPair Chapter11/CoveringEquivalenceDiagram · PC.Examples.isEquivalence_fibredFst Chapter11/CoveringEquivalenceDiagram · PC.Examples.diagram'.

Proposition 11.6.8 (2017: 11.8) (V11.41) Chapter11/CoveringEquivalenceDiagram · PC.proposition_11_6_8. Given (H, H') for d, d' and (K, K') for d', d'', take the weak fibred product $A'' \times_{A'} B''$ of H' and K (Theorem 11.6.6) with the equivalences $H \circ \pi_{A''}$ and $K' \circ \pi_{B''}$. For the Γ's: $\Gamma \circ H \circ \pi_{A''} \simeq \Gamma' \circ H' \circ \pi_{A''} \simeq \Gamma' \circ K \circ \pi_{B''} \simeq \Gamma'' \circ K' \circ \pi_{B''}$, the middle step by Theorem 11.6.6 (3). For the ε's the text's chain ends at $\varepsilon' \circ K \circ \pi_{B''}$; the last step to $\varepsilon'' \circ K' \circ \pi_{B''}$ uses $\varepsilon' \circ K \simeq \varepsilon'' \circ K'$ and is added. Reflexivity and symmetry are immediate Chapter11/CoveringEquivalenceDiagram · PC.Diagram.hasCommonRefinement_equivalence. Witnesses Chapter11/CoveringEquivalenceDiagram · PC.Examples.hasCommonRefinement_diagram Chapter11/CoveringEquivalenceDiagram · PC.Examples.hasCommonRefinement_diagram_diagram.

The localization and Remark 11.6.9. $\mathcal{PC}(E^{-1})$ and I, whose composition the text leaves to the reader, are §XI.21 (V11.43; G.134). Remark 11.6.9 (V11.42) recalls Definition 8.5.9 Chapter8/Book2021 · EpStrongBundle.

`IsProperStrongBundleCovering2021` and sketches the category of proper strong bundle coverings and its localization, which is §XI.22 (G.135).

XI.16 Witnesses and the hypotheses

Hypotheses taken. *§XI.8.* Chapter 9's independence of the tangent sums of Theorem 9.2.1 was assumed for the branched ep⁺-subgroupoid being moved and proved for its images. Chapter 9 now proves it for every Θ `Chapter9/GeneralCorollaries · EpGroupoid.hasIndependentTangentSums`, so it is no longer a restriction; it holds on $\mathbb{R}^n$ `Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_rn`. The boundary-transfer hypothesis of Theorem 11.3.9 for a representative holds along the identity, on every M-polyfold and on $\mathbb{R}^2$. Theorem 11.3.9 for the intrinsic boundary orientation needs no hypothesis. The boundary-lift form of Theorem 11.3.9 asks that f carry tame tangent sets and that $f_*\widehat{\Theta}$ be tame; both hold along the identity and for Example 9.3.5. *§XI.9.* The first forms of Theorems 11.4.4 and 11.4.5 carry the class hypothesis `hcov` of Chapter 9; it is derived from the text's hypotheses (§IX.36), and the forms with `hloc` take none beyond the text's. The local regularity property in Propositions 11.2.8, 11.2.9 and the local unique continuation property in Theorems 11.4.2, 11.4.5 are the text's. *Elsewhere.* The branched statements of §11.3 and the category $\mathcal{PC}$ take their ep-groupoids in one universe; the tangent functor on $\mathcal{EP}(E^{-1})$ lives where Chapter 8's T does. In Remark 11.6.9 the class E_{SBPC}, which the text does not define, is read as "both components strong bundle equivalences" (§XI.22). Several statements need less than the text assumes: Lemma 11.3.2 (§XI.5), Proposition 11.1.6 (§XI.3), Theorem 11.6.6 (1) (§XI.14), and Proposition 11.2.8 with the property on one side (§XI.4).

Every definition has a witness. Faces and (weak) face structure: the quadrant with and without the swap (§XI.2). Push-forwards, pull-backs, branched ep⁺-subgroupoids along an equivalence that is not injective on objects and along a generalized isomorphism: $1_{\{0\}}$ on $\mathbb{Z}/2 \ltimes \mathbb{R}$ (§XI.7). Section functors and oriented branched ep⁺-subgroupoids: Example 9.3.5 on $\mathbb{R}^2$ (§XI.8). Auxiliary norms and multisection functors: the constant functions (§§XI.10, XI.11). Functors, equivalences and natural transformations of proper coverings, Theorem 11.6.6, diagrams and common refinements: the pair groupoid of $\mathbb{Z}$ over the point (§§XI.13–XI.15). Two faces meeting in more than one face: $([0,\infty)^3 \oplus W) \setminus K$ (§XI.20). The local regularity property in Propositions 11.2.8 and 11.2.9: the point and $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (§XI.4). The class hypothesis of Theorems 11.4.4 and 11.4.5: compact boundary sheets and the closed upper hemisphere (§XI.9). sc⁺-multisections: the zero section (§XI.11). $\mathcal{PC}(E^{-1})$: pair $\to$ point (§XI.21). Proper strong bundle coverings and their weak fibred products: $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ (§XI.22).

XI.17 What is not formalized

All eight gaps of Chapter 11 are closed.

- Closed: §11.4 (G.131, §XI.9). Theorems 11.4.1, 11.4.2 and Remark 11.4.3 are proved with no added hypothesis, and Theorems 11.4.4 and 11.4.5 with the text's hypotheses for every $n \geq 1$.
- Closed: §11.1, Proposition 11.1.6 with the second claim of Proposition 11.1.8 (G.128, §§XI.3, XI.19, XI.20; the intersection clause does not hold as we read it, and holds in a union form); §11.2 (G.129, §XI.4); §11.3, Theorem 11.3.9 for the intrinsic boundary orientation (G.130, §XI.8); §11.5, conditions (1) and (2) in Proposition 11.5.1 (G.132, §XI.10) and the local section structures of sc⁺-multisections (G.133, §XI.11); §11.6, $\mathcal{PC}(E^{-1})$ and I (G.134, §XI.21) and the category of Remark 11.6.9 with its localization (G.135, §XI.22).

- Within closed items, the boundary-lift form of Theorem 11.3.9 keeps two hypotheses that the intrinsic form does not need: that sc-diffeomorphisms carry tame induced structures on submanifolds to tame induced structures, and that $f_*\widehat{\Theta}$ is tame. The ordered face structures of Remark 11.1.5 have no formal content.

XI.18 What the formalization taught us

Most of Chapter 11 is Chapter 10's machinery applied to one more kind of object, and the formal work went into the two places where an object is carried along an equivalence by a choice. For functors Θ the choice of x and $F(x) \to y$ does not matter as soon as F is full, which is all that the well-definedness in Lemma 11.3.2 uses; the statements for generalized maps then reduce, through one base change formula in the weak fibred product, to statements for equivalences, and the pull-back is defined for every generalized map. For branched ep$^+$-subgroupoids the text's proof moves a local branching structure by $\widehat{\psi} \circ F$; written out, the sheets must first be cut down to the domain of that local sc-diffeomorphism, and the paracompactness of the support, part of 2021 Definition 9.1.2, must be carried along too. Theorem 11.3.3 (6) refers to the uniqueness of dimensional decompositions, which does not hold as we read it (Chapter 9); the statement for every decomposition is what the proof gives. The rest of §11.3, whose proofs the text leaves to the reader, needed Chapter 9's notions between two M-polyfolds rather than on one, Chapter 9's independence hypothesis carried along, and, for Theorem 11.3.9, a condition on the boundary operations that Chapter 9 takes as data. Faces are a question about $|X|$ alone: once the faces of $|X|$ are Chapter 2's faces for $d_{|X|}$, Remark 11.1.5 and the bijections of Proposition 11.1.6 are statements about homeomorphisms preserving an index, and hold for every generalized isomorphism; only the ep-groupoid structure of a face needs more. In §11.6 the fibre condition of Definition 11.6.4 (2) comes for free for proper coverings, by unique lifting; and the chain of natural equivalences in Proposition 11.6.8 needs one more step than it displays.

What the closing of the gaps shows. Proposition 11.1.6 splits: its first and last claims hold, the faces being tame ep-groupoids because near each of its points X_θ is one face of the object M-polyfold, and the faces of a generalized isomorphism being related by the restricted diagram (11.1); but two faces can meet in several components, so the intersection is a union of faces of each, not one face. In §11.2 the forms behave best when F_* is defined as the inverse of F^*: the bijectivity of F^* is a descent along the projection $X \times_Y Y \to Y$, which is a surjective local sc-diffeomorphism, and the text's formula, composition rules, naturality and independence of the representative all follow by inverting statements about pull-backs. Theorem 11.3.9 needed no condition once the boundary orientation is taken intrinsically: Definition 9.3.8's operation is built from presented charts, which sc-diffeomorphisms transport, and "$d_M(x)$ is preserved" is read for the reduced codimension, which needs no tameness. §11.4 follows the text's proof, structures moved along transport charts and the sums (11.12), (11.13) compared, with two additions: $|f|$ preserves the σ-algebra $\mathcal{L}$, which the text uses silently, and in the effective case $\sharp G^{eff}$ is preserved because the transport isomorphism carries T_x onto T_y. The boundary versions first inherited Chapter 9's class condition near corners; once Chapter 9 derived it, and covered $n = 1$, they hold with the text's hypotheses. In §11.5 conditions (1) and (2) need a local lift of a sequence through local inverses of t and ϕ, and the push-forward of local section structures needs the explicit domain on which the text's sections s_i' are defined. In §11.6 the localizations of $\mathcal{PC}$ and of proper strong bundle coverings follow §10.3 once Theorem 11.6.6 is stated with the equivalence on the left; for strong bundles the weak fibred product is a pull-back, and conditions (2) and (3) of Definition 8.5.9 come for free over a proper covering.

XI.19 Proposition 11.1.6 (2017: 11.2): faces as ep-groupoids, and Proposition 11.1.8 (2017: 11.3) for $[d]_F$

Faces are tame ep-groupoids (V11.6) `Chapter11/FaceEpGroupoids · EpGroupoid.proposition_11_1_6_faces`. The text: for a face structured tame X, the object and morphism sets of a face X_θ are sub-M-polyfolds with tame induced structures, and X_θ is an ep-groupoid. The proof (pp. 524–525) passes through the faces of the object M-polyfold, and so does the formal one. *Step 1, local pieces.* π maps the face of the object M-polyfold through p into the face of $|X|$ through $|p|$ `Chapter11/FaceEpGroupoids · EpGroupoid.image_faceAt_subset`. For the local pieces of $\{d_X = 1\}$ at x, the faces of $|X|$ through $|x|$ are those through the images of the pieces `Chapter11/FaceEpGroupoids · EpGroupoid.orbitFacesThrough_eq_range`, and for face structured X the map is injective `Chapter11/FaceEpGroupoids · EpGroupoid.injective_orbitFaceAt_pt`. This is the text's count: if x lay in fewer global faces of the object M-polyfold, fewer faces of $|X|$ would pass through $|x|$. So the object M-polyfold is weakly face structured `Chapter11/FaceEpGroupoids · EpGroupoid.IsEpFaceStructured.isWeaklyFaceStructured`. *Step 2, one face near each point.* Every $x \in X_\theta$ lies in exactly one face of the object M-polyfold contained in X_θ `Chapter11/FaceEpGroupoids · EpGroupoid.IsEpFaceStructured.existsUnique_face_subset`, and near x, X_θ is that face `Chapter11/FaceEpGroupoids · EpGroupoid.IsEpFaceStructured.exists_nhds_eq_face`. The text's "disjoint union" is read in this local form, which is what is used. *Step 3, sub-M-polyfold.* A set that near each of its points agrees with a sub-M-polyfold is one `Chapter11/FaceEpGroupoids · MPolyfold.isSubMPolyfold_of_forall_nhds`, and faces are sub-M-polyfolds with tame induced structure (Theorem 2.5.17); so X_θ is a closed `Chapter11/FaceEpGroupoids · EpGroupoid.isClosed_faceObj` saturated sub-M-polyfold `Chapter11/FaceEpGroupoids · EpGroupoid.isSubMPolyfold_faceObj`. *Step 4, the ep-groupoid.* With the morphisms between its points it is an ep-subgroupoid `Chapter11/FaceEpGroupoids · EpGroupoid.isEpSubgroupoid_faceObj` (Proposition 7.1.22) `Chapter11/FaceEpGroupoids · EpGroupoid.faceEpGroupoid`, tame on objects and on morphisms `Chapter11/FaceEpGroupoids · EpGroupoid.isTame_faceEpGroupoid` `Chapter11/FaceEpGroupoids · EpGroupoid.isTame_faceEpGroupoid_mor`, with $d_X(x) = d_{X_\theta}(x) + 1$ `Chapter11/FaceEpGroupoids · EpGroupoid.degIndex_faceEpGroupoid`.

The generalized isomorphisms $[d]_F$ of (11.1) `Chapter11/FaceGeneralizedIso · EPLocalization.EPLoc.proposition_11_1_6_faceIso`. The text: for $X \xleftarrow{\Phi} A \xrightarrow{\Psi} Y$ and a face F of X, $B = \Phi^{-1}(F)$ is a face of A, the saturation C of $\Psi(B)$ is a face of Y, and $F \leftarrow B \rightarrow C$ is a generalized isomorphism by Theorem 10.3.10. *Step 1, restriction.* An sc-smooth functor restricts to ep-subgroupoids $A = G^{-1}(B)$ `Chapter11/FaceGeneralizedIso · EpFunctor.restrictSub`, and so do natural transformations `Chapter11/FaceGeneralizedIso · EpNatTrans.restrictSub`. For an equivalence and saturated B the restriction is full, faithful, essentially surjective and a local sc-diffeomorphism, hence an equivalence by Lemma 10.1.2 `Chapter11/FaceGeneralizedIso · EpFunctor.IsEquivalence.restrictSub`; this is the text's argument. Face structure passes both ways along equivalences `Chapter11/FaceGeneralizedIso · EpFunctor.IsEquivalence.isEpFaceStructured_iff`. *Step 2, the diagram (11.1).* The face as an object of $\mathcal{EP}$ `Chapter11/FaceGeneralizedIso · EPLocalization.faceEP` and the restricted diagram `Chapter11/FaceGeneralizedIso · EPLocalization.Diagram.restrictFace`, with $C = Y_{|\Psi|(\theta_A)}$ (Remark 11.1.5). Refinements and equivalent diagrams restrict to refinements and equivalent diagrams `Chapter11/FaceGeneralizedIso · EPLocalization.Diagram.IsRefinement.restrictFace` `Chapter11/FaceGeneralizedIso · EPLocalization.Diagram.Equiv.restrictFace`, and $|d_F|$ is the restriction of $|d|$ `Chapter11/FaceGeneralizedIso · EPLocalization.Diagram.orbitMap_restrictFace`. *Step 3, the class.* $[d]_F$ `Chapter11/FaceGeneralizedIso · EPLocalization.EPLoc.faceIso` is the class of the restriction of every representative `Chapter11/FaceGeneralizedIso · EPLocalization.EPLoc.homMk_restrictFace_eq_faceIso`. It is a generalized isomorphism `Chapter11/FaceGeneralizedIso · EPLocalization.EPLoc.isGeneralizedIso_faceIso` whose orbit map is the restriction of $|[d]|$ `Chapter11/FaceGeneralizedIso · EPLocalization.EPLoc.orbitMap_faceIso`. The target face is

passed as a set with a proof that it is $f_*\theta$, so that $[f]_F$ and $[g]_F$ have the same type when $f_* = g_*$.

Proposition 11.1.8, the second claim (V11.8) `Chapter11/FaceGeneralizedIso · EPLocalization.EPLoc.proposition_11_1_8_faceIso`. If $|f| = |g|$, then $f_* = g_*$ (§XI.3), and the generalized isomorphisms $F \to f_*F$ and $F \to g_*F$ induce the same map $|F| \to |f_*F|$, both being restrictions of $|f| = |g|$.

XI.20 Proposition 11.1.6 (2017: 11.2): the intersection clause

The clause as printed does not hold as we read it (V11.6) `Chapter11/FaceIntersectionCounterexample · EpGroupoid.FacesMeetInFaces` `Chapter11/FaceIntersectionCounterexample · FaceMeetExample.not_facesMeetInFaces`. The text: if two faces F, G meet, $F \cap G$ is a face in F as well as in G. We read "face in F" in the ep-groupoid X_θ of §XI.19. *The example.* An open subset of a tame M-polyfold, with Part I's open-subset structure, has the same degeneracy index, is tame, and is face structured when the open set is `Chapter11/FaceIntersectionCounterexample · MPolyfold.isTame_restrictOpen` `Chapter11/FaceIntersectionCounterexample · MPolyfold.isFaceStructuredOpen_restrictOpen`; its trivial ep-groupoid is then face structured `Chapter11/FaceIntersectionCounterexample · EpGroupoid.isEpFaceStructured_trivial_restrictOpen`. Take $V = ([0,\infty)^3 \oplus W) \setminus K$ with $K = \{p_0 + p_1 \le 1/4,\ 1 \le p_2 \le 2\}$, a closed block cutting the edge $p_0 = p_1 = 0$ at heights $[1,2]$ `Chapter11/FaceIntersectionCounterexample · FaceMeetExample.V` `Chapter11/FaceIntersectionCounterexample · FaceMeetExample.isEpFaceStructured_XV`. *Step 1.* $\theta_0 = \{p_0 = 0\}$ and $\theta_1 = \{p_1 = 0\}$ are distinct faces `Chapter11/FaceIntersectionCounterexample · FaceMeetExample.isOrbitFace_θ₀` `Chapter11/FaceIntersectionCounterexample · FaceMeetExample.θ₀_ne_θ₁`, connected by paths in them `Chapter11/FaceIntersectionCounterexample · FaceMeetExample.mem_face_of_path`. *Step 2.* They meet in the two pieces $p_2 \in [0,1)$ and $p_2 \in (2,\infty)$ of the edge. A face of X_{θ_0} is the closure of one connected component of its index-one set, and the continuous function p_2 takes values below 1 or above 2 on the edge `Chapter11/FaceIntersectionCounterexample · FaceMeetExample.cF_lt_or_gt`, so the intersection is not one face `Chapter11/FaceIntersectionCounterexample · FaceMeetExample.not_isEpFace_inter`. With $W = 0$ this is a 3-manifold with boundary and corners.

The union form `Chapter11/FaceIntersectionUnion · EpGroupoid.IsEpFaceStructured.faceObj_inter_eq_sUnion` `Chapter11/FaceIntersectionUnion · EpGroupoid.IsEpFaceStructured.facesMeetInUnions`. For distinct faces $\theta \ne \theta'$ of $|X|$, $X_\theta \cap X_{\theta'}$ is the union of the faces of X_θ it contains, and likewise for $X_{\theta'}$ `Chapter11/FaceIntersectionUnion · EpGroupoid.FacesMeetInUnionsOfFaces`. *Step 1, a ray in a tame chart.* A point x of two distinct faces is a limit of points of $F_1 \cap F_2$ with $d_X = 2$: in a tame chart the faces are those of two distinct indices $i \ne j$ of Lemma 4.C, and the points $r(o+ty)$ with $\lambda_i(y) = \lambda_j(y) = 0$ and $\lambda_l(y) = 1$ otherwise have index 2 and lie in both closures `Chapter11/FaceIntersectionUnion · MPolyfold.mem_closure_inter_faces`. *Step 2.* Near a point $y \in X_{\theta'}$ with $d_X(y) = 2$, every point with $d_X = 2$ lies in $X_{\theta'}$ `Chapter11/FaceIntersectionUnion · EpGroupoid.IsEpFaceStructured.exists_nhds_degIndex_two`. *Step 3.* Those points lie in a local piece of $\{d_{X_\theta} = 1\}$ at x. The face of X_θ through that piece contains x, and it lies in $X_{\theta'}$ because $X_{\theta'} \cap \{d_{X_\theta} = 1\}$ is open and closed in $\{d_{X_\theta} = 1\}$.

XI.21 The localization $\mathcal{PC}(E^{-1})$ and I (2021 pp. 561–563)

$\mathcal{PC}(E^{-1})$ (V11.43) `Chapter11/ProperCoveringLocalization · PCLoc` `Chapter11/ProperCoveringLocalization · PCLoc.I`. The text sketches the localization of $\mathcal{PC}$ at E_{PC} "as in the ep-groupoid case" and leaves the details to the reader. *Step 1, maps into weak fibred products.* For functors $L : P \to X$, $R : P \to Z$ and $\alpha : F \circ L \to G \circ R$, $p \mapsto (L(p), \alpha(p), R(p))$ is a functor into $X \times_Y Z$ with $\pi_X \circ \langle L, R\rangle =$

L and $\pi_Z \circ \langle L, R\rangle = R$ on the nose Chapter11/ProperCoveringLocalization · WeakFibredProduct.pairFunctor. Its version for proper coverings, with the square (11.20) for α Chapter11/ProperCoveringLocalization · PC.fibredLift, is an equivalence by two out of three Chapter11/ProperCoveringLocalization · PC.IsEquivalence.of_comp_left Chapter11/ProperCoveringLocalization · PC.isEquivalence_fibredLift, the projection to the side of the equivalence being one (Theorem 11.6.6 (2)). *Step 2, composition.* The composite of $d : F \leftarrow A \rightarrow F'$ and $d' : F' \leftarrow A' \rightarrow F''$ is taken through the weak fibred product $A' \times_{F'} A$ Chapter11/ProperCoveringLocalization · PC.Diagram.comp. The text takes $A \times_{F'} A'$; Theorem 11.6.6 is stated here with the equivalence on the left, and the two products are isomorphic. Its class does not depend on the representatives Chapter11/ProperCoveringLocalization · PC.Diagram.hasCommonRefinement_comp: a single equivalence M with $\Gamma' \circ M \simeq \Gamma$ and $\varepsilon' \circ M \simeq \varepsilon$ gives a common refinement Chapter11/ProperCoveringLocalization · PC.Diagram.hasCommonRefinement_of_map, and Step 1 gives such maps. Associativity and the units hold up to common refinement Chapter11/ProperCoveringLocalization · PC.Diagram.hasCommonRefinement_assoc. *Step 3, I.* I is the identity on objects with $I(\varepsilon) = [F \xleftarrow{1} F \xrightarrow{\varepsilon} F']$. Equivalences go to isomorphisms with inverse $[F' \xleftarrow{\Gamma} F \xrightarrow{1} F]$ Chapter11/ProperCoveringLocalization · PCLoc.isIso_I_map Chapter11/ProperCoveringLocalization · PCLoc.inv_I_map, $[d] = I(\Gamma)^{-1} \circ I(\varepsilon)$ Chapter11/ProperCoveringLocalization · PCLoc.homMk_eq, and the inverse of $[F \xleftarrow{\Gamma} A \xrightarrow{\Gamma'} F']$ for an equivalence Γ' is the reversed class Chapter11/ProperCoveringLocalization · PCLoc.isIso_homMk Chapter11/ProperCoveringLocalization · PCLoc.inv_homMk (G.134). *Witness.* pair $\rightarrow$ point is not an isomorphism of $\mathcal{PC}$, and its image under I is one Chapter11/ProperCoveringLocalization · PCLoc.Examples.not_isIso_pairToPoint_isIso_I_map.

XI.22 Remark 11.6.9 (2017: 11.4): proper strong bundle coverings up to equivalence

The category $\mathcal{SBPC}$ (V11.42) Chapter11/ProperStrongBundleCoverings · SBPC Chapter11/ProperStrongBundleCoverings · SBPC.Hom Chapter11/ProperStrongBundleCoverings · instCategorySBPC. The remark views a strong bundle as a functor, recalls Definition 8.5.9 and describes the category of proper strong bundle coverings: its objects are pairs (A, F) with A a strong bundle functor covering a proper covering F, its morphisms commutative cubes, and its localization is left to the reader. A morphism is a pair of morphisms of $\mathcal{SEP}$ with $A' \circ a = b \circ A$, the top and bottom faces of the cube; the side faces commute because morphisms of $\mathcal{SEP}$ are strong bundle functors. There is a functor to $\mathcal{PC}$ Chapter11/ProperStrongBundleCoverings · SBPC.forget. *The class of equivalences.* The text does not define it. We take the morphisms whose two components are strong bundle equivalences Chapter11/ProperStrongBundleCoverings · SBPC.equivalences; the condition on the base face is then automatic Chapter11/StrongBundleCoveringFibredProduct · SBPC.equivalences_iff. Mathlib's localization gives $\mathcal{SBPC}[E^{-1}]$ with its universal property Chapter11/ProperStrongBundleCoverings · SBPC.Loc Chapter11/ProperStrongBundleCoverings · SBPC.isLocalization_Q Chapter11/ProperStrongBundleCoverings · SBPC.lift Chapter11/ProperStrongBundleCoverings · SBPC.uniq, with functors to $\mathcal{PC}(E^{-1})$ and to $\mathcal{SEP}(F^{-1})$ Chapter11/ProperStrongBundleCoverings · SBPC.toPCLoc Chapter11/ProperStrongBundleCoverings · SBPC.toSEPLocW Chapter11/ProperStrongBundleCoverings · SBPC.toSEPLocV. The identity covering of a strong bundle is an object Chapter11/ProperStrongBundleCoverings · SBPC.ofSEP.

Theorem 11.6.6 for strong bundles Chapter11/StrongBundleCoveringFibredProduct · SBPC.theorem_11_6_6. *Step 1, Definition 8.5.9 over a proper covering.* Condition (3) holds for every strong bundle and every proper covering: $(\psi, w) \mapsto ((F(\psi), s(\psi)), w)$ is a map of pull-backs along the sc-diffeomorphism of Lemma 11.6.1 Chapter11/StrongBundleCoveringPullback · EpStrongBundle.isStrongBundleIso_fibredLift_of_

`isProperCovering`. Condition (2) follows from the local strong bundle isomorphisms of (1) and the even covering `Chapter11/StrongBundleCoveringPullback · StrongBundle.evenlyCovered_of_isStrongBundleIsoOn`. So over a proper covering Definition 8.5.9 is (0) and (1) `Chapter11/StrongBundleCoveringPullback · EpStrongBundle.isProperStrongBundleCovering2021_of`. *Step 2, pull-backs.* The weak fibred product of strong bundles is a pull-back (Chapter 10). The pull-back $(p, w) \mapsto (H(p), A(w))$ of a proper strong bundle covering along an equivalence of proper coverings is one `Chapter11/StrongBundleCoveringPullback · SEP.pullbackCovering` `Chapter11/StrongBundleCoveringPullback · SEP.isProperStrongBundleCovering2021_pullbackCovering`. *Step 3, the theorem.* With natural transformations of morphisms of $\mathcal{SBPC}$, strong bundle natural transformations on both levels with (11.20) for the bases `Chapter11/StrongBundleCoveringFibredProduct · SBPC.NatTrans` `Chapter11/StrongBundleCoveringFibredProduct · SBPC.NaturallyEquivalent`: the weak fibred product `Chapter11/StrongBundleCoveringFibredProduct · SBPC.fibred` is a proper strong bundle covering covering H, its second projection is an equivalence, and $\Gamma \circ \mathrm{pr}_1 \simeq \Lambda \circ \mathrm{pr}_2$ `Chapter11/StrongBundleCoveringFibredProduct · SBPC.fibredNatTrans`.

$\mathcal{SBPC}[E^{-1}]$ by diagrams `Chapter11/StrongBundleCoveringLocalization · SBPCLoc`. The procedure of §XI.21, with Step 3 in place of Theorem 11.6.6: maps into weak fibred products `Chapter11/StrongBundleCoveringLocalization · SEP.wfpLift` `Chapter11/StrongBundleCoveringLocalization · SBPC.fibredLift`, diagrams and common refinements `Chapter11/StrongBundleCoveringLocalization · SBPC.Diagram`, Proposition 11.6.8 for them `Chapter11/StrongBundleCoveringLocalization · SBPC.proposition_11_6_8`, composition `Chapter11/StrongBundleCoveringLocalization · SBPC.Diagram.comp` `Chapter11/StrongBundleCoveringLocalization · SBPC.Diagram.hasCommonRefinement_comp`, and I with equivalences going to isomorphisms `Chapter11/StrongBundleCoveringLocalization · SBPCLoc.isIso_I_map` `Chapter11/StrongBundleCoveringLocalization · SBPCLoc.homMk_eq` `Chapter11/StrongBundleCoveringLocalization · SBPCLoc.inv_homMk`. Mathlib's localization maps to it compatibly with I `Chapter11/StrongBundleCoveringLocalization · SBPCLoc.fromLoc` (G.135). *Witness.* Theorem 11.6.6 for the identity covering of $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$, all conclusions, with a nonempty weak fibred product `Chapter11/StrongBundleCoveringLocalization · SBPC.Examples.theorem_11_6_6_ep`.

Part XII. Sc-Fredholm sections on ep-groupoids (HWZ Chapter 12, §§12.1–12.5)

Chapter 12 opens Part III of the book (Springer 2021, pp. 569–597; the numbers below are the 2021 ones, with the 2017 text's in parentheses). It carries the Fredholm package of Part I (Chapters 5 and 6) from strong bundles over M-polyfolds to strong bundles $(P : W \to X, \mu)$ over ep-groupoids. Everything in Part I lives on one M-polyfold; an ep-groupoid has two, the objects and the morphisms, and a strong bundle over it is a strong bundle over the objects together with an action μ of the morphisms on the fibres. The chapter asks of every Part I notion whether it can be made compatible with the morphisms, and answers by one construction, repeated: build the object on a neighbourhood $U(x)$ carrying the natural representation of the isotropy group G_x, average it over G_x, carry it to the saturation of $U(x)$ by morphisms, and glue the pieces with a partition of unity of the orbit space $|X|$. §12.1 defines sc-Fredholm section functors (Definition 12.1.1 (2017: 12.1)). §12.2 builds auxiliary norms (Definition 12.2.1, Theorem 12.2.2, Definitions 12.2.3, 12.2.4, Proposition 12.2.5, Theorems 12.2.6, 12.2.7 (2017: Definition 12.2, Theorem 12.1, Definitions 12.3, 12.4, Proposition 12.1, Theorems 12.2, 12.3)) and their reflexive variant (Definitions 12.2.8, 12.2.10, Remarks 12.2.9, 12.2.12, Theorems 12.2.11, 12.2.13 (2017: Definitions 12.5, 12.6, Remarks 12.1, 12.2, Theorems 12.4, 12.5)). §12.3 supplies sc$^+$-section functors (Definition 12.3.1, Theorems 12.3.2, 12.3.3, Remarks 12.3.4, 12.3.5, Definition 12.3.6, Proposition 12.3.7, Corollary 12.3.8, Theorem 12.3.9 (2017: Definition 12.7, Theorems 12.6, 12.7, Remarks 12.3, 12.4, Definition 12.8, Proposition 12.2, Corollary 12.1, Theorem 12.8)). §12.4 treats compactness (Definition 12.4.1, Theorem 12.4.2, Definition 12.4.3, Lemma 12.4.4, Proposition 12.4.5, Theorem 12.4.6 (2017: Definition 12.9, Theorem 12.9, Definition 12.10, Lemma 12.1, Proposition 12.3, Theorem 12.10)), and §12.5 the orientation bundle O_f (Lemma 12.5.1, Definition 12.5.2, Theorem 12.5.3, Definition 12.5.4, Remark 12.5.5, Proposition 12.5.6, Theorem 12.5.7 (2017: Lemma 12.2, Definition 12.11, Theorem 12.11, Definition 12.12, Remark 12.5, Proposition 12.4, Theorem 12.12)). The sections of this Part follow the book: §XII.1 is §12.1; §§XII.2–XII.9 are §12.2; §§XII.10–XII.13 are §12.3; §§XII.14–XII.15 are §12.4; §§XII.16–XII.22 are §12.5; §XII.23 collects the hypotheses and their witnesses, §XII.24 what is not formalized and §XII.25 what the formalization taught us.

What is proved, and what is not. Every numbered item of Chapter 12 is formalized, and no statement of Chapter 12 was found not to hold as we read it. What remains open is §12.5's inheritance of Part I's hypotheses (gap G.140) and Theorem 12.5.7 restated for Chapter 10's classes of diagrams (gap G.192).

§12.1. Definition 12.1.1 and the remark after it (regularizing on objects and on morphisms, sc-Fredholm germs) are formalized as printed (§XII.1).

§12.2. Definition 12.2.1 is Definition 8.3.5, and Theorem 12.2.2, Definitions 12.2.3 and 12.2.4 with the remark between them, Proposition 12.2.5 and the good neighbourhood Theorem 12.2.6 are proved as printed (§§XII.2–XII.5). Theorem 12.2.7, the extension of an auxiliary norm from ∂X, is proved as printed for every tame X with paracompact $|X|$ (§XII.6): its global part is the averaging–transport–gluing construction, and its local step is the norm $N'(w) = N(\text{transport of } w \text{ to } \rho(Pw)) + \|h - A(u)h\|_{F_1}$ built from a continuous retraction ρ onto the boundary. Definitions 12.2.8 and 12.2.10 are formalized, Remark 12.2.9 is proved in a stronger form (morphisms preserve mixed convergence) (§XII.7); Theorem 12.2.11 is proved as printed with its three clauses, and Remark 12.2.12 as printed, its clause (3) in a stronger form (§XII.8). Theorem 12.2.13 is proved with the text's hypotheses and with the paracompactness of $|X|$ that its proof takes from Theorem 12.2.11 made explicit; the local step the text settles by inspection is a lemma on the lower semicontinuity of N' under mixed convergence (§XII.9).

§12.3. Definition 12.3.1 is formalized as printed and in a pointwise reading. Theorem 12.3.2 (1) is proved in each of the two consistent readings, Theorem 12.3.2 (2) in the pointwise reading (§XII.10); Theorem 12.3.3 in the pointwise reading, Remarks 12.3.4 and 12.3.5 in explicit forms (§XII.11). Pairing Definition 12.3.1 as printed with Part I's pointwise bump functions would need Part I's open weak-to-pointwise question; this is a reading, not formalization work. Definition 12.3.6 is formalized in its local form and its ep-groupoid form, Proposition 12.3.7 in a stronger form (only the face germs at the point are used) and Corollary 12.3.8 explicitly (§XII.12). Theorem 12.3.9, the extension of sc$^+$-section functors from ∂X with the bound $N(\tilde{s}) \leq \delta$, is proved as printed (§XII.13).

§12.4. Definition 12.4.1, Theorem 12.4.2 and Lemma 12.4.4 are proved (§XII.14). Definition 12.4.3 is formalized, and Proposition 12.4.5 and Theorem 12.4.6 are proved in stronger forms, several of the text's hypotheses being unused (§XII.15).

§12.5. The action of a morphism on linearizations goes through a local strong bundle isomorphism built from its sheet (§XII.16). Lemma 12.5.1 and the action φ_* on orientations are proved with §12.5's standing hypothesis, inherited from Part I, that sc$^+$-sections pass through smooth fibre points (§XII.17). Definition 12.5.2 is Part I's O_f (§XII.18). Theorem 12.5.3 is proved under Part I's hypotheses (tame X, bump functions on the objects, tame germ data), including its last sentence: the local isomorphism of a morphism carries a continuous local orientation near x to one near x', with value $\sigma_* o_x$ at x' (§XII.19). Definition 12.5.4 is formalized, and its continuity clause, read for the topology of O_f, agrees with Part I's continuity of orientations along paths (§XII.20). Remark 12.5.5 is prose. Proposition 12.5.6 is proved in both directions, (□) with the continuity of the orientation (§XII.21). Theorem 12.5.7 is proved for one representative $W \leftarrow V \rightarrow W'$ of a generalized strong bundle isomorphism: (1) with homeomorphisms $|O_f| \rightarrow |O_{f'}|$ over $|X_\infty| \rightarrow |X'_\infty|$, and (2) with continuity in the sense of Definition 12.5.4; its independence of the representative and its functoriality, which the text leaves to the reader, are proved for the data of the diagrams with no further hypothesis (§XII.22).

Narrower items, and why. Theorems 12.3.2 (2) and 12.3.3: the pointwise reading of bump functions, the project's reading of Part I's Definition 5.3.1. Theorem 12.3.2 (1) is restated as two equivalences, one per reading. Theorem 12.5.3, Proposition 12.5.6 (□), Theorem 12.5.7: Part I's hypotheses of Theorem 6.6.17 and the hypothesis that sc$^+$-sections pass through smooth fibre points, both inherited (G.140); Theorem 12.5.7 also by being stated for the data of one diagram rather than for Chapter 10's classes (G.192).

Open gaps. G.140 (Part I's hypotheses carried by §12.5) and G.192 (Theorem 12.5.7 for `SDiagram` and `GenSBMap`), with estimates in §XII.24.

Decisions that shape the chapter.

(1) *Part I on the object M-polyfold.* An sc-Fredholm section functor is an sc-smooth section functor that is a Part I sc-Fredholm section of the object bundle $W \to X$; an auxiliary norm is a Part I auxiliary norm invariant under μ (Definition 8.3.5, Part VIII). Every Part I theorem applies to the objects without restatement, and the chapter's work is the passage through the morphisms.

(2) *Norms are built from Part I norms, and completeness is automatic.* The local norms the text takes from a chart are Part I auxiliary norms (Proposition 5.1.3, Part V). Completeness of a glued fibre norm is not checked by hand: a Part I lemma shows that a continuous fibrewise seminorm satisfying condition (2) of an auxiliary norm is an auxiliary norm (§XII.2). This lemma is used in Theorems 12.2.2, 12.2.7, 12.2.11 and 12.2.13.

(3) *Bump functions in two readings.* Definition 12.3.1 asks for an sc-smooth function "not identically" zero; Part I's Definition 5.3.1 is read pointwise in this project (Part V). Both readings are kept for ep-groupoids and related (§XII.10); the theorems that need a function nonzero at a given point use the pointwise one.

(4) *Orientations live in chart points.* $\mathrm{Lin}(f, x)$ and Or_x^f are Part I's, read in a chosen chart point at x. A morphism φ acts through a local strong bundle isomorphism built from the sheet of φ (`morIso`, the inverse of the text's $\hat{\Phi}$), along which chart points are pulled back and compared (§XII.16). A strong bundle equivalence acts in the same way, through the local isomorphisms it is near each object (§XII.22).

(5) *Continuity of an orientation in two readings.* Part I defines continuity of an orientation section over an open set by propagation along sc-smooth paths (`IsContOrOn`); Definition 12.5.4 asks continuity of $x \mapsto (x, o_x)$ for the topology of O_f. The proofs of §12.5 use the first, and the two are shown to agree (§XII.20), so the statements are given in the text's form.

(6) *Theorem 12.5.7 for the data of diagrams.* A generalized strong bundle isomorphism is a class of diagrams $W \leftarrow V \to W'$ of strong bundle equivalences (Chapter 10, Part X). The theorem is proved for one representative, with the independence of the representative stated through common refinements (Chapter 10's equivalence of diagrams) and the functoriality through the shape of Chapter 10's weak fibred product.

(7) *Hypotheses of Part I are inherited, not added.* Where §12.5 rests on Part I's Theorem 6.6.17, it carries that theorem's hypotheses; they are listed once, in §XII.23, with witnesses.

Map.

Fredholm section functors and compactness. - `Chapter12/CompactnessEp`: Definition 12.1.1 and the remark after it; Definition 12.4.1, Theorem 12.4.2, Lemma 12.4.4; the local step of Part I's Theorem 5.2.2 used there. - `Chapter12/ReflexiveCompactnessEp`: Definition 12.4.3, compact closures from orbit sets, Proposition 12.4.5, Theorem 12.4.6.

Auxiliary norms. - `Chapter12/AuxNormEp`: the Part I lemma on automatic completeness; the lifted action on $W_{0,1}$; averaging, transport and gluing of norms; Definitions 12.2.3, 12.2.4, Proposition 12.2.5; Theorem 12.2.2. - `Chapter12/GoodNeighborhood`: Theorem 12.2.6. - `Chapter12/AuxNormExtension`: auxiliary norms over a subset; the global part of Theorem 12.2.7. - `Chapter12/AuxNormLocalExtension`: the retraction onto the boundary of a tame M-polyfold, the norm N', the local step and Theorem 12.2.7.

Reflexive auxiliary norms. - `Chapter12/ReflexiveAuxNormEp`: reflexive 1-fibres, Definitions 12.2.8, 12.2.10, Remark 12.2.9, mixed lower semicontinuity and its algebra, Theorem 12.2.11, Remark 12.2.12. - `Chapter12/ReflexiveAuxNormExtension`: reflexive norms over a subset; the global part of Theorem 12.2.13.

- `Chapter12/ReflexiveLocalExtension`: the fibre maps of a retraction under weak convergence, the lower semicontinuity of N', the local step and Theorem 12.2.13.

Bump functions and sc⁺-section functors. - `Chapter12/BumpEp`: Definition 12.3.1 in two readings, averages and invariant extensions of functions, Theorem 12.3.2. - `Chapter12/ScPlusSectionFunctor`: the action of morphisms on sections, equivariant averages, the natural extension, Theorem 12.3.3, Remarks 12.3.4, 12.3.5. - `Chapter12/LocalBoundaryExtension`: Definition 12.3.6 in a local model, the inclusion–exclusion extension, Proposition 12.3.7, local bundle actions, Corollary 12.3.8. - `Chapter12/ScPlusExtension`: Definition 12.3.6 on ep-groupoids, bump functions from partitions of unity, gluing of sections, the local and global extensions, Theorem 12.3.9.

Morphisms acting on linearizations and orientations. - `Chapter12/MorphismIso`: the local strong bundle isomorphism of a morphism. - `Chapter12/ChartPull`: chart points pulled back along a local strong bundle isomorphism of a bundle to itself. - `Chapter12/OrientationBundleEp`: the generic orientation map of a local isomorphism; $T\varphi$ and $\mu(\varphi,\cdot)$ in chart points; Lemma 12.5.1; φ_* on orientations. - `Chapter12/OrientationOrbit`: orbit spaces of X_∞ and O_f, Definition 12.5.2, the topological part of Theorem 12.5.3, Definition 12.5.4, Proposition 12.5.6 (□) and the invariant lift of (□).

Orientations carried along local isomorphisms. - `Chapter12/ChartPullTwoBundles`: chart points pulled back along a local strong bundle isomorphism between two bundles, and its orientation map. - `Chapter12/SectionPull`, `Chapter12/StabilizedKernelPull`, `Chapter12/PushDataPull`: Part I's push-forward of orientations carried along a local isomorphism: the stabilizing section, the stabilized kernels, the push-forward. - `Chapter12/DyadicPathInduction`: dyadic induction on sc-smooth paths; continuity of orientations is local. - `Chapter12/OrientationLocalCommute`: the transport commutes with a local isomorphism; carried continuous orientations are continuous. - `Chapter12/OrientationGerm`: the orientation map of a local isomorphism depends only on its germ. - `Chapter12/OrientationMorphismContinuity`: the last sentence of Theorem 12.5.3.

Orientability and continuity. - `Chapter12/OrientationFromSection`: Proposition 12.5.6 (□) with continuity. - `Chapter12/ContinuousOrientationSection`: the two readings of continuity of an orientation agree.

Naturality (Theorem 12.5.7). - `Chapter12/OrientationEquivalenceStep`: the local isomorphisms of a strong bundle equivalence, its orientation bijections and their naturality. - `Chapter12/OrientationNaturality`: invariant orientations, the orbit maps, pull-back and push-forward of orientations, Theorem 12.5.7 without topology. - `Chapter12/OrientationEquivalenceContinuity`: pull-back and push-forward keep continuity; Theorem 12.5.7 (2) with Part I's continuity. - `Chapter5/SmoothLevelContinuity`: sc-smooth maps are continuous on the levels and on X_∞. - `Chapter12/OrientationOrbitHomeo`: open orbit maps, the homeomorphisms of one equivalence. - `Chapter12/OrientationTransferTheorem`: Theorem 12.5.7 (1) and (2) as printed. - `Chapter12/OrientationEquivalenceNatural`, `Chapter12/OrientationEquivalenceComp`, `Chapter12/OrientationRepresentativeIndep`: natural equivalences, composition, independence of the representative and functoriality.

Notation. Throughout, $(P : W \to X, \mu)$ is a strong bundle over an ep-groupoid X with source and target maps s, t; B is the underlying Part I strong bundle on objects, $W_{0,1}$ its level-$(0,1)$ part, $\pi : X \to |X|$ the orbit map, G_x the isotropy group, and a natural representation of G_x on $U(x)$ is the one of Theorem 7.1.13 `Chapter7/NaturalRep · theorem_7_1_13`: a group homomorphism $\Phi : G_x \to \mathrm{Diff}_{sc}(U(x))$ and a map $\Gamma : G_x \times U(x) \to \mathbf{X}$ with $s(\Gamma(g,y)) = y$, $t(\Gamma(g,y)) = \Phi(g)(y)$, such that every morphism

between points of $U(x)$ is $\Gamma(g, y)$ for exactly one g. The saturation of a set A of objects is the set of objects joined by a morphism to a point of A; A is saturated if it equals its saturation. ∂X is the set of objects of degeneracy index at least 1; it is saturated, since morphisms are local sc-diffeomorphisms and keep the degeneracy index.

XII.1 Definition 12.1.1 (2017: 12.1) and the remark after it

Definition 12.1.1 (same) `Chapter12/CompactnessEp · EpStrongBundle.definition_12_1_1`, p. 569. Let $(P : W \to X, \mu)$ be a strong bundle over an ep-groupoid. An sc-smooth section functor f (an sc-smooth section of the object bundle with $f(t(\varphi)) = \mu(\varphi, f(s(\varphi)))$ for every morphism φ) is *sc-Fredholm* when it is an sc-Fredholm section of the object bundle in Part I's sense (Definition 3.4.17 as read in Part III): regularizing, and with an sc-Fredholm germ at every smooth point. The Lean declaration is the definition unfolded: it states that the project's predicate `IsScFredholmSectionFunctor` is the conjunction of the section functor property and Part I's sc-Fredholm property for the bundle B. Nothing is added; the text's phrase "for the strong bundle $W \to X$" is the object bundle, since the morphisms carry no further bundle.

The remark after Definition 12.1.1 (explicit), p. 569. The text observes, without proof, that the regularizing property holds on objects and on morphisms, and that the germs are sc-Fredholm. Each is a separate statement.

On objects `Chapter12/CompactnessEp · EpStrongBundle.remark_12_1_1_regularizing_obj`. If $x \in X_m$ and $f(x) \in W_{m,m+1}$, then $x \in X_{m+1}$. This is Part I's regularizing property of f on the object bundle, read off the definition.

On morphisms `Chapter12/CompactnessEp · EpStrongBundle.remark_12_1_1_regularizing_mor`. The text writes $f(\varphi)$ for a morphism φ. A section functor gives no vector over a morphism directly; the formal statement reads $f(\varphi)$ as the section $\varphi \mapsto (\varphi, f(s(\varphi)))$ of the pull-back bundle $\mathbf{X}\,{}_s\!\times_P W \to \mathbf{X}$, the bundle over the morphism space whose fibre over φ is $W_{s(\varphi)}$. The statement: if $\varphi \in \mathbf{X}_m$ and $f(\varphi)$ lies on level $(m, m+1)$ of the pull-back bundle, then $\varphi \in \mathbf{X}_{m+1}$. *Step 1.* The projection of the pull-back bundle to W is a strong bundle map, so it keeps the bi-levels; $f(s(\varphi))$ lies in $W_{m,m+1}$. *Step 2.* By the regularizing property on objects, $s(\varphi) \in X_{m+1}$. *Step 3.* The levels of the morphism M-polyfold are those of the sources, since s is a local sc-diffeomorphism; so $\varphi \in \mathbf{X}_{m+1}$.

Germs `Chapter12/CompactnessEp · EpStrongBundle.remark_12_1_1_germ`. At a smooth object x, the germ of f at x is an sc-Fredholm germ. This is the second clause of Part I's definition, read at x.

No hypothesis is added. The definition has a witness in every Part I sc-Fredholm section of a trivial ep-groupoid (an M-polyfold with only identity morphisms), where the functor property is empty.

XII.2 Definition 12.2.1 (2017: 12.2): auxiliary norms, and completeness for free

Definition 12.2.1 (same) `Chapter8/Book2021 · EpStrongBundle.IsAuxNorm`, p. 570. The text recalls Definition 8.3.5, and the formalization uses that declaration unchanged: an auxiliary norm is a Part I auxiliary norm $N : W_{0,1} \to [0, \infty)$ of the object bundle (Definition 5.1.1, Part V: continuous, a complete norm on each fibre, and condition (2): if $P(h_k) \to x$ in X and $N(h_k) \to 0$, then $h_k \to 0_x$ in $W_{0,1}$) with $N(\mu(\varphi, k)) = N(k)$

for every morphism φ and every $k \in W_{0,1}$ over $s(\varphi)$. In the project N is a function on all of W, and "$N(k) \le C$" is Part I's relation, which fails off $W_{0,1}$ (equivalently, $N = \infty$ there).

Completeness is automatic (extra, Part I) `Chapter12/AuxNormEp · StrongBundle.isAuxNorm_of_tendsto`. The text asserts, of each glued norm it builds, that it is complete on the fibres. The formal proof replaces these assertions by one lemma about Part I's Definition 5.1.1. Call $N : W \to [0,\infty)$ a *fibre seminorm* if its restriction to each fibre $W_{0,1}|_x$ is a seminorm `Chapter12/AuxNormEp · StrongBundle.IsFiberSeminorm`, and say it satisfies *condition (2)* if $P(h_k) \to x$, $N(h_k) \to 0$ imply $h_k \to 0_x$ in $W_{0,1}$ `Chapter12/AuxNormEp · StrongBundle.TendstoZeroOne`. The lemma: a continuous fibre seminorm on $W_{0,1}$ satisfying condition (2) is an auxiliary norm.

Proof. Fix x and a strong bundle chart Φ around x with fibre F, and write $|w|_1 = \|\mathrm{pr}_2 \Phi[1](w)\|_{F_1}$ for w over points near x; on each fibre this is a complete norm, because $\Phi[1]$ is a linear isomorphism of the fibre onto a closed subspace (the image of the fibre projection $A(u)$) of F_1.

Step 1, an upper bound. By continuity of N at 0_x there is $\delta > 0$ and a neighbourhood O of x with $N(w) \le 1$ when $P(w) \in O$ and $|w|_1 \le \delta$. By homogeneity, $N(w) \le \delta^{-1}|w|_1$ for $P(w) \in O$. (In Lean: `StrongBundle.exists_le_mul_norm_liftOne`.)

Step 2, a lower bound `Chapter12/AuxNormEp · StrongBundle.exists_norm_liftOne_le_of_tendsto`. Suppose no $c > 0$ and neighbourhood of x satisfy $|w|_1 \le c\,N(w)$. Then there are w_k over points $P(w_k) \to x$ with $|w_k|_1 > k\,N(w_k)$; rescale so that $|w_k|_1 = 1$. Then $N(w_k) < 1/k \to 0$, and condition (2) gives $w_k \to 0_x$ in $W_{0,1}$, hence $|w_k|_1 \to 0$ by continuity of the chart on level 1, against $|w_k|_1 = 1$.

Step 3, positivity and completeness. By Step 2, $N(w) = 0$ forces $|w|_1 = 0$, so N is a norm on each fibre. By Steps 1 and 2 it is equivalent on each fibre near x to the complete norm $|\cdot|_1$, hence complete (`complete_of_bounds`). The other two clauses of Definition 5.1.1, continuity and condition (2), are hypotheses. □

The same lemma in local form: a *local auxiliary norm* on an open U asks continuity on $W_{0,1}|U$, the seminorm property over U and condition (2) for limits $x \in U$ only `Chapter12/AuxNormEp · StrongBundle.IsLocalAuxNorm`. The local norms of all constructions below are local auxiliary norms, and the glued norms are proved to be auxiliary norms by the lemma. The step is a strengthening of Part I's toolkit: it applies equally to Part I's Proposition 5.1.3, though Part I proves that proposition directly.

XII.3 Theorem 12.2.2 (2017: 12.1): auxiliary norms exist

Theorem 12.2.2 (same) `Chapter12/AuxNormEp · EpStrongBundle.theorem_12_2_2`, p. 570: every strong bundle $(P : W \to X, \mu)$ over an ep-groupoid with paracompact orbit space $|X|$ admits an auxiliary norm. The same for a bundled object of the category $\mathcal{SEP}$ of strong bundles over ep-groupoids `Chapter12/AuxNormEp · SEP.theorem_12_2_2`, a one-line wrapper. The hypotheses are the text's; the conclusion is Definition 12.2.1 in full.

The proof follows the text's three steps, average, transport, glue; the formal construction is a chain of definitions in `AuxNormEp`, each with its lemma.

Step 0: the lifted action on $W_{0,1}$. The strong bundle action μ is defined on the pull-back $\mathbf{X}_s \times_P W$. For a function α from objects to morphisms with $s(\alpha(y)) = y$, define $\alpha * w = \mu(\alpha(P(w)), w)$ `Chapter12/AuxNormEp · EpStrongBundle.actOne`. When α is sc-smooth on an open set, $\alpha*$ is an sc-smooth map of $W[1]$ there, so it preserves $W_{0,1}$ and is continuous on it. Along a natural representation, $g * w = \mu(\Gamma(g, P(w)), w)$ is a group action of G_x on $W|U(x)$: $1 * w = w$ `Chapter12/AuxNormEp · EpStrongBundle.`

actOne_Γ_one because $\Gamma(1,y) = 1_y$, and $(gh) * w = g * (h * w)$ Chapter12/AuxNormEp · EpStrongBundle.actOne_Γ_mul because $\Gamma(gh,y) = \Gamma(g,\Phi(h)y) \circ \Gamma(h,y)$ by the uniqueness clause of Theorem 7.1.13 and μ respects composition. Restricted to one fibre, $g*$ is the linear map $\mu(\Gamma(g,y),\cdot) : W_y \to W_{\Phi(g)y}$ Chapter12/AuxNormEp · EpStrongBundle.actFiber.

Step 1: average. Fix x, a natural representation on $U(x)$ and a local auxiliary norm M on $U(x)$. Put

$$\bar{M}(w) = \frac{1}{|G_x|} \sum_{g \in G_x} M(g * w)$$

Chapter12/AuxNormEp · EpStrongBundle.avgNorm. *Invariance* Chapter12/AuxNormEp · EpStrongBundle.avgNorm_actOne: $\bar{M}(h * w) = \bar{M}(w)$, by reindexing the sum through $g \mapsto gh$ and Step 0. *A local auxiliary norm* Chapter12/AuxNormEp · EpStrongBundle.isLocalAuxNorm_avgNorm. Continuity: each term $M \circ (g*)$ is continuous, as a composite of continuous maps over $U(x)$, and $\Phi(g)$ maps $U(x)$ into itself. Seminorm: each term is a seminorm composed with a linear fibre map. Condition (2): if $P(w_k) \to y \in U(x)$ and $\bar{M}(w_k) \to 0$, then the term with $g = 1$ satisfies $M(w_k) \le |G_x|\,\bar{M}(w_k) \to 0$, since all terms are nonnegative, and condition (2) for M gives $w_k \to 0_y$.

Step 2: transport to the saturation. Let $S = \mathrm{sat}(U(x))$ be the set of objects y admitting a morphism φ with $s(\varphi) = y$ and $t(\varphi) \in U(x)$ Chapter12/AuxNormEp · EpGroupoid.satSet. Choose one such morphism φ_y for every $y \in S$ Chapter12/AuxNormEp · EpGroupoid.selMor (a choice function; no regularity is asked) and put

$$T(w) = \bar{M}(\mu(\varphi_{P(w)}, w)) \text{ for } P(w) \in S, \qquad T(w) = 0 \text{ otherwise}$$

Chapter12/AuxNormEp · EpStrongBundle.transNorm.

Independence of the choice Chapter12/AuxNormEp · EpStrongBundle.transNorm_eq_of. If ψ is another morphism from y into $U(x)$, then $\psi \circ \varphi_y^{-1}$ is a morphism between points of $U(x)$, hence equals $\Gamma(g, t(\varphi_y))$ for a unique $g \in G_x$; so $\mu(\psi, w) = g * \mu(\varphi_y, w)$ and $\bar{M}(\mu(\psi,w)) = \bar{M}(\mu(\varphi_y,w))$ by invariance. At this point the text invokes the good neighbourhood property (5) of Theorem 12.2.6; the invariance of $\bar{M}$ makes it unnecessary, because what has to be compared are two morphisms into $U(x)$, and any two differ by an element of G_x.

Continuity Chapter12/AuxNormEp · EpStrongBundle.continuousOn_transNorm. Fix w_0 over $y_0 \in S$ and its chosen morphism φ_0. Since s is a local sc-diffeomorphism, there is a local inverse ρ of s on a neighbourhood O of y_0 with $\rho(y_0) = \varphi_0$ (`X.isLocalScDiffeo_src`); shrinking O, $t(\rho(y)) \in U(x)$ for $y \in O$, so $O \subset S$. For w over O, $\rho(P(w))$ is a morphism from $P(w)$ into $U(x)$, so by independence of the choice $T(w) = \bar{M}(\rho * w)$ near w_0, and this is continuous by Step 0.

Condition (2) Chapter12/AuxNormEp · EpStrongBundle.tendsto_transNorm. Let $P(w_k) \to y_0 \in S$ and $T(w_k) \to 0$. With ρ as above, $v_k = \rho * w_k$ lies over $t(\rho(P(w_k))) \to t(\varphi_0) \in U(x)$ and $\bar{M}(v_k) = T(w_k) \to 0$ eventually; condition (2) for $\bar{M}$ gives $v_k \to 0$. To bring this back, let τ be a local inverse of t near φ_0, and $\alpha' = \mathrm{inv} \circ \tau$, an sc-smooth local section of s near $t(\varphi_0)$; then $w_k = \alpha' * v_k$ eventually, by μ_inv_μ and the identity $\tau(t(\rho(y))) = \rho(y)$, and $\alpha'*$ is continuous with $\alpha' * 0 = 0$. So $w_k \to 0_{y_0}$.

Seminorm Chapter12/AuxNormEp · EpStrongBundle.isFiberSeminorm_transNorm: on the fibre over $y \in S$, T is the seminorm $\bar{M}$ composed with the linear map $\mu(\varphi_y, \cdot)$; off S it is 0. So T is a local auxiliary norm on S Chapter12/AuxNormEp · EpStrongBundle.isLocalAuxNorm_transNorm, and it is *invariant under every morphism* Chapter12/AuxNormEp · EpStrongBundle.transNorm_μ: for $\psi : y \to y'$, $y \in S$ iff $y' \in S$, and $\varphi_{y'} \circ \psi$ is a morphism from y into $U(x)$, so $T(\mu(\psi, w)) = \bar{M}(\mu(\varphi_{y'} \circ \psi, w)) = T(w)$ by independence of the choice.

Step 3: glue. *Partitions of unity of X* (Definition 12.2.3, §XII.4) are pulled back from $|X|$. For each object x the set $\pi(S_x)$, $S_x = \mathrm{sat}(U(x))$, is open in $|X|$ because π is open (Proposition 7.1.8), and these sets cover $|X|$. Since $|X|$ is paracompact and Hausdorff, there is a partition of unity $(\beta_i)_{i\in I}$ of $|X|$ subordinate to a subcover $(\pi(S_{x_i}))$. Put

$$N(w) = \sum_{i\in I} \beta_i(\pi(P(w)))\, T_i(w)$$

`Chapter12/AuxNormEp · EpStrongBundle.glueNorm`, a locally finite sum. A term vanishes where $\pi(P(w)) \notin \operatorname{supp}\beta_i$ `Chapter12/AuxNormEp · EpStrongBundle.partition_eq_zero_of_not_mem` and all terms are nonnegative `Chapter12/AuxNormEp · EpStrongBundle.glue_term_nonneg`.

N is an auxiliary norm `Chapter12/AuxNormEp · EpStrongBundle.isAuxNorm_glueNorm`. By the lemma of §XII.2 three things must be checked, and invariance.

- *Continuity.* Near any w_0, only finitely many $\beta_i \circ \pi \circ P$ are nonzero (local finiteness, pulled back through the continuous $\pi \circ P$), so N is a finite sum there (`continuous_finsum`). Each term is continuous: where $\pi(P(w)) \in \operatorname{supp}\beta_i \subset \pi(S_{x_i})$, $P(w)$ lies in the open saturated S_{x_i}, on which T_i is continuous; outside the closed set $\pi^{-1}(\operatorname{supp}\beta_i)$ the term is identically 0 nearby.
- *Seminorm.* A nonnegative combination of seminorms on each fibre.
- *Condition (2).* Let $P(w_k) \to y$ and $N(w_k) \to 0$. Choose i with $\beta_i(\pi(y)) > 0$. Then $\beta_i(\pi(P(w_k))) \geq \beta_i(\pi(y))/2$ eventually, so $T_i(w_k) \leq 2N(w_k)/\beta_i(\pi(y)) \to 0$, the term being squeezed below the sum by nonnegativity. Since $\pi(y) \in \operatorname{supp}\beta_i \subset \pi(S_{x_i})$, $y \in S_{x_i}$, and condition (2) for T_i gives $w_k \to 0_y$.
- *Invariance.* $\beta_i \circ \pi$ is constant along morphisms (`π_src_eq_π_tgt`), and each T_i is invariant.

The ingredients are all prior results. The local norms M_i are the restrictions to $U(x_i)$ of one Part I auxiliary norm of the object bundle `Chapter5/AuxNorm · proposition_5_1_3_exists`, which Part I provides on the object M-polyfold. The natural representations come from Theorem 7.1.13 at every object, with $U(x)$ inside X. The partition of unity exists because $|X|$ is paracompact (the hypothesis) and Hausdorff (Proposition 7.1.8; `PartitionOfUnity.exists_isSubordinate`); Theorem 7.3.1 `Chapter7/OrbitTopology · theorem_7_3_1_metrizable` supplies the regularity that the subordinate partition needs. □

Where the formal proof departs from the text. (a) The text takes the local norm in a strong bundle chart as $\|\mathrm{pr}_2\Psi(\cdot)\|_{F_1}$; a Part I auxiliary norm restricted to $U(x)$ serves equally and comes with condition (2). (b) Completeness of $\bar{M}$, T and N is not checked separately; the lemma of §XII.2 gives it. (c) Property (5) of Theorem 12.2.6 is not used; invariance under G_x replaces it. (d) The text's "N is continuous since the sum is locally finite" needs that each term is continuous where its coefficient is nonzero, which is the continuity of T_i on the open saturation; this is proved through local inverses of s.

XII.4 Definitions 12.2.3, 12.2.4 (2017: 12.3, 12.4) and Proposition 12.2.5 (2017: 12.1): partitions of unity

Definition 12.2.3 (same) `Chapter12/AuxNormEp · EpGroupoid.IsContPartitionOfUnity`, p. 570. A continuous partition of unity of the ep-groupoid X, indexed by Λ, is a family of continuous functions $\beta_\lambda : X \to [0,1]$ on the objects with (1) $\beta_\lambda(s(\varphi)) = \beta_\lambda(t(\varphi))$ for every morphism φ; (2) local finiteness through saturated sets: every object has a saturated open neighbourhood meeting the supports of only finitely many β_λ; (3) $\sum_\lambda \beta_\lambda = 1$. Supports are closures in X. The text's preamble has $|X|$ paracompact; the definition does not use it, and it is not a field, but a hypothesis of Proposition 12.2.5.

The remark after Definition 12.2.3 (same), p. 570, has two directions.

From $|X|$ to X `Chapter12/AuxNormEp · EpGroupoid.isContPartitionOfUnity_comp_π`. If $(\hat\beta_\lambda)$ is a continuous partition of unity of $|X|$, then $\beta_\lambda = \hat\beta_\lambda \circ \pi$ is one of X. (1) holds because $\pi(s(\varphi)) = \pi(t(\varphi))$. Continuity: π is continuous. (2): a point $\pi(x)$ has an open O meeting finitely many $\operatorname{supp}\hat\beta_\lambda$; $\pi^{-1}(O)$ is open and saturated, and $\operatorname{supp}(\hat\beta_\lambda \circ \pi) \subset \pi^{-1}(\operatorname{supp}\hat\beta_\lambda)$, so it meets finitely many supports. (3) is pointwise.

From X to $|X|$ `Chapter12/AuxNormEp · EpGroupoid.IsContPartitionOfUnity.descend`. A continuous partition of unity of X descends: by (1) each β_λ is constant on orbits, so $\beta_\lambda = \hat\beta_\lambda \circ \pi$ for a unique $\hat\beta_\lambda$ on $|X|$. Continuity of $\hat\beta_\lambda$ holds because π is an open surjection, hence a quotient map. Local finiteness on $|X|$: for a saturated open U meeting finitely many supports, $\pi(U)$ is open, by openness of π (Proposition 7.1.8), and $\pi(U)$ meets $\operatorname{supp}\hat\beta_\lambda$ only if U meets $\operatorname{supp}\beta_\lambda$; for the latter, a point of $\pi(U) \cap \{\hat\beta_\lambda \neq 0\}$ lifts into U because U is saturated, and $\pi^{-1}(\operatorname{cl}\{\hat\beta_\lambda \neq 0\}) = \operatorname{cl}\pi^{-1}\{\hat\beta_\lambda \neq 0\}$ for the open map π.

Definition 12.2.4 (same) `Chapter12/AuxNormEp · EpGroupoid.IsContPartitionOfUnity.IsSubordinate`, p. 571. For a family (U_λ) of saturated open sets covering X, a continuous partition of unity (β_λ) is subordinate if $\operatorname{supp}\beta_\lambda \subset U_\lambda$ for every λ, the supports taken in X.

Proposition 12.2.5 (same) `Chapter12/AuxNormEp · EpGroupoid.proposition_12_2_5`, p. 571. If $|X|$ is paracompact, every cover (U_λ) of X by saturated open sets has a subordinate continuous partition of unity, indexed by the same Λ.

Proof, the text's. *Step 1.* The sets $\pi(U_\lambda)$ are open in $|X|$ because π is open, and cover $|X|$ because π is surjective. *Step 2.* $|X|$ is Hausdorff (Proposition 7.1.8) and paracompact, hence normal, and admits a partition of unity $(\hat\beta_\lambda)$ subordinate to $(\pi(U_\lambda))$, with the same index set (`PartitionOfUnity.exists_isSubordinate`). *Step 3.* By the first direction of the remark, $\beta_\lambda = \hat\beta_\lambda \circ \pi$ is a continuous partition of unity of X. *Step 4, subordination* `Chapter12/AuxNormEp · EpGroupoid.isSubordinate_comp_π`: $\operatorname{supp}\beta_\lambda \subset \pi^{-1}(\operatorname{supp}\hat\beta_\lambda) \subset \pi^{-1}(\pi(U_\lambda)) = U_\lambda$, the last equality because U_λ is saturated, and the first because π^{-1} of a closed set is closed and contains $\{\beta_\lambda \neq 0\}$. □

No hypothesis is added, and nothing in the text's argument is replaced. The construction of §XII.3 uses Proposition 12.2.5 in the form of Steps 1–4 directly, with U_λ the saturations $\operatorname{sat}(U(x))$.

XII.5 Theorem 12.2.6 (2017: 12.2): good neighbourhoods

Theorem 12.2.6 (same) `Chapter12/GoodNeighborhood · EpGroupoid.theorem_12_2_6`, pp. 571–572. Let $(P : W \to X, \mu)$ be a strong bundle over an ep-groupoid and x an object. There are open sets $U'(x) \subset U''(x)$ containing x and a natural representation R of G_x with domain $U''(x)$ such that:

1. $\overline{U'(x)} \subset U''(x)$;
2. $U'(x)$ is invariant under the natural representation: $\Phi(g)(U'(x)) = U'(x)$ for all $g \in G_x$;
3. $U''(x)$ lies in the base of one strong bundle chart of the atlas, which restricted to $P^{-1}(U''(x))$ is a strong bundle isomorphism onto a local model;
4. for every sequence of morphisms φ_k with $s(\varphi_k) \in \overline{U''(x)}$ and $t(\varphi_k) \to z$, a subsequence converges to a morphism φ with $s(\varphi) \in \overline{U''(x)}$ and $t(\varphi) = z$;
5. $(\pi^{-1}(\pi(U'(x))) \setminus U'(x)) \cap \overline{U''(x)} = \emptyset$: a point of $\overline{U''(x)}$ in the saturation of $U'(x)$ lies in $U'(x)$.

The text proves the theorem by pointing to Chapter 7; the formal proof assembles it in four steps.

Step 1, properness. Since X is an ep-groupoid, x has an open neighbourhood V_0 such that $t : s^{-1}(\overline{V_0}) \to X$ is proper (`X.exists_isProperMap`, the properness clause of the definition of an ep-groupoid). A strong

bundle chart c of the atlas contains x in its base. Theorem 7.1.13 gives a natural representation R_0 of G_x on a domain inside $V_0 \cap \operatorname{dom} c$ `Chapter7/NaturalRep · theorem_7_1_13`.

Step 2, two shrinkings. Choose a closed neighbourhood t_1 of x inside $R_0.U$, and shrink the natural representation to an invariant domain inside the interior of t_1 `Chapter7/NaturalRep · exists_naturalRep_shrink`; this is R_1, with domain $U''(x) = R_1.U$ and $\overline{R_1.U} \subset t_1 \subset R_0.U$. Repeat with R_1 in place of R_0 to get R_2 with $\overline{R_2.U} \subset R_1.U$, and put $U'(x) = R_2.U$. The shrinking lemma keeps the group homomorphism Φ (restricted), so $\Phi_2(g) = \Phi_1(g)|U'$ and (1) holds. *(2).* $U'(x)$ is the domain of the natural representation R_2, which is invariant under its own Φ; the image of the domain under $\Phi(g)$ is the domain `Chapter12/GoodNeighborhood · EpGroupoid.NaturalRep.image_Φ_eq`, because $\Phi(g)$ maps the domain into itself and so does $\Phi(g^{-1}) = \Phi(g)^{-1}$. *(3).* $U''(x) \subset R_0.U \subset \operatorname{dom} c$; the chart is a member of the atlas, so its restriction is a strong bundle isomorphism onto its local model.

Step 3, (4) `Chapter12/GoodNeighborhood · EpGroupoid.exists_subseq_of_isProperMap`. The general lemma: if $t : s^{-1}(\overline{V_0}) \to X$ is proper, $A \subset \overline{V_0}$ is closed, $s(\varphi_k) \in A$ and $t(\varphi_k) \to z$, then some subsequence $\varphi_{\sigma(k)}$ converges to a ψ with $s(\psi) \in A$, $t(\psi) = z$. *Proof.* $K = \{z\} \cup \{t(\varphi_k)\}$ is compact, as a convergent sequence with its limit; its preimage under the proper map is compact and contains every φ_k. The morphism space is metrizable, so compact sets are sequentially compact, and a subsequence converges to some ψ in the preimage. Continuity of s and closedness of A give $s(\psi) \in A$; continuity of t and uniqueness of limits give $t(\psi) = z$. Applied with $A = \overline{U''(x)} \subset \overline{V_0}$, this is (4).

Step 4, (5). Let $y \in \overline{U''(x)}$ with $\pi(y) = \pi(z)$ for some $z \in U'(x)$. There is a morphism $\varphi : z \to y$. Both $z \in U'(x) \subset R_0.U$ and $y \in \overline{U''(x)} \subset R_0.U$, so by the uniqueness clause of Theorem 7.1.13, $\varphi = \Gamma(g, z)$ for a unique $g \in G_x$, and $y = t(\varphi) = \Phi(g)(z)$. By (2), $\Phi(g)(z) \in U'(x)$. So $y \in U'(x)$. □

The first shrinking serves (5): it puts $\overline{U''(x)}$ inside the domain $R_0.U$ of a natural representation, on which every morphism is a $\Gamma(g, \cdot)$. No hypothesis is added. Property (5) is not used in the construction of §XII.3 (§XII.3, Step 2); Theorem 12.2.6 is used through (3) and (4) in the compactness arguments of §§XII.14–XII.15 and in later chapters.

XII.6 Theorem 12.2.7 (2017: 12.3): extension of auxiliary norms from the boundary

Theorem 12.2.7 (same) `Chapter12/AuxNormLocalExtension · EpStrongBundle.theorem_12_2_7`, pp. 574–575. Let $(P : W \to X, \mu)$ be a strong bundle over a tame ep-groupoid X with paracompact $|X|$, and N an auxiliary norm of $W_{0,1}|\partial X$ (Definition 12.2.1 over ∂X). Then there is an auxiliary norm $\tilde{N}$ of W with $\tilde{N} = N$ over ∂X. The hypotheses are the text's.

Auxiliary norms over a subset. For a set A of objects, *an auxiliary norm over* A `Chapter12/AuxNormExtension · EpStrongBundle.IsAuxNormOn` is a function N on $W_{0,1}$ that is continuous on $W_{0,1}|A$, a complete norm on each fibre over A, satisfies condition (2) for sequences w_k over A whose base points converge to a point of A, and is invariant where defined: $N(\mu(\varphi, k)) = N(k)$ when $P(k) = s(\varphi) \in A$. The text's N is a function on $W_{0,1}|\partial X$; a function on all of $W_{0,1}$ with conditions over ∂X only is the same datum, its values off ∂X being irrelevant. A global auxiliary norm restricts to one over any A `Chapter12/AuxNormExtension · EpStrongBundle.IsAuxNorm.isAuxNormOn`. Morphisms preserve ∂X in both directions: $t(\varphi) \in \partial X$ iff $s(\varphi) \in \partial X$ `Chapter12/AuxNormExtension · EpGroupoid.tgt_mem_boundary_iff`, because $t \circ (s|_{\text{sheet}})^{-1}$ is a local sc-diffeomorphism and keeps the degeneracy index; so $X \setminus \partial X$ is open `Chapter12/`

AuxNormExtension · EpGroupoid.isOpen_compl_boundary (the degeneracy index is upper semicontinuous) and saturated.

The proof has two parts: a global part, the construction of §XII.3 with local extensions near the boundary as input, and a local step that constructs those local extensions.

The global part

The local extensions Chapter12/AuxNormExtension · EpStrongBundle.HasLocalExtensions: say N *has local extensions* if every $x \in \partial X$ has an open $V \ni x$ and a local auxiliary norm M on V (§XII.2) with $M = N$ on $W_{0,1}|(V \cap \partial X)$.

The global part Chapter12/AuxNormExtension · EpStrongBundle.theorem_12_2_7_hyp: if $|X|$ is paracompact and N is an auxiliary norm over ∂X with local extensions, then there is an auxiliary norm $\tilde{N}$ of W with $\tilde{N} = N$ over ∂X. Tameness is not used here.

Step 1, data at each point. For every object x choose a natural representation R_x and a local auxiliary norm M_x on its domain: - if $x \in \partial X$: the local extension M on $V \ni x$, and R_x from Theorem 7.1.13 with domain inside V; $M_x = M$ restricted Chapter12/AuxNormExtension · StrongBundle.IsLocalAuxNorm.mono, still equal to N over $R_x.U \cap \partial X$; - if $x \notin \partial X$: R_x with domain inside the open set $X \setminus \partial X$, and $M_x = N_0$, the restriction of a fixed Part I auxiliary norm N_0 of the object bundle (Proposition 5.1.3).

Step 2, average and transport, as in §XII.3: $\bar{M}_x$ is G_x-invariant on $R_x.U$, and $T_x = \mathrm{transNorm}(R_x, \bar{M}_x)$ is a local auxiliary norm on $S_x = \mathrm{sat}(R_x.U)$, invariant under morphisms.

Step 3, glue. A partition of unity (β_x) of $|X|$ subordinate to $(\pi(S_x))_x$, which cover $|X|$ (every $\pi(y)$ lies in $\pi(S_y)$), and $\tilde{N} = \sum_x (\beta_x \circ \pi \circ P)\, T_x$. By §XII.3, $\tilde{N}$ is an auxiliary norm of W.

Step 4, $\tilde{N} = N$ over ∂X. Fix w over $y \in \partial X$. The claim is that every term is $\beta_x(\pi(y))\, N(w)$; then $\tilde{N}(w) = (\sum_x \beta_x(\pi(y)))N(w) = N(w)$. If $\beta_x(\pi(y)) = 0$ there is nothing to prove. Otherwise $\pi(y) \in \mathrm{supp}\, \beta_x \subset \pi(S_x)$, so $y \in S_x$ (S_x is saturated), and there is a morphism $\varphi : y \to y'$ with $y' \in R_x.U$. - *x is a boundary point.* Otherwise $R_x.U \subset X \setminus \partial X$, but $y' = t(\varphi) \in \partial X$ because $y \in \partial X$; contradiction. - *The transported vector.* $T_x(w) = \bar{M}_x(v)$ with $v = \mu(\varphi_y, w)$ for the selected morphism φ_y; v lies over a point of $R_x.U \cap \partial X$, and $N(v) = N(w)$ by the invariance of N over ∂X. - *Each term of the average is $N(w)$.* For $g \in G_x$, $g * v$ lies over $\Phi(g)(P(v)) \in R_x.U$, again in ∂X; there $M_x = N$, so $M_x(g * v) = N(g*v) = N(v) = N(w)$, by invariance of N under the morphism $\Gamma(g, P(v))$. - *The average of constants.* $\bar{M}_x(v) = |G_x|^{-1} \cdot |G_x| \cdot N(w) = N(w)$; G_x is finite and nonempty.

So $\beta_x(\pi(y))T_x(w) = \beta_x(\pi(y))N(w)$ for every x, and $\tilde{N}(w) = N(w)$. □

The text's argument for Step 4 is the same in substance: its local norms near boundary points restrict to N, and its interior pieces are supported away from ∂X. The formal proof records why interior pieces do not reach ∂X: their saturations miss ∂X because ∂X is saturated.

A property of this Lean statement. The global part takes the local extensions as the hypothesis `HasLocalExtensions`; it is valid as stated and serves as the global half of Theorem 12.2.7, the hypothesis being discharged below for every tame X. Its own satisfiability witness Chapter12/AuxNormExtension · EpStrongBundle.theorem_12_2_7_hyp_witness: over every ep-groupoid with paracompact $|X|$ some N satisfies both hypotheses, the restriction of a norm of Theorem 12.2.2, since a global auxiliary norm has itself as local extension Chapter12/AuxNormExtension · EpStrongBundle.hasLocalExtensions_of_isAuxNorm. This witness is a consistency check only: in it the extension problem is solved by the global norm itself.

The local step

The local step is a statement about one strong bundle over a tame M-polyfold, the object bundle; it does not see the morphisms. The text takes a tame local strong bundle model $K = R(U \lhd F) \to O$ at a boundary point, the tame retraction r of its base, and defines on $K|O$ a norm by the formula $N(rq(u), A(rq(u))w) + \|w - A(rq(u))w\|_{F_1}$, with q a shift onto the boundary. The formal proof splits this into a retraction onto the boundary, which lives in a tame chart of the base, and a norm formula, which can be evaluated in any strong bundle chart.

The retraction onto the boundary (extra, Part I) `Chapter12/AuxNormLocalExtension · MPolyfold.exists_retraction_boundary`. Let X be a tame M-polyfold and $x_0 \in \partial X$. There are an open $V \ni x_0$ and a map $\rho : X \to X$, continuous on V, with $\rho(V) \subset \partial X$ and $\rho(y) = y$ for $y \in V \cap \partial X$.

The coordinates. Let C be a partial quadrant with a presentation $T : C \cong [0,\infty)^n \oplus H$ (a linear isomorphism of the ambient space carrying C onto the standard partial quadrant), and write points as (t, h), $t \in [0,\infty)^n$. Define

$$\tau(t,h) = \min(t_1, \dots, t_n), \qquad q(t,h) = (t,h) - \tau(t,h)\,(1, \dots, 1, 0)$$

`Chapter12/AuxNormLocalExtension · minCoord` `Chapter12/AuxNormLocalExtension · boundaryShift`, read through T, for $n \geq 1$. The properties: - τ and q are continuous `Chapter12/AuxNormLocalExtension · continuous_minCoord` `Chapter12/AuxNormLocalExtension · continuous_boundaryShift`, as a minimum of finitely many coordinates and an affine expression in it; - the coordinates of q are $t_i - \tau$ `Chapter12/AuxNormLocalExtension · coord_boundaryShift`, and $\tau \leq t_i$ for all i `Chapter12/AuxNormLocalExtension · minCoord_le`, so q takes values in C `Chapter12/AuxNormLocalExtension · mem_quadrant_boundaryShift`; - the minimum is attained at some i `Chapter12/AuxNormLocalExtension · exists_coord_eq_minCoord`, where the coordinate of q is 0, so q takes values in ∂C `Chapter12/AuxNormLocalExtension · exists_coord_boundaryShift_eq_zero`; - if some $t_i = 0$, the minimum is 0 and q is the identity `Chapter12/AuxNormLocalExtension · boundaryShift_eq_self`.

A point of C has degeneracy index at least 1 iff some half-line coordinate vanishes `Chapter12/AuxNormLocalExtension · one_le_degIndexWith_iff`.

The retraction. By tameness, x_0 lies in a tame chart ψ (`IsTame.exists_chart`) whose model is a tame retract $O = r(U)$, $U \subset C$ relatively open, r a tame sc-smooth retraction. Choose a presentation T of C. *(a) The degeneracy index is read in the chart.* Because r is tame, the degeneracy index of X at $z \in \operatorname{dom}\psi$ equals that of C at $\psi(z)$ (`degIndex_eq_chartDegIndex_of_isTameWith`). So $z \in \partial X$ iff some coordinate t_i of $T\psi(z)$ vanishes. *(b) $n \geq 1$:* $x_0 \in \partial X$ forces a vanishing coordinate. *(c) The domain.* $U = V_0 \cap C$ for an open V_0 of the ambient space; put $V = \operatorname{dom}\psi \cap \psi^{-1}(q^{-1}(V_0))$, open by continuity of ψ and q. It contains x_0: $q(\psi(x_0)) = \psi(x_0) \in O \subset U \subset V_0$ by (a) and the last property of q. *(d) The map.* $\rho = \psi^{-1} \circ r \circ q \circ \psi$ on V. For $y \in V$, $q(\psi(y)) \in V_0 \cap C = U$, so r applies and $r(q(\psi(y))) \in O$, the image of the chart. *(e) Continuity* on V: a composite of continuous maps, r being continuous on level 0 since it is sc-smooth. *(f) Values in ∂X.* Tameness of r means r keeps the degeneracy index: $d_C(r(p)) = d_C(p)$ for $p \in U$ (`degIndex_apply`); $q(\psi(y)) \in \partial C$, so $r(q(\psi(y))) \in \partial C$, and by (a) $\rho(y) \in \partial X$. *(g) The identity on $V \cap \partial X$.* For $y \in V \cap \partial X$, $q(\psi(y)) = \psi(y)$ by (a) and the last property of q; $\psi(y) \in O$ and r is the identity on its image (`apply_eq_self_of_mem_retract`); so $\rho(y) = y$. □

The text takes $x_0 = (0, h_0)$ in the chart. This is not needed: q maps into ∂C and fixes ∂C whatever the coordinates of x_0.

The norm N' (extra, Part I) `Chapter12/AuxNormLocalExtension · StrongBundle.retractNorm`. Fix a strong bun-

dle chart c of the atlas, with base chart φ, model $K = R(U \lhd F)$, and its level-1 form $\Phi[1] : W_{0,1}|\operatorname{dom}\varphi \to (U \lhd F)[1]$. The fibre part of the retraction R of the chart at a base point u is a map $A(u) : F_1 \to F_1$, $h \mapsto \mathrm{pr}_2 R(u,h)$ `Chapter12/AuxNormLocalExtension · StrongBundle.chartFibMap`. Its properties: - $A(u)$ is linear in h `Chapter12/AuxNormLocalExtension · StrongBundle.isLinearMap_chartFibMap`, since R is a strong bundle retraction, linear on fibres; - for $u \in O$, $(u, A(u)h) \in K[1]$ `Chapter12/AuxNormLocalExtension · StrongBundle.mem_K_chartFibMap` and $A(u)h = h$ when $(u,h) \in K[1]$ `Chapter12/AuxNormLocalExtension · StrongBundle.chartFibMap_of_mem`, since R is a retraction onto K; points of O lie in U `Chapter12/AuxNormLocalExtension · StrongBundle.mem_U_of_mem_O`; - $A(u)h$ is the fibre part of the shifted retraction $R[1]$ `Chapter12/AuxNormLocalExtension · StrongBundle.shiftFiber_eq_chartFibMap`, and $(u,h) \mapsto R[1](u,h)$ is continuous on $U \oplus F_1$ `Chapter12/AuxNormLocalExtension · StrongBundle.continuousOn_shiftFiber_R`, as level 0 of the sc-smooth map $R : (U \lhd F)[1] \to (U \lhd F)[1]$.

The chart $\Phi[1]$ reads $W_{0,1}$ over the chart domain: projections commute with it `Chapter12/AuxNormLocalExtension · StrongBundle.proj_oneChartOf_symm` and it agrees with the level-1 lift of the chart `Chapter12/AuxNormLocalExtension · StrongBundle.oneChartOf_toOne`. For a vector w over y with chart coordinates $(u,h) = \Phi[1](w)$ and a map ρ with $\rho(y)$ in the chart domain, the *transport* of w to the fibre over $\rho(y)$ is

$$\Phi[1]^{-1}(\varphi(\rho(y)),\ A(\varphi(\rho(y)))h),$$

a linear map of fibres $W_y \to W_{\rho(y)}$ `Chapter12/AuxNormLocalExtension · StrongBundle.retractTransport`, compatible with the level-1 lift `Chapter12/AuxNormLocalExtension · StrongBundle.liftOneFiber_retractTransport`. Then

$$N'(w) = N(\text{transport of } w) + \|h - A(\varphi(\rho(y)))h\|_{F_1},$$

the text's formula read in $\Phi[1]$ `Chapter12/AuxNormLocalExtension · StrongBundle.retractNorm_toOne`. It is a fibre seminorm over each y in the chart domain with $\rho(y)$ in the chart domain and N a seminorm over $\rho(y)$ `Chapter12/AuxNormLocalExtension · StrongBundle.isFiberSeminorm_retractNorm`: the first term is a seminorm composed with a linear map, the second the norm of a linear expression in h.

N' is a local auxiliary norm `Chapter12/AuxNormLocalExtension · StrongBundle.isLocalAuxNorm_retractNorm`. The lemma is stated for a general set A of base points (not only ∂X), a function N that is continuous on $W_{0,1}|A$, a seminorm on the fibres over A and satisfies condition (2) for sequences over A converging to points of A, and a map ρ continuous on an open V inside the chart domain, with $\rho(V) \subset A \cap \operatorname{dom}\varphi$. Then N' is a local auxiliary norm on V. By §XII.2, three checks.

Write $S = W_{0,1}|V$, $u(w) = \varphi(\rho(P(w)))$, $z(w) = (u(w), A(u(w))\,\mathrm{pr}_2\Phi[1](w)) \in K[1]$ and $T(w) = \Phi[1]^{-1}(z(w))$, which lies over $\rho(P(w))$.

Continuity. $\Phi[1]$ is continuous on S; u is continuous on S (continuity of ρ on V and of φ); z is continuous as $R[1]$ applied to $(u(w), \mathrm{pr}_2\Phi[1](w))$; T is continuous as $\Phi[1]^{-1}$ after z, whose values lie in $K[1]$. Then $N \circ T$ is continuous on S because N is continuous on $W_{0,1}|A$ and $T(w)$ lies over $\rho(P(w)) \in A$; the second term is continuous as a norm of a difference of continuous maps.

Seminorm. Over $y \in V$: the lemma above, with $\rho(y) \in A$.

Condition (2). Let $P(w_k) \to y \in V$ and $N'(w_k) \to 0$. Eventually $w_k \in S$. - Both terms are nonnegative and bounded by $N'(w_k)$, so $N(T(w_k)) \to 0$ and $\|h_k - A(u_k)h_k\| \to 0$, where $h_k = \mathrm{pr}_2\Phi[1](w_k)$, $u_k = u(w_k)$. - $T(w_k)$ lies over $\rho(P(w_k)) \in A$, which converges to $\rho(y) \in A$ by continuity of ρ. Condition (2) for N, applied to the tail of the sequence that lies over V, gives $T(w_k) \to 0_{\rho(y)}$ in $W_{0,1}$. - Reading this in the chart, $\Phi[1](T(w_k)) = z(w_k) \to (\varphi(\rho(y)), 0)$, so $A(u_k)h_k \to 0$ in F_1. - Then

$h_k = (h_k - A(u_k)h_k) + A(u_k)h_k \to 0$ in F_1, and the base coordinates $\varphi(P(w_k)) \to \varphi(y)$. - Convergence in the chart gives convergence in $W_{0,1}$: $w_k \to 0_y$ `Chapter12/AuxNormLocalExtension · MChart.tendsto_of_tendsto_apply`. □

$N' = N$ where ρ is the identity `Chapter12/AuxNormLocalExtension · StrongBundle.retractNorm_eq`: if $y \in V \cap A$ and $\rho(y) = y$, then $N'(w) = N(w)$ for w over y. Indeed $\Phi[1](w) = (\varphi(y), h)$ lies in $K[1]$, so $A(\varphi(y))h = h$; the transport is $\Phi[1]^{-1}(\varphi(y), h) = w$, and the second term is $\|h - h\| = 0$ `Chapter12/AuxNormLocalExtension · StrongBundle.retractNorm_eq_of_fix`.

The local step `Chapter12/AuxNormLocalExtension · StrongBundle.exists_localExtension_of_isTame`. Let X be a tame M-polyfold, and N continuous on $W_{0,1}|\partial X$, a seminorm on the fibres over ∂X, with condition (2) for sequences over ∂X. Every $x_0 \in \partial X$ has an open $V \ni x_0$ and a local auxiliary norm M on V with $M = N$ over $V \cap \partial X$.

Proof. Take ρ and V_1 from the retraction lemma, a strong bundle chart c of the atlas at x_0, and

$$V = V_1 \cap \operatorname{dom}\varphi \cap \rho^{-1}(\operatorname{dom}\varphi),$$

open because ρ is continuous on the open $V_1 \cap \operatorname{dom}\varphi$; it contains x_0 because $\rho(x_0) = x_0$. On V, ρ is continuous with values in $\partial X \cap \operatorname{dom}\varphi$. Put $M = N'$ with $A = \partial X$. By the lemma, M is a local auxiliary norm on V; by the identity clause of the retraction lemma, $\rho(y) = y$ on $V \cap \partial X$, so $M = N$ there. □

For an ep-groupoid, the local step applied to the object M-polyfold, which is tame when X is, discharges the hypothesis of the global part `Chapter12/AuxNormLocalExtension · EpStrongBundle.hasLocalExtensions_of_isTame`: an auxiliary norm over ∂X in the sense above is continuous, a seminorm and satisfies condition (2) over ∂X, which is what the local step asks.

Theorem 12.2.7 is then one line: the global part with the local step `Chapter12/AuxNormLocalExtension · EpStrongBundle.theorem_12_2_7`.

Where the formal proof departs from the text. (a) The text works in a tame local strong bundle model, a strong bundle chart whose base retraction is tame. The formal proof places the retraction onto the boundary in a tame chart of the base, which tameness of X provides, and evaluates the norm formula in an arbitrary strong bundle chart of the atlas. Only the continuity of ρ and its two properties enter the norm formula, so no tame strong bundle chart is needed; the result is the text's statement under the text's hypotheses. (b) The text's formula composes A with $rq(u)$ in the same chart where the fibre lives; the formal N' composes with $\varphi(\rho(y))$, which is the same expression read through ψ and φ. (c) The invariance of N under morphisms is not used in the local step; it enters through Step 4 of the global part, where each term of the average over G_x is identified with N. (d) Tameness is used once, in the retraction lemma, to read the degeneracy index in the chart and to keep it under r.

XII.7 Definition 12.2.8 (2017: 12.5) and Remark 12.2.9 (2017: 12.1): mixed convergence

The second half of §12.2 repeats the first for *reflexive* auxiliary norms, the norms of Part I's Definition 5.1.7 (Part V), which are lower semicontinuous along sequences that converge on the base and weakly on the fibres. They are needed in Chapter 15 for compactness in the presence of perturbations. The

new ingredient is one fact: morphisms preserve mixed convergence (Remark 12.2.9 in the stronger form below). With it, every step of §XII.3 carries lower semicontinuity.

Reflexive 1-fibres (restated) `Chapter12/ReflexiveAuxNormEp · EpStrongBundle.HasReflexiveFibers`, p. 576. The text's standing assumption before Definition 12.2.8 is that the strong bundle has reflexive $(0,1)$-fibres. The formalization reads it on the object bundle in the sense of Part I's Definition 5.1.4: every object lies in the base of a strong bundle chart of the atlas whose fibre F has F_1 reflexive.

Definition 12.2.8 (same) `Chapter12/ReflexiveAuxNormEp · EpStrongBundle.MixedTendsto` `Chapter12/ReflexiveAuxNormEp · EpStrongBundle.definition_12_2_8`, p. 576. A sequence $w_k \in W_{0,1}$ converges *mixed* to $w \in W_{0,1}$ if $P(w_k) \to P(w)$ in X and, in some strong bundle chart around $P(w)$ with reflexive F_1, the fibre coordinates $\mathrm{pr}_2\Phi[1](w_k)$ converge weakly in F_1 to $\mathrm{pr}_2\Phi[1](w)$. It is Part I's mixed convergence on the object bundle; the declaration `definition_12_2_8` records the unfolding.

Remark 12.2.9 (stronger) `Chapter12/ReflexiveAuxNormEp · EpStrongBundle.remark_12_2_9`, pp. 576–577. The text remarks that mixed convergence does not depend on the chart and is compatible with the morphisms. The formal statement has three clauses: for $w_k \to w$ mixed,

- (i) in *every* chart of the atlas around $P(w)$ with reflexive F_1 the fibre coordinates converge weakly;
- (ii) $w_k \to w$ in W (on level 0);
- (iii) for every sc-smooth local section α of the source map on an open $S \ni P(w)$ ($s(\alpha(y)) = y$ for $y \in S$), $\alpha * w_k \to \alpha * w$ mixed `Chapter12/ReflexiveAuxNormEp · EpStrongBundle.mixedTendsto_actOne`.

(i) is Part I's Lemma 5.1.5 (Part V), and (ii) Part I's observation that weak convergence in F_1 implies convergence in F_0 by the compactness of $F_1 \to F_0$. Clause (iii) is the stronger part: the text states compatibility with morphisms; the formal statement gives it along any sc-smooth family of morphisms, which is what the averaging and transport of §XII.3 need. Through a given morphism φ with $s(\varphi) = P(w)$ one takes for α the local inverse of s on the sheet of φ, and gets $\mu(\alpha(P(w_k)), w_k) \to \mu(\varphi, w)$ mixed `Chapter12/ReflexiveAuxNormEp · EpStrongBundle.exists_mixedTendsto_μ`.

Proof of (iii). The text's argument for chart independence uses that a transition map is bounded on the fibres and continuous on level 0. The formal proof applies the same argument to the fibre maps of μ.

Step 1, level 0. The map $w \mapsto \alpha * w$ is sc-smooth on $W|S$ (`isScSmoothOn_sectionAct`), hence continuous on level 0; by (ii), $\alpha * w_k \to \alpha * w$ in W.

Step 2, a bound. Fix a Part I auxiliary norm N_0 of the object bundle. Near any point, two auxiliary norms are comparable: if N, N' are a local auxiliary norm and a continuous fibre seminorm, there are a neighbourhood V and $C > 0$ with $N'(v) \le C\,N(v)$ over V `Chapter12/ReflexiveAuxNormEp · StrongBundle.exists_le_mul_auxNorm`. (This is Part I's Lemma 5.1.2, in local form, proved through the upper and lower bounds of §XII.2.) Applied to N_0 and $N_0 \circ (\alpha*)$, which is a continuous fibre seminorm near $P(w)$: $N_0(\alpha * v) \le C\,N_0(v)$ for $P(v)$ near $P(w)$. A mixed convergent sequence is eventually bounded in N_0 `Chapter12/ReflexiveAuxNormEp · StrongBundle.MixedTendsto.exists_eventually_le`: its fibre coordinates converge weakly, hence are bounded in F_1 by the uniform boundedness principle, and N_0 is bounded by a multiple of the chart norm near $P(w)$. So $N_0(\alpha * w_k)$ is eventually bounded.

Step 3, bounded and convergent on level 0 *implies mixed* `Chapter12/ReflexiveAuxNormEp · StrongBundle.mixedTendsto_of_bounded`. Let $v_k \to v$ in W with $N_0(v_k) \le r$ eventually. Take a chart around $P(v)$ with reflexive F_1 (reflexive fibres). Near $P(v)$, N_0 dominates a multiple of the chart norm (the lower bound of §XII.2, ex–

ists_bounds), so the fibre coordinates are eventually bounded in F_1. A bounded sequence in a reflexive space whose limit on level 0 is known converges weakly to that limit (weakTendsto_liftOneSnd_of_tendsto, Part I): every subsequence has a weakly convergent subsequence (weak sequential compactness in reflexive spaces, Part V), its weak limit equals the level-0 limit because $F_1 \to F_0$ is continuous and limits in F_0 are unique, so all subsequential weak limits agree, and the whole sequence converges weakly. Hence $v_k \to v$ mixed.

Steps 1–3 give (iii). □

Lower semicontinuity along mixed convergence (extra) Chapter12/ReflexiveAuxNormEp · StrongBundle.MixedLscAt. A function M on $W_{0,1}$ is *mixed lower semicontinuous at* w if for every sequence $w_k \to w$ mixed and every $\varepsilon > 0$, eventually $M(w) - \varepsilon < M(w_k)$. This implies $M(w) \le \liminf_k M(w_k)$ in the extended reals Chapter12/ReflexiveAuxNormEp · StrongBundle.MixedLscAt.le_liminf, the form in Definition 12.2.10. The pointwise ε-form is used because it composes well:

- *sums*: if M_1, M_2 are mixed lower semicontinuous at w, so is $M_1 + M_2$, with $\varepsilon/2$ for each Chapter12/ReflexiveAuxNormEp · StrongBundle.MixedLscAt.add, and so are finite sums Chapter12/ReflexiveAuxNormEp · StrongBundle.mixedLscAt_finset_sum;
- *nonnegative constants*: cM for $c \ge 0$, with ε/c (trivially for $c = 0$) Chapter12/ReflexiveAuxNormEp · StrongBundle.MixedLscAt.const_mul;
- *products with functions of the base point* Chapter12/ReflexiveAuxNormEp · StrongBundle.MixedLscAt.mul_comp: if $g \ge 0$ is continuous at $P(w)$ and $M \ge 0$ over an open $V \ni P(w)$, then $w \mapsto g(P(w))M(w)$ is mixed lower semicontinuous at w. With $a = g(P(w))$, $b = M(w)$ and $\delta = \varepsilon/(a + b + 1)$: eventually $g(P(w_k)) > a - \delta$ (continuity of g along $P(w_k) \to P(w)$), $M(w_k) > b - \delta$, and both are nonnegative; then $g(P(w_k))M(w_k) \ge ab - \delta(a + b) > ab - \varepsilon$;
- *functions that vanish at the point* Chapter12/ReflexiveAuxNormEp · StrongBundle.mixedLscAt_of_nonpos: if $M(w) \le 0$ and $M \ge 0$ everywhere, M is mixed lower semicontinuous at w;
- *local agreement* Chapter12/ReflexiveAuxNormEp · StrongBundle.MixedLscAt.congr: if $M = M'$ over an open $V \ni P(w)$, the property passes, because $P(w_k) \in V$ eventually;
- *constants* Chapter12/ReflexiveAuxNormEp · StrongBundle.mixedLscAt_const;
- *through morphisms* Chapter12/ReflexiveAuxNormEp · EpStrongBundle.mixedLscAt_comp_actOne: if M is mixed lower semicontinuous at $\alpha * w$, then $M \circ (\alpha*)$ is at w, by (iii) of Remark 12.2.9.

XII.8 Definition 12.2.10, Theorem 12.2.11 and Remark 12.2.12 (2017: Definition 12.6, Theorem 12.4, Remark 12.2): reflexive auxiliary norms exist

Definition 12.2.10 (same) Chapter12/ReflexiveAuxNormEp · EpStrongBundle.IsReflexiveAuxNorm, p. 577. A reflexive auxiliary norm is an auxiliary norm N (Definition 12.2.1: continuous, invariant under morphisms, a complete norm on each fibre, condition (2)) with the additional property (3): $N(w) \le \liminf_k N(w_k)$ whenever $w_k \to w$ mixed, the liminf taken in the extended reals. The text prints the liminf as $k \to 0$; it is presumably $k \to \infty$, as in Part I's Definition 5.1.7 (Part V), and is read so. A reflexive auxiliary norm of the ep-groupoid is one of Part I's on the object bundle Chapter12/ReflexiveAuxNormEp · EpStrongBundle.IsReflexiveAuxNorm.toStrong, and it is mixed lower semicontinuous at every point Chapter12/ReflexiveAuxNormEp · StrongBundle.IsReflexiveAuxNorm.mixedLscAt Chapter12/ReflexiveAuxNormEp · EpStrongBundle.IsReflexiveAuxNorm.mixedLscAt (the liminf inequality gives the ε-form).

Conversely, an auxiliary norm that is mixed lower semicontinuous at every point is reflexive Chapter12/ReflexiveAuxNormEp · EpStrongBundle.isReflexiveAuxNorm_of_mixedLscAt. The constructions below prove the ε-form and convert at the end.

Theorem 12.2.11 (same) Chapter12/ReflexiveAuxNormEp · EpStrongBundle.theorem_12_2_11, p. 577. Let $(P : W \to X, \mu)$ be a strong bundle over an ep-groupoid with paracompact $|X|$ and reflexive $(0,1)$-fibres. Then (1) reflexive auxiliary norms exist; (2) if N is one and $f : X \to (0,\infty)$ is continuous and constant along morphisms, then $h \mapsto f(P(h))N(h)$ is one; (3) for every auxiliary norm N' there are reflexive auxiliary norms $N_1 \le N' \le N_2$. The hypotheses are the text's; the formal statement is the conjunction of the three clauses.

(1) Existence Chapter12/ReflexiveAuxNormEp · EpStrongBundle.theorem_12_2_11_exists. The construction is that of Theorem 12.2.2 (§XII.3), with lower semicontinuity carried through each step.

The local norms. The text takes on $U(x)$ the norm $\|\mathrm{pr}_2 \circ \Psi(g * h)\|_{F_1}$ in a chart with reflexive F_1. The formal proof takes the restriction of one reflexive auxiliary norm N_0 of the object bundle, which Part I's Proposition 5.1.8 (Part V) provides under reflexive fibres; it is mixed lower semicontinuous everywhere.

Average Chapter12/ReflexiveAuxNormEp · EpStrongBundle.mixedLscAt_avgNorm. Each term $M(g * w)$ of $\bar{M}(w) = |G_x|^{-1} \sum_g M(g * w)$ is mixed lower semicontinuous at w over $U(x)$: $g*$ is the action of the sc-smooth family of morphisms $\Gamma(g, \cdot)$, so the property passes through it; the finite sum and the factor $|G_x|^{-1}$ keep it.

Transport Chapter12/ReflexiveAuxNormEp · EpStrongBundle.mixedLscAt_transNorm. Near a point w_0 over the saturation, §XII.3 showed $T = \bar{M} \circ (\rho*)$ for a local inverse ρ of s; this is an sc-smooth local section of s, so the property passes through $\rho*$, and T agrees with $\bar{M} \circ (\rho*)$ on a neighbourhood, so it passes to T by local agreement.

Glue Chapter12/ReflexiveAuxNormEp · EpStrongBundle.mixedLscAt_glueNorm. Fix w over y. By local finiteness, $\pi(y)$ has a neighbourhood n in $|X|$ meeting only finitely many supports; on $O = \pi^{-1}(\mathrm{int}\, n)$, open and containing y, $\tilde{N} = \sum_{i \in F} (\beta_i \circ \pi \circ P) T_i$ for a finite set F. For $i \in F$: - if $y \in S_i$ (the saturation carrying T_i): T_i is mixed lower semicontinuous at w and nonnegative over the open S_i, and $\beta_i \circ \pi$ is continuous and nonnegative, so the product rule applies; - if $y \notin S_i$: then $\beta_i(\pi(y)) = 0$, since $\mathrm{supp}\, \beta_i \subset \pi(S_i)$; the term vanishes at w and is nonnegative, so it is mixed lower semicontinuous at w.

The finite sum is mixed lower semicontinuous, and so is $\tilde{N}$ by local agreement on O. The text's argument for (3) at this point is "since the sum is locally finite"; the formal proof shows that this suffices, the terms with vanishing coefficient at the limit being harmless because they are nonnegative.

Conclusion Chapter12/ReflexiveAuxNormEp · EpStrongBundle.isReflexiveAuxNorm_glueNorm: $\tilde{N}$ is an auxiliary norm by §XII.3 and mixed lower semicontinuous everywhere, hence reflexive.

(2) Scaling Chapter12/ReflexiveAuxNormEp · EpStrongBundle.theorem_12_2_11_mul, which is **Remark 12.2.12 (1)** (same) Chapter12/ReflexiveAuxNormEp · EpStrongBundle.remark_12_2_12_1, p. 577. Let N be a reflexive auxiliary norm and $f > 0$ continuous with $f(s(\varphi)) = f(t(\varphi))$. Then $(f \circ P)N$ is: continuous as a product; a complete norm on each fibre, being a positive multiple of one; condition (2): if $P(h_k) \to x$ and $f(P(h_k))N(h_k) \to 0$, then $N(h_k) \to 0$ because $f(P(h_k)) \to f(x) > 0$; invariant because f is constant along morphisms and N invariant; and mixed lower semicontinuous by the product rule.

Remark 12.2.12 (2) (same) Chapter12/ReflexiveAuxNormEp · EpStrongBundle.remark_12_2_12_2, p. 577. For auxiliary norms N_1, N_2 and paracompact $|X|$ (which the remark's context has), there is a continuous

$f : X \to (0,\infty)$, constant along morphisms, with $N_1 \le (f \circ P)N_2$.

Step 1, local constants. For every object y, the local comparison lemma of §XII.7 gives an open $V_y \ni y$ and $C_y > 0$ with $N_1(w) \le C_y N_2(w)$ over V_y.

Step 2, saturation. The bound holds over $\mathrm{sat}(V_y)$: for w over z with a morphism $\varphi : z \to z'$, $z' \in V_y$, the vector $k = \mu(\varphi, w)$ `Chapter8/Book2021 · EpStrongBundle.exists_oneVal_eq_μ` lies over V_y, and $N_i(w) = N_i(k)$ by invariance of both norms.

Step 3, glue the constants. Let (β_y) be a partition of unity of $|X|$ subordinate to the open cover $(\pi(\mathrm{sat}(V_y)))_y$, and $g = \sum_y \beta_y C_y$ on $|X|$, continuous by local finiteness and positive because at each point some $\beta_y > 0$. Put $f = g \circ \pi$, continuous and constant along morphisms.

Step 4, the estimate. For w over z with $\pi(z) = \zeta$:

$$N_1(w) = \sum_y \beta_y(\zeta) N_1(w) \le \sum_y \beta_y(\zeta) C_y N_2(w) = f(z) N_2(w),$$

the inequality termwise: where $\beta_y(\zeta) \neq 0$, $z \in \mathrm{sat}(V_y)$ by subordination and Step 2 applies. □

Remark 12.2.12 (3) (stronger) `Chapter12/ReflexiveAuxNormEp · EpStrongBundle.remark_12_2_12_3`, p. 578. The text: for a saturated open U, a function $N \ge 0$ on $W_{0,1}|U$ satisfying the liminf inequality there, and a continuous functorial $f \ge 0$ with support in U, the function $\tilde{N} = f \cdot N$ on U, 0 off U, satisfies the liminf inequality on all of $W_{0,1}$. The formal statement drops what the estimate does not use: U open (not necessarily saturated), $N \ge 0$ and mixed lower semicontinuous over U, $f \ge 0$ continuous (not necessarily functorial or supported in U). *Proof.* Fix $w_k \to w$ mixed. If $P(w) \in U$: the product rule gives lower semicontinuity of $(f \circ P)N$ at w, and $\tilde{N}$ agrees with it over the open U, so local agreement applies. If $P(w) \notin U$: $\tilde{N}(w) = 0$ and $\tilde{N} \ge 0$ everywhere. In both cases the liminf inequality follows. □ The saturation of U, the functoriality of f and $\mathrm{supp}\, f \subset U$ serve the continuity and the invariance of $\tilde{N}$, which the text uses when it applies the remark, not the estimate.

(3) The sandwich `Chapter12/ReflexiveAuxNormEp · EpStrongBundle.theorem_12_2_11_sandwich`. Given an auxiliary norm N', take a reflexive N from (1). Remark 12.2.12 (2), applied twice, gives functorial continuous $f, g > 0$ with $N \le (f \circ P)N'$ and $N' \le (g \circ P)N$. Put $N_1 = (f \circ P)^{-1}N$ and $N_2 = (g \circ P)N$; both are reflexive by (2) (with $1/f$ and g), and $N_1 \le N' \le N_2$. □

XII.9 Theorem 12.2.13 (2017: 12.5): extension of reflexive auxiliary norms

Theorem 12.2.13 (explicit) `Chapter12/ReflexiveLocalExtension · EpStrongBundle.theorem_12_2_13`, p. 578. Let X be a tame ep-groupoid with $|X|$ paracompact, $(P : W \to X, \mu)$ a strong bundle with reflexive $(0,1)$-fibres, and N a reflexive auxiliary norm of $W_{0,1}|\partial X$. Then there is a reflexive auxiliary norm $\tilde{N}$ of W with $\tilde{N} = N$ over ∂X.

Explicit: the paracompactness of $|X|$ is not printed in the statement. The proof refers to "partitions of unity, as in Theorem 12.2.11", which takes it from Theorem 12.2.11; Theorem 12.2.7, whose construction is reused, states it. *A reading*: a reflexive auxiliary norm "for $W_{0,1}|\partial X$" asks the liminf inequality only for mixed convergent sequences over ∂X. The proof needs no more, because the first term of the local norm N' is evaluated over $\rho(\cdot) \in \partial X$.

The proof has the two parts of §XII.6, each with lower semicontinuity added.

Reflexive norms over a subset `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.IsReflexiveAuxNormOn`: an auxiliary norm over A (§XII.6) with $N(w) \le \liminf N(w_k)$ for mixed convergent sequences over A with limit over A. A global reflexive norm restricts `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.IsReflexiveAuxNorm.isReflexiveAuxNormOn`.

The global part

Reflexive local extensions `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.HasReflexiveLocalExtensions`: every $x \in \partial X$ has an open $V \ni x$ and a local auxiliary norm M on V, mixed lower semicontinuous at every vector over V, with $M = N$ over $V \cap \partial X$. They are in particular local extensions in the sense of §XII.6 `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.HasReflexiveLocalExtensions.hasLocalExtensions`.

The global part `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.theorem_12_2_13_hyp`: for paracompact $|X|$, reflexive $(0,1)$-fibres and a reflexive auxiliary norm N over ∂X with reflexive local extensions, there is a reflexive auxiliary norm $\tilde{N}$ of W with $\tilde{N} = N$ over ∂X. *Proof.* Steps 1–4 of §XII.6, with these choices: at boundary points the reflexive local extensions; at interior points the restriction of a reflexive auxiliary norm N_0 of the object bundle (Part I's Proposition 5.1.8). The result is an auxiliary norm equal to N over ∂X by §XII.6. Its lower semicontinuity: the averages, transports and the glued sum keep mixed lower semicontinuity by the three lemmas of §XII.8 (`mixedLscAt_avgNorm`, `mixedLscAt_transNorm`, `mixedLscAt_glueNorm`), which hold for any local norms that are mixed lower semicontinuous. So $\tilde{N}$ is reflexive. □

A property of this Lean statement. The global part takes the reflexive local extensions as a hypothesis; it is valid as stated and serves as the global half of Theorem 12.2.13, the hypothesis being discharged below for every tame X. Its satisfiability witnesses: the restriction of a global reflexive norm has reflexive local extensions `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.hasReflexiveLocalExtensions_of_isReflexiveAuxNorm`, so every bundle with reflexive fibres and paracompact $|X|$ satisfies all hypotheses of the global part `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.theorem_12_2_13_hyp_witness`; concretely the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ over the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter12/ReflexiveAuxNormExtension · TrivialBundle.epBundle_reflexive_witness`. These are consistency checks: in the first the extension is the global norm itself, and in the second $\mathbb{R}^n$ is modelled on all of $\mathbb{R}^n$ with a single chart, so $\partial X = \emptyset$ and the boundary condition is empty.

The local step: the inspection

The text says that the local extension N' of Theorem 12.2.7 is well behaved under mixed convergence "by inspection". The formal proof supplies the inspection, in three lemmas.

The setting. Near $x_0 \in \partial X$, in a tame chart of the base with the retraction ρ onto ∂X (§XII.6), and in a strong bundle chart Φ with fibre F, F_1 reflexive (which reflexive fibres provide around every point), write $u(w) = \varphi(\rho(P(w)))$ and $A(u)$ for the fibre map of the retraction of the chart; then

$$N'(w) = N\big(\Phi[1]^{-1}(u, A(u)h)\big) + \|h - A(u)h\|_{F_1},$$

where h is the fibre coordinate of w.

Two facts about the fibre maps.

A uniform bound Chapter12/ReflexiveLocalExtension · StrongBundle.exists_norm_chartFibMap_le: near a point u_0 of the chart domain, $\|A(u)\|_{L(F_1)} \le C$ uniformly in u. *Proof.* $(u,h) \mapsto A(u)h$ is continuous at $(u_0,0)$ on level 1 (§XII.6), so for u near u_0 and $\|h\| \le \delta$, $\|A(u)h\| \le 1$; by linearity $\|A(u)\| \le 1/\delta$.

Weak convergence Chapter12/ReflexiveLocalExtension · StrongBundle.weakTendsto_chartFibMap: if $u_k \to u$ and $h_k \rightharpoonup h$ in the reflexive F_1, then $A(u_k)h_k \rightharpoonup A(u)h$. *Proof.* (a) h_k is bounded (uniform boundedness), so $A(u_k)h_k$ is bounded by the uniform bound. (b) The inclusion $F_1 \to F_0$ is compact, so $h_k \to h$ in F_0; the retraction is continuous on level 0, so $A(u_k)h_k \to A(u)h$ in F_0. (c) A bounded sequence in the reflexive F_1 that converges in F_0 converges weakly in F_1 to the same limit, as in Step 3 of Remark 12.2.9. The fibre map lies over the base point Chapter12/ReflexiveLocalExtension · StrongBundle.toBase_chartFibMap, and weak limits are compatible with differences Chapter12/ReflexiveLocalExtension · WeakTendsto.sub.

The inspection Chapter12/ReflexiveLocalExtension · StrongBundle.mixedLscAt_retractNorm: let N be mixed lower semicontinuous at the vectors over a set A containing $\rho(V)$; then N' is mixed lower semicontinuous at every w with $P(w) \in V$. *Proof.* Let $w_k \to w$ mixed, and read in the chart with reflexive F_1: $u_k = u(w_k) \to u(w)$ by continuity of ρ, and $h_k \rightharpoonup h$ in F_1 (Remark 12.2.9 (i), for this chart). - *Second term.* By the weak convergence fact, $h_k - A(u_k)h_k \rightharpoonup h - A(u)h$; the norm of a Banach space is weakly lower semicontinuous, so $\|h - A(u)h\| \le \liminf \|h_k - A(u_k)h_k\|$. - *First term.* The vectors $T(w_k) = \Phi[1]^{-1}(u_k, A(u_k)h_k)$ lie over $\rho(P(w_k)) \in A$; their base points converge to $\rho(P(w))$ and their fibre coordinates converge weakly in F_1 by the same fact. So $T(w_k) \to T(w)$ mixed (Definition 12.2.8 in this chart), and $N(T(w)) \le \liminf N(T(w_k))$ by the hypothesis on N. - The ε-form for the sum follows from those of the two terms.

The set A is a parameter, so the lemma serves any extension step from a closed subset with a continuous retraction, not only $A = \partial X$. □

The local step Chapter12/ReflexiveLocalExtension · StrongBundle.exists_reflexiveLocalExtension_of_isTame: on a tame M-polyfold with a strong bundle with reflexive $(0,1)$-fibres, a function N that is continuous over ∂X, a fibre seminorm over ∂X, satisfies condition (2) for sequences over ∂X and is mixed lower semicontinuous at vectors over ∂X for sequences over ∂X has, at every $x_0 \in \partial X$, a local auxiliary norm M on an open $V \ni x_0$, mixed lower semicontinuous at every vector over V, with $M = N$ over $V \cap \partial X$. *Proof.* Choose the strong bundle chart of §XII.6 among those with reflexive F_1. Then $M = N'$; that it is a local auxiliary norm equal to N on the boundary is §XII.6, and its lower semicontinuity is the inspection with $A = \partial X$; the mixed convergent sequences that the inspection feeds to N lie over $\rho(V) \subset \partial X$, so the hypothesis on N over ∂X suffices. □ Tameness enters only through the retraction onto the boundary.

On the ep-groupoid, the local step in the object M-polyfold discharges the hypothesis of the global part for every tame X Chapter12/ReflexiveLocalExtension · EpStrongBundle.hasReflexiveLocalExtensions_of_isTame, and Theorem 12.2.13 is the global part with this hypothesis discharged.

Witness. The hypotheses together are satisfiable Chapter12/ReflexiveLocalExtension · TrivialBundle.theorem_12_2_13_witness: the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k$ over the tame trivial ep-groupoid of $\mathbb{R}^n$, with a reflexive auxiliary norm over its boundary. This witness is degenerate: the M-polyfold $\mathbb{R}^n$ of the project (`rnPolyfold`) has the single chart onto all of $\mathbb{R}^n$, so $\partial X = \emptyset$ and the boundary data are vacuous. Since Theorem 12.2.13 adds no hypothesis beyond the text's (paracompactness of $|X|$ being explicit), the witness checks only that the hypotheses are consistent; a non-degenerate case would take a tame ep-groupoid with boundary, such as the quadrant witness of Proposition 12.3.7 (§XII.12) seen as a trivial ep-groupoid, and a reflexive norm over its boundary, restricted from a global one by Theorem 12.2.11.

XII.10 Definition 12.3.1 (2017: 12.7) and Theorem 12.3.2 (2017: 12.6): bump functions

§12.3 supplies sc⁺-section functors, the perturbations of the later chapters. A section functor must commute with the morphisms; the sc⁺-sections of Part I, built with bump functions in charts, do not. The remedy is the pattern of §12.2 once more: bump functions and sections are built near a point with the natural action, averaged over the isotropy group and extended to the saturation by the morphisms.

Definition 12.3.1, as printed (same) `Chapter12/BumpEp · EpGroupoid.AdmitsBumpFunctions`, p. 579. An ep-groupoid X admits sc-smooth bump functions if for every object x and every saturated open $U \ni x$ there is an sc-smooth $f : X \to \mathbb{R}$ on the object M-polyfold, not identically 0, with $\operatorname{supp} f \subset U$ and $f(s(\varphi)) = f(t(\varphi))$ for every morphism φ.

The pointwise reading `Chapter12/BumpEp · EpGroupoid.AdmitsBumpFunctionsPointwise` asks $f(x) \neq 0$ instead of $f \neq 0$. It is the project's reading of Part I's Definition 5.3.1 (Part V), where the bump function at x is read as nonzero at x. The pointwise reading implies the printed one `Chapter12/BumpEp · EpGroupoid.AdmitsBumpFunctionsPointwise.weak`. The two readings are kept side by side because Part I has a matching pair: `MPolyfold.AdmitsBumpFunctions` (pointwise) and `MPolyfold.AdmitsBumpFunctionsWeak` (not identically zero), and whether the second implies the first is Part I's open weak-to-pointwise question (Part V).

Tools on saturations. The saturation of an open set is open `Chapter12/BumpEp · EpGroupoid.isOpen_saturation_of_isOpen`, because $\operatorname{sat}(V) = t(s^{-1}(V))$ and t is a local homeomorphism, hence open. The saturation is saturated `Chapter12/BumpEp · EpGroupoid.isSaturated_saturation`, and is contained in any saturated set containing V `Chapter12/BumpEp · EpGroupoid.saturation_subset_of_isSaturated`. Two objects with the same orbit are joined by a morphism `Chapter12/BumpEp · EpGroupoid.eq_of_π_eq`. Around every x in an open U there are a natural representation R of G_x with $R.U \subset U$ and an open $V \ni x$ invariant under R with $\overline{V} \subset R.U$ and $\overline{V}$ proper (Chapter 7, `exists_invariant_nhds_of_isOpen`); "proper" is Chapter 7's notion: the restriction of t to $s^{-1}(\overline{V})$ is proper, so the saturation of a closed subset of $\overline{V}$ is closed `Chapter7/ProperSubsets · isClosed_saturation_of_isProperSubset`.

Averaging a function `Chapter12/BumpEp · EpGroupoid.exists_natAverage`. Let R be a natural representation on $U(x)$, V invariant with $\overline{V} \subset U(x)$, and g sc-smooth with $\operatorname{supp} g \subset V$. Then there is an sc-smooth h with $\operatorname{supp} h \subset V$, invariant under $\Phi(k)$ on $U(x)$, and equal to $|G_x|^{-1} \sum_k g(\Phi(k)y)$ on $U(x)$. *Proof.* Each $g_k = \mathbf{1}_{U(x)} \cdot (g \circ \Phi(k))$ is sc-smooth: on $U(x)$ it is $g \circ \Phi(k)$, with $\Phi(k)$ an sc-diffeomorphism of $U(x)$; near a point $y \notin U(x)$ it vanishes, because $\operatorname{supp}(g \circ \Phi(k)) \subset \Phi(k)^{-1}(\overline{V}) = \overline{V}$ by invariance, a closed set inside $U(x)$, so a neighbourhood of y misses it. The average of the g_k is h; its support lies in V by invariance of V; it is Φ-invariant by reindexing.

Extending an invariant function `Chapter12/BumpEp · EpGroupoid.exists_invariant_extension`. With $\overline{V}$ proper in addition, an sc-smooth h supported in V and invariant under Φ on $U(x)$ extends to an sc-smooth β on X, constant along morphisms, equal to h on $U(x)$, with support in $\operatorname{sat}(\overline{V})$ and values in $h(U(x)) \cup \{0\}$. *Proof.* $\beta = \bar{h} \circ \pi$, where $\bar{h}$ on $|X|$ is h on $\pi(U(x))$ and 0 elsewhere (Chapter 7's `orbitExtension`); it is well defined because h is invariant under all morphisms between points of $U(x)$, which are the $\Gamma(k, \cdot)$. It is sc-smooth (`isScSmoothMap_orbitExtension`): on the saturation of $U(x)$, near y with a morphism $\varphi : y \to y'$, $\beta = h \circ t \circ \sigma$ for a local inverse σ of s at φ; off $\operatorname{sat}(\operatorname{supp} h)$, which is closed because $\operatorname{supp} h \subset \overline{V}$ is proper, β vanishes on a neighbourhood. The support is the closure of $\operatorname{sat}(\{h \neq 0\})$, inside the closed $\operatorname{sat}(\overline{V})$.

Theorem 12.3.2 (1) (restated), p. 579: X admits sc-smooth bump functions iff the object M-polyfold does. It is proved in each consistent reading: pointwise on both sides `Chapter12/BumpEp · EpGroupoid.theorem_12_3_2_1` and "not identically 0" on both sides `Chapter12/BumpEp · EpGroupoid.theorem_12_3_2_1_weak`. The printed statement pairs Definition 12.3.1 ("not identically 0") with Part I's Definition 5.3.1, which this project reads pointwise; its direction (□) would then be Part I's weak-to-pointwise question, which is open. This is a reading of the statement, recorded as such, not a gap in the formalization.

(□), from ep bump functions to object bump functions `Chapter12/BumpEp · EpGroupoid.exists_objBump_of_epBump`. Let $x \in U$, U open (not necessarily saturated). *Step 1.* Choose R and V as above, with $\overline{V} \subset R.U \subset U$. *Step 2.* Apply the hypothesis to the saturated open set $\mathrm{sat}(V) \ni x$: an invariant sc-smooth f with $\mathrm{supp}\, f \subset \mathrm{sat}(V)$, nonzero at x (pointwise reading) or not identically zero. *Step 3, the cut-off* $\mathbf{1}_V f$. It equals f on V and has support in $\overline{V} \subset U$. It is sc-smooth: on the open V it is f; near a point outside $\overline{V}$ it is 0; at a point $y \in \overline{V} \setminus V$, $y \notin \mathrm{supp}\, f$, because a point of $\mathrm{supp}\, f \subset \mathrm{sat}(V)$ that lies in $\overline{V} \subset R.U$ is joined by a morphism to a point of V, hence is $\Phi(k)$ of a point of V (Theorem 7.1.13), hence lies in V by invariance; so f vanishes near y, and so does $\mathbf{1}_V f$. *Step 4, nonvanishing.* Pointwise: $(\mathbf{1}_V f)(x) = f(x) \neq 0$. Weak: if $f(z) \neq 0$, then $z \in \mathrm{sat}(V)$, so z is joined to some $y \in V$ and $f(y) = f(z) \neq 0$ by invariance of f.

(□), from object bump functions to ep bump functions `Chapter12/BumpEp · EpGroupoid.exists_epBump_of_objBump`. Let $x \in U$, U saturated and open. *Step 1.* R and V as above. *Step 2.* An object bump function g at x with support in V. *Step 3, average the square.* Average g^2 over G_x (the averaging lemma) and extend (the extension lemma) to an invariant sc-smooth β with support in $\mathrm{sat}(\overline{V}) \subset U$ (the last because U is saturated and contains $\overline{V}$), and with $\beta \geq |G_x|^{-1} g^2$ on $R.U$: the average of the nonnegative $g^2 \circ \Phi(k)$ is at least its $k = 1$ term divided by $|G_x|$. *Step 4, nonvanishing.* Pointwise: $\beta(x) \geq |G_x|^{-1} g(x)^2 > 0$. Weak: if $g(y) \neq 0$ then $y \in V \subset R.U$ and $\beta(y) > 0$.

Where the formal proof departs from the text. The text averages f itself. The average of a function over a finite group can vanish identically even if the function does not (for example g and $g \circ \Phi(k) = -g$). The formal proof averages g^2, which is presumably what is intended, since the bump functions of Part I's Proposition 5.5.3 are nonnegative, and with nonnegative g the text's average is already fine.

Theorem 12.3.2 (2) (narrower) `Chapter12/BumpEp · EpGroupoid.theorem_12_3_2_2`, p. 579. If X admits sc-smooth bump functions in the pointwise reading, then for every x and saturated open $U \ni x$ there is an sc-smooth $f : X \to [0,1]$, constant along morphisms, with $f(x) = 1$ and $\mathrm{supp}\, f \subset U$.

Proof, the text's. *Step 1.* R and V as above. *Step 2.* By (1), the object M-polyfold admits bump functions, and Part I's Proposition 5.5.3 `Chapter5/BumpFunctions · proposition_5_5_3` gives $g : X \to [0,1]$, sc-smooth, with $g \equiv 1$ near x and $\mathrm{supp}\, g \subset V$. *Step 3.* Average g (not its square: $g \geq 0$ here) to h and extend h to f. *Step 4, the values.* On $R.U$, $f = h = |G_x|^{-1} \sum_k g \circ \Phi(k)$ lies in $[0,1]$, as an average of numbers in $[0,1]$; elsewhere f takes values in $h(R.U) \cup \{0\} \subset [0,1]$. At x, $\Phi(k)(x) = x$ for every k, so $f(x) = |G_x|^{-1} \cdot |G_x| \cdot 1 = 1$. □

Narrower only by the pointwise reading of the hypothesis. Witnesses of the pointwise reading: the trivial ep-groupoid on $\mathbb{R}^n$ `Chapter12/BumpEp · EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`, and in general every ep-groupoid whose object M-polyfold admits bump functions `Chapter12/BumpEp · EpGroupoid.admitsBumpFunctionsPointwise_of_objects`, which is (1) read backwards.

XII.11 Theorem 12.3.3 (2017: 12.7), Remarks 12.3.4 and 12.3.5 (2017: 12.3, 12.4): sc⁺-section functors through a point

The text proves Theorem 12.3.3 with a construction that it states afterwards, in Remark 12.3.5. This section gives the tools first, then the theorem, then the two remarks in the book's order; the proof of the theorem cites the remark's construction.

Morphisms acting on sections. For a section σ of the object bundle and a morphism φ, put $\varphi \cdot \sigma = \mu(\varphi, \sigma(s(\varphi)))$, a vector over $t(\varphi)$ `Chapter12/ScPlusSectionFunctor · EpStrongBundle.secAct`. It respects composition, $(\psi \circ \varphi)\cdot\sigma = \mu(\psi, \varphi\cdot\sigma)$ `Chapter12/ScPlusSectionFunctor · EpStrongBundle.secAct_comp`, and units, $1_x \cdot \sigma = \sigma(x)$ `Chapter12/ScPlusSectionFunctor · EpStrongBundle.secAct_unit`; a section is functorial iff $\varphi\cdot\sigma = \sigma(t(\varphi))$ for all φ `Chapter12/ScPlusSectionFunctor · EpStrongBundle.secAct_eq_of_isFunctorialSection`.

Smoothness along a family of morphisms `Chapter12/ScPlusSectionFunctor · EpStrongBundle.exists_isScSmoothOn_one_secAct`. If σ is an sc⁺-section and τ is an sc-smooth family of morphisms on an open set (for instance the local inverse of t on a sheet), then $y \mapsto \mu(\tau(y), \sigma(s(\tau(y))))$ is sc-smooth into $W[1]$ there. *Proof.* An sc⁺-section is an sc-smooth map into $W[1]$ `Chapter12/ScPlusSectionFunctor · StrongBundle.IsScPlusSection.exists_one`. The map $y \mapsto (\tau(y), \sigma(s(\tau(y))))$ into the pull-back $\mathbf{X}_s \times_P W[1]$ is sc-smooth by the universal property of the pull-back at level $[1]$ (`isScSmoothOn_to_pullback_one`), and μ is a strong bundle map, so sc-smooth from the pull-back of $W[1]$ to $W[1]$ (`isStrongBundleMap.isScSmoothMap_one`). *Locality* `Chapter12/ScPlusSectionFunctor · StrongBundle.isScPlusSection_of_locally`: a section that agrees near every point with an sc-smooth map into $W[1]$ is an sc⁺-section.

Sums and bounds. Sums of sections are taken fibrewise; at the base point of $\mu(\varphi, \cdot)$ the sum is carried termwise `Chapter12/ScPlusSectionFunctor · EpStrongBundle.sectionSum_tgt`, and a complete fibre norm is subadditive on finite sums `Chapter12/ScPlusSectionFunctor · StrongBundle.IsCompleteFiberNorm.sum_le`. A section σ satisfies the *bound* b for N if $N(\sigma(y)) \le b$ wherever $\sigma(y) \in W_{0,1}$ `Chapter12/ScPlusSectionFunctor · EpStrongBundle.NormBound`; for an auxiliary norm, $N(\varphi\cdot\sigma) = N(\sigma(s(\varphi)))$ `Chapter12/ScPlusSectionFunctor · EpStrongBundle.IsAuxNorm.secAct_eq`, by invariance.

The equivariant average `Chapter12/ScPlusSectionFunctor · EpStrongBundle.exists_equivariant_average`. Let R be a natural representation on $U(x)$, V invariant with $\overline{V} \subset U(x)$, and s an sc⁺-section with support in V. For $k \in G_x$ define

$$h_k(y) = \mu(\Gamma(k, \Phi(k^{-1})y),\ s(\Phi(k^{-1})y)) \text{ for } y \in U(x), \qquad h_k(y) = 0 \text{ otherwise}$$

`Chapter12/ScPlusSectionFunctor · EpStrongBundle.natTerm`, the translate of s by k. Each h_k is an sc⁺-section `Chapter12/ScPlusSectionFunctor · EpStrongBundle.isScPlusSection_natTerm`: on $U(x)$ by the smoothness lemma along the sc-smooth family $y \mapsto \Gamma(k, \Phi(k^{-1})y)$, and near points outside $U(x)$ it vanishes, since its support lies in $\Phi(k)(\overline{V}) = \overline{V}$. The translates are permuted by morphisms `Chapter12/ScPlusSectionFunctor · EpStrongBundle.natTerm_tgt`: for a morphism $\psi = \Gamma(\gamma, y)$ between points of $U(x)$, $h_{\gamma k}(t(\psi)) = \mu(\psi, h_k(s(\psi)))$, by the composition law $\Gamma(\gamma k, \cdot) = \Gamma(\gamma, \cdot) \circ \Gamma(k, \cdot)$ (`Γ_mul`) and the composition rule for $\cdot$. Put $\sigma = |G_x|^{-1} \sum_k h_k$. Then: - σ is an sc⁺-section, a finite sum of sc⁺-sections times a constant; - *equivariance*: $\sigma(t(\psi)) = \psi \cdot \sigma$ for every morphism ψ between points of $U(x)$, by reindexing $k \mapsto \gamma k$ and linearity of $\mu(\psi, \cdot)$; - its support lies in $\overline{V}$; - if $s(x)$ is fixed by G_x, then $\sigma(x) = s(x)$: $\Phi(k^{-1})x = x$, so $h_k(x) = \mu(k, s(x)) = s(x)$; - every bound $N(s) \le b$ for an auxiliary norm passes to σ: $N(h_k(y)) = N(s(\Phi(k^{-1})y)) \le b$ by invariance of N, and the average of vectors of norm at most b has norm at most b by subadditivity and homogeneity.

The natural extension Chapter12/ScPlusSectionFunctor · EpStrongBundle.natExtension. For a section σ and y an object, if some morphism φ has $t(\varphi) = y$ and $s(\varphi) \in U(x)$, choose one and put $\tilde{\sigma}(y) = \varphi \cdot \sigma$; otherwise $\tilde{\sigma}(y) = 0$. It does not depend on the choice when σ is equivariant on $U(x)$ Chapter12/ScPlusSectionFunctor · EpStrongBundle.natExtension_eq: two choices φ, φ' differ by the morphism $\varphi^{-1}\varphi'$ between points of $U(x)$, and equivariance moves σ along it. On $U(x)$ it is σ Chapter12/ScPlusSectionFunctor · EpStrongBundle.natExtension_of_mem (choose the identity), it is functorial Chapter12/ScPlusSectionFunctor · EpStrongBundle.natExtension_isFunctorial (for $\psi : y \to y'$ and a choice φ at y, $\psi\varphi$ is a choice at y'), and it vanishes off the saturation of supp σ Chapter12/ScPlusSectionFunctor · EpStrongBundle.natExtension_eq_zero.

Theorem 12.3.3 (narrower) Chapter12/ScPlusSectionFunctor · EpStrongBundle.theorem_12_3_3, p. 581. Hypotheses: X admits sc-smooth bump functions (pointwise reading); $|X|$ is paracompact; N is an auxiliary norm; x is a smooth object; $e \in W_x$ is smooth with $\mu(g, e) = e$ for all $g \in G_x$; $\hat{O} \ni |x|$ is open in $|X|$; $\varepsilon > 0$. Conclusion: $e \in W_{0,1}$, and there is an sc$^+$-section functor σ with supp $\sigma \subset \pi^{-1}(\hat{O})$, $\sigma(x) = e$ and $N(\sigma(y)) \le N(e) + \varepsilon$ for all y.

Proof.

Step 1, a good neighbourhood. Apply the Chapter 7 lemma to the open $\pi^{-1}(\hat{O}) \ni x$: R on $U(x) \subset \pi^{-1}(\hat{O})$, $V \ni x$ invariant, $\overline{V} \subset U(x)$, $\overline{V}$ proper.

Step 2, a local section through e. By Theorem 12.3.2 (1) the object M-polyfold admits bump functions, and Part I's Lemma 5.3.3 Chapter5/BumpFunctions · lemma_5_3_3 gives: $e \in W_{0,1}$, and an sc$^+$-section s of the object bundle with $s(x) = e$, support in V and $N(s(y)) \le N(e) + \varepsilon$. The text builds this section as the constant section e in a chart times a cut-off β and controls its norm by continuity; Lemma 5.3.3 is that construction in Part I, so no separate cut-off is needed.

Step 3, average. The equivariant average σ' of s: an sc$^+$-section, equivariant on $U(x)$, with support in $\overline{V}$; since $s(x) = e$ is fixed by G_x, $\sigma'(x) = e$; and the bound $N(e) + \varepsilon$ passes. The text's average is the same sum, written $\frac{1}{|G|} \sum_g g * (\beta s_0)$.

Step 4, extend by Remark 12.3.5 (below): the natural extension of σ' is an sc$^+$-section functor, equal to σ' on $U(x)$, so its value at x is e; its support lies in sat(supp σ') $\subset$ sat$(U(x)) \subset \pi^{-1}(\hat{O})$; and it keeps the bound. □

Narrower by the pointwise reading of bump functions, which Step 2 needs (Part I's Lemma 5.3.3 is stated with pointwise bump functions). *Kept and unused*: the paracompactness of $|X|$. The text needs it only for the existence of an auxiliary norm, which is here a hypothesis.

Remark 12.3.4 (explicit) Chapter12/ScPlusSectionFunctor · EpStrongBundle.remark_12_3_4, p. 584. For a functorial section σ and $g \in G_x$, $\mu(g, \sigma(x)) = \sigma(x)$: functoriality at the morphism $g : x \to x$. So the invariance of e in Theorem 12.3.3 cannot be dropped, and sc$^+$-section functors cannot achieve transversality at points with isotropy, which is the remark's point. The rest of the remark is commentary.

Remark 12.3.5 (explicit) Chapter12/ScPlusSectionFunctor · EpStrongBundle.remark_12_3_5, p. 584. Let R be a natural representation on $U(x)$, $\overline{V} \subset U(x)$ with $\overline{V}$ proper, and σ an sc$^+$-section with support in $\overline{V}$, equivariant under the morphisms between points of $U(x)$. Then the natural extension $\tilde{\sigma}$ is an sc$^+$-section functor, equal to σ on $U(x)$, with support in sat(supp σ), and for every auxiliary norm N and $b \ge 0$, $N(\sigma) \le b$ implies $N(\tilde{\sigma}) \le b$.

Proof. Functoriality and the value on $U(x)$ are the properties of the natural extension above.

sc⁺, near a point y_0 of the saturation. There is a morphism φ_0 with $t(\varphi_0) = y_0$, $s(\varphi_0) \in U(x)$. Let τ be the local inverse of t on the sheet of φ_0, on a neighbourhood O of y_0 with $s(\tau(O)) \subset U(x)$. For $y \in O$, $\tau(y)$ is a morphism into y from a point of $U(x)$, so $\tilde\sigma(y) = \tau(y) \cdot \sigma$ by independence of the choice; this is sc-smooth into $W[1]$ by the smoothness lemma.

sc⁺, near a point y_0 off the saturation of $\operatorname{supp}\sigma$. This saturation is closed, because $\operatorname{supp}\sigma \subset \overline{V}$ is proper `Chapter7/ProperSubsets · isClosed_saturation_of_isProperSubset`; so y_0 has a neighbourhood off it, where $\tilde\sigma = 0$, the zero section, which is sc⁺. The two cases cover all y_0, since a point of the saturation of $\operatorname{supp}\sigma$ lies in the saturation of $U(x)$. Locality concludes.

Support. Off the closed set $\operatorname{sat}(\operatorname{supp}\sigma)$, $\tilde\sigma = 0$.

Bound. $N(\tilde\sigma(y)) = N(\varphi \cdot \sigma) = N(\sigma(s(\varphi))) \le b$ by invariance of N, and $N(0) = 0 \le b$. □

Explicit: the text's section is $f \cdot s$ with a cut-off f, and its "controlled" norm is a bound. In the formal statement the section is already cut off (its support lies in $\overline{V}$), and the control is the bound b. The properness of $\overline{V}$, which the text has from Theorem 12.2.6, is a hypothesis, supplied in every application by the Chapter 7 lemma.

XII.12 Definition 12.3.6 (2017: 12.8), Proposition 12.3.7 (2017: 12.2) and Corollary 12.3.8 (2017: 12.1): sc⁺-sections over the boundary, and their local extension

Theorem 12.3.9 extends an sc⁺-section functor given on ∂X to all of X. The text first says what an sc⁺-section over the boundary is (Definition 12.3.6), extends it locally in a tame local model by an inclusion–exclusion formula (Proposition 12.3.7), and makes the local extension equivariant under a finite group (Corollary 12.3.8). The formalization keeps all three, in a local model; Part I's Theorem 5.4.2 (Part V) is assembled from Proposition 12.3.7, and Theorem 12.3.9 (§XII.13) uses Part I's local extension.

The local model. Let E be an sc-Banach space with a partial quadrant C and a tame presentation $T : E \cong \mathbb{R}^n \oplus W$ with $C = T^{-1}([0,\infty)^n \oplus W)$. A tame local strong bundle model is $K = R(U \lhd F) \to O$: $U \subset C$ relatively open, R a strong bundle retraction of $U \lhd F$, linear on fibres, whose base retraction r is tame with respect to T, and $O = r(U)$. For $z \in O$, $I(z) = \{i : t_i(z) = 0\}$ is the set of vanishing coordinates (`zeroSet`), and ∂O is the set of $z \in O$ of nonzero degeneracy index `Chapter12/LocalBoundaryExtension · retractBoundary`, equivalently with $I(z) \neq \emptyset$ `Chapter12/LocalBoundaryExtension · mem_retractBoundary_iff`. A section of K over a subset is read as a map $s : E_0 \to E_0 \times F_1$ with first component the base point.

Definition 12.3.6, local form (restated) `Chapter12/LocalBoundaryExtension · IsBoundaryScPlusSectionOn`, p. 584. A section s over $D \subset \partial O$ is an sc⁺-section over the boundary when for every $x \in D$ and $i \in I(x)$, there is an open $V \ni x$ such that the face germ $O \cap V \cap \{t_i = 0\}$ is an sc-retract of C and s restricted to it is an sc⁺-section of K `Chapter12/LocalBoundaryExtension · IsScPlusOnFaceNear`. The text says "an sc⁺-section on each local face"; the faces of a tame retract are sc-retracts near the point, by Chapter 2's corner germs, and the formal definition carries that as part of the clause, so that "sc⁺ on the face" makes sense. Witness of the quantifier over models and presentations: the zero section is an sc⁺-section over the boundary in every tame model `Chapter12/LocalBoundaryExtension · isBoundaryScPlusSectionOn_zero`, since 0 is sc⁺ on every sc-retract `Chapter12/LocalBoundaryExtension · isScPlusSection_zero_local`.

Definition 12.3.6, ep-groupoid form (restated) Chapter12/ScPlusExtension · EpStrongBundle.IsBoundaryScPlusSection. A section s of $W|\partial X$ is an sc$^+$-section over the boundary when it is one for the object bundle in the sense of Part I's Definition 5.4.1 (Part V): $P(s(x)) = x$, $s(x) \in W_{0,1}$ for $x \in \partial X$, and in every tame strong bundle chart and every tame presentation, the local representative is an sc$^+$-section over the boundary in the local form at every boundary point of the chart. The local faces are read chart-wise, as in Definition 5.4.1; the definition does not involve μ. **An sc$^+$-section functor of** $W|\partial X$, as Theorem 12.3.9 uses it Chapter12/ScPlusExtension · EpStrongBundle.IsBoundaryScPlusSectionFunctor, adds $s(t(\varphi)) = \mu(\varphi, s(s(\varphi)))$ for every morphism φ with $s(\varphi) \in \partial X$; then $t(\varphi) \in \partial X$ as well, because morphisms keep the degeneracy index Chapter12/ScPlusExtension · EpGroupoid.tgt_mem_boundary_iff_p3x. Witness: the zero section is an sc$^+$-section functor of $W|\partial X$ in every strong bundle over an ep-groupoid Chapter12/ScPlusExtension · EpStrongBundle.isBoundaryScPlusSectionFunctor_fiberZero, and of W Chapter12/ScPlusExtension · EpStrongBundle.isScPlusSectionFunctor_fiberZero_p3x.

The inclusion–exclusion extension. Fix $x_0 \in \partial O$ and $\sigma = I(x_0)$. For $I \subseteq \sigma$ and z near x_0, let z_I be z with the coordinates t_i, $i \in I$, set to 0 (`cornerKill`), and then retracted by r. Then

$$\tilde{s}(z) = R\Big(z,\ \sum_{\emptyset \neq I \subseteq \sigma} (-1)^{|I|+1}\, s(z_I)\Big)$$

Chapter12/LocalBoundaryExtension · boundaryExtension, the text's sum written with the retraction applied once at the end. Its base component is z Chapter12/LocalBoundaryExtension · boundaryExtension_fst; its fibre component is the text's sum projected by $A(z)$ Chapter12/LocalBoundaryExtension · boundaryExtension_toBase_snd; it lies in $K[1]$ Chapter12/LocalBoundaryExtension · boundaryExtension_mem.

The corner kills. The coordinates of z_I before retraction are 0 on I and $t_i(z)$ elsewhere Chapter12/LocalBoundaryExtension · coord_cornerKill; killing the empty set changes nothing Chapter12/LocalBoundaryExtension · cornerKill_empty; and if $t_{i_0}(z) = 0$, killing $I \cup \{i_0\}$ is killing I Chapter12/LocalBoundaryExtension · cornerKill_insert_of_coord_eq_zero.

Each term is sc-smooth Chapter12/LocalBoundaryExtension · exists_isScSmooth_faceTerm. For $\emptyset \neq I \subseteq \sigma$, $z \mapsto s(z_I)$ is sc-smooth into $E \times F_1$ near x_0 on C: $z \mapsto z_I$ is the corner retraction of the face $\{t_i = 0, i \in I\}$ Chapter2/CornerGerms · corner_retraction followed by r, an sc-smooth map of a neighbourhood of x_0 in C into the face germ at x_0; and s is sc$^+$ on that face germ by hypothesis (for $|I| > 1$ the intersection of faces is a face of a face, and the hypothesis on single faces gives sc$^+$ on it by restriction). The scalar multiples and sums are sc-smooth (`ScBanach.p3bSmulOp`, `IsScSmooth.p3bConstSmul`) Chapter12/LocalBoundaryExtension · ScBanach.p3bSmulOp Chapter12/LocalBoundaryExtension · IsScSmooth.p3bConstSmul.

The cancellation Chapter12/LocalBoundaryExtension · sum_powerset_erase_empty_alternating. Let v be a function on subsets of σ and $i_0 \in \sigma$ with $v(I \cup \{i_0\}) = v(I)$ for all $I \subseteq \sigma$. Then

$$\sum_{\emptyset \neq I \subseteq \sigma} (-1)^{|I|+1} v(I) = v(\emptyset).$$

Proof. The full alternating sum $\sum_{I \subseteq \sigma} (-1)^{|I|} v(I)$ vanishes: split the subsets into those not containing i_0 and their unions with $\{i_0\}$ (`sum_powerset_insert`); paired terms have equal v and opposite signs. Splitting off $I = \emptyset$ and changing sign gives the claim. □

Proposition 12.3.7 (stronger) Chapter12/LocalBoundaryExtension · proposition_12_3_7_of_faces_at, p. 585. Let $K \to O$ be a tame local strong bundle model, $x_0 \in \partial O$, and s a section that is sc$^+$ on the face germs

at x_0 itself: for every $i \in I(x_0)$, s is sc$^+$ on $O \cap V \cap \{t_i = 0\}$ for some open $V \ni x_0$. Then there is an open $V \ni x_0$ such that $O \cap V$ is an sc-retract of C, $\tilde{s}$ is an sc$^+$-section of K over $O \cap V$, and $\tilde{s} = s$ on $\partial O \cap V$.

Proof. Step 1, the neighbourhood. V is the intersection of: an open $Z \ni x_0$ on which $I(z) \subseteq \sigma$ (coordinates that are positive at x_0 stay positive nearby, `eventually_zeroSet_subset`); for each nonempty $I \subseteq \sigma$, an open $N_I \ni x_0$ on which z_I stays in the domain where its term is sc-smooth; and the domains V_i of the face germs. Finitely many open sets, so V is open. *Step 2, sc-retract.* $O \cap V$ is an sc-retract of C, with retraction r restricted (`r.restrict`). *Step 3, sc$^+$.* By `isScSmoothRetractMap_iff_exists` it suffices that $z \mapsto \tilde{s}(z)$ is sc-smooth into $(U \lhd F)[1]$ on $O \cap V$: the alternating sum is sc-smooth (`IsScSmooth.finsetSum`), paired with the identity (`scPair`), and R is sc-smooth on level $[1]$ (`isScSmooth_one` of the strong bundle retraction). *Step 4, equality on the boundary.* Let $z \in \partial O \cap V$. Then $I(z) \neq \emptyset$ and $I(z) \subseteq \sigma$ by Step 1; pick $i_0 \in I(z)$. Since $t_{i_0}(z) = 0$, $z_{I \cup \{i_0\}} = z_I$ for all I; with $v(I) = s(z_I)$, the cancellation gives $\sum_{\emptyset \neq I \subseteq \sigma} (-1)^{|I|+1} s(z_I) = s(z_\emptyset) = s(z)$, the last because $z_\emptyset = r(z) = z$ for $z \in O$. Finally $R(z, s(z)) = s(z)$ because $s(z) \in K[1]$ and R fixes its image. □

Stronger: the text asks s to be sc$^+$ on the local faces at every boundary point near x_0; the proof uses only the face germs at x_0, because every face through a nearby boundary point is a face through x_0 (Step 1). **The text's form** (same) `Chapter12/LocalBoundaryExtension · proposition_12_3_7`: for an open V_0 and s an sc$^+$-section over $\partial O \cap V_0$ in the sense of Definition 12.3.6, every $x_0 \in \partial O \cap V_0$ has an open $V \ni x_0$ and an sc$^+$-section s' over $O \cap V$ with $s' = s$ on $\partial O \cap V$. It follows by applying the stronger form at x_0. The text writes the model on $\mathbb{R}^n \oplus H$; here it is any tame presentation T.

Non-degenerate witness `Chapter12/LocalBoundaryExtension · proposition_12_3_7_nonvacuous`: for every sc-Banach spaces W, F, the partial quadrant $C = [0, \infty) \oplus W$ with the identity presentation, $U = C$, the identity strong bundle retraction `Chapter12/LocalBoundaryExtension · isStrongBundleRetraction_id_p3b` (tame), the point $0 \in \partial O$, and the zero section, which is an sc$^+$-section over all of ∂O. Here $\partial O = \{0\} \oplus W \neq \emptyset$, so the hypotheses hold in a case with boundary.

Local bundle actions (explicit) `Chapter12/LocalBoundaryExtension · LocalBundleAction`. Corollary 12.3.8 assumes that a finite group "acts" on O with a lift to K. The formalization makes this a structure: a finite group G acting on E_0 by maps $x \mapsto g * x$ that are sc-smooth maps of the retract O, with $1 * x = x$ and $(gh) * x = g * (h * x)$ on O; and a lift $p \mapsto g * p$ on $E_0 \times F_1$, sc-smooth on $K[1]$, covering the action, linear on fibres over O, with the same unit and composition laws on $K[1]$. Witness: the trivial action of any group `Chapter12/LocalBoundaryExtension · LocalBundleAction.trivial`. In the applications (§XII.13) the action is the natural representation of G_x and its lift through μ.

The average `Chapter12/LocalBoundaryExtension · LocalBundleAction.average`: for a section s',

$$\hat{s}(x) = \Big(x,\ \frac{1}{|G|} \sum_{g \in G} \mathrm{pr}_2\big(g^{-1} * s'(g * x)\big)\Big).$$

A sum of vectors of $K[1]$ over the same point, with real coefficients, stays in $K[1]$ `Chapter12/LocalBoundaryExtension · mem_strongShiftOne_sum_smul`, because $K[1]$ is the image of a fibrewise linear retraction.

Corollary 12.3.8 (explicit) `Chapter12/LocalBoundaryExtension · corollary_12_3_8`, p. 585. Let $K \to O$ be a tame local model with a local bundle action of a finite G, $x_0 \in \partial O \cap V_0$ fixed by G, $\partial O \cap V_0$ invariant, and s an sc$^+$-section over $\partial O \cap V_0$ that is equivariant: $s(g * x) = g * s(x)$. Then there are an open $V \ni x_0$

with $O \cap V$ invariant and an sc⁺-section $\hat{s}$ over $O \cap V$, equivariant, with $\hat{s} = s$ on $\partial O \cap V$.

Proof. Step 1. Proposition 12.3.7 at x_0 gives V_1 and an extension $\tilde{s}$ on $O \cap V_1$. *Step 2, an invariant neighbourhood.* $V = \bigcap_g (g*)^{-1}(V_1 \cap V_0)$ near x_0, open by continuity of the action and containing x_0 because x_0 is fixed; $O \cap V$ is invariant. *Step 3, average.* $\hat{s}$ is the average of $\tilde{s}$. It is sc⁺: each term $g^{-1} * \tilde{s}(g * x)$ is a composite of sc-smooth maps into $K[1]$, and the sum stays in $K[1]$. *Step 4, equivariance*: reindexing $g \mapsto gh$, with the composition law of the lift. *Step 5, extension.* On $\partial O \cap V$, $g * x \in \partial O \cap V_1$ by invariance, so $\tilde{s}(g * x) = s(g * x) = g * s(x)$, and each term is $g^{-1} * g * s(x) = s(x)$; the average of $|G|$ copies of $s(x)$ is $s(x)$. □

Explicit: (a) "acts with a lift" is the structure above; (b) that G fixes x_0, true of an isotropy group, is a hypothesis; (c) the text's "invariant" section is read as equivariant, the only reading under which the average extends s.

XII.13 Theorem 12.3.9 (2017: 12.8): extension of sc⁺-section functors from the boundary

Theorem 12.3.9 (same) `Chapter12/ScPlusExtension · EpStrongBundle.theorem_12_3_9`, p. 586. Let $(P : W \to X, \mu)$ be a strong bundle over an ep-groupoid whose object bundle is tame (every object lies in a tame strong bundle chart, as in Part I's Theorem 5.4.2), with $|X|$ paracompact and X admitting sc-smooth partitions of unity (Definition 7.5.5: for every cover by saturated open sets, an sc-smooth partition of unity subordinate to it and constant along morphisms). Let s be an sc⁺-section functor of $W|\partial X$ and $U \supset \partial X$ a saturated open set. Then (1) there is an sc⁺-section functor $\tilde{s}$ of W with $\operatorname{supp} \tilde{s} \subset U$ and $\tilde{s} = s$ on ∂X; (2) if N is an auxiliary norm and $N(s(x)) < \delta$ for all $x \in \partial X$, then $\tilde{s}$ can be chosen with $N(\tilde{s}(x)) \le \delta$ for all x. The Lean statement takes the object and morphism types in the lowest universe, where Definition 7.5.5 is stated.

The proof has a local step, a gluing and a bound.

Bump functions from partitions of unity (extra) `Chapter12/ScPlusExtension · EpGroupoid.admitsBumpFunctionsPointwise_of_hasScPartitionsOfUnity`. An ep-groupoid with sc-smooth partitions of unity admits sc-smooth bump functions in the pointwise reading. *Proof.* For x in a saturated open U, take R and an invariant $V \ni x$ with $\overline{V} \subset R.U \subset U$, $\overline{V}$ proper (§XII.10). The saturation K of $\overline{V}$ is closed (properness) and contained in U (saturated). The two saturated open sets U and $X \setminus K$ cover X (complements of saturated sets are saturated `Chapter12/ScPlusExtension · EpGroupoid.isSaturated_compl_p3x`). A subordinate partition of unity $\{\rho_U, \rho_{X\setminus K}\}$ has $\rho_{X\setminus K} = 0$ on $K \ni x$, so $\rho_U(x) = 1$; ρ_U is sc-smooth, invariant and supported in U. □ The text uses this tacitly when it says "using Theorem 12.3.2 (2)"; by Theorem 12.3.2 (1) `Chapter12/BumpEp · EpGroupoid.theorem_12_3_2_1` the object M-polyfold then admits bump functions, which Part I's local extension needs. A two-set partition of unity is indexed by the two values of `Bool`, and one of its functions is 1 where the other vanishes `Chapter12/ScPlusExtension · EpGroupoid.partitionOfUnity_true_eq_one_p3x`.

Gluing sections (extra). For a partition of unity (ρ_i) of X and sections σ_i, the section $\sum_i \rho_i \sigma_i$ `Chapter12/ScPlusExtension · glueSection` is computed pointwise `Chapter12/ScPlusExtension · glueSection_apply`, is a section `Chapter12/ScPlusExtension · isSection_glueSection`, is locally a finite sum `Chapter12/ScPlusExtension · glueSection_finsum_eq_sum`, and: - takes the value v at y when every σ_i with $\rho_i(y) \neq 0$ takes the value v there `Chapter12/ScPlusExtension · glueSection_eq_of`, since $\sum_i \rho_i(y) = 1$; - has support in $\bigcup_i (\operatorname{supp} \rho_i \cap \operatorname{supp} \sigma_i)$ `Chapter12/ScPlusExtension · sectionSupport_glueSection_subset`; - is an

sc⁺-section when the σ_i are Chapter12/ScPlusExtension · isScPlusSection_glueSection: near each point it is a finite sum of products of sc-smooth functions with sc⁺-sections; - is functorial when the σ_i are and the ρ_i are constant along morphisms Chapter12/ScPlusExtension · EpStrongBundle.isFunctorialSection_glueSection, by linearity of $\mu(\varphi,\cdot)$; an invariant function times a functorial section is functorial Chapter12/ScPlusExtension · EpStrongBundle.isFunctorialSection_sectionSmul_p3x; together, an sc⁺-section functor Chapter12/ScPlusExtension · EpStrongBundle.isScPlusSectionFunctor_glueSection.

The local step Chapter12/ScPlusExtension · EpStrongBundle.exists_local_functor_extension. Assume the object bundle tame and bump functions on the objects. For $x \in \partial X$ in a saturated open U, there are a saturated open $U' \ni x$ and an sc⁺-section functor σ with $\operatorname{supp}\sigma \subset U$ and $\sigma = s$ on $U' \cap \partial X$.

Step 1, a good neighbourhood. R on $R.U \subset U$, $V \ni x$ invariant, $\overline{V} \subset R.U$, $\overline{V}$ proper.

Step 2, Part I's local extension. In a tame strong bundle chart at x, Part I's local extension Chapter5/BoundaryExtension · StrongBundle.exists_local_extension (Proposition 12.3.7 read in the chart, Part V) gives an open $U_1 \ni x$ and an sc⁺-section σ_0 of the object bundle with $\sigma_0 = s$ on $U_1 \cap \partial X$. A bump function $\beta : X \to [0,1]$ of Part I's Proposition 5.5.3 Chapter5/BumpFunctions · proposition_5_5_3, $\equiv 1$ near x and supported in $V \cap U_1$, cuts it off: $\sigma_1 = \beta\sigma_0$ is an sc⁺-section supported in V, equal to s on $W_1 \cap \partial X$ where $W_1 = \operatorname{int}\{\beta = 1\} \cap U_1 \ni x$.

Step 3, average. The equivariant average σ_2 of σ_1 over G_x (§XII.11) Chapter12/ScPlusExtension · EpStrongBundle.exists_equivariant_average_p3x is equivariant on $R.U$, supported in $\overline{V}$, and takes the value w at every point where all translates take the value w. Put $W_2 = \bigcap_k R.U \cap \Phi(k)^{-1}(W_1)$, an open neighbourhood of x ($\Phi(k)$ fixes x). At $y \in W_2 \cap \partial X$, each translate is $\mu(\Gamma(k,y'),\sigma_1(y'))$ with $y' = \Phi(k^{-1})y \in W_1 \cap \partial X$ (morphisms preserve ∂X), hence $\mu(\Gamma(k,y'),s(y')) = s(\Phi(k)y') = s(y)$ by functoriality of s on ∂X. So $\sigma_2 = s$ on $W_2 \cap \partial X$. This is Corollary 12.3.8 in its natural-representation form, with the lift given by μ.

Step 4, extend by Remark 12.3.5 Chapter12/ScPlusSectionFunctor · EpStrongBundle.remark_12_3_5: the natural extension σ of σ_2 is an sc⁺-section functor supported in $\operatorname{sat}(\overline{V}) \subset U$, equal to σ_2 on $R.U$. Put $U' = \operatorname{sat}(W_2)$, open and saturated. At $y \in U' \cap \partial X$, y is joined by φ to a point $y' \in W_2$, also in ∂X; $\sigma(y) = \mu(\varphi^{-1},\sigma(y')) = \mu(\varphi^{-1},s(y')) = s(y)$ by functoriality of σ and of s. □

The extension Chapter12/ScPlusExtension · EpStrongBundle.exists_extension_of_tameCharts: under the hypotheses of Theorem 12.3.9 without the norm, clause (1). *Step 1.* Bump functions on the objects, from the partitions of unity (above). *Step 2, a cover.* For $x \in \partial X$, the local step gives U'_x and σ_x supported in U. For $x \notin \partial X$, take the saturated open $X \setminus \partial X$ Chapter12/ScPlusExtension · EpGroupoid.isSaturated_compl_boundary_p3x with the zero section. These sets cover X. *Step 3, glue.* Take an sc-smooth partition of unity (ρ_x) subordinate to the cover, constant along morphisms, and $\tilde{s} = \sum_x \rho_x\sigma_x$, an sc⁺-section functor. *Step 4, properties.* On ∂X: the interior pieces have $\rho = 0$ there (their support lies in $X \setminus \partial X$), and every boundary piece with $\rho_x(y) \neq 0$ has $y \in U'_x$, so $\sigma_x(y) = s(y)$; hence $\tilde{s}(y) = s(y)$. Support: every σ_x is supported in U (the zero section trivially). □

The bound Chapter12/ScPlusExtension · EpStrongBundle.exists_extension_bounded_of_tameCharts, clause (2). *Step 1.* $\tilde{s}$ from (1). *Step 2, the open set.* $V = \{y : N(\tilde{s}(y)) < \delta\}$: $\tilde{s}$ is an sc⁺-section, so $y \mapsto \tilde{s}(y) \in W_{0,1}$ is continuous and $N \circ \tilde{s}$ is continuous; V is open, saturated by functoriality of $\tilde{s}$ and invariance of N, and contains ∂X by the hypothesis. *Step 3, a cut-off.* A partition of unity for the two-set cover $\{V, X \setminus \partial X\}$ gives β, sc-smooth, invariant, with $\operatorname{supp}\beta \subset V$ and $\beta = 1$ on ∂X (the other function vanishes there). *Step 4.* $\beta\tilde{s}$ is an sc⁺-section functor, equal to s on ∂X, supported in U, and $N(\beta\tilde{s}(y)) = \beta(y)N(\tilde{s}(y)) \leq \delta$: where $\beta(y) \neq 0$, $y \in V$; $\beta \leq 1$. □

Where the formal proof departs from the text. (a) The text obtains bump functions from Theorem 12.3.2 (2); here they come from the assumed partitions of unity, so no hypothesis is added. (b) The text's bound uses a second neighbourhood V' with $\overline{V'} \subset V$; the two-set partition of unity makes it unnecessary. (c) The local extension is Part I's (Theorem 5.4.2's local step), which is Proposition 12.3.7 in a chart; the averaging of Corollary 12.3.8 is done with the natural representation directly.

Hypotheses. All are the text's. The paracompactness of $|X|$ is kept and unused, since partitions of unity are assumed. **Witness for all hypotheses together** `Chapter12/ScPlusExtension · EpStrongBundle.theorem_12_3_9_hypotheses_rn`: the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ over the trivial ep-groupoid of $\mathbb{R}^n$, the zero section and $U = X$. This witness is degenerate: $\mathbb{R}^n$ is modelled on all of $\mathbb{R}^n$, so $\partial X = \emptyset$ and the boundary data are vacuous. With boundary, the data of the statement are inhabited on every X (the zero section functor of $W|\partial X$ above, $U = X$), and the local construction has the non-degenerate witness of Proposition 12.3.7 on the quadrant $[0, \infty) \oplus W$ (§XII.12). Since Theorem 12.3.9 adds no hypothesis to the text's, the witness serves only as a consistency check.

XII.14 Definition 12.4.1, Theorem 12.4.2 and Lemma 12.4.4 (2017: Definition 12.9, Theorem 12.9, Lemma 12.1): compactness

§12.4 states the compactness notions of the Fredholm theory on ep-groupoids. All are statements about orbit sets $|A| = \pi(A)$ in $|X|$ and their closures, and the formal proofs consist of Part I's local compactness at zeros (Theorem 5.2.2) and two facts on closures in $|X|$, which the text states as Lemma 12.4.4 after it uses them. The lemma is given first here.

Lemma 12.4.4 (same), p. 589. (1) `Chapter12/CompactnessEp · EpGroupoid.lemma_12_4_4_1`: for a saturated set Σ of objects, $\mathrm{cl}_{|X|}\, \pi(\Sigma) = \pi(\mathrm{cl}_X\, \Sigma)$. (2) `Chapter12/CompactnessEp · EpGroupoid.lemma_12_4_4_2`: the same for any Σ with compact closure in X.

Proof of (1). π is continuous, surjective and open (Proposition 7.1.8). For an open map, preimages commute with closures: $\pi^{-1}(\mathrm{cl}\, \pi(\Sigma)) = \mathrm{cl}\, \pi^{-1}(\pi(\Sigma))$. Since Σ is saturated, $\pi^{-1}(\pi(\Sigma)) = \Sigma$. Applying π and using surjectivity, $\mathrm{cl}\, \pi(\Sigma) = \pi(\pi^{-1}(\mathrm{cl}\, \pi(\Sigma))) = \pi(\mathrm{cl}\, \Sigma)$. *Proof of (2).* $\pi(\mathrm{cl}\, \Sigma)$ is compact, hence closed in the Hausdorff $|X|$, and contains $\pi(\Sigma)$; so $\mathrm{cl}\, \pi(\Sigma) \subset \pi(\mathrm{cl}\, \Sigma)$. The reverse inclusion is continuity of π. □

A consequence `Chapter12/CompactnessEp · EpGroupoid.isClosed_image_π_of_isSaturated`: the image of a closed saturated set is closed, by (1) with $\mathrm{cl}\, \Sigma = \Sigma$.

Definition 12.4.1 (same) `Chapter12/CompactnessEp · EpStrongBundle.HasCompactSolutionSetEp` `Chapter12/CompactnessEp · EpStrongBundle.IsProperFunctor` `Chapter12/CompactnessEp · EpStrongBundle.IsProperLtInftyFunctor`, p. 587. For a section functor f with solution set $S_f = \{x : f(x) = 0\}$: 1. f has *compact solution set* if $|S_f|$ is compact in $|X|$; 2. f is *proper* if $|S_f|$ is compact and for every auxiliary norm N there is a saturated open $U \supset S_f$ with $\mathrm{cl}_{|X|}\, |\{x \in U : N(f(x)) \le 1\}|$ compact; 3. f is *proper*$_{<\infty}$ if $|S_f|$ is compact and for every auxiliary norm N there is one saturated open $U \supset S_f$ with $\mathrm{cl}_{|X|}\, |\{x \in U : N(f(x)) \le C\}|$ compact for every $0 < C < \infty$.

"$N(f(x)) \le C$" is Part I's relation `AuxLe`, which does not hold when $f(x) \notin W_{0,1}$. It passes along morphisms `Chapter12/CompactnessEp · EpStrongBundle.auxLe_tgt_of_auxLe_src`: if $N(f(s(\varphi))) \le C$ then $N(f(t(\varphi))) \le C$, since $f(t(\varphi)) = \mu(\varphi, f(s(\varphi)))$, μ preserves $W_{0,1}$ and N is invariant. So the sublevel sets are saturated when U is.

Theorem 12.4.2 (same) Chapter12/CompactnessEp · EpStrongBundle.theorem_12_4_2, pp. 587–588. For an sc-Fredholm section functor f over an ep-groupoid with paracompact $|X|$, the three properties are equivalent. The Lean statement is a list of equivalent propositions [proper, compact solution set, proper$_{<\infty}$].

The order of the implications. The text's proof labels its three steps (3)□ (2), (2)□ (1), (1)□ (3). Their arguments are, as we read them, those of (3)□ (1), (1)□ (2), (2)□ (3) in the numbering of the definition: the first is "take $C = 1$", the second "drop the second clause", the third the construction below. The labels are presumably permuted; the formal proof follows the arguments, and the cycle closes either way.

(3)□ (1) Chapter12/CompactnessEp · EpStrongBundle.IsProperLtInftyFunctor.isProperFunctor: proper$_{<\infty}$ implies proper, with $C = 1$. *(1)□ (2)* Chapter12/CompactnessEp · EpStrongBundle.IsProperFunctor.hasCompactSolutionSetEp: the first clause.

(2)□ (3) Chapter12/CompactnessEp · EpStrongBundle.isProperLtInftyFunctor_of_hasCompactSolutionSetEp: let $|S_f|$ be compact and N an auxiliary norm.

Step 1, local compactness at zeros Chapter12/CompactnessEp · exists_nhds_isCompact_closure_auxLe. For an sc-Fredholm section f of a strong bundle over an M-polyfold (here the object bundle), an auxiliary norm N and a zero x, there is an open $V(x) \ni x$ such that $\mathrm{cl}\{y \in V(x) : N(f(y)) \le C\}$ is compact for every $C > 0$. *Proof.* Part I's local compactness at a zero (the local step of Theorem 5.2.2 Chapter5/Book2021 · theorem_5_2_2, exists_nbhd_isCompact_closure_bdd): in a chart c at x and a neighbourhood V, the set of $y \in V$ with $\|\mathrm{pr}_2 \Phi[1](f(y))\|_{F_1} \le b$ has compact closure for every b, by the contraction germ normal form of the sc-Fredholm germ at x. Near x, N dominates a multiple of the chart norm: $|w|_1 \le A\,N(w)$ (the lower bound of §XII.2). So $N(f(y)) \le C$ gives $|f(y)|_1 \le AC$, and the sublevel set of N lies in a set with compact closure.

Step 2, a finite cover. The open sets $\pi(V(x))$, $x \in S_f$, cover the compact $|S_f|$; choose finitely many, $x_1, \dots, x_m$, and put $U = \pi^{-1}(\bigcup_i \pi(V(x_i)))$, open and saturated, containing S_f.

Step 3, compactness. For every $C > 0$, $|\{x \in U : N(f(x)) \le C\}| \subset \bigcup_i \pi(\mathrm{cl}\{y \in V(x_i) : N(f(y)) \le C\})$: a point of U is joined by a morphism to a point y of some $V(x_i)$, and the sublevel condition passes along morphisms. The right side is a finite union of compact sets, hence compact and closed, so the closure of the left side is compact. □

Where the formal proof departs from the text. The text shrinks the neighbourhoods to ones with the natural action; this is not needed, because Step 3 only uses that sublevel sets are saturated. *Kept and unused*: the paracompactness of $|X|$, since all three properties quantify over the auxiliary norms, and their existence (Theorem 12.2.2) is not needed for the equivalence.

XII.15 Definition 12.4.3, Proposition 12.4.5 and Theorem 12.4.6 (2017: Definition 12.10, Proposition 12.3, Theorem 12.10): reflexive local compactness

Definition 12.4.3 (same) Chapter12/ReflexiveCompactnessEp · EpStrongBundle.HasReflexiveLocalCompactnessEp, pp. 588–589. A section functor f has the *ep-groupoid reflexive local compactness property* if for every reflexive auxiliary norm N and every object x there is an open $U(x) \ni x$ with $\mathrm{cl}_X\{y \in U(x) : N(f(y)) \le 1\}$ compact. The text's standing assumptions (reflexive $(0,1)$-fibres, f sc-smooth and regularizing) are hypotheses of the statements that use the definition, not part of it. Part I's Definition 5.2.6 (Part V) on

the object bundle implies it Chapter12/ReflexiveCompactnessEp · EpStrongBundle.hasReflexiveLocalCompactnessEp_of_strong: a reflexive auxiliary norm of the ep-groupoid is one of the object bundle (§XII.8).

Two tools.

Saturated sublevel sets Chapter12/ReflexiveCompactnessEp · EpStrongBundle.isSaturated_setOf_auxLe: for a functorial f and an auxiliary norm N, $\{y : N(f(y)) \le C\}$ is saturated (§XII.14); the intersection of saturated sets is saturated Chapter12/ReflexiveCompactnessEp · EpGroupoid.isSaturated_inter.

Compact closures from orbit sets Chapter12/ReflexiveCompactnessEp · EpGroupoid.isCompact_closure_of_isProperMap: if $t : s^{-1}(\overline{V_0}) \to X$ is proper and $A \subset V_0$ has $\mathrm{cl}_{|X|}\,|A|$ compact, then $\mathrm{cl}_X A$ is compact. *Proof.* X is metrizable, so it suffices that every sequence a_k in $\mathrm{cl}\,A$ has a subsequence converging in $\mathrm{cl}\,A$. *Step 1.* $\pi(a_k)$ lies in $\mathrm{cl}\,|A|$ (continuity of π), which is compact, so a subsequence converges in $|X|$ to some $\pi(q)$. *Step 2, lifting* Chapter12/ReflexiveCompactnessEp · EpGroupoid.exists_lift_tendsto: if $\pi(u_k) \to \pi(q)$ in $|X|$, there are objects v_k with $\pi(v_k) = \pi(u_k)$ and $v_k \to q$ in X. Take a decreasing neighbourhood basis V_n of q; since π is open, $\pi(V_n)$ is a neighbourhood of $\pi(q)$, so $\pi(u_k) \in \pi(V_n)$ for $k \ge K_n$; for each k let $m(k)$ be the largest $n \le k$ with $K_n \le k$, and choose $v_k \in V_{m(k)}$ over $\pi(u_k)$; $m(k) \to \infty$, so $v_k \to q$. *Step 3.* Morphisms $\varphi_k : a_k \to v_k$ have sources in $\mathrm{cl}\,A \subset \overline{V_0}$ and convergent targets $v_k \to q$; by properness (the lemma of §XII.5, `exists_subseq_of_isProperMap`), a subsequence converges to a morphism ψ; then $a_k = s(\varphi_k) \to s(\psi)$ along it, and $s(\psi) \in \mathrm{cl}\,A$, which is closed. □

Proposition 12.4.5 (stronger) Chapter12/ReflexiveCompactnessEp · EpStrongBundle.proposition_12_4_5, p. 590. For an sc-smooth section functor f that is regularizing, with reflexive fibres, the following are equivalent: 1. the reflexive local compactness property (Definition 12.4.3); 2. for every reflexive N and every $z \in |X|$ there is a saturated open $\tilde{U}$ with $z \in |\tilde{U}|$ and $|\,\mathrm{cl}_X\{y \in \tilde{U} : N(f(y)) \le 1\}|$ compact; 3. the same with $\mathrm{cl}_{|X|}\,|\{y \in \tilde{U} : N(f(y)) \le 1\}|$ compact.

Stronger: the equivalence holds for every functorial section Chapter12/ReflexiveCompactnessEp · EpStrongBundle.proposition_12_4_5_of_functorial; reflexive fibres, sc-smoothness and the regularizing property are hypotheses of the book's form that the proof does not use.

(2)□ (3). The sublevel set $\{y \in \tilde{U} : N(f(y)) \le 1\}$ is saturated (intersection of two saturated sets), so by Lemma 12.4.4 (1) its closure in $|X|$ is the image of its closure in X; the two compactness conditions are the same.

(1)□ (3). Given $z = \pi(x)$, take $U(x)$ from (1) and $\tilde{U} = \mathrm{sat}(U(x))$, open and saturated. Its sublevel set has the same orbit set as that of $U(x)$ (sublevel sets are saturated); by Lemma 12.4.4 (2), the closure in $|X|$ of the orbit set of the sublevel set of $U(x)$ is the image of its compact closure, hence compact.

(3)□ (1), which the text says is easily obtained. Given x, take $V_0 \ni x$ on which t is proper over $s^{-1}(\overline{V_0})$ (the properness clause of an ep-groupoid), $\tilde{U}$ from (3) for $z = \pi(x)$, and a point of $\tilde{U}$ over z, which may be taken to be x since $\tilde{U}$ is saturated. Put $U(x) = \tilde{U} \cap V_0$, open and containing x. Its sublevel set A lies in V_0, and $|A| \subset |\{y \in \tilde{U} : N(f(y)) \le 1\}|$ has compact closure in $|X|$; by the second tool, $\mathrm{cl}_X A$ is compact. □ The step is short in the formal proof only because the second tool, a lifting argument through the properness of t, carries it; that argument is what "easily obtained" leaves out.

Theorem 12.4.6 (stronger) Chapter12/ReflexiveCompactnessEp · EpStrongBundle.theorem_12_4_6, pp. 590–592. Hypotheses, as printed: X tame, $|X|$ paracompact, reflexive $(0,1)$-fibres, N a reflexive auxiliary norm, f an sc-Fredholm section functor with $|S_f|$ compact and the reflexive local compactness property, and U_∂ a saturated subset of ∂X, open in ∂X, containing $S_f \cap \partial X$, with $\mathrm{cl}_{|X|}\,|\{y \in U_\partial : N(f(y)) \le 1\}|$ compact. Conclusion: a saturated open $U \supset S_f$ with $U \cap \partial X = U_\partial$ and $\mathrm{cl}_{|X|}\,|\{x \in U : N(f(x)) \le 1\}|$ com-

pact. The text writes "U_∂ open in ∂X"; it is read as $U_\partial = W' \cap \partial X$ for an open W' of X, the subspace topology.

Stronger: the conclusion holds for every continuous section functor f with local compactness at the given N `Chapter12/ReflexiveCompactnessEp · EpStrongBundle.theorem_12_4_6_of_continuous`. Tameness, paracompactness and the Fredholm property beyond continuity are unused; reflexive fibres and the reflexivity of N are used once, for Step 0.

Proof, the text's, carried out in $|X|$.

Step 0, the sublevel set is closed. $A = \{x : N(f(x)) \le 1\}$ is closed in X, by Part I's Proposition 5.2.7 (2) (`StrongBundle.isClosed_setOf_auxLe`): a limit of points with $N(f(x_k)) \le 1$ has $f(x_k) \to f(x)$ on level 0 with $N(f(x_k))$ bounded, hence mixed convergent (Remark 12.2.9, Step 3), and $N(f(x)) \le \liminf N(f(x_k)) \le 1$ by reflexivity of N. A is saturated, so $\pi(A)$ is closed (§XII.14), and so is ∂X, which is closed and saturated.

Step 1, the boundary part. For each object q, local compactness gives an open $U(q) \ni q$ with $\mathrm{cl}\{y \in U(q) : N(f(y)) \le 1\}$ compact. The open sets $|U(q)|$ cover the compact $\mathrm{cl}\,|U_\partial \cap A|$; take finitely many, $q_1, \dots, q_m$, with union O_1. Put $O_t = O_1 \cup (|X| \setminus |A|)$, open.

Step 2, the interior part. The compact $|S_f|$ minus the open O_t is compact; it lies in $|X \setminus \partial X|$, because a zero in ∂X lies in $U_\partial \cap A$ and so in O_1. It is covered by the open sets $|U(x) \setminus \partial X|$, $x \in S_f \setminus \partial X$; take finitely many, with union O_2.

Step 3, the set U. Put $O_V = |W'| \cup |X \setminus \partial X|$, open, and $U = \pi^{-1}((O_t \cup O_2) \cap O_V)$, open and saturated. - $S_f \subset U$: a zero x lies in $O_t \cup O_2$ by Steps 1–2, and in O_V: if $x \in \partial X$, then $x \in U_\partial \subset W'$. - $U \cap \partial X = U_\partial$: if $x \in U \cap \partial X$, then $\pi(x) \in |W'|$ (it is not in $|X \setminus \partial X|$, since ∂X is saturated), so x is joined to a point of $W' \cap \partial X = U_\partial$, and $x \in U_\partial$ because U_∂ is saturated. Conversely, points of U_∂ lie in O_V, and in O_t (either in A, then in O_1 by Step 1, or not in A). - *Compactness*: $|\{x \in U : N(f(x)) \le 1\}| = |U \cap A|$ lies in $O_t \cup O_2$ intersected with $|A|$, hence in $O_1 \cup O_2$ intersected with $|A|$, which lies in the finite union T of the compact sets $\pi(\mathrm{cl}\{y \in U(q_i) : N(f(y)) \le 1\})$ and the corresponding sets for the interior points. T is compact and closed, so the closure of $|U \cap A|$ is compact. □

Hypotheses. None added. *Kept and unused* in the book's form: tameness, paracompactness, the Fredholm property beyond continuity. Witness of Definition 12.4.3: any section functor with Part I's reflexive local compactness property (Definition 5.2.6) on the object bundle, by the implication above; every hypothesis of Proposition 12.4.5 and Theorem 12.4.6 is a hypothesis of the text, so no witness beyond the text's is owed.

XII.16 The local strong bundle isomorphism of a morphism, and chart points carried along it (§12.5, pp. 592–594)

§12.5 orients sc-Fredholm section functors. Part I attaches to an sc-Fredholm section f of a strong bundle over an M-polyfold, at each smooth point x, the set $\mathrm{Lin}(f, x)$ of linearizations $(f - s_0)'(x)$, s_0 an sc$^+$-section with $s_0(x) = f(x)$, and the two-point set Or_x^f of orientations of the determinant line of any of them, all identified by the canonical isomorphisms of determinant lines along the convex set $\mathrm{Lin}(f, x)$ (Chapter 6, Part VI). The chapter's task is to make the morphisms act on these sets. The text does it in one sentence: a morphism $\varphi : x \to x'$ extends to an sc-diffeomorphism $\widehat{\varphi} : U(x) \to U(x')$ with a lift $\widehat{\Phi}$ to the bundle, and $\varphi_* S = \mu(\varphi, \cdot) \circ S \circ (T\varphi)^{-1}$ maps $\mathrm{Lin}(f, x)$ to $\mathrm{Lin}(f, x')$. The formalization reads

this sentence in Part I's chart points, which is where Lin lives, and the comparison of chart points is the bulk of §12.5's formal content. This section collects the tools; Lemma 12.5.1 is §XII.17.

Part I's chart points. In the project, $\mathrm{Lin}(f,x)$ is read in a *chart point* at x (`ChartPoint`, Part VI): a strong bundle chart around x with a level-1 point z of its model, through which T_xX is the tangent space T_zO of the retract and the fibre W_x is the fibre K_z of the model. $\mathrm{Lin}(f,x)$ is then a set of operators $T_zO \to K_z$ (`linSet`), and two chart points at the same x are compared by Part I's transfer maps of tangent spaces and fibres (`transferT`, `transferK`) and the induced bijection of orientation sets (`orTransfer`), which form a cocycle. At each smooth object of an ep-groupoid one chart point is chosen once `Chapter12/OrientationBundleEp · orPt`, and Or^f_x is read in it; restricting its chart to an open $V \ni x$ gives a chart point inside V `Chapter12/OrientationBundleEp · restrPt`. An sc-Fredholm section functor is functorial `Chapter12/OrientationBundleEp · EpStrongBundle.functorial_of_isScFredholmSectionFunctor`.

The isomorphism of a morphism `Chapter12/MorphismIso · EpStrongBundle.morIso`. Let $\sigma : x \to x'$ be a morphism, and choose a sheet of σ, an open neighbourhood of σ in the morphism space on which s and t are sc-diffeomorphisms onto open sets $s(\mathrm{sheet}) \ni x$ and $t(\mathrm{sheet}) \ni x'$ (`sheetAt`). Write $\tau = (t|_{\mathrm{sheet}})^{-1}$. Then

$$\mathrm{morIso}(\sigma) : W|t(\mathrm{sheet}) \to W|s(\mathrm{sheet}), \qquad y \mapsto s(\tau(y)), \qquad w \mapsto \mu(\tau(P(w))^{-1}, w)$$

is a local strong bundle isomorphism (`StrongBundle.LocalIso`: an sc-diffeomorphism of the bases with a strong bundle isomorphism over it, both with inverses). The ingredients: - the local section $y \mapsto \tau(y)^{-1}$ of s near x' `Chapter12/MorphismIso · EpStrongBundle.morInvSec`, which starts at y `Chapter12/MorphismIso · EpStrongBundle.src_morInvSec` and is sc-smooth `Chapter12/MorphismIso · EpStrongBundle.isScSmoothOn_morInvSec` (inversion is sc-smooth on an ep-groupoid); - the base map $s \circ \tau$, an sc-diffeomorphism $t(\mathrm{sheet}) \to s(\mathrm{sheet})$ with inverse $t \circ (s|_{\mathrm{sheet}})^{-1}$ `Chapter12/MorphismIso · EpStrongBundle.isScDiffeoOn_morBase`; - the bundle map is the action of the sc-smooth family $\tau(\cdot)^{-1}$ of morphisms, with inverse the action of $(s|_{\mathrm{sheet}})^{-1}$, both strong bundle maps (`sectionAct`).

The text's $\hat{\Phi}$ goes from $U(x)$ to $U(x')$; $\mathrm{morIso}(\sigma)$ is its inverse, in the direction in which charts at x are carried to charts at x' (Part I's transported chart, `LocalIso.transportChart`). Its properties: x' lies in its domain `Chapter12/MorphismIso · EpStrongBundle.tgt_mem_morIso` and goes to x `Chapter12/MorphismIso · EpStrongBundle.morIso_ϕ_tgt` (because $\tau(x') = \sigma$), with $\tau(x') = \sigma$ `Chapter12/MorphismIso · EpStrongBundle.tgtInv_tgt` and $x \in s(\mathrm{sheet})$ `Chapter12/MorphismIso · EpStrongBundle.src_mem_morIso`; on the fibre over x' the bundle map is $\mu(\sigma^{-1}, \cdot)$ `Chapter12/MorphismIso · EpStrongBundle.morIso_Φ_eq`; every functorial section f is invariant under it, $f = \Psi \circ f \circ \phi$ on $t(\mathrm{sheet})$ `Chapter12/MorphismIso · EpStrongBundle.eqOn_morIso`, because $f(y) = \mu(\tau(y), f(s(\tau(y))))$; and for composable $\varphi : x \to y$, $\psi : y \to z$, the composite of the isomorphisms of φ and ψ agrees near z with the isomorphism of $\psi \circ \varphi$ `Chapter12/MorphismIso · EpStrongBundle.germAgree_morIso_comp`: on the base because $\tau_\psi(y') \circ \tau_\varphi(\ldots)$ and $\tau_{\psi\varphi}(y')$ are both local inverses of t through $\psi\varphi$, hence agree near z, and on the bundle by $\mu(\psi\gamma, w) = \mu(\psi, \mu(\gamma, w))$. The construction is that of Chapter 10's `exists_localIso_mor` (Part X), with the data kept.

Chart points carried along a local isomorphism of a bundle to itself. Let $L : W|U \to W|U'$ be a local strong bundle isomorphism over $\phi : U \to U'$, and p a chart point at $x = \phi(x')$ whose chart lies in U'. The *carried chart point* $\mathrm{pullPt}(L,p)$ at x' is the chart point of the transported chart $\Psi' \circ \Phi$ (Part I's chart composed with the bundle map of L), with the same model, the same retraction and the same level-1 point z `Chapter12/ChartPull · StrongBundle.LocalIso.pullPt`. Its chart, point, tangent space and fibre are those of p `Chapter12/ChartPull · StrongBundle.LocalIso.pullPt_c Chapter12/ChartPull · StrongBundle.LocalIso.`

pullPt_z Chapter12/ChartPull · StrongBundle.LocalIso.pullPt_T Chapter12/ChartPull · StrongBundle.LocalIso.pullPt_K.

Linearizations agree Chapter12/ChartPull · StrongBundle.LocalIso.linSet_pullPt: if $f = \Psi \circ f \circ \phi$ on U, then $\mathrm{Lin}(f, x')$ read in $\mathrm{pullPt}(L, p)$ is $\mathrm{Lin}(f, x)$ read in p. *Proof.* The principal part of f in the transported chart is the principal part of $\Psi \circ f \circ \phi$ in the chart of p, which is that of f, so the two are the same map near z. For an sc⁺-section s_1 through $f(x')$, $(f - s_1)'$ read in the transported chart differs from $(f - s_0)'$ read in p (for any s_0 through $f(x)$) by the derivative of the difference of the two local sc⁺-representatives, which is an sc⁺-operator Chapter12/ChartPull · StrongBundle.LocalIso.exists_isScPlusOp_pullPt; and Lin is closed under adding sc⁺-operators (Part VI). So each set contains the other. This is the text's identity $\mu(\varphi, (f - s_0)'(x) \circ T\varphi^{-1}) = (f - s_1)'(x')$, read in the transported chart, where $T\varphi$ and $\mu(\varphi, \cdot)$ become identities.

A reading. The sections in Lin are global sc⁺-sections in the project, while the text uses a local section $s_1 = \hat{\Phi}_* s_0$. Any global sc⁺-section through $f(x')$ serves, and their existence is §12.5's standing hypothesis B.`HasScPlusSectionsThrough` (an sc⁺-section through every smooth fibre point), under which Part I proves $\mathrm{Lin}(f, x) \neq \emptyset$. Bump functions on the objects give it (Part III) Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough.

Transfers commute with carrying Chapter12/ChartPull · StrongBundle.LocalIso.transferT_pullPt Chapter12/ChartPull · StrongBundle.LocalIso.transferK_pullPt: Part I's transfer between two carried chart points is the transfer between the original ones, since it is defined from the derivative of the transition map of the charts and the transported charts have the same transition. The derivative of the base transition and the fibre transition carry over Chapter12/ChartPull · StrongBundle.LocalIso.baseDeriv_pullPt Chapter12/ChartPull · StrongBundle.LocalIso.fiberMap_pullPt. *Germs* Chapter12/ChartPull · StrongBundle.LocalIso.GermAgree: two pairs of composable local isomorphisms *agree near* x'' if their composites have the same base and bundle maps on a neighbourhood of x''. Then the transfer between the two doubly carried chart points is the identity Chapter12/ChartPull · StrongBundle.LocalIso.transferT_pullPt_pullPt Chapter12/ChartPull · StrongBundle.LocalIso.transferK_pullPt_pullPt, since the two transported charts coincide near the point; the derivative forms Chapter12/ChartPull · StrongBundle.LocalIso.baseDeriv_pullPt_pullPt Chapter12/ChartPull · StrongBundle.LocalIso.fiberMap_pullPt_pullPt.

The orientation map of a local isomorphism. For chart points q at x and q' at x', with an intermediate chart point a at x inside U': - the identifications of T_x and $T_{x'}$, and of the fibres, are cast between a chart point and its carried chart point Chapter12/OrientationBundleEp · StrongBundle.LocalIso.castT Chapter12/OrientationBundleEp · StrongBundle.LocalIso.castK; - Or^f_x read in a and $\mathrm{Or}^f_{x'}$ read in the carried chart point are the same set, since the two Lin agree Chapter12/OrientationBundleEp · StrongBundle.LocalIso.orPull; - the maps $T_x X \to T_{x'} X$ and $W_x \to W_{x'}$ of L read in q, q' are Part I's transfer from q to a, the cast to the carried point, and the transfer to q' Chapter12/OrientationBundleEp · StrongBundle.LocalIso.morT Chapter12/OrientationBundleEp · StrongBundle.LocalIso.morK; - the orientation map $\mathrm{orVia} : \mathrm{Or}^f_x(q) \to \mathrm{Or}^f_{x'}(q')$ is orTransfer from q to a, then orPull, then orTransfer to q' Chapter12/OrientationBundleEp · StrongBundle.LocalIso.orVia, and it is the conjugation of a line orientation by morT and morK Chapter12/OrientationBundleEp · StrongBundle.LocalIso.orVia_apply.

All three are independent of the intermediate chart point a Chapter12/OrientationBundleEp · StrongBundle.LocalIso.morT_eq Chapter12/OrientationBundleEp · StrongBundle.LocalIso.morK_eq Chapter12/OrientationBundleEp · StrongBundle.LocalIso.orVia_eq (the cocycle of transfers and the commutation of transfers with carrying), and composing the maps of L and L_2 gives the map of L_3 when $L \circ L_2$ and L_3 agree near the point Chapter12/OrientationBundleEp · StrongBundle.LocalIso.morT_comp Chapter12/

OrientationBundleEp · StrongBundle.LocalIso.morK_comp, by the germ lemma above.

Between two bundles (extra) Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.mpullPt. Theorem 12.5.7 compares two different strong bundles $Y \to U$ and $Y' \to U'$, through a strong bundle equivalence, and Theorem 12.5.3's last sentence carries orientations along a local isomorphism. The same construction works for a local strong bundle isomorphism $L : Y|U \to Y'|U'$ over an sc-diffeomorphism ϕ between different M-polyfolds: a chart point p of Y' at $\phi(b)$ inside U' is carried to the chart point $\mathrm{mpullPt}(L, p)$ of Y at b, in the transported chart. Under $f = \Psi \circ f' \circ \phi$ on U (the text's "$\hat{\hat{\Phi}}$ maps f to f'"): - $\mathrm{Lin}(f, b)$ in the carried chart point is $\mathrm{Lin}(f', \phi(b))$ in p Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.linSet_mpullPt, through an sc$^+$-operator difference Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.exists_isScPlusOp_mpullPt; - transfers commute with carrying Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.baseDeriv_mpullPt Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.fiberMap_mpullPt Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.transferT_mpullPt Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.transferK_mpullPt; - for two composites with the same germ, the four-term form Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.GermAgree₄ Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.baseDeriv_mpullPt₄ Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.fiberMap_mpullPt₄ Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.transferT_mpullPt₄ Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.transferK_mpullPt₄; - the cast Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.mcastT, the identification of orientation sets Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.morPull, the maps of tangent spaces and fibres Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.mmorT Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.mmorK, and the orientation map $\mathrm{Or}^{f'}_{\phi(b)} \to \mathrm{Or}^{f}_{b}$ Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.morVia, a conjugation Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.morVia_apply, independent of the intermediate chart point Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.mmorT_eq Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.morVia_eq.

For a bundle to itself the carried chart point is the one above Chapter12/ChartPullTwoBundles · StrongBundle.LocalIso.mpullPt_eq_pullPt.

The orientation map depends only on the germ (extra) Chapter12/OrientationGerm · StrongBundle.LocalIso.GermEq Chapter12/OrientationGerm · StrongBundle.LocalIso.morVia_germ. Two local isomorphisms L_1, L_3 *have the same germ at* b if their base maps and bundle maps agree on an open neighbourhood of b. Then they act alike on orientations at b. *Proof.* The chart points carried along L_1 and L_3 have, at b, base transition derivative $Dr(z)$ and fibre transition the identity on K_z, as from a chart point to itself Chapter12/OrientationGerm · StrongBundle.LocalIso.baseDeriv_mpullPt_germ Chapter12/OrientationGerm · StrongBundle.LocalIso.fiberMap_mpullPt_germ, because the two transported charts agree near b; so the transfers between them are identities Chapter12/OrientationGerm · StrongBundle.LocalIso.transferT_mpullPt_germ Chapter12/OrientationGerm · StrongBundle.LocalIso.transferK_mpullPt_germ, the maps of tangent spaces and fibres agree Chapter12/OrientationGerm · StrongBundle.LocalIso.mmorT_germ Chapter12/OrientationGerm · StrongBundle.LocalIso.mmorK_germ, and so do the conjugations. The text passes silently from one sheet of a morphism to another, and from the local isomorphism of an equivalence at one object to that at a nearby object; this lemma is what justifies it. Two sheets through one morphism σ have the same local inverse of t near $t(\sigma)$ Chapter12/OrientationGerm · EpGroupoid.Sheet.tgtInv_eq_near_or, because t is injective on each sheet and both inverses are continuous through σ.

XII.17 Lemma 12.5.1 (2017: 12.2): morphisms act on linearizations and on orientations

$T\varphi$ and $\mu(\varphi,\cdot)$ in the chosen chart points `Chapter12/OrientationBundleEp · EpStrongBundle.morTx` `Chapter12/OrientationBundleEp · EpStrongBundle.morKx`. For a morphism $\sigma : x \to x'$ between smooth objects, $T\sigma : T_xX \to T_{x'}X$ and $\mu(\sigma,\cdot) : W_x \to W_{x'}$ are read in the chosen chart points at x and x' as the composite of three maps: Part I's transfer from the chosen chart point at x to the chart point restricted to $s(\text{sheet})$, the identity to its chart point carried along $\text{morIso}(\sigma)$ at x', and the transfer from there to the chosen chart point at x'; they are morT and morK of $\text{morIso}(\sigma)$ `Chapter12/OrientationBundleEp · EpStrongBundle.morTx_eq` `Chapter12/OrientationBundleEp · EpStrongBundle.morKx_eq`. Lin in a chart point inside $s(\text{sheet})$ and in its carried chart point coincide `Chapter12/OrientationBundleEp · EpStrongBundle.linSet_pullPt_mor`, by the invariance of the functor f under $\text{morIso}(\sigma)$.

Functoriality in chart points. - *Composition* `Chapter12/OrientationBundleEp · EpStrongBundle.morTx_comp` `Chapter12/OrientationBundleEp · EpStrongBundle.morKx_comp`: $T(\psi\varphi) = T\psi \circ T\varphi$ and $\mu(\psi\varphi,\cdot) = \mu(\psi,\cdot) \circ \mu(\varphi,\cdot)$. The proof works on the open set $s(\text{sheet}\,\varphi) \cap s(\text{sheet}\,\psi\varphi) \cap \widehat{\varphi}^{-1}(s(\text{sheet}\,\psi))$ around x `Chapter12/OrientationBundleEp · EpStrongBundle.compSet`: there the isomorphisms of φ, ψ compose to that of $\psi\varphi$ as germs, and `morT_comp` applies. - *Units* `Chapter12/OrientationBundleEp · EpStrongBundle.morTx_unit` `Chapter12/OrientationBundleEp · EpStrongBundle.morKx_unit`: $T(1_x) = \text{Id}$, since $1_x \circ 1_x = 1_x$ and composition gives an idempotent invertible map. - *Inverses* `Chapter12/OrientationBundleEp · EpStrongBundle.morTx_inv` `Chapter12/OrientationBundleEp · EpStrongBundle.morKx_inv`: $T(\varphi^{-1}) \circ T\varphi = \text{Id}$, from composition and units; so $T(\varphi^{-1}) = (T\varphi)^{-1}$ `Chapter12/OrientationBundleEp · EpStrongBundle.morTx_inv_eq_symm` `Chapter12/OrientationBundleEp · EpStrongBundle.morKx_inv_eq_symm`.

The action on linearizations `Chapter12/OrientationBundleEp · EpStrongBundle.linMor`: $\varphi_*S = \mu(\varphi,\cdot) \circ S \circ (T\varphi)^{-1}$ (p. 592), read in the chosen chart points `Chapter12/OrientationBundleEp · EpStrongBundle.linMor_apply`.

Lemma 12.5.1 (explicit), p. 592. Let f be an sc-Fredholm section functor and assume §12.5's standing hypothesis that sc$^+$-sections pass through every smooth fibre point (`hsec`).

(1) `Chapter12/OrientationBundleEp · EpStrongBundle.lemma_12_5_1_one`: for $\varphi : x \to x'$ between smooth objects and $S \in \text{Lin}(f,x)$, $\varphi_*S \in \text{Lin}(f,x')$. *Proof.* Read in the chosen chart point at x', φ_*S is the transfer of S read in the chart point carried along $\text{morIso}(\varphi)$. By `linSet_pullPt_mor`, S read there lies in $\text{Lin}(f,x')$ read there. Part I's transfers map Lin to Lin (Part VI). The conjugate in the other direction, $\mu(\varphi,\cdot)^{-1} \circ S' \circ T\varphi$ for $S' \in \text{Lin}(f,x')$, lies in $\text{Lin}(f,x)$ by the same argument `Chapter12/OrientationBundleEp · EpStrongBundle.morTx_mem`.

(2) `Chapter12/OrientationBundleEp · EpStrongBundle.lemma_12_5_1_two`: $\varphi_* : \text{Lin}(f,x) \to \text{Lin}(f,x')$ is a bijection. *Proof.* Injective, as conjugation by two isomorphisms `Chapter12/OrientationBundleEp · EpStrongBundle.linMor_injective`; surjective by the converse membership of (1). The text derives (2) from (1) and (3) (apply (1) to φ^{-1} and compose); the formal proof gives it directly.

(3) `Chapter12/OrientationBundleEp · EpStrongBundle.lemma_12_5_1_three_unit` `Chapter12/OrientationBundleEp · EpStrongBundle.lemma_12_5_1_three_comp`: $(1_x)_* = \text{Id}$ and $(\psi\varphi)_* = \psi_* \circ \varphi_*$, from the unit and composition laws of T and $\mu(\cdot,\cdot)$ in chart points.

Explicit: the hypothesis `hsec` is not stated in the text; it is Part I's standing hypothesis for $\text{Lin}(f,x) \neq \emptyset$, which §12.5 uses through Part I's Lin, and bump functions on the objects give it. The text calls the lemma obvious; the formal proof is about 1,100 lines, most of it the chart-point comparisons of §XII.16.

On orientations (p. 593). $\varphi_* : \mathrm{Or}_x^f \to \mathrm{Or}_{x'}^f$ sends an orientation o of $\mathrm{DET}(f,x)$ to the orientation whose value on $\varphi_* S$ is the conjugate of $o(S)$ by $T\varphi$ and $\mu(\varphi,\cdot)$ `Chapter12/OrientationBundleEp · EpStrongBundle.orMor` `Chapter12/OrientationBundleEp · EpStrongBundle.orMor_apply`: conjugation by isomorphisms maps the kernel and cokernel of S to those of $\varphi_* S$, hence determinant lines to determinant lines (Part I's `LineOrientation.conj`), compatibly with the identifications along Lin. It is functorial: $(\psi\varphi)_* = \psi_*\varphi_*$ `Chapter12/OrientationBundleEp · EpStrongBundle.orMor_comp`, $(1_x)_* = \mathrm{Id}$ `Chapter12/OrientationBundleEp · EpStrongBundle.orMor_unit`, $(\varphi^{-1})_* = (\varphi_*)^{-1}$ `Chapter12/OrientationBundleEp · EpStrongBundle.orMor_inv`.

XII.18 Definition 12.5.2 (2017: 12.11): the orientation bundle and the orbit spaces

Definition 12.5.2 (restated) `Chapter12/OrientationOrbit · EpStrongBundle.definition_12_5_2`, pp. 593–594. The text defines O_f as the set of pairs (x,o), x a smooth object and $o \in \mathrm{Or}_x^f$, with the projection $\sigma : O_f \to X_\infty$. The formalization takes Part I's orientation space of f as a section of the object bundle, `OrSpace` (Part VI), which is this set. Under the standing hypothesis, σ is surjective with exactly two points over each x `Chapter12/OrientationOrbit · orProj_two`, the given orientation and the other `Chapter12/OrientationOrbit · exists_other_or`, since $\mathrm{Lin}(f,x) \neq \emptyset$ and a real line has two orientations. Part I's topology on O_f (the one of Theorem 6.6.17, Part VI) is the one the text then characterizes in Theorem 12.5.3.

The orbit spaces. The text uses $|X_\infty|$ and $|O_f|$ in Proposition 12.5.6 and Theorem 12.5.7 without defining them. The formalization defines: - the relation on smooth objects "joined by a morphism" `Chapter12/OrientationOrbit · EpGroupoid.SmoothRel`, an equivalence `Chapter12/OrientationOrbit · EpGroupoid.smoothRel_equivalence`, with quotient $|X_\infty|$; smooth points go to smooth points under morphisms, because s and t are local sc-diffeomorphisms; - the relation on O_f: $(x,o) \sim (x',o')$ if some $\varphi : x \to x'$ has $\varphi_* o = o'$ `Chapter12/OrientationOrbit · EpStrongBundle.OrRel`, an equivalence `Chapter12/OrientationOrbit · EpStrongBundle.orRel_equivalence` by the functoriality of φ_* (units, inverses, composition), with quotient $|O_f|$; - the projection $|\sigma| : |O_f| \to |X_\infty|$, $|(x,o)| \mapsto |x|$ `Chapter12/OrientationOrbit · EpStrongBundle.orbitProj`, well defined because related pairs have related base points.

X_∞ carries the topology of Part I's Theorem 2.3.10 (Part II), O_f Part I's topology, and the orbit spaces the quotient topologies.

XII.19 Theorem 12.5.3 (2017: 12.11): the topology of O_f, and morphisms keep orientations continuous

Theorem 12.5.3 (narrower), p. 594. The text: for an sc-Fredholm section functor f over a tame ep-groupoid, O_f has exactly one topology for which it is Hausdorff, $\sigma : O_f \to X_\infty$ is a local homeomorphism, there are exactly two continuous sections near each point, and they propagate as in Part I; and for a morphism $\varphi : x \to x'$ with lift $\hat{\Phi}$, a continuous local section of σ near x is carried by $\hat{\Phi}$ to a continuous local section near x'. The formalization proves the two halves separately.

The topology

The topological part `Chapter12/OrientationOrbit · EpStrongBundle.theorem_12_5_3_hyp`: for X tame, bump functions on the object M-polyfold and tame germ data for f (`IsScFredholmSectionTame`), there is exactly

one topology on O_f with the properties of Part I's Theorem 6.6.17 (`IsOrientationTopology`), and every fibre of σ has exactly two points. *Proof.* Part I's Theorem 6.6.17 `Chapter6/OrientationMainClean · theorem_6_6_17` applied to f as a section of the object bundle, and the two-point fibres of §XII.18 with the standing hypothesis, which bump functions give. The morphisms play no role in this half: the topology of O_f is Part I's. The Lean statement takes Part I's paracompactness instances for the products $\mathbb{R}^n \times X$ and the pull-back bundles, which hold for metrizable spaces and are discharged inside the statement (`paracompactSpace_rn_prod`, `paracompactSpace_rn_pullback`).

Narrower, by the hypotheses inherited from Theorem 6.6.17: the text assumes only a tame X, and Theorem 6.6.17 needs bump functions and tame germ data (gap G.140). The fibre bijections φ_* and their functoriality, which the text recalls in the statement, are §XII.17.

The last sentence

The last sentence is proved in four layers: Part I's push-forward of orientations carried along a local isomorphism in one chart; locality of continuity; the commutation of the transport with a local isomorphism along short paths and then all paths; and the specialization to the isomorphism of a morphism.

Part I's transport, recalled. Part I propagates orientations of f along sc-smooth paths $\gamma : [0,1] \to X$ (Theorem 6.6.13, Part VI): in one chart, the linearizations along the path are stabilized by an sc$^+$-section S of the pull-back bundle over $\mathbb{R}\times$ (the path's parameter space), so that the stabilized kernels form a vector bundle over $[0,1]$; an orientation at $\gamma(0)$ is moved along this bundle to $\gamma(1)$ (a *push-forward*, `IsPush`). The local transport is defined by property (□): along short paths it is a push-forward in one chart (`orLocalTransport_transport_eq`), and along any path the composite over dyadic pieces. An orientation section o over an open U is *continuous* (`IsContOrOn`) if for every sc-smooth path in U between smooth points, the transport carries o at the start to o at the end.

Layer 1: push-forwards carried along a local isomorphism. Let L be a local strong bundle isomorphism from $Y \to U$ to $Y' \to U'$ over ϕ, with $f = \Psi \circ f' \circ \phi$ on U. Each ingredient of a push-forward of f' along $\phi \circ \gamma$ in a chart c of Y' is carried to the transported chart c_T of Y along γ.

- *The stabilizing section* `Chapter12/SectionPull · StrongBundle.LocalIso.exists_pullSection`: an sc$^+$-section S' of the pull-back of Y' gives an sc$^+$-section S of the pull-back of Y whose principal part in c_T is $\beta \cdot S'$ read in c, for a cut-off β supported in the domain of c_T. The cut-off is needed because S' composed with L is defined only over the image of the chart, and an sc$^+$-section must be global. The products with the chart domain are read through the transported chart `Chapter12/SectionPull · StrongBundle.LocalIso.ofProd_transportChart_0` `Chapter12/SectionPull · StrongBundle.LocalIso.Kg_ofProd_transportChart`.
- *Principal parts* `Chapter12/StabilizedKernelPull · StrongBundle.LocalIso.principal_transportChart_eq`: where $\beta = 1$, the principal part of f in c_T is that of f' in c, and also for $f + S$ and $f' + S'$ `Chapter12/StabilizedKernelPull · StrongBundle.LocalIso.principal_transportChart_sectionAdd`; so are the linearizations `Chapter12/StabilizedKernelPull · StrongBundle.LocalIso.linAt_transportChart_sectionAdd`.
- *Stabilized kernels* `Chapter12/StabilizedKernelPull · StrongBundle.LocalIso.isStabKernels_mpull`: if (f', S', s') stabilizes along $\phi \circ \gamma$ in c, the carried data stabilize along γ in c_T; the kernels are the same subspaces of the same model.
- *Orientations* `Chapter12/StabilizedKernelPull · StrongBundle.LocalIso.morPull_apply`: the carried chart points of §XII.16 act on orientations by conjugation; they map the orientation rays of the stabilized kernels to each other `Chapter12/StabilizedKernelPull · StrongBundle.LocalIso.orRay_mpull`, and commute with Part I's push-forward `Chapter12/StabilizedKernelPull · StrongBundle.LocalIso.push_mpull`: pushing

forward and then carrying is carrying and then pushing forward. Conjugation by identities fixes line orientations `Chapter12/StabilizedKernelPull · LineOrientation.conj_refl_apply`.

- *Zeros* `Chapter12/PushDataPull · StrongBundle.LocalIso.isZeroAt_of_transportChart`: zeros of $f' + S'$ at $\phi(x)$ are zeros of $f + S$ at x.

The chart-level statement `Chapter12/PushDataPull · StrongBundle.LocalIso.isPush_mpull`: if e' is a push-forward of f' along $\phi \circ \gamma$ in a chart c_0 of Y' restricted to an open W, then the conjugate of e' by the carried chart points at the two ends is a push-forward of f along γ in the transported chart. The hypotheses are the standing hypothesis for both bundles, $f = \Psi \circ f' \circ \phi$, Part I's paracompactness instances, and proof data: the chart c_0, the open W, a cut-off β equal to 1 on the transported $c_0|W$, and the path γ, parametrized on $[0,1] \subset \mathbb{R}^1$. The proof assembles the five items: the carried section stabilizes, the carried kernels are the stabilized kernels, and the push-forward commutes with the conjugation.

Layer 2: continuity of an orientation is local (extra) `Chapter12/DyadicPathInduction · isContOrOn_of_local_or`. For any of Part I's local transports L and an open U: if every smooth point of U has an open neighbourhood V over which o is continuous, then o is continuous over U. Part I uses this without stating it; it is not immediate, because a path in U need not lie in one V.

Dyadic induction `Chapter12/DyadicPathInduction · scPath_induction_or`. Let Q be a property of paths carried to both halves of a path, and R a property of paths that passes from the two halves to the path. If every point $\gamma(t)$ of a path with Q has an open neighbourhood V such that every sc-smooth path with Q and image in V has R, then every sc-smooth path with Q has R. *Proof.* The preimages $\gamma^{-1}(V_t)$ cover the compact $[0,1]$; take a Lebesgue number λ, and n with $2^{-n} < \lambda$. Every dyadic piece $[k2^{-n}, (k+1)2^{-n}]$ lies in one $\gamma^{-1}(V_t)$, so its reparametrized path has R; induction on n over the halves gives R for γ. *The halves* are reparametrized by the smooth transition function $\chi : \mathbb{R} \to [0,1]$, $\chi(0) = 0$, $\chi(1) = 1$, constant near the ends: $s \mapsto \gamma(\chi(s))$ `Chapter12/DyadicPathInduction · smoothTransPath_or`, sc-smooth `Chapter12/DyadicPathInduction · isScSmoothMap_smoothTransPath_or` and with values in $\gamma([0,1])$ `Chapter12/DyadicPathInduction · smoothTransPath_or_mem`. Part I's transport along a path is the composite of the transports along its two reparametrized halves (`transport_halves`).

Proof of locality. Apply the induction with Q = "lies in U" and R = "the transport carries o from start to end"; R passes from halves to the whole by `transport_halves`, and small paths in a V have R by hypothesis. Continuity passes to open subsets `Chapter12/DyadicPathInduction · IsContOrOn.mono_or` and to sections with the same values `Chapter12/DyadicPathInduction · IsContOrOn.congr_or`.

Layer 3: the transport commutes with a local isomorphism. Let L be as in Layer 1. At a smooth $a \in U$, the orientation map of §XII.16, through a chart point at $\phi(a)$ inside U' (the chosen chart restricted to U' `Chapter12/OrientationLocalCommute · StrongBundle.LocalIso.innerPt`), is a bijection $\mathrm{orxVia}_a : \mathrm{Or}^{f'}_{\phi(a)} \to \mathrm{Or}^{f}_{a}$ `Chapter12/OrientationLocalCommute · StrongBundle.LocalIso.orxVia`, equal to morVia through any chart point inside U' `Chapter12/OrientationLocalCommute · StrongBundle.LocalIso.orxVia_eq_morVia`. An orientation section o' of f' is carried to $a \mapsto \mathrm{orxVia}_a(o'_{\phi(a)})$ on U `Chapter12/OrientationLocalCommute · StrongBundle.LocalIso.pullOrx`. For a path γ in U, $\phi \circ \gamma$ is sc-smooth `Chapter12/OrientationLocalCommute · isScSmoothMap_comp_or`.

On short paths `Chapter12/OrientationLocalCommute · StrongBundle.LocalIso.orxVia_comm_local`: every smooth $a_0 \in U$ has an open $V \ni a_0$ such that for every sc-smooth path γ in U with $\gamma([0,1]) \subset V$, orxVia at the start followed by Part I's transport of f along γ equals the transport of f' along $\phi \circ \gamma$ followed by orxVia at the end. *Proof.* 1. Part I's Theorem 6.6.11 gives a propagation set of f' at $\phi(a_0)$ in a chart c' and one of f at a_0 in a chart c_B: on them, the transport along short paths is a push-forward in that chart (property (□)). 2. Choose V inside both propagation sets (pulled back by ϕ for the first). For γ with image in

V, Part I gives a push-forward e' of f' along $\phi \circ \gamma$ in c', which by (□) represents the transport of f'. 3. Transfer e' to $c'|_{U' \cap W}$, with $W = g^{-1}(N)$ for an open $N \supset \gamma([0,1])$ on which a cut-off β equals 1 (`exists_isScSmoothMap_eq_one_nhdsSet`, from bump functions on the compact image); push-forwards transfer between charts (`IsPush.transfer`). 4. Layer 1 carries it to a push-forward of f along γ in the transported chart; transfer that to c_B, where by (□) it represents the transport of f. 5. The two sides are now composites of orTransfer maps, the conjugation by carried chart points and the two push-forwards; Part I's cocycle identities for orTransfer regroup them (`equiv_comm_square_or`, an identity of composites of bijections `Chapter12/OrientationLocalCommute · equiv_comm_square_or`), and the square commutes.

Along every path in U `Chapter12/OrientationLocalCommute · StrongBundle.LocalIso.orxVia_comm`: dyadic induction with Q = "lies in U" and R = "the square commutes", which passes from halves to the whole because both transports are composites over halves.

Continuity `Chapter12/OrientationLocalCommute · StrongBundle.LocalIso.isContOrOn_pullOrx`: if o' is continuous over an open V', then the carried section is continuous over $U \cap \phi^{-1}(V')$. For a path γ in $U \cap \phi^{-1}(V')$, the commutation turns the transport of the carried section along γ into the transport of o' along $\phi \circ \gamma$, conjugated, and o' propagates along $\phi \circ \gamma$ because it lies in V'. Paths in $U \cap \phi^{-1}(V')$ are first reparametrized by χ, so that all their values (not only those on $[0,1]$) lie in U, as the commutation lemma needs. The converse holds as well `Chapter12/OrientationEquivalenceContinuity · StrongBundle.LocalIso.isContOrOn_of_pullOrx`: if the carried section is continuous over $U \cap \phi^{-1}(V')$, then o' is continuous over $U' \cap V'$, since paths in U' are ϕ of paths in U (ϕ is an sc-diffeomorphism) and the commutation runs backwards.

Layer 4: the morphism. For a morphism σ, $L = \mathrm{morIso}(\sigma)$ maps f to f `Chapter12/OrientationMorphismContinuity · EpStrongBundle.eqOn_morIso_of_functor` (functoriality, §XII.16).

- *The carried value at x'* `Chapter12/OrientationMorphismContinuity · EpStrongBundle.pullOrx_morIso_tgt`: for $\sigma : x \to x'$ and any orientation section o, the section carried by $\mathrm{morIso}(\sigma)$ has the value $\sigma_* o_x$ at x': both are the conjugation by morT and morK of $\mathrm{morIso}(\sigma)$ at x' (`morTx_eq`).
- *The carried value along the sheet* `Chapter12/OrientationGerm · EpStrongBundle.pullOrx_morIso_eq_orMor`: at every y' in $t(\mathrm{sheet})$, the carried value is $\sigma'_* o_{s(\sigma')}$ for the morphism $\sigma' = \tau(y')$ of the sheet ending at y', because the isomorphism of σ and that of σ' have the same germ at y' `Chapter12/OrientationGerm · EpStrongBundle.germEq_morIso_sheet` (the sheet of σ is a sheet through σ', and two sheets through σ' have the same local inverse of t near y') and the germ lemma applies.

Theorem 12.5.3, last sentence (narrower) `Chapter12/OrientationMorphismContinuity · EpStrongBundle.theorem_12_5_3_push`, p. 594. Let X be tame with bump functions on the objects, f an sc-Fredholm section functor with tame germ data, $\sigma : x \to x'$ a morphism between smooth objects, and o an orientation section continuous over an open $V \ni x$ (Part I's `IsContOrOn`). Then the section carried by $\mathrm{morIso}(\sigma)$ is continuous over the open set $t(\mathrm{sheet}) \cap \phi^{-1}(V)$, which contains x', and its value at x' is $\sigma_* o_x$. *Proof.* Continuity is Layer 3 with $L = \mathrm{morIso}(\sigma)$ and $f' = f$ `Chapter12/OrientationMorphismContinuity · EpStrongBundle.theorem_12_5_3_push_hyp`, the form with Part I's paracompactness instances; $x' \in t(\mathrm{sheet})$ and $\phi(x') = x \in V$; the value is the first item of Layer 4. The form without instance arguments discharges them for metrizable spaces. □

Readings. (a) The text's $\hat{\Phi}$ is read as the local strong bundle isomorphism of the sheet of σ (§XII.16), from the target side to the source side; carrying a section from near x to near x' is pulling it back along this isomorphism. (b) "Continuous local section" is read as Part I's continuity of orientations over an open set; its section $y \mapsto (y, o_y)$ is then continuous for Part I's topology (`continuousOn_orSec`), and the converse is §XII.20. *Narrower*: by the hypotheses of Theorem 6.6.17, inherited as in the topological part (G.140),

and the standing hypothesis, which bump functions give.

XII.20 Definition 12.5.4 (2017: 12.12): orientations, and the two readings of continuity

Definition 12.5.4 (same) `Chapter12/OrientationOrbit · EpStrongBundle.definition_12_5_4`, p. 594. Given a topology t on O_f, an *orientation* of f is a section $o : x \mapsto o_x \in \mathrm{Or}_x^f$ over the smooth objects such that $x \mapsto (x, o_x)$ is continuous $X_\infty \to O_f$ for t and $\varphi_* o_{s(\varphi)} = o_{t(\varphi)}$ for every morphism φ between smooth objects. The topology is a parameter; the text's is Part I's, the unique one of Theorem 12.5.3. The second clause alone is *invariance* `Chapter12/OrientationNaturality · EpStrongBundle.IsOrInvariant`, and the definition is continuity and invariance `Chapter12/OrientationNaturality · EpStrongBundle.definition_12_5_4_iff`.

Remark 12.5.5, p. 595, is prose pointing to Chapter 15 (orientations of solution sets) and is not formalized.

The two readings agree (extra). Part I has two notions of a continuous orientation of f over an open set: continuity of $x \mapsto (x, o_x)$ for the topology of O_f (`Continuous (orSec o)`), which is how Definition 12.5.4 says it, and propagation along every sc-smooth path (`IsContOrOn`). Part I proves that the second implies the first (`continuousOn_orSec`), and for Part I's own topology also the converse at points with two orientations (`continuous_orSec_iff` `Chapter6/OrientationBundle · continuous_orSec_iff`). The proofs of §12.5 use the second; the following makes the two interchangeable for every topology with the properties of Theorem 6.6.17.

Agreement is open `Chapter12/ContinuousOrientationSection · isOpen_setOf_orSec_eq`: if $x \mapsto (x, o_x)$ is continuous on $U \cap X_\infty$ and o'' is continuous in Part I's sense over an open $V \supset U$, then the set of smooth points of U where $o = o''$ is open in $U \cap X_\infty$. Indeed, a basic open set of the topology of O_f around (y, o''_y) consists of points (z, o''_z) of the continuous orientation o'', and its preimage under the continuous $z \mapsto (z, o_z)$ is a neighbourhood of y on which $o = o''$.

A continuous section is a continuous orientation `Chapter12/ContinuousOrientationSection · isContOrOn_univ_of_continuous_orSec`: for any local transport of orientations with exactly two orientations at every smooth point, on an M-polyfold whose smooth points have neighbourhoods with preconnected smooth part (`HasConnectedSmoothNbhds`), a section o with $x \mapsto (x, o_x)$ continuous on X_∞ is continuous in Part I's sense over all of X. *Proof.* Near a smooth x, Part I gives a continuous orientation o_1 over some $V \ni x$ with $o_1(x) = o(x)$ (`exists_isContOrOn`); the other orientation o'_1, pointwise the opposite of o_1, is continuous as well. On a neighbourhood $U \subset V$ of x with preconnected smooth part, the sets where $o = o_1$ and where $o = o'_1$ are open by the lemma, disjoint, and cover $U \cap X_\infty$ (two orientations at each point); the first contains x; by preconnectedness, $o = o_1$ on $U \cap X_\infty$. So o is continuous near every smooth point, and locality (Layer 2 of §XII.19) concludes. □ Neither hypothesis restricts: two orientations at every point hold under the standing hypothesis, and every M-polyfold has neighbourhoods with preconnected smooth part `Chapter6/OrientationMainClean · MPolyfold.hasConnectedSmoothNbhds` (Part VI).

The two readings agree (extra) `Chapter12/ContinuousOrientationSection · isContOrOn_univ_iff_continuous`, the form used for Theorem 6.6.17 and Definition 12.5.4: with Part I's hypotheses (tame, bump functions on the objects, tame germ data, sc⁺-sections through smooth points), for every topology t on O_f with the properties of Theorem 6.6.17, $x \mapsto (x, o_x)$ is t-continuous iff o is continuous in Part I's sense over all of X. *Proof.* By the uniqueness in Theorem 6.6.17, t is Part I's topology (`eq_orTopology_of`); then the forward direction is the lemma above and the backward direction is Part I's `continuousOn_orSec`. □ So

Definition 12.5.4 may be read with either notion of continuity, and the statements of §§XII.21–XII.22 are given in the text's.

XII.21 Proposition 12.5.6 (2017: 12.4): orientability through the orbit spaces

Proposition 12.5.6 (narrower), p. 595. The text: f is orientable iff $|\sigma| : |O_f| \to |X_\infty|$ is $(2:1)$ and has a continuous global section.

(□) (same) `Chapter12/OrientationOrbit · EpStrongBundle.proposition_12_5_6_mp`. If o is an orientation in the sense of Definition 12.5.4 (for any topology t on O_f), then $|\sigma|$ is $(2:1)$ `Chapter12/OrientationOrbit · EpStrongBundle.IsTwoToOne` (over every class $|x|$ exactly two classes) and $|x| \mapsto |(x, o_x)|$ is a continuous section of $|\sigma|$. *Proof. Two-to-one.* Over $|x|$ lie $|(x, o_x)|$ and $|(x, \bar{o}_x)|$, $\bar{o}_x$ the other orientation (§XII.18). They differ: if $\varphi : x \to x$ had $\varphi_* o_x = \bar{o}_x$, invariance would give $\varphi_* o_x = o_x$. Any class over $|x|$ is $|(y, \omega)|$ with a morphism $\psi : y \to x$; $\psi_* \omega$ is o_x or $\bar{o}_x$, so the class is one of the two. *The section* is well defined on $|X_\infty|$ because o is invariant, and continuous because $x \mapsto (x, o_x)$ is continuous and the quotient maps are continuous, $|X_\infty|$ carrying the quotient topology.

(□), the invariant lift (narrower) `Chapter12/OrientationOrbit · EpStrongBundle.proposition_12_5_6_mpr_invariant`. If $|\sigma|$ is $(2:1)$, every section s of $|\sigma|$ (continuous or not) lifts to an invariant section o of σ with $|(x, o_x)| = s(|x|)$. *Proof. Existence at each x.* $s(|x|) = |(y, \omega)|$ for some (y, ω) with $|y| = |x|$; a morphism $\varphi : y \to x$ gives $\varphi_* \omega$, with $|(x, \varphi_* \omega)| = s(|x|)$. *Uniqueness at each x.* If two orientations at x have the same class, they are equal `Chapter12/OrientationFromSection · EpStrongBundle.eq_of_quot_eq_of_isTwoToOne`: otherwise the two points of $\sigma^{-1}(x)$ would have the same class, and the fibre of $|\sigma|$ over $|x|$ would have one point. So o_x is determined. *Invariance.* For $\varphi : x \to x'$, $\varphi_* o_x$ has the class of o_x, which is $s(|x|) = s(|x'|)$, so $\varphi_* o_x = o_{x'}$ by uniqueness. □

The rest of (□) is continuity: for Part I's topology on O_f, if s is continuous, then $x \mapsto (x, o_x)$ is continuous.

The isotropy fixes orientations `Chapter12/OrientationFromSection · EpStrongBundle.orMor_isotropy_of_isTwoToOne`: if $|\sigma|$ is $(2:1)$, then $g_* \omega = \omega$ for $g \in G_x$ and $\omega \in \mathrm{Or}^f_x$, since $g_* \omega$ and ω have the same class. And $\sigma_* o_x$ depends only on σ, its source and target being read off it `Chapter12/OrientationFromSection · EpStrongBundle.orMor_eq_of_eq`.

Morphisms near a point fix a continuous orientation `Chapter12/OrientationFromSection · EpStrongBundle.exists_nhds_orMor_eq_hyp`. Let $|\sigma|$ be $(2:1)$ and $\hat{o}$ continuous over an open $U_0 \ni x_0$. There is an open $V \ni x_0$ in U_0 such that every morphism $\varphi : z \to y$ between smooth points of V has $\varphi_* \hat{o}_z = \hat{o}_y$. *Proof.* 1. A natural representation of G_{x_0} on $U(x_0) \subset U_0$ (Theorem 7.1.13): every morphism between points of $U(x_0)$ is $\Gamma(g, z)$ for a unique g, and $\Gamma(g, \cdot)$ is the local inverse of s on a sheet of g near x_0. 2. For each of the finitely many $g \in G_{x_0}$, the last sentence of Theorem 12.5.3 (§XII.19) carries $\hat{o}$ along the sheet of g to a section o^g, continuous over an open $V_g \ni x_0$, with $o^g_{x_0} = g_* \hat{o}_{x_0} = \hat{o}_{x_0}$, by the isotropy fact. 3. o^g and $\hat{o}$ are both continuous near x_0 and agree at x_0; two continuous orientations that agree at a point agree on a neighbourhood with preconnected smooth part (the argument of §XII.20: the agreement set and its complement are both open). Shrink V_g accordingly. 4. Let V be $U(x_0)$ intersected with all V_g and with the points z for which $\Gamma(g, z)$ lies in the sheet of g used in step 2. 5. For $\varphi : z \to y$ between smooth points of V: $\varphi = \Gamma(g, z)$ lies in the sheet of g, so by the second item of Layer 4 (§XII.19) the carried value at y is $\varphi_* \hat{o}_z$; it equals $o^g_y = \hat{o}_y$ by step 3.

Openness `Chapter12/OrientationFromSection · EpStrongBundle.isOpen_orSat_hyp`: for $\hat{o}$ continuous over an open V, the preimage in O_f of $\{|(z, \hat{o}_z)| : z \in V\}$ is open for Part I's topology. It is the set of $(y, \varphi_* \hat{o}_z)$ for morphisms $\varphi : z \to y$ with $z \in V$. Near such a point, the section carried along the sheet of φ is continuous (§XII.19), so its graph over a neighbourhood of y is a basic open set of Part I's topology; its points are $(y', \tau(y')_* \hat{o}_{s(\tau(y'))})$ with $s(\tau(y')) \in V$ near y, again of the given form.

(□) with continuity (narrower) `Chapter12/OrientationFromSection · EpStrongBundle.proposition_12_5_6_mpr_hyp`, p. 595. Let X be tame with bump functions on the objects, f an sc-Fredholm section functor with tame germ data. If $|\sigma|$ is $(2 : 1)$ and has a continuous global section s, for Part I's topology on O_f and the quotient topologies on $|O_f|$ and $|X_\infty|$, then f is orientable: there is an orientation o in the sense of Definition 12.5.4 with $|(x, o_x)| = s(|x|)$.

Proof. Let o be the invariant lift. It remains to show that $x \mapsto (x, o_x)$ is continuous at every smooth x_0. 1. Part I gives a continuous orientation $\hat{o}$ over an open $U_0 \ni x_0$ with $\hat{o}_{x_0} = o_{x_0}$ (`exists_isContOrOn`). 2. The lemma gives $V \ni x_0$ on which morphisms fix $\hat{o}$. 3. The set $A = \{|(z, \hat{o}_z)| : z \in V\}$ is open in $|O_f|$, by openness and the definition of the quotient topology, and contains $s(|x_0|) = |(x_0, o_{x_0})| = |(x_0, \hat{o}_{x_0})|$. 4. By continuity of s and of $X_\infty \to |X_\infty|$, every smooth y near x_0 (inside V) has $s(|y|) \in A$: $s(|y|) = |(w, \hat{o}_w)|$ for some $w \in V$. 5. Since also $s(|y|) = |(y, o_y)|$, some morphism $\varphi : w \to y$ has $\varphi_* \hat{o}_w = o_y$; by step 2, $\varphi_* \hat{o}_w = \hat{o}_y$. So $o_y = \hat{o}_y$ near x_0. 6. $\hat{o}$ is continuous in Part I's sense near x_0, so its section is continuous for Part I's topology (`continuousOn_orSec`), and so is that of o, which agrees with it near x_0. □

The form with Part I's instance arguments `Chapter12/OrientationFromSection · EpStrongBundle.proposition_12_5_6_mpr_hyp'` is the one proved; the form without them discharges the instances for metrizable spaces. *Narrower* only by Part I's hypotheses (G.140). The topology of O_f is Part I's, which Theorem 12.5.3 shows to be the only one with its properties, so the statement is the text's under those hypotheses.

XII.22 Theorem 12.5.7 (2017: 12.12): naturality of the orientation bundle

The setting. A generalized strong bundle isomorphism between strong bundles $W \to X$ and $W' \to X'$ over ep-groupoids is represented by a diagram $W \xleftarrow{\Phi_1} V \xrightarrow{\Phi_2} W'$ of strong bundle equivalences (Definition 10.4.1, Part X) over $\phi_1 : X'' \to X$, $\phi_2 : X'' \to X'$, together with sc-Fredholm section functors f, f', f'' with $f'' = \Phi_1^* f = \Phi_2^* f'$. The text writes f' as the push-forward of f; any f' with the same pull-back serves, and the push-forward is one `Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward_witness`: for every sc-Fredholm section functor f of W' and equivalence Φ, $\Phi^* f$ and f satisfy the hypothesis, and $\Phi^* f$ is sc-Fredholm (Theorem 10.4.5 `Chapter10/FredholmEquivalence · theorem_10_4_5_fred`).

Theorem 12.5.7 (narrower), pp. 595–597. (1) There is a homeomorphism $|O_f| \to |O_{f'}|$ covering the homeomorphism $|X_\infty| \to |X'_\infty|$ induced by the diagram. (2) An orientation o of f determines a unique orientation o' of f' corresponding to it over V. The text proves (1) and (2) for one representative and leaves to the reader that the maps do not depend on the representative and are functorial.

One equivalence

Let $\Phi : W \to W'$ be a strong bundle equivalence over $\phi : X \to X'$, with $f = \Phi^* f'$.

The local isomorphisms Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.locIso. By Definition 10.4.1 (2), every object a has an open $U(a) \ni a$ Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.locSet such that ϕ maps it sc-diffeomorphically onto an open $U'(a)$ Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.locSet' and Φ is a strong bundle isomorphism over it: a local strong bundle isomorphism $W|U(a) \to W'|U'(a)$ with base map ϕ and bundle map Φ. a is smooth iff $\phi(a)$ is Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.smooth_obj Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.smooth_of_obj. Near a, $f = \Psi \circ f' \circ \phi$ Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.eqOn_locIso Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.eqOn_loc, which is the pull-back formula.

The orientation bijection Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.orEqv. With the chosen chart point at $\phi(a)$ restricted to $U'(a)$ Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.locPt, the maps $T_{\phi(a)}X' \to T_aX$ and $W'_{\phi(a)} \to W_a$ of Φ^{-1}, read in the chosen chart points Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.eT Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.eK, are mmorT, mmorK of the local isomorphism Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.eT_eq Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.eK_eq. $\mathrm{Lin}(f, a)$ read in the carried chart point is $\mathrm{Lin}(f', \phi(a))$ Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.linSet_loc (§XII.16, `linSet_mpullPt`), and conjugation by eT, eK is a bijection $\mathrm{orEqv}_a : \mathrm{Or}^{f'}_{\phi(a)} \to \mathrm{Or}^{f}_{a}$ Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.orEqv_apply.

Naturality Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.orMor_orEqv: for a morphism $\alpha : a \to b$ between smooth objects, $\alpha_* \circ \mathrm{orEqv}_a = \mathrm{orEqv}_b \circ \phi(\alpha)_*$. *Proof.* (a) The composites $\Phi \circ \mathrm{morIso}(\alpha)$ and $\mathrm{morIso}(\phi(\alpha)) \circ \Phi$ agree near b Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.germAgree₄_mor: on the base because ϕ maps the sheet of α into the sheet of $\phi(\alpha)$ near α (ϕ is a functor and continuous on morphisms), on the bundle because $\Phi(\mu(\gamma, w)) = \mu'(\phi(\gamma), \Phi(w))$. (b) On the set of points where all these isomorphisms are defined Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.natSet, the four-term germ lemma of §XII.16 gives $T\alpha \circ \mathrm{eT}_a = \mathrm{eT}_b \circ T(\phi(\alpha))$ and its fibre form Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.morTx_eT Chapter12/OrientationEquivalenceStep · EpStrongBundle.IsStrongBundleEquivalence.morKx_eK. (c) Conjugations compose. The text calls these the canonical lifts (p. 597).

The orbit maps. ϕ on smooth objects induces $|\phi| : |X_\infty| \to |X'_\infty|$ (`smoothOrbitMap`), a bijection Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_bijective: surjective because an equivalence is essentially surjective and every smooth x' is isomorphic to $\phi(a)$ for a smooth a; injective because it is fully faithful. The map $(a, o) \mapsto (\phi(a), \mathrm{orEqv}_a^{-1} o)$ of $O_f \to O_{f'}$ (`orSpaceMap`) preserves the orbit relation, by naturality (`orRel_orSpaceMap`), and induces $|\Phi|_* : |O_f| \to |O_{f'}|$ (`orbitOrMap`), a bijection Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_bijective over $|\phi|$: $|\sigma'| \circ |\Phi|_* = |\phi| \circ |\sigma|$ Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.orbitProj_orbitOrMap.

Pull-back and push-forward of orientations. *Pull-back* (`pullOr`): $a \mapsto \mathrm{orEqv}_a(o'_{\phi(a)})$; it is invariant when o' is Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.isOrInvariant_pullOr, by naturality (and its inverse form, `orMor_orEqv_symm`). *Push-forward* Chap-

ter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.pushOr of an invariant o: at a smooth x', choose a smooth a and a morphism $\psi : \phi(a) \to x'$, and put $o'_{x'} = \psi_* \mathrm{orEqv}_a^{-1}(o_a)$. It does not depend on the choice Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.pushOr_eq: for two choices (a,ψ), (b,ψ'), $\psi'^{-1}\psi : \phi(a) \to \phi(b)$ is $\phi(\beta)$ for a unique $\beta : a \to b$ (full faithfulness), and $\beta_* o_a = o_b$ with naturality give the same value. At $\phi(a)$ it is $\mathrm{orEqv}_a^{-1}(o_a)$ Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.pushOr_obj, it is invariant Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.isOrInvariant_pushOr, it is the only invariant section corresponding to o at the points $\phi(a)$ Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.pushOr_unique (an invariant section is determined by its values at one point of each orbit), and pull-back undoes it Chapter12/OrientationNaturality · EpStrongBundle.IsStrongBundleEquivalence.pullOr_pushOr.

Continuity of pull-back and push-forward. By the germ lemma, orEqv_b is the orientation map of the local isomorphism at any a with $b \in U(a)$ Chapter12/OrientationGerm · EpStrongBundle.IsStrongBundleEquivalence.orEqv_eq_orxVia, since the local isomorphisms at a and at b have the same germ at b (both are ϕ and Φ) Chapter12/OrientationGerm · EpStrongBundle.IsStrongBundleEquivalence.germEq_locIso. So the pull-back of orientations is, on $U(a)$, the section carried by the local isomorphism at a (Layer 3 of §XII.19) Chapter12/OrientationEquivalenceContinuity · EpStrongBundle.IsStrongBundleEquivalence.pullOr_eq_pullOrx. - *Pull-back* Chapter12/OrientationEquivalenceContinuity · EpStrongBundle.IsStrongBundleEquivalence.isContOrOn_pullOr_hyp: if o' is continuous over an open V', then $\mathrm{pullOr}(o')$ is continuous over $\phi^{-1}(V')$: near each a by Layer 3, and locality joins the neighbourhoods. - *Push-forward* Chapter12/OrientationEquivalenceContinuity · EpStrongBundle.IsStrongBundleEquivalence.isContOrOn_pushOr_hyp: if o is invariant and continuous over all of X, then $\mathrm{pushOr}(o)$ is continuous over all of X'. Near a point $\phi(a)$: the pull-back of $\mathrm{pushOr}(o)$ is o, continuous, so $\mathrm{pushOr}(o)$ is continuous over $U'(a)$ by the converse of Layer 3's continuity Chapter12/OrientationEquivalenceContinuity · StrongBundle.LocalIso.isContOrOn_of_pullOrx: a section is continuous where the section it carries is, since an sc-smooth path in U' is the image under ϕ of its pull-back by the inverse of ϕ, and the commutation of Layer 3 applies to that pull-back. Near an arbitrary x': choose $\psi : \phi(a) \to x'$; the last sentence of Theorem 12.5.3 carries $\mathrm{pushOr}(o)$, continuous near $\phi(a)$, along the sheet of ψ to a section continuous near x' whose values are $\tau(y')_* \mathrm{pushOr}(o)_{s(\tau(y'))}$ (§XII.19, Layer 4), which are the values of $\mathrm{pushOr}(o)$ by invariance. Locality concludes.

The homeomorphisms of one equivalence. Topologies: Part I's on O_f and $O_{f'}$, Theorem 2.3.10's on X_∞, quotient topologies on the orbit spaces. - *A Part I fact* (extra) Chapter5/SmoothLevelContinuity · MPolyfold.IsScSmoothOn.continuousOn_level_comp Chapter5/SmoothLevelContinuity · MPolyfold.IsScSmoothOn.continuousOn_smoothLevel Chapter5/SmoothLevelContinuity · MPolyfold.IsScSmoothMap.continuous_smoothLevel: an sc-smooth map is continuous on every level X_m and on X_∞; the text uses it without comment. *Proof.* A chart is continuous on level m near every point of its domain Chapter5/SmoothLevelContinuity · MPolyfold.continuousAt_levelLift (the generating sets of the topology of X_m are chart preimages); in charts, $\psi \circ f \circ \varphi^{-1}$ is sc^0, hence continuous on level m; so f lifted to level m is continuous near every point; and X_∞ carries the initial topology of the levels. - *The smooth levels of a local sc-diffeomorphism* Chapter12/OrientationOrbitHomeo · MPolyfold.IsScDiffeoOn.smoothFwd Chapter12/OrientationOrbitHomeo · MPolyfold.IsScDiffeoOn.smoothBwd: for an sc-diffeomorphism $U \to U'$, the maps $U_\infty \to U'_\infty$ and back (extended by constants outside), continuous on the smooth parts of U, U' by the Part I fact; so the image of an open subset of U_∞ is open in X'_∞. - *ϕ on X_∞* is continuous and open Chapter12/OrientationOrbitHomeo · EpStrongBundle.IsStrongBundleEquivalence.continuous_smoothMap Chapter12/OrientationOrbitHomeo · EpStrongBundle.IsStrongBundleEquivalence.isOpenMap_smoothMap: on $U(a)_\infty$ it is the forward map of the local sc-diffeomorphism. - *The map of orientation spaces* is continuous and open

Chapter12/OrientationOrbitHomeo · EpStrongBundle.IsStrongBundleEquivalence.continuous_orSpaceMap Chapter12/OrientationOrbitHomeo · EpStrongBundle.IsStrongBundleEquivalence.isOpenMap_orSpaceMap. A basic open set of Part I's topology is the graph of a continuous local orientation over an open set. On the graph of a pulled-back continuous orientation, $(a, o) \mapsto (\phi(a), \mathrm{orEqv}_a^{-1} o)$ is $y \mapsto (\phi(y), o'_{\phi(y)})$ (`orSpaceMap_orSec_pullOr`); preimages and images of graphs are graphs of continuous orientations by the continuity of pull-back, and of the carried sections of Layer 3. - *Open orbit maps.* The saturation of an open subset of O_f is open Chapter12/OrientationOrbitHomeo · EpStrongBundle.isOpen_orSat: near a point $(y_0, \varphi_*\omega)$ with (x, ω) in the open set S, the continuous orientation through (x, ω) whose graph lies in S is carried along the sheet of φ (Theorem 12.5.3, last sentence) to one through $(y_0, \varphi_*\omega)$, whose graph lies in the saturation (`orMor_tgtInv_eq_pullOrx`). So $O_f \to |O_f|$ is open Chapter12/OrientationOrbitHomeo · EpStrongBundle.isOpenMap_quotMk_or; likewise $X_\infty \to |X_\infty|$ Chapter12/OrientationOrbitHomeo · EpGroupoid.isOpenMap_quotMk_smooth, since near $t(\varphi)$ the map $y \mapsto s(\tau(y))$ is continuous on the smooth level. - *Homeomorphisms* Chapter12/OrientationOrbitHomeo · EpStrongBundle.IsStrongBundleEquivalence.orbitOrHomeo Chapter12/OrientationOrbitHomeo · EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitHomeo: a bijection of quotients induced by a continuous open map between spaces whose quotient maps are open is continuous and open, hence a homeomorphism.

The diagram

Without topology. *(1)* Chapter12/OrientationNaturality · theorem_12_5_7_one_hyp: for the diagram, $F = |\Phi_2|_* \circ |\Phi_1|_*^{-1} : |O_f| \to |O_{f'}|$ and $G = |\phi_2| \circ |\phi_1|^{-1}$ are bijections with $|\sigma'| \circ F = G \circ |\sigma|$. *(2)* Chapter12/OrientationNaturality · theorem_12_5_7_two_invariant: an invariant section o of σ has exactly one invariant section o' of σ' with $\mathrm{orEqv}_v^{(2)}(o'_{\phi_2(v)}) = \mathrm{orEqv}_v^{(1)}(o_{\phi_1(v)})$ at every smooth object v of V: pull back along Φ_1, push forward along Φ_2, and uniqueness by `pushOr_unique`. These two statements are valid as stated and are the set-level halves of the theorem; the topology is added below.

With Part I's continuity (narrower) Chapter12/OrientationEquivalenceContinuity · theorem_12_5_7_two_hyp: with Part I's hypotheses on the three bases, if o is invariant and continuous over all of X in Part I's sense, then the unique invariant o' of (2) is continuous over all of X': the pull-back along Φ_1 is continuous, and so is its push-forward along Φ_2. The form without instance arguments Chapter12/OrientationEquivalenceContinuity · theorem_12_5_7_two_cont.

Theorem 12.5.7 (1) (narrower) Chapter12/OrientationTransferTheorem · theorem_12_5_7_one_homeo_hyp: with Part I's hypotheses on X, X' and the base of V, there are homeomorphisms $F : |O_f| \to |O_{f'}|$ and $G : |X_\infty| \to |X'_\infty|$ with $|\sigma'| \circ F = G \circ |\sigma|$, and they are the maps of the diagram: $F \circ |\Phi_1|_* = |\Phi_2|_*$, $G \circ |\phi_1| = |\phi_2|$. *Proof.* The homeomorphisms of the two legs, one inverted, composed. The form without instance arguments Chapter12/OrientationTransferTheorem · theorem_12_5_7_one_homeo.

Theorem 12.5.7 (2), as printed (narrower) Chapter12/OrientationTransferTheorem · theorem_12_5_7_two_book_hyp: if o is an orientation of f in the sense of Definition 12.5.4 (continuous section of σ for Part I's topology, invariant), there is exactly one o' that is an orientation of f' in the sense of Definition 12.5.4 and corresponds to o over V. *Proof.* Part I's comparison `continuous_orSec_iff` turns the continuity of Definition 12.5.4 into Part I's continuity over all of X and back, at points with two orientations, which the standing hypothesis gives (§XII.20 proves the same for every topology of Theorem 6.6.17); then the previous statement. The form without instance arguments Chapter12/OrientationTransferTheorem · theorem_12_5_7_two_book.

Narrower, in (1) and (2): by Part I's hypotheses on X, X' and the base of V (tame, bump functions, tame

germ data) and the standing hypothesis, inherited as in Theorem 12.5.3 (G.140); and by being stated for one representative's data, not for Chapter 10's classes (G.192).

Independence of the representative, and functoriality

Both are left to the reader in the text. They are proved for the data of diagrams (equivalences, natural transformations of strong bundle functors, Definition 10.4.8, and section functors on the middles) and need none of Part I's hypotheses, since they are statements about the bijections.

Natural equivalences act through morphisms. Let $\Phi, \Psi : W \to W'$ be strong bundle equivalences over ϕ, ψ with $f = \Phi^* f' = \Psi^* f'$, and $T : \Phi \Rightarrow \Psi$ a natural transformation covering $\tau : \phi \Rightarrow \psi$; its object map is continuous `Chapter12/OrientationEquivalenceNatural · EpStrongBundle.IsStrongBundleEquivalence.continuous_natApp`. *Germs* `Chapter12/OrientationEquivalenceNatural · EpStrongBundle.IsStrongBundleEquivalence.germ_nat`: near a, $\mathrm{morIso}(\tau(a)) \circ \Psi = \Phi$ on base and bundle: on the base because $\tau(y)$ lies in the sheet of $\tau(a)$ for y near a, so $s(\tau_{\mathrm{sheet}}^{-1}(\psi(y))) = \phi(y)$; on the bundle because $T(w) = (\tau(P(w)), \Phi(w))$ is a morphism from $\Phi(w)$ to $\Psi(w)$, i.e. $\Psi(w) = \mu'(\tau(Pw), \Phi(w))$. On the comparison set `Chapter12/OrientationEquivalenceNatural · EpStrongBundle.IsStrongBundleEquivalence.natSet'` the germ lemma gives $\mathrm{eT}_\Psi(a) \circ T(\tau(a)) = \mathrm{eT}_\Phi(a)$ and the fibre form `Chapter12/OrientationEquivalenceNatural · EpStrongBundle.IsStrongBundleEquivalence.eT_nat` `Chapter12/OrientationEquivalenceNatural · EpStrongBundle.IsStrongBundleEquivalence.eK_nat`, hence `Chapter12/OrientationEquivalenceNatural · EpStrongBundle.IsStrongBundleEquivalence.orEqv_nat`

$$\mathrm{orEqv}_\Psi(a) \circ \tau(a)_* = \mathrm{orEqv}_\Phi(a).$$

On orbits, (a, o) goes to $(\phi(a), \ldots)$ and $(\psi(a), \ldots)$, related by $\tau(a)$ (`orRel_orSpaceMap_nat`), so $|\Phi|_* = |\Psi|_*$ `Chapter12/OrientationEquivalenceNatural · EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_nat` and $|\phi| = |\psi|$ `Chapter12/OrientationEquivalenceNatural · EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_nat`.

Composition. For equivalences $\Phi : W \to W'$ over ϕ and $\Psi : W' \to W''$ over ψ with $f = \Phi^* f'$, $f' = \Psi^* f''$: $f = (\Psi \circ \Phi)^* f''$ `Chapter12/OrientationEquivalenceComp · EpStrongBundle.IsStrongBundleEquivalence.pullback_comp_of` (Proposition 10.4.4). Near a, the local isomorphism of $\Psi \circ \Phi$ is the composite of those of Ψ at $\phi(a)$ and Φ at a (`germ_comp`, on the comparison set `compLocSet`). A three-term form of the transfer lemma `Chapter12/OrientationEquivalenceComp · StrongBundle.LocalIso.transferT_mpullPt₃` `Chapter12/OrientationEquivalenceComp · StrongBundle.LocalIso.transferK_mpullPt₃` (a chart point carried along L then L_2 and one carried along M agree to first order when $L \circ L_2 = M$ near the point) gives $\mathrm{eT}_{\Psi\Phi}(a) = \mathrm{eT}_\Phi(a) \circ \mathrm{eT}_\Psi(\phi a)$ and the fibre form `Chapter12/OrientationEquivalenceComp · EpStrongBundle.IsStrongBundleEquivalence.eT_comp` `Chapter12/OrientationEquivalenceComp · EpStrongBundle.IsStrongBundleEquivalence.eK_comp`, hence $\mathrm{orEqv}_{\Psi\Phi}(a) = \mathrm{orEqv}_\Phi(a) \circ \mathrm{orEqv}_\Psi(\phi a)$ `Chapter12/OrientationEquivalenceComp · EpStrongBundle.IsStrongBundleEquivalence.orEqv_comp`, and on orbits $|\Psi\Phi|_* = |\Psi|_* \circ |\Phi|_*$ `Chapter12/OrientationEquivalenceComp · EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_comp`, $|\psi\phi| = |\psi| \circ |\phi|$ `Chapter12/OrientationEquivalenceComp · EpStrongBundle.IsStrongBundleEquivalence.smoothOrbitMap_comp`.

Independence of the representative (restated). *Refinement* `Chapter12/OrientationRepresentativeIndep · EpStrongBundle.IsStrongBundleEquivalence.orbitOrMap_refine`: if H is an equivalence and $\Phi \circ H \simeq K$ (a natural equivalence), then $|K|_* = |\Phi|_* \circ |H|_*$, by composition and the natural-equivalence lemma; on bases likewise (`smoothOrbitMap_refine`). So a diagram and its refinement through H (legs $\Phi_1 \circ H \simeq K_1$, $\Phi_2 \circ H \simeq K_2$)

give the same map $|K_2|_* \circ |K_1|_*^{-1} = |\Phi_2|_* \circ |\Phi_1|_*^{-1}$ Chapter12/OrientationRepresentativeIndep · EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_refinement Chapter12/OrientationRepresentativeIndep · EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_refinement_base. Two diagrams with a common refinement, which is the relation `SDiagram.Equiv` of Chapter 10's localization, give the same maps $|O_f| \to |O_{f'}|$ and $|X_\infty| \to |X'_\infty|$ Chapter12/OrientationRepresentativeIndep · EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_rep_indep Chapter12/OrientationRepresentativeIndep · EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_rep_indep_base; the homeomorphisms of (1), having the same underlying maps, agree as well.

Functoriality (restated) Chapter12/OrientationRepresentativeIndep · EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_functorial Chapter12/OrientationRepresentativeIndep · EpStrongBundle.IsStrongBundleEquivalence.theorem_12_5_7_functorial_base: for diagrams $W \leftarrow V_1 \to W'$ and $W' \leftarrow V_2 \to W''$, a middle V_3 with equivalences $\Pi_1 : V_3 \to V_1$, $\Pi_2 : V_3 \to V_2$ and a natural equivalence between the two composites to W' (the shape of Chapter 10's weak fibred product), the map of the composite diagram $W \leftarrow V_3 \to W''$ is the composite of the two maps. *Proof.* Composition turns each leg of the composite into a composite of legs, and the natural equivalence identifies the two routes through W'.

Restated: the independence is stated through refinements and the composite through its data, rather than for Chapter 10's classes `SDiagram`, `GenSBMap` and their composition; feeding those in is gap G.192 (§XII.24).

XII.23 Witnesses and the hypotheses

This section lists, for the whole chapter, every hypothesis that a formal statement carries beyond the text's, whether it is added, made explicit or inherited, and its witness. It also lists the text's hypotheses that the proofs do not use, and a witness for every definition. Degenerate witnesses, those whose boundary data are empty because $\partial X = \emptyset$ or whose construction is solved by its own input, are marked as such, with the non-degenerate case named where one exists.

Hypotheses of §12.2

Theorem 12.2.2 (§XII.3). None added. The paracompactness of $|X|$ is the text's. The good neighbourhood property (5) of Theorem 12.2.6, which the text invokes for the transport, is not used: the transport is well defined by invariance under G_x, two morphisms from y into $U(x)$ differing by a unique element of G_x. Witness: the theorem itself gives an auxiliary norm on every strong bundle over an ep-groupoid with paracompact $|X|$.

Theorem 12.2.7 (§XII.6). None added to the book statement Chapter12/AuxNormLocalExtension · EpStrongBundle.hasLocalExtensions_of_isTame: for every tame X with paracompact $|X|$ an auxiliary norm of $W|\partial X$ has local extensions, and the global part then gives the extension. The global part Chapter12/AuxNormExtension · EpStrongBundle.theorem_12_2_7_hyp takes the local extensions as a hypothesis and needs no tameness; its witness Chapter12/AuxNormExtension · EpStrongBundle.theorem_12_2_7_hyp_witness is degenerate in the second sense: the local extension is the global norm of Theorem 12.2.2 restricted Chapter12/AuxNormExtension · EpStrongBundle.hasLocalExtensions_of_isAuxNorm. The non-degenerate case is the book statement itself, whose local step builds the norm N' from a continuous retraction onto the boundary in a tame chart Chapter12/AuxNormLocalExtension · StrongBundle.exists_localExtension_of_isTame. *Kept and unused.* Tameness in the global part; the local step uses only a tame chart of the base, the fibre formula being evaluated in any strong bundle chart of the atlas.

Theorems 12.2.11 and Remark 12.2.12 (§XII.8). None added. Reflexive fibres and paracompactness of $|X|$ are the text's. Witness: the theorem itself, for every bundle with reflexive fibres and paracompact $|X|$.

Theorem 12.2.13 (§XII.9). None added; the paracompactness of $|X|$ is made explicit. The text's proof takes it from Theorem 12.2.11, and Theorem 12.2.7, of which Theorem 12.2.13 is the reflexive variant, states it. Witness of all hypotheses together `Chapter12/ReflexiveLocalExtension · TrivialBundle.theorem_12_2_13_witness`: degenerate, $\partial X = \emptyset$ for the project's $\mathbb{R}^n$ (`rnPolyfold`, a single chart onto all of $\mathbb{R}^n$), so the boundary norm is empty data. The book statement adds nothing to the text's hypotheses, so the witness serves only as a consistency check; a non-degenerate case would take the quadrant $[0,\infty)\oplus W$ of Proposition 12.3.7's witness, seen as a trivial ep-groupoid, with a reflexive norm over its boundary restricted from one of Theorem 12.2.11. The global part `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.theorem_12_2_13_hyp` takes the reflexive local extensions as a hypothesis; for tame X it is discharged `Chapter12/ReflexiveLocalExtension · EpStrongBundle.hasReflexiveLocalExtensions_of_isTame` `Chapter12/ReflexiveLocalExtension · StrongBundle.exists_reflexiveLocalExtension_of_isTame`. Its witnesses are degenerate: the restriction of a global reflexive norm `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.hasReflexiveLocalExtensions_of_isReflexiveAuxNorm` `Chapter12/ReflexiveAuxNormExtension · EpStrongBundle.theorem_12_2_13_hyp_witness` (the extension is its own input), and the trivial bundle over $\mathbb{R}^n$ `Chapter12/ReflexiveAuxNormExtension · TrivialBundle.epBundle_reflexive_witness` ($\partial X = \emptyset$).

Hypotheses of §12.3

Theorem 12.3.2 and Theorem 12.3.3 (§§XII.10–XII.11). (2) of Theorem 12.3.2 and Theorem 12.3.3 take ep bump functions in the pointwise reading: for every object x and open $U \ni x$, an sc-smooth invariant function supported in the saturation of U and nonzero at x. This is the project's reading of Part I's Definition 5.3.1 (Part V), not an added hypothesis. Witnesses: the trivial ep-groupoid on $\mathbb{R}^n$ `Chapter12/BumpEp · EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`, and every ep-groupoid whose object M-polyfold admits pointwise bump functions and $|X|$ is paracompact (Theorem 12.3.2 (2) itself). The first is degenerate in a third sense, which does not concern the boundary: its isotropy groups are trivial, so the averaging step of the construction is the identity. The second is the general case. Theorem 12.3.2 (1) is restated as two equivalences, one per reading. *Kept and unused.* Paracompactness of $|X|$ in Theorem 12.3.3, where it is used only through the pointwise bump functions that the hypothesis already provides.

Definition 12.3.6, Proposition 12.3.7 and Corollary 12.3.8 (§XII.12). Corollary 12.3.8 makes explicit two things that the text leaves implicit: the finite group fixes x_0, and "acts with a lift" is a structure, a group action on the retract and a linear lift to the fibres compatible with the bundle structure. Witness of the local bundle action: the trivial action of any finite group. Witness of Definition 12.3.6 and Proposition 12.3.7, non-degenerate `Chapter12/LocalBoundaryExtension · proposition_12_3_7_nonvacuous`: the partial quadrant $[0,\infty)\oplus W$ with the identity strong bundle retraction, the point 0 on the boundary, and a section of the boundary that is sc$^+$ on the face. The zero section gives a witness on every tame retract `Chapter12/LocalBoundaryExtension · isBoundaryScPlusSectionOn_zero`. *Needs less than the text.* Proposition 12.3.7 uses only the face germs at x_0, so the proposition holds for sections that are sc$^+$ on each face near x_0 without a compatibility condition along the corners.

Theorem 12.3.9 (§XII.13). None added. The bump functions on the objects that the proof takes from Theorem 12.3.2 (2) come from the partitions of unity that the theorem assumes. Witness of all hypotheses together `Chapter12/ScPlusExtension · EpStrongBundle.theorem_12_3_9_hypotheses_rn`: degenerate, $\partial X = \emptyset$. With boundary the data are inhabited on every X (the zero section functor of $W|\partial X$), and the local construc-

tion has the non-degenerate witness of Proposition 12.3.7. *Kept and unused.* Paracompactness of $|X|$, since partitions of unity are assumed. The second neighbourhood V' of the text's bound is not needed.

Hypotheses of §12.4

Theorem 12.4.2 (§XII.14). None added. *Kept and unused.* Paracompactness of $|X|$.

Proposition 12.4.5 and Theorem 12.4.6 (§XII.15). None added. *Kept and unused* in the book's forms: reflexive fibres, sc-smoothness of the section functor and the regularizing property in Proposition 12.4.5, which uses only continuity on level 0 and the compactness properties; tameness and paracompactness in Theorem 12.4.6. Witness of Definition 12.4.3: every section functor with Part I's reflexive local compactness property (Definition 5.2.6) on the object bundle, by Proposition 12.4.5.

Hypotheses of §12.5

§12.5 carries two kinds of hypotheses, both inherited from Part I and both part of gap G.140.

Sc⁺-sections through smooth fibre points (`hsec`). Lemma 12.5.1, the action φ_* on orientations, Theorem 12.5.3, Proposition 12.5.6 and Theorem 12.5.7 (for the three bundles of the diagram) assume that for every smooth point w of a fibre over a smooth object there is an sc⁺-section through w. Part I uses this to make $\mathrm{Lin}(f,x)$ nonempty; the text uses $\mathrm{Lin}(f,x)$ without comment. Witness: bump functions on the objects give it `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`. In the tame case with partitions of unity it would also follow from Theorem 12.3.9, but that adds partitions of unity, which is no improvement.

Part I's hypotheses of Theorem 6.6.17. Theorem 12.5.3, Proposition 12.5.6 (□), the continuity in Theorem 12.5.7 and its homeomorphisms assume, for X (and X' and the base of V in Theorem 12.5.7): X tame, bump functions on the objects, tame germ data for f. These are exactly the hypotheses under which Part I builds the topology of O_f and propagates orientations along paths; the text's §12.5 assumes only a tame X. Witnesses: those of Theorem 6.6.17 in Part VI, with the bump function witness above. Part I's paracompactness instances for its transports hold for metrizable spaces and are discharged inside the forms without instance arguments `Chapter12/OrientationMorphismContinuity · EpStrongBundle.theorem_12_5_3_push` `Chapter12/OrientationFromSection · EpStrongBundle.proposition_12_5_6_mpr_hyp` `Chapter12/OrientationEquivalenceContinuity · theorem_12_5_7_two_hyp`.

Hypotheses of step lemmas, not of book statements. - $f = \Psi \circ f' \circ \phi$ on U: the text's "$\hat{\Phi}$ maps f to f'". It is discharged for the isomorphisms of morphisms by functoriality `Chapter12/OrientationMorphismContinuity · EpStrongBundle.eqOn_morIso_of_functor` and for equivalences by the pull-back formula; in general it holds for f defined by the formula. - The chart, the open set and the cut-off in the comparison of push-forwards (§XII.19, Layer 1): proof data, constructed in Layer 3. - Continuity hypotheses on orientations: satisfiable by Part I's existence of continuous local orientations near every smooth point. - In the comparison of the two readings of continuity (§XII.20), the general form assumes two orientations at every smooth point and neighbourhoods with preconnected smooth part. Both hold in every M-polyfold under §12.5's standing hypothesis `Chapter6/OrientationMainClean · MPolyfold.hasConnectedSmoothNbhds`, so the specialized form `Chapter12/ContinuousOrientationSection · isContOrOn_univ_iff_continuous` has only Part I's hypotheses.

No hypothesis of Chapter 12 quantifies over all charts or all presentations in a way that would make it unsatisfiable. Definition 12.3.6 asks for a property in every tame chart and presentation; the zero section

and the quadrant witness satisfy it.

The independence of the representative and the functoriality of Theorem 12.5.7 need none of these hypotheses: they are statements about the bijections of orbit spaces, proved for the data of the diagrams.

Superseded forms that remain as component statements

Several statements of the formalization are halves of a book statement, valid as stated, used in the proof of the full form, and kept. Each is described once, where it is used: - the global parts of Theorems 12.2.7 and 12.2.13, with the local extensions as a hypothesis (§§XII.6, XII.9); - the topological part of Theorem 12.5.3, without its last sentence `Chapter12/OrientationOrbit · EpStrongBundle.theorem_12_5_3_hyp` (§XII.19); - the invariant lift of Proposition 12.5.6 (□), without continuity (§XII.21); - Theorem 12.5.7 (1) and (2) without topology `Chapter12/OrientationNaturality · theorem_12_5_7_one_hyp`, and (2) with Part I's continuity in place of Definition 12.5.4's (§XII.22).

A witness for every definition

Definition	Witness	Kind
sc-Fredholm section functor (Def 12.1.1)	every Part I sc-Fredholm section of a trivial ep-groupoid	general
auxiliary norm (Def 12.2.1)	Theorem 12.2.2, every bundle with paracompact $\lvert X\rvert$	general
norm over ∂X with local extensions	restriction of a global norm	degenerate (solved by its input); the book form discharges it for tame X
reflexive auxiliary norm (Def 12.2.10)	Theorem 12.2.11	general
mixed convergence (Def 12.2.8)	constant sequences, and sequences converging on level 1	general
ep bump function, pointwise (Def 12.3.1)	trivial ep-groupoid on $\mathbb{R}^n$; Theorem 12.3.2 (2)	trivial isotropy; general
sc$^+$-section over the boundary, local (Def 12.3.6)	quadrant $[0,\infty)\oplus W$	non-degenerate
sc$^+$-section functor of $W\lvert\partial X$ (Def 12.3.6)	the zero section functor	general, data only
local bundle action (Cor 12.3.8)	the trivial action	general
reflexive local compactness (Def 12.4.3)	Part I's Definition 5.2.6 on the object bundle	general
local isomorphism of a morphism	every morphism of an ep-groupoid	general
orientation bundle O_f (Def 12.5.2)	Part I's O_f under `hsec`	general
continuous local orientation	Part I's, near every smooth point (Theorem 6.6.17)	general
germ equality of local isomorphisms	reflexivity	general

Definition	Witness	Kind
equivalence data of Theorem 12.5.7	the identity diagram; `pullback_pushforward_witness`	general

XII.24 What is not formalized

Gap G.140: §12.5 with a tame X only. Theorem 12.5.3 with its last sentence, Proposition 12.5.6 (□) and the continuity and homeomorphisms of Theorem 12.5.7 carry the hypotheses of Part I's Theorem 6.6.17 (bump functions on the objects, tame germ data). Lemma 12.5.1, Proposition 12.5.6 and Theorem 12.5.7 carry the hypothesis that sc⁺-sections pass through smooth fibre points. The text assumes only that X is tame. These hypotheses are inherited from Part I, and Chapter 12 adds nothing to them. Removing them is Part I work: - Part I's propagation of orientations along sc-smooth paths would have to be built without bump functions on the objects; Part I uses them to construct the stabilizing sc⁺-sections of its push-forward. - The tame germ data would have to be derived from the tameness of X and the Fredholm property; Part I takes it as a separate hypothesis. - `hsec` would then follow, in the tame case, from the bump functions or from Theorem 12.3.9 with partitions of unity.

Where the hypotheses enter. `hsec` enters through Part I's $\mathrm{Lin}(f, x)$: without it the set of linearizations at a point may be empty, and the action of a morphism on orientations (§XII.17) is a map between possibly empty sets. Bump functions and tame germ data enter through Part I's Theorem 6.6.17 only: the topology of O_f, the existence of a continuous local orientation through every point of O_f, and the propagation along sc-smooth paths that defines continuity. The germ lemma, Lemma 12.5.1, the invariant lift of Proposition 12.5.6, the set-level halves of Theorem 12.5.7, and the independence and functoriality statements do not use Theorem 6.6.17's hypotheses; they carry only `hsec`, or nothing. So a Part I theorem without these hypotheses would remove them from Theorem 12.5.3, Proposition 12.5.6 (□) and the continuity in Theorem 12.5.7 without changing their proofs.

No line estimate is recorded for G.140. Its size is that of the corresponding Part I work, and Chapter 12's statements would follow from the Part I theorem by dropping the hypotheses, since every proof of §12.5 uses them only through Part I's lemmas.

Gap G.192: Theorem 12.5.7 for Chapter 10's classes of diagrams (`SDiagram`, `GenSBMap`). Everything is proved for the data of the diagrams (§XII.22). Two pieces are missing. - (a) *The restatement for a diagram of Chapter 10*, about 80–150 lines. Both legs are in the class of equivalences; the section functors on the middle are the pull-backs. Chapter 10's equivalence of diagrams, a common refinement with a natural equivalence, is unpacked into the independence statement. The result is a map on generalized strong bundle isomorphisms. - (b) *Chapter 10's composite diagram fed into the functoriality statement*, about 70–150 lines. This needs two facts: the right leg of the composite is an equivalence when the given one is, and the cone of the weak fibred product supplies the natural equivalence of the two composites.

Together about 150–300 lines. Neither part involves orientation theory; both are bookkeeping between Chapter 10's localization and the statements of §XII.22.

What (a) involves, step by step. A generalized strong bundle isomorphism is a class of diagrams under Chapter 10's equivalence. To define the map $|O_f| \to |O_{f'}|$ of a class, one picks a representative diagram, checks that its legs are strong bundle equivalences in the sense used by §XII.22 (the class of equivalences of Chapter 10 is defined through Definition 10.4.1, so this is an unfolding), and takes $|\Phi_2|_* \circ |\Phi_1|_*^{-1}$. The

section functor on the middle is not part of Chapter 10's data; it is $\Phi_1^* f$, and the hypothesis $\Phi_1^* f = \Phi_2^* f'$ is the condition that the class maps f to f'. Two representatives related by Chapter 10's equivalence have a common refinement with natural equivalences on both legs, which is the hypothesis of `theorem_12_5_7_rep_indep`; so the map is well defined on the class. The homeomorphisms of (1) have the same underlying maps, so they too descend. Nothing new is proved in (a); its size comes from unpacking the quotient and the existential data of Chapter 10's relation.

What (b) involves. Chapter 10 composes two classes through the weak fibred product of the two middles over the shared bundle. Its legs are composites of a projection of the weak fibred product with a leg of a given diagram. That the projection is an equivalence when the opposite leg is, is a statement of Chapter 10 about weak fibred products of equivalences, needed here in the strong bundle form. The cone of the weak fibred product supplies the natural equivalence between the two routes to the shared bundle, which is the hypothesis of `theorem_12_5_7_functorial`. With these two facts the composite's map is the composite of the maps, and functoriality on classes follows from (a).

A reading, not formalization work: Theorem 12.3.2 (1) in the printed pairing. Theorem 12.3.2 (1) says that an ep-groupoid admits sc-smooth bump functions iff its object M-polyfold does. With Definition 12.3.1 as printed (functions "not identically" zero) on both sides, and with the pointwise reading on both sides, it is proved (§XII.10). Pairing Definition 12.3.1 as printed for the ep-groupoid with Part I's pointwise bump functions on the objects, its (□) is Part I's open question whether weak bump functions give pointwise ones (Part V). This is a question about the reading of Definition 5.3.1, which Chapter 12 inherits; it needs no Chapter 12 formalization.

Remark 12.5.5 is prose. It comments on the choices behind Definition 12.5.4 and makes no claim that is not formalized elsewhere in §12.5.

No numbered item is unformalized. Every definition, lemma, proposition, theorem, corollary and remark of Chapter 12 has a formal counterpart, as listed in the Map; no statement was found not to hold as we read it.

XII.25 What the formalization taught us

One pattern carries the chapter. Chapter 12 is Part I seen through the morphisms, and most of its formal content is one pattern used again and again: build an object on a neighbourhood $U(x)$ with the natural action of G_x, average it over G_x, transport it to the saturation by a choice of morphisms, and glue with a partition of unity of $|X|$. Norms (Theorems 12.2.2, 12.2.7), reflexive norms (Theorems 12.2.11, 12.2.13), bump functions (Theorem 12.3.2) and sc$^+$-sections (Theorems 12.3.3, 12.3.9, Remark 12.3.5) all follow it. Formalizing the pattern once, as averaging, transport and gluing lemmas with the property to be preserved as a parameter, made each of these theorems a matter of checking that its property survives the three steps.

Invariance, not property (5), makes the transport well defined. The text invokes the good neighbourhood property (5) of Theorem 12.2.6 for the transport. What the proof needs is that the averaged object on $U(x)$ is invariant under G_x: two morphisms from y into $U(x)$ differ by a unique element of G_x, so the transported value does not depend on the choice. Property (5) is proved (§XII.5) but plays no role in the constructions.

The average of a square. The average of a bump function over a finite group can vanish identically: positive and negative values may cancel. The average of its square cannot, since it is nonnegative and

positive where the function is nonzero. The text's functions are nonnegative anyway, so this changes nothing in the text's argument, but the formal construction squares before averaging (§XII.10).

Completeness comes free. Completeness of glued fibre norms is not a separate check. A continuous fibrewise seminorm satisfying condition (2) of an auxiliary norm is squeezed, near every point, between two multiples of the chart norm, so completeness follows (§XII.2). This lemma is Part I's as much as Chapter 12's, and it removes a step from each of the four norm theorems.

Morphisms preserve mixed convergence. For the reflexive variant one fact carries the pattern: morphisms preserve mixed convergence, because they act boundedly on the fibres and continuously on level 0, and a bounded, level-0 convergent sequence in a reflexive fibre is mixed convergent. This is a stronger form of Remark 12.2.9 (§XII.7). With it, averaging, transport and gluing all keep lower semicontinuity. The gluing step needs one more observation: near a point only finitely many terms of the partition of unity matter, and the others are nonnegative.

The inspection. The local step of Theorem 12.2.13, which the text settles by inspection, rests on the same fact used once more inside one chart. The fibre maps $A(u)$ of the retraction are uniformly bounded near a point and continuous on level 0. So $A(u_k)h_k$ is bounded and convergent in F_0, hence weakly convergent in the reflexive F_1, and the norm N' of the local step is lower semicontinuous under mixed convergence (§XII.9).

Inclusion–exclusion uses only the face germs. The inclusion–exclusion formula of Proposition 12.3.7 restricts to s on the boundary because an alternating sum over subsets collapses when one index is already on its face. Written out, the face germs at x_0 are all that is used. So the proposition holds in a stronger form, without compatibility along the corners, and Part I's ∂-extension theorem is assembled from it (Part V).

Less than the tame local model. The local step of Theorem 12.2.7 needs less than the text's tame local model. The retraction onto the boundary is built in a tame chart of the base. The fibre formula can be evaluated in any strong bundle chart of the atlas, since only three properties of the retraction enter: it is continuous, its values lie in ∂X, and it is the identity on ∂X.

Hypotheses that are not used. - Paracompactness of $|X|$ in Theorems 12.3.3, 12.3.9 (where partitions of unity are assumed), 12.4.2 and 12.4.6. - Tameness in the global part of Theorem 12.2.7 and in Theorem 12.4.6. - Reflexivity, sc-smoothness and the regularizing property in Proposition 12.4.5. - The second neighbourhood V' in the bound of Theorem 12.3.9.

Theorem 12.3.9 uses bump functions on the objects, which the text takes from Theorem 12.3.2 (2); they come from the assumed partitions of unity, so no hypothesis is added.

A hypothesis that is used but not printed. Theorem 12.2.13 needs paracompactness of $|X|$. Its proof takes it from Theorem 12.2.11, and Theorem 12.2.7, its non-reflexive counterpart, states it. The formal statement makes it explicit (§XII.9).

The labels of Theorem 12.4.2. The proof of Theorem 12.4.2 labels its three implications in an order that its arguments presumably permute. The formal proof follows the arguments: each implication is proved by the argument that establishes it (§XII.14).

A lifting argument in Proposition 12.4.5. The step (3)$\square$ (1) of Proposition 12.4.5, "easily obtained" in the text, needs a lifting argument through the properness of t. If the classes $\pi(u_k)$ converge to $\pi(q)$ in $|X|$, there are objects v_k in the same classes with $v_k \to q$: along a decreasing neighbourhood basis V_n of q, v_k is chosen in the basic set of the largest index already reached by $\pi(u_k)$, using that π is open.

The properness of t then gives a convergent subsequence of the morphisms $a_k \to v_k$, and with it of the sources (§XII.15).

The obvious lemma. In §12.5 the text's "obvious" Lemma 12.5.1 took about 1,100 lines. A morphism acts on $\mathrm{Lin}(f, x)$ through a local strong bundle isomorphism built from its sheet. Part I's linearizations live in chart points, so they have to be pulled back along it and compared. Theorem 12.5.7 repeats this between two different bundles (§§XII.16–XII.17, XII.22).

Three ingredients of orientation transport. The orientation bundle is Part I's, and Part I's Theorem 6.6.17 supplies its topology. What §12.5 adds is one statement, used three times (Theorem 12.5.3's last sentence, Proposition 12.5.6 (□), Theorem 12.5.7 (2)): Part I's propagation of orientations along paths agrees with the action of local strong bundle isomorphisms. It took about 2,100 lines and three ingredients the text does not mention. 1. *The stabilizing sc⁺-section is carried along.* The stabilizing sc⁺-section of Part I's push-forward must itself be carried along the isomorphism. This needs a cut-off, and then stabilized kernels, orientation rays and push-forwards correspond term by term (§XII.19, Layer 1). 2. *Continuity is local.* Part I never states that continuity of an orientation is a local property. It is, by a dyadic induction on sc-smooth paths with a Lebesgue number of the cover by chart domains, and this locality lemma turns the chart-level comparison into continuity (§XII.19, Layer 2). 3. *The germ lemma.* The orientation map of a local isomorphism depends only on its germ. So the isomorphisms of two sheets through one morphism, or of an equivalence at two nearby objects, act alike. The text uses this silently in passing from one sheet to another (§XII.16).

With these, Proposition 12.5.6 (□) is short. Over a two-to-one $|\sigma|$ the isotropy fixes the orientations, so near a point every morphism maps a continuous local orientation to itself, and the lifted section is that orientation there (§XII.21). What Theorem 12.5.7 needs beyond this is the topology of quotients and the comparison of representatives, not orientation theory.

The connectedness argument. The two readings of continuity of an orientation agree by a connectedness argument (§XII.20). With exactly two orientations at each smooth point, two sets are open: where a continuous section agrees with a continuous orientation, and where it agrees with the opposite one. They are complementary and cover a neighbourhood with connected smooth part, so one of them is all of it.

Levels. The homeomorphism of Theorem 12.5.7 (1) needs one fact that Part I uses without comment: an sc-smooth map is continuous on every level and on X_∞. It also needs the openness of the orbit maps of O_f, which is once more the last sentence of Theorem 12.5.3 (§XII.22).

The natural-equivalence lemma. The independence of the representative and the functoriality, which the text leaves to the reader, rest on one lemma. A natural transformation between two equivalences changes their orientation maps only by the action of morphisms, which the orbit relation identifies. It is the germ lemma applied to the local isomorphism of a morphism (§XII.22).

What Chapter 12 asks of Part I. Beyond the statements of Part I that it restates on the objects, the chapter uses Part I in four places, and each is a point where a Part I lemma had to be stated in a form Part I itself did not need: - the automatic completeness lemma for glued fibre norms (§XII.2), used by every norm theorem; - mixed convergence and its lower semicontinuity in a pointwise ε-form that composes under averaging, transport and gluing (§XII.7); - the ∂-extension of Part I's Definition 5.4.1, assembled from the inclusion–exclusion construction of Proposition 12.3.7 (§XII.12); - Part I's orientation transport, through the comparison with local strong bundle isomorphisms, the locality of continuity, and the level continuity of sc-smooth maps (§§XII.19–XII.22).

Each of these is stated once and used throughout; the chapter's own proofs are the passage through the morphisms.

How the statements depend on each other. The four norm theorems rest on Theorem 12.2.2's construction and the completeness lemma; Theorems 12.2.7 and 12.2.13 add a local step near the boundary, the second under mixed convergence. The bump function theorems feed Theorem 12.3.3 and, through the partitions of unity, Theorem 12.3.9, whose local step is Proposition 12.3.7. The compactness results of §12.4 use only the properness of the morphism maps and Part I's compactness notions. In §12.5, Lemma 12.5.1 and the germ lemma feed Theorem 12.5.3; its last sentence feeds Proposition 12.5.6 (□), the openness of the orbit maps, and the continuity of push-forwards in Theorem 12.5.7; the natural-equivalence lemma and composition feed the independence of the representative and functoriality. No statement of §§12.2–12.4 is used in §12.5 except through Part I, so the open gaps G.140 and G.192 touch §12.5 only.

Equivalent representatives are Chapter 10's. Chapter 10's equivalence of diagrams, a common refinement (`SDiagram.Equiv`), is exactly the notion of equivalent representatives that Theorem 12.5.7 needs. What is left (G.192) is to unpack Chapter 10's classes into the statements of §XII.22.

Part XIII. Sc⁺-multisections (HWZ Chapter 13, §§13.1–13.5)

Chapter 13 (Springer 2021, pp. 599–636; 2017 numbers in parentheses) builds the perturbations used by the Fredholm theory on ep-groupoids. The perturbations are sc⁺-multisections: functors $\Lambda : W \to \mathbb{Q}^+$ on a strong bundle $P : W \to X$ that near every point are given by finitely many sc⁺-sections with rational weights. The chapter has five sections. §13.1 constructs *good systems of neighbourhoods*. In such a system every point x has a neighbourhood $U(x)$ that carries the natural action of the isotropy group G_x. The target map is proper on the morphisms leaving $\mathrm{cl}\,U(x)$. And the morphisms from $U(x)$ to $U(x')$ split into components on which source and target are sc-diffeomorphisms onto their images. The section's statements are Lemma 13.1.1, Definition 13.1.2, Proposition 13.1.3, Definitions 13.1.4 and 13.1.5, Lemma 13.1.6, Theorems 13.1.7 and 13.1.8, and Proposition 13.1.9 (2017: Lemma 13.1, Definition 13.1, Proposition 13.1, Definitions 13.2 and 13.3, Lemma 13.2, Theorems 13.1 and 13.2, Proposition 13.2). §13.2 defines the supports, pointwise norms and triviality of sc⁺-multisections. It also defines symmetric section structures, decorated sc⁺-multisections, and the algebra of sums, scalar multiples, products with functions and locally finite sums: Definitions 13.2.1–13.2.5, Lemma 13.2.6, Definitions 13.2.7–13.2.10, Lemma 13.2.11, Definition 13.2.12 and Lemma 13.2.13 (2017: Definitions 13.4–13.8, Lemma 13.3, Definitions 13.9–13.12, Lemma 13.4, Definition 13.13, Lemma 13.5). §13.3 introduces correspondences between index sets. With them it defines structurable and structured sc⁺-multisections, proves that sums, multiples and locally finite sums of structurable ones are structurable, and introduces the tight structures (Definitions 13.3.1–13.3.11, Proposition 13.3.12, Lemma 13.3.14, Proposition 13.3.15, Definition 13.3.17, Corollary 13.3.18, Proposition 13.3.19; 2017: Definitions 13.14–13.21, Propositions 13.3–13.5, Lemma 13.6, Corollary 13.1). §13.4 moves structured sc⁺-multisections along strong bundle equivalences and proper strong bundle coverings, extends them from an open set that meets every orbit, and defines commensurable sums (Theorems 13.4.1, 13.4.3, 13.4.4, Proposition 13.4.2, Definition 13.4.5; 2017: Theorems 13.3–13.5, Proposition 13.6, Definition 13.22). §13.5 defines atomic sc⁺-multisections, the ones built from a single point. It proves that they are structurable and constructs them from bump functions. It also shows that every structured sc⁺-multisection is a locally finite commensurable sum of atomic ones (Definition 13.5.1, Theorems 13.5.3–13.5.7; 2017: Definition 13.23, Theorems 13.6–13.10).

What is proved, and what is not. Every numbered statement of Chapter 13 is formalized and proved, and no gap of the chapter is open.

§13.1. - Lemma 13.1.1 is proved in a stronger form, without the invariance of V (§XIII.1). - Definition 13.1.2 is restated: "carries the natural G_x-action" is read as an invariant open subset of the domain of a natural representation (§XIII.2). - Proposition 13.1.3 and Definition 13.1.4 hold as printed in that reading

(§XIII.2). - Definition 13.1.5, the natural bi-action, is written as the formula its proofs use, with its action laws and its sc-smoothness. Lemma 13.1.6 is proved in a stronger form (§XIII.3). - Theorem 13.1.7 is proved in two forms. - A stronger form reads properness in X and needs neither Q open nor any condition on the closures of the new neighbourhoods (§XIII.4). - The form as printed reads properness in the full subcategory on the open set Q (§XIII.5). - Theorem 13.1.8 is proved as printed (§XIII.6). - Proposition 13.1.9 is proved in a stronger form, with uniqueness among all maps (§XIII.7).

§13.2. - Definition 13.2.1 is Chapter 11's. - Definitions 13.2.2 and 13.2.4 and the claims after them hold as printed. Definition 13.2.3 is restated as a supremum that is shown to be attained (§XIII.8). - Definition 13.2.5 is restated on the domain of a natural representation. Lemma 13.2.6 is proved as printed. The symmetric structures on neighbourhoods with the natural action, which the later definitions need, are formalized with it (§XIII.9). - Definitions 13.2.7 and 13.2.8 hold as printed (§XIII.10). - Definition 13.2.9 is restated. When the intersection $U'(x) \cap U''(x)$ carries no single natural action for both structures, it is read as a neighbourhood inside it on which the two actions agree. The remark after it, that there are in general no inverses, is proved by an example (§XIII.11). - Definition 13.2.10 and Lemma 13.2.11 are proved as printed (§XIII.12). - Definition 13.2.12 is as printed. Lemma 13.2.13 is restated, the infinite sum being written out (§XIII.13).

§13.3. - The correspondences of Definitions 13.3.1–13.3.4, Example 13.3.3 and Remark 13.3.5 are formalized. Definition 13.3.2 also gets the action laws, which the text does not state (§XIII.14). - Definitions 13.3.6–13.3.11 and Remark 13.3.7 are formalized (§XIII.15). - Proposition 13.3.12 is restated: "we may assume $\mathcal{U}_1 = \mathcal{U}_2$" is carried out (§XIII.16). - Remark 13.3.13 is proved in its positive half; the negative half has no formal statement. Lemma 13.3.14 is as printed (§XIII.16). - The construction before Proposition 13.3.15 is proved in a stronger form, with a compatibility the text does not check (§XIII.17). - Proposition 13.3.15 has bijections in place of equalities of index sets (§XIII.17). - Definition 13.3.17 and Corollary 13.3.18 are as printed (§XIII.17). - Proposition 13.3.19 is restated. It holds as printed for a nonempty family, and for every family once a good system exists. The product of correspondences that the text leaves to the reader is written out (§XIII.18). - Remarks 13.3.9 and 13.3.16 are prose.

§13.4. - Theorem 13.4.1 is proved in a stronger form: the canonical structure is a map of structured classes, independent of all choices (§XIII.19). - Proposition 13.4.2 is restated. The project has no strong bundle over an open sub-ep-groupoid, so "Λ structured on $W|V$" is read as structured data relative to V, and "canonical" as uniqueness among the functors on W that extend Λ (§XIII.21). - Theorem 13.4.3 is proved as printed, without Proposition 13.4.2 (§XIII.22). - Theorem 13.4.4 is proved as printed, with the same canonicity as Theorem 13.4.1 (§XIII.23). - Definition 13.4.5 is formalized, with the well-definedness the text calls evident proved (§XIII.24).

§13.5. - Definition 13.5.1 is restated on the domain of a natural representation, and read also on neighbourhoods with the natural action, an extra (§XIII.25). - Theorem 13.5.3 is narrower by one added hypothesis, that a good system exists. It is necessary, since structurability includes a good system, and paracompactness of $|X|$ supplies it (§XIII.26). - Theorems 13.5.4 and 13.5.6 are narrower only through the pointwise reading of "admits sc-smooth bump functions"; in other respects their conclusions are stronger (§§XIII.27, XIII.29). - Theorem 13.5.5 is narrower by $I \neq \emptyset$, which the printed formula presupposes. Its last clause carries the same good-system hypothesis as Theorem 13.5.3 (§XIII.28). - Theorem 13.5.7 is proved in a stronger form: for every structured lift, with partitions of unity and without the paracompactness of $|X|$ (§XIII.30). - Remark 13.5.2 is prose.

Gaps. The chapter's gaps G.141 (Theorem 13.1.7 as printed), G.142 (decorated sc$^+$-multisections and their algebra), G.143 (no inverses), G.148 (Proposition 13.3.19), G.149 (Theorem 13.5.3 and the last clause of

Theorem 13.5.5) and G.163 (Proposition 13.4.2) are all closed. No statement of Chapter 13 was found that does not hold as we read it.

Four decisions that shape the chapter. 1. *Neighbourhoods with the natural action.* The project's natural representation of G_x (Theorem 7.1.13, Part VII) has a connected domain. The invariant subsets of §13.1 need not be connected. A neighbourhood with the natural action is therefore a natural representation together with an open $U \ni x$ inside its domain, invariant under every $\Phi(g)$ `Chapter13/GoodSystem · EpGroupoid.NatNbhd`. In this reading every claim of §13.1 that an invariant open subset carries the natural action holds as printed, and the good systems of §13.1 are made of such neighbourhoods. Definitions 13.2.5 and 13.5.1, which predate the good systems in the formalization, ask for the connected domain of a natural representation, the reading of Chapter 11. The structures on good systems (§§XIII.9–XIII.10) and the atomic pieces of Theorem 13.5.7 (§XIII.30) are therefore formalized on neighbourhoods with the natural action. 2. *Components named by a point.* A component of $\mathcal{U}(x, x')$ is the connected component in $\mathcal{U}(x, x')$ of one of its morphisms, and it is empty if the morphism is not in $\mathcal{U}(x, x')$. No statement needs a separate index set of components, and many statements need no membership hypothesis. 3. *Multisections as functors, structures as data.* An sc$^+$-multisection is a functor $\Lambda : W \to \mathbb{Q}^+$ satisfying Chapter 11's predicate, with local section structures on open sets `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsScPlusMultisection` `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn`. Decorations, structures and structured tuples are data attached to it. The decorated and structured multisections of the text are quotients of the data by the equivalences of Definitions 13.2.8 and 13.3.8. 4. *Index sets are types.* The text writes equalities of index sets: $I'_x = \{*\}$ and $I'_x = I_x$ in Proposition 13.3.15, $I_x = I'_x$ in Definition 13.3.8, $K_x = \prod_l K^l_x$ in Proposition 13.3.19. The formalization has types with bijections between them, and compares correspondences after relabelling along the bijections. Where the text asks two index sets to be equal (Definitions 13.3.8 and 13.4.5), the formalization asks for an equality of types and compares the correspondences through it.

Hypotheses added, and their witnesses. Each added hypothesis is not derived from the text. - *A good system exists.* Added in Theorem 13.5.3, in the last clause of Theorem 13.5.5, and in Proposition 13.3.19 for the empty family (§§XIII.18, XIII.26, XIII.28). It is necessary. Definition 13.3.6 includes a good system in every structure, so for atomic Λ structurability is equivalent to it. Witnesses: - Theorem 13.1.8 for paracompact $|X|$. - In particular every M-polyfold seen as a trivial ep-groupoid `Chapter13/GoodSystemExistence · EpGroupoid.nonempty_goodSystem_trivial` `Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial`, and the bundles $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ used in Chapter 15 `Chapter15/GeneralPositionAssembly · SBLocalization.nonempty_goodSystem_epSEP_zero`. - A nonempty family in Proposition 13.3.19 needs nothing, since any member's structure carries a good system. - *$I \neq \emptyset$ in Theorem 13.5.5* (§XIII.28). An sc$^+$-multisection has weights summing to 1, and with $I = \emptyset$ the printed formula gives $\Lambda = 0$ over $U(x)$. Witness: the families of Construction I, whose index set is G_x (§XIII.27). - *Pointwise bump functions in Theorems 13.5.4 and 13.5.6* (§§XIII.27, XIII.29). "Admits sc-smooth bump functions" is read as Part XII's pointwise reading of Theorem 12.3.3. Witness: the trivial ep-groupoids $\mathbb{R}^n$ `Chapter12/BumpEp · EpGroupoid.admitsBumpFunctionsPointwise_trivial_rn`. - *$c \neq 0$ and $k \geq 1$ in the example for the remark that there are no inverses* (§XIII.11). These are parameters of an example, not hypotheses of a statement. - *Paracompactness of $|X|$ in the existence of decorations and in the structurability of the zero multisection* (§§XIII.10, XIII.15). This is the text's own setting, through Theorem 13.1.8. Witness: as for good systems.

The hypotheses the text states and the formalization keeps without using: - the properness property in Theorem 13.5.4, and the smoothness of x in its part (2); - bump functions and the smoothness of x in Theorem 13.5.5; - the paracompactness of $|X|$ in Theorems 13.5.6 and 13.5.7.

Map. All files are in `Polyfold/Chapter13/`. They are grouped by topic.

Good systems (§13.1). - `GoodSystem`: neighbourhoods with the natural action and their invariant cores; Lemma 13.1.1; the sets $\mathcal{U}(x, x')$ and their components; Definition 13.1.2, Proposition 13.1.3, Definition 13.1.4; families good on a set. - `BiAction`: the bi-action of Definition 13.1.5 with its laws; the components of a good system; Lemma 13.1.6; stabilizers, D_Σ, the partner homomorphism, Proposition 13.1.9. - `GoodSystemExistence`: transports; Theorem 13.1.7 in the stronger form; Theorem 13.1.8; the witness on trivial ep-groupoids. - `GoodSystemRelative`: the full subcategory on Q, properness relative to Q, properness by ultrafilters, Theorem 13.1.7 as printed.

Sc⁺-multisections as functors (§13.2). - `ScPlusSectionOn`: sc⁺-sections on open sets, through chart representatives; sums, multiples, finite sums and extensions by 0. - `MultisectionSupport`: Definitions 13.2.2–13.2.4 and the claims after them. - `MultisectionSymmetric`: transported sections, equal weights, Definition 13.2.5, Lemma 13.2.6. - `MultisectionConvolution`: the convolution of finitely many functors. - `MultisectionAlgebra`: Definitions 13.2.9 and 13.2.10, Lemma 13.2.11, for functors. - `MultisectionLocallyFinite`: Definition 13.2.12, Lemma 13.2.13.

Decorated sc⁺-multisections (§13.2). - `SymmetricOn`: the relation "same natural action" between neighbourhoods; moving vectors along morphisms; symmetric structures on a neighbourhood with the natural action and their sums, multiples and products. - `DecoratedMultisection`: Definitions 13.2.7 and 13.2.8; Definitions 13.2.9, 13.2.10 and Lemmas 13.2.11, 13.2.13 for decorations and on classes. - `MultisectionNoInverse`: the remark after Definition 13.2.9.

Correspondences and structurable sc⁺-multisections (§13.3). - `Correspondence`: Definitions 13.3.1 and 13.3.2, Example 13.3.3, Remark 13.3.5, products of two correspondences. - `StructurableMultisection`: Definitions 13.3.4, 13.3.6, 13.3.8, 13.3.10, 13.3.11; Proposition 13.3.12; Lemma 13.3.14; the zero multisection as witness. - `TightMultisection`: T_Λ; the construction before Proposition 13.3.15; Proposition 13.3.15, Definition 13.3.17, Corollary 13.3.18. - `PiCorrespondence`: products of families of correspondences; reindexing symmetric structures. - `TightOn`: tight versions over a given adapted good system. - `LocallyFiniteStructurable`: the data of the proof of Proposition 13.3.19, the product structure, Proposition 13.3.19. - `LocallyFiniteSupports`: from locally finite domain supports to Definition 13.2.12; Proposition 13.3.19 with the printed hypothesis.

Pull-backs, push-forwards, extension and commensurable sums (§13.4). - `StrongBundlePullMap`: the common data of a pull-back; equivalences and proper coverings as such data; pulled-back sections. - `MultisectionPullback`: adapted neighbourhoods; the pulled-back structure; canonicity; Theorems 13.4.1 and 13.4.4. - `SymmetricStructurePush`: charts along a morphism; push neighbourhoods; pushed sections and structures. - `MultisectionPushforward`: lifted morphisms; the pushed good system and structure; Theorem 13.4.3; generalized strong bundle isomorphisms. - `StructuredExtensionData`: functors and sc⁺-multisections on $W|V$; structured data on V; the extension functor; unit charts; extension charts. - `StructuredExtension`: the extended structure, Proposition 13.4.2. - `Commensurable`: common data c; count transport along correspondences; Definition 13.4.5 and the locally finite commensurable sum.

Atomic sc⁺-multisections (§13.5). - `AtomicMultisection`: Definition 13.5.1; the multisection of atomic data; its local structure. - `AtomicConstruction`: symmetric families; Theorems 13.5.4, 13.5.5 (existence and uniqueness) and 13.5.6. - `AtomicStructurableLocal`: the local alternative of the proof of Theorem 13.5.3, and the form of Theorems 13.5.3 and 13.5.5 with structurability as a parameter. - `AtomicStructurable`: the data and the structure of the proof of Theorem 13.5.3; Theorem 13.5.3; the last clause of Theorem 13.5.5. - `AtomicSum`: atomic data on a neighbourhood with the natural action; Theorem 13.5.7.

Conventions. Throughout, X is an ep-groupoid with source and target maps s, t, composition $\circ$, inversion $\varphi \mapsto \varphi^{-1}$ and orbit map $\pi : X \to |X|$. The isotropy group G_x is finite (Proposition 7.1.12, Part VII). A natural representation of G_x (Theorem 7.1.13, Part VII) on an open connected $U \ni x$ consists of sc-diffeomorphisms $\Phi(g) : U \to U$ and morphisms $\Gamma(g, z) : z \to \Phi(g)(z)$. Every morphism between two points of U is $\Gamma(g, z)$ for exactly one g. We write $g * z = \Phi(g)(z)$. For $U, U' \subset X$,

$$\mathcal{U}(U, U') = \{\varphi : s(\varphi) \in U,\ t(\varphi) \in U'\}$$

`Chapter13/GoodSystem · EpGroupoid.morBetween`, and $\mathcal{U}(x, x') = \mathcal{U}(U(x), U(x'))$ for a family of neighbourhoods. A subset $A \subset X$ has *proper closure*, or the properness property, if $t : s^{-1}(\mathrm{cl}\, A) \to X$ is proper. From §XIII.8 on, $(P : W \to X, \mu)$ is a strong bundle over X, 0_x the zero of the fibre W_x, and $\Lambda : W \to \mathbb{Q}^+$.

XIII.1 Neighbourhoods with the natural action, and Lemma 13.1.1 (2017: Lemma 13.1)

Definition `Chapter13/GoodSystem · EpGroupoid.NatNbhd`. A *neighbourhood of x with the natural action* is a pair $N = (R, U)$: - R is a natural representation of G_x on the connected open domain U_R, with maps Φ, Γ; - U is an open subset with $x \in U \subset U_R$ and $\Phi(g)(U) \subset U$ for all $g \in G_x$.

All the laws of R restrict to U. On U: - $\Phi(g)$ maps U into U; - $\Gamma(g, z)$ has source z and target $g * z$; - $\Gamma(1, z)$ is the unit; - $\Gamma(gh, z) = \Gamma(g, h * z) \circ \Gamma(h, z)$; - $\Gamma(g^{-1}, g * z) = \Gamma(g, z)^{-1}$; - $\Gamma(\cdot, z)$ is injective in g; - every morphism between two points of U is $\Gamma(g, \cdot)$ for a unique g; - $\pi(g * z) = \pi(z)$.

Reading. The text says "$U(x)$ is G_x-invariant and carries the natural action". The project's natural representations live on connected domains. Proposition 13.1.3 shrinks the neighbourhoods of a good system to arbitrary invariant open subsets, which need not be connected, so the neighbourhood is an invariant open subset of such a domain. Since the domain is open and U lies inside it, the uniqueness clause for morphisms between points of U holds as for the domain itself. Nothing in §13.1 uses connectedness of $U(x)$.

Invariant cores. For a neighbourhood N with the natural action and an open set $V \ni x$, the *invariant core* is

$$U \cap \bigcap_{g \in G_x} \Phi(g)^{-1}(V)$$

`Chapter13/GoodSystem · EpGroupoid.NatNbhd.invariantCore`. It is open because G_x is finite and each $\Phi(g)$ is continuous. It contains x, because $\Phi(g)(x) = x$. It is invariant, because $\Phi(h)$ maps it into itself by the action law $\Phi(g) \circ \Phi(h) = \Phi(gh)$. And it lies in $U \cap V$, taking $g = 1$. So every neighbourhood with the natural action has, inside any open $V \ni x$, a smaller one with the same natural representation `Chapter13/GoodSystem · EpGroupoid.NatNbhd.exists_restrict_subset`. This is the step "we may assume $U(x)$ is invariant" that the text uses throughout.

Existence with the properness property `Chapter13/GoodSystem · EpGroupoid.exists_natNbhd_isProperSubset`. Inside every open $W \ni x$ there is a neighbourhood N of x with the natural action and with $t : s^{-1}(\mathrm{cl}\, U) \to X$ proper. - *Step 1.* Chapter 7 gives, inside any open set around x, an open neighbourhood $U_0 \subset W$ of x that is the domain of a natural representation and whose closure has the properness property (the invariant neighbourhoods of Theorem 7.1.13, Part VII). - *Step 2.* The neighbourhood is that representation

with $U = U_0$. It is invariant because it is the domain.

This is the source of all the neighbourhoods in Chapter 13. Every construction below starts from it and shrinks by invariant cores.

Lemma 13.1.1 (stronger), p. 599 `Chapter13/GoodSystem · EpGroupoid.lemma_13_1_1`. Let N be a neighbourhood of x with the natural action, with $t : s^{-1}(\operatorname{cl} U) \to X$ proper, and let $V \subset U$ be any subset. Then

$$\operatorname{cl}_X V \subset U \iff \operatorname{cl}_{|X|} \pi(V) \subset \pi(U).$$

The text assumes V invariant. The form with that assumption follows at once `Chapter13/GoodSystem · EpGroupoid.lemma_13_1_1_book`. The two directions use different halves of the hypotheses: (□) uses only properness, and (□) uses only the natural action.

Proof of (□). Suppose $\operatorname{cl} V \subset U$. Then $\operatorname{cl} V \subset \operatorname{cl} U$, so $\operatorname{cl} V$ is a closed subset of a set with proper closure. 1. For a closed set A with t proper on $s^{-1}(A)$, the orbit map satisfies $\pi(\operatorname{cl} A) = \operatorname{cl} \pi(A)$ (Proposition 7.1.10, Part VII). 2. Apply this to $\operatorname{cl} V$: $\operatorname{cl}_{|X|} \pi(V) \subset \operatorname{cl}_{|X|} \pi(\operatorname{cl} V) = \pi(\operatorname{cl} V) \subset \pi(U)$.

Proof of (□). Suppose $\operatorname{cl} \pi(V) \subset \pi(U)$, and let $y \in \operatorname{cl} V$. 1. $\pi(y) \in \operatorname{cl} \pi(V) \subset \pi(U)$, so there is a point $z \in U$ and a morphism $\varphi : z \to y$. 2. The target map is a local sc-diffeomorphism, so near φ there is a sheet: a local inverse τ of t on an open $V' \ni y$ with $\tau(y) = \varphi$, continuous, with $t(\tau(y')) = y'$. Set $O = V' \cap \tau^{-1}(s^{-1}(U))$. It is an open neighbourhood of y, since $s(\tau(y)) = z \in U$ and U is open. 3. For $y' \in O \cap V$, $\tau(y')$ is a morphism from $s(\tau(y')) \in U$ to $y' \in V \subset U$. It is therefore $\Gamma(g, s(\tau(y')))$ for a unique g, and $y' = g * s(\tau(y'))$. 4. Since $y \in \operatorname{cl} V$, the filter $\mathcal{N}_V(y)$ of neighbourhoods of y within V is nontrivial, and $O \cap V$ belongs to it. The index g of step 3 takes only finitely many values, since G_x is finite (Proposition 7.1.12, Part VII). So some g_0 occurs frequently along $\mathcal{N}_V(y)$: every neighbourhood of y contains points $y' \in V$ with index g_0. 5. Along these points $y' = \Phi(g_0)(s(\tau(y')))$. As $y' \to y$ along the filter, $s(\tau(y')) \to s(\tau(y)) = z$ by continuity, so the right side tends to $\Phi(g_0)(z)$. By uniqueness of limits in the Hausdorff space X, $y = \Phi(g_0)(z) \in U$, since U is invariant.

Where the formal proof departs from the text. The text takes V invariant and argues with sequences. The formal proof works with filters, and with "occurs frequently" in place of the pigeonhole over a sequence. It never uses that V is invariant: only $V \subset U$ enters, through step 3.

XIII.2 Definition 13.1.2 (2017: Definition 13.1), Proposition 13.1.3 (2017: Proposition 13.1) and Definition 13.1.4 (2017: Definition 13.2)

Components and diffeo-components. For a set S of morphisms and $\varphi \in S$, the component of φ is the connected component of φ in S. For $\varphi \notin S$ it is empty.

A set Σ of morphisms is a *diffeo-component* `Chapter13/GoodSystem · EpGroupoid.IsDiffeoComponent` if $s|_\Sigma$ and $t|_\Sigma$ are sc-diffeomorphisms onto open subsets of the objects.

For an open Σ this is equivalent to s and t being injective on Σ `Chapter13/GoodSystem · EpGroupoid.isDiffeoComponent_iff_injOn`, by the following argument. 1. s and t are local sc-diffeomorphisms (Definition 7.1.1 (1), Part VII). 2. An injective local sc-diffeomorphism on an open set is an sc-diffeomorphism onto its image, which is open `Chapter13/GoodSystem · MPolyfold.IsLocalScDiffeo.isScDiffeoOn_invFunOn`. - The image is open, being a union of the open images of the local sheets. - The inverse is the inverse function on the image. Near each image point it agrees with the inverse of one sheet, by injectivity, so it is sc-smooth

there.

The components of $\mathcal{U}(x,x')$ are open, because $\mathcal{U}(x,x')$ is open (s, t continuous, $U(x)$, $U(x')$ open) and the morphism space is locally connected.

Definition 13.1.2 (restated), p. 600 `Chapter13/GoodSystem · EpGroupoid.GoodSystem`. A *good system of neighbourhoods* is a family $(N(x))_{x\in X}$ of neighbourhoods $U(x)$ with the natural action of G_x (condition (2) of the text) such that: - (1) $t : s^{-1}(\operatorname{cl} U(x)) \to X$ is proper for every x; - (3) for all x, x' and every $\varphi \in \mathcal{U}(x,x')$, the component of φ in $\mathcal{U}(x,x')$ is a diffeo-component.

By the equivalence above, (3) is the injectivity of s and t on every component. The text's "the images are open" is automatic.

On a good system the inverses $(s|_\Sigma)^{-1}$, $(t|_\Sigma)^{-1}$ are functions on the objects (defined arbitrarily off $s(\Sigma)$, resp. $t(\Sigma)$) `Chapter13/GoodSystem · EpGroupoid.GoodSystem.srcInv`. They are sc-smooth on the images. *Restated*: "carries the natural G_x-action" is the reading of §XIII.1, and components are named by a point.

Proposition 13.1.3 (same), p. 600 `Chapter13/GoodSystem · EpGroupoid.proposition_13_1_3`. Let $(U(x))$ be a good system and $\tilde{U}(x) \subset U(x)$ open invariant neighbourhoods of x. Then $(\tilde{U}(x))$ is a good system with the same natural representations `Chapter13/GoodSystem · EpGroupoid.GoodSystem.restrict`.

Proof. 1. *(2).* $\tilde{U}(x)$ is open, contains x, lies in the domain of the representation of $U(x)$, and is invariant. So it is a neighbourhood with the natural action, with the same representation. This is where decision (1) matters: in the reading "the connected domain of a natural representation", $\tilde{U}(x)$ could fail to be such a domain, and the proposition would not hold as we read it. 2. *(1).* $\operatorname{cl}\tilde{U}(x)$ is a closed subset of $\operatorname{cl} U(x)$. A proper map restricted to a closed subset of its domain is proper, so $t : s^{-1}(\operatorname{cl}\tilde{U}(x)) \to X$ is proper. 3. *(3).* $\tilde{\mathcal{U}}(x,x') \subset \mathcal{U}(x,x')$, so the component of φ in the smaller set lies in its component in the larger set, on which s and t are injective. Injectivity passes to subsets, and the smaller component is open, so it is a diffeo-component `Chapter13/GoodSystem · EpGroupoid.GoodSystem.isDiffeoComponent_of_subset`.

The step after the proposition. The text uses the proposition to shrink a good system into given open neighbourhoods $V(x)$. Inside $U(x) \cap V(x)$ the invariant core of §XIII.1 is an invariant open neighbourhood. So every good system has a refinement inside any family of open neighbourhoods `Chapter13/GoodSystem · EpGroupoid.GoodSystem.exists_refinement_subset`.

Definition 13.1.4 (same), p. 600 `Chapter13/GoodSystem · EpGroupoid.GoodSystem.IsRefinement` `Chapter13/GoodSystem · EpGroupoid.GoodSystem.IsSubordinate`. - A good system $\mathcal{W}$ refines $\mathcal{U}$ if $W(x) \subset U(x)$ for all x. - $\mathcal{U}$ is subordinate to an open cover (O_a) if each $U(x)$ lies in some O_a. - The restrictions of Proposition 13.1.3 are refinements `Chapter13/GoodSystem · EpGroupoid.GoodSystem.isRefinement_restrict`.

Families good on a set `Chapter13/GoodSystem · EpGroupoid.IsGoodOn`. Theorems 13.1.7 and 13.1.8 build good systems from partial ones. A family $(N(q))$ of neighbourhoods with the natural action, indexed by all points, is *good on* $Q \subset X$ if for $q, q' \in Q$: - $U(q) \subset Q$; - $t : s^{-1}(\operatorname{cl}_X U(q)) \to X$ is proper; - condition (3) holds on $\mathcal{U}(q,q')$.

Every good system is good on X `Chapter13/GoodSystem · EpGroupoid.GoodSystem.isGoodOn`. A family good on X is a good system. The relation between "good on Q" and the text's good system of the sub-ep-groupoid Q is §XIII.5.

XIII.3 Definition 13.1.5 (2017: Definition 13.3) and Lemma 13.1.6 (2017: Lemma 13.2)

Definition 13.1.5 (restated), p. 601 `Chapter13/BiAction · EpGroupoid.biact`. The text names a map $G_{x'} \times \mathcal{U}(x,x') \times G_x \to \mathcal{U}(x,x')$, $(g',\varphi,g) \mapsto g' * \varphi * g$, "the natural bi-action". The formalization writes the formula that the text's proofs use. For neighbourhoods $N = (R,U)$ at x and $N' = (R',U')$ at x', with $\varphi \in \mathcal{U}(U,U')$:

$$g' * \varphi * g = \Gamma'(g', t(\varphi)) \circ \varphi \circ \Gamma(g, \Phi(g^{-1})(s(\varphi))).$$

- *Source and target.* The source is $\Phi(g^{-1})(s(\varphi))$ `Chapter13/BiAction · EpGroupoid.src_biact` and the target is $\Phi'(g')(t(\varphi))$ `Chapter13/BiAction · EpGroupoid.tgt_biact`. The right factor $\Gamma(g,\cdot)$ is taken at $g^{-1} * s(\varphi)$, so that its target is $s(\varphi)$ and the composite is defined. - *It stays in $\mathcal{U}$.* By invariance of U and U', $g' * \varphi * g \in \mathcal{U}(U,U')$ `Chapter13/BiAction · EpGroupoid.biact_mem`. - *Action laws.* $g' * (h' * \varphi * h) * g = (g'h') * \varphi * (hg)$ `Chapter13/BiAction · EpGroupoid.biact_biact`. Write out both sides and use associativity of composition with $\Gamma'(g'h', z) = \Gamma'(g', h' * z) \circ \Gamma'(h', z)$ on the left and the corresponding identity for Γ on the right. With $1 * \varphi * 1 = \varphi$, from $\Gamma(1,\cdot) = 1$, this is an action of $G_{x'} \times G_x^{\mathrm{op}}$ `Chapter13/BiAction · EpGroupoid.biactMulAction`. - *Cancellations.* $g' * \varphi * g = g' * \psi * g$ implies $\varphi = \psi$, and $g' * \varphi = h' * \varphi$ implies $g' = h'$. The second holds by right cancellation of φ and the injectivity of $\Gamma'(\cdot, t\varphi)$. - *Sc-smoothness* `Chapter13/BiAction · EpGroupoid.isScSmoothOn_biact`. Each $\varphi \mapsto g' * \varphi * g$ is sc-smooth on $\mathcal{U}(U,U')$. - The left action $\varphi \mapsto \Gamma'(g', t\varphi) \circ \varphi$ is sc-smooth because $\Gamma'(g',\cdot)$ is, t is, and composition is sc-smooth on the fibred product. - The right action is reduced to a left action by inversion: $\varphi * g = (g^{-1} * \varphi^{-1})^{-1}$ in the bi-action for $\mathcal{U}(U',U)$. - Inversion is an sc-diffeomorphism.

The text defines the map before Lemma 13.1.6 and says it permutes components. On a good system the bi-action is $\mathcal{U}(x,x') \to \mathcal{U}(x,x')$ `Chapter13/BiAction · EpGroupoid.GoodSystem.biact`, and it maps components onto components `Chapter13/BiAction · EpGroupoid.GoodSystem.image_biact_comp`. 1. The bi-action by (g',g) is continuous, so it maps the connected component of φ into a connected subset of $\mathcal{U}(x,x')$ containing $g' * \varphi * g$. That subset lies in the component of $g' * \varphi * g$. 2. The bi-action by (g'^{-1}, g^{-1}) maps that component back. So the image is the whole component.

Inversion maps components of $\mathcal{U}(x,x')$ onto components of $\mathcal{U}(x',x)$ by the same argument `Chapter13/BiAction · EpGroupoid.GoodSystem.image_inv_comp`. The components cover $\mathcal{U}(x,x')$ and are pairwise disjoint or equal `Chapter13/BiAction · EpGroupoid.GoodSystem.components`.

Two tools. - *Same source, left factor* `Chapter13/BiAction · EpGroupoid.GoodSystem.exists_biact_left_of_src`. Let $\psi_1, \psi_2 \in \mathcal{U}(x,x')$ have the same source. Then $\chi = \psi_2 \circ \psi_1^{-1}$ is a morphism from $t(\psi_1)$ to $t(\psi_2)$, two points of $U(x')$. So $\chi = \Gamma'(g', t\psi_1)$ for a unique g', and $\psi_2 = g' * \psi_1 * 1$. - *Same target, right factor* `Chapter13/BiAction · EpGroupoid.GoodSystem.exists_biact_right_of_tgt`. Morphisms with the same target differ by a right factor, by the same argument with $\psi_2^{-1} \circ \psi_1$.

Lemma 13.1.6 (stronger), p. 602. Let $\mathcal{U}$ be a good system and Σ a component of $\mathcal{U}(x,x')$.

(1) If $\psi \in \Sigma$ and $g' * \psi * 1 \in \Sigma$, then $g' = 1$ `Chapter13/BiAction · EpGroupoid.GoodSystem.eq_one_of_biact_left_mem`.

- $g' * \psi * 1$ has source $\Phi(1)(s\psi) = s\psi$.
- s is injective on Σ by condition (3), so $g' * \psi * 1 = \psi$.
- Left cancellation gives $\Gamma'(g', t\psi) = 1_{t\psi} = \Gamma'(1, t\psi)$, and injectivity of Γ' in the group element gives $g' = 1$.

The text's (1), $g' * \Sigma = \Sigma \Rightarrow g' = 1$, follows by applying this to one $\psi \in \Sigma$ `Chapter13/BiAction · EpGroupoid.GoodSystem.lemma_13_1_6_one`. The stronger form needs one morphism, not the whole component, and Proposition 13.1.9 uses it in that form.

(2) The same on the right: $\psi \in \Sigma$ and $1 * \psi * g \in \Sigma$ imply $g = 1$, using injectivity of t `Chapter13/BiAction · EpGroupoid.GoodSystem.eq_one_of_biact_right_mem` `Chapter13/BiAction · EpGroupoid.GoodSystem.lemma_13_1_6_two`.

(3) For two components Σ_1, Σ_2, the images $s(\Sigma_1)$ and $s(\Sigma_2)$ are equal or disjoint `Chapter13/BiAction · EpGroupoid.GoodSystem.lemma_13_1_6_three_src`, and so are $t(\Sigma_1)$ and $t(\Sigma_2)$ `Chapter13/BiAction · EpGroupoid.GoodSystem.lemma_13_1_6_three_tgt`.

- Suppose $s(\varphi_1) = s(\varphi_2)$ with $\varphi_i \in \Sigma_i$. By the first tool, $\varphi_2 = g' * \varphi_1 * 1$.
- The bi-action by $(g', 1)$ maps the component of φ_1 onto the component of φ_2, so $\Sigma_2 = g' * \Sigma_1$.
- Since $s(g' * \psi * 1) = s(\psi)$, this gives $s(\Sigma_2) = s(\Sigma_1)$.

No membership hypotheses on the components are needed: the component of a morphism outside $\mathcal{U}(x, x')$ is empty (decision (2)).

Stronger refers to (1) and (2), which need one morphism, not the whole component. The text proves (1) through the sc-diffeomorphism $s|_\Sigma$. The formal proof uses only injectivity, which is all that (3) of Definition 13.1.2 gives directly.

XIII.4 Theorem 13.1.7 (2017: Theorem 13.1), the stronger form: extending a family good on Q

The text states Theorem 13.1.7 (p. 602) for an open $Q \subset X$ with $|Q| = |X|$ and a good system of the full sub-ep-groupoid on Q. It extends that system to a good system of X that keeps the given neighbourhoods at the points of Q and has the new neighbourhoods inside given open sets. The formalization proves it twice: - a stronger form, here, for families good on Q in the sense of §XIII.2, with properness read in X; - the printed form, in §XIII.5, by showing that the printed hypothesis implies the stronger form's.

Transports `Chapter13/GoodSystemExistence · EpGroupoid.IsTransport`. Let $U \subset X$ be open and V another open set. A transport from U into V is a map σ from objects to morphisms with: - σ continuous and injective on U; - $t(\sigma(z)) = z$ for $z \in U$; - $s(\sigma(z)) \in V$ for $z \in U$.

The units $z \mapsto 1_z$ are a transport from U into U.

Condition (3) passes back along transports `Chapter13/GoodSystemExistence · EpGroupoid.injOn_comp_of_transport`. Let σ be a transport from U into V and σ' one from U' into V', with $s \circ \sigma$ injective on U and $s \circ \sigma'$ injective on U'. Suppose s and t are injective on every component of $\mathcal{U}(V, V')$. Then s and t are injective on every component of $\mathcal{U}(U, U')$. 1. Define $F(\varphi) = \sigma'(t\varphi)^{-1} \circ \varphi \circ \sigma(s\varphi)$ on $M = \mathcal{U}(U, U')$. The composite is defined because $t(\sigma(s\varphi)) = s\varphi$ and $t(\sigma'(t\varphi)) = t\varphi$. 2. F has source $s(\sigma(s\varphi)) \in V$ and target $s(\sigma'(t\varphi)) \in V'$, so F maps M into $\mathcal{U}(V, V')$. 3. F is continuous on M: it is built from σ, σ', s, t, composition and inversion, each continuous, through the composition lemma for continuous maps into composable pairs. 4. So F maps the component C of φ_0 in M, a connected set, into the component of $F(\varphi_0)$ in $\mathcal{U}(V, V')$, on which s and t are injective. 5. *Injectivity of s on C.* Let $\varphi_1, \varphi_2 \in C$ with $s\varphi_1 = s\varphi_2$. - $F(\varphi_1)$ and $F(\varphi_2)$ have the same source $s(\sigma(s\varphi_1))$, so by step 4 they are equal. -

Their targets agree, so $s(\sigma'(t\varphi_1)) = s(\sigma'(t\varphi_2))$, and injectivity of $s \circ \sigma'$ gives $t\varphi_1 = t\varphi_2$. - Cancelling the equal factors $\sigma'(t\varphi_i)^{-1}$ and $\sigma(s\varphi_i)$ in $F(\varphi_1) = F(\varphi_2)$ gives $\varphi_1 = \varphi_2$. 6. *Injectivity of t.* Symmetric, through σ.

A sheet around a morphism `Chapter13/GoodSystemExistence · EpGroupoid.exists_isOpen_injOn_src_tgt`. Every morphism ψ has an open neighbourhood on which s and t are both injective. Each is a local sc-diffeomorphism, so injective near ψ; intersect the two neighbourhoods.

A transport at each point `Chapter13/GoodSystemExistence · EpGroupoid.exists_transport_of_isGoodOn`. Let the family $(N(q))$ be good on Q, with $\pi(Q) = |X|$, and let $W \ni x$ be open. Then there are: - a point $q \in Q$; - a neighbourhood M of x with the natural action and proper closure, $M \subset W$; - a transport from M into $U(q)$ with $s \circ \sigma$ injective on M.

Case $x \in Q$. Take $q = x$, $M = N(x)$ (inside W by restricting to an invariant core, §XIII.1) and the unit transport.

Case $x \notin Q$. 1. Since $\pi(Q) = |X|$ there is $q \in Q$ and a morphism $\psi : q \to x$. 2. Take an open $O_0 \ni \psi$ on which s and t are injective, and set $O = O_0 \cap s^{-1}(U(q))$. 3. $t|_O$ is an injective local sc-diffeomorphism on the open set O, so it is an sc-diffeomorphism onto the open set $t(O) \ni x$ (§XIII.2). Its inverse on $t(O)$ is $\sigma = (t|_O)^{-1}$. 4. Choose M with the natural action and proper closure inside $W \cap t(O)$ (§XIII.1). 5. Then σ is continuous and injective on M, $t(\sigma(z)) = z$, and $s(\sigma(z)) \in U(q)$ since $\sigma(z) \in O$. And $s \circ \sigma$ is injective because s is injective on O and σ is injective.

Theorem 13.1.7, stronger form (stronger), p. 602 `Chapter13/GoodSystemExistence · EpGroupoid.theorem_13_1_7`. Let $Q \subset X$ with $\pi(Q) = |X|$, let $(N(q))$ be good on Q (§XIII.2), and for $y \notin Q$ let $W(y) \ni y$ be open. Then there is a good system $\mathcal{U}$ of X with: - $N_{\mathcal{U}}(q) = N(q)$ for $q \in Q$; - $U(y) \subset W(y)$ for $y \notin Q$.

Proof. 1. *Choices.* For every y, choose the data of the previous lemma with $W = W(y)$ for $y \notin Q$ (and $W = X$ for $y \in Q$): - a point $q_y \in Q$; - a neighbourhood $M(y)$; - a transport σ_y from $M(y)$ into $U(q_y)$.

For $y \in Q$ the lemma's first case keeps $M(y) = N(y)$. 2. *The family is good on X.* - Each $M(y)$ has the natural action and proper closure. - Condition (3) for $\mathcal{U}(y, y')$ follows from condition (3) for $\mathcal{U}(q_y, q_{y'})$, which holds since the family is good on Q. The step is the transport lemma with $\sigma = \sigma_y$, $\sigma' = \sigma_{y'}$. 3. *A family good on all of X is a good system* `Chapter13/GoodSystem · EpGroupoid.GoodSystem.ofIsGoodOn`.

Where the formal proof departs from the text, and why stronger. The text's hypothesis is a good system of the full sub-ep-groupoid Q, with properness read in Q, and Q open. Here properness is read in X and Q is any set meeting every orbit. The text also asks the new $U(y)$ to have closures with a further smallness condition; it is not used, and the conclusion holds without it. The printed form is derived from this one in §XIII.5.

XIII.5 Theorem 13.1.7 (2017: Theorem 13.1) as printed: properness in the full subcategory on Q

The full subcategory on Q. For $Q \subset X$, the morphisms of the full subcategory are the morphisms with source and target in Q. Its source and target maps are the restrictions `Chapter13/GoodSystemRelative · EpGroupoid.subSrc` `Chapter13/GoodSystemRelative · EpGroupoid.subTgt`, valued in Q with the subspace topology. - A subset $A \subset Q$ is *proper in Q* if $t : s^{-1}(A) \to Q$ is proper in the subcategory `Chapter13/GoodSystemRelative · EpGroupoid.IsProperSubsetIn`. - A family $(N(q))_{q \in Q}$ is a *good system for Q* `Chapter13/`

GoodSystemRelative · EpGroupoid.IsGoodIn if for $q, q' \in Q$: - $U(q) \subset Q$; - $\mathrm{cl}_Q U(q)$ is proper in Q (Definition 13.1.2 (1) read in Q); - condition (3) holds on $\mathcal{U}(q, q')$.

Conditions (2) and (3) are the same in Q and in X: - a natural representation of G_q whose domain lies in Q is one for the ep-groupoid Q; - $\mathcal{U}(q, q')$ is the same set of morphisms in both, since the subcategory is full.

No sub-ep-groupoid structure is built. The subcategory is its set of morphisms with the restricted maps, which is all the printed hypothesis refers to.

Every good system of X is a good system for $Q = X$ Chapter13/GoodSystemRelative · EpGroupoid.GoodSystem.isGoodIn_univ. With Theorem 13.1.8 this is the witness of the notion.

Properness by ultrafilters (extra) Chapter13/GoodSystemRelative · EpGroupoid.isProperMap_iff_ultrafilter_of_isInducing. Let f be a continuous map defined on a subspace S of a space T. Then f is proper iff the following holds. For every ultrafilter $\mathcal{F}$ on T that contains S and whose image under f converges to some b, $\mathcal{F}$ converges to some $a \in S$ with $f(a) = b$. This is Mathlib's characterization of proper maps by ultrafilters, transported along the inclusion of the subspace. It turns the two properness notions into statements about ultrafilters of morphisms: - in X Chapter13/GoodSystemRelative · EpGroupoid.isProperSubset_iff_ultrafilter; - in Q Chapter13/GoodSystemRelative · EpGroupoid.isProperSubsetIn_iff_ultrafilter.

Two tools. - Chapter13/GoodSystemRelative · EpGroupoid.exists_mor_tgt_mem: if $\pi(Q) = |X|$, every object z has a morphism $z \to q''$ with $q'' \in Q$. It is the inverse of a morphism from a point of Q to z. - Chapter13/GoodSystemRelative · EpGroupoid.exists_localSection: through a morphism χ with target in an open O there is a local section σ of s near $s(\chi)$, continuous, with: - $\sigma(s\chi) = \chi$; - $s(\sigma(z)) = z$; - $t(\sigma(z)) \in O$.

It is the local inverse of the local homeomorphism s at χ, cut down by the continuity of t to the preimage of O.

Step 1: the closure stays in Q Chapter13/GoodSystemRelative · EpGroupoid.closure_subset_of_isProperSubsetIn. Let Q be open with $\pi(Q) = |X|$, $U \subset Q$, and $\mathrm{cl}_Q U$ proper in Q. Then $\mathrm{cl}_X U \subset Q$. 1. Let $z \in \mathrm{cl}_X U$. Choose $\chi : z \to q''$ with $q'' \in Q$, and a local section σ of s through χ, defined on an open $V \ni z$, with targets in Q. 2. Let $\mathcal{F}$ be an ultrafilter finer than the filter of neighbourhoods of z within $U \cap V$. It is nontrivial because $z \in \mathrm{cl}\, U$ and V is a neighbourhood of z. 3. Push $\mathcal{F}$ forward by σ. For $y' \in U \cap V$, $\sigma(y')$ is a morphism with source $y' \in U \subset Q$ and target in Q, a morphism of the subcategory with source in $\mathrm{cl}_Q U$. Its targets $t(\sigma(y'))$ converge to $t(\chi) = q'' \in Q$, by continuity. 4. By properness in Q, read through ultrafilters, the pushed ultrafilter converges in the subcategory to a morphism ρ with $t(\rho) = q''$. 5. The pushed ultrafilter also converges in X to $\sigma(z) = \chi$, by continuity of σ. Limits in the Hausdorff space of morphisms are unique, so $\rho = \chi$. 6. Since ρ is a morphism of the subcategory, its source z lies in Q.

Step 2: properness in X Chapter13/GoodSystemRelative · EpGroupoid.isProperSubset_of_isProperSubsetIn. Under the same hypotheses, $t : s^{-1}(\mathrm{cl}_X U) \to X$ is proper. 1. By Step 1, $\mathrm{cl}_X U = \mathrm{cl}_Q U$. 2. Let $\mathcal{F}$ be an ultrafilter of morphisms with sources in $\mathrm{cl}_X U$ whose targets converge to $y \in X$. 3. Choose $\chi : y \to q''$ with $q'' \in Q$, and a local section ρ of s through χ with targets in Q. 4. $F(\varphi) = \rho(t\varphi) \circ \varphi$ is defined eventually along $\mathcal{F}$. It is a morphism of the subcategory with source $s\varphi \in \mathrm{cl}_Q U$ and target $t(\rho(t\varphi)) \in Q$, and these targets converge to $t(\chi) = q''$. 5. By properness in Q the image ultrafilter converges to some morphism ψ. 6. Then $\varphi = \rho(t\varphi)^{-1} \circ F(\varphi)$ converges to $\chi^{-1} \circ \psi$, by continuity of inversion and composition. This is a limit of $\mathcal{F}$ with target y.

The converse Chapter13/GoodSystemRelative · EpGroupoid.isProperSubsetIn_of_isProperSubset. Properness in X to-

gether with $\mathrm{cl}_X U \subset Q$ gives properness in Q. An ultrafilter in the subcategory whose targets converge in Q has, by properness in X, a limit, whose source lies in $\mathrm{cl}_X U \subset Q$ and whose target is the limit in Q, so it is a morphism of the subcategory.

Consequence `Chapter13/GoodSystemRelative · EpGroupoid.isGoodIn_iff`. For open Q with $\pi(Q) = |X|$, a family is a good system for Q iff it is good on Q in the sense of §XIII.2 and $\mathrm{cl}_X U(q) \subset Q$ for all $q \in Q$.

Theorem 13.1.7, as printed (stronger), p. 602 `Chapter13/GoodSystemRelative · EpGroupoid.theorem_13_1_7_book`. Let Q be open with $|Q| = |X|$ and $(N(q))_{q\in Q}$ a good system for the ep-groupoid Q. Given open $W(y) \ni y$ for $y \notin Q$, there is a good system of X that keeps $N(q)$ for $q \in Q$ and has $U(y) \subset W(y)$ for $y \notin Q$.

Proof. 1. A good system for Q is good on Q `Chapter13/GoodSystemRelative · EpGroupoid.IsGoodIn.isGoodOn`, by Step 2. 2. The stronger form of §XIII.4 applies `Chapter13/GoodSystemExistence · EpGroupoid.theorem_13_1_7`.

No hypothesis is added. The text writes $|Q| = |Y|$ for $\pi(Q) = |X|$. The verdict is "stronger" because the conclusion omits nothing and the smallness condition on the new closures is not required. That condition is a property of the text's construction, not used afterwards in the chapter.

XIII.6 Theorem 13.1.8 (2017: Theorem 13.2): good systems exist on paracompact orbit spaces

Finitely many morphisms into a point, and one sheet for a connected set `Chapter13/GoodSystemExistence · EpGroupoid.exists_nhds_injOn_of_isProperSubset`. Let $A_1, \dots, A_n$ be closed sets with proper closure (the properness property), $A = \bigcup_i A_i$, and $y \in X$. Then there is an open $V \ni y$ such that, for every preconnected set C of morphisms from A to V, s and t are injective on C. 1. Θ *is finite.* Let $\Theta = \{\varphi : s\varphi \in A,\ t\varphi = y\}$. For each i, the morphisms with source in A_i and target y form a fibre of the proper map t on $s^{-1}(A_i)$. The map t is locally injective, so this fibre is discrete and compact, hence finite. A finite union of finite sets is finite. 2. *Separating sheets.* For each $\theta \in \Theta$ choose an open sheet $O'(\theta) \ni \theta$ on which s and t are injective (§XIII.4). Shrink them to be pairwise disjoint, by Hausdorff separation of the finitely many points of Θ. Let $\Omega = \bigcup_\theta O'(\theta)$, an open set. 3. *Closed images.* $B_i = t(s^{-1}(A_i) \setminus \Omega)$ is closed, since $s^{-1}(A_i) \setminus \Omega$ is closed in $s^{-1}(A_i)$ and the proper map t is closed. Also $y \notin B_i$, since every morphism from A_i to y is in $\Theta \subset \Omega$. 4. *The neighbourhood.* $V = \bigcap_i B_i^c$ intersected with $\bigcap_\theta t(O'(\theta))$ is open and contains y. 5. *A connected set lies in one sheet.* Let C be preconnected with sources in A and targets in V. Every $\varphi \in C$ lies in Ω, since otherwise $t\varphi \in B_i$ for some i. The $O'(\theta)$ are disjoint and open, so the preconnected C lies in one of them, on which s and t are injective.

Theorem 13.1.8 (same), p. 604 `Chapter13/GoodSystemExistence · EpGroupoid.theorem_13_1_8`. If $|X|$ is paracompact, then for every family of open $W(x) \ni x$ there is a good system with $U(x) \subset W(x)$ for all x.

Reading. The text's "for all $\in X$" is read as "for all $x \in X$".

Proof, following the text's steps. 1. *Neighbourhoods with proper closure.* For each x choose a neighbourhood $N_0(x)$ with the natural action and proper closure inside $W(x)$ (§XIII.1). 2. *A locally finite shrinking in* $|X|$. The sets $\pi(U_0(x))$ are open in $|X|$, because π is open (Proposition 7.1.8, Part VII), and cover $|X|$. - Paracompactness gives a locally finite open refinement $(\tilde v_a)_a$ with $\tilde v_a \subset \pi(U_0(a))$. - The shrinking lemma for point-finite covers of a normal space gives open $\tilde v_a'$ with $\mathrm{cl}\,\tilde v_a' \subset \tilde v_a$, still covering. The paracompact Hausdorff space $|X|$ is normal (Part VII). - The closures $\mathrm{cl}\,\tilde v_a'$ are locally finite. 3. *The set*

V. Let $V_a = \pi^{-1}(\tilde{v}'_a) \cap U_0(a)$ and $V = \bigcup_a V_a$. - V is open and $\pi(V) = |X|$. - By Lemma 13.1.1 (□), $\operatorname{cl} V_a \subset U_0(a)$, since $\operatorname{cl} \pi(V_a) \subset \operatorname{cl} \tilde{v}'_a \subset \pi(U_0(a))$. So each $\operatorname{cl} V_a$ has the properness property. - The V_a are locally finite, and $\operatorname{cl} V = \bigcup_a \operatorname{cl} V_a$. 4. *Finitely many pieces near each point.* For each x there are an open $G \ni x$ and a finite set F of indices such that every morphism φ with $s\varphi \in \operatorname{cl} V$ and $t\varphi \in G$ has $s\varphi \in \operatorname{cl} V_a$ for some $a \in F$. - Take $G = \pi^{-1}(\tilde{N})$, with $\tilde{N}$ a neighbourhood of $\pi(x)$ meeting only finitely many $\operatorname{cl} \tilde{v}'_a$. - $\pi(s\varphi) = \pi(t\varphi) \in \tilde{N}$, and $s\varphi \in \operatorname{cl} V_a$ for some a; that $\operatorname{cl} \tilde{v}'_a$ meets $\tilde{N}$, so $a \in F$. 5. *The neighbourhoods at points of V.* For $x \in V$: - take G and F from step 4, and V' from the sheet lemma for $A_a = \operatorname{cl} V_a$, $a \in F$, and $y = x$; - let $N_1(x)$ be the invariant core of $N_0(x)$ in $V \cap G \cap V'$.

Then $N_1(x) \subset W(x)$, it has proper closure (a closed subset of $\operatorname{cl} U_0(x)$), and $U_1(x) \subset V$. For a preconnected set C of morphisms from $\operatorname{cl} V$ to $U_1(x)$, steps 4 and the sheet lemma give s, t injective on C. For $x \notin V$ keep $N_1(x) = N_0(x)$. 6. *Good on V.* For $q, q' \in V$ and a component Σ of $\mathcal{U}(q, q')$: - the sources of Σ lie in $U_1(q) \subset V \subset \operatorname{cl} V$ and its targets in $U_1(q')$; - Σ is connected, so by step 5 at q' the maps s and t are injective on it; - Σ is open, so it is a diffeo-component.

The other two conditions of "good on V" are in step 5. 7. *Extension.* The stronger Theorem 13.1.7 (§XIII.4), with $Q = V$, gives a good system that keeps $N_1(q)$ on V and has $U(y) \subset W(y)$ off V.

Where the formal proof departs from the text. The steps are the text's. The separation of the finitely many morphisms in the sheet lemma uses only Hausdorff separation and the properness of t, and metrizability of $|X|$ is not used anywhere. The neighbourhoods off V come from the stronger Theorem 13.1.7, so the smallness of their closures that the printed Theorem 13.1.7 arranges is not needed.

The witness `Chapter13/GoodSystemExistence · EpGroupoid.nonempty_goodSystem_trivial`. For every paracompact Hausdorff M-polyfold Z, the trivial ep-groupoid on Z (only units) has paracompact orbit space $|X| = Z$ `Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial`. Theorem 13.1.8 gives it a good system. So Definition 13.1.2 is not vacuous.

XIII.7 Proposition 13.1.9 (2017: Proposition 13.2)

Stabilizers and D_Σ. Let $\mathcal{U}$ be a good system and Σ a component of $\mathcal{U}(x, x')$. - $G_{x,\Sigma}$ is the set of $g \in G_x$ with $g * s(\Sigma) = s(\Sigma)$ `Chapter13/BiAction · EpGroupoid.GoodSystem.stabSrc`. - $G_{x',\Sigma}$ is the set of $g' \in G_{x'}$ with $g' * t(\Sigma) = t(\Sigma)$ `Chapter13/BiAction · EpGroupoid.GoodSystem.stabTgt`.

Both are subgroups. The map

$$D_\Sigma = t \circ (s|_\Sigma)^{-1} : s(\Sigma) \to t(\Sigma)$$

`Chapter13/BiAction · EpGroupoid.GoodSystem.D` is an sc-diffeomorphism with inverse $s \circ (t|_\Sigma)^{-1}$ `Chapter13/BiAction · EpGroupoid.GoodSystem.isScDiffeoOn_D`. It is a composite of the sc-diffeomorphisms $(s|_\Sigma)^{-1}$ and $t|_\Sigma$ of §XIII.2.

Proposition 13.1.9 (stronger), p. 607 `Chapter13/BiAction · EpGroupoid.GoodSystem.proposition_13_1_9`. There is a group isomorphism $G_{x,\Sigma} \to G_{x',\Sigma}$, $g \mapsto g'(g)$, with

$$(s|_\Sigma)^{-1}(g * z) = g'(g) * (s|_\Sigma)^{-1}(z) * g^{-1} \quad \text{for } z \in s(\Sigma),$$

and every map $G_{x,\Sigma} \to G_{x',\Sigma}$ with this property equals it.

Proof. 1. *The partner exists.* Let $g \in G_{x,\Sigma}$ and $\varphi \in \Sigma$. The morphism $1 * \varphi * g^{-1}$ has source $g * s(\varphi)$, which lies in $s(\Sigma)$ since g stabilizes $s(\Sigma)$. So there is $\psi \in \Sigma$ with the same source, and by the first tool of §XIII.3,

$\psi = g' * (1 * \varphi * g^{-1}) * 1 = g' * \varphi * g^{-1}$ for some g'. Thus there is g' with $g' * \varphi * g^{-1} \in \Sigma$. 2. *It is unique* `Chapter13/BiAction · EpGroupoid.GoodSystem.partner`. - *Left uniqueness.* If $g'_1 * \varphi * g^{-1}$ and $g'_2 * \varphi * g^{-1}$ both lie in Σ, they have the same source, so they are equal by injectivity of s on Σ. Left cancellation in the bi-action gives $g'_1 = g'_2$. - *Right uniqueness.* If $g' * \varphi * g_1$ and $g' * \varphi * g_2$ both lie in Σ, injectivity of t on Σ and right cancellation give $g_1 = g_2$. 3. *It does not depend on φ.* The bi-action by (g', g^{-1}) maps the component of φ, which is Σ, onto the component of $g' * \varphi * g^{-1}$, which is Σ again. So $g' * \psi * g^{-1} \in \Sigma$ for every $\psi \in \Sigma$, with the same g'. 4. *It lies in $G_{x',\Sigma}$.* $t(g' * \psi * g^{-1}) = g' * t(\psi)$, so $g' * t(\Sigma) \subset t(\Sigma)$, with equality by the same argument for the inverse. And it satisfies the identity: for $z = s(\psi)$, $(s|_\Sigma)^{-1}(g * z) = g' * \psi * g^{-1}$, since the right side lies in Σ and has source $g * z$. 5. *A homomorphism* `Chapter13/BiAction · EpGroupoid.GoodSystem.partnerHom`. For g, h: $(g'(g)g'(h)) * \varphi * (gh)^{-1} = g'(g) * (g'(h) * \varphi * h^{-1}) * g^{-1}$ by the action law, and it lies in Σ. Left uniqueness gives $g'(gh) = g'(g)g'(h)$. 6. *Bijective.* - Injective: if $g'(g_1) = g'(g_2)$ then $g'(g_1) * \varphi * g_1^{-1}$ and $g'(g_1) * \varphi * g_2^{-1}$ lie in Σ, and right uniqueness gives $g_1 = g_2$. - Surjective: for $k \in G_{x',\Sigma}$, the second tool of §XIII.3 applied to $k * \varphi * 1$ and an element of Σ with the same target gives h with $k * \varphi * h \in \Sigma$. One checks $h^{-1} \in G_{x,\Sigma}$, and its partner is k. 7. *Uniqueness among all maps.* If $f : G_{x,\Sigma} \to G_{x',\Sigma}$ satisfies the identity, then at $z = s(\varphi)$ the identity says $f(g) * \varphi * g^{-1} = (s|_\Sigma)^{-1}(g * z) \in \Sigma$. Left uniqueness gives $f(g) = g'(g)$.

Equivariance of D_Σ `Chapter13/BiAction · EpGroupoid.GoodSystem.proposition_13_1_9_D`: $\Phi'(g'(g)) \circ D_\Sigma = D_\Sigma \circ \Phi(g)$ on $s(\Sigma)$. Apply t to the identity of the proposition: $t(g' * \psi * g^{-1}) = g' * t(\psi)$.

Where the formal proof departs from the text, and why stronger. The text proves the identity through the smoothness of a map Θ_g and a connectedness argument on it, and states uniqueness for homomorphisms. The formal proof uses only Lemma 13.1.6 in its one-morphism form, with the action laws and the cancellations. The uniqueness it proves is among all maps, not only homomorphisms.

XIII.8 Definition 13.2.1 recalled, sc⁺-sections on open sets, and Definitions 13.2.2–13.2.4 (2017: Definitions 13.4–13.7): supports, norms, triviality

Definition 13.2.1 (p. 607). $\Lambda : W \to \mathbb{Q}^+$ is an sc⁺-multisection in Chapter 11's sense `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsScPlusMultisection` (Definition 13.2.1, Part XI) if two conditions hold. - Λ is a functor: $\Lambda(\mu(\varphi, w)) = \Lambda(w)$ whenever $P(w) = s(\varphi)$. - Every point x has a *local section structure*: an open $O \ni x$, finitely many sc⁺-sections $s_1, \dots, s_k$ on O, and weights $\sigma_1, \dots, \sigma_k \in \mathbb{Q}^+$, $\sigma_i > 0$, with $\sum_i \sigma_i = 1$ and

$$\Lambda(w) = \sum_{i : s_i(P(w)) = w} \sigma_i \quad \text{for } P(w) \in O$$

`Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn`.

This is the definition the whole chapter builds on. The structures of §§13.2–13.5 are additional data on such a functor.

Sc⁺-sections on an open set. An sc⁺-section on an open U is a section of P over U with values in $W_{0,1}$ that is sc-smooth as a map into $W_{0,1}$ (Chapter 11). Chapter 13 needs sums, multiples and cut-offs of such sections. The formalization proves the following. - *Characterization by charts.* Being sc⁺ on U can be tested on chart representatives near each point: the representative of the section in a strong bundle chart, read on

the $(0,1)$-level Chapter13/ScPlusSectionOn · SBLocalization.isScPlusSectionOn_of_localRepOne. - *Values in $W_{0,1}$.* The values of an sc⁺-section lie in $W_{0,1}$ Chapter13/ScPlusSectionOn · SBLocalization.IsScPlusSectionOn.mem_oneSet. - *Sums and multiples.* $s_1 + s_2$ is sc⁺ on U Chapter13/ScPlusSectionOn · SBLocalization.IsScPlusSectionOn.sectionAdd. So is βs for an sc-smooth $\beta : U \to \mathbb{R}$ Chapter13/ScPlusSectionOn · SBLocalization.IsScPlusSectionOn.sectionSmul, and in particular as for a constant a Chapter13/ScPlusSectionOn · SBLocalization.IsScPlusSectionOn.sectionSmul_const. Finite sums are sc⁺ Chapter13/ScPlusSectionOn · SBLocalization.isScPlusSectionOn_sectionSum, and the zero section is sc⁺ Chapter13/ScPlusSectionOn · SBLocalization.isScPlusSectionOn_fiberZero. In charts the fibres are linear and the operations are sc-smooth maps of the fibre coordinates, sc⁺ preserved. - *Extension by zero.* A section defined on U is extended by 0 off U Chapter13/ScPlusSectionOn · SBLocalization.extendZero, agrees with the original on U Chapter13/ScPlusSectionOn · SBLocalization.extendZero_eqOn, and is again sc⁺ on U Chapter13/ScPlusSectionOn · SBLocalization.IsScPlusSectionOn.extendZero. A local section structure can be replaced by the extended sections Chapter13/ScPlusSectionOn · SBLocalization.IsLocalRepOn.extendZero. This makes all local sections functions on all objects, which the transports of Lemma 13.2.6 need.

Definition 13.2.2 (same), p. 608 Chapter13/MultisectionSupport · SBLocalization.domSupp Chapter13/MultisectionSupport · SBLocalization.supp Chapter13/MultisectionSupport · SBLocalization.suppAt. - The *domain support* is $\mathrm{dom\text{-}supp}(\Lambda) = \mathrm{cl}\{x : \exists w \in W_x,\ w \neq 0_x,\ \Lambda(w) > 0\}$. - The *support* is $\mathrm{supp}(\Lambda) = \{w : \Lambda(w) > 0\}$, with fibres $\mathrm{supp}(\Lambda)(x)$.

The text then claims several facts. Each is proved. Let Λ be an sc⁺-multisection and (s_i, σ_i) a local section structure on O. 1. *On O the support is the set of section values* Chapter13/MultisectionSupport · SBLocalization.IsLocalRepOn.suppAt_eq: $\mathrm{supp}(\Lambda)(x) = \{s_1(x), \dots, s_k(x)\}$ for $x \in O$. If $w = s_i(x)$ then $\Lambda(w) \geq \sigma_i > 0$. Conversely, if $\Lambda(w) > 0$ the sum defining $\Lambda(w)$ is nonempty, so $w = s_i(x)$ for some i. 2. *Finite* Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.finite_suppAt: from 1, at most k elements. 3. *The weights over a point sum to* 1 Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.sum_suppAt Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.finsum_suppAt. Group the indices i by the value $s_i(x)$: $\sum_{w \in \mathrm{supp}(\Lambda)(x)} \Lambda(w) = \sum_w \sum_{i : s_i(x) = w} \sigma_i = \sum_i \sigma_i = 1$. 4. *Nonempty* Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.suppAt_nonempty: an empty sum is $0 \neq 1$. 5. *Domain support as a closure* Chapter13/MultisectionSupport · SBLocalization.domSupp_eq_closure_image: $\mathrm{dom\text{-}supp}(\Lambda) = \mathrm{cl}\, P(\mathrm{supp}(\Lambda) \setminus 0_W)$, which is the definition rewritten.

Definition 13.2.3 (restated), p. 608 Chapter13/MultisectionSupport · SBLocalization.pointwiseNorm. For an auxiliary norm N on $W_{0,1}$ (Definition 8.3.5, Part VIII), the pointwise norm is $N(\Lambda)(x) = \max\{N(w) : w \in \mathrm{supp}(\Lambda)(x)\}$. - N is read on all of W by setting it 0 off $W_{0,1}$ Chapter13/MultisectionSupport · SBLocalization.normW. The values of sc⁺-sections, and so the support, lie in $W_{0,1}$ Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.suppAt_subset_oneSet, where N is the given norm. - The maximum is written as a supremum. It is attained Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.isGreatest_pointwiseNorm, because the support over x is finite and nonempty (2 and 4 above). - For an auxiliary norm, $\mathrm{dom\text{-}supp}(\Lambda) = \mathrm{cl}\{x : N(\Lambda)(x) > 0\}$ Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.domSupp_eq_closure_pointwiseNorm. $N(\Lambda)(x) > 0$ iff some $w \in \mathrm{supp}(\Lambda)(x)$ has $N(w) > 0$, iff some such w is not 0_x, since an auxiliary norm vanishes exactly at 0 Chapter13/MultisectionSupport · SBLocalization.normW_eq_zero_iff.

Restated: the maximum of the text is a supremum shown to be a maximum, and N is extended by 0 off $W_{0,1}$, where it is never evaluated on the support.

Definition 13.2.4 (same), p. 609 Chapter13/MultisectionSupport · SBLocalization.IsTrivialOn. Λ is trivial on V if $\Lambda(0_x) = 1$ for all $x \in V$. - The text's "and hence $\Lambda(w) = 0$ for $w \neq 0_x$" is proved: $\Lambda(0_x) = 1$ iff $\Lambda(w) = 0$ for all $w \neq 0_x$ in W_x Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.apply_

`fiberZero_eq_one_iff`, because the weights over x sum to 1 (3 above). - dom-supp(Λ) is the smallest closed set off which Λ is trivial `Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.isLeast_domSupp`. - Off dom-supp(Λ) no $w \neq 0$ has positive weight, so Λ is trivial there. - If Λ is trivial off a closed C, then every x with some $w \neq 0_x$, $\Lambda(w) > 0$, lies in C, and so does the closure of the set of such x. - The *zero multisection* has $\Lambda(0_x) = 1$ and $\Lambda(w) = 0$ for $w \neq 0$. It is an sc⁺-multisection, with the zero section of weight 1, and it is trivial everywhere `Chapter13/MultisectionSupport · SBLocalization.isTrivialOn_zeroMultisection`. It is the witness for every notion of this section.

XIII.9 Definition 13.2.5 (2017: Definition 13.8) and Lemma 13.2.6 (2017: Lemma 13.3): symmetric structures

Transported sections `Chapter13/MultisectionSymmetric · SBLocalization.transportSection`. For a map τ from objects to morphisms and a map b from objects to W with $P(b(z)) = s(\tau(z))$, the transported section is $z \mapsto \mu(\tau(z), b(z))$. It is an sc⁺-section on U when: - τ is sc-smooth on U; - b is sc⁺ along $s \circ \tau$; - $t(\tau(z)) = z$

`Chapter13/MultisectionSymmetric · SBLocalization.isScPlusSectionOn_transportSection`. The action μ is an sc-smooth strong bundle map on the fibred product (Chapter 8, Part VIII), and it preserves the $(0,1)$-level. Every construction in this chapter that moves a section along morphisms is a transported section.

Definition 13.2.5 (restated), p. 610 `Chapter13/MultisectionSymmetric · SBLocalization.SymmetricStructure`. A symmetric structure of Λ at x consists of: - the domain $U(x)$ of a natural representation of G_x with the properness property; - a finite index set I_x with an action of G_x; - sc⁺-sections s_i on $U(x)$, $i \in I_x$, with - (1) $s_{g(i)}(g * y) = \mu(\Gamma(g, y), s_i(y))$ for $y \in U(x)$; - (2) $\Lambda(w) = |I_x|^{-1} \#\{i : s_i(P(w)) = w\}$ for $P(w) \in U(x)$.

Restated: "G_x-invariant with the natural action" is read as the domain of a natural representation, the reading of Chapter 11. The index set lives in the universe of the isotropy groups. The version on neighbourhoods with the natural action, which the good systems need, is below.

Equal weights `Chapter13/MultisectionSymmetric · SBLocalization.IsLocalRepOn.exists_equal_weights`. A local section structure $(s_i, \sigma_i)_{i \le k}$ on O can be rewritten with all weights equal. 1. Write $\sigma_i = n_i/d_i$ and $p = \prod_j d_j$. 2. Set $m_i = n_i \prod_{j \neq i} d_j$, so $\sigma_i = m_i/p$. 3. Take the index type $\coprod_i \{1, \dots, m_i\}$, the section s_i on each of the m_i copies, and the common weight $1/p$. 4. The count is unchanged: $\sum_{(i,r):s_i=w} 1/p = \sum_{i:s_i=w} m_i/p$.

The total number of indices is $\sum_i m_i = p \sum_i \sigma_i = p$. This is the text's $k_j q$.

Lemma 13.2.6 (same), p. 610 `Chapter13/MultisectionSymmetric · SBLocalization.lemma_13_2_6`. Every sc⁺-multisection has a symmetric structure at every point. The proof is the text's, before Definition 13.2.5. 1. *A local structure on a natural domain.* - Take a local section structure (s_i, σ_i) on an open $O \ni x$. - Take a natural representation of G_x with the properness property on a domain $U \subset O$ (Theorem 7.1.13 with the properness property, Part VII). - Restrict the structure to U and extend the sections by 0 off U (§XIII.8). 2. *Equal weights.* Rewrite with equal weights $1/p$ on an index type I_1 with $|I_1| = p$. 3. *Symmetrize.* - Let $\hat{I} = I_1 \times G_x$, with $h \cdot (j, g) = (j, hg)$ `Chapter13/MultisectionSymmetric · SBLocalization.prodIsotropyAction`. - Let $S_{(j,g)}(z) = \mu(\Gamma(g, g^{-1} * z), t_j(g^{-1} * z))$, the section t_j moved by g: the transported section with $\tau(z) = \Gamma(g, g^{-1} * z)$, whose target is z. - These are sc⁺ on U `Chapter13/MultisectionSymmetric · SBLocalization.isScPlusSectionOn_transportSection`, since $\Phi(g^{-1})$ and $\Gamma(g, \cdot)$ are sc-smooth. 4. *Equivariance (1).* We need $S_{(j,hg)}(h * z) = \mu(\Gamma(h, z), S_{(j,g)}(z))$. - The left side is

$\mu(\Gamma(hg, g^{-1} * z), t_j(g^{-1} * z))$. - By $\Gamma(hg, u) = \Gamma(h, g * u) \circ \Gamma(g, u)$ with $u = g^{-1} * z$, and the composition law $\mu(\psi \circ \chi, w) = \mu(\psi, \mu(\chi, w))$, this is the right side. 5. *The count (2).* Fix w with $P(w) = z \in U$. - For each g, let $w_g = \mu(\Gamma(g, g^{-1} * z)^{-1}, w)$, a vector over $g^{-1} * z$. - $\mu(\Gamma(g, \cdot), \cdot)$ is a bijection of fibres, so $S_{(j,g)}(z) = w$ iff $t_j(g^{-1} * z) = w_g$. - Λ is a functor, so $\Lambda(w_g) = \Lambda(w)$. By the equal-weight structure, $\#\{j : t_j(g^{-1} * z) = w_g\} = p\Lambda(w)$. - Summing over the $|G_x|$ values of g: $\#\{(j, g) : S_{(j,g)}(z) = w\} = |G_x| p\Lambda(w) = |\hat{I}|\Lambda(w)$.

Neighbourhoods with the same action `Chapter13/SymmetricOn · EpGroupoid.NatNbhd.Le`. For two neighbourhoods N_1, N_2 of x with the natural action, $N_1 \leq N_2$ means: - $U_1 \subset U_2$; - $\Gamma_1(g, z) = \Gamma_2(g, z)$ for $z \in U_1$, hence also $\Phi_1(g) = \Phi_2(g)$ on U_1.

It is reflexive and transitive. Two neighbourhoods with the same natural representation are comparable.

A common refinement `Chapter13/SymmetricOn · EpGroupoid.NatNbhd.exists_le_of_finite`. Given a neighbourhood N_0, finitely many neighbourhoods $N_1, \dots, N_m$ of x with the natural action, and an open $V \ni x$, there is $M \leq N_0$, $M \leq N_i$ for all i, with $U_M \subset V$ and the representation of N_0. 1. Two natural representations of G_x agree on the connected component of x in the intersection of their domains, by the uniqueness of natural representations (Theorem 7.1.13, Part VII). 2. The component of x in the open set $U_{R_0} \cap \bigcap_i U_{R_i}$ is open, by local connectedness. 3. On it all the Γ_i agree with Γ_0. 4. The invariant core (§XIII.1) of N_0 inside this component, intersected with V and all U_i, is the required M.

This lemma is what makes the intersections in Definitions 13.2.9 and 13.2.12 meaningful.

Moving vectors along morphisms `Chapter13/SymmetricOn · SBLocalization.fibAct`. $\mu(\psi, v)$ is written for v over $s(\psi)$, and set to v otherwise, so that it is a total function. It lands over $t(\psi)$ and maps 0 to 0. It is additive and homogeneous on sections: - $\mu(\psi, s_1 + s_2) = \mu(\psi, s_1) + \mu(\psi, s_2)$ `Chapter13/SymmetricOn · SBLocalization.fibAct_sectionAdd`; - $\mu(\psi, \beta s) = \beta\mu(\psi, s)$ `Chapter13/SymmetricOn · SBLocalization.fibAct_sectionSmul`; - $\mu(\psi, \sum_l s_l) = \sum_l \mu(\psi, s_l)$ `Chapter13/SymmetricOn · SBLocalization.fibAct_sectionSum`,

because $\mu(\psi, \cdot)$ is a linear isomorphism of fibres (Definition 8.3.1, Part VIII). These identities make every new family below equivariant and compatible with correspondences.

Symmetric structures on a neighbourhood with the natural action `Chapter13/SymmetricOn · SBLocalization.SymmetricStructureOn`. This is Definition 13.2.5 on a neighbourhood N of §XIII.1 instead of a natural domain. It consists of: - a finite index type I with a G_x-action; - sc$^+$-sections s_i on U, equivariant through Γ of N; - the counting formula over U.

Its first properties: - $I \neq \emptyset$ `Chapter13/SymmetricOn · SBLocalization.SymmetricStructureOn.card_pos`. Over x the support of Λ is nonempty (§XIII.8), and a nonempty count needs an index. - $|I|^{-1}$ summed over I is 1. - *Where $\Lambda(0_y) = 1$ all sections vanish at y* `Chapter13/SymmetricOn · SBLocalization.SymmetricStructureOn.s_eq_fiberZero`. The count of 0_y is $|I|^{-1}\#\{i : s_i(y) = 0_y\} = 1$, so every i counts. The tight structures (§XIII.17) and the atomic structures (§XIII.26) use this. - *Restriction* `Chapter13/SymmetricOn · SBLocalization.SymmetricStructureOn.restrict`. A structure on N restricts to any $M \leq N$, with the same index set and sections. Equivariance needs that Γ of M is Γ of N on U_M, which is what $\leq$ says. - The structures of Definition 13.2.5 are structures on the neighbourhood given by their domain `Chapter13/SymmetricOn · SBLocalization.SymmetricStructure.toOn`. So by Lemma 13.2.6 they exist at every point `Chapter13/SymmetricOn · SBLocalization.exists_symmetricStructureOn`.

The algebra of structures is §§XIII.11–XIII.13.

XIII.10 Definitions 13.2.7 and 13.2.8 (2017: Definitions 13.9 and 13.10): decorated sc⁺-multisections

Good systems with smaller sets. The decorations need good systems whose neighbourhoods are smaller than a given system's and whose natural actions may be different representations. `Chapter13/DecoratedMultisection · EpGroupoid.GoodSystem.ofSubset`: if $(M(x))$ are neighbourhoods with the natural action, with $U_M(x) \subset U(x)$ for a good system $\mathcal{U}$, then $(M(x))$ is a good system. - Properness passes to closed subsets. - Condition (3) passes to subsets, as in Proposition 13.1.3.

The proof of Proposition 13.1.3 does not use that the representations are the same.

For two good systems there is a third inside both `Chapter13/DecoratedMultisection · EpGroupoid.GoodSystem.exists_subset_inter`. At each x the invariant core of $U_1(x)$ in $U_1(x) \cap U_2(x)$ is a neighbourhood with the natural action of the first system, and the previous lemma applies.

Definition 13.2.7 (same), p. 610 `Chapter13/DecoratedMultisection · SBLocalization.Decoration` `Chapter13/DecoratedMultisection · SBLocalization.DecoratedTuple`. A decoration of an sc⁺-multisection Λ is a pair $(\mathcal{U}, S)$: - $\mathcal{U}$ is a good system; - for every x, $S(x)$ is a symmetric structure of Λ on the neighbourhood $U(x)$.

A decorated tuple is $(\Lambda, \mathcal{U}, S)$ with Λ an sc⁺-multisection. The properness clause of Definition 13.2.5 is Definition 13.1.2 (1), and the natural action is the good system's (§XIII.1).

Existence `Chapter13/DecoratedMultisection · SBLocalization.exists_decoration`. Every sc⁺-multisection over an ep-groupoid with paracompact orbit space has a decoration. 1. At each x take a symmetric structure $S_0(x)$ on a neighbourhood $N(x)$ (Lemma 13.2.6). 2. Theorem 13.1.8 gives a good system $\mathcal{U}_0$ with $U_0(x) \subset U_{N(x)}$. 3. At each x take, by the common refinement lemma of §XIII.9, $M(x) \leq N_0(x)$ and $M(x) \leq N(x)$. On $M(x)$ the actions of the good system and of the structure agree. 4. $(M(x))$ is a good system by the lemma above, and $S_0(x)$ restricts to $M(x)$.

The paracompactness is the text's: the text's §13.2 assumes it through Theorem 13.1.8. It is witnessed as in the intro.

Definition 13.2.8 (same), p. 611 `Chapter13/DecoratedMultisection · SBLocalization.Decoration.Rel`. $(\Lambda, \mathcal{U}, S) \sim (\Lambda', \mathcal{U}', S')$ if: - $\Lambda = \Lambda'$; - there is a good system $\mathcal{U}''$ with $U''(x) \subset U(x) \cap U'(x)$ for all x; - there are G_x-equivariant bijections $b_x : I_x \to I'_x$ with $s'_{b_x(i)} = s_i$ on $U''(x)$.

Equivariance of b_x is with respect to the actions of the two structures. On $U''(x)$ these are the actions of the respective neighbourhoods. The definition does not ask that the two representations agree; the bijection and the sections carry the comparison.

An equivalence relation. - Reflexive with $\mathcal{U}'' = \mathcal{U}$ and the identity `Chapter13/DecoratedMultisection · SBLocalization.Decoration.Rel.refl`. - Symmetric with the inverse bijections `Chapter13/DecoratedMultisection · SBLocalization.Decoration.Rel.symm`. - Transitive `Chapter13/DecoratedMultisection · SBLocalization.Decoration.Rel.trans`: given $\mathcal{U}''_{12}$ and $\mathcal{U}''_{23}$, take a good system inside both (the lemma above) and compose the bijections. On the common system both equalities of sections hold.

So it is an equivalence `Chapter13/DecoratedMultisection · SBLocalization.DecoratedTuple.equivalence`. Its classes are the decorated sc⁺-multisections `Chapter13/DecoratedMultisection · SBLocalization.DecoratedMultisection`, with the map to the underlying functor `Chapter13/DecoratedMultisection · SBLocalization.DecoratedMultisection.forget`, whose image consists of sc⁺-multisections.

XIII.11 Definition 13.2.9 (2017: Definition 13.11) and the remark that there are no inverses: sums

Convolution Chapter13/MultisectionConvolution · SBLocalization.convAt Chapter13/MultisectionConvolution · SBLocalization.conv. For finitely many functions $\Lambda_1, \dots, \Lambda_n : W \to \mathbb{Q}^+$ and $w \in W_x$,

$$(*_i \Lambda_i)(w) = \sum_{(w_1,\dots,w_n)\in W_x^n,\ \sum_i w_i = w} \prod_i \Lambda_i(w_i).$$

The sum is formally over all tuples in the fibre, as a sum with finite support. It is finite when each Λ_i has finite support over x, which is the case for sc⁺-multisections (§XIII.8). The algebra of §13.2 consists of this one operation.

Expansion of a product of sums. If each Λ_i is given on O by a local structure $(s^i_j, \sigma^i_j)_{j\in J_i}$, then for $P(w) \in O$

$$(*_i \Lambda_i)(w) = \sum_{(j_1,\dots,j_n):\sum_i s^i_{j_i}(P(w))=w} \prod_i \sigma^i_{j_i}$$

Chapter13/MultisectionConvolution · SBLocalization.conv_eq_sum_of_isLocalRepOn. 1. Substitute $\Lambda_i(w_i) = \sum_{j:s^i_j=w_i} \sigma^i_j$ into the convolution. 2. Expand the product of sums as a sum over tuples $(j_1, \dots, j_n)$, distributivity over a product of finite sums. 3. Exchange the sums. For a fixed tuple (j_i) the inner sum over (w_i) has the single term $w_i = s^i_{j_i}$, present iff $\sum_i s^i_{j_i} = w$.

A local structure for the convolution Chapter13/MultisectionConvolution · SBLocalization.isLocalRepOn_conv. On O, the sections $\sum_i s^i_{j_i}$, indexed by tuples, are sc⁺ (sums, §XIII.8), and the weights $\prod_i \sigma^i_{j_i}$ are positive and sum to $\prod_i \sum_j \sigma^i_j = 1$.

A functor Chapter13/MultisectionConvolution · SBLocalization.isWFunctor_conv. $\mu(\varphi, \cdot)$ is a linear isomorphism $W_x \to W_{x'}$, so it maps the decompositions of w bijectively to the decompositions of $\mu(\varphi, w)$. Each factor keeps its value, the Λ_i being functors.

An sc⁺-multisection Chapter13/MultisectionConvolution · SBLocalization.isScPlusMultisection_conv. At each point, intersect the domains of local structures of the Λ_i and use the local structure above.

Definition 13.2.9 for functors. The sum $\Lambda' \oplus \Lambda''(w) = \sum_{w'+w''=w} \Lambda'(w')\Lambda''(w'')$ is the text's formula (1) Chapter13/MultisectionAlgebra · SBLocalization.multisectionAdd. It is the convolution of the pair, through the identification of pairs with functions on a two-element set Chapter13/MultisectionAlgebra · SBLocalization.multisectionAdd_eq_conv. - So it is an sc⁺-multisection Chapter13/MultisectionAlgebra · SBLocalization.definition_13_2_9. - Locally it is given by the sections $s'_i + s''_j$ with weights $\sigma'_i\sigma''_j$ on a common domain Chapter13/MultisectionAlgebra · SBLocalization.multisectionAdd_eq_sum_of_isLocalRepOn, which is the text's (2)–(3) at the level of local structures. - *Commutative* for all functions Chapter13/MultisectionAlgebra · SBLocalization.multisectionAdd_comm, by swapping the two summands of a decomposition. - *Associative* on sc⁺-multisections Chapter13/MultisectionAlgebra · SBLocalization.multisectionAdd_assoc. Both sides are sums over decompositions $w = w_1+w_2+w_3$ with weight $\Lambda_1(w_1)\Lambda_2(w_2)\Lambda_3(w_3)$, exchanged as finite sums because the fibre supports are finite. - *The zero multisection is neutral* Chapter13/MultisectionNoInverse · SBLocalization.multisectionAdd_zeroMultisection. In $\Lambda \oplus \Lambda_0(w)$ only the decomposition $w = w + 0$ contributes.

Definition 13.2.9 for decorations (restated), p. 611 Chapter13/DecoratedMultisection · SBLocalization.definition_13_2_9_decorated. Given decorations $(\mathcal{U}', S')$ of Λ' and $(\mathcal{U}'', S'')$ of Λ'', the decoration

of $\Lambda' \oplus \Lambda''$ Chapter13/DecoratedMultisection · SBLocalization.Decoration.add has: - *Neighbourhoods* inside $U'(x) \cap U''(x)$ Chapter13/DecoratedMultisection · SBLocalization.Decoration.add_U_subset, chosen by the common refinement lemma of §XIII.9 so that the actions of $U'(x)$ and $U''(x)$ agree on them Chapter13/DecoratedMultisection · SBLocalization.Decoration.exists_common. They form a good system, by the lemma of §XIII.10. - *Index sets* $I'_x \times I''_x$ with the diagonal action Chapter13/DecoratedMultisection · SBLocalization.Decoration.add_I. - *Sections* $s_{(i,i')} = s'_i + s''_{i'}$ Chapter13/DecoratedMultisection · SBLocalization.Decoration.add_s_apply.

Reading. The text takes $U(x) = U'(x) \cap U''(x)$. The two natural actions on it need not agree. A symmetric structure needs one action for which both families are equivariant, and Definition 13.1.2 needs one natural representation. So $U(x)$ is an open neighbourhood of x inside the intersection on which the two actions agree. The class does not depend on this choice, since Definition 13.2.8 allows shrinking.

The symmetric structure of the sum Chapter13/SymmetricOn · SBLocalization.SymmetricStructureOn.add. - *Sc⁺:* sums of sc⁺-sections. - *Equivariance:* $s'_{gi}(g*y) + s''_{gi'}(g*y) = \mu(\Gamma, s'_i(y)) + \mu(\Gamma, s''_{i'}(y)) = \mu(\Gamma, s'_i(y) + s''_{i'}(y))$, by additivity of $\mu(\psi, \cdot)$. - *The count:* $|I' \times I''|^{-1} \#\{(i,i') : s'_i + s''_{i'} = w\}$ is the expansion of the convolution with weights $|I'|^{-1}|I''|^{-1}$, so it equals $\Lambda' \oplus \Lambda''(w)$.

Well defined on classes Chapter13/DecoratedMultisection · SBLocalization.Decoration.add_rel. If $(\Lambda', \mathcal{U}'_1, S'_1) \sim (\Lambda', \mathcal{U}'_2, S'_2)$ with bijections b'_x on $\mathcal{U}'_{12}$, and likewise for Λ'', then the sums are equivalent. Use a good system inside all four neighbourhood systems and the two common ones, and the product bijections $b'_x \times b''_x$.

So $\oplus$ is defined on decorated sc⁺-multisections Chapter13/DecoratedMultisection · SBLocalization.DecoratedMultisection.add, compatibly with forgetting Chapter13/DecoratedMultisection · SBLocalization.DecoratedMultisection.forget_add. It is commutative Chapter13/DecoratedMultisection · SBLocalization.DecoratedMultisection.add_comm and associative Chapter13/DecoratedMultisection · SBLocalization.DecoratedMultisection.add_assoc there, through the swap bijection $I' \times I'' \to I'' \times I'$ Chapter13/DecoratedMultisection · SBLocalization.Decoration.add_comm_rel and the reassociation bijection of triple products Chapter13/DecoratedMultisection · SBLocalization.Decoration.add_assoc_rel.

The remark after Definition 13.2.9: no inverses (same), p. 611. "Inverse" is read with respect to the neutral element Λ_0, the zero multisection. 1. *The example.* - Take the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ over the trivial ep-groupoid $\mathbb{R}^n$ Chapter13/MultisectionNoInverse · SBLocalization.NoInverse.A. - For a constant $c \in \mathbb{R}^k$, let Λ_c give weight $\frac{1}{2}$ to the zero section and $\frac{1}{2}$ to the constant section c Chapter13/MultisectionNoInverse · SBLocalization.NoInverse.twoValued. When $c \neq 0$, $\Lambda_c(0) = \Lambda_c(c) = \frac{1}{2}$. - It is an sc⁺-multisection Chapter13/MultisectionNoInverse · SBLocalization.NoInverse.isScPlusMultisection_twoValued, with that two-section local structure everywhere. Constant sections of a trivial bundle are sc⁺. 2. *Two values have total weight at most* 1 Chapter13/MultisectionNoInverse · SBLocalization.NoInverse.add_le_one_of_ne. For an sc⁺-multisection Λ' and $u \neq v$ in one fibre, $\Lambda'(u) + \Lambda'(v) \leq 1$, because the weights over a point sum to 1 (§XIII.8). 3. *No inverse* Chapter13/MultisectionNoInverse · SBLocalization.NoInverse.remark_13_2_9_no_inverse. For $c \neq 0$ and every sc⁺-multisection Λ', $\Lambda_c \oplus \Lambda' \neq \Lambda_0$. - Over $x = 0$, the decompositions $0 = w' + w''$ with $\Lambda_c(w') > 0$ are $w' = 0$ and $w' = c$. - So $(\Lambda_c \oplus \Lambda')(0_x) = \frac{1}{2}\Lambda'(0) + \frac{1}{2}\Lambda'(-c) \leq \frac{1}{2}$, by step 2 with $0 \neq -c$. - But $\Lambda_0(0_x) = 1$. 4. *Such a* Λ_c *exists* when $k \geq 1$ Chapter13/MultisectionNoInverse · SBLocalization.NoInverse.exists_no_inverse, with c the constant vector with all entries 1.

XIII.12 Definition 13.2.10 (2017: Definition 13.12) and Lemma 13.2.11 (2017: Lemma 13.4): multiples and products with functions

Definition 13.2.10 for functors (same), p. 611 `Chapter13/MultisectionAlgebra · SBLocalization.multisectionSmul`. For $a \in \mathbb{R}$: - $(0 \cdot \Lambda)(0_x) = 1$ and $(0 \cdot \Lambda)(w) = 0$ for $w \neq 0_x$; - $(a\Lambda)(w) = \Lambda(a^{-1}w)$ for $a \neq 0$.

Facts: - $0 \cdot \Lambda$ is the zero multisection `Chapter13/MultisectionAlgebra · SBLocalization.multisectionSmul_zero`. - $a\Lambda$ is a functor: $\mu(\varphi, \cdot)$ is linear, so $\mu(\varphi, a^{-1}w) = a^{-1}\mu(\varphi, w)$. - If Λ is given on O by (s_i, σ_i), then $a\Lambda$ is given on O by (as_i, σ_i) `Chapter13/MultisectionAlgebra · SBLocalization.multisectionSmul_eq_sum_of_isLocalRepOn`. - For $a \neq 0$: $as_i(P(w)) = w$ iff $s_i(P(w)) = a^{-1}w$. - For $a = 0$: all sections as_i are 0. The sum of all weights is 1 at $w = 0$ and the empty sum is 0 at $w \neq 0$, which is the zero multisection. - The sections as_i are sc⁺ (constant multiples, §XIII.8). So $a\Lambda$ is an sc⁺-multisection `Chapter13/MultisectionAlgebra · SBLocalization.definition_13_2_10`.

Lemma 13.2.11 for functors (same), p. 612 `Chapter13/MultisectionAlgebra · SBLocalization.lemma_13_2_11`. Let $\beta : X \to \mathbb{R}$ be an sc-smooth functor: sc-smooth on the objects and constant along morphisms, $\beta(t\varphi) = \beta(s\varphi)$. Define $(\beta\Lambda)(w) = (\beta(P(w))\Lambda)(w)$ `Chapter13/MultisectionAlgebra · SBLocalization.multisectionSmulFun`. Then $\beta\Lambda$ is an sc⁺-multisection. 1. *A functor.* For φ with $P(w) = s\varphi$: $(\beta\Lambda)(\mu(\varphi, w)) = (\beta(t\varphi)\Lambda)(\mu(\varphi, w)) = (\beta(s\varphi)\Lambda)(\mu(\varphi, w)) = (\beta(s\varphi)\Lambda)(w)$, using the invariance of β and the functoriality of $\beta(s\varphi)\Lambda$. 2. *Local structure.* Near a point, take (s_i, σ_i) for Λ on O. The sections βs_i are sc⁺ on O (β sc-smooth, §XIII.8). With the weights σ_i they represent $\beta\Lambda$, by the pointwise formula of Definition 13.2.10 at $a = \beta(P(w))$. 3. No case distinction is needed between $\beta \neq 0$ and $\beta = 0$. The local formula of Definition 13.2.10 holds for every a, including 0, and is applied pointwise.

Decorated versions (same). The decoration of $a\Lambda$ `Chapter13/DecoratedMultisection · SBLocalization.Decoration.smul` `Chapter13/DecoratedMultisection · SBLocalization.definition_13_2_10_decorated` keeps the good system and index sets and multiplies the sections by a. - Equivariance is homogeneity of $\mu(\psi, \cdot)$ (§XIII.9). - The count is the local formula above, with all weights $|I_x|^{-1}$. - For $a = 0$ the sections are 0 and the count gives the zero multisection.

The decoration of $\beta\Lambda$ `Chapter13/DecoratedMultisection · SBLocalization.Decoration.smulFun` `Chapter13/DecoratedMultisection · SBLocalization.lemma_13_2_11_decorated` has the sections βs_i. Equivariance of βs_i uses $\beta(g * y) = \beta(y)$, which is the invariance of β along $\Gamma(g, y)$, together with homogeneity of μ.

Both are compatible with Definition 13.2.8 `Chapter13/DecoratedMultisection · SBLocalization.Decoration.smulFun_rel`, with the same bijections. So they are defined on decorated classes `Chapter13/DecoratedMultisection · SBLocalization.DecoratedMultisection.smul`.

XIII.13 Definition 13.2.12 (2017: Definition 13.13) and Lemma 13.2.13 (2017: Lemma 13.5): locally finite sums

Definition 13.2.12 (same), p. 612 `Chapter13/MultisectionLocallyFinite · SBLocalization.IsLocallyFiniteFamily`. A family $(\Lambda_j)_{j \in J}$ of functions $W \to \mathbb{Q}^+$ is locally finite if every x has an open saturated $O(x) \ni x$ for which the set

$$J_{O(x)} = \{j : \exists w \neq 0,\ P(w) \in O(x),\ \Lambda_j(w) > 0\}$$

`Chapter13/MultisectionLocallyFinite · SBLocalization.nontrivIdx` is finite.

The remark after it Chapter13/MultisectionLocallyFinite · SBLocalization.IsLocallyFiniteFamily.exists_stable. $O(x)$ can be chosen so that the index set does not change on smaller open saturated neighbourhoods of x. 1. Among the open saturated $O \ni x$ with J_O finite, choose one whose J_O has the fewest elements, a minimum of natural numbers. 2. For a smaller open saturated $O' \ni x$, $J_{O'} \subset J_O$, and $J_{O'}$ is finite. By minimality it is not smaller, so $J_{O'} = J_O$.

Lemma 13.2.13 for functors (restated), p. 612 Chapter13/MultisectionLocallyFinite · SBLocalization.lemma_13_2_13. The sum of a locally finite family of sc⁺-multisections is an sc⁺-multisection.

The sum written out Chapter13/MultisectionLocallyFinite · SBLocalization.lfSum. The text writes $\bigoplus_{j\in J} \Lambda_j$ without a formula for infinite J. It is read as

$$\Big(\bigoplus_j \Lambda_j\Big)(w) = \sum_{(w_j)\text{ finitely supported},\ \sum_j w_j = w} \prod_j \Lambda_j(w_j),$$

where the product over j is over the finitely many j with $w_j \neq 0$, each other factor $\Lambda_j(0)$ being read as 1 near the point. For finite J this is the convolution of §XIII.11 Chapter13/MultisectionLocallyFinite · SBLocalization.lfSum_eq_conv: a finitely supported family on a finite set is a function, and the identification is a bijection of index sets of the two sums.

Proof. 1. *Trivial members.* For $j \notin J_{O(x)}$ and $y \in O(x)$, Λ_j is trivial at y. Its only positive value over y is at 0_y, with weight 1 (§XIII.8). 2. *Over $O(x)$ the sum is a finite convolution* Chapter13/MultisectionLocallyFinite · SBLocalization.lfSum_eq_conv_of_mem. - In a decomposition (w_j) of w over $y \in O(x)$, every $j \notin J_{O(x)}$ with $w_j \neq 0$ makes the product 0. - So only decompositions supported in the finite set $J_{O(x)}$ contribute, and the factors of the other indices are $\Lambda_j(0) = 1$. - These are exactly the terms of the finite convolution over $J_{O(x)}$. 3. *A functor.* The saturated $O(x)$ contains both ends of every morphism leaving it. Along φ the formula of step 2 is the finite convolution, which is a functor (§XIII.11). 4. *Local structure.* On $O(x)$ the finite convolution has the local structure of §XIII.11, intersected over the finitely many local structures of the Λ_j, $j \in J_{O(x)}$.

Restated: the infinite sum, left implicit in the text, is written out.

Decorated version (restated) Chapter13/DecoratedMultisection · SBLocalization.lemma_13_2_13_decorated. Given decorations $(\mathcal{U}_j, S_j)$ of a locally finite family, and a good system $\mathcal{U}_0$, the decoration of $\bigoplus_j \Lambda_j$ Chapter13/DecoratedMultisection · SBLocalization.Decoration.lfSum has the following data at x. - *Neighbourhood*: inside $O(x) \cap \bigcap_{j\in J_x} U_j(x)$, with $J_x = J_{O(x)}$. It is chosen by the common refinement lemma of §XIII.9 so that the natural actions of $\mathcal{U}_0$ and of all $U_j(x)$, $j \in J_x$, agree on it Chapter13/DecoratedMultisection · SBLocalization.exists_lfData. - *Index set* $\prod_{j\in J_x} I^j_x$, with sections $\sum_{j\in J_x} s^{j,x}_{i_j}$ Chapter13/DecoratedMultisection · SBLocalization.lfSumStructure. - This is the symmetric structure of a finite product Chapter13/SymmetricOn · SBLocalization.SymmetricStructureOn.pi, whose count is the expansion of §XIII.11. - It is then transferred to $\bigoplus_j \Lambda_j$ by step 2 above, since the count formula only depends on Λ through its values on the neighbourhood.

Reading. The intersection is read as for Definition 13.2.9. When $J_x = \emptyset$, the text leaves the natural action on $O(x)$ open. It is taken from $\mathcal{U}_0$, an argument of the construction: any $\mathcal{U}_j$, or Theorem 13.1.8. With $J_x = \emptyset$ the index set is the empty product, a one-point set, and the section is the empty sum 0, which represents the trivial sum.

XIII.14 Definitions 13.3.1 and 13.3.2, Example 13.3.3, Definition 13.3.4 and Remark 13.3.5 (2017: Definitions 13.14 and 13.15, Example 13.1, Definition 13.16, Remark 13.1): correspondences

§13.3 opens with the display (13.5) on p. 613. It names, for two points x, x' with symmetric structures, the finite index sets I_x and $I_{x'}$ that the correspondences relate. The formalization has no separate statement for it. Its content is the setting of Definition 13.3.1.

Definition 13.3.1 (same), p. 613 `Chapter13/Correspondence · SBLocalization.IndexDiagram`. A diagram $I_x \xleftarrow{a} I \xrightarrow{b} I_{x'}$ of finite sets with: - a and b surjective; - all fibres of a of one size; - all fibres of b of one size.

The fibre sizes are compared through cardinalities of subtypes. Two diagrams are equivalent if a bijection $c : I \to I'$ makes both triangles commute `Chapter13/Correspondence · SBLocalization.IndexDiagram.Equiv`. This is an equivalence relation `Chapter13/Correspondence · SBLocalization.IndexDiagram.equivalence`, and $J(x, x')$ is the set of classes, the *correspondences* `Chapter13/Correspondence · SBLocalization.IndexCorr`.

Constructions on classes: - A bijection $e : I_x \to I_{x'}$ gives $[I_x \xleftarrow{\mathrm{id}} I_x \xrightarrow{e} I_{x'}]$. The special case $e : I_x \to I_x$ a permutation by $g \in G_x$ is written $[\mathrm{id}, g]$ `Chapter13/Correspondence · SBLocalization.IndexCorr.ofPerm`. - Relabelling the two outer sets by bijections acts on classes `Chapter13/Correspondence · SBLocalization.IndexCorr.map`. It is well defined because an equivalence c of diagrams is also one of the relabelled diagrams.

Definition 13.3.2 (stronger), p. 614. - *The pseudo-inverse* $\sharp$ reverses a diagram, $[a, b]^\sharp = [b, a]$ `Chapter13/Correspondence · SBLocalization.IndexCorr.swap`. It is an involution `Chapter13/Correspondence · SBLocalization.IndexCorr.swap_swap`, so a bijection $J(x, x') \to J(x', x)$ `Chapter13/Correspondence · SBLocalization.IndexCorr.swapEquiv`. - *The bi-action* is $g' * [a, b] * g = [g^{-1} \circ a, g' \circ b]$ `Chapter13/Correspondence · SBLocalization.IndexCorr.biact`, written as the relabelling by the permutations g^{-1} and g'. - *Stronger*: the text defines the maps; the formalization also proves the action laws, which the text does not state: - $1 * \tau * 1 = \tau$ `Chapter13/Correspondence · SBLocalization.IndexCorr.biact_one_one`; - $g' * (h' * \tau * h) * g = (g'h') * \tau * (hg)$ `Chapter13/Correspondence · SBLocalization.IndexCorr.biact_biact`.

This is an action of $G_{x'} \times G_x^{\mathrm{op}}$, the same shape as the bi-action on morphisms (§XIII.3). It is what Definition 13.3.4 (2) compares with.

Example 13.3.3 (same), p. 614 `Chapter13/Correspondence · SBLocalization.IndexCorr.example_13_3_3`. If $I_x = \{*\}$, then $g' * \tau * g = \tau$ for all g, g'. 1. a is constant, so $g^{-1} \circ a = a$. 2. For $g' \circ b$, build a bijection $c : I \to I$ with $b \circ c = g' \circ b$ fibre by fibre. The fibre $b^{-1}(j)$ and the fibre $b^{-1}(g'^{-1}j)$ have the same size, so there is a bijection between them. Assembling these bijections gives a bijection of I, with $a \circ c = a$ trivially.

Definition 13.3.4 (same), p. 614 `Chapter13/StructurableMultisection · SBLocalization.Decoration.IsCorrespondence`. For a decoration $(\mathcal{U}, S)$ and points x, x', a map τ from morphisms to $J(x, x')$ is a correspondence if on $\mathcal{U}(x, x')$: - (1) it is locally constant and has finitely many values on $\mathcal{U}(x, x')$; - (2) it is equivariant: $\tau(g' * \varphi * g) = g' * \tau(\varphi) * g$, with the bi-action of the good system (§XIII.3) on the left and that of Definition 13.3.2 on the right, for the actions of the symmetric structures on the index sets; - (3) it is compatible with the sections: for every representative $I_x \xleftarrow{a} I \xrightarrow{b} I_{x'}$ of $\tau(\varphi)$, every $k \in I$ and every ψ in the component of φ in $\mathcal{U}(x, x')$,

$$s^{x'}_{b(k)}(t\psi) = \mu(\psi, s^{x}_{a(k)}(s\psi))$$

`Chapter13/StructurableMultisection · SBLocalization.Decoration.Compat`.

τ is a function on all morphisms; only its values on $\mathcal{U}(x, x')$ enter.

Remark 13.3.5 (same), p. 615 `Chapter13/Correspondence · SBLocalization.IndexCorr.sat_iff`. For a relation R on $I_x \times I_{x'}$, "$R(a(k), b(k))$ for all $k \in I$" is invariant under equivalence of diagrams `Chapter13/Correspondence · SBLocalization.IndexDiagram.sat_iff_of_equiv`: the bijection c only renames k. So it is a property of the class `Chapter13/Correspondence · SBLocalization.IndexCorr.Sat`, held by τ iff by one representative iff by every representative. Condition (3) is of this form. The text's remark, that (3) does not depend on the representative, is this lemma.

The saturation property has the expected closure properties, used below: - it passes to products, relabellings and $\sharp$; - it is monotone in R; - it holds for every R that holds everywhere.

XIII.15 Definitions 13.3.6–13.3.11 and Remarks 13.3.7, 13.3.9 (2017: Definitions 13.17–13.20, Remarks 13.2, 13.3): structurable and structured sc⁺-multisections

Definition 13.3.6 (same), p. 615 `Chapter13/StructurableMultisection · SBLocalization.Decoration.Tau`. A *structure* on a decoration $(\mathcal{U}, S)$ is a family of correspondences $\tau_{x,x'}$, for all x, x', with: - (1) $\tau_{x,x}(\Gamma(g, y)) = [\mathrm{id}, g]$ for $y \in U(x)$, $g \in G_x$; - (2) $\tau_{x',x}(\varphi) = \tau_{x,x'}(\varphi^{-1})^\sharp$ for $\varphi \in \mathcal{U}(x', x)$.

A function Λ is *structurable* if it is an sc⁺-multisection and some decoration of it carries a structure `Chapter13/StructurableMultisection · SBLocalization.IsStructurable`. A decorated class is structurable if a representative carries one `Chapter13/StructurableMultisection · SBLocalization.DecoratedMultisection.IsStructurable`. The two notions agree: Λ is structurable iff it underlies a structurable decorated class `Chapter13/StructurableMultisection · SBLocalization.isStructurable_iff`.

The witness, the zero multisection `Chapter13/StructurableMultisection · SBLocalization.isStructurable_zeroMultisection`, over an ep-groupoid with a good system: - at every point the index set $\{*\}$ with the zero section, which is a symmetric structure of the zero multisection; - every $\tau_{x,x'}$ the unit diagram $[\{*\} \leftarrow \{*\} \rightarrow \{*\}]$.

Conditions (1)–(2) hold because, on one-point sets, every correspondence is the unit diagram (Example 13.3.3), and (3) holds because both sides are zero. The good system is from Theorem 13.1.8, so the witness is stated over a paracompact orbit space.

Remark 13.3.7 and Definition 13.3.8 (same), p. 616. The tuples $(\Lambda, \mathcal{U}, S, \tau)$ `Chapter13/StructurableMultisection · SBLocalization.StructuredTuple` are compared `Chapter13/StructurableMultisection · SBLocalization.StructuredTuple.Equiv` as follows. $(\Lambda, \mathcal{U}, S, \tau) \sim (\Lambda', \mathcal{U}', S', \tau')$ if: - $\Lambda = \Lambda'$; - there is a good system $\mathcal{U}''$ with $U''(x) \subset U(x) \cap U'(x)$; - $I_x = I'_x$ for all x; - $\tau = \tau'$ on $\mathcal{U}''(x, x')$.

The equality of index sets is an equality of types, and τ' is compared with τ after transport along it. This is an equivalence relation `Chapter13/StructurableMultisection · SBLocalization.StructuredTuple.equivalence`. For transitivity, use a good system inside both intermediate ones (§XIII.10) and compose the type equalities. Its classes are the *structured sc⁺-multisections* `Chapter13/StructurableMultisection · SBLocalization.StructuredMultisection`.

Reading. As printed, Definition 13.3.8 puts no condition on the sections S, S'. It is formalized as printed. Nothing below uses the sections of an equivalent tuple: every construction on classes is shown independent of the representative through Λ, the good systems and τ. Remark 13.3.7 says the structures are what makes sums well defined; this is §XIII.16.

Remark 13.3.9 (nothing), p. 616: prose on the role of the structures. It has no formal counterpart.

Definition 13.3.10 (same), p. 616: structured sc⁺-multisections are the classes above, and the forgetful map to sc⁺-multisections is $[\Lambda, \mathcal{U}, S, \tau] \mapsto \Lambda$.

Definition 13.3.11 (same), p. 617. - $\Gamma^+_m(P)$, the sc⁺-multisections `Chapter13/StructurableMultisection · SBLocalization.Γm`; - $\Gamma^+_s(P)$, the structurable ones `Chapter13/StructurableMultisection · SBLocalization.Γs`; - $\Gamma^+_{st}(P)$, the structured ones `Chapter13/StructurableMultisection · SBLocalization.Γst`.

$\Gamma^+_s \subset \Gamma^+_m$, and the forgetful map sends Γ^+_{st} onto Γ^+_s `Chapter13/StructurableMultisection · SBLocalization.StructuredMultisection.range_forget`. This is the text's statement that every structurable Λ lifts to a structured one: a structurable Λ has a decoration with a structure, whose class lies over Λ.

XIII.16 Proposition 13.3.12 (2017: Proposition 13.3), Remark 13.3.13 (2017: Remark 13.4) and Lemma 13.3.14 (2017: Lemma 13.6): sums and multiples of structurable sc⁺-multisections

The product of two correspondences `Chapter13/Correspondence · SBLocalization.IndexCorr.prod`. $[a_1, b_1] \times [a_2, b_2] = [a_1 \times a_2, b_1 \times b_2]$, a diagram $I^1_x \times I^2_x \leftarrow I_1 \times I_2 \rightarrow I^1_{x'} \times I^2_{x'}$. - The fibres of a product map are products of fibres `Chapter13/Correspondence · SBLocalization.IndexDiagram.card_fiber_prodMap`, so they have equal sizes. Products of surjections are surjective. - The product is well defined on classes, by the product of the equivalences. - It commutes with $\sharp$ `Chapter13/Correspondence · SBLocalization.IndexCorr.prod_swap`, with the bi-action for the diagonal actions `Chapter13/Correspondence · SBLocalization.IndexCorr.prod_biact`, and with $[\mathrm{id}, g]$ `Chapter13/Correspondence · SBLocalization.IndexCorr.prod_ofPerm`. - A product saturates the product relation when each factor saturates its relation.

Proposition 13.3.12 (restated), p. 617 `Chapter13/StructurableMultisection · SBLocalization.proposition_13_3_12`. The sum of two structurable sc⁺-multisections is structurable.

Proof. 1. *"We may assume $\mathcal{U}_1 = \mathcal{U}_2$."* Let Λ_1, Λ_2 carry structures τ^1, τ^2 over decorations $(\mathcal{U}_1, S_1)$, $(\mathcal{U}_2, S_2)$. The decorated sum (§XIII.11) has neighbourhoods $M(x)$ with $M(x) \leq N_1(x)$ and $M(x) \leq N_2(x)$. - Both structures restrict to the decorations on $(M(x))$ `Chapter13/StructurableMultisection · SBLocalization.Decoration.Tau.restrict`. - On smaller neighbourhoods, $\mathcal{U}_M(x, x') \subset \mathcal{U}_i(x, x')$ and the components are smaller, so local constancy and compatibility restrict. - The bi-action does not change on a smaller neighbourhood with the same action `Chapter13/StructurableMultisection · EpGroupoid.GoodSystem.biact_congr_of_le`, because it is built from the Γ's, which agree there. So equivariance restricts. - The diagonal condition (1) holds because Γ is the same. 2. *The structure of the sum* `Chapter13/StructurableMultisection · SBLocalization.Decoration.Tau.prod` `Chapter13/StructurableMultisection · SBLocalization.Decoration.Tau.add` is $\tau = \tau^1 \times \tau^2$ on the index sets $I^1_x \times I^2_x$ with sections $s^1_i + s^2_j$. - *Local constancy:* both factors are locally constant near φ. - *Finiteness:* the values lie in the product of two finite sets of values. - *Equivariance:* factorwise, through the product lemma for the bi-action. - *(1) and (2) of Definition 13.3.6:* factorwise, through the product lemmas for $[\mathrm{id}, g]$

and ♯. - *Compatibility (3):* the text's computation. For representatives with $s^1_{b_1(k)} = \mu(\psi, s^1_{a_1(k)})$ and $s^2_{b_2(l)} = \mu(\psi, s^2_{a_2(l)})$, $\mu(\psi, s^1_{a_1(k)} + s^2_{a_2(l)}) = s^1_{b_1(k)} + s^2_{b_2(l)}$, by additivity of $\mu(\psi, \cdot)$ (§XIII.9). 3. For structured tuples the sum is again a structured tuple `Chapter13/StructurableMultisection · SBLocalization.StructuredTuple.add`, over $\Lambda_1 \oplus \Lambda_2$ `Chapter13/StructurableMultisection · SBLocalization.StructuredTuple.add_Λ`.

Restated: "we may assume $\mathcal{U}_1 = \mathcal{U}_2$" is carried out by restriction to the neighbourhoods of the decorated sum, on which the two natural actions agree.

Remark 13.3.13 (narrower), p. 618. The structured sums $\Lambda_1 \oplus \Lambda_2$ and $\Lambda_2 \oplus \Lambda_1$ have the same underlying functor. This is `StructuredTuple.add_Λ` with the commutativity of §XIII.11. The remark's negative half is that the two differ as structured classes, since the index sets are $I^1_x \times I^2_x$ and $I^2_x \times I^1_x$. That is a statement that two types are unequal, and it has no formal statement (§XIII.31).

Lemma 13.3.14 (same), p. 618 `Chapter13/StructurableMultisection · SBLocalization.lemma_13_3_14` `Chapter13/StructurableMultisection · SBLocalization.lemma_13_3_14_fun`. If Λ is structurable, so is $r\Lambda$ for every $r \in \mathbb{R}$, including 0, and so is $\beta\Lambda$ for every sc-smooth functor β. - The same τ serves `Chapter13/StructurableMultisection · SBLocalization.Decoration.Tau.smul` `Chapter13/StructurableMultisection · SBLocalization.Decoration.Tau.smulFun`, on the decorations of §XIII.12. - Compatibility (3) is homogeneity: $\mu(\psi, rs_{a(k)}) = r\mu(\psi, s_{a(k)}) = rs_{b(k)}$. - For β, additionally $\beta(t\psi) = \beta(s\psi)$. - Conditions (1), (2) and equivariance do not involve the sections.

XIII.17 The construction before Proposition 13.3.15, Proposition 13.3.15, Remark 13.3.16, Definition 13.3.17 and Corollary 13.3.18 (2017: Proposition 13.4, Remark 13.5, Definition 13.21, Corollary 13.1): tight structures

The trivial set T_Λ `Chapter13/TightMultisection · SBLocalization.trivialSet`. The set of points y that have an open neighbourhood on which Λ is trivial (Definition 13.2.4). - It is open `Chapter13/TightMultisection · SBLocalization.isOpen_trivialSet`. - Λ is trivial on it `Chapter13/TightMultisection · SBLocalization.isTrivialOn_trivialSet`. - Triviality at $s(\psi)$ and at $t(\psi)$ are the same: $\Lambda(0_{t\psi}) = \Lambda(\mu(\psi, 0_{s\psi})) = \Lambda(0_{s\psi})$, Λ being a functor `Chapter13/TightMultisection · SBLocalization.apply_fiberZero_tgt`. For a neighbourhood, transport by a sheet of t.

The construction (stronger), pp. 618–619. Let $(\Lambda, \mathcal{U}, S, \tau)$ be a structured tuple. 1. *Neighbourhoods.* For $x \in T_\Lambda$, shrink $U(x)$ to a neighbourhood with the same action inside $U(x) \cap T_\Lambda$ `Chapter13/TightMultisection · SBLocalization.Decoration.exists_tightNbhd` `Chapter13/TightMultisection · SBLocalization.Decoration.tightNbhd`. This is the common refinement lemma of §XIII.9 with an empty family and $V = T_\Lambda$. For $x \notin T_\Lambda$ keep $U(x)$. 2. *Index sets.* One type that is $\{*\}$ when $x \in T_\Lambda$ and I_x otherwise `Chapter13/TightMultisection · SBLocalization.TightIdx`: the sum of $\{*\}$ restricted to "$x \in T_\Lambda$" and I_x restricted to "$x \notin T_\Lambda$". There are bijections $\{*\} \simeq I'_x$ when $x \in T_\Lambda$ `Chapter13/TightMultisection · SBLocalization.TightIdx.eT` and $I_x \simeq I'_x$ when $x \notin T_\Lambda$ `Chapter13/TightMultisection · SBLocalization.TightIdx.eN`. The text's equalities $I'_x = \{*\}$, $I'_x = I_x$ are these bijections. The G_x-action and the sections are carried along them: the zero section on $\{*\}$, the old ones on I_x. Counts are invariant under such relabelling. 3. *The decoration* `Chapter13/TightMultisection · SBLocalization.Decoration.tight`. On $\{*\}$ the zero section represents Λ on $U'(x) \subset T_\Lambda$, because Λ is trivial there. 4. *The correspondences* `Chapter13/TightMultisection · SBLocalization.Decoration.tightτ`, by the four cases for (x, x'): - both in T_Λ: the unit diagram, relabelled; - $x \in T_\Lambda$,

$x' \notin T_\Lambda$: $[\{*\} \leftarrow I_{x'} \xrightarrow{\mathrm{id}} I_{x'}]$, relabelled `Chapter13/TightMultisection · SBLocalization.Decoration.tightτ_TN`; - $x \notin T_\Lambda$, $x' \in T_\Lambda$: $[I_x \xleftarrow{\mathrm{id}} I_x \to \{*\}]$, relabelled; - neither: the old $\tau_{x,x'}$, relabelled by e_x, $e_{x'}$ `Chapter13/TightMultisection · SBLocalization.Decoration.tightτ_NN`.

It is again structured (stronger) `Chapter13/TightMultisection · SBLocalization.Decoration.Tau.tight`. 1. *Constant correspondences are correspondences*, as far as (1)–(2) of Definition 13.3.4 go `Chapter13/TightMultisection · SBLocalization.Decoration.isCorrespondence_const`. A constant is locally constant with one value. For equivariance, a diagram with one outer set a point and the other mapped by the identity is fixed by the bi-action, by Example 13.3.3 and its mirror image. 2. *Relabelling commutes with $\sharp$ and the bi-action* `Chapter13/TightMultisection · SBLocalization.IndexCorr.swap_map` `Chapter13/TightMultisection · SBLocalization.IndexCorr.biact_map`, so the case "neither" inherits (1)–(2) and conditions (1)–(2) of Definition 13.3.6 from τ. Relabelling the unit diagram by $\{*\} \simeq I'_x$ commutes with permutations. 3. *Compatibility (3)* is the step the text does not check. At a morphism ψ in a component of $\mathcal{U}'(x,x')$ with $x \in T_\Lambda$ or $x' \in T_\Lambda$: - $s\psi \in U'(x)$ and $t\psi \in U'(x')$, and one of these lies in T_Λ. Triviality passes along ψ, so $\Lambda(0) = 1$ at both ends. - Every section of a symmetric structure vanishes where $\Lambda(0_y) = 1$ (§XIII.9). So all sections at both ends are 0 at the relevant points. - Both sides of (3) are 0, since $\mu(\psi, 0) = 0$. 4. *Diagonal (1) of Definition 13.3.6* for $x \in T_\Lambda$: $[\mathrm{id}, g]$ on a point is the unit diagram. For $x \notin T_\Lambda$ it is τ's, relabelled.

Proposition 13.3.15 (restated), p. 619 `Chapter13/TightMultisection · SBLocalization.proposition_13_3_15`. From a structured representative $(\Lambda, \mathcal{U}, S, \tau)$ one gets a structured $(\Lambda, \mathcal{U}', S', \tau')$ with: - (1) for $x \in T_\Lambda$: $I'_x = \{*\}$ with the zero section, and $U'(x) \subset T_\Lambda$ with the action of $U(x)$; - (2) for $x \notin T_\Lambda$: $U'(x) = U(x)$, and equivariant bijections $e_x : I_x \to I'_x$ with $s'_{e_x(i)} = s_i$; - (3) τ' is τ relabelled by e_x, $e_{x'}$, for $x, x' \notin T_\Lambda$; - (4)–(5) the unit-type diagrams otherwise.

It is the construction above `Chapter13/TightMultisection · SBLocalization.Decoration.isTight_tight`. Condition (3) holds for all morphisms, not only on $\mathcal{U}'(x,x')$. *Restated*: the equalities of index sets are the bijections of step 2.

Remark 13.3.16 (nothing), p. 619: prose. No formal counterpart.

Definition 13.3.17 (same), p. 620 `Chapter13/TightMultisection · SBLocalization.Decoration.IsTight` `Chapter13/TightMultisection · SBLocalization.StructuredMultisection.IsTight`. A structured representative is tight if: - (1) $I_x = \{*\}$ with the zero section for $x \in T_\Lambda$; - (2) $\tau_{x,x'}$ is $[* \leftarrow I_{x'} \xrightarrow{\mathrm{id}} I_{x'}]$, resp. $[I_x \xleftarrow{\mathrm{id}} I_x \to *]$, when exactly one of x, x' lies in T_Λ; - (3) $[\{*\} \leftarrow \{*\} \to \{*\}]$ when both do.

"$I_x = \{*\}$" is a bijection $\{*\} \simeq I_x$, and (2)–(3) are stated for every such bijection. All are equal, since a one-point set has one bijection to a given one-point set. A structured class is tight if one representative is.

Corollary 13.3.18 (same), p. 620 `Chapter13/TightMultisection · SBLocalization.corollary_13_3_18` `Chapter13/TightMultisection · SBLocalization.corollary_13_3_18_structured`. Every structurable Λ has a tightly structured version, and every element of Γ_s^+ underlies a tightly structured class. This is Proposition 13.3.15 applied to a structure of Λ.

XIII.18 Proposition 13.3.19 (2017: Proposition 13.5): locally finite sums of structurable sc⁺-multisections

The text proves Proposition 13.3.19 (p. 620) in a paragraph, and leaves to the reader the check that the product of infinitely many correspondences is well defined. The formalization writes that check out. It needs three tools: products of families of correspondences, tight versions over a common good system, and the bookkeeping that near two given points only finitely many factors are not the unit.

Products of families of correspondences `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.pi`. Let $(\tau^l)_{l\in L}$ be correspondences, $\tau^l \in J(I^l, I'^l)$. - *On diagrams* `Chapter13/PiCorrespondence · SBLocalization.IndexDiagram.pi`, the product of $I^l \xleftarrow{a_l} M^l \xrightarrow{b_l} I'^l$ is $\prod_l I^l \leftarrow \prod_l M^l \to \prod_l I'^l$. It is a diagram when $\prod_l M^l$ is finite. The fibres of a product map are products of fibres, so all fibres have one size `Chapter13/PiCorrespondence · SBLocalization.IndexDiagram.card_fiber_pi`. - *On classes*, the product is taken through chosen representatives `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.out`. Any family of representatives gives the same class `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.pi_eq_mk`: the product of the equivalences is an equivalence. So equal factors give equal products `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.pi_congr`. - *Finiteness.* A product of finite types that are one-point outside a finite set is finite `Chapter13/PiCorrespondence · SBLocalization.finite_pi_of_subsingleton`. A correspondence equal to a diagram with one-point middle set has one-point middle set in its chosen representative `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.subsingleton_out_I`, since equivalent diagrams have bijective middle sets. - *Laws.* - The product saturates the conjunction of relations when every factor saturates its relation `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.sat_pi`. - It commutes with $\sharp$ `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.swap_pi` and with the bi-action of the isotropy groups acting factorwise `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.biact_pi`. - The product of the diagrams $[\mathrm{id}, g]$ is $[\mathrm{id}, g]$ for the product action `Chapter13/PiCorrespondence · SBLocalization.IndexCorr.pi_ofPerm`. - *Reindexing.* A symmetric structure can be relabelled along an equivariant bijection of index sets `Chapter13/PiCorrespondence · SBLocalization.SymmetricStructureOn.reindex`. - *Additivity.* $\mu(\psi, \cdot)$ is additive over finite sums of vectors `Chapter13/PiCorrespondence · SBLocalization.fibAct_sum`.

Tight versions over a given good system. The tight version of §XIII.17 lives on the decoration's own good system. The sum needs tight versions of all members over one common good system. A good system $\mathcal{U}$ is *adapted* to a structured decoration $(\mathcal{U}_\Lambda, S, \tau)$ of Λ `Chapter13/TightOn · SBLocalization.Decoration.IsAdapted` if: - $U(x) \subset T_\Lambda$ for $x \in T_\Lambda$; - for $x \notin T_\Lambda$, the neighbourhood $U(x)$ is $\leq U_\Lambda(x)$, the same action on a smaller set.

Over an adapted system: - *The tight structure* `Chapter13/TightOn · SBLocalization.SymmetricStructureOn.tightRel` has index set $\{*\}$ with the zero section at $x \in T_\Lambda$, and I_x with the restricted sections at $x \notin T_\Lambda$. The decoration is `Chapter13/TightOn · SBLocalization.Decoration.tightOn`. - *The correspondences* `Chapter13/TightOn · SBLocalization.Decoration.tightOnτ` are the given τ between points outside T_Λ, and the unit-type diagrams otherwise, as in §XIII.17. They form a structure `Chapter13/TightOn · SBLocalization.Decoration.Tau.tightOn` by the argument of §XIII.17. Compatibility at the trivial points is again the vanishing of all sections where $\Lambda(0) = 1$. - *It is tight* (Definition 13.3.17) `Chapter13/TightOn · SBLocalization.Decoration.isTight_tightOn`.

The data of the proof `Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData` consist of: - the members Λ_l, $l \in L$, sc⁺-multisections; - the local finiteness of Definition 13.2.12; - decorations $(\mathcal{U}^l, S^l)$ with structures τ^l; - a good system $\mathcal{U}_0$, which provides the natural actions at points where no member is

nontrivial.

The index type L lives in the universe of the bundle.

The common good system. 1. $O(x)$ *and* $F(x)$ Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.O Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.F. For each x, $O(x)$ is the open saturated set of Definition 13.2.12. $F(x)$ is the finite set of members nontrivial over $O(x)$; it contains the text's J_x. For $l \notin F(x)$, $O(x) \subset T_{\Lambda_l}$ Chapter13/LocallyFiniteStructurable · SBLocalization.subset_trivialSet_of_not_mem_nontrivIdx: an open set over which no $w \neq 0$ has positive weight is a set on which Λ_l is trivial (§XIII.8). 2. $V(x)$ Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.V. $V(x) = O(x) \cap \bigcap_{l\in F(x),\, x\in T_{\Lambda_l}} T_{\Lambda_l}$, open, containing x. This is the text's "$V(x) \subset T_l$". 3. $M(x)$ Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.exists_M Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.M. A neighbourhood with the representation of $\mathcal{U}_0$ at x, inside $V(x)$, with $M(x) \leq U_0(x)$ and $M(x) \leq U^l(x)$ for all $l \in F(x)$. This is the common refinement lemma of §XIII.9 for the finitely many $U^l(x)$, $l \in F(x)$. It is the text's $U(x) = V(x) \cap \bigcap_{l\in J_x} U^l(x)$, shrunk so that the natural actions agree. 4. *The good system* Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.□ is $(M(x))$, by the lemma of §XIII.10 applied inside $\mathcal{U}_0$. 5. *Adapted to every member* Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.isAdapted. These are properties (3)–(4) of the text's proof. - If $x \in T_{\Lambda_l}$: either $l \in F(x)$, and $M(x) \subset V(x) \subset T_{\Lambda_l}$ by step 2; or $l \notin F(x)$, and $M(x) \subset O(x) \subset T_{\Lambda_l}$ by step 1. - If $x \notin T_{\Lambda_l}$: then $l \in F(x)$, by step 1, and $M(x) \leq U^l(x)$ by step 3. 6. *Tight versions* Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.Dt Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.tt of all members over $\mathcal{U}$. For $l \notin F(x)$, the tight index set at x is a point and its section is 0.

The symmetric structure of the sum. 1. *The index set* $K_x = \prod_{l\in L} K_x^l$, with K_x^l the tight index set of Λ_l at x Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.K. It is finite Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.finite_K: all factors with $l \notin F(x)$ are points. 2. *The base structure* Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.baseS is the locally finite sum structure of §XIII.13 over the finite set $F(x)$, with index set $\prod_{l\in F(x)} K_x^l$. 3. *Restriction to $F(x)$ is a bijection* $K_x \to \prod_{l\in F(x)} K_x^l$ Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.res_bijective. It is injective because the other factors are points, and surjective by extending with the unique point. 4. *The structure s^x* Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.Sx is the base structure relabelled along that bijection. The decoration of $\bigoplus_l \Lambda_l$ is $(\mathcal{U}, s)$ Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.Dsum. 5. *The text's formula* Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.Sx_s_apply: $s_k^x(y) = \sum_{l\in T} s_{k(l)}^{l,x}(y)$ for every finite $T \supset F(x)$. The terms with $l \notin F(x)$ are zero sections Chapter13/LocallyFiniteStructurable · SBLocalization.tightIdx_elim_zero. This independence of T is what the compatibility below needs: at a morphism from $U(x)$ to $U(x')$ both structures are written over $T = F(x) \cup F(x')$.

The correspondences. 1. *Unit factors.* For $l \notin F(x) \cup F(x')$, both x and x' lie in T_{Λ_l}, so $\tau_{x,x'}^l$ is the relabelled unit diagram, the same for every morphism Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.τ_TT. 2. *The product is defined* Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.finite_out. The middle sets of the chosen representatives are points outside the finite $F(x) \cup F(x')$, so their product is finite. 3. *The correspondence* $\tau_{x,x'}(\varphi) = \prod_{l\in L} \tau_{x,x'}^l(\varphi)$ Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.τ. It depends only on the finitely many factors in $F(x) \cup F(x')$ Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.τ_congr.

Definition 13.3.4 for the product Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.

isCorrespondence. 1. *Locally constant.* Each of the finitely many relevant factors is locally constant near φ, so they are simultaneously constant near φ, and the product is too. 2. *Finitely many values.* If $\mathcal{U}(x, x')$ is empty there is nothing to show. Otherwise the tuple of relevant factors takes values in a finite product of finite sets. The product correspondence is a function of that tuple, so it takes finitely many values. 3. *Equivariance.* Factorwise, by the product law for the bi-action. 4. *Compatibility.* The product of the factors' saturations saturates the conjunction (the product law for saturation). Given the conjunction for indices k, k', the sum formula with $T = F(x) \cup F(x')$ and additivity of $\mu(\psi, \cdot)$ give $s^{x'}_{k'}(t\psi) = \sum_l s^{l,x'}_{k'(l)}(t\psi) = \sum_l \mu(\psi, s^{l,x}_{k(l)}(s\psi)) = \mu(\psi, s^{x}_{k}(s\psi))$.

Definition 13.3.6 for the product Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.tau. - *Diagonal:* the product of $[\mathrm{id}, g]$ is $[\mathrm{id}, g]$. - *Swap:* the product of the swaps is the swap of the product; the finiteness of the swapped product is again by the unit factors.

This is the step left to the reader.

Proposition 13.3.19 (restated), p. 620 Chapter13/LocallyFiniteStructurable · SBLocalization.proposition_13_3_19. For a nonempty family of structurable sc⁺-multisections, locally finite in the sense of Definition 13.2.12, the sum $\bigoplus_l \Lambda_l$ is structurable. The data above give the structure Chapter13/LocallyFiniteStructurable · SBLocalization.LFStructData.isStructurable, with Lemma 13.2.13 for the sc⁺-multisection property. For a nonempty family the good system $\mathcal{U}_0$ is taken from any member's structure. - For any family, including the empty one, given a good system Chapter13/LocallyFiniteStructurable · SBLocalization.proposition_13_3_19_of_goodSystem. - For any family over a paracompact $|X|$, through Theorem 13.1.8 Chapter13/LocallyFiniteStructurable · SBLocalization.proposition_13_3_19_of_paracompact.

The sum over the empty family is the zero multisection, which is structurable exactly when a good system exists, as Definition 13.3.6 requires. *Added hypothesis*: for the empty family, the existence of a good system. It is not derived from the text, it is necessary, and its witness is Theorem 13.1.8.

Locally finite domain supports Chapter13/LocallyFiniteSupports · SBLocalization.isLocallyFiniteFamily_of_locallyFinite. The text's hypothesis is that the domain supports form a locally finite family of subsets of X. It implies Definition 13.2.12 for families of functors. 1. For x take an open $V \ni x$ meeting finitely many dom-supp(Λ_l), and $O = \pi^{-1}(\pi(V))$. It is open by Proposition 7.1.8 (Part VII), and saturated. 2. Suppose $\Lambda_l(w) > 0$ for some $w \neq 0$ over $y \in O$. Choose a morphism $\varphi : y \to v$ with $v \in V$. 3. $\mu(\varphi, w) \neq 0$ Chapter13/LocallyFiniteSupports · SBLocalization.μ_ne_fiberZero_lf, since $\mu(\varphi, \cdot)$ is a linear isomorphism. Also $\Lambda_l(\mu(\varphi, w)) = \Lambda_l(w) > 0$. 4. So $v \in$ dom-supp$(\Lambda_l) \cap V$, and l is one of the finitely many indices.

The saturation of the domain supports is not needed. So Proposition 13.3.19 holds with the hypothesis as printed Chapter13/LocallyFiniteSupports · SBLocalization.proposition_13_3_19_locallyFinite, and for any family over a paracompact $|X|$ Chapter13/LocallyFiniteSupports · SBLocalization.proposition_13_3_19_locallyFinite_of_paracompact.

Restated: the index sets K_x are products of types relabelled by a bijection, and the structure is built over one adapted good system rather than "the same $\mathcal{U}$" the text takes. "Locally finite" is proved from the printed hypothesis.

XIII.19 Theorem 13.4.1 (2017: Theorem 13.3): pull-back along a strong bundle equivalence

The common data of a pull-back `Chapter13/StrongBundlePullMap · SBLocalization.PullMap`. Both Theorem 13.4.1 and Theorem 13.4.4 pull local sc⁺-section structures back along a map $\Phi : W \to W'$ covering a functor $F : X \to X'$. What the two proofs use of Φ is the same: - Φ commutes with the actions: $\Phi(\mu(\varphi, w)) = \mu'(F(\varphi), \Phi(w))$; - every point has an open neighbourhood over which Φ is a strong bundle isomorphism onto its image, over a local sc-diffeomorphism of the bases; - F is faithful: two morphisms with the same source, the same target and the same image agree.

Instances: - A strong bundle equivalence (Definition 10.4.1, Part X) gives such data `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.ofEquivalence`. Its base is an equivalence, faithful by definition. - A proper strong bundle covering, 2021 Definition 8.5.9, gives such data `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.ofCovering`. Faithfulness comes from Definition 8.5.1 (3): the map $\varphi \mapsto (F(\varphi), s(\varphi))$ is an sc-diffeomorphism onto the fibred product, so injective.

Consequences of the data. - $P'(\Phi(w)) = F(P(w))$. - On fibres, Φ is a linear homeomorphism $W_y \to W'_{F(y)}$ `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.fiberEquiv`. It is injective on fibres `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.eq_of_obj_eq`. - F is a local sc-diffeomorphism on objects `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.isLocalScDiffeo_obj`. - Φ intertwines the actions in the total form of §XIII.9 `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.obj_fibAct`.

Pulled-back sections `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.pullSec`: $z \mapsto \Phi_z^{-1}(s'(F(z)))$, with Φ_z the fibre map. For an sc⁺-section s' on U' and an open U with $F(U) \subset U'$, it is an sc⁺-section on U `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.isScPlusSectionOn_pullSec`: near each point it is s' composed with a local strong bundle isomorphism and the local sc-diffeomorphism F. It satisfies $\Phi(\text{pull } s'(z)) = s'(F(z))$, and pull $s'(z) = w$ iff $s'(F(z)) = \Phi(w)$ `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.pullSec_eq_iff`.

The pulled-back functor $\Phi^*\Lambda' = \Lambda' \circ \Phi$ `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.pull` is a functor `Chapter13/StrongBundlePullMap · SBLocalization.PullMap.isWFunctor_pull`: $\Lambda'(\Phi(\mu(\varphi, w))) = \Lambda'(\mu'(F\varphi, \Phi w)) = \Lambda'(\Phi w)$.

$\Phi^*\Lambda'$ is an sc⁺-multisection `Chapter13/MultisectionPullback · SBLocalization.PullMap.isScPlusMultisection_pull`. A local structure (s'_i, σ_i) of Λ' on $O' \ni F(x)$ pulls back to $(\text{pull } s'_i, \sigma_i)$ on $F^{-1}(O')$ `Chapter13/MultisectionPullback · SBLocalization.PullMap.exists_isLocalRepOn_pull`. By the last property of pulled-back sections, pull $s'_i(P(w)) = w$ iff $s'_i(F(P(w))) = \Phi(w)$, so the counts agree.

Adapted neighbourhoods `Chapter13/MultisectionPullback · SBLocalization.PullMap.IsPullNbhd`. For a neighbourhood N' of $F(x)$ with the natural action, a neighbourhood N of x is adapted if: - F is injective on U; - $F(U) \subset U'$; - $F(\Gamma(g, z)) = \Gamma'(F(g), F(z))$ for $z \in U$, with $F(g) \in G'_{F(x)}$ the image of g under the isotropy homomorphism.

Then $F(g * z) = F(g) * F(z)$ `Chapter13/MultisectionPullback · SBLocalization.PullMap.IsPullNbhd.Φ_eq`, applying t to the identity.

Existence inside every open $V \ni x$, with the properness property `Chapter13/MultisectionPullback · SBLocalization.PullMap.exists_isPullNbhd`. 1. F is a local sc-diffeomorphism at x: there is an open $U_0 \ni x$ on which it is injective with a local inverse. 2. Let $O = V \cap U_0 \cap F^{-1}(U')$ and N_0 a neighbourhood of x with the natural action and proper closure inside O (§XIII.1). 3. Let E be the set of $y \in U_0(N_0)$ with $F(\Gamma(g, y)) =$

$\Gamma'(F(g), F(y))$ for all g. - For each g, the two sides are continuous lifts of $y \mapsto F(y)$ through s: both have source $F(y)$. - Two continuous lifts through the local homeomorphism s agree on an open set `Chapter13/StrongBundlePullMap · EpGroupoid.isOpen_setOf_eq_of_src`. - E is a finite intersection of such sets, so open. It contains x, since $\Gamma(g, x)$ is g and F maps it to $F(g)$. 4. Restrict N_0 to its invariant core in E (§XIII.1). Properness passes to the closed subset.

The text arranges $F(U(x)) = U'(F(x))$ "without loss of generality". The inclusion with injectivity is all that is used.

Adapted good systems `Chapter13/MultisectionPullback · SBLocalization.PullMap.PullData` `Chapter13/MultisectionPullback · SBLocalization.PullMap.exists_pullData`. For a good system $\mathcal{U}'$ of X' and open sets $V(x)$, there is a good system of X with $U(x) \subset V(x)$ and each $N(x)$ adapted to $N'(F(x))$. - Properness is from the existence lemma. - Condition (3) is the text's argument `Chapter13/MultisectionPullback · SBLocalization.PullMap.isDiffeoComponent_of_isPull`. 1. F maps $\mathcal{U}(x, x')$ into $\mathcal{U}'(F(x), F(x'))$ `Chapter13/MultisectionPullback · SBLocalization.PullMap.mem_morBetween`, and maps components into components `Chapter13/MultisectionPullback · SBLocalization.PullMap.mapsTo_connectedComponentIn`, by continuity. 2. Let ψ_1, ψ_2 in a component Σ have the same source. Their images lie in one component of $\mathcal{U}'$, where s is injective, so $F(\psi_1) = F(\psi_2)$. 3. Then $F(t\psi_1) = F(t\psi_2)$, so $t\psi_1 = t\psi_2$ by injectivity of F on $U(x')$. 4. Faithfulness gives $\psi_1 = \psi_2$. The same for t. - No paracompactness is used: the system is built point by point, and the good-system conditions are checked directly.

F intertwines the bi-actions `Chapter13/MultisectionPullback · SBLocalization.PullMap.IsPullNbhd.mor_biact`: $F(g' * \varphi * g) = F(g') * F(\varphi) * F(g)$, from the identity for Γ and functoriality.

The pulled-back structure. Let $(\Lambda', \mathcal{U}', S', \tau')$ be a structured tuple and take an adapted good system. - *Index sets* $I_x = I'_{F(x)}$, with G_x acting through $F : G_x \to G'_{F(x)}$. - *Sections* pull $s_i'^{F(x)}$ on $U(x)$. The symmetric structure is `Chapter13/MultisectionPullback · SBLocalization.PullMap.pullStructure`, and the decoration `Chapter13/MultisectionPullback · SBLocalization.PullMap.pullDecoration`. - Equivariance: apply Φ^{-1} to the equivariance of s' at $F(g)$, with $F(g*z) = F(g)*F(z)$ and $\Phi\mu = \mu'\Phi$. - Count: the count of s' at $\Phi(w)$. - *Correspondences* $\tau_{x,y}(\varphi) = \tau'_{F(x),F(y)}(F(\varphi))$ `Chapter13/MultisectionPullback · SBLocalization.PullMap.pullTau`. - Local constancy and finiteness come from those of τ', through the continuity of F and the inclusion $F(\mathcal{U}(x, y)) \subset \mathcal{U}'(F(x), F(y))$. - Equivariance: the bi-action on correspondences for actions through a homomorphism is the bi-action of the images `Chapter13/MultisectionPullback · SBLocalization.PullMap.biact_compHom`. With the intertwining of the bi-actions on morphisms, it is that of τ'. - The diagonal $[\mathrm{id}, g]$ for the action through F is $[\mathrm{id}, F(g)]$ `Chapter13/MultisectionPullback · SBLocalization.PullMap.ofPerm_compHom`. - Compatibility: Φ^{-1} applied to that of τ', since F maps components into components. - The structured tuple is `Chapter13/MultisectionPullback · SBLocalization.PullMap.pullTuple`, so $\Phi^*\Lambda'$ is structurable `Chapter13/MultisectionPullback · SBLocalization.PullMap.isStructurable_pull`.

Canonicity `Chapter13/MultisectionPullback · SBLocalization.PullMap.pullTuple_equiv`. Equivalent structured tuples, pulled back over any two adapted good systems, give equivalent tuples. 1. Take an adapted good system inside both pulled-back systems and adapted to the common system of the two given tuples. 2. On it the index sets are equal, $I'_{F(x)}$ for both, and the correspondences agree because the given ones agree on the common system and F maps into it.

So the pull-back is a map of structured classes `Chapter13/MultisectionPullback · SBLocalization.PullMap.pullStructured` `Chapter13/MultisectionPullback · SBLocalization.PullMap.pullStructured_mk`, over the pull-back of functors `Chapter13/MultisectionPullback · SBLocalization.PullMap.forget_pullStructured`.

Theorem 13.4.1 (stronger), pp. 621–622 `Chapter13/MultisectionPullback · SBLocalization.theorem_13_4_1` `Chap-`

ter13/MultisectionPullback · SBLocalization.theorem_13_4_1_structured. For a strong bundle equivalence Φ over the equivalence F: - $\Phi^*\Lambda'$ is structurable when Λ' is; - there is a map of structured classes over Φ^*, given on every representative and every adapted good system by the pulled-back structure:

$$I_x = I'_{F(x)}, \qquad \Phi(s_i^x(z)) = s_i'^{F(x)}(F(z)), \qquad \tau_{x,y}(\varphi) = \tau'_{F(x),F(y)}(F(\varphi)).$$

Stronger: the text's "canonical way" is read as this map. Its independence of all choices is asserted in the text and proved here. No hypothesis is added; paracompactness is not used.

XIII.20 Charts along a morphism: the tool of Proposition 13.4.2 and Theorem 13.4.3

Proposition 13.4.2 and Theorem 13.4.3 both move a structure from a point to an isomorphic point. The text does it in two ways: through $F|_{U(x_y)}$ for points in $F(X)$, and through $t \circ (s|_{U(\psi)})^{-1}$ for points reached by a morphism ψ. The formalization does both moves at once, by one chart along a morphism. This subsection sets the chart up, for a strong bundle equivalence Φ over $F : X \to X'$. Proposition 13.4.2 uses it for F the identity.

The chart at $y \in X'$ Chapter13/SymmetricStructurePush · SBLocalization.PushChart. - *Choices.* F is essentially surjective, so there are $x_y \in X$ and a morphism $\sigma_y : F(x_y) \to y$. - *The transport chart* (Chapter 10, Part X) is

$$Q = t \circ (s|_{\text{sheet of } \sigma_y})^{-1} \circ F$$

near x_y. It comes with morphisms $\psi_u : F(u) \to Q(u)$, ψ_u the morphism of the sheet through σ_y with source $F(u)$. Q is an sc-diffeomorphism from an open $W \ni x_y$ onto an open $W' \ni y$, with inverse Q^{-1}. - A chart is chosen at every y Chapter13/SymmetricStructurePush · SBLocalization.pushChart. - For $y = F(x)$ and σ_y the unit, Q is the text's $F|_{U(x_y)}$.

The isotropy isomorphism Chapter13/SymmetricStructurePush · SBLocalization.conjEquiv. $g \mapsto \psi \circ F(g) \circ \psi^{-1}$, $G_{x_y} \to G'_y$, with $\psi = \psi_{x_y}$, is a group isomorphism (Remark 10.1.12, Part X). It is bijective because F is an equivalence.

Push neighbourhoods Chapter13/SymmetricStructurePush · SBLocalization.IsPushNbhd. For a neighbourhood N of x_y with the natural action, a neighbourhood N' of y with the natural action is a push neighbourhood if: - $U' \subset W'$ and $Q^{-1}(U') \subset U$; - $Q^{-1}(U')$ is invariant under G_{x_y} within the chart; - the natural actions are intertwined (2021 (10.2)–(10.3)): for $u = Q^{-1}(z)$,

$$\Gamma'(\psi F(g)\psi^{-1}, z) = \psi_{g*u} \circ F(\Gamma(g,u)) \circ \psi_u^{-1}.$$

Consequences: - $(\psi F(g)\psi^{-1}) * z = Q(g * Q^{-1}(z))$ Chapter13/SymmetricStructurePush · SBLocalization.IsPushNbhd.Φ_eq, applying t. - The identity $\Gamma'(\cdot, z) \circ \psi_u = \psi_{g*u} \circ F(\Gamma(g,u))$ Chapter13/SymmetricStructurePush · SBLocalization.IsPushNbhd.Γ_comp_ψ.

Existence, with the properness property Chapter13/SymmetricStructurePush · SBLocalization.exists_isPushNbhd. 1. Inside $Q(W_0)$, with W_0 the invariant core of U in the chart domain, take a neighbourhood of y with the natural action and proper closure (§XIII.1). 2. The two lifts $z \mapsto \Gamma'(\psi F(g)\psi^{-1}, z)$ and $z \mapsto \psi_{g*u} \circ F(\Gamma(g,u)) \circ \psi_u^{-1}$ through s are continuous, have source z, and agree at y. So they agree on

an open set, as in §XIII.19. 3. Restrict to the invariant core of that open set.

Pushed sections Chapter13/SymmetricStructurePush · SBLocalization.pushSec: $z \mapsto \mu'(\psi_u, \Phi(s(u)))$ with $u = Q^{-1}(z)$, the transported section of §XIII.9 along $\psi \circ Q^{-1}$. For an sc⁺-section s on U it is an sc⁺-section on U' Chapter13/SymmetricStructurePush · SBLocalization.IsPushNbhd.isScPlusSectionOn_pushSec: - Q^{-1} is sc-smooth; - Φ is a strong bundle map; - $z \mapsto \psi_{Q^{-1}(z)}$ is sc-smooth, a composite of the sheet inverse with F.

The pushed symmetric structure Chapter13/SymmetricStructurePush · SBLocalization.pushStructure. A symmetric structure of Λ on N gives one of $\Phi_*\Lambda$ on N': - $I'_y = I_{x_y}$, with G'_y acting through the inverse of the isotropy isomorphism; - $s'_i =$ push s_i.

Equivariance is the intertwining identity together with $\Phi\mu = \mu'\Phi$ and the composition law of μ'. The count: push $s_i(z) = w'$ iff $s_i(u) = \Phi^{-1}(\mu'(\psi_u^{-1}, w'))$. $\Phi_*\Lambda$ at w' is Λ at that vector, Chapter 11's push-forward being defined through any preimage.

Lifting morphisms Chapter13/MultisectionPushforward · SBLocalization.PushChart.liftMor. For charts T at y and T' at y' and a morphism φ' from W' to $W'_{T'}$, the lift is the unique $\varphi : u \to u'$, $u = Q^{-1}(s\varphi')$, $u' = Q'^{-1}(t\varphi')$, with

$$\psi'_{u'} \circ F(\varphi) = \varphi' \circ \psi_u$$

Chapter13/MultisectionPushforward · SBLocalization.PushChart.comp_liftMor. - *Existence:* F is full, so the morphism $\psi'^{-1}_{u'} \circ \varphi' \circ \psi_u : F(u) \to F(u')$ is $F(\varphi)$ for some φ. - *Uniqueness:* F is faithful.

Properties of the lift: - *Continuous* Chapter13/MultisectionPushforward · SBLocalization.PushChart.continuousOn_liftMor. A map into the morphisms of X is continuous at a point when its source, target and F-image are Chapter13/MultisectionPushforward · SBLocalization.PushChart.continuousAt_of_mor. F on morphisms is a local sc-diffeomorphism, so near $f(a)$ a local inverse recovers f from $F \circ f$, provided f stays in the sheet. That is forced by faithfulness, and by the continuity of $s \circ f$ and $t \circ f$, which pin the sheet. For the lift, source and target are Q^{-1} and Q'^{-1} of continuous functions, and the F-image is $\psi' \circ \varphi' \circ \psi^{-1}$, continuous. - *Injective* Chapter13/MultisectionPushforward · SBLocalization.PushChart.injOn_liftMor: φ' is recovered as $\psi' \circ F(\varphi) \circ \psi^{-1}$. - *Functorial* Chapter13/MultisectionPushforward · SBLocalization.PushChart.liftMor_comp Chapter13/MultisectionPushforward · SBLocalization.PushChart.liftMor_inv: by uniqueness of lifts. - *Compatible with* Γ Chapter13/MultisectionPushforward · SBLocalization.PushChart.liftMor_Γ: the lift of $\Gamma'(\psi F(g)\psi^{-1}, z)$ is $\Gamma(g, Q^{-1}z)$, by the intertwining identity. - *Compatible with the bi-action* Chapter13/MultisectionPushforward · SBLocalization.PushChart.liftMor_biact: from functoriality and compatibility with Γ. - *Carries components to components* Chapter13/MultisectionPushforward · SBLocalization.PushChart.isDiffeoComponent_push. A component Σ' of $\mathcal{U}'(y, y')$ is mapped by the continuous lift into a component Σ of $\mathcal{U}(x_y, x_{y'})$, where s and t are injective. The lift is injective, and source and target of the lift determine those of φ' through Q, Q'. So condition (3) of good systems is carried forward. - *The pushed sections are compatible along the lift* Chapter13/MultisectionPushforward · SBLocalization.PushChart.pushSec_compat: $\mu'(\varphi', \text{push } s(s\varphi')) = \mu'(\psi'_{u'}, \Phi(\mu(\varphi, s(u))))$, by $\varphi' \circ \psi_u = \psi'_{u'} \circ F(\varphi)$ and $\Phi\mu = \mu'\Phi$.

XIII.21 Proposition 13.4.2 (2017: Proposition 13.6): extension from an open V with $|V| = |X|$

Structured data on V. The text takes the strong bundle $W|V$ over the open full sub-ep-groupoid V and a structured sc⁺-multisection functor on it. The project has no strong bundle over an open sub-ep-groupoid, so the data are read relative to V inside W. - *A functor on $W|V$* Chapter13/

StructuredExtensionData · SBLocalization.IsWFunctorOn is a $\Lambda : W \to \mathbb{Q}^+$ invariant along morphisms with both ends in V. Its values over $X \setminus V$ are irrelevant. Two vectors over V in one orbit have the same value Chapter13/StructuredExtensionData · SBLocalization.IsWFunctorOn.eq_of_π_eq, since the morphism joining them has both ends in V. - *An sc⁺-multisection on $W|V$* Chapter13/StructuredExtensionData · SBLocalization.IsScPlusMultisectionOn is such a functor with a local section structure around every $q \in V$ on the domain of a natural representation inside V. The natural representations of the ep-groupoid V are those of X with domain in V. - *Structured data on V* Chapter13/StructuredExtensionData · SBLocalization.StructuredOn are Definitions 13.3.4 and 13.3.6 with every index in V: - neighbourhoods $N(q)$, $q \in V$, with $U(q) \subset V$, good for V in the sense of Theorem 13.1.7 as printed (§XIII.5); - symmetric structures (I_q, s_i^q) representing Λ on $U(q)$; - correspondences $\tau_{q,q'}$ on $\mathcal{U}(q, q')$, locally constant with finitely many values, with the bi-action, the compatibility with the sections, the diagonal and the swap.

Each I_q is nonempty, by the argument of §XIII.9. - *Witness.* Every structured tuple on X gives structured data with $V = X$ Chapter13/StructuredExtensionData · SBLocalization.StructuredTuple.toStructuredOn, for instance the zero multisection's on a paracompact orbit space.

The extension functor Chapter13/StructuredExtensionData · SBLocalization.extendMult. $\bar{\Lambda}(w) = \Lambda(w_0)$ for some w_0 over V in the orbit of w in W. - Such w_0 exists because $\pi(V) = |X|$ Chapter13/StructuredExtensionData · SBLocalization.exists_mem_of_π_image. A morphism $\varphi : q \to P(w)$ with $q \in V$ gives $w_0 = \mu(\varphi^{-1}, w)$. - The value does not depend on the choice, by the previous paragraph, and depends only on the orbit of w Chapter13/StructuredExtensionData · SBLocalization.extendMult_congr. - It is a functor on W Chapter13/StructuredExtensionData · SBLocalization.isWFunctor_extendMult, because w and $\mu(\varphi, w)$ have the same orbit. - It equals Λ over V Chapter13/StructuredExtensionData · SBLocalization.extendMult_eq: for w over V, take $w_0 = w$. - Every functor on W that agrees with Λ over V is $\bar{\Lambda}$ Chapter13/StructuredExtensionData · SBLocalization.eq_extendMult, since every orbit meets V.

This uniqueness is how "canonical" is read.

Unit charts. At a point $q \in V$ the extension keeps the given data, through the *unit chart* of the identity equivalence Chapter13/StructuredExtensionData · SBLocalization.unitChart: $Q = \mathrm{id}$ and $\psi_u = 1_u$. - The identity is a strong bundle equivalence Chapter13/StructuredExtensionData · SBLocalization.equivalences_id_sep, and push-forward along it is the identity on functors Chapter13/StructuredExtensionData · SBLocalization.pushMult_id. - A neighbourhood is a push neighbourhood of itself for the unit chart Chapter13/StructuredExtensionData · SBLocalization.isPushNbhd_unit, because the isotropy isomorphism is the identity. - Pushed sections are the given sections Chapter13/StructuredExtensionData · SBLocalization.pushSec_unit and lifted morphisms are the given morphisms Chapter13/StructuredExtensionData · SBLocalization.liftMor_unit, since $\mu(1, w) = w$. - Push neighbourhoods pass to smaller neighbourhoods with the same action Chapter13/StructuredExtensionData · SBLocalization.IsPushNbhd.of_le.

Extension charts Chapter13/StructuredExtensionData · SBLocalization.ExtChart. At a point x, an extension chart consists of: - a point $q_x \in V$ and a morphism $\psi_x : q_x \to x$; - the transport chart of the identity along ψ_x, $Q = t \circ (s|_{\text{sheet of } \psi_x})^{-1}$, which is the text's D_x; - a push neighbourhood $U(x)$ of $U(q_x)$ with the properness property.

Charts are chosen as follows. - At $x \in V$ the unit chart is used Chapter13/StructuredExtensionData · SBLocalization.unitExt. - At $x \notin V$ a chart exists inside any open $O \ni x$ Chapter13/StructuredExtensionData · SBLocalization.exists_extChart: §XIII.20 with $F = \mathrm{id}$, then restriction to O. - A choice of all charts with these properties Chapter13/StructuredExtensionData · SBLocalization.ExtSetup Chapter13/StructuredExtensionData · SBLocalization.extChart gives unit charts on V Chapter13/

StructuredExtensionData · SBLocalization.extChart_of_mem.

The extended structure. 1. *At each x* Chapter13/StructuredExtension · SBLocalization.ExtChart.sym: the structure at q_x pushed along the chart, $I_x = I_{q_x}$. For the identity equivalence, $\Phi_*\bar\Lambda = \bar\Lambda$, and $\bar\Lambda = \Lambda$ over $U(q_x) \subset V$. So the pushed structure represents $\bar\Lambda$ on $U(x)$. Hence $\bar\Lambda$ is an sc⁺-multisection on W Chapter13/StructuredExtension · SBLocalization.isScPlusMultisection_extendMult: a symmetric structure gives a local section structure Chapter13/StructuredExtension · SBLocalization.hasLocalSectionStructure_of_symmetricOn. 2. *Correspondences* Chapter13/StructuredExtension · SBLocalization.ExtChart.tau: $\tau_{x,x'}(\varphi) = \tau_{q_x,q_{x'}}(\mathrm{lift}\,\varphi)$, with the lift of §XIII.20 through the two charts. - The text distinguishes three cases: both ends in V, one end, neither, with maps A and C and a pseudo-inverse. The lift covers them uniformly. For the identity, the lift is the text's C; it is A when $x \in V$ and its inverse when $x' \in V$. 3. *The good system* Chapter13/StructuredExtension · SBLocalization.extGood of the chart neighbourhoods. Condition (3) is carried from $(U(q))_{q\in V}$ by the lift Chapter13/StructuredExtension · SBLocalization.ExtChart.isDiffeoComponent, as in §XIII.20. 4. *The decoration, the structure and the tuple* Chapter13/StructuredExtension · SBLocalization.extDecoration Chapter13/StructuredExtension · SBLocalization.extTau Chapter13/StructuredExtension · SBLocalization.extTuple. The conditions of Definitions 13.3.4 and 13.3.6 for τ are those of the given $\tau_{q,q'}$, through the properties of the lift: continuity, functoriality, compatibility with Γ and the bi-action, and compatibility of pushed sections. 5. *The given data are kept at the points of V.* There the charts are unit charts, so: - the actions on the index sets are the given ones Chapter13/StructuredExtension · SBLocalization.extSym_smul_of_eq; - the sections are the given ones Chapter13/StructuredExtension · SBLocalization.extSym_s_of_eq; - the correspondences on $\mathcal{U}(q,q')$ are the given ones Chapter13/StructuredExtension · SBLocalization.extTau_of_eq.

All three hold through the identifications of the index types.

Proposition 13.4.2 (restated), pp. 623–625 Chapter13/StructuredExtension · SBLocalization.proposition_13_4_2. Let V be open with $\pi(V) = |X|$, and structured data on V for Λ be given. - (1) $\bar\Lambda$ is an sc⁺-multisection on W, equal to Λ over V, and the only functor on W with this property. - (2) For every family of open $O(x) \ni x$ there is a structured tuple for $\bar\Lambda$ with the following properties: - its neighbourhoods at $q \in V$ are the given $U(q)$, and $U(x) \subset O(x)$ for $x \notin V$; - its index sets with their G_q-actions, its sections on $U(q)$ and its correspondences on $\mathcal{U}(q,q')$ are the given ones at the points of V.

Structurable Chapter13/StructuredExtension · SBLocalization.proposition_13_4_2_structurable: $\bar\Lambda$ is structurable.

Restated: the structured data relative to V replace a structure on the bundle $W|V$. The good-system condition on V is that of Theorem 13.1.7 as printed for the open V. "Canonical" is uniqueness among functors. No hypothesis is added.

XIII.22 Theorem 13.4.3 (2017: Theorem 13.4): push-forward along a strong bundle equivalence, and generalized strong bundle isomorphisms

The pushed good system Chapter13/MultisectionPushforward · SBLocalization.pushGood. Let Φ be a strong bundle equivalence over $F : X \to X'$ and $(\Lambda, \mathcal{U}, S, \tau)$ a structured tuple. - At each $y \in X'$, take the chart of §XIII.20 and a push neighbourhood of $U(x_y)$ Chapter13/MultisectionPushforward · SBLocalization.pushNbhd Chapter13/MultisectionPushforward · SBLocalization.isPushNbhd_pushNbhd. - Properness holds by construction. - Condition (3) holds by the lift.

The pushed structure. - *Decoration* Chapter13/MultisectionPushforward · SBLocalization.pushDecoration:

the pushed symmetric structures of §XIII.20, $I'_y = I_{x_y}$. - *Correspondences* Chapter13/MultisectionPushforward · SBLocalization.pushTau: $\tau'_{y,y'}(\varphi') = \tau_{x_y,x_{y'}}(\mathrm{lift}\,\varphi')$. - Local constancy: τ is locally constant and the lift continuous. - Finiteness: values of τ. - Equivariance: compatibility of the lift with the bi-action, and the isotropy isomorphisms on index sets. - Diagonal: the lift of Γ' is Γ. - Swap: the lift of an inverse is the inverse of the lift. - Compatibility: the pushed sections are compatible along the lift, and the lift maps components into components. - *The tuple* Chapter13/MultisectionPushforward · SBLocalization.pushTuple.

Theorem 13.4.3 (same), pp. 625–626 Chapter13/MultisectionPushforward · SBLocalization.theorem_13_4_3 Chapter13/MultisectionPushforward · SBLocalization.theorem_13_4_3_structured. $\Phi_*\Lambda$, Chapter 11's push-forward, is structurable when Λ is. A structured tuple pushes to a structured tuple with $I'_y = I_{x_y}$ and the pushed sections and correspondences.

Where the formal proof departs from the text. The text structures $\Phi_*\Lambda$ on $F(X)$ through $F|_{U(x_y)}$, then extends by Proposition 13.4.2 from $F(X)$, which has $|F(X)| = |X'|$. The formal proof gives every y, in $F(X)$ or not, a chart along σ_y. This does both steps at once and does not use Proposition 13.4.2. No canonicity is claimed, as in the text. No hypothesis is added.

Generalized strong bundle isomorphisms. A diagram $W \xleftarrow{\Phi} V \xrightarrow{\Psi} W'$ of strong bundle equivalences gives push-forwards $[D]_* = \Psi_*\Phi^*$ and pull-backs $[D]^* = \Phi_*\Psi^*$ of functors (Chapter 11, Part XI). Composing Theorems 13.4.1 and 13.4.3: - both preserve structurability Chapter13/MultisectionPushforward · SBLocalization.SDiagram.isStructurable_pushMult Chapter13/MultisectionPushforward · SBLocalization.SDiagram.isStructurable_pullMult; - so do push-forwards and pull-backs along a generalized strong bundle isomorphism Chapter13/MultisectionPushforward · SBLocalization.GenSBMap.isStructurable_pushMult Chapter13/MultisectionPushforward · SBLocalization.GenSBMap.isStructurable_pullMult, through any representing diagram.

The text says this "follows immediately" at the opening of §16.5. The interior case of Proposition 14.1.8 uses it (Part XIV).

XIII.23 Theorem 13.4.4 (2017: Theorem 13.5): pull-back along a proper strong bundle covering

Theorem 13.4.4 (same), pp. 626–628 Chapter13/MultisectionPullback · SBLocalization.theorem_13_4_4 Chapter13/MultisectionPullback · SBLocalization.theorem_13_4_4_structured. For a proper strong bundle covering $\Phi : W \to W'$ over $F : X \to X'$, the 2021 Definition 8.5.9 (Part VIII): - $\Phi^*\Lambda'$ is structurable when Λ' is; - a structured Λ' gives a structured class on $\Phi^*\Lambda'$, with $I_y = I'_{F(y)}$, G_y acting through F, $\Phi(s_i^y(z)) = s_i'^{F(y)}(F(z))$, and $\tau_{y,y'}(\varphi) = \tau'_{F(y),F(y')}(F(\varphi))$; - the same canonicity as Theorem 13.4.1 holds.

Proof. The covering gives pull-back data (§XIII.19) Chapter13/StrongBundlePullMap · SBLocalization.PullMap.ofCovering. 1. Φ intertwines the actions and is a local strong bundle isomorphism: Definition 8.5.9. 2. F is faithful: by Definition 8.5.1 (3) the map $\varphi \mapsto (F(\varphi), s(\varphi))$ is an sc-diffeomorphism onto the fibred product, so two morphisms with the same source and the same image agree. 3. The rest is the construction of §XIII.19, word for word. - The text's identity $F(\Gamma_Y(g,z)) = \Gamma_X(F(g),F(z))$ is the third clause of an adapted neighbourhood. It is obtained by the same open-set argument. - The text's disjoint decomposition $F^{-1}(U(x)) = \bigsqcup V(z)$ is not needed: adapted neighbourhoods suffice, since only injectivity of

F on $U(y)$ and $F(U(y)) \subset U'(F(y))$ enter.

No hypothesis is added; paracompactness is not used.

XIII.24 Definition 13.4.5 (2017: Definition 13.22): commensurable structured sc⁺-multisections and their sums

A choice of common data `Chapter13/Commensurable · SBLocalization.CommData`. A choice c consists of: - a good system $\mathcal{U}$; - nonempty finite index types I_x with G_x-actions; - maps $\tau_{x,x'}$ from morphisms to $J(I_x, I_{x'})$ satisfying the conditions of Definitions 13.3.4 (1)–(2) and 13.3.6 (1)–(2), which do not involve sections.

The actions are determined by τ `Chapter13/Commensurable · SBLocalization.CommData.smul_eq_of_diag`. By Definition 13.3.6 (1) at $y = x$, $\tau_{x,x}(\Gamma(g,x)) = [\mathrm{id}, g]$, and $[\mathrm{id}, e] = [\mathrm{id}, e']$ only if $e = e'$ `Chapter13/Commensurable · SBLocalization.IndexCorr.ofEquiv_injective`: an equivalence c of the two diagrams satisfies $c = \mathrm{id}$, from the first leg, and then $e = e'$. So including the actions in c adds nothing to the text's "$\tau_1 = \tau_2$".

Sections over c `Chapter13/Commensurable · SBLocalization.CommData.Sections` are sections s_i^x, $i \in I_x$, on every $U(x)$ that are: - sc⁺; - equivariant; - compatible with τ in the sense of Definition 13.3.4 (3) `Chapter13/Commensurable · SBLocalization.CommData.Compat`.

No functor is given yet; the counting formula defines one: - $\mathrm{count}_x(w) = |I_x|^{-1}\#\{i : s_i^x(P(w)) = w\}$ `Chapter13/Commensurable · SBLocalization.CommData.countAt`; - the functor takes the count at $x = P(w)$ `Chapter13/Commensurable · SBLocalization.CommData.sumMult`.

Data over c for a given Λ `Chapter13/Commensurable · SBLocalization.CommData.Over` are sections over c with the counting formula for Λ on every $U(x)$. They give a structured tuple and a structured class `Chapter13/Commensurable · SBLocalization.CommData.Over.toStructured`. Every structured tuple T is of this form, over its own choice `Chapter13/Commensurable · SBLocalization.CommData.ofTuple Chapter13/Commensurable · SBLocalization.CommData.Over.ofTuple_toTuple`. The index sets are nonempty there because the index sets of every symmetric structure are (§XIII.9).

Count transport along a correspondence `Chapter13/Commensurable · SBLocalization.IndexCorr.count_eq_of_sat`. Let $I_x \xleftarrow{a} I \xrightarrow{b} I_{x'}$ represent τ, with $s'_{b(k)} = f(s_{a(k)})$ for all k and f injective on the relevant values. Then

$$|I_x|^{-1}\#\{i : s_i = v\} = |I_{x'}|^{-1}\#\{j : s'_j = f(v)\}.$$

1. If I_x is empty, so are I and $I_{x'}$, by surjectivity, and both sides are 0. 2. Let p be the common fibre size of a and q that of b, both positive. Then $|I| = p|I_x| = q|I_{x'}|$ `Chapter13/Commensurable · SBLocalization.IndexCorr.card_filter_comp_eq`, by counting I fibrewise. 3. Count the $k \in I$ with $s_{a(k)} = v$ in two ways. - Fibrewise over a: $p\#\{i : s_i = v\}$. - The condition is equivalent to $s'_{b(k)} = f(v)$, by the relation and injectivity of f. Fibrewise over b: $q\#\{j : s'_j = f(v)\}$. 4. Divide by the equal numbers $p|I_x| = q|I_{x'}|$.

This is why the counting formula of (2) defines a functor and does not depend on x: - *Along a morphism* `Chapter13/Commensurable · SBLocalization.CommData.Sections.countAt_fibAct`: for $\varphi \in \mathcal{U}(x,x')$, $\mathrm{count}_x(v) = \mathrm{count}_{x'}(\mu(\varphi, v))$. Use compatibility at φ itself and injectivity of $\mu(\varphi,\cdot)$. - *Independent of x* `Chapter13/Commensurable · SBLocalization.CommData.Sections.countAt_eq`: $\mathrm{count}_x(w) = \mathrm{count}_{P(w)}(w)$ for $P(w) \in U(x)$. Apply the previous item to the unit $1_{P(w)} \in \mathcal{U}(x, P(w))$. - *A functor* `Chapter13/Commensurable · SBLocalization.CommData.Sections.isWFunctor_sumMult`: apply the first item to $\varphi \in \mathcal{U}(s\varphi, t\varphi)$. -

An sc⁺-multisection `Chapter13/Commensurable · SBLocalization.CommData.Sections.isScPlusMultisection_sumMult`: on each $U(x)$ the sections s_i^x with weights $|I_x|^{-1}$ are a local structure, by independence of x.

Definition 13.4.5 (1) (restated), p. 628 `Chapter13/Commensurable · SBLocalization.Commensurable`. Two structured classes are commensurable if they have representatives over one common c. The relation is reflexive `Chapter13/Commensurable · SBLocalization.Commensurable.refl`, by the choice of the tuple itself, and symmetric `Chapter13/Commensurable · SBLocalization.Commensurable.symm`. The paracompactness of $|X|$ in the text's setting is not needed for the definition. *Restated*: "representatives can be taken with $\mathcal{U}_1 = \mathcal{U}_2$, $\tau_1 = \tau_2$, $I_x^1 = I_x^{2}$" is "data over one c".

The locally finite commensurable sum (same), p. 629 `Chapter13/Commensurable · SBLocalization.CommData.definition_13_4_5_lf`. Let $(\Lambda_j)_{j\in J}$ be locally finite (Definition 13.2.12) with data over one c. Then: - the sections $s_i^x = \sum_j s_i^{j,x}$ are defined; - they are sections over c; - $\bigoplus_j^c \Lambda_j$, their count, is an sc⁺-multisection with these data, and $\Lambda(w) = |I_x|^{-1}\#\{i : s_i^x(P(w)) = w\}$ for $P(w) \in U(x)$.

Proof. 1. *Finitely many nonzero summands near each point* `Chapter13/Commensurable · SBLocalization.CommData.exists_finite_support`. Over $O(y)$ of Definition 13.2.12, every j outside the finite set has Λ_j trivial, and its sections vanish there (§XIII.9). 2. So near each point $\sum_j s_i^{j,x}$ is a finite sum of sections extended by zero `Chapter13/Commensurable · SBLocalization.CommData.sumSec_eq_sectionSum`. Being sc⁺ on an open set is local `Chapter13/Commensurable · SBLocalization.IsScPlusSectionOn.of_locally`, so the sum is sc⁺. 3. Equivariance and compatibility are additivity of $\mu(\psi, \cdot)$ over finitely supported sums `Chapter13/Commensurable · SBLocalization.fibAct_finsum`. Compatibility of the sum needs one representative that works for all j at once: the conjunction of the relations is saturated when each is `Chapter13/Commensurable · SBLocalization.IndexCorr.sat_iInter`, since all j share the same τ. 4. The sections over c are `Chapter13/Commensurable · SBLocalization.CommData.lfSections`, and the count is an sc⁺-multisection by the previous paragraph.

"Pairwise commensurable" is read as "over one common c", which is what the text's notation $[\Lambda_j, \mathcal{U}, S_j, \tau]$ writes. "Locally finite" is Definition 13.2.12; the text's condition after Definition 13.4.5 is the same with "open". The text calls the construction evident; its well-definedness is the count transport above.

Definition 13.4.5 (2) (same), p. 628 `Chapter13/Commensurable · SBLocalization.CommData.definition_13_4_5`. Over a common c, the sum $\Lambda_1 \oplus_c \Lambda_2$ has sections $s_i^x = s_i^{1,x} + s_i^{2,x}$, and $\Lambda(w) = |I_x|^{-1}\#\{i : s_i^x(P(w)) = w\}$ on $U(x)$. It is an sc⁺-multisection with these data. It is the locally finite sum over a two-element index set `Chapter13/Commensurable · SBLocalization.CommData.commSum₂`. A finite family is locally finite, with $O(x) = X$.

The index set of $\Lambda_1 \oplus_c \Lambda_2$ is I_x, not $I_x^1 \times I_x^2$. The commensurable sum is a different operation from the sum of §XIII.11. In general the two functors differ, since the commensurable sum adds sections with the same index only.

XIII.25 Definition 13.5.1 (2017: Definition 13.23) and Remark 13.5.2 (2017: Remark 13.6): atomic sc⁺-multisections

Atomic data `Chapter13/AtomicMultisection · SBLocalization.AtomicData` are the data of Definition 13.5.1 (1)–(2), p. 629: - a point x; - a natural representation of G_x on $U(x)$, with the properness property: $t : s^{-1}(\mathrm{cl}\, U(x)) \to X$ is proper; - an open $V \ni x$, invariant under every $\Phi(g)$, with $\mathrm{cl}\, V \subset U(x)$; - a finite index type I with a G_x-action; - sc⁺-sections s_i on $U(x)$ that are equivariant, $s_{g(i)}(g{*}y) = \mu(\Gamma(g,y), s_i(y))$ for $y \in U(x)$, and vanish on $U(x) \setminus V$.

The text's "G_x-invariant open neighbourhood $U(x)$" is read here as the domain of a natural representation (Theorem 7.1.13, Part VII). A second reading, on neighbourhoods with the natural action, follows below. Both readings are used.

The count, and condition (3). - *The count* `Chapter13/AtomicMultisection · SBLocalization.AtomicData.count` of a vector w' is $|I|^{-1}\#\{i : s_i(P(w')) = w'\}$. - *Reachable points* `Chapter13/AtomicMultisection · SBLocalization.AtomicData.Reach`: y is reachable if some morphism φ has $s(\varphi) = y$ and $t(\varphi) \in U(x)$. The reachable points are the saturation of $U(x)$ `Chapter13/AtomicMultisection · SBLocalization.AtomicData.reach_iff_mem_saturation`. - *Condition (3)* `Chapter13/AtomicMultisection · SBLocalization.AtomicData.IsRelated`. A functor Λ is related to the data if two things hold: - $\Lambda(w)$ is the count of $\mu(\varphi, w)$ for every morphism φ from $P(w)$ into $U(x)$; - Λ agrees with the zero multisection over the points that are not reachable. - *Atomic* (restated), p. 629 `Chapter13/AtomicMultisection · SBLocalization.IsAtomic`: an sc⁺-multisection related to some atomic data.

The count does not depend on the morphism. The text writes condition (3) for any φ with $t(\varphi) \in U(x)$, and does not say that the value is independent of the choice. It is. 1. *Equivariance permutes the indices* `Chapter13/AtomicMultisection · SBLocalization.AtomicData.card_filter_μ_Γ`. For $g \in G_x$ and $y \in U(x)$, the permutation $i \mapsto g(i)$ is a bijection from $\{i : s_i(y) = w\}$ to $\{i : s_i(g * y) = \mu(\Gamma(g, y), w)\}$. This is the equivariance of the sections, together with injectivity of $\mu(\Gamma(g, y), \cdot)$ on the fibre. 2. *So the count is invariant under the* $\Gamma(g, \cdot)$ `Chapter13/AtomicMultisection · SBLocalization.AtomicData.count_μ_Γ`. 3. *Two morphisms from* $P(w)$ *into* $U(x)$ *give the same count* `Chapter13/AtomicMultisection · SBLocalization.AtomicData.count_μ_eq`. - For φ, φ' from $P(w)$ into $U(x)$, the morphism $\psi = \varphi' \circ \varphi^{-1}$ joins two points of $U(x)$. So it is $\Gamma(g, t\varphi)$ for some $g \in G_x$: the natural representation is onto the morphisms between points of $U(x)$. - Then $\mu(\varphi', w) = \mu(\psi, \mu(\varphi, w))$, and step 2 applies.

The multisection of the data `Chapter13/AtomicMultisection · SBLocalization.AtomicData.mult`. It is the count of $\mu(\varphi, w)$ for some φ from $P(w)$ into $U(x)$ when $P(w)$ is reachable, and the zero multisection otherwise. - It is computed by any such φ `Chapter13/AtomicMultisection · SBLocalization.AtomicData.mult_eq`, by the previous paragraph. - Off the reachable points it is trivial `Chapter13/AtomicMultisection · SBLocalization.AtomicData.mult_of_not_reach`. - Over $U(x)$ it is the count itself `Chapter13/AtomicMultisection · SBLocalization.AtomicData.mult_of_mem`, by the unit morphism. - It is related to the data `Chapter13/AtomicMultisection · SBLocalization.AtomicData.isRelated_mult`. - *Every functor related to the data is this one* `Chapter13/AtomicMultisection · SBLocalization.AtomicData.IsRelated.eq_mult`. So an atomic Λ is determined by its data. - *It is a functor* `Chapter13/AtomicMultisection · SBLocalization.AtomicData.isWFunctor_mult`. For a morphism ψ and w over $s(\psi)$: if $s(\psi)$ is reachable through φ, then $t(\psi)$ is reachable through $\varphi \circ \psi^{-1}$, and the two counts are those of the same vector. If not, neither is reachable, and the zero multisection is a functor.

Condition (2) as printed `Chapter13/AtomicMultisection · SBLocalization.AtomicData.IsRelated.symmetricStructure`. For a Λ related to the data, the data are a symmetric structure of Λ on $U(x)$ in the sense of Definition 13.2.5 (§XIII.9). The representation on $U(x)$ is the count. So the reading of (2) as "sections with a G_x-action and equivariance" agrees with the text's "symmetric sc⁺-section structure".

The index set is nonempty `Chapter13/AtomicMultisection · SBLocalization.AtomicData.IsRelated.nonempty`. For Λ an sc⁺-multisection related to the data, $I \neq \emptyset$. - The weights of Λ over x sum to 1, so some vector over x has positive weight. - With $I = \emptyset$ the count of every vector is 0.

The text does not say this. It matters for Theorem 13.5.5 (§XIII.28).

The local structure of the multisection of the data. 1. *Properness passes to* $\mathrm{cl}\, V$ `Chapter13/`

AtomicMultisection · SBLocalization.AtomicData.isProperSubset_closure_V: a closed subset of the proper $\mathrm{cl}\,U(x)$. 2. *The saturation of* $\mathrm{cl}\,V$ *is closed* Chapter13/AtomicMultisection · SBLocalization.AtomicData.isClosed_saturation_closure_V. This is the closedness of saturations of proper sets (Part VII). It lies in the saturation of $U(x)$ Chapter13/AtomicMultisection · SBLocalization.AtomicData.saturation_closure_V_subset. 3. *Off the saturation of* $\mathrm{cl}\,V$ *the multisection is trivial* Chapter13/AtomicMultisection · SBLocalization.AtomicData.mult_eq_zeroMultisection. At a reachable point outside it, the transported point $t(\varphi)$ lies in $U(x) \setminus V$, where all sections vanish. The count at a point where all sections vanish is the zero multisection Chapter13/AtomicMultisection · SBLocalization.AtomicData.count_eq_zeroMultisection: it is 1 at 0 and 0 elsewhere. 4. *So the domain support lies in the saturation of* $\mathrm{cl}\,V$ Chapter13/AtomicMultisection · SBLocalization.AtomicData.domSupp_mult_subset, which is closed. 5. *Near a reachable point* y_0 Chapter13/AtomicMultisection · SBLocalization.AtomicData.exists_isLocalRepOn_transport. - Choose φ_0 from y_0 into $U(x)$. The target map is a local sc-diffeomorphism at φ_0^{-1}, so there are an open $V_2 \ni y_0$ and an sc-smooth $\tau : V_2 \to \mathbf{X}$ with $t(\tau(z)) = z$ and $\tau(y_0) = \varphi_0^{-1}$. - On $O = V_2 \cap (s \circ \tau)^{-1}(U(x))$, which is open and contains y_0, the sections $s_i'(z) = \mu(\tau(z), s_i(s(\tau(z))))$ are sc$^+$, being transported sections in the sense of §XIII.9. - With weights $|I|^{-1}$ they represent the multisection. At w over $z \in O$, use the morphism $\tau(z)^{-1}$ from z into $U(x)$ in the count. 6. *Local section structures.* A local section structure at every reachable point Chapter13/AtomicMultisection · SBLocalization.AtomicData.hasLocalSectionStructure_of_reach. At every point outside the saturation of $\mathrm{cl}\,V$, the single zero section on the complement Chapter13/AtomicMultisection · SBLocalization.AtomicData.hasLocalSectionStructure_of_not_mem. These two cases cover all points. 7. *Conclusion.* The multisection of the data is an sc$^+$-multisection when $I \neq \emptyset$ Chapter13/AtomicMultisection · SBLocalization.AtomicData.isScPlusMultisection_mult, and it is atomic Chapter13/AtomicMultisection · SBLocalization.AtomicData.isAtomic_mult.

Witnesses. - *Atomic neighbourhoods exist inside every open set* Chapter13/AtomicMultisection · SBLocalization.exists_atomicNbhd. For x in an open W there are a natural representation on $U(x) \subset W$ with the properness property and an invariant open $V \ni x$ with $\mathrm{cl}\,V \subset U(x)$. This uses Theorem 7.1.13 and the shrinking in the proof of Theorem 7.5.7 (Part VII). - *The zero section at any point* Chapter13/AtomicMultisection · SBLocalization.AtomicData.zero. One index, the zero section, on such a neighbourhood. Its multisection is the zero multisection Chapter13/AtomicMultisection · SBLocalization.AtomicData.zero_mult. - So *the zero multisection is atomic*, at every point Chapter13/AtomicMultisection · SBLocalization.isAtomic_zeroMultisection.

Atomic on a neighbourhood with the natural action (extra) Chapter13/AtomicSum · SBLocalization.AtomicDataOn Chapter13/AtomicSum · SBLocalization.IsAtomicOn. This is Definition 13.5.1 with $U(x)$ a neighbourhood with the natural action (§XIII.1) and proper closure, in place of the domain of a natural representation. - The sets of a good system are of this kind, and they need not be connected. Theorem 13.5.7 produces atomic pieces on such sets, so it needs this reading (§XIII.30). - The count Chapter13/AtomicSum · SBLocalization.AtomicDataOn.count, reachability Chapter13/AtomicSum · SBLocalization.AtomicDataOn.Reach and condition (3) Chapter13/AtomicSum · SBLocalization.AtomicDataOn.IsRelated are as above. - Atomic data on the domain of a natural representation are atomic data in this sense Chapter13/AtomicSum · SBLocalization.AtomicData.toOn: the domain with the representation is a neighbourhood with the natural action. So atomic implies atomic in this sense Chapter13/AtomicSum · SBLocalization.IsAtomic.isAtomicOn, and the zero multisection is a witness.

Restated: $U(x)$ is the domain of a natural representation, and V is invariant under the $\Phi(g)$. The count is shown to be independent of φ. Condition (2) is the data of a symmetric structure.

Remark 13.5.2 (p. 630) is prose. It says that atomic multisections are built from one section moved around

by G_x, and that bump functions suffice, not partitions of unity. Theorems 13.5.4–13.5.6 below use bump functions only, as the remark says. Theorem 13.5.7 uses partitions of unity.

XIII.26 Theorem 13.5.3 (2017: Theorem 13.6): atomic sc⁺-multisections are structurable

The text constructs a structure for an atomic Λ in four cases: - points of $U(x_0)$, with index set I_{x_0}; - points near which Λ is trivial, with $\{*\}$; - the remaining points, through a morphism φ_x into $\operatorname{cl} V(x_0)$; - and correspondences between all pairs.

It closes with "one easily verifies" that the sections are compatible. The formalization reorganizes the cases around one closed saturated set Z and one function H, and verifies all compatibilities from two properties of H.

General tools. - *The full correspondence* `Chapter13/AtomicStructurable · SBLocalization.IndexCorr.full`. $[I \leftarrow I \times I' \to I']$ for nonempty finite I, I', built on the diagram `Chapter13/AtomicStructurable · SBLocalization.IndexDiagram.full`. - Its fibres have the sizes $|I'|$ and $|I|$ `Chapter13/AtomicStructurable · SBLocalization.IndexDiagram.card_fst_fiber` `Chapter13/AtomicStructurable · SBLocalization.IndexDiagram.card_snd_fiber`. - It is fixed by $\sharp$ up to the order of factors `Chapter13/AtomicStructurable · SBLocalization.IndexCorr.swap_full`, and by the bi-action `Chapter13/AtomicStructurable · SBLocalization.IndexCorr.biact_full`. - *Correspondences of bijections.* - The bi-action on $[\mathrm{id}, e]$ is $[\mathrm{id}, g'eg^{-1}]$ `Chapter13/AtomicStructurable · SBLocalization.IndexCorr.biact_ofEquiv`. - $\sharp$ of $[\mathrm{id}, e]$ is $[\mathrm{id}, e^{-1}]$ `Chapter13/AtomicStructurable · SBLocalization.IndexCorr.swap_ofEquiv`. - $[\mathrm{id}, e]$ saturates a relation R iff $R(i, e(i))$ for all i `Chapter13/AtomicStructurable · SBLocalization.IndexCorr.sat_ofEquiv`. - *Locally constant choices* `Chapter13/AtomicStructurable · SBLocalization.eventually_eq_of_existsUnique`. Let f and g_h, for h in a finite set, be continuous maps into a Hausdorff space on an open set, and suppose $f(z) = g_h(z)$ for exactly one h at every z. Then the h with $f(z_0) = g_h(z_0)$ still works near z_0. For each of the finitely many other h', $g_{h'}(z_0) \neq f(z_0)$, and this persists near z_0 by Hausdorffness and continuity. The equality for h itself is then forced by existence. - *Locally constant functions are constant on components* `Chapter13/AtomicStructurable · SBLocalization.eq_of_mem_connectedComponentIn`. The level sets are open in the open set, so each component lies in one of them. - *Composition rules for μ*, in the total form of §XIII.9: composition `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.fibAct_comp`, units `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.fibAct_unit`, and inverses `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.fibAct_inv_fibAct` `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.fibAct_fibAct_inv`. - The transported section of §XIII.9 is the action at each point `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.transportSection_eq_fibAct`. - $\varphi \circ (\varphi^{-1} \circ \chi) = \chi$ `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.comp_inv_comp`.

The data of the proof `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData`. - Atomic data D related to Λ, at x_0, with $I \neq \emptyset$. - A good system $\mathcal{U}_0$. - For every y in $Z = \pi^{-1}\pi(\operatorname{cl} V(x_0))$: an open $O(y) \ni y$ and an sc-smooth $\tau_y : O(y) \to \mathbf{X}$ with $t(\tau_y(z)) = z$ and $s(\tau_y(z)) \in U(x_0)$, along which the atomic sections transport to sc⁺-sections.

Existence `Chapter13/AtomicStructurable · SBLocalization.exists_atomicStructData`: for every atomic Λ with data D and every good system. A point of Z is reachable, and the local structure of §XIII.25 (step 5) gives $O(y)$ and τ_y. τ_y is the text's $(s|_{U(\varphi_x)})^{-1}$, inverted.

The set Z. Z `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.Z` is the saturation of $\operatorname{cl} V(x_0)$. - It is closed `Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.isClosed_Z`, by §XIII.25. - It is sat-

urated: the source of a morphism is in Z iff its target is Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.mem_Z_iff_of_mor. - Off Z, Λ is trivial (§XIII.25).

The text's four cases become two: $y \in Z$ and $y \notin Z$. The text's points of $U(x_0)$ outside Z, and its points with a neighbourhood where Λ is trivial, fall into the second case. Its points of $U(x_0)$ in Z use τ_y like every other point of Z, with no special treatment.

The partner function H. Fix $y, y' \in Z$. Write: - $\sigma_y(z) = s(\tau_y(z)) \in U(x_0)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.σ; - $\mathrm{Dom}(y, y')$ for the morphisms from $O(y)$ to $O(y')$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.Dom, an open set.

For $\psi \in \mathrm{Dom}(y, y')$ the conjugate

$$\mathrm{conj}(\psi) = \tau_{y'}(t\psi)^{-1} \circ \psi \circ \tau_y(s\psi)$$

Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.conj runs from $\sigma_y(s\psi)$ to $\sigma_{y'}(t\psi)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.src_conj Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.tgt_conj. These are two points of $U(x_0)$, so: 1. $\mathrm{conj}(\psi) = \Gamma(h, \sigma_y(s\psi))$ *for a unique* $h \in G_{x_0}$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.existsUnique_rel. Existence is the surjectivity of the natural representation. Uniqueness is its injectivity: $\Gamma(\cdot, z)$ is injective for each z. 2. $H(\psi)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H is this h, and 1 when $y \notin Z$, $y' \notin Z$ or $\psi \notin \mathrm{Dom}(y, y')$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_far. It is characterized by its defining equation Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_spec, and any h satisfying it is $H(\psi)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_eq_of_rel. 3. *Functorial* Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_comp Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_unit Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_inv: - $H(\psi_1 \circ \psi_2) = H(\psi_1)H(\psi_2)$, because the conjugates compose, the inner τ's cancelling, and $\Gamma(h_1, \cdot) \circ \Gamma(h_2, \cdot) = \Gamma(h_1 h_2, \cdot)$; - $H(1_z) = 1$; - $H(\psi^{-1}) = H(\psi)^{-1}$.

Each follows by uniqueness in step 1. 4. *Locally constant* Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_eventually. The conjugate depends continuously on ψ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.continuousOn_conj, since τ_y, $\tau_{y'}$, composition and inversion are continuous. Each $\psi \mapsto \Gamma(h, \sigma_y(s\psi))$ is continuous. G_{x_0} is finite, and the space of morphisms is Hausdorff. Apply the tool on locally constant choices.

The homomorphisms γ_y. For $y \in Z$, the morphism $\Gamma_0(g, y)$ of the natural representation of $\mathcal{U}_0$ at y lies in $\mathrm{Dom}(y, y)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.Γ₀_mem_Dom. Define

$$\gamma_y(g) = H(\Gamma_0(g, y))$$

Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.γ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.γ_apply. - It is a homomorphism $G_y \to G_{x_0}$: units by $H(1) = 1$, products by functoriality. - It is trivial for $y \notin Z$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.γ_far. - It is the text's γ_x in both of the text's cases. For $y \in U(x_0)$ it is defined by $\Gamma^y(g, y) = \Gamma^{x_0}(\gamma_y(g), y)$ when τ_y is the unit. For the other points of Z it is the conjugation $\varphi_x \circ g \circ \varphi_x^{-1}$.

The good system. The neighbourhoods must be small enough that H is constant along the natural action. 1. $W_n(y)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.Wn: the points z of $U_0(y) \cap O(y)$ with $g * z \in O(y)$ and $H(\Gamma_0(g, z)) = \gamma_y(g)$ for all g. 2. *It is a neighbourhood of y* Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.eventually_Wn. For each of the finitely many

g, local constancy of H at $\Gamma_0(g,y)$, with continuity of $z \mapsto \Gamma_0(g,z)$, gives a neighbourhood. Intersect them. 3. $W(y)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.W is the interior of $W_n(y)$ for $y \in Z$, and Z^c for $y \notin Z$. It is open Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.isOpen_W and contains y Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.mem_W. 4. $M(y)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.M is the invariant core of $U_0(y)$ in $W(y)$ (§XIII.1). It has the action of $\mathcal{U}_0$ at y. 5. *The good system* Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.□ is $(M(y))$, by the lemma of §XIII.10. Its properties: - $U(y) \subset W(y)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.U_subset_W; - $U(y) \subset W_n(y)$ for $y \in Z$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.mem_Wn, hence $U(y) \subset O(y)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.U_subset_0; - $U(y) \cap Z = \emptyset$ for $y \notin Z$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.not_mem_Z; - $H(\Gamma(g,z)) = \gamma_y(g)$ for $z \in U(y)$, $y \in Z$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.W_cond; - morphisms between $U(y)$ and $U(y')$ lie in $\mathrm{Dom}(y,y')$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.mor_subset_Dom.

The sections Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.sec. The index set is $I = I_{x_0}$ at every point. - For $y \in Z$, the transported sections $z \mapsto \mu(\tau_y(z), s_i(\sigma_y(z)))$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.sec_of_mem Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.sec_apply. They lie over the right points Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.P_s_σ. - For $y \notin Z$, the zero sections Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.sec_of_not_mem. - *Compatibility through H* Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.sec_compat. For $y, y' \in Z$ and $\psi \in \mathrm{Dom}(y,y')$:

$$\mathrm{sec}_{y'}(H(\psi)\cdot i)(t\psi) = \mu(\psi, \mathrm{sec}_y(i)(s\psi)).$$

Proof: 1. Write $\psi = \tau_{y'}(t\psi) \circ \Gamma(H(\psi), \sigma_y(s\psi)) \circ \tau_y(s\psi)^{-1}$. 2. Apply the composition rule for μ. 3. Apply the equivariance of the atomic sections at $\sigma_y(s\psi)$. - *Vanishing off Z* Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.sec_eq_zero. At a point $z \notin Z$ every section is zero. For $y \in Z$, $\sigma_y(z) \in U(x_0)$ lies outside $V(x_0)$, since $z \notin Z$, so the atomic sections vanish there.

The symmetric structures Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.S have index set I, with G_y acting through γ_y, and the sections above. - *Sc⁺:* transported sections are sc⁺ by the data. Zero sections are sc⁺. - *Equivariance:* for $y \in Z$, compatibility at $\Gamma(g,z)$ with $H(\Gamma(g,z)) = \gamma_y(g)$, which is the role of W_n. For $y \notin Z$, the zero sections are equivariant for any action. - *Representation by the count* (Definition 13.2.5): - for $y \in Z$, the count of Λ at w over $z \in U(y)$ is that of $\mu(\tau_y(z)^{-1}, w)$ over $\sigma_y(z) \in U(x_0)$, by Definition 13.5.1 (3), and $\mu(\tau_y(z), \cdot)$ carries the one count to the other; - for $y \notin Z$, Λ is trivial on $U(y)$ and the zero sections represent it, whatever $|I| \geq 1$ is.

The decoration is $(\mathcal{U}, S)$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.dec.

The correspondences Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.corr. For φ from $U(y)$ to $U(y')$:

$$\tau_{y,y'}(\varphi) = \begin{cases} [I \xleftarrow{\mathrm{id}} I \xrightarrow{H(\varphi)} I] & \text{if } y \in Z \Leftrightarrow y' \in Z, \\ [I \leftarrow I \times I \rightarrow I] & \text{otherwise} \end{cases}$$

Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.corr_of_iff Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.corr_of_not_iff.

When exactly one of y, y' lies in Z, the morphisms between $U(y)$ and $U(y')$ join points outside Z. One of the two sets misses Z, and Z is saturated. There all sections vanish, and the full correspondence is

compatible.

Definition 13.3.4 Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.isCorrespondence. 1. *Locally constant.* H is locally constant on the morphisms between $U(y)$ and $U(y')$ Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_eventually'. The full correspondence is constant. 2. *Finitely many values.* G_{x_0} is finite. 3. *Equivariance* Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.H_biact. $H(g' * \varphi * g) = \gamma_{y'}(g')H(\varphi)\gamma_y(g)^{-1}$, by functoriality and the condition of W_n. With the law for the bi-action on $[\mathrm{id}, e]$ this is Definition 13.3.4 (2). The full correspondence is fixed by the bi-action. 4. *Compatibility.* On a component Σ of $\mathcal{U}(y, y')$, H is constant, since it is locally constant. So for $\varphi \in \Sigma$ the relation saturated by $[\mathrm{id}, H(\varphi)]$ holds at every $\psi \in \Sigma$, by the compatibility of the sections through $H(\psi) = H(\varphi)$. Across Z and Z^c, all sections vanish.

Definition 13.3.6 Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.tau. - *Diagonal:* $\tau_{y,y}(\Gamma(g, z)) = [\mathrm{id}, H(\Gamma(g, z))] = [\mathrm{id}, \gamma_y(g)]$, by the condition of W_n. This is $[\mathrm{id}, g]$ for the action through γ_y. - *Swap:* $[\mathrm{id}, H(\varphi^{-1})] = [\mathrm{id}, H(\varphi)^{-1}] = [\mathrm{id}, H(\varphi)]^\sharp$, and the full correspondence is its own swap up to the order of factors.

So Λ is structurable Chapter13/AtomicStructurable · SBLocalization.AtomicStructData.isStructurable.

Where the formal proof departs from the text. The text uses $\{*\}$ with the zero section where Λ is trivial, and correspondences $[\{*\} \leftarrow I \rightarrow I]$ to such points. The formalization uses I with zero sections and the full correspondence. Both are structures in the sense of Definition 13.3.6. This is a different witness, not a different statement.

Theorem 13.5.3 (narrower), p. 630 Chapter13/AtomicStructurable · SBLocalization.theorem_13_5_3_of_goodSystem. If X has a good system, every atomic sc⁺-multisection is structurable.

The added hypothesis, that a good system exists, is not derived from the text, whose statement assumes nothing on X. It is necessary: for atomic Λ, structurable is equivalent to the existence of a good system Chapter13/AtomicStructurable · SBLocalization.theorem_13_5_3, since Definition 13.3.6 includes one. Its witness is Theorem 13.1.8 for paracompact $|X|$ (§XIII.6) Chapter13/GoodSystemExistence · EpGroupoid.theorem_13_1_8, which gives the form Chapter13/AtomicStructurable · SBLocalization.theorem_13_5_3_of_paracompact.

As printed, the theorem puts no hypothesis on X, but a structurable sc⁺-multisection contains a good system by Definition 13.3.6, so the statement needs one; Chapter 13 has no standing assumption that supplies it, and the formal statement names it (narrower, with the added hypothesis shown necessary). The text's proof begins by choosing neighbourhoods with the natural action and the properness property at every point, which is a good system in all but condition (3).

The local alternative, and a parametric form. A first Lean form of the theorem isolates the local part of the proof. - *The local alternative* Chapter13/AtomicStructurableLocal · SBLocalization.AtomicData.IsLocallyTransported Chapter13/AtomicStructurableLocal · SBLocalization.theorem_13_5_3_local. For atomic data of Λ, at every point y one of two things holds: - on an open $O \ni y$, Λ is represented by the atomic sections transported along an sc-smooth τ into $U(x_0)$, with weights $|I|^{-1}$; - Λ is trivial on an open $O \ni y$.

This is §XIII.25 (steps 3 and 5), split by reachability. - *Remark on the parametric form.* The same file states Theorem 13.5.3 for any predicate "structurable" that holds for every sc⁺-multisection with atomic data whose local alternatives hold everywhere Chapter13/AtomicStructurableLocal · SBLocalization.theorem_13_5_3_hyp. That hypothesis is satisfiable, with "structurable" read as "is an sc⁺-multisection" Chapter13/AtomicStructurableLocal · SBLocalization.theorem_13_5_3_hyp_witness. The last clause of Theorem 13.5.5 is stated in the same form Chapter13/AtomicStructurableLocal · SBLocalization.theorem_13_5_5_structurable_hyp.

The proof above supersedes the hypothesis for the predicate of Definition 13.3.6. The parametric clause is still the route by which the last clause of Theorem 13.5.5 is derived (§XIII.28), with Theorem 13.5.3 substituted for the parameter.

XIII.27 Theorem 13.5.4 (2017: Theorem 13.7): Construction I

Symmetric families `Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily`. These are the input of Theorem 13.5.5 (p. 633): for a natural representation of G_x on $U(x)$, - a finite index type I with a G_x-action; - sc⁺-sections s_i of $W \to X$, global, viewing X as an M-polyfold, with support in $U(x)$; - equivariance $s_{g(i)}(g * z) = \mu(\Gamma(g, z), s_i(z))$ for $z \in U(x)$.

Construction I produces such families. They vanish off V, which is stronger than "on $U(x) \setminus V$", since they are global and supported in V.

Auxiliary facts on μ and the auxiliary norm. - $\mu(\varphi, \cdot)$ preserves the level-1 part of the fibres `Chapter13/AtomicConstruction · SBLocalization.mem_oneSet_μ` `Chapter13/AtomicConstruction · SBLocalization.mem_oneSet_of_μ`, in both directions, since it is a strong bundle isomorphism of fibres. - An auxiliary norm N (Definition 8.3.5, 2021 numbering; Part VIII) is invariant: $N(\mu(\varphi, w)) = N(w)$ `Chapter13/AtomicConstruction · SBLocalization.normW_μ`. This is the compatibility of auxiliary norms with the structure maps. - The zero vector is at level 1 `Chapter13/AtomicConstruction · SBLocalization.fiberZero_mem_oneSet` and has norm 0 `Chapter13/AtomicConstruction · SBLocalization.normW_fiberZero`.

The family of one seed `Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed`. Let s_0 be an sc⁺-section supported in an invariant open V with $\mathrm{cl}\, V \subset U(x)$. Take: - $I = G_x$, acting on itself by left multiplication; - $s_g(z) = \mu(\Gamma(g, g^{-1} * z), s_0(g^{-1} * z))$ on $U(x)$, and 0 off $U(x)$.

These are the terms of the average of Part XII. They are global sc⁺-sections supported in $\Phi(g)(\mathrm{supp}\, s_0) \subset V$. Their values `Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed_s`: - $s_1 = s_0$ on $U(x)$ `Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed_one`; - the family vanishes off V `Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed_vanish`; - if $N(s_0(y)) \le b$ everywhere then $N(s_g(y)) \le b$ everywhere `Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed_normW_le`, by invariance of N.

Equivariance, the text's (13.8), is $s_{gh}(g * z) = \mu(\Gamma(g, z), s_h(z))$. It follows from $\Gamma(g, h * z) \circ \Gamma(h, z) = \Gamma(gh, z)$ and the composition law of μ. The text's map $\tau : G_x \to \mathrm{Per}(G_x)$ is the left multiplication action.

Theorem 13.5.4 (1) (narrower), p. 632 `Chapter13/AtomicConstruction · SBLocalization.theorem_13_5_4_1`. Assume: - X admits bump functions in the pointwise reading; - x is smooth; - $U(x)$ is the domain of a natural representation with the properness property; - $V \ni x$ is open, invariant, with $\mathrm{cl}\, V \subset U(x)$; - $w \in W_x$ is smooth.

Then there is a symmetric family vanishing off V with $s_{i_0}(x) = w$ for some i_0.

Proof. 1. An auxiliary norm of Part I exists, by Proposition 5.1.3 (Part V). 2. Part I's Lemma 5.3.3 gives an sc⁺-section s_0 with $s_0(x) = w$, supported in V `Chapter5/BumpFunctions · lemma_5_3_3`. Its bump-function hypothesis on the object M-polyfold comes from the pointwise reading on the ep-groupoid, by Theorem 12.3.2 (1) (Part XII). This is the text's "construct in local coordinates an sc⁺-section … multiply by a bump function". 3. The family of the seed s_0, with $i_0 = 1$.

The cut-off Chapter13/AtomicConstruction · SBLocalization.exists_cutoff. For an sc⁺-section s on an open $O \ni x$ and an open $V \ni x$, there is an sc⁺-section s_0 of $W \to X$ with these properties: - it is supported in V; - it equals s near x; - it vanishes off $V \cap O$; - $N(s_0(y)) \le N(s(y))$ on O for every auxiliary norm N.

Proof. 1. Proposition 5.5.3 (Part V) gives an sc-smooth $\beta : X \to [0,1]$ with $\beta = 1$ near x and $\operatorname{supp}\beta \subset V \cap O$. 2. $s_0 = \beta \cdot s$, with s extended by zero off O. It vanishes where $\beta = 0$ Chapter13/AtomicConstruction · SBLocalization.sectionSmul_eq_fiberZero. 3. It is sc⁺: on O as a product of an sc-smooth function and an sc⁺-section, and off the closed support of β as the zero section. These two open sets cover X. 4. The norm bound: $N(\beta(y)s(y)) = |\beta(y)|N(s(y)) \le N(s(y))$.

The text asks for $\beta(x) = 1$ "with arbitrarily small support". The conclusion "$s_{i_0} = s$ near x" needs $\beta = 1$ near x, which Proposition 5.5.3 provides.

Theorem 13.5.4 (2) (narrower), p. 632 Chapter13/AtomicConstruction · SBLocalization.theorem_13_5_4_2. For an sc⁺-section s on an open $O \ni x$, there is a symmetric family vanishing off V with $s_{i_0} = s$ near x. *Proof*: the family of the seed given by the cut-off, with $i_0 = 1$.

Narrower only through the pointwise reading of bump functions, as in Theorem 12.3.3 (Part XII). The conclusions are stronger than printed: global sections, vanishing off V. The properness property in both parts, and the smoothness of x in (2), are kept and not used.

XIII.28 Theorem 13.5.5 (2017: Theorem 13.8): Construction II

Step 1: an invariant V Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily.exists_vanishing_nbhd. For a symmetric family on $U(x)$ there is an open invariant $V \ni x$ with $\operatorname{cl} V \subset U(x)$ off which every section vanishes. 1. $K = \{x\} \cup \bigcup_i \operatorname{supp} s_i$ is closed, and lies in $U(x)$ since the supports do. 2. The object space is Hausdorff and paracompact, hence normal. There is an open $V_0 \supset K$ with $\operatorname{cl} V_0 \subset U(x)$. 3. $V = \bigcap_{g \in G_x} (U(x) \cap \Phi(g)^{-1}(V_0))$ is open, contains x, and is invariant. Its closure lies in $\operatorname{cl} V_0 \subset U(x)$. 4. Every section vanishes off V. Equivariance gives $s_{g(i)}(g * z) = 0$ iff $s_i(z) = 0$ Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily.eq_zero_iff, since $\mu(\Gamma(g,z),\cdot)$ is injective and maps 0 to 0. A point $z \in U(x) \setminus V$ has some g with $g * z \notin V_0$. Then $g * z \notin \operatorname{supp} s_{g(i)}$, so $s_{g(i)}(g * z) = 0$ and $s_i(z) = 0$. A point outside $U(x)$ is outside every support.

This is condition (ii) of the text's proof, which takes such a $V(x)$ as given.

Step 2: the atomic data Chapter13/AtomicConstruction · SBLocalization.AtomicData.ofFamily. The family restricted to $U(x)$, with this V, is atomic data in the sense of §XIII.25. Global sc⁺-sections are sc⁺ on $U(x)$.

Theorem 13.5.5 (narrower), p. 633 Chapter13/AtomicConstruction · SBLocalization.theorem_13_5_5. For a symmetric family on $U(x)$, with $U(x)$ having the properness property and $I \neq \emptyset$, there is a unique Λ with these properties: - it is atomic; - it is trivial over the points outside the saturation of $U(x)$; - over $U(x)$, $\Lambda(w) = |I|^{-1}\#\{i : s_i(P(w)) = w\}$.

Proof. 1. *Existence.* The multisection of the data of step 2. It is atomic, by §XIII.25. It is trivial off the saturation of $\operatorname{cl} V$, which lies in the saturation of $U(x)$. Over $U(x)$ it is the count. 2. *Uniqueness.* Let Λ have the three properties, and w a vector. - If $P(w)$ is reachable through φ, then $\Lambda(w) = \Lambda(\mu(\varphi, w))$, since Λ is a functor. That is the count, by the third property. - Otherwise $P(w)$ is outside the saturation of $U(x)$, and Λ is trivial there by the second property.

In both cases $\Lambda(w)$ is the value of the multisection of the data.

Narrower by $I \neq \emptyset$. The text does not state it, and the printed formula presupposes it. With $I = \emptyset$ the formula gives $\Lambda = 0$ on all vectors over $U(x)$, which is not an sc⁺-multisection, since the weights over a point sum to 1. For atomic Λ it is automatic (§XIII.25), and the families of Construction I have $I = G_x$ (§XIII.27).

"Supported in the saturation of $U(x)$" is read as triviality over the points not isomorphic to a point of $U(x)$. This is the text's definition of Λ off the reachable points. Bump functions and the smoothness of x are kept and not used.

The last clause: Λ is structurable (narrower), p. 633 `Chapter13/AtomicStructurable · SBLocalization.theorem_13_5_5_structurable`. Given a good system, the atomic Λ of Construction II is structurable. It is the parametric form of §XIII.26 `Chapter13/AtomicStructurableLocal · SBLocalization.theorem_13_5_5_structurable_hyp` with Theorem 13.5.3 substituted, as the text derives it ("We have already established that they are structurable"). The added hypothesis is that of Theorem 13.5.3, with the same witness.

XIII.29 Theorem 13.5.6 (2017: Theorem 13.9): atomic multisections through a vector, and with a bound

Two facts about the multisection of atomic data. - *A norm bound* `Chapter13/AtomicConstruction · SBLocalization.AtomicData.pointwiseNorm_mult_le`. If $N(s_i(z)) \leq b$ for all i and $z \in U(x)$, with $b \geq 0$, then the pointwise norm of Definition 13.2.3 (§XIII.8) satisfies $N(\Lambda)(y) \leq b$ for every y. Every vector with positive weight falls in one of two cases: - over a reachable point, it is $\mu(\varphi^{-1}, s_i(t\varphi))$ for some i, and N is invariant; - otherwise, it is 0. - *The local structure on $U(x)$* `Chapter13/AtomicConstruction · SBLocalization.AtomicData.isLocalRepOn_mult`. Over $U(x)$ the sections of the data, enumerated, with weights $|I|^{-1}$, are a local section structure of the multisection.

Theorem 13.5.6, first part (narrower), p. 634 `Chapter13/AtomicConstruction · SBLocalization.theorem_13_5_6_1`. Let $w \neq 0$ be smooth over x, and $U(x)$ the domain of a natural representation. Then there is an atomic Λ with: 1. dom-supp(Λ) in the saturation of $U(x)$; 2. $\Lambda(\mu(g, w)) > 0$ for every $g \in G_x$.

Proof. 1. Inside $U(x)$ choose a natural representation $U'(x)$ with the properness property and an invariant V with $\mathrm{cl}\, V \subset U'(x)$ (§XIII.25). 2. x is smooth, since w is. 3. Construction I (1) gives a family through w, vanishing off V. 4. Its atomic data give Λ, the multisection of the data, with $I = G_x$. 5. The domain support lies in the saturation of $\mathrm{cl}\, V$, so in that of $U(x)$. 6. $\Lambda(\mu(g, w)) = \Lambda(w)$, since Λ is a functor. That is the count at w, which is positive because $s_1(x) = w$.

The text's "$\pi^{-1}(U(x))$" is read as the saturation of $U(x)$, and "$\Lambda(g * w)$" as $\Lambda(\mu(g, w))$.

Theorem 13.5.6, second part (narrower), p. 634 `Chapter13/AtomicConstruction · SBLocalization.theorem_13_5_6_2`. Let N be an auxiliary norm, s an sc⁺-section on $U(x)$, and $\varepsilon > 0$. Then there is an atomic Λ with: 1. domain support in the saturation of $U(x)$; 2. $N(\Lambda)(y) \leq N(s(x)) + \varepsilon$ for all y; 3. a local section structure on an open $O \ni x$ one of whose sections equals s on O, with positive weight.

Proof. 1. Let $b = N(s(x)) + \varepsilon$. The set $O_1 = U(x) \cap \{N(s(\cdot)) < b\}$ is open and contains x: N is continuous on the level-1 part and s is sc⁺. 2. Inside O_1 choose $U'(x)$ and V as in the first part. 3. The cut-off of s on $U'(x)$ gives a seed s_0 equal to s near x. Its norm is $\leq N(s(y)) < b$ on $U'(x)$, and 0 elsewhere, so $\leq b$ everywhere. 4. The family of the seed keeps the bound, and its atomic data give Λ, with the norm bound by the first fact above. 5. Near x, $s_1 = s_0 = s$. The local structure on $U'(x)$, restricted to $U'(x) \cap O'$ with O' the set where $s_0 = s$, contains s with weight $|G_x|^{-1} > 0$.

"Whose local section structure near x contains s with positive weight" is read as the third property.

Both parts `Chapter13/AtomicConstruction · SBLocalization.theorem_13_5_6`. *Narrower* only through the pointwise reading of bump functions. The paracompactness of $|X|$ is kept and not used. The text uses it for the existence of N (Theorem 12.2.2), and here N is given.

XIII.30 Theorem 13.5.7 (2017: Theorem 13.10): decomposition into commensurable atomic pieces

Tools. - *The pieces as data over c* `Chapter13/AtomicSum · SBLocalization.CommData.smulFunOver`. For a structured tuple T and an sc-smooth function β invariant along morphisms, the sections βs_i^x are data over the choice c of T (§XIII.24) for $\beta \odot \Lambda$. This is Lemma 13.3.14 (§XIII.16), with the same index sets and correspondences. - *$\beta \odot \Lambda$ is trivial where $\beta = 0$* `Chapter13/AtomicSum · SBLocalization.multisectionSmulFun_of_eq_zero`. - *Shrinking in the orbit space* `Chapter13/AtomicSum · SBLocalization.exists_orbit_nbhd`. For a good system and a point x there is an open $O_x \ni |x|$ in $|X|$ with $\operatorname{cl} O_x \subset |U(x)|$. $|U(x)|$ is open, by Proposition 7.1.8, and $|X|$ is regular, by Lemma 7.3.5 (Part VII).

Theorem 13.5.7 (stronger), p. 635 `Chapter13/AtomicSum · SBLocalization.theorem_13_5_7`. If X has sc-smooth partitions of unity, then for every structured tuple $T = (\Lambda, \mathcal{U}, S, \tau)$ there are: - a locally finite family $(\Lambda_j)_{j \in X}$, each atomic on a neighbourhood with the natural action (§XIII.25); - data over the choice c of T for each Λ_j;

such that - the locally finite commensurable sum (§XIII.24) is Λ; - its sections are those of T on every $U(x)$; - its structured class is that of T.

Proof. 1. *A partition of unity.* For each x take O_x by the shrinking. The sets $\pi^{-1}(O_x)$ form an open saturated cover of X. Take an sc-smooth partition of unity of functors $(\beta_j)_{j \in X}$ subordinate to it, with $\operatorname{supp} \beta_j \subset \pi^{-1}(O_j)$. 2. *The pieces.* $\Lambda_j = \beta_j \odot \Lambda$, with data $\beta_j s_i^x$ over c. 3. *Locally finite* (Definition 13.2.12). Near each point only finitely many β_j are nonzero, and Λ_j is trivial where $\beta_j = 0$. The neighbourhoods can be taken saturated, because the β_j are functors. 4. *The sections.* $\sum_j \beta_j(y) s_i^x(y) = s_i^x(y)$, since $\sum_j \beta_j = 1$. The finite sum is that of §XIII.24, with the zero summands removed. 5. *The sum is Λ.* The counts of the same sections agree, and the count over c is Λ on every $U(x)$. 6. *Each piece is atomic* at $x_j = j$: - $U(j)$ with its natural action, and proper closure (good systems); - $V = U(j) \cap \pi^{-1}(O_j)$, open and invariant, with $\operatorname{cl} V \subset U(j)$ by Lemma 13.1.1 (§XIII.1); - sections $\beta_j s_i^j$, which vanish on $U(j) \setminus V$, where $\beta_j = 0$; - condition (3) at reachable points is the count of Λ_j transported along φ; - off the saturation of $U(j)$, $\beta_j = 0$, since $\operatorname{cl} O_j \subset |U(j)|$, so Λ_j is trivial. 7. *The classes are equal.* The commensurable sum's tuple and T have the same multisection, the same good system and the same index sets. Their correspondences agree after the identity relabelling. So they are equivalent in the sense of Definition 13.3.8, and their classes are equal.

The text's form `Chapter13/AtomicSum · SBLocalization.theorem_13_5_7_book`. For structurable Λ on X with paracompact $|X|$ and sc-smooth partitions of unity, Λ has a structured lift that is the locally finite commensurable sum of atomic pieces. Paracompactness is kept and not used.

Stronger: - the decomposition holds for every structured lift, not only one; - the class equality is literal, through the quotient by Definition 13.3.8; - the paracompactness of $|X|$ is not used, since regularity of $|X|$, which every ep-groupoid has, suffices for the shrinking.

The text's refinement $(V_j)_{j \in J}$ with points x_j is replaced by the index set X itself, with $V_j = O_j$. The

atomic pieces are atomic on neighbourhoods with the natural action, the extra reading of Definition 13.5.1. The sets of a good system need not be domains of natural representations.

XIII.31 What is not formalized

- No numbered statement of Chapter 13 is open. The chapter's gaps G.141, G.142, G.143, G.148, G.149 and G.163 are closed:
  - G.141: Theorem 13.1.7 as printed (§XIII.5);
  - G.142: decorated sc$^+$-multisections and their algebra (§§XIII.10–XIII.13);
  - G.143: no inverses (§XIII.11);
  - G.148: Proposition 13.3.19 (§XIII.18);
  - G.149: Theorem 13.5.3 and the last clause of Theorem 13.5.5 (§§XIII.26, XIII.28);
  - G.163: Proposition 13.4.2, in the reading of structured data relative to the open V (§XIII.21).
- Remark 13.3.13, the negative half. The claim is that the structured sums $\Lambda_1 \oplus \Lambda_2$ and $\Lambda_2 \oplus \Lambda_1$ differ, because $I_x^1 \times I_x^2$ and $I_x^2 \times I_x^1$ differ. It is a statement about inequality of types and has no formal counterpart. The positive half, that the two are related by the relabelling, is in §XIII.16.
- Remarks 13.3.9, 13.3.16 and 13.5.2 are prose.
- The text's sub-ep-groupoid $W|V$ in Proposition 13.4.2 is read as data relative to V inside W (§XIII.21). No strong bundle over an open full sub-ep-groupoid is built.

XIII.32 What the formalization taught us

§13.1 is a study of the bi-action. Once the bi-action is written as a formula, almost everything follows from two facts: - morphisms between points of a neighbourhood with the natural action are $\Gamma(g, \cdot)$ for a unique g; - s and t are injective on each component.

Lemma 13.1.6 (1) then needs one morphism with $g' * \psi \in \Sigma$, not $g' * \Sigma = \Sigma$. Proposition 13.1.9 follows from Lemma 13.1.6 with the action laws, without the smoothness argument of the text, and with uniqueness among all maps.

The phrase "carries the natural G_x-action" needed a decision. Proposition 13.1.3 shrinks to invariant subsets that need not be connected, so a neighbourhood with the natural action cannot be the connected domain of a natural representation. It is an invariant open subset of one. With that reading the section holds as printed.

In Lemma 13.1.1 the two directions use different halves of the hypotheses: properness one way, the natural action the other. The invariance of V is used by neither.

Theorem 13.1.7 holds in two forms. - With properness read in X, it needs neither the openness of Q nor the smallness of the closures of $U(y)$. - As printed, with properness read in the full subcategory on the open Q, the hypothesis is equivalent to properness in X together with $\mathrm{cl}_X U(q) \subset Q$. A limit of points of $U(q)$ outside Q is moved into Q by a local section of s, and properness in Q then places it in Q.

No sub-ep-groupoid structure is needed: the full subcategory is its set of morphisms, with the restricted source and target.

In §13.2 the multisection algebra is convolution. $\oplus$, scalar multiples and locally finite sums are all sums over decompositions of a vector in a fibre, and their local structures are sums of sections with products

of weights. A locally finite sum over infinitely many indices, which the text does not write out, is a sum over finitely supported families. Over $O(x)$ it is a finite convolution.

The decorations add one subtlety. The intersection $U'(x) \cap U''(x)$ of two neighbourhoods with natural actions need not carry one natural action for which both structures are symmetric. So the sum is decorated on a smaller neighbourhood inside it on which the two actions agree, which Definition 13.2.8 allows.

The absence of inverses is a one-line computation once a neutral element is fixed: a two-valued multisection convolved with anything gives weight at most $\frac{1}{2}$ to the zero vector.

§13.3 is mostly combinatorics of finite index sets, and two points needed care. - The text writes equalities of index sets ($I'_x = \{*\}$, $I'_x = I_x$) where a formalization has bijections. The correspondences are compared after transport along them. - The tight version of Proposition 13.3.15 needs a compatibility the text does not check. It holds because every section of a structure vanishes where Λ is trivial.

Definition 13.3.8 as printed puts no condition on the sections. Presumably they are meant to agree as in Definition 13.2.8; nothing proved depends on it.

Products of infinitely many correspondences are well defined only because, near any two points, all but finitely many factors are the unit diagram $[\{*\} \leftarrow \{*\} \rightarrow \{*\}]$. The text's "left to the reader" in Proposition 13.3.19 is this bookkeeping. With the tight versions over a common good system, it is about five hundred lines.

§13.4 is simpler than printed. - The pull-back needs only F injective on $U(x)$ with $F(U(x)) \subset U'(F(x))$, not the equality the text arranges. The canonicity it asserts is a short argument once the pull-back is written as a map of structured classes. - The push-forward does not need Proposition 13.4.2. Every point of X', in $F(X)$ or not, gets a chart along a morphism $F(x_y) \to y$, which merges the text's two steps. The lifted morphisms are continuous because F is faithful and a local sc-diffeomorphism. - Proposition 13.4.2 is then the same chart for the identity equivalence. The text's three cases for the correspondences, with the maps A, C and a pseudo-inverse, are one formula: the correspondence of the lifted morphism. - The commensurable sum of Definition 13.4.5 is well defined because the count is transported along correspondences: $p|I_x| = q|I_{x'}| = |I|$ for the fibre sizes p, q of the two legs.

In §13.5 the atomic multisection is determined by its data. The count in Definition 13.5.1 (3) does not depend on the morphism used, and its local structure near any point is the given sections transported by a local inverse of t. What the text does not say is that the index set must be nonempty, since the weights of a multisection sum to 1.

In Theorem 13.5.3 the action of G_y on the index set comes from a single function H. It assigns to a morphism the element of G_{x_0} the morphism becomes after both ends are transported into $U(x_0)$. H is functorial, and locally constant because G_{x_0} is finite and X Hausdorff. Every compatibility follows from these two properties. Using the index set I_{x_0} everywhere, with zero sections where Λ is trivial and the full correspondence across the frontier of the saturation, avoids the case distinction of the text's $\{*\}$. The theorem needs a good system and nothing else: structurability is equivalent to the existence of one.

Theorem 13.5.7 needs regularity of $|X|$ for its shrinking, which every ep-groupoid has, and partitions of unity, not paracompactness. Its atomic pieces are atomic on neighbourhoods with the natural action, since the sets of a good system need not be connected domains of natural representations.

Part XIV. Extension of sc⁺-multisections (HWZ Chapter 14, §§14.1–14.6)

Chapter 14 (Springer 2021, pp. 637–682; 2017 numbers in parentheses) proves the ∂-extension theorem: a structurable sc⁺-multisection functor $\Lambda : W|\partial X \to \mathbb{Q}^+$ over the boundary of a tame ep-groupoid extends to a structurable $\Lambda' : W \to \mathbb{Q}^+$ on the whole bundle, with prescribed support and with its size bounded by a given function. It is the tool that makes the transversality theory of Chapter 15 work relative to the boundary, and through Chapter 16 relative to the boundary of a polyfold. §14.1 introduces good neighbourhoods of boundary points, sc⁺-multisections over ∂X and their structured versions, and states the theorem (Proposition 14.1.1, Definitions 14.1.2, 14.1.3, Proposition 14.1.4, Definition 14.1.5, Theorem 14.1.6, Definition 14.1.7, Proposition 14.1.8, Theorem 14.1.9, Remark 14.1.10 (2017: the discussion before Definition 14.1, Definitions 14.1, 14.2, Remark 14.2, Definition 14.3, Theorem 14.1, Definition 14.4, Proposition 14.1, Theorem 14.2, Remark 14.3)). §14.2 replaces a structured representative by one whose correspondences have separated level sets (Lemmas 14.2.1, 14.2.2, Proposition 14.2.3 (2017: the text before Lemma 14.1, Lemma 14.1, Proposition 14.2)). §14.3 extends the correspondences from the boundary morphisms to open sets of morphisms and covers ∂X by sets $\widehat{V}_a$ on which the extensions are defined (Lemma 14.3.1, Definition 14.3.2, Lemma 14.3.3, Proposition 14.3.4, Remark 14.3.5, Lemmas 14.3.6, 14.3.7, Proposition 14.3.8 (2017: Lemma 14.2, the text before Lemma 14.3, Lemma 14.3, Proposition 14.3, Remark 14.4, Lemmas 14.4, 14.5, Proposition 14.4)). §14.4 records, through z-loops, which boundary sections are forced to agree, and extends the boundary sections near each $\widehat{V}_a$ respecting these identifications (Remark 14.4.1, Definitions 14.4.2–14.4.4, Lemma 14.4.5, Proposition 14.4.6, Lemma 14.4.7, Definition 14.4.8, Lemma 14.4.9, Proposition 14.4.10, Definition 14.4.11, Remark 14.4.12, Lemma 14.4.13, Proposition 14.4.14 (2017: Remark 14.5, Definitions 14.5–14.7, Lemma 14.6, Proposition 14.5, Lemma 14.7, Definition 14.8, Lemma 14.8, Proposition 14.6, Definition 14.9, Remark 14.6, Lemma 14.9, Proposition 14.7)). §14.5 cuts the local extensions off, transports them along morphisms, sums them and assembles the extension, using a structurability criterion (Lemmas 14.5.1, 14.5.2, Remark 14.5.3, Proposition 14.5.4, Theorem 14.5.5 and the end of the proof of Theorem 14.1.9 (2017: Lemmas 14.10, 14.11, Remark 14.7, Proposition 14.8, Theorem 14.3)). §14.6 builds boundary data face by face for inductive constructions (Definitions 14.6.1, 14.6.2, Theorem 14.6.3 (2017: Definitions 14.10, 14.11, Theorem 14.4)). This Part follows the book: §14.1 in §§XIV.1–XIV.7, §14.2 in §XIV.8, §14.3 in §§XIV.9–XIV.12, §14.4 in §§XIV.13–XIV.20, §14.5 in §§XIV.21–XIV.24, the proof of Theorem 14.1.9 in §XIV.25 with a witness with boundary in §XIV.26, §14.6 in §§XIV.27–XIV.28; then what is not formalized (§XIV.29) and what the formalization taught us (§XIV.30).

What is proved, and what is not. The ∂-extension theorem, Theorem 14.1.9, is proved with two readings made explicit: the strong bundle has a tame strong bundle chart at every object, which is how "a strong

bundle over a tame ep-groupoid" is read throughout Part XII and here, and Λ charges every boundary fibre, which is what the factor $|I_x|^{-1}$ of Definition 14.1.3 presumes; the second condition is also necessary (§§XIV.7, XIV.25). Every step of the construction of §§14.2–14.5 is proved, for the objects the construction itself produces, with no hypothesis beyond the text's; the half-space bundle $[0,\infty)\times\mathbb{R}^m\times\mathbb{R}^k$ with a constant non-zero section over its boundary satisfies all hypotheses of the theorem jointly (§XIV.26). *§14.1.* Proposition 14.1.1 is proved as printed, by Part I's abstract face calculus instead of a chart computation (§XIV.1); Definition 14.1.2 is formalized, and the remark after it, that the faces $\bar{\theta}_j$ are tame M-polyfolds, is proved (§XIV.1). Definition 14.1.3 is formalized with condition (1) read through the tame strong bundle charts of X (§XIV.2). Proposition 14.1.4 is proved, with the two standing assumptions named (§XIV.3). Theorem 14.1.6 is proved with tameness of X and paracompactness of $|X|$, which its proof uses, in place of the printed paracompactness of $|\partial X|$ (§XIV.4). Definition 14.1.7 is formalized (§XIV.5). Proposition 14.1.8 is proved in a stronger form: in its interior case and over ∂X, along generalized strong bundle isomorphisms and their classes, without the paracompactness of the orbit spaces (§XIV.6). Remark 14.1.10 is prose. *§14.2.* Lemmas 14.2.1, 14.2.2 and Proposition 14.2.3 are proved as printed (§XIV.8). *§14.3.* Lemma 14.3.1 is proved in a stronger form, the extended sets being symmetric under inversion; Definition 14.3.2 is formalized; Lemma 14.3.3 and Proposition 14.3.4 are proved as printed, the diagonal property of Proposition 14.3.4 by an argument the text does not give (§§XIV.9–XIV.10). Lemma 14.3.6 and Proposition 14.3.8 are proved with the facts their proofs use made explicit, and Lemma 14.3.7 in a stronger form (§§XIV.11–XIV.12). *§14.4.* The setting of §14.4 is built from the objects of §14.3 (§XIV.13). Definitions 14.4.2–14.4.4 and 14.4.8 are formalized, with the equivalence-relation properties the text asserts proved; Lemmas 14.4.5 and 14.4.7 are proved (§§XIV.14–XIV.15). Proposition 14.4.6 is proved, (1) without Lemma 14.4.7, (2) in a stronger form, for loops at points of $\widehat{V}_a$ and limit points in $\mathrm{cl}_X(\widehat{V}_a)\cap\partial X$ (§§XIV.15–XIV.16). Lemma 14.4.9, Proposition 14.4.10 and the existence in Definition 14.4.11 are proved, (3) of the lemma in a stronger form (§§XIV.17–XIV.18). Lemma 14.4.13 is proved, its (2) for points of $\mathrm{cl}_X(V_a)$ (§XIV.19). Proposition 14.4.14 is proved by a construction different from the text's (gluing by an ordinary partition of unity, then averaging over G_a), for the objects of §14.3 (§XIV.20). *§14.5.* Lemma 14.5.1 is proved in a stronger form, without tameness, and (14.53) holds for the cover of §14.3 itself, without shrinking (§XIV.21). Lemma 14.5.2 is proved in the form the proof of Theorem 14.1.9 uses it, an identity for the summed sections (§XIV.22). Proposition 14.5.4 is proved, its input carrying the property of the cut-offs that Lemma 14.5.1 (1) gives (§XIV.23). Theorem 14.5.5 is proved in a stronger form, without its properness and closure conditions (§XIV.24). *§14.6.* Faces are Part XI's ep-faces; Definitions 14.6.1 and 14.6.2 are formalized, sc^+ on a face and on a corner read through the charts of X; the restriction of Definition 14.6.1 is sc^+ on the corner; every structured sc^+-multisection on a face has a representative whose neighbourhoods satisfy (1) and (3) of a good system inside the face (§XIV.27). Theorem 14.6.3 is proved for representatives that agree on the intersections of faces, its extension part from Theorem 14.1.9 (§XIV.28). *Open* (§XIV.29): G.172 (the multisection form of Lemma 14.5.2, about 120 lines, unused), G.173 (Theorem 14.6.3 from a class-level compatible assignment), G.185 (condition (2) of a good system on a face, unused) and G.186 (the two readings of Theorem 14.1.9 built into the definitions or derived). No statement of Chapter 14 was found not to hold as we read it. One property of our own formalization is recorded in a remark (§XIV.7): the formal Definition 14.1.3 admits empty index sets, and a Lean statement of Theorem 14.1.9 without the charging condition does not hold when $\partial X\neq\emptyset$.

Decisions that shape the chapter.

- *The boundary.* ∂X is the set of objects with degeneracy index $d_X\geq 1$. It is closed, and saturated because the index is invariant along morphisms; sets open in ∂X are written $O\cap\partial X$ with O open in X. Good neighbourhoods are neighbourhoods of the object space, with the natural action of

Part XIII, and $U(x) = V(x) \cap \partial X$.

- *Sc⁺ on faces through the charts of X.* "$s|\bar{\theta}_j$ is sc⁺" in Definition 14.1.3 (1) is read as: in every tame strong bundle chart of W, at every point of $U(x)$, the local representative of s is sc⁺ on every local face germ $\{t_i = 0\}$ of the model, the condition of Part I's Definition 5.4.1 with ∂X replaced by $U(x)$. In a tame chart the germs of the $\bar{\theta}_j$ at a point are exactly these local faces, so no sub-M-polyfold structure on the $\bar{\theta}_j$ is needed. §14.6 reads sc⁺ on a face F, and on a corner $F \cap F'$, in the same way.
- *The standing assumptions.* "A strong bundle over a tame ep-groupoid" supplies tame strong bundle charts; it is formalized as the hypothesis that every object lies in a tame strong bundle chart, as in Part XII. "X admits sc-smooth partitions of unity" is Definition 5.5.1 for the object M-polyfold. Tameness of X and paracompactness of $|X|$ are assumed where a proof uses them; each result says which it uses.
- *Index sets.* The formal Definition 14.1.3 allows $I_x = \emptyset$. Theorem 14.1.9 is stated with the condition that Λ charges every boundary fibre, which for a structured Λ is the same as $I_x \neq \emptyset$ and is necessary for the conclusion.
- *Abstract steps, concrete instance.* Each step of §§14.3–14.5 is proved over a structure that names the data and the facts of the previous steps it uses (the z-loop setting, the boundary sections, the limit setting, the local extension data, the transport input, the assembly input, the criterion data), and then instantiated by the objects the construction produces from a structured representative. The final theorem takes no hypothesis on these structures.
- *z-loops with their arcs.* A z-loop and a z-arc along it are one inductive predicate, built by appending one morphism and one index step at a time. Concatenation, reversal, the action of G_a, the deformation of Proposition 14.4.6 and the transport of Proposition 14.5.4 are inductions on it.
- *Averaging.* Equivariance is obtained by averaging over the whole isotropy group, $\bar{s}_j(y) = |G_x|^{-1} \sum_g \mu(\Gamma(g^{-1}, g * y), s_{g(j)}(g * y))$, in Proposition 14.1.4 and in Proposition 14.4.14, instead of the text's averaging over a stabilizer and transport along an orbit.
- *Faces in §14.6* are Part XI's ep-faces X_θ (Definition 11.1.3). Theorem 14.6.3 takes chosen representatives that agree on the intersections of faces.

Map. All files are in `Polyfold/Chapter14`.

- *Good neighbourhoods of boundary points.* `GoodBoundary`: faces of an open set at a boundary point, good neighbourhoods, Proposition 14.1.1, Definition 14.1.2 (§XIV.1). `FaceTame`: the faces $\bar{\theta}_j$ of a good neighbourhood as faces of a face-structured open set, sub-M-polyfolds, tame M-polyfolds (§XIV.1).
- *Sc⁺-multisections over ∂X.* `BoundaryMultisection`: face-wise sc⁺ sections, Definition 14.1.3, Definition 14.1.5, Theorem 14.1.6, the averaging of Proposition 14.1.4 (§§XIV.2–XIV.4). `BoundaryStructured`: decorations, correspondences, structured tuples and their equivalence, Definition 14.1.7, the boundary domain support, the uncharged Lean statement of Theorem 14.1.9 (§§XIV.5, XIV.7). `EmptyIndex`: charging, empty index sets, necessity of charging (§§XIV.2, XIV.7).
- *Proposition 14.1.4.* `LocalExtension`: the local extension at a point and the gluing over an open set with closure inside V. `OpenExtension`: the extension on the open set itself, Proposition 14.1.4, the witness $\mathbb{R}^n \times \mathbb{R}^k$ (§XIV.3).
- *Proposition 14.1.8 over ∂X.* `BoundaryPull`, `BoundaryPush`: pull-backs and push-forwards of boundary structures, conditional on the transport of Definition 14.1.3 (1). `FaceTransport`, `FacePush`: that transport, and Proposition 14.1.8 for representatives and for classes (§XIV.6).
- *§14.2.* `GoodStructured`: Lemmas 14.2.1, 14.2.2, (P4), Proposition 14.2.3 (§XIV.8).
- *§14.3.* `ExtendCorrespondences`: (P5), Lemma 14.3.1, Definition 14.3.2, Lemma 14.3.3, Proposition

14.3.4 (§§XIV.9–XIV.10). `BoundaryCover`: saturated sets, Lemmas 14.3.6, 14.3.7, and Lemma 14.5.1 (§§XIV.11, XIV.21). `CoverVHat`: (P6), Proposition 14.3.8 (§XIV.12).

- *§14.4.* `ZLoops`: the z-loop setting, z-loops with arcs, Definitions 14.4.2–14.4.4, Lemma 14.4.5 (§XIV.14). `FiniteRepresentation`: Lemma 14.4.7 (§XIV.15). `ZLoopDeform`: the setting with topology, Proposition 14.4.6 (1) (§XIV.15). `ZRelationClosure`: equivariant neighbourhoods, Definition 14.4.8, Lemma 14.4.9 (1), (2), (4) (§XIV.17). `ZLoopCoherent`: boundary sections, Proposition 14.4.6 (2) at points of V_a and as a statement, Lemma 14.4.9 (3), Proposition 14.4.10, Definition 14.4.11, Lemma 14.4.13 (§§XIV.16–XIV.19). `ZLoopLimits`, `ZLoopLimitExists`: the limit argument, Proposition 14.4.6 (2) and its consequences (§§XIV.16–XIV.18, XIV.20). `CoherentExtension`: Definition 14.4.11 in the form Proposition 14.4.14 consumes, Proposition 14.4.14 (§XIV.20). `ZDataLocalExtension`: Proposition 14.4.14 for the boundary sections (§XIV.20). `ZLoopSetting`: the setting of §14.4 from the objects of §14.3, and all of §14.4 for them (§§XIV.13, XIV.16–XIV.20).
- *§14.5 and Theorem 14.1.9.* `TransportStatement`: the input and the statement of Proposition 14.5.4. `TransportProof`: Proposition 14.5.4 (§XIV.23). `TransportedSum`: the assembly input, the summed sections, the identity of Lemma 14.5.2 (§XIV.22). `StructurabilityCriterion`: Theorem 14.5.5 (§XIV.24). `ExtensionAssembly`: the functor Λ', the criterion data, the extension (§XIV.25). `Norm–Control`: norms of multisections, the size control (§XIV.25). `AssemblyInputInstance`: (14.53) for the cover of §14.3, Lemma 14.5.1 for it, the data of §§14.3–14.5 for every charging Λ (§§XIV.21, XIV.25). `BoundaryExtensionTheorem`: Theorem 14.1.9 and the extension part of Theorem 14.6.3 (§§XIV.7, XIV.25, XIV.28).
- *A witness with boundary.* `HalfSpaceBundle`: the half-space bundle as a strong bundle over an ep-groupoid, with all standing hypotheses. `HalfSpaceConstSection`: sc$^+$-sections satisfy Definition 14.1.3 (1); the multisection of a constant section; Theorem 14.1.9 applied (§§XIV.2, XIV.26).
- *§14.6.* `FaceGerm`: faces through a boundary point, sc$^+$ relative to a face, the gluing criterion. `FaceStructured`: structured data over a set, Definitions 14.6.1, 14.6.2. `CornerScPlus`: sc$^+$ on a corner. `FaceGoodSystem`: good systems on a face (§XIV.27). `FaceAssembly`: Theorem 14.6.3 (§XIV.28).

Throughout, $(P : W \to X, \mu)$ is a strong bundle over an ep-groupoid X (an object of the category of strong bundles over ep-groupoids of Part X) with source and target maps s, t, orbit map $\pi : X \to |X|$, isotropy groups G_x, and $\mu(\varphi, \cdot) : W_{s(\varphi)} \to W_{t(\varphi)}$ the fibre action. A neighbourhood with the natural action, $(U(x), \Phi, \Gamma)$, is Part XIII's: an open $U(x) \ni x$ with an action $\Phi_g = g * \cdot$ of G_x by sc-diffeomorphisms and morphisms $\Gamma(g, y) : y \to g * y$ that are all the morphisms between points of $U(x)$. A set B of objects is saturated if $s(\varphi) \in B$ iff $t(\varphi) \in B$. For sets U, U' of objects, $\mathcal{U}(U, U') = \{\varphi : s(\varphi) \in U,\ t(\varphi) \in U'\}$, and $\mathcal{U}(x, x')$ when the sets are those of x and x'. Index correspondences $J(x, x')$, their bi-action $h * [d] * g$, the swap $[d]^\sharp$ and the correspondences $[I \xleftarrow{\mathrm{id}} I \xrightarrow{g} I]$ are Part XIII's (Definition 13.3.4); a *pair* of a correspondence $[I \xleftarrow{a} K \xrightarrow{b} I']$ is $(a(k), b(k))$ for some $k \in K$ `Chapter14/ZLoops · SBLocalization.IndexCorr.Pairs`, and a relation holds for a correspondence when it holds on all its pairs `Chapter14/TransportProof · SBLocalization.IndexCorr.sat_of_pairs_t9` `Chapter14/TransportProof · SBLocalization.IndexCorr.rel_of_sat_t9`; every index of the right-hand set has a partner `Chapter14/TransportProof · SBLocalization.IndexCorr.exists_pairs_left_t9`. The pairs of $[d]^\sharp$ are the reversed pairs of $[d]$ `Chapter14/ZLoops · SBLocalization.IndexCorr.pairs_swap`, and those of $[\mathrm{id}, g]$ are the $(i, g(i))$ `Chapter14/ZLoops · SBLocalization.IndexCorr.pairs_ofPerm`. Two fibre-action identities are used everywhere: $\mu(\psi \circ \gamma, v) = \mu(\psi, \mu(\gamma, v))$ `Chapter14/BoundaryMultisection · SBLocalization.fibAct_comp_p3w3` and $\mu(1_y, v) = v$ `Chapter14/BoundaryMultisection · SBLocalization.fibAct_unit_p3w3`.

XIV.1 Proposition 14.1.1, Definition 14.1.2 and the remark after it (2017: the discussion before Definition 14.1, Definition 14.1): good neighbourhoods of boundary points

Faces at a point. For $x \in \partial X$ and an open set $V \ni x$, *faces of V at x* are sets $\theta_1, \dots, \theta_{d_X(x)}$ `Chapter14/GoodBoundary · EpGroupoid.HasBoundaryFaces` that are pairwise distinct connected components of $\{y \in V : d_X(y) = 1\}$, whose closures $\bar{\theta}_i = V \cap \operatorname{cl}\theta_i$ in V contain x, and with the counting formula $d_X(y) = \#\{i : y \in \bar{\theta}_i\}$ for $y \in V$. The formula gives (4)(c) of the proposition, $V \cap \partial X = \bigcup_i \bar{\theta}_i$ `Chapter14/GoodBoundary · EpGroupoid.HasBoundaryFaces.inter_boundary_eq`: a point of V lies in ∂X iff its count is positive.

Proposition 14.1.1 (same) `Chapter14/GoodBoundary · EpGroupoid.proposition_14_1_1`, p. 638. *Statement*: let X be tame, $x \in \partial X$ and $W \ni x$ open. There is an open $V(x) \subset W$ with the natural G_x-action such that (1) $V(x)$ and $V(x) \cap \partial X$ are connected, (2) the natural action is defined on $V(x)$, (3) $t : s^{-1}(\operatorname{cl}_X V(x)) \to X$ is proper, and (4) there are faces $\theta_1, \dots, \theta_{d_X(x)}$ of $V(x)$ at x, with $V(x) \cap \partial X = \bigcup_i \bar{\theta}_i$. "Arbitrarily small" is read as "inside every open $W \ni x$". The 2017 text states this inside the discussion before Definition 14.1; the 2021 edition gives a proof in a tame chart.

Proof. The Lean proof is the relative form `Chapter14/GoodBoundary · EpGroupoid.exists_isGoodBoundaryNbhd_le`: given a neighbourhood N of x with the natural action and t proper on $s^{-1}(\operatorname{cl} N)$, and an open $W \ni x$, there is a good $V \subset N \cap W$ carrying the action of N. The proposition follows by taking N from Part XIII (Theorem 7.1.13 with the properness of Lemma 13.1.2), and the relative form is what refinements of good systems need (§§XIV.4, XIV.8). It does not follow the text's chart computation; it uses Part I's abstract face calculus, which works for any function d satisfying the corner axioms, and the degeneracy index of a tame M-polyfold does `Chapter2/TameBoundary · MPolyfold.isCornerIndex_degIndex`.

1. *A neighbourhood all of whose open subsets are face-structured.* Part I provides an open $D \ni x$ such that every open subset of D is a face-structured open set in the sense of Definition 2.7.6 `Chapter2/TameBoundary · TameBoundary.exists_nbhd_forall_isFaceStructuredOpen`.
2. *Shrink inside $W \cap D$.* Restrict N to an invariant neighbourhood $N' \subset N \cap W \cap D$ with the same action `Chapter13/GoodSystem · EpGroupoid.NatNbhd.exists_restrict_subset`; the properness passes to subsets.
3. *The centred part.* Let $V_1 = \operatorname{reach}(N', x)$, the set of $y \in N'$ such that every face of N' through y contains x `Chapter2/TameBoundary · TameBoundary.reach`. It is open `Chapter2/TameBoundary · TameBoundary.isOpen_reach` and centred at x, every face of V_1 containing x `Chapter2/TameBoundary · TameBoundary.isCenteredAt_reach`. It is invariant under the natural action: each Φ_g is a homeomorphism of N' preserving d_X and fixing x, so it maps the faces through y onto the faces through $g * y$ `Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.facesThrough_image`, and the condition "every face through y contains x" is carried to $g * y$. That the natural action preserves d_X is `Chapter14/GoodBoundary · EpGroupoid.NatNbhd.degIndex_Φ`.
4. *The component.* Let V be the connected component of x in V_1. It is open (components of open sets in the locally connected object space are open) and invariant: $\Phi_g(V)$ is connected, contains $\Phi_g(x) = x$, and lies in V_1, hence in V.
5. *V is centred and face-structured.* The component of a centred set is centred `Chapter14/GoodBoundary · TameBoundary.isCenteredAt_connectedComponentIn`, and the faces of V are the traces of those of V_1: for $p \in V$, the component of p in $V \cap \{d = 1\}$ is its component in $V_1 \cap \{d = 1\}$ `Chapter14/GoodBoundary · TameBoundary.connectedComponentIn_inter_deg_of_mem_cc`. V is face-structured by step 1.

6. *The faces θ_i.* Since V is centred and face-structured, the faces of V through x are $d_X(x)$ in number; enumerate them through an equivalence with $\{1, \dots, d_X(x)\}$ and let θ_i be the component of $V \cap \{d = 1\}$ at a chosen point of the i-th face. The counting formula is the face-structured property of V, and $x \in \bar{\theta}_i$ is centredness.
7. *Connectedness of $V \cap \partial X$.* By (4)(c), $V \cap \partial X = \bigcup_i \bar{\theta}_i$ (together with x); each face is preconnected `Chapter14/GoodBoundary · TameBoundary.isPreconnected_faceAt` and all contain x, so the union is preconnected.
8. *Properness.* t is proper on $s^{-1}(\operatorname{cl} V)$ because $\operatorname{cl} V \subset \operatorname{cl} N$ is closed and t is proper on $s^{-1}(\operatorname{cl} N)$ `Chapter7/ProperMaps · isProperMap_domRestrict_of_subset_of_isClosed`.

The text's argument builds the set Q in a tame chart with $\partial C \cap Q = \Theta$; the face calculus replaces it and gives the same four properties.

Definition 14.1.2 (restated) `Chapter14/GoodBoundary · EpGroupoid.IsGoodBoundaryNbhd`, p. 641. A neighbourhood $V(x)$ with the natural G_x-action is a *good open neighbourhood of x in X* if (1) $V(x)$ and $V(x) \cap \partial X$ are connected, (3) t is proper on $s^{-1}(\operatorname{cl}_X V(x))$, and (4) faces at x exist; the *good open neighbourhood in ∂X* is $U(x) = V(x) \cap \partial X$ `Chapter14/GoodBoundary · EpGroupoid.NatNbhd.bdry`. *Restated*: the text phrases the definition through $U(x)$; the formal one is on $V(x)$, of which $U(x)$ is the trace, because every later use (Definition 14.1.5, Proposition 14.1.4, §14.3) needs $V(x)$. "Admits the natural G_x-action" is read as carrying one, which is how Part XIII reads the same words. The remarks after the definition are proved: $U(x) = \bigcup_i \bar{\theta}_i$ `Chapter14/GoodBoundary · EpGroupoid.IsGoodBoundaryNbhd.bdry_eq`; $U(x)$ is invariant under the natural action `Chapter14/GoodBoundary · EpGroupoid.NatNbhd.mapsTo_Φ_bdry`, since the action preserves d_X; and only boundary points have good neighbourhoods `Chapter14/GoodBoundary · EpGroupoid.IsGoodBoundaryNbhd.mem_boundary`: for $d_X(x) = 0$ the counting formula forces $V \cap \partial X = \emptyset$, which is not connected in the sense that requires non-emptiness. Good neighbourhoods exist inside every open $W \ni x$ `Chapter14/GoodBoundary · EpGroupoid.exists_isGoodBoundaryNbhd`, the witness of the definition at every boundary point of a tame X.

The remark after Definition 14.1.2 (same) `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.face_isTame`, p. 641: for X tame, the $\bar{\theta}_j$ are tame M-polyfolds. *Proof.* Fix V open with faces $\theta_1, \dots, \theta_{d_X(x)}$ at x.

1. *The $\bar{\theta}_i$ are the faces of V.* A point of V with $d_X = 1$ in $\operatorname{cl} \theta_i$ lies in θ_i, a component of $V \cap \{d = 1\}$ being closed in it `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.mem_of_mem_closure`; so $\bar{\theta}_i$ is the face of V through any point of θ_i `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.eq_faceAt`, hence a face of V in the sense of Definition 2.7.6 `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.isFaceIn`, and every face of V is one of them `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.isFaceIn_iff`. They are pairwise distinct `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.injective_closure`, and the faces of V through y are the $\bar{\theta}_i \ni y$ `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.facesThrough_eq`.
2. *V is face-structured.* The counting formula of (4) is the face-structured condition, so V is a face-structured open set `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.isFaceStructuredOpen`. Two bridge lemmas identify Definition 2.5.10 (faces of an M-polyfold) with Definition 2.7.6 (faces of an open set) on the whole space `Chapter14/FaceTame · MPolyfold.isFace_iff_isFaceIn_univ` `Chapter14/FaceTame · MPolyfold.isFaceStructured_iff_univ`; with them, the M-polyfold V (Part I's restriction to an open set) is face-structured `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.isFaceStructured_restrictOpen` and each $\bar{\theta}_i$ is one of its faces `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.isFace_restrictOpen`.
3. *Tameness.* Part I's Theorem 2.5.17 (a face of a tame face-structured M-polyfold is a tame M-polyfold, with $d = d_F + 1$ on it) applied to $\bar{\theta}_i \subset V$ gives the remark.

Each $\bar{\theta}_i$ is also a sub-M-polyfold of X `Chapter14/FaceTame · EpGroupoid.HasBoundaryFaces.isSubMPolyfold`, stated for good neighbourhoods `Chapter14/FaceTame · EpGroupoid.IsGoodBoundaryNbhd.faces_isSubMPolyfold`. That this

structure agrees with the one as a face of V is not stated, though both come from the same local face charts. Only the tameness of X is assumed. Nothing in §§14.1–14.5 uses the remark: Definition 14.1.3 (1) is read in the charts of X (§XIV.2), and so is sc$^+$ on a face in §14.6 (§XIV.27).

XIV.2 Definition 14.1.3 (2017: Definition 14.2): sc$^+$-multisections over ∂X

Face-wise sc$^+$ sections `Chapter14/BoundaryMultisection · SBLocalization.IsFaceScPlusSectionOn`. Condition (1) of Definition 14.1.3 asks that "$s_i^x|\bar\theta_j$ is an sc$^+$-section of $W|\bar\theta_j$" for every face $\bar\theta_j$ of $U(x)$. Formally, a map s from objects to W is *face-wise sc$^+$ on a set U* if (a) for $y \in U$, $P(s(y)) = y$ and $s(y) \in W_{0,1}$, and (b) for every tame strong bundle chart c of W, every tame presentation T of its corner, $C = T^{-1}(\text{quadrant})$, every $y \in U$ in the chart's domain and every index i with $t_i(T\varphi(y)) = 0$, the local representative of s is sc$^+$ on the local face germ $\{t_i = 0\}$ near $\varphi(y)$: some neighbourhood of $\varphi(y)$ meets the face in an sc-retract, and the representative restricted to it is an sc$^+$-section there (Part XII's notion `IsScPlusOnFaceNear`). This is Part I's Definition 5.4.1 (an sc$^+$-section over ∂X) with ∂X replaced by U. *Reading*: at a point of $U(x)$ the germs of the $\bar\theta_j$ through it are, in a tame chart, exactly the local faces $\{t_i = 0\}$ through its image, so (b) says that the restriction to each $\bar\theta_j$ is sc$^+$ for the structure the $\bar\theta_j$ inherit from the charts, without building that structure. The condition is monotone in U `Chapter14/BoundaryMultisection · SBLocalization.IsFaceScPlusSectionOn.mono` and pointwise in U `Chapter14/FaceTransport · SBLocalization.isFaceScPlusSectionOn_iff_singleton`; it depends only on the values of s on ∂X near U, because a local face of a tame chart is mapped by the chart inverse into ∂X `Chapter14/BoundaryPull · SBLocalization.symm_mem_boundary_fm` `Chapter14/BoundaryPull · SBLocalization.IsFaceScPlusSectionOn.congr_fm`, and since the representative near a point only sees the section on a neighbourhood `Chapter14/BoundaryPull · IsScPlusOnFaceNear.congr_open_fm`. Three sources of examples: an sc$^+$-section over ∂X in the sense of Definition 5.4.1 is face-wise sc$^+$ on every $U \subset \partial X$ `Chapter14/BoundaryMultisection · SBLocalization.isFaceScPlusSectionOn_of_isBoundaryScPlusSection`; the zero section is `Chapter14/BoundaryMultisection · SBLocalization.isFaceScPlusSectionOn_fiberZero`; and every sc$^+$-section of W on all of X is face-wise sc$^+$ on every set `Chapter14/HalfSpaceConstSection · SBLocalization.isFaceScPlusSectionOn_of_isScPlusSectionOn_w1` (extra). The last is proved in a tame chart and a tame presentation: the local face germ $O \cap V \cap \{t_i = 0\}$ is a sub-retract of O (Part I's corner retraction), the chart representative of s is sc-smooth on O as a map into F_1, and an sc-smooth map between retracts restricts to a sub-retract of its source. It is what makes the multisection of a constant section an example with boundary (§XIV.26).

Definition 14.1.3 (restated) `Chapter14/BoundaryMultisection · SBLocalization.BoundarySymmetricStructure` `Chapter14/BoundaryMultisection · SBLocalization.IsBoundaryScPlusMultisection`, p. 642. *Statement as formalized*: let V be a neighbourhood of x with the natural action and $U(x) = V \cap \partial X$. A *symmetric sc$^+$-section structure for Λ on $U(x)$* is a finite index set I_x with an action of G_x, and maps s_i^x, $i \in I_x$, with (1) s_i^x face-wise sc$^+$ on $U(x)$; (2) $s^x_{g(i)}(g * y) = \mu(\Gamma(g, y), s_i^x(y))$ for $y \in U(x)$; (3) $\Lambda(w) = |I_x|^{-1} \cdot \#\{i \in I_x : s_i^x(P(w)) = w\}$ for $P(w) \in U(x)$. Each s_i^x is a section over $U(x)$ `Chapter14/BoundaryMultisection · SBLocalization.BoundarySymmetricStructure.P_s`, and a structure restricts to a smaller neighbourhood with the same action `Chapter14/BoundaryMultisection · SBLocalization.BoundarySymmetricStructure.restrict`. $\Lambda : W \to \mathbb{Q}^+$ is a *functor on $W|\partial X$* if $\Lambda(\mu(\varphi, w)) = \Lambda(w)$ for every morphism φ with $s(\varphi) \in \partial X$ and $P(w) = s(\varphi)$ `Chapter14/BoundaryMultisection · SBLocalization.IsBoundaryWFunctor` (then $t(\varphi) \in \partial X$ as well), and an *sc$^+$-multisection over ∂X* if it is such a functor and every $x \in \partial X$ has a good neighbourhood $V(x)$ with a symmetric sc$^+$-section structure on $U(x)$. Only the values of Λ over ∂X enter. *Restated*, in three respects. The text's standing assumption that $|X|$ is paracompact is not part of the definition, which

does not use it. The text indexes the structure by $U(x)$; the formal one is attached to $V(x)$, with $U(x)$ its trace. And the index set I_x is not required to be non-empty; with $I_x = \emptyset$ condition (3) reads $\Lambda(w) = 0^{-1} \cdot 0 = 0$ in $\mathbb{Q}^+$, a reading the text's $|I_x|^{-1}$ does not intend. This last point is a property of the formal definition, discussed with Theorem 14.1.9 (§XIV.7); it is gap G.186 (a) (§XIV.29). *Witness*: for X tame, the zero multisection ($\Lambda(w) = 1$ iff $w = 0_{P(w)}$) is an sc⁺-multisection over ∂X `Chapter14/BoundaryMultisection · SBLocalization.isBoundaryScPlusMultisection_zero`, with one index and the zero section on every $U(x)$ `Chapter14/BoundaryMultisection · SBLocalization.zeroBoundaryStructure`, good neighbourhoods coming from Proposition 14.1.1. A non-degenerate witness, with boundary and non-empty support, is the multisection of a constant section on the half-space bundle (§XIV.26).

Charging the boundary `Chapter14/EmptyIndex · SBLocalization.ChargesBoundary` (extra). Λ *charges* ∂X if over every $x \in \partial X$ there is some w with $\Lambda(w) \neq 0$. For the text's multisections, whose local weights $|I_x|^{-1}\#\{\dots\}$ sum to 1 on every fibre, this holds automatically. For a structure it is the same as non-empty index sets: a structure at a point that Λ charges has $I_x \neq \emptyset$ `Chapter14/EmptyIndex · SBLocalization.BoundarySymmetricStructure.nonempty_I_a6`, since with $I_x = \emptyset$ (3) gives $\Lambda = 0$ over $U(x)$; conversely, with $i \in I_x$ the point $w = s_i^x(x)$ has $\Lambda(w) \geq |I_x|^{-1} > 0$ `Chapter14/EmptyIndex · SBLocalization.chargesBoundary_of_nonempty_a6`. The zero multisection charges ∂X, $\Lambda(0_x) = 1$ `Chapter14/EmptyIndex · SBLocalization.chargesBoundary_zeroMultisection_a6`.

XIV.3 Proposition 14.1.4 (2017: Remark 14.2): the local extension

Proposition 14.1.4 (explicit) `Chapter14/OpenExtension · SBLocalization.proposition_14_1_4_full`, p. 642. *Statement as formalized*: suppose every object lies in a tame strong bundle chart and the object M-polyfold admits sc-smooth partitions of unity. Let $(s_i)_{i \in I_x}$ be a symmetric sc⁺-section structure for some Λ on $U(x) = V(x) \cap \partial X$. Then there are sc⁺-sections $\bar{s}_i$ on $V(x)$ with $\bar{s}_i = s_i$ on $U(x)$ and $\bar{s}_{g(i)}(g * y) = \mu(\Gamma(g, y), \bar{s}_i(y))$ for all $y \in V(x)$ (Definition 13.2.5 (1)). *Explicit*: the two hypotheses are the readings of the text's standing assumptions, "a strong bundle over a tame ep-groupoid" (tame strong bundle charts) and "X admits sc-smooth partitions of unity" (Definition 5.5.1 for the object M-polyfold), the same readings as in Theorem 12.3.9 (Part XII). The paracompactness of $|X|$ and the goodness of $V(x)$ are not used. The text calls the $\bar{s}_i$ "sc⁺-multisections"; sc⁺-sections are meant, as its proof shows. *Witness of the hypotheses*: the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ over the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter14/OpenExtension · SBLocalization.proposition_14_1_4_hypotheses_rn`, which has $\partial X = \emptyset$; the half-space bundle of §XIV.26 witnesses them with boundary.

Proof. The text extends each s_i to an sc⁺-section on $V(x)$ "using an sc-smooth partition of unity argument" (Proposition 12.3.7 locally), averages over the stabilizer H of an index, and transports the result along the orbit of the index. The Lean proof has the same two stages, with a different gluing and a different averaging.

Stage 1: the non-equivariant extension. For each i, find an sc⁺-section on $V(x)$ equal to s_i on $V(x) \cap \partial X$ `Chapter14/OpenExtension · SBLocalization.exists_scPlusSectionOn_extension_of_face` (for any open V and any face-wise sc⁺ s on $V \cap \partial X$).

1. *The local step* `Chapter14/LocalExtension · StrongBundle.exists_local_extension_of_face`. Let y lie in a tame strong bundle chart c, let $G \ni y$ be open, and let s be a section with values in $W_{0,1}$ on $G \cap \partial X$ that is sc⁺ on every local face germ at y (for $y \notin \partial X$ there are none). Then there are an open U with $y \in U \subset G$ and an sc⁺-section σ of W with $\sigma = s$ on $U \cap \partial X$. This is Part I's local step of the ∂-extension (Proposition 12.3.7 in a tame chart, Part XII) with the hypothesis made local: the

face condition only at y, the section property only on $G \cap \partial X$. The proof applies the inclusion–exclusion construction at the faces through $\varphi(y)$ `Chapter12/LocalBoundaryExtension · proposition_12_3_7_of_faces_at`, which gives an sc⁺ extension e of the local representative on a neighbourhood V' of $\varphi(y)$; restricts the chart to $S = \mathrm{dom} \cap \varphi^{-1}(V')$; takes an sc-smooth bump function β at y supported in $S \cap G$ `Chapter5/BumpFunctions · proposition_5_5_3`; and sets $\sigma =$ the section cut off by β, $\beta \cdot e$ in the chart and 0 outside, which is sc⁺ `Chapter3/IndexAlongPathsLocalBump · isScPlusSection_cutoffSection_of_isChart` and equals s on the interior of $\{\beta = 1\}$ intersected with $S \cap G \cap \partial X$.

2. *Gluing over a set with closure inside V* `Chapter14/LocalExtension · SBLocalization.exists_scPlusSection_extension_of_face`. Let G be open with $\mathrm{cl}\, G \subset V$. Every object y has a local piece: for $y \in \partial X \cap V$ the local extension of step 1 with $G' = V$; for $y \in \partial X \setminus V$, hence $y \notin \mathrm{cl}\, G$, the zero section on $X \setminus \mathrm{cl}\, G$; for $y \notin \partial X$ the zero section on $X \setminus \partial X$. These open sets cover X; an sc-smooth partition of unity subordinate to them glues the pieces to a global section `Chapter12/ScPlusExtension · glueSection`, which is sc⁺ `Chapter12/ScPlusExtension · isScPlusSection_glueSection`, and at a point of $G \cap \partial X$ every piece whose weight is non-zero there equals s, so the glued section does. A single partition of unity cannot give the extension on all of V: it would have to cover the frontier of V, where the face data may be unbounded and no local extension exists.
3. *Exhaustion.* The object space is metrizable. Let $G_n = \{y : \mathrm{dist}(y, X \setminus V) > 1/(n+1)\}$; then $\mathrm{cl}\, G_n \subset G_{n+1}$ and $\bigcup_n G_n = V$. By step 2, σ_n is a global sc⁺-section equal to s on $G_{n+1} \cap \partial X$.
4. *Cut-offs.* For each n an sc-smooth $\chi_n : X \to [0,1]$ equals 1 on $\mathrm{cl}\, G_n$ and has support in G_{n+1} `Chapter14/OpenExtension · SBLocalization.exists_scSmooth_cutoff_p3w4` (a partition of unity for the two-set cover $\{G_{n+1}, X \setminus \mathrm{cl}\, G_n\}$). With $c_0 = 0$ and $c_{n+1} = \chi_n$, the weights $w_n = c_{n+1} - c_n$ vanish off G_{n+1}, are locally finite on V (on G_m, $c_n = 1$ for $n > m$, so only $w_0, \dots, w_m$ are non-zero there), and sum to 1 on V.
5. *The extension* is $\sum_n w_n \sigma_n$. On each G_m it is a finite sum of sc⁺-sections times sc-smooth functions, hence sc⁺ there; being sc⁺ on an open set is local `Chapter14/OpenExtension · SBLocalization.IsScPlusSectionOn.of_locally_p3w4`, so it is sc⁺ on V. At $y \in V \cap \partial X$, every σ_n with $w_n(y) \neq 0$ has $y \in G_{n+1}$ and so $\sigma_n(y) = s(y)$; the weights sum to 1.

Stage 2: averaging. Given sc⁺-sections σ_k on V (indexed by I_x), the average is `Chapter14/BoundaryMultisection · SBLocalization.avgSection`

$$\bar{s}_j(y) = |G_x|^{-1} \sum_{g \in G_x} \mu\big(\Gamma(g^{-1}, g * y),\ \sigma_{g(j)}(g * y)\big),$$

the sum of the translates `Chapter14/BoundaryMultisection · SBLocalization.avgTerm` taken in the fibre over y. The morphism $\Gamma(g^{-1}, g * y)$ goes from $g * y$ to y `Chapter14/BoundaryMultisection · SBLocalization.src_Γ_inv_Φ` `Chapter14/BoundaryMultisection · SBLocalization.tgt_Γ_inv_Φ`, so each translate is a section over V `Chapter14/BoundaryMultisection · SBLocalization.P_avgTerm`. Each translate is sc⁺ on V `Chapter14/BoundaryMultisection · SBLocalization.isScPlusSectionOn_avgTerm`: $y \mapsto g * y$ is an sc-diffeomorphism of V and $y \mapsto \Gamma(g^{-1}, g*y)$ an sc-smooth family of morphisms, along which the fibre action preserves sc⁺-sections (Part XIII). A finite sum of sc⁺-sections is one `Chapter14/BoundaryMultisection · SBLocalization.isSection_sumSection_p3w3`, so the average is sc⁺ `Chapter14/BoundaryMultisection · SBLocalization.isScPlusSectionOn_avgSection`; its value in the fibre is the displayed sum `Chapter14/BoundaryMultisection · SBLocalization.avgSection_apply`.

1. *The key identity* `Chapter14/BoundaryMultisection · SBLocalization.avgTerm_equivariant`: $\mu(\Gamma(h, y), \mathrm{term}_{gh,j}(y)) = \mathrm{term}_{g,h(j)}(h * y)$. Both sides are μ of a composite morphism applied to $\sigma_{gh(j)}(gh * y)$, and the two composites, $\Gamma(h, y) \circ \Gamma((gh)^{-1}, gh * y)$ and $\Gamma(g^{-1}, g * (h * y))$,

are equal by the multiplicativity of the natural representation; the fibre action is functorial.

2. *Equivariance* `Chapter14/BoundaryMultisection · SBLocalization.avgSection_equivariant`: $\bar{s}_{h(j)}(h * y) = \mu(\Gamma(h,y), \bar{s}_j(y))$. Reindex the sum defining $\bar{s}_j(y)$ along $g \mapsto gh$ and apply the key identity term by term; $\mu(\Gamma(h,y),\cdot)$ is linear on fibres.
3. *The average extends what is already equivariant* `Chapter14/BoundaryMultisection · SBLocalization.avgSection_eq_of_equivariant`: if $\sigma_{g(j)}(g*y) = \mu(\Gamma(g,y),\sigma_j(y))$ on an invariant $B \subset V$, then every translate equals σ_j on B, since $\mu(\Gamma(g^{-1}, g*y)) \circ \mu(\Gamma(g,y)) = \mu(1_y) = \mathrm{id}$, and the average of $|G_x|$ equal terms is that term.
4. *Conclusion* `Chapter14/BoundaryMultisection · SBLocalization.proposition_14_1_4_hyp`: for any sc$^+$-sections σ_i on $V(x)$ with $\sigma_i = s_i$ on $U(x)$, the averages are sc$^+$, equivariant, and equal to s_i on $U(x)$, by 3 with $B = U(x)$, which is invariant (§XIV.1) and on which the s_i are equivariant by Definition 14.1.3 (2). For the zero structure the zero sections are such σ_i `Chapter14/BoundaryMultisection · SBLocalization.proposition_14_1_4_hyp_witness`.

The text averages over the stabilizer of one index of each orbit and transports along the orbit; averaging over all of G_x produces the same symmetry and needs no choice of orbit representatives. Two variants are proved on the way: the extension to any $V' \le V(x)$ with the same action and $\mathrm{cl}\, V' \subset V(x)$, from step 2 alone `Chapter14/LocalExtension · SBLocalization.proposition_14_1_4_of_closure`, and the extension to a good $V'(x) \le V(x)$ with $\mathrm{cl}\, V'(x) \subset V(x)$ provided by Lemma 14.2.1 `Chapter14/LocalExtension · SBLocalization.proposition_14_1_4`.

XIV.4 Definition 14.1.5 and Theorem 14.1.6 (2017: Definition 14.3, Theorem 14.1): good systems for ∂X

Definition 14.1.5 (restated) `Chapter14/BoundaryMultisection · EpGroupoid.BoundaryGoodSystem`, p. 643. A *good system for ∂X* assigns to every $x \in \partial X$ a good neighbourhood $V(x)$ with its natural action (Definition 14.1.2) such that for all $x, x' \in \partial X$ and every connected component Σ of $V(x,x') = \mathcal{U}(V(x), V(x'))$, the maps $s : \Sigma \to V(x)$ and $t : \Sigma \to V(x')$ are sc-diffeomorphisms onto open subsets. It carries $U(x) = V(x) \cap \partial X$ `Chapter14/BoundaryMultisection · EpGroupoid.BoundaryGoodSystem.U` and the boundary morphism sets $\mathcal{U}(x,x') = \mathcal{U}(U(x), U(x'))$ `Chapter14/BoundaryMultisection · EpGroupoid.BoundaryGoodSystem.mor`. *Restated*: the text says that for all x, x' "there exist" good $V(x)$, $V(x')$ restricting to $U(x)$, $U(x')$ with the component condition; one $V(x)$ per point is fixed, which is how the proof of Theorem 14.1.6 chooses them and how every later use takes them. The component condition passes to open subsets `Chapter14/BoundaryMultisection · EpGroupoid.BoundaryGoodSystem.isDiffeoComponent_of_subset`: a component of the smaller morphism set lies in a component of the larger one, on which s and t are sc-diffeomorphisms onto open sets, and their restrictions to an open subset remain so. *Refinement* `Chapter14/BoundaryMultisection · EpGroupoid.BoundaryGoodSystem.exists_refinement`: for tame X and open $W(x) \ni x$, a good system has a refinement inside the $W(x)$ whose neighbourhoods carry the same natural actions; at each x, Proposition 14.1.1 in its relative form (§XIV.1) shrinks $V(x)$ inside $W(x)$ keeping the action, and the component condition passes to the smaller sets. This is the boundary analogue of Proposition 13.1.3.

Theorem 14.1.6 (explicit) `Chapter14/BoundaryMultisection · EpGroupoid.theorem_14_1_6_hyp`, p. 643. *Statement as formalized*: let X be tame with $|X|$ paracompact, and let $O(x) \ni x$ be open in X for $x \in \partial X$. There is a good system for ∂X with $V(x) \subset O(x)$, so $U(x) \subset O(x) \cap \partial X$. A cover of ∂X by sets open in ∂X is written $(O(x) \cap \partial X)$ with $O(x)$ open in X.

Proof, the text's. 1. Put $W(y) = O(y)$ for $y \in \partial X$ and $W(y) = X$ otherwise, and apply Theorem 13.1.8 to X (Part XIII) `Chapter13/GoodSystemExistence · theorem_13_1_8`: there is a good system of X, with neighbourhoods $N(y) \subset W(y)$ carrying the natural action, t proper on $s^{-1}(\mathrm{cl}\, N(y))$, and the component condition. 2. At each $x \in \partial X$, Proposition 14.1.1 in its relative form shrinks $N(x)$ to a good $V(x) \subset N(x)$ with the same action (§XIV.1). 3. The component condition passes from the $N(x)$ to the smaller open $V(x)$, as above. *Explicit*: the text assumes $|\partial X|$ paracompact; its proof applies Theorem 13.1.8 to X, which needs $|X|$ paracompact, and Proposition 14.1.1, which needs X tame. Both are standing assumptions of the chapter, and the formal statement takes them; whether paracompactness of $|\partial X|$ alone suffices is not examined. *Witness*: the trivial ep-groupoid of a tame quadrant `Chapter7/TameBoundaryEp · EpGroupoid.trivial_quadrant_witness`, whose orbit space is paracompact `Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial`, and the half-space ep-groupoid of §XIV.26.

XIV.5 Definition 14.1.7 (2017: Definition 14.4): structured sc⁺-multisections over ∂X

Decorations and correspondences. A *decoration* of Λ `Chapter14/BoundaryStructured · SBLocalization.BoundaryDecoration` is a good system $\mathcal{U}$ for ∂X (Definition 14.1.7 (1)) with a symmetric sc⁺-section structure $S_x = (I_x, (s_i^x))$ for Λ on each $U(x)$ (Definition 14.1.7 (3)). For φ with $s(\varphi) \in U(x)$, $t(\varphi) \in U(x')$ and indices $i \in I_x$, $j \in I_{x'}$, the identity of (4) is $s_j^{x'}(t(\varphi)) = \mu(\varphi, s_i^x(s(\varphi)))$ `Chapter14/BoundaryStructured · SBLocalization.BoundaryDecoration.Compat`. *Correspondences* for a decoration `Chapter14/BoundaryStructured · SBLocalization.BoundaryDecoration.Tau`, p. 645, are maps $\tau_{x,x'} : \mathcal{U}(x,x') \to J(x,x')$, $x, x' \in \partial X$, with (2) $\tau_{x,x'}$ locally constant on $\mathcal{U}(x,x')$ with finitely many values; (a) $\tau_{x',x}(\varphi^{-1}) = \tau_{x,x'}(\varphi)^\sharp$; (b) $\tau_{x,x}(\Gamma(g,z)) = [I_x \xleftarrow{\mathrm{id}} I_x \xrightarrow{g} I_x]$; (c) $\tau_{x,x'}(g' * \varphi * g) = g' * \tau_{x,x'}(\varphi) * g$ for the bi-action of $G_{x'} \times G_x$; and (4) the identity above for every pair (i,j) of $\tau_{x,x'}(\varphi)$. Condition (4) is read pointwise on $\mathcal{U}(x,x')$, as printed; Definition 13.3.4 asks it along the component of φ, and on all of $\mathcal{U}(x,x')$ the two are the same. A *structured tuple* is $(\Lambda, \mathcal{U}, S, \tau)$ `Chapter14/BoundaryStructured · SBLocalization.BoundaryStructuredTuple` with Λ an sc⁺-multisection functor over ∂X (Definition 14.1.3, whose structures are those of the decoration).

Definition 14.1.7 (restated) `Chapter14/BoundaryStructured · SBLocalization.BoundaryStructuredMultisection`, p. 644. A *structured sc⁺-multisection over ∂X* is an equivalence class of structured tuples; Λ is *structurable over ∂X* if some class lies over it `Chapter14/BoundaryStructured · SBLocalization.IsBoundaryStructurable`. The text defines the equivalence as "similar to the X-case" (Definition 13.3.8). *The equivalence as formalized* `Chapter14/BoundaryStructured · SBLocalization.BoundaryStructuredTuple.Equiv`: $T \sim T'$ if $\Lambda = \Lambda'$ over ∂X, and there are open $O(x) \ni x$ with $O(x) \cap \partial X \subset U(x) \cap U'(x)$, $I_x = I'_x$, and $\tau_{x,x'} = \tau'_{x,x'}$, transported along these equalities, on the morphisms from $O(x) \cap \partial X$ to $O(x') \cap \partial X$. It is an equivalence relation `Chapter14/BoundaryStructured · SBLocalization.BoundaryStructuredTuple.equivalence` `Chapter14/BoundaryStructured · SBLocalization.BoundaryStructuredTuple.setoid`: transitivity intersects the open sets, which needs no tameness. *Agreement with the literal analogue* `Chapter14/BoundaryStructured · SBLocalization.BoundaryStructuredTuple.equiv_iff_goodSystem`: for X tame, $T \sim T'$ iff $\Lambda = \Lambda'$ over ∂X and some good system $\mathcal{U}''$ for ∂X with $U''(x) \subset U(x) \cap U'(x)$ carries equal index sets and equal correspondences. One direction takes $O(x) = V''(x)$; the other refines the good system of T inside the $O(x)$ (§XIV.4), whose boundary traces lie in $O(x) \cap \partial X \subset U(x) \cap U'(x)$. The underlying functor of a class `Chapter14/BoundaryStructured · SBLocalization.BoundaryStructuredMultisection.forget` is well defined over ∂X `Chapter14/BoundaryStructured · SBLocalization.BoundaryStructuredMultisection.forget_mk`, and Λ is structurable iff it

is the functor of a structured tuple Chapter14/BoundaryStructured · SBLocalization.isBoundaryStructurable_iff. *Restated*: as in Definition 14.1.3, the paracompactness of $|X|$ is not part of the definition. *Witness*: for X tame with $|X|$ paracompact, the zero multisection is structurable over ∂X Chapter14/BoundaryStructured · SBLocalization.isBoundaryStructurable_zero, on a good system from Theorem 14.1.6, with the zero structures and $\tau \equiv [\{*\} \leftarrow \{*\} \rightarrow \{*\}]$ Chapter14/BoundaryStructured · SBLocalization.BoundaryDecoration.zeroTau. A non-degenerate witness is the multisection of a constant non-zero section on the half-space bundle (§XIV.26).

XIV.6 Proposition 14.1.8 (2017: Proposition 14.1): generalized strong bundle isomorphisms

Proposition 14.1.8 (stronger), p. 645. *The text*: a generalized strong bundle isomorphism $\bar{f} : W \to W'$ between strong bundles over tame ep-groupoids carries structurable sc⁺-multisections to structurable ones, on the whole bundle and over the boundary; the proof is "similar to" those of Theorems 13.4.1 and 13.4.3, for a representative $W \xleftarrow{\Phi} V \xrightarrow{\Psi} W'$ by strong bundle equivalences with V over a tame ep-groupoid: first pull back along Φ, then push forward along Ψ.

The interior case (same). For sc⁺-multisections of the whole bundle, the push-forward and the pull-back along a generalized strong bundle isomorphism preserve structurability Chapter13/MultisectionPushforward · SBLocalization.GenSBMap.isStructurable_pushMult Chapter13/MultisectionPushforward · SBLocalization.GenSBMap.isStructurable_pullMult; this is Theorems 13.4.1 and 13.4.3 composed along a representative, proved in Part XIII. Nothing in it is specific to the boundary.

The boundary case. For Λ structurable over ∂X (Definition 14.1.7), $\bar{f}_*\Lambda$ is structurable over $\partial X'$, and for Λ' structurable over $\partial X'$, $\bar{f}^*\Lambda'$ is structurable over ∂X: for a representative with tame middle and tame target Chapter14/FacePush · SBLocalization.proposition_14_1_8 Chapter14/FacePush · SBLocalization.proposition_14_1_8_pullMult; for a representative between tame ep-groupoids, the middle being then tame Chapter14/FacePush · SBLocalization.proposition_14_1_8_tame Chapter14/FacePush · SBLocalization.proposition_14_1_8_pullMult_tame; and at the level of classes Chapter14/FacePush · SBLocalization.GenSBMap.proposition_14_1_8 Chapter14/FacePush · SBLocalization.GenSBMap.proposition_14_1_8_pullMult, through the chosen representative of the class (Part X), the pull-back needing only X tame. *Stronger*: the paracompactness of the orbit spaces is not used, and the two steps hold for every pull map (strong bundle equivalences and proper strong bundle coverings) and every strong bundle equivalence.

Proof. The construction is Part XIII's for Theorems 13.4.1 and 13.4.3, carried out over the boundary; what is new is that the transported sections stay sc⁺ on the faces, which the text takes for granted.

1. *Boundaries correspond.* A pull map F is a local sc-diffeomorphism on objects, so it preserves the degeneracy index (Proposition 2.4.4) Chapter14/BoundaryPull · SBLocalization.PullMap.degIndex_obj and $F^{-1}(\partial X') = \partial X$ Chapter14/BoundaryPull · SBLocalization.PullMap.mem_boundary_iff; F restricts to a map of boundary points Chapter14/BoundaryPull · SBLocalization.PullMap.bdryObj. The index is also invariant along morphisms Chapter14/BoundaryPush · SBLocalization.mem_boundary_src_iff_fm, so isomorphic points of W lie over ∂X together Chapter14/BoundaryPush · SBLocalization.mem_boundary_of_π_eq_fm, and Part XIII's transport charts $x \mapsto y$ of an equivalence preserve the boundary Chapter14/BoundaryPush · SBLocalization.transportChart_mem_boundary_iff_fm Chapter14/BoundaryPush · SBLocalization.Q_mem_boundary_iff_fm.
2. *The functors.* A functor over ∂X takes equal values on isomorphic points over ∂X Chapter14/

BoundaryPush · SBLocalization.IsBoundaryWFunctor.eq_of_π_eq_fm. Hence $\Phi^*\Lambda'$ is a functor over ∂X Chapter14/BoundaryPull · SBLocalization.PullMap.isBoundaryWFunctor_pull, and $\Phi_*\Lambda(w') = \Lambda(w)$ whenever w lies over ∂X and $\Phi(w) \cong w'$ Chapter14/BoundaryPush · SBLocalization.pushMult_eq_bdry_fm, so $\Phi_*\Lambda$ is a functor over $\partial X'$ Chapter14/BoundaryPush · SBLocalization.isBoundaryWFunctor_pushMult_fm: Lemma 11.5.2 over the boundary, where only values over ∂X enter.

3. *Pull-back.* For a good system $\mathcal{U}'$ for $\partial X'$, Part XIII's adapted neighbourhoods $N(x)$ (with $F(N(x)) \subset N'(F(x))$ and Φ a local strong bundle isomorphism over $N(x)$) map $U(x)$ into $U'(F(x))$ Chapter14/BoundaryPull · SBLocalization.PullMap.IsPullNbhd.mapsTo_bdry, and the component condition is carried back Chapter14/BoundaryPull · SBLocalization.PullMap.isDiffeoComponent_of_isPull_fm. Shrinking each $N(x)$ to a good neighbourhood with the same action by Proposition 14.1.1 gives a good system for ∂X adapted to $\mathcal{U}'$ Chapter14/BoundaryPull · SBLocalization.PullMap.BdryPullData Chapter14/BoundaryPull · SBLocalization.PullMap.exists_bdryPullData, for X tame. On it, the pulled-back structure has $I_x = I'_{F(x)}$, the action through $F : G_x \to G_{F(x)}$, and sections defined by $\Phi(s_i^x(z)) = s_i'^{F(x)}(F(z))$ Chapter14/BoundaryPull · SBLocalization.PullMap.bdryPullStructure; the decoration Chapter14/BoundaryPull · SBLocalization.PullMap.bdryPullDecoration and correspondences $\tau_{x,y}(\varphi) = \tau'_{F(x),F(y)}(F(\varphi))$ Chapter14/BoundaryPull · SBLocalization.PullMap.bdryPullTau form a structured tuple Chapter14/BoundaryPull · SBLocalization.PullMap.bdryPullTuple; a functor over ∂X with a decoration is an sc⁺-multisection over ∂X Chapter14/BoundaryPull · SBLocalization.BoundaryDecoration.isBoundaryScPlusMultisection_fm. Conditions (2)–(4) and (a)–(c) are Part XIII's arguments; condition (1) of Definition 14.1.3 for the pulled-back sections is the transport property `FacePull_hyp` Chapter14/BoundaryPull · SBLocalization.PullMap.FacePull_hyp: for open O, O' with $F(O) \subset O'$, a section face-wise sc⁺ on $O' \cap \partial X'$ pulls back to one face-wise sc⁺ on $O \cap \partial X$. The pull-back step is proved under this property Chapter14/BoundaryPull · SBLocalization.PullMap.isBoundaryStructurable_pull_hyp, for equivalences in particular Chapter14/BoundaryPull · SBLocalization.proposition_14_1_8_pull_hyp, and the property is a theorem (step 5).
4. *Push-forward.* For an equivalence $F : W \to W'$ with X' tame and a decoration of Λ, Part XIII's push neighbourhoods $N'(y)$ around a transport chart $x_y \mapsto y$ are shrunk to good neighbourhoods with the same action Chapter14/BoundaryPush · SBLocalization.exists_isPushNbhd_bdry_fm; the inverse of the chart maps $U'(y)$ into $U(x_y)$ Chapter14/BoundaryPush · Qinv_mem_bdry_fm, and the component condition is carried forward Chapter14/BoundaryPush · isDiffeoComponent_push_fm. With the chosen point x_y Chapter14/BoundaryPush · SBLocalization.bdryX and neighbourhood Chapter14/BoundaryPush · SBLocalization.bdryPushNbhd this is a good system for $\partial X'$ Chapter14/BoundaryPush · SBLocalization.bdryPushGood. The pushed structure has $I'_y = I_{x_y}$ and sections $s'_i(z) = \mu(\psi_u, \Phi(s_i(u)))$, $u = Q^{-1}(z)$ Chapter14/BoundaryPush · SBLocalization.bdryPushStructure, by Part XIII's pointwise formulas with the section property needed only at the points used Chapter14/BoundaryPush · P_obj_fm Chapter14/BoundaryPush · pushSec_eq_fm Chapter14/BoundaryPush · pushSec_compat_fm; the correspondences are $\tau'_{y,y'}(\varphi') = \tau_{x_y,x_{y'}}(\tilde{\varphi})$ for the lift $\tilde{\varphi}$ of φ', which maps boundary morphisms to boundary morphisms Chapter14/BoundaryPush · liftMor_mem_bdry_fm Chapter14/BoundaryPush · SBLocalization.bdryPushDecoration Chapter14/BoundaryPush · SBLocalization.bdryPushTau. Condition (1) for the pushed sections is the property `FacePush_hyp` Chapter14/BoundaryPush · SBLocalization.FacePush_hyp; the push step is proved under it Chapter14/BoundaryPush · SBLocalization.isBoundaryStructurable_pushMult_hyp, and so is the proposition for a representative Chapter14/BoundaryPush · SBLocalization.proposition_14_1_8_hyp Chapter14/BoundaryPush · SBLocalization.proposition_14_1_8_pullMult_hyp.
5. *The transport of face-wise sc⁺ sections* Chapter14/FaceTransport · SBLocalization.isFaceScPlusSectionOn_of_

localIso, the key lemma. Let $L : W|U \to W'|U'$ be a local strong bundle isomorphism over $\phi : U \to U'$ with fibre map Ψ, $G \subset U$ open around $x \in \partial X$, and $s = \Psi \circ s' \circ \phi$ on $G \cap \partial X$. If s' is face-wise sc$^+$ at $\phi(x)$, then s is at x. *Proof*: given a tame strong bundle chart c of W at x, restrict it to $\operatorname{dom}\varphi \cap G$; the restriction is again a tame chart, and its retract is $O \cap N$ for an open N. Transport the restricted chart along L^{-1}: the result is a tame strong bundle chart of W' with the same model, by Part I's lemma on charts transported along local strong bundle isomorphisms Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.isChart_transportChart Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.isTameChart_transportChart, proved for Theorem 3.6.8. In these two charts the local representatives of s and of s' agree at every point of a local face, since such points are mapped into ∂X, where $s = \Psi \circ s' \circ \phi$. The face condition for s' at $\phi(x)$ in the transported chart is therefore the face condition for s at x in the restricted chart, and a face germ on the smaller retract $O \cap N$ is one on O Chapter14/FaceTransport · IsScPlusOnFaceNear.of_restrict_p6. Since the condition is pointwise Chapter14/FaceTransport · SBLocalization.isFaceScPlusSectionOn_iff_singleton, this proves it on sets.

6. *Both transport properties hold.* `FacePull_hyp` holds for every pull map Chapter14/FaceTransport · SBLocalization.PullMap.facePull_hyp: near every x the pulled-back section is $\Psi \circ s' \circ F$ for the local strong bundle isomorphism of Φ at x, and step 5 applies. `FacePush_hyp` holds for every strong bundle equivalence Chapter14/FacePush · SBLocalization.facePush_hyp_of_equivalence: near $z_0 = Q(u_0)$ the pushed section $z \mapsto \mu(\psi_u, \Phi(s(u)))$ is $\Psi_2 \circ s_1 \circ \phi_2$, where $s_1 = \Phi \circ s \circ F^{-1}$ is s transported along the local strong bundle isomorphism of Φ at u_0, and (ϕ_2, Ψ_2) is the action of the morphisms ψ_u near $\sigma = \psi_{u_0}$, itself a local strong bundle isomorphism with an explicit formula: on an open $\mathcal{T} \ni \sigma$ of morphisms, $\phi_2(t\psi) = s\psi$ and $\Psi_2 = \mu(\psi, \cdot)$ Chapter14/FacePush · SBLocalization.exists_localIso_act_p6 (Part X's local isomorphism of a morphism, with its formula exposed). Step 5 applies twice. Two direct checks of the pull property, superseded by this general one, remain in the formalization: it holds for the identity of every strong bundle Chapter14/BoundaryPull · SBLocalization.PullMap.facePull_hyp_id, and vacuously when $\partial X = \emptyset$ Chapter14/BoundaryPull · SBLocalization.PullMap.facePull_hyp_of_boundary_eq_empty; the push property holds vacuously when $\partial X' = \emptyset$ Chapter14/BoundaryPush · SBLocalization.facePush_hyp_of_boundary_eq_empty.
7. *Assembly.* Pull-backs along every pull map Chapter14/FacePush · SBLocalization.PullMap.isBoundaryStructurable_pull and push-forwards along every equivalence Chapter14/FacePush · SBLocalization.isBoundaryStructurable_pushMult preserve structurability over the boundary; composing them along $W \xleftarrow{\Phi} V \xrightarrow{\Psi} W'$ gives $\bar{f}_* = \Psi_*\Phi^*$ and $\bar{f}^* = \Phi_*\Psi^*$. The middle V is tame when X is Chapter14/FacePush · SBLocalization.SDiagram.isTame_B_p6, because Φ covers an equivalence of ep-groupoids and an ep-groupoid equivalent to a tame one is tame (Part X).

XIV.7 Theorem 14.1.9 and Remark 14.1.10 (2017: Theorem 14.2, Remark 14.3): the ∂-extension theorem

The boundary domain support Chapter14/BoundaryStructured · SBLocalization.boundaryDomSupp: Definition 13.2.14 read over ∂X, the closure of the set of $x \in \partial X$ over which Λ charges a non-zero point, $\Lambda(w) > 0$ for some $w \neq 0_x$. It is saturated Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.isSaturated_boundaryDomSupp, and empty for the zero multisection Chapter14/BoundaryStructured · SBLocalization.boundaryDomSupp_zero.

Theorem 14.1.9 (narrower) Chapter14/BoundaryExtensionTheorem · SBLocalization.theorem_14_1_9_charged_a6 Chap-

ter14/BoundaryExtensionTheorem · SBLocalization.theorem_14_1_9_charged_statement, p. 646. *Statement as formalized*: let $(P : W \to X, \mu)$ be a strong bundle over a tame ep-groupoid X with $|X|$ paracompact, whose object M-polyfold admits sc-smooth partitions of unity, and such that every object lies in a tame strong bundle chart. Let $\Lambda : W|\partial X \to \mathbb{Q}^+$ be structurable over ∂X and charge every boundary fibre, N an auxiliary norm (Definition 8.3.5), $\tilde{U}$ an open saturated set containing the boundary domain support of Λ, and $f : X \to [0, \infty)$ a continuous functor with $\mathrm{cl}\{f \neq 0\} \subset \tilde{U}$ and $N(\Lambda)(x) < f(x)$ on the boundary domain support. Then there is a structurable sc$^+$-multisection functor $\Lambda' : W \to \mathbb{Q}^+$ (Definition 13.3.6) with (1) $N(\Lambda')(x) \leq f(x)$ for all x, (2) dom-supp$(\Lambda') \subset \tilde{U}$, (3) $\Lambda' = \Lambda$ over ∂X. *Narrower*, by the two readings named in the decisions: the tame strong bundle charts at every object, the reading of "a strong bundle over a tame ep-groupoid" used for Proposition 14.1.4 (§XIV.3); and charging, the reading of the factor $|I_x|^{-1}$ of Definition 14.1.3 (3). Both are gap G.186 (§XIV.29). Two readings of the hypotheses on f: "supported in $\tilde{U}$" is read as $\mathrm{cl}\{f \neq 0\} \subset \tilde{U}$; and the text writes $N(\Lambda)(x) < f(x)$ "for $x \in \partial X$" in the theorem, while what is needed and assumed is the inequality on the boundary domain support (off it, $N(\Lambda) = 0$ and $f \geq 0$ would allow $f = 0$). The proof, the whole of §§14.2–14.5, is §XIV.25.

Charging is necessary Chapter14/EmptyIndex · SBLocalization.chargesBoundary_of_structurable_eq_a6: if a structurable Λ' on W agrees with Λ over ∂X, then Λ charges every boundary fibre. *Proof*: a structurable Λ' has local section structures at every point (Definition 13.2.1), whose local weights sum to 1 on every fibre; so over each x some w has $\Lambda'(w) > 0$, and over ∂X this is $\Lambda(w)$. Hence the exact form Chapter14/BoundaryExtensionTheorem · SBLocalization.theorem_14_1_9_iff_a6: under the other hypotheses, a structurable Λ over ∂X has an extension as in the theorem if and only if it charges every boundary fibre. For a structured tuple, charging is the same as non-empty index sets at all boundary points Chapter14/EmptyIndex · SBLocalization.BoundaryStructuredTuple.nonempty_I_a6 (§XIV.2).

Remark: the uncharged Lean statement. The Lean statement of Theorem 14.1.9 without the charging condition and without the tame strong bundle charts Chapter14/BoundaryStructured · SBLocalization.theorem_14_1_9_statement, whose hypotheses are jointly satisfiable ($\Lambda = 0$, $\tilde{U} = \emptyset$, $f = 0$) Chapter14/BoundaryStructured · SBLocalization.theorem_14_1_9_statement_hyp_witness, does not hold when $\partial X \neq \emptyset$ Chapter14/EmptyIndex · SBLocalization.not_theorem_14_1_9_statement_a6, for X tame with $|X|$ paracompact and sc-smooth partitions of unity. Because the formal Definition 14.1.3 admits $I_x = \emptyset$, the function $\Lambda \equiv 0$ is structurable over ∂X in the formal sense Chapter14/EmptyIndex · SBLocalization.isBoundaryStructurable_zeroFun_a6: on a good system from Theorem 14.1.6, with the empty index set at every boundary point Chapter14/EmptyIndex · SBLocalization.emptyBoundaryStructure_a6 under the trivial action on the empty type Chapter14/EmptyIndex · SBLocalization.pemptyAction_a6, and the unique correspondence between empty index sets Chapter14/EmptyIndex · SBLocalization.IndexCorr.eq_of_isEmpty_a6 Chapter14/EmptyIndex · SBLocalization.emptyTau_a6, which satisfies (a)–(c) and (4) vacuously. Its boundary domain support is empty Chapter14/EmptyIndex · SBLocalization.boundaryDomSupp_zeroFun_a6, so $\tilde{U} = \emptyset$ and $f = 0$ satisfy the hypotheses, with N from Theorem 12.2.2; an extension would be structurable and 0 over a boundary point, which necessity excludes. The statement holds when $\partial X = \emptyset$ Chapter14/BoundaryExtensionTheorem · SBLocalization.theorem_14_1_9_statement_of_boundary_eq_empty_a6. This is a property of that Lean statement and of the formal Definition 14.1.3, not of the text, whose $|I_x|^{-1}$ presumes $I_x \neq \emptyset$; the statement with charging above is the one proved. The forms derived from that statement are recorded where they occur (§§XIV.25, XIV.28).

Remark 14.1.10 (2017: Remark 14.3), p. 646, says that the proof needs preparation and is lengthy; nothing to formalize.

XIV.8 Lemmas 14.2.1, 14.2.2 and Proposition 14.2.3 (2017: the text before Lemma 14.1, Lemma 14.1, Proposition 14.2): a good structured representative

§14.2 fixes a representative $(\Lambda, \mathcal{U}^0, S^0, \tau^0)$ of a structured sc⁺-multisection over ∂X; the text names its properties (P1) (the good system), (P2) (the structures) and (P3) (the correspondences), which are the fields of a structured tuple (§XIV.5). It then shrinks the good system so that the level sets of the correspondences acquire disjoint closures, property (P4), which §14.3 needs to separate them by open sets.

Lemma 14.2.1 (same) `Chapter14/GoodStructured · EpGroupoid.lemma_14_2_1`, p. 648. *Statement*: let X be tame, $x \in \partial X$ and $V^0(x)$ a good neighbourhood, $U^0(x) = V^0(x) \cap \partial X$. There is a good neighbourhood $V(x) \le V^0(x)$ (same natural action, restricted) with $\mathrm{cl}_X V(x) \subset V^0(x)$, $\mathrm{cl}_X U(x) \subset U^0(x)$, and $\mathrm{cl}_{|X|} |U(x)| \subset |U^0(x)|$ (14.15). The text states (14.15) with the closure in $|\partial X|$; since $|\partial X| = \pi(\partial X)$ is closed in $|X|$ (§XIV.11), the closure in $|X|$ is the same. The faces, the connectedness and the invariance of $U(x)$ are those of a good neighbourhood (Definition 14.1.2).

Proof. 1. Choose a closed neighbourhood C of x with $C \subset V^0(x)$ (the object space is regular). 2. Apply Proposition 14.1.1 in its relative form inside the interior of C (§XIV.1): this gives a good $V(x) \le V^0(x)$ with $V(x) \subset C$, so $\mathrm{cl}\, V(x) \subset C \subset V^0(x)$, and $\mathrm{cl}\, U(x) \subset \mathrm{cl}\, V(x) \cap \partial X \subset U^0(x)$ because ∂X is closed. 3. For the orbit closure, Proposition 7.1.10 for the proper set $\mathrm{cl}\, V^0(x)$ `Chapter7/Basic · proposition_7_1_10_of_isProperMap` gives $\mathrm{cl}_{|X|}\, \pi(A) = \pi(\mathrm{cl}_X A)$ for $A \subset V^0(x)$ with $\mathrm{cl}\, A$ inside; applied to $A = U(x)$ it gives (14.15). For all boundary points at once `Chapter14/GoodStructured · EpGroupoid.lemma_14_2_1_system`: a good system $\mathcal{U} \le \mathcal{U}^0$ for ∂X with these properties; the component condition passes to the smaller sets (§XIV.4).

Lemma 14.2.2 (same) `Chapter14/GoodStructured · SBLocalization.lemma_14_2_2`, p. 648. *Statement*: let $\mathcal{U}$, $\mathcal{U}^0$ be good systems for ∂X with $\mathrm{cl}_X U(x) \subset U^0(x)$ for all x, and let τ be locally constant on $\mathcal{U}^0(x, x')$. For $[d] \neq [d']$ the closures in the morphism space of $\tau^{-1}([d]) \cap \mathcal{U}(x, x')$ and $\tau^{-1}([d']) \cap \mathcal{U}(x, x')$ are disjoint. *Proof*: $\mathrm{cl}\, \mathcal{U}(x, x') \subset \mathcal{U}^0(x, x')$ `Chapter14/GoodStructured · SBLocalization.closure_mor_subset`, since s and t are continuous and $\mathrm{cl}\, U(x) \subset U^0(x)$. A common limit point φ of the two sets therefore lies in $\mathcal{U}^0(x, x')$, where τ is locally constant; near φ in $\mathcal{U}^0(x, x')$, τ is constant, and the two sets both meet that neighbourhood, so $[d] = \tau(\varphi) = [d']$. *A reading*: the text defines $\mathcal{U}(x, x')$ with $t(\varphi) \in U(x)$; presumably $U(x')$ is meant, as in (P3), and that is what is formalized.

Restriction. A decoration restricts to a smaller good system with the same actions `Chapter14/GoodStructured · SBLocalization.BoundaryDecoration.restrictTo`, the boundary morphism sets shrink `Chapter14/GoodStructured · SBLocalization.BoundaryDecoration.mor_mono`, and the correspondences restrict `Chapter14/GoodStructured · SBLocalization.BoundaryDecoration.Tau.restrict`: local constancy, finiteness of the values, (a)–(c) and (4) are inherited by subsets, and the bi-action is the same because the actions are. The restricted tuple `Chapter14/GoodStructured · SBLocalization.BoundaryStructuredTuple.restrict` is equivalent to the original `Chapter14/GoodStructured · SBLocalization.BoundaryStructuredTuple.equiv_restrict`, with $O(x)$ the smaller neighbourhoods.

Proposition 14.2.3 (same) `Chapter14/GoodStructured · SBLocalization.BoundaryStructuredTuple.proposition_14_2_3`, p. 649. *Statement*: for X tame, every structured tuple $(\Lambda, \mathcal{U}, S, \tau)$ is equivalent to one with the same Λ, the same index sets and the same correspondences on a smaller good system with restricted actions, which has (P4) `Chapter14/GoodStructured · SBLocalization.BoundaryStructuredTuple.HasP4`: for $[d] \neq [d']$ the closures of $\tau^{-1}_{x,x'}([d])$ and $\tau^{-1}_{x,x'}([d'])$ (preimages in $\mathcal{U}(x, x')$) are disjoint. *Proof*: Lemma 14.2.1

for the good system of the tuple, restriction to the smaller system, and Lemma 14.2.2 with $\mathcal{U}^0$ the original system, on which τ is locally constant by (2) of Definition 14.1.7. (P4) persists under further shrinking `Chapter14/GoodStructured · SBLocalization.BoundaryStructuredTuple.HasP4.restrict` (the text: "it will persist when we replace these sets by even smaller ones"), since the preimages only shrink. On classes `Chapter14/GoodStructured · SBLocalization.BoundaryStructuredTuple.proposition_14_2_3_class`: every structured sc⁺-multisection over ∂X of a tame X has a representative with (P4). The hypotheses of Theorem 14.1.9 enter only as tameness.

XIV.9 (P5) and Lemma 14.3.1 (2017: Lemma 14.2): open neighbourhoods of the level sets

(P5) `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds`, p. 649. Fix a tuple with (P1)–(P4). Neighbourhoods $\tilde{U}'(x)$, $x \in \partial X$, with (P5) are open sets of objects with (i) the natural G_x-action and t proper on $s^{-1}(\mathrm{cl}_X \tilde{U}'(x))$, and (ii) $\tilde{U}'(x) \cap \partial X = U^0(x)$. *Reading*: "admits the natural G_x-action" is read as carrying the natural representation of the good system $\mathcal{U}^0$ itself, extended to $\tilde{U}'(x)$; this makes the bi-action on $\tilde{U}'(x,x') = \mathcal{U}(\tilde{U}'(x), \tilde{U}'(x'))$ (14.19) `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds.mor` `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds.isOpen_mor`, `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds.biact`, agree with the bi-action of Definition 14.1.7 (c) on the boundary morphisms `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds.biact_eq`, the natural bi-action depending only on the representations `Chapter14/ExtendCorrespondences · EpGroupoid.biact_congr_rep_p3w4`. The bi-action preserves $\tilde{U}'(x,x')$ `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds.biact_mem`, and that of $\mathcal{U}^0$ preserves $U^0(x,x')$ `Chapter14/ExtendCorrespondences · EpGroupoid.BoundaryGoodSystem.biact_mem_mor_p3w4`. *Boundary morphisms* are those with source and target in ∂X `Chapter14/ExtendCorrespondences · SBLocalization.boundaryMor`; by (ii), the boundary morphisms of $\tilde{U}'(x,x')$ are exactly $U^0(x,x')$ `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds.mor_inter_boundaryMor` `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds.mor_subset`, and inversion maps $U^0(x,x')$ to $U^0(x',x)$ `Chapter14/ExtendCorrespondences · EpGroupoid.BoundaryGoodSystem.inv_mem_mor_p3w4`. *Witness*: the open sets $V(x)$ of the good system itself `Chapter14/ExtendCorrespondences · SBLocalization.P5Nbhds.ofGood`, which have $V(x) \cap \partial X = U^0(x)$ by definition.

Two identities of the bi-actions are used: $(g' * [d] * g)^\sharp = g^{-1} * [d]^\sharp * g'^{-1}$ `Chapter14/ExtendCorrespondences · SBLocalization.IndexCorr.swap_biact_p3w4` for correspondences, and $(g' * \varphi * g)^{-1} = g^{-1} * \varphi^{-1} * g'^{-1}$ `Chapter14/ExtendCorrespondences · EpGroupoid.inv_biact_p3w4` for morphisms (Definition 13.1.5).

Lemma 14.3.1 (stronger) `Chapter14/ExtendCorrespondences · SBLocalization.lemma_14_3_1` `Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension`, p. 650. *Statement*: under (P1)–(P5), for all $x, x' \in \partial X$ and every $[d] \in J(x,x')$ there is an open $\tilde{O}_{x,x',[d]} \supset \tau^{0-1}([d])$ with (1) $\tilde{O}_{x,x',[d]} \subset \tilde{U}'(x,x')$; (2) pairwise disjoint closures for $[d] \neq [d']$ `Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.disjoint`, so that a morphism lies in at most one of them `Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.eq_of_mem`; (3) $h * \tilde{O}_{[d]} * g = \tilde{O}_{h*[d]*g}$ `Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.image_biact`; (4) the union $\tilde{O}_{x,x'}$ `Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.Ot` meets ∂X exactly in $U^0(x,x')$ `Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.iUnion_inter_boundaryMor`; and (14.20), $\tilde{O}_{[d]} \cap \partial X = \tau^{0-1}([d])$ `Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.inter_boundaryMor`. *Stronger*: in addition, $\varphi \in \tilde{O}_{x,x',[d]}$ iff $\varphi^{-1} \in \tilde{O}_{x',x,[d]^\sharp}$. The printed form, (1)–(4) alone, is `Chapter14/ExtendCorrespondences · SBLocalization.lemma_14_3_1_book`.

Proof, the text's with one step added.

1. *Separation* Chapter14/ExtendCorrespondences · SBLocalization.CorrPre.exists_of_hasP4 Chapter14/ExtendCorrespondences · SBLocalization.CorrPre. Fix x, x'. The correspondence $\tau^0_{x,x'}$ has finitely many values (Definition 14.1.7 (2)), and by (P4) the closures of its level sets are pairwise disjoint closed sets. The morphism space is metrizable, hence normal, and in a normal space finitely many pairwise disjoint closed sets have open neighbourhoods with pairwise disjoint closures Chapter14/ExtendCorrespondences · P3W4S3.exists_open_disjoint_closure (an induction on the finite set, the empty level sets getting the empty set). Intersecting with the open set $\tilde{U}'(x,x')$ gives open $\tilde{O}'_{x,x',[d]} \supset \tau^{0-1}([d])$ inside $\tilde{U}'(x,x')$ with disjoint closures. The level set lies in $\tilde{U}'(x,x')$ because $U^0(x,x') \subset \tilde{U}'(x,x')$.
2. *Invariance under the bi-action* Chapter14/ExtendCorrespondences · SBLocalization.CorrPre.biInv. Replace $\tilde{O}'_{[d]}$ by $\bigcap_{h,g}(h * \cdot * g)^{-1}(\tilde{O}'_{h*[d]*g})$, the intersection over the finite group $G_{x'} \times G_x$ of the preimages under the bi-action, which is a homeomorphism of $\tilde{U}'(x,x')$. This is the text's $\bigcap h * \tilde{O}'_{[d']} * g$ over the $(h,[d'],g)$ with $h * [d'] * g = [d]$, written through preimages. The result is open, still contains $\tau^{0-1}([d])$ by the equivariance (c) of τ^0 and the invariance of $U^0(x,x')$, keeps (1) and (2) (it is smaller), and satisfies $h * \varphi * g \in \tilde{O}_{h*[d]*g} \iff \varphi \in \tilde{O}_{[d]}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrPre.biInv_biact_mem_iff, which is (3).
3. *Symmetry under inversion* (added) Chapter14/ExtendCorrespondences · SBLocalization.CorrPre.symm. Replace $\tilde{O}_{x,x',[d]}$ by $\tilde{O}_{x,x',[d]} \cap \iota^{-1}(\tilde{O}_{x',x,[d]^\sharp})$, ι the inversion. This is open, contains $\tau^{0-1}([d])$ by (a) of Definition 14.1.7, keeps (1) and (2), and has the symmetry Chapter14/ExtendCorrespondences · SBLocalization.CorrPre.symm_inv_mem_iff, since $[d]^{\sharp\sharp} = [d]$; it keeps (3) Chapter14/ExtendCorrespondences · SBLocalization.CorrPre.symm_biact_mem_iff by the two identities of the bi-actions above.
4. *(4) and (14.20).* A boundary morphism in $\tilde{O}_{[d]}$ lies in $U^0(x,x')$ by (1) and (P5) (ii), so it has some value $\tau^0(\varphi) = [d']$ and lies in $\tilde{O}_{[d']}$; by (2), $[d'] = [d]$. Conversely the level sets are inside.

The inversion symmetry is what makes the extended correspondences satisfy Definition 14.1.7 (a) (§XIV.10), which the z-loops of §14.4 need to be reversible. Only finitely many $[d]$ have non-empty sets.

XIV.10 Definition 14.3.2, Lemma 14.3.3, Proposition 14.3.4 and Remark 14.3.5 (2017: the text before Lemma 14.3, Lemma 14.3, Proposition 14.3, Remark 14.4): the extended correspondences

Definition 14.3.2 (restated) Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.Ohat Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat, p. 651. $\hat{O}_{x,x'}$ is the union of the connected components of $\tilde{O}_{x,x'}$ that meet ∂X. For $\varphi \in \tilde{O}_{x,x'}$, $\hat{\tau}_{x,x'}(\varphi)$ is the unique $[d]$ with $\varphi \in \tilde{O}_{x,x',[d]}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_eq Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.mem_O_tauHat; off $\tilde{O}_{x,x'}$ the value is irrelevant. $\hat{O}_{x,x',[d]} = \{\varphi \in \hat{O}_{x,x'} : \hat{\tau}(\varphi) = [d]\}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.OhatD. *Restated*: the text defines $[d]_\varphi = \tau^0(\psi)$ for a boundary morphism ψ in the component of φ; the formal definition reads the value off the decomposition of Lemma 14.3.1, and the two agree:

1. $\tilde{O}_{x,x'}$ is open Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.isOpen_Ot, lies in $\tilde{U}'(x,x')$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.Ot_subset_mor and contains $U^0(x,x')$

Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.mor_subset_Ot.

2. The $\tilde{O}_{[d]}$ are pairwise disjoint open sets with union $\tilde{O}_{x,x'}$, so the component in the union of a point of one of them lies in it Chapter14/ExtendCorrespondences · P3W4S3.connectedComponentIn_iUnion_subset Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.component_subset: the component is connected and the other sets separate it.
3. Hence $\hat{\tau}$ is constant on the components of $\tilde{O}_{x,x'}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_eq_of_mem_component, and, by (14.20), $\hat{\tau}(\varphi) = \tau^0(\psi)$ for every boundary morphism ψ in the component of φ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_eq_tau_of_mem_component_Ot, which is the text's definition.

(14.23): $\hat{O}_{x,x'} \subset \tilde{O}_{x,x'}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.Ohat_subset_Ot, so $\hat{O}_{x,x'} \subset \tilde{U}'(x,x')$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.Ohat_subset_mor; $U^0(x,x') \subset \hat{O}_{x,x'}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.mor_subset_Ohat, each of its points lying in a component that meets ∂X at that point; and $\hat{O}_{x,x'} \cap \partial X = U^0(x,x')$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.Ohat_inter_boundaryMor by (4). The component in $\tilde{O}_{x,x'}$ of a point of $\hat{O}_{x,x'}$ lies in $\hat{O}_{x,x'}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.component_subset_Ohat, so $\hat{O}_{x,x'}$, a union of components of an open set in a locally connected space, is open Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.isOpen_Ohat. And $\hat{O}_{[d]} = \hat{O} \cap \tilde{O}_{[d]}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.OhatD_eq, with $\hat{O}_{[d]} \cap \partial X = \tau^{0-1}([d])$ (14.24) Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.OhatD_inter_boundaryMor.

Lemma 14.3.3 (same) Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.lemma_14_3_3, p. 651: two boundary morphisms in one component of $\hat{O}_{x,x'}$ have the same τ^0. *Proof*: a component of $\hat{O}_{x,x'}$ lies in a component of $\tilde{O}_{x,x'}$, which lies in one $\tilde{O}_{[d]}$ (item 2 above); by (14.20) both morphisms have $\tau^0 = [d]$. So $\varphi \mapsto [d]_\varphi$ is well defined, and $\hat{\tau}(\varphi) = \tau^0(\psi)$ for every boundary ψ in the component of φ in $\hat{O}_{x,x'}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_eq_tau_of_mem_component. Every component of $\hat{O}_{x,x'}$ contains a boundary morphism, indeed one in the component in $\tilde{O}_{x,x'}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.exists_boundary_mem_component, by definition of $\hat{O}$.

Proposition 14.3.4 (same) Chapter14/ExtendCorrespondences · SBLocalization.proposition_14_3_4, p. 652. *Statement*: under (P1)–(P5) there are open sets $\hat{O}_{x,x'}$ of morphisms with (1) $U^0(x,x') \subset \hat{O}_{x,x'} \subset \tilde{U}'(x,x')$ and $\hat{O}_{x,x'} \cap \partial X = U^0(x,x')$; (2) $\hat{O}_{x,x'}$ invariant under the bi-action of $G_{x'} \times G_x$; (3) a locally constant extension $\hat{\tau}_{x,x'}$ of $\tau^0_{x,x'}$ to $\hat{O}_{x,x'}$, unique among locally constant extensions, with $\hat{\tau}(g' * \varphi * g) = g' * \hat{\tau}(\varphi) * g$ and $\hat{\tau}_{x,x}(\Gamma(g,y)) = \tau^0_{x,x}(\Gamma(g,z))$ for $y \in \tilde{U}'(x)$ with $\Gamma(g,y) \in \hat{O}_{x,x}$ and $z \in U^0(x)$; and the decomposition (14.24). *Proof.*

1. *(1)* is (14.23) above.
2. *Local constancy* Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_locallyConstant: near $\varphi \in \tilde{O}_{[d]}$ the open set $\tilde{O}_{[d]}$ is a neighbourhood on which $\hat{\tau} = [d]$.
3. *Extension* Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_eq_tau: on $U^0(x,x')$, $\varphi \in \tilde{O}_{\tau^0(\varphi)}$ because $\tilde{O}_{[d]} \supset \tau^{0-1}([d])$.
4. *Uniqueness* Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_unique: a function locally constant on $\hat{O}_{x,x'}$ is constant on its components Chapter14/ExtendCorrespondences · P3W4S3.eq_of_mem_connectedComponentIn; every component contains a boundary morphism ψ, where the function equals $\tau^0(\psi) = \hat{\tau}(\psi)$, and $\hat{\tau}$ is constant on the component too. The values of $\hat{\tau}$ on $\hat{O}$ are values of τ^0 Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_image_subset, hence finitely many

Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.finite_tauHat (the text's "taking the same values as τ^0").

5. *Equivariance and (2)* Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_biact: by (3) of Lemma 14.3.1, $\varphi \in \tilde{O}_{[d]}$ iff $h * \varphi * g \in \tilde{O}_{h*[d]*g}$. The bi-action maps $\tilde{O}_{x,x'}$ to itself Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.biact_mem_Ot, and, being a homeomorphism preserving ∂X, maps components meeting ∂X to components meeting ∂X Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.biact_mem_Ohat.
6. *The diagonal property* Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_Γ: $\hat{\tau}_{x,x}(\Gamma(g,y)) = [I_x \xleftarrow{\mathrm{id}} I_x \xrightarrow{g} I_x]$ for $y \in \tilde{U}'(x)$ with $\Gamma(g,y) \in \hat{O}_{x,x}$. The text states it without proof. *Sheets.* For a natural representation on R, every morphism φ between points of R is $\Gamma(g, s(\varphi))$ for exactly one g, its *sheet label* Chapter14/ExtendCorrespondences · EpGroupoid.sheetLabel_p3w4 Chapter14/ExtendCorrespondences · EpGroupoid.Γ_sheetLabel_p3w4 Chapter14/ExtendCorrespondences · EpGroupoid.sheetLabel_eq_p3w4. The sheet $\{\Gamma(g,z)\}$ is open: if $\varphi_0 = \Gamma(g, s(\varphi_0))$, then $\varphi = \Gamma(g, s(\varphi))$ for all φ near φ_0 Chapter14/ExtendCorrespondences · EpGroupoid.eventually_Γ_src_eq_p3w4, because s is a local homeomorphism and $z \mapsto \Gamma(g,z)$ is a continuous local inverse of s through φ_0. So the sheet label is locally constant on the morphisms between points of R Chapter14/ExtendCorrespondences · EpGroupoid.sheetLabel_locallyConstant_p3w4. *The argument.* The component of $\Gamma(g,y)$ in $\tilde{O}_{x,x}$ meets ∂X (as $\Gamma(g,y) \in \hat{O}$), say in ψ; the sheet label is constant on that component, so $\psi = \Gamma(g, s(\psi))$ with $s(\psi) \in U^0(x)$; then $\hat{\tau}(\Gamma(g,y)) = \hat{\tau}(\psi) = \tau^0(\psi) = [\mathrm{id}, g]$ by (b) of Definition 14.1.7.
7. *Inversion* (added): $\varphi \in \hat{O}_{x,x'}$ iff $\varphi^{-1} \in \hat{O}_{x',x}$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.inv_mem_Ohat, inversion being a homeomorphism that preserves ∂X and, by the symmetry of Lemma 14.3.1, maps $\tilde{O}_{x,x'}$ onto $\tilde{O}_{x',x}$; and $\hat{\tau}_{x',x}(\varphi^{-1}) = \hat{\tau}_{x,x'}(\varphi)^\sharp$ Chapter14/ExtendCorrespondences · SBLocalization.CorrExtension.tauHat_inv, which is Definition 14.1.7 (a) for $\hat{\tau}$.

Remark 14.3.5 (2017: Remark 14.4), p. 653, observes that $\hat{O}_{x,x'}$ need not be of the form $\tilde{U}(x,x')$ for neighbourhoods of objects; it states nothing to prove, and nothing formal depends on that form.

XIV.11 Lemmas 14.3.6 and 14.3.7 (2017: Lemmas 14.4, 14.5): a locally finite cover of the boundary

The rest of §14.3 chooses the index set A of the construction and the sets $\hat{V}_a$ on which the local extensions of §14.4 live. Both lemmas are formalized in an abstract setting that the objects of Chapter 14 instantiate (§XIV.12): the boundary is any set B of objects (with $|B|$ closed for Lemma 14.3.6, saturated for Lemma 14.3.7), the neighbourhoods are natural neighbourhoods in the sense of §13.1 (Part XIII), and the sets $\hat{O}_{a,b}$ are any open sets of morphisms that contain the boundary morphisms $U'(a,b)$. The proofs use only these properties; isolating them makes visible which facts of §§14.1–14.3 the lemmas need.

Closed saturated sets. Four small facts about saturated sets are used throughout §§14.3–14.5. A saturated set contains the target of every morphism with source in it Chapter14/BoundaryCover · EpGroupoid.IsSaturated.tgt_mem (that is the definition, read through the inverse morphism); the complement of a saturated set is saturated Chapter14/BoundaryCover · EpGroupoid.isSaturated_compl (a morphism from the complement into the set would, inverted, leave the set); every preimage $\pi^{-1}(S)$ of a set of orbits is saturated Chapter14/BoundaryCover · EpGroupoid.isSaturated_preimage_π; and the saturation of an open set is open Chapter14/BoundaryCover · EpGroupoid.isOpen_saturation, which is Proposition 7.1.8 (Part VII). From these: *a closed saturated set B has closed image $|B|$ in $|X|$* Chapter14/BoundaryCover · EpGroupoid.isClosed_π_image_of_isSaturated.

Proof: the complement of $|B|$ is $\pi(X \setminus B)$, because B is saturated (an orbit meets B iff it lies in B); $X \setminus B$ is open and saturated, so $\pi(X \setminus B)$ is open by Proposition 7.1.8. For $B = \partial X$: the degeneracy index is invariant under morphisms, so ∂X is saturated `Chapter14/CoverVHat · EpGroupoid.isSaturated_boundary_p3w4`, and it is closed; hence $|\partial X|$ is closed in $|X|$, which the text uses tacitly when it treats $|\partial X|$ as a closed subset.

Lemma 14.3.6 (explicit) `Chapter14/BoundaryCover · EpGroupoid.lemma_14_3_6`, p. 654. *Statement*: let $|X|$ be paracompact, B a set of objects with $|B|$ closed in $|X|$, and $\tilde{U}(x)$, for every object x, an open neighbourhood with the natural G_x-action. There are $A \subset B$ and nonempty open sets $\tilde{V}_a \subset \tilde{U}(a)$, $a \in A$, with (1) $|\tilde{V}_a|$ open and $\mathrm{cl}_{|X|}\,|\tilde{V}_a| \subset |\tilde{U}(a)|$, (2) $\tilde{V}_a$ invariant under the natural G_a-action, (3) $(|\tilde{V}_a|)_{a\in A}$ locally finite in $|X|$ and covering $|B|$. *Explicit*: the closedness of $|\partial X|$, which the text's proof uses, is a hypothesis here, discharged for $B = \partial X$ as above.

Proof, the text's.

1. *The cover.* The sets $|\tilde{U}(x)|$, $x \in B$, and $|X| \setminus |B|$ form an open cover of $|X|$, indexed by B plus one extra index: $|\tilde{U}(x)| = \pi(\tilde{U}(x))$ is open by Proposition 7.1.8 (Part VII), and $|X| \setminus |B|$ is open because $|B|$ is closed. It covers: an orbit meeting B at x lies in $|\tilde{U}(x)|$.
2. *Refinement.* $|X|$ is paracompact, so the cover has a locally finite open refinement (v_i) indexed by the same set, each v_i inside the i-th member (precise refinement). A paracompact Hausdorff space is normal, and a point-finite open cover of a normal space can be shrunk to an open cover (w_i) with $\mathrm{cl}\, w_i \subset v_i$; (w_i) is again locally finite.
3. *The sets $\tilde{V}_a$.* For $a \in B$ put $\tilde{V}_a = \tilde{U}(a) \cap \pi^{-1}(w_a)$, and let A be the set of $a \in B$ with $\tilde{V}_a \neq \emptyset$. Then $\tilde{V}_a$ is open, contained in $\tilde{U}(a)$, and invariant under the natural G_a-action, because $\tilde{U}(a)$ is and $\pi^{-1}(w_a)$ is saturated. Its image is $|\tilde{V}_a| = w_a$: $w_a \subset v_a \subset |\tilde{U}(a)|$, so every orbit in w_a meets $\tilde{U}(a)$. So $|\tilde{V}_a|$ is open and $\mathrm{cl}\,|\tilde{V}_a| = \mathrm{cl}\, w_a \subset v_a \subset |\tilde{U}(a)|$.
4. *Local finiteness and cover.* $(|\tilde{V}_a|)_{a\in A}$ is a subfamily of (w_i), re-indexed injectively, hence locally finite. A point $\pi(x)$ with $x \in B$ lies in some w_i; not in the extra one, since $\mathrm{cl}\, w_{\text{extra}} \subset |X| \setminus |B|$; so $\pi(x) \in w_a$ for some $a \in B$, and then $\tilde{V}_a \neq \emptyset$ and $a \in A$.

Lemma 14.3.7 (stronger) `Chapter14/BoundaryCover · EpGroupoid.lemma_14_3_7`, p. 655. *Setting*: B saturated; $a \in A$ fixed; $\tilde{U}'(a)$ a neighbourhood of a with the natural G_a-action; for $b \in A$, $\mathrm{cl}_X\, \tilde{U}(b) \subset \tilde{U}'(b)$ and $\mathrm{cl}_X\, \tilde{U}(b)$ proper (Definition 7.1.13: t is proper on $s^{-1}(\mathrm{cl}_X\, \tilde{U}(b))$); the sets $\hat{O}_{a,b}$ open and containing every φ with $s(\varphi) \in \tilde{U}'(a) \cap B$, $t(\varphi) \in \tilde{U}'(b) \cap B$; and $(|\tilde{V}_b|)_{b\in A}$ locally finite. *Statement*: every $y \in U'(a) = \tilde{U}'(a) \cap B$ has a finite $o(y) \subset A$ and an open $\tilde{O}(y) \ni y$ inside $\tilde{U}'(a)$ with (1) $|\tilde{O}(y)| \cap |\tilde{V}_b| \neq \emptyset$ only for $b \in o(y)$; (2) $g * \tilde{O}(y) = \tilde{O}(g * y)$ and $o(g * y) = o(y)$ for $g \in G_a$; (3) every φ with $s(\varphi) \in \tilde{O}(y)$ and $t(\varphi) \in \tilde{U}(b)$, $b \in o(y)$, lies in $\hat{O}_{a,b}$. *Stronger*: the text asks (3) for $s(\varphi) \in \tilde{O}(y) \cap \tilde{U}(a)$; it holds without the intersection, at no cost.

Proof, the text's, in four steps.

0. *The domain.* $U'(a) = \tilde{U}'(a) \cap B$ is invariant under the natural action: $g * y$ is the target of $\Gamma(g, y)$, whose source y lies in the saturated B.
1. *Finitely many b near each orbit.* By local finiteness every orbit p has an open $W(p) \ni p$ in $|X|$ meeting only finitely many $|\tilde{V}_b|$; let $F(p)$ be that finite set of b. Put $o(y) = F(|y|)$. It depends only on the orbit, so $o(g * y) = o(y)$, which is the second half of (2).
2. *Properness* `Chapter14/BoundaryCover · EpGroupoid.exists_nhds_of_isProperSubset`. For $b \in o(y)$: every morphism φ with $s(\varphi) = y$ and $t(\varphi) \in \mathrm{cl}_X\, \tilde{U}(b)$ lies in $\hat{O}_{a,b}$, because $y \in \tilde{U}'(a) \cap B$, $t(\varphi) \in \mathrm{cl}\, \tilde{U}(b) \subset \tilde{U}'(b)$, and $t(\varphi) \in B$ by saturation `Chapter14/BoundaryCover · EpGroupoid.IsSaturated.tgt_mem`. The gen-

eral lemma: if Q is proper, S open, and every φ with $s(\varphi) = y$, $t(\varphi) \in Q$ lies in S, then the same holds for all φ with $s(\varphi)$ in an open $W \ni y$. Its proof: the inverses of the bad morphisms, $\{\psi \in s^{-1}(Q) : \psi^{-1} \notin S\}$, form a closed subset T of $s^{-1}(Q)$; by properness of t on $s^{-1}(Q)$, $t(T)$ is closed; $y \notin t(T)$ by hypothesis; W is the complement of $t(T)$. So each $b \in o(y)$ gives an open $W_b(y) \ni y$.

3. $O_0(y)$. $O_0(y) = \tilde{U}'(a) \cap \pi^{-1}(W(|y|)) \cap \bigcap_{b \in o(y)} W_b(y)$, a finite intersection of open sets containing y. It has (1), as $|O_0(y)| \subset W(|y|)$, and (3), by step 2 and $\tilde{U}(b) \subset \operatorname{cl} \tilde{U}(b)$.
4. *Symmetrization.* $\tilde{O}(y) = \bigcap_{g \in G_a} \{z \in \tilde{U}'(a) : g * z \in O_0(g * y)\}$. It is open (finite intersection, the actions continuous), contains y (as $g*y \in O_0(g*y)$), lies in $O_0(y)$ (take $g = 1$), so keeps (1) and (3). For (2): $z \in \tilde{O}(y)$ iff $hz \in O_0(hy)$ for all h; then for $g*z$ and $g*y$, $h*(g*z) = (hg)*z \in O_0((hg)*y)$, so $g*\tilde{O}(y) \subset \tilde{O}(g*y)$, and the reverse inclusion is the same argument for g^{-1}, since $g^{-1}*(g*z) = z$ for the natural action.

The G_y-invariance `Chapter14/BoundaryCover · EpGroupoid.lemma_14_3_7_stab`: the text asks $\tilde{O}(y)$ to be G_y-invariant. It is read as invariance under the elements of G_a fixing y, whose natural action near y is that of G_y (every morphism between points of $\tilde{U}'(a)$ is a $\Gamma(g, \cdot)$); by (2), $g * \tilde{O}(y) = \tilde{O}(g * y) = \tilde{O}(y)$ for such g.

The hypotheses of the setting are what the proof uses, named: $\operatorname{cl} \tilde{U}(b) \subset \tilde{U}'(b)$ and its properness come from (P6) (§XIV.12); $\hat{O}_{a,b}$ open with $U'(a,b) \subset \hat{O}_{a,b}$ is Proposition 14.3.4 (1); local finiteness is Lemma 14.3.6 (3); saturation of ∂X is shown above.

XIV.12 (P6) and Proposition 14.3.8 (2017: Proposition 14.4): the sets $\hat{V}_a$

(P6) `Chapter14/CoverVHat · SBLocalization.P5Nbhds.P6Nbhds`, p. 653. Given (P5) neighbourhoods $\tilde{U}'(x)$, $x \in \partial X$, (P6) neighbourhoods are open $\tilde{U}(x) \ni x$ with (i) the natural G_x-action and the properness property, (ii) $\operatorname{cl}_{|X|} |\tilde{U}(x)| \subset |\tilde{U}'(x)|$. *Reading*: "the natural G_x-action" is read as the restriction of that of $\tilde{U}'(x)$, the same representation with $\tilde{U}(x) \subset \tilde{U}'(x)$, so that the bi-actions and the sets $\hat{O}$ of §XIV.10 are the same for both. *The "obvious ramification"* `Chapter14/CoverVHat · SBLocalization.P5Nbhds.P6Nbhds.closure_subset`: $\operatorname{cl}_X \tilde{U}(x) \subset \tilde{U}'(x)$. *Proof*: a point $z \in \operatorname{cl}_X \tilde{U}(x)$ has $|z| \in \operatorname{cl} |\tilde{U}(x)| \subset |\tilde{U}'(x)|$, so z is isomorphic to a point of $\tilde{U}'(x)$; Lemma 13.1.1 (Part XIII), with the properness of $\tilde{U}'(x)$ from (P5) (i), says that a limit of points of $\tilde{U}(x) \subset \tilde{U}'(x)$ isomorphic to points of $\tilde{U}'(x)$ lies in $\tilde{U}'(x)$ itself. *Witness* `Chapter14/CoverVHat · SBLocalization.P5Nbhds.P6Nbhds.exists_`: for any (P5) data, without tameness: choose a closed neighbourhood C of x inside $\tilde{U}'(x)$ and an open neighbourhood with the properness property inside its interior (Proposition 7.1.10 and Corollary 7.1.17, Part VII), intersect over the finite group to make it invariant, and use Proposition 7.1.10 again for (ii).

Off ∂X the neighbourhoods are extended by arbitrary natural neighbourhoods, irrelevant since $A \subset \partial X$; the sets $\hat{O}_{a,b}$ are extended by $\emptyset$ `Chapter14/CoverVHat · SBLocalization.CorrExtension.OhatObj`, so that the abstract lemmas, indexed by objects, apply.

Proposition 14.3.8 (explicit), p. 656. *Abstract form* `Chapter14/CoverVHat · EpGroupoid.proposition_14_3_8`. *Setting*: B saturated; $\tilde{U}'(a)$, $\tilde{U}(a)$ natural neighbourhoods with the same representation, $\operatorname{cl}_X \tilde{U}(b) \subset \tilde{U}'(b)$ proper; $\tilde{V}_a \subset \tilde{U}(a)$ open and invariant, $(|\tilde{V}_a|)$ locally finite and covering $|B|$; $\hat{O}_{a,b}$ open with $U'(a,b) \subset \hat{O}_{a,b}$. *Statement*: the sets (14.32) $\hat{V}_a = (\bigcup_{y \in U'(a)} \tilde{O}(y)) \cap \tilde{V}_a$, with $\tilde{O}(y)$ from Lemma 14.3.7, satisfy (1) $\hat{V}_a$ open, G_a-invariant, $\hat{V}_a \subset \tilde{V}_a \subset \tilde{U}(a)$; (2) $\hat{V}_a \cap B = \tilde{V}_a \cap B$; (3) $(|\hat{V}_a|)$ a locally finite cover of $|B|$ by

open subsets of $|X|$; (4) $\hat{V}_{a,a'} \subset \hat{O}_{a,a'}$, where $\hat{V}_{a,a'}$ is the set of morphisms from $\hat{V}_a$ to $\hat{V}_{a'}$. *Explicit*: the hypotheses are named.

Proof.

1. *(1).* $\hat{V}_a$ is a union of open sets intersected with an open set. Invariance: $g*\tilde{O}(y) = \tilde{O}(g*y)$ (Lemma 14.3.7 (2)) and $g * y \in U'(a)$, so the union is invariant, and $\tilde{V}_a$ is invariant.
2. *(2).* $\tilde{U}(a) \subset \tilde{U}'(a)$ (same representation and (P6)), so a point $y \in \tilde{V}_a \cap B$ lies in $U'(a)$ and in its own $\tilde{O}(y)$; hence $\tilde{V}_a \cap B \subset \hat{V}_a$.
3. *(3).* $|\hat{V}_a|$ is open (Proposition 7.1.8) and contained in $|\tilde{V}_a|$, so the family is locally finite; it covers $|B|$ because $(|\tilde{V}_a|)$ does and, by (2), $\hat{V}_a$ keeps the boundary points of $\tilde{V}_a$.
4. *(4),* the text's argument. Let φ go from $z \in \hat{V}_a$ to $z' \in \hat{V}_{a'}$, and $z \in \tilde{O}(y)$, $y \in U'(a)$. Then $|z| = |z'| \in |\tilde{O}(y)| \cap |\tilde{V}_{a'}|$, so $a' \in o(y)$ by Lemma 14.3.7 (1); and $t(\varphi) = z' \in \tilde{V}_{a'} \subset \tilde{U}(a')$, so $\varphi \in \hat{O}_{a,a'}$ by Lemma 14.3.7 (3). This is where the strengthening of (3) is convenient: $s(\varphi) \in \tilde{O}(y)$ suffices.

Instantiated `Chapter14/CoverVHat · SBLocalization.CorrExtension.proposition_14_3_8`: for a representative with (P1)–(P4), (P5) neighbourhoods, the extension of Lemma 14.3.1, (P6) neighbourhoods, and $|X|$ paracompact, there are $A \subset \partial X$, $\tilde{V}_a$ with the conclusions of Lemma 14.3.6, and $\hat{V}_a$ with (1)–(4) for the $\hat{O}_{a,a'}$ of Definition 14.3.2. Every hypothesis of the abstract form is discharged: saturation of ∂X; closedness of $|\partial X|$; the closure and properness conditions by (P6); $U'(a,b) \subset \hat{O}_{a,b}$ by Proposition 14.3.4 (1). The text's summary (14.33)–(14.38) is contained in this statement.

Definition 14.1.7 (a), (b) for $\hat{\tau}$ on $\hat{V}$. §14.4 needs two properties of $\hat{\tau}$ on the sets $\hat{V}_a$, which Proposition 14.3.4 gives on $\hat{O}$: (b) $\hat{\tau}_{a,a}(\Gamma(g,w)) = [I_a \xleftarrow{\mathrm{id}} I_a \xrightarrow{g} I_a]$ for $w \in \hat{V}_a$ `Chapter14/CoverVHat · SBLocalization.CorrExtension.tauHat_Γ_of_VHat`, since $\Gamma(g,w)$ is a morphism of $\hat{V}_{a,a} \subset \hat{O}_{a,a}$ by (4) and invariance; and (a) $\hat{\tau}_{a',a}(\varphi) = \hat{\tau}_{a,a'}(\varphi^{-1})^\sharp$ on $\hat{V}_{a',a}$ `Chapter14/CoverVHat · SBLocalization.CorrExtension.tauHat_inv_of_VHat`, from step 7 of Proposition 14.3.4. They are the conditions *diag* and *swap* of the setting of §XIV.13.

XIV.13 Remark 14.4.1 and the setting of §14.4 (2017: Remark 14.5, (14.33)–(14.38)): from the objects of §14.3

§14.4 works with the objects of §14.3 through a short list of their properties, (14.33)–(14.38). The formalization does the same: the results of §14.4 are proved in an abstract setting that carries exactly those properties, and the objects of §14.3 are then shown to form an instance of it. **Remark 14.4.1** (2017: Remark 14.5), p. 657, explains the plan of the section (the local extension near ∂X must respect the identifications of indices forced by loops of morphisms) and has nothing to prove.

The combinatorial setting `Chapter14/ZLoops · EpGroupoid.ZSetup`, p. 659: an index set A of objects; for $a \in A$ an open neighbourhood $\tilde{U}(a)$ with the natural G_a-action, a G_a-invariant set $\hat{V}_a \subset \tilde{U}(a)$, and a finite index set I_a with a G_a-action; and maps $\hat{\tau}_{a,a'}$ from morphisms to correspondences $J(a,a')$ with two properties: *diag*, $\hat{\tau}_{a,a}(\Gamma(g,w)) = [I_a \xleftarrow{\mathrm{id}} I_a \xrightarrow{g} I_a]$ for $w \in \hat{V}_{a'}$ and *swap*, $\hat{\tau}_{a',a}(\varphi) = \hat{\tau}_{a,a'}(\varphi^{-1})^\sharp$ for φ from $\hat{V}_{a'}$ to $\hat{V}_a$. These are Definition 14.1.7 (a), (b) for the extended correspondences; the text uses them in the proof of Lemma 14.4.5 without naming them. *Witness* `Chapter14/ZLoops · EpGroupoid.ZSetup.trivialSetup`: every ep-groupoid, with A all objects, $\hat{V}_a = \tilde{U}(a)$ a neighbourhood with the natural action (Theorem 7.1.13, Part VII), one-point index sets and constant $\hat{\tau}$.

The setting with topology Chapter14/ZLoopDeform · EpGroupoid.ZData, pp. 657–664: the combinatorial setting together with the three facts of §14.3 that §14.4 uses: the $\hat{V}_a$ are open (Proposition 14.3.8 (1)); $\mathrm{cl}_X \hat{V}_a \subset \tilde{U}(a)$, which is (14.33); and $\hat{\tau}_{a,a'}$ is locally constant at every morphism from $\hat{V}_a$ to $\hat{V}_{a'}$ (Proposition 14.3.4 through Proposition 14.3.8 (4)). *Witness* Chapter14/ZLoopDeform · EpGroupoid.ZData.trivial: the trivial setting with smaller $\hat{V}_a$, $\mathrm{cl}\,\hat{V}_a \subset \tilde{U}(a)$ (two natural neighbourhoods one inside the other with the same action, by Proposition 7.1.10 and Theorem 7.1.13); in it every a lies in its $\hat{V}_a$ and $i \sim_a i$ Chapter14/ZLoopDeform · EpGroupoid.ZData.trivial_zRel.

The boundary sections Chapter14/ZLoopCoherent · SBLocalization.ZSections, p. 665: over a setting with topology, sections s_i^{0a}, $i \in I_a$, with $P(s_i^{0a}(y)) = y$ and (14.45), $s_{g(i)}^{0a}(g * y) = \mu(\Gamma(g,y), s_i^{0a}(y))$, on $U(a) = \tilde{U}(a) \cap \partial X$, and Definition 14.1.7 (4) for $\hat{\tau}$: $s_j^{0a'}(t(\varphi)) = \mu(\varphi, s_i^{0a}(s(\varphi)))$ whenever (i,j) is a pair of $\hat{\tau}_{a,a'}(\varphi)$ and φ is a morphism from $\hat{V}_a$ to $\hat{V}_{a'}$ with source in ∂X (there $\hat{\tau} = \tau^0$). A morphism has source in ∂X iff its target is Chapter14/ZLoopCoherent · SBLocalization.tgt_mem_boundary_iff_z4, so such a φ is a boundary morphism. *Witness* Chapter14/ZLoopCoherent · SBLocalization.ZSections.zero: the zero sections over the trivial setting.

The objects of §14.3 (explicit) Chapter14/ZLoopSetting · SBLocalization.ZConcrete: for a tuple T with (P4), (P5) neighbourhoods, an extension E of the correspondences (Lemma 14.3.1) and (P6) neighbourhoods, the data of Proposition 14.3.8: the set $A \subset \partial X$, the sets $\tilde{V}_a$ with the conclusions of Lemma 14.3.6, the sets $\hat{V}_a$ with (1)–(4) of Proposition 14.3.8, and a base point $x_0 \in \partial X$ that fixes irrelevant values off ∂X. They exist when $|X|$ is paracompact and $\partial X \neq \emptyset$ Chapter14/ZLoopSetting · SBLocalization.ZConcrete.exists_, by the instantiated Proposition 14.3.8 (§XIV.12); the whole chain from (P4) Chapter14/ZLoopSetting · SBLocalization.exists_zConcrete_of_hasP4: (P5) from the good system of T (§XIV.9), the extension from Lemma 14.3.1, (P6) from its witness, and the data from Proposition 14.3.8; and for tame X with $|X|$ paracompact and a boundary point, from the zero tuple with (P4) by Proposition 14.2.3 Chapter14/ZLoopSetting · SBLocalization.exists_zConcrete_of_tame, a nonvacuity check.

Index transport. The objects of §14.3 are indexed by boundary points (a subtype of the objects), those of §14.4 by objects. A retraction r of the objects onto ∂X, the identity on ∂X and x_0 elsewhere Chapter14/ZLoopSetting · SBLocalization.bdryRetr_z5 Chapter14/ZLoopSetting · SBLocalization.bdryRetr_z5_of_mem, with the homomorphism $G_a \to G_{r(a)}$ that is the identity on morphisms for $a \in \partial X$ Chapter14/ZLoopSetting · SBLocalization.isoRetr_z5, carries the index sets, actions and correspondences over without casts; the natural action at $r(a)$ through this homomorphism is that at a, and $[\mathrm{id}, g]$ is unchanged by the relabelling. The base point only fixes values at objects off ∂X, where no $a \in A$ lies.

The instance. Chapter14/ZLoopSetting · SBLocalization.ZConcrete.toZData: $\tilde{U}(a)$ the (P6) neighbourhood, $\hat{V}_a$ from Proposition 14.3.8, I_a the index set of T at a, and $\hat{\tau}_{a,a'}$ the extended correspondence of Definition 14.3.2. Every field is a proved property: *diag* and *swap* are the last two properties of §XIV.12; $\hat{V}_a$ open and invariant is Proposition 14.3.8 (1); local constancy of $\hat{\tau}$ at the morphisms of $\hat{V}_{a,a'}$ is Proposition 14.3.4 with $\hat{V}_{a,a'} \subset \hat{O}_{a,a'}$; and $\mathrm{cl}_X \hat{V}_a \subset \tilde{U}(a)$ Chapter14/ZLoopSetting · SBLocalization.ZConcrete.closure_Vh_subset comes from $\mathrm{cl}_{|X|} |\tilde{V}_a| \subset |\tilde{U}(a)|$ (Lemma 14.3.6 (1)), Lemma 13.1.1 (Part XIII) and the properness of $\tilde{U}(a)$, exactly as the text derives (14.33). *The boundary sections* Chapter14/ZLoopSetting · SBLocalization.ZConcrete.toZSections are those of T: (14.45) is Definition 14.1.3 (2); Definition 14.1.7 (4) for $\hat{\tau}$ along the boundary morphisms Chapter14/ZLoopSetting · SBLocalization.ZConcrete.compatNbhd holds because $\hat{\tau} = \tau^0$ on the boundary morphisms (Proposition 14.3.4 (3)) and $\tilde{U}(a) \cap \partial X \subset U^0(a)$, where the tuple's sections live. A pair of a correspondence satisfies every condition the correspondence satisfies, which is how Definition 14.1.7 (4), stated for all representatives, applies to a pair. *The facts used by the limit ar-*

gument (§XIV.16) `Chapter14/ZLoopSetting · SBLocalization.ZConcrete.toLimit`: the sets $\widehat{O}_{b,c}$, which contain the boundary morphisms from $\tilde{U}(b)$ to $\mathrm{cl}_X \tilde{U}(c)$ `Chapter14/ZLoopSetting · SBLocalization.ZConcrete.morBetween_subset_Ohat` (since $\mathrm{cl}\,\tilde{U}(c) \subset \tilde{U}'(c)$ by (P6) and $U'(b,c) \subset \widehat{O}_{b,c}$); the properness of $\mathrm{cl}_X \tilde{U}(b)$ `Chapter14/ZLoopSetting · SBLocalization.ZConcrete.isProper`; and the local finiteness of Proposition 14.3.8 (3).

Verdict: explicit. The setting is the text's; the identification of objects with boundary points and the three facts are named.

XIV.14 Definitions 14.4.2–14.4.4 and Lemma 14.4.5 (2017: Definitions 14.5–14.7, Lemma 14.6): z-loops and the relations $\sim_z$

Pairs of a correspondence `Chapter14/ZLoops · SBLocalization.IndexCorr.Pairs`: for a correspondence $\tau = [I \xleftarrow{\alpha} K \xrightarrow{\beta} I']$, (i,j) is a pair if $(i,j) = (\alpha(k), \beta(k))$ for some k; this does not depend on the representative, since equivalent representatives differ by a bijection of K. The pairs of $\tau^\sharp$ are the reversed pairs of τ `Chapter14/ZLoops · SBLocalization.IndexCorr.pairs_swap`, and the pairs of $[\mathrm{id}, g]$ are the $(i, g(i))$ `Chapter14/ZLoops · SBLocalization.IndexCorr.pairs_ofPerm`.

Definitions 14.4.2 and 14.4.3 (restated) `Chapter14/ZLoops · EpGroupoid.ZSetup.Arc`, pp. 659–660. The text: for $a \in A$ and $z \in \widehat{V}_a$, a z-loop is a sequence $a = a_0, \dots, a_n = a$ in A with composable morphisms φ_j from $\widehat{V}_{a_j}$ to $\widehat{V}_{a_{j+1}}$, $s(\varphi_0) = z$, composing to 1_z; a z-arc along it is a sequence of indices $i_j \in I_{a_j}$ with each (i_j, i_{j+1}) of the form $(\alpha_j(k_j), \beta_j(k_j))$ for a representative of $\hat{\tau}_{a_j, a_{j+1}}(\varphi_j)$. *As formalized*, loop and arc are one inductive predicate $\mathrm{Arc}(a, b, i, j, \varphi)$: "there are $a = a_0, \dots, a_m = b$ in A, composable morphisms through the $\widehat{V}_{a_j}$ with composite φ, and indices $i = i_0, \dots, i_m = j$ with (i_j, i_{j+1}) a pair of $\hat{\tau}_{a_j, a_{j+1}}(\varphi_j)$". It is generated by two rules: *start*, $\mathrm{Arc}(a, a, i, i, 1_z)$ for $a \in A$, $z \in \widehat{V}_a$; and *append*, from $\mathrm{Arc}(a, b, i, j, \varphi)$, a morphism ψ with $s(\psi) = t(\varphi)$, $t(\psi) \in \widehat{V}_c$, $c \in A$, and a pair (j,k) of $\hat{\tau}_{b,c}(\psi)$, conclude $\mathrm{Arc}(a, c, i, k, \psi \circ \varphi)$. A z-loop with a z-arc from i to i' is $\mathrm{Arc}(a, a, i, i', 1_z)$: the composite 1_z fixes the start and the end at z. Every arc has $a, b \in A$, $s(\varphi) \in \widehat{V}_a$ and $t(\varphi) \in \widehat{V}_b$, by induction. *Restated*: the text's loop is a sequence and the arc a further sequence along it; the predicate records only what the later proofs use, the composite and the end indices.

Concatenation `Chapter14/ZLoops · EpGroupoid.ZSetup.Arc.trans`: from $\mathrm{Arc}(a, b, i, j, \varphi)$ and $\mathrm{Arc}(b, c, j, k, \psi)$ with $s(\psi) = t(\varphi)$, $\mathrm{Arc}(a, c, i, k, \psi \circ \varphi)$. *Proof*: induction on the second arc; its start $1_{t(\varphi)}$ composes to φ, and each appended step is appended to the concatenation, by associativity. *Reversal* `Chapter14/ZLoops · EpGroupoid.ZSetup.Arc.symm`: from $\mathrm{Arc}(a, b, i, j, \varphi)$, $\mathrm{Arc}(b, a, j, i, \varphi^{-1})$. *Proof*: induction; for an appended step ψ with pair (j,k) of $\hat{\tau}_{b,c}(\psi)$, the reversed chain starts with ψ^{-1} from $\widehat{V}_c$ to $\widehat{V}_b$, whose correspondence is $\hat{\tau}_{c,b}(\psi^{-1}) = \hat{\tau}_{b,c}(\psi)^\sharp$ by *swap*, with the reversed pair (k,j); it is followed by the reversal of the rest, and $(\psi \circ \varphi)^{-1} = \varphi^{-1} \circ \psi^{-1}$.

Definition 14.4.4 (same) `Chapter14/ZLoops · EpGroupoid.ZSetup.ZRel`, p. 660: $i \sim_z i'$ if there is a z-loop for a with a z-arc from i to i', that is, $\mathrm{Arc}(a, a, i, i', 1_z)$; the class (14.43) is $[i]_{\sim_z} = \{i' : i' \sim_z i\}$ `Chapter14/ZLoops · EpGroupoid.ZSetup.zClass`. $i \sim_z i'$ forces $z \in \widehat{V}_a$ `Chapter14/ZLoops · EpGroupoid.ZSetup.ZRel.mem_V`. *$\sim_z$ is an equivalence relation* `Chapter14/ZLoops · EpGroupoid.ZSetup.zRel_equivalence` for $a \in A$, $z \in \widehat{V}_a$, as the text calls it without proof: reflexivity is the start rule; symmetry is reversal, $1_z^{-1} = 1_z$; transitivity is concatenation, $1_z \circ 1_z = 1_z$.

Loops of length zero. The start rule gives loops with no morphism, which the text does not list. They

add nothing `Chapter14/ZLoops · EpGroupoid.ZSetup.zRel_iff_unit_step`: $i \sim_z i'$ iff a loop with at least one step relates them. *Proof*: append the step $1_z = \Gamma(1, z)$ from $\hat{V}_a$ to $\hat{V}_a$, whose correspondence is $[\mathrm{id}, 1] = [\mathrm{id}]$ by *diag*, with pairs (i', i'); conversely a loop with a last step is a loop.

Lemma 14.4.5 (explicit) `Chapter14/ZLoops · EpGroupoid.ZSetup.lemma_14_4_5`, p. 660. *Statement*: for $a \in A$, $z \in \hat{V}_a$, $i, i' \in I_a$ and $g \in G_a$, $i \sim_z i'$ iff $g(i) \sim_{g*z} g(i')$; hence $[g(i)]_{\sim_{g*z}} = g([i]_{\sim_z})$ `Chapter14/ZLoops · EpGroupoid.ZSetup.lemma_14_4_5_class`. *Explicit*: *diag* is a hypothesis of the setting. *Proof*, the text's `Chapter14/ZLoops · EpGroupoid.ZSetup.zRel_smul`. Let $w = g * z$, which lies in $\hat{V}_a$ by invariance.

1. *The first step.* $\Gamma(g^{-1}, w)$ is a morphism from w to $g^{-1} * w = z$, between points of $\hat{V}_a$, with correspondence $[\mathrm{id}, g^{-1}]$ by *diag*; its pairs are $(k, g^{-1}(k))$, so $(g(i), i)$ is one. Start at w with index $g(i)$ and append this step: $\mathrm{Arc}(a, a, g(i), i, \Gamma(g^{-1}, w))$.
2. *The loop.* Concatenate the given $\mathrm{Arc}(a, a, i, i', 1_z)$: the composite is $\Gamma(g^{-1}, w)$ again.
3. *The last step.* Append $\Gamma(g, z)$ from z to w, correspondence $[\mathrm{id}, g]$, pair $(i', g(i'))$. The composite is $\Gamma(g, z) \circ \Gamma(g^{-1}, w) = 1_w$, by the functoriality of the natural representation ($\Gamma(g, g^{-1} * w) \circ \Gamma(g^{-1}, w) = \Gamma(gg^{-1}, w)$). So $g(i) \sim_w g(i')$.
4. *The converse* applies this to g^{-1} and w: $g^{-1}(g(i)) = i$, $g^{-1} * w = z$. The class identity follows from the equivalence at each i'.

XIV.15 Proposition 14.4.6 (1) and Lemma 14.4.7 (2017: Proposition 14.5 (1), Lemma 14.7): deforming z-loops

Proposition 14.4.6 (1) (same) `Chapter14/ZLoopDeform · EpGroupoid.ZData.proposition_14_4_6_1`, p. 661. *Statement*: if $i \sim_{z_0} i'$, then $i \sim_z i'$ for all z near z_0 (all of them in $\hat{V}_a$). Open-set form `Chapter14/ZLoopDeform · EpGroupoid.ZData.proposition_14_4_6_1_open`: an open $O \ni z_0$, $O \subset \hat{V}_a$, with $i \sim_z i'$ on O. Class form `Chapter14/ZLoopDeform · EpGroupoid.ZData.eventually_zClass_subset`: $[i]_{\sim_{z_0}} \subset [i]_{\sim_z}$ for z near z_0, because I_a is finite and finitely many "eventually" statements hold together.

The text's proof moves the morphisms of a z_0-loop along with z, using the finite natural representation of Lemma 14.4.7, and concludes by the local constancy of $\hat{\tau}$. *The Lean proof* moves them along sheets (Definition 7.1.4, Part VII), the open sets of morphisms on which s is an sc-diffeomorphism onto an open set, and does not need Lemma 14.4.7.

Sheets. For a sheet $\mathbf{U}$, the local inverse $(s|\mathbf{U})^{-1}$ is defined on the open set $s(\mathbf{U})$ and continuous; $\mathbf{U}$ is open; $(s|\mathbf{U})^{-1}(s(\varphi)) = \varphi$ for $\varphi \in \mathbf{U}$; and $(s|\mathbf{U})^{-1}(z) \to \varphi$ as $z \to s(\varphi)$ `Chapter14/ZLoopDeform · EpGroupoid.Sheet.tendsto_srcInv`. If $1_x \in \mathbf{U}$, then $(s|\mathbf{U})^{-1}(z) = 1_z$ for z near x `Chapter14/ZLoopDeform · EpGroupoid.Sheet.eventually_srcInv_eq_unit`: the unit map is continuous, so $1_z \in \mathbf{U}$ for z near x, and 1_z has source z, where $s|\mathbf{U}$ is injective. Composition is continuous along composable pairs `Chapter14/ZLoopDeform · EpGroupoid.tendsto_comp_of_eventually`: if $a_n \to a_0$, $b_n \to b_0$ and a_n, b_n are eventually composable, then $a_n \circ b_n \to a_0 \circ b_0$, since the multiplication is continuous on the fibre product, which carries the subspace topology.

Key step `Chapter14/ZLoopDeform · EpGroupoid.ZData.Arc.eventually_srcInv`: if $\mathrm{Arc}(a, b, i, j, \varphi)$ and $\mathbf{U} \ni \varphi$ is a sheet, then $\mathrm{Arc}(a, b, i, j, (s|\mathbf{U})^{-1}(z))$ for all z near $s(\varphi)$. *Proof*, by induction on the arc.

1. *Start*: $\varphi = 1_{z_0}$, $z_0 \in \hat{V}_a$. For z near z_0, $(s|\mathbf{U})^{-1}(z) = 1_z$ and $z \in \hat{V}_a$ ($\hat{V}_a$ open), so the start rule applies at z.
2. *Append*: $\varphi = \psi \circ \varphi_1$ with $\mathrm{Arc}(a, b', i, j', \varphi_1)$, $s(\psi) = t(\varphi_1)$, $t(\psi) \in \hat{V}_b$, (j', j) a pair of $\hat{\tau}_{b',b}(\psi)$. Choose sheets $\mathbf{U}_1 \ni \varphi_1$ and $\mathbf{U}_2 \ni \psi$ (sheets cover the morphisms, s being a local sc-

diffeomorphism). For z near $s(\varphi_1) = s(\varphi)$: $\varphi_1' = (s|\mathbf{U}_1)^{-1}(z)$ carries an arc from i to j', by induction, and is near φ_1; so $t(\varphi_1')$ is near $s(\psi)$ and $\psi' = (s|\mathbf{U}_2)^{-1}(t(\varphi_1'))$ is defined and near ψ.

3. *The step survives*: $t(\psi') \in \widehat{V}_b$ for ψ' near ψ ($\widehat{V}_b$ open, t continuous); ψ is a morphism from $\widehat{V}_{b'}$ to $\widehat{V}_b$ (its source is $t(\varphi_1) \in \widehat{V}_{b'}$), where $\widehat{\tau}_{b',b}$ is locally constant, so $\widehat{\tau}_{b',b}(\psi') = \widehat{\tau}_{b',b}(\psi)$ and (j', j) is still a pair. Hence $\mathrm{Arc}(a, b, i, j, \psi' \circ \varphi_1')$.
4. *Identification*: $\psi' \circ \varphi_1' \to \psi \circ \varphi_1 = \varphi$ by continuity of composition, so it eventually lies in the open sheet $\mathbf{U}$; it has source z, so it equals $(s|\mathbf{U})^{-1}(z)$.

Proof of (1). $i \sim_{z_0} i'$ is $\mathrm{Arc}(a, a, i, i', 1_{z_0})$. Take a sheet $\mathbf{U} \ni 1_{z_0}$: by the key step, $\mathrm{Arc}(a, a, i, i', (s|\mathbf{U})^{-1}(z))$ for z near z_0, and $(s|\mathbf{U})^{-1}(z) = 1_z$ for z near z_0. So $i \sim_z i'$.

Lemma 14.4.7 (restated) `Chapter14/FiniteRepresentation · EpGroupoid.lemma_14_4_7`, pp. 661–662. The text restates Proposition 7.1.15 and Corollary 7.1.17 (Part VII) for a finite set Y of isomorphic objects and open $Q(x) \ni x$. Part VII has the proposition as a structure (a natural representation of Y inside one open set): open, connected, mutually disjoint $O(x)$, $x \in Y$, a functor f from the morphisms between points of Y to sc-diffeomorphisms $f(\psi) : O(s(\psi)) \to O(t(\psi))$, the morphisms between points of $O = \bigcup O(x)$ the disjoint union of components $\mathbf{O}(\psi)$, and items (1)–(3). The lemma adds:

1. *Per-object neighbourhoods and (7)* `Chapter14/FiniteRepresentation · EpGroupoid.exists_naturalRepFinite_subset`: $O(x) \subset Q(x)$, and each $\mathrm{cl}\, O(x)$ proper. *Proof*: choose pairwise disjoint open neighbourhoods of the finitely many points of Y inside the $Q(x)$ with the properness property (Hausdorff, Corollary 7.1.17), apply Proposition 7.1.15 inside their union, and note that the connected $O(x) \ni x$ lies in the member containing x.
2. *(4)* `Chapter14/FiniteRepresentation · EpGroupoid.NaturalRepFinite.F_isotropy`: for $g \in G_x$, $f(g)(z) = t(\Gamma(g, z)) = g * z$, with a natural representation of G_x on $O(x)$: the restriction of the representation to the isotropy group is one.
3. *(5)* `Chapter14/FiniteRepresentation · EpGroupoid.NaturalRepFinite.F_equivariant`: $f(\varphi)(g * z) = \gamma_\varphi(g) * f(\varphi)(z)$ with $\gamma_\varphi(g) = \varphi \circ g \circ \varphi^{-1}$: functoriality, $f(\varphi) \circ f(g) = f(\varphi g) = f(\gamma_\varphi(g)\varphi) = f(\gamma_\varphi(g)) \circ f(\varphi)$.
4. *(6)*: the multiplication $\mathbf{O}(\psi) \times_{s,t} \mathbf{O}(\varphi) \to \mathbf{O}(\psi \circ \varphi)$. *The key identity* `Chapter14/FiniteRepresentation · EpGroupoid.NaturalRepFinite.comp_srcInv_eq`: for $\varphi' \in \mathbf{O}(\varphi)$, $(s|\mathbf{O}(\psi))^{-1}(t(\varphi')) \circ \varphi' = (s|\mathbf{O}(\psi \circ \varphi))^{-1}(s(\varphi'))$. *Proof* by connectedness: the map $\varphi' \mapsto (s|\mathbf{O}(\psi))^{-1}(t(\varphi')) \circ \varphi'$ is continuous on the connected $\mathbf{O}(\varphi)$ and takes φ to $\psi \circ \varphi$; its image is connected and consists of morphisms between points of O, so it lies in one component, $\mathbf{O}(\psi \circ \varphi)$; there s is injective. *Bijection* `Chapter14/FiniteRepresentation · EpGroupoid.NaturalRepFinite.bijOn_comp`: every pair (ψ', φ') of the fibre product has $\psi' = (s|\mathbf{O}(\psi))^{-1}(t(\varphi'))$, so $m(\psi', \varphi') = (s|\mathbf{O}(\psi \circ \varphi))^{-1}(s(\varphi'))$, injective in φ'; surjectivity uses the inverse $\chi \mapsto (\ldots, (s|\mathbf{O}(\varphi))^{-1}(s(\chi)))$. *sc-diffeomorphism* `Chapter14/FiniteRepresentation · EpGroupoid.NaturalRepFinite.isScDiffeoOn_comp`: parametrizing the fibre product by its second factor, the multiplication is $(s|\mathbf{O}(\psi \circ \varphi))^{-1} \circ s : \mathbf{O}(\varphi) \to \mathbf{O}(\psi \circ \varphi)$, with inverse $(s|\mathbf{O}(\varphi))^{-1} \circ s$, both composites of sc-diffeomorphisms. *Restated*: the text's fibre product is an M-polyfold; the parametrization by the second factor avoids building that structure.

Lemma 14.4.7 is not used by the proof of Proposition 14.4.6 (1) above; it is used by the proof of (2) (§XIV.16).

XIV.16 Proposition 14.4.6 (2) (2017: Proposition 14.5 (2)): limits of z-loops

Transport along chains. Over boundary sections (§XIV.13), the sections are transported along every arc that starts in ∂X `Chapter14/ZLoopCoherent · SBLocalization.ZSections.arc_transport`: from $\mathrm{Arc}(a,b,i,j,\varphi)$ with $s(\varphi)\in\partial X$, $s_j^{0b}(t(\varphi))=\mu(\varphi,s_i^{0a}(s(\varphi)))$. *Proof* by induction: at the start $\mu(1_z,w)=w$; for an appended ψ with pair (j',j) of $\hat\tau(\psi)$, the source $t(\varphi_1)$ of ψ is in ∂X by saturation, so Definition 14.1.7 (4) for $\hat\tau$ gives $s_j(t\psi)=\mu(\psi,s_{j'}(s\psi))$, and $\mu(\psi,\mu(\varphi_1,\cdot))=\mu(\psi\circ\varphi_1,\cdot)$ (Definition 8.3.5, Part VIII). For a loop, $\varphi=1_z$: if $i\sim_z i'$ with $z\in V_a=\hat V_a\cap\partial X$, then $s_{i'}^{0a}(z)=s_i^{0a}(z)$ `Chapter14/ZLoopCoherent · SBLocalization.ZSections.zRel_sections_eq`.

Proposition 14.4.6 (2) at points of V_a `Chapter14/ZLoopCoherent · SBLocalization.ZSections.proposition_14_4_6_2_of_mem`: if $z_0\in V_a$ and $i\sim_{z_0} i'$, then $s_i^{0a}=s_{i'}^{0a}$ on a neighbourhood of z_0 in ∂X. *Proof*: by (1), $i\sim_z i'$ for z near z_0, and at those z in ∂X the transport gives the equality.

Proposition 14.4.6 (2) (same, and stronger), p. 661. *The printed form* `Chapter14/ZLoopCoherent · SBLocalization.proposition_14_4_6_2_statement`: if $z_k\in V_a$, $i\sim_{z_k} i'$ and $z_k\to z_0\in \mathrm{cl}_X(V_a)$, then $s_i^{0a}=s_{i'}^{0a}$ on a neighbourhood of z_0 in $U(a)$. The sequences are read as closure: $z_0\in\mathrm{cl}\{y\in V_a : i\sim_y i'\}$. *The form with $z_k\in\hat V_a$* `Chapter14/ZLoopCoherent · SBLocalization.HasThetaLocalEq`: for $(i,i')\in\Theta_{z_0}$ (Definition 14.4.8, §XIV.17), that is, $z_k\in\hat V_a$, and $z_0\in\mathrm{cl}_X(V_a)$, the same conclusion. It implies the printed form `Chapter14/ZLoopCoherent · SBLocalization.HasThetaLocalEq.proposition_14_4_6_2`, since $V_a\subset\hat V_a$. *A reading*: Lemma 14.4.9 (3) cites (2) for the pairs of Θ_z, whose sequences lie in $\hat V_a$, while (2) is printed for $z_k\in V_a$. The text's proof works for $z_k\in\hat V_a$, and that is the form proved.

The setting of the proof `Chapter14/ZLoopLimits · SBLocalization.ZLimitSetting`, the facts of §14.3 it uses beyond the setting with topology: open sets $M_{b,c}$ of morphisms (the $\hat O_{b,c}$ of Proposition 14.3.4) on which $\hat\tau_{b,c}$ is locally constant and which contain every boundary morphism from $\tilde U(b)$ to $\mathrm{cl}_X\tilde U(c)$; the properness of $\mathrm{cl}_X\tilde U(b)$ ((P6)); and the local finiteness of $(|\hat V_b|)_{b\in A}$ (Proposition 14.3.8 (3)). And Definition 14.1.7 (4) for $\hat\tau$ along the boundary morphisms from $\tilde U(b)$ to $\tilde U(c)$ `Chapter14/ZLoopLimits · SBLocalization.ZSections.CompatNbhd`, slightly more than along those of $\hat V_{b,c}$, since the limit loop runs through the closures. All of these hold for the objects of §14.3 (§XIV.13).

The general statement `Chapter14/ZLoopLimitExists · SBLocalization.sections_eq_of_theta_closure` (stronger): for every $z_0\in\mathrm{cl}_X(\hat V_a)\cap\partial X$ and $(i,i')\in\Theta_{z_0}$, $s_i^{0a}=s_{i'}^{0a}$ on a neighbourhood of z_0 in ∂X. From it, the form with $z_k\in\hat V_a$ `Chapter14/ZLoopLimitExists · SBLocalization.hasThetaLocalEq_of_limit` ($\mathrm{cl}_X(V_a)\subset\mathrm{cl}_X(\hat V_a)\cap\partial X$, ∂X being closed) and the printed form `Chapter14/ZLoopLimitExists · SBLocalization.proposition_14_4_6_2_of_limit`. *Stronger*: $z_k\in\hat V_a$ instead of V_a, and every $z_0\in\mathrm{cl}_X(\hat V_a)\cap\partial X$ instead of $z_0\in\mathrm{cl}_X(V_a)$.

Proof, the text's, in Lean's order.

1. *The finite representation at z_0* `Chapter14/ZLoopLimitExists · SBLocalization.ZLimitSetting.exists_limitRep` `Chapter14/ZLoopLimits · SBLocalization.ZLimitRep`. (a) By local finiteness, a neighbourhood of $|z_0|$ meets only the $|\hat V_b|$ with b in a finite set $o\subset A$. (b) $K=\bigcup_{b\in o}\mathrm{cl}_X\hat V_b$ lies in the proper set $\bigcup_{b\in o}\mathrm{cl}_X\tilde U(b)$, a finite union of proper sets; so only finitely many objects of K are isomorphic to z_0 (Lemma 7.1.11, Part VII); let $Y\ni z_0$ be that finite set. (c) For $y\in Y$ choose neighbourhoods $Q(y)$: for each pair $b,c\in o$ with $y\in\mathrm{cl}\,\hat V_b\cap\partial X$, inside $\tilde U(b)$ and with every morphism from $Q(y)$ into $\mathrm{cl}_X\tilde U(c)$ in $M_{b,c}$ `Chapter14/ZLoopLimitExists · SBLocalization.ZLimitSetting.exists_nbhd_near` (by the properness step of Lemma 14.3.7, §XIV.11, since at y itself all such morphisms are boundary mor-

phisms from $\tilde{U}(b)$ to $\operatorname{cl}\tilde{U}(c)$, which lie in $M_{b,c}$); for each $b \in o$ with $y \notin \operatorname{cl}\hat{V}_b$, disjoint from $\operatorname{cl}\hat{V}_b$ `Chapter14/ZLoopLimitExists · SBLocalization.ZLimitSetting.exists_nbhd_yb`; and the intersection over the finitely many conditions. (d) Lemma 14.4.7 on Y inside the $Q(y)$ gives the sheets $\mathbf{O}(\psi)$. (e) Properness of K gives a neighbourhood $W \subset O(z_0)$ of z_0 such that every morphism from W into a $\hat{V}_c$ has $c \in o$ `Chapter14/ZLoopLimits · SBLocalization.ZLimitRep.mem_o_of_mem_W` and ends in $\bigcup_y O(y)$: otherwise morphisms with sources tending to z_0 and targets in K outside $\bigcup O(y)$ would have a convergent subsequence, with limit a morphism from z_0 to a point of K isomorphic to z_0, so a point of Y, inside the open $O(y)$.

2. *Sheets and correspondences.* The morphisms between points of the $O(y)$ lie in sheets, and the sheet containing a given morphism is unique `Chapter14/ZLoopLimits · EpGroupoid.NaturalRepFinite.eq_of_mem_sheet_z5` (the sheets are disjoint components). For a morphism $\omega : y \to y'$ of Y with $y \in \operatorname{cl}\hat{V}_b$, $y' \in \operatorname{cl}\hat{V}_c$, the sheet $\mathbf{O}(\omega)$ lies in $M_{b,c}$ `Chapter14/ZLoopLimits · SBLocalization.ZLimitRep.sheet_subset_M`, by the choice of $Q(y)$ (its morphisms go into $O(y') \subset \tilde{U}(c)$). A map locally constant at every point of a set is constant on every preconnected subset `Chapter14/ZLoopLimits · EpGroupoid.eq_of_isPreconnected_z5`, so $\hat{\tau}_{b,c}$ is constant on the connected sheet $\mathbf{O}(\omega)$ `Chapter14/ZLoopLimits · SBLocalization.ZLimitRep.tau_eq`.
3. *The limit loop* `Chapter14/ZLoopLimits · SBLocalization.ZLimitRep.arc_transport`. Claim: for an arc $\mathrm{Arc}(a, b, i, j, \varphi)$ with $s(\varphi) \in W$ and $z_0 \in \operatorname{cl}\hat{V}_a$, there is a morphism $\psi : z_0 \to y$ of Y with $y \in \operatorname{cl}\hat{V}_b$ and $\varphi \in \mathbf{O}(\psi)$, and along the deformed morphisms $(s|\mathbf{O}(\psi))^{-1}(y')$, $y' \in O(z_0) \cap \partial X$, the sections are transported: $s_j^{0b}(t((s|\mathbf{O}(\psi))^{-1}(y'))) = \mu((s|\mathbf{O}(\psi))^{-1}(y'), s_i^{0a}(y'))$. *Proof* by induction. At the start, $\varphi = 1_z$ with $z \in W \subset O(z_0)$ lies in the sheet of 1_{z_0}, and the deformed morphisms are units. For an appended step χ (from $\hat{V}_{b'}$ to $\hat{V}_b$, pair (j', j)): by induction $\varphi_1 \in \mathbf{O}(\psi_1)$, $\psi_1 : z_0 \to y_1$; χ starts at $t(\varphi_1) \in O(y_1)$ and ends in $\hat{V}_{b'}$, and the source of $\chi \circ \varphi_1$ is in W, so by 1 (e) it ends in some $O(y)$ and lies in a sheet $\mathbf{O}(\omega)$, $\omega : y_1 \to y$, with $y \in \operatorname{cl}\hat{V}_b$ (since $O(y)$ meets $\hat{V}_b$, and $O(y)$ misses the closures not containing y). By Lemma 14.4.7 (6), $\chi \circ \varphi_1 \in \mathbf{O}(\omega \circ \psi_1)$; put $\psi = \omega \circ \psi_1$. For $y' \in O(z_0) \cap \partial X$, the deformed $(s|\mathbf{O}(\psi))^{-1}(y')$ is the composite of the deformed φ_1 and the morphism of $\mathbf{O}(\omega)$ from its target (the key identity of Lemma 14.4.7 (6)); the latter is a boundary morphism from $\tilde{U}(b')$ to $\operatorname{cl}\tilde{U}(b)$, its correspondence is $\hat{\tau}_{b',b}(\chi)$ by step 2, so (j', j) is a pair of it, and Definition 14.1.7 (4) along it, composed with the induction hypothesis, gives the transport.
4. *Conclusion* `Chapter14/ZLoopLimits · SBLocalization.ZLimitRep.sections_eq`. Let $(i, i') \in \Theta_{z_0}$: there is $z \in W \cap \hat{V}_a$ with $i \sim_z i'$, an arc with composite 1_z. By 3, $1_z \in \mathbf{O}(\psi)$ for a morphism ψ from z_0; the unit 1_z lies in the sheet of 1_{z_0} too, so $\psi = 1_{z_0}$ by uniqueness of the sheet, and the deformed morphisms are the units $1_{y'}$. The transport gives $s_{i'}^{0a}(y') = s_i^{0a}(y')$ for $y' \in O(z_0) \cap \partial X$.

How Lean departs from the text. The text builds the limit z_0-loop (the limits of the morphisms of the z_k-loops, which exist by properness) and deforms it along ∂X. Here each step of one z_k-loop, k large (a $z \in W$), is located in a sheet of the finite representation directly; the limit loop is the chain of sheet centres ψ, and the identity of its composite is read off from the sheet of the unit. The two arguments use the same facts. Gap G.168 is closed by this proof. For the objects of §14.3 (§XIV.13), where every fact of the setting is proved, both forms hold with no hypothesis beyond the text's data: the printed one `Chapter14/ZLoopSetting · SBLocalization.ZConcrete.proposition_14_4_6_2` and the one with $z_k \in \hat{V}_a$ `Chapter14/ZLoopSetting · SBLocalization.ZConcrete.hasThetaLocalEq`.

XIV.17 Definition 14.4.8 and Lemma 14.4.9 (2017: Definition 14.8, Lemma 14.8): the relations $\approx_z$ on the closure

At points of $\mathrm{cl}_X \hat{V}_a$ outside $\hat{V}_a$ there are no z-loops; the relation $\approx_z$ records which indices are identified by loops at nearby points.

Definition 14.4.8 (restated), p. 663. Θ_z `Chapter14/ZRelationClosure · EpGroupoid.ZData.Theta`: $(i,i') \in \Theta_z$ if $z \in \mathrm{cl}\{y : i \sim_y i'\}$. The text asks for a sequence $z_k \in \hat{V}_a$ with $i \sim_{z_k} i'$ and $z_k \to z$; the closure is the same in a Fréchet–Urysohn space `Chapter14/ZRelationClosure · EpGroupoid.ZData.theta_iff_seq`, for instance a metrizable one, which the object space is. $\approx_z$ `Chapter14/ZRelationClosure · EpGroupoid.ZData.Approx`: the reflexive-transitive closure of Θ_z, that is, $i \approx_z i'$ if there is a chain $i = i_0, \dots, i_k = i'$ with $(i_p, i_{p+1}) \in \Theta_z$ (the text's last index is printed $i_p = i'$; $i_k = i'$ is meant). (14.44): $[i]_{\approx_z} = \{i' : i' \approx_z i\}$ `Chapter14/ZRelationClosure · EpGroupoid.ZData.approxClass`. *Restated*: closure in place of sequences, and the reflexive closure included, so that $\approx_z$ is reflexive also where Θ_z is empty (off $\mathrm{cl}\,\hat{V}_a$, where nothing is asked).

$\approx_z$ is an equivalence relation `Chapter14/ZRelationClosure · EpGroupoid.ZData.approx_equivalence`, as the text calls it. Θ_z is symmetric `Chapter14/ZRelationClosure · EpGroupoid.ZData.theta_symm`, because $\sim_y$ is (§XIV.14); so the reflexive-transitive closure is symmetric (reverse the chain), and it is reflexive and transitive by construction. On $\mathrm{cl}_X \hat{V}_a$, Θ_z is itself reflexive `Chapter14/ZRelationClosure · EpGroupoid.ZData.theta_refl`: $i \sim_y i$ for every $y \in \hat{V}_a$, so $z \in \mathrm{cl}\,\hat{V}_a = \mathrm{cl}\{y : i \sim_y i\}$.

Lemma 14.4.9 (same; (3) stronger), pp. 663–664.

(1) `Chapter14/ZRelationClosure · EpGroupoid.ZData.lemma_14_4_9_1`: for $z \in \hat{V}_a$, $i \sim_z i'$ implies $i \approx_z i'$. *Proof*: by Proposition 14.4.6 (1), $i \sim_y i'$ for y near z, so z is in the closure of that set and $(i,i') \in \Theta_z$.

(2) `Chapter14/ZRelationClosure · EpGroupoid.ZData.lemma_14_4_9_2`: $i \approx_z i'$ implies $g(i) \approx_{g*z} g(i')$. *Proof*: for a single step `Chapter14/ZRelationClosure · EpGroupoid.ZData.theta_smul`, the set $\{y : g(i) \sim_y g(i')\}$ contains $g * \{y : i \sim_y i'\}$ by Lemma 14.4.5, and $z \mapsto g * z$ is continuous on $\tilde{U}(a)$, which contains $\mathrm{cl}\,\hat{V}_a$ (14.33); so $g * z$ is in the closure of the image. Chains map to chains. *Classes* `Chapter14/ZRelationClosure · EpGroupoid.ZData.lemma_14_4_9_2_class`: $[g(i)]_{\approx_{g*z}} = g([i]_{\approx_z})$ for $z \in \mathrm{cl}_X \hat{V}_a$; the reverse inclusion applies (2) to g^{-1} and uses $g^{-1} * (g * z) = z$, valid on $\tilde{U}(a) \supset \mathrm{cl}\,\hat{V}_a$.

(4) `Chapter14/ZRelationClosure · EpGroupoid.ZData.lemma_14_4_9_4`: for $i \in I_a$ and $z_0 \in \mathrm{cl}_X \hat{V}_a$ there is an open $\tilde{Q}(z_0) \ni z_0$ in $\tilde{U}(a)$ with $[i]_{\sim_z} \subset [i]_{\approx_{z_0}}$ for $z \in \hat{V}_a \cap \tilde{Q}(z_0)$. *Proof*: for each of the finitely many $i' \notin [i]_{\approx_{z_0}}$, $(i',i) \notin \Theta_{z_0}$ (as $\Theta_{z_0} \subset \approx_{z_0}$), so z_0 has an open neighbourhood missing $\{y : i' \sim_y i\}$. Intersect these with $\tilde{U}(a)$. At z in the intersection, every $i' \sim_z i$ is in $[i]_{\approx_{z_0}}$. For all i at once `Chapter14/ZRelationClosure · EpGroupoid.ZData.lemma_14_4_9_4_all`, intersect over the finite I_a. The text argues by contradiction with sequences; this is the same argument with neighbourhoods.

(3), p. 663: for $z \in \mathrm{cl}_X(V_a)$ and $i \approx_z i'$, $s_i^{0a} = s_{i'}^{0a}$ on $U(a)$ near z. *Proof*, the text's: chain Proposition 14.4.6 (2), in the form with $z_k \in \hat{V}_a$, along the steps of Θ_z, intersecting the finitely many neighbourhoods. With that form as a hypothesis `Chapter14/ZLoopCoherent · SBLocalization.lemma_14_4_9_3_hyp`, and, the form being proved (§XIV.16), unconditionally and for every $z \in \mathrm{cl}_X(\hat{V}_a) \cap \partial X$ `Chapter14/ZLoopLimitExists · SBLocalization.lemma_14_4_9_3_of_limit`, from the general statement of §XIV.16. *Stronger*: $z \in \mathrm{cl}_X(\hat{V}_a) \cap \partial X$, which contains $\mathrm{cl}_X(V_a)$.

XIV.18 Proposition 14.4.10, Definition 14.4.11 and Remark 14.4.12 (2017: Proposition 14.6, Definition 14.9, Remark 14.6): coherent neighbourhood systems

The natural action on open sets. For a natural neighbourhood N of x and $O \subset N$: $g*O = N \cap \Phi_{g^{-1}}^{-1}(O)$, so $g * O$ is open when O is `Chapter14/ZRelationClosure · EpGroupoid.NatNbhd.isOpen_image_Φ`, and $(gh) * O = g * (h * O)$ `Chapter14/ZRelationClosure · EpGroupoid.NatNbhd.image_Φ_mul`. The action preserves ∂X `Chapter14/ZLoopCoherent · SBLocalization.Φ_mem_boundary_z4` (the degeneracy index is invariant) and maps $\mathrm{cl}_X \hat{V}_a$ and $\mathrm{cl}_X(V_a)$ into themselves `Chapter14/ZLoopCoherent · SBLocalization.mapsTo_Φ_closure_z4` (continuity on $\tilde{U}(a) \supset \mathrm{cl}\,\hat{V}_a$).

Equivariant shrinking `Chapter14/ZRelationClosure · EpGroupoid.NatNbhd.exists_equivariant_nbhds`, the step of Proposition 14.4.10 (2) the text leaves to the reader. *Statement*: let $C \subset N$ be G_x-invariant and $P(z, O)$ a property of neighbourhoods of $z \in C$ that passes to smaller sets and is carried along by the action ($P(z,O) \Rightarrow P(g*z, g*O)$); if every $z \in C$ has an open $O \ni z$ with $P(z,O)$, there are open $O(z) \ni z$, $O(z) \subset N$, with $P(z, O(z))$, $O(g*z) = g*O(z)$, and $O(z) \cap O(g*z) = \emptyset$ whenever $g*z \neq z$. *Proof.*

1. *At one point $w \in C$.* Start from O with $P(w,O)$, intersected with N. The finitely many points $g*w \neq w$ are distinct from w; the object space is Hausdorff, so shrink O so that $O \cap g*O = \emptyset$ for each such g (choose O inside a neighbourhood U_g of w with $U_g \cap g*U_g = \emptyset$, which exists by continuity of the action and $g*w \neq w$). Then intersect over the stabilizer: $O_w = \bigcap_{h*w=w} h*O$. It is open, contains w, is smaller than O (so $P(w, O_w)$ and the separation persist), and is invariant under the stabilizer.
2. *Transport from orbit representatives.* Choose one w in each G_x-orbit of C and put $O(g*w) = g*O_w$. This is well defined: if $g*w = g'*w$, then $g^{-1}g'$ fixes w and $g'*O_w = g*(g^{-1}g')*O_w = g*O_w$. P holds by the equivariance of P; $O(h*z) = h*O(z)$ by construction.
3. *Separation.* If $z = h*w$ and $g*z \neq z$, then $k = h^{-1}gh$ moves w, so $O_w \cap k*O_w = \emptyset$; applying h, $O(z) \cap O(g*z) = \emptyset$.

Proposition 14.4.10 (same) `Chapter14/ZLoopCoherent · SBLocalization.proposition_14_4_10_hyp`, p. 664. *Statement*: for $a \in A$, every $z \in \mathrm{cl}_X \hat{V}_a$ has an open $\tilde{O}(z) \ni z$ with (1) $\tilde{O}(z) \subset \tilde{U}(a)$; (2) $\tilde{O}(g*z) = g*\tilde{O}(z)$, and $\tilde{O}(z) \cap \tilde{O}(g*z) = \emptyset$ if $g*z \neq z$; (3) $[i]_{\sim_y} \subset [i]_{\approx_z}$ for $y \in \tilde{O}(z) \cap \hat{V}_a$; (4) for $z \in \mathrm{cl}_X(V_a)$ and $i \approx_z i'$, $s_i^{0a} = s_{i'}^{0a}$ on $\tilde{O}(z) \cap U(a)$. *Proof.* The text calls it immediate from Lemma 14.4.9. Apply the equivariant shrinking with $C = \mathrm{cl}_X \hat{V}_a$ (invariant, inside $\tilde{U}(a)$ by (14.33)) and $P(z,O)$ the conjunction of (3) at z and (4) at z.

1. *Existence at each z*: (3) holds on the $\tilde{Q}(z)$ of Lemma 14.4.9 (4) for all i; (4) holds on a neighbourhood by Lemma 14.4.9 (3), for each of the finitely many pairs $i \approx_z i'$; intersect.
2. *Shrinking*: both are statements about all points of O.
3. *Action*: (3) at $g*z$ on $g*O$: for $y \in O \cap \hat{V}_a$, $[i]_{\sim_{g*y}} = g([g^{-1}(i)]_{\sim_y}) \subset g([g^{-1}(i)]_{\approx_z}) = [i]_{\approx_{g*z}}$ by Lemma 14.4.5 and Lemma 14.4.9 (2), with $\hat{V}_a$ invariant. (4) at $g*z$: if $i \approx_{g*z} i'$, then $g^{-1}(i) \approx_z g^{-1}(i')$, so $s^{0a}_{g^{-1}(i)} = s^{0a}_{g^{-1}(i')}$ on $O \cap U(a)$; by (14.45), $s_i^{0a}(g*y) = \mu(\Gamma(g,y), s^{0a}_{g^{-1}(i)}(y))$, and the same for i', so they agree on $g*(O \cap U(a)) = g*O \cap U(a)$, the action preserving ∂X.

The hypothesis of this form, Proposition 14.4.6 (2) with $z_k \in \hat{V}_a$, is used only in step 1 through Lemma 14.4.9 (3). Unconditionally `Chapter14/ZLoopLimitExists · SBLocalization.proposition_14_4_10_of_limit`, by §XIV.16; for the objects of §14.3, with no hypothesis beyond the text's data `Chapter14/`

ZLoopSetting · SBLocalization.ZConcrete.proposition_14_4_10.

Definition 14.4.11 (restated) Chapter14/ZLoopCoherent · SBLocalization.ZCoherentNbhdSystem, p. 664. A coherent neighbourhood system for $\mathrm{cl}_X(\hat V_a) \subset \tilde U(a)$ (the 2017 text calls it a good neighbourhood system) is a family of open $\tilde O(z) \ni z$, $z \in \mathrm{cl}_X \hat V_a$, with (1), (2), (3) for $z \in \hat V_a$ as printed, and (4). *Existence*, "in view of Proposition 14.4.10": from the form with $z_k \in \hat V_a$ Chapter14/ZLoopCoherent · SBLocalization.exists_zCoherentNbhdSystem_hyp; unconditionally Chapter14/ZLoopLimitExists · SBLocalization.exists_zCoherentNbhdSystem_of_limit; for the objects of §14.3 Chapter14/ZLoopSetting · SBLocalization.ZConcrete.exists_zCoherentNbhdSystem. For the zero data the hypothesis holds trivially Chapter14/ZLoopCoherent · SBLocalization.hasThetaLocalEq_zero, and a system exists Chapter14/ZLoopCoherent · SBLocalization.exists_zCoherentNbhdSystem_zero.

The abstract form used by Proposition 14.4.14. The local extension is proved over data for one fixed a Chapter14/CoherentExtension · SBLocalization.LocalExtensionBase: $\tilde U(a)$ with the natural G_a-action, an open invariant $\hat V_a \subset \tilde U(a)$, a finite I_a with a G_a-action, boundary sections face-wise sc⁺ on $U(a) = \tilde U(a) \cap \partial X$ with (14.45), relations $\sim_y$ with Lemma 14.4.5, and relations $\approx_z$ as parameters. Over them, a coherent neighbourhood system Chapter14/CoherentExtension · SBLocalization.CoherentNbhdSystem has (1), (3) for $z \in \mathrm{cl}_X \hat V_a$, and (4) as fields; (2) is a separate predicate Chapter14/CoherentExtension · SBLocalization.CoherentNbhdSystem.IsEquivariant, since the proof of Proposition 14.4.14 below does not use it. *A reading*: (3) is printed for $z \in \hat V_a$; Proposition 14.4.10 (3) states it for $z \in \mathrm{cl}_X \hat V_a$, and the proof of Proposition 14.4.14 uses it at points of the closure; we read $z \in \mathrm{cl}_X \hat V_a$, presumably what is meant. The data come from the setting of §XIV.13 Chapter14/CoherentExtension · SBLocalization.LocalExtensionBase.ofZSetup, with $\sim_y$ the relation of Definition 14.4.4 and Lemma 14.4.5 discharged; and from the boundary sections Chapter14/ZDataLocalExtension · SBLocalization.ZSections.toLocalExtensionBase, with $\approx_z$ that of Definition 14.4.8 and the face-wise sc⁺ condition of Definition 14.1.3 (1) as input. The system in this form, with (2), comes from Proposition 14.4.10 Chapter14/ZDataLocalExtension · SBLocalization.coherentNbhdSystem_hyp, and unconditionally Chapter14/ZLoopLimitExists · SBLocalization.coherentNbhdSystem_of_limit. *Witness* of the abstract form at every boundary point of a tame X Chapter14/CoherentExtension · SBLocalization.exists_coherentNbhdSystem_zero: a good V_0 at x contains a good V with $\mathrm{cl}\, V \subset V_0$ (Lemma 14.2.1), and $\tilde U(x) = V_0$, $\hat V_x = V \ni x$ with zero sections, one index and trivial relations carry the system $\tilde O(z) = V_0$.

Remark 14.4.12 (2017: Remark 14.6), p. 664, explains the meaning of the conditions (sections constant on the classes near z can be extended compatibly) and has nothing to prove.

XIV.19 Lemma 14.4.13 (2017: Lemma 14.9): the sections near an orbit

Fix a coherent neighbourhood system and $z \in \mathrm{cl}_X(\hat V_a) \cap \partial X$ with orbit $[z] = G_a * z$. The text sets $\tilde O([z]) = \bigcup_{z' \in [z]} \tilde O(z')$, a disjoint union Chapter14/ZLoopCoherent · SBLocalization.ZCoherentNbhdSystem.disjoint_orbit: distinct points of one orbit have disjoint neighbourhoods, by (2).

Lemma 14.4.13 (narrower in (2)), p. 666. The $\approx_{z'}$-classes partition I_a Chapter14/ZLoopCoherent · SBLocalization.approxClass_eq_or_disjoint, $\approx_{z'}$ being an equivalence relation: two classes are equal or disjoint.

(1) Chapter14/ZLoopCoherent · SBLocalization.lemma_14_4_13_1: an element g of the stabilizer $G_a(z')$ maps $[i]_{\approx_{z'}}$ onto $[g(i)]_{\approx_{z'}}$, so it permutes the classes. *Proof*: Lemma 14.4.9 (2) with $g * z' = z'$.

(2) Chapter14/ZLoopCoherent · SBLocalization.lemma_14_4_13_2: for $z' \in \mathrm{cl}_X(V_a)$ and $g \in G_a(z')$ fixing the class $[i]_{\approx_{z'}}$, $s_i^{0a}(g * y) = \mu(\Gamma(g, y), s_i^{0a}(y))$ for $y \in \tilde O(z') \cap \partial X$. *Proof*: within one class the sections

agree on $\tilde{O}(z')\cap\partial X$ `Chapter14/ZLoopCoherent · SBLocalization.lemma_14_4_13_classSection`, by (4) of the system (the text recalls this as the definition of the sections $s_j^{z'}$ of the proof). $g(i)$ is in the class of i; by (14.45), $s^{0a}_{g(i)}(g*y)=\mu(\Gamma(g,y),s_i^{0a}(y))$; and $g*y\in\tilde{O}(g*z')\cap\partial X=\tilde{O}(z')\cap\partial X$, where $s^{0a}_{g(i)}=s_i^{0a}$. *Narrower*: the text states (2) for every $z\in\mathrm{cl}_X(\hat{V}_a)\cap\partial X$. The proof uses Definition 14.4.11 (4), which covers only points of $\mathrm{cl}_X(V_a)$, and a point of $\mathrm{cl}_X(\hat{V}_a)\cap\partial X$ need not lie in $\mathrm{cl}_X(V_a)$. Presumably $z\in\mathrm{cl}_X(V_a)$ is meant; it is what Proposition 14.4.14 uses. The printed (2) at the other points would need a system with (4) there, which the stronger Lemma 14.4.9 (3) of §XIV.17 permits (about 90 lines, not done, not needed).

(3) `Chapter14/ZLoopCoherent · SBLocalization.lemma_14_4_13_3`: for $g\in G_a(z')$ and every i, $s^{0a}_{g(i)}(g*y)=\mu(\Gamma(g,y),s_i^{0a}(y))$ for $y\in\tilde{O}(z')\cap\partial X$. This is (14.45) on $\tilde{O}(z')\cap\partial X\subset U(a)$; it relates the class of i to the class $g([i])=[g(i)]$ when they differ.

(4) `Chapter14/ZLoopCoherent · SBLocalization.lemma_14_4_13_4`: for $z''=g*z'$, the sections over $\tilde{O}(z'')\cap\partial X$ are determined by those over $\tilde{O}(z')\cap\partial X$. *Proof*: every point of $\tilde{O}(z'')\cap\partial X=g*\tilde{O}(z')\cap\partial X$ is $g*y$ with $y\in\tilde{O}(z')\cap\partial X$, and there $s^{0a}_{g(i)}(g*y)=\mu(\Gamma(g,y),s_i^{0a}(y))$ by (14.45).

The lemma serves the text's construction of Proposition 14.4.14 orbit by orbit. The Lean proof of Proposition 14.4.14 (§XIV.20) does not go orbit by orbit and uses none of (1)–(4); they are proved as printed (with the reading of (2)) for completeness.

XIV.20 Proposition 14.4.14 (2017: Proposition 14.7): the local extension on $\hat{V}_a$

Proposition 14.4.14 (restated) `Chapter14/CoherentExtension · SBLocalization.proposition_14_4_14`, pp. 667–671. *Statement*: let every object lie in a tame strong bundle chart and let the object M-polyfold admit sc-smooth partitions of unity (the hypotheses of Proposition 14.1.4, §XIV.3). For the data of a fixed a and a coherent neighbourhood system (abstract form, (1), (3), (4)) there are sc$^+$-sections $\hat{s}_i^a$, $i\in I_a$, on $\hat{V}_a$ with $\hat{s}^a_{g(i)}(g*y)=\mu(\Gamma(g,y),\hat{s}_i^a(y))$ on $\hat{V}_a$, (1) $\hat{s}_i^a=s_i^{0a}$ on $V_a=\hat{V}_a\cap\partial X$, and (2) $\hat{s}_i^a(z)=\hat{s}_{i'}^a(z)$ whenever $i\sim_z i'$, $z\in\hat{V}_a$. *Restated*: "a local sc$^+$ symmetric section structure on $\hat{V}_a$" is read as sc$^+$-sections with the symmetry of Definition 13.2.5 (1) (Part XIII); the weights and the functor $\hat{\Lambda}_a$ are defined from them in §14.5.

The text's proof builds, on each $\tilde{O}([z])$, sections constant on the $\approx_{z'}$-classes and equivariant under G_a, class by class and orbit by orbit (using Lemma 14.4.13), and glues them with a G_a-invariant partition of unity subordinate to the $\tilde{O}([z])$. *The Lean proof* glues non-equivariant pieces with an ordinary partition of unity and then averages over G_a, as in Proposition 14.1.4.

1. *(4) for the equivalence closure* `Chapter14/CoherentExtension · SBLocalization.CoherentNbhdSystem.eqvGen_s0`: for $z\in\mathrm{cl}_X(V_a)$, if i and i' are related by the equivalence relation generated by $\approx_z$, then $s_i^{0a}=s_{i'}^{0a}$ on $\tilde{O}(z)\cap U(a)$; equality is an equivalence relation, so (4) passes to the generated one. Thus $\approx_z$ need not be an equivalence relation for this proof.
2. *Local pieces* `Chapter14/CoherentExtension · SBLocalization.CoherentNbhdSystem.exists_local_piece`: every object y has an open $U\ni y$ with sc$^+$-sections σ_i of P that equal s_i^{0a} on $U\cap\hat{V}_a\cap\partial X$ and are constant on the $\sim_z$-classes at every $z\in U\cap\hat{V}_a$. Three cases.
   (a) $y\in\mathrm{cl}_X(V_a)$. Take an open $G\ni y$ with $\mathrm{cl}\,G\subset\tilde{O}(y)$ (regularity). Choose a representative $r(i)$ of the class of each i under the equivalence relation generated by $\approx_y$. The section $s^{0a}_{r(i)}$ is face-wise sc$^+$ on $\tilde{O}(y)\cap U(a)$, so it has an sc$^+$ extension $\bar{s}_{r(i)}$ equal to it on $G\cap\partial X$ (the partition-

of-unity step of Proposition 14.1.4, §XIV.3). Put $U = G$ and $\sigma_i = \bar{s}_{r(i)}$. On $G \cap \hat{V}_a \cap \partial X$, $\sigma_i = s^{0a}_{r(i)} = s^{0a}_i$ by step 1. At $z \in G \cap \hat{V}_a$, $i \sim_z i'$ implies $i' \in [i]_{\approx_y}$ by (3), so $r(i) = r(i')$ and $\sigma_i = \sigma_{i'}$.

(b) $y \in \mathrm{cl}_X(\hat{V}_a) \setminus \mathrm{cl}_X(V_a)$. $U = \tilde{O}(y) \setminus \mathrm{cl}_X(V_a)$, $\sigma_i = 0$. $U \cap \hat{V}_a \cap \partial X \subset U \cap V_a = \emptyset$, so nothing is to be matched, and zero sections are constant on every class.

(c) $y \notin \mathrm{cl}_X \hat{V}_a$. U the complement of $\mathrm{cl}_X \hat{V}_a$, $\sigma_i = 0$; nothing is asked.

3. *Gluing.* Choose an sc-smooth partition of unity (ρ_k) subordinate to the cover by the sets U of step 2, and put $\sigma_i = \sum_k \rho_k \sigma^k_i$, an sc⁺-section (a locally finite sum of products of sc-smooth functions with sc⁺-sections). On $\hat{V}_a \cap \partial X$ each piece with $\rho_k \neq 0$ equals s^{0a}_i, so the sum does. A glued section takes equal values for i, i' wherever every piece with non-zero weight does `Chapter14/CoherentExtension · SBLocalization.glueSection_congr_at`; so $\sigma_i(z) = \sigma_{i'}(z)$ when $i \sim_z i'$.
4. *Averaging over G_a.* $\hat{s}^a_i$ is the average of §XIV.3 applied to (σ_i) on the invariant $\hat{V}_a$. It is sc⁺ and equivariant (§XIV.3, stage 2). It still extends s^{0a}, because s^{0a} is equivariant on V_a by (14.45), and the average of an equivariant family is itself. It still satisfies (2): if $i \sim_z i'$, then $g(i) \sim_{g*z} g(i')$ by Lemma 14.4.5, so each translate $\mu(\Gamma(g,z)^{-1}, \sigma_{g(i)}(g * z))$ agrees for i and i'.

Consequences of the departure. Definition 14.4.11 (2) is not needed, nor Lemma 14.4.13, nor a G_a-invariant partition of unity; $\approx_z$ need not be an equivalence relation; and (4) is used only at points of $\mathrm{cl}_X(V_a)$, as printed, since the other points of $\mathrm{cl}_X \hat{V}_a$ get neighbourhoods missing $\mathrm{cl}_X(V_a)$.

For the setting of §14.4. With Proposition 14.4.6 (2) as a hypothesis `Chapter14/ZDataLocalExtension · SBLocalization.proposition_14_4_14_of_hyp`; unconditionally, from the facts of §14.3 and Definition 14.1.7 (4) `Chapter14/ZLoopLimitExists · SBLocalization.proposition_14_4_14_of_limit`; and for the objects of §14.3 `Chapter14/ZLoopSetting · SBLocalization.ZConcrete.proposition_14_4_14`, with $\hat{s}^a_i$ equal to the sections of the tuple on $\hat{V}_a \cap \partial X$, under the hypotheses of Proposition 14.1.4 only. There the boundary sections are face-wise sc⁺ on $U(a)$ `Chapter14/ZLoopSetting · SBLocalization.ZConcrete.isFaceScPlus`, by Definition 14.1.3 (1) for the tuple, restricted to $\tilde{U}(a) \cap \partial X \subset U^0(a)$.

XIV.21 Lemma 14.5.1 and (14.53) (2017: Lemma 14.10): the cut-offs

§14.5 transports the local extensions of §14.4 from the $\hat{V}_a$ to every $\hat{V}_b$, cut off by functors β_a, sums them, and assembles the result into a structurable multisection on X.

Lemma 14.5.1 (stronger) `Chapter14/BoundaryCover · EpGroupoid.lemma_14_5_1`, p. 671. *Statement*: let X admit sc-smooth partitions of unity (Definition 7.5.5, Part VII), B a closed saturated set of objects, $A \subset B$, and $\hat{V}_a$, $a \in A$, open with $B \subset \bigcup_{a \in A} \pi^{-1}(\pi(\hat{V}_a))$. There are sc-smooth functors $\beta_a : X \to [0,1]$ ($\beta_a(s\varphi) = \beta_a(t\varphi)$ for every morphism) with (1) $\mathrm{supp}\,\beta_a \subset \pi^{-1}(\pi(\hat{V}_a))$; (2) every object has a saturated open neighbourhood on which only finitely many β_a are not identically 0; (3) $\sum_{a \in A} \beta_a = 1$ on B. *Stronger*: the tameness of X, part of the text's setting, is not used.

Proof, the text's.

1. *The cover.* For each object x, let $U_x = \pi^{-1}(\pi(\hat{V}_x))$, the saturation of $\hat{V}_x$, if $x \in A$, and $U_x = X \setminus B$ otherwise. Each U_x is open (the saturation of an open set is open, §XIV.11) and saturated (a preimage under π, resp. the complement of the saturated B). They cover X: a point of B lies in some saturation by hypothesis, and a point off B lies in $X \setminus B = U_y$ for any $y \notin A$, in particular for itself (it is not in $B \supset A$).

2. *The partition.* Definition 7.5.5 for ep-groupoids gives an sc-smooth partition of unity (ρ_x) by functors, subordinate to a saturated open cover, locally finite on saturated open sets. Put $\beta_a = \rho_a$ for $a \in A$.
3. *(1)* is subordination. *(2)* is the local finiteness of (ρ_x). *(3)*: at $y \in B$, every ρ_x with $x \notin A$ vanishes, since its support lies in $X \setminus B$; so $\sum_{a\in A} \beta_a(y) = \sum_x \rho_x(y) = 1$.

(14.53), p. 672: $\mathrm{cl}_X\{y \in \tilde{U}(a) : \beta_a(y) \neq 0\} \subset \hat{V}_a$. The text obtains it by shrinking the cover. For the objects of §14.3 no shrinking is needed: it follows from (1), by a property of natural neighbourhoods.

Closure points of a natural neighbourhood `Chapter14/AssemblyInputInstance · SBLocalization.tgt_mem_of_closure_a6`: for a natural neighbourhood $\tilde{U}(x)$ and an invariant $V \subset \tilde{U}(x)$, a morphism φ from a point of V to a point of $\mathrm{cl}_X \tilde{U}(x)$ ends in V. *Proof*: t is a local homeomorphism, hence open, so φ is a limit of morphisms ψ with $s(\psi) \in V$ (open condition near φ along the sheet) and $t(\psi) \in \tilde{U}(x)$; such ψ are $\Gamma(g_\psi, s(\psi))$ for some $g_\psi \in G_x$, by the naturality of the representation. G_x is finite, so one g occurs for ψ arbitrarily close to φ. The condition $\Gamma(g, s(\psi)) = \psi$ is closed near φ (both sides continuous, the morphism space Hausdorff), so $\varphi = \Gamma(g, s(\varphi))$, whose target $g * s(\varphi)$ lies in V by invariance.

(14.53) from (1) `Chapter14/AssemblyInputInstance · SBLocalization.closure_support_subset_a6`: if $\operatorname{supp}\beta \subset \pi^{-1}(\pi(V))$ for an invariant open $V \subset \tilde{U}(x)$, then $\mathrm{cl}_X\{y \in \tilde{U}(x) : \beta(y) \neq 0\} \subset V$. *Proof*: a point y of the closure lies in $\mathrm{cl}\,\tilde{U}(x)$ and in $\operatorname{supp}\beta$, so it is the target of a morphism from a point of V `Chapter14/AssemblyInputInstance · SBLocalization.exists_mor_of_ne_zero_a6` (that is what $y \in \pi^{-1}(\pi(V))$ says); by the previous lemma, $y \in V$.

For the objects of §14.3 `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.exists_betaSpec`: Lemma 14.5.1 with $B = \partial X$ (closed and saturated, §XIV.11), A and the $\hat{V}_a$ of Proposition 14.3.8 (whose orbits cover $|\partial X|$ by (3)), and $\beta_a = 0$ for $a \notin A$, gives cut-offs with all the properties the transport needs `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.BetaSpec`: sc-smooth functors into $[0,1]$, (1)–(3), and (14.53) for the $\hat{V}_a$ themselves, by the lemma above with $V = \hat{V}_a \subset \tilde{U}(a)$ invariant.

XIV.22 Lemma 14.5.2 and Remark 14.5.3 (2017: Lemma 14.11, Remark 14.7): the cut-offs over ∂X

Lemma 14.5.2, p. 672, says that for Λ structurable over ∂X and the β_a of Lemma 14.5.1, the family $(\beta_a|\partial X) \odot \Lambda$ is commensurable (Definition 13.3.11, Part XIII) and its sum is Λ. *As formalized* (narrower): the identity the proof of Theorem 14.1.9 uses, for the summed sections, $\tilde{s}^b_j = s^{0b}_j$ on $\hat{V}_b \cap \partial X$ `Chapter14/TransportedSum · SBLocalization.AssemblyInput.usec_boundary`, proved below once the summed sections are defined. The multisection form, the commensurable family over ∂X and its sum, is not built: it needs the boundary analogue of the commensurability data of §13.3 (about 120 lines), and nothing uses it (gap G.172, §XIV.29).

The data of §§14.3–14.5 `Chapter14/TransportedSum · SBLocalization.AssemblyInput`, for Λ structurable over ∂X, as the assembly uses them: a setting of §14.4 with an input of Proposition 14.5.4 (§XIV.23) whose cut-offs satisfy Lemma 14.5.1 (1)–(3); the values of $\hat{\tau}$ finite in number and equivariant for the bi-action (Proposition 14.3.4); non-empty index sets I_b; the representation of Λ by the boundary sections s^{0b} on $\hat{V}_b \cap \partial X$ (Definition 14.1.3 (3)); and the functor property of Λ over ∂X.

A closed set off which everything vanishes. $K_a = \mathrm{cl}\{y \in \tilde{U}(a) : \beta_a(y) \neq 0\} \subset \hat{V}_a$ (14.53), and $C = \bigcup_{a\in A} t(s^{-1}(K_a))$ `Chapter14/TransportedSum · SBLocalization.AssemblyInput.C`, the set of

points isomorphic to a point where some cut-off is non-zero near its own $\widehat{V}_a$. *C is closed* `Chapter14/TransportedSum · SBLocalization.AssemblyInput.isClosed_C`: each $t(s^{-1}(K_a))$ is closed, K_a being a closed subset of the proper cl $\widetilde{U}(a)$ ((P6)), and the family is locally finite by Lemma 14.5.1 (2) (a saturated neighbourhood on which only finitely many β_a are non-zero meets only finitely many of these saturated sets). *C is saturated* `Chapter14/TransportedSum · SBLocalization.AssemblyInput.isSaturated_C`. *Every transported section vanishes off C* `Chapter14/TransportedSum · SBLocalization.AssemblyInput.tsec_eq_zero_of_not_mem_C`: a value $\mu(\varphi, \beta_a(s\varphi)\widehat{s}^a_i(s\varphi))$ with $t(\varphi) \notin C$ has $s(\varphi) \notin K_a$, so $\beta_a(s\varphi) = 0$. This is how the text's points $|x| \notin \bigcup_b |\widehat{V}_b|$, where "all sections vanish", are handled.

The sum `Chapter14/TransportedSum · SBLocalization.AssemblyInput.usec`: $\widetilde{s}^b_j(y) = \sum_{a\in A} \widetilde{s}^a_{b,j}(y)$ in the fibre over y. *Locally finite* `Chapter14/TransportedSum · SBLocalization.AssemblyInput.exists_finite_tsec`: every y_0 has an open $Q \ni y_0$ and a finite $F \subset A$ with $\widetilde{s}^a_{b,j} = 0$ on Q for $a \notin F$, by Lemma 14.5.1 (2) (if β_a vanishes on a saturated neighbourhood, every transport of $\beta_a \widehat{s}^a$ vanishes there). On Q the sum is a finite section sum `Chapter14/TransportedSum · SBLocalization.AssemblyInput.usec_eq_sectionSum`, so *sc⁺ on $\widehat{V}_b$* `Chapter14/TransportedSum · SBLocalization.AssemblyInput.isScPlusSectionOn_usec`, a finite sum of sc⁺-sections (Proposition 14.5.4) being sc⁺. *Equivariant* `Chapter14/TransportedSum · SBLocalization.AssemblyInput.usec_equivariant` and *compatible with $\widehat{\tau}$* `Chapter14/TransportedSum · SBLocalization.AssemblyInput.usec_compat`: $\mu(\varphi, \cdot)$ is linear on the fibres and commutes with the finite sum, and each summand has the property by Proposition 14.5.4 (4) and the equivariance there.

Proof of the sum identity (Lemma 14.5.2 as used). For $y \in \widehat{V}_b \cap \partial X$: $\widetilde{s}^a_{b,j}(y) = \beta_a(y) s^{0b}_j(y)$ by Proposition 14.5.4 (3), so $\widetilde{s}^b_j(y) = (\sum_a \beta_a(y)) s^{0b}_j(y) = s^{0b}_j(y)$ by Lemma 14.5.1 (3).

The cut-off of the size control, folded into the sections: for an sc-smooth functor σ, $\sigma\widetilde{s}^b_j$ `Chapter14/TransportedSum · SBLocalization.AssemblyInput.ssec` is again sc⁺ on $\widehat{V}_b$, equivariant and compatible with $\widehat{\tau}$ (σ is invariant under morphisms), and zero off C.

Remark 14.5.3 (2017: Remark 14.7), p. 672, comments on the role of the cut-offs and states nothing to prove.

XIV.23 Proposition 14.5.4 (2017: Proposition 14.8): the transport

The input `Chapter14/TransportStatement · SBLocalization.TransportInput`, over a combinatorial setting of §14.4: the open sets $\widehat{V}_a$; the properness of cl $\widetilde{U}(a)$; $\widehat{\tau}$ locally constant; cut-offs β_a, sc-smooth functors with (14.53); boundary sections s^{0a} with Definition 14.1.7 (4) for $\widehat{\tau}$ over ∂X; and local extensions $\widehat{s}^a$, the output of Proposition 14.4.14: sc⁺ on $\widehat{V}_a$, G_a-equivariant, equal to s^{0a} on V_a, and equal on $\sim_z$-related indices.

Proposition 14.5.4 (same) `Chapter14/TransportStatement · SBLocalization.proposition_14_5_4_statement`, p. 673. *Statement*: for every $a \in A$ there are sections $\widetilde{s}^a_{b,j}$, $b \in A$, $j \in I_b$, sc⁺ and G_b-equivariant on $\widehat{V}_b$, with (1) $\widetilde{s}^a_{a,i} = \beta_a \widehat{s}^a_i$ on $\widehat{V}_a$; (2) the support of $\widetilde{s}^a_{a,i}$ closed in X and contained in $\widehat{V}_a$, read as: $\widetilde{s}^a_{a,i}$ vanishes off a closed subset of X contained in $\widehat{V}_a$; (3) $\widetilde{s}^a_{b,j} = \beta_a s^{0b}_j$ on $V_b = \widehat{V}_b \cap \partial X$; (4) $\widetilde{s}^a_{b',\beta(k)}(t\varphi) = \mu(\varphi, \widetilde{s}^a_{b,\alpha(k)}(s\varphi))$ for $\varphi \in \widehat{V}_{b,b'}$ and every representative $[I_b \xleftarrow{\alpha} K \xrightarrow{\beta} I_{b'}]$ of $\widehat{\tau}_{b,b'}(\varphi)$. "Symmetric sc⁺-section structure on $\widehat{V}_b$" is read as sc⁺ and G_b-equivariant (Definition 13.2.5 (1)). *The statement is consistent* `Chapter14/TransportStatement · SBLocalization.proposition_14_5_4_statement_of_zero`: for vanishing data it holds with $\widetilde{s} = 0$, multiples of zero sections being zero `Chapter14/TransportStatement · SBLocalization.sectionSmul_fiberZero_gl`.

Proved Chapter14/TransportProof · SBLocalization.TransportInput.proposition_14_5_4, for every input whose cut-offs satisfy Lemma 14.5.1 (1) Chapter14/TransportProof · SBLocalization.TransportInput.SuppInSat: $\beta_a(y) \neq 0$ only if y is the target of a morphism with source in $\hat{V}_a$. The input records Lemma 14.5.1 only through (14.53); (1) itself is needed for (3), at boundary points of $\hat{V}_b$ not reached from $\hat{V}_a$. The cut-offs of §XIV.21 have it; so does $\beta \equiv 0$ Chapter14/TransportProof · SBLocalization.TransportInput.suppInSat_of_zero. Gap G.170 is closed by this proof.

Proof, the text's, with the transport defined through arcs.

1. *The cut-off local extensions.* $\beta_a \hat{s}^a_i$, extended by 0 off $\hat{V}_a$, is a global section, sc$^+$ on $\hat{V}_a$ Chapter14/TransportProof · SBLocalization.TransportInput.isScPlusSectionOn_bsec (product of an sc-smooth function with an sc$^+$-section).
2. *Definition* Chapter14/TransportProof · SBLocalization.TransportInput.tsec: $\tilde{s}^a_{b,j}(y) = \mu(\varphi, \beta_a \hat{s}^a_i(s\varphi))$ for any arc $\mathrm{Arc}(a,b,i,j,\varphi)$ (§XIV.14) with $t(\varphi) = y$ Chapter14/TransportProof · SBLocalization.TransportInput.tsec_of_arc, and 0_y if there is none Chapter14/TransportProof · SBLocalization.TransportInput.tsec_of_not. A morphism from $\hat{V}_a$ to $\hat{V}_b$ with a pair (i,j) of $\hat{\tau}_{a,b}(\varphi)$ is an arc of length one Chapter14/TransportProof · SBLocalization.TransportInput.arc_of_pairs, so the definition agrees with the text's one-step form Chapter14/TransportProof · SBLocalization.TransportInput.tsec_of_pairs; and the section vanishes at points of $\hat{V}_b$ that are targets of no morphism from $\hat{V}_a$ Chapter14/TransportProof · SBLocalization.TransportInput.tsec_of_not_reach. *Why arcs*: with arcs the equivariance and (4) are immediate (append a step); the text defines the value along one morphism and proves independence by z-loops, which is the next step.
3. *Independence of the arc* Chapter14/TransportProof · SBLocalization.TransportInput.fibAct_bsec_eq, the text's z-loop argument. Let $\mathrm{Arc}(a,b,i,j,\varphi)$ and $\mathrm{Arc}(a,b,i',j,\varphi')$ end at the same point. Concatenating the second with the reversal of the first gives $\mathrm{Arc}(a,a,i',i,\varphi^{-1}\circ\varphi')$, a chain from $s(\varphi') \in \hat{V}_a$ to $s(\varphi) \in \hat{V}_a$. Both ends lie in $\tilde{U}(a)$, so $\varphi^{-1}\circ\varphi' = \Gamma(g, s\varphi')$ for some $g \in G_a$. Append $\Gamma(g^{-1}, s\varphi)$, with pair $(i, g^{-1}(i))$ by *diag*: the composite is $1_{s\varphi'}$, a z-loop, so $i' \sim_{s\varphi'} g^{-1}(i)$. By Proposition 14.4.14 (2), $\hat{s}^a_{i'}(s\varphi') = \hat{s}^a_{g^{-1}(i)}(s\varphi')$; by equivariance, $\mu(\Gamma(g, s\varphi'), \hat{s}^a_{g^{-1}(i)}(s\varphi')) = \hat{s}^a_i(s\varphi)$; β_a is invariant. Hence $\mu(\varphi^{-1}\circ\varphi', \beta_a \hat{s}^a_{i'}(s\varphi')) = \beta_a \hat{s}^a_i(s\varphi)$, and applying $\mu(\varphi, \cdot)$ gives the claim.
4. *sc$^+$ on $\hat{V}_b$* Chapter14/TransportProof · SBLocalization.TransportInput.isScPlusSectionOn_tsec. Two cases at $y_0 \in \hat{V}_b$. (a) $y_0 = t(\varphi_0)$ for a morphism φ_0 with source in $\hat{V}_a$. Take a local sc-smooth inverse τ of t through φ_0 (Definition 7.1.4), and i with (i,j) a pair of $\hat{\tau}_{a,b}(\varphi_0)$ (the right leg of a representative is onto). For z near y_0: $\tau(z)$ goes from $\hat{V}_a$ to $\hat{V}_b$, and $\hat{\tau}_{a,b}(\tau(z)) = \hat{\tau}_{a,b}(\varphi_0)$ by local constancy, so $\tilde{s}^a_{b,j}(z) = \mu(\tau(z), \beta_a \hat{s}^a_i(s\tau(z)))$, the transport of an sc$^+$-section along an sc-smooth family of morphisms, which is sc$^+$ (Part XIII). (b) y_0 is not such a target. Then $y_0 \notin t(s^{-1}(K_a))$, $K_a = \mathrm{cl}\{y \in \tilde{U}(a) : \beta_a \neq 0\} \subset \hat{V}_a$, a closed set by the properness of $\mathrm{cl}\,\tilde{U}(a)$; on the open complement intersected with $\hat{V}_b$, every arc ending there starts at a point with $\beta_a = 0$, so the section is 0 near y_0.
5. *Equivariance* Chapter14/TransportProof · SBLocalization.TransportInput.tsec_equivariant: for $y \in \hat{V}_b$ and $g \in G_b$, append the step $\Gamma(g,y)$, with pair $(j, g(j))$ by *diag*, to an arc ending at (y,j); if there is none at (y,j), there is none at $(g*y, g(j))$ either (append $\Gamma(g^{-1}, g*y)$), and both values are zero.
6. *(4)* Chapter14/TransportProof · SBLocalization.TransportInput.tsec_compat: for $\varphi \in \hat{V}_{b,b'}$ and a pair (j,j') of $\hat{\tau}_{b,b'}(\varphi)$, append φ to an arc ending at $(s\varphi, j)$; if there is none, there is none at $(t\varphi, j')$ (append φ^{-1}, with the reversed pair by *swap*), and both sides are zero.
7. *(3)* Chapter14/TransportProof · SBLocalization.TransportInput.tsec_boundary: for $y \in \hat{V}_b \cap \partial X$. If an arc from (x,i), $x \in \hat{V}_a$, reaches (y,j), then $x \in \partial X$ (saturation), $\hat{s}^a_i(x) = s^{0a}_i(x)$ by Proposition 14.4.14 (1), and Definition 14.1.7 (4) along the arc (§XIV.16) gives $\mu(\varphi, s^{0a}_i(x)) = s^{0b}_j(y)$; with the

invariance of β_a, $\tilde{s}^a_{b,j}(y) = \beta_a(y) s^{0b}_j(y)$. If no arc reaches (y, j), then no morphism from $\widehat{V}_a$ reaches y (with any pair, as the right leg is onto), so $\beta_a(y) = 0$ by Lemma 14.5.1 (1), and both sides are zero.

8. *(1)*: for $y \in \widehat{V}_a$ the start arc $\mathrm{Arc}(a, a, i, i, 1_y)$ ends at (y, i), so $\tilde{s}^a_{a,i}(y) = \mu(1_y, \beta_a \hat{s}^a_i(y)) = \beta_a(y)\hat{s}^a_i(y)$; by step 3 every other arc ending at (y, i) gives the same value.
9. *(2)*: $\tilde{s}^a_{a,i}$ vanishes off $K_a \cap \widehat{V}_a$, by (1) and the definition of K_a, and K_a is closed in X and contained in $\widehat{V}_a$ by (14.53).

XIV.24 Theorem 14.5.5 (2017: Theorem 14.3): a structurability criterion

Theorem 14.5.5 (stronger) `Chapter14/StructurabilityCriterion · SBLocalization.theorem_14_5_5`, p. 678. *The data* `Chapter14/StructurabilityCriterion · SBLocalization.CriterionData`, indexed by a type with points $\mathrm{pt}(a)$: (a) neighbourhoods $U(a)$ with the natural G_a-action; (b) invariant open $V_a \subset U(a)$; (c) $\bigcup_a |V_a| = |X|$; (d) symmetric sc⁺-section structures $(s^a_i)_{i \in I_a}$ on V_a representing Λ; (e) correspondences $\tau_{a,a'}$ on $V(a, a') = \{\varphi : s\varphi \in V_a, t\varphi \in V_{a'}\}$. *Statement*: over an ep-groupoid with $|X|$ paracompact, an sc⁺-multisection functor Λ with such data is structurable. With a good system in place of paracompactness `Chapter14/StructurabilityCriterion · SBLocalization.theorem_14_5_5_of_goodSystem`; paracompactness gives one by Theorem 13.1.8 (Part XIII). *Stronger*: the text's (a) also asks t proper on $s^{-1}(\mathrm{cl}\, U(a))$, and (b) asks $\mathrm{cl}\,|V_a| \subset |U(a)|$; neither is used, so neither is assumed. *A reading*: "correspondence" in (e) is read as in Definitions 13.3.4 and 13.3.6 (Part XIII): locally constant with finitely many values, equivariant for the bi-actions, relating the sections pointwise on $V(a, a')$, with $\tau_{a,a}(\Gamma(g, y)) = [\mathrm{id}, g]$ and $\tau_{a',a}(\varphi) = \tau_{a,a'}(\varphi^{-1})^\sharp$; the text's proof uses the first of the last two for τ_{a_x,a_x}.

The text's proof restricts Λ to the full subcategory on $V = \bigcup V_a$, an equivalence by Theorem 13.4.3 (Part XIII), chooses a_x with $x \in V_{a_x}$ and small $Q(x) \subset V_{a_x}$, and restricts the data. *The Lean proof* does not pass through the restriction; it pulls the structures back along local transports, as in Theorem 13.5.3 (Part XIII).

1. *Local transports* `Chapter14/StructurabilityCriterion · SBLocalization.CriterionData.exists_transport`: every object y has an index a_y, an open $O(y) \ni y$ and an sc-smooth $\tau_y : O(y) \to$ morphisms with $t(\tau_y(z)) = z$ and $s(\tau_y(z)) \in V_{a_y}$. By (c), y is the target of a morphism φ_0 from some V_a; t is a local sc-diffeomorphism, and its local inverse through φ_0, restricted to where the source stays in the open V_a, is τ_y. With a good system $\mathcal{U}_0$ these data form the structure of the proof `Chapter14/StructurabilityCriterion · SBLocalization.CriterionData.nonempty_transport`.
2. *Conjugation.* Put $\sigma_y(z) = s(\tau_y(z)) \in V_{a_y}$. For ψ from $O(y)$ to $O(y')$, $\mathrm{conj}(\psi) = \tau_{y'}(t\psi)^{-1} \circ \psi \circ \tau_y(s\psi)$, a morphism from $\sigma_y(s\psi)$ to $\sigma_{y'}(t\psi)$, continuous in ψ and functorial: $\mathrm{conj}(\psi_1 \circ \psi_2) = \mathrm{conj}(\psi_1) \circ \mathrm{conj}(\psi_2)$, $\mathrm{conj}(1_z) = 1_{\sigma_y(z)}$, $\mathrm{conj}(\psi^{-1}) = \mathrm{conj}(\psi)^{-1}$.
3. *The homomorphisms* $\gamma_y : G_y \to G_{a_y}$. For ψ between points of $O(y)$ near the isotropy, $\mathrm{conj}(\psi)$ is a morphism between points of $V_{a_y} \subset U(a_y)$, hence $\Gamma(h, \sigma_y(s\psi))$ for a unique $h = H(\psi) \in G_{a_y}$, functorial in ψ. $\gamma_y(g) = H(\Gamma(g, y))$, with Γ the representation of $\mathcal{U}_0$, is a homomorphism. *The good system*: inside $\mathcal{U}_0(y) \cap O(y)$, a neighbourhood with the restricted natural action on which $\mathrm{conj}(\Gamma(g, z)) = \Gamma_{a_y}(\gamma_y(g), \sigma_y(z))$, so $\sigma_y(g * z) = \gamma_y(g) * \sigma_y(z)$ (this holds near y by continuity of H, which is locally constant).
4. *The structures.* At y: index set I_{a_y}, G_y acting through γ_y, and sections $s^y_i(z) = \mu(\tau_y(z), s^{a_y}_i(\sigma_y(z)))$, sc⁺ as transports along an sc-smooth family. They are symmetric, by step 3 and the symmetry of s^{a_y}, and represent Λ, because Λ is a functor and s^{a_y} represents it at $\sigma_y(z)$.

5. *The correspondences* $\bar{\tau}_{y,y'}(\varphi) = \tau_{a_y,a_{y'}}(\mathrm{conj}(\varphi))$: locally constant (conj continuous, τ locally constant), finitely many values, equivariant because conj of the bi-action is the bi-action of conj through γ_y, $\gamma_{y'}$, and the bi-action of correspondences pulls back along homomorphisms `Chapter14/StructurabilityCriterion · SBLocalization.IndexCorr.biact_compHom_gl`; they relate the sections, because a relation $s_j^{a'}(t\chi) = \mu(\chi, s_i^a(s\chi))$ for $\chi = \mathrm{conj}(\psi)$ transports to $s_j^{y'}(t\psi) = \mu(\psi, s_i^y(s\psi))$; the diagonal and swap conditions follow from those of τ.
6. *Conclusion* `Chapter14/StructurabilityCriterion · SBLocalization.CriterionData.Transport.isStructurable`: the decoration and the correspondences form a structured tuple (Definition 13.3.6), so Λ is structurable.

Use. The assembly applies the criterion to Λ' with the data built from the summed sections (§XIV.25).

XIV.25 The proof of Theorem 14.1.9 (2017: Theorem 14.2), pp. 656–678

The statement is in §XIV.7. The Lean proof `Chapter14/BoundaryExtensionTheorem · SBLocalization.theorem_14_1_9_charged_a6` has three lines: the data of §§14.3–14.5 exist (step A); they give, for every saturated open set containing the boundary domain support, a structurable extension with domain support inside it (step B); and the size control turns such extensions into one with $N(\Lambda') \le f$ (step C). This is the text's proof, §§14.2–14.5 and the end of p. 677.

Step A: the data exist (explicit) `Chapter14/AssemblyInputInstance · SBLocalization.exists_assemblyInput_a6`. *Statement*: for X tame with $|X|$ paracompact, sc-smooth partitions of unity on the objects, a tame strong bundle chart at every object, and Λ structurable over ∂X and charging every boundary fibre, there are data of §§14.3–14.5 for Λ (§XIV.22). *Proof.*

1. $\partial X = \emptyset$. The empty data `Chapter14/ExtensionAssembly · SBLocalization.emptyAssemblyInput`: $A = \emptyset$, $\beta \equiv 0$, zero sections; every condition is vacuous or about the empty boundary.
2. *A good representative.* Λ is the functor of a structured tuple (§XIV.5, structurability), and by Proposition 14.2.3 the tuple may be taken with (P4) (§XIV.8).
3. *The objects of §14.3* from (P4) and a boundary point (§XIV.13): (P5) neighbourhoods from the good system, the extension of the correspondences of Lemma 14.3.1, (P6) neighbourhoods, and the cover of Proposition 14.3.8.
4. *The cut-offs*: Theorem 7.5.7 (Part VII) gives sc-smooth partitions of unity on $|X|$ from those of the objects and paracompactness; Lemma 14.5.1 for the cover of ∂X by the $\widehat{V}_a$, with (14.53) (§XIV.21).
5. *The local extensions*: Proposition 14.4.14 for the objects of §14.3 (§XIV.20), one choice $\hat{s}^a$ for each $a \in A$ and zero for $a \notin A$ `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.shA6` `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.shA6_spec`.
6. *The input of Proposition 14.5.4* `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.transportInput_a6`: $\widetilde{U}(a)$, $\widehat{V}_a$ and $\hat{\tau}$ from the setting of §XIV.13, the boundary sections of the tuple, the properness of (P6), the local constancy of $\hat{\tau}$, the cut-offs of step 4 and the $\hat{s}^a$ of step 5. Lemma 14.5.1 (1) for these cut-offs is part of their specification.
7. *The data of §§14.3–14.5* `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.assemblyInput_a6`. The remaining fields: the values of $\hat{\tau}$ are finitely many (Proposition 14.3.4); the bi-action `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.biact_τ_a6`, from the equivariance of $\hat{\tau}$ in Proposition 14.3.4, through the identification of the bi-actions `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.biact_eq_a6`, since Γ and the natural action of the setting at $a \in \partial X$ are those of the tuple's good system `Chapter14/AssemblyInputInstance · SBLocalization.ZConcrete.Γ_eq_a6` `Chapter14/`

`AssemblyInputInstance · SBLocalization.ZConcrete.Φ_eq_a6` (the retraction of §XIV.13 is the identity there); the representation of Λ by the boundary sections and the functor property, from the tuple (Definition 14.1.3 (3)); and the non-empty index sets I_a, from charging (§XIV.2). *Explicit*: charging and the tame strong bundle charts are named.

Step B: the extension `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.extension`. *Statement*: for data of §§14.3–14.5, $|X|$ paracompact, sc-smooth partitions of unity, and a saturated open $\tilde{U}'$ containing the boundary domain support of Λ, there is a structurable Λ'' on W with dom-supp$(\Lambda'') \subset \tilde{U}'$ and $\Lambda'' = \Lambda$ over ∂X. *Proof.*

1. *The cut-off* `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.exists_cutoff`: an sc-smooth functor $\sigma : X \to [0,1]$ with $\sigma = 1$ on the boundary domain support and $\mathrm{cl}\{\sigma \neq 0\} \subset \tilde{U}'$. The boundary domain support is closed and saturated (§XIV.7), so its image K in $|X|$ is closed (§XIV.11), and $|\tilde{U}'|$ is open (Proposition 7.1.8). $|X|$ is paracompact Hausdorff, hence normal: choose an open u with $K \subset u$, $\mathrm{cl}\, u \subset |\tilde{U}'|$. The sc-smooth Urysohn lemma on the orbit space `Chapter7/PartitionsOfUnity · EpGroupoid.exists_isScSmoothMap_orbit_urysohn` (Theorem 7.5.7, Part VII) gives $F : |X| \to [0,1]$ sc-smooth, $F = 1$ on K, $F = 0$ off u; $\sigma = F \circ \pi$ is a functor, and its support lies in $\pi^{-1}(\mathrm{cl}\, u) \subset \tilde{U}'$, $\tilde{U}'$ being saturated.
2. *The functor* $\Lambda'' = \Lambda'_\sigma$ `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.mult`, with the cut-off folded into the sections (§XIV.22): for w with a morphism φ from $P(w)$ into some $\hat{V}_b$, $\Lambda''(w) = |I_b|^{-1}\#\{j : \sigma\tilde{s}^b_j(t\varphi) = \mu(\varphi, w)\}$; where there is no such φ, $\Lambda''(w) = 1$ if w is the zero of its fibre and 0 otherwise. *Independent of the chart* `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.mult_eq`: for two charts $\varphi : P(w) \to \hat{V}_b$, $\varphi' : P(w) \to \hat{V}_{b'}$, the morphism $\chi = \varphi' \circ \varphi^{-1}$ goes from $\hat{V}_b$ to $\hat{V}_{b'}$; the summed sections are compatible with $\hat{\tau}_{b,b'}(\chi)$ (§XIV.22), and a correspondence $[I_b \xleftarrow{\alpha} K \xrightarrow{\beta} I_{b'}]$ with bijective legs (Definition 13.3.1) matches the indices j with $\sigma\tilde{s}^b_j(t\varphi) = \mu(\varphi, w)$ to the indices j' with $\sigma\tilde{s}^{b'}_{j'}(t\varphi') = \mu(\varphi', w)$, and $|I_b| = |K| = |I_{b'}|$. Off C, Λ'' is the trivial functor (the sections vanish there). *A functor* `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.isWFunctor_mult`: a chart for w composed with ψ^{-1} is a chart for $\mu(\psi, w)$.
3. *Criterion data for* Λ'' `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.crit`, indexed by $A \sqcup X$. For $b \in A$: $\tilde{U}(b)$, $\hat{V}_b$, I_b, the sections $\sigma\tilde{s}^b_j$. For an object x: a natural neighbourhood of x and inside it the invariant set of its points off C `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.critV` (open, since C is closed, and invariant, since C is saturated), with one index and the zero section. The correspondences are the text's: $\hat{\tau}_{b,b'}$; $[I_b \leftarrow I_b \to \{1\}]$ from $\hat{V}_b$ to a set off C, where all $\sigma\tilde{s}^b_j$ vanish, so they all relate to the zero section; its reverse; and $[\{1\} \leftarrow \{1\} \to \{1\}]$. The covering condition (c): a point of C is isomorphic to a point of some $K_a \subset \hat{V}_a$, and a point x off C lies in its own set. The sections represent Λ'' on each set by the definition of Λ'' (resp. its triviality off C); the diagonal, swap and equivariance conditions for the correspondences involving $\{1\}$ are identities of correspondences with a one-point set.
4. Λ'' *is an sc⁺-multisection* `Chapter14/ExtensionAssembly · SBLocalization.CriterionData.hasLocalSectionStructure_t9`: criterion data with non-empty index sets for a functor give local section structures at every point, the structure of some set of the data transported by a local inverse of t (as in step 1 of Theorem 14.5.5, §XIV.24). The non-empty index sets are where charging enters: the weights $|I_b|^{-1}$ must sum to 1.
5. *Structurable* `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.isStructurable_mult`, by Theorem 14.5.5 (§XIV.24).
6. $\Lambda'' = \Lambda$ *over* ∂X `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.mult_boundary`, when $\sigma = 1$

on the boundary domain support. A boundary point y has a chart φ into some V_b, by Proposition 14.3.8 (3) and (2). There $\sigma\tilde{s}_j^b = \sigma s_j^{0b}$ by the sum identity (§XIV.22); the boundary sections vanish off the boundary domain support of Λ, so $\sigma s_j^{0b} = s_j^{0b}$ at all boundary points; and the count with the s_j^{0b} is $\Lambda(\mu(\varphi, w)) = \Lambda(w)$, by Definition 14.1.3 (3) and the functor property of Λ.

7. *The domain support* `Chapter14/ExtensionAssembly · SBLocalization.AssemblyInput.domSupp_mult_subset`: dom-supp$(\Lambda'') \subset \mathrm{cl}\{\sigma \neq 0\} \subset \tilde{U}'$, since where $\sigma = 0$ every section is zero and Λ'' is trivial.

How the assembly departs from the text. The text forms Λ' from the summed sections on the $\hat{V}_b$, gives the points $|x| \notin \bigcup |\hat{V}_b|$ a neighbourhood with the zero section, applies Theorem 14.5.5, and multiplies by the cut-off afterwards (Lemma 13.3.14). Here the cut-off σ, which only ensures the domain support lies in $\tilde{U}'$, is folded into the sections before Theorem 14.5.5 is applied, and the points off the closed set C (which contains every point where some section is non-zero) are covered by sets around all objects. The size control (step C) multiplies by a second cut-off afterwards, as the text does.

Step C: size control `Chapter14/BoundaryExtensionTheorem · SBLocalization.sizeControl_a6`, the end of p. 677. *Statement*: if for every saturated open $\tilde{U}' \supset$ the boundary domain support there is a structurable extension with domain support in $\tilde{U}'$, then, for N, $\tilde{U}$, f as in Theorem 14.1.9, there is a structurable Λ' with $N(\Lambda') \leq f$, dom-supp$(\Lambda') \subset \tilde{U}$, $\Lambda' = \Lambda$ over ∂X. *Tools*: the pointwise norm $N(\Lambda)(x) = \max\{N(w) : \Lambda(w) > 0, P(w) = x\}$ (Definition 13.2.3) is non-negative `Chapter14/NormControl · SBLocalization.pointwiseNorm_nonneg` and a functor `Chapter14/NormControl · SBLocalization.pointwiseNorm_tgt` ($\mu(\varphi, \cdot)$ is an isometry for an auxiliary norm and preserves Λ); on the domain of an sc⁺-section $y \mapsto N(s(y))$ is continuous `Chapter14/NormControl · SBLocalization.IsScPlusSectionOn.continuousOn_normW`, so $\{N(\Lambda) < f\}$ is open `Chapter14/NormControl · SBLocalization.isOpen_pointwiseNorm_lt` for continuous f, since on the domain of a local representation $N(\Lambda)(y) = \max_i N(s_i(y))$; $N(w) \leq |a| N(\Lambda)(P(w))$ on the support of $a\Lambda$ `Chapter14/NormControl · SBLocalization.normW_le_of_multisectionSmul_pos` (Definition 13.2.10), hence $N(\beta \odot \Lambda) \leq |\beta| N(\Lambda)$ `Chapter14/NormControl · SBLocalization.pointwiseNorm_smulFun_le` (Lemma 13.2.11); and $\beta \odot \Lambda = \Lambda$ where $\beta = 1$ `Chapter14/NormControl · SBLocalization.multisectionSmulFun_of_eq_one`. Saturation: ∂X is saturated `Chapter14/NormControl · SBLocalization.isSaturated_boundary_gl`, and closures `Chapter14/NormControl · SBLocalization.isSaturated_closure_gl` (saturations of open sets are open) and unions `Chapter14/NormControl · SBLocalization.isSaturated_union_gl` of saturated sets are saturated. The zero multisection has empty domain support `Chapter14/NormControl · SBLocalization.domSupp_zeroMultisection_gl`. *Proof.*

1. *A smaller target.* $U' = \{f > 0\}$ is open, saturated (f a functor), contains the boundary domain support (there $0 \leq N(\Lambda) < f$), and lies in $\tilde{U}$ (as $\mathrm{cl}\{f \neq 0\} \subset \tilde{U}$). Take the extension Λ'' with dom-supp$(\Lambda'') \subset U'$.
2. *The good set.* Put $h = N(\Lambda'')$ and $K =$ dom-supp$(\Lambda'') = \mathrm{cl}\{h > 0\}$ (Definition 13.2.14 through the norm); $h = 0$ off K. $G = \{h < f\} \cup (X \setminus K)$ is open and saturated (h, f functors; K the closure of a saturated set).
3. $\partial X \subset G$. Let $x \in \partial X \cap K$. Over x, Λ'' and Λ have the same support. If x is in the boundary domain support, $h(x) = N(\Lambda)(x) < f(x)$. If not, every w over x with $\Lambda(w) > 0$ is the zero (else x would be in the boundary domain support), so $h(x) = 0$; and $f(x) > 0$ because $x \in K \subset U'$. Either way $x \in \{h < f\}$.
4. *The cut-off.* $|\partial X|$ and $|X \setminus G|$ are closed (closed saturated sets, §XIV.11) and disjoint (by 3 and saturation). The sc-smooth Urysohn lemma `Chapter7/PartitionsOfUnity · EpGroupoid.exists_isScSmoothMap_orbit_urysohn` gives $F : |X| \to [0, 1]$ with $F = 1$ on $|\partial X|$ and $F = 0$ on $|X \setminus G|$; $\sigma = F \circ \pi$ is an sc-smooth functor.
5. $\Lambda' = \sigma \odot \Lambda''$ is structurable, by Lemma 13.3.14 (Part XIII) for functions.
6. *The bound.* $N(\Lambda') \leq \sigma h$. On $\{h < f\}$: $\sigma h \leq h < f$. On $X \setminus K$: $h = 0 \leq f$. Off G: $\sigma = 0 \leq f$.

7. *Domain support.* Where $N(\Lambda') > 0$, $\sigma h > 0$, so $h > 0$; hence dom-supp$(\Lambda') \subset \mathrm{cl}\{h > 0\} = K \subset U' \subset \tilde{U}$.
8. *Over* ∂X, $\sigma = 1$, so $\Lambda' = \Lambda'' = \Lambda$.

Readings in the size control, as in the statement (§XIV.7): the text uses $N(\Lambda)(x) < f(x)$ for $x \in \partial X$; the hypothesis gives it on the boundary domain support. At boundary points outside it where $f = 0$, the argument needs Λ'' trivial nearby; asking the extension to have domain support in $\{f > 0\}$ (step 1), which step B allows, provides this.

The chain, in the book's terms. §14.2: Proposition 14.2.3 (§XIV.8). §14.3: Lemma 14.3.1, Proposition 14.3.4, Lemmas 14.3.6, 14.3.7 and Proposition 14.3.8 (§§XIV.9–XIV.12). §14.4: the setting from §14.3, Lemma 14.4.5, Proposition 14.4.6, Lemmas 14.4.7, 14.4.9, Proposition 14.4.10, Definition 14.4.11 and Proposition 14.4.14 (§§XIV.13–XIV.20). §14.5: Lemma 14.5.1 with (14.53), the sum identity of Lemma 14.5.2, Proposition 14.5.4, Theorem 14.5.5, the assembly and the size control (§§XIV.21–XIV.25). No step assumes a hypothesis beyond the theorem's, read with the tame strong bundle charts and charging.

Remark on the superseded route. The size control and the assembly were first proved for the Lean statement of Theorem 14.1.9 without charging (§XIV.7). That route has four statements, kept in the formalization as properties of that statement. (i) The extension statement `Chapter14/NormControl · SBLocalization.theorem_14_1_9_extension_statement` (for every Λ structurable over ∂X and saturated open $\tilde{U}$ containing its boundary domain support, a structurable extension with domain support in $\tilde{U}$) implies that statement `Chapter14/NormControl · SBLocalization.theorem_14_1_9_hyp`, by the size control above for all Λ at once; it holds when $\partial X = \emptyset$ `Chapter14/NormControl · SBLocalization.theorem_14_1_9_extension_statement_of_boundary_eq_empty`. (ii) The hypothesis that every Λ structurable over ∂X has the data of §§14.3–14.5 `Chapter14/ExtensionAssembly · SBLocalization.AssemblyHyp` implies the extension statement `Chapter14/ExtensionAssembly · SBLocalization.theorem_14_1_9_extension_statement_of_assembly` and the uncharged statement `Chapter14/ExtensionAssembly · SBLocalization.theorem_14_1_9_of_assembly`, and holds when $\partial X = \emptyset$ `Chapter14/ExtensionAssembly · SBLocalization.assemblyHyp_of_boundary_eq_empty`. (iii) For X tame with $|X|$ paracompact and partitions of unity, that hypothesis holds only when $\partial X = \emptyset$ `Chapter14/BoundaryExtensionTheorem · SBLocalization.boundary_eq_empty_of_assemblyHyp_a6`, since it implies the uncharged statement, which does not hold when $\partial X \neq \emptyset$; step A is its form with charging. The proof above uses none of (i)–(iii).

XIV.26 A witness with boundary: the half-space bundle

The hypotheses of Theorem 14.1.9 are jointly satisfiable with $\partial X \neq \emptyset$, non-zero fibres and a non-trivial Λ; the theorem is applied to such data.

The object `Chapter14/HalfSpaceBundle · HalfSpaceBundle.halfSEP`: Part I's trivial strong bundle $H \times \mathbb{R}^k \to H$ over the half-space $H = [0,\infty) \times \mathbb{R}^m$, over the trivial ep-groupoid of H (objects H, only identity morphisms) `Chapter14/HalfSpaceBundle · HalfSpaceBundle.epBundleW1`. *The hypotheses of the theorem*: H is a tame quadrant, so the ep-groupoid is tame `Chapter14/HalfSpaceBundle · HalfSpaceBundle.isTame_halfSEP`; its orbit space, H itself, is metrizable, hence paracompact `Chapter14/HalfSpaceBundle · HalfSpaceBundle.hasParacompactOrbitSpace_halfSEP`; it admits sc-smooth partitions of unity `Chapter14/HalfSpaceBundle · HalfSpaceBundle.admitsScPartitionsOfUnity_halfSEP`, by Theorem 5.5.10 (Part V) with its one chart, whose model is finite-dimensional (Proposition 5.5.11); and Part I's re-centred charts are tame strong bundle charts at every object `Chapter14/HalfSpaceBundle · HalfSpaceBundle.hB_halfSEP`. *Non-degenerate*: $\partial H = \{x_1 = 0\}$ `Chapter14/HalfSpaceBundle · HalfSpaceBundle.mem_boundary_`

halfSEP_iff (degeneracy index 1 there), which contains 0 Chapter14/HalfSpaceBundle · HalfSpaceBundle.zero_mem_boundary_halfSEP; and the fibres $\mathbb{R}^k$ are non-zero for $k \geq 1$ Chapter14/HalfSpaceBundle · HalfSpaceBundle.nontrivial_fibre_halfSEP. All hypotheses hold jointly with the zero multisection, an auxiliary norm from Theorem 12.2.2 (Part XII), $\tilde{U} = X$ and $f \equiv 1$ Chapter14/HalfSpaceBundle · HalfSpaceBundle.theorem_14_1_9_charged_hypotheses_halfSEP, and the theorem applies Chapter14/HalfSpaceBundle · HalfSpaceBundle.theorem_14_1_9_halfSEP.

A non-trivial Λ. For $v \in \mathbb{R}^k$, $v \neq 0$, the constant section $x \mapsto (x, v)$ Chapter14/HalfSpaceConstSection · HalfSpaceBundle.constSec is sc⁺ Chapter14/HalfSpaceConstSection · HalfSpaceBundle.isScPlusSectionOn_constSec (constant in a finite-dimensional fibre). The function $\Lambda_v(w) = 1$ if $w = (P(w), v)$ and 0 otherwise Chapter14/HalfSpaceConstSection · HalfSpaceBundle.constMulti is structurable over ∂H Chapter14/HalfSpaceConstSection · HalfSpaceBundle.isBoundaryStructurable_constMulti: on a good system for ∂H, one index, the constant section Chapter14/HalfSpaceConstSection · HalfSpaceBundle.constStructure, and the trivial correspondence Chapter14/HalfSpaceConstSection · HalfSpaceBundle.constTau; Definition 14.1.3 (1) holds by the general lemma of §XIV.2 (an sc⁺-section on X is face-wise sc⁺ on every set). It charges ∂H Chapter14/HalfSpaceConstSection · HalfSpaceBundle.chargesBoundary_constMulti; its boundary domain support contains 0 Chapter14/HalfSpaceConstSection · HalfSpaceBundle.boundaryDomSupp_constMulti_nonempty, since $v \neq 0$; and $N(\Lambda_v)(x) = N(x, v)$ Chapter14/HalfSpaceConstSection · HalfSpaceBundle.pointwiseNorm_constMulti, a continuous function Chapter14/HalfSpaceConstSection · HalfSpaceBundle.continuous_normW_constSec. *The theorem applied* Chapter14/HalfSpaceConstSection · HalfSpaceBundle.theorem_14_1_9_constSec: with $\tilde{U} = X$ and $f(x) = N(x, v) + 1$ (a continuous functor, $> N(\Lambda_v)$ everywhere), there is a structurable Λ' on the bundle with $N(\Lambda')(x) \leq N(x, v) + 1$ and $\Lambda' = \Lambda_v$ over ∂H; every hypothesis is discharged.

Scope of the witness: the ep-groupoid has no isotropy and one chart, so it shows that the hypotheses are jointly satisfiable in a non-degenerate case, not that the isotropy part of the construction is exercised. The trivial bundle over $\mathbb{R}^n$, the witness of the tame strong bundle charts in §XIV.3, has $\partial X = \emptyset$ and so cannot serve here.

XIV.27 §14.6 and Definitions 14.6.1, 14.6.2 (2017: §14.6, Definitions 14.10, 14.11): structured sc⁺-multisections on the faces

§14.6, pp. 680–682, assumes X tame and face-structured. Each face F is saturated and a sub-M-polyfold, hence an ep-groupoid with the morphisms between its points. One is given a structured sc⁺-multisection on each $W|F$, compatible on the intersections of faces, and Theorem 14.6.3 glues them into one over ∂X, which then extends to W by Theorem 14.1.9. This subsection treats the preamble of p. 680 and the two definitions; the next one treats the theorem.

Faces. The faces of §14.6 are Part XI's ep-faces X_θ (Definition 11.1.3, Part XI): for a face θ of the orbit space, the set of objects whose orbit lies in θ. They are closed and saturated, and tame ep-groupoids with the morphisms between their points (Proposition 11.1.6, Part XI). The proof of Theorem 14.6.3 uses five properties of them near a boundary point, each proved separately.

1. *Faces consist of boundary points* Chapter14/FaceGerm · EpGroupoid.IsEpFace.subset_boundary. *Proof*: X_θ is the preimage of the closure of a component of $\{d = 1\}$ in $|X|$, and since the orbit map is open (Proposition 7.1.8, Part VII), the preimage of the closure is the closure of the preimage. Let y be in that closure and suppose $d(y) = 0$. Near y the degeneracy index is at most $d(y)$ (Lemma 2.4.3, Part II, upper semicontinuity), so a point y' of the component close to y has $1 = d(y') \leq d(y) = 0$,

which is impossible. Hence $d(y) \geq 1$.

2. *Every boundary point lies on a face* `Chapter14/FaceGerm · EpGroupoid.exists_isEpFace_mem`. *Proof*: face-structured means that $|x|$ lies on exactly $d_X(x)$ faces of $|X|$; for $x \in \partial X$ this number is at least 1, so the set of faces through $|x|$ is non-empty, and the ep-face of any of them contains x. *And on finitely many* `Chapter14/FaceGerm · EpGroupoid.finite_isEpFace_mem`: the faces of $|X|$ through $|x|$ number $d_X(x)$, and every ep-face through x is the image of one of them.
3. *Near x, every face through a point contains x* `Chapter14/FaceGerm · EpGroupoid.eventually_mem_of_isEpFace`. *Proof*: the degeneracy index of a tame M-polyfold is a corner index (Part II), so at x there are local pieces: an open $B \ni x$ such that each face of X through a point of B, read in B, is one of the $d(x)$ local pieces of $\{d = 1\}$ at x, each of which has x in its closure (Proposition 2.7.5 (3), Part II). For $y \in B$ and $X_\theta \ni y$, the unique face of the object M-polyfold through y inside X_θ is such a piece, so it contains x, and so does X_θ.
4. *Faces are saturated* `Chapter14/FaceAssembly · EpGroupoid.IsEpFace.tgt_mem_iff`: $s(\varphi) \in F$ iff $t(\varphi) \in F$, because $s(\varphi)$ and $t(\varphi)$ have the same orbit. Hence a face is invariant under every natural action at its points `Chapter14/FaceAssembly · EpGroupoid.IsEpFace.Φ_mem`: $\Phi(g)(y) = t(\Gamma(g, y))$ and $s(\Gamma(g, y)) = y$. This is the text's $g * (F \cap U(x)) = F \cap U(x)$.
5. *A local face of a tame chart lies in one face* `Chapter14/FaceGerm · EpGroupoid.exists_isEpFace_germ`: in a tame chart φ, if $t_i(\varphi(y)) = 0$, there are $\varepsilon > 0$ and a face X_θ such that every point e' of the retract within ε of $\varphi(y)$ with $t_i(e') = 0$ has $\varphi^{-1}(e') \in X_\theta$. *Proof*: by Part I's local face structure, there are ε and a point p of $\{d_O = 1\}$ such that, within ε, the face $\{t_i = 0\}$ of the retract lies in the closure of the component S of $\{d_O = 1\}$ through p. In a tame chart the degeneracy index of X is that of the model, so $\varphi^{-1}(S) \subset \{d_X = 1\}$; it is preconnected, as a continuous image of a preconnected set, and therefore lies in the component of $\{d_X = 1\}$ through $q = \varphi^{-1}(p)$. Let θ be the face of $|X|$ through $|q|$. A point e' as above lies in cl S, so $\varphi^{-1}(e')$ lies in the closure of $\varphi^{-1}(S)$, by continuity of φ^{-1} within S, hence in the face of the object M-polyfold through q, whose image in $|X|$ lies in θ.

A sixth fact concerns natural actions, not faces: *two natural representations of G_x agree near x* `Chapter14/FaceGerm · EpGroupoid.NaturalRep.eventually_Γ_eq`. For each g, $\Gamma(g, \cdot)$ and $\Gamma'(g, \cdot)$ are continuous with value g at x, so near x both lie in an open set around g on which s is injective; there $s(\Gamma(g, y)) = y = s(\Gamma'(g, y))$ gives $\Gamma(g, y) = \Gamma'(g, y)$. The natural actions agree as well `Chapter14/FaceGerm · EpGroupoid.NaturalRep.eventually_Φ_eq`. This is why the text can speak of "the" natural action on $F \cap U(x)$: it does not depend on $U(x)$ near x.

Sc⁺ relative to a face (restated) `Chapter14/FaceGerm · SBLocalization.IsFaceRelScPlusSectionOn`, p. 680. The text's sections of $W|F$ are sc⁺ for the induced M-polyfold structure of F. *As formalized*: s is a section over $U \subset F$ with values in $W_{0,1}$, and in every tame strong bundle chart, at every $y \in U$, on every local face germ $\{t_i = 0\}$ at $\varphi(y)$ that lies in F `Chapter14/FaceGerm · SBLocalization.FaceGermIn` (for some $\varepsilon > 0$, the points of the retract within ε of $\varphi(y)$ with $t_i = 0$ map into F), the local representative is sc⁺ on that germ in the sense of Part I. *Reading*: the local face germs lying in F are the local models of the sub-M-polyfold F at its points (Theorem 2.5.17, Part II; Proposition 11.1.6, Part XI), so this is "$s|F$ is an sc⁺-section of $W|F$" read in the charts of X, as Definition 14.1.3 (1) is read (§XIV.2); only the values of s on F near U enter. Definition 14.1.3 (1) implies it for every face `Chapter14/FaceGerm · SBLocalization.IsFaceScPlusSectionOn.isFaceRelScPlusSectionOn`, since it asks the same on every local face germ, whether in F or not; the property passes to subsets of U `Chapter14/FaceGerm · SBLocalization.IsFaceRelScPlusSectionOn.mono`. *Witness*: the zero section, on every $U \subset \partial X$ and for every face `Chapter14/FaceGerm · SBLocalization.isFaceRelScPlusSectionOn_fiberZero`.

Locality. Being sc⁺ on a face germ at a point depends only on the section near the point on the germ

Chapter14/FaceGerm · IsScPlusOnFaceNear.congr_nhds: if $s = s'$ on $O \cap N \cap \{t_i = 0\}$ for an open $N \ni x$, and s is sc⁺ on the germ $O \cap V \cap \{t_i = 0\}$, then s' is sc⁺ on $O \cap (V \cap N) \cap \{t_i = 0\}$: this set is an sc-retract, as an open subset of one (Proposition 2.3.1 (1), Part II), and s' agrees there with the restriction of s. The local representative of a section at z depends only on the section at $\varphi^{-1}(z)$ Chapter14/FaceGerm · SBLocalization.localRepOne_congr.

The gluing criterion Chapter14/FaceGerm · SBLocalization.isFaceScPlusSectionOn_of_local. *Statement*: let s be a section over U with values in $W_{0,1}$ such that for every $y \in U$ and every face $F \ni y$ there are a section σ, sc⁺ relative to F at y, and an open $N \ni y$ with $s = \sigma$ on $F \cap N$. Then s satisfies Definition 14.1.3 (1) on U. *Proof.* 1. Fix a tame strong bundle chart, $y \in U$ in its domain, and i with $t_i(\varphi(y)) = 0$. By fact 5, there are ε and a face F with the local face germ $\{t_i = 0\}$ at $\varphi(y)$, within ε, mapping into F; in particular $y \in F$ (take $e' = \varphi(y)$). 2. The hypothesis gives σ and N; since the germ lies in F, σ is sc⁺ on it at $\varphi(y)$. 3. The set of model points over N is $N' \cap O$ for an open N', by continuity of φ^{-1} on O. On $O \cap (N' \cap B_\varepsilon) \cap \{t_i = 0\}$, a point z has $\varphi^{-1}(z) \in F \cap N$, so $s(\varphi^{-1}(z)) = \sigma(\varphi^{-1}(z))$ and the local representatives agree at z. 4. Locality transfers sc⁺ from σ to s on the germ.

Structured data over a set Chapter14/FaceStructured · SBLocalization.RelTuple. For K a face or an intersection of faces, and an index family Chapter14/FaceStructured · SBLocalization.IndexFamily (finite sets I_x with G_x-actions, one for every object), the data of a structured sc⁺-multisection over K are:

- a function $\Lambda : W \to \mathbb{Q}^+$ that is a functor over K, $\Lambda(\mu(\varphi, w)) = \Lambda(w)$ for $s(\varphi) \in K$;
- for $x \in K$ a neighbourhood $U(x)$ with the natural G_x-action of X, with t proper on $s^{-1}(\mathrm{cl}(U(x) \cap K))$, condition (1) of Definition 13.1.4 (Part XIII) for the trace on K;
- sections s_i^x, $i \in I_x$, over $U(x) \cap K$, equivariant under the natural action, $s^x_{g(i)}(g * y) = \mu(\Gamma(g, y), s_i^x(y))$, and representing Λ on $U(x) \cap K$ by the counting formula of Definition 14.1.3 (3);
- correspondences $\tau_{x,x'}$ on the morphisms between $U(x) \cap K$ and $U(x') \cap K$, locally constant with finitely many values, with (a) inversion, (b) the diagonal, (c) the bi-action, and (4) the compatibility of the sections of Definition 14.1.7.

The regularity of the sections is not part of these data; it is added for faces below. The traces are written $U(x) \cap K$ Chapter14/FaceStructured · SBLocalization.RelTuple.U, and the morphism sets between them $\mathcal{U}(x, x')$ Chapter14/FaceStructured · SBLocalization.RelTuple.mor.

The equivalence Chapter14/FaceStructured · SBLocalization.RelTuple.Equiv, "similar to the X-case" (Definition 13.3.8, Part XIII), in the form of §XIV.5: $T \sim T'$ if $\Lambda = \Lambda'$ over K and there are open $O(x) \ni x$ with $O(x) \cap K \subset U(x) \cap U'(x)$, $I_x = I'_x$, and $\tau'_{x,x'} = \tau_{x,x'}$, transported along these equalities, on the morphisms between the $O(x) \cap K$. Reflexive Chapter14/FaceStructured · SBLocalization.RelTuple.Equiv.refl with $O(x) = U(x)$; symmetric Chapter14/FaceStructured · SBLocalization.RelTuple.Equiv.symm by reversing the equalities; transitive Chapter14/FaceStructured · SBLocalization.RelTuple.Equiv.trans with $O(x) = O_1(x) \cap O_2(x)$, a morphism between these sets being one for both pairs. The classes are the structured multisections over K Chapter14/FaceStructured · SBLocalization.RelStructured.

Reading: the text's "good neighbourhood $U_F(x) \subset F$" is read as it explains it on p. 680, $U(x) \cap F$ for a neighbourhood of X with the natural action and the properness of Definition 13.1.4 (1). Since F is closed, $\mathrm{cl}(U(x) \cap F)$ lies in F, and the properness in X is the properness inside the face (shown below).

Restriction to a smaller set Chapter14/FaceStructured · SBLocalization.RelTuple.restrictTo, for $K' \subset K$: the same Λ, index sets, sections and correspondences, on the traces $U(x) \cap K'$ for $x \in K'$ Chapter14/FaceStructured · SBLocalization.RelTuple.restrictTo_U. Properness passes to the smaller closed set;

the remaining conditions are inherited by subsets. Restriction respects the equivalence `Chapter14/FaceStructured · SBLocalization.RelTuple.Equiv.restrictTo`: the same open sets $O(x)$, for $x \in K'$, witness the equivalence of the restrictions, since $O(x) \cap K' \subset O(x) \cap K$. Restricting twice is restricting once `Chapter14/FaceStructured · SBLocalization.RelTuple.restrictTo_restrictTo_equiv`, by reflexivity, the two tuples being equal; this is the text's remark that $\tilde{\Lambda}_{F\cap F'\cap F''}$ can be formed for any number of faces. On classes `Chapter14/FaceStructured · SBLocalization.RelStructured.restrictTo`.

A structured sc⁺-multisection on $W|F$ `Chapter14/FaceStructured · SBLocalization.FaceStructuredTuple`, p. 680: data over a face F whose sections s_i^x are sc⁺ relative to F on $U(x) \cap F$. The equivalence is that of the underlying data `Chapter14/FaceStructured · SBLocalization.FaceStructuredTuple.setoid`, and the classes `Chapter14/FaceStructured · SBLocalization.FaceStructuredMultisection` are the text's $\tilde{\Lambda}_F = [\Lambda_F, \mathcal{U}_F, S_F, \tau_F]$. *Witness* `Chapter14/FaceStructured · SBLocalization.FaceStructuredTuple.zero`: the zero multisection on any face, on the index family $I_x = \{*\}$ with the trivial action `Chapter14/FaceStructured · SBLocalization.IndexFamily.unit`, with the zero section, the trivial correspondence $[\{*\} \leftarrow \{*\} \rightarrow \{*\}]$, and neighbourhoods with the properness property, which Part XIII supplies at every point.

Definition 14.6.1 (same) `Chapter14/FaceStructured · SBLocalization.FaceStructuredTuple.restrictFace`, p. 681: the restriction $\tilde{\Lambda}_F|F'$ to $W|(F \cap F')$ is the restriction of the data to $K' = F \cap F'$: it keeps Λ_F, the index sets, the sections and the correspondences, with neighbourhoods $U_F(x) \cap F \cap F'$ for $x \in F \cap F'$ and the correspondences restricted to the morphisms between them. Since restriction respects the equivalence, it is well defined on classes `Chapter14/FaceStructured · SBLocalization.FaceStructuredMultisection.restrictFace`, as the text's "obtained by taking a representative" asks. The text's $\tilde{\Lambda}_{F'}|F$, whose base is $F' \cap F$, is read as a class over $F \cap F'$ `Chapter14/FaceStructured · SBLocalization.FaceStructuredMultisection.restrictFace'`, the two sets being equal. The sections of the restriction are sc⁺ relative to F `Chapter14/FaceStructured · SBLocalization.FaceStructuredTuple.restrictFace_isFaceRelScPlus`, being restrictions of such sections.

Sc⁺ on the corner (same). The definition says that $\tilde{\Lambda}_F|F'$ is a structured sc⁺-multisection on $W|(F \cap F')$, so its sections must be sc⁺ on the corner $F \cap F'$, not only relative to F. *Locally* `Chapter14/CornerScPlus · IsScPlusOnCornerNear`: near x, for an open $V \ni x$, the corner germ $O \cap V \cap \{t_i = 0\} \cap \{t_j = 0\}$ is an sc-retract and s restricted to it is an sc⁺-section; the condition is symmetric in i and j `Chapter14/CornerScPlus · IsScPlusOnCornerNear.symm`, the two hyperplanes being intersected in either order. *From a face germ to a corner germ* `Chapter14/CornerScPlus · IsScPlusOnFaceNear.isScPlusOnCornerNear`: in a tame local model, sc⁺ on the face germ $\{t_i = 0\}$ at x gives sc⁺ on every corner germ $\{t_i = t_j = 0\}$ at x. *Proof*: by Part I's corner retraction for the pair $\{i, j\}$ of a tame retraction, the corner germ is an sc-retract; it is a subset of the face germ, and an sc-smooth map on a retract restricts to an sc-smooth map on a sub-retract of its source; the fibre condition of an sc⁺-section is pointwise. *On X* `Chapter14/CornerScPlus · SBLocalization.IsCornerRelScPlusSectionOn`: the sc⁺-sections of $W|(F \cap F')$, read in the tame strong bundle charts of X on the corner germs whose two face germs lie in F and in F'. Sc⁺ relative to F gives it `Chapter14/CornerScPlus · SBLocalization.IsFaceRelScPlusSectionOn.isCornerRelScPlusSectionOn`, by the local statement applied to the face germ in F (relative to F' it follows by the symmetry). The zero section has it `Chapter14/CornerScPlus · SBLocalization.isCornerRelScPlusSectionOn_fiberZero`. *Hence* `Chapter14/CornerScPlus · SBLocalization.FaceStructuredTuple.restrictFace_isCornerRelScPlus`: the sections of $\tilde{\Lambda}_F|F'$ are sc⁺ on the corner $F \cap F'$, and the restriction is a structured sc⁺-multisection on $W|(F \cap F')$ in the sense of the definition. This closes gap G.184.

The neighbourhoods $U_F(x)$ inside the face (restated, up to equivalence). The text asks the $U_F(x)$ to be good neighbourhoods for the ep-groupoid F, that is, the conditions of Definitions 13.1.2 and 13.1.4 inside F. Three steps.

1. *Conditions (1) and (3) of Definition 13.1.4 pass from X to a face.* Properness `Chapter14/FaceGoodSystem · EpGroupoid.isProperSubset_face`: if t is proper on $s^{-1}(\mathrm{cl}(U \cap X_\theta))$ in X, then it is proper on s^{-1} of the closure of the trace in the face ep-groupoid. The proof is by ultrafilters: an ultrafilter of face morphisms with sources in the closure and targets converging to $z \in X_\theta$ is an ultrafilter of morphisms of X with the same property, so it converges in X to some φ with $s(\varphi) \in \mathrm{cl}(U \cap X_\theta) \subset X_\theta$, the face being closed, and $t(\varphi) = z$; then φ is a face morphism, and the ultrafilter converges to it in the face. Components `Chapter14/FaceGoodSystem · EpGroupoid.isDiffeoComponent_face`: if every component of the morphisms between U and U' in X has s and t sc-diffeomorphisms onto open sets, so does every component of the face morphisms between $U \cap X_\theta$ and $U' \cap X_\theta$. A component C of the face morphism set is preconnected in X, so it lies in the component of X through any of its points; on that component s and t are injective, hence on C; and on an open set of morphisms of an ep-groupoid, injectivity of s and t is the component condition (the morphisms of the face form a sub-M-polyfold, with s, t local sc-diffeomorphisms). So every good system of X restricts to neighbourhoods of X_θ with (1) `Chapter14/FaceGoodSystem · EpGroupoid.GoodSystem.isProperSubset_face` and (3) `Chapter14/FaceGoodSystem · EpGroupoid.GoodSystem.isDiffeoComponent_face`.
2. *Shrinking a tuple.* For open $W(x) \ni x$, the invariant core of $U(x) \cap W(x)$ under the natural action is an open neighbourhood of x carrying the same action `Chapter14/FaceGoodSystem · EpGroupoid.NatNbhd.shrink`. Replacing the neighbourhoods of a tuple by these cores, and keeping Λ, the sections and the correspondences, gives a tuple `Chapter14/FaceGoodSystem · SBLocalization.RelTuple.shrink` equivalent to the original `Chapter14/FaceGoodSystem · SBLocalization.RelTuple.equiv_shrink`, with $O(x)$ the cores. Sections that were sc$^+$ relative to F stay so on the smaller sets `Chapter14/FaceGoodSystem · SBLocalization.FaceStructuredTuple.shrink`.
3. *Choosing the shrinking.* For $|X|$ paracompact, Theorem 13.1.8 (Part XIII) gives a good system $\mathcal{G}$ of X; shrinking inside its neighbourhoods $G(x)$ gives an equivalent tuple whose neighbourhoods satisfy (3) in X `Chapter14/FaceGoodSystem · SBLocalization.RelTuple.exists_equiv_isDiffeoComponent`, since (3) passes to smaller open sets (§XIV.4). By step 1, they satisfy (1) and (3) inside the face.

Conclusion `Chapter14/FaceGoodSystem · SBLocalization.FaceStructuredTuple.exists_equiv_faceGood`: for X tame and face-structured with $|X|$ paracompact, every face tuple over X_θ is equivalent to one with the same Λ, index sets, sections and correspondences whose $U_F(x)$ satisfy (1) and (3) inside the face ep-groupoid and are invariant under the natural G_x-action of X, $\Phi(g)(U(x) \cap F) \subset U(x) \cap F$. On classes `Chapter14/FaceGoodSystem · SBLocalization.FaceStructuredMultisection.exists_rep_faceGood`: every structured sc$^+$-multisection on $W|F$ has such a representative. *Restated*: the text takes good neighbourhoods inside F for every representative; the formalization shows that every class has a representative with them, which is all the gluing needs. *Not built*: condition (2) of Definition 13.1.4 inside the face, a natural representation of the ep-groupoid F itself with a connected domain inside F (Theorem 7.1.13, Part VII, for F). The natural action used is that of X restricted to $U(x) \cap F$, which is how the text describes it on p. 680. Nothing in §14.6 uses (2) (gap G.185, §XIV.29).

Definition 14.6.2 (same) `Chapter14/FaceStructured · SBLocalization.IsCompatibleAssignment`, p. 681: an assignment $F \mapsto \tilde{\Lambda}_F$, one structured sc$^+$-multisection on $W|F$ for each face, is compatible if $\tilde{\Lambda}_F|F' = \tilde{\Lambda}_{F'}|F$ as classes over $F \cap F'$ for all faces F, F'; the common class is the text's $\tilde{\Lambda}_{F\cap F'}$ `Chapter14/FaceStructured · SBLocalization.IsCompatibleAssignment.inter`. Equal classes have equivalent representatives `Chapter14/FaceStructured · SBLocalization.IsCompatibleAssignment.equiv`: the restrictions of the chosen representatives of $\tilde{\Lambda}_F$ and $\tilde{\Lambda}_{F'}$ to $F \cap F'$ are equivalent. The text draws two consequences, both proved from this: "$I_x^F = I_x^{F'}$" for $x \in F \cap F'$ `Chapter14/FaceStructured · SBLocalization.IsCompatibleAssignment.`

`index_eq`, since equivalent tuples have equal index sets; and $\Lambda_F = \Lambda_{F'}$ on $W|(F \cap F')$ `Chapter14/FaceStructured · SBLocalization.IsCompatibleAssignment.Λ_eq`, since equivalent tuples have equal functors. What the equivalence does not give, the equality of the sections, is the subject of the next subsection. *Witness* `Chapter14/FaceStructured · SBLocalization.isCompatibleAssignment_zero`: the zero multisection on every face, on any choice of neighbourhoods; all restrictions are the zero data over $F \cap F'$, equivalent with $O(x)$ the intersections of the two neighbourhoods.

XIV.28 Theorem 14.6.3 (2017: Theorem 14.4): gluing the faces, and the extension

Theorem 14.6.3 (narrower) `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.theorem_14_6_3_hyp`, p. 682. *Input* `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData`, the chosen representatives of the text's proof:

- one index family I_x, the text's "$I_x := I_x^F$";
- for every face F a face tuple $(\Lambda_F, \mathcal{U}_F, S_F, \tau_F)$ on this index family;
- $\Lambda_F = \Lambda_{F'}$ on $W|(F \cap F')$;
- $s^x_{F,i} = s^x_{F',i}$ on $U_F(x) \cap U_{F'}(x) \cap F \cap F'$, for $x \in F \cap F'$;
- $\tau_{F,x,x'} = \tau_{F',x,x'}$ on the morphisms between the sets $U_F(x) \cap U_{F'}(x) \cap F \cap F'$ and $U_F(x') \cap U_{F'}(x') \cap F \cap F'$.

Such data form a compatible assignment `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.isCompatibleAssignment`: the restrictions of the two tuples to $F \cap F'$ are equivalent, with $O(x) = U_F(x) \cap U_{F'}(x)$, equal index sets, and equal correspondences on the morphisms between these sets. *Statement*: for X tame and face-structured with $|X|$ paracompact, the glued function $\Lambda(w) = \Lambda_F(w)$ for $P(w) \in F$ `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.Λg` is structurable over ∂X and equals Λ_F over each face F `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.Λg_eq`; its structure restricts to the given ones.

Narrower: the input is representatives that agree, not a compatible assignment of classes. The text says that the compatibility gives $s^x_{F,i} = s^x_{F',i}$ on $U_F(x) \cap U_{F'}(x)$. As we read Definition 14.6.2, this does not follow: the equivalence of Definition 13.3.8, and its analogue here, compares Λ, the index sets and the correspondences on smaller neighbourhoods, not the sections, so two equivalent tuples may have different sections at the same index. *An illustration* (not formalized): on a face with trivial isotropy along which the only morphisms are identities, take $I_x = \{1, 2\}$ at every point, sections s^x_1, s^x_2 that agree across the neighbourhoods but $s_1 \neq s_2$, and correspondences that pair each index with itself. Exchanging s^x_1 and s^x_2 at every x gives a tuple with the same Λ (the counting formula is symmetric in the indices), the same index sets, and the same correspondences, which still satisfy (4), since the pair (i, i) now relates the exchanged sections, which again agree. The two tuples are equivalent, and their sections at the index 1 differ. Over one face such a difference can be undone by relabelling; the text's step asks that it can be undone on all faces through x at once, consistently with the correspondences. Presumably the text means that representatives can be chosen so that they agree; we neither prove this nor claim the contrary. The two routes to the class-level statement are gap G.173 (§XIV.29).

Proof, the text's, as the Lean proof goes.

1. *The glued functor.* $\Lambda(w) = \Lambda_F(w)$ for any face $F \ni P(w)$, and 0 off ∂X. It is well defined by the agreement of the Λ_F on $F \cap F'$, and defined on all of $W|\partial X$ because every boundary point lies

on a face (fact 2 of §XIV.27). It is a functor over ∂X: for $s(\varphi) \in F$, also $t(\varphi) \in F$ (fact 4), and Λ_F is a functor over F.

2. *A face selection.* For $x \in \partial X$ and objects y, x', choose a face through y, x and x' if there is one, and otherwise any face through x `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.faceSel`; when one exists, the chosen face contains all three `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.faceSel_spec`. The glued sections and correspondences are read from the selected face: $s_i^x(y) = s_{F,i}^x(y)$ with F selected for (x, y, x) `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.sg`, and $\tau_{x,x'}(\varphi) = \tau_{F,x,x'}(\varphi)$ with F selected for $(x, s(\varphi), x')$ `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.τg`. That the choice does not matter is proved in step 4.
3. *An adapted good system* `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.exists_adapted`, the text's choice of $V(x)$ with $U(x) \cap F \subset U_F(x)$. A good system $\mathcal{W}$ for ∂X is *adapted* `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.Adapted` if for every $x \in \partial X$ and $y \in U(x)$: every face through y contains x; and for every face $F \ni x$, $y \in U_F(x)$ and $\Gamma(g, y) = \Gamma_F(g, y)$ for all $g \in G_x$. *Construction*: (a) a good system $\mathcal{U}_0$ for ∂X from Theorem 14.1.6 (§XIV.4); (b) at each x, an open $O(x) \ni x$ on which every face through a point contains x (fact 3); (c) at each x, a neighbourhood $M(x) \subset O(x)$ with the natural action, below $U_0(x)$ and below each of the finitely many $U_F(x)$, $F \ni x$ (fact 2), with the restricted actions: the common refinement of finitely many natural neighbourhoods of x (Part XIII); (d) a refinement of $\mathcal{U}_0$ inside the $M(x)$ (§XIV.4). The natural action on $V(x)$ is that of $U_0(x)$, which on $M(x)$ is that of $M(x)$, which is that of each $U_F(x)$. Two consequences: on $U(x)$ the natural actions of $V(x)$ and $V_F(x)$ agree `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.Φ_eq`, as targets of the same morphisms $\Gamma(g, y)$; and $U(x) \cap F \subset U_F(x)$ for the faces $F \ni x$ `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.glue_U_inter_subset`.
4. *The glued data agree with the face data.* On $U(x) \cap F$, $s_i^x = s_{F,i}^x$ `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.sg_eq`: for $y \in U(x) \cap F$, F contains x (adapted), so a face through y and x exists and the selected face G contains both; $y \in U_G(x) \cap U_F(x) \cap G \cap F$ (adapted), where $s_{G,i}^x = s_{F,i}^x$ by the input. A morphism φ between $U(x)$ and $U(x')$ with source on F is one between $U_F(x) \cap F$ and $U_F(x') \cap F$ `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.mem_mor_face`: $t(\varphi) \in F$ (saturation), $x, x' \in F$ (adapted), and the traces lie in U_F (adapted). Hence on such morphisms $\tau_{x,x'} = \tau_{F,x,x'}$ `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.τg_eq`, by the input's agreement of the correspondences for the selected face and F.
5. *The symmetric structure on $U(x)$* `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.S`, with index set I_x and sections s_i^x.
   - *Face-wise sc⁺* (Definition 14.1.3 (1)), by the gluing criterion of §XIV.27: the values lie in $W_{0,1}$ because at y, on a face $F \ni y$, $s_i^x(y) = s_{F,i}^x(y)$, which is sc⁺ relative to F; and near y, on each face $F \ni y$, s_i^x agrees on $F \cap U(x)$ with $s_{F,i}^x$, which is sc⁺ relative to F at y.
   - *Equivariant*: for $y \in U(x)$ on a face F, also $g * y \in F$ (fact 4); both sides are read in F by step 4, the natural actions of $V(x)$ and $V_F(x)$ agree there by step 3, and S_F is equivariant.
   - *Represents* Λ: for $P(w) \in U(x) \cap F$, $\Lambda(w) = \Lambda_F(w)$, which is $|I_x|^{-1}$ times the number of i with $s_{F,i}^x(P(w)) = w$, the same as the number with $s_i^x(P(w)) = w$ by step 4.

   So the glued Λ is an sc⁺-multisection over ∂X `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.isBoundaryScPlusMultisection`, with the decoration $(\mathcal{W}, S)$ `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.dec`.
6. *The correspondences* `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.tau`, $\tau_{x,x'}(\varphi) = \tau_{F,x,x'}(\varphi)$ for $s(\varphi) \in F$.
   - *Locally constant*. Near φ, every face through the source of a nearby morphism contains $s(\varphi)$ (fact 3, pulled back by the continuous s). For each of the finitely many faces $F \ni s(\varphi)$, τ_F

is locally constant at φ. Intersecting these finitely many neighbourhoods, a nearby ψ in the morphism set has its source on some face G, which contains $s(\varphi)$, and $\tau(\psi) = \tau_G(\psi) = \tau_G(\varphi) = \tau(\varphi)$ by step 4.

- *Finitely many values*: every value is a value of some τ_F with F through x and x', a finite union of finite sets.
- *(a) Inversion*: $s(\varphi^{-1}) = t(\varphi) \in F$; read both sides in F.
- *(b) The diagonal*: $\Gamma(g, z)$ has source z; for z on F, $\Gamma(g, z) = \Gamma_F(g, z)$ by step 3, and τ_F has (b).
- *(c) The bi-action*: the source of $g' * \varphi * g$ is $\Phi(g^{-1})(s(\varphi))$, on F by fact 4. The bi-action computed with the natural actions of $V(x)$, $V(x')$ equals that computed with those of $V_F(x)$, $V_F(x')$, by step 3; then τ_F has (c).
- *(4) Compatibility*: read in the face of $s(\varphi)$, with $s_j^{x'}(t(\varphi)) = s_{F,j}^{x'}(t(\varphi))$ and $s_i^x(s(\varphi)) = s_{F,i}^x(s(\varphi))$ by step 4.

7. *The structured tuple* `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.glue`: $(\Lambda, \mathcal{W}, S, \tau)$, so Λ is structurable over ∂X. It restricts to the face data: $s_i^x = s_{F,i}^x$ on $U(x) \cap F$ `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.glue_s`, and $\tau_{x,x'} = \tau_{F,x,x'}$ on the morphisms of $\mathcal{U}(x, x')$ with source on F `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.glue_τ`.

Witness `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.zero`: the zero multisection on every face, on the index family $I_x = \{*\}$; all agreements are equalities of zero data, and the glued Λ is the zero multisection over ∂X `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.zero_Λg`. It charges every boundary fibre (the zero vector has weight 1), so it also satisfies the charging hypothesis of the extension part below. The witness is degenerate in the sections, all zero, but not in the boundary: tame face-structured ep-groupoids with $|X|$ paracompact and $\partial X \neq \emptyset$ exist (Part XI).

The extension part (narrower) `Chapter14/BoundaryExtensionTheorem · SBLocalization.CompatibleFaceData.theorem_14_6_3_extension_a6`. *Statement*: let X be face-structured (hence tame) with $|X|$ paracompact, with sc-smooth partitions of unity on the object M-polyfold and a tame strong bundle chart at every object, and let face data with agreeing representatives be given whose glued Λ charges every boundary fibre. For N, $\tilde{U}$, f as in Theorem 14.1.9 (with the boundary domain support of the glued Λ), there is a structurable Λ' on W with $N(\Lambda') \leq f$, dom-supp$(\Lambda') \subset \tilde{U}$, and $\Lambda' = \Lambda_F$ over every face F. *Proof*: 1. The glued Λ is structurable over ∂X (above). 2. Theorem 14.1.9 with charging (§XIV.7, proved in §XIV.25) gives Λ' with the size and support properties and $\Lambda' = \Lambda$ over ∂X. 3. For $P(w) \in F$, $P(w) \in \partial X$ (fact 1), so $\Lambda'(w) = \Lambda(w) = \Lambda_F(w)$. *Narrower*: the agreeing representatives, and the two readings of Theorem 14.1.9 (§XIV.7). Charging is necessary here too, since the conclusion makes Λ the restriction of a structurable functor (§XIV.7).

Remark: the form with the uncharged statement. A form taking the uncharged Lean statement of Theorem 14.1.9 as a hypothesis `Chapter14/FaceAssembly · SBLocalization.CompatibleFaceData.theorem_14_6_3_extension_hyp` remains in the formalization, with the same proof. That hypothesis holds only when $\partial X = \emptyset$ (§XIV.7), in which case there are no faces and the statement says nothing; the form above does not use it.

XIV.29 What is not formalized

Every numbered statement of Chapter 14 is formalized, either as printed or in a form recorded in its subsection. What remains open is listed here by gap number, with what is missing and an estimate in lines of Lean. None of the open gaps lies on the path of Theorem 14.1.9: the theorem, with its two readings, is proved from the construction of §§14.2–14.5, and its extension part over the faces from Theorem 14.6.3

for agreeing representatives.

- **Gap G.172**, the multisection form of Lemma 14.5.2 (formalized in part, §XIV.22). *Formalized*: the identity for the summed sections, $\sum_a \tilde{s}^a_{b,j} = s^{0b}_j$ over ∂X, which is the only use the proof of Theorem 14.1.9 makes of the lemma. *Missing*: that the structured functors $\Lambda_a = (\beta_a|\partial X) \odot \Lambda$ form a commensurable, locally finite family over ∂X whose sum is Λ. This is a boundary version of the commensurable sums of Part XIII (Definitions 13.3.11 and 13.4.5): the index sets and correspondences of Λ would be carried to each Λ_a, the commensurability checked on the common good system of §XIV.21, and the sum computed fibrewise from the section identity. About 120 lines. Nothing formalized uses it. Remark 14.5.3, which explains the lemma, is prose.
- **Gap G.173**, Theorem 14.6.3 from a class-level compatible assignment (§XIV.28). *Formalized*: faces as Part XI's ep-faces with their local properties; sc$^+$ relative to a face and on a corner; Definitions 14.6.1 and 14.6.2, with the two consequences the text draws; good systems on a face up to equivalence; Theorem 14.6.3 for face data whose chosen representatives agree on the intersections of faces; and its extension part, from Theorem 14.1.9 with charging. *Missing*: the passage from a compatible assignment of classes, $\tilde{\Lambda}_F|F' = \tilde{\Lambda}_{F'}|F$, to representatives that agree. The equivalence of structured multisections on a face compares Λ, the index sets and the correspondences on smaller neighbourhoods, not the sections, so two routes are open. (a) *A re-choice of representatives*: near each $x \in \partial X$, relabellings $I^{F'}_x \to I^F_x$ compatible with the correspondences and with the G_x-action, carrying the sections of F' to those of F on $F \cap F'$, chosen consistently over the finitely many faces through x (at a corner of index d the d relabellings must commute) and locally constant in x; possibly with a hypothesis on the local constancy of the relabelling where two sections cross. About 350–550 lines. (b) *An example*: two equivalent representatives over a corner whose sections differ by an index swap, such that no re-choice agrees on two faces at once, which would show that the text's step needs the agreeing representatives as an assumption. About 150–250 lines. We do not know which of the two holds.
- **Gap G.185**, condition (2) of a good system on a face (formalized in part, §XIV.27). *Formalized*: every structured sc$^+$-multisection on $W|F$ has a representative whose neighbourhoods $U_F(x)$ satisfy (1), the properness of t over the closure, and (3), the component condition, inside the face ep-groupoid, and are invariant under the natural action of X restricted to $U(x) \cap F$. *Missing*: condition (2) of Definition 13.1.4 in the strict sense, a natural representation of the ep-groupoid F itself with a connected domain inside F (Theorem 7.1.13, Part VII, applied to F), and its identification with the restricted action of X near x. About 150–250 lines. Nothing in §14.6 uses it: the gluing of Theorem 14.6.3 reads the neighbourhoods $U_F(x)$ as $U(x) \cap F$, as the text explains them on p. 680.
- **Gap G.186**, the readings of the standing hypotheses of Theorem 14.1.9 (§§XIV.3, XIV.7, XIV.26). (a) *Non-empty index sets in the formal Definition 14.1.3.* The theorem is stated with charging, which for a structured Λ is the same as $I_x \neq \emptyset$ at every boundary point, and which is necessary. Building $I_x \neq \emptyset$ into the definition itself is an edit of an existing definition, with about 20–60 lines of follow-up where structures with empty index sets serve as witnesses; charging would then follow from structurability. (b) *A witness with $\partial X \neq \emptyset$* in which all hypotheses of the theorem hold jointly, tame strong bundle charts included: done, the half-space bundle with a constant non-zero section (§XIV.26). (c) *The tame strong bundle charts at every object derived from the tameness of X*, if that holds: a strong bundle over a tame ep-groupoid would be shown to have, at every object, a strong bundle chart whose base retraction is tame, by Part I's construction of tame retractions applied to the base of a strong bundle retraction and lifted to the bundle. About 300–600 lines.
- **Lemma 14.4.13 (2) at every point of** $\mathrm{cl}_X(\widehat{V}_a) \cap \partial X$ (§XIV.19). Proved for points of $\mathrm{cl}_X(V_a)$, which

is what Proposition 14.4.14 uses. The printed range would need a coherent system with (4) at the other points, which the stronger Lemma 14.4.9 (3) of §XIV.17 permits. About 90 lines; not needed.

- **Prose.** Remarks 14.1.10, 14.3.5, 14.4.1, 14.4.12 and 14.5.3 are explanatory and have nothing to formalize beyond what their subsections record.

Closed gaps, for reference: G.164, Proposition 14.1.8 over ∂X (§XIV.6); G.165, the faces $\bar{\theta}_j$ as tame M-polyfolds (§XIV.1); G.168, Proposition 14.4.6 (2) in general (§XIV.16); G.169, the setting of §14.4 from the objects of §14.3 (§XIV.13); G.170, the proof of Proposition 14.5.4 (§XIV.23); G.171, the data of §§14.3–14.5 for every charging Λ (§XIV.25); G.184, sc$^+$ on the corners $F \cap F'$ (§XIV.27).

XIV.30 What the formalization taught us

No statement of Chapter 14 was found not to hold as we read it. Two statements are proved with readings that are hypotheses of their Lean form (Theorem 14.1.9: tame strong bundle charts and charging; Theorem 14.6.3: agreeing representatives), and one has a printed range wider than its proof covers (Lemma 14.4.13 (2)). The rest is proved as printed, or in a stronger form.

Two kinds of formal content. Chapter 14 is the longest construction of the book, and what it needs splits in two.

The first kind is *local topology of the morphism space*: shrinking good neighbourhoods so that closures stay inside (Lemma 14.2.1, §XIV.8), separating the values of a locally constant correspondence by open sets (Lemma 14.3.1, §XIV.9), extending it by components (Proposition 14.3.4, §XIV.10), and choosing neighbourhoods compatible with the action and with local finiteness (Lemmas 14.3.6, 14.3.7, Proposition 14.4.10, §§XIV.11, XIV.18). All of it rests on three facts used again and again. First, t is proper on s^{-1} of the closure of a good neighbourhood, so a property of the morphisms from a point persists near it. Second, s and t are local homeomorphisms on sheets, so the sheet through a morphism is locally constant. Third, a locally constant function on an open set is constant on its components. The diagonal property of Proposition 14.3.4, for which the text gives no argument, is the second fact applied through sheet labels: the component of $\Gamma(g, y)$ stays in the sheet of g and reaches the boundary there (§XIV.10). The inversion symmetry of the extended correspondences, which the text does not state, costs one more intersection in Lemma 14.3.1 and is what the z-loops of §14.4 need to be reversible (§§XIV.9, XIV.14).

The second kind is *combinatorics of index sets along chains of morphisms*. Encoding a z-loop together with its arc as one inductive predicate, built by appending one morphism and one index step at a time, made the equivalence-relation properties of $\sim_z$, which the text asserts, and Lemma 14.4.5 short (§XIV.14). It made Proposition 14.4.6 (1) a deformation of the chain along sheets, one morphism at a time, without Lemma 14.4.7 (§XIV.15). It carried the limit argument of Proposition 14.4.6 (2): with Lemma 14.4.7 around the finitely many isomorphs of the limit point, each step of a nearby loop moves along its own sheet to the boundary, so the text's limit loop need not be built, and the result holds for loops at points of $\hat{V}_a$, the form Lemma 14.4.9 (3) uses (§XIV.16). It also carried Proposition 14.5.4: defining the transported section through chains rather than single morphisms gives its equivariance and its compatibility with $\hat{\tau}$ directly, and its well-definedness is the text's closing of two chains into a loop (§XIV.23).

Steps that needed less than printed.

- *Theorem 14.5.5* needs neither its properness nor its closure condition: local transports into the sets V_a and the conjugation homomorphisms of Theorem 13.5.3 (Part XIII) carry the structure (§XIV.24).

- *Lemma 14.5.1* does not use tameness, and (14.53) holds for the cover of §14.3 itself, without the shrinking the text performs: a morphism from an invariant subset of a natural neighbourhood to a point of its closure ends in that subset, because the isotropy group is finite and the morphism space Hausdorff (§XIV.21).
- *Lemma 14.5.2* enters the proof of Theorem 14.1.9 only through its identity for the summed sections; the commensurable family over ∂X is not needed (§XIV.22, gap G.172).
- *Proposition 14.1.8* holds along generalized strong bundle isomorphisms and their classes without the paracompactness of the orbit spaces (§XIV.6).
- *Definition 14.4.11 (2)* is not needed once Proposition 14.4.14 is proved by averaging (§§XIV.18, XIV.20).
- *Proposition 14.4.6 (1)* does not need Lemma 14.4.7 (§XIV.15).

Steps that needed more, or a different argument.

- *Proposition 14.1.4.* The local extension cannot be glued with one partition of unity of X, because the boundary data on an open $V(x)$ may be unbounded near its frontier. An exhaustion of $V(x)$ by the sets at distance $> 1/n$ from its complement, glued by sc-smooth cut-offs, does it (§XIV.3).
- *Averaging.* Averaging over the whole isotropy group, $\bar{s}_j(y) = |G_x|^{-1} \sum_g \mu(\Gamma(g^{-1}, g*y), s_{g(j)}(g*y))$, replaces the text's average over a stabilizer together with transport along an orbit, both in Proposition 14.1.4 and in Proposition 14.4.14. In Proposition 14.4.14 it also replaces the class-by-class, orbit-by-orbit construction with an invariant partition of unity (§§XIV.3, XIV.20).
- *Theorem 14.1.6* is proved from tameness of X and paracompactness of $|X|$, which its proof uses, in place of the paracompactness of $|\partial X|$ printed in the statement (§XIV.4).
- *The size control of Theorem 14.1.9* hides one choice: the extension must be taken with support in $\{f > 0\}$, not merely in $\tilde{U}$, or the cut-off cannot reach the boundary points where $f = 0$; and the inequality $N(\Lambda) < f$ is needed, and assumed, on the boundary domain support rather than on all of ∂X (§§XIV.7, XIV.25).
- *Proposition 14.5.4* needs one property of the cut-offs that its statement does not list, that β_a vanishes off the saturation of $\hat{V}_a$; Lemma 14.5.1 (1) gives it (§XIV.23).
- *Index sets on two domains.* The index sets of §14.1 live on boundary points and those of §14.4 on all objects; a map to boundary points that is the identity on ∂X, with the identity on morphisms as a homomorphism of isotropy groups, joins them (§XIV.13).
- *Proposition 14.1.8 over ∂X* carries everything along except Definition 14.1.3 (1), which is read in every tame strong bundle chart; its transport needs that tame charts compose with local strong bundle isomorphisms, a fact Part I proved for Theorem 3.6.8 (§XIV.6).
- *Sc⁺ on corners.* The restriction of Definition 14.6.1 is sc⁺ on the corner $F \cap F'$ because, in a tame model, the corner germ $\{t_i = t_j = 0\}$ is itself an sc-retract by Part I's corner retraction, and an sc⁺ map on a face germ restricts to it (§XIV.27).

Readings. Each is recorded in its subsection; together:

- Lemma 14.2.2: the morphism sets $\mathcal{U}(x, x')$ are read with $t(\varphi) \in U(x')$, as in (P3), where the text prints $U(x)$ (§XIV.8).
- Definition 14.4.8: the last index of an $\approx_z$-chain is i_k (§XIV.17).
- Proposition 14.4.6 (2): the loops are taken at points $z_k \in \hat{V}_a$ (§XIV.16).
- Definition 14.4.11 (3): read for the closure, as Lemma 14.4.9 provides it (§XIV.18).
- Lemma 14.4.13 (2): stated for $z \in \mathrm{cl}_X(V_a)$, the range Definition 14.4.11 (4) covers and Proposition 14.4.14 uses (§XIV.19).

- Definition 14.1.3 (1): "$s|\bar{\theta}_j$ is sc$^+$" read through the tame strong bundle charts of X, on every local face germ (§XIV.2); likewise sc$^+$ relative to a face and on a corner (§XIV.27).
- Theorem 14.1.9: "a strong bundle over a tame ep-groupoid" read as tame strong bundle charts at every object; the factor $|I_x|^{-1}$ read as charging; "supported in $\tilde{U}$" read as $\mathrm{cl}\{f \neq 0\} \subset \tilde{U}$ (§XIV.7).
- §14.6: "good neighbourhood $U_F(x) \subset F$" read as $U(x) \cap F$ with the restricted natural action (§XIV.27); Theorem 14.6.3 read with representatives that agree on $F \cap F'$ (§XIV.28).

A property of our own formalization. Definition 14.1.3 (3) writes $\Lambda(w) = |I_x|^{-1} \cdot \#\{i : s_i^x(P(w)) = w\}$, which presumes $I_x \neq \emptyset$. The formal definition does not ask for it, so the zero function, with empty index sets, counts as structurable over ∂X, and no structurable extension can vanish over a boundary fibre. A Lean statement of Theorem 14.1.9 without the charging condition, and every hypothesis that asks for its conclusion for all Λ, therefore holds only when $\partial X = \emptyset$ (§XIV.7); witnesses on spaces without boundary cannot reveal this, since every condition on ∂X is then vacuous. Two lessons follow for the formalization rather than for the text. A witness on a space without boundary says nothing about a statement on ∂X; the half-space bundle (§XIV.26) is the witness that does. And a statement about structurable functors should be checked against the zero function with empty index sets, which the formal definitions admit and the printed formulas exclude. Stated with charging, the theorem is proved, and charging is necessary: the theorem is an equivalence (§XIV.7).

§14.6. The step from compatible classes to agreeing representatives is where the text is shortest. The equivalence of structured multisections compares Λ, the index sets and the correspondences, not the sections, so the text's identification $s^x_{F,i} = s^x_{F',i}$ is a choice of representatives, which Theorem 14.6.3 takes as its input (§XIV.28, gap G.173). Everything else in §14.6 is local: faces are closed and saturated, every boundary point lies on finitely many, near a point every face through a nearby point contains it, and a local face of a tame chart lies in one face (§XIV.27). These four facts, with the gluing criterion for sc$^+$ on faces, are the whole proof of the gluing.

How the chapter is used later. Theorem 14.1.9 is the tool by which the transversality results of Chapter 15 and the cobordism results of Chapter 16 control a multisection near the boundary. Theorem 15.3.9 (Part XV) uses it in its charged form, with the same two readings, tame strong bundle charts at every object and charging; the cut-off step of its proof uses the extension of face-wise sc$^+$ sections of Proposition 14.1.4 (§XIV.3) and Definition 14.1.3 (1) read through the charts (§XIV.2). Theorem 16.5.7 (Part XVI) uses it in one representative of a strong polyfold bundle, and the transport between representatives is Proposition 14.1.8 over ∂X (§XIV.6). In both places the remark of §XIV.7 has a counterpart: a Lean statement that takes the uncharged form of Theorem 14.1.9 as a hypothesis, or asks its conclusion for every structurable Λ, holds only without boundary, and the charged form replaces it. What these later chapters need from Chapter 14 is therefore exactly what is proved here: the charged theorem, its necessity, the transport over ∂X, and the half-space witness with boundary.

A note on how the proof is organized. The text proves Theorem 14.1.9 in one pass, introducing the objects of each section from those of the previous one. The formalization splits each step into an abstract statement over a structure that names its inputs, and an instance built from the objects of the construction (§§XIV.13, XIV.25). The split made three things visible. First, which properties of the previous objects each step actually uses: Theorem 14.5.5 uses neither properness nor closures, Lemma 14.5.1 not tameness, Proposition 14.4.14 not Definition 14.4.11 (2). Second, where a step uses a property its statement does not list: Proposition 14.5.4 and the support of the cut-offs. Third, where each standing hypothesis enters: paracompactness of $|X|$ through the good systems of Theorem 14.1.6 and Part XIII and through the size control; tameness through Proposition 14.1.1 and the faces; the sc-smooth partitions of unity and the

tame strong bundle charts through the local extensions of Propositions 14.1.4 and 14.4.14, the charts also through Definition 14.1.3 (1) and its transport (§XIV.6). Each of these is named in the subsection where it is used.

Part XV. Transversality and invariants (HWZ Chapter 15, §§15.1–15.4)

Chapter 15 (Springer 2021, pp. 683–703; the 2017 numbers in parentheses) is the perturbation theory of an sc-Fredholm section functor f of a strong bundle $(P : W \to X, \mu)$ over an ep-groupoid by sc$^+$-multisections Λ, and the invariants obtained from the perturbed solution sets $\Lambda \circ f$. §15.1 defines the linearization $T(f, \Lambda)$ of $\Lambda \circ f$, a section functor with values in the formal sums GrF of sc-Fredholm operators, and proves that it is well defined through a separation argument for finitely many sections (Theorem 15.1.1, Remark 15.1.2, Proposition 15.1.3, Lemmas 15.1.4, 15.1.5, Remark 15.1.6, Definition 15.1.7 (2017: Theorem 15.1, Remark 15.1, Proposition 15.1, Lemmas 15.1, 15.2, Definition 15.1; Remark 15.1.6 is new in 2021)). §15.2 defines transversality and good position and describes the local solution sets, which make $\Lambda \circ f$ a branched ep$^+$-subgroupoid (Definitions 15.2.1, 15.2.4, Theorems 15.2.2, 15.2.5, Remarks 15.2.3, 15.2.6). §15.3 controls compactness by an auxiliary norm, defines admissible perturbations and general position, and constructs perturbations: Theorem 15.3.7 (general position), Theorem 15.3.9 (transversal extension of boundary data) and Corollary 15.3.10 (regular perturbations exist, and any two are joined over the cylinder $[0, 1] \times X$). §15.4 orients $\Lambda \circ f$ by an orientation of f (Theorem 15.4.1, Definition 15.4.2) and integrates sc-differential forms over it: Stokes' theorem (Theorem 15.4.3) and the invariant on de Rham cohomology (Corollary 15.4.4). This Part retells the formal proofs in the book's order: §15.1 in §§XV.1–XV.11, §15.2 in §§XV.12–XV.16, §15.3 in §§XV.17–XV.33 (the definitions in §§XV.17–XV.20, Theorem 15.3.7 in §§XV.21–XV.26, Theorem 15.3.9 in §§XV.27–XV.29, Corollary 15.3.10 in §§XV.30–XV.33), §15.4 in §§XV.34–XV.37; then what is not formalized (§XV.38) and what the formalization taught us (§XV.39).

What is proved, and what is not. *§15.1. Theorem 15.1.1* (narrower) is proved in its substance in a stronger form: the formal sum $\sum_{f(x)=s_i(x)} \sigma_i \cdot (f - s_i)'(x)$ does not depend on the local sc$^+$-section structure or on the chart, at every level-1 point, for any sc-smooth f, with no use of positivity of the weights or of the functoriality of Λ (§§XV.2, XV.7); the value $T(f, \Lambda)(x)$ is Definition 15.1.7 (§XV.8); the functor property $T(f, \Lambda)(t(\varphi)) = \varphi_* T(f, \Lambda)(s(\varphi))$ holds at all level-1 points, with no added hypothesis (§XV.9, gap G.144 closed). The row is narrower for one reason: the summands are proved sc-Fredholm in the chart reading and for f with Part I's tame germ data (§XV.11, gap G.145 closed). *Remark 15.1.2* and *Definition 15.1.7* are restated (§§XV.2, XV.8). *Lemmas 15.1.4 and 15.1.5* are proved in a stronger, purely topological form (§§XV.4, XV.5), *Remark 15.1.6* with "slightly" made explicit (§XV.6). *Proposition 15.1.3* is proved with the added hypothesis that the base point is not isolated; at an isolated point, for instance for the one-point local model of a zero-dimensional X, it does not hold as we read it, since no sequence $x_k \neq x_0$ converges to x_0. The case is degenerate and does not reach Theorem 15.1.1, whose formal proof does not use the proposition (§§XV.3, XV.7). The proof of Theorem 15.1.1 is read in two places: the separation and the limit run in the level-1 topology, the "weak topology" being pointwise convergence of operators; and

the final sum is presumably over the i with $f(0) = s_i(0)$ (§XV.7).

§15.2. Definitions 15.2.1 and 15.2.4 are restated, the first with "nonzero" and "sc-Fredholm" automatic (§§XV.12, XV.15). Theorems 15.2.2, 15.2.5 and Remark 15.2.3 (narrower) are proved with Part I's tame germ data for f, inherited from Part I's Theorem 3.6.8, on which they rest; this is the only hypothesis they add. Their "tame" submanifolds and tame branched ep$^+$-subgroupoids (Definition 9.3.6) are proved (§§XV.14, XV.16, gap G.150 closed). "The ep-groupoid $U(x)$" of Remark 15.2.3 is read as the translation groupoid $G_x \ltimes U(x)$; on it the statements hold as printed, the paracompactness of the support included (§XV.14, gap G.152 closed). Their global forms on X take a paracompact orbit space of the support, Definition 9.1.2 (1), which is new in 2021 and which the remarks do not address. The closing sentence of Theorem 15.2.5, under general position, is proved for tame X (§XV.16, gap G.151 closed). Remark 15.2.6 is prose.

§15.3. The paragraph before Definition 15.3.1, Definition 15.3.3 with its example, and the first sentence of Remark 15.3.6 are proved as stated; Definitions 15.3.1 and 15.3.5 are restated (§§XV.17, XV.18, XV.20). The rest of Remark 15.3.6 is proved at every point of the support, not only on ∂X, with the inherited tame germ data and, for the branched ep$^+$-subgroupoid, a paracompact support (§XV.20, gap G.161 closed). The products remark after Definition 15.3.3: product ep-groupoids, product strong bundles, products of sections and section functors and the auxiliary norm $\max(N, N')$ are formalized; $f \times f'$ sc-Fredholm and $\Lambda \times \Lambda'$ (Remark 15.3.4) are not (§XV.19, gap G.160, in part). *Theorem 15.3.7* (explicit) is proved with the text's hypotheses, the inherited tame germ data being the only one not printed, with no good system and no paracompactness of $|X|$, as Remark 15.3.8 says; the addendum "structurable" needs a good system, which is necessary for it and which exists for paracompact $|X|$; its "in particular" holds with "tame" (§§XV.21–XV.26; gap G.162 closed for Theorem 15.3.7). *Theorem 15.3.9* (narrower) is proved in the charged form, the boundary multisection having non-empty index sets, which the conclusion forces, from tame strong bundle charts and one open proposition, the relative local families of steps (c)–(d) (gap G.178); its face condition is asked for every sc$^+$ extension (gap G.179); step (a) is proved (gap G.177 closed), step (b) is Theorem 14.1.9 in its charged form (Part XIV), and the Sard and compactness parts of steps (c)–(d) are proved (§§XV.27–XV.29). Its witnesses have no boundary and zero fibres. *Corollary 15.3.10* (1) follows from Theorem 15.3.7 for paracompact $|X|$, with bump functions and the tame germ data, the tameness of X derived from $d_X \equiv 0$ (§§XV.30, XV.31, gap G.183 closed); (2) is proved with an explicit interpolated auxiliary norm (§XV.31); (3) is reduced to the charged Theorem 15.3.9 on $[0, 1] \times X$, with the partitions of unity of the cylinder proved from those of X (gap G.181 closed), the structurability of the glued boundary data proved given that lifted sc$^+$-sections are sc$^+$ on the faces (gap G.193), and the face condition of the glued data and tame strong bundle charts on the cylinder as hypotheses (gaps G.180, G.189) (§§XV.32, XV.33). The last paragraph of the proof of the corollary is proved as printed (§XV.33). Remarks 15.3.2 and 15.3.11 are prose.

§15.4. Theorem 15.4.1 and Definition 15.4.2 (narrower) are proved at a single point, in Part I's chart reading: the operators $(f - s_i)'(y)$ lie in Part I's $\mathrm{Lin}(f, y)$, the canonical lift exists, has the text's formula, is unique and does not depend on the local structure, and the kernel orientations are defined; their assembly into an orientation of $\Lambda \circ f$ is open (§XV.34, gap G.159, in part). Theorem 15.4.3 (narrower) is proved with the orientation as data, by Part IX's Stokes theorem (§XV.35). Corollary 15.4.4 (narrower) is proved by the text's cobordism argument from oriented cobordisms between regular perturbations; on the cylinder the ends and the splitting of the boundary measure are proved, and the oriented end identities are a hypothesis (gaps G.182, G.190); the effective version adds the local unique continuation property on the cylinder (gap G.191) (§§XV.36, XV.37). Remark 15.4.5 is prose.

Open: G.159 and G.160 in part; G.162 for Theorem 15.3.9 and Corollary 15.3.10 (3); G.178–G.180, G.182, G.189–G.191, G.193 (§XV.38). *Closed:* G.144, G.145, G.150–G.152, G.161, G.177, G.181, G.183. Proposition 15.1.3 at an isolated point is the one statement of the chapter that does not hold as we read it, a degenerate case; no statement of §§15.2–15.4 was found not to hold as we read it.

The decisions that shape the chapter. (1) *Formal sums are finitely supported functions.* $\mathrm{GrF}(x)$ is the type of finitely supported functions from the linear maps $T_xX \to W_x$ to $\mathbb{Q}^{\geq 0}$, defined at every level-1 point `Chapter15/LinearizationIntrinsic · SBLocalization.GrF`; that the operators are sc-Fredholm is proved for the operators that occur, not built into the type. On a strong bundle model the operators are $T_zO \to F_0$ `Chapter15/LinearizationModel · Linearization15.GrFModel`. (2) *The separation argument is pure topology.* Proposition 15.1.3, Lemmas 15.1.4, 15.1.5 and Remark 15.1.6 are proved for finitely many continuous maps into a Hausdorff space and applied on the level-1 subspace of the model, where the sections and their derivatives are continuous. (3) *Pairs, not operators.* The proof of Theorem 15.1.1 compares finite formal sums of pairs (value, derivative) over all branches, $\mathrm{wsum}(a,\sigma)(q) = \sum_{i:a_i=q}\sigma_i$ `Chapter15/SeparationLemmas · Linearization15.wsum` with masses of sets `Chapter15/SeparationLemmas · Linearization15.mass` `Chapter15/SeparationLemmas · Linearization15.mass_eq_of_wsum_eq`; the sum of the theorem is the part with first entry $f(y)$. (4) *A branch is a section problem of Part I.* A local sc$^+$-section s_i on an open O is a global sc$^+$-section of the restricted bundle $W|_O$, and $(f-s_i)|_O$ is an sc-Fredholm section there (§XV.10); every analytic statement of §§15.2–15.4 is Part I's theorem applied branch by branch. The price is Part I's hypothesis: *tame germ data* for f, inherited from Part I's Theorems 3.1.18 and 3.6.8, which are proved for it; with the untamed notion of 2021 Theorem 3.6.8 does not hold as we read it (Part III). (5) *Operators are read in charts.* T_xX and W_x carry no sc-structure in the project; "sc-Fredholm", "onto on T^R_xX" and "$\in \mathrm{Lin}(f,y)$" are read in a strong bundle chart, through the identifications that define the intrinsic operators, and shown independent of the chart where it matters. (6) *The cylinder is a product with Part I's cube.* $[0,1]\times X$ is built as $[0,1]^n \times X$ with $n=1$, so that Part I's parametrized Fredholm theory applies to $\tilde{f}$ (§XV.30). (7) *Boundary data are charged.* A multisection over ∂X is taken with non-empty index sets, as the text's "sc$^+$-multisection over ∂X" has them; Theorem 15.3.9 needs it, and the uncharged Lean statement does not hold when $\partial X \neq \emptyset$ (§XV.27). (8) *Open steps are named propositions.* Where a theorem rests on a step that is not proved, the step is a named statement (for instance the relative local families of Theorem 15.3.9, the oriented end identities of Corollary 15.4.4), taken as a hypothesis with a witness, and the theorem is proved from it.

Map.

- *The linearization (§15.1).* `Chapter15/SeparationLemmas`: formal sums of points, Proposition 15.1.3 with its witness and the isolated point, Lemmas 15.1.4, 15.1.5, Remark 15.1.6, the limit step and the core lemma (§§XV.1, XV.3–XV.7). `LinearizationModel`: GrF on a strong bundle model, level-1 continuity and locality of derivatives, Theorem 15.1.1 and Remark 15.1.2 on the model (§§XV.2, XV.7). `Linearization`: principal parts and counts in a strong bundle chart, Theorem 15.1.1 in a chart, the chart form of Definition 15.1.7 (§§XV.7, XV.8). `LinearizationIntrinsic`: intrinsic operators, the change of chart, Theorem 15.1.1 with intrinsic operators, Definition 15.1.7 (§§XV.1, XV.7, XV.8). `LinearizationFunctor`: the action of morphisms on GrF and the functor property (§XV.9).
- *Branches as section problems, and §15.2.* `LocalScPlus`, `LocalIFT`: local sc$^+$-sections on $W|_O$, the sections $f-s_i$, the chart linearization, local solution sets (§XV.10). `FredholmSummands`: the sc-Fredholm summands (§XV.11). `Transversality`: Definition 15.2.1, Theorem 15.2.2, Remark 15.2.3 (§§XV.12–XV.14). `BoundaryPerturbation`: Definition 15.2.4, Theorem 15.2.5 (§§XV.15, XV.16). `InteriorPerturbationWitness`, `GoodPositionWitness`, `TameWitness`: witnesses for §15.2. `Tame–`

`Branched`: tame local solution sets and the tame forms of Remarks 15.2.3, 15.3.6, Theorems 15.2.5, 15.3.7 (§§XV.14, XV.16, XV.20, XV.21). `RestrictedEp`: the translation groupoid $G_x \ltimes U(x)$ and the statements on it (§§XV.14, XV.16). `GeneralPositionLocal`: general position at a point, its relation to good position and to Part I, Theorem 15.2.5 under general position (§XV.16).

- *Definitions of §15.3 and products.* `ControllingPairs`: Definitions 15.3.1, 15.3.3, 15.3.5, the paragraph before Definition 15.3.1, the example after Definition 15.3.3, the first sentence of Remark 15.3.6 (§§XV.17, XV.18, XV.20). `ProductEp`: product ep-groupoids (§XV.19). `ProductStrongBundle`, `ProductBundle`, `ProductBundleLevels`, `ProductBundleMap`, `ProductEpBundle`, `ProductAuxNorm`: product strong bundles, their levels and sections, the product over $X \times X'$, $\max(N, N')$ (§XV.19). `GoodPositionBranches`, `GeneralPositionBranched`: the rest of Remark 15.3.6, the statement of Theorem 15.3.7 and its "in particular" (§§XV.20, XV.21).
- *Theorem 15.3.7.* `ParamFamily`: linear families, (15.1), sums of families, norm and support bounds (§XV.22). `GeneralPositionAssembly`: the assembly from the local families and the parametric step (§XV.22). `ParamOntoTransport`, `LocalFamilies`, `LocalSeeds`: (15.1) with vectors and along morphisms, seeds, the local families (§XV.23). `ParamOntoBridge`: (15.1) and Part I's conditions (i)–(ii) (§XV.24). `LocalSlice`, `BranchChart`, `ParametricGeneralPosition`: the local Sard step, a branch as a Part I parametrized section, achieving global transversality (§XV.25). `GeneralPositionPointwise`, `GeneralPositionGoodSystem`, `GeneralPositionExistence`: the theorem with the text's hypotheses, the structurable form, Remark 15.3.8 (§XV.26).
- *Theorem 15.3.9.* `TransversalExtension`: the face condition and the reduction to the steps of the proof (§§XV.27, XV.29). `CutoffFunctor`: step (a) (§XV.28). `RelativeFamilies`, `RelativeBranch`, `RelativeParametric`, `RelativeReduction`: relative (15.1), relative families, the relative parametric step (§XV.29). `ChargedExtensionFibres`: the charged statement and the theorem (§§XV.27, XV.29). `ChargedExtensionCylinder`: the uncharged Lean statement with boundary (§XV.27) and Corollary 15.3.10 (3) with the charged theorem (§XV.33).
- *The cylinder and Corollary 15.3.10.* `CubeEp`: $[0,1]^n \times X$, $\widetilde{W}$, $\tilde{f}$ (§XV.30). `Chapter5/DegeneracyZeroTame`: $d_X \equiv 0$ implies tame (§XV.30). `CubeAuxNorm`: Corollary 15.3.10 (2) (§XV.31). `RegularPerturbations`, `Cylinder`: regular perturbations, Corollary 15.3.10 (1), the orbit space of the cylinder (§§XV.30, XV.31, XV.33). `CylinderGlue`: the glued boundary data, the last paragraph of the proof (§§XV.32, XV.33). `Chapter5/CubePartitions`, `CylinderPartitions`: partitions of unity of the cylinder (§XV.32). `CubeLift`, `GlueStructurable`: lifted structures and the structurability of the glued data (§XV.32).
- *§15.4.* `OrientedLinearizations`, `CanonicalLift`: oriented formal sums, the canonical lift, Theorem 15.4.1 and Definition 15.4.2 at a chart point (§XV.34). `PerturbedStokesInvariants`: Theorem 15.4.3, the cobordism argument, Corollary 15.4.4 (§§XV.35, XV.36). `CylinderEnds`, `CylinderCobordism`: the ends of the cylinder, the boundary measure, Corollary 15.4.4 on the cylinder and its effective version (§XV.37).

Throughout, $(P : W \to X, \mu)$ is a strong bundle over an ep-groupoid, f an sc-Fredholm section functor, and Λ an sc$^+$-multisection with local sc$^+$-section structures (s_i, σ_i), the local representations of Definition 13.2.1 on open sets `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn` (Part XIII). "Part I's" refers to Part I of the book (Chapters 1–6, M-polyfolds), whose statements Parts I–VI of this book retell.

XV.1 The preamble of §15.1: $\mathrm{GrF}(x)$ and finite formal sums

The text (p. 683) fixes, at a smooth point x, the set of sc-Fredholm operators $L : T_xX \to W_x$, and lets $\mathrm{GrF}(x)$ consist of the finite formal sums $\sum \sigma_L \cdot L$ with non-negative rational coefficients; a morphism $\varphi : x \to y$ acts by $\varphi_* L \circ T\varphi = \mu(\varphi, L(\cdot))$, and $\mathrm{GrF}(X) \to X_\infty$ becomes a category over X. Three objects of the formalization stand for this preamble.

$\mathrm{GrF}(x)$ (restated) `Chapter15/LinearizationIntrinsic · SBLocalization.GrF`. For a level-1 point x, $\mathrm{GrF}(x)$ is the type of finitely supported functions from the linear maps $T_xX \to W_x$ to $\mathbb{Q}^{\geq 0}$, where T_xX is the project's tangent fibre of the object M-polyfold (Part II of the book, with the linear structure of Chapter 2) and $W_x = P^{-1}(x)$ is the fibre with its vector space structure. Two departures from the text are deliberate. First, the point ranges over X_1, not only over X_∞: the linearization of a section at a zero is defined at level-1 points already, and everything proved below holds there. Second, the operators are all linear maps, not only the sc-Fredholm ones. The sc-Fredholm property is a property of the operators that actually occur, proved for them in §XV.11; it is not built into the type, so that $\mathrm{GrF}(x)$ is a plain free commutative monoid on which sums, push-forwards and supports behave without side conditions. The text's $\mathrm{GrF}(X)$ and the category structure are not built as such; the action of morphisms is defined in §XV.9, where it is needed.

On a strong bundle model `Chapter15/LinearizationModel · Linearization15.GrFModel`. On a strong bundle model $U \lhd F$ with retraction R and base retraction r, $O = r(U)$, the same object at a level-1 point z is the type of finitely supported functions on the bounded operators $T_zO \to F_0$, where $T_zO = \operatorname{range} Dr(z)$ is the tangent space of the retract, as in Part I's linearization in a chart `Chapter3/LinearizationCharts · linAt`. This is the form in which the analysis is done.

Intrinsic operators from chart operators. An operator of a strong bundle chart $c = (\varphi, \Phi)$ around x becomes an element of $\mathrm{GrF}(x)$ in two steps. First $L : E_0 \to F_0$ is read along T_xX `Chapter15/LinearizationIntrinsic · SBLocalization.chartOp`: $\alpha \mapsto L(T\varphi\,\alpha)$, where $T\varphi$ identifies T_xX with the tangent space $T_{\varphi(x)}O$ of the model `Chapter15/LinearizationIntrinsic · SBLocalization.chartOp_apply`. Then the values are pulled back to W_x through the fibre part $\Phi_x : W_x \to F_0$ of the chart, which is linear and injective `Chapter15/LinearizationIntrinsic · SBLocalization.toIntr`: if every value $L(T\varphi\,\alpha)$ lies in the range of Φ_x, the operator is $\Phi_x^{-1} \circ L \circ T\varphi : T_xX \to W_x$, and otherwise it is 0. The default value is never used: at a zero of a section that is sc-smooth in the chart, the chart derivative takes values in $\Phi_x(W_x)$ (§XV.7, step (f)), and then $\Phi_x \circ (\text{intrinsic operator}) = L \circ T\varphi$ `Chapter15/LinearizationIntrinsic · SBLocalization.fiberLin_toIntr`.

Finite formal sums of points. The proof of Theorem 15.1.1 compares formal sums of *points*, not of operators, and the formalization makes these first-class. For finitely many points a_i of a set Q with weights σ_i in a commutative monoid, the coefficient of q in $\sum \sigma_i \cdot a_i$ is

$$\mathrm{wsum}(a, \sigma)(q) = \sum_{i : a_i = q} \sigma_i$$

`Chapter15/SeparationLemmas · Linearization15.wsum`, and the mass of a set $N \subseteq Q$ is $\mathrm{mass}(a, \sigma)(N) = \sum_{i : a_i \in N} \sigma_i$ `Chapter15/SeparationLemmas · Linearization15.mass`. The mass is the sum of the coefficients over any finite set S containing all a_i, restricted to N `Chapter15/SeparationLemmas · Linearization15.mass_eq_sum_wsum` (exchange the two finite sums), so two families with the same coefficients have the same mass on every set: take S the union of the two finite images `Chapter15/SeparationLemmas · Linearization15.mass_eq_of_wsum_eq`. The coefficient of q' in the image $\sum \sigma_i \cdot \psi(a_i)$ is the mass of $\psi^{-1}(q')$ `Chapter15/SeparationLemmas · Linearization15.wsum_map`, so equal formal sums have equal images under every map `Chapter15/SeparationLemmas · Linearization15.wsum_map_eq_of_wsum_eq`. This last remark is the formal content of "se-

lect the terms with $s_i(y) = f(y)$, then forget the value": both are images, and images respect equality.

XV.2 Theorem 15.1.1 (2017: 15.1) and Remark 15.1.2 (2017: 15.1): the statements as formalized

The text. For a strong bundle $(P : W \to X, \mu)$ over an ep-groupoid, an sc-Fredholm section f and an sc$^+$-multisection Λ, there is a section functor $T(f,\Lambda)$ of $\mathrm{GrF}(X) \to X_\infty$, uniquely characterized by (1) $T(f,\Lambda)(x) = 0$ off $\mathrm{supp}(\Lambda \circ f)$, and (2) for x in the support and any local sc$^+$-section structure $(s_i,\sigma_i)_{i\in I}$ describing Λ around x, $T(f,\Lambda)(y) = \sum_{\{i: f(y)=s_i(y)\}} \sigma_i \cdot (f - s_i)'(y)$ at smooth y near x (p. 684). The substance is that the right side of (2) does not depend on the structure; existence and uniqueness then follow from the covering of X by structures. A local sc$^+$-section structure is the local representation of Definition 13.2.1 (Part XIII) on an open set: sc$^+$-sections $s_1,\dots,s_k$ on O with positive rational weights summing to 1 such that $\Lambda(w) = \sum_{i: w = s_i(P(w))} \sigma_i$ for w over O `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn`, with s_i an sc$^+$-section on O `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsScPlusSectionOn`, and an sc$^+$-multisection is a functor $W \to \mathbb{Q}^{\geq 0}$ with such a structure on the domain of a natural representation around every object `Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsScPlusMultisection`.

The formalization states the theorem three times, on three levels of the reduction "without loss of generality $X = O$", and then once more for the value of Definition 15.1.7 (§XV.8) and for the functor property (§XV.9).

On a strong bundle model (stronger than the text's local statement) `Chapter15/LinearizationModel · Linearization15.theorem_15_1_1_model`. Let $U \lhd F$ be a strong bundle model (U relatively open in a partial quadrant, R a strong bundle retraction, $O = r(U)$), and let $\hat f$, $\hat s_1,\dots,\hat s_k$, $\hat t_1,\dots,\hat t_l$ be principal parts $E_0 \to F_0$ with $\hat f \circ r$, $\hat s_i \circ r$, $\hat t_j \circ r$ sc-smooth on U. The formal sum at a level-1 point z is

$$\sum_{i:\hat s_i(z)=\hat f(z)} \sigma_i \cdot (\hat f - \hat s_i)'(z) \in \mathrm{GrF}_{\mathrm{model}}(z), \qquad (\hat f - \hat s_i)'(z) = D((\hat f - \hat s_i)\circ r)(z)|_{T_zO}$$

`Chapter15/LinearizationModel · Linearization15.linSumModel`, with its coefficient of an operator ℓ the sum of the σ_i with $\hat s_i(z) = \hat f(z)$ and $(\hat f - \hat s_i)'(z) = \ell$ `Chapter15/LinearizationModel · Linearization15.linSumModel_apply`. If the two structures give the same weighted count $\sum_{i:\hat s_i(y)=v} \sigma_i = \sum_{j:\hat t_j(y)=v} \tau_j$ of every value v at every $y \in O \cap N$, N open, then the two formal sums agree at every level-1 point of $O \cap N$. The same holds after replacing each operator $D((\hat f - \hat s_i)\circ r)(z)$ by its image under any map ψ of operators `Chapter15/LinearizationModel · Linearization15.linSumModelMap` `Chapter15/LinearizationModel · Linearization15.linSumModelMap_apply` `Chapter15/LinearizationModel · Linearization15.theorem_15_1_1_model_map`; the map that matters is the passage to intrinsic operators. The statement is stronger than what the proof of the text needs in two ways: all level-1 points instead of the smooth ones, and no condition on Λ at all, only that the two structures have the same counts. Neither positivity of the weights nor functoriality is used.

In a strong bundle chart (stronger) `Chapter15/Linearization · SBLocalization.theorem_15_1_1`. For an sc-Fredholm section functor f and two local sc$^+$-section structures (s_i,σ_i) on O and (t_j,τ_j) on O' of the same Λ, the formal sums $\sum_{i:f(y)=s_i(y)} \sigma_i \cdot (f - s_i)'(y)$, read in a strong bundle chart c with domain in $O \cap O'$ `Chapter15/Linearization · SBLocalization.linSumChart`, agree at every level-1 point of the chart. The

same holds for any sc-smooth section f `Chapter15/Linearization · SBLocalization.linSumChart_eq`; this is what the proof uses.

With intrinsic operators (narrower, as the row of the theorem; stronger in its substance) `Chapter15/LinearizationIntrinsic · SBLocalization.theorem_15_1_1_intrinsic`. The formal sum in $\mathrm{GrF}(x)$ through a chart c,

$$\sum_{i:f(x)=s_i(x)} \sigma_i \cdot (f-s_i)'(x), \qquad (f-s_i)'(x) = \Phi_x^{-1} \circ D((\hat{f}-\hat{s}_i)\circ r)(\varphi(x)) \circ T\varphi$$

`Chapter15/LinearizationIntrinsic · SBLocalization.linSumIntr`, is the same for two structures of Λ on open O, O', computed through any chart c with domain in O and any chart c' with domain in O', at every $x \in X_1$ in both domains. This is the text's (2) with the operators in their intrinsic form, and the formulation in which Definition 15.1.7 is made.

Why the row is narrower. What Theorem 15.1.1 claims beyond the independence is (a) that $T(f,\Lambda)$ is a *section functor*, which is proved at all level-1 points in §XV.9, and (b) that its values lie in GrF, i.e. that the summands are sc-Fredholm. (b) is proved in §XV.11 only in a chart reading, since T_xX and W_x carry no sc-structure of their own in the project, and only for f with Part I's tame germ data, the hypothesis of Part I's implicit function theorem on which the proof rests. That inherited hypothesis is the only reason the row is narrower; the independence itself is proved in a stronger form. The text's "uniquely characterized" is the remark that (1) and (2) determine the value at every point, which in the formalization is Definition 15.1.7 with $T(f,\Lambda)(x)$ equal to the sum of *any* local data (§XV.8).

Remark 15.1.2 (restated). The text observes that $(f-s_i)'(y)$ is undefined where $f(y) \neq s_i(y)$ but is not counted, and that with the empty sum equal to 0, (1) follows from (2). In the formalization the operator $(\hat{f}-\hat{s}_i)'(z)$ is defined for every i, and the sum is filtered by $\hat{s}_i(z) = \hat{f}(z)$; so the first half of the remark is the definition of the formal sum. The second half is proved on each level. *On the model* `Chapter15/LinearizationModel · Linearization15.remark_15_1_2_model`: if the weights are positive and the weighted count of $\hat{f}(z)$ is 0, then each term of that count is 0, so no i has $\hat{s}_i(z) = \hat{f}(z)$ (a term with $\hat{s}_i(z) = \hat{f}(z)$ would contribute $\sigma_i > 0$), every summand of the formal sum is the empty case, and the formal sum is 0. The same counting gives the description of the support used by Definition 15.2.1: with positive weights an operator ℓ has a nonzero coefficient iff $\ell = (\hat{f}-\hat{s}_i)'(z)$ for some i with $\hat{s}_i(z) = \hat{f}(z)$ `Chapter15/LinearizationModel · Linearization15.mem_support_linSumModel_iff`. *In a chart* `Chapter15/Linearization · SBLocalization.remark_15_1_2`: where $\Lambda(f(y)) = 0$, the weighted count of $\hat{f}(\varphi(y))$ in the chart is $\Lambda(\Phi^{-1}(\varphi(y), \hat{f}(\varphi(y)))) = \Lambda(f(y)) = 0$ (the count formula of §XV.7, step (e)), so the model remark applies. *Intrinsically*, Theorem 15.1.1 (1) `Chapter15/LinearizationIntrinsic · SBLocalization.linearization_eq_zero`: $T(f,\Lambda)(x) = 0$ where $\Lambda(f(x)) = 0$, by the same count read through the data that define $T(f,\Lambda)(x)$ (§XV.8).

XV.3 Proposition 15.1.3 (2017: 15.1): separating sequences

The separation property at a point z for a partition $e : I \to \{1,\dots,p\}$ of the index set `Chapter15/SeparationLemmas · Linearization15.IsSeparatedAt`: (1) $s_i(z) = s_{i'}(z)$ iff $e(i) = e(i')$, and (2) if $e(i) = e(i')$ then $s_i = s_{i'}$ on a neighbourhood of z. This is the conclusion of the proposition at one point x_k.

Statement (narrower) `Chapter15/SeparationLemmas · Linearization15.proposition_15_1_3_hyp`. Let Z be a first countable T_1 space, s_i, $i \in I$ finite, maps continuous on an open $V \ni x_0$ with values in a Hausdorff space, and suppose that x_0 is not isolated. Then there are $x_k \in V$ with $x_k \neq x_0$ and $x_k \to x_0$, and

one surjective partition $e : I \to \{1, \dots, p\}$, such that the separation property holds at every x_k. The text states it on the local model O, with the metric of E_0, $x_0 = 0$ and the sections of a local sc$^+$-section structure. The formal statement is more general in its setting (topological, no sc-structure, the maps only continuous) and narrower by one hypothesis.

Proof. *Step 1, points near x_0 where the maps are dichotomous.* Call z *locally dichotomous* for the family if every pair of the maps either coincides on a neighbourhood of z or differs at z `Chapter15/SeparationLemmas · Linearization15.IsLocDich`. Every nonempty open set on which the maps are continuous contains such a point `Chapter15/SeparationLemmas · Linearization15.exists_isLocDich_mem`: Lemma 15.1.5 in its topological form (§XV.5) gives a nonempty open W inside it on which each pair either coincides or differs everywhere, and any point of W will do (coincidence on W is coincidence near the point, since W is open). Now take a decreasing countable basis (u_k) of neighbourhoods of x_0, which exists by first countability. For each k the set $\mathrm{int}(u_k) \cap V \setminus \{x_0\}$ is open, because Z is T_1 so that $\{x_0\}$ is closed; and it is nonempty, because it belongs to the punctured neighbourhood filter of x_0, which is not trivial since x_0 is not isolated. So it contains a locally dichotomous point z_k; $z_k \neq x_0$, $z_k \in V$, and $z_k \to x_0$ since $z_k \in u_k$ and the u_k form a decreasing basis.

Step 2, the subsequence `Chapter15/SeparationLemmas · Linearization15.exists_isSeparatedAt_subseq`. At each z_k the *pattern* of coincidences is the relation $\{(i,j) : s_i(z_k) = s_j(z_k)\}$ on $I \times I$. There are finitely many such relations, so one of them, P, occurs for infinitely many k (a map from $\mathbb{N}$ to a finite set has an infinite fibre), and an extraction $k \mapsto \phi(k)$, strictly increasing, picks those k. The partition e is taken from the values at $z_{\phi(0)}$: $e(i) = e(j)$ iff $s_i(z_{\phi(0)}) = s_j(z_{\phi(0)})$ `Chapter15/SeparationLemmas · Linearization15.exists_fin_partition`, the classes of equal values numbered by a finite set. Since the pattern is P at every $z_{\phi(k)}$, (1) holds at every point of the subsequence. For (2): if $e(i) = e(j)$ then $s_i(z_{\phi(k)}) = s_j(z_{\phi(k)})$, and since $z_{\phi(k)}$ is locally dichotomous, the pair cannot differ there, so it coincides near $z_{\phi(k)}$. The sequence of the proposition is $x_k = z_{\phi(k)}$.

How this departs from the text's proof. The text picks $y_k \to 0$, $y_k \neq 0$, applies Lemma 15.1.5 on small balls around y_k to get z_k with partitions $I = I_1^k \sqcup \cdots \sqcup I_{p_k}^k$, and then takes successive subsequences making I_1^k, I_2^k, … constant. The formal proof takes a single subsequence on which the whole pattern is constant, by the pigeonhole principle on the finite set of relations on I; the result is the same. The points come from a neighbourhood basis rather than from a sequence chosen in advance, so no metric is needed.

The added hypothesis, and the isolated point. That x_0 is not isolated is not derived from the text; it is what makes the punctured neighbourhoods nonempty in Step 1. Its witness is two maps $z \mapsto 0$ and $z \mapsto z$ on $\mathbb{R}$ at 0 `Chapter15/SeparationLemmas · Linearization15.proposition_15_1_3_hyp_witness`, where all hypotheses hold and the proposition applies. **At an isolated point, Proposition 15.1.3 does not hold as we read it.** If $\{x_0\}$ is open, no sequence $x_k \neq x_0$ converges to x_0: convergence would put some x_k in the open set $\{x_0\}$ `Chapter15/SeparationLemmas · Linearization15.not_proposition_15_1_3_of_isolated`. This happens for the one-point local model $O = \{0\}$ of a zero-dimensional X `Chapter15/SeparationLemmas · Linearization15.not_proposition_15_1_3_point`. The case is degenerate, and nothing depends on it: the formal proof of Theorem 15.1.1 does not use the proposition, because its core lemma needs separated points in every neighbourhood of x_0, and x_0 itself may be one of them (§XV.7). At an isolated point the theorem therefore holds without change.

XV.4 Lemma 15.1.4 (2017: 15.1): finite closed covers of a ball

Statement (stronger) `Chapter15/SeparationLemmas · Linearization15.lemma_15_1_4`. In any pseudo-metric space, if a ball $B_\varepsilon(y)$, $\varepsilon > 0$, is the union of finitely many sets A_i that are closed in $B_\varepsilon(y)$, then one of the A_i has nonempty interior. The text states it for O with the metric of E_0 (p. 685); the formal statement holds in every pseudo-metric space.

Proof. *The topological core* `Chapter15/SeparationLemmas · Linearization15.interior_biUnion_eq_empty`: in any topological space, a finite union of closed sets with empty interiors has empty interior. By induction over the finite index set: the empty union is empty; and if A is closed with empty interior and B (the union so far) has empty interior, then $A \cup B$ has empty interior, Mathlib's fact for the union of a closed set and a set with empty interior. *The cover form* `Chapter15/SeparationLemmas · Linearization15.exists_nonempty_interior_of_finite_closed_cover`: a nonempty open set V covered by finitely many closed sets meets the interior of one of them. Otherwise all interiors are empty, so the union has empty interior by the core; but V is open and contained in the union, so every point of V lies in the interior of the union. *The lemma* is the cover form in the subspace $B_\varepsilon(y)$, which is nonempty (it contains y) and open in itself: the preimages of the A_i in the subspace are closed by hypothesis and cover it, so one of them has a point x in its interior in the subspace. The image of that interior under the inclusion is open in the ambient space, because the ball is open, and it lies in A_i; so x lies in the interior of A_i.

How this departs from the text's proof. The text moves a point z step by step, each step adding to the set of indices i with $z \notin \Sigma_i$, until z lies in the interior of some Σ_i. The formal proof is the induction on the index set above. Neither uses the Baire property, which a finite union does not need. The lemma is not used by the formal proof of Lemma 15.1.5 (§XV.5); it is formalized as stated.

XV.5 Lemma 15.1.5 (2017: 15.2): local dichotomy of finitely many maps

Statement (stronger) `Chapter15/SeparationLemmas · Linearization15.lemma_15_1_5`. In a pseudo-metric space, let s_i, $i \in I$ finite, be maps continuous on a ball $B_\varepsilon(y)$, $\varepsilon > 0$, with values in a Hausdorff space. Then there are z with $d(z, y) < \varepsilon$, $\delta > 0$ and a surjective partition $e : I \to \{1, \dots, p\}$ such that (1) $B_\delta(z) \subseteq B_\varepsilon(y)$, (2) $s_i = s_{i'}$ on $B_\delta(z)$ when $e(i) = e(i')$, and (3) $s_i(w) \neq s_j(w)$ for $w \in B_\delta(z)$ when $e(i) \neq e(j)$. The text (pp. 685–686) states it for the sections of a local sc$^+$-section structure on O; the formal maps need only be continuous, with no linear or sc-structure.

The topological form `Chapter15/SeparationLemmas · Linearization15.exists_isOpen_dichotomy`: for finitely many maps continuous on a nonempty open set V of a topological space, with values in a Hausdorff space, there is a nonempty open $W \subseteq V$ on which every pair of maps either coincides everywhere or differs everywhere.

Proof. By induction over a finite set S of pairs (i, j), the claim being that some nonempty open $W \subseteq V$ has the dichotomy for the pairs in S. For $S = \emptyset$ take $W = V$. For the step, let W serve the pairs of S and add the pair (i, j). The map $w \mapsto (s_i(w), s_j(w))$ is continuous on W, and the complement of the diagonal is open because the target is Hausdorff, so

$$W' = \{w \in W : s_i(w) \neq s_j(w)\}$$

is open. If W' is empty, then $s_i = s_j$ on W, and W serves $S \cup \{(i, j)\}$. If W' is nonempty, then W' serves the new pair (the maps differ everywhere on it) and still serves the pairs of S, since coincidence

or difference everywhere on W passes to the subset W'. Applying this to the set of all pairs gives the claim.

The lemma from the topological form. Apply it to the ball, open and nonempty: it gives W and a point $z \in W$; $d(z, y) < \varepsilon$ since W lies in the ball. As W is open, some $B_\delta(z) \subseteq W$, which gives (1). The partition is that of the values at z `Chapter15/SeparationLemmas · Linearization15.exists_fin_partition`: $e(i) = e(j)$ iff $s_i(z) = s_j(z)$. For (2), if $e(i) = e(i')$ the pair cannot differ everywhere on W (it agrees at z), so it coincides on $W \supseteq B_\delta(z)$. For (3), if $e(i) \neq e(j)$ the pair cannot coincide on W (it differs at z), so it differs everywhere on W.

How this departs from the text's proof. The text builds the pieces one at a time: it fixes i_1, considers the closed sets $\Sigma_j = \{s_j = s_{i_1}\}$, uses Lemma 15.1.4 when they cover the ball, moves the point slightly, and repeats with $i_2 \notin I_1$ on a ball of a quarter of the radius. The formal proof goes through the pairs instead of the pieces and shrinks the open set at each step to where the new pair differs, if that set is not empty. The argument for a single pair is the one observation that the set where two continuous maps into a Hausdorff space differ is open; if it is empty on an open set, the maps agree there. So Lemma 15.1.4 is not needed. The same form gives the locally dichotomous points used in §§XV.3, XV.6 and XV.7 `Chapter15/SeparationLemmas · Linearization15.exists_isLocDich_mem`.

XV.6 Remark 15.1.6: modifying the sequence

The remark, new in the 2021 edition (p. 686), makes two observations: a sequence with the properties of Proposition 15.1.3 can be obtained from any sequence $x'_k \to 0$ by modifying it "slightly" and passing to a subsequence; and modifying such a sequence slightly keeps the properties. Both are formalized, with "slightly" made explicit (restated).

First part `Chapter15/SeparationLemmas · Linearization15.remark_15_1_6`. In a metric space, for finitely many maps continuous on an open V with values in a Hausdorff space and any sequence $x'_k \to x_0$ in V, there are $x''_k \in V$ with $d(x'_k, x''_k) < 1/(k+1)$ and $x''_k \to x_0$, a strictly increasing ϕ, and a partition e with the separation property at every $x''_{\phi(k)}$. *Proof.* For each k the set $B_{1/(k+1)}(x'_k) \cap V$ is open and nonempty (it contains x'_k), so it contains a locally dichotomous point x''_k (§XV.3, Step 1). Then $d(x''_k, x_0) \le d(x''_k, x'_k) + d(x'_k, x_0)$, and both terms tend to 0, the first being below $1/(k+1)$; by squeezing, $x''_k \to x_0$. The subsequence and the partition come from the pigeonhole step of §XV.3 `Chapter15/SeparationLemmas · Linearization15.exists_isSeparatedAt_subseq`. "Slightly" is the bound $1/(k+1)$, which gives the text's $d(x'_k, x''_k) \to 0$. The modified points need not differ from x_0; if $x'_k \neq x_0$ is wanted as well, one removes x_0 from the open sets as in §XV.3.

Second part `Chapter15/SeparationLemmas · Linearization15.IsSeparatedAt.eventually`. If the separation property holds at z for a partition e and the maps are continuous at z, it holds at every z' in a neighbourhood of z. *Proof*, pair by pair, finitely many conditions each holding on a neighbourhood. For a pair in one piece, $s_i = s_j$ on a neighbourhood N of z; on the open interior of N both clauses hold at every point, (1) because the values agree and (2) because the germs agree. For a pair in different pieces, $s_i(z) \neq s_j(z)$, so $(s_i(z), s_j(z))$ lies off the diagonal, which is closed in a Hausdorff space; by continuity of $w \mapsto (s_i(w), s_j(w))$ at z, the values differ on a neighbourhood, and (1) holds there while (2) is vacuous for the pair. So small moves of the points of a separating sequence keep it separating, which is the text's second observation; "slightly" means "inside this neighbourhood".

XV.7 Proof of Theorem 15.1.1: the core lemma, the model, the chart, the change of chart

The proof follows the text's reduction to a local model and its limit argument, with three changes: the separation runs in the level-1 topology, the two structures are separated together without halving weights, and the formal sums compared are of pairs (value, derivative) over all branches. The steps are (a) a limit lemma for finite formal sums, (b) the comparison at a separated point, (c) the core lemma in a topological space, (d) the model, (e) the chart, (f) the change of chart and the intrinsic statement.

(a) The limit step `Chapter15/SeparationLemmas · Linearization15.wsum_eq_of_frequently`. Let $a_i(z)$, $b_j(z)$ be finitely many points of a Hausdorff space Q, depending on z in a topological space and continuous at x_0, with weights σ_i, τ_j. If the formal sums $\sum \sigma_i \cdot a_i(z)$ and $\sum \tau_j \cdot b_j(z)$ agree for z frequently near x_0 (that is, in every neighbourhood of x_0 there is such a z), they agree at x_0. *Proof.* Fix $q \in Q$. For each point $p \neq q$ choose disjoint neighbourhoods $N_p \ni q$ and $V_p \ni p$ (Hausdorff); for $p = q$ take $N_q = V_q = Q$. Let N_0 be the intersection of the finitely many N_p for p among the values $a_i(x_0)$, $b_j(x_0)$: a neighbourhood of q. For z near x_0, and each i: if $a_i(x_0) = q$, then N_0 is a neighbourhood of $a_i(x_0)$ and $a_i(z) \in N_0$ by continuity; if $a_i(x_0) = p \neq q$, then $a_i(z) \in V_p$, which is disjoint from $N_p \supseteq N_0$. So eventually $a_i(z) \in N_0 \iff a_i(x_0) = q$, for all i, and likewise for the b_j. Choose z with this property at which the two formal sums agree. Then the mass of N_0 under the first sum at z is the coefficient of q in the first sum at x_0, the mass under the second is that of the second, and the two masses are equal because the formal sums at z are (§XV.1) `Chapter15/SeparationLemmas · Linearization15.mass_eq_of_wsum_eq`. This is the text's convergence "in the weak topology" of the formal sums: the points cluster near their limits, and a neighbourhood that separates one limit from the others carries the limit's coefficient.

(b) At a locally dichotomous point `Chapter15/SeparationLemmas · Linearization15.wsum_pair_eq_at_of_isLocDich`. Let the branches c_α, $\alpha \in I \sqcup K$, of two families with weights σ on I and τ on K give the same weighted count of every value at u, let u be locally dichotomous for the combined family, and let the "derivatives" $D_\alpha(u)$ depend only on germs: $c_\alpha = c_\beta$ near u implies $D_\alpha(u) = D_\beta(u)$. Then the formal sums of the pairs $(c_\alpha(u), D_\alpha(u))$ over I and over K agree. *Proof.* Fix a pair (v, L). Two branches through v at u cannot differ at u, so by dichotomy they coincide near u, and by locality they have the same D. Hence there is one L_v with $D_\alpha(u) = L_v$ for every branch through v (any L if there is none). For such a family, the coefficient of (v, L) is the coefficient of v if $L = L_v$ and 0 otherwise `Chapter15/SeparationLemmas · Linearization15.wsum_pair_of_const`: the pairs with first entry v are exactly (v, L_v). Applying this to both families, with the same L_v since it was chosen for the combined family, the coefficients are the counts of v, which agree.

(c) The core lemma `Chapter15/SeparationLemmas · Linearization15.wsum_pair_eq_of_isLocal`. Let both families be continuous on an open V, with equal weighted counts of every value at every point of V, and let the D_α be continuous on V and depend only on germs at every point of V. Then at every $x_0 \in V$ the formal sums $\sum \sigma_i \cdot (c_i, D_i)(x_0)$ and $\sum \tau_j \cdot (c_j, D_j)(x_0)$ agree. *Proof.* The maps $z \mapsto (c_\alpha(z), D_\alpha(z))$ are continuous at x_0, so the limit step (a) applies once the sums agree frequently near x_0. Given a neighbourhood N of x_0, the set $\mathrm{int}(N) \cap V$ is open and contains x_0, so it contains a point u locally dichotomous for the combined family (§XV.3, Step 1); by (b) the sums agree at u. *Departure from the text.* The text forms the union of the two structures with halved weights, applies Proposition 15.1.3 to it, and compares multiplicities piece by piece along the sequence. Here the dichotomy is applied to the combined family $I \sqcup K$ directly, which needs no new structure and no weights. The points u may include x_0 itself, so no sequence $x_k \neq x_0$ is needed and isolated points are covered; this is why Proposition 15.1.3 is not used and its failure at isolated points (§XV.3) does not reach the theorem.

(d) On the model. Two facts of the sc-calculus at level 1 come first. *Continuity of values* `Chapter15/LinearizationModel · Linearization15.continuousOn_comp_toBase_one`: an sc^0 map is continuous on U_1 in the level-1 topology, read in F_0 (it is continuous $U_1 \to F_1$, and $F_1 \to F_0$ is continuous). *Continuity of derivatives* `Chapter15/LinearizationModel · Linearization15.continuousOn_scDeriv_coe`: for an sc^1 map g, $u \mapsto Dg(u)$ is continuous on U_1 (level-1 topology) into the operators $E_0 \to F_0$ with the topology of pointwise convergence. This is Definition 1.1.15 (3), the continuity of $(u,h) \mapsto Dg(u)h$ on $U_1 \times E_0$, read one h at a time. *Locality of the derivative* `Chapter15/LinearizationModel · Linearization15.scDeriv_eq_of_eventuallyEq_lvl`: if two sc^1 maps g, g' on an sc-domain agree near a level-1 point x in the level-1 topology, then $Dg(x) = Dg'(x)$. Both derivatives are Fréchet derivatives of $g \circ \iota$, $g' \circ \iota$ within U_1 at x; the two composites agree near x within U_1, so they have the same derivatives there, and within an sc-domain the derivative within U_1 is unique; finally an operator on E_0 is determined by its values on the dense E_1.

The comparison on the model `Chapter15/LinearizationModel · Linearization15.wsum_model_eq`. Let the two structures have the same counts at every $y \in O \cap N$. Work on the space $Z = U_1$, the level-1 points of U with the level-1 topology, and on its open subset $V = \{u : r(u) \in N\}$, open because $u \mapsto r(u)$ is continuous on U_1 (r is sc-smooth, so sc^0). The combined family has branches $c_\alpha(u) = \hat{s}_\alpha(r(u))$ (the $\hat{s}_i$ and the $\hat{t}_j$ together) and derivatives $D_\alpha(u) = D((\hat{f} - \hat{s}_\alpha) \circ r)(u)$, regarded as functions $E_0 \to F_0$. The hypotheses of the core lemma hold. Continuity: the c_α by the first fact, since $\hat{s}_\alpha \circ r$ is sc-smooth; the D_α by the second, since $(\hat{f} - \hat{s}_\alpha) \circ r$ is sc-smooth, a difference of sc-smooth maps. Counts: at $u \in V$ they are the counts at $r(u) \in O \cap N$, equal by hypothesis. Locality: if $c_\alpha = c_\beta$ near u in U_1, then $(\hat{f} - \hat{s}_\alpha) \circ r$ and $(\hat{f} - \hat{s}_\beta) \circ r$ agree near u in U_1 (both are $\hat{f} \circ r$ minus the branch), so their derivatives agree at u by the locality fact. The core lemma gives equal formal sums of pairs at the point z, which lies in U_1 because it lies over $O \subseteq U$, and in V because $r(z) = z$ for $z \in O$.

Theorem 15.1.1 on the model, from the pairs. The coefficient of an operator $\ell : T_zO \to F_0$ in the formal sum of a structure is the mass, under the formal sum of pairs, of the set

$$N_\ell = \{(v, D) : v = \hat{f}(z),\ D(h) = \ell(h) \text{ for all } h \in T_zO\}.$$

Indeed, the summand i contributes σ_i iff $\hat{s}_i(z) = \hat{f}(z)$ and $(\hat{f} - \hat{s}_i)'(z) = \ell$, and the latter, an equality of operators on T_zO, says that the full derivative $D((\hat{f} - \hat{s}_i) \circ r)(z)$ agrees with ℓ on T_zO, which is the membership of the pair in N_ℓ (with $r(z) = z$). Equal formal sums of pairs give equal masses (§XV.1), so the coefficients of ℓ agree for the two structures; this proves `theorem_15_1_1_model`. The form with an arbitrary map ψ of operators is the same argument with the set $\{(v, D) : v = \hat{f}(z),\ \psi(L) = q' \text{ for every operator } L \text{ with } L = D\}$: the derivative is recorded as a function, and ψ is applied to the operator it determines.

Two readings of the text's proof (Findings, Chapter 15, items 1 and 2). (a) The text takes the points x_k of Proposition 15.1.3 in the metric of E_0 and lets $T(f - s_i)(x_k)$ converge "in the weak topology". The derivative of an sc^1 map is continuous only on level 1, and there only pointwise as an operator $E_0 \to F_0$. So the separation is run in the level-1 topology, where the branches are continuous and the separation lemmas hold equally (they are topological), and the "weak topology" is read as pointwise convergence of operators, which is the topology in which the derivatives are continuous. (b) The last display of the text sums $\sigma_i \cdot T(f - s_i)(0)$ over all i and then uses $f(0) = s_i(0)$; this holds only for some i, and the sum is presumably over $\{i : f(0) = s_i(0)\}$. The formal proof compares the formal sums of the pairs $(s_i, (f - s_i)')$ over all i, where no such equation is needed, and selects the first entry $f(0)$ only at the end, by the mass of N_ℓ.

(e) In a strong bundle chart. Let c be a strong bundle chart of P with base chart φ, model O and fibre model F, and let s be a section over $O_X \supseteq \mathrm{dom}(c)$. Its principal part $\hat{s}$ is the fibre component of $c \circ s \circ \varphi^{-1}$: $c(s(\varphi^{-1}(z))) = (z, \hat{s}(z))$ for $z \in O$ `Chapter15/Linearization · SBLocalization.apply_localSection`. *Principal parts are sc-smooth.* A local sc⁺-section on O_X has $\hat{s} \circ r$ sc-smooth on U in every chart of the atlas with domain in O_X `Chapter15/Linearization · SBLocalization.IsScPlusSectionOn.isScSmooth_principal_comp_r`. The sc⁺-section is an sc-smooth map into $W_{0,1}$; composed with the sc-smooth inclusion $W_{0,1} \to W$ it is sc-smooth into W on O_X; read in the chart, $z \mapsto (z, \hat{s}(z))$ is an sc-smooth retract map $O \to K$; composed with the base retraction $r : U \to O$ and projected to the second factor, $\hat{s} \circ r$ is sc-smooth. The same holds for an sc-smooth global section `Chapter15/Linearization · SBLocalization.isScSmooth_principal_comp_r`. *Counts in the chart* `Chapter15/Linearization · SBLocalization.IsLocalRepOn.count_eq`: for a local section structure of Λ on $O_X \supseteq \mathrm{dom}(c)$ and $y \in O$, the weighted count $\sum_{i:\hat{s}_i(y)=v} \sigma_i$ equals $\Lambda(\Phi^{-1}(y, v))$ if $(y, v) \in K$, and 0 otherwise. If $(y, v) \in K$, let $w = \Phi^{-1}(y, v)$, which lies over $\varphi^{-1}(y) \in O_X$; the defining formula of the structure gives $\Lambda(w) = \sum_{i:w=s_i(Pw)} \sigma_i$, and $w = s_i(\varphi^{-1}(y))$ iff $\hat{s}_i(y) = v$, by injectivity of the chart and the formula for $\hat{s}_i$. If $(y, v) \notin K$, no branch can have $\hat{s}_i(y) = v$, since $(y, \hat{s}_i(y))$ is the chart image of $s_i(\varphi^{-1}(y))$ and lies in K. *The statement in a chart* `Chapter15/Linearization · SBLocalization.linSumChart_eq`: for two structures of Λ on O and O' and a chart with domain in $O \cap O'$, both counts at every y of the model equal $\Lambda(\Phi^{-1}(y, v))$ or 0, so the model theorem applies with N the whole space; Theorem 15.1.1 in a chart is this for sc-Fredholm section functors, which are sc-smooth `Chapter15/Linearization · SBLocalization.theorem_15_1_1`. Every object lies in such a chart `Chapter15/Linearization · SBLocalization.exists_chart_isLocalRepOn`: an sc⁺-multisection has a local section structure on the domain $U(x)$ of a natural representation around x, and any atlas chart around x restricted to its domain intersected with $U(x)$ is a chart with domain in $U(x)$.

(f) The change of chart, and the statement with intrinsic operators. *Independence of the structure in* $\mathrm{GrF}(x)$ `Chapter15/LinearizationIntrinsic · SBLocalization.linSumIntr_eq_of_isLocalRepOn`: in one chart with domain in $O \cap O'$, the intrinsic formal sums of two structures agree. This is the model theorem with the map ψ that sends a chart operator to its intrinsic operator (§XV.1), at the level-1 point over $\varphi(x)$, with the counts of (e).

The text's "without loss of generality $X = O$" is, formally, the independence of the intrinsic operator $(f - s)'(x)$ from the chart. The proof makes $f - s$ a global section to which Part I's chain rule applies. *The section $f - s$ on O_X* `Chapter15/LinearizationIntrinsic · SBLocalization.sectionSubOn`: at $y \in O_X$ the difference $f(y) - s(y)$ in the fibre, and $f(y)$ elsewhere; a section `Chapter15/LinearizationIntrinsic · SBLocalization.isSection_sectionSubOn` whose principal part in a chart with domain in O_X is $\hat{f} - \hat{s}$ `Chapter15/LinearizationIntrinsic · SBLocalization.principal_sectionSubOn`, because the chart is linear on fibres. In such a chart $z \mapsto (z, \hat{f}(z) - \hat{s}(z))$ is an sc-smooth retract map $O \to K$ `Chapter15/LinearizationIntrinsic · SBLocalization.isScSmoothRetractMap_sectionSubOn`: for every sc-smooth retraction R' onto O, $\hat{f} \circ R'$ and $\hat{s} \circ R'$ are sc-smooth (as in (e)), and so is the pair $(R', \hat{f} \circ R' - \hat{s} \circ R')$; this is the chart-wise form of sc-smoothness that Part I's statements take. The derivative of $(\hat{f} - \hat{s}) \circ r$ at a level-1 point equals that of the principal part of $f - s$ composed with r `Chapter15/LinearizationIntrinsic · SBLocalization.scDeriv_sub_eq_sectionSubOn`, since the two maps agree on U (derivatives of sc¹ maps agreeing on a relatively open set agree on its level-1 points).

The values lie in the fibre `Chapter15/LinearizationIntrinsic · SBLocalization.chartOp_mem_range`. At a zero x of a global section g whose principal part is sc-smooth in the chart, every value $D(\hat{g} \circ r)(z)(T\varphi\,\alpha)$ lies in $\Phi_x(W_x)$. Part I's lemma on the derivative at a zero `Chapter3/LinearizationCharts · rho_scDeriv_comp_r` says that the bundle projection $\rho(z)$ onto the fibre K_z fixes $D(\hat{g} \circ r)(z)$ on T_zO when $\hat{g}(z) = 0$; so the values

lie in $K_z = \Phi_x(W_x)$. Hence the default branch of the intrinsic operator is never taken at the points of interest.

The chain rule `Chapter15/LinearizationIntrinsic · SBLocalization.toIntr_chart_independent`. At a zero $x \in X_1$ of a section g whose principal parts are sc-smooth in two charts c, c' around x, the intrinsic operators through c and through c' are equal. Part I's chain rule with the operators named `Chapter3/LinearizationChartChange · linAt_chart_change_key` provides a retraction r_W onto the base overlap and the identity

$$D(\hat{g}' \circ r')(z')(D(\tau \circ r_W)(z)v) = \Gamma(z)(D(\hat{g} \circ r)(z)v),$$

with τ the base transition and $\Gamma(z)$ the fibre transition operator. Two further identities translate this to intrinsic operators: the chart vector of $\alpha \in T_xX$ in c' is $D(\tau \circ r_W)(z)$ applied to its chart vector in c (the transition of tangent fibres), and the fibre part of c' is $\Gamma(z)$ composed with that of c. With these, Φ'_x applied to the operator through c' equals Φ'_x applied to the operator through c, and Φ'_x is injective. This is the argument of Part I's chart independence of $f'(x)$ `Chapter3/LinearizationIntrinsic · StrongBundle.linIntr_chart_independent`, with chart-wise hypotheses only, so that it applies to $f - s$, which is sc-smooth only over O_X.

The formal sum does not depend on the chart `Chapter15/LinearizationIntrinsic · SBLocalization.linSumIntr_chart_independent`. For one structure on O_X and two charts with domains in O_X, the formal sums agree term by term. The selection condition is the same through both charts: $\hat{s}_i(\varphi(y)) = \hat{f}(\varphi(y))$ iff $s_i(y) = f(y)$ `Chapter15/LinearizationIntrinsic · SBLocalization.principal_eq_iff`, by injectivity of the chart on the fibre. For a selected i, the operator $(f - s_i)'(x)$ through either chart is the intrinsic operator of the global section $f - s_i$ at its zero x (by the derivative identity above), and the chain rule gives equality.

Proof of the intrinsic statement. Given structures on O, O' and charts c, c' with domains in O, O' around $x \in X_1$, let c'' be c restricted to $\mathrm{dom}(c) \cap O \cap O'$, a chart around x with domain in $O \cap O'$. Then: the sum of (s, σ) through c equals that through c'' (chart independence, structure s); through c'' it equals the sum of (t, τ) (independence of the structure); and that equals the sum of (t, τ) through c' (chart independence, structure t). Only the sc-smoothness of f enters; the theorem is stated for sc-Fredholm section functors, as in the text.

XV.8 Definition 15.1.7 (2017: 15.1): the linearization $T(f, \Lambda)$

Definition (restated) `Chapter15/LinearizationIntrinsic · SBLocalization.linearization`. *Local data* at $x \in X_1$ `Chapter15/LinearizationIntrinsic · SBLocalization.LocalLinData` are an open set O, a local sc⁺-section structure $(s_i, \sigma_i)_{i<k}$ of Λ on O, and a strong bundle chart c of the atlas around x with domain in O; their formal sum is the intrinsic sum through c `Chapter15/LinearizationIntrinsic · SBLocalization.LocalLinData.sum`. $T(f, \Lambda)(x) \in \mathrm{GrF}(x)$ is the sum of some local data, chosen, and 0 if there are none. The text calls $T(f, \Lambda)$ the linearization of $\Lambda \circ f$ "at the points in $\mathrm{supp}(\Lambda \circ f)$" (p. 687); the formal definition is made at every level-1 point, with value 0 off the support (§XV.2).

Well-defined `Chapter15/LinearizationIntrinsic · SBLocalization.LocalLinData.sum_eq`: any two local data at x give the same sum, by the intrinsic Theorem 15.1.1 (§XV.7 (f)). **Data exist** `Chapter15/LinearizationIntrinsic · SBLocalization.nonempty_localLinData` at every level-1 point of an sc⁺-multisection, from the chart of §XV.7 (e). **Theorem 15.1.1 (2)** `Chapter15/LinearizationIntrinsic · SBLocalization.linearization_eq`: $T(f, \Lambda)(x)$ is the sum of *any* local data at x; so the value at y may be computed from any structure around x and any chart around y with domain inside it, which is the

text's (2) for all y near x, smooth or of level 1. **Theorem 15.1.1 (1) and Remark 15.1.2** Chapter15/LinearizationIntrinsic · SBLocalization.linearization_eq_zero: where $\Lambda(f(x)) = 0$, $T(f,\Lambda)(x) = 0$. *Proof.* Take the chosen data (O, s, σ, c). The point $(\varphi(x), \widehat{f}(\varphi(x)))$ lies in K, and Φ^{-1} of it is $f(x)$; so by the count formula (§XV.7 (e)) the weighted count of $\widehat{f}(\varphi(x))$ is $\Lambda(f(x)) = 0$. With positive weights, no branch has $\widehat{s}_i(\varphi(x)) = \widehat{f}(\varphi(x))$, and every summand of the formal sum vanishes.

In a chart. The formalization also keeps the chart-level definition: in a chart c, the formal sum of any structure on an open set containing the domain of c, and 0 if there is none Chapter15/Linearization · SBLocalization.linearizationIn, equal to the sum of every such structure at every level-1 point Chapter15/Linearization · SBLocalization.linearizationIn_eq, and 0 where $\Lambda(f(y)) = 0$ Chapter15/Linearization · SBLocalization.remark_15_1_2. It is the form used where an argument stays inside one chart.

The definition is the value of a section; that this section is a *functor* is the next subsection.

XV.9 Theorem 15.1.1: the functor property

The text calls $T(f,\Lambda)$ a section functor of $\mathrm{GrF}(X) \to X_\infty$, i.e. $T(f,\Lambda)(t(\varphi)) = \varphi_* T(f,\Lambda)(s(\varphi))$ for every morphism, and gives no separate argument for it. The formalization proves it at every level-1 point (stronger than the text's smooth points), with no added hypothesis.

The action of a morphism on GrF**.** For $\varphi : x \to y$, $\mu(\varphi,\cdot) : W_x \to W_y$ is a linear map Chapter15/LinearizationFunctor · SBLocalization.muLin Chapter15/LinearizationFunctor · SBLocalization.muLin_apply, linear because the action of a strong bundle over an ep-groupoid is linear on fibres. If x is of level 1 then so is φ Chapter15/LinearizationFunctor · SBLocalization.mem_levelSet_one_of_src (a morphism lies in $\mathbf{X}_m$ iff its source lies in X_m), and then Chapter 7's tangent map $T\varphi : T_xX \to T_yX$ is a bounded linear isomorphism Chapter7/TangentMorphismLinear · EpGroupoid.tangentMorCLE. An operator $L : T_xX \to W_x$ is carried to

$$\varphi_* L = \mu(\varphi,\cdot) \circ L \circ (T\varphi)^{-1} : T_yX \to W_y$$

Chapter15/LinearizationFunctor · SBLocalization.conjOp Chapter15/LinearizationFunctor · SBLocalization.conjOp_apply, which is the text's $\varphi_* L \circ T\varphi = \mu(\varphi, L(\cdot))$, and a formal sum to the push-forward of its coefficients along $L \mapsto \varphi_* L$ Chapter15/LinearizationFunctor · SBLocalization.grfPush.

Statement (stronger, for this clause) Chapter15/LinearizationFunctor · SBLocalization.theorem_15_1_1_functor. For an sc-Fredholm section functor f, an sc$^+$-multisection Λ, a morphism φ and level-1 points $x = s(\varphi)$, $y = t(\varphi)$: $T(f,\Lambda)(y) = \varphi_* T(f,\Lambda)(x)$ in $\mathrm{GrF}(y)$.

Proof. Since $T(f,\Lambda)(y)$ may be computed from *any* local data at y (§XV.8), it suffices to carry local data at x to local data at y and to compare the two sums. The carrying uses the local strong bundle isomorphism attached to a morphism by its sheet, built in Chapter 12 (Part XII) Chapter12/MorphismIso · EpStrongBundle.morIso: over the image $t(\text{sheet})$ of the sheet of φ it consists of the base sc-diffeomorphism $\phi = s \circ (t|_{\text{sheet}})^{-1}$ onto $s(\text{sheet})$, with $\phi(y) = x$, and the fibre maps Φ, Ψ given by the action μ of the morphisms of the sheet.

Step 1, data at x inside the sheet. Take local data (O, s, σ, c) at x, restrict the chart to $\mathrm{dom}(c) \cap s(\text{sheet})$ and the structure to $O \cap s(\text{sheet})$ (a local structure restricts to open subsets Chapter11/MultisectionLocalStructure2021 · SBLocalization.IsLocalRepOn.mono). Call the result (O, s, σ, c) again.

Step 2, carrying the sections Chapter15/LinearizationFunctor · SBLocalization.transSec. A section s near x becomes $z \mapsto \Psi(s(\phi(z)))$ on $t(\text{sheet}) \cap \phi^{-1}(O)$ Chapter15/LinearizationFunctor · SBLocalization.transSet, an

open set Chapter15/LinearizationFunctor · SBLocalization.isOpen_transSet (the preimage of an open set under the continuous ϕ), and the carried section is a section there Chapter15/LinearizationFunctor · SBLocalization.P_transSec because Ψ covers ϕ^{-1}. An sc⁺-section stays sc⁺ Chapter15/LinearizationFunctor · SBLocalization.isScPlusSectionOn_transSec: its level-1 lift is the composite of the lift of s with ϕ and with the level-1 part of Ψ, all sc-smooth. A local section structure of a *functor* Λ is carried to one of Λ, with the same weights Chapter15/LinearizationFunctor · SBLocalization.isLocalRepOn_transSec: for w over the new domain, $\Lambda(w) = \Lambda(\Phi(w))$ because Φ acts by μ of a morphism and Λ is a functor Chapter15/LinearizationFunctor · SBLocalization.isWFunctor_morIso_Φ; the formula of the structure at $\Phi(w)$, which lies over $\phi(P(w)) \in O$, counts the i with $\Phi(w) = s_i(\phi(Pw))$; and this holds iff $w = \Psi(s_i(\phi(Pw)))$, since Φ and Ψ are inverse on the relevant fibres.

Step 3, carrying the chart. The chart c at x is carried to the transported chart $c \circ \Phi$ at y Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.transportChart, with base chart $\varphi \circ \phi$ and the same model; it is a chart of the atlas Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.isChart_transportChart. In it the principal parts of the carried sections are those of the given sections in c Chapter15/LinearizationFunctor · SBLocalization.principal_transportChart_transSec (unwind $\Phi \circ \Psi = \mathrm{id}$ and $\phi \circ \phi^{-1} = \mathrm{id}$), and the principal part of f is that of f in c Chapter3/FredholmPullbackIso · StrongBundle.LocalIso.principal_transportChart, because a functorial section satisfies $f = \Psi \circ f \circ \phi$ on $t(\text{sheet})$ Chapter12/MorphismIso · EpStrongBundle.eqOn_morIso, which is $f(t(\psi)) = \mu(\psi, f(s(\psi)))$ for the morphisms ψ of the sheet.

Step 4, the chain rule in the carried chart. Two identities at the point. $T\varphi$ is the identity on chart vectors Chapter15/LinearizationFunctor · SBLocalization.chartVec_transport: the chart vector of $T\varphi(\beta)$ in $c \circ \Phi$ equals the chart vector of β in c. Indeed $T\varphi = T\hat{\varphi}$ at x (Chapter 7's tangent map of a morphism is that of the local diffeomorphism $\hat{\varphi} = t \circ (s|_{\text{sheet}})^{-1}$), $T(\phi) \circ T(\hat{\varphi}) = \mathrm{id}$ near x since $\phi \circ \hat{\varphi} = \mathrm{id}$, and the chart vector of $Tg(\alpha)$ in c is that of α in $c \circ g$. And $\mu(\varphi, \cdot)$ followed by the fibre part of the carried chart at y is the fibre part of c at x Chapter15/LinearizationFunctor · SBLocalization.fiberLin_transport_muLin, because on W_x the fibre map Ψ is $\mu(\varphi, \cdot)$ and Φ undoes it; also the fibre part of the carried chart at y takes values in that of c at x Chapter15/LinearizationFunctor · SBLocalization.fiberLin_transport_mem_range. Together Chapter15/LinearizationFunctor · SBLocalization.toIntr_transportChart: for every chart operator L, the intrinsic operator through the carried chart is $\mu(\varphi, \cdot) \circ (\text{intrinsic operator through } c) \circ (T\varphi)^{-1}$. If the values of $L \circ T\varphi$ lie in the fibre for c, those through the carried chart lie in the fibre (the preimages are the $\mu(\varphi, \cdot)$-images), and the two sides agree after applying the injective fibre part; if not, neither do the carried ones, and both sides are 0.

Step 5, the sums correspond Chapter15/LinearizationFunctor · SBLocalization.linSumIntr_transportChart. The level-1 point of the carried chart over y is the level-1 point of c over x, since $\phi(y) = x$. Term by term: the selection condition is the same (the principal parts agree), and the derivative $D((\hat{f} - \hat{s}_i) \circ r)(z)$ is the same in both charts, the composites agreeing on U. So each term of the sum at y is φ_* of the corresponding term at x by Step 4, and push-forward of formal sums is additive. With the carried data at y and the given data at x, §XV.8 identifies both sides with $T(f, \Lambda)(y)$ and $\varphi_* T(f, \Lambda)(x)$.

Not formalized. That $\varphi \mapsto \varphi_*$ is itself functorial, $(\psi\varphi)_* = \psi_*\varphi_*$ and $1_* = \mathrm{id}$, which the text uses to speak of $\mathrm{GrF}(X)$ as a category, is not formalized and not needed for the theorem; it would follow from Chapter 7's functoriality of T and the composition law of μ, in about 60–100 lines.

XV.10 Local sc⁺-sections, the sections $f - s_i$ on $W|_O$, and local solution sets (prerequisite for §15.2)

The proofs of Theorems 15.2.2 and 15.2.5 and of the sc-Fredholm property of the summands (§XV.11) apply Part I's implicit function theorem to $f - s_i$ near x, where s_i is defined only on an open O. The formalization makes $f - s_i$ a section problem of Part I on the restricted strong bundle $W|_O$ (extra).

Local sc⁺-sections in compatible charts. Chapter 11 defines a local sc⁺-section on O as a section over O that is sc-smooth into $W_{0,1}$. Two chart facts are needed for charts that are only compatible with the atlas, such as restricted charts. In every such chart with domain in O, the principal part takes values in F_1 `Chapter15/LocalScPlus · SBLocalization.IsScPlusSectionOn.principal_mem_range_of_isChart`: compare with an atlas chart around $s(x)$, in which this holds by Chapter 13's local form `Chapter13/ScPlusSectionOn · SBLocalization.IsScPlusSectionOn.isScSmoothRetractMap_localRepOne`, through the transition, which preserves level 1 in the fibre. And the level-1 representative $z \mapsto (z, \hat{s}_1(z))$ `Chapter15/LocalScPlus · SBLocalization.localRepOne_eq_of_sub` is an sc-smooth retract map $O \to K[1]$ `Chapter15/LocalScPlus · SBLocalization.IsScPlusSectionOn.isScSmoothRetractMap_localRepOne_of_isChart`. The proof is local: near each point, the representative is the composite of the base transition into an atlas chart c' around $s(x)$, the representative in c' (sc-smooth by Chapter 13), and the level-1 part of the bundle transition back to c; the composite agrees with the representative on the overlap.

Restriction to O `Chapter15/LocalScPlus · SBLocalization.IsScPlusSectionOn.restrictOpen`. A local sc⁺-section on O is a global sc⁺-section of the restricted strong bundle $W|_O$, Part I's pull-back along the inclusion $O \to X$ `Chapter3/RestrictOpen · StrongBundle.restrictOpen`: the charts of $W|_O$ are restrictions of charts with domain in O, and in them the two chart facts above are the conditions of an sc⁺-section. (The restricted bundle needs the paracompactness of O and of the pulled-back total space as type-class data; they hold for open subsets of M-polyfolds and are supplied in the proofs.)

The section $f - s$ `Chapter15/LocalScPlus · SBLocalization.subSec`: $f + (-1) \cdot \bar{s}$, where $\bar{s}$ is s extended by 0 off O; a global section `Chapter15/LocalScPlus · SBLocalization.isSection_subSec` with principal part $\hat{f} - \hat{s}$ in every chart with domain in O `Chapter15/LocalScPlus · SBLocalization.principal_subSec` and with zeros in O exactly where $f = s$ `Chapter15/LocalScPlus · SBLocalization.isZeroAt_subSec_iff`. Its restriction to $W|_O$ `Chapter15/LocalScPlus · SBLocalization.subRestrict` is a tame sc-Fredholm section when f has Part I's tame germ data `Chapter15/LocalScPlus · SBLocalization.isScFredholmSectionTame_subRestrict`. *Proof.* Restriction to an open set keeps tame germ data (Part I), so $f|_O$ is tame sc-Fredholm on $W|_O$; $-\bar{s}$ is an sc⁺-section on O (Chapter 13: sc⁺-sections on O are closed under constant multiples), hence a global sc⁺-section of $W|_O$ by the restriction above; and an sc⁺-perturbation of a tame sc-Fredholm section is one, by Part I's Theorem 3.1.18 in its tame form `Chapter3/TameGermsAddG · theorem_3_1_18_tame`. Finally, the restriction of a sum of sections is the sum of the restrictions.

The linearization in a chart. At a level-1 point z of a chart with domain in O, the chart operator of $(f - s)'$ is $D((\hat{f} - \hat{s}) \circ r)(z) : E_0 \to F_0$ `Chapter15/LocalIFT · SBLocalization.subDeriv`, the derivative of an sc¹ map `Chapter15/LocalIFT · SBLocalization.isSc1_sub_comp_r`, and "onto" means that it maps T_zO onto the fibre $K_z = \{k : (z,k) \in K\}$ `Chapter15/LocalIFT · SBLocalization.ChartOnto`, Part I's "$f'(x)$ onto" read in the chart. *The intrinsic and the chart readings agree* `Chapter15/LocalIFT · SBLocalization.surjective_toIntr_subDeriv_iff`: at $x \in X_1$ with $f(x) = s(x)$, the intrinsic operator $(f - s)'(x) : T_xX \to W_x$ of §XV.1 is onto iff the chart operator maps T_zO onto K_z, z over $\varphi(x)$. The chart operator equals that of the global section $f - s$, whose values at its zero lie in the fibre (§XV.7 (f)); the fibre part of the chart maps W_x bijectively onto $K_{\varphi(x)}$, and $T\varphi$ maps T_xX onto T_zO, so surjectivity transfers in both directions. *And they are Part*

I's surjectivity on $W|_O$. In the restricted chart of $W|_O$ the principal part of $(f-s)|_O$ composed with r has the same derivative as $(\widehat{f}-\widehat{s})\circ r$ `Chapter15/LocalIFT · SBLocalization.scDeriv_restrictTo_subRestrict`; so chart surjectivity gives Part I's surjective linearization of $(f-s)|_O$ at x `Chapter15/LocalIFT · SBLocalization.isLinSurjAt_subRestrict`, and conversely, at a zero, Part I's surjectivity gives chart surjectivity in every chart with domain in O and at every level-1 point over $\varphi(x)$ `Chapter15/LocalIFT · SBLocalization.chartOnto_of_isLinSurjAt_subRestrict`, by the chart independence of Part I's notion.

Local solution sets. *At a point of good position* `Chapter15/LocalIFT · SBLocalization.exists_solutionSet_of_isGoodPosition`. If $(f-s)|_O$ is in good position at x in the sense of Part I's Definition 3.6.6 in a tame chart, there is an open $V \ni x$ inside the domain of c such that $\{y \in V : f(y) = s(y)\}$ is a smooth finite-dimensional submanifold of X (Definition 4.2.1), and at each of its points y: y is smooth, $(f-s)|_O$ is in good position at y, and the chart operator is onto at every level-1 point over $\varphi(y)$. *Proof.* Part I's Theorem 3.6.8 `Chapter3/GlobalIFTNearbyOntoG · theorem_3_6_8'` for the tame sc-Fredholm section $(f-s)|_O$ at its zero x gives an open $V_x \subseteq O$ on which every zero is in good position; on $W = V_x \cap \mathrm{dom}(c)$ the solution set is a smooth submanifold of O by Part I's lemma on solution sets in good position `Chapter5/GeneralPosition · StrongBundle.isSmoothSubmanifold_solutionSet_inter`, and its image under the sc-diffeomorphism $O \to X$ (the inclusion) is a smooth submanifold of X. Zeros of an sc-Fredholm section are smooth points; good position implies surjective linearization; and the converse direction above gives chart surjectivity. *At an interior point* `Chapter15/LocalIFT · SBLocalization.exists_solutionSet_interior`: if $d_X(x)=0$, $f(x)=s(x)$ and the chart operator is onto at x, the same conclusion holds without "good position" in it. The restriction $(f-s)|_O$ has degeneracy index 0 at x (the index of an open subset is that of X), it has surjective linearization at x by the transfer above, and at a point of index 0 Part I's surjectivity gives good position `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`; then the first case applies.

All of this rests on Part I's Theorem 3.6.8 for tame germ data; with the 2021 untamed notion that theorem does not hold as we read it (Part III), which is why f carries the tame germ data here and in §15.2 (§XV.11).

XV.11 Theorem 15.1.1 (2): the summands are sc-Fredholm

The text puts sc-Fredholm operators into $\mathrm{GrF}(x)$ and asserts that $T(f,\Lambda)(x)$ is an element of it: each $(f-s_i)'(x)$ with $f(x)=s_i(x)$ is sc-Fredholm, because $f-s_i$ is an sc$^+$-perturbation of an sc-Fredholm section. The formalization proves this for the operators that occur, in a chart (narrower).

The chart operator is sc-Fredholm `Chapter15/FredholmSummands · SBLocalization.subDeriv_isScFredholm`. Let f have Part I's tame germ data, s be sc$^+$ on an open O containing the domain of the chart c, $f(x)=s(x)$, and z a level-1 point over $\varphi(x)$ in U. Then $D((\widehat{f}-\widehat{s})\circ r)(z)$ is an sc-operator, T_zO and K_z are sc-subspaces, the operator maps T_zO into K_z, and its restriction $T_zO \to K_z$ is sc-Fredholm, Fredholm on every level with the index of level 0. *Proof.* By §XV.10, $(f-s)|_O$ is a tame sc-Fredholm section of $W|_O$ with a zero at x, and the restricted chart of c is a chart of $W|_O$. Part I's statement that the linearization of an sc-Fredholm section at a zero is sc-Fredholm in every chart `Chapter3/LinearizationChartsScG · StrongBundle.linAt_isScFredholm` gives all the conclusions for the derivative of the restricted principal part composed with r, which is $D((\widehat{f}-\widehat{s})\circ r)(z)$ by the derivative identity of §XV.10.

Theorem 15.1.1 (2), the summands (restated, with the inherited hypothesis) `Chapter15/FredholmSummands · SBLocalization.linearization_summand_isScFredholm`. For $x \in X_1$ and any local data (O,s,σ,c) (§XV.8), every operator L in the support of $T(f,\Lambda)(x)$ is $(f-s_i)'(x)$, the intrinsic operator of $D((\widehat{f}-\widehat{s}_i)\circ r)(z)$, for some i with $f(x)=s_i(x)$, and that chart operator is sc-Fredholm as above.

Proof. $T(f,\Lambda)(x)$ is the sum of the given data (§XV.8); an operator in the support of that sum is the intrinsic operator of a branch through $f(x)$ `Chapter15/Transversality · SBLocalization.mem_support_linSumIntr_iff` (positive weights, as in Remark 15.1.2); and the chart statement applies to that branch.

Reading. T_xX and W_x carry no sc-structure in the project: they are a tangent fibre and a fibre with their linear structures. "$(f-s_i)'(x) : T_xX \to W_x$ is sc-Fredholm" is therefore read in a chart, through the identifications $T_xX \cong T_zO$ and $W_x \cong K_z$ by which the intrinsic operator is defined. The sc-structure on T_zO and K_z is that of sc-subspaces of E and F; that it does not depend on the chart is not formalized, and not needed, since the statement holds in every chart. $\mathrm{GrF}(x)$ is not cut down to sc-Fredholm operators (§XV.1); the lemma says what lies in the support.

The hypothesis taken. *Tame germ data for f* (Part I's notion, an sc-Fredholm section whose germs are conjugate to basic germs in tame charts) is not stated in §15.1, where f is "an sc-Fredholm section". It is not new: it is inherited from Part I's Theorems 3.1.18 and 3.6.8, which are proved for tame germ data and on which the proof rests; with the 2021 untamed notion, Theorem 3.6.8 does not hold as we read it `Chapter3/UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not` (Part III). The same hypothesis is carried by all of §15.2 (§§XV.12–XV.16) and by the cylinder of Corollary 15.3.10 (§XV.30). Its witness is the graph section of 0 over $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, which has tame germ data by Part I `Chapter3/NonVacuityG · isScFredholmSectionTame_graphSection`, with the zero multisection; there all hypotheses of the §15.2 theorems hold together `Chapter15/InteriorPerturbationWitness · SBLocalization.theorem_15_2_2_hyp_witness`. The witness has zero fibres, so it shows that the hypotheses are consistent, not that the summands are non-trivially sc-Fredholm; the sc-Fredholm statement itself is a theorem for every f with tame germ data.

XV.12 Definition 15.2.1 (2017: Definition 15.2): transversality

§15.2 (pp. 688–691) is the local theory: what $T(f,\Lambda)$ says about $\mathrm{supp}(\Lambda \circ f)$ near one point. Its tool throughout is the passage of §XV.10 from a branch s_i of a local sc⁺-section structure, a section defined only on an open O, to the sc-Fredholm section $(f-s_i)|_O$ of the restricted strong bundle $W|_O$, to which Part I's implicit function theorems apply. The definitions come first.

Statement (restated). Let f be an sc-Fredholm section functor and Λ an sc⁺-multisection. At a level-1 point x, $T(f,\Lambda)(x)$ *is onto* if every operator in the support of the formal sum $T(f,\Lambda)(x) \in \mathrm{GrF}(x)$ (Definition 15.1.7, §XV.8) is surjective onto W_x `Chapter15/Transversality · SBLocalization.IsOntoAtLevel`. For an arbitrary object x the same is said with the clause $x \in X_1$ built in, since $T(f,\Lambda)$ is defined only on X_1 `Chapter15/Transversality · SBLocalization.IsOntoAt`. (f,Λ) is *transversal at x* if $\Lambda(f(x)) > 0$ and $T(f,\Lambda)(x)$ is onto `Chapter15/Transversality · SBLocalization.IsTransversalAt` (the text's (2)), and *transversal* if it is transversal at every point of $\mathrm{supp}(\Lambda \circ f)$ `Chapter15/Transversality · SBLocalization.IsTransversal` (the text's (1)).

The text phrases the condition twice: as "a nonzero combination of surjective sc-Fredholm operators", and, after "i.e.", as the surjectivity of all $(f-s_i)'(x)$ with $f(x) = s_i(x)$ for a local sc⁺-section structure (s_i). The formal definition is the first phrasing minus the two words that turn out to be automatic; the two lemmas below recover the second phrasing and the word "nonzero", and §XV.11 the word "sc-Fredholm".

The operators of the formal sum. Fix local data at x (§XV.8): a local sc⁺-section structure $(s_i,\sigma_i)_{i\in I}$ of Λ on an open O, with positive weights, and a strong bundle chart c around x with domain in O. Then an operator $L : T_xX \to W_x$ lies in the support of $\sum_{i:f(x)=s_i(x)} \sigma_i \cdot (f-s_i)'(x)$ iff $L = (f-s_i)'(x)$ for some i with $s_i(x) = f(x)$ `Chapter15/Transversality · SBLocalization.mem_support_linSumIntr_iff`. *Proof.* The

formal sum through c is, by definition, the push-forward of the chart-level sum, whose filter is "the principal parts of s_i and f agree at $\varphi(x)$" (§XV.7). For sections over the chart domain this is the same as $s_i(x) = f(x)$, the chart being injective on fibres; this rewrites the filter. The coefficient of L is then $\sum_i [s_i(x) = f(x) \wedge (f - s_i)'(x) = L]\,\sigma_i$, a finite sum of non-negative rationals; it is nonzero iff some term is nonzero, and a term is nonzero iff its condition holds, because $\sigma_i > 0$. So positivity of the weights is exactly what makes "occurs in the sum" the same as "is the derivative of some branch through $f(x)$".

The text's reading `Chapter15/Transversality · SBLocalization.isOntoAtLevel_iff`: for f an sc-Fredholm section functor and any local data at $x \in X_1$, $T(f,\Lambda)(x)$ is onto iff every $(f - s_i)'(x)$ with $s_i(x) = f(x)$, computed through the chart of the data, is surjective. *Proof.* By Definition 15.1.7, $T(f,\Lambda)(x)$ is the formal sum of these data, whichever data are taken `Chapter15/LinearizationIntrinsic · SBLocalization.linearization_eq` (this is Theorem 15.1.1, §XV.7); the support is described by the previous lemma; the two directions are then immediate. The text's "if (s_i) is a local sc⁺-section structure" is thereby read "for any", which Theorem 15.1.1 makes the same as "for some".

"Nonzero" is automatic `Chapter15/Transversality · SBLocalization.linearization_ne_zero`: at $x \in X_1$ with $\Lambda(f(x)) > 0$, $T(f,\Lambda)(x) \neq 0$. *Proof.* Local data exist at every level-1 point `Chapter15/LinearizationIntrinsic · SBLocalization.nonempty_localLinData`. The count formula of the local structure, evaluated at the point $f(x)$ of the fibre over x, gives $\Lambda(f(x)) = \sum_{i : s_i(x) = f(x)} \sigma_i$; if no branch passed through $f(x)$ the sum would be empty and $\Lambda(f(x)) = 0$. So some i has $s_i(x) = f(x)$, the operator $(f - s_i)'(x)$ lies in the support of the formal sum, and a formal sum with nonempty support is not 0.

"sc-Fredholm" is automatic as well, for f with Part I's tame germ data: every operator of the support is, read in the chart, sc-Fredholm (§XV.11). So Definition 15.2.1 asks only for surjectivity, and that is what the formal definition says.

Where the support lives. $T(f,\Lambda)$ is defined on X_1, so the formal definition carries "$x \in X_1$". Under the tame germ data this costs nothing: every point of the support is smooth. *One branch* `Chapter15/Transversality · SBLocalization.mem_smoothSet_of_eq`: if s is sc⁺ on an open $O \ni y$ and $f(y) = s(y)$, then y is a zero of the section $(f - s)|_O$ of $W|_O$ `Chapter15/LocalScPlus · SBLocalization.isZeroAt_subSec_iff`, which is a tame sc-Fredholm section (§XV.10) `Chapter15/LocalScPlus · SBLocalization.isScFredholmSectionTame_subRestrict`; zeros of sc-Fredholm sections are smooth points (Part I) `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`, and a point of the open M-polyfold O is smooth there iff it is smooth in X. *The support* `Chapter15/Transversality · SBLocalization.mem_smoothSet_of_pos`: if $\Lambda(f(y)) > 0$, the count formula of a local structure around y produces a branch with $f(y) = s_i(y)$, as in the previous proof, and the one-branch lemma applies. The same fact without tame germ data, $\operatorname{supp}(\Lambda \circ f) \subset X_1$, is proved for Definition 15.3.5 in §XV.20 `Chapter15/ControllingPairs · SBLocalization.mem_levelSet_one_of_pos`.

A witness. Over the trivial ep-groupoid of $\mathbb{R}^n$, the strong bundle $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ with the graph section of 0 and the trivial multisection Λ_0 (Chapter 13; $\Lambda_0(w) = 1$ on the zero vectors, 0 elsewhere) is transversal, with $\Lambda_0(f(x)) = 1$ at every point `Chapter15/InteriorPerturbationWitness · SBLocalization.theorem_15_2_2_hyp_witness`. The fibres are $\{0\}$, so every operator into a fibre is onto. The example has zero fibres: it shows that the definition is satisfiable with nonempty support, not more.

XV.13 Theorem 15.2.2 (2017: Theorem 15.2): interior local perturbation

Statement (narrower) `Chapter15/Transversality · SBLocalization.theorem_15_2_2_hyp`. Let f be an sc-Fredholm section functor *with Part I's tame germ data*, Λ an sc⁺-multisection, $d_X(x) = 0$, and (f,Λ) transversal

at x. Then there are a natural representation $U(x)$ of G_x (Theorem 7.1.13), a local sc$^+$-section structure $(s_i, \sigma_i)_{i\in I}$ of Λ on an open O, and a strong bundle chart c with $U(x) \subset \mathrm{dom}(c) \subset O$, such that, with $M_i = \{y \in U(x) : f(y) = s_i(y)\}$:

1. $d_X(y) = 0$ for all $y \in U(x)$;
2. $T(f, \Lambda)(y)$ is onto at every $y \in U(x)$ with $\Lambda(f(y)) > 0$;
3. (a) every M_i is a smooth finite-dimensional submanifold of X (Definition 4.2.1); (b) the inclusion $M_i \hookrightarrow U(x)$ is a proper map; (c) $\{y \in U(x) : \Lambda(f(y)) > 0\} = \bigcup_i M_i$; (d) $\Lambda(f(y)) = \sum_{i:f(y)=s_i(y)} \sigma_i$ on $U(x)$; (e) $T(f, \Lambda)(y) = \sum_{i:f(y)=s_i(y)} \sigma_i \cdot (f - s_i)'(y)$, the operators read in c, at every level-1 point y of $\mathrm{dom}(c)$;

and, in addition, a branch meets f in $U(x)$ only if it passes through $f(x)$: $f(y) = s_i(y)$ for some $y \in U(x)$ implies $s_i(x) = f(x)$.

Narrower only by the tame germ data. The text's hypothesis $x \in \mathrm{supp}(\Lambda \circ f)$ is part of transversality at x.

Proof. The text's proof is four sentences: choose $U'(x)$ and a local structure on it; by assumption $f(x) = s_i(x)$ for all i; apply the implicit function theorem to each $f - s_i$; intersect. The Lean proof follows it, with two additions: the branches that do not pass through $f(x)$ are handled separately, and every step is read through one chart inside the domain of the structure, so that §§XV.8 and XV.10 apply.

Step 1, a structure and a chart inside it. Since Λ is an sc$^+$-multisection, there is a natural representation $U_0(x)$ with a local sc$^+$-section structure $(s_i, \sigma_i)_{i\in I}$ on $U_0(x)$ (Definition 13.2.5 in its 2021 reading, Part XIII). Choose any strong bundle chart c_0 around x in the atlas and restrict it to $\mathrm{dom}(c_0) \cap U_0(x)$, which is open; the restriction is a chart of the atlas, and its domain lies in the domain of the structure. Call it c. The structure and c are local data at x in the sense of Definition 15.1.7 `Chapter15/LinearizationIntrinsic · SBLocalization.LocalLinData`.

Step 2, from transversality to the chart. Transversality at x gives $x \in X_1$ and $T(f, \Lambda)(x)$ onto; by the text's reading (§XV.12) `Chapter15/Transversality · SBLocalization.isOntoAtLevel_iff`, every intrinsic $(f - s_i)'(x)$ with $s_i(x) = f(x)$ is surjective. The intrinsic operator is onto W_x iff the chart operator $D((\widehat{f} - \widehat{s}_i) \circ r)(z)$ maps $T_z O$ onto the fibre K_z at the level-1 chart point z over $\varphi(x)$ `Chapter15/LocalIFT · SBLocalization.surjective_toIntr_subDeriv_iff` (§XV.10).

Step 3, one branch `Chapter15/Transversality · SBLocalization.exists_branch_interior`. For each i there is an open $V_i \ni x$ inside $\mathrm{dom}(c)$ such that $\{y \in V_i : f(y) = s_i(y)\}$ is a smooth submanifold, every point y of it is smooth and has the chart operator of $f - s_i$ onto at every chart point over $\varphi(y)$, and, if some $y \in V_i$ has $f(y) = s_i(y)$, then $s_i(x) = f(x)$. Two cases.

- *The branch passes through $f(x)$.* Then the chart operator is onto at x (Step 2), and the interior local solution lemma of §XV.10 `Chapter15/LocalIFT · SBLocalization.exists_solutionSet_interior` gives V_i: it is Part I's implicit function theorem (Theorem 3.6.8) for the tame sc-Fredholm section $(f - s_i)|_O$ of $W|_O$ at its zero x, where $d_X(x) = 0$ makes the good position automatic, read back in X through the inclusion $O \to X$. The last clause holds trivially.
- *The branch does not pass through $f(x)$.* Then $V_i = \{y \in \mathrm{dom}(c) : f(y) \neq s_i(y)\}$. It is open `Chapter15/Transversality · SBLocalization.isOpen_sep_ne`: f is continuous, s_i is continuous on O `Chapter15/Transversality · SBLocalization.IsScPlusSectionOn.continuousOn` (an sc$^+$-section is the composite of a continuous lift to $W_{0,1}$ with the continuous inclusion $W_{0,1} \to W$), and the set where two continuous maps into a Hausdorff space differ is the preimage of the complement of the closed diagonal. It

contains x. The solution set in V_i is empty, so it is a submanifold, and the remaining clauses hold vacuously.

The second case is where the formal statement departs from the text's proof, which says "by assumption, $f(x) = s_i(x)$ for $i \in I$". That holds only for the branches through $f(x)$; presumably the text means those, the others having empty solution sets near x. The formal statement keeps the whole index set, and the last clause of the theorem says that the other branches contribute nothing on $U(x)$.

Step 4, the neighbourhood. The set $\mathrm{dom}(c) \cap \bigcap_i V_i \cap \{d_X = 0\}$ is open, the last set being open because the degeneracy index is upper semicontinuous, and it contains x. A natural representation can be shrunk into any open neighbourhood of x `Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_shrink` (Theorem 7.1.13); this gives $U(x) \subset U_0(x)$. Property (1) holds by construction.

Step 5, the count. For $y \in U(x)$ the count formula of the structure at the point $f(y)$ gives (d), $\Lambda(f(y)) = \sum_{i:f(y)=s_i(y)} \sigma_i$. Since the weights are positive, $\Lambda(f(y)) > 0$ iff some branch passes through $f(y)$; this gives (c).

Step 6, (2). Let $y \in U(x)$ with $\Lambda(f(y)) > 0$. By Step 5 some i has $f(y) = s_i(y)$; then y lies in the solution set of Step 3 for that branch, hence is smooth and in particular at level 1. The same structure with the chart c is local data at y, so by the text's reading it suffices that each $(f - s_j)'(y)$ with $s_j(y) = f(y)$ is onto; Step 3 for the branch j gives the chart operator onto at y, and §XV.10 turns this into the intrinsic surjectivity `Chapter15/LocalIFT · SBLocalization.surjective_toIntr_subDeriv_iff`.

Step 7, (a) and (b). $M_i = \{y \in V_i : f(y) = s_i(y)\} \cap U(x)$, the intersection of a smooth submanifold with an open set, which is again one. For (b), an inclusion of a subset is a proper map iff the subset is closed; M_i is the set of points of $U(x)$ where the two continuous maps f and s_i agree, which is closed in $U(x)$.

Step 8, (e). At a level-1 point y of $\mathrm{dom}(c)$, the structure and c are local data at y, and Definition 15.1.7 gives $T(f, \Lambda)(y)$ as their formal sum `Chapter15/LinearizationIntrinsic · SBLocalization.linearization_eq`. The last clause is the last clause of Step 3.

Readings. (i) "$d_{M_i} \equiv 0$ and therefore a natural (classical) smooth manifold structure" in (a) is read as Definition 4.2.1 inside $\{d_X = 0\}$, which (1) provides; Definition 4.2.1 allows boundary and corners and here has none. (ii) The text's (e) "for $y \in U(x)$" uses the structure on $U(x)$; the formal (e) is read in the chart c, which is no restriction, since the formal sum does not depend on the chart (§XV.7). (iii) The text's "$f_i(y) = s_i(y)$" in (e) is read $f(y) = s_i(y)$. (iv) The index set is that of the structure; $M_i = \emptyset$ for the branches not through $f(x)$.

The added hypothesis. *Tame germ data for f* (Part I's sc-Fredholm sections with germ data in tame charts), not stated in §15.2, which says only "f an sc-Fredholm section functor". It is inherited, not new: the proof is Part I's Theorem 3.6.8 applied to $(f - s_i)|_O$, and Part I proves that theorem for tame germ data; with the untamed 2021 notion of an sc-Fredholm germ Theorem 3.6.8 does not hold as we read it `Chapter3/UntamedBundleTheorem · UntamedCorner.theorem_3_6_8_untamed_not` (Part III). The restriction to O keeps tame germ data, and the sc$^+$-perturbation by $-s_i$ keeps them by Part I's Theorem 3.1.18 (§XV.10). No ep-level counterexample to the untamed §15.2 was built. *Witness* of all hypotheses together `Chapter15/InteriorPerturbationWitness · SBLocalization.theorem_15_2_2_hyp_witness`: the example of §XV.12, where the graph section of 0 has tame germ data by Part I `Chapter3/NonVacuityG · isScFredholmSectionTame_graphSection`, Λ_0 is an sc$^+$-multisection, $d_X \equiv 0$ on $\mathbb{R}^n$, $\Lambda_0(f(x)) = 1$ and (f, Λ_0) is transversal everywhere. The example has zero fibres: it witnesses consistency, and the solution sets are all of $\mathbb{R}^n$.

XV.14 Remark 15.2.3 (2017: Remark 15.2): $\Lambda \circ f$ is locally branched; tame; the ep-groupoid $U(x)$

The remark says that on the ep-groupoid $U(x)$ the functor $z \mapsto \Lambda(f(z))$ is a branched, tame ep⁺-subgroupoid with support $M_{U(x)}$. The formalization proves it in three forms: as a local branching structure at a point of X, as a global statement on X, and on the ep-groupoid $U(x)$ itself, each also with "tame".

Local form (narrower) `Chapter15/Transversality · SBLocalization.remark_15_2_3_hyp`. With tame germ data for f: at a point y with $d_X(y) = 0$ where (f, Λ) is transversal, $z \mapsto \Lambda(f(z))$ has a local branching structure at y in the sense of Definition 9.1.2 (2)–(4) (Part IX): an open $U \ni y$, finitely many smooth submanifolds $M_j \subset U$ through y with proper inclusions, and positive rational weights σ_j with $\Lambda(f(z)) = \sum_{j : z \in M_j} \sigma_j$ on U. *Proof.* Theorem 15.2.2 at y gives $U(y)$, the structure $(s_i, \sigma_i)_{i \in I}$, the sets M_i and (a)–(d). The index set of the branching structure is the set of branches through $f(y)$, $\{i \in I : s_i(y) = f(y)\}$, so that every M_j contains y, as Definition 9.1.2 asks; the submanifold property and properness are (a) and (b); the weights are positive. The count is computed in three rewrites: by (d), $\Lambda(f(z)) = \sum_{i \in I} [f(z) = s_i(z)]\, \sigma_i$; by the last clause of Theorem 15.2.2 every branch with $f(z) = s_i(z)$ for some $z \in U(y)$ passes through $f(y)$, so the sum may be restricted to the branches through $f(y)$; and that sum is the sum over the index set of the structure, after reindexing by the subtype.

Global form (narrower) `Chapter15/Transversality · SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp`. With tame germ data: if (f, Λ) is transversal, $d_X = 0$ on $\operatorname{supp}(\Lambda \circ f)$, and the orbit space $|\operatorname{supp}(\Lambda \circ f)|$ is paracompact, then $\Lambda \circ f$ is a branched ep⁺-subgroupoid of X (Definition 9.1.2). *Proof*, field by field. The values are non-negative rationals. Functoriality: for a morphism φ, $f(t(\varphi)) = \mu(\varphi, f(s(\varphi)))$ because f is a section functor, and $\Lambda(\mu(\varphi, w)) = \Lambda(w)$ because Λ is a functor on W; so $\Lambda(f(t(\varphi))) = \Lambda(f(s(\varphi)))$. The support consists of smooth points (§XV.12) `Chapter15/Transversality · SBLocalization.mem_smoothSet_of_pos`. The paracompactness of the orbit space of the support, Definition 9.1.2 (1), is the hypothesis. The local branching structures at the support points are the local form. *Narrower* also by that hypothesis: Definition 9.1.2 (1) is new in the 2021 edition, and the remark, being local, does not address it. *Witness* `Chapter15/InteriorPerturbationWitness · SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp_witness`: the example of §XV.12, with the paracompactness of the orbit space of any subset of the trivial ep-groupoid of $\mathbb{R}^n$ from Part IX `Chapter9/Propositions · EpGroupoid.paracompactSpace_orbitSupp_trivial_rn`.

"Tame". A tame local branching structure (Definition 9.3.6) is a local branching structure whose domain is a natural representation with the properness property (Definition 7.5.8) and whose submanifolds M_j have tame induced structures (Definition 2.5.18). At an interior point of transversality (f, Λ) is in good position (Definition 15.2.4; the lemma is in §XV.15), and at a point of good position Theorem 15.2.5 produces a tame local branching structure (§XV.16). Hence *the local tame form* (narrower) `Chapter15/TameBranched · SBLocalization.remark_15_2_3_tame_hyp`: at y with $d_X(y) = 0$ and (f, Λ) transversal at y, $z \mapsto \Lambda(f(z))$ has a tame local branching structure at y; and *the global tame form* (narrower) `Chapter15/TameBranched · SBLocalization.remark_15_2_3_isTameBranched_hyp`: under the hypotheses of the global form, $\Lambda \circ f$ is a branched ep⁺-subgroupoid and tame, by the global form of Theorem 15.2.5 (§XV.16) applied with good position at every support point. *Witness* `Chapter15/TameWitness · SBLocalization.remark_15_2_3_isTameBranched_hyp_witness`: the example of §XV.12, with support all of $\mathbb{R}^n$.

The ep-groupoid $U(x)$. *Reading.* "The ep-groupoid $U(x)$" is read as the translation groupoid $G_x \ltimes U(x)$ of the natural action of the isotropy group on a natural representation, whose objects are the points of $U(x)$ and whose morphisms are the pairs (g, y) from y to $g{*}y$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.`

translationEpGroupoid. Part VII proves that it is an ep-groupoid and that the functor $\Gamma : G_x \ltimes U(x) \to X$ is fully faithful onto the full subcategory of X on $U(x)$ (Theorem 7.1.13 (3)), with the same orbit space (Proposition 7.1.19). So the full subcategory on $U(x)$ is an ep-groupoid in this form. The restriction of a functor Θ on X to it is $\Theta_U(y) = \Theta(y)$, and "with support $M_{U(x)}$" is then the definition of the support of Θ_U.

The generic step (extra) Chapter15/RestrictedEp · EpGroupoid.NaturalRep.isTameBranched_translation. For any ep-groupoid X, natural representation $U(x)$ and functor $\Theta \geq 0$ on X with closed support, such that every point of $\operatorname{supp}\Theta \cap U(x)$ is smooth and carries a tame local branching structure of Θ, the restriction Θ_U is a branched ep⁺-subgroupoid of $G_x \ltimes U(x)$, and tame. Unlike the global form on X, no paracompactness is assumed: on $G_x \ltimes U(x)$ it is proved. *Proof*, field by field.

- *Functoriality.* A morphism (g, y) of the translation groupoid is carried by Γ to a morphism of X with source y and target $g * y$, so $\Theta(g * y) = \Theta(y)$ by functoriality of Θ on X.
- *Smoothness.* A point of the open sub-M-polyfold $U(x)$ is smooth there iff it is smooth in X.
- *Paracompactness of the orbit space of the support* Chapter15/RestrictedEp · EpGroupoid.NaturalRep.paracompactSpace_orbitSupp_translation. Let $S = \operatorname{supp}\Theta_U$, closed in $U(x)$, hence paracompact, and let $p : S \to |S|$ be the orbit map restricted to it, continuous and onto. Two facts about the orbit map π of $G_x \ltimes U(x)$. It is closed Chapter15/RestrictedEp · EpGroupoid.NaturalRep.isClosedMap_translation_π: every closed subset C of $U(x)$ is a proper subset of the translation groupoid in the sense of Definition 7.5.8 Chapter15/RestrictedEp · EpGroupoid.NaturalRep.translation_isProperSubset, because Part VII proves that the restriction of the source–target map to the morphisms over the closure of any subset is proper Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isProperMap, and the orbit of a proper subset is closed (Proposition 7.5.10 (1)) Chapter7/ProperSubsets · EpGroupoid.proposition_7_5_10_one. Its fibres are finite Chapter15/RestrictedEp · EpGroupoid.NaturalRep.finite_fiber_translation_π: two points of $U(x)$ have the same orbit iff they differ by the action of an element of G_x, so a fibre is the image of the finite group G_x under $g \mapsto g * y$. Restricted to the closed set S, p is again closed with finite fibres. Now Michael's criterion (Lemma 2.7.3) Chapter7/Michael · paracompactSpace_of_closed_refinement: a regular space is paracompact if every open cover has a locally finite closed refinement that covers. Given an open cover of $|S|$, pull it back to S, take a locally finite closed precise refinement there Chapter7/Michael · exists_closed_precise_refinement, and push it forward by p. The images are closed (p closed), cover (p onto), refine the cover, and form a locally finite family by the next lemma.
- *Closed maps with finite fibres* Chapter15/RestrictedEp · locallyFinite_image_of_isClosedMap: a closed map p with finite fibres carries a locally finite family (v_i) to a locally finite family $(p(v_i))$. *Proof.* Fix y. Each point z of the finite fibre $p^{-1}(y)$ has a neighbourhood N_z meeting only finitely many v_i. The union W of the interiors of the N_z is open and contains the fibre, so $y \notin p(W^c)$, and $(p(W^c))^c$ is an open neighbourhood of y, p being closed. If $p(v_i)$ meets it, at a point $p(z)$ with $z \in v_i$, then $z \notin W^c$, so z lies in some $N_{z'}$ and v_i meets $N_{z'}$; there are finitely many such i for each of the finitely many z'.
- *Tame local branching structures move into $U(x)$* Chapter15/RestrictedEp · EpGroupoid.NaturalRep.nonempty_tameBranching_translation. Let C be a tame local branching structure of Θ at $y \in U(x)$, with domain $C.U$. Theorem 7.1.13 for the translation groupoid Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13 gives a natural representation U' of $G_x \ltimes U(x)$ at y inside the preimage of $C.U$; its image in X is open and inside $C.U \cap U(x)$. The structure C is restricted to this image (the M_j intersected with an open set stay submanifolds with tame induced structures), and moved into the M-polyfold $U(x)$ by the inclusion $U(x) \to X$, an sc-diffeomorphism onto its open image Chapter8/TangentLocal · MPolyfold.

`isScDiffeoOn_subtypeVal`; images of tame induced structures under sc-diffeomorphisms are tame. The properness property of U' inside the translation groupoid holds for every open subset of $U(x)$ `Chapter15/RestrictedEp · EpGroupoid.NaturalRep.translation_hasPropernessProperty`, since every closed subset is proper there.

Remark 15.2.3 on the ep-groupoid $U(x)$ (narrower) `Chapter15/RestrictedEp · SBLocalization.remark_15_2_3_restricted_hyp`. With tame germ data: at x with $d_X(x) = 0$ and (f, Λ) transversal at x, there is a natural representation $U(x)$ with $d_X \equiv 0$ on it such that $z \mapsto \Lambda(f(z))$ is a branched ep$^+$-subgroupoid of $G_x \ltimes U(x)$, and tame. *Proof.* Theorem 15.2.2 at x gives $U(x)$ with (1) and (2). Apply the generic step to $\Theta = \Lambda \circ f$: it is ≥ 0 and a functor `Chapter15/GeneralPositionBranched · SBLocalization.isFunctorial_comp`; its support is closed in X `Chapter15/GeneralPositionBranched · SBLocalization.isClosed_supp_comp`, being locally the finite union of the closed sets $\{f = s_i\}$ (§XV.21); its support points are smooth (§XV.12); and at a support point $y \in U(x)$, $d_X(y) = 0$ by (1) and (f, Λ) is transversal at y by (2), so the local tame form above gives a tame local branching structure at y. *No paracompactness* is assumed: the remark holds on $U(x)$ exactly as stated, with the inherited tame germ data. *Witness* `Chapter15/TameWitness · SBLocalization.remark_15_2_3_restricted_hyp_witness`: all hypotheses together at $0 \in \mathbb{R}^1$ in the example of §XV.12, with the conclusion.

XV.15 Definition 15.2.4 (2017: Definition 15.3): good position

At a point with $d_X(x) \geq 1$ transversality is not enough: the kernels must also sit well against the boundary. The text calls Definition 15.2.4 "a version of Definition 3.6.6"; the formalization makes it exactly that, branch by branch.

Statement (restated). *One branch* `Chapter15/BoundaryPerturbation · SBLocalization.IsBranchGoodPosition`: for s sc$^+$ on an open $O \ni x$, the branch $f - s$ is in good position at x if the section $(f - s)|_O$ of $W|_O$ is in good position at x in a tame chart, Part I's Definition 3.6.6 in its tame germ reading: germ data at x in a tame chart of $W|_O$, a linearization at x that is onto, and a kernel in good position to the partial quadrant (Definition 3.1.24). *The pair* `Chapter15/BoundaryPerturbation · SBLocalization.IsGoodPositionAt`: (f, Λ) is in good position at x if $\Lambda(f(x)) > 0$, $T(f, \Lambda)(x)$ is onto (Definition 15.2.1), and there is a local sc$^+$-section structure (s_i, σ_i) of Λ on an open $O \ni x$ such that every branch with $s_i(x) = f(x)$ is in good position at x.

Readings. (i) The text's (1), "X is tame near x", and (3), "the kernels of the L_i are in good position to $C_x X$", are read together per branch as Part I's good position of $(f - s_i)|_O$, which carries a tame chart at x (giving (1)) and the kernel condition (giving (3)). (ii) The condition is asked for one local structure; the text does not say whether it depends on the structure chosen. Theorem 15.2.5 below propagates it with the same structure. (iii) The text assumes $x \in \partial X$; the formal definition does not, and everything below holds at any point where the condition holds.

At an interior point, transversality is good position (extra) `Chapter15/TameBranched · SBLocalization.isGoodPositionAt_of_interior`: if $d_X(x) = 0$ and (f, Λ) is transversal at x (f with tame germ data), then (f, Λ) is in good position at x. *Proof.* Take the structure (s_i, σ_i) on a natural representation $U_0(x)$ and a chart c inside it, as in Step 1 of §XV.13. For a branch through $f(x)$, transversality and the text's reading give the chart operator onto at x (Step 2 of §XV.13); this is Part I's surjectivity of the linearization of $(f - s_i)|_{U_0(x)}$ at x `Chapter15/LocalIFT · SBLocalization.isLinSurjAt_subRestrict` (§XV.10). The degeneracy index of x in the open M-polyfold $U_0(x)$ is that in X, so 0. Part I: at a zero with $d = 0$, an onto linearization is good position in a tame chart `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`, since at an interior point the partial quadrant is the whole space and the kernel condition is empty. So

every branch through $f(x)$ is in good position, which is Definition 15.2.4.

This lemma is what lets the tame forms of Remark 15.2.3 (§XV.14) be derived from those of Theorem 15.2.5 (§XV.16).

Witness `Chapter15/GoodPositionWitness · SBLocalization.theorem_15_2_5_hyp_witness`: in the example of §XV.12, (f, Λ_0) is in good position at every point of $\mathbb{R}^n$. *Proof.* There every pair is in general position (Definition 15.3.5), the fibres being 0 `Chapter15/GeneralPositionBranched · SBLocalization.isInGeneralPosition_epSEP_zero`, and general position gives good position at every point of the support (Remark 15.3.6, §XV.20) `Chapter15/GoodPositionBranches · SBLocalization.remark_15_3_6_goodPosition_hyp`. Zero fibres: a consistency witness.

XV.16 Theorem 15.2.5 (2017: Theorem 15.3) and Remark 15.2.6: boundary local perturbation, tame, and under general position

Statement (narrower) `Chapter15/BoundaryPerturbation · SBLocalization.theorem_15_2_5_hyp`. Let f be an sc-Fredholm section functor with tame germ data, Λ an sc$^+$-multisection, and (f, Λ) in good position at x. Then there are a natural representation $U(x)$, a local sc$^+$-section structure (s_i, σ_i) on an open O and a chart c with $U(x) \subset \mathrm{dom}(c) \subset O$ such that, with $M_i = \{y \in U(x) : f(y) = s_i(y)\}$: every M_i is a smooth submanifold (Definition 4.2.1); (1) the inclusions $M_i \hookrightarrow U(x)$ are proper; $\{y \in U(x) : \Lambda(f(y)) > 0\} = \bigcup_i M_i$; (2) (f, Λ) is in good position, with the same structure, at every $y \in U(x)$ with $\Lambda(f(y)) > 0$, so $T(f, \Lambda)(y)$ is onto with the branches in good position; (3) $\Lambda(f(y)) = \sum_{i : y \in M_i} \sigma_i$ on $U(x)$; $T(f, \Lambda)(y) = \sum_{i : f(y) = s_i(y)} \sigma_i \cdot (f - s_i)'(y)$ in c; and only the branches through $f(x)$ meet f in $U(x)$. "Tame submanifolds" and "tame branched" are the tame form below. *Narrower* by the inherited tame germ data; $x \in \partial X$ is not required.

Proof. The text: "along the lines of the proof of Theorem 15.2.2", with Part I's implicit function theorem in good position. The Lean proof is that proof with the good-position lemma in place of the interior one.

Step 1. Good position at x supplies the structure (s_i, σ_i) on an open $O \ni x$ in which the branches through $f(x)$ are in good position. Take a chart c_0 around x and restrict it to $\mathrm{dom}(c_0) \cap O$: a chart c with domain in O. A natural representation $U_0(x)$ is taken from the multisection property of Λ; only its existence is used, for the shrinking in Step 3.

Step 2, one branch `Chapter15/BoundaryPerturbation · SBLocalization.exists_branch_goodPosition`: for each i an open $V_i \ni x$ in $\mathrm{dom}(c)$ with $\{y \in V_i : f(y) = s_i(y)\}$ a smooth submanifold, each of its points smooth, with the branch in good position there and the chart operator onto there, and only meeting f if $s_i(x) = f(x)$. If the branch passes through $f(x)$ this is the good-position local solution lemma of §XV.10 `Chapter15/LocalIFT · SBLocalization.exists_solutionSet_of_isGoodPosition`: Part I's Theorem 3.6.8 for $(f - s_i)|_O$ at x in good position, which gives an open set of the zeros of which each is in good position, read back in X; the chart surjectivity at those zeros comes from the good position there. If it does not, $V_i = \{y \in \mathrm{dom}(c) : f(y) \neq s_i(y)\}$, as in §XV.13 `Chapter15/Transversality · SBLocalization.isOpen_sep_ne`.

Step 3. Shrink $U_0(x)$ into the open set $\mathrm{dom}(c) \cap \bigcap_i V_i$ `Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_shrink`. No condition on d_X enters.

Step 4. The count (3), the description of the support, the submanifolds M_i, the properness (1) and the formula for $T(f, \Lambda)$ are proved word for word as Steps 5, 7 and 8 of §XV.13.

Step 5, (2). At $y \in U(x)$ with $\Lambda(f(y)) > 0$: some branch passes through $f(y)$, so y is smooth, hence at

level 1, by Step 2. The structure on O and the chart c are local data at y; every branch j through $f(y)$ has its chart operator onto at y (Step 2), so $T(f,\Lambda)(y)$ is onto by the text's reading of §XV.12 `Chapter15/LocalIFT · SBLocalization.surjective_toIntr_subDeriv_iff`; and every branch j through $f(y)$ is in good position at y (Step 2). These are the three clauses of Definition 15.2.4 at y, with the same structure.

Tame local solution sets (extra) `Chapter15/TameBranched · SBLocalization.exists_tame_solutionSet_of_isBranchGoodPosition`. If a branch $f - s$ with $f(x) = s(x)$ is in good position at x, there is an open $V \ni x$ inside O such that $\{y \in V : f(y) = s(y)\}$ is a smooth submanifold of X with a tame induced structure (Definition 2.5.18). *Proof.* Part I's Theorem 3.1.26, the implicit function theorem in the boundary case, for the tame sc-Fredholm section $(f-s)|_O$ at its zero x in good position `Chapter3/GlobalIFTNearbyOntoG · theorem_3_1_26'`, gives an open $U \subset O$ around x whose zeros are smooth and in good position and whose zero set is a sub-M-polyfold with a tame induced structure. The zero set in U is a smooth submanifold of O `Chapter5/GeneralPosition · StrongBundle.isSmoothSubmanifold_solutionSet_inter`, the zeros of $(f-s)|_O$ being the points with $f = s$ `Chapter15/LocalScPlus · SBLocalization.isZeroAt_subSec_iff`. Both properties are carried to X by the sc-diffeomorphism $O \to X$ onto the open set O `Chapter8/TangentLocal · MPolyfold.isScDiffeoOn_subtypeVal`; V is the image of U.

Theorem 15.2.5, tame (narrower) `Chapter15/TameBranched · SBLocalization.theorem_15_2_5_tame_hyp`. In good position at x, $z \mapsto \Lambda(f(z))$ has a tame local branching structure at x (Definition 9.3.6): a natural representation $U(x)$ with the properness property, the sets $M_j = \{z \in U(x) : f(z) = s_j(z)\}$ over the branches through $f(x)$, each a smooth submanifold with a tame induced structure and a proper inclusion, the weights σ_j, and the count. *Proof.* For every branch choose V_i: the tame local solution set above if the branch passes through $f(x)$, and $\{y \in O : f(y) \neq s_i(y)\}$ otherwise. Every object has a neighbourhood Q with the properness property (Definition 7.5.8) `Chapter7/EpGroupoid · EpGroupoid.exists_hasPropernessProperty`. Theorem 7.1.13 `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_13` gives a natural representation $U(x)$ inside the open set $Q \cap O \cap \bigcap_i V_i$; it has the properness property, its closure lying in that of Q. The M_j are the intersections of the tame solution sets with the open $U(x)$, which keeps both the submanifold property and the tame induced structure; properness and the count are as in the local form of Remark 15.2.3 (§XV.14), with the index set the branches through $f(x)$. This is the text's "by definition $\Lambda \circ f$ … is a tame branched ep⁺-subgroupoid, since it is represented by tame submanifolds".

The global tame form (narrower) `Chapter15/TameBranched · SBLocalization.theorem_15_2_5_isTameBranched_hyp`: if (f,Λ) is in good position at every point of the support and $|\operatorname{supp}(\Lambda \circ f)|$ is paracompact, then $\Lambda \circ f$ is a branched ep⁺-subgroupoid of X and tame; the fields are those of the global form of Remark 15.2.3 (§XV.14), with functoriality `Chapter15/GeneralPositionBranched · SBLocalization.isFunctorial_comp` and the tame local branching structures just proved. The same assembly gives the global tame form of Remark 15.2.3 (§XV.14), and, through good position from general position (Remark 15.3.6), the tame forms of Remark 15.3.6 `Chapter15/TameBranched · SBLocalization.remark_15_3_6_isTameBranched_hyp` and of the "in particular" of Theorem 15.3.7 `Chapter15/TameBranched · SBLocalization.theorem_15_3_7_in_particular_tame_hyp` (§§XV.20–XV.21).

On the ep-groupoid $U(x)$ (narrower) `Chapter15/RestrictedEp · SBLocalization.theorem_15_2_5_restricted_hyp`: in good position at x, there is a natural representation $U(x)$ on whose translation groupoid $G_x \ltimes U(x)$ the functor $z \mapsto \Lambda(f(z))$ is a branched ep⁺-subgroupoid, and tame. *Proof.* Theorem 15.2.5 gives $U(x)$ with good position at every support point in it; the generic step of §XV.14 applies, the tame local branching structures at those points coming from the tame form above. No paracompactness is assumed.

The closing sentence: general position. The text: the same conclusion holds if (2) is replaced by the requirement that the restrictions of $T(f,\Lambda)(y)$ to the reduced tangent spaces $T^R_y X$ are surjective, pointing

ahead to Definition 15.3.5.

Definition 15.3.5 at one point Chapter15/GeneralPositionLocal · SBLocalization.IsReducedOntoAtLevel Chapter15/GeneralPositionLocal · SBLocalization.IsInGeneralPositionAt: (f, Λ) is in general position at x if $x \in X_1$ and every operator L of $T(f, \Lambda)(x)$, restricted to $T^R_x X$, is onto W_x, with $T^R_x X$ read in every chart around x and every presentation of its partial quadrant (Part I's reduced tangent fibre). Definition 15.3.5 (§XV.20) is this condition at every point of the support Chapter15/GeneralPositionLocal · SBLocalization.isInGeneralPosition_iff_forall_at, the support lying in X_1 Chapter15/ControllingPairs · SBLocalization.mem_levelSet_one_of_pos.

Reading. "(2) replaced by general position" is read: the hypothesis "good position at x" is replaced by "general position at x", and the conclusion (2) by "general position at every point of $M_{U(x)}$". Good position at x and on $M_{U(x)}$ still holds, so the formal conclusion gives both forms of (2).

Four lemmas connect general position of the pair to Part I's general position of the branches (Definition 5.3.9 for $(f - s_i)|_O$: the linearization at the zero is onto, with kernel transversal to T^R).

- *Part I, in every chart* (extra) Chapter15/GeneralPositionLocal · StrongBundle.linKer_sup_reduced_eq_of_isGeneralPositionAt: if a smooth submanifold A of the solution set of a section f is in general position at y (Definition 4.2.10), then in every strong bundle chart around y, at every level-1 point z over $\varphi(y)$ and for every presentation of the partial quadrant, $\ker \mathrm{Lin}(z) + T^R_z O = T_z O$. *Proof.* General position of A gives, in the chart, a splitting $T_z O = N \oplus B$ with N the tangent space of A and $B \subset T^R_z O$. The tangent space of a submanifold of zeros lies in the kernel of the linearization, so $N \subset \ker \mathrm{Lin}(z)$; hence $T_z O = N + B \subset \ker \mathrm{Lin}(z) + T^R_z O \subset T_z O$. This is the converse of Part I's chart criterion Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_of_sup_eq, and it holds in every chart, not only in the chart of the germ data.
- *From the pair to the branches* Chapter15/GeneralPositionLocal · SBLocalization.isGeneralPositionGerm_branch_at: if (f, Λ) is in general position at x, every branch $(f - s_i)|_O$ through $f(x)$ of any local structure on an open $O \ni x$ is in Part I's general position at x. *Proof.* The point x is smooth (§XV.12) Chapter15/Transversality · SBLocalization.mem_smoothSet_of_eq, so the tame germ data of f provide a tame chart centred at x; restrict it to O. With the structure it is local data at x, so $(f - s_i)'(x)$ through this chart lies in the support of $T(f, \Lambda)(x)$ Chapter15/Transversality · SBLocalization.mem_support_linSumIntr_iff and is onto on $T^R_x X$ by hypothesis. The chart step of Remark 15.3.6 (§XV.20) Chapter15/GoodPositionBranches · SBLocalization.isGeneralPositionGerm_subRestrict turns this into Part I's general position of $(f - s_i)|_O$: in the transported chart of $W|_O$ the intrinsic operator is the chart derivative of the branch, onto from $T_0 O \cap T^R_0$ means onto with $\ker + T^R_0 O = T_0 O$, which is Part I's chart criterion.
- *Good position from general position* Chapter15/GeneralPositionLocal · SBLocalization.isGoodPositionAt_of_isInGeneralPositionAt: general position at x with $\Lambda(f(x)) > 0$ gives good position at x. *Proof.* $T(f, \Lambda)(x)$ is onto, because each operator is already onto on the subspace $T^R_x X$, read in some chart with a presentation Chapter15/ControllingPairs · SBLocalization.exists_chart_presentation. Take any local structure on an open $O \ni x$ Chapter13/MultisectionSupport · SBLocalization.IsScPlusMultisection.exists_isLocalRepOn; each branch through $f(x)$ is in Part I's general position (previous lemma), hence in good position in a tame chart, by Part I's Lemma 5.26 Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGermTame, the branch being a tame sc-Fredholm section of $W|_O$ with x a zero.
- *From the branches back to the pair* Chapter15/GeneralPositionLocal · SBLocalization.isReducedOntoAtLevel_of_isGeneralPositionGerm: if, for a local structure on $O \ni y$ and a chart c around y with domain in

O, every branch through $f(y)$ is in Part I's general position at y, then (f,Λ) is in general position at y. *Proof.* The reduced tangent fibre does not depend on the chart and presentation `Chapter2/ReducedTangentIntrinsic · reducedTangentFiber_eq`, so it suffices to work in c with one presentation. An operator of the support is $(f-s_j)'(y)$ for a branch j through $f(y)$ (§XV.12). Its reduced restriction is onto `Chapter15/GeneralPositionLocal · SBLocalization.surjective_reduced_toIntr_of_isGeneralPositionGerm`, by the chart lemma: *every vector k of the fibre K_z has a preimage under the chart operator in $T^R_zO \cap T_zO$* `Chapter15/GeneralPositionLocal · SBLocalization.exists_reduced_preimage_of_isGeneralPositionGerm`. For this lemma, Part I's general position of $(f-s_j)|_O$ gives good position, hence a smooth solution submanifold in general position at y `Chapter5/GeneralPosition · exists_isGeneralPositionGerm_iff`; the first lemma gives $\ker +T^R = T$ in the restricted chart; the linearization is onto, so k has a preimage $v = a+b$ with a in the kernel and $b \in T^R$, and b is a preimage; and the derivative of the restricted chart is that of c `Chapter15/LocalIFT · SBLocalization.scDeriv_restrictTo_subRestrict`. Passing from the chart operator to the intrinsic one has two cases, as the intrinsic operator is defined by cases: if the chart operator takes values in the image of the fibre identification, the preimage b is carried to a reduced tangent vector with the right image; otherwise the intrinsic operator is 0, and the full operator being onto (§XV.10) forces $W_y = 0$, where any operator is onto.

Theorem 15.2.5, closing sentence (narrower) `Chapter15/GeneralPositionLocal · SBLocalization.theorem_15_2_5_generalPosition_hyp`. Let X be tame, f with tame germ data, $\Lambda(f(x)) > 0$ and (f,Λ) in general position at x. Then all conclusions of Theorem 15.2.5 hold on a natural representation $U(x)$, and (f,Λ) is in general position and in good position at every $y \in U(x)$ with $\Lambda(f(y)) > 0$. *Proof. Step 1.* General position at x gives good position at x, and Theorem 15.2.5 gives $U_1(x)$, the structure on O, the chart and all conclusions. *Step 2, openness of general position.* O is tame, an open subset of the tame X `Chapter11/FaceIntersectionCounterexample · MPolyfold.isTame_restrictOpen`. For each branch through $f(x)$, Part I's general position of $(f-s_i)|_O$ at x (the second lemma) is open along the solution set: Part I's Theorem 4.2.11 (3), in the form that a zero in general position of a tame sc-Fredholm section over a tame M-polyfold has an open neighbourhood W_i at whose zeros the section is in general position `Chapter5/GeneralPosition · exists_isOpen_isGeneralPositionGerm`. For the other branches $W_i = X$. *Step 3.* Shrink $U_1(x)$ into $\bigcap_i W_i$ `Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_shrink`; the conclusions of Theorem 15.2.5 survive the shrinking, as in Step 7 of §XV.13. *Step 4.* At a support point $y \in U(x)$, every branch through $f(y)$ passes through $f(x)$ (last clause of Theorem 15.2.5) and $y \in W_i$, so it is in Part I's general position at y; the fourth lemma gives general position of the pair at y, and good position is clause (2) of Theorem 15.2.5. *The hypothesis of tameness* is that of Definition 15.3.5, which the text states over tame ep-groupoids, and it is what Part I's openness of general position needs. *Witness* `Chapter15/GoodPositionWitness · SBLocalization.theorem_15_2_5_generalPosition_hyp_witness`: the example of §XV.12, where the trivial ep-groupoid of $\mathbb{R}^n$ is tame `Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn` and every pair is in general position `Chapter15/GeneralPositionBranched · SBLocalization.isInGeneralPosition_epSEP_zero`. Zero fibres: a consistency witness.

Witnesses for Theorem 15.2.5. The hypotheses in good position hold together at every point of the example of §XV.12 (§XV.15) `Chapter15/GoodPositionWitness · SBLocalization.theorem_15_2_5_hyp_witness`, and the conclusion is obtained there, with good position on all of $U(x)$ `Chapter15/GoodPositionWitness · SBLocalization.theorem_15_2_5_hyp_applies`. The example has no boundary and zero fibres, so it does not exercise the corner geometry for which Theorem 15.2.5 is made; Part I's Theorem 3.1.26, on which the tame form rests, has its own witnesses in Part III.

Remark 15.2.6 (pp. 690–691; nothing to formalize). Prose: with an oriented (f,o) the branches M_i inherit orientations $o_{i,y}$ from the surjective oriented $(f-s_i)'(y)$, recorded in the formal sum $\widehat{T}_{((f,o),\Lambda)}(y) = \sum \sigma_i \cdot (T_yM_i, o_{i,y})$. The local data it needs are in place: the M_i of Theorems 15.2.2 and 15.2.5 and the

surjective $(f - s_i)'(y)$ of (2) with the text's reading of §XV.12. The orientations themselves are Theorem 15.4.1 and Definition 15.4.2, formalized at a single point in §XV.34.

XV.17 The paragraph before Definition 15.3.1, Definition 15.3.1 and Remark 15.3.2 (2017: Definition 15.4, Remark 15.4)

§15.3 opens (p. 691) with the standing data of a perturbation problem: a strong bundle $(P : W \to X, \mu)$ over an ep-groupoid, an sc-Fredholm section functor f, and an auxiliary norm N (Definition 12.2.1, which is Definition 8.3.5). An auxiliary norm is defined on $W_{0,1}$; the text extends it by $N(h) = +\infty$ off $W_{0,1}$, so that "$N(f(x)) \le 1$" makes sense for every x. The formalization does not extend N. It keeps N on the level-$(0,1)$ part $W[1]$ and reads "$N(h) \le r$" as Part I's relation AuxLe: $h \in W_{0,1}$ and the value of N at h is at most r `Chapter5/Compactness · StrongBundle.AuxLe`. This is exactly the text's convention, a point with $N(h) = +\infty$ satisfying no inequality $N(h) \le r$.

Definition 15.3.1 (restated) `Chapter15/ControllingPairs · SBLocalization.ControlsCompactness`. *(N, U) controls the compactness of f* if

1. N is an auxiliary norm of (W, μ) (Definition 8.3.5: an auxiliary norm of the strong bundle, invariant under μ);
2. $U \subset X$ is open;
3. U is saturated;
4. $S_f = \{x : f(x) = 0_x\} \subset U$;
5. the closure in $|X|$ of $\pi(\{x \in U : N(f(x)) \le 1\})$ is compact, $\pi : X \to |X|$ the orbit map.

Readings. The orbit space "of a set" $A \subset X$ is its image $\pi(A)$. The text places the definition over an ep-groupoid with paracompact $|X|$. Paracompactness is not part of the formal definition, which is therefore more general; the statements that use paracompactness (the structurable addendum of Theorem 15.3.7, Theorem 15.3.9, Corollary 15.3.10) carry it as a hypothesis of their own. The text calls f an sc-Fredholm section functor; the formal predicate takes any f, and the theorems that use it supply the Fredholm property.

Monotonicity `Chapter15/ControllingPairs · SBLocalization.ControlsCompactness.mono`. If (N, U) controls the compactness of f and $U' \subset U$ is a smaller saturated open set still containing S_f, then (N, U') controls it. *Proof.* Clauses (1)–(4) for U' are the hypotheses. For (5): $\{x \in U' : N(f(x)) \le 1\} \subset \{x \in U : N(f(x)) \le 1\}$, so the closure of its image lies in the closure of the larger image, a compact set; a closed subset of a compact set is compact. The lemma is used wherever a proof shrinks the controlling neighbourhood.

The paragraph before Definition 15.3.1 (same) `Chapter15/ControllingPairs · SBLocalization.exists_controlsCompactness`. *Statement.* For an sc-Fredholm section functor f with $|S_f| = \pi(S_f)$ compact and any auxiliary norm N, there is a U with (N, U) controlling the compactness of f. The text says this follows from "the discussion in Section 12.4".

Proof. It is Theorem 12.4.2, the implication (2) $\Rightarrow$ (3), of Part XII: an sc-Fredholm section functor with compact solution set is proper$_{<\infty}$, that is, for every auxiliary norm N there is a saturated open $U \supset S_f$ such that $\mathrm{cl}_{|X|}\,\pi(\{x \in U : N(f(x)) \le C\})$ is compact for every $0 < C < \infty$ `Chapter12/CompactnessEp · EpStrongBundle.isProperLtInftyFunctor_of_hasCompactSolutionSetEp` `Chapter12/CompactnessEp · EpStrongBundle.IsProperLtInftyFunctor`. The formal proof then takes $C = 1$. The proof of the

implication, as Part XII retells it, runs in four steps, and we recall it because it is the whole content of the paragraph.

Step 1, the local step. At each zero x of f the local step of Theorem 5.2.2 gives an open $V(x) \ni x$ such that $\mathrm{cl}\{y \in V(x) : N(f(y)) \le C\}$ is compact for every $C > 0$ `Chapter12/CompactnessEp · exists_nhds_isCompact_closure_auxLe`; this uses only that f, as a section of the object M-polyfold, is sc-Fredholm.

Step 2, a finite cover. The orbit map is open (Proposition 7.1.8, Part VII) `Chapter7/Basic · EpGroupoid.proposition_7_1_8`, so the sets $\pi(V(x))$, $x \in S_f$, are an open cover of the compact $|S_f|$; finitely many, $\pi(V(x_1)), \dots, \pi(V(x_m))$, suffice. Let O be their union and $U = \pi^{-1}(O)$: it is open (continuity of π), saturated (a preimage under π), and contains S_f.

Step 3, invariance. If $y \in U$ with $N(f(y)) \le C$, then $\pi(y) = \pi(y')$ for some $y' \in V(x_i)$, that is, there is a morphism $\varphi : y \to y'$. Since f is a section functor and N is μ-invariant, $N(f(y')) = N(\mu(\varphi, f(y))) = N(f(y)) \le C$ `Chapter12/CompactnessEp · EpStrongBundle.auxLe_tgt_of_auxLe_src`.

Step 4, compactness. Hence $\pi(\{y \in U : N(f(y)) \le C\})$ lies in $K = \bigcup_i \pi(\mathrm{cl}\{y \in V(x_i) : N(f(y)) \le C\})$, a finite union of continuous images of compact sets, compact, and closed because $|X|$ is Hausdorff. The closure of the image lies in K and is compact.

Departure from the text. The text's proof of Theorem 12.4.2 shrinks to neighbourhoods with the natural action and uses $\mathrm{cl}_{|X|}|U| = |\mathrm{cl}_X U|$; neither is needed, because the closure lies in the finite union K. The paracompactness of $|X|$, which Theorem 12.4.2 assumes, is not used `Chapter12/CompactnessEp · EpStrongBundle.theorem_12_4_2`.

Remark 15.3.2 (nothing to formalize). The remark says that the construction of perturbations needs sc-smooth bump functions in general, and that geometric constructions may replace them at the price of checking compactness by hand. It is a comment. In the formalization bump functions enter as a hypothesis exactly where perturbations are built: Theorem 15.3.7 assumes that X admits sc-smooth bump functions in the pointwise reading (§XV.21), and the cylinder $[0,1] \times X$ of Corollary 15.3.10 inherits them from X `Chapter15/CubeEp · EpGroupoid.admitsBumpFunctionsPointwise_cubeProd` (§XV.30). The sentence after Definition 15.3.1, "we also assume that X admits sc-smooth bump functions", is likewise read as a hypothesis of the theorems that construct perturbations, not as part of any definition.

XV.18 Definition 15.3.3 (2017: 15.5) and the example after it

Definition 15.3.3 (same) `Chapter15/ControllingPairs · SBLocalization.IsAdmissible`. An (N, U)-*admissible perturbation* for f is a function $\Lambda : W \to \mathbb{Q}^+$ such that

1. Λ is an sc$^+$-multisection (Definition 13.2.1, in the 2021 form with local section structures, Part XIII);
2. $N(\Lambda)(x) \le 1$ for every $x \in X$, where $N(\Lambda)(x)$ is the pointwise norm of Definition 13.2.3, the largest value of N on the finite set $\mathrm{supp}(\Lambda)(x) = \{w \in W_x : \Lambda(w) > 0\}$ `Chapter13/MultisectionSupport · SBLocalization.pointwiseNorm`;
3. $\{x \in X : \exists t \in [-1,1],\ (t\Lambda)(f(x)) > 0\} \subset U$, where $t\Lambda$ is the scalar multiple of Definition 13.2.10: $(0\Lambda)(w) = 1$ if w is a zero vector and 0 otherwise, and $(t\Lambda)(w) = \Lambda(w/t)$ for $t \neq 0$ `Chapter13/MultisectionAlgebra · SBLocalization.multisectionSmul`.

Readings. The text's standing data, f sc-Fredholm with compact $|S_f|$ and (N, U) controlling compactness, are not part of the predicate; they are hypotheses of the theorems that use it (Theorem 15.3.9, the

boundary conditions of Corollary 15.3.10). The pointwise norm is defined as a supremum; for an sc$^+$-multisection it is attained, the support over a point being finite and nonempty, and the points of the support lie in $W_{0,1}$, where N is defined (Part XIII). Condition (3) with $t = 0$ asks that $S_f \subset U$, since $(0\Lambda)(f(x)) > 0$ exactly when $f(x) = 0_x$; with $t = \pm 1$ it asks that the support of $\Lambda \circ f$ and of $(-\Lambda) \circ f$ lie in U. The interval $[-1, 1]$ is what the cobordism argument of Corollary 15.3.10 (3) needs: perturbations $t\Lambda$ along a path stay inside U.

The example after Definition 15.3.3 (same) `Chapter15/ControllingPairs · SBLocalization.isAdmissible_of_domSupp_subset`. *Statement.* Let f be a section with $S_f \subset U$, and Λ an sc$^+$-multisection with $N(\Lambda) \leq 1$ whose domain support lies in U. Then Λ is (N, U)-admissible. The domain support is dom-supp$(\Lambda) = \mathrm{cl}_X\{x : \exists w \in W_x \setminus \{0_x\},\ \Lambda(w) > 0\}$ (Definition 13.2.2 (1)) `Chapter13/MultisectionSupport · SBLocalization.domSupp`.

Proof. Conditions (1) and (2) are hypotheses; condition (3) is the content, and it goes in three steps.

Step 1, off the domain support Λ sees only the zero vector `Chapter15/ControllingPairs · SBLocalization.eq_zero_of_not_mem_domSupp`. If $x \notin$ dom-supp(Λ) and $w \in W_x$ is not 0_x, then $\Lambda(w) = 0$: otherwise x would lie in the set whose closure is the domain support.

Step 2, the same for every scalar multiple `Chapter15/ControllingPairs · SBLocalization.eq_fiberZero_of_multisectionSmul_pos`. If $P(w) \notin$ dom-supp(Λ) and $(t\Lambda)(w) > 0$ for some real t, then w is the zero vector of its fibre. For $t = 0$ this is the definition of 0Λ. For $t \neq 0$, $(t\Lambda)(w) = \Lambda(t^{-1}w)$, with $t^{-1}w$ computed in the fibre; if $w \neq 0$ then $t^{-1}w \neq 0$ (multiply back by t), so $\Lambda(t^{-1}w) = 0$ by Step 1, a contradiction.

Step 3. Let $x \notin U$ with $(t\Lambda)(f(x)) > 0$ for some $t \in [-1, 1]$. Since dom-supp$(\Lambda) \subset U$, $x \notin$ dom-supp(Λ), and $P(f(x)) = x$ because f is a section; by Step 2, $f(x) = 0_x$, so $x \in S_f \subset U$, a contradiction.

The text's argument is Step 1 and Step 3 for $t = 1$; Step 2 makes the argument cover every $t \in [-1, 1]$, including $t = 0$, where 0Λ is the multisection of the zero section and the claim is $S_f \subset U$. Only the section property of f and $S_f \subset U$ are used: neither the Fredholm property nor the control of compactness enters. Witness of the definition: the zero multisection, whose domain support is empty and whose pointwise norm is 0 (§XV.21), is admissible for every controlling pair by this example.

XV.19 The products remark after Definition 15.3.3, and Remark 15.3.4: product ep-groupoids and product strong bundles

After Definition 15.3.3 (pp. 692–693) the text observes that admissible perturbations "behave well under products": for two problems (W, f, N, U, Λ) and $(W', f', N', U', \Lambda')$ it takes the product $f \times f'$ as an sc-Fredholm section of $W \times W' \to X \times X'$, the auxiliary norm $(N \times N')(h, h') = \max\{N(h), N'(h')\}$, the product multisection $(\Lambda \times \Lambda')(h, h') = \Lambda(h)\Lambda'(h')$, and checks that $\Lambda \times \Lambda'$ is admissible for $(N \times N', U \times U')$. Remark 15.3.4 adds that the domain support of $\Lambda \times \Lambda'$ lies in $(U \times X') \cup (X \times U')$, not necessarily in $U \times U'$. The text takes the product ep-groupoid, the product strong bundle and its structure map for granted. The formalization builds them: the product ep-groupoid (extra, row V15.17), the product strong bundle of Part I (restated, V15.53), products of sections (same, V15.54), the product over the product ep-groupoid with $\mu \times \mu'$ (same, V15.55), and $N \times N'$ as an auxiliary norm (same, V15.56). What remains of the remark, the sc-Fredholm property of $f \times f'$ and the multisection $\Lambda \times \Lambda'$ with its admissibility and domain support, is gap G.160 (§XV.38). The product ep-groupoid is also the

base of the cylinder $[0,1]^n \times X$ of Corollary 15.3.10 (§XV.30).

The product ep-groupoid (extra) Chapter15/ProductEp · EpGroupoid.prod. *Statement.* For ep-groupoids X, X', the groupoid with objects $X \times X'$ and morphisms $\mathbf{X} \times \mathbf{X}'$, all structure maps (source, target, composition, unit, inverse) taken componentwise Chapter15/ProductEp · GroupoidStr.prod, and the product M-polyfold structures on objects and morphisms (Exposition Lemma 4.D, Part II), is an ep-groupoid (Definition 7.1.3, Part VII). The groupoid axioms hold componentwise. *Proof* of the four conditions of Definition 7.1.3.

(1)–(2), the étale property and the smooth structure maps. Products of sc-smooth maps are sc-smooth on products of open sets: $f \times g = (f \circ \mathrm{pr}_1, g \circ \mathrm{pr}_2)$, a pair of composites with the sc-smooth projections Chapter15/ProductEp · MPolyfold.IsScSmoothOn.prodMap. Hence a product of sc-diffeomorphisms between open sets is one, with the product of the inverses as inverse Chapter15/ProductEp · MPolyfold.IsScDiffeoOn.prodMap, and a product of local sc-diffeomorphisms is one: at (p_1, p_2) take the product of the two local inverses Chapter15/ProductEp · MPolyfold.IsLocalScDiffeo.prodMap. So $s \times s'$ and $t \times t'$ are local sc-diffeomorphisms, and the unit and inverse maps, products of sc-smooth maps on univ $\times$ univ, are sc-smooth.

(3), properness. At (x, x') take the neighbourhoods $V \ni x$, $V' \ni x'$ of condition (3) for X and X', with $t : s^{-1}(\mathrm{cl}\, V) \to X$ and $t' : s'^{-1}(\mathrm{cl}\, V') \to X'$ proper. For $V \times V'$, the closure of a product is the product of the closures, so $(s \times s')^{-1}(\mathrm{cl}(V \times V')) = s^{-1}(\mathrm{cl}\, V) \times s'^{-1}(\mathrm{cl}\, V')$, and $t \times t'$ restricted to it is proper: a product of proper maps, composed with the homeomorphism between a product of subsets and the subset of the product Chapter15/ProductEp · isProperMap_domRestrict_prod Chapter15/ProductEp · EpGroupoid.prod_exists_isProperMap.

(4), sc-smooth composition. Condition (4) asks that composition be sc-smooth on the fibred product $\mathbf{X}\,{}_s\!\times_t \mathbf{X}$; Part VII proves it equivalent to its sheet form: for every sheet σ of s (a local inverse of s on an open V), the map $\psi \mapsto m(\sigma(t(\psi)), \psi)$ is sc-smooth on $t^{-1}(V)$ Chapter7/EpGroupoid · isScSmoothComp_iff_sheet. The formal proof checks the sheet form for the product Chapter15/ProductEp · EpGroupoid.prod_isScSmoothCompSheet. Let σ be a sheet of $s \times s'$ over V, and $\psi_0 \in (t \times t')^{-1}(V)$. Take sheets h_1 of s near the first component of $\sigma(t(\psi_0))$, defined on U_1, and h_2 of s' near the second, on U_2; let O be the open set of ψ near ψ_0 with $\sigma(t(\psi)) \in U_1 \times U_2$. On O, $s(\sigma(t(\psi))_1) = t(\psi_1)$, and since h_1 is a left inverse of s on U_1, $\sigma(t(\psi))_1 = h_1(t(\psi_1))$; likewise for the second component. So on O the map is $\psi \mapsto (m(h_1(t\psi_1), \psi_1), m'(h_2(t'\psi_2), \psi_2))$, a pair of maps that are sc-smooth by the sheet form of (4) for X and X', composed with the projections. Sc-smoothness is local, which concludes.

Consequences. The projections are sc-smooth functors Chapter15/ProductEp · EpGroupoid.prodFst Chapter15/ProductEp · EpGroupoid.prodSnd. If X and X' are tame, so is $X \times X'$, with $d_{X\times X'}(x, x') = d_X(x) + d_{X'}(x')$ Chapter15/ProductEp · EpGroupoid.isTame_prod Chapter15/ProductEp · EpGroupoid.degIndex_prod, from the M-polyfold statement Chapter2/FibredProduct · MPolyfold.isTame_prod. Two objects are isomorphic iff their components are: a morphism of the product is a pair of morphisms Chapter15/ProductEp · EpGroupoid.prod_π_eq_π_iff. *Reading.* The type of ep-groupoids requires paracompact object and morphism spaces; on the products these are instances that always hold, M-polyfolds being metrizable Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod.

The model level: products of strong bundle maps and retracts. A strong bundle chart of W takes values in $E \lhd F$, pairs (base, fibre); for the product the natural model is $(E \oplus E') \lhd (F \oplus F')$, base first. The two are related by the swap $((a, a'), (b, b')) \mapsto ((a, b), (a', b'))$ Chapter15/ProductStrongBundle · bundleSwap, which on every level is a continuous linear isomorphism, an sc-operator Chapter15/ProductStrongBundle · ScBanach.bundleSwapOp. The product of local maps Φ, Φ' is $\Phi \times \Phi' :=$ swap $\circ$

$(\Phi \times \Phi') \circ \mathrm{swap}$ Chapter15/ProductStrongBundle · prodBundleMap Chapter15/ProductStrongBundle · prodBundleMap_eq, sc-smooth on $(U \times U') \lhd (F \oplus F')$ when Φ, Φ' are, as a composite of the product of sc-smooth maps with two sc-operators Chapter15/ProductStrongBundle · isScSmooth_prodBundleMap.

Products of strong bundle maps (Definition 2.6.1) Chapter15/ProductStrongBundle · IsStrongBundleMap.prod. Each clause of the definition is checked componentwise. $\Phi \times \Phi'$ maps $(U \times U')\lhd$ into $(V \times V')\lhd$; its base part does not depend on the fibre variable; it is linear in the fibre; it preserves the double filtration, because a pair lies on level m of $E \oplus E'$ iff both entries do Chapter15/ProductStrongBundle · ScBanach.mem_range_prod_toBase; it is sc-smooth on $\lhd$ by the previous paragraph. For the shifted bundle $\lhd[1]$ one needs that the $[1]$-version of the product is the product of the $[1]$-versions: the lift to level 1 of a pair of level-1 points is the pair of the lifts Chapter15/ProductStrongBundle · ScBanach.invFun_prod_toBase_one Chapter15/ProductStrongBundle · shiftFiber_prodBundleMap, and the $[1]$-maps are sc-smooth by the same composition.

Retractions and retracts (Definitions 2.6.1, 2.6.2). For strong bundle retractions R, R', $R \times R'$ is idempotent componentwise, so it is a strong bundle retraction Chapter15/ProductStrongBundle · IsStrongBundleRetraction.prod; its base retraction is $r \times r'$ Chapter15/ProductStrongBundle · baseRetraction_prod, with retract the product of the base retracts Chapter15/ProductStrongBundle · retract_baseRetraction_prod; and its image is the swap preimage of $K \times K'$, $K = R(U \lhd F)$ Chapter15/ProductStrongBundle · image_prodBundleMap. Hence $\mathrm{swap}^{-1}(K \times K')$ is a local strong bundle retract of $(C \times C') \lhd (F \oplus F')$ Chapter15/ProductStrongBundle · IsStrongBundleRetract.prod. Sc-smooth maps between retracts multiply in the same way: $T \times T'$ is the swap, the product $T \times T'$ of retract maps, and the swap back, three sc-smooth maps between retracts Chapter15/ProductStrongBundle · IsScSmoothRetractMap.prodBundleMap. The $[1]$-retract of $\mathrm{swap}^{-1}(K \times K')$ is $\mathrm{swap}^{-1}(K[1] \times K'[1])$ Chapter15/ProductStrongBundle · strongShiftOne_bundleSwap_preimage.

The product strong bundle (restated) Chapter15/ProductBundle · StrongBundle.prod. *Statement.* For strong bundles $P : W \to X$ and $P' : W' \to X'$ over M-polyfolds, $P \times P' : W \times W' \to X \times X'$ is a strong bundle (Definition 2.6.5), with the atlas of products of charts. *Construction.*

The fibres. The fibre over (x, x') is in bijection with $P^{-1}(x) \times P'^{-1}(x')$ Chapter15/ProductBundle · prodFiberEquiv; the vector space structure is transported along the bijection, which then is a topological linear isomorphism Chapter15/ProductBundle · prodFiberCLE. A product of Banachable spaces is Banachable Chapter15/ProductBundle · IsBanachable.prod, and Banachability passes along topological linear isomorphisms Chapter15/ProductBundle · IsBanachable.of_equiv.

The charts. For charts Φ of W over φ and Φ' of W' over φ', the product chart is $y \mapsto \mathrm{swap}(\Phi(y_1), \Phi'(y_2))$ over $\varphi \times \varphi'$, with fibre model $F \oplus F'$, domain $U \times U'$ and strong bundle retraction $R \times R'$ Chapter15/ProductBundle · StrongBundleChart.prod; its inverse is the pair of inverses after the swap Chapter15/ProductBundle · StrongBundleChart.prodPartialEquiv, it covers $\varphi \times \varphi'$, and it is linear on fibres, componentwise. Its image is $\mathrm{swap}^{-1}(K \times K')$ Chapter15/ProductBundle · StrongBundleChart.prod_K. *Compatibility.* The overlap of two product charts is the swap preimage of the product of the overlaps Chapter15/ProductBundle · StrongBundleChart.overlap_prod, and the transition map is the product of the transition maps Chapter15/ProductBundle · StrongBundleChart.transition_prod. Compatibility (Definition 2.6.4) asks four sc-smoothness statements, the transition maps and their inverses on $[0]$ and on $[1]$, and two level-preservation statements. The first four are the product of retract maps of the previous paragraph, on $[1]$ after identifying the $[1]$-map of the product with the product of the $[1]$-maps; the last two hold componentwise Chapter15/ProductBundle · StrongBundleChart.Compatible.prod. *The atlas* consists of the products $\Phi \times \Phi'$ of charts of the two atlases; any two are compatible by what was just shown, and they cover $W \times W'$ Chapter15/ProductBundle · StrongBundle.prodAtlas Chapter15/ProductBundle · StrongBundle.prod_mem_charts. $W \times W'$ is

paracompact, $W[0]$ and $W'[0]$ being metrizable Chapter15/ProductBundle · StrongBundle.paracompactSpace_prod. *Restated*, because the text does not describe the structure; the atlas of products of charts is the one the M-polyfold product of Part II uses.

The levels. As M-polyfolds, $(W \times W')[0]$ and $W[0] \times W'[0]$ coincide: the identity is sc-smooth in both directions, being the swap in the charts $\Phi \times \Phi'$ and $\Phi[0] \times \Phi'[0]$ Chapter15/ProductBundleLevels · StrongBundle.isScSmoothMap_id_prod_zero Chapter15/ProductBundleLevels · StrongBundle.isScSmoothMap_id_prod_zero_symm. The $(0,1)$-part is the product, $(W \times W')_{0,1} = W_{0,1} \times W'_{0,1}$ Chapter15/ProductBundleLevels · StrongBundle.mem_oneSet_prod, since in a product chart the level-1 lift is the swap of the pair of lifts Chapter15/ProductBundleLevels · StrongBundleChart.liftOne_prod. With their own topologies, $(W \times W')[1]$ and $W[1] \times W'[1]$ are homeomorphic Chapter15/ProductBundleLevels · StrongBundle.prodOneHomeomorph, the right side paracompact Chapter15/ProductBundleLevels · StrongBundle.paracompactSpace_prod_one, and the homeomorphism and its inverse are sc-smooth, again the swap in the $[1]$-charts Chapter15/ProductBundleLevels · StrongBundle.isScSmoothMap_prodOneHomeomorph Chapter15/ProductBundleLevels · StrongBundle.isScSmoothMap_prodOneHomeomorph_symm.

Products of strong bundle maps over maps of the bases (§8.3's strong bundle maps). Products of sc-smooth maps of total spaces are sc-smooth on $[0]$ Chapter15/ProductBundleMap · StrongBundle.isScSmoothMap_prodMap_zero and, through the homeomorphism of the $[1]$-levels, on $[1]$ Chapter15/ProductBundleMap · StrongBundle.isScSmoothMap_prodMap_one Chapter15/ProductBundleMap · StrongBundle.oneRestrict_prodMap. If Φ covers g and Φ' covers g', then $\Phi \times \Phi'$ is a strong bundle map covering $g \times g'$: it covers componentwise, is linear on fibres, maps $(W_1 \times W_1')_{0,1}$ into $(W_2 \times W_2')_{0,1}$ by the product formula for the $(0,1)$-parts, and is sc-smooth on $[0]$ and $[1]$ Chapter15/ProductBundleMap · StrongBundle.IsStrongBundleMap.prod.

Products of sections (same). $f \times f'$ is a section of $P \times P'$ Chapter15/ProductBundle · IsSection.prodMap. *Sc-smooth* Chapter15/ProductBundle · StrongBundle.IsScSmoothSection.prod: near any point p, an sc-smooth section has a chart representative: a chart φ of the base around p_1, a strong bundle chart Φ around $f(p_1)$, and an open $V \ni p_1$ on which $\Phi \circ f \circ \varphi^{-1}$ is an sc-smooth map between retracts Chapter15/ProductBundle · StrongBundle.exists_chart_rep. Do the same for f' at p_2; on $V \times V'$ the representative of $f \times f'$ in the product charts is the swap composed with the product of the two representatives, sc-smooth between retracts. *Sc⁺* Chapter15/ProductBundle · StrongBundle.IsScPlusSection.prod: in a product chart the fibre values lie on level 1 because both components do, and the representative into the $[1]$-retract $\mathrm{swap}^{-1}(K[1] \times K'[1])$ is again a swap composed with the product of the $[1]$-representatives.

Over the product ep-groupoid (same) Chapter15/ProductEpBundle · EpStrongBundle.prod. *Statement.* For strong bundles (W, μ) over X and (W', μ') over X', $(W \times W', \mu \times \mu')$ is a strong bundle over $X \times X'$ (Definition 8.3.1), with $(\mu \times \mu')((\varphi, \varphi'), (w, w')) = (\mu(\varphi, w), \mu'(\varphi', w'))$ Chapter15/ProductEpBundle · EpStrongBundle.prod_μ. *Proof.* The structure map is defined on the pull-back $(\mathbf{X} \times \mathbf{X}')_{s \times s'} \times_{P \times P'} (W \times W')$, a strong bundle over the morphism space; $\mu \times \mu'$ is a map on the product of the two pull-backs. The first step identifies the two. The splitting $((\varphi, \varphi'), (w, w')) \mapsto ((\varphi, w), (\varphi', w'))$ Chapter15/ProductEpBundle · prodPullbackSplit, with inverse the joining map Chapter15/ProductEpBundle · prodPullbackJoin, is a strong bundle map covering the identity of $\mathbf{X} \times \mathbf{X}'$, on $[0]$ and on $[1]$ Chapter15/ProductEpBundle · isStrongBundleMap_prodPullbackSplit: each of its two components is sc-smooth into its pull-back by the universal property of the pull-back (a map into a pull-back is sc-smooth when its two projections are), the projections being the projections of the product composed with the identification $(W \times W')[0] = W[0] \times W'[0]$ above, and likewise on $[1]$ with the homeomorphism of the $[1]$-levels. Its inverse is sc-smooth by the same argument, so it is an sc-diffeomorphism and in particular a local sc-diffeomorphism Chapter15/ProductEpBundle · isLocalScDiffeo_prodPullbackSplit. Then $\mu \times \mu' := (\mu \times \mu') \circ \mathrm{split}$

`Chapter15/ProductEpBundle · EpStrongBundle.prodμ` is: a strong bundle map covering $t \times t'$, as the composite of the splitting with the product of the strong bundle maps μ, μ'; surjective, by joining preimages; a local sc-diffeomorphism, as the composite of the splitting, the identity between the product strong bundle and the product M-polyfold on $[0]$, the product of the local sc-diffeomorphisms μ, μ', and the identity back; and it satisfies the unit and composition laws componentwise.

Section functors. If f and f' satisfy the functor condition $f(t\varphi) = \mu(\varphi, f(s\varphi))$, so does $f \times f'$, componentwise `Chapter15/ProductEpBundle · EpStrongBundle.IsFunctorialSection.prod`; products of sc-smooth section functors are sc-smooth section functors `Chapter15/ProductEpBundle · EpStrongBundle.IsScSmoothSectionFunctor.prod`, and products of sc⁺-section functors are sc⁺ `Chapter15/ProductEpBundle · EpStrongBundle.IsScPlusSectionFunctor.prod` (Definition 8.3.4 (1), (2)). The paracompactness instances on the pull-backs and products that the types require are produced by the lemmas above; they always hold.

$N \times N' = \max(N, N')$ **is an auxiliary norm** (same). *Definition* `Chapter15/ProductAuxNorm · StrongBundle.prodAuxNorm`: on $(W \times W')[1]$, $(N \times N')(w, w') = \max(N(w), N'(w'))$, read through the homeomorphism $(W \times W')[1] \cong W[1] \times W'[1]$; on a fibre, the level-$(0,1)$ fibre of the product is the product of the fibres `Chapter15/ProductAuxNorm · StrongBundle.prodOneFiberFst` `Chapter15/ProductAuxNorm · StrongBundle.prodOneFiberMk`, and the norm is the maximum of the norms of the components `Chapter15/ProductAuxNorm · StrongBundle.prodAuxNorm_toOne`. *Statement* `Chapter15/ProductAuxNorm · StrongBundle.isAuxNorm_prodAuxNorm`: if N, N' are auxiliary norms (Definition 5.1.1), so is $N \times N'$. *Proof*, clause by clause. *Continuity*: a maximum of continuous functions. *On each fibre a complete norm*: (i) if $\max(N(v_1), N'(v_2)) = 0$, both are 0, since both are non-negative, so $v_1 = 0$, $v_2 = 0$, and $v = 0$ `Chapter15/ProductAuxNorm · StrongBundle.prodOneFiber_ext`; (ii) homogeneity, $\max(|t|N(v_1), |t|N'(v_2)) = |t| \max(N(v_1), N'(v_2))$; (iii) the triangle inequality, each component's inequality bounded by the sum of the maxima; (iv) completeness: a sequence Cauchy for the maximum is Cauchy for each component, the components converge to a and b by completeness of N and N', and the sequence converges to the pair (a, b) for the maximum. *Condition (2) of Definition 5.1.1*: if the base points $P(w_k)$ converge to x and $(N \times N')(w_k) \to 0$, then both $N(w_{k,1}) \to 0$ and $N'(w_{k,2}) \to 0$ by squeezing between 0 and the maximum, condition (2) for N and N' gives $w_{k,1} \to 0_{x_1}$ in $W[1]$ and $w_{k,2} \to 0_{x_2}$ in $W'[1]$, and the inverse of the homeomorphism of the $[1]$-levels carries the pair to $w_k \to 0_x$ in $(W \times W')[1]$. *Over the ep-groupoid* `Chapter15/ProductAuxNorm · EpStrongBundle.isAuxNorm_prodAuxNorm` (Definition 8.3.5): in addition $N \times N'$ is invariant under $\mu \times \mu'$, because each component is invariant under its own structure map.

What is not formalized of the remark (gap G.160, §XV.38): (c) $f \times f'$ is sc-Fredholm with tame germ data; Part I has no product of sc-Fredholm germs, only the case of the cube $[0,1]^n$ as a parameter, which the cylinder uses (§XV.30); (d) $\Lambda \times \Lambda'$ as an sc⁺-multisection, from products of local section structures with weights $\sigma_i \sigma'_j$, the identity $(N \times N')(\Lambda \times \Lambda') = \max(N(\Lambda), N'(\Lambda'))$, the identity $t(\Lambda \times \Lambda') = (t\Lambda) \times (t\Lambda')$ and the admissibility for $(N \times N', U \times U')$, and Remark 15.3.4 on the domain support. The ingredient that the sections of the local structures multiply to sc⁺-sections is `IsScPlusSection.prod` above.

XV.20 Definition 15.3.5 and Remark 15.3.6 (2017: Definition 15.6, Remark 15.6): general position

Definition 15.3.5 (restated) `Chapter15/ControllingPairs · SBLocalization.IsInGeneralPosition`. *Statement.* (f, Λ) is *in general position* if for every $x \in X_1$ with $\Lambda(f(x)) > 0$, every operator L in the support of the formal

sum $T(f,\Lambda)(x) \in \mathrm{GrF}(x)$ (Definition 15.1.7, §XV.8), every chart φ of X around x and every presentation T of the partial quadrant C of its model ($C = T^{-1}([0,\infty)^k \times V)$), the restriction of $L : T_xX \to W_x$ to the reduced tangent space T^R_xX is surjective. T^R_xX is Part II's reduced tangent fibre (Definition 2.5.8), read in the chart φ and the presentation T `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber`.

Readings. (a) The text quantifies over $x \in \mathrm{supp}(\Lambda \circ f)$; the formal definition takes x in X_1, where $T(f,\Lambda)$ is defined. Nothing is lost, because the support lies in X_1 (below). (b) The text writes $T(f,\Lambda)(x) = \sum_i \sigma_i L_i$ with $\sigma_i > 0$; these L_i are the elements of the support of the formal sum, the operators with positive coefficient. (c) "T^R_xX (Definition 2.5.8)" is read in every chart and every presentation; since T^R_xX does not depend on them, one suffices (below). (d) The text works over a tame ep-groupoid; tameness is not part of the formal definition, and it enters the statements that need T^R_xX to be the right space, through Part I's results on tame M-polyfolds.

The support lies in X_1 `Chapter15/ControllingPairs · SBLocalization.mem_levelSet_one_of_pos`. *Statement.* For an sc-Fredholm section functor f and an sc$^+$-multisection Λ, $\Lambda(f(x)) > 0$ implies $x \in X_1$. *Proof.* Take a local section structure (s_i, σ_i) of Λ on an open $O \ni x$ (the 2021 form, Part XI). The count formula gives $\Lambda(f(x)) = \sum_{i : f(x) = s_i(x)} \sigma_i$; the sum is positive, so some i has $f(x) = s_i(x)$. The value $s_i(x)$ of an sc$^+$-section lies in $W_{0,1}$, and an sc-Fredholm section functor is regularizing: $x \in X_0$ and $f(x) \in W_{0,1}$ give $x \in X_1$ `Chapter12/CompactnessEp · EpStrongBundle.remark_12_1_1_regularizing_obj` (the remark after Definition 12.1.1, Part XII).

One chart suffices. Every level-1 point lies in a chart with a presentation of its partial quadrant `Chapter15/ControllingPairs · SBLocalization.exists_chart_presentation`. The reduced tangent fibre does not depend on the chart and the presentation (Proposition 2.4.10, Part II) `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber_eq`, so surjectivity on it, checked in one chart and presentation at each point, holds in all `Chapter15/ControllingPairs · SBLocalization.isInGeneralPosition_iff_exists`: given the check in (φ, T) and another pair (ψ, T'), rewrite the reduced fibre of (ψ, T') as that of (φ, T).

The reduced linearization. The text calls the restriction $T^R_{(f,\Lambda)}(x) = \sum_i \sigma_i L_i | T^R_xX$ the reduced linearization. Formally it is the push-forward of the formal sum along $L \mapsto L|T^R_xX$, in a chart and presentation `Chapter15/ControllingPairs · SBLocalization.reducedLinearization`. Two distinct operators may have the same restriction; their coefficients then add. With coefficients in $\mathbb{Q}^+$ nothing cancels, so the support of a push-forward is the image of the support `Chapter15/ControllingPairs · Finsupp.support_mapDomain_nnrat`: for $b = g(a)$ with a in the support, the coefficient of b is a sum of non-negative terms one of which is positive. Hence general position is the surjectivity of every operator of the reduced linearization `Chapter15/ControllingPairs · SBLocalization.isInGeneralPosition_iff_reducedLinearization`.

Witness `Chapter15/ControllingPairs · SBLocalization.isInGeneralPosition_witness`: over the trivial ep-groupoid of $\mathbb{R}^n$ with the bundle $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, the zero section and the zero multisection (the multisection of the zero section, value 1 at zero vectors) are in general position, and $\mathrm{supp}(\Lambda \circ f)$ is all of $\mathbb{R}^n$. The fibres are 0, so every operator onto them is surjective. The witness shows that the all-charts form is satisfiable with nonempty support; it has zero fibres and is **weak** as a test of the condition.

Remark 15.3.6, first sentence (same) `Chapter15/ControllingPairs · SBLocalization.IsInGeneralPosition.surjective`. *Statement.* In general position, every operator of the support of $T(f,\Lambda)(x)$, $x \in \mathrm{supp}(\Lambda \circ f) \cap X_1$, is surjective. *Proof.* Choose a chart and a presentation at x; for $w \in W_x$ general position gives $\alpha \in T^R_xX$ with $L\alpha = w$, and $\alpha \in T_xX$. "Onto" is read operator by operator, as in Definition 15.2.1 (§XV.12); so general position implies transversality, which the proof of Theorem 15.3.7 uses (§XV.22).

Remark 15.3.6, second sentence (narrower; stronger in its range) Chapter15/GoodPositionBranches · SBLocalization.remark_15_3_6_goodPosition_hyp. *Statement.* Let f be an sc-Fredholm section functor with Part I's tame germ data, Λ an sc$^+$-multisection, (f,Λ) in general position and $\Lambda(f(x)) > 0$. Then (a) (f,Λ) is in good position at x (Definition 15.2.4, §XV.15); (b) for every local section structure (s_i,σ_i) of Λ on an open $O \ni x$ and every i with $s_i(x) = f(x)$, the section $(f - s_i)|_O$ of the restricted bundle $W|_O$ (§XV.10) is in general position at x in Part I's sense (Definition 5.3.9). The text states this for $x \in \partial X$, with "good position, in fact even in general position"; (b) is the reading of "even in general position", Definition 15.3.5 being "a natural extension" of Definition 5.3.9. The formal statement holds at every point of the support, on the boundary or not: stronger in that respect.

Proof. Four steps, the first two in a chart.

Step 1, the reduced surjectivity read in a chart Chapter15/GoodPositionBranches · SBLocalization.exists_reduced_preimage. Let c be a strong bundle chart around x with base chart φ, retraction r on U, fibre retract K, and z the level-1 point over $\varphi(x)$. Let L be a chart operator $E_0 \to F_0$ whose intrinsic operator $\Phi_x^{-1} \circ L \circ T\varphi$ (§XV.7) is onto W_x on $T_x^R X$. Then every k with $(\varphi(x),k) \in K$ is Lw for some w in the chart's reduced tangent space at z that also lies in the range of $Dr(z)$, the chart tangent space T_zO. The intrinsic operator is built from L only when all chart vectors are mapped into the range of the fibre identification; in that case the preimage $\alpha \in T_x^R X$ of $\Phi_x^{-1}(k)$ gives w, the chart vector of α, which lies in T_zO and, by the chart description of the reduced fibre, in the reduced tangent space. Otherwise the intrinsic operator is 0 by definition, surjectivity forces $W_x = 0$, and $w = 0$ serves.

Step 2, the chart step Chapter15/GoodPositionBranches · SBLocalization.isGeneralPositionGerm_subRestrict. *Statement.* Let s be sc$^+$ on an open O, $f(x) = s(x)$, and c a **tame** strong bundle chart of W **centred** at x ($\varphi(x) = 0$) with domain in O. If the intrinsic operator $(f - s)'(x)$ (§XV.7) is onto W_x on $T_x^R X$, then $(f - s)|_O$ is in general position at x in Part I's sense. *Proof.* Restrict c to a chart c' of $W|_O$, tame because c is Chapter15/LocalIFT · SBLocalization.scDeriv_restrictTo_subRestrict: in it the derivative at 0 of the principal part of $(f - s)|_O$ composed with the retraction is the chart operator $D((\hat{f} - \hat{s}) \circ r)(0)$ of W. The section $(f - s)|_O$ is tame sc-Fredholm (§XV.10) and vanishes at x. By Step 1 the linearization $\mathrm{Lin}(0)$ of $(f - s)|_O$ in c' maps the reduced tangent space onto the fibre K_0, in particular it is onto. Next, $\ker \mathrm{Lin}(0) + T_0^R = T_0O$: for $v \in T_0O$, $k = \mathrm{Lin}(0)v$ lies in the fibre of K over 0, because the derivative of the principal part composed with r is fixed by the fibre projection ρ at a zero; Step 1 gives a reduced w with $\mathrm{Lin}(0)w = k$; and $v = (v - w) + w$ with $v - w \in \ker \mathrm{Lin}(0)$. These two facts are Part I's chart criterion for general position, proved in Part I's chapter on general position Chapter5/CylinderPerturbation2021 · isGeneralPositionGerm_of_sup_eq, which concludes.

Step 3, every branch through $f(x)$ Chapter15/GoodPositionBranches · SBLocalization.isGeneralPositionGerm_branch. Let (s_i,σ_i) be a local structure on $O \ni x$ and $s_i(x) = f(x)$. The point x is smooth, being a zero of the regularizing section $f - s_i$ Chapter15/Transversality · SBLocalization.mem_smoothSet_of_eq, in particular in X_1. The tame germ data of f give a tame chart centred at x; restricted to O it stays tame and centred with domain in O. With the local structure and this chart as local data (§XV.8), $T(f,\Lambda)(x)$ is their formal sum, so the intrinsic operator $(f - s_i)'(x)$ is in its support, the branch passing through $f(x)$ Chapter15/Transversality · SBLocalization.mem_support_linSumIntr_iff Chapter15/LinearizationIntrinsic · SBLocalization.linearization_eq. General position makes it onto on $T_x^R X$ in this chart, and Step 2 applies.

Step 4, good position. Definition 15.2.4 asks three things at x. $\Lambda(f(x)) > 0$ is the hypothesis. $T(f,\Lambda)(x)$ is onto: $x \in X_1$ by the lemma above, and every operator of the support is onto by the first sentence. For some local structure on an open $O \ni x$, given by the sc$^+$-multisection, every branch through

$f(x)$ is in Part I's good position (Definition 3.6.6): by Step 3 it is in general position, and general position implies good position for a tame sc-Fredholm section at a zero, Part I's Lemma 5.26 `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGermTame`, applied to $(f - s_i)|_O$, which is tame sc-Fredholm `Chapter15/LocalScPlus · SBLocalization.isScFredholmSectionTame_subRestrict` and vanishes at x `Chapter15/LocalScPlus · SBLocalization.isZeroAt_subSec_iff`.

Narrower by the inherited hypothesis: Part I's tame germ data of f, the hypothesis of all of §15.2 (§§XV.13–XV.16), inherited from Part I's Theorems 3.1.18 and 3.6.8, without which Theorem 3.6.8 does not hold as we read it (Part III). Here it supplies the tame chart centred at x of Step 2. Witness of all hypotheses together: the zero section of $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ with the zero multisection, which has tame germ data `Chapter15/InteriorPerturbationWitness · SBLocalization.theorem_15_2_2_hyp_witness` and is in general position `Chapter15/GeneralPositionBranched · SBLocalization.isInGeneralPosition_epSEP_zero`; it has zero fibres (weak). No $d_X = 0$ is needed: unlike Remark 15.2.3 (§XV.14), the boundary points are reached through good position.

Remark 15.3.6, last sentence (narrower). The text concludes that $\Lambda \circ f$ is a tame branched ep⁺-subgroupoid. Three formal statements carry this.

Local branching `Chapter15/GeneralPositionBranched · SBLocalization.remark_15_3_6_localBranching_hyp`. *Statement.* Under the hypotheses of the second sentence, at every y with $\Lambda(f(y)) > 0$ the function $z \mapsto \Lambda(f(z))$ has a local branching structure in the sense of Definition 9.1.2 (2)–(4) (Part IX). *Proof.* By the second sentence (f, Λ) is in good position at y, so Theorem 15.2.5 (§XV.16) applies at y: it gives a natural representation $U(y)$, a local section structure (s_i, σ_i) and a chart, such that each $M_i = \{z \in U(y) : f(z) = s_i(z)\}$ is a smooth submanifold properly embedded in $U(y)$, the count $\Lambda(f(z)) = \sum_{i : f(z) = s_i(z)} \sigma_i$ holds on $U(y)$, and only branches through $f(y)$ meet f in $U(y)$. The branching structure is indexed by the branches through $f(y)$, with the sets M_j, which contain y, and the weights $\sigma_j > 0$. The count identity of Definition 9.1.2 is the count formula with the sum restricted to these branches: the terms of the other branches vanish on $U(y)$ by the last clause of Theorem 15.2.5.

Branched `Chapter15/GeneralPositionBranched · SBLocalization.remark_15_3_6_isBranchedEpSubgroupoid_hyp`. *Statement.* If moreover the orbit space $|\operatorname{supp}(\Lambda \circ f)|$ is paracompact, $\Lambda \circ f$ is a branched ep⁺-subgroupoid of X (Definition 9.1.2). *Proof.* Each clause of the definition: the values are non-negative rationals; $\Lambda \circ f$ is a functor, because f is a section functor and Λ is a functor, $\Lambda(f(t\varphi)) = \Lambda(\mu(\varphi, f(s\varphi))) = \Lambda(f(s\varphi))$ `Chapter15/GeneralPositionBranched · SBLocalization.isFunctorial_comp`; the support consists of smooth points, the tame germ data making $f - s_i$ regularizing `Chapter15/Transversality · SBLocalization.mem_smoothSet_of_pos`; the paracompactness of $|\operatorname{supp}|$ is the hypothesis; and the local branching structures are the previous statement. *The added hypothesis*, the paracompactness of the orbit space of the support, is Definition 9.1.2 (1), new in the 2021 edition, which the remark does not address; it is not derived from the text. It is automatic when $|\operatorname{supp}(\Lambda \circ f)|$ is compact, the situation of Theorem 15.3.7 (§XV.21); witness for $\mathbb{R}^n$ `Chapter9/Propositions · EpGroupoid.paracompactSpace_orbitSupp_trivial_rn`.

Tame `Chapter15/TameBranched · SBLocalization.remark_15_3_6_isTameBranched_hyp`. *Statement.* Under the same hypotheses $\Lambda \circ f$ is a branched ep⁺-subgroupoid and tame (Definition 9.3.6). *Proof.* Good position holds at every point of the support (second sentence), and the global form of Theorem 15.2.5 (§XV.16) `Chapter15/TameBranched · SBLocalization.theorem_15_2_5_isTameBranched_hyp` concludes: at each support point Theorem 15.2.5 with "tame" gives a tame local branching structure, a natural representation with the properness property whose sheets M_j carry a tame induced structure (Definition 2.5.18); forgetting the tameness gives the local branching structures of Definition 9.1.2, and the tame structures are Definition 9.3.6. Witness of the tame global form on $\mathbb{R}^n \times \mathbb{R}^0$, with support everything `Chapter15/TameWitness · SBLocalization.remark_15_2_3_isTameBranched_hyp_witness` (zero fibres, weak).

Narrower in all three only through the inherited tame germ data and, for the global forms, the paracompact support of Definition 9.1.2 (1). The text writes "tame branched ep⁺-groupoid", read as "ep⁺-subgroupoid". The compact form, under the hypotheses of Theorem 15.3.7, is its "in particular" (§XV.21), where compactness supplies the paracompactness.

XV.21 Theorem 15.3.7 (2017: 15.4): the statement and its "in particular"

Theorem 15.3.7 (pp. 693–698) is the existence theorem for perturbations in general position. Over a tame ep-groupoid with sc-smooth bump functions, it takes an sc-Fredholm section functor f with compact solution set $|S_f|$, a pair (N, U) controlling compactness, and a continuous functor $h : X \to (0, 1]$. It produces an sc⁺-multisection functor Λ with four properties, and then adds two claims: "in particular" $\Lambda \circ f$ is a compact, tame, branched ep⁺-subgroupoid, and Λ "can be taken to be structurable". The formalization states the theorem as two propositions, one with and one without "structurable", and proves both. The proof is told in §§XV.22–XV.25; the final theorems are in §XV.26.

Statement (explicit) `Chapter15/GeneralPositionExistence · SBLocalization.theorem_15_3_7_existence_statement`. Suppose: - X is tame; - X admits sc-smooth bump functions, in the pointwise reading of Part XII, which Theorem 12.3.2 (1) shows equivalent to bump functions on the object M-polyfold `Chapter12/BumpEp · EpGroupoid.theorem_12_3_2_1`; - f is an sc-Fredholm section functor with the tame germ data of §15.2, and $|S_f|$ is compact; - (N, U) controls compactness (Definition 15.3.1, §XV.17) `Chapter15/ControllingPairs · SBLocalization.ControlsCompactness`. That is, N is an auxiliary norm, U is an open saturated set containing S_f, and the orbit set of $\{x \in U : N(f(x)) \le 1\}$ has compact closure in $|X|$; - h is continuous, takes the same value at the source and the target of every morphism, and satisfies $0 < h \le 1$.

Then there is an sc⁺-multisection Λ with: - (1) $N(\Lambda)(x) < h(x)$ for every x, where $N(\Lambda)(x)$ is the pointwise norm of Part XIII (Definition 13.2.3), the largest $N(w)$ over the $w \in W_x$ with $\Lambda(w) > 0$; - (2) domain support in U; - (3) (f, Λ) transversal (Definition 15.2.1, §XV.12) `Chapter15/Transversality · SBLocalization.IsTransversal`; - (4) (f, Λ) in general position (Definition 15.3.5, §XV.20) `Chapter15/ControllingPairs · SBLocalization.IsInGeneralPosition`.

The structurable form `Chapter15/GeneralPositionBranched · SBLocalization.theorem_15_3_7_statement` is the same proposition with Λ structurable (Definition 13.3.6). The theorem with its addendum pairs the two: (1)–(4) under the hypotheses above, and the structurable form whenever X has a good system `Chapter15/GeneralPositionExistence · SBLocalization.theorem_15_3_7`.

Readings. - The text's "U … a saturated subset" is read as open and saturated, as in Definition 15.3.1. - A "continuous functor $h : X \to (0, 1]$" is read as a continuous function on objects that is constant along morphisms. This is a functor into the trivial category on $(0, 1]$. - (3) is "$T(f, \Lambda)$ is surjective on $\mathrm{supp}(\Lambda \circ f)$", i.e. every operator of the linearization of Definition 15.1.7 (§XV.8) is onto at every support point. - (4) is formalized as the reduced linearization being onto at every support point. This follows the first sentence of the proof: "Point (4) is equivalent to" that requirement. - "Structurable" is Definition 13.3.6.

Explicit: the only hypothesis not printed in the theorem is the tame germ data of f. It is inherited from §15.2 (§XV.16), where the boundary local perturbation needs it. The theorem itself uses no good system and no paracompactness of $|X|$.

"In particular", compact and branched (narrower) `Chapter15/GeneralPositionBranched · SBLocalization.`

theorem_15_3_7_in_particular_hyp. Let f be an sc-Fredholm section functor with tame germ data, and let (N, U) control compactness. Let Λ be an sc$^+$-multisection with $N(\Lambda) \leq 1$, domain support in U, and (f, Λ) in general position. Then $\Lambda \circ f$ is a branched ep$^+$-subgroupoid (Definition 9.1.2) with compact support (Definition 9.1.6 (2)).

Proof. 1. f is continuous, being an sc^0 section. 2. *The support is closed* Chapter15/GeneralPositionBranched · SBLocalization.isClosed_supp_comp. Near any point x_0, Λ has a local representation $(s_i, \sigma_i)_{i \in I}$ on an open $O \ni x_0$, with positive weights. On O, $\Lambda(w) = \sum_{i : s_i(y) = w} \sigma_i$ for $w \in W_y$. Since the weights are positive, $\Lambda(f(y)) > 0$ if and only if $f(y) = s_i(y)$ for some i Chapter15/GeneralPositionBranched · SBLocalization.pos_iff_exists_eq. So the complement of the support meets O in $O \cap \bigcap_i \{y : f(y) \neq s_i(y)\}$. Each set $\{f \neq s_i\}$ is open in O, because two continuous sections into a Hausdorff bundle disagree on an open set. Hence the complement is open. 3. *Its orbit set is closed* Chapter15/GeneralPositionBranched · SBLocalization.isClosed_orbitSupp_comp. The support is saturated: $f(\mathrm{tgt}\,\varphi) = \mu(\varphi, f(\mathrm{src}\,\varphi))$, and Λ is a functor, so $\Lambda \circ f$ is constant along morphisms. Hence the complement of $|\mathrm{supp}(\Lambda \circ f)|$ is the image of the complement of the support under the orbit map. The orbit map is open (Proposition 7.1.8) Chapter7/Basic · EpGroupoid.proposition_7_1_8, so this image is open. 4. *The support lies in the controlled set* Chapter15/GeneralPositionBranched · SBLocalization.supp_comp_subset_controlled. Let $\Lambda(f(x)) > 0$. - If $f(x) = 0$, then $x \in S_f \subseteq U$. - Otherwise Λ is positive at a non-zero vector over x, so x is in the domain support, hence in U.

In both cases $f(x)$ lies in $\mathrm{supp}(\Lambda)(x)$, so $N(f(x)) \leq N(\Lambda)(x) \leq 1$. 5. *Compactness* Chapter15/GeneralPositionBranched · SBLocalization.isCompact_orbitSupp_comp. $|\mathrm{supp}(\Lambda \circ f)|$ is a closed subset of the compact closure that (N, U) provides. 6. *Branched.* A compact space is paracompact. This gives the paracompact support that Definition 9.1.2 (1) asks for, and Remark 15.3.6 (§XV.20) supplies the local branching structures Chapter15/GeneralPositionBranched · SBLocalization.remark_15_3_6_isBranchedEpSubgroupoid_hyp.

Only (1) (with $h \leq 1$), (2) and (4) are used. (3) follows from (4) Chapter15/ControllingPairs · SBLocalization.IsInGeneralPosition.surjective. *Narrower*: the tame germ data are inherited from §15.2, and here they enter the conclusion, through good position at every support point (§XV.16).

"In particular", tame (narrower) Chapter15/TameBranched · SBLocalization.theorem_15_3_7_in_particular_tame_hyp. Under the same hypotheses, $\Lambda \circ f$ is moreover a tame branched ep$^+$-subgroupoid (Definition 9.3.6).

Proof. The first two conclusions are the previous theorem. Tameness is the tame form of Remark 15.3.6 (§XV.20) Chapter15/TameBranched · SBLocalization.remark_15_3_6_isTameBranched_hyp, applied with the paracompactness that the compact support gives.

The zero multisection. Suppose (f, Λ_0) is already in general position, where Λ_0 is the trivial multisection (weight 1 at the zero vector). Then Λ_0 has all the conclusions of the structurable form, for every $h > 0$ and every U, provided $|X|$ is paracompact (needed for "structurable") Chapter15/GeneralPositionBranched · SBLocalization.theorem_15_3_7_of_isInGeneralPosition_zero: - its pointwise norm is 0 Chapter15/GeneralPositionBranched · SBLocalization.pointwiseNorm_zeroMultisection; - its domain support is empty Chapter15/GeneralPositionBranched · SBLocalization.domSupp_zeroMultisection.

On the bundle $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, every pair is in general position, because every operator onto the zero fibre is surjective Chapter15/GeneralPositionBranched · SBLocalization.isInGeneralPosition_epSEP_zero. So the structurable form holds there with $\Lambda = \Lambda_0$ Chapter15/GeneralPositionBranched · SBLocalization.theorem_15_3_7_statement_epSEP_zero. This instance shows that the statement is not vacuous. It is weak: the fibres are zero and there is no boundary.

XV.22 Proof of Theorem 15.3.7, I: linear families, (15.1), and the assembly

The text's proof: - builds a family Λ_t, $t \in \mathbb{R}^{\bar\ell}$, of multisections from local pieces at the points of S_f; - shows that the parametrized problem is onto on reduced tangent spaces at $\{0\} \times S_f$ ((15.1)); - uses the compactness (15.2) to stay near S_f for small $|t|$; - picks a small regular value t_0 of the projection to the parameter space, by Sard's theorem on the intersections of faces.

The formalized proof of (1)–(4) `Chapter15/GeneralPositionExistence · SBLocalization.theorem_15_3_7_existence` follows this plan in five steps: - (a) at every point of S_f, small sections with (15.1) near that point ("seeds"; §§XV.23–XV.24); - (b) from the seeds, a family good on a saturated neighbourhood of each point of S_f (§XV.23); - (c) by compactness of $|S_f|$, one family good on a neighbourhood of S_f (this subsection); - (d) for this family, general position for almost every small t (§XV.25); - (e) the choice of t (this subsection).

The structurable form `Chapter15/GeneralPositionGoodSystem · SBLocalization.theorem_15_3_7_of_goodSystem` has the same five steps with structurable families. This subsection gives the objects the proof works with and steps (c) and (e).

Linear combinations of sections `Chapter15/ParamFamily · SBLocalization.paramSec`. For sections $s_1, \dots, s_\ell$ and $t \in \mathbb{R}^\ell$, the section $\sum_j t_j s_j$ is taken fibrewise, in the linear structure of W_y: - it is a section, and an sc⁺-section when each s_j is one; - concatenating the parameters adds the combinations: $\sum_j t_j s_j + \sum_k t'_k s'_k$ is the combination of the concatenated sections at the concatenated parameter `Chapter15/ParamFamily · SBLocalization.paramSec_append`; - $N(\sum_j t_j s_j(y)) \le \sum_j |t_j|\, N(s_j(y))$, by subadditivity and homogeneity of the auxiliary norm on the fibres `Chapter15/ParamFamily · SBLocalization.normW_paramSec_le`.

Linear families `Chapter15/ParamFamily · SBLocalization.IsLinearFamily`. A family $t \mapsto \Lambda_t$ of multisections, $t \in \mathbb{R}^\ell$, is *linear* if around every point x_0 there are: - an open $O \ni x_0$; - sections s_{ij} on O ($i \in I$ finite, $j \le \ell$); - weights $\sigma_i > 0$ with $\sum_i \sigma_i = 1$;

such that, for every t, the sections $\sum_j t_j s_{ij}$ with the weights σ_i form a local representation of Λ_t on O. So one domain, one set of sections and one set of weights serve all t. This is how the text's Λ_t and $\tilde\Lambda(t, w) = \Lambda_t(w)$ are formalized. The text asks that $\tilde\Lambda$ be an sc⁺-multisection of the pull-back of W to $\mathbb{R}^{\bar\ell} \times X$; a linear family gives that representation directly. At $t = 0$ every branch is the zero section, so Λ_0 is the trivial multisection.

The pointwise norm of a linear family is controlled by its sections: - the norm $N(\Lambda)(x)$ is attained at a branch value of any local representation `Chapter15/ParamFamily · SBLocalization.pointwiseNorm_le_of_isLocalRepOn`; - so if $N(s_{ij}) \le b$ then $N(\Lambda_t)(x) \le |t|_1\, b$, where $|t|_1 = \sum_j |t_j|$ `Chapter15/ParamFamily · SBLocalization.pointwiseNorm_le_of_linear`.

(15.1) at a point `Chapter15/ParamFamily · SBLocalization.IsParamOntoAt`. Let x be a level-1 point. The family satisfies (15.1) at x if it has a linear representation around x with the following property. For every operator L of the linearization $T(f, \Lambda_0)(x)$ (§XV.8), every chart presentation of the reduced tangent space $T^R_x X$, every branch i and every $w \in W_x$, there are $v \in T^R_x X$ and $\tau \in \mathbb{R}^\ell$ with

$$L v + \textstyle\sum_j \tau_j s_{ij}(x) = w.$$

The text writes (15.1) as "$T^R_{(\tilde f, \tilde\Lambda)}(0, x)$ is onto". Its local constructions phrase it as: the image of $f'(x)$ on

$T^R_x X$, together with the vectors $s_{ij}(x)$, spans W_x. Two readings are made: - $f'(x)$ is read as the operator of $T(f, \Lambda_0)(x)$, which at a zero of f is the linearization of f (§XV.24); - the t-derivative of $f - \sum_j t_j s_{ij}$ is $\tau \mapsto -\sum_j \tau_j s_{ij}(x)$, and its sign is immaterial for surjectivity.

At points with $d_X(x) = 0$, $T^R_x X = T_x X$, and (15.1) is the text's interior condition. So the interior and boundary constructions of the text become one construction.

Sums of families `Chapter15/ParamFamily · SBLocalization.famAdd`. For families Λ over $\mathbb{R}^\ell$ and Λ' over $\mathbb{R}^{\ell'}$, the sum over $\mathbb{R}^{\ell+\ell'}$ is $(\Lambda \oplus \Lambda')_{(t,t')} = \Lambda_t \oplus \Lambda'_{t'}$, the sum of multisections of Definition 13.2.9. This is the text's $\Lambda_t = \bigoplus_i \Lambda_{t_i}^{x_i}$. - *Local representations of a sum* `Chapter15/ParamFamily · SBLocalization.isLocalRepOn_famAdd`. If (s_i, σ_i) represents Λ and $(s'_{i'}, \sigma'_{i'})$ represents Λ' on O, then the sections $s_i + s'_{i'}$ with weights $\sigma_i \sigma'_{i'}$ represent the sum. The pairs (i, i') are re-enumerated as one finite index set; the weights sum to 1; the sums are sc⁺ on O. The value of the sum at w is $\sum\{\sigma_i \sigma'_{i'} : s_i(y) + s'_{i'}(y) = w\}$, which is how Definition 13.2.9 computes a sum on a common representation domain. - *Linearity* `Chapter15/ParamFamily · SBLocalization.IsLinearFamily.famAdd`. With linear representations, the branches $\sum_j t_j s_{ij} + \sum_k t'_k s'_{i'k}$ are the combinations of the concatenated sections (`paramSec_append` above). So the sum of linear families is linear, on $O \cap O'$. - *(15.1) for a sum* `Chapter15/ParamFamily · SBLocalization.IsParamOntoAt.famAdd_left` `Chapter15/ParamFamily · SBLocalization.IsParamOntoAt.famAdd_right`. (15.1) for either summand at x gives it for the sum. At $t = 0$ the sum is $\Lambda_0 \oplus \Lambda_0 = \Lambda_0$, so the operators L are the same. For a branch (i, i') of the sum, solve with the summand that has (15.1), and put $\tau = 0$ on the parameters of the other summand. - *Two inequalities the text uses silently.* - $N(\Lambda \oplus \Lambda')(x) \le N(\Lambda)(x) + N(\Lambda')(x)$ `Chapter15/ParamFamily · SBLocalization.pointwiseNorm_multisectionAdd_le`. A vector in the support of the sum is $a + b$ with $\Lambda(a) > 0$ and $\Lambda'(b) > 0$ `Chapter15/ParamFamily · SBLocalization.exists_of_multisectionAdd_pos`, and $N(a+b) \le N(a) + N(b)$ `Chapter15/ParamFamily · SBLocalization.normW_add_le`. - The domain support of a sum lies in the union of the domain supports `Chapter15/ParamFamily · SBLocalization.domSupp_multisectionAdd_subset`. Off both, each summand is trivial, and so is the sum; the domain support is a closure, and the closure of a union of two sets is the union of their closures.

Families good on a set `Chapter15/GeneralPositionExistence · SBLocalization.ScPlusGPFamily`. For a set Q, a family *good on* Q is a linear family Λ over some $\mathbb{R}^\ell$ with: - every Λ_t an sc⁺-multisection; - $N(\Lambda_t)(x) \le |t|_1 \, h(x)$ for all t and x; - domain supports in U; - (15.1) at every level-1 point of $S_f \cap Q$.

The structurable twin `Chapter15/ParamFamily · SBLocalization.GPFamily` asks every Λ_t to be structurable instead.

The operations on good families: - A family good on Q is good on every subset of Q `Chapter15/ParamFamily · SBLocalization.GPFamily.mono` `Chapter15/GeneralPositionExistence · SBLocalization.ScPlusGPFamily.mono`. - *The empty family* `Chapter15/GeneralPositionExistence · SBLocalization.ScPlusGPFamily.zero` has $\ell = 0$ and $\Lambda_t = \Lambda_0$, and is good on $\emptyset$. It is linear with one branch, the zero section of weight 1 `Chapter15/ParamFamily · SBLocalization.isLocalRepOn_zeroMultisection`. - The structurable empty family `Chapter15/ParamFamily · SBLocalization.GPFamily.zero` needs Λ_0 to be structurable. With a good system it is, with the zero decoration `Chapter15/ParamFamily · SBLocalization.isStructurable_zeroMultisection_of_goodSystem`. - *Sums* `Chapter15/GeneralPositionExistence · SBLocalization.ScPlusGPFamily.add`. Families good on Q and Q' add to a family good on $Q \cup Q'$: - the sum of sc⁺-multisections is one (Definition 13.2.9) `Chapter13/MultisectionAlgebra · SBLocalization.definition_13_2_9`; - the norm bound adds, $|t|_1 h + |t'|_1 h = |(t, t')|_1 h$; - the domain supports stay in U; - (15.1) holds on Q by `famAdd_left` and on Q' by `famAdd_right`.

The structurable sum `Chapter15/ParamFamily · SBLocalization.GPFamily.add` is the same, with Proposition 13.3.12 for the structurability of a sum `Chapter13/StructurableMultisection · SBLocalization.proposition_13_3_`

12. - By induction on a finite index set, with the empty family as the empty sum, families good on $Q_1, \dots, Q_k$ give a family good on $\bigcup_i Q_i$ Chapter15/GeneralPositionExistence · SBLocalization.nonempty_scPlusGPFamily_biUnion Chapter15/GeneralPositionAssembly · SBLocalization.nonempty_gpFamily_biUnion. - So without a good system, sums of sc⁺-families need nothing; with structurable families, the empty sum needs a good system. This is the only place where the two forms of the theorem differ.

Step (c), from local to global Chapter15/GeneralPositionExistence · SBLocalization.exists_scPlusGPFamily_of_compact Chapter15/GeneralPositionAssembly · SBLocalization.exists_gpFamily_of_compact. Suppose every $x \in S_f$ lies in an open saturated Q_x carrying a good family. In the structurable form this hypothesis is named Chapter15/GeneralPositionAssembly · SBLocalization.HasLocalGPFamilies. Then there is one family good on an open $Q \supseteq S_f$. - The sets $\pi(Q_x)$ are open in $|X|$, since π is open (Proposition 7.1.8), and they cover the compact $|S_f|$. Choose finitely many $x_1, \dots, x_k$ and put $Q = \bigcup_i Q_{x_i}$. - $S_f \subseteq Q$: a point of S_f has its orbit in some $\pi(Q_{x_i})$, and Q_{x_i} is saturated. - The family on Q is the sum of the k families.

The sc⁺ version first saturates the open neighbourhoods it is given (§XV.23). This is the text's "From local to global", with $\bar{\ell} = \ell_{x_1} + \cdots + \ell_{x_k}$.

Steps (d) and (e), the choice of t. Step (d) (§XV.25) gives $\varepsilon > 0$ and a Lebesgue-null set off which (f, Λ_t) is in general position for $\|t\| < \varepsilon$ (sup norm on $\mathbb{R}^\ell$). The hypothesis $N(\Lambda_t) \le |t|_1$ it needs follows from $N(\Lambda_t) \le |t|_1 h$ and $h \le 1$; (15.1) holds on S_f because $S_f \subseteq Q$. - Put $r = \min(\varepsilon, 1/(\ell+1)) > 0$. A ball of positive radius has positive measure, so it is not contained in a null set. Hence there is t with $\|t\| < r$ and (f, Λ_t) in general position Chapter15/GeneralPositionAssembly · SBLocalization.exists_norm_lt_of_ae. - From $\|t\| < 1/(\ell+1)$: $|t|_1 \le \ell \|t\| < 1$ Chapter15/GeneralPositionAssembly · SBLocalization.sum_abs_lt_one. - (1): $N(\Lambda_t)(x) \le |t|_1 h(x) < h(x)$, since $h(x) > 0$. - (2): the family's domain supports lie in U. - (4) is the choice of t. - (3) follows from (4) Chapter15/GeneralPositionAssembly · SBLocalization.isTransversal_of_isInGeneralPosition. A support point is a level-1 point, and an operator that is onto on the reduced tangent space $T^R_x X \subseteq T_x X$ is onto.

Readings. - The text's ε_0, "so that $N(\Lambda_t)(x) < 1$ … provided $|t| \le \varepsilon_0$", is $1/(\ell+1)$ in the sup norm. The bound actually used is $N(\Lambda_t) \le |t|_1 h$, which gives (1) with h and not only with 1. - "A set of measure zero … for t in its complement" is "for almost every t". - The text's remark that the complement "contains points arbitrarily close to 0" is the positive measure of balls.

Departures from the text's assembly. - The text starts from "a large parameter space $\mathbb{R}^h$, i.e. $h >> 0$". In the formalization the number of parameters is whatever the construction produces, the sum $\ell_{x_1} + \cdots + \ell_{x_k}$ of the local ones. - The text views $\tilde{\Lambda}$ as an sc⁺-multisection over $\mathbb{R}^{\bar{\ell}} \times X$ and $\tilde{f}(t,x) = (t, f(x))$ as an sc-Fredholm section there. The formalization never forms this pair on the product ep-groupoid. Everything it needs from $\tilde{\Lambda}$ is in the linear representations: (15.1) at the points of S_f, the norm bound, and the domain supports. The product appears only branch by branch, on $\mathbb{R}^\ell \times O$ with O open in the object M-polyfold (§XV.25), where Part I applies.

The assembly with the two analytic steps as hypotheses Chapter15/GeneralPositionAssembly · SBLocalization.theorem_15_3_7_hyp. Steps (c) and (e) for structurable families, as one statement. Its hypotheses: - a good system; - compact $|S_f|$; - (N, U) controlling compactness; - $0 < h \le 1$; - the local families of step (b) Chapter15/GeneralPositionAssembly · SBLocalization.HasLocalGPFamilies; - the parametric general position of step (d) Chapter15/GeneralPositionAssembly · SBLocalization.HasParametricGeneralPosition. This is stated for every linear family of sc⁺-multisections with $N(\Lambda_t) \le |t|_1$, domain supports in U and (15.1) on S_f.

Its conclusion is a structurable Λ with (1)–(4). The structurable form of the theorem then follows from a good system and the two steps Chapter15/GeneralPositionAssembly · SBLocalization.theorem_15_3_7_of_hyp. Both

steps are proved from the theorem's hypotheses in §§XV.23–XV.25, so these statements are the skeleton of the final proof, not additional assumptions of it.

XV.23 Proof of Theorem 15.3.7, II: the local constructions

The text's "Local constructions at $x \in S_f$": - choose vectors $w_1, \dots, w_\ell \in W_x$ completing the image of $f'(x)$, on $T^R_x X$ at boundary points; - extend them to sc⁺-sections s^1_j supported in an invariant $V(x)$ with closure in $U(x)$; - symmetrize over G_x; - take the atomic multisection; - invoke the implicit function theorem for a neighbourhood $Q(x)$ on which the parametrized problem stays onto.

The formalization follows this order. It needs one more tool, the transport of (15.1) along morphisms, because the symmetrized branches are moved by the isotropy group, and because the text passes to saturations "by equivariance".

(15.1) with given vectors `Chapter15/ParamOntoTransport · SBLocalization.IsParamOntoVec`. At a level-1 point x and for vectors $v_1, \dots, v_\ell \in W_x$: for every operator L of $T(f, \Lambda_0)(x)$, every chart presentation of $T^R_x X$ and every $w \in W_x$, there are $u \in T^R_x X$ and τ with $Lu + \sum_j \tau_j v_j = w$. (15.1) for a family at x is this property for the values $s_{ij}(x)$ of every branch of a linear representation `Chapter15/ParamOntoTransport · SBLocalization.isParamOntoAt_iff_vec`.

Transport along morphisms `Chapter15/ParamOntoTransport · SBLocalization.IsParamOntoVec.of_mor`. If $\varphi : x \to y$ and (15.1) holds at x with v_j, it holds at y with $\mu(\varphi, v_j)$.

Proof. 1. *The operators at y.* Every operator L' of $T(f, \Lambda_0)(y)$ is the conjugate $\mu(\varphi, \cdot) \circ L \circ (T\varphi)^{-1}$ of an operator L of $T(f, \Lambda_0)(x)$. This is the functor property of the linearization (§XV.9) `Chapter15/LinearizationFunctor · SBLocalization.theorem_15_1_1_functor`, read on the support of the pushed-forward formal sum. 2. *The reduced tangent space at y.* Take a chart at x, restricted to the sheet of φ, and transport it along φ. Then $T\varphi$ maps $T^R_x X$ into the reduced tangent space at y computed in the transported chart `Chapter15/ParamOntoTransport · SBLocalization.tangentMorCLE_mem_reducedTangentFiber_transport`. By chart independence of T^R (Part I) `Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber_eq`, this is the reduced tangent space of every chart presentation at y. 3. *Solving.* $\mu(\varphi, \cdot) : W_x \to W_y$ is a linear bijection with inverse $\mu(\varphi^{-1}, \cdot)$ `Chapter15/ParamOntoTransport · SBLocalization.muLin_surjective`. Write $w = \mu(\varphi, w_0)$ and solve $Lu + \sum_j \tau_j v_j = w_0$ at x. Then $T\varphi\, u \in T^R_y X$ and

$$L'(T\varphi\, u) + \textstyle\sum_j \tau_j \mu(\varphi, v_j) = \mu(\varphi, Lu + \sum_j \tau_j v_j) = w.$$

Linear representations are carried too `Chapter15/ParamOntoTransport · SBLocalization.exists_linearRep_of_mor`. Suppose Λ_t is a functor for every t and $(\sum_j t_j s_{ij}, \sigma_i)$ represents it on $O \ni x$. Transport each section along the local diffeomorphism of φ; this gives sections on a neighbourhood of y `Chapter15/ParamOntoTransport · SBLocalization.transSec_eq_muLin`. Since μ is fibrewise linear, the transported combinations are the combinations of the transported sections `Chapter15/ParamOntoTransport · SBLocalization.paramSec_transSec_eqOn`. They represent Λ_t there because Λ_t is a functor. So (15.1) for a family passes along morphisms in both directions `Chapter15/ParamOntoTransport · SBLocalization.IsParamOntoAt.of_mor` `Chapter15/ParamOntoTransport · SBLocalization.IsParamOntoAt.of_mor_inv`.

Saturation ("the same assertion also holds for points … in the saturation of $Q(x)$"). - S_f is saturated: if $\varphi : a \to z$ and $f(a) = 0$, then $f(z) = \mu(\varphi, 0) = 0$ `Chapter15/LocalFamilies · SBLocalization.mem_solutionSet_tgt`. - A family good on an open V is good on the saturation of V `Chapter15/LocalFamilies · SBLocalization.`

GPFamily.saturate Chapter15/GeneralPositionExistence · SBLocalization.ScPlusGPFamily.saturate. For $z \in S_f$ in the saturation there are $a \in V$ and $\varphi : a \to z$. Then $a \in S_f$, so (15.1) holds at a and passes to z along φ. - The saturation of an open set is open, because π is open. So it suffices to produce, at each $x \in S_f$, an open $V \ni x$ carrying a good family Chapter15/LocalFamilies · SBLocalization.hasLocalGPFamilies_of_nhds.

Seed data Chapter15/LocalFamilies · SBLocalization.SeedData. Seed data consist of: - a point x; - a natural representation of G_x on an open $U(x)$ with the properness property; - an open $V(x) \ni x$ invariant under the action, with $\operatorname{cl} V(x) \subseteq U(x)$; - sc⁺-sections $s_1, \dots, s_\ell$ of W supported in $V(x)$.

Such $U(x)$, $V(x)$ exist inside any open neighbourhood of x Chapter13/AtomicMultisection · SBLocalization.exists_atomicNbhd (Theorem 7.1.13 and the proof of Theorem 7.5.7).

The atomic family Chapter15/LocalFamilies · SBLocalization.SeedData.Λ. For each t, Λ_t is Part XIII's atomic multisection generated by the seed $\sum_j t_j s_j$. Its branches on $U(x)$ are the translates

$$(\textstyle\sum_j t_j s_j)^g(y) = \mu(\Gamma(g, g^{-1} * y), \sum_j t_j s_j(g^{-1} * y)), \qquad g \in G_x,$$

which is the text's $s_t^{x,g}$. Each Λ_t is an sc⁺-multisection. - *Linear branches on $U(x)$* Chapter15/LocalFamilies · SBLocalization.SeedData.isLocalRepOn_branch. Since μ is linear on fibres, $(\sum_j t_j s_j)^g = \sum_j t_j s_j^g$ Chapter15/LocalFamilies · SBLocalization.natTerm_paramSec. So on $U(x)$ the family has the linear representation with sections s_j^g and weights $1/|G_x|$. - *Away from the seed* Chapter15/LocalFamilies · SBLocalization.SeedData.isLocalRepOn_zero. Off the saturation of $\operatorname{cl} V(x)$, Λ_t is the trivial multisection, represented by the zero section. - *Linear everywhere* Chapter15/LocalFamilies · SBLocalization.SeedData.isLinearFamily. There are two cases for a point x_0. - If x_0 is the target of a morphism φ from a point of $U(x)$, the representation on $U(x)$ is carried along φ (`exists_linearRep_of_mor`). - Otherwise x_0 lies off the closed saturation of $\operatorname{cl} V(x)$, and the zero representation serves. - *Structurable* Chapter15/LocalFamilies · SBLocalization.SeedData.isStructurable_Λ. Given a good system, every Λ_t is structurable: an atomic multisection is structurable (Theorem 13.5.3) Chapter13/AtomicStructurable · SBLocalization.theorem_13_5_3_of_goodSystem. This is the text's "for fixed t it is structurable". - *Domain support* Chapter15/LocalFamilies · SBLocalization.SeedData.domSupp_Λ_subset. It lies in the saturation of $U(x)$. - *Norm* Chapter15/LocalFamilies · SBLocalization.SeedData.norm_le. If $N(s_j) \le c$ and $c \le h$ on the saturation of $U(x)$, then $N(\Lambda_t) \le |t|_1 h$. - At a point reached from $U(x)$, every branch value is $\mu(\Gamma, \sum_j t_j s_j(\cdot))$. Its norm is at most $|t|_1 c$, since N is invariant under μ Chapter13/AtomicConstruction · SBLocalization.SymmetricFamily.ofSeed_normW_le Chapter13/AtomicConstruction · SBLocalization.AtomicData.pointwiseNorm_mult_le. And $c \le h$ there. - At other points the pointwise norm is 0. - *(15.1)* Chapter15/LocalFamilies · SBLocalization.SeedData.isParamOntoAt. Let $W \ni x$ be open such that the vectors $s_j(y)$ satisfy (15.1) at every point y of $S_f \cap W$. Put

$$Q_0 = \bigcap_{g \in G_x} (U(x) \cap \Phi_g^{-1}(W)).$$

This set is open (a finite intersection) and contains x. At $y \in Q_0 \cap S_f$, the branch g takes the values $\mu(m, s_j(\operatorname{src} m))$, where $m = \Gamma(g, \Phi_{g^{-1}}(y))$ is a morphism from $\Phi_{g^{-1}}(y) \in W$ to y. Its source lies in S_f, since S_f is saturated, and in W. So (15.1) holds there with s_j, and `IsParamOntoVec.of_mor` carries it to y for the branch g. This is the text's "the linearization … at $(0, x)$ is also surjective for every g", extended from x to the points of S_f near x. - *Together* Chapter15/LocalFamilies · SBLocalization.SeedData.gpFamily Chapter15/GeneralPositionExistence · SBLocalization.SeedData.scPlusGPFamily. The atomic family is good on Q_0: structurable given a good system, and sc⁺ without one.

Local families from local seeds Chapter15/LocalFamilies · SBLocalization.hasLocalGPFamilies_of_seeds Chap-

ter15/GeneralPositionExistence · SBLocalization.exists_scPlusGPFamily_nhds. The analytic input is named *local seeds* Chapter15/LocalFamilies · SBLocalization.HasLocalParamSeeds. It says: at every $x \in S_f$, for every open $O \ni x$ and every $\varepsilon > 0$, there are - finitely many sc⁺-sections s_j supported in O with $N(s_j) \le \varepsilon$, and - an open $W \ni x$ such that the $s_j(y)$ satisfy (15.1) at every level-1 point y of $S_f \cap W$.

Given local seeds, every $x \in S_f$ has an open neighbourhood carrying a good family.

Proof. 1. Put $c = h(x)/2 > 0$ and $W_1 = U \cap \{y : h(y) > c\}$, an open neighbourhood of x. 2. Choose $U(x) \subseteq W_1$ and $V(x)$ as above, and seeds with $O = V(x)$ and $\varepsilon = c$. 3. $c \le h$ *on the saturation of* $U(x)$. A point y there is isomorphic to some $a \in U(x)$, and $h(y) = h(a) > c$ because h is constant along morphisms. 4. *The saturation of* $U(x)$ *lies in* U, because U is saturated. 5. So the atomic family has $N(\Lambda_t) \le |t|_1 h$, domain support in U, and (15.1) on Q_0.

The text asks $N(s_j^1(y)) \le \min\{1, h(y)\}$ on $U(x)$. The formalization takes the constant bound $h(x)/2$ and shrinks $U(x)$ so that $h > h(x)/2$ on it. This gives the bound on the saturation directly.

With local seeds, the structurable form is proved in one statement Chapter15/LocalFamilies · SBLocalization.theorem_15_3_7_hyp_seeds Chapter15/LocalFamilies · SBLocalization.theorem_15_3_7_of_seeds. These are intermediate steps of the final proof.

Seeds from two facts at single points Chapter15/LocalSeeds · SBLocalization.hasLocalParamSeeds_of. Local seeds follow from bump functions, the smoothness of the points of S_f, and two facts: - *small vectors* Chapter15/LocalSeeds · SBLocalization.HasPointwiseParamOnto: at every level-1 point $x \in S_f$ and for every $\varepsilon > 0$, there are smooth vectors $w_j \in W_x$ with $N(w_j) < \varepsilon$ satisfying (15.1) at x; - *openness* Chapter15/LocalSeeds · SBLocalization.HasParamOntoOpen: if the values of sc⁺-sections s_j satisfy (15.1) at $x \in S_f$, there is an open $W \ni x$ such that they satisfy it at every level-1 point of $S_f \cap W$.

Proof. 1. At $x \in S_f$, take the small vectors with $N(w_j) < \varepsilon/2$. 2. Part I's Lemma 5.3.3, on the object M-polyfold with its bump functions (Theorem 12.3.2 (1)), extends each smooth w_j to an sc⁺-section s_j with $s_j(x) = w_j$, support in O, and $N(s_j) \le \varepsilon$ Chapter5/BumpFunctions · lemma_5_3_3. 3. Then (15.1) holds at x with the $s_j(x)$, and openness gives W.

Points of S_f are smooth by the regularity that the tame germ data provide Chapter15/Transversality · SBLocalization.mem_smoothSet_of_eq. The structurable form from the two facts and parametric general position Chapter15/LocalSeeds · SBLocalization.theorem_15_3_7_of_pointwise is the next intermediate step.

The two facts are the text's choice of the w_j ("we can pick the w_i as small as we wish") and its appeal to the implicit function theorem for $Q(x)$. Both are proved from the theorem's hypotheses in §XV.24.

Departures from the text's local constructions. - The text's two cases, $d_X(x) = 0$ and $d_X(x) \ge 1$, are one construction, because (15.1) is stated on $T_x^R X$. - The text's "(Interior)" and "(Boundary)" conclusions assert that the parametrized solution set over $\{|t| < \varepsilon_x\} \times Q(x)$ is a tame branched ep⁺-subgroupoid. The formalization does not use this structure: what the assembly needs from $Q(x)$ is (15.1) at the points of $S_f \cap Q(x)$. The local structure of the parametrized solution set enters only in §XV.25, branch by branch, through Part I.

XV.24 Proof of Theorem 15.3.7, III: (15.1) and Part I's conditions (i)–(ii)

The two facts of §XV.23 concern (15.1), stated on the ep-groupoid with reduced tangent spaces. Part I's proof of Theorem 5.3.10 works with conditions (i)–(ii) for a parametrized section in an M-polyfold chart,

with the full tangent space. This subsection proves the two conditions equivalent at a zero of f, and then derives both facts from Part I.

Part I's conditions Chapter5/GeneralPosition · StrongBundleChart.IsParamGP. Take a strong bundle chart centred at y (chart point 0, retraction r, model fibre K_0) and a presentation T of its partial quadrant, whose minimal subspace E_0 is the chart's reduced tangent space. Let $F(a,u) = \mathrm{Lin}_g(0)u + \sum_i a_i \hat{t}_i(0)$ be the linearization of $g + \sum_i \lambda_i t_i$ at $(0,y)$. The conditions are: - (i) F maps $\mathbb{R}^m \times T_0O$ onto K_0; - (ii) the kernel of F is transversal to $\mathbb{R}^m \oplus T^R$: every $w \in T_0O$ is $u + e$ with $e \in E_0$ and $F(a,u) = 0$ for some a.

The bridge (extra) Chapter15/ParamOntoBridge · SBLocalization.isParamOntoVec_iff_isParamGP. Let x be a level-1 zero of f, let a strong bundle chart of W be centred at x with a presentation T, and let s_j be sc⁺-sections. Then (15.1) at x with the vectors $s_j(x)$ holds if and only if Part I's (i)–(ii) hold in that chart for $f + \sum_j t_j s_j$.

Proof. Both sides are read in the chart. 1. *The operator.* At a zero of f, the only operator of $T(f,\Lambda_0)(x)$ is the intrinsic form of the chart linearization $D((\hat{f} - \hat{0}) \circ r)(0)$ Chapter15/ParamOntoBridge · SBLocalization.mem_support_linearization_zero_iff Chapter15/ParamOntoBridge · SBLocalization.linAt_principal_eq_subDeriv. It is taken at the chart point of x, which is 0 Chapter15/ParamOntoBridge · SBLocalization.levelLift_eq_zero_of_centred. 2. *(15.1) in the chart* Chapter15/ParamOntoBridge · SBLocalization.onto_chart_iff. (15.1) says that $(v,\tau) \mapsto D((\hat{f} - \hat{0}) \circ r)(0)v + \sum_j \tau_j \hat{s}_j(0)$ maps $(T_0O \cap E_0) \times \mathbb{R}^\ell$ onto K_0. The fibre W_x is K_0 read through the chart, and $T^R_x X$ is $T_0O \cap E_0$. 3. *(i)–(ii) are equivalent to this reading* Chapter15/ParamOntoBridge · SBLocalization.isParamGP_iff_onto_chart. Write L for the chart linearization and $\hat{s}_j$ for $\hat{s}_j(0)$. - *Forward.* Given $k \in K_0$, (i) gives (a,u) with $Lu + \sum a_j \hat{s}_j = k$. Then (ii), applied to u, gives (a',u') with $Lu' + \sum a'_j \hat{s}_j = 0$ and $u - u' \in E_0$. So $(a - a', u - u')$ hits k with $u - u'$ reduced. - *Backward.* (i) is immediate. For (ii), take $w \in T_0O$. Then $Lw \in K_0$; solve $Lu + \sum \tau_j \hat{s}_j = Lw$ with u reduced. The pair $(-\tau, w - u)$ is in the kernel, and $w - (w - u) = u \in E_0$. 4. *Every chart presentation.* (15.1) quantifies over all chart presentations, while the right side uses one. The two agree because the reduced tangent space does not depend on the chart (Part I) Chapter2/ReducedTangentIntrinsic · MPolyfold.reducedTangentFiber_eq.

Extra: the text states (15.1) with $T^R_x X$, while Part I's proof of Theorem 5.3.10 uses (i)–(ii) with the full tangent space. The text presumably takes their equivalence for granted when it cites "the corresponding implicit function theorem".

Tame strong bundle charts centred at a point of S_f, with a presentation, exist by the tame germ data Chapter15/GeneralPositionGoodSystem · SBLocalization.exists_tameChart_centred.

Small vectors (explicit) Chapter15/GeneralPositionGoodSystem · SBLocalization.hasPointwiseParamOnto. With bump functions and the tame germ data, the small-vectors fact of §XV.23 holds.

Proof. Let $x \in S_f$ be a level-1 point and $\varepsilon > 0$. 1. Take a tame chart centred at x. 2. Part I's proof of Theorem 5.3.10 provides sc⁺-sections s_j of the object M-polyfold bundle for which $f + \sum_j t_j s_j$ satisfies (i)–(ii) in this chart. They can be chosen with norm at most 1 for any given auxiliary norm Chapter5/GeneralPosition · exists_sections_isParamGP. 3. Apply this to the auxiliary norm $(2/\varepsilon)N$; a positive multiple of an auxiliary norm is one. This gives $N(s_j) \le \varepsilon/2 < \varepsilon$. 4. The bridge turns (i)–(ii) into (15.1) for the vectors $w_j = s_j(x)$. 5. The w_j are smooth, because x is smooth and sc⁺-sections map smooth points to smooth vectors.

Explicit: this is the text's choice of $w_1, \dots, w_\ell$ "as small as we wish", made in a chart.

Openness (explicit) Chapter15/GeneralPositionGoodSystem · SBLocalization.hasParamOntoOpen. For tame X and f with tame germ data, the openness fact of §XV.23 holds.

Proof. Let sc⁺-sections s_j satisfy (15.1) at $x \in S_f$. 1. The bridge, in a tame chart centred at x, gives (i)–(ii). 2. Part I turns (i)–(ii) into general position of $F(t,y) = f(y) + \sum_j t_j s_j(y)$ on $\mathbb{R}^\ell \times X$ at $(0,x)$ `Chapter5/GeneralPosition · isGeneralPositionGerm_paramSection_of_isParamGP`. 3. $\mathbb{R}^\ell \times X$ is tame `Chapter3/ParamPerturbation · prod_rn_isTame` and F is a tame sc-Fredholm section `Chapter3/ParamPerturbationG · theorem_3_1_21_tame`. So general position at a zero spreads to the zeros in an open $W_0 \ni (0,x)$ `Chapter5/GeneralPosition · exists_isOpen_isGeneralPositionGerm`. 4. Put $W = \{y : (0,y) \in W_0\}$. At a point $y \in S_f \cap W$, $(0,y)$ is a zero of F in general position. 5. In a tame chart centred at y, Part I recovers (i)–(ii) `Chapter5/GeneralPosition · isParamGP_of_isGeneralPositionGerm`, and the bridge at y gives (15.1) there.

Explicit: this is the text's "from the local implicit function theorem we deduce", made precise as the openness of Part I's general position.

Both facts are needed only at points of S_f, which are smooth, so centred tame charts are available.

XV.25 Proof of Theorem 15.3.7, IV: achieving global transversality

Parametric general position (explicit) `Chapter15/ParametricGeneralPosition · SBLocalization.hasParametricGeneralPosition`. Suppose X is tame, f is an sc-Fredholm section functor with tame germ data, $|S_f|$ is compact, and (N,U) controls compactness. Let $t \mapsto \Lambda_t$, $t \in \mathbb{R}^\ell$, be a family of sc⁺-multisections with: - $N(\Lambda_t) \le |t|_1$; - domain supports in U; - (15.1) at every level-1 point of S_f.

Then there is $\varepsilon > 0$ such that (f, Λ_t) is in general position for almost every t with $\|t\| < \varepsilon$. This is step (d) of §XV.22. The proof does not use that the family is linear everywhere (the linear representations inside (15.1) at the points of S_f suffice), nor bump functions, nor h.

The text proves this in "Achieving global transversality": - it uses Theorem 15.2.5 for an ε below which the parametrized solution set is a tame branched ep⁺-subgroupoid with surjective reduced linearizations; - around each of its points (t_0, x_0) it writes the branches as manifolds with corners M_i; - it applies Sard's theorem to $F \cap M_i \to \mathbb{R}^{\bar\ell}$ for every intersection F of local faces; - it covers the set (15.3) by countably many, or by compactness finitely many, such neighbourhoods.

The formalization keeps the plan but makes three changes. - It centres the local neighbourhoods only at points $(0,x)$ with $x \in S_f$. The form of (15.2) proved below shows that for small t the support of $\Lambda_t \circ f$ stays near S_f. - It works branch by branch, where each branch is a section of a strong bundle over an open subset of the object M-polyfold, so that Part I's results apply directly. - Part I's Theorem 4.3.1 in local form carries the faces. This replaces the text's "obvious branched ep⁺-groupoid version of Theorem 4.3.1".

(i) Part I's Sard step, locally (extra) `Chapter15/LocalSlice · StrongBundle.exists_null_isGeneralPositionGerm_paramSlice_near`. Let X be a tame M-polyfold, g a tame sc-Fredholm section, and $t_1, \dots, t_m$ sc⁺-sections. Suppose $F(\lambda, y) = g(y) + \sum_i \lambda_i t_i(y)$ is in general position at a zero z_0. Then there are an open $W \ni z_0$ in $\mathbb{R}^m \times X$ and a null set $C \subseteq \mathbb{R}^m$ such that, for $\lambda \notin C$, the slice $F(\lambda, \cdot)$ is in general position at its zeros y with $(\lambda, y) \in W$.

Proof. 1. F is a tame sc-Fredholm section on the tame $\mathbb{R}^m \times X$ `Chapter3/ParamPerturbationG · theorem_3_1_21_tame Chapter3/ParamPerturbation · prod_rn_isTame`. 2. General position at z_0 gives an open W_0 on which the zero set A of F is in general position `Chapter5/GeneralPosition · exists_isOpen_isGeneralPositionGerm`. 3. *Faces.* Part I's Theorem 4.3.1 in local form gives a null set C_1 off which the fibres $\{y : (\lambda, y) \in A\}$ are submanifolds in general position to the faces `Chapter4/Families · exists_local_family`. 4. *Surjectivity.*

Part I's Sard step gives a null set C_2 off which the linearizations of the slice are onto at its zeros near z_0 Chapter5/Transversality · exists_null_isLinSurjAt_paramSlice_near. 5. *Assembling general position of the slice.* - The slice is a tame sc-Fredholm section Chapter3/TameGermsAddG · theorem_3_1_18_tame. - Near a zero, its zero set is the fibre piece of A, a submanifold Chapter15/LocalSlice · MPolyfold.IsGeneralPositionAt.exists_inter. - Part I's characterization of general position by good position of a tame germ applies Chapter5/GeneralPosition · isGoodPositionGermTame_of_isGeneralPositionAt. - So the slice is in general position at every zero in W, with $C = C_1 \cup C_2$.

Extra: Part I's Theorem 5.3.10 is stated globally, under compactness Chapter5/GeneralPosition · theorem_5_3_10_of_family. The proof of Theorem 15.3.7 needs its Sard step near one point.

(ii) A branch as a Part I section. Let $(\sum_j t_j s_{ij}, \sigma_i)$ be a linear representation of the family on O, and fix a branch i. - *The parametrized section.* On the open subset O of the object M-polyfold (tame, since an open subset of a tame M-polyfold is tame), take $g = (f-0)|_O$, the section of $W|_O$ of §XV.10, and $t_j = (-s_{ij})|_O$ Chapter15/BranchChart · SBLocalization.negRestrict. - *Its slices are the branches* Chapter15/BranchChart · SBLocalization.paramSlice_branch_eq. The slice at t of $g + \sum_j t_j(-s_{ij})|_O$ is $(f - \sum_j t_j s_{ij})|_O$. Its zeros are the points where f meets branch i of Λ_t. - *(15.1) gives Part I's (i)–(ii)* Chapter15/BranchChart · SBLocalization.isParamGP_restrictTo_of_onto. Take a chart of W centred at x with domain in O, transported to $W|_O$. The chart operator of g is the chart linearization of f Chapter15/BranchChart · SBLocalization.principal_eq_fiberLin, and the principal part of $(-s)|_O$ is $-\hat{s}$ Chapter15/BranchChart · SBLocalization.principal_restrictTo_negRestrict. - For (i): solve with (15.1) and change the sign of τ. - For (ii): for w, solve $Lv + \sum_j \tau_j \hat{s}_{ij} = Lw$ with v reduced. Then $(\tau, w - v)$ is in the kernel and $w - (w - v) = v \in E_0$.

This is the argument of the bridge (§XV.24), in the form needed here. - *General position at* $(0, y)$ Chapter15/ParametricGeneralPosition · SBLocalization.isGeneralPositionGerm_branchParam. Suppose (15.1) holds at $y \in S_f \cap O$. 1. y is smooth Chapter15/Transversality · SBLocalization.mem_smoothSet_of_eq. 2. The tame germ data give a tame chart centred at y, restricted into O. 3. The operator of $T(f, \Lambda_0)(y)$ is obtained from a local linearization datum of Λ_0. 4. Then (15.1) gives (i)–(ii) for the branch, and Part I gives general position of the parametrized branch at $(0, y)$ Chapter5/GeneralPosition · isGeneralPositionGerm_paramSection_of_isParamGP.

The s_{ij} are sc$^+$ on O, as the branches of Λ_{e_j} Chapter15/ParametricGeneralPosition · SBLocalization.isScPlusSectionOn_of_paramSec.

(iii) Local general position Chapter15/ParametricGeneralPosition · SBLocalization.exists_local_paramGP. Suppose X is tame, f has tame germ data, the Λ_t are sc$^+$, $x_0 \in S_f$ is a level-1 point, and (15.1) holds at x_0. Then there are an open $V \ni x_0$, a $\delta > 0$ and a null set C such that (f, Λ_t) is in general position at every $y \in V$ with $\Lambda_t(f(y)) > 0$, whenever $\|t\| < \delta$ and $t \notin C$.

Proof. 1. Take the linear representation on $O \ni x_0$ from (15.1). 2. For each branch i, (ii) and (i), applied on $\mathbb{R}^\ell \times O$, give an open $W_i \ni (0, x_0)$ and a null set C_i. 3. Shrink each W_i to a product $u_i \times v_i$. Take δ with the δ-ball inside every u_i, put $v' = \bigcap_i v_i$, let V be its image in X, and put $C = \bigcup_i C_i$. 4. *General position at the support points in* V. Let $y \in V$ with $\Lambda_t(f(y)) > 0$. The operators of $T(f, \Lambda_t)(y)$ are the linearizations at y of $f - \sum_j t_j s_{ij}$ over the branches through $f(y)$ (Definition 15.1.7, §XV.8). For each such branch, y is a zero of the slice at t, so the slice is in general position at y. 5. In a chart at y restricted to O, this gives the reduced surjectivity of that operator (§XV.16) Chapter15/GeneralPositionLocal · SBLocalization.isReducedOntoAtLevel_of_isGeneralPositionGerm. Hence general position at y in the sense of Definition 15.3.5 at one point Chapter15/GeneralPositionLocal · SBLocalization.IsInGeneralPositionAt.

(iv) Invariance under morphisms (extra) Chapter15/ParametricGeneralPosition · SBLocalization.

isInGeneralPositionAt_of_mor. General position at a point passes along morphisms: if $\varphi : y \to x$ and (f, Λ) is in general position at y, it is so at x. The proof is that of the transport of (15.1) in §XV.23 without the parameters Chapter15/ParametricGeneralPosition · SBLocalization.isReducedOntoAtLevel_of_mor: - the operators at x are conjugates of those at y (§XV.9); - $T\varphi$ maps reduced tangent vectors to reduced tangent vectors; - $\mu(\varphi, \cdot)$ is onto.

Extra: the text uses this silently when it passes to saturations "by equivariance" ("the properties of $(\tilde{f}, \tilde{\Lambda})$ at $(0, x')$ are the same as those of $(0, x)$").

(v) The form of (15.2) the proof uses (extra) Chapter15/ParametricGeneralPosition · SBLocalization.exists_pos_auxLe_subset. Suppose f is an sc-Fredholm section functor, (N, U) controls compactness, and $V \supseteq S_f$ is open and saturated. Then there is $\varepsilon > 0$ with $\{x \in U : N(f(x)) \le \varepsilon\} \subseteq V$.

Proof. By contradiction. 1. Suppose that for every n there is $x_n \in U \setminus V$ with $N(f(x_n)) \le 1/(n+1)$. The orbits $\pi(x_n)$ lie in the compact closure that (N, U) provides, so they have a cluster point $q = \pi(z)$. 2. The object space is metrizable, so z has a decreasing neighbourhood basis B_k. Since π is open and q is a cluster point, there are $n_k \ge k$ and $y_k \in B_k$ with $\pi(y_k) = \pi(x_{n_k})$. 3. Because U and V are saturated, $y_k \in U \setminus V$. 4. $N(f(y_k)) \le 1/(k+1)$, since $N \circ f$ is invariant along morphisms: $f(\mathrm{tgt}\,\varphi) = \mu(\varphi, f(\mathrm{src}\,\varphi))$, and N is invariant under μ. 5. So $y_k \to z$ and $N(f(y_k)) \to 0$. The third property of an auxiliary norm (Part I's Definition 5.1.1 (3)) gives $f(y_k) \to 0_z$. 6. Since f is continuous, $f(z) = 0$. Then $z \in S_f \subseteq V$, and V is open, so $y_k \in V$ for large k, a contradiction.

On the support of $\Lambda \circ f$, $N(f(x)) \le N(\Lambda)(x)$, and $x \in U$ once the domain support lies in U Chapter15/ParametricGeneralPosition · SBLocalization.auxLe_of_pos. This is the argument of step 4 of the "in particular" (§XV.21).

Extra in this form: the text states (15.2) as the compactness of $\{(t, x) : |t| \le \varepsilon_0, \Lambda_t(f(x)) > 0\}$. It uses that compactness to keep the parametrized solution set within finitely many local neighbourhoods. The formalization proves exactly that consequence.

Proof of parametric general position. 1. *Local neighbourhoods.* Every $z \in S_f$ is smooth, hence a level-1 point, and (15.1) holds there. So (iii) gives V_z, δ_z, C_z. 2. *Finite cover.* The open sets $\pi(V_z)$ cover the compact $|S_f|$. Choose a finite set T of such z, and let V be the saturation of $\bigcup_{z \in T} V_z$. It is open, saturated, and contains S_f. 3. *Staying near S_f.* (v) gives ε_0 for this V. Choose $\varepsilon > 0$ so that $\|t\| < \varepsilon$ implies $\|t\| < \delta_z$ for $z \in T$ and $|t|_1 \le \varepsilon_0$ (recall $|t|_1 \le \ell \|t\|$). 4. *The null set* is $C = \bigcup_{z \in T} C_z$, a finite union. 5. *General position.* Let $\|t\| < \varepsilon$ and $t \notin C$. General position is checked point by point Chapter15/GeneralPositionLocal · SBLocalization.isInGeneralPosition_iff_forall_at. Let $\Lambda_t(f(x)) > 0$. - By auxLe_of_pos with $N(\Lambda_t) \le |t|_1 \le \varepsilon_0$, $x \in U$ and $N(f(x)) \le \varepsilon_0$, so $x \in V$. - So there are $z \in T$, $y \in V_z$ and $\varphi : y \to x$. - $\Lambda_t(f(y)) = \Lambda_t(f(x)) > 0$, because f and Λ_t are functors. - So (iii) gives general position at y, and (iv) carries it to x.

Explicit: this is the step "Achieving global transversality", with four readings: - Sard "on the intersections of faces" is read as Part I's local Theorem 4.3.1 with its Sard step, branch by branch on $\mathbb{R}^\ell \times O$; - the passage to saturations is (iv); - "using the compactness … a finite number of sets … suffices" is the finite cover of $|S_f|$ together with (v); - the text's appeal to Theorem 15.2.5 for the structure of the parametrized solution set is replaced by Part I's results, through (ii) and (i).

With parametric general position, the structurable form follows from the local families alone Chapter15/ParametricGeneralPosition · SBLocalization.theorem_15_3_7_hyp_local Chapter15/ParametricGeneralPosition · SBLocalization.theorem_15_3_7_of_local. These are intermediate steps of the

final proof.

XV.26 Theorem 15.3.7 with the text's hypotheses, the structurable addendum, and Remark 15.3.8

Theorem 15.3.7, (1)–(4) (explicit) `Chapter15/GeneralPositionExistence · SBLocalization.theorem_15_3_7_existence`. Under the hypotheses of §XV.21 there is an sc⁺-multisection Λ with (1)–(4).

Proof, as the Lean proof goes. 1. N is an auxiliary norm, being part of "(N, U) controls compactness". 2. *Local seeds* (§XV.23) come from: - the bump functions; - the smoothness of the points of S_f; - the small-vectors fact (§XV.24); - the openness fact (§XV.24). 3. *Local families* (§XV.23). Each $x \in S_f$ has an open neighbourhood carrying a family of sc⁺-multisections good for (N, U, h). This uses the continuity of h, its invariance along morphisms, and $h > 0$. 4. *One family* on a neighbourhood Q of S_f, by saturation and the compactness of $|S_f|$ (§§XV.22–XV.23). 5. *Parametric general position* for this family (§XV.25), with the bound $N(\Lambda_t) \le |t|_1 h \le |t|_1$ (from $h \le 1$), domain supports in U, and (15.1) on $S_f \subseteq Q$. This gives ε. 6. *The choice of* t with $\|t\| < \min(\varepsilon, 1/(\ell+1))$ (§XV.22). $\Lambda = \Lambda_t$ has (1)–(4).

Explicit: the hypotheses are the text's, plus the tame germ data inherited from §15.2. - The tame germ data are used for the smoothness of the points of S_f, the tame charts of the bridge, and the branches as Part I sections. - Tameness of X is used in the openness fact and in the local Sard step. - Bump functions are used for the seeds. - No good system and no paracompactness of $|X|$ are used.

The structurable addendum (narrower) `Chapter15/GeneralPositionGoodSystem · SBLocalization.theorem_15_3_7_of_goodSystem`. If X has a good system, the structurable form of §XV.21 holds. The proof has the same five steps, with structurable families (§XV.22): the atomic families are structurable by Theorem 13.5.3, sums by Proposition 13.3.12, and the empty family with the good system. The last step is the intermediate statement from the two facts at single points (§XV.23). For paracompact $|X|$, Theorem 13.1.8 gives a good system `Chapter13/GoodSystemExistence · EpGroupoid.theorem_13_1_8`, so the structurable form holds `Chapter15/GeneralPositionGoodSystem · SBLocalization.theorem_15_3_7_of_paracompact`.

Narrower: the good system is not in the text's hypotheses. It is necessary for this part, since a structurable multisection (Definition 13.3.6) carries a good system, so the conclusion implies the hypothesis. The text's proof opens with "We shall construct a structurable perturbation", and the step "for fixed t it is structurable" is Theorem 13.5.3, which in the formalization needs a good system (Part XIII). The text presumably has in mind the paracompact situation that Remark 15.3.8 calls usual in applications.

The theorem with its addendum `Chapter15/GeneralPositionExistence · SBLocalization.theorem_15_3_7`. (1)–(4) always, and Λ structurable whenever a good system exists.

The intermediate statements are steps of these two proofs. Each one proves the structurable form from a hypothesis that another step of the proof discharges: - `theorem_15_3_7_hyp` and `theorem_15_3_7_of_hyp` (local families and parametric general position; §XV.22); - `theorem_15_3_7_hyp_seeds` and `theorem_15_3_7_of_seeds` (local seeds; §XV.23); - `theorem_15_3_7_of_pointwise` (the two facts at single points; §XV.23); - `theorem_15_3_7_hyp_local` and `theorem_15_3_7_of_local` (local families alone; §XV.25); - the form from the two facts and a good system `Chapter15/GeneralPositionPointwise · SBLocalization.theorem_15_3_7_of_param0nto`, which the structurable addendum specializes by proving the two facts.

Each has its hypotheses satisfied on $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$: - a good system there, by Theorem 13.1.8 on $\mathbb{R}^n$ `Chapter15/GeneralPositionAssembly · SBLocalization.nonempty_goodSystem_epSEP_zero`; - lo-

cal families with $\ell = 0$ for every f, N, U, h Chapter15/GeneralPositionAssembly · SBLocalization.hasLocalGPFamilies_epSEP_zero; - parametric general position, since every pair is in general position there Chapter15/GeneralPositionAssembly · SBLocalization.hasParametricGeneralPosition_epSEP_zero; - local seeds Chapter15/LocalFamilies · SBLocalization.hasLocalParamSeeds_epSEP_zero; - small vectors with $\ell = 0$ Chapter15/LocalSeeds · SBLocalization.hasPointwiseParamOnto_epSEP_zero and openness Chapter15/LocalSeeds · SBLocalization.hasParamOntoOpen_epSEP_zero, both resting on (15.1) being automatic with zero fibres Chapter15/LocalSeeds · SBLocalization.isParamOntoVec_epSEP_zero.

The conclusion holds there in each form: - Chapter15/GeneralPositionAssembly · SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_hyp - Chapter15/LocalFamilies · SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_seeds - Chapter15/LocalSeeds · SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_pointwise - Chapter15/ParametricGeneralPosition · SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_local - Chapter15/GeneralPositionPointwise · SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_paramOnto - Chapter15/GeneralPositionGoodSystem · SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_goodSystem

These witnesses are weak: the fibres are zero and there is no boundary, so every pair is in general position. They show only that the intermediate statements are not vacuous. The final theorems need no such witness, since all their added hypotheses are discharged, except the inherited tame germ data and, for the addendum, the necessary good system.

Remark 15.3.8 (nothing to formalize). The remark says the theorem does not assume a paracompact orbit space, only an auxiliary norm, and that constructing auxiliary norms usually needs paracompactness of $|X|$. - In the formalization, (1)–(4) need neither paracompactness nor a good system. The auxiliary norm enters only through "(N, U) controls compactness". - The addendum "structurable" needs a good system. Paracompactness supplies one (Theorem 13.1.8), and so does any structurable multisection.

So the remark holds as read for (1)–(4). For the addendum, paracompactness, or at least a good system, does enter. The part of gap G.162 that concerns Theorem 15.3.7 is closed.

XV.27 Theorem 15.3.9 (2017: 15.5): the statement, the face condition, and charging

Theorem 15.3.9, the Transversal Extension Theorem (p. 698), extends a perturbation given over the boundary to a perturbation over all of X that is transversal, without changing it over ∂X. It is the tool for the cobordism part of Corollary 15.3.10 (§XV.33), where the boundary data are two regular perturbations sitting at the ends of the cylinder $[0, 1] \times X$. Its proof (pp. 698–699) has five steps: (a) a cut-off functor h_0 between $N(\Lambda)$ and h; (b) Theorem 14.1.9, which extends Λ to a structurable Λ_0 with $N(\Lambda_0) \le h_0$; (c) transversality of (f, Λ_0) near ∂X, from the hypothesis on the faces; (d) the argument of Theorem 15.3.7 away from ∂X, with families Λ_t whose domain support avoids ∂X; (e) $\tilde{\Lambda} = \Lambda_t \oplus \Lambda_0$ for a small regular t. The formalization follows these steps in this order. This subsection gives the definitions, the statement as formalized and the property of the uncharged Lean statement; step (a) is §XV.28, steps (b)–(e) are §XV.29.

General position at one point. For a level-1 point x, (f, Λ) is *reducedly onto at* x if every operator L in the support of $T(f, \Lambda)(x)$ (Definition 15.1.7, §XV.8) restricts to a surjection $T^R_x X \to W_x$, the reduced tangent space $T^R_x X$ being read in every chart of X around x and every presentation of the partial quadrant of its model Chapter15/TransversalExtension · SBLocalization.IsReducedOntoAtPt. This is Definition 15.3.5 at one point: (f, Λ) is in general position iff it is reducedly onto at every level-1 point of $\mathrm{supp}(\Lambda \circ f)$, by def-

inition Chapter15/TransversalExtension · SBLocalization.isInGeneralPosition_iff_forall_reducedOntoAtPt (it is the same predicate as the pointwise general position of §XV.16, Chapter15/GeneralPositionLocal · SBLocalization.IsInGeneralPositionAt, there with the level-1 membership as a separate clause). In general position (f, Λ) is transversal: every operator of the support is onto on the larger space $T_xX \supseteq T^R_xX$, and the support points are at level 1 Chapter15/TransversalExtension · SBLocalization.IsInGeneralPosition.isTransversal Chapter15/ControllingPairs · SBLocalization.mem_levelSet_one_of_pos.

The boundary data. Chapter 14's objects over ∂X are used as Part XIV formalizes them. An *sc⁺-multisection over ∂X* (Definition 14.1.3) is a function $\Lambda : W \to \mathbb{Q}^+$ of which only the values over ∂X enter: it is a functor on $W|\partial X$, and every $x \in \partial X$ has a good neighbourhood $U(x) = V(x) \cap \partial X$ carrying a *symmetric sc⁺-section structure*, a finite index set I_x with a G_x-action and sections s_i that are sc⁺ on the faces (Definition 14.1.3 (1)), equivariant (2), and represent Λ by $\Lambda(w) = |I_x|^{-1}\#\{i : s_i(P(w)) = w\}$ (3) Chapter14/BoundaryMultisection · SBLocalization.IsBoundaryScPlusMultisection Chapter14/BoundaryMultisection · SBLocalization.BoundarySymmetricStructure. It is *structurable over ∂X* (Definition 14.1.7) if it carries in addition a decoration and a family of index correspondences Chapter14/BoundaryStructured · SBLocalization.IsBoundaryStructurable. Its *boundary domain support* is the closure of the set of $x \in \partial X$ over which Λ charges a non-zero vector Chapter14/BoundaryStructured · SBLocalization.boundaryDomSupp. Finally, Λ *charges* ∂X if over every $x \in \partial X$ some w has $\Lambda(w) \neq 0$ Chapter14/EmptyIndex · SBLocalization.ChargesBoundary. For a symmetric structure at a charged point the index set is not empty, because the formula (3) with $I_x = \emptyset$ gives $\Lambda = 0$ on the fibre Chapter14/EmptyIndex · SBLocalization.BoundarySymmetricStructure.nonempty_I_a6. The formal Definition 14.1.3 admits $I_x = \emptyset$, while the text's $|I_x|^{-1}$ presumes $I_x \neq \emptyset$; charging is the reading of the text's definition that excludes the empty index sets (Theorem 14.1.9, Part XIV).

The face condition Chapter15/TransversalExtension · SBLocalization.IsBoundaryFaceOnto. The text asks that $T_{(f|\partial X,\Lambda)}(x)$ be "surjective with respect to intersections of local faces" at every smooth $x \in \partial X$. It is read as follows: for every sc⁺-multisection Λ' on W that agrees with Λ over ∂X, and every smooth level-1 point $x \in \partial X$ with $\Lambda(f(x)) > 0$, (f, Λ') is reducedly onto at x. Two readings are in this. (i) The intersections of the tangent spaces of the local faces through x form a family whose smallest member is T^R_xX (Part I, Definition 2.5.8), and surjectivity on a subspace implies surjectivity on every larger one; so the condition for all intersections is the condition on T^R_xX, which is general position at x. (ii) $T(f|\partial X, \Lambda)$ is not defined by the text: Λ lives only over ∂X, and its linearization needs local sections near x, not only on the faces. The restriction of $(f - s_i)'(x)$ to T^R_xX sees only the values of s_i on the faces through x, so it should not depend on how Λ is extended; the formal condition is therefore asked for every sc⁺ extension Λ'. That it does not depend on the extension is not proved (gap G.179, §XV.38); in the proof the condition is used for exactly one extension, the Λ_0 of step (b) (§XV.29).

Statement (narrower) Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_statement. Let $(P : W \to X, \mu)$ be a strong bundle over a tame ep-groupoid with $|X|$ paracompact, whose object M-polyfold admits sc-smooth partitions of unity, and every object of which lies in a tame strong bundle chart. Let f be an sc-Fredholm section functor with Part I's tame germ data and compact $|S_f|$, and (N, U) control compactness (Definition 15.3.1, §XV.17; U open and saturated). Let Λ be structurable over ∂X and charge ∂X, with boundary domain support in U and $N(\Lambda) \leq 1$ on ∂X, and let $h : X \to (0, 1]$ be a continuous functor with $N(\Lambda)(x) < h(x)$ for $x \in \partial X$. Assume the face condition. Then there is a structurable sc⁺-multisection $\tilde{\Lambda}$ with

1. $\tilde{\Lambda} = \Lambda$ over ∂X;
2. domain support in U, $N(\tilde{\Lambda}) < h$ everywhere, and $\tilde{\Lambda}$ (N, U)-admissible (Definition 15.3.3, §XV.18);

3. $(f, \tilde{\Lambda})$ transversal (Definition 15.2.1, §XV.12);

and $\tilde{\Lambda} \circ f$ is a branched ep⁺-subgroupoid with compact support.

Readings. "U a saturated subset" is read as open and saturated, which Definition 15.3.1 contains. "Λ is (U, N)-admissible" over ∂X is read as boundary domain support in U and $N(\Lambda) \leq 1$ on ∂X, the two parts of Definition 15.3.3 that make sense over ∂X. "Onto on $\operatorname{supp}(\tilde{\Lambda} \circ f)$" is transversality. The word "tame" of the "moreover" is not in the formal conclusion: the $\tilde{\Lambda}$ that the proof produces is in general position (§XV.29), and for such a $\tilde{\Lambda}$ the tame form of the "in particular" of Theorem 15.3.7 (§XV.21) gives a tame branched ep⁺-subgroupoid, but the statement as formalized exports only the text's (1)–(3) with "branched" and "compact".

Narrower, by four hypotheses, of which two are readings and two are open. **Tame strong bundle charts** at every object: the reading of "a strong bundle over a tame ep-groupoid" that Part XIV uses for Theorem 14.1.9; it is needed in step (a) (§XV.28), not derived from the text, and it has a witness with boundary and non-zero fibres, the trivial bundle $[0,\infty) \times \mathbb{R}^m \times \mathbb{R}^k \to [0,\infty) \times \mathbb{R}^m$ of Part XIV `Chapter14/HalfSpaceBundle · HalfSpaceBundle.hB_halfSEP`, as well as one without boundary `Chapter14/OpenExtension · SBLocalization.proposition_14_1_4_hypotheses_rn`. **Charging** of ∂X by Λ: the reading $I_x \neq \emptyset$ of "sc⁺-multisection over ∂X". It is necessary: if a structurable $\tilde{\Lambda}$ agrees with Λ over ∂X, then Λ charges ∂X `Chapter15/ChargedExtensionFibres · SBLocalization.chargesBoundary_of_theorem_15_3_9_r6`, because a local representation of $\tilde{\Lambda}$ at $x \in \partial X$ (Definition 13.2.1) has weights summing to 1, so it has a branch s_i, and $\tilde{\Lambda}(s_i(x)) \geq \sigma_i > 0$ `Chapter14/EmptyIndex · SBLocalization.chargesBoundary_of_structurable_eq_a6`. The zero multisection charges every fibre `Chapter14/EmptyIndex · SBLocalization.chargesBoundary_zeroMultisection_a6`, and the boundary hypotheses of Theorem 14.1.9 with charging hold together on the half-space bundle, whose boundary is not empty `Chapter14/HalfSpaceBundle · HalfSpaceBundle.theorem_14_1_9_charged_hypotheses_halfSEP`. The **tame germ data** of f are inherited from §15.2 and Part I's Theorem 3.6.8, as everywhere in this Part. The **face condition for every extension** is the reading (ii) above (gap G.179). Partitions of unity on the objects and the paracompactness of $|X|$ are the text's.

The theorem (narrower) `Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_of_local_rel`. The charged statement holds for every strong bundle for which the *relative local families* exist `Chapter15/ChargedExtensionFibres · SBLocalization.relative_local_families_statement`: under the hypotheses of steps (c)–(d), every point of $\operatorname{supp}(\Lambda_0 \circ f)$ lies in a saturated open set carrying a parameter family with domain support in $U \setminus \partial X$, the bound $N(\Lambda_t) \leq |t|_1(h - N(\Lambda_0))$ and the relative form of (15.1) (§XV.29). This is the one analytic input of the proof that is not formalized, gap G.178: everything else, the cut-off, Theorem 14.1.9, the relative parametric transversality, the compactness and the assembly, is proved. *Witnesses*: the relative local families and the charged statement hold on the trivial bundle $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ `Chapter15/ChargedExtensionFibres · SBLocalization.relative_local_families_statement_epSEP_zero` `Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_statement_epSEP_zero`, where $\partial X = \emptyset$ `Chapter15/TransversalExtension · SBLocalization.boundary_epSEP_eq_empty` and the fibres are zero, so every pair is in general position. These witnesses are **weak**: no strong bundle with boundary is known to satisfy the relative local families, and the theorem has no non-degenerate witness.

Remark: the uncharged Lean statement. The uncharged Lean statement `Chapter15/TransversalExtension · SBLocalization.theorem_15_3_9_statement` is the charged statement without the tame strong bundle charts and without charging; the charged statement follows from it `Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_of_statement`. As a Lean statement it does not hold when $\partial X \neq \emptyset$ and one sc-Fredholm f with tame germ data, compact $|S_f|$ and a pair (N, U) controlling compactness exists `Chapter15/ChargedExtensionCylinder · SBLocalization.not_theorem_15_`

3_9_statement_w2. *Proof.* Take $\Lambda \equiv 0$. It is structurable over ∂X in the formal sense, with empty index sets Chapter14/EmptyIndex · SBLocalization.isBoundaryStructurable_zeroFun_a6; its boundary domain support is empty Chapter14/EmptyIndex · SBLocalization.boundaryDomSupp_zeroFun_a6; its pointwise norm is 0, its support over every point being empty Chapter15/ChargedExtensionCylinder · SBLocalization.pointwiseNorm_zeroFun_w2; with $h \equiv 1$ the norm conditions hold, and the face condition holds vacuously since $\Lambda(f(x)) = 0$. The statement would give a structurable $\tilde{\Lambda}$ vanishing over ∂X, and no structurable multisection vanishes over a point (the necessity of charging above). On the cylinder $[0,1] \times X$ of §XV.30, with $d_X \equiv 0$ and X nonempty, the hypotheses of Corollary 15.3.10 provide all of this: the cylinder is tame since X is (§XV.30), its orbit space is paracompact, $(0,x)$ lies on its boundary, $\tilde{f}$ is sc-Fredholm with tame germ data and compact solution set, and Corollary 15.3.10 (2) gives a controlling pair; so the uncharged statement fails on every such cylinder Chapter15/ChargedExtensionCylinder · SBLocalization.not_theorem_15_3_9_statement_cube_w2. This is a property of our formalization, not of the text, whose Λ has non-empty index sets; it has the same source as the failure of the uncharged Lean statement of Theorem 14.1.9 (Part XIV). *Consequences.* The Lean forms that conclude the uncharged statement are correct implications whose hypotheses fail when $\partial X \neq \emptyset$; they are vacuous there and are used only as intermediate steps for $\partial X = \emptyset$: the reduction to Theorem 14.1.9 and two proof steps Chapter15/TransversalExtension · SBLocalization.theorem_15_3_9_hyp, its form with step (a) proved Chapter15/CutoffFunctor · SBLocalization.theorem_15_3_9_of_extension_ty, and its forms with the relative parametric step proved Chapter15/RelativeReduction · SBLocalization.theorem_15_3_9_of_rel_hyp Chapter15/RelativeParametric · SBLocalization.theorem_15_3_9_of_local_rel, which take the uncharged extension statement of Theorem 14.1.9 as a hypothesis. The forms of Corollary 15.3.10 (3) that take the uncharged statement on the cylinder are vacuous for nonempty X (§XV.33). The charged statement, built on Theorem 14.1.9 in its charged form (Part XIV), replaces all of them, and the rest of this subsection and §§XV.28–XV.29 tell its proof. The uncharged statement does hold on the boundaryless example, where its three propositional inputs hold as well Chapter15/TransversalExtension · SBLocalization.theorem_15_3_9_statement_epSEP_zero Chapter15/TransversalExtension · SBLocalization.lemma_15_3_9_cutoff_statement_epSEP_zero Chapter15/TransversalExtension · SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero: there $\Lambda_0 = 0$ has every conclusion, being in general position Chapter15/GeneralPositionBranched · SBLocalization.isInGeneralPosition_epSEP_zero, and the boundary domain support is empty Chapter15/TransversalExtension · SBLocalization.boundaryDomSupp_epSEP.

The shape of the proof Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_hyp. The charged statement follows from the *relative statement*, the proposition that steps (c)–(d) produce Chapter15/TransversalExtension · SBLocalization.theorem_15_3_7_relative_statement: given a structurable Λ_0 with domain support in U, $N(\Lambda_0) < h$, and (f, Λ_0) reducedly onto at the points of $\operatorname{supp}(\Lambda_0 \circ f) \cap \partial X$, there is a structurable $\tilde{\Lambda}$ equal to Λ_0 over ∂X, with domain support in U, $N(\tilde{\Lambda}) < h$, transversal and in general position. The relative statement follows from the relative local families alone Chapter15/RelativeParametric · SBLocalization.theorem_15_3_7_relative_of_local, the parametric step being proved. §XV.28 proves step (a); §XV.29 retells steps (b)–(e) and the proof of the relative statement.

XV.28 Proof of Theorem 15.3.9, step (a): the cut-off functor h_0

Statement (explicit) Chapter15/CutoffFunctor · SBLocalization.lemma_15_3_9_cutoff_hyp. If every object lies in a tame strong bundle chart, then step (a) holds Chapter15/TransversalExtension · SBLocalization.lemma_15_3_9_cutoff_statement: for X tame with $|X|$ paracompact and sc-smooth partitions of unity, Λ an sc$^+$-multisection over ∂X, N an auxiliary norm, U open and saturated containing the boundary domain

support of Λ, and $h > 0$ a continuous functor with $N(\Lambda) < h$ on ∂X, there is a continuous functor $h_0 \geq 0$ with tsupp $h_0 \subset U$, $h_0 < h$ everywhere, and $N(\Lambda) < h_0$ on the boundary domain support of Λ. These are the hypotheses that Theorem 14.1.9 asks of its function f with $\tilde{U} = U$, and the margin $h_0 < h$ is used at the end of the proof.

A reading. The text asks for h_0 with $N(\Lambda)(x) < h_0(x) < h(x)$ for every $x \in \partial X$ and h_0 supported in U. Off the boundary domain support, over ∂X, the support of Λ is the zero vector and $N(\Lambda) = 0$; at a point of ∂X outside U the two requirements would ask $0 < h_0 = 0$. So the strict lower bound is presumably meant on the domain support, which is what Theorem 14.1.9 asks; the formal statement says so. The text's $h_0 \leq 1$ is not needed, $h_0 < h \leq 1$ holding anyway in the application.

Proof. Write C for the boundary domain support of Λ. Four facts about Λ come first.

The support over a boundary point. If S is a symmetric structure of Λ on $V(x)$ and $y \in V(x) \cap \partial X$, then w lies in the support of Λ over y iff $w = s_i(y)$ for some i `Chapter15/CutoffFunctor · SBLocalization.BoundarySymmetricStructure.mem_suppAt_iff_ty`: by the representation formula (3), $\Lambda(w) > 0$ iff the count $\#\{i : s_i(y) = w\}$ is positive, the index set being non-empty as soon as it contains one i. In particular the support over ∂X lies in $W_{0,1}$, each s_i being sc$^+$ on the faces `Chapter15/CutoffFunctor · SBLocalization.IsBoundaryScPlusMultisection.suppAt_subset_oneSet_ty`, so the pointwise norm $N(\Lambda)(y) = \sup\{N(w) : \Lambda(w) > 0\}$ is the maximum of the finitely many $N(s_i(y))$.

$N(\Lambda)$ is a functor on ∂X `Chapter15/CutoffFunctor · SBLocalization.pointwiseNorm_tgt_of_boundary_ty`. For a morphism φ with source in ∂X, $\mu(\varphi, \cdot)$ carries the support over $s(\varphi)$ into the support over $t(\varphi)$, since Λ is a functor on $W|\partial X$, and it preserves N, since an auxiliary norm is invariant under μ `Chapter13/AtomicConstruction · SBLocalization.normW_μ`; so the set of norms over $s(\varphi)$ lies in that over $t(\varphi)$ `Chapter15/CutoffFunctor · SBLocalization.normW_image_suppAt_subset_ty`, and the inverse morphism, whose source again lies in the saturated set ∂X, gives the other inclusion.

C is saturated `Chapter15/CutoffFunctor · SBLocalization.isSaturated_boundaryDomSupp_ty`. The set of $x \in \partial X$ over which Λ charges a non-zero w is saturated: $\mu(\varphi, w)$ is charged over $t(\varphi)$, and it is not zero, since $N(\mu(\varphi, w)) = N(w) > 0$, an auxiliary norm vanishing only at zero vectors. The closure of a saturated set is saturated. And $C \subset \partial X$, ∂X being closed `Chapter15/CutoffFunctor · SBLocalization.boundaryDomSupp_subset_boundary_ty`.

$N(\Lambda)$ is upper semicontinuous along ∂X `Chapter15/CutoffFunctor · SBLocalization.exists_nhds_pointwiseNorm_lt_ty`: for $x \in \partial X$ and $c > N(\Lambda)(x)$ there is an open $O \ni x$ with $N(\Lambda) < c$ on $O \cap \partial X$. Take a symmetric structure on $V(x)$. Its sections s_i are sc$^+$ only on the faces; with tame strong bundle charts at every object and sc-smooth partitions of unity, each extends to an sc$^+$-section $\bar{s}_i$ on the open set $V(x)$ with $\bar{s}_i = s_i$ on $V(x) \cap \partial X$ `Chapter14/OpenExtension · SBLocalization.exists_scPlusSectionOn_extension_of_face` (the extension lemma of Part XIV behind Proposition 14.1.4). The functions $y \mapsto N(\bar{s}_i(y))$ are continuous on $V(x)$. With $c' = (N(\Lambda)(x) + c)/2$, the set $O = V(x) \cap \bigcap_i \{N \circ \bar{s}_i < c'\}$ is open, contains x because $N(s_i(x)) \leq N(\Lambda)(x) < c'$, and on $O \cap \partial X$ every norm $N(s_i(y)) = N(\bar{s}_i(y))$ is below c', so $N(\Lambda)(y) \leq c' < c$. This is the only place where the tame strong bundle charts enter. Without them Definition 14.1.3 (1) constrains the s_i only face by face, and nothing prevents $N(\Lambda)$ from jumping up along ∂X; the text uses the semicontinuity without comment.

Construction of g on $|X|$. The orbit space $|X|$ is paracompact and Hausdorff. For an orbit q let

$$t(q) = \{y \geq 0 : y < h(x) \text{ and, if } x \in C,\ y > N(\Lambda)(x),\ \text{for every } x \text{ with } \pi(x) = q\},$$

an intersection of intervals, hence convex. Since h and $N(\Lambda)|_{\partial X}$ are constant on orbits and C is saturated, $t(q)$ is described by any one x over q. Each $q = \pi(x)$ has a constant c that lies in $t(q')$ for all q' near q. If $x \in C$, take $c = (N(\Lambda)(x) + h(x))/2$: by the upper semicontinuity there is an open $O \ni x$ with $N(\Lambda) < c$ on $O \cap \partial X$, and $h > c$ on an open neighbourhood by continuity; if $x \notin C$, take $c = h(x)/2$ on $(X \setminus C) \cap \{h > h(x)/2\}$, open since C is closed. In both cases the open set maps to an open neighbourhood of q, the orbit map being open (Proposition 7.1.8) `Chapter7/Basic · EpGroupoid.proposition_7_1_8`, and the invariance of the data transfers the inequalities from the chosen representatives to all points of the orbits. Mathlib's selection lemma for convex-valued maps with local constant selections on a paracompact space `Mathlib/Analysis/Convex/PartitionOfUnity · exists_continuous_forall_mem_convex_of_local_const`, a partition of unity argument, gives a continuous $g : |X| \to \mathbb{R}$ with $g(q) \in t(q)$ for all q: $0 \le g < h$, and $g > N(\Lambda)$ on $\pi(C)$.

The Urysohn factor. $\pi(C)$ is closed in $|X|$, C being closed and saturated `Chapter14/BoundaryCover · EpGroupoid.isClosed_π_image_of_isSaturated`, and $\pi(U)$ is open and contains it. By normality there is an open V with $\pi(C) \subset V$ and $\operatorname{cl} V \subset \pi(U)$, and an Urysohn function $\psi : |X| \to [0,1]$ with $\psi = 1$ on $\pi(C)$ and $\psi = 0$ off V.

The functor. $h_0 = (\psi g) \circ \pi$. It is continuous, non-negative, constant on orbits, hence a functor. Its topological support lies in $\pi^{-1}(\operatorname{cl} V) \subset \pi^{-1}(\pi(U)) = U$, U being saturated. $h_0 \le g \circ \pi < h$, since $0 \le \psi \le 1$. On C, $\psi = 1$ and $h_0 = g \circ \pi > N(\Lambda)$. □

The added hypothesis, tame strong bundle charts at every object, is not stated in the theorem; it is the reading of the standing assumption used for Theorem 14.1.9 in Part XIV. Witnesses: the trivial bundle over $\mathbb{R}^n$ `Chapter14/OpenExtension · SBLocalization.proposition_14_1_4_hypotheses_rn`, with $\partial X = \emptyset$, and the half-space bundle, with boundary and non-zero fibres `Chapter14/HalfSpaceBundle · HalfSpaceBundle.hB_halfSEP`. On $\mathbb{R}^n \times \mathbb{R}^0$ the cut-off statement holds trivially with $h_0 = 0$, the boundary domain support being empty `Chapter15/TransversalExtension · SBLocalization.lemma_15_3_9_cutoff_statement_epSEP_zero`. Gap G.177 is closed.

XV.29 Proof of Theorem 15.3.9, steps (b)–(e): relative general position and the assembly

This subsection follows the proof of the charged theorem `Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_hyp` after the cut-off of §XV.28, and then the proof of the relative statement that steps (c)–(d) supply.

Step (b): the extension Λ_0. Apply Theorem 14.1.9 in its charged form (Part XIV) `Chapter14/BoundaryExtensionTheorem · SBLocalization.theorem_14_1_9_charged_a6` to Λ, with $\tilde{U} = U$ and the function h_0 of step (a). Its hypotheses are those of Theorem 15.3.9 together with the properties of h_0: X tame, $|X|$ paracompact, partitions of unity, tame strong bundle charts, Λ structurable over ∂X and charging ∂X, N an auxiliary norm, U open, saturated and containing the boundary domain support, $h_0 \ge 0$ a continuous functor with topological support in U, and $N(\Lambda) < h_0$ on the boundary domain support `Chapter14/BoundaryExtensionTheorem · SBLocalization.theorem_14_1_9_charged_statement`. It gives a structurable Λ_0 with $N(\Lambda_0) \le h_0$ everywhere, domain support in U, and $\Lambda_0 = \Lambda$ over ∂X. In particular $N(\Lambda_0) \le h_0 < h$ everywhere. *A reading*: after applying Theorem 14.1.9 the text writes "$N(\Lambda_0)(x) \le h(x)$"; presumably $\le h_0(x)$ is meant, since the margin $h_0 < h$ is what makes the final $N(\tilde{\Lambda}) < h$ strict.

Step (c): general position of Λ_0 on the boundary. For a level-1 point $x \in \partial X$ with $\Lambda_0(f(x)) > 0$, (f, Λ_0) is reducedly onto at x. Indeed $f(x)$ lies over $x \in \partial X$, where $\Lambda_0 = \Lambda$, so $\Lambda(f(x)) > 0$; x is a

smooth point, every point of $\operatorname{supp}(\Lambda_0 \circ f)$ being smooth under the tame germ data, since $f(x)$ is then the value of an sc⁺-section and $f - s$ is regularizing `Chapter15/Transversality · SBLocalization.mem_smoothSet_of_pos`; and Λ_0 is an sc⁺-multisection on W that agrees with Λ over ∂X, so the face condition applies to it. The text says at this point that the boundary assumption gives, for every boundary point of the support, a neighbourhood $Q(x)$ on which the reduced linearization of (f, Λ_0) is onto; the formal proof takes the condition at the boundary support points only and leaves the neighbourhood to the local families (below).

Steps (c)–(d): the relative statement. Its hypotheses hold for Λ_0: structurable, domain support in U, $N(\Lambda_0) < h$, reducedly onto at the boundary support points. It gives a structurable $\tilde{\Lambda}$ with $\tilde{\Lambda} = \Lambda_0$ over ∂X, domain support in U, $N(\tilde{\Lambda}) < h$, $(f, \tilde{\Lambda})$ transversal and in general position. Its proof is the rest of this subsection.

Step (e): the conclusions. $\tilde{\Lambda} = \Lambda_0 = \Lambda$ over ∂X. Since $h \le 1$, $N(\tilde{\Lambda}) \le 1$, and with the domain support in $U \supseteq S_f$ the example after Definition 15.3.3 (§XV.18) makes $\tilde{\Lambda}$ admissible `Chapter15/ControllingPairs · SBLocalization.isAdmissible_of_domSupp_subset`. The "moreover": general position, $N(\tilde{\Lambda}) \le 1$ and domain support in U are the hypotheses of the "in particular" of Theorem 15.3.7 (§XV.21) `Chapter15/GeneralPositionBranched · SBLocalization.theorem_15_3_7_in_particular_hyp`, which gives a branched ep⁺-subgroupoid with compact support. □

The proof of the relative statement is "argue as in Theorem 15.3.7" (p. 698), made precise. The text takes parameter families Λ_t with domain support away from ∂X and asks of $\Lambda_t \oplus \Lambda_0$ near $(0, x)$, x in the support, that the reduced linearization of the parametrized problem be onto. The formalization reads this as (15.1) relative to the branches of Λ_0, and then repeats the three parts of the proof of Theorem 15.3.7 (§§XV.22–XV.25): the assembly of local families, the parametric transversality, and the choice of t.

Relative (15.1) `Chapter15/RelativeFamilies · SBLocalization.IsParamOntoAtRel`. Let $t \mapsto \Lambda_t$, $t \in \mathbb{R}^\ell$, be a family of sc⁺-multisections and x a level-1 point. The family satisfies (15.1) relative to Λ_0 at x if it has, on an open $O \ni x$, a linear local representation by the sections $\sum_j t_j s_{ij}$ with weights σ_i independent of t (as in §XV.22, `Chapter15/ParamFamily · SBLocalization.paramSec`), and for every operator L in the support of $T(f, \Lambda_0)(x)$, every chart presentation of $T^R_x X$ and every branch i, the map

$$T^R_x X \times \mathbb{R}^\ell \to W_x, \quad (v, \tau) \mapsto Lv + \sum_j \tau_j s_{ij}(x)$$

is onto. The operators of $T(f, \Lambda_0)(x)$ are the $(f - s^0_a)'(x)$ for the branches s^0_a of Λ_0 through $f(x)$ (§XV.12); the text's (15.1) has the operators of $T(f, 0)(x)$, which is the case $\Lambda_0 = 0$ `Chapter15/RelativeFamilies · SBLocalization.isParamOntoAtRel_zero_iff`. Two properties. *Where (f, Λ_0) is already reducedly onto*, relative (15.1) holds for every linear family, with $\tau = 0$ `Chapter15/RelativeFamilies · SBLocalization.IsParamOntoAtRel.of_reducedOnto`; this is how step (c) feeds steps (c)–(d) at the boundary support points. *Sums*: if one summand satisfies relative (15.1) and the other is linear, the sum family $\Lambda_t \oplus \Lambda'_{t'}$ on $\mathbb{R}^{\ell+\ell'}$ does `Chapter15/RelativeFamilies · SBLocalization.IsParamOntoAtRel.famAdd_left` `Chapter15/RelativeFamilies · SBLocalization.IsParamOntoAtRel.famAdd_right`. The sum family is linear with the branches $\sum_j t_j s_{ij} + \sum_k t'_k s'_{i'k}$ on the intersection of the two domains `Chapter15/ParamFamily · SBLocalization.famAdd` `Chapter15/ParamFamily · SBLocalization.isLocalRepOn_famAdd`; given (v, τ) for the branch i of the first family, $(v, (\tau, 0))$ serves for every branch (i, i') of the sum.

Relative families `Chapter15/RelativeFamilies · SBLocalization.RelGPFamily`. For a function g and a set Q: a linear family `Chapter15/ParamFamily · SBLocalization.IsLinearFamily` of structurable sc⁺-multisections

with $N(\Lambda_t)(x) \le |t|_1\, g(x)$, domain support in a given U', and relative (15.1) at every level-1 point of $\operatorname{supp}(\Lambda_0 \circ f) \cap Q$. In the application $g = h - N(\Lambda_0)$ and $U' = U \setminus \partial X$. The family of Theorem 15.3.7 (§XV.22) is the case $\Lambda_0 = 0$, $g = h$. A family on Q is one on every smaller set `Chapter15/RelativeFamilies · SBLocalization.RelGPFamily.mono`. Given a good system, the parameter-free family $\Lambda_t = 0$ ($\ell = 0$), which is linear `Chapter15/RelativeFamilies · SBLocalization.isLinearFamily_zero` and structurable `Chapter15/ParamFamily · SBLocalization.isStructurable_zeroMultisection_of_goodSystem`, is a relative family on every set on which (f, Λ_0) is reducedly onto at the support points `Chapter15/RelativeFamilies · SBLocalization.RelGPFamily.zeroOn`, in particular on $\emptyset$ `Chapter15/RelativeFamilies · SBLocalization.RelGPFamily.zero`. Two families on Q and Q' add to one on $Q \cup Q'$ `Chapter15/RelativeFamilies · SBLocalization.RelGPFamily.add`: the sum of structurable multisections is structurable (Proposition 13.3.12) `Chapter13/StructurableMultisection · SBLocalization.proposition_13_3_12`; the norm of a sum is at most the sum of the norms `Chapter15/ParamFamily · SBLocalization.pointwiseNorm_multisectionAdd_le`, so $N \le (|t|_1 + |t'|_1)g = |(t,t')|_1\, g$; the domain support of a sum lies in the union `Chapter15/ParamFamily · SBLocalization.domSupp_multisectionAdd_subset`; and relative (15.1) passes to sums from either summand.

Off the domain support. An sc⁺-multisection is the zero multisection over every point outside its domain support `Chapter15/RelativeFamilies · SBLocalization.eq_zeroMultisection_of_notMem_domSupp`: in a local representation at that point every branch passes through the zero vector, for otherwise the point would be in the domain support, so $\Lambda(0_x) = \sum_i \sigma_i = 1$ and $\Lambda = 0$ elsewhere on the fibre. Hence $\Lambda_0 \oplus \Lambda = \Lambda_0$ over every point outside the domain support of Λ `Chapter15/RelativeFamilies · SBLocalization.multisectionAdd_eq_left_of_notMem_domSupp`, the zero multisection being neutral for $\oplus$ (Part XIII) `Chapter13/MultisectionNoInverse · SBLocalization.multisectionAdd_zeroMultisection`. This is what "domain support away from ∂X" achieves: $\Lambda_0 \oplus \Lambda_t = \Lambda_0$ over ∂X.

The two analytic steps, relative to Λ_0, as propositions. *The relative local families* `Chapter15/RelativeReduction · SBLocalization.HasLocalGPFamiliesRel`: every x with $\Lambda_0(f(x)) > 0$ lies in an open saturated Q carrying a relative family. *The relative parametric transversality* `Chapter15/RelativeReduction · SBLocalization.HasParametricGeneralPositionRel`: for every linear family of sc⁺-multisections with $N(\Lambda_t) \le |t|_1(1 - N(\Lambda_0))$, domain support in U and relative (15.1) at every point of $\operatorname{supp}(\Lambda_0 \circ f)$, there is $\varepsilon > 0$ such that $(f, \Lambda_0 \oplus \Lambda_t)$ is in general position for almost every t with $\|t\| < \varepsilon$.

The relative statement from the two steps `Chapter15/RelativeReduction · SBLocalization.theorem_15_3_7_relative_hyp`. Let Λ_0 be structurable with domain support in U and $N(\Lambda_0) < h \le 1$, and assume the relative local families with domain support in $U \setminus \partial X$ and $g = h - N(\Lambda_0)$, and the relative parametric transversality. *Step 1, a good system.* A structurable Λ_0 carries a decoration, whose good system is all that the empty family needs. *Step 2, compactness of the support.* $|\operatorname{supp}(\Lambda_0 \circ f)|$ is compact `Chapter15/RelativeReduction · SBLocalization.isCompact_image_supp_comp`: with $N(\Lambda_0) \le 1$ and domain support in U this is the compactness of the "in particular" of Theorem 15.3.7 (§XV.21) `Chapter15/GeneralPositionBranched · SBLocalization.isCompact_orbitSupp_comp`, read as a statement about the image of $\{x : \Lambda_0(f(x)) > 0\}$ in $|X|$. *Step 3, from local to global* `Chapter15/RelativeReduction · SBLocalization.exists_relGPFamily_of_compact`. The images $\pi(Q_x)$ of the sets of the local families are open (Proposition 7.1.8) and cover the compact set of Step 2; finitely many suffice; the families on them add, by induction over the finite set starting from the empty family `Chapter15/RelativeReduction · SBLocalization.nonempty_relGPFamily_biUnion`, to one relative family F on a set Q that contains $\operatorname{supp}(\Lambda_0 \circ f)$, the Q_x being saturated. *Step 4, a regular parameter.* F satisfies the hypotheses of the relative parametric transversality: its domain support lies in $U \setminus \partial X \subset U$, and $|t|_1(h - N(\Lambda_0)) \le |t|_1(1 - N(\Lambda_0))$ since $h \le 1$. So for some

$\varepsilon > 0$, almost every t with $\|t\| < \varepsilon$ gives general position. A ball has positive measure, so there is such a t with $\|t\| < \min(\varepsilon, 1/(\ell+1))$ Chapter15/GeneralPositionAssembly · SBLocalization.exists_norm_lt_of_ae; in the sup norm of $\mathbb{R}^\ell$ this gives $|t|_1 < 1$ Chapter15/GeneralPositionAssembly · SBLocalization.sum_abs_lt_one. *Step 5, the perturbation* $\tilde{\Lambda} = \Lambda_0 \oplus F_t$. It is structurable (Proposition 13.3.12). Over ∂X it equals Λ_0: a point of ∂X is not in the domain support of F_t, which lies in $U \setminus \partial X$. Its domain support lies in the union of the two, inside U. Its norm: with $m = h - N(\Lambda_0) > 0$ at x, $N(F_t)(x) \le |t|_1 m < m$, so $N(\tilde{\Lambda})(x) \le N(\Lambda_0)(x) + N(F_t)(x) < h(x)$, pointwise, without a uniform margin. It is in general position by the choice of t, hence transversal Chapter15/GeneralPositionAssembly · SBLocalization.isTransversal_of_isInGeneralPosition. □ Under the hypotheses of the relative statement the two steps give the relative statement Chapter15/RelativeReduction · SBLocalization.theorem_15_3_7_relative_of_hyp; the relative parametric transversality is proved (below), so the relative local families alone suffice Chapter15/RelativeParametric · SBLocalization.theorem_15_3_7_relative_of_local.

Readings. "Supported away from ∂X" is domain support in $U \setminus \partial X$ (∂X is closed and saturated). The text bounds the family by the norm of the perturbation without saying by what; the bound $N(\Lambda_t) \le |t|_1(h - N(\Lambda_0))$ is the one that keeps $N(\Lambda_0 \oplus \Lambda_t) < h$ for $|t|_1 < 1$ pointwise, and the parametric step needs only the weaker $|t|_1(1 - N(\Lambda_0))$. The text writes $\Lambda_t \oplus \Lambda_0$; the formalization adds in the other order, which changes nothing.

The relative parametric transversality (explicit) Chapter15/RelativeParametric · SBLocalization.hasParametricGeneralPositionRel. For X tame, f an sc-Fredholm section functor with tame germ data, (N, U) controlling compactness and Λ_0 an sc⁺-multisection with $N(\Lambda_0) \le 1$ and domain support in U, the relative parametric transversality holds. The compactness of $|S_f|$ is not used; that of $|\operatorname{supp}(\Lambda_0 \circ f)|$ comes from the control of compactness. The proof is that of the parametric transversality of Theorem 15.3.7 (§XV.25) with a branch s_a^0 of Λ_0 in place of the zero section, in four parts: one branch, the local step, the relative (15.2), and the assembly.

One branch of $\Lambda_0 \oplus \Lambda_t$. On a common open set O, local representations (s_a^0, σ_a^0) of Λ_0 and (s_i, σ_i) of Λ give the representation of $\Lambda_0 \oplus \Lambda$ by the sections $s_a^0 + s_i$ with the weights $\sigma_a^0 \sigma_i$, reindexed by a finite type (Definition 13.2.9 (3)) Chapter15/RelativeBranch · SBLocalization.isLocalRepOn_multisectionAdd: the sums are sc⁺ on O, the weights sum to $\sum_a \sigma_a^0 \sum_i \sigma_i = 1$, and the count formula is that of the sum of Part XIII Chapter13/MultisectionAlgebra · SBLocalization.multisectionAdd_eq_sum_of_isLocalRepOn. For a linear family the branch through s_a^0 and $\sum_j t_j s_{ij}$ is the slice at t of Part I's parametrized section on $\mathbb{R}^\ell \times O$,

$$F_{a,i}(t, y) = (f - s_a^0)|_O(y) + \sum_j t_j (-s_{ij})|_O(y),$$

that is, the slice at t is $(f - (s_a^0 + \sum_j t_j s_{ij}))|_O$ Chapter15/RelativeBranch · SBLocalization.paramSlice_branch_eq_rel, a fibrewise identity in W_y (the restricted sections being f plus (-1) times the section extended by zero off O).

Relative (15.1) gives Part I's conditions (i)–(ii) Chapter15/RelativeBranch · SBLocalization.isParamGP_restrictTo_of_onto_rel. Let c be a strong bundle chart of W with domain in O, centred at x ($c(x) = 0$), with $f(x) = s^0(x)$, T a presentation of its model quadrant, and assume that $(v, \tau) \mapsto (f - s^0)'(x)v + \sum_j \tau_j s_j(x)$ maps $T_x^R X \times \mathbb{R}^\ell$ onto W_x. Then in the transported chart of $W|_O$ the parametrized section F has Part I's properties (i)–(ii) of the proof of Theorem 5.3.10 Chapter5/GeneralPosition · StrongBundleChart.IsParamGP. The proof reads everything in the chart. The linearization at 0 of the principal part of $(f - s^0)|_O$ in the transported chart is the chart derivative $D((\widehat{f} -$

$\widehat{s}^0) \circ r)(0)$ Chapter15/LocalIFT · SBLocalization.scDeriv_restrictTo_subRestrict; the principal part of $-s_j$ at 0 is minus the chart image of $s_j(x)$ Chapter15/BranchChart · SBLocalization.principal_eq_fiberLin Chapter15/BranchChart · SBLocalization.principal_restrictTo_negRestrict; and the chart image of the intrinsic operator applied to v is the chart derivative applied to the chart vector of v Chapter15/BranchChart · SBLocalization.fiberLin_toIntr_subDeriv. So the hypothesis says: for every w there are $v \in T^R_x X$ and τ with $D(v) + \sum_j \tau_j \widehat{s}_j(0) = \widehat{w}$. (i), $F'(0,x)$ onto: given k with $(0,k) \in K$, write $k = \widehat{w}$ and take $u = v$ and $a = -\tau$, the sign coming from the $-s_j$. (ii), the kernel of $F'(0,x)$ is transversal to $\mathbb{R}^\ell \oplus T^R$: given a tangent vector w at 0, apply the hypothesis to the image $D(w)$; it gives $v \in T^R$ and τ with $D(v) + \sum_j \tau_j \widehat{s}_j(0) = D(w)$; then $u = w - v$ has $D(u) - \sum_j \tau_j \widehat{s}_j(0) = 0$, so (τ, u) is in the kernel, and $w - u = v \in T^R$. These are the two lines of linear algebra of the bridge of §XV.24, with s^0 for 0.

The parametrized branch is in general position Chapter15/RelativeBranch · SBLocalization.isGeneralPositionGerm_branchParam_rel. Let (s^0_a, σ^0_a) represent Λ_0 on O, $s^0_a(x) = f(x)$, and assume relative (15.1) at x for the sections s_j. Then $F_a(t,y) = (f - s^0_a)|_O + \sum_j t_j(-s_j)|_O$ is in general position at $(0,x)$ (Definition 5.3.9 on $\mathbb{R}^\ell \times O$). *Proof.* x is smooth, being a zero of $f - s^0_a$ Chapter15/Transversality · SBLocalization.mem_smoothSet_of_eq, so the tame germ data give a tame strong bundle chart centred at x; restrict it to O, and take a presentation T of its quadrant. With this chart and the local structure (s^0, σ^0) as local data, the intrinsic operator $(f - s^0_a)'(x)$ lies in the support of $T(f, \Lambda_0)(x)$ Chapter15/LinearizationIntrinsic · SBLocalization.linearization_eq Chapter15/Transversality · SBLocalization.mem_support_linSumIntr_iff, so relative (15.1) applies to it in this chart. The previous step gives (i)–(ii) in the transported chart; $(f - s^0_a)|_O$ is a tame sc-Fredholm section of $W|_O$ Chapter15/LocalScPlus · SBLocalization.isScFredholmSectionTame_subRestrict vanishing at x; and Part I's lemma that (i)–(ii) at a zero give general position of the parametrized section Chapter5/GeneralPosition · isGeneralPositionGerm_paramSection_of_isParamGP concludes.

The local step Chapter15/RelativeParametric · SBLocalization.exists_local_paramGP_rel. Let X be tame, Λ_0 an sc$^+$-multisection, Λ_t sc$^+$-multisections with relative (15.1) at x_0. Then there are an open $V \ni x_0$, $\delta > 0$ and a Lebesgue null set $C_V \subset \mathbb{R}^\ell$ such that $(f, \Lambda_0 \oplus \Lambda_t)$ is in general position at every $y \in V$ with $(\Lambda_0 \oplus \Lambda_t)(f(y)) > 0$, for $\|t\| < \delta$, $t \notin C_V$. *Proof.* Relative (15.1) provides a linear representation $\sum_j t_j s_{ij}$ of the family on O_1; take a representation (s^0_a, σ^0_a) of Λ_0 on O_2, its sections extended by zero, and $O = O_1 \cap O_2$; O is a tame M-polyfold, an open subset of a tame one. For every pair (a,i) there are an open $W_{a,i} \ni (0, x_0)$ in $\mathbb{R}^\ell \times O$ and a null set $C_{a,i}$ such that, for $(t,y) \in W_{a,i}$ with $f(y) = s^0_a(y) + \sum_j t_j s_{ij}(y)$ and $t \notin C_{a,i}$, the branch $(f - (s^0_a + \sum_j t_j s_{ij}))|_O$ is in Part I's general position at y. Two cases. If $f(x_0) = s^0_a(x_0)$, then $(0, x_0)$ is a zero of $F_{a,i}$, the slice at $t = 0$ being $f - s^0_a$ Chapter15/RelativeParametric · SBLocalization.sectionAdd_paramSec_zero, $F_{a,i}$ is in general position there by the previous lemma, and Part I's local Sard step, the local form of Theorem 5.3.10 without compactness Chapter15/LocalSlice · StrongBundle.exists_null_isGeneralPositionGerm_paramSlice_near, gives the neighbourhood and the null set; its slices are the branches. If $f(x_0) \neq s^0_a(x_0)$, the zero set of $F_{a,i}$, closed because $F_{a,i}$ is sc-smooth (Part I's Theorem 3.1.21) Chapter3/ParamPerturbationG · Polyfold.theorem_3_1_21_tame Chapter5/Transversality · isClosed_solutionSet, misses $(0, x_0)$; its complement is $W_{a,i}$ and $C_{a,i} = \emptyset$. The finitely many $W_{a,i}$ contain a product $B_\delta(0) \times V'$, and $C_V = \bigcup_{a,i} C_{a,i}$ is null. For $y \in V'$, $\|t\| < \delta$, $t \notin C_V$ and $(\Lambda_0 \oplus \Lambda_t)(f(y)) > 0$: y is at level 1; restrict a chart around y to O; the sum representation has the branches $s^0_a + \sum_j t_j s_{ij}$; each branch through $f(y)$ is in Part I's general position at y by the choice of $W_{a,i}$; and Part I's general position of the branches through $f(y)$ is Definition 15.3.5 at y Chapter15/GeneralPositionLocal · SBLocalization.isReducedOntoAtLevel_of_isGeneralPositionGerm (§XV.16).

Relative (15.2) Chapter15/RelativeParametric · SBLocalization.exists_pos_supp_rel_subset. If (N, U) controls

compactness, $N(\Lambda_0) \le 1$, the domain supports lie in U, $N(\Lambda_t) \le |t|_1(1-N(\Lambda_0))$, and V is a saturated open set containing $\operatorname{supp}(\Lambda_0 \circ f)$, then for some $\varepsilon > 0$ every t with $|t|_1 < \varepsilon$ has $\operatorname{supp}((\Lambda_0 \oplus \Lambda_t) \circ f) \subset V$. The text's (15.2) has S_f; relative to Λ_0 the support of $\Lambda_0 \circ f$ takes its place. *Proof*, by contradiction. Suppose there are t_n with $|t_n|_1 < 1/(n+1)$ and $x_n \notin V$ with $(\Lambda_0 \oplus \Lambda_{t_n})(f(x_n)) > 0$. For $|t|_1 \le 1$, $N(\Lambda_0 \oplus \Lambda_t) \le N(\Lambda_0) + |t|_1(1 - N(\Lambda_0)) \le 1$, so $x_n \in U$ and $N(f(x_n)) \le 1$ `Chapter15/ParametricGeneralPosition · SBLocalization.auxLe_of_pos`, using the sum of multisections (Definition 13.2.9) `Chapter13/MultisectionAlgebra · SBLocalization.definition_13_2_9`. The orbits $\pi(x_n)$ lie in the compact closure that the control of compactness provides; they have a cluster point $\pi(z)$. The object space is metrizable; along a decreasing basis B_k of neighbourhoods of z pick $n_k \ge k$ and $y_k \in B_k$ in the orbit of x_{n_k} (the orbit map is open, so $\pi(B_k)$ is a neighbourhood of $\pi(z)$ met infinitely often). Then $y_k \to z$, $y_k \notin V$ (V saturated), and $(\Lambda_0 \oplus \Lambda_{t_{n_k}})(f(y_k)) > 0$, since f and the sum are functors. Decompose $f(y_k) = a_k + b_k$ with $\Lambda_0(a_k) > 0$ and $\Lambda_{t_{n_k}}(b_k) > 0$ `Chapter15/RelativeParametric · SBLocalization.exists_of_multisectionAdd_pos_at` `Chapter15/ParamFamily · SBLocalization.exists_of_multisectionAdd_pos`; then $N(b_k) \le N(\Lambda_{t_{n_k}})(y_k) \le |t_{n_k}|_1 \le 1/(k+1)$. Take a local representation (s_i^0, σ_i^0) of Λ_0 on $O_2 \ni z$; for large k, $y_k \in O_2$ and $a_k = s_i^0(y_k)$ for some branch i; some i occurs infinitely often, and along that subsequence $(f - s_i^0)(y_k) = b_k$ `Chapter15/RelativeParametric · SBLocalization.subSec_eq_of_add`. The section $f - s_i^0$ is continuous on O_2, being a tame sc-Fredholm section of $W|_{O_2}$, so $(f - s_i^0)(y_k) \to (f - s_i^0)(z)$; and $b_k \to 0_z$, because the b_k lie in $W_{0,1}$, their base points converge to z and their norms to 0, which for an auxiliary norm forces convergence to the zero vector. Hence $f(z) = s_i^0(z)$, so $\Lambda_0(f(z)) > 0$ and $z \in V$; V being open, $y_k \in V$ for large k, a contradiction. The text's analogue for $\Lambda_0 = 0$ is the statement (15.2) of §XV.25; the new point is the branch of Λ_0 met infinitely often.

The assembly. Let $S = \operatorname{supp}(\Lambda_0 \circ f)$. Every $z \in S$ is at level 1 (§XV.20), so the local step applies at z and gives V_z, δ_z, C_z. $|S|$ is compact (Step 2 above); finitely many $\pi(V_z)$, $z \in T$, cover it. Let V_s be the saturation of $\bigcup_{z \in T} V_z$, open and saturated, containing S. Relative (15.2) gives ε_0 with $\operatorname{supp}((\Lambda_0 \oplus \Lambda_t) \circ f) \subset V_s$ for $|t|_1 < \varepsilon_0$. Choose $\varepsilon > 0$ so that $\|t\| < \varepsilon$ implies $\|t\| < \delta_z$ for $z \in T$ and $|t|_1 < \varepsilon_0$. The union of the C_z, $z \in T$, is null. For t off it with $\|t\| < \varepsilon$, and x in the support of $(\Lambda_0 \oplus \Lambda_t) \circ f$: $x \in V_s$, so there are $y \in V_z$ for some $z \in T$ and a morphism $\varphi : y \to x$; the support is saturated, so y is in it; general position holds at y by the local step; and general position at a point is carried along morphisms `Chapter15/ParametricGeneralPosition · SBLocalization.isInGeneralPositionAt_of_mor` (§XV.25). Pointwise general position everywhere on the support is general position `Chapter15/GeneralPositionLocal · SBLocalization.isInGeneralPosition_iff_forall_at`. □ The text discusses a countable cover and then remarks that a finite one suffices; the formal proof uses the finite cover directly, as in §XV.25.

What remains: the relative local families (gap G.178). Everything in steps (c)–(d) is proved except the proposition `Chapter15/ChargedExtensionFibres · SBLocalization.relative_local_families_statement` that every point of $\operatorname{supp}(\Lambda_0 \circ f)$ has a relative family with domain support in $U \setminus \partial X$ and bound $|t|_1(h - N(\Lambda_0))$, under the hypotheses of the relative statement. It splits in two `Chapter15/RelativeReduction · SBLocalization.hasLocalGPFamiliesRel_of_boundary`: at the points of $\operatorname{supp}(\Lambda_0 \circ f) \cap \partial X$ the empty family suffices as soon as (f, Λ_0) is reducedly onto at the support points of a saturated open neighbourhood, by the empty family of a set on which this holds; at the other points a relative family is needed. The first is the openness of general position near the boundary part of the support, which the text gets from the implicit function theorem (the neighbourhoods $Q(x)$ of its step (c)); the second is the text's "for $x \in U \setminus Q$ … we note that $d_X(x) = 0$ and we can argue as in Theorem 15.3.7": seeds at interior support points as in §XV.23, now making all the operators $(f - s_a^0)'(x)$ of $T(f, \Lambda_0)(x)$ onto at once, with domain support in a small neighbourhood that avoids ∂X. What each needs, with estimates, is in §XV.38. Wit-

nesses of the two relative steps: on $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ every pair is in general position, so the empty family covers everything Chapter15/RelativeReduction · SBLocalization.hasLocalGPFamiliesRel_epSEP_zero Chapter15/RelativeReduction · SBLocalization.hasParametricGeneralPositionRel_epSEP_zero, and the relative statement holds there from them Chapter15/RelativeReduction · SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero_of_hyp Chapter15/RelativeParametric · SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero_of_local; these witnesses have no boundary and zero fibres, so they are **weak**.

XV.30 The cylinder $[0,1]^n \times X$, and $d_X \equiv 0$ implies tame (prerequisites for Corollary 15.3.10)

Corollary 15.3.10 (p. 699) and its proof work on "the tame ep-groupoid $[0,1] \times X$" with the pulled-back strong bundle $\widetilde{W} \to [0,1] \times X$, whose fibre over (t,x) is W_x, and the section $\tilde{f}(t,x) = (t, f(x))$. The text takes all of this for granted, and it assumes $d_X \equiv 0$ where Theorems 15.3.7 and 15.3.9, which it applies, ask for a tame X. This subsection builds the cylinder and proves the two facts the corollary uses silently: that the cylinder is a tame ep-groupoid with the expected boundary, orbit space and bundle, and that $d_X \equiv 0$ makes X tame.

The cube instead of the interval. The formalization builds $[0,1]^n \times X$ for every n, with Part I's cube $[0,1]^n$ (the M-polyfold of maps $\{1,\dots,n\} \to [0,1]$); the text's $[0,1] \times X$ is $n = 1$. The reason is that Part I's parametrized Fredholm theory, which gives the sc-Fredholm property of $\tilde{f}$, is stated for the cube. Where the boundary or the time coordinate matters, $n = 1$ is fixed.

The ep-groupoid (extra) Chapter15/CubeEp · EpGroupoid.cubeProd. $[0,1]^n \times X$ is the product (§XV.19) of the trivial ep-groupoid of the cube, whose only morphisms are identities, with X: objects (q,x), morphisms (q,φ), with source $(q, s(\varphi))$ and target $(q, t(\varphi))$. Two functors connect it with X. The projection $(q,x) \mapsto x$, $(q,\varphi) \mapsto \varphi$ is the second projection of the product Chapter15/CubeEp · EpGroupoid.cubeProj. The inclusion $i_q : x \mapsto (q,x)$, $\varphi \mapsto (q,\varphi)$ at a point q of the cube Chapter15/CubeEp · EpGroupoid.cubeIncl is sc-smooth because a constant map with a smooth value is sc-smooth Chapter15/CubeEp · MPolyfold.isScSmoothMap_const_of_mem_smoothSet_poly and every point of the finite-dimensional cube is smooth Chapter15/CubeEp · cubePolyfold_mem_smoothSet; and $\mathrm{pr} \circ i_q = \mathrm{id}$ Chapter15/CubeEp · EpGroupoid.cubeProj_comp_cubeIncl.

Tameness and the boundary. If X is tame, so is $[0,1]^n \times X$ Chapter15/CubeEp · EpGroupoid.isTame_cubeProd, by Part I's tameness of products with the cube Chapter6/CubeSections · prod_cube_isTame. The degeneracy index is additive, $d(q,x) = \#\{i : q_i \in \{0,1\}\} + d_X(x)$ Chapter15/CubeEp · EpGroupoid.degIndex_cubeProd: the first term is the degeneracy index of the cube at q, and the sum is Part I's degeneracy index of a product of tame M-polyfolds. For $n = 1$ and $d_X \equiv 0$ the count runs over a one-element index set, so $d(t,x) = 1$ if $t \in \{0,1\}$ and $d(t,x) = 0$ otherwise Chapter15/CubeEp · EpGroupoid.degIndex_cubeProd_one_of_degIndex_eq_zero. The boundary is the set of points of positive degeneracy index, so $\partial([0,1] \times X) = \{0,1\} \times X$ Chapter15/RegularPerturbations · SBLocalization.mem_boundary_cubeSEP_iff. This is the fact behind "Λ over $i = 0,1$": the boundary data of Theorem 15.3.9 on the cylinder are data over the two ends.

Isomorphism classes and the orbit space. Since the cube contributes only identities, (q,x) and (q',x') are isomorphic iff $q = q'$ and x, x' are isomorphic Chapter15/CubeEp · EpGroupoid.cubeProd_π_eq_π_iff; hence $S \times U$ is saturated whenever U is Chapter15/CubeEp · EpGroupoid.isSaturated_prod_cubeProd. The map $(q, |x|) \mapsto |(q,x)|$ from $[0,1]^n \times |X|$ to the orbit space of the cylinder is well defined by the same criterion Chapter15/CubeEp · EpGroupoid.cubeOrbitMap. It is continuous Chapter15/CubeEp · EpGroupoid.continuous_cubeOrbitMap: $\mathrm{id} \times \pi$ is an open quotient map (the orbit map of X is open, Proposition 7.1.8), and

composed with it the map is the orbit map of the cylinder, which is continuous. It carries $[0,1]^n \times |A|$ onto $|[0,1]^n \times A|$ `Chapter15/CubeEp · EpGroupoid.image_π_cubeProd`, so $|[0,1]^n \times A|$ is compact when $|A|$ is `Chapter15/CubeEp · EpGroupoid.isCompact_image_π_cubeProd`. More is true. The map is injective `Chapter15/Cylinder · EpGroupoid.injective_cubeOrbitMap` (the isomorphism criterion again), surjective `Chapter15/Cylinder · EpGroupoid.surjective_cubeOrbitMap` (every orbit has a representative (q,x)), and open `Chapter15/Cylinder · EpGroupoid.isOpenMap_cubeOrbitMap`: the image of an open $O \subset [0,1]^n \times |X|$ is the image under the orbit map of the cylinder of the open set $(\mathrm{id} \times \pi)^{-1}(O)$, and the orbit map of the cylinder is open by Proposition 7.1.8. So it is a homeomorphism $[0,1]^n \times |X| \cong |[0,1]^n \times X|$ `Chapter15/Cylinder · EpGroupoid.cubeOrbitHomeomorph`. A product of a compact space with a paracompact space is paracompact, and paracompactness is carried by homeomorphisms; so the orbit space of the cylinder is paracompact when $|X|$ is `Chapter15/Cylinder · EpGroupoid.hasParacompactOrbitSpace_cubeProd`. This is the hypothesis "paracompact orbit space" of Theorem 15.3.9 on the cylinder, derived from the corollary's own hypothesis on $|X|$.

Bump functions. If X admits sc-smooth bump functions (Definition 12.3.1, read pointwise), so does $[0,1]^n \times X$ `Chapter15/CubeEp · EpGroupoid.admitsBumpFunctionsPointwise_cubeProd`: Theorem 12.3.2 (1) passes between the ep-groupoid and its object M-polyfold, and Part I proves bump functions on products with the cube `Chapter6/CubeSections · admitsBumpFunctions_prod_cube`.

The bundle and the section (narrower). $\widetilde{W}$ is the pull-back of W along the projection functor `Chapter15/CubeEp · SEP.cubeSEP`, Chapter 10's pull-back of a strong bundle over an ep-groupoid along a functor. A point of $\widetilde{W}$ is a pair $((q,x),w)$ with $w \in W_x$ `Chapter15/CubeEp · SEP.cubeVal`, and the morphism (q,φ) acts by $\tilde{\mu}((q,\varphi),((q,x),w)) = ((q,t(\varphi)),\mu(\varphi,w))$ `Chapter15/CubeLift · SBLocalization.cubeVal_fibAct`. The section is $\tilde{f}(q,x) = ((q,x),f(x))$ `Chapter15/CubeEp · SEP.cubeLiftEp`, a section of $\widetilde{W}$ `Chapter15/CubeEp · SEP.isSection_cubeLiftEp`. The text writes $\tilde{f}(t,x) = (t,f(x))$ in the statement and $\tilde{f}(t,x) = f(x)$ in the proof; both mean this section.

It is a functor `Chapter15/CubeEp · SEP.isFunctorialSection_cubeLiftEp`: at a morphism (q,φ) the identity $\tilde{f}(t(q,\varphi)) = \tilde{\mu}((q,\varphi),\tilde{f}(s(q,\varphi)))$ has first component $(q,t(\varphi))$ on both sides, and its second component is $f(t(\varphi)) = \mu(\varphi,f(s(\varphi)))$, the functor property of f. *It is sc-smooth* `Chapter15/CubeEp · SEP.isScSmoothSectionFunctor_cubeLiftEp`, by Part I's pull-back of sc-smooth sections along the sc-smooth projection. *It is sc-Fredholm when f has Part I's tame germ data* `Chapter15/CubeEp · SEP.isScFredholmSectionFunctor_cubeLiftEp`: the germ of $\tilde{f}$ at (q,x) is the germ of f at x with the cube coordinates as parameters, and Part I proves that this parametrized germ has tame germ data again `Chapter6/CubeFredholm · StrongBundle.IsScFredholmSectionTame.paramCube`, which gives the sc-Fredholm property; the tame germ data of $\tilde{f}$ are recorded separately, for the theorems that need them on the cylinder `Chapter15/RegularPerturbations · SBLocalization.isScFredholmSectionTame_cubeLiftEp`. *Its solution set* is $S_{\tilde{f}} = [0,1]^n \times S_f$ `Chapter15/CubeEp · SEP.solutionSet_cubeLiftEp`, since the second component of $\tilde{f}(q,x)$ is $f(x)$ and the zero of $\widetilde{W}_{(q,x)}$ is $((q,x),0_x)$; so $|S_{\tilde{f}}|$ is compact when $|S_f|$ is `Chapter15/CubeEp · SEP.hasCompactSolutionSetEp_cubeLiftEp`, by the compactness of $|[0,1]^n \times A|$ above.

The added hypothesis. The row is narrower only through the tame germ data of f, Part I's hypothesis for its parametrized Fredholm theory (Part I's untamed parametrized case is not formalized; with the 2021 untamed notion Part I's Theorem 3.6.8 does not hold as we read it, Part III). It is inherited, as in all of §15.2 (§XV.10), not new; its witness is the zero section of $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ with its graph section's tame germ data `Chapter15/InteriorPerturbationWitness · SBLocalization.theorem_15_2_2_hyp_witness` `Chapter3/NonVacuityG · isScFredholmSectionTame_graphSection`. The rest (tameness, boundary, orbit space, bump functions, compact solution set) holds as the text uses it.

$d_X \equiv 0$ **implies tame** (extra) `Chapter5/DegeneracyZeroTame · EpGroupoid.isTame_of_degIndex_eq_zero`. *Statement.* An ep-groupoid whose degeneracy index vanishes at every object is tame; it is the statement for its object M-polyfold `Chapter5/DegeneracyZeroTame · MPolyfold.isTame_of_degIndex_eq_zero`. Corollary 15.3.10 assumes $d_X \equiv 0$ and applies Theorems 15.3.7 and 15.3.9, which are stated for tame X; the text passes from one to the other without comment.

Proof, the local step `Chapter5/DegeneracyZeroTame · isTameRetract_inter_of_degIndex_eq_zero_ty`. Let $r : U \to U$ be an sc-smooth retraction of a partial quadrant $C \subset E$, presented by an sc-isomorphism $T : E \cong \mathbb{R}^k \times W'$ with $C = T^{-1}([0,\infty)^k \times W')$, and let W be open with $d_C = 0$ on $r(U) \cap W$. The claim is that $r(U) \cap W$ is a tame retract (Definition 2.5.2). (i) For $z \in C$, $d_C(z) = 0$ iff all k half-line coordinates of $T(z)$ are positive `Chapter5/DegeneracyZeroTame · degIndexWith_eq_zero_iff_ty` `Chapter5/DegeneracyZeroTame · degIndex_eq_zero_iff_pos_ty`, and the set D of such z is open `Chapter5/DegeneracyZeroTame · isOpen_setOf_pos_coord_ty`, a finite intersection of open half-spaces. (ii) Restrict r to $U' = U \cap r^{-1}(W) \cap D$. This is still an sc-smooth retraction: U' is relatively open, r maps U' into $r(U) \cap W$, where $d_C = 0$, hence into D, and $r \circ r = r$. Its retract is $r(U) \cap W$: a point of $r(U) \cap W$ is fixed by r and lies in D. (iii) Definition 2.5.2 (1), $d_C(r(x)) = d_C(x)$ on U', holds because both sides are 0 there. (iv) Definition 2.5.2 (2) asks, at a smooth point x of the retract, for an sc-complement of T_xO inside the minimal subspace E_x of E spanned by the directions in which no vanishing coordinate is tied down. Since no half-line coordinate of x vanishes, $E_x = E$, and the range of $Dr(x)$ is sc-complemented in E by $\ker Dr(x)$, Part I's sc-splitting of the range of the derivative of a retraction. Hence the restricted retract is tame.

Proof, the M-polyfold. At a point x take a chart φ whose degeneracy index at x attains $d_X(x) = 0$ (Definition 2.4.1 takes a minimum over charts, and it is attained). The points of its domain with chart degeneracy index 0 form an open set s, by the upper semicontinuity of the chart degeneracy index; the restriction of φ to s is again a chart of the structure, and its image is $r(U) \cap W$ for an open W. On it $d_C = 0$, so it is a tame retract by the local step, and the M-polyfold is tame by Part I's criterion that every point lie in a chart on a tame retract `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart`. The ep-groupoid is tame when its object M-polyfold is. Gap G.183 is closed.

XV.31 Corollary 15.3.10 (1) and (2) (2017: Corollary 15.1 (1), (2))

The setting. Corollary 15.3.10 (p. 699) assumes a strong bundle over an ep-groupoid with $d_X \equiv 0$ and paracompact orbit space $|X|$, and an sc-Fredholm section functor f with $|S_f|$ compact. Part (1) produces a perturbation for one controlling pair, (2) an auxiliary norm on the cylinder that interpolates two controlling pairs, (3) a perturbation on the cylinder that interpolates two perturbations (§§XV.32–XV.33).

Definition: regular perturbations `Chapter15/RegularPerturbations · SBLocalization.IsRegularPerturbation`. (N, U, Λ) is a structurable regular perturbation of f if (N, U) controls the compactness of f (Definition 15.3.1, §XV.17), Λ is structurable (Definition 13.3.6, Part XIII), its domain support lies in U, $N(\Lambda)(x) < 1$ for all x, and (f, Λ) is transversal (Definition 15.2.1, §XV.12). This is the text's definition in (1): "$T(f,\Lambda)(x)$ onto for all $x \in \operatorname{supp}(\Lambda \circ f)$" is transversality, and the clauses about Θ are a consequence, not part of the definition.

Corollary 15.3.10 (1) (narrower) `Chapter15/Cylinder · SBLocalization.corollary_15_3_10_1_ty`. *Statement.* Let $d_X \equiv 0$, $|X|$ paracompact, X admit sc-smooth bump functions, f be an sc-Fredholm section functor with Part I's tame germ data and $|S_f|$ compact, and (N, U) control the compactness of f. Given the statement of Theorem 15.3.7 for the bundle, there is a Λ such that (N, U, Λ) is a structurable reg-

ular perturbation, (f, Λ) is in general position (Definition 15.3.5), and $\Theta = \Lambda \circ f$ is a branched ep⁺-subgroupoid with compact $|\operatorname{supp}\Theta|$. The statement of Theorem 15.3.7 holds for every bundle whose orbit space is paracompact `Chapter15/GeneralPositionGoodSystem · SBLocalization.theorem_15_3_7_of_paracompact` (§XV.26), which the corollary assumes; so the hypothesis is discharged under the corollary's own assumptions. The two declarations compose; the composite is not stated as one declaration.

Proof. (i) *Tameness.* X is tame because $d_X \equiv 0$ (§XV.30). (ii) *Theorem 15.3.7 with $h \equiv 1$.* The constant function 1 is continuous, constant along morphisms and satisfies $0 < h \leq 1$; the statement of Theorem 15.3.7 (§XV.21) then gives a structurable Λ with $N(\Lambda) < 1$, domain support in U, (f, Λ) transversal and in general position. These are the clauses of a regular perturbation, with general position in addition. (iii) *The "in particular".* With $N(\Lambda) \leq 1$, domain support in U and general position, Theorem 15.3.7's "in particular" `Chapter15/GeneralPositionBranched · SBLocalization.theorem_15_3_7_in_particular_hyp` (§XV.21) makes Θ a branched ep⁺-subgroupoid with compact support; its tame form `Chapter15/TameBranched · SBLocalization.theorem_15_3_7_in_particular_tame_hyp` gives the text's "tame branched". (iv) *Where the hypotheses enter.* $d_X \equiv 0$ is used only for (i), and paracompactness only through Theorem 15.3.7's structurable addendum, which needs a good system, supplied by Theorem 13.1.8 for paracompact $|X|$ (§XV.26). The text's proof says "a special case of Theorem 15.3.7"; the special case is $h \equiv 1$.

Intermediate form. The same conclusion with "X tame" as an extra hypothesis in place of its derivation `Chapter15/RegularPerturbations · SBLocalization.corollary_15_3_10_1_hyp`; the final form calls it with the tameness of §XV.30.

Added hypotheses. Sc-smooth bump functions, which Theorem 15.3.7 needs and the corollary does not state; Remark 15.3.2 says that the theory needs them. And the tame germ data of f, inherited from §15.2 (§XV.10). Neither is derived from the corollary's statement; they hold together with the corollary's own hypotheses on the example of §XV.13 over $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ `Chapter15/InteriorPerturbationWitness · SBLocalization.theorem_15_2_2_hyp_witness`, where $d_X \equiv 0$ and $|X| = \mathbb{R}^n$ is paracompact; there the zero section with the zero multisection is a regular perturbation, a degenerate one (zero fibres).

Corollary 15.3.10 (2) (same) `Chapter15/CubeAuxNorm · SEP.corollary_15_3_10_2`. *Statement.* If (N^0, U^0) and (N^1, U^1) control the compactness of f, there are an auxiliary norm $\widetilde{N}$ of $\widetilde{W} \to [0,1] \times X$ and a saturated open $\widetilde{U} \supset S_{\tilde{f}}$ such that $(\widetilde{N}, \widetilde{U})$ controls the compactness of $\tilde{f}$ and restricts to (N^i, U^i) over $t = i$. The text states (2) for regular perturbations (N^i, U^i, Λ^i); the Λ^i play no role in it, and only the control of compactness is assumed. The text does not give $\widetilde{N}$; its proof pulls back a single N, which is the first step below. The formalization proves both.

The proof's step: one pair, pulled back. $\widetilde{N}(q, x, h) = N(h)$ `Chapter15/CubeAuxNorm · SEP.cubeAuxNorm` is Part I's pull-back of an auxiliary norm along a map of the base `Chapter5/CylinderPerturbation2021 · StrongBundle.pullbackAuxNorm`; its value at a point of $\widetilde{W}_{0,1}$ is N of the W-component `Chapter15/CubeAuxNorm · SEP.cubeAuxNorm_eq`. It is an auxiliary norm of the strong bundle $\widetilde{W}$ (Definition 5.1.1) `Chapter15/CubeAuxNorm · SEP.isAuxNorm_cubeAuxNorm'`, by Part I's lemma that pull-backs of auxiliary norms are auxiliary norms `Chapter5/CylinderPerturbation2021 · StrongBundle.IsAuxNorm.pullback`. It is invariant under $\tilde{\mu}$ (Definition 8.3.5 (3)) `Chapter15/CubeAuxNorm · SEP.isAuxNorm_cubeAuxNorm`: if $h = \tilde{\mu}((q, \varphi), k)$, then the W-component of h is $\mu(\varphi, k_2)$ and the cube coordinate is unchanged `Chapter15/CubeAuxNorm · SEP.cubeVal_μ`, so $\widetilde{N}(h) = N(\mu(\varphi, k_2)) = N(k_2) = \widetilde{N}(k)$ by the invariance of N. At the values of $\tilde{f}$, "$\widetilde{N}(\tilde{f}(q, x)) \leq r$" is "$N(f(x)) \leq r$" `Chapter15/CubeAuxNorm · SEP.auxLe_cubeAuxNorm_iff`. Then $(\widetilde{N}, [0,1]^n \times U)$ controls the compactness of $\tilde{f}$ whenever (N, U) controls that of f `Chapter15/CubeAuxNorm · SEP.controlsCompactness_cube`: $[0,1]^n \times U$ is open, saturated

(§XV.30) and contains $S_{\tilde f} = [0,1]^n \times S_f$; and its sublevel set $\{z \in [0,1]^n \times U : \widetilde N(\tilde f(z)) \le 1\}$ is $[0,1]^n$ times the sublevel set S of (N, U). The orbit set of $[0,1]^n \times S$ lies in the image of $[0,1]^n \times \mathrm{cl}\,|S|$ under the continuous map $(q, |x|) \mapsto |(q, x)|$, a compact set, so its closure is compact `Chapter15/CubeAuxNorm · SEP.isCompact_closure_image_π_cube`.

The weights. $\alpha(s) = \min(1, 3(1-s))$ `Chapter15/CubeAuxNorm · SEP.cubeWeight₀` and $\beta(s) = \min(1, 3s)$ `Chapter15/CubeAuxNorm · SEP.cubeWeight₁` are continuous and non-negative on $[0,1]$, $\alpha = 1$ on $[0, \frac{2}{3}]$, $\beta = 1$ on $[\frac{1}{3}, 1]$, $\alpha(1) = 0$, $\beta(0) = 0$, and at every s one of them is positive `Chapter15/CubeAuxNorm · SEP.cubeWeight_pos` (for $s \le \frac{2}{3}$, $\alpha(s) = 1$; otherwise $\beta(s) = 1$). With the time coordinate $t(w)$ of a point of $\widetilde W_{0,1}$ `Chapter15/CubeAuxNorm · SEP.cubeTime`, a continuous function, the interpolated norm is

$$\widetilde N(w) = \max\left(\alpha(t(w))\,\widetilde N^0(w),\ \beta(t(w))\,\widetilde N^1(w)\right)$$

`Chapter15/CubeAuxNorm · SEP.interpAuxNorm`, with $\widetilde N^i$ the pull-back of N^i, and the interpolated set is $\widetilde U = ([0, \frac{2}{3}) \times U^0) \cup ((\frac{1}{3}, 1] \times U^1)$ `Chapter15/CubeAuxNorm · SEP.interpSet`.

Step 1, a lemma on maxima. For auxiliary norms N_0, N_1 of a strong bundle and constants $a, b \ge 0$, not both 0, $\max(aN_0, bN_1)$ is an auxiliary norm `Chapter15/CubeAuxNorm · StrongBundle.isAuxNorm_max_mul`. If both are positive, Part I's lemmas that positive multiples and maxima of auxiliary norms are auxiliary norms apply; if $b = 0$, the maximum is aN_0, since $aN_0 \ge 0$, and symmetrically for $a = 0$.

Step 2, $\widetilde N$ is an auxiliary norm of $(\widetilde W, \tilde\mu)$ `Chapter15/CubeAuxNorm · SEP.isAuxNorm_interpAuxNorm`. Definition 5.1.1 has three conditions and Definition 8.3.5 adds the invariance. (0) Continuity on $\widetilde W_{0,1}$: products and maxima of continuous functions. (1) On each fibre a complete norm: on the fibre over z the time is the constant $t(z)$ `Chapter15/CubeAuxNorm · SEP.cubeTime_toOne`, so there $\widetilde N$ agrees with $\max(\alpha(t(z))\widetilde N^0, \beta(t(z))\widetilde N^1)$ for constants, which is a complete fibre norm by Step 1; completeness of a fibre norm depends only on the values on that fibre `Chapter15/CubeAuxNorm · StrongBundle.IsCompleteFiberNorm.congr`. (2) The convergence condition, that $w_k \in \widetilde W_{0,1}$ with $\widetilde P(w_k) \to z$ and $\widetilde N(w_k) \to 0$ converge to 0_z. Let t_0 be the time of z; one of $\alpha(t_0)$, $\beta(t_0)$ is positive, say $\alpha(t_0)$. By continuity $\alpha(t(w_k)) > \alpha(t_0)/2$ for large k, so $\widetilde N^0(w_k) \le (2/\alpha(t_0))\,\widetilde N(w_k) \to 0$, and condition (2) for $\widetilde N^0$ gives $w_k \to 0_z$; symmetrically if $\beta(t_0) > 0$. (3) Invariance: both $\widetilde N^i$ are invariant, and $\tilde\mu$ does not change the time (the same lemma on the cube coordinate as above).

Step 3, the restrictions. Over $t = 0$, $\alpha = 1$ and $\beta = 0$, so $\widetilde N = \widetilde N^0$ `Chapter15/CubeAuxNorm · SEP.interpAuxNorm_of_cubeTime_eq_zero`, and $(0, x) \in \widetilde U$ iff $x \in U^0$ `Chapter15/CubeAuxNorm · SEP.mem_interpSet_zero`; over $t = 1$, $\widetilde N = \widetilde N^1$ `Chapter15/CubeAuxNorm · SEP.interpAuxNorm_of_cubeTime_eq_one` and $(1, x) \in \widetilde U$ iff $x \in U^1$ `Chapter15/CubeAuxNorm · SEP.mem_interpSet_one`. This is "restricts over $i = 0, 1$ to N^i, U^i".

Step 4, $(\widetilde N, \widetilde U)$ controls the compactness of $\tilde f$. $\widetilde U$ is open, a union of two products of open sets. It is saturated: each piece is a product of a set of times with a saturated set (§XV.30), and images and preimages under the orbit map commute with unions. It contains $S_{\tilde f} = [0,1] \times S_f$: a point (t, x) with $x \in S_f$ lies in the first piece if $t < \frac{2}{3}$, since $S_f \subset U^0$, and in the second otherwise, since then $t > \frac{1}{3}$ and $S_f \subset U^1$. For the compactness, let $(t, x) \in \widetilde U$ with $\widetilde N(\tilde f(t, x)) \le 1$; the value $f(x)$ then lies in $W_{0,1}$, the level of the pull-back being that of W. If (t, x) is in the first piece, $\alpha(t) = 1$, so $N^0(f(x)) \le \widetilde N(\tilde f(t, x)) \le 1$ and x lies in the sublevel set S_0 of (N^0, U^0); in the second piece, x lies in the sublevel set S_1 of (N^1, U^1). So the sublevel set of $(\widetilde N, \widetilde U)$ lies in $([0,1] \times S_0) \cup ([0,1] \times S_1)$, whose orbit set has compact closure, by the pulled-back step applied to each piece. A closed subset of a compact set is compact.

Reading. The text's (2) asks for one pair restricting to the two given ones; it does not say how. The

interpolation is ours; any partition of the time interval into two overlapping pieces works, and the margin between $\frac{1}{3}$ and $\frac{2}{3}$ makes $\widetilde{U}$ open while each end sees only one of the two pairs. With §XV.30 ($\tilde{f}$ sc-Fredholm with compact solution set, under the inherited tame germ data), all of (2) is covered.

XV.32 Corollary 15.3.10 (3) (2017: Corollary 15.1 (3)): the glued boundary data and the partitions of unity of the cylinder

The proof of (3) (pp. 699–700) applies Theorem 15.3.9 on $[0,1]\times X$ to the multisection over $\partial([0,1]\times X) = \{0,1\}\times X$ that is Λ^i over $\{i\}\times X$, and concludes from its output. Theorem 15.3.9 in its charged form (§XV.27) asks, of the space, that it be tame with paracompact orbit space, sc-smooth partitions of unity and tame strong bundle charts at every object; and of the boundary data, that they be structurable over the boundary (Definition 14.1.7), charge every boundary fibre, have boundary domain support in $\widetilde{U}$, satisfy $\widetilde{N}(\Lambda) < h$ on the boundary, and satisfy the face condition. The text checks none of this explicitly. This subsection proves what can be proved about these hypotheses: tameness and paracompactness were done in §XV.30; here are the glued data with their values, support, norm and charge, the partitions of unity of the cylinder, and the structurability of the glued data up to one analytic statement.

The glued data `Chapter15/CylinderGlue · SBLocalization.cubeGlue`. $\Lambda_\partial(w) = \Lambda^1(w_2)$ if the time of w is 1, and $\Lambda^0(w_2)$ otherwise, w_2 being the W-component; so $\Lambda_\partial = \Lambda^0$ over $t = 0$ `Chapter15/CylinderGlue · SBLocalization.cubeGlue_of_zero` and Λ^1 over $t = 1$ `Chapter15/CylinderGlue · SBLocalization.cubeGlue_of_one`. The values at times strictly between 0 and 1 are irrelevant: Theorem 15.3.9 looks at its boundary data only over $\partial([0,1]\times X)$, and Λ there is a function on all of W of which only these values enter (§XV.27).

Boundary domain support `Chapter15/CylinderGlue · SBLocalization.boundaryDomSupp_cubeGlue_subset`. If the domain support of Λ^i lies in U^i, the boundary domain support of Λ_∂, the closure of the set of boundary points z over which some nonzero w has positive weight, lies in $\widetilde{U}$. *Proof.* Let z be such a point, $w \neq 0_z$ with $\Lambda_\partial(w) > 0$. By §XV.30, z has time 0 or 1; say 0. Then $\Lambda^0(w_2) > 0$ and $w_2 \neq 0$ in W_{z_2}, since w and $((z, 0_{z_2}))$ differ only in the W-component `Chapter15/CylinderGlue · SBLocalization.cubeVal_snd_ne_fiberZero`; so z_2 lies in the domain support of Λ^0. Hence the set lies in $(\{0\}\times\text{dom-supp }\Lambda^0)\cup(\{1\}\times\text{dom-supp }\Lambda^1)$, which is closed, so its closure does too; and a point of $\{i\}\times\text{dom-supp }\Lambda^i$ lies in $\widetilde{U}$ by the restriction of $\widetilde{U}$ over $t = i$ (§XV.31).

The norm on the boundary. $\widetilde{N}(\Lambda_\partial)(0,x) \le N^0(\Lambda^0)(x)$ `Chapter15/CylinderGlue · SBLocalization.pointwiseNorm_cubeGlue_le_zero` and $\widetilde{N}(\Lambda_\partial)(1,x) \le N^1(\Lambda^1)(x)$ `Chapter15/CylinderGlue · SBLocalization.pointwiseNorm_cubeGlue_le_one`. *Proof.* $N(\Lambda)(x)$ is the supremum of the norms of the points of the fibre over x with positive weight (Definition 13.2.3). A point w over $(0,x)$ with $\Lambda_\partial(w) > 0$ has W-component w_2 over x with $\Lambda^0(w_2) > 0$, and its norm is at most that of w_2 `Chapter15/CylinderGlue · SBLocalization.normW_interpAuxNorm_le`: on level 1, $\widetilde{N} = \widetilde{N}^0$ over $t = 0$ and $\widetilde{N}^0(w) = N^0(w_2)$; off level 1 both sides are the convention value. For an sc⁺-multisection the supremum is attained, so it is at least $N^0(w_2)$. Hence $\widetilde{N}(\Lambda_\partial) < 1$ on the boundary when $N^i(\Lambda^i) < 1$, which holds for regular perturbations.

The glued data charge the boundary. *An sc⁺-multisection charges every fibre* `Chapter15/ChargedExtensionCylinder · SBLocalization.exists_ne_zero_of_isScPlusMultisection`: near x it has a local representation by k sections with positive weights summing to 1, so $k > 0$; the first section gives $w = s_0(x)$ over x, and $\Lambda(w)$, the sum of the weights of the sections through w, is at least $\sigma_0 > 0$. So the glued data charge $\partial([0,1]\times X)$ `Chapter15/ChargedExtensionCylinder · SBLocalization.chargesBoundary_`

cubeGlue in the sense of Part XIV's charging condition Chapter14/EmptyIndex · SBLocalization.ChargesBoundary: at a boundary point (i, x) take $((i, x), v)$ with $\Lambda^i(v) \neq 0$. This is the hypothesis that the charged form of Theorem 15.3.9 adds (§XV.27), discharged for the cylinder.

Partitions of unity on $[0, 1] \times X$ (explicit). Theorem 15.3.9 asks for sc-smooth partitions of unity on the M-polyfold of objects; the text does not say why the cylinder has them. *Step 1, from Urysohn functions to partitions of unity* Chapter5/CubePartitions · admitsScPartitionsOfUnity_of_urysohn. If any two disjoint closed sets of an M-polyfold are separated by an sc-smooth function with values in $[0, 1]$, the M-polyfold admits sc-smooth partitions of unity: this is the second half of the proof of Theorem 5.5.10. Given an open cover, a bump covering subordinate to it by such functions f_i exists by a general construction in paracompact spaces (Mathlib's subordinate bump coverings with functions from a prescribed class), and $\rho_i = f_i \prod_{j<i}(1 - f_j)$ is the partition of unity; locally only finitely many factors are nontrivial, so each ρ_i is locally a finite product of sc-smooth real functions, hence sc-smooth.

Step 2, a local separation. Let s, t be disjoint closed sets of $[0, 1]^n \times X$ and $x \in X$. The slices $S = \{q : (q, x) \in s\}$, $T = \{q : (q, x) \in t\}$ are closed in the compact cube, hence compact, and disjoint. A smooth Urysohn function $g_x : \mathbb{R}^n \to [0, 1]$ is 0 near S and 1 near T. The points of the cube where $g_x \neq 0$ form a compact set disjoint from S, and the tube lemma gives an open $V_x \ni x$ such that $g_x(q) = 0$ whenever $(q, y) \in s$ and $y \in V_x$; likewise $g_x(q) = 1$ whenever $(q, y) \in t$ and $y \in V_x$, after shrinking V_x.

Step 3, gluing over X Chapter5/CubePartitions · exists_cubeProd_urysohn_w5. Let (ρ_x) be an sc-smooth partition of unity of X subordinate to (V_x), which exists by hypothesis, and

$$F(q, y) = \sum_x \rho_x(y)\, g_x(q).$$

The sum is locally finite, since ρ is, and each term is sc-smooth (an sc-smooth function of y times a smooth function of the cube coordinates); so F is sc-smooth. On s, every term with $\rho_x(y) \neq 0$ has $y \in V_x$, so $g_x(q) = 0$, and $F = 0$; on t, every such term has $g_x(q) = 1$, so $F = \sum_x \rho_x(y) = 1$; and F takes values in $[0, 1]$ as a convex combination of values in $[0, 1]$. *Conclusion.* $[0, 1]^n \times X$ admits sc-smooth partitions of unity when X does Chapter5/CubePartitions · admitsScPartitionsOfUnity_cubeProd, and so does the object M-polyfold of the cylinder bundle Chapter15/CylinderPartitions · SBLocalization.admitsScPartitionsOfUnity_cubeSEP. Gap G.181 is closed. What remains is a hypothesis on X: sc-smooth partitions of unity, which Corollary 15.3.10 does not state and Theorem 15.3.9 on the cylinder needs; witness $\mathbb{R}^n$ Chapter7/LocalDiffeo · rnPolyfold_admitsScPartitionsOfUnity. For $X = \mathbb{R}^n$ the cylinder's partitions of unity have also a direct proof, from Theorem 5.5.10 with the product charts of the cube and $\mathbb{R}^n$, whose finite-dimensional models have smooth partitions of unity (Proposition 5.5.11) Chapter15/Cylinder · SBLocalization.admitsScPartitionsOfUnity_cube_rn.

The glued data are structurable over the boundary (narrower: given one analytic statement). The text treats the data over $\{0, 1\} \times X$ as structurable because the Λ^i are. The formalization makes this precise by lifting the structures of Λ^i along the projection $[0, 1] \times X \to X$.

Lifted sections. For a section s of W (corrected to the zero of the fibre where $s(y)$ does not lie over y, a formality), the lift is $\tilde{s}(q, y) = ((q, y), s(y))$ Chapter15/CubeLift · SBLocalization.cubeLiftSec, a section of $\widetilde{W}$ Chapter15/CubeLift · SBLocalization.P_cubeLiftSec, and $\tilde{s}(p) = w$ iff $s(p_2) = w_2$ for w over p Chapter15/CubeLift · SBLocalization.cubeLiftSec_eq_iff. Along a morphism, $\tilde{\mu}$ acts on the W-component by μ (§XV.30), so the equivariance of lifted sections reduces to that of the sections: if $s'(t(\varphi_2)) = \mu(\varphi_2, s(s(\varphi_2)))$ then $\tilde{s}'(t(\varphi)) = \tilde{\mu}(\varphi, \tilde{s}(s(\varphi)))$ Chapter15/GlueStructurable · SBLocalization.cubeLiftSec_compat. The isotropy

group of $z = (q, x)$ maps to that of x by $(q, g) \mapsto g$ Chapter15/CubeLift · SBLocalization.cubeIso; it is an isomorphism, since the cube contributes only identities.

The key step: natural representations on a connected neighbourhood Chapter15/CubeLift · SBLocalization.cube_Γ_snd. Let V be a natural neighbourhood of z in the cylinder (Theorem 7.1.13), with V connected and lying over the domain of a natural neighbourhood N of z_2 in X. Then for $g \in G_z$ and $y \in V$, the X-part of the morphism $\Gamma_V(g, y)$ is $\Gamma_N(g_2, y_2)$. *Proof.* The maps $a(p) = \Gamma_V(g, p)_2$ and $b(p) = \Gamma_N(g_2, p_2)$ are continuous on V with values in the morphism space of X and have the same source p_2. Two such maps agree on an open set, by the étale property Chapter13/StrongBundlePullMap · isOpen_setOf_eq_of_src, and disagree on an open set, the morphism space being Hausdorff. These two open sets cover V; they agree at z, where both are g (the natural representation at its centre is the given isotropy element); and V is connected, so they agree on all of V. In particular the action of g on V is $\Phi_V(g)(y) = (y_1, \Phi_N(g_2)(y_2))$ Chapter15/CubeLift · SBLocalization.cube_Φ_eq, and the same holds for the bi-action of isotropy groups on morphisms between two such neighbourhoods Chapter15/GlueStructurable · SBLocalization.cube_biact_snd. The text uses this compatibility of the natural representations of $[0, 1] \times X$ and of X without comment; formally it is the uniqueness of continuous lifts through the source map.

Lifting one structure Chapter15/CubeLift · SBLocalization.liftStructure. A symmetric sc⁺-section structure of Λ on N (Definition 13.2.5, Part XIII) — an index set I with an action of G_{z_2}, sc⁺-sections s_i on N and weights σ_i, equivariant, representing Λ — gives a boundary symmetric structure (Definition 14.1.3, Part XIV) of any $\widetilde{\Lambda}$ with $\widetilde{\Lambda}(w) = \Lambda(w_2)$ over $V \cap \partial$: the same index set and weights, G_z acting through $G_z \cong G_{z_2}$, and the lifted sections $\tilde{s}_i$. Equivariance follows from that of the s_i with the key step: $\tilde{s}_{g\cdot i}(\Phi_V(g)(y))$ has W-component $s_{g_2\cdot i}(\Phi_N(g_2)(y_2)) = \mu(\Gamma_N(g_2, y_2), s_i(y_2))$, the W-component of $\tilde{\mu}(\Gamma_V(g, y), \tilde{s}_i(y))$. The representation: $\widetilde{\Lambda}(w) = \Lambda(w_2) = \sum_{i:s_i(y_2)=w_2} \sigma_i = \sum_{i:\tilde{s}_i(y)=w} \sigma_i$. Condition (1) of Definition 14.1.3, that the sections be sc⁺ on the local faces, is taken as given (below). The index set is that of a structure of an sc⁺-multisection, so it is not empty Chapter13/TightMultisection · SBLocalization.Decoration.nonempty_I.

The two sides. A point p of the cylinder is on the side "$t > \frac{1}{2}$" or "$t < \frac{1}{2}$" Chapter15/GlueStructurable · SBLocalization.cubeSide; both sides are open Chapter15/GlueStructurable · SBLocalization.isOpen_sideSet, every boundary point lies on its side Chapter15/GlueStructurable · SBLocalization.mem_sideSet_cubeSide, morphisms keep the time and hence the side Chapter15/GlueStructurable · SBLocalization.cubeSide_src, and on the boundary Λ_∂ is Λ^{side} Chapter15/GlueStructurable · SBLocalization.cubeGlue_eq_side. A function that is $\Lambda^{\text{side}}(w_2)$ over the boundary is a functor there when the Λ^i are Chapter15/GlueStructurable · SBLocalization.isBoundaryWFunctor_of_sides, since morphisms keep the side and act by μ on the W-component.

The glued structure. Let $(\mathcal{U}^b, S^b)$ be decorations of Λ^b with correspondences τ^b (Definition 13.3.6, Λ^b structurable). At a boundary point z let $O(z) = \{\text{side of } z\} \times \{p : p_2 \in U^{\text{side}}(z_2)\}$ Chapter15/GlueStructurable · SBLocalization.glue0, an open set containing z Chapter15/GlueStructurable · SBLocalization.isOpen_glue0 Chapter15/GlueStructurable · SBLocalization.mem_glue0. Theorem 14.1.6 (Part XIV) gives a good system for the boundary of the tame cylinder with paracompact orbit space whose neighbourhoods lie in the $O(z)$ Chapter14/BoundaryMultisection · EpGroupoid.theorem_14_1_6_hyp; its neighbourhoods are connected. Two facts make everything reduce to X. (i) A morphism of the good system from z to z' stays on one side, and its X-part is a morphism of $\mathcal{U}^b(z_2, z_2')$ Chapter15/GlueStructurable · SBLocalization.mor_glue, because its source and target lie in $O(z)$ and $O(z')$. (ii) The structure at z is the lift of the structure of $\Lambda^{\text{side}(z)}$ at z_2 Chapter15/GlueStructurable · SBLocalization.glueS. The correspondences are $\tilde{\tau}_{z,z'}(\varphi) = \tau^b_{z_2,z_2'}(\varphi_2)$ Chapter15/GlueStructurable · SBLocalization.glueTau (between different sides the full correspondence is put in, but there are no morphisms between them, so it is never used). The conditions of Definition 14.1.7 on correspon-

dences follow from those of τ^b: local constancy and finiteness of the image along the X-parts; symmetry under inversion; the diagonal condition at isotropy elements, by the key step $\Gamma_V(g,y)_2 = \Gamma_N(g_2,y_2)$; equivariance under the bi-action of isotropy groups, by its form for morphisms; and the compatibility with the sections, by the equivariance of lifted sections above.

Structurability `Chapter15/GlueStructurable · SBLocalization.isBoundaryStructurable_of_sides`. For tame X with $d_X \equiv 0$ and paracompact $|X|$, structurable Λ^0, Λ^1, and the analytic statement below, any $\widetilde{\Lambda}$ equal to $\Lambda^{\mathrm{side}}(w_2)$ over the boundary is structurable over the boundary: functor by the side lemma, a structure at every boundary point by the lift, the decoration and correspondences as just built. In particular the glued data are `Chapter15/GlueStructurable · SBLocalization.isBoundaryStructurable_cubeGlue_w5`. Their index sets are those of the Λ^i, hence not empty, in agreement with the charging above.

The analytic statement `Chapter15/GlueStructurable · SBLocalization.CubeLiftFace`: for an sc$^+$-section s on an open $U \subset X$, the lift $\tilde{s}$ is sc$^+$ on the local faces through the points of $([0,1]\times U)\cap\partial([0,1]\times X)$, condition (1) of Definition 14.1.3. It is a statement in the text's setting, which the text uses without comment (that the lift of an sc$^+$-section along the projection is sc$^+$), but it is not proved: Part I's lemma on sc$^+$-sections in charts is available only for charts with domain inside the open set, and the pull-back lemma of Part I is stated for sc-smooth, not sc$^+$, sections. It is taken as a hypothesis and has no witness; a witness would be the general proof. This is gap G.193.

XV.33 Corollary 15.3.10 (3): the statement with the charged Theorem 15.3.9, the last paragraph of the proof, and Remark 15.3.11

Corollary 15.3.10 (3) (narrower) `Chapter15/ChargedExtensionCylinder · SBLocalization.corollary_15_3_10_3_charged`. *Statement.* Let $d_X \equiv 0$, $|X|$ paracompact, f an sc-Fredholm section functor with Part I's tame germ data and $|S_f|$ compact, and (N^i, U^i, Λ^i), $i = 0, 1$, structurable regular perturbations of f. Assume on the cylinder: the charged statement of Theorem 15.3.9 `Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_statement` (§XV.27), a tame strong bundle chart at every object, sc-smooth partitions of unity, and, of the glued data Λ_∂ of §XV.32, structurability over the boundary and the face condition. Then there is $\widetilde{\Lambda}$ such that $(\widetilde{N},\widetilde{U},\widetilde{\Lambda})$ is a structurable regular perturbation of $\tilde{f}$, with $(\widetilde{N},\widetilde{U})$ of (2), $\widetilde{\Lambda} = \Lambda^i$ over $\{i\}\times X$, and $\widetilde{\Theta} = \widetilde{\Lambda}\circ\tilde{f}$ a branched ep$^+$-subgroupoid with compact $|\operatorname{supp}\widetilde{\Theta}|$: the text's "compact cobordism $\widetilde{\Theta}$ between Θ_0 and Θ_1". The text's (3) does not repeat the pairs; it is stated for different (N^i, U^i), as in (2).

Proof, as the text's, with every input named. (i) X is tame (§XV.30), hence so is the cylinder, and $\partial([0,1]\times X)$ lies over $t \in \{0,1\}$. (ii) $(\widetilde{N},\widetilde{U})$ controls the compactness of $\tilde{f}$, by (2) (§XV.31). (iii) The data on the cylinder: $\tilde{f}$ is an sc-Fredholm section functor with tame germ data and compact solution set, and $|[0,1]\times X|$ is paracompact (§XV.30). (iv) The boundary data: Λ_∂ is structurable over the boundary (hypothesis), charges every boundary fibre, has boundary domain support in $\widetilde{U}$, and satisfies $\widetilde{N}(\Lambda_\partial) \le 1$ and < 1 on the boundary (§XV.32, the last from $N^i(\Lambda^i) < 1$). (v) Apply the charged Theorem 15.3.9 on the cylinder with $h \equiv 1$. It returns $\widetilde{\Lambda}$ structurable, equal to Λ_∂ over the boundary, with domain support in $\widetilde{U}$, $\widetilde{N}(\widetilde{\Lambda}) < 1$, $(\tilde{f},\widetilde{\Lambda})$ transversal, and $\widetilde{\Lambda}\circ\tilde{f}$ branched with compact support. (vi) These are the clauses of a regular perturbation; and over $t = i$ the points lie in the boundary (§XV.30), where $\widetilde{\Lambda} = \Lambda_\partial = \Lambda^i(w_2)$ by §XV.32. The declaration concludes "branched and compact"; the text's "tame" for $\widetilde{\Theta}$ would come from general position, which the statement of Theorem 15.3.9 does not conclude.

The hypotheses, discharged and open. Of the hypotheses that the declaration takes, two are proved for

the corollary's data in §XV.32: the partitions of unity of the cylinder from those of X, and the structurability of Λ_∂ from the structurability of the Λ^i, given the analytic statement on lifted sections; the declarations compose, and the composite is not stated as one declaration. The remaining inputs are: (a) the charged Theorem 15.3.9 on the cylinder, which holds given the relative local families on the cylinder `Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_of_local_rel` (§XV.29; gap G.178); (b) tame strong bundle charts at every object of the cylinder, the product of a tame strong bundle chart of W with the tame charts of $[0,1]$, which is not formalized (gap G.189); (c) the analytic statement on lifted sections (gap G.193); (d) the face condition of Λ_∂ (gap G.180); (e) sc-smooth partitions of unity on X, which the corollary does not state. *Reading of (d).* The text's proof says only $\widetilde{U} = [0,1] \times U$ and $\widetilde{f}(t,x) = f(x)$ and does not say how the face condition of the glued data is checked; presumably from the transversality of (f, Λ^i), since $T^R_{(i,x)}([0,1] \times X) = \{0\} \times T_x X$ when $d_X \equiv 0$, so that the reduced surjectivity on the face is the surjectivity of $T(f, \Lambda^i)(x)$.

Remark: the forms with the uncharged statement. The same conclusion is also proved from the uncharged Lean statement of Theorem 15.3.9 on the cylinder: with six hypotheses on arbitrary boundary data Λ_b `Chapter15/RegularPerturbations · SBLocalization.corollary_15_3_10_3_hyp`; the same with tameness and paracompactness of the cylinder derived `Chapter15/Cylinder · SBLocalization.corollary_15_3_10_3_ty`; for the glued data of two regular perturbations `Chapter15/CylinderGlue · SBLocalization.corollary_15_3_10_3_glue_hyp`; with the partitions of unity of X `Chapter15/CylinderPartitions · SBLocalization.corollary_15_3_10_3_pu_w5` `Chapter15/CylinderPartitions · SBLocalization.corollary_15_3_10_3_glue_w5`; and with the analytic statement on lifted sections in place of structurability `Chapter15/GlueStructurable · SBLocalization.corollary_15_3_10_3_struct_w5`. That statement, which admits boundary data with empty index sets, does not hold for the cylinder of a nonempty X under the corollary's hypotheses `Chapter15/ChargedExtensionCylinder · SBLocalization.not_theorem_15_3_9_statement_cube_w2`: the point $(0,x)$ is a boundary point, $\widetilde{f}$ and the pair $(\widetilde{N}, \widetilde{U})$ of (2) for the pair (N,U) taken twice satisfy its hypotheses, and the general counterexample of §XV.27 applies. So these forms are vacuous for nonempty X; this is a property of our formalization, not of the text, whose boundary data are sc$^+$-multisections with nonempty index sets. They are the steps at which the lemmas of §§XV.30–XV.32 were proved, and those lemmas discharge the corresponding hypotheses of the charged form above.

Witnesses, and their limits. On $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$ with $\Lambda^0 = \Lambda^1 = 0$, every hypothesis of the forms above on the boundary data holds together with tameness, $d_X \equiv 0$ and the paracompactness of the cylinder `Chapter15/Cylinder · SBLocalization.corollary_15_3_10_3_boundary_witness`, and the glued data are structurable over the boundary and satisfy the face condition `Chapter15/CylinderGlue · SBLocalization.corollary_15_3_10_3_glue_witness`. There the fibres are 0, so the projection $W \to X$ is injective, every operator onto a fibre is onto `Chapter15/Cylinder · SBLocalization.isReducedOntoAtPt_of_injective` and the face condition is automatic `Chapter15/Cylinder · SBLocalization.isBoundaryFaceOnto_of_injective`; the zero multisection is 1 at the only point of each fibre `Chapter15/Cylinder · SBLocalization.zeroMultisection_eq_one_of_injective`, the glued data are the zero multisection of the cylinder, and that is structurable over the boundary `Chapter14/BoundaryStructured · SBLocalization.isBoundaryStructurable_zero`. These witnesses are **weak**: zero fibres, and they show only that the hypotheses are consistent. That $\Lambda^i = 0$ are themselves regular perturbations on that example is not rechecked. The analytic statement on lifted sections and the tame strong bundle charts of the cylinder have no witness.

The last paragraph of the proof (same). The text closes with: if (N,U) and (N',U') control compactness, so does $(N'',U'') = (\max\{N,N'\}, U \cap U')$, and a perturbation for it is controlled by both pairs, so one can connect Λ with Λ'' and Λ'' with Λ'. *Proof.* (i) $\max\{N,N'\}$ is an auxiliary norm of

(W,μ) Chapter15/CylinderGlue · SBLocalization.isAuxNorm_max_ty: Part I's lemma for the maximum of two auxiliary norms, and invariance from that of N and N'. (ii) (N'',U'') controls compactness Chapter15/CylinderGlue · SBLocalization.controlsCompactness_max: $U\cap U'$ is open, saturated (a morphism with source in both saturated sets has its target in both) and contains S_f; and its sublevel set $\{x\in U\cap U' : \max\{N,N'\}(f(x))\le 1\}$ lies in the sublevel set of (N,U), since $N\le\max\{N,N'\}$, whose orbit set has compact closure. (iii) For an sc⁺-multisection, $N(\Lambda)\le\max\{N,N'\}(\Lambda)$ pointwise Chapter15/CylinderGlue · SBLocalization.pointwiseNorm_le_max_left: the supremum defining $N(\Lambda)(x)$ is attained at some w, and $N(w)\le\max\{N,N'\}(w)\le\max\{N,N'\}(\Lambda)(x)$. (iv) Hence a regular perturbation for (N'',U'') is one for (N,U) Chapter15/CylinderGlue · SBLocalization.IsRegularPerturbation.of_max_left and for (N',U') Chapter15/CylinderGlue · SBLocalization.IsRegularPerturbation.of_max_right: domain support in $U\cap U'\subset U$, norm $N(\Lambda)\le N''(\Lambda)<1$, structurability and transversality unchanged, and the control of compactness by (N,U) given. *A remark on its role.* The formal (3) takes the two regular perturbations with different pairs (N^0,U^0), (N^1,U^1) from the start, through the interpolated pair of (2), so it does not need this paragraph; the paragraph is proved as printed, and the two-step connection it describes, (3) applied twice, is not stated as a separate declaration.

Remark 15.3.11 (nothing to formalize) is prose: without orientations the perturbation theory gives only rational chains, and with f oriented, $\Lambda\circ\tilde f$ carries a natural orientation and gives invariants, which is §15.4 (§§XV.34–XV.37).

XV.34 Oriented formal sums, Theorem 15.4.1 and Definition 15.4.2 (2017: Theorem 15.6, Definition 15.7)

§15.4 opens (p. 700) with a list of standing data, Assumption 15.4: a strong bundle over a tame ep-groupoid, an oriented sc-Fredholm section functor (f,o) with compact $|S_f|$, and an sc⁺-multisection Λ with $T(f,\Lambda)$ onto on $\mathrm{supp}(\Lambda\circ f)$, kernels in good position at the boundary, and $\Theta=\Lambda\circ f$ a branched ep⁺-subgroupoid. Theorem 15.4.1 then lifts the tangent functor $T\Theta$ to oriented tangent spaces, using o. The text's route has two layers. The first is linear algebra at one point y: every summand $(f-s_i)'(y)$ of $T(f,\Lambda)(y)$ inherits an orientation $o_{i,y}$ from o, and the oriented formal sum $\hat T_{((f,o),\Lambda)}(y)=\sum_{f(y)=s_i(y)}\sigma_i\cdot((f-s_i)'(y),o_{i,y})$ does not depend on the local section structure. The second layer is geometry: for y in the support the operators are onto, $o_{i,y}$ orients $\ker(f-s_i)'(y)$, these orientations vary continuously along the branch $M_i=\{f=s_i\}$, and $\hat T_{(\Theta,o)}(y)=\sum\sigma_i\cdot T_y(M_i,o_i)$ is an orientation of Θ in Chapter 9's sense. The formalization proves the first layer completely, at a chart point, and the identification of the kernel orientation; the second layer is gap G.159 (§XV.38).

The setting: chart points and $\mathrm{Lin}(f,y)$**.** Part I's orientation theory of sc-Fredholm sections (Chapter 6) is formalized at a *chart point*: a strong bundle chart c around x together with a level-1 point z of the model over $\varphi(x)$ Chapter6/LinSet · ChartPoint. There T_xX is read as $T_zO=\mathrm{im}\,Dr(z)$ and the fibre W_x as $K_z=\mathrm{im}\,\rho(z)$. Part I's $\mathrm{Lin}(f,x)$ (Definition 6.2.1) is the set of operators $T_zO\to K_z$ of the form $(f+s)'(x)+a$, with s an sc⁺-section through $-f(x)$ and a an sc⁺-operator Chapter6/LinSet · ChartPoint.linSet; for one fixed s it is $(f+s)'(x)+\{\text{sc⁺-operators}\}$ Chapter6/LinSet · ChartPoint.mem_linSet_iff, it is convex Chapter6/LinSet · ChartPoint.convex_linSet, and at a smooth point it is not empty when sc⁺-sections pass through every smooth point of the fibre Chapter6/LinSet · ChartPoint.linSet_nonempty, the standing hypothesis of §12.5. Its members are Fredholm Chapter6/LinOrientation · ChartPoint.isFredholm_of_mem_linSet, so the determinant lines form a line bundle $\mathrm{DET}(f,x)$ over $\mathrm{Lin}(f,x)$ Chapter6/LinOrientation · ChartPoint.detLin, and Or_x^f is the set of its orientations (Definition 6.6.2) Chapter6/LinOrientation · ChartPoint.Or. Since $\mathrm{Lin}(f,x)$

is convex, hence contractible, an orientation is determined by its value at any one member: $o \mapsto o(L)$ is a bijection of Or_x^f with the two rays of $\det L$ Chapter6/LinOrientation · ChartPoint.orEquivRay, because two orientations of a line bundle over a preconnected base that agree on one fibre agree everywhere Chapter6/LineBundleOrientation · LineOrientation.eq_of_apply_eq.

Oriented formal sums (restated). At a chart point p, an *oriented member of* $\mathrm{Lin}(f,x)$ is a pair (L,r) with $L \in \mathrm{Lin}(f,x)$ and r a ray of $\det L$ Chapter15/OrientedLinearizations · ChartPoint.OrLin. The text's $\widehat{\mathrm{Gr}}_F(x)$ (p. 701) is the set of finite formal sums of oriented operators with weights in $\mathbb{Q}^+$, read here as finitely supported functions on oriented members Chapter15/OrientedLinearizations · ChartPoint.GrFHat, as $\mathrm{GrF}(x)$ was read in §XV.1. The forgetful map f_F sends $\sum \sigma \cdot (L,r)$ to $\sum \sigma \cdot L$ Chapter15/OrientedLinearizations · ChartPoint.forgetOr. *The reading*: the text's oriented operators are all sc-Fredholm operators $T_xX \to W_x$; here they are the members of $\mathrm{Lin}(f,x)$, which are exactly the operators that o can orient, and all operators that occur in $T(f,\Lambda)(x)$ are of this kind (below).

f_F is 2–1 Chapter15/OrientedLinearizations · ChartPoint.exists_two_orLin: over every $L \in \mathrm{Lin}(f,x)$ there are exactly two oriented members. *Proof.* $\det L$ is a real line, since L is Fredholm Chapter6/FredholmDet · finrank_fredholmDet, and a real line has exactly two rays, a nonzero vector v and $-v$ spanning them Chapter15/OrientedLinearizations · ChartPoint.exists_two_rays_of_finrank_eq_one. Two oriented members over the same L are equal iff their rays are.

The canonical lift of a formal sum. For $o \in \mathrm{Or}_x^f$ and $L \in \mathrm{Lin}(f,x)$, $(L, o(L))$ is the oriented member L inherits from o Chapter15/OrientedLinearizations · ChartPoint.orLin, and $L \mapsto (L,o(L))$ is injective Chapter15/OrientedLinearizations · ChartPoint.orLin_injective. The canonical lift of a formal sum $w = \sum \sigma_L \cdot L$ of members of $\mathrm{Lin}(f,x)$ is its push-forward along this map, $\sum \sigma_L \cdot (L, o(L))$ Chapter15/OrientedLinearizations · ChartPoint.canonLift. Four facts follow, each a short computation with push-forwards of finitely supported functions.

- *f_F undoes the lift*: $f_F(\mathrm{lift}_o(w)) = w$, read as a sum of operators Chapter15/OrientedLinearizations · ChartPoint.forgetOr_canonLift, since $f_F \circ (L \mapsto (L,o(L)))$ is the inclusion of $\mathrm{Lin}(f,x)$. The lift is additive and commutes with finite sums and single terms Chapter15/OrientedLinearizations · ChartPoint.canonLift_add Chapter15/OrientedLinearizations · ChartPoint.canonLift_sum Chapter15/OrientedLinearizations · ChartPoint.canonLift_single.
- *Every summand carries the orientation induced by o* Chapter15/OrientedLinearizations · ChartPoint.snd_eq_of_mem_support_canonLift: the support of a push-forward lies in the image of the support.
- *Uniqueness* Chapter15/OrientedLinearizations · ChartPoint.canonLift_unique: a formal sum W of oriented members whose image under f_F is w and all of whose summands carry the orientation induced by o equals $\mathrm{lift}_o(w)$. *Proof.* On the support of W the identity agrees with $q \mapsto (q_1, o(q_1))$, so W is the push-forward of itself along that map; this push-forward factors as the push-forward along the first projection, which is w, followed by the lift.
- *The lift sees o* Chapter15/OrientedLinearizations · ChartPoint.canonLift_injective_o: two orientations o, o' that give the same lift of a nonzero sum w are equal. *Proof.* Take L in the support of w. Then $(L,o(L))$ lies in the support of the lift, hence of the lift by o', so by the previous point it is $(L,o'(L))$, and $o(L) = o'(L)$. Since $\mathrm{Lin}(f,x)$ is convex, hence preconnected, two orientations of $\mathrm{DET}(f,x)$ that agree at L agree everywhere (Part I, as above).

Since operators of $\mathrm{Lin}(f,x)$ are determined by their values, formal sums of members with the same image as sums of operators are equal Chapter15/OrientedLinearizations · ChartPoint.mapDomain_val_injective, and so have the same lift Chapter15/OrientedLinearizations · ChartPoint.canonLift_eq_of_mapDomain_eq.

The kernel orientation of an onto member. The text identifies $\Lambda^{\max}\ker L$ with $\det L = \Lambda^{\max}\ker L \otimes (\Lambda^{\max}\operatorname{coker} L)^*$ when L is onto (p. 702). Formally, for $L \in \mathrm{Lin}(f,x)$ onto and $N \subset E_0$ the image of $\ker L$ under the inclusion $T_zO \subset E_0$, the ray $o(L)$ is carried to a ray of $\Lambda^{\max}N$ `Chapter15/OrientedLinearizations · ChartPoint.orKer` by Part I's map $\det L \cong \Lambda^{\max}\ker L \cong \Lambda^{\max}N$ `Chapter6/SolutionOrientation · surjRay`, built from the determinant of an onto operator (the cokernel being $\{0\}$) and the isomorphism $\ker L \cong N$ `Chapter6/SolutionOrientation · kerEquivOfMap`. Composed with $o \mapsto o(L)$ this is a bijection $\mathrm{Or}^f_x \cong \mathrm{Ray}(\Lambda^{\max}N)$ `Chapter15/OrientedLinearizations · ChartPoint.orKerEquiv Chapter15/OrientedLinearizations · ChartPoint.orKerEquiv_apply`: the orientation of f at x and the induced orientation of the kernel determine each other. The kernel is finite-dimensional `Chapter15/OrientedLinearizations · ChartPoint.finiteDimensional_of_mem_linSet`, and the ray can be read as an orientation of the subspace N in Mathlib's sense, through Part I's pull-back of rays to orientations `Chapter6/SolutionSetOrientation · pullOr Chapter15/OrientedLinearizations · ChartPoint.orKerOrientation`: a basis $b_1,\dots,b_k$ of N is positive iff $b_1 \wedge \cdots \wedge b_k$ lies in the ray `Chapter15/OrientedLinearizations · ChartPoint.orientation_eq_orKerOrientation_iff`. This orientation still determines o `Chapter15/OrientedLinearizations · ChartPoint.orKerOrientation_injective`. *Proof.* Fix a basis b of N; its orientation is either the orientation from o or its negative. In the first case the characterization above says that the ray from o is the ray of $b_1 \wedge \cdots \wedge b_k$, and the same holds for o', whose orientation is the same by hypothesis. In the second case the ray from o is not that of the wedge (the characterization would make the orientations equal, while an orientation is never its negative), so, $\Lambda^{\max}N$ being a line, it is the opposite ray `Chapter6/SolutionSetOrientation · ray_eq_or_eq_neg_of_finrank_eq_one`; again the same for o'. In both cases the two rays agree, and the bijection $\mathrm{Or}^f_x \cong \mathrm{Ray}(\Lambda^{\max}N)$ gives $o = o'$.

The operators $(f-s)'(y)$ as members of $\mathrm{Lin}(f,y)$. The bridge between §15.1 and Part I is the chart point at which the linearization of §§XV.7–XV.8 is read: for a chart c around $y \in X_1$, the point z is the level-1 lift of $\varphi(y)$ `Chapter15/CanonicalLift · SBLocalization.liftPt`. Let s be an sc$^+$-section on an open O containing the domain of c, with $f(y) = s(y)$. Then $\hat f(\varphi(y)) = \hat s(\varphi(y))$ `Chapter15/CanonicalLift · SBLocalization.principal_eq_of_eq`, and the derivative $D((\hat f - \hat s)\circ r)(z)$ of §XV.10 takes values in K_z `Chapter15/CanonicalLift · SBLocalization.subDeriv_mem_K`: the principal part of the section $f-s$ maps $r(U)$ into the fibres K of the retract, and at a zero the derivative of such a map lies in K_z (Part I's lemma on derivatives of sections of a strong bundle retract). Restricted to T_zO it is an operator $(f-s)'(y) : T_zO \to K_z$ `Chapter15/CanonicalLift · SBLocalization.subOp Chapter15/CanonicalLift · SBLocalization.coe_subOp`; composed with $K_z \subset F$ `Chapter15/CanonicalLift · SBLocalization.toF`, an injective operation `Chapter15/CanonicalLift · SBLocalization.toF_injective`, it is the operator of the chart sum of Theorem 15.1.1 `Chapter15/CanonicalLift · SBLocalization.toF_subOp`.

Lemma (extra) `Chapter15/CanonicalLift · SBLocalization.subOp_mem_linSet`: at a smooth y, under §12.5's standing hypothesis, $(f-s)'(y) \in \mathrm{Lin}(f,y)$. This is what lets o_y orient every summand of $T(f,\Lambda)(y)$; the text takes it from §12.5 (p. 701, "see Section 12.5"). *Proof.* By the standing hypothesis there is an sc$^+$-section s_0 with $(f+s_0)(y) = 0$ `Chapter6/LinSet · ChartPoint.linSet_nonempty`, and by Part I's description for one section it suffices to write $(f-s)'(y) = (f+s_0)'(y) + a$ with a an sc$^+$-operator. In the chart, both s and s_0 are sc$^+$, so their principal parts lift to the shifted fibre F^1, and these lifts composed with r are sc-smooth into the shift `Chapter3/IndexLocallyConstantLocal · StrongBundle.HasScPlusSectionsThrough` (for s through Part I's chart lemma for local sc$^+$-sections, §XV.10). Put $h = (\hat s_1 + \hat s_{0,1})\circ r$. On U,

$$(f \hat{+} s_0)\circ r = (\hat f - \hat s)\circ r + \iota \circ h,$$

$\iota : F^1 \to F^0$ the inclusion, an identity checked pointwise from the principal part of a sum of sections

and the lifts. Differentiating at z gives $D((f \mathbin{\hat{+}} s_0)\circ r)(z) = D((\hat f - \hat s)\circ r)(z) + \iota \circ Dh(z)$. Here $Dh(z)$ is an sc-operator into the shifted space, because h is sc-smooth and z is smooth; its values composed with ι lie in K_z, being the difference of two derivatives with values in K_z. So $b = \iota \circ Dh(z)$ restricted to $T_zO \to K_z$ is an sc⁺-operator, by Part I's criterion for operators that factor through the shift `Chapter6/LinSet · ChartPoint.isScPlusOp_of_shift`, and $(f-s)'(y) = (f+s_0)'(y) + (-1)b$. The standing hypothesis holds whenever X has sc-smooth bump functions; it is the hypothesis Part XII carries for §12.5.

The formal sum with members of $\mathrm{Lin}(f,y)$**.** Let (s_i,σ_i) be a local section structure of Λ on an open O (Definition 13.2.1) and c a chart with domain in O. At a smooth y in the domain, $\sum_{i:s_i(y)=f(y)} \sigma_i \cdot (f-s_i)'(y)$ is a formal sum of members of $\mathrm{Lin}(f,y)$ `Chapter15/CanonicalLift · SBLocalization.linW`. Its image as a sum of operators $T_zO \to F$ is the chart sum $\sum_{\hat s_i = \hat f} \sigma_i(\hat f - \hat s_i)'(z)$ of Theorem 15.1.1 in a chart (§XV.7) `Chapter15/CanonicalLift · SBLocalization.mapDomain_toF_linW`. *Proof.* Term by term. The condition $s_i(y) = f(y)$ is equivalent to $\hat s_i(\varphi(y)) = \hat f(\varphi(y))$ `Chapter15/CanonicalLift · SBLocalization.eq_iff_principal_eq`, because the chart is injective on the fibre over y and maps $s_i(y)$, $f(y)$ to $(\varphi(y), \hat s_i(\varphi(y)))$, $(\varphi(y), \hat f(\varphi(y)))$; where it holds, the operator of the term is that of the chart sum (above), and where it fails both terms are 0.

Theorem 15.4.1 (narrower) `Chapter15/CanonicalLift · SBLocalization.theorem_15_4_1_lift`. Let f be an sc-Fredholm section, (s_i,σ_i) a local sc⁺-section structure of Λ on an open O, c a chart with domain in O, y a smooth point of the domain, and $o \in \mathrm{Or}^f_y$ read at the chart point of y. Then $\hat T = \mathrm{lift}_o(\sum_{s_i(y)=f(y)} \sigma_i \cdot (f-s_i)'(y))$ satisfies (1) $f_F(\hat T) = T(f,\Lambda)(y)$, the chart sum of Theorem 15.1.1; (2) every summand of $\hat T$ carries the orientation induced by o; and $\hat T$ is the only element of $\widehat{\mathrm{Gr}}_F(y)$ with (1) and (2). *Proof.* The general characterization `Chapter15/CanonicalLift · SBLocalization.canonLift_spec`: for any formal sum w of members of $\mathrm{Lin}(f,y)$ whose image as operators is a given T, the lift of w maps to T (by $f_F \circ \mathrm{lift}_o = \mathrm{id}$), its summands carry o, and any W with these two properties has $f_F(W) = w$, because the passage from formal sums of members to formal sums of operators $T_zO \to F$ is injective (inclusion of $\mathrm{Lin}(f,y)$, then the injective $L \mapsto \iota \circ L$), so the uniqueness of the lift applies. With w the formal sum above and T the chart sum, the hypothesis of the characterization is the previous paragraph. *The formula* is the text's, $\hat T = \sum_{f(y)=s_i(y)} \sigma_i \cdot ((f-s_i)'(y), o((f-s_i)'(y)))$ `Chapter15/CanonicalLift · SBLocalization.canonLift_linW_eq_sum`, by additivity of the lift on the finite sum over i.

Independence of the local structure `Chapter15/CanonicalLift · SBLocalization.theorem_15_4_1_indep`: two local section structures (s_i,σ_i) on O and (t_j,τ_j) on O', both containing the domain of c, give the same $\hat T$. This is the text's argument (p. 701), that the occurring operators do not depend on the choice. *Proof.* By Theorem 15.1.1 in a chart (§XV.7) the two chart sums agree at the point `Chapter15/Linearization · SBLocalization.linSumChart_eq`. Both are the images of the corresponding formal sums of members of $\mathrm{Lin}(f,y)$, and that passage is injective, so the two formal sums of members agree, and so do their lifts.

Definition 15.4.2 (narrower) `Chapter15/CanonicalLift · SBLocalization.canonicalLift`: the canonical lift $\hat T_{((f,o),\Lambda)}(y) \in \widehat{\mathrm{Gr}}_F(y)$, read at the chart point. *The kernel orientations.* For an onto summand, surjectivity onto K_z is the chart surjectivity of §XV.10 `Chapter15/CanonicalLift · SBLocalization.surjective_subOp_iff` (onto K_z means: every k in the range of $\rho(z)$ is hit, and K_z is that range), and $o_{i,y}$ is the orientation of $\ker(f-s_i)'(y)$ induced by o_y `Chapter15/CanonicalLift · SBLocalization.branchOr`, the kernel-orientation construction above applied to the member $(f-s_i)'(y)$. It determines o_y `Chapter15/CanonicalLift · SBLocalization.branchOr_injective`, by the bijection $\mathrm{Or}^f_y \cong \mathrm{Ray}(\Lambda^{\max} N)$.

Narrower. Theorem 15.4.1 lifts $T\Theta$, a section functor of oriented tangent planes of Θ, to $\widehat{\mathrm{Gr}}(X)$ of Chapter 9. What is proved is the pointwise layer, in $\widehat{\mathrm{Gr}}_F(y)$ at a chart point: the lift of $T(f,\Lambda)(y)$, its formula, its uniqueness, its independence of the local structure, and the kernel orientations $o_{i,y}$. Not formalized

(gap G.159, §XV.38): that the tangent space of M_i at y is the kernel of $(f - s_i)'(y)$, the continuity of $y \mapsto o_{i,y}$ along M_i as an orientation of a submanifold in Chapter 9's sense, the naturality under morphisms ($\varphi_*(L, o) = (\varphi_* L, \varphi_* o)$ in the text's category $\widehat{\mathrm{Gr}}_F(X)$), and the assembly into an orientation of Θ. So Theorem 15.4.3 and Corollary 15.4.4 below take the orientation of Θ as data. *Hypotheses.* §12.5's standing hypothesis (sc⁺-sections through smooth points), which bump functions give; and y smooth, where $\mathrm{Lin}(f, y)$ is defined. Neither is new: the text works at smooth points and refers to §12.5 for the orientations. The chart-point reading of $\widehat{\mathrm{Gr}}_F$ is the reading of GrF in §XV.1, with the operators of $\mathrm{Lin}(f, y)$.

XV.35 Theorem 15.4.3 (2017: 15.7): branched, compact, Stokes

The theorem (pp. 702–703) has two parts: under (1) $N(\Lambda) \leq 1$, (2) $T(f, \Lambda)$ onto on the support and (3) good position at the support points in ∂X, $\Theta = \Lambda \circ f$ is a compact, tame, branched ep⁺-subgroupoid, oriented by o; and Stokes' theorem holds for it, with the effective version under the local unique continuation property. The first part is a statement of §§15.2–15.3, proved here from them. The second is Part IX's Stokes theorem applied to Θ, once Θ carries an orientation by oriented tame structures.

The local branching at a point of good position `Chapter15/PerturbedStokesInvariants · SBLocalization.localBranching_of_isGoodPositionAt`. If (f, Λ) is in good position at y (Definition 15.2.4, §XV.15) and f has tame germ data, then $z \mapsto \Lambda(f(z))$ has a local branching structure at y in the sense of Definition 9.1.2 (2)–(4) (Part IX). *Proof.* Theorem 15.2.5 at y (§XV.16) `Chapter15/BoundaryPerturbation · SBLocalization.theorem_15_2_5_hyp` gives a natural representation $U(y)$, a local section structure $(s_i, \sigma_i)_{i \leq k}$ and the sets $M_i = \{z \in U(y) : f(z) = s_i(z)\}$, each a smooth submanifold properly embedded in $U(y)$, the count $\Lambda(f(z)) = \sum_{f(z) = s_i(z)} \sigma_i$ on $U(y)$, and the clause that only the branches through $f(y)$ meet f in $U(y)$. The branching structure is indexed by the branches through $f(y)$, $\{i : s_i(y) = f(y)\}$, with the sheets M_i and the weights σ_i; each sheet contains y, is a submanifold and properly embedded. The count is a chain of four rewritings: the count formula as a sum over all i of σ_i or 0; restriction of the sum to the branches through $f(y)$, the others contributing 0 at points of $U(y)$ by the last clause; and two changes of the index type (a subtype, then its lift to the universe of X) which do not change the sum.

Branched and compact `Chapter15/PerturbedStokesInvariants · SBLocalization.theorem_15_4_3_isBranched_hyp` (narrower). Let f be an sc-Fredholm section functor with tame germ data, (N, U) control the compactness of f (Definition 15.3.1, §XV.17), and let Λ be an sc⁺-multisection with domain support in U, $N(\Lambda) \leq 1$, transversal (Definition 15.2.1, §XV.12), and in good position at every support point in ∂X. Then $\Theta = \Lambda \circ f$, read with values in $\mathbb{Q}$, is a branched ep⁺-subgroupoid with compact support. *Proof.* The compactness of $|\operatorname{supp} \Theta|$ is that of the "in particular" of Theorem 15.3.7 (§XV.21) `Chapter15/GeneralPositionBranched · SBLocalization.isCompact_orbitSupp_comp`: the support lies in the sublevel set $\{x \in U : N(f(x)) \leq 1\}$, whose orbit set has compact closure, and is closed. Compactness gives the paracompactness of the orbit space of the support, Definition 9.1.2 (1). Θ is nonnegative, a functor (because f and Λ are) `Chapter15/GeneralPositionBranched · SBLocalization.isFunctorial_comp`, and its support consists of smooth points under the tame germ data `Chapter15/Transversality · SBLocalization.mem_smoothSet_of_pos`. The local branching at a support point z is a case split. If $z \in \partial X$, (3) and the previous paragraph. If not, $d_X(z) = 0$, and Remark 15.2.3 at an interior point of transversality (§XV.14) `Chapter15/Transversality · SBLocalization.remark_15_2_3_hyp` gives the structure. *Readings.* The text's (2) is transversality in the sense of Definition 15.2.1 (the operators of $T(f, \Lambda)$ are onto on the support); its (3), "in good position to ∂X", is Definition 15.2.4; "tame" for Θ is the tame form of §XV.21. The tameness of X, which the text assumes, is not used: good position carries what is needed near the boundary points.

Narrower only through the tame germ data of f, inherited from §15.2 (§XV.13).

Theorem 15.4.3 (narrower) `Chapter15/PerturbedStokesInvariants · SBLocalization.theorem_15_4_3_hyp`. Under the hypotheses above, together with sc-smooth partitions of unity on X (Definition 7.5.5) and an orientation $\hat{T}$ of Θ (Definition 9.3.x of Part IX: a lift of $T\Theta$ to oriented tangent planes) represented at every support point by an oriented tame structure of dimension $m+1$, $m \geq 1$: Θ is a compact branched ep⁺-subgroupoid, and for every $\omega \in \Omega^m_{\mathrm{ep},\infty}(X)$

$$\int_{(\Theta,\hat{T})} d\omega = \mu^{\partial\Theta}_{\omega}(\partial S),$$

the boundary measure of Theorem 9.5.15 of the whole boundary $\partial S = |\operatorname{supp} \partial\Theta|$, which is the text's $\int_{\partial(\Theta,o)} \omega$. *Proof.* The first part is the lemma above; the identity is Part IX's Theorem 9.5.17 `Chapter9/StokesTextHypotheses · EP.theorem_9_5_17`, whose hypotheses are exactly these: partitions of unity on X, Θ compact and branched, oriented, and tame of dimension $m+1$. The boundary measure `Chapter9/StokesTextHypotheses · EP.boundaryMeasure` is defined on the σ-algebra $L(\partial S)$ of Definition 9.5.14 over the orbit space of the support of $\partial\Theta$ (Definition 9.2.16) `Chapter9/Boundary · EpGroupoid.boundaryTheta`.

The effective version `Chapter15/PerturbedStokesInvariants · SBLocalization.theorem_15_4_3_eff_hyp`: if moreover X has the local unique continuation property (Definition 7.2.18) `Chapter7/ReducedCriterion · EpGroupoid.IsLUC`, then $\oint_{(\Theta,\hat{T})} d\omega$ is the effective boundary measure of ∂S (Theorem 9.5.16), by Part IX's Theorem 9.5.18 `Chapter9/StokesTextHypotheses · EP.theorem_9_5_18`; the local structures then have the property needed for the effective integral `Chapter9/StokesTextHypotheses · EP.hloc_hasLUC`. *Dimension* 1 `Chapter15/PerturbedStokesInvariants · SBLocalization.theorem_15_4_3_dim1_hyp`: for Θ of dimension 1 and ω a function, $\int_{\Theta} d\omega$ is the signed count over the finitely many oriented boundary points `Chapter9/StokesDimOne · EP.boundaryBranchedMeasureDim1`, Part IX's Stokes theorem in dimension 1 `Chapter9/StokesDimOne · EP.theorem_9_5_17_dim1`. Together these cover the text's "every sc-differential form of degree $k-1$" for every index $k \geq 1$, $k = m+1$.

Narrower. (a) The orientation is data: the text takes the canonical orientation of Theorem 15.4.1, and with it the induced orientation of $\partial\Theta$ (Definition 9.3.10). Theorem 15.4.1 is formalized only pointwise (§XV.34, gap G.159), so the orientation $\hat{T}$ and its oriented tame structures are hypotheses; "tame" and the dimension $m+1=k$ are carried by these structures, which is how Part IX's Theorem 9.5.17 reads its hypotheses. (b) Sc-smooth partitions of unity on X, a hypothesis of Theorem 9.5.17, inherited; the text's Theorem 15.4.3 does not state it, and Part IX has witnesses for it. (c) The tame germ data of f, inherited (§XV.13). The Fredholm index k of the text enters only through the dimension of the oriented tame structures.

XV.36 Corollary 15.4.4 (2017: 15.2): the invariant and the cobordism argument

The corollary (p. 703) asserts, for $d_X \equiv 0$, paracompact $|X|$, compact $|S_f|$ and an orientation o of f, a well-defined map $\Phi_{(f,o)} : H^*_{\mathrm{dR}}(X) \to \mathbb{R}$ with $\Phi_{(f,o)}([\omega]) = \int_{(\Lambda\circ f,o)} \omega$ for any regular perturbation (N,U,Λ), $\Lambda \circ f$ carrying the canonical orientation; and the effective version under the local unique continuation property. Its proof is left to the reader. Read degree by degree, "well defined" says two things: the integral of an exact form over $(\Lambda \circ f, o)$ vanishes, so the integral descends to cohomology; and a closed form has the same integral over any two regular perturbations. The first is Stokes with

$\partial\Theta = 0$. The second is the cobordism argument: two regular perturbations are joined by a compact oriented cobordism $\tilde{\Theta}$ over $[0,1] \times X$ (Corollary 15.3.10 (3)), and Stokes on $\tilde{\Theta}$ for the pull-back of a closed form. The formalization proves both steps for an abstract cobordism (this subsection) and then builds the cobordism on the cylinder in the text's order (§XV.37).

Oriented regular perturbations. An oriented regular perturbation of dimension $m+1$ `Chapter15/PerturbedStokesInvariants · SBLocalization.OrientedRegularPerturbation` is a regular perturbation (N, U, Λ) of f (Corollary 15.3.10 (1), §XV.31: (N, U) controls compactness, Λ structurable, domain support in U, $N(\Lambda) < 1$, transversal) `Chapter15/RegularPerturbations · SBLocalization.IsRegularPerturbation`, with $\Theta = \Lambda \circ f$ compact and branched, and an orientation $\hat{T}$ of Θ represented by oriented tame structures of dimension $m+1$, the data that Theorem 15.4.1 would provide. Its integral is $\int_{(\Theta,\hat{T})} \omega$ for $\omega \in \Omega^{m+1}_{\mathrm{ep},\infty}(X)$ `Chapter15/PerturbedStokesInvariants · SBLocalization.OrientedRegularPerturbation.integral`, the branched integral of Definition 9.5.6 (1) `Chapter9/TameOpen · EpGroupoid.FormEval.tameBranchedIntegral`.

The statement (narrower) `Chapter15/PerturbedStokesInvariants · SBLocalization.corollary_15_4_4_statement`. For a predicate Can on oriented regular perturbations, standing for "the orientation is the canonical one from (f, o)": (i) for every P with $\mathrm{Can}(P)$ and every $\eta \in \Omega^m_{\mathrm{ep},\infty}(X)$, $\int_P d\eta = 0$; (ii) for any P_0, P_1 with Can and every closed $\omega \in \Omega^{m+1}_{\mathrm{ep},\infty}(X)$, $\int_{P_0} \omega = \int_{P_1} \omega$. Then $\Phi_{(f,o)}([\omega]) = \int_P \omega$ is well defined on H^{m+1}_{dR}, in the degree $m+1 = \dim\Theta$; on the other degrees the text's Φ is 0 by the degree of the integrand, which the formal statement does not spell out. Can is a parameter because the canonical orientation is formalized only pointwise (§XV.34).

Exact forms. *The integral of the zero form vanishes* `Chapter15/PerturbedStokesInvariants · SBLocalization.tameBranchedIntegral_zero`, by linearity of the sheet integrals. *If $\partial X \cap \operatorname{supp}\Theta = \emptyset$, every signed measure on ∂S gives 0 to ∂S* `Chapter15/PerturbedStokesInvariants · SBLocalization.signedMeasure_univ_eq_zero_of_boundary`: $\operatorname{supp}\partial\Theta \subset \partial X \cap \operatorname{supp}\Theta$ `Chapter9/Boundary · EpGroupoid.supp_boundaryTheta_subset`, so the orbit set of $\operatorname{supp}\partial\Theta$ is empty and its whole space is the empty set; in particular the boundary measure vanishes `Chapter15/PerturbedStokesInvariants · SBLocalization.boundaryMeasure_univ_eq_zero`. **Lemma** `Chapter15/PerturbedStokesInvariants · SBLocalization.integral_d_eq_zero_of_degIndex_eq_zero`: if $d_X \equiv 0$ and X has sc-smooth partitions of unity, then $\int_{(\Theta,\hat{T})} d\eta = 0$ for every compact, branched, oriented, tame Θ of dimension $m+1 \geq 2$. *Proof.* Theorem 9.5.17 turns the left side into the boundary measure of ∂S; $\partial X = \{d_X \geq 1\}$ is empty, so the previous fact applies. In dimension 1 the same with the dimension-1 Stokes theorem `Chapter15/PerturbedStokesInvariants · SBLocalization.integral_d_eq_zero_of_degIndex_eq_zero_dim1`.

The cobordism argument `Chapter15/PerturbedStokesInvariants · SBLocalization.integral_eq_of_cobordism`. Let $F : Y \to X$ be an sc-smooth functor between ep-groupoids (in the text, the projection $[0,1] \times X \to X$), Y with sc-smooth partitions of unity, $\tilde{\Theta}$ a compact, branched, oriented, tame ep$^+$-subgroupoid of Y of dimension $m+2$, and I_0, I_1 two functions on $\Omega^{m+1}_{\mathrm{ep},\infty}(X)$ such that for every ω

$$\mu^{\partial\tilde{\Theta}}_{F^*\omega}(\partial\tilde{S}) = I_1(\omega) - I_0(\omega).$$

Then $I_0(\omega) = I_1(\omega)$ for every closed ω. *Proof.* Stokes (Theorem 9.5.17) on Y for the form $F^*\omega$ of degree $m+1$: $\int_{\tilde{\Theta}} dF^*\omega = \mu^{\partial\tilde{\Theta}}_{F^*\omega}(\partial\tilde{S})$. The exterior derivative commutes with the pull-back along a functor, $dF^* = F^*d$ on $\Omega^*_{\mathrm{ep},\infty}$ `Chapter8/EpFormsFunctor · EP.epFormsInftyD_comp_epFormsInftyPullback` (Theorem 11.2.1 (2), the functor case, Part XI; on the direct limit it is checked level by level from the same identity for forms of a fixed level). So $dF^*\omega = F^*d\omega = 0$, the left side is the integral of the zero form, 0, and

$I_1(\omega) - I_0(\omega) = 0$. The text's form, with $\partial\tilde{\Theta} = \Theta_1 \sqcup -\Theta_0$, is the case $I_i = \int_{\Theta_i}$; the lemma separates the analytic content (Stokes and $dF^* = F^*d$) from the identification of the boundary, which is §XV.37.

Cobordant perturbations `Chapter15/PerturbedStokesInvariants · SBLocalization.OrientedRegularPerturbation.HasOrientedCobordism`. P_0 and P_1 are joined by an oriented compact cobordism if there are Y, F, $\tilde{\Theta}$ as in the lemma with $\mu^{\partial\tilde{\Theta}}_{F^*\omega}(\partial\tilde{S}) = \int_{P_1}\omega - \int_{P_0}\omega$ for all ω. **Corollary 15.4.4** (narrower) `Chapter15/PerturbedStokesInvariants · SBLocalization.corollary_15_4_4_hyp`: for $d_X \equiv 0$ and X with sc-smooth partitions of unity, in every dimension $m+1 \geq 1$, the statement holds for every Can whose members are pairwise cobordant. *Proof.* (i) is the exact-form lemma, split by $m = 0$ (dimension 1) or $m \geq 1$. (ii): the cobordism between P_0 and P_1 and the cobordism argument with $I_i = \int_{P_i}$.

The added hypothesis, that any two perturbations of the class are cobordant, is not derived here; on the cylinder it is reduced to the oriented end identities (§XV.37). It is satisfiable with a nonempty class: every oriented regular perturbation is cobordant to itself `Chapter15/PerturbedStokesInvariants · SBLocalization.OrientedRegularPerturbation.hasOrientedCobordism_refl`, with $Y = X$, F the identity and $\tilde{\Theta} = 0$, which is branched `Chapter9/Basic · EpGroupoid.isBranchedEpSubgroupoid_zero`, compact (its support is empty) and oriented by the zero lift `Chapter9/OrientationBasic · EpGroupoid.isOrientation_zero`, with vanishing boundary measure. This witness is **weak**: it shows that the hypothesis can hold, not that it holds for two different perturbations; no oriented regular perturbation with nonempty support exists in the project. *Further hypotheses*: partitions of unity on X, for Stokes; the paracompactness of $|X|$, which the corollary assumes, is not used in this form (it enters through the cylinder, §XV.37).

XV.37 Corollary 15.4.4 on the cylinder: the ends, the boundary measure, the effective version; Remark 15.4.5

The text's argument, "in view of Corollary 15.3.10", has four steps: (a) two regular perturbations (N^i, U^i, Λ^i) of f with $\Theta_i = \Lambda^i \circ f$; (b) Theorem 15.3.9 on $[0,1] \times X$ gives $\tilde{\Lambda}$ with $\tilde{\Lambda} = \Lambda^i$ over $\{i\} \times X$, and $\tilde{\Theta} = \tilde{\Lambda} \circ \tilde{f}$ is a compact cobordism between Θ_0 and Θ_1 (Corollary 15.3.10 (3)); (c) $\tilde{\Theta}$ carries the canonical orientation from $(\tilde{f}, \tilde{o})$, and with the boundary orientation of Definition 9.3.10, $\partial\tilde{\Theta} = \Theta_1 \sqcup -\Theta_0$; (d) Stokes on $\tilde{\Theta}$ (§XV.36). The formalization follows this order. Proved: (a), (b) with Theorem 15.3.9 in its charged form (§§XV.27–XV.33), the restriction of $\tilde{\Theta}$ to the ends as sets and as orbits, the splitting of the boundary measure over the two ends, and (d). Step (c), the oriented identification of each end, is a hypothesis (gap G.182).

Throughout, the cylinder is $[0,1] \times X$, the case $n = 1$ of $[0,1]^n \times X$ (§XV.30), with the pulled-back bundle and $\tilde{f}(t,x) = (t, f(x))$; $\tilde{\Lambda}$ is any function on the cylinder bundle with $\tilde{\Lambda}(w) = \Lambda^0(w_2)$ over $t = 0$ and $\tilde{\Lambda}(w) = \Lambda^1(w_2)$ over $t = 1$, w_2 the W-component.

$\tilde{\Theta}$ on the ends (same; V15.48). Over $t = 0$, $\tilde{\Lambda}(\tilde{f}(t,x)) = \Lambda^0(f(x))$ `Chapter15/CylinderEnds · SBLocalization.theta_cyl_of_time_zero`, and over $t = 1$, $\Lambda^1(f(x))$ `Chapter15/CylinderEnds · SBLocalization.theta_cyl_of_time_one`: the W-component of $\tilde{f}(t,x)$ is $f(x)$. Hence $\tilde{\Theta} \circ i_0 = \Theta_0$ and $\tilde{\Theta} \circ i_1 = \Theta_1$ for the inclusion functors $i_q : X \to [0,1] \times X$, $x \mapsto (q,x)$ (§XV.30) `Chapter15/CylinderEnds · SBLocalization.theta_cubeIncl_zero Chapter15/CylinderEnds · SBLocalization.theta_cubeIncl_one`, the text's "restricts to $\{i\} \times X = X$ as Θ_i" (p. 700). **The boundary part of the support** `Chapter15/CylinderEnds · SBLocalization.mem_boundary_inter_supp_cyl_iff`: for $d_X \equiv 0$, $(t,x) \in \partial([0,1] \times X) \cap \operatorname{supp}\tilde{\Theta}$ iff $t = 0$ and $x \in \operatorname{supp}\Theta_0$, or $t = 1$ and $x \in \operatorname{supp}\Theta_1$. *Proof.* $d_X \equiv 0$ makes X tame (§XV.30) `Chapter5/DegeneracyZeroTame · EpGroupoid.isTame_of_`

degIndex_eq_zero, and then $\partial([0,1]\times X)=\{0,1\}\times X$ Chapter15/RegularPerturbations · SBLocalization.mem_boundary_cubeSEP_iff; on each end the value of $\tilde\Theta$ is that of Θ_i, by the two identities above.

The ends on orbit spaces. The orbit space of the cylinder is $[0,1]\times|X|$ (the homeomorphism of §XV.30) Chapter15/Cylinder · EpGroupoid.cubeOrbitHomeomorph. Its time coordinate, $|(t,x)|\mapsto t$, is defined through the inverse of this homeomorphism Chapter15/CylinderEnds · EpGroupoid.cylOrbitTime, takes the value t on the orbit of (t,x) Chapter15/CylinderEnds · EpGroupoid.cylOrbitTime_π and is continuous Chapter15/CylinderEnds · EpGroupoid.continuous_cylOrbitTime. For Θ on the cylinder, the *end at time i* of $\partial S=|\operatorname{supp}\partial\Theta|$ is the set of its points of time i Chapter15/CylinderEnds · EpGroupoid.cylEnd. It is closed, as the preimage of a point under a continuous map Chapter15/CylinderEnds · EpGroupoid.isClosed_cylEnd, hence Borel, hence in the σ-algebra $L(\partial S)$ of Definition 9.5.14, which contains the Borel sets Chapter9/BoundaryNullSigmaAlgebra · EpGroupoid.borel_le_lebesgueSigmaBoundaryH Chapter15/CylinderEnds · EpGroupoid.measurableSet_cylEnd. The map $|(i,x)|\mapsto|x|$ from the end at time i to $|X|$ Chapter15/CylinderEnds · EpGroupoid.cylEndMap is $|x|$ on the orbit of (i,x) Chapter15/CylinderEnds · EpGroupoid.cylEndMap_π, continuous Chapter15/CylinderEnds · EpGroupoid.continuous_cylEndMap and injective Chapter15/CylinderEnds · EpGroupoid.injective_cylEndMap. *Proof of injectivity.* Two points of the end at time i come from objects (q,x), (q',x') of the support with the same time i, so $q=q'$ (the cube $[0,1]^1$ has one coordinate); if $|x|=|x'|$, the two objects are isomorphic in $[0,1]\times X$, because two objects of the product are isomorphic iff their cube coordinates agree and their X-components are isomorphic (§XV.30), and so the two orbits agree. For $d_X\equiv 0$ the end at time i of $|\operatorname{supp}\partial\tilde\Theta|$ maps into $|\operatorname{supp}\Theta_i|$ Chapter15/CylinderEnds · SBLocalization.cylEndMap_mem_orbitSupp_zero Chapter15/CylinderEnds · SBLocalization.cylEndMap_mem_orbitSupp_one: a point of $\operatorname{supp}\partial\tilde\Theta$ lies in $\partial([0,1]\times X)\cap\operatorname{supp}\tilde\Theta$ (Definition 9.2.16), so by the description of the boundary part of the support its X-component lies in $\operatorname{supp}\Theta_i$ for its time i; the other end is excluded because $0\neq 1$.

The boundary measure splits Chapter15/CylinderEnds · SBLocalization.boundaryMeasure_cyl_univ. For $d_X\equiv 0$ and any compact, branched, oriented, tame $\tilde\Theta$ of dimension $m+2$ on the cylinder, $\mu_\omega^{\partial\tilde\Theta}(\partial S)=\mu_\omega^{\partial\tilde\Theta}(\text{end }0)+\mu_\omega^{\partial\tilde\Theta}(\text{end }1)$; the same for the effective boundary measure Chapter15/CylinderEnds · SBLocalization.boundaryMeasureEff_cyl_univ. *Proof.* A general lemma Chapter15/CylinderEnds · EpGroupoid.signedMeasure_univ_eq_cylEnds: if every boundary object of the cylinder has time 0 or 1, then for every σ-algebra on ∂S containing the Borel sets and every signed measure μ on it, $\mu(\partial S)=\mu(\text{end }0)+\mu(\text{end }1)$. Every point of ∂S has time 0 or 1 Chapter15/CylinderEnds · EpGroupoid.cylOrbitTime_eq_zero_or_one, because it is the orbit of a point of $\operatorname{supp}\partial\Theta\subset\partial([0,1]\times X)$; so ∂S is the union of the two ends, which are disjoint and measurable, and μ is additive on them. For $d_X\equiv 0$ the boundary objects have time 0 or 1 Chapter15/CylinderEnds · SBLocalization.boundary_cyl_time, and the boundary measures of Theorems 9.5.15 and 9.5.16 live on $L(\partial S)$, which contains the Borel sets.

Step (c) as a proposition. *The end identities* Chapter15/CylinderCobordism · SBLocalization.IsCylinderBoundaryOf: for oriented regular perturbations P_0, P_1 of dimension $m+1$ and a compact, branched, oriented, tame $\tilde\Theta$ of dimension $m+2$ on the cylinder, for every $\omega\in\Omega^{m+1}_{\mathrm{ep},\infty}(X)$,

$$\mu^{\partial\tilde\Theta}_{pr^*\omega}(\text{end }1)=\int_{P_1}\omega,\qquad \mu^{\partial\tilde\Theta}_{pr^*\omega}(\text{end }0)=-\int_{P_0}\omega,$$

\$pr = \$ the projection functor to X (§XV.30) Chapter15/CubeEp · EpGroupoid.cubeProj. This is the text's "$\partial\tilde\Theta=\Theta_1\sqcup-\Theta_0$", read on the level of the boundary measure, which is what Stokes uses: with the outward normal first (Definition 9.3.10), the outward normal is $+\partial_t$ at $t=1$ and $-\partial_t$ at $t=0$, whence the sign. *The orientation statement* Chapter15/CylinderCobordism · SBLocalization.CylinderOrientationStatement: every

$\tilde\Lambda$ that step (b) can produce, that is, every regular perturbation of $\tilde f$ for the interpolated pair $(\tilde N, \tilde U)$ of Corollary 15.3.10 (2) (§XV.31) Chapter15/CubeAuxNorm · SEP.interpAuxNorm Chapter15/CubeAuxNorm · SEP.interpSet with $\tilde\Lambda = \Lambda^i$ over $t = i$ and $\tilde\Theta$ compact and branched, carries an orientation by oriented tame structures of dimension $m + 2$ with the end identities. In the text this orientation is the canonical one from $(\tilde f, \tilde o)$; the statement asks only for the properties of it that the argument uses.

The cobordism from the end identities Chapter15/CylinderCobordism · SBLocalization.hasOrientedCobordism_of_cylinder: for $d_X \equiv 0$ and the cylinder with sc-smooth partitions of unity, a $\tilde\Theta$ with the end identities joins P_0 and P_1 in the sense of §XV.36, with $Y = [0,1] \times X$ and $F = pr$. *Proof.* $\mu^{\partial\tilde\Theta}_{pr^*\omega}(\partial S)$ is the sum over the two ends, and the end identities make it $\int_{P_1}\omega - \int_{P_0}\omega$. *Partitions of unity on the cylinder* Chapter15/CylinderCobordism · SBLocalization.hasScPartitionsOfUnity_cyl: in the ep-groupoid sense of Definition 7.5.5, from Theorem 7.5.7 Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_5_7, given the paracompactness of the orbit space of the cylinder, which follows from that of $|X|$ Chapter15/Cylinder · EpGroupoid.hasParacompactOrbitSpace_cubeProd, and partitions of unity on the object M-polyfold of the cylinder, which follow from those of X (§XV.32) Chapter15/CylinderPartitions · SBLocalization.admitsScPartitionsOfUnity_cubeSEP.

Steps (b) and (c) together Chapter15/CylinderCobordism · SBLocalization.hasOrientedCobordism_cylinder_hyp: for $d_X \equiv 0$, $|X|$ paracompact, f sc-Fredholm with tame germ data and compact $|S_f|$, if the charged Theorem 15.3.9 holds on the cylinder, the cylinder bundle has tame strong bundle charts at every object, its object M-polyfold has sc-smooth partitions of unity, the glued boundary data of Λ^0, Λ^1 are structurable over the boundary and satisfy the face condition (§XV.32), and step (c) holds, then P_0 and P_1 are cobordant. *Proof.* Corollary 15.3.10 (3) in its charged form (§XV.33) Chapter15/ChargedExtensionCylinder · SBLocalization.corollary_15_3_10_3_charged produces $\tilde\Lambda$: a regular perturbation of $\tilde f$ for $(\tilde N, \tilde U)$, equal to Λ^i over $t = i$, with $\tilde\Theta$ compact and branched. Step (c) gives the orientation and the end identities, and the previous paragraph the cobordism.

Corollary 15.4.4 along the text's argument (narrower) Chapter15/CylinderCobordism · SBLocalization.corollary_15_4_4_cylinder_hyp: for $d_X \equiv 0$, $|X|$ paracompact, sc-smooth partitions of unity on X and on the object M-polyfold of the cylinder, f sc-Fredholm with tame germ data and compact $|S_f|$, and a class Can: if the charged Theorem 15.3.9 holds on the cylinder, the cylinder bundle has tame strong bundle charts, and for any two members of Can the glued data are structurable with the face condition and step (c) holds, then the statement of §XV.36 holds. *Proof.* Corollary 15.4.4 from cobordisms (§XV.36), the cobordisms from the previous paragraph. *With Theorem 15.3.9 on the cylinder reduced* Chapter15/CylinderCobordism · SBLocalization.corollary_15_4_4_cylinder_of_rel: the charged Theorem 15.3.9 on the cylinder follows from the relative statement of steps (c)–(d) there (§XV.29) Chapter15/ChargedExtensionFibres · SBLocalization.theorem_15_3_9_charged_hyp Chapter15/TransversalExtension · SBLocalization.theorem_15_3_7_relative_statement, so the corollary rests on that relative statement in place of the theorem.

Narrower. Formalized in the text's order: the two regular perturbations, the cylinder perturbation from the charged Theorem 15.3.9, its restriction to the ends, the splitting of the boundary measure, and Stokes with $dF^* = F^*d$. The hypotheses are: (1) step (c), the oriented end identities (gap G.182), which contain the weights on the ends ($\partial\tilde\Theta(i,x) = \Theta_i(x)$, gap G.190; proved here are the values of $\tilde\Theta$, the support and the injectivity of the end maps, not that the end maps are onto $|\operatorname{supp}\Theta_i|$ nor the weights of $\partial\tilde\Theta$) and the canonical orientation on the cylinder (Theorem 15.4.1, gap G.159), so Can stays a parameter; (2) the charged Theorem 15.3.9 on the cylinder, hence the relative local families there (gap G.178); (3) tame strong bundle charts on the cylinder (gap G.189); (4) the structurability of the glued data, proved given

that lifted sc⁺-sections are sc⁺ on the faces (gap G.193, §XV.32), and their face condition (gap G.180); (5) partitions of unity on X, used for Stokes and for those of the cylinder, which the corollary does not state. The paracompactness of $|X|$, which the corollary assumes, is used for the cylinder's partitions of unity. *Witnesses.* Step (c) has **no witness in the non-degenerate case**: the project has no oriented regular perturbation with nonempty support. The remaining hypotheses have the weak witnesses listed with §§XV.29, XV.32–XV.33 (on $\mathbb{R}^n \times \mathbb{R}^0 \to \mathbb{R}^n$, no boundary, zero fibres); tame strong bundle charts on the cylinder have none of their own.

The effective version (narrower; V15.49). *Definitions*: the effective integral $\oint_P \omega$ over an oriented regular perturbation, for X with the local unique continuation property (Definition 9.5.6 (2)) `Chapter15/CylinderCobordism · SBLocalization.OrientedRegularPerturbation.integralEff` `Chapter9/TameOpen · EpGroupoid.FormEval.tameBranchedIntegralEff`, and effective cobordisms, as in §XV.36 with Y having the local unique continuation property and the effective boundary measure `Chapter15/CylinderCobordism · SBLocalization.OrientedRegularPerturbation.HasOrientedCobordismEff`. *The statement* `Chapter15/CylinderCobordism · SBLocalization.corollary_15_4_4_eff_statement`: exact forms have effective integral 0 over every member of Can, and closed forms have the same effective integral over any two, so $\Phi^{\mathrm{eff}}_{(f,o)}([\omega]) = \oint_{(\Lambda\circ f,o)} \omega$ is well defined in degree $m+1$. *The lemmas*, as in §XV.36: the effective integral of the zero form vanishes `Chapter15/CylinderCobordism · SBLocalization.tameBranchedIntegralEff_zero`; the cobordism argument with Theorem 9.5.18 on Y in place of Theorem 9.5.17 `Chapter15/CylinderCobordism · SBLocalization.integralEff_eq_of_cobordism`; and exact forms `Chapter15/CylinderCobordism · SBLocalization.integralEff_d_eq_zero_of_degIndex_eq_zero`, for $d_X \equiv 0$, by Theorem 9.5.18 and, in dimension 1, its dimension-1 form `Chapter9/EffectiveDimOneLUC · EP.theorem_9_5_18_dim1_isLUC`, the boundary measure vanishing because $\partial X = \emptyset$. *Proof of the statement* `Chapter15/CylinderCobordism · SBLocalization.corollary_15_4_4_eff_hyp`, for $d_X \equiv 0$, partitions of unity and the local unique continuation property on X, and pairwise effective cobordisms: the two lemmas, in every dimension $m+1 \geq 1$. The cobordism hypothesis is satisfiable with a nonempty class `Chapter15/CylinderCobordism · SBLocalization.OrientedRegularPerturbation.hasOrientedCobordismEff_refl` (self-cobordism by $\tilde{\Theta} = 0$; **weak**, as in §XV.36).

On the cylinder: the effective end identities `Chapter15/CylinderCobordism · SBLocalization.IsCylinderBoundaryEffOf`, with the effective boundary measure and $\oint_{P_i}$, give an effective cobordism `Chapter15/CylinderCobordism · SBLocalization.hasOrientedCobordismEff_of_cylinder` through the splitting of the effective boundary measure; step (c) in effective form `Chapter15/CylinderCobordism · SBLocalization.CylinderOrientationEffStatement`; and the corollary along the text's argument `Chapter15/CylinderCobordism · SBLocalization.corollary_15_4_4_eff_cylinder_hyp`, from the charged Theorem 15.3.9 on the cylinder and the hypotheses of the non-effective form, with the local unique continuation property on X and on $[0,1]\times X$. The latter presumably follows from that on X, natural representations of the cylinder at (t,x) being products of those of X at x with the trivial groupoid of $[0,1]$; it is not derived (gap G.191) and has no witness.

Remark 15.4.5 (nothing to formalize). The remark mentions another route to invariants, triangulating the weighted solution set into a rational chain, and says it is not carried out; there is nothing to formalize.

XV.38 What is not formalized

The open gaps of the chapter, with what is proved of each, what is missing, and an estimate of the Lean it needs.

- **Gap G.159** (formalized in part): Theorem 15.4.1 and Definition 15.4.2 as an orientation of $\Theta = \Lambda \circ f$. Proved at single points in Part I's chart reading: $(f - s_i)'(y) \in \mathrm{Lin}(f, y)$, the existence, formula and uniqueness of the canonical lift, its independence of the local structure, and the kernel orientations $o_{i,y}$ (§XV.34). Missing: (a) the tangent spaces of the branches $M_i = \{f = s_i\}$ in good position are the kernels of $(f - s_i)'(y)$ in the chart, 150–250 lines; (b) the continuity of $y \mapsto o_{i,y}$ along M_i as an orientation of a submanifold in Chapter 9's sense, with a bridge from Part I's ray orientations to frames, 300–450; (c) naturality under morphisms, 150–250; (d) the assembly into an orientation of Θ with independent tangent sums, after which the class of canonically oriented perturbations that Corollary 15.4.4 takes as a parameter, and the orientation that Theorem 15.4.3 takes as data, are defined; 150–250. Together 750–1,200 lines.
- **Gap G.160** (formalized in part): the products remark after Definition 15.3.3 and Remark 15.3.4. Formalized: product ep-groupoids, the product strong bundle and the product over $X \times X'$ with $\mu \times \mu'$, products of sections and section functors, sc-smooth and sc$^+$, products of strong bundle maps, the level 1 of the product, and $\max(N, N')$ as an auxiliary norm (§XV.19). Missing: (c) $f \times f'$ sc-Fredholm with tame germ data, for which Part I has only the cube parameter case: products of filling germs, of basic germs after reordering coordinates, and of the regularizing property, 300–500 lines; (d) $\Lambda \times \Lambda'$ as an sc$^+$-multisection, from products of local section structures with weights $\sigma_i \sigma'_j$, its norm max for $N \times N'$, admissibility for $(N \times N', U \times U')$ and the domain-support inclusion of Remark 15.3.4, 150–300 lines. Together 450–800 lines.
- **Gap G.162** (closed for Theorem 15.3.7): what remains is Theorem 15.3.9 and Corollary 15.3.10 (3), that is, the gaps G.178–G.180, G.189 and G.193 below. Corollary 15.3.10 (1) is proved from Theorem 15.3.7 for paracompact $|X|$ (§XV.31), though the composite is not stated as one declaration.
- **Gap G.178**: Theorem 15.3.9, steps (c)–(d), the relative local families, the one proposition from which the charged Theorem 15.3.9 is proved (§XV.29). Proved: everything else in steps (c)–(d), the relative Sard and compactness steps included, and the trivial families where the support meets only ∂X. Missing: (a) seeds at interior support points: one family making all $(f - s_a^0)'(x)$ onto at once, by the bridge of §XV.24 applied to each $f - s_a^0$ and complements of the finitely many cokernels, then openness for each pair of branches, and a seed family with domain support in a small neighbourhood avoiding ∂X; (b) openness at boundary support points: the family vanishes near x, so the reduced surjectivity of (f, Λ_0) at x must persist near x, from Part I's openness with $\ell = 0$ and the passage from Part I's general position to reduced surjectivity, about 60–100 lines. Together 400–700 lines.
- **Gap G.179**: the face condition of Theorem 15.3.9 does not depend on the extension. It is asked for every sc$^+$ extension of the boundary data (§XV.27). Missing: (a) the restriction of $(f - s_i)'(x)$ to $T_x^R X$ depends only on s_i on the faces through x, a germ argument in a tame chart, about 80 lines; (b) two sc$^+$ extensions have local representations whose branches agree on the faces up to reindexing, a statement of the kind of Theorem 15.1.1 on faces, with crossing branches and the charts of boundary symmetric structures, 170–320 lines; (b) needs the non-empty index sets that charging gives. Together 250–400 lines.
- **Gap G.180**: Corollary 15.3.10 (3), the face condition of the glued boundary data on $\{0, 1\} \times X$. Their values, boundary domain support, norm and charging are proved, and their structurability given G.193 (§XV.32). Missing: restriction of sc$^+$ extensions to the slices $\{i\} \times X$, $T^R_{(i,x)}([0, 1] \times X) = \{0\} \times T_x X$ when $d_X \equiv 0$, and the transversality of (f, Λ^i); 150–250 lines after G.179 (b). Its only witness has zero fibres.
- **Gap G.182**: Corollary 15.4.4, the oriented end identities $\mu^{\partial\tilde{\Theta}}_{pr^*\omega}(\text{end } i) = \pm \int_{\Theta_i} \omega$ on the cylinder, assumed through the orientation statement of step (c) (§XV.37). Missing: (a) the weights on the

ends, gap G.190; (b) oriented tame structures of $\tilde{\Theta}$ at (i,x) whose boundary sheets are $\{i\}\times M_j$, with the weights and isotropy of Θ_i; (c) the boundary orientation of $\{i\}\times M_j$ inside $\tilde{M}_j$, outer normal first (Definition 9.3.10), equal to $(-1)^{i+1}$ times the canonical orientation of M_j, Part I's sign bookkeeping of Theorem 6.7.1 for the kernel of $(\tilde{f}-\tilde{s})'$ with ∂_t, which needs Theorem 15.4.1 on the cylinder (G.159); (d) the sheet integral of $pr^*\omega$ over $\{i\}\times M$ equals that of ω over M; (e) the transfer of measurability along the end maps and the uniqueness of Theorem 9.5.4 for the push-forward. 1,200–2,200 lines, plus (c) after G.159; a non-degenerate witness, an oriented regular perturbation with nonempty support, 300–600 lines more.

- **Gap G.189**: tame strong bundle charts on $[0,1]\times X$, a hypothesis of Corollaries 15.3.10 (3) and 15.4.4 on the cylinder (§§XV.33, XV.37), without a witness. The cylinder bundle is a pull-back whose atlas contains the charts over the identity of a cube-times-chart model; missing is that the products of a tame strong bundle chart of W with the tame charts of the cube are modelled on relatively open subsets of a partial quadrant. 80–250 lines.
- **Gap G.190**: the weights on the ends of the cylinder, $\partial\tilde{\Theta}(i,x)=\Theta_i(x)$. Proved: $\tilde{\Theta}(i,x)=\Theta_i(x)$, the boundary part of the support is the two ends, and each end maps injectively into $|\operatorname{supp}\Theta_i|$ (§XV.37). Missing: every local sheet $\tilde{M}_j$ through (i,x) has (i,x) on its boundary. $T\tilde{M}_j$ has dimension $k+1$, and its intersection with $T^R=\{0\}\times T_xX$ is the kernel of an onto operator of index k by the face condition, so $T\tilde{M}_j\not\subseteq T^R$ and Remark 9.2.14 applies; then the end map is onto. 300–500 lines.
- **Gap G.191**: the local unique continuation property on $[0,1]\times X$, a hypothesis of the effective cylinder forms of Corollary 15.4.4 (§XV.37): natural representations of the cylinder at (t,x) from those of X at x, as a product with the trivial groupoid of $[0,1]$. 80–150 lines.
- **Gap G.193**: lifted sc$^+$-sections are sc$^+$ on the faces of $[0,1]\times X$, the statement from which the glued data are structurable (§XV.32). Missing: a local form of Part I's lemma on sc$^+$-sections in charts, for charts whose domain is not inside the open set, 80–120 lines; the lift $(q,y)\mapsto((q,y),s(y))$ is sc$^+$ on $[0,1]\times U$, the level-$(0,1)$ analogue of Part I's pull-back of sc-smooth sections, 120–200; the face lemma of Chapter 14 (Part XIV) for an open set in place of X, 40–80. Together 250–400 lines. It has no witness: a witness would be the statement for all tame charts of the cube, which is the general proof.

Not formalized, and not needed. The functoriality of $\varphi\mapsto\varphi_*$ on GrF, $(\psi\varphi)_*=\psi_*\varphi_*$ and $1_*=\mathrm{id}$, and $\mathrm{GrF}(X)$ as a category over X, which the text uses without a numbered statement; about 60–100 lines from Chapter 7, and not used by the functor property (§XV.9). The independence of the sc-structures of T_zO and K_z from the chart, not used since the sc-Fredholm property holds in every chart (§XV.11). "Tame" in the conclusions of Theorem 15.3.9 and Corollary 15.3.10 (3): the formal conclusions say branched and compact; the perturbations built are in general position, so the tame form of the "in particular" of Theorem 15.3.7 (§XV.21) applies to them, but the combination is not stated.

Hypotheses without a non-degenerate witness. The relative local families of Theorem 15.3.9 and the charged theorem itself are witnessed only on $\mathbb{R}^n\times\mathbb{R}^0\to\mathbb{R}^n$, with no boundary and zero fibres (§XV.29); the face condition of the glued data only with zero fibres (§XV.33); the lifted sections on faces (G.193), the tame bundle charts of the cylinder (G.189) and the local unique continuation property on the cylinder (G.191) have none; the orientation statement on the cylinder (G.182) has none with nonempty support, and the cobordism hypothesis of Corollary 15.4.4 only the self-cobordism with $\tilde{\Theta}=0$ (§§XV.36, XV.37). Every other hypothesis of the Part has a witness, named in its subsection.

Prose. Remarks 15.2.6, 15.3.2, 15.3.11 and 15.4.5 have nothing to formalize; Remark 15.3.8 holds in the

form of §XV.26.

XV.39 What the formalization taught us

§15.1 is topology plus Part I's chain rule. The proof of Theorem 15.1.1 is a statement about finitely many continuous maps and their germs, and it holds in that generality: two finite families of branches with the same weighted count of every value at every point have the same formal sums of (value, germ invariant) pairs at every point, as soon as the germ invariants vary continuously (§XV.7). Lemmas 15.1.4, 15.1.5, Proposition 15.1.3 and Remark 15.1.6 are pure topology, and Lemma 15.1.5 needs only the observation that two continuous maps into a Hausdorff space differ on an open set, so it is proved pair by pair without Lemma 15.1.4 (§XV.5). Written this way, two places of the text needed a reading. The derivative of an sc¹ map is continuous only on level 1, and there pointwise; so the argument runs in the level-1 topology, where it holds equally, and the text's "weak topology" is pointwise convergence. And the final display of the text sums over all branches, as if every $s_i(0)$ equalled $f(0)$; comparing pairs (value, derivative) for all branches and selecting the value $f(0)$ at the end avoids this. Separating the two families together, rather than their union with halved weights, gives the comparison in one step, and it lets x_0 itself serve as a separated point. So Proposition 15.1.3, which asks for $x_k \neq 0$ converging to 0 and does not hold as we read it at an isolated point, is not needed. The step "without loss of generality $X = O$" is the independence of the intrinsic operator $(f - s)'(x)$ from the chart, which Part I's chain rule gives at a zero of $f - s$ made a global section (§XV.7). The functor property, for which the text gives no argument, is the same chain rule once more: a local structure near $s(\varphi)$ is carried to one near $t(\varphi)$ by the local strong bundle isomorphism of the sheet of φ, and in the carried chart $T\varphi$ is the identity on chart vectors, so the intrinsic operators are conjugate by $\mu(\varphi, \cdot)$ (§XV.9).

§15.2 is Part I branch by branch, and so inherits Part I's hypothesis. A local sc⁺-section on O is a global sc⁺-section of $W|_O$, and $(f - s_i)|_O$ is an sc-Fredholm section there, so Part I's Theorems 3.1.18, 3.1.26 and 3.6.8 apply to each branch (§XV.10). This gives the sc-Fredholm summands of $T(f, \Lambda)$ in the chart reading, the local solution sets, their tameness, and good position along them. The price is Part I's tame germ data, without which Theorem 3.6.8 does not hold as we read it; it is inherited, not new, and it is the same hypothesis that §12.5 and the cylinder carry. Definition 15.2.1 asks for less than it says: "nonzero" and "sc-Fredholm" are automatic, so transversality is the surjectivity of every summand (§XV.12). The proof of Theorem 15.2.2, which writes $f(x) = s_i(x)$ for all i, presumably means the branches through $f(x)$; the others have empty solution sets near x (§XV.13). "The ep-groupoid $U(x)$" is the translation groupoid $G_x \ltimes U(x)$, and on it the paracompactness of the support, which Definition 9.1.2 (1) asks of a branched ep⁺-subgroupoid since 2021, is automatic, its orbit map being closed with finite fibres; on X it is a hypothesis that the remarks do not address (§XV.14).

General position is Part I's general position of each branch. On $T^R_x X$ the intrinsic operator of a branch is the derivative of the branch in a transported chart of $W|_O$, so surjectivity there is Part I's general position of $(f - s_i)|_O$; that gives good position by Part I and persists near x. This is why Remark 15.3.6 holds at every point of the support and not only at ∂X, and why it needs no $d_X = 0$ (§§XV.16, XV.20). General position at a point is invariant under morphisms, which the text uses silently when it passes to saturations "by equivariance" (§XV.25).

The existence theorems separate into an assembly and Part I. The assembly of Theorem 15.3.7 is finite covers of $|S_f|$, sums of linear families, the bounds $N(\Lambda \oplus \Lambda') \leq N(\Lambda) + N(\Lambda')$ and on domain supports that the text uses silently, and a small parameter off a null set (§XV.22). The local families are Part XIII's

atomic multisections built from seeds, linear in the parameter and carried along morphisms (§XV.23). Parametric transversality is Part I's local Sard step branch by branch: in a tame chart a branch of Λ_t is the slice at t of a Part I parametrized section, and (15.1) for the branch is Part I's general position of that section at $(0, x)$ (§XV.25). The one bridge the text takes for granted is between (15.1), which uses $T^R_x X$, and Part I's conditions (i)–(ii), which use the full tangent space; in a tame chart centred at the zero the two are equivalent by two lines of linear algebra in each direction, and the text's appeal to the implicit function theorem for the openness of (15.1) is Part I's openness of general position on the product with $\mathbb{R}^\ell$ (§XV.24). "Structurable" contains a good system, so the addendum of Theorem 15.3.7 needs one; (1)–(4) do not, with the sums of Definition 13.2.9 in place of structurable ones, which is Remark 15.3.8 (§XV.26).

Boundary data must charge the boundary. Theorem 15.3.9 is Theorem 14.1.9 followed by Theorem 15.3.7 relative to the boundary. Stated with the index sets of the formal Definition 14.1.3, which admits empty ones, it fails with boundary: the zero function with empty index sets is structurable over ∂X, and no structurable extension vanishes there (§XV.27). The text's boundary multisection has non-empty index sets; with it, the charged statement, the theorem is proved from one open proposition. A witness on a space without boundary could not reveal this, which is why every hypothesis here is asked for a witness in the non-degenerate case. Step (a), the cut-off h_0, uses the upper semicontinuity of $N(\Lambda)$ along ∂X, which holds when the bundle has tame charts at every object and which the text uses without comment (§XV.28). Steps (c)–(d) are Theorem 15.3.7 with the operators of $T(f, \Lambda_0)$ in place of those of $T(f, 0)$, and the parametric part carries over branch by branch (§XV.29).

The cylinder needs more than the text says. The product with $[0, 1]$ is built as the product with Part I's cube, whose parametrized Fredholm theory is available; $d_X \equiv 0$ makes X tame, a step the text takes without comment (§XV.30); the auxiliary norm of Corollary 15.3.10 (2), which the text does not give, is the interpolation $\max(\alpha(t)N^0, \beta(t)N^1)$ (§XV.31). The partitions of unity of $[0, 1] \times X$ are a product construction, $\sum_x \rho_x(y) g_x(q)$, and the structurability of the glued boundary data comes from lifting the structures of the Λ^i along $[0, 1] \times X \to X$, where two continuous lifts that agree at one point agree on a connected neighbourhood (§XV.32). The text's proof does not say how the face condition of the glued data is checked; presumably from the transversality of (f, Λ^i), since $T^R_{(i,x)}([0, 1] \times X) = \{0\} \times T_x X$ when $d_X \equiv 0$ (§XV.33).

In §15.4 the argument is short and the data are hard. The canonical lift is pointwise linear algebra once the operators $(f - s_i)'(y)$ are seen to be in Part I's $\mathrm{Lin}(f, y)$, because $f - s_i$ differs from f by an sc^+ term (§XV.34). The cobordism argument for the invariants is Stokes' theorem with $dF^* = F^* d$ (§XV.36), and Theorem 15.4.3 does not use the tameness of X, good position carrying what is needed at boundary points (§XV.35). What is hard is the data the argument takes: the continuity of the kernel orientations along the branches, and on the cylinder the weights and signs on the ends, where Part I's orientation bookkeeping has to meet Chapter 9's boundary orientation (§XV.37, gaps G.159, G.182, G.190).

Part XVI. Polyfolds (HWZ Chapter 16, §§16.1–16.6)

Chapter 16 (Springer 2021, pp. 705–740; 2017 numbers in parentheses) passes from ep-groupoids to polyfolds. A polyfold is a topological space Z together with an equivalence class of ep-groupoid structures on it, and every notion of the earlier chapters (bump functions, partitions of unity, the degeneracy index, tameness, the tangent, sc-differential forms, strong bundles, auxiliary norms, sections, branched ep$^+$-subgroupoids, sc$^+$-multisections, sc-Fredholm sections, orientations and the invariants of Chapter 15) is carried to the class. The method is uniform: a notion of a polyfold is a notion of one structure in the class, and a theorem of invariance under equivalences of ep-groupoids, or under generalized isomorphisms, shows that it does not depend on the structure. §16.1 defines polyfold structures, their equivalence and polyfolds (Definitions 16.1.1–16.1.3 (2017: Definitions 16.1–16.3)), shows that bump functions and partitions of unity are invariant (Proposition 16.1.4, Definition 16.1.5 (2017: Proposition 16.2, Definition 16.4)), defines the maps between polyfolds (Definitions 16.1.6, 16.1.8, 16.1.10, 16.1.11, Remarks 16.1.7, 16.1.9 (2017: Definitions 16.5–16.8, Remarks 16.1, 16.2)), transports the degeneracy index, the boundary and tameness (Lemma 16.1.13, Definitions 16.1.14, 16.1.17, Lemma 16.1.19, Definition 16.1.20 (2017: Lemma 16.1, Definitions 16.9, 16.11, Lemma 16.2, Definition 16.12)), and introduces the raised polyfolds $(Z,c)^i$. It also treats the lifts Z_i, subpolyfolds and faces (Proposition 16.1.12, Definition 16.1.15, Remark 16.1.16, Example 16.1.18, Definition 16.1.21, Proposition 16.1.22), which are not formalized. §16.2 builds the tangent $T(Z,c)$, the tangent functor and the complex of sc-differential forms. §16.3 defines strong polyfold bundles, their maps, auxiliary norms, reflexivity, mixed convergence and sections. §16.4 defines branched finite-dimensional suborbifolds of a polyfold, their orientations and integrals, and proves Stokes' theorem. §16.5 defines sc$^+$-multisections of strong polyfold bundles and proves the extension theorem 16.5.7. §16.6 defines sc-Fredholm sections, their compactness properties and orientations, and states the integration theorem 16.6.6.

This Part follows the book's order. §§XVI.1–XVI.6 treat §16.1: polyfold structures and polyfolds with the metrizability remark (§XVI.1), the uniqueness statement after Definition 16.1.2 and Remark 16.1.7 (§XVI.2), Proposition 16.1.4 and Definition 16.1.5 (§XVI.3), the maps of polyfolds (§XVI.4), the degeneracy index, boundary and tameness (§XVI.5) and the raised polyfolds (§XVI.6). §§XVI.7–XVI.10 treat §16.2: Lemma 16.2.1 (§XVI.7), the tangent $T(Z,c)$ and the projection τ_Z (§XVI.8), Theorem 16.2.4 (§XVI.9) and sc-differential forms with Theorem 16.2.6 (§XVI.10). §§XVI.11–XVI.16 treat §16.3: strong polyfold bundles (§XVI.11), their maps (§XVI.12), auxiliary norms (§XVI.13), reflexivity and mixed convergence (§XVI.14), Theorems 16.3.15, 16.3.17 and Proposition 16.3.16 (§XVI.15), and sections with the zero section (§XVI.16). §§XVI.17–XVI.20 treat §16.4: branched suborbifolds (§XVI.17), Proposition 16.4.3 (§XVI.18), orientations and integrals (§XVI.19) and Stokes' theorem (§XVI.20). §§XVI.21–XVI.23 treat §16.5: sc$^+$-

multisections and (16.8) (§XVI.21), structurability (§XVI.22) and Theorem 16.5.7 (§XVI.23). §§XVI.24–XVI.26 treat §16.6: sc-Fredholm sections and Theorem 16.6.4 (§XVI.24), orientations (§XVI.25) and Theorem 16.6.6 (§XVI.26). Then come the witnesses (§XVI.27), what is not formalized (§XVI.28) and what the formalization taught us (§XVI.29).

What is proved, and what is not. All six sections of Chapter 16 are formalized, except Proposition 16.1.12, the subpolyfolds and faces of §16.1 and Remark 16.2.7.

Formalized as printed. Definitions 16.1.1, 16.1.2, 16.1.3, 16.1.5, 16.1.6, 16.1.8, 16.1.14, 16.1.17 and 16.1.20, Remark 16.1.9 and Lemma 16.1.13 (§§XVI.1, XVI.3–XVI.5). The remark after Definition 16.1.1 that Z is metrizable (§XVI.1). The construction of $T(Z,c)$ with the maps $T\alpha$ and (16.5), Lemma 16.2.2 and Definition 16.2.3 (§XVI.8). Definition 16.2.5 with the remark before it, and Theorem 16.2.6 in both parts, the complex and the pull-back by sc-smooth maps (§XVI.10). In §16.3: Definitions 16.3.1, 16.3.3, 16.3.4, 16.3.6, 16.3.7, 16.3.8, 16.3.12, 16.3.13, 16.3.14 and 16.3.18, Lemma 16.3.2, Proposition 16.3.10, Theorem 16.3.15, Proposition 16.3.16 (2), Theorem 16.3.17 (1), the parenthetical of Definition 16.3.12 that the model does not matter, the independence of the model of mixed convergence and of reflexive auxiliary norms, and the zero section (§§XVI.11–XVI.16). In §16.4: Definitions 16.4.1, 16.4.2, 16.4.4 and 16.4.5, Proposition 16.4.3 and the integrals of p. 732 on classes (§§XVI.17–XVI.19). In §16.5: Definitions 16.5.1, 16.5.4 and 16.5.6, Remark 16.5.3 and (16.8) with the continuity of $n(\lambda)$ (§§XVI.21–XVI.22). In §16.6: the claim before Definition 16.6.1, Definitions 16.6.1, 16.6.2 and 16.6.3, and Theorem 16.6.4 (§XVI.24). Where the text asserts that a relation is an equivalence relation, or lists a condition that the others imply, the relation is proved and the condition derived.

Restated. The raised polyfolds $(Z,c)^i$ and the projection τ_Z, with $(Z,c)^1$ a space injecting into Z with image Z_1 (§§XVI.6, XVI.8). Definitions 16.2.5, 16.4.1 and 16.4.4, whose representatives are made explicit as pairs (§§XVI.10, XVI.17, XVI.19). Definition 16.3.5 (2), (3) through representatives, Definition 16.1.21 not being formalized (§XVI.11). Definition 16.3.9 with n a function on Y whose values on $Y_{0,1}$ matter (§XVI.13). Definition 16.3.19 as the sc-smooth sections with an sc$^+$ functor (§XVI.16). Definition 16.5.2 with $y \neq 0_z$, since the set as printed is all of Z; Definition 16.5.5 with the pull-back over ∂X in orbit form (§§XVI.21–XVI.22).

Stronger. Proposition 16.1.4 with the text's sketch of part (2) written out (§XVI.3); Lemma 16.1.19 as an equivalence (§XVI.5); Proposition 16.3.16 (1) for every auxiliary norm, without reflexivity (§XVI.15). The metrizability of Z is proved without the Nagata–Smirnov theorem (§XVI.1).

Narrower. Definition 16.1.11: the polyfold category is built on effective polyfolds, since as we read Definition 16.1.10 the composite is not defined on all of them (§XVI.4). Theorem 16.2.4: a functor on the category of polyfold structures, well defined on sc-smooth maps of all polyfolds, and a functor on the category of effective polyfolds, the tangent of an effective polyfold being effective (§XVI.9). Theorem 16.3.17 (2), with the reflexive local step of Theorem 12.2.13 as a hypothesis, which Part XII discharges for tame bases (§XVI.15). Definition 16.6.5: an orientation on one representative, carried to every other along any diagram under the standing hypotheses of the M-polyfold theory on the bundles of the diagram (§XVI.25). Theorem 16.6.6: the integrals live on the polyfold and depend only on $[\tau]$; their independence of λ is proved from cobordisms between perturbations with the inherited orientation, both taken as inputs (§XVI.26). Proposition 16.4.3 and the integrals on classes are proved as printed; only the boundary integrals on classes are not wrapped (§XVI.19).

With an added hypothesis. Theorem 16.4.6, Stokes' theorem, with sc-smooth partitions of unity, which the Chapter 9 Stokes theorem it reduces to assumes (§XVI.20); Theorem 16.6.6 likewise (§XVI.26). Theorem

16.5.7, with tame strong bundle charts in the models and λ with non-empty index sets, which is the text's notion (§XVI.23).

Does not hold as we read it. Five statements, all by one mechanism: Theorem 10.3.19 (Part X) does not hold as we read it, on the ep-groupoid $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action, which satisfies (ZHOU) and has a paracompact orbit space, so it is a polyfold structure in the exact sense of Definition 16.1.1. (1) The sentence after Definition 16.1.2, that the generalized isomorphism f with $\alpha = \beta \circ |f|$ is unique as a consequence of (ZHOU) (§XVI.2). (2) Remark 16.1.7, that for a homeomorphism h there is at most one generalized isomorphism f with $h \circ \alpha = \beta \circ |f|$, together with the sentence on pp. 709–710 that an isomorphism $(f, \mathfrak{f})$ of polyfold structures is determined by f (§XVI.2). (3) Lemma 16.2.1 (§XVI.7), which holds on orbit spaces, the form §16.2 uses. (4) The claim of Definition 16.1.10 that the composite does not depend on the choices, on the middle structure $(\mathbb{Z}/3)^2 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action (§XVI.4). (5) The uniqueness of f and F after Definition 16.3.3, on the strong bundle $\mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$ over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (§XVI.11). All five hold when the target, or the middle structure, is effective, and each such form is proved. The class-level notions of §§16.3–16.6 do not depend on the non-unique isomorphisms: a section is determined by its map f, a multisection by its function λ, a branched suborbifold by its function θ (§§XVI.16, XVI.21, XVI.26).

Not formalized. The second sentence of Remark 16.1.7, on continuous maps, which has no formal content beyond an example; Remark 16.3.11, which is prose. *Not yet formalized:* Proposition 16.1.12, Definition 16.1.15, Remark 16.1.16, Example 16.1.18, Definition 16.1.21, Proposition 16.1.22 and Remark 16.2.7; the boundary integrals of p. 732 as functions on classes; weak face structure under generalized isomorphisms.

Remaining gaps (§XVI.28). G.153: Stokes' theorem on polyfolds without partitions of unity. G.175, in part: the orientation of $\lambda \circ f$ inherited from (f, o) on the polyfold, and the cobordism of Corollary 15.4.4 on the polyfold. G.176, in part: the independence of the transported orientation of the diagram, and the transfer of tameness, bump functions and tame germ data along strong bundle equivalences. G.187: general position of (f, λ) does not depend on the model. G.188: Theorem 16.6.6 in index 0.

The decisions that shape the chapter.

(1) *Polyfolds as quotients.* A polyfold structure is a Lean structure holding the ep-groupoid, its two properties (paracompact orbit space, (ZHOU)) and the homeomorphism; a polyfold on Z is an element of the quotient of the polyfold structures on Z by the equivalence of Definition 16.1.2. Every notion of a polyfold is "some structure in c has the property", and it comes with the theorem that then every structure in c has it. The same pattern gives strong polyfold bundles (quotients of strong polyfold bundle structures), sc-smooth maps (quotients of morphisms of PS), forms, sections, branched suborbifolds and multisections (quotients of representatives).

(2) *Bump functions on the objects.* "Admits sc-smooth bump functions" in Proposition 16.1.4 (1) and Definition 16.1.5 (1) is read for the object M-polyfold of the ep-groupoid, pointwise (Definition 5.3.1), as the text's proof uses it; by Theorem 12.3.2 (1) (Part XII) this is equivalent to Definition 12.3.1 in the pointwise reading. Partitions of unity are read in the ep-groupoid sense of Definition 7.5.5, and shown equivalent to partitions of unity of the orbit space whose pull-backs to the objects are sc-smooth.

(3) *A base structure instead of a gluing.* The text builds $T(Z, c)$, and the raised polyfold $(Z, c)^1$, by gluing over all structures of c along canonical maps. The formalization chooses one structure (X_0, α_0) of c (the representative chosen by the quotient) and takes $T(Z, c) = |TX_0|$, $(Z, c)^1 = |X_0^1|$; every

other structure is attached to it by the maps $T\alpha$, α^1, which are proved independent of every choice. This avoids the gluing and makes visible that what the gluing needs is a canonical map on orbit spaces, not a canonical generalized isomorphism.

(4) *Effective polyfolds for composition.* As we read it, Definition 16.1.10 does not give a well-defined composite on all polyfolds (§XVI.4). The formalization keeps the category PS of polyfold structures for all structures, keeps sc-smooth maps of all polyfolds and everything that is defined map by map (the tangent of a map, the pull-back of forms), and builds the category of polyfolds on the effective ones, where Theorem 10.3.19 holds and the composite is well defined.

(5) *Functions determine the class-level objects.* In §§16.3, 16.5 and 16.6 the relation between representatives is stated with a generalized strong bundle isomorphism, which need not be unique. The formalization proves in each case that the relation holds iff the functions on Y or Z agree (the section f, the multisection λ, the function $\theta = |\Theta| \circ \alpha^{-1}$), so the non-uniqueness of the isomorphisms does not reach these notions.

(6) *Added hypotheses are named and witnessed.* Where the text's proof reduces to a statement of an earlier chapter that carries a hypothesis the polyfold statement does not print (partitions of unity in Stokes' theorem; tame strong bundle charts in Theorem 14.1.9; the reflexive local step of Theorem 12.2.13), the hypothesis is added to the polyfold statement, named, and witnessed (§XVI.27).

Map.

Polyfold structures and polyfolds (§16.1). - `Chapter16/PolyfoldStructure`: Definition 16.1.1, the equivalence of Definition 16.1.2, Definition 16.1.3, uniqueness for effective targets, Lemma 16.1.13, Definitions 16.1.14, 16.1.17, Lemma 16.1.19, Definition 16.1.20 (§§XVI.1, XVI.2, XVI.5). - `Chapter16/ZhouUniquenessCheck`: the polyfold structure on $\mathbb{R}$ given by $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$; the uniqueness after Definition 16.1.2 and Remark 16.1.7 as statements, with the example on which they do not hold as we read them (§§XVI.1–XVI.2). - `Chapter16/PolyfoldBump`: bump points along equivalences, Proposition 16.1.4 (1), Definition 16.1.5 (1) (§XVI.3). - `Chapter16/PolyfoldPartitions`: partitions of unity read on $|X|$, along equivalences, Proposition 16.1.4 (2), Definition 16.1.5 (2) (§XVI.3). - `Chapter16/PolyfoldExamples`: the witnesses on the polyfold $\mathbb{R}$ (§XVI.27). - `Chapter16/PolyfoldMaps`: the category PS, Definitions 16.1.6, 16.1.8, Remark 16.1.9, the composition for an effective middle structure, Definition 16.1.11 on effective polyfolds (§XVI.4). - `Chapter16/CompositionCounterexample`: the claim of Definition 16.1.10 and the example on which it does not hold as we read it (§XVI.4). - `Chapter16/RaisedPolyfold`: the raise of generalized maps, the raised polyfolds $(Z, c)^i$, and the tangent projection τ_Z (§§XVI.6, XVI.8).

The tangent and forms (§16.2). - `Chapter16/TangentLemma`: tangents of generalized maps, Lemma 16.2.1 as a statement, its example, its orbit form and its form for an effective target (§XVI.7). - `Chapter16/TangentPolyfold`: metrizability of polyfolds, $T(Z, c)$, (16.5), Lemma 16.2.2, Definition 16.2.3 (§§XVI.1, XVI.8). - `Chapter16/TangentFunctor`: the tangent functor on PS and on sc-smooth maps (§XVI.9). - `Chapter16/TangentEffective`: the tangent of an effective ep-groupoid with (ZHOU) is effective; the tangent functor on effective polyfolds (§XVI.9). - `Chapter16/FormsIndependence`, `Chapter16/PolyfoldForms`: the remark before Definition 16.2.5, Definition 16.2.5, Theorem 16.2.6, the complex (§XVI.10). - `Chapter16/FormsPullback`: Theorem 16.2.6, the pull-back by sc-smooth maps (§XVI.10).

Strong polyfold bundles (§16.3). - `Chapter16/StrongPolyfoldBundle`: Definitions 16.3.1, 16.3.3, 16.3.4, 16.3.5, Lemma 16.3.2, the induced polyfold (§XVI.11). - `Chapter16/PolyfoldBundleUniqueness`: the uniqueness after Definition 16.3.3, its example and its form for an effective target (§XVI.11). - `Chapter16/PolyfoldBundleCategory`: the category $\mathcal{SPBS}$, Definitions 16.3.6–16.3.8 (§XVI.12). - `Chapter16/`

`PolyfoldAuxNorm`: Definition 16.3.9 to Theorem 16.3.17 (§§XVI.13–XVI.15). - `Chapter16/ReflexivePull`: reflexive 1-fibres pull back along local strong bundle isomorphisms and equivalences (§XVI.14). - `Chapter16/ReflexivePush`: reflexive 1-fibres push forward along equivalences; the parenthetical of Definition 16.3.12 (§XVI.14). - `Chapter16/MixedTransfer`: mixed convergence and reflexive auxiliary norms along equivalences; Proposition 16.3.16 (2) (§§XVI.14–XVI.15). - `Chapter16/PolyfoldSections`: the relation before Definition 16.3.18, Definitions 16.3.18, 16.3.19 (§XVI.16). - `Chapter16/FibreZero`, `Chapter16/ZeroSection`: the zero of a fibre and the zero section (§XVI.16). - `Chapter16/PolyfoldBundleExamples`: the witnesses on the $\mathbb{Z}/3$ bundle (§XVI.27).

Branched suborbifolds (§16.4). - `Chapter16/BranchedSuborbifold`: Definitions 16.4.1, 16.4.2, 16.4.4, 16.4.5, Proposition 16.4.3 on one representative, the integrals on representatives, Theorem 16.4.6 (§§XVI.17–XVI.20). - `Chapter16/BranchedTame`: tameness along equivalences and generalized isomorphisms, Definition 16.4.2 (5), the integrals on classes (§§XVI.17, XVI.19). - `Chapter16/SuppNaturalityStatement`, `Chapter16/SuppNaturality`: the transition between the supports of two representatives, and Proposition 16.4.3 (§XVI.18).

Sc^+-multisections (§16.5). - `Chapter16/PolyfoldMultisections`: Definitions 16.5.1, 16.5.2, Remark 16.5.3, (16.8), Definitions 16.5.4–16.5.6, and the Lean form of Theorem 16.5.7 from the uncharged Theorem 14.1.9 (§§XVI.21–XVI.23). - `Chapter16/StructurablePullback`: Definition 16.5.4, "and therefore all" (§XVI.22). - `Chapter16/RepresentativeChecks`: the multisection relation on the $\mathbb{Z}/3$ example (§XVI.21). - `Chapter16/NormLamContinuous`: the continuity of $N(\Lambda)$ and of $n(\lambda)$ in (16.8) (§XVI.21). - `Chapter16/PolyfoldBoundaryExtension`: Theorem 16.5.7 from the charged Theorem 14.1.9 (§XVI.23).

Sc-Fredholm sections (§16.6). - `Chapter16/PolyfoldFredholmSections`: the claim before Definition 16.6.1, Definitions 16.6.1–16.6.3, Theorem 16.6.4, Definition 16.6.5 on a representative, the reduction and the choice of λ in Theorem 16.6.6 (§§XVI.24–XVI.26). - `Chapter16/OrientationTransport`: the transport of orientations in Definition 16.6.5 (§XVI.25). - `Chapter16/PerturbedSuborbifold`: $\lambda \circ f$ as a branched suborbifold (§XVI.26). - `Chapter16/PerturbedIntegralIndependence`: the integrals of Theorem 16.6.6 on the polyfold, cobordisms, Theorem 16.6.6 (§XVI.26).

Throughout, a generalized map $\mathfrak{f} = [X \xleftarrow{F} A \xrightarrow{\Phi} Y]$ is a morphism of Chapter 10's localization of the category $\mathcal{EP}$ of ep-groupoids at the equivalences; its orbit map is $|\mathfrak{f}| = |\Phi| \circ |F|^{-1} : |X| \to |Y|$, and it is a generalized isomorphism if it is invertible in the localization. (ZHOU) is Zhou's condition of Chapter 7, (LR) the local regularity property of Chapter 10, and an ep-groupoid is effective if the only isotropy element acting trivially near its object is the identity (Definition 7.2.1).

XVI.1 Definitions 16.1.1–16.1.3 (2017: 16.1–16.3): polyfold structures and polyfolds, pp. 705–707

Statement (same). *Definition 16.1.1* (p. 705) `Chapter16/PolyfoldStructure · PolyfoldStructure`: a polyfold structure on a topological space Z is a pair (X, β) of an ep-groupoid X and a homeomorphism $\beta : |X| \to Z$ such that (1) the orbit space $|X|$ is paracompact and (2) X satisfies (ZHOU). *Definition 16.1.2* (p. 707) `Chapter16/PolyfoldStructure · PolyfoldStructure.Equiv`: two polyfold structures (X, α), (Y, β) on Z are equivalent if there is a generalized isomorphism $f : X \to Y$ with $\alpha = \beta \circ |f|$. *Definition 16.1.3* (p. 707) `Chapter16/PolyfoldStructure · PolyfoldClass`: a polyfold is a pair (Z, c) with c an equivalence class of polyfold structures on Z. *The remark after Definition 16.1.1* (same) `Chapter16/TangentPolyfold · PolyfoldStructure.metrizableSpace`: if Z carries a polyfold structure, Z is metrizable.

Readings. Z lives in the universe of the object and morphism sets of X, so that β is a homeomorphism between types of one universe; a polyfold on Z is therefore a class of structures whose ep-groupoids live in that universe. The 2017 Definition 16.1 had neither (1) nor (2); both are part of the formal structure, and every theorem of this Part that needs (LR), (LUC), metrizability or the existence of auxiliary norms obtains it from them. The pair (X, β) is an orbit chart in the sense of Remark 10.3.18 `Chapter16/PolyfoldStructure · PolyfoldStructure.toOrbitChart`, and Definition 16.1.2 is the relation of Remark 10.3.18 on orbit charts `Chapter16/PolyfoldStructure · PolyfoldStructure.equiv_iff_rel`.

Proof that Definition 16.1.2 is an equivalence relation. The text calls the relation an equivalence relation without proof; it is proved in three steps.

Step 1, reflexivity `Chapter16/PolyfoldStructure · PolyfoldStructure.Equiv.refl`. The identity 1_X of the localization is a generalized isomorphism and $|1_X| = \mathrm{id}_{|X|}$, so $\alpha = \alpha \circ |1_X|$.

Step 2, symmetry `Chapter16/PolyfoldStructure · PolyfoldStructure.Equiv.symm`. If $f : X \to Y$ is a generalized isomorphism with $\alpha = \beta \circ |f|$, its inverse f^{-1} in the localization is a generalized isomorphism with $|f^{-1}| = |f|^{-1}$ (Proposition 10.3.11, Part X). Then $\beta = \alpha \circ |f|^{-1} = \alpha \circ |f^{-1}|$.

Step 3, transitivity `Chapter16/PolyfoldStructure · PolyfoldStructure.Equiv.trans`. For $f : X \to Y$ and $g : Y \to V$ with $\alpha = \beta \circ |f|$ and $\beta = \gamma \circ |g|$, the composite $g \circ f$ is a generalized isomorphism with $|g \circ f| = |g| \circ |f|$ (Proposition 10.3.11), so $\alpha = \gamma \circ |g \circ f|$.

The quotient by this relation `Chapter16/PolyfoldStructure · PolyfoldStructure.setoid` is the type of polyfolds on Z. A consequence used throughout: if $(X, \alpha) \sim (Y, \beta)$ through f, then $|f| = \beta^{-1} \circ \alpha$ `Chapter16/PolyfoldStructure · PolyfoldStructure.orbitMap_eq_of_equiv`, since β is a bijection. So the orbit map of a realizing generalized isomorphism is determined by the two structures, even though (§XVI.2) the generalized isomorphism itself is not.

(ZHOU) gives (LR) `Chapter16/PolyfoldStructure · PolyfoldStructure.isLR`. *Proof.* Proposition 7.4.22 (Part VII) `Chapter7/SmoothRegularAssembly · proposition_7_4_22_of_isZhou` shows that an ep-groupoid with (ZHOU) has the local regularity property. This is the hypothesis the text has in mind when it cites Theorem 10.3.19 after Definition 16.1.2, and the hypothesis of Proposition 10.1.10 and Proposition 11.2.9, which carry §16.2 (§§XVI.7–XVI.10).

Proof of the remark: polyfolds are metrizable. *Step 1.* $|X|$ is paracompact by Definition 16.1.1 (1). *Step 2.* Theorem 7.3.1 (Part VII) gives that $|X|$ is Hausdorff, regular and locally metrizable; with paracompactness, $|X|$ is metrizable `Chapter7/OrbitTopology · theorem_7_3_1_metrizable`. *Step 3, the metrization step.* It is the theorem of Chapter 2 (Part II) that a paracompact Hausdorff space which is locally metrizable is metrizable `Chapter2/Metrizable · metrizableSpace_of_locally`, proved through the pull-back of a metrizable topology along an embedding `Chapter2/Metrizable · metrizableSpace_of_pullback`. The Nagata–Smirnov theorem, which the text's appeal to metrization theorems suggests, is not needed. *Step 4.* β is a homeomorphism, so the metrizable topology of $|X|$ carries over to Z. The text's 2017 Proposition 16.1 is this remark.

The polyfold $\mathbb{R}$ with isotropy $\mathbb{Z}/3$. This example is the witness of every definition of §16.1 and the counterexample of §XVI.2, so it is built with care. Let $X = \mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ be the action groupoid of the trivial action of $\mathbb{Z}/3$ on $\mathbb{R}$: objects $\mathbb{R}$, morphisms $\mathbb{Z}/3 \times \mathbb{R}$, source and target the projection.

Step 1, the orbit space. The orbit map $\pi : \mathbb{R} \to |X|$ is injective, since every orbit is a point `Chapter16/ZhouUniquenessCheck · EpGroupoid.Examples.injective_π_trivialZ3`. It is continuous and open (Proposition 7.1.8, Part VII), and surjective, hence a homeomorphism `Chapter16/ZhouUniquenessCheck · EpGroupoid.Examples.`

isHomeomorph_π_trivialZ3. Its inverse gives $|X| \cong \mathbb{R}$ Chapter16/ZhouUniquenessCheck · EpGroupoid.Examples.trivialZ3OrbitHomeomorph.

Step 2, paracompactness. $\mathbb{R}$ is paracompact, hence so is $|X|$ Chapter16/ZhouUniquenessCheck · EpGroupoid.Examples.hasParacompactOrbitSpace_trivialZ3.

Step 3, (ZHOU). Every isotropy element acts trivially, so for every x the natural representation of G_x is trivial and the conditions of (ZHOU) hold with the whole of $\mathbb{R}$ as the neighbourhood Chapter7/Zhou · EpGroupoid.Examples.isZhou_trivialZ3.

Step 4, the structure. With β the inverse of the homeomorphism of Step 1, (X, β) is a polyfold structure on $\mathbb{R}$ in the exact sense of Definition 16.1.1 Chapter16/ZhouUniquenessCheck · PolyfoldStructure.z3PolyfoldStructure, and its class is a polyfold Chapter16/ZhouUniquenessCheck · PolyfoldStructure.z3Polyfold.

Step 5, not effective. The non-trivial elements of $\mathbb{Z}/3$ act as the identity near every object, so X is not effective Chapter16/ZhouUniquenessCheck · PolyfoldStructure.not_isEffective_z3PolyfoldStructure. This is what makes the example a counterexample to the uniqueness statements, and why their effective forms hold.

Where Lean departs from the text. The text states Definitions 16.1.1–16.1.3 and treats the relation of Definition 16.1.2 as evidently an equivalence relation; the formalization proves the three properties from Proposition 10.3.11, the only input being that orbit maps of generalized maps compose and invert. The universe in which Z lives is implicit in the text; the formalization fixes it to be that of the object and morphism sets of X, which costs nothing because every construction of the chapter stays inside one universe or moves up by one (the tangent). The metrizability remark is proved by a different route from the one the 2017 text suggests: instead of a general metrization theorem for the orbit space, Theorem 7.3.1 provides local metrizability, and the Chapter 2 theorem for paracompact Hausdorff spaces finishes. The polyfold $\mathbb{R}$ with isotropy $\mathbb{Z}/3$ is not in the text; it is the example of Part X carried to polyfolds, and it serves both as the universal witness of the definitions and as the example of §XVI.2.

XVI.2 The uniqueness after Definition 16.1.2, Remark 16.1.7 (2017: Remark 16.1), and the sentence on pp. 709–710

Statement (does not hold as we read it; holds for an effective target). After Definition 16.1.2 (p. 707) the text says that the generalized isomorphism f with $\alpha = \beta \circ |f|$ is necessarily unique as a consequence of (ZHOU), citing Theorem 10.3.19: for generalized isomorphisms $f, f' : X \to Y$, $|f| = |f'|$ implies $f = f'$ if one of the ep-groupoids has (LR). The claim in Lean Chapter16/ZhouUniquenessCheck · PolyfoldStructure.Uniqueness_16_1_2: for polyfold structures (X, α), (Y, β) on Z, at most one generalized isomorphism f has $\alpha = \beta \circ |f|$. Remark 16.1.7 (p. 711) Chapter16/ZhouUniquenessCheck · PolyfoldStructure.Remark_16_1_7: for a homeomorphism $h : Z \to W$, a polyfold structure (X, α) on Z and one (Y, β) on W, at most one generalized isomorphism f has $h \circ \alpha = \beta \circ |f|$. The sentence on pp. 709–710: an isomorphism $(f, \mathfrak{f})$ of polyfold structures in the category PS is determined by f.

Verdict. The uniqueness after Definition 16.1.2 does not hold as we read it Chapter16/ZhouUniquenessCheck · PolyfoldStructure.not_uniqueness_16_1_2, nor does Remark 16.1.7 Chapter16/ZhouUniquenessCheck · PolyfoldStructure.not_remark_16_1_7, nor the sentence on pp. 709–710. All three hold when the target ep-groupoid is effective Chapter16/PolyfoldStructure · PolyfoldStructure.uniqueness_16_1_2_of_isEffective Chapter16/PolyfoldStructure · PolyfoldStructure.remark_16_1_7_of_isEffective Chapter16/ZhouUniquenessCheck · PolyfoldStructure.uniqueness_16_1_2_of_forall_isEffective.

Proof that the uniqueness does not hold as we read it. *Step 1, the structure.* Take the polyfold structure (X, β) on $\mathbb{R}$ of §XVI.1, with $X = \mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ and the trivial action, as both (X, α) and (Y, β). All hypotheses of Definition 16.1.1 hold: paracompact orbit space, (ZHOU), and a homeomorphism to $\mathbb{R}$ `Chapter16/ZhouUniquenessCheck · PolyfoldStructure.uniqueness_16_1_2_counterexample`.

Step 2, two generalized isomorphisms. Let $\mathrm{Inv} : X \to X$ be the functor that is the identity on objects and inverts morphisms; since $\mathbb{Z}/3$ is abelian and acts trivially, Inv is a functor, and it is an equivalence. Take $f = [1_X]$ `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.genId` and $f' = [X \xleftarrow{1} X \xrightarrow{\mathrm{Inv}} X]$ `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.genInv`. Both are generalized isomorphisms `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.isGeneralizedIso_genInv`.

Step 3, both realize the equivalence. Both functors are the identity on objects, so both orbit maps are the identity of $|X|$, and $\beta = \beta \circ |f|$ `Chapter16/ZhouUniquenessCheck · PolyfoldStructure.β_eq_β_comp_orbitMap_genId` and $\beta = \beta \circ |f'|$ `Chapter16/ZhouUniquenessCheck · PolyfoldStructure.β_eq_β_comp_orbitMap_genInv`.

Step 4, they differ `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.genInv_ne_genId`. Two morphisms of the localization represented by functors $X \to X$ through the identity leg agree only if the functors are connected by a natural transformation after a refinement. A natural transformation from 1 to Inv would be a family $\tau_x \in \mathbb{Z}/3$ with $\tau_x g = g^{-1}\tau_x$, that is $g = g^{-1}$, for every g; this fails for a generator. The argument is the one of Part X, where it shows that Theorem 10.3.19 does not hold as we read it `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.not_theorem_10_3_19`.

Step 5, conclusion. (ZHOU) gives (LR) (§XVI.1), so the text's citation of Theorem 10.3.19 applies to this X, and the example shows that (LR) does not give the uniqueness. For Remark 16.1.7, take $h = \mathrm{id}_{\mathbb{R}}$ in the same example. For the sentence on pp. 709–710, the pairs (id, f) and (id, f') are two isomorphisms of PS with the same continuous map.

Proof of the effective form. *Step 1.* If Y is effective, Theorem 10.3.19 holds `Chapter10/LocalizationLRCounterexample · EPLocalization.EPLoc.theorem_10_3_19_of_isEffective` (Part X): an effective target rules out isotropy acting trivially, which is what Step 4 above uses. *Step 2.* Two generalized isomorphisms f, f' with $\beta \circ |f| = h \circ \alpha = \beta \circ |f'|$ have $|f| = |f'|$, because β is injective. *Step 3.* By Step 1, $f = f'$. The form for $h = \mathrm{id}$ is the uniqueness after Definition 16.1.2; the form quantified over all effective polyfold structures is a corollary.

What depends on the uniqueness. Definitions 16.1.2 and 16.1.3 do not: an equivalence of structures asks only that some generalized isomorphism exist. Nor do Definition 16.1.6, which carries its generalized isomorphisms as data, the invariance statements of §16.1, or any of the class-level definitions of §§16.3–16.6 (decision (5)). The uniqueness is used in the text in three places: Lemma 16.2.1 and the gluing of $T(Z, c)$ (§§XVI.7–XVI.8), where the orbit form suffices; the claim of Definition 16.1.10 (§XVI.4), which does not hold as we read it; and the uniqueness after Definition 16.3.3 (§XVI.11), which does not hold as we read it either. The second sentence of Remark 16.1.7, that for a continuous map there may be none or many generalized maps, is an observation without formal content beyond an example and is not formalized.

Where Lean departs from the text. The text's argument is one sentence: (ZHOU), then Theorem 10.3.19. The formalization splits it into its two steps and tests each. The first, (ZHOU) gives (LR), is Proposition 7.4.22 and holds. The second, (LR) gives the uniqueness, is Theorem 10.3.19 and does not hold as we read it; the example of Part X satisfies every hypothesis of Definition 16.1.1, so the polyfold statement inherits the gap exactly. The effective form keeps the text's two steps and replaces Theorem 10.3.19 by

its effective form. The formalization does not attempt to strengthen (ZHOU) or (LR) to a condition that would restore uniqueness without effectiveness; effectiveness is the condition under which Part X proves Theorem 10.3.19, and it is the one the rest of this Part uses.

XVI.3 Proposition 16.1.4 (2017: Proposition 16.2) and Definition 16.1.5 (2017: Definition 16.4), p. 708

Statement. *Proposition 16.1.4 (1)* (same) `Chapter16/PolyfoldBump · PolyfoldStructure.proposition_16_1_4_one`: for equivalent polyfold structures $(X, \alpha) \sim (Y, \beta)$, the object M-polyfold of X admits sc-smooth bump functions iff that of Y does. *Proposition 16.1.4 (2)* (explicit) `Chapter16/PolyfoldPartitions · PolyfoldStructure.proposition_16_1_4_two`: X admits sc-smooth partitions of unity in the sense of Definition 7.5.5 iff Y does. Both together `Chapter16/PolyfoldPartitions · PolyfoldStructure.proposition_16_1_4`. *Definition 16.1.5* (same) `Chapter16/PolyfoldBump · PolyfoldClass.AdmitsBumpFunctions` `Chapter16/PolyfoldPartitions · PolyfoldClass.AdmitsScPartitionsOfUnity`: a polyfold admits sc-smooth bump functions, or sc-smooth partitions of unity, if some structure in c does; then every structure does `Chapter16/PolyfoldBump · PolyfoldClass.definition_16_1_5_one` `Chapter16/PolyfoldPartitions · PolyfoldClass.definition_16_1_5_two`. *Verdict* for (2): "explicit", because the text's proof of (2) is a sketch, calling a step an easy exercise; the steps below fill it.

Proof of (1). *Step 0, reduction to one equivalence.* An equivalence of structures is realized by a diagram $X \xleftarrow{F} A \xrightarrow{G} Y$ of equivalences of ep-groupoids, so it suffices to show that bump functions pass in both directions along one equivalence $F : A \to X$ `Chapter16/PolyfoldBump · EpFunctor.IsEquivalence.admitsBumpFunctions_iff`.

Step 1, bump functions are local. Call a point a bump point if there are sc-smooth bump functions at it, in every neighbourhood. An M-polyfold admits bump functions iff every point is a bump point. Sc-smoothness of a real function is local `Chapter16/PolyfoldBump · MPolyfold.isScSmoothMap_real_of_locally`, so a function that is sc-smooth near its support and 0 elsewhere is sc-smooth.

Step 2, bump points along sc-diffeomorphisms of open sets `Chapter16/PolyfoldBump · MPolyfold.IsScDiffeoOn.hasBumpAt`. Let $\varphi : U \to U'$ be an sc-diffeomorphism of open sets with $\varphi(a) = b$, and b a bump point. For a neighbourhood V of a, choose a bump function g at b supported in a small neighbourhood inside $\varphi(V \cap U)$ whose closure is inside U'; then $g \circ \varphi$ on U, extended by 0, is sc-smooth by Step 1 and is a bump function at a supported in V.

Step 3, along an equivalence `Chapter16/PolyfoldBump · EpFunctor.IsEquivalence.hasBumpAt_of_hasBumpAt_obj` `Chapter16/PolyfoldBump · EpFunctor.IsEquivalence.hasBumpAt_obj_of_hasBumpAt`. An equivalence is a local sc-diffeomorphism on objects (Chapter 10), so Step 2 carries bump points from $F(a)$ to a and from a to $F(a)$.

Step 4, essential surjectivity `Chapter16/PolyfoldBump · EpGroupoid.hasBumpAt_tgt`. Every $y \in X$ is the target of a morphism $\psi : F(a) \to y$. The local sc-diffeomorphism $\hat{\psi} = t \circ (s|_{U(\psi)})^{-1}$ of the morphism, defined near $F(a)$, sends $F(a)$ to y, so by Step 2 bump points are invariant under $\hat{\psi}$, and y is a bump point if $F(a)$ is. With Step 3 every object of X is a bump point if every object of A is; the converse is Step 3 alone.

Proof of (2). *Step 1, read on the orbit space.* Definition 7.5.5 holds for X iff every open cover of $|X|$ has a subordinate partition of unity (σ_λ) of $|X|$ with every $\sigma_\lambda \circ \pi$ sc-smooth `Chapter16/PolyfoldPartitions · EpGroupoid.HasOrbitScPartitionsOfUnity` `Chapter16/PolyfoldPartitions · EpGroupoid.hasScPartitionsOfUnity_iff_orbit`. The proof has three ingredients. An invariant continuous function on the objects descends to a continuous

function on $|X|$ `Chapter16/PolyfoldPartitions · EpGroupoid.descendP3Z`, so an ep-groupoid partition of unity, which is invariant, gives one of $|X|$. Closures commute with π^{-1}, because π is open `Chapter16/PolyfoldPartitions · EpGroupoid.preimage_closure_π`, so subordination to a saturated cover upstairs is subordination to the image cover downstairs. And a partition of unity of $|X|$ pulls back along the continuous map π to a family that is again locally finite with sum 1 `Chapter16/PolyfoldPartitions · PartitionOfUnity.comapP3Z`.

Step 2, the text's "easy exercise" `Chapter16/PolyfoldPartitions · EpFunctor.IsEquivalence.isScSmoothMap_comp_π_of_comp_obj`. For an equivalence $F : A \to X$ and $g : |X| \to \mathbb{R}$: if $g \circ \pi_X \circ F$ is sc-smooth on the objects of A, then $g \circ \pi_X$ is sc-smooth on the objects of X. Near a point $F(a)$, $g \circ \pi_X = (g \circ \pi_X \circ F) \circ F_{\mathrm{loc}}^{-1}$ with F_{loc}^{-1} a local inverse of F, which exists because F is a local sc-diffeomorphism. Near a point y isomorphic to $F(a)$ through ψ, $g \circ \pi_X = (g \circ \pi_X) \circ \widehat{\psi}^{-1}$ near y, because π_X is invariant under $\widehat{\psi}$; so sc-smoothness near $F(a)$ gives it near y. The converse direction, that $g \circ \pi_X$ sc-smooth gives $g \circ \pi_X \circ F$ sc-smooth, is the chain rule.

Step 3, transport. $|F| : |A| \to |X|$ is a homeomorphism (Chapter 10, Part X), so partitions of unity of $|X|$ and of $|A|$ correspond under composition with $|F|$, and $\sigma \circ \pi_X \circ F = \sigma \circ |F| \circ \pi_A$. With Step 2 the property of Step 1 passes from A to X and back `Chapter16/PolyfoldPartitions · EpFunctor.IsEquivalence.hasScPartitionsOfUnity_iff`, and along a diagram by two applications.

Proof of Definition 16.1.5. By Proposition 16.1.4 applied to any two structures of c. *Where Lean departs from the text.* The text's proof of (1) goes through local inverses and $\widehat{\psi}$ as here. Its proof of (2) indicates the reduction to the orbit space; Steps 1–3 are the details. The reading of bump functions on the object M-polyfold is decision (2).

Why the reading on the objects. Proposition 16.1.4 (1) could be read for the ep-groupoid in the sense of Definition 12.3.1, with bump functions that are themselves invariant. The text's proof works with bump functions on the object M-polyfold and moves them by local inverses and by $\widehat{\psi}$, which does not preserve invariance; so the reading on the objects is the one the proof supports. By Theorem 12.3.2 (1) the two readings agree pointwise, so nothing is lost. For partitions of unity the opposite holds: the text's Definition 7.5.5 is the ep-groupoid notion, and the formalization keeps it, proving the orbit-space form as a lemma rather than replacing the definition.

XVI.4 Definitions 16.1.6–16.1.11 (2017: Definitions 16.5–16.8) and Remark 16.1.9 (2017: Remark 16.2): maps of polyfolds, pp. 709–711

Statement. *The category PS* (p. 709, same) `Chapter16/PolyfoldMaps · PSCat`: objects are polyfold structures; a morphism $(X, \alpha) \to (Y, \beta)$ is a pair $(f, \mathfrak{f})$ of a continuous map $f : Z \to W$ and a generalized map $\mathfrak{f} : X \to Y$ with $f \circ \alpha = \beta \circ |\mathfrak{f}|$ `Chapter16/PolyfoldMaps · PSHom`. *Definition 16.1.6* (p. 710, same) `Chapter16/PolyfoldMaps · PSHom.Equiv`: $(f, \mathfrak{f}) \sim (f', \mathfrak{f}')$ if $(f', \mathfrak{f}') \circ (h, \mathrm{Id}_Z) = (k, \mathrm{Id}_W) \circ (f, \mathfrak{f})$ for isomorphisms (h, Id_Z), (k, Id_W) of PS. *Definition 16.1.8 and Remark 16.1.9* (p. 711, same) `Chapter16/PolyfoldMaps · PolyfoldClass.ScMap` `Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.remark_16_1_9`: an sc-smooth map $(Z, c) \to (W, d)$ is an equivalence class $[(f, \mathfrak{f})]$ of morphisms between structures of c and d, and its underlying continuous map is f. *Definition 16.1.10* (p. 711, does not hold as we read it; holds for an effective middle structure) `Chapter16/PolyfoldMaps · PSHom.Definition_16_1_10` `Chapter16/CompositionCounterexample · CompositionCounterexample.not_definition_16_1_10` `Chapter16/PolyfoldMaps · PSHom.Equiv.comp_of_isEffective`: $\widehat{g} \circ \widehat{f} = [(g \circ f, \mathfrak{g} \circ \mathfrak{f})]$ for representatives with a common middle structure, and

the text states that this does not depend on the choices. *Definition 16.1.11* (p. 711, narrower) `Chapter16/PolyfoldMaps · EffPolyfold`: the category of effective polyfolds and sc-smooth maps.

Proof that PS is a category. *Step 1, the continuous map is determined* `Chapter16/PolyfoldMaps · PSHom.f_eq` `Chapter16/PolyfoldMaps · PSHom.continuous_f`: from $f\circ\alpha = \beta\circ|\mathfrak{f}|$, $f = \beta\circ|\mathfrak{f}|\circ\alpha^{-1}$, a composite of continuous maps, since orbit maps of generalized maps are continuous (Part X). So the text's requirement that f be continuous is a consequence, and a morphism is determined by its generalized map `Chapter16/PolyfoldMaps · PSHom.ext`; every generalized map defines a morphism `Chapter16/PolyfoldMaps · PSHom.ofGen`.

Step 2, identities and composition `Chapter16/PolyfoldMaps · PSHom.id` `Chapter16/PolyfoldMaps · PSHom.comp`: $(\mathrm{id}_Z, 1_X)$, and $(g,\mathfrak{g})\circ(f,\mathfrak{f}) = (g\circ f, \mathfrak{g}\circ\mathfrak{f})$, which is a morphism because $|\mathfrak{g}\circ\mathfrak{f}| = |\mathfrak{g}|\circ|\mathfrak{f}|$ (Proposition 10.3.11). *Step 3, the laws* `Chapter16/PolyfoldMaps · PSHom.comp_assoc`: componentwise, from those of the localization.

Step 4, isomorphisms over the identity `Chapter16/PolyfoldMaps · PSHom.IsIdIso`: a morphism (h, Id_Z) between structures on one space Z with h a generalized isomorphism. They have inverses of the same kind `Chapter16/PolyfoldMaps · PSHom.IsIdIso.inv`, compose `Chapter16/PolyfoldMaps · PSHom.IsIdIso.comp` and include the identities `Chapter16/PolyfoldMaps · PSHom.isIdIso_id`; for one of them, $\alpha = \beta\circ|h|$ `Chapter16/PolyfoldMaps · PSHom.IsIdIso.β_eq`, so two structures are related by one iff they are equivalent in the sense of Definition 16.1.2, and one exists between any two equivalent structures `Chapter16/PolyfoldMaps · PSHom.exists_isIdIso`. For two structures of one class the formalization fixes one such isomorphism `Chapter16/PolyfoldMaps · PolyfoldClass.transport` `Chapter16/PolyfoldMaps · PolyfoldClass.transport_isIdIso`; it is used to define composites of representatives below.

Proof of Definition 16.1.6 and Definition 16.1.8. *Step 1, the continuous maps agree* `Chapter16/PolyfoldMaps · PSHom.Equiv.f_eq`: comparing the continuous components of the square, $f'\circ\mathrm{Id}_Z = \mathrm{Id}_W\circ f$. *Step 2, an equivalence relation* `Chapter16/PolyfoldMaps · PSHom.Equiv.refl` `Chapter16/PolyfoldMaps · PSHom.Equiv.symm` `Chapter16/PolyfoldMaps · PSHom.Equiv.trans`: reflexive with $h = 1$, $k = 1$; symmetric with the inverses h^{-1}, k^{-1}, multiplying the square by them; transitive by pasting two squares, with the composites $h'\circ h$, $k'\circ k$ of isomorphisms over the identity. The text states the relation without proof. *Step 3, the representatives* `Chapter16/PolyfoldMaps · PolyfoldClass.MapRep` `Chapter16/PolyfoldMaps · PolyfoldClass.MapRep.setoid`: morphisms of PS from a structure of c to a structure of d, with the relation; composing a representative with isomorphisms over the identity on either side stays in its class `Chapter16/PolyfoldMaps · PSHom.equiv_comp_left` `Chapter16/PolyfoldMaps · PSHom.equiv_comp_right`. *Step 4, Definition 16.1.8.* An sc-smooth map is a class `Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.mk`. Every class has a representative between any two given structures of c and d `Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.exists_rep`: compose a representative $(f,\mathfrak{f}) : (X',\alpha') \to (Y',\beta')$ with the chosen isomorphisms $(X,\alpha)\to(X',\alpha')$ and $(Y',\beta')\to(Y,\beta)$ and use Step 3. *Step 5, Remark 16.1.9* `Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.toFun`: the continuous map of the class is $\beta\circ|\mathfrak{f}|\circ\alpha^{-1}$ for every representative, by Step 1 and Step 1 of PS. *Step 6, identities* `Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.id` `Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.id_eq`: $1_{(Z,c)} = [(\mathrm{Id}_Z, [1_X])]$, the same class for every $(X,\alpha)\in c$, since two such representatives are related by the chosen isomorphism on both sides.

Proof that the claim of Definition 16.1.10 does not hold as we read it. *The claim in Lean:* if $(f,\mathfrak{f})\sim(f',\mathfrak{f}')$ and $(g,\mathfrak{g})\sim(g',\mathfrak{g}')$, and both pairs have a common middle structure, then the composites are equivalent. The example uses only the printed hypotheses: polyfold structures and morphisms of PS.

Step 1, the structures. $Z = W = V = \mathbb{R}$. $(X,\alpha) = (U,\gamma)$ is the polyfold structure $\mathbb{Z}/3\times\mathbb{R} \rightrightarrows \mathbb{R}$ with the trivial action (§XVI.1), with $\gamma\circ\pi = \mathrm{id}$ `Chapter16/CompositionCounterexample · CompositionCounterexample.z3_β_π`. (Y,β) is $(\mathbb{Z}/3)^2\times\mathbb{R}\rightrightarrows\mathbb{R}$, the action groupoid

of the trivial action of $K = (\mathbb{Z}/3)^2$ Chapter16/CompositionCounterexample · CompositionCounterexample.K2 Chapter16/CompositionCounterexample · CompositionCounterexample.k2EpGroupoid. As in §XVI.1, its orbit map is injective Chapter16/CompositionCounterexample · CompositionCounterexample.injective_π_k2 and a homeomorphism Chapter16/CompositionCounterexample · CompositionCounterexample.isHomeomorph_π_k2, $|Y| \cong \mathbb{R}$ Chapter16/CompositionCounterexample · CompositionCounterexample.k2OrbitHomeomorph, it satisfies (ZHOU) Chapter16/CompositionCounterexample · CompositionCounterexample.isZhou_k2, and (Y, β) is a polyfold structure Chapter16/CompositionCounterexample · CompositionCounterexample.k2Structure with $\beta \circ \pi = \mathrm{id}$ Chapter16/CompositionCounterexample · CompositionCounterexample.k2Structure_β_π. All continuous maps are the identity of $\mathbb{R}$.

Step 2, the functors. All are the identity on objects: $\iota : \mathbb{Z}/3 \to K$, $\iota(g) = (g, 1)$ Chapter16/CompositionCounterexample · CompositionCounterexample.ι; $\sigma : K \to K$, the swap of the two factors Chapter16/CompositionCounterexample · CompositionCounterexample.σ, with $\sigma \circ \sigma = 1$ Chapter16/CompositionCounterexample · CompositionCounterexample.σ_σ; $\mathrm{pr}_1 : K \to \mathbb{Z}/3$ Chapter16/CompositionCounterexample · CompositionCounterexample.pr₁; and $T : \mathbb{Z}/3 \to \mathbb{Z}/3$, $T(g) = 1$ Chapter16/CompositionCounterexample · CompositionCounterexample.T. The orbit map of the generalized map of a functor that is the identity on objects is the identity on orbits Chapter16/CompositionCounterexample · CompositionCounterexample.orbitMap_map_π, so every pair $(\mathrm{Id}, [\Phi])$ below is a morphism of PS.

Step 3, two equivalent representatives of $\hat{f}$. For $k \in \{1, \sigma\}$ let $a_k = (\mathrm{Id}, [\iota] \circ [k]) : (X, \alpha) \to (Y, \beta)$ Chapter16/CompositionCounterexample · CompositionCounterexample.a, and $b = (\mathrm{Id}, [\mathrm{pr}_1]) : (Y, \beta) \to (U, \gamma)$ Chapter16/CompositionCounterexample · CompositionCounterexample.b. $[\sigma]$ is a generalized isomorphism, its own inverse Chapter16/CompositionCounterexample · CompositionCounterexample.isGeneralizedIso_σ, so $(\mathrm{Id}, [\sigma])$ Chapter16/CompositionCounterexample · CompositionCounterexample.kσ is an isomorphism over the identity Chapter16/CompositionCounterexample · CompositionCounterexample.kσ_isIdIso. The square of Definition 16.1.6 with $h = 1$ and $k = [\sigma]$ commutes, so $a_1 \sim a_\sigma$ Chapter16/CompositionCounterexample · CompositionCounterexample.a_equiv.

Step 4, the composites. $b \circ a_1$ has generalized map $[\iota] \circ [\mathrm{pr}_1] = [\mathrm{pr}_1 \circ \iota] = [1_X]$ Chapter16/CompositionCounterexample · CompositionCounterexample.ι_pr₁ Chapter16/CompositionCounterexample · CompositionCounterexample.gen_comp_one, and $b \circ a_\sigma$ has $[\mathrm{pr}_1 \circ \sigma \circ \iota] = [T]$, since $\mathrm{pr}_1(\sigma(g, 1)) = \mathrm{pr}_1(1, g) = 1$ Chapter16/CompositionCounterexample · CompositionCounterexample.ι_σ_pr₁ Chapter16/CompositionCounterexample · CompositionCounterexample.gen_comp_σ.

Step 5, not equivalent Chapter16/CompositionCounterexample · CompositionCounterexample.not_equiv_comp. Suppose $(\mathrm{Id}, [1_X]) \sim (\mathrm{Id}, [T])$ through h, k. Then $[T] \circ h = k \circ [1_X]$, so $[T] = k \circ h^{-1}$ is a generalized isomorphism (cancellation of isomorphisms in the localization). But $[T]$ is the class of a functor, and by Theorem 10.3.10 (Part X) the class of a functor is invertible iff the functor is an equivalence; T is not, not being faithful: it sends the non-trivial isotropy elements to 1 Chapter16/CompositionCounterexample · CompositionCounterexample.not_isEquivalence_T Chapter16/CompositionCounterexample · CompositionCounterexample.not_isIso_T.

Step 6, on classes Chapter16/CompositionCounterexample · CompositionCounterexample.not_exists_comp: no composition of sc-smooth maps agrees with the composite of representatives for all choices, since it would have to equal both classes of Step 5.

The mechanism. It is that of §XVI.2: in Definition 16.1.6 the isomorphism (k, Id_W) of the middle structure is not determined by Id_W, because two generalized isomorphisms of Y with the identity orbit map need not agree. The middle structure is not effective Chapter16/CompositionCounterexample · CompositionCounterexample.not_isEffective_k2.

Proof of the effective form. *Step 1* Chapter16/PolyfoldMaps · PSHom.Equiv.comp_of_isEffective. Let the middle structure (Y', β') be effective, $a \sim a'$ through (h, k) and $b \sim b'$ through (k', l), where k and k' are isomorphisms over the identity $(Y, \beta) \to (Y', \beta')$. By the uniqueness for an effective target (§XVI.2), $k = k'$. Then the two squares paste to one square for the composites, with the outer isomorphisms h and l. *Step 2, composites of representatives* Chapter16/PolyfoldMaps · PolyfoldClass.MapRep.comp: for representatives $(f, \mathfrak{f}) : (X, \alpha) \to (Y, \beta)$ and $(g, \mathfrak{g}) : (Y', \beta') \to (V, \gamma)$ whose middle structures differ, the composite is $(g \circ f, \mathfrak{g} \circ k \circ \mathfrak{f})$ with the chosen isomorphism k of Step 4 of PS. For an effective d it respects the relation Chapter16/PolyfoldMaps · PolyfoldClass.MapRep.comp_equiv, by Step 1 and the uniqueness, which makes the chosen k the only one. *Step 3, effectiveness on classes* Chapter16/PolyfoldMaps · PolyfoldClass.IsEffective Chapter16/PolyfoldMaps · PolyfoldClass.isEffective_iff: a polyfold is effective if one structure is, and then every structure is, since effectiveness is invariant under equivalences of ep-groupoids Chapter16/PolyfoldMaps · EpFunctor.IsEquivalence.isEffective_iff (Remark 10.1.12, through the bijections of isotropy groups) and so under generalized isomorphisms Chapter16/PolyfoldMaps · EPLoc.isEffective_iff_of_isGeneralizedIso. *Step 4, the composite of classes* Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.comp Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.comp_mk: for effective d, the lift of Step 2 to the quotients. *Step 5, the laws* Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.id_comp Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.comp_id Chapter16/PolyfoldMaps · PolyfoldClass.ScMap.comp_assoc: on representatives, with the chosen isomorphisms absorbed by Step 3 of Definition 16.1.6.

Definition 16.1.11 (narrower) Chapter16/PolyfoldMaps · EffPolyfold. Objects are effective polyfolds, morphisms sc-smooth maps, identities $[(\mathrm{Id}_Z, [1_X])]$, and composition from Step 4. *Narrower* because, as we read Definition 16.1.10, the composition is not defined on all polyfolds. Presumably the text intends effective polyfolds, or morphisms carrying a chosen isomorphism of the middle structure; the formalization takes the first. Witness: the one-point polyfold Chapter16/PolyfoldMaps · PolyfoldStructure.pointStructure Chapter16/PolyfoldMaps · EffPolyfold.point, whose ep-groupoid has trivial isotropy and so is effective.

Where Lean departs from the text. The text gives Definitions 16.1.6–16.1.11 without proofs. Three points are made explicit. First, the laws of PS, which hold componentwise once the continuous map is seen to be determined by the generalized map. Second, the composite of two classes: the text composes representatives "with a common middle structure", which leaves open how two representatives are brought to a common middle structure; the formalization does it with a chosen isomorphism over the identity between the two middle structures, and it is exactly there that the choice matters, since another isomorphism over the identity can give an inequivalent composite. Third, the answer to the question "which polyfolds form a category": the effective ones, where Theorem 10.3.19 makes the isomorphism over the identity unique, so that the choice disappears. The counterexample is minimal in the sense that all continuous maps are the identity and all functors are the identity on objects; the only data that differ are the isotropy homomorphisms.

XVI.5 Lemma 16.1.13 (2017: Lemma 16.1), Definitions 16.1.14, 16.1.17 (2017: 16.9, 16.11), Lemma 16.1.19 (2017: Lemma 16.2), Definition 16.1.20 (2017: 16.12), pp. 712–714

Statement. *The degeneracy index of a structure* Chapter16/PolyfoldStructure · PolyfoldStructure.degIndex: $d_{(X,\beta)}(z) = d_{|X|}(\beta^{-1}(z))$. *Lemma 16.1.13* (p. 712, same) Chapter16/PolyfoldStructure · PolyfoldStructure.lemma_16_1_13: equivalent structures have the same degeneracy index. *Definition 16.1.14* (p. 712, same) Chapter16/PolyfoldStructure · PolyfoldClass.degIndex Chapter16/PolyfoldStructure · PolyfoldClass.definition_16_

1_14: $d_Z(z) = d_X(x)$ for every $(X,\beta) \in c$ and $\beta(|x|) = z$. *Definition 16.1.17* (p. 713, same) Chapter16/PolyfoldStructure · PolyfoldClass.boundary: $\partial Z = \{z : d_Z(z) \geq 1\}$. *Lemma 16.1.19* (p. 714, stronger) Chapter16/PolyfoldStructure · PolyfoldStructure.lemma_16_1_19: for equivalent structures, X is tame iff Y is; the text states one direction. *Definition 16.1.20* (p. 714, same) Chapter16/PolyfoldStructure · PolyfoldClass.IsTame Chapter16/PolyfoldStructure · PolyfoldClass.definition_16_1_20: a polyfold is tame if one, and then every, structure in c has X tame.

Proof of Lemma 16.1.13. *Step 1, the index on the orbit space.* $d_{(X,\beta)}(z) = d_X(x)$ for any x with $\beta(|x|) = z$ Chapter16/PolyfoldStructure · PolyfoldStructure.degIndex_eq, because d_X is invariant under morphisms and so descends to $|X|$. *Step 2, invariance.* For a generalized isomorphism $f : X \to Y$, $d_{|Y|} \circ |f| = d_{|X|}$: Proposition 11.1.2 (Part XI), which reduces to the invariance of the degeneracy index under local sc-diffeomorphisms (Part IV) through the legs of a diagram. *Step 3.* If $(X,\alpha) \sim (Y,\beta)$ through f, then $|f| = \beta^{-1} \circ \alpha$ (§XVI.1), so $d_{|Y|}(\beta^{-1}(z)) = d_{|Y|}(|f|(\alpha^{-1}(z))) = d_{|X|}(\alpha^{-1}(z))$.

Proof of Definitions 16.1.14 and 16.1.17. The index of the class is that of its chosen structure, and by Lemma 16.1.13 that of every structure in c. ∂Z is computed in any representative Chapter16/PolyfoldStructure · PolyfoldClass.mem_boundary_iff: $z \in \partial Z$ iff $x \in \partial X$ for one, or every, x with $\beta(|x|) = z$, ∂X being the objects of positive degeneracy index.

Proof of Lemma 16.1.19 and Definition 16.1.20. *Step 1.* Tameness of an ep-groupoid is invariant under equivalences in both directions (Chapter 11, Part XI): an equivalence is a local sc-diffeomorphism on objects, and tameness of an M-polyfold is local and invariant under sc-diffeomorphisms (Part II). *Step 2.* Along a diagram $X \leftarrow A \to Y$, X tame iff A tame iff Y tame. The text states "if X is tame then Y is"; the other direction follows by symmetry of the relation, so the formal statement is the equivalence. *Step 3.* Definition 16.1.20 is "the chosen structure is tame", which by Step 2 is "some, equivalently every, structure is tame".

Not formalized from §16.1. Proposition 16.1.12 (the lifts Z_i and their topology, which the raised polyfolds of §XVI.6 partly cover), Definition 16.1.15 and Remark 16.1.16 (subpolyfolds), Example 16.1.18, Definition 16.1.21 (faces of a polyfold) and Proposition 16.1.22 (§XVI.28).

Where Lean departs from the text. The text proves Lemma 16.1.13 by the invariance of the degeneracy index under generalized isomorphisms, as here; the formalization makes the orbit-space form of the index explicit, $d_{|X|}$, because the structure map β lives on $|X|$. For Lemma 16.1.19 the text proves one direction; the formal statement is an equivalence, which costs one application of the symmetry of Definition 16.1.2. Definition 16.1.17 is stated through the index as printed; the characterization of ∂Z through any representative is a lemma, used in §§XVI.15, XVI.22 and XVI.26 to translate boundary conditions between Z and a model.

XVI.6 The raised polyfolds $(Z,c)^i$ (p. 715)

Statement (restated) Chapter16/RaisedPolyfold · PolyfoldClass.raise. The text defines $(Z,c)^1$ as Z_1 with the structures $(X^1,\alpha|)$. The formalization: $(Z,c)^1$ is the space $|X_0^1|$ for the chosen structure (X_0,α_0) of c Chapter16/RaisedPolyfold · PolyfoldClass.RaisedSpace, with the polyfold class of the structures (X^1,α^1) for $(X,\alpha) \in c$ Chapter16/RaisedPolyfold · PolyfoldClass.raiseStructure, where $\alpha^1 : |X^1| \to |X_0^1|$ Chapter16/RaisedPolyfold · PolyfoldClass.raiseMap is induced by the raise of a generalized isomorphism realizing $(X,\alpha) \sim (X_0,\alpha_0)$. An injective continuous map $(Z,c)^1 \to Z$ has image $Z_1 = \beta(|X|_1)$ for every $(X,\beta) \in c$ Chapter16/RaisedPolyfold · PolyfoldClass.range_raiseIncl. $(Z,c)^i$ is the i-fold raise Chapter16/

RaisedPolyfold · PolyfoldClass.raiseN. *Restated* because $(Z,c)^1$ is a space with an injection into Z, not the subspace Z_1: its topology is that of $|X^1|$, which is finer than the subspace topology. The text means the former (the structure of X^1 carries its own topology); the formalization makes that explicit.

Proof. *Step 1, raising functors.* The index raise $X \mapsto X^1$ of Chapter 7 is functorial on equivalences and functors Chapter16/RaisedPolyfold · EpFunctor.liftOne_comp Chapter16/RaisedPolyfold · EpFunctor.liftOne_id, and the orbit map of Φ^1 is the restriction of that of Φ to $|X^1| \subseteq |X|$ Chapter16/RaisedPolyfold · EpFunctor.orbitIncl_orbitMap_liftOne.

Step 2, raising diagrams. A diagram $X \xleftarrow{F} A \xrightarrow{\Phi} Y$ raises to $X^1 \xleftarrow{F^1} A^1 \xrightarrow{\Phi^1} Y^1$ Chapter16/RaisedPolyfold · EPLocalization.Diagram.liftOne, the raise of an equivalence being an equivalence. Refinements raise to refinements Chapter16/RaisedPolyfold · EPLocalization.Diagram.IsRefinement.liftOne, including the identity refinement and composites Chapter16/RaisedPolyfold · EPLocalization.Diagram.isRefinement_liftOne_id Chapter16/RaisedPolyfold · EPLocalization.Diagram.isRefinement_liftOne_comp, because the weak fibre product of Chapter 10 commutes with the raise; so equivalent diagrams raise to equivalent diagrams Chapter16/RaisedPolyfold · EPLocalization.Diagram.Equiv.liftOne, and the orbit maps restrict Chapter16/RaisedPolyfold · EPLocalization.Diagram.orbitIncl_orbitMap_liftOne.

Step 3, raising generalized maps Chapter16/RaisedPolyfold · EPLocalization.EPLoc.liftOneHom: by Step 2 the raise is well defined on the localization, with $1^1 = 1$ and $(\mathfrak{g}\circ\mathfrak{f})^1 = \mathfrak{g}^1 \circ \mathfrak{f}^1$ Chapter16/RaisedPolyfold · EPLocalization.EPLoc.liftOneHom_id Chapter16/RaisedPolyfold · EPLocalization.EPLoc.liftOneHom_comp, so it keeps generalized isomorphisms Chapter16/RaisedPolyfold · EPLocalization.EPLoc.isGeneralizedIso_liftOneHom, and $|\mathfrak{f}^1| = |\mathfrak{f}|$ on $|X^1|$ Chapter16/RaisedPolyfold · EPLocalization.EPLoc.orbitIncl_orbitMap_liftOneHom. Two generalized maps with the same orbit map have raises with the same orbit map Chapter16/RaisedPolyfold · EPLocalization.EPLoc.orbitMap_liftOneHom_eq, because $|X^1| \to |X|$ is injective.

Step 4, the raised structure is a polyfold structure. X^1 has a paracompact orbit space Chapter16/RaisedPolyfold · PolyfoldStructure.liftOne_paracompact: $|X^1|$ is metrizable by Proposition 7.5.3 (Part VII), since X^1 is again an ep-groupoid. X^1 satisfies (ZHOU) Chapter16/RaisedPolyfold · PolyfoldStructure.liftOne_isZhou, by Propositions 7.4.8 and 7.4.9 (Part VII): the natural representations of X^1 are restrictions of those of X, and the sets Fix and their interiors behave under the raise as the propositions state.

Step 5, the maps α^1. For $(X,\alpha) \in c$, $\alpha^1 = |\mathfrak{f}^1|$ for a generalized isomorphism $\mathfrak{f}$ with $\alpha_0 \circ |\mathfrak{f}| = \alpha$; it does not depend on $\mathfrak{f}$ by Step 3 Chapter16/RaisedPolyfold · PolyfoldClass.raiseMap_eq, is compatible with composition Chapter16/RaisedPolyfold · PolyfoldClass.raiseMap_comp, and is a homeomorphism Chapter16/RaisedPolyfold · PolyfoldClass.isHomeomorph_raiseMap Chapter16/RaisedPolyfold · PolyfoldClass.raiseHomeomorph, being the orbit map of a generalized isomorphism. So (X^1,α^1) is a polyfold structure on $(Z,c)^1$, and all of them are equivalent through the raised isomorphisms Chapter16/RaisedPolyfold · PolyfoldClass.raiseStructure_equiv; the class does not depend on the representative it is built from Chapter16/RaisedPolyfold · PolyfoldClass.raise_eq.

Step 6, the inclusion. $(Z,c)^1 \to Z$ is $\alpha_0 \circ \iota$ with $\iota : |X_0^1| \to |X_0|$ Chapter16/RaisedPolyfold · PolyfoldClass.raiseIncl; it is injective Chapter16/RaisedPolyfold · PolyfoldClass.raiseIncl_injective (orbits of X^1 are orbits of X through level-1 points) and continuous Chapter16/RaisedPolyfold · PolyfoldClass.continuous_raiseIncl, it satisfies $\mathrm{incl} \circ \alpha^1 = \alpha \circ \iota$ for every structure Chapter16/RaisedPolyfold · PolyfoldClass.raiseIncl_raiseMap, and its image is Z_1 by Step 3. *Step 7, iteration.* $(Z,c)^0 = (Z,c)$ Chapter16/RaisedPolyfold · PolyfoldClass.raiseN_zero and $(Z,c)^{i+1} = ((Z,c)^i)^1$ Chapter16/RaisedPolyfold · PolyfoldClass.raiseN_succ, with the space and class carried together Chapter16/RaisedPolyfold · PolyfoldClass.PolyfoldPair. The text's index notation mI is not introduced; $Z_i = \alpha(|X_i|)$ for every structure.

Where Lean departs from the text. The text defines the raised polyfold by restricting each structure (X, α) to X^1 and gluing; that the restricted pairs are polyfold structures, and that they are equivalent, is left implicit. The formalization proves both: paracompactness through Proposition 7.5.3, (ZHOU) through Propositions 7.4.8 and 7.4.9, and the equivalence through the raise of generalized isomorphisms, which needs the raise to commute with weak fibre products so that refinements raise to refinements. The topology is the second point: the text writes $(Z, c)^1$ for a space whose underlying set is Z_1, and the formalization keeps the topology of $|X^1|$ rather than the subspace topology of $Z_1 \subseteq Z$, since only the former makes α^1 a homeomorphism; the inclusion into Z is continuous and injective but in general not an embedding.

XVI.7 Lemma 16.2.1 (2017: Lemma 16.3), p. 716: isomorphisms of polyfold structures and their tangents

Statement (does not hold as we read it; its orbit form and its form for an effective target hold). The text: if two isomorphisms $(f, \mathfrak{f}), (g, \mathfrak{g}) : (X, \alpha) \to (Y, \beta)$ of polyfold structures have $f = g$, then $\mathfrak{f} = \mathfrak{g}$ and $T\mathfrak{f} = T\mathfrak{g}$. Its proof: $|\mathfrak{f}| = |\mathfrak{g}|$, then Theorem 10.3.19. The statement in Lean `Chapter16/TangentLemma · PolyfoldStructure.Lemma_16_2_1`: f, g homeomorphisms, $\mathfrak{f}$, $\mathfrak{g}$ generalized isomorphisms with $f \circ \alpha = \beta \circ |\mathfrak{f}|$ and $g \circ \alpha = \beta \circ |\mathfrak{g}|$, and $f = g$; conclusion $\mathfrak{f} = \mathfrak{g}$ and $T\mathfrak{f} = T\mathfrak{g}$. It does not hold as we read it `Chapter16/TangentLemma · PolyfoldStructure.not_lemma_16_2_1`. *The orbit form* `Chapter16/TangentLemma · PolyfoldStructure.lemma_16_2_1_orbit`, under exactly the text's hypotheses: $|\mathfrak{f}| = |\mathfrak{g}|$ and $|T\mathfrak{f}| = |T\mathfrak{g}|$. *The effective form* `Chapter16/TangentLemma · PolyfoldStructure.lemma_16_2_1_of_isEffective`: if Y is effective, the text's conclusion holds.

The tangent of a generalized map. For $\mathfrak{f} = [X \xleftarrow{F} A \xrightarrow{\Phi} Y]$, $T\mathfrak{f}$ is the class of $[TX \xleftarrow{TF} TA \xrightarrow{T\Phi} TY]$ `Chapter16/TangentLemma · EPLocalization.EPLoc.tangentHom`. *Step 1.* The tangent of an ep-groupoid (Theorem 8.1.1, Part VIII) is taken in any universe `Chapter16/TangentLemma · EP.tangentAny`, and agrees with Chapter 8's where both are defined. *Step 2.* The tangent of a diagram is a diagram, since the tangent of an equivalence is an equivalence (Theorem 10.1.7, Part X) `Chapter16/TangentLemma · EPLocalization.Diagram.tangentAny`. *Step 3.* Refinements go to refinements `Chapter16/TangentLemma · EPLocalization.Diagram.IsRefinement.tangentAny`, compatibly with composition of refinements `Chapter16/TangentLemma · EPLocalization.Diagram.isRefinement_tangentAny_comp`, so equivalent diagrams have equivalent tangents `Chapter16/TangentLemma · EPLocalization.Diagram.Equiv.tangentAny`. *Step 4.* The result is a copy of Chapter 11's tangent functor on the localization `Chapter16/TangentLemma · EPLocalization.EPLoc.tangentHom_eq_tangentFunctor_map`, compatible with identities and composition `Chapter16/TangentLemma · EPLocalization.EPLoc.tangentHom_id` `Chapter16/TangentLemma · EPLocalization.EPLoc.tangentHom_comp`, and it preserves generalized isomorphisms `Chapter16/TangentLemma · EPLocalization.EPLoc.isGeneralizedIso_tangentHom`.

Proof that the lemma does not hold as we read it. *The example* `Chapter16/TangentLemma · PolyfoldStructure.lemma_16_2_1_counterexample` is the pair of §XVI.2. On the polyfold structure $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ of $\mathbb{R}$ with the trivial action, take $f = g = \mathrm{id}$, $\mathfrak{f} = [X \xleftarrow{1} X \xrightarrow{\mathrm{Inv}} X]$ and $\mathfrak{g} = [1_X]$. Both squares commute, since both orbit maps are the identity; the orbit space is paracompact; (ZHOU) holds; $f = g$; and $\mathfrak{f} \neq \mathfrak{g}$ (§XVI.2, Step 4). The last step of the text's proof is Theorem 10.3.19, which does not hold as we read it `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.not_theorem_10_3_19` (Part X). Whether $T\mathfrak{f} \neq T\mathfrak{g}$ in this example is not decided; their orbit maps agree, by the orbit form.

Proof of the orbit form. *Step 1* `Chapter16/TangentLemma · PolyfoldStructure.orbitMap_eq_of_comp`: $|\mathfrak{f}| = \beta^{-1} \circ f \circ \alpha$,

so $f = g$ gives $|\mathfrak{f}| = |\mathfrak{g}|$.

Step 2, reduction to an automorphism Chapter16/TangentLemma · EPLocalization.EPLoc.orbitMap_tangentHom_eq: if X or Y has (LR) and $|\mathfrak{f}| = |\mathfrak{g}|$, then $|T\mathfrak{f}| = |T\mathfrak{g}|$. With $\mathfrak{g}'$ an inverse of $\mathfrak{f}$, $e = \mathfrak{g}' \circ \mathfrak{g}$ is a generalized automorphism of X with $|e| = |\mathfrak{f}|^{-1} \circ |\mathfrak{g}| = 1$ (Proposition 10.3.11), and $\mathfrak{g} = \mathfrak{f} \circ e$; since T is a functor, $|T\mathfrak{g}| = |T\mathfrak{f}| \circ |Te|$, and it suffices that $|Te| = 1$. If only Y has (LR), the same argument runs with $\mathfrak{g} \circ \mathfrak{f}^{-1}$, an automorphism of Y.

Step 3, the automorphism Chapter16/TangentLemma · EPLocalization.EPLoc.orbitMap_tangentHom_eq_id. Represent e by a diagram $X \xleftarrow{F} A \xrightarrow{\Phi} X$ of equivalences; $|e| = 1$ says $|F| = |\Phi|$. A has (LR) by Proposition 10.1.9 (Part X), being equivalent to X. By Proposition 10.1.10 (1), two functors $A \to X$ with the same orbit map from an ep-groupoid with (LR) are connected locally by families of morphisms, and by Proposition 10.1.10 (2) these give $|TF| = |T\Phi|$, that is $|Te| = 1$.

Step 4. (ZHOU) gives (LR) for X (§XVI.1), so Steps 2–3 apply to every polyfold structure.

Proof of the effective form. If Y is effective, then $\mathfrak{f} = \mathfrak{g}$ by Remark 16.1.7 for an effective target (§XVI.2), and so $T\mathfrak{f} = T\mathfrak{g}$. Its hypotheses are satisfiable Chapter16/TangentLemma · PolyfoldStructure.lemma_16_2_1_of_isEffective_hypotheses: the point ep-groupoid is effective and satisfies (ZHOU).

What the rest of §16.2 needs. Only the orbit form: the gluing of $T(Z, c)$ uses the maps $|T\mathfrak{f}|$ (§XVI.8), and the remark before Definition 16.2.5 uses the analogous statement for forms (§XVI.10).

Where Lean departs from the text. The text's proof has two steps: $|\mathfrak{f}| = |\mathfrak{g}|$, then Theorem 10.3.19 for $\mathfrak{f} = \mathfrak{g}$, from which $T\mathfrak{f} = T\mathfrak{g}$ is immediate. The formalization keeps the first step, replaces the second by Proposition 10.1.10 at the level of tangent orbit maps, and proves the conclusion on orbit spaces only. For the tangent conclusion this is a genuinely different route: it never compares $\mathfrak{f}$ with $\mathfrak{g}$, only their effect on orbits, and it needs only (LR), which (ZHOU) gives. The full conclusion is recovered for an effective target, through the effective Theorem 10.3.19.

XVI.8 The tangent $T(Z, c)$, (16.5), Lemma 16.2.2, Definition 16.2.3 (2017: Lemma 16.4, Definition 16.14), and the projection τ_Z, pp. 716–719

Statement. *The construction* (pp. 716–718, same, with a base structure in place of the gluing; decision (3)): $T(Z, c) = |TX_0|$ for the chosen structure (X_0, α_0) of c Chapter16/TangentPolyfold · PolyfoldClass.TangentSpace, with the orbit space topology Chapter16/TangentPolyfold · PolyfoldClass.tangentSpaceTopology; for $(X, \alpha) \in c$, $T\alpha = |T\mathfrak{f}| : |TX| \to T(Z, c)$ for a generalized isomorphism $\mathfrak{f} : X \to X_0$ with $\alpha_0 \circ |\mathfrak{f}| = \alpha$ Chapter16/TangentPolyfold · PolyfoldClass.tangentMap. *(16.5)* (same) Chapter16/TangentPolyfold · PolyfoldClass.tangentMap_comp: $T\beta \circ |Th| = T\alpha$ for every generalized isomorphism $h : X \to Y$ with $\beta \circ |h| = \alpha$. *Lemma 16.2.2* (p. 718, same) Chapter16/TangentPolyfold · PolyfoldClass.lemma_16_2_2: (1) every $T\alpha$ is a homeomorphism; (2) the topology of $T(Z, c)$ is the unique one for which one, and then every, $T\alpha$ is a homeomorphism; (3) $T(Z, c)$ is metrizable. *Definition 16.2.3* (p. 719, restated) Chapter16/TangentPolyfold · PolyfoldClass.tangent Chapter16/TangentPolyfold · PolyfoldClass.definition_16_2_3: the tangent $(T(Z, c), Tc)$ is the polyfold whose class contains every $(TX, T\alpha)$, $(X, \alpha) \in c$. *The projection* (p. 717, restated) Chapter16/RaisedPolyfold · PolyfoldClass.tangentProj: $\tau_Z : T(Z, c) \to (Z, c)^1$ with $\tau_Z \circ T\alpha = \alpha^1 \circ |\tau_X|$.

Proof of the construction. *Step 1, $T\alpha$ does not depend on $\mathfrak{f}$* Chapter16/TangentPolyfold · PolyfoldClass.

`tangentMap_eq`. Two generalized isomorphisms $\mathfrak{f}, \mathfrak{f}'$ with $\alpha_0 \circ |\mathfrak{f}| = \alpha = \alpha_0 \circ |\mathfrak{f}'|$ have the same orbit map $\alpha_0^{-1} \circ \alpha$, hence the same $|T\mathfrak{f}|$ by the orbit form of Lemma 16.2.1 (§XVI.7, Steps 2–3). The text says "the unique $\mathfrak{f}$"; $\mathfrak{f}$ need not be unique (§XVI.2), but $|T\mathfrak{f}|$ is, and it is all that the construction uses. *Step 2, (16.5).* Let $\mathfrak{f}_Y : Y \to X_0$ realize (Y, β). Then $\mathfrak{f}_Y \circ h$ realizes (X, α), since $\alpha_0 \circ |\mathfrak{f}_Y| \circ |h| = \beta \circ |h| = \alpha$. By Step 1, $T\alpha = |T(\mathfrak{f}_Y \circ h)| = |T\mathfrak{f}_Y| \circ |Th| = T\beta \circ |Th|$, using that T is a functor on the localization (§XVI.7). *Step 3, homeomorphisms* `Chapter16/TangentPolyfold · PolyfoldClass.isHomeomorph_tangentMap` `Chapter16/TangentPolyfold · PolyfoldClass.tangentHomeomorph`: the orbit map of a generalized isomorphism is a homeomorphism `Chapter16/TangentPolyfold · EPLocalization.EPLoc.isHomeomorph_orbitMap_of_isGeneralizedIso` (its inverse is the orbit map of the inverse, by Proposition 10.3.11), and $T\mathfrak{f}$ is a generalized isomorphism.

Proof of Lemma 16.2.2. (1) is Step 3. (2): if τ is a topology on the set $T(Z, c)$ for which some $T\alpha$ is a homeomorphism, then the open sets of τ are exactly the sets whose preimage under $T\alpha$ is open in $|TX|$; the same holds for the given topology by (1); so the two agree. Since all $T\alpha$ are homeomorphisms for the given topology, "one" and "every" agree. (3): Theorem 8.1.15 (Part VIII) makes $|TX_0|$ metrizable `Chapter8/TangentOrbit · metrizableSpace_tangent_orbitSpace`.

Proof of Definition 16.2.3. *Step 1, $(TX, T\alpha)$ is a polyfold structure* `Chapter16/TangentPolyfold · PolyfoldClass.tangentStructure`. $|TX|$ is paracompact by Theorem 8.1.15 `Chapter16/TangentPolyfold · PolyfoldStructure.tangent_paracompact` `Chapter8/TangentOrbit · theorem_8_1_15`, and TX satisfies (ZHOU) by Proposition 8.1.16 in its 2021 form `Chapter16/TangentPolyfold · PolyfoldStructure.tangent_isZhou` `Chapter8/TangentZhouReflect2021 · proposition_8_1_16_2021` (Part VIII). *Step 2, they are equivalent* `Chapter16/TangentPolyfold · PolyfoldClass.tangentStructure_equiv`: for $h : X \to Y$ with $\beta \circ |h| = \alpha$, Th is a generalized isomorphism with $T\beta \circ |Th| = T\alpha$ by (16.5). *Step 3.* Tc is the class of $(TX_0, T\alpha_0)$, and by Step 2 every $(TX, T\alpha)$ lies in it. *Readings.* $T(Z, c)$ lives one universe up, as TX does in Chapter 8. "Restated" because the text defines $T(Z, c)$ as a disjoint union modulo identifications; the base structure gives a canonically homeomorphic space (Lemma 16.2.2 (2)).

Proof of the projection τ_Z. *Step 1, naturality of τ for functors* `Chapter16/RaisedPolyfold · EpFunctor.orbitMap_tangentProj_tangent`: for a functor Φ, $|\tau_Y| \circ |T\Phi| = |\Phi^1| \circ |\tau_X|$, from the pointwise identity $\tau \circ T\Phi = \Phi^1 \circ \tau$ on objects. *Step 2, for diagrams and generalized maps* `Chapter16/RaisedPolyfold · EPLocalization.Diagram.orbitMap_tangentProj_tangentAny` `Chapter16/RaisedPolyfold · EPLocalization.EPLoc.orbitMap_tangentProj_tangentHom`: through both legs, with the raised generalized maps of §XVI.6. *Step 3, the definition*: $\tau_Z = |\tau_{X_0}| : |TX_0| \to |X_0^1|$; it is continuous `Chapter16/RaisedPolyfold · PolyfoldClass.continuous_tangentProj`, and for every $(X, \alpha) \in c$, $\tau_Z \circ T\alpha = \alpha^1 \circ |\tau_X|$ `Chapter16/RaisedPolyfold · PolyfoldClass.tangentProj_tangentMap` by Step 2 applied to a realizing $\mathfrak{f}$. Composed with the inclusion $(Z, c)^1 \to Z$, it is the text's square into Z `Chapter16/RaisedPolyfold · PolyfoldClass.raiseIncl_tangentProj_tangentMap`.

Where Lean departs from the text. The text constructs $T(Z, c)$ as the disjoint union of the $|TX|$ over all $(X, \alpha) \in c$, modulo the identifications by $|T\mathfrak{f}|$ for the unique $\mathfrak{f}$, and Lemma 16.2.2 then puts a topology on the quotient. The formalization takes one structure as base point and defines $T\alpha$ as a map into it; the identifications of the text become the identity (16.5), and Lemma 16.2.2 (2) shows that the base point does not matter topologically. The two constructions give canonically homeomorphic spaces. The uniqueness of $\mathfrak{f}$, which the text uses to make the identifications consistent, is replaced by the uniqueness of $|T\mathfrak{f}|$, which is what consistency of the identifications needs. The projection τ_Z is built the same way, as $|\tau_{X_0}|$ with a naturality square for every structure.

XVI.9 Theorem 16.2.4 (2017: Theorem 16.1), p. 719: the tangent functor

Statement (narrower). *On PS* `Chapter16/TangentFunctor · PSCat.tangentFunctor`: $(X,\alpha) \mapsto (TX, T\alpha)$ and $(f,\mathfrak{f}) \mapsto (Tf, T\mathfrak{f})$ is a functor from PS to PS one universe up. *On sc-smooth maps* `Chapter16/TangentFunctor · PolyfoldClass.theorem_16_2_4`: $T\hat{f} = [(Tf, T\mathfrak{f})]$ is well defined for every sc-smooth map between any polyfolds, with $Tf \circ T\alpha = T\beta \circ |T\mathfrak{f}|$, and $T1 = 1$. *On effective polyfolds* `Chapter16/TangentEffective · theorem_16_2_4_eff`: T is a functor on the category of effective polyfolds `Chapter16/TangentEffective · EffPolyfold.tangentFunctor`. *Narrower* because there is no category of all polyfolds as we read Definition 16.1.10 (§XVI.4); on effective polyfolds, where the composite is defined, it is the text's statement with no added hypothesis.

Proof on PS. *Step 1.* A generalized map alone determines a morphism of PS one universe up `Chapter16/TangentFunctor · PSHom.ofGen'`, compatibly with composition `Chapter16/TangentFunctor · PSHom.ofGen'_comp`: its continuous map is forced by Step 1 of PS (§XVI.4). *Step 2.* $T(X,\alpha)$ is the structure $(TX, T\alpha)$ on $|TX|$ itself, with $T\alpha$ the identity of $|TX|$ for the structure taken as its own base `Chapter16/TangentFunctor · PolyfoldStructure.tangentSelf`, and the functor laws are those of T on the localization (§XVI.7). *Step 3.* Isomorphisms over the identity go to isomorphisms over the identity `Chapter16/TangentFunctor · PolyfoldClass.tangent_isIdIso`, by (16.5).

Proof on sc-smooth maps. *Step 1, representatives* `Chapter16/TangentFunctor · PolyfoldClass.MapRep.tangent`: for a representative $(f,\mathfrak{f}) : (X,\alpha) \to (Y,\beta)$, $(Tf, T\mathfrak{f}) : (TX, T\alpha) \to (TY, T\beta)$ with $Tf = T\beta \circ |T\mathfrak{f}| \circ (T\alpha)^{-1}$. *Step 2, well defined* `Chapter16/TangentFunctor · PolyfoldClass.MapRep.tangent_equiv`: a square $(f',\mathfrak{f}') \circ (h, \mathrm{Id}) = (k, \mathrm{Id}) \circ (f,\mathfrak{f})$ goes under T to a square with the isomorphisms over the identity (Th, Id) and (Tk, Id), by Step 3 on PS. No effectiveness is used, because Definition 16.1.6 carries h and k as data. *Step 3, the class* `Chapter16/TangentFunctor · PolyfoldClass.ScMap.tangent` with its map determined by $Tf \circ T\alpha = T\beta \circ |T\mathfrak{f}|$ `Chapter16/TangentFunctor · PolyfoldClass.ScMap.tangent_toFun` `Chapter16/TangentFunctor · PolyfoldClass.ScMap.tangent_mk`. *Step 4.* $T1 = 1$ `Chapter16/TangentFunctor · PolyfoldClass.ScMap.tangent_id`, since $T[1_X] = [1_{TX}]$; and the tangent of a composite of representatives is the composite of the tangents `Chapter16/TangentFunctor · PolyfoldClass.ScMap.tangent_mk_comp`. *Step 5, composites of classes.* For an effective middle polyfold d whose tangent is effective, $T(\hat{g} \circ \hat{f}) = T\hat{g} \circ T\hat{f}$ `Chapter16/TangentFunctor · PolyfoldClass.ScMap.tangent_comp`: the composite on the right is defined only if Td is effective, which the next proof supplies.

The tangent of an effective ep-groupoid with (ZHOU) is effective `Chapter16/TangentEffective · EpGroupoid.isEffective_tangentEpGroupoid`. *Statement:* for X effective with (ZHOU), the tangent ep-groupoid TX (Theorem 8.1.1) is effective (Definition 7.2.1). This is not printed; the text uses it implicitly when it calls T a functor on the polyfold category.

Proof. Suppose an isotropy element g of TX at a tangent vector a, $g \neq 1$, acts trivially near a: there is an open $W \ni a$ in the object space of TX on which the local diffeomorphism of g is the identity.

Step 1, the base morphism is not a unit. Let $\varphi \in X$ be the base morphism of g (the morphism of X under the tangent morphism g, which acts on $T_{s(\varphi)}X$ by $T\varphi$). If φ were a unit, then g would be a unit `Chapter16/TangentEffective · EpGroupoid.tangent_isotropy_eq_one`, since a tangent morphism over a unit with the same source and target is the unit. So $\varphi \neq 1$.

Step 2, separation. The morphism space is Hausdorff, so choose disjoint open sets $A \ni \varphi$ and $B \ni 1_{s(\varphi)}$.

Step 3, a sheet of the source map. s is a local sc-diffeomorphism at φ; let σ be its local inverse on a neighbourhood V of $s(\varphi)$, with $\sigma(s(\varphi)) = \varphi$, and $T\sigma$ its tangent.

Step 4, an open set of tangent vectors. Let $W_0 \subseteq W$ be the tangent vectors whose base morphism lies in A, in the sheet, and has source whose unit lies in B, and let O be the set of tangent vectors b over V with $T\sigma(b) \in W_0$. O is open in TX and contains a.

Step 5, a smooth point. O is open in TX, whose topology is that of level 1; smooth points are dense, so pick $b \in O$ over a smooth point. Its base point $x = \tau(b)$ lies in X_∞ `Chapter16/TangentEffective · EpGroupoid.levelVal_tangentProj_mem_smoothSet`, because the projection $TX \to X^1$ is sc-smooth and $(X^1)_\infty = X_\infty$.

Step 6, the morphism at x. Put $\psi = \sigma(x)$, a morphism of X from x, lying in A. By the choice of W_0, the tangent morphism $T\psi$ acts as the identity on an open set of T_xX around b, through the triviality of g's action on W transported along the sheet.

Step 7, linearity. $T\psi : T_xX \to T_{t(\psi)}X$ is linear and equals the identity on an open subset of the fibre, so $T\psi = \mathrm{id}$ on the whole fibre and $t(\psi) = x$ `Chapter16/TangentEffective · linearMap_eq_self_of_eqOn_isOpen`.

Step 8, (ZHOU) and effectiveness. $\psi \in G_x$ with $T_x\psi = \mathrm{id}$ at the smooth point x; (ZHOU) says that then ψ acts as the identity near x, and effectiveness of X gives $\psi = 1_x$. But $\psi \in A$ and $1_x \in B$, with $A \cap B = \emptyset$. This contradiction shows $g = 1$.

Why both hypotheses. Effectiveness of X is a statement at level 0, while open sets of TX see only level 1; an isotropy element acting trivially near a tangent vector must first be moved to a smooth point, where (ZHOU) converts the triviality of the linearized action into triviality near the point. **For polyfolds** `Chapter16/TangentEffective · PolyfoldClass.IsEffective.tangent`: the tangent of an effective polyfold is effective, since (ZHOU) is part of every polyfold structure (Definition 16.1.1 (2)); no hypothesis is added.

Proof on effective polyfolds. *Step 1* `Chapter16/TangentEffective · EffPolyfold.tangent`: an effective polyfold has an effective tangent, an object of the category. *Step 2* `Chapter16/TangentEffective · PolyfoldClass.ScMap.tangent_comp_of_isEffective`: $T(\hat{g} \circ \hat{f}) = T\hat{g} \circ T\hat{f}$ whenever the middle polyfold is effective, from Step 5 of the previous proof with the effectiveness of Td supplied by the theorem above. *Step 3*: with $T1 = 1$ this is a functor, and with $Tf \circ T\alpha = T\beta \circ |T\mathfrak{f}|$ it is Theorem 16.2.4.

Where Lean departs from the text. The text states Theorem 16.2.4 for the polyfold category of Definition 16.1.11 and treats functoriality as evident from the functoriality of T on generalized maps. The formalization proves it at three levels: on PS, where it is the functoriality on the localization; on sc-smooth maps of all polyfolds, where well-definedness needs only that Definition 16.1.6 carries its isomorphisms as data; and on the category of effective polyfolds, where composites are defined. At the third level a statement the text does not print is needed, that the tangent of an effective polyfold is effective, since otherwise the tangent of a morphism of effective polyfolds would not be a morphism of the category. Its proof is new, and it uses (ZHOU), part of every polyfold structure, in an essential way.

XVI.10 The remark before Definition 16.2.5, Definition 16.2.5 and Theorem 16.2.6 (2017: Definition 16.15, Theorem 16.2), pp. 720–721: sc-differential forms

Statement. *The remark* (p. 720, same) `Chapter16/FormsIndependence · EPLocalization.EPLoc.formsPush_eq_of_orbitMap_eq_or`: for generalized isomorphisms $h, h' : X \to Y$ with $|h| = |h'|$, $h_* = h'_*$ on $\Omega^k_{ep,\infty}$ if X or Y has (LR); likewise for pull-backs `Chapter16/FormsIndependence · EPLocalization.EPLoc.formsPullback_eq_of_orbitMap_eq_or`. *Definition 16.2.5* (p. 720, restated) `Chapter16/PolyfoldForms · PolyfoldClass.ScForm`: $\Omega^k_{sc}(Z, c)$

is the set of classes of pairs $(\omega,(X,\alpha))$. *Theorem 16.2.6* (p. 721, same), in two parts: the complex Chapter16/PolyfoldForms · PolyfoldClass.theorem_16_2_6_complex and the pull-back by sc-smooth maps Chapter16/FormsPullback · PolyfoldClass.theorem_16_2_6_pullback.

Proof of the remark. It is Proposition 11.2.9 (Part XI) Chapter11/OrbitForms · formsPush_eq_of_orbitMap_eq, which takes (LR) on the source; with (LR) on the target, Proposition 10.3.16 moves it to the source, being invariant under equivalences. For polyfold structures (ZHOU) gives (LR) (§XVI.1), so $h_*\omega$ depends only on α and β, since $|h| = \beta^{-1}\circ\alpha$. The mechanism is that of §XVI.7: generalized isomorphisms with one orbit map differ by an automorphism with trivial orbit map, which acts trivially on forms by Proposition 10.1.10. Pull-back by a generalized isomorphism is push-forward by its inverse Chapter16/PolyfoldForms · EPLocalization.EPLoc.formsPush_eq_formsPullback.

Proof of Definition 16.2.5. *Step 1, representatives* Chapter16/PolyfoldForms · PolyfoldClass.FormRep: pairs $(\omega,(X,\alpha))$ with $(X,\alpha)\in c$ and $\omega\in\Omega^k_{ep,\infty}(X)$. *Step 2, the relation* Chapter16/PolyfoldForms · PolyfoldClass.FormRep.Rel: $(\omega,(X,\alpha))\sim(\omega',(Y,\beta))$ if $\omega' = h_*\omega$ for some, equivalently (by the remark) every, generalized isomorphism h with $\beta\circ|h| = \alpha$. *Step 3, an equivalence relation*: reflexive by $1_* = 1$ Chapter16/PolyfoldForms · EPLocalization.EPLoc.formsPush_id Chapter16/PolyfoldForms · PolyfoldClass.FormRep.Rel.refl (Theorem 11.2.1 (3), Part XI), symmetric by $(h^{-1})_* = (h_*)^{-1}$ Chapter16/PolyfoldForms · PolyfoldClass.FormRep.Rel.symm, transitive by $(h'\circ h)_* = h'_*\circ h_*$ Chapter16/PolyfoldForms · EPLocalization.EPLoc.formsPush_comp, with composites and identities of generalized isomorphisms again generalized isomorphisms Chapter16/PolyfoldForms · EPLocalization.EPLoc.isGeneralizedIso_comp Chapter16/PolyfoldForms · EPLocalization.EPLoc.isGeneralizedIso_id. *Step 4, classes* Chapter16/PolyfoldForms · PolyfoldClass.ScForm.mk, injective on each structure Chapter16/PolyfoldForms · PolyfoldClass.ScForm.mk_injective, and related through push-forward Chapter16/PolyfoldForms · PolyfoldClass.ScForm.mk_eq_mk_formsPush. *Step 5, the representative on a structure* Chapter16/PolyfoldForms · PolyfoldClass.ScForm.toStructure: for $(X,\alpha)\in c$ and a class τ, the unique ω with $\tau = [(\omega,(X,\alpha))]$, obtained by pushing any representative along a realizing isomorphism Chapter16/PolyfoldForms · PolyfoldClass.ScForm.exists_iso Chapter16/PolyfoldForms · PolyfoldClass.ScForm.formsPush_eq Chapter16/PolyfoldForms · PolyfoldClass.ScForm.repOn. This gives a bijection $\Omega^k_{sc}(Z,c)\cong\Omega^k_{ep,\infty}(X)$ Chapter16/PolyfoldForms · PolyfoldClass.ScForm.equivStructure, and representatives on related structures are related by pull-back Chapter16/BranchedSuborbifold · PolyfoldClass.ScForm.toStructure_eq_formsPullback. *Step 6, the vector space structure.* It is transported from the chosen structure Chapter16/PolyfoldForms · PolyfoldClass.ScForm.out and satisfies the text's formulas for every representative: $[(\omega,q)]+[(\omega',q)] = [(\omega+\omega',q)]$ Chapter16/PolyfoldForms · PolyfoldClass.ScForm.mk_add and $r[(\omega,q)] = [(r\omega,q)]$ Chapter16/PolyfoldForms · PolyfoldClass.ScForm.mk_smul Chapter16/PolyfoldForms · PolyfoldClass.ScForm.mk_add_smul, because h_* is linear. So each bijection of Step 5 is linear Chapter16/PolyfoldForms · PolyfoldClass.ScForm.toStructure Chapter16/PolyfoldForms · PolyfoldClass.ScForm.equivStructure. *Restated* because the text's representatives are written $(\omega,(X,\alpha))$ informally and the vector space operations on classes are defined by these formulas; the formalization defines them on one structure and proves the formulas.

Proof of Theorem 16.2.6, the complex. *Step 1, d on representatives* Chapter16/PolyfoldForms · PolyfoldClass.ScForm.dRep Chapter16/PolyfoldForms · PolyfoldClass.ScForm.dFun: $d[(\omega,(X,\alpha))] = [(d\omega,(X,\alpha))]$. *Step 2, well defined* Chapter16/PolyfoldForms · PolyfoldClass.ScForm.formsPush_d: h_* commutes with d for a generalized isomorphism (Theorem 11.2.1, Part XI), so related representatives have related differentials. *Step 3, the operator* Chapter16/PolyfoldForms · PolyfoldClass.ScForm.d Chapter16/PolyfoldForms · PolyfoldClass.ScForm.d_apply is linear, and the formula holds for every $(X,\alpha)\in c$ Chapter16/PolyfoldForms · PolyfoldClass.ScForm.d_mk. *Step 4, $d\circ d = 0$* Chapter16/PolyfoldForms · PolyfoldClass.ScForm.d_d Chapter16/PolyfoldForms · PolyfoldClass.ScForm.d_comp_d, from $d\circ d = 0$ on $\Omega^*_{ep,\infty}(X)$ (Part XI). *Step 5.* $\omega\mapsto[(\omega,(X,\alpha))]$ is a linear isomorphism of complexes.

Proof of Theorem 16.2.6, the pull-back. *Statement:* an sc-smooth map $\widehat{f} : (Z, c) \to (W, d)$ pulls back forms by $\widehat{f}^*[(\omega', (Y, \beta))] = [(\mathfrak{f}^*\omega', (X, \alpha))]$ for a representative $(f, \mathfrak{f}) : (X, \alpha) \to (Y, \beta)$ `Chapter16/FormsPullback · PolyfoldClass.ScMap.pullback Chapter16/FormsPullback · PolyfoldClass.ScMap.pullback_mk`, linearly, commuting with d, functorially. *Step 1, on a representative* `Chapter16/FormsPullback · PolyfoldClass.MapRep.pullback`: $\omega' \mapsto \mathfrak{f}^*\omega'$ through the representatives on the structures of the representative (Step 5 above), a linear map. *Step 2, independence of the representative* `Chapter16/FormsPullback · PolyfoldClass.MapRep.pullback_eq_of_equiv`, with no effectiveness: for related representatives with the square $\mathfrak{f}' \circ h = k \circ \mathfrak{f}$, the key identity is $\mathfrak{f} = k^{-1} \circ \mathfrak{f}' \circ h$, and pull-back is contravariantly functorial on the localization (Theorem 11.2.1), so $\mathfrak{f}^* = h^* \circ \mathfrak{f}'^* \circ (k^{-1})^*$; but forms on a class are identified exactly along h and k (Definition 16.2.5), whichever k the square uses. So the non-uniqueness of §XVI.4 does not reach pull-backs. *Step 3, d* `Chapter16/FormsPullback · PolyfoldClass.ScMap.d_comp_pullback`: $d \circ \widehat{f}^* = \widehat{f}^* \circ d$, because generalized maps commute with d (Theorem 11.2.1 (2), Part XI). *Step 4, functoriality.* $1^* = 1$ `Chapter16/FormsPullback · PolyfoldClass.ScMap.pullback_id`; $(\widehat{g} \circ \widehat{f})^* = \widehat{f}^* \circ \widehat{g}^*$ on representatives `Chapter16/FormsPullback · PolyfoldClass.ScMap.pullback_mk_comp`, and on classes for an effective middle polyfold, where the composite is defined `Chapter16/FormsPullback · PolyfoldClass.ScMap.pullback_comp`. With the complex this is Theorem 16.2.6 in full.

Not formalized from §16.2. Remark 16.2.7 (§XVI.28).

Where Lean departs from the text. The remark before Definition 16.2.5 cites Proposition 11.2.9; the formalization states it with (LR) on either side, since in a polyfold both ep-groupoids have (LR) and the comparison is sometimes made from the target. Definition 16.2.5 is restated: the text defines the vector space operations on classes by the formulas $[(\omega, q)] + [(\omega', q)] = [(\omega + \omega', q)]$ and $r[(\omega, q)] = [(r\omega, q)]$, which presupposes that any two classes have representatives on a common structure and that the result does not depend on it; the formalization transports the structure from one structure of c and proves the formulas for all. The pull-back half of Theorem 16.2.6 is printed with a one-line proof; the formalization isolates the one point that needs care, independence of the representative of $\widehat{f}$, and shows it uses no effectiveness.

XVI.11 Definitions 16.3.1, 16.3.3–16.3.5 and Lemma 16.3.2 (2017: Definitions 16.16–16.19), pp. 721–723: strong polyfold bundles

Statement. *Definition 16.3.1* (p. 721, same) `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure`: a strong polyfold bundle structure for a continuous surjection $p : Y \to Z$ is a strong bundle $(P : W \to X, \mu)$ over an ep-groupoid X with (ZHOU), paracompact $|X|$ and the local regularity property of Definition 8.4.2, together with homeomorphisms $\Gamma : |W| \to Y$ and $\gamma : |X| \to Z$ with $p \circ \Gamma = \gamma \circ |P|$. *Lemma 16.3.2* (p. 722, same; new in 2021) `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.lemma_16_3_2`: Y and Z are metrizable. *Definition 16.3.3* (p. 722, same) `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.Equiv`: two structures are equivalent if a generalized strong bundle isomorphism F covering a generalized isomorphism f has $\gamma' \circ |f| = \gamma$ and $\Gamma' \circ |F| = \Gamma$. *The uniqueness after Definition 16.3.3* (p. 723, does not hold as we read it; holds for an effective target base) `Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.Uniqueness_16_3_3 Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.not_uniqueness_16_3_3 Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.uniqueness_16_3_3_of_isEffective`. *The induced polyfold* (p. 723, same) `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.toPolyfoldStructure`. *Definition 16.3.4* (p. 723, same) `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundle`: a strong polyfold bundle

is p with an equivalence class $\bar{c}$ of structures. *Definition 16.3.5* (p. 723, (1) same, (2)–(3) restated) `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundle.IsTame` `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundle.IsWeaklyFaceStructured` `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundle.IsFaceStructured`: tame, weakly face-structured, face-structured.

Proof of Definition 16.3.1's consequences. *Step 1, p is determined* `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.p_eq`: $p = \gamma \circ |P| \circ \Gamma^{-1}$, and in orbit form $p(\Gamma|w|) = \gamma(|P(w)|)$ `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.p_Γ`. *Step 2* `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.continuous_p` `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.surjective_p`: p is continuous, as $|P|$ is (Part VIII), and surjective, as P is surjective on objects. The text assumes both; they follow.

Proof of Lemma 16.3.2. Z: $|X|$ is paracompact, and Theorem 7.3.1 with Chapter 2's metrization theorem make it metrizable (§XVI.1); γ is a homeomorphism. Y: Proposition 8.4.1 (Part VIII) shows that $|W|$ is paracompact and metrizable for a strong bundle over an ep-groupoid with paracompact orbit space and the local regularity property, through the metrizability of orbit spaces `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.lemma_16_3_2`; Γ is a homeomorphism.

Proof that Definition 16.3.3 is an equivalence relation. *Step 1, orbit maps of generalized strong bundle maps compose.* This is the strong bundle form of Proposition 10.3.11. On a diagram of strong bundle equivalences the orbit maps compose `Chapter16/StrongPolyfoldBundle · SBLocalization.SDiagram.orbitMap_comp'`, the proof running on the weak fibre product $W'' \times_{W'} V''$ of the middle bundles through the action μ of Chapter 10, and the identity diagram has the identity orbit map `Chapter16/StrongPolyfoldBundle · SBLocalization.SDiagram.orbitMap_id'`. So for generalized strong bundle maps `Chapter16/StrongPolyfoldBundle · SBLocalization.GenSBMap.orbitMap_comp` `Chapter16/StrongPolyfoldBundle · SBLocalization.SEPLoc.orbitMap_comp` `Chapter16/StrongPolyfoldBundle · SBLocalization.SEPLoc.orbitMap_id`, also for inverses `Chapter16/StrongPolyfoldBundle · SBLocalization.SEPLoc.orbitMap_inv_comp` `Chapter16/StrongPolyfoldBundle · SBLocalization.SEPLoc.orbitMap_comp_inv`, and for the base maps `Chapter16/StrongPolyfoldBundle · SBLocalization.GenSBMap.base_comp` `Chapter16/StrongPolyfoldBundle · SBLocalization.GenSBMap.baseOrbitMap_comp` `Chapter16/StrongPolyfoldBundle · SBLocalization.SEPLoc.baseOrbitMap_comp`. *Step 2* `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.Equiv.refl` `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.Equiv.symm` `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.Equiv.trans`: as in §XVI.1, by the identity, inverses and composites, with Step 1 for the two conditions. The quotient `Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.setoid` is the type of strong polyfold bundles for p. *Step 3, the base condition is redundant* `Chapter16/PolyfoldSections · StrongPolyfoldBundleStructure.Γ_base` `Chapter16/PolyfoldSections · StrongPolyfoldBundleStructure.equiv_iff_exists_Γ`: $\Gamma' \circ |F| = \Gamma$ implies $\gamma' \circ |f| = \gamma$, because $\gamma \circ |P| = p \circ \Gamma = p \circ \Gamma' \circ |F| = \gamma' \circ |P'| \circ |F| = \gamma' \circ |f| \circ |P|$ and $|P|$ is surjective.

Proof that the uniqueness after Definition 16.3.3 does not hold as we read it. The text says that for equivalent structures f and F are uniquely determined, by $|f| = \gamma'^{-1} \circ \gamma$, (ZHOU), Theorem 10.3.19 and Proposition 10.4.7. The claim in Lean: any two generalized strong bundle isomorphisms satisfying both conditions agree, on the bases and on the bundles. *Step 1, the example* `Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.z3Structure`: the trivial bundle $\mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$ over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$, with $\mathbb{Z}/3$ acting trivially on base and fibres, and $\Gamma = \gamma = \mathrm{id}$ after the identifications of §XVI.1. It satisfies (ZHOU), has a paracompact orbit space and the local regularity property `Chapter10/LocalizationLRCounterexample · LocalizationLRCounterexample.isLR_z3EpBundle`, so it is a strong polyfold bundle structure in the exact sense of Definition 16.3.1; its class is a strong polyfold bundle

Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.z3Bundle. *Step 2, two isomorphisms.* The generalized strong bundle isomorphisms $[1_W]$ and $[W \xleftarrow{1} W \xrightarrow{\mathrm{Inv}} W]$ of Remark 10.5.5 (Part X) induce the identity on both orbit spaces Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.z3_orbitMap_id Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.z3_baseOrbitMap_id, so both satisfy the conditions of Definition 16.3.3. *Step 3, they differ, already on the bases* Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.uniqueness_16_3_3_counterexample: the base maps are the pair of §XVI.2. The base is not effective Chapter16/PolyfoldBundleUniqueness · StrongPolyfoldBundleStructure.not_isEffective_z3Structure. This is the bundle form of §XVI.2.

Proof of the effective form. For an effective target base, f is unique by Theorem 10.3.19 for an effective target (§XVI.2), and F is then unique by Theorem 10.5.7 for an effective base (Part X), which determines a generalized strong bundle isomorphism by its orbit map and base. Nothing later in §16.3 uses the uniqueness: the relations of §§16.3, 16.5 and 16.6 are proved independent of the isomorphisms (decision (5)).

Proof of the induced polyfold and Definition 16.3.4. *Step 1* Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.toPolyfoldStructure: (X, γ) is a polyfold structure on Z, its two conditions being part of Definition 16.3.1. *Step 2* Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundleStructure.toPolyfoldStructure_equiv: equivalent bundle structures give equivalent polyfold structures, through the base f of F (Remark 10.5.3). *Step 3* Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundle.polyfold: a strong polyfold bundle has a polyfold (Z, c), the class of the bases of its structures.

Proof of Definition 16.3.5. (1) A strong polyfold bundle is tame if (Z, c) is (Definition 16.1.20), equivalently if the base of every structure is tame Chapter16/StrongPolyfoldBundle · StrongPolyfoldBundle.isTame_iff, by Lemma 16.1.19 (§XVI.5). (2), (3): through representatives Chapter16/StrongPolyfoldBundle · PolyfoldClass.IsWeaklyFaceStructured Chapter16/StrongPolyfoldBundle · PolyfoldClass.IsFaceStructured, since Definition 16.1.21 (faces of a polyfold) is not formalized. Face structure holds for one representative iff for all Chapter16/StrongPolyfoldBundle · PolyfoldClass.isFaceStructured_iff, by Remark 11.1.7 (Part XI); for weak face structure the invariance under generalized isomorphisms is not formalized, and "some representative" is the reading (§XVI.28).

Where Lean departs from the text. The text lists the continuity and surjectivity of p among the data; the formalization derives them from the structure. In Definition 16.3.3 the text imposes both $\gamma' \circ |f| = \gamma$ and $\Gamma' \circ |F| = \Gamma$; the second implies the first, which the formalization proves and then uses to state every class-level relation of §§16.3, 16.5, 16.6 through Γ alone. For the uniqueness after Definition 16.3.3, the text's argument (Theorem 10.3.19 for f, then Proposition 10.4.7 for F) is tested step by step, as in §XVI.2; the first step does not hold as we read it on the $\mathbb{Z}/3$ bundle, and the effective form uses Theorem 10.5.7 in place of the text's second step. Definition 16.3.5 (2), (3) are restated through representatives, because the faces of a polyfold (Definition 16.1.21) are not formalized.

XVI.12 Definitions 16.3.6–16.3.8 (2017: Definitions 16.20–16.22), pp. 724–726: the category $\mathcal{SPBS}$ and strong polyfold bundle maps

Statement (same). *Definition 16.3.6* (p. 724) Chapter16/PolyfoldBundleCategory · SPBS: the category $\mathcal{SPBS}$ of strong polyfold bundle structures. A morphism Chapter16/PolyfoldBundleCategory · SPBS.Hom from (W, Γ, γ) over p to (W', Γ', γ') over p' is $(A, a, \mathfrak{A}, \mathfrak{a})$ with continuous $A : Y \to Y'$, $a : Z \to Z'$, a generalized

strong bundle map $\mathfrak{A}$ covering $\mathfrak{a}$, $\Gamma' \circ |\mathfrak{A}| = A \circ \Gamma$ and $\gamma' \circ |\mathfrak{a}| = a \circ \gamma$. *Definition 16.3.7* (p. 725) `Chapter16/PolyfoldBundleCategory · SPBS.MorEquiv`: two morphisms are equivalent if they differ by invertible morphisms of the special form $(\mathfrak{H}, \mathrm{Id}_Y)$ on both sides. *Definition 16.3.8* (p. 726) `Chapter16/PolyfoldBundleCategory · SPBS.StrongPolyfoldBundleMap`: an sc-smooth strong polyfold bundle map is a class of morphisms between structures of $\bar{c}$ and $\bar{c}'$.

Proof of Definition 16.3.6. *Step 1* `Chapter16/PolyfoldBundleCategory · SPBS.Hom.p_comm`: $p' \circ A = a \circ p$ follows, since $p' \circ A \circ \Gamma = p' \circ \Gamma' \circ |\mathfrak{A}| = \gamma' \circ |P'| \circ |\mathfrak{A}| = \gamma' \circ |\mathfrak{a}| \circ |P| = a \circ \gamma \circ |P| = a \circ p \circ \Gamma$, and Γ is surjective. The text lists it as a condition. *Step 2* `Chapter16/PolyfoldBundleCategory · SPBS.Hom.id Chapter16/PolyfoldBundleCategory · SPBS.Hom.comp Chapter16/PolyfoldBundleCategory · SPBS.category`: identities and componentwise composition, with Step 1 of §XVI.11 for the orbit maps; the laws hold componentwise. A constructor from the bundle data alone `Chapter16/PolyfoldBundleCategory · SPBS.mk'` derives the base conditions. *Step 3, invertibility* `Chapter16/PolyfoldBundleCategory · SPBS.Hom.IsInvertible Chapter16/PolyfoldBundleCategory · SPBS.isIso_iff_isInvertible`: a morphism is invertible in the text's sense (A, a homeomorphisms, $\mathfrak{A}$ a generalized strong bundle isomorphism) iff it is an isomorphism of the category, the inverse being $(A^{-1}, a^{-1}, \mathfrak{A}^{-1}, \mathfrak{a}^{-1})$ `Chapter16/PolyfoldBundleCategory · SPBS.Hom.inv Chapter16/PolyfoldBundleCategory · SPBS.Hom.comp_inv Chapter16/PolyfoldBundleCategory · SPBS.Hom.inv_comp`. *Step 4, the text before Definition 16.3.7* `Chapter16/PolyfoldBundleCategory · SPBS.exists_isInvertible_iff_equiv`: $\bar{c}$ consists of the structures related to a given one by an invertible morphism $(\mathfrak{A}, \mathrm{Id}_Y)$, and for such a morphism $A = \mathrm{id}$ forces $a = \mathrm{id}$ `Chapter16/PolyfoldBundleCategory · SPBS.Hom.f_eq_id Chapter16/PolyfoldBundleCategory · SPBS.inv_F_eq_id`, by Step 1 and the surjectivity of p.

Proof of Definitions 16.3.7 and 16.3.8. *Step 1* `Chapter16/PolyfoldBundleCategory · SPBS.MorEquiv.F_eq Chapter16/PolyfoldBundleCategory · SPBS.MorEquiv.f_eq`: equivalent morphisms have $A_1 = A_0$ and $a_1 = a_0$, comparing the continuous components of the square. *Step 2* `Chapter16/PolyfoldBundleCategory · SPBS.MorEquiv.refl Chapter16/PolyfoldBundleCategory · SPBS.MorEquiv.symm Chapter16/PolyfoldBundleCategory · SPBS.MorEquiv.trans`: an equivalence relation, by the identity, inverses and pasting squares, as for Definition 16.1.6. *Step 3* `Chapter16/PolyfoldBundleCategory · SPBS.MorRep Chapter16/PolyfoldBundleCategory · SPBS.MorRep.Rel Chapter16/PolyfoldBundleCategory · SPBS.MorRep.setoid`: representatives are morphisms between structures of $\bar{c}$ and $\bar{c}'$, and the maps are the quotient. *Step 4, the continuous maps* `Chapter16/PolyfoldBundleCategory · SPBS.StrongPolyfoldBundleMap.toFun Chapter16/PolyfoldBundleCategory · SPBS.StrongPolyfoldBundleMap.baseFun Chapter16/PolyfoldBundleCategory · SPBS.StrongPolyfoldBundleMap.continuous_toFun Chapter16/PolyfoldBundleCategory · SPBS.StrongPolyfoldBundleMap.p_comm`: A and a are well defined on classes by Step 1, continuous, and commute with the projections. *Step 5* `Chapter16/PolyfoldBundleCategory · SPBS.StrongPolyfoldBundleMap.id Chapter16/PolyfoldBundleCategory · SPBS.StrongPolyfoldBundleMap.toFun_id`: the identity map. The composition of strong polyfold bundle maps on classes would meet the question of §XVI.4; the text does not define it, and it is not formalized.

Where Lean departs from the text. The text defines $\mathcal{SPBS}$, its equivalence of morphisms and the strong polyfold bundle maps without verifying the category laws or the relation; the formalization proves them, and derives the commutation $p' \circ A = a \circ p$ and the base condition $\gamma' \circ |\mathfrak{a}| = a \circ \gamma$ from the bundle conditions. The text's remark before Definition 16.3.7, that $\bar{c}$ consists of the structures related by invertible morphisms over the identity of Y, is proved as an equivalence with Definition 16.3.3. The text does not define the composition of strong polyfold bundle maps on classes; by the mechanism of §XVI.4 it would need an effective middle base, and it is not formalized.

XVI.13 Definition 16.3.9 (2017: Definition 16.23), Proposition 16.3.10, Remark 16.3.11, p. 726: auxiliary norms

Statement. *Definition 16.3.9* (restated) `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.IsAuxNorm`: a function $n : Y \to \mathbb{R}$ is an auxiliary norm for $\bar{c}$ if, for every structure of $\bar{c}$, the function $N(e) = n(\Gamma(|e|))$ on $W_{0,1}$ `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.normOf` is an auxiliary norm of W in the sense of Definition 12.2.1 `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.IsAuxNormFor`. The text says one structure suffices `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.definition_16_3_9`. *(16.7)* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.normOf_pull`: $N = [D]^* N'$ for related structures. *Proposition 16.3.10* (same) `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.proposition_16_3_10`: auxiliary norms exist. *Remark 16.3.11* (nothing to formalize): prose. *Restated* because the text's n is a function on $Y_{0,1}$, the image of the level-$(0,1)$ part; the formalization takes a function on Y whose values on $Y_{0,1}$ matter, so that no subtype of Y needs a topology.

Background: auxiliary norms on a strong bundle. Over an ep-groupoid, an auxiliary norm of a strong bundle $P : W \to X$ (Definition 12.2.1, Part XII) is a continuous function $N : W_{0,1} \to [0,\infty)$ on the bi-level $(0,1)$ part, invariant under the action of the morphisms, whose restriction to every fibre is a complete norm on the level-1 fibre, and which has the compactness property: if $P(h_k) \to x$ and $N(h_k) \to 0$, then $h_k \to 0_x$ in $W[1]$. Invariance is what lets N descend to $|W[1]|$ and hence, through Γ, to a function on $Y_{0,1}$; the compactness property is what Proposition 16.3.16 (1) reads on Y. Auxiliary norms are the device by which Chapter 12 makes sense of "small" perturbations of an sc-Fredholm section: a multisection is controlled if its pointwise norm $N(\Lambda)$ is at most 1 (§XVI.26), and properness (Definition 12.4.1) is a compactness statement for sublevel sets of $N \circ F$. The polyfold definitions of §16.3 are these notions moved to Y along the homeomorphisms Γ; the only substantive points are that the move does not depend on the structure, and that the reflexive variants of §XVI.14, whose definitions involve sequences and weak limits rather than pointwise values, move as well.

The set $Y_{0,1}$. *Step 1* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.Y01`: for a structure, $Y_{0,1} = \Gamma(|W_{0,1}|)$, with the map $|W[1]| \to Y$ `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.oneToY`. *Step 2* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.Y01_subset_of_equiv` `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.Y01_eq_of_equiv`: it does not depend on the structure, since generalized strong bundle isomorphisms preserve the bi-level $(0,1)$ (Chapter 10) and commute with Γ. So $Y_{0,1}$ is defined for the bundle `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.Y01` `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.Y01_eq`. *Step 3* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.oneToY_pull`: the maps $|W[1]| \to Y$ of related structures are compatible with the orbit map of the realizing isomorphism.

Proof of Definition 16.3.9, "one structure suffices". *Step 1, invariance of N* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.isOneInvariant_normOf`: $N = n \circ \Gamma \circ \pi$ is invariant under the morphisms of W, since π is. *Step 2, (16.7).* For related structures through F, $N = N' \circ F$ on orbits, from $\Gamma' \circ |F| = \Gamma$ `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.normOf_pull`. *Step 3* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.isAuxNormFor_of_equiv`: auxiliary norms pull back along generalized strong bundle isomorphisms, Proposition 11.5.1 (Part XI). So if N' is an auxiliary norm, so is N. *Step 4.* The definition quantifies over all structures; Step 3 gives the equivalence with one. *Continuity and nonnegativity.* The continuity of n on $Y_{0,1}$ is read in the topology of $|W[1]|$ carried by Γ, where it is the continuity of the descended function `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.normOrbit` `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.continuous_normOrbit`, and nonnegativity `Chapter16/`

PolyfoldAuxNorm · StrongPolyfoldBundle.IsAuxNorm.nonneg follows from that of N.

Proof of Proposition 16.3.10. *Step 1.* Take a structure of $\bar{c}$. Theorem 12.2.2 (Part XII) gives an auxiliary norm N of W, using the paracompactness of $|X|$, which is part of Definition 16.3.1. *Step 2, descend* Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.descend Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.normOf_descend Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundleStructure.descend_oneToY: N is invariant, so $n(\Gamma|e|) = N(e)$ defines n on $Y_{0,1}$, extended by 0, and N is recovered as $n \circ \Gamma \circ \pi$. *Step 3* Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.isAuxNorm_descend: n is an auxiliary norm by Definition 16.3.9 for one structure.

Where Lean departs from the text. The text writes $n : Y_{0,1} \to [0, \infty)$; the formalization takes $n : Y \to \mathbb{R}$, with only its values on $Y_{0,1}$ entering the definition. This avoids giving the subset $Y_{0,1}$ its own topology: continuity is read in the topology of $|W[1]|$ carried by Γ, which is the topology the text intends (the bi-level $(0, 1)$ topology). The text's reason that one structure suffices is (16.7); the formalization makes it Proposition 11.5.1 applied to a realizing isomorphism, (16.7) being the identity $N = [D]^* N'$ that the proposition uses. Proposition 16.3.10 is proved as printed, by Theorem 12.2.2 in a representative, with the paracompactness that Definition 16.3.1 now includes.

XVI.14 Definitions 16.3.12–16.3.14 (2017: Definitions 16.24–16.26), p. 727: reflexive 1-fibres, mixed convergence, reflexive auxiliary norms, and their independence of the model

Statement. *Definitions 16.3.12, 16.3.13, 16.3.14* (same) Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.HasReflexiveFibers Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.MixedTendsto Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.IsReflexiveAuxNorm: $\bar{c}$ has reflexive 1-fibres if some model of $\bar{c}$ has them (Definition 12.2.6, with compatible charts whose fibre F_1 is reflexive); $y_k \to y$ in the mixed sense if this holds in some model (Definition 12.2.8); n is a reflexive auxiliary norm if it is represented in some model by a reflexive auxiliary norm (Definition 12.2.10). Mixed convergence in a given model Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.MixedTendstoIn. *The parenthetical of Definition 16.3.12* (same) Chapter16/ReflexivePush · StrongPolyfoldBundle.hasReflexiveFibers_iff_forall: then every model has reflexive 1-fibres. *The model does not matter for mixed convergence* (same; the remark before Theorem 16.3.17) Chapter16/MixedTransfer · StrongPolyfoldBundle.mixedTendsto_iff_forall, *nor for reflexive auxiliary norms* (same) Chapter16/MixedTransfer · StrongPolyfoldBundle.isReflexiveAuxNorm_iff_forall.

Proof, the pull direction for reflexive fibres. *Step 1, charts* Chapter16/ReflexivePull · StrongBundle.LocalIso.exists_reflexive_chart. Let $\Phi : W|U \to W'|U'$ be a local strong bundle isomorphism over ϕ, and $x \in U$. A compatible chart of W' at $\phi(x)$ whose fibre F_1 is reflexive gives one of W at x: restrict the chart to U', then transport it along Φ^{-1}; the transported chart is compatible with the strong bundle structure of W because Φ is a strong bundle isomorphism, and its fibre model is the same Banach space, so it is reflexive. *Step 2, bundles* Chapter16/ReflexivePull · StrongBundle.hasReflexiveFibers_of_isStrongBundleIsoOn: if the objects of X are covered by the domains of such isomorphisms, reflexive 1-fibres of W' give reflexive 1-fibres of W. *Step 3, equivalences* Chapter16/ReflexivePull · EpStrongBundle.IsStrongBundleEquivalence.hasReflexiveFibers_of: for a strong bundle equivalence $W \to W'$ (Definition 10.4.1), the local isomorphisms covering the objects are part (2) of the definition, so reflexive 1-fibres of W' give those of W, with no hypothesis added.

Proof, the push direction for reflexive fibres Chapter16/ReflexivePush · EpStrongBundle.

IsStrongBundleEquivalence.hasReflexiveFibers_push. For a strong bundle equivalence $L : W \to W'$ over ϕ and reflexive 1-fibres of W: *Step 1.* For $x' \in X'$ pick x and a morphism $\sigma : \phi(x) \to x'$ of X', which exist because ϕ is essentially surjective. *Step 2.* L is a local strong bundle isomorphism near x; its inverse L_0^{-1} and Step 1 of the pull direction give a reflexive chart of W' at $\phi(x)$. *Step 3.* The action μ of W' along σ is a local strong bundle isomorphism from a neighbourhood of $\phi(x)$ to one of x' (Chapter 10, the morphism form of the local isomorphisms of a strong bundle); Step 1 of the pull direction for its inverse moves the reflexive chart to x'. *Step 4, along generalized isomorphisms* Chapter16/ReflexivePush · EpStrongBundle.IsStrongBundleEquivalence.hasReflexiveFibers_iff Chapter16/ReflexivePush · SBLocalization.SDiagram.hasReflexiveFibers_of Chapter16/ReflexivePush · SBLocalization.GenSBMap.hasReflexiveFibers_iff: with both directions, reflexivity passes along the two legs of a diagram. *Step 5, structures and the class* Chapter16/ReflexivePush · StrongPolyfoldBundleStructure.hasReflexiveFibers_of_equiv Chapter16/ReflexivePush · StrongPolyfoldBundle.HasReflexiveFibers.of_mem: equivalent structures have reflexive 1-fibres together, so "some model" and "every model" agree, which is the parenthetical.

Proof for mixed convergence. *Step 1, along a local strong bundle isomorphism* Chapter16/MixedTransfer · StrongBundle.LocalIso.mixedTendsto. Let Φ be a local strong bundle isomorphism into a bundle with reflexive fibres, and $w_k \to w$ in the mixed sense: the base points converge, (w_k) is bounded on level 1 in local charts, and $w_k \rightharpoonup w$ weakly. In charts, Φ is linear and bounded on the fibres, uniformly near the base point: Proposition 5.1.3 (Part V) gives the chart form, and a bound of the fibre norms of the image by a multiple of the auxiliary norm holds near the point. Bounded sequences stay bounded, and a bounded linear map preserves weak limits; the fibre seminorms compose Chapter16/MixedTransfer · StrongBundle.LocalIso.isFiberSeminorm_comp, and the property is stable under eventual equality Chapter16/MixedTransfer · StrongBundle.MixedTendsto.congr_eventually. *Step 2, along a diagram* Chapter16/MixedTransfer · SBLocalization.SDiagram.exists_mixedTendsto_transfer. For $W \xleftarrow{L} W'' \xrightarrow{L'} W'$ and $w_k \to w$ in W: move the sequence along a morphism σ into the image of L by the action, which is a local strong bundle isomorphism (with the identity $\mu(1,-)$ Chapter16/MixedTransfer · SBLocalization.mk_actOne); lift through a local inverse of L at the base point; push through L', a local isomorphism; Step 1 at each stage. The images represent the orbits of the images of w_k and converge in the mixed sense. *Step 3, structures* Chapter16/MixedTransfer · StrongPolyfoldBundleStructure.mixedTendstoIn_of_equiv: for equivalent structures, mixed convergence of $y_k \to y$ read in one gives it in the other, through a sequence patched to the representatives that Step 2 provides, compatible with the maps $|W[1]| \to Y$ Chapter16/MixedTransfer · StrongPolyfoldBundleStructure.oneToY_of_oneOrbitMap. *Step 4* Chapter16/MixedTransfer · StrongPolyfoldBundle.mixedTendsto_iff_forall: the model does not matter.

Proof for reflexive auxiliary norms. *Step 1* Chapter16/MixedTransfer · SBLocalization.SDiagram.isReflexiveAuxNorm_pullNorm Chapter16/MixedTransfer · SBLocalization.SDiagram.isReflexiveAuxNorm_pushNorm: along a diagram, a reflexive auxiliary norm pulls back and pushes forward to one. Reflexivity of a norm (Definition 12.2.10) is lower semicontinuity along mixed convergent sequences; by the mixed-convergence transfer of the previous proof, applied in both directions (the push direction through the inverse diagram), it moves with the norm. *Step 2* Chapter16/MixedTransfer · SBLocalization.GenSBMap.isReflexiveAuxNorm_pullNorm Chapter16/MixedTransfer · SBLocalization.GenSBMap.isReflexiveAuxNorm_pushNorm: for generalized strong bundle isomorphisms, through representative diagrams. *Step 3* Chapter16/MixedTransfer · StrongPolyfoldBundleStructure.isReflexiveAuxNorm_of_equiv Chapter16/MixedTransfer · StrongPolyfoldBundle.IsReflexiveAuxNorm.of_mem Chapter16/MixedTransfer · StrongPolyfoldBundle.isReflexiveAuxNorm_iff_forall: for related structures, $N = [D]^* N'$ by (16.7), so N is reflexive iff N' is, and Definition 16.3.14 does not depend on the structure.

Hypotheses. The text's standing hypothesis in Definitions 16.3.13 and 16.3.14, reflexive 1-fibres, is the context; no hypothesis is added. Gap G.166 is closed.

Where Lean departs from the text. The text asserts in a parenthesis that reflexivity of the 1-fibres does not depend on the model, and before Theorem 16.3.17 that mixed convergence does not either; no proof is given. The formalization proves both, together with the independence of reflexive auxiliary norms, which the text uses implicitly when it speaks of a reflexive auxiliary norm of $\bar{c}$. The proofs are transports along the two legs of a diagram, with one asymmetry: along the leg that is an equivalence into the target, reflexivity pulls back by transporting a chart, while in the other direction it must also be moved along the morphisms of the target, through the action μ. Mixed convergence additionally needs that a local strong bundle isomorphism is bounded and linear on the fibres, uniformly near a point, which the chart form of Proposition 5.1.3 provides.

XVI.15 Theorem 16.3.15, Proposition 16.3.16, Theorem 16.3.17 (2017: Theorems 16.3, 16.4, Proposition 16.5), pp. 727–728

Statement. *Theorem 16.3.15* (same) `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.theorem_16_3_15_one` `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.theorem_16_3_15_two`: if $\bar{c}$ has reflexive 1-fibres, (1) a reflexive auxiliary norm exists, and (2) every auxiliary norm n admits reflexive auxiliary norms $n_1 \le n \le n_2$ on $Y_{0,1}$. *Proposition 16.3.16 (1)* (stronger) `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.proposition_16_3_16_one`: if $p(y_k) \to z$ and $n(y_k) \to 0$, then $y_k \to 0_z$; the formal statement is for every auxiliary norm, with the convergence in $|W[1]|$ of a model, and does not use reflexivity. *Proposition 16.3.16 (2)* (same) `Chapter16/MixedTransfer · StrongPolyfoldBundle.proposition_16_3_16_two`: for a reflexive auxiliary norm n and $y_k \to y$ in the mixed sense, $n(y) \le \liminf n(y_k)$. *Theorem 16.3.17 (1)* (same) `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.theorem_16_3_17_one`: over a tame polyfold, an auxiliary norm of $Y_{0,1}|\partial Z$ extends to an auxiliary norm. *Theorem 16.3.17 (2)* (narrower) `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.theorem_16_3_17_two_hyp`: if the boundary norm is reflexive, the extension can be taken reflexive, with the reflexive local step of Theorem 12.2.13 as an added hypothesis.

Proof of Theorem 16.3.15. *Step 1.* By the parenthetical of Definition 16.3.12 (§XVI.14), the chosen structure of $\bar{c}$ has reflexive 1-fibres. *Step 2, (1).* Theorem 12.2.11 (Part XII) gives a reflexive auxiliary norm N of W; descend it as in Proposition 16.3.10 (§XVI.13). *Step 3, (2).* For an auxiliary norm n, its representative N on the structure is an auxiliary norm (Definition 16.3.9), and the sandwich form of Theorem 12.2.11 gives reflexive $N_1 \le N \le N_2$; they descend to $n_1 \le n \le n_2$ on $Y_{0,1}$, because the inequality is pointwise on orbits.

Proof of Proposition 16.3.16 (1). *Step 1, lift the sequence.* $p(y_k) \to z$ in Z; through the open quotient map $X \to |X| \to Z$, pick x over z and lift a tail of the sequence to objects $x_k \to x$, and w_k over x_k representing y_k (via Γ and the surjectivity of $W_{0,1} \to Y_{0,1}$ on orbits). *Step 2, align the representatives.* For each k a morphism moves the chosen representative to one over x_k; the value of the action at level 1 is the representative needed, so $N(w_k) = n(y_k) \to 0$ for N the representative of n. *Step 3, the compactness property.* Theorem 12.2.2's auxiliary norm property, which holds for every auxiliary norm (Definition 12.2.1 (2)), states that $x_k \to x$ and $N(w_k) \to 0$ imply $w_k \to 0_x$ in $W[1]$. *Step 4.* Taking orbits and Γ, $y_k \to 0_z$ in the topology of $|W[1]|$ carried by Γ. *Stronger* because the text states it for reflexive auxiliary norms and the proof does not use reflexivity.

Proof of Proposition 16.3.16 (2). *Step 1, the same-model form* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.`

`proposition_16_3_16_two_of_mem`: if the mixed convergence is read in the model in which n is represented by a reflexive auxiliary norm N, the inequality is the reflexivity of N (Definition 12.2.10): $N(w) \leq \liminf N(w_k)$ for $w_k \to w$ in the mixed sense, read on representatives and transported by Γ. *Step 2, the text's form.* The mixed convergence may be given in any model, and n reflexive in any model; by §XVI.14 both notions are independent of the model, so the mixed convergence holds in the model where n is reflexive, and Step 1 applies.

Proof of Theorem 16.3.17 (1). *Step 1, the boundary norm* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.IsAuxNormBoundary`: an auxiliary norm of $Y_{0,1}|\partial Z$ is read as in Definition 16.3.9 over ∂X, in all models. The set $p^{-1}(\partial Z)$ corresponds to $W|\partial X$: $p(\Gamma|e|) \in \partial Z$ iff $P(e) \in \partial X$ `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.mem_boundary_iff`, by the computation of ∂Z in a representative (§XVI.5) and $p \circ \Gamma = \gamma \circ |P|$. *Step 2.* Over a tame base, Theorem 12.2.7 (Part XII) extends an auxiliary norm of $W|\partial X$ to one of W; apply it in the chosen structure, whose base is tame by Definition 16.3.5 (1). *Step 3.* Descend the extension (§XVI.13); it agrees with n over ∂Z because the representative of n over ∂X was extended.

Proof of Theorem 16.3.17 (2). *Step 1, the reading.* "n reflexive over ∂Z" is read as reflexive 1-fibres and a reflexive representative over ∂X in one model. *Step 2.* Theorem 12.2.13 (Part XII), in the form that takes its reflexive local step as a hypothesis `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.theorem_16_3_17_two_hyp`, gives a reflexive extension in that model, which descends. *The added hypothesis*, not derived from the text: the reflexive local step, that near every boundary point a reflexive auxiliary norm over the boundary has a reflexive local extension. *Witness* `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.theorem_16_3_17_two_hyp_witness`: on every bundle with reflexive 1-fibres, for the boundary norm that is the restriction of a global reflexive auxiliary norm. **This witness covers only a degenerate case**: when n is already the restriction of a global reflexive norm, the local step is satisfied by restricting that norm, and nothing is extended. The non-degenerate case is supplied by Part XII, which proves the reflexive local step over tame bases `Chapter12/ReflexiveLocalExtension · EpStrongBundle.hasReflexiveLocalExtensions_of_isTame`; Theorem 16.3.17 assumes a tame base, so the unconditional form is a one-line corollary, not stated as a separate Lean declaration. Tameness is not otherwise used.

Where Lean departs from the text. The text proves Theorem 16.3.15, Proposition 16.3.16 and Theorem 16.3.17 by the corresponding statements of Chapter 12 in a representative, and the formalization follows this. Three differences. Proposition 16.3.16 (1) is proved for every auxiliary norm, since the Chapter 12 property it rests on holds for all of them. Proposition 16.3.16 (2) needs the independence of the model of §XVI.14, which the text takes for granted, to bring the mixed convergence into the model where n is reflexive. Theorem 16.3.17 (2) takes the reflexive local step of Theorem 12.2.13 as a hypothesis, discharged over tame bases by Part XII; since Theorem 16.3.17 is stated over tame polyfolds, the text's statement holds, but the composite is not packaged as one declaration.

XVI.16 Definitions 16.3.18, 16.3.19 (2017: Definitions 16.27, 16.28), pp. 728–729: sections, the zero of a fibre and the zero section

Statement. *The relation before Definition 16.3.18* (same) `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep` `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep.Rel`: a representative is (f, Γ, W, F) with f a continuous section of p and F an sc-smooth section functor of a structure with $\Gamma \circ |F| = f \circ \gamma$; two are related if $f = f'$ and a generalized strong bundle isomorphism $\mathfrak{A}$ has $\Gamma' \circ |\mathfrak{A}| = \Gamma$ and $\mathfrak{A}_* F = F'$. *Definition 16.3.18* (same) `Chapter16/PolyfoldSections · StrongPolyfoldBundle.ScSection`: an sc-smooth section

is a class. *Definition 16.3.19* (restated) `Chapter16/PolyfoldSections · StrongPolyfoldBundle.ScSection.IsScPlus` `Chapter16/PolyfoldSections · StrongPolyfoldBundle.ScPlusSection`: an sc⁺-section is an sc-smooth section whose functor in one, equivalently every, representative is sc⁺. *The zero of a fibre* (p. 728, same) `Chapter16/FibreZero · StrongPolyfoldBundle.zeroAt`: $0_z = \Gamma(|0_x|)$ for $\gamma(|x|) = z$. *The zero section* (p. 728, same) `Chapter16/ZeroSection · StrongPolyfoldBundle.zeroSection`: $z \mapsto 0_z$ is an sc⁺-section.

Proof that the relation is an equivalence relation and depends only on f. *Step 1* `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep.Rel.refl` `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep.Rel.symm` `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep.Rel.trans` `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep.setoid`: by the identity, the inverse with $(\mathfrak{A}^{-1})_* = (\mathfrak{A}_*)^{-1}$, and composites with $(\mathfrak{B} \circ \mathfrak{A})_* = \mathfrak{B}_* \circ \mathfrak{A}_*$ (Chapter 10). The text states "this defines an equivalence relation". *Step 2, the push-forward condition is redundant* `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep.pushforward_eq_of_Γ`. If $\Gamma' \circ |\mathfrak{A}| = \Gamma$ and $f = f'$, then $\mathfrak{A}_* F$ and F' are section functors of W' with the same orbit map $\Gamma'^{-1} \circ f \circ \gamma$; a section functor is determined by its values on orbits. Indeed, if $F_1(x)$ and $F_2(x)$ lie in one orbit of W, a morphism of W joins them, lying over an isotropy element $g \in G_x$; then $F_2(x) = \mu(g, F_1(x))$, and $\mu(g, F_1(x)) = F_1(x)$ because F_1 is a functor, so $F_1(x) = F_2(x)$. The orbit of $(\mathfrak{A}_* F)(x)$ is computed from $|\mathfrak{A}|$ and $|F|$, which gives the equality of orbits. *Step 3* `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep.rel_iff` `Chapter16/PolyfoldSections · StrongPolyfoldBundle.SectionRep.rel_of_f_eq`: two representatives are related iff $f = f'$ and their structures are equivalent. So the non-uniqueness of §XVI.11 does not reach sections.

Proof of Definitions 16.3.18 and 16.3.19. *Step 1* `Chapter16/PolyfoldSections · StrongPolyfoldBundle.ScSection.toFun` `Chapter16/PolyfoldSections · StrongPolyfoldBundle.ScSection.p_toFun` `Chapter16/PolyfoldSections · StrongPolyfoldBundle.ScSection.continuous_toFun`: the section f of a class is well defined, continuous, and a section of p. *Step 2* `Chapter16/PolyfoldSections · StrongPolyfoldBundle.ScSection.ext_toFun`: by Step 3 above, a section is determined by f (two classes with the same f have representatives with equivalent structures, hence related). *Step 3, sc⁺* `Chapter16/PolyfoldSections · StrongPolyfoldBundle.ScSection.isScPlus_iff`: being sc⁺ is invariant under push-forward by generalized strong bundle isomorphisms, Theorem 10.5.4 (Part X), so one representative is sc⁺ iff every one is. *Restated* because the text phrases Definition 16.3.19 through the continuity of f into $Y_{0,1}$; in a representative that is the continuity of F into $W_{0,1}$, which is what sc⁺ asks.

Proof of the zero of a fibre. *Step 1* `Chapter16/FibreZero · StrongPolyfoldBundleStructure.zeroAt` `Chapter16/FibreZero · StrongPolyfoldBundleStructure.zeroAt_eq`: for a structure, $0_z = \Gamma(|0_x|)$ for any x with $\gamma(|x|) = z$; it does not depend on x, because a morphism $x \to x'$ acts linearly on fibres and sends 0_x to $0_{x'}$ `Chapter16/FibreZero · SBLocalization.π_fiberZero_eq_of_π_eq`, and the action of a morphism moves a non-zero vector to a non-zero vector `Chapter16/FibreZero · SBLocalization.μ_ne_fiberZero`, so the orbits of zeros are exactly the zeros of orbits `Chapter16/FibreZero · SBLocalization.π_eq_fiberZero_iff`. *Step 2* `Chapter16/FibreZero · StrongPolyfoldBundleStructure.p_zeroAt` `Chapter16/FibreZero · StrongPolyfoldBundle.p_zeroAt`: $p(0_z) = z$. *Step 3, independence of the structure* `Chapter16/FibreZero · StrongPolyfoldBundleStructure.zeroAt_eq_of_equiv`: diagrams and generalized strong bundle maps send orbits of zeros to orbits of zeros `Chapter16/FibreZero · SBLocalization.SDiagram.orbitMap_π_fiberZero` `Chapter16/FibreZero · SBLocalization.GenSBMap.orbitMap_π_fiberZero`, since each leg is linear on fibres. *Step 4* `Chapter16/FibreZero · StrongPolyfoldBundleStructure.Γ_π_eq_zeroAt_iff`: $\Gamma(|w|) = 0_z$ iff w is the zero of its fibre and lies over z. *Auxiliary norms extended by 0* `Chapter16/FibreZero · StrongPolyfoldBundle.normY`: off $Y_{0,1}$ an auxiliary norm is set to 0, and $n(\Gamma|w|) = N(w)$ for $w \in W_{0,1}$ `Chapter16/FibreZero · StrongPolyfoldBundle.normY_Γ`; this is the reading of $n(f(z))$ in Definition 16.6.3 (§XVI.24).

Proof of the zero section. *Step 1, continuity* Chapter16/ZeroSection · StrongPolyfoldBundle.continuous_fiberZero_W Chapter16/ZeroSection · StrongPolyfoldBundle.continuous_zeroAt: the zero section functor of a model is continuous; $z \mapsto 0_z$ is continuous because $x \mapsto \Gamma(|0_x|)$ is, and $x \mapsto \gamma(|x|)$ is a quotient map Chapter16/NormLamContinuous · StrongPolyfoldBundle.isQuotientMap_γπ (§XVI.21). *Step 2, a representative in every model* Chapter16/ZeroSection · StrongPolyfoldBundle.zeroRepOf: the zero functor of a structure, which is an sc-smooth (indeed sc$^+$) section functor, with $\Gamma \circ |0| = 0_{(\cdot)} \circ \gamma$ by Step 1 of the zero of a fibre. *Step 3, the class* Chapter16/ZeroSection · StrongPolyfoldBundle.zeroSection Chapter16/ZeroSection · StrongPolyfoldBundle.zeroSection_eq_mk: all these representatives have the same map f, so they give one class (Step 3 of the relation). Its map is $z \mapsto 0_z$ Chapter16/ZeroSection · StrongPolyfoldBundle.toFun_zeroSection. *Step 4* Chapter16/ZeroSection · StrongPolyfoldBundle.isScPlus_zeroSection Chapter16/ZeroSection · StrongPolyfoldBundle.zeroPlusSection: it is an sc$^+$-section, since the zero functor is sc$^+$. *Step 5* Chapter16/ZeroSection · StrongPolyfoldBundle.rep_F_eq_fiberZero Chapter16/ZeroSection · StrongPolyfoldBundle.ScSection.eq_zeroSection_iff Chapter16/ZeroSection · StrongPolyfoldBundle.ScSection.eq_zeroSection_iff_F: a section is the zero section iff $f(z) = 0_z$ for all z, iff its functor is the zero functor in some, equivalently every, representative. *Step 6* Chapter16/ZeroSection · StrongPolyfoldBundle.solutionSet_zeroSection: the solution set of the zero section is Z. Gap G.167 is closed, and Definitions 16.3.18 and 16.3.19 have a witness on every strong polyfold bundle.

Where Lean departs from the text. The text states the relation on representatives of sections and calls it an equivalence relation; the formalization proves this and the stronger fact that the push-forward condition is implied by the others, so that a section is determined by its map f. This is what protects §§16.3–16.6 from the non-uniqueness of §XVI.11: whichever isomorphism realizes an equivalence of structures, it pushes a representative to the unique representative with the same map. The zero of a fibre is defined in the text by a representative; the formalization proves its independence of the representative and of the structure. The zero section is mentioned in the text as an example; the formalization builds it as an element of Definition 16.3.18, with representatives in every model, and characterizes it.

XVI.17 Definitions 16.4.1 and 16.4.2 (2017: Definitions 16.29, 16.30), pp. 729–730: branched suborbifolds and their properties

Statement. *Definition 16.4.1* (restated) Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold: a branched suborbifold of (Z, c) is a class $\theta = [\Theta, (X, \alpha)]$ of representatives $(\Theta, (X, \alpha))$ with $(X, \alpha) \in c$ and Θ a branched ep$^+$-subgroupoid of X Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedRep, where $(\Theta, (X, \alpha)) \sim (\Theta', (X', \alpha'))$ if a generalized isomorphism $f : X \to X'$ has $\alpha' \circ |f| = \alpha$ and $f^*\Theta' = \Theta$ Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedRep.Rel. The function $\theta = |\Theta| \circ \alpha^{-1} : Z \to \mathbb{Q}^+$ is well defined Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.toFun Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_1_wellDefined. *Definition 16.4.2* (p. 730; (1)–(4) same, (5) same) Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.IsClosed Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.IsCompact Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.IsManifoldType Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.IsOrbifoldType Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.IsTame: θ is closed, compact, of manifold type, of orbifold type, or tame, if a representative Θ is; then every representative is Chapter16/BranchedTame · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_tame. *Restated* for Definition 16.4.1 because the text's class notation is made into explicit pairs and the relation is proved to be an equivalence relation.

Background: branched ep⁺-subgroupoids. A branched ep⁺-subgroupoid of X (Definition 9.1.1, Part IX) is a functor $\Theta : X \to \mathbb{Q}^+$, with values in the nonnegative rationals and supported in X_∞, that near every point of its support is described by a local branching structure: finitely many finite-dimensional submanifolds $M_1, \dots, M_k$ of an open set U invariant under the natural representation, with positive rational weights $\sigma_1, \dots, \sigma_k$, such that $\Theta(y) = \sum_{i : y \in M_i} \sigma_i$ on U. It is of manifold type or of orbifold type when its support carries the corresponding structure of Proposition 9.1.9, and tame if the local branching structures can be chosen tame in the sense of Definition 9.3.6. An orientation $\widehat{T}_\Theta$ (Chapter 11) is a choice of orientations of the tangent spaces of the branches, compatible with the weights. Pull-backs along functors are composition, and push-forwards along equivalences are read on orbits; Chapter 11 extends both to generalized isomorphisms, which is the operation f^* of Definition 16.4.1. So a branched suborbifold of a polyfold is a rational-weighted, locally finitely branched, finite-dimensional subset of Z, presented in one structure and moved to any other by f^*.

Pull-backs of branched ep⁺-subgroupoids. *Step 1* `Chapter16/BranchedSuborbifold · EPLocalization.EPLoc.pullTheta_eq_of_orbitMap`: for a generalized isomorphism f and a functorial Θ', $(f^*\Theta')(x) = \Theta'(y)$ whenever $|f|(|x|) = |y|$. This is the text's $|f^*\Theta'| = |\Theta'| \circ |f|$ (p. 730). The proof goes through a representative diagram $X \xleftarrow{F} A \xrightarrow{\Phi} X'$: $f^*\Theta'$ is the push-forward along F of $\Phi^*\Theta'$, the pull-back along a functor is composition, and the push-forward along an equivalence of a functorial function is read on orbits. *Step 2* `Chapter16/BranchedSuborbifold · EPLocalization.EPLoc.pull_congr`: two generalized isomorphisms with the same orbit map pull back alike, by Step 1. *Step 3* `Chapter16/BranchedSuborbifold · EPLocalization.EPLoc.pullTheta_id_of_isFunctorial`: $1^*\Theta = \Theta$ for functorial Θ.

Proof of Definition 16.4.1. *Step 1, an equivalence relation.* Reflexive by Step 3 above `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedRep.Rel.refl`. Symmetric `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedRep.Rel.symm`: if $f^*\Theta' = \Theta$, then $(f^{-1})^*\Theta = \Theta'$, by Step 1 on orbits, since $|f^{-1}| = |f|^{-1}$. Transitive `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedRep.Rel.trans`: $(g \circ f)^* = f^* \circ g^*$ on functorial functions, again by Step 1. The quotient is the type of branched suborbifolds. *Step 2, the function on a representative* `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedRep.toFun` `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedRep.toFun_eq`: $z \mapsto \Theta(x)$ for any x with $\alpha(|x|) = z$, well defined because Θ is functorial. *Step 3, related representatives give the same function* `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedRep.toFun_eq_of_rel`: for $z = \alpha(|x|) = \alpha'(|y|)$ with $|y| = |f|(|x|)$, $\Theta(x) = (f^*\Theta')(x) = \Theta'(y)$ by Step 1 of the pull-backs. So θ is defined on classes, with $\theta(\alpha(|x|)) = \Theta(x)$ for every representative and every x (the well-definedness statement), and $\theta \geq 0$ `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.toFun_nonneg`. *Step 4, θ determines the class* `Chapter16/PerturbedSuborbifold · PolyfoldClass.BranchedRep.rel_of_toFun_eq` `Chapter16/PerturbedSuborbifold · PolyfoldClass.BranchedSuborbifold.toFun_injective` `Chapter16/PerturbedSuborbifold · PolyfoldClass.BranchedSuborbifold.toFun_mk`: if two representatives have the same function, any generalized isomorphism f with $\alpha' \circ |f| = \alpha$ (which exists, both structures being in c) has $f^*\Theta' = \Theta$, because both sides are read on orbits and the orbit values agree. So the non-uniqueness of §XVI.2 does not reach branched suborbifolds: the relation is "the functions agree".

Proof of Definition 16.4.2 (1)–(4) and the dimension. *Step 1, the pattern* `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.HasRepProperty` `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.hasRepProperty_iff`: a property holds for θ if it holds for some representative; for a property stable under pull-backs by generalized isomorphisms, some representative has it iff all have it, since any two representatives are related by a generalized isomorphism. *Step 2.* Closed, compact, of manifold type and of orbifold type are stable, by Theorem 11.3.3 (3)–(5)

(Part XI), so they hold for every representative iff for one Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_closed Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_compact Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_manifold Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_orbifold. *Step 3.* So is the dimension n Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.IsNDimensional Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_nDim, by Theorem 11.3.3 (6); it is used in Theorem 16.4.6.

Proof of Definition 16.4.2 (5), tameness. The text says on p. 729 that tameness of branched ep$^+$-subgroupoids is preserved by f_* and f^*. *Step 1, restriction* Chapter16/BranchedTame · EpGroupoid.TameBranchingStructure.restrictRep: a tame local branching structure of Definition 9.3.6 at x restricts to any smaller natural representation at x. *Step 2, moving* Chapter16/BranchedTame · EpGroupoid.TameBranchingStructure.move Chapter16/BranchedTame · EpGroupoid.TameBranchingStructure.mapU_eq: it moves along an sc-diffeomorphism of open sets $W \to W'$ that intertwines the natural representations, the branches and the tame sets being carried with it, since tameness of a set (Part IV) is invariant under sc-diffeomorphisms. *Step 3, pull-back along an equivalence* Chapter16/BranchedTame · EpFunctor.isTameBranched_pullbackTheta. For an equivalence $\Phi : X \to X'$, $x \in X$ and a tame Θ': take a transport chart of Theorem 11.4.1 at x (Part XI), which identifies natural representations at x and $\Phi(x)$ through local sc-diffeomorphisms; take a tame branching structure D of Θ' at $\Phi(x)$ and a natural representation R_0 at x with the properness property (Theorem 7.1.13, Part VII); push R_0 to a representation R' inside D's domain; restrict D to R' (Step 1) and move it back along the chart (Step 2). The branches of $\Phi^*\Theta'$ near x are the pulled-back branches, so the moved structure is a tame branching structure of $\Phi^*\Theta'$. *Step 4, push-forward* Chapter16/BranchedTame · EpFunctor.isTameBranched_pushforwardTheta Chapter16/BranchedTame · EpFunctor.isTameBranched_pullbackTheta_iff: the same argument in the other direction, using that $\Phi_*\Phi^*\Theta' = \Theta'$ for functorial Θ'; so $\Phi^*\Theta'$ is tame iff Θ' is. *Step 5, generalized isomorphisms* Chapter16/BranchedTame · EPLocalization.EPLoc.isTameBranched_pullTheta: through the two legs of a diagram. *Step 6* Chapter16/BranchedTame · PolyfoldClass.BranchedSuborbifold.definition_16_4_2_tame: tameness is stable, so Step 1 of the pattern applies. Gap G.154 is closed.

Where Lean departs from the text. The text writes $|f^*\Theta'| = |\Theta'| \circ |f|$ and draws from it the well-definedness of θ; the formalization proves the identity through a representative diagram and draws, besides well-definedness, the converse: the function θ determines the class. For Definition 16.4.2 the text states that the properties hold for one representative iff for all, citing Theorem 11.3.3 for (1)–(4) and the remark on p. 729 for tameness. For (1)–(4) the formalization follows the text. For tameness the remark on p. 729 is stated without proof; the formalization proves it through the transport charts of Theorem 11.4.1, with the restriction and the moving of tame branching structures as the two lemmas that the text's remark presupposes.

XVI.18 Proposition 16.4.3 (2017: Proposition 16.4), p. 730: the structure on $\operatorname{supp}\theta$

Statement (same). For a branched suborbifold θ of manifold type (orbifold type), $\operatorname{supp}\theta = \{z : \theta(z) > 0\}$ Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.supp carries a natural manifold (orbifold) structure. In Lean, in three parts. *The set and the topology* Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_supp Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_homeomorph: for every representa-

tive, $\operatorname{supp}\theta = \alpha(|\operatorname{supp}\Theta|)$, and α restricts to a homeomorphism $|\operatorname{supp}\Theta| \cong \operatorname{supp}\theta$. *The structure on one representative* `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_manifold` `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_orbifold`: Proposition 9.1.9 (1), (2) on $|\operatorname{supp}\Theta|$. *Naturality* `Chapter16/SuppNaturality · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_naturality`: for any two representatives, and any structures induced by Proposition 9.1.9 (1) on their supports, the transition $|\operatorname{supp}\Theta| \to |\operatorname{supp}\Theta'|$ over $\operatorname{supp}\theta$ is an sc-diffeomorphism `Chapter16/SuppNaturality · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_naturality_all`. The word "natural" of the text is read as this independence of the representative, the statement being `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_naturality_statement`. The formal form is stronger than needed in one respect: the induced structures need not be chosen, since any two give the same comparison.

Proof of the set and the topology. *Step 1.* $\theta(\alpha(|x|)) = \Theta(x)$ (§XVI.17), so $\theta(z) > 0$ iff $\Theta(x) > 0$ for one, equivalently every, x over z; that is $z \in \alpha(|\operatorname{supp}\Theta|)$. *Step 2.* α is a homeomorphism $|X| \to Z$, and $|\operatorname{supp}\Theta|$ carries the subspace topology of $|X|$; so α restricts to a homeomorphism onto $\operatorname{supp}\theta$ with its subspace topology `Chapter16/BranchedSuborbifold · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_homeomorph_apply`.

Proof of the structure on one representative. *Manifold type.* With Proposition 9.1.9 (1) (Part IX), $|\operatorname{supp}\Theta|$ carries an M-polyfold structure, finite-dimensional, for which $\operatorname{supp}\Theta \to |\operatorname{supp}\Theta|$, $a \mapsto |a|$, is a local sc-diffeomorphism; it is carried to $\operatorname{supp}\theta$ by the homeomorphism above. *Orbifold type.* With Proposition 9.1.9 (2), $|\operatorname{supp}\Theta|$ is the orbit space of the ep-groupoid that X induces on $\operatorname{supp}\Theta$.

Proof of naturality. *Step 1, the transition* `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.suppTransition` `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.β_suppTransition` `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.suppTransition_eq_iff`: for two representatives Θ, Θ' of θ, the homeomorphism $t = \alpha'^{-1} \circ \alpha$ restricted to $|\operatorname{supp}\Theta| \to |\operatorname{supp}\Theta'|$, characterized by $\alpha' \circ t = \alpha$. *Step 2, the cocycle identities* `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.suppTransition_self` `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.suppTransition_trans` `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.suppTransition_symm`: $t_{aa} = 1$, $t_{a'a''} \circ t_{aa'} = t_{aa''}$, $t_{a'a} = t_{aa'}^{-1}$. *Step 3, the induced structures* `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.IsInducedSuppStructure`: a structure on $|\operatorname{supp}\Theta|$ is induced if $a \mapsto |a|$ is a local sc-diffeomorphism from $\operatorname{supp}\Theta$ with its M-polyfold structure (Proposition 9.1.9 (1)). For one representative the transition is the identity, an sc-diffeomorphism between any two induced structures `Chapter16/SuppNaturalityStatement · PolyfoldClass.BranchedSuborbifold.proposition_16_4_3_naturality_self`. *Step 4, the transition through a diagram* `Chapter16/SuppNaturality · PolyfoldClass.BranchedSuborbifold.suppTransition_orbitSuppProj`. For a generalized isomorphism $X \xleftarrow{F} A \xrightarrow{\Phi} X'$ between the representatives ($\alpha' \circ |f| = \alpha$, $f^*\Theta' = \Theta$), the transition sends $|F(b)|$ to $|\Phi(b)|$ for every b in the support of $\Phi^*\Theta'$. *Step 5, sc-smoothness* `Chapter16/SuppNaturality · PolyfoldClass.BranchedSuborbifold.isScSmoothOn_suppTransition`. The legs F and Φ restrict to equivalences of the sub-ep-groupoids on the supports (the argument of Proposition 11.1.6, Part XI), which are local sc-diffeomorphisms of the finite-dimensional M-polyfolds supp. Near $|F(b)|$, the transition is the composite $p' \circ \Phi|_{\text{supp}} \circ (F|_{\text{supp}})^{-1} \circ p^{-1}$, with p, p' the local sc-diffeomorphisms $a \mapsto |a|$ of the induced structures and $(F|_{\text{supp}})^{-1}$ a local inverse; so it is sc-smooth. Only Θ' of orbifold type is used, for the sub-ep-groupoid on the support of $\Phi^*\Theta'$; manifold type, the text's case, gives orbifold type. *Step 6, inverse.* By the cocycle identities the inverse is the transition in the other direction, sc-smooth by Step 5. Gap G.155 is closed.

Where Lean departs from the text. The text's proof of Proposition 16.4.3 is the reference to Proposition 9.1.9,

which gives the structure on $|\operatorname{supp}\Theta|$ for one representative; that the structure is natural, that is, independent of the representative, is in the word "natural". The formalization proves the independence explicitly through the transition maps between supports, and in a form stronger than a choice: any structures induced by Proposition 9.1.9 on the two supports are related by an sc-diffeomorphism. The proof needs the restriction of an equivalence to the sub-ep-groupoids on the supports, which is the argument of Proposition 11.1.6 in Part XI.

XVI.19 Definitions 16.4.4 and 16.4.5 (2017: Definitions 16.31, 16.34), pp. 730–732: oriented branched suborbifolds and their integrals

Statement. *Definition 16.4.4* (restated) `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedSuborbifold`: an oriented branched suborbifold is a class $\hat\theta = [(\Theta,\hat T_\Theta),(X,\alpha)]$ of representatives with $(\Theta,\hat T_\Theta)$ an oriented branched ep$^+$-subgroupoid of X (Chapter 11) `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep`, related by $f^*(\Theta',\hat T_{\Theta'}) = (\Theta,\hat T_\Theta)$ and $\alpha'\circ|f| = \alpha$ `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.Rel`, with an underlying branched suborbifold `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedSuborbifold.toBranched`. *Definition 16.4.5* (p. 731, same) `Chapter16/BranchedSuborbifold · PolyfoldClass.definition_16_4_5_wellDefined`: $\partial\hat\theta = [\partial(\Theta,\hat T_\Theta),(X,\alpha)]$ does not depend on the representative. *The integrals* (p. 732, same) `Chapter16/BranchedTame · PolyfoldClass.OrientedSuborbifold.integral` `Chapter16/BranchedTame · PolyfoldClass.OrientedSuborbifold.integralEff`: $\int_{\hat\theta}\tau = \mu_\omega^{\hat\Theta}(|\operatorname{supp}\Theta|)$ and $\oint_{\hat\theta}\tau$, for a tame compact oriented class, equal to the integrals on every representative `Chapter16/BranchedTame · PolyfoldClass.OrientedSuborbifold.integral_eq` `Chapter16/BranchedTame · PolyfoldClass.OrientedSuborbifold.integralEff_eq`.

Proof of Definition 16.4.4. *Step 1, the normalization.* The text's $\hat T_\Theta$ lives over X_∞, and f^* only reads it there; the formalization makes $\hat T_\Theta$ zero off X_∞ `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.mem_grHatSections`. Reflexivity needs it: $1^*\hat T = \hat T$ holds for such $\hat T$ `Chapter16/BranchedSuborbifold · EPLocalization.EPLoc.pullSection_id_of_mem` `Chapter16/BranchedSuborbifold · EPLocalization.EPLoc.pull_id_of_mem`, and fails otherwise, since the pull-back is 0 off X_∞. Every pull-back of Chapter 11 has this property, so it is a normalization, not an added hypothesis. *Step 2, an equivalence relation* `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.Rel.refl` `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.Rel.symm` `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.Rel.trans`: as in §XVI.17, with the pull-back of oriented branched ep$^+$-subgroupoids of Chapter 11 (Part XI), which is functorial on generalized isomorphisms. *Step 3* `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.toBranchedRep` `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.Rel.toBranchedRep`: forgetting the orientation respects the relations, which defines the underlying θ `Chapter16/BranchedTame · PolyfoldClass.OrientedSuborbifold.toBranched_mk`.

Proof of Definition 16.4.5. *Step 1.* If two representatives are related through f, then $f^*\partial\Theta' = \partial\Theta$ and $f^*\hat T_{\partial\Theta'} = \hat T_{\partial\Theta}$, by Theorem 11.3.9 (Part XI) `Chapter11/BoundaryOrientationEquiv · theorem_11_3_9_pull_boundaryOrientation`. So $\partial\hat\theta$ does not depend on the representative. *Step 2* `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryFun` `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryFun_eq_of_rel`: the boundary function $\partial\theta = |\partial\Theta|\circ\alpha^{-1}$ does not either, so neither does $\partial S = \operatorname{supp}\partial\theta$. *Readings.* The compatibility holds without tameness, which the text assumes for $\partial\hat\theta$ to be a branched suborbifold. $\partial(\Theta,\hat T_\Theta)$ is formalized componentwise, as in Chapter 11.

Proof of the integrals. *Step 1, on a representative.* A representative is tame of dimension n in the sense of Chapter 9 if oriented tame structures describe $\hat T_\Theta$ at every point of $\operatorname{supp}\Theta$ `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.IsTameOriented`. For such a representative that

is compact, $\int = \mu_\omega^{\hat\Theta}(|\operatorname{supp}\Theta|)$ Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.integral and $\oint = \mu_\omega^{\hat\Theta,\mathrm{eff}}(|\operatorname{supp}\Theta|)$ Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.integralEff, with ω the representative of τ on (X,α) (§XVI.10). For the effective integral, (ZHOU) gives the local uniqueness condition (LUC) by Proposition 7.4.10 (Part VII). *Step 2, the boundary integrals on a representative* Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegral Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralEff: the measures of Theorems 9.5.15 and 9.5.16 (Part IX), with separate forms in dimension 1, where the boundary is 0-dimensional Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralDim1 Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralEffDim1. *Step 3, independence of the representative* Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.integral_eq_of_rel Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.integralEff_eq_of_rel: if two tame compact representatives are related through f, the representative of τ on the first structure is f^* of that on the second Chapter16/BranchedSuborbifold · PolyfoldClass.ScForm.toStructure_eq_formsPullback, and the tame integrals are invariant under pull-back, Theorems 11.4.1 and 11.4.2 with $K = |\operatorname{supp}\Theta|$ (Part XI) Chapter16/BranchedSuborbifold · EPLocalization.EPLoc.tameBranchedIntegral_pull Chapter16/BranchedSuborbifold · EPLocalization.EPLoc.tameBranchedIntegralEff_pull. *Step 4, tameness on classes.* Related oriented representatives are tame together Chapter16/BranchedTame · PolyfoldClass.OrientedRep.Rel.isTameOriented Chapter16/BranchedTame · PolyfoldClass.OrientedRep.Rel.isTameOriented_iff: the oriented tame structures move along the transport charts as in §XVI.17, with the orientations pulled back. So tameness of an oriented class Chapter16/BranchedTame · PolyfoldClass.OrientedSuborbifold.IsTameOriented holds in every representative Chapter16/BranchedTame · PolyfoldClass.OrientedSuborbifold.isTameOriented_iff, and compactness is that of any representative Chapter16/BranchedTame · PolyfoldClass.OrientedSuborbifold.isCompactBranched_of_mk. *Step 5, on classes.* The integral of the class is computed on the chosen representative, and by Steps 3–4 equals the integral on every representative. *Step 6, the boundary integrals.* Their independence of the representative is obtained from Stokes' theorem (§XVI.20) Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegral_eq_of_rel Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralEff_eq_of_rel Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.boundaryIntegralDim1_eq_of_rel: both sides equal the integral of $d\tau$, which is independent by Step 3. This needs the hypothesis of §XVI.20, sc-smooth partitions of unity. The boundary integrals are not wrapped as functions on classes (§XVI.28); they would follow from Step 6 as the interior integrals follow from Step 3.

Where Lean departs from the text. The text defines the integrals on p. 732 through a representative and says that they do not depend on the choice; for the interior integrals the formalization proves this through Theorems 11.4.1 and 11.4.2, as the text intends. For the boundary integrals the direct route would be the boundary form of Theorem 11.4.1, which is not formalized; the formalization instead derives their independence from Stokes' theorem, which costs the hypothesis of sc-smooth partitions of unity. Definition 16.4.4 adds the normalization of $\hat T_\Theta$ off X_∞, which the text does not need to state because it only ever reads $\hat T_\Theta$ on X_∞, but which a formal reflexivity proof does need.

XVI.20 Theorem 16.4.6 (2017: Theorem 16.5), p. 732: Stokes' theorem on polyfolds

Statement (with an added hypothesis) Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_hyp. Let (Z,c) admit sc-smooth partitions of unity (Definition 16.1.5), $\hat\theta = [(\Theta,\hat T_\Theta),(X,\alpha)]$ oriented, compact and tame of dimension $n = m+1 \geq 2$, and $\tau \in \Omega^m_{sc}(Z,c)$. Then

$\int_{\hat\theta} d\tau = \int_{\partial\hat\theta} \tau$. The effective form Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_eff_hyp: $\oint_{\hat\theta} d\tau = \oint_{\partial\hat\theta} \tau$. Dimension 1 Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_dim1_hyp Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_eff_dim1_hyp. Both identities together Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_both_hyp.

Proof. *Step 1, to the representative.* The representative of $d\tau$ on (X,α) is d of the representative ω of τ Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.toStructure_d (§XVI.10, the formula $d[(\omega,(X,\alpha))] = [(d\omega,(X,\alpha))]$). So both sides are integrals over Θ on X: $\mu^{\tilde\Theta}_{d\omega}(|\operatorname{supp}\Theta|)$ and the boundary measure of ω. *Step 2, partitions of unity on X.* By Definition 16.1.5 (2) Chapter16/PolyfoldPartitions · PolyfoldClass.definition_16_1_5_two (§XVI.3), X admits sc-smooth partitions of unity, since (Z,c) does. *Step 3, Stokes on X.* Theorem 9.5.17 (Part IX) Chapter9/StokesTextHypotheses · theorem_9_5_17 gives the identity on X; for the effective form, Theorem 9.5.18 Chapter9/StokesTextHypotheses · theorem_9_5_18, with (LUC) from (ZHOU). In dimension 1, the dimension-1 forms of both theorems. *Step 4.* By §XVI.19, Steps 3 and 6, the result does not depend on the representative.

The added hypothesis. Theorem 16.4.6 does not state that (Z,c) admits sc-smooth partitions of unity, but its proof reduces to Theorem 9.5.17, which assumes them for X; the polyfold theorem is proved with the same hypothesis, in the form of Definition 16.1.5. It is not derived from the text. *Witness* Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_hyp_witness: the polyfold $\mathbb{R}$ of §XVI.1 admits sc-smooth partitions of unity Chapter16/PolyfoldExamples · PolyfoldStructure.z3Polyfold_admitsScPartitionsOfUnity. Without the hypothesis the theorem is gap G.153 (§XVI.28). *Reading.* "Tame" is Chapter 9's, on the representative used; by §XVI.19, Step 4, it holds on every representative.

Where Lean departs from the text. None in the argument: the text's proof is the reduction to Theorem 9.5.17 in a representative, and the formalization is that reduction with each translation step made explicit. The departure is in the statement, which carries the hypothesis of Theorem 9.5.17 that the polyfold statement does not print. Dimension 1 is stated separately because there the boundary of $\hat\theta$ is a weighted finite set of points and the boundary measure of Chapter 9 is a weighted sum of values rather than an integral over a manifold; Theorems 9.5.17 and 9.5.18 have separate forms for it, and the polyfold theorem inherits the split. The four Lean statements together cover every dimension $n \geq 1$ and both integrals.

XVI.21 Definitions 16.5.1, 16.5.2, Remark 16.5.3 and (16.8) (2017: Definitions 16.35, 16.36, Remark 16.5), p. 733: sc⁺-multisections and $n(\lambda)$

Statement. *Definition 16.5.1* (same) Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ScPlusMultisection: an sc⁺-multisection $\lambda : Y \to \mathbb{Q}^+$ of a strong polyfold bundle is a class of representatives $(\lambda,\Gamma,W,\Lambda)$ Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.MultisectionRep, Λ an sc⁺-multisection functor of a structure of $\bar c$ (Definition 13.1.4) with $\lambda(\Gamma|w|) = \Lambda(w)$; two are related if $\lambda = \lambda'$ and $\Lambda = \bar f^*\Lambda'$ for a generalized strong bundle isomorphism $\bar f$ with $\Gamma' \circ |\bar f| = \Gamma$ Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.MultisectionRep.Rel. *Definition 16.5.2* (restated) Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ScPlusMultisection.domSupp: the domain support of λ is the closure of the set of z over which some $y \neq 0_z$ has $\lambda(y) > 0$. *Remark 16.5.3* (same, with γ for the printed Γ) Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.remark_16_5_3: for a representative, $\gamma(|\text{dom-supp }\Lambda|) = \text{dom-supp }\lambda$. *(16.8)* (same) Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.normLam Chapter16/NormLamContinuous · StrongPolyfoldBundle.ScPlusMultisection.continuous_normLam: for an auxiliary norm n, $n(\lambda)(z) = \max\{n(y) : p(y) = z, \lambda(y) \neq 0\}$ is a continuous function $Z \to [0,\infty)$.

Background: sc⁺-multisection functors. Over an ep-groupoid, an sc⁺-multisection functor (Definition 13.1.4, Part XIII) is a functor $\Lambda : W \to \mathbb{Q}^+$ which near every object x has a local section structure: an invariant open neighbourhood O of x, finitely many sc⁺-sections $s_1, \dots, s_k$ of P over O and positive rational weights $\sigma_1, \dots, \sigma_k$ with $\sum_i \sigma_i = 1$, such that $\Lambda(w) = \sum_{i : s_i(P(w)) = w} \sigma_i$ for w over O. Thus over every object the values of Λ on the fibre are finitely many positive weights summing to 1, carried by the values of the local sections, and Λ vanishes elsewhere. The domain support (Definition 13.2.2) is the closure of the objects over which Λ charges a non-zero vector, and for an auxiliary norm N the pointwise norm $N(\Lambda)(x)$ (Definition 13.2.3) is the largest $N(w)$ over the charged vectors w of the fibre. A multisection is structurable (Definition 13.3.6) if it admits local section structures that fit together under the morphisms, the condition under which the perturbation theory of Chapters 14 and 15 works. An sc⁺-multisection of a strong polyfold bundle is the function on Y that such a functor induces through Γ.

Proof of Definition 16.5.1. *Step 1, the map $x \mapsto \gamma(|x|)$* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundleStructure.continuous_γπ` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundleStructure.surjective_γπ` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundleStructure.isSaturated_preimage_γπ`: it is continuous, surjective, and preimages of sets of Z are saturated; and $p(\Gamma(|w|)) = \gamma(|P(w)|)$ `Chapter16/PolyfoldMultisections · StrongPolyfoldBundleStructure.p_Γ_π`. These are the translations used throughout §§16.5–16.6. *Step 2, the relation does not depend on $\bar f$* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.pullMult_eq_of_Γ`. By §XVI.11 the isomorphism $\bar f$ need not be unique; but for every $\bar f$ with $\Gamma' \circ |\bar f| = \Gamma$ and every functor Λ' represented by λ, $(\bar f^* \Lambda')(w) = \lambda(\Gamma|w|)$. The pull-back of a multisection functor along a generalized strong bundle isomorphism is read on orbits (Part XIII), Λ' is $\lambda \circ \Gamma' \circ \pi$, and $\Gamma' \circ |\bar f| = \Gamma$. *Step 3* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.MultisectionRep.rel_iff` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.MultisectionRep.rel_of_lam_eq`: so two representatives are related iff their functions agree and their structures are equivalent; and it is an equivalence relation `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.MultisectionRep.setoid`. *Step 4, the check on the $\mathbb{Z}/3$ example* `Chapter16/RepresentativeChecks · StrongPolyfoldBundle.pullMult_z3_independent`: on the strong bundle of §XVI.11, the two generalized isomorphisms $[1_W]$ and $[W \xleftarrow{1} W \xrightarrow{\mathrm{Inv}} W]$ pull every multisection functor back to the same functor, as Step 2 predicts. *Step 5, a multisection is its function* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ScPlusMultisection.toFun` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ScPlusMultisection.ext_toFun`. *Step 6, representatives in every structure* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.exists_rep_of_mem`: every structure of $\bar c$ carries a representative of every λ, by pulling back a representative along a realizing isomorphism and Proposition 11.5.3 (Part XI), which keeps sc⁺-multisection functors. *Step 7* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ofFunctor`: an sc⁺-multisection functor in one structure of $\bar c$ defines an sc⁺-multisection. *Witness* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.zero`: the trivial multisection, $\lambda = 1$ on the zero section and 0 elsewhere.

Proof of Definition 16.5.2 and Remark 16.5.3. *Step 1, the printed set is all of Z* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ScPlusMultisection.domSuppText` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ScPlusMultisection.domSuppText_eq_univ`. As printed, dom-supp(λ) is the closure of $\{z : \exists y \in p^{-1}(z),\ \lambda(y) > 0\}$. Over every z the weights of λ sum to 1 (Definition 13.1.4, through a representative), so some y has $\lambda(y) > 0$, and the set is Z. With $y \neq 0_z$, as in Definition 13.2.2 (Part XIII), it is the intended set: the domain support of the ep-groupoid notion is defined with $w \neq 0$. *Step 2, the set before the closure* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.nzSet` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.isSaturated_nzSet` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.image_nzSet`: in a representative, the set $\{x : \exists w \in W_x \setminus \{0\},\ \Lambda(w) > 0\}$ is saturated (because Λ is a functor and morphisms act linearly on

fibres, so they preserve non-zero vectors) and its image under $x \mapsto \gamma(|x|)$ is the set before the closure in Definition 16.5.2, using that $\Gamma(|w|) = 0_z$ iff w is a zero (§XVI.16). *Step 3, Remark 16.5.3.* The image of the closure of a saturated set under the open quotient map $x \mapsto \gamma(|x|)$ is the closure of its image; with Step 2 this is $\gamma(|\,\text{dom-supp}\,\Lambda|) = \text{dom-supp}\,\lambda$. The text writes Γ; both sets lie in $|X|$ and Z, so we read γ.

Proof of (16.8), the formula. *Step 1, the definition* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.normLam`: $n(\lambda)(z)$ is the supremum of $n(y)$ over y with $p(y) = z$ and $\lambda(y) \neq 0$, where n is extended by 0 off $Y_{0,1}$ (§XVI.16). *Step 2, in a representative* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.image_normY_eq` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.normLam_eq`: the set of values $\{n(y)\}$ over $\gamma(|x|)$ equals the set $\{N(w) : w \in W_x,\ \Lambda(w) \neq 0\}$, because Γ maps the fibre W_x onto $p^{-1}(\gamma|x|)$ up to the isotropy action, which preserves N and Λ. So $n(\lambda)(\gamma(|x|)) = N(\Lambda)(x)$, the function of Definition 13.2.3 (Part XIII). *Step 3* `Chapter16/NormLamContinuous · StrongPolyfoldBundle.ScPlusMultisection.isGreatest_normLam`: the maximum is attained, since Λ has finitely many non-zero values over each x (finitely many local sections in a local section structure). *Step 4* `Chapter16/NormLamContinuous · StrongPolyfoldBundle.ScPlusMultisection.normLam_nonneg`: $n(\lambda) \geq 0$, as n is.

Proof of (16.8), the continuity. *Step 1, $N(\Lambda)$ is continuous on X* `Chapter16/NormLamContinuous · SBLocalization.IsScPlusMultisection.continuous_pointwiseNorm`. Let Λ be an sc$^+$-multisection functor and N continuous on $W_{0,1}$. Near each x there is a local section structure: sc$^+$-sections $s_1, \dots, s_k$ on an open $O \ni x$ with weights $\sigma_i > 0$ summing to 1, with $\Lambda(w) = \sum_{s_i(y)=w} \sigma_i$. On O, $N(\Lambda)(y) = \max_i N(s_i(y))$ `Chapter16/NormLamContinuous · SBLocalization.IsLocalRepOn.pointwiseNorm_eq`, over a non-empty index set `Chapter16/NormLamContinuous · SBLocalization.IsLocalRepOn.univ_nonempty` (the weights sum to 1); every s_i is continuous into $W_{0,1}$, being sc$^+$, so $N \circ s_i$ is continuous, and a finite maximum of continuous functions is continuous. *Step 2, a quotient map* `Chapter16/NormLamContinuous · StrongPolyfoldBundle.isQuotientMap_γπ`: $x \mapsto \gamma(|x|)$ is a quotient map, π being an open quotient map (Proposition 7.1.8, Part VII) and γ a homeomorphism. *Step 3.* $n(\lambda) \circ \gamma \circ \pi = N(\Lambda)$ by the formula, with N the representative of n (continuous by Definition 16.3.9), so $n(\lambda)$ is continuous by Step 2. The text calls $n(\lambda)$ continuous; this is the proof. Gap G.174 is closed.

Where Lean departs from the text. The text defines sc$^+$-multisections by representatives and their relation with a chosen $\bar{f}$; the formalization proves that the choice of $\bar{f}$ does not matter, which the text does not address, and which is needed because $\bar{f}$ is not unique (§XVI.11). The text asserts that $n(\lambda)$ is continuous; the formalization proves it, through the local section structures of Chapter 13 and a quotient map. Definition 16.5.2 is read with $y \neq 0_z$, and Remark 16.5.3 with γ.

XVI.22 Definitions 16.5.4–16.5.6 (2017: Definitions 16.37–16.39), p. 734: structurable multisections, and multisections over ∂Z

Statement. *Definition 16.5.4* (same) `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ScPlusMultisection.IsStructurable`: λ is structurable if one representative is (Definition 13.3.6); "and therefore all" `Chapter16/StructurablePullback · StrongPolyfoldBundle.ScPlusMultisection.definition_16_5_4`. *Definition 16.5.5* (restated) `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisection`: sc$^+$-multisections of $Y|\partial Z$, classes of representatives with Λ a multisection functor on $W|\partial X$ (Definition 14.1.3) `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisectionRep`. *Definition 16.5.6* (same) `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisection.IsStructurable`: λ over ∂Z is structurable if one representative is (Definition 14.1.7). The text writes $\lambda : Z \to \mathbb{Q}^+$ in Definition 16.5.4; we read $\lambda : Y \to \mathbb{Q}^+$, the domain of Definition 16.5.1.

Proof of Definition 16.5.4. *Step 1, the pull-back proposition* `Chapter16/PolyfoldMultisections · StructurablePullback`: pull-backs of structurable sc⁺-multisection functors along generalized strong bundle isomorphisms are structurable. *Step 2, from it the class statement* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.ScPlusMultisection.definition_16_5_4_hyp`: if one representative is structurable, every other is its pull-back along a realizing isomorphism, by Step 2 of §XVI.21, and so is structurable. *Step 3, the proposition holds* `Chapter16/StructurablePullback · fnStructurablePullback`: by Theorems 13.4.1 and 13.4.3 for generalized isomorphisms (Part XIII), which pull back structured representatives along the two legs of a diagram. So Definition 16.5.4 holds as printed `Chapter16/StructurablePullback · StrongPolyfoldBundle.ScPlusMultisection.definition_16_5_4`. *Witness* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.zero_isStructurable`: the zero multisection is structurable.

Proof of Definition 16.5.5. *Step 1, the boundary* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.mem_boundary_iff_obj`: $\gamma(|x|) \in \partial Z$ iff $x \in \partial X$ (§XVI.5). *Step 2, the relation* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisectionRep.Rel`: $\lambda = \lambda'$ over ∂Z, and a generalized strong bundle isomorphism $\bar{f}$ with $\Gamma' \circ |\bar{f}| = \Gamma$ has $\Lambda(w) = \Lambda'(w')$ whenever $|\bar{f}|(|w|) = |w'|$ over ∂X. *Restated* because the pull-back of multisection functors over the boundary is read in this orbit form: a pull-back over ∂X along generalized isomorphisms is part of the gap of Proposition 14.1.8 over ∂X (Part XIV). *Step 3* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisectionRep.rel_iff` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisectionRep.rel_of_lam_eq`: as for Definition 16.5.1, two representatives are related iff their functions agree over $p^{-1}(\partial Z)$ and their structures are equivalent. *Step 4, the function* `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisection.toFun` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisection.toFun_mk_of_mem`: λ on $Y|\partial Z$, read as 0 off $p^{-1}(\partial Z)$, is well defined.

Proof of Definition 16.5.6. Structurability over ∂Z is that of one representative, as printed. The domain support over ∂Z `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisection.domSupp` is Definition 16.5.2 read over ∂Z, as Theorem 16.5.7 uses it, and $|\partial\text{-dom-supp}(\Lambda)| \subseteq \partial\text{-dom-supp}(\lambda)$ through $x \mapsto \gamma(|x|)$ `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.boundaryDomSupp_subset`, the image of the closure lying in the closure of the image by continuity. The text adds no "and therefore all" here; the invariance would need the boundary pull-back of Step 2 above.

Where Lean departs from the text. The text's "and therefore all" in Definition 16.5.4 is left unproved; the formalization isolates the proposition it needs, that structurability pulls back along generalized strong bundle isomorphisms, and proves it from Theorems 13.4.1 and 13.4.3. Definition 16.5.5 is restated, since the pull-back of multisection functors over the boundary along generalized isomorphisms is not available as formalized; its orbit form suffices for the relation, by the same argument as for Definition 16.5.1.

XVI.23 Theorem 16.5.7 (2017: Theorem 16.6), p. 735: extension of sc⁺-multisections from the boundary

Statement (with added hypotheses) `Chapter16/PolyfoldBoundaryExtension · theorem_16_5_7_a6`. Let $p : Y \to Z$ be a tame strong polyfold bundle whose models admit sc-smooth partitions of unity on the objects and tame strong bundle charts at every object, n an auxiliary norm, λ an sc⁺-multisection over ∂Z that is structurable with non-empty index sets `Chapter16/PolyfoldBoundaryExtension · BoundaryMultisection.IsStructurableNe`, $\tilde{U}$ an open set containing dom-supp(λ), and $f : Z \to [0, \infty)$ continuous with closed support in $\tilde{U}$ and $n(\lambda) < f$ on dom-supp(λ). Then there is a structurable sc⁺-multisection λ' with (1) $n(\lambda') \le f$, (2) dom-supp$(\lambda') \subseteq \tilde{U}$, and (3) $\lambda' = \lambda$ over ∂Z.

Readings and added hypotheses. (a) *Non-empty index sets* `Chapter16/PolyfoldBoundaryExtension · BoundaryMultisection.IsStructurableNe`: λ has a representative structurable over ∂X whose Λ charges every boundary fibre; this implies Definition 16.5.6 `Chapter16/PolyfoldBoundaryExtension · BoundaryMultisection.IsStructurableNe.isStructurable`. The text's λ is an sc$^+$-multisection over ∂Z, whose local section structures have non-empty index sets (the weights sum to 1 over each boundary fibre), so this is the text's notion, made explicit because Definition 14.1.3 as formalized admits empty index sets. (b) *Tame strong bundle charts* at every object of every model: an added hypothesis, not derived from the text; it is the reading of Theorem 14.1.9 in Part XIV, where the derivation of such charts from tameness is a gap of that Part. (c) *Partitions of unity* on the objects of every model, the form Theorem 14.1.9 uses; for a polyfold this is Definition 16.1.5 (2) in the object form. (d) "Support" is the closed support.

Proof. *Step 1, choose the model.* Take a representative $a = (\lambda, \Gamma, W, \Lambda)$ over ∂X in a structure S of $\bar{c}$, with Λ structurable and charging every boundary fibre. Let $g = \gamma \circ \pi : X \to Z$; it is continuous (§XVI.21).

Step 2, translate the hypotheses to X. The open set $g^{-1}(\tilde{U})$ is open and saturated `Chapter16/PolyfoldMultisections · StrongPolyfoldBundleStructure.isSaturated_preimage_γπ`. $f \circ g$ is continuous, non-negative, invariant under morphisms (since g is), and its closed support lies in $g^{-1}(\tilde{U})$ `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.tsupport_comp_subset_preimage'`, the closed support of a composite with a continuous map lying in the preimage of the closed support. The boundary domain support of Λ maps into that of λ (§XVI.22), hence lies in $g^{-1}(\tilde{U})$. For x in the boundary domain support of Λ, $x \in \partial X$ (it lies in the closure of a set of boundary points, and ∂X is closed), and $N(\Lambda)(x) = n(\lambda)(g(x))$ by (16.8) in the representative (§XVI.21), with the value of λ over the boundary read through the representative: for w over x, $p(\Gamma|w|) = g(x) \in \partial Z$ `Chapter16/PolyfoldMultisections · StrongPolyfoldBundleStructure.p_Γ_π` `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.mem_boundary_iff_obj`, so $\lambda(\Gamma|w|) = \Lambda(w)$ `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.BoundaryMultisection.toFun_mk_of_mem`. So $N(\Lambda) < f \circ g$ on the boundary domain support of Λ.

Step 3, Theorem 14.1.9 in the model. The charged form of Theorem 14.1.9 (Part XIV) applies in S: X is tame (Definition 16.3.5 (1)), $|X|$ paracompact (Definition 16.3.1), partitions of unity and tame strong bundle charts by (b), (c), Λ structurable over ∂X and charging every boundary fibre by (a), N an auxiliary norm by Definition 16.3.9, and the data of Step 2. It gives a structurable sc$^+$-multisection functor Λ' with $N(\Lambda') \le f \circ g$, dom-supp$(\Lambda') \subseteq g^{-1}(\tilde{U})$ and $\Lambda' = \Lambda$ over ∂X.

Step 4, back to Y. Let λ' be the sc$^+$-multisection of Λ' (§XVI.21, Step 7); it is structurable by Definition 16.5.4. (1): every z is $g(x)$ `Chapter16/PolyfoldMultisections · StrongPolyfoldBundleStructure.surjective_γπ`, and $n(\lambda')(g(x)) = N(\Lambda')(x) \le f(g(x))$ by (16.8). (2): by Remark 16.5.3, dom-supp$(\lambda') = g(\text{dom-supp}\,\Lambda') \subseteq \tilde{U}$. (3): every y over ∂Z is $\Gamma(|w|)$ with $P(w) \in \partial X$, and $\lambda'(y) = \Lambda'(w) = \Lambda(w) = \lambda(y)$.

Where Lean departs from the text. The text's proof is "Theorem 14.1.9 in a representative"; the formalization carries the translation of each hypothesis and conclusion explicitly, through (16.8) and Remark 16.5.3.

A remark on a superseded Lean statement. The declaration `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.theorem_16_5_7_hyp` proves the same conclusion with "Theorem 14.1.9 holds in every model of $\bar{c}$" as a hypothesis, where Theorem 14.1.9 is the Lean statement in which multisections over ∂X may have empty index sets (Definition 14.1.3 as formalized admits them). That Lean statement of Theorem 14.1.9 does not hold when $\partial X \neq \emptyset$ (Part XIV), so this declaration is vacuous for polyfolds with boundary, the case the theorem is about; its other hypotheses are jointly satisfiable on every tame bundle `Chapter16/PolyfoldMultisections · StrongPolyfoldBundle.theorem_16_5_7_hyp_witness`, with the zero multisection over ∂Z,

$\tilde{U} = \emptyset$ and $f = 0$. The statement above, with non-empty index sets, is the one that formalizes Theorem 16.5.7.

Witness of the hypotheses. The ep-groupoid hypotheses of the charged Theorem 14.1.9 hold, with boundary and non-zero fibres, on the half-space bundle of Part XIV. No strong polyfold bundle is built on it, so at the level of polyfolds the tame strong bundle charts and the non-empty index sets have no witness of their own; the remaining hypotheses are witnessed as above, and partitions of unity by the polyfold $\mathbb{R}$ (§XVI.27).

XVI.24 The claim before Definition 16.6.1, Definitions 16.6.1–16.6.3 and Theorem 16.6.4 (2017: Definitions 16.40, 16.41, Theorem 16.7), pp. 735–737: sc-Fredholm sections

Statement. *The claim* (pp. 735–736, same) `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.isFredholm_iff`: if one representative of an sc-smooth section is an sc-Fredholm section functor (Definition 12.1.2), every other is; a section is sc-Fredholm if one representative is `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.IsFredholm`. *Definition 16.6.1* (p. 736, same) `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.FredholmSection`: the sc-Fredholm sections. *Definition 16.6.2* (p. 736, same) `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.HasCompactSolutionSet`: f has a compact solution set if $S_f = \{z : f(z) = 0_z\}$ `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.solutionSet` is compact. *Definition 16.6.3* (pp. 736–737, same; new in 2021) `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.IsProper` `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.IsProperLtInfty`: (1) f is proper if for every auxiliary norm n there is an open $U \supseteq S_f$ with $\{z \in U : n(f(z)) \le 1\}$ having compact closure; (2) f is proper$< \infty$ if the same holds for every C in place of 1, with U depending on C; (n, U) controls compactness if U works with $C = 1$ `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.ControlsCompactness`. *Theorem 16.6.4* (p. 737, same) `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.theorem_16_6_4`: for an sc-Fredholm section, a compact solution set, proper and proper$< \infty$ are equivalent.

Proof of the claim and Definition 16.6.1. *Step 1.* Two representatives are related by a generalized strong bundle isomorphism pushing one functor to the other (§XVI.16). *Step 2.* Theorem 10.5.4 (Part X): the push-forward of an sc-Fredholm section functor along a generalized strong bundle isomorphism is sc-Fredholm, since the legs are local strong bundle isomorphisms and the sc-Fredholm property is local and invariant under them (Part XII). *Step 3* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.exists_rep_of_mem`: every structure of $\bar{c}$ carries a representative of every sc-smooth section, by pushing a representative along a realizing isomorphism. So the property is that of any representative in any structure.

The dictionary between Z and a representative. For a representative $a = (f, \Gamma, W, F)$ write $g = \gamma \circ \pi : X \to Z$. *Step 1* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.isSection_F`: $P \circ F = \mathrm{id}$. *Step 2* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.Γ_mem_Y01_iff`: $\Gamma(|w|) \in Y_{0,1}$ iff $w \in W_{0,1}$. *Step 3, the solution set* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.solutionSet_eq`: $S_f = g(S_F)$, because $f(g(x)) = \Gamma(|F(x)|)$ and $\Gamma(|w|) = 0_z$ iff w is a zero (§XVI.16). *Step 4, sublevel sets* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScSection.sublevel` `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.sublevel_eq`: $\{z \in V : f(z) \in Y_{0,1},\ n(f(z)) \le C\} = g(\{x \in g^{-1}(V) : N(F(x)) \le C\})$ with N

the representative of n; the text's "$n(f(z)) \le C$" is read as $f(z) \in Y_{0,1}$ and $n(f(z)) \le C$, since n is defined on $Y_{0,1}$. *Step 5, topology* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.isOpen_g_image` `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.preimage_g_image` `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.isCompact_closure_g_image_iff`: g is open, a saturated set is the preimage of its image, and a saturated set has compact closure in X's orbit space iff its image has compact closure in Z (the image of a compact set under the continuous g is compact; conversely, γ is a homeomorphism, so compactness in Z is compactness in $|X|$, which is the meaning of compact closure for saturated sets in Chapter 12). *Step 6, the U-conditions* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.forall_iff_of_saturated` `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.forall_iff_of_open` `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.solutionSet_subset_iff` `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.solutionSet_subset_preimage_iff`: an open $U \supseteq S_f$ in Z corresponds to the saturated open $g^{-1}(U) \supseteq S_F$, and conversely a saturated open $U' \supseteq S_F$ to $g(U')$, with the sublevel conditions corresponding for each C. *Step 7, every auxiliary norm* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.forall_auxNorm_iff`: "for every auxiliary norm n of Y" may be read as "for every auxiliary norm N of W", since auxiliary norms correspond by descent and representation (§XVI.13).

Proof of Definitions 16.6.2, 16.6.3 in a representative. *Definition 16.6.2* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.hasCompactSolutionSet_iff`: S_f is compact iff $|S_F|$ is, by Steps 3 and 5; that is Definition 12.4.1 (1). *Definition 16.6.3 (1)* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.isProper_iff`: by Steps 4–7 it is Definition 12.4.1 (2). *Definition 16.6.3 (2)* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.isProperLtInfty_iff`: Definition 12.4.1 (3). *Controlling compactness* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.SectionRep.controlsCompactness_iff`: (n, U) controls compactness iff $(N, g^{-1}(U))$ does in the sense of Definition 15.3.1 (Part XV).

Proof of Theorem 16.6.4. Take a representative in any structure (Step 3 of the claim). By the translations above, the three conditions are those of Definition 12.4.1 for F, and Theorem 12.4.2 (Part XII) shows that they are equivalent for an sc-Fredholm section functor over an ep-groupoid with paracompact orbit space, which every structure of $\bar{c}$ has.

Where Lean departs from the text. The text proves the claim before Definition 16.6.1 by Theorem 10.5.4, and Theorem 16.6.4 by Theorem 12.4.2 in a representative; the formalization follows both. What it adds is the dictionary between conditions on Z, Y and conditions on a model, which the text leaves implicit: solution sets, sublevel sets, open neighbourhoods of the solution set, compact closures and the quantifier over auxiliary norms. Each translation is a separate lemma, and together they show that every notion of Definitions 16.6.2 and 16.6.3 is, in every representative, the corresponding notion of Definition 12.4.1.

XVI.25 Definition 16.6.5 (2017: Definition 16.42), p. 739: orientations of sc-Fredholm sections

Statement (narrower). *The definition* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.OrientationRep` `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.FredholmSection.IsOrientable`: an orientation of an sc-Fredholm section f is an orientation of a representative (W, F): sc$^+$-sections through every point, a topology on the orientation bundle O_F as in Theorem 12.5.3, and an orientation in the sense of Definition 12.5.4; f is orientable if one exists. *The transport* `Chapter16/OrientationTransport · StrongPolyfoldBundle.`

`OrientationRep.transport_hyp`: an orientation of one representative gives one of every other representative, the text's "an orientation of one representative determines orientations of all". *Narrower* in two ways: the standing hypotheses of the M-polyfold theory (tameness, sc-smooth bump functions, tame sc-Fredholm germs) are taken on the bundles of the diagram along which the orientation is carried, as in Theorem 12.5.7; and that the carried orientation does not depend on the diagram is not formalized. Both are gap G.176, formalized in part (§XVI.28).

Background: orientations of sc-Fredholm section functors. For an sc-Fredholm section functor F of a strong bundle over an ep-groupoid, Chapter 12 attaches to every smooth point x the set of linearizations of F at x, which differ by sc$^+$-operators, and to each linearization its determinant line; Theorem 12.5.3 shows that these lines form a bundle O_F over the smooth points with a topology for which local sc$^+$-sections through a point give continuous local trivializations. An orientation (Definition 12.5.4) is a choice of one of the two orientations of each determinant line that is continuous in this topology and invariant under the morphisms of X, which act on linearizations by conjugation with the local strong bundle isomorphisms of the action. Theorem 12.5.7 shows that a strong bundle equivalence transports orientations, because it identifies linearizations at corresponding points. On a polyfold an orientation is therefore an orientation of one representative together with the promise, made by the text and proved here in part, that the transports to every other representative are consistent.

Proof of the transport along a diagram `Chapter16/OrientationTransport · SBLocalization.SDiagram.orientation_transport_hyp`. *Statement:* for a diagram $D = [W \xleftarrow{L} W'' \xrightarrow{L'} W']$ of strong bundle equivalences with $[D]_*F = F'$ and an orientation o of F in the sense of Definition 12.5.4, for any topology on O_F as in Theorem 12.5.3, there is exactly one invariant orientation o' of F' compatible with o over W'', and it satisfies Definition 12.5.4 for every topology on $O_{F'}$ as in Theorem 12.5.3. *Step 1, pull back along L.* Theorem 12.5.7 (Part XII) carries o to an orientation of L^*F on W'', the linearizations of F and L^*F being identified by the local strong bundle isomorphisms of L; the standing hypotheses on W and W'' are those of Theorem 12.5.7. *Step 2, push forward along L'.* The same theorem in the other direction carries it to F', using $[D]_*F = F'$, that is $L'_*L^*F = F'$ up to the identification of the middle. *Step 3, uniqueness and invariance.* An invariant orientation of F' is determined on orbits, and over the image of L' it is forced by compatibility; since L' is essentially surjective, every orbit meets the image. *Step 4, continuity for every topology* `Chapter16/OrientationTransport · SBLocalization.SDiagram.exists_orientation_of_eq`: Theorem 12.5.3 allows several topologies on O_F, all with the same continuous sections in the sense used by Definition 12.5.4; Part XII identifies the two readings of continuity, and an orientation for every such topology carries over along an equality $F_1 = F_2$.

Proof on classes. *Step 1* `Chapter16/OrientationTransport · StrongPolyfoldBundle.OrientationRep.exists_diagram`: for an orientation of a representative (W, F) and any other representative (W', F') of the same section, a diagram D with $[D]_*F = F'$ exists, by §XVI.16 (related representatives are related by a generalized strong bundle isomorphism pushing one functor to the other) and a representative diagram of that isomorphism. *Step 2* `Chapter16/OrientationTransport · StrongPolyfoldBundle.OrientationRep.transport_hyp`: the transport along D gives an orientation of F'.

What is not formalized. (i) Two diagrams give the same transported orientation; this needs a common refinement and the compatibility of the orientation maps of Theorem 12.5.7 with composition and with natural equivalences, which Part XII proves for the data of diagrams. (ii) The standing hypotheses on the middle bundle W'' of a diagram follow from those on W and W'; this needs the transfer of tameness, bump functions and tame germ data along strong bundle equivalences. On the $\mathbb{Z}/3$ example the two isomorphisms of §XVI.11 are the identity on objects and transport alike; no example was found where

the diagram matters.

Where Lean departs from the text. The text says that an orientation of one representative determines orientations of all, by Theorem 12.5.7, and defines the orientation of f as the resulting class. The formalization proves the first half, the transport along any diagram, and stops short of the second: that the result does not depend on the diagram, which the text does not argue, and which would be needed for the orientation of f to be a well-defined class rather than a choice on one representative. The formal Definition 16.6.5 therefore records the orientation on one representative, which is what Theorem 16.6.6 uses.

XVI.26 Theorem 16.6.6 (2017: Theorem 16.8), p. 739: the invariant of an oriented sc-Fredholm section

Statement (narrower) `Chapter16/PerturbedIntegralIndependence · StrongPolyfoldBundle.theorem_16_6_6_hyp`. The text: for $\partial Z = \emptyset$, f an oriented sc-Fredholm section with compact solution set, n an auxiliary norm and U open with (n, U) controlling compactness, there are structurable sc$^+$-multisections λ controlled by (n, U) in general position; $\lambda \circ f$ is a compact branched suborbifold with an orientation inherited from (f, o), and $\Phi_{(f,o)}([\tau]) = \int_{(\lambda\circ f,o)} \tau$ defines a linear map on cohomology that does not depend on λ. *The statement in Lean* `Chapter16/PerturbedIntegralIndependence · StrongPolyfoldBundle.theorem_16_6_6_statement`, degree-wise in the dimension $m + 1$ of $\lambda \circ f$, with a parameter IsInherited, a predicate on pairs of an orientation of f and an oriented perturbation: under the text's hypotheses, for every orientation o of f, (i) for every admissible oriented perturbation with the inherited orientation, $\int$ and $\oint$ depend only on $[\tau]$, and (ii) for closed τ, any two such perturbations give the same $\int$ and the same $\oint$. *Proved* given sc-smooth partitions of unity on Z and cobordisms between any two such perturbations. *Narrower*: "inherited from (f, o)" is a parameter; the cobordisms are a hypothesis; "general position" is read in one model; index 0 is not covered.

How the pieces fit. Theorem 16.6.6 combines four ingredients, which the formalization keeps separate so that each can be checked against its source. The perturbation λ comes from Theorem 15.3.7 in one model, transported to the polyfold by (16.8) and Remark 16.5.3. The branched suborbifold $\lambda \circ f$ comes from Theorem 15.4.3 in that model, transported by the fact that a branched suborbifold is its function. The integral of a form over it comes from Chapter 9 in that model, transported by Theorems 11.4.1 and 11.4.2, and its dependence only on the cohomology class comes from Stokes' theorem on the polyfold. The independence of λ comes from a cobordism, which Chapter 15 constructs on a cylinder over one model. Of these, the first three are proved here for the polyfold; the fourth and the orientation that it must respect are the open part.

Controlled multisections and admissible perturbations. *Controlled* `Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.ScPlusMultisection.IsControlledBy`: λ is controlled by (n, U) if $n(\lambda) \leq 1$ and dom-supp$(\lambda) \subseteq U$ (the example after Definition 15.3.3). *General position* `Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.ScSection.IsInGeneralPositionWith`: (f, λ) is in general position if, in some model, the representatives (F, Λ) are (Definitions 15.2.1, 15.3.4). *An oriented perturbation of dimension* $m + 1$ `Chapter16/PerturbedIntegralIndependence · StrongPolyfoldBundle.OrientedPerturbation`: a tame compact oriented branched suborbifold whose function is $\lambda \circ f$ for an sc$^+$-multisection λ; it is *admissible* `Chapter16/PerturbedIntegralIndependence · StrongPolyfoldBundle.OrientedPerturbation.IsAdmissible` if λ is controlled by (n, U) and (f, λ) is in general position. The text's $Z_{0,1}$ for the domain of n is read as $Y_{0,1}$.

Proof, $\lambda\circ f$ as a branched suborbifold. *Step 1, the function* Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.perturbedFun Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.IsPerturbedSuborbifold: $\theta = \lambda \circ f : Z \to \mathbb{Q}^+$; a branched suborbifold "is $\lambda \circ f$" if its function is. *Step 2, in a model* Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.toPolyfoldStructure_mem Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.lam_f_eq: the base (X,γ) of a model $S \in \bar{c}$ is a structure of (Z,c), and $\lambda(f(\gamma|x|)) = \Lambda(F(x))$ for representatives F of f and Λ of λ in S. *Step 3, the representative* Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.perturbedRep: $(\Lambda \circ F,(X,\gamma))$, for $\Lambda \circ F$ a branched ep⁺-subgroupoid, which every model admits since every model carries a functor representing λ Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.exists_functor_rep. Its function is $\lambda \circ f$ in every model Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.perturbedRep_toFun Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.perturbedRep_toFun_ofFunctor. *Step 4* Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.isPerturbedSuborbifold_iff Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.isPerturbedSuborbifold_unique: since a branched suborbifold is determined by its function (§XVI.17), $\theta = \lambda \circ f$ iff θ is the class of $(\Lambda \circ F,(X,\gamma))$ in any model, and such θ is unique. *Step 5, existence* (explicit) Chapter16/PerturbedSuborbifold · StrongPolyfoldBundle.exists_isPerturbedSuborbifold_hyp: for $\partial Z = \emptyset$, f sc-Fredholm, (n,U) controlling compactness and λ controlled by (n,U) and in general position in some model, $\lambda \circ f$ is a compact branched suborbifold. In the model, Theorem 15.4.3 (Part XV) makes $\Lambda \circ F$ a compact branched ep⁺-subgroupoid; good position is vacuous without boundary. The tame germ data of the model are inherited as in Part XV.

Proof, the reduction to a model Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.theorem_16_6_6_reduction. For $\partial Z = \emptyset$, f sc-Fredholm with compact solution set, (n,U) controlling compactness and λ controlled by (n,U): in every structure of $\bar{c}$, the representatives F, Λ satisfy the hypotheses of the ep-groupoid statements of §§15.3–15.4. *Step 1.* $\partial X = \emptyset$, since ∂Z is computed in any representative (§XVI.5) Chapter16/PerturbedIntegralIndependence · PolyfoldClass.not_mem_boundary_of_eq_empty. *Step 2.* $|S_F|$ is compact (§XVI.24). *Step 3.* $(N, g^{-1}(U))$ controls compactness (§XVI.24). *Step 4.* Λ is controlled: $N(\Lambda) = n(\lambda) \circ g \le 1$ by (16.8), and dom-supp$(\Lambda) \subseteq g^{-1}(U)$ by Remark 16.5.3.

Proof, the choice of λ Chapter16/PolyfoldFredholmSections · StrongPolyfoldBundle.exists_generalPosition_hyp. From the statement of Theorem 15.3.7 in a representative, with its standing hypotheses (tameness, sc-smooth bump functions, tame germ data), there is a structurable λ controlled by (n,U) with $n(\lambda) < 1$ whose representative is transversal and in general position. *Step 1.* By the reduction, the hypotheses of Theorem 15.3.7 hold in the model. *Step 2.* Theorem 15.3.7 gives Λ. *Step 3.* λ is the multisection of Λ (§XVI.21), structurable (§XVI.22), controlled by (16.8) and Remark 16.5.3. *The hypothesis.* Theorem 15.3.7 in the representative is not an assumption of the text; it is discharged by Part XV's proof of Theorem 15.3.7 for paracompact orbit spaces, which every strong polyfold bundle structure has (Definition 16.3.1). The unconditional form is a one-line corollary, not stated as a separate declaration.

Proof, the integrals on the polyfold. *Step 1, additivity* Chapter16/PerturbedIntegralIndependence · PolyfoldClass.ScForm.toStructure_add Chapter16/PerturbedIntegralIndependence · PolyfoldClass.ScForm.toStructure_zero Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedRep.integral_add Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedRep.integralEff_add Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedSuborbifold.integral_add Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedSuborbifold.integralEff_add: the representative of a sum is the sum of the representatives, and the branched integrals of Chapter 9 are additive in the form. *Step 2, exact forms* Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedRep.integral_d_eq_zero_of_boundary Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedRep.integralEff_d_eq_zero_of_boundary Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedSuborbifold.integral_d_eq_zero Chapter16/

PerturbedIntegralIndependence · PolyfoldClass.OrientedSuborbifold.integralEff_d_eq_zero: for $\partial Z = \emptyset$ with sc-smooth partitions of unity, $\int_{\hat\theta} d\eta = 0$ and $\oint_{\hat\theta} d\eta = 0$ for every tame compact oriented branched suborbifold of dimension $m+1 \geq 1$. By Stokes (§XVI.20), $\int_{\hat\theta} d\eta = \int_{\partial\hat\theta} \eta$, and $\partial\Theta = 0$ because no object lies in ∂X. *Step 3* Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedSuborbifold.integral_add_d Chapter16/PerturbedIntegralIndependence · PolyfoldClass.OrientedSuborbifold.integralEff_add_d: so the integrals depend only on $[\tau]$. *Step 4, compact oriented suborbifolds* Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.integral Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.integralEff: tame compact oriented branched suborbifolds of dimension $m+1$ bundled with their proofs; their integral is, in every model, the integral of Definition 9.5.6 Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.integral_mk_rep (§XVI.19).

Proof, cobordisms. *The notion* Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.HasCobordism Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.HasCobordismEff: P_0 and P_1 are joined by an oriented compact cobordism if there are an ep-groupoid Y with sc-smooth partitions of unity (and, for the effective form, the local unique continuation property), a compact oriented tame branched ep$^+$-subgroupoid $\tilde\Theta$ of Y of dimension $m+2$, and an sc-smooth F from Y into a model of Z, with $\int_{\partial\tilde\Theta} F^*\omega = \int_{P_1}\omega - \int_{P_0}\omega$ for every form ω. This is the shape of the proof of Corollary 15.4.4 (Part XV). *Closed forms* Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.integral_eq_of_hasCobordism Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.integralEff_eq_of_hasCobordismEff: for $d\omega = 0$, $\int_{P_1}\omega - \int_{P_0}\omega = \int_{\partial\tilde\Theta} F^*\omega = \int_{\tilde\Theta} dF^*\omega = \int_{\tilde\Theta} F^* d\omega = 0$, by Theorem 9.5.17 on Y (Theorem 9.5.18 for the effective form) and the commutation of d with pull-back. *Satisfiable* Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.hasCobordism_refl Chapter16/PerturbedIntegralIndependence · PolyfoldClass.CompactOrientedSuborbifold.hasCobordismEff_refl: every P is cobordant to itself, with $Y = X$ and $\tilde\Theta = 0$.

Proof of Theorem 16.6.6. Let o be an orientation of f. (i) For an admissible perturbation with the inherited orientation, $\int_P(\tau + d\eta) = \int_P \tau$ and the same for $\oint$, by the integrals on the polyfold, Step 3. (ii) For two such perturbations P_0, P_1 and closed τ, the cobordism hypothesis gives $\tilde\Theta$, and the closed-forms identity gives $\int_{P_0}\tau = \int_{P_1}\tau$, and likewise for $\oint$. The well-definedness of the integral on the class (§XVI.19) uses no cobordism.

Added hypotheses and parameters. (a) Sc-smooth partitions of unity on Z, the hypothesis of Stokes' theorem (§XVI.20); not derived from the text. Witness: the polyfold $\mathbb{R}$ of §XVI.1 (§XVI.27). (b) The cobordisms between any two admissible perturbations with the inherited orientation: Corollary 15.4.4 on the polyfold, which is gap G.175 in part. *Witness* Chapter16/PerturbedIntegralIndependence · StrongPolyfoldBundle.theorem_16_6_6_hyp_hcob_witness: for the predicate IsInherited that singles out one perturbation P, the cobordism hypotheses hold through the self-cobordism of P. **This witness covers only a degenerate case**: with one perturbation there is nothing to compare, and no strong polyfold bundle with an admissible oriented perturbation of non-empty support is built. (c) "Inherited from (f, o)" is a parameter, any predicate on oriented perturbations; its definition needs the canonical orientation of Theorem 15.4.1 on the polyfold (gap G.159 of Part XV and gap G.175 in part). (d) "Generic" is read as general position in some model; its independence of the model is gap G.187. (e) The degree is the dimension $m+1 \geq 1$ of $\lambda \circ f$; index 0, where the integral is a weighted count of points, is gap G.188, the branched integrals of Chapter 9 being defined in dimension at least 1.

Where Lean departs from the text. The text's proof of Theorem 16.6.6 refers to Chapter 15 in a representative:

Theorem 15.3.7 for the choice of λ, Theorem 15.4.1 for the orientation of $\lambda \circ f$, and Corollary 15.4.4 for the independence of λ. The formalization does the first unconditionally, and isolates the second and third: the orientation as a parameter, the cobordism as a hypothesis. It also separates what does not need Chapter 15 at all: that the integrals are defined on the polyfold and depend only on $[\tau]$ follows from Stokes' theorem alone, because $\partial Z = \emptyset$. The text's statement is in cohomological degree k; the formalization is degree-wise in the dimension $m + 1$ of $\lambda \circ f$, which equals the Fredholm index, and does not cover index 0.

XVI.27 Witnesses and the hypotheses

Hypotheses taken, by section. Each is named, is not derived from the text unless said, and has a witness.

- *§XVI.2, §XVI.7.* The effective forms of the uniqueness after Definition 16.1.2, Remark 16.1.7 and Lemma 16.2.1 take an effective target. Witness: the point ep-groupoid `Chapter16/TangentLemma · PolyfoldStructure.lemma_16_2_1_of_isEffective_hypotheses`. Without it the statements do not hold as we read them; the orbit form of Lemma 16.2.1 needs no added hypothesis.
- *§XVI.4.* An effective middle polyfold for the composition of sc-smooth maps and for Definition 16.1.11. The text claims the composition in general; as we read it, it does not hold without effectiveness. Witness: the one-point polyfold `Chapter16/PolyfoldMaps · EffPolyfold.point`.
- *§XVI.9.* The functor on effective polyfolds takes only effectiveness; (ZHOU), which the effectiveness of the tangent needs, is part of every polyfold structure. The form on classes with an effective middle polyfold whose tangent is effective has that second hypothesis discharged by the theorem that tangents of effective polyfolds are effective, so it holds whenever the composite is defined.
- *§XVI.10.* The pull-back of forms by a composite of classes takes an effective middle polyfold, where the composite is defined; the pull-back by one sc-smooth map takes nothing.
- *§XVI.11.* The effective form of the uniqueness after Definition 16.3.3 takes an effective target base; satisfiable, since trivial ep-groupoids are effective (Chapter 7).
- *§XVI.15.* Theorem 16.3.17 (2): the reflexive local step of Theorem 12.2.13. Witness `Chapter16/PolyfoldAuxNorm · StrongPolyfoldBundle.theorem_16_3_17_two_hyp_witness` on every bundle with reflexive 1-fibres, **degenerate**: the boundary norm is the restriction of a global reflexive norm. The non-degenerate case is Part XII's proof of the local step over tame bases, which Theorem 16.3.17 assumes.
- *§XVI.20.* Sc-smooth partitions of unity, in Theorem 16.4.6 and in the independence of the boundary integrals (§XVI.19). Witness: the polyfold $\mathbb{R}$ `Chapter16/BranchedSuborbifold · PolyfoldClass.OrientedRep.theorem_16_4_6_hyp_witness`.
- *§XVI.23.* Tame strong bundle charts in every model, and sc-smooth partitions of unity on the objects of every model. The partitions of unity are witnessed on the polyfold $\mathbb{R}$; the ep-groupoid hypotheses of the charged Theorem 14.1.9 hold non-degenerately, with boundary and non-zero fibres, on the half-space bundle of Part XIV; at the level of polyfolds, the charts and the non-empty index sets have no witness of their own. The remaining hypotheses (structurable λ over ∂Z, $\tilde{U}$, f) are jointly satisfiable on every tame bundle with the zero multisection, $\tilde{U} = \emptyset$, $f = 0$, which is degenerate.
- *§XVI.25.* The standing hypotheses of the M-polyfold theory on the three bundles of the diagram, as in Theorem 12.5.7; Part XII's witnesses for Theorem 12.5.7 apply.
- *§XVI.26.* Sc-smooth partitions of unity on Z (witness: the polyfold $\mathbb{R}$); the cobordisms (witness **degenerate**: one perturbation, self-cobordism); the predicate "inherited" (a parameter, not a hy-

pothesis on data); the inherited tame germ data in the existence of $\lambda \circ f$.

No hypothesis of this Part quantifies over all charts of a bundle, and none is unsatisfiable: each has a witness above, degenerate where said.

Readings, not hypotheses. Bump functions on the object M-polyfold (decision (2)); the normalization $\widehat{T}_\Theta = 0$ off X_∞ (§XVI.19); $y \neq 0_z$ in Definition 16.5.2 (§XVI.21); γ for Γ in Remark 16.5.3; $\lambda : Y \to \mathbb{Q}^+$ in Definition 16.5.4; $n(f(z)) \leq C$ read with $f(z) \in Y_{0,1}$ in Definition 16.6.3; $Y_{0,1}$ for the printed $Z_{0,1}$ in Theorem 16.6.6; "n reflexive" read in one model in Theorem 16.3.17 (2); the boundary pull-back in orbit form in Definition 16.5.5.

Every definition has a witness.

Polyfolds (on the polyfold $\mathbb{R}$ given by $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$, §XVI.1). Polyfold structure and polyfold; tame `Chapter16/PolyfoldExamples · PolyfoldStructure.z3Polyfold_isTame`; admits sc-smooth bump functions `Chapter16/PolyfoldExamples · PolyfoldStructure.z3Polyfold_admitsBumpFunctions` and sc-smooth partitions of unity `Chapter16/PolyfoldExamples · PolyfoldStructure.z3Polyfold_admitsScPartitionsOfUnity`; degeneracy index $\equiv 0$ `Chapter16/PolyfoldExamples · PolyfoldStructure.z3Polyfold_degIndex` and empty boundary `Chapter16/PolyfoldExamples · PolyfoldStructure.z3Polyfold_boundary`. The same polyfold is the example of §XVI.2, so the hypotheses of Definition 16.1.1 are satisfiable there in a non-effective way. Its tangent and its raised polyfolds are witnesses of Definition 16.2.3 and of $(Z, c)^i$, τ_Z.

Maps. The identity sc-smooth map on every polyfold; the category PS, with identities; effective polyfolds, the one-point polyfold. *Forms.* $\Omega^k_{sc}(Z, c)$ is a vector space isomorphic to $\Omega^k_{ep,\infty}(X)$ for every structure in c (§XVI.10).

Strong polyfold bundles (on the $\mathbb{Z}/3$ bundle $\mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$, §XVI.11, with non-trivial isotropy). Structure and bundle; tame `Chapter16/PolyfoldBundleExamples · StrongPolyfoldBundle.z3Bundle_isTame`; an invertible morphism of $\mathcal{SPBS}$ `Chapter16/PolyfoldBundleExamples · StrongPolyfoldBundle.z3_id_isInvertible`; reflexive 1-fibres `Chapter16/PolyfoldBundleExamples · StrongPolyfoldBundle.z3Bundle_hasReflexiveFibers`, since $\mathbb{R}^k$ is reflexive; an auxiliary norm `Chapter16/PolyfoldBundleExamples · StrongPolyfoldBundle.z3Bundle_exists_auxNorm` and a reflexive one `Chapter16/PolyfoldBundleExamples · StrongPolyfoldBundle.z3Bundle_exists_reflexiveAuxNorm`; auxiliary norms over the boundary, by restriction `Chapter16/PolyfoldBundleExamples · StrongPolyfoldBundle.IsAuxNorm.isAuxNormBoundary`, and the hypotheses of Theorem 16.3.17 (1) `Chapter16/PolyfoldBundleExamples · StrongPolyfoldBundle.theorem_16_3_17_one_witness`. *Sections.* The zero section on every strong polyfold bundle (§XVI.16), an sc$^+$-section. *Multisections.* The zero multisection, structurable; over ∂Z, the zero multisection of the witness of §XVI.23. *Sc-Fredholm sections and their orientations* are not separately witnessed at the polyfold level; Part XII's examples of sc-Fredholm section functors and orientations give them in any model.

Branched suborbifolds. A class exists as soon as a representative does; $\lambda \circ f$ gives one in every model (§XVI.26). No separate polyfold-level example of a compact tame oriented one is built; the hemisphere of Part IX is the nearest ep-level example.

XVI.28 What is not formalized

- **Gap G.153**: Stokes' theorem on polyfolds without partitions of unity (Theorem 16.4.6). Proved with sc-smooth partitions of unity, as Theorem 9.5.17 assumes (§XVI.20). Missing: Stokes on ep-groupoids without partitions of unity, using a finite cut-off from the compactness of $\operatorname{supp}\Theta$, so

that only finitely many local charts are needed and the partition can be built by hand from bump functions. Estimate: 300–600 lines. A candidate for the notes to the authors: whether the text intends partitions of unity here.

- **Gap G.175 (formalized in part).** Proved: the integrals of Theorem 16.6.6 on the polyfold, their dependence only on $[\tau]$, and their independence of λ given cobordisms (§XVI.26). Missing: (a) the orientation of $\lambda \circ f$ inherited from (f, o), on the polyfold (p. 739, Definition 15.4.2), which is a parameter of Theorem 16.6.6: Theorem 15.4.1 in a model beyond single points (gap G.159 of Part XV); the lift of its orientation to an oriented representative on the model, with independent tangent sums and $\widehat{T} = 0$ off X_∞; its independence of the model, by the transport of o (G.176) and the pull-back of $\widehat{T}$ along generalized isomorphisms; 250–450 lines after G.159. (b) The cobordism between two admissible perturbations, on the polyfold, a hypothesis of Theorem 16.6.6: Corollary 15.4.4 on the cylinder of a model, with the gaps of Part XV; the transfer of its cobordism to the polyfold, the integral side being done by the comparison of integrals in a model; about 80–150 lines; and the effective form, with the local unique continuation property on the cylinder.
- **Gap G.176 (formalized in part)**: orientations of sc-Fredholm sections do not depend on the representative (Definition 16.6.5). Proved: the transport along any diagram with $[D]_* F = F'$, with continuity, and an orientation on every other representative (§XVI.25). Missing: (i) two diagrams give the same transported orientation, through a common refinement and the compatibility of the orientation maps with composition and natural equivalences, which Part XII proves for the data of diagrams; 200–300 lines. (ii) The transfer of tameness, bump functions and tame germ data along strong bundle equivalences, so that the hypotheses on the middle bundle can be dropped; 100–200 lines.
- **Gap G.187**: general position of (f, λ) does not depend on the model; §XVI.26 reads it in one model. Missing: the invariance of general position under generalized strong bundle isomorphisms, the linearizations being transported by the local sc-diffeomorphisms as for Fredholmness in Theorem 10.5.4; 200–350 lines.
- **Gap G.188**: Theorem 16.6.6 in Fredholm index 0, where the integral is a weighted count; the branched integrals of Chapter 9 are defined in dimension at least 1 (§XVI.26); 150–300 lines.
- *Closed gaps of this chapter.* G.146 (metrizability, §XVI.1), G.154 (tameness of branched suborbifolds, §XVI.17), G.155 (Proposition 16.4.3, §XVI.18), G.156 (raised polyfolds and τ_Z, §§XVI.6, XVI.8), G.157 (the tangent of an effective polyfold, §XVI.9), G.166 (reflexivity, mixed convergence and reflexive norms along equivalences, §XVI.14), G.167 (the zero section, §XVI.16), G.174 (continuity of $n(\lambda)$, §XVI.21).
- *Not formalized, small.* The boundary integrals of p. 732 as functions on classes; they follow from the independence among representatives (§XVI.19, Step 6) as the interior integrals do; about 40 lines. Weak face structure under generalized isomorphisms, which Definition 16.3.5 (2) would need in its class form (§XVI.11). The composition of strong polyfold bundle maps on classes, which the text does not define (§XVI.12). The invariance of structurability over ∂Z (§XVI.22), which needs the boundary pull-back of Proposition 14.1.8 (Part XIV).
- *Not formalized, no formal content.* The second sentence of Remark 16.1.7, on continuous maps; Remark 16.3.11, which is prose.
- *Not yet formalized.* Proposition 16.1.12 (the lifts Z_i and their topology), Definition 16.1.15 and Remark 16.1.16 (subpolyfolds), Example 16.1.18, Definition 16.1.21 and Proposition 16.1.22 (faces of a polyfold); Remark 16.2.7.
- *Open question, not a gap.* In the example of §XVI.7, $|T\mathfrak{f}| = |T\mathfrak{g}|$ is proved; whether $T\mathfrak{f} \neq T\mathfrak{g}$ is not decided.

XVI.29 What the formalization taught us

(ZHOU) does not give uniqueness of isomorphisms. A polyfold structure in the 2021 sense adds two conditions to the 2017 definition, paracompactness of $|X|$ and (ZHOU), and the text derives from (ZHOU) that the generalized isomorphism realizing an equivalence of structures is unique. Formalized, (ZHOU) gives (LR), and (LR) is the hypothesis of Theorem 10.3.19. But Theorem 10.3.19 does not hold as we read it (Part X), and its example, $\mathbb{Z}/3$ acting trivially on $\mathbb{R}$, satisfies (ZHOU) and has a paracompact orbit space. It is therefore a polyfold structure, and the two generalized isomorphisms $[1_X]$ and $[X \overset{1}{\leftarrow} X \overset{\mathrm{Inv}}{\longrightarrow} X]$ both realize the identity of $\mathbb{R}$. Five statements of the chapter do not hold as we read them for this reason: the uniqueness after Definition 16.1.2, Remark 16.1.7 with the sentence on pp. 709–710, Lemma 16.2.1, the claim of Definition 16.1.10 and the uniqueness after Definition 16.3.3. All five hold for an effective target or middle structure, where Theorem 10.3.19 holds. The isotropy that acts trivially is exactly what (ZHOU) allows and effectiveness forbids.

Where the uniqueness bites, and where it does not. None of the definitions of §§16.1 and 16.3 uses the uniqueness: an equivalence of structures only asks that some generalized isomorphism exist. The polyfold category is where it bites. Definition 16.1.6 lets the isomorphism (k, Id_W) of the middle structure be any generalized isomorphism over Id_W; when the middle structure has isotropy acting trivially, two equivalent representatives of $\hat{f}$ composed with one representative of $\hat{g}$ give inequivalent composites. Presumably the text intends effective polyfolds, or morphisms with a chosen isomorphism; the formalization builds the category of effective polyfolds, while the category PS of polyfold structures, the sc-smooth maps of all polyfolds, their tangents and the pull-back of forms need no restriction, because they are defined map by map and Definition 16.1.6 carries its isomorphisms as data.

Canonical on orbit spaces is enough for §16.2. The text glues $T(Z,c)$ along "the unique" $\mathfrak{f}$ with $\beta \circ |\mathfrak{f}| = \alpha$, and Lemma 16.2.1 is there to make the gluing maps canonical. What the gluing uses is only $|T\mathfrak{f}|$, and that is unique: (LR) gives, for functors with the same orbit map, local families of morphisms between them (Proposition 10.1.10 (1)), and these give the same tangent orbit maps (Proposition 10.1.10 (2)). So $T(Z,c)$, $T\alpha$, (16.5), Lemma 16.2.2 and Definition 16.2.3 go through as printed, with one structure of c as base point instead of the gluing over all. The same argument, Proposition 11.2.9, makes $h_*\omega$ independent of h, which is the remark before Definition 16.2.5. The raised polyfolds follow the same pattern, through the raise of generalized maps.

The tangent of an effective polyfold needs (ZHOU) too. The tangent functor lives on effective polyfolds, because the tangent of an effective polyfold is effective. The proof uses (ZHOU) as well as effectiveness: effectiveness is a statement at level 0, while the open sets of TX see only level 1, so an isotropy element acting trivially near a tangent vector is first moved to a smooth point, where (ZHOU) turns the triviality of the linearized action into triviality near the point.

Invariance statements are statements about equivalences. The invariance statements of §16.1 are statements about equivalences of ep-groupoids, transported to classes. Bump functions are local; partitions of unity are best read on $|X|$; and the text's "easy exercise" in Proposition 16.1.4 (2) is the descent of sc-smoothness along an equivalence. The metrizability of a polyfold follows from Theorem 7.3.1 and the metrization theorem for locally metrizable paracompact Hausdorff spaces, without the Nagata–Smirnov theorem.

§16.4 is Chapters 9 and 11 on classes, with two implicit points. An orientation $\hat{T}_\Theta$ is read only on X_∞, so a representative fixes it to be 0 elsewhere, without which $1^*\hat{T} = \hat{T}$ fails. And Theorem 16.4.6 reduces to Theorem 9.5.17, which is stated for ep-groupoids with sc-smooth partitions of unity; the polyfold version

needs the same hypothesis, which by Definition 16.1.5 holds for every representative as soon as it holds for one. The tameness of branched ep$^+$-subgroupoids moves along equivalences through the transport charts of Theorem 11.4.1, so the integrals live on classes. Proposition 16.4.3 is the same transport once more: the two legs of a generalized isomorphism restrict to equivalences of the sub-ep-groupoids on the supports, and the transition between the induced structures is, near each point, a composite of local sc-diffeomorphisms. The structures of Proposition 9.1.9, which are existential, need not be chosen, since any two give the same comparison.

Functions on Y and Z determine the class-level objects. §§16.3, 16.5 and 16.6 survive the non-uniqueness because their relations are determined by functions. Two representatives of a section are related as soon as their maps f agree: the condition $\mathfrak{A}_* F = F'$ follows from $\Gamma' \circ |\mathfrak{A}| = \Gamma$, since a functor is determined on orbits. Likewise the pull-back of a multisection functor along any generalized strong bundle isomorphism compatible with Γ is $\lambda \circ \Gamma$, whichever isomorphism is chosen; on the $\mathbb{Z}/3$ example both isomorphisms are checked to pull back every functor alike. A branched suborbifold is determined by its function θ, which makes $\lambda \circ f$ a well-defined branched suborbifold. Several conditions the text lists are consequences: the continuity and surjectivity of p, $p' \circ A = a \circ p$ for morphisms, the base condition of Definition 16.3.3.

Small readings. Read literally, the domain support of Definition 16.5.2 is all of Z, since the weights over each fibre sum to 1, so that some y over every z has $\lambda(y) > 0$; with $y \neq 0_z$, as in Definition 13.2.2, it is the intended set. The continuity of $n(\lambda)$ in (16.8) is local: in a local section structure $N(\Lambda)$ is the maximum of the continuous functions $N \circ s_i$, and it descends to Z through a quotient map.

Reflexivity is a transport too. Reflexivity of the 1-fibres pulls back along a local strong bundle isomorphism by transporting a chart, whose fibre model does not change; the push direction needs the same for the action μ along a morphism, which Chapter 10 provides. Mixed convergence moves along a local strong bundle isomorphism because the isomorphism is linear and bounded on the fibres, so bounded sequences stay bounded and weak limits are preserved; the independence of the model for the reflexive notions is again a transport.

§16.6 is a transport, and what remains is ep-level. Once the definitions are on classes, every notion of §16.6 equals the ep-groupoid notion in every representative, and Theorem 16.6.4 is Theorem 12.4.2 there. Theorem 16.5.7 follows from Theorem 14.1.9 in one model, provided the multisections over ∂X have non-empty index sets, which the text's notion has: a Lean form of Theorem 14.1.9 without that condition fails with boundary, and a polyfold theorem built on it would be vacuous exactly in the case it is about. For Theorem 16.6.6 the integrals live on the polyfold and depend only on the class of the form by Stokes alone; what the independence of λ needs is the canonical orientation and a cobordism between two perturbations, both ep-level statements of Chapter 15. Orientations of sc-Fredholm sections follow the same pattern: Theorem 12.5.7 carries one along any diagram, and what is left is that two diagrams carry it alike, for which Part XII's independence of the representative is the tool.

Concordance

Every declaration cited in this book, by file, with its line in the source. 13009 declarations in 989 files. The web edition also shows each exact formal statement.

Polyfold/Chapter1/Book2021.lean. `remark_1_1_2` (63); `remark_1_1_2_openMapping` (82); `proposition_1_1_11` (107).

Polyfold/Chapter1/ChainRule.lean. `IsSc1With.norm_sub_le_of_segment` (68); `IsSc1With.exists_delta` (95); `IsSc1With.hasFDerivWithinAt_comp` (125); `IsSc1With.tangentMapWith_comp` (245); `IsSc1With.comp` (256); `theorem_1_3_1'` (269); `theorem_1_3_1` (281); `IsSc1.tangentMap_congr` (289); `IsSck.congr` (298); `IsSck.comp` (306); `IsScSmooth.comp` (317); `IsSck.comp_of_isRelOpenInQuadrant` (323).

Polyfold/Chapter1/CylinderCompactInclusions.lean. `GapP1Sobolev.totallyBounded_of_isometry` (23); `GapP1Sobolev.dvCLM_mem_H1ball` (64); `GapP1Sobolev.cutWt` (81); `GapP1Sobolev.abs_cutWt_le` (83); `GapP1Sobolev.tail_le` (94); `GapP1Sobolev.isCompactOperator_inclH` (106).

Polyfold/Chapter1/CylinderCutoff.lean. `GapP1Sobolev.chi` (119); `GapP1Sobolev.tendsto_cut` (232).

Polyfold/Chapter1/CylinderFourier.lean. `GapP1Sobolev.bc` (67); `GapP1Sobolev.hasSum_bc` (115); `GapP1Sobolev.bc_ds` (365); `GapP1Sobolev.bc_dt` (433); `GapP1Sobolev.ae_eq_zero_of_forall_test` (499); `GapP1Sobolev.HasWeakDeriv.ae_eq` (573).

Polyfold/Chapter1/CylinderIntegrationByParts.lean. `GapP1Sobolev.SmPer.hasWeakDeriv` (146); `GapP1Sobolev.IsTest.mem_Hsub` (177).

Polyfold/Chapter1/CylinderLeibniz.lean. `GapP1Sobolev.leibniz` (134).

Polyfold/Chapter1/CylinderMultipliers.lean. `GapP1Sobolev.mulH` (128); `GapP1Sobolev.dv_mulH` (131).

Polyfold/Chapter1/CylinderRellich.lean. `GapP1Sobolev.totallyBounded_of_approx` (30); `GapP1Sobolev.totallyBounded_image_of_coords` (45); `GapP1Sobolev.exists_cutoff'` (90); `GapP1Sobolev.summable_freq_bc` (122); `GapP1Sobolev.cutR` (205); `GapP1Sobolev.H1ball` (208); `GapP1Sobolev.norm_cutR_sq_le` (213); `GapP1Sobolev.tsum_sub_le` (266); `GapP1Sobolev.exists_N_eta` (313); `GapP1Sobolev.rellich` (342).

Polyfold/Chapter1/CylinderScBanach.lean. `GapP1Sobolev.nonneg_of_strictMono` (26); `GapP1Sobolev.scaleH` (30); `GapP1Sobolev.toBase_toH` (37); `GapP1Sobolev.example_1_1_3` (49); `GapP1Sobolev.example_1_1_3_lvl` (58); `GapP1Sobolev.level0_isometry` (64); `GapP1Sobolev.level0_surjective` (82).

Polyfold/Chapter1/CylinderSobolevComplete.lean. `GapP1Sobolev.isClosed_range_ΦH` (42);

GapP1Sobolev.completeSpace_H (95); GapP1Sobolev.inclH (154); GapP1Sobolev.valCLM_inclH (159); GapP1Sobolev.inclH_injective (163).

Polyfold/Chapter1/CylinderSobolevSpaces.lean. GapP1Sobolev.wt (28); GapP1Sobolev.MemH (108); GapP1Sobolev.Hsub (123); GapP1Sobolev.H (233); GapP1Sobolev.norm_sq_H (257); GapP1Sobolev.isometry_ΦH (264).

Polyfold/Chapter1/CylinderTestDensity.lean. GapP1Sobolev.approx_supp (80); GapP1Sobolev.dense_test (281).

Polyfold/Chapter1/CylinderTrigPolynomials.lean. GapP1Sobolev.trig (94); GapP1Sobolev.pD_trig (129); GapP1Sobolev.bc_eexp (165); GapP1Sobolev.tendsto_box_sq (228).

Polyfold/Chapter1/CylinderWeakDerivatives.lean. GapP1Sobolev.cyl (39); GapP1Sobolev.IsTest (133); GapP1Sobolev.HasWeakDeriv (286).

Polyfold/Chapter1/Degeneracy.lean. quadrantDeg (60); degIndexWith (65); coordMap (77); coordMap_surjective (82); finrank_quotient_ker_coordMap (91); mem_ker_coordMap (99); eq_zero_of_eventually_nonneg_mul (110); eventually_add_smul_mem_quadrant (122); map_ker_coordMap_le (141); quadrantDeg_eq_of_equiv (158); quadrantDeg_zero (177); lemma_1_3_4 (184); lemma_1_3_4_dim (204); degIndex (226); degIndex_eq (229); degIndex_quadrant (237).

Polyfold/Chapter1/ExpositionLemmas.lean. ScScale.finiteDimensional_of_surjective_incl (75); ScBanach.smooth_eq_top_of_finiteDimensional (96); ScBanach.toBase_bijective_of_finiteDimensional (102); ScBanach.isoConstOfFiniteDimensional (111); ScBanach.isoConstOfFiniteDimensional_apply (117); ScBanach.IsScSplitting.isTopCompl (128); ScBanach.IsScSplitting.scProj (136); ScBanach.IsScSplitting.scProj_apply (152); ScBanach.IsScSplitting.scProj_idem (156); ScBanach.IsScSplitting.range_scProj (161); ScBanach.IsScSplitting.ker_scProj (169); ScOperator.idem_lvl (177); ScOperator.levelSub_range_eq (182); ScBanach.isScSplitting_of_idem (198); ScBanach.isScSplitting_iff_exists_scProj (236); ScBanach.inclBound (256); ScBanach.one_le_inclBound (258); ScBanach.norm_incl_le (264); ScBanach.norm_toBase_le (267); ScBanach.norm_inclK_le (277); ScBanach.exists_smooth_mem_coset (295); ScBanach.mk_mem_quotient_smooth_iff (365); continuousOn_apply_iff_uniformOnCompact (392); IsSc1With.deriv_uniformOnCompact (442); ScBanach.lvlSet_const (458); isSc1_iff_contDiffOn_const (469); isSc1_iff_fderivWithin_const (483); isScSmooth_iff_contDiffOn_const (494); isSc1_iff_contDiffOn_of_isRelOpenInQuadrant (508); isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant (514).

Polyfold/Chapter1/FredholmIndex.lean. linIndex (93); HasFiniteIndex (97); finite_of_finite_quotient (103); finite_of_finite_ker_range (111); finiteDimensional_ker_of_comp (137); finiteDimensional_coker_of_comp (143); HasFiniteIndex.comp_and_linIndex (153); HasFiniteIndex.comp (215); linIndex_comp (218); HasFiniteIndex.equiv (231); linIndex_equiv (238); HasFiniteIndex.of_finiteRange_sub_id_and_linIndex (250); linIndex_eq_zero_of_finiteRange_sub_id (306); HasFiniteIndex.of_bijective_add_and_linIndex (313); HasFiniteIndex.of_comp_of_comp (333); fredholmIndex (351); IsFredholm.hasFiniteIndex (355); finiteDimensional_range_sub_id_of_equiv (362); bijective_one_add_of_norm_lt (369); hasFiniteIndex_comp_add_of_norm_lt (384); fredholmIndex_add_of_norm_lt (401); fredholmIndex_add_isCompactOperator (439); ScBanach.finrank_levelSub_of_le_smooth (468); ScFredholmSplitting.C_le_smooth (480); ScFredholmSplitting.fredholmIndex_lvl (488); IsScFredholm.fredholmIndex_lvl (498); scIndex_eq_fredholmIndex (505); proposition_1_1_11_index (509); proposition_1_1_11_index_lvl (517).

Polyfold/Chapter1/FredholmPerturbation.lean. isStrictMap_of_isClosed_range (62); isFredholm_of_banach (73); isCompactOperator_of_finiteDimensional_range (84); finiteDimensional_ker_one_add (103); exists_bound_below_on_compl (130); isClosed_range_one_add (181); exists_unit_far (215); pick (243); pick_notMem (246); chain (255); chain_mono (260); chain_eq_sup (265); chain_ne_top (276); chain_isClosed (285); coFG_range_one_add (292); isFredholm_one_add (333); isCompactOperator_sub_one_of_equiv (344); isFredholm_add_isCompactOperator (354).

Polyfold/Chapter1/Levels.lean. exists_delta_of_continuousOn (117); continuousOn_clm_comp_of_isCompactOperator (144); contDiffOn_clm_comp_of_isCompactOperator (170); ScBanach.shift_lvlSet (310); ScBanach.shift_tangentSet (320); ScBanach.shift_inclK_mem_lvlSet (331); ScBanach.incl_shift_inclK (337); ScBanach.tangent_inclK (343); IsSc0.levelMap_shift (354); IsSc0.shift (364); IsScDomain.shift (377); levelDeriv (389); toBase_levelDeriv (394); levelDeriv_zero (400); IsSc1With.exists_levelDeriv (409); IsSc1With.toBase_levelDeriv (432); IsSc1With.levelMap_tangentMapWith (439); IsSc1With.continuousOn_levelDeriv (455); IsSc1With.continuousOn_levelDeriv_clm (468); IsSc1With.hasFDerivWithinAt_level (483); IsSc1With.contDiffOn_level (592); IsSc1With.levelDeriv_eq (604); IsSc1Levels (644); IsSc1Levels.continuousOn_clm (656); IsSc1Levels.contDiffOn (662); IsSc1Levels.toBase_apply (674); IsSc1Levels.isSc1With (704); IsSc1With.isSc1Levels (734); proposition_1_2_1 (744); IsSc1With.shift (752); IsSck.shift (769); IsScSmooth.shift (789); IsSck.contDiffOn_levels' (802); IsSck.contDiffOn_levels (856); IsSck.contDiffOn_levels_of_le (862); IsSck.contDiffOn_base (868); IsScSmooth.contDiffOn_levels (874); proposition_1_2_4 (884); corollary_1_2_5 (919); corollary_1_2_5_base (932); IsScSmooth.of_contDiffOn_levels (943); IsScSmooth.of_contDiffOn_finiteDimensional (954).

Polyfold/Chapter1/LevelsWeak.lean. ScBanach.castCLE (96); ScBanach.inclLE (111); ScBanach.toBase_inclLE (114); ScBanach.preimage_inclLE_lvlSet (127); ScBanach.shiftAdd (174); ScBanach.shiftAddZeroIso (185); ScBanach.shiftAddShiftIso (197); ScBanach.toBase_shiftAddShiftIso_symm (209); ScOperator.zeroOp (230); ScOperator.addOp (235); IsSck.zero (244); IsSck.add (248); IsSck.sum (254); cmm_eval_eq (272); hasFDerivAt_cmm_eval (279); IsDerivTower (322); levelDerivTower (336); baseDerivTower (342); contDiffOn_levels_min (348); levelDerivTower_toBase (368); baseDerivTower_toBase (405); isDerivTower_of_contDiffOn_levels (419); IsDerivTower.eq_of_toBase (450); ElemData (467); ElemData.Good (483); ElemData.dom (489); ElemData.map (492); ElemData.mem_lvlSet_dom (496); ElemData.ρ₀ (502); ElemData.ρ (505); ElemData.ρ₀_incl (515); ElemData.map_toBase (532); ElemData.levelMap_eq (552); ElemData.levelMap_incl_eq (565); ElemData.isSc0 (584); ElemData.deriv (629); ElemData.deriv_apply (641); ElemData.deriv_toBase (651); ElemData.hasFDerivWithinAt_level (679); ElemData.continuousOn_deriv (719); ElemData.isSc1Levels (740); ElemData.isSc1With (746); IsScDomain.of_eq (753); IsSck.of_eq_set (763); ScOperator.tangentX (774); ScOperator.tangentH (779); ElemData.shiftData (809); ElemData.derivData (813); ElemData.slotData (818); ElemData.Good.shiftData (823); ElemData.Good.derivData (828); ElemData.Good.slotData (836); ElemData.tangentSet_dom (845); ElemData.tangentMap_eqOn (854); ElemData.isSck (900); ElemData.base (935); ElemData.base_dom (938); ElemData.base_good (942); proposition_1_2_4_text (949); IsScSmooth.of_contDiffOn_levels_text (973); corollary_1_2_5_text (984); IsScSmooth.of_contDiffOn_levels_finiteDimensional_text (1005).

Polyfold/Chapter1/PartialCones.lean. IsPartialCone (40); exists_toBase_mem_of_isOpen (47); interior_lvlSet_nonempty_of_interior (54); IsPartialCone.isScDomain_of_convex (68); IsPartialCone.isScDomain (81); IsPartialCone.tangentSet (87); IsPartialQuadrant.isPartialCone (106); remark_1_1_17_chainRule (128).

Polyfold/Chapter1/Properties.lean. IsSc1With.exists_delta_deriv (69); IsSc1With.continuousOn_deriv_clm (93); IsSc1With.contDiffOn_base (120); IsSc1With.scOperator_deriv (130); proposition_1_2_4_zero (165); corollary_1_2_5_zero (233).

Polyfold/Chapter1/ScBanach.lean. ScScale (84); ScScale.toBase (104); ScScale.toBase_injective (112); ScScale.inclK (117); ScScale.toBase_inclK (126); ScScale.smooth (135); ScScale.mem_smooth_iff (137); ScScale.smoothLvl (141); ScScale.exists_inclK_of_mem_smoothLvl (148); ScScale.smoothLvl_submodule_closed_sub (162); ScBanach (171); ScBanach.denseRange_inclK (181); ScBanach.denseRange_incl (184); ScBanach.constScale (191); ScBanach.constScale_toBase (200); ScBanach.constScale_smoothLvl (206); ScBanach.const (213); ScBanach.const_toBase (221); ScBanach.prodScale (227); ScBanach.prodScale_toBase (238); ScBanach.prodScale_smoothLvl (244); ScBanach.prod (267); ScBanach.prod_smoothLvl (279); ScBanach.shiftScale (285); ScBanach.shiftScale_toBase (291); ScBanach.shiftScale_smoothLvl (297); ScBanach.shift (315); ScBanach.shift_toBase (323); ScBanach.shift_smoothLvl (326); ScBanach.shiftK (330); ScOperator (341); ScOperator.toBase_comm (352); ScOperator.ext_base (359); ScOperator.id (370); ScOperator.comp (375); ScOperator.add (383); ScOperator.mapsTo_smoothLvl (393); IsScOperator (405); ScOperator.isScOperator (408); IsScOperator.toScOperator (412); IsScOperator.toScOperator_zero (419); ScIso (429); ScIso.toScOperator (441); ScIso.symm (449); ScIso.trans (459); ScIso.toBase_comm (464); ScIso.symm_toBase_comm (467); ScIso.refl (471); ScPlus (479); ScBanach.shiftIncl (484); ScPlus.toScOperator (495); ScPlus.toScOperator_apply (497); ScPlus.isCompactOperator (501); ScBanach.levelSub (513); ScBanach.mem_levelSub (516); ScBanach.levelSub_zero (519); ScBanach.incl_mem_levelSub (522); ScBanach.isClosed_levelSub (525); ScBanach.IsScSubspace (531); ScBanach.IsScSubspace.isClosed (535); ScBanach.ofSubspaceScale (539); ScBanach.ofSubspaceScale_coe_toBase (554); ScBanach.mem_ofSubspaceScale_smoothLvl (561); ScBanach.ofSubspaceScale_smoothLvl (578); ScBanach.ofSubspace (584); ScBanach.IsScSplitting (598); ScBanach.IsScSplitting.closedComplemented (601); ScBanach.IsScSplitting.symm (606); ScBanach.proposition_1_1_7_iff (613); ScBanach.levelRestrict (633); ScBanach.levelRestrict_bijective (636); ScBanach.levelEquiv (643); ScBanach.finProj (652); ScBanach.finProj_apply_of_mem (659); ScBanach.finProj_eq_zero_iff (672); ScBanach.proposition_1_1_7 (680); ScBanach.quotIncl (723); ScBanach.quotIncl_mk (728); ScBanach.quotIncl_injective (731); ScBanach.quotIncl_isCompactOperator (740); ScBanach.quotientScale (752); ScBanach.quotientScale_toBase_mk (759); ScBanach.quotient (768); ScBanach.quotientMap (781); ScBanach.proposition_1_1_8 (787).

Polyfold/Chapter1/ScFredholm.lean. ScOperator.ker_lvl (70); ScOperator.range_lvl_le (80); ScPlus.toScOperator_mem_range_toBase_succ (89); ScBanach.range_toBase_succ_le (96); ScFredholmSplitting (106); IsScFredholm (124); ScFredholmSplitting.index (127); scIndex (131); ScFredholmSplitting.index_eq (140); ScFredholmSplitting.K_le_smooth (148); ScFredholmSplitting.ker_lvl (152); ScFredholmSplitting.range_lvl (155); ScFredholmSplitting.isFredholm (161); proposition_1_1_10 (184); ScPlus.regularizing_add (208); proposition_1_1_11_preprint (226).

Polyfold/Chapter1/ScFredholmCriterion.lean. ScOperator.IsRegularizing (69); ScOperator.IsRegularizing.ker_le_smooth (73); ScOperator.IsRegularizing.surjective (82); IsScFredholm.isRegularizing (92); isScFredholm_of_isRegularizing (105); isScFredholm_of_isRegularizing_index (210); isScFredholm_iff (217); isScOperator_of_mapsTo (228); ScBanach.ofSubspace_mem_range_toBase_iff (253); ScOperator.restrict (272); ScOperator.IsRegularizing.restrict (284); ScOperator.IsRegularizing.isScFredholm_restrict (296); IsScFredholm.restrict (307).

Polyfold/Chapter1/ScOneNotNormContinuous.lean. ScOneExample.c0 (64); ScOneExample.diag

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Polyfold/Chapter15/ParametricGeneralPosition.lean. SBLocalization.isScPlusSectionOn_of_paramSec (71); SBLocalization.isGeneralPositionGerm_branchParam (101); SBLocalization.isReducedOntoAtLevel_of_mor (195); SBLocalization.isInGeneralPositionAt_of_mor (232); SBLocalization.exists_local_paramGP (257); SBLocalization.auxLe_of_pos (360); SBLocalization.exists_pos_auxLe_subset (381); SBLocalization.hasParametricGeneralPosition (468); SBLocalization.theorem_15_3_7_hyp_local (543); SBLocalization.theorem_15_3_7_of_local (554); SBLocalization.theorem_15_3_7_statement_epSEP_zero_of_local (569).

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Polyfold/Chapter15/ProductAuxNorm.lean. StrongBundle.prodOneFiberFst (43); StrongBundle.prodOneFiberMk (51); StrongBundle.prodOneFiber_ext (67); StrongBundle.prodAuxNorm (112); StrongBundle.prodAuxNorm_toOne (117); StrongBundle.isAuxNorm_prodAuxNorm (124); EpStrongBundle.isAuxNorm_prodAuxNorm (200).

Polyfold/Chapter15/ProductBundle.lean. prodFiberEquiv (51); prodFiberCLE (80); IsBanachable.prod (97); IsBanachable.of_equiv (105); StrongBundleChart.prodPartialEquiv (128); StrongBundleChart.prod (147); StrongBundleChart.prod_K (196); StrongBundleChart.overlap_prod (200); StrongBundleChart.transition_prod (212); StrongBundleChart.Compatible.prod (219); IsSection.prodMap (271); StrongBundle.paracompactSpace_prod (292); StrongBundle.prodAtlas (297); StrongBundle.prod (313); StrongBundle.prod_mem_charts (323); StrongBundle.exists_chart_rep (333); StrongBundle.IsScSmoothSection.prod (350); StrongBundle.IsScPlusSection.prod (378).

Polyfold/Chapter15/ProductBundleLevels.lean. StrongBundle.isScSmoothMap_id_prod_zero (45); StrongBundle.isScSmoothMap_id_prod_zero_symm (75); StrongBundleChart.liftOne_prod (107); StrongBundle.mem_oneSet_prod (116); StrongBundle.prodOneHomeomorph (224); StrongBundle.paracompactSpace_prod_one (240); StrongBundle.isScSmoothMap_prodOneHomeomorph (256); StrongBundle.isScSmoothMap_prodOneHomeomorph_symm (285).

Polyfold/Chapter15/ProductBundleMap.lean. StrongBundle.isScSmoothMap_prodMap_zero (48); StrongBundle.isScSmoothMap_prodMap_one (61); StrongBundle.oneRestrict_prodMap (77); StrongBundle.IsStrongBundleMap.prod (87).

Polyfold/Chapter15/ProductEp.lean. GroupoidStr.prod (53); MPolyfold.IsScSmoothOn.prodMap (98); MPolyfold.IsScDiffeoOn.prodMap (109); MPolyfold.IsLocalScDiffeo.prodMap (121); isProperMap_domRestrict_prod (134); EpGroupoid.prod_exists_isProperMap (156); EpGroupoid.prod_isScSmoothCompSheet (169); EpGroupoid.prod (224); EpGroupoid.prodFst (256); EpGroupoid.prodSnd (267); EpGroupoid.isTame_prod (282); EpGroupoid.degIndex_prod (285); EpGroupoid.prod_π_eq_π_iff (290).

Polyfold/Chapter15/ProductEpBundle.lean. prodPullbackSplit (47); prodPullbackJoin (54); isStrongBundleMap_prodPullbackSplit (103); isLocalScDiffeo_prodPullbackSplit (164); EpStrongBundle.prodμ (188); EpStrongBundle.prod (194); EpStrongBundle.prod_μ (234); EpStrongBundle.IsFunctorialSection.prod (242); EpStrongBundle.IsScSmoothSectionFunctor.prod (249); EpStrongBundle.IsScPlusSectionFunctor.prod (257).

Polyfold/Chapter15/ProductStrongBundle.lean. bundleSwap (41); ScBanach.bundleSwapOp (54); ScBanach.mem_range_prod_toBase (63); ScBanach.invFun_prod_toBase_one (74); prodBundleMap (90); prodBundleMap_eq (104); isScSmooth_prodBundleMap (115); shiftFiber_prodBundleMap (135); IsStrongBundleMap.prod (148); IsStrongBundleRetraction.prod (200); baseRetraction_prod (213); retract_baseRetraction_prod (218); image_prodBundleMap (226); IsStrongBundleRetract.prod

(242); IsScSmoothRetractMap.prodBundleMap (261); strongShiftOne_bundleSwap_preimage (286).

Polyfold/Chapter15/RegularPerturbations.lean. SBLocalization.IsRegularPerturbation (63); SBLocalization.corollary_15_3_10_1_hyp (75); SBLocalization.isScFredholmSectionTame_cubeLiftEp (92); SBLocalization.mem_boundary_cubeSEP_iff (101); SBLocalization.corollary_15_3_10_3_hyp (121).

Polyfold/Chapter15/RelativeBranch.lean. SBLocalization.isLocalRepOn_multisectionAdd (51); SBLocalization.paramSlice_branch_eq_rel (80); SBLocalization.isParamGP_restrictTo_of_onto_rel (153); SBLocalization.isGeneralPositionGerm_branchParam_rel (273).

Polyfold/Chapter15/RelativeFamilies.lean. SBLocalization.IsParamOntoAtRel (62); SBLocalization.isParamOntoAtRel_zero_iff (76); SBLocalization.IsParamOntoAtRel.of_reducedOnto (84); SBLocalization.IsParamOntoAtRel.famAdd_left (96); SBLocalization.IsParamOntoAtRel.famAdd_right (115); SBLocalization.RelGPFamily (141); SBLocalization.RelGPFamily.mono (157); SBLocalization.isLinearFamily_zero (168); SBLocalization.RelGPFamily.zeroOn (176); SBLocalization.RelGPFamily.zero (189); SBLocalization.RelGPFamily.add (195); SBLocalization.eq_zeroMultisection_of_notMem_domSupp (219); SBLocalization.multisectionAdd_eq_left_of_notMem_domSupp (238).

Polyfold/Chapter15/RelativeParametric.lean. SBLocalization.sectionAdd_paramSec_zero (63); SBLocalization.subSec_eq_of_add (73); SBLocalization.exists_of_multisectionAdd_pos_at (92); SBLocalization.exists_local_paramGP_rel (108); SBLocalization.exists_pos_supp_rel_subset (240); SBLocalization.hasParametricGeneralPositionRel (405); SBLocalization.theorem_15_3_7_relative_of_local (481); SBLocalization.theorem_15_3_9_of_local_rel (506); SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero_of_local (526).

Polyfold/Chapter15/RelativeReduction.lean. SBLocalization.HasLocalGPFamiliesRel (78); SBLocalization.HasParametricGeneralPositionRel (88); SBLocalization.nonempty_relGPFamily_biUnion (103); SBLocalization.exists_relGPFamily_of_compact (117); SBLocalization.isCompact_image_supp_comp (138); SBLocalization.theorem_15_3_7_relative_hyp (157); SBLocalization.theorem_15_3_7_relative_of_hyp (199); SBLocalization.theorem_15_3_9_of_rel_hyp (224); SBLocalization.hasLocalGPFamiliesRel_of_boundary (249); SBLocalization.hasLocalGPFamiliesRel_epSEP_zero (267); SBLocalization.hasParametricGeneralPositionRel_epSEP_zero (283); SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero_of_hyp (294).

Polyfold/Chapter15/RestrictedEp.lean. locallyFinite_image_of_isClosedMap (70); EpGroupoid.NaturalRep.translation_isProperSubset (113); EpGroupoid.NaturalRep.translation_hasPropernessProperty (120); EpGroupoid.NaturalRep.isClosedMap_translation_π (125); EpGroupoid.NaturalRep.finite_fiber_translation_π (129); EpGroupoid.NaturalRep.paracompactSpace_orbitSupp_translation (148); EpGroupoid.NaturalRep.nonempty_tameBranching_translation (186); EpGroupoid.NaturalRep.isTameBranched_translation (228); SBLocalization.remark_15_2_3_restricted_hyp (271); SBLocalization.theorem_15_2_5_restricted_hyp (295).

Polyfold/Chapter15/SeparationLemmas.lean. Linearization15.wsum (85); Linearization15.mass (88); Linearization15.mass_eq_sum_wsum (90); Linearization15.mass_eq_of_wsum_eq (99); Linearization15.wsum_map (108); Linearization15.wsum_map_eq_of_wsum_eq (112); Linearization15.wsum_pair_of_const (119); Linearization15.wsum_eq_of_frequently (134); Linearization15.interior_biUnion_eq_empty (186); Linearization15.exists_nonempty_interior_of_finite_

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Polyfold/Chapter15/TameBranched.lean. SBLocalization.exists_tame_solutionSet_of_isBranchGoodPosition (73); SBLocalization.isGoodPositionAt_of_interior (126); SBLocalization.theorem_15_2_5_tame_hyp (171); SBLocalization.remark_15_2_3_tame_hyp (263); SBLocalization.theorem_15_2_5_isTameBranched_hyp (275); SBLocalization.remark_15_2_3_isTameBranched_hyp (297); SBLocalization.remark_15_3_6_isTameBranched_hyp (311); SBLocalization.theorem_15_3_7_in_particular_tame_hyp (324).

Polyfold/Chapter15/TameWitness.lean. SBLocalization.remark_15_2_3_isTameBranched_hyp_witness (34); SBLocalization.remark_15_2_3_restricted_hyp_witness (50).

Polyfold/Chapter15/TransversalExtension.lean. SBLocalization.IsReducedOntoAtPt (84); SBLocalization.isInGeneralPosition_iff_forall_reducedOntoAtPt (92); SBLocalization.IsBoundaryFaceOnto (101); SBLocalization.theorem_15_3_9_statement (120); SBLocalization.lemma_15_3_9_cutoff_statement (143); SBLocalization.theorem_15_3_7_relative_statement (162); SBLocalization.IsInGeneralPosition.isTransversal (180); SBLocalization.theorem_15_3_9_hyp (195); SBLocalization.boundary_epSEP_eq_empty (221); SBLocalization.boundaryDomSupp_epSEP (231); SBLocalization.lemma_15_3_9_cutoff_statement_epSEP_zero (239); SBLocalization.theorem_15_3_7_relative_statement_epSEP_zero (248); SBLocalization.theorem_15_3_9_statement_epSEP_zero (256).

Polyfold/Chapter15/Transversality.lean. SBLocalization.mem_support_linSumIntr_iff (68); SBLocalization.IsOntoAtLevel (103); SBLocalization.IsOntoAt (108); SBLocalization.IsTransversalAt (114); SBLocalization.IsTransversal (119); SBLocalization.isOntoAtLevel_iff (125); SBLocalization.linearization_ne_zero (141); SBLocalization.IsScPlusSectionOn.continuousOn (165); SBLocalization.isOpen_sep_ne (171); SBLocalization.mem_smoothSet_of_eq (178); SBLocalization.mem_smoothSet_of_pos (196); SBLocalization.exists_branch_interior (216); SBLocalization.theorem_15_2_2_hyp (249); SBLocalization.remark_15_2_3_hyp (360); SBLocalization.remark_15_2_3_isBranchedEpSubgroupoid_hyp (399).

Polyfold/Chapter16/BranchedSuborbifold.lean. EPLocalization.EPLoc.pullTheta_id_of_isFunctorial (85); EPLocalization.EPLoc.pullSection_id_of_mem (93); EPLocalization.EPLoc.pullTheta_eq_of_orbitMap (103); EPLocalization.EPLoc.pull_congr (123); EPLocalization.EPLoc.pull_id_of_mem (130); EPLocalization.EPLoc.tameBranchedIntegral_pull (139); EPLocalization.EPLoc.tameBranchedIntegralEff_pull (156); PolyfoldClass.ScForm.toStructure_eq_formsPullback (189); PolyfoldClass.BranchedRep (203); PolyfoldClass.BranchedRep.Rel (217); PolyfoldClass.BranchedRep.Rel.refl (221); PolyfoldClass.BranchedRep.Rel.symm (225); PolyfoldClass.BranchedRep.Rel.trans (234); PolyfoldClass.BranchedRep.toFun (247); PolyfoldClass.BranchedRep.toFun_eq (249); PolyfoldClass.BranchedRep.toFun_eq_of_rel (258); PolyfoldClass.BranchedSuborbifold (274); PolyfoldClass.BranchedSuborbifold.toFun (279);

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Polyfold/Chapter16/BranchedTame.lean. EpGroupoid.TameBranchingStructure.restrictRep (60); EpGroupoid.TameBranchingStructure.mapU_eq (74); EpGroupoid.TameBranchingStructure.move (86); EpFunctor.isTameBranched_pullbackTheta (115); EpFunctor.isTameBranched_pushforwardTheta (137); EpFunctor.isTameBranched_pullbackTheta_iff (156); EPLocalization.EPLoc.isTameBranched_pullTheta (171); PolyfoldClass.BranchedSuborbifold.definition_16_4_2_tame (191); PolyfoldClass.OrientedRep.Rel.isTameOriented (207); PolyfoldClass.OrientedRep.Rel.isTameOriented_iff (220); PolyfoldClass.OrientedSuborbifold.IsTameOriented (232); PolyfoldClass.OrientedSuborbifold.isTameOriented_iff (237); PolyfoldClass.OrientedSuborbifold.toBranched_mk (244); PolyfoldClass.OrientedSuborbifold.isCompactBranched_of_mk (247); PolyfoldClass.OrientedSuborbifold.integral (254); PolyfoldClass.OrientedSuborbifold.integral_eq (261); PolyfoldClass.OrientedSuborbifold.integralEff (268); PolyfoldClass.

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Polyfold/Chapter16/CompositionCounterexample.lean. CompositionCounterexample.K2 (54); CompositionCounterexample.k2EpGroupoid (67); CompositionCounterexample.isZhou_k2 (73); CompositionCounterexample.injective_π_k2 (77); CompositionCounterexample.isHomeomorph_π_k2 (81); CompositionCounterexample.k2OrbitHomeomorph (86); CompositionCounterexample.k2Structure (90); CompositionCounterexample.k2Structure_β_π (93); CompositionCounterexample.z3_β_π (96); CompositionCounterexample.ι (104); CompositionCounterexample.pr₁ (117); CompositionCounterexample.σ (128); CompositionCounterexample.T (140); CompositionCounterexample.ι_pr₁ (156); CompositionCounterexample.ι_σ_pr₁ (157); CompositionCounterexample.σ_σ (158); CompositionCounterexample.not_isEquivalence_T (161); CompositionCounterexample.not_isIso_T (167); CompositionCounterexample.orbitMap_map_π (175); CompositionCounterexample.a (184); CompositionCounterexample.b (194); CompositionCounterexample.kσ (204); CompositionCounterexample.isGeneralizedIso_σ (213); CompositionCounterexample.kσ_isIdIso (218); CompositionCounterexample.a_equiv (221); CompositionCounterexample.gen_comp_one (227); CompositionCounterexample.gen_comp_σ (234); CompositionCounterexample.not_equiv_comp (241); CompositionCounterexample.not_definition_16_1_10 (257); CompositionCounterexample.not_exists_comp (263); CompositionCounterexample.not_isEffective_k2 (287).

Polyfold/Chapter16/FibreZero.lean. SBLocalization.π_fiberZero_eq_of_π_eq (43); SBLocalization.μ_ne_fiberZero (50); SBLocalization.π_eq_fiberZero_iff (65); SBLocalization.SDiagram.orbitMap_π_fiberZero (89); SBLocalization.GenSBMap.orbitMap_π_fiberZero (115); StrongPolyfoldBundleStructure.zeroAt (134); StrongPolyfoldBundleStructure.zeroAt_eq (138); StrongPolyfoldBundleStructure.p_zeroAt (147); StrongPolyfoldBundleStructure.zeroAt_eq_of_equiv (157); StrongPolyfoldBundleStructure.Γ_π_eq_zeroAt_iff (171); StrongPolyfoldBundle.zeroAt (193); StrongPolyfoldBundle.p_zeroAt (199); StrongPolyfoldBundle.normY (203); StrongPolyfoldBundle.normY_Γ (208).

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Polyfold/Chapter16/FormsPullback.lean. PolyfoldClass.MapRep.pullback (55); PolyfoldClass.MapRep.pullback_eq_of_equiv (65); PolyfoldClass.ScMap.pullback (97); PolyfoldClass.ScMap.pullback_mk (103); PolyfoldClass.ScMap.d_comp_pullback (111); PolyfoldClass.ScMap.pullback_id (123); PolyfoldClass.ScMap.pullback_mk_comp (133); PolyfoldClass.ScMap.pullback_comp (143); PolyfoldClass.theorem_16_2_6_pullback (159).

Polyfold/Chapter16/MixedTransfer.lean. StrongBundle.MixedTendsto.congr_eventually (67); StrongBundle.LocalIso.isFiberSeminorm_comp (75); StrongBundle.LocalIso.mixedTendsto (99); SBLocalization.mk_actOne (131); SBLocalization.SDiagram.exists_mixedTendsto_transfer (140); SBLocalization.SDiagram.isReflexiveAuxNorm_pullNorm (206); SBLocalization.SDiagram.isReflexiveAuxNorm_pushNorm (221); SBLocalization.GenSBMap.isReflexiveAuxNorm_pullNorm (233); SBLocalization.GenSBMap.isReflexiveAuxNorm_pushNorm (240); StrongPolyfoldBundleStructure.isReflexiveAuxNorm_of_equiv (256); StrongPolyfoldBundleStructure.oneToY_of_oneOrbitMap (264); StrongPolyfoldBundleStructure.mixedTendstoIn_of_equiv (277); StrongPolyfoldBundle.IsReflexiveAuxNorm.of_mem (310); StrongPolyfoldBundle.isReflexiveAuxNorm_iff_forall (318); StrongPolyfoldBundle.mixedTendsto_iff_forall (327); StrongPolyfoldBundle.proposition_16_3_16_two (340).

Polyfold/Chapter16/NormLamContinuous.lean. SBLocalization.IsLocalRepOn.pointwiseNorm_eq (44); SBLocalization.IsLocalRepOn.univ_nonempty (58); SBLocalization.IsScPlusMultisection.continuous_pointwiseNorm (68); StrongPolyfoldBundle.isQuotientMap_γπ (93); StrongPolyfoldBundle.ScPlusMultisection.continuous_normLam (101); StrongPolyfoldBundle.ScPlusMultisection.isGreatest_normLam (113); StrongPolyfoldBundle.ScPlusMultisection.normLam_nonneg (122).

Polyfold/Chapter16/OrientationTransport.lean. SBLocalization.SDiagram.orientation_transport_hyp (64); SBLocalization.SDiagram.exists_orientation_of_eq (123); StrongPolyfoldBundle.OrientationRep.exists_diagram (155); StrongPolyfoldBundle.OrientationRep.transport_hyp (166).

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Polyfold/Chapter4/ScFormsPullback2021.lean. MPolyfold.tangentFiberMap (75); MPolyfold.formPullback (89); MPolyfold.chartForm_formPullback (162); MPolyfold.formPullback_mem_scForms (217); MPolyfold.formPullback_id (225); MPolyfold.formPullback_comp (234); MPolyfold.formLower_formPullback (246); MPolyfold.extD_formPullback (306); BundledMPolyfold.formsPullback (418); BundledMPolyfold.formsInftyPullback (444); BundledMPolyfold.formsInftyD_comp_formsInftyPullback (461); BundledMPolyfold.formsInftyPullback_comp (499); BundledMPolyfold.scDeRhamPullback (531).

Polyfold/Chapter4/ScFormsSquare2021.lean. sum_removeNth_symm_eq_zero (36); scDeriv_ambientD_whitneyMk (143); ambientD_ambientD (168); retractD_shift_retractD (198); MPolyfold.extD_extD (264).

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